

3D machine learning-based complexity variability and fluidity quantification of preterm and writhing general movements

Item Type Journal Article

Author A. Soualmi

Author O. Alata

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Abstract General Movement Assessment (GMA) is a reliable non-invasive method for the early detection of neurodevelopmental disorders in infants, based on the qualitative evaluation of three key parameters: complexity, variability, and fluidity. Current automated approaches primarily use composite indices on two-dimensional data, limiting parameter-specific evaluations and failing to fully exploit three-dimensional kinematic information. This study addresses three critical gaps in automated GMA research: (1) the lack of publicly available datasets for preterm and writhing movements, (2) the need for parameter-specific quantification methods aligned with clinical assessment criteria, and (3) the limited exploration of deep learning approaches for 3D skeletal data analysis. We introduce the AGMA dataset containing 264 three-dimensional movement trajectories from 126 preterm infants (born < 33 weeks gestational age), with expert annotations for complexity, variability, and fluidity. We propose novel handcrafted features for individually quantifying each GMA parameter, achieving area under the curve (AUC) values of 0.75, 0.77, and 0.86 for complexity, variability, and fluidity classification, respectively, using Random Forest classifiers. Notably, spectral entropy emerged as a highly discriminative feature for fluidity assessment (AUC = 0.86, recall = 0.83), representing the first application of this measure in the GMA context. We also evaluate graph convolutional networks (STGCN and DeGCN) for movement classification, though their performance was limited by the dataset size. The publicly released AGMA dataset, containing 264 trajectories from 126 infants, along with our parameter-specific classification methods, establishes a reproducible baseline for future automated GMA research and clinical validation studies. © 2026 The Authors.

Date 2026

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Archive Scopus

Language English

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Publication Computers in Biology and Medicine

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Journal Abbr Comput. Biol. Med.

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Tags:

Article, Automation, female, Female, human, Humans, male, Male, Deep learning, Classification (of information), Learning systems, machine learning, Trajectories, Learning algorithms, spatiotemporal analysis, Complex networks, entropy, Entropy, Machine Learning, procedures, data analysis, deep learning, artificial neural network, random forest, Diagnosis, area under the curve, data accuracy, clinical assessment, cross validation, Clinical research, infant, newborn, physiology, Imaging, Three-Dimensional, three-dimensional imaging, graph neural network, 3d infant pose, 3D infant pose, AGMA infant dataset, Early neurodevelopmental screening, Fluidity, General movement assessment, General movements assessment, Infant, Newborn, Infant, Premature, Motion analysis, Movement, movement (physiology), Movement trajectories, Movement trajectory analysis, Movement trajectory analyze, Noninvasive medical procedures, Parameter-specific classification, prematurity, Preterm infant movement, Preterm infant movements, Preterm infants, Spectral entropy, Trajectory analysis

Notes:

EID: 2-s2.0-105042659092

Export Date: 20 September 2026; Cited By: 0; Funding details: École Doctorale 488 SIS; Agence Nationale de la Recherche, ANR, (ANR-23-SSAI-0005-02); Correspondence Address: A. Giraud; INSERM, U1059 SAINBIOSE, Université Jean Monnet, Saint-Étienne, France; email: antoine.giraud@univ-st-etienne.fr; CODEN: CBMDA

3rd International Conference on Machine Learning for Networking, MLN 2020

Item Type Journal Article

Editor Renault E.

Editor Boumerdassi S.

Editor Mühlethaler P.

Abstract The proceedings contain 24 papers. The special focus in this conference is on Machine Learning for Networking. The topics include: Three Quantum Machine Learning Approaches for Mobile User Indoor-Outdoor Detection; learning Resource Allocation Algorithms for Cellular Networks; Enhanced Pub/Sub Communications for Massive IoT Traffic with SARSA Reinforcement Learning; deep Learning-Aided Spatial Multiplexing with Index Modulation; A Self-gated Activation Function SINSIG Based on the Sine Trigonometric for Neural Network Models; spectral Analysis for Automatic Speech Recognition and Enhancement; road Sign Identification with Convolutional Neural Network Using TensorFlow; a Semi-automated Approach for Identification of Trends in Android Ransomware Literature; Towards Machine Learning in Distributed Array DBMS: Networking Considerations; using Machine Learning to Quantify the Robustness of Network Controllability; deep Learning Environment Perception and Self-tracking for Autonomous and Connected Vehicles; remote Sensing Scene Classification Based on Effective Feature Learning by Deep Residual Networks; identifying Device Types for Anomaly Detection in IoT; a Novel Heuristic Optimization Algorithm for Solving the Delay-Constrained Least-Cost Problem; terms Extraction from Clustered Web Search Results; Configuration Faults Detection in IP Virtual Private Networks Based on Machine Learning;

improving Android Malware Detection Through Dimensionality Reduction Techniques; A Regret Minimization Approach to Frameless Irregular Repetition Slotted Aloha: IRSA-RM; mobility Based Genetic Algorithm for Heterogeneous Wireless Networks; geographical Information Based Clustering Algorithm for Internet of Vehicles; active Probing for Improved Machine-Learned Recognition of Network Traffic; a Dynamic Time Warping and Deep Neural Network Ensemble for Online Signature Verification.

Date 2021

URL <https://www.scopus.com/pages/publications/85103569737?origin=resultslist>

Archive Scopus

Language English

Volume 12629 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Journal Abbr Lect. Notes Comput. Sci.

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Notes:

EID: 2-s2.0-85103569737

Export Date: 20 September 2026; Cited By: 0; Conference name: 3rd International Conference on Machine Learning for Networking, MLN 2020; Conference date: 24 November 2020 through 26 November 2020; Conference code: 256129

4th International Conference on Science of Cyber Security, SciSec 2022

Item Type Journal Article

Editor Su C.

Editor Sakurai K.

Editor Liu F.

Abstract The proceedings contain 36 papers. The special focus in this conference is on Science of Cyber Security. The topics include: Hybrid Routing for Efficient Fine-Grained Management of Specific Services in SDN; atNet: A Novel Anti-tracking Network with Multi-Party Judgement Capability Based on Cross-Domain Small-World Topology; A Two-Stage Method for Fine-Grained DNS Covert Tunnel Behavior Detection; analysis and Detection Against Overlapping Phenomenon of Behavioral Attribute in Network Attacks; integration of Cybersecurity Related Development Processes by Using a Quantification Method; zoomPass: A Zoom-Based Android Unlock Scheme on Smart Devices; metasploit for Cyber-Physical Security Testing with Real-Time Constraints; Passive User Authentication Utilizing Consecutive Touch Action Features for IIoT Systems; malware Classification Based on Semi-Supervised Learning; BlockRAT: An Enhanced Remote Access Trojan Framework via Blockchain; malware Detected and Tell Me Why: An Verifiable Malware Detection Model with Graph Metric Learning; malware Detection Using Automated

Generation of Yara Rules on Dynamic Features; question Answering Models for Privacy Policies of Mobile Apps: Are We There Yet?; Design of End-To-End Security for MQTT 5.0; DroidFP: A Zero-Permission Detection Framework for Android Devices Based on Gated Recurrent Unit; an Intrusion Detection System Based on Deep Belief Networks; HINCDG: Multi-Meta-Path Graph Auto-Encoders for Mining of Weak Association Malicious Domains; AttVAE: A Novel Anomaly Detection Framework for Multivariate Time Series; BASNEA: Threat Hunting for Ethereum Smart Contract Based on Backtrackless Aligned-Spatial Network Entity Alignment; Multi-region SRAM-Based TCAM for Longest Prefix; Adapted PBFT Consensus Protocol for Sharded Blockchain; a Solution for the Offline Double-Spending Issue of Digital Currencies; a Hierarchical Macroeconomic Copula Model for Cyber Damages Based on Current Cyber Insurance Prices; cyber Social Engineering Kill Chain; towards Practical Personalized Security Nudge Schemes: Investigating the Moderation Effects of Behavioral Features on Nudge Effects.

Date 2022

URL <https://www.scopus.com/pages/publications/85140473153?origin=resultslist>

Archive Scopus

Language English

Volume 13580 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Journal Abbr Lect. Notes Comput. Sci.

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EID: 2-s2.0-85140473153

Export Date: 20 September 2026; Cited By: 0; Conference name: 4th International Conference on Science of Cyber Security, SciSec 2022; Conference date: 10 August 2022 through 12 August 2022; Conference code: 284459

5-Hydroxymethylcytosine as a clinical biomarker: Fluorescence-based assay for high-throughput epigenetic quantification in human tissues

Item Type Journal Article

Author S. Margalit

Author S. Avraham

Author T. Shahal

Author Y. Michaeli

Author N. Gilat

Author P. Magod

Author M. Caspi

Author S. Loewenstein

Author G. Lahat

Author D. Friedmann-Morvinski

Author R. Kariv

Author R. Rosin-Arbesfeld

Author S. Zirkin

Author Y. Ebenstein

Abstract Epigenetic transformations may provide early indicators for cancer and other disease. Specifically, the amount of genomic 5-hydroxymethylcytosine (5-hmC) was shown to be globally reduced in a wide range of cancers. The integration of this global biomarker into diagnostic workflows is hampered by the limitations of current 5-hmC quantification methods. Here we present and validate a fluorescence-based platform for high-throughput and cost-effective quantification of global genomic 5-hmC levels. We utilized the assay to characterize cancerous tissues based on their 5-hmC content, and observed a pronounced reduction in 5-hmC level in various cancer types. We present data for glioblastoma, colorectal cancer, multiple myeloma, chronic lymphocytic leukemia and pancreatic cancer, compared to corresponding controls. Potentially, the technique could also be used to follow response to treatment for personalized treatment selection. We present initial proof-of-concept data for treatment of familial adenomatous polyposis. © 2019 UICC

Date 2020

DOI 10.1002/ijc.32519

URL <https://www.scopus.com/pages/publications/85068520753?origin=resultslist>

Archive Scopus

Language English

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Publisher Wiley-Liss Inc.

Pages 115-122

Publication International Journal of Cancer

ISSN 00207136 (ISSN)

Issue 1

Journal Abbr Int. J. Cancer

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Tags:

Article, human, Humans, quantitative analysis, procedures, controlled study, classification, cost effectiveness analysis, economics, genetics, genomics, human tissue, animal, Animals, biological marker, metabolism, Biomarkers, Tumor, tumor marker, Mice, mouse, multiple myeloma, colorectal cancer, cancer, pancreas cancer, neoplasm, Epigenesis, Genetic, epigenetics, genetic epigenesis, proof of concept, glioblastoma, fluorescence, Fluorescence, cancer tissue, Neoplasms, Proof of Concept Study, priority journal, 5 hydroxymethylcytosine, 5 methylcytosine, 5-hmC, 5-hydroxymethylcytosine, 5-Methylcytosine, chronic lymphatic leukemia, cost benefit analysis, Cost-Benefit Analysis, epigenetic biomarker, fluorescence analysis, high throughput screening, High-Throughput Screening Assays

Notes:

EID: 2-s2.0-85068520753

Export Date: 20 September 2026; Cited By: 28; Funding details: BeyondSeq Consortium; EU-Horizon2020 program; European Research Council Proof of Concept; Joint Israeli German R&D Nanotechnology, (61976); Seventh Framework Programme, FP7, (337830, 767931); Seventh Framework Programme, FP7; European Commission, EU, (63489); European Commission, EU; European Research Council, ERC; Correspondence Address: S. Zirkin; School of Chemistry, Center for Nanoscience and Nanotechnology, Center for Light-Matter Interaction, Raymond and Beverly Sackler Faculty of Exact Sciences, Tel Aviv University, Tel Aviv, Israel; email: shaharzirkin@gmail.com; CODEN: IJCNA

5th International Conference on Financial Technology, ICFT 2024

Item Type Journal Article

Editor Huang K.-W.

Editor Cao Q.

Editor Su R.

Abstract The proceedings contain 17 papers. The special focus in this conference is on Financial Technology. The topics include: Improved Machine Learning Algorithms for Fraud Detection in Fintech Companies; A Cognitive Analysis of CEO Speeches and Their Effects on Stock Markets; does E-Money Mediate the Effect of Fundamental Factors on the Stock Price Index?; Multi-Attribute Decision Making (MADM) Model Based on Bayesian Neural Network Classification with Comparative Scoring Quantification Method Can Solve Intangible Value Valuation Issues in Benefit-Cost Analysis (BCA); Exploring Vulnerabilities in Near Field Communication (NFC) Devices: A Comprehensive Investigation; a Novel Electronic Payment System Based on Zero-Knowledge Proof and Blockchain; A Comprehensive AI and Blockchain Framework for Detecting and Preventing Money Laundering in Bangladesh Financial Systems; detecting Persuasion in Financial Short Texts: A Computational Approach; The Impact of Colored Noise on the CIR Model; A Secure NFC Cardless Cash Withdrawal System; FinTech Digital Transformation: Generative AI, Humanoid Robots, Metaverse, Human-AI Collaboration, and Industry 5.0; Artificial Intelligence (AI) and Virtual Reality Convergence in Financial Services: The Power of Digital Twin Robo-Advisers; CITRONN: A Convolutional Neural Network for Crypto Image-Based Trading; can the Government's Distribution of Consumption Vouchers Stimulate Adoption of Digital Payment Channels? Insights from a Social Learning Perspective; financial Time Series Simulation with Transformer-Based Generative Models Under Continuous Conditions.

Date 2025

URL <https://www.scopus.com/pages/publications/105002368921?origin=resultslist>

Archive Scopus

Language English

Volume 2437 CCIS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Communications in Computer and Information Science

ISSN 18650929 (ISSN); 978-981963810-9 (ISBN)

Journal Abbr Commun. Comput. Info. Sci.

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EID: 2-s2.0-105002368921

Export Date: 20 September 2026; Cited By: 0; Conference name: 5th International Conference on Financial Technology, ICFT 2024; Conference date: 23 September 2024 through 25 September 2024; Conference code: 329639

13th International Conference of the Cross-Language Evaluation Forum for European Languages, CLEF 2022

Item Type Journal Article

Editor Barrón-Cedeño A.

Editor Da San Martino G.

Editor Faggioli G.

Editor Ferro N.

Editor Degli Esposti M.

Editor Sebastiani F.

Editor Macdonald C.

Editor Pasi G.

Editor Hanbury A.

Editor Potthast M.

Abstract The proceedings contain 31 papers. The special focus in this conference is on Cross-Language Evaluation Forum for European Languages. The topics include: Evaluating Research Dataset Recommendations in a Living Lab; analysing Moral Beliefs for Detecting Hate Speech Spreaders on Twitter; query Expansion, Argument Mining and Document Scoring for an Efficient Question Answering System; transformer-Encoder-Based Mathematical Information Retrieval; ImageCLEF 2021 Best of Labs: The Curious Case of Caption Generation for Medical Images; data-Centric and Model-Centric Approaches for Biomedical Question Answering; did I Miss Anything? A Study on Ranking Fusion and Manual Query Rewriting in Consumer Health Search; overview of eRisk 2022: Early Risk Prediction on the Internet; Overview of LifeCLEF 2022: An Evaluation of Machine-Learning Based Species Identification and Species Distribution Prediction; tracking News Stories in Short Messages in the Era of Infodemic; Overview of ARQMath-3 (2022): Third CLEF Lab on Answer Retrieval for Questions on Math; overview of Touché 2022: Argument Retrieval; Overview of BioASQ 2022: The Tenth BioASQ Challenge on Large-Scale Biomedical Semantic Indexing and Question Answering; A Concise Overview of LeQua@CLEF 2022: Learning to Quantify; Overview of PAN 2022: Authorship Verification, Profiling Irony and Stereotype Spreaders, and Style Change Detection; Intelligent Disease Progression Prediction: Overview of iDPP@CLEF 2022; Overview of HIPE-2022: Named Entity Recognition and Linking in Multilingual Historical Documents; Overview of JOKER@CLEF 2022: Automatic Wordplay and Humour Translation Workshop; Overview of the CLEF 2022 SimpleText Lab: Automatic Simplification of Scientific Texts; Overview of the CLEF-2022 CheckThat! Lab on Fighting the COVID-19 Infodemic and Fake

News Detection; leveraging Wikipedia Knowledge for Distant Supervision in Medical Concept Normalization; Overview of ChEMU 2022 Evaluation Campaign: Information Extraction in Chemical Patents; Overview of the ImageCLEF 2022: Multimedia Retrieval in Medical, Social Media and Nature Applications; noise-Reduction for Automatically Transferred Relevance Judgments; the Effect of Prolonged Exposure to Online Education on a Classroom Search Companion; rhythmic and Psycholinguistic Features for Authorship Tasks in the Spanish Parliament: Evaluation and Analysis.

Date 2022

URL <https://www.scopus.com/pages/publications/85137975521?origin=resultslist>

Archive Scopus

Language English

Volume 13390 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-303113642-9 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

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EID: 2-s2.0-85137975521

Export Date: 20 September 2026; Cited By: 0; Conference name: 13th International Conference of the CLEF Association, CLEF 2022; Conference date: 5 September 2022 through 8 September 2022; Conference code: 282409

13th International Conference on Flexible Query Answering Systems, FQAS 2019

Item Type Journal Article

Editor Cuzzocrea A.

Editor Greco S.

Editor Saccà D.

Editor Larsen H.L.

Editor Andreassen T.

Editor Christiansen H.

Abstract The proceedings contain 37 papers. The special focus in this conference is on Flexible Query Answering Systems. The topics include: WeLink: A Named Entity Disambiguation Approach for a QAS over Knowledge Bases; a Heuristic Pruning Technique for Dialectical Trees on Argumentation-Based Query-Answering Systems; a Method for Efficient Argument-Based Inquiry; DAQAP: Defeasible Argumentation Query Answering Platform; an Efficient Algorithm for Computing the Set of Semi-stable Extensions; using Word Embeddings and Deep Learning for Supervised Topic Detection in Social Networks; generalized Association Rules for Sentiment Analysis in Twitter; Data Exploration in the HIFUN Language; reducing Skyline Query

Results: An Approach Based on Fuzzy Satisfaction of Concepts; Personal Big Data, GDPR and Anonymization; quantify the Variability of Time Series of Imprecise Data; semantic Understanding of Natural Language Stories for Near Human Question Answering; deductive Querying of Natural Logic Bases; clustering of InterCriteria Analysis Data Using a Health-Related Quality of Life Data; a Flexible Query Answering System for Movie Analytics; can Blockchain Technology Provide Information Systems with Trusted Database? The Case of HyperLedger Fabric; anticipating Depression Based on Online Social Media Behaviour; method for Modeling and Simulation of Parallel Data Integration Processes in Wireless Sensor Networks; find the Right Peers: Building and Querying Multi-IoT Networks Based on Contexts; handling Veracity of Nominal Data in Big Data: A Multipolar Approach; tutorial: Supervised Learning for Prevalence Estimation; interCriteria Analysis with Interval-Valued Intuitionistic Fuzzy Evaluations; business Dynamism and Innovation Capability in the European Union Member States in 2018 Through the Prism of InterCriteria Analysis.

Date 2019

URL <https://www.scopus.com/pages/publications/85072857828?origin=resultslist>

Archive Scopus

Language English

Volume 11529 LNAI

Publisher Springer Verlag

Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Journal Abbr Lect. Notes Comput. Sci.

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EID: 2-s2.0-85072857828

Export Date: 20 September 2026; Cited By: 0; Conference name: 13th International Conference on Flexible Query Answering Systems, FQAS 2019; Conference date: 2 July 2019 through 5 July 2019; Conference code: 231829

37th International Conference on Machine Learning, ICML 2020

Item Type Journal Article

Editor Daumé III H.

Editor Singh A.

Abstract The proceedings contain 1084 papers. The special focus in this conference is on Machine Learning. The topics include: Efficient Optimistic Exploration in Linear-Quadratic Regulators via Lagrangian Relaxation; super-efficiency of automatic differentiation for functions defined as a minimum; a Geometric Approach to Archetypal Analysis via Sparse Projections; context-Aware Local Differential Privacy; efficient Intervention Design for Causal Discovery with Latents; the Neural Tangent Kernel in High Dimensions: Triple Descent and a Multi-Scale Theory of

Generalization; rank Aggregation from Pairwise Comparisons in the Presence of Adversarial Corruptions; boosting for Control of Dynamical Systems; an Optimistic Perspective on Offline Reinforcement Learning; optimal Bounds between f-Divergences and Integral Probability Metrics; LazyIter: A Fast Algorithm for Counting Markov Equivalent DAGs and Designing Experiments; learning What to Defer for Maximum Independent Sets; invariant Risk Minimization Games; why bigger is not always better: on finite and infinite neural networks; discriminative Jackknife: Quantifying Uncertainty in Deep Learning via Higher-Order Influence Functions; frequentist Uncertainty in Recurrent Neural Networks via Blockwise Influence Functions; random extrapolation for primal-dual coordinate descent; a new regret analysis for Adam-type algorithms; restarted Bayesian Online Change-point Detector achieves Optimal Detection Delay; maximum Likelihood with Bias-Corrected Calibration is Hard-To-Beat at Label Shift Adaptation; the Implicit Regularization of Stochastic Gradient Flow for Least Squares; structural Language Models of Code; LowFER: Low-rank Bilinear Pooling for Link Prediction; discount Factor as a Regularizer in Reinforcement Learning; neuro-Symbolic Visual Reasoning: Disentangling “Visual” from “Reasoning”; the Differentiable Cross-Entropy Method; Customizing ML Predictions For Online Algorithms; fairwashing Explanations with Off-Manifold Detergent; quantum Boosting.

Date 2020

URL <https://www.scopus.com/pages/publications/105021315438?origin=resultslist>

Archive Scopus

Language English

Volume 119

Publisher ML Research Press

Publication Proceedings of Machine Learning Research

ISSN 26403498 (ISSN)

Journal Abbr Proc. Mach. Learn. Res.

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Export Date: 20 September 2026; Cited By: 0; Conference name: 37th International Conference on Machine Learning, ICML 2020; Conference date: 13 July 2020 through 18 July 2020; Conference code: 340739

2013 2nd International Conference on Mechatronics and Computational Mechanics, ICMCM 2013

Item Type Journal Article

Abstract The proceedings contain 62 papers. The special focus in this conference is on Applied Mechanics, Design, Modelling and Simulation, Control Systems and Materials Research. The topics include: Carbon fiber reinforced the cracking framework top side columns caused by roof temperature deformation; nanowelding of single walled carbon nanotubes onto electrodes using molecular dynamics method; preparation

process of long fiber reinforced impact polypropylene copolymer composites; the mechanical properties of cement reinforced loess and pore microstructure characteristics; the study for non-destructive quantification method of reinforcement corrosion degree based on electrochemical detection and finite analysis technology; computational analysis of a magnetohydrodynamic flow in an electrically conducting hairpin duct; flow regime identification for wet gas flow based on kurtosis feature of flow-induced pipeline vibration; numerical modeling on low velocity impact on a foam core sandwich panel; research on the aero dynamicity of steady wing of non-parachute terminal sensitive projectile; stress analysis and applications of sucker rod in deviated wells; study on aftershocks triggering by static stress changes of the minxian-zhangxian 6.6 earthquake; study on the production decline laws for vertical fracture well in low-permeability gas reservoirs; the numerical simulation and performance analysis of the vortex pump for solid-liquid two phase medium; analysis of modal and seismic property of oil storage tanks; calculation of kill probability for kinetic energy rod against TBM warhead; design for disassembly based on substance-field analysis; overview of visualization tool as an experimental method in iterative participatory design; research for calculation model of allotypic pumping unit load; solving the deflection problem of laptop casing structure by finite difference method; numerical studies on the performance of saccardo ventilation system in emergency fire scenarios; design and optimization of urban signal fuzzy controller based on IQPSO; error modeling and simulation of serial-parallel mechanism based on screw theory; missile trajectory simulation in vertical plane; experimental verification of the suppression effect of ship shaft-rate electric field; modal analysis of a electric tool; an efficient quadratic conversion architecture for real-time wideband DDC; digital structural design and research of optical microscope; real-time suppression research of shaft-rate electric field based on the compensated negative potential; design of rate gyro stabilized platform control system; robotic fish technology and its applications to space mechatronics; design of automatic control system of floating dock; design of new intelligent redundant serial bus with isolated driver in the distributed control system; design of pneumatic and control system for quartz crystal tin dipping machine; study and application of sliding mode control strategy for high-power current source inverter; design of a communication interface between the controller and the galvanometer; reliability evaluation of main electrical connection of substation based on graph theory; mechanical disassembly of end-of-life products; influencing factors and evaluation standards of vibration on human comfort; capability based weapon system-of-systems evaluation strategies design space and exploratory analysis; risk evaluation in marine traffic systems based on fuzzy catastrophe theory; approximate evaluation of exact solutions of measurements inverse problems; scene classification based on improved spatial partition model and superconvergence analysis of finite element method for onlinear Klein-Gordon equation.

Date 2014

URL <https://www.scopus.com/pages/publications/84895155420?origin=resultslist>

Archive Scopus

Language English

Volume 527

Publication Applied Mechanics and Materials

ISSN 16627482 (ISSN); 978-303835029-3 (ISBN)

Journal Abbr Appl. Mech. Mater.

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EID: 2-s2.0-84895155420

Export Date: 20 September 2026; Cited By: 0; Conference name: 2013 2nd International Conference on Mechatronics and Computational Mechanics, ICMCM 2013; Conference date: 30 December 2013 through 31 December 2013; Conference code: 102946

2018 IEEE Statistical Signal Processing Workshop, SSP 2018**Item Type** Journal Article

Abstract The proceedings contain 173 papers. The topics discussed include: diffusion-based Bayesian cluster enumeration in distributed sensor networks; a distributed transmission scheduling algorithm for wireless networks based on the ising model; on convergence analysis of gradient based primal-dual method of multipliers; dempster-shafer theory based robust sequential detection in distributed sensor networks; generalized adaptive weighted recursive least squares dictionary learning for retinal vessel inpainting; the statistical restricted isometry property for Gabor systems; a network compatibility condition for compressed sensing over complex networks; coupled compressive sensing; sequential reinforcement approach; least-squares based layerwise pruning of convolutional neural networks; learning DAGs using multiclass support vector machines; edge consensus computing for heterogeneous data sets; differential privacy for positive and unlabeled learning with known class priors; weakly supervised learning of multiple-scale dictionaries; wireless communication links as opportunistic IoT for near ground rain monitoring; tomographic reconstruction of rainfall fields using heterogeneous frequency microwave links; learning-based rainfall estimation via communication satellite links; a geometrical study of the bivariate fractional gaussian noise; and secrecy capacity analysis of transmit-receive diversity systems.

Date 2018**URL** <https://www.scopus.com/pages/publications/85053834166?origin=resultslist>**Archive** Scopus**Language** English**Publisher** Institute of Electrical and Electronics Engineers Inc.**Publication** 2018 IEEE Statistical Signal Processing Workshop, SSP 2018**ISSN** 978-153861570-6 (ISBN)**Journal Abbr** IEEE Stat. Signal Process. Workshop, SSP**Date Added** 20/09/2026, 9:52:48 pm**Modified** 20/09/2026, 9:52:48 pm**Notes:**

EID: 2-s2.0-85053834166

Export Date: 20 September 2026; Cited By: 0; Conference name: 20th IEEE Statistical Signal Processing Workshop, SSP 2018; Conference date: 10 June 2018 through 13 June 2018; Conference code: 139091

A Bayesian approach for classification of buried objects using non-parametric prior model

Item Type Conference Paper

Author A. Aliamiri

Author J. Stalnaker

Author E.L. Miller

Abstract We used kernel density estimation to build a-priori probability distributions on the vector of features used to characterize unexploded ordnance from electromagnetic induction sensor data. This priori information is then used in a Bayesian framework to develop a new suite of estimation and classification algorithms. Based on this prior information several classification algorithms are developed in feature and signal space. Results using real field data show more robust estimation and significant improvement in classification performance for signal space classifiers comparing to conventional Gaussian approximation to the density of the features.

Date 2006

DOI 10.1109/IGARSS.2006.1003

URL <https://www.scopus.com/pages/publications/33847732377?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Dig Int Geosci Remote Sens Symp (IGARSS)

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Pages 3894-3897

Proceedings Title Dig Int Geosci Remote Sens Symp (IGARSS)

Conference Name International Geoscience and Remote Sensing Symposium (IGARSS)

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Tags:

Feature extraction, Gaussian distribution, Sensors, Bayesian networks, Data reduction, Approximation theory, Bayesian frameworks, Classification algorithms, Electromagnetic induction, Signal space

Notes:

EID: 2-s2.0-33847732377

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: A. Aliamiri; Center for Subsurface Sensing and Imaging Systems, Department of Electrical and Computer Engineering, Northeastern University, Boston, MA 02115-5000, United States; email: aaliamir@ece.neu.edu; Conference name: 2006 IEEE International Geoscience and Remote Sensing Symposium, IGARSS 2006; Conference date: 31 July 2006 through 4 August 2006; Conference code: 70348; CODEN: IGRSE

A Bayesian approach for maintenance action recommendation

Item Type Journal Article

Author V. Katsouros

Author V. Papavassiliou

Author C. Emmanouilidis

Abstract This paper presents a Bayesian approach for maintenance action recommendation tested on the PHM 2013 Data Challenge dataset. The Challenge focused on maintenance action recommendation based on historical cases and the algorithms were evaluated on their ability to recommend confirmed problem types. The proposed approach is based on a Bayesian inference methodology and deals with recommending an already known problem type for each case. The recommender can be viewed as a classifier among the confirmed problem types. For each such problem type class the a priori probabilities for the events which characterize the problem type from the training data are estimated. When testing cases are presented, the recommender calculates the a posteriori probabilities for each of the confirmed problem types and suggests the type of problem that corresponds to the maximum a posteriori (MAP) probability.

Date 2013

URL <https://www.scopus.com/pages/publications/84892385211?origin=resultslist>

Archive Scopus

Language English

Volume 4

Publisher Prognostics and Health Management Society

Publication International Journal of Prognostics and Health Management

ISSN 21532648 (ISSN)

Issue 2

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Export Date: 20 September 2026; Cited By: 9

A Bayesian framework for discovering interpretable Lagrangian of dynamical systems from data

Item Type Journal Article

Author T. Tripura

Author S. Chakraborty

Abstract Learning and predicting the dynamics of physical systems requires a profound understanding of the underlying physical laws. Recent works on learning physical

laws involve the extension of the equation discovery frameworks to the discovery of Hamiltonian and Lagrangian of physical systems. While the existing methods parameterize the Lagrangian using neural networks, we propose an alternate framework for learning interpretable Lagrangian descriptions of physical systems from limited data using the sparse Bayesian approach. Unlike existing neural network-based approaches, the proposed approach (a) yields an interpretable description of Lagrangian, (b) exploits Bayesian learning to quantify the epistemic uncertainty due to limited data, (c) automates the distillation of Hamiltonian from the learned Lagrangian using Legendre transformation, and (d) provides ordinary (ODE) and partial differential equation (PDE) based descriptions of the observed systems. Six different examples involving both discrete and continuous systems illustrate the efficacy of the proposed approach. © 2024 Elsevier Ltd

Date 2024

DOI 10.1016/j.ymssp.2024.111737

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Language English

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Journal Abbr Mech Syst Signal Process

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Machine-learning, Machine learning, Uncertainty analysis, Distillation, Metadata, Bayesian networks, Dynamical systems, Bayesian learning, Lagrange multipliers, Explainable machine learning, Conservation law, Hamiltonians, Lagrangian, Lagrangian discovery, Ordinary differential equations, Probabilistic machine learning, Probabilistic machines, Sparse bayesian, Sparse bayesian learning, Sparse Bayesian learning

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Ministry of Education, MOE; Science and Engineering Research Board, SERB, (SRG/2021/000467); Science and Engineering Research Board, SERB; Ministry of Port and Shipping, (ST-14011/74/MT, 356529); Correspondence Address: T. Tripura; Department of Applied Mechanics, Indian Institute of Technology Delhi, Hauz Khas, Delhi, 110016, India; email: tapas.t@am.iitd.ac.in; CODEN: MSSPE

A Bayesian non-parametric Potts model with application to pre-surgical FMRI data

Item Type Conference Paper

Author T.D. Johnson

Author Z. Liu

Author A.J. Bartsch

Author T.E. Nichols

Abstract The Potts model has enjoyed much success as a prior model for image segmentation. Given the individual classes in the model, the data are typically modeled as Gaussian random variates or as random variates from some other parametric distribution. In this article, we present a non-parametric Potts model and apply it to a functional magnetic resonance imaging study for the pre-surgical assessment of peritumoral brain activation. In our model, we assume that the Z-score image from a patient can be segmented into activated, deactivated, and null classes, or states. Conditional on the class, or state, the Z-scores are assumed to come from some generic distribution which we model non-parametrically using a mixture of Dirichlet process priors within the Bayesian framework. The posterior distribution of the model parameters is estimated with a Markov chain Monte Carlo algorithm, and Bayesian decision theory is used to make the final classifications. Our Potts prior model includes two parameters, the standard spatial regularization parameter and a parameter that can be interpreted as the a priori probability that each voxel belongs to the null, or background state, conditional on the lack of spatial regularization. We assume that both of these parameters are unknown, and jointly estimate them along with other model parameters. We show through simulation studies that our model performs on par, in terms of posterior expected loss, with parametric Potts models when the parametric model is correctly specified and outperforms parametric models when the parametric model is misspecified. © The Author(s) 2012 Reprints and permissions: sagepub.co.uk/journalsPermissions.nav.

Date 2013

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Proceedings Title Stat. Methods Med. Res.

Conference Name Statistical Methods in Medical Research

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adult, female, human, Humans, simulation, algorithm, Algorithms, Computer Simulation, probability, statistical analysis, image processing, Bayes theorem, Bayes Theorem, Monte Carlo method, Monte Carlo Method, image analysis, Image Interpretation, Computer-Assisted, Magnetic Resonance Imaging, statistical model, functional magnetic resonance imaging, electroencephalogram, Normal Distribution, decision theory, Brain Neoplasms, cancer surgery, case report, preoperative evaluation, statistical parameters, Biostatistics, conference paper, Decision Theory, Dirichlet process, FMRI, functional neuroimaging, hidden Markov random field, Markov Chains, non-parametric Bayes, nonparametric test, oligodendroglioma, Potts model, Statistics, Nonparametric

Notes:

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A bibliometric analysis of research on remote sensing-based monitoring of soil organic matter conducted between 2003 and 2023

Item Type Journal Article

Author X. Chen

Author F. Yuan

Author S.T. Ata-Ul-Karim

Author X. Liu

Author Y. Tian

Author Y. Zhu

Author W. Cao

Author Q. Cao

Abstract Soil organic matter (SOM) is a key metric for assessing soil quality and crop yield potential. It plays a vital role in maintaining the ecological balance environment and promoting sustainable farming practices. This review examines the evolving trends in remote sensing (RS)-based SOM monitoring by analyzing 739 scholarly publications from the Web of Science database from 2003 to 2023 using a bibliometric approach. The study reveals that research on RS-based SOM monitoring has entered a rapid growth phase since 2018, with China and the United States as the main contributors and an extensive international cooperation network. In model construction, high frequency covariates such as soil pH, precipitation, temperature, and topography significantly improved the prediction accuracy. Data preprocessing methods such as Standard Normal Variables (SNV), Principal Component Analysis (PCA), and Multiple Scattering Correction (MSC) enhanced data consistency. Traditional statistical models are gradually being replaced by nonlinear machine learning and deep learning methods (CNN, XGBoost, and Stacking), which are particularly good at handling complex high-dimensional data. Regional spectral libraries (OzSoil and

AfSIS) excel in local accuracy, while global spectral libraries (ISRIC and LUCAS) are more suitable for cross-region modeling, and the migration learning technique effectively improves the model generalization ability in low data regions. Integrated models (CNN-LSTM and GAN) have significant advantages in capturing the spatial and temporal dynamics of SOMs, and uncertainty quantification methods (Bayesian inference, Monte Carlo simulation) enhance the reliability of the models in multi-source data and data-scarce scenarios. Future research should focus on further optimization of multi-source data fusion and uncertainty quantification to promote the development and applicability of RS-based SOM monitoring techniques for precision soil management and sustainable agriculture. © 2025 The Authors

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Archive Scopus

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Transfer learning, Remote-sensing, Information management, Data fusion, Data reliability, Data accuracy, Data consistency, Model method, Multi-Sources, Uncertainty quantifications, Bibliometrics analysis, Modeling methods, Soil organic matters, Source data, Spatial mapping, Spectral libraries

Notes:

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A bottom-up building stock quantification methodology for construction minerals using Earth Observation. The case of Hanoi

Item Type Journal Article**Author** T. Bide**Author** A. Novellino**Author** E. Petavratzi**Author** C.S. Watson

Abstract Increasing demand for significant volumes of construction materials, especially sand for use in concrete, in rapidly developing urban environments is becoming a significant socio-economic and environmental issue. The consumption of concrete (comprised of sand, aggregates and cement) is especially concerning on a city level as vast volumes of materials are extracted within the urban hinterland, causing direct impacts locally and the potential for supply issues directly impacting city level metabolism. Excessive consumption and poor management of these materials make it increasingly hard for society to ensure new urban development and infrastructure projects, essential for maintaining the health of cities, meet sustainable development objectives. However, it is difficult to implement suitable resource management policies without first understanding how materials are produced and consumed at an appropriate spatial level. For many areas, especially on a city level, such data is absent, especially so for sand and aggregates which can further exacerbate these local supply issues and environmental impacts. This study attempts to address this data gap via combining earth observation datasets with estimates of materials contained within urban infrastructure (material intensities) to calculate the rapid increase of construction material stocks in Hanoi. Spatial data on buildings have been gathered using, producing, and collating a variety of spaceborne open-source datasets on built up areas (GlobalMLBuildingFootprint, World Settlement Footprint 3D, Open Street Map) and land use classification maps. Linking this spatial data with estimated quantities of sand, gravel, cement and concrete in typical buildings in Hanoi enables quantification of building stocks for a range of building types over a time series. The results show that for every new km² of urban infrastructure approximately 520,000 tonnes of concrete, or 360,000 tonnes of sand, 580,000 tonnes of gravel and 115,000 tonnes of cement are required. If the Hanoi Masterplan is to be achieved by 2030, then the material demand is likely to be for 106 million tonnes of concrete or 73 million tonnes of sand, 118 million tonnes of gravel and 24 million tonnes of cement. These all exceed historical consumption trends and are far in excess of current extraction rates and therefore careful planning is required to ensure access to sustainable resources into the future. © 2023 The Authors

Date 2023**DOI** 10.1016/j.cesys.2023.100109**URL** <https://www.scopus.com/pages/publications/85148689336?origin=resultslist>**Archive** Scopus**Language** English**Volume** 8**Publisher** Elsevier Ltd**Publication** Cleaner Environmental Systems**ISSN** 26667894 (ISSN)**Journal Abbr** Clean. Environ. Syst.**Date Added** 20/09/2026, 9:52:11 pm**Modified** 20/09/2026, 9:52:11 pm**Tags:**

Environmental impact, Classification (of information), Sustainable development, Economics, Extraction, Buildings, Urban growth, Socio-economics, Land use, Urban environments, Building stocks, Bottom up, Cements, Concrete aggregates, Construction materials, Earth observation, Earth observations, Economic issues, Environmental issues, Gravel, Material intensity, Materials intensity, Observatories, Sand, Spatial data, Urban infrastructure

Notes:

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A cascade framework for on-device uncertainty-aware event detection on microcontrollers

Item Type Journal Article

Author H. Jia

Author Y.D. Kwon

Author D. Ma

Author N. Pham

Author L. Qendro

Author T. Vu

Author C. Mascolo

Abstract Pervasive sensing enables diverse wearable event detection (WED) applications, but deploying machine learning models on resource-constrained microcontrollers (MCUs) poses significant challenges, particularly in ensuring prediction reliability under data shifts or out-of-distribution (OOD) inputs. While Uncertainty quantification methods offer a way to assess this reliability, many are computationally prohibitive for MCUs, and detecting multiple events concurrently further exacerbates resource constraints. Addressing these combined challenges, this paper presents an uncertainty and resource-aware framework designed for reliable and efficient multi-event WED on MCUs, significantly extending our preliminary work. The proposed framework achieves this by integrating Evidential Deep Learning (EDL) for efficient, single-pass uncertainty estimation with a novel cascade learning architecture. This architecture promotes resource efficiency via: (i) intra-event sharing using uncertainty-aware early exits within a staged model (shallow, medium, deep), allowing simpler samples to terminate inference earlier; and (ii) inter-event sharing using a multi-head design where multiple event detectors share a common backbone, minimizing overhead. System efficiency is further enhanced through MCU-specific optimizations, including targeted architecture search, quantization, efficient uncertainty operator implementation using standard TensorFlow Lite Micro (TFLM) operations, and library footprint reduction. We conducted extensive experiments on four distinct wearable datasets (Oesense, KWS, ECG5000, and HHAR) and two MCU platforms (STM32F446ZE, STM32H747XI), comparing the proposed

framework against strong baselines including Deep Ensembles and Vanilla EDL. Results demonstrate the proposed framework's effectiveness, achieving competitive accuracy and uncertainty performance (e.g., up to 22% lower NLL than data augmentation) while drastically reducing resource consumption, offering up to 8.64× faster inference, up to 8.57× lower energy use, and 55% smaller memory footprint compared to ensemble methods. The proposed framework enables the deployment of reliable, uncertainty-aware multi-event detection on a wider range of low-power MCUs. © 2026

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Journal Abbr Pervasive Mob. Comput.

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Notes:

EID: 2-s2.0-105033235539

Export Date: 20 September 2026; Cited By: 1; Funding details: Nokia; European Research Council, ERC, (833296); Correspondence Address: H. Jia; University of Cambridge, Cambridge, United Kingdom; email: h.jia.cam@gmail.com

A case study of SVM extension techniques on classification of imbalanced data

Item Type Journal Article

Author K.K. Lee

Author C.J. Harris

Author S.R. Gunn

Author P.A.S. Reed

Abstract In many classification problems the data is imbalanced, that is the class priors are different. Here, we consider the classification problem of fatigue crack initiation in automotive camshafts, where this imbalance is significant. The extension techniques

of Support Vector Machine (SVM) - the Control Sensitivity (CSSVM) and Adaptive Margin (AMSVM) - which offer different ways of dealing with imbalanced data was investigated. Geometric mean was used to evaluate the performance of the model. The CSSVM has outperformed the AMSVM. The use of different kernels did not produce significant changes in the results. The ratio between the misclassification cost and the training size for each class is very similar, indicating a strong relationship between them.

Date 2001

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Classification (of information), Data acquisition, Computer science, Probability, Sensitivity analysis, SVM, Imbalanced data, Problem solving, Support Vector Machine (SVM), Crack initiation, Austempered Ductile Iron (ADI), Camshafts, Finite Body Tessellation (FBT), Geometric Mean, Receiver Operating Characteristic (ROC) analysis

Notes:

EID: 2-s2.0-4944237255

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K.K. Lee; Department of Electronics, University of Southampton, Southampton SO17 1BJ, Highfield Road, United Kingdom; email: kkl98r@ecs.soton.ac.uk

A classification and quantitative assessment method for internal and external surface defects in pipelines based on ASTC-Net

Item Type Journal Article

Author J. Yuan

Author M. Qiao

Author C. Hu

Author Y. Cheng

Author Z. Wang

Author D. Zheng

Abstract Classification and size quantification of defects on both the internal and external surfaces of pipelines are critical to pipeline integrity assessment. However, defect

classification is challenging because of the similarities of defect signals on the internal and external surfaces. In addition, most existing size quantification methods are not sufficiently accurate. To solve these problems, this paper proposes a classification and quantitative evaluation method based on an asymmetric student–teacher with a classifier network (ASTC-Net). First, a novel approach for expanding defect magnetic flux leakage (MFL) data is validated through experiments and simulations. Second, ASTC-Net is built to address the problem of defect classification and quantification. Finally, the superiority of the method is verified by experiments. The results show that this approach pioneers the accurate classification of defects on both internal and external surfaces by achieving an accuracy of 99.41%. Furthermore, a high-precision quantitative assessment of defect size is realized, with length, width, and depth errors of only 0.35 mm, 0.34 mm, and 0.41% of the wall thickness, respectively. These experimental results clearly demonstrate that this method has exceptionally high accuracy in defect classification and quantification, offering vast prospects for its application in pipeline MFL evaluation. © 2024 Elsevier Ltd

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Pipelines, Classification, Student teachers, Magnetic flux leakage, Magnetic leakage, Quantification, Defect classification, Magnetic shielding, ASTC-Net, Asymmetric student–teacher with a classifier network, External surfaces, Internal surfaces, Leak detection, Leaks detections, Pipeline leak detection, Pipeline leaks, Surface defects

Notes:

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A classification for macroscopic charcoal morphologies found in Holocene lacustrine sediments

Item Type Journal Article

Author C.J.C. Mustaphi

Author M.F. Pisaric

Abstract Macroscopic charcoal analysis of lake sediment stratigraphies is a widely used approach to reconstruct past biomass burning patterns of ecosystems. The development of fire records often relies on a single quantification method of charcoal in a sediment subsample; however, recent studies have shown that additional paleoecological information can be obtained by classifying charcoal morphologies. The morphologies and diagnostic features of charcoal yields information about fuel sources, fire type, and charcoal taphonomy, and can aid in calibrating sediment records to known historical fires. This additional information enhances paleoecological inferences by providing more paleoenvironmental information than studies of total charcoal as the only metric. Here we present a classification of 27 macroscopic charcoal morphologies observed in Holocene sediments of lakes located in the mixed-conifer forests of southeastern British Columbia, Canada. This classification system builds on other morphological classifications that have been previously utilized, but is more inclusive of the morphological variability observed and is flexible to modification for use when applied to other study settings. The morphological classification presented here was developed following the observation of >100,000 macroscopic charcoal fragments >150 µm. This paper focuses on the observed morphological classes, their identification, potential fuel sources, and the morphotype assemblage stratigraphy from one site as an example. The charcoal assemblages varied throughout the mid-to-late Holocene contemporaneously with known regional scale hydroclimatic changes in British Columbia. Major changes in fire frequency were also concomitant with morphotype assemblage changes. Future work focusing on linking fuel types with charcoal morphotypes, post-fire observations of charcoal taphonomy, and the analysis of multiple attribute charcoal data sets from a variety of ecosystems will improve our understanding of biomass burning and long-term fire ecology. © The Author(s) 2014.

Date 2014

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morphology, Canada, wildfire, climate variation, fire, fire history, biomass burning, anthracology, British Columbia, charcoal, Coniferophyta, disturbance, geological record, Holocene, lacustrine deposit, lake sediments, morphotypes, paleobotany, paleoclimate, paleoecology, paleoenvironment, Quaternary, stratigraphy, taphonomy

Notes:

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A Classification for Synthetic Inertia Quantification Methods with Practical Recommendations

Item Type	Conference Paper
Author	M. Mohammadi
Author	W. Ripley
Author	T. Hsu
Author	M.G. Dozein
Abstract	With ongoing advancements in grid-forming (GFM) technologies and inverter design, future power systems will likely operate with a mix of synthetic inertia, provided by inverter-based resources (IBRs) and large-scale loads, and synchronous inertia from traditional rotating machines such as synchronous condensers and conventional power plants. In this context, there is an essential need to quantify synthetic inertia to ensure its commercial viability as a service, add further certainty to GFM system interconnection and business cases of GFM projects, and their integration into emerging ancillary service market mechanisms. This paper presents a classification for synthetic inertia quantification methods, including large disturbance methods, RoCoF emulation methods, probing signal injection strategies, and conceptual approaches that may become practical in the future. The mathematical and implementation process for these synthetic inertia quantification methods are presented, as well as the issues and requirements. A rigorous comparison between different methods are presented considering accuracy, perturbation type, damping decoupling, as well as the issues with implementation and the complexity involved. Finally, practical recommendations and future research directions are presented along with industry specifications on equipment capable of providing synthetic inertial responses. © 2025 IEEE.
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Archive	Scopus
Language	English
Extra	Journal Abbreviation: IEEE Kiel PowerTech, PowerTech
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	979-833154397-6 (ISBN)

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Conference Name 2025 IEEE Kiel PowerTech, PowerTech 2025
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Large-scales, Industrial research, Power, Quantification methods, Electric inverters, Forming technology, Inverter-based, Low-inertia power system, Low-Inertia Power Systems, Practical recommendation, Quantification Methods, Rotating machine, Synthetic inertia, Synthetic Inertia, Virtual inertia, Virtual Inertia

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Mohammadi; Australian Energy Market Operator (AEMO), Melbourne, Australia; email: mohammad.mohammadi@aemo.com.au; Conference name: 2025 IEEE Kiel PowerTech, PowerTech 2025; Conference date: 29 June 2025 through 3 July 2025; Conference code: 213431

A Cluster-based Regrouping approach for Imbalanced data distributions

Item Type Conference Paper

Author W. Yu

Author S. Jiang

Abstract In real-world applications, it has been observed that class imbalance (significant differences in class prior probabilities) may produce an important deterioration of the classifier performance, in particular with patterns belonging to the less represented classes. In this paper, we propose a Cluster-based Regrouping approach (CR) which divides the whole training data into positive group and negative group by clustering through the outlier factor. As a result, the similar samples will be in the same group while the dissimilar samples will be in the different groups. Then the basic classifier is employed to build the models on both the positive group and the negative group respectively. When classifying the new object, the model used to evaluate will be chosen according to the type of the group which the new object is nearest. The experimental results demonstrate that our approach achieved promising performance in some cases by directly or indirectly reducing the class distribution skewness. © 2012 TSI Press.

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Training data, Class imbalance, Imbalanced data, Prior probability, Class distributions, Classifier performance, C4.5, Cluster-based, Imbalanced data classification, Naïve-bayes, One-pass, One-pass clustering, Real-world application

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: W. Yu; School of Management, Guangdong University of Foreign Studies, Guangzhou 510006, China; email: yuyu_588@163.com; Conference name: 2012 World Automation Congress, WAC 2012; Conference date: 24 June 2012 through 28 June 2012; Conference code: 94214

A combined machine learning and numerical approach for evaluating the uncertainty of 3D angle-interlock woven composites

Item Type Journal Article

Author Z. Tang

Author L. Guo

Author T. Zheng

Author Z. Li

Author R. Sun

Author K. Huang

Abstract The characterization and evaluation of uncertainties caused by the automated manufacturing procedure are essential for the early structural design and application of 3D woven composites. To account for the inherent uncertainties, numerical simulations must be integrated with statistical uncertainty quantification and propagation methods. However, in some cases, it is very challenging and computationally time consuming. In this paper, a multiscale uncertainty quantification method combining finite element analysis, wide & deep neural network and sensitivity analysis is proposed to probabilistically evaluate the tensile response of 3D angle-interlock woven composites. The required dataset for training and validating the model is created by a two-step numerical simulation. With the present model, a framework of the importance of each uncertainty in determining the macroscopic properties' variance can be established with fewer computational resources. The results indicate that the effect of uncertainties on the material tensile response is significant and the sensitivity information can serve as a guide for reducing uncertainties at the macroscale. © 2022 Elsevier Ltd

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Tags:

Deep neural networks, Uncertainty analysis, Uncertainty, Sensitivity analysis, Machine learning approaches, Quantification methods, Composite materials, Structural design, Backpropagation, Numerical models, Numerical methods, 3-D angle-interlock woven composites, 3D woven composites, Automated Manufacturing, Multi-scale simulation, Multiscale simulation, Numerical approaches, Tensile response, Wide & deep neural network

Notes:

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A Comfort Quantification Method Based on Semi-Supervised Learning for Automated Vehicle at Lane Change Scenarios

Item Type Journal Article

Author W. Zhu

Author X. Zhang

Author C. Hu

Author B. Zhao

Author S. Peng

Author H. Yang

Abstract This paper systematically investigates a passenger comfort quantification model for automated vehicle by integrating attention mechanism-based bidirectional long short-term memory (AM-Bi-LSTM) network, semi-supervised learning strategy, and passenger heterogeneity. Firstly, a comfort quantification dataset is gathered from a physical vehicle. More than thirty volunteers are recruited. Passenger classification based on volunteer heterogeneity is introduced for label reconstruction. Secondly, this

paper innovatively develops a comfort degree quantification model based on the combination of AM-Bi-LSTM and semi-supervised learning. The semi-supervised learning strategy is developed to generate pseudo-labels and reduce dataset noise caused by individual variances of volunteers. The final model precision increases by 22.7% compared with the model trained on raw dataset. Finally, the deployment of model on physical automated vehicle is accomplished. The outputs of onboard real-time comfort quantification system are consistent with the driving intensity of automated vehicle. The results illustrate that the proposed model can accurately identify different comfort degrees of lane change scenarios, which can enhance the evaluation capacity of automated vehicle and optimize the decision-making and path planning modules of automated driving framework with more personalized driving styles. © 2016 IEEE.

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Behavioral research, Intelligent vehicle highway systems, Vehicles, Automation, Intelligent systems, Automated vehicles, Passengers comfort, Decision making, Heart, Supervised learning, Psychology, automated vehicle, Long short-term memory, Attention mechanisms, Semi-supervised learning, Learning strategy, Electrophysiology, Mechanism-based, AM-Bi-LSTM, Attention mechanism-based bidirectional long short-term memory, Galvanic skin response, Heart rate variability, heart rate variability and galvanic skin response, Heart rate variability and galvanic skin response, passenger comfort quantification, Passenger comfort quantification, passenger heterogeneity, Passenger heterogeneity, Self-reported score, self-reported scores, semi-supervised learning strategy, Semi-supervised learning strategy, Vehicle's dynamics

Notes:

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A Comparative Approach to Threshold Optimization for Classifying Imbalanced Data

Item Type Conference Paper

Author J. Hancock

Author J.M. Johnson

Author T.M. Khoshgoftaar

Abstract For the practical application of a classifier, it is necessary to select an optimal output probability threshold to obtain the best classification results. There are many criteria one may employ to select a threshold. However, selecting a threshold will often involve trading off performance in terms of one metric for performance in terms of another metric. In our literature review of studies involving selecting thresholds to optimize classification of imbalanced data, we find there is an opportunity to expand on previous work for an in-depth study of threshold selection. Our contribution is to present a systematic method for selecting the best threshold value for a given classification task and its desired performance constraints. Just as a machine learning algorithm is optimized on some training data set, we demonstrate how a user-defined set of performance metrics can be utilized to optimize the classification threshold. In this study we use four popular metrics to optimize thresholds: precision, Matthews' Correlation Coefficient, f-measure and geometric mean of true positive rate, and true negative rate. Moreover, we compare classification results for thresholds optimized for these metrics with the commonly used default threshold of 0.5, and the prior probability of the positive class (also known as the minority to majority class ratio). Our results show that other thresholds handily outperform the default threshold of 0.5. Moreover, we show that the positive class prior probability is a good benchmark for finding classification thresholds that perform well in terms of multiple metrics. © 2022 IEEE.

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A Comparative Evaluation of Quantification Methods

Item Type Journal Article

Author T. Schumacher

Author M. Strohmaier

Author F. Lemmerich

Abstract Quantification represents the problem of estimating the distribution of class labels on unseen data. It also represents a growing research field in supervised machine learning, for which a large variety of different algorithms has been proposed in recent years. However, a comprehensive empirical comparison of quantification methods that supports algorithm selection is not available yet. In this work, we close this research gap by conducting a thorough empirical performance comparison of 24 different quantification methods on in total more than 40 datasets, considering binary as well as multiclass quantification settings. We observe that no single algorithm generally outperforms all competitors, but identify a group of methods that perform best in the binary setting, including the threshold selection-based median sweep and TSMax methods, the DyS framework including the HDy method, Forman's mixture model, and Friedman's method. For the multiclass setting, we observe that a different, broad group of algorithms yields good performance, including the HDx method, the generalized probabilistic adjusted count, the readme method, the energy distance minimization method, the EM algorithm for quantification, and Friedman's method. We also find that tuning the underlying classifiers has in most cases only a limited impact on the quantification performance. More generally, we find that the performance on multiclass quantification is inferior to the results obtained in the binary setting. Our results can guide practitioners who intend to apply quantification algorithms and help researchers identify opportunities for future research. ©2025

Tobias Schumacher, Markus Strohmaier, and Florian Lemmerich.

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A Comparative Study of Landslide Susceptibility Mapping Using Bagging PU Learning in Class-Prior Probability Shift Datasets

Item Type Journal Article

Author L. Zhao

Author H. Ma

Author J. Dong

Author X. Wu

Author H. Xu

Author R. Niu

Abstract Landslide susceptibility mapping is typically based on binary prediction probabilities. However, non-landslide samples in modeling datasets are often unlabeled data, and the phenomenon of class-prior shift, that is, the proportion of landslide samples frequently deviates from real-world scenarios and is spatially heterogeneous. By comparing the classification performance and predicted probability distributions across multiple unbalanced datasets with known and unknown sample proportions, this study assesses the landslide susceptibility model's generalization ability in the context of class-prior shifts. The study investigates the potential of Bagging PU

Learning, a semi-supervised learning approach, in improving the generalization performance of landslide susceptibility models and proposes the Bagging PU-GDBT algorithm. Our findings highlight the effectiveness of Bagging PU Learning in enhancing the recall of landslides and the generalization capabilities of models on unbalanced datasets. This method reduces prediction uncertainties, especially in high and very high susceptibility zones. Furthermore, results emphasize the superiority of models trained on balanced datasets with 1:1 sample ratio for landslide susceptibility mapping over those trained on unbalanced datasets. © 2023 by the authors.

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A Comparison of Data-Driven Shape Quantification Methods for 2D Microscopy Images

Item Type Conference Paper

Author A.F. Romero

Author A. Krull

Author V. Uhlmann

Abstract Shape is a fundamental property of cells and nuclei that informs on their function and state, and is one of the key readouts of 2D microscopy image data. Quantifying shape in an unbiased manner however remains challenging, especially in exploratory contexts where discovering phenotypes with as little prior as possible is highly desirable. In this work, we demonstrate the practical value of data-driven shape quantification methods on three publicly available microscopy image datasets of cells and nuclei. We compare the information that can be gathered from shape representations obtained with state-of-the-art self-supervised learning models and contrast it against the performance of the widely-used baseline consisting of describing shape with a collection of summary statistics. Our results identify the recently-published ShapeEmbed model as a powerful option, prompting us to propose a simpler variant, coined ShapeEmbedLite, which reduces the complexity of the approach without sacrificing performance and even achieve gains in some metrics. Through our experiments, we demonstrate that data-driven representations obtained through self-supervised learning can facilitate the visual and quantitative exploration of shape differences in 2D microscopy image data with minimal bias, and that the variations they capture align with true signal originating from different experimental conditions or biological states. These findings outline the promising outlook of such methods to drive biology discovery and to help identify rare or unknown shape phenotypes. © 2025 IEEE.

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A comparison of three different methods for the quantification of the in vitro wear of dental materials

Item Type Journal Article

Author S.D. Heintze

Author A. Cavalleri

Author M. Forjanic

Author G. Zellweger

Author V. Rousson

Abstract Objective: Different approaches are utilized to quantify the wear generated on flat specimens with a wear simulator. However, there are no systematic studies comparing different wear quantification methods with a series of materials that exhibit different wear rates. Methods: Sixteen restorative materials, including 14 composites (BelleGlass, Chromasit, Estenia, Esthet-X, Four Seasons, Heliomolar RO, Heliomolar HB, Herculite XRV, InTen-S, Point 4, SureFil, Targis cured at 95 and 130 °C, Tetric Ceram) as well as an amalgam (Amalcap) and a ceramic (Empress) material, were subjected to attrition wear against standardized Empress antagonists in the Willytec wear simulator (120,000 cycles, 5 kg, 1.6 Hz). The volume and maximal vertical loss were quantified directly on the specimens with a profilometry device (Perthometer) and the FRT MicroProf optical sensor. After the fabrication of plaster replicas, the loss was also determined with a 3D laser scanning device. For the statistical analysis, the data were subjected to a logarithmic transformation. Intraclass correlation was calculated to measure the agreement among all three methods, while limits of agreement were used to compare one method against another. Results: There was a very good agreement between all three quantification methods for both volume and vertical loss. The mechanical sensor measured consistently higher values compared to the optical sensors for the volume loss (correction factor 0.95), whereas for the vertical loss, consistently lower values were obtained (correction factor 1.17). However, the ranking of the materials was only marginally influenced by the quantification method. Significance: All three sensors are suitable for the quantification of wear facets. Due to speed and simplicity, the laser sensor has greater advantages over the two other sensors. © 2005 Academy of Dental Materials.

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A Comparison of Uncertainty Quantification Methods for Active Learning in Image Classification

Item Type Conference Paper**Author** A. Hein**Author** S. Rohrl**Author** T. Grobel**Author** M. Lengl**Author** N. Hafez**Author** M. Knopp**Author** C. Klenk

Author D. Heim

Author O. Hayden

Author K. Diepold

Abstract The adoption of machine learning (ML) technology in real-world settings like medical imaging is currently hampered by a lack of trust in ML models and a lack of labeled data. These two issues are currently addressed in parallel by two subdisciplines of ML: uncertainty quantification (UQ) is concerned with obtaining reliable estimates of a model's confidence in its outputs, and active learning (AL) deals with efficiently training models in low-data regimes. To date, the usefulness of the new methods emerging from the field of UQ for AL remains under-explored. We here take a step to address this by comparing seven different UQ methods on three image classification data sets. Our experiments confirm previous indications in the AL literature that the ranking of sampling strategies can vary greatly across models and data sets. We find that Concrete Dropout, Least Confidence, Smallest Margin, and Entropy sampling consistently outperform Random sampling across data sets, whereas Ensembles, Monte-Carlo Dropout, and Bayes-by-Backprop do not. We also observe that AL training stability is sensitive to data quality. © 2022 IEEE.

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A comprehensive strategy using chromatographic profiles combined with chemometric methods: Application to quality control of *Polygonum cuspidatum* Sieb. et Zucc

Item Type Journal Article

Author F. Gao

Author Z. Xu

Author W. Wang

Author G. Lu

Author Y. Vander Heyden

Author T. Zhou

Author G. Fan

Abstract For the strict quality control of herbs, a comprehensive strategy based on chromatographic profiles and chemometric methods which could reliably select quantitative indices, robustly quantitate multi-markers and systematically compare different chemometric methods was proposed and successfully applied to the quality analysis of *P. cuspidatum*. Based on the construction of chromatographic profiles by an efficient accelerated solvent extraction (ASE) and reliable high-performance liquid chromatography-ultraviolet (HPLC-UV) methods, different chemometric methods were employed, namely similarity analyses (SA), hierarchical clustering analysis (HCA) and linear discriminant analysis (LDA). The differences in classification of herb samples were studied for the first time. To reasonably determine the quality of herbs and evaluate different chemometric methods, a comprehensive strategy containing three key steps was performed including selection of quantitative indices, development of a reliable quantification method and adoption of an easily calculated and visible parameter. The quantitative method which was acceptable with good linearity with correlation coefficients >0.9995 and satisfactory repeatability ($\text{RSD} < 1.5\%$), precision ($\text{RSD} < 2.84\%$), reproducibility ($\text{RSD} < 2.88\%$), stability ($\text{RSD} < 2.85\%$) and recoveries (91.5%–105.6%, $\text{RSD} < 2.83\%$) was applied to quality evaluation of fourteen batches of the *P. cuspidatum* samples through simultaneous quantitative determination of fifteen marker compounds. The limits of quantitation of fifteen compounds ranged from 1 to 60 $\mu\text{g}/\text{ml}$. From the results of the quality evaluation, it was found that the different calculation theories of the chemometric methods resulted in the variation of classifiers of samples: SA classified samples through the mean values and HCA & LDA classified similar objects. © 2016 Elsevier B.V.

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A computer vision-based crack quantification of reinforced concrete shells using graph theory measures

Item Type Conference Paper

Author P. Bazrafshan

Author T. On

Author A. Ebrahimkhanlou

Editor Zonta D.

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Editor Su Z.

Abstract SHM and NDE/NDT are developed to evaluate the amount of damage to a structure. Since human inspections are prone to misdiagnosis and/or misinterpretations, the damage assessment and identification methods are shifting from human-based to computer-based. In this article, a computer vision method is used to detect and quantify the cracks on a concrete surface. After processing the

crack images, cracks are modeled as graphs for feature extraction. To study the proposed method, concrete surface crack images from a reinforced concrete shell under quasi-linear load at each load step are used. Pearson Correlation Analysis is implemented to evaluate the relationship between graph measures at the north and south faces of the wall. All correlation values are above 50%, and 8 out of 10 graph measures demonstrated a correlation above 85%. The average correlation value between both sides of the wall reports 84% compatibility. High correlation values between the two sides of the wall attribute to the fidelity of graph features to the crack patterns on the two sides of the wall. Moreover, the monotonic increases in different graph measures and damage indexes indicate that the extracted features from the converted crack patterns to graph are meaningful information. Results of this study reveal that the proposed crack quantification method has the potential of translating crack patterns for further machine learning applications. This novel idea of using different graph measures introduces more features and can act as a base to study the fundamental and mathematical relation between crack patterns and graph theory. © 2022 SPIE.

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A Conceptual Model to Estimate Microplastics (MPs) Mass Flux Discharged from Urban Areas: Identification of Relevant Pathways and Uncertainties

Item Type Journal Article

Author T. Nasrabadi

Author D. Renner

Author H. Ruegner

Author P. Grathwohl

Abstract Many studies concerning the occurrence of MPs in urban areas so far are prone to a variety of uncertainties due to significant deficiencies in temporal/spatial resolution of sampling, representativeness of samples, sampled volume/mass adequacy, size classification, and characterization. Even more important, a remarkable lack exists in assessing the suitability of methods/concepts needed for quantification of MP fluxes. In this study, quantification methods based on mass loadings (flux) analysis rather than details on number and size/shape are suggested for the estimation of MPs occurrence and release from urban areas, which is considered one of the most significant sources of MPs in the environment. MP sources are categorized in different classes e.g.: houses/offices/workshops (textile washing, kitchen work, bathroom activities, municipal waste management, carwash) and traffic, which is mainly responsible for the generation of tire and road wear particles. Release of MPs considers various pathways such as compost/sludge, (un)treated wastewater/surface runoffs/baseflows and wind-driven transport. A conceptual model is proposed to allow quantification of total discharged MP mass flux in relevant pathways normalized by population density. This has the advantage of comparing MP fluxes with other relevant fluxes, such as particulate organic carbon or ubiquitous pollutants. A detailed classification of existing urban wastewater and solid waste collection and disposal networks is presented, which is a prerequisite for the potential implementation of improved waste management strategies. Finally, to achieve an unbiased/ fair share of different cities to global MPs generation, a scheme including factors of tourism intensity, economic well-being, and plastic waste trade is suggested. © University of Tehran 2026.

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A Concise Overview of LeQua@CLEF 2022: Learning to Quantify

Item Type Conference Paper

Author A. Esuli

Author A. Moreo

Author F. Sebastiani

Author G. Sperduti

Editor Barrón-Cedeño A.

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Editor Faggioli G.

Editor Ferro N.

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Editor Sebastiani F.

Editor Macdonald C.

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Abstract LeQua 2022 is a new lab for the evaluation of methods for “learning to quantify” in textual datasets, i.e., for training predictors of the relative frequencies of the classes of interest $Y = \{y_1, \dots, y_n\}$ in sets of unlabelled textual documents. While these predictions could be easily achieved by first classifying all documents via a text classifier and then counting the numbers of documents assigned to the classes, a growing body of literature has shown this approach to be suboptimal, and has proposed better methods. The goal of this lab is to provide a setting for the comparative evaluation of methods for learning to quantify, both in the binary setting and in the single-label multiclass setting; this is the first time that an evaluation exercise solely dedicated to quantification is organized. For both the binary setting and the single-label multiclass setting, data were provided to participants both in ready-made vector form and in raw document form. In this overview article we describe the structure of the lab, we report the results obtained by the participants in the four proposed tasks and subtasks, and we comment on the lessons that can be learned from these results. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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A continual test-time domain adaptation method for online machinery fault diagnosis under dynamic operating conditions

Item Type Journal Article

Author J. Tian

Author Y. Yu

Author H.R. Karimi

Author F. Gao

Author J. Lin

Abstract In practical industrial scenarios, monitoring data is collected in a streaming fashion under dynamic changes in operating conditions of mechanical systems, with continual covariate shift and label shift occurring in the collected data. Traditional transfer learning-based fault diagnosis methods typically involve pre-collecting substantial monitoring data for offline training and testing under static conditions. These approaches cannot adjust the model in real-time to continuous data shifts caused by dynamically changing conditions, resulting in a lack of adaptability and generalization. To overcome this practical challenge, a continual test-time domain

adaptation (CTDA) approach with a teacher-student framework is developed for online machinery fault diagnosis under dynamic operating conditions in this study. Firstly, a class-balanced sampling mechanism is proposed to eliminate the impact of continual condition label shift by enforcing the model to learn from a uniform label distribution. Secondly, a joint positive-negative learning strategy is employed to guide model optimization and reduce the interference from pseudo-label noise. Lastly, the continual covariate shift is mitigated by performing the knowledge alignment between the teacher and student models. Comprehensive experiments on four rotating machinery datasets demonstrate that the proposed method improves average diagnosis accuracy by 3.78% in handling dynamic industrial streaming data compared to existing fault diagnosis methods. © 2025

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A corpus for multilingual document classification in eight languages

Item Type Conference Paper

Author H. Schwenk

Author X. Li

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Editor Tokunaga T.

Abstract Cross-lingual document classification aims at training a document classifier on resources in one language and transferring it to a different language without any additional resources. Several approaches have been proposed in the literature and the current best practice is to evaluate them on a subset of the Reuters Corpus Volume 2. However, this subset covers only few languages (English, German, French and Spanish) and almost all published works focus on the the transfer between English and German. In addition, we have observed that the class prior distributions differ significantly between the languages. We argue that this complicates the evaluation of the multilinguality. In this paper, we propose a new subset of the Reuters corpus with balanced class priors for eight languages. By adding Italian, Russian, Japanese and Chinese, we cover languages which are very different with respect to syntax, morphology, etc. We provide strong baselines for all language transfer directions using multilingual word and sentence embeddings respectively. Our goal is to offer a freely available framework to evaluate cross-lingual document classification, and we hope to foster by these means, research in this important area. © LREC 2018 - 11th International Conference on Language Resources and Evaluation. All rights reserved.

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'current, Cross-lingual, Multilinguality, Information retrieval systems, Document Classification, Cross-lingual document classification, Eight languages, Evaluation framework, Multilingual documents, Reuter corpus volume 2, Reuters, Reuters Corpus Volume 2 (RCV2)

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A Critical Reassessment of the Saerens-Latinne-Decaestecker Algorithm for Posterior Probability Adjustment

Item Type Journal Article

Author A. Esuli

Author A. Molinari

Author F. Sebastiani

Abstract We critically re-examine the Saerens-Latinne-Decaestecker (SLD) algorithm, a well-known method for estimating class prior probabilities ("priors") and adjusting posterior probabilities ("posteriors") in scenarios characterized by distribution shift, i.e., difference in the distribution of the priors between the training and the unlabelled documents. Given a machine learned classifier and a set of unlabelled documents for which the classifier has returned posterior probabilities and estimates of the prior probabilities, SLD updates them both in an iterative, mutually recursive way, with the goal of making both more accurate; this is of key importance in downstream tasks such as single-label multiclass classification and cost-sensitive text classification. Since its publication, SLD has become the standard algorithm for improving the quality of the posteriors in the presence of distribution shift, and SLD is still considered a top contender when we need to estimate the priors (a task that has become known as "quantification"). However, its real effectiveness in improving the quality of the posteriors has been questioned. We here present the results of systematic experiments conducted on a large, publicly available dataset, across multiple amounts of distribution shift and multiple learners. Our experiments show that SLD improves the quality of the posterior probabilities and of the estimates of the prior probabilities, but only when the number of classes in the classification scheme is very small and the classifier is calibrated. As the number of classes grows, or as we use non-calibrated classifiers, SLD converges more slowly (and often does not converge at all), performance degrades rapidly, and the impact of SLD on the quality of the prior estimates and of the posteriors becomes negative rather than

positive. © 2020 ACM.

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A cumulative, pipeline-oriented framework for extreme imbalanced classification: Large-scale benchmarking of 4,100 rare-event models

Item Type Journal Article

Author J. Villanueva-García

Author I. Moral-Arce

Author L.J. García Villalba

Abstract Extreme class imbalance remains a long-standing challenge in supervised learning, particularly when the minority class corresponds to rare but high-impact events such as fraud, equipment failure, or retirement transitions. Although a wide range of imbalance-handling techniques has been proposed-including penalization, resampling, loss shaping, threshold calibration, optimization, and ensemble strategies-the field largely lacks an operational understanding of how these

components should be combined, calibrated, and consolidated within end-to-end pipelines. This paper introduces a cumulative, pipeline-oriented framework for imbalanced classification under extreme skew. Rather than categorizing learning algorithms or paradigms, the framework reorganizes the design space of imbalanced learning around where and how imbalance-aware interventions act within the modeling pipeline. By progressively integrating data-level adjustments, objective and decision-rule shaping, and higher-level consolidation mechanisms, the framework exposes the structural logic underlying effective rare-event classifiers. We validate this framework through a large-scale benchmark comprising 4109 pipeline configurations applied to administrative microdata from the Spanish Social Security system, where monthly retirement detection exhibits a positive-class prevalence of 2.34%. The results reveal clear and interpretable performance regimes as pipeline complexity increases. Explicit pipeline-level ensemble architectures achieve the strongest recall and balance-sensitive metrics, while well-calibrated and optimized intermediate pipelines frequently attain comparable or superior performance in selected metrics—such as F1-score and Matthews correlation coefficient—relative to ensemble baselines, at substantially lower computational cost and with improved interpretability. Distributional and robustness analyses further show that consolidation mechanisms act primarily by stabilizing the upper tail of the performance distribution rather than by improving isolated best-case outcomes. Beyond this application, the proposed framework provides a reproducible benchmarking protocol and a unifying operational lens for understanding when additional methodological sophistication yields robust, stable, and practically meaningful performance gains under extreme class imbalance. © 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license. <http://creativecommons.org/licenses/by/4.0/>

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: J. Villanueva-García; Group of Analysis, Security and Systems (GASS), Department of Software Engineering and Artificial Intelligence (DISIA), Faculty of Computer Science and Engineering, Universidad Complutense de Madrid (UCM), Madrid, Office 431 Calle Profesor José García Santesmases 9 Ciudad Universitaria, 28040, Spain; email: jaimevil@ucm.es; CODEN: ESAPE

A deep learning framework for non-invasive quantification of wing and foot phenotypes in live broiler chickens

Item Type Journal Article

Author Y. Yang

Author X. Ge

Author W. Wu

Author Q. Shen

Author X. Kou

Author X. Hu

Author L. Wang

Author H. Xue

Author L. Li

Abstract Accurate phenotypic quantification in poultry is essential for precision breeding and genetic improvement. Recent advances in deep learning have substantially improved the automation and accuracy of phenotyping. However, non-invasive and precise measurement of specific anatomical traits, particularly wings and feet, remains challenging. To address this challenge, we propose a CT imaging-based framework combined with deep learning to quantify wing and foot characteristics in live chickens. Specifically, we develop the DGCUNet for estimating the volumes of eight phenotypic targets associated with wings and feet. The encoder incorporates dilation transformer modules and stage-specific learning strategies to capture hierarchical representations, while the decoder employs a multi-scale fusion architecture to enhance feature integration. Four CT datasets covering multiple chicken breeds are constructed to validate the proposed framework. On the WB2024 dataset, five-fold cross-validation results demonstrate that DGCUNet achieves a Dice similarity coefficient of 86.41%, a 95th percentile Hausdorff distance of 3.28 mm, and an average symmetric surface distance of 0.69 mm. Furthermore, regression models establish relationships between segmented volumes and ground truth weights, and Bland-Altman analysis illustrates the agreement between estimated and actual weights. Overall, our framework enables fine-grained quantification of individual phenotypic traits rather than coarse whole-body measurements, demonstrating the potential of CT imaging and deep learning for advancing poultry phenotyping. © 2026 Elsevier B.V.

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A deep learning method for extensible microstructural quantification of DP steel enhanced by physical metallurgy-guided data augmentation

Item Type Journal Article

Author C. Shen

Author X. Wei

Author C. Wang

Author W. Xu

Abstract The small sample problem caused by time-consuming annotation has greatly restricted the development of deep learning (DL) methods with high generality and extensibility, hindering the wide application of DL-based microstructural analysis. To address this problem, an extensible microstructural quantification method is proposed by combining physical metallurgy (PM)-guided data augmentation and DL. In this method, PM-guided data augmentation, which combines key PM information highly related to microstructural evaluation and random image transformation, is used to generate microstructures with other processes and therefore construct a comprehensive dataset. The proposed method is successfully applied and validated to dual-phase (DP) steels to classify martensite/ferrite phases and quantify their fractions and grain sizes at various temperatures based on only two SEM images of

annealing temperatures of 750 and 780 °C, in which a comprehensive dataset covering the entire dual phase regime is constructed with guidance from the content and morphology of the phase and dissolved carbide during annealing. Moreover, the good extensibility and data independency of the present method is demonstrated by various DP steels with different compositions and processes collected from the literature. © 2021 Elsevier Inc.

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A Deep Learning Pipeline For Segmentation of Proteus Mirabilis Colony Patterns

Item Type Conference Paper

Author A. Doshi

Author M. Shaw

Author R. Tonea

Author R. Minyety

Author S. Moon

Author A. Laine

Author J. Guo

Author T. Danino

Abstract The motility mechanisms of microorganisms are critical virulence factors, enabling their spread and survival during infection. Motility is frequently characterized by qualitative analysis of macroscopic colonies, yet the standard quantification method has mainly been limited to manual measurement. Recent studies have applied deep learning for classification and segmentation of specific microbial species in microscopic images, but less work has focused on macroscopic colony analysis. Here, we advance computational tools for analyzing colonies of *Proteus mirabilis*, a bacterium that produces a macroscopic bullseye-like pattern via periodic swarming, a process implicated in its virulence. We present a dual-task pipeline for segmenting (1) the macroscopic colony including faint outer swarm rings, and (2) internal ring boundaries, unique features of oscillatory swarming. Our convolutional neural network for patch-based colony segmentation and U-Net with a VGG-11 encoder for ring boundary segmentation achieved test Dice scores of 93.28% and 83.24%, respectively. The predicted masks at times improved on the ground truths from our automated annotation algorithms. We demonstrate how application of our pipeline to a typical swarming assay enables ease of colony analysis and precise measurements of more complex pattern features than those which have been historically quantified. An implementation of our work can be found on <https://github.com/daninolab/proteus-mirabilis>. © 2022 IEEE.

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Notes:

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A Deep Learning-Based Assessment of the Synergy Between New Energy Policies and New Quality Productive Forces: An Integrated Goal-Instrument-Value Framework for Sustainable Development

Item Type Journal Article

Author J. Cao

Author R. Pan

Abstract China has shifted from a stage of rapid growth to a stage of high-quality development. This highlights the critical need for policy frameworks that synergistically align technological innovation with sustainable development. To address the research gap in systematically assessing the collaboration between New Energy Industry (NEI) policies and New Quality Productive Forces (NQPF), this study proposes a three-dimensional “Goal-Instrument-Value” framework. Methodologically, we employ a combination of deep learning models (including LDA topic modeling, LSTM networks, and the Soft EDA algorithm) and policy quantification methods, analyzing 135 NQPF policies and nearly 800 NEI policies. The findings reveal a significant and strengthening synergy between the two policy domains. Notably, a misalignment exists in the goal dimension, where the weight of science and technology in NEI policies remains modest at 20%, indicating substantial potential for enhancement. In the instrument dimension, there is a predominant reliance on economically driven instruments, along with a notable underutilization of environmental instruments. Nevertheless, the overall synergy in policy value, as measured by a specialized New-Force Dictionary and the BM25 model, exhibits a consistent upward trend. Based on these findings, we recommend strengthening investment in NEI technology R&D, increasing the deployment of environment-oriented policy instruments, and establishing a cross-departmental policy synergy mechanism. These measures are crucial to fully harness the synergistic potential of NEI and NQPF policies for accelerating China’s green industrial transformation and achieving its sustainable development goals. © 2025 by the authors.

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A Deep Learning-Based Intelligent Quantification Method for Wellbore Flow Rate

Item Type Conference Paper

Author Y. Li

Author X. Huang

Author T. Zhou

Author Z. Huang

Author F. Li

Author H. Wang

Abstract This paper proposes a classification and recognition method for Distributed Acoustic Sensing (DAS) signals based on a two-dimensional attention-enhanced YOLOv9 network, aimed at intelligent monitoring and identification of downhole flow in wells. Addressing the issues of low accuracy and poor real-time performance of traditional flow monitoring technologies in complex downhole environments, this invention transforms the one-dimensional time-series vibration signals collected by DAS into two-dimensional time-frequency spectra through Short-Time Fourier Transform (STFT). It then combines the YOLOv9 network with its advanced detection capabilities to adaptively extract temporal, frequency, and spatial features of the signals, achieving precise classification of low, medium, and high flow rate vibration patterns. Experimental results demonstrate that this method significantly enhances the accuracy and robustness of signal recognition, resolving the issues of low precision and poor real-time performance encountered by traditional methods in complex downhole environments. The application of this study can provide reliable technical support for monitoring liquid profiles during fracturing processes and interpreting production profiles in the production phase, thereby facilitating real-time control and efficiency improvement in unconventional oil and gas development. Copyright © 2025, Society of Petroleum Engineers

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A deep learning-based pipeline for mosquito detection and classification from wingbeat sounds

Item Type Journal Article**Author** M.S. Yin**Author** P. Haddawy**Author** T. Ziemer**Author** F. Wetjen**Author** A. Supratak**Author** K. Chiamsakul**Author** W. Siritanakorn**Author** T. Chantanalertvilai**Author** P. Sriwichai**Author** C. Sa-ngamuang**Abstract** Mosquito vector-borne diseases such as malaria and dengue constitute some of the most serious public health burdens in tropical and sub-tropical countries. Effective

targeting of disease control efforts requires accurate estimates of mosquito vector population density. The traditional, and still most common, approach to this involves the use of traps along with manual counting and classification of mosquito species. This process is costly and labor-intensive, which hinders its widespread use. In this paper we present a software pipeline for detection and classification of mosquito wingbeat sounds. Since our target platform is low-cost IoT devices, we explore the tradeoff between accuracy and efficiency. When a fast binary mosquito detector precedes the classifier, we can reduce the computational demand compared with use of the classifier alone by a factor of 10. While the accuracy of traditional machine learning model drops from 90% to 64% when reducing the sample rate from 96 kHz to 8 kHz, our deep-learning models maintain an accuracy of almost 83%, even when additionally reducing the bit depth from 24 to 16 bits. We conclude that the combination of an efficient mosquito detector with a convolutional neural network provides for an excellent trade-off between accuracy and efficiency to detect, classify and count mosquitoes. © 2022, The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature.

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A Deep-Learning Phase Picker with Calibrated Bayesian-Derived Uncertainties for Earthquakes in the Yellowstone Volcanic Region

Item Type Journal Article

Author A.D. Armstrong

Author Z. Claerhout

Author B. Baker

Author K.D. Koper

Abstract Traditional seismic phase pickers perform poorly during periods of elevated seismicity due to inherent weakness when detecting overlapping earthquake waveforms. This weakness results in incomplete seismic catalogs, particularly deficient in earthquakes that are close in space and time. Supervised deep-learning (DL) pickers allow for improved detection performance and better handle the overlapping waveforms. Here, we present a DL phasepicking procedure specifically trained on Yellowstone seismicity and designed to fit within the University of Utah Seismograph Stations (UUSS) real-time system. We modify and combine existing DL models to label the seismic phases in continuous data and produce better phase arrival times. We use transfer learning to achieve consistency with UUSS analysts while maintaining robust models. To improve the performance during periods of enhanced seismicity, we develop a data augmentation strategy to synthesize waveforms with two nearly coincident P arrivals. We also incorporate a model uncertainty quantification method, Multiple Stochastic Weight Averaging-Gaussian (MultiSWAG), for arrival-time estimates and compare it to dropout—a more standard approach. We use an efficient, model-agnostic method of empirically calibrating the uncertainties to produce meaningful 90% credible intervals. The credible intervals are used downstream in association, location, and quality assessment. For an in-depth evaluation of our automated method, we apply it to continuous data recorded from 25 March to 3 April 2014, on 20 three-component stations and 14 vertical-component stations. This 10-day period contains an Mw 4.8 event, the largest earthquake in the Yellowstone region since 1980. A seismic analyst manually examined more than 1000 located events, including ~ 855 previously unidentified, and concluded that only two were incorrect. Finally, we present an analyst-created, high-resolution arrival-time data set, including 651 new arrival times, for one hour of data from station WY.YNR for robust evaluation of missed detections before association. Our method identified 60% of the analyst P picks and 81% of the S picks. © Seismological Society of America.

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A design-centered review of in-situ tracer tests for HLW repositories: an overview of radioactive tracer cases, comparison with non-radioactive tracers, and a Korea-tailored strategy

Item Type Journal Article

Author Y. Lee

Author J.-S. Kim

Author J.-D. Kim

Abstract This review is organized around a design-centered framework that links experimental objectives to design premises, test configurations, tracer selection, and interpretation strategies. On this basis, a strategy tailored to the Korean context is proposed. The contents are as follows: (1) An overview of in-situ tracer tests is presented, including testing methodologies, design strategies, and prerequisites to ensure the reliability of safety assessments. (2) The classification criteria for tracers and the significance of radioactive tracer use are clarified. Tracers are categorized according to sorption properties and quantification methods, and the advantages of radioactive tracers—in terms of feasibility, cost-effectiveness, and improved reliability of safety assessments—are described. (3) Case studies of in-situ radioactive tracer injections conducted in Sweden (two sites), Switzerland, Finland, and Belgium (one site each) are introduced and compared with non-radioactive tracer cases to characterize the comparative advantages of radioactive tracers. (4) The results of non-radioactive in-situ tracer tests in Korea are reviewed. Considering the utility of radioactive tracers alongside the social climate in Korea—where in-situ injection of radioactive materials is virtually impossible—a strategic direction based on a dual approach is proposed, combining in-situ non-radioactive field tests with laboratory radioactive tracer experiments as the most feasible pathway. © 2026 Korean Nuclear Society, Published by Elsevier Korea LLC. This is an open access article under the CC BY-NC-ND license. <http://creativecommons.org/licenses/by-nc-nd/4.0/>

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A Detailed Overview of LeQua@CLEF 2022: Learning to Quantify

Item Type Conference Paper

Author A. Esuli

Author A. Moreo

Author F. Sebastiani

Author G. Sperduti

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Editor Augustusplatz 10, Leipzig University of Leipzig

Abstract LeQua 2022 is a new lab for the evaluation of methods for “learning to quantify” in textual datasets, i.e., for training predictors of the relative frequencies of the classes of interest $Y = \{y_1, \dots, y_n\}$ in sets of unlabelled textual documents. While these predictions could be easily achieved by first classifying all documents via a text classifier and then counting the numbers of documents assigned to the classes, a growing body of literature has shown this approach to be suboptimal, and has proposed better methods. The goal of this lab is to provide a setting for the comparative evaluation of methods for learning to quantify, both in the binary setting and in the single-label multiclass setting; this is the first time that an evaluation exercise solely dedicated to quantification is organized. For both the binary setting and the single-label multiclass setting, data were provided to participants both in ready-made vector form and in raw document form. In this overview article we describe the structure of the lab, we report the results obtained by the participants in the four proposed tasks and subtasks, and we comment on the lessons that can be learned from these results. © 2022 Copyright for this paper by its authors.

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A Diffusion-Based Uncertainty Quantification Method to Advance E3SM Land Model Calibration

Item Type Journal Article

Author D. Lu

Author Y. Liu

Author Z. Zhang

Author F. Bao

Author G. Zhang

Abstract Calibrating land surface models and accurately quantifying their uncertainty are crucial for improving the reliability of simulations of complex environmental processes. This, in turn, advances our predictive understanding of ecosystems and supports climate-resilient decision-making. Traditional calibration methods, however, face challenges of high computational costs and difficulties in accurately quantifying parameter uncertainties. To address these issues, we develop a diffusion-based uncertainty quantification (DBUQ) method. Unlike conventional generative diffusion methods, which are computationally expensive and memory-intensive, DBUQ innovates by formulating a parameterized generative model and approximates this model through supervised learning, which enables quick generation of parameter posterior samples to quantify its uncertainty. DBUQ is effective, efficient, and general-purpose, making it suitable for site-specific ecosystem model calibration and broadly applicable for parameter uncertainty quantification across various earth system models. In this study, we applied DBUQ to calibrate the Energy Exascale Earth System Model land model at the Missouri Ozark AmeriFlux forest site. Results indicated that DBUQ produced accurate parameter posterior distributions similar to those from Markov Chain Monte Carlo sampling but with 30 times less computing time. This significant improvement in efficiency suggests that DBUQ can enable rapid, site-level model calibration at a global scale, enhancing our predictive understanding of climate impacts on terrestrial ecosystems. © 2024 Oak Ridge National Laboratory, managed by UT–Battelle, LLC and The Author(s). Journal of Geophysical Research: Machine Learning and Computation published by Wiley Periodicals LLC on behalf of American Geophysical Union.

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A discrete fracture network approach to study the variability of the geological strength index

Item Type Conference Paper

Author T. Miyoshi

Author D. Elmo

Author S. Rogers

Abstract One of the important applications of DFN models is the quantification of rock mass quality through rock mass classification systems such as Geological Strength Index (GSI, Hoek et al., 1995). For instance, Cai et al. (2004) presented a quantitative method to assist in the use of the GSI system for rock mass classification, introducing the concept of equivalent block volume and considering the impact of fracture persistence. However, the analysis was limited to a simple conceptual rock mass and did not account for the complex configurations in term of fracture intersections and connectivity that would arise when considering natural fracture networks. In this context, DFN models allow more realistic representation of 3D network of fractures and better prediction of block fragmentation of rock mass. This paper discusses an alternative method of quantifying the variation of GSI by using the quantification method of Cai et al. (2004) in combination with the block volume cumulative density function (CDF) plots obtained from DFN models. Since DFN approaches exploits the use of fracture data collected from mapping of exposed surfaces and boreholes, including direct physical mapping and indirect mapping by remote sensing techniques, the authors believe the proposed method would provide a better and more objective quantification of GSI variability. Copyright 2018 ARMA.

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Publisher American Rock Mechanics Association (ARMA)

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A fast in situ non-invasive approach to classify mortars from a construction of high historical value

Item Type Journal Article

Author C. García-Florentino

Author M. Maguregui

Author H. Morillas

Author I. Marcaida

Author J.M. Madariaga

Abstract The use of hand-held X-ray fluorescence (HH-XRF) spectrometers has increased in the last years due to its easily portable capacity and its rapid and non-destructive analysis. In this work, HH-XRF is proposed as a fast in situ non-invasive technique to classify mortars from a construction of high historical value. The current HH-XRF devices are able to provide large amounts of information almost instantaneously based on Fundamental Parameters quantification methodology that especially for the analyses of samples with very complex matrixes such as building materials is not very reliable. The accuracy of the semi-quantitative information provided by the HH-XRF spectrometer was compared with a previously optimized WD-XRF quantification method and a complete spectral data treatment followed by Principal Component Analysis is also proposed to classify easily and quickly the mortars used in this construction, without a previous sampling procedure, according to the similarities/differences in their elemental composition. © 2017 Elsevier B.V.

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Principal Component Analysis, Materials classification, Mortars, Non-invasive technique, X-ray fluorescence

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A fast method for simultaneous reconstruction and segmentation in X-ray CT application

Item Type Journal Article**Author** Y. Dong**Author** C. Wu**Author** S. Yan

Abstract In this paper, we propose a new method to solve the minimization problem in a simultaneous reconstruction and segmentation (SRS) model for X-ray computed tomography (CT). The SRS model uses Bayes' rule and the maximum a posteriori (MAP) estimate on the hidden Markov measure field model (HMMFM). The original method [Romanov M, Dahl AB, Dong Y, Hansen PC. Simultaneous tomographic reconstruction and segmentation with class priors. *Inverse Problems Sci Eng.* 2016;24(8):1432–1453] includes a subproblem with logarithmic-summation (log-sum) term, which is non-separable to the classification index. This subproblem was solved by Frank–Wolfe algorithm, which is very time consuming especially when dealing with large-scale CT problems. The starting point of this paper is the commutativity of log-sum operations, where the log-sum problem could be transformed into a sum-log problem by introducing an auxiliary variable. The corresponding sum-log problem for the SRS model is separable. By applying the primal-dual algorithm, the sum-log problem turns into several easy-to-solve convex subproblems. In addition, we introduce an improved model by adding Tikhonov regularization on the SRS model, and give some convergence results for the proposed methods. Experimental results demonstrate that the proposed methods produce comparable results compared with the original SRS method with much less CPU time. © 2021 Informa UK Limited, trading as Taylor & Francis Group.

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A Feature Quantification method based on Binary Tree Oriented for Constructional Frameset of Heterogeneous Models

Item Type Conference Paper

Author N. Shen

Author C. Zhao

Author W. Liu

Author Y. Liu

Author H. Yang

Editor Pei Z.

Abstract To meet the more demanding requirements of the modeling and simulation technology in various fields, more diverse modeling languages and methods are emerging, which in turn makes it more challenging to integrate and reuse heterogeneous models for field applications. Therefore, it is necessary to extract a

constructional frameset of heterogeneous models based on a unified modeling language. This paper presents a method of constructional frameset extraction and feature quantization to address the issues of matching, classification, and composition of multi-domain heterogeneous models. Firstly, heterogeneous models are converted into unified models, which are described by X language. Then, the structural features of each X model are extracted to be incorporated into a predefined template. Then the template can be quantified in the form of a binary tree. The sequence obtained by inorder transversal of the binary tree can be a quantified feature of the template. All the model templates make up the constructional frameset of multi-domain heterogeneous models. The results show that the proposed method can abstract the features of multi-domain heterogeneous models and support a constructional frameset library to rapid modeling, integration, and reuse. © 2022 SPIE.

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A Fetal Fraction Optimized 106-Plex Digital PCR Assay for Non-Invasive Prenatal Testing of Fetal Trisomy

Item Type Journal Article

Author S. Chen

Author X. Luan

Author X. Cao

Author C. Liu

Author L. Zhang

Author W. Shang

Author Y. Zhang

Author Z. Yang

Author C. Xu

Abstract Background/Objectives: Non-invasive prenatal testing (NIPT) for fetal aneuploidy requires accurate trisomy detection together with reliable fetal fraction assessment. This study evaluated the clinical feasibility of a 106-plex digital PCR (dPCR) NIPT assay for trisomies 13, 18, and 21 with internal fetal fraction quantification. Methods: We consecutively recruited 470 women with high-risk singleton pregnancies. Fetal trisomies were detected using dPCR-NIPT and confirmed by invasive prenatal diagnosis. Pregnancies with negative prenatal diagnostic results were followed to birth. Analytical performance and quality control were assessed using trisomic DNA. The euploid cut-off and diagnostic performance were evaluated in two independent maternal plasma sample sets, using invasive diagnosis and clinical outcome as the reference standard. Results: dPCR-NIPT measured fetal fraction irrespective of fetal sex and detected trisomies at fetal fractions $\geq 3\%$ using 5 ng DNA. A total of 12 of 470 plasma samples failed cell-free DNA quality control and were excluded before dPCR testing. Of the remaining 458 samples, 5 had fetal fractions below 3% and were classified as failed tests, yielding a nonreportable rate of 1.1%. Using a cut-off of 6.9 established in 103 training samples, no false-positive or false-negative trisomy calls were observed in the 350-sample testing set, corresponding to 100% sensitivity (95% confidence interval [CI], 85.18–100%) and 100% specificity (95% CI, 98.88–100%) for 23 confirmed trisomies. Conclusions: This proof-of-principle study supports the feasibility of fetal fraction-informed dPCR-NIPT for trisomy detection in high-risk singleton pregnancies. Larger prospective studies in average-risk and earlier-gestation populations are required. © 2026 by the authors.

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A fixed-point algorithm to minimax learning with neural networks

Item Type Journal Article

Author A. Guerrero-Curieses

Author R. Alaiz-Rodríguez

Author J. Cid-Sueiro

Abstract In some real applications, such as medical diagnosis or remote sensing, available training data do not often reflect the true a priori probabilities of the underlying data distribution. The classifier designed from these data may be suboptimal. Building classifiers that are robust against changes in prior probabilities is possible by applying a minimax learning strategy. In this paper, we propose a simple fixed-point algorithm that is able to train a neural minimax classifier [i.e., a classifier minimizing the worst (maximum) possible risk]. Moreover, we present a new parametric family of loss functions that is able to provide the most accurate estimates for the posterior class probabilities near the decision regions, and we also discuss the application of these functions together with a minimax learning strategy. The results of the experiments carried out on different real databases point out the ability of the proposed algorithm to find the minimax solution and produce a robust classifier when the real a priori

probabilities differ from the estimated ones. © 2004 IEEE.

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A Framework for Deep Quantification Learning

Item Type Conference Paper

Author L. Qi

Author M. Khaleel

Author W. Tavanapong

Author A. Sukul

Author D. Peterson

Editor Hutter F.

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Editor Lijffijt J.

Editor Valera I.

Abstract A quantification learning task estimates class ratios or class distribution given a test set. Quantification learning is useful for a variety of application domains such as commerce, public health, and politics. For instance, it is desirable to automatically estimate the proportion of customer satisfaction in different aspects from product reviews to improve customer relationships. We formulate the quantification learning problem as a maximum likelihood problem and propose the first end-to-end Deep Quantification Network (DQN) framework. DQN

jointly learns quantification feature representations and directly predicts the class distribution. Compared to classification-based quantification methods, DQN avoids three separate steps: classification of individual instances, calculation of the predicted class ratios, and class ratio adjustment to account for classification errors. We evaluated DQN on four public datasets, ranging from movie and product reviews to multi-class news. We compared DQN against six existing quantification methods and conducted a sensitivity analysis of DQN performance. Compared to the best existing method in our study, (1) DQN reduces Mean Absolute Error (MAE) by about 35%. (2) DQN uses around 40% less training samples to achieve a comparable MAE. © 2021, Springer Nature Switzerland AG.

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A framework for monitoring classifiers' performance: When and why failure occurs?

Item Type Journal Article

Author D.A. Cieslak

Author N.V. Chawla

Abstract Classifier error is the product of model bias and data variance. While understanding the bias involved when selecting a given learning algorithm, it is similarly important to understand the variability in data over time, since even the One True Model might perform poorly when training and evaluation samples diverge. Thus, it becomes the ability to identify distributional divergence is critical towards pinpointing when fracture points in classifier performance will occur, particularly since contemporary methods such as tenfolds and hold-out are poor predictors in divergent circumstances. This article implement a comprehensive evaluation framework to proactively detect breakpoints in classifiers' predictions and shifts in data distributions through a series of statistical tests. We outline and utilize three scenarios under which data changes: sample selection bias, covariate shift, and shifting class priors. We evaluate the framework with a variety of classifiers and datasets. © Springer-Verlag London Limited 2008.

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A Fuzzy Neural Network-Based System for Alleviating Students' Boredom in English Learning

Item Type Journal Article**Author** L. Yang**Author** X. Wu

Abstract In order to explore students' boredom in English learning, a recognition algorithm based on fuzzy neural network is proposed. The algorithm selects Gaussian membership function and initializes the clustering center obtained by fuzzy c-means algorithm to the center of Gaussian function, and the width of Gaussian function is obtained by the membership and center of fuzzy c-means clustering algorithm. In the construction of base classifier, diversity strategy is adopted to increase its diversity and complementarity. In the selection of base classifiers, the combination of contour coefficient and clustering algorithm is used to determine the number of classifiers to be fused, and the inconsistency measurement method is used to evaluate their differences. In the combination strategy, we learn from the Bayesian thought and dynamically adapt the weight of learning base classifier based on its a priori probability and class conditional probability. The experimental results show that 10 times of cross-validation are carried out, respectively, and the accuracy of each algorithm is given. The algorithm based on tree structure obviously has better performance, followed by the rule-based algorithm, and finally the fuzzy neural network algorithm based on neural network, while the accuracy of SVM and logistic regression algorithm LR is lower. It is proved that the fuzzy neural network can effectively identify students' boredom in English learning. © 2022 Liuhui Yang and Xiuying Wu.

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A Gaussian adaptive resonance theory neural network classification algorithm applied to supervised land cover mapping using multitemporal vegetation index data

Item Type Journal Article

Author D. Muchoney

Author J. Williamson

Abstract Neural network classifiers have been shown to provide supervised classification results that significantly improve on traditional classification algorithms such as the Bayesian (maximum likelihood [ML]) classifier. While the predominant neural network architecture has been the feedforward multilayer perceptron known as backpropagation, Adaptive resonance theory (ART) neural networks offer advantages to the classification of optical remote sensing data for vegetation and land cover mapping. A significant advantage is that it does not require prior specification of the neural net structure, creating as many internal nodes as are needed to represent the calibration (training) data. The Gaussian ARTMAP classification algorithm bases the probability that input training samples belong to specific classes on the parameters of its Gaussian distributions: the means, standard deviations, and a priori probabilities. The performance of the Gaussian ARTMAP classification algorithm in terms of classification accuracy using independent validation data indicated was over 70% accurate when applied to an annual series of monthly 1-km advanced very high resolution radiometer (AVHRR) satellite normalized difference vegetation index (NDVI) data. The accuracies were comparable to those of fuzzy ARTMAP and a univariate decision tree, and significantly higher than a Bayesian classification algorithm. Algorithm testing is based on calibration and validation data developed and applied to Central America to map the International Geosphere-Biosphere Programmed (IGBP) land cover classification system. Thus, it provides a realistic test of the algorithms for operational classification of a regional remote sensing and site dataset.

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A Generalized Approach to Label Shift: The Conditional Probability Shift Model

Item Type Conference Paper

Author P. Teisseyre

Author J. Mielniczuk

Editor Lynce I.

Editor Murano N.

Editor Vallati M.

Editor Villata S.

Editor Chesani F.

Editor Milano M.

Editor Omicini A.

Editor Dastani M.

Abstract In many practical applications of machine learning, a discrepancy often arises between a source distribution from which labeled training examples are drawn and a target distribution for which only unlabeled data is observed. Traditionally, two main scenarios have been considered to address this issue: covariate shift (CS), where only the marginal distribution of features changes, and label shift (LS), which involves a change in the class variable's prior distribution. However, these frameworks do not encompass all forms of distributional shift. This paper introduces a new setting, Conditional Probability Shift (CPS), which captures the case when the conditional distribution of the class variable given some specific features changes while the distribution of remaining features given the specific features and the class is preserved. For this scenario we present the Conditional Probability Shift Model (CPSM) based on modeling the class variable's conditional probabilities using multinomial regression. Since the class variable is

not observed for the target data, the parameters of the multinomial model for its distribution are estimated using the Expectation-Maximization algorithm. The proposed method is generic and can be combined with any probabilistic classifier. The effectiveness of CPSM is demonstrated through experiments on synthetic datasets and a case study using the MIMIC medical database, revealing its superior balanced classification accuracy on the target data compared to existing methods, particularly in situations of conditional distribution shift and no prior distribution shift, which are not detected by LS-based methods. © 2025 The Authors.

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A generalized shapelet-based method for analysis of nanostructured surface imaging

Item Type Journal Article

Author T.J. Akdeniz

Author D.J. Lizotte

Author N.M. Abukhdeir

Abstract The determination of quantitative structure-property relations is a vital but challenging task for nanostructured materials research due to the presence of large-scale spatially varying patterns resulting from nanoscale processes such as self-assembly and nano-lithography. Focusing on nanostructured surfaces, recent advances have been made in automated quantification methods for orientational and translational order using shapelet functions, originally developed for analysis of images of galaxies, as a reduced-basis for surface pattern structure. In this work, a method combining shapelet functions and machine learning is developed and applied to a representative set of images of self-assembled surfaces from experimental characterization techniques including scanning electron microscopy, atomic force microscopy and transmission electron microscopy. The method is shown to be computationally efficient and able to quantify salient pattern features including deformation, defects, and grain boundaries from a broad range of patterns typical of self-assembly processes. © 2018 IOP Publishing Ltd.

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A generalized spatially adaptive sparse grid combination technique with dimension-wise refinement

Item Type Journal Article

Author M. Obersteiner

Author H.-J. Bungartz

Abstract Today, high-dimensional calculations can be found in almost all scientific disciplines. The application of machine learning and uncertainty quantification methods are common examples where high-dimensional problems appear. Typically, these problems are computationally expensive or even infeasible on current machines due to the curse of dimensionality. The sparse grid combination technique is one method to mitigate this effect, but it still does not generate optimal grids for many application scenarios. In such cases, adaptivity strategies are applied to further optimize the grid generation. Generally, adaptive grid generation strategies can be classified as spatially or dimensionally adaptive. One of the most prominent examples is the dimension-adaptive combination technique, which is easy to implement and suitable for cases where dimensions contribute in different magnitudes to the accuracy of the solution. Unfortunately, spatial adaptivity is not possible in the standard combination technique due to the required regular structure of the grids. We therefore propose a new algorithmic variant of the combination technique that is based on the combination of rectilinear grids which can adapt themselves locally to the target function using 1-dimensional refinements. We further increase the efficiency by adjusting point levels with tree rebalancing. Results for numerical quadrature and interpolation show that we can significantly improve upon the standard combination technique and compete with or even surpass common spatially adaptive implementations with sparse grids. At the same time our method keeps the black-box property of the combination technique which makes it possible to apply it to black-box solvers. © 2021 Michael Obersteiner and Hans-Joachim Bungartz

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A generic high-throughput microstructure classification and quantification method for regular SEM images of complex steel microstructures combining EBSD labeling and deep learning

Item Type Journal Article**Author** C. Shen**Author** C. Wang**Author** M. Huang**Author** N. Xu**Author** S. van der Zwaag**Author** W. Xu

Abstract We present an electron backscattered diffraction (EBSD)-trained deep learning (DL) method integrating traditional material characterization informatics and artificial intelligence for a more accurate classification and quantification of complex microstructures using only regular scanning electron microscope (SEM) images. In this method, EBSD analysis is applied to produce accurate ground truth data for guiding the DL model training. An U-Net architecture is used to establish the correlation between SEM input images and EBSD ground truth data using only small experimental datasets. The proposed method is successfully applied to two engineering steels with complex microstructures, i.e., a dual-phase (DP) steel and a quenching and partitioning (Q&P) steel, to segment different phases and quantify phase content and grain size. Alternatively, once properly trained the method can also produce quasi-EBSD maps by inputting regular SEM images. The good generality of the trained models is demonstrated by using DP and Q&P steels not associated with the model training. Finally, the method is applied to SEM images with various states, i.e., different imaging modes, image qualities and magnifications, demonstrating its good robustness and strong application ability. Furthermore, the visualization of feature maps during the segmenting process is utilised to explain the mechanism of this method's good performance. © 2021

Date 2021**DOI** 10.1016/j.jmst.2021.04.009**URL** <https://www.scopus.com/pages/publications/85106960149?origin=resultslist>**Archive** Scopus**Language** English**Volume** 93**Publisher** Chinese Society of Metals**Pages** 191-204**Publication** Journal of Materials Science and Technology**ISSN** 10050302 (ISSN)**Journal Abbr** J. Mater. Sci. Technol.

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Deep learning, Learning systems, Image classification, Quantification methods, Scanning electron microscopy, Ground truth data, Dual phase steel, Electron backscatter diffraction, Microstructure, Complex microstructures, Electron back-scattered diffraction, Microstructure quantification, NET architecture, Quenching and partitioning, Small sample problem, Steel microstructure, Traditional materials

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A highly sensitive and quantitative telomerase activity assay with pancreatic juice is useful for diagnosis of pancreatic carcinoma without problems due to polymerase chain reaction inhibitors: Analysis of 100 samples of pancreatic juice from consecutive patients

Item Type Journal Article

- Author** K. Ohuchida
- Author** K. Mizumoto
- Author** N. Ishikawa
- Author** N. Sato
- Author** E. Nagai
- Author** K. Yamaguchi
- Author** H. Takaishi
- Author** T. Ide
- Author** M. Tanaka

Abstract BACKGROUND. Early detection of pancreatic carcinoma is difficult even with current diagnostic tools. Novel biomarkers and detection techniques are urgently needed. Telomerase activity is a promising diagnostic marker. However, the conventional telomeric repeat amplification protocol (TRAP) assay is not suitable for clinical application because of its complexity, time-consuming nature, and the effects of polymerase chain reaction (PCR) inhibitors in samples leading to difficulties in quantification. METHODS. The authors used a hybridization protection assay in combination with TRAP (TRAP/HPA) to investigate the effects of PCR inhibitors in

pancreatic juice on quantification of telomerase activity. They analyzed 117 consecutive samples of pancreatic juice to determine the feasibility of TRAP/HPA for diagnosis of pancreatic carcinoma. RESULTS. The authors found that TRAP/HPA was 1000-fold more sensitive than the conventional TRAP assay, and that the effects of PCR inhibitors could be avoided by diluting samples. In a large analysis of pancreatic juice samples with TRAP/HPA, 17 samples were excluded from the final analysis because of insufficient follow-up periods or inadequate treatment of the samples. Relative telomerase activity (RTA) in samples from patients with pancreatic carcinoma was significantly higher in comparison to samples from patients with pancreatitis and 13 (61.9%) of 21 samples from patients with pancreatic carcinoma showed high RTA (> 4 U). Meanwhile, high RTAs were observed in 4 of 35 (11.4%) samples from patients with intraductal papillary mucinous tumor and in 1 of 40 samples (2.5%) from patients without malignant disease. CONCLUSIONS. TRAP/HPA accurately evaluated weak telomerase activity in pancreatic juice samples without the problem due to PCR inhibitors. This large analysis of nonselected pancreatic juice samples suggested that TRAP/HPA is a promising approach for the diagnosis of pancreatic carcinoma. © 2004 American Cancer Society.

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Notes:

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A hippocampal-based biosensor for neurotoxins detection and classification using a novel short-term plasticity quantification method

Item Type Journal Article

Author G. Gholmieh

Author S. Courellis

Author V. Marmarelis

Author M. Baudry

Author T.W. Berger

Abstract Current sensors such as the handheld chemical agent monitor (CAM by Graseby Dynamics) and the M90-D1-C (EnviroNics, Finland) are capable of detecting and identifying nerve agents but come short of identifying unknown chemical compounds. Hence, there is a need to develop a new type of sensor that can detect newly uncharacterized chemical compounds. In this chapter, we present a newly developed biosensor for rapid neurotoxins screening based on the quantification of short-term plasticity (STP) of CA1 hippocampal system in vitro using a newly developed analytical approach. The hippocampal slice with its preserved trisynaptic pathway (Andersen et al., 1971) has been a very popular experimental preparation for investigating memory mechanisms and drug effects using long-term potentiation (LTP) and short-term plasticity (Bliss and Lomo, 1973; Fountain and Teyler, 1995; Xie et al., 1997; Buonomano, 1999, 2000). Short-term plasticity (STP)-based methods are typically chosen to assess drug effects of rapidly acting chemical compounds inasmuch as they require less experimental time. Current experimental methods for measuring STP are mostly based on the analysis of paired pulse (Dobrunz et al., 1997; Leung et al., 1994; Creager et al., 1980) or fixed frequency train stimulation (Papatheodoropoulos et al., 2000; Buonomano 1999; Pananceau et al., 1998; Castro-Alamancos et al., 1997). These methods lack the ability to test all the possible time intervals in a time-efficient manner. Based on a variant of the Volterra approach, we have developed and implemented an efficient method to characterize and quantify STP using Poisson distributed random train (RIT) stimulation (Gholmieh et al., 2002). It can be viewed as a hybrid between the paired pulse and the fixed frequency train approaches. It is, however, more time efficient than both of them. Responses to a large variety of interimpulse intervals can be evaluated in three minutes in contrast to several hours when using the paired pulse approach (Gholmieh et al., 2001, 2002). The computed STP descriptors of the underlying nonlinear dynamics are the first- and second-order Volterra kernels that describe the state of the system. The second-order kernel can be further linearly decomposed into nine Laguerre functions. The first-order kernel and the Laguerre coefficients of the second-order kernel expansion can be viewed as the characteristics of the system. These characteristics are speculated to change under different chemical environments, giving each drug a unique signature. The proposed biosensor is comprised of a multi-electrode array, a hippocampal slice, and MATLAB software for data acquisition and online nonlinear STP analysis. It was tested using chemical

compounds belonging to six different classes: DNQX and CNQX (AMPA receptor antagonists), DAP5 (an NMDA receptor antagonist), carbachol (a cholinergic agonist), trimethylopropane phosphate (TMPP) and picrotoxin (GABAA antagonists), tetraethylammonium (a potassium channel blocker), and valproate (an antiepileptic medication). The current chapter summarizes our successful efforts for the last four years to build a hippocampal-based biosensor for neurotoxin detection and classification. Section 12.2 describes the novel modeling approach in detail. Section 12.3 presents the nonlinear characteristics of the CA1 region obtained through the described nonlinear modeling approach. Section 12.4 proves that the novel method can detect changes in STP. Section 12.5 tests our hypothesis that each drug can have a unique signature, and Section 12.6 presents a proof of principle solution for long-term monitoring. © 2006 Springer Science+Business Media, Inc. All rights reserved.

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A hybrid ensemble learning method for tourist route recommendations based on geo-tagged social networks

Item Type Journal Article

Author L. Wan

Author Y. Hong

Author Z. Huang

Author X. Peng

Author R. Li

Abstract Geo-tagged travel photos on social networks often contain location data such as points of interest (POIs), and also users' travel preferences. In this paper, we propose a hybrid ensemble learning method, BAYes-Knn, that predicts personalized tourist routes for travelers by mining their geographical preferences from these location-tagged data. Our method trains two types of base classifiers to jointly predict the next travel destination: (1) The K-nearest neighbor (KNN) classifier quantifies users'

location history, weather condition, temperature and seasonality and uses a feature-weighted distance model to predict a user's personalized interests in an unvisited location. (2) A Bayes classifier introduces a smooth kernel function to estimate a-priori probabilities of features and then combines these probabilities to predict a user's latent interests in a location. All the outcomes from these subclassifiers are merged into one final prediction result by using the Borda count voting method. We evaluated our method on geo-tagged Flickr photos and Beijing weather data collected from 1 January 2005 to 1 July 2016. The results demonstrated that our ensemble approach outperformed 12 other baseline models. In addition, the results showed that our framework has better prediction accuracy than do context-aware significant travel-sequence-patterns recommendations and frequent travel-sequence patterns. © 2018, © 2018 Informa UK Limited, trading as Taylor & Francis Group.

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A Hybrid Time-Series Simulation Framework for Provincial Carbon Emissions Using Multi-Factor Decomposition and Deep Learning

Item Type Journal Article

Author L. Zhang

Author Y. Ye

Author X. Ren

Author X. Qiu

Author Z. Sun

Author W. Zhou

Author D. Han

Abstract Accurate time-series simulation of carbon emissions for both the whole society and the electricity industry is pivotal for realizing China's "Dual Carbon" goals. This research constructs a hybrid simulation architecture integrating factor decomposition with deep learning to quantify emission trajectories for both the whole society and the electricity industry in Anhui Province. First, the extended Kaya identity and Logarithmic Mean Divisia Index (LMDI) are employed to analyze socioeconomic drivers. The decomposition analysis indicates that per capita income is the primary driver of carbon emissions, whereas energy intensity exerts the strongest inhibitory effect. Subsequently, Variational Mode Decomposition (VMD) is applied to the nonstationary emission series to produce multi-scale sub-signals, which are then fed into a predictive model comprising a Bayesian-optimized (BO) Transformer coupled with Long Short-Term Memory (LSTM) networks. The study establishes three distinct evolution scenarios: Moderate Sustainability (MS), Business as Usual (BAU), and Strong Economic Growth (SEG). Simulation results indicate that under MS, carbon emissions from the whole society and the electricity industry peak in 2029 at 435.2 Mt and 2030 at 281.2 Mt, respectively. Conversely, the SEG scenario delays the peak of the whole society to 2034, while the electricity industry fails to peak before 2035. These findings reveal significant risks of temporal asynchrony between the whole society and the electricity industry peaks, providing robust methodological support for regional decarbonization planning. © 2026 by the authors.

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A Joint Training-Calibration Framework for Test-Time Personalization with Label Shift in Federated Learning

Item Type Conference Paper

Author J. Xu

Author S.-L. Huang

Abstract The data heterogeneity has been a challenging issue in federated learning in both training and inference stages, which motivates a variety of approaches to learn either personalized models for participating clients or test-time adaptations for unseen clients. One such approach is employing a shared feature representation and a customized classifier head for each client. However, previous works either neglect the global head with rich knowledge or assume the new clients have enough labeled data, which significantly limit their broader practicality. In this work, we propose a lightweight framework to tackle with the label shift issue during the model deployment by test priors estimation and model prediction calibration. We also demonstrate the importance of training a balanced global model in FL so as to guarantee the general effectiveness of prior estimation approaches. Evaluation results on benchmark datasets demonstrate the superiority of our framework for model adaptation in unseen clients with unknown label shifts. © 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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A learning-based automatic segmentation and quantification method on left ventricle in gated myocardial perfusion SPECT imaging: A feasibility study

Item Type Journal Article**Author** T. Wang**Author** Y. Lei**Author** H. Tang**Author** Z. He**Author** R. Castillo**Author** C. Wang**Author** D. Li**Author** K. Higgins**Author** T. Liu**Author** W.J. Curran**Author** W. Zhou**Author** X. Yang

Abstract Background: The performance of left ventricular (LV) functional assessment using gated myocardial perfusion SPECT (MPS) relies on the accuracy of segmentation. Current methods require manual adjustments that are tedious and subjective. We propose a novel machine-learning-based method to automatically segment LV myocardium and measure its volume in gated MPS imaging without human intervention. Methods: We used an end-to-end fully convolutional neural network to segment LV myocardium by delineating its endocardial and epicardial surface. A novel compound loss function, which encourages similarity and penalizes discrepancy between prediction and training dataset, is utilized in training stage to

achieve excellent performance. We retrospectively investigated 32 normal patients and 24 abnormal patients, whose LV myocardial contours automatically segmented by our method were compared with those delineated by physicians as the ground truth. Results: The results of our method demonstrated very good agreement with the ground truth. The average DSC metrics and Hausdorff distance of the contours delineated by our method are larger than 0.900 and less than 1 cm, respectively, among all 32 + 24 patients of all phases. The correlation coefficient of the LV myocardium volume between ground truth and our results is 0.910 ± 0.061 ($P < 0.001$), and the mean relative error of LV myocardium volume is $-1.09 \pm 3.66\%$. Conclusion: These results strongly indicate the feasibility of our method in accurately quantifying LV myocardium volume change over the cardiac cycle. The learning-based segmentation method in gated MPS imaging has great promise for clinical use. © 2019, American Society of Nuclear Cardiology.

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adult, female, Female, human, Humans, male, Male, middle aged, Middle Aged, machine learning, aged, Aged, segmentation, Heart, article, reproducibility, Reproducibility of Results, algorithm, Algorithms, Machine Learning, quantitative analysis, prediction, procedures, controlled study, Probability, Neural Networks, Computer, correlation coefficient, probability, clinical article, feasibility study, image processing, Retrospective Studies, retrospective study, diagnostic imaging, Image Processing, Computer-Assisted, physician, pathology, Feasibility Studies, heart ventricle, Heart Ventricles, heart left ventricle, heart, perfusion, single photon emission computed tomography, SPECT, Tomography, Emission-Computed, Single-Photon, cardiac muscle, Myocardium, loss of function mutation, heart cycle, automated pattern recognition, Pattern Recognition, Automated, endocardium, heart muscle perfusion, Myocardial perfusion, myocardial perfusion imaging, Myocardial Perfusion Imaging, Perfusion

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EID: 2-s2.0-85061041289

Export Date: 20 September 2026; Cited By: 114; Funding details: National Institutes of Health, NIH; National Cancer Institute, NCI, (R01CA215718); National Cancer Institute, NCI; American Heart Association, AHA, (17AIREA33700016); American Heart Association, AHA; Winship Cancer Institute; Correspondence Address: X. Yang; Department of Radiation Oncology and Winship Cancer Institute, Emory University, Atlanta, 30322, United States; email: xiaofeng.yang@emory.edu; W. Zhou; School of Computing, University of Southern Mississippi, Long Beach, 39560, United States; email: weihua.zhou@usm.edu; CODEN: JNCAE

A learning-based optimal uncertainty quantification method and its application to ballistic impact problems

Item Type Journal Article

Author X. Sun

Author B. Liu

Abstract This paper concerns the study of optimal (supremum and infimum) uncertainty bounds for systems where the input (or prior) probability measure is only partially/imperfectly known (e.g., with only statistical moments and/or on a coarse topology) rather than fully specified. Such partial knowledge provides constraints on the input probability measures. The theory of Optimal Uncertainty Quantification allows us to convert the task into a constraint optimization problem where one seeks to compute the least upper/greatest lower bound of the system's output uncertainties by finding the extremal probability measure of the input. Such optimization requires repeated evaluation of the system's performance indicator (input to performance map) and is high-dimensional and non-convex by nature. Therefore, it is difficult to find the optimal uncertainty bounds in practice. In this paper, we examine the use of machine learning, especially deep neural networks, to address the challenge. We achieve this by introducing a neural network classifier to approximate the performance indicator combined with the stochastic gradient descent method to solve the optimization problem. We demonstrate the learning-based framework on the uncertainty quantification of the impact of magnesium alloys, which are promising light-weight structural and protective materials. Finally, we show that the approach can be used to construct maps for the performance certificate and safety design in engineering practice. © 2023

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DOI 10.1016/j.mechmat.2023.104727

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Publication Mechanics of Materials

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Journal Abbr Mech Mater

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Notes:

EID: 2-s2.0-85162982504

Export Date: 20 September 2026; Cited By: 5; Funding details: Granta Design; University of Kentucky, UK; Correspondence Address: B. Liu; University of Cambridge, Cambridge, United Kingdom; email: bl377@cam.ac.uk; CODEN: MSMSD

A Machine Learning based uncertainty quantification for compressive strength of high-performance concrete

Item Type Journal Article

Author N. Vu-Bac

Author T. Le-Anh

Author T. Rabczuk

Abstract High performance concrete (HPC) properties depend on both its constituent materials and their interaction. This study presents a machine learning framework to quantify the effects of constituents on HPC compressive strength. We first develop a stochastic constitutive model using experimental data and subsequently employ an uncertainty quantification method to identify key parameters in relation to the compressive strength of HPC. The resultant sensitivity indices indicate that fly ash content has the strongest influence on compressive strength, followed by concrete age at test and blast surface slag content. © Higher Education Press 2025.

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Journal Abbr Front. Struct. Civ. Eng.

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machine learning, Machine-learning, Convolutional neural networks, Neural-networks, Learning frameworks, Uncertainty quantifications, Stochastic models, uncertainty quantification, Artificial Neural Networks, compressive strength of concrete, Compressive strength of concrete, Concrete compressive strength, Concrete properties, Constituent materials, dependent variables, Dependent variables, High performance concrete, High-performance concrete

Notes:

EID: 2-s2.0-105006888300

Export Date: 20 September 2026; Cited By: 10; Funding details: University of Science Ho Chi Minh City, HCMUS; Correspondence Address: N. Vu-Bac; Faculty of Civil Engineering, Ho Chi Minh University of Technology (HCMUT), Ho Chi Minh City, 740500, Viet Nam; email: vbnam@hcmut.edu.vn

A Machine Learning-Based Image Segmentation Method to Quantify In Vitro Osteoclast Culture Endpoints

Item Type Journal Article

Author B.K. Davies

Author A.P. Hibbert

Author S.J. Roberts

Author H.C. Roberts

Author J.C. Tickner

Author G. Holdsworth

Author T.R. Arnett

Author I.R. Orriss

Abstract Quantification of in vitro osteoclast cultures (e.g. cell number) often relies on manual counting methods. These approaches are labour intensive, time consuming and result in substantial inter- and intra-user variability. This study aimed to develop and validate an automated workflow to robustly quantify in vitro osteoclast cultures. Using ilastik, a machine learning-based image analysis software, images of tartrate resistant acid phosphatase-stained mouse osteoclasts cultured on dentine discs were used to train the ilastik-based algorithm. Assessment of algorithm training showed that osteoclast numbers strongly correlated between manual- and automatically quantified values ($r = 0.87$). Osteoclasts were consistently faithfully segmented by the model when visually compared to the original reflective light images. The ability of this method to detect changes in osteoclast number in response to different treatments was validated using zoledronate, ticagrelor, and co-culture with MCF7 breast cancer cells. Manual and automated counting methods detected a 70% reduction ($p < 0.05$) in osteoclast number, when cultured with 10 nM zoledronate and a dose-dependent decrease with 1–10 μM ticagrelor ($p < 0.05$). Co-culture with MCF7 cells increased osteoclast number by $\geq 50\%$ irrespective of quantification method. Overall, an automated image segmentation and analysis workflow, which consistently and sensitively identified in vitro osteoclasts, was developed. Advantages of this

workflow are (1) significantly reduction in user variability of endpoint measurements (93%) and analysis time (80%); (2) detection of osteoclasts cultured on different substrates from different species; and (3) easy to use and freely available to use along with tutorial resources. © 2023, The Author(s).

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DOI 10.1007/s00223-023-01121-z

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Export Date: 20 September 2026; Cited By: 9; Funding details: UCB Pharma; Biotechnology and Biological Sciences Research Council, BBSRC; UK Research and Innovation, UKRI, (BB/M009513/1); UK Research and Innovation, UKRI; Correspondence Address: I.R. Orriss; Department of Comparative Biomedical Sciences, Royal Veterinary College, London, Royal College Street, NW1 0TU, United Kingdom; email: iorriss@rvc.ac.uk; CODEN: CTIND

A MALDI-TOF mass spectrometry-based haemoglobin chain quantification method for rapid screen of thalassaemia

Item Type Journal Article

Author J. Zhang

Author Z. Liu

Author R. Chen

Author Q. Ma

Author Q. Lyu

Author S. Fu

Author Y. He

Author Z. Xiao

Author Z. Luo

Author J. Luo

Author X. Wang

Author X. Liu

Author P. An

Author W. Sun

Abstract Background: Thalassaemia is one of the most common inherited monogenic diseases worldwide with a heavy global health burden. Considering its high prevalence in low and middle-income countries, a cheap, accurate and high-throughput screening test of thalassaemia prior to a more expensive confirmatory diagnostic test is urgently needed. Methods: In this study, we constructed a machine learning model based on MALDI-TOF mass spectrometry quantification of haemoglobin chains in blood, and for the first time, evaluated its diagnostic efficacy in 674 thalassaemia (including both asymptomatic carriers and symptomatic patients) and control samples collected in three hospitals. Parameters related to haemoglobin imbalance (α -globin, β -globin, γ -globin, α/β and $\alpha-\beta$) were used for feature selection before classification model construction with 8 machine learning methods in cohort 1 and further model efficiency validation in cohort 2. Results: The logistic regression model with 5 haemoglobin peak features achieved good classification performance in validation cohort 2 (AUC 0.99, 95% CI 0.98–1, sensitivity 98.7%, specificity 95.5%). Furthermore, the logistic regression model with 6 haemoglobin peak features was also constructed to specifically identify β -thalassaemia (AUC 0.94, 95% CI 0.91–0.97, sensitivity 96.5%, specificity 87.8% in validation cohort 2). Conclusions: For the first time, we constructed an inexpensive, accurate and high-throughput classification model based on MALDI-TOF mass spectrometry quantification of haemoglobin chains and demonstrated its great potential in rapid screening of thalassaemia in large populations. Key messages Thalassaemia is one of the most common inherited monogenic diseases worldwide with a heavy global health burden. We constructed a machine learning model based on MALDI-TOF mass spectrometry quantification of haemoglobin chains to screen for thalassaemia. © 2022 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

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Journal Abbr Ann. Med.

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adult, Article, human, Humans, machine learning, correlation analysis, controlled study, major clinical study, area under the curve, receiver operating characteristic, clinical practice, cohort analysis, blood sampling, sensitivity and specificity, feature selection, blood, hemoglobin, asymptomatic carrier, beta thalassemia, beta-Thalassemia, genotyping, haemoglobin, hemoglobin alpha chain, hemoglobin beta chain, hemoglobin chain, hemoglobin F, hemoglobin gamma chain, Hemoglobins, heterozygote, Heterozygote, MALDI-TOF, matrix assisted laser desorption ionization time of flight mass spectrometry, matrix-assisted laser desorption-ionization mass spectrometry, molecular diagnostics, Spectrometry, Mass, Matrix-Assisted Laser Desorption-Ionization, thalassemia, Thalassemia

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A Markov random field approach for sidescan sonar change detection

Item Type Journal Article

Author S. Wei

Author H. Leung

Abstract One of the main challenges in mine detection using sidescan sonar images is the high density of mine-like objects (MLOs) in a clutter environment. This paper proposes an image change detection technique for bitemporal images which suppresses false alarms efficiently without involving large training data sets. The proposed approach uses the spatial dependence of a stationary object between bitemporal images to eliminate the differences caused by a position error. Bayes theory is then employed to classify the changed and unchanged objects. In particular, the a priori probabilities are formulated by the Markov random field (MRF). The likelihood functions are modeled using the coarseness difference of objects as the test statistics, and the parameters are estimated using the expectation-maximization (EM) method. Real sidescan sonar data are used to validate the proposed method. Results show that the proposed MRF change detection method is robust to the poor quality of object boundaries due to speckle noise, and outperforms the conventional pixel-level change detection methods. © 1976-2012 IEEE.

Date 2012

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Tags:

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Notes:

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Export Date: 20 September 2026; Cited By: 15; Correspondence Address: S. Wei; Department of Electrical and Computer Engineering, University of Calgary, Calgary, AB T2N 1N4, Canada; email: wishuang@gmail.com; CODEN: IJOED

A method for early diagnosis of lung cancer from tumor originated DNA fragments using plasma cfDNA methylome and fragmentome profiles

Item Type Journal Article

Author Y.J. Kim

Author H. Jeon

Author S. Jeon

Author S.-H. Lee

Author C. Kim

Author J.-H. Ahn

Author H. Um

Author Y.J. Woo

Author S.-H. Jeong

Author Y. Kim

Author H.-Y. Park

Author H.-J. Oh

Author H.-J. Cho

Author J.-H. Bae

Author J.-H. Kim

Author S. An

Author S.-B. Kang

Author S. Jho

Author O. Biro

Author D. Kis

Author B.C. Kim

Author Y. Kim

Author J.H. Kim

Author B.-C. Kim

Author J. Bhak

Author I.-J. Oh

Abstract Early detection is critical for minimizing mortality from cancer. Plasma cell-free DNA (cfDNA) contains the signatures of tumor DNA, allowing us to quantify the signature and diagnose early-stage tumors. Here, we report a novel tumor fragment quantification method, TOF (Tumor Originated Fragment) for the diagnosis of lung cancer by quantifying and analyzing both the plasma cfDNA methylation patterns and fragmentomic signatures. TOF utilizes the amount of ctDNA predicted from the methylation density information of each cfDNA read mapped on 6243 lung-tumor-specific CpG markers. The 6243 tumor-specific markers were derived from lung tumor tissues by comparing them with corresponding normal tissues and healthy blood from public methylation data. TOF also utilizes two cfDNA fragmentomic signatures: 1) the short fragment ratio, and 2) the 5' end-motif profile. We used 298 plasma samples to analyze cfDNA signatures using enzymatic methyl-sequencing data from 201 lung cancer patients and 97 healthy controls. The TOF score showed 0.98 of the area under the curve in correctly classifying lung cancer from normal samples. The TOF score resolution was high enough to clearly differentiate even the early-stage non-small cell lung cancer patients from the healthy controls. The same was true for small cell lung cancer patients. © 2022 Elsevier Ltd

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Notes:

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A method of evidence association mining on target identification

Item Type Conference Paper

Author X. Wang

Author H. Wang

Abstract in the target identification system, it is difficult to get the target identification result because of massive and heterogeneous data obtaining from various information sensors. In this paper, a probability statistical method is used to analyze the quantification method through the association analysis of support evidences and the fact of the evidence. Using this novel algorithm, a large number of evidences which are not related to the facts could be filtered out quickly, and the evidences of different aspects which reflect the comprehensive identification of targets could be extracted. Meanwhile, the evidence history information classification model and the evidence-related intensity matrix are established, and the model is updated iteratively according to the new evidences contained in multiple information sources in the target identification system. Based on this method, this paper carries out a simulation experiment of evidence correlation of target identification, which verifies the feasibility and validity of the method. © 2017 IEEE.

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A Methodology for Disaster Construction &Demolition Debris Quantification

Item Type Journal Article

Author E. Mansour

Author I. Srouf

Author S. Nassar

Abstract Post disaster Construction and Demolition (C&D) waste quantification is critical in contexts where recurrent conflicts and natural disasters have led to significant infrastructure damage and large-scale debris accumulation. Debris quantification is a natural first step for efficient recovery planning, removal and management, and circular economy strategies. However, existing quantification methodologies lack national or regional standardization and fail to combine both rapid assessment as well as material characterization. Therefore, this study proposes a structured six-step debris quantification framework in post-disaster settings that integrates field data collection, material classification, density-based calculations, and demographic normalization to estimate per capita C&D waste generation. The framework was integrated in an automated decision-making tool using Visual Basic for Applications

(VBA) macros in Microsoft Excel. The tool was applied to fully collapsed buildings, amid the 2023–2024 war in Lebanon. Based on an average household size of 4 to 5 persons and two apartments per floor in these regions, results indicated that the calculated C&D waste generation rate was approximately 55–69 tons per capita—an order of magnitude higher than typical demolition or municipal solid waste figures reported globally. Additionally, the study showed the tool's effectiveness in producing rapid, replicable, and verifiable estimates, enabling faster operational decision-making compared to traditional, resource-intensive post-conflict assessments. Findings highlight the importance of standardized debris quantification protocols in enhancing disaster waste management readiness, with implications for contingency planning, resource allocation, and integration of remote sensing in damage assessment. © 2026 The Author(s)

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A microfluidic approach to study variations in *Chlamydomonas reinhardtii* alkaline phosphatase activity in response to phosphate availability

Item Type Journal Article

Author A. Rahnama

Author M. Vaithiyanathan

Author L. Briceno-Mena

Author T.M. Dugas

Author K.L. Yates

Author J.A. Romagnoli

Author A.T. Melvin

Abstract Algal growth depends strongly on phosphorus (P) as a key nutrient, underscoring the significance of monitoring P levels. Algal species display a sensitive response to fluctuations in P availability, notably through the expression of alkaline phosphatase (AP) when challenged with P-depletion. As such, alkaline phosphatase activity (APA) serves as a valuable metric for P availability, offering insights into how algae utilize and fix available P resources. However, current APA quantification methods lack single cell resolution, while also being time- and reagent consuming. Microfluidics offers a promising cost-effective solution to these limitations, providing a platform for precise single-cell analysis. In this study, a trap-based microfluidic device was integrated with a commercially available AP live stain to study the single cell APA response of a model algae strain, *Chlamydomonas reinhardtii*, when exposed to different exogenous P levels. A three-step culture-starve-spike process was used to induce APA in cells cultured under two different basal P levels (1 and 21 mM). When challenged with different spiked P levels (ranging from 0.1-41 mM), *C. reinhardtii* cells demonstrated a highly heterogeneous APA response. Two-way ANOVA confirmed that this response is influenced by both spiked and basal P levels. Utilizing an unsupervised machine learning approach (HDBSCAN), distinct subpopulations of *C. reinhardtii* cells were identified exhibiting varying levels of APA at the single-cell level. These subpopulations encompass significant groups of individual cells with either notably high or low APA, contributing to the overall behavior of the cohorts. Considerable intrapopulation differences in APA were observed across cohorts with similar average behavior. For instance, while some cohorts exhibited a concentrated distribution around the overall average APA, others displayed subpopulations dispersed across a wider range of APA levels. This underscores the potential bias introduced by analyzing a small number of cells in bulk, which may skew results by overrepresenting extreme behavioral subpopulations. The findings of this study highlight the need for analytical approaches that account for single cell heterogeneity in APA and demonstrate the utility of microfluidics as a well-suited means for such investigations. This study illuminates the complexities of APA regulation at the single cell level, providing crucial insights that advance our understanding of algal phosphorus metabolism and environmental responses. © 2024 The Royal Society of Chemistry.

Date 2024

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Notes:

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A miRNA signature for defining aggressive phenotype and prognosis in gliomas

Item Type Journal Article

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Author B. Pasculli

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Author V. D'Angelo

Author N. Icolaro

Author M. Coco

Author L. Dimitri

Author P. Graziano

Author M. Copetti

Author V.M. Valori

Author E. Maiello

Author M. Carella

Author V.M. Fazio

Author P. Parrella

Abstract Gliomas represent a disparate group of tumours for which there are to date no cure. Thus, there is a recognized need for new diagnostic and therapeutic approaches based on increased understanding of their molecular nature. We performed the comparison of the microRNA (miRNA) profile of 8 WHO grade II gliomas and 24 higher grade tumours (2 WHO grade III and 22 glioblastomas) by using the Affymetrix GeneChip miRNA Array v. 1.0. A relative quantification method (RT-qPCR) with standard curve was used to confirm the 22 miRNA signature resulted by array analysis. The prognostic performances of the confirmed miRNAs were estimated on the Tumor

Cancer Genome Atlas (TCGA) datasets. We identified 22 miRNAs distinguishing grade II gliomas from higher grade tumours. RT-qPCR confirmed the differential expression in the two patients' groups for 13 out of the 22 miRNAs. The analysis of the Glioblastoma Multiforme (GBM) and Lower Grade Glioma (LGG) datasets from TCGA demonstrated the association with prognosis for 6 of those miRNAs. Moreover, in the GBM dataset miR-21 and miR-210 were predictors of worse prognosis in both univariable and multivariable Cox regression analyses (HR 1.19, $p = 0.04$, and HR 1.18, $p = 0.029$ respectively). Our results support a direct contribution of miRNAs to glioma cancerogenesis and suggest that miR-21 and miR-210 may play a role in the aggressive clinical behaviour of glioblastomas. © 2014 Barbano et al. This is an open-access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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A Mobile Application for Electronic Parts Counting and Identification in Physics Laboratory

Item Type Conference Paper

Author T. Leelawattananon

Author S. Chittayasothorn

Abstract Counting of small objects such as resistors, capacitors, and inductors in physics laboratories are tedious and challenging tasks because of their numbers and small size. In this paper we present the development of a mobile application for counting and identifying small electronics parts. The system employs object edges detection using the difference of gaussian operator and classify the objects using template matching. Pictures taken using mobile phone camera are used as the input to the mobile application which correctly classify and count the objects. Electronics objects used in the counting and classification include resistors, mylar capacitors, and ceramic capacitors. The mobile application helps reduce the work required by laboratory personnel for intermediate physics laboratory management and is now deployed in the physics laboratory of the School of Science, King Mongkut's Institute of Technology Ladkrabang. © 2025 IEEE.

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Tags:

Object recognition, Object detection, Mobile edge computing, Mobile computing, Laboratories, Matchings, Similarity measurements, Edge detection, Human resource management, Ceramic capacitors, Difference of Gaussians, Edge Detection, Electronic parts, Mobile app, Mobile applications, Object edge, Physics laboratories, Resistor capacitors, Similarity Measurement, Small objects, Template matching, Template Matching

Notes:

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Export Date: 20 September 2026; Cited By: 0; Conference name: 13th International Conference on Information and Education Technology, ICIET 2025; Conference date: 18 April 2025 through 20 April 2025; Conference code: 210017

A model-free Bayesian classifier

Item Type Journal Article

Author Z. Geng

Author Q. Meng

Author J. Bai

Author J. Chen

Author Y. Han

Author Q. Wei

Author Z. Ouyang

Abstract When using the naïve Bayesian classifier (NBC), it is necessary to assume that the attributes are independent. Although the relationship between the attributes can be analyzed by the Bayesian network (BN), the network structure and parameters need to be trained. Meanwhile, each continuous attribute is given a priori probability distribution for both the NBC and the BN. If the actual probability distribution of the samples is inconsistent with the priori probability distribution, the classification result is poor. In this paper, a model-free Bayesian classifier (MFBC) is proposed to cope with the drawbacks of the NBC and the BN. The information of the sample space and the joint probability density can be obtained based on the nearest neighbor (NN) strategy. The proposed MFBC is able to cope with discrete or continuous attributes even if there is no a priori probability distribution of the attributes. It is also not necessary to build a network structure to obtain a unitive calculation framework for the attributes. Moreover, the probability distribution for limited amounts of samples is approximated using the NN method. Then, the classification label of a sample can be predicted by the MFBC. The sensitivity analysis of the MFBC is obtained using the UCI data sets. Compared with some classical and ensemble classifiers, the analysis results verify the effectiveness and convergence performance of the MFBC. © 2019 Elsevier Inc.

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Bayesian networks, Sensitivity analysis, Probability distributions, Classifiers, Classification results, Classifier, Classification labels, Bayesian theorem, Bayesian Theorem, Convergence performance, Ensemble classifiers, Model free, Model-free, Nearest neighbor, Nearest neighbors, Unitive framework

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A modified MRF segmentation of brain MR images

Item Type Conference Paper

Author Y. Zhang

Author P. Lu

Author X. Liu

Author S. Zhou

Editor Li Q.

Editor Wang L.

Editor Zhou M.

Editor Sun L.

Editor Qiu S.

Editor Liu H.

Abstract Image noise seriously affects the accuracy of brain magnetic resonance (MR) image classification. The traditional Gaussian mixture model (GMM) is widely applied in image segmentation, which uses only image intensity information thus is susceptible to image noise. Markov random field (MRF) model overcome the problem by using a priori probability for local smoothing and denoising, which depends on the pixel class labelling with the Gibbs distribution. However, the MRF model still lacks accurate classification of brain tissue. To improve the segmentation of brain MR image, the paper proposes a joint segmentation scheme based on combination of GMM and MRF model to classify the brain tissue into three parts: white matter (WM), grey matter (GM) and cerebrospinal fluid (CSF). During the implementation, both image intensity and local neighbouring voxels are considered. The main steps are as follows: 1) we use K-means estimation to acquire the initial distribution parameters of the three brain tissues; then use Expectation maximization (EM) estimation to obtain GMM parameters. 2) For the spatial voxels, the energy function of joint probability is change in terms of their corresponding GMM class information; then the intensity-related energy

function's component is controlled. Therefore, the parameter of the joint model is estimated adaptively according to the local intensity and spatial information. The proposed method performs well in the experiments, for example, it is not sensitive to image noise, and has good robustness and segmentation accuracy as well as high computational efficiency. © 2017 IEEE.

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Conference Name Proceedings - 2017 10th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics, CISP-BMEI 2017

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Markov processes, Risk assessment, Gaussian distribution, Gaussian noise (electronic), Image segmentation, Image enhancement, Images segmentations, K-means clustering, Magnetic resonance imaging, Gaussian Mixture Model, Maximum likelihood estimation, Images classification, Energy functions, Markov Random Fields, Markov random field, Brain tissue, Markov Random Field model, brain magnetic resonance image, Brain magnetic resonance images, Gaussian mixture model, Image noise, joint energy function, Joint energy function

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A multi-analytical approach to investigate the retention of tyre-road wear particles, tyre-derived chemicals and metals in tunnel wash water

Item Type Journal Article

Author E.S. Rødland
Author J. Rausch
Author D. Jaramillo-Vogel
Author J.T. Rundberget
Author S. Meland
Author G. Granheim
Author L. Jacobsen
Author L.S. Heier

Abstract Environmental context: Tunnel wash water contains high concentrations of tyre-road wear particles, tyre-derived chemicals, and metals, which can threaten aquatic ecosystems if untreated. This study applies a multi-analytical approach to assess pollutant retention and transformation during sedimentation, revealing treatment limitations and emphasising the need for advanced strategies to reduce pollution from enclosed road infrastructures. Rationale: Tunnel wash water (TWW) accumulates tyre-road wear particles (TRWPs), tyre-derived chemicals (TDCs) and metals at high concentrations, posing risks to aquatic ecosystems if untreated. Understanding pollutant retention and transformation during treatment is essential for effective mitigation. Methodology: We investigated TWW from the Vålereng tunnel (Oslo, Norway) using a multi-analytical approach. TRWPs were quantified by pyrolysis–gas chromatography–mass spectrometry (PYR-GC/MS) with a marker mix (M4), a single marker (4-vinylcyclohexene, 4-VCH), and automated scanning electron microscopy–energy dispersive X-ray spectroscopy (SEM-EDX) single particle analysis with machine-learning classification (automated single particle analysis, ASPA, MC2). Metals were analysed by inductively coupled plasma–mass spectrometry (ICP-MS) and TDCs by ultraperformance liquid chromatography–tandem mass spectrometry (UPLC-MS/MS). Samples were collected before, during and after sedimentation treatment over 21 days. Results: Zinc (Zn) was the most abundant metal (274–2300 µg L⁻¹), reduced by up to 93% post-treatment. TRWP concentrations ranged from 15–160 mg L⁻¹ (M4), 13–122 mg L⁻¹ (4-VCH) and 13–240 mg L⁻¹ (ASPA MC2), with treatment efficiencies of 95–99.7%. Despite reductions, fine particles (<20 µm) dominated both untreated and treated water (up to 85%). For TDCs, N-(1,3-dimethylbutyl)-N-phenyl-p-phenylenediamine (6PPD) showed 85% reduction, whereas 6PPD-quinone (6PPD-Q) (–26%) and N,N-diphenylguanidine (DPG) (–3.7%) exhibited negative retention, likely due to leaching and transformation. 1-hydroxybenzotriazole hydrate (OHBT) increased by up to 167% during sedimentation. Discussion: Sedimentation effectively removed particles and metals but was insufficient for several dissolved TDCs, releasing them at ecotoxicologically relevant concentrations (e.g. 6PPD-Q exceeded Environment Protection Authority freshwater limits). This first cross-validation of three TRWP quantification methods highlights the need for advanced treatment technologies and regulatory thresholds for tyre-derived pollutants. © 2026 The Author(s) (or their employer(s)). Published by CSIRO Publishing.

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Issue 2 Collection: Tyre road wear particles – chemistry and impacts

Journal Abbr Environ. Chem.

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machine learning, particle size, mass spectrometry, ASPA, automated SEM-EDX single particle analysis, ICP-MS, liquid chromatography, metals, morpho-chemical based machine-learning classification, organic tyre compounds, Oslo, PYR-GC/MS, road, road pollution, sedimentation, tire, tunnel wash water, tyre-road wear particles, wear, X-ray spectroscopy

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A multi-resolution hidden markov model for optimal detection, tracking, separation, and classification of marine mammal vocalizations

Item Type Conference Paper

Author B.F. Harrison

Author P.M. Baggenstoss

Abstract We employ the multi-resolution hidden Markov model (MRHMM) to develop an improved algorithm for modeling marine mammal wandering tone vocalizations (whistles). A vocalization is modeled by a series of time segments in which the signal has a constant frequency rate (chirps). Rather than using chirps of uniform length, the segments are allowed to be of variable size, thus adapting to both short rapid changes in frequency rate as well as long segments of constant rate. The method supports the goals of finding the single best segmentation or the average joint probability density function of the data over all possible segmentations, weighted by the a priori probability of each segmentation. The probability density function (PDF) projection theorem is used to allow likelihood comparisons in the raw data domain. Simulated data and recorded marine mammal vocalizations are used to demonstrate the technique.

Date 2008

DOI 10.1109/OCEANS.2008.5151894

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Extra Journal Abbreviation: OCEANS

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Tags:

Probability density function, Markov processes, Hidden-Markov models, Chirp modulation, Optimal classification, Marine mammals, Constant frequency, Frequency rate, Improved * algorithm, Optimal detection, Optimal tracking, Time segments, Variable sizes

Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: B. F. Harrison; Naval Undersea Warfare Center, Newport RI, 02841, United States; email: harrison_bf@ieee.org; Conference name: 2008 Edition of Marine Technology Society and the Institute of Electrical and Electronics Engineers Oceans Conference, MTS/IEEE OCEANS 2006; Conference date: 15 September 2008 through 18 September 2008; Conference code: 77663; CODEN: OCNSD

A multi-source comprehensive risk quantification method for cascade reservoirs based on Monte Carlo simulation, HEC-RAS and Bayesian networks: Development and application in the Dadu river Basin

Item Type Journal Article

Author T. Wang

Author H. Yuan

Author Y. Li

Author T. Zhang

Author Y. Zhang

Author J. Chai

Author Y. Liu

Abstract The failure of cascade dams is prone to triggering the superposition, transmission, and amplification of risks in the reservoir group of a river basin. Accurate assessment of the comprehensive risks of cascade reservoirs is of decisive significance for the safe operation of water conservancy systems and the protection of people's lives and property. At present, the traditional comprehensive risk analysis methods based on flood disaster risk transmission have obvious limitations. They only focus on single risk factors like flooding, and the flood routing analysis mostly adopts empirical formulas, resulting in insufficient assessment accuracy. To address this issue, we propose a novel multi-source comprehensive risk quantification method for cascade dams integrating Monte Carlo simulation, the hydrodynamic model and Bayesian networks. Firstly, Monte Carlo simulations are applied to the hydrodynamic model simulated in HEC-RAS by combining with self-developed programs using Visual Basic for Applications. This application enables automated analysis of actual river flood routing, considering random flood frequencies and random initial reservoir

water levels. Meanwhile, machine learning algorithms including N-M, Adam, and DE are employed to conduct inverse analysis of river roughness, which is determined as 0.0833. Then, based on the analysis of overtopping damage from flood evolution after dam failures in cascade reservoirs, we quantitatively analyze the inter-dam transmission risks by applying the theory of comprehensive influence coefficients. Further, we combine the risk analysis results of floods, earthquakes and landslides for each dam in the system, and use Bayesian networks to assess the failure risk of each individual dam. Finally, by integrating the transmission risk analysis, we develop a multi-source comprehensive risk analysis method for cascade reservoir systems. This method is applied to four cascade dams in the Dadu River Basin of Sichuan Province, China. The results show that the main cause of the single dam failure is the rare high-energy floods, and earthquakes have a significant impact on dam losses. The Changheba Dam has the highest reliability and can block risks, while the Huangjinping Dam and the Luding Dam require focused attention due to their relatively high comprehensive risks. We recommend prioritizing enhanced monitoring and structural reinforcement for the critical dams including Huangjinping and Luding. This method is expected to accurately quantify the risk transmission between dams in cascade reservoir groups under multi-disaster scenarios, thereby contributing to risk management and plan formulation for the water conservancy system. © 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Notes:

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A Multivariate Drought Index for Seasonal Agriculture Drought Classification in Semiarid Regions

Item Type Journal Article

Author K. Bageshree

Author T. Kinouchi

Abstract Drought assessment in any region primarily hinges on precipitation deficiency, which is subsequently propagated to various components and sectors, leading to different drought types. In countries such as India, an intricate relationship between various governing factors, drought types, and their quantification methodologies make it elusive to timely initiate government relief measures. This also prevents comprehensive inclusion of the integrated effect of the principal drivers of drought, resulting in ambiguous categorization of severity, where groundwater storage variability is often neglected despite its significant role in irrigation. Here, we developed a multivariate Joint Drought Index (JDI) combining satellite and model-based standardized indices of precipitation and evapotranspiration (SPEI), soil moisture (SSI), groundwater (SGI), and surface runoff (SRI) with different temporal scales by employing two robust methods, principal component analysis (PCA) and Gaussian copula, and applied the index to highly drought-prone Marathwada region from central India. Our novel approach of using different scale combinations of integrated indices for two primary seasons (Kharif and Rabi) provides more realistic drought intensities than multiple univariate indices, by incorporating the response from each index, representing the seasonal drought conditions corroborating with the seasonal crop yields. JDI, with both methods, successfully identified two major drought events in 2015 and 2018, while effectively capturing the groundwater drought. Moreover, despite the high correlation between JDI using PCA and copula, we observed a significant difference in the intensities reported by these methods, where copula detected exceptional drought conditions more frequently than PCA. JDI effectively detected the onset, duration, and termination of drought, where the improved accuracy of drought detection can play a critical role in policy formation and socioeconomic security of the related stakeholders. Seasonal agriculture drought categorization for holistic quantification of drought conditions as presented in this study should provide broad methodological implications on drought monitoring and mitigation measures, especially for agriculture-dominated regions in semiarid climates. © 2022 by the authors.

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Agriculture, Drought, Principal component analysis, PCA, Principal-component analysis, Soil moisture, Groundwater, Arid regions, agricultural drought, Agricultural drought, Agriculture droughts, copula, Copula, drought classification, Drought classifications, Drought conditions, GLDAS, Integrated effects, multivariate drought index, Multivariate drought index, Semi-arid region

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A neuro-fuzzy approach to virtual screening in molecular bioinformatics

Item Type Journal Article

Author J. Paetz

Author G. Schneider

Abstract Molecular bioinformatics is a transdisciplinary working area. One hot topic is the design of drugs using computers and intelligent algorithms. This is known as in silico approach. We use a new in silico approach for separating active ligand molecules from inactive ones for different drug targets. This kind of retrospective virtual screening is performed by using encoded molecule data and a neuro-fuzzy methodology for classification, feature selection, and rule generation. We generate rules in a retrospective screening process that identify regions, where clearly more active compounds can be found compared to their a priori probability. We show that our approach is superior to a common descriptor-based standard technique. © 2004 Elsevier B.V. All rights reserved.

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A new active learning approach for adsorbate–substrate structural elucidation in silico

Item Type Journal Article

Author M.P. Lourenço

Author L.B. Herrera

Author J. Hostaš

Author P. Calaminici

Author A.M. Köster

Author A. Tchagang

Author D.R. Salahub

Abstract Adsorbate interactions with substrates (e.g. surfaces and nanoparticles) are fundamental for several technologies, such as functional materials, supramolecular chemistry, and solvent interactions. However, modeling these kinds of systems in silico, such as finding the optimum adsorption geometry and energy, is challenging, due to the huge number of possibilities of assembling the adsorbate on the surface. In the current work, we have developed an artificial intelligence (AI) approach based on an active learning (AL) method for adsorption optimization on the surface of materials. AL uses machine learning (ML) regression algorithms and their uncertainties to make a decision (based on a policy) for the next unexplored structures to be computed, increasing, though, the probability of finding the global minimum with a small number of calculations. The methodology allows an accurate and automated structural elucidation of the adsorbate on the surface, based on the

minimization of the total electronic energy. The new AL method for adsorption optimization was developed and implemented in the quantum machine learning software/agent for material design and discovery (QMLMaterial) program and was applied for C60@TiO₂ anatase (101). It marks another software extension with a new feature in addition to the automatic structural elucidation of defects in materials and of nanoparticles as well. SCC-DFTB calculations were used to build the complex search surfaces with a reasonably low computational cost. An artificial neural network (NN) was employed in the AL framework evaluated together with two uncertainty quantification methods: K-fold cross-validation and non-parametric bootstrap (BS) resampling. Also, two different acquisition functions for decision-making were used: expected improvement (EI) and the lower confidence bound (LCB). Graphical abstract: [Figure not available: see fulltext.] © 2022, The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature.

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A new active learning approach for global optimization of atomic clusters

Item Type Journal Article

Author M.P. Lourenço

Author B.R.L. Galvão

Author L. Barrios Herrera

Author J. Hostaš

Author A. Tchagang

Author M.X. Silva

Author D.R. Salahub

Abstract In catalysis, an accurate structural elucidation of molecules, atomic clusters, nanoparticles and solid surfaces is required to understand chemical processes. Therefore, an efficient and automatic structure determination for these systems is of great benefit since it requires a global search within huge chemical spaces. In this work, we propose a new active learning (AL) method intended for atomic clusters that uses different supervised machine learning techniques and their uncertainties to decide the promising non-observed (virtual) structures to be evaluated from quantum calculations. The method was developed for structural elucidation of (I) clusters where all atomic coordinates are allowed to change in a continuous chemical space and (II) doped ones where the atoms exchange in a sufficiently rigid structure, thus, a discrete search space where all cluster descriptors are known. Particularly for case I, a genetic algorithm operator was used to create the unknown virtual structures (and then their descriptors) from the observed ones to improve the performance of the AL search. The proposed AL was applied to the global optimization of heteronuclear (Al_4Si_7 and $4\text{Al}@ \text{Si}_{11}$) and homonuclear (Na_{20}) clusters using self-consistent charge density-functional tight-binding (SCC-DFTB), where a new repulsion parameter was developed to reproduce isomers evaluated from high-level calculations. The performance of the Gaussian process and artificial neural network algorithms was evaluated together with several uncertainty quantification methods: from Gaussian process, K-fold cross-validation and nonparametric bootstrap (BS) resampling. The efficiency of the AL was compared to conventional global optimization methods, such as random search and a genetic algorithm. The results show that the AL can find efficiently the global minimum of atomic clusters. © 2021, The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature.

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A new approach for EEG feature extraction for detecting error-related potentials

Item Type Conference Paper

Author Z. Pang

Author J. Li

Author H. Ji

Author M. Li

Abstract Error-related negativity (ERN) is one of the electroencephalographical (EEG) traces related to the subject perception of erroneous responses. Methods for detecting the ERN from EEG signals have been widely investigated. Many known techniques use the ERN component of event-related potentials (ERP) by extracting relevant features and feeding those features to a classifier. In these approaches, feature extraction becomes the key point. In this paper, we used the t-CWT method which was based on the continuous wavelet transform (CWT) and Student's t-test for extracting ERN features. In the experiment, EEG was recorded from 5 electrode sites (Fz, FC1, Cz, FCz and FC2) during a sustained attention experiment in 10 subjects. We used a support vector machine for classification. In

addition, the algorithm provides fully automated detection and quantification methods for extracting the most discriminative ERP components between two cognitive states and are particular suitable for classifying single-trial ERPs. The features extracted by the algorithm can be interpreted in terms of signal characteristics that are contributing to the efficiency of classification, giving a new method for brain activity investigation. © 2016 IEEE.

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Feature extraction, Classification (of information), Brain, Signal detection, Fully automated, Extraction, Biomedical signal processing, Wavelet transforms, Relevant features, Continuous wavelet transforms, Electrode sites, Error related potentials, Event-related potentials, Signal characteristic, Student's t tests

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A new approach to estimate a priori probabilities in remote sensing digital image classification

Item Type Journal Article

Author C. Liczbinski

Author V. Haertel

Abstract An adaptive classifier designed for high-dimensional image data classification has recently been proposed. The main feature in this proposed classifier consists of a novel method to acquire semilabeled samples in an iterative way during the

classification process itself. The semilabeled samples are then added to the set of available labeled samples, thus increasing the total number of training samples and consequently the reliability of the estimated values for the parameters in a parametric classifier. In this study we propose the insertion of an additional step into the iterative procedure implemented in the adaptive classifier to produce more realistic estimates of the a priori probability of the classes. The proposed procedure to estimate the a priori probability of the classes makes use of the same weighted approach used by the adaptive classifier, is computationally efficient, and can be easily implemented. Tests using hyperspectral image data have shown that the proposed approach can contribute to increased accuracies in the classification process. © 2008, Taylor & Francis Group, LLC. All rights reserved.

Date 2008

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: V. Haertel; Center for Remote Sensing, Federal University at Rio Grande do Sul-UFRGS, Porto Alegre, RS, 91501-970, Caixa Postal 15052, Brazil; email: victor.haertel@ufrgs.br

A new approach to identify, classify and count drug-related events

Item Type Journal Article

Author T. Bürkle

Author F. Müller
Author A. Patapovas
Author A. Sonst
Author B. Pfistermeister
Author B. Plank-Kiegele
Author H. Dormann
Author R. Maas

Abstract Aims: The incidence of clinical events related to medication errors and/or adverse drug reactions reported in the literature varies by a degree that cannot solely be explained by the clinical setting, the varying scrutiny of investigators or varying definitions of drug-related events. Our hypothesis was that the individual complexity of many clinical cases may pose relevant limitations for current definitions and algorithms used to identify, classify and count adverse drug-related events. Methods: Based on clinical cases derived from an observational study we identified and classified common clinical problems that cannot be adequately characterized by the currently used definitions and algorithms. Results: It appears that some key models currently used to describe the relation of medication errors (MEs), adverse drug reactions (ADRs) and adverse drug events (ADEs) can easily be misinterpreted or contain logical inconsistencies that limit their accurate use to all but the simplest clinical cases. A key limitation of current models is the inability to deal with complex interactions such as one drug causing two clinically distinct side effects or multiple drugs contributing to a single clinical event. Using a large set of clinical cases we developed a revised model of the interdependence between MEs, ADEs and ADRs and extended current event definitions when multiple medications cause multiple types of problems. We propose algorithms that may help to improve the identification, classification and counting of drug-related events. Conclusions: The new model may help to overcome some of the limitations that complex clinical cases pose to current paper- or software-based drug therapy safety. © 2013 The British Pharmacological Society.

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A new approach to robustness-based optimization using uncertainty set constructed through machine learning

Item Type Journal Article

Author R.M. Shahbab

Author K. Zaman

Abstract This paper proposes three machine learning-based uncertainty set construction methods and a novel uncertainty quantification method for robustness-based optimization. The proposed methods are capable of capturing all possible types of uncertainties in the form of sparse point and/or interval data and efficiently quantify uncertainty in robustness-based optimization. In contrast to traditional approaches that rely on expert opinion or assumptions to construct uncertainty sets, our methods leverage machine learning algorithms to extract uncertainty patterns from historical data and effectively capture epistemic uncertainty in input variables. Another challenge in existing robustness-based optimization under epistemic uncertainty is the high computational cost associated with the iterative method of epistemic analysis. Moreover, these methods struggle with uncertainty sets containing a mixture of point and interval data. To overcome these limitations, we propose a unified probabilistic approach that utilizes maximum likelihood estimation to efficiently quantify uncertainty, regardless of the data form. Using the proposed methods, we introduce new approaches for robustness-based portfolio optimization (RPO) and multidisciplinary design optimization (RO/MDO), namely the worst-case maximum likelihood estimation (WMLE)-based single-loop RPO and WMLE-based RO/MDO approach, respectively. Illustrated through applications to minimum variance portfolios, our methods consistently demonstrate significant improvements over established formulations in terms of both return and risk. Additionally, when applied to a complex multidisciplinary engineering design, the WMLE-based RO/MDO approach showcases a computationally efficient solving technique, yielding more realistic results than the existing methodologies. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2024.

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Optimisations, Machine-learning, Machine learning, Uncertainty analysis, Learning algorithms, Uncertainty, Iterative methods, Optimization, Financial data processing, Epistemic uncertainties, New approaches, Maximum likelihood estimation, Design aids, Epistemic uncertainty, Interval data, Maximum-likelihood estimation, Multi-disciplinary design optimisation, Multidisciplinary design optimization, Portfolio optimization, Robust optimization

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: K. Zaman; Department of Industrial and Production Engineering, Bangladesh University of Engineering and Technology, Dhaka, 1000, Bangladesh; email: kaiszaman@yahoo.com; CODEN: SMOTB

A New Classifier by Remixing Features and Adjusting Logits for Long-Tailed Visual Data

Item Type Conference Paper

Author X. Deng

Author Y. Wang

Author M. Li

Abstract The natural visual recognition for long-tailed distribution data is a difficult problem because the classifier is more biased towards the head classes, which often results in incorrect prediction. To solve this problem, we propose a simple one-stage network by remixing features and adjusting logits (RFAL) to handle long-tailed recognition. Our method enhances the expression ability of the tail class through the effective re-organization of deep features, and reassigns the label in favor of the tail class by providing a higher weight to the tail class. Extensive experiments on three long-tailed visual recognition datasets are conducted and the comparison of the proposed network with the existing state-of-the-art networks is made. The results show the better performance of the proposed network in long-tailed data recognition. © 2021 IEEE.

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A new laboratory SCAL device and processing method for OBM vug density quantification

Item Type Conference Paper

Author O. Al-Mutwali

Author H. Khemissa

Author S. Dama

Author T. Zhang

Author C. Shrivastava

Author S. Yang

Author L. Comparon

Author S. Al-Ofi

Abstract Vugular porosity, per this work is defined as a set of secondary porosity development in carbonate rocks that could have developed through alteration of rock-fabric, commonly by processes such as dolomitization and dissolution (including mouldic porosity). It is one of the important secondary porosity types and its characterization can influence rock model petro-physical behavior. Traditionally vugular porosity can be quantified on water-based mud (WBM) high resolution electrical imaging logs by counting the image pixels that has electrical resistivity lower than the rock background. With the popularity of non-conductive drilling fluid (OBM) and improvement of electrical image logging technology, it is possible nowadays to acquire similar high resolution electrical images in OBM environment. Vugs can be identified visually on these new types of electrical images. However open vugs filled with OBM can appear resistive on those images and invalidate the basic assumptions of secondary porosity quantification software that open vugs should have lower electrical resistivity. At the same time, the new generation of electrical imaging tools uses megahertz logging frequencies to decrease the capacitance of the OBM to acquire formation images. With this frequency range, both electrical conductivity and dielectric permittivity impact the image log acquired. This challenges our understanding of the tool response in terms of image color contrast and opens the question of how to integrate such OBM electrical images into current geologic interpretation software that commonly uses the image resistivity value thresholds. In this case, a new laboratory device that features the same logging frequency has been developed and presented in this paper. The device was used to investigate images of secondary porosity features acquired in OBM in controlled laboratory conditions. It is demonstrated that by integrating both resistivity image and dielectric permittivity image, it is possible to overcome the OBM filled resistive vug limitation and account for all fluid-filled effective vugs. A new post processing method is introduced by combining the effects of resistivity and dielectric permittivity and generating a new image called the rock Hayman factor image. On the Hayman factor image, it is possible to differentiate fluid-filled vugs from cemented vugs. Vug density quantification in WBM is based on resistivity threshold values to characterize conductive and resistive heterogeneity patches and can be highly user dependent. In OBM environment, mud filled features will appear resistive and may decrease the contrast with rock resistive matrix. This can make current vug quantification software miss parts of open vugs. Two popular vug separation and quantification methods (Gray-scale reconstruction transform plus watershed transform method and Sobel gradient edge detection plus Otsu's image thresholding method) are investigated first on a vuggy limestone outcrop and then on a downhole log as a case study to derive a vug density quantification workflow. The new Hayman factor image has proven on the laboratory data its ability to quantify both vugs beneath the rock surface and on the rock surface filled with OBM. Integrating the Hayman factor image, the procedure to set heterogeneity patches can be well-defined and not rely on user settings. The two image class separation methods can give very similar vug density quantified. Best practices of parameter adjustments on these two image classification methods are provided that can achieve the best results based on both laboratory data and downhole data. In this paper investigation results including both laboratory measurements and downhole logging data analysis of the OBM electrical image vug density quantification are presented. A new post-processing Hayman factor image is designed to quantify rock resistivity and the dielectric permittivity relationship and help vug density quantification. Application of this new image and best practices of how to set processing parameters using this image are investigated and summarized. Using the new vug characterization workflow involving Hayman image can improve OBM well vug quantification

accuracy and repeatability. Copyright © 2017 Society of Petroleum Engineers.

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A new method for acquiring images of meiobenthic images using the FlowCAM

Item Type Journal Article

Author T. Kitahashi

Author H.K. Watanabe

Author M. Tsuchiya

Author H. Yamamoto

Author H. Yamamoto

Abstract The purpose of this study was to develop a new method for investigating sediment-inhabiting meiobenthos using the Flow Cytometer And Microscope (FlowCAM). Meiobenthos are widely recognized as a useful indicator for assessing the effects of anthropogenic and natural disturbances in both shallow and deep ocean ecosystems. These small benthic invertebrates are traditionally investigated by individually

counting and identifying specimens under a microscope, which is labor intensive and time consuming. However, FlowCAM, which was originally developed to semiautomatically analyze microplankton, has the potential to resolve these challenges. Meiobenthic specimens were extracted from sediment using the centrifugal separation method and were then pipetted into the FlowCAM system and imaged. The images were then used to classify and count the specimens at high taxonomic levels to verify the effectiveness of this method compared with traditional methods. We found that FlowCAM system: • Enabled sufficient meiobenthic images to be obtained to allow the identification and classification of specimens at high taxonomic levels. • Obtained comparable numbers of individuals to traditional methods. • Has the potential to rapidly process large the volumes of meiobenthos samples that are required when monitoring seasonal and spatial variation in ocean ecosystems and conducting long-term environmental impact assessments. © 2018 The Author(s)

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A new method for assessment of regional drought risk: information diffusion and interval mapping adjustment based on k-means cluster points

Item Type Journal Article

Author W. Xu

Author M. Tang

Author Y. Li

Abstract Aiming at the defect that there is no ability for the conventional weighted comprehensive assessment method (WCA) to grade drought risk directly, a method based on k-means cluster points to realize the classification of drought risk is proposed in this paper. On the basis of calculating the drought risk value of cluster points, the inverse distance weight interpolation method (IDWI) and multidimensional normal diffusion method (MND) were used to quantify the drought risk value, and the discrimination between the risk value and grade was improved by interval mapping adjustment (IMA). In this paper, the drought risk of Anhui Province from 2000 to 2020 was calculated to verify the above method. The results show that: (1) The drought risk quantification method based on information redistribution of k-means cluster point can not only realize automatic risk classification, but also re-quantify the risk value of the assessment object in the same risk grade, which makes up for the defects that the conventional WCA cannot carry out grade division and the conventional clustering method cannot assign the risk value of the assessment object. (2) The result of information redistribution based on MND is closer to the actual drought situation and more reasonable than IDWI. (3) The dispersion effect of risk value obtained by information redistribution based on k-means cluster point can be improved by the IMA of drought risk. It improves the discrimination degree of risk value, so that the grades can be displayed more intuitively. The defect of the WCA is overcome by the new method proposed in this paper, the follow-up utilization space is widened, and the thinking of risk quantification in drought risk assessment is broadened. © 2022 The Authors.

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A new quantitative 3D gap area measurement of fracture displacement of intra-articular distal radius fractures: Reliability and clinical applicability

Item Type Journal Article

Author L.J.M. Roelofs

Author A.M.L. Meesters

Author N. Assink

Author J. Kraeima

Author T.D. Van der Meulen

Author J.N. Doornberg

Author J.-P.P.M. De Vries

Author J. Hoekstra

Author K.T. Duis

Author F.F.A. IJpma

Abstract Introduction Gap and step-off measurements are generally used in the surgical decision-making process of distal radius fractures. Unfortunately, there is no consensus on treatment choice as these measurements are prone to inter- and intraobserver variability. In this study, we aim to introduce a new 3D fracture quantification method and compare it to conventional fracture analysis. Methods Forty patients with a minimally displaced intra-articular distal radius fracture that was treated nonoperatively between 2008–2015 were included. 2D-CT images were reassessed by three orthopedic trauma surgeons who performed gap and step-off measurements. Subsequently, 3D models were created and a 3D measurement method for fracture displacement was developed. For each fracture, the ‘3D gap area’ (3D surface between all fracture fragments) was determined by three observers. Interobserver agreements were calculated for all measurements, and the intraobserver agreement was calculated for the new 3D measurement. All patients completed two questionnaires in order to link our measurements to functional outcome. Results The interobserver agreement of the 2D measurements was fair (ICC = 0.54) for the gap and poor (ICC = 0.21) for the step-off. The median gap was 2.8 (IQR: 1.9–3.5) mm

and step-off was 0.9 (IQR: 0.0–1.6) mm. Interobserver agreement on 3D gap area measurements was excellent (ICC = 0.81), with a median difference between measurements of 6.0 (IQR: 2.0–19.0) mm², which indicates reliable assessment of 3D fracture displacement. Intraobserver agreement was also excellent (ICC = 0.98), with a median difference of 4.0 (IQR: 1.5–5.5) mm². No significant differences in clinical outcome were found between the above and below 2mm displacement groups. The score of the DASH was 3.4 (IQR: 0.4–8.8) versus 4.2 (IQR: 0.0–11.6) respectively. Results from the PRWE questionnaire shows a similar result of 3.5 (IQR: 0.0–12.6) versus 5.0 (IQR: 0.0–25.5). Conclusion 3D gap area is a more objective measurement method compared to the conventional gap and step-off measurements to quantify the level of fracture displacement of distal radius fractures. 3D fracture assessment can be used in addition to the currently used classification systems of distal radius fractures. Copyright: © 2022 Roelofs et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: F.F.A. IJpma; Department of Surgery, Subdivision of Trauma Surgery, University Medical Center Groningen, University of Groningen, Groningen, Netherlands; email: f.f.a.ijpma@umcg.nl; CODEN: POLNC

A new quantitative method for pivot shift grading

Item Type Journal Article

Author S. Kopf

Author R. Kauert

Author J. Halfpaap

Author T. Jung

Author R. Becker

Abstract Purpose: The purposes of the study were to evaluate and to quantify the pivot shift phenomenon by using a small and easy to handle measuring device for pivot shift quantification. Methods: Twenty patients (forty knees) with primary torn anterior cruciate ligaments (ACL) were tested under anesthesia, graded by the examiner and by the device according to the IKDC classification [normal (0), glide (1), clunk (2), and gross (3)]. For the grading by the device, a femoral and a tibial miniature inertial sensor measured the acceleration and the angular velocities. Three parameters were used for pivot shift identification and quantification: (1) difference between the positive and negative acceleration peak value (a diff), (2) the maximum jerk (j max), and (3) the standard deviation (SD a) of the acceleration. The ratio between the ACL-deficient and the intact knees was calculated in order to normalize the data. Results: The pivot shift phenomenon could be identified, and all three parameters showed significant higher values in the ACL-deficient knees compared to the intact knees ($p < 0.05$). The grading by examiner did not significantly correlate with a diff ($p = 0.38$; $r = 0.21$), j max ($p = 0.36$; $r = -0.22$), SD a ($p = 0.65$; $r = 0.11$), and grading by the device ($p = 0.62$; $r = 0.12$). Conclusions: The present study has shown that the quantification of the pivot shift test is practicable when inertial sensors are used. The results have shown that the subjective grading of the pivot shift test does not correlate well with objective quantification. © 2012 Springer-Verlag.

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A Neyman-Pearson approach to statistical learning

Item Type Journal Article

Author C. Scott

Author R. Nowak

Abstract The Neyman-Pearson (NP) approach to hypothesis testing is useful in situations where different types of error have different consequences or a priori probabilities are unknown. For any $\alpha > 0$, the NP lemma specifies the most powerful test of size α , but assumes the distributions for each hypothesis are known or (in some cases) the likelihood ratio is monotonic in an unknown parameter. This paper investigates an extension of NP theory to situations in which one has no knowledge of the underlying distributions except for a collection of independent and identically distributed (i.i.d.) training examples from each hypothesis. Building on a "fundamental lemma" of Cannon et al., we demonstrate that several concepts from statistical learning theory have counterparts in the NP context. Specifically, we consider constrained versions of empirical risk minimization (NP-ERM) and structural risk minimization (NP-SRM), and prove performance guarantees for both. General conditions are given under which NP-SRM leads to strong universal consistency. We also apply NP-SRM to (dyadic) decision trees to derive rates of convergence. Finally, we present explicit algorithms to implement NP-SRM for histograms and dyadic decision trees. © 2005 IEEE.

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A Non-intrusive Quantification Method for Biochar Water Retention Capacity Using A Portable Microwave Sensor and Machine Learning

Item Type Conference Paper

Author D. An

Author H. Niu

Author Y. Chen

Abstract Biomass from agriculture and forestry can be converted to biochar by pyrolysis gasification process and when applied to soil, this becomes a carbon negative technology. How to quantify the effect of biochar applied into soil in terms of water retention capacity accurately, quickly, cost-effectively and in large scale mapping are still a challenging problem. This work for the first time, reported extensive lab tests to demonstrate that a non-intrusive quantification method for biochar water retention capacity using a portable microwave sensor and machine learning is feasible. Scaling up can be achieved by putting the sensor on a mobility platform, either ground or aerial. We reported in this paper details of our materials and methods, and machine learning workflow and results. The significance of this work lays the foundation for future carbon negative technology applications assessment such as soil carbon accounting. We evaluated our nonintrusive quantification method with SVM and LDA classifier algorithm on the data collected from real-world experiments. Results show that our method

achieved a 74.0% recognition accuracy over six different kinds of biochar footprints. © 2021 IEEE.

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DOI 10.1109/ICCMA54375.2021.9646210

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Language English

Extra Journal Abbreviation: Int. Conf. Control, Mechatronics Autom., ICCMA

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Antennas, Machine-learning, Machine learning, Forestry, Quantification methods, Carbon, Soils, A-carbon, Non-intrusive, Biochar, Carbon negative technology, Carbon Negative Technology, Gasification process, Microwave, Microwave sensors, Microwaves, Pyrolysis gasifications, Walabot, Water retention capacity

Notes:

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A note on a priori forecasting and simplicity bias in time series

Item Type Journal Article

Author K. Dingle

Author R. Kamal

Author B. Hamzi

Abstract To what extent can we forecast a time series without fitting to historical data? Can universal patterns of probability help in this task? Deep relations between pattern Kolmogorov complexity and pattern probability have recently been used to make a priori probability predictions in a variety of systems in physics, biology and engineering. Here we study simplicity bias (SB) – an exponential upper bound decay in pattern probability with increasing complexity – in discretised time series extracted from the World Bank Open Data collection. We predict upper bounds on the probability of discretised series patterns, without fitting to trends in the data. Thus we

perform a kind of ‘forecasting without training data’, predicting time series shape patterns a priori, but not the actual numerical value of the series. Additionally we make predictions about which of two discretised series is more likely with accuracy of ~80%, much higher than a 50% baseline rate, just by using the complexity of each series. These results point to a promising perspective on practical time series forecasting and integration with machine learning methods. © 2022 The Author(s)

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Learning systems, Forecasting, Computational complexity, Exponentials, Time series, Times series, Upper Bound, A-priori probabilities, Open Data, Algorithmic probability, Parallel processing systems, Historical data, Kolmogorov complexity, Simplicity bias, Universal patterns, World bank

Notes:

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A novel approach for quantification and analysis of the color doppler twinkling artifact with application in noninvasive surface roughness characterization: An in vitro phantom study

Item Type Journal Article

Author A. Jamzad

Author S.K. Setarehdan

Abstract Objectives-The twinkling artifact is an undesired phenomenon within color Doppler sonograms that usually appears at the site of internal calcifications. Since the appearance of the twinkling artifact is correlated with the roughness of the calculi,

noninvasive roughness estimation of the internal stones may be considered as a potential twinkling artifact application. This article proposes a novel quantitative approach for measurement and analysis of twinkling artifact data for roughness estimation. Methods-A phantom was developed with 7 quantified levels of roughness. The Doppler system was initially calibrated by the proposed procedure to facilitate the analysis. A total of 1050 twinkling artifact images were acquired from the phantom, and 32 novel numerical measures were introduced and computed for each image. The measures were then ranked on the basis of roughness quantification ability using different methods. The performance of the proposed twinkling artifact-based surface roughness quantification method was finally investigated for different combinations of features and classifiers. Results-Eleven features were shown to be the most efficient numerical twinkling artifact measures in roughness characterization. The linear classifier outperformed other methods for twinkling artifact classification. The pixel count measures produced better results among the other categories. The sequential selection method showed higher accuracy than other individual rankings. The best roughness recognition average accuracy of 98.33% was obtained by the first 5 principle components and the linear classifier. Conclusions-The proposed twinkling artifact analysis method could recognize the phantom surface roughness with average accuracy of 98.33%. This method may also be applicable for noninvasive calculi characterization in treatment management. © 2014 by the American Institute of Ultrasound in Medicine.

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Export Date: 20 September 2026; Cited By: 18; Correspondence Address: A. Jamzad; Control and Intelligent Processing Center of Excellence, School of Electrical and Computer Engineering, University of Tehran, Tehran 14395-515, North Kargar Street, Iran; email: a.jamzad@ut.ac.ir; CODEN: JUMED

A novel approach to maximize g-mean in nonstationary data with recurrent imbalance shifts

Item Type Journal Article

Author R. Kulkarni

Author S. Revathy

Author S. Patil

Abstract One of the noteworthy difficulties in the classification of nonstationary data is handling data with class imbalance. Imbalanced data possess the characteristics of having a lot of samples of one class than the other. It, thusly, results in the biased accuracy of a classifier in favour of a majority class. Streaming data may have inherent imbalance resulting from the nature of dataspace or extrinsic imbalance due to its nonstationary environment. In streaming data, timely varying class priors may lead to a shift in imbalance ratio. The researchers have contemplated ensemble learning, online learning, issue of class imbalance and cost-sensitive algorithms autonomously. They have scarcely ever tended to every one of these issues mutually to deal with imbalance shift in nonstationary data. This correspondence shows a novel methodology joining these perspectives to augment G-mean in no stationary data with Recurrent Imbalance Shifts (RIS). This research modifies the state-of-the-art boosting algorithms, 1) AdaC2 to get G-mean based Online AdaC2 for Recurrent Imbalance Shifts (GOA-RIS) and AGOA-RIS (Ageing and G-mean based Online AdaC2 for Recurrent Imbalance Shifts), and 2) CSB2 to get G-mean based Online CSB2 for Recurrent Imbalance Shifts (GOC-RIS) and Ageing and G-mean based Online CSB2 for Recurrent Imbalance Shifts (AGOCRIS). The study has empirically and statistically analysed the performances of the proposed algorithms and Online AdaC2 (OA) and Online CSB2 (OC) algorithms using benchmark datasets. The test

outcomes demonstrate that the proposed algorithms globally beat the performances of OA and OC. © 2021, Zarka Private University. All rights reserved.

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Imbalanced data, Online learning, Cost-sensitive algorithms, Data stream classification, Population shift, Skewed data stream

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Export Date: 20 September 2026; Cited By: 4

A novel approach to probabilistic biomarker-based classification using functional near-infrared spectroscopy

Item Type Journal Article

Author T. Hahn

Author A.F. Marquand

Author M.M. Plichta

Author A.-C. Ehlis

Author M.W. Schecklmann

Author T. Dresler

Author T.A. Jarczok

Author E. Eirich

Author C. Leonhard

Author A. Reif

Author K.-P. Lesch

Author M.J. Brammer

Author J. Mourao-Miranda

Author A.J. Fallgatter

Abstract Pattern recognition approaches to the analysis of neuroimaging data have brought new applications such as the classification of patients and healthy controls within reach. In our view, the reliance on expensive neuroimaging techniques which are not well tolerated by many patient groups and the inability of most current biomarker algorithms to accommodate information about prior class frequencies (such as a disorder's prevalence in the general population) are key factors limiting practical application. To overcome both limitations, we propose a probabilistic pattern recognition approach based on cheap and easy-to-use multi-channel near-infrared spectroscopy (fNIRS) measurements. We show the validity of our method by applying it to data from healthy controls ($n = 14$) enabling differentiation between the conditions of a visual checkerboard task. Second, we show that high-accuracy single subject classification of patients with schizophrenia ($n = 40$) and healthy controls ($n = 40$) is possible based on temporal patterns of fNIRS data measured during a working memory task. For classification, we integrate spatial and temporal information at each channel to estimate overall classification accuracy. This yields an overall accuracy of 76% which is comparable to the highest ever achieved in biomarker-based classification of patients with schizophrenia. In summary, the proposed algorithm in combination with fNIRS measurements enables the analysis of sub-second, multivariate temporal patterns of BOLD responses and high-accuracy predictions based on low-cost, easy-to-use fNIRS patterns. In addition, our approach can easily compensate for variable class priors, which is highly advantageous in making predictions in a wide range of clinical neuroimaging applications. © 2012 Wiley Periodicals, Inc.

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Notes:

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Export Date: 20 September 2026; Cited By: 28; Correspondence Address: T. Hahn; Department of Cognitive Psychology II, Johann Wolfgang Goethe University Frankfurt, 60325 Frankfurt am Main, Main Mertonstraße 17, Germany; email: TimHahn@gmx.net; CODEN: HBMAE

A novel discriminative feature extraction for acoustic scene classification using RNN based source separation

Item Type Conference Paper

Author S. Mun

Author S. Shon

Author W. Kim

Author D.K. Han

Author H. Ko

Abstract Various types of classifiers and feature extraction methods for acoustic scene classification have been recently proposed in the IEEE Detection and Classification of Acoustic Scenes and Events (DCASE) 2016 Challenge Task 1. The results of the final evaluation, however, have shown that even top 10 ranked teams, showed extremely low accuracy performance in particular class pairs with similar sounds. Due to such sound classes being difficult to distinguish even by human ears, the conventional deep learning based feature extraction methods, as used by most DCASE participating teams, are considered facing performance limitations. To address the low performance problem in similar class pair cases, this letter proposes to employ a recurrent neural network (RNN) based source separation for each class prior to the classification step. Based on the fact that the system can effectively extract trained sound components using the RNN structure, the mid-layer of the RNN can be considered to capture discriminative information of the trained class. Therefore, this letter proposes to use this mid-layer information as novel discriminative features. The proposed feature shows an average classification rate improvement of 2.3% compared to the conventional method, which uses additional classifiers for the similar class pair issue. Copyright © 2017 The Institute of Electronics, Information and Communication Engineers.

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Extra Journal Abbreviation: IEICE Trans Inf Syst
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Feature extraction, Classification (of information), Transfer learning, Extraction, Recurrent neural networks, Recurrent neural network, Recurrent neural network (RNN), Source separation, Feature extraction methods, Acoustic scene classification, Bottleneck feature, Bottleneck features, Discriminative feature extraction, Learning-based feature extractions, Performance limitations, Scene classification

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A novel high-entropy TiVCrMoC₃T_x assisted LDI MS for serum metabolic fingerprint in rheumatoid arthritis

Item Type Journal Article

Author Z. Chu

Author X. Yu

Author X. Wang

Author W. Zhang

Author Y. Li

Author X. Yan

Author W. Yan

Author H. Meng

Author L. Li
Author G. Zhang
Author R. Wang
Author M. Li
Author J. Li
Author W. Liang
Author C. Ma

Abstract Worldwide, rheumatoid diseases account for approximately 37.57/100,000 disability-adjusted life years (DALYs). Early diagnosis of rheumatoid arthritis (RA) can effectively reduce associated risks and improve quality of life. To achieve this, accurate and rapid advanced tools are required. Herein, TiVCrMoC₃, a novel high-entropy two-dimensional (2D) carbide MXene that exhibits excellent electrical conductivity, is prepared and applied to surface-assisted laser desorption ionization mass spectrometry (SALDI-MS) analysis for the first time. TiVCrMoC₃ demonstrates superior performance in small-molecule detection compared with typical inorganic and organic substrates. Moreover, it exhibits ultrahigh sensitivity (limit of detection (LOD) at the 10 pg/mL level), excellent repeatability (coefficient of variation (CV) < 4%), excellent quantitative detection capability (coefficient of determination = 0.99987), clean background, and broad analyte coverage. Results showed that TiVCrMoC₃-assisted laser desorption ionization MS (LDI-MS) is efficient in screening RA diseases, with the advantages of high throughput and low cost. It enables accurate and quantitative analysis of various low-concentration metabolites in 1 µL of biological fluid within seconds. Combined with machine learning, TiVCrMoC₃-assisted LDI-MS is used to accurately diagnose multiple RA samples. A total of 8 potential different small-molecule metabolites between individuals with osteoarthritis (OA) and RA were identified based on relative quantification methods, and their associated metabolic changes were discussed. TiVCrMoC₃-assisted LDI-MS provides a potential tool for the rapid diagnosis of RA and may pave the way for precision medicine. © 2025 The Authors. Published by Elsevier B.V. on behalf of Xi'an Jiaotong University. This is an open access article under the CC BY-NC-ND license. <http://creativecommons.org/licenses/by-nc-nd/4.0/>

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A Novel Hyperspectral Image Classification Method Using Class-Weighted Domain Adaptation Network

Item Type Journal Article

Author H. Wang

Author Y. Cheng

Author X. Wang

Abstract With the development of science and technology, hyperspectral image (HSI) classification has been studied in depth by researchers as one of the important means of human cognition in living environments and the exploration of surface information. Nevertheless, the shortage of labeled samples is a major difficulty in HSI classification. To address this issue, we propose a novel HSI classification method called class-weighted domain adaptation network (CWDAN). First, the convolutional domain adaption network (ConDAN) is designed to align the marginal distributions and second-order statistics, respectively, of both domains via multi-kernel maximum mean discrepancy (MK-MMD) and CORAL loss. Then, the class-weighted MMD (CWMMD) is defined to simultaneously consider the conditional distribution discrepancy and changes of class prior distributions, and the CWMMD-based domain adaptation term is incorporated into the classical broad learning system (BLS) to construct the weighted conditional broad network (WCBN). The WCBN is applied to reduce the conditional distribution discrepancy and class weight bias across domains, while performing breadth expansion on domain-invariant features to further enhance representation ability. In comparison with several existing mainstream methods, CWDAN has excellent classification performance on eight real

HSI data pairs when only using labeled source domain samples. © 2023 by the authors.

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A novel machine learning application: Water quality resilience prediction Model

Item Type Journal Article

Author M. Imani

Author M.M. Hasan

Author L.F. Bittencourt

Author K. McClymont

Author Z. Kapelan

Abstract Resilience-informed water quality management embraces the growing environmental challenges and provides greater accuracy by unpacking the systems' characteristics in response to failure conditions in order to identify more effective opportunities for intervention. Assessing the resilience of water quality requires complex analysis of influential parameters which can be challenging, time consuming and costly to compute. It may also require building detailed conceptual and/or physically process-based models that are difficult to build, calibrate and validate. This study utilises Artificial Neural Network (ANN) to develop a novel application to predict water quality resilience to simplify resilience evaluation. The Fuzzy Analytic Hierarchy Process method is used to rank water basins based on their level of resilience and to identify the ones that demand prompt restoration strategies. The commonly used 'magnitude * duration of being in failure state' quantification method has been used to formulate and evaluate resilience. A 17-years long water quality dataset from the 22 water basins in the State of São Paulo, Brazil, was used to train and test the ANN model. The overall agreement between the measured and simulated WQI resilience values is satisfactory and hence, can be used by planners and decision makers for improved water management. Moreover, comparative analyses show similarities and differences between the 'level of criticalities' reported in each zone by Environment Agency of the state of São Paulo (CETESB) and by the resilience model in this study. © 2021 Elsevier B.V.

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Article, Decision making, machine learning, Machine learning, Neural networks, Predictive analytics, Statistical tests, decision making, ground water, water management, Water management, water supply, Analytic hierarchy process, prediction, controlled study, artificial neural network, Resilience, Fuzzy logic, Quality control, numerical model, Computer aided instruction, Quantification methods, Brazil, fuzzy mathematics, surface water, Water conservation, Machine learning applications, analytic method, Comparative analysis, priority journal, analytical hierarchy process, Artificial neural network, Columns (structural), ecosystem restoration, Environmental challenges, environmental resilience, fresh water, Fuzzy analytic hierarchy process, mathematical computing, Novel applications, Process-based models, Quality management, Restoration strategies, Triangular fuzzy number, water quality, Water quality

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A novel method and classification criteria for analyzing urine turbidity and its relationship with urine dry chemical parameters

Item Type Journal Article

Author J. Liu

Author Y. Cheng

Author Z. Zhang

Author L. Zhu

Author L. Pan

Author H. Zhou

Author H. Zhao

Author X. Ren

Abstract Background Urine turbidity is a significant diagnostic marker for early screening of urinary tract infections, kidney stones, and other related conditions. However, current methods for analyzing urine turbidity often rely on subjective assessments. This study aims to investigate the relationship between urine turbidity and the urine color values measured by spectrophotometer, providing an objective quantification method for both urine turbidity and urine color, while also exploring the underlying causes of urine turbidity. Methods A cross-sectional study was conducted among newly enrolled university students undergoing physical examination in Beijing. Basic demographic information and morning urine samples were collected. Urine turbidity was assessed using human visual evaluation and a urine chemical analyzer, while urine color CIE $L^*a^*b^*$ (International Commission on illumination) was measured using a spectrophotometer. Routine urine chemical examination was also performed. Correlations among urine turbidity, urine color, and urine dry chemical parameters were analyzed. Results A total of 1220 participants (68.7% female, mean age: 23.66 years) were included in the study. Spearman correlation analysis showed that urine turbidity was significantly negatively correlated with L^* (lightness) and significantly positively correlated with a^* (redness) and b^* (yellowness). Regression analysis identified L^* as the most affected parameter by urine turbidity (standardized coefficient $\beta = -1.030$, $p < 0.05$). Receiver operating characteristic (ROC) analysis showed that L^* was highly effective in distinguishing different urine turbidity levels, with $L^* < 89.165$ achieving excellent sensitivity and specificity (AUC=0.984) and 96% accuracy in identifying turbid urine. In addition, urine turbidity was positively correlated with urine specific gravity, protein, and urine color ($p < 0.05$), while its relationship with pH was nonlinear. These findings suggest that multiple factors collectively influence urine turbidity. Conclusion This study provides a novel and objective approach for assessing urine turbidity, advancing the modernization of urine diagnostic practices in traditional medicine. © 2025 Liu et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium,

provided the original author and source are credited. [grant numbers 82374299, 81973697]. This was a government funded project. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

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A novel method for forest spatial structure heterogeneity evaluation of plantation utilizing point-wise vector network and neighborhood index

Item Type Journal Article

Author H. Qiu

Author H. Zhang

Author K. Lei

Author J. Wang

Author H. Zhang

Author L. Yu

Abstract Existing forest structure heterogeneity studies focus on single-perspective analysis and field investigation, lacking top-down perspective and automated forest heterogeneity quantification methods. Quantifying and elucidating the stand structural heterogeneity differences is challenging in uneven-aged forests. This paper addresses such challenges by presenting a point-wise stand spatial structure heterogeneity evaluation approach through deep learning and multi-dimensional structure index. This method aims to fill this gap of micro-region forest structure heterogeneity evaluation, and optimize the priority region selection of forest management. This research contributes to decreasing the costs of labor-intensive forest management. The approach was applied in a subtropical plantation in Southern China. Results showed that point clouds exhibit great potential in depicting fine-scale stand spatial structure heterogeneity. The precision and the F-score of the individual tree extraction were more than 95 % and 92 % in different forest types, respectively. The analysis of the tree- and plot-level stand structure indicated that the spatial heterogeneity of tree height and crown are at a relatively low level. Mixed forests with pronounced vertical layering are considered to have a more heterogeneous structural composition. Large-scale structure mapping showed that spatial distribution pattern is predominantly characterized by uniform distribution, and the overall degree of competition is relatively low. Priority forest management area were found to be in the northwest and southeast with a relatively high stem density distribution. The top-down 3D perspective of UAV-LiDAR makes up for the limitations of the 2D perspective of UAV images and traditional field measured-based structure calculation. Our method provides a novel and feasible solution for micro area stand structure heterogeneity mapping across various forest types. © 2024 Elsevier B.V.

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A novel model to predict O-glycosylation sites using a highly unbalanced dataset

Item Type Journal Article

Author K. Zhou

Author C. Ai

Author P. Dong

Author X. Fan

Author L. Yang

Abstract In silico approaches have become an alternative method to study O-glycosylation. In this paper, we developed a linear interpretable model for O-glycosylation prediction based on an unbalanced dataset, analyzing the underlying biological knowledge of glycosylation. A training set of 4446 sites involving 468 positive sites and 3978 negative sites was developed during this research. The sites were encoded using the amino acid index (AAindex), and the forward stepwise procedure utilized for feature selection. The linear discriminant analysis with an equal a priori probability (PP-LDA) was employed to develop the interpretable model. Performance of the model was verified using both the internal leave-one-out cross-validation and external validation methods. Two non-linear algorithms, the supervised support vector machine and the unsupervised self-organizing competitive neural network, were used as comparisons. The PP-LDA model exhibited improved classification results with accuracy of 82.1 % for crossvalidations and 80.3 % for external prediction. Further analysis of this linear model indicated that the properties at position R1 and the properties relative to hydrophobicity contributed more to the glycosylation prediction. However, the alpha and turn propensities at the C-Terminal, together with physicochemical properties at the N-Terminal, are also relative to the glycosylation activity. This model is not only capable of predicting the possibility of glycosylation using an unbalanced dataset, but is also helpful to understand the underlying biological mechanisms of glycosylation. Considering the publicly accessibility of our prediction model, a downloadable program is provided in our supply materials. ©

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article, prediction, controlled study, artificial neural network, Software, accuracy, support vector machine, validation process, intermethod comparison, mathematical model, probability, Feature selection, discriminant analysis, Databases, Protein, priority journal, Models, Genetic, glycosylation, Neural Networks (Computer), Amino acid index, amino terminal sequence, carboxy terminal sequence, Glycoproteins, Glycosylation, hydrophobicity, PP-LDA, Protein glycosylation prediction, Protein Structure, Tertiary, Sequence Analysis, Protein

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A novel multi-featured decision system for multi-classification tasks

Item Type Journal Article

Author T. Xu

Abstract Feature engineering is a difficult task, and for real signal data, it is difficult to find a certain feature that can easily distinguish all classes. Multiple features can provide more information, which means the fusion of multi-feature learning strategies has potential significant advantages. Based on this premise, this paper proposes a multi-class framework based on the multi-featured decision to distinguish all the different classes, and takes automatic dependent surveillance-broadcast (ADS-B) signal data as

an example, first extracts the phase features and wavelet decomposition features of the signal data, then selects the features with high discrimination between classes, then proposes a one-dimensional residual neural network based on 16 convolutional layers to learn the unique features of different features and classes separately, and finally proposes a novel multi-featured decision method based on voting method and a priori probability. Results show that the proposed one-dimensional residual neural network has better performance metrics on the test set compared to some machine learning-based and neural network-based algorithms, with classification accuracies of 86.1%, 84.6% and 83.6% on wavelet decomposition features, raw features and phase features, respectively, on ADS-B preamble signals. The proposed feature decision framework based on the voting method and a priori probability has a recall, precision and F1 value of 80.24%, 89.89% and 84.79% on ADS-B preamble signals, respectively. © 2023 IOP Publishing Ltd.

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A novel non-destructive assay methodology for the disposal of the LHC beam dumping system at CERN

Item Type Journal Article

Author N. Mena

Author P. Dyrce

Author A.-P. Bernardes

Author M. Calviani

Author G. Dumont

Author A. Infantino

Author M. Magistris

Author C. Theis

Abstract During the Long Shut-down (LS2), the beam stopper of the Large Hadron Collider (LHC) was replaced. It was decided to eliminate the extracted two LHC beam dumps, also known as Target Dump for Ejected Beam External (TDE), as radioactive waste. The radiological characterization of the dumps is a vital and a necessary step for the proper classification and subsequent acceptance by the radioactive waste repository. The acceptance criteria require accurate declaration of the activity values of the Easy-to-Measure (ETM) radionuclides. This paper provides an overview of the challenges related to the high contact dose rates (reaching 2 mSv/h) and the heterogeneous distribution of the ETM radionuclides. This work proposes a novel quantification methodology that addresses those challenges. The developed methodologies followed can serve as a guide for addressing similar challenges in other accelerator and nuclear facilities. © 2025 Elsevier Ltd

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Article, simulation, article, nonhuman, Nuclear energy, software, methodology, feasibility study, Monte Carlo method, Monte Carlo methods, calibration, Nondestructive examination, 1,1 dichloro 2,2 bis(4 chlorophenyl)ethane, Activation, activity concentration, activity distribution, Activity distribution, Actual geometry model optimization, analytical parameters, averaged specific activity, CERN monte carlo, Chemical activation, chemical composition, chemical elemental composition, cobalt 57, cobalt 60, Colliding beam accelerators, contact dose rate, conversion coefficient, declaration threshold, dumping, dumping system, electron beam, elemental composition, encapsulation threshold, Fluka calculation, FLUKA., FLUKA.CERN Monte Carlo, gamma spectrometry, Gamma spectrometry, Geological repositories, geometry, Geometry model, germanium, graphite, hadron, heterogeneity, High energy physics, immobilization matrix, InSitu Counting Object System, irradiation, Laboratory Sourceless Calibration Software, Large Hadron Collider, large hadron collider beam, Large-hadron colliders, magnetic and electromagnetic equipment, manganese 54, maximum acceptance limit, Microstrip devices, Model optimization, nondestructive assay, Particle accelerator, Plant shutdowns, proton energy, radiation dose, radiation exposure, radioactive waste, Radioactive waste, radioactive waste disposal, Radioactive waste disposal, Radioactive waste storage, Radioactive wastes, Radioactivity, radioisotope, Radiological characterization, radiological parameters, relative activity concentration, relative efficiency, scintiscanning, sodium 22, stainless steel, titanium 44, tritium, unclassified drug, uniform geometry

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A one-step biomarker quantification methodology for DCE-MRI of adnexal masses: Capturing kinetic pattern from early to late enhancement

Item Type Journal Article

Author A. Fathi Kazerooni

Author M. Nabil

Author H. Haghighat Khah

Author S. Parviz

Author M. Gity

Author H. Saligheh Rad

Abstract Purpose: To develop a one-step quantification approach that accounts for joint preprocessing and quantification of whole-range kinetics (early and late-phase washout) of dynamic contrast-enhanced (DCE) MRI of indeterminate adnexal

masses. Methods: Preoperative DCE-MRI of 43 (24 benign, 19 malignant) sonographically indeterminate adnexal masses were analyzed prospectively. A five-parameter sigmoid function was implemented to model the enhancement curves calculated within regions of interest. Diagnostic performance of five-parameter sigmoid model parameters (P1 through P5) was compared with pharmacokinetic (PK) modeling, semiquantitative analysis, and three-parameter sigmoid. Statistical analysis was performed using two-tailed student's t-test. Results: The results revealed that P2, representing the enhancement amplitude, is significantly higher, and P5, indicating the terminal phase, is generally negative in malignant lesions ($P < 0.001$). P2 (sensitivity = 79%, specificity = 87.5%, accuracy = 84%, area under the receiver operating characteristic curve = 91%) outperforms classification performances of PK and semiquantitative parameters. A combination of P2 and P5 shows comparable performance (sensitivity = 79%, specificity = 87.5%, accuracy = 84%, area under the receiver operating characteristic curve = 92%) to that of the combination of PK parameters, whereas the five-parameter sigmoid function maintains fewer assumptions than PK. Conclusions: The presented one-step quantification approach is helpful for accurate discrimination of benign from malignant indeterminate adnexal masses. Accordingly, P2 has considerably high diagnostic performance and terminal slope (P5), as a previously overlooked feature, contributes more than widely accepted early-enhancement kinetic features. *Magn Reson Med* 79:1165–1171, 2018. © 2017 International Society for Magnetic Resonance in Medicine. © 2017 International Society for Magnetic Resonance in Medicine

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A physics-informed uncertainty quantification framework for deep learning-based real-time lane-change intention prediction

Item Type Journal Article

Author Z. Cao

Author Z. Zheng

Author M. Yildirimoglu

Author S.M.M. Haque

Abstract Lane-change intention prediction represents a significant challenge in autonomous driving, particularly within mixed traffic environments where autonomous vehicles (AV) must interact and share road space with human-driven vehicles. While deep learning has advanced lane-change intention prediction, the absence of real-time uncertainty quantification limits operational robustness. We propose a physics-informed uncertainty quantification framework that integrates seamlessly with prediction models, enabling reliable and interpretable lane-change assessment for autonomous vehicles. We present a framework that advances lane-change intention prediction by overcoming two long-standing challenges: bias from class imbalance in naturalistic data and dependence on precise maneuver labels. It does so through empirical probability distribution maps and a Scenario-Aware Dynamic Distribution Integration (SADI) method that adaptively captures temporal driving patterns while embedding vehicle physics. This physics-informed approach aligns uncertainty

estimates with plausible behavior, yielding reliable and interpretable predictions. Tests on the highD dataset show that, unlike traditional uncertainty quantification methods such as dropout or ensemble methods, our framework delivers real-time reliability scores that remain robust across diverse and evolving traffic scenarios. To validate generalizability across diverse traffic conditions, we further evaluate the framework on the AD4CHE dataset, demonstrating zero-shot transferability without hyperparameter retuning. By embedding vehicle physics into uncertainty estimation, our framework transforms lane-change prediction from brittle heuristics into reliable, real-time assessments that adapt to complex traffic dynamics. © 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license. <http://creativecommons.org/licenses/by/4.0/>

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A Positive-Unlabeled Learning Approach With Self-Correcting Regularized Risk

Item Type Journal Article

Author S. Saha

Author A. Hassan

Author J.H. Paik

Abstract The primary objective of positive unlabeled learning (PU learning) is to train a binary classifier with positively labeled data and unlabeled data. An inherent aspect of this approach involves incorporating the positive class prior of unlabeled data directly into the classification process, which is typically challenging in real-world scenarios. Moreover, existing studies often lack evaluations of PU classifiers without involving the positive class prior (true or estimated) of the unlabeled data, representing a significant research gap. In this article, we introduce a robust, two-step PU learning algorithm by incorporating a potential negative sampler in step 1 (warm start) and minimizing a self-correcting regularized risk function in step 2. The risk function possesses a self-correcting property that attempts to mitigate the weakness of the potential negative sampler in the warm start step. The risk function enables us to enhance robustness in the presence of mislabeled candidate negative samples. We demonstrate the effectiveness of our method on image as well as text benchmarks. Results show that the proposed method consistently outperforms the state-of-the-art (SOTA) PU learning algorithms. © 2020 IEEE.

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A positive/unlabeled approach for the segmentation of medical sequences using point-wise supervision

Item Type Journal Article

Author L. Lejeune

Author R. Sznitman

Abstract The ability to quickly annotate medical imaging data plays a critical role in training deep learning frameworks for segmentation. Doing so for image volumes or video sequences is even more pressing as annotating these is particularly burdensome. To alleviate this problem, this work proposes a new method to efficiently segment medical imaging volumes or videos using point-wise annotations only. This allows annotations to be collected extremely quickly and remains applicable to numerous segmentation tasks. Our approach trains a deep learning model using an appropriate Positive/Unlabeled objective function using sparse point-wise annotations. While most methods of this kind assume that the proportion of positive samples in the data is known a-priori, we introduce a novel self-supervised method to estimate this prior efficiently by combining a Bayesian estimation framework and new stopping criteria. Our method iteratively estimates appropriate class priors and yields high segmentation quality for a variety of object types and imaging modalities. In addition, by leveraging a spatio-temporal tracking framework, we regularize our predictions by leveraging the complete data volume. We show experimentally that our approach outperforms state-of-the-art methods tailored to the same problem. © 2021

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A practical study on the feasibility of alpha and gamma-tocopherol quantification for distinguishing Iberian pig feeding systems

Item Type Journal Article

Author A.I. Rey

Author D. Amazán

Author J. García-Casco

Abstract This study was undertaken to test the feasibility of using alpha and gamma-tocopherol quantification in fat and muscle to establish a correct classification of Iberian pigs according to their feeding background in practical situations. Samples were obtained over three different seasons from the four categories of pigs described in the Industry Quality Policy (FREE-OUT "bellota"; FREE-OUT-FEED "recebo"; FEEDOUT "campo"; and FEED-IN "cebo"). Linear discriminant functions were calculated with data obtained from seasons 1 and 2 and validated. The classification of fat and muscle samples from season 3 in the four feeding categories according to the calculated discriminant functions achieved an average of 76% success rate in distinguishing the true origin of pig samples. Quantification in muscle seemed to have higher prediction ability. Regression equations to quantify weight gained depending on the gamma-tocopherol concentration had higher R² values for muscle than for fat (R² = 0.81 vs. 0.62). Merging of the feeding categories established by the Quality Policy could raise the accuracy of the alpha and gamma-tocopherol quantification method to up to between 89% and 98%.

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A preference learning framework for multiple criteria sorting with diverse additive value models and valued assignment examples

Item Type Journal Article

Author J. Liu

Author M. Kadziński

Author X. Liao

Author X. Mao

Author Y. Wang

Abstract We present a preference learning framework for multiple criteria sorting. We consider sorting procedures applying an additive value model with diverse types of marginal value functions (including linear, piecewise-linear, splined, and general monotone ones) under a unified analytical framework. Differently from the existing sorting methods that infer a preference model from crisp decision examples, where each reference alternative is assigned to a unique class, our framework allows considering valued assignment examples in which a reference alternative can be classified into multiple classes with respective credibility degrees. We propose an optimization model for constructing a preference model from such valued examples by maximizing the credible consistency among reference alternatives. To improve the predictive ability of the constructed model on new instances, we employ the regularization techniques. Moreover, to enhance the capability of addressing large-scale datasets, we introduce a state-of-the-art algorithm that is widely used in the machine learning community to solve the proposed optimization model in a computationally efficient way. Using the constructed additive value model, we determine both crisp and valued assignments for non-reference alternatives. Moreover, we allow the Decision Maker to prioritize the importance of classes and give the method a flexibility to adjust classification performance across classes according to the specified priorities. The practical usefulness of the analytical framework is demonstrated on a real-world dataset by comparing it to several existing sorting methods. © 2020 Elsevier B.V.

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A preregistered, open pipeline for early cerebral palsy risk assessment from infant videos

Item Type Journal Article

Author M. Segado

Author L.A. Prosser

Author A.F. Duncan

Author M.J. Johnson

Author K.P. Kording

Abstract Cerebral palsy (CP), affecting approximately 1 in 500 children due to abnormal brain development, impacts movement control. Early risk assessment via the general movements assessment (GMA) at 3–4 months is highly predictive for CP but relies on trained clinicians. Machine-learning-based approaches for predicting GMA score from video have shown considerable promise, but typically rely on dataset-specific preprocessing, custom feature sets, and manually designed model pipelines, which make external benchmarking more difficult. This, combined with strict privacy constraints on sharing data, makes it challenging to train and evaluate models across

datasets, which is important for assessing clinical utility. There is therefore a need to develop approaches that will work across different datasets to enable multi-site dataset aggregation and model training. To address this gap, we developed an end-to-end pipeline that uses off-the-shelf pose estimation, general-purpose feature extraction, and automated machine learning—none of which are tuned to a specific dataset. We applied this approach to a newly generated large dataset of 1053 infants (with approximately 10–12% positive class for adverse GMA outcome, drawn from a high-risk clinical cohort) within a preregistered study design. Model performance was evaluated on a strict “lock-box” test set, which remained untouched during any phase of model development or preprocessing optimization, and only used for evaluation once the final model and pipeline had been preregistered. The developed model achieved moderate predictive accuracy for clinician-assessed GMA scores (area under the receiver operating characteristic curve, ROC-AUC = 0.77; area under the precision-recall curve, PR-AUC = 0.41). The moderate accuracy is noteworthy given the 10–12% positive class prevalence, and power-law scaling of ROC-AUC as a function of increasing dataset size. By releasing de-identified feature data and open-source code, and simplifying the training pipeline using AutoML, our work establishes essential groundwork for future robust, globally relevant CP screening tools suitable for low-resource settings. © The Author(s) 2026. Published by Oxford University Press on behalf of GigaScience. This is an Open Access article distributed under the terms of the Creative Commons Attribution License (<https://creativecommons.org/licenses/by/4.0/>), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.

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Article, female, Female, human, Humans, male, Male, risk assessment, Risk Assessment, machine learning, spatiotemporal analysis, Machine Learning, kinematics, predictive model, procedures, feature extraction, diagnosis, predictive modeling, benchmarking, prevalence, Video Recording, videorecording, computer vision, area under the curve, diagnostic test accuracy study, receiver operating characteristic, child, cross validation, infant, resource limited setting, clinical evaluation, ROC Curve, sensitivity and specificity, pathophysiology, Infant, child development, brain development, cerebral palsy, Cerebral Palsy, clinician, infant development, motor dysfunction, movement analysis, movement disorders, pediatrics

Notes:

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A Prevalence Estimation of Exstrophy and Epispadias in Germany From Public Health Insurance Data

Item Type Journal Article

Author A.-K. Ebert

Author N. Zwink

Author H.M. Reutter

Author E. Jenetzky

Abstract Introduction: The prevalence of rare diseases is very important for health care research. According to the European Surveillance of Congenital Anomalies (EUROCAT) registers, the live prevalence for exstrophy and/or epispadias (grades 1–3) is reported with 1:23,255 (95% CI: 1:26,316; 1:20,000). A Europe-wide prevalence evaluation based on reports from excellence centers estimates a prevalence for exstrophies of 1:32,200 and for isolated epispadias of 1:96,800 in 2010. However, the frequency of exstrophy [International Statistical Classification of Diseases and Related Health Problems revision 10 (ICD-10): Q64.1] and epispadias (ICD-10: Q64.0) treated in different age groups in Germany remains unclear. Material and Method: Public health insurance data from 71 million people (approximately 87% of the population) were provided by the German Institute for Medical Documentation and Information (DIMDI) in accordance to the German Social Insurance Code for this research purpose. DIMDI analyzed the data source for the ICD diagnoses exstrophy and epispadias between 2009 and 2011. As provided data were robust over the years, averaged data are mentioned. Detailed subgroup analysis of small numbers was forbidden due to privacy protection. Results: Annually, 126 persons of all ages with epispadias and 244 with exstrophy are treated as inpatients. In the observed population, 34 infants (<1 year of age) with epispadias and 19 with exstrophy (58% male) are treated as outpatients each year. This corresponds to an estimated live prevalence of 1:11,000 (95% CI: 1:14,700; 1:8,400) for EEC (exstrophy–epispadias complex), more specifically a prevalence of 1:17,142 for epispadias and of 1:30,675 for exstrophy. The male-to-female ratio for exstrophy is 1.4:1 for infants and 1.6:1 for all minors. In children and adolescents, 349 epispadias and 393 exstrophies (up to the age of 17) are treated annually, whereas adults with exstrophy and even more with epispadias make comparatively less use of medical care. Conclusion: With the help of DIMDI data, the live prevalence of bladder exstrophy and epispadias in Germany could be estimated. The prevalence of epispadias was higher than in previous reports, in which milder epispadias phenotypes (grade 1 or 2) may not have been included. These analyses might enlighten knowledge about nationwide incidence and treatment numbers of rare diseases such as the EEC. Copyright © 2021 Ebert, Zwink, Reutter and Jenetzky.

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adult, Article, female, human, male, adolescent, public health, controlled study, major clinical study, prevalence, Germany, child, infant, sex ratio, incidence, adolescence, adulthood, bladder exstrophy, bladder exstrophy epispadias complex (EEC), epispadias, health insurance, information security, public health insurance

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A Property of the CHAID Partitioning Method for Dichotomous Randomized Response Data and Categorical Predictors

Item Type Journal Article

Author P.F. Perri

Author P.G.M. van der Heijden

Abstract In this paper, we present empirical and theoretical results on classification trees for randomized response data. We considered a dichotomous sensitive response variable with the true status intentionally misclassified by the respondents using rules prescribed by a randomized response method. We assumed that classification trees are grown using the Pearson chi-square test as a splitting criterion, and that the randomized response data are analyzed using classification trees as if they were not perturbed. We proved that classification trees analyzing observed randomized response data and estimated true data have a one-to-one correspondence in terms of ranking the splitting variables. This is illustrated using two real data sets. © 2011 Springer Science+Business Media, LLC.

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A Prospective Study of Hla-Oriented Therapy: An Attempt to Predict the Response to Anti-Cancer Therapy in Gastric Cancer

Item Type Journal Article

Author K. Ogoshi

Author K. Iwata

Author M. Miyaji

Author K. Nakamura

Author Y. Kajiura

Author K. Nabeshima

Author M. Morita

Author J. Soeda

Author Y. Kondoh

Author T. Tajima

Author H. Makuuchi

Abstract There are no strategies to predict a patient's response to therapy. In a previous report, we classified Japanese patients into four groups according to the incidence of HLA antigens, by using quantification method III. In that study, we examined the patients' incidence of HLA antigens before surgery and evaluated their outcome according to the HLA classification by quantification method III. The aim of the present study is to evaluate whether HLA classification should be used in the treatment of individual

cancer patients. A consecutive series of 76 Japanese patients who had undergone gastrectomy between July 1998 and January 2000 was evaluated to compare the patients HLA classifications, derived from quantification method III, with their prognoses following gastrectomy. Retrospective analysis of 193 gastric cancer patients revealed that HLA type I, type II, and type IV patients who had received postoperative adjuvant chemotherapy concomitant with PSK survived longer than those who had received postoperative adjuvant chemotherapy, whereas HLA type III patients who had received postoperative adjuvant chemotherapy showed significantly longer survival than those with chemotherapy + PSK. Following these results, we started a prospective study; we treated patients with HLA type I, type II, and type IV for postoperative chemotherapy + PSK, and those with HLA type III for postoperative chemotherapy following gastrectomy. The prospective study showed that patients who had been examined for HLA antigens showed significantly longer survival than those who had not been examined for HLA antigens. From the results of our retrospective and prospective studies, it can be inferred that it is necessary to examine gastric cancer patients for HLA antigens, and that HLA-oriented therapy is a very promising strategy for cancer treatment. We are planning to conduct an important study with control subjects. © 2000, The Japanese Society of Strategies for Cancer Research and Therapy. All rights reserved.

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A qPCR Assay for the Quantification of Selected Genotypic Variants of *Spodoptera frugiperda* Multiple Nucleopolyhedrovirus (Baculoviridae)

Item Type Journal Article

Author C.S. Molina-Ruiz

Author J.A. Zamora-Briseño

Author O. Simón

Author R. Lasa

Author T. Williams

Abstract Alphabaculoviruses are lethal dsDNA viruses of Lepidoptera that have high genetic diversity and are transmitted in aggregates within proteinaceous occlusion bodies. This mode of transmission has implications for their efficacy as biological insecticides. A Nicaraguan isolate of *Spodoptera frugiperda* multiple nucleopolyhedrovirus (SfMNPV-NIC) comprising nine genotypic variants has been the subject of considerable study due to the influence of variant interactions on the insecticidal properties of mixed-variant occlusion bodies. As part of a systematic study on the replication and transmission of variant mixtures, a tool for the accurate quantification of a selection of genotypic variants was developed based on the quantitative PCR technique (qPCR). First, primer pairs were designed around a region of high variability in four variants named SfNic-A, SfNic-B, SfNic-C and SfNic-E to produce amplicons of 103–150 bp. Then, using cloned purified amplicons as standards, amplification was demonstrated over a dynamic range of 108–10¹ copies of each target. The assay was efficient (mean $\pm$ SD: 98.5 $\pm$ 0.8%), reproducible, as shown by low inter- and intra-assay coefficients of variation (<5%), and specific to the target variants (99.7–100% specificity across variants). The quantification method was validated on mixtures of genotype-specific amplicons and demonstrated accurate quantification. Finally, mixtures of the four variants were quantified based on mixtures of budded virions and mixtures of DNA extracted from occlusion-derived virions. In both cases, mixed-variant preparations compared favorably to total viral genome numbers by quantification of the polyhedrin (polh) gene that is present in all variants. This technique should prove invaluable in elucidating the influence of variant diversity on the transmission and insecticidal characteristics of this pathogen. © 2024 by the authors.

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Article, nonhuman, procedures, classification, DNA extraction, gene amplification, genetics, genomic DNA, isolation and purification, genotype, animal, Animals, gene sequence, genetic variability, virology, Genotype, real time polymerase chain reaction, genetic variation, Genetic Variation, Sanger sequencing, Lepidoptera, alphabaculovirus, Baculoviridae, centrifugation, copy number determination, DNA isolation, DNA, Viral, efficiency, fall armyworm, genotypic diversity, Nucleopolyhedroviruses, polyhedrin, Real-Time Polymerase Chain Reaction, *Spodoptera*, *Spodoptera frugiperda*, virus DNA

Notes:

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: T. Williams; Instituto de Ecología AC (INECOL), Xalapa, Veracruz, 91073, Mexico; email: trevor.williams@inecol.mx

A Quantification Method for Environmental Exposure Based on Computer Image Processing Technology and Its Applications

Item Type Conference Paper**Author** R. Zheng**Editor** Trajkovic L.**Editor** Agaian S.S.**Editor** Avola D.**Editor** He J.

Abstract With the advancement of computer image recognition technology, traditional exposure measurement methods urgently need improvements in effectiveness and measurement accuracy. This study employs a method based on image processing and computer vision technology, combined with a wearable device aiming to measure individual exposure to geographic environments. A spatiotemporal weighting algorithm was applied to dynamically adjust exposure data, and comparisons with traditional methods were conducted for validation. Experimental results demonstrate that the proposed method can accurately measure exposure across different environments, particularly achieving higher classification accuracy in natural and built environments. Compared with traditional questionnaire-based approaches and geographic information system methods, the proposed method exhibits significant advantages in real-time assessment, accuracy, and detail capture of exposure. In natural environments, the study revealed significantly higher exposure levels to green vegetation and water bodies compared to urban roads. In cultural environments, commercial areas showed higher exposure levels, reflecting the influence of population density and social activities. Additionally, the stability and applicability of the method in real-world scenarios were validated, consistently providing reliable measurement results across different geographical regions and environmental conditions. This study empirically demonstrates an innovative application of computer vision technology, with significant prospects for public health applications. It advances the value of big data in interdisciplinary research. © 2025 IOS Press.

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Pages 297-307**Proceedings Title** Adv. Transdiscipl. Eng.**Conference Name** Advances in Transdisciplinary Engineering**Date Added** 20/09/2026, 9:51:42 pm**Modified** 20/09/2026, 9:51:42 pm**Tags:**

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A quantitative study of geometric characteristics of urban space based on the correlation with microclimate

Item Type Journal Article**Author** H. Ji**Author** Y. Peng**Author** W. Ding

Abstract With the sustainability of contemporary cities gaining more and more attention, interest in the correlation between urban geometry and urban microclimate is increasing. On this basis, this paper aims to investigate the quantification of geometric characteristics of urban space. Based on a combination of easily accessible software packages, a quantitative method composed of spatial partition, spatial characteristic indices (area, shape, and openness), and a spatial classification chart is proposed for the study of the correlation between urban spatial geometry and urban microclimate. Two blocks with different spatial geometric characteristics of the Xinjiekou central area in Nanjing are selected as the cases to verify the operability and effectiveness of this method. The results reveal that complex real urban space can be quantitatively described and classified by this spatial quantification method. In addition, a possible correlation between urban spatial geometry and urban wind environment is demonstrated by using the method, which may also be applicable to the correlation study between urban spatial geometry and other environmental issues.

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A RAIR–ATC transcriptional axis and multimodal drug-response modelling reveal class-level vulnerabilities in thyroid cancer

Item Type Journal Article

Author S. Wang

Author D. Zhang

Author W. Liu

Author M. Zhu

Author R. Li

Author Z. Liu

Author P. Wu

Author T. Liu

Abstract Introduction – Radioiodine-refractory papillary thyroid carcinoma (RAIR PTC) and anaplastic thyroid carcinoma (ATC) are clinically challenging thyroid cancer states, yet the molecular continuum between radioiodine refractoriness, anaplastic transformation and targeted drug sensitivity remains unclear. We hypothesised that

RAIR and ATC share a common transcriptional axis that can be distilled into a transferable signature to prioritise targeted therapies in RAIR-like models. **Methods** – We integrated RNA-seq and microarray cohorts (TCGA-THCA, one RAIR vs. radioiodine-avid PTC cohort, and five ATC cohorts) to derive RAIR/ATC gene modules and annotate pathway activity (MSigDB Hallmark). A single-sample RAIR signature was applied to TCGA tumours and DepMap thyroid cell lines. We combined RAIR scores with PRISM drug-response AUC data (18 thyroid cell lines) to quantify differential sensitivity (Δ AUC) and trained an interpretable multimodal ridge model integrating cell and drug features to predict AUC. External transferability and pharmacology-oriented mechanisms were assessed in an independent thyroid RNA-seq cohort (GSE126698) using ssGSEA and mechanism-relevant pathways. **Results** – RAIR and ATC upregulated modules converged on an EMT–angiogenesis–inflammatory/interferon axis, with ATC representing a more extreme state. The RAIR signature stratified TCGA tumours with graded activation of these programs and was largely independent of BRAF and TERT promoter status, while preserving pathway associations when transferred to DepMap. In PRISM, VEGFR/KDR and RAF/BRAF inhibitor classes showed the most negative class-averaged Δ AUC values, indicating preferential activity in RAIR-high thyroid lines. The multimodal ridge model improved AUC prediction over cell-only or drug-only baselines and was concordant with class-level sensitivity patterns. In GSE126698, RAIR scores retained robust associations with hypoxia and interferon/inflammatory programs and aligned with drug-mechanism pathways including KRAS signaling up and PI3K–AKT–mTOR signaling. Initial in vitro validation in a single RAIR-high/ATC-like thyroid cancer model provided proof-of-concept support for these predictions, with sorafenib showing dose-dependent cytotoxicity and suppression of RAF–MAPK signalling. **Conclusion** – A module-based RAIR signature captures a disease-focused component of a broader RAIR–ATC axis, transfers to cell-line models and can be embedded into an interpretable multimodal framework for drug-response prediction and targeted drug-class prioritisation, prioritising VEGFR/KDR and RAF/BRAF inhibitor classes as candidates for further translational evaluation in RAIR-like thyroid models. Copyright © 2026 Wang, Zhang, Liu, Zhu, Li, Liu, Wu and Liu.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: T. Liu; Department of Geriatrics, Changde Hospital, Xiangya School of Medicine, Central South University (The First People's Hospital of Changde City), Changde, Hunan, China; email: lty19840519@163.com

A rapid and feasible¹ H-NMR quantification method of ephedrine alkaloids in ephedra herbal preparations

Item Type Journal Article

Author H.-Y. Hung

Author S.-M. Lin

Author C.-Y. Li

Author S.-H. Lam

Author Y.-Y. Chan

Author M.-J. Liou

Author P.-C. Shieh

Author F.-A. Chen

Author P.-C. Kuo

Author T.-S. Wu

Abstract A highly specific and sensitive proton nuclear magnetic resonance (¹H-NMR) method has been developed for the quantification of ephedrine alkaloid derivatives in Ephedra herbal commercial prescriptions. At the region of δ 4.0 to 5.0 ppm in the ¹H NMR spectrum, the characteristic signals are separated well from each other, and six analogues in total, methylephedrine (ME), ephedrine (EP), norephedrine (NE), norpseudoephedrine (NP), pseudoephedrine (PE), and methylpseudoephedrine (MP) could be identified. The quantities of these compounds are calculated by the relative ratio of the integral values of the target peak for each compound to the known concentrations of the internal standard anthracene. The present method allows for a

rapid and simple quantification of ephedrine alkaloid derivatives in Ephedra-related commercial prescriptions without any preliminary purification steps and standard compounds, and accordingly it can be a powerful tool to verify different Ephedra species. In comparison to conventional chromatographic methods, the advantages of this method include the fact that no standard compounds are required, the quantification can be directly performed on the crude extracts, a better selectivity for various ephedrine alkaloid derivatives, and the fact that a very significant time-gain may be achieved. © 2021 by the authors. Licensee MDPI, Basel, Switzerland.

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A Rapid, Accurate and Machine-Agnostic Segmentation and Quantification Method for CT-Based COVID-19 Diagnosis

Item Type Journal Article**Author** L. Zhou**Author** Z. Li**Author** J. Zhou**Author** H. Li**Author** Y. Chen**Author** Y. Huang**Author** D. Xie**Author** L. Zhao**Author** M. Fan**Author** S. Hashmi**Author** F. Abdelkareem**Author** R. Eiada**Author** X. Xiao**Author** L. Li**Author** Z. Qiu**Author** X. Gao

Abstract COVID-19 has caused a global pandemic and become the most urgent threat to the entire world. Tremendous efforts and resources have been invested in developing diagnosis, prognosis and treatment strategies to combat the disease. Although nucleic acid detection has been mainly used as the gold standard to confirm this RNA virus-based disease, it has been shown that such a strategy has a high false negative rate, especially for patients in the early stage, and thus CT imaging has been applied as a major diagnostic modality in confirming positive COVID-19. Despite the various, urgent advances in developing artificial intelligence (AI)-based computer-aided systems for CT-based COVID-19 diagnosis, most of the existing methods can only perform classification, whereas the state-of-the-art segmentation method requires a high level of human intervention. In this paper, we propose a fully-automatic, rapid, accurate, and machine-agnostic method that can segment and quantify the infection regions on CT scans from different sources. Our method is founded upon two innovations: 1) the first CT scan simulator for COVID-19, by fitting the dynamic change of real patients' data measured at different time points, which greatly alleviates the data scarcity issue; and 2) a novel deep learning algorithm to solve the large-scene-small-object problem, which decomposes the 3D segmentation problem into three 2D ones, and thus reduces the model complexity by an order of magnitude and, at the same time, significantly improves the segmentation accuracy. Comprehensive experimental results over multi-country, multi-hospital, and multi-machine datasets demonstrate the superior performance of our method over the existing ones and suggest its important application value in combating the disease. © 1982-2012 IEEE.

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A Rapid, in-Situ Minimally-Invasive Technique to Assess Infections with *Pseudogymnoascus destructans* in Bats

Item Type Journal Article

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Author S.J. Puechmaille
Author J. Fickel
Author G.A. Czirják
Author C.C. Voigt

Abstract Emerging infectious diseases may become serious threats to wildlife, a prominent example being the white-nose disease (WND). In case of WND, the cold-loving fungus *Pseudogymnoascus destructans* colonizes bats during hibernation, invades the skin and has already lead to the death of millions of bats in North America. *P. destructans* most likely originated from Europe, where it also causes lesions but without associated mortalities. However, it is still unclear how European bats cope with the fungus. Hence, it is important to have tools that precisely characterise disease progression. Because hibernation is a physiological state during which bats are vulnerable to disturbance, in-situ assessments of the clinical status should be carried out minimal-invasively to avoid detrimental impacts on bats. However, currently available disease assessment methods require handling/touching bats and are therefore invasive: i) UV-light trans-illumination of wing membranes to detect lesions and ii) a qPCR-based quantification of fungal material from wing membrane swabs. Since *P. destructans* (‘Pd’) becomes visible on all furless skin with distinct distribution patterns, we investigated the use of visible symptoms to assess levels of infections without handling/touching bats. We introduce a technique which we termed ‘Visual Pd-score’ (a visual classification scheme), which can be applied without disturbing the animals. To assess its reliability, we used *P. destructans* infected greater mouse-eared bats (*Myotis myotis*) to compare the novel method with the two existing golden-standard techniques. Our results show that infection levels obtained from all three techniques are correlated. Importantly, the information carried by the Visual Pd-score is most similar to a composite index combining the information from the qPCR-based and UV-light quantification methods. We conclude that the Visual Pd-score represents a promising index to better characterise disease severity as it is simultaneously representative for fungal colonization and wing damage. We discuss advantages and disadvantages of the applied techniques and conclude that the Visual Pd-score is particularly useful for routine hibernacula counts or large-scale *P. destructans*-surveillance. In combination with the lesion detection technique the new method is also applicable to immunological studies where both fungal colonization and associated damage have to be investigated, while qPCRs from swabs of all body parts are especially useful if it is necessary to detect cryptic infections, e.g. during the early hibernation period. © 2021 Museum and Institute of Zoology PAS. All rights reserved.

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A recurrent neural network for sentiment quantification

Item Type Conference Paper

Author A. Esuli

Author A. Moreo Fernandez

Author F. Sebastiani

Editor Paton N.

Editor Candan S.

Editor Wang H.

Editor Allan J.

Editor Agrawal R.

Editor Labrinidis A.

Editor Cuzzocrea A.

Editor Zaki M.

Editor Srivastava D.

Editor Broder A.

Editor Schuster A.

Abstract Quantification is a supervised learning task that consists in predicting, given a set of classes C and a set D of unlabelled items, the prevalence (or relative frequency) $p_c(D)$ of each class $c \in C$ in D . Quantification can in principle be solved by classifying all the unlabelled items and counting how many of them have been attributed to each class. However, this classify and count approach has been shown to yield suboptimal quantification accuracy; this has established quantification as a task of its own, and given rise to a number of methods specifically devised for it. We propose a recurrent neural network architecture for quantification (that we call QuaNet) that observes the classification predictions to learn higher-order quantification embeddings, which are then refined by incorporating quantification predictions of simple classify-and-count-like methods. We test QuaNet on sentiment quantification on text, showing that it substantially outperforms several state-of-the-art baselines. © 2018 Copyright

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A reliability-noise-delay quantification method for APUF physical modeling and CRPs screening

Item Type Journal Article

Author Z. Huang

Author R. Liu

Author Z. Wang

Author Y. Zhu

Author J. Bian

Author Z. Yang

Author Y. Lu

Author Y. Ouyang

Author H. Liang

Author T. Ni

Abstract Arbiter Physical Unclonable Functions (APUFs), widely used strong PUFs, are highly sensitive to environmental noise, yet the quantitative link between noise and reliability remains unclear. This paper introduces a Reliability-Noise-Delay (RND) quantification method grounded in statistical analysis of the APUF delay model. By modeling noise, we uncover CRP instability patterns under varying noise levels and derive a closed-form relationship between reliability and noise. Leveraging RND, the delay difference between last-stage paths is calculated without extra hardware by using environmental noise measurements and CRP reliability. We develop an efficient n-stages APUF modeling approach requiring only n+1 CRPs with known reliability, significantly reducing computational complexity and resource demands compared to conventional machine-learning techniques. Furthermore, we propose a screening strategy to identify high- and low-reliability CRPs: at a noise level of 0.15, low-reliability CRP reliability drops from 75 % (random) to 57 %, while high-reliability CRPs reach 99 %. Our method offers an effective tool for APUF modeling and optimization in noisy environments. © 2025 Elsevier Ltd

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A Review of Precipitation Use Efficiency: Integrative Analysis of Ecological Connotation, Quantification Methods, and Driving Factors

Item Type Journal Article

Author S. Zou

Author L. Cao

Author F. Meng

Author E. Zheng

Author T. Li

Author G. Li

Author M. Li

Abstract Precipitation Use Efficiency (PUE) is a key ecological indicator for evaluating how vegetation converts precipitation into biomass or productivity. A thorough analysis of its quantification methods and driving mechanisms is of great significance for improving regional precipitation use efficiency and ensuring agricultural and ecological water security. In this study, we conducted a comprehensive literature search without time restrictions in the Web of Science and China National Knowledge Infrastructure (CNKI) databases, using “Precipitation Use Efficiency” and “PUE” as core keywords. After retrieval, a strict “independent dual-screening plus cross-checking” procedure was adopted with unified inclusion and exclusion criteria to ensure literature quality. Only highly relevant and methodologically rigorous studies were retained, resulting in a final set of 80 eligible publications. Key information was systematically extracted using content analysis, followed by integrated summarization and inductive analysis. This paper systematically illustrates the ecological connotation of PUE, compares diverse quantification and research methods with their applicable conditions, analyzes spatiotemporal differentiation characteristics and multidimensional driving mechanisms, summarizes practical approaches for PUE improvement, and reviews current research limitations. It represents a systematic integration and refinement of the research framework of precipitation use efficiency. The results can provide targeted theoretical support for revealing the driving mechanisms of PUE and promoting the efficient utilization of precipitation resources. © 2026 by the authors.

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A review of quantification methodologies for multi-hazard interrelationships

Item Type Journal Article

Author A. Tilloy

Author B.D. Malamud

Author H. Winter

Author A. Joly-Laugel

Abstract Globally and yearly, individual hazards and hazard interrelations have the potential to result in socio-economic losses. Here, in this critical review, we use grey- and peer-review literature to identify and compare current research available for the quantification of hazard interrelations, focussing on 14 different natural hazards. We first provide a historical context for quantitative single hazard and multi-hazard assessment. We then construct a literature database with 146 references related to multi-hazard interrelations. We use our literature database to identify trends for hazard interrelation and multi-hazard and from these group hazard interrelations into five types: triggering, change condition, compound, independence and mutually exclusive. Our critical review identifies 19 different modelling methods to quantify natural hazard interrelationships which we cluster into three broad modelling approaches: stochastic, empirical, and mechanistic. We then synthesize results of our classification of quantification methods for hazard interrelationships and using two matrices illustrate this in practice for 24 different interrelations between 14 natural hazards, one for cascading hazards (temporal order in the multi-hazard event) and one for compound hazards (two or more hazards acting together). Finally, we provide examples of applications for each the three quantitative modelling approaches defined. We believe that this review will lead to a better understanding of quantification methodologies for hazard interrelations between different sub-disciplines that focus on natural hazards, thus aiding cross-disciplinary approaches for better understanding potential risk related to multi-hazard events. © 2019 The Authors

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Export Date: 20 September 2026; Cited By: 296; Correspondence Address: A. Tilloy; Department of Geography, King's College London, London, Bush House (North East Wing), 30 Aldwych, WC2B 4BG, United Kingdom; email: alois.tilloy@kcl.ac.uk

A Review of Super-Resolution Single-Molecule Localization Microscopy Cluster Analysis and Quantification Methods

Item Type Journal Article

Author I.M. Khater

Author I.R. Nabi

Author G. Hamarneh

Abstract Single-molecule localization microscopy (SMLM) is a relatively new imaging modality, winning the 2014 Nobel Prize in Chemistry, and considered as one of the key super-resolution techniques. SMLM resolution goes beyond the diffraction limit of light microscopy and achieves resolution on the order of 10–20 nm. SMLM thus enables imaging single molecules and study of the low-level molecular interactions at the subcellular level. In contrast to standard microscopy imaging that produces 2D pixel or 3D voxel grid data, SMLM generates big data of 2D or 3D point clouds with millions of localizations and associated uncertainties. This unprecedented breakthrough in imaging helps researchers employ SMLM in many fields within biology and medicine, such as studying cancerous cells and cell-mediated immunity and accelerating drug discovery. However, SMLM data quantification and interpretation methods have yet to keep pace with the rapid advancement of SMLM imaging. Researchers have been actively exploring new computational methods for SMLM data analysis to extract biosignatures of various biological structures and functions. In this survey, we describe the state-of-the-art clustering methods adopted to analyze and quantify SMLM data and examine the capabilities and shortcomings of the surveyed methods. We classify the methods according to (1) the biological application (i.e., the imaged molecules/structures), (2) the data acquisition (such as imaging modality, dimension, resolution, and number of localizations), and (3) the analysis details (2D versus 3D, field of view versus region of interest, use of

machine-learning and multi-scale analysis, biosignature extraction, etc.). We observe that the majority of methods that are based on second-order statistics are sensitive to noise and imaging artifacts, have not been applied to 3D data, do not leverage machine-learning formulations, and are not scalable for big-data analysis. Finally, we summarize state-of-the-art methodology, discuss some key open challenges, and identify future opportunities for better modeling and design of an integrated computational pipeline to address the key challenges. Recent developments in super-resolution SMLM imaging techniques enable researchers to study macromolecular structures at the nanometer scale. However, SMLM data quantification and interpretation methods have yet to keep pace with the rapid advancement of SMLM imaging. This article provides a balanced and comprehensive review of state-of-the-art SMLM image analysis methods and ties disparate approaches together in a cohesive manner. Researchers are actively exploring new computational methods to analyze SMLM data, including recent approaches to use data-driven and machine-learning approaches. However, the validation of the SMLM clustering methods remains an open challenge. Potential future directions using multi-modality imaging (e.g., SMLM and electron microscopy) might help validate quantitative SMLM image analysis methods. Super-resolution single-molecule localization microscopy (SMLM) enables localization of components of macromolecular complexes at the nanometer scale. However, determining a complex structure from SMLM data-clustering analysis faces challenges of imaging artifacts, big data, 2D versus 3D data, and so forth. In this Review, we provide a holistic overview of state-of-the-art computational methods leveraged to quantify SMLM data. We classify the methods and list their pros and cons to help the researcher optimally consider the most appropriate quantification method. Finally, we show how the field is growing and draw conclusions about the applicability of data-driven approaches as well as methods validation and benchmarking. © 2020 The Author(s)

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A review of uncertainty quantification in deep learning: Techniques, applications and challenges

Item Type Journal Article

Author M. Abdar

Author F. Pourpanah

Author S. Hussain

Author D. Rezazadegan

Author L. Liu

Author M. Ghavamzadeh

Author P. Fieguth

Author X. Cao

Author A. Khosravi

Author U.R. Acharya

Author V. Makarenekov

Author S. Nahavandi

Abstract Uncertainty quantification (UQ) methods play a pivotal role in reducing the impact of uncertainties during both optimization and decision making processes. They have been applied to solve a variety of real-world problems in science and engineering. Bayesian approximation and ensemble learning techniques are two widely-used types of uncertainty quantification (UQ) methods. In this regard, researchers have proposed different UQ methods and examined their performance in a variety of applications

such as computer vision (e.g., self-driving cars and object detection), image processing (e.g., image restoration), medical image analysis (e.g., medical image classification and segmentation), natural language processing (e.g., text classification, social media texts and recidivism risk-scoring), bioinformatics, etc. This study reviews recent advances in UQ methods used in deep learning, investigates the application of these methods in reinforcement learning, and highlights fundamental research challenges and directions associated with UQ. © 2021 The Authors

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A review of uncertainty quantification in medical image analysis: Probabilistic and non-probabilistic methods

Item Type Journal Article

Author L. Huang

Author S. Ruan

Author Y. Xing

Author M. Feng

Abstract The comprehensive integration of machine learning healthcare models within clinical practice remains suboptimal, notwithstanding the proliferation of high-performing solutions reported in the literature. A predominant factor hindering widespread adoption pertains to an insufficiency of evidence affirming the reliability of the aforementioned models. Recently, uncertainty quantification methods have been proposed as a potential solution to quantify the reliability of machine learning models and thus increase the interpretability and acceptability of the results. In this review, we offer a comprehensive overview of the prevailing methods proposed to quantify the uncertainty inherent in machine learning models developed for various medical image tasks. Contrary to earlier reviews that exclusively focused on probabilistic methods, this review also explores non-probabilistic approaches, thereby furnishing a more holistic survey of research pertaining to uncertainty quantification for machine learning models. Analysis of medical images with the summary and discussion on medical applications and the corresponding uncertainty evaluation protocols are presented, which focus on the specific challenges of uncertainty in medical image analysis. We also highlight some potential future research work at the end. Generally, this review aims to allow researchers from both clinical and technical backgrounds to gain a quick and yet in-depth understanding of the research in uncertainty quantification for medical image analysis machine learning models. © 2024 The Authors

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A review on rice plant phenotyping traits estimation for disease and growth management using modern ML techniques

Item Type Journal Article**Author** B. Patel**Author** A. Sharaff

Abstract Over the past decades, rice crops have been crucially acknowledged as one of the most powerful energy streams for the production of resources. Plant phenotyping trait estimation includes the external feature evaluation of the plants for production growth. Phenotyping using machine learning techniques outperforms the other imaging techniques for the analysis of traits including leaf, seed, branch, panicle, flower root, shoot, etc. Rice plants, categorized by multiple traits such as growth analysis and disease management, are considered a contributing factor to the agricultural, economic, and communal losses in the upcoming development of the agricultural field. The last 15 years' diagnosis of plant disease in relation to image processing techniques has remained an area of interest among researchers. Several disease detections, identification, and quantification methods have been developed and applied to a wide variety of crops. This paper reviews the related research papers from the period between 2007 and 2023, with a focus on the development of the state of the art. The related studies are compared based on image segmentation, feature extraction, feature selection, and classification. This paper also outlines the current achievements, limitations, and suggestions for future research associated with the diagnosis of rice plant growth analysis and disease identification using machine learning techniques. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2023.

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A review on quantification learning

Item Type Journal Article**Author** P. González**Author** A. Castaño**Author** N.V. Chawla**Author** J.J. Del Coz

Abstract The task of quantification consists in providing an aggregate estimation (e.g., the class distribution in a classification problem) for unseen test sets, applying a model that is trained using a training set with a different data distribution. Several real-world applications demand this kind of method that does not require predictions for individual examples and just focuses on obtaining accurate estimates at an aggregate level. During the past few years, several quantification methods have been proposed from different perspectives and with different goals. This article presents a unified review of the main approaches with the aim of serving as an introductory tutorial for newcomers in the field. © 2017 ACM.

Date 2018**DOI** 10.1145/3117807**URL** <https://www.scopus.com/pages/publications/85030687694?origin=resultslist>**Archive** Scopus**Language** English**Volume** 50**Publisher** Association for Computing Machinery**Publication** ACM Computing Surveys**ISSN** 03600300 (ISSN)**Issue** 5**Journal Abbr** ACM Comput Surv**Date Added** 20/09/2026, 9:52:39 pm

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A risk-based comparison of classification systems

Item Type Conference Paper

Author S. Thorsen

Author S. Wagenman

Author M.E. Oxley

Author K.W. Bauer

Abstract Performance measures for families of classification system families that rely upon the analysis of receiver operating characteristics (ROCs), such as area under the ROC curve (AUC), often fail to fully address the issue of risk, especially for classification systems involving more than two classes. For the general case, we denote matrices of class prevalence, costs, and class-conditional probabilities, and assume costs are subjectively fixed, acceptable estimates for expected values of class-conditional probabilities exist, and mutual independence between a variable in one such matrix and those of any other matrix. The ROC Risk Functional (RRF), valid for any finite number of classes, has an associated parameter argument, which specifies a member of a family of classification systems, and for which there is an associated classification system minimizing Bayes risk over the family. We typify joint distributions for class prevalences over standard simplices by means of uniform and beta distributions, and create a family of classification systems using actual data, testing independence assumptions under two such class prevalence distributions. Examples are given where the risk is minimized under two different sets of costs. © 2009 SPIE.

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Volume 7336**ISBN** 0277786X (ISSN); 978-081947602-9 (ISBN)**Proceedings Title** Proc SPIE Int Soc Opt Eng**Conference Name** Proceedings of SPIE - The International Society for Optical Engineering**Date Added** 20/09/2026, 9:52:42 pm**Modified** 20/09/2026, 9:52:42 pm**Tags:**

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Thorsen; Department of Mathematical Sciences, US Air Force Academy, USAFA, CO 80840, United States; email: steven.thorsen@usafa.edu; Conference name: Signal Processing, Sensor Fusion, and Target Recognition XVIII; Conference date: 13 April 2009 through 15 April 2009; Conference code: 76809; CODEN: PSISD

A Robotic-Based Framework for Quantifying Surface Cracks of Concrete Shear Walls

Item Type Conference Paper**Author** P. Bazrafshan**Author** A. Ebrahimkhanlou**Editor** Farhangdoust S.**Editor** Guemes A.**Editor** Chang F.-K.

Abstract The inherently subjective and manual nature of current structural assessment procedures undermines the reliability of such practices. As a result, researchers are using artificial intelligence and robots to encourage autonomy and create methods that are more objective and less biased. Robots are equipped with cameras and sensors to collect information about the structure, such as images of the crack patterns of concrete structures. In this paper, a robotic framework is proposed to utilize the collected images of reinforced concrete shear walls (RCSWs) with crack patterns, and then those images are used to quantify the level of damage. Crack pattern images are from the load steps of a quasi-static test on an RCSW. To quantify the extent of damage, crack patterns are first converted to a mathematical representation, a graph. Next, a machine learning algorithm is trained to predict the energy dissipated during each load cycle based on the graphs. Results reveal that the presented graph-based damage quantification method is able to quantify the level of structural damage. High R2 scores of the machine learning regression (above 0.98) attest to the success of the

proposed robotic-based framework. © 2023 by DEStech Publications, Inc. All rights reserved

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A robust and interpretable machine learning approach using multimodal biological data to predict future pathological tau accumulation

Item Type Journal Article

Author J. Giorgio

Author W.J. Jagust

Author S. Baker

Author S.M. Landau

Author P. Tino

Author Z. Kourtzi

Abstract The early stages of Alzheimer's disease (AD) involve interactions between multiple pathophysiological processes. Although these processes are well studied, we still lack robust tools to predict individualised trajectories of disease progression. Here, we employ a robust and interpretable machine learning approach to combine multimodal biological data and predict future pathological tau accumulation. In particular, we use machine learning to quantify interactions between key pathological markers (β -amyloid, medial temporal lobe atrophy, tau and APOE 4) at mildly impaired and asymptomatic stages of AD. Using baseline non-tau markers we derive a prognostic index that: (a) stratifies patients based on future pathological tau accumulation, (b) predicts individualised regional future rate of tau accumulation, and (c) translates predictions from deep phenotyping patient cohorts to cognitively normal individuals. Our results propose a robust approach for fine scale stratification and prognostication with translation impact for clinical trial design targeting the earliest stages of AD. © 2022, The Author(s).

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A Scalable Framework for Multidimensional Quality Assessment and Benchmarking of EEG Datasets

Item Type Journal Article

Author M.Q. Perez

Author S. Lopez Bernal

Author E.H. Prat

Author L.M.D. Campo

Author L.F. Maimo

Author A.H. Celdran

Abstract Ensuring the quality of datasets containing electroencephalography signals remains challenging due to heterogeneous acquisition protocols, diverse preprocessing practices, and inconsistent annotation standards. Existing toolboxes mainly focus on signal cleaning and pipeline checks, while generic data quality frameworks lack domain specificity and do not provide an interpretable dataset-level score for electroencephalography datasets used in tasks such as affect recognition and event-related paradigms. This work introduces a platform for dataset-level quality assessment of electroencephalography collections tailored to these downstream tasks. The platform computes a set of indicators that capture session completeness, consistency of session length, class balance, annotation timing, signal-to-noise ratio, outlier prevalence, feature information content, and class separability. It then combines these indicators into a normalized global Quality Score ranging from 0 to 100. The aggregation relies on order-preserving mappings and subject-wise summarization to enable robust comparisons across datasets and to prevent longer recordings from dominating the score. The evaluation uses six public datasets from affect recognition, event-related paradigm and motor imagery, all processed under identical protocols. Classification performance is reported using Balanced Accuracy and Area Under the ROC Curve. The global score reflects differences in data integrity, temporal consistency, and separability while remaining interpretable across heterogeneous collections. In addition, correlation analyses indicate positive associations between separability-oriented indicators and downstream performance, and negative associations with a higher prevalence of outliers. Targeted training adjustments guided by these indicators, including the removal of highly contaminated windows and loss reweighting based on class priors, consistently improve

performance. Implementation and configuration files are publicly available to support reproducibility and adoption. © 2013 IEEE.

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A seismic risk classification framework for non-structural elements

Item Type Journal Article

Author G.J. O'Reilly

Author G.M. Calvi

Abstract The ability to quantify the seismic risk associated with structural and non-structural elements is a critical aspect of earthquake engineering. While methods to improve the understanding of structural response to earthquake shaking and how to quantify their risk have been studied, non-structural elements (NSEs) have more recently emerged as a crucial aspect to address given their pertinence in overall building performance and loss-related issues. This article describes the development of a risk quantification methodology for NSEs whereby the mean annual frequency of exceeding an NSE's

damage state is computed and rated as part of a risk classification scheme using recently developed approaches. The basis of the methodology is described in detail followed by an example implementation, where the details surrounding hazard, structural and non-structural response are quantified consistently, ensuring that uncertainties are also incorporated to be in line with modern performance-based earthquake engineering. Discussion is provided surrounding the potential future use of such an NSE risk classification scheme for both structural engineers looking to improve the performance of their buildings via NSE performance and also manufacturers. © 2021, The Author(s), under exclusive licence to Springer Nature B.V.

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A self-supervised representation learning method for detecting and classifying fishing behaviour from AIS data in the St. Anns Bank MPA

Item Type Journal Article

Author N. Beaulieu

Author B. MacInnis

Author C. Evans

Author A. Czenze

Author M. Mehlmann

Author M. Sutherland

Author C. Whidden

Author L. Morrison

Abstract Marine Protected Areas (MPA) increasingly rely on automatic identification system (AIS) data to monitor fishing activity. Enforcement remains difficult because AIS is self-reported, labels of fishing behaviour are scarce, and traffic patterns differ strongly between regions. This study develops a hierarchical machine learning framework that estimates the probability of fishing from AIS position time series using only latitude and longitude. First, a self-supervised temporal encoder is trained on millions of unlabeled AIS messages to learn motion embeddings that summarize 75-minute vessel trajectories. Supervised classifiers are then trained on these embeddings to predict fishing vessel type, active fishing behaviour, and fishing gear type. Finally, a label-shift calibration step combines these outputs with operational assumptions about local vessel composition, fishing activity levels, and gear usage. This step adapts the globally trained models to the specific conditions of the St. Anns Bank MPA to produce locally calibrated fishing probabilities. On held-out data, the vessel-type and behaviour classifiers achieve areas under the receiver operating characteristic curve of 0.91 and 0.94, and average precision scores of 0.86 and 0.88, respectively, while the gear classifier obtained moderate overall performance, performing best for common gears. When applied to two years of AIS data from St. Anns Bank, the framework successfully highlights a sparse set of trajectories, allowing for more focused inspection. These results show that self-supervised motion embeddings, combined with probabilistic fusion and operational calibration, offer a label-efficient and adaptable approach to AIS-based monitoring in marine protected areas. Copyright © 2026 Beaulieu, MacInnis, Evans, Czenze, Mehlmann, Sutherland, Whidden and Morrison.

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A Semi-Supervised Transfer Approach Based on Pairwise Learning for Tensile Strength Prediction of Hot-Rolled Cord Steel

Item Type Journal Article**Author** Z. Zhang**Author** Z. Lv**Author** F. He**Author** Y. Zhou

Abstract The tensile strength prediction of cord steel is critically importance for ensuring product quality and mitigating failure risks; however, the scarcity of labeled data, particularly for customized steel grades, presents a significant challenge. To address this challenge, a multi-source dataset is introduced, comprising labeled data from source steel grades and unlabeled data from target steel grades, to improve predictive accuracy. Accordingly, a Semi-Supervised Transfer Learning approach based on Pairwise Deviation Regression Network, termed PDRN-SSTL, is proposed. First, the Pairwise Deviation Regression Network reformulates regression as learning label deviations between sample pairs, thereby alleviating cross-domain label shift and capturing relative sample relationships. Second, an unsupervised domain adaptation strategy is employed to construct a pre-trained model that extracts common features while reducing distributional discrepancies between source and target domains. Subsequently, the shallow-layer parameters are frozen, and the model is fine-tuned using limited labeled target-domain data together with a consistency constraint on unlabeled data, thereby further improving the predictive accuracy. Experiments on two customized steel grades, SWRH72A-C and SWRH72A-D, demonstrate the superiority of the proposed method, achieving MAPE values of 1.369% and 1.447%, and 3% Relative Error Hit Rate of 94.23% and 88.24%, respectively. Comparative and ablation studies further confirm its effectiveness. © 2026 Wiley-VCH GmbH.

Date 2026**DOI** 10.1002/srin.202501000**URL** <https://www.scopus.com/pages/publications/105034319894?origin=resultslist>**Archive** Scopus**Language** English**Publisher** John Wiley and Sons Inc**Publication** Steel Research International**ISSN** 16113683 (ISSN)**Journal Abbr** Steel Res. Int.**Date Added** 20/09/2026, 9:51:43 pm**Modified** 20/09/2026, 9:51:43 pm

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A Sensing-to-Sorting Multimodal Robotic Pipeline for Waste Classification Using Low-Cost RGB-D Vision and Spectral Sensors With Adaptive Bayesian Fusion

Item Type Journal Article

Author P. Jitngernmadan

Author P. Boonmee

Author T. Nupim

Author P. Wannapanon

Author P.K.N. Waengnoi

Author A. Klayburee

Author P. Chophuk

Abstract The increasing volume of municipal solid waste has created a demand for reliable waste- sorting systems with a low-cost sensing configuration. However, the RGB and RGB-depth (RGB-D) vision methods remain sensitive to illumination variation, occlusion, deformation, and visual similarity between recyclable materials, such as transparent plastic and glass. This study addresses these issues by presenting a sensing-to-sorting multimodal robotic waste-sorting pipeline integrating RGB-D vision, low-cost AS7265x visible–near-infrared spectral sensing, adaptive Bayesian fusion, and robotic pick-and-place execution. The vision module uses YOLOv8s instance segmentation for detection and localization, whereas the spectral module uses CatBoost to classify the 18-channel reflectance signatures. The fusion strategy combines vision and spectral probabilities using confidence-aware weighting, contact validation, uncertainty indicators, cross-modal agreement, class priors, and confusion-matrix likelihoods. The experiments used 2,000 RGB images with 24,848 annotated instances and 1,000 spectral samples across four classes: glass, metal, paper, and plastic. YOLOv8s achieved 0.960 mAP50 and 0.819 mAP50–95, whereas CatBoost achieved 73.0% accuracy. Real-world validation using a 600-object evaluation set tested under three lighting conditions showed that the proposed method achieved an accuracy and macro-F1 scores of 0.831 ± 0.023 and 0.829 ± 0.024 , respectively, improving the average accuracy by approximately 10.9 and 11.8

percentage points over the vision- and spectral-only approaches, respectively. These results demonstrate improved robotic recyclable waste classification under practical lighting conditions with stable performance across lighting levels, thereby indicating the potential of the proposed system for small- and medium-scale recycling applications under laboratory conveyor-belt conditions. © 2013 IEEE.

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A Simple Method for Classifier Accuracy Prediction Under Prior Probability Shift

Item Type Conference Paper

Author L. Volpi

Author A. Moreo

Author F. Sebastiani

Editor Pedreschi D.

Editor Monreale A.

Editor Guidotti R.

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Editor Pellungrini R.

Abstract The standard technique for predicting the accuracy that a classifier will have on unseen data (classifier accuracy prediction – CAP) is cross-validation (CV). However, CV relies on the assumption that the training data and the test data are sampled from the same distribution, an assumption that is often violated in many real-world scenarios. When such violations occur (i.e., in the presence of dataset shift), the estimates returned by CV are unreliable. In this paper we propose a CAP method specifically designed to address prior probability shift (PPS), an instance of dataset shift in which the training and test distributions are characterized by different class priors. By solving a system of n^2 independent linear equations, with n the number of classes, our method estimates the n^2 entries of the contingency table of the test data, and thus allows estimating any specific evaluation measure. Since a key step in this method involves predicting the class priors of the test data, we further observe a connection between our method and the field of “learning to quantify”. Our experiments show that, when combined with state-of-the-art quantification techniques, under PPS our method tends to outperform existing CAP methods. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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A simple segmentation and quantification method for numerical quantitative analysis of cells and tissues

Item Type Journal Article

Author H.-K. Kang

Author K.-H. Kim

Author J.-S. Ahn

Author H.-B. Kim

Author J.-H. Yi

Author H.-S. Kim

Abstract BACKGROUND: Microscopic image analysis based on image processing is required for quantitative evaluation of decellularization. Existing methods are not widely used because of expensive commercial software, and machine learning-based techniques lack generality for decellularization because many high-resolution image data has to be processed. OBJECTIVE: In this study, we developed an image processing algorithm for quantitative analysis of tissues and cells in a general microscopic image. METHODS: The proposed method extracts the color images obtained by the microscope into reference images consisting of grayscale, red (R), green (G), and blue (B) information and transforms each into a binary image. The transformed images were extracted by separating the cells and tissues through outlier noise elimination, logical multiplication and labeling. In order to verify the method, decellularization of porcine aortic valve was performed by the electrical method. Slice samples were obtained by time and the proposed method was applied. RESULTS: The experimental results show that the segmentation of cells and tissues, and quantitative analysis of the number of cells and changes in tissue area during the decellularization process was possible. CONCLUSIONS: The proposed method shows that cell and tissue extraction and quantitative numerical analysis were possible in different brightness of microscopic images. © 2020-IOS Press and the authors. All rights reserved.

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A Soil Carbon Content Quantification Method Using A Miniature Millimeter Wave Radar Sensor and Machine Learning

Item Type Conference Paper

Author D. An

Author Y.Q. Chen

Abstract Soil carbon content plays an essential role in combating climate change, water cycling, and sustaining soil biodiversity. However, the conventional way of quantifying soil carbon content is labor intensive, lack of precision, slow, and costly. On large spatial scale, assessment of the effect of carbon (biochar) applied to the soil for soil health conditioning, remains to be very difficult. This paper for the first time demonstrates the viability using a millimeter-wave sensing method for quantifying soil carbon content. It can also distinguish biochar types from different biomass species. Furthermore, soil moisture monitoring, and biochar water retention capacity can also be quantified by utilizing the same miniature millimeter wave radar sensor empowered by machine learning. Specifically, in this study, we present our research materials, methodology, machine learning workflow, results, and the explanation and interpretation based on the physical principles of the millimeter wave radar array sensor in the context of soil carbon content. We validated our quantification method with supervised machine learning algorithm using real soil data collected in the field mixed with known biochar contents. The results show that our technique achieved a 95.7 per cent recognition accuracy across seven different biochar types. The work laid the foundation for future real-time, large spatial-scale evaluation and assessment of soil carbon content using biochar amendments or other related carbon-negative technologies. Thus, soil carbon content site-specific management can be made possible. © 2022 IEEE.

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A standardized framework for robust fragmentomic feature extraction from cell-free DNA sequencing data

Item Type Journal Article

Author H. Wang

Author P.D. Mennea

Author Y.K.E. Chan

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Author A. Vijayaraghavan

Author E.-J. Ditter

Author R. Bowers

Author M.D. Eldridge

Author D.S. Shcherbo

Author C.G. Smith

Author F. Markowetz

Author W.N. Cooper

Author T. Kaplan

Author N. Rosenfeld

Author H. Zhao

Abstract Fragmentomics features of cell-free DNA represent promising non-invasive biomarkers for cancer diagnosis. A lack of systematic evaluation of biases in feature quantification hinders the adoption of such applications. We compare features derived from whole-genome sequencing of ten healthy donors using nine library kits and ten data-processing routes and validated in 1182 plasma samples from published studies. Our results clarify the variations from library preparation and feature quantification methods. We design the Trim Align Pipeline and cfDNAPro R package as unified interfaces for data pre-processing, feature extraction, and visualization to standardize multi-modal feature engineering and integration for machine learning. © The Author(s) 2025.

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Article, human, Humans, Feature extraction, machine learning, Machine Learning, normal human, quality control, procedures, feature extraction, controlled study, Software, software, gene expression, clinical article, bioinformatics, DNA extraction, DNA sequencing, genetics, high throughput sequencing, High-Throughput Nucleotide Sequencing, Sequence Analysis, DNA, blood sampling, Cancer genomics, cell free nucleic acid, Cell-Free Nucleic Acids, CfDNA, circulating free DNA, fragmentomics, Fragmentomics, gene ontology, genomics, genotype, human cell, molecular biology, omics, oncogenomics, polymerase chain reaction, sequence alignment, whole genome sequencing, Whole Genome Sequencing

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A statistical approach to the confusion matrix for classification problems using machine learning

Item Type Journal Article

Author R. Sanchez-Marquez

Author J. Jabaloyes Vivas

Abstract The kappa coefficient and accuracy are widely utilised metrics for evaluating classification systems in machine learning and quality inspections. However, traditional kappa suffers a significant drawback: its value depends on class prevalence (the proportion of instances in each category), which can misrepresent a system's actual performance. Furthermore, relying solely on point estimates for both kappa and accuracy overlooks the crucial impact of sample size and estimation error. This paper addresses these critical limitations by providing a comprehensive and practical method for estimating the lower bound of the intrinsic kappa coefficient and accuracy, generalised to classification problems with any number of categories. Unlike its traditional counterpart, the “intrinsic” kappa is robust to sample prevalence, offering a more reliable measure of a classifier's inherent ability. We develop both exact and approximate methods for estimating these lower bounds. Our results demonstrate that the approximate methods yield estimations remarkably close to those from the exact method across a wide range of sample sizes and misclassified instances. This reliability is further validated through extensive Monte Carlo simulations and benchmarking on diverse machine learning datasets with varying degrees of class imbalance. These findings confirm the proposed framework's practical utility for providing a stable, uncertainty-aware performance measure in high-stakes applications. Real-life examples further illustrate the straightforward application of these robust estimation methods for practitioners. © 2026 Elsevier B.V.

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Notes:

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A stochastic gradient descent algorithm for structural risk minimisation

Item Type Conference Paper
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Editor Gavalda R.
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Editor Jantke K.P.
Editor Im Stadtwald, Geb. 43.8, Saarbrücken, 66125 Deutsches Forschungszentrum für Künstliche Intelligenz GmbH
Editor Takimoto E.
Abstract Structural risk minimisation (SRM) is a general complexity regularization method which automatically selects the model complexity that approximately minimises the misclassification error probability of the empirical risk minimiser. It does so by adding a complexity penalty term $\epsilon(m, k)$ to the empirical risk of the candidate hypotheses and then for any fixed sample size m it minimises the sum with respect to the model complexity variable k . When learning multicategory classification there are M subsamples m_i , corresponding to the M pattern classes with a priori probabilities p_i , $1 \leq i \leq M$. Using the usual representation for a multi-category classifier as M individual boolean classifiers, the penalty becomes $\sum_{i=1}^M p_i \epsilon(m_i, k_i)$. If the m_i are given then the standard SRM trivially applies here by minimizing the penalised empirical risk with respect to k_i , $1, \dots, M$. However, in situations where the total sample size $\sum_{i=1}^M m_i$ needs to be minimal one needs to also minimize the penalised empirical risk with respect to the variables m_i , $i = 1, \dots, M$. The obvious problem is that the empirical risk can only

be defined after the subsamples (and hence their sizes) are given (known). Utilising an on-line stochastic gradient descent approach, this paper overcomes this difficulty and introduces a sample-querying algorithm which extends the standard SRM principle. It minimises the penalised empirical risk not only with respect to the k_i , as the standard SRM does, but also with respect to the m_i , $i = 1, \dots, M$. The challenge here is in defining a stochastic empirical criterion which when minimised yields a sequence of subsample-size vectors which asymptotically achieve the Bayes' optimal error convergence rate. © Springer-Verlag Berlin Heidelberg 2003.

Date 2003

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A Study on Threat Analysis and Risk Assessment Based on the 'Asset Container' Method and CWSS

Item Type Journal Article**Author** Y. Kawanishi**Author** H. Nishihara**Author** H. Yoshida**Author** H. Yamamoto**Author** H. Inoue

Abstract In recent years, legislation and standardization of cyber security management for cyber-physical systems such as automotive systems have been progressing steadily. ISO/SAE 21434, published in 2021, addresses the management and analysis of electrical systems within road vehicles from a cybersecurity perspective. It also recommends some methods for the threat analysis and risk assessment (TARA) process. However, there are two problems in the evaluation methods derived from conventional security analysis approaches. One problem is related to the insufficient evaluation of attack feasibilities for cyber-physical systems by the CVSS-based approach. Another problem is the unclear relationship between damage factors in analyzing the impact of damage to each asset. In this paper, we focus on the TARA process, and apply an 'asset container' method for threat classification, proposed by the authors at DECSoS 2017, and a CWSS-based risk quantification method. Moreover, we can also add some perspective to improve risk evaluation suitable for automotive systems. Following our past studies on methodologies to evaluate the risk of such special cyber-physical systems, we can quantify risks limited to some cyber-physical systems, such as direct access attacks to in-vehicle networks. © 2013 IEEE.

Date 2023**DOI** 10.1109/ACCESS.2023.3246497**URL** <https://www.scopus.com/pages/publications/85149182868?origin=resultslist>**Archive** Scopus**Language** English**Volume** 11**Publisher** Institute of Electrical and Electronics Engineers Inc.**Pages** 18148-18156**Publication** IEEE Access**ISSN** 21693536 (ISSN)**Journal Abbr** IEEE Access**Date Added** 20/09/2026, 9:52:20 pm**Modified** 20/09/2026, 9:52:20 pm**Tags:**

Cybersecurity, Risk assessment, Vehicles, Risks assessments, Vehicle security, Cyber Physical System, risk analysis, Cybe-physical systems, Cyber-physical systems, Embedded systems, Risk analysis, Threat analyse and risk assessment, Threats analysis, ISO/SAE 21434, TARA, Containers, CWSS, In-vehicle security, security design, Security design

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A survey on forensic age estimation through DNA methylation: recent progress and future directions towards legal age estimation

Item Type Journal Article

Author R. Gonzalez-Romero

Author V. Suarez-Ulloa

Author O. Ibañez

Abstract DNA methylation (DNAm) has emerged as one of the most promising molecular biomarkers for age estimation (AE), offering a powerful complement to traditional anthropological and radiological methods. Its potential spans forensic, biomedical, and humanitarian domains, particularly in the identification of deceased individuals and the estimation of legal age in living persons. Despite its rapid development, the forensic implementation of DNAm-based AE models remains challenged by issues of validation, dataset diversity, and tissue-specific variability. This survey provides an up-to-date overview of DNAm-based AE models developed between 2019 and 2025. We catalogued and analyzed published models according to their data origin, tissue type, methylation quantification method, and mathematical approach. We also distinguish between models designed to estimate chronological age in living individuals and those focused on age-at-death estimation in deceased persons. Our analysis highlights trends toward tissue-specific model optimization, targeted approaches with a small number of markers, and the integration of machine learning to improve predictive performance. Special attention is given to the development of pediatric and young-adult epigenetic clocks tailored to the estimation of legal age, an area with growing societal and humanitarian relevance, particularly in migration and child protection contexts. By identifying the main methodological advances, limitations, and challenges, this review outlines a roadmap for the next generation of DNAm-based AE models, emphasizing the need for standardized validation protocols, increased population diversity, and interdisciplinary approaches that integrate molecular, radiological, and computational tools. © 2026 The Authors.

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A survey on state of art approaches in handling imbalance, positive and unlabelled data

Item Type Conference Paper

Author N. Deepa

Author R. Sumathi

Abstract Machine learning algorithm extends its horizon in various application domains. Real time applications in various sectors which can be effectively handled with machine learning techniques lack from it due to the unavailability of the required data. In such cases the data may be imbalanced, with a single class labeled and other unlabelled. However, this will be the prevailing scenario in future and has to be addressed. This work concentrates on studying the various approaches that are in practice to handle these kinds of data. In specific, it concentrates on approaches used for handling imbalance data sets and the recent works in it, methods used for handling positive unlabelled data. Though there are few solutions in practice for handling imbalance data and positive unlabelled data. Approaches that work at the data level and algorithm level approach for handling the former and techniques such as biased learning and class prior models for handling the later are in practice. One class classifier is another approach that can be used for addressing the problem that arises with the class imbalance in the dataset and

related issues. It has also been noted from the literature that one class classifier will be more suitable but it has not been widely used in addressing the problems such as class rarity. It can be inferred from this study that the one class classifier would play a significant role in various domains in future. © 2022 IEEE.

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A Survey on Uncertainty Quantification Methods for Deep Learning

Item Type Journal Article

Author W. He

Author Z. Jiang

Author T. Xiao

Author Z. Xu

Author Y. Li

Abstract Deep neural networks (DNNs) have achieved tremendous success in computer vision, natural language processing, and scientific and engineering domains. However, DNNs can make unexpected, incorrect, yet overconfident predictions, leading to serious consequences in high-stakes applications such as autonomous driving,

medical diagnosis, and disaster response. Uncertainty quantification (UQ) estimates the confidence of DNN predictions in addition to their accuracy. In recent years, many UQ methods have been developed for DNNs. It is valuable to systematically categorize these methods and compare their strengths and limitations. Existing surveys mostly categorize UQ methodologies by neural network architecture or Bayesian formulation, while overlooking the uncertainty sources each method addresses, making it difficult to select an appropriate approach in practice. To fill this gap, this article presents a taxonomy of UQ methods for DNNs based on uncertainty sources (e.g., data versus model uncertainty). We summarize the advantages and disadvantages of each category, and illustrate how UQ can be applied to machine learning problems (e.g., active learning, out-of-distribution robustness, and deep reinforcement learning). We also identify future research directions, including UQ for large language models (LLMs), AI-driven scientific simulations, and DNNs with structured outputs. © 2026 Copyright held by the owner/author(s).

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A theoretical treatment of the sources of variability in the output of pattern classifiers

Item Type Conference Paper

Author S. Gupta

Author M.K. Markey

Abstract Previously, several instances of variability in the output of pattern classifiers that have the same Receiver Operating Characteristic (ROC) curve have been observed. We present a theoretical framework for understanding some sources of this variability, which result in classifiers with monotonically related outputs. We restrict our analysis to pattern classifiers that discriminate between two linearly separable classes. We show that variability in the output of pattern classifiers can arise due to differences in the functional mappings between their inputs and outputs. We further identify some practical situations wherein such variability in the output of such pattern classifiers arises. These include situations in which there are differences in (a) the datasets employed for training and evaluation of classifiers, (b) the a priori probabilities of the two classes, or (c) the stochastic processes employed for training the different pattern classifiers. Previously, we proposed a technique based on the matching of the histograms of differently distributed classifier output to reduce the variability in their diagnostic performance and their output values. Here, we prove theoretically and demonstrate empirically on simulated data, that for monotonically related classifier outputs, this technique successfully learns the true monotonic transformation function that exists between different pattern classifier outputs.

Date 2009

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A toolkit for quantifying individual response to herbal extracts in metabolic and inflammatory stress

Item Type Journal Article**Author** S.-Y. Park**Author** O. Kwon**Author** T. van den Broek**Author** J. Bouwman**Author** J.Y. Kim

Abstract This study developed a health assessment tool to analyze dynamic stress responses and resilience with the PhenFlex challenge. This study integrated a health space model and machine learning to quantify and visualize the impact of herbal extracts on inflammatory and metabolic health at the individual level. Two randomized, double-blind, placebo-controlled crossover trials were conducted involving participants with PhenFlex challenge after overnight fasting. Blood samples were collected, and a machine learning algorithm was used to predict health estimation scores based on metabolic and inflammatory responses. The resulting health space model visually represents individuals' health status in a 2-D space. Intervention with herbal extracts (e.g., *Angelica keiskei*, AK, and *Capsosiphon fulvescens*, CF) resulted in higher health scores in the health space, indicating improved health. This research emphasizes the quantification of phenotypic changes for personalized nutrition and health optimization. Overall, this study provides a valuable toolkit for validating herbal extract efficacy and extends its application to personalized nutrition. © The Author(s) 2025.

Date 2025**DOI** 10.1038/s41538-024-00354-y**URL** <https://www.scopus.com/pages/publications/85218187684?origin=resultslist>**Archive** Scopus**Language** English**Volume** 9**Publisher** Nature Research**Publication** npj Science of Food**ISSN** 23968370 (ISSN)**Issue** 1**Journal Abbr** npj Sci. Food**Date Added** 20/09/2026, 9:51:30 pm**Modified** 20/09/2026, 9:51:30 pm**Notes:**

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A two dimensional accuracy-based measure for classification performance

Item Type Journal Article

Author M. Carbonero-Ruz

Author F.J. Martínez-Estudillo

Author F. Fernández-Navarro

Author D. Becerra-Alonso

Author A.C. Martínez-Estudillo

Abstract Accuracy has been used traditionally to evaluate the performance of classifiers. However, it is well known that accuracy is not able to capture all the different factors that characterize the performance of a multiclass classifier. In this manuscript, accuracy is studied and analyzed as a weighted average of the classification rate of each class. This perspective allows us to propose the dispersion of the classification rate of each class as its complementary measure. In this sense, a graphical performance metric, which is defined in a two dimensional space composed by accuracy and dispersion, is proposed to evaluate the performance of classifiers. We show that the combined values of accuracy and dispersion must fall within a clearly bounded two dimensional region, different for each problem. The nature of this region depends only on the a priori probability of each class, and not on the classifier used. Thus, the performance of multiclassifiers is represented in a two dimensional space where the models can be compared in a more fair manner, providing greater awareness of the strategies that are more accurate when trying to improve the performance of a classifier. Furthermore we experimentally analyze the behavior of seven different performance metrics based on the computation of the confusion matrix values in several scenarios, identifying clusters and relationships between measures. As shown in the experimentation, the graphical metric proposed is specially suitable in challenging, highly imbalanced and with a high number of classes datasets. The approach proposed is a novel point of view to address the evaluation of multiclassifiers and it is an alternative to other evaluation measures used in machine learning. © 2016 Elsevier Inc.

Date 2017

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Notes:

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Export Date: 20 September 2026; Cited By: 16; Correspondence Address: F. Fernández-Navarro; Department of Quantitative Methods, Universidad Loyola Andalucia, c/ Escritor Aguayo, 4, Córdoba, Spain; email: i22fenaf@uco.es; CODEN: ISIJB

A two-step anomaly detection based method for PU classification in imbalanced data sets

Item Type Journal Article

Author C. Ortega Vázquez

Author S. vanden Broucke

Author J. De Weerd

Abstract Several machine learning applications, including genetics and fraud detection, suffer from incomplete label information. In such applications, a classifier can only train from positive and unlabeled (PU) examples in which the unlabeled data consist of both positive and negative examples. Despite a substantial presence of PU learning in the literature, few works have considered a class imbalance setting. Hence, we propose a novel two-step method that exploits anomaly detection to identify hidden positives within the unlabeled data. Our method allows the end-user to choose the anomaly detector depending on preference or domain knowledge. Moreover, we introduce Nearest-Neighbor Isolation Forest (NNIF), a novel semi-supervised anomaly detector based on the Isolation Forest. In contrast to unsupervised anomaly detectors, NNIF can utilize all available label information. Empirical analysis shows that our method generally outperforms, using NNIF as the anomaly detector, state-of-the-art PU learning methods for imbalanced data sets under different labeling mechanisms. Further experiments suggest that our two-step method shows strong robustness to wrong class prior estimates. © 2023, The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature.

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Publisher Springer**Pages** 1301-1325**Publication** Data Mining and Knowledge Discovery**ISSN** 13845810 (ISSN)**Issue** 3**Journal Abbr** Data Min. Knowl. Discov.**Date Added** 20/09/2026, 9:52:14 pm**Modified** 20/09/2026, 9:52:14 pm**Tags:**

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Notes:

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A Unified Framework for Multi-view Multi-class Object Pose Estimation

Item Type Conference Paper**Author** C. Li**Author** J. Bai**Author** G.D. Hager**Editor** Weiss Y.**Editor** Ferrari V.**Editor** Sminchisescu C.**Editor** Hebert M.

Abstract One[NOSPACE] [NOSPACE][SPACE]core challenge in object pose estimation is to ensure accurate and robust performance for large numbers of diverse foreground objects amidst complex background clutter. In this work, we present a scalable framework for accurately inferring six Degree-of-Freedom (6-DoF) pose for a large number of object classes from single or multiple views. To learn discriminative pose features, we integrate three new capabilities into a deep Convolutional Neural Network (CNN): an inference scheme that combines both classification and pose regression based on a uniform tessellation of the Special Euclidean group in three dimensions (SE(3)), the fusion of class priors into the training process via a tiled class map, and an additional regularization using deep supervision with an object mask. Further, an efficient multi-view framework is formulated to address single-view ambiguity. We show that this framework consistently improves the performance of the single-view network. We evaluate

our method on three large-scale benchmarks: YCB-Video, JHUScene-50 and ObjectNet-3D. Our approach achieves competitive or superior performance over the current state-of-the-art methods. © 2018, Springer Nature Switzerland AG.

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A Unified View of Double-Weighting for Marginal Distribution Shift

Item Type Journal Article

Author J.I. Segovia-Martín

Author S. Mazuelas

Author A. Liu

Abstract Supervised classification traditionally assumes that training and testing samples are drawn from the same underlying distribution. However, practical scenarios are often affected by distribution shifts, such as covariate and label shifts. Most existing

techniques for correcting distribution shifts are based on a reweighted approach that weights training samples, assigning lower relevance to the samples that are unlikely at testing. However, these methods may achieve poor performance when the weights obtained take large values at certain training samples. In addition, in multi-source cases, existing methods do not exploit complementary information among sources, and equally combine sources for all instances. In this paper, we establish a unified learning framework for distribution shift adaptation. We present a double-weighting approach to deal with distribution shifts, considering weight functions associated with both training and testing samples. For the multi-source case, the presented methods assign source-dependent weights for training and testing samples, where weights are obtained jointly using information from all sources. We also present generalization bounds for the proposed methods that show a significant increase in the effective sample size compared with existing approaches. Empirically, the proposed methods achieve enhanced classification performance in both synthetic and empirical experiments. © 2025, Transactions on Machine Learning Research. All rights reserved.

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A unified view of label shift estimation

Item Type Conference Paper

Author S. Garg

Author Y. Wu

Author S. Balakrishnan

Author Z.C. Lipton

Abstract Under label shift, the label distribution $p(y)$ might change but the class-conditional distributions $p(x|y)$ do not. There are two dominant approaches for estimating the label marginal. BBSE, a moment-matching approach based on confusion matrices, is provably consistent and provides interpretable error

bounds. However, a maximum likelihood estimation approach, which we call MLLS, dominates empirically. In this paper, we present a unified view of the two methods and the first theoretical characterization of MLLS. Our contributions include (i) consistency conditions for MLLS, which include calibration of the classifier and a confusion matrix invertibility condition that BBSE also requires; (ii) a unified framework, casting BBSE as roughly equivalent to MLLS for a particular choice of calibration method; and (iii) a decomposition of MLLS's finite-sample error into terms reflecting miscalibration and estimation error. Our analysis attributes BBSE's statistical inefficiency to a loss of information due to coarse calibration. Experiments on synthetic data, MNIST, and CIFAR10 support our findings. © 2020 Neural information processing systems foundation. All rights reserved.

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A variational approach for learning from positive and unlabeled data

Item Type Conference Paper

Author H. Chen

Author F. Liu

Author Y. Wang

Author L. Zhao

Author H. Wu

Abstract Learning binary classifiers only from positive and unlabeled (PU) data is an important and challenging task in many real-world applications, including web text classification, disease gene identification and fraud detection, where negative samples are difficult to verify experimentally. Most recent PU learning methods are developed based on the misclassification risk of the supervised learning type, and they may suffer from inaccurate estimates of class prior probabilities. In this paper, we introduce a variational principle for PU learning that allows us to quantitatively evaluate the modeling error of the Bayesian classifier directly from given data. This leads to a loss function which can be efficiently calculated without involving class prior estimation or any other intermediate estimation problems, and the variational learning method can then be employed to optimize the classifier under general conditions. We illustrate the effectiveness of the proposed variational method on a number of benchmark examples. © 2020 Neural information processing systems foundation. All rights reserved.

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A vertical profile imaging method for quantifying rock fragments in gravelly soil

Item Type Journal Article

Author Z.-D. Jiang

Author Q.-B. Wang

Author K. Adhikari

Author K.R. Brye

Author Z.-X. Sun

Author F.-J. Sun

Author P.R. Owens

Abstract Rock fragments (RFs) play an important role in soils that can influence land use and management decisions. Rock fragment concentrations most often are visually estimated (VE) in the field, for which accuracy strongly depends on experience and practice of soil classifiers. The direct measurement (DM) method is more accurate, but is costly and labor-intensive. Due to its importance, a simple and inexpensive method for estimating RF concentrations without destructive soil sampling is necessary. The development of camera technology and digital image processing provide an opportunity for soil descriptions and quantitative analyses using an alternative soil profile imaging (PI) method. Therefore, the objective of this study was to explore the potential for RF concentration estimation using a PI method as compared to the VE and DM methods. Apart from VE and DM, different image processing procedures inside the PI method were compared in a natural, gravelly, alluvial profile (Entisol) with low contrast among RFs in order to develop an alternative and efficient RF quantification method. Results showed that the hue-saturation-value (HSV) color model performed better than the red-green-blue color model for identifying RFs. The proposed profile imaging method accurately segmented 84.9% of the RFs from the soil matrix. The PI had a greater correlation with DM ($r = 0.81$) compared to the VE method ($r = 0.78$). Results showed that the alternative PI method could be used to accurately and efficiently estimate RF concentrations in a natural, gravelly soil profile with low contrast among RFs. © 2020 Elsevier B.V.

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A vision-based hybrid classifier for weeds detection in precision agriculture through the bayesian and fuzzy k-Means paradigms

Item Type Conference Paper
Author A. Tellaeche
Author X.-P. Burgosartizzu
Author G. Pajares
Author A. Ribeiro
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Trondheim, O.S. Bragstads plass 2E Norwegian Univ. of Sci. and Technol.
Abstract One objective in Precision Agriculture is to minimize the volume of herbicides that are applied to the fields through the use of site-specific weed management systems. This paper outlines an automatic computer vision system for the

detection and differential spraying of *Avena sterilis*, a noxious weed growing in cereal crops. With such purpose we have designed a hybrid decision making system based on the Bayesian and Fuzzy k-Means (FkM) classifiers, where the a priori probability required by the Bayes framework is supplied by the FkM. This makes the main finding of this paper. The method performance is compared against other available strategies. © 2007 Springer-Verlag Berlin Heidelberg.

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Absolute Quantification of Bacteria in the Microbiome and Its Application

Item Type Journal Article

Author J.-X. Guo

Author Y.-Z. Gao

Abstract The advent of genomics and deep sequencing technologies has facilitated the development of absolute quantification techniques, which offer researchers more objective and precise sequencing outcomes. Unlike traditional relative quantification methods, which provide comparative data, absolute quantification delivers definitive measurements of genes or taxa. This analytical approach mitigates the potential for extraneous influences when comparing disparate samples, thereby reducing analytical errors. The implementation of absolute quantification techniques enhances our comprehension of microbial community structures, ecological dynamics, and their associations with host health or disease conditions. This chapter emphasizes a straightforward and broadly applicable method for genomic quantification, which necessitates the incorporation of a specified amount of internal standard DNA into the

samples, eliminating the need for subsequent adjustments during library construction and sequencing. By assessing the proportion of internal standard DNA across various samples, sequencing data can be transformed into absolute quantification metrics. The internal standard method for absolute quantification is versatile and can be effectively utilized across multiple domains, including disease diagnosis, microbial ecology research, the fermentation industry, and environmental monitoring. Overall, absolute quantification methods furnish a more accurate and holistic perspective for microbiome research. © The Author(s), under exclusive license to Springer Science+Business Media, LLC, part of Springer Nature 2026.

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Absolute quantitative metagenomic analysis reveals unique gut bacteria underlying berberine and metformin's anti-metabolic disorders effects

Item Type Journal Article**Author** J. Zhan**Author** B. Cheng**Author** K. Guo**Author** X. Tao**Author** X. Cai**Author** Z. Li**Author** Z. Tang**Author** J. Zhan**Author** C. Wu

Abstract This study aimed to evaluate the potential functional bacteria in the efficacy of two drugs in ameliorating diet-induced metabolic disorder model using absolute and relative quantification methods and to evaluate the benefit of absolute quantification compared with relative quantification. The gut microbiota is implicated in the pathogenesis of various chronic diseases, including diet-induced metabolic disorder model. Berberine (BBR) and metformin (MET) are commonly used in the clinical management of metabolic disorder, yet their effects on gut microbiota regulation differ. This study employs both relative and absolute quantitative methods to assess the differential impacts of these drugs on the modulation of gut microbiota in metabolic disorder mice. Both BBR and MET effectively ameliorate the condition of metabolic disorder. While some relative quantitative sequencing results contradicted the absolute sequencing data, the latter was more consistent with the actual microbial community composition. Absolute quantitative sequencing provides a more accurate reflection of the drug's effects. Notably, both absolute and relative quantitative sequencing demonstrated an upregulation of Akkermansia. **IMPORTANCE** Our study underscores the importance of absolute quantitative analysis in accurately representing the true microbial counts in a sample and evaluating the modulatory effects of drugs on the microbiome, which plays a vital role in the study of the microbiome. Copyright © 2025 Zhan et al.

Date 2025**DOI** 10.1128/spectrum.00084-25**URL** <https://www.scopus.com/pages/publications/105014954637?origin=resultslist>**Archive** Scopus**Language** English**Volume** 13**Publisher** American Society for Microbiology**Publication** Microbiology Spectrum**ISSN** 21650497 (ISSN)**Issue** 9**Journal Abbr** Microbiol. Spectr.**Date Added** 20/09/2026, 9:51:36 pm**Modified** 20/09/2026, 9:51:36 pm**Tags:**

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ABSTENTION IN REGRESSION

Item Type Conference Paper
Author C. Garcia-Cardona
Author J. Mohd-Yusof
Author T. Bhattacharya
Editor Papadrakakis M.
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Abstract Identifying noisy samples and outliers at training time is crucial to achieve good performance and improve robustness of deep learning models. However, this process can be difficult and excessively time consuming if attempted by hand, especially for large data sets. Moreover, in realistic situations, there are gradations in our judgement about noisy samples. In this work, we extend the Deep Abstention Classifier (DAC) framework to regression problems and devise a deep abstaining regressor (DAR) model. This is a forgiving strategy that limits the detrimental effect of noisy samples on the performance of the predictive model, while, at the same time, retaining information in ambiguous cases. We apply the new DAR formulation to synthetic and real data sets and demonstrate that the model is able to identify noisy samples, reducing their influence during training, while learning to quantify the uncertainty in the predictions via a heteroscedastic criterion. DAR constitutes a new tool for the machine learning practitioner, producing actionable uncertainty quantification predictions and helping with the automatic curation of data. This may be an efficient way to reduce the burden of

manual labelling for evaluation by a human-in-the-loop. © 2023 UNCECOMP Proceedings. All rights reserved.

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Accelerated Bayesian Calibration and Uncertainty Quantification of RANS Turbulence Model Parameters for Stratified Atmospheric Boundary Layer Flows

Item Type Journal Article

Author E.Y. Shin

Author M.F. Howland

Abstract In operational weather models, the effects of turbulence in the atmospheric boundary layer (ABL) on the resolved flow are modeled using turbulence parameterizations. These parameterizations typically use a predetermined set of model parameters that are tuned to limited data from canonical flows. Using these fixed parameters results in deterministic predictions that neglect uncertainty in the unresolved turbulence processes. In this study, we perform a machine learning-accelerated Bayesian inversion of a single-column model of the ABL. This approach is used to calibrate

and quantify uncertainty in model parameters of Reynolds-averaged Navier–Stokes turbulence models. To verify the data-driven uncertainty quantification methodology, we test in an idealized setup in which a prescribed but unobserved set of parameters is learned from noisy approximations of the model output. Following this verification, we learn the parameters and their uncertainties in two different turbulence models conditioned on scale-resolving large-eddy simulation data over a range of ABL stabilities. We show how Bayesian inversion of a numerical model improves flow predictions by investigating the underlying mean momentum budgets. Further, we show that uncertainty quantification based on neutral ABL surface layer data recovers the relationships between parameters that have been predicted using theoretical modeling, but that learning the parameters based on stable ABL data or data from outside the surface layer can lead to different parameter relationships than neutral surface layer theory. Efforts to systematically reduce parameter uncertainty reveal that (1) sampling wind speed up to the ABL height can reduce uncertainty in key model parameters by up to, and (2) assimilating fluid flow quantities beyond first-order moment statistics can further reduce uncertainty in ways that baseline wind speed assimilation alone cannot achieve. The parameters learned using Bayesian uncertainty quantification generally yield lower error than standard deterministic parameters in out-of-sample tests and also provide uncertainty intervals on predictions. © The Author(s) 2025.

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Accelerating ensemble uncertainty estimates in supervised materials property regression models

Item Type Journal Article

Author V. Agrawal

Author S. Zhang

Author L.E. Schultz

Author D. Morgan

Abstract A method is proposed to accelerate ensemble uncertainty quantification methods by replacing the ensemble method inference with a single model. An ensemble of N models is approximately N times more computationally demanding compared to a similar complexity single model when it is used for inference. We focus on regression models for material properties. Three material data sets were studied which include property data related to dilute solute diffusion, perovskite stability, and super conduction critical temperatures. We generated uncertainties from calibrated ensemble models for the aforementioned datasets. We explored fitting a single model to replace ensemble uncertainty estimates, which allows us to estimate uncertainties without the need for a full ensemble. Our approach is based on three models: Model A for predictive accuracy, Model AE for ensemble-based uncertainty prediction, and Model B, fit to data from Model AE, to be used for predicting the values of AE but with only one model evaluation. Model B leverages synthetic data augmentation to estimate uncertainties efficiently. This approach offers a highly flexible method of uncertainty quantification that can approximate that of ensemble methods but only requires a single extra model evaluation over Model A during inference. We evaluate the performance of this approach on a set of problems in materials sciences using modest-sized datasets with fewer than 100 features, demonstrating its potential generalizability to similar models and datasets. © 2024 Elsevier B.V.

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Accounting for Label Shift of Positive Unlabeled Data under Selection Bias

Item Type Journal Article

Author J. Mielniczuk

Author A. Wawrzęńczyk

Abstract We consider the scenario when two samples of positive unlabeled (PU) data are available and for the second sample the change in prior probability of classes occurs while distributions of predictors in classes remain the same (label shift setting). The selection of positive elements may be object-dependent. We study the properties of the underlying probabilistic structure under the novel augmented PU scenario, proving in particular that label shift occurs also for unlabeled populations. We introduce and investigate an estimator of prior probability for label-shifted population. Furthermore, in this case we construct and analyze behavior of Bayes classifier in this setting. It turns out to be a Bayes classifier for the unlabeled class with a modified threshold. This gives rise to its three empirical counterparts which are compared on benchmark data sets. © 2025 Jan Mielniczuk et al., published by Sciendo.

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Accuracy assessment in convolutional neural network-based deep learning remote sensing studies—part 1: Literature review

Item Type Journal Article

Author A.E. Maxwell

Author T.A. Warner

Author L.A. Guillén

Abstract Convolutional neural network (CNN)-based deep learning (DL) is a powerful, recently developed image classification approach. With origins in the computer vision and image processing communities, the accuracy assessment methods developed for CNN-based DL use a wide range of metrics that may be unfamiliar to the remote sensing (RS) community. To explore the differences between traditional RS and DL RS methods, we surveyed a random selection of 100 papers from the RS DL literature. The results show that RS DL studies have largely abandoned traditional RS accuracy assessment terminology, though some of the accuracy measures typically used in DL papers, most notably precision and recall, have direct equivalents in traditional RS terminology. Some of the DL accuracy terms have multiple names, or are equivalent to another measure. In our sample, DL studies only rarely reported a complete confusion matrix, and when they did so, it was even more rare that the confusion matrix estimated population properties. On the other hand, some DL studies are increasingly paying attention to the role of class prevalence in designing accuracy assessment approaches. DL studies that evaluate the decision boundary threshold over a range of values tend to use the precision-recall (P-R) curve, the associated area under the curve (AUC) measures of average precision (AP) and mean average precision (mAP), rather than the traditional receiver operating characteristic (ROC) curve and its AUC. DL studies are also notable for testing the generalization of their models on entirely new datasets, including data from new areas, new acquisition times, or even new sensors. © 2021 by the authors. Licensee MDPI, Basel, Switzerland.

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Export Date: 20 September 2026; Cited By: 325; Funding details: National Science Foundation, NSF, (2046059); National Science Foundation, NSF; Correspondence Address: A.E. Maxwell; Department of Geology and Geography, West Virginia University, Morgantown, 26505, United States; email: Aaron.Maxwell@mail.wvu.edu

Accuracy of Estimated Ratios as Affected by Dynamic Classification Errors

Item Type Journal Article

Author A. Van Delden

Author S. Scholtus

Author J. Burger

Author Q. Meertens

Abstract Within the field of official statistics, we are often interested in the relative change of statistical indicators through time. A typical example is economic growth measured as the ratio of gross domestic product at two consecutive time periods. In this article, we investigate the accuracy of such ratios when the numerator and denominator describe a subpopulation. Subpopulations are always identified by a classification variable, which might contain classification errors. Our main objective is to quantify the bias and variance of statistical estimates of ratios of statistical indicators that are affected by classification errors. Previous studies have already shown how to estimate the bias and variance of statistical indicators for a single subpopulation as affected by classification errors. For estimated ratios, such results are not yet available in the

academic literature. In this article, we will provide those results. More specifically, we will investigate three situations: the classification errors for the numerator and denominator are (1) completely dependent, (2) completely independent, or (3) partially dependent. For these three cases, we derive estimators of the bias and variance using a Taylor expansion and a bootstrap approach. By means of a simulation study, we show that the Taylor expansion is valid unless the distribution of the target variable of the numerator or the denominator is highly skewed. The methods are applied to two widely differing case studies: one on classification errors made by a machine learning classifier (independent errors) and one on classification errors in register data (partially dependent errors). This illustrates that our method can be useful for a wide range of applications. © 2022 The Author(s). Published by Oxford University Press on behalf of the American Association for Public Opinion Research. All rights reserved.

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Misclassifications, Quantification learning, Analytical approximation, Bootstrap, Nonsampling errors

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Accuracy of whole slide image based image analysis is adversely affected by preanalytical factors such as stained tissue slide and paraffin block age

Item Type Journal Article

Author N. Shaker

Author R. Sardana

Author S. Hamasaki

Author D.G. Nohle

Author L.W. Ayers

Author A.V. Parwani

Abstract Background: Personalized medicine and accurate quantification of tumor and biomarker expression have become the cornerstone of cancer diagnostics. This requires Quality Control (QC) of research tissue samples to confirm adequate targeted tumor tissue sampling. Digitalization of stained tissue slides offer a precious way to archive, preserve, and retrieve necessary information when needed. This study is aimed to assess the most significant pre-analytic and analytic factors that might contribute to the efficacy of obtaining accurate whole slide images (WSIs) interpretation. Various studies are needed to identify such factors to allow for appropriate AI application and adequate tumor area/percentage quantification. Methods: Hematoxyline and Eosine (H&E) stained WSIs collected from tissue specimens provided by the Cooperative Human Tissue Network (CHTN) Midwestern Division (CHTNMWD) were analyzed. Tissue specimens were processed, fixed, stained, and scanned contemporaneously (within 1 month). Two cohorts of malignant, colorectal cancer, 20X WSI (ScanscopeXT, Leica Biosystems, Illinois), were assembled. The study identified a “recent cohort” that included 76 WSIs created on 2018 or later. “Aged cohort” included 73 WSIs from specimens procured in the period of (2012-2014). Twenty recent WSIs of adenocarcinoma cases were used to construct WSIs analysis algorithms (VIS, Visiopharm A/S, Denmark) using machine learning to produce morphometric maps and calculate tissue and tumor areas. Results: Algorithmic analysis of 69 WSIs from rescanned aged slides vs. that of contemporaneous WSIs concluded 18 (28%) similar finding in tumor areas (within 10%), 56 (82%) had identical tissue areas, and 54 (79%) had similar tumor percentages. Conclusion: WSIs of aged H&E slides and stained paraffin block re-cuts produce different tumor quantification compared to those of original scanned slides most likely due to pre-analytical factors. The difference in tumor area detected between original and rescanned WSIs trended upward in the period between 2012 and 2014. Less tumor area was detected as the slides age. Recut and H&E-stained tissues from stored paraffin blocks may detect more tumor due to excess eosinophilia. These results highlights the value of documenting archives of H&E WSIs collected at the procurement time. Such images provide a superior archive over glass slides and Formalin-Fixed Paraffin-Embedded (FFPE) blocks and contribute better to WSIs analysis application. © 2022 The Authors

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Journal Abbr J. Pathol. Inform.

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Whole slide images, AI algorithm, Overall tumor percentage, Slide age, Tumor area identification

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Export Date: 20 September 2026; Cited By: 8; Funding details: National Institutes of Health, NIH, (UM1CA239749); National Institutes of Health, NIH; National Cancer Institute, NCI; Correspondence Address: N. Shaker; Department of Pathology, The Ohio State University Wexner Medical Center, Columbus, United States; email: nada.shaker@osumc.edu

Accurate Classification of Pathological Whole-Slide Images for Out-of-Distribution Generalization

Item Type Journal Article

Author K. Sun

Author K. Huang

Author J. Huang

Author M. Zhou

Author G. Yu

Abstract WSI-based classification often suffers from out-of-distribution (OOD) generalization due to the distribution mismatch between training on mixed patches from multiple WSIs and testing on individual WSIs with varying tissue compositions. This prior shift impairs model generalization and degrades performance. To address this issue, we propose two distribution alignment strategies: intra-WSI rearrange and inter-WSI rearrange, which, respectively, regulate patch distribution within individual WSIs and across different WSIs. These strategies are embedded into a transformer-based multi-instance learning (MIL) framework enabling more accurate and robust classification. Our method achieves excellent AUC scores of 0.959 and 0.963 on the CAMELYON16 and TCGA-NSCLC datasets, respectively. Moreover, it reaches an average AUC of 0.974 in 5-fold cross-validation on a private CRC dataset, matching the performance of patch-based approaches. Ablation studies further validate the effectiveness of our proposed strategies in mitigating the OOD challenge in WSI classification. Overall, these strategies enhance the robustness and accuracy of WSI-based models in handling OOD challenges. Copyright © 2025 Kai Sun et al. International Journal of Intelligent Systems published by John Wiley & Sons Ltd.

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Issue 1

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Deep learning, Intelligent systems, Performance, Classification (of information), Learning systems, Copyrights, Generalisation, deep learning, Image classification, Learning frameworks, Model generalization, Whole slide images, Out-of-distribution, image classification, Images classification, Multi-instance learning, out-of-distribution, Silicon compounds, Tissue composition, Tungsten compounds, whole-slide image, WSI circuits

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Accurate Graph-Based PU Learning without Class Prior

Item Type Conference Paper

Author J. Yoo

Author J. Kim

Author H. Yoon

Author G. Kim

Author C. Jang

Author U. Kang

Editor Bailey J.

Editor Miettinen P.

Editor Koh Y.S.

Editor Tao D.

Editor Wu X.

Abstract How can we classify graph-structured data only with positive labels? Graph-based positive-unlabeled (PU) learning is to train a binary classifier given only the positive labels when the relationship between examples is given as a graph. The problem is of great importance for various tasks such as detecting malicious accounts in a social network, which are difficult to be modeled by supervised learning when the true negative labels are absent. Previous works for graph-based PU learning assume that the prior distribution of positive nodes is known in advance, which is not true in many real-world cases. In this work, we propose GRAB (Graph-based Risk minimization with iterAtive Belief propagation), a

novel end-to-end approach for graph-based PU learning that requires no class prior. GRAB models a given graph as a Markov network and runs the marginalization and update steps iteratively. The marginalization step estimates the marginals of latent variables, while the update step trains a classifier network utilizing the computed priors in the objective function. Extensive experiments on five datasets show that GRAB achieves state-of-the-art accuracy, even compared with previous methods that are given the true prior. © 2021 IEEE.

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Extra Journal Abbreviation: Proc. IEEE Int. Conf. Data Min. ICDM

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Conference Name Proceedings - IEEE International Conference on Data Mining, ICDM

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Tags:

Markov processes, Computer vision, Graph algorithms, graph neural networks, Graph neural networks, Iterative methods, Graphic methods, Graph-based, Positive-unlabeled learning, Graph structured data, Binary classifiers, Backpropagation, PU learning, Belief propagation, Loopy belief propagation, Marginalization, Markov networks, Accurate graphs, loopy belief propagation, True negative

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Accurate quantification of choline-to-creatine ratio as a biomarker to distinguish osteosarcoma patients from normal subjects employing proton magnetic resonance spectroscopy imaging at 3 tesla

Item Type Journal Article

Author S.K. Alavijeh

Author F. Pashaei

Author M. Malek

Author H.S. Rad

Abstract Purpose: This study focused on accurate quantification of a maximum of Choline-to-Creatine ratio (Max (Cho/Cr)) in 10 Osteosarcoma patients, in comparison with 5 healthy volunteers as our control group using proton Magnetic Resonance Spectroscopy Imaging (1H-MRSI). Materials and Methods: Max (Cho/Cr) were obtained in 10 patients with Osteosarcoma over their corresponding ratio maps containing diseased tissue, to be compared with Cho/Cr in 5 healthy volunteers at 3T, employing MRSI (Performed Employing Pointed-resolved Spectroscopy (PRESS), TR/TE: 2500s /135 ms) with water-suppression. An extra unsuppressed water Single-Voxel Spectroscopy (SVS) was acquired to provide phase information for further Eddy Current Correction (ECC). Multi-stage preprocessing was applied. Subtract QUEST MRSI as a time-domain technique was employed to accurately quantify the metabolites' ratios and to estimate the baseline. Results: An optimal database for Subtract QUEST was achieved based on multiple trials evaluated by acceptable peak-fitting and Cramer-Rao-Bound (CRB). Lipids at frequencies of 0.94 and 1.33ppm were combined to increase the accuracy of the Lipid estimation. Conclusion: Estimation of Max (Cho/Cr) evaluated over Cho/Cr spatial maps to distinguish Osteosarcoma patients from normal subjects suggested that the proposed quantification method leads to high power and linear classifier with a high degree of reproducibility, considering 1H-MRSI at 3T machine as a high efficacy diagnostic tool for musculoskeletal radiology. Copyright © 2021 Tehran University of Medical Sciences.

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Magnetic Resonance Spectroscopy, Magnetic Resonance Spectroscopy Imaging, Metabolite, Osteosarcoma, Subtract Quantum Estimation

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H.S. Rad; Department of Medical Physics and Biomedical Engineering, School of Medicine, Tehran University of Medical Sciences, Tehran, Iran; email: h-salighehrad@tums.ac.ir

Accurate quantification of soil organic matter content using VNIR-SWIR spectra: The role of straw and spectrally active materials

Item Type Journal Article

Author C. Tan

Author H. Luan

Author Q. He

Author S. Yu

Author M. Zheng

Author L. Wang

Abstract Soil organic matter (SOM) is crucial for carbon sequestration and sustainable agriculture, yet traditional quantification methods are challenging to apply at large scales. Hyperspectral technology combined with machine-learning offers promising prospects for rapid quantification. This study explores the impact of using VNIR-SWIR spectra on SOM quantification in regions characterized by distinctive soil properties and agricultural activity. Specifically, we propose an innovative approach using 105 soil samples from Yueyang City, China, to refine the range of spectrally active materials and evaluate the effectiveness of iron oxides and straw on SOM quantification. Three feature construction methods (conventional (VNIR-SWIR spectra), optimal (information spectrum subset, ISS), and straw-merged ISS (SISS)) and seven models were employed to evaluate the contributions of iron oxides and straw in SOM quantification. The results indicate that the SISS improved the generalization (RPD and R²) of nonlinear and linear models by approximately 9 % and 4 %, respectively. The relative contributions of straw and iron oxides in modelling are approximately 35 % and 10 %, respectively. Our research successfully developed the SISS by refining the range of spectrally active materials and considering the background formed by the soil properties of the study area. We used it to evaluate the impact of straw on SOM quantification and demonstrated that the spectroscopic characterization of SOM can assess the carbon sequestration benefits of agricultural activities. This approach can be applied to regions with similar soil properties globally, offering a new perspective for SOM quantification. © 2024 Elsevier B.V.

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Machine learning, Hyperspectral, Anthrosols, Gleysols, Soil organic matter, Spectrally active materials, Straw

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Accurately quantifying a billion instances per second

Item Type Conference Paper**Author** W. Hassan**Author** A. Maletzke**Author** G. Batista**Editor** Webb G.**Editor** Zhang Z.**Editor** Tseng V.S.**Editor** Williams G.**Editor** Vlachos M.**Editor** Cao L.

Abstract Quantification is a thriving research area that develops methods to estimate the class prior probabilities in an unlabelled set of observations. Quantification and classification share several similarities. For instance, the most straightforward quantification method, Classify & Count (CC), directly counts the output of a classifier. However, CC has a systematic bias that makes it increasingly misestimate the counts as the class distribution drifts away from a distribution it perfectly quantifies. This issue has motivated the development of more reliable quantification methods. Such newer methods can consistently outperform CC at the cost of a significant increase in processing requirements. Yet, for a large number of applications, quantification speed is an additional criterion that must be considered. Frequently, quantification methods need to deal with large amounts of data or fast-paced streams, as it is the case of news feeding, tweets and sensor data. In this paper, we propose Sample Mean Matching (SMM), a highly efficient algorithm able to quantify billions of data instances per second. We compare SMM to a set of 14 established and state-of-the-art quantifiers in an empirical analysis comprising 25 benchmark and real-world datasets. We show

that SMM is competitive with state-of-the-art methods with no statistical difference in counting accuracy, and it is orders of magnitude faster than the vast majority of the algorithms. © 2020 IEEE.

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Extra Journal Abbreviation: Proc. - IEEE Int. Conf. Data Sci. Adv. Anal., DSAA

Publisher Institute of Electrical and Electronics Engineers Inc.

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Conference Name Proceedings - 2020 IEEE 7th International Conference on Data Science and Advanced Analytics, DSAA 2020

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Machine-learning, Machine learning, Computer vision, Machine Learning, Research areas, Matchings, Quantification methods, Prior probability, Quantification, Counts-as, Mixture method, Mixture Methods, Sample means, Systematic bias

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EID: 2-s2.0-85097975776

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Achieving Reliable and Fair Skin Lesion Diagnosis via Unsupervised Domain Adaptation

Item Type Conference Paper

Author J. Wang

Author Y. Zhang

Author Z. Ding

Author J. Hamm

Abstract The development of reliable and fair diagnostic systems is often constrained by the scarcity of labeled data. To address this challenge, our work explores the feasibility of unsupervised domain adaptation (UDA) to integrate large external datasets for developing reliable classifiers. The adoption of UDA with multiple sources can simultaneously enrich the training set and bridge the domain gap between different skin lesion datasets, which vary due to distinct acquisition

protocols. Particularly, UDA shows practical promise for improving diagnostic reliability when training with a custom skin lesion dataset, where only limited labeled data are available from the target domain. In this study, we investigate three UDA training schemes based on source data utilization: single-source, combined-source, and multi-source UDA. Our findings demonstrate the effectiveness of applying UDA on multiple sources for binary and multi-class classification. A strong correlation between test error and label shift in multi-class tasks has been observed in the experiment. Crucially, our study shows that UDA can effectively mitigate bias against minority groups and enhance fairness in diagnostic systems, while maintaining superior classification performance. This is achieved even without directly implementing fairness-focused techniques. This success is potentially attributed to the increased and well-adapted demographic information obtained from multiple sources. © 2024 IEEE.

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Archive Scopus

Language English

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Conference Name IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops

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Tags:

Labeled data, Domain adaptation, Domain Adaptation, Fairness, Data assimilation, Training sets, Acquisition protocols, Skin lesion, Multiple source, Diagnostic systems, Lesion analysis, Skin Lesion Analysis, Skin lesion analyze

Notes:

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Export Date: 20 September 2026; Cited By: 11; Correspondence Address: J. Wang; Tulane University, United States; email: swang47@tulane.edu; Conference name: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, CVPRW 2024; Conference date: 16 June 2024 through 22 June 2024; Conference code: 202970

ACML 2015 - 7th Asian Conference on Machine Learning

Item Type Journal Article

Abstract The proceedings contain 28 papers. The topics discussed include: detecting accounting frauds in publicly traded U.S. firms: a machine learning approach; improving Sybil detection via graph pruning and regularization techniques; class-prior estimation for learning from positive and unlabeled data; curriculum learning of Bayesian network structure; expectation propagation for rectified linear Poisson regression; preference relation-based Markov random fields for recommender systems; surrogate regret bounds for generalized classification performance metrics; streaming variational inference for Dirichlet process mixtures; continuous target shift adaptation in supervised learning; proximal average approximated incremental gradient method for composite penalty regularized empirical risk minimization; a unified framework for jointly learning distributed representations of word and attributes; regularized policy gradients: direct variance reduction in policy gradient estimation; budgeted bandit problems with continuous random costs; statistical unfolded logic learning; and integration of single-view graphs with diffusion of tensor product graphs for multi-view spectral clustering.

Date 2015

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Archive Scopus

Language English

Publisher Asian Conference on Machine Learning

Publication ACML 2015 - 7th Asian Conference on Machine Learning

Journal Abbr ACML - Asian Conf. Mach. Learn.

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ACon2: Adaptive Conformal Consensus for Provable Blockchain Oracles

Item Type Conference Paper

Author S. Park

Author O. Bastani

Author T. Kim

Abstract Blockchains with smart contracts are distributed ledger systems that achieve block-state consistency among distributed nodes by only allowing deterministic operations of smart contracts. However, the power of smart contracts is enabled by interacting with stochastic off-chain data, which in turn opens the possibility to undermine the block-state consistency. To address this issue, an oracle smart contract is used to provide a single consistent source of external data; but, simultaneously, this introduces a single point of failure, which is called the oracle problem. To address the oracle problem, we propose an adaptive conformal consensus (ACon2) algorithm that derives a consensus set of data from multiple

oracle contracts via the recent advance in online uncertainty quantification learning. Interesting, the consensus set provides a desired correctness guarantee under distribution shift and Byzantine adversaries. We demonstrate the efficacy of the proposed algorithm on two price datasets and an Ethereum case study. In particular, the Solidity implementation of the proposed algorithm shows the potential practicality of the proposed algorithm, implying that online machine learning algorithms are applicable to address security issues in blockchains. © USENIX Security 2023. All rights reserved.

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Language English

Extra Journal Abbreviation: USENIX Secur. Symp., USENIX Secur.

Volume 5

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Proceedings Title USENIX Secur. Symp., USENIX Secur.

Conference Name 32nd USENIX Security Symposium, USENIX Security 2023

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Tags:

Block-chain, Blockchain, Machine learning, Distributed ledger, Smart contract, Learning algorithms, Stochastic systems, Stochastics, E-learning, Power, Deterministics, Oracle problem, Block state, Distributed nodes, Electric power distribution, Is-enabled, Multiple oracles, Single point

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Acoustic classification of mosquitoes using convolutional neural networks combined with activity circadian rhythm information

Item Type Journal Article

Author J. Kim

Author J. Oh

Author T.-Y. Heo

Abstract Many researchers have used sound sensors to record audio data from insects, and used these data as inputs of machine learning algorithms to classify insect species. In image classification, the convolutional neural network (CNN), a well-known deep learning algorithm, achieves better performance than any other machine learning algorithm. This performance is affected by the characteristics of the convolution filter (ConvFilter) learned inside the network. Furthermore, CNN performs well in sound classification. Unlike image classification, however, there is little research on suitable ConvFilters for sound classification. Therefore, we compare the performances of three convolution filters, 1D-ConvFilter, 3×1 2D-ConvFilter, and 3×3 2D-ConvFilter, in two different network configurations, when classifying mosquitoes using audio data. In insect sound classification, most machine learning researchers use only audio data as input. However, a classification model, which combines other information such as activity circadian rhythm, should intuitively yield improved classification results. To utilize such relevant additional information, we propose a method that defines this information as a priori probabilities and combines them with CNN outputs. Of the networks, VGG13 with 3×3 2D-ConvFilter showed the best performance in classifying mosquito species, with an accuracy of 80.8%. Moreover, adding activity circadian rhythm information to the networks showed an average performance improvement of 5.5%. The VGG13 network with 1D-ConvFilter achieved the highest accuracy of 85.7% with the additional activity circadian rhythm information. © 2021, Universidad Internacional de la Rioja. All rights reserved.

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Notes:

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Acoustic language model classes for large vocabulary continuous speech recognizer

Item Type Conference Paper

Author V. Fischer

Author S.J. Kunzmann

Abstract In a maximum a posteriori probability approach to speech recognition stochastic n-gram language models are used for the estimation of a word sequence's a priori probability. In any practical implementation of a large vocabulary speech recognition system the language model acts as a hypotheses filter that has to differ between candidate words with similar acoustic evidence. For that purpose, the combination of word based and class based language models is attractive, because it allows to fall back to the more reliable estimates of the class based model in case of sparse training data. However, class language models can differ between words from the same class only in terms of a priori probability. To improve the discriminative power for words with similar acoustic score, it is therefore useful to put similar sounding words into different classes. Based on the above considerations, the paper presents an automatic procedure for the optimal classification of a large vocabulary into classes with acoustic dissimilar words. In combination with a standard word based trigram model the so created acoustic class language model provides a relative reduction in word error rate of up to 16 percent and performs slightly better than a perplexity minimizing automatically created class language model.

Date 2000

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Language English

Extra Journal Abbreviation: Int. Conf. Spok. Lang. Process., ICSLP

Publisher International Speech Communication Association

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Conference Name 6th International Conference on Spoken Language Processing, ICSLP 2000

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Computational linguistics, Speech recognition, Stochastic systems, Probability, Probability distributions, Stochastic models, Discriminative power, A-priori probabilities, Optimal classification, Maximum A posteriori probabilities, Automatic procedures, Class-based language model, Large vocabulary speech recognition, N-gram language models, Vocabulary control

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Active Learning and the Saerens-Latinne-Decaestecker Algorithm: An Evaluation

Item Type Conference Paper

Author A. Molinari

Author A. Esuli

Author F. Sebastiani

Editor Tamine L.

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Abstract The Saerens-Latinne-Decaestecker (SLD) algorithm is a method whose goal is improving the quality of the posterior probabilities (or simply “posteriors”) returned by a probabilistic classifier in scenarios characterized by prior probability shift (PPS) between the training set and the unlabelled (“test”) set. This is an important task, (a) because posteriors are of the utmost importance in downstream tasks such as, e.g., multiclass classification and cost-sensitive classification, and (b) because PPS is ubiquitous in many applications. In this paper we explore whether using SLD can indeed improve the quality of posteriors returned by a classifier trained via active learning (AL), a class of machine learning (ML) techniques that indeed tend to generate substantial PPS. Specifically, we target AL via relevance sampling (ALvRS) and AL via uncertainty sampling (ALvUS), two AL techniques that are very well-known especially because, due to their low computational cost, are suitable to being applied in scenarios characterized by large datasets. We present experimental results obtained on the RCV1-v2 dataset, showing that SLD fails to deliver better-quality posteriors with both ALvRS and ALvUS, thus contradicting previous findings in the literature, and that this is due not to the amount of PPS that these techniques generate, but to how the examples they prioritize for annotation are distributed. © 2022 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

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Active learning for outdoor obstacle detection

Item Type Conference Paper

Author C. Dima

Author M. Hebert

Editor Thrun S.

Editor Stanford University, Gates Hall 1A; 355 Serra Mall, Stanford, 94305-9010, CA
Stanford Artificial Intelligence Laboratory

Editor Sukhatme G.

Editor Schaal S.

Editor Brock O.

Abstract Real-world applications of mobile robotics call for increased autonomy, requiring reliable perception systems. Since manually tuned perception algorithms are difficult to adapt to new operating environments, systems based on supervised learning are necessary for future progress in autonomous navigation. Data labeling is a major concern when supervised learning is applied to the large-scale problems occurring in realistic robotics applications. We believe that algorithms for automatically selecting important data for labeling are necessary, and propose to employ active learning techniques to reduce the amount of labeling required to learn from a data set. In this paper we show that several standard active learning algorithms can be adapted to meet specific constraints characteristic to our domain, such as the need to learn from data with severely unbalanced class priors. We validate the solutions we propose by extensive experimentation on multiple realistic data sets captured with a robotic vehicle. Based on our results for the task of obstacle detection, we conclude that active learning techniques are applicable to our domain, and they can lead to significant reductions in the labeling effort required to use supervised learning in outdoor perception. © 2005 Massachusetts Institute of Technology.

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Active Sequential Two-Sample Testing

Item Type Journal Article

Author W. Li

Author P. Kadambi

Author P. Saidi

Author K.N. Ramamurthy

Author G. Dasarathy

Author V. Berisha

Abstract A two-sample hypothesis test is a statistical procedure used to determine whether the distributions generating two samples are identical. We consider the two-sample testing problem in a new scenario where the sample measurements (or sample features) are inexpensive to access, but their group memberships (or labels) are costly. To address the problem, we devise the first active sequential two-sample testing framework that not only sequentially but also actively queries. Our test statistic is a likelihood ratio where one likelihood is found by maximization over all class priors, and the other is provided by a probabilistic classification model. The classification model is adaptively updated and used to predict where the (unlabelled) features have a high dependency on labels; labeling the “high-dependency” features leads to the increased power of the proposed testing framework. In theory, we provide the proof that our framework produces an anytime-valid p-value. In addition, we characterize the proposed framework’s gain in testing power by analyzing the mutual information between the feature and label variables in asymptotic and finite-sample scenarios. In practice, we introduce an instantiation of our framework and evaluate it

using several experiments; the experiments on the synthetic, MNIST, and application-specific datasets demonstrate that the testing power of the instantiated active sequential test significantly increases while the Type I error is under control. © 2024, Transactions on Machine Learning Research. All rights reserved.

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Adapt, Generate, and Supervise: Geometry-Aware Diffusion-Guided SAM Framework for Remote Sensing Semantic Segmentation

Item Type Journal Article
Author W. Zhou
Author J. Xie
Author C. Xu
Author Y. Liu
Author Y. Wang
Abstract Foundation models such as the segment anything model (SAM) have remarkable generalization capabilities in natural image segmentation. However, their application to remote sensing (RS) semantic segmentation faces significant challenges. The single-modal architecture of SAM cannot effectively utilize multimodal RS data; its vision Transformer encoder lacks sensitivity to multiscale spatial structures that are characteristic of RS imagery, and its dependence on manual prompts hinders large-scale automation. To address these challenges, we propose a geometry-aware diffusion-guided SAM framework (GeoSAM) that transforms SAM into fully automated multimodal semantic segmentation through three synergistic innovations. First, we introduce a multiscale geometric-aware adaptation module (GeoAdapter) that hierarchically integrates RGB images with normalized digital surface model (nDSM) data within the encoder. GeoAdapter incorporates a novel class-prior generator that combines geometric convolution, prototype similarity matching, and statistical modeling to produce structure-aware guidance for semantic–geometric feature fusion. Second, we present a diffusion prompt module (DPM) that pioneers

the use of conditional diffusion models for automatic prompt generation in RS applications, eliminating manual interactions through semantic-feature-guided denoising diffusion implicit model (DDIM) sampling. Third, we propose a prompt-level supervision strategy (PLSS) that mitigates training–inference distribution discrepancies through constraints on semantic consistency and structural alignment, ensuring robust prompt generation across different phases. Extensive experiments demonstrate state-of-the-art performance: 90.92% mean accuracy (mAcc) and 82.71% mean intersection over union (mIoU) on Vaihingen, and 85.28% mAcc and 75.49% mIoU on Potsdam, with improvements of 16.52% mAcc and 16.29% mIoU over original SAM on Vaihingen. © 1980-2012 IEEE.

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Adapting a classification rule to local and global shift when only unlabelled data are available

Item Type Journal Article**Author** V. Hofer

Abstract For evolving populations the training data and the test data need not follow the same distribution. Thus, the performance of a prediction model will deteriorate over the course of time. This requires the re-estimation of the prediction model after some time. However, in many applications e.g. credit scoring, new labelled data are not available for re-estimation due to verification latency, i.e. label delay. Thus, methods which enable a prediction model to adapt to distributional changes by using only unlabelled data are highly desirable. A shift adaptation method for binary classification is presented here. The model is based on mixture distributions. The conditional feature distributions are determined at the time where labelled data are available, and the unconditional feature distribution is determined at the time where new unlabelled data are accessible. These mixture distributions provide information on the old and the new positions of subpopulations. A transition model then describes how the subpopulations of each class have drifted to form the new unconditional feature distribution. Assuming that the conditional distributions are reorganised using a minimum of energy, a two-step estimation procedure results. First, for a given class prior distribution the transfer of probability mass is estimated such that the energy required to obtain the new unconditional distribution by a local transfer of the old conditional distributions is a minimum. Since the optimal solution of the resulting transportation problem measures the distance between the old and the new distributions, the change of the class prior distribution is found in a second step by solving the transportation problem for varying class prior distributions and selecting the value for which the objective function is a minimum. Using the solution of the transportation problem and the component parameters of the unconditional feature distribution, the new conditional feature distribution can be determined. This thus allows for a shift adaptation of the classification rule. The performance of the proposed model is investigated using a large real-world dataset on default rates in Danish companies. The results show that the shift adaptation improves classification results. © 2014 Published by Elsevier B.V.

Date 2015**DOI** 10.1016/j.ejor.2014.11.022**URL** <https://www.scopus.com/pages/publications/84921438319?origin=resultslist>**Archive** Scopus**Language** English**Volume** 243**Publisher** Elsevier B.V.**Pages** 177-189**Publication** European Journal of Operational Research**ISSN** 03772217 (ISSN)**Issue** 1**Journal Abbr** Eur J Oper Res**Date Added** 20/09/2026, 9:52:56 pm**Modified** 20/09/2026, 9:52:56 pm**Tags:**

Forecasting, Population statistics, Large dataset, Predictive analytics, Probability distributions, Concept drifts, Dataset shifts, Dataset shift, Concept drift, Mixtures, Global drift, Local drift, Verification latencies, Verification latency

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Adapting a standardised international 24 h dietary recall methodology (GloboDiet software) for research and dietary surveillance in Korea

Item Type Journal Article**Author** M.K. Park**Author** J.Y. Park**Author** G. Nicolas**Author** H.Y. Paik**Author** J. Kim**Author** N. Slimani

Abstract During the past decades, a rapid nutritional transition has been observed along with economic growth in the KOR. Since this dramatic change in diet has been frequently associated with cancer and other non-communicable diseases, dietary monitoring is essential to understand the association. Benefiting from pre-existing standardised dietary methodologies, the present study aimed to evaluate the feasibility and describe the development of a Korean version of the international computerised 24 h dietary recall method (GloboDiet software) and its complementary tools, developed at the International Agency for Research on Cancer (IARC), WHO. Following established international Standard Operating Procedures and guidelines, about seventy common and country-specific databases on foods, recipes, dietary supplements, quantification methods and coefficients were customised and translated. The main results of the present study highlight the specific adaptations made to adapt the GloboDiet software for research and dietary surveillance in Korea. New (sub-) subgroups were added into the existing common food classification, and new descriptors were added to the facets to classify and describe specific Korean foods. Quantification methods were critically evaluated and adapted considering the foods and food packages available in the Korean market. Furthermore, a picture book of foods/dishes was prepared including new pictures and food portion sizes relevant to Korean diet. The development of the Korean version of GloboDiet demonstrated that it was possible to adapt the IARC-WHO international dietary tool to an Asian context without compromising its concept of standardisation and software structure. It, thus, confirms that this international dietary methodology, used so far only in Europe, is flexible and robust enough to be customised for other regions worldwide. © 2015 The Authors.

Date 2015**DOI** 10.1017/S0007114515000987**URL** <https://www.scopus.com/pages/publications/84930932068?origin=resultslist>**Archive** Scopus**Language** English**Volume** 113**Publisher** Cambridge University Press**Pages** 1810-1818

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Adapting Classifiers to Changing Class Priors During Deployment

Item Type Conference Paper

Author N. Daba

Author B. McIntosh

Author A. Mahalanobis

Editor Meo R.

Editor Silvestri F.

Abstract Conventional classifiers are trained and evaluated using “balanced” data sets in which all classes are equally present. Classifiers are now trained on large data sets such as ImageNet, and are now able to classify hundreds (if not thousands) of different classes. On one hand, it is desirable to train such general-purpose classifier on a very large number of classes so that it performs well regardless of the settings in which it is deployed. On the other hand, it is unlikely that all classes known to the classifier will occur in every deployment scenario, or that they will occur with the same prior probability. In reality, only a relatively small subset of the known classes may be present in a particular setting or environment. For example, a classifier will encounter mostly animals if its deployed in a zoo or for monitoring wildlife, aircraft and service vehicles at an airport, or various types

of automobiles and commercial vehicles if is used for monitoring traffic. Furthermore, the exact class priors are generally unknown and can vary over time. In this paper, we explore different methods for estimating the class priors based on the output of the classifier itself. We then show that incorporating the estimated class priors in the overall decision scheme enables the classifier to increase its run-time accuracy in the context of its deployment scenario. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Adapting the standardised computer- and interview-based 24 h dietary recall method (GloboDiet) for dietary monitoring in Latin America

Item Type Journal Article

Author S. Bel-Serrat

Author V. Knaze

Author G. Nicolas

Author D.M. Marchioni

Author J. Steluti
Author A. Mendes
Author S.P. Crispim
Author R.M. Fisberg
Author R.A. Pereira
Author M.C. Araujo
Author R. Sichieri
Author E.M. Yokoo
Author T. Sánchez-Pimienta
Author T.C. Aburto
Author L.S. Pedraza
Author N. Slimani

Abstract Objective The present study describes the procedure and approaches needed to adapt and harmonise the GloboDiet methodology, a computer- and interview-based 24 h dietary recall, for use in two Latin American pilot countries, Brazil and Mexico. Design About seventy common and country-specific databases on foods, recipes, dietary supplements, quantification methods and coefficients were customised and translated following standardised guidelines, starting from existing Spanish and Portuguese versions. Setting Brazil and Mexico. Subjects Not applicable. Results New subgroups were added into the existing common food classification together with new descriptors required to better classify and describe specific Brazilian and Mexican foods. Quantification methods were critically evaluated and adapted considering types and quantities of food consumed in these two countries, using data available from previous surveys. Furthermore, the photos to be used for quantification purposes were identified for compilation in country-specific but standardised picture booklets. Conclusions The completion of the customisation of the GloboDiet Latin America versions in these two pilot countries provides new insights into the adaptability of this dietary international tool to the Latin American context. The ultimate purpose is to enable dietary intake comparisons within and between Latin American countries, support building capacities and foster regional and international collaborations. The development of the GloboDiet methodology could represent a major benefit for Latin America in terms of standardised dietary methodologies for multiple surveillance, research and prevention purposes. Copyright © The Authors 2017.

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Adapting to Online Label Shift with Provable Guarantees

Item Type Conference Paper

Author Y. Bai

Author Y.-J. Zhang

Author P. Zhao

Author M. Sugiyama

Author Z.-H. Zhou

Editor Koyejo S.

Editor Mohamed S.

Editor Agarwal A.

Editor Belgrave D.

Editor Cho K.

Editor Oh A.

Abstract The standard supervised learning paradigm works effectively when training data shares the same distribution as the upcoming testing samples. However, this stationary assumption is often violated in real-world applications, especially when testing data appear in an online fashion. In this paper, we formulate and investigate the problem of online label shift (OLaS): the learner trains an initial model from the labeled offline data and then deploys it to an unlabeled online environment where the underlying label distribution changes over time but the label-conditional density does not. The non-stationarity nature and the lack of supervision make the problem challenging to be tackled. To address the difficulty,

we construct a new unbiased risk estimator that utilizes the unlabeled data, which exhibits many benign properties albeit with potential non-convexity. Building upon that, we propose novel online ensemble algorithms to deal with the non-stationarity of the environments. Our approach enjoys optimal dynamic regret, indicating that the performance is competitive with a clairvoyant who knows the online environments in hindsight and then chooses the best decision for each round. The obtained dynamic regret bound scales with the intensity and pattern of label distribution shift, hence exhibiting the adaptivity in the OLaS problem. Extensive experiments are conducted to validate the effectiveness and support our theoretical findings. © 2022 Neural information processing systems foundation. All rights reserved.

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Adapting to the Unknown: Training-Free Audio-Visual Event Perception with Dynamic Thresholds

Item Type Conference Paper

Author E. Shaar

Author A. Shaulov

Author G. Chechik

Author L. Wolf

Abstract In the domain of audio-visual event perception, which focuses on the temporal localization and classification of events across distinct modalities (audio and visual), existing approaches are constrained by the vocabulary available in their training data. This limitation significantly impedes their capacity to generalize to novel, unseen event categories. Furthermore, the annotation process for this task is labor-intensive, requiring extensive manual labeling across modalities and temporal segments, limiting the scalability of current methods. Current state-of-the-art models ignore the shifts in event distributions over time, reducing their ability to adjust to changing video dynamics. Additionally, previous methods rely on late fusion to combine audio and visual information. While straightforward, this approach results in a significant loss of multimodal interactions. To address these challenges, we propose AudioVisual Adaptive Video Analysis (AV2A), a model-agnostic approach that requires no further training and integrates a score-level fusion technique to retain richer multimodal interactions. AV2A also includes a within-video label shift algorithm, leveraging input video data and predictions from prior frames to dynamically adjust event distributions for subsequent frames. Moreover, we present the first training-free, open-vocabulary baseline for audio-visual event perception, demonstrating that AV2A achieves substantial improvements over naive training-free baselines. We demonstrate the effectiveness of AV2A on both zero-shot and weakly-supervised state-of-the-art methods, achieving notable improvements in performance metrics over existing approaches. Our code is available on Github. © 2025 IEEE.

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Adaptive Conformal Prediction for Quantum Machine Learning

Item Type Journal Article

Author D. Spencer

Author S. Nicholls

Author M. Caprio

Abstract Quantum machine learning seeks to leverage quantum computers to improve upon classical machine learning algorithms. Currently, robust uncertainty quantification methods remain underdeveloped in the quantum domain, despite the critical need for reliable and trustworthy predictions. Recent work has introduced quantum conformal prediction, a framework that produces prediction sets that are guaranteed to contain the true outcome with a user-specified probability. In this work, we formalise how the time-varying noise inherent in quantum processors can undermine conformal guarantees, even when calibration and test data are exchangeable. To address this challenge, we draw on Adaptive Conformal Inference, a method which maintains validity over time via repeated recalibration. We introduce Adaptive Quantum Conformal Prediction (AQCP), an algorithm which provides asymptotic average coverage guarantees under arbitrary hardware noise conditions. Empirical studies on an IBM quantum processor demonstrate that AQCP achieves the target coverage level and exhibits greater stability than quantum conformal prediction. © 2026, Transactions on Machine Learning Research. All rights reserved.

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Adaptive Conformal Prediction Intervals Using Data-Dependent Weights With Application to Seismic Response Prediction

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Author P. Hajibabae

Author F. Pourkamali-Anaraki

Author M.A. Hariri-Ardebili

Abstract Machine learning often lacks transparent performance indicators, especially in generating point predictions. This paper addresses this limitation through conformal prediction, a non-parametric forecasting technique seamlessly integrating with regression algorithms to produce prediction intervals at specified confidence levels. A crucial element in conformal prediction is the non-conformity score, traditionally based on absolute residual errors. In this work, we propose a novel approach, introducing data-dependent weights for computing non-conformity scores. This enhancement, considering the distances of training instances from the test sample, aims to improve overall algorithm performance. Empirical investigations across various real-world regression data sets, including scientific data, evaluate the efficiency and validity of prediction intervals from different uncertainty quantification methods. Results show that prediction intervals computed with data-dependent weights adapt to estimator uncertainty, offering more precise predictions in certain scenarios and appropriately conservative predictions in high uncertainty situations. Additionally, we compare predictive regions generated by conformal prediction with those from Gaussian Process Regression (GPR) for scientific data in structural engineering. To augment conformal prediction, we explore Conformalized Quantile Regression (CQR), a recent innovation combining conformal prediction with classical quantile regression, claiming full adaptability to heteroscedasticity. Our findings indicate that conformal prediction methods using data-dependent non-conformity scores achieve a 1% higher effective coverage level and a 15% reduction in prediction interval widths compared to other methods. The comparative analysis against GPR and CQR underscores the practical value of our approach in providing accurate prediction intervals in scientific and engineering domains. © 2013 IEEE.

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Export Date: 20 September 2026; Cited By: 15; Correspondence Address: P. Hajibabaei; Florida Polytechnic University, Department of Data Science and Business Analytics, Lakeland, 33805, United States; email: phajibabaei@floridapoly.edu

Adaptive continual botnet detection with self-supervised prototypical networks and few-shot learning for evolving botnet detection

Item Type Journal Article

Author N. Junar

Author K. Arockiasamy

Abstract Introduction – One of the problems faced by botnet intrusion detection systems is the detection of unknown attacks or zero-day attacks. Zero-day attacks give security analysts very little time for action since there is only little information that is known about the new attack that is launched, given very few samples available for the training process. Class imbalance issues faced by many benchmark datasets also reduce the ability to detect a zero-day attack. It leads to very few samples of a particular class, which makes it difficult to trace different types of the same attack. The rapid evolution of botnets also poses a challenge to the existing models, which often fail to adapt to the new attack patterns without catastrophic forgetting of previously learned knowledge. Methods – The objective of this study was to develop a model for adaptive botnet detection in cases of zero-day attacks and imbalanced dataset availability. The study integrates self-supervised learning (SSL), enhanced prototypical networks with attention mechanisms, and few-shot learning (FSL) to overcome the existing issues. Wasserstein generative adversarial network with gradient penalty (WGAN-GP) generates synthetic samples to handle class imbalance. An adaptive continual learning module is used, which combines elastic weight consolidation (EWC) and an experience replay buffer which enables the model to accommodate new botnet behaviors while preserving performance on previously seen samples. Moreover, a drift detector using the KS test, Wasserstein distance, and class-prior shift triggers model adaptation when concept drift is identified. Results – The performance of the proposed model was evaluated on the CIC IoT dataset 2023, which demonstrated an exceptional accuracy of 98.50%, outperforming the traditional baselines. Few-shot learning evaluations show strong generalization with as few as 1–5 shots per class. This result shows the overall achievement of the model

in zero-day attack detection and real-time adaptation to the evolving botnet threats. Discussion – The findings demonstrate that the proposed framework provides an effective approach for zero-day botnet attack detection while maintaining knowledge of previously observed attacks. The combination of few-shot learning and adaptive continual learning improves the model's ability to respond to emerging botnet behaviours without requiring extensive retraining. Thus, the proposed approach shows potential for adaptive and real-time botnet detection in evolving IoT network environments. Copyright © 2026 Junar and Arockiasamy.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K. Arockiasamy; School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, India; email: karmel.a@vit.ac.in

Adaptive learning of bayesian networks for the qualification of traffic data by contaminated dirichlet density functions

Item Type Conference Paper

Author M. Junghans

Author H.-J. Jentschel

Abstract The concept of Bayesian Networks (BNs) is an established method to model data fusion in sensor networks of several equal or different sensors. Although the method is powerful, there is a particular need for accurate sensors, the consideration of the affecting external, e.g. environmental conditions, and internal influences, e.g. the physical life of the sensor, in the sensor model and an accurate a-priori knowledge about the underlying process. In this paper an adaptive algorithm for learning BNs is introduced, which is applied to update the time-variant a-priori probabilities in sensor networks. This algorithm makes use of

Contaminated Dirichlet Density Functions (CDDFs). The effectiveness of adaptive learning is demonstrated for vehicle classification in traffic surveillance. © IEEE.

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Sensor networks, Sensor data fusion, Image segmentation, Learning algorithms, Sensors, Intelligent networks, Bayesian networks, Inference engines, Mathematical models, Speech analysis, Dirichlet, Environmental conditions, Distributed parameter networks, Adaptive algorithms, Adaptive learnings, Density functions, External-, Information technology, Priori knowledges, Priori probabilities, Sensor models, Traffic datums, Traffic surveillances, Vehicle classifications

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M. Junghans; Institute for Traffic Communications Engineering, Dresden University of Technology, Germany; email: marek.junghans@tu-dresden.de; Conference name: 2008 1st International Conference on Information Technology, IT 2008; Conference date: 18 May 2008 through 21 May 2008; Conference code: 74807

Adaptive learning of minority class prior to minority oversampling

Item Type Journal Article

Author P. Sadhukhan

Author S. Palit

Abstract The minority oversampling techniques have substantiated their appropriateness and utility in the domain of class-imbalance learning. However, this does not affirm the true class of the synthetic minority points. In this work, Adaptive Learning of Minority Class prior to Minority Oversampling (ALMCMO), we work towards bridging this gap by estimating the minority set before oversampling the synthetic points. We estimate a varying and adaptive volume of minority space around the minority points. We aim to guarantee the class-memberships of the synthetic minority points by sampling them from the estimated minority spaces. In our empirical study,

we have used six comparing methods, 23 datasets and two classifiers. The results indicate the certain superiority of the proposed method over the six competing schemes. © 2020

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Adaptive real time traffic prediction using deep neural networks

Item Type Journal Article

Author P.R. Iyer

Author S.R. Iyer

Author R. Ramesh

Author M.R. Anala

Author K.N. Subramanya

Abstract The ever-increasing sale of vehicles and the steady increase in population density in metropolitan cities have raised many growing concerns, most importantly commute time, air and noise pollution levels. Traffic congestion can be alleviated by opting adaptive traffic light systems, instead of fixed-time traffic signals. In this paper, a system is proposed which can detect, classify and count vehicles passing through any traffic junction using a single camera (as opposed to multi-sensor approaches). The detection and classification are done using SSD Neural Network object detection algorithm. The count of each class (2-wheelers, cars, trucks, buses etc.) is used to

predict the signal green-time for the next cycle. The model self-adjusts every cycle by utilizing weighted moving averages. This system works well because the change in the density of traffic on any given road is gradual, spanning multiple traffic stops throughout the day. © 2019 Institute of Advanced Engineering and Science. All rights reserved.

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Adaptive skew-sensitive ensembles for face recognition in video surveillance

Item Type Journal Article

Author M. De-La-Torre

Author E. Granger

Author R. Sabourin

Author D.O. Gorodnichy

Abstract Decision support systems for surveillance rely more and more on face recognition (FR) to detect target individuals of interest captured with video cameras. FR is a challenging problem in video surveillance due to variations in capture conditions, to camera interoperability, and to the limited representativeness of target facial models used for matching. Although adaptive classifier ensembles have been applied for robust face matching, it is often assumed that the proportions of faces captured for target and non-target individuals are balanced, known a priori, and do not change

over time. Recently, some techniques have been proposed to adapt the fusion function of an ensemble according to class imbalance of the input data stream. For instance, Skew-Sensitive Boolean combination (SSBC) is a active approach that estimates target vs. non-target proportions periodically during operations using Hellinger distance, and adapts its ensemble fusion function to operational class imbalance. Beyond the challenges of estimating class imbalance, such techniques commonly generate diverse pools of classifiers by selecting balanced training data, limiting the potential diversity produced using the abundant non-target data. In this paper, adaptive skew-sensitive ensembles are proposed to combine classifiers trained by selecting data with varying levels of imbalance and complexity, to sustain a high level the performance for video-to-video FR. Faces captured for each person in the scene are tracked and regrouped into trajectories. During enrollment, captures in a reference trajectory are combined with selected non-target captures to generate a pool of 2-class classifiers using data with various levels of imbalance and complexity. During operations, the level of imbalance is periodically estimated from the input trajectories using the HDx quantification method, and pre-computed histogram representations of imbalanced data distributions. This approach allows one to adapt pre-computed histograms and ensemble fusion functions based on the imbalance and complexity of operational data. Finally, the ensemble scores are accumulated of trajectories for robust spatio-temporal recognition. Results on synthetic data show that adapting the fusion function of ensemble trained with different complexities and levels of imbalance can significantly improve performance. Results on the Face in Action video data show that the proposed method can outperform reference techniques (including SSBC and meta-classification) in imbalanced video surveillance environments. Transaction-based analysis shows that performance is consistently higher across operational imbalances. Individual-specific analysis indicates that goat- and lamb-like individuals can benefit the most from adaptation to the operational imbalance. Finally, trajectory-based analysis shows that a video-to-video FR system based on the proposed approach can maintain, and even improve overall system discrimination. © 2015 Elsevier Ltd. All rights reserved.

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ADBSCSL: adaptive DBSCAN-SMOTE with cost-sensitive learning to enhance diagnostic accuracy for imbalanced medical datasets

Item Type Journal Article

Author M. Kavitha

Author M. Kasthuri

Abstract Medical diagnosis is complicated by imbalanced datasets, in which biased models cannot distinguish minority class cases like rare diseases. To improve diagnosis accuracy, this research introduces ADBSCSL, which stands for adaptive DBSCAN-SMOTE with cost-sensitive learning. Adaptive DBSCAN, SMOTE, and cost-sensitive learning handle skewed data well. Adaptive DBSCAN clusters minority class occurrences. It changes parameters to dataset density change. The diversity of the density condition cannot have caused the minority class to misidentify. SMOTE is then applied to these clusters to increase synthetic examples and class balance. It reduces misclassification costs using cost-sensitive learning. This pushes the model toward minority class priority and avoids majority class bias. The approach was evaluated on brain stroke, cerebral stroke, and autism spectrum disorder datasets. ADBSCSL F1-scores of 91.8% and 90.6% indicate accuracy over 90% on brain stroke and cerebral stroke datasets. On ASD datasets, it had 100% accuracy, precision, recall, and F1-score. Results show that the ADBSCSL increases classification performance, making it a powerful and efficient tool for medicine diagnosis with highly imbalanced datasets. Copyright © 2025 Inderscience Enterprises Ltd.

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ADC-derived spatial features can accurately classify adnexal lesions

Item Type Journal Article

Author A. Fathi Kazerooni

Author M. Nabil

Author H. Haghighat Khah

Author M. Alviri

Author M. Heidari-Sooreshjaani

Author M. Gity

Author M. Malek

Author H. Saligheh Rad

Abstract Background: The role of quantitative apparent diffusion coefficient (ADC) maps in differentiating adnexal masses is unresolved. Purpose/Hypothesis: To propose an objective diagnostic method devised based on spatial features for predicting benignity/malignancy of adnexal masses in ADC maps. Study Type: Prospective. Population: In all, 70 women with sonographically indeterminate and histopathologically confirmed adnexal masses (38 benign, 3 borderline, and 29 malignant) were considered for this study. Field Strength/Sequence: Conventional and diffusion-weighted magnetic resonance (MR) images (b-values = 50, 400, 1000 s/mm²) were acquired on a 3T scanner. Assessment: For each patient, two radiologists in consensus manually delineated lesion borders in whole ADC map volumes, which were consequently analyzed using spatial models (first-order histogram [FOH], gray-

level co-occurrence matrix [GLCM], run-length matrix [RLM], and Gabor filters). Two independent radiologists were asked to identify the attributed (benign/malignant) classes of adnexal masses based on morphological features on conventional MRI. Statistical Tests: Leave-one-out cross-validated feature selection followed by cross-validated classification were applied to the feature space to choose the spatial models that best discriminate benign from malignant adnexal lesions. Two schemes of feature selection/classification were evaluated: 1) including all benign and malignant masses, and 2) scheme 1 after excluding endometrioma, hemorrhagic cysts, and teratoma (14 benign, 29 malignant masses). The constructed feature subspaces for benign/malignant lesion differentiation were tested for classification of benign/borderline/malignant and also borderline/malignant adnexal lesions. Results: The selected feature subspace consisting of RLM features differentiated benign from malignant adnexal masses with a classification accuracy of ~92%. The same model discriminated benign, borderline, and malignant lesions with 87% and borderline from malignant with 100% accuracy. Qualitative assessment of the radiologists based on conventional MRI features reached an accuracy of 80%. Data Conclusion: The spatial quantification methodology proposed in this study, which works based on cellular distributions within ADC maps of adnexal masses, may provide a helpful computer-aided strategy for objective characterization of adnexal masses. Level of Evidence: 1. Technical Efficacy: Stage 2. J. Magn. Reson. Imaging 2018;47:1061–1071. © 2017 International Society for Magnetic Resonance in Medicine

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Addressing class distribution issues of the drawing vs writing classification in an ink stroke sequence

Item Type Conference Paper**Author** X. Wang**Author** M. Biswas**Author** S. Raghupathy

Abstract Complicated by temporal correlations among the strokes and varying distributions of the underlying classes, the drawing/writing classification of ink strokes in a digital ink file poses interesting challenges. In this paper, we present our efforts in addressing some of the issues. First, we describe how we adjust the outputs of the neural network to a priori probabilities of new observations to produce more accurate estimates of the posterior probabilities. Second, we describe how to adapt the parameters of the HMM to new data sets. Albeit the fact that the emission probabilities of the HMM are computed indirectly from the outputs of the neural network, our modified Baum-Welch algorithm still finds the correct estimates for the HMM's parameters. We also present experimental results of our new algorithms on 6 real-world data sets. The results show that our methods increase the F-Measures of both the drawing and the writing classes on the more "drawing-intensive" data sets which have stronger temporal correlations. But they do not perform well on the more "writing-intensive" data sets. Copyright © 2007.

Date 2007**DOI** 10.1145/1384429.1384458**URL** <https://www.scopus.com/pages/publications/57349164862?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Sketch-Based Interfaces Model. - ACM SIGGRAPH/Eurographics Symp. Proc.**Pages** 139-146**Proceedings Title** Sketch-Based Interfaces Model. - ACM SIGGRAPH/Eurographics Symp. Proc.**Conference Name** Sketch-Based Interfaces and Modeling 2007 - ACM SIGGRAPH/Eurographics Symposium Proceedings**Date Added** 20/09/2026, 9:53:07 pm**Modified** 20/09/2026, 9:53:07 pm**Tags:**

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Addressing Dataset Shift for Trustworthy Deep Learning Diagnostic Ultrasound Decision Support

Item Type Journal Article

Author C. Zhu

Author M.D. Noseworthy

Author T.E. Doyle

Abstract Ultrasound (US) is the most widely used medical imaging modality due to its low cost, portability, real time imaging ability and use of non-ionizing radiation. However, unlike other imaging modalities such as CT or MRI, it is a heavily operator dependent, requiring trained expertise to leverage these benefits. Recently there has been an explosion of interest in AI across the medical community and many are turning to the growing trend of deep learning (DL) models to assist in diagnosis. However, due to possible differences in training and deployment, model performance suffers which can lead to misdiagnosis and operator hesitancy. This issue is known as dataset shift. Two aims to address dataset shift were proposed. The first was to quantify how US operator skill and hardware affects acquired images. The second was to use this skill quantification method to screen and match data to deep learning models to improve performance. A CAE Healthcare BLUE phantom with mock lesions was scanned by three operators using three different US systems (Siemens S3000, Clarius L15, and Ultrasonix SonixTouch) producing 39013 images. DL models were trained on a specific set to classify the presence of a simulated tumour and tested with data from differing sets. Principle Component Analysis (PCA) for dimension reduction was applied, then K-Means clustering was used to separate images generated by operator and hardware into clusters. This clustering algorithm was then used to screen incoming images during deployment to best match input to an appropriate DL model which is trained specifically to classify that type of operator or hardware. Results showed a noticeable difference when models were given data from differing datasets with the largest accuracy drop being 81.26% to 31.26%. Overall, operator differences more significantly affected DL model performance. Clustering models had much higher success separating hardware data compared to operator data. The proposed method reflects this result with a much higher accuracy across the hardware test set compared to the operator data. © 2023, The Author(s), under exclusive license to Springer-Verlag GmbH, DE, part of Springer Nature.

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Addressing Measurement Error in Random Forests Using Quantitative Bias Analysis

Item Type Journal Article

Author T. Jiang

Author J.L. Gradus

Author T.L. Lash

Author M.P. Fox

Abstract Although variables are often measured with error, the impact of measurement error on machine-learning predictions is seldom quantified. The purpose of this study was to assess the impact of measurement error on the performance of random-forest models and variable importance. First, we assessed the impact of misclassification (i.e., measurement error of categorical variables) of predictors on random-forest model performance (e.g., accuracy, sensitivity) and variable importance (mean decrease in accuracy) using data from the National Comorbidity Survey Replication (2001-2003). Second, we created simulated data sets in which we knew the true

model performance and variable importance measures and could verify that quantitative bias analysis was recovering the truth in misclassified versions of the data sets. Our findings showed that measurement error in the data used to construct random forests can distort model performance and variable importance measures and that bias analysis can recover the correct results. This study highlights the utility of applying quantitative bias analysis in machine learning to quantify the impact of measurement error on study results. © 2021 The Author(s). Published by Oxford University Press on behalf of the Johns Hopkins Bloomberg School of Public Health. All rights reserved.

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Adjusting the outputs of a classifier to new a Priori probabilities: A simple procedure

Item Type Journal Article

Author M. Saerens

Author P. Latinne

Author C. Decaestecker

Abstract It sometimes happens (for instance in case control studies) that a classifier is trained on a data set that does not reflect the true a priori probabilities of the target classes on real-world data. This may have a negative effect on the classification accuracy obtained on the real-world data set, especially when the classifier's decisions are based on the a posteriori probabilities of class membership. Indeed, in this case, the trained classifier provides estimates of the a posteriori probabilities that are not valid for this real-world data set (they rely on the a priori probabilities of the training set). Applying the classifier as is (without correcting its outputs with respect to these new conditions) on this new data set may thus be suboptimal. In this note, we present a simple iterative procedure for adjusting the outputs of the trained classifier with respect to these new a priori probabilities without having to refit the model, even when these probabilities are not known in advance. As a by-product, estimates of the new a priori probabilities are also obtained. This iterative algorithm is a straightforward instance of the expectation-maximization (EM) algorithm and is shown to maximize the likelihood of the new data. Thereafter, we discuss a statistical test that can be applied to decide if the a priori class probabilities have changed from the training set to the real-world data. The procedure is illustrated on different classification problems involving a multilayer neural network, and comparisons with a standard procedure for a priori probability estimation are provided. Our original method, based on the EM algorithm, is shown to be superior to the standard one for a priori probability estimation. Experimental results also indicate that the classifier with adjusted outputs always performs better than the original one in terms of classification accuracy, when the a priori probability conditions differ from the training set to the real-world data. The gain in classification accuracy can be significant.

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Publication Neural Computation

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Tags:

article, Probability, classification, Classification, methodology, probability, Probability Theory, Support, Non-U.S. Gov't, system analysis

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Export Date: 20 September 2026; Cited By: 358; Correspondence Address: M. Saerens; IRIDIA Laboratory, Université Libre de Bruxelles, B-1050 Brussels, cp 194/6, Belgium; email: saerens@ulb.ac.be

Advanced Data Mining and Applications - 7th International Conference, ADMA 2011, Proceedings

Item Type Journal Article

Abstract The proceedings contain 61 papers. The topics discussed include: retrieval in CBR using a combination of similarity and association knowledge; a clustering approach using weighted similarity majority margins; a method for finding groups of related herbs in traditional Chinese medicine; a new hybrid clustering method for reducing very large spatio-temporal dataset; a normal distribution-based over-sampling approach to imbalanced data classification; a probabilistic topic model with social tags for query reformulation in informational search; bi-clustering gene expression data using co-similarity; generating syntactic tree templates for feature-based opinion mining; handling concept drift via ensemble and class distribution estimation technique; hybrid artificial immune algorithm and CMAC neural network classifier for supporting business and medical decision making; and individual doctor recommendation model on medical social network.

Date 2011

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Volume 7120 LNAI

Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Advanced Data Mining and Applications - 7th International Conference, ADMA 2011, Proceedings

Item Type Journal Article

Abstract The proceedings contain 61 papers. The topics discussed include: retrieval in CBR using a combination of similarity and association knowledge; a clustering approach using weighted similarity majority margins; a method for finding groups of related herbs in traditional Chinese medicine; a new hybrid clustering method for reducing very large spatio-temporal dataset; a normal distribution-based over-sampling approach to imbalanced data classification; a probabilistic topic model with social tags for query reformulation in informational search; bi-clustering gene expression data using co-similarity; generating syntactic tree templates for feature-based opinion mining; handling concept drift via ensemble and class distribution estimation technique; hybrid artificial immune algorithm and CMAC neural network classifier for supporting business and medical decision making; and individual doctor recommendation model on medical social network.

Date 2011

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Export Date: 20 September 2026; Cited By: 0; Conference name: 7th International Conference on Advanced Data Mining and Applications, ADMA 2011; Conference date: 17 December 2011 through 19 December 2011; Conference code: 87846

Advanced data-driven interpretable analysis for predicting resistant starch content in rice using NIR spectroscopy

Item Type Journal Article

Author Q. Zhu

Author Y. Gao

Author B. Yang

Author K. Zhao

Author Z. Wang

Author F. Huang

Author F. Cheng

Author Q. Zhao

Author J. Huang

Abstract Resistant starch (RS) is a vital dietary component with notable health benefits, but tradition quantification methods are labor-intensive, costly, and unsuitable for large-scale applications. This study introduced an innovative data-driven framework integrating Near-Infrared (NIR) spectroscopy with Convolutional Neural Networks (CNN) and data augmentation to achieve rapid, cost-effective RS prediction. Achieving exceptional accuracy ($R_p^2 = 0.992$), the CNN model outperformed traditional methods like Partial Least Squares Regression (PLSR) and Support Vector Machine Regression (SVMR). To overcome the “black-box” limitation of deep learning, SHapley Additive exPlanations (SHAP) were innovatively employed, pinpointing critical wavelengths (2000–2500 nm), significantly narrowing the spectral range while providing meaningful insights into the contribution of specific wavelengths to RS prediction. This optimized spectral enhanced data acquisition efficiency, reduces analytical costs, and simplifies operational complexity, establishing a practical and scalable solution for deploying NIR spectroscopy in food quality assessment and production-line applications. © 2025 Elsevier Ltd

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Advances in machine learning for the detection and characterization of microplastics in the environment

Item Type Journal Article

Author M.M. Khanam

Author M.K. Uddin

Author J.U. Kazi

Abstract Microplastics are increasingly recognized as a pervasive pollutant in both aquatic and terrestrial environments, raising pressing concerns about their ecological impacts and implications for human health. Traditional detection and quantification methods—including manual microscopy and standalone spectroscopic techniques—offer reliable accuracy but are limited by labor-intensive procedures and low throughput. Recent advances in machine learning (ML) have revolutionized the field of microplastic research by automating and enhancing detection processes. In particular, algorithms such as support vector machines, random forests, and convolutional neural networks have demonstrated considerable success in classifying microplastics based on chemical signatures and visual characteristics. This review offers a comprehensive overview of ML approaches utilized for monitoring microplastic contamination across diverse aquatic settings. Spectral techniques, including infrared and Raman spectroscopy, leverage molecular vibrations to facilitate highly specific identification of polymer types, even within heterogeneous matrices. Image-based methods make use of sophisticated computer vision techniques to classify microplastics by shape, size, and color, reducing the subjectivity inherent in manual counting. Extending these capabilities further, hyperspectral imaging combines spatial and spectral data to generate comprehensive chemical maps, enabling the simultaneous assessment of polymer composition and distribution. Integrating ML algorithms into these various approaches has improved sensitivity, speed, and scalability, thereby addressing critical challenges in high-throughput and real-time monitoring. Despite these advances, key obstacles remain, including the need for larger, higher-quality datasets and the development of robust models capable of handling complex environmental conditions. Nevertheless, ongoing improvements in imaging hardware and ML methodologies hold significant promise for establishing more effective, automated, and accurate strategies for microplastic detection. By providing a comprehensive overview of current technologies and future opportunities, this review aims to guide researchers and stakeholders in developing science-based solutions for mitigating the global threat of microplastic pollution. Copyright © 2025 Khanam, Uddin and Kazi.

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Export Date: 20 September 2026; Cited By: 65; Correspondence Address: M.K. Uddin; Department of Environmental Sciences, Jahangirnagar University, Dhaka, Bangladesh; email: khabir88@juniv.edu; J.U. Kazi; Division of Translational Cancer Research, Department of Laboratory Medicine, Lund University, Lund, Sweden; email: kazi.uddin@med.lu.se

Advances in theoretical calculation of single-atom catalysts for electrochemical nitrogen reduction to ammonia

Item Type Journal Article

Author W. Gan

Author X. Zhang

Author Z. Ji

Author Y. Xue

Abstract Theoretical calculation is one of the important means to study the mechanism of catalytic reactions and design high-performance catalysts, which can not only visualize the electronic structure of the active sites and the adsorption configurations of reactants at atomic level, but also can quickly assess catalytic performance by simulating reaction paths, and significantly reduce development cycles and costs. In this paper, we systematically introduced the application of theoretical calculations in the research and development of electrochemical nitrogen reduction of ammonia synthesis (eNRR) with single-atom catalysts (SACs), including the study of the catalytic mechanisms, high-throughput screening of catalysts, and theoretical design of catalysts. The theoretical activities of eNRR single-atom catalysts on different supports were summarized, and the existing problems and development prospects in current research were discussed. Meanwhile multiple strategic approaches, such as optimizing high-throughput calculations, introducing multiscale simulations and

advanced quantification methods, and integrating machine learning techniques, were proposed to improve the screening efficiency and performance prediction of electrochemical catalysts, accelerate the precise construction and development of catalysts, and promote the industrialization process of eNRR. © (2025), (Huagong Jinzhan/Chemical Industry and Engineering Progress). All rights reserved.

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Short Title 电化学氮还原合成氨单原子催化剂的理论计算研究进展

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: X. Zhang; Beijing Low Carbon and Clean Energy Institute, Beijing, 102200, China; email: xiaofang.zhang.h@chnenergy.com.cn

Advancing Chronic Liver Disease Diagnoses: Targeted Proteomics for the Non-Invasive Detection of Fibrosis

Item Type Journal Article

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Author M. Akyildiz
Author M. Dayangac
Author H. Kirimlioglu
Author M. Zeybel
Author F. Edfors

Abstract Chronic liver disease poses significant challenges to healthcare systems, which frequently struggle to meet the needs of end-stage liver disease patients. Early detection and management are essential because liver damage and fibrosis are potentially reversible. However, the implementation of population-wide screenings is hindered by the asymptomatic nature of early chronic liver disease, along with the risks and costs associated with traditional diagnostics, such as liver biopsies. This study pioneers the development of innovative, minimally invasive methods capable of improving the outcomes of liver disease patients by identifying liver disease biomarkers using quantification methods with translational potential. A targeted mass spectrometry assay based on stable isotope standard protein epitope signature tags (SIS-PrESTs) was employed for the absolute quantification of 108 proteins in just two microliters of plasma. The plasma profiles were derived from patients of various liver disease stages and etiologies, including healthy controls. A set of potential biomarkers for stratifying liver fibrosis was identified through differential expression analysis and supervised machine learning. These findings offer promising alternatives for improved diagnostics and personalized treatment strategies in liver disease management. Moreover, our approach is fully compatible with existing technologies that facilitate the robust quantification of clinically relevant protein targets via minimally disruptive sampling methods. © 2025 by the authors.

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mass spectrometry, chronic liver disease (CLD), fibrosis biomarkers, plasma proteome profiling, targeted proteomics

Notes:

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Advancing Fusion with Machine Learning Research Needs Workshop Report

Item Type Journal Article

Author D. Humphreys

Author A. Kupresanin

Author M.D. Boyer

Author J. Canik

Author C.S. Chang

Author E.C. Cyr

Author R. Granetz

Author J. Hittinger

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Author E. Lawrence

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Author A. Patra

Author D. Schissel

Abstract Machine learning and artificial intelligence (ML/AI) methods have been used successfully in recent years to solve problems in many areas, including image recognition, unsupervised and supervised classification, game-playing, system identification and prediction, and autonomous vehicle control. Data-driven machine learning methods have also been applied to fusion energy research for over 2 decades, including significant advances in the areas of disruption prediction, surrogate model generation, and experimental planning. The advent of powerful and dedicated computers specialized for large-scale parallel computation, as well as advances in statistical inference algorithms, have greatly enhanced the capabilities of these computational approaches to extract scientific knowledge and bridge gaps between theoretical models and practical implementations. Large-scale commercial success of various ML/AI applications in recent years, including robotics, industrial processes, online image recognition, financial system prediction, and autonomous vehicles, have further demonstrated the potential for data-driven methods to produce dramatic transformations in many fields. These advances, along with the urgency of need to bridge key gaps in knowledge for design and operation of reactors such as ITER, have driven planned expansion of efforts in ML/AI within the US government and around the world. The Department of Energy (DOE) Office of Science programs in Fusion Energy Sciences (FES) and Advanced Scientific Computing Research (ASCR) have organized several activities to identify best strategies and approaches for applying ML/AI methods to fusion energy research. This paper describes the results of a joint FES/ASCR DOE-sponsored Research Needs Workshop on Advancing Fusion with Machine Learning, held April 30–May 2, 2019, in Gaithersburg, MD (full report available at https://science.osti.gov/-/media/fes/pdf/workshop-reports/FES_ASCR_Machine_Learning_Report.pdf). The workshop drew on broad representation from both FES and ASCR scientific communities, and

identified seven Priority Research Opportunities (PRO's) with high potential for advancing fusion energy. In addition to the PRO topics themselves, the workshop identified research guidelines to maximize the effectiveness of ML/AI methods in fusion energy science, which include focusing on uncertainty quantification, methods for quantifying regions of validity of models and algorithms, and applying highly integrated teams of ML/AI mathematicians, computer scientists, and fusion energy scientists with domain expertise in the relevant areas. © 2020, The Author(s).

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Advancing Gastrointestinal Cancer Risk Prediction With Patient-Centered
Machine Learning: Machine Learning Modeling Study

Item Type Journal Article

Author D. Baublyte

Author J. Lee

Author M. Gunathilake

Author J. Kim

Abstract Background: Gastrointestinal (GI) cancers are a significant health concern in South Korea. Recently, machine learning (ML) models have emerged as powerful tools to support early screening efforts and identify people at risk before disease onset. However, the low incidence of GI malignancies in prospective cohorts leads to severe class imbalance, often causing ML models to favor the majority “healthy” class at the expense of clinical sensitivity. Objective: This study aimed to evaluate class imbalance mitigation strategies and develop ML-based GI cancer risk prediction models using noninvasive and minimally invasive predictors linked to modifiable behavioral and metabolic risk factors. Methods: We analyzed a prospective cohort (n=7652) with 156 incident GI cancer cases (2%) identified over a 14-year follow-up period. The data were randomly split into training (5356/7652, 70%) and testing (2296/7652, 30%) sets. To address class imbalance while preserving observed population structure, we developed a patient-centered undersampling technique (PCUSTe) based on the logic of frequency-matched case-control studies. PCUSTe was compared with commonly used resampling approaches, including synthetic minority oversampling (SMOTE), adaptive synthetic sampling (ADASYN), and SMOTE with edited nearest neighbors (ENN). Six classifiers were implemented, including both batch and incremental training variants. To account for the prior shift introduced by resampling, probability correction was applied. Model performance was evaluated on the independent test set using a classification threshold equal to the observed event proportion (cumulative incidence) in the training data and then across thresholds reflecting incidence values between 1% and 5%. Primary performance metrics included sensitivity, specificity, Matthews correlation coefficient, and area under the receiver operating characteristic curve (AUC). Results: Models trained using PCUSTe demonstrated improved sensitivity compared with standard resampling techniques, particularly for more complex classifiers. The incrementally trained stochastic gradient descent model achieved the highest overall performance trained on PCUSTe data with a sensitivity of 0.77 (95% CI 0.64-0.89), specificity of 0.65 (95% CI 0.63-0.67), AUC of 0.77 (95% CI 0.70-0.84), and Matthews correlation coefficient of 0.12 (95% CI 0.08-0.16). In contrast, logistic regression achieved balanced performance without resampling (sensitivity 0.70, 95% CI 0.57-0.83; specificity 0.71, 95% CI 0.69-0.72; AUC 0.75, 95% CI 0.68-0.82). Our results showed that PCUSTe primarily enhanced sensitivity in more complex models at the expense of specificity. Conclusions: Integrating epidemiological principles, including covariate frequency matching and threshold selection based on the observed cumulative incidence in the training data, improved minority class detection in GI cancer risk prediction. However, model performance varied by algorithm, and in some cases, decision threshold adjustment alone achieved comparable or superior results to data resampling. These findings highlight the importance of carefully selecting imbalance mitigation strategies based on modeling objectives. The resulting models achieved sensitivity levels that may be suitable for early risk identification in cohort settings and could contribute to personalized risk stratification and targeted prevention or screening strategies. © Daina Baublyte, Jeonghee Lee, Madhawa Gunathilake, Jeongseon Kim.

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Correspondence Address: J. Kim; Department of Cancer Biomedical Science, National Cancer Center
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Advancing Imbalanced Domain Adaptation: Cluster-Level Discrepancy Minimization With a Comprehensive Benchmark

Item Type Journal Article

Author J. Yang

Author J. Yang

Author S. Wang

Author S. Cao

Author H. Zou

Author L. Xie

Abstract Unsupervised domain adaptation methods have been proposed to tackle the problem of covariate shift by minimizing the distribution discrepancy between the feature embeddings of source domain and target domain. However, the standard evaluation protocols assume that the conditional label distributions of the two domains are invariant, which is usually not consistent with the real-world scenarios such as long-tailed distribution of visual categories. In this article, the imbalanced domain adaptation (IDA) is formulated for a more realistic scenario where both label shift and covariate shift occur between the two domains. Theoretically, when label shift exists, aligning the marginal distributions may result in negative transfer. Therefore, a novel cluster-level discrepancy minimization (CDM) is developed. CDM proposes cross-domain similarity learning to learn tight and discriminative clusters, which are utilized for both feature-level and distribution-level discrepancy minimization, palliating the negative effect of label shift during domain transfer. Theoretical

justifications further demonstrate that CDM minimizes the target risk in a progressive manner. To corroborate the effectiveness of CDM, we propose two evaluation protocols according to the real-world situation and benchmark existing domain adaptation approaches. Extensive experiments demonstrate that negative transfer does occur due to label shift, while our approach achieves significant improvement on imbalanced datasets, including Office-31, Image-CLEF, and Office-Home. © 2022 IEEE.

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Advancing non-target analysis of emerging environmental contaminants with machine learning: Current status and future implications

Item Type Journal Article

Author A. Canchola

Author L.N. Tran

Author W. Woo

Author L. Tian

Author Y.-H. Lin

Author W.-C. Chou

Abstract Emerging environmental contaminants (EECs) such as pharmaceuticals, pesticides, and industrial chemicals pose significant challenges for detection and identification due to their structural diversity and lack of analytical standards. Traditional targeted screening methods often fail to detect these compounds, making non-target analysis (NTA) using high-resolution mass spectrometry (HRMS) essential for identifying unknown or suspected contaminants. However, interpreting the vast datasets generated by HRMS is complex and requires advanced data processing techniques. Recent advancements in machine learning (ML) models offer great potential for enhancing NTA applications. As such, we reviewed key developments, including optimizing workflows using computational tools, improved chemical structure identification, advanced quantification methods, and enhanced toxicity prediction capabilities. It also discusses challenges and future perspectives in the field, such as refining ML tools for complex mixtures, improving inter-laboratory validation, and further integrating computational models into environmental risk assessment frameworks. By addressing these challenges, ML-assisted NTA can significantly enhance the detection, quantification, and evaluation of EECs, ultimately contributing to more effective environmental monitoring and public health protection. © 2025 The Authors

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Advancing radiotherapy with deep Learning: A review of dose prediction models

Item Type Journal Article

Author P. Darkhor

Author B. Hashemi

Author A. Farajollahi

Author A. Jodeiri

Author A. Ghasemi

Abstract Background Artificial intelligence (AI), especially deep learning algorithms, holds significant promise in transforming radiation therapy by enabling more precise, personalized, and efficient treatment planning. In particular, deep learning models have demonstrated their utility in critical tasks such as dose prediction, helping to streamline complex processes and improve patient outcomes. Methods This paper provides a comprehensive review of recent research on deep learning applications for radiotherapy dose prediction, highlighting the architectures, datasets, and methodologies used. Results The review highlights the advancements in accuracy, efficiency, and robustness achieved by deep learning-based dose prediction models. These approaches help reduce treatment planning time and minimize human error, thereby improving clinical workflows. Conclusion Deep learning offers promising solutions to enhance radiotherapy dose delivery. Future research should focus on developing the generalized dose prediction models, multi-institutional validation frameworks, explainable AI approaches for clinical trust, real-time adaptation capabilities, uncertainty quantification methods, and strategies for handling rare anatomical variations. © 2026 Associazione Italiana di Fisica Medica e Sanitaria.

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Advancing Skyborne Technologies and High-Resolution Satellites for Pasture Monitoring and Improved Management: A Review

Item Type Journal Article

Author M.G. Ogungbuyi

Author C. Mohammed

Author I. Ara

Author A.M. Fischer

Author M.T. Harrison

Abstract The timely and accurate quantification of grassland biomass is a prerequisite for sustainable grazing management. With advances in artificial intelligence, the launch of new satellites, and perceived efficiency gains in the time and cost of the quantification of remote methods, there has been growing interest in using satellite imagery and machine learning to quantify pastures at the field scale. Here, we systematically reviewed 214 journal articles published between 1991 to 2021 to determine how vegetation indices derived from satellite imagery impacted the type and quantification of pasture indicators. We reveal that previous studies have been limited by highly spatiotemporal satellite imagery and prognostic analytics. While the number of studies on pasture classification, degradation, productivity, and management has increased exponentially over the last five years, the majority of vegetation parameters have been derived from satellite imagery using simple linear regression approaches, which, as a corollary, often result in site-specific parameterization that become spurious when extrapolated to new sites or production systems. Few studies have successfully invoked machine learning as retrievals to understand the relationship between image patterns and accurately quantify the biophysical variables, although many studies have purported to do so. Satellite imagery has contributed to the ability to quantify pasture indicators but has faced the barrier of monitoring at the paddock/field scale (20 hectares or less) due to (1) low

sensor (coarse pixel) resolution, (2) infrequent satellite passes, with visibility in many locations often constrained by cloud cover, and (3) the prohibitive cost of accessing fine-resolution imagery. These issues are perhaps a reflection of historical efforts, which have been directed at the continental or global scales, rather than at the field level. Indeed, we found less than 20 studies that quantified pasture biomass at pixel resolutions of less than 50 hectares. As such, the use of remote sensing technologies by agricultural practitioners has been relatively low compared with the adoption of physical agronomic interventions (such as ‘no-till’ practices). We contend that (1) considerable opportunity for advancement may lie in fusing optical and radar imagery or hybrid imagery through the combination of optical sensors, (2) there is a greater accessibility of satellite imagery for research, teaching, and education, and (3) developers who understand the value proposition of satellite imagery to end users will collectively fast track the advancement and uptake of remote sensing applications in agriculture. © 2023 by the authors.

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Adversarial confound regression and uncertainty measurements to classify heterogeneous clinical MRI in Mass General Brigham

Item Type Journal Article

Author M. Leming

Author S. Das

Author H. Im

Abstract In this work, we introduce a novel deep learning architecture, MUCRAN (Multi-Confound Regression Adversarial Network), to train a deep learning model on clinical brain MRI while regressing demographic and technical confounding factors. We trained MUCRAN using 17,076 clinical T1 Axial brain MRIs collected from Massachusetts General Hospital before 2019 and demonstrated that MUCRAN could successfully regress major confounding factors in the vast clinical dataset. We also applied a method for quantifying uncertainty across an ensemble of these models to automatically exclude out-of-distribution data in AD detection. By combining MUCRAN and the uncertainty quantification method, we showed consistent and significant increases in the AD detection accuracy for newly collected MGH data (post-2019; 84.6% with MUCRAN vs. 72.5% without MUCRAN) and for data from other hospitals (90.3% from Brigham and Women's Hospital and 81.0% from other hospitals). MUCRAN offers a generalizable approach for deep-learning-based disease detection in heterogenous clinical data. © 2023 Leming et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Adversarial Open-Set Domain Adaptation for Cross-Condition Fault Diagnosis of Marine Gearboxes

Item Type Conference Paper

Author M. Xie

Author Y. Hao

Author Q. Yuan

Author X. Lu

Editor Linlin T.

Abstract The gearbox is a critical component of the ship propulsion system, and its health directly impacts operational safety and reliability. Conventional deep-learning-based diagnostic methods, however, struggle under cross-condition scenarios due to domain shift and the presence of unseen fault types. To address these challenges, we propose an Adversarial Open-Set Domain Adaptation (AOSDA) framework. AOSDA employs a dual-path adversarial training scheme: one path constructs and refines decision boundaries for known faults in both source and target domains, driving target-domain features into alignment; the other path actively pushes unknown-fault features away from these boundaries. We further incorporate evidential deep learning to quantify domain uncertainty and adaptively adjust the threshold for unknown-class detection, thereby enhancing robustness in high-noise environments. A ResNet-based time-frequency feature extractor captures both local and global signal characteristics. We evaluate AOSDA on six cross-condition transfer tasks drawn from two public bearing and gearbox vibration datasets, systematically comparing against four state-of-the-art baselines. Results show that AOSDA achieves the highest classification accuracy for both known and unknown faults (93.2% and 91.6%, respectively). t-SNE visualizations illustrate its synergistic effect on feature alignment and unknown-fault separation, and confusion-matrix analysis confirms high recall for unknown faults alongside high precision for known faults. The proposed method thus offers an effective solution for intelligent operation and maintenance of marine gearboxes under dynamic conditions and open set fault scenarios. © 2025 IEEE.

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ISBN 29945100 (ISSN)**Pages** 429-434**Proceedings Title** Int. Conf. Electron. Meas. Instruments**Conference Name** International Conference on Electronic Measurement and Instruments**Issue** 2025**Date Added** 20/09/2026, 9:52:01 pm**Modified** 20/09/2026, 9:52:01 pm**Tags:**

Safety engineering, Condition, Deep learning, Learning systems, Failure analysis, Domain adaptation, Target domain, Computer aided diagnosis, domain adaptation, Electric fault currents, Fault detection, Faults diagnosis, open-set, Open-set, Critical component, Gearbox fault diagnose, gearbox fault diagnosis, Gears, Marine engineering, Marine gearbox, Operational safety, Ship propulsion, Ship propulsion system

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Adversarial Positive-Unlabeled Learning-Based Invasive Plant Detection in Alpine Wetland Using Jilin-1 and Sentinel-2 Imageries

Item Type Journal Article**Author** E. Zhu**Author** A. Samat**Author** E. Li**Author** R. Xu**Author** W. Li**Author** W. Li

Abstract Invasive plants (IPs) pose a significant threat to local ecosystems. Recent advances in remote sensing (RS) and deep learning (DL) significantly improved the accuracy of IP detection. However, mainstream DL methods often require large, high-quality labeled data, leading to resource inefficiencies. In this study, a deep learning framework called adversarial positive-unlabeled learning (APUL) was proposed to achieve high-precision IP detection using a limited number of target plant samples. APUL employs a dual-branch discriminator to constrain the class prior-free classifier, effectively harnessing information from positive-unlabeled data through the adversarial process and enhancing the accuracy of IP detection. The framework was

tested on very high-resolution Jilin-1 and Sentinel-2 imagery of Bayinbuluke grasslands in Xinjiang, where the invasion of *Pedicularis kansuensis* has caused serious ecological and livestock damage. Results indicate that the adversarial structure can significantly improve the performance of positive-unlabeled learning (PUL) methods, and the class prior-free approach outperforms traditional PUL methods in IP detection. APUL achieved an overall accuracy of 92.2% and an F1-score of 0.80, revealing that *Pedicularis kansuensis* has invaded 4.43% of the local plant population in the Bayinbuluke grasslands, underscoring the urgent need for timely control measures. © 2025 by the authors.

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Adversarial Unsupervised Domain Adaptation with Conditional and Label Shift: Infer, Align and Iterate

Item Type Conference Paper

Author X. Liu
Author Z. Guo
Author S. Li
Author F. Xing
Author J. You
Author C.-C.J. Kuo
Author G. El Fakhri
Author J. Woo

Abstract In this work, we propose an adversarial unsupervised domain adaptation (UDA) method under inherent conditional and label shifts, in which we aim to align the distributions w.r.t. both $p(x|y)$ and $p(y)$. Since labels are inaccessible in a target domain, conventional adversarial UDA methods assume that $p(y)$ is invariant across domains and rely on aligning $p(x)$ as an alternative to the $p(x|y)$ alignment. To address this, we provide a thorough theoretical and empirical analysis of the conventional adversarial UDA methods under both conditional and label shifts, and propose a novel and practical alternative optimization scheme for adversarial UDA. Specifically, we infer the marginal $p(y)$ and align $p(x|y)$ iteratively at the training stage, and precisely align the posterior $p(y|x)$ at the testing stage. Our experimental results demonstrate its effectiveness on both classification and segmentation UDA and partial UDA. © 2021 IEEE

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Agnostically Learning Multi-Index Models with Queries

Item Type Conference Paper

Author I. Diakonikolas

Author D.M. Kane

Author V. Kotonis

Author C. Tzamos

Author N. Zarifis

Abstract We study the power of query access for the fundamental task of agnostic learning under the Gaussian distribution. In the agnostic model, no assumptions are made on the labels of the examples and the goal is to compute a hypothesis that is competitive with the best-fit function in a known class, i.e., it achieves error $\text{opt} + \epsilon$, where opt is the error of the best function in the class. We focus on a general family of Multi-Index Models (MIMs), which are d -variate functions that depend only on few relevant directions, i.e., have the form $G(Wx)$ for an unknown link function G and a $k \times d$ matrix W . Multi-index models cover a wide range of commonly studied function classes, including real-valued function classes such as constant-depth neural networks with ReLU activations, and Boolean concept classes such as intersections of halfspaces. Our main result shows that query access gives significant runtime improvements over random examples for agnostically learning both real-valued and Boolean-valued MIMs. Under standard regularity assumptions for the link function (namely, bounded variation or surface area), we give an agnostic query learner for MIMs with running time $O(k) \text{poly}(1/\epsilon) \text{poly}(d)$. In contrast, algorithms that rely only on random labeled examples inherently require $d \text{poly}(1/\epsilon)$ samples and runtime, even for the basic problem of agnostically learning a single ReLU or a halfspace. As special cases of our general approach, we obtain the following results: •For the class of depth- ℓ , width- S ReLU networks on $\mathbb{R}^d$, our agnostic query learner runs in time $\text{poly}(d) 2^{\text{poly}(\ell S/\epsilon)}$. This bound qualitatively matches the runtime of an algorithm by [1] for the realizable PAC setting with random examples. •For the class of arbitrary intersections of k halfspaces on $\mathbb{R}^d$, our agnostic query learner runs in time $\text{poly}(d) 2^{\text{poly}(\log(k)/\epsilon)}$. Prior to our work, no improvement over the agnostic PAC model complexity (without queries) was known, even for the case of a single halfspace. In both these settings, we provide evidence that the $2^{\text{poly}(1/\epsilon)}$ runtime dependence is required for proper query learners, even for agnostically learning a single ReLU or halfspace. Our algorithmic result establishes a strong computational separation between the agnostic PAC and the agnostic PAC+Query models under the Gaussian distribution for a range of natural function classes. Prior to our work, no such separation was known for any natural concept class - even for the case of a single halfspace, for which it was an

open problem posed by Feldman [2]. Our results are enabled by a general dimension-reduction technique that leverages query access to estimate gradients of (a smoothed version of) the underlying label function. © 2024 IEEE.

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AI-Based Hybrid Models for Predicting Loan Risk in the Banking Sector

Item Type	Journal Article
Author	V. Kumar
Author	S.S. Saheb
Author	A. Ghayas
Author	S. Kumari
Author	J.K. Chandel
Author	S.K. Pandey

Author S. Kumar

Abstract Every real-world scenario is now digitally replicated in order to reduce paperwork and human labor costs. Machine Learning (ML) models are also being used to make predictions in these applications. Accurate forecasting requires knowledge of these machine learning models and their distinguishing features. The datasets we use as input for each of these different types of ML models, yielding different results. The choice of an ML model for a dataset is critical. A loan risk model is used to show how ML models for a dataset can be linked together. The purpose of this study is to look into how we could use machine learning to quantify or forecast mortgage credit risk. This phrase refers to the process of evaluating massive amounts of data in order to derive useful information for making decisions in a variety of fields. If credit risk is considered, a method based on an examination of what caused and how mortgage credit risk affected credit defaults during the still-current economic crisis of 2021 will be tried. Various approaches to credit risk calculation will be examined, ranging from the most basic to the most complex. In addition, we will conduct a case study on a sample of mortgage loans and compare the results of three different analytical approaches, logistic regression, decision tree, and gradient boost to see which one produced the most commercially useful insights. © 2018 Tsinghua University Press.

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AI-Powered Digital Twins for Building Energy Management: Modeling Frameworks, Validation and Uncertainty Quantification, Smart Grid Integration, and Deployment Roadmap

Item Type Journal Article

Author Ł. Łach

Abstract The global buildings and construction sector remains a dominant contributor to anthropogenic climate change, and deep decarbonization has positioned digital twin technology as a transformative pathway for intelligent building energy management. Despite considerable research momentum, the field lacks a coherent synthesis mapping AI capabilities onto the full digital twin lifecycle—from sensor-driven calibration through real-world deployment to district-scale operation. This review addresses this gap through six objectives: analyzing AI-enhanced modeling approaches for building digital twins; examining data infrastructure and interoperability requirements; evaluating validation, calibration, and uncertainty quantification practices; synthesizing real-world implementation evidence across diverse building typologies; assessing integration with renewable energy systems and smart grids; and identifying challenges, research gaps, and a strategic deployment roadmap. Physics-based, data-driven, and hybrid modeling strategies occupy distinct and complementary roles. Physics-informed surrogate models preserve thermodynamic interpretability while reducing computational overhead; deep learning architectures—including recurrent networks and reinforcement learning agents—deliver adaptive control; and federated learning frameworks enable privacy-preserving optimization across distributed building portfolios. Rigorous multi-metric validation aligned with established calibration standards proves essential for trustworthy deployment, while Bayesian and ensemble-based uncertainty quantification methods emerge as indispensable components of operationally credible digital twins. Evidence from real-world deployments in residential, commercial, healthcare, and industrial facilities confirms that AI-powered digital twins consistently deliver substantial energy savings and measurable improvements in occupant comfort. Scaling to district and urban levels introduces challenges in data governance, computational architecture, and multi-stakeholder coordination, yet federated digital twin frameworks are beginning to demonstrate viable pathways. The paper concludes with a decade-long strategic roadmap spanning technological maturation, market development, regulatory alignment, and decarbonization impact—positioning AI-enhanced digital twins not as incremental optimization tools, but as the foundational infrastructure for the coordinated transformation of the global building stock. © 2026 by the author.

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ALEX: Towards Effective Graph Transfer Learning with Noisy Labels

Item Type Conference Paper**Author** J. Yuan**Author** X. Luo**Author** Y. Qin**Author** Z. Mao**Author** W. Ju**Author** M. Zhang

Abstract Graph Neural Networks (GNNs) have garnered considerable interest due to their exceptional performance in a wide range of graph machine learning tasks. Nevertheless, the majority of GNN-based approaches have been examined using well-annotated benchmark datasets, leading to suboptimal performance in real-world graph learning scenarios. To bridge this gap, the present paper investigates the problem of graph transfer learning in the presence of label noise, which transfers knowledge from a noisy source graph to an unlabeled target graph. We introduce a novel technique termed Balance Alignment and Information-aware Examination (ALEX) to address this challenge. ALEX first employs singular value decomposition to generate different views with crucial structural semantics, which help provide robust node representations using graph contrastive learning. To mitigate both label shift and domain shift, we estimate a prior distribution to build subgraphs with balanced label distributions. Building on this foundation, an adversarial domain discriminator is incorporated for the implicit domain alignment of complex multi-modal distributions. Furthermore, we project node representations into a different space, optimizing the mutual information between the projected features and labels. Subsequently, the inconsistency of similarity structures is evaluated to identify noisy samples with potential overfitting. Comprehensive experiments on various benchmark datasets substantiate the outstanding superiority of the proposed ALEX in different settings. © 2023 ACM.

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Algorithmic statistics, prediction and machine learning

Item Type Conference Paper
Author A. Milovanov
Editor Vollmer H.
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Editor LIFO, Cedex 2, Orleans Universite d'Orleans
Abstract Algorithmic statistics considers the following problem: given a binary string x (e.g., some experimental data), find a "good" explanation of this data. It uses algorithmic information theory to define formally what is a good explanation. In

this paper we extend this framework in two directions. First, the explanations are not only interesting in themselves but also used for prediction: we want to know what kind of data we may reasonably expect in similar situations (repeating the same experiment). We show that some kind of hierarchy can be constructed both in terms of algorithmic statistics and using the notion of a priori probability, and these two approaches turn out to be equivalent (Theorem 5). Second, a more realistic approach that goes back to machine learning theory, assumes that we have not a single data string x but some set of "positive examples" $x_1, \dots, x_l$ that all belong to some unknown set A , a property that we want to learn. We want this set A to contain all positive examples and to be as small and simple as possible. We show how algorithmic statistic can be extended to cover this situation (Theorem 11). © Alexey Milovanov; licensed under Creative Commons License CC-BY.

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Aligning Evaluation with Clinical Priorities: Calibration, Label Shift, and Error Costs

Item Type Conference Paper

Author G.A. Flores

Author A.H. Smith

Author J.A. Fukuyama

Author A.C. Wilson

Abstract Machine learning-based decision support systems are increasingly deployed in clinical settings, where probabilistic scoring functions are used to inform and prioritize patient management decisions. However, widely used scoring rules, such as accuracy and AUC-ROC, fail to adequately reflect key clinical priorities, including calibration, robustness to distributional shifts, and sensitivity to asymmetric error costs. In this work, we propose a principled yet practical evaluation framework for selecting calibrated thresholded classifiers that explicitly accounts for uncertainty in class prevalences and domain-specific cost asymmetries. Building on the theory of proper scoring rules, particularly the Schervish representation, we derive an adjusted variant of cross-entropy (log score) that averages cost-weighted performance over clinically relevant ranges of class balance. The resulting evaluation is simple to apply, sensitive to clinical deployment conditions, and designed to prioritize models that are both calibrated and robust to real-world variations. © 2025 Neural information processing systems foundation. All rights reserved.

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Conference Name Advances in Neural Information Processing Systems

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ALM-PU: positive and unlabeled learning with constrained optimization

Item Type Journal Article**Author** J. Wei**Author** Y. Wu**Author** B. Shi**Author** K. Li**Author** B. Dong

Abstract Positive and Unlabeled Learning (PUL) is a special semi-supervised learning paradigm trained on datasets comprising positive and unlabeled samples, and it is widely applied in many real-world applications. Due to the lack of reliable labeled negative samples, PUL is much more challenging than traditional semi-supervised learning. Previous works primarily focus on two fronts: applying varying loss weights for different samples or leveraging algorithms to construct auxiliary datasets with reliable labels. Despite the impressive performance the previous works have already achieved, they are still faced with the challenge of negative classification bias, while struggling to get rid of the need for class prior knowledge. In this paper, we have transformed the PUL task into a constrained optimization problem, and propose a new PUL framework to solve it, namely ALM-PU. More detailed, our ALM-PU aims to minimize the classifier's classification loss on reliable labeled set as the ultimate optimization goal, while subject to the constraint of reducing the cost on the unlabeled set to a certain extent. Subsequently, ALM-PU integrates the primary objective with the constraints to construct the PUL-constrained optimization framework, and implement it in the neural network structure. During the training process, our approach corrects the model's negative classification bias, achieving superior classification performance compared to previous methods. Additionally, a prediction sequence-based algorithm is utilized to aid the classifier in better distinguishing positive from negative samples with training results. We conducted extensive experiments on multiple PUL benchmarks. ALM-PU achieves an average improvement of 2% in key metrics, attaining state-of-the-art performance. These findings validate the effectiveness of our ALM-PU approach. Complete code and more experimental details can be found at ALM-PU. © The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature 2025.

Date 2025**DOI** 10.1007/s10994-025-06849-3**URL** <https://www.scopus.com/pages/publications/105013682325?origin=resultslist>**Archive** Scopus**Language** English**Volume** 114**Publisher** Springer**Publication** Machine Learning**ISSN** 08856125 (ISSN)**Issue** 9**Journal Abbr** Mach Learn**Date Added** 20/09/2026, 9:51:29 pm**Modified** 20/09/2026, 9:51:29 pm**Tags:**

Optimisations, Learning systems, Learning algorithms, Benchmarking, Real-world, Labeled data, Semi-supervised learning, Constrained optimization, Augmented Lagrange method, Augmented-Lagrange method, Classification bias, Learning paradigms, Negative samples, Plutonium alloys, Positive and unlabeled learning, PU-learning, Unlabeled samples

Notes:

EID: 2-s2.0-105013682325

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: B. Shi; School of Computer Science and Technology, Xi'an Jiaotong University, Xi'an, China; email: shibin@xjtu.edu.cn; CODEN: MALEE

Alterations of Bile Acids and Gut Microbiota in Obesity Induced by High Fat Diet in Rat Model

Item Type Journal Article

Author H. Lin

Author Y. An

Author H. Tang

Author Y. Wang

Abstract Obesity has become a worldwide health issue and has attracted much public attention. In the current study, we aim to elucidate the roles of bile acids and their associations with gut microbiota during obesity development, employing high fat diet (HFD)-induced obesity in a rat model. We collected feces and plasma, liver tissues, and segments of intestinal tissues and a developed bile acids quantification method by employing an ultraperformance liquid chromatography coupled with mass spectrometry detection (UPLC-MS) strategy. We then assessed bile acids fluxes in the biological matrixes collected. We found that, irrespective of dietary regimes, taurine-conjugated bile acids were the dominant species in the liver whereas unconjugated bile acids were in plasma. However, HFD caused slight increases in the total bile acids pool and particularly the increases in the levels of deoxycholic acid (DCA) (138.67 ± 37.225 nmol/L in control group, 242.61 ± 43.16 nmol/L in HFD group, $p = 0.014$) and taurodeoxycholic acid (TDCA) (2.8 ± 0.247 nmol/g in control group, 4.5 ± 0.386 nmol/g in HFD group, $p = 0.0018$) in plasma and liver tissues, respectively, which were consistent with the increased levels of DCA in intestinal tissues and feces. These changes are correlated to an increase in abundance of genera *Blautia*, *Coprococcus*, *Intestinimonas*, *Lactococcus*, *Roseburia*, and *Ruminococcus*. Our investigation revealed the fluxes of bile acids and their association with gut microbiota during obesity development and explicated unfavorable impact of HFD on health. © 2019 American Chemical Society.

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Export Date: 20 September 2026; Cited By: 222; Funding details: National Key R&D Program of China, (2017YFC0906800); National Natural Science Foundation of China, NSFC, (21675169); National Natural Science Foundation of China, NSFC; Correspondence Address: H. Tang; State Key Laboratory of Genetic Engineering, Zhongshan Hospital and School of Life Sciences, Metabolomics and Systems Biology Laboratory, Human Phenome Institute, Fudan University, Shanghai, 200433, China; email: huiru_tang@fudan.edu.cn; CODEN: JAFCA

An accurate quantitative method for NdFeB magnetism based on laser-induced breakdown spectroscopy

Item Type Journal Article

Author G. Chen

Author J. Chen

Author D. Qu

Author G. Yang

Author H. Sun

Abstract NdFeB magnetic materials are widely used in daily life, such as in permanent magnet motors, loudspeakers and computer disks. The NdFeB magnetic material has excellent magnetic properties, and its magnetic properties are also the key to judge the production quality of NdFeB. Therefore, the precise quantification of the magnetic properties of NdFeB magnetic materials is crucial. Laser induced breakdown spectroscopy (LIBS) is a technique to obtain the spectrum of chemical

elements by excitation of plasma on the surface of a sample with a high energy laser. In this paper, a precise classification and magnetic quantification method for NdFeB magnetic materials based on laser-induced breakdown spectroscopy is designed, which is different from the traditional direct magnetic property detection method and uses element detection to quantitatively analyze the magnetic properties indirectly. A laser-induced breakdown spectroscopy system was used to collect the characteristic spectrum of NdFeB magnetic materials, and the sliding window minimum removal base method was independently designed to further optimize the detection accuracy. A classification model and quantitative analysis method model were further established and optimized. The random forest method was used to preliminarily classify NdFeB magnetic materials, and the GA-ELM method was used to conduct quantitative analysis of magnetic properties. Quantitative magnetic properties include Br, Hcj, Hcb and (BH)max. The error analysis of the final quantitative analysis is as follows: RMSE of Br reaches 0.0001526, RMSE of Hcj reaches 0.0001937, RMSE of Hcb reaches 0.00197, and RMSE of (BH)max reaches 0.00785. It is verified that the magnetic quantification method for NdFeB magnetic materials based on laser-induced breakdown spectroscopy can effectively conduct accurate quantitative analysis of the magnetic properties of NdFeB magnetic materials and provide a fast, convenient, accurate and economical detection method for the quality control of magnetic materials workshops. © 2025 The Royal Society of Chemistry.

Date 2024

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Issue 1

Journal Abbr J Anal At Spectrom

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An Adapted Deep Convolutional Neural Network for Automatic Measurement of Pancreatic Fat and Pancreatic Volume in Clinical Multi-Protocol Magnetic Resonance Images: A Retrospective Study with Multi-Ethnic External Validation

Item Type Journal Article

Author J.Z. Yang

Author J. Zhao

Author R. Nemati

Author X. Yin

Author K.H. He

Author L. Plank

Author R. Murphy

Author J. Lu

Abstract Pancreatic volume and fat fraction are critical prognoses for metabolic diseases like type 2 diabetes (T2D). Magnetic Resonance Imaging (MRI) is a required non-invasive quantification method for the pancreatic fat fraction. The dramatic development of deep learning has enabled the automatic measurement of MR images. Therefore, based on MRI, we intend to develop a deep convolutional neural network (DCNN) that can accurately segment and measure pancreatic volume and fat fraction. This retrospective study involved abdominal MR images from 148 diabetic patients and 246 healthy normoglycemic participants. We randomly separated them into training and testing sets according to the proportion of 80:20. There were 2364 recognizable pancreas images labeled and pre-treated by an upgraded superpixel algorithm for a discernible pancreatic boundary. We then applied them to the novel DCNN model, mimicking the most accurate and latest manual pancreatic segmentation process. Fat phantom and erosion algorithms were employed to increase the accuracy. The results were evaluated by dice similarity coefficient (DSC). External validation datasets included 240 MR images from 10 additional patients. We assessed the pancreas and pancreatic fat volume using the DCNN and compared them with those of specialists. This DCNN employed the cutting-edge idea of manual pancreas segmentation and achieved the highest DSC (91.2%) compared with any reported models. It is the first framework to measure intra-pancreatic fat volume and fat deposition. Performance validation reflected by regression R2 value between manual operation and trained DCNN segmentation on the pancreas and pancreatic fat volume were 0.9764 and 0.9675, respectively. The performance of the novel DCNN enables accurate pancreas segmentation, pancreatic fat volume, fraction measurement, and calculation. It achieves the same segmentation level of experts. With further training, it may well surpass any expert and provide accurate

measurements, which may have significant clinical relevance. © 2022 by the authors.

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machine learning, deep convolutional neural network (DCNN), diabetes prognosis, organ fat measurement, pancreas segmentation

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EID: 2-s2.0-85149474314

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An Adaptive Asymmetric Loss Function for Positive Unlabeled Learning

Item Type Conference Paper

Author K. Jaskie

Author N. Vaughn

Author V. Narayanaswamy

Author S. Zaare

Author J. Marvin

Author A. Spanias

Editor Hammoud R.I.

Editor Overman T.L.

Editor Mahalanobis A.

Abstract We introduce a new and efficient solution to the Positive and Unlabeled (PU) problem which is tailored specifically for a deep learning framework. We demonstrate the merit of this method using image classification. When only positive and unlabeled images are available for training, our custom loss function,

paired with a simple linear transform of the output, results in an inductive classifier where no estimate of the class prior is required. This algorithm, known as the aaPU (Adaptive Asymmetric Positive Unlabeled) algorithm, provides near supervised classification accuracy with very low levels of labeled data on several image benchmark sets. aaPU demonstrates significant performance improvements over current state-of-the-art positive unlabeled learning algorithms. © 2023 SPIE.

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Deep learning, Classification (of information), Computer vision, Learning algorithms, Supervised learning, Deep Learning, Learning frameworks, Classification, Semi-supervised learning, Semi-Supervised Learning, Loss functions, Supervised classification, Images classification, Asymmetric loss function, Linear transform, Neural Nets, Positive unlabeled, Positive Unlabeled, Simplest linear

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: K. Jaskie; SenSIP Center, ECEE, Arizona State University, Tempe, United States; email: Kristen.Jaskie@psg-inc.net; Conference name: Automatic Target Recognition XXXIII 2023; Conference date: 1 May 2023 through 4 May 2023; Conference code: 189955; CODEN: PSISD

An Adaptive Video-based Vehicle Detection, Classification, Counting, and Speed-measurement System for Real-time Traffic Data Collection

Item Type Conference Paper

Author A. Ghosh

Author M.S. Sabuj

Author H.H. Sonet

Author S. Shatabda

Author D.M. Farid

Abstract Intelligent Transportation System (ITS) is an integral part for efficiently and effectively managing road-transport network in metros and smart cities. ITS provides several important features including public transportation management, route information, safety and vehicle control, electronic timetable and payment system etc. In this paper, we have designed and developed an adaptive video-based vehicle detection, classification, counting, and speed-measurement tool using Java programming language and OpenCV for real-time traffic data collection. It can be used for traffic flow monitoring, planning, and controlling to manage transport network as part of implementing intelligent transport management system in smart cities. The proposed system can detect, classify, count, and measure the speed of vehicles that pass through on a particular road. It can extract traffic data in csv/xml format from real-time video and recorded video, and then send the data to the central data-server. The proposed system extracts image frames from video and apply a filter based on the user-defined threshold value. We have applied MOG2 background subtraction algorithm for subtracting background from the object, which separates foreground objects from the background in a sequence of image frames. The proposed system can detect, classify, and count vehicles of different types and size as a plug play system. We have tested the proposed system at six locations under different traffic and environmental conditions in Dhaka city, which is the capital of Bangladesh. The overall average accuracy is above 80% for classifying all types of vehicles in Dhaka city. © 2019 IEEE.

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Highway administration, Motor transportation, Road vehicles, Intelligent systems, Roads and streets, Classification (of information), Intelligent transportation systems, Advanced public transportation systems, Smart city, Data acquisition, Intelligent Transportation System, Traffic data collection, Adaptive control systems, Transportation routes, Information management, Control system synthesis, Smart City, Computer programming, Vehicle classification, Vehicle Classification, Environmental conditions, Background subtraction algorithms, Public transportation management, Real-time traffic datum, Road transport networks, Traffic Data Collection

Notes:

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Export Date: 20 September 2026; Cited By: 21; Conference name: 2019 IEEE Region 10 Symposium, TENSYP 2019; Conference date: 7 June 2019 through 9 June 2019; Conference code: 157262

An AI Framework for Estimating Obesity Levels Using Randomly Optimized Machine-Learning Models

Item Type Conference Paper

Author V. Eswarakrishnan

Author P. Singla

Editor In C.S.

Editor Londhe N.S.

Editor Bhatt N.

Editor Kitsing M.

Abstract Obesity is another common health issue that doesn't care about a person's gender and has gained attention in recent years. Future prevalence estimations made by the WHO indicate that threshold levels of obesity and overweight will impact more than 50% of the world's population by 2030. This increasing health issue illustrates the importance of preventive measures and the timely discovery of any possible causes. The present instruments are restricted mainly to simple BMI computation and exclude vital factors such as family history, exercise regimen, genealogy, and other related factors. This work introduces an original computational intelligence model for integrated assessment and prediction of obesity threats. Our method utilizes high-level algorithms, namely SVM, RF, and XGB algorithms, that have been enriched with precise pre-processing strategies such as removing outliers, handling missing and duplicate data, and scaling. In addition, measures such as univariate and multivariate analyses were conducted on the different variables to understand the obesity factors better. Random searches for the hyperparameters further enhanced the capability of the proposed method in optimizing the models with substantial improvements in the spheres of predictive precision. The tuned XGB model achieved the highest accuracy of 97.61%. This research is helpful for the fight against obesity as it employs computerized analyses to shed light on the diverse aspects of obesity and enhance approaches to its early diagnosis. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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An analysis modality for vascular structures combining tissue-clearing technology and topological data analysis

Item Type Journal Article

Author K. Takahashi

Author K. Abe

Author S.I. Kubota

Author N. Fukatsu

Author Y. Morishita

Author Y. Yoshimatsu

Author S. Hirakawa

Author Y. Kubota

Author T. Watabe

Author S. Ehata

Author H.R. Ueda

Author T. Shimamura

Author K. Miyazono

Abstract The blood and lymphatic vasculature networks are not yet fully understood even in mouse because of the inherent limitations of imaging systems and quantification methods. This study aims to evaluate the usefulness of the tissue-clearing technology for visualizing blood and lymphatic vessels in adult mouse. Clear, unobstructed brain/body imaging cocktails and computational analysis (CUBIC) enables us to capture the high-resolution 3D images of organ- or area-specific vascular structures. To evaluate these 3D structural images, signals are first classified from the original captured images by machine learning at pixel base. Then, these classified target signals are subjected to topological data analysis and non-homogeneous Poisson process model to extract geometric features. Consequently, the structural difference of vasculatures is successfully evaluated in mouse disease models. In conclusion, this study demonstrates the utility of CUBIC for analysis of vascular structures and

presents its feasibility as an analysis modality in combination with 3D images and mathematical frameworks. © 2022, The Author(s).

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An analysis of 45 large-scale wastewater sites in England to estimate SARS-CoV-2 community prevalence

Item Type Journal Article

Author M. Morvan

Author A.L. Jacomo

Author C. Souque

Author M.J. Wade

Author T. Hoffmann

Author K. Pouwels

Author C. Lilley

Author A.C. Singer

Author J. Porter

Author N.P. Evens

Author D.I. Walker

Author J.T. Bunce

Author A. Engeli

Author J. Grimsley

Author K.M. O'Reilly

Author L. Danon

Abstract Accurate surveillance of the COVID-19 pandemic can be weakened by under-reporting of cases, particularly due to asymptomatic or pre-symptomatic infections, resulting in bias. Quantification of SARS-CoV-2 RNA in wastewater can be used to infer infection prevalence, but uncertainty in sensitivity and considerable variability has meant that accurate measurement remains elusive. Here, we use data from 45 sewage sites in England, covering 31% of the population, and estimate SARS-CoV-2 prevalence to within 1.1% of estimates from representative prevalence surveys (with 95% confidence). Using machine learning and phenomenological models, we show that differences between sampled sites, particularly the wastewater flow rate, influence prevalence estimation and require careful interpretation. We find that SARS-CoV-2 signals in wastewater appear 4–5 days earlier in comparison to clinical testing data but are coincident with prevalence surveys suggesting that wastewater surveillance can be a leading indicator for symptomatic viral infections. Surveillance for viruses in wastewater complements and strengthens clinical surveillance, with significant implications for public health. © 2022, Crown.

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An Analytical Review of Construction and Demolition Waste Management and Quantification Methods Using a Science Mapping Approach

Item Type	Journal Article
Author	W. Sun
Author	Q. Tushar
Author	G. Zhang
Author	A. Song
Author	L. Hou
Author	J. Zhang
Author	S. Wang
Abstract	Construction and demolition waste (CDW) management remains a pressing challenge in the construction industry, contributing significantly to environmental degradation and resource depletion. Accurate waste measurement is essential for improving resource recovery and circular economy adoption. However, existing research lacks standardised estimation methods, the integration of digital technologies, and comprehensive lifecycle analysis approaches, limiting the effectiveness of waste prediction and management strategies. This study addresses the gap by conducting a scientometric analysis using CiteSpace and SciMAT, examining research trends, thematic clusters, and knowledge evolution in CDW quantification and management from 2014 to 2024. It establishes a conceptual framework for integrating digital systems and sustainable practices in CDW, focusing on waste generation rate, carbon emission, and phase-based waste management analysis. Network cluster analysis

reveals the integral role of estimation tools and modelling techniques in refining waste generation quantification for building constructions. It also examines the interplay of digital tools, their influence on environmental cost reduction, and factors affecting waste production and environmental protection across project phases. This conjugate approach highlights the importance of the successful implementation of waste quantification and the imperative of machine learning for further investigation. This review offers an evidence-based framework to identify key stakeholders, guide future research, and implement sustainable waste management policies. © 2025 by the authors.

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: G. Zhang; School of Engineering, RMIT University, Melbourne, GPO Box 2476, 3001, Australia; email: kevin.zhang@rmit.edu.au

An Application-Oriented Process Model for Selecting Uncertainty Quantification Methods in Machine Learning

Item Type Conference Paper

Author N. Holderied

Author S. Hörtling

Author K. Bause

Author T. Düser

Abstract Uncertainty quantification (UQ) is increasingly recognized as essential when machine learning (ML) is employed in domains that are safety-relevant, cost-intensive, or legally binding, such as the product engineering of battery electric vehicle (BEV) energy systems. UQ methods aim to estimate the aleatoric, epistemic or both uncertainties associated with the predictions of a machine learning model. However, the landscape of UQ methods is diverse and rapidly evolving, with no single approach proving optimal across all tasks. Consequently, the selection of methods in practice is often driven by experience, constrained by limited comprehensive knowledge, time, and implementation capacity. This paper introduces an application-oriented process model supporting data scientists in selecting UQ methods in ML by adapting the SPALTEN [1] problem-solving methodology and the Algorithm Selection Process Model (ASPM) into an Algorithm Selection Process Model for Uncertainty Quantification (UQ-ASPM). This model can be integrated into the modeling phase of a data mining process, such as the Cross Industry Standard Process for Data Mining (CRISP-DM). Ethnographic observations and expert interviews conducted within the research environment of BEV energy system development were analyzed using inductive qualitative content analysis to identify practical barriers, motivations, and requirements. The resulting process translates task requirements and boundary conditions into UQ-specific criteria, primarily including the source of uncertainty, integration depth, and output type. It employs a funnel-like narrowing from method families to candidate algorithms and utilizes a transparent evaluation matrix with weighted criteria, consequence analysis, and learning through a continuous information pool. An illustrative predictive-maintenance example demonstrates the instantiation of the process when an existing deterministic ML model must be retained. The contribution is made at a meta-level, facilitating structured navigation of the method space rather than providing a direct comparison of individual UQ algorithms. © 2026 SAE International. All Rights Reserved.

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An assessment of adding value of traffic information and other attributes as part of its classifiers in a data mining tool set for predicting surface ozone levels

Item Type Journal Article

Author A. Nawahda

Abstract This study seeks to examine to what extent traffic information can improve the prediction of surface ozone levels from mobile sources when coupled with a state of the art air quality monitoring system and the application of data mining tools. For the purpose of the experiment an open-path Differential Optical Absorption Spectroscopy (DOAS) instrument is used and 10 min video samples obtained from Sohar's main highway (SHW) (Sultanate of Oman). This traffic information is collated to recognize, classify, and count three types of vehicles passenger car; light duty vehicle; and heavy duty vehicle. The DOAS is deployed to measure the following gases; ambient nitrogen dioxide (NO₂); ozone (O₃); sulfur dioxide (SO₂); and BTX (benzene, toluene, xylene) across SHW. The ambient concentrations of these gases are measured in situ at time resolutions that vary from 30 s to 1 min along with simultaneous measurements of meteorological parameters. The Waikato Environment for Knowledge Analysis (WEKA) (Witten and Frank, 2005) software was used for the data mining part of the study. To identify which classifiers in WEKA would be the most suitable in predicting surface O₃ levels the following five indexes were used: correlation coefficient (CC); mean absolute error (MAE); root mean square error (RMSE); relative absolute error (RAE); and root relative squared error (RRSE). It was found that the Bagging and M5P classifiers were the most robust when compared to others within the software when measured against the five indexes. It was identified that with the additions of time and day of the week as well as changing of the parameters as part of the classifiers in WEKA the robustness of the predictions was not enhanced significantly. However, the findings did illustrate that the analysis of traffic information does improve the robustness of the prediction of surface O₃ levels. © 2015 The Institution of Chemical Engineers. Published by Elsevier B.V. All rights reserved.

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An automatic segmentation and quantification method for austenite and ferrite phases in duplex stainless steel based on deep learning

Item Type Journal Article

Author L. Che

Author Z. He

Author K. Zheng

Author X. Xu

Author F. Zhao

Abstract Traditional microstructural analysis and phase identification largely rely on manual efforts, inevitably affecting the consistency and accuracy of the results. Historically, the identification of ferrite and retained austenite phases and the extraction of grain information have predominantly been conducted by experts based on their experience. This process is not only time-consuming and labor-intensive but also prone to discrepancies due to the subjective nature of expert judgment. With the continuous advancement of deep learning technologies, new solutions for the classification and analysis of microstructures have emerged. This study proposes a microstructural segmentation method for dual-phase steel based on the Mask R-CNN deep learning model, which can quickly and accurately segment ferrite and retained austenite phases in dual-phase steel subjected to different heat treatment temperatures, enabling quantitative analysis of grain information. First, lightweight dual-phase steel is subjected to heat treatments at five different temperatures, and electron microscope images are obtained as training and testing data for the network. Through data preprocessing, annotation, and augmentation, a microstructural image dataset is constructed. Subsequently, the Mask R-CNN deep learning model is employed to recognize and segment the microstructural dataset of dual-phase steel. From the mask images output by the model, quantitative parameters such as the volume fraction and average grain size of ferrite and retained austenite are successfully extracted. Furthermore, the approach demonstrates high portability and

applicability, particularly relying on a small sample dataset. © 2025 The Royal Society of Chemistry.

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An Efficient Arrhythmia Classifier Using Convolutional Neural Network with Incremental Quantification

Item Type Conference Paper

Author J. Huang

Author Z. Liu

Author H. Liu

Abstract Cardiovascular disease (CVD) is a dangerous disease, which can be effectively prevented by detecting arrhythmia early. In order to detect arrhythmia accurately, more and more researches use artificial intelligence methods to realize the classification and detection of electrocardiogram (ECG) signals. However, most of these designs are unfriendly to the hardware design due to too many parameters which lead to large calculation power and data accessing power. In

this paper, we present an efficient arrhythmia classifier based on the convolutional neural network with the incremental quantification. This more efficient design can classify ECG signals accurately with lower capacity of parameters. The simulation results show that the recognition rate of the network with the incremental quantification has reached 92.76% with 39.34KB memory footprint, which is beneficial to hardware design and has better accuracy than other advanced quantification methods. © Published under licence by IOP Publishing Ltd.

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: J. Huang; Microelectronics, Southeast University, Nanjing, Jiang Su, 214000, China; email: huangjunguang@seu.edu.cn; Conference name: 2021 International Conference on Artificial Intelligence and Industrial Technology Applications, AIITA 2021; Conference date: 18 June 2021 through 20 June 2021; Conference code: 170475

An Efficient Framework for Detecting Model Inversion Attacks in Connected Healthcare Systems

Item Type Journal Article

Author A. Malik

Author S.M. Qaisar

Author M.Z. Khan

Author M.N. Khan

Author T.H.H. Aldhyani

Author R.H. Alsisi

Abstract The rapid adoption of artificial intelligence (AI) in healthcare systems and consumer electronics is enhancing clinical decision-making and reshaping patient care. However, these advancements also raise serious security and privacy concerns, particularly model inversion attacks (MIAs), which reconstruct sensitive patient data from trained AI models. Addressing MIAs is essential to ensure trustworthiness and effectiveness of AI-driven healthcare solutions. To address these challenges, a novel four-stage detection framework is proposed to safeguard healthcare AI models against MIAs. Firstly, the data preprocessing and normalization layer eliminates the input noise and stabilizes the incoming query patterns. Onward, the behavioral feature extraction is carried out. It seeks to detect adversarial intent by profiling the temporal patterns as well as the irregularities found in the query behavior. Afterward, statistical anomaly scoring is applied using Mahalanobis distance learning to quantify deviations in query behavior. Lastly, an ensemble-based decision module, built on support vector machine and random forest classifiers, is used for the identification of MIAs. It combines anomaly scores and utilizes a threshold-based decision to accurately separate benign and possible MIA sessions while enhancing robustness against sophisticated attacks. Extensive experimental results on the NIH ChestXray14 and MIT-BIH datasets show that the proposed framework improves best-performing benchmarks in terms of precision by 1.5%, recall by 2.8%, F1-score by 1.18%, and AUROC by 2.23%, while reducing overhead by 9.09%. © 2026 IEEE.

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An Efficient Intelligent Edge Diagnosis System Based on WDCNN in the Application of Equipment Fault Classification

Item Type Conference Paper

Author Q. Wu

Author X. Ding

Author Q. He

Author H. Xiang

Author H. Tan

Author Y. Shao

Abstract Edge computing has been extensively used in industries to solve problems such as data explosion. However, in the field of mechanical equipment fault diagnosis, there are still problems such as the large amount of data transmission and storage and the difficulty of applying intelligent algorithms to equipment edge fault diagnosis. Therefore, an intelligent edge diagnosis system (IEDS) is designed and applied to mechanical equipment fault diagnosis, which includes an edge diagnosis unit (EDU) and an edge network model. The EDU uses the STM32F405 chip as the core and acquires vibration signals by MEMS acceleration sensor. Using Deep convolutional neural networks with wide first-layer kernels (WDCNN) as the edge network model. Due to the limitation of hardware resources, this paper uses a fixed-point quantification method to minimize the hardware resource usage of WDCNN. The proposed method can efficiently implement the edge fault diagnosis of mechanical equipment, which is reflected by the fact that IEDS can be deployed directly at the edge of equipment to accomplish efficient data reduction and accurate fault diagnosis results output. The proposed method provides a new solution for intelligent fault diagnosis of mechanical equipment. © 2022 IEEE.

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Conference Name 2022 International Conference on Sensing, Measurement and Data Analytics in the Era of Artificial Intelligence, ICSMD 2022 - Proceedings

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An Efficient Quantification Method Based on Feature Selection for High-Dimensional Uncertainties of Multistage Compressors

Item Type Journal Article**Author** J. Wang**Author** X. Zheng**Author** H. Yang**Author** Z. Sun**Author** Z. Song**Author** Y. Fu

Abstract Multistage compressors are widely used in aero-engines, gas turbines, and industrial compressors. However, many inevitable geometric/condition uncertainties will lead to low manufacturing qualified rate and severe performance degradation of multistage compressors. There might be hundreds of uncertainty variables simultaneously, so the "curse of dimensionality" problem has seriously constrained the uncertainty quantification (UQ) study in multistage compressors. In this paper, a feature selection method based on interpretable machine learning and bidirectional search is developed to reduce the dimensionality in a multistage compressor UQ study. The method is applied to a threestage compressor and reduces the uncertainties dimensionality from 292 to 41 and 66 for mass flow rate and efficiency prediction. Consequently, the required sample size is reduced from 1674 to 160 with the model accuracy almost unchanged. For the first time, this study realizes the UQ modeling study on the high dimensional problem of a multistage compressor. In addition, significant stage-by-

stage uncertainty propagation is found in the compressor. The last stage has the most significant efficiency deviation, which is significantly affected by the geometric uncertainty of the first two stages. Therefore, traditional studies, which usually simplify the multistage compressor UQ problem to a single cascade/row level, may seriously underestimate the uncertainty effect due to the neglect of stage-by-stage propagation. This study not only reveals the necessity for direct UQ study of multistage compressors but also reduces the computational cost, which lays a foundation for the uncertainty control of multistage compressors in engineering practice. © 2023 American Society of Mechanical Engineers (ASME). All rights reserved.

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An Efficient Uncertainty Quantification Method Based on Inter-Blade Decoupling for Compressors

Item Type Journal Article

Author Y. Fu
Author J. Wang
Author D. Shi
Author K. Tang
Author J. Zhao

Abstract A compressor usually contains multiple blade rows, and its uncertainty dimensionality grows proportionally with the number of blade rows, leading to a rapid increase in required sample size for uncertainty quantification analysis. This paper proposes a new model decomposition method based on inter-blade decoupling, by analyzing uncertainty propagation in compressors. Traditionally, a surrogate model with uncertainty variables of all blade rows as input is directly established and the dimensionality is high. To solve this problem, this study decomposes the compressor domain into subdomains, each containing one blade row. For each subdomain, a submodel introduces aerodynamic uncertainties at the interfaces connecting different subdomains. The dimensionality of a submodel is roughly equal to the uncertainty factors in a single row, significantly reducing the required sample size. The uncertainties in the rotor and stator blade rows of a one-stage compressor are investigated to verify this method. Using principal component analysis and machine learning, the projection amplitudes of the interface aerodynamic flow field onto the principal modes are extracted, and submodels are established. Results show that the original 25-dimensional model can be decoupled into a 13-dimensional submodel for the rotor and a 16-dimensional submodel for the stator, reducing the required sample size from 600 to 90 with similar accuracy. This model decomposition method greatly reduces the cost of predicting compressor performance with uncertainty, laying a foundation for comprehensive analysis and effective control of uncertainty factors in engineering applications. © 2025 by ASME.

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An Empirical Evaluation of Out-of-Distribution Detection Using Pretrained Language Models

Item Type Conference Paper

Author B. Yoon

Author J. Kim

Abstract In Natural Language Processing (NLP) tasks, detecting out-of-distribution (OOD) samples is essential to safely deploy a language model in real-world problems. Recently, several studies report that pre-trained language models (PLMs) accurately detect OOD data compared to LSTM, but we empirically find that PLMs show sub-par OOD detection performance when (1) OOD samples have similar semantic representation to indistribution (IND) samples and (2) PLMs are finetuned under data scarcity settings. To alleviate above issues, state-of-the-art uncertainty quantification (UQ) methods can be used, but the comprehensive analysis of UQ methods with PLMs has received little consideration. In this work, we investigate seven UQ methods with PLMs and show their effectiveness in the text classification task. © 2023 IEEE.

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An Enhanced Fault Diagnosis Method with Uncertainty Quantification Using Bayesian Convolutional Neural Network

Item Type Conference Paper

Author Q. Fang

Author G. Xiong

Author X. Shang

Author S. Liu

Author B. Hu

Author Z. Shen

Abstract Fault diagnosis is a vital technique to pinpoint the machine malfunctions in manufacturing systems. In recent years, the deep learning techniques greatly improve the fault detection accuracy, but there still remain some problems. If one fault is absent in the training data or the fault signal is disturbed by severe noise interference, the fault classifier may misjudge the health state. This problem limits the reliability of the fault diagnosis in real applications. In this paper, we enhance the fault diagnosis method by using Bayesian Convolutional Neural Network (BCNN). A Shannon entropy-based method is presented to quantify the prediction uncertainty. The BCNN turns the deterministic predictions to probabilistic distributions and enhances the robustness of the fault diagnosis. The uncertainty quantification method helps to indicate the wrong predictions, detect unknown faults, and discover the strong disturbances. Then, a fine-tuning strategy is applied to enhance the model performance further. The potential usability of the proposed method in monitoring the motors of 3D printers is studied. And the experiment is conducted on a motor bearing dataset provided by Case Western Reserve University. The proposed BCNN achieves 99.82% fault classification accuracy over nine health conditions. Its robustness is verified by comparing the testing accuracy with three other methods on the noisy datasets. And the uncertainty quantification method successfully detects the outlier inputs. © 2020 IEEE.

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An Equivalence Analysis of Binary Quantification Methods

Item Type Conference Paper

Author A. Castaño

Author J. Alonso

Author P. González

Author J.J. del Coz

Editor Williams B.

Editor Chen Y.

Editor Neville J.

Abstract Quantification (or prevalence estimation) algorithms aim at predicting the class distribution of unseen sets (or bags) of examples. These methods are useful for two main tasks: 1) quantification applications, for instance when we need to track the proportions of several groups of interest over time, and 2) domain adaptation problems, in which we usually need to adapt a previously trained classifier to a different –albeit related– target distribution according to the estimated prevalences. This paper analyzes several binary quantification algorithms showing that not only do they share a common framework but are, in fact, equivalent. Inspired by this study, we propose a new method that extends one of the approaches analyzed. After an empirical evaluation of all these methods using synthetic and benchmark datasets, the paper concludes recommending three of them due to their precision, efficiency, and diversity. Copyright © 2023, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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An evaluation of the performance of stopping rules in AI-aided screening for psychological meta-analytical research

Item Type Journal Article

Author L. König

Author S. Zitzmann

Author T. Fütterer

Author D.G. Campos

Author R. Scherer

Author M. Hecht

Abstract Several AI-aided screening tools have emerged to tackle the ever-expanding body of literature. These tools employ active learning, where algorithms sort abstracts based on human feedback. However, researchers using these tools face a crucial dilemma: When should they stop screening without knowing the proportion of relevant studies? Although numerous stopping rules have been proposed to guide users in this decision, they have yet to undergo comprehensive evaluation. In this study, we evaluated the performance of three stopping rules: the knee method, a data-driven heuristic, and a prevalence estimation technique. We measured performance via sensitivity, specificity, and screening cost and explored the influence of the prevalence of relevant studies and the choice of the learning algorithm. We curated a dataset of abstract collections from meta-analyses across five psychological research domains. Our findings revealed performance differences between stopping rules regarding all performance measures and variations in the performance of stopping rules across different prevalence ratios. Moreover, despite the relatively minor impact of the learning algorithm, we found that specific combinations of stopping rules and learning algorithms were most effective for certain prevalence ratios of relevant abstracts. Based on these results, we derived practical recommendations for users of AI-aided screening tools. Furthermore, we discuss possible implications and offer suggestions for future research. © 2024 The Author(s). Research Synthesis Methods published by John Wiley & Sons Ltd.

Date 2024

DOI 10.1002/jrsm.1762

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Language English

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Journal Abbr Res. Synth. Methods

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artificial intelligence, human, Humans, machine learning, Artificial Intelligence, reproducibility, Reproducibility of Results, algorithm, Algorithms, methodology, sensitivity and specificity, meta-analysis, meta analysis (topic), Sensitivity and Specificity, literature screening, Meta-Analysis as Topic, Research Design, stopping rules, systematic reviews

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An Evaluation of Valuable Lands towards Landslide Susceptibility at Heritage Site, Sangiran, Central Java, Indonesia

Item Type Conference Paper

Author S. Afifah

Author A.U. Putra

Author A. Christianawati

Author K.P. Prastika

Author M.I. Thoban

Author R. Lukafiardi

Author S.E. Siahaan

Author U. Widyanarko

Author D.R. Hizbaron

Abstract Sangiran Early Man Site is a world heritage located in a peri-urban area in Sragen Regency, Central Java, Indonesia. Due to its significance, this 59.21 km² area is recognized as a valuable land. However, landslides persistently threaten its existence. The research intended to evaluate 1) the landslide susceptibility and 2) the vulnerability of the valuable lands in Sangiran Site. It applied the weights-of-evidence method to produce a landslide susceptibility map, which was further validated with field survey. The analysis output included five classes of susceptibility, namely, very low, low, moderate, high, and very high. The classification aimed to generate matrices containing indications of valuable lands in each class. The field survey found no landslides in very-low susceptible areas and several small ones in low susceptible areas. Landslides were found in valuable lands endangered with a moderate degree of landslides. Large but non-risky landslides were detected in highly susceptible areas, while large and risky ones were in very highly susceptible areas. Sangiran consists of 19 valuable lands that help to narrate geological and human evolution through fossils, scarcity, stratigraphy, and other unique features. For each land, a number is assigned to it to mark priority based its importance clues. Priority 13 was included in the low susceptibility class. While priorities 1, 5, 6, 7, 8, 12, 17, and 18 belonged to moderate susceptibility class, priorities 2, 3, 4, 9, 10, 11, 14, 15, 16, and 19 were classified into high susceptibility class. Therefore, Sangiran Site comprises

heterogeneous characteristics that make its valuable lands the subjects of protection against landslide occurrences. © Published under licence by IOP Publishing Ltd.

Date 2020

DOI 10.1088/1755-1315/412/1/012019

URL <https://www.scopus.com/pages/publications/85079001245?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: IOP Conf. Ser. Earth Environ. Sci.

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Publisher Institute of Physics Publishing

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Conference Name IOP Conference Series: Earth and Environmental Science

Issue 1

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Tags:

Surveys, Urban growth, Landslides, Stratigraphy, Landslide susceptibility, Heritage sites, Heterogeneous characteristic, Human evolution, Peri-urban areas, Priority-based, Unique features, Weights of evidences

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: S. Afifah; Department of Environmental Geography, Faculty of Geography, Universitas Gadjah Mada, Indonesia; email: sani.afifah@mail.ugm.ac.id; Conference name: 2nd International Conference on Geography and Education, ICGE 2019; Conference code: 156936

An extensible software platform for interdisciplinary cardiovascular imaging research

Item Type Journal Article

Author M. Huellebrand

Author D. Messroghli

Author L. Tautz

Author T. Kuehne

Author A. Hennemuth

Abstract Background and objective: Cardiovascular imaging is an exponentially growing field with aspects ranging from image acquisition and analysis to disease characterization, and evaluation of therapy approaches. The transfer of innovative new technological

and algorithmic solutions into clinical practice is still slow. In addition to the verification of solutions, their integration in the clinical processing workflow must be enabled for the assessment of clinical impact and risks. The goal of our software platform for cardiac image processing – CAIPI – is to support researchers from different specialties such as imaging physics, computer science, and medicine by a common extensible platform to address typical challenges and hurdles in interdisciplinary cardiovascular imaging research. It provides an integrated solution for method comparison, integrated analysis, and validation in the clinical context. The interface concept enables a combination with existing frameworks that address specific aspects of the pipeline, such as modeling (e.g., OpenCMISS, CARP) or image reconstruction (Gadgetron). **Methods:** In our platform, we developed a concept for import, integration, and management of cardiac image data. The integration approach considers the spatiotemporal properties of the beating heart through a specific data model. The solution is based on MeVisLab and provides functionalities for data retrieval and storage. Two types of plugins can be added. While ToolPlugins usually provide processing algorithms such as image correction and segmentation, AnalysisPlugins enable interactive data exploration and reporting. GUI integration concepts are presented for both plugin types. We developed domain-specific reporting and visualization tools (e.g., AHA segment model) to enable validation studies by clinical experts. The platform offers plugins for calculating and reporting quantitative parameters such as cardiac function, which can be used to, e.g., evaluate the effect of processing algorithms on clinical parameters. Export functionalities include quantitative measurements to Excel, image data to PACS, and STL models to modeling and simulation tools. **Results:** To demonstrate the applicability of this concept both for method development and clinical application, we present use cases representing different problems along the innovation chain in cardiac MR imaging. Validation of an image reconstruction method (MRI T1 mapping) Validation of an image correction method for real-time 2D-PC MRI Comparison of quantification methods for blood flow analysis Training and integration of machine learning solutions with expert annotations Clinical studies with new imaging techniques (flow measurements in the carotid arteries and peripheral veins as well as cerebral spinal fluid). **Conclusion:** The presented platform can be used in interdisciplinary teams, in which engineers or data scientists perform the method validation, followed by clinical research studies in patient collectives. The demonstrated use cases show how it enables the transfer of innovations through validation in the cardiovascular application context. © 2019

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Publication Computer Methods and Programs in Biomedicine

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Journal Abbr Comput. Methods Programs Biomed.

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Risk assessment, Article, automation, Digital storage, human, Humans, machine learning, Heart, Image segmentation, reproducibility, Reproducibility of Results, Data integration, algorithm, Algorithms, quantitative analysis, image segmentation, Information management, Medical imaging, procedures, Medical applications, Image processing, Medical image processing, Software, Segmentation, convolutional neural network, software, information processing, mathematical model, Image analysis, Clinical research, information retrieval, clinical research, image processing, Disease control, diagnostic imaging, image analysis, computer assisted diagnosis, Image Interpretation, Computer-Assisted, image reconstruction, Magnetic resonance imaging, Magnetic Resonance Imaging, nuclear magnetic resonance imaging, MRI, Medical image analysis, Image reconstruction, Imaging systems, segmentation algorithm, digital imaging and communications in medicine, Cerebrospinal fluid, Cardiology, image registration, blood flow velocity, Blood Flow Velocity, cardiac imaging, Cardiovascular applications, cardiovascular system, Cardiovascular System, Image acquisition and analysis, Image reconstruction methods, Information Storage and Retrieval, Interactive data exploration, interdisciplinary research, Interdisciplinary teams, Modeling and simulation tools, Quantitative measurement, Solution mining, Spatio-temporal properties

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An improved classification scheme for chromosomes with missing data

Item Type Conference Paper

Author E. Poletti

Author A. Ruggeri

Author E. Grisan

Abstract Karyotyping, or the automatic classification of human chromosomes, is mostly based on the analysis of the chromosome specific banding pattern. Unfortunately, the most informative phases of the cell division cycle are composed of long chromosomes that easily overlap: the involved banding pattern information is corrupted, resulting in a drastic increase of the classification error. Assuming the availability of a probabilistic classifier, the improvement of the classification of chromosomes with corrupted data would require the additional estimation of the joint probability density of the observed and missing data for each chromosome class. Given the number of classes, the possible position and extension of the corrupted data within a chromosome, and the dimensionality of the feature space,

a reliable estimation would need an impossible number of training samples. We chose to circumvent the estimation problem by developing a statistical generative model of the pattern of each class, so that the corrupted part can be substituted with a partial pattern synthetically generated from the model. This allows to obtain a Monte Carlo estimate of the maximum a posteriori probability for the class given the observation and the missing data, which reduces to a simple voting scheme if the a priori probability for each class is equal. Moreover, this Monte Carlo classification is superior to the voting scheme based on the simple imputation of the classes mean to the missing data. © 2011 IEEE.

Date 2011

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Language English

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Conference Name Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS

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Tags:

Probability density function, article, reproducibility, Reproducibility of Results, algorithm, Algorithms, Estimation, Probability, Generative model, methodology, Training sample, genetics, Monte Carlo methods, sensitivity and specificity, artifact, Artifacts, Sensitivity and Specificity, Feature space, A-priori probabilities, Classification errors, Probabilistic classifiers, Estimation problem, automated pattern recognition, Pattern Recognition, Automated, Classification scheme, sample size, Sample Size, Maximum A posteriori probabilities, Automatic classification, Banding patterns, Cell division cycle, Cell proliferation, chromosome, Chromosomes, Corrupted data, Human chromosomes, Joint probability, karyotyping, Karyotyping, Missing data, MONTE CARLO, Monte Carlo estimates, Partial patterns, Voting schemes

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An Improved Method of Automated Nonparametric Content Analysis for Social Science

Item Type Journal Article

Author C.T. Jerzak

Author G. King

Author A. Strezhnev

Abstract Some scholars build models to classify documents into chosen categories. Others, especially social scientists who tend to focus on population characteristics, instead usually estimate the proportion of documents in each category - using either parametric classify-and-count methods or direct nonparametric estimation of proportions without individual classification. Unfortunately, classify-and-count methods can be highly model-dependent or generate more bias in the proportions even as the percent of documents correctly classified increases. Direct estimation avoids these problems, but can suffer when the meaning of language changes between training and test sets or is too similar across categories. We develop an improved direct estimation approach without these issues by including and optimizing continuous text features, along with a form of matching adapted from the causal inference literature. Our approach substantially improves performance in a diverse collection of 73 datasets. We also offer easy-to-use software that implements all ideas discussed herein. © 2022 The Author(s).

Date 2023

DOI 10.1017/pan.2021.36

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natural language processing, quantification, non-parametric statistics

Notes:

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Export Date: 20 September 2026; Cited By: 11

An improvement on learning with local and global consistency

Item Type Conference Paper

Author L. Gui

Author D.-S. Huang

Author Z. You

Abstract A modified version for semi-supervised learning algorithm with local and global consistency was proposed in this paper. The new method adds the label information, and adopts the geodesic distance rather than Euclidean distance as the measure of the difference between two data points when conducting calculation. In addition we add class prior knowledge. It was found that the effect of class prior knowledge was different between under high label rate and low label rate. The experimental results show that the changes attain the satisfying classification performance better than the original algorithms. © 2008 IEEE.

Date 2008

DOI 10.1109/ICPR.2008.4761295

URL <https://www.scopus.com/pages/publications/77955366231?origin=resultslist>

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Language English

Extra Journal Abbreviation: Proc. Int. Conf. Pattern Recognit.

Publisher Institute of Electrical and Electronics Engineers Inc.

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Proceedings Title Proc. Int. Conf. Pattern Recognit.

Conference Name Proceedings - International Conference on Pattern Recognition

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Tags:

Federated learning, Adversarial machine learning, Contrastive Learning, Prior-knowledge, Self-supervised learning, Semi-supervised learning, Classification performance, Datapoints, Label information, Geodesic distances, Euclidean distance, Global consistency, Local consistency, Original algorithms

Notes:

EID: 2-s2.0-77955366231

Export Date: 20 September 2026; Cited By: 11; CODEN: PICRE

An incremental learning framework combining sample confidence and discrimination with an application to automatic image annotation

Item Type Conference Paper

Author B. Byun

Author C.-H. Lee

Abstract We propose an incremental classifier learning framework that starts with a small amount of labeled training data to create an initial set of classifiers, and gradually incorporates unlabeled data into the incremental learning process to improve the models. A key to the effectiveness of the proposed framework is to judiciously select a good incremental learning subset from all remaining unlabeled samples by computing a confidence measure and a margin-like discrimination score that measures potential contributions of the selection set to enhancing the existing models. To further refine the above selection set, class prior densities were also exploited. The proposed framework was tested on an automatic image annotation application using a subset of the Corel image set. When all data, including both initially labeled and incrementally learned samples, were used, the final model was shown to achieve a significant improvement over the initial set of classifiers in terms of micro-averaging F1 even when only a small number of images were initially labeled. Furthermore, when 30% of the images were initially labeled the incrementally learned models achieved comparable results to the case when models were created with all training data labeled. ©2009 IEEE.

Date 2009

DOI 10.1109/ICIP.2009.5414560

URL <https://www.scopus.com/pages/publications/77951957024?origin=resultslist>

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Language English

Extra Journal Abbreviation: Proc. Int. Conf. Image Process. ICIP

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Tags:

Data reduction, Training data, Image annotation, Classifiers, Image analysis, Unlabeled data, Unlabeled samples, Imaging systems, Incremental learning, Classifier learning, Automatic image annotation, Confidence measure, Confidence Measure, Data selection, Image sets, Labeled training data, Spectral clustering

Notes:

EID: 2-s2.0-77951957024

Export Date: 20 September 2026; Cited By: 3; Correspondence Address: B. Byun; School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, United States; email: yorke3@ece.gatech.edu; Conference name: 2009 IEEE International Conference on Image Processing, ICIP 2009; Conference date: 7 November 2009 through 10 November 2009; Conference code: 79910

An industrial equipment fault diagnosis method based on deep belief rule base fusing indicator uncertainty

Item Type Journal Article

Author X. Liu

Author H. Huang

Author Y. Li

Author X. Xie

Author R. Dai

Abstract Precise fault diagnosis under complex working conditions is critical for ensuring the safety of industrial equipment. The belief rule base (BRB), characterized by its interpretability and capability to handle nonlinear, small-sample data, is inherently suitable for such diagnostic tasks. However, traditional BRB methods are impeded by the combinatorial rule explosion derived from high-dimensional features and the uncertainty introduced by environmental noise. To address structural complexity and environmental uncertainty simultaneously, this paper proposes a deep BRB fault diagnosis method integrated with indicator uncertainty deep BRB uncertainty. First, the extreme gradient boosting algorithm is employed to evaluate feature importance and select key features, constructing a hierarchical deep inference topology. Second, a quantification method based on the average distance of monitoring data is proposed to extract the uncertainty of key indicators, establishing an inference mechanism that fuses this uncertainty. Third, the projected covariance matrix adaptation evolution strategy is utilized to jointly optimize the full parameter set of the model, thereby enhancing generalization performance. Finally, experimental validation is conducted using bearing and gearbox datasets. Results indicate that the proposed method outperforms traditional approaches in terms of comprehensive diagnostic performance, achieving higher fault classification accuracy and stronger robustness, thus validating its superiority in handling interference and uncertain environments. © 2026 IOP Publishing Ltd. All rights, including for text and data mining, AI training, and similar technologies, are reserved. This article is available under the terms of the <https://publishingsupport.iopscience.iop.org/iop-standard/v1>.

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An innovative methodology for detection and quantification of cracks through incorporation of depth perception

Item Type Journal Article

Author M.R. Jahanshahi

Author S.F. Masri

Author C.W. Padgett

Author G.S. Sukhatme

Abstract Visual inspection of structures is a highly qualitative method in which inspectors visually assess a structure's condition. If a region is inaccessible, binoculars must be used to detect and characterize defects. Although several Non-Destructive Testing methods have been proposed for inspection purposes, they are nonadaptive and cannot quantify crack thickness reliably. In this paper, a contact-less remote-sensing crack detection and quantification methodology based on 3D scene reconstruction (computer vision), image processing, and pattern recognition concepts is introduced. The proposed approach utilizes depth perception to detect cracks and quantify their thickness, thereby giving a robotic inspection system the ability to analyze images captured from any distance and using any focal length or resolution. This unique adaptive feature is especially useful for incorporating mobile systems, such as unmanned aerial vehicles, into structural inspection methods since it would allow inaccessible regions to be properly inspected for cracks. Guidelines are presented for optimizing the acquisition and processing of images, thereby enhancing the quality and reliability of the damage detection approach and allowing the capture of even the slightest cracks (e.g., detection of 0.1 mm cracks from a distance of 20 m), which are routinely encountered in realistic field applications where the camera-object distance and image contrast are not controllable. © 2011 Springer-Verlag.

Date 2013

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Tags:

Computer vision, Pattern recognition, Damage detection, Remote sensing, Mobile systems, Depth perception, Image processing, Detection approach, Cracks, Three dimensional computer graphics, Mathematical morphology, Crack detection, Structural inspections, Visual inspection, Field application, Pattern classification, Morphological operations, 3D scene reconstruction, Adaptive features, Focal lengths, Image contrasts, Morphological operation, Nondestructive testing method, Qualitative method, Robotic inspection system, Thickness quantification, Three dimensional

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EID: 2-s2.0-84872284342

Export Date: 20 September 2026; Cited By: 280; Correspondence Address: M.R. Jahanshahi; Sonny Astani Department of Civil and Environmental Engineering, University of Southern California, Los Angeles, CA 90089, United States; email: jahansha@usc.edu; CODEN: MVAPE

An integrated hierarchical Gaussian mixture model to estimate vigilance level based on EEG recordings

Item Type Conference Paper

Author J.-N. Gu

Author H.-J. Liu

Author H.-T. Lu

Author B.-L. Lu

Abstract Effective vigilance level estimation can be used to prevent disastrous accident occurred frequently in high-risk tasks. Brain Computer Interface(BCI) based on ElectroEncephalo-Graph(EEG) is a relatively reliable and convenient mechanism to reflect one's psychological phenomena. In this paper we propose a new integrated approach to predict human vigilance level, which incorporate an automatically artifact removing pre-process, a novel vigilance quantification method and finally a hierarchical Gaussian Mixed Model(hGMM) to discover the underlying pattern of EEG signals. A reasonable high classification performance

(88.46% over 12 data sets) is obtained using this integrated approach. The hGMM is proved to be more powerful against Support Vector Machine(SVM) and Linear Discriminant Analysis(LDA) under complicated probability distributions. © 2011 Springer-Verlag.

Date 2011

DOI 10.1007/978-3-642-24955-6_46

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Language English

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Notes:

EID: 2-s2.0-81855226571

Export Date: 20 September 2026; Cited By: 5; Correspondence Address: J.-N. Gu; MOE-Microsoft Laboratory for Intelligent Computing and Intelligent Systems, Department of Computer Science and Engineering, Shanghai Jiao Tong University, Shanghai 200240, China; email: albertgu@sjtu.edu.cn; Conference name: 18th International Conference on Neural Information Processing, ICONIP 2011; Conference date: 13 November 2011 through 17 November 2011; Conference code: 87406

An Integrated, Explainable and Uncertainty-Aware Satellite-Based Precipitation Product for the Mediterranean Region

Item Type Conference Paper

Author P. Kossieris

Author I. Tsoukalas

Author L. Brocca

Author H. Mosaffa

Author C. Makropoulos

Author A. Anghelea

Abstract Precipitation is a key variable of earth observation systems, involved in a wide spectrum of applications, from the optimal water resources management to early warning systems for flood/drought events. During the last decade, significant research effort has been given in the development of satellite-based precipitation products (SPPs), with improved accuracy and high spatio-temporal coverage. This work presents a methodological framework that couples machine learning and explainability techniques, along with uncertainty quantification methods, to develop a new SPP for the Mediterranean region. Specifically, the methodology couples: a) ensemble (tree-based) machine learning algorithms to merge multiple satellite-based observations (i.e., both precipitation and soil moisture-based products), b) global and local explainable AI techniques to provide explanations at the model and instance level, respectively, and c) a copula-based approach to quantify the predictive uncertainty of the final integrated product. The new product fuses data from two widely known low-latency SPPs, i.e., the SM2RAIN-ASCAT and GPM Late Run dataset, while its performance is assessed against state-of-art products, i.e., the EMO5, GPM Final Run, ERA5 and PERSIANN CDR. © 2024 IEEE.

Date 2024

DOI 10.1109/IGARSS53475.2024.10641931

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Language English

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Proceedings Title Dig Int Geosci Remote Sens Symp (IGARSS)

Conference Name International Geoscience and Remote Sensing Symposium (IGARSS)

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Machine Learning, Explainable AI, Precipitation data merging, Precipitation products, predictive uncertainty

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An inter-observer Ki67 reproducibility study applying two different assessment methods: on behalf of the Danish Scientific Committee of Pathology, Danish breast cancer cooperative group (DBCG)

Item Type Journal Article

Author A.-V. Laenkholm

Author D. Grabau

Author M.-L. Møller Talman

Author E. Balslev

Author A.M. Bak Jylling

Author T.P. Tabor

Author M. Johansen

Author A. Brüggemann

Author G. Lelkaitis

Author T. Di Caterino

Author H. Mygind

Author T. Poulsen

Author H. Mertz

Author G. Søndergaard

Author B. Bruun Rasmussen

Abstract Introduction: In 2011, the St. Gallen Consensus Conference introduced the use of pathology to define the intrinsic breast cancer subtypes by application of immunohistochemical (IHC) surrogate markers ER, PR, HER2 and Ki67 with a specified Ki67 cutoff (>14%) for luminal B-like definition. Reports concerning impaired reproducibility of Ki67 estimation and threshold inconsistency led to the initiation of this quality assurance study (2013–2015). The aim of the study was to investigate inter-observer variation for Ki67 estimation in malignant breast tumors by two different quantification methods (assessment method and count method) including measure of agreement between methods. Material and methods: Fourteen experienced breast pathologists from 12 pathology departments evaluated 118 slides from a consecutive series of malignant breast tumors. The staining interpretation was performed according to both the Danish and Swedish guidelines. Reproducibility was quantified by intra-class correlation coefficient (ICC) and Lights Kappa with dichotomization of observations at the larger than (>) 20% threshold. The agreement between observations by the two quantification methods was evaluated by Bland–Altman plot. Results: For the fourteen raters the median ranged from 20% to 40% by the assessment method and from 22.5% to 36.5% by the count method. Light's Kappa was 0.664 for observation by the assessment method and 0.649 by the count method. The ICC was 0.82 (95% CI: 0.77–0.86) by the assessment method vs. 0.84 (95% CI: 0.80–0.87) by the count method. Conclusion: Although the study in general showed a moderate to good inter-observer agreement according to both ICC and Lights Kappa, still major discrepancies were identified in especially the mid-range of observations. Consequently, for now Ki67 estimation is not implemented in the DBCG treatment algorithm. © 2017 Acta Oncologica Foundation.

Date 2018

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Article, female, Female, human, Humans, reproducibility, Reproducibility of Results, procedures, practice guideline, standards, medical society, correlation coefficient, histopathology, human tissue, consensus development, pathology, biological marker, Biomarkers, metabolism, Breast Neoplasms, breast tumor, immunohistochemistry, Immunohistochemistry, Practice Guidelines as Topic, breast cancer, pathologist, staining, observer variation, cancer classification, Ki 67 antigen, Ki-67 Antigen, Staining and Labeling, Denmark, priority journal, Consensus Development Conferences as Topic, kappa statistics, Pathology, Clinical

Notes:

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Export Date: 20 September 2026; Cited By: 40; Correspondence Address: A.-V. Laenkholtm; Department of Surgical Pathology, Zealand University Hospital, Slagelse, Slagelse, Ingemannsvej 48, 4200, Denmark; email: anlac@regionsjaelland.dk; CODEN: ACTOE

An interpretable CNN model for NMR-based whole metabolomic profiling of sepsis

Item Type Journal Article

Author Z. Wang

Author X. Liu

Author Y. Shi

Author R. Wu

Author M. Jiang

Author B. Feng

Author C. Shen

Abstract NMR metabolomics faces critical sensitivity limitations when analyzing whole spectral profiles, particularly in clinical biomarker discovery where weak metabolite signals often elude detection. To address this challenge, we present NMR metabolomics convolutional neural network(NmCNN) as an interpretable 1D-

convolutional neural network framework that innovatively integrates spectral translation invariance with explainable AI. Our approach presents a novel concept distinct from traditional machine learning methods, and the incorporation of an interpreter exhibits a higher sensitivity in detecting differential metabolites. The model incorporates translation invariance from CNN, facilitating efficient processing of spectra and effectively addressing issues pertaining to deviations in spectral peak alignment. Validated on sepsis biomarker discovery for whole blood and blood platelet samples, NmCNN achieved 90–95 % sensitivity, 80 %-85 % specificity and area under curve 0.83–0.96. The interpretability framework with SHapley Additive exPlanations (SHAP) identified previously undetected metabolic features, demonstrating sensitivity improvement over conventional quantification methods through its whole-spectrum utilization. Overall, our NmCNN model offers a distinctive approach that enhances the utilization of raw NMR-based spectral metabolomics profiles in AI-driven clinical research, thereby improving interpretability and sensitivity in metabolomics analysis. © 2025

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: C. Shen; Senior Department of Burns and Plastic Surgery, Chinese PLA Hospital, Beijing, China; email: shenchuanan@301hospital.com.cn

An Uncertainty Quantitative Method for Cell Image Classification under Label Shift

Item Type Conference Paper

Author L. Zhao

Author J. Yang

Abstract With the development of data-driven deep classification models for cervical cancer diagnosis, results with uncertain cell data are not credible. Uncertainty data limits the application of the deep model in medical scenarios. Cervical cells have the label shift problem between source and target domain images and this uncertainty problem puts forward new requirements for the robustness of the classification model. In this paper, we focus on the label shift uncertainty problem for different cervical cell image classification datasets and propose a control method that performs robust classification under the label shift uncertainty situation. Experiments show that the proposed method cannot reduce the classification performance while improving the robustness of the cell classification model. © 2024 Copyright held by the owner/author(s).

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Conference Name ACM International Conference Proceeding Series

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Uncertainty, Image classification, Classification models, Uncertainty quantifications, Label shift, Data uncertainty, Uncertainty quantification, Images classification, Cancer cells, Cell image classification, Cell images, Cervical cells, Uncertainty problems

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An uncertainty-based decision support tool to evaluate the southern king crab (*Lithodes santolla*) fishery in a scarce information context

Item Type Journal Article**Author** F. Bozzeda**Author** S.L. Marín**Author** L. Nahuelhual

Abstract Decision-making in fishery management depends on the reliable quantification of uncertainty, which emerges from a variety of sources. For small-scale fisheries (SSF) in developing countries, the main sources of uncertainty are the available data, which are often untrustworthy, and poor institutional controls, which allow illegal fishing activities. Despite the global spread of SSF, the management of these socio-ecological resources is usually based on insufficient assessments estimated using scarce data. In this study, we identify and quantify the uncertainty associated with the harvested biomass of southern king crab (SKC) (*Lithodes santolla*), as declared by artisanal fishers of the Magallanes Region, Chile, in order to assess the harvestable biomass under a precautionary (conservative) approach to its exploitation in a scarce information context. This analysis includes the application of (i) fuzzy-Bayesian machine learning to quantify the uncertainty related to the harvested biomass per vessel per fishing cruise (C), and (ii) nonlinear polynomial regression models to project the spatial distribution of C according to the estimated uncertainty limits (minimum and maximum). The uncertainty analysis shows a 2000 kg threshold for C; beyond that, the declared harvest does not follow the estimated relationship between biomass and habitat characteristics (temperature, turbidity). The maps created from this analysis convey crucial information that would allow a more rational exploitation of the SKC stock and could be used to estimate a sustainable fishing quota, i.e., allocating effort according to the harvestable biomass. Our results suggest that the proposed tool can reliably inform management faced with uncertainty in stock exploitation by SSF, particularly when information is scarce. © 2018

Date 2019**DOI** 10.1016/j.pocean.2018.10.013**URL** <https://www.scopus.com/pages/publications/85056287298?origin=resultslist>**Archive** Scopus**Language** English**Volume** 174**Publisher** Elsevier Ltd**Pages** 64-71**Publication** Progress in Oceanography**ISSN** 00796611 (ISSN)**Journal Abbr** Prog. Oceanogr.**Date Added** 20/09/2026, 9:53:01 pm**Modified** 20/09/2026, 9:53:01 pm**Tags:**

Decision support systems, sustainability, Decision making, Uncertainty analysis, Uncertainty, Information management, decision making, decision support system, Harvesting, biomass, Biomass, uncertainty analysis, Fisheries, assessment method, Developing countries, Polynomial regression, Chaceon affinis, Chile, crab, crab fishery, Decision support tool, Decision support tools, developing world, fishery management, fishing, *Lithodes santolla*, Magallanes, Precautionary principle, Shellfish, Small-scale fisheries, Southern king crab, Southern king crabs, stock assessment

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An unsupervised multi-class ensemble classifier for identifying equipment failure mechanisms from maintenance records

Item Type Journal Article**Author** A.S. Bhardwaj**Author** D. Veeramani

Abstract Unstructured data in equipment maintenance records contain valuable information regarding failures. The ability to classify failure incidents into contributing failure mechanisms can help in improving equipment design and maintenance plans to achieve higher equipment uptime. Supervised learning approaches for automated extraction of failure mechanisms from unstructured data are impractical due to the manual labeling effort. Further, due to the complexities inherent in unstructured data, it can be beneficial to utilize multiple base classifier algorithms for analyzing the maintenance records from different perspectives. In this paper, we propose a novel unsupervised multi-class ensemble classifier (UMEC) model to automatically extract failure mechanisms from unstructured maintenance records by leveraging continuous scores generated by multiple base classifiers. The model decomposes the unsupervised multi-class classification problem into multiple binary classifiers using error-correcting output codes (ECOC). We encode the multi-class classification problems to multiple binary classes by using maximum as an order statistic to reduce multi-class scores to binary-classes. We also address the issue of unbalanced datasets in unsupervised classification. We study the influence of different types of noise structure (including feature noise and class mislabeling noise) over the classifiers and demonstrate the effectiveness of our approach using simulated and real-world industrial data. © 2025

Date 2025**DOI** 10.1016/j.ress.2025.111410**URL** <https://www.scopus.com/pages/publications/105010843670?origin=resultslist>**Archive** Scopus**Language** English**Volume** 264**Publisher** Elsevier Ltd**Publication** Reliability Engineering and System Safety**ISSN** 09518320 (ISSN)**Journal Abbr** Reliab Eng Syst Saf**Date Added** 20/09/2026, 9:51:51 pm**Modified** 20/09/2026, 9:51:51 pm

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Safety engineering, Codes (symbols), Failure (mechanical), Outages, Classification (of information), Maintenance, Classifiers, Label noise, Unsupervised learning, Unstructured data, Class prevalence, Ensemble-classifier, Error correcting output code, Error-correcting output codes, Failure mechanism, Feature noise, Maintenance records, Spectral decomposition, Unsupervised classification

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An unsupervised shapelet-based method for quantification of nanostructured surface imaging

Item Type Journal Article

Author C. Chin

Author M.P. Tino

Author N.M. Abukhdeir

Abstract Computational methods to quantify structure–property relationships of materials are lacking in comparison to the widespread availability of nanostructure image-based characterization techniques. There are several existing quantification methods to characterize self-assembled nanomaterials, however many techniques are infeasible for use in processing large-scale spatial variation and/or require significant manual image pre-processing. The use of shapelet functions to computationally quantify nanostructured materials is a promising approach that is generalized for arbitrary pattern types but the current state of the art, the response distance method, requires user/researcher supervision. This need for supervision, along with the computational complexity of the method, make it infeasible for use in fully-automated large-scale computational analysis of nanostructured surface images. The development of a fully-automated unsupervised method for quantification of nanostructured surfaces would enable the development of broadly applicable structure–property relationships for nanomaterials, leveraging large amounts of previously captured data (so-called meta-analyses). In this work, these issues with the shapelet-based response distance method are resolved through the development and validation of a multi-step machine learning process which enables unsupervised (fully-automated) analysis of nanostructure imaging. The presented method removes the need for any researcher knowledge regarding order symmetries and enables truly large-scale analysis (e.g., meta-analyses) of the vast amount of nanostructure imaging data currently available. The presented unsupervised shapelet-based response distance method also results in at least an order-of-magnitude reduction in computational complexity compared to the existing method, at the cost of some degree of generality. A software implementation of the presented unsupervised method is provided in the open-source shapelets Python package. © 2025 World Scientific Publishing Europe Ltd.

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Analysis classes for space exploration reliability assessments

Item Type Conference Paper

Author B.J. Franzini

Author B.F. Putney

Abstract Understanding and effectively communicating the spectrum of risk and reliability analysis classes is critical for an analyst to contribute to a team working on developmental space exploration architecture, mission, vehicle, or technology design. The proper characterization of class prior to the risk and reliability analysis can help calibrate stakeholder expectations with regard to required data inputs as well as anticipated outputs and insights. It can also help in the elicitation of data and cooperative participation from technical experts. This analysis class spectrum is bound on the bottom by low fidelity analyses conducted early in design definition cycles when designers and decision makers are looking for qualitative differences in design options to help discriminate, from a risk standpoint, and incorporate inherent safety into a design. At the other end of this spectrum, classic reliability assessment tools are applied to well defined or even operational hardware. These component level reliability analyses use Probabilistic Risk Assessment (PRA), Failure Mode and Effects Analysis (FMEA), and Hazard Analysis (HA) methodologies to quantify system level reliability values with relatively high confidence. For example, the risk and reliability analysis performed during the early definition of exploration architectures such as the Mars Architecture is significantly different from the analysis performed in support of an existing vehicle such as the Space Shuttle. Levels of design definition, requirement refinement, speed of design iterations,

and available system data determine where on this spectrum an analysis will fall as shown in Figure 1. The proper classification and corresponding analysis type is critical to successfully adding value to design iterations. While most R&M practitioners are familiar with reliability assessment techniques for existing hardware, analysis methodologies at the other end of the analysis spectrum are not as defined. This paper explores two examples representative of the range of analysis classes and associated insights proffered by each class in an attempt to establish a framework of analysis classes with associated inputs and anticipated outputs. Examinations of the risk and reliability assessments performed for NASA's Lunar Surface System Reliability Study and Mars Architecture Study provide a perspective of the unique insights each of these varied assessments can provide. © 2009 IEEE.

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Risk assessment, Assessment, Risk, Risk management, Reliability, Risk analysis, System levels, Spectrum analysis, Design, Hazard analysis, Reliability analysis, Failure mode and effects analysis, Manned space flight, Failure analysis, Occupational risks, Technology designs, Component levels, High confidence, Decision makers, Quality assurance, Reliability assessments, Probabilistic risk assessment, Analysis spectrum, Data input, Design definition, Design iteration, Design option, Exploration, Inherent safety, Low fidelity analysis, Lunar surface systems, Maintainability, Mars architectures, NASA, Qualitative differences, Random access storage, Space, Space exploration architecture, Space explorations, Space shuttles, Team working, Technical experts

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Analysis of Interacting Factors Contributing to the Onset of Diabetes Type 2 in the Female Pima Population: A Step towards Understanding the Links

Item Type Conference Paper

Author V.I. Dobromyslin

Author D.B. Megherbi

Abstract One of the biggest challenges in building predictive models relates to proper feature selection. As it is well known, good feature selection will result in accurate prediction independently of the classifier. We propose a novel processing pipeline consisting of unsupervised clustering to balance the class prevalence in the training set, which enables the development of stable models with less overfitting. In this paper, we use the proposed pipeline to predict the onset of Type 2 Diabetes (T2D) in the female Pima American Indian population. The Pima population, living near Phoenix, Arizona in the United States, has been known to have the highest T2D prevalence. While it generally remains unclear how various physiological variables and interacting risk factors contribute to the onset of T2D, here for the first time, we quantify the contribution of each of the physiological variables and their interactions to the predicted probability of developing T2D. Our findings in the adult Pima female T2D cohort point to the existence of two distinct population sub-groups, separated primarily by age and the number of pregnancies. For the first time in this dataset, we show several factor interactions: a) subjects with lower BMI (26.0 ± 2.2 vs. 41.5 ± 4.9) experienced a 57-78% reduction in T2D conversion rate at the high plasma glucose (>133.7 mg/dL) and high serum insulin (>168.0 pmol/L) levels respectively, and b) low serum insulin (70.1 ± 16.7 vs. 294.7 ± 136.6 pmol/L) was associated with a 90% lower risk of T2D conversion rate in individuals with high triceps skin thickness (>34 mm), and interestingly c) non-hypotensive (85.6 ± 7.2 vs. 54.3 ± 7.7 mm Hg) women with ≤ 1 pregnancy had 81% lower risk of T2D conversion. With an increasing number of pregnancies, the trend reversed, and non-hypotensive women experienced a 41% higher rate of T2D conversion. Through the evaluation of multiple feature selection methods and classifiers, we show the potential value of the proposed methodology for feature selection and analysis of interacting factors contributing to the onset of T2D in the female Pima Population. © 2020 IEEE.

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Feature extraction, Intelligent computing, Factor analysis, Predictive analytics, Pipelines, Predictive models, Physiology, unsupervised machine learning, Obstetrics, clustering, Bioinformatics, Accurate prediction, Insulin, American Indians, cardiovascular predictors for post-pregnancy chronic disease development in the Pima population, Conversion rates, factor interaction analysis, Multiple features, Novel processing, Pima American Indians diabetes population, Potential values, Type 2 Diabetes, Unsupervised clustering

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Analysis of learning from positive and unlabeled data

Item Type Conference Paper

Author M.C. Du Plessis

Author G. Niu

Author M. Sugiyama

Editor Ghahramani Z.

Editor Welling M.

Editor Lawrence N.D.

Editor Cortes C.

Editor Weinberger K.Q.

Abstract Learning a classifier from positive and unlabeled data is an important class of classification problems that are conceivable in many practical applications. In this paper, we first show that this problem can be solved by cost-sensitive learning between positive and unlabeled data. We then show that convex surrogate loss functions such as the hinge loss may lead to a wrong classification boundary due to an intrinsic bias, but the problem can be avoided by using non-convex loss functions such as the ramp loss. We next analyze the excess risk when the class prior is estimated from data, and show that the classification accuracy is not sensitive to class prior estimation if the unlabeled data is dominated by the positive data (this is naturally satisfied in inlier-based outlier detection because inliers are dominant in the unlabeled dataset). Finally, we provide generalization error bounds and show that, for an equal number of labeled and unlabeled samples, the generalization error of learning only from positive and unlabeled samples is no worse than $2\sqrt{2}$ times the fully supervised case. These theoretical findings are also validated through experiments.

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Risk perception, Classification (of information), Information science, Error analysis, Classification accuracy, Unlabeled samples, Outlier Detection, Cost-sensitive learning, Generalization Error, Generalization error bounds, Classification boundary, Non-convex loss function

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Analysis of tetraterpenes and tetraterpenoids (carotenoids)

Item Type Journal Article

Author M. Da Silveira Vasconcelos

Author L.M.N. De Oliveira

Author D. Nunes-Pinheiro

Author F.R. Da Silva Mendes

Author F.D. De Sousa

Author L. De Siqueira Oliveira

Author A.C. De Aquino

Author T. De Fátima Goebel De Souza

Author A.S. Silva

Author S.M. Nabavi

Author D.F. De Melo

Abstract The chemistry of natural products has been achieved through the improvement of physical and chemical techniques for the isolation and purification of secondary metabolism products such as carotenoids. Carotenoids, also called tetraterpenoids, produce the particular pigmentation found in nature such as red, yellow, and orange colors. These pigments have gained attention due their physiological and ecological roles. Several challenges in carotenoid analysis have promoted news techniques of extraction and purification considering chemical and physical properties of these

compounds. In this chapter, the phytochemistry, classification, and biosynthesis of carotenoids were addressed, including their recent analytical advances. Throughout this chapter, their chemistry is also presented in detail and intrinsically related to the several health benefits and different industrial applications. Thus, the need to further develop the identification and quantification methods of such molecules as well as novel and green analytical techniques was emphasized. © 2020 Elsevier Inc. All rights reserved.

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Analysis of Vegetables

Item Type Journal Article

Author W. He

Author B. Xu

Abstract Vegetables serve an important role in human nutrition. The flavor quality of vegetables determines consumers' preferences, consumer acceptance, and repeat purchases. Aroma compounds (volatile compounds) and flavor compounds (non-volatile) both are essential to create the sensory perception of vegetables. The aroma and flavor compounds will be influenced by the species of vegetables, the biosynthesis pathways in vegetables, and the combined response of olfaction and taste perceptions. Therefore, this chapter presents a description of the factors that influence the flavor profile of vegetables, including the classification of vegetables, primary and secondary metabolites of vegetables, and primary and secondary flavor compounds of vegetables, the key flavor compounds, and their biosynthesis pathways in vegetables. In addition, the identification and quantification methods of both

volatile and non-volatile compounds in vegetables using both instrumental analysis and sensory evaluation methods will be discussed. © 2024 selection and editorial matter, Leo M. L. Nollet and Matteo Bordiga; individual chapters, the contributors.

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Analytical model for performance-evaluation of multistage interconnection networks supporting multi-class priority service

Item Type Journal Article

Author J. Garofalakis

Author E. Stergiou

Abstract In this paper, a discrete-time, single server with parallel multi-buffers, switching element (SE) is presented and analysed. This SE is used in the construction of multistage interconnection networks which have the ability to service the multi-priority traffic-supporting good quality of service. Through a queuing analysis and under various values of multi-traffic classifications, a performance evaluation is determined. This analytical model is based on the steady-state description of the system behaviour. An utilisation's formula is extracted by considering the steady state of the discrete-time multistage network. Consequently, we estimated the rest of the performance metrics. The results are validated using simulation. Apart from that our multi-priority architecture, validated in marginal cases, is compared to some existing single and two class priority models of multistage networks. This comparison is made by gathering metrics and comparing the total packet throughput. Also, the gain of the high-priority packets against the lower-priority packets under different configuration schemas is demonstrated. Eventually, this proposed analytical model is very accurate for various network sizes, with various values of offered multi-priority traffic on fabric's inputs and, therefore, can be used in performance prediction of multistage networks which supports quality of service. © 2011 Taylor & Francis.

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Analyzing infant cries using a committee of neural networks in order to detect hypoxia related disorder

Item Type Journal Article

Author M. Poel

Author T. Ekkel

Abstract Based on the hypothesis that the sound of the infant cry contains information on the infant's health status, research has been done on how to improve classification of neonate crying sounds into categories called 'normal' and 'abnormal' - the latter referring to some hypoxia-related disorder. Research in this field is hindered by lack of test cases and limited understanding of feature relevance. The research described here combines various ways of dealing with the small data set problem. First, feature pre-selection is done using sequential backwards elimination of possible combinations where the performance of the set of features is tested by a Probabilistic Neural Network which has the advantage of fast learning. Using these features a neural network committee, consisting of Radial Basis Function Neural Networks, was trained on the data, using bootstrapping. This construction yields a multi-classifier system with an overall classification performance of 85% on the so-called "All Cry Units" (ACU) data set, an increase of 34% with respect to the a priori probability of 51%. Several leave-1-out experiments for Linear Discriminant Analysis (LDA), Support Vector Machines (SVM) and Neural Networks (NN) have been conducted in

order to compare the performance of the multi-classifier system. © World Scientific Publishing Company.

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Analyzing Uncertainty Quantification in Statistical and Deep Learning Models for Probabilistic Electricity Price Forecasting

Item Type Journal Article

Author A. Lebedev

Author A. Das

Author S. Pappert

Author S. Schlüter

Abstract Precise probabilistic forecasts are fundamental for energy risk management, and there is a wide range of both statistical and machine learning models for this purpose. Inherent to these probabilistic models is some form of uncertainty quantification. However, most models do not capture the full extent of uncertainty, which arises not only from the data itself but also from model and distributional choices. In this study, we examine uncertainty quantification in state-of-the-art statistical and deep learning probabilistic forecasting models for electricity price forecasting in the German market. In particular, we consider distributional deep neural networks (DDNNs) and augment them with an ensemble approach, Monte Carlo (MC) dropout, deep evidential regression (DER), and conformal prediction to account for model uncertainty. Additionally, we consider the extreme gradient boosting (XGBoost) and

the LASSO-estimated autoregressive (LEAR) model, each combined with quantile regression averaging (QRA), generalized autoregressive conditional heteroskedasticity (GARCH), and conformal prediction. Across a range of performance metrics, we find that the XGBoost-based models perform well in terms of probabilistic forecasting, irrespective of the uncertainty quantification method. Furthermore, we find that DDNNs benefit from incorporating both data and model uncertainty, improving accuracy of both point and probabilistic forecasts by 7.4% and 7.7% in terms of MAE and CRPS. Uncertainty itself appears to be best captured by the models using conformal prediction. Overall, our extensive study shows that all models under consideration perform competitively. However, their relative performance depends on the choice of metrics for point and probabilistic forecasting. © 2013 IEEE.

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Export Date: 20 September 2026; Cited By: 3; Funding details: Deutsche Forschungsgemeinschaft, DFG; Correspondence Address: A. Das; Faculty of Mathematics and Economics, Ulm University, Ulm, 89081, Germany; email: abhinav.das@uni-ulm.de

Anchoring Public Health Forecasts: High-Precision Obesity Prevalence
Estimation Via Hybrid Geodemographic Ensemble Regression

Item Type Journal Article**Author** T. Ward**Author** I. Osunmakinde

Abstract Obesity remains a major driver of chronic disease and rising healthcare costs in the United States. Yet, most existing predictive models rely on single-model classifiers and rarely incorporate spatial structure or ensemble-based regression, thereby limiting their ability to capture geodemographic variation. This study introduces a hybrid unsupervised–supervised ensemble framework that integrates K-Means spatial profiling with a Stacking Regressor to address these limitations. A geodemographic cluster feature is engineered from latitude, longitude, and population density to capture latent regional health patterns. The supervised ensemble, comprising HistGradientBoosting, Random Forest, and Multi-Layer Perceptron base learners with a RidgeCV meta learner, is trained on CDC BRFSS data cleaned to 19,078 detailed records to forecast obesity prevalence as a continuous regression target. The final ensemble achieves an R^2 of 0.8045, an accuracy of 91.95%, a correlation (r) of 0.897, and a Mean Absolute Error of 2.20%, outperforming all individual models and related literature. To evaluate robustness, the framework is stress-tested on secondary datasets, where it consistently maintains superior accuracy. These findings demonstrate the value of hybrid spatial ensemble modeling for high-precision public health forecasting. © 2026 Trenton Ward and Isaac Osunmakinde. This open-access article is distributed under a Creative Commons Attribution (CC-BY) 4.0 license.

Date 2026**DOI** 10.3844/jcssp.2026.2139.2155**URL** <https://www.scopus.com/pages/publications/105048659845?origin=resultslist>**Archive** Scopus**Language** English**Volume** 22**Publisher** Science Publications**Pages** 2139-2155**Publication** Journal of Computer Science**ISSN** 15493636 (ISSN)**Issue** 7**Journal Abbr** J. Comput. Sci.**Date Added** 20/09/2026, 9:51:44 pm**Modified** 20/09/2026, 9:51:44 pm**Tags:**

Ensemble Learning, Public Health, Geodemographic Analysis, K-Means Clustering, Obesity Prediction, Stacking Regressor

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: I. Osunmakinde; Department of Computer Science, Norfolk State University, Norfolk, 23504, United States; email: ioosunmakinde@nsu.edu

Antibiotic monitoring in dogs, cats, and horses – how can practices be made comparable?

Item Type Journal Article

Author R. Merle

Author L. Feuer

Author K. Frenzer

Author J.-L. Plenio

Author A. Bethe

Author A. Lübke-Becker

Author W. Bäumer

Abstract The use of antibiotics in companion animals is a key One Health concern, as antimicrobial-resistant bacteria can be transmitted between animals and humans, limiting treatment options in both veterinary and human medicine. This study aimed to assess whether routine data from veterinary practices are suitable for establishing an antibiotic monitoring system for dogs, cats, and horses. We analyzed over 3.7 million consultation records from 106 practices (2018–2023) and compared three quantification methods: (a) the proportion of consultations involving antibiotic treatment, (b) the number of Used Daily Doses (nUDD), and (c) the number of Defined Daily Doses (nDDD). Cats had the highest median proportion of antibiotic-treated consultations, while horses had the lowest. Overall, practice classifications into low, medium, or high antibiotic use were largely consistent across the three indicators. The nDDD could be calculated for most cases but required numerous assumptions regarding dosage and standard animal weights. The nUDD relied on standard treatment durations due to the lack of documented therapy lengths. In contrast, the proportion of antibiotic-treated consultations was fully calculable from routine data and closely aligned with the other indicators. Given its simplicity, transparency, and full data availability, this indicator appears most suitable – at least during the implementation phase of a monitoring system. However, for comprehensive surveillance and antimicrobial stewardship, future systems should also integrate diagnostic information and antimicrobial classes to support prudent use and resistance prevention more effectively. © 2025 Schlütersche Fachmedien GmbH, ein Unternehmen der Schlüterschen Mediengruppe

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Antibody-supervised deep learning for quantification of tumor-infiltrating immune cells in hematoxylin and eosin stained breast cancer samples

Item Type Journal Article

Author R. Turkki

Author N. Linder

Author P.E. Kovanen

Author T. Pellinen

Author J. Lundin

Abstract Background: Immune cell infiltration in tumor is an emerging prognostic biomarker in breast cancer. The gold standard for quantification of immune cells in tissue sections is visual assessment through a microscope, which is subjective and semi-quantitative. In this study, we propose and evaluate an approach based on antibody-guided annotation and deep learning to quantify immune cell-rich areas in hematoxylin and eosin (H&E) stained samples. Methods: Consecutive sections of formalin-fixed paraffin-embedded samples obtained from the primary tumor of twenty breast cancer patients were cut and stained with H&E and the pan-leukocyte CD45 antibody. The stained slides were digitally scanned, and a training set of immune cell-rich and cell-poor tissue regions was annotated in H&E whole-slide images using the CD45-expression as a guide. In analysis, the images were divided into small homogenous regions, superpixels, from which features were extracted using a pretrained convolutional neural network (CNN) and classified with a support of vector machine. The CNN approach was compared to texture-based classification and to visual assessments performed by two pathologists. Results: In a set of 123,442 labeled superpixels, the CNN approach achieved an F-score of 0.94 (range: 0.92-0.94) in discrimination of immune cell-rich and cell-poor regions, as compared to an F-score of 0.88 (range: 0.87-0.89) obtained with the texture-based classification. When compared to visual assessment of 200 images, an agreement of 90% ($k = 0.79$) to quantify immune infiltration with the CNN approach was achieved while the inter-observer agreement between pathologists was 90% ($k = 0.78$). Conclusions: Our findings indicate that deep learning can be applied to quantify immune cell infiltration in breast cancer samples using a basic morphology staining only. A good discrimination of immune cell-rich areas was achieved, well in concordance with both leukocyte antigen expression and pathologists' visual assessment. © 2016 Journal of Pathology Informatics | Published by Wolters Kluwer -Medknow.

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Anticipative Bayesian classification for data streams with verification latency

Item Type Journal Article

Author V. Hofer

Author G. Kreml

Author D. Lang

Abstract Most of the existing adaptive classification algorithms in non-stationary data streams require recent labelled data for their updates. Such recent labels are often missing. For stream classification under verification latency only few approaches exist. Most of them assume clustered data or homogeneous drift in all features, which limits their applicability. We address this by proposing Anticipative Bayesian stream Classifier (ABClass), an approach that is capable of integrating and automatically selecting from different components. In its Bayesian classification framework, ABClass combines density estimation techniques, extended to extrapolate drift patterns over time, with unsupervised parameter tuning and unsupervised model selection. ABClass allows for multivariate density estimation and extrapolation techniques. In this work, we assume conditional independence between features given the class label for modelling feature-specific drift patterns. ABClass is generative and can also be used for explaining and visualising concept drift patterns. It is generic, making it easy to include further types of drift models, both for the class-conditional feature distribution and for the class prior distribution. The experimental evaluation on several real-world data streams shows its competitiveness compared to other state-of-the-art approaches. ABClass is in most cases ten- to hundred-times faster than its competitors, both for model fitting and for prediction. © 2024 The Author(s).

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Apple Watercore Grade Classification Method Based on ConvNeXt and Visible/Near-Infrared Spectroscopy

Item Type Journal Article

Author C. Zhao

Author Z. Yin

Author Y. Tan

Author W. Zhang

Author P. Guo

Author Y. Ma

Author H. Wu

Author D. Hu

Author Q. Lu

Abstract To address the issues of insufficient rigor in existing methods for quantifying apple watercore severity and the complexity and low accuracy of traditional classification models, this study proposes a method for watercore quantification and a classification model based on a deep convolutional neural network. Initially, visible/near-infrared transmission spectral data of apple samples were collected. The apples were then sliced into 4.5 mm thick sections using a specialized tool, and image data of each slice were captured. Using BiSeNet and RIFE algorithms, a three-dimensional model of the watercore regions was constructed from the apple slices to calculate the watercore severity, which was subsequently categorized into five distinct levels. Next, methods such as the Gramian Angular Summation Field (GASF), Gram Angular Difference Field (GADF), and Markov Transition Field (MTF) were applied to transform the one-dimensional spectral data into two-dimensional images. These images served as input for training and prediction using the ConvNeXt deep convolutional neural network. The results indicated that the GADF method yielded the best performance, achieving a test set accuracy of 98.73%. Furthermore, the study contrasted the classification and prediction of watercore apples using traditional methods with the existing quantification approaches for watercore levels. The comparative results demonstrated that the proposed GADF-ConvNeXt model is more straightforward and efficient, achieving superior performance in classifying watercore grades. Furthermore, the newly proposed quantification method for watercore levels proved to be more effective. © 2025 by the authors.

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Application and comparison of geomorphological and hydrological pool and riffle quantification methods

Item Type Journal Article

Author A.M. Krueger

Author K.M. Frothingham

Abstract Pools and riffles have long been recognized as fundamental geomorphological elements of meandering streams. As such, the quantification of pool and riffle characteristics and development of techniques to identify these bed forms has been an important research field for decades. The definition and identification of pools and riffles, however, can vary amongst disciplines. This research compares methods used in two different disciplines, geomorphology and hydrology, and determines the level of agreement between them. Initially, we identified pools and riffles using the bed differencing technique (bdt) in fifteen reaches of Ransom Creek, Clarence, New York. We then calculated the hydrological index, using the Froude Number (Fr) method, for each bed form identified by the bdt, and investigated the agreement between the two indices. Eleven bed forms were identified using the bdt and eight showed agreement geomorphologically (bdt) and hydrologically (Fr). In the three cases where the classifications did not agree, the bdt appeared to be a more appropriate technique to use to identify pools and riffles. © 2007 by Gamma Theta Upsilon.

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Application of Artificial Intelligence in Hydrological Modeling for Streamflow Prediction in Ungauged Watersheds: A Review

Item Type Journal Article

Author J.G. Gacu

Author C.E.F. Monjardin

Author R.G.T. Mangulabnan

Author J.C.F. Mendez

Abstract Streamflow prediction in ungauged watersheds remains a critical challenge in hydrological science due to the absence of in situ measurements, particularly in remote, data-scarce, and developing regions. This review synthesizes recent advancements in artificial intelligence (AI) for streamflow modeling, focusing on machine learning (ML), deep learning (DL), and hybrid modeling frameworks. Three core methodological domains are examined: regionalization techniques that transfer models from gauged to ungauged basins using physiographic similarity and transfer learning; synthetic data generation through proxy variables such as NDVI, soil moisture, and digital elevation models; and model performance evaluation using both deterministic and probabilistic metrics. Findings from recent literature consistently demonstrate that AI-based models, especially Long Short-Term Memory (LSTM) networks and hybrid attention-based architectures, outperform traditional conceptual and physically based models in capturing nonlinear hydrological responses across diverse climatic and physiographic settings. The integration of AI with remote sensing enhances generalizability, particularly in ungauged and human-impacted basins. This review also addresses several persistent research gaps, including inconsistencies in model evaluation protocols, limited transferability across heterogeneous regions, a lack of reproducibility and open-source tools, and insufficient integration of physical hydrological knowledge into AI models. To bridge these gaps, future research should prioritize the development of physics-informed AI frameworks, standardized benchmarking datasets, uncertainty quantification methods, and interpretable modeling tools to support robust, scalable, and operational streamflow forecasting in ungauged watersheds. © 2025 by the authors.

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Application of chemometric methods for classification of atmospheric particles based on thin-window electron probe microanalysis data

Item Type Conference Paper

Author J. Osán

Author J. De Hoog

Author A. Worobiec

Author C.-U. Ro

Author K.-Y. Oh

Author I. Szalóki

Author R. Van Grieken

Abstract Conventional single-particle electron probe microanalysis (EPMA) is widely used for evaluating the sources of atmospheric aerosol. The method is capable of simultaneously detecting the chemical composition and the morphology of each particle. Computer-controlled automatic EPMA allows the analysis of huge numbers of individual particles. Cluster as well as factor analysis are used for the classification of particles based on the obtained data set. However, the method is not able to detect low-Z elements (C, N, O), therefore, e.g. organic particles can only be identified by their typical inorganic content and high background. Using a thin-window X-ray detector, the capabilities of EPMA can be extended to determine low-Z elements. The recently developed quantification method based

on Monte Carlo simulations is capable to evaluate elemental concentrations in single microscopic particles, including C, N and O. It was shown that also chemical species can be determined from the obtained concentrations. Hierarchical and non-hierarchical cluster analysis, as well as principal component analysis were applied for the classification of particles based on low-Z EPMA data. A mixture of standard particles as well as atmospheric aerosol samples were used to test the classification methods. Different input data (X-ray intensities or elemental concentrations) and scaling functions were used for the chemometric methods. Cluster and factor analysis appear to be efficient tools for classification of particles based on low-Z EPMA data. As an example, atmospheric ammonium sulphate and organic sulphur were classified in separate groups, which was not possible by conventional EPMA. © 2001 Elsevier Science B.V. All rights reserved.

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Application of Machine Learning and Multi-Dimensional Perception in Urban Spatial Quality Evaluation: A Case Study of Shanghai Underground Pedestrian Street

Item Type Journal Article

Author T. Yao

Author Y. Xu

Author L. Sun

Author P. Liao

Author J. Wang

Abstract The exploitation of urban subsurface space in urban inventory planning is closely connected to the quality of urban environments. Currently, the construction of underground pedestrian streets is characterised by inefficiency and traffic congestion, making them insufficient for fulfilling the demand for well-designed and human-centred spaces. In the study of spatial quality, traditional evaluation methods, such as satellite remote sensing and street maps, often suffer from low accuracy and slow updating rates, and they frequently overlook human perceptual evaluations. Consequently, there is a pressing need to develop a set of spatial quality evaluation methods incorporating pedestrian perspectives, thereby addressing the neglect of subjective human experiences in spatial quality research. This study first quantifies and clusters the characteristics of underground pedestrian spaces using spatial syntax. It then gathers multidimensional perception data from selected locations and ultimately analyses and predicts the results employing machine learning techniques, specifically Random Forest and XGBoost. The research results indicate variability in pedestrians' evaluations of spatial quality across different functionally oriented spaces. Key factors influencing these evaluations include Gorgeous, Warm, Good Ventilation, and Flavour indicators. The study proposes a comprehensive and applicable spatial quality evaluation model integrating spatial quantification methods, machine learning algorithms, and multidimensional perception measurements. The development of this model offers valuable scientific guidance for the planning and construction of high-quality urban public spaces. © 2024 by the authors.

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Application of Machine Learning to Quantify the Role of the Dryline in the Severe Thunderstorm Climatology of the U.S. Southern Great Plains

Item Type Journal Article

Author M.J. Hosek

Author K.A. Hoogewind

Author A.J. Clark

Author A.D. Justin

Abstract The dryline is well understood to be an important mechanism for convection initiation in the U.S. Southern Great Plains (SGP) during its peak severe weather season (March–June). However, the role of the dryline in the severe climatology of the SGP has not yet been quantified. In this study, a machine learning front-identification model (FrontFinder AI) is used to construct a climatology of March–June 0000 UTC drylines over 26 years (1995–2020). Severe storm reports between 2100 and 0300 UTC within the expected region of dryline convection initiation are attributed to the dryline to determine the fraction of the SGP severe climatology associated with drylines. From southwest Texas through western Oklahoma, over 50% of March–June severe storm days are associated with a dryline. During May__the month with the most severe reports__drylines are associated with over 75% of severe thunderstorm days in the Texas Panhandle and northwest Oklahoma. Across the entire SGP, drylines are associated with approximately 41% of severe storm reports and 48% of significant severe storm reports. © 2026 American Meteorological Society.

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Application of relative quantification TaqMan real-time polymerase chain reaction technology for the identification and quantification of *Thunnus alalunga* and *Thunnus albacares*

Item Type Journal Article

Author I. Lopez

Author M.A. Pardo

Abstract A novel one-step methodology based on real-time Polymerase Chain Reaction (PCR) technology has been developed for the identification of two of the most valuable tuna species. Nowadays, species identification of seafood products has a major concern due to the importing to Europe of new species from other countries. To achieve this aim, two specific TaqMan systems were devised to identify *Thunnus alalunga* and *Thunnus albacares*. Another system specific to Scombroidei species was devised as a consensus system. In addition, a relative quantification methodology was carried out to quantify *T. alalunga* and *T. albacares* in mixtures after the relative amount of the target was compared with the consensus. This relative quantification methodology does not require a known amount of standard, allowing the analysis of many more samples together and saving costs and time. The utilization of real-time PCR does not require sample handling, preventing contamination and resulting in much faster and higher throughput results. © 2005 American Chemical Society.

Date 2005

DOI 10.1021/jf0500841

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article, comparative study, classification, methodology, genetics, polymerase chain reaction, sequence alignment, chemistry, sensitivity and specificity, animal, Animals, Polymerase Chain Reaction, DNA, nucleotide sequence, Sensitivity and Specificity, Species identification, Base Sequence, molecular genetics, Molecular Sequence Data, Sequence Alignment, Real-time PCR, Canned tuna, Scombridae gen. sp., Scombroides, Thunnus alalunga, Thunnus albacares, tuna, Tuna

Notes:

EID: 2-s2.0-20744434408

Export Date: 20 September 2026; Cited By: 91; Correspondence Address: M.A. Pardo; Department of Food Technology, AZTI-Fisheries and Food Technological Institute, E-48395 Sukarrieta (Bizkaia), Txatxarramendi ugarte a z/g, Spain; email: mpardo@suk.azti.es; CODEN: JAFCA

Application of supervised learning to quantify uncertainties in turbulence and combustion modeling

Item Type Conference Paper

Author B. Tracey

Author K. Duraisamy

Author J.J. Alonso

Abstract The accuracy of low-fidelity models of turbulent flow such as those based on the Reynolds Averaged Navier-Stokes (RANS) equations can be questionable, especially when these models are applied in situations different from those in which the models were calibrated. At present, there is no general method to quantify structural uncertainties in such models. Greater accuracy and a reliable quantification of modeling errors is much needed to expand the use of affordable simulation models in engineering design and analysis. In this paper, we introduce a methodology aimed at improving low-fidelity models of turbulence and combustion and obtaining error bounds. Towards this end, we first develop a new machine learning algorithm to construct a stochastic model of the error of low-fidelity models using information from higher-fidelity data sets. Then, by applying this error model to the low-fidelity result, we obtain better approximations of uncertain model outputs and generate confidence intervals on

the prediction of simulation outputs. We apply this technique to two representative flow problems. The first application is in flamelet-based simulations to model combustion in a turbulent mixing layer; and the second application is in the prediction of the anisotropy of turbulence in a non-equilibrium boundary layer flow. We demonstrate that our methodology can be used to improve aspects of predictive modeling while offering a route towards obtaining error bounds. © 2013 by Brendan Tracey.

Date 2013

DOI 10.2514/6.2013-259

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Archive Scopus

Language English

Extra Journal Abbreviation: 51st AIAA Aerosp. Sci. Meet. New Horiz. Forum Aerosp. Exp. 2013

Publisher American Institute of Aeronautics and Astronautics Inc.

ISBN 978-162410181-6 (ISBN)

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Conference Name 51st AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition 2013

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Machine learning, Uncertainty analysis, Learning algorithms, Stochastic systems, Predictive modeling, Error analysis, Confidence interval, Stochastic models, Boundary layer flow, Atmospheric thermodynamics, Turbulent flow, Boundary layers, Engineering design, Aerospace engineering, Combustion, Low-fidelity models, Navier Stokes equations, Nonequilibrium boundaries, Reynolds averaged Navier Stokes (RANS) equations, Structural uncertainty, Turbulence, Turbulent mixing layers

Notes:

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Application of three different methods to determine the prevalence, the abundance and the environmental drivers of culturable *Vibrio cholerae* in fresh and brackish bathing waters

Item Type Journal Article

Author A.K.T. Kirschner

Author S. Pleininger

Author S. Jakwerth
Author S. Rehak
Author A.H. Farnleitner
Author S. Huhulescu
Author A. Indra

Abstract Aims: Three cultivation methods were used to study the prevalence and abundance of *Vibrio cholerae* in Eastern Austrian bathing waters and to elucidate the main factors controlling their distribution. Methods and Results: *Vibrio cholerae* abundance was monitored at 36 inland bathing sites with membrane filtration (MF), a standard most probable number (MPN) approach and direct plating (DP). Membrane filtration yielded the most reliable and sensitive results and allowed *V. cholerae* detection at 22 sites with concentrations up to 39 000 CFU per 100 ml, all belonging to serogroups other than O1 and O139 and not coding for cholera toxin and toxin coregulated pilus. Direct plating turned out as an easy method for environments with high *V. cholerae* abundances, conductivity was the only significant predictor of *V. cholerae* abundance in the bathing waters at warm water temperatures. Conclusions: *Vibrio cholerae* nonO1/nonO139 are widely prevalent in Eastern Austrian bathing waters. Instead of the standard MPN approach, MF and DP are recommended for *V. cholerae* monitoring. Conductivity can be used as a first easy-to-measure parameter to identify potential bathing waters at risk. Significance and Impact of the Study: *Vibrio cholerae* nonO1/nonO139 infections associated with bathing activities are an increasing public health issue in many countries of the northern hemisphere. However, there are only limited data available on the prevalence and abundance of *V. cholerae* in coastal and inland bathing waters. For monitoring *V. cholerae* prevalence and abundance, reliable and simple quantification methods are needed. Moreover, prediction of *V. cholerae* abundance from environmental parameters would be a helpful tool for risk assessment. This study identified the best culture-based quantification methods and a first quick surrogate parameter to attain these aims. © 2018 The Authors. Journal of Applied Microbiology published by John Wiley & Sons Ltd on behalf of The Society for Applied Microbiology.

Date 2018

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Risk assessment, Article, environmental factor, risk assessment, comparative study, nonhuman, procedures, prevalence, classification, methodology, Filtration, filtration, bacterium, environmental monitoring, genetics, isolation and purification, Quantification methods, microbiology, Prevalence, growth, development and aging, electric conductivity, membrane, infection, devices, Water Microbiology, Bacteriological Techniques, microbiological examination, fresh water, population abundance, Austria, abundance, bacterial count, bacterium detection, bath, Bathing water, Baths, brackish water, concentration (composition), conductivity, Conductivity, cultivation, Culturable, direct plating, Direct plating, Fresh Water, hot water, Infection, infectivity, membrane filtration, Membrane filtrations, Microfiltration, most probable number, Most probable number, Northern Hemisphere, parasite prevalence, Plating, serotype, species cultivation, Vibrio cholera nono1/nono139, Vibrio cholerae, Vibrio cholerae nonO1/nonO139, water temperature, Water temperature

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Application of two-step quantitative reverse-transcription PCR to bacterial diagnostics.

Item Type Journal Article

Author B. Glynn

Abstract Use of high copy number bacterial RNA offers several advantages in a diagnostics context compared with current deoxyribonucleic acid-based assays. The opportunity to only detect viable cells by targeting labile RNA transcripts may create an opportunity for "real-time" monitoring of pathogen load in response to a treatment regimen, while the natural amplification provided by the relative abundance of the RNA target compared with its corresponding gene opens a door to potential nonamplified direct detection technologies. In this chapter, a method is described to accurately quantify specific RNA transcripts and thus determine their potential utility as "high-copy" targets. The quantification method described also has application in gene-expression analysis.

Date 2006

DOI 10.1385/1-59745-143-6:97

URL <https://www.scopus.com/pages/publications/33750602703?origin=resultslist>

Archive Scopus

Language English

Volume 345

Pages 97-109**Publication** Methods in molecular biology (Clifton, N.J.)**ISSN** 10643745 (ISSN)**Journal Abbr** Methods Mol. Biol.**Date Added** 20/09/2026, 9:53:05 pm**Modified** 20/09/2026, 9:53:05 pm**Tags:**

article, classification, methodology, evaluation, Bacteria, bacterium, gene dosage, genetics, metabolism, reverse transcription polymerase chain reaction, bacterial RNA, Reverse Transcriptase Polymerase Chain Reaction, complementary RNA, Gene Dosage, RNA, Bacterial, RNA, Complementary

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EID: 2-s2.0-33750602703

Export Date: 20 September 2026; Cited By: 3

Applications of Machine Learning to in Silico Quantification of Chemicals without Analytical Standards

Item Type Journal Article**Author** D. Panagopoulos Abrahamsson**Author** J.-S. Park**Author** R.R. Singh**Author** M. Sirota**Author** T.J. Woodruff

Abstract Non-targeted analysis provides a comprehensive approach to analyze environmental and biological samples for nearly all chemicals present. One of the main shortcomings of current analytical methods and workflows is that they are unable to provide any quantitative information constituting an important obstacle in understanding environmental fate and human exposure. Herein, we present an in silico quantification method using machine-learning for chemicals analyzed using electrospray ionization (ESI). We considered three data sets from different instrumental setups: (i) capillary electrophoresis electrospray ionization-mass spectrometry (CE-MS) in positive ionization mode (ESI+), (ii) liquid chromatography quadrupole time-of-flight mass spectrometry (LC-QTOF/MS) in ESI+ and (iii) LC-QTOF/MS in negative ionization mode (ESI-). We developed and applied two different machine-learning algorithms: A random forest (RF) and an artificial neural network (ANN) to predict the relative response factors (RRFs) of different chemicals based on their physicochemical properties. Chemical concentrations can then be calculated by dividing the measured abundance of a chemical, as peak area or peak height, by its corresponding RRF. We evaluated our models and tested their predictive power using 5-fold cross-validation (CV) and y randomization. Both the RF and the ANN models showed great promise in predicting RRFs. However, the accuracy of the

predictions was dependent on the data set composition and the experimental setup. For the CE-MS ESI+ data set, the best model predicted measured RRFs with a mean absolute error (MAE) of 0.19 log units and a cross-validation coefficient of determination (Q²) of 0.84 for the testing set. For the LC-QTOF/MS ESI+ data set, the best model predicted measured RRFs with an MAE of 0.32 and a Q² of 0.40. For the LC-QTOF/MS ESI-data set, the best model predicted measured RRFs with a MAE of 0.50 and a Q² of 0.20. Our findings suggest that machine-learning algorithms can be used for predicting concentrations of nontargeted chemicals with reasonable uncertainties, especially in ESI+, while the application on ESI-remains a more challenging problem. © 2020 American Chemical Society.

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human, Humans, machine learning, Machine learning, Forecasting, Learning algorithms, Neural networks, Machine Learning, Statistical tests, computer simulation, Computer Simulation, Decision trees, Physicochemical properties, Mass spectrometry, liquid chromatography, Chromatography, Liquid, Liquid chromatography, Chemicals, Capillary electrophoresis, Chemical concentrations, Coefficient of determination, Electrospray ionization, Electrospray ionization (ESI), Electrospray ionization mass spectrometry, electrospray mass spectrometry, Environmental and biological samples, Liquid chromatography-quadrupole time-of-flight mass spectrometries, Negative ions, Quantitative information, Relative response factors, Spectrometry, Mass, Electrospray Ionization

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Applying deep learning to quantify empty lacunae in histologic sections of osteonecrosis of the femoral head

Item Type Journal Article

Author E. Lui

Author M. Maruyama

Author R.A. Guzman

Author S. Moeinzadeh

Author C.-C. Pan

Author A.K. Pius

Author M.S.V. Quig

Author L.E. Wong

Author S.B. Goodman

Author Y.P. Yang

Abstract Osteonecrosis of the femoral head (ONFH) is a disease in which inadequate blood supply to the subchondral bone causes the death of cells in the bone marrow. Decalcified histology and assessment of the percentage of empty lacunae are used to quantify the severity of ONFH. However, the current clinical practice of manually counting cells is a tedious and inefficient process. We utilized the power of artificial intelligence by training an established deep convolutional neural network framework, Faster-RCNN, to automatically classify and quantify osteocytes (healthy and pyknotic) and empty lacunae in 135 histology images. The adjusted correlation coefficient between the trained cell classifier and the ground truth was $R = 0.98$. The methods detailed in this study significantly reduced the manual effort of cell counting in ONFH histological samples and can be translated to other fields of image quantification. © 2021 Orthopaedic Research Society. Published by Wiley Periodicals LLC.

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Notes:

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Applying the Landscape Services Concept in Landscape Research: A Review

Item Type Journal Article

Author S. Zhang

Abstract Considerable attention has been given to the concept of landscape services since the development of research on ecosystem services. However, from the landscape perspective, there are numerous challenges and unanswered issues regarding landscape services. The objective of this paper is to provide a comprehensive review about the state-of-the-art concept of landscape services and its corresponding approaches while trying to set up a foundation for further discussions. First, the definitions and classifications of ecosystem services and landscape services are presented. Second, the relationship between ecosystem services and landscape services, and the studies that integrate ecosystem services into landscape research, are discussed. Finally, an overview is provided concerning research conducted at multiple scales, and the various assessment and quantification methods used for mapping and studying landscape services, even though there are many challenges in establishing a comprehensive approach for such quantification and mapping. © 2022 by author(s).

Date 2022

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Approach to an answer to “How dangerous microplastics are to the human body”: A systematic review of the quantification of MPs and simultaneously exposed chemicals

Item Type Journal Article

Author J. Lee

Author S. Jeong

Abstract This review aims to facilitate future research on microplastics (MPs) in the environment using systematic and analytical protocols, ultimately contributing to assessment of the risk to human health due to continuous daily exposure to MPs. Despite extensive studies on MP abundance in environment, identification, and treatment, their negative effects on human health remain unknown due to the lack of proof from clinical studies and limited technology on the MP identification. To assess the risk of MPs to human health, the first step is to estimate MP intake via ingestion, inhalation, and dermal contact under standardized exposure conditions in daily life. Furthermore, rather than focusing on the sole MPs, migrating chemicals from plastic products should be quantified and their health risk be assessed concurrently with MP release. The critical factors influencing MP release and simultaneously exposed chemicals (SECs) must be investigated using a standardized identification method. This review summarises release sources, factors, and possible routes of MPs from the environment to the human body, and the quantification methods used in risk assessment. We also discussed the issues encountered in MP release and SEC migration. Consequently, this review provides directions for future MP studies that can answer questions about MP toxicity to human health. © 2023 Elsevier B.V.

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Risk perception, Risk assessment, environmental factor, human, Humans, risk assessment, Risk Assessment, Risks assessments, Health risks, Systematic Review, Technology, quantitative analysis, technology, nonhuman, temperature, Review, classification, polymer, pH, Fourier transform infrared spectroscopy, plastic waste, limit of detection, FTIR, FTIR spectroscopy, microplastic, Microplastic, Microplastics, mouse, health risk, volatile organic compound, plastic, Plastics, human activities, toxicity, analytic method, Raman spectrometry, Analytical protocols, butylated hydroxyanisole, butylcresol, coloring agent, environmental exposure, food chain, health hazard, health impact, health risk assessment, Human bodies, Human Body, Human health, ingestion, inhalation, Microplastic exposure, monomer, photoaging, pollution exposure, Raman spectroscopy, Risk to human health, SECs, Simultaneously exposed chemical

Notes:

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Approaches to improve accuracy of neural network classification of images dominated by mixed pixels

Item Type Conference Paper

Author M.A. Ibrahim

Author M.K. Arora

Author S.K. Ghosh

Author H.-M. Chen

Abstract Often, previous knowledge about the classes on the ground may assist in the improvement of a land cover classification from remote sensing data. There are many ways in which the previous information in the form of a priori probabilities can be included in the training stage of a supervised classification process. The inclusion of a priori probabilities may be useful to classify the images dominated by mixed pixels. In this paper, an approach to incorporate a priori probabilities in back propagation neural network (BPNN) algorithm, by way of replicating the

training data (consisting of pure pixels) of a class according to its proportional area coverage has been implemented. By doing this, the abundant classes are assigned more weights than the other classes in the image. The results show a significant improvement in classification accuracy by 20% over the case when the a priori probabilities are not included. However, the major limitation of this approach is that it depends on the use of pure pixels, which may often be hard to find in images dominated by mixed pixels. We therefore suggest an alternative approach that incorporates mixed pixels in the training stage of BPNN algorithm. The results from this approach also reflect a significant improvement in classification accuracy by 14%. Further, both the approaches produce significantly higher accuracy than the most widely used maximum likelihood classifier (MLC). However, the second approach, which does not depend on identification of pure pixels in the image and their replication, appears more attractive to produce meaningful and accurate land cover classification from remote sensing data.

Date 2004

DOI 10.1109/IGARSS.2004.1369090

URL <https://www.scopus.com/pages/publications/15944375869?origin=resultslist>

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Language English

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Conference Name International Geoscience and Remote Sensing Symposium (IGARSS)

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Learning algorithms, Neural networks, Remote sensing, Image classification, Mathematical models, Probability, Image analysis, Classification accuracy, Matrix algebra, Maximum likelihood estimation, Backpropagation, Land cover mapping, Mixed pixels

Notes:

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Architectures, algorithms, and applications using bayesian networks

Item Type Conference Paper

Author T. Kingsbury

Abstract A Bayesian network is a tree structure where each branch represents a classification candidate. The leaves of the tree represent observable target features such as frequency or length. An optimized tree groups similar features together, e.g. frequency and pulse width, while collecting dissimilar or disparate information, e.g. spectral and kinematics, all within the same unifying structure. A vehicular track then is a subset of the a priori candidate library and contains only feasible branches. The algorithm for updating the confidence of each feasible candidate according to Bayes' rule is embedded in each track, as is the ability of a track to learn, apply a priori probability distributions, switch modes, switch among kinematics models, apply tracking history to classification and apply classification history to tracking, and support multisensor correlation and sensor fusion. © 2011 SPIE.

Date 2011

DOI 10.1117/12.882173

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Sensor fusion, Network architecture, Algorithms, Sensors, Intelligent networks, Bayesian networks, Inference engines, Mathematical models, Tracking, Information fusion, Trees (mathematics), Classification, Probability distributions, Bayesian, Bayes' rule, Kinematics, Distributed parameter networks, Tree structures, Plant extracts, Kinematics models, Multi sensor, Pulse width, Target feature

Notes:

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Export Date: 20 September 2026; Cited By: 0; Conference name: Multisensor, Multisource Information Fusion: Architectures, Algorithms, and Applications 2011; Conference date: 27 April 2011 through 28 April 2011; Conference code: 85453; CODEN: PSISD

Are Cramér-Rao lower bounds an accurate estimate for standard deviations in in vivo magnetic resonance spectroscopy?

Item Type Journal Article

Author K. Landheer

Author C. Juchem

Abstract Due to inherent time constraints for in vivo experiments, it is infeasible to repeat multiple MRS scans to estimate standard deviations on the desired measured parameters. As such, the Cramér-Rao lower bounds (CRLBs) have become the routine method to approximate standard deviations for in vivo experiments. Cramér-Rao lower bounds, however, as the name suggests, are theoretically a lower bound on the standard deviation and it is not clear if and under what circumstances this approximation is valid. Realistic synthetic 3 T spectra were used to investigate the relationship between estimated CRLBs, true CRLBs and standard deviations. Here we demonstrate that, although the CRLBs are theoretically truly a lower bound on the standard deviation (not an equality) for the problem typically encountered in quantification, they are still an adequate approximation to standard deviation as long as the model perfectly characterizes the data. In the case when the macromolecule basis deviates from the measured macromolecules it was shown that the CRLBs can deviate from standard deviations by approximately 50% for N-acetylaspartic acid, creatine and glutamate and of the order of 100% or more for myo-inositol and γ -aminobutyric acid. In the case when the model perfectly reflects the data the CRLBs are within approximately 10% of standard deviations for all metabolites. The result of the CRLB being within 10% of standard deviations means that, for an accurate model, novel quantification methods such as machine learning or deep learning will not be able to obtain substantially more precise estimates for the desired parameters than traditional maximum-likelihood estimation. © 2021 John Wiley & Sons, Ltd.

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Article, human, Humans, Deep learning, Statistics, algorithm, Algorithms, theoretical study, deep learning, controlled study, computer simulation, Computer Simulation, Accurate modeling, Monte Carlo method, Monte Carlo Method, Quantification methods, Magnetic resonance, Linear Models, statistical model, Metabolites, measurement, Standard deviation, Magnetic Resonance Spectroscopy, nuclear magnetic resonance spectroscopy, Maximum likelihood estimation, maximum likelihood method, in vivo study, Monte Carlo, priority journal, 4 aminobutyric acid, Amino acids, Cramer Rao lower bound, Cramér-Rao lower bounds, creatine, Gamma-aminobutyric acids, glutamic acid, in vivo MRS, In-vivo experiments, inositol, macromolecule, Macromolecules, Magnetic resonance spectroscopy, Measured parameters, MRS, n acetylaspartic acid, N-Acetylaspartic acid, standard deviations, Time constraints

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Are Your Fairness Metrics Accurate? A Semi-Supervised Approach to Improving Fairness Estimates Under Sample Selection Bias

Item Type Conference Paper

Author M.C. De Paolis Kaluza

Author T. Tholeti

Author Y. Chen

Author R. Baeza-Yates

Author P. Radivojac

Author S. Jain

Abstract A key challenge impeding the widespread deployment of machine learning is overcoming the impact of statistical biases in the data. Models trained on unrepresentative data can perform worse than anticipated and differentially affect cross-sections of the population. Therefore, evaluating and vetting models based on an appropriate notion of fairness is often indispensable, making accurate estimation of fairness metrics a critical step to safeguard against deployment of unfair algorithms. It is often assumed that a fairness metric computed from the observed data is accurate. However, in presence of selection bias, also referred to as distributional shifts, fairness metric estimates too can have systematic application-specific errors. In this work, we demonstrate this phenomenon and, relying on access to an unbiased unlabeled data, derive a semi-supervised approach to mitigate estimation errors emerging from the biased labeled data. Specifically, we introduce a novel selection bias model called “sub-class-conditional invariance” (SCC-invariance), that offers a flexible framework to

effectively capture distributional shifts in the real-world data, particularly compared to traditional models such as label shift and covariate shift. Assuming a finite Gaussian mixture form for each class-conditional distribution, we then derive an Expectation-Maximization algorithm to estimate model parameters and correction weights necessary for computing unbiased estimates. We focus on three widely used fairness metrics—equal opportunity, predictive equality, and predictive parity—and demonstrate the effectiveness of our approach in improving their estimates on synthetic data. Finally, we apply our bias mitigation approach to clinical genetics and study the fairness of pathogenicity predictors across ancestral groups. © 2025 Copyright held by the owner/author(s)

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Conference Name Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining

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Model-based OPC, Learning systems, Machine-learning, Gaussian distribution, Human computer interaction, Population statistics, Labeled data, Systematic errors, Semi-supervised, Semi-supervised learning, Fairness, Expectation Maximization, Expectation maximization algorithm, Maximum principle, bias mitigation, Bias mitigation, expectation maximization, gaussian mixture model, Gaussian Mixture Model, importance sampling, Importance sampling, sample selection bias, Sample selection bias, Selection bias, Statistical bias

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ARGs-OAP v3.0: Antibiotic-Resistance Gene Database Curation and Analysis Pipeline Optimization

Item Type Journal Article

Author X. Yin

Author X. Zheng

Author L. Li

Author A.-N. Zhang

Author X.-T. Jiang

Author T. Zhang

Abstract Antibiotic resistance, which is encoded by antibiotic-resistance genes (ARGs), has proliferated to become a growing threat to public health around the world. With technical advances, especially in the popularization of metagenomic sequencing, scientists have gained the ability to decipher the profiles of ARGs in diverse samples with high accuracy at an accelerated speed. To analyze thousands of ARGs in a high-throughput way, standardized and integrated pipelines are needed. The new version (v3.0) of the widely used ARGs online analysis pipeline (ARGs-OAP) has made significant improvements to both the reference database—the structured ARG (SARG) database—and the integrated analysis pipeline. SARG has been enhanced with sequence curation to improve annotation reliability, incorporate emerging resistance genotypes, and determine rigorous mechanism classification. The database has been further organized and visualized online in the format of a tree-like structure with a dictionary. It has also been divided into sub-databases for different application scenarios. In addition, the ARGs-OAP has been improved with adjusted quantification methods, simplified tool implementation, and multiple functions with user-defined reference databases. Moreover, the online platform now provides a diverse biostatistical analysis workflow with visualization packages for the efficient interpretation of ARG profiles. The ARGs-OAP v3.0 with an improved database and analysis pipeline will benefit academia, governmental management, and consultation regarding risk assessment of the environmental prevalence of ARGs. © 2022 THE AUTHORS

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Artificial Intelligence (AI) approach for the quantification of C-phycocyanin in *Spirulina platensis*: Hybrid stacking-ensemble model based on machine learning and deep learning

Item Type Journal Article

Author J.W.R. Chong

Author K.S. Khoo

Author H.-Y. Ting

Author I. Koji

Author Z. Ma

Author P.L. Show

Abstract The rising demand for natural pigments in nutraceuticals, pharmaceuticals, and cosmetics has highlighted the need for efficient, non-destructive methods to predict C-phycocyanin (CPC) concentrations in *Spirulina platensis*. Conventional extraction and quantification methods are labour-intensive, environmentally harmful, and time-consuming. This study proposes a hybrid stacking-ensemble model integrating convolutional neural networks (CNN) for automated feature extraction with both Support Vector Machine (SVM) and eXtreme gradient boosting (XGBoost) as base models and multiple meta-regressor models. The meta-regressors include Ridge regression with built-in cross-validation (RidgeCV), Linear Regression (LR), Support Vector Regressor (SVR), Decision Tree (DT), Random Forest (RF), and XGBoost. The datasets comprising 11,000 images of biomass and extracted CPC were captured under varying lighting conditions and device setups to reflect real-world conditions accurately. While digital cameras achieved higher accuracy, smartphones provide competitive results under both covered and light-disturbed conditions, demonstrating smartphones' scalability for real-time applications. While XGBoost as a meta-regressor (standard deviation (Std) = 0.0052, R2Train = 0.9985, R2Val = 0.9983 with 95 % confident interval (CI) [0.9979, 0.9986]), demonstrates marginal improvements over individual models such as SVM (Std = 0.0065, R2Train = 0.9975, R2Val = 0.9973 with 95 % CI [0.9970, 0.9976]) and XGBoost (Std = 0.0053, R2Train = 0.9987, R2Val = 0.9982 with 95 % CI [0.9978, 0.9986]), yet ensemble approach

offers lower variability, reduced overfitting, enhanced stability, and generalisation. CPC datasets delivered better accuracy but were competitive against biomass datasets. Thus, biomass datasets are more feasible in real-world applications by excluding the need for extraction steps, enabling rapid, reliable, and accurate CPC concentration predictions. © 2025 The Authors

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Artificial Intelligence and Big Data Analytics for Seismic Hazard Assessment: Methodological Advances and Computational Frameworks for the Marmara Region, Türkiye

Item Type Journal Article

Author P. Lemenkova

Author A.C. Zülfikar

Abstract The Marmara region of Türkiye, situated along the North Anatolian Fault Zone (NAFZ), constitutes one of the most seismically active and densely monitored zones globally. Given the region's high vulnerability and the catastrophic impacts of historical events—notably the 1999 İzmit and 2023 Kahramanmaraş sequences—there is a critical need for advanced seismic hazard risk assessment (SHRA) methods that move beyond static models. This review examines the paradigm shift from traditional geophysics to big data seismology, characterized by the “Five Vs”: volume, velocity, variety, veracity, and value. Critically, we distinguish between two fundamentally different problems: Earthquake Early Warning (EEW), which operates on sub-second timescales after rupture initiation, and probabilistic earthquake forecasting, which operates on timescales of years to decades. The study discusses how cloud-native platforms such as Azure Databricks, combined with data pipelines using Apache Kafka (version 3.5.1) and Apache Spark (version 4.1.2), enable the real-time processing of petabyte-scale seismic sensor streams. Key technological tools, including Physics-Informed Neural Networks (PINNs) and deep learning models such as PhaseNet, are analyzed for their demonstrated ability to enhance EEW systems through sub-second phase picking and automated event detection. Seismic tomography is also undergoing AI-enabled transformation, yielding higher-resolution subsurface imaging. We present statistical validation metrics and uncertainty quantification methods essential for credible hazard assessment. By addressing computational bottlenecks through hybrid computing architectures and edge computing, this framework aims to improve the warning lead time for Istanbul's critical infrastructure. This work provides a structured roadmap for bridging the gap between traditional seismic data analysis and operational predictive analytics in the Marmara region. © 2026 by the authors.

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Artificial intelligence-based Raynaud's quantification index (ARTIX): an objective mobile-based tool for patient-centered assessment of Raynaud's phenomenon

Item Type Journal Article

Author M. Di Battista

Author S. Colak

Author A. Howard

Author F. Donadoni

Author C. Owen-Smith

Author A. Rindone

Author S. Di Donato

Author C. Hartley

Author L.-A. Bissell

Author F. Del Galdo

Abstract Background: We aimed to develop an artificial intelligence algorithm able to assess Raynaud's phenomenon (RP) from mobile phone photography, ensuring as a patient-centered, image-based method for RP quantification. Methods: ARTIX (artificial intelligence-based Raynaud's quantification index) score was developed as a multi-step process of segmentation, decomposition and filters application to mobile phone pictures of the hand. ARTIX was validated by the ability to assess finger response to standardised cold challenge in patients with primary and secondary RP and healthy controls (HC) and compared with thermography as a reference. Results: Forty-five RP patients (91.1% female, mean age 52.2 years, 75.5% secondary RP) were enrolled, along with 22 HC comparable for age and gender. RP patients presented significantly lower ARTIX values than HC both at baseline ($p < 0.001$) and across all timepoints of the cold challenge ($p < 0.01$ for all), paralleling a similarly significant difference observed by thermography. ARTIX score was higher in males and in patients taking vasoactive drugs, whereas lower values were obtained in patients with late capillaroscopic pattern, diffuse cutaneous skin subset, or negative for anti-centromere antibodies. ARTIX showed also good ability to discriminate between RP and HC response to cold challenge. Conclusion: We developed and validated ARTIX, a novel machine learning-driven method for the objective quantification of RP. Real-life longitudinal studies in patients with RP will determine the value of ARTIX to complement patient self-assessment surrogate measures of RP activity and severity. © The Author(s) 2025.

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Artificial neural networks for classifying olfactory signals

Item Type Conference Paper
Author R. Linder
Author S.J. Pöppel

Abstract For practical applications, artificial neural networks have to meet several requirements: Mainly they should learn quick, classify accurate and behave robust. Programs should be user-friendly and should not need the presence of an expert for fine tuning diverse learning parameters. The present paper demonstrates an approach using an oversized network topology, adaptive propagation (APROP), a modified error function, and averaging outputs of four networks described for the first time. As an example, signals from different semiconductor gas sensors of an electronic nose were classified. The electronic nose smelt different types of edible oil with extremely different a-priori-probabilities. The fully-specified neural network classifier fulfilled the above mentioned demands. The new approach will be helpful not only for classifying olfactory signals automatically but also in many other fields in medicine, e.g. in data mining from medical databases.

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ASAP: Unsupervised Post-training with Label Distribution Shift Adaptive Learning Rate

Item Type Conference Paper

Author H. Park

Author M. Joe

Author M. Kim

Author M. Kwon

Abstract In real-world applications, machine learning models face online label shift, where label distributions change over time. Effective adaptation requires careful learning rate selection: too low slows adaptation and too high causes instability. We propose ASAP (Adaptive Shift Aware Post-training), which dynamically adjusts the learning rate by computing the cosine distance between current and previous unlabeled outputs and mapping it within a bounded range. ASAP requires no labels, model ensembles, or past inputs, using only the previous softmax output for fast, lightweight adaptation. Experiments across multiple datasets and shift scenarios show ASAP consistently improves accuracy and efficiency, making it practical for unsupervised model adaptation. © 2025 Copyright held by the owner/author(s).

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Aspect-Based Sentiment Quantification

Item Type Journal Article

Author V. Matsiako

Author F. Frasincar

Author D. Boekestijn

Abstract In the current literature, many methods have been devised for sentiment quantification. In this work, we propose AspEntQuaNet, one of the first methods for aspect-based sentiment quantification. It extends the state-of-the-art QuaNet deep learning method for sentiment quantification in two ways. First, it considers aspects and ternary sentiment quantification concerning these aspects instead of binary sentiment quantification. Second, it improves on the results of QuaNet with an entropy-based sorting procedure instead of multisorting. Other sentiment quantification methods have also been adapted for ternary sentiment quantification instead of binary sentiment quantification. Using the modified version of the SemEval 2016 dataset for aspect-based sentiment quantification, we show that AspEntQuaNet is superior to all other considered existing methods based on obtained results for various aspect categories. In particular, AspEntQuaNet outperforms QuaNet often by a factor of 2 on all considered evaluation measures. © 2010-2012 IEEE.

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Assessing cropland disagreement in Tanzania using machine learning methods with Sentinel-2 and Planet Scope imagery

Item Type Journal Article

Author V.W. Ogwen

Author N. Moore

Author D. Wanyama

Abstract East Africa faces major land use pressures arising from the harsh effects of climate change and fast-increasing human populations. Recent efforts to increase food production in this region have focused on cropland mapping and approaches to improve yield. Generating accurate cropland maps provides critical support to inform policy, investment, and logistical decisions that address food security, but cropland disagreement assessment studies are lacking mostly due to the unavailability of fine-scale data. Understanding the inconsistent nature of pre-existing global cropland mapping products necessitates disagreement assessment as the most valuable urgent tool to align agricultural resource allocation and policy realignment to tackle food security challenges. This study evaluates cropland disagreement in Kilombero and Ulanga districts in Morogoro, Tanzania based on land use land cover (LULC) maps generated using machine learning methods and high-resolution PlanetScope and Sentinel-2 data sets acquired from 1 January 2017 to 31 December 2018. We created time-variant median composites and performed a three-step assessment; (i) generating multi-class LULC maps in Google Earth Engine, (ii) generating croplands and non-cropland LULC classes from the multi-class images, and (iii) assessing cropland disagreement based on GIS conditional statements. Results show minimal disagreement in mapping cropland based on higher-resolution PlanetScope and Sentinel-2. Results based on the Random Forests and Support Vector Machines relatively outperform those from Classification and Regression Trees achieving 93%, 89%, and 83% with PlanetScope and 91%, 86%, and 76% with Sentinel-2, respectively. On average, 73% of the pixels were consistently classified while 27% were misclassified between cropland and non-cropland classes. Prior unsupervised land cover cluster analysis may substantiate the quality of sampling for accurate land cover classification. Our results indicate that in data-sparse regions where confirmatory ground truth data are lacking, remotely sensed cropland identification

may be more successful using Support Vector Machines on newer high-resolution imagery, thereby improving crop forecasts. © 2023 Informa UK Limited, trading as Taylor & Francis Group.

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Assessing Internet Addiction Using the Parsimonious Internet Addiction Components Model-A Preliminary Study

Item Type Journal Article

Author D.J. Kuss

Author G.W. Shorter

Author A.J. van Rooij

Author M.D. Griffiths

Author T.M. Schoenmakers

Abstract Internet usage has grown exponentially over the last decade. Research indicates that excessive Internet use can lead to symptoms associated with addiction. To date, assessment of potential Internet addiction has varied regarding populations studied and instruments used, making reliable prevalence estimations difficult. To overcome the present problems a preliminary study was conducted testing a parsimonious Internet addiction components model based on Griffiths' addiction components (Journal of Substance Use, 10, 191-197, 2005), including salience, mood modification, tolerance, withdrawal, conflict, and relapse. Two validated measures of Internet addiction were used (Compulsive Internet Use Scale [CIUS], Meerkerk et al. in Cyberpsychology & Behavior, 12(1), 1-6, 2009, and Assessment for Internet and Computer Game Addiction Scale [AICA-S], Wölfling et al. 2010) in two independent samples ($n_s = 3,105$ and $2,257$). The fit of the model was analysed using Confirmatory Factor Analysis. Results indicate that the Internet addiction components model fits the data in both samples well. The two sample/two instrument approach provides converging evidence concerning the degree to which the components model can organize the self-reported behavioural components of Internet addiction. Recommendations for future research include a more detailed assessment of tolerance as addiction component. © 2013 Springer Science+Business Media New York.

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Assessing Membership Inference Attacks under Distribution Shifts

Item Type Conference Paper**Author** S. Yichuan**Author** O. Kotevska**Author** V. Reshniak**Author** A. Sadovnik**Editor** Ding W.**Editor** Lu C.-T.**Editor** Wang F.**Editor** Di L.**Editor** Wu K.**Editor** Huan J.**Editor** Nambiar R.**Editor** Li J.**Editor** Ilievski F.**Editor** Baeza-Yates R.**Editor** Hu X.

Abstract Membership inference attacks (MIAs) exploit machine learning models to infer whether a data point was in the training set, posing significant privacy risks even with limited black-box access. These attacks rely on the attacker approximating the target model's training distribution, yet the impact of distribution shifts between target and shadow models on MIA success remains underexplored. We systematically evaluate five types of distribution shifts - -cutout, jitter, Gaussian noise, label shift, and attribute shift - - at varying intensities. Our results reveal that these shifts affect MIA effectiveness in nuanced ways, with some reducing attack success while others exacerbate vulnerabilities, and the same shift can have opposite effects depending on the type of MIA. This highlights the complex interplay between distributional differences and attack performance, offering critical insights for improving model defenses against MIAs. © 2024 IEEE.

Date 2024**DOI** 10.1109/BigData62323.2024.10825580**URL** <https://www.scopus.com/pages/publications/85218033896?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc. - IEEE Int. Conf. Big Data, BigData**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 26391589 (ISSN); 979-835036248-0 (ISBN)**Pages** 4127-4131**Proceedings Title** Proc. - IEEE Int. Conf. Big Data, BigData**Conference Name** Proceedings - 2024 IEEE International Conference on Big Data, BigData 2024**Date Added** 20/09/2026, 9:52:05 pm**Modified** 20/09/2026, 9:52:05 pm**Tags:**

Security, Privacy, privacy, security, Gaussian distribution, Gaussian noise (electronic), Differential privacy, Adversarial machine learning, Contrastive Learning, Privacy risks, Machine learning models, Black boxes, Training sets, Datapoints, Distribution shift, Inference attacks, Target model, distribution shift

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Assessing operational accuracy of CNN-based image classifiers using an oracle surrogate

Item Type Journal Article

Author A. Guerriero

Author M.R. Lyu

Author R. Pietrantuono

Author S. Russo

Abstract Context: Assessing the accuracy in operation of a Machine Learning (ML) system for image classification on arbitrary (unlabeled) inputs is hard. This is due to the oracle problem, which impacts the ability of automatically judging the output of the classification, thus hindering the accuracy of the assessment when unlabeled previously unseen inputs are submitted to the system. Objective: We propose the Image Classification Oracle Surrogate (ICOS), a technique to automatically evaluate the accuracy in operation of image classifiers based on Convolutional Neural Networks (CNNs). Method: To establish whether the classification of an arbitrary image is correct or not, ICOS leverages three knowledge sources: operational input data, training data, and the ML algorithm. Knowledge is expressed through likely invariants - properties which should not be violated by correct classifications. ICOS infers and filters invariants to improve the correct detection of misclassifications, reducing the number of false positives. We evaluate ICOS experimentally on twelve CNNs – using the popular MNIST, CIFAR10, CIFAR100, and ImageNet datasets. We compare it to two alternative strategies, namely cross-referencing and self-checking. Results: Experimental results show that ICOS exhibits performance comparable to the other strategies in terms of accuracy, showing higher stability over a variety of CNNs and datasets with different complexity and size. Conclusions: ICOS likely invariants are shown to be effective in automatically detecting misclassifications by CNNs used in image classification tasks when the expected output is unknown; ICOS ultimately yields faithful assessments of their accuracy in operation. Knowledge about input data can also be manually incorporated into ICOS, to increase robustness against unexpected phenomena in operation, like label shift. © 2022 The Author(s)

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Tags:

Machine-learning, Machine learning, Convolution, Convolutional neural network, Convolutional neural networks, Convolutional Neural Networks, Machine Learning, Image classification, Misclassifications, Network-based, Images classification, Accuracy assessment, Image Classifiers, Input datas, Input output programs, Likely invariants, Oracle problem, Oracle Problem

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Assessing rare diseases prevalence using literature quantification

Item Type Journal Article

Author S. Jason

Author W. Maxime

Author J. Anne-Sophie

Abstract Introduction: Estimating the prevalence of diseases is crucial for the organization of healthcare. The amount of literature on a rare pathology could help differentiate between rare and very rare diseases. The objective of this work was to evaluate to what extent the number of publications can be used to predict the prevalence of a given pathology. Methods: We queried Orphanet for the global prevalence class for all conditions for which it was available. For these pathologies, we cross-referenced the Orphanet, MeSH, and OMIM vocabularies to assess the number of publication available on Pubmed using three different query strategies (one proposed in the literature, and two built specifically for this study). We first studied the association of the number of publications obtained by each of these query strategies with the prevalence class, then their predictive ability. Results: Class prevalence was available for 3128 conditions, 2970 had a prevalence class < 1/1,000,000, 41 of 1–9/1,000,000, 84 of 1–9/100,000, and 33 of 1–9/10,000. We show a significant association and

excellent predictive performance of the number of publication, with an AUC over 94% for the best query strategy. Conclusion: Our study highlights the link and the excellent predictive performance of the number of publications on the prevalence of rare diseases provided by Orphanet. © 2021, The Author(s).

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Tags:

Article, automation, human, Humans, prediction, prevalence, classification, area under the curve, health care delivery, Delivery of Health Care, publication, disease association, Prevalence, rare disease, Rare diseases, Rare Diseases, Kruskal Wallis test, Bibliometrics

Notes:

EID: 2-s2.0-85103212600

Export Date: 20 September 2026; Cited By: 31; Correspondence Address: S. Jason; Department of Medical Informatics, Hôpital Européen Georges Pompidou, Paris, AP-HP, 20 Rue Leblanc, 75015, France; email: jason.shourick@aphp.fr

Assessing the Impact of the COVID-19 Pandemic on Electricity Consumption: A Machine Learning Approach

Item Type Journal Article

Author X. Qian

Author H. Hong

Author X. Zhan

Author Y. Wang

Author Y. Chen

Author K. Xiao

Author Y. Huang

Abstract This study aims to quantitatively assess the impact of the COVID-19 pandemic on societal electricity consumption behavior, particularly considering the interference of special events. Currently, there is insufficient quantitative analysis of the extent to which the COVID-19 pandemic affects electricity consumption behavior. Therefore, we endeavor to introduce new theories and methods to comprehensively understand the potential impact of the pandemic on electricity usage in this field. Our proposed machine learning methods demonstrate significant results and advantages in two key aspects. Firstly, we innovatively introduce an abnormality detection algorithm based on the lunar calendar, thereby establishing a detection system capable of accurately identifying abnormal fluctuations in electricity consumption, to precisely determine the onset of the pandemic. Secondly, we design an evaluation system that integrates CNN-LSTM predictive models and controlled variable strategies, enabling us to reconstruct electricity usage patterns less affected by the pandemic. By comparing actual electricity consumption with reconstructed patterns, we deeply evaluate the impact of the pandemic on different regions and industries and introduce the pandemic impact percentage as a quantification method to precisely assess the extent of the impact. The findings indicate a significant impact of the COVID-19 pandemic on commercial and industrial electricity consumption. Furthermore, our methods are not limited to pandemic research but can also be applied to investigate the effects of other events such as holidays, typhoons, and special production schedules on electricity consumption. In summary, by delving into the complex associations underlying electricity consumption, this study provides a solid theoretical and methodological foundation for future similar research and offers crucial decision support for the electricity industry and emergency management. © 2024 The Authors.

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: Y. Wang; School of Mathematical Sciences, Xiamen University, Xiamen, 361005, China; email: wymzs_xd@163.com

Assessing the Performance of Remaining Time Prediction Methods for Business Processes

Item Type Journal Article

Author J. Roider

Author A. Nguyen

Author D. Zanca

Author B.M. Eskofier

Abstract The prediction of the remaining time for business processes is a major task in predictive process monitoring (PPM). In the last years, various machine learning methods were introduced which reduced error levels steadily. However, the commonly applied metric for optimization and evaluation, the Mean Absolute Error (MAE), has limitations regarding its interpretability. In this work we introduce and evaluate the normalized Mean Absolute Error (nMAE) as an interpretable metric for model evaluation. It accounts for different kinds of label shifts, which are a special type of concept drift that can distort remaining time results. We investigate these concepts in a thorough benchmark study and use them to assess the current state of remaining time prediction for business processes. This includes the evaluation of four different baseline models, identifying the most accurate one. Furthermore, our study compares three different state-of-the-art methods, namely XGBoost, DA-LSTM, and PGT-Net. In contrary to prior studies we find that there is no significant difference in the performance between these models. Additionally, using the nMAE as evaluation metric we find that these models do not perform reasonably well on a range of event logs. Initial ideas for this behaviour are discussed and consolidated along with other findings from the case study into a comprehensive list motivating future research directions. © 2013 IEEE.

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Commerce, machine learning, Machine-learning, Prediction models, Graph neural networks, LSTM, Xgboost, XGBoost, graph neural network, Business process, Business Process, Mean absolute error, Predictive process, predictive process monitoring, Predictive process monitoring, process mining, Process mining, remaining time, Remaining time

Notes:

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Assessing Uncertainty in Machine Learning for Polymer Property Prediction: A Benchmark Study

Item Type Journal Article

Author H. Tang

Author T. Yue

Author Y. Li

Abstract Machine learning (ML) has emerged as a transformative tool in material science, enabling accelerated discovery and design of novel molecules while reducing experimental costs. Uncertainty quantification (UQ) is crucial for enhancing the reliability of ML predictions, particularly in high-stakes applications, such as functional polymer discovery. In this study, we present a comprehensive evaluation of nine UQ methods in ML—ensemble, Gaussian Process Regression (GPR), Monte Carlo Dropout (MCD), mean-variance estimation (MVE), Bayesian Neural Network based on Variational Inference (BNN-VI) and Markov Chain Monte Carlo (BNN-MCMC), evidential deep learning (EDL), quantile regression (QR), natural gradient boosting (NGBoost)—for predicting key polymer properties, including glass transition temperature (T_g), band gap (E_g), melting temperature (T_m) and decomposition temperature (T_d). The models are assessed using three independent metrics, including prediction accuracy (R^2), Spearman's rank correlation coefficient and calibration area, offering a robust framework for evaluating both mean predictions and uncertainty estimates. Our analysis spans data sets of four properties, out-of-distribution (OOD) experimental and molecular dynamics (MD)-derived data, high- T_g polymers and diverse polymer types, providing a holistic perspective on model performance. Our findings reveal that optimal UQ method selection is highly context-dependent. Ensemble method consistently excelled for general in-distribution predictions across four properties. For challenging OOD scenarios, BNN-MCMC offered a strong balance of predictive accuracy and reliable UQ. NGBoost emerged as the top-performing method for high- T_g polymers, effectively balancing accuracy and uncertainty characterization, with Ensemble method also providing excellent accuracy in this case. Furthermore, BNN-VI demonstrated superior and consistent performance across the nine distinct polymer classes evaluated. This comprehensive benchmark underscores the critical importance of selecting tailored UQ strategies to enhance the trustworthiness of ML predictions, optimize experimental validation efforts, and ultimately accelerate the discovery of advanced functional polymers. ©

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Assessment and quantification methods of urbanization patterns using geospatial techniques and spatial metrics in Tirunelveli city, Tamil Nadu, India

Item Type Journal Article

Author A.V.T. Annamalai

Author S.R. Kolanuvada

Author N. Radhakrishnan

Abstract Urban expansion has become a major challenge in recent years, threatening environmental sustainability and urban planning. Methods to quantify urban growth are limited, but the combination of geospatial techniques and spatial metrics offers significant advantages in quantifying growth and detecting early signs of sprawl for sustainable development. This study evaluated the effectiveness of the entropy model, landscape metrics, and urban metrics in quantifying urban growth. We focused on Tirunelveli City, the sixth-largest corporation in Tamil Nadu, India, by its evolution, spatial patterns and urban expansion trends. Landsat images and the CART classifier were used to prepare LULC maps for the years 1997, 2005, 2015 and 2022. Built-up areas of the city were extracted, and Shannon Entropy, five landscape metrics, four urban metrics, and the urban sprawl matrix were applied. Landscape metrics and Shannon Entropy methods indicated that the city was experiencing sprawl. Urban metrics were applied to five concentric rings, each with a 2 km radius from the city center, revealing a moderate growth rate. The urban sprawl matrix analyzed the city's spatial pattern, confirming the presence of a multi-nuclei structure and development along major roads, known as ribbon development. We conclude that the urban sprawl matrix is an effective method for analyzing city spatial patterns. Shannon Entropy and landscape metrics provide robust results for quantifying sprawl, and urban metrics are reliable for assessing growth rate. Therefore, local planners should focus on Concentric Ring 04, where significant growth has occurred, to ensure sustainable urban development. © The Japan Section of the Regional Science Association International 2025.

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Landscape metrics, Tirunelveli city, Urban metrics, Urban sprawl matrix

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Assessment of Classified Regulation Capability of New Energy Storage for Urban Grid Flexibility Demands

Item Type Conference Paper

Author S. Wang

Author J. Yu

Author S. Chen

Author J. Yang

Author X. Dai

Author X. Yuan

Abstract With the increasing proportion of external power supply and the high penetration of renewable energy in Shanghai's power system, the demand for flexibility resources in urban grids continues to grow. Novel energy storage systems (NESS) feature fast response and high precision; however, systematic evaluation methods for their adjustable capability across diverse scenarios, such as peak-shaving and frequency regulation, are currently lacking. To address this gap, this paper proposes a flexibility demand quantification method based on load variation to characterize the grid's requirements. On this basis, an operational and economic model for energy storage is established, incorporating round-trip efficiency and battery degradation costs. Furthermore, a multi-dimensional evaluation framework is constructed, encompassing energy-oriented, power/ramp-rate-oriented, and economic indicators to enable standardized scoring and classification of various configurations. Using a 50 MW/200 MWh energy storage station as a case study, simulation results demonstrate that the integration of energy storage significantly reduces the peak-valley difference and ramp rate of the urban grid, effectively meeting flexibility requirements. These findings provide a robust foundation for optimizing energy storage configurations and assessing adjustable capabilities in different application contexts. © 2026 IEEE.

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Assessment of multiresolution segmentation for delimiting drumlins in digital elevation models

Item Type Journal Article**Author** C. Eisan**Author** M. Smith**Author** J. Hillier

Abstract Mapping or "delimiting" landforms is one of geomorphology's primary tools. Computer-based techniques such as land-surface segmentation allow the emulation of the process of manual landform delineation. Land-surface segmentation exhaustively subdivides a digital elevation model (DEM) into morphometrically-homogeneous irregularly-shaped regions, called terrain segments. Terrain segments can be created from various land-surface parameters (LSP) at multiple scales, and may therefore potentially correspond to the spatial extents of landforms such as drumlins. However, this depends on the segmentation algorithm, the parameterization, and the LSPs. In the present study we assess the widely used multiresolution segmentation (MRS) algorithm for its potential in providing terrain segments which delimit drumlins. Supervised testing was based on five 5-m DEMs that represented a set of 173 synthetic drumlins at random but representative positions in the same landscape. Five LSPs were tested, and four variants were computed for each LSP to assess the impact of median filtering of DEMs, and logarithmic transformation of LSPs. The testing scheme (1) employs MRS to partition each LSP exhaustively into 200 coarser scales of terrain segments by increasing the scale parameter (SP), (2) identifies the spatially best matching terrain segment for each reference drumlin, and (3) computes four segmentation accuracy metrics for quantifying the overall spatial match between drumlin segments and reference drumlins. Results of 100 tests showed that MRS tends to perform best on LSPs that are regionally derived from filtered DEMs, and then log-transformed. MRS delineated 97% of the detected drumlins at SP values between 1 and 50. Drumlin delimitation rates with values up to 50% are in line with the success of manual interpretations. Synthetic DEMs are well-suited for assessing landform quantification methods such as MRS, since subjectivity in the reference data is avoided which increases the reliability, validity and applicability of results. © 2014 The Authors.

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Assessment of neurocognitive disorder in studies of cognitive impairment due to substance use disorder: A systematic review

Item Type Journal Article

Author A. Toledo-Fernández

Author A. Brzezinski-Rittner

Author C. Roncero

Author C. Benjet

Author J. Salvador-Cruz

Author R. Marín-Navarrete

Abstract Background: Cognitive impairments induced by substance use disorders (SUDs) are broadly documented. Evidence for these effects ought to be considered with regard to the new classification of neurocognitive disorders (NCDs). Failing to assess diagnostic criteria for NCD could be a limitation of studies addressing the effects of SUDs on cognition, resulting in misdiagnosis and inaccurate prevalence estimation. Methods: A systematic search of original articles (2000–2016) was conducted in Web of Knowledge and Science Direct. Key terms were: NCD and associated terms, and SUDs or alcohol, marijuana, cocaine and inhalants. Results: 59 cross-sectional and five prospective studies were reviewed. Criterion A.1 (evaluation of subjective concern of cognitive decline relative to a baseline), and criterion B (evaluation of impairment in daily life activities due to cognitive impairment) were not effectively evaluated in any of the studies. All studies addressed criterion A.2 (objective evidence of cognitive decline) via heterogeneous neuropsychological testing. Criteria C (absence of delirium) and D (absence of other possible etiologies) were frequently

considered for control of confounding effects, mostly via methodological procedures (e.g. abstinence before evaluation, and exclusion of participants with comorbid disorders). Conclusion: These findings highlight the need to develop and disseminate standard procedures for assessment of substance-induced NCD. © 2017, © 2017 Taylor & Francis Group, LLC.

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Assessment of problematic internet use by the compulsive internet use scale and the internet addiction test: A sample of problematic and pathological gamblers

Item Type Journal Article

Author D. Guertler

Author H.-J. Rumpf

Author A. Bischof

Author N. Kastirke

Author K.U. Petersen

Author U. John

Author C. Meyer

Abstract This study aims to analyze psychometric properties and validity of the Compulsive Internet Use Scale (CIUS) and the Internet Addiction Test (IAT) and, second, to determine a threshold for the CIUS which matches the IAT cut-off for detecting problematic Internet use. A total of 292 subjects with problematic or pathological gambling (237 men, 55 women) aged 14-63 years and with private Internet use for at least 1 h per working or weekend day were recruited via different recruitment channels. Results include that both scales were internally consistent (Cronbach's = 0.9) and had satisfactory convergent validity ($r = 0.75$; 95% CI 0.70-0.80). The correlation with duration of private Internet use per week was significantly higher for the CIUS ($r = 0.54$) compared to the IAT ($r = 0.40$). Among all participants, 25.3% were classified as problematic Internet users based on the IAT with a cut-off ≥ 40 . The highest proportion of congruent classified cases results from a CIUS cut-off ≥ 18 (sensitivity 79.7%, specificity 79.4%). However, a higher cut-off (≥ 21) seems to be more appropriate for prevalence estimation of problematic Internet use. © 2013 S. Karger AG, Basel.

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Assessment of simplified methods for quantification of [18F]-DPA-714 using 3D whole-brain TSPO immunohistochemistry in a non-human primate

Item Type Journal Article**Author** N. Van Camp**Author** Y. Balbastre**Author** A.-S. Herard**Author** S. Lavisse**Author** C. Tauber**Author** C. Wimberley**Author** M. Guillermier**Author** A. Berniard**Author** P. Gipchtein**Author** C. Jan**Author** R.A. Badin**Author** T. Delzescaux**Author** P. Hantraye**Author** G. Bonvento

Abstract The 18 kDa translocator protein (TSPO) is the main molecular target to image neuroinflammation by positron emission tomography (PET). However, TSPO-PET quantification is complex and none of the kinetic modelling approaches has been validated using a voxel-by-voxel comparison of TSPO-PET data with the actual TSPO levels of expression. Here, we present a single case study of binary classification of in vivo PET data to evaluate the statistical performance of different TSPO-PET quantification methods. To that end, we induced a localized and adjustable increase of TSPO levels in a non-human primate brain through a viral-vector strategy. We then performed a voxel-wise comparison of the different TSPO-PET quantification approaches providing parametric [18F]-DPA-714 PET images, with co-registered in vitro three-dimensional TSPO immunohistochemistry (3D-IHC) data. A data matrix was extracted from each brain hemisphere, containing the TSPO-IHC and TSPO-PET data for each voxel position. Each voxel was then classified as false or true, positive or negative after comparison of the TSPO-PET measure to the reference 3D-IHC method. Finally, receiver operating characteristic curves (ROC) were calculated for each TSPO-PET quantification method. Our results show that standard uptake value ratios using cerebellum as a reference region (SUVCBL) has the most optimal ROC score amongst all non-invasive approaches. © The Author(s) 2019.

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adult, Article, male, Male, brain, Brain, comparative study, image segmentation, nonhuman, procedures, information processing, area under the curve, diagnostic test accuracy study, receiver operating characteristic, multimodality, image processing, blood sampling, sensitivity and specificity, image reconstruction, Imaging, Three-Dimensional, nuclear magnetic resonance imaging, three-dimensional imaging, neuroimaging, Neuroimaging, animal, Animals, heart rhythm, immunohistochemistry, Immunohistochemistry, breathing rate, unclassified drug, 4 aminobutyric acid receptor, positron emission tomography, Positron-Emission Tomography, radiopharmaceutical agent, Radiopharmaceuticals, Receptors, GABA, animal experiment, animal model, animal tissue, quantification, primate, pyrimidine derivative, Pyrimidines, carrier protein, in vivo study, standardized uptake value, protein expression, fluorine 18, priority journal, 18 kDa translocator protein, carbon dioxide tension, confocal microscopy, electron spin resonance, fluorine, Fluorine Radioisotopes, lentivirus vector, Macaca fascicularis, n,n diethyl 2 [4 (2 fluoroethoxy)phenyl] 5,7 dimethylpyrazolo[1,5 a]pyrimidine 3 acetamide, N,N-diethyl-2-(2-(4-(2-fluoroethoxy)phenyl)-5,7-dimethylpyrazolo(1,5-a)pyrimidin-3-yl)acetamide, non-human primate, oxygen tension, pyrazole derivative, Pyrazoles, radiochemistry, radiochromatography, radioligand, reference methods, supervised cluster analysis, three-dimensional immunohistochemistry

Notes:

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: N. Van Camp; Commissariat à l'Energie Atomique et aux Energies Alternatives (CEA), Département de la Recherche Fondamentale, Institut François Jacob, Molecular Imaging, Center (MIRCen), CNRS UMR 9199, Université Paris-Sud, Université Paris-Saclay, Fontenay-aux-Roses, France; email: Nadja.van-camp@cea.fr; CODEN: JCBMD

Assessment of spatiotemporal dynamics of diurnal fog occurrence in
subtropical montane cloud forests

Item Type Journal Article**Author** H.-J. Li**Author** M.-H. Lo**Author** J.-Y. Juang**Author** J. Wang**Author** C.-Y. Huang

Abstract Montane cloud forests (MCFs) are frequently immersed in low-altitude clouds or fog. The diurnal (defined as 07:00–16:50 local time) cycle of fog is particularly important for regulating the carbon, water and energy cycles of these ecosystems. Elevated temperatures may alter the spatiotemporal dynamics of fog and have cascading impacts on MCFs. Therefore, systematic monitoring of fog occurrence is essential for understanding the ramifications of climate change on these unique forests. This study aims to quantify three years (2018–2020) of diurnal fog occurrence with high spatiotemporal (5 km, 10 minutes) resolutions in subtropical MCF in northeast Taiwan. Four open-sky observation stations were installed along an elevation gradient (1151, 1514, 1670 and 1811 m a.s.l.) within the cloud band to record meteorological data including rainfall, air temperature and relative humidity. We also acquired spatiotemporally-corresponding photosynthetic photon flux density from the geostationary Himawari-8 satellite and derived solar zenith angle for each station. We utilized these ground and satellite meteorological attributes to model fog occurrence using seven machine learning methods. By referring to time-lapse images, the performance of random forests was determined to be superior compared to other approaches and was therefore selected to quantify spatiotemporal dynamics of fog occurrence. Fog was determined to be more abundant in terms of probability, frequency and duration in mid-elevations when compared to the lower and higher ends. Temporal analysis demonstrated that overall seasonality was pronounced with higher fog abundance in the afternoons and cold months but varied from station to station. In addition, three-year fog duration and event variability for each month were notable for all stations; the foggiest station was at 1670 m a.s.l. This study demonstrates the feasibility of using machine learning to quantify spatiotemporal dynamics of fog using cross-scale meteorological attributes, which may facilitate monitoring the impact of climate change on MCFs. © 2022 Elsevier B.V.

Date 2022**DOI** 10.1016/j.agrformet.2022.108899**URL** <https://www.scopus.com/pages/publications/85126128098?origin=resultslist>**Archive** Scopus**Language** English**Volume** 317**Publisher** Elsevier B.V.**Publication** Agricultural and Forest Meteorology**ISSN** 01681923 (ISSN)**Journal Abbr** Agric. For. Meteorol.**Date Added** 20/09/2026, 9:52:26 pm**Modified** 20/09/2026, 9:52:26 pm**Tags:**

Machine learning, spatiotemporal analysis, Taiwan, climate change, subtropical region, Temperature, air temperature, assessment method, cloud forest, diurnal variation, fog, Himawari-8, montane forest, Precipitation, Relative humidity, Solar zenith angle

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Assigning Test Priority to Modules Using Code-Content and Bug History

Item Type Journal Article

Author E. Daka

Author E. Mustafa

Abstract Regression testing is a process that is repeated after every change in the program. Prioritization of test cases is an important process during regression test execution. Nowadays, there exist several techniques that decide which of the test cases will run first as per their priority levels, while increasing the probability of finding bugs earlier in the test life cycle. However, sometimes algorithms used to select important test cases may stop searching in local minima while missing the rest of the tests that might be important for a given change. To address this limitation further, we propose a domain-specific model that assigns testing priority to classes in applications based on developers' judgments for priority. Moreover, our technique which takes into consideration applications' code content and bug history, relates these features to overall class priority for testing. In the end, we test the proposed approach with a new (unknown) dataset of 20 instances. The predicted results are compared with developers' priority score and saw that this metric can prioritize correctly 70% of classes under test. © 2022 CCIS

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AUC Optimization from Multiple Unlabeled Datasets

Item Type Conference Paper
Author Z. Xie
Author Y. Liu
Author M. Li
Editor Wooldridge M.
Editor Dy J.
Editor Natarajan S.
Abstract Weakly supervised learning aims to make machine learning more powerful when the perfect supervision is unavailable, and has attracted much attention from researchers. Among the various scenarios of weak supervision, one of the most challenging cases is learning from multiple unlabeled (U) datasets with only a little knowledge of the class priors, or Um learning for short. In this paper, we study the problem of building an AUC (area under ROC curve) optimal model from multiple unlabeled datasets, which maximizes the pairwise ranking ability of the classifier. We propose Um-AUC, an AUC optimization approach that converts the Um data into a multi-label AUC optimization problem, and can be trained efficiently. We show that the proposed Um-AUC is effective theoretically and empirically. Copyright © 2024, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.
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AUC_μ: A performance metric for multi-class machine learning models

Item Type Conference Paper**Author** R.S. Kleiman**Author** D. Page

Abstract The area under the receiver operating characteristic curve (AUC) is arguably the most common metric in machine learning for assessing the quality of a two-class classification model. As the number and complexity of machine learning applications grows, so too does the need for measures that can gracefully extend to classification models trained for more than two classes. Prior work in this area has proven computationally intractable and/or inconsistent with known properties of AUC, and thus there is still a need for an improved multi-class efficacy metric. We provide in this work a multi-class extension of AUC that we call AUC_μ that is derived from first principles of the binary class AUC. AUC_μ has similar computational complexity to AUC and maintains the properties of AUC critical to its interpretation and use. Copyright © 2019 ASME

Date 2019**URL** <https://www.scopus.com/pages/publications/85078263278?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Int. Conf. Mach. Learn., ICML**Volume** 2019-June**Publisher** International Machine Learning Society (IMLS)**ISBN** 978-151088698-8 (ISBN)**Pages** 5972-5983**Proceedings Title** Int. Conf. Mach. Learn., ICML**Conference Name** 36th International Conference on Machine Learning, ICML 2019**Date Added** 20/09/2026, 9:52:43 pm

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Machine learning, Learning algorithms, Machine learning models, Performance metrics, Classification models, Machine learning applications, Receiver operating characteristic curves, Class extension, First principles

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AUC_μ: A Performance Metric for Multi-Class Machine Learning Models

Item Type Conference Paper**Author** R.S. Kleiman**Author** D. Page**Editor** Chaudhuri K.**Editor** Salakhutdinov R.

Abstract The area under the receiver operating characteristic curve (AUC) is arguably the most common metric in machine learning for assessing the quality of a two-class classification model. As the number and complexity of machine learning applications grows, so too does the need for measures that can gracefully extend to classification models trained for more than two classes. Prior work in this area has proven computationally intractable and/or inconsistent with known properties of AUC, and thus there is still a need for an improved multi-class efficacy metric. We provide in this work a multi-class extension of AUC that we call AUC_μ that is derived from first principles of the binary class AUC. AUC_μ has similar computational complexity to AUC and maintains the properties of AUC critical to its interpretation and use. © 2019 by the author(s).

Date 2019**URL** <https://www.scopus.com/pages/publications/105020263084?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc. Mach. Learn. Res.**Volume** 97**Publisher** ML Research Press**ISBN** 26403498 (ISSN)**Pages** 3439-3447

Proceedings Title Proc. Mach. Learn. Res.
Conference Name Proceedings of Machine Learning Research
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Authenticity Identification and Quantitative Analysis of *Dendrobium officinale* Based on Near-Infrared Spectroscopy Combined with Chemometrics

Item Type Journal Article

Author Z.-L. Fan

Author Q. Li

Author Z.-T. Zhang

Author L. Bai

Author X. Pu

Author T.-W. Shi

Author Y.-H. Chai

Abstract *Dendrobium officinale* is a valuable medicinal and edible homologous health food. It has immunomodulatory, antioxidant, and metabolism-regulating properties. However, its adulteration is widespread, seriously compromising product quality and safety. Traditional adulteration detection methods are complex, costly, and time-consuming, making it urgent to establish a rapid and non-destructive detection approach. This study developed a rapid identification and quantification method for adulterated *D. officinale*. The method combined near-infrared (NIR) spectroscopy with data-driven soft independent modeling of class analogy (DD-SIMCA) and partial least squares regression (PLSR) models. PCA, PLS-DA, and OPLS-DA were first used to visualize sample clustering and group differences. DT, SVM, ANN, and NB were used for classification. DD-SIMCA and PLSR were used for one-class modeling and quantitative analysis. Raw spectral data were preprocessed using multiplicative scatter correction (MSC), the standard normal variate (SNV), the first derivative, and

Savitzky–Golay smoothing. In the identification analysis, the DD-SIMCA model achieved 100% sensitivity and 100% specificity in the validation set. Its overall accuracy in the independent test set was 99.2%, demonstrating excellent discrimination performance. In addition, SVM combined with NIR also achieved good accuracy. In the quantitative analysis of adulteration, the PLSR model predicted different adulteration levels. Most calibration and validation sets showed R^2 values above 0.99 and RMSE values below 0.05, indicating excellent predictive performance. The results indicate that NIR combined with DD-SIMCA and PLSR can achieve rapid identification and accurate quantification of adulterated *D. officinale* samples. This approach provides strong support for quality control and regulatory supervision of high-value health foods. © 2026 by the authors.

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Notes:

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Export Date: 20 September 2026; Cited By: 8; Correspondence Address: Y.-H. Chai; School of Basic Chinese Medicine, Guizhou University of Traditional Chinese Medicine, Guizhou, 550000, China; email: chaiyihui736@gzy.edu.cn

AutoFoci, an automated high-throughput foci detection approach for analyzing low-dose DNA double-strand break repair

Item Type Journal Article

Author N. Lengert

Author J. Mirsch
Author R.N. Weimer
Author E. Schumann
Author P. Haub
Author B. Drossel
Author M. Löbrich

Abstract Double-strand breaks (DSBs) are the most lethal DNA damages induced by ionising radiation (IR) and their efficient repair is crucial to limit genomic instability. The cellular DSB response after low IR doses is of particular interest but its examination requires the analysis of high cell numbers. Here, we present an automated DSB quantification method based on the analysis of γ H2AX and 53BP1 foci as markers for DSBs. We establish a combination of object properties, combined in the object evaluation parameter (OEP), which correlates with manual object classification. Strikingly, OEP histograms show a bi-modal distribution with two maxima and a minimum in between, which correlates with the manually determined transition between background signals and foci. We used algorithms to detect the minimum, thus separating foci from background signals and automatically assessing DSB levels. To demonstrate the validity of this method, we analyzed over 600.000 cells to verify results of previous studies showing that DSBs induced by low doses are less efficiently repaired compared with DSBs induced by higher doses. Thus, the automated foci counting method, called AutoFoci, provides a valuable tool for high-throughput image analysis of thousands of cells which will prove useful for many biological screening approaches. © 2018, The Author(s).

Date 2018

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Automated classification of local patches in colon histopathology

Item Type	Conference Paper
Author	H. Kalkan
Author	M. Nap
Author	R.P.W. Duin
Author	M. Loog
Abstract	An automated histology analysis is proposed for classification of local image patches of colon histopathology images into four principle classes: normal, cancer, adenomatous and inflamed classes. Shape features based on stroma, lumen and imperfectly segmented nuclei are combined with texture features for classification. The classification is analyzed under the three scenarios: normal vs. abnormal, cancer vs. non-cancer and four-class classification on a labeled dataset consisting of 2000 patches per class which were collected from 55 different slices. The proposed method achieves 79.28% mean accuracy between normal and abnormal; 87.67% accuracy between cancer and non-cancer and 75.15% between the four classes with equal class priories. © 2012 ICPR Org Committee.
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URL	https://www.scopus.com/pages/publications/84874562835?origin=resultslist
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Language	English
Extra	Journal Abbreviation: Proc. Int. Conf. Pattern Recognit.
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Image classification, Labeled dataset, Feature-based, Texture features, Lung cancer, Automated classification, Image patches, Shape features

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Export Date: 20 September 2026; Cited By: 32; Correspondence Address: H. Kalkan; Pattern Recognition Laboratory, Delft University of Technology, Netherlands; email: habilkalkan@sdu.edu.tr; Conference name: 21st International Conference on Pattern Recognition, ICPR 2012; Conference date: 11 November 2012 through 15 November 2012; Conference code: 95857; CODEN: PICRE

Automated classification of self-grooming in mice using open-source software

Item Type Journal Article

Author B.J.G. van den Boom

Author P. Pavlidi

Author C.J.H. Wolf

Author A.H. Mooij

Author I. Willuhn

Abstract Background Manual analysis of behavior is labor intensive and subject to inter-rater variability. Although considerable progress in automation of analysis has been made, complex behavior such as grooming still lacks satisfactory automated quantification. New method We trained a freely available, automated classifier, Janelia Automatic Animal Behavior Annotator (JAABA), to quantify self-grooming duration and number of bouts based on video recordings of SAPAP3 knockout mice (a mouse line that self-grooms excessively) and wild-type animals. Results We compared the JAABA classifier with human expert observers to test its ability to measure self-grooming in three scenarios: mice in an open field, mice on an elevated plus-maze, and tethered mice in an open field. In each scenario, the classifier identified both grooming and non-grooming with great accuracy and correlated highly with results obtained by human observers. Consistently, the JAABA classifier confirmed previous reports of excessive grooming in SAPAP3 knockout mice. Comparison with existing methods Thus far, manual analysis was regarded as the only valid quantification method for self-grooming. We demonstrate that the JAABA classifier is a valid and reliable scoring tool, more cost-efficient than manual scoring, easy to use, requires minimal effort, provides high throughput, and prevents inter-rater variability. Conclusion We introduce the JAABA classifier as an efficient analysis tool for the assessment of rodent self-grooming with expert quality. In our “how-to” instructions, we provide all information necessary to implement behavioral classification with JAABA. © 2017 Elsevier B.V.

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Export Date: 20 September 2026; Cited By: 40; Correspondence Address: I. Willuhn; Netherlands Institute for Neuroscience, Amsterdam, Meibergdreef 47, 1105 BA, Netherlands; email: i.willuhn@nin.knaw.nl; CODEN: JNMED

Automated Eosinophil Quantification Using Deep Learning to Predict Therapy Escalation in Pediatric Ulcerative Colitis

- Item Type** Journal Article
- Author** J. Reigle
- Author** X. Liu
- Author** O. Lopez-Nunez
- Author** E. Drysdale
- Author** D. Qtaish
- Author** T.D. Walters
- Author** D. Mack
- Author** B.M. Boyle
- Author** S. Kugathasan
- Author** N. Denson
- Author** L. Erdman
- Author** S. Prasath
- Author** M.H. Collins
- Author** J.S. Hyams
- Author** L.A. Denson

Author A.M. Griffiths

Author I. Siddiqui

Author J. Dhaliwal

Abstract INTRODUCTION: – Emerging evidence implicates eosinophils as important modulators of disease activity and therapeutic response in ulcerative colitis (UC). Automated image analysis provides a scalable and reproducible approach to their evaluation, overcoming the limitations of manual quantification. METHODS: – We developed a U-Net-based deep learning model to automate eosinophil detection, using whole-slide images from 2 pediatric sites. Pathologist-annotated 500×500 -pixel patches served as training data; model performance was evaluated using area under the receiver operator curve (AUROC), precision, recall, and F1 score and benchmarked against SAU-Net, YOLO-v8, and HoVer-Next. The validated algorithm was applied to rectal biopsies from treatment-naïve children with moderate-to-severe UC in the PROTECT inception cohort ($n = 221$). Eosinophil metrics (density in cells/mm² and peak high-power field counts) were analyzed for associations with baseline clinical, histologic, and endoscopic features and 1-year clinical outcomes including anti-tumor necrosis factor α (TNF α) escalation and corticosteroid-free clinical remission. Predictive models were evaluated with logistic regression and machine learning approaches using 5-fold cross-validation. RESULTS: – The segmentation model achieved high accuracy (test AUROC = 0.94, F1 score = 0.86) and strong concordance with expert counts ($p = 0.87$ – 0.89). Higher eosinophil density and counts correlated with basal lymphoid aggregates ($P < 0.001$) and predicted week 4 and corticosteroid-free remission ($P = 0.03$ and $P = 0.01$, respectively). Lower eosinophil counts predicted escalation to anti-TNF α therapy ($P = 0.04$). Incorporation of eosinophil metrics into Predicting Response to Standardized Pediatric Colitis Therapy (PROTECT) clinical models maintained predictive accuracy for anti-TNF α escalation (AUROC = 0.79; accuracy = 0.76). DISCUSSION: – Deep learning-based eosinophil quantification has the potential to transform pediatric UC care by serving as a reproducible biomarker that informs risk stratification and to guide therapy. © 2026 The Author(s). Published by Wolters Kluwer Health, LLC. on behalf of The American College of Gastroenterology

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artificial intelligence, female, Female, human, Humans, male, Male, adolescent, Adolescent, Deep Learning, procedures, deep learning, diagnosis, predictive modeling, drug therapy, child, Child, histopathology, ulcerative colitis, diagnostic imaging, eosinophil, severity of illness index, Severity of Illness Index, pathology, clinical trial, biopsy, multicenter study, predictive learning (machine learning), Predictive Learning Models, treatment outcome, Biopsy, blood, Colitis, Ulcerative, deep learning model, eosinophils, Eosinophils, immunology, leukocyte count, Leukocyte Count, Pediatric ulcerative colitis, rectum, Rectum, remission, Remission Induction, Treatment Outcome, tumor necrosis factor, Tumor Necrosis Factor-alpha, U-Net

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK); Crohn's and Colitis Foundation of America, CCFA; Children's Liver Disease Foundation, CLDF; Crohn's and Colitis Foundation Clinical Research Investigator-Initiated Awards (CRIA), (879083); National Institutes of Health, NIH, (P30 DK078392); Canadian Institutes of Health Research, IRSC, (27862); National Institute of Health Sciences, NIHS, (P30 DK078392); Correspondence Address: J. Dhaliwal; Division of Gastroenterology, Hepatology and Nutrition, Cincinnati Children's Hospital Medical Center, Cincinnati, United States; email: Jasbir.dhaliwal@cchmc.org

Automated graded prognostic assessment for patients with hepatocellular carcinoma using machine learning

Item Type Journal Article

Author M. Gross

Author S.P. Haider

Author T. Ze'evi

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Author B. Gebauer

Author F. Fleckenstein

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Author A. Jaffe

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Author J. Chapiro

Author J.A. Onofrey

Abstract Background: Accurate mortality risk quantification is crucial for the management of hepatocellular carcinoma (HCC); however, most scoring systems are subjective. Purpose: To develop and independently validate a machine learning mortality risk quantification method for HCC patients using standard-of-care clinical data and liver

radiomics on baseline magnetic resonance imaging (MRI). Methods: This retrospective study included all patients with multiphasic contrast-enhanced MRI at the time of diagnosis treated at our institution. Patients were censored at their last date of follow-up, end-of-observation, or liver transplantation date. The data were randomly sampled into independent cohorts, with 85% for development and 15% for independent validation. An automated liver segmentation framework was adopted for radiomic feature extraction. A random survival forest combined clinical and radiomic variables to predict overall survival (OS), and performance was evaluated using Harrell's C-index. Results: A total of 555 treatment-naïve HCC patients (mean age, 63.8 years $\pm$ 8.9 [standard deviation]; 118 females) with MRI at the time of diagnosis were included, of which 287 (51.7%) died after a median time of 14.40 (interquartile range, 22.23) months, and had median followed up of 32.47 (interquartile range, 61.5) months. The developed risk prediction framework required 1.11 min on average and yielded C-indices of 0.8503 and 0.8234 in the development and independent validation cohorts, respectively, outperforming conventional clinical staging systems. Predicted risk scores were significantly associated with OS ($p < .00001$ in both cohorts). Conclusions: Machine learning reliably, rapidly, and reproducibly predicts mortality risk in patients with hepatocellular carcinoma from data routinely acquired in clinical practice. Clinical relevance statement: Precision mortality risk prediction using routinely available standard-of-care clinical data and automated MRI radiomic features could enable personalized follow-up strategies, guide management decisions, and improve clinical workflow efficiency in tumor boards. Key Points: • Machine learning enables hepatocellular carcinoma mortality risk prediction using standard-of-care clinical data and automated radiomic features from multiphasic contrast-enhanced MRI. • Automated mortality risk prediction achieved state-of-the-art performances for mortality risk quantification and outperformed conventional clinical staging systems. • Patients were stratified into low, intermediate, and high-risk groups with significantly different survival times, generalizable to an independent evaluation cohort. © The Author(s) 2024.

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Automated measurement of penile curvature using deep learning-based novel quantification method

Item Type Journal Article

Author S.B. Baray

Author M. Abdelmoniem

Author S. Mahmud

Author S. Kabir

Author M.A.A. Faisal

Author M.E.H. Chowdhury

Author T.O. Abbas

Abstract Objective: Develop a reliable, automated deep learning-based method for accurate measurement of penile curvature (PC) using 2-dimensional images. Materials and methods: A set of nine 3D-printed models was used to generate a batch of 913 images of penile curvature (PC) with varying configurations (curvature range 18° to 86°). The penile region was initially localized and cropped using a YOLOv5 model, after which the shaft area was extracted using a UNet-based segmentation model. The penile shaft was then divided into three distinct predefined regions: the distal zone, curvature zone, and proximal zone. To measure PC, we identified four distinct locations on the shaft that reflected the mid-axes of proximal and distal segments, then trained an HRNet model to predict these landmarks and calculate curvature

angle in both the 3D-printed models and masked segmented images derived from these. Finally, the optimized HRNet model was applied to quantify PC in medical images of real human patients and the accuracy of this novel method was determined. Results: We obtained a mean absolute error (MAE) of angle measurement $<5^\circ$ for both penile model images and their derivative masks. For real patient images, AI prediction varied between 1.7° (for cases of $\sim 30^\circ$ PC) and approximately 6° (for cases of 70° PC) compared with assessment by a clinical expert. Discussion: This study demonstrates a novel approach to the automated, accurate measurement of PC that could significantly improve patient assessment by surgeons and hypospadiology researchers. This method may overcome current limitations encountered when applying conventional methods of measuring arc-type PC. 2023 Baray, Abdelmoniem, Mahmud, Kabir, Faisal, Chowdhury and Abbas.

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Article, artificial intelligence, human, male, machine learning, image segmentation, deep learning, controlled study, YOLO, deep neural network, convolutional neural network, mean absolute error, chordee, HRNet, hypospadias, male genital tract parameters, penile curvature, penile shaft, penis, stereolithography, three dimensional printing, two-dimensional imaging, UNET

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Automated Morphological Classification of Neural Cells growing in Culture

Item Type Conference Paper

Author J. Kim

Author D.F. Gillies

Abstract Studies of developing neural cells are widely used in fundamental research into the mechanisms of nervous diseases, such as multiple sclerosis. These studies normally require researchers to estimate the populations of different classes of cells in microscope images of either tissue or cultures. The estimation process is carried out by human visual inspection, and is time consuming and subjective in nature. As an attack on this problem, we have been investigating the use of computer vision to classify and count the neural cells automatically. We apply various image processing techniques and then use measures such as the fractal dimension and 2nd moment to classify cells according to their spatial growth characteristics seen during their development. The measures are then combined using a Bayesian classifier, and a decision is taken as to which class a cell belongs. The results on isolated cell images showed that more than 80% agreement between the computer analysis and human judgement. Our latest work concerns the problems to be tackled on mixed cell images where dendrites of different cells overlap. We have been investigating how to apply the shape measures in this case and results are promising.

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Automated MPI segmentation in X-space and calibration to quantify iron concentration in a tracer distribution

Item Type Journal Article**Author** Z. Sun**Author** J. Tian**Author** Y. Du

Abstract Magnetic particle imaging (MPI) is a new, radiation-free medical imaging modality that relies on the non-linear magnetization response of superparamagnetic iron oxide nanoparticles (SPIONs) to reconstruct their concentration distribution with high sensitivity and medical safety. Current quantification methods for region of interest (ROI) are inadequate and are usually outlined manually or using deep learning methods. We propose two new models for ROI selection based on machine learning, one is the K-means++-based threshold-inflated image segmentation model and the other is the image segmentation model based on MPI simulation and SVM. We have developed an accurate quantification of 2D MPI images and established the calibration curve to predict the corresponding iron content based on the MPI image. © 2023 Infinite Science Publishing.

Date 2023**DOI** 10.18416/IJMPI.2023.2303037**URL** <https://www.scopus.com/pages/publications/85152656260?origin=resultslist>**Archive** Scopus**Language** English**Volume** 9**Publisher** Infinite Science Publishing**Publication** International Journal on Magnetic Particle Imaging**ISSN** 23659033 (ISSN)**Journal Abbr** Int. J. Magn. Particle Imaging.**Date Added** 20/09/2026, 9:52:11 pm**Modified** 20/09/2026, 9:52:11 pm**Notes:**

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Automated MRI assessment confirms cartilage thickness modification in patients with knee osteoarthritis: post-hoc analysis from a phase II sprifermin study

Item Type Journal Article**Author** A. Brett**Author** M.A. Bowes**Author** P.G. Conaghan

Author C. Ladel

Author H. Guehring

Author F. Moreau

Author F. Eckstein

Abstract Background: Sprifermin is under investigation as a potential disease-modifying osteoarthritis drug. Previously, 2-year results from the FORWARD study showed significant dose-dependent modification of cartilage thickness in the total femorotibial joint (TFTJ), medial and lateral femorotibial compartments (MFTC, LFTC), and central medial and lateral TFTJ subregions, by quantitative magnetic resonance imaging (qMRI) using manual segmentation. Objective: To determine whether qMRI findings from FORWARD could be reproduced by an independent method of automated segmentation using an identical dataset and similar anatomical regions in a post-hoc analysis. Method: Cartilage thickness was assessed at baseline and 6, 12, 18 and 24 months, using automated cartilage segmentation with active appearance models, a supervised machine learning method. Images were blinded for treatment and timepoint. Treatment effect was assessed by observed and adjusted changes using a linear mixed model for repeated measures. Results: Based on automated segmentation, statistically significant, dose-dependent structural modification of cartilage thickness was observed over 2 years with sprifermin vs placebo for TFTJ (overall treatment effect and dose response, both $P < 0.001$), MFTC ($P = 0.004$ and $P = 0.044$), and LFTC (both $P < 0.001$) regions. For highest dose, in the central medial tibial ($P = 0.008$), central lateral tibial ($P < 0.001$) and central lateral femoral ($P < 0.001$) regions. Conclusions: Cartilage thickness assessed by automated segmentation provided a consistent dose response in structural modification compared with manual segmentation. This is the first time that two independent quantification methods of image analysis have reached the same conclusions in an interventional trial, strengthening the conclusions that sprifermin modifies structural progression in knee osteoarthritis. © 2020 Osteoarthritis Research Society International

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Notes:

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Automated multiclass structural damage detection and quantification using augmented reality

Item Type Journal Article

Author O. Awadallah

Author A. Sadhu

Abstract Civil infrastructure worldwide is ageing and enduring increasingly adverse weather conditions. Traditional structural health monitoring (SHM) involves the expensive and time-consuming installation of contact sensors. For example, inspectors use costly large-scale equipment to reach a certain area of the structure and at different heights to inspect it, which can pose a risk to the inspector's safety. Moreover, the inspectors rely only on the batch data acquired during the inspection period, which are analyzed by engineers at a later time due to the limited availability of a real-time visualization approach for structural inspection within the traditional mode of SHM. To address these timely challenges, an Augmented Reality (AR)-based automated multiclass damage identification and quantification methodology is proposed in this paper. The interactive visualization framework of AR is integrated with the autonomous decision-making of Artificial Intelligence (AI) in a unified fashion to incorporate human-sensor interaction. The proposed system uses an AI model that is trained and optimized using the YOLOv5 architecture to detect and classify four different types of anomalies/damages (i.e., cracks, spalls, pittings, and joints). The AI model is then updated to quantify the length, area, and perimeter of any damage using segmentation to further assess its severity. Once the model is developed, the model is embedded with the AR device and tested through its interactive environment for SHM of various structures. The paper concludes that the proposed approach

successfully classifies four types of damage with an accuracy of more than 90% for up to 2 m, and it also quantifies the length, area, and perimeter with less than 2% of error. © 2022 The Author(s)

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Notes:

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Automated Quality-Controlled Cardiovascular Magnetic Resonance Pericardial Fat Quantification Using a Convolutional Neural Network in the UK Biobank

Item Type Journal Article

Author A. Bard

Author Z. Raisi-Estabragh

Author M. Ardissino

Author A.M. Lee

Author F. Pugliese

Author D. Dey

Author S. Sarkar

Author P.B. Munroe

Author S. Neubauer

Author N.C. Harvey

Author S.E. Petersen

Abstract Background: Pericardial adipose tissue (PAT) may represent a novel risk marker for cardiovascular disease. However, absence of rapid radiation-free PAT quantification methods has precluded its examination in large cohorts. Objectives: We developed a fully automated quality-controlled tool for cardiovascular magnetic resonance (CMR) PAT quantification in the UK Biobank (UKB). Methods: Image analysis comprised contouring an en-bloc PAT area on four-chamber cine images. We created a ground truth manual analysis dataset randomly split into training and test sets. We built a neural network for automated segmentation using a Multi-residual U-net architecture with incorporation of permanently active dropout layers to facilitate quality control of the model's output using Monte Carlo sampling. We developed an in-built quality control feature, which presents predicted Dice scores. We evaluated model performance against the test set (n = 87), the whole UKB Imaging cohort (n = 45,519), and an external dataset (n = 103). In an independent dataset, we compared automated CMR and cardiac computed tomography (CCT) PAT quantification. Finally, we tested association of CMR PAT with diabetes in the UKB (n = 42,928). Results: Agreement between automated and manual segmentations in the test set was almost identical to inter-observer variability (mean Dice score = 0.8). The quality control method predicted individual Dice scores with Pearson $r = 0.75$. Model performance remained high in the whole UKB Imaging cohort and in the external dataset, with medium–good quality segmentation in 94.3% (mean Dice score = 0.77) and 94.4% (mean Dice score = 0.78), respectively. There was high correlation between CMR and CCT PAT measures (Pearson $r = 0.72$, p -value 5.3×10^{-18}). Larger CMR PAT area was associated with significantly greater odds of diabetes independent of age, sex, and body mass index. Conclusions: We present a novel fully automated method for CMR PAT quantification with good model performance on independent and external datasets, high correlation with reference standard CCT PAT measurement, and expected clinical associations with diabetes. Copyright © 2021 Bard, Raisi-Estabragh, Ardissino, Lee, Pugliese, Dey, Sarkar, Munroe, Neubauer, Harvey and Petersen.

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Automated quantification of 18F-flutemetamol PET activity for categorizing scans as negative or positive for brain amyloid: Concordance with visual image reads

Item Type Journal Article

Author L. Thurfjell

Author J. Lilja

Author R. Lundqvist

Author C. Buckley

Author A. Smith

Author R. Vandenberghe

Author P. Sherwin

Abstract Clinical trials of the PET amyloid imaging agent 18F-flutemetamol have used visual assessment to classify PET scans as negative or positive for brain amyloid. However, quantification provides additional information about regional and global tracer uptake and may have utility for image assessment over time and across different centers. Using postmortem brain neuritic plaque density data as a truth standard to derive a standardized uptake value ratio (SUVR) threshold, we assessed a fully automated quantification method comparing visual and quantitative scan categorizations. We also compared the histopathology-derived SUVR threshold with one derived from healthy controls.; Methods: Data from 345 consenting subjects enrolled in 8 prior clinical trials of 18F-flutemetamol injection were used. We grouped subjects into 3 cohorts: an autopsy cohort (n = 68) comprising terminally ill patients with postmortem confirmation of brain amyloid status; a test cohort (n = 172) comprising 33 patients with clinically probable Alzheimer disease, 80 patients with mild cognitive impairment, and 59 healthy volunteers; and a healthy cohort of 105 volunteers, used to define a reference range for SUVR. Visual image categorizations for comparison were from a previous study. A fully automated PET-only

quantification method was used to compute regional neocortical SUVRs that were combined into a single composite SUVR. An SUVR threshold for classifying scans as positive or negative was derived by ranking the PET scans from the autopsy cohort based on their composite SUVR and comparing data with the standard of truth based on postmortem brain amyloid status for subjects in the autopsy cohort. The derived threshold was used to categorize the 172 scans in the test cohort as negative or positive, and results were compared with categorization using visual assessment. Different reference and composite region definitions were assessed. Threshold levels were also compared with corresponding thresholds derived from the healthy group.; Results: Automated quantification (using pons as the reference region) demonstrated 91% sensitivity and 88% specificity and gave 3 false-positive and 4 false-negative scans. All 3 false-positive cases were either borderline-normal by standard of truth or had moderate to heavy cortical diffuse plaque burden. In the test cohort, the concordance between quantitative and visual read categorization ranged from 97.1% to 99.4% depending on the selection of reference and composite regions. The threshold derived from the healthy group was close to the histopathology-derived threshold.; Conclusion: Categorization of 18F-flutemetamol amyloid imaging data using an automated PET-only quantification method showed good agreement with histopathologic classification of neuritic plaque density and a strong concordance with visual read results. COPYRIGHT © 2014 by the Society of Nuclear Medicine and Molecular Imaging, Inc.

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Automated quantification of epicardial adipose tissue (EAT) in coronary CT angiography; comparison with manual assessment and correlation with coronary artery disease

Item Type Journal Article

Author C. Muhl

Author D. Loeffen

Author M.O. Versteilen

Author R.A.P. Takx

Author P.J. Nelemans

Author E.C. Nijssen

Author F. Vega-Higuera

Author J.E. Wildberger

Author M. Das

Abstract Background: Epicardial adipose tissue (EAT) is emerging as a risk factor for coronary artery disease (CAD). Objective: The aim of this study was to determine the applicability and efficiency of automated EAT quantification. Methods: EAT volume was assessed both manually and automatically in 157 patients undergoing coronary CT angiography. Manual assessment consisted of a short-axis-based manual measurement, whereas automated assessment on both contrast and non-contrast-

enhanced data sets was achieved through novel prototype software. Duration of both quantification methods was recorded, and EAT volumes were compared with paired samples t test. Correlation of volumes was determined with intraclass correlation coefficient; agreement was tested with Bland-Altman analysis. The association between EAT and CAD was estimated with logistic regression. Results: Automated quantification was significantly less time consuming than automated quantification (17 ± 2 seconds vs 280 ± 78 seconds; $P < .0001$). Although manual EAT volume differed significantly from automated EAT volume (75 ± 33 cm³ vs 95 ± 45 cm³; $P < .001$), a good correlation between both assessments was found ($r = 0.76$; $P < .001$). For all methods, EAT volume was positively associated with the presence of CAD. Stronger predictive value for the severity of CAD was achieved through automated quantification on both contrast-enhanced and non-contrast-enhanced data sets. Conclusion: Automated EAT quantification is a quick method to estimate EAT and may serve as a predictor for CAD presence and severity. © 2014 Society of Cardiovascular Computed Tomography.

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Export Date: 20 September 2026; Cited By: 33; Correspondence Address: M. Das; Department of Radiology, Maastricht University Medical Center, 6202 AZ Maastricht, P. Debyelaan 25, PO Box 5800, Netherlands; email: m.das@mumc.nl

Automated quantification of photoreceptor outer segments in developing and degenerating retinas on microscopy images across scales

Item Type Journal Article

Author S. Seidemann

Author F. Salomon

Author K.B. Hoffmann

Author T. Kurth

Author I.F. Sbalzarini

Author R. Haase

Author M. Ader

Abstract The functionality of photoreceptors, rods, and cones is highly dependent on their outer segments (POS), a cellular compartment containing highly organized membranous structures that generate biochemical signals from incident light. While POS formation and degeneration are qualitatively assessed on microscopy images, reliable methodology for quantitative analyses is still limited. Here, we developed methods to quantify POS (QuaPOS) maturation and quality on retinal sections using automated image analyses. POS formation was examined during the development and in adulthood of wild-type mice via light microscopy (LM) and transmission electron

microscopy (TEM). To quantify the number, size, shape, and fluorescence intensity of POS, retinal cryosections were immunostained for the cone POS marker S-opsin. Fluorescence images were used to train the robust classifier QuaPOS-LM based on supervised machine learning for automated image segmentation. Characteristic features of segmentation results were extracted to quantify the maturation of cone POS. Subsequently, this quantification method was applied to characterize POS degeneration in “cone photoreceptor function loss 1” mice. TEM images were used to establish the ultrastructural quantification method QuaPOS-TEM for the alignment of POS membranes. Images were analyzed using a custom-written MATLAB code to extract the orientation of membranes from the image gradient and their alignment (coherency). This analysis was used to quantify the POS morphology of wild-type and two inherited retinal degeneration (“retinal degeneration 19” and “rhodopsin knock-out”) mouse lines. Both automated analysis technologies provided robust characterization and quantification of POS based on LM or TEM images. Automated image segmentation by the classifier QuaPOS-LM and analysis of the orientation of membrane stacks by QuaPOS-TEM using fluorescent or TEM images allowed quantitative evaluation of POS formation and quality. The assessments showed an increase in POS number, volume, and membrane coherency during wild-type postnatal development, while a decrease in all three observables was detected in different retinal degeneration mouse models. All the code used for the presented analysis is open source, including example datasets to reproduce the findings. Hence, the QuaPOS quantification methods are useful for in-depth characterization of POS on retinal sections in developmental studies, for disease modeling, or after therapeutic interventions affecting photoreceptors. Copyright © 2024 Seidemann, Salomon, Hoffmann, Kurth, Sbalzarini, Haase and Ader.

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Automated quantification of spikes**Item Type** Journal Article**Author** V. Chavakula**Author** I. Sánchez Fernández**Author** J.M. Peters**Author** G. Popli**Author** W. Bosl**Author** S. Rakhade**Author** A. Rotenberg**Author** T. Loddenkemper

Abstract Methods for rapid and objective quantification of interictal spikes in raw, unprocessed electroencephalogram (EEG) samples are scarce. We evaluated the accuracy of a tailored automated spike quantification algorithm. The automated quantification was compared with the quantification by two board-certified clinical neurophysiologists (gold-standard) in five steps: 1) accuracy in a single EEG channel (5 EEG samples), 2) accuracy in multiple EEG channels and across different stages of the sleep-wake cycles (75 EEG samples), 3) capacity to detect lateralization of spikes (6 EEG samples), 4) accuracy after application of a machine-learning mechanism (11 EEG samples), and 5) accuracy during wakefulness only (8 EEG samples). Our method was accurate during all stages of the sleep-wake cycle and improved after the application of the machine-learning mechanism. Spikes were correctly lateralized in all cases. Our automated method was accurate in quantifying and detecting the lateralization of interictal spikes in raw unprocessed EEG samples. © 2012 Elsevier Inc.

Date 2013**DOI** 10.1016/j.yebeh.2012.11.048**URL** <https://www.scopus.com/pages/publications/84872039599?origin=resultslist>**Archive** Scopus**Language** English**Volume** 26**Pages** 143-152**Publication** Epilepsy and Behavior**ISSN** 15255069 (ISSN)**Issue** 2

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Humans, machine learning, article, algorithm, Algorithms, accuracy, Diagnosis, Signal Processing, Computer-Assisted, electroencephalogram, Electroencephalography, Sleep, Epilepsy, automated quantification, Cerebral Cortex, Automated quantification methods, Electrical status epilepticus in sleep, sleep waking cycle, Sleep-wake cycle, Spike detection, Spike focus lateralization, spike wave, Spikes, Wavelet Analysis, Wavelet transform

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Automated quantification of vesicoureteral reflux using machine learning with advancing diagnostic precision

Item Type Journal Article

Author M. Alqaraleh

Author M. Al-Batah

Author M. Salem Alzboon

Author E. Alzaghouli

Abstract This article uses machine learning to quantify vesicoureteral reflux (VUR). VCUGs in pediatric urology are used to diagnose VUR. The goal is to increase diagnostic precision. Various machine learning models categorize VUR grades (Grade 1 to Grade 5) and are evaluated using performance metrics and confusion matrices. Study datasets come from internet repositories with repository names and accession numbers. Machine learning models performed well across several measures. KNN, Random Forest, AdaBoost, and CN2 Rule Induction consistently scored 100 % in AUC, CA, F1-score, precision, recall, MCC, and specificity. These models classified grades well individually and collectively. In contrast, the Constant model performed poorly across all criteria, suggesting its inability to categorize VUR grades reliably. With the most excellent average performance ratings, the CN2 Rule Induction model excelled at grade categorization. Confusion matrices demonstrate that machine

learning models predict VUR grades. The large diagonal numbers of the matrices show that the models are regularly predicted effectively. However, the Constant model's constant Grade 5 forecast reduced its differentiation. This study shows that most machine learning methods automate VUR measurement. The findings aid objective pediatric urology grading and radiographic evaluation. The CN2 Rule Induction model accurately classifies VUR grades. Machine learning-based diagnostic techniques may increase diagnostic precision, clinical decision-making, and patient outcomes. © 2025; Los autores.

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Short Title Cuantificación automatizada del reflujo vesicoureteral mediante aprendizaje automático para mejorar la precisión diagnóstica

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Automated skin lesion detection and prevalence estimation in Tamanend's bottlenose dolphins

Item Type Journal Article

Author C.J. Murphy

Author M.A. Collier

Author A.-M. Jacoby

Author E.M. Patterson

Author M.M. Wallen

Author J. Mann

Author S. Bansal

Abstract Anthropogenic global change is occurring at alarming rates, leading to increased urgency in the ability to monitor wildlife health in real time. Monitoring sentinel marine species, such as bottlenose dolphins, is particularly important due to extensive anthropogenic modifications to their habitats. The most common non-invasive method of monitoring cetacean health is documentation of skin lesions, often associated with poor health or disease, but the current methodology is inefficient and imprecise. Recent advancements in technology, such as machine learning, can provide researchers with more efficient ecological monitoring methods to address health questions at both the population and the individual levels. Our work develops a machine learning model to classify skin lesions on the understudied Tamanend's bottlenose dolphins (*Tursiops erebennus*) of the Chesapeake Bay, using manual estimates of lesion presence in photographs. We assess the model's performance and find that our best model performs with a high mean average precision (65.6 %–86.8 %), and generally increased accuracy with improved photo quality. We also demonstrate the model's ability to address ecological questions across scales by generating model-based estimates of lesion prevalence and testing the effect of gregariousness on health status. At the population level, our model accurately estimates a prevalence of 72.1 % spot and 27.3 % fringe ring lesions, with a slight underprediction compared to manual estimates (82.2 % and 32.1 %). On the other hand, we find that individual-level analyses from the model predictions may be more sensitive to data quality, and thus, some individual scale questions may not be feasible to address if data quality is inconsistent. Manually, we do find that lesion presence in individuals suggests a positive relationship between lesion presence and gregariousness. This work demonstrates that object detection models on photographic data are reasonably successful, highly efficient, and provide initial estimates on the health status of understudied populations of bottlenose dolphins. © 2024 The Authors

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Automated Sorting System for Tahiti Lemons Using Raspberry PI

- Item Type** Conference Paper
- Author** C. Janampa-Paitan
- Author** J.E. Flores-Llallico
- Author** B.F. Orellana-Garcia
- Author** H.A.V. Baca
- Abstract** According to FreshFruit in the period 2023 the production of Tahitian lemons was 13 515 tons for USD 14 million being one of the most important citrus fruits in the Peruvian economy. This work develops an automated system for the classification of Tahiti lemons by size and maturity grade from green to yellow. For the simulation of the classification of Tahiti lemons by size from 3 to 6 cm, the bottleneck was determined by direct observation and then the mechanical system was programmed for the classification by size and degree of maturity through the detection of color by HSV to classify Tahiti lemons greater than 10% damage or by maturity, and then obtain the simulation in Factory IO and TIA PORTAL with connection to PLC S7-1200 1214 DC/DC/DC and a HMI TP700. Finally, the grading proposal was implemented in which 100% of the Tahiti lemons were recognized and packed for export through the interactive HMI screen, being able to classify and count them automatically, which has a graphical environment for the operator to manipulate them. © 2023 IEEE.
- Date** 2023
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- Language** English
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Automated Visual Large Scale Monitoring of Faunal Biodiversity

Item Type Journal Article

Author B. Radig

Author P. Bodesheim

Author D. Korsch

Author J. Denzler

Author T. Haucke

Author M. Klasen

Author V. Steinhage

Abstract Abstract: To observe biodiversity, the variety of plant and animal life in the world or in a particular habitat, human observers make the most common examinations, often assisted by technical equipment. Measuring objectively the number of different species of animals, plants, fungi, and microbes that make up the ecosystem can be difficult. In order to monitor changes in biodiversity, data have to be compared across space and time. Cameras are an essential sensor to determine the species range, abundance, and behavior of animals. The millions of recordings from camera traps set up in natural environments can no longer be analyzed by biologists. We started research on doing this analysis automatically without human interaction. The focus of our present sensor is on image capture of wildlife and moths. Special hardware elements for the detection of different species are designed, implemented, tested, and improved, as well as the algorithms for classification and counting of samples from images and image sequences, e.g., to calculate presence, absence, and abundance values or the duration of characteristic activities related to the spatial mobilities. For this purpose, we are developing stereo camera traps that allow spatial reconstruction of the observed animals. This allows three-dimensional coordinates to be recorded and the shape to be characterized. With this additional feature data, species identification and movement detection are facilitated. To classify and count moths, they are attracted to an illuminated screen, which is then photographed at intervals by a high-resolution color camera. To greatly reduce the volume of data, redundant elements and elements that are consistent from image to image are eliminated. All design decisions take into account that at remote sites and in fully autonomous operation, power supply on the one hand and possibilities for data exchange with central servers on the other hand are limited. Installation at hard-to-reach locations requires a sophisticated and demanding system design with an optimal balance

between power requirements, bandwidth for data transmission, required service and operation in all environmental conditions for at least ten years. © 2021, Pleiades Publishing, Ltd.

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Automatic 1D 1H NMR metabolite quantification for bioreactor monitoring

Item Type Journal Article

Author R.C.C. Wang

Author D.A. Campbell

Author J.R. Green

Author M. Čuperlović-Culf

Abstract High-throughput metabolomics can be used to optimize cell growth for enhanced production or for monitoring cell health in bioreactors. It has applications in cell and gene therapies, vaccines, biologics, and bioprocessing. NMR metabolomics is a method that allows for fast and reliable experimentation, requires only minimal sample preparation, and can be set up to take online measurements of cell media for bioreactor monitoring. This type of application requires a fully automated metabolite quantification method that can be linked with high-throughput measurements. In this review, we discuss the quantifier requirements in this type of application, the existing methods for NMR metabolomics quantification, and the performance of three existing quantifiers in the context of NMR metabolomics for bioreactor monitoring. © 2021 by the authors. Licensee MDPI, Basel, Switzerland.

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Automatic 3D Postoperative Evaluation of Complex Orthopaedic Interventions

Item Type Journal Article

Author J. Ackermann

Author A. Hoch

Author J.G. Snedeker

Author P.O. Zingg

Author H. Esfandiari

Author P. Färnstahl

Abstract In clinical practice, image-based postoperative evaluation is still performed without state-of-the-art computer methods, as these are not sufficiently automated. In this study we propose a fully automatic 3D postoperative outcome quantification method for the relevant steps of orthopaedic interventions on the example of Periacetabular Osteotomy of Ganz (PAO). A typical orthopaedic intervention involves cutting bone, anatomy manipulation and repositioning as well as implant placement. Our method includes a segmentation based deep learning approach for detection and quantification of the cuts. Furthermore, anatomy repositioning was quantified through a multi-step registration method, which entailed a coarse alignment of the pre- and postoperative CT images followed by a fine fragment alignment of the repositioned anatomy. Implant (i.e., screw) position was identified by 3D Hough transform for line detection combined with fast voxel traversal based on ray tracing. The feasibility of our approach was investigated on 27 interventions and compared against manually performed 3D outcome evaluations. The results show that our method can accurately assess the quality and accuracy of the surgery. Our evaluation of the fragment repositioning showed a cumulative error for the coarse and fine alignment of 2.1 mm. Our evaluation of screw placement accuracy resulted in a distance error of 1.32 mm for screw head location and an angular deviation of 1.1° for screw axis. As a next step we will explore generalisation capabilities by applying the method to different interventions. © 2023 by the authors.

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Automatic Classification and Quantification of Cell Adhesion Locations on the Endothelium

Item Type Journal Article

Author J. Wei

Author B. Cai

Author L. Zhang

Author B.M. Fu

Abstract To target tumor hematogenous metastasis and to understand how leukocytes cross the microvessel wall to perform immune functions, it is necessary to elucidate the adhesion location and transmigration pathway of tumor cells and leukocytes on/ across the endothelial cells forming the microvessel wall. We developed an algorithm to classify and quantify cell adhesion locations from photomicrographs taken from the experiments of tumor cell/leukocyte adhesion in individual microvessels. The first step is to identify the microvessel by a novel gravity-field dynamic programming (DP) procedure. Next, an anisotropic image smoothing suppresses noises without unduly mitigating crucial visual features. After an adaptive thresholding process further tackles uneven lighting conditions during the imaging process, a series of local mathematical morphological operators and eigenanalysis identify tumor cells or leukocytes. Finally, a novel double component labeling procedure categorizes the cell adhesion locations. This algorithm has generated consistently encouraging performances on photomicrographs obtained from in vivo experiments for tumor cell and leukocyte adhesion locations on the endothelium forming the microvessel wall. Compared with human experts, this algorithm used 1/500–1/200 of the time without having the errors due to human subjectivity. Our automatic classification and quantification method provides a reliable and cost effective approach for biomedical image processing. © 2014, Biomedical Engineering Society.

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Automatic classified system for customs export cargo

Item Type Conference Paper

Author M.-H. Hsu

Author Y.-F. Wang

Author D.-A. Chiang

Author Y.-L. Chuang

Abstract Test Classification is functioned to categorized the unclassified text documents into the pre-defined category. We can take the customs export cargo for example and we chiefly make the comparison for the subjective cargo description of the texts and terms categorized by the typists. Most of the said fields are composed of free content, namely, the contents categorized by identical typists. During various periods, it will also exist in distinct cargo descriptions for the same cargo commodity. This situation cause poor classification results. It is necessary for users to lots of time and effort for analysis, filter and determination with the categorized data finished. Thus, we propose an improved term algorithm allowable for the notion of multi-classification and class priority. It is helpful to enhance the accuracy of existent classification system with the ultimate goal that it is available for users to perform rapid an accurate information inquiry.

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Volume 1

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Text mining, Text categorization, Multi classification

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Automatic complex concrete crack detection and quantification based on point clouds and deep learning

Item Type Journal Article

Author S. Chen

Author G. Fan

Author J. Li

Author H. Hao

Abstract The inspection of cracks on structural surface is an essential component of structural condition assessment. Recent advancements in image-based techniques aim to automate this process and reduce the associated costs. However, these methods typically focus on localized planar cracks, neglecting the more severe and complex multi-planar cracks. Moreover, accurate global positioning and quantification of these complex cracks pose challenges due to image distortions. To tackle these challenges, this paper proposes a point cloud-based framework for the automatic detection and quantification of complex cracks on concrete surfaces. The proposed framework involves generating a semantic point cloud for crack positioning and quantification by integrating a pixel-level crack segmentation network with 3D reconstruction algorithms. With the establishment of the semantic point clouds, a crack skeletonization based crack length quantification method and a crack width quantification algorithm are proposed to facilitate the automated and effective crack quantification. Experimental evaluations are conducted on a test beam with five characteristic flexural cracks. The results demonstrate the superior performance of the proposed framework over conventional methods with a mean relative error of merely 1.3 % in length quantification and a median of absolute errors less than 0.15 mm in crack width measurements. The efficacy and precision of the proposed framework enhance the potential applications of vision techniques for crack inspection. © 2025 The Authors

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automation, Point-clouds, machine learning, Semantic Segmentation, algorithm, quantitative analysis, data processing, detection method, Crack quantification, Detection and quantifications, Cloud processing, concrete, Concrete structural inspection, Crack detection, Crack skeletonization, cracking (fracture), pixel, Planar cracks, Point cloud processing, Point clouds processing, Skeletonization, Strain measurement, Structural inspections, three-dimensional modeling

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Export Date: 20 September 2026; Cited By: 34; Funding details: National Natural Science Foundation of China, NSFC, (52178279, 52108277); Correspondence Address: G. Fan; Earthquake Engineering Research & Test Center, Guangzhou University, Guangzhou, 510006, China; email: gao.fan@gzhu.edu.cn; J. Li; Center of Infrastructural Monitoring and Protection, School of Civil and Mechanical Engineering, Curtin University, Bentley, 6102, Australia; email: junli@curtin.edu.au; CODEN: ENSTD

Automatic coronary calcium scoring in low-dose chest computed tomography

Item Type Journal Article**Author** I. Išgum**Author** M. Prokop**Author** M. Niemeijer**Author** M.A. Viergever**Author** B. Van Ginneken

Abstract The calcium burden as estimated from non-ECG-synchronized computed tomography (CT) exams acquired in screening of heavy smokers has been shown to be a strong predictor of cardiovascular events. We present a method for automatic coronary calcium scoring with low-dose, non-contrast-enhanced, non-ECG-synchronized chest CT. First, a probabilistic coronary calcium map was created using multi-atlas segmentation. This map assigned an a priori probability for the presence of coronary calcifications at every location in a scan. Subsequently, a statistical pattern recognition system was designed to identify coronary calcifications by texture, size, and spatial features; the spatial features were computed using the coronary calcium map. The detected calcifications were quantified in terms of volume and Agatston score. The best results were obtained by merging the results of three different supervised classification systems, namely direct classification with a nearest neighbor classifier, and two-stage classification with nearest neighbor and support vector machine classifiers. We used a total of 231 test scans containing 45 674 mm³ of coronary calcifications. The presented method detected on average 157/198 mm³ (sensitivity 79.2%) of coronary calcium volume with on average 4 mm³ false positive volume. Calcium scoring can be performed automatically in low-dose, noncontrast enhanced, non-ECG-synchronized chest CT in screening of heavy smokers to identify subjects who might benefit from preventive treatment. © 2012 IEEE.

Date 2012**DOI** 10.1109/TMI.2012.2216889**URL** <https://www.scopus.com/pages/publications/84870488848?origin=resultslist>**Archive** Scopus**Language** English**Volume** 31**Pages** 2322-2334**Publication** IEEE Transactions on Medical Imaging**ISSN** 02780062 (ISSN)**Issue** 12

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human, Humans, middle aged, Middle Aged, aged, Aged, Heart, Pattern recognition systems, article, risk factor, Risk Factors, Electrocardiography, support vector machine, methodology, Synchronization, image processing, False positive, Image Processing, Computer-Assisted, coronary artery disease, pathology, smoking, Computerized tomography, computer assisted tomography, Tomography, X-Ray Computed, Supervised classification, Bone, A-priori probabilities, radiography, Radiography, Thoracic, thorax radiography, Lung cancer screening, automated pattern recognition, Pattern Recognition, Automated, Support Vector Machines, Smoking, Coronary Artery Disease, Calcium, Nearest Neighbor classifier, Nearest neighbors, Biomineralization, calcinosis, Calcinosis, Agatston score, Automatic coronary calcium scoring, Cardiovascular event, Cardiovascular risk, cardiovascular risk assessment, chest computed tomography (CT), Chest CT, Computed Tomography, Coronary calcifications, Coronary calcium, Low dose, lung cancer screening, Preventive treatments, Spatial features, Statistical pattern recognition

Notes:

EID: 2-s2.0-84870488848

Export Date: 20 September 2026; Cited By: 118; Correspondence Address: I. Išgum; Image Sciences Institute, University Medical Center Utrecht, 3584 CX Utrecht, Netherlands; email: ivana@isi.uu.nl; CODEN: ITMID

Automatic extraction of land cover statistics from satellite imagery by deep learning

Item Type Journal Article**Author** E. Bernasconi**Author** F. De Fausti**Author** F. Pugliese**Author** M. Scannapieco**Author** D. Zardetto

Abstract In this paper, we address the challenge of producing fully automated land cover estimates from satellite imagery through Deep Learning algorithms. We developed our system according to a tile-based, classify-and-count design. To implement the classification engine of the system, we adopted a cutting-edge Convolutional Neural Network model named Inception-V3, which we customized and trained for scene classification on the EuroSAT dataset. We tested and validated our system on two Sentinel-2 images representing quite different Italian territories (with an area of 751

km² and 443 km², respectively). Because no genuine ground-truth is available for the land cover of these sub-regional territories, we built a pseudo ground-truth by integrating land cover information from flagship European projects CORINE and LUCAS. A critical and careful analysis shows that our automatic land cover estimates are in good agreement with the pseudo ground-truth and offers extensive evidence of the remarkable segmentation ability of our system. The limits of our approach are also critically discussed in the paper and possible countermeasures are illustrated. When compared with traditional projects like CORINE and LUCAS, our automatic land cover estimation system exhibits three fundamental advantages: it can dramatically reduce production costs; it can allow delivering very timely and frequent land cover statistics; it can enable land cover estimation for very small territorial areas, well beyond the NUTS-2 level. As an additional outcome of land cover estimation, our system also automatically generates moderate resolution land cover maps that might be used in cartography projects as an agile first-level tool for map update or change detection purposes. © 2022 - IOS Press. All rights reserved.

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Convolutional Neural Networks, deep learning, computer vision, Land cover, satellite imagery, scene classification

Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: D. Zardetto; Istat - Italian National Institute of Statistics, Rome, Via Cesare Balbo 16, 00184, Italy; email: zardetto@istat.it

Automatic facial spirit classification for traditional Chinese medicine based on mutiple facial features

Item Type Conference Paper

Author C. Sun

Author H. Zhang

Author X. Qiu

Author Q. Peng

Author Z. Sun

Author W. Zhang

Author F. Li

Editor Zheng H.

Editor Hu X.T.

Editor Berrar D.

Editor Wang Y.

Editor Dubitzky W.

Editor Hao J.-K.

Editor Cho K.-H.

Editor Gilbert D.

Abstract This paper presents a new automatically quantitative facial features classification system for TCM spirit diagnosis based on facial features. Facial diagnosis is an important diagnostic method in TCM (Traditional Chinese Medicine) and has been used for a long time. However, this traditional diagnostic method is mainly based on observation by TCM doctors and their personal experience. To develop quantification methods for TCM is very important. First, we capture facial features from video images through some algorithms, and then we use some classification models to analyze them. Finally, IG (Information Gain) is used to analyse which kind of features has good performance. Experiment results show that our system has high classification accuracy. © 2014 IEEE.

Date 2014

DOI 10.1109/BIBM.2014.6999314

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Language English

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Publisher Institute of Electrical and Electronics Engineers Inc.

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Tags:

artificial intelligence, eye detection, spirit diagnosis, TCM objectification

Notes:

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Automatic Monitoring System for Seed Germination Test Based on Deep Learning

Item Type Journal Article

Author Q. Peng

Author L. Tu

Author Y. Wu

Author Z. Yu

Author G. Tang

Author W. Song

Abstract Germination test is an irreplaceable step in seed selection and breeding. The current traditional germination test method must rely on experienced professional technicians to repeatedly classify and count the germination status of seeds and count the germination rate at different moments during the whole test process (usually takes 2 to 10 days). Currently, only the German seed germination detection system (Germination Scanalyzer) can solve this problem, but it is so expensive that it has not been practically promoted. In order to improve breeding efficiency, an automatic monitoring system for seed germination tests based on deep learning was designed. It includes a modified germination thermostat, connected with a three-dimensional movable camera bin with built-in camera; a multifunctional software system capable of online, offline, and sentinel mode monitoring; a dense distributed small target detection algorithm (DDST-CenterNet) for seed germination monitoring systems. The system test results show that the seed germination test automatic monitoring system is low cost, does not depend on the seed background, light, camera model, and other usage environments, and has high scalability. The DDST-CenterNet algorithm proposed in this paper can still maintain high accuracy and good stability in the process of seed target detection and classification as the number and density of seeds increase, which is suitable for a special application scenario of seed germination test. In addition, the algorithm has high computational efficiency and can give detection results at a frame rate of not less than 10fps, which can be used in practical applications. © 2022 Qi Peng et al.

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'current, Deep learning, Computational efficiency, Test method, Cameras, Online systems, Monitoring, Detection system, Software-systems, Cultivation, Automatic monitoring-system, Germination, Germination rates, Germination test, Seed germination, Seed-selection, Test process

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Automatic MRI Quantifying Methods in Behavioral-Variant Frontotemporal Dementia Diagnosis

Item Type Journal Article

Author A. Cajanus

Author A. Hall

Author J. Koikkalainen

Author E. Solje

Author A. Tolonen

Author T. Urhema

Author Y. Liu

Author R.M. Haanpää

Author P. Hartikainen

Author S. Helisalmi

Author V. Korhonen

Author D. Rueckert

Author S. Hasselbalch

Author G. Waldemar

Author P. Mecocci

Author R. Vanninen

Author M. Van Gils

Author H. Soininen

Author J. Lötjönen

Author A.M. Remes

Abstract Aims: We assessed the value of automated MRI quantification methods in the differential diagnosis of behavioral-variant frontotemporal dementia (bvFTD) from Alzheimer disease (AD), Lewy body dementia (LBD), and subjective memory complaints (SMC). We also examined the role of the C9ORF72-related genetic status in the differentiation sensitivity. Methods: The MRI scans of 50 patients with bvFTD (17 C9ORF72 expansion carriers) were analyzed using 6 quantification methods as follows: voxel-based morphometry (VBM), tensor-based morphometry, volumetry (VOL), manifold learning, grading, and white-matter hyperintensities. Each patient was then individually compared to an independent reference group in order to attain diagnostic suggestions. Results: Only VBM and VOL showed utility in correctly identifying bvFTD from our set of data. The overall classification sensitivity of bvFTD with VOL + VBM achieved a total sensitivity of 60%. Using VOL + VBM, 32% were misclassified as having LBD. There was a trend of higher values for classification sensitivity of the C9ORF72 expansion carriers than noncarriers. Conclusion: VOL, VBM, and their combination are effective in differential diagnostics between bvFTD and AD or SMC. However, MRI atrophy profiles for bvFTD and LBD are too similar for a reliable differentiation with the quantification methods tested in this study. © 2018 The Author(s) Published by S. Karger AG, Basel.

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Journal Abbr Dement. Geriatr. Cogn. Dis. Extra

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adult, Article, female, human, male, Machine learning, aged, controlled study, learning, Dementia, sensitivity analysis, clinical article, Alzheimer disease, nuclear magnetic resonance imaging, differential diagnosis, MRI, volumetry, Neuroimaging, priority journal, heterozygote, voxel based morphometry, diffuse Lewy body disease, Frontotemporal lobar degeneration, frontal variant frontotemporal dementia, Frontotemporal dementia, guanine nucleotide exchange C9orf72, memory disorder, morphometry, subjective memory complaint, tensor based morphometry

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Automatic music mood classification by learning cross-media relevance between audio and lyrics

Item Type Conference Paper

Author Y. Xiong

Author F. Su

Author Q. Wang

Abstract Automatic analysis of the mood of a piece of music is of great value in music searching, understanding, recommendation and some other music-related applications. Different from most of previous methods that adopted a discriminative mood classification scheme, in this paper, we propose a generative multimodal method for automatically classifying the mood of a piece of music based on effective learning of the relevance (i.e. the joint distribution) between the audio and the lyrics modalities of music. We present effective algorithms for computing the word-to-audio and word-to-word relations in the music as well as a priori probability of lyrics words, which altogether form the multimodal joint distribution that distinctively captures the intrinsic characteristic of one specific music mood. A piece of music is then classified to the mood category that maximizes this joint probability of different modalities of music data. The experiment results demonstrated the effectiveness of the proposed method. © 2017 IEEE.

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Language English

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Multi-modal, Effective learning, Data description, Automatic analysis, Classification scheme, Joint distributions, Cross-media, Effective algorithms, Music mood classification, Relevance model, Relevance models, SVDD

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Automatic quantification of HER2 gene amplification in invasive breast cancer from chromogenic in situ hybridization whole slide images

Item Type Journal Article

Author M.S. Hossain

Author M.G. Hanna

Author N. Uraoka

Author T. Nakamura

Author M. Edelweiss

Author E. Brogi

Author M.R. Hameed

Author M. Yamaguchi

Author D.S. Ross

Author Y. Yagi

Abstract Human epidermal growth factor receptor 2 (HER2), a transmembrane tyrosine kinase receptor encoded by the ERBB2 gene on chromosome 17q12, is a predictive and prognostic biomarker in invasive breast cancer (BC). Approximately 20% of BC are HER2-positive as a result of ERBB2 gene amplification and overexpression of the HER2 protein. Quantification of HER2 is performed routinely on all invasive BCs, to assist in clinical decision making for prognosis and treatment for HER2-positive BC patients by manually counting gene signals. We propose an automated system to quantify the HER2 gene status from chromogenic in situ hybridization (CISH) whole slide images (WSI) in invasive BC. The proposed method selects untruncated and nonoverlapped singular nuclei from the cancer regions using color unmixing and machine learning techniques. Then, HER2 and chromosome enumeration probe 17 (CEP17) signals are detected based on the RGB intensity and counted per nucleus. Finally, the HER2-to-CEP17 signal ratio is calculated to determine the HER2 amplification status following the ASCO/CAP 2018 guidelines. The proposed method reduced the labor and time for the quantification. In the experiment, the correlation coefficient between the proposed automatic CISH quantification method and pathologist manual enumeration was 0.98. The p-values larger than 0.05 from the one-sided paired t-test ensured that the proposed method yields statistically

indifferent results to the reference method. The method was established on WSI scanned by two different scanners. Through the experiments, the capability of the proposed system has been demonstrated. © 2019 Society of Photo-Optical Instrumentation Engineers (SPIE).

Date 2019

DOI 10.1117/1.JMI.6.4.047501

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Publisher SPIE

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Article, Automation, human, Decision making, Learning systems, machine learning, Medical imaging, controlled study, false positive result, Diagnosis, Diseases, noise reduction, Patient treatment, signal detection, diagnostic accuracy, Digital pathologies, digital pathology, false negative result, gene amplification, feature selection, fluorescence microscopy, epidermal growth factor receptor 2, breast cancer, cell differentiation, Breast Cancer, Genes, image registration, Amino acids, tumor invasion, whole slide imaging, Amplification, automatic CISH quantification, chromogenic in situ hybridization, chromosome 17, Gene amplification, HER2 gene amplification, HER2-positive breast cancer, in situ hybridization, watershed algorithm, Whole slide imaging (WSI)

Notes:

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Export Date: 20 September 2026; Cited By: 29; Correspondence Address: M.S. Hossain; Tokyo Institute of Technology, School of Engineering, Department of Information and Communications Engineering, Yokohama, Japan; email: shakhawat.h.aa@m.titech.ac.jp

Automatic quantification of living cells via a non-invasive achromatic colorimetric sensor through machine learning-assisted image analysis using a smartphone

Item Type Journal Article

Author D.H. Jung

Author Y. Kim

Author H.H. Cho

Author B. Lee

Author S.-J. Suh

Author J.H. Heo

Author J.H. Lee

Abstract Several cell quantification techniques including optical and electrochemical methods have been developed. However, these techniques have certain limitations such as the requirement of considerable effort and time or insufficient feasibility and accessibility. Moreover, cell quantification is generally performed in an invasive manner, which makes performing continuous cell experiments difficult and leads to the requirement of large number of samples. In this study, a simple, rapid, and accurate in-situ cell quantification method using an achromatic colorimetric sensor system is proposed. A non-invasive achromatic colorimetric cell quantitation sensor (Ach-CQ sensor) is developed based on the subtractive color model. The color of the sensor changes from black to orange owing to the pH change occurring due to the release of acid molecules such as proton donors and lactate during cellular respiration. The Ach-CQ sensor exhibits a higher color contrast difference and effectively determines the number of cells through color changes within 20 s compared with that of a conventional pH indicator such as bromothymol blue. Moreover, machine learning-assisted, saturation-based image analysis is applied for the facile quantification of living cells to achieve automatic quantification of living cells in a non-invasive manner with a limit of detection of 51×10^4 cells in the case of HeLa cells using a smartphone. Additionally, a 3D-printed Ach-CQ sensor kit is fabricated for on-site cell quantification. © 2022 Elsevier B.V.

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Automatic quantification of lymphocyte vacuolization in peripheral blood smears of patients with Batten's disease (CLN3 disease)

Item Type Journal Article**Author** L.J.P. Nonkes**Author** W.F.E. Kuper**Author** K. Berrens-Hogenbirk**Author** R.E.A. Musson**Author** P.M. van Hasselt**Author** A. Huisman

Abstract Quantifying lymphocyte vacuolization in peripheral blood smears (PBSs) serves as a measure for disease severity in CLN3 disease—a lysosomal storage disorder of childhood-onset. However, thus far quantification methods are based on labor-intensive manual assessment of PBSs. As machine learning techniques like convolutional neural networks (CNNs) have been deployed quite successfully in detecting pathological features in PBSs, we explored whether these techniques could be utilized to automate quantification of lymphocyte vacuolization. Here, we present and validate a deep learning pipeline that automates quantification of lymphocyte vacuolization. By using two CNNs in succession, trained for cytoplasm-segmentation and vacuolization-detection, respectively, we obtained an excellent correlation with manual quantification of lymphocyte vacuolization ($r = 0.98$, $n = 40$). These results show that CNNs can be utilized to automate the otherwise cumbersome task of manually quantifying lymphocyte vacuolization, thereby aiding prompt clinical decisions in relation to CLN3 disease, and potentially beyond. © 2021 The Authors. JIMD Reports published by John Wiley & Sons Ltd on behalf of SSIEM.

Date 2021**DOI** 10.1002/jmd2.12191**URL** <https://www.scopus.com/pages/publications/85102257035?origin=resultslist>**Archive** Scopus**Language** English**Volume** 58**Publisher** John Wiley and Sons Inc**Pages** 100-103**Publication** JIMD Reports**ISSN** 21928304 (ISSN)**Issue** 1**Journal Abbr** JIMD Reports

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Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: P.M. van Hasselt; Department of Metabolic Diseases, Wilhelmina Children's Hospital, University Medical Center Utrecht, Utrecht University, Utrecht, Netherlands; email: p.vanhasselt@umcutrecht.nl

Automatic recognition of mitotic HEp-2 cells

Item Type Conference Paper**Author** G. Percannella**Author** P. Soda**Author** M. Vento

Abstract Indirect immunofluorescence (IIF) is the recommended method to diagnose the presence of antinuclear autoantibodies in patient serum, but it suffers from several issues limiting its reliability and reproducibility. Diagnostic procedure consists of the following four steps: image acquisition, mitosis detection, fluorescence intensity classification and staining pattern recognition. The detection of mitotic cells, which rarely occur in a well in comparison to other cells, permits to assess the correctness of slide preparation process, providing also information on image staining pattern. In order to support IIF diagnostic procedure, recent researches have been directed towards the development of computer-aided-diagnosis systems, focusing on image acquisition, fluorescence intensity classification and staining pattern recognition. In this respect, we first present a set of features specifically tailored for this application and then we propose mitotic cells classification system able to handle their low a priori probability based on a class skew learning method. The approach has been evaluated on an annotated dataset of mitotic cells and successfully compared to five learning methods. Copyright © 2011 ibai-publishing.

Date 2011**URL** <https://www.scopus.com/pages/publications/84896896284?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Adv. Mass Data Anal. Images Signals Med., Biotechnol., Chem. Food Ind. - Int. Conf., MDA, Proc.**ISBN** 978-394295202-6 (ISBN)**Pages** 91-105**Proceedings Title** Adv. Mass Data Anal. Images Signals Med., Biotechnol., Chem. Food Ind. - Int. Conf., MDA, Proc.

Conference Name Advances in Mass Data Analysis of Images and Signals in Medicine, Biotechnology, Chemistry and Food Industry - 6th International Conference, MDA 2011, Proceedings
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Tags:

Pattern recognition systems, Signal detection, Computer aided diagnosis, Classification system, Image acquisition, Cells, Cytology, A-priori probabilities, Fluorescence, Automatic recognition, Chemical analysis, Antinuclear auto-antibodies, Diagnostic procedure, Indirect immunofluorescence, Biotechnology, Fluorescence intensities, Preparation process

Notes:

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Balanced knowledge distillation for long-tailed learning

Item Type Journal Article

Author S. Zhang

Author C. Chen

Author X. Hu

Author S. Peng

Abstract Deep models trained on long-tailed datasets exhibit unsatisfactory performance on tail classes. Existing methods usually modify the classification loss to increase the learning focus on tail classes, which unexpectedly sacrifice the performance on head classes. In fact, this scheme leads to a contradiction between the two goals of long-tailed learning, i.e., learning generalizable representations and facilitating learning for tail classes. In this work, we explore knowledge distillation in long-tailed scenarios and propose a novel distillation framework, named Balanced Knowledge Distillation (BKD), to disentangle the contradiction between the two goals and achieve both simultaneously. Specifically, given a teacher model, we train the student model by minimizing the combination of an instance-balanced classification loss and a class-balanced distillation loss. The former benefits from the sample diversity and learns generalizable representation, while the latter considers the class priors and facilitates learning for tail classes. We conduct extensive experiments on several long-tailed benchmark datasets and demonstrate that the proposed BKD is an effective knowledge distillation framework in long-tailed scenarios, as well as a competitive method for long-tailed learning. Our source code is available: <https://github.com/EricZsy/BalancedKnowledgeDistillation>. © 2023 Elsevier B.V.

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Tags:

human, Performance, Classification (of information), Learning systems, article, Distillation, Knowledge distillation, vision, Text classification, Text processing, learning, Benchmark datasets, Student Modeling, distillation, Learn+, Long-tailed learning, Source codes, teacher, Teacher models, Vision and text classification

Notes:

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Bamboo phase quantification using thermogravimetric analysis: deconvolution and machine learning

Item Type Journal Article

Author F. de Campos Vitorino

Author M. Nazarkovsky

Author A. Azadeh

Author C. Martins

Author B.M. da Cunha Gomes

Author J. Dweck

Author R.D.T. Filho

Author H. Savastano

Abstract The focus of this paper is to provide a fast and reliable quantification method of bamboo's main chemical components. Therefore, thermogravimetric analysis was used to determine holocellulose and lignin content in different bamboo specimens.

The influence of nitrogen vs. air atmospheres was investigated on the thermal degradation behavior of *Phyllostachys edulis* (Moso), *Bambusa vulgaris* (BV) and Iranian *Phyllostachys* (IR) bamboos. Due to peaks overlapping, the deconvolution process was carried out to resolve hidden peaks and to allow adequate phase quantification. Also, a set of machine learning (ML) algorithms was applied to predict the composition of the studied bamboos within the 200–500 °C range in their TGA-DTG profiles. The ensembles of the ML models at $R^2 > 0.99$ proved a connection between the features in thermogravimetric curves with two concentrations of the main components, which were preliminarily established by means of chemical extraction from the respective samples. © 2022, The Author(s), under exclusive licence to Springer Nature B.V.

Date 2023

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Journal Abbr Cellulose

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Tags:

Machine-learning, Machine learning, Quantification methods, Deconvolutions, Air atmosphere, Bamboo, Chemical component, Chemicals, Components, Degradation behavior, Gravimetry, Holocellulose, Lignin contents, Machinery, Phase quantification, *Phyllostachys*, Thermal Analysis, Thermal degradation', Thermogravimetric analysis

Notes:

EID: 2-s2.0-85143597667

Export Date: 20 September 2026; Cited By: 6; Funding details: Fundação de Amparo à Pesquisa do Estado de São Paulo, FAPESP, (2018/25011–9, 307723/2017–8); Hebrew University of Jerusalem, HUIJ; Conselho Nacional de Desenvolvimento Científico e Tecnológico, CNPq; Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro, FAPERJ; Fundação Coordenação de Projetos, Pesquisas e Estudos Tecnológicos, COPPETEC Foundation; Pontifícia Universidade Católica do Rio de Janeiro, PUC-Rio; Correspondence Address: M. Nazarkovsky; Chemistry Department, Pontifical Catholic University of Rio de Janeiro, Rio de Janeiro, 225 Marquês de São Vicente Str., RJ, 22451-900, Brazil; email: nazarkovsky_m@esp.puc-rio.br; CODEN: CELLE

Bayes covariant multi-class classification

Item Type Journal Article

Author O. Šuch

Author S. Barreda

Abstract We consider multi-class classification models built from complete sets of pairwise binary classifiers. The Bradley–Terry model is often used to estimate posterior distributions in this setting. We introduce the notion of Bayes covariance, which holds if the multi-class classifier respects multiplicative group action on class priors. As a consequence, a Bayes covariant method yields the same result whether new priors are considered before or after combination of the individual classifiers, which has several practical advantages for systems with feedback. In the paper, we construct a Bayes covariant combining method and compare it with previously published methods in both Monte Carlo simulations as well as on a practical speech frame recognition task. © 2016 Elsevier B.V.

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DOI 10.1016/j.patrec.2016.08.014

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Journal Abbr Pattern Recogn. Lett.

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Tags:

Intelligent systems, Classification (of information), Learning systems, Speech recognition, Classifiers, Multi-class classification, Monte Carlo methods, Bins, Bayes classifier, Bayes Classifier, Bradley–Terry model, Combining binary classifiers, TIMIT, Vowel classification

Notes:

EID: 2-s2.0-84986889095

Export Date: 20 September 2026; Cited By: 10; Funding details: Agentúra na Podporu Výskumu a Vývoja, APVV, (APVV-0219-12, APVV-14-0560); Correspondence Address: O. Šuch; University of Žilina, Žilina, Slovakia; email: ondrej.such@fri.uniza.sk

BayesAHDD: A new Bayesian rule-based adaptive hypersphere data description for few-shot one-class classification

Item Type Journal Article**Author** Y. Ren**Author** X. Liu**Author** Y. Pei**Author** Y. Li**Author** Y. Wei

Abstract Few-shot one-class classification (FS-OCC) is a challenging classification problem that involves learning from a very limited number of training samples, all from a single class. Recently, several data description methods have been proposed to address the FS-OCC problem. Unlike conventional one-class classification problems, the few-shot setting requires the model to generalize to novel tasks with previously unseen positive classes. Most existing methods learn decision boundaries in the feature space without explicitly modeling the underlying data distributions, which limits the generalization ability of the learned representations. To address this issue, we propose Bayesian Rule-based Adaptive Hypersphere Data Description (BayesAHDD), a probabilistic framework that represents data with multivariate Gaussian distributions and performs classification according to the Bayes decision rule. Based on the assumption that negative samples are more dispersed in the feature space, BayesAHDD models the negative class by scaling the positive class variance vector element-wise using a learnable vector. To address the challenges of exploding gradients and numerical overflow, we impose a lower bound on the positive class variance vector and introduce a trainable parameter that integrates the class prior probability ratio with the normalization constants of the Gaussian class-conditional densities. Experimental results on both benchmark and domain-specific datasets show that BayesAHDD consistently outperforms existing baselines and state-of-the-art FS-OCC methods. Moreover, quantitative analysis demonstrates that the learned feature representations exhibit superior discriminative ability compared to those produced by previous approaches. © 2025 Published by Elsevier Ltd.

Date 2026**DOI** 10.1016/j.eswa.2025.129647**URL** <https://www.scopus.com/pages/publications/105020587390?origin=resultslist>**Archive** Scopus**Language** English**Volume** 298**Publisher** Elsevier Ltd**Publication** Expert Systems with Applications**ISSN** 09574174 (ISSN)**Journal Abbr** Expert Sys Appl**Date Added** 20/09/2026, 9:51:45 pm**Modified** 20/09/2026, 9:51:45 pm**Tags:**

Classification (of information), Gaussian distribution, Bayesian networks, Sampling, Barium compounds, Classifiers, Vector spaces, Rule based, Training sample, Bayes decision rule, Bayes decision rules, Bayesian rule, Data description, Feature space, Few-shot one-class classification, Hyper-spheres, Multivariate Gaussian distribution, Multivariate Gaussian Distributions, One-class classification, One-class Classification

Notes:

EID: 2-s2.0-105020587390

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: X. Liu; School of Computer Science, Beijing Institute of Technology, Beijing, 100081, China; email: liuxiabi@bit.edu.cn; CODEN: ESAPE

Bayesian classification of the ice cover of the Arctic seas**Item Type** Journal Article**Author** N.Y. Zakhvatkina**Author** I.A. Bychkova

Abstract A classification of sea ice in the Arctic by age (multiyear; first-year; and first-year deformed ice, nilas, etc.) is developed based on the Bayesian approach using satellite radar data and taking into account regional peculiarities of these types of ice for different sectors of the Arctic. Estimations of a priori probabilities for each ice type, which are required for the use of the Bayesian classification, are obtained by the analysis of ice charts in the Arctic seas developed at the AARI in 2008–2013 using satellite data. A posterior probabilities are estimated visually by an expert. Types of sea ice distinguished by the expert on satellite images make it possible to create sample values of the radar-scattering cross section (RSCS). Examples of the proposed Bayesian classification of ice in the Laptev Sea according to Envisat satellite data are given. © 2015, Pleiades Publishing, Ltd.

Date 2015**DOI** 10.1134/S0001433815090212**URL** <https://www.scopus.com/pages/publications/84954320762?origin=resultslist>**Archive** Scopus**Language** English**Volume** 51**Publisher** Maik Nauka-Interperiodica Publishing**Pages** 883-888**Publication** Izvestiya - Atmospheric and Ocean Physics**ISSN** 00014338 (ISSN)**Issue** 9**Journal Abbr** Izv. Atmos. Ocean Phys.**Date Added** 20/09/2026, 9:52:59 pm**Modified** 20/09/2026, 9:52:59 pm**Tags:**

satellite data, Bayesian analysis, Bayesian classification, synthetic aperture radar, satellite, a priori probability estimation, Arctic Ocean, Arctic Seas, Envisat, ice cover, Laptev Sea, sea ice

Notes:

EID: 2-s2.0-84954320762

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: I.A. Bychkova; Arctic and Antarctic Research Institute, St. Petersburg, Russian Federation; email: natalia.piotrovskaya@niersc.spb.ru

Bayesian classifier based on the multivariate normal distribution

Item Type Journal Article

Author I.F. Iatan

Abstract The aim of this paper is the Bayesian estimation techniques to obtain the form of the a posteriori density $p(\mu|D)$ and the desired probability density $p(X|D)$ in the case when $p(X|\mu) \sim N(\mu, \Sigma)$. The treatment of the multivariate case in which Σ is known but μ is not represents the generalization of the univariate case (see [5]). This approach is then used in Bayesian classification. COPYRIGHT 2008 EUDOXUS PRESS, LLC.

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Archive Scopus

Language English

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ISSN 15211398 (ISSN)

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Journal Abbr J. Comput. Anal. Appl.

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Tags:

Bayes theorem, Conditional distribution, Bayesian estimation, A priori probability, A posteriori probability, Posterior distribution

Notes:

EID: 2-s2.0-45949090337

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: I. F. Iatan; Department of Mathematics and Informatics, Technical University of Civil Engineering, Bucharest, Romania; email: iuliafi@yahoo.com

Bayesian classifier for multi-oriented video text recognition system

Item Type Journal Article

Author S. Roy

Author P. Shivakumara

Author P.P. Roy

Author U. Pal

Author C.L. Tan

Author T. Lu

Abstract Developing an automatic system for recognizing video texts such as signboards, street names, room numbers, building names and hotels names is challenging due to low resolution, complex background, font or font size variations, and multiple orientations of texts. In this paper, we develop a new system to recognize video texts through binarization by introducing a Bayesian classifier. We explore wavelet decomposition and gradient sub-bands to enhance text information in video. The enhanced information is used in different ways to calculate the requirement of Bayesian classifier, such as a priori probability and conditional probabilities of text pixels to estimate the posterior probability automatically, which results in text components. Connected component analysis is then applied to restore missing text information before sending it to an OCR engine if any disconnection exists in the text components. Experimental results on video data, the benchmark ICDAR scene character data (camera images) and arbitrary orientation data (camera images) show that the proposed method outperforms existing baseline methods in terms of recognition rates at both character and pixel levels. ©2015 Elsevier Ltd. All rights reserved.

Date 2015

DOI 10.1016/j.eswa.2015.02.030

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Journal Abbr Expert Sys Appl

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Classification (of information), Image segmentation, Wavelet decomposition, Cameras, Probability, Character recognition, Pixels, Bayesian classifier, Subbands, Text binarization, Text lines, Video text binarization, Video text lines, Video text recognition, Wavelet and gradient sub-bands

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Bayesian Embeddings for Few-Shot Open World Recognition

Item Type Journal Article

Author J. Willes

Author J. Harrison

Author A. Harakeh

Author C. Finn

Author M. Pavone

Author S.L. Waslander

Abstract As autonomous decision-making agents move from narrow operating environments to unstructured worlds, learning systems must move from a closed-world formulation to an open-world and few-shot setting in which agents continuously learn new classes from small amounts of information. This stands in stark contrast to modern machine learning systems that are typically designed with a known set of classes and a large number of examples for each class. In this work we extend embedding-based few-shot learning algorithms to the open-world recognition setting. We combine Bayesian non-parametric class priors with an embedding-based pre-training scheme to yield a highly flexible framework which we refer to as few-shot learning for open world recognition (FLOWR). We benchmark our framework on open-world extensions of the common MiniImageNet and TieredImageNet few-shot learning datasets. Our results show, compared to prior methods, strong classification accuracy performance and up to a 12% improvement in H-measure (a measure of novel class detection) from our non-parametric open-world few-shot learning scheme. © 1979-2012 IEEE.

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DOI 10.1109/TPAMI.2022.3201541

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Issue 3

Journal Abbr IEEE Trans Pattern Anal Mach Intell

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Notes:

EID: 2-s2.0-85137900454

Export Date: 20 September 2026; Cited By: 25; Correspondence Address: J. Willes; University of Toronto, Toronto, M5T 2S8, Canada; email: john.willes@mail.utoronto.ca; CODEN: ITPID

Bayesian Physics Informed Neural Networks for Reliable Transformer Prognostics

Item Type Conference Paper

Author I. Ramirez

Author J. Alcibar

Author J. Pino

Author M. Sanz

Author D. Pardo

Author J.I. Aizpurua

Editor Kulkarni C.S.

Editor Orchard M.E.

Abstract Scientific Machine Learning (SciML) integrates physics and data into the learning process, offering improved generalization compared with purely data-driven models. Despite its potential, applications of SciML in prognostics remain limited, partly due to the complexity of incorporating partial differential equations (PDEs) for ageing physics and the scarcity of robust uncertainty quantification methods. This work introduces a Bayesian Physics-Informed Neural Network (B-PINN) framework for probabilistic prognostics estimation. By embedding Bayesian Neural Networks into the PINN architecture, the proposed approach produces principled, uncertaintyaware predictions. The method is applied to a transformer ageing case study, where insulation degradation is primarily driven by thermal stress. The heat diffusion PDE is used as the physical residual, and different prior distributions are investigated to examine their impact on predictive posterior distributions and their ability to encode a priori physical knowledge. The framework is validated against a finite element model developed and tested with real measurements from a solar power plant. Results, benchmarked against a dropout-PINN baseline, show that the proposed B-PINN delivers more reliable prognostic predictions by accurately quantifying predictive uncertainty. This capability is crucial for supporting robust and informed maintenance decision-making in critical power assets. © 2025, Prognostics and Health Management Society. All rights reserved.

Date 2025

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Tags:

Decision making, Learning systems, Machine-learning, Machine learning, Uncertainty analysis, Neural-networks, Neural networks, Bayesian networks, Solar energy, Electric transformers, Reliable transformers, Bayesian, Scientific machine learning, Bayesian neural networks, Learning process, Bayesian Neural Networks, Electrical transformers, Electrical Transformers, Physic informed neural network, Physics Informed Neural Networks, Prognostic, Prognostics, Scientific Machine Learning

Notes:

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Bayesian Population-Based Prevalence Estimation of KCNV2-Associated Retinopathy: A Comparative Analysis of Disease-Associated Variant Frequencies in Russian and Global Populations

Item Type Journal Article

Author A.V. Rozhkova

Author E.A. Rutkovskaya

Author V.V. Kadyshev

Author A.A. Esibov

Author A. Alsalloum

Author E.S. Kuznetsova

Author J.A. Krupinova

Author O.N. Mityaeva

Author K.K. Shefer

Author E.V. Boiko

Author O.V. Sagaydak

Author N.A. Doroshchuk

Author M.V. Makarova

Author M. Woroncow

Author V.P. Bogdanov

Author P.Y. Volchkov

Abstract Cone dystrophy with supernormal rod responses (CDSRR) is a rare autosomal recessive hereditary retinal dystrophy caused by biallelic KCNV2 variants. Accurate prevalence data remain limited because current estimates rely on clinically ascertained cases. This study aimed to estimate its population prevalence by integrating curated variant evidence and large-scale population genomics resources. The key methodological feature of this study is a multi-tiered pathogenicity framework combined with estimation of a biologically plausible prevalence range: Known pathogenic variants define conservative estimates, whereas probably pathogenic variants and strong variants of uncertain significance define broader estimates. Allele frequencies were analyzed in 807,162 individuals from the gnomAD database and 144,127 Russian genomes from GDB and EvogenDB. A Bayesian framework was applied to calculate disease prevalence from aggregated carrier frequencies across predefined variant tiers. Conservative estimates based on known pathogenic variants, were 1 in 343,926 globally and 1 in 1,911,347 in Russia. Including broader tiers increased estimates to 1 in 14,620 and 1 in 32,764, respectively. Compared to previous clinically ascertained prevalence of 1 in 865,000, these estimates define a wider biologically plausible prevalence range and are consistent with possible clinical underascertainment. This tier-based approach accounts for uncertainty in variant classification when estimating rare disease prevalence. © 2026 by the authors.

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Bayesian Quantification with Black-Box Estimators

Item Type Journal Article

Author A. Ziegler

Author P. Czyż

Abstract Understanding how different classes are distributed in an unlabeled data set is important for the calibration of probabilistic classifiers and uncertainty quantification. Methods like adjusted classify and count, black-box shift estimators, and invariant ratio estimators use an auxiliary and potentially biased black-box classifier trained on a different data set to estimate the class distribution on the current data set and yield asymptotic guarantees under weak assumptions. We demonstrate that these algorithms are closely related to the inference in a particular probabilistic graphical model approximating the assumed ground-truth generative process, and we propose a Bayesian estimator. Then, we discuss an efficient Markov chain Monte Carlo sampling scheme for the introduced model and show an asymptotic consistency guarantee in the large-data limit. We compare the introduced model against the established point estimators in a variety of scenarios, and show it is competitive, and in some cases superior, with the non-Bayesian alternatives. © 2024, Transactions on Machine Learning Research. All rights reserved.

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Volume 2024

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Notes:

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Bayesian statistical analysis for spams

Item Type Conference Paper
Author Y. Begriche
Author A. Serhrouchni
Abstract This paper presents a Bayesian statistical analysis applied to the spam problem. In most anti-spam related research, generally it is assumed that the probability of a spam occurrence is equal to 0.5, which is in our opinion unrealistic. It is also assumed that in the spam message, words are considered as an independent family of words. This makes us look at how the posterior probability behaves when the a priori probability is different from 0.5 and derive the consequences of the assumption of independent words on the posterior probability. The first assumption pushes us to define a prior and find a posterior probability laws to enhance the spam detection and increase the reliability decision. This analysis differs from previous results, that used the Bayesian approach to the anti-spam issue, especially through refinement and enhancement of various probability laws. © 2010 IEEE.
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Language English
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Bayesian networks, Statistical methods, Internet, Probability, Classification, Bayesian statistical model, Binomial law, Conditional density, Distribution attachment, Ham(H), Spam(S)

Notes:

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: Y. Begriche; Institut Telecom., Telecom. ParisTech, France; email: Youcef.Begriche@telecom-paristech.fr; Conference name: 35th Annual IEEE Conference on Local Computer Networks, LCN 2010; Conference date: 10 October 2010 through 14 October 2010; Conference code: 84547

Bayesian Statistics Applied to the Recovery of Copper and Cobalt-Bearing Ores: Establishment of the Probability Laws of Leaching Parameters

Item Type	Conference Paper
Author	B. Mbuya
Author	E. Fosso-Kankeu
Author	J.C. Bongaerts
Author	A.F. Mulaba-Bafubiandi
Editor	Samia N.
Abstract	Currently, industries and researchers recognize the need for new ways to incorporate prior knowledge into processes. With the increasing accessibility of machine learning techniques through software packages, data-driven modeling applications are becoming more prevalent, incorporating more advanced techniques and analytics. Bayesian methods utilize advanced concepts of conditional probability for their inferences. This is crucial for developing robust and understandable solutions that can be used for prediction, optimization, or decision-making. In this paper, we discuss the implementation of Bayesian modeling in hydrometallurgy, focusing specifically on its application in the leaching process. Bayesian statistics provides the opportunity to incorporate prior knowledge into a model, making it a valuable approach in this field. Therefore, parameters that influence leaching, such as mineralogy, particle size, redox potential (pH), acidity, and oxido-reduction potential (ORP), were included as random variables in the model. This enables the designing of Bayesian networks with precisely defined a priori probability laws. © 2024, Avestia Publishing. All rights reserved.
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DOI	10.11159/icsta24.140
URL	https://www.scopus.com/pages/publications/85205698789?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: Proc. Int. Conf. Stat.
Publisher	Avestia Publishing
ISBN	25627767 (ISSN); 978-199080042-9 (ISBN)
Proceedings Title	Proc. Int. Conf. Stat.
Conference Name	Proceedings of the International Conference on Statistics
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Tags:

modeling, leaching, Bayesian statistics, Conditional probability, hydrometallurgy, ore

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Bayesian uncertainty quantification with multi-fidelity data and gaussian processes for impedance cardiography of aortic dissection

Item Type Journal Article

Author S. Ranftl

Author G.M. Melito

Author V. Badeli

Author A. Reinbacher-Köstinger

Author K. Ellermann

Author W. von der Linden

Abstract In 2000, Kennedy and O'Hagan proposed a model for uncertainty quantification that combines data of several levels of sophistication, fidelity, quality, or accuracy, e.g., a coarse and a fine mesh in finite-element simulations. They assumed each level to be describable by a Gaussian process, and used low-fidelity simulations to improve inference on costly high-fidelity simulations. Departing from there, we move away from the common non-Bayesian practice of optimization and marginalize the parameters instead. Thus, we avoid the awkward logical dilemma of having to choose parameters and of neglecting that choice's uncertainty. We propagate the parameter uncertainties by averaging the predictions and the prediction uncertainties over all the possible parameters. This is done analytically for all but the nonlinear or inseparable kernel function parameters. What is left is a low-dimensional and feasible numerical integral depending on the choice of kernels, thus allowing for a fully Bayesian treatment. By quantifying the uncertainties of the parameters themselves too, we show that "learning" or optimising those parameters has little meaning when data is little and, thus, justify all our mathematical efforts. The recent hype about machine learning has long spilled over to computational engineering but fails to acknowledge that machine learning is a big data problem and that, in computational engineering, we usually face a little data problem. We devise the fully Bayesian uncertainty quantification method in a notation following the tradition of E.T. Jaynes and find that generalization to an arbitrary number of levels of fidelity and parallelisation becomes rather easy. We scrutinize the method with mock data and demonstrate its advantages in its natural application where high-fidelity data is little but low-fidelity data is not. We then apply the method to quantify the uncertainties in finite element simulations of impedance cardiography of aortic dissection. Aortic dissection is a cardiovascular disease that frequently requires immediate surgical treatment and, thus, a fast diagnosis before. While traditional medical imaging techniques such as computed tomography, magnetic resonance tomography, or echocardiography

certainly do the job, Impedance cardiography too is a clinical standard tool and promises to allow earlier diagnoses as well as to detect patients that otherwise go under the radar for too long. © 2019 by the authors.

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BC-DMKDE: Bayesian Classifier Based on Distance-Corrected Multivariate Kernel Density Estimation

Item Type Conference Paper

Author G. Chen

Author X. Wu

Author Z. Li

Author R. Zhao

Author J. Jiang

Author K. Yang

Abstract In practical classification problems, imbalanced class distribution is very common. Traditional classification methods usually aim at optimizing the overall accuracy, which leads to classification decisions biased towards categories with a large number of samples. Similar to most traditional classifiers, Naive Bayes (NB) is sensitive to the class prior distribution and has poor recognition

performance for minority class samples, which restricts its application in imbalanced classification. Therefore, we propose a Bayesian classifier based on distance-corrected multivariate kernel density estimation (BC-DMKDE). BC-DMKDE relaxes the assumptions regarding conditional independence and the normality of continuous variables in NB classifiers, by incorporating the multivariate kernel density estimation (MKDE). We also incorporate a distance-correction mechanism utilizing kernel density estimation, specifically to alleviate the impact of class imbalance and overlapping data samples. Across 18 real-world benchmarks, BC-DMKDE exhibits superior classification performance and robustness, outperforming existing methods at both the algorithm and data levels. © 2025 IEEE.

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Classification (of information), Benchmarking, Statistics, Bayesian networks, Naive Bayes, Signal processing, Class imbalance, Classification, Classifiers, Naive bayes, Class Imbalance, Classification methods, Kernel density estimation, Kernel Density Estimation, Class distributions, Bayesian classifier, Imbalanced class, Multivariate kernel densities, Overall accuracies, Overlap

Notes:

EID: 2-s2.0-105034847055

Export Date: 20 September 2026; Cited By: 0; Funding details: National Natural Science Foundation of China, NSFC, (72471238, 72231011); National Natural Science Foundation of China, NSFC; Correspondence Address: X. Wu; National University of Defense Technology, College of Systems Engineering, Changsha, China; email: wuxinyang525@126.com; Conference name: 11th International Conference on Big Data and Information Analytics, BigDIA 2025; Conference date: 8 November 2025 through 11 November 2025; Conference code: 220522

Behavior Knowledge Space-Based Fusion for Copy-Move Forgery Detection

Item Type Journal Article

Author A. Ferreira
Author S.C. Felipussi
Author C. Alfaro
Author P. Fonseca
Author J.E. Vargas-Munoz
Author J.A. Dos Santos
Author A. Rocha

Abstract The detection of copy-move image tampering is of paramount importance nowadays, mainly due to its potential use for misleading the opinion forming process of the general public. In this paper, we go beyond traditional forgery detectors and aim at combining different properties of copy-move detection approaches by modeling the problem on a multiscale behavior knowledge space, which encodes the output combinations of different techniques as a priori probabilities considering multiple scales of the training data. Afterward, the conditional probabilities missing entries are properly estimated through generative models applied on the existing training data. Finally, we propose different techniques that exploit the multi-directionality of the data to generate the final outcome detection map in a machine learning decision-making fashion. Experimental results on complex data sets, comparing the proposed techniques with a gamut of copy-move detection approaches and other fusion methodologies in the literature, show the effectiveness of the proposed method and its suitability for real-world applications. © 1992-2012 IEEE.

Date 2016

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Artificial intelligence, Decision making, Learning systems, Information analysis, Data handling, Conditional probabilities, A-priori probabilities, Multi-scale datum, Behavior knowledge spaces, behaviour knowledge space, Copy-move detections, Copy-move forgery detection, Copy-move forgery detections, Facsimile, fusion, Fusion methodology, Fusion reactions, Knowledge spaces, multi-direction data analysis, multi-scale data analysis

Notes:

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Benchmarking Energy Quantification Methods to Predict Heating Energy Performance of Residential Buildings in Germany

Item Type Journal Article**Author** S. Wenninger**Author** C. Wiethe

Abstract To achieve ambitious climate goals, it is necessary to increase the rate of purposeful retrofit measures in the building sector. As a result, Energy Performance Certificates have been designed as important evaluation and rating criterion to increase the retrofit rate in the EU and Germany. Yet, today's most frequently used and legally required methods to quantify building energy performance show low prediction accuracy, as recent research reveals. To enhance prediction accuracy, the research community introduced data-driven methods which obtained promising results. However, there are no insights in how far Energy Quantification Methods are particularly suited for energy performance prediction. In this research article the data-driven methods Artificial Neural Network, D-vine copula quantile regression, Extreme Gradient Boosting, Random Forest, and Support Vector Regression are compared with and validated by real-world Energy Performance Certificates of German residential buildings issued by qualified auditors using the engineering method required by law. The results, tested for robustness and systematic bias, show that all data-driven methods exceed the engineering method by almost 50% in terms of prediction accuracy. In contrast to existing literature favoring Artificial Neural Networks and Support Vector Regression, all tested methods show similar prediction accuracy with marginal advantages for Extreme Gradient Boosting and Support Vector Regression in terms of prediction accuracy. Given the higher prediction accuracy of data-driven methods, it seems appropriate to revise the current legislation prescribing engineering methods. In addition, data-driven methods could support different organizations, e.g., asset management, in decision-making in order to reduce financial risk and to cut expenses. © 2021, The Author(s).

Date 2021**DOI** 10.1007/s12599-021-00691-2**URL** <https://www.scopus.com/pages/publications/85102462165?origin=resultslist>**Archive** Scopus**Language** English**Volume** 63**Publisher** Springer Gabler**Pages** 223-242**Publication** Business and Information Systems Engineering**ISSN** 23637005 (ISSN)**Issue** 3**Journal Abbr** Busin. Info. Sys. Eng.**Date Added** 20/09/2026, 9:52:21 pm**Modified** 20/09/2026, 9:52:21 pm

Tags:

Machine learning algorithms, Benchmarking, Data analytics, Energy quantification methods, Building energy, Energy performance certificates, Data-driven methods, Energy informatics

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Benchmarking Uncertainty and Its Disentanglement in Multi-label Chest X-Ray Classification

Item Type Conference Paper

Author S. Baur

Author W. Samek

Author J. Ma

Editor Sudre C.H.

Editor Hoque M.I.

Editor Mehta R.

Editor Qin C.

Editor Ouyang C.

Editor Rakic M.

Editor Wells W.M.

Abstract Reliable uncertainty quantification is crucial for trustworthy decision-making and the deployment of AI models in medical imaging. While prior work has explored the ability of neural networks to quantify predictive, epistemic, and aleatoric uncertainties using an information-theoretical approach in synthetic or well-defined data settings like natural image classification, its applicability to real-life medical diagnosis tasks remains underexplored. In this study, we provide an extensive uncertainty quantification benchmark for multi-label chest X-ray classification using the MIMIC-CXR-JPG dataset. We evaluate 13 uncertainty quantification methods for convolutional (ResNet) and transformer-based (Vision Transformer) architectures across a wide range of tasks. Additionally, we extend Evidential Deep Learning, HetClass NNs, and Deep Deterministic Uncertainty to the multi-label setting. Our analysis provides insights into uncertainty estimation effectiveness and the ability to disentangle epistemic and aleatoric uncertainties, revealing method- and architecture-specific strengths and limitations. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

Date 2026**DOI** 10.1007/978-3-032-06593-3_18**URL** <https://www.scopus.com/pages/publications/105019231360?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Lect. Notes Comput. Sci.**Volume** 16166 LNCS**Publisher** Springer Science and Business Media Deutschland GmbH**ISBN** 03029743 (ISSN); 978-303206592-6 (ISBN)**Pages** 193-203**Proceedings Title** Lect. Notes Comput. Sci.**Conference Name** Lecture Notes in Computer Science**Date Added** 20/09/2026, 9:51:51 pm**Modified** 20/09/2026, 9:51:51 pm**Tags:**

Deep learning, Decisions makings, Classification (of information), Decision making, Network architecture, Computer vision, Neural-networks, Uncertainty, Medical imaging, Image classification, Diagnosis, Multi-labels, Uncertainty quantifications, Uncertainty Quantification, Images classification, Data settings, Disentanglement, Natural images, Radiology, Theoretical approach

Notes:

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Benchmarking uncertainty quantification for protein engineering

Item Type Journal Article**Author** K.P. Greenman**Author** A.P. Amini**Author** K.K. Yang

Abstract Machine learning sequence-function models for proteins could enable significant advances in protein engineering, especially when paired with state-of-the-art methods to select new sequences for property optimization and/or model improvement. Such methods (Bayesian optimization and active learning) require calibrated estimations of model uncertainty. While studies have benchmarked a variety of deep learning uncertainty quantification (UQ) methods on standard and molecular machine-learning

datasets, it is not clear if these results extend to protein datasets. In this work, we implemented a panel of deep learning UQ methods on regression tasks from the Fitness Landscape Inference for Proteins (FLIP) benchmark. We compared results across different degrees of distributional shift using metrics that assess each UQ method's accuracy, calibration, coverage, width, and rank correlation. Additionally, we compared these metrics using one-hot encoding and pretrained language model representations, and we tested the UQ methods in retrospective active learning and Bayesian optimization settings. Our results indicate that there is no single best UQ method across all datasets, splits, and metrics, and that uncertainty-based sampling is often unable to outperform greedy sampling in Bayesian optimization. These benchmarks enable us to provide recommendations for more effective design of biological sequences using machine learning. © 2025 Greenman et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Markov processes, Article, human, Deep learning, machine learning, Machine-learning, Benchmarking, Federated learning, Uncertainty, Deep Learning, Adversarial machine learning, Contrastive Learning, Machine Learning, procedures, data analysis, deep learning, State-of-the-art methods, convolutional neural network, benchmarking, Active learning, Active Learning, evaluation study, Bayesian learning, bioinformatics, Computational Biology, Bayesian optimization, molecular biology, Bayes theorem, Bayes Theorem, chemistry, Quantification methods, uncertainty, Uncertainty quantifications, protein, calibration, root mean squared error, mean absolute error, Proteins, Function modelling, amino acid sequence, Databases, Protein, immunoglobulin binding factor, Learning sequences, Properties optimizations, protein database, protein engineering, Protein engineering, Protein Engineering, Streptococcus

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Benchmarking Uncertainty Quantification Methods for Large Language Models with LM-Polygraph

Item Type Journal Article

Author R. Vashurin

Author E. Fadeeva

Author A. Vazhentsev

Author L. Rvanova

Author D. Vasilev

Author A. Tsvigun

Author S. Petrakov

Author R. Xing

Author A. Sadallah

Author K. Grishchenkov

Author A. Panchenko

Author T. Baldwin

Author P. Nakov

Author M. Panov

Author A. Shelmanov

Abstract The rapid proliferation of large language models (LLMs) has stimulated researchers to seek effective and efficient approaches to deal with LLM hallucinations and low-quality outputs. Uncertainty quantification (UQ) is a key element of machine learning applications in dealing with such challenges. However, research to date on UQ for LLMs has been fragmented in terms of techniques and evaluation methodologies. In this work, we address this issue by introducing a novel benchmark that implements a collection of state-of-the-art UQ baselines and offers an environment for controllable and consistent evaluation of novel UQ techniques over various text generation tasks. Our benchmark also supports the assessment of confidence normalization methods in terms of their ability to provide interpretable scores. Using our benchmark, we conduct a large-scale empirical investigation of UQ and normalization techniques across eleven tasks, identifying the most effective approaches. © 2025 Association for Computational Linguistics.

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Notes:

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Export Date: 20 September 2026; Cited By: 28

Benchmarking Uncertainty Quantification on Biosignal Classification Tasks Under Dataset Shift

Item Type Journal Article

Author T. Xia

Author J. Han

Author C. Mascolo

Abstract A biosignal is a signal that can be continuously measured from human bodies, such as respiratory sounds, heart activity (ECG), brain waves (EEG), etc., based on which, machine learning models have been developed with very promising performance for automatic disease detection and health status monitoring. However, dataset shift, i.e., data distribution of inference varies from the distribution of the training, is not uncommon for real biosignal-based applications. To improve the robustness, probabilistic models with uncertainty quantification are adapted to capture how reliable a prediction is. Yet, assessing the quality of the estimated uncertainty remains a challenge. In this work, we propose a framework to evaluate the capability of the estimated uncertainty in capturing different types of biosignal dataset shifts with various degrees. In particular, we use three classification tasks based on respiratory sounds and electrocardiography signals to benchmark five representative uncertainty quantification methods. Extensive experiments show that, although Ensemble and Bayesian models could provide relatively better uncertainty estimations under dataset shifts, all tested models fail to meet the promise in trustworthy prediction and model calibration. Our work serves as a benchmark evaluation for any future biosignal classifiers without requiring additional datasets. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Notes:

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Export Date: 20 September 2026; Cited By: 8; Correspondence Address: T. Xia; Department of Computer Science and Technology, University of Cambridge, Cambridge, United Kingdom; email: tx229@cam.ac.uk

Beneficial effects of adjunctive Aripiprazole in major depressive disorder are not dependent on antidepressant therapy history: A post hoc analysis of 3 randomized, double-blind, placebo-controlled trials

Item Type Journal Article

Author D.L. Dunner

Author K.K. Laubmeier

Author G. Manos

Author R.A. Forbes

Author R.A. Baker

Author R.M. Berman

Abstract Objective: To determine whether switching within or between antidepressant therapy (ADT) classes prior to the use of adjunctive antipsychotic treatment is associated with different outcomes in major depressive disorder (MDD). Method: This was a post hoc analysis of pooled data from 3 similar, multicenter, randomized, double-blind, placebo-controlled registrational studies of aripiprazole adjunctive to ADT conducted between September 2004 and April 2008. The trials comprised the following 3 phases: a 7- to 28-day screening phase, an 8-week single-blind prospective treatment phase, and a 6-week double-blind, randomized phase. Patients were aged 18-65 years and met DSM-IV-TR criteria for MDD. Patients with an inadequate response to ADT during the screening phase entered the prospective treatment phase, during which

they were switched to another ADT medication of either the same or a different class. Those patients with an inadequate response were then randomized to double-blind adjunctive aripiprazole or adjunctive placebo and followed for 6 weeks. Results: Mean improvement in Montgomery-Asberg Depression Rating Scale total score was significantly greater with adjunctive aripiprazole versus adjunctive placebo for both between-class (-9.2 vs -6.2, $P < .001$) and within-class (-9.8 vs -6.6, $P < .001$) switch groups. Relative risks for response were 1.6 (95% CI = 1.3-2.1) for those who switched between classes and 1.7 (95% CI = 1.2-2.2) for those who switched within class. Conclusions: Augmentation with aripiprazole, after either a between-class or within-class switch following initial ADT failure, is an effective option for patients with nonresponsive MDD. In contrast to current strategies employed in clinical practice, these results suggest that adjunctive aripiprazole is a logical strategy in patients unresponsive to ADT. Trial Registration: ClinicalTrials.gov identifiers: NCT00105196, NCT00095758, NCT00095823. © 2012 Physicians Postgraduate Press, Inc.

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Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: D. L. Dunner; FACPsych, Center for Anxiety and Depression, Mercer Island, WA, 7525 SE 24th St, Ste 400, United States; email: dldunner@comcast.net

Beyond burden metrics: Wearable photoplethysmography-derived spatiotemporal progression of atrial fibrillation linked to clinical outcomes

Item Type Journal Article

Author Y. Guo

Author H. Wang

Author H. Wang

Author H. Zhang

Author Z. Jin

Abstract Background Traditional atrial fibrillation (AF) classification lacks dynamic quantification. Current AF burden assessment is constrained by intermittent monitoring and simplistic metrics. Objective This study aimed to establish a continuous, multidimensional AF progression model using wearable photoplethysmography (PPG) for real-world, dynamic burden quantification. Methods In this prospective cohort, 110 patients with paroxysmal AF underwent synchronized PPG (Huawei Watch GT3) and 24-hour Holter monitoring. We developed a multiscale fusion AF algorithm and a 5-dimensional spatiotemporal progression model quantifying episode frequency, duration, clustering, circadian rhythm, and tachycardia burden. Results The fusion algorithm achieved an accuracy of 0.97. The 5-dimensional model showed strong concordance with Holter monitoring, with near-perfect correlation for episode duration ($r = 0.97$) and high interchangeability (intraclass correlation coefficient >0.75). It demonstrated excellent diagnostic performance for burden trajectories (area under the curve 0.98). A composite AF burden score of ≥ 0.59 identified patients at high risk of AF-related symptoms or heart rate issues. Clinically, AF burden increased with worsening European Heart Rhythm Association symptoms ($P = .04$), varied by risk profile ($P = .03$), and differed between ablation- and drug-treated patients (episode duration $P < .001$). Conclusion Continuous PPG-based spatiotemporal modeling robustly quantified dynamic AF progression, enabled precise phenotyping, and may support early intervention in high-risk patients. © 2026 Heart Rhythm Society.

Date 2026

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Beyond compliance: Patient risk, ERAS adherence, and postoperative outcomes through explainable machine learning

Item Type Journal Article

Author J. Ripollés-Melchor

Author Á.V. Espinosa

Author A. Abad-Motos

Author A. Abad-Gurumeta

Author P. Galán-Menéndez

Author A. Zorrilla-Vaca

Author R. Navarro-Pérez

Author A. Ruiz-Escobar

Author J. Fernanz-Antón

Author A. Suárez-de-la-Rica

Author C. Aldecoa

Abstract Background: Enhanced Recovery After Surgery (ERAS) protocols improve outcomes after colorectal surgery, but adherence remains variable and may interact with patient risk. Traditional compliance scores lack granularity to explore these dynamics. We

aimed to use interpretable machine learning to quantify the contribution of individual ERAS items and clinical features to postoperative complications, and to identify data-driven ERAS phenotypes. Methods: This was a secondary analysis of the EuroPOWER cohort (NCT04889798), a prospective European study including 2,841 adults undergoing elective colorectal surgery. Two Extreme Gradient Boosting models were trained to predict in-hospital complications: a complete model (clinical variables + 23 ERAS items) and an ERAS-only model. Both were interpreted using Shapley Additive Explanations (SHAP). In the complete model, SHAP matrices were clustered to derive phenotypes. Feature importance, adherence, and complication rates were compared descriptively. Results: The complete model achieved an AUC of 0.627. SHAP analysis identified frailty, ASA class, BMI, and age as leading predictors, followed by early mobilisation, nutritional care, and thromboprophylaxis. Three phenotypes were identified, with complication rates of 17.7%, 27.1%, and 41.1%, corresponding to robust, intermediate, and frail profiles. The ERAS-only model showed similar discrimination (area under the curve 0.642), but reduced interpretability. SHAP redundancy analysis supported inclusion of all ERAS items. Conclusions: The clinical effect of ERAS adherence appears to be modulated by baseline vulnerability and implementation patterns. SHAP-based models enable transparent risk attribution and phenotype identification, supporting more targeted ERAS strategies and future development of automated quality monitoring tools. © 2025 Sociedad Española de Anestesiología, Reanimación y Terapéutica del Dolor

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Short Title Más allá del cumplimiento: riesgo del paciente, adherencia a ERAS y resultados postoperatorios mediante aprendizaje automatizado explicable

Language English

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Machine learning, Colorectal surgery, Enhanced recovery after surgery, Frailty, Postoperative complications

Notes:

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Beyond engagement: an EEG-based methodology for assessing user's confusion in an educational game

Item Type Journal Article

Author Y. Zhou

Author T. Xu

Author S. Li

Author R. Shi

Abstract Confusion is an emotion, which may occur when the learner is confronting inconsistency between new knowledge and existing cognitive structure, or reasoning for solving the puzzle and problem. Although confusion is not pleasant, it is necessary for the learner to engage in understanding and deep learning. Consequently, confusion assessment has attracted increased interest in e-learning. However, current studies have targeted no further than engagement detection and measurement, while there is lack of studies in investigating cognitive and emotional aspects beyond engagement in the context of game-based learning. To quantify confused states in logic reasoning in game-based learning, we propose an EEG-based methodology for assessing the user's confusion using the OpenBCI device with 8 channels. In the complicated context of game play, it is difficult, and sometimes impossible, to collect the ground truth of the data in real tasks. To solve this issue, this work leverages cross-task and cross-subject methods to build a classifier, that is, training on the data of one standardized cognitive test paradigm (Raven's test) and testing on the data of real tasks in game play (Sokoban Game). It provides a new possibility to create a classifier based on a small dataset. We also employ the end-to-end algorithm of deep learning in machine learning. Results showed the feasibility of this proposal in the task variation of the classifier, with an accuracy of 91.04%. The proposed EEG-based methodology is suitable to analyze learners' confusion on the long game-play duration and has a good adaption in real tasks. © 2019, Springer-Verlag GmbH Germany, part of Springer Nature.

Date 2019

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Beyond Kd: Next Generation Performance Assessment Framework through Machine Learning

Item Type Conference Paper

Author H. Wainwright

Author E. Chang

Author D. Ermakova

Author H. Lu

Author L. Beverly

Author M. Nole

Author M. Zavarin

Author O. Omosebi

Author L. Zheng

Abstract In this study, we have developed a framework to propagate the uncertainty from the sorption experiments to the THC model, and then to the PA model. To quantify the uncertainty in SCM parameters, we developed a Python package for the Bayesian inference and general UQ. In addition, we developed a surrogate model for the THC model to quantify the buffer-averaged Kd. The Kd values were then transferred to the PA model (PFLOTRAN) for UQ. Although the results are preliminary, they demonstrate multi-model connections and associated uncertainty quantification methodology to maximize the use of existing experimental datasets as well as more complex geochemical models than Kd-based retardation coefficients. © 2022 Proceedings of the International High-Level Radioactive Waste Management Conference, IHLRWM 2022, Embedded with the 2022 ANS Winter Meeting. All rights reserved.

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Beyond point predictions: Quantifying uncertainty in E. coli ML-based monitoring

Item Type Journal Article

Author D. Abert-Fernández

Author E. Aguilera

Author P. Emiliano

Author F. Valero

Author H. Monclús

Abstract Machine learning regression models are increasingly used to improve management, decision-making, and monitoring of drinking water quality, leveraging growing data from real-time sensors and laboratory analyses. However, most models provide only point predictions, ignoring inherent uncertainty caused by unobserved factors that can produce varying outcomes under similar conditions. This study benchmarks state-of-the-art regression algorithms and uncertainty quantification methods for predicting E. coli concentrations in a drinking water catchment. Gradient-boosted decision trees (GBDT) proved effective for real-time tracking, with CatBoost achieving the lowest error (RMSLE = 0.877), improving on the naïve baseline (1.160) and outperforming Random Forest by 5 %. Uncertainty quantification techniques successfully generated

valid prediction intervals to identify high-risk contamination events, with Conformalized Quantile Regression emerging as the most reliable method. By combining accurate GBDT predictions with well-calibrated uncertainty estimates, this approach enhances microbial water quality forecasting, offering improved risk assessment and supporting more robust decision-making in drinking water management. © 2025 The Author(s)

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Beyond pointwise supervision: A pairwise-augmented approach for positive-unlabeled learning

Item Type Journal Article

Author L. Tang

Author J. Wang

Author Y. Tian

Author P.M. Pardalos

Abstract Positive-unlabeled (PU) learning trains binary classifiers using a limited subset of labeled positives alongside vast unlabeled data. Existing methods predominantly rely on pointwise supervision, thereby neglecting the rich latent supervisory signals inherent in the unlabeled data. To address this, we propose a novel paradigm that

augments traditional instance-level supervision with a pairwise comparison (Pcomp) perspective. By constructing positive-unlabeled sample pairs, this paradigm learns ranking relationships where a labeled positive instance is expected to exhibit a higher positive confidence than an unlabeled one. This formulates PU learning as a Pcomp classification problem subject to one-sided noise from hidden positives. To render the Pcomp loss tractable under unknown class priors, we devise a Bayesian strategy that iteratively estimates the prior by explicitly modeling instance-dependent labeling probabilities. Furthermore, to mitigate the noise inherent in constructed sample pairs, we introduce a theoretically grounded loss decomposition mechanism. This strategy decouples the Pcomp loss into cyclically symmetric and asymmetric components, exploiting the intrinsic noise-invariance of the former while applying targeted corrections exclusively to the latter. Consequently, our method achieves strong robustness, even in label-scarce regimes. Extensive benchmark experiments demonstrate that our approach significantly outperforms state-of-the-art algorithms, substantiating both its effectiveness and robustness. © 2026 Elsevier Inc.

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Big data driven Cyber-Physical System based Lithium-ion battery modeling method with battery aging considered

Item Type Conference Paper

Author S. Li

Author H. He

Author J. Li

Author H. Wang

Abstract As the bottleneck technology of electric vehicles (EVs), the battery has complex and hardly observable inside chemical reactions. Therefore, a precise mathematical model is crucial for the battery management system to ensure the secure and stable operation of the battery. Aiming at achieving a flexible, self-configuring, reliable Battery Management System (BMS), this paper mainly focuses on the following research points: Firstly, a Cyber-Physical system (CPS) based BMS is presented for a better use of battery data. Next, a data cleaning method based on machine learning algorithm is applied to the big data of batteries in electric vehicles. Finally, a rain-flow cycle counting algorithm-based battery degradation quantification method is proposed and a Stacked Denoising Autoencoders-Extreme Learning Machine (SDAE-ELM) algorithm-based battery modeling method is also built to deal with the influence of battery aging phenomenon. Using the battery data extracted from electric buses, model effectiveness and accuracy are validated. The error of the estimated battery terminal voltage estimator is within 2.5%. © 2019, Scanditale AB. All rights reserved.

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Big data driven Internet of Things for credit evaluation and early warning in finance

Item Type Journal Article

Author C. Wen

Author J. Yang

Author L. Gan

Author Y. Pan

Abstract The big data technology framework has been successfully used in the Internet of Things, and the financial industry also hopes to use the advanced technology of big data to integrate and improve internal and external data related to credit risks. Relying on more efficient machine learning algorithms to get a reasonable prediction of credit risk can reduce the self-generated losses of the Internet of Things finance and increase profits. This article uses distributed search engine technology to customize web crawlers to obtain the required bank card and transaction data from the multi-source heterogeneous data of the Internet of Things financial industry, design the corresponding Spark parallel algorithm to preprocess the data, and establish an inverted table and two Level index file provides data source for big data analysis platform. After the data source is determined, the Mutually Exclusive Collectively Exhaustive (MECE) analysis method is combined with the scores of many financial business experts in the industry to obtain a set of candidate indicators and quantification methods for the financial credit risk evaluation of the Internet of Things, and analyze the correlation of indicators and risk grading. The random forest algorithm in the big data machine learning library is used to select the feature of the candidate index set, and a multi-level spatial association rule algorithm based on the Hash structure is designed to mine the financial risk information of the Internet of Things, and build a credit risk assessment and intelligent early warning model. This paper selects 26 indicators of Internet of Things finance as the research objects, uses SPSS26.0 software to perform sample Kaiser–Meyer–Olkin (KMO) test and Bartlett sphere test on the original data, and describes the results of factor analysis in detail. The particle swarm algorithm is introduced into the parameter optimization of random forest, and the financial credit risk assessment model of the Internet of Things is established. The results show that this method can significantly reduce the probability of banks making the first and second error rates when evaluating the credit risk of financing the Internet of Things Finance. This is conducive to the smooth development of the Internet of Things financial business for banks, which enables banks to enhance their own profitability while effectively reducing losses due to incorrect credit provision. © 2021 Elsevier B.V.

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DOI 10.1016/j.future.2021.06.003**URL** <https://www.scopus.com/pages/publications/85107772971?origin=resultslist>**Archive** Scopus**Language** English**Volume** 124**Publisher** Elsevier B.V.**Pages** 295-307**Publication** Future Generation Computer Systems**ISSN** 0167739X (ISSN)**Journal Abbr** Future Gener Comput Syst**Date Added** 20/09/2026, 9:52:34 pm**Modified** 20/09/2026, 9:52:34 pm**Tags:**

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Binary Classification from Multiple Unlabeled Datasets via Surrogate Set Classification

Item Type Conference Paper**Author** N. Lu**Author** S. Lei**Author** G. Niu**Author** I. Sato**Author** M. Sugiyama**Abstract** To cope with high annotation costs, training a classifier only from weakly supervised data has attracted a great deal of attention these days. Among various approaches, strengthening supervision from completely unsupervised classification is a promising direction, which typically employs class priors as the

only supervision and trains a binary classifier from unlabeled (U) datasets. While existing risk-consistent methods are theoretically grounded with high flexibility, they can learn only from two U sets. In this paper, we propose a new approach for binary classification from m U sets for $m \geq 2$. Our key idea is to consider an auxiliary classification task called surrogate set classification (SSC), which is aimed at predicting from which U set each observed sample is drawn. SSC can be solved by a standard (multi-class) classification method, and we use the SSC solution to obtain the final binary classifier through a certain linear-fractional transformation. We built our method in a flexible and efficient end-to-end deep learning framework and prove it to be classifier-consistent. Through experiments, we demonstrate the superiority of our proposed method over state-of-the-art methods. Copyright © 2021 by the author(s)

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Binary Classification only from Unlabeled Data by Iterative Unlabeled-unlabeled Classification

Item Type Conference Paper

Author H. Kaji

Author M. Sugiyama

Abstract Unlabeled-unlabeled (UU) classification (du Plessis et al. 2013) allows us to train a binary classifier from two sets of unlabeled data with different class priors. In this paper, we go beyond this scenario and try to train a binary classifier only from a single set of unlabeled data. Our key idea is to iteratively perform UU classification: We initially split the original single unlabeled dataset into two disjoint datasets and perform UU classification. We then split the original unlabeled dataset in a different way based on the obtained classifier, perform UU classification, and repeat this process until convergence. We numerically show that the classification accuracy tends to be improved over iterations. Finally, we apply our iterative UU classification method to a realworld drowsiness prediction problem and demonstrate its usefulness. © 2019 IEEE.

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Binary classification: Counterbalancing class imbalance by applying regression models in combination with one-sided label shifts

Item Type Conference Paper

Author P. Bellmann

Author H. Hihn

Author D.A. Braun

Author F. Schwenker

Editor Rocha A.P.

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Editor van den Herik J.

Abstract In many real-world pattern recognition scenarios, such as in medical applications, the corresponding classification tasks can be of an imbalanced nature. In the current study, we focus on binary, imbalanced classification tasks, i.e. binary classification tasks in which one of the two classes is under-represented (minority class) in comparison to the other class (majority class). In the literature, many different approaches have been proposed, such as under- or oversampling, to counter class imbalance. In the current work, we introduce a novel method, which addresses the issues of class imbalance. To this end, we first transfer the binary classification task to an equivalent regression task. Subsequently, we generate a set of negative and positive target labels, such that the corresponding regression task becomes balanced, with respect to the redefined target label set. We evaluate our approach on a number of publicly available data sets in combination with Support Vector Machines. Moreover, we compare our proposed method to one of the most popular oversampling techniques (SMOTE). Based on the detailed discussion of the presented outcomes of our experimental evaluation, we provide promising ideas for future research directions. © 2021 by SCITEPRESS - Science and Technology Publications, Lda.

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Binary quantification and dataset shift: an experimental investigation

Item Type Journal Article

Author P. González

Author A. Moreo

Author F. Sebastiani

Abstract Quantification is the supervised learning task that consists of training predictors of the class prevalence values of sets of unlabelled data, and is of special interest when the labelled data on which the predictor has been trained and the unlabelled data are not IID, i.e., suffer from dataset shift. To date, quantification methods have mostly been tested only on a special case of dataset shift, i.e., prior probability shift; the relationship between quantification and other types of dataset shift remains, by and large, unexplored. In this work we carry out an experimental analysis of how current quantification algorithms behave under different types of dataset shift, in order to identify limitations of current approaches and hopefully pave the way for the development of more broadly applicable methods. We do this by proposing a fine-grained taxonomy of types of dataset shift, by establishing protocols for the generation of datasets affected by these types of shift, and by testing existing quantification methods on the datasets thus generated. One finding that results from this investigation is that many existing quantification methods that had been found robust to prior probability shift are not necessarily robust to other types of dataset shift. A second finding is that no existing quantification method seems to be robust enough to dealing with all the types of dataset shift we simulate in our experiments. The code needed to reproduce all our experiments is publicly available at https://github.com/pglez82/quant_datasetshift. © The Author(s) 2024.

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Binocular Video-Based Automatic Pixel-Level Crack Detection and Quantification Using Deep Convolutional Neural Networks for Concrete Structures

Item Type Journal Article

Author L. Liu

Author B. Shen

Author S. Huang

Author R. Liu

Author W. Liao

Author B. Wang

Author S. Diao

Abstract Crack detection and quantification play crucial roles in assessing the condition of concrete structures. Herein, a novel real-time crack detection and quantification method that leverages binocular vision and a lightweight deep learning model is proposed. In this methodology, the proposed method based on the following four modules is adopted: a lightweight classification algorithm, a high-precision segmentation algorithm, a semi-global block matching algorithm (SGBM), and a crack quantification technique. Based on the crack segmentation results, a framework is developed for quantitative analysis of the major geometric parameters, including crack length, crack width, and crack angle of orientation at the pixel level. Results indicate that, by incorporating channel attention and spatial attention mechanisms in the MBConv module, the detection accuracy of the improved EfficientNetV2 increased by 1.6% compared with the original EfficientNetV2. Results indicate that using the proposed quantification method can achieve low quantification errors of 2%, 4.5%, and 4% for the crack length, width, and angle of orientation, respectively. The proposed method can contribute to crack detection and quantification in practical use by being deployed on smart devices. © 2025 by the authors.

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Biosensors in drug discovery and drug analysis

Item Type Journal Article

Author E.B. Aydin

Author M. Aydin

Author M.K. Sezginturk

Abstract Background: The determination of drugs in pharmaceutical formulations and human biologic fluids is important for pharmaceutical and medical sciences. Successful analysis requires low sensitivity, high selectivity and minimum interference effects. Current analytical methods can detect drugs at very low levels but these methods require long sample preparation steps, extraction prior to analysis, highly trained technical staff and high-cost instruments. Biosensors offer several advantages such as short analysis time, high sensitivity, real-time analysis, low-cost instruments, and short pretreatment steps over traditional techniques. Biosensors allow quantification not only of the active component in pharmaceutical formulations, but also the

degradation products and metabolites in biological fluids. The present review gives comprehensive information on the application of biosensors for drug discovery and analysis. Moreover, this review focuses on the fabrication of these biosensors.

Methods: Biosensors can be classified as the utilized bioreceptor and the signal transduction mechanism. The classification based on signal transductions includes electrochemical optical, thermal or acoustic. Electrochemical and optic transducers are mostly utilized transducers used for drug analysis. There are many biological recognition elements, such as enzymes, antibodies, cells that have been used in fabricating of biosensors. Aptamers and antibodies are the most widely used recognition elements for the screening of the drugs. Electrochemical sensors and biosensors have several advantages such as low detection limits, a wide linear response range, good stability and re-productibility. Optical biosensors have several advantages such as direct, real-time and label-free detection of many biological and chemical substances, high specificity, sensitivity, small size and low cost. Modified electrodes enhance sensitivity of the electrodes to develop a new biosensor with desired features. Chemically modified electrodes have gained attention in drug analysis owing to low background current, wide potential window range, simple surface renewal, low detection limit and low cost. Modified electrodes produced by modifying of a solid surface electrode via different materials (carbonaceous materials, metal nanoparticles, polymer, biomolecules) immobilization. Recent advances in nanotechnology offer opportunities to design and construct biosensors. Unique features of nanomaterials provide many advantages in the fabrication of biosensors. Nanomaterials have controllable chemical structures, large surface to volume ratios, functional groups on their surface. To develop protein-inorganic hybrid nanomaterials, four preparation methods have been used. These methods are immobilization, conjugation, crosslinking and self-assembly. In the present manuscript, applications of different biosensors, fabricated by using several materials, for drug analysis are reviewed. The biosensing strategies are investigated and discussed in detail.

Results: Several analytical techniques such as chromatography, spectroscopy, radiometry, immunoassays and electrochemistry have been used for drug analysis and quantification. Methods based on chromatography require time-consuming procedure, long sample-preparation steps, expensive instruments and trained staff. Compared to chromatographic methods, immunoassays have simple protocols and lower cost. Electrochemical measurements have many advantages over traditional chemical analyses and give information about drug quantity, metabolic fate of drugs, and pharmacological activity. Moreover, the electroanalytical methods are useful to determine drugs sensitively and selectivity. Additionally, these methods decrease analysis cost and require low-cost instruments and simple sample pretreatment steps.

Conclusion: In recent years, drug analyses are performed using traditional techniques. These techniques have a good detection limit, but they have some limitations such as long analysis time, expensive device and experienced personnel requirement. Increased demand for practical and low-cost analytical techniques biosensor has gained interest for drug determinations in medical sciences. Biosensors are unique and successful devices when compared to traditional techniques. For drug determination, different electrode modification materials and different biorecognition elements are used for biosensor construction. Several biosensor construction strategies have been developed to enhance the biosensor performance. With the considerable progress in electrode surface modification, promotes the selectivity of the biosensor, decreases the production cost and provides miniaturization. In the next years, advances in technology will provide low cost, sensitive, selective biosensors for drug analysis in drug formulations and biological samples. © 2019 Bentham Science Publishers.

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Tags:

Diagnosis, Cost benefit analysis, Drug discovery, Metabolites, Chemical detection, Low-costs, Electrodes, Biosensors, Antibodies, Bacteriophages, Biorecognition element, Biorecognition elements, Crosslinking, Degradation, Drug analysis, Electrochemical biosensor, Electrochemical biosensors, Enzymes, Fabrication, Medical science, Metal nanoparticles, Molecular biology, Nanostructured materials, Optical bio-sensors, Optical biosensors, Pharmaceutical formulation, Sample preparation, Self assembly, Signal transduction, Traditional techniques, Transducers

Notes:

EID: 2-s2.0-85071041515

Export Date: 20 September 2026; Cited By: 27; Correspondence Address: M.K. Sezginturk; Çanakkale Onsekiz Mart University, Faculty of Engineering, Bioengineering Department, Çanakkale, Turkey; email: msezginturkcomu.edu.tr

Biospectroscopy, biospectrometry and imaging of: *Ilex paraguariensis*. Basis for non-destructive quality evaluation using artificial vision

Item Type Journal Article

Author T.L. Parapugna

Author G. Petroselli

Author R. Erra-Basells

Author M.G. Lagorio

Abstract Yerba mate (YM) is massively produced and consumed as an infusion in South America and is gaining popularity all over the world. This product is obtained from the dried leaves of *Ilex paraguariensis* Saint Hilaire, mixed with fragments of its dried branches. For its commercialization YM must have a minimum percentage of leaves according to a standard classification. Until now, composition quantification has been mechanically performed, thus development of new methods is still pending. In this work a quantification method using solid-phase molecular fluorescence, alternately

diffuse reflectance spectroscopy and an imaging technique using a scanner have been proposed. Strong differences between the spectroscopic properties of leaves and sticks were observed and in all the cases linear correlations between the processed signals and composition were observed. Interesting differences in chemical composition of YM leaves and sticks were additionally obtained in this work by means of total phenol content quantification, ultraviolet matrix assisted laser-desorption ionization mass spectrometry and ultraviolet laser-desorption ionization mass spectrometry. This journal is © The Royal Society of Chemistry and Owner Societies 2016.

Date 2016

DOI 10.1039/c6pp00054a

URL <https://www.scopus.com/pages/publications/84978952365?origin=resultslist>

Archive Scopus

Language English

Volume 15

Publisher Royal Society of Chemistry

Pages 879-888

Publication Photochemical and Photobiological Sciences

ISSN 1474905X (ISSN)

Issue 7

Journal Abbr Photochem. Photobiol. Sci.

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plant leaf, chemistry, signal noise ratio, Signal-To-Noise Ratio, metabolism, spectrofluorometry, Spectrometry, Fluorescence, electrospray mass spectrometry, Spectrometry, Mass, Electrospray Ionization, plant extract, chlorophyll, Chlorophyll, chlorophyll a, Ilex paraguariensis, Plant Extracts, Plant Leaves, polyphenol, Polyphenols, ultraviolet radiation, Ultraviolet Rays

Notes:

EID: 2-s2.0-84978952365

Export Date: 20 September 2026; Cited By: 5; Correspondence Address: M.G. Lagorio; INQUIMAE-CONICET/Dpto. de Química Inorgánica, Analítica y Química Física, Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires, Buenos Aires, Pabellón II 1er piso C1428EHA, Argentina; email: mgl@qi.fcen.uba.ar; CODEN: PPSHC

Birdsnap: Large-scale fine-grained visual categorization of birds

Item Type Conference Paper

Author T. Berg

Author J. Liu

Author S.W. Lee

Author M.L. Alexander

Author D.W. Jacobs

Author P.N. Belhumeur

Abstract We address the problem of large-scale fine-grained visual categorization, describing new methods we have used to produce an online field guide to 500 North American bird species. We focus on the challenges raised when such a system is asked to distinguish between highly similar species of birds. First, we introduce 'one-vs-most classifiers.' By eliminating highly similar species during training, these classifiers achieve more accurate and intuitive results than common one-vs-all classifiers. Second, we show how to estimate spatio-temporal class priors from observations that are sampled at irregular and biased locations. We show how these priors can be used to significantly improve performance. We then show state-of-the-art recognition performance on a new, large dataset that we make publicly available. These recognition methods are integrated into the online field guide, which is also publicly available. © 2014 IEEE.

Date 2014

DOI 10.1109/CVPR.2014.259

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Language English

Extra Journal Abbreviation: Proc IEEE Comput Soc Conf Comput Vision Pattern Recognit

Publisher IEEE Computer Society

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Conference Name Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition

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Tags:

Pattern recognition, Large dataset, State of the art, Improve performance, Recognition methods, recognition, Species identification, birds, Birds, Fine-grained visual categorization, Large scale classifications, large-scale classification, Spatio temporal, species identification, Visual categorization

Notes:

EID: 2-s2.0-84911399752

Export Date: 20 September 2026; Cited By: 343; Conference name: 27th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2014; Conference date: 23 June 2014 through 28 June 2014; Conference code: 114804; CODEN: PIVRE

Black-Box Modeling of Electromagnetic Interference Effects and Key Port Determination Method in Electronic Systems Based on Bayesian Networks

Item Type Conference Paper

Author Z. Han

Author Z. Zhou

Abstract The concept of modeling electromagnetic interference effects in electronic systems with Bayesian networks (BNs) is proposed to transform the complex mapping problem between system port interference response and system effects into a classification problem solved by Bayesian networks. For a specific use case, network structure learning and parameter learning are performed on the basis of discrete processing of raw continuous data. Through the validation of test data, it is demonstrated that the learned network structure meets our expectation of the relationship between electromagnetic environment, ports, and system effects, and has a high accuracy in predicting the system electromagnetic interference effects. Based on this, a Bayesian formula-based port risk quantification method is proposed to assist in identifying critical ports. © 2023 IEEE.

Date 2023

DOI 10.1109/EEI59236.2023.10212544

URL <https://www.scopus.com/pages/publications/85170103027?origin=resultslist>

Archive Scopus

Language English

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Pages 204-208

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Conference Name 2023 5th International Conference on Electronic Engineering and Informatics, EEI 2023

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Tags:

Complex networks, Bayesia n networks, Bayesian networks, Black box modelling, black box models, Complex mapping, complex systems, Determination methods, Electromagnetic compatibility, Electromagnetic interference effect, electromagnetic interference effects, Electromagnetic pulse, Electromagnetic wave interference, Electronics system, Mapping problem, Network structure learning, Signal interference, Structure parameter, System effects

Notes:

EID: 2-s2.0-85170103027

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: Z. Han; School of Software Engineering, Southeast University, Suzhou, China; email: 220205680@seu.edu.cn; Conference name: 5th International Conference on Electronic Engineering and Informatics, EEI 2023; Conference date: 30 June 2023 through 2 July 2023; Conference code: 191847

Boosting GNN-Based Link Prediction via PU-AUC Optimization

Item Type Journal Article

Author Y. Mao

Author Y. Hao

Author X. Cao

Author Y. Gao

Author C. Yao

Author X. Lin

Abstract Link prediction, which aims to predict the existence of a link between two nodes in a network, has various applications ranging from friend recommendation to protein interaction prediction. Recently, Graph Neural Network (GNN)-based link prediction has demonstrated its advantages and achieved the state-of-the-art performance. Typically, GNN-based link prediction can be formulated as a binary classification problem. However, in link prediction, we only have positive data (observed links) and unlabeled data (unobserved links), but no negative data. Therefore, Positive Unlabeled (PU) learning naturally fits the link prediction scenario. Unfortunately, the unknown class prior and data imbalance of networks impede the use of PU learning in link prediction. To deal with these issues, this paper proposes a novel model-agnostic PU learning algorithm for GNN-based link prediction by means of Positive-Unlabeled Area Under the Receiver Operating Characteristic Curve (PU-AUC) optimization. The proposed method is free of class prior estimation and able to handle the data imbalance. Moreover, we propose an accelerated method to reduce the operational complexity of PU-AUC optimization from quadratic to approximately linear. Extensive experiments back up our theoretical analysis and validate that the proposed method is capable of boosting the performance of the state-of-the-art GNN-based link prediction models. © 1989-2012 IEEE.

Date 2025

DOI 10.1109/TKDE.2025.3525490

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Journal Abbr IEEE Trans Knowl Data Eng

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Optimisations, Prediction models, Graph neural networks, State-of-the-art performance, graph neural network, Network-based, Data imbalance, Friend recommendations, Interaction prediction, Link prediction, Positive unlabeled-AUC optimization, Protein interaction, PU-AUC optimization

Notes:

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Boosting positive and unlabeled learning for anomaly detection with multi-features

Item Type Journal Article

Author J. Zhang

Author Z. Wang

Author J. Meng

Author Y.-P. Tan

Author J. Yuan

Abstract One of the key challenges of machine learning-based anomaly detection relies on the difficulty of obtaining anomaly data for training, which is usually rare, diversely distributed, and difficult to collect. To address this challenge, we formulate anomaly detection as a Positive and Unlabeled (PU) learning problem where only labeled positive (normal) data and unlabeled (normal and anomaly) data are required for learning an anomaly detector. As a semi-supervised learning method, it does not require providing labeled anomaly data for the training, thus it is easily deployed to various applications. As the unlabeled data can be extremely unbalanced, we introduce a novel PU learning method, which can tackle the situation where an unlabeled data set is mostly composed of positive instances. We start by using a linear model to extract the most reliable negative instances followed by a self-learning process to add reliable negative and positive instances with different speeds based on the estimated positive class prior. Furthermore, when feedback is available, we adopt boosting in the self-learning process to advantageously exploit the instability characteristic of PU learning. The classifiers in the self-learning process are weighted combined based on the estimated error rate to build the final classifier. Extensive experiments on six real datasets and one synthetic dataset show that our methods have better results under different conditions compared to existing methods. © 1999-2012 IEEE.

Date 2019

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Language English
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Journal Abbr IEEE Trans Multimedia
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Tags:

Reliability, Learning algorithms, Supervised learning, Anomaly detection, Reliability analysis, Error analysis, Personnel training, semi-supervised learning, Positive and unlabeled learning, Negative instances, PU learning, Semi-supervised learning, Hidden Markov models, Boosting, Pu learning, boosting, Instability characteristics, Semi-supervised learning methods

Notes:

EID: 2-s2.0-85053610972

Export Date: 20 September 2026; Cited By: 42; Correspondence Address: J. Zhang; School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, 642273, Singapore; email: jzhang069@e.ntu.edu.sg; CODEN: ITMUF

Bootstrap Latent Prototypes for graph positive-unlabeled learning

Item Type Journal Article

Author C. Liang

Author Y. Tian

Author D. Zhao

Author M. Li

Author S. Pan

Author H. Zhang

Author J. Wei

Abstract Graph positive-unlabeled (GPU) learning aims to learn binary classifiers from only positive and unlabeled (PU) nodes. The state-of-the-art methods rely on provided class prior probabilistic and their performance lags far behind the fully labeled counterparts. To bridge the gap, we propose Bootstrap Latent Prototypes (BLP), a framework that consists of a graph representation learning module and a two-step strategy algorithm. The learning module bootstraps previous versions of node

representations to serve as targets and learns enhanced representations by predicting the latent prototypes respectively for the P set and each individual in the U set. It eliminates the requirement for a class prior, while capturing positive similarity information, as well as the low-level semantic similarity and uniformity information, thereby producing closely aligned and discriminated representations for positive nodes. The algorithm module utilizes the obtained representations to select reliable negative nodes and train a binary classifier with both labeled positives and selected reliable negatives. Experimental results on diverse real-life datasets demonstrate that our proposed BLP method not only outperforms state-of-the-art approaches but also surpasses fully labeled classification models in most cases. The source code is available at <https://github.com/lcq411/BLP>. © 2024 Elsevier B.V.

Date 2024

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Performance, Classification (of information), Semantics, State-of-the-art methods, Prototypes, Graph theory, Learn+, Graph representation, Probabilistics, Prototype, Learning modules, Binary classifiers, Bootstrapping, Graph positive-unlabeled learning

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Bootstrap Your Own Prior: Towards Distribution-Agnostic Novel Class Discovery

Item Type Conference Paper

Author M. Yang

Author L. Wang

Author C. Deng

Author H. Zhang

Abstract Novel Class Discovery (NCD) aims to discover unknown classes without any annotation, by exploiting the transferable knowledge already learned from a base set of known classes. Existing works hold an impractical assumption that the novel class distribution prior is uniform, yet neglect the imbalanced nature of real-world data. In this paper, we relax this assumption by proposing a new challenging task: distribution-agnostic NCD, which allows data drawn from arbitrary unknown class distributions and thus renders existing methods useless or even harmful. We tackle this challenge by proposing a new method, dubbed 'Boot-strapping Your Own Prior (BYOP)', which iteratively estimates the class prior based on the model prediction it-self. At each iteration, we devise a dynamic temperature technique that better estimates the class prior by encouraging sharper predictions for less-confident samples. Thus, BYOP obtains more accurate pseudo-labels for the novel samples, which are beneficial for the next training iteration. Extensive experiments show that existing methods suffer from imbalanced class distributions, while BYOP1 out-performs them by clear margins, demonstrating its effectiveness across various distribution scenarios. © 2023 IEEE.

Date 2023

DOI 10.1109/CVPR52729.2023.00337

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Archive Scopus

Language English

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Conference Name Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition

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Transfer learning, Federated learning, Adversarial machine learning, Contrastive Learning, Iterative methods, Zero-shot learning, Class distributions, Unknown class, Long tail, Transfer, Boot-strapping, continual, Continual, low-shot, Low-shot, meta, Meta, or long-tail learning, Or long-tail learning, Transferable knowledge

Notes:

EID: 2-s2.0-85209563903

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BottleEdgeNet: A Calibrated Edge-Convolutional Neural Network (CNN) Framework for PET Bottle Quality Inspection in Deposit Return Schemes (DRS)

Item Type Conference Paper

Author L. Duma

Author M.D.C. Mendoza

Abstract Ensuring the cleanliness and structural integrity of PET bottles is a key requirement for the effective operation of Deposit Return Schemes (DRS) in circular economy frameworks. Traditional inspection systems and commercial Reverse Vending Machines (RVMs) often fail to detect subtle defects, leading to high False Acceptance and Rejection Rates. This paper presents BottleEdgeNet, a Calibrated Edge-Convolutional Neural Network (CNN) framework for PET bottle quality inspection that integrates EfficientNet-B0, ResNet-50, and a Google Teachable Machine backbone within a unified, low-power inference pipeline implemented in TensorFlow Lite (FP32/INT8) for CPU-oriented edge inference. The proposed system introduces a class-wise power reweighting calibration layer to stabilize confidence scores and mitigate label-shift effects under heterogeneous lighting and background conditions. A hybrid dataset comprising 7,849 training, 972 validation, and 972 testing images (plus 80 real-world samples) was used to evaluate performance across four operational classes: Clean Intact, Clean Damaged, Dirty Intact, and Dirty Damaged. Internal tests revealed that EfficientNetB0 (FP32) achieved the best accuracy (87.96%), outperforming ResNet-50 (84.47%) and Teachable Machine (67.45%), with negligible degradation after INT8 quantization. External validation confirmed EfficientNet-B0's generalization stability (87.5%), while ResNet-50 prioritized contamination detection and Teachable Machine provided a lightweight, no-code alternative for rapid deployment. To enhance field robustness, improvements should focus on data diversity and controlled illumination rather than model complexity. We report operational decision metrics (FAR/FRR) under argmax decision logic to connect model-level classification behavior with accept/reject decisions. In addition, edge-oriented efficiency is discussed through proxy-based considerations across FP32 and INT8 inference. Together, these analyses strengthen the practical relevance of the proposed framework by linking accuracy and confidence stabilization to DRS throughput, quality control, and deployment feasibility. The results demonstrate that calibrated Edge-CNNs can enable accurate, efficient, and scalable inspection of returnable PET bottles, supporting sustainable and decentralized DRS operations. © 2026 IEEE.

Date 2026

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Language English

Extra Journal Abbreviation: IEEE Conf. Technol. Sustain., SusTech

Publisher Institute of Electrical and Electronics Engineers Inc.

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Proceedings Title IEEE Conf. Technol. Sustain., SusTech

Conference Name IEEE Conference on Technologies for Sustainability, SusTech

Issue 2026

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Google+, Acceptance tests, Computer vision, Computer Vision, Convolution, Convolutional neural network, Convolutional neural networks, Statistical tests, Edge AI, Inspection, Errors, Calibration, Bottle inspection, Circular economy, Circular Economy, Convolutional Neural Networks (CNN), Deposit return scheme, Deposit Return Schemes (DRS), Deposits, Efficientnet-b0, EfficientNet-B0, Google teachable machine, Google Teachable Machine, Label-shift adaptation, Label-Shift Adaptation, PET bottle inspection, PET Bottle Inspection, PET bottles, Plastic bottles, Post-training calibration, Post-Training Calibration, Resnet-50, ResNet-50, Tensorflow lite (INT8), TensorFlow Lite (INT8)

Notes:

EID: 2-s2.0-105041690327

Export Date: 20 September 2026; Cited By: 0; Conference name: 13th IEEE Conference on Technologies for Sustainability, SusTech 2026; Conference date: 19 April 2026 through 22 April 2026; Conference code: 224411

Bound on the probability of HMM misclassification

Item Type Conference Paper

Author C. Keroglou

Author C.N. Hadjicostis

Abstract In this paper we consider the problem of classifying a sequence of observations among two known hidden Markov models (HMMs). We use a classifier that minimizes the probability of error (i.e., the probability of misclassification), and we are interested in assessing its performance by computing the a priori probability of error (before any observations are made). This probability (that the classifier makes an incorrect decision) can be obtained, as a function of the length of the sequence of observations, by summing up the probability of misclassification over all possible observation sequences, weighted by their corresponding probabilities. In an effort to avoid the high complexity associated with the computation of the exact probability of error, we establish an upper bound on the probability of error, and we find the necessary and sufficient conditions for this bound to tend to zero exponentially with the number of observation steps. We focus on classification among two HMMs that have the same language, which is the most difficult case to characterize; our approach can easily be applied to classification among any two arbitrary HMMs. The bound we obtain can also be used to approximate the dissimilarity between the two given HMMs. © 2011 IEEE.

Date 2011

DOI 10.1109/MED.2011.5983107

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Language English

Extra Journal Abbreviation: Mediterr. Conf. Control Autom., MED

ISBN 978-145770125-2 (ISBN)

Pages 449-454**Proceedings Title** Mediterr. Conf. Control Autom., MED**Conference Name** 2011 19th Mediterranean Conference on Control and Automation, MED 2011**Date Added** 20/09/2026, 9:52:49 pm**Modified** 20/09/2026, 9:52:49 pm**Tags:**

Probability, classification, Misclassifications, hidden Markov model, Upper Bound, A-priori probabilities, Hidden Markov models, Probability of misclassification, probability of error, Probability of errors, stochastic diagnosis, Sufficient conditions

Notes:

EID: 2-s2.0-80052342475

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: C.N. Hadjicostis; 1678 Nicosia, 75 Kallipoleos Avenue, Cyprus; email: chadjic@ucy.ac.cy; Conference name: 2011 19th Mediterranean Conference on Control and Automation, MED 2011; Conference date: 20 June 2011 through 23 June 2011; Conference code: 86335

Bounds on the probability of misclassification among hidden Markov models

Item Type Conference Paper**Author** C. Keroglou**Author** C.N. Hadjicostis

Abstract Given a sequence of observations, classification among two known hidden Markov models (HMMs) can be accomplished with a classifier that minimizes the probability of error (i.e., the probability of misclassification) by enforcing the maximum a posteriori probability (MAP) rule. For this MAP classifier, we are interested in assessing the a priori probability of error (before any observations are made), something that can be obtained (as a function of the length of the sequence of observations) by summing up the probability of error over all possible observation sequences of the given length. To avoid the high complexity of computing the exact probability of error, we devise techniques for merging different observation sequences, and obtain corresponding upper bounds by summing up the probabilities of error over the merged sequences. We show that if one employs a deterministic finite automaton (DFA) to capture the merging of different sequences of observations (of the same length), then Markov chain theory can be used to efficiently determine a corresponding upper bound on the probability of misclassification. The result is a class of upper bounds that can be computed with polynomial complexity in the size of the two HMMs and the size of the DFA. © 2011 IEEE.

Date 2011**DOI** 10.1109/CDC.2011.6161128**URL** <https://www.scopus.com/pages/publications/84860663711?origin=resultslist>**Archive** Scopus

Language English**Extra** Journal Abbreviation: Proc IEEE Conf Decis Control**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 07431546 (ISSN); 978-161284800-6 (ISBN)**Pages** 385-390**Proceedings Title** Proc IEEE Conf Decis Control**Conference Name** Proceedings of the IEEE Conference on Decision and Control**Date Added** 20/09/2026, 9:52:56 pm**Modified** 20/09/2026, 9:52:56 pm**Tags:**

Stochastic systems, Stochastics, classification, Hidden-Markov models, hidden Markov model, Markov chains, Stochastic models, Upper Bound, Hidden Markov models, Maximum A posteriori probabilities, Maximum a posteriori probability, Probabilities of error, Probability of misclassification, probability of error, Diagnosers, probabilistic diagnosis, Probabilistic diagnosis, stochastic diagnoser, Stochastic diagnoser

Notes:

EID: 2-s2.0-84860663711

Export Date: 20 September 2026; Cited By: 9; Correspondence Address: C. Keroglou; Department of Electrical and Computer Engineering, University of Cyprus, Nicosia, Cyprus; email: keroglou.christoforos@ucy.ac.cy; Conference name: Joint 50th IEEE Conference on Decision and Control and 2011 European Control Conference, CDC-ECC 2011; Conference date: 12 December 2011 through 15 December 2011; Conference code: 89653; CODEN: PCDCD

Brain tissue segmentation in neurosurgery: a systematic analysis for quantitative tractography approaches

Item Type Journal Article**Author** P.R. Kumar**Author** R.K. Jha**Author** A. Katti

Abstract Diffusion magnetic resonance imaging (dMRI) is a cutting-edge imaging method that provides a macro-scale in vivo map of the white matter pathways in the brain. The measurement of brain microstructure and the enhancement of tractography rely heavily on dMRI tissue segmentation. Anatomical MRI technique (e.g., T1- and T2-weighted imaging) is the most widely used method for segmentation in dMRI. In comparison to anatomical MRI, dMRI suffers from higher image distortions, lower image quality, and making inter-modality registration more difficult. The dMRI tractography study of brain connectivity has become a major part of the neuroimaging landscape in recent years. In this research, we provide a high-level overview of the methods used to segment several brain tissues types, including grey and white matter and cerebrospinal fluid, to enable quantitative studies of structural connectivity in the

brain in health and illness. In the first part of our review, we discuss the three main phases in the quantitative analysis of tractography, which are correction, segmentation, and quantification. Methodological possibilities are described for each phase, along with their popularity and potential benefits and drawbacks. After that, we will look at research that used quantitative tractography approaches to examine the white and grey matter of the brain, with an emphasis on neurodevelopment, ageing, neurological illnesses, mental disorders, and neurosurgery as possible applications. Even though there have been substantial advancements in methodological technology and the spectrum of applications, there is still no consensus regarding the "optimal" approach in the quantitative analysis of tractography. As a result, researchers should tread carefully when interpreting the findings of quantitative analysis of tractography. © The Author(s) under exclusive licence to Belgian Neurological Society 2023.

Date 2024

DOI 10.1007/s13760-023-02170-9

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human, Humans, machine learning, brain, Brain, aging, algorithm, quantitative analysis, image segmentation, procedures, surgery, Review, Segmentation, support vector machine, mental disease, schizophrenia, image processing, diagnostic imaging, Image Processing, Computer-Assisted, image reconstruction, nuclear magnetic resonance imaging, neuroimaging, white matter, White Matter, functional magnetic resonance imaging, functional connectivity, principal component analysis, brain cortex, discriminant analysis, false discovery rate, maximum likelihood method, Diffusion Magnetic Resonance Imaging, diffusion weighted imaging, anisotropic analytical algorithm, anisotropy, brain tissue segmentation, cerebrospinal fluid, Cerebrospinal fluid, dMRI, Fibre tractography, gray matter, Gray matter, Gray Matter, neurologic examination, neurosurgery, Neurosurgery, tractography, White matter, white matter tractography

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Export Date: 20 September 2026; Cited By: 13; Correspondence Address: P.R. Kumar; Department of Electronics and Communication Engineering, IcfaiTech (Faculty of Science and Technology), IFHE University, Hyderabad, 501203, India; email: revanth123451.rk@gmail.com; CODEN: ANUBB

Bread and porridge at Early Neolithic Göbekli Tepe: A new method to recognize products of cereal processing using quantitative functional analyses on grinding stones

Item Type Journal Article

Author L. Dietrich

Author M. Haibt

Abstract The Early Neolithic site Göbekli Tepe which has delivered the largest corpus of grinding stones known by now in Northern Mesopotamia is used as a case study to test the possibility of wear classification and the application of quantification methods in use-wear caused by the processing of cereals. The research design is based on combined optical and tactile analyses as well as geometric analyses of shape deformation and 3D-modelling of surface roughness on original grinding stones and experimentally used replicas. This approach can be used to differentiate between fine and coarse flour as products of cereal processing as they produce different wear markers. Also, it can be applied to quantify use-lives of handstones. At Göbekli Tepe, handstones with markers for fine flour production, which can be possibly linked to bread-like products, are in the minority while handstones with markers for coarse flour clearly predominate. The social implications of these results have to be discussed with a view towards Göbekli Tepe's interpretation as a ritual site. We propose fine flour as a special product within feasting, in accordance with previous interpretations of the site. © 2020 Elsevier Ltd

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3D-modeling, Fine and coarse flour, Göbekli Tepe, Grinding stones, Neolithic, Quantitative methods, Use-life, Use-wear

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Breast cancer classification by a new approach to assessing deep neural network-based uncertainty quantification methods

Item Type Journal Article**Author** F. Hamedani-KarAzmoodehFar**Author** R. Tavakkoli-Moghaddam**Author** A.R. Tajally**Author** S.S. Aria

Abstract Deep learning-based approaches have become widespread in medical fields and have achieved profound success in recent years. Nonetheless, most of these approaches cannot provide the certainty of their output numerically, and their output often has too many confidence levels. Therefore, they cannot be efficient and may even lead to irreparable damage. Accordingly, to tackle such a challenge, the Bayesian approximation and Ensemble learning techniques are deliberated as uncertainty quantification (UQ) methods. We apply and evaluate three UQ models regarding the classification of breast tumor tissue types. These methods are Mont Carlo-dropout (MCD), Bayesian Ensemble, and MCD Ensemble. Furthermore, to increase the classification's precision and eliminate the adverse effects of the smallness of the data collection in Wisconsin Diagnostic Breast Cancer (WDBC) used in the present research, transfer learning technique and pre-trained Convolutional Neural Network (DenseNet121) are considered. Three proposed models are compared based on their ability to quantify the reliability of classification according to novel performance metrics employed to assess estimated uncertainty. Our quantitative and qualitative examinations in the study demonstrate that models show high uncertainty in misclassifications, which is vital in determining the rate of medical diagnosis risks. As a result, by using these new assessment criteria, we intend to ascertain when can trust the output of the deep neural network. Moreover, in the examination, we witness more reliability in uncertainty quantification of the Bayesian Ensemble model. © 2022 Elsevier Ltd

Date 2023**DOI** 10.1016/j.bspc.2022.104057**URL** <https://www.scopus.com/pages/publications/85135906398?origin=resultslist>**Archive** Scopus**Language** English**Volume** 79**Publisher** Elsevier Ltd**Publication** Biomedical Signal Processing and Control**ISSN** 17468094 (ISSN)

Journal Abbr Biomed. Signal Process. Control

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Article, human, Deep neural network, Deep neural networks, Learning systems, Uncertainty analysis, Convolutional neural networks, Learning algorithms, Transfer learning, entropy, Uncertainty, quantitative analysis, decision making, prediction, deep learning, controlled study, Bayesian network, deep neural network, convolutional neural network, performance indicator, Diagnosis, Diseases, transfer of learning, Quantification methods, Uncertainty quantifications, Bayesian, Uncertainty quantification, Network-based, breast cancer, New approaches, Learning techniques, Breast cancer classification, Breast cancer classifications, cancer classification, cancer mortality, cancer risk, Mont carlo-dropout, Mont Carlo-dropout

Notes:

EID: 2-s2.0-85135906398

Export Date: 20 September 2026; Cited By: 42; Correspondence Address: R. Tavakkoli-Moghaddam; School of Industrial Engineering, College of Engineering, University of Tehran, Tehran, Iran; email: tavakoli@ut.ac.ir

Bridging expert knowledge and machine intelligence: a hybrid spatial indicator framework for ecotourism suitability

Item Type Journal Article

Author E. Karadeniz

Author S. Er

Author M. Aydogdu

Author F. Sunbul

Abstract Ecotourism suitability assessments increasingly rely on ecological indicators that capture spatial heterogeneity, ecosystem sensitivity, and biodiversity value. This study develops a hybrid indicator-based framework that combines fuzzy expert knowledge with ensemble machine learning to quantify ecotourism suitability in Malatya Province, Türkiye. Fifteen ecological and socio-environmental predictors, including elevation, slope, climate variables, river proximity, biodiversity richness, endemic species distributions, and land-cover patterns, were incorporated into a GIS-based analytical environment. Expert-derived fuzzy weights were computed using the Fuzzy Logarithmic Methodology of Additive Weights (F-LMAW) to generate an Ecological Suitability Indicator (ESI). Complementarily, K-Means clustering was used to derive data-driven suitability classes, which were modeled using Random Forest and XGBoost with spatial-block cross-validation. XGBoost demonstrated superior classification performance (accuracy = 66.8%; $\kappa = 0.585$). Across all models, biodiversity richness, endemic species presence, slope gradients, and river corridors consistently emerged as key ecological determinants. While the ESI

produced conservative suitability zones, ensemble learning identified broader high-quality ecological landscapes, including the Levent Valley corridor, Nemrut foothills, and river-based habitats. The convergence between expert- and data-driven indicators demonstrates the reliability of hybrid ecological assessment. The proposed framework offers a transparent and transferable approach for constructing ecological suitability indicators in data-scarce, biodiversity-rich regions. Copyright © 2026. Published by Elsevier Ltd.

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Ecosystems, Learning systems, machine learning, Machine-learning, Machine learning, cluster analysis, Hotspots, Forestry, climate change, K-means clustering, biodiversity, K-means++ clustering, Biodiversity, Biodiversity hotspot, Biodiversity hotspots, Bitlis, ecological modeling, Ecological modeling, Ecological suitability, Ecological suitability indicator, Ecological suitability indicators, endemic species, Ensemble machine learning, environmental quality, Forest ecology, Fuzzy LMAW, Geographic information systems, GIS, GIS-based assessment, Malatya, Nemrut, Rivers, Spatial ecological modeling, Turkey

Notes:

EID: 2-s2.0-105029052228

Export Date: 20 September 2026; Cited By: 6; Correspondence Address: S. Er; Inonu University, Faculty of Arts and Sciences, Department of Geography, Malatya, Turkey; email: selman.er@inonu.edu.tr

Bridging the gaps in cyber risk assessment: a comprehensive systematic review of standards, frameworks and quantification methods

Item Type Journal Article

Author P. Sampson

Author B. Sheehan

Author D. Shannon
Author A. Cartwright
Author J. Seifert
Author E. Cartwright
Author T. Meurs

Abstract The increasing frequency of cyber threats poses substantial challenges for organizations in both the private and public sectors. This systematic literature review evaluates and categorizes current cyber risk assessment methodologies and frameworks, supporting the selection of suitable approaches for practical and academic applications. Utilizing the PRISMA framework, 712 relevant studies were filtered from an initial pool of 1900 academic publications and subsequently analyzed and organized into a structured database, providing an overview of the advantages and limitations of widely cited approaches in this domain. The findings indicate a strong preference for established risk management frameworks, including the ISO 27000 family, OCTAVE, and NIST Special Publications, as well as mathematical approaches such as Bayesian networks, fuzzy logic, and multi-criteria decision-making techniques. The 217 approaches identified were grouped into two primary categories: Standards, Frameworks, and Guidelines and Risk Assessment Methods, with further classification by the application sectors addressed in the literature. Analysis suggests that no single approach offers universal applicability. The choice of methodology should therefore be informed by an organization's specific resources, size, and sectoral requirements. A cross-analysis of methods and sectors reveals gaps in sector-specific coverage, particularly for healthcare, finance, and small and medium-sized enterprises. The review identifies a trend toward hybrid approaches that combine organizational frameworks with quantitative methods and documents persistent barriers to adoption, including cost, data scarcity, and insufficient management engagement. Based on these findings, a conceptual framework is developed to evaluate approaches across five dimensions and to derive a typology of governance-oriented, quantitative, and hybrid methods. Implications for practitioners, regulators, and researchers are discussed in relation to current regulatory frameworks, including the NIS-2 Directive and the Digital Operational Resilience Act (DORA). © The Author(s) 2026.

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Language English

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Building a dynamic classifier for large text data collections

Item Type Conference Paper

Author P. Kalinov

Author B. Stantic

Author A. Sattar

Abstract Due to the lack of in-built tools to navigate the web, people have to use external solutions to find information. The most popular of these are search engines and web directories. Search engines allow users to locate specific information about a particular topic, whereas web directories facilitate exploration over a wider topic. In the recent past, statistical machine learning methods have been successfully exploited in search engines. Web directories remained in their primitive state, which resulted in their decline. Exploration however is a task which answers a different information need of the user and should not be neglected. Web directories should provide a user experience of the same quality as search engines. Their development by machine learning methods however is hindered by the noisy nature of the web, which makes text classifiers unreliable when applied to web data. In this paper we propose Stochastic Prior Distribution Adjustment (SPDA) - a variation of the Multinomial Naive Bayes (MNB) classifier which makes it more suitable to classify real-world data. By stochastically adjusting class prior distributions we achieve a better overall success rate, but more importantly we also significantly improve error distribution across classes, making the classifier equally reliable for all classes and therefore more usable. © 2010, Australian Computer Society, Inc.

Date 2010

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Language English

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Conference Name Conferences in Research and Practice in Information Technology Series

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Export Date: 20 September 2026; Cited By: 9; Correspondence Address: P. Kalinov; Institute for Integrated and Intelligent Systems, Griffith University, Brisbane, Australia; email: P.Kalinov@griffith.edu.au; Conference name: 21st Australasian Database Conference, ADC 2010; Conference date: 18 January 2010 through 21 January 2010; Conference code: 93772

Building a model to predict classifier accuracy

Item Type Journal Article

Author S.S. Aubakirov

Author P. Trigo

Author D.Z. Ahmed-zaki

Abstract In this paper, we propose an optimization workflow to predict classifiers accuracy based on the exploration of the space composed of different data features and the configurations of the classification algorithms. The overall process is described considering the text classification problem. We take three main features that affect text classification and therefore the accuracy of classifiers. The first feature considers the words that comprise the input text; here we use the N-gram concept with different N values. The second feature considers the adoption of textual pre-processing steps such as the stop-word filtering and stemming techniques. The third feature considers the classification algorithms hyperparameters. In this paper, we take the well-known classifiers K-Nearest Neighbors (KNN) and Naive Bayes (NB) where K (from KNN) and a-priori probabilities (from NB) are hyperparameters that influence accuracy. As a result, we explore the feature space (correlation among textual and classifier aspects) and we present an approximation model that is able to predict classifiers accuracy.

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Building carbon emissions (2016–2025): A PRISMA-based systematic review of definitions, quantification methods and policies

Item Type Journal Article

Author X. Gu

Author L. Fan

Author R. Mahabir

Abstract Managing carbon emissions from buildings requires integrated approaches across the entire lifecycle. Establishing such approaches depends on a clear understanding of key definitions for conceptual grounding, reliable quantification methods for measurement frameworks, and well-designed policies to guide effective reduction strategies. This paper analysed 111 publications on building carbon emissions from 2016 to 2025 using the PRISMA method. Key definitions have evolved from emission source perspectives to life cycle frameworks based on ISO 21930, which divides a building's life cycle into standardized stages. Past research has primarily focused on the B1-B7 stage, followed by the A1-A5 stage, while the C1-D remains understudied. Our review covers diverse quantification methods, including traditional estimation methods, data-driven prediction models (statistical, shallow machine learning, deep learning and hybrid), and emerging large language models. Our review reveals significant regional differences in policy approaches: European frameworks emphasize systematic standards, Chinese approaches centre on mandatory requirements, and American strategies focus on economic incentives. Research contributions are concentrated in China (29 %), Europe (14 %), and the United States (12 %). Drawing on the review, we also propose various reduction measures from a life cycle perspective. This review provides critical guidance for researchers, practitioners, and policymakers working to advance carbon neutrality goals. © 2025 Elsevier B.V.

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Policy, Life cycle assessment, Carbon emissions, Building, Carbon reduction, Emission quantification

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Export Date: 20 September 2026; Cited By: 6; Correspondence Address: L. Fan; Department of Civil Engineering, Xi'an Jiaotong-Liverpool University, Suzhou, 215123, China; email: Lei.Fan@xjtlu.edu.cn

Building feature-based machine learning regression to quantify urban material stocks: A Hong Kong study

Item Type Journal Article

Author L. Yuan

Author W. Lu

Author F. Xue

Author M. Li

Abstract Urban material stock (UMS) represents elegant thinking by perceiving cities as a repository of construction materials that can be reused in the future, rather than a burdensome generator of construction and demolition waste. Many studies have attempted to quantify UMS but they often fall short in accuracy, primarily owing to the lack of proper quantification methods or good data available at a micro level. This research aims to develop a simple but satisfactory model for UMS quantification by focusing on individual buildings. Generally, it is a “bottom-up” approach that uses building features to proximate the material stocks of individual buildings. The research benefits from a set of valuable, “post-mortem” ground truth data related to 71 buildings that have been demolished in Hong Kong. By comparing a series of machine learning-based models, a multiple linear regression model with six building features, namely building type, building year, height, perimeter, total floor area, and total floor number, is found to yield a satisfactory estimate of building material stocks with a mean absolute percentage error of 9.1%, root-mean-square error of 474.13, and R-square of 0.93. The major contribution of this research is to predict a building's material stock based on several easy-to-obtain building features. The methodology of machine learning regression is novel. The model provides a useful reference for quantifying UMS in other regions. Future explorations are recommended to calibrate the model when data in these regions is available. © 2022 by the International Society for Industrial Ecology.

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machine learning, Machine-learning, Machine learning, Sustainable development, China, Learning algorithms, Hong-kong, sustainable development, Feature-based, Mean square error, Linear regression, regression analysis, Floors, Circular economy, building, error correction, Construction and demolition waste, Demolition, Building materials, Buildings materials, circular economy, construction material, construction materials, demolition, Hong Kong, industrial ecology, Industrial ecology, machine learning regression, Machine learning regression, repository, urban economy, Urban material stock, urban material stocks, Urban materials

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CADSim: Robust and Scalable in-the-wild 3D Reconstruction for Controllable Sensor Simulation

Item Type Conference Paper
Author J. Wang
Author S. Manivasagam
Author Y. Chen
Author Z. Yang
Author I.A. Bârsan
Author A.J. Yang

Author W.-C. Ma

Author R. Urtasun

Editor Liu K.

Editor Kulic D.

Editor Ichnowski J.

Abstract Realistic simulation is key to enabling safe and scalable development of self-driving vehicles. A core component is simulating the sensors so that the entire autonomy system can be tested in simulation. Sensor simulation involves modeling traffic participants, such as vehicles, with high quality appearance and articulated geometry, and rendering them in real time. The self-driving industry has typically employed artists to build these assets. However, this is expensive, slow, and may not reflect reality. Instead, reconstructing assets automatically from sensor data collected in the wild would provide a better path to generating a diverse and large set with good real-world coverage. Nevertheless, current reconstruction approaches struggle on in-the-wild sensor data, due to its sparsity and noise. To tackle these issues, we present CADSim, which combines part-aware object-class priors via a small set of CAD models with differentiable rendering to automatically reconstruct vehicle geometry, including articulated wheels, with high-quality appearance. Our experiments show our method recovers more accurate shapes from sparse data compared to existing approaches. Importantly, it also trains and renders efficiently. We demonstrate our reconstructed vehicles in several applications, including accurate testing of autonomy perception systems. © 2023 Proceedings of Machine Learning Research. All rights reserved.

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Autonomous vehicles, Real- time, Self drivings, Machine learning, Sensors data, High quality, Computer aided design, 3D modeling, 3D reconstruction, Image reconstruction, Three dimensional computer graphics, 3D Reconstruction, Autonomy systems, CAD models, Core components, Realistic simulation, Self-Driving, Sensor Simulation, Sensor Simulations

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Calibrating Pseudo-Labeling with Class Distribution for Semi-supervised Text Classification

Item Type Conference Paper

Author W. Yang

Author R. Zhang

Author J. Chen

Author J. Sheng

Editor Christodoulopoulos C.

Editor Chakraborty T.

Editor Rose C.

Editor Peng V.

Abstract Semi-supervised text classification (SSTC) aims to train text classification models with few labeled data and massive unlabeled data. Existing studies develop effective pseudo-labeling methods, but they can struggle with unlabeled data that have imbalanced classes mismatched with the labeled data, making the pseudo-labeling biased towards majority classes, resulting in catastrophic error propagation. We believe it is crucial to explicitly estimate the overall class distribution, and use it to calibrate pseudo-labeling to constrain majority classes. To this end, we formulate the pseudo-labeling as an optimal transport (OT) problem, which transports the unlabeled sample distribution to the class distribution. With a memory bank, we dynamically collect both the high-confidence pseudo-labeled data and true labeled data, thus deriving reliable (pseudo-) labels for class distribution estimation. Empirical results on 3 commonly used benchmarks demonstrate that our model is effective and outperforms previous state-of-the-art methods. © 2025 Association for Computational Linguistics.

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Conference Name EMNLP 2025 - 2025 Conference on Empirical Methods in Natural Language Processing, Proceedings of the Conference

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Can Class-Priors Help Single-Positive Multi-Label Learning?

Item Type Conference Paper**Author** B. Liu**Author** N. Xu**Author** J. Wang**Author** X. Geng

Abstract Single-positive multi-label learning (SPMLL) is a weakly supervised multi-label learning problem, where each training example is annotated with only one positive label. Existing SPMLL methods typically assign pseudo-labels to unannotated labels with the assumption that prior probabilities of all classes are identical. However, the class-prior of each category may differ significantly in real-world scenarios, which makes the predictive model not perform as well as expected due to the unrealistic assumption on real-world application. To alleviate this issue, a novel framework named CRISP, i.e., Class-pRiors Induced Single-Positive multi-label learning, is proposed. Specifically, a class-priors estimator is introduced, which can estimate the class-priors that are theoretically guaranteed to converge to the ground-truth class-priors. In addition, based on the estimated class-priors, an unbiased risk estimator for classification is derived, and the corresponding risk minimizer can be guaranteed to approximately converge to the optimal risk minimizer on fully supervised data. Experimental results on ten MLL benchmark datasets demonstrate the effectiveness and superiority of our method over existing SPMLL approaches. © 2025 Neural information processing systems foundation. All rights reserved.

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Can gut microbiota composition predict response to dietary treatments?

Item Type Journal Article

Author J.R. Biesiekierski

Author J. Jalanka

Author H.M. Staudacher

Abstract Dietary intervention is a challenge in clinical practice because of inter-individual variability in clinical response. Gut microbiota is mechanistically relevant for a number of disease states and consequently has been incorporated as a key variable in personalised nutrition models within the research context. This paper aims to review the evidence related to the predictive capacity of baseline microbiota for clinical response to dietary intervention in two specific health conditions, namely, obesity and irritable bowel syndrome (IBS). Clinical trials and larger predictive modelling studies were identified and critically evaluated. The findings reveal inconsistent evidence to support baseline microbiota as an accurate predictor of weight loss or glycaemic response in obesity, or as a predictor of symptom improvement in irritable bowel syndrome, in dietary intervention trials. Despite advancement in quantification methodologies, research in this area remains challenging and larger scale studies are needed until personalised nutrition is realistically achievable and can be translated to clinical practice. © 2019 by the authors. Licensee MDPI, Basel, Switzerland.

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human, Humans, nonhuman, prediction, Review, classification, alanine aminotransferase, body weight loss, obesity, Obesity, clinical practice, Bacteria, bacterium, isolation and purification, Microbiota, microbiology, physiology, body mass, treatment response, animal, Animals, genetic variability, fermentation, Gastrointestinal Microbiome, intestine flora, Mice, mouse, treatment outcome, Treatment Outcome, Akkermansia muciniphila, physical activity, real time polymerase chain reaction, glucose blood level, diet, Diet, hyperglycemia, abdominal pain, Bacteroides, Blood Glucose, diet restriction, diet therapy, Dietary intervention, fecal microbiota transplantation, Firmicutes, food composition, food intake, Gastrointestinal symptoms, immunomodulation, Irritable bowel syndrome, Irritable Bowel Syndrome, irritable colon, maltodextrin, microbiome, Personalised nutrition, probiotic agent, randomized controlled trial (topic), short chain fatty acid, Weight Loss

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Can Land Cover Classification Models Benefit From Distance-Aware Architectures?

Item Type Journal Article

Author C. Koller

Author P. Jung

Author X.X. Zhu

Abstract The quantification of predictive uncertainties helps to understand where the existing models struggle to find the correct prediction. A useful quality control tool is the task of detecting out-of-distribution (OOD) data by examining the model's predictive uncertainty. For this task, deterministic single forward pass frameworks have recently been established as deep learning models and have shown competitive performance in certain tasks. The unique combination of spectrally normalized weight matrices and residual connection networks with an approximate Gaussian process (GP) output layer can here offer the best trade-off between performance and complexity. We utilize this framework with a refined version that adds spectral batch normalization and an inducing points approximation of the GP for the task of OOD detection in remote sensing image classification. This is an important task in the field of remote sensing, because it provides an evaluation of how reliable the model's predictive uncertainty estimates are. By performing experiments on the benchmark datasets Eurosat and So2Sat LCZ42, we can show the effectiveness of the proposed adaptations to the residual networks (ResNets). Depending on the chosen dataset, the proposed methodology achieves OOD detection performance up to 16% higher than previously considered distance-aware networks. Compared with other uncertainty quantification methodologies, the results are on the same level and exceed them in certain experiments by up to 2%. In particular, spectral batch normalization, which normalizes the batched data as opposed to normalizing the network weights by the spectral normalization (SN), plays a crucial role and leads to performance gains of up to 3% in every single experiment. For reproducibility, the code can be found here: https://github.com/ChrisKo94/DUE_Land_Cover. © 2004-2012 IEEE.

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Economic and social effects, Deep learning, Uncertainty analysis, Gaussian distribution, Gaussian noise (electronic), Remote sensing, quality control, remote sensing, classification, Classification models, Normalisation, Quality control, detection method, land cover, Uncertainty quantifications, uncertainty quantification, Predictive uncertainty, performance assessment, uncertainty analysis, Out-of-distribution, satellite imagery, Gaussian Processes, correction, Distance awareness, Distance-awareness, land cover classification, Land cover classification, out-of-distribution (OOD), Spectral normalization, spectral normalization (SN), Useful qualities

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Can Lymph Transportation Capacity Predict Treatment Efficacy of Lower Extremity Lymphedema by LVA?

Item Type Journal Article

Author Y. Sowa

Author T. Kodama

Author T. Numajiri

Abstract Treatment outcomes for lower extremity lymphedema (LEL) using multiple lymphaticovenular anastomoses (LVA) are still uncertain. Classification of progression of lymphedema by disease staging is a potential preoperative predictor of the efficacy of treatment, but it is difficult to judge progression of lymphedema objectively. Recent studies have indicated that lymph pump dysfunction, which reflects lymph transportation capacity, is associated with lymphedema progression. Indocyanine green (ICG) lymphography, a minimally invasive modality for pathophysiological assessment of lymphedema, can be used for rapid and objective measurement of ICG velocity (ICGv) and transit time to the knee (TTk), which are parameters of lymph transportation capacity, over a certain period. In the current study, we analyzed the relationship between these parameters and outcomes for LEL treated by multiple LVA. Thirty-four consecutive patients who underwent multiple LVA and ICG lymphography were enrolled in the study. The relationship of ICGv and TTk with the efficacy of treatment by LVA (LEL index reduction) was investigated using Pearson correlation coefficient analysis. LEL index reduction was more strongly correlated with ICGv than with TTk, whereas it was weakly correlated with both quantification methods of lymph pump function ($r > 0.6$). Both ICGv and TTk are objective and simple parameters that can measure lymph pump functions quickly. Lymph pump function, especially calculated with ICGv, might help predict the treatment efficacy and objective evaluation after therapeutic intervention using multiple LVA. © 2021 Lippincott Williams and Wilkins. All rights reserved.

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Capturing uncertainty in black-box chromatography modelling using conformal prediction and Gaussian processes

Item Type Journal Article

Author T.D. Pham

Author R. Bassett

Author U. Aickelin

Abstract We demonstrate that conformal predictors – specifically conformalised quantile regression (CQR) and locally adaptive conformal predictors (LACP) – outperform the commonly used Gaussian Process Regression (GPR) in uncertainty quantification of machine learning surrogate models for chromatography modelling. CQR excelled in black-box scenarios, effectively estimating challenging target variable distributions, while LACP provided extremely informative intervals when kinetic parameters were included. Incorporating kinetic data significantly reduced epistemic uncertainty and increased model accuracy, supporting the hypothesis that adding mechanistic data to black-box models improves prediction uncertainty. This study represents the first application of conformal methods in chromatography modelling, indicating high applicability of this new uncertainty quantification methodology. Our findings offer a promising direction for advancing uncertainty quantification methods in data-driven bioprocess modelling and optimisation. © 2025

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CASUAL: Conditional Support Alignment for Domain Adaptation with Label Shift

Item Type Conference Paper

Author A.T. Nguyen

Author L. Tran

Author A. Tong

Author T.-D.H. Nguyen

Author T. Tran

Editor Walsh T.

Editor Shah J.

Editor Kolter Z.

Abstract Unsupervised domain adaptation (UDA) refers to a domain adaptation framework in which a learning model is trained based on the labeled samples on the source domain and unlabelled ones in the target domain. The dominant existing methods in the field that rely on the classical covariate shift assumption to learn domain-invariant feature representation have yielded suboptimal performance under label distribution shift. In this paper, we propose a novel Conditional Adversarial SUPport ALignment (CASUAL) whose aim is to minimize the conditional symmetric support divergence between the source's and target domain's feature representation distributions, aiming at a more discriminative representation for the classification task. We also introduce a novel theoretical target risk bound, which justifies the merits of aligning the supports of conditional feature distributions compared to the existing marginal support alignment approach in the UDA settings. We then provide a complete training process for learning in which the objective optimization functions are precisely based on the proposed target risk bound. Our empirical results demonstrate that CASUAL outperforms other state-of-the-art methods on different UDA benchmark tasks under different label shift conditions. Copyright © 2025, Association for the Advancement of

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EID: 2-s2.0-105003904140

551 of 2790

Cataract quantification using swept-source optical coherence tomography

Item Type Journal Article

Author J.E.G. Brás

Author W. Sickenberger

Author N. Hirnschall

Author O. Findl

Abstract Purpose: To develop and evaluate a cataract quantification method using a swept-source optical coherence tomography (SS-OCT) device (IOLMaster 700). Setting: Hanusch Hospital, Vienna, Austria. Design: Prospective multicenter case series. Methods: This study included patients with cataract in at least 1 eye. Two independent examiners performed Lens Opacities Classification System II (LOCS II) grading at the slitlamp independently. Corrected distance visual acuity, contrast sensitivity, and SS-OCT measurements were also performed. In addition, the phacoemulsification energy and time were recorded. To develop an objective SS-OCT-based cataract quantification system, all SS-OCT scans were segmented and the local pixel intensity unit of the lens was analyzed using Matlab's grayscale counting. The pixel intensity unit of the lens nucleus (OCT-based cataract quantification system score) was equated to the clinician's subjective nuclear opalescence grading. Results: The study evaluated 186 eyes (113 patients). The correlation between the independent examiners' LOCS grading was good (0.91) ($P < .01$). The correlation between the LOCS grading and OCT-based cataract quantification system score was 0.86 for examiner 1 and 0.76 for examiner 2 (both $P < .01$). In 24 patients, phacoemulsification time, power, and energy; visual acuity; and contrast sensitivity were available and included in the study. The OCT-based cataract quantification system scores correlated significantly with phacoemulsification time (0.71) and energy (0.64) (both $P < .01$). Conclusions: Cataract density could be evaluated using an SS-OCT device, and the OCT-based cataract quantification system values correlated with the conventional LOCS II classification. Swept-source OCT measurements allowed quantification and documentation cataract density. © 2018 ASCRS and ESCRS

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Export Date: 20 September 2026; Cited By: 23; Correspondence Address: O. Findl; Hanusch Hospital, Department of Ophthalmology, Vienna, Heinrich-Collin-Straße 30, 1140, Austria; email: oliver@findl.at; CODEN: JCSUE

Categorization of multiple objects in a scene using a biased sampling strategy

Item Type Journal Article

Author L. Yang

Author N. Zheng

Author M. Chen

Author Y. Yang

Author J. Yang

Abstract Recently, various bag-of-features (BoF) methods show their good resistance to within-class variations and occlusions in object categorization. In this paper, we present a novel approach for multi-object categorization within the BoF framework. The approach addresses two issues in BoF related methods simultaneously: how to avoid scene modeling and how to predict labels of an image when multiple categories of objects are co-existing. We employ a biased sampling strategy which combines the bottom-up, biologically inspired saliency information and loose, top-down class prior information for object class modeling. Then this biased sampling component is further integrated with a multi-instance multi-label learning and classification algorithm. With the proposed biased sampling strategy, we can perform multiobject categorization within an image without semantic segmentation. The experimental results on PASCAL VOC2007 and SUN09 show that the proposed method significantly improves the discriminative ability of BoF methods and achieves good performance in multi-object categorization tasks. © Springer Science+Business Media New York 2013.

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Artificial intelligence, Software engineering, Semantic segmentation, Classification algorithm, Discriminative ability, Bag-of-Features, Bag-of-features method, Biased sampling, Biased sampling strategy, Biologically inspired, Multi-instance multi-label learning, Object categorization

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Categorizing and mining concept drifting data streams

Item Type Conference Paper

Author P. Zhang

Author X. Zhu

Author Y. Shi

Abstract Mining concept drifting data streams is a defining challenge for data mining research. Recent years have seen a large body of work on detecting changes and building prediction models from stream data, with a vague understanding on the types of the concept drifting and the impact of different types of concept drifting on the mining algorithms. In this paper, we first categorize concept drifting into two scenarios: Loose Concept Drifting (LCD) and Rigorous Concept Drifting (RCD), and then propose solutions to handle each of them separately. For LCD data streams, because concepts in adjacent data chunks are sufficiently close to each other, we apply kernel mean matching (KMM) method to minimize the discrepancy of the data chunks in the kernel space. Such a minimization process will produce weighted instances to build classifier ensemble and handle concept drifting data streams. For RCD data streams, because genuine concepts in adjacent data chunks may randomly and rapidly change, we propose a new Optimal Weights Adjustment (OWA) method to determine the optimum weight values for classifiers trained from the most recent (up-to-date) data chunk, such that those classifiers can form an accurate classifier ensemble to predict instances in the yet-to-come data chunk. Experiments on synthetic and real-world datasets will show that weighted instance approach is preferable when the concept drifting is mainly caused by the changing of the class prior probability; whereas the

weighted classifier approach is preferable when the concept drifting is mainly triggered by the changing of the conditional probability. © 2008 ACM.

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Export Date: 20 September 2026; Cited By: 87; Correspondence Address: P. Zhang; Research Center on Fictitious Economy and Data Sciences, Chinese Academy of Sciences, Beijing 100190, China; email: zhangpeng04@mails.gucas.ac.cn; Conference name: 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD 2008; Conference date: 24 August 2008 through 27 August 2008; Conference code: 77094

Category-specific unlabeled data risk minimization for ultrasound semi-supervised segmentation

Item Type Journal Article

Author L. Xu

Author M. Liu

Author B. Wei

Author Y. He

Author Z. Gao

Author H. Han

Author J. Zhang

Abstract Achieving accurate computer-aided analysis of ultrasound images is challenging, since not only its image artifacts but also the difficulties in collecting large-scale pixel-wise annotations from experts for training. Semi-supervised segmentation is a solution for learning from labeled and unlabeled data, which mainly focuses on generating pseudo annotations for unlabeled data or learning consistent features in enhanced views of images to enhance model generalization. However, anatomically, diverse learning difficulties across tissues are overlooked, and, technically, the estimation and minimization of empirical risk for unlabeled training data are largely ignored. Motivated by them, this work proposes a semi-supervised segmentation model, named CSUDRM, with two modules. The former is called category-specific distribution alignment (CSDA), which learns more consistent feature representations of the same class across labeled and unlabeled data. Moreover, it enhances feature space intra-class compactness and inter-class discrepancy and provides category-specific penalties for more robust learning. The latter one is Unlabeled Data Risk Minimization (UDRM). It minimizes the risk on the entire training data, which distinguishes it from most existing works that merely optimize labels. The risk of unlabeled data is estimated by a novel learnable class prior estimator, with the help of distributional hints from CSDA. This design could reinforce the robustness of the model and achieve stable segmentation. CSUDRM achieves state-of-the-art performances on four ultrasound datasets. Extensive ablation studies, including quantitative comparisons, feature space visualization, and robustness analysis, demonstrate the superiority of our designs. Copyright © 2025. Published by Elsevier B.V.

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Causal Analysis of Factors for Li Ionic Conductivity in Olivine-Type LiMXO₄ Materials Using LiNGAM

Item Type Journal Article

Author K. Gocho

Author M. Hamaie

Author N. Tanibata

Author H. Takeda

Author M. Nakayama

Author M. Karasuyama

Author R. Kobayashi

Abstract All-solid-state batteries are increasingly examined to improve the safety of lithium-ion batteries. Considerable efforts are being made to develop solid electrolyte materials with high ionic conductivity to realize such batteries. Recently, materials informatics (MI), which uses machine learning to quantify the relationships between descriptors that represent the composition, structure, and properties of materials, has been increasingly utilized. However, MI has primarily focused on evaluating the correlations between descriptors, with few examples analyzing hierarchical cause-and-effect relationships. This study analyzes the causal relationships among the descriptors of lithium-ion conductivity in olivine-type LiMXO₄ materials using linear non-Gaussian acyclic model (LiNGAM), a method of causal discovery. We previously compiled a data set for these materials using first-principles calculations and performed machine learning regression analyses. A key aspect of this research is the systematization of knowledge regarding the composition, structure, and ionic conductivity of materials. Clarifying the causal relationships can lead to the discovery of factors that truly control ionic conductivity. Therefore, we qualitatively interpret the causal relationship diagram and provide specific guidelines for controlling the lithium-ion conductivity by comparing the results of previous machine learning regression analyses with those of LiNGAM. © 2025 The Authors. Published by American Chemical Society.

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Causal Language Model Aided Sequential Decoding With Natural Redundancy

Item Type Journal Article

Author Z. Zhu

Author H. Yu

Author C. Shen

Author J. Du

Author Z. Shen

Author Z. Wang

Abstract A high-performance communication receiver desires a sufficient and accurate recognition of transmitted sources. In this paper, we propose a sequential decoding algorithm for the robust reception of sources with natural redundancy (NR) over the AWGN channel. To fully exploit the abundant NR in the exemplified English text sources, a causal language modeling (CLM) with a powerful Transformer decoder neural network (NN) structure is adopted for modeling the joint probability distribution. For versatility of byte-level tokenization, the UTF-8 encoding scheme is considered for the Wikipedia corpus, namely the English edition of the Wiki-40B dataset. The tree search-based M-algorithm (MA) is integrated with CLM, denoted as CLM-MA algorithm, to synthesize the a priori probability of information sequence and likelihood values of polluted symbols. Both simulation experiments and hardware platform evaluations are conducted to analyze the performance of the CLM-MA algorithm. With an adequately large M value, tremendous performance profit is achievable, especially for high-efficiency modulation levels and extremely noisy conditions. This mechanism of adopting the pre-trained NN model merely desires self-supervised learning architecture and eliminates the requirement of high-cost and accurate labels, which paves a new way for communication receiver design. © 1972-2012 IEEE.

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Issue 5**Journal Abbr** IEEE Trans Commun**Date Added** 20/09/2026, 9:51:52 pm**Modified** 20/09/2026, 9:51:52 pm**Tags:**

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Causation analysis of collision accidents using Chi-squared test and Bayesian network

Item Type Conference Paper**Author** X. Liu**Author** X. Shen**Author** Q. Yu**Editor** Le Sourne H.**Editor** Soares C.G.

Abstract Maritime accident research conducted at conventions often suffers from insufficient quantitative descriptions of the correlations between factors. This paper aims to conduct new research by incorporating maritime accident data from eight coastal areas into a data-driven Bayesian network model (BN) to analyze the relationship between the severity of marine accidents and relevant Accident Influential Factors (AIFs). To achieve this aim, the factors contributing to maritime accidents are identified, and each of them is subjected to a chi-squared test against the severity of the accident. The results of the chi-square test are then compiled into a comparison matrix, which is imported into the Bayesian network as a priori probability. Next, a Bayesian search algorithm is used to establish a data-driven BN model, and the AIFs of the database for eight different coastal areas are analyzed through data training and machine learning based on the prior probability. This study found that “Type of accident,” “Weather,” and “Sea condition” have the highest correlation with accident severity. Additionally, factors such as visibility, ship size, and age significantly impact serious accidents. The research findings are beneficial for the analysis of marine accidents and

provide insights to stakeholders such as safety authorities and ship owners. ©
2024 The Author(s).

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CDFNet: Cross-Domain Feature Fusion Network for PolSAR Terrain Classification

Item Type Journal Article
Author S. Wang
Author Z. Sun
Author T. Bian
Author Y. Guo
Author L. Dai
Author Y. Guo
Author L. Jiao

Abstract The scarcity of labeled data and domain shift among polarimetric synthetic aperture radar (PolSAR) images degrades the performance of the supervised-learning-based algorithm. Some unsupervised domain adaptation (UDA) algorithms have been proposed to address this problem and achieve good performance. The existing UDA algorithms for PolSAR terrain classification focus on the feature distribution shift problem but ignore the label shift problem in UDA task. In addition, feature alignment-based algorithms generate pseudo labels for target domain which introduce label noise and compromising the UDA performance. To alleviate the problems above, we present a cross-domain feature fusion network (CDFNet) for PolSAR terrain classification. Specifically, a domain-balanced sampling (DBS) module is proposed to obtain a nearly balanced training dataset to alleviate the label shift problem. Then, a cross-domain feature fusion (CDF) module is presented to achieve class-wise feature alignment with no additional label noise introduction. Experimental results on four PolSAR datasets demonstrate that our algorithm outperforms state-of-the-art UDA algorithms in terms of target domain performance. © 1980-2012 IEEE.

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Centerline-based quantification of true Lumen helical Morphology in Type B aortic dissection: Unlocking the potential of helicity as a geometric biomarker

Item Type Journal Article

Author F.O. Kolawole

Author J. Bondesson

Author B.W. Ullery

Author C.P. Cheng

Abstract The helical morphology of Type B aortic dissections (TBAD) represents a potentially important geometric biomarker that may influence dissection progression. While three-dimensional surface-based quantification methods provide accurate TBAD helicity assessment, their clinical adoption remains limited by significant processing time. We developed and validated a clinically practical centerline-based helicity quantification method using routine imaging software (TeraRecon) against an extensively validated surface-based method (SimVascular). In 87 TBAD patients, we semi-automatically extracted aortic, true lumen, and branch vessel centerlines from CT imaging. Helical parameters, including true lumen helical angle and peak helical twist, were computed relative to a standardized anatomical reference, enabling classification of patients into four distinct helicity categories: left-chiral, right-chiral, non-helical, and mixed-chiral patterns. The centerline method demonstrated 92% classification accuracy with excellent agreement with surface-based measurements (Cohen's $\kappa=0.88$, $p<0.001$). Wilcoxon signed-rank tests revealed a median difference of -0.4° ($z=-1.08$, $p=0.28$), indicating no statistically significant systematic bias between methods. This centerline approach we have developed provides clinically feasible TBAD helicity classification while maintaining excellent agreement with the gold-standard surface-based method. This technique can integrate seamlessly with existing clinical workflows, enabling practical assessment of TBAD helical

morphology for enhanced risk stratification and personalized treatment planning. ©
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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: F.O. Kolawole; Stanford, 1201 Welch Rd, 94305, United States; email: fikunwa@stanford.edu; CODEN: JBMCB

centroIDA: Cross-domain class discrepancy minimization based on
accumulative class-centroids for Imbalanced Domain Adaptation

Item Type Journal Article

Author X. Sun
Author Z. Wu
Author Y. Ji
Author Z. Zhan

Abstract Unsupervised Domain Adaptation (UDA) is a critical task in transfer learning, which seeks to apply the knowledge acquired in one domain (the source domain) to a different but related domain (the target domain). In this context, the two domains exhibit Out-Of-Distribution(OOD), and the label information of the target domain is unavailable. Although numerous UDA methods have made significant strides in mitigating the discrepancies in feature distributions across domains, they often neglect a more prevalent real-world scenario. This scenario, referred to as Imbalanced Domain Adaptation (IDA) task, is characterized by the coexistence of differences in both feature and label distributions. Our empirical analysis suggests that overlooking the disparities in label distributions can severely compromise the performance of the model, thereby limiting the generalizability of domain adaptation models. Consequently, our objective is to introduce a model capable of concurrently addressing differences in both feature and label distributions, thereby fostering a broader application of domain adaptation methodologies. In this paper, we propose a cross-domain class discrepancy minimization method based on accumulative class-centroids for IDA, termed as centroIDA. The strategies of accumulative class-centroids alignment and class-wise feature alignment pay heed to the relationship between cross-domain intra-class features and inter-class features, thereby facilitating conditional feature alignment to overcome both feature and label distribution differences simultaneously. Our experimental evaluation on OfficeHome and DomainNet datasets reveals that, in comparison to TIToK (a novel method also designed to tackle the IDA problem), our method enhances the average accuracy by 5.6% and 2.7%, respectively. Furthermore, as the label shift intensifies, our method consistently outperforms other methods, demonstrating a robust resistance to label shift. © 2024 Elsevier Ltd

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Certifiable Active Class Selection in Multi-Class Classification

Item Type Conference Paper

Author M. Senz

Author M. Bunse

Author K. Morik

Editor Kottke D.

Editor Krempl G.

Editor Holzinger A.

Editor Hammer B.

Editor CITEC Center of Excellence, Inspiration 1, Bielefeld University of Bielefeld

Abstract Active class selection (ACS) requires the developer of a classifier to actively choose the class proportions of the training data. This freedom of choice puts the trust in the trained classifier at risk if the true class proportions, which occur during deployment, are subject to uncertainties. This issue has recently motivated a certificate for ACS-trained classifiers, which builds trust by proving that a classifier is sufficiently correct within a specific set of class proportions and with a high probability. However, this certificate was only developed in the context of binary classification. In this paper, we employ Hölder's inequality to extend the binary ACS certificate to multi-class settings. We demonstrate that our extension indeed provides correct and tight upper bounds of the classifier's error. We conclude with several directions for future work. © 2022 for this paper by its authors.

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Certification of Model Robustness in Active Class Selection

Item Type Conference Paper

Author M. Bunse

Author K. Morik

Editor Oliver N.

Editor Pérez-Cruz F.

Editor Kramer S.

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Abstract Active class selection provides machine learning practitioners with the freedom to actively choose the class proportions of their training data. While this freedom can improve the model performance and decrease the data acquisition cost, it also puts the practical value of the trained model into question: is this model really appropriate for the class proportions that are handled during deployment? What if the deployment class proportions are uncertain or change over time? We address these questions by certifying supervised models that are trained through active class selection. Specifically, our certificate declares a set of class proportions for which the certified model induces a training-to-deployment gap that is small with a high probability. This declaration is theoretically justified by PAC bounds. We apply our proposed certification method in astro-particle physics, where a simulation generates telescope recordings from actively chosen particle classes. © 2021, Springer Nature Switzerland AG.

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Challenge of validation in whole-cell spike-in amplicon sequencing to comprehensively quantify food lactic acid bacteriota

Item Type Journal Article

Author M. Oshiro

Author K. Nakamura

Author Y. Tashiro

Abstract Lactic acid bacteria (LAB) shape diverse communities in fermented foods. Developing comprehensive quantification methods for community structure will revolutionize our understanding of food LAB microbiome. For this purpose, 16S rRNA gene amplicon-based quantification, using spiked exogenous bacterial cells as an internal standard, shows potential for comprehensiveness and accuracy. We validated cell spike-in amplicon sequencing for quantifying LAB communities in food. Low efficiency of LAB DNA extraction underscores the importance of compensating for DNA loss by spiking internal standard cells. Quantitative equations generated using 15 selected LAB mock species showed positive relationships between the ratio of MiSeq read counts and the expected 16S rRNA gene copy numbers, with coefficients of determination (R^2) ≥ 0.6823 . The fold differences between observed and expected 16S copy numbers were within the range of 1/3 to 3-fold. Our validation highlights that accurate preparation of the LAB mock community is crucial for cell spike-in amplicon sequencing accuracy. © The Author(s) 2024. Published by Oxford University Press on behalf of Japan Society for Bioscience, Biotechnology, and Agrochemistry.

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procedures, classification, bacterial DNA, DNA sequencing, DNA, Bacterial, genetics, high throughput sequencing, High-Throughput Nucleotide Sequencing, isolation and purification, Microbiota, microflora, Sequence Analysis, DNA, Quantification methods, RNA 16S, RNA, Ribosomal, 16S, 16S rRNA gene, Amplicons, Diverse community, DNA, fermented food, Fermented foods, Food chemistry, food control, Food microbiology, Food Microbiology, Gene encoding, Internal standards, lactic acid bacteria, Lactic acid bacteria, Lactobacillales, mock community, Mock community, quantitative amplicon sequencing, Quantitative amplicon sequencing, RNA, Whole cell

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Challenges in projecting clustering results across gene expression-profiling datasets

Item Type Journal Article
Author L. Lusa
Author L.M. McShane

Author J.F. Reid

Author L. De Cecco

Author F. Ambroggi

Author E. Biganzoli

Author M. Gariboldi

Author M.A. Pierotti

Abstract Background: Gene expression microarray studies for several types of cancer have been reported to identify previously unknown subtypes of tumors. For breast cancer, a molecular classification consisting of five subtypes based on gene expression microarray data has been proposed. These subtypes have been reported to exist across several breast cancer microarray studies, and they have demonstrated some association with clinical outcome. A classification rule based on the method of centroids has been proposed for identifying the subtypes in new collections of breast cancer samples; the method is based on the similarity of the new profiles to the mean expression profile of the previously identified subtypes. Methods: Previously identified centroids of five breast cancer subtypes were used to assign 99 breast cancer samples, including a subset of 65 estrogen receptor-positive (ER+) samples, to five breast cancer subtypes based on microarray data for the samples. The effect of mean centering the genes (i.e., transforming the expression of each gene so that its mean expression is equal to 0) on subtype assignment by method of centroids was assessed. Further studies of the effect of mean centering and of class prevalence in the test set on the accuracy of method of centroids classifications of ER status were carried out using training and test sets for which ER status had been independently determined by ligand-binding assay and for which the proportion of ER+ and ER- samples were systematically varied. Results: When all 99 samples were considered, mean centering before application of the method of centroids appeared to be helpful for correctly assigning samples to subtypes, as evidenced by the expression of genes that had previously been used as markers to identify the subtypes. However, when only the 65 ER+ samples were considered for classification, many samples appeared to be misclassified, as evidenced by an unexpected distribution of ER+ samples among the resultant subtypes. When genes were mean centered before classification of samples for ER status, the accuracy of the ER subgroup assignments was highly dependent on the proportion of ER+ samples in the test set; this effect of subtype prevalence was not seen when gene expression data were not mean centered. Conclusions: Simple corrections such as mean centering of genes aimed at microarray platform or batch effect correction can have undesirable consequences because patient population effects can easily be confused with these assay-related effects. Careful thought should be given to the comparability of the patient populations before attempting to force data comparability for purposes of assigning subtypes to independent subjects. © The Author 2007. Published by Oxford University Press.

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Chance-corrected utility-weighted metrics for binary classification and information retrieval: A conceptual and analytical framework

Item Type Journal Article

Author W.-C. Lee

Abstract Standard confusion-matrix metrics (e.g., AUROC, Youden's index, F1, Jaccard) often do not explicitly account for class imbalance, asymmetric utilities, or performance relative to a no-information baseline. These limitations can obscure the utility structure relevant to model choice. We propose a unified utility-weighted evaluation framework for binary classification and information retrieval, together with a kappa-like chance correction. For classification, we define the utility-weighted accuracy rate (UWAR) by normalizing expected utility to the perfect-model utility, yielding a weighted average of sensitivity and specificity controlled by a single utility-weighting parameter that combines positive-class prevalence with the relative benefit of a true positive compared with a true negative. We then introduce the chance-corrected UWAR (ccUWAR) to quantify utility-weighted performance above the no-information baseline, with interpretable special cases that connect to familiar indices such as AUROC and Youden's index. For information retrieval, we develop parallel measures in the positivity-conditioned space: the conditional utility-weighted accuracy rate (CUWAR) and its chance-corrected version (ccCUWAR), parameterized by the relative cost of missed relevant items versus retrieved non-relevant items and the prevalence of relevant items; CUWAR reduces to the Jaccard

index and is directly related to the F1 score when these error costs are equal. We further derive better-than-chance regions, indifference points, and regions of dominance. Through controlled analytical examples, we illustrate that rankings that appear fixed under symmetric metrics can reverse once utility structure and chance correction are made explicit. The framework is intended as a conceptual and analytical tool for utility-aligned evaluation rather than as a universal replacement for existing metrics. © 2026 Elsevier B.V.

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Changes in mouse gastrointestinal microbial ecology with ingestion of kale

Item Type Journal Article

Author Y. Uyeno

Author S. Katayama

Author S. Nakamura

Abstract Kale, a cultivar of *Brassica oleracea*, has attracted a great deal of attention because of its health-promoting effects, which are thought to be exerted through modulation of the intestinal microbiota. The present study was performed to investigate the effects of kale ingestion on the gastrointestinal microbial ecology of mice. 21 male

C57BL/6J mice were divided into three groups and housed in a specific pathogen-free facility. The animals were fed either a control diet or experimental diets supplemented with different commercial kale products for 12 weeks. Contents of the caecum and colon of the mice were processed for the determination of active bacterial populations by a bacterial rRNA-based quantification method and short-chain fatty acids by HPLC. rRNAs of *Bacteroides-Prevotella*, the *Clostridium coccoides*-*Eubacterium* rectale group, and *Clostridium leptum* subgroup constituted the major fraction of microbiota regardless of the composition of the diet. The ratio of Firmicutes to Bacteroidetes was higher in the colon samples of one of the kale diet groups than in the control. The colonic butyrate level was also higher with the kale-supplemented diet. Overall, the ingestion of kale tended to either increase or decrease the activity of specific bacterial groups in the mouse gastrointestinal tract, however, the effect might vary depending on the nutritional composition. © 2014 Wageningen Academic Publishers.

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CHANNEL-SELECTIVE NORMALIZATION FOR LABEL-SHIFT ROBUST TEST-TIME ADAPTATION

Item Type Conference Paper

Author P. Vianna

Author M. Chaudhary

Author P. Mehrbod

Author A. Tang

Author G. Cloutier

Author G. Wolf

Author M. Eickenberg

Author E. Belilovsky

Editor Lomonaco V.

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Editor Chandar S.

Editor Pascanu R.

Abstract Deep neural networks have useful applications in many different tasks, however their performance can be severely affected by changes in the input distribution. For example, in the biomedical field, their performance can be affected by changes in the data (different machines, populations) between training and test datasets. To ensure robustness and generalization to real-world scenarios, test-time adaptation has been recently studied as an approach to adjust models to a new data distribution during inference. Test-time batch normalization is a simple and popular method that achieved compelling performance on domain shift benchmarks. It is implemented by recalculating batch normalization statistics on test batches. Although initial work focused on analysis with test data that has the same label distribution as the training data, in many practical applications this technique is vulnerable to label distribution shifts, sometimes causing catastrophic failure. This presents a risk in applying test time adaptation methods in deployment. We propose to tackle this challenge by only selectively adapting channels in a deep network, minimizing drastic adaptation that is sensitive to label shifts. Our selection scheme is based on two principles that we empirically motivate: (1) later layers of networks are more sensitive to label shift; (2) individual features can be sensitive to specific classes. We apply the proposed technique to three classification tasks, including CIFAR10-C, Imagenet-C, and diagnosis of fatty liver, where we explore both covariate and label distribution shifts. We find that our method allows to bring the benefits of TTA while significantly reducing the risk of failure common in imbalanced scenarios. © 2024 CoLLAs. All Rights Reserved.

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Characterization of Impact Damage Uncertainty of Composite Materials Based on Bayesian Neural Network

Item Type Conference Paper
Author Z. Song
Author Y. Feng
Author C. Lu
Author W. Liu
Author Y. Liu
Author H. Hu
Abstract To solve the problem of multi-source parameter coupling and damage dispersion quantification in the uncertainty characterization of impact damage in composite materials, this paper proposes a confidence interval quantification method based on a Bayesian neural network to realize the propagation analysis of material parameters and impact energy uncertainty to damage state. Firstly, based on Bayesian inference, the probability distribution of weight and threshold parameters was constructed, and the probability distribution of the predicted value was directly output through the weight posterior distribution sampling, and

the traditional point estimation was transformed into probabilistic interval prediction. Secondly, a hierarchical Monte Carlo sampling strategy was designed, and the input material parameters were dispersed to inject the uncertainty of the damage state to generate a probability envelope covering 95% of the measured data points. Finally, the confidence interval curve of pit depth and damage area is predicted, including the confidence interval curve of pit depth and damage area, and the covariant relationship between the two under energy change is revealed, and the results show that the pit depth and damage area increase nonlinearly with the increase of impact energy, but show different sensitivities, the pit depth increases slowly in the low-energy segment (<50J), and the growth rate of the high-energy segment increases significantly, while the damage area expands exponentially before 50J. © 2025 IEEE.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Z. Song; School of Aeronautics, Northwestern Polytechnical University, Xi'an, China; email: songzhicen@mail.nwpu.edu.cn; Conference name: 2025 IEEE 5th International Conference on Applied Mathematics, Modeling and Computer Simulation, AMMCS 2025; Conference date: 23 August 2025 through 24 August 2025; Conference code: 222075

Characterization of spatio-temporal patterns of grassland utilization intensity in the Selinco watershed of the Qinghai-Tibetan Plateau from 2001 to 2019 based on multisource remote sensing and artificial intelligence algorithms

Item Type Journal Article

Author C. Ma

Author Y. Xie

Author S.-B. Duan

Author W. Qin

Author Z. Guo

Author G. Xi

Author X. Zhang

Author Q. Bie

Author H. Duan

Author L. He

Abstract Due to the limitations of spatial quantification methods, the spatio-temporal patterns of grassland utilization intensity (GUI) in the Selinco watershed (SLCW), the core region of ecological security on the Qinghai-Tibetan Plateau, is unclear under multiple utilization modes. This paper quantified GUI by constructing the association between the potential and actual Enhanced Vegetation Index (EVI) of grasslands in terms of interannual variability. To obtain an accurate spatio-temporal dataset of potential EVI, the following two components were considered on. Firstly, the temporal lag effects of each raw climate factor were investigated to determine the optimal climate variables affecting vegetation productivity. Secondly, four machine learning (ML) algorithms, including an artificial neural network, random forest, support vector machine, and gradient boosting regression tree combined with the Bayesian model average, were used to construct grassland potential EVI models involving EVI, grassland type, and environmental factors (topography, soil, raw climate, and bioclimatic). Meanwhile, to maximize the performance of ML models, variable selection, variable transformation, and hyperparameter optimization were systematically implemented, where the hyperparameter optimization algorithms employ the grid search algorithm, Bayesian optimization, genetic algorithm, and particle swarm optimization. Then, the spatio-temporal dataset of GUI in the SLCW from 2001 to 2019 was established by using the above quantification method based on multisource remote sensing and artificial intelligence algorithms. The analysis of spatio-temporal variation in GUI showed that the implementation of ecological restoration projects leads to a significant and rapid decline in the overall GUI of the SLCW after 2010 (declining by 4.8%), which is more obvious in the non-nature reserve (declining by 9.3%). In the Qiangtang Nature Reserve within the SLCW, although the GUI shows a declining trend after 2010 because of the implementation of ecological restoration projects, it shows an insignificant increase from 2001 to 2019 due to the recovery increase of wildlife populations in recent decades. Besides, by exploring the effects of elevation and slope on the GUI, it is found that grasslands on higher slopes at lower elevations are at a greater risk of degradation due to more intensive grassland utilization. © 2022 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

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Characterization of Taxonomic and Functional Dynamics Associated with Harmful Algal Bloom Formation in Recreational Water Ecosystems

Item Type Journal Article
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Author R. Atrache
Author J.L. Jiang
Author K.L. Tran
Author E. Li
Author A. Paschos
Author T.A. Edge
Author H.E. Schellhorn

Abstract Harmful algal bloom (HAB) formation leads to the eutrophication of water ecosystems and may render recreational lakes unsuitable for human use. We evaluated the applicability and comparison of metabarcoding, metagenomics, qPCR, and ELISA-based methods for cyanobacteria/cyanotoxin detection in bloom and non-bloom sites for the Great Lakes region. DNA sequencing-based methods robustly identified differences between bloom and non-bloom samples (e.g., the relative prominence of *Anabaena* and *Planktothrix*). Shotgun sequencing strategies also identified the enrichment of metabolic genes typical of cyanobacteria in bloom samples, though toxin genes were not detected, suggesting deeper sequencing or PCR methods may be needed to detect low-abundance toxin genes. PCR and ELISA indicated microcystin levels and microcystin gene copies were significantly more abundant in bloom sites. However, not all bloom samples were positive for microcystin, possibly due to bloom development by non-toxin-producing species. Additionally, microcystin levels were significantly correlated (positively) with microcystin gene copy number but not with total cyanobacterial 16S gene copies. In summary, next-generation sequencing-based methods can identify specific taxonomic and functional targets, which can be used for absolute quantification methods (qPCR and ELISA) to augment conventional water monitoring strategies. © 2024 by the authors.

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Characterizing forecastability of wind sites in the United States

Item Type Journal Article

Author C. Feng

Author M. Sun

Author M. Cui

Author E.K. Chartan

Author B.-M. Hodge

Author J. Zhang

Abstract With the rapid growth of wind power, managing its uncertainty and variability becomes critical in power system operations. Wind forecasting is one of the enablers to partially tackle challenges associated with wind power uncertainty. To improve the 'forecasting ability' defined as forecastability, different forecasting methods have been developed to assist grid integration of wind energy. However, forecasting performance not only relies on the power of forecasting models, but is also related to local weather conditions and (known as wind characteristics) wind farm properties. In this study, geospatial and instance spatial distributions of six wind characteristics and two forecasting error metrics are first analyzed based on 126,000 + wind sites in the United States. Forecasts in different look-ahead times are generated by using a machine learning based multi-model forecasting framework and the Weather Research and Forecasting model. A forecastability quantification method is developed by characterizing the relationship between forecastability and wind series entropy using three regression methods, i.e., linear approximation, locally weighted scatterplot smoother nonlinear nonparametric regression, and quantile regression. It is found that the forecastability of a wind site can be successfully characterized by wind series characteristics, thereby providing valuable information at different stages of wind energy projects. © 2018 Elsevier Ltd

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Characterizing Neural Radiance Field Models for 3D Scene Understanding of Hyperspectral Imagery

Item Type Conference Paper

Author S. Jarman

Author S. Zorek

Author K. Moon

Author A. Carr

Author Z. Hampel-Arias

Abstract Hyperspectral imaging (HSI) is crucial for remote detection of various targets. However, detection becomes challenging when targets are obscured by other objects in the scene. To address this, we apply 3D scene understanding methods to create spatially-relevant representations of HSI data. 3D scene understanding with hyperspectral data has the ability to improve target characterization, as it allows for differentiation between materials that may appear similar in standard visual imagery and provide better spatial discrimination compared to single-image captures of a scene. Our approach enables the incorporation of multiple observations from different angles, enhancing the ability to 'look around' targets or obstructions by rendering new views of a scene. We apply the 3D methods of neural radiance fields (NeRFs) to Long Wave IR hyperspectral imagery. We will present results from our analysis pipeline applying machine learning 3D scene understanding methods to physics-driven synthetic images from DIRSIG to quantify accuracy tradeoffs with number of collected view angles, view geometry, and material composition of the scene. We demonstrate the capability of scaling the spectral dimensionality of NeRF models from data comprising single channel to multi-channel imagery, while also including uncertainty quantification methods to measure associated confidence in modeling spatial-spectral components of the scene. © 2026 IEEE.

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Characterizing Out-of-Distribution Error via Optimal Transport

Item Type Conference Paper

Author Y. Lu
Author Y. Qin
Author R. Zhai
Author A. Shen
Author K. Chen
Author Z. Wang
Author S. Kolouri
Author S. Stepputtis

Author J. Campbell

Author K. Sycara

Editor Oh A.

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Abstract Out-of-distribution (OOD) data poses serious challenges in deployed machine learning models, so methods of predicting a model's performance on OOD data without labels are important for machine learning safety. While a number of methods have been proposed by prior work, they often underestimate the actual error, sometimes by a large margin, which greatly impacts their applicability to real tasks. In this work, we identify pseudo-label shift, or the difference between the predicted and true OOD label distributions, as a key indicator of this underestimation. Based on this observation, we introduce a novel method for estimating model performance by leveraging optimal transport theory, Confidence Optimal Transport (COT), and show that it provably provides more robust error estimates in the presence of pseudo label shift. Additionally, we introduce an empirically-motivated variant of COT, Confidence Optimal Transport with Thresholding (COTT), which applies thresholding to the individual transport costs and further improves the accuracy of COT's error estimates. We evaluate COT and COTT on a variety of standard benchmarks that induce various types of distribution shift - synthetic, novel subpopulation, and natural - and show that our approaches significantly outperform existing state-of-the-art methods with up to 3x lower prediction errors. Our code can be found at <https://github.com/luyuzhe111/COT>. © 2023 Neural information processing systems foundation. All rights reserved.

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Chemical features consistently associated with kinase inhibitor activity: a cross-target SHAP analysis under scaffold-split evaluation

Item Type Journal Article

Author A.S. Salihu

Author M.S. Rahama

Author W.M.N.H.W. Salleh

Author N.S. Gwaram

Author A. Salisu

Author S.S. Yusuf

Abstract Interpretable machine learning approaches to quantitative structure–activity relationship (QSAR) modelling are increasingly applied in drug discovery, yet most studies remain confined to single targets and report feature attributions without translating them into chemically meaningful insights. We introduce a cross-target SHAP entropy framework for quantifying shared versus target-specific structure–activity relationships across protein families, applied to 31 human kinase targets from BindingDB under scaffold-based train-test evaluation. Random Forest classifiers trained on Morgan ECFP4 fingerprints achieved a median AUROC of 0.994, AUPRC of 0.9998, and MCC of 0.674, confirming genuine SAR learning beyond class prevalence exploitation. Pairwise Spearman rank correlation of mean absolute SHAP profiles across targets yielded moderate cross-target consistency (mean $r = 0.332$; 465 pairs). Shannon entropy-based classification of the top 200 fingerprint bits identified 15 consensus features dominated by aromatic N-heterocycles, aliphatic rings, and hydrogen bond environments, and 50 divergent features showing 2.8-fold higher SHAP magnitude than consensus features. SAR validation confirmed genuine enrichment of two top consensus fragments in active compounds. These findings indicate that kinase QSAR models share a low-magnitude consensus descriptor signal across the kinase family, while target-specific features dominate predictive decision boundaries. All SHAP attributions describe model decision behavior and should not be interpreted as causal binding mechanisms. The entropy decomposition framework is generalisable to other protein families and provides a transferable workflow for converting SHAP outputs into chemically actionable insights. All code and data are publicly available. © The Author(s), under exclusive licence to Springer Nature Switzerland AG 2026.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A.S. Salihu; Department of Pure and Industrial Chemistry, Faculty of Natural and Applied Sciences, Umaru Musa Yar'adua University, Katsina, 832101, Nigeria; email: abusiddiq.salihu@umyu.edu.ng; CODEN: MODIF

Chemical fingerprinting and quantification of Chinese cinnamomi cortex by ultra high performance liquid chromatography coupled with chemometrics methods

Item Type Journal Article**Author** N. Ma**Author** Y. Ding**Author** Y. Zhang**Author** T. Zhang**Author** Y. Yi**Author** B. Wang

Abstract To rapidly clarify and quantify the chemical profiling of Cinnamomi cortex a reliable and feasible strategy of chromatographic fingerprinting with a suite of chemometrics methods was developed and validated by ultra-high performance liquid chromatography coupled with diode array detection. Furthermore, to identify more meaningful chemical markers, the chemometrics methods including hierarchical cluster analysis (HCA), principal component analysis (PCA) and similarity, which all generate quality evaluations and correlation classifications of Cinnamomi cortex, were used to improve the Cinnamomi cortex quality control standards. A total of 12 characteristic peaks were confirmed, seven of which were identified by comparing their retention times, UV and MS spectra with authentic compounds. Moreover, 11 analytes were accurately determined, as a complementary quantification method of chromatographic fingerprinting. For quantitative analyses, selective detection was performed at 254, 280 and 340 nm. The tested samples were separated and determined using UPLC and a series of methodologies including linearity, precision, accuracy, limit of detection and quantification and extraction recoveries were validated. Meanwhile the method bias for all the analytes did not exceed 5%. A total of 42 samples were acquired in China and analyzed. The results demonstrated that chromatographic fingerprinting in combination with chemometrics methods provides a promising and practical method to more effectively and comprehensively control the quality of Cinnamomi cortex from various sources, which would be a useful

reference for the development and further study of Cinnamomi cortex and related formulations. © 2018 by the authors.

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Chemometric classification and quantification of olive oil in blends with any edible vegetable oils using FTIR-ATR and Raman spectroscopy

Item Type Journal Article

Author A.M. Jiménez-Carvelo

Author M.T. Osorio

Author A. Koidis

Author A. González-Casado

Author L. Cuadros-Rodríguez

Abstract Samples of olive oils (n = 67) from different qualities and samples of other vegetable edible oils (including soybean, sunflower, rapeseed, corn oil etc; n = 79) were used in this study as pure oils. Previous to spectroscopy analysis, a transesterification step was applied to the pure vegetable oil samples and all the different oil blends were then prepared to create in-house blended samples. Spectral acquisition was performed with typical parameters to collect the FTIR and Raman fingerprints. For the olive/non-olive classification model, three classification strategies have been applied: (i) one input-class (1iC) classification; (ii) two input-class (2iC) classification; and (iii) one input-class plus one 'dummy' class classification (or pseudo two input-class (p2iC) classification). The multivariate classification methods used were k-nearest neighbours (kNN), partial least squared-discriminant analysis (PLS-DA), one-class partial least squares (OCPLS), support vector machine classification (SVM-C), and soft independent modelling of class analogies (SIMCA). The multivariate quantification method used was partial least square-regression (PLS-R). FTIR fingerprints showed excellent classification ability to distinguish pure olive from non-olive oil. When PLS-DA or SVM-C techniques are applied, 100% of olive oil samples and 92% of other vegetable edible oils are correctly classified. In general FTIR fingerprints were more discriminative than Raman's in both classification and regression scenarios. © 2017 Elsevier Ltd

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Notes:

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Export Date: 20 September 2026; Cited By: 149; Correspondence Address: A.M. Jiménez-Carvelo; Department of Analytical Chemistry, University of Granada, Granada, c/ Fuentenueva, s.n., E-18071, Spain; email: amariajc@ugr.es; CODEN: LBWTA

Chemometric classification and quantification of sesame oil adulterated with other vegetable oils based on fatty acids composition by gas chromatography

Item Type Journal Article

Author C. Xing

Author X. Yuan

Author X. Wu

Author X. Shao

Author J. Yuan

Author W. Yan

Abstract The detection of adulteration of sesame oil with four vegetable oils was investigated by measuring fatty acid composition. Adulteration based fatty acid composition was evaluated with 73 sesame seed samples, 57 rapeseed seed samples, 103 soybean seed samples, 11 brands of sunflower oil and 14 brands of maize oil. Discriminate analysis and principal component analysis were used to study the data distribution patterns based on the fatty acid composition and their ratios obtained from pure sesame oils and blends. Pure vegetable oils were gathered in a tendency in principal component analysis score diagram. Although the parameters calculated by fatty acid composition were simple and effective in detection of oil mixture of single vegetable edible oil or different vegetable edible oil, this method could only be used for qualitative determination. Linear discriminant analysis applied and the obtained overall accuracies were between 97.27% and 100%. Partial least-squares regression, the multivariate quantification method, have been successfully applied and quantitatively recognized the type and the level of extra added vegetable oils into the sesame oils. In general, the fatty acid profile of vegetable oils obtained by GC was useful in both qualitative and quantitative detection of sesame oil blends. © 2019 Elsevier Ltd

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Chemometrics and digital image colorimetry approaches applied to paper-based analytical devices: A review

Item Type Journal Article**Author** I. Berasarte**Author** G. Albizu**Author** W.F. Santos**Author** L.F. de Lima**Author** M. Ostra**Author** M. Vidal**Author** W.R. de Araujo

Abstract Colorimetric paper-based analytical devices (CPADs) are cost-efficient and high-throughput technologies that use readily available materials for point-of-need (PON) applications by leveraging color changes in response to target analytes. However, the complexity of samples can limit the precision and accuracy of CPAD applications. Therefore, CPADs have been combined with chemometric approaches to enhance analytical performance and provide simple solutions to complex systems. The integration of formal optimization techniques, such as Design of Experiments (DoE), classification tools, quantification methods, and other advanced algorithms enables optimal experimental analytical conditions and extracting meaningful information from the complex colorimetric data generated by CPADs. These approaches facilitate robust calibration and prediction models, enabling reliable quantifications or sample differentiation. In addition, chemometrics combined with CPADs contributes to Green Analytical Chemistry once they have the potential to minimize the number of experiments, provide optimal designs and consumption of reagents, and decrease waste generation. The synergy between digital colorimetry using CPADs and smart devices with chemometric techniques holds great promise for portable analysis in resource-limited settings, with applications ranging from environmental monitoring to point-of-care diagnostics. Herein, we review recent advances in the development of CPADs, ranging from manufacturing methods to extraction of color patterns and data treatment using several chemometric tools, performance assessment, and potential transfer to onsite applications for relevant analytical problems. © 2024 Elsevier B.V.

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Citizen science coupled with machine learning to quantify green-blue infrastructure cooling potential in Maricopa County, Arizona

Item Type Journal Article

Author A. Molla
Author K. Lamer
Author D.J. Sailor

Abstract This study investigates the spatiotemporal cooling performance of green and blue infrastructure (GBI) in the Dobson Ranch urban neighborhood in Phoenix, Arizona. We leveraged citizen science near-surface (2 m) air temperature (T_{air}) measurements to train a highly accurate T_{air} predicting LightGBM machine learning model (R²: 0.986, MAE: 0.251 °C, RMSE: 0.585 °C). On June 16, 2024, the park area exhibited approximately 1 °C cooling effect (relative to the neighborhood mean) during both day and night. In contrast, the nearby artificial lake exhibited a stronger cooling effect of 2.4 °C during the day but a slight warming of 0.3 °C at night. At 00:00, locations 50 m downwind of the park were 0.3 °C warmer than the park, while locations 50 m upwind were 0.8 °C warmer. At 11:00, we observed that the downwind area is 0.8 °C cooler and the upwind area is 0.6 °C warmer—at the same 50 m distances relative to the park. We also observed 1 °C cooler and warmer effects respectively at the same 50 m downwind and upwind locations at 19:00 on June 17, 2024. Our data-driven analysis highlights potential limitations of car-traverse measurements, showing that failure to account for temporal variations during the traverse can lead to overestimation of T_{air} at night and underestimation during the day. Our analysis also showed only a weak correlation (coefficient: 0.48) between Landsat-derived land surface temperature (LST) and model predicted T_{air} at the time of the local Landsat overpass (~11.00). This highlights the potential error of relying solely on LST for human thermal exposure analysis—particularly within the heterogenous built-environment. © 2026

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Clarifying trust of materials property predictions using neural networks with distribution-specific uncertainty quantification

Item Type Journal Article

Author C.J. Gruich

Author V. Madhavan

Author Y. Wang

Author B.R. Goldsmith

Abstract It is critical that machine learning (ML) model predictions be trustworthy for high-throughput catalyst discovery approaches. Uncertainty quantification (UQ) methods allow estimation of the trustworthiness of an ML model, but these methods have not been well explored in the field of heterogeneous catalysis. Herein, we investigate different UQ methods applied to a crystal graph convolutional neural network to predict adsorption energies of molecules on alloys from the Open Catalyst 2020 dataset, the largest existing heterogeneous catalyst dataset. We apply three UQ methods to the adsorption energy predictions, namely k-fold ensembling, Monte Carlo dropout, and evidential regression. The effectiveness of each UQ method is assessed based on accuracy, sharpness, dispersion, calibration, and tightness. Evidential regression is demonstrated to be a powerful approach for rapidly obtaining tunable, competitively trustworthy UQ estimates for heterogeneous catalysis applications when using neural networks. Recalibration of model uncertainties is shown to be essential in practical screening applications of catalysts using uncertainties. © 2023 The Author(s). Published by IOP Publishing Ltd.

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Class and subclass probability re-estimation to adapt a classifier in the presence of concept drift

Item Type Journal Article

Author R. Alaiz-Rodríguez

Author A. Guerrero-Curieses

Author J. Cid-Sueiro

Abstract We consider the problem of classification in environments where training and test data may come from different probability distributions. When the fundamental stationary distribution assumption made in supervised learning (and often not satisfied in practice) does not hold, the classifier performance may significantly deteriorate. Several proposals have been made to deal with classification problems where the class priors change after training, but they may fail when the class conditional data densities also change. To cope with this problem, we propose an algorithm that uses unlabeled test data to adapt the classifier outputs to new operating conditions, without re-training it. The algorithm is based on a posterior probability model with two main assumptions: (1) the classes may be decomposed in several (unknown) subclasses, and (2) all changes in data distributions arise from changes in prior subclass probabilities. Experimental results with a neural network model on synthetic and remote sensing practical settings show that the adaptation at the subclass level can get a better adjustment to the new operational conditions than the methods based on class prior changes. © 2011 Elsevier B.V.

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DOI 10.1016/j.neucom.2011.03.019

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article, Neural networks, Algorithms, Remote sensing, remote sensing, controlled study, artificial neural network, Probability distributions, classification algorithm, intermethod comparison, probability, Concept drifts, Supervised classification, Class distributions, Concept drift, priority journal, Posterior probability, mathematical computing, Operational conditions, Changing operational conditions, Imprecise class distribution, Imprecise data, Imprecise data distribution, Posterior probability estimation, statistical distribution

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Class distribution estimation based on the Hellinger distance

Item Type Journal Article

Author V. González-Castro

Author R. Alaiz-Rodríguez

Author E. Alegre

Abstract Class distribution estimation (quantification) plays an important role in many practical classification problems. Firstly, it is important in order to adapt the classifier to the operational conditions when they differ from those assumed in learning. Additionally, there are some real domains where the quantification task is itself valuable due to the high variability of the class prior probabilities. Our novel quantification approach for two-class problems is based on distributional divergence measures. The mismatch between the test data distribution and validation distributions generated in a fully controlled way is measured by the Hellinger distance in order to estimate the prior probability that minimizes this divergence. Experimental results on several binary classification problems show the benefits of this approach when compared to such approaches as counting the predicted class labels and other methods based on the classifier confusion matrix or on posterior probability estimations. We also illustrate these techniques as well as their robustness against the base classifier performance (a neural network) with a boar semen quality

control setting. Empirical results show that the quantification can be conducted with a mean absolute error lower than 0.008, which seems very promising in this field. © 2012 Published by Elsevier Inc.

Date 2013

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Class distribution estimation in imprecise domains based on supervised learning

Item Type Journal Article

Author V. González-Castro

Author R. Alaiz-Rodríguez

Author E. Alegre

Abstract Quantification-or proportion estimation-plays an important role in many practi{reversed not sign}cal classification problems. On the one hand, a machine that automatically classifies an element into a group of predefined classes will make suboptimal decisions if the class distribution in the test (real) domain differs from the one assumed in learning. Estimating the new class distribution is necessary in order to adapt the classifier to the new operational conditions. On the other hand, there are some real domains where the quantification task itself is the main goal. Some fields,

such as quality control, direct marketing, tendency study or some textual recognition tasks, require methods that can reliably estimate the proportion of elements within each category without any concerns about how each element has been classified individually. We describe several quantification techniques that rely on supervised learning and provide these estimations based on: (a) the classifier confusion matrix, (b) the posterior probability estimations, and (c) distributional divergence measures. We illustrate these techniques, as well as their robustness against the base classifier performance in a practical seminal quality control setting where the ultimate goal is to quantify the proportion of sperm cells with damaged/intact acrosome. © 2012 by Nova Science Publishers, Inc. All rights reserved.

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Class Information-Guided Reconstruction for Automatic Modulation Open-Set Recognition

Item Type Journal Article

Author Z. Zhang

Author M. Zhu

Author J. Liu

Author Y. Li

Author S. Wang

Abstract Automatic Modulation Recognition (AMR) is vital for radar and communication systems. Traditional AMR operates under closed-set scenarios where all modulation types are pre-defined. However, in practical settings, unknown modulation types may emerge due to technological advancements. Closed-set training poses the risk of misclassifying unknown modulations into existing known classes, leading to serious implications for situation awareness and threat assessment. To tackle this challenge, this paper presents a Class Information guided Reconstruction (CIR) framework that

can simultaneously achieve Known Class Classification (KCC) and Unknown Class Identification (UCI). The CIR leverages reconstruction losses to differentiate between known and unknown classes, utilizing Class Conditional Vectors (CCVs) and a Mutual Information (MI) loss function to fully exploit class information. The CCVs offer class-specific guidance for reconstruction process, ensuring accurate reconstruction for known samples while producing subpar results for unknown ones. Moreover, to enhance distinguishability, an MI loss function is introduced to capture class-discriminative semantics in latent space, enabling closer alignment with CCVs during reconstruction. The synergistic relationship between CCVs and MI facilitates optimal UCI performance without compromising KCC accuracy. The CIR is evaluated on simulated, public and real-world datasets, demonstrating its effectiveness and robustness, particularly in low SNR and high unknown class prevalence scenarios. © 2015 IEEE.

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Class prior estimation from positive and unlabeled data

Item Type Journal Article

Author M.C. Du Plessis

Author M. Sugiyama

Abstract We consider the problem of learning a classifier using only positive and unlabeled samples. In this setting, it is known that a classifier can be successfully learned if the class prior is available. However, in practice, the class prior is unknown and thus must be estimated from data. In this paper, we propose a new method to estimate the class prior by partially matching the class-conditional density of the positive class to the input density. By performing this partial matching in terms of the Pearson divergence, which we estimate directly without density estimation via lower-bound maximization, we can obtain an analytical estimator of the class prior. We further show that an existing class prior estimation method can also be interpreted as performing partial matching under the Pearson divergence, but in an indirect manner. The superiority of our direct class prior estimation method is illustrated on several benchmark datasets. © 2014 The Institute of Electronics, Information and Communication Engineers.

Date 2014

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Class prior estimation in active positive and unlabeled learning

Item Type Conference Paper

Author L. Perini

Author V. Vercruyssen

Author J. Davis

Editor Bessiere C.

Abstract Estimating the proportion of positive examples (i.e., the class prior) from positive and unlabeled (PU) data is an important task that facilitates learning a classifier from such data. In this paper, we explore how to tackle this problem when the observed labels were acquired via active learning. This introduces the challenge that the observed labels were not selected completely at random, which is the primary assumption underpinning existing approaches to estimating the class prior from PU data. We analyze this new setting and design an algorithm that is able to estimate the class prior for a given active learning strategy. Empirically, we show that our approach accurately recovers the true class prior on a benchmark of anomaly detection datasets and that it does so more accurately than existing methods. © 2020 Inst. Sci. inf., Univ. Defence in Belgrade. All rights reserved.

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Conference Name IJCAI International Joint Conference on Artificial Intelligence

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Class prior estimation with biased positives and unlabeled examples

Item Type Conference Paper

Author S. Jain

Author J.D. Delano

Author H. Sharma

Author P. Radivojac

Abstract Positive-unlabeled learning is often studied under the assumption that the labeled positive sample is drawn randomly from the true distribution of positives. In many application domains, however, certain regions in the support of the positive class-conditional distribution are over-represented while others are under-represented in the positive sample. Although this introduces problems in all aspects of positive-unlabeled learning, we begin to address this challenge by focusing on the estimation of class priors, quantities central to the estimation of posterior probabilities and the recovery of true classification performance. We start by making a set of assumptions to model the sampling bias. We then extend the identifiability theory of class priors from the unbiased to the biased setting. Finally, we derive an algorithm for estimating the class priors that relies on clustering to decompose the original problem into subproblems of unbiased positive-unlabeled learning. Our empirical investigation suggests feasibility of the correction strategy and overall good performance. © 2020, Association for the Advancement of Artificial Intelligence.

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Artificial intelligence, Classification performance, Under-represented, Conditional distribution, Posterior probability, Empirical investigation, Identifiability, Sub-problems

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Class Prior-Free Positive-Unlabeled Learning with Taylor Variational Loss for Hyperspectral Remote Sensing Imagery

Item Type Conference Paper

Author H. Zhao

Author X. Wang

Author J. Li

Author Y. Zhong

Abstract Positive-unlabeled learning (PU learning) in hyperspectral remote sensing imagery (HSI) is aimed at learning a binary classifier from positive and unlabeled data, which has broad prospects in various earth vision applications. However, when PU learning meets limited labeled HSI, the unlabeled data may dominate the optimization process, which makes the neural networks overfit the unlabeled data. In this paper, a Taylor variational loss is proposed for HSI PU learning, which reduces the weight of the gradient of the unlabeled data by Taylor series expansion to enable the network to find a balance between overfitting and underfitting. In addition, the self-calibrated optimization strategy is designed to stabilize the training process. Experiments on 7 benchmark datasets (21 tasks in total) validate the effectiveness of the proposed method. Code is at: <https://github.com/Hengwei-Zhao96/T-HOneCls>. © 2023 IEEE.

Date 2023

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Class prioritization and server dedication in queueing systems: Discrimination and fairness aspects

Item Type Journal Article

Author D. Raz

Author H. Levy

Author B. Avi-Itzhak

Abstract Customer classification and prioritization are commonly utilized in applications to provide queue preferential service. Their fairness aspects, which are inherent to any preferential system and highly important to customers, have not been fully studied and quantified to date. We use the recently proposed Resource Allocation Queueing Fairness Measure (RAQFM), and a newly introduced metric called Class Discrimination, which is based on RAQFM, to analyze such systems and derive their relative fairness values as well as the discrimination experienced by the various classes. Specifically, we study two practices, commonly used in public facilities as well as in computer systems: class prioritization and dedication of servers to classes. © 2009 Elsevier B.V. All rights reserved.

Date 2010

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CLASS PROBABILITY MATCHING WITH CALIBRATED NETWORKS FOR LABEL SHIFT ADAPTATION

Item Type Conference Paper

Author H. Wen

Author A. Betken

Author H. Hang

Abstract We consider the domain adaptation problem in the context of label shift, where the label distributions between source and target domain differ, but the conditional distributions of features given the label are the same. To solve the label shift adaptation problem, we develop a novel matching framework named class probability matching (CPM). It is inspired by a new understanding of the source domain's class probability, as well as a specific relationship between class probability ratios and feature probability ratios between the source and target domains. CPM is able to maintain the same theoretical guarantees as the existing feature probability matching framework, while significantly improving the computational efficiency due to directly matching the probabilities of the label variable. Within the CPM framework, we propose an algorithm named class probability matching with calibrated networks (CPMCN) for target domain classification. From the theoretical perspective, we establish a generalization bound of the CPMCN method in order to explain the benefits of introducing calibrated networks. From the experimental perspective, real data comparisons show that CPMCN outperforms existing matching-based and EM-based algorithms. © 2024 12th International Conference on Learning Representations, ICLR 2024. All rights reserved.

Date 2024

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Class quantification of aerial images using Maximum Likelihood Estimation

Item Type Conference Paper

Author S.S. Wanarse

Author T.G. Patil

Author S.S. Patankar

Author J.V. Kulkarni

Editor Negi A.

Editor Kishore K.V.K.

Editor Kumar K.H.

Editor Venkatramaphanikumar S.

Abstract Class quantification of aerial images plays a vital role in remote sensing. One of the class quantification method is discussed in this paper. Proposed method uses Maximum Likelihood Estimation based classifier for class quantification. Algorithm is trained by the sample classes derived from parent image. Feature space is estimated from each training sample. Different classes are labeled in test image by maximizing the likelihood function. The experimentation is done on aerial images obtained by Geo eye satellite at the elevation of 0.6km. The percentage area covered by the labeled classes is computed for all test images. © 2014 IEEE.

Date 2014

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Conference Name 1st International Conference on Networks and Soft Computing, ICNSC 2014 - Proceedings

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Antennas, Remote sensing, Different class, Training sample, Quantification methods, Parameter estimation, Feature space, Likelihood functions, Maximum likelihood estimation, Test images, Soft computing, Aerial images, class quantification, Maximum Likelihood Estimation, parameter estimation

Notes:

EID: 2-s2.0-84908608848

Export Date: 20 September 2026; Cited By: 0; Conference name: 1st International Conference on Networks and Soft Computing, ICNSC 2014; Conference date: 19 August 2014 through 20 August 2014; Conference code: 114651

Class treatment in queueing systems: Discrimination and fairness aspects

Item Type Conference Paper

Author D. Raz

Author H. Levy

Author B. Avi-Itzhak

Abstract Customer classification and prioritization are commonly utilized in applications to provide queue preferential service. Their fairness aspects, which are inherent to any preferential system and highly important to customers, have not been fully studied and quantified to date. We use the recently proposed Resource Allocation Queueing Fairness Measure (RAQFM), and a newly introduced metric called class discrimination, which is based on RAQFM, to analyze such systems and derive their relative fairness values as well as the discrimination experienced by the various classes. Specifically, we study two practices, commonly used in public facilities as well as in computer systems: class prioritization and dedication of resources to classes. © 2008 ICST ISBN.

Date 2008

DOI 10.4108/ICST.VALUETOOLS2008.4344

URL <https://www.scopus.com/pages/publications/84896981949?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: VALUETOOLS - Int. Conf. Perform. Eval. Methodol. Tools

Publisher ICST

Proceedings Title VALUETOOLS - Int. Conf. Perform. Eval. Methodol. Tools

Conference Name VALUETOOLS 2008 - 3rd International Conference on Performance Evaluation Methodologies and Tools

Date Added 20/09/2026, 9:53:06 pm

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Tags:

Resource allocation, Tools, Fairness, Prioritization, Discrimination, Job scheduling, Multiple class, Multiple classes, Unfairness

Notes:

EID: 2-s2.0-84896981949

Export Date: 20 September 2026; Cited By: 1; Conference name: 3rd International Conference on Performance Evaluation Methodologies and Tools, VALUETOOLS 2008; Conference date: 20 October 2008 through 24 October 2008; Conference code: 103411

Class-Aware Feature Perturbation for Long-Tailed Visual Recognition

Item Type Conference Paper

Author X. Chen

Author H. Ye

Author F. Zhou

Abstract The distribution of data in the real world is often imbalanced, with a small number of classes having a large number of instances, while most other classes have relatively few sample instances, resulting in a long-tailed distribution. Supervised contrastive learning still tends to favor classes with more samples, making it challenging to learn discriminative and robust features in imbalanced settings. In our work, we theoretically analyze the advantages of hard samples in contrastive learning and propose an end-to-end training framework based on class-aware perturbation. Specifically, we introduce different levels of perturbation based on class priors to construct hard samples, improve the disadvantaged position of tail classes, balance the feature space distribution, and dynamically adjust the classifier output with a combination of local batch priors and global priors. Experimental results show that our method achieves competitive performance across multiple long-tailed benchmark datasets. © 2024 IEEE.

Date 2024

DOI 10.1109/ICME57554.2024.10687966

URL <https://www.scopus.com/pages/publications/85206559970?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. IEEE Int. Conf. Multimedia Expo

Publisher IEEE Computer Society

ISBN 19457871 (ISSN); 979-835039015-5 (ISBN)

Proceedings Title Proc. IEEE Int. Conf. Multimedia Expo

Conference Name Proceedings - IEEE International Conference on Multimedia and Expo

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Tags:

Real-world, Federated learning, Adversarial machine learning, Contrastive Learning, Self-supervised learning, contrastive learning, Training framework, Learn+, End to end, Long-tailed distributions, Feature space, feature perturbation, Feature perturbation, Long-tailed learning, Number of class, visual recognition, Visual recognition

Notes:

EID: 2-s2.0-85206559970

Export Date: 20 September 2026; Cited By: 0; Funding details: National Key Research and Development Program of China, NKRDP, (2020AAA0107000); National Key Research and Development Program of China, NKRDP; National Natural Science Foundation of China, NSFC, (62222605); National Natural Science Foundation of China, NSFC; Correspondence Address: H. Ye; Nanjing University of Aeronautics and Astronautics, College of Computer Science and Technology, China; email: yhb@nuaa.edu.cn; Conference name: 2024 IEEE International Conference on Multimedia and Expo, ICME 2024; Conference date: 15 July 2024 through 19 July 2024; Conference code: 203042

Class-Aware Federated Early Exit Networks for Efficient Distributed Inference

Item Type	Conference Paper
Author	M. Ayyat
Author	T. Nadeem
Abstract	Early-exit models improve efficiency by allowing inputs to terminate inference at shallow layers, while class-aware variants further prioritize critical classes for faster processing. Yet existing solutions assume centralized training and cannot accommodate distributed settings where devices may have distinct class priorities. We introduce CAFED (Class-Aware Federated Early-Exit Network), the first framework that integrates class-aware early exits into federated learning. CAFED combines clustered aggregation with layer freezing: shared lower layers capture general features, while personalized upper layers host exits aligned with local priorities. Experiments show that CAFED accelerates inference for prioritized classes by up to 40% and improves accuracy by up to 30% over conventional federated and early-exit baselines, establishing a new path for privacy-preserving collaborative learning with class prioritization. © 2025 IEEE.
Date	2025
DOI	10.1109/GLOBECOM59602.2025.11432532
URL	https://www.scopus.com/pages/publications/105036275006?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: Proc. - IEEE Glob. Commun. Conf., GLOBECOM
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	23340983 (ISSN); 979-833157781-0 (ISBN)
Pages	3951-3957
Proceedings Title	Proc. - IEEE Glob. Commun. Conf., GLOBECOM
Conference Name	Proceedings - IEEE Global Communications Conference, GLOBECOM

Date Added 20/09/2026, 9:51:55 pm**Modified** 20/09/2026, 9:51:55 pm**Tags:**

Artificial intelligence, Distributed computer systems, Collaborative learning, Privacy preserving, Centralised, Privacy-preserving techniques, Federated Learning, Edge AI, Class-aware classification, Prioritization, Distributed inference, Fast Processing, Shallowest layers, Upper layer

Notes:

EID: 2-s2.0-105036275006

Export Date: 20 September 2026; Cited By: 2; Funding details: National Science Foundation, NSF, (OAC-2212424); Correspondence Address: M. Ayyat; Virginia Commonwealth University, Department of Computer Science, Richmond, United States; email: ayyatma@vcu.edu; Conference name: 2025 IEEE Global Communications Conference, GLOBECOM 2025; Conference date: 8 December 2025 through 12 December 2025; Conference code: 221854

Class-Conditional Conformal Prediction for Reliable Anomaly Detection Under Extreme Class Imbalance

Item Type Journal Article**Author** B. Althani

Abstract Anomaly detection systems deployed in critical applications require not only high accuracy but also reliable uncertainty quantification and coverage guarantees. This paper is an empirical study—rather than a contribution of new conformal-prediction machinery—of class-conditional (Mondrian) conformal prediction for anomaly detection under extreme class imbalance, characterizing where standard conformal prediction fails and how class-conditional calibration restores valid coverage. Class-conditional conformal prediction constructs prediction sets that, under exchangeability, contain the true label with user-specified confidence (e.g., 90%), enabling systems to abstain on uncertain predictions. Unlike standard conformal prediction that fails catastrophically under extreme imbalance—achieving only 52.94% anomaly coverage at a 1:345 imbalance ratio—class-conditional calibration maintains 90.59% anomaly coverage by computing quantiles separately for each class. We apply the standard softmax-based nonconformity score (Formula presented.) within each class, ensuring valid coverage for both normal and anomalous instances with coverage gaps ranging from 0.50% to 5.18% depending on dataset characteristics. Extensive experiments on three real-world datasets (Microsoft Azure KPI, Yahoo, NAB) demonstrate that the method achieves empirical coverage within 0.06–0.33% of theoretical targets at confidence levels (Formula presented.) ; on the most imbalanced benchmark (Microsoft Azure KPI at a 1:345 ratio and (Formula presented.)), this corresponds to a 37.65 percentage point improvement in anomaly coverage over standard conformal prediction. We restate finite-sample coverage bounds and exchangeability conditions in the binary anomaly detection setting and validate them empirically through Monte Carlo simulation. Multi-model evaluation across XGBoost, Random Forest, and Neural Networks demonstrates the model-agnostic property of the framework, while also identifying conditions (poor base-

classifier discrimination, small minority calibration sets) under which coverage may be marginally violated. Comparison with alternative uncertainty quantification methods (isotonic probability calibration, Monte Carlo dropout) shows that only conformal prediction provides formal guarantees while maintaining 90.59% anomaly coverage versus 76.47% and 84.71% for alternatives. The abstention mechanism identifies 34–66% of predictions as uncertain at high confidence levels (99%), enabling safety-critical systems to defer difficult cases to human experts while preserving baseline discrimination (ROC-AUC unchanged). © 2026 by the author.

Date 2026

DOI 10.3390/make8070190

URL <https://www.scopus.com/pages/publications/105045793022?origin=resultslist>

Archive Scopus

Language English

Volume 8

Publisher Multidisciplinary Digital Publishing Institute (MDPI)

Publication Machine Learning and Knowledge Extraction

ISSN 25044990 (ISSN)

Issue 7

Journal Abbr Mach. Learn. Knowl. Extr.

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Tags:

machine learning, anomaly detection, class imbalance, XGBoost, time series, uncertainty quantification, conformal prediction, coverage guarantees

Notes:

EID: 2-s2.0-105045793022

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: B. Althani; Department of Management Information Systems, College of Business Administration, Imam Abdulrahman Bin Faisal University, Dammam, 31451, Saudi Arabia; email: bfalthani@iau.edu.sa

Class-prior probability estimation using density ratio between unlabeled instances and positively labeled noisy instances

Item Type Journal Article

Author T. Yoshida

Author E. Shin'ya

Author T. Washio

Abstract In this paper, the authors propose a new approach for determining the mixture proportions of positive and negative instances in both a small set of positively labeled instances contaminated with an unknown proportion of negative instances and an unlabeled instance set with an unknown mixture of positive and negative instances.

The unique advantage of this method lies in its ability to derive these proportions with a high level of accuracy by directly leveraging probability density ratio estimation. Experiments conducted on synthetic and UCI repository data demonstrate that the proposed technique can robustly estimate class-prior probabilities with greater speed and better accuracy compared to existing methods, particularly in the presence of instance number and imbalance challenges. © 2024 Elsevier B.V.

Date 2025

DOI 10.1016/j.neucom.2024.129074

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Archive Scopus

Language English

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ISSN 09252312 (ISSN)

Journal Abbr Neurocomputing

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Tags:

article, learning, Semi-supervised learning, probability, Class prior probability estimations, Class-prior probability estimation, density, Density ratio, Density ratio estimation, Density-ratio estimations, Mixture proportions, Negative instances, New approaches, Positive instances, Probability densities, UCI repository, velocity

Notes:

EID: 2-s2.0-85211367044

Export Date: 20 September 2026; Cited By: 0; Funding details: Japan Science and Technology Agency, JST, (CREST JPMJCR1666); Japan Science and Technology Agency, JST; Japan Society for the Promotion of Science, JSPS, (JP20K21815); Japan Society for the Promotion of Science, JSPS; Correspondence Address: T. Yoshida; The Institute of Scientific and Industrial Research Osaka University, Ibaraki-shi, 8-1 Mihogaoka, Osaka-fu, 567-0047, Japan; email: tyoshida@ar.sanken.osaka.u.ac.jp; CODEN: NRCGE

Class-prioritized compression of multispectral imagery data

Item Type Journal Article

Author J.A. Saghri

Author A.G. Tescher

Author M.G.H. Omran

Abstract A joint classification-compression scheme that provides the user with added capability to prioritize classes of interest in the compression process is proposed. The dual compression system includes a primary unit for conventional coding of a

multispectral image set followed by an auxiliary unit to code the resulting error induced on pixel vectors that represent classes of interest. This technique effectively allows classes of interest in the scene to be coded at a relatively higher level of precision than nonessential classes. Prioritized classes are selected from a thematic map or directly specified by their unique spectral signatures. Using the specified spectral signatures of the prioritized classes as end members, a modified linear spectral unmixing procedure is applied to the original data as well as to the decoded data. The resulting two sets of concentration maps, which represent classes prioritized before and after compression, are compared and the differences between them are coded via an auxiliary compression unit and transmitted to the receiver along with a conventionally coded image set. At the receiver, the differences found are blended back into the decoded data for enhanced restoration of the prioritized classes. The utility of this approach is that it works with any multispectral compression scheme. This method has been applied to test the imagery from various platforms including NOAA's AVHRR (1.1 km GSD), and LANDSAT 5 TM (30 m GSD), LANDSAT 5 MSS (79 m GSD). © 2002 SPIE and IS&T.

Date 2002

DOI 10.1117/1.1455008

URL <https://www.scopus.com/pages/publications/0036530216?origin=resultslist>

Archive Scopus

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Volume 11

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ISSN 10179909 (ISSN)

Issue 2

Journal Abbr J Electron Imaging

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Tags:

Algorithms, Image enhancement, Image quality, Computational complexity, Image coding, Mathematical models, Vectors, C (programming language), Image analysis, Image reconstruction, Imaging techniques, Computer simulation, Image compression, Class prioritized compression, Joint classification compression, Multispectral imagery data, Pixel vector, Thematic map

Notes:

EID: 2-s2.0-0036530216

Export Date: 20 September 2026; Cited By: 4

Class-Specific Reconstruction Transfer Learning for Visual Recognition across Domains

Item Type Journal Article**Author** S. Wang**Author** L. Zhang**Author** W. Zuo**Author** B. Zhang

Abstract Subspace learning and reconstruction have been widely explored in recent transfer learning work. Generally, a specially designed projection and reconstruction transfer functions bridging multiple domains for heterogeneous knowledge sharing are wanted. However, we argue that the existing subspace reconstruction based domain adaptation algorithms neglect the class prior, such that the learned transfer function is biased, especially when data scarcity of some class is encountered. Different from those previous methods, in this article, we propose a novel class-wise reconstruction-based adaptation method called Class-specific Reconstruction Transfer Learning (CRTL), which optimizes a well modeled transfer loss function by fully exploiting intra-class dependency and inter-class independency. The merits of the CRTL are three-fold. 1) Using a class-specific reconstruction matrix to align the source domain with the target domain fully exploits the class prior in modeling the domain distribution consistency, which benefits the cross-domain classification. 2) Furthermore, to keep the intrinsic relationship between data and labels after feature augmentation, a projected Hilbert-Schmidt Independence Criterion (pHSIC), that measures the dependency between data and label, is first proposed in transfer learning community by mapping the data from raw space to RKHS. 3) In addition, by imposing low-rank and sparse constraints on the class-specific reconstruction coefficient matrix, the global and local data structure that contributes to domain correlation can be effectively preserved. Extensive experiments on challenging benchmark datasets demonstrate the superiority of the proposed method over state-of-the-art representation-based domain adaptation methods. The demo code is available in <https://github.com/wangshanshanCQU/CRTL>. © 1992-2012 IEEE.

Date 2020**DOI** 10.1109/TIP.2019.2948480**URL** <https://www.scopus.com/pages/publications/85078293107?origin=resultslist>**Archive** Scopus**Language** English**Volume** 29**Publisher** Institute of Electrical and Electronics Engineers Inc.**Pages** 2424-2438**Publication** IEEE Transactions on Image Processing**ISSN** 10577149 (ISSN)**Journal Abbr** IEEE Trans Image Process**Date Added** 20/09/2026, 9:52:30 pm**Modified** 20/09/2026, 9:52:30 pm**Tags:**

Machine learning, Supervised learning, Transfer learning, Image classification, semi-supervised learning, Domain distribution, Matrix algebra, image classification, cross-domain learning, Cross-domain learning, Heterogeneous Knowledge, Hilbert-schmidt independence criterions, Reconstruction matrix, Semi-supervised learning, Subspace reconstruction, Transfer functions

Notes:

EID: 2-s2.0-85078293107

Export Date: 20 September 2026; Cited By: 111; Funding details: National Science Fund of China, (61771079); Fundamental Research Funds for the Key Research Program of Chongqing Science and Technology Commission, (cstc2018jcyjAX0250); Fundamental Research Funds for the Key Research Program of Chongqing Science and Technology Commission; Huxiang Youth Talent Support Program; Correspondence Address: L. Zhang; School of Computer Science and Technology, Harbin Institute of Technology, Harbin, 150001, China; email: leizhang@cqu.edu.cn; CODEN: IIPRE

Class-Specific Reconstruction Transfer Learning via Sparse Low-Rank Constraint

Item Type Conference Paper**Author** S. Wang**Author** L. Zhang**Author** W. Zuo

Abstract Subspace learning and reconstruction have been widely explored in recent transfer learning work and generally a specially designed projection and reconstruction transfer matrix are wanted. However, existing subspace reconstruction based algorithms neglect the class prior such that the learned transfer function is biased, especially when data scarcity of some class is encountered. Different from those previous methods, in this paper, we propose a novel reconstruction-based transfer learning method called Class-specific Reconstruction Transfer Learning (CRTL), which optimizes a well-designed transfer loss function without class bias. Using a class-specific reconstruction matrix to align the source domain with the target domain which provides help for classification with class prior modeling. Furthermore, to keep the intrinsic relationship between data and labels after feature augmentation, a projected Hilbert-Schmidt Independence Criterion (pHSIC), that measures the dependency between two sets, is first proposed by mapping the data from original space to RKHS in transfer learning. In addition, combining low-rank and sparse constraints on the class-specific reconstruction coefficient matrix, the global and local data structures can be effectively preserved. Extensive experiments demonstrate that the proposed method outperforms conventional representation-based domain adaptation methods. © 2017 IEEE.

Date 2018**DOI** 10.1109/ICCVW.2017.116**URL** <https://www.scopus.com/pages/publications/85046252749?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc. - IEEE Int. Conf. Comput. Vis. Workshops, ICCVW**Volume** 2018-January**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 978-153861034-3 (ISBN)**Pages** 949-957**Proceedings Title** Proc. - IEEE Int. Conf. Comput. Vis. Workshops, ICCVW

Conference Name Proceedings - 2017 IEEE International Conference on Computer Vision Workshops, ICCVW 2017

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Tags:

Transfer matrixes, Transfer learning, Federated learning, Adversarial machine learning, Contrastive Learning, Data scarcity, Target domain, Loss functions, Reconstruction matrix, Subspace reconstruction, Rank constraints, Subspace learning, Transfer learning methods

Notes:

EID: 2-s2.0-85046252749

Export Date: 20 September 2026; Cited By: 12; Correspondence Address: L. Zhang; College of Communication Engineering, Chongqing University, Chongqing, China; email: leizhang@cqu.edu.cn; Conference name: 16th IEEE International Conference on Computer Vision Workshops, ICCVW 2017; Conference date: 22 October 2017 through 29 October 2017; Conference code: 134301

Classification and knowledge discovery in protein databases

Item Type Journal Article

Author P. Radivojac

Author N.V. Chawla

Author A.K. Dunker

Author Z. Obradovic

Abstract We consider the problem of classification in noisy, high-dimensional, and class-imbalanced protein datasets. In order to design a complete classification system, we use a three-stage machine learning framework consisting of a feature selection stage, a method addressing noise and class-imbalance, and a method for combining biologically related tasks through a prior-knowledge based clustering. In the first stage, we employ Fisher's permutation test as a feature selection filter. Comparisons with the alternative criteria show that it may be favorable for typical protein datasets. In the second stage, noise and class imbalance are addressed by using minority class over-sampling, majority class under-sampling, and ensemble learning. The performance of logistic regression models, decision trees, and neural networks is systematically evaluated. The experimental results show that in many cases ensembles of logistic regression classifiers may outperform more expressive models due to their robustness to noise and low sample density in a high-dimensional feature space. However, ensembles of neural networks may be the best solution for large datasets. In the third stage, we use prior knowledge to partition unlabeled data such that the class distributions among non-overlapping clusters significantly differ. In our experiments, training classifiers specialized to the class distributions of each cluster resulted in a further decrease in classification error. © 2004 Elsevier Inc. All rights reserved.

Date 2004

DOI 10.1016/j.jbi.2004.07.008

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Language English
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Journal Abbr J. Biomed. Informatics
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Tags:

Artificial Intelligence, Cluster Analysis, article, Reproducibility of Results, algorithm, Algorithms, artificial neural network, model, Class imbalance, learning, logistic regression analysis, Classification, bioinformatics, Feature selection, Proteins, density, Sensitivity and Specificity, Databases, Protein, protein database, Noise, machine, priority journal, Clustering, Pattern Recognition, Automated, Information Storage and Retrieval, Sequence Analysis, Protein, Sequence Alignment, Class-distribution estimation, Database Management Systems, Fisher exact test

Notes:

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Export Date: 20 September 2026; Cited By: 84; Funding details: National Institutes of Health, NIH; National Science Foundation, NSF, (0219736, IIS-0219736); U.S. National Library of Medicine, NLM, (R01LM007688); Correspondence Address: Ctr. for Info. Sci. and Technology, Temple University, USA, United States; email: zoran@ist.temple.edu; CODEN: JBI0B

Classification and Measurement of Land Cover of Wildfires in Australia Using Remote Sensing

Item Type Journal Article

Author A.H. Thary Al-Ghrai

Author E.K. Hassan

Author H.M. Saeed

Abstract Remote sensing techniques used in many studies for classfying and measuring of wildfires. Satellite Landsat8(OLI) imagery is used in the presented work. The satellite is considered as a near-polar orbit, with a high multispectral resolution for covering Wollemi National Park in Australia. The work aims to study and measure wildfire natural resources prior to and throughout fire breakout which occurred in Wollemi National Park in Australia for a year (October, 2019), as well as analyzing the harm resulting from such wildfires and their effects on earth and environment through recognizing satellite images for studied region prior to and throughout wildfires. A

discussion of methods for computing the affected area is covered regarding each one of the classes and lessening or limiting the quickly-spreading wildfires damage. This paper propose a 2-phases techniques: Training and classifying. In the training phase, the number of clustering is computed by using C# Programming Language and feature extracted and clustered as a group and stored in the dataset. The classification used the moments with (K-Means) classification approach in RS (Remote Sensing) for classified image. The results of classification showed 5 distinctive classes (trees, rivers, bare earth, buildings with no trees, and buildings with trees) in which it might be indicates that the region is secured via each one of the classes prior to and throughout wildfires as well as the changed pixels with regard to all the classes. Also, the classification experimental methods results indicate an excellent performance recision with a good classifying and result analysis about the harms caused by fires in the study area. © 2022 University of Baghdad-College of Science. All rights reserved.

Date 2022

DOI 10.24996/ij.s.2022.63.1.38

URL <https://www.scopus.com/pages/publications/85124980965?origin=resultslist>

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Language English

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Journal Abbr Iraqi J. Sci.

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Tags:

Landsat-8(OLI), Moment feature, Remote sensing Techniques, Unsupervised Classifying Image, Wollemi National Park

Notes:

EID: 2-s2.0-85124980965

Export Date: 20 September 2026; Cited By: 6; Correspondence Address: E.K. Hassan; Department of Computer Science, College of Science, Unversity of Baghdad, Baghdad, Iraq; email: Enas.mkhazal@gmail.com

Classification and quantification based on image analysis for sperm samples with uncertain damaged/intact cell proportions

Item Type Conference Paper

Author L. Sánchez

Author V. González

Author E. Alegre

Author R. Alaiz

Abstract Classifying damaged-intact cells in a semen sample presents the peculiarity that the test class distribution is unknown. This paper studies under which design conditions the misclassification rate is minimum for the uncertainty region of interest (ratio of damaged cells lower than 20%) and (b) deals with quantifying the proportion of damaged/intact sperm cells in a given sample based on computer vision and supervised learning. We have applied a discrete wavelet transform to the spermatozoa head images and computed the mean and standard deviation (WSF) and four Haralick descriptors (WCF). Using a backpropagation neural network, the error rate averaged over distributions in the region of interest is 4.85% with WCF. The assessment of several quantification methods shows the conditions under which the Adjusted Count method leads to an overall mean absolute error of 3.2 and the Classify & Count method yields 2.4, both with WCF features. Deviations of this order are considered reasonable for this field. © 2008 Springer-Verlag Berlin Heidelberg.

Date 2008

DOI 10.1007/978-3-540-69812-8_82

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Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

Volume 5112 LNCS

Publisher Springer Verlag

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Pages 827-836

Proceedings Title Lect. Notes Comput. Sci.

Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Tags:

Supervised learning, Neural networks, Discrete wavelet transforms, Image analyze, Image-analysis, Class distributions, Region-of-interest, Damaged cells, Design condition, Intact cells, Misclassification rates, Regions of interest, Uncertainty regions

Notes:

EID: 2-s2.0-47749100121

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Classification and quantification of sesame oil in edible oils and adulterated mixtures using ¹H NMR spectroscopy combined with multivariate, machine learning, and deep learning models

Item Type Journal Article

Author H. Lim

Author H. Cho

Author J.Y. Kim

Author Y.J. Shin

Author H.S. Chun

Author B.H. Kim

Author S. Ahn

Abstract Sesame oil is often adulterated with cheaper oils, necessitating accurate authentication and quantification methods. This study investigates the performance of AI-based models using ¹H NMR spectral data for edible oil classification and sesame oil quantification in adulterated mixtures. All classification models—PCA-LDA, SVM, and 1D-CNN—achieved 100 % accuracy, with 1D-CNN additionally capturing both lignan and fatty acid signals. For regression, PLSR and SVR models achieved RMSEP values of 1.94 and 1.40 ($R^2 = 0.998$), while the 1D-CNN regression model demonstrated superior performance (RMSEP = 1.03, $R^2 = 0.999$) with broader spectral feature integration. External test samples incorporating previously unused oil types further validated the robustness of the CNN model, which accurately predicted sesame oil content within a 2 % error margin. These findings highlight the potential of explainable deep learning integrated with NMR spectroscopy for reliable detection and quantification of adulterated sesame oil. © 2025 Elsevier Ltd

Date 2025

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URL <https://www.scopus.com/pages/publications/105013781468?origin=resultslist>

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Publication Food Chemistry

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Artificial intelligence, Article, Performance, Deep neural networks, Classification (of information), Learning systems, machine learning, Machine-learning, Convolutional neural network, Convolutional neural networks, Learning models, Deep Learning, Machine Learning, quantitative analysis, Regression analysis, procedures, deep learning, controlled study, multivariate analysis, Convolutional neural networks (CNN), explainable artificial intelligence, convolutional neural network, classification, support vector machine, evaluation study, measurement accuracy, chemistry, Quantification methods, partial least squares regression, food contamination, Food Contamination, machine prediction model, Nuclear magnetic resonance spectroscopy, regression model, Multivariate Analysis, H NMR spectroscopy, Magnetic Resonance Spectroscopy, nuclear magnetic resonance spectroscopy, proton nuclear magnetic resonance, fatty acid, ¹H NMR, Authentication methods, edible oil, Edible oil, lignan, Sesame oil, Sesame Oil, sesame seed oil

Notes:

EID: 2-s2.0-105013781468

Export Date: 20 September 2026; Cited By: 8; Funding details: Chung-Ang University, CAU; Ministry of Food and Drug Safety, MFDS, (22193MFD471); Ministry of Food and Drug Safety, MFDS; Correspondence Address: S. Ahn; Department of Chemistry, Chung-Ang University, Seoul, 06974, South Korea; email: sangdoo@cau.ac.kr; CODEN: FOCHD

Classification based Integration of Quantifications for LC-MS Analysis

Item Type Conference Paper

Author T. Li

Author L. Chen

Author H. Wei

Abstract A classification based integration of quantification method for the Liquid Chromatography - Mass Spectrometry (LC-MS) analysis is described in this paper. Typically, one biological tissue may be sent to the LC-MS many times in practice to generate multiple LC-MS data. Due to the precise level or the profile of the search engine, these multiple individual quantitative results of the multiple LC-MS data may be partially identical. Here we proposed a method to integrate the quantitative results for the case where there are multiple individual measurements but the results are only partially identical. This proposed method applies a classifier to the peptides and treats the predicted probabilities of the classification as the weights to combine these multiple individual quantitative results into a better one. Experimental results show that in the task of quantitative LC-MS, the results generated by this integration method perform better than the ones produced by other individual measurements. © 2018 IEEE.

Date 2018

DOI 10.1109/SPAC46244.2018.8965613

URL <https://www.scopus.com/pages/publications/85079175561?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Int. Conf. Secur., Pattern Anal., Cybernetics, SPAC

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-172810551-2 (ISBN)

Pages 97-102

Proceedings Title Int. Conf. Secur., Pattern Anal., Cybernetics, SPAC

Conference Name 2018 International Conference on Security, Pattern Analysis, and Cybernetics, SPAC 2018

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Tags:

Integration, Search engines, Quantitative result, Quantification methods, Mass spectrometry, Liquid chromatography, Biological tissues, Integration method, Lc-ms analysis, Liquid chromatography-Mass spectrometry, Precise levels

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Classification Error Bound for Low Bayes Error Conditions in Machine Learning

Item Type Conference Paper

Author Z. Yang

Author V. Eminyan

Author R. Schlüter

Author H. Ney

Abstract In statistical classification and machine learning, classification error is an important performance measure, which is minimized by the Bayes decision rule. In practice, the unknown true distribution is usually replaced with a model distribution estimated from the training data in the Bayes decision rule. This substitution introduces a mismatch between the Bayes error and the model-based classification error. In this work, we apply classification error bounds to study the relationship between the error mismatch and the Kullback-Leibler divergence in machine learning. Motivated by recent observations of low model-based classification errors in many machine learning tasks, bounding the Bayes error to be lower, we propose a linear approximation of the classification error bound for low Bayes error conditions. Then, the bound for class priors are discussed.

Moreover, we extend the classification error bound for sequences. Using automatic speech recognition as a representative example of machine learning applications, this work analytically discusses the correlations among different performance measures with extended bounds, including cross-entropy loss, language model perplexity, and word error rate. © 2025 IEEE.

Date 2025

DOI 10.1109/ICASSP49660.2025.10890759

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Language English

Extra Journal Abbreviation: ICASSP IEEE Int Conf Acoust Speech Signal Process Proc

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Proceedings Title ICASSP IEEE Int Conf Acoust Speech Signal Process Proc

Conference Name ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings

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Tags:

Computational linguistics, Speech communication, Condition, Classification (of information), Learning systems, machine learning, Machine-learning, Speech recognition, Machine learning, Signal processing, Error analysis, Barium compounds, Bayes decision rules, Bayes' errors, classification error bound, Classification error bound, Classification errors, Error bound, Error condition, mismatch condition, Mismatch condition, Performance measure, speech recognition

Notes:

EID: 2-s2.0-105009586156

Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing, ICASSP 2025; Conference date: 6 April 2025 through 11 April 2025; Conference code: 208116; CODEN: IPROD

Classification from only positive and unlabeled functional data

Item Type Journal Article

Author Y. Terada

Author I. Ogasawara

Author K. Nakata

Abstract In various fields, data recorded continuously during a time interval and curve data, such as spectral data, become common. These kinds of data can be interpreted as functional data. In this paper we have studied binary classification from only positive and unlabeled functional data (PU classification for functional data). Our first contribution is to present a simple classification algorithm for this problem. The key

feature of the algorithm is that it is not required an estimation of the unknown class prior (or the constant probability that a positive object is labeled). It is worth noting that the idea of our method can be applied to kernel linear discriminant analysis for general data. Our second contribution is to prove that, under mild regularity conditions similar to those in a supervised context, the proposed algorithm can achieve perfect asymptotic classification in the context of PU classification. In fact, we show that the proposed algorithm works well not only in numerical experiments but also for real data examples. Moreover, as an important practical application, we have used the proposed algorithm to identify handball players at risk for anterior cruciate ligament (ACL) injury based on ground reaction force data. © Institute of Mathematical Statistics, 2020.

Date 2020

DOI 10.1214/20-AOAS1404

URL <https://www.scopus.com/pages/publications/85098235419?origin=resultslist>

Archive Scopus

Language English

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Publisher Institute of Mathematical Statistics

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Publication Annals of Applied Statistics

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Issue 4

Journal Abbr Ann. Appl. Statistics

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Tags:

Binary classification, PU learning, Functional data analysis

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Export Date: 20 September 2026; Cited By: 2; Funding details: Japan Society for the Promotion of Science, KAKEN, (20K19756, 24700716, 16K16024)

Classification from positive and unlabeled data based on likelihood invariance for measurement

Item Type Journal Article

Author T. Yoshida

Author T. Washio

Author T. Ohshiro

Author M. Taniguchi

Abstract We propose novel approaches for classification from positive and unlabeled data (PUC) based on maximum likelihood principle. These are particularly suited to measurement tasks in which the class prior of the target object in each measurement is unknown and significantly different from the class prior used for training, while the likelihood function representing the observation process is invariant over the training and measurement stages. Our PUCs effectively work without estimating the class priors of the unlabeled objects. First, we present a PUC approach called Naive Likelihood PUC (NL-PUC) using the maximum likelihood principle in a nontrivial but rather straightforward manner. The extended version called Enhanced Likelihood PUC (EL-PUC) employs an algorithm iteratively improving the likelihood estimation of the positive class. This is advantageous when the availability of the labeled positive data is limited. These characteristics are demonstrated both theoretically and experimentally. Moreover, the practicality of our PUCs is demonstrated in a real application to single molecule measurement. © 2021 - IOS Press. All rights reserved.

Date 2021

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Publication Intelligent Data Analysis

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Issue 1

Journal Abbr Intell. Data Anal.

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Classification (of information), Target object, Iterative methods, Semi-supervised learning, Unlabeled data, Likelihood functions, Maximum likelihood estimation, Application to measurement, Classification from positive and unlabeled data, Extended versions, Likelihood estimation, Likelihood invariance, Maximum likelihood Principle, Real applications, Single molecule measurements

Notes:

EID: 2-s2.0-85100537218

Export Date: 20 September 2026; Cited By: 3; Funding details: , (19H00852); Correspondence Address: T. Washio; Institute of Scientific and Industrial Research, Osaka University, Ibarakishi, Osaka, 8-1, Mihogaoka, 567-0047, Japan; email: washio@ar.sanken.osaka-u.ac.jp

Classification methods and models for automatic determination of goods code by foreign economic activity goods nomenclature

Item Type Conference Paper

Author J.U. Sevinov

Author G.R. Khamroev

Editor Tanaino I.

Editor Dzholdosheva T.

Abstract Classification methods and models for automatic determination of goods code according to the commodity nomenclature of foreign economic activity (CN FEA) are considered. In the classification procedure, the object is reduced to one or more classes based on the results of the comparison to evaluate the proximity and make a conclusion. The relationship between vectors calculated by scalar multiplication was used as a measure of proximity. In order to determine the most informative symbols for classes, linguostatistical methods based on the information about the a priori probability of occurrence of terms and symbols were used. In doing so, macro-average and micro-average methods, which are considered effective in assessing the quality of classification for several classes, were used. Also, on the basis of syntactic and linguostatistical analysis, a generalized scheme of the process of automatic classification of goods in column 31_1 of the goods declaration graph is proposed. This recommended method of automatic identification of the CN FEA code allows participants of foreign economic activity to ease the processes of filling out declarations for goods. © The Authors, published by EDP Sciences. This is an open access article distributed under the terms of the Creative Commons Attribution License 4.0 (<https://creativecommons.org/licenses/by/4.0/>).

Date 2023

DOI 10.1051/e3sconf/202340203036

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Language English

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Volume 402

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Conference Name E3S Web of Conferences

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: J.U. Sevinov; Tashkent State Technical University, Tashkent, Uzbekistan; email: sevinovjasur@gmail.com; Conference name: 2023 International Scientific Siberian Transport Forum, TransSiberia 2023; Conference date: 16 May 2023 through 19 May 2023; Conference code: 190756

Classification of 3D structures based on an object detection for facade elements in multiple views during the reconstruction process

Item Type Conference Paper

Author R. Baumann

Author J.-F. Evers-Senne

Author M. Strand

Abstract The objective of this paper is to propose an approach which combines the multi view 3D reconstruction process of a building's facade with semantic information obtained by a deep neural network on the 2D observations of the considered scene. The object detector was used to annotate precise 2D features with semantic labels based on the object proposals. In a next step, spatially connected 2D features were evaluated to extend the initially computed labels to the whole facade element. The set of annotated 2D precise features was used to derive a classification of reconstructed 3D features. The a priori probabilities of the object detection model were used to compensate for biased predictions. The described approach was evaluated thoroughly with a view to different strategies for the annotation correction in 2D. Several failure patterns occurring in the considered application of facade reconstruction were observed and suggested solutions were given. © 2019 IEEE.

Date 2019

DOI 10.1109/ICPHYS.2019.8780181

URL <https://www.scopus.com/pages/publications/85070875268?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - IEEE Int. Conf. Ind. Cyber Phys. Syst., ICPS

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-153868500-6 (ISBN)

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Conference Name Proceedings - 2019 IEEE International Conference on Industrial Cyber Physical Systems, ICPS 2019

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Tags:

Deep learning, Deep neural networks, Object recognition, Classification (of information), Computer vision, Object detection, Embedded systems, Semantics, Information fusion, 3D reconstruction, Image reconstruction, A-priori probabilities, Stereo image processing, Facade elements, Facades, Failure patterns, Object detectors, Reconstruction process, Semantic information, Semantic labels, Stereo vision

Notes:

EID: 2-s2.0-85070875268

Export Date: 20 September 2026; Cited By: 3; Conference name: 2019 IEEE International Conference on Industrial Cyber Physical Systems, ICPS 2019; Conference date: 6 May 2019 through 9 May 2019; Conference code: 150221

Classification of Breast Tumors Using Elastographic and B-mode Features: Comparison of Automatic Selection of Representative Slice and Physician-Selected Slice of Images

Item Type Journal Article

Author W.K. Moon

Author S.-C. Chang

Author J.M. Chang

Author N. Cho

Author C.-S. Huang

Author J.-W. Kuo

Author R.-F. Chang

Abstract Inter-observer variability and image quality are two key factors that can affect the diagnostic performance of elastography and B-mode ultrasound for breast tumor characterization. The purpose of this study is to use an image quantification method that automatically chooses a representative slice and then segments the tumor contour to evaluate the diagnostic features for tumor characterization. First, the representative slice is selected based on either the stiffness inside the tumor (the signal-to-noise ratio on the elastogram [SNRe]) or the contrast between the tumor and the surrounding normal tissue (the contrast-to-noise ratio on the elastogram [CNRe]). Next, the level set method is used to segment the tumor contour. Finally, the B-mode and elastographic features related to the segmented tumor are extracted for tumor characterization. The performance of the representative slice selected using the proposed methods is compared to that of the physician-selected slice in 151 biopsy-proven lesions (89 benign and 62 malignant). The diagnostic accuracies using elastographic features are 82.1% (124/151) for the slice with the maximum CNRe value, 82.1% (124/151) for the slice with the maximum SNRe value and 82.8% (125/151) for the physician-selected slice, whereas the diagnostic accuracies using B-mode features are 80.8% (122/151) for the slice with the maximum CNRe value, 87.4% (132/151) for the slice with the maximum SNRe value and 84.1% (127/151) for the physician-selected slice. When using both the B-mode and elastographic features to characterize the tumor, the accuracy of diagnosis is 86.1% (130/151) for the slice with the maximum CNRe value, 90.1% (136/151) for the slice with the maximum SNRe value and 89.4% (135/151) for the physician-selected slice. Our results show that the representative slice selected by SNRe and CNRe could be used to reduce the observer variability and to increase the diagnostic performance by the B-mode and elastographic features. © 2013 World Federation for Ultrasound in Medicine & Biology.

Date 2013

DOI 10.1016/j.ultrasmedbio.2013.01.017

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Tags:

adult, automation, female, human, aged, article, Medical imaging, controlled study, false positive result, major clinical study, interrater reliability, diagnostic test accuracy study, receiver operating characteristic, predictive value, diagnostic accuracy, Computer aided diagnosis, sensitivity and specificity, computer assisted diagnosis, image quality, signal noise ratio, breast tumor, radiologist, elastography, Elastography, Signal to noise ratio, Diagnostic performance, Ultrasonics, Interobserver variability, B scan, Tumors, priority journal, Characterization, B-mode ultrasound, Breast tumor, Computer-aided diagnosis, contrast to noise ratio, Contrast to noise ratio, Image quantification, Level set segmentation, Tumor characterization, tumor classification, ultrasound scanner

Notes:

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Export Date: 20 September 2026; Cited By: 11; Funding details: Innovative Research Institute for Cell Therapy, (A062260); National Science Council, NSC, (NSC 101-2221-E-002-068-MY3); Ministry of Education, MOE, (AE-00-00-06); National Research Foundation of Korea, NRF, (2012010846); Ministry of Economic Affairs, MOEA, (100-EC-17-A-19-S1-164); Correspondence Address: R.-F. Chang; Department of Computer Science and Information Engineering, National Taiwan University, Taipei 106, Taiwan; email: rfchang@csie.ntu.edu.tw; CODEN: USMBA

Classification of collective actions for rural community vitalization in Chugoku Mountainous region, Japan: applying multiple correspondence analysis

Item Type Journal Article

Author N. Yasunaga

Abstract This study quantitatively examines the classification of collectivities in rural communities and the factors affecting the collectivities in hilly and mountainous areas. Using the collective action data of rural communities in the Chugoku region (Tottori, Shimane, Okayama, Hiroshima, and Yamaguchi Prefectures) obtained from the Census of Agriculture and Forestry in view of social, economic, and environmental aspects. We quantitatively examined the typological characteristics and the factors of collectivities in rural communities based on the multiple

correspondence analysis (quantification method, type III). We made three main findings. First, we extracted the dimensions of weak-strong independence of community (there are relatively strong and weak communities linked with other communities in terms of collectivity), high-low collective behavior of community (there are strong and weak collectivity on social, economic, and environmental collective activities), and economic-social nature of collective behavior of community. Second, the forms of collectivities in rural communities were classified into three or four forms from the cluster analysis using the sample scores obtained from the multiple correspondence analysis. Locational factors such as closeness to DID, road station and expressway partially influence the differences among the forms. Third, as well as locational factors, the elements of human capital, bonding social capital, bridging social capital have an influence on the levels of collectivities. These findings indicate that it is critical to take measures to consider the rural vitalization conditions across the rural communities for the sustainability of rural resources. © 2020, The Japan Section of the Regional Science Association International.

Date 2020

DOI 10.1007/s41685-020-00145-9

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Archive Scopus

Language English

Volume 4

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Pages 553-592

Publication Asia-Pacific Journal of Regional Science

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Issue 2

Journal Abbr Asia-Pac. J. Reg. Sci.

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Classification, Collective actions, Hilly and mountainous areas, Multiple correspondence analysis, Rural community, Rural community vitalization

Notes:

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Classification of dynamical diffusion states in single molecule tracking microscopy

Item Type Journal Article

Author P.J. Bosch

Author J.S. Kanger

Author V. Subramaniam

Abstract Single molecule tracking of membrane proteins by fluorescence microscopy is a promising method to investigate dynamic processes in live cells. Translating the trajectories of proteins to biological implications, such as protein interactions, requires the classification of protein motion within the trajectories. Spatial information of protein motion may reveal where the protein interacts with cellular structures, because binding of proteins to such structures often alters their diffusion speed. For dynamic diffusion systems, we provide an analytical framework to determine in which diffusion state a molecule is residing during the course of its trajectory. We compare different methods for the quantification of motion to utilize this framework for the classification of two diffusion states (two populations with different diffusion speed). We found that a gyration quantification method and a Bayesian statistics-based method are the most accurate in diffusion-state classification for realistic experimentally obtained datasets, of which the gyration method is much less computationally demanding. After classification of the diffusion, the lifetime of the states can be determined, and images of the diffusion states can be reconstructed at high resolution. Simulations validate these applications. We apply the classification and its applications to experimental data to demonstrate the potential of this approach to obtain further insights into the dynamics of cell membrane proteins. © 2014 Biophysical Society.

Date 2014

DOI 10.1016/j.bpj.2014.05.049

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Language English

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procedures, statistical analysis, fluorescence microscopy, metabolism, motion, Motion, Diffusion, Data Interpretation, Statistical, diffusion, Microscopy, Fluorescence, membrane protein, Membrane Proteins, molecular imaging, Molecular Imaging, protein transport, Protein Transport

Notes:

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Classification of Left Atrial Diseased Tissue Burden Determined by Automated Voltage Analysis Predicts Outcomes after Ablation for Atrial Fibrillation

Item Type Journal Article

Author S. Herczeg

Author J.J. Keaney

Author E. Keelan

Author C. Howard

Author K. Walsh

Author L. Geller

Author G. Szeplaki

Author J. Galvin

Abstract Background. The burden and persistence of atrial fibrillation (AF) have been associated with the presence and extent of left atrial (LA) fibrosis. Recent reports have implicated an association between the extent of LA fibrosis and the outcome of pulmonary vein isolation (PVI). We aimed to analyse the value of an automated scar quantification method in the prediction of success following PVI. Methods. One hundred and nine consecutive patients undergoing PVI for paroxysmal or persistent AF were included in our observational study with a 2-year follow-up. Prior to PVI, patients underwent high-definition LA electroanatomical mapping, and scar burden was quantified by automated software (Voltage Histogram Analysis, CARTO 3, Biosense Webster), then classified into 4 subgroups (Dublin Classes I-IV). Recurrence rates were analysed on and off antiarrhythmic drug therapy (AAD), respectively. Results. The overall success rate was 74% and 67% off AAD at 1- and 2-year follow-up, respectively. Patients with Dublin Class IV had significantly lower success rates ($p=0.008$, off AAD). Dublin Class IV ($OR=2.27$, $p=0.022$, off AAD) and the presence of arrhythmia in the blanking period ($OR=3.28$, $p=0.001$, off AAD) were the only significant predictors of recurrence. The use of AAD did not affect these results. Conclusions. We propose a classification of low voltage areas based on automated quantification by software during 3D mapping prior to PVI. Patients with high burden of low voltage areas ($>31\%$ of <0.5 mV, Dublin Class IV) have a higher risk of recurrence following PVI. Information gathered during electroanatomical mapping may have important prognostic value. © 2021 Szilvia Herczeg et al.

Date 2021

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Volume 2021

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adult, Adult, Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, aged, Aged, Aged, 80 and over, very elderly, Electrocardiography, controlled study, major clinical study, observational study, follow up, Retrospective Studies, retrospective study, disease burden, comorbidity, atrial fibrillation, electrocardiogram, hypertension, pathology, electrocardiography, catheter ablation, prognosis, heart disease, echocardiography, Echocardiography, proportional hazards model, transesophageal echocardiography, general anesthesia, computed tomographic angiography, histogram, treatment outcome, Treatment Outcome, heart left atrium, ablation therapy, vascular disease, Fibrosis, fibrosis, heart left ventricle ejection fraction, Kaplan Meier method, Follow-Up Studies, Computed Tomography Angiography, antiarrhythmic agent, anticoagulant therapy, Atrial Fibrillation, blood clotting time, Catheter Ablation, CHA2DS2-VASc score, Clinical Decision Rules, Dublin classification, electrical potential parameters, electroanatomical mapping, electrophysiological procedures, esophagus injury, Heart Atria, heart atrium, heart palpitation, heart supraventricular arrhythmia, heparin, Kaplan-Meier Estimate, paroxysmal atrial fibrillation, pathological tissue, patient acuity, Patient Acuity, persistent atrial fibrillation, Proportional Hazards Models, proton pump inhibitor, pulmonary vein isolation, Recurrence, recurrence risk, recurrent disease, thrombosis prevention, thrombus, voltage histogram analysis tool

Notes:

EID: 2-s2.0-85109341660

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: G. Szeplaki; Atrial Fibrillation Institute, Mater Private Hospital, Dublin 7, Eccles Street, Ireland; email: szeplaki.gabor@gmail.com; CODEN: DMARD

Classification of Micromobility Vehicles in Thermal-Infrared Images Based on Combined Image and Contour Features Using Neuromorphic Processing

Item Type Journal Article**Author** B. Stahl**Author** J. Apfelbeck**Author** R. Lange

Abstract Trends of environmental awareness, combined with a focus on personal fitness and health, motivate many people to switch from cars and public transport to micromobility solutions, namely bicycles, electric bicycles, cargo bikes, or scooters. To accommodate urban planning for these changes, cities and communities need to know how many micromobility vehicles are on the road. In a previous work, we proposed a concept for a compact, mobile, and energy-efficient system to classify and count micromobility vehicles utilizing uncooled long-wave infrared (LWIR) image sensors and a neuromorphic co-processor. In this work, we elaborate on this concept by focusing on the feature extraction process with the goal to increase the classification accuracy. We demonstrate that even with a reduced feature list compared with our early concept, we manage to increase the detection precision to more than 90%. This is achieved by reducing the images of 160×120 pixels to only 12×18 pixels and combining them with contour moments to a feature vector of only 247 bytes. © 2023 by the authors.

Date 2023

DOI 10.3390/app13063795

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Issue 6

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urban planning, machine learning, machine vision, long-wave infrared, micromobility, neuromorphic processing, thermal imaging, traffic data

Notes:

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Classification of natural areas in Northern Finland using optical remote sensing images and data fusion

Item Type Conference Paper

Author M. Törmä

Abstract The aim of this study was to make CORINE 2000 classification better in Northern Finland by making the classes of natural areas more separable using optical remote sensing images and decision based data fusion methods. Phenological time-series derived from MODIS images and ETM-image mosaic were classified using Maximum Likelihood classifier. Classification results were merged using different data fusion methods and their result compared. The used methods were not particularly successful, because the overall accuracy of CORINE 2000-classification, 52%, was better than the overall accuracies of spectral classifications and data fusion methods. The best data fusion methods were maximum joint a posteriori probability-classification and classification where a priori probabilities have been acquired from lower resolution classification. © 2007 IEEE.

Date 2007

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Archive Scopus

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Conference Name International Geoscience and Remote Sensing Symposium (IGARSS)

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Tags:

Classification (of information), Decision making, Remote sensing, Data fusion, Probability, Time series analysis, Classification, Image fusion, Landsat, Maximum likelihood, MODIS, CORINE 2000, ETM-image mosaic, Natural areas

Notes:

EID: 2-s2.0-79956108669

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: M. Törmä; Institute of Photogrammetry and Remote Sensing, Helsinki University of Technology, Finland; email: Markus.Torma@tkk.fi; Conference name: 2007 IEEE International Geoscience and Remote Sensing Symposium, IGARSS 2007; Conference date: 23 June 2007 through 28 June 2007; Conference code: 71398; CODEN: IGRSE

Classification of nonstationary cardiac signals based on their spectral and probabilistic properties

Item Type Conference Paper

Author S.M. Koval

Author E.I. Sokol
Author P.F. Shchapov
Author R.S. Tomashevskyi
Author O.V. Soltmann
Author W. Wójcik
Author K. Dassibekov
Editor Wójcik W.
Editor Pavlov S.
Editor Kalimoldayev M.

Abstract This research presents a quantification method for the estimation of normalised correlation-spectral cardiac signal parameters that applies a probabilistic model of the cardiac signal as a random and a priori nonstationary process. The high sensitivity and informative significance of the proposed parameters of nonstationarity are confirmed by statistical cardiac signal classification (significance testing) and multi-alternative discrimination by types of cardiac state. The possibility is shown for the building of the absolute metric scales of the cardiac states. © 2019 Taylor & Francis Group, London.

Date 2019

DOI 10.1201/9780429057618-31

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Language English

Extra Journal Abbreviation: Inf. Technol. Med. Diagn. II - Proc. Int. Sci. Internet Conf. Comput. Gr. Image Proc. Int. Sci. Pract. Conf. Appl. Lasers Med. Biol.

Publisher CRC Press/Balkema

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Pages 267-274

Proceedings Title Inf. Technol. Med. Diagn. II - Proc. Int. Sci. Internet Conf. Comput. Gr. Image Proc. Int. Sci. Pract. Conf. Appl. Lasers Med. Biol.

Conference Name Information Technology in Medical Diagnostics II - Proceedings of the International Scientific Internet Conference on Computer Graphics and Image Processing and 48th International Scientific and Practical Conference on Application of Lasers in Medicine and Biology, 2018

Date Added 20/09/2026, 9:53:02 pm

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Tags:

Heart, Medical image processing, Diagnosis, Quantification methods, Computer graphics, High sensitivity, Non-stationarities, Probabilistic modeling, Cardiac signals, Laser applications, Nonstationary process, Probabilistic properties, Significance testing

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Export Date: 20 September 2026; Cited By: 2; Conference name: International Scientific Internet Conference on Computer Graphics and Image Processing, 2018 and 48th International Scientific and Practical Conference on Application of Lasers in Medicine and Biology, 2018; Conference date: 30 May 2018 through 31 May 2018; Conference code: 223989

Classification of pollen species using autofluorescence image analysis

Item Type Journal Article

Author K. Mitsumoto

Author K. Yabusaki

Author H. Aoyagi

Abstract A new method to classify pollen species was developed by monitoring autofluorescence images of pollen grains. The pollens of nine species were selected, and their autofluorescence images were captured by a microscope equipped with a digital camera. The pollen size and the ratio of the blue to red pollen autofluorescence spectra (the B/R ratio) were calculated by image processing. The B/R ratios and pollen size varied among the species. Furthermore, the scatter-plot of pollen size versus the B/R ratio showed that pollen could be classified to the species level using both parameters. The pollen size and B/R ratio were confirmed by means of particle flow image analysis and the fluorescence spectra, respectively. These results suggest that a flow system capable of measuring both scattered light and the autofluorescence of particles could classify and count pollen grains in real time. © 2008 The Society for Biotechnology, Japan. All rights reserved.

Date 2009

DOI 10.1016/j.jbiosc.2008.10.001

URL <https://www.scopus.com/pages/publications/58149289462?origin=resultslist>

Archive Scopus

Language English

Volume 107

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Publication Journal of Bioscience and Bioengineering

ISSN 13891723 (ISSN)

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Journal Abbr J. Biosci. Bioeng.

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Real time systems, Image enhancement, Cameras, Image processing, Image analysis, Pollen, Image Processing, Computer-Assisted, Models, Statistical, Fluorescence, Spectrometry, Fluorescence, Particle size analysis, Flow Cytometry, Microscopy, Fluorescence, Autofluorescence, Fluorescent Dyes, Hay fever, Microscopic image analysis, Particle Size, Pollen classification, Pollen counting, Scattering, Radiation

Notes:

EID: 2-s2.0-58149289462

Export Date: 20 September 2026; Cited By: 56; Correspondence Address: K. Yabusaki; Electronics and Optics Research Laboratories, Kowa Company, Ltd., Chofu, Tokyo, 182-0021, 3-3-1 Chofugaoka, Japan; email: yabusaki@kowa.co.jp; CODEN: JBBIF

Classification of Width 1 Lattice Tetrahedra by Their Multi-Width

Item Type	Journal Article
Author	G. Hamm
Abstract	We introduce the multi-width of a lattice polytope and use this to classify and count all lattice tetrahedra with multi-width $(1, w_2, w_3)$. The approach used in this classification can be extended into a computer algorithm to classify lattice tetrahedra of any given multi-width. We use this to classify tetrahedra with multi-width $(2, w_2, w_3)$ for small w_2 and w_3 and make conjectures about the function counting lattice tetrahedra of any multi-width. © The Author(s) 2024.
Date	2025
DOI	10.1007/s00454-024-00659-5
URL	https://www.scopus.com/pages/publications/105001089510?origin=resultslist
Archive	Scopus
Language	English
Volume	73
Publisher	Springer
Pages	859-885
Publication	Discrete and Computational Geometry
ISSN	01795376 (ISSN)
Issue	3
Journal Abbr	Discrete Comput. Geom.
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Tags:

Classification algorithm, Lattice polytope, Multi-width, Simplex

Notes:

EID: 2-s2.0-105001089510

Export Date: 20 September 2026; Cited By: 0; Funding details: Heilbronn Institute for Mathematical Research, HIMR; Correspondence Address: G. Hamm; Mathematical Sciences, University of Nottingham, Nottingham, NG7 2RD, United Kingdom; email: girtrude.hamm@nottingham.ac.uk

Classification on data with biased class distribution

Item Type Conference Paper

Author S. Vucetic

Author Z. Obradovic

Editor de Raedt L.

Editor Department of Computer Science, Georges Köhler-Allee, Geb. 079, Freiburg, 79110 Albert-Ludwigs University Freiburg

Editor Flach P.

Editor Department of Computer Science, Merchant Ventures Bldg., Woodland Road, Bristol, BS8 1UB University of Bristol

Abstract Labeled data for classification could often be obtained by sampling that restricts or favors choice of certain classes. A classifier trained using such data will be biased, resulting in wrong inference and sub-optimal classification on new data. Given an unlabeled new data set we propose a bootstrap method to estimate its class probabilities by using an estimate of the classifier's accuracy on training data and an estimate of probabilities of classifier's predictions on new data. Then, we propose two methods to improve classification accuracy on new data. The first method can be applied only if a classifier was designed to predict posterior class probabilities where predictions of an existing classifier are adjusted according to the estimated class probabilities of new data. The second method can be applied to an arbitrary classification algorithm, but it requires retraining on the properly resampled data. The proposed bootstrap algorithm was validated through experiments with 500 replicates calculated on 1,000 realizations for each of 16 choices of data set size, number of classes, prior class probabilities and conditional probabilities describing a classifier's performance. Applications of the proposed methodology to a benchmark data set with various class probabilities on unlabeled data and balanced class probabilities on the training data provided strong evidence that the proposed methodology can be successfully used to significantly improve classification on unlabeled data. © Springer-Verlag Berlin Heidelberg 2001.

Date 2001

DOI 10.1007/3-540-44795-4_45

URL <https://www.scopus.com/pages/publications/42749092337?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

Volume 2167

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Proceedings Title Lect. Notes Comput. Sci.
Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)
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Tags:

Classification (of information), Machine learning, Forecasting, Benchmarking, Classification algorithm, Classification accuracy, Conditional probabilities, Class distributions, Class probabilities, Optimal classification, Bootstrap algorithms, Posterior class probabilities

Notes:

EID: 2-s2.0-42749092337

Export Date: 20 September 2026; Cited By: 37; Conference name: 12th European Conference on Machine Learning, ECML 2001; Conference date: 5 September 2001 through 7 September 2001; Conference code: 134049

Classification under Nuisance Parameters and Generalized Label Shift in Likelihood-Free Inference

Item Type Conference Paper

Author L. Masserano

Author A. Shen

Author M. Doro

Author T. Dorigo

Author R. Izbicki

Author A.B. Lee

Editor Salakhutdinov R.

Editor Kolter Z.

Editor Heller K.

Editor Weller A.

Editor Oliver N.

Editor Scarlett J.

Editor Berkenkamp F.

Abstract An open scientific challenge is how to classify events with reliable measures of uncertainty, when we have a mechanistic model of the data-generating process but the distribution over both labels and latent nuisance parameters is different between train and target data. We refer to this type of distributional shift as generalized label shift (GLS). Direct classification using observed data X as covariates leads to biased predictions and invalid uncertainty estimates of labels Y . We overcome these biases by proposing a new method for robust uncertainty quantification that casts classification as a hypothesis testing problem under nuisance parameters. The key idea is to estimate the classifier's receiver operating

characteristic (ROC) across the entire nuisance parameter space, which allows us to devise cutoffs that are invariant under GLS. Our method effectively endows a pretrained classifier with domain adaptation capabilities and returns valid prediction sets while maintaining high power. We demonstrate its performance on two challenging scientific problems in biology and astroparticle physics with data from realistic mechanistic models. Copyright 2024 by the author(s)

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URL <https://www.scopus.com/pages/publications/85203805703?origin=resultslist>

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Language English

Extra Journal Abbreviation: Proc. Mach. Learn. Res.

Volume 235

Publisher ML Research Press

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Proceedings Title Proc. Mach. Learn. Res.

Conference Name Proceedings of Machine Learning Research

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Tags:

Uncertainty quantifications, Measure of uncertainty, Uncertainty estimates, Covariates, Parameter spaces, Hypothesis testing, Mechanistic models, Nuisance parameter, Observed data, Receiver operating characteristics

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Classification with the pot–pot plot

Item Type Journal Article

Author O. Pokotylo

Author K. Mosler

Abstract We propose a procedure for supervised classification that is based on potential functions. The potential of a class is defined as a kernel density estimate multiplied by the class's prior probability. The method transforms the data to a potential–potential (pot–pot) plot, where each data point is mapped to a vector of potentials.

Separation of the classes, as well as classification of new data points, is performed on this plot. For this, either the α -procedure (α -P) or k-nearest neighbors (k-NN) are employed. For data that are generated from continuous distributions, these classifiers prove to be strongly Bayes-consistent. The potentials depend on the kernel and its bandwidth used in the density estimate. We investigate several variants of bandwidth selection, including joint and separate pre-scaling and a bandwidth regression approach. The new method is applied to benchmark data from the literature, including simulated data sets as well as 50 sets of real data. It compares favorably to known classification methods such as LDA, QDA, max kernel density estimates, k-NN, and DD-plot classification using depth functions. © 2016, Springer-Verlag Berlin Heidelberg.

Date 2019

DOI 10.1007/s00362-016-0854-8

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Journal Abbr Stat. Pap.

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Bandwidth choice, DD-plot, DD α -classifier, k-Nearest-neighbors classification, Kernel density estimates, Potential functions, α -Procedure

Notes:

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Export Date: 20 September 2026; Cited By: 1; Funding details: Cologne Graduate School of Management, Economics and Social Sciences; Correspondence Address: O. Pokotylo; Statistics and Econometrics, Universität zu Köln, Cologne, Albertus Magnus Platz, 50923, Germany; email: pokotylo@wiso.uni-koeln.de

Classification, deconstruction and evaluation of frost heave models: How modeling methods cause simulation error

Item Type Journal Article

Author S.-H. Cheng

Author B.A. Engel

Author H.-X. Wu

Author P.-Z. Duan

Author Y.-B. Wang

Abstract Frost heave models are important tools to study engineering damage in cold regions. However, multi-physical-field-coupling and different methods of coupling components bring difficulties to compare the models and to optimize their error sources. Therefore, classifying and deconstructing of different frost heave models and further evaluating them are significant for model development. We reviewed frost heave models for the past 50 years and built frost heave model families to show the classification and inheritance relationships. Then, the models were deconstructed by the thermal–hydraulic framework (THF) and frost heave quantification method (FHQM). Finally, based on simulated and experimental data in cited papers, the model errors caused by different THFs and FHQMs were analyzed. The results show: (a) Present frost heave models can be classified as porosity increase based (P) and ice lens increase based (I) models and they are tending to merge with each other; (b) THFs of P models regarding freezing soil as porous medium are the main reason for water migration error in the region of multi-ice layers, which also results high uncertainty of hydraulic parameters under different simulation conditions; (c) The most used two FHQMs of P models cause error exceeding 20 % because inaccurate relationship between pore strain and frost heave displacement; (d) The errors of THFs and FHQMs offset each other, which will hidden the error sources of the models. This paper can provide a reference for the improvement of frost heave models used in engineering. © 2022

Date 2022

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simulation, Classification (of information), Error analysis, classification, Classification, Model method, error analysis, Cold regions, Porous materials, coupling, Deconstruction, Error sources, freezing, frost heave, Frost heave, Frost heave model, Ice, Method evaluation, Multi-physical fields, P-model, Simulation error

Notes:

EID: 2-s2.0-85141273858

Export Date: 20 September 2026; Cited By: 15; Funding details: Science and Technology Integrated Innovation Project; Shaanxi Province of China, (2016KTZDNY-01-01); National Key Research and Development Program of China, NKRDPC, (2016YFC0400205); Higher Education Discipline Innovation Project, (B12007); Correspondence Address: Y.-B. Wang; Key Lab. of Agricultural Soil and Water Engineering in Arid and Semiarid Areas, Ministry of Education, Institute of Water Saving Agriculture in Arid Regions of China, Northwest A&F Univ., Yangling, Shaanxi, 712100, China; email: wangyb@nwfau.edu.cn; CODEN: JHYDA

Classifier Guidance Enhances Diffusion-Based Adversarial Purification by Preserving Predictive Information

Item Type Conference Paper

Author M. Zhang

Author J. Li

Author W. Chen

Author J. Guo

Author X. Cheng

Editor Endriss U.

Editor Melo F.S.

Editor Bach K.

Editor Bugarin-Diz A.

Editor Alonso-Moral J.M.

Editor Barro S.

Editor Heintz F.

Abstract Adversarial purification is one of the promising approaches to defend neural networks against adversarial attacks. Recently, methods utilizing diffusion probabilistic models have achieved great success for adversarial purification in image classification tasks. However, such methods fall into the dilemma of balancing the needs for noise removal and information preservation. This paper points out that existing adversarial purification methods based on diffusion models gradually lose sample information during the core denoising process, causing occasional label shift in subsequent classification tasks. As a remedy, we suggest to suppress such information loss by introducing guidance from the classifier confidence. Specifically, we propose Classifier-cOnfidence gUided Purification (COUP) algorithm, which purifies adversarial examples while keeping away from the classifier decision boundary. Experimental results show that COUP can achieve better adversarial robustness under strong attack methods.
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Date 2024

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Archive Scopus

Language English

Extra Journal Abbreviation: Front. Artif. Intell. Appl.

Volume 392**Publisher** IOS Press BV**ISBN** 09226389 (ISSN); 978-164368548-9 (ISBN)**Pages** 2234-2241**Proceedings Title** Front. Artif. Intell. Appl.**Conference Name** Frontiers in Artificial Intelligence and Applications**Date Added** 20/09/2026, 9:52:07 pm**Modified** 20/09/2026, 9:52:07 pm**Tags:**

Neural-networks, Adversarial machine learning, Generative adversarial networks, Classification tasks, Diffusion model, Probabilistic models, Images classification, Information preservations, Noise information, Noises removal, Predictive information, Purification method

Notes:

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Classifying cadmium contaminated leafy vegetables using hyperspectral imaging and machine learning

Item Type Journal Article**Author** A. Souza**Author** M.Z. Rojas**Author** Y. Yang**Author** L. Lee**Author** L. Hoagland

Abstract Cadmium (Cd) is a toxic element that can accumulate in edible plant tissues and negatively impact human health. Traditional Cd quantification methods are time-consuming, expensive, and generate a lot of toxic waste, slowing development of methods to reduce uptake. The objective of this study was to determine whether hyperspectral imaging (HSI) and machine learning (ML) can be used to predict Cd concentrations in plants using kale (*Brassica oleracea*) and basil (*Ocimum basilicum*) as model crops. The experiments were conducted in an automated phenotyping facility where all environmental conditions except soil Cd concentration were kept constant. Cd concentrations were determined at harvest using traditional methods and used to train the ML models with data collected from the imaging sensor. Visible/near infrared (VNIR) images were also collected at harvest and processed to calculate reflectance at 473 bands between 400 to 998 nm. All reflectance spectra were subject

to the feature selection algorithm ReliefF and Principal Component Analysis (PCA) to generate data and provide input to evaluate three ML classification models: artificial neural network (ANN), ensemble learning (EL), and support vector machine (SVM). Plants were categorized according to Cd concentrations higher or lower than the safety threshold of 0.2 mg kg⁻¹ Cd. Wavelengths with the highest ranks for Cd detection were between 519 and 574, and 692 and 732 nm, indicating that Cd content likely altered the plants' chlorophyll content and altered leaf internal structure. All models were able to sort the plants into groups, though the model with the best F1 score was the ANN for the validation subset that utilized reflectance from all wavelengths. This study demonstrates that HSI and ML are promising technologies for the fast and precise diagnosis of Cd in leafy green plants, though additional studies are needed to adapt this approach for more complex field environments. © 2022 The Author(s)

Date 2022

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Tags:

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Export Date: 20 September 2026; Cited By: 21; Funding details: NIFA-HATCH, (1015999); Purdue Libraries Open Access Fund; Correspondence Address: L. Hoagland; Horticulture and Landscape Architecture Department, Purdue University, West Lafayette, United States; email: lhoaglan@purdue.edu

Classifying Image Series with a Reoccurring Concept Drift Using a Markov Chain Predictor as a Feedback

Item Type Conference Paper

Author M. Skoczeń

Author W. Rafajłowicz

Author E. Rafajłowicz

Editor Rutkowski L.
Editor Scherer R.
Editor Korytkowski M.
Editor Pedrycz W.
Editor Tadeusiewicz R.
Editor Zurada J.M.

Abstract We consider the problem of classifying images, ordered in a sequence (series), to several classes. A distinguishing feature of this problem is the presence of a slowly varying, reoccurring concept drift which results in the existence of relatively long subsequences of similar images from the same class. It seems useful to incorporate this knowledge into a classifier. To this end, we propose a novel idea of constructing a Markov chain (MC) that predicts a class label for the next image to be recognized. Then, the feedback is used in order to modify slightly a priori probabilities of class memberships. In particular, the a priori probability of the class predicted by the MC is increased at the expense of decreasing the a priori probabilities of other classes. The idea of applying an MC predictor of classes with the feedback to a priori probabilities is rather general. Thus, it can be applied not only to images but also to vectors of features arising in other applications. Additionally, this idea can be combined with any classifier that is able to take into account a priori probabilities. We provide an analysis when this approach leads to the reduction of the classification errors in comparison to classifying each image separately. As a vehicle to elucidate the idea we selected a recently proposed classifier (see [23, 25]) that is based on assuming matrix normal distribution (MND) of classes. We use a specific structure of the MND's covariance matrix that can be estimated directly from images without feature extraction. The proposed approach was tested on simulated images as well as on an example of the sequence of images from an industrial gas burner's flames. The aim of the classification is to decide whether natural gas contains a sufficient amount of methane (the combustion process is proper, lower air pollution) or not. © 2020, Springer Nature Switzerland AG.

Date 2020

DOI 10.1007/978-3-030-61534-5_7

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Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

Volume 12416 LNAI

Publisher Springer Science and Business Media Deutschland GmbH

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Pages 71-84

Proceedings Title Lect. Notes Comput. Sci.

Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Tags:

Artificial intelligence, Image classification, Markov chains, Concept drifts, Covariance matrix, Normal distribution, A-priori probabilities, Classification errors, Classification of ordered images, Classifier with feedback, Combustion pro-cess, Generalized matrix normal distribution, Image series, Markov chain predictor, Reoccurring concept drift, Sequence of images, Similar image, Simulated images, Soft computing

Notes:

EID: 2-s2.0-85096621872

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: E. Rafajłowicz; Faculty of Electronics, Wrocław University of Science and Technology, Wrocław, Poland; email: ewaryst.rafajlowicz@pwr.edu.pl; Conference name: 19th International Conference on Artificial Intelligence and Soft Computing, ICAISC 2020; Conference date: 12 October 2020 through 14 October 2020; Conference code: 250569

Classifying incomplete vectors using decision trees

Item Type Conference Paper

Author B. Twala

Author R. Pillay

Author R. Nkoana

Editor Merelo J.J.

Editor Garibaldi J.

Editor Wagner C.

Editor Back T.

Editor Madani K.

Editor Warwick K.

Abstract An attempt is made to address the problem of classifying incomplete vectors using decision trees. The essence of the approach is the proposal that in supervised learning classification of incomplete vectors can be improved in probabilistic terms. This approach, which is based on the a priori probability of each value determined from the instances at that node of the tree that has specified values, first exploits the total probability and Bayes' theorems and then the probit and logit model probabilities. The proposed approach (developed in three versions) is evaluated using 21 machine learning datasets from its effect or tolerance of incomplete test data. Experimental results are reported, showing the effectiveness of the proposed approach in comparison with multiple imputation and fractioning of instances strategy. Copyright © 2020 by SCITEPRESS - Science and Technology Publications, Lda. All rights reserved

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Language English

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Publisher SciTePress
ISBN 978-989758475-6 (ISBN)
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Proceedings Title IJCCI - Proc. Int. Jt. Conf. Comput. Intell.
Conference Name IJCCI 2020 - Proceedings of the 12th International Joint Conference on Computational Intelligence
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Tags:

Logit models, Learning systems, Machine-learning, Supervised learning, Forestry, Decision trees, Probability, Classification accuracy, Probabilistics, A-priori probabilities, Total probabilities, Baye's theorem, Incomplete data, Model probabilities, Probit and Logit models

Notes:

EID: 2-s2.0-85107293393

Export Date: 20 September 2026; Cited By: 0; Funding details: Faculty of Engineering; Durban University of Technology, DUT; Correspondence Address: B. Twala; Faculty of Engineering and the Built Environment, Durban University of Technology, Durban, 4000, South Africa; email: bhekisiphot@dut.ac.za; Conference name: 12th International Joint Conference on Computational Intelligence, IJCCI 2020; Conference date: 2 November 2020 through 4 November 2020; Conference code: 165037

Classifying Incomplete Vectors using Decision Trees

Item Type Conference Paper

Author B. Twala

Author R. Pillay

Author R. Nkoana

Editor Merele J.J.

Editor Garibaldi J.

Editor Wagner C.

Editor Bäck T.

Editor Madani K.

Editor Warwick K.

Abstract An attempt is made to address the problem of classifying incomplete vectors using decision trees. The essence of the approach is the proposal that in supervised learning classification of incomplete vectors can be improved in probabilistic terms. This approach, which is based on the a priori probability of each value determined from the instances at that node of the tree that has specified values, first exploits the total probability and Bayes' theorems and then the probit and logit model probabilities. The proposed approach (developed in

three versions) is evaluated using 21 machine learning datasets from its effect or tolerance of incomplete test data. Experimental results are reported, showing the effectiveness of the proposed approach in comparison with multiple imputation and fractioning of instances strategy. © 2020 by SCITEPRESS – Science and Technology Publications, Lda. All rights reserved.

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Extra Journal Abbreviation: Int. Jt. Conf. Comput. Intell.

Volume 1

Publisher Science and Technology Publications, Lda

ISBN 21843236 (ISSN); 978-989758475-6 (ISBN)

Pages 455-463

Proceedings Title Int. Jt. Conf. Comput. Intell.

Conference Name International Joint Conference on Computational Intelligence

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Tags:

Logit models, Machine-learning, Computer vision, Decision trees, Supervised Learning, Classification accuracy, Probabilistics, A-priori probabilities, Total probabilities, Baye's theorem, Incomplete data, Model probabilities, Probit and Logit models, Classification Accuracy, Decision Trees, Incomplete Data, Logit Models

Notes:

EID: 2-s2.0-85190672919

Export Date: 20 September 2026; Cited By: 0; Funding details: Fakultas Teknik Universitas Indonesia, FTUI; Durban University of Technology, DUT; Conference name: 12th International Joint Conference on Computational Intelligence, IJCCI 2020; Conference date: 2 November 2020 through 4 November 2020; Conference code: 310789

Classifying interpersonal interaction in virtual reality: sensor-based analysis of human interaction with pre-recorded avatars

Item Type Journal Article

Author Y. Arima

Author Y. Harada

Author M. Okada

Abstract This study investigates human engagement with a non-responsive, pre-recorded avatar in VR environments. Rather than bidirectional collaboration, we focus on unidirectional synchrony from human participants to the avatar and evaluate its detectability using sensor-based machine learning. Using a random forest model, we classified interactions into cooperation, conformity, and competition, achieving an F1 score of 0.89. Feature importance analysis identified hand rotation and head position as key predictors of interaction states. We compared human-human and human interaction with a non-responsive avatar (pre-recorded motion replay) during a joint Simon task by covertly switching collaborators between humans and non-responsive avatars. Using the classification model, a synchrony index was derived from VR motion data to quantify behavioral coordination patterns during joint actions. The classification indexes were associated with higher cooperation in human-human interactions (Formula presented.) and greater conformity in human interaction with a non-responsive avatar (Formula presented.). The synchrony index was significantly lower in the non-responsive avatar condition (Formula presented.), indicating reduced interpersonal synchrony with non-responsive avatars. These findings demonstrate the feasibility of using VR sensor data and machine learning to quantify social interaction dynamics. This study aimed to explore the feasibility of a sensor-based machine learning model for classifying interpersonal interactions in VR, based on preliminary data from small-sample experiments. Copyright © 2025 Arima, Harada and Okada.

Date 2025

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Tags:

virtual reality, machine learning, human activity recognition, interpersonal synchrony, joint simon effect

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Japan Society for the Promotion of Science, KAKENHI, (21K02988, 23K22343, 23K25727); Japan Society for the Promotion of Science, KAKENHI; Correspondence Address: Y. Arima; Department of Psychology, Center for Social and Psychological Research of Metaverse, Kyoto University of Advanced Science, Kyoto, Japan; email: arima.yoshiko@kuas.ac.jp

Classifying microwave radar images using decision based data fusion

Item Type Conference Paper

Author M. Törmä

Author M. Engdahl

Editor Kerle N.

Editor Skidmore A.

Abstract Four different data fusion methods for classifying SAR images with different spatial resolutions were tested and compared. Test area was Helsinki region in Southern Finland and test data 14 ERS-1/2 Tandem pairs. The best method was a method where a posteriori probabilities of lower spatial resolution classification are used as a priori probabilities of higher resolution classification. The increase of overall accuracy was 7-14 %-units depending on date and 10.4% on average when compared to original Tandem pairs. Median filtering increased classification accuracy, but not that much when data fusion methods were used. This means that the need of spatial filtering can be at least partially compensated using data fusion of different spatial resolution images. © 2006 International Society for Photogrammetry and Remote Sensing. All rights reserved.

Date 2006

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Language English

Extra Journal Abbreviation: Int. Arch. Photogramm., Remote Sens. Spat. Inf. Sci. - ISPRS Arch.

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Publisher International Society for Photogrammetry and Remote Sensing

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Conference Name International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives

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Classification (of information), Remote sensing, Synthetic aperture radar, Image classification, Fusion, Image resolution, Classification, Image fusion, Classification accuracy, Land cover, Pixels, Radar imaging, Overall accuracies, A-priori probabilities, SAR, A-posteriori probabilities, Fusion reactions, Data fusion methods, Land Cover, Median filters, Radar signal processing, Spatial filterings, Spatial resolution images

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Ecological Society of America, ESA, (A03-277); Conference name: 7th Symposium on Remote Sensing: From Pixels to Processes 2006; Conference date: 8 May 2006 through 11 May 2006; Conference code: 155661

Climate change poses risks to water retention and carbon sequestration capacity in the source area of the Yangtze River

Item Type Journal Article

Author H. Liu

Author H. Liu

Author H. Peng

Author T. Zheng

Author Y. Zhao

Author S. Wang

Author Y. Jia

Author Q. Zeng

Author Z. Yang

Author L. Yan

Author Q. Yang

Author B. Zhao

Author J. Wang

Author P. Hu

Author J. Luo

Abstract The source area of the Yangtze River (SAYR), part of the Tibetan Plateau, is an ecologically fragile alpine region sensitive to climate change. Current research has predominantly examined hydrological and ecological responses as isolated systems, failing to address the coupled mechanisms through which permafrost degradation mediates water-carbon interactions. In this study, we used a fully coupled eco-hydrological model that integrates permafrost processes, along with multi-source remote sensing data, experimental monitoring, and machine learning, to quantify the water retention and carbon sequestration capacity over the past 20 years. The region was categorized into three risk zones based on changes in soil moisture, net ecosystem productivity (NEP), and dissolved organic carbon (DOC) fluxes in streams. We evaluated eight factors, including precipitation, temperature, vegetation phenology and cover, and their contributions to changes of water retention and carbon sequestration using an interpretable machine learning approach. Results show that the central and eastern regions of the study area face the highest risk of declining water retention and carbon sequestration capacity. The changes of temperatures and precipitation have led to depletion of soil water and carbon reserves. This depletion raises concerns about the potential shift from a carbon sink to a carbon source considering land-to-river carbon loss. Our study provides critical insights into the water and carbon flux dynamics and offers valuable guidance for water resource and ecological management in alpine river systems. © 2025 Elsevier Ltd

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'current, Article, risk assessment, Ecosystems, Learning systems, machine learning, Environmental Monitoring, Machine learning, China, Carbon cycle, Statistics, Remote sensing, ecosystem, Ecosystem, climate change, precipitation, remote sensing, Stream flow, temperature, Climate Change, carbon, environmental monitoring, chemistry, vegetation, Rivers, Carbon, soil, Soil, environmental risk, soil moisture, Soil moisture, organic carbon, permafrost, river, Carbon Economy, Organic carbon, carbon sequestration, A-carbon, Alpine regions, Carbon capture, Carbon sequestration, Carbon Sequestration, carbon source, dissolved organic matter, Eco-hydrology, ecohydrology, Ecohydrology, hydrological model, Hydrology, Permafrost, phenology, river water, runoff, soil water, Source area, The tibetan plateau, The Tibetan Plateau, water retention, Water retention, yangtze river, Yangtze River

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Climate warming increases extreme daily wildfire growth risk in California

Item Type Journal Article

Author P.T. Brown

Author H. Hanley

Author A. Mahesh

Author C. Reed

Author S.J. Strenfel

Author S.J. Davis

Author A.K. Kochanski

Author C.B. Clements

Abstract California has experienced enhanced extreme wildfire behaviour in recent years 1–3, leading to substantial loss of life and property 4,5. Some portion of the change in wildfire behaviour is attributable to anthropogenic climate warming, but formally quantifying this contribution is difficult because of numerous confounding factors 6,7 and because wildfires are below the grid scale of global climate models. Here we use machine learning to quantify empirical relationships between temperature (as well as the influence of temperature on aridity) and the risk of extreme daily wildfire growth (>10,000 acres) in California and find that the influence of temperature on the risk is primarily mediated through its influence on fuel moisture. We use the uncovered relationships to estimate the changes in extreme daily wildfire growth risk under anthropogenic warming by subjecting historical fires from 2003 to 2020 to differing background climatological temperatures and aridity conditions. We find that the influence of anthropogenic warming on the risk of extreme daily wildfire growth varies appreciably on a fire-by-fire and day-by-day basis, depending on whether or not climate warming pushes conditions over certain thresholds of aridity, such as 1.5 kPa of vapour-pressure deficit and 10% dead fuel moisture. So far, anthropogenic warming has enhanced the aggregate expected frequency of extreme daily wildfire growth by 25% (5–95 range of 14–36%), on average, relative to preindustrial conditions. But for some fires, there was approximately no change, and for other fires, the enhancement has been as much as 461%. When historical fires are subjected to a range of projected end-of-century conditions, the aggregate expected frequency of extreme daily wildfire growth events increases by 59% (5–95 range of 47–71%) under a low SSP1–2.6 emissions scenario compared with an increase of 172% (5–95 range of 156–188%) under a very high SSP5–8.5 emissions scenario, relative to preindustrial conditions. © 2023, The Author(s), under exclusive licence to Springer Nature Limited.

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California, risk assessment, United States, machine learning, article, climate change, moisture, temperature, Climate Change, wildfire, extreme event, aridity, Temperature, climate, Climate, climate warming, forest, global climate, warming, fire, fire behavior, fire history, Fires, Forests, moisture content, vapor pressure, Wildfires

Notes:

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Climate-Chemistry Synergy Is Reversing Global Urban–Rural Ozone Differences

Item Type Journal Article

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Author Y. Zhang

Author Y. Yang

Author S. Wang

Author Y. Wang

Author S. Wang

Author F. Fan

Author Y. Wang

Author S. Wang

Author X. He

Author H. Zhang

Abstract Health-damaging surface ozone tends to be lower in populated urban areas than in the surroundings, yet this urban–rural difference is narrowing. While evidence from observations have attributed this narrowing to the reduced inhibition effect of urban nitrogen oxides, we lack clarity on how intensified humid-heat modulates precursor sensitivity in ways that reconfigure the urban–rural gradient under climate warming. Here, we integrated high-resolution air pollutant reanalysis with interpretable machine learning to quantify the global spatial patterns and temporal evolution of paired urban–rural ozone gradients. We show that meteorology–chemistry synergy in urban areas, encompassing weakened inhibition effects of NO2 and dewpoint temperature and an intensified positive effect of radiation, contributes significantly to the reversal of urban–rural ozone disparities in tropical regions and to the long-term global convergence (0.1 μg m^{−3} year^{−1}, 2005–2023). Although heatwaves have become more frequent, ozone anomalies have diminished driven by moist atmospheric conditions and stringent emission cuts. Yet urban anomalies exceed rural

ones, with the differences during moist heatwaves nearly 2-fold those observed in dry events. Our study urges targeted and proactive policies to prevent climate warming synergies from disproportionately worsening urban ozone burdens. © 2026 American Chemical Society

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Article, machine learning, spatiotemporal analysis, Rural areas, Urban areas, Cities, city, climate change, nonhuman, rural area, atmospheric pollution, urban area, chemistry, explainable machine learning, Climate warming, air pollutant, Air Pollutants, Air pollution, binocular convergence, climate, Climate, climate warming, dew point temperature, heat wave, heatwave extreme, Heatwave extreme, Heatwaves, Inhibition effect, meteorology, nitrogen dioxide, Nitrogen oxides, ozone, Ozone, ozone layer, radiation, Radiation effects, surface ozone, Surface ozone, temporal evolution, Trend, trends, tropical region, Urban-rural, Urban-rural gradients, urban-rural migration, urban–rural differences, Urban/rural differences

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Clinical features, etiology and outcomes of community-acquired pneumonia in patients with chronic obstructive pulmonary disease

Item Type Journal Article

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Author D. Viasus

Author P. Millat-Martínez

Author A. Simonetti

Author M. Salud Santos

Author C. Ardanuy

Author J. Dorca

Author J. Carratalà

Abstract Background: Community-acquired pneumonia (CAP) is a frequent complication of chronic obstructive pulmonary disease (COPD), but previous studies are often contradictory. Objectives: We aimed to ascertain the characteristics and outcomes of CAP in patients with COPD as well as to determine the risk factors for mortality and *Pseudomonas aeruginosa* pneumonia in COPD patients with CAP. We also describe the etiology and outcomes of CAP in COPD patients receiving chronic oxygen therapy at home and those receiving inhaled steroids. Methods: An observational analysis of a prospective cohort of hospitalized adults with CAP (1995-2011) was performed. Results: We documented 4121 CAP episodes, of which 983 (23.9%) occurred in patients with COPD; the median FEV1 value was 50%, and 57.8% were classified as stage III or IV in the GOLD classification. Fifty-eight per cent of patients were receiving inhaled steroids, and 14.6% chronic oxygen therapy at home. Patients with COPD presented specific clinical features. *S. pneumoniae* was the leading causative organism overall, but *P. aeruginosa* was more frequent in COPD (3.4 vs. 0.5%; $p < 0.001$). Independent risk factors for case-fatality rate in patients with COPD were multilobar pneumonia, *P. aeruginosa* pneumonia, and high-risk PSI classes. Prior pneumococcal vaccination was found to be protective. FEV1 was an independent risk factor for *P. aeruginosa* pneumonia. Conclusions: CAP in patients with COPD presents specific characteristics and risk factors for mortality. Prior pneumococcal vaccine has a beneficial effect on outcomes. *P. aeruginosa* pneumonia is associated with low FEV1 values and poor prognosis. © 2014 Gómez-Junyent et al.

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female, Female, human, Humans, male, Male, middle aged, Middle Aged, aged, Aged, Aged, 80 and over, article, very elderly, risk factor, Risk Factors, controlled study, age, major clinical study, complication, observational study, outcome assessment, Prospective Studies, prospective study, clinical feature, cohort analysis, disease classification, disease association, high risk patient, mortality, prognosis, Prognosis, Pseudomonas aeruginosa, chronic obstructive lung disease, treatment outcome, Treatment Outcome, pneumonia, Pulmonary Disease, Chronic Obstructive, Age Factors, community acquired pneumonia, Community-Acquired Infections, Oxygen Inhalation Therapy, oxygen therapy, Pneumococcus vaccine, Pneumonia, Pseudomonas pneumonia

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Clinical implications of changing thyroglobulin and antithyroglobulin antibodies analytical methods in the follow-up of patients with differentiated thyroid carcinoma

Item Type Journal Article

Author S. Deza

Author J. Maroto

Author O. Tellechea

Author N. Orbegozo

Author J. Merino

Author J.C. Galofré

Author E. Alegre

Author Á. González

Abstract Background and aims: Patients' response to treatment in differentiated thyroid cancer (DTC) is classified according to serum thyroglobulin concentrations (Tg), usually using the American Thyroid Association guidelines and considering potential interfering anti-thyroglobulin antibodies (Ab-Tg). We aim to evaluate the clinical implications of changing Tg and Ab-Tg quantification method. Material and methods: Tg and Ab-Tg were quantified in 82 serum samples (60 from DTC patients) by Elecsys and Access immunoassays. Results: Elecsys immunoassay rendered higher values of Tg than Access: mean bias 5.03 ng/mL (95%CI:-14.14–24.21). In DTC patients, there was an almost perfect agreement for response classification (kappa index = 0.833). Discrepancies appeared in patients with undetermined response, with a more tendency to subclassification with Access. Ab-Tg showed a poor correlation ($r = 0.5394$). When Elecsys cut-off was reduced to 43 IU/mL, agreement for positive/negative classification improved from a kappa index of 0.607 to 0.650. Prospective study with personalized follow-up showed that only 6.3% of Tg results required an analytical confirmation, being confirmed 93% of them. Conclusions: Despite the biases observed, clinical impact of an analytical change is minimal in patients' management. However, cautious and personalized follow-up

period after the change is still mandatory, especially in patients with Tg levels between 0.2 and 1 ng/mL. © 2023

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Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: Á. González; autor at: Service of Biochemistry. Clínica, Universidad de Navarra Av, Pamplona, Pío XII 36, 31008, Spain; email: agonzaleh@unav.es; CODEN: CCATA

Clinician-in-the-Loop Smart Home System to Detect Urinary Tract Infection Flare-Ups via Uncertainty-Aware Decision Support

Item Type Conference Paper

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Author R. Fritz

Author D.J. Cook

Author J. Doppa

Editor Koenig S.

Editor Jenkins C.

Editor Taylor M.E.

Abstract Urinary tract infection (UTI) flare-ups pose a significant health risk for older adults with chronic conditions. These infections often go unnoticed until they become severe, making early detection through innovative smart home technologies crucial. Traditional machine learning (ML) approaches relying on simple binary classification for UTI detection offer limited utility to nurses and practitioners as they lack insight into prediction uncertainty, hindering informed clinical decision-making. This paper presents a clinician-in-the-loop (CIL) smart home system that leverages ambient sensor data to extract meaningful behavioral markers, train robust predictive ML models, and calibrate them to enable uncertainty-aware decision support. The system incorporates a statistically valid uncertainty quantification method called Conformal-Calibrated Interval (CCI), which quantifies uncertainty and abstains from making predictions (“I don’t know”) when the ML model’s confidence is low. Evaluated on real-world data from eight smart homes, our method outperforms baseline methods in recall and other classification metrics while maintaining the lowest abstention proportion and interval width. A survey of 42 nurses confirms that our system’s outputs are valuable for guiding clinical decision-making, underscoring their practical utility in improving informed decisions and effectively managing UTIs and other condition flare-ups in older adults. © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Artificial intelligence, Decision support systems, Automation, Decision making, Learning systems, Uncertainty analysis, Health risks, Data mining, Intelligent buildings, Uncertainty, Machine learning models, Older adults, Smart homes, Decision supports, Machine learning approaches, Chronic conditions, Clinical decision making, Nursing, Smart Home Technology, Smart-home system, Urinary tract infections

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ClipMix for Domain Generalization

Item Type Journal Article

Author A.-A. Liu

Author H.-C. Li

Author W.-H. Li

Author D. Song

Author H.-S. Tian

Author L.-J. Wang

Abstract Domain Generalization (DG) is a growing field in machine learning that aims to train the model across multiple source domains, thereby enabling effective generalization to new, unseen target domains. Recent studies suggest that data augmentation, which enhances the diversity of the source domain, might be a promising solution to address this task. Current data augmentation methods use random fusion coefficients or local regional fusion, which cannot adaptively design the weights based on data, or preserve the integrity of original semantics. Inspired by the pre-trained model CLIP, which contains extensive multimodal knowledge, we propose ClipMix to address these limitations. Firstly, we use the CLIP model as the external knowledge to adaptively evaluate the alignment between images and their labels, using this alignment to assess the complexity of learning each image and guide adaptive augmentation. Secondly, we implement a label shift mechanism to dynamically assign soft labels to fused images, helping the model focus on hard-to-learn patterns and also gather domain-agnostic representation. Furthermore, we enhance the diversity of fused images at both the pixel and feature levels. Experimental results across sixteen domains from four databases verify the effectiveness of our method. © 1991-2012 IEEE.

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Machine-learning, Generalisation, Data augmentation, Target domain, Label shift, Domain generalization, label shift, CLIP-guided mixing, Current data, Fused images, Multiple source

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Cluster cooperation and risk level in small and medium-sized enterprises

Item Type Journal Article

Author K. Havierníková

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Author A. Lemańska-Majdzik

Abstract One of the most important stages in the process of risk management is the risk identification and classification, which are required in each entity located and operating in the environment that is: dynamic, influenced by various external and internal factors and exposed to some types of risks. The main research aim is to identify the most important categories of risk and to develop knowledge about risk management in cluster cooperation. The proposal of the methodology is based on the results of the survey conducted in small and medium sized enterprises in Slovakia in 2016 and the literature review. Risk categories will be analyzed and evaluated with the use of quantification method – non-parametric multi-selective Kruskal-Wallis test. The contribution of the paper consists in pointing out the types of risk the entities have to consider in the framework of cluster cooperation. The results of the paper show that the most important categories of risk are mainly market, political and technical risks. © 2016 Czestochowa University of Technology. All rights reserved.

Date 2016

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Short Title Współpraca klastrowa a poziom ryzyka w małych i średnich przedsiębiorstwach

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Issue 2

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Risk management, Cluster, Cooperation, SMEs

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Export Date: 20 September 2026; Cited By: 16; Correspondence Address: K. Havierníková; Alexander Dubček University of Trenčín, Slovakia; email: katarina.haviernikova@tnuni.sk

Cluster headache and rare-disease classification in Brazil: a systematic review and policy perspective

Item Type Journal Article

Author I.L. Riso

Author O.A.O. Franco

Author A.B. Oliveira

Author V. Grassi

Author W.S. Oliveira

Author E.R.R. Santos

Author C.V.M.G. Simioni

Author E.M. Sarmento

Author M.E. Jurno

Author M. Franco

Author M.F.P. Peres

Author V. Acar

Abstract Background Cluster headache is a highly disabling trigeminal autonomic cephalalgia associated with substantial diagnostic delay, severe pain-related disability, and limited access to evidence-based treatments. In Brazil, epidemiological data are scarce, and CH is not formally recognized within rare-disease policy frameworks. Clarifying the available prevalence evidence may help guide surveillance, research priorities, and health-policy planning. Objective To analyze the prevalence of cluster headache (CH) in Brazil and assess whether the currently available Brazilian population-based evidence is compatible with national rare-disease criteria, while discussing clinical and public-health implications. Methods We conducted a systematic review of population-based epidemiological studies on CH in Brazilian adults, reported according to PRISMA 2020. Searches were performed in PubMed/MEDLINE, LILACS, and SciELO on March 1, 2026, covering publications from January 1, 2010, to March 1, 2026, with no language restrictions. Eligible studies were original Brazilian population-based studies in adults reporting CH prevalence or providing a denominator allowing prevalence estimation. Diagnosis was required to be based on

ICHD-3, ICHD-3 beta, ICHD-2, or clearly described clinical criteria compatible with CH diagnosis. Brazilian Ministry of Health and CONITEC documents were reviewed to identify regulatory definitions and policy criteria for rare-disease classification. Methodological quality and risk of bias were assessed using the JBI Critical Appraisal Checklist for Studies Reporting Prevalence Data. Results The search identified 19 records. After removal of 12 duplicates, 7 records were screened, 4 full-text articles were assessed for eligibility, and 1 population-based Brazilian study met inclusion criteria. The eligible study, conducted in Barbacena, Minas Gerais, identified 15 CH cases among 36,145 adults, corresponding to a prevalence of 41.4 per 100,000 adults. This estimate is below the Brazilian rare-disease threshold of 65 per 100,000 inhabitants. The best available Brazilian population-based estimate is therefore compatible with CH meeting the Brazilian rare-disease prevalence threshold, although national confirmation requires additional population-based studies. Conclusion Current Brazilian population-based evidence is compatible with cluster headache meeting the national rare-disease prevalence threshold. However, this conclusion is based on a single population-based study, and further multicenter epidemiological research is required. Formal recognition may be scientifically justified and could support public-health planning, epidemiological surveillance, professional training, and development of national care pathways for diagnosis and treatment. © 2026, Brazilian Headache Society. All rights reserved.

Date 2026

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Clustered Federated Learning Based on Client's Prototypes

Item Type Conference Paper

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Author Z. Xu

Author Q. Yan

Editor Shen W.

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Editor Zhou X.

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Editor Zhu H.

Editor Peng K.

Editor Xu T.

Editor Chen N.

Abstract Federated learning empowers multiple parties to train machine learning models collaboratively while preserving data within local confines. However, due to the intrinsic non-independent and identically distributed (Non-IID) nature of client-side data, aggregating these diverse local models into a global model remains a significant challenge. Clustered Federated Learning (CFL) offers a compelling solution to mitigate the impact of data heterogeneity by organizing clients into clusters. Existing CFL methods often involve unstable, time-consuming cluster identity estimation during training. In this work, we propose an innovative federated learning approach that efficiently identifies similarities among client data distributions by analyzing client's prototypes. Our method accurately identifies client cluster identities during the initialization phase. Clients employ the same randomly initialized model to compute the client's prototypes for their local data and provide it to the server. The server performs one-shot client clustering by comparing the similarity of the client's prototypes. We further design a unique inter-cluster aggregation strategy that adapts inter-cluster aggregation weights based on the similarity of the client's prototypes, thereby enhancing model convergence. We evaluate two mixed datasets with three Non-IID settings, and our approach outperforms several popular baseline methods. Compared to IFCA and FlexCFL, our approach yields more reasonable clustering results. It achieves an average test accuracy improvement of over 11.51% in the feature and label shift Non-IID setting of the Digits-5 dataset. © 2024 IEEE.

Date 2024

DOI 10.1109/CSCWD61410.2024.10580076

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Language English

Extra Journal Abbreviation: Proc. Int. Conf. Comput. Support. Coop. Work Des., CSCWD

Publisher Institute of Electrical and Electronics Engineers Inc.

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Pages 909-914

Proceedings Title Proc. Int. Conf. Comput. Support. Coop. Work Des., CSCWD

Conference Name Proceedings of the 2024 27th International Conference on Computer Supported Cooperative Work in Design, CSCWD 2024

Date Added 20/09/2026, 9:52:15 pm**Modified** 20/09/2026, 9:52:15 pm**Tags:**

Learning systems, Federated learning, Machine learning models, Clustering algorithms, Statistical tests, Clusterings, Global models, Local model, Data heterogeneity, Client clustering, Client sides, Cluster aggregation, Inter clusters

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Export Date: 20 September 2026; Cited By: 7; Funding details: National Natural Science Foundation of China, NSFC, (61976142); Shenzhen Science and Technology Plan Project, (JCYJ20210324093609025); Correspondence Address: Q. Yan; ShenZhen University, College of Computer Science & Software Engineering, Shenzhen, 518060, China; email: yanq@szu.edu.cn; Conference name: 27th International Conference on Computer Supported Cooperative Work in Design, CSCWD 2024; Conference date: 8 May 2024 through 10 May 2024; Conference code: 200937

COLING/ACL 2006 - 21st International Conference on Computational Linguistics and 44th Annual Meeting of the Association for Computational Linguistics, Proceedings of the Conference

Item Type Journal Article

Abstract The proceedings contain 65 papers. The topics discussed include: combination of Arabic preprocessing schemes for statistical machine translation; kernel-based pronoun resolution with structured syntactic knowledge; estimating class priors in domain adaptation for word sense disambiguation; meaningful clustering of senses helps boost word sense disambiguation performance; relation extraction using label propagation based semi-supervised learning; dependency parsing of Japanese spoken monologue based on clause boundaries; dependencies between student state and speech recognition problems in spoken tutoring dialogues; semi-supervised conditional random fields for improved sequence segmentation and labeling; efficient unsupervised discovery of word categories using symmetric patterns and high frequency words; models for sentence compression: a comparison across domains, training requirements and evaluation measures; and learning event durations from event descriptions.

Date 2006**URL** <https://www.scopus.com/pages/publications/84860528743?origin=resultslist>**Archive** Scopus**Language** English**Volume** 1

Publication COLING/ACL 2006 - 21st International Conference on Computational Linguistics and 44th Annual Meeting of the Association for Computational Linguistics, Proceedings of the Conference

ISSN 1932432655 (ISBN); 978-193243265-7 (ISBN)

Journal Abbr COLING/ACL - Int. Conf. Comput. Linguist. Annu. Meet. Assoc. Comput. Linguist., Proc. Conf.

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Export Date: 20 September 2026; Cited By: 0; Conference name: 21st International Conference on Computational Linguistics and 44th Annual Meeting of the Association for Computational Linguistics, COLING/ACL 2006; Conference date: 17 July 2006 through 21 July 2006; Conference code: 89520

Combination of active learning and self-paced learning for deep answer selection with bayesian neural network

Item Type Conference Paper

Author Q. Wang

Author W. Wu

Author Y. Qi

Author Z. Xin

Editor De Giacomo G.

Editor Catala A.

Editor Dilkina B.

Editor Milano M.

Editor Barro S.

Editor Bugarin A.

Editor Lang J.

Abstract Answer Selection is an important subtask of Question Answering tasks. For this learning-to-rank problem, deep learning methods have outperformed traditional methods. To train a high-quality deep answer selection model, it often requires large amounts of labeled data, which is a costly and noise-prone process. Active learning and semi-supervised learning are usually applied in the modelling training procedure to achieve optimal accuracy with fewer labeled training samples. However, traditional active learning methods rely on good uncertainty estimates that are hard to obtain with standard neural networks. And the performance of semi-supervised learning methods are always affected adversely by the quality of the pseudo-labeled data. In this work, we propose a new framework integrating active learning and self-paced learning in training deep answer selection models. This framework proposes an uncertainty quantification method based on Bayesian neural network, which can guide active learning and self-paced learning in the same iterative process of model training. Experiments were conducted on two kinds of deep answer selection models with real-world datasets including YahooCQA and SemiEvalCQA. The results reveal that the proposed method can significantly reduce the labeled samples for model training.
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Language English
Extra Journal Abbreviation: Front. Artif. Intell. Appl.
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Publisher IOS Press BV
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Conference Name Frontiers in Artificial Intelligence and Applications
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Tags:

Deep neural networks, Learning systems, Uncertainty analysis, Model training, High quality, Learning methods, Iterative methods, Labeled data, Active Learning, Subtask, Bayesian neural networks, Question Answering Task, Selection model, Self-paced learning

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Combined effects of class imbalance and class overlap on instance-based classification

Item Type Conference Paper

Author V. García

Author R. Alejo

Author J.S. Sánchez

Author J.M. Sotoca

Author R.A. Mollineda

Abstract In real-world applications, it has been often observed that class imbalance (significant differences in class prior probabilities) may produce an important deterioration of the classifier performance, in particular with patterns belonging

to the less represented classes. This effect becomes especially significant on instance-based learning due to the use of some dissimilarity measure. We analyze the effects of class imbalance on the classifier performance and how the overlap has influence on such an effect, as well as on several techniques proposed in the literature to tackle the class imbalance. Besides, we study how these methods affect to the performance on both classes, not only on the minority class as usual. © Springer-Verlag Berlin Heidelberg 2006.

Date 2006

DOI 10.1007/11875581_45

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Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

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Combining accuracy and prior sensitivity for classifier design under prior uncertainty

Item Type Conference Paper

Author T. Landgrebe

Author R.P.W. Duin

Abstract Considering the classification problem in which class priors or misallocation costs are not known precisely, receiver operator characteristic (ROC) analysis has become a standard tool in pattern recognition for obtaining integrated performance measures to cope with the uncertainty. Similarly, in situations in

which priors may vary in application, the ROC can be used to inspect performance over the expected range of variation. In this paper we argue that even though measures such as the area under the ROC (AUC) are useful in obtaining an integrated performance measure independent of the priors, it may also be important to incorporate the sensitivity across the expected prior-range. We show that a classifier may result in a good AUC score, but a poor (large) prior sensitivity, which may be undesirable. A methodology is proposed that combines both accuracy and sensitivity, providing a new model selection criterion that is relevant to certain problems. Experiments show that incorporating sensitivity is very important in some realistic scenarios, leading to better model selection in some cases. © Springer-Verlag Berlin Heidelberg 2006.

Date 2006

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Extra Journal Abbreviation: Lect. Notes Comput. Sci.

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Pattern recognition, Mathematical models, Sensitivity analysis, Problem solving, Model selection, Area under the ROC (AUC), Logic design, Receiver operator characteristic (ROC)

Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: T. Landgrebe; Elect. Eng., Maths and Comp. Sc., Delft University of Technology, Delft, Netherlands; email: t.c.w.landgrebe@ewi.tudelft.nl; Conference name: Joint IAPR International Workshops on Structural, Syntactic, and Statistical Pattern Recognition, SSPR 2006 and SPR 2006; Conference date: 17 August 2006 through 19 August 2006; Conference code: 68288

Combining adaptive local aggregation average and test-time energy adaptation for federated learning

Item Type Journal Article

Author J. Liao

Author C. Yi

Author K. Chen

Author Q. Peng

Abstract Federated test-time adaptation (FTTA) aims to adapt knowledge from diverse source models to different but related unlabeled target data in an online and privacy-aware manner. However, existing FTTA methods struggle with decreased adaptation performance caused by data distribution shifts among clients. In this paper, we propose an FTTA method (ALAA-TTEA) called adaptive local aggregation average and test-time energy adaptation. The method consists of two continuous stages. First, in the federated aggregation stage, clients adaptively aggregate the global model and local model through adaptive local aggregation (ALA) to initialize the client model. Then, they obtain the global model through personalized training and model averaging. Second, in the test-time adaptation stage, test-time energy adaptation (TTEA) uses an energy function to transform the global model into an energy-based model. It aligns the model distribution with the test data distribution, thereby enhancing the model's ability to adapt to the test distribution and improving overall performance. Extensive experiments demonstrate that ALAA-TTEA effectively handles data distribution shifts, including feature shift, label shift, hybrid shift, and domain shift. Moreover, it consistently outperforms existing FTTA methods under most conditions. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2025.

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Combining instance selection and self-training to improve data stream quantification

Item Type Journal Article

Author A.G. Maletzke

Author D.M. dos Reis

Author G.E.A.P.A. Batista

Abstract In the last years, learning from data streams has attracted the attention of researchers and practitioners due to its large number of applications. These applications have motivated the research community to propose a significant amount of methods to solve problems in diverse tasks, more prominently in classification, clustering, and anomaly detection. However, a relevant task known as quantification has remained mostly unexplored. The quantification goal is to provide an estimate of the class prevalence in an unlabeled set. Recently, we proposed the SQSI algorithm to quantify data streams with concept drifts. SQSI uses a statistical test to identify concept drifts and retrain the classifiers. However, the retraining involves requiring the labels for all newly arrived instances. In this paper, we extend SQSI algorithm by exploring instance selection techniques allied to semi-supervised learning. The idea is to request the classes of a smaller subset of recent examples. Our experiments demonstrate that SQSI's extension significantly reduces the dependency on actual labels while maintaining or improving the quantification accuracy. © 2018, The Author(s).

Date 2018

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Journal Abbr J. Braz. Comput. Soc.

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Combining similarity and transformer methods for case law entailment

Item Type Conference Paper**Author** J. Rabelo**Author** M.-Y. Kim**Author** R. Goebel

Abstract We tackle the complex problem of determining entailment relationships between case law documents, one of the tasks in the Competition on Legal Information Extraction and Entailment (COLIEE). With input of an entailed fragment from a case coupled with a candidate entailing paragraph from a noticed case, our approach relies on four main components: (1) extraction of similarity measures between the two pieces of text; (2) application of a transformer-based technique on the input text; (3) applying a threshold-based classifier; and (4) post-processing the results considering the a priori probability determined by the data distribution on the training samples and combining the results of (1) and (2). Our experiments achieved an F-score of 0.70 on the official COLIEE test dataset, ranking first among all competitors for that task in the 2019 competition. © 2019 Association for Computing Machinery.

Date 2019**DOI** 10.1145/3322640.3326741**URL** <https://www.scopus.com/pages/publications/85071254934?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc. Int. Conf. Artif. Intell. Law, ICAIL**Publisher** Association for Computing Machinery, Inc

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Tags:

Artificial intelligence, Classification (of information), Statistical tests, Probability distributions, Information retrieval systems, Binary classification, Textual entailment, Data distribution, A-priori probabilities, Imbalanced Data-sets, Document similarity, Imbalanced datasets, Legal textual entailment, Similarity measure, Threshold-based classifiers

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CoMBO: Conflict Mitigation via Branched Optimization for Class Incremental Segmentation

Item Type Conference Paper
Author K. Fang
Author A. Zhang
Author G. Gao
Author J. Jiao
Author C.H. Liu
Author Y. Wei
Abstract Effective Class Incremental Segmentation (CIS) requires simultaneously mitigating catastrophic forgetting and ensuring sufficient plasticity to integrate new classes. The inherent conflict above often leads to a back-and-forth, which turns the objective into finding the balance between the performance of previous (old) and incremental (new) classes. To address this conflict, we introduce a novel approach, Conflict Mitigation via Branched Optimization (CoMBO). Within this approach, we present the Query Conflict Reduction module, designed to explicitly refine queries for new classes through lightweight, class-specific adapters. This module provides an additional branch for the acquisition of new classes while preserving the original queries for distillation. Moreover, we develop two strategies to further mitigate the conflict following the branched structure, i.e., the Half-Learning Half-Distillation (HDHL) over classification probabilities, and the Importance-Based Knowledge Distillation (IKD) over query

features. HDHL selectively engages in learning for classification probabilities of queries that match the ground truth of new classes, while aligning unmatched ones to the corresponding old probabilities, thus ensuring retention of old knowledge while absorbing new classes via learning negative samples. Meanwhile, IKD assesses the importance of queries based on their matching degree to old classes, prioritizing the distillation of important features and allowing less critical features to evolve. Extensive experiments in Class Incremental Panoptic and Semantic Segmentation settings have demonstrated the superior performance of CoMBO. Project page: <https://guangyu-ryan.github.io/CoMBO>. © 2025 IEEE.

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knowledge distillation, class incremental panoptic segmentation, class incremental segmentation, class incremental semantic segmentation

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Comment on: “A comprehensive landscape of AI applications in broad-spectrum drug interaction prediction: a systematic review” (Marzouk et al., 2025)

Item Type Journal Article

Author A. Kargar

Author M.A. Zamani

Author G. Mohammadnezhad

Abstract Marzouk et al. reviewed 147 studies on artificial intelligence (AI) applications for predicting drug–drug, drug–disease, and drug–nutrient interactions, providing a broad overview of current machine learning and deep-learning approaches. However, several methodological and conceptual limitations reduce the reproducibility and interpretability of the review. The search strategy appears largely restricted to PubMed with title- and abstract-level filtering, while manual record removal is reported without explicit criteria defining “irrelevant” studies, limiting transparency and reproducibility. Protocol registration, duplicate independent screening, standardized extraction procedures, and formal bias assessment using established frameworks such as ROBIS, PROBAST+AI, and TRIPOD+AI were not clearly reported. The review reports performance metrics such as area under the receiver operating characteristic curve (AUROC), but does not provide a structured framework for interpreting or comparing metrics across heterogeneous datasets, prediction tasks, and evaluation protocols. Because the interpretation of AUROC and precision–recall metrics depends on class prevalence, outcome definition, and the intended prediction task, future reviews should report complementary discrimination metrics, calibration, uncertainty estimates, and external validation rather than assuming that any single metric is universally preferable. Claims of superior model performance should be supported by confidence intervals and statistical comparisons appropriate to the evaluation design, such as paired DeLong testing when applicable. Claims of superior model performance should also be supported by appropriate statistical testing, including methods such as the nonparametric DeLong test. Several conceptual clarifications are also warranted. AI models may prioritize hypotheses but do not replace experimental or clinical validation under current regulatory standards. Furthermore, AUROC should not be conflated with pharmacokinetic area under the curve, and SciBERT should not be characterized as a three-dimensional molecular graph framework. Future reviews should adopt transparent multi-database searches, structured bias assessment, and reproducible reporting practices. © The Author(s) 2026.

Date 2026

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: G. Mohammadnezhad; Department of Clinical Pharmacy, School of Pharmacy, Shahid Beheshti University of Medical Sciences, Tehran, Iran; email: awati19991378@gmail.com

Comparative analysis of the predictive capabilities of some machine learning models: A case study using wind speed data

Item Type Journal Article

Author K. Makubyane

Author C. Sigauke

Abstract The widespread use of fossil fuels for global energy production significantly contributes to global warming. This study presents a comparative analysis of various machine learning models, which are the long short-term memory (LSTM) network, support vector regression (SVR), and gradient boosting method (GBM). Gaussian process regression (GPR) is a benchmark model across different forecasting horizons. The study uses South African wind speed data from 1 January 2018 to 31 December 2021, sourced from the Western Cape province. The dataset underwent preprocessing, and diverse feature selection techniques were implemented to enhance model accuracy. Performance evaluation of the models was done using mean absolute error (MAE), root mean squared error (RMSE), and mean absolute scaled error (MASE). Results indicate that SVR exhibits superior accuracy to other models for two distinct forecast horizons ($h = 670$ and $h = 1339$), respectively. Additionally, GPR surpasses other models for the forecasting horizon $h = 224$. This study provides insights into the comparative strengths and weaknesses of different machine learning models for wind speed prediction, which could be useful in selecting an appropriate model for future applications in renewable energy and weather forecasting. Potential areas for future research include improving prediction accuracy via ensemble deep learning algorithms and incorporating additional meteorological variables. Moreover, investigating temporal dynamics, broadening geographical coverage and integrating uncertainty quantification methods can improve wind speed prediction, thereby facilitating more effective renewable energy planning and decision-making processes

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Renewable energy, LSTM, Predictive capability, Skill score, SVR

Notes:

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Comparative analysis of wastewater sample processing methods for antimicrobial resistance surveillance

Item Type Journal Article
Author H.F. Leal
Author A. Ouradou
Author N. Koukene
Author A. Barka
Author C. Quach
Author É. Bédard

Abstract Antimicrobial resistance monitoring in wastewater represents a key challenge, especially in light of the multiple methods available. Our study assessed the impact of selected sample processing techniques on microbial diversity and antimicrobial resistance gene recovery from wastewater. We analyzed 38 subsamples derived from 12 wastewater samples collected across three wastewater treatment plants over 4 months. We compared the effects of different centrifugation volumes (50, 250 mL), speeds (3,400 x g vs 15,000 x g), and pellet versus filter concentration methods, as well as the performance of two DNA extraction kits—DNeasy PowerSoil Pro Kit and DNeasy PowerLyzer Kit (Qiagen, DE)—on wastewater samples. Additionally, we compared the performance of two DNA quantification instruments, NanoDrop One and Invitrogen Qubit 4 (Thermo Scientific, US). The analysis involved 16S rRNA gene sequencing on an Illumina MiSeq platform (PE250, 10M reads) and quantitative PCR (qPCR) of 11 selected antimicrobial resistance genes (ARGs). Higher centrifugation speeds and larger sample volumes lead to improved DNA yield. Pellets generally exhibited higher (1.32 times) ARG recoveries compared to filters. DNA quantification methods showed strong positive correlation (Pearson R = 0.78, Spearman ρ = 0.81, P < 0.01) with each other and with qPCR data. The approach

implemented in this study (centrifuging 200 mL samples at 15,000 x g, followed by extraction with the DNeasy PowerSoil Pro Kit) shows promise for antimicrobial resistance (AMR) monitoring. IMPORTANCE To monitor the prevalence of antimicrobial resistance genes (ARGs) is crucial for anticipating and mitigating the spread of antimicrobial resistance (AMR). Urban wastewater provides community-level data to inform timely interventions. By establishing a method that balances efficiency and feasibility, this protocol represents a step toward a standardized wastewater surveillance (WS) framework for AMR, enhancing our ability to safeguard community health. Copyright © 2026 Leal et al. This is an open-access article distributed under the terms of the Creative Commons Attribution 4.0 International license.

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Article, comparative study, nonhuman, procedures, controlled study, classification, Bacteria, bacterial DNA, bacterium, DNA extraction, DNA, Bacterial, genetics, isolation and purification, polymerase chain reaction, microbiology, Anti-Bacterial Agents, antibiotic resistance, antiinfective agent, antimicrobial resistance, antimicrobial resistance gene, DNA quantification, drug effect, Drug Resistance, Bacterial, gene sequence, microbial diversity, nanodrop, quantum dot, Qubit, rain, RNA 16S, RNA, Ribosomal, 16S, sample concentration, sewage, specimen handling, Specimen Handling, waste water management, waste water treatment plant, wastewater, Wastewater, wastewater treatment plant, wastewater-based epidemiology, water

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Comparative Analysis Yolov5 and Yolov8 for Vehicle Classification

Item Type Conference Paper

Author M. Jain

Author R. Sharma

Editor Dwivedi R.K.

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Editor Bhardwaj S.

Editor Gupta R.

Abstract The research describes a system it use adaptive background reduction and object tracking to count vehicles on the highways and roads. Background subtraction that adapts, picture vehicle counting, segmentation, and vehicle tracking are its four primary phases of operation. Video streams from stationary cameras placed on roads and highways are used. This system's precise vehicle tracking and counting is meant to improve traffic surveillance. Its capacity to efficiently count automobiles and produce extremely accurate findings are among its key qualities. The study analyses traffic recordings for vehicle detection, classification, and counting using sort algorithms based on artificial intelligence. Vehicles in three classes-cars, trucks, buses-were identified and categorised with an average accuracy of using YOLOv5 and YOLOv8, a deep learning technique. After detection, the number of passing vehicles was counted using the SORT algorithm, which yielded an average accuracy rate of 95%. Based on the experimental results, it is possible to successfully recognise, classify, and count vehicles in flowing traffic films by utilising the YOLOv5 and YOLOv8 to SORT algorithms. © 2024 IEEE.

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Automobiles, Deep learning, Intelligent systems, % reductions, Image segmentation, Image coding, Comparative analyzes, Accuracy, Trucks, Video streaming, Adaptive backgrounds, SORT, Vehicle classification, Vehicle Classification, Vehicle Count, Vehicle counting, Vehicle counts, Yolov5, Yolov8

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Comparative assessment of statistical brain MR image segmentation algorithms and their impact on partial volume correction in PET

Item Type Journal Article

Author H. Zaidi

Author T. Ruest

Author F. Schoenahl

Author M.-L. Montandon

Abstract Magnetic resonance imaging (MRI)-guided partial volume effect correction (PVC) in brain positron emission tomography (PET) is now a well-established approach to compensate the large bias in the estimate of regional radioactivity concentration, especially for small structures. The accuracy of the algorithms developed so far is, however, largely dependent on the performance of segmentation methods partitioning MRI brain data into its main classes, namely gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF). A comparative evaluation of three brain MRI segmentation algorithms using simulated and clinical brain MR data was performed, and subsequently their impact on PVC in 18F-FDG and 18F-DOPA brain PET imaging was assessed. Two algorithms, the first is bundled in the Statistical Parametric Mapping (SPM2) package while the other is the Expectation Maximization Segmentation (EMS) algorithm, incorporate a priori probability images derived from MR images of a large number of subjects. The third, here referred to as the HBSA algorithm, is a histogram-based segmentation algorithm incorporating an Expectation Maximization approach to model a four-Gaussian mixture for both global and local histograms. Simulated under different combinations of noise and intensity non-uniformity, MR brain phantoms with known true volumes for the different brain classes were generated. The algorithms' performance was checked by calculating the kappa index assessing similarities with the "ground truth" as well as multiclass type I and type II errors including misclassification rates. The impact of image segmentation algorithms on PVC was then quantified using clinical data. The segmented tissues of patients' brain MRI were given as input to the region of interest (RoI)-based geometric transfer matrix (GTM) PVC algorithm, and quantitative comparisons were made. The results of digital MRI phantom studies suggest that the use of HBSA produces the best performance for WM classification. For GM classification, it is suggested to use the EMS. Segmentation performed on clinical MRI data show quite substantial differences, especially when lesions are present. For the particular case of PVC, SPM2 and EMS algorithms show very similar results and may be used interchangeably. The use of HBSA is not recommended for PVC. The partial volume corrected activities in some regions of the brain show quite large relative differences when performing paired analysis on 2 algorithms, implying a careful choice of the segmentation algorithm for GTM-based PVC. © 2006 Elsevier Inc. All rights reserved.

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Comparative Evaluation of 24 LDL-C Estimation Equations Against Direct Assays in Two Independent Cohorts

Item Type Journal Article
Author I. Györfi
Author O.R. Oprea
Author I.B. Mănescu
Author A. Curici
Author M. Dobreanu

Abstract Background: Low-density lipoprotein cholesterol (LDL-C) is essential in diagnosing and managing dyslipidemias. While direct assays are faster than the reference beta-quantification method, many labs continue using the Friedewald (FW) equation, despite its limitations. Methods: Two large datasets were analyzed: 10,174 hospital samples (Cobas/Roche) and 21,091 private lab samples (Alinity/Abbott). Various literature-based LDL-C equations were compared, focusing on FW, Sampson (SN), and Martin–Hopkins (MH). Direct LDL-C served as the reference. Evaluation metrics included bias and classification accuracy. Results: In samples with triglycerides < 400 mg/dL, several lesser-known equations showed acceptable bias ($\pm 5\%$), outperforming FW, SN, and MH, which had biases from -7.4% to -4.9% . Classification accuracy was higher with equations like Vujovic (up to 82.5%), compared to FW (65.8%), SN (73.1%), and MH (72.4%). The Vujovic equation showed minimal bias across triglyceride levels and the highest net gain in correct classification (3.4% and 1.57%). Conclusions: Multiple lesser-known LDL-C formulas outperformed widely used ones. The Vujovic equation yielded the best results, but the limited net clinical improvement suggests that replacing Friedewald may not be urgently necessary. © 2025 by the authors.

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Comparative performance of Chenomx and MagMet in NMR-based metabolite profiling with validation by liquid chromatography coupled with tandem mass spectrometry

Item Type Journal Article

Author W.A. Ali

Author A.A. Thomas

Author L. Belkacem

Author H.E. Mustafa

Author H. Alsafar

Author S.E. Richards

Abstract In this work, two computational approaches for metabolite quantification in serum samples using ¹H NMR spectroscopy were evaluated: the spectral matching method (MSM) implemented in MagMet and the non-linear least squares method (MNLLS) implemented in Chenomx. The comparison focused on their underlying methodologies, including deconvolution algorithms and user workflows, to assess their relative performance and suitability for metabolomics data analysis. As various analyses (e.g. pattern recognition, classification, biomarker discovery, and pathway analysis) rely on the precision and consistency of input features (e.g., metabolite concentrations), selecting a robust quantification method is essential. Variability in quantification can introduce noise and impact the stability and comparability of analytical outputs. To validate performance, MSM (MagMet) and MNLLS (Chenomx) were benchmarked against quantitative NMR (qNMR) and liquid chromatography–tandem mass spectrometry (LC-MS/MS), the latter serving as the primary reference due to its high sensitivity and broad metabolite coverage (Gika et al., 2014) [1]. Although LC–MS/MS may be affected by matrix effects and ion suppression; these factors are well characterized and routinely mitigated through isotope-labeled internal standards and validated analytical workflows. Moreover, LC–MS offers substantially higher sensitivity than NMR, typically by two to three orders of magnitude, enabling the detection and quantification of hundreds to thousands of metabolites within a single analysis (Nagana Gowda and Raftery, 2022) [2]. qNMR was included as a complementary technique to provide orthogonal validation rather than serving as the sole benchmark. Ten independent serum control samples from a healthy reference group were analyzed to account for natural biological variability, enhancing the generalizability of the findings. The comparison was structured around four criteria: (i) quantitative performance, (ii) computational stability, (iii) usability and processing time, and (iv) method-based similarity via partial least squares–discriminant analysis (PLS-DA). This work differs from prior studies by integrating statistical validation, repeatability testing, and practical usability assessment, and by benchmarking computational quantification pipelines against experimentally grounded methods such as qNMR and LC-MS/MS [3–5]. The selected approach is expected to demonstrate improved consistency in quantification relative to the alternative, contributing to more reliable biological interpretations and more reproducible analytical outcomes across datasets. © 2026 Elsevier B.V.

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Comparative Prevalence Estimation and Phylogenetic Analysis of Novel Porcine Parvoviruses (PPV2–7) in Hungarian Pig Herds

Item Type Journal Article
Author B. Igriczi
Author L. Dénes

Author K. Schönhardt

Author A. Woźniak

Author T. Stadejek

Author G. Balka

Abstract To date, seven novel parvoviruses have been identified in pigs and designated as porcine parvovirus 2–7 (PPV2–7). The presence of these emerging viruses has been reported in several countries around the world, although their pathogenic role and clinical and economical relevance are largely unknown. Here, we report the estimated prevalence and genetic diversity of novel PPV2–7 in Hungarian pig herds and the detection of these viruses in two Slovakian pig farms. For the comparative prevalence estimation, 2505 serum samples from different age groups, 218 oral fluid samples, and 111 processing fluid samples were collected from 26 large-scale Hungarian farms according to a systematic, cross-sectional sampling protocol. All samples were tested by real-time quantitative polymerase chain reaction (qPCR), and the presence of at least one PPV was detected in 24 of the 26 (92%) Hungarian and both Slovakian farms, suggesting high levels of subclinical circulation in most herds. The estimated PPV2–7 prevalence in Hungary varied from 50% to 89%, with PPV4 being the least and PPV2 being the most prevalent virus. The highest detection rates were observed in oral fluid samples, indicating that this sample type is most suitable for screening PPVs, but all viruses were also detected in serum samples and processing fluids. All novel PPVs were most frequently detected in the serum samples of weaned pigs and fatteners, with slightly higher viral burden in the younger age groups. These results may suggest an age-related susceptibility, which could play a significant role in the epidemiology of these viruses, impacting herd health and productivity. Copyright © 2024 Barbara Igriczi et al.

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Comparing Machine Learning and Binary Thresholding Methods for Quantification of Callose Deposits in the Citrus Phloem

Item Type Journal Article**Author** S. Welker**Author** A. Levy

Abstract Callose is a polysaccharide that can be fluorescently stained to study many developmental and immune functions in plants. High-throughput methods to accurately gather quantitative measurements of callose from confocal images are useful for many applications in plant biology. Previous callose quantification methods relied upon binary local thresholding, which had the disadvantage of not being able to differentiate callose in conditions with low contrast from background material. Here, a measurement approach that utilizes the Ilastik supervised machine learning imagery data collection software is described. The Ilastik software method provided superior efficiency for acquiring counts of callose deposits. We also determined the accuracy of these methods as compared to manual counts. We demonstrate that the automated software methods are both good predictors of manual counts, but that the Ilastik counts are significantly closer. Researchers can use this information to guide their choice of method to quantify callose in their work. © 2022 by the authors. Licensee MDPI, Basel, Switzerland.

Date 2022**DOI** 10.3390/plants11050624**URL** <https://www.scopus.com/pages/publications/85125252042?origin=resultslist>**Archive** Scopus**Language** English**Volume** 11**Publisher** MDPI**Publication** Plants**ISSN** 22237747 (ISSN)**Issue** 5**Journal Abbr** Plants**Date Added** 20/09/2026, 9:52:37 pm**Modified** 20/09/2026, 9:52:37 pm**Tags:**

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Comparing the quality of neural network uncertainty estimates for classification problems

Item Type Conference Paper**Author** D. Ries**Author** J. Michalenko**Author** T. Ganter**Author** R.I.-F. Baiyasi**Author** J. Adams**Editor** Wani M.A.**Editor** Kantardzic M.**Editor** Palade V.**Editor** Neagu D.**Editor** Yang L.**Editor** Chan K.-Y.

Abstract Traditional deep learning (DL) models are powerful classifiers, but many approaches do not provide uncertainties for their estimates. Uncertainty quantification (UQ) methods for DL models have received increased attention in the literature due to their usefulness in decision making, particularly for high-consequence decisions. However, there has been little research done on how to evaluate the quality of such methods. We use statistical methods of frequentist interval coverage and interval width to evaluate the quality of credible intervals, and expected calibration error to evaluate classification predicted confidence. These metrics are evaluated on Bayesian neural networks (BNN) fit using Markov Chain Monte Carlo (MCMC) and variational inference (VI), bootstrapped neural networks (NN), Deep Ensembles (DE), and Monte Carlo (MC) dropout. We apply these different UQ for DL methods to a hyperspectral image target detection problem and show the inconsistency of the different methods' results and the necessity of a UQ quality metric. To reconcile these differences and choose a UQ method that appropriately quantifies the uncertainty, we create a simulated data set with fully parameterized probability distribution for a two-class classification problem. The gold standard MCMC performs the best overall, and the bootstrapped NN is a close second, requiring the same computational expense as DE. Through this comparison, we demonstrate that, for a given data set, different models can produce uncertainty estimates of markedly different quality. This in turn points to a great need for principled assessment methods of UQ quality in DL applications. © 2022 IEEE.

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Deep learning, Classification (of information), Decision making, Uncertainty analysis, Neural-networks, Learning models, Neural networks, Uncertainty, Target tracking, deep learning, Bayesian networks, Probability distributions, Quality control, Monte Carlo methods, Quantification methods, Uncertainty quantifications, Hyperspectral imaging, Uncertainty estimates, Bayesian neural networks, Bayesian neural network, Deep ensemble, Deep Ensembles, un-certainty quantification, Un-certainty quantification

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Comparing three body mass index classification systems to assess overweight and obesity in children and adolescents

Item Type Journal Article
Author I. Gonzalez-Casanova
Author O.L. Sarmiento
Author J.A. Gazmararian
Author S.A. Cunningham
Author R. Martorell
Author M. Pratt

Author A.D. Stein

Abstract Objective. To compare the International Obesity Task Force (IOTF) 2005, Centers for Disease Control and Prevention (CDC) 2000, and World Health Organization (WHO) 2007 body mass index (BMI) classification systems in terms of prevalence estimation and association with demographic factors. Methods. The 18 265 children and adolescents ages 5 to 18 years (mean = 11.2 years, standard deviation = 3.9 years) in the nationally representative Colombian National Nutrition Survey of 2005 were classified as overweight or obese according to IOTF, CDC, and WHO criteria. Prevalence estimates were compared according to each system and associations with age, sex, socioeconomic status, and population density were tested. Results. Prevalence estimates of combined overweight and obesity differed by system (males: IOTF = 8.5%, CDC = 10.8%, WHO = 14.1%; females: IOTF = 14.6%, CDC = 13.8%, WHO = 17.1%; $P < 0.001$). The association between combined overweight and obesity and age and sex varied by system. The odds of having overweight and obesity in children (5 to 10 years) compared with adolescents (11 to 18 years) were: IOTF, odds ratio (OR) = 0.87 and 95% confidence interval (CI) = 0.77-0.98; CDC, OR = 1.27 and CI = 1.14-1.42; WHO, OR = 1.21 and CI = 1.08-1.35. The values for females compared with males were: IOTF, OR = 1.84 and CI = 1.6-2.10; CDC, OR = 1.33 and CI = 1.17-1.51; WHO, OR = 1.25 and CI = 1.12-1.41. Conclusions. There is a lack of consistency among the three main international systems in assessing overweight and obesity in children and adolescents. Appreciably different estimates of prevalence and associations with age and sex are obtained depending on which system is used. Future studies should assess how well each system reflects valid measures of body composition.

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Comparison of conventional and alternative determination methods for copper, iron, manganese and zinc in mehlich-1 extracted soil solution

Item Type Journal Article

Author R. Soares

Author V. Escaleira

Author D.V.B. de Campos

Author M.I.C. Monteiro

Author R.E. Santelli

Author A.C.C. Bernardi

Abstract In Brazilian laboratories, the determination of micronutrient contents for soil fertility analyses is based on extraction with Mehlich-1 solution (M-1), at a 1:5 w/v soil:extractant ratio, with subsequent quantification by atomic absorption spectrometry (AAS). However, many laboratories also use a 1:10 w/v soil:extractant M-1 ratio for the determination of macronutrients, and/or quantification by optical emission spectrometry with inductively coupled plasma (ICP OES), instead of AAS. As comparative studies between the micronutrient concentrations obtained by these alternative methods are scarce, the aim of this work was to investigate significant differences in results from these different extraction and quantification methods. In this work, statistical analysis showed that the Fe and Mn concentrations using both soil:extractant M-1 ratios (1:5 and 1:10) and Zn concentrations at 1:10 w/v soil:extractant M-1 ratio were similar regardless the quantification equipment. However, the Cu concentrations using both soil:extractant M-1 ratios and the Zn concentrations at 1:5 w/v soil:extractant M-1 ratio differed according to the equipment used to quantify these nutrients. Also, all Cu, Fe, Mn and Zn concentrations determined at 1:5 w/v differed from those at 1:10 w/v soil:extractant M-1 ratio, regardless of the quantification technique. Most of these concentrations were higher at a ratio of 1:5 w/v and affected the classification of the contents. It is therefore not recommended to use a soil:extractant ratio of 1:5 for the extraction of the micronutrients Cu, Fe, Mn and Zn from soil samples.

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Short Title Comparação de métodos convencional e alternativo para determinar Cu, Fe, Mn e Zn Em extratos de solos obtidos com solução de mehlich-1

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Comparison of current state of control valve stiction detection and quantification techniques

Item Type Journal Article**Author** V. Akavalappil**Author** T.K. Radhakrishnan

Abstract The process control and automation system is one of the significant features in any process plant. The control valve is considered a fundamental part of the process control system. It regulates the fluid flow by changing the size of the flow passage as directed by the controller. If properly maintained, the control valves can help to reduce process variability and improve product quality, which in turn increases the overall efficiency of the plant. But control valve stiction is an enduring concern within process industries which eventually affects the efficiency of plant operation. Hence, it is extremely essential to detect and quantify stiction at the earliest opportunity, to determine the correct maintenance measures. Numerous methods related to stiction detection and estimation have been carried out so far by researchers. A literature study of the latest publications in stiction detection and quantification discloses that research in the conventional methods of stiction detection and quantification such as limit cycle pattern-based, waveform shape-based, nonlinearity-based, and model-based methods is gradually declining. At the same time, most stiction detection/quantification methods reported in the recent literature are based on artificial neural network and convolutional neural network. This implies that future stiction detection and quantification research will be focused towards machine learning (ML) algorithms. Taking this into consideration, this paper aims to provide a literature review of stiction detection and quantification methods by focusing on the latest research ideas and innovative approaches. This review also aims to compare recent techniques based on ML algorithms with conventional methods by pointing out the relationship among different methods, differences, and possible points of weakness. © The Author(s) 2021.

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Export Date: 20 September 2026; Cited By: 18; Correspondence Address: T.K. Radhakrishnan; Department of Chemical Engineering, National Institute of Technology, Trichy, India; email: radha@nitt.edu; CODEN: TICOD

Comparison of different approaches for estimation of prevalence of type 2 diabetes mellitus in the Czech Republic

Item Type Journal Article

Author P.H.M. S

Author M. Škodová

Author P. Kučová

Author J. Votápková

Abstract Prevalence of type 2 diabetes in the Czech Republic can be estimated from data reported by health care providers either to the statistical office or to health insurers. The latter include both diagnosis and prescribed drugs. Patient classification to a certain chronic condition based on consumption of drugs (Pharmacy-based Cost Groups classification) has been used in the Czech Republic for the purpose of redistribution of collected funds among health insurers since the beginning of 2018. This article compares prevalence of diabetes estimated from statistical data and from data reported to health insurers, and discusses current and future advantages and disadvantages of all methodologies. Diagnoses reporting in outpatient care are considered to be inaccurate, on the other hand peroral antidiabetics do not seem to be prescribed to all indicated patients. The appropriate approach to prevalence

estimation from data reported to health insurers is becoming even more important after the creation of National Diabetes Register based on insurance data and the related abatement of direct reporting to the statistical office.

Date 2020

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Short Title Srovnání různých přístupů k určení prevalence diabetu 2. typu v ČR

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Comparison of different criteria for rheumatic heart disease screening: an empirical study in Sierra Leone

Item Type Journal Article

Author B. Fernández-de-Toro

Author L. Cobarro

Author J. Pérez-Martín

Author F.J. Díez

Abstract Background: Rheumatic heart disease (RHD) affects approximately 40 million people, primarily in low- and middle-income countries. Transthoracic echocardiography is the gold standard for early detection of RHD. Yet, no single echocardiographic finding or combination of findings unequivocally confirms RHD, especially in subclinical stages. As a result, multiple diagnostic guidelines have been proposed. Objectives: To characterise the echocardiographic RHD findings and criteria included in five of those guidelines; to assess how these diagnostic criteria differ when applied to a concrete population, namely a cohort of high-school students

in Sierra Leone; and, as a secondary objective, to provide an initial estimate of the burden of RHD in this country. **Methods:** Our study has two parts. First, we performed a qualitative comparison of five major echocardiographic guidelines, mapping every diagnostic criterion into four roles (required, sufficient, combined, or not considered) for each guideline. Second, we conducted a school-based screening in Sierra Leone. A cohort of 604 asymptomatic female students (aged 10–22 years) underwent a two-phase echocardiographic protocol. We quantified regurgitant jet characteristics and morphological signs according to the criteria of the compared guidelines. **Results:** Guideline discrepancies were substantial, particularly in jet length thresholds and the diagnostic role of morphological features. These differences shifted individual classifications across categories. In our cohort, 13 students (2.2%) had findings beyond normal under at least one guideline, and positive cases ranged from 0.99% to 2.15% depending on the guideline. Using the World Heart Federation (WHF) 2023 criteria, 1.16% (95% CI: 0.47–2.37) were classified as positive. **Conclusions:** Variability in echocardiographic criteria influences RHD classification and prevalence estimation. Our empirical study shows that the use of rigid echocardiographic cut-offs may lead to discarding some cases suspicious of RHD, especially in the WHF 2023 screening guideline, because one unfulfilled criterion automatically leads to a negative diagnosis, regardless of the number and relevance of positive findings. We argue that a probabilistic causal model that ponders the numerical measurements and considers correlations among findings would be more appropriate for screening and diagnosis. © The Author(s) 2026.

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Comparison of inverse uncertainty quantification methods for critical flow test

Item Type Journal Article

Author P. Domitr

Author M. Włostowski

Author R. Laskowski

Author R. Jurkowski

Abstract A problem of epistemic uncertainties introduced by code input parameters that cannot be estimated other than by user expertise has existed in nuclear reactor systems analyses since first uncertainty analyses were conducted. Inverse Uncertainty Quantification (IUQ) methods aim at providing an estimation of the distributions of such parameters. This paper compares and assesses two approaches to quantify the uncertainty of critical flow model parameters available in the TRACE code. First novel approach is based on a machine-learning algorithm, using a Random Forest classifier to assign the results of calculations to one of the defined classes of prediction accuracy with respect to experimental data. The second approach is based on Markov Chain Monte Carlo sampling and Bayesian inference. Both methods allows to assign probability distribution functions to TRACE internal variables, which is a goal for IUQ methods. Both approaches were used to estimate uncertainties for four critical flow parameters available in the TRACE code based on the Marviken critical flow experiment. In case of two parameters both methods produced similar results of the probability density functions. Based on the results of machine learning two additional parameters were deemed noninfluential, while for the second method such definitive assessment could not be made. © 2022 Elsevier Ltd

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Notes:

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Export Date: 20 September 2026; Cited By: 9; Correspondence Address: P. Domitr; Warsaw University of Technology, Institute of Heat Engineering, Warsaw, Nowowiejska 21/25, 00-665, Poland; email: pawel.domitr.dokt@pw.edu.pl; CODEN: ENEYD

Comparison of machine learning and semi-quantification algorithms for (I123)FP-CIT classification: the beginning of the end for semi-quantification?

Item Type Journal Article

Author J.C. Taylor

Author J.W. Fenner

Abstract Background: Semi-quantification methods are well established in the clinic for assisted reporting of (I123) Ioflupane images. Arguably, these are limited diagnostic tools. Recent research has demonstrated the potential for improved classification performance offered by machine learning algorithms. A direct comparison between methods is required to establish whether a move towards widespread clinical adoption of machine learning algorithms is justified. This study compared three machine learning algorithms with that of a range of semi-quantification methods, using the Parkinson's Progression Markers Initiative (PPMI) research database and a locally derived clinical database for validation. Machine learning algorithms were based on support vector machine classifiers with three different sets of features: Voxel intensities Principal components of image voxel intensities Striatal binding ratios from the putamen and caudate. Semi-quantification methods were based on striatal binding ratios (SBRs) from both putamina, with and without consideration of the caudates. Normal limits for the SBRs were defined through four different methods: Minimum of age-matched controls Mean minus 1/1.5/2 standard deviations from age-matched controls Linear regression of normal patient data against age (minus 1/1.5/2 standard errors) Selection of the optimum operating point on the receiver operator characteristic curve from normal and abnormal training data Each machine learning and semi-quantification technique was evaluated with stratified, nested 10-fold cross-validation, repeated 10 times. Results: The mean accuracy of the semi-quantitative methods for classification of local data into Parkinsonian and non-Parkinsonian groups varied from 0.78 to 0.87, contrasting with 0.89 to 0.95 for classifying PPMI data into healthy controls and Parkinson's disease groups. The machine learning algorithms gave mean accuracies between 0.88 to 0.92 and 0.95 to 0.97 for local and PPMI data respectively. Conclusions: Classification performance was lower for the local database than the research database for both semi-quantitative and machine

learning algorithms. However, for both databases, the machine learning methods generated equal or higher mean accuracies (with lower variance) than any of the semi-quantification approaches. The gain in performance from using machine learning algorithms as compared to semi-quantification was relatively small and may be insufficient, when considered in isolation, to offer significant advantages in the clinical context. © 2017, The Author(s).

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Notes:

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Comparison of Pakistani CKD-EPI, new Asian-modified CKD-EPI and revised Lund–Malmö study equations in a South Asian CKD population: a study from a Pakistani CKD cohort

Item Type Journal Article

Author A. Safdar

Author W. Akram

Author M.A. Khan

Author M.N. Alvi

Abstract Background: Newly proposed estimating glomerular filtration rate equations need to be studied, evaluated and compared for chronic kidney disease staging, diagnosis and medication dosing in South Asians. The objectives of the study were (1) to assess the performance of the CKD-EPIPK, CKD-EPIAsian-Modified, and LMRevised equations in the Pakistani chronic kidney disease population, and (2) to investigate prospective implications on chronic kidney disease classification and end-stage kidney disease prevalence. Methods: We conducted a cross-sectional analysis on a chronic kidney disease cohort of 385 participants 18 years of age or above. Results: CKD-EPIPK showed the lowest bias (-1.33 ml/min/ 1.73 m 2), highest precision [IQR, 2.33 (-2.36 , -0.03)] and enhanced P30 accuracy (89.35%) compared to the CKD-EPIAsian-Modified and LMRevised equations. The mean difference (ml/min/ 1.73 m 2), 95% limit of agreement (ml/min/ 1.73 m 2) of the equations were; CKD-EPIAsian-Modified: -5.98 , -13.03 , LMRevised: -4.06 , -8.13 and CKD-EPIPK: -1.18 , -6.14 ($P < 0.001$). CKD-EPIAsian-Modified and LMRevised showed upward re-classification of the GFR categories compared to the CKD-EPIPK equation except in the G5 category where the highest count (217, 56.36%) was noted for the CKD-EPIPK equation. End-stage kidney disease prevailed in all age groups according to all equations, and the prevalence was high in females in all equations. Conclusion: CKD-EPIPK showed the best performance, whereas both CKD-EPIAsian-Modified and LMRevised showed poor performance and did not offer a sufficient advantage in chronic kidney disease classification and end-stage kidney disease prevalence estimation over CKD-EPIPK. Hence, CKD-EPIPK seems ideal for South Asians, thus appropriate measures should be taken for its implementation, at least in Pakistani laboratories. Graphical abstract: (Figure presented.) © The Author(s) under exclusive licence to Italian Society of Nephrology 2023.

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: A. Safdar; Faculty of Pharmaceutical Sciences, University of Central Punjab, Lahore, Avenue 1, Khayaban-e-Jinnah Road, Johar Town, Punjab, 54000, Pakistan; email: 11f20mpch0005@ucp.edu.pk; CODEN: JLNEE

Comparison of rapid eye movement without atonia quantification methods to diagnose rapid eye movement sleep behavior disorder: a systematic review

Item Type Journal Article

Author J.-I. Byun

Author T.-W. Yang

Author J.-S. Sunwoo

Author W.C. Shin

Author O.-Y. Kwon

Author K.-Y. Jung

Abstract Study Objectives: Rapid eye movement (REM) sleep without atonia (RWA) is essential for diagnosing REM sleep behavior disorder (RBD). Manual and automatic quantifications of RWA that use different criteria have been validated. This study compared the RWA quantification methods for diagnosing RBD. Methods: The PubMed, EMBASE, Web of Science, and Cochrane Library databases were systemically searched for studies published from inception to December 2021. The inclusion criteria were cohort, cross-sectional, and case-control studies assessing the sensitivity and specificity of RWA quantification methods. Pooled estimates of the sensitivity, specificity, diagnostic odds ratio (DOR), and area under the curve (AUC) were determined. Risk of bias and certainty of evidence was assessed using the Quality Assessment of Diagnostic Accuracy Studies tool and the Grading of Recommendations, Assessment, Development, and Evaluations framework, respectively. Results: Fourteen articles including 402 patients with RBD met the inclusion criteria. Manual methods evaluating any chin and phasic flexor digitorum superficialis (FDS) activity had the highest DOR (138.8, 95% CI = 21.8% to 881.7%) and AUC (0.9686). The automatic REM atonia index (RAI) showed similar or higher sensitivity (89.1%, 95% CI = 84.6% to 92.7%) but a lower specificity (73.5%), DOR (43.1), and AUC (0.9369) than the manual techniques. Conclusions: In this meta-

analysis, manual RWA quantification that employed chin or phasic FDS activity had the best RBD diagnostic performance. The automatic RAI method may be useful for screening patients with RBD. The results should be interpreted carefully because of the high risk of bias in patient selection and significant heterogeneity among the studies. PROSPERO Registration number: CRD42021276445. © 2022 The Author(s).

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Comparison of the performance of partial least squares and support vector regressions for predicting fatty acids and fatty acid classes in marine oil dietary supplements by using vibrational spectroscopic data

Item Type Journal Article

Author S.R. Karunathilaka

Author B.J. Yakes

Author S.H. Choi

Author L. Brückner

Author M.M. Mossoba

Abstract Simple, fast, and accurate analytical techniques for verifying the accuracy of label declarations for marine oil dietary supplements containing eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) are required because of the increased consumption of these products. We recently developed broad-based partial least squares regression (PLS-R) models to quantify six fatty acids (FAs) and FA classes by using the spectroscopic data from a portable Fourier transform infrared (FTIR) device and a benchtop Fourier transform near infrared (FT-NIR) spectrometer. We developed an improved quantification method for these FAs and FA classes by incorporating a nonlinear calibration approach based on the machine learning technique support vector machines. For the two spectroscopic methods, high accuracy in prediction was indicated by low root mean square error of prediction and by correlation coefficients (R^2) close to 1, indicating excellent model performance. The percent accuracy of the support vector regression (SV-R) model predicted values for EPA and DHA in the reference material was 90 to 110%. In comparison to PLS-R, SV-R accuracy for prediction of FA and FA class concentrations was up to 2.4 times higher for both ATRFTIR and FT-NIR spectroscopic data. The SV-R models also provided closer agreement with the certified and reference values for the prediction of EPA and DHA in the reference standard. Based on our findings, the SV-R methods had superior accuracy and predictive quality for predicting the FA concentrations in marine oil dietary supplements. The combination of SV-R with ATR-FTIR and/or FT-NIR spectroscopic data can potentially be applied for the rapid screening of marine oil products to verify the accuracy of label declarations. © 2020 International Association for Food Protection. All rights reserved.

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Comparison of Total Kidney Volume Quantification Methods in Autosomal Dominant Polycystic Disease for a Comprehensive Disease Assessment

Item Type Journal Article

Author D. Turco

Author M. Busutti

Author R. Mignani

Author R. Magistroni

Author C. Corsi

Abstract Background: In recent times, the scientific community has been showing increasing interest in the treatments aimed at slowing the progression of the autosomal dominant polycystic kidney disease (ADPKD). Therefore, in this paper, we test and evaluate the performance of several available methods for total kidney volume (TKV) computation in ADPKD patients - from echography to MRI - in order to optimize patient classification. Methods: Two methods based on geometric assumptions (mid-slice [MS], ellipsoid [EL]) and a third one on true contour detection were tested on 40 ADPKD patients at different disease stage using MRI. The EL method was also tested using ultrasound images in a subset of 14 patients. Their performance was compared against TKVs derived from reference manual segmentation of MR images. Patient clinical classification was also performed based on computed volumes. Results: Kidney volumes derived from echography significantly underestimated reference volumes. Geometric-based methods applied to MR images had similar acceptable results. The highly automated method showed better performance. Volume assessment was accurate and reproducible. Importantly, classification resulted in 79, 13, 10, and 2.5% of misclassification using kidney volumes obtained from echo and MRI applying the EL, the MS and the highly automated method respectively. Conclusion: Considering the fact that the image-based technique is the only approach providing a 3D patient-specific kidney model and allowing further analysis including

cyst volume computation and monitoring disease progression, we suggest that geometric assumption (e.g., EL method) should be avoided. The contour-detection approach should be used for a reproducible and precise morphologic classification of the renal volume of ADPKD patients. © 2017 S. Karger AG, Basel. Copyright: All rights reserved.

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Comparison of two-class and three-class Bayesian artificial neural networks in estimation of observations drawn from simulated bivariate normal distributions

Item Type Conference Paper

Author N. Bhooshan

Author D. Edwards

Author M. Giger

Abstract The development and application of multi-class BANN classifiers in computer-aided diagnosis methods motivated this study in which we compared estimates produced by two-class and three-class BANN classifiers to true observations drawn from simulated distributions. Observations were drawn from three Gaussian bivariate distributions with distinct means and variances to generate G1, G2, and G3 simulated datasets. A two-class BANN was trained on each training dataset for a total of ten different trained BANNs. The same testing dataset was run on each trained BANN. The average and standard deviation of the resulting ten sets of BANN outputs were then calculated. This process was repeated with three-class BANNs. Different sample numbers and values of a priori probabilities were investigated. The relationship between the average BANN output and true distribution was measured using Pearson and Spearman coefficients, R-squared and mean square error for two-class and three-class BANNs. There was significantly high correlation between the average BANN output and true distribution for two-class and three-class BANNs; however, subtle non-linearities and spread were found in comparing the true and estimated distributions. The standard deviations of two-class and three-class BANNs were comparable, demonstrating that three-class BANNs can perform as reliably as two-class BANN classifiers in estimating true distributions and that the observed non-linearities and spread were not simply due to statistical uncertainty but were valid characteristics of the BANN classifiers. In summary, three-class BANN decision variables were similar in performance to those of two-class BANNs in estimating true observations drawn from simulated bivariate normal distributions. © 2011 SPIE.

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Comparison of WHO, CDC and IOTF growth references in relation to overweight and obesity in school going adolescents of Delhi, India

Item Type Journal Article**Author** B. Deepika**Author** S. Kalyani**Author** S. Manisha

Abstract Background and Objective: Obesity has become a serious public health problem. Accurate measurement of this health issue is crucial in defining its burden. Thus the objective of the study was to compare the body mass index (BMI) classification systems for World Health Organization (WHO), Centers for Disease Control and Prevention (CDC) and International Obesity Task Force (IOTF) prevalence estimation of overweight and obesity and in terms of level of agreement between the classification systems. Method: In a cross sectional survey, 877 school going adolescents of age 12 to 15 years (mean = 13.04 years) were included in the study. Height and weight of subjects were measured to calculate the BMI. Subjects were classified as overweight and obese according to IOTF, CDC, and WHO criteria and Kappa statistics was used to calculate the level of agreement between the three classification systems. Results: Prevalence estimates of combined overweight and obesity differed by system (males WHO = 33.16%, CDC=29.78%, IOTF=28.55%; females WHO = 33.54%, CDC=29.71%, IOTF = 29.39%). A substantial agreement was observed between CDC vs IOTF and WHO vs CDC (Kappa=0.82) and a moderate agreement between WHO vs IOTF (kappa=0.70). Conclusion: IOTF is a frequently used system for BMI classification of children and adolescents in India but it reports the lowest combined prevalence in comparison to CDC and WHO. Although the agreement between CDC vs IOTF and WHO vs CDC is substantial but the system which is most suitable to a country like India is WHO, 2007 since the sample includes children from India among other countries. © 2016, Indian Journal of Public Health Research and Development. All rights reserved.

Date 2016**DOI** 10.5958/0976-5506.2016.00134.0**URL** <https://www.scopus.com/pages/publications/84977137941?origin=resultslist>**Archive** Scopus**Language** English**Volume** 7**Publisher** Institute of Medico-Legal Publications**Pages** 83-90**Publication** Indian Journal of Public Health Research and Development**ISSN** 09760245 (ISSN)**Issue** 3

Journal Abbr Indian J. Public Health Res. Dev.

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Tags:

Article, female, human, male, adolescent, comparative study, controlled study, age, major clinical study, prevalence, Adolescents, Obesity, child, body weight, public health service, disease classification, sex difference, body mass, sex ratio, school child, India, body height, reference value, adolescent obesity, Body mass index (BMI), child nutrition, growth reference, Growth reference, health care organization, International Obesity Task Force, Overweight, world health organization

Notes:

EID: 2-s2.0-84977137941

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: B. Deepika; Lady Irwin College, University of Delhi, 110001, India; email: bahl.deepa@gmail.com

Complete group-type quantification of petroleum middle distillates based on comprehensive two-dimensional gas chromatography time-of-flight mass spectrometry (GCxGC-TOFMS) and visual basic scripting

Item Type Journal Article

Author M.K. Jennerwein

Author M. Eschner

Author T. Gröger

Author T. Wilharm

Author R. Zimmermann

Abstract The subject of the presented work was the development of a two-dimensional GCxGC - time-of-flight mass spectrometric method (GCxGC-TOFMS) for the complete group-type quantification of petroleum middle distillates. The development of this method was possible due to the inherent features of GCxGC-TOFMS, namely the structured arrangement of compound groups and the mass fragmentation pattern, which provide the possibility of using Visual Basic Scripts as an analytical tool and thereby the classification of several thousand different compounds. The analysis method was focused on common petroleum based fuels from light to heavier middle distillation fractions. For the implementation of an absolute quantification method, a set of standard substances representing the main substance classes and carbon numbers within middle distillates and well-separated and recognizable internal standards were thoroughly chosen in order to obtain individual response factors and group specific response curves. The results of the qualitative and quantitative analysis were compared to well-established standard methods used in the petrochemical industry. The quantification of aromatic hydrocarbons was compared to EN 12916, a high performance liquid chromatography (HPLC) method that provides a rough separation between mono-, di-, and higher aromatics. Further, the quantification of

the fatty acid methyl ester (FAME) content in diesel fuel was compared to EN 14078 and the distribution of single FAMES was compared to EN 14103. It could be stated that an absolute quantification as the here presented method has not been reported before and the results were in good agreement with the reference methods. Furthermore, the here presented GCxGC-TOFMS quantification method is able to itemize according substance classes and carbon number. The detection limit of the method allows accurate and sensitive quantification for different limiting values of middle distillates with a single method. © 2014 American Chemical Society.

Date 2014

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Archive Scopus

Language English

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Distillation, Quantification methods, High performance liquid chromatography, Carbon, Diesel fuels, Mass spectrometry, Spectrometric methods, Absolute quantification, Fatty acids, Gas chromatography, Aromatic hydrocarbons, Comprehensive two-dimensional gas chromatography-time of flight mass spectrometries, Distillation fractions, Fatty acid methyl ester, Petrochemical industry, Petroleum distillates, Qualitative and quantitative analysis, Visual BASIC

Notes:

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Complexity Evaluation of Test Scenarios for Autonomous Vehicle Safety Validation Using Information Theory

Item Type Journal Article

Author M. Issler

Author Q. Goss

Author M.İ. Akbaş

Abstract The validation of autonomous vehicles remains a vexing challenge for the automotive industry's goal of fully autonomous driving. The systematic hierarchization of the test scenarios would provide valuable insights for the development, testing, and verification of autonomous vehicles, enabling nuanced performance evaluations based on scenario complexity. In this paper, an information entropy-based quantification method is proposed to evaluate the complexity of autonomous vehicle validation scenarios. The proposed method addresses the dynamic uncertainties within driving scenarios in a comprehensive way which includes the unpredictability of dynamic agents such as autonomous vehicles, human-driven vehicles, and pedestrians. The numerical complexity calculation of the approach and the ranking of the scenarios are presented through sample scenarios. To automate processes and assist with the calculations, a novel software tool with a user-friendly interface is developed. The performance of the approach is also evaluated through six example driving scenarios, then through extensive simulation using an open-source microscopic traffic simulator. The performance evaluation results confirm the numerical classification and demonstrate the method's adaptability to diverse scenarios with a comparison of complexity calculation ranking to the ratio of collision, near collision, and normal operation tests observed during simulation testing. The proposed quantification method contributes to the improvement of autonomous vehicle validation procedures by addressing the multifaceted nature of scenario complexities. Beyond advancing the field of validation, the approach also aligns with the broad and active drive of the industry for the widespread deployment of fully autonomous driving. © 2024 by the authors.

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DOI 10.3390/info15120772

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Autonomous vehicles, Autonomous Vehicles, autonomous vehicle, Autonomous driving, scenario-based testing, Scenario-based testing, Test scenario, Automotive industry, entropy, Validation, Automobile testing, Automobile driver simulators, Quantification methods, Performances evaluation, Complexity calculation, Evaluation of test, information theory, scenario complexity, Scenario complexity, validation

Notes:

EID: 2-s2.0-85213079590

Export Date: 20 September 2026; Cited By: 7; Correspondence Address: M.İ. Akbaş; Department of Electrical Engineering and Computer Science, Embry-Riddle Aeronautical University, Daytona Beach, 32119, United States; email: akbasmi@erau.edu

Composition, Abundance, and Diversity of the Soil Microbiome Associated with the Halophytic Plants *Tamarix aphylla* and *Halopeplis perfoliata* on Jeddah Seacoast, Saudi Arabia

Item Type Journal Article

Author N.N. Baeshen

Author L. Baz

Author A.Y. Shami

Author R.A. Ashy

Author R.S. Jalal

Author A.A. Abulfaraj

Author M. Refai

Author M.A. Majeed

Author S.S. Abuzahrah

Author H. Abdelkader

Author N.A. Baeshen

Author M.N. Baeshen

Abstract The coast of the Red Sea in Jeddah City is home to a unique microbial community that has adapted to extreme environmental conditions. Therefore, it is essential to characterize the microbial community in this unique microbiome to predict how environmental changes will affect it. The aim of this study was to conduct metagenomic sequencing of 16S rRNA and ITS rRNA genes for the taxonomic classification of the microbial community in soil samples associated with the halophytic plants *Tamarix aphylla* and *Halopeplis perfoliata*. Fifteen soil samples were collected in triplicate to enhance robustness and minimize sampling bias. Firstly, to identify novel microbial candidates, the gDNAs were isolated from the saline soil samples surrounding each plant, and then bacterial 16S (V3–V4) and fungal ITS1 regions were sequenced utilizing a high-throughput approach (next-generation sequencing; NGS) on an Illumina MiSeq platform. Quality assessment of the constructed amplicon libraries was conducted using Agilent Bioanalyzer and fluorometric quantification methods. The raw data were processed and analyzed using the Pipeline (Nova Lifetech, Singapore) for bioinformatics analysis. Based on the total number of readings, it was determined that the phylum Actinobacteriota was the most prevalent in the soil samples examined, followed by the phylum Proteobacteria. Based on ITS rRNA gene analysis, the alpha and beta fungal diversity in the studied soil samples revealed that the fungal population is structured into various groups according to the crust (c) and/or rhizosphere (r) plant parts. Fungal communities in the soil samples indicated that Ascomycota and Basidiomycota were the two most abundant phyla based on the total amount of sequence reads. Secondly, heat-map analysis of the diversity indices showed that the bacterial alpha diversity, as measured by Shannon, Simpson, and InvSimpson, was associated with soil crust (Hc and Tc

enclosing *H. perfoliata* and *T. aphylla*, respectively) and that the soil rhizosphere (Hr and Tr) was strongly correlated with bacterial beta diversity. Finally, fungal-associated Tc and Hc samples clustered together, according to observations made using the Fisher and Chao1 methods, and Hr and Tr samples clustered together according to Shannon, Simpson, and InvSimpson analyses. As a result of the soil investigation, potential agents that have been identified could lead to innovative agricultural, medical, and industrial applications. © 2023 by the authors.

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Journal Abbr Plants

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Tags:

16S rRNA gene, alpha diversity, beta diversity, *Haloepelis perfoliata*, ITS rRNA gene, microbial communities, *Tamarix aphylla*

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Comprehensive marine substrate classification applied to Canada's Pacific shelf

Item Type Journal Article

Author E.J. Gegr

Author D.R. Haggarty

Author S.C. Davies

Author C. Fields

Author J. Lessard

Abstract Maps of bottom type are essential to the management of marine resources and biodiversity because of their foundational role in characterizing species' habitats. They are also urgently needed as countries work to define marine protected areas.

Current approaches are time consuming, focus largely on grain size, and tend to overlook shallow waters. Our random forest classification of almost 200,000 observations of bottom type is a timely alternative, providing maps of coastal substrate at a combination of resolution and extents not previously achieved. We correlated the observations with depth, depth-derivatives, and estimates of energy to predict marine substrate at 100 m resolution for Canada's Pacific shelf, a study area of over 135,000 km². We built five regional models with the same data at 20 m resolution. In addition to standard tests of model fit, we used three independent data sets to test model predictions. We also tested for regional, depth, and resolution effects. We guided our analysis by asking: 1) does weighting for prevalence improve model predictions? 2) does model resolution influence model performance? And 3) is model performance influenced by depth? All our models fit the build data well with true skill statistic (TSS) scores ranging from 0.56 to 0.64. Weighting models with class prevalence improved fit and the correspondence with known spatial features. Class-based metrics showed differences across both resolutions and spatial regions, indicating non-stationarity across these spatial categories. Predictive power was lower (TSS from 0.10 to 0.36) based on independent data evaluation. Model performance was also a function of depth and resolution, illustrating the challenge of accurately representing heterogeneity. Our work shows the value of regional analyses to assessing model stationarity and how independent data evaluation and the use of error metrics can improve understanding of model performance and sampling bias. Copyright: © 2021 Gregr et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Date 2021

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Language English

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article, prediction, prevalence, random forest, biodiversity, writing, Canada, physiology, statistical model, Biodiversity, biomass, Biomass, Models, Statistical, sea water, Seawater, Aquatic Organisms, aquatic species, Geologic Sediments, Pacific Ocean, sampling bias, sediment, skill

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EID: 2-s2.0-85118277466

Export Date: 20 September 2026; Cited By: 18; Correspondence Address: E.J. Gregr; SciTech Environmental Consulting, Vancouver, Canada; email: ed@scitechconsulting.com; CODEN: POLNC

Computationally efficient class-prior estimation under class balance change using energy distance

Item Type Journal Article

Author H. Kawakubo

Author M.C. Du Plessis

Author M. Sugiyama

Abstract In many real-world classification problems, the class balance often changes between training and test datasets, due to sample selection bias or the non-stationarity of the environment. Naive classifier training under such changes of class balance systematically yields a biased solution. It is known that such a systematic bias can be corrected by weighted training according to the test class balance. However, the test class balance is often unknown in practice. In this paper, we consider a semi-supervised learning setup where labeled training samples and unlabeled test samples are available and propose a class balance estimator based on the energy distance. Through experiments, we demonstrate that the proposed method is computationally much more efficient than existing approaches, with comparable accuracy. Copyright © 2016 The Institute of Electronics, Information and Communication Engineers.

Date 2016

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ISSN 09168532 (ISSN)

Issue 1

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Classification (of information), Supervised learning, Sample selection bias, Non-stationarities, Systematic bias, Computationally efficient, Semi-supervised learning, Class balance change, Class-prior estimation, Energy distance, Naive classifiers

Notes:

EID: 2-s2.0-84953218773

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Computer aided characterization of the solitary pulmonary nodule using volumetric and contrast enhancement features

Item Type Journal Article**Author** S.K. Shah**Author** M.F. McNitt-Gray**Author** S.R. Rogers**Author** J.G. Goldin**Author** R.D. Suh**Author** J.W. Sayre**Author** I. Petkovska**Author** H.J. Kim**Author** D.R. Aberle

Abstract Rationale and Objectives. To investigate the utility of a computer-aided diagnosis (CAD) in the task of differentiating malignant nodules from benign nodules based on quantitative features extracted from volumetric thin section CT image data acquired before and after the injection of contrast media. Materials and Methods. 35 volumetric CT datasets of solitary pulmonary nodules (SPN) with proven diagnoses (19 malignant/16 benign) were contoured by a thoracic radiologist. All studies had at least a baseline series obtained without contrast media and at least one series following an intravenous contrast injection at 45, 90, 180, and 360 seconds. Two separate contours were created for each nodule: one including only the solid portion and another including the ground-glass component, if any, of the nodule. For each contour 31 features were calculated that measured the attenuation, shape, and enhancement of the nodule due to the injection of contrast. These features were input into a feature selection step and three different classifiers to determine if the diagnosis could be predicted from the resulting feature vector. In addition, observer input was introduced to two of the classifiers as an a priori probability of malignancy and the resulting performance was compared. Training and testing was conducted in a resubstitution and leave-one-out fashion and performance was evaluated using ROC analysis. Results. In a leave-one-out testing methodology, the classifiers achieved areas under the ROC curves (AZ) that ranged from 0.69 to 0.92. A classifier based on logistic regression performed the best with an AZ of 0.92 while a classifier based on quadratic discriminant analysis performed the poorest (AZ, 0.69). The AZ increased when using a priori observer input in most cases reaching a maximum of 0.95. Conclusion. Based on this initial work with a limited number of nodules in our dataset, it appears that CAD using volumetric and contrast-enhanced data has the potential to assist radiologists in the task of differentiating solitary pulmonary nodules and in the management of these patients. Further studies with an increased number of patients are required to validate these results. © AUR, 2005.

Date 2005**DOI** 10.1016/j.acra.2005.06.005**URL** <https://www.scopus.com/pages/publications/25144483323?origin=resultslist>**Archive** Scopus

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adult, Adult, female, Female, human, Humans, male, Male, Middle Aged, Artificial Intelligence, Cluster Analysis, aged, Aged, Aged, 80 and over, article, Reproducibility of Results, Algorithms, quantitative analysis, prediction, calculation, controlled study, logistic regression analysis, receiver operating characteristic, probability, clinical article, computer assisted diagnosis, Contrast Media, contrast medium, Imaging, Three-Dimensional, volumetry, diagnostic procedure, computer assisted tomography, radiologist, imaging, Sensitivity and Specificity, Radiographic Image Interpretation, Computer-Assisted, Lung cancer, thickness, Lung Neoplasms, lung nodule, injection, tumor diagnosis, contrast enhancement, thorax radiography, priority journal, Pattern Recognition, Automated, Information Storage and Retrieval, time, Computer-aided diagnosis, Neoplasm Staging, Lung nodule, Radiographic Image Enhancement, Coin Lesion, Pulmonary, Enhancement, ROC

Notes:

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Computer Aided Diagnosis Based Hand Thermal Image Analysis: A Potential Tool for the Evaluation of Rheumatoid Arthritis

Item Type Journal Article

Author S. Umapathy

Author S. Vasu

Author N. Gupta

Abstract The aim of the study was to evaluate the potential of the thermogram in diagnosing rheumatoid arthritis and to compare the implementation of k-means algorithm and fuzzy c means algorithm using a computer aided diagnostic tool for classification of rheumatoid arthritis (RA) and normal based on the feature extracted from the

segmented thermal image. The skin surface temperature measurement, thermal image segmentation based on k-means and fuzzy c-means algorithm and the feature extraction were used in this study. The average skin surface temperature measured at the second meta carpo-phalangeal (MCP) and MCP3 in RA patients ($35.40 \pm 0.6^\circ\text{C}$ and $35.52 \pm 0.7^\circ\text{C}$, respectively) were significantly higher ($p < 0.01$) than those measured in normal subjects ($33.66 \pm 0.2^\circ\text{C}$ and $33.74 \pm 0.2^\circ\text{C}$, respectively). The mean difference in temperature between RA patients and healthy controls in the MCP2 and MCP3 region was found to be 1.74 and 1.78°C respectively. The receiver operating characteristics (ROC) curve depicted a sensitivity of 86.6% and specificity of 79% achieved in the MCP region of the hand thermal image. Thermal image segmentation using the k-means algorithm provided better segmentation results compared to the fuzzy c-means algorithm in diagnosing the disease. Therefore, the computer aided diagnostic based thermography method could be used as a validated quantification method for interpreting and evaluating arthritis. © 2017, Taiwanese Society of Biomedical Engineering.

Date 2018

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Journal Abbr J. Med. Biol. Eng.

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Tags:

adult, Article, female, human, male, Feature extraction, Image segmentation, comparative study, algorithm, image segmentation, Clustering algorithms, Fuzzy systems, Extraction, feature extraction, controlled study, Diseases, Fuzzy clustering, receiver operating characteristic, clinical article, diagnostic accuracy, Computer aided diagnosis, disease duration, clinical evaluation, Quantification methods, thermal analysis, sensitivity and specificity, image analysis, Atmospheric temperature, Surface properties, Computer aided analysis, rheumatoid arthritis, Rheumatoid arthritis, thermography, C reactive protein, diagnostic test, fuzzy system, skin temperature, computer aided design, computer aided diagnosis, Computer aided diagnostics, Copying, Fuzzy c-means algorithm, Fuzzy C-means algorithms, hand thermal image analysis, immunoglobulin M, k-Means algorithm, K-means algorithm, metacarpophalangeal joint, Receiver operating characteristics (ROC) curve, Receiver operating characteristics curves (ROC), rheumatoid factor, skin surface, Skin surface temperature, temperature measurement, Temperature measurement, Thermal image analysis

Notes:

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Export Date: 20 September 2026; Cited By: 32; Correspondence Address: S. Umapathy; Department of Biomedical Engineering, SRM University, Chennai, Kattankulathur, 603203, India; email: sneha_samuma@yahoo.co.in

Computer aided quantification for retinal lesions in patients with moderate and severe non-proliferative diabetic retinopathy: A retrospective cohort study

Item Type Journal Article

Author H. Wu

Author X. Zhang

Author X. Geng

Author J. Dong

Author G. Zhou

Abstract Background: Detection of retinal lesions like micro-aneurysms and exudates are important for the clinical diagnosis of diabetes retinopathy. The traditional subjective judgments by clinicians are dependent on their experience and can be subject to lack of consistency and therefore a quantification method is worthwhile. Methods: In this study, 10 moderate non-proliferative diabetes retinopathy (NPDR) patients and 10 severe NPDR ones were retrospectively selected as a cohort. Mathematical morphological methods were used for automatic segmentation of lesions. For exudates detection, images were pre-processed with adaptive histogram equalization to enhance contrast, then binary images for area calculation were obtained by threshold classification. For micro-aneurysms detection, the images were pre-processed by top-hat and bottom-hat transformation, then Otsu method and Hough transform were used to classify micro-aneurysms. Post-processing morphological methods were used to preclude the false positive noise. Results: After segmentation, the area of exudates divided by optic disk area (exudates/disk ratio) and counts of microaneurysms were quantified and compared between the moderate and severe non-proliferative diabetic retinopathy groups, which had significant difference ($P < 0.05$). Conclusions: In conclusion, morphological features of lesion might be an image marker for NPDR grading and computer aided quantification of retinal lesion could be a practical way for clinicians to better investigate diabetic retinopathy. © 2014 Wu et al.

Date 2014

DOI 10.1186/1471-2415-14-126

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Tags:

female, Female, human, Humans, male, Male, middle aged, Middle Aged, aged, Aged, cohort analysis, Retrospective Studies, retrospective study, computer assisted diagnosis, pathology, Diagnosis, Computer-Assisted, diabetic retinopathy, Cohort Studies, retina blood vessel, subretinal fluid, Subretinal Fluid, aneurysm, Aneurysm, Diabetic retinopathy, Diabetic Retinopathy, exudate, Exudates and Transudates, Exudation, Fundus lesions, Microaneurysms, Retinal Vessels

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Export Date: 20 September 2026; Cited By: 14; Funding details: National Natural Science Foundation of China, NSFC, (81271668); National Natural Science Foundation of China, NSFC; Correspondence Address: J. Dong; Department of Medical Informatics, Medical School of Nantong University, Nantong, 226001, China; email: dongjc@ntu.edu.cn

Computer Aided System for Human Blood Cell Identification, Classification and Counting

Item Type Conference Paper

Author N. Christy Evangeline

Author M. Annalatha

Abstract Evaluation of Blood Smear is one of the most basic laboratory procedures in Hematology as deviations of blood cell count from their reference values can be indicators of various disease conditions. Digital Image Processing can help the technologists in the analysis and classification of blood cells and diagnosis of certain health conditions based on the CBC values and their morphology. This project aims at generating software that can identify, classify and count various blood cells from the peripheral blood smear samples. Locating, identifying and counting of Blood cells manually can be tedious, time consuming and inaccurate. On the other hand, Automatic Hematology Analyzers that work on Flow cytometry principle for classification are expensive and can perform only quantitative analysis. Thus taking into consideration, the wide scope of Image processing in the field of medicine, this proposed work aims to generate a preliminary framework of Automatic Analysis of Human Blood smear for identification, classification and counting of blood cells by using various Image processing. © 2018 IEEE.

Date 2018

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Tags:

Image processing, Blood, Image analysis, Computer aided diagnosis, Cells, Automatic analysis, Cytology, Blood cells, Computer-aided systems, Automatic Analysis, Blood cell counting, Blood Cell counting, Blood cell Identification, Blood cell identifications, Classification of Blood Cells, Digital Image Processing, Human blood cells, Laboratory procedures, Peripheral blood smears

Notes:

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Export Date: 20 September 2026; Cited By: 15; Conference name: 4th International Conference on Biosignals, Images and Instrumentation, ICBSII 2018; Conference date: 22 March 2018 through 24 March 2018; Conference code: 142165

Computer image analysis and classification of giant ragweed seeds

Item Type Journal Article

Author Y. Sako

Author E.E. Regnier

Author T. Daoust

Author K. Fujimura

Author S.K. Harrison

Author M.B. McDonald

Abstract Giant ragweed exhibits a high degree of polymorphism among individual plants in seed size, shape, spininess, and color. These features may play an important role in giant ragweed seed survival and predation avoidance; however, they are difficult to evaluate because of lack of quantification methods. A computer imaging technique was developed for describing and classifying giant ragweed seeds using digital images of the seed top and side views. Seed samples collected from 20 different giant ragweed plants (classes) were mounted and digitally scanned. Quantitative features were extracted from the seed images, including color, width, height, area, and seed perimeter. A polygon (convex hull) of the seed image based on the seed outline was constructed, from which spininess indices were developed. Fisher's linear discriminant with normalized nearest neighbor classification was used to classify randomly selected images of individual seeds according to class (maternal origin), using the extracted features as a database. The best classification rate achieved was 99%, with 138 out of 140 seeds correctly matched using data from both the top and side views. Seed features were easily extracted and varied from 1.2- to 4.5-fold

among classes. Area and perimeter measurements varied least within classes but varied most among classes, suggesting that these features discriminate effectively among seeds from different plants in giant ragweed. Convex hull area : seed area ratio, using the seed top view images, was the best index of seed spininess, aligning well with visual assessment and providing greatest discrimination among classes. This experiment shows that in the case of giant ragweed, seeds from different plants are distinguishable in an objective and quantitative manner. This imaging technique can be applied to identification of seeds from different species and to studies on variable seed morphology within a species.

Date 2001

DOI 10.1614/0043-1745(2001)049[0738:CIAACO]2.0.CO;2

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Issue 6

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Tags:

FLD, Fruit morphology, Fruit shape, Seed classification, Seed identification, Seed morphology, Seed polymorphism, Seed shape

Notes:

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Computer vision and machine learning to quantify microstructure

Item Type Journal Article

Author E.A. Holm

Author R. Cohn

Author N. Gao

Author A.R. Kitahara

Author B. Lei

Author S.R. Yarasi

Author T.P. Matson
Abstract Computer vision and machine learning systems for microstructural characterization and analysis are used for a variety of image analysis tasks, including image classification, semantic segmentation, object detection, and instance segmentation, leading to accurate, autonomous, objective, repeatable results in an indefatigable and permanently available manner. © 2021, ASM International. All rights reserved.
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Machine learning, Computer vision, Object detection, Image segmentation, Semantics, Semantic segmentation, Micro-structural characterization, Repeatable results

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Computer-aided assessment of regional abdominal fat with food residue removal in CT

Item Type Journal Article

Author S. Makrogiannis

Author G. Caturegli

Author C. Davatzikos

Author L. Ferrucci

Abstract Rationale and Objectives: Separate quantification of abdominal subcutaneous and visceral fat regions is essential to understand the role of regional adiposity as risk factor in epidemiological studies. Fat quantification is often based on computed tomography (CT) because fat density is distinct from other tissue densities in the

abdomen. However, the presence of intestinal food residues with densities similar to fat may reduce fat quantification accuracy. We introduce an abdominal fat quantification method in CT with interest in food residue removal. **Materials and Methods:** Total fat was identified in the feature space of Hounsfield units and divided into subcutaneous and visceral components using model-based segmentation. Regions of food residues were identified and removed from visceral fat using a machine learning method integrating intensity, texture, and spatial information. Cost-weighting and bagging techniques were investigated to address class imbalance. **Results:** We validated our automated food residue removal technique against semimanual quantifications. Our feature selection experiments indicated that joint intensity and texture features produce the highest classification accuracy at 95%. We explored generalization capability using k-fold cross-validation and receiver operating characteristic (ROC) analysis with variable k. Losses in accuracy and area under ROC curve between maximum and minimum k were limited to 0.1% and 0.3%. We validated tissue segmentation against reference semimanual delineations. The Dice similarity scores were as high as 93.1 for subcutaneous fat and 85.6 for visceral fat. **Conclusions:** Computer-aided regional abdominal fat quantification is a reliable computational tool for large-scale epidemiological studies. Our proposed intestinal food residue reduction scheme is an original contribution of this work. Validation experiments indicate very good accuracy and generalization capability. © 2013.

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Computer-assisted lip diagnosis on traditional Chinese medicine using multi-class support vector machines

Item Type Journal Article

Author F. Li

Author C. Zhao

Author Z. Xia

Author Y. Wang

Author X. Zhou

Author G.-Z. Li

Abstract Background: In Traditional Chinese Medicine (TCM), the lip diagnosis is an important diagnostic method which has a long history and is applied widely. The lip color of a person is considered as a symptom to reflect the physical conditions of organs in the body. However, the traditional diagnostic approach is mainly based on observation by doctor's nude eyes, which is non-quantitative and subjective. The non-quantitative approach largely depends on the doctor's experience and influences accurate the diagnosis and treatment in TCM. Developing new quantification methods to identify the exact syndrome based on the lip diagnosis of TCM becomes urgent and important. In this paper, we design a computer-assisted classification model to provide an automatic and quantitative approach for the diagnosis of TCM based on the lip images. Methods: A computer-assisted classification method is designed and applied for syndrome diagnosis based on the lip images. Our purpose is to classify the lip images into four groups: deep-red, red, purple and pale. The proposed scheme consists of four steps including the lip image preprocessing, image feature extraction, feature selection and classification. The extracted 84 features contain the lip color space component, texture and moment features. Feature subset selection is performed by using SVM-RFE (Support Vector Machine with recursive feature elimination), mRMR (minimum Redundancy Maximum Relevance) and IG (information gain). Classification model is constructed based on the collected lip image features using multi-class SVM and Weighted multi-class SVM (WSVM). In addition, we compare SVM with k-nearest neighbor (kNN) algorithm, Multiple Asymmetric Partial Least Squares Classifier (MAPLSC) and Naïve Bayes for the diagnosis performance comparison. All displayed faces image have obtained consent from the participants. Results: A total of 257 lip images are collected for the modeling of lip diagnosis in TCM. The feature selection method SVM-RFE selects 9 important features which are composed of 5 color component features, 3 texture features and 1 moment feature. SVM, MAPLSC, Naïve Bayes, kNN showed better classification results based on the 9 selected features than the results obtained from all the 84 features. The total classification accuracy of the five methods is 84%, 81%, 79% and 81%, 77%, respectively. So SVM achieves the best classification accuracy. The classification accuracy of SVM is 81%, 71%, 89% and 86% on Deep-red, Pale

Purple, Red and lip image models, respectively. While with the feature selection algorithm mRMR and IG, the total classification accuracy of WSVM achieves the best classification accuracy. Therefore, the results show that the system can achieve best classification accuracy combined with SVM classifiers and SVM-REF feature selection algorithm. Conclusions: A diagnostic system is proposed, which firstly segments the lip from the original facial image based on the Chan-Vese level set model and Otsu method, then extracts three kinds of features (color space features, Haralick co-occurrence features and Zernike moment features) on the lip image. Meanwhile, SVM-REF is adopted to select the optimal features. Finally, SVM is applied to classify the four classes. Besides, we also compare different feature selection algorithms and classifiers to verify our system. So the developed automatic and quantitative diagnosis system of TCM is effective to distinguish four lip image classes: Deep-red, Purple, Red and Pale. This study puts forward a new method and idea for the quantitative examination on lip diagnosis of TCM, as well as provides a template for objective diagnosis in TCM. © 2012 Li et al.; licensee BioMed Central Ltd.

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Conditional adaptation deep networks for unsupervised cross domain image classification

Item Type Conference Paper

Author Y. Chen

Author C. Yang

Author Y. Zhang

Author Y. Li

Abstract Unsupervised domain adaptation aims at improving the performance on an unknown target domain by utilizing the knowledge learned from a related source domain. In this paper, we proposed a conditional adaptation deep network (CADN) to deal with the unsupervised domain adaptation problem. CADN is implemented based on a deep neural network and attempts to learn domain invariant features. We present a conditional adaptation strategy to reduce the domain distribution discrepancy and to solve the category mismatch and class prior bias, which are usually ignored in marginal adaptation approaches. We perform a set of comparative experiments on the standard domain adaptation benchmarks. The results demonstrate that the proposed CADN outperforms previous adaptation methods. © 2019 IEEE.

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Conditional Calibrated Explanations: Finding a Path Between Bias and Uncertainty

Item Type Conference Paper

Author H. Löfström

Author T. Löfström

Editor Longo L.

Editor Lapuschkin S.

Editor Seifert C.

Abstract While Artificial Intelligence and Machine Learning models are becoming increasingly prevalent, it is essential to remember that they are not infallible or inherently objective. These models depend on the data they are trained on and the inherent bias of the chosen machine learning algorithm. Therefore, selecting and sampling data for training is crucial for a fair outcome of the model. A model predicting, e.g., whether an applicant should be taken further in the job application process, could create heavily biased predictions against women if the data used to train the model mostly contained information about men. The well-known concept of conditional categories used in Conformal Prediction can be utilised to address this type of bias in the data. The Conformal Prediction framework includes uncertainty quantification methods for classification and regression. To help meet the challenges of data sets with potential bias, conditional categories were incorporated into an existing explanation method called Calibrated Explanations, relying on conformal methods. This approach allows users to try out different settings while simultaneously having the possibility to study how the uncertainty in the predictions is affected on an individual level. Furthermore, this paper evaluated how the uncertainty changed when using conditional categories based on attributes containing potential bias. It showed that the uncertainty significantly increased, revealing that fairness came with a cost of increased uncertainty. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Pages 332-355**Proceedings Title** Commun. Comput. Info. Sci.**Conference Name** Communications in Computer and Information Science**Date Added** 20/09/2026, 9:51:54 pm**Modified** 20/09/2026, 9:51:54 pm**Tags:**

Artificial intelligence learning, Machine learning, Forecasting, Uncertainty analysis, Learning algorithms, Uncertainty, Bias, Machine learning models, XAI, Fairness, Uncertainty quantifications, Uncertainty Quantification, Conformal predictions, Calibrated explanation, Calibrated Explanations, Post-hoc explanation, Post-hoc Explanations

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Confidence bounds of petrophysical predictions from conventional neural networks

Item Type Journal Article**Author** P.M. Wong**Author** A.G. Bruce**Author** T.D. Gedeon

Abstract Neural networks are powerful tools for solving the complex regression problems which abound in geosciences. Estimation of prediction confidence from neural networks is an important area. Many procedures are available to date, but it is often tedious for practitioners to implement such procedures without significant modification of the existing learning algorithms. In many cases, the procedures are also computationally intensive. This paper presents a practical solution using conventional backpropagation networks with simple data pre-processing and post-processing algorithms. The methodology involves conversions of the target outputs into linguistic variables (classes) prior to learning. When the classification network converges, minimum and maximum predictions are derived from the output activations using a simple averaging algorithm. Two examples from petroleum reservoirs are used to demonstrate the proposed methodology. The results show that the confidence bounds of the petrophysical predictions are realistic in both cases. The proposed methodology is generally useful, and can be implemented in simple spreadsheets without altering any existing neural network code.

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Confidence Intervals for Class Prevalences under Prior Probability Shift

Item Type Journal Article

Author D. Tasche

Abstract Point estimation of class prevalences in the presence of dataset shift has been a popular research topic for more than two decades. Less attention has been paid to the construction of confidence and prediction intervals for estimates of class prevalences. One little considered question is whether or not it is necessary for practical purposes to distinguish confidence and prediction intervals. Another question so far not yet conclusively answered is whether or not the discriminatory power of the classifier or score at the basis of an estimation method matters for the accuracy of the estimates of the class prevalences. This paper presents a simulation study aimed at shedding some light on these and other related questions. © 2019 by the author.

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prior probability shift, prediction interval, confidence interval, class prevalence

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Conformal prediction based on principal component analysis for high-dimensional outlier detection

Item Type Conference Paper

Author X. Qian

Author J. Wu

Author Y. Lin

Editor Kong X.

Editor Dai W.

Abstract Current outlier detection methods exhibit many limitations in high-dimensional settings. Traditional statistical approaches rely on strong assumptions and lack practicality and generality. Meanwhile, despite the better performance, machine learning methods suffer from low interpretability and reliability due to their complex mechanisms and the absence of confidence estimation. Although outlier detection methods based on principal component analysis (PCA) have shown some advantages by extracting important features from high-dimensional data, they have not entirely solved these problems. Conformal prediction, a finite-sample distribution-free uncertainty quantification method recently applied to outlier detection, produces a set-valued prediction with a Type-I error guarantee and false discovery rate control. This paper proposes a new distribution-free method for high-dimensional outlier detection, PCA-CP, combining principal component analysis and conformal prediction. PCA-CP overcomes the shortcomings of previous methods, demonstrating high generality and reliability while achieving high performance. Experiments on simulation and real data show that PCA-CP outperforms previous methods, achieving higher power and lower False Discovery Rate, thus proving its significant advantages. © 2024 SPIE.

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Performance, Prediction models, Principal component analysis, Detection methods, High-dimensional, Higher-dimensional, Principal-component analysis, Conformal predictions, Outlier Detection, Conformal prediction, False discovery rate, High dimensional data, High-dimensional data, Outlier detection, Prediction-based

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Conformal Prediction for Federated Uncertainty Quantification Under Label Shift

- Item Type** Conference Paper
- Author** V. Plassier
- Author** M. Makni
- Author** A. Rubashevskii
- Author** E. Moulines
- Author** M. Panov
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Editor Scarlett J.

Abstract Federated Learning (FL) is a machine learning framework where many clients collaboratively train models while keeping the training data decentralized. Despite recent advances in FL, the uncertainty quantification topic (UQ) remains partially addressed. Among UQ methods, conformal prediction (CP) approaches provides distribution-free guarantees under minimal assumptions. We develop a new federated conformal prediction method based on quantile regression and take into account privacy constraints. This method takes advantage of importance weighting to effectively address the label shift between agents and provides theoretical guarantees for both valid coverage of the prediction sets and differential privacy. Extensive experimental studies demonstrate that this method outperforms current competitors. © 2023 Proceedings of Machine Learning Research. All rights reserved.

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Conformal prediction for stochastic decision-making of PV power in electricity markets

Item Type Journal Article**Author** Y. Renkema**Author** N. Brinkel**Author** T. Alskauf

Abstract This paper studies the use of conformal prediction (CP), an emerging probabilistic forecasting method, for day-ahead photovoltaic power predictions to enhance participation in electricity markets. First, machine learning models are used to construct point predictions. Thereafter, several variants of CP are implemented to quantify the uncertainty of those predictions by creating CP intervals and cumulative distribution functions. Optimal quantity bids for the electricity market are estimated using several bidding strategies under uncertainty, namely: trust-the-forecast, worst-case, Newsvendor and expected utility maximization (EUM). Results show that CP in combination with k-nearest neighbors and/or Mondrian binning outperforms its corresponding linear quantile regressors. Using CP in combination with certain bidding strategies can yield high profit with minimal energy imbalance. In concrete, using conformal predictive systems with k-nearest neighbors and Mondrian binning after random forest regression yields the best profit and imbalance regardless of the decision-making strategy. Combining this uncertainty quantification method with the EUM strategy with conditional value at risk (CVaR) can yield up to 93% of the potential profit with minimal energy imbalance. © 2024 The Author(s)

Date 2024**DOI** 10.1016/j.epsr.2024.110750**URL** <https://www.scopus.com/pages/publications/85196662535?origin=resultslist>**Archive** Scopus**Language** English**Volume** 234**Publisher** Elsevier Ltd**Publication** Electric Power Systems Research**ISSN** 03787796 (ISSN)**Journal Abbr** Electr Power Syst Res**Date Added** 20/09/2026, 9:52:08 pm**Modified** 20/09/2026, 9:52:08 pm**Tags:**

Distribution functions, Nearest neighbor search, Decision making, Machine-learning, Machine learning, Forecasting, Uncertainty analysis, Uncertainty, Power markets, Profitability, Conformal predictions, Conformal prediction, Bidding strategy, Electricity markets, Expected utility, Minimal energy, Mondrian, Motion compensation, Photovoltaic power, Stochastic optimization, Stochastic optimizations, Utility maximizations, Value engineering

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Conformalized Semi-supervised Random Forest for Classification and Abnormality Detection

Item Type Conference Paper

Author Y. Han

Author M. Xu

Author L. Guan

Editor Dasgupta S.

Editor Mandt S.

Editor Li Y.

Abstract The Random Forests classifier, a widely utilized off-the-shelf classification tool, assumes training and test samples come from the same distribution as other standard classifiers. However, in safety-critical scenarios like medical diagnosis and network attack detection, discrepancies between the training and test sets, including the potential presence of novel outlier samples not appearing during training, can pose significant challenges. To address this problem, we introduce the Conformalized Semi-Supervised Random Forest (CSForest), which couples the conformalization technique Jackknife+aB with semi-supervised tree ensembles to construct a set-valued prediction $C(x)$. Instead of optimizing over the training distribution, CSForest employs unlabeled test samples to enhance accuracy and flag unseen outliers by generating an empty set. Theoretically, we establish CSForest to cover true labels for previously observed inlier classes under arbitrarily label-shift in the test data. We compare CSForest with state-of-the-art methods using synthetic examples and various real-world datasets, under different types of distribution changes in the test domain. Our results highlight CSForest's effective prediction of inliers and its ability to detect outlier samples unique to the test data. In addition, CSForest shows persistently good performance as the sizes of the training and test sets vary. Codes of CSForest are available at <https://github.com/yujinhan98/CSForest>. Copyright 2024 by the author(s).

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Conformalized-KANs: uncertainty quantification with coverage guarantees for Kolmogorov–Arnold Networks (KANs) in scientific machine learning

Item Type Journal Article

Author A. Mollaali

Author C. Bolivar Moya

Author A.A. Howard

Author A. Heinlein

Author P. Stinis

Author G. Lin

Abstract This paper explores uncertainty quantification (UQ) methods in the context of Kolmogorov–Arnold Networks (KANs). We apply an ensemble approach to KANs to obtain a heuristic measure of UQ, enhancing interpretability and robustness in modeling complex functions. However, particularly in data-limited and complex scientific settings, ensemble-based uncertainty estimates, as is the case for many UQ methods, are sensitive to implicit distributional assumptions and do not provide formal coverage guarantees. Building on this, we develop conformalized-KANs, a conformal UQ framework for KANs that combines ensemble-based uncertainty estimates with split conformal prediction to generate calibrated prediction intervals with formal coverage guarantees. Numerical experiments are conducted to evaluate the effectiveness of these methods, focusing particularly on the robustness and accuracy of the prediction intervals under various hyperparameter settings. We further show that this framework extends naturally to recent KAN variants, including finite basis KANs and multi-fidelity KANs. The results demonstrate the potential of our approaches to improve the reliability and applicability of KANs in scientific machine learning. © 2026 The Author(s). Published by IOP Publishing Ltd. Original content from this work may be used under the terms of the <https://creativecommons.org/licenses/by/4.0/>. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

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Considerations for point-of-care diagnostics: Evaluation of acridine orange staining and postprocessing methods for a three-part leukocyte differential test

Item Type Journal Article

Author A.J. Powless

Author R.J. Conley

Author K.A. Freeman

Author T.J. Muldoon

Abstract There exists a broad range of techniques that can be used to classify and count white blood cells in a point-of-care (POC) three-part leukocyte differential test. Improvements in lenses, light sources, and cameras for image-based POC systems have renewed interest in acridine orange (AO) as a contrast agent, whereby subpopulations of leukocytes can be differentiated by colorimetric analysis of AO fluorescence emission. We evaluated the effect on test accuracy using different AO staining and postprocessing methods in the context of an image-based POC

colorimetric cell classification scheme. Thirty blood specimens were measured for percent cell counts using our POC system and a conventional hematology analyzer for comparison. Controlling the AO concentration used during whole-blood staining, the incubation time with AO, and the colorimetric ratios among the three population of leukocytes yielded a percent deviation of 0.706%, -1.534%, and -0.645% for the lymphocytes, monocytes, and granulocytes, respectively. Overall, we demonstrated that a redshift in AO fluorescence was observed at elevated AO concentrations, which lead to reproducible inaccuracy of cell counts. This study demonstrates there is a need for a strict control of the AO staining and postprocessing methods to improve test accuracy in these POC systems. © 2017 The Authors. Published by SPIE under a Creative Commons Attribution 3.0 Unported License. Distribution or reproduction of this work in whole or in part requires full attribution of the original publication, including its.

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CONSISTENT MULTI-CLASS CLASSIFICATION FROM MULTIPLE UNLABELED DATASETS

Item Type Conference Paper

Author Z. Wei

Author S. Shu

Author Y. Cao

Author H. Wei

Author B. An

Author L. Feng

Abstract Weakly supervised learning aims to construct effective predictive models from imperfectly labeled data. The recent trend of weakly supervised learning has focused on how to learn an accurate classifier from completely unlabeled data, given little supervised information such as class priors. In this paper, we consider a newly proposed weakly supervised learning problem called multi-class classification from multiple unlabeled datasets, where only multiple sets of unlabeled data and their class priors (i.e., the proportion of each class) are provided for training the classifier. To solve this problem, we first propose a classifier-consistent method (CCM) based on a probability transition function. However, CCM cannot guarantee risk consistency and lacks of purified supervision information during training. Therefore, we further propose a risk-consistent method (RCM) that progressively purifies supervision information during training by importance weighting. We provide comprehensive theoretical analyses for our methods to demonstrate the statistical consistency. Experimental results on multiple benchmark datasets across various settings demonstrate the superiority of our proposed methods. © 2024 12th International Conference on Learning Representations, ICLR 2024. All rights reserved.

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Consistent prompt learning for vision-language models

Item Type Journal Article

Author Y. Zhang

Author X. Tian

Abstract Pre-trained vision-language models, such as CLIP, have shown remarkable capabilities across various downstream tasks by learning prompts that consist of context concatenated with a class name; for example, ‘a photo of a [dog]’ with [dog] as a class prior. Advanced prompt-learning methods typically initialize and optimize the context; for example, ‘a photo of a’ for downstream task adaptation. However, context optimization typically leads to poor generalization performance over novel classes or datasets sampled from different distributions. This may be attributed to prompt inconsistency; namely, prompts optimized using one image distribution may differ from those optimized using a different image distribution. To improve the generalization performance of optimized prompts, we propose the novel consistent prompt learning (CPL) approach that identifies and addresses the image distribution that causes prompt inconsistency by performing distributional exploration. CPL identifies and mitigates prompt inconsistency in an adversarial training scheme, in which prompt inconsistency is measured as the similarity discrepancy between images and two different prompts. Specifically, CPL calculates two similarities between a query image and two prompts, and determines the prompt inconsistency through the discrepancy between these two similarities. Subsequently, CPL performs distributional exploration to enlarge the discrepancy and uses an adversarial-training approach to mitigate the discrepancy. Consequently, the model predictions are insensitive to prompt changes. The optimized prompt performs well under various image distributions. Comprehensive experiments show that the proposed CPL method performs favorably on four types of representative tasks across 11 datasets, which improves on existing prompt-learning methods, achieving state-of-the-art performance. © 2025 Elsevier B.V.

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Language model, Visual languages, Federated learning, Adversarial machine learning, Contrastive Learning, Generalisation, Learning methods, Prompt learning, Down-stream, Domain adaptation, Vision-language models, Vision-language model, Digital elevation model, Domain generalization, Generalization performance, Image distributions

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Constraint-based, transductive learning for distributed ensemble classification

Item Type Conference Paper

Author D.J. Miller

Author S. Pal

Author Y. Wang

Abstract We consider ensemble classification when there is no common labeled data for designing the function which aggregates classifier decisions. In recent work, we dubbed this problem distributed ensemble classification, addressing e.g. when local classifiers are trained on different (e.g. proprietary, legacy) databases or operate on different sensing modalities. Typically, fixed (untrained) rules of classifier combination such as voting methods are used in this case. However, these may perform poorly, especially when the local class priors, used in training, differ from the true (test batch) priors. Alternatively, we proposed a transductive strategy, optimizing the combining rule for an objective function measured on the test batch. We proposed both maximum likelihood (ML) and information-theoretic (IT) objectives and found that IT achieved superior performance. Here, we identify that the fundamental advantage of the IT method is its ability to properly account for statistical redundancy in the ensemble. We also develop an extension of IT that improves its performance. Experiments are conducted on the UC Irvine machine learning repository. © 2006 IEEE.

Date 2006

DOI 10.1109/MLSP.2006.275514

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Language English

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Publisher IEEE Computer Society

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Conference Name Proceedings of the 2006 16th IEEE Signal Processing Society Workshop on Machine Learning for Signal Processing, MLSP 2006

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: D.J. Miller; Penn State University, Department of EE, University Park, PA 16802-2701, United States; email: millerdj@ee.psu.edu; Conference name: 16th IEEE Signal Processing Society Workshop on Machine Learning for Signal Processing, MLSP 2006; Conference date: 6 September 2006 through 8 September 2006; Conference code: 71402

CONTINUAL MOMENTUM FILTERING ON PARAMETER SPACE FOR ONLINE TEST-TIME ADAPTATION

Item Type Conference Paper

Author J.-H. Lee

Author J.-H. Chang

Abstract Deep neural networks (DNNs) have revolutionized tasks such as image classification and speech recognition but often falter when training and test data diverge in distribution. External factors, from weather effects on images to varied speech environments, can cause this discrepancy, compromising DNN performance. Online test-time adaptation (OTTA) methods present a promising solution, recalibrating models in real-time during the test stage without requiring historical data. However, the OTTA paradigm is imperfect, often falling prey to issues such as catastrophic forgetting due to its reliance on noisy, self-trained predictions. Although some contemporary strategies mitigate this by tying adaptations to the static source model, this restricts model flexibility. This paper introduces a continual momentum filtering (CMF) framework, leveraging the Kalman filter (KF) to strike a balance between model adaptability and information retention. The CMF intertwines optimization via stochastic gradient

descent with a KF-based inference process. This methodology not only aids in averting catastrophic forgetting but also provides high adaptability to shifting data distributions. We validate our framework on various OTTA scenarios and real-world situations regarding covariate and label shifts, and the CMF consistently shows superior performance compared to state-of-the-art methods. © 2024 12th International Conference on Learning Representations, ICLR 2024. All rights reserved.

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Continuous Sweep for Binary Quantification Learning

Item Type Journal Article

Author K. Kloos

Author J.D. Karch

Author Q.A. Meertens

Author S. Scholtus

Author M. de Rooij

Abstract A quantifier is a supervised machine learning algorithm that focuses on estimating the class prevalence in a data set rather than labeling its individual observations. We introduce continuous sweep, a new parametric binary quantifier inspired by the well-performing median sweep, which is an ensemble method based on adjusted count estimators. We modified two aspects of median sweep: (1) using parametric class distributions instead of empirical distributions for the true and false positive rates; (2) using the mean instead of the median of a set of adjusted count estimates. These two modifications allow for a theoretical analysis of the bias and variance of the continuous sweep. Furthermore, the expressions of bias and variance can be used to define optimal decision boundaries of the set of adjusted count estimates to be used in the ensemble. In three simulation studies, we show that continuous sweep outperforms the quantifiers in the group classify, count, and correct, including median sweep, and is competitive with the two best quantifiers in the group distribution matchers. Also, an empirical data set is analyzed with these quantifiers, showing similar performances. © The Author(s) 2026.

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Continuous target shift adaptation in supervised learning

Item Type Conference Paper

Author T.D. Nguyen

Author M.C. Du Plessis

Author M. Sugiyama

Abstract Supervised learning in machine learning concerns inferring an underlying relation between covariate x and target y based on training covariate-target data. It is traditionally assumed that training data and test data, on which the generalization performance of a learning algorithm is measured, follow the same probability distribution. However, this standard assumption is often violated in many real-world applications such as computer vision, natural language processing, robot control, or survey design, due to intrinsic non-stationarity of the environment or inevitable sample selection bias. This situation is called dataset shift and has attracted a great deal of attention recently. In the paper, we consider supervised learning problems under the target shift scenario, where the target marginal distribution $p(y)$ changes between the training and testing phases, while the target-conditioned covariate distribution $p(x|y)$ remains unchanged. Although various methods for mitigating target shift in classification (a.k.a. class prior change) have been developed so far, few methods can be applied to continuous targets. In this paper, we propose methods for continuous target shift adaptation in regression and conditional density estimation. More specifically, our contribution is a novel importance weight estimator for continuous targets. Through experiments, the usefulness of the proposed method is demonstrated. © 2015 T.D. Nguyen, M.C. du Plessis & M. Sugiyama.

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Contrastive Conditional Alignment Based on Label Shift Calibration for Imbalanced Domain Adaptation

Item Type Conference Paper

Author X. Sun

Author Z. Wu

Author Z. Zhan

Author Y. Ji

Editor Antonacopoulos A.

Editor Chaudhuri S.

Editor Chellappa R.

Editor Liu C.-L.

Editor Bhattacharya S.

Editor Pal U.

Abstract Many existing unsupervised domain adaptation (UDA) methods primarily focus on covariate shift, limiting their effectiveness in imbalanced domain adaptation (IDA) where both covariate shift and label shift coexist. Recent IDA methods have achieved promising results based on self-training using target pseudo labels. However, under the IDA scenarios, the classifier learned in the source domain will exhibit different decision bias from the target domain. It will potentially make target pseudo labels unreliable, and will further lead to error accumulation with incorrect class alignment. Thus, we propose contrastive conditional alignment based on label shift calibration (CCA-LSC) for IDA, to address both covariate shift and label shift. Initially, our contrastive conditional alignment resolve covariate shift to learn representations with domain invariance and class discriminability, which include domain adversarial learning, sample-weighted moving average centroid alignment and discriminative feature alignment. Subsequently, we estimate the probability distribution of the target domain, and calibrate target sample classification predictions based on label shift metrics to encourage labeling pseudo-labels more consistently with the distribution of real target data. Extensive experiments are conducted and demonstrate that our method outperforms existing UDA and IDA methods on benchmarks with both label shift and covariate shift. Our code is available at <https://github.com/ysxcj-hub/CCA-LSC>. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Contrastive Diffusion Generative Adversarial Network for Generalized Zero-Shot Learning

Item Type Conference Paper

Author Z. Zhang

Author K. Qiu

Author Y. Wang

Author M. Hong

Author L. You

Author S. Huang

Editor Mahmud M.

Editor Doborjeh M.

Editor Wong K.

Editor Leung A.C.S.

Editor Doborjeh Z.

Editor Tanveer M.

Abstract Generative models have been frequently used as a sample augmentation tool to tackle Generalized Zero-Shot Learning (GZSL). Generative ZSL methods aim to produce diverse and high-fidelity visual features. They do this from limited category semantics. However, on one hand, mainstream generative model families, such as GANs, VAEs, and diffusion models, cannot balance sampling speed, quality, and diversity effectively, which reduces their utility in GZSL. On the other hand, the extreme information shift between visual and semantic spaces severely restricts the discriminative and generalization abilities of the generated features. To address these challenges, we propose a novel hybrid generative model named Contrastive Diffusion Generative Adversarial Network (CD-GAN) for GZSL. As a hybrid model of the diffusion model and Generative Adversarial

Network (GAN), CD-GAN incorporates the merits of the diffusion model in terms of the diversity and quality of generated features, as well as the merits of GAN in terms of feature generation efficiency. CD-GAN potentially aligns visuals and semantics by unifying real and generated features into a common embedding space and then conducting contrastive learning guided by between-class prior knowledge in this space. Moreover, we strengthen visual-semantic alignment by introducing a classification regularization. This technique further guarantees that the embedding encodes corresponding semantics. Compared to baseline models, CD-GAN achieves promising performance, as demonstrated by extensive experiments on four popular datasets. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Notes:

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Contrastive Machine Learning to Quantify Hypertensive Multiorgan Damage and Identify New Disease Phenotypes: A Multinational Multimodal Study

Item Type Journal Article

Author M. Alkhodari

Author W. Lapidaire

Author T. Kart

Author Z. Xiong

Author S. Krasner

Author A. Fletcher

Author S. Bibi

Author N. Savage

Author K. Suriano

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Author E. Ohuma

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Author R.J. McManus

Author N.A.B. Ntusi

Author A.H. Khandoker

Author L.J. Hadjileontiadis

Author A.J. Lewandowski

Author A. Banerjee

Author P. Leeson

Abstract BACKGROUND: – Hypertension induces structural and functional damage in multiple organs. Evidence of subclinical damage increases risk of vascular events and death but can be difficult to identify in the clinic. We developed a novel machine learning approach that quantifies current hypertension-associated multiorgan damage, mapping progression from health to advanced disease, in a pseudotemporal manner and predicts organ-specific disease progression trajectories. METHODS: – We analyzed 566 multimodal imaging and nonimaging variables from 27 099 participants in the UK Biobank imaging substudy to develop a semisupervised contrastive trajectory inference (cTI) framework that models multiorgan alterations associated with hypertension exposure, including heart, brain, kidneys, vasculature, lungs, liver, and metabolic information. Model stability was validated through multiple internal validation steps, and external validity was tested on 5507 participants from the Atherosclerosis Risk in Communities study (ARIC). Clinical relevance was evaluated against existing risk scores and through ability to predict survival and incident multiorgan disease for up to 7 years, across both UK Biobank and ARIC. RESULTS: – In the UK Biobank (mean age 63.27 ± 7.48 years; 53.4% women) our global organ damage score (HyperScore) achieved an area under the curve of 0.964 (0.941–0.987) for identification of individuals with severe end-organ disease and robust stability in cross-validation with a mean root mean square error of 0.104 ± 0.084 . Survival odds differed significantly across HyperScore stages ($P < 0.001$), whereas stratification by blood pressure was nonsignificant. We further revealed 6 hypertensive disease phenotypes (HyperTrajectory), characterized by predominant cardiac, lipoprotein, atherothrombosis, brain, cardiorenal, and liver features, respectively. External testing

in ARIC confirmed stability of the model, with Jensen-Shannon distances as low as 0.10 for HyperScore distributions, without significant deviation in organ damage progression patterns ($P>0.05$) and consistent end-organ and outcome characteristics between ARIC and UK Biobank across HyperTrajectories. **CONCLUSIONS:** – Machine learning–derived global organ damage scores are feasible in hypertension and enable identification of distinct hypertension-associated organ-disease phenotypes. New frameworks for hypertension assessment and monitoring using imaging to derive personalized risk assessment and phenotype-specific intervention may be achievable. © 2026 The Authors.

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Contributions of strength and body composition to the gender difference in anaerobic power

Item Type Journal Article

Author J.L. Mayhew

Author K. Hancock

Author L. Rollison

Author T.E. Ball

Author J.C. Bowen

Abstract Background: The purpose of this study was to determine the significant factors contributing to the difference between men and women in anaerobic power (AP) generation. Methods: Participants: college men (n=113) and women (n=175) were randomly selected for evaluation from a fitness class prior to training. Measures: AP was determined from the Margaria-Kalamen stair run. Dynamic strength was evaluated from the 1-RM single-leg extension. Body composition was estimated from gender-specific skinfold prediction equations. Results: Discriminant analysis identified %fat, fat-free mass (FFM), and leg extension strength as significant components differentiating between men and women and allowed proper gender classification in 99% of the cases. When both strength and body composition variables were held constant by covariance, there was no significant difference between men and women in AP ($F=1.25$). A multiple regression equation using %fat, FFM, and leg extension strength predicted AP ($R=0.91$) with an error of 150 W. When the multiple regression equations was applied to a cross-validation sample, AP could be accurately predicted ($r=0.91$, $t=0.17$) with an average error of $3\% \pm 21\%$. Conclusions: Body size and dynamic strength are the major factors explaining the explosive leg power difference between men and women without a specific or unique gender quality.

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Controlled False Negative Reduction of Minority Classes in Semantic Segmentation

Item Type Conference Paper**Author** R. Chan**Author** M. Rottmann**Author** F. Huger**Author** P. Schlicht**Author** H. Gottschalk

Abstract In semantic segmentation datasets, classes of high importance are oftentimes underrepresented, e.g., humans in street scenes. Neural networks are usually trained to reduce the overall number of errors, attaching identical loss to errors of all kinds. However, this is not necessarily aligned with human intuition. For instance, an overlooked pedestrian seems more severe than an incorrectly detected one. One possible remedy is to deploy different decision rules by introducing class priors that assign more weight to underrepresented classes. While reducing the false negatives of the underrepresented class, at the same time this leads to a considerable increase of false positive indications. In this work, we combine decision rules with methods for false positive detection. Therefore, we fuse false negative detection with uncertainty based false positive meta classification. We present the efficiency of our method for the semantic segmentation of street scenes on the Cityscapes dataset based on predicted instances of the 'human' class. In the latter we employ an advanced false positive detection method using uncertainty measures aggregated over instances. We, thereby, achieve improved trade-offs between false negative and false positive samples of the underrepresented classes. © 2020 IEEE.

Date 2020**DOI** 10.1109/IJCNN48605.2020.9207104**URL** <https://www.scopus.com/pages/publications/85093846173?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc Int Jt Conf Neural Networks**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 978-172816926-2 (ISBN)**Proceedings Title** Proc Int Jt Conf Neural Networks**Conference Name** Proceedings of the International Joint Conference on Neural Networks**Date Added** 20/09/2026, 9:52:28 pm**Modified** 20/09/2026, 9:52:28 pm

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CONTROLLING DIRECTIONS ORTHOGONAL TO A CLASSIFIER

Item Type Conference Paper

Author Y. Xu

Author H. He

Author T. Shen

Author T. Jaakkola

Abstract We propose to identify directions invariant to a given classifier so that these directions can be controlled in tasks such as style transfer. While orthogonal decomposition is directly identifiable when the given classifier is linear, we formally define a notion of orthogonality in the non-linear case. We also provide a surprisingly simple method for constructing the orthogonal classifier (a classifier utilizing directions other than those of the given classifier). Empirically, we present three use cases where controlling orthogonal variation is important: style transfer, domain adaptation, and fairness. The orthogonal classifier enables desired style transfer when domains vary in multiple aspects, improves domain adaptation with label shifts and mitigates the unfairness as a predictor. The code is available at https://github.com/Newbeeer/orthogonal_classifier. © 2022 ICLR 2022 - 10th International Conference on Learning Representations. All rights reserved.

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Controlling factors for the distribution of sapric peat soils extracted by GIS analysis

Item Type Journal Article

Author M. Morishita

Author M. Kawahigashi

Abstract This study aimed to clarify the distribution trend of sapric peat soils in Japanese agricultural fields from the view point of drainage and landform conditions using the Geographic Information System (GIS). To this end, the environmental conditions in the distribution area of organic soils were divided by combining existing GIS databases of environmental factors that consist of groundwater level, drainage condition, landform classification, and grain size of sediment. In order to evaluate the contribution of these environmental factors to the distribution trend of sapric peat soils, we applied quantification method type II to construct the discrimination model that enables the differentiation of the distribution area of sapric and non-sapric peat soils. In the case of organic soils without mineral surface cover, our discrimination model clearly suggests that sapric peat soils tend to be distributed under good drainage condition on floodplains, and with coarse sediments. This indicates that the effect of drainage conditions on peat decomposition is determined by landform condition. On the other hand, in the case of the land covered with mineral soils on the surface, sapric peat soil distribution was not well estimated. Determining the relationships between the distribution trend of sapric peat soils and the environmental factors by analyzing the existing GIS database is a new application. This was achieved by the combination of GIS and quantification method type II, which contributes to the description of the peat decomposition process from the viewpoint of geographical conditions. © 2018, © 2018 Japanese Society of Soil Science and Plant Nutrition.

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Cultivated land, geospatial analysis, Histosols, lowland, quantification method type II

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Cooperative Multi-Model Training for Personalized Federated Learning over Heterogeneous Devices

Item Type Journal Article

Author J. Xu

Author S. Wan

Author Y. Li

Author S. Luo

Author Z. Chen

Author Y. Shao

Author Z. Chen

Author S.-L. Huang

Author L. Song

Abstract Federated learning (FL) is an increasingly popular paradigm for protecting data privacy in machine learning systems. However, the data heterogeneity and high computation cost/latency are challenging barriers for employing FL in real-world applications with heterogeneous devices. In this paper, we propose a novel personalized FL framework named CompFL allowing cooperative training of models with varied structures to mitigate those issues. First, CompFL initializes a set of expert models in varied sizes and allows each client to choose one or multiple expert models for training according to their capacities. Second, CompFL combines the model decoupling strategy and local-global feature alignment to mitigate the adverse impact of label heterogeneity, where clients only share the feature extractor part for each model architecture. Third, to encourage mutual enhancement of various models, knowledge distillation in local training is further applied to improve the overall performance. To make our framework workable in real systems, we implement it in both centralized settings

with server-coordinated parallel training, and decentralized settings with newly developed device-to-device training-forwarding schemes. Extensive experiments on benchmark datasets are conducted to verify the potential of our framework for personalized FL over heterogeneous devices. © 2024 IEEE.

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COPING WITH LABEL SHIFT VIA DISTRIBUTIONALLY ROBUST OPTIMISATION

Item Type Conference Paper

Author J. Zhang

Author A.K. Menon

Author A. Veit

Author S. Bhojanapalli

Author S. Kumar

Author S. Sra

Abstract The label shift problem refers to the supervised learning setting where the train and test label distributions do not match. Existing work addressing label shift usually assumes access to an unlabelled test sample. This sample may be used to estimate the test label distribution, and to then train a suitably re-weighted classifier. While approaches using this idea have proven effective, their scope is limited as it is not always feasible to access the target domain; further, they require repeated retraining if the model is to be deployed in multiple test environments. Can one instead learn a single classifier that is robust to arbitrary label shifts from a broad family? In this paper, we answer this question by proposing a model that minimises an objective based on distributionally robust optimisation (DRO). We then design and analyse a gradient descent-proximal mirror ascent algorithm tailored for large-scale problems to optimise the proposed objective. Finally, through experiments on CIFAR-100 and ImageNet, we show that our technique can significantly improve performance over a number of baselines in settings where label shift is present. © 2021 ICLR 2021 - 9th International Conference on Learning Representations. All rights reserved.

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Correcting a class of complete selection bias with external data based on importance weight estimation

Item Type Conference Paper

Author V.-T. Tran

Author A. Aussem

Editor Arik S.

Editor Huang T.

Editor Lai W.K.

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Abstract We present a practical bias correction method for classifier and regression models learning under a general class of selection bias. The method hinges on two assumptions: (1) a feature vector, X_s , exists such that S , the variable that controls the inclusion of the samples in the training set, is conditionally independent of (X, Y) given X_s ; (2) one has access to some external samples drawn from the population as a whole in order to approximate the unbiased distribution of X_s . This general framework includes covariate shift and prior probability shift as special cases. We first show how importance weighting can remove this bias. We also discuss the case where our key assumption about X_s is not valid and where X_s is only partially observed in the test set. Experimental results on synthetic and real-world data demonstrate that our method works well in practice. © Springer International Publishing Switzerland 2015.

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- Author** N. Pawar
- Author** T.T.J.M. Berendschot
- Author** N.J.C. Bauer
- Author** M. R
- Author** D. Maheshwari
- Author** B.D. Giap
- Author** N. Nallasamy

Abstract In the original version of this article, Tables 1 and 2 were published with an incorrect format. These tables contain only 6 columns; they should instead consist of 12 columns to include the full dataset. The corrected tables are provided below: (Table presented.) (Table presented.) Studies implementing artificial intelligence (AI) for visual screening, prevalence estimation, and myopia classification in pediatric population S.No Author(Year) Country Age(Mean,Range) Study Sample Myopia Definition/CR AI Model Data Modality Concept Result(AUC,Sensitivity,Specificity) Conclusion 1 Piotr R, et al. 2024 Republic of Poland 7–9 yrs Prospective cohort 1049 children Sph -0.75 (D) in at least one eye, CR in clinic CNN Model Combination of standardized VA tests, AR, and assessment of UWF fundus images by CNN model Screening by CNN Model at school and comparing to clinic based pediatric ophthalmologist diagnosis AI model performance- Dice coefficient- 93.3% Reported AI based visual screening.Myopia prevalence 3.7% 2 Shu et al., 2024 China 6–12 yrs Cross sectional 476 children NCR A ConvNeXt Face,ocular images with Mobile Home Screening and Identification of myopia using mobile photos An accuracy of 0.80 (95% CI, 0.78–0.81) and an AUC of 0.84(95% CI, 0.83—0.85) Smartphone images can screen and predict myopia 3 Yang Y et al.,2020 China 6–18 yrs Prospective Clinical trial 2350 eyes SE less than and equal to-0.5D, NCR DCNN- Facial recognition,VGG -face Ocular appearance images captured by camera,phone Screening AUC, sensitivity, and specificity of the DLS—0.9270 (95% CI, 0.8580–0.9610), 81.13% (95% CI, 76.86–95.39%), and 86.42% (95% CI, 82.30–90.54%) respectively, DLS can screen myopia through ocular features 4 Chun et al., 2020 China 6 months to 8 yrs Cross sectional 305 children CR CNN Model(Res Net-50) Photorefracton images Screening andPrediction of myopia by Photorefracton images Accuracy of DL—81.6 Photorefracton images by smart phone can predict myopia 5 Xiao J et al.,2023 China 6-18yrs Prospective 965 school age children NA SVM,Logistic Regression,XG Boost Clinical,Questionnaire set Myopia classification XGBoost best for prediction-AUC 0.764 The four risk factors that have the greatest impact on myopia in children and adolescent are age, vision status of parents, and the

attention paid to children, eye hygiene respectively. Overall prevalence 39.054% 6 Tong ET al., 2023 China Grade 1–12 Retrospective cross sectional 7472 NCR Decision Tree, Random Forest, SVM, KNN, XGBoost, AdaBoost Clinical (Vision screening), questionnaire Myopia classification Overall school RF -AUC 0.752, Junior High school SVM -AUC 0.672 Genetics and eye use behavior essential role in myopia and differences exist in between grades. The prevalence of myopia was 48.51% in high school 7 Tu Y et al., 2022 China 6–10 yrs Cross sectional 16653 students At least one eye has a poor VA (logMAR VA of > 0.0), SE of less than or equal to -0.5 D, CR Five machine-learning models -GBDT, Logistic Regression, DT, RF, and XGBoost Big data obtained from 15013 valid questionnaires with clinical refraction data Myopia classification. Input variables related to myopia. The response variable whether the student displayed myopia Tree-based model, GBDT, ROC 0.92 Higher overall prevalence of myopia 21.52% and older students, urban students, and early education are top 3 contributing factors 8 Guan J et al., 2023 China 11.80(3.07) yrs Retrospective 12,85,609 children SER less than or equal -0.50 D, NCR Random Forest, SVM, KNN, Logistic regression Clinical data and anthropometric data Screen and Predict onset of high myopia Random forest algorithm had high accuracy of 0.948, (AUC)-0.97, External validation AUC 0.97 Age, uncorrected distance visual acuity, and spherical equivalents were predictive factors for high myopia onset in schoolchildren. Overall prevalence of high myopia was 4.48% (2019), 4.88% (2020), and 3.17% (2021) 9 Van Eenwyk ME et al., 2008 USA 6 months -6 years Prospective 610 children NA Four AI techniques-case-based reasoning, case-based fuzzy logic, artificial neural networks, and decision tree Bruckner pupil reflex and eccentric photorefraction Screening of amblyopia due to high myopia Accuracy 45–66 for all models for myopia. Overall DT Accuracy-77 Acceptable cost benefits for detecting amblyogenic factors in children aged 6 months to 6 years, mainly in high myopia Summary of machine learning (ML) and DL studies predicting myopia onset, progression, and axial length estimation, and metabolomic data S.No Author (Year) Country Age (Mean, Range) Study Sample Myopia Definition/CR AI Model Data Modality Concept Result (AUC, Sensitivity, Specificity) Conclusion 1 Lin H et al., 2018 China 6–20 yrs Retrospective 129,242 from 8 ophthalmic centers High myopia SE ≤ -6.0 D, CR Random Forest Clinical data-EHR data Prediction of the development of high myopia The algorithm provided accuracy over 3 years (the AUC 0.940 to 0.985), 5 years (AUC 0.856 to 0.901) ML Improved the prediction of myopia progression 2 Barraza-Bernal JM et al., 2023 China Cross-sectional 5–16 yrs Cross-sectional, Longitudinal 12780 (12554 -cross-sectional, 226-longitudinal) CR SE-0.93, SD = 1.85 girls, -0.88, SD = 1.83 boys Cross-sectional data: GPR- Sq -exponential, GPR- Matern, Longitudinal data: SVM-Quadratic, SVR- linear Clinical from EMR Prediction of spherical power progression with a smaller number of inputs A combination of linear SVR and Matern Gaussian process regression delivered the best results in terms of bias and limits of agreement GPR -Matern R^2 : 0.78, RMSE: 0.87 D SVR-Linear R^2 : 0.77, RMSE: 0.97 D Best performance algorithm uses age, sphere and gender as inputs 3 Yang X et al., 2020 China grade 1–6 Retrospective 3112 NCR GBRT, SVM Clinical, behavioural Prediction model AUC 0.98 Prediction model based on measurement and behavior 4 Li M Set al., 2023 China 7.1(0.4) yrs Prospective 2740 baseline followed for 5 years CR Random Forest Clinical Prediction When the absolute error between predicted and actual myopia progression was set at 0.50 D, the prediction accuracy was 80% UDVA and SE were best predictor 5 Lingham G et al., 2024 China (Data from Ireland, Northern Ireland and China) 4–22 yrs Population based -Ireland Eye Study, NICER, refractive error study from Zhejiang 8135 SER less than or equal -0.50 D, CR Linear Regression, symbolic regression, gradient boosting and multilayer perceptron models Clinical To estimate AL by MLM by using age, sex, SER The model better at identifying high-risk eyes (axial length > 98 th centile) than SER alone (AUC 0.89 vs 0.79, respectively)

Superior axial length estimation with a linear regression model 6 Peng W, et al. 2023 China 9.06(1.67) yrs Prospective cohort Out of 2221,260 developed myopia SE less than $-0.5D$, CR RF, SVM, GB, Catboost, Logistic Regression Clinical, Anthropometric, behavioural Prediction of onset of myopia Catboost algorithm - highest AUC of 0.951 Daily information is reliable predictors for childhood myopia 7 Huang J et al., 2023 Chengdu, China 6–20 yrs Retrospective 37,586 school children and adolescents SER less than or equal $-0.50D$, NCR RNN, (T-LSTM, LSTM) Random Forest, Logistic Regression Clinical Quantitative prediction of school going children adolescents, SE based on EMR records T-LSTM MAE-0.103(0.140) Time-Aware Long Short-Term Memory captured real data with higher applicability 8 Hou XW et al., 2023 China 4–18 yrs 4–18 yrs Part of Large-scale refractive eye development study, Training set = 516, Validation set = 60 HM $> 6.0 D$ for 6-8 yrs (myopic) $> 8.0 D$ for 9 yrs and older CR Lasso, Random Forest, XGBoost Clinical and Serum metabolic data To characterize serum metabolic profile of myopic retinopathy Logistic regression (AUC 0.842, 95% CI 0.724–0.96) and XGBoost (AUC 0.897, 95% CI 0.807–0.986) Good predictive performance of myopia based on metabolic profiles 9 Ying et al., 2024 China 5–18 yrs Cross sectional, 1938 training, 1476 validation CR RF, XGBoost, MLP, SVM, Linear, LASSO for SER Clinical data To develop and validate ML to predict SE XG Boost best for SER (Sensitivity 91.1%, Specificity 97.2%) Non cycloplegic parameters can predict cycloplegic SER 10 Du et al., 2023 China 6–18 yrs Retrospective N = 2,467 students NCR SVR, Random Forest, DNN Clinical data To develop and validate ML to predict SE The model for prediction of SE had R^2 0.8892 to 0.927, MSE 0.250–0.380, MAE 0.37–0.436, accuracy—80.32–81.7% and F1 Score 80.3–81.7% The difference between before and after cycloplegia can be predicted in children 11 Zhao et al., 2024 China 3–18 yrs Retrospective CR = 88,250 $\leq -0.5D$ (CR) RF, XGBoost Clinical data To predict the SE and onset of myopia and high myopia XGBoost outperformed RF over 15 years with R^2 more than 0.79 and MAE from 0.078 to 1.802 in the test set Myopia onset showed AUC 0.845 to 0.953 and high myopia onset AUC 0.807–0.997 XGBoost model had high accuracy in predicting SE—onset of myopia and high myopia 12 Li et al., 2024 China 8.30 SD 0.94 Longitudinal 50512 SE less than $-0.5D$, CR XGBoost, KNN, DT, Logistic Regression, Gaussian NB model Clinical data and Questionnaire To develop the model to predict myopia progression and risk of developing high myopia XGBoost AUC = 0.96 for predicting Annual Myopia Progression Baseline refraction is the most important factor in predicting high myopia at an early age. Myopia progression is affected by age, sex, and myopia severity 13 Mu et al., 2024 China 6–12 yrs Case control study 2947 elementary school student NCR SE $\leq -0.5D$ RF, XGBoost, DT, SVM, Logistic Regression Clinical data and Questionnaire To evaluate myopia prediction model AUC of the ROC curve > 0.8 for all models except the DT (AUC = 0.791). SVM had highest accuracy (0.844, Precision 0.851, Recall 0.812, F1-Score 0.831 in testing set AL, Sex, Age maternal myopia and feeding pattern were most important myopia risk factor 14 Bushra et al., 2020 Sydney Retrospective Data form OLSM 618 samples NCR Logistic, MLP, NB, KNN, DT, Random Forest, SVM Clinical data Classification of myopia in children MLP had highest accuracy among all models Optimization of prediction accuracy by selecting important features through RFE with tree based classifier can be done 15 Hao et al., 2025 China school-age Retrospective N = 2,483 eyes CR Random Forest Clinical Data Prediction of SE, Classification of myopia The regression model internal validation showed an out-of-bag validation R^2 of 0.86, RMSE of 0.66, and MAE of 0.49. In the external validation, R^2 —0.88, RMSE—0.63, and the MAE was 0.48 The regression model accurately predicts post-cycloplegia myopic correction 16 Varosanec et al., 2024 Croatia 4–18 yrs Retrospective 895 CR Time aware LSTM Clinical Data To predict SE/Variable length historical vision records Testing set had MAE of 0.10 SD 0.10 Extended gate time aware LSTM can be used to quantitatively

predict children's and adolescents' SE within 7 years 17 Zhu et al.,2023 China 6–9 yrs Retrospective 179 NCR MLP, KR Extra tree, OMPL Clinical Data To predict changes in SER and AL MLP, R2 0.899, for predicting AL In predicting SER, OMP model perform best; in predicting AL, theKR and MLP models better than other models 18 Xu S et al., 2024 China Preschoolers Prospective 650,671 NCR XG Boost,RF,DBT,CatBoost,KNN,SVM Pupildiameter and clinical data Association of pupil diameter with myopia AUC XG Boost-0.972,RF-0.967 Preschoolers with myopia had a reduced pupil diameter 19 Wei R et al.,2024 China 7–9 yrs School based cohort 1997, Analysis1667 SE less than or equal-0.50D 4 DLM (Two UNet-varying training seeds,UNet-residual block, SegResNet FP(45degree view),Hand held fundus camera Association of tessellation density with AL and SE,quantification of TD by AI Dice coefficient—UNet-1 0.4601,UNet-2 0.4704,UNet-Res 0.4848,SegResNet 0.4884 TD potential indicator of rapid elongation of AL 20 Foo et al.,2023 Singapore 6–12 yrs School cohort 965 SE less than or equal -6D,CR DenseNet- 121 FP Image, Clinical, and Mix(FP) Single-time point objective data and fundus imaging to predict the 5-year development of high myopia Primary dataset AUC 0.93; Test dataset 0.91,0.93), clinical models (Primary dataset AUC 0.90,0.97; Test dataset 0.93,0.94) and mixed (image + clinical) models (Primary dataset AUC 0.97; Test dataset 0.97,0.98.) Baseline fundus image can predict at risk of high myopia early 21 Qi et al.,2024 China 4–14 yrs DeepMyopia Study,Multicenter cohort 1,638,315 participants,14,802 participants with images SE and AL thresholds, NCR ResNet -50 and Ensemble ML Fundus images + biometric data (AL, SE, age, sex) Predicting myopia onset, high myopia, and progression Myopi onset prediction: Internal AUC 0.81- 0.91, For 1–3 Year prediction, External AUC 0.77–0.81, Progression prediction > 0.9 Multimodal imaging and biometry predict onset, improve long-term myopia prediction and risk stratification 22 Gong et al., 2024 China 4–18 yrs Cross-sectional 14,324 SE less than or equal-0.50D,NCR ResNet-FCN Fundus Photography (macular region 6 mm) Validation of FTD as a predictive biomarker for myopia progression AUC for macular FTD segmentation: 0.774 Quantification of fundus tessellation detect early myopic maculopathy 23 Huang Y et al.,2025 China 6–15 yrs Cross-sectional 1942 SE less than or equal to -0.50D,CR ResNet-FCN Fundus Photography (macular region) Quantification of macular and peripapillary FTD as a predictive biomarker for myopia progression AUC for macular FTD segmentation: 0.774 Macular FTD is a clinically useful biomarker associated with large AL and high SE 24 Liu F et al.,2023 China 6–12 yrs Retrospective case controlled 226 eyes (113 children) SE less than or equal-0.50D NCR DLS-ResNet-101-UNet FP Explore PPA changes The accuracy of feature recognition for each result—0.96, sensitivity 0.85, and specificity 0.96 In school-age children with high myopia, the nasal and temporal neuroretinal rims were narrowed and even lost, with high sensitivity 25 Huang D et al., 2023 China 7.44(0.28)yrs Population based cohort,Cross -sectional 577 NCR DLS-ResNet-18 FP,45 degree non mydriatic Fundus photograph,OCTA (Optovue RTVueXR) To explore association of FTD with various factors(SFCT,RNFL,PPA,DFD,vessel density inside optic disc) The accuracy—0.9652, the sensitivity -0.7247, and the specificity- 0.9605 FTD can be applied as a quick quantitative biomarker to estimate SFCT and myopia in primary school children 26 Li et al.,2020 China 6.3 (0.5) years at baseline School cohort 382 NCR SVM, RBF kernel implemented FP PPA detection and quantification and to assess association between PPA and myopia The accuracy of PPA identification— 90.78%. F1-score of PPA segmentation—0.67 The width of PPA predict progression of children myopia 27 Huang D et al., 2023 China 7 yrs Population based cohort,Cross- sectional 1062 NCR DL based Envision AI FP To investigate the prevalence of FTD FTD can be applied for rapid screening The original article has been corrected. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026.

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Correlation of HER2 levels expression through HER2 mRNA PCR vs immunohistochemistry (IHC) in hormone receptor positive (HR+)/HER2 negative (HER2-) early breast cancer

Item Type Journal Article**Author** A.L. de Sá**Author** J. Tejerina**Author** M. Amann**Author** P. Ballestín**Author** B. González**Author** A. Pascual**Author** C.D. Del Arco**Author** V. García-Barberán**Author** M. Paz-Cabezas**Author** A. De Luna**Author** J.Á. García-Sáenz**Author** F. Moreno

Abstract Purpose: To evaluate the correlation between HER2 expression levels measured by HER2 mRNA using Oncotype DX and by immunohistochemistry (IHC) in hormone receptor-positive (HR+) and HER2-negative (HER2-) early breast cancer. In addition, we assessed whether low HER2 expression is associated with distinct clinicopathological characteristics and prognosis in our series. Methods: We conducted a retrospective study that included 500 patients diagnosed with stage I–III HR+/HER2- breast cancer who underwent surgery and had Oncotype DX recurrence score determined between 2009 and 2023 at Hospital Clínico San Carlos, Madrid, Spain. HER2 mRNA levels obtained through Oncotype DX were compared across IHC groups (HER2 0+, HER2 1+, HER2 2+/ISH-negative). Event-free survival (EFS) was analyzed according to HER2 expression. Results: Although HER2 mRNA levels increased with higher IHC HER2 categories, variability and overlap were observed between subgroups. Median Oncotype DX recurrence scores also rose slightly across HER2 IHC groups but did not reach statistical significance. EFS did not differ between HER2 expression levels. Conclusions: We found that HER2 mRNA measurement by Oncotype DX provides a quantitative approach to assess HER2 expression. However, its results overlap within traditional IHC categories. While HER2-low classification may have therapeutic implications for new antibody–drug conjugates, its prognostic relevance appears limited. Further studies are needed to improve HER2 quantification methods for improved clinical decision-making. © The Author(s) 2025.

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adult, Adult, Article, female, Female, human, Humans, middle aged, Middle Aged, aged, Aged, Aged, 80 and over, very elderly, comparative study, major clinical study, follow up, cohort analysis, Retrospective Studies, retrospective study, genetics, polymerase chain reaction, human tissue, pathology, metabolism, anthracycline, Antibody–drug conjugates, Biomarkers, Tumor, Breast Neoplasms, breast tumor, cancer chemotherapy, cancer prognosis, Early breast cancer, early cancer, epidermal growth factor receptor 2, Erb-b2 Receptor Tyrosine Kinases, ERBB2 protein, human, estrogen receptor, event free survival, gene overexpression, HER2 low, hormone receptor-positive, HER2-negative breast cancer, immunohistochemistry, Immunohistochemistry, invasive lobular breast carcinoma, messenger RNA, mortality, Neoplasm Recurrence, Local, Oncotype Dx, Polymerase Chain Reaction, progesterone receptor, prognosis, Prognosis, Receptors, Estrogen, Receptors, Progesterone, RNA, Messenger, tumor marker, tumor recurrence, tumor volume

Notes:

EID: 2-s2.0-105013782260

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: F. Moreno; Servicio de Oncología Médica, Hospital Clínico San Carlos, Instituto de Investigación Sanitaria Hospital Clinico San Carlos (IdISSC), Madrid, C/Professor Martin Lagos S/N, 28040, Spain; email: laventi2002@yahoo.es

Corrosion degree detection of ferromagnetic materials using a hybrid machine learning approach and self-magnetic flux leakage technology

Item Type Journal Article

Author H. Zhang

Author H. Li

Author R. Xia

Author T. Hu

Author J. Qiu

Author J. Zhou

Abstract Corrosion of ferromagnetic materials can lead to performance degradation or even failure in engineering structures. Therefore, detecting the degree of corrosion in these materials is crucial. Traditional detection methods are primarily manual and lack precision, highlighting the need for developing automated, high-precision detection techniques. In this paper, we developed a novel self-magnetic flux leakage (SMFL) scanner. We enhanced the Marine Predator Algorithm (MPA) by integrating Chaos Opposition (CO) and Group Learning (GL), resulting in the COGL-MPA-XGB model. This model optimizes Extreme Gradient Boosting parameters for accurate corrosion prediction, achieving a 96% coefficient of determination (R^2) and a mean error of 1.87. We tested the method on 170 rod-shaped steel samples with varying degrees of corrosion, evaluating its effectiveness through ablation experiments and comparisons with three other machine learning models. The results show that the combined SMFL and hybrid model strategy reduces the maximum detection error by

73.7% and offers a new perspective for quantitative corrosion detection. © 2024 Elsevier Ltd

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Cost Volume Aggregation in Stereo Matching Revisited: A Disparity Classification Perspective

Item Type Journal Article

Author Y. Wang

Author L. Wang

Author K. Li

Author Y. Zhang

Author D.O. Wu

Author Y. Guo

Abstract Cost aggregation plays a critical role in existing stereo matching methods. In this paper, we revisit cost aggregation in stereo matching from disparity classification and propose a generic yet efficient Disparity Context Aggregation (DCA) module to improve the performance of CNN-based methods. Our approach is based on an insight that a coarse disparity class prior is beneficial to disparity regression. To obtain such a prior, we first classify pixels in an image into several disparity classes and treat pixels within the same class as homogeneous regions. We then generate homogeneous region representations and incorporate these representations into the cost volume to suppress irrelevant information while enhancing the matching ability for cost aggregation. With the help of homogeneous region representations, efficient and informative cost aggregation can be achieved with only a shallow 3D CNN. Our DCA module is fully-differentiable and well-compatible with different network architectures, which can be seamlessly plugged into existing networks to improve performance with small additional overheads. It is demonstrated that our DCA module can effectively exploit disparity class priors to improve the performance of cost aggregation. Based on our DCA, we design a highly accurate network named DCANet, which achieves state-of-the-art performance on several benchmarks. © 1992-2012 IEEE.

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human, Performance, article, controlled study, Depth Estimation, major clinical study, benchmarking, classification, Matchings, Improve performance, Cost aggregations, cost volume, Cost volume, depth estimation, disparity classification, Disparity classification, Homogeneous regions, pharmacoeconomics, Stereo image processing, Stereo matching, Stereo matching method, Stereo-matching

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Cost-sensitive support vector machines

Item Type Journal Article

Author A. Iranmehr

Author H. Masnadi-Shirazi

Author N. Vasconcelos

Abstract Many machine learning applications involve imbalance class prior probabilities, multi-class classification with many classes (often addressed by one-versus-rest strategy), or “cost-sensitive” classification. In such domains, each class (or in some cases, each sample) requires special treatment. In this paper, we use a constructive procedure to extend SVM's standard loss function to optimize the classifier with respect to class imbalance or class costs. By drawing connections between risk minimization and probability elicitation, we show that the resulting classifier guarantees Bayes consistency. We further analyze the primal and the dual objective functions and derive the objective function in a regularized risk minimization framework. Finally, we extend the classifier to the with cost-sensitive learning with example dependent costs. We perform experimental analysis on class imbalance, cost-sensitive learning with given class and example costs and show that proposed algorithm provides superior generalization performance, compared to conventional methods. © 2019

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DOI 10.1016/j.neucom.2018.11.099

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Tags:

human, human experiment, Support vector machines, Classification (of information), Learning systems, article, Class imbalance, learning, support vector machine, Classification, SVM, Cost benefit analysis, probability, Multi-class classification, Generalization performance, Cost-sensitive learning, Machine learning applications, loss of function mutation, Bayes consistency, Cost-sensitive support vector machines, drawing, Dual objective function

Notes:

EID: 2-s2.0-85061674625

Export Date: 20 September 2026; Cited By: 211; Correspondence Address: A. Iranmehr; Department of Electrical and Computer Engineering, University of California San Diego, La Jolla, 92039, United States; email: airanmehr@ucsd.edu; CODEN: NRCGE

Counting dense object of multiple types based on feature enhancement

Item Type Journal Article

Author Q. Fu

Author W. Min

Author W. Sheng

Author C. Peng

Abstract Introduction: Accurately counting the number of dense objects in an image, such as pedestrians or vehicles, is a challenging and practical task. The existing density map regression methods based on CNN are mainly used to count a class of dense objects in a single scene. However, in complex traffic scenes, objects such as vehicles and pedestrians usually exist at the same time, and multiple classes of dense objects need to be counted simultaneously. Methods: To solve the above issues, we propose a new multiple types of dense object counting method based on feature enhancement, which can enhance the features of dense counting objects in complex traffic scenes to realize the classification and regression counting of dense vehicles and people. The counting model consists of the regression subnet and the classification subnet. The regression subnet is primarily used to generate two-channel predicted density maps, mainly including the initial feature layer and the feature enhancement layer, in which the feature enhancement layer can enhance the classification features and regression counting features of dense objects in complex traffic scenes. The classification subnet mainly supervises classifying dense vehicles and people into two feature channels to assist the regression counting task of the regression subnets. Results: Our method is compared on VisDrone+ datasets, ApolloScape+ datasets, and UAVDT+ datasets. The experimental results show that the method counts two kinds of dense objects simultaneously and outputs a high-quality two-channel predicted density map. The counting performance is better than the state-of-the-art counting network in dense people and vehicle counting. Discussion: In future work, we will further improve the feature extraction ability of the model in complex traffic scenes to classify and count a variety of dense objects such as cars, pedestrians, and non-motor vehicles. Copyright © 2024 Fu, Min, Sheng and Peng.

Date 2024

DOI 10.3389/fnbot.2024.1383943**URL** <https://www.scopus.com/pages/publications/85194764916?origin=resultslist>**Archive** Scopus**Language** English**Volume** 18**Publisher** Frontiers Media SA**Publication** Frontiers in Neurorobotics**ISSN** 16625218 (ISSN)**Journal Abbr** Front. Neurorobotics**Date Added** 20/09/2026, 9:51:54 pm**Modified** 20/09/2026, 9:51:54 pm**Tags:**

Vehicles, Article, car, Classification (of information), traffic, pedestrian, Pedestrian safety, Complex networks, image segmentation, Traffic scene, Regression analysis, correlation analysis, training, feature extraction, artificial neural network, convolutional neural network, classification, unsupervised machine learning, computer model, learning algorithm, sensitivity analysis, regression analysis, probability, motor vehicle, root mean squared error, mean absolute error, spatial analysis, residual neural network, ablation therapy, Vehicle counting, clasubnet unsupervised machine learning, crowd counting, Crowd counting, Dense object of multiple type, dense object of multiple types, density map regression, Density map regression, Density maps, Enhancement Layers, feature enhancement, Feature enhancement, kernel method, regsubnet unsupervised machine learning, Subnets, Two channel, vehicle counting

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Coupling positive matrix factorization with machine learning to quantify contamination sources and map health risks of heavy metals in karst soils

Item Type Journal Article**Author** Z. Liu**Author** Y. Zeng**Author** W. Wang**Author** B. Zhang

Author J. Liu

Author X. Zhao

Abstract Heavy metal contamination in karst ecosystems poses great environmental risks due to the interplay between high geochemical backgrounds and anthropogenic activities. This study performed an integrated source-to-risk assessment of soil metals based on 674 topsoil samples from Guangxi karst region in China, combining three machine learning models including positive matrix factorization (PMF), random forest (RF), and multiscale geographically weighted regression (MGWR). Geochemical characterization identified SiO₂ (63%), Al₂O₃ (18%), and Fe₂O₃ (6.4%) as dominant oxides in tested soils, with higher organic matter (3.3% content) than regional background value. A majority of 36 examined metals displayed concentrations exceeding local background values, of which Cd and Hg exhibited the highest enrichment levels (3.83 and 6.75 folds of background values, respectively). PMF revealed distinct source patterns across land-use types, with natural weathering dominating in forest land, and agricultural inputs prevailing in grasslands. Riparian land was primarily influenced by traffic-related anthropogenic activities, while construction land was dominated by industrial and mining/smelting sources. PMF-RF-MGWR modeling demonstrated pronounced spatial heterogeneity with contamination hotspots being mainly concentrated in industrial- and mining-intensive areas of central-western Guangxi. Notably, karst weathering increased soil metal mobility, while secondary carbonate mineral formation contributed to immobilization of Cd and Pb. Health risk assessment identified As, Cr and Ni as priority contaminants, and their presence in industrial zones posed unacceptable carcinogenic risks. Our findings underscore the role of the interplay between geological processes and anthropogenic activities in shaping metal distribution, providing valuable references for targeted soil management and contaminated land remediation in karst regions. © 2026 The Authors.

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Tags:

risk assessment, machine learning, Machine learning, China, heterogeneity, Cd, Hg, As, Cr, and Ni, contaminated land, environmental risk, Guangxi Zhuangzu, health risk, Health risk, heavy metal, Heavy metals, High geological background, karst, Karst soils, Metal enrichment, PMF-RF-MGWR model, Positive matrix factorization, Soil health, soil management, source apportionment, Source apportionment, Spatial heterogeneity

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Courtyard design for energy efficiency and thermal comfort: machine learning insights across hot and warm climates

Item Type Journal Article**Author** M.A. Aloshan**Author** K.M. Aldali

Abstract Courtyards, long used as a passive climate-control element in vernacular housing, are increasingly reinterpreted in modern residential design, yet their quantified impact on energy use and comfort remains poorly documented. This study investigates how courtyard design affects energy efficiency and thermal comfort in villas across five hot and warm climates: Cairo, Riyadh, Gizan, Miami, and Sevilla. Using nearly 29,000 parametric simulations, design variables (geometry, window-to-wall ratio (WWR), glazing, insulation, and orientation) were systematically tested. Random Forest models achieved $R^2 > 0.99$ for both annual cooling demand and discomfort hours, with Shapley Additive explanations (SHAP) analysis revealing variable importance. Results show that compact courtyards with low WWR, external insulation, and high-performance glazing reduce cooling demand under 75 kWh/m² year. In contrast, elongated courtyards with higher WWR reduce discomfort to under 70 h annually in hot-dry climates, though often at the cost of increased cooling demand in hot-humid regions. Courtyard dimensions accounted for less than 3% of outcome variance, compared with > 40% for WWR and insulation. This study combines large-scale simulation with interpretable machine learning to quantify courtyard performance across climates. The findings provide climate-specific, evidence-based design guidance and demonstrate a pathway toward performance-based building codes that integrate courtyard design for hot regions. © The Author(s) 2025.

Date 2025**DOI** 10.1038/s41598-025-32297-z**URL** <https://www.scopus.com/pages/publications/105024724615?origin=resultslist>**Archive** Scopus**Language** English**Volume** 15**Publisher** Nature Research**Publication** Scientific Reports**ISSN** 20452322 (ISSN)

Issue 1**Journal Abbr** Sci. Rep.**Date Added** 20/09/2026, 9:51:51 pm**Modified** 20/09/2026, 9:51:51 pm**Tags:**

adult, human, male, simulation, machine learning, Energy efficiency, Machine learning, article, normal human, controlled study, comfort, random forest, housing, explainable machine learning, Shapley additive explanation, climate, cooling, energy, geometry, productivity, Climate-responsive, construction work and architectural phenomena, Courtyard design, Residential villa, Thermal comfort

Notes:

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Covid-19 Pneumonia Chest X-Ray Pattern Synthesis by Stable Diffusion

Item Type Conference Paper**Author** Z. Liang**Author** Z. Xue**Author** S. Rajaraman**Author** S. Antani

Abstract In this study, we fine-tuned a stable diffusion model to synthesize high resolution chest X-ray images (512x512) with bilateral lung edema caused by COVID-19 pneumonia using the class-specific prior preservation strategy. 300 positive images were selected from the MIDRC dataset as subject instances with an additional 400 negative images for class prior preservation. We synthesized images respectively using the new technique and the conventional technique for comparison. The synthetic images by the stable diffusion fine-tuned by the prior preservation technique have the Frechet inception distance (FID) of 9.2158 and kernel inception distance (KID) 0.0818 computed with the real positive images, which is superior to the synthetic images using the conventional methods such as WGAN and DDIM. The classification accuracy is 0.9975 with precision of 1.0 and recall of 0.9950 when the synthetic positive images with the real negative images were classified by a trained vision transformer (ViT). We conclude that the stable diffusion model can synthesize high-quality and high-resolution chest x-ray images using the prior preservation strategy with a small number of real images as subject instances and text prompt as guidance for the designated patterns. © 2024 IEEE.

Date 2024

DOI 10.1109/SSIAI59505.2024.10508671**URL** <https://www.scopus.com/pages/publications/85192554706?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc IEEE Southwest Symp Image Anal Interpret**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 15505782 (ISSN); 979-835036011-0 (ISBN)**Pages** 21-24**Proceedings Title** Proc IEEE Southwest Symp Image Anal Interpret**Conference Name** Proceedings of the IEEE Southwest Symposium on Image Analysis and Interpretation**Date Added** 20/09/2026, 9:52:05 pm**Modified** 20/09/2026, 9:52:05 pm**Tags:**

COVID-19, Diffusion, Diffusion model, High resolution, chest x-ray, Chest X-ray image, Chest x-rays, image synthesis, Images synthesis, latent diffusion model, Latent diffusion model, Negative image, Preservation strategies, prior preservation, Prior preservation, stable diffusion, Stable diffusion

Notes:

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cPU: Consistent Risk Estimator for Positive-Unlabeled Learning

Item Type Journal Article**Author** Y. Zhang**Author** K. Wang**Author** J. Tang**Author** Y. Zhang**Author** D. Liang

Abstract As one of the classical strategies for positive-unlabeled (PU) learning, the cost-sensitive methods achieve binary classification by minimizing the overall risk. To achieve this goal, the class prior is usually exploited to guide model learning in many state-of-the-art cost-sensitive approaches. However, how to effectively mine the underlying supervision information using the class prior remains a major challenging problem. It is known that the distribution of positive and negative samples is fixed in unlabeled samples when the class prior is available. Based on this knowledge, we derive a novel supervision formulation from a risk perspective: if the class prior is known, the ratio between the positive risk distribution of negative samples and the

negative risk distribution of positive samples converges to a fixed value in unlabeled samples. By fully exploiting this brand-new constraint, we construct a consistent risk estimator to impose consistency between the negative expected risk of the positive samples and the positive expected risk of the negative samples. Moreover, we introduce Mixup regularization to mitigate the confirmation bias of the learned model, and meanwhile employ the entropy minimization technique to improve the separability of samples, which can further enhance classification performance. Experimental results on four benchmark datasets demonstrate the advantage of our model over several baseline methods. © 2012 IEEE.

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Risk perception, Classification (of information), Learning systems, Costs, Benchmarking, Negative samples, Unlabeled samples, Cost-sensitive, Cost-sensitive learning, Expected risk, generalization bound, Generalization bound, Positive-unlabeled learning, positive-unlabeled (PU) learning, Risk distribution, risk estimator, Risk estimators, Sensitive method

Notes:

EID: 2-s2.0-105046073471

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: J. Tang; Anhui University, School of Electronic and Information Engineering, Hefei, 230039, China; email: tangjunahu@163.com

Cross-domain transfer fault diagnosis by class-imbalanced deep subdomain adaptive network

Item Type Journal Article

Author J. Zhou

Author X. Zhang

Author H. Jiang

Author J. Li

Author Z. Shao

Abstract In variable working conditions and class imbalances, fault diagnosis in gearbox systems becomes difficult due to the difficulty in learning the classification boundaries of the minority class. Moreover, the diversity in fault mode distributions under various working conditions poses significant challenges for domain adaptation. To this end, this work proposed a Class-imbalanced Deep Subdomain Adaptive Network (CIDSAN) for complex cross-domain fault diagnosis scenarios involving feature and label shifts. We design a convolutional neural network embedded with SimAM (SimAM-CNN) to extract global feature information from time-domain vibration signals. Subsequently, a novel cost-sensitive classifier is presented for class re-weighting, along with a Joint Supervised Classifier that refines intra-class and inter-class distances by re-weighting class weights in the latent space using cost-sensitive classifiers while leveraging LDAM regularization to adjust classification boundaries. Diagnostic case studies on two datasets under various working conditions validate the effectiveness and superiority of the proposed approach. The code <https://github.com/Zjy0121/CIDSAN>. © 2024 Elsevier Ltd

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Notes:

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Cross-Lingual Sentiment Quantification

Item Type Journal Article

Author A. Esuli
Author A. Moreo
Author F. Sebastiani
Author E. Cambria

Abstract Sentiment Quantification is the task of estimating the relative frequency of sentiment-related classes-such as $\{\text{Positive}\}$ Positive and $\{\text{Negative}\}$ Negative-in a set of unlabeled documents. It is an important topic in sentiment analysis, as the study of sentiment-related quantities and trends across a population is often of higher interest than the analysis of individual instances. In this article, we propose a method for cross-lingual sentiment quantification, the task of performing sentiment quantification when training documents are available for a source language $\mathcal{S}$, but not for the target language $\mathcal{T}$, for which sentiment quantification needs to be performed. Cross-lingual sentiment quantification (and cross-lingual text quantification in general) has never been discussed before in the literature; we establish baseline results for the binary case by combining state-of-the-art quantification methods with methods capable of generating cross-lingual vectorial representations of the source and target documents involved. Experiments on publicly available datasets for cross-lingual sentiment classification show that the presented method performs cross-lingual sentiment quantification with high accuracy. © 2001-2011 IEEE.

Date 2020

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Tags:

Classification (of information), Sentiment analysis, Sentiment classification, State of the art, Target language, Quantification methods, Relative frequencies, Baseline results, Training documents, Unlabeled documents

Notes:

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CROSS-LINGUAL TRANSFER WITH CLASS-WEIGHTED LANGUAGE-INVARIANT REPRESENTATIONS

Item Type Conference Paper

Author R. Xian

Author H. Ji

Author H. Zhao

Abstract Recent advances in neural modeling have produced deep multilingual language models capable of extracting cross-lingual knowledge from non-parallel texts and enabling zero-shot downstream transfer. While their success is often attributed to shared representations, quantitative analyses are limited. Towards a better understanding, through empirical analyses, we show that the invariance of feature representations across languages-an effect of shared representations-strongly correlates with transfer performance. We also observe that distributional shifts in class priors between source and target language task data negatively affect performance, a largely overlooked issue that could cause negative transfer with existing unsupervised approaches. Based on these findings, we propose and evaluate a method for unsupervised transfer, called importance-weighted domain alignment (IWDA), that performs representation alignment with prior shift estimation and correction using unlabeled target language task data. Experiments demonstrate its superiority under large prior shifts, and show further performance gains when combined with existing semi-supervised learning techniques. © 2022 ICLR 2022 - 10th International Conference on Learning Representations. All rights reserved.

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Language English

Extra Journal Abbreviation: ICLR - Int. Conf. Learn. Represent.

Publisher International Conference on Learning Representations, ICLR

Proceedings Title ICLR - Int. Conf. Learn. Represent.

Conference Name ICLR 2022 - 10th International Conference on Learning Representations

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Language model, Empirical analysis, Learning systems, Supervised learning, Modeling languages, Cross-lingual, Zero-shot learning, Down-stream, Target language, Neural modelling, Feature representation, Invariant representation, Parallel text, Shared representations

Notes:

EID: 2-s2.0-85144973495

Export Date: 20 September 2026; Cited By: 4; Conference name: 10th International Conference on Learning Representations, ICLR 2022; Conference date: 25 April 2022 through 29 April 2022; Conference code: 186704

Cross-Modal Learning for CTC-Based ASR: Leveraging CTC-Bertscore and Sequence-Level Training

Item Type Conference Paper

Author M.-H. Lee

Author S.-E. Lee

Author J.-E. Choi

Author J.-H. Chang

Abstract Due to the nature of neural networks that easily overfit the training set, neural network-based speech recognition models are vulnerable to prior shifts in data distribution or unseen words. Therefore, studies have been conducted to overcome this problem by using language models trained with a relatively easy-to-obtain unpaired corpus. In this paper, we present a new training method that uses BERT to improve the performance of a connectionist temporal classification (CTC)-based ASR model. The proposed method follows a cross-modal learning scenario and induces the CTC model to better embed contextual information by utilizing an auxiliary objective function operating at the sequence level. We applied the proposed method to fine-tune the pre-trained wav2vec 2.0 model with CTC loss and confirmed that the proposed method improves the generalization performance of the ASR model. © 2023 IEEE.

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CSNF-TimeGAN: Class-prior spatiotemporal nonlinear feature-based time series generative adversarial network for blast furnace fault diagnosis

Item Type Journal Article**Author** Y. Yang**Author** C. Yang**Author** S. Lou**Author** D. Gao**Author** X. Zhang**Author** W. Wang

Abstract Fault diagnosis for the blast furnace ironmaking process (BFIP) is vital to ensuring production stability and efficiency. However, it is challenged by the complex interplay of nonlinearity, dynamic behavior, spatial correlations, and a significant scarcity of labeled fault data. To overcome these limitations, a novel fault diagnosis method based on class-prior spatiotemporal nonlinear feature-based time series generative adversarial network (CSNF-TimeGAN) is proposed. First, we develop the class-prior spatial improved mixed kernel canonical variate analysis (CSI-MKCVA) to extract class-prior spatiotemporal nonlinear features (CSNF). Unlike existing kernel CVA variants, CSI-MKCVA introduces a novel mixed kernel architecture integrating radial basis, polynomial, and cosine kernels to capture local, global, and temporally correlated nonlinearities in a unified manner. The architecture is further enhanced by incorporating class-prior information and Laplacian regularization into covariance estimation, preserving the spatiotemporal dynamics while maximizing fault separability. An efficient alternating Lagrangian optimization algorithm ensures robust convergence. Second, CSNF-TimeGAN is proposed as a diagnosis-oriented generative framework that bridges advanced feature representation and synthetic data generation. Operating a Wasserstein GAN with gradient penalty in the CSNF space, the model achieves stable distribution matching. It innovates theoretically through a temporal consistency loss and an adaptive loss informed by classifier feedback, as formalized in new equations that mitigate mode collapse and enhance discriminability beyond standard GANs. Subsequently, an integrated fault diagnosis mechanism is designed, synergizing feature representation, data generation, and classification into a cohesive workflow. In deployment, real-time diagnosis leverages the unified CSNF feature extractor and optimized classifier, ensuring computational efficiency. Finally, CSNF-TimeGAN demonstrates leading-edge performance through validation on real BFIP operational data. © 2025 Elsevier Ltd.

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D-optimal candexch algorithm-enhanced machine learning UV-spectrophotometry for five-analyte determination in novel anti-glaucoma formulations and ocular fluids: four-color sustainability framework with NQS assessment and UN-SDG integration

Item Type Journal Article

Author O. Al kamaly

Author L.A. Al-Khateeb

Author M.K. Halim

Author N.S. katamesh

Author G. Magdy

Author A.E.F. Abbas

Abstract The novel anti-glaucoma ophthalmic preparation containing latanoprost, netarsudil, and benzalkonium chloride has posed a significant challenge due to its complexity and the lack of environmentally sustainable quantification methods, with only a single published method available for its quantification that lacks environmental

consideration. This study aims to address this crucial gap by presenting a novel and sustainable approach using machine learning-enhanced UV-spectrophotometric chemometric models for the concurrent quantification of latanoprost, netarsudil, benzalkonium chloride, and two related compounds in ophthalmic preparations and aqueous humour. A strategic multi-level, multi-factor experimental design creates a 25-mixture calibration set for four models (PLS, GA-PLS, PCR, and MCR-ALS). The key novelty was using the D-optimal design generated by MATLAB's candexch algorithm to construct a robust validation set, overcoming random data splitting limitations in machine learning chemometric methods and ensuring unbiased evaluation across concentrations. The optimized MCR-ALS model outperforms in predictive ability, with recovery percentages of 98–102%, low root mean square errors of calibration and prediction, favorable bias-corrected mean square error of prediction, relative root mean square error within acceptable limits, and adequate limits of detection for pharmaceutical analysis. The Greenness Index Spider Charts and the Green Solvents Selection Tool were applied to replace hazardous solvents. A total of seven advanced evaluation tools were employed to assess the method's greenness, blueness, violetness, and whiteness, highlighting its eco-friendly profile, practical relevance, and innovation potential. Additionally, the method's environmental and societal benefits were further validated using the Need, Quality, Sustainability (NQS) index. Overall, this machine learning-based framework contributes meaningfully to ten United Nations Sustainable Development Goals (UN-SDGs), underscoring its value for future-oriented pharmaceutical research. © The Author(s) 2025.

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DACL-IDA: a dynamic alignment and compactness learning framework for imbalanced domain adaptation in bearing fault diagnosis

Item Type Journal Article

Author X. Peng

Author Y. Tao

Author L. Hua

Author K. Zhu

Abstract Existing unsupervised domain adaptation methods for bearing fault diagnosis under varying operating conditions generally assume identical class priors across domains, overlooking the label shift that arises in practice, where healthy-state samples far outnumber fault samples and fault-type frequencies vary with speed, load, and operating history. When feature shift and label shift coexist, prematurely introduced high-intensity alignment can severely degrade minority-class discriminability. We propose DACL-IDA (Dynamic alignment and compactness learning for imbalanced domain adaptation), built on an alignment-scheduling principle rather than a new alignment loss: global adversarial alignment is delayed until classification warmup and black-box shift estimation (BBSE) have stabilized, and is kept restrained thereafter. The BBSE estimate both re-weights the source loss and drives the scheduler, supported by intra-class compactness regularization and long-tailed optimization. A diagnostic experiment separating global adversarial from local class-conditional alignment shows that sustained class-conditional matching (local maximum mean discrepancy) progressively degrades the classifier under severe (reversed-prior) label shift; the final configuration therefore retains only the scheduled global adversarial alignment. On the Paderborn University dataset, under a strictly leakage-free protocol (fixed last-epoch reporting, no target-label model selection; 4 protocols $\times$ 12 transfer tasks $\times$ 5 seeds), DACL-IDA attains an overall mean balanced accuracy of 89.63% versus 81.30% for the strongest published competitor, DIDA (a deep imbalanced-domain-adaptation framework), winning three of the four protocols by more than 13 percentage points each (paired-t $p \leq 5.23 \times 10^{-4}$); on the remaining protocol (I2B), where scheduled alignment is counter-productive, it underperforms both DIDA and the source-only baseline—a failure reported and analyzed rather than omitted. DACL-IDA is also the most convergence-stable method in the stability comparison: outside I2B, its gap between best transient and final-epoch accuracy stays within 2.4 pp, versus 8.2–11.2 pp for DIDA. Cross-dataset experiments on the Case Western Reserve University dataset, a second backbone, imbalance ratios up to $\rho = 50$, and injected measurement noise support the reported behavior. © 2026 IOP Publishing Ltd. All rights, including for text and data mining, AI training, and similar technologies, are reserved. This article is available under the terms of the IOP-Standard License.

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Damage State Recognition and Quantification Method for Shield Machine Hob Based on Deep Forest

Item Type Journal Article**Author** H. Wang**Author** Q. Gao**Author** S. Liu**Author** P. Liu**Author** X. Wang**Author** Y. Tian

Abstract The damage status of shield machine disc cutters directly impacts the safety and efficiency of tunnelling projects. Current manual inspection methods involve high risks and low efficiency, while existing detection methods suffer from low accuracy and poor real-time performance in complex environments, often lacking quantitative analysis capabilities. To address these issues, this paper proposes an intelligent identification and quantitative assessment method for disc cutter damage based on the Deep Forest (DF) model. First, an eddy current sensor calibration platform was established, and a mapping relationship between output voltage and actual wear was developed through piecewise fitting to achieve precise wear quantification. In the data preprocessing stage, signal quality was improved via filtering, and typical damage features such as edge chipping, cracks, and eccentric wear were extracted using pulse edge detection. These feature segments were then resampled to construct the model training dataset. The DF model utilizes a hierarchical ensemble structure to mine data correlations, enabling accurate identification of four states: normal, edge

chipping, eccentric wear, and cracks. Simultaneously, a DF regression model was employed to provide continuous quantitative predictions of damage size. Experimental results show that the classification model achieved accuracies of 98%, 96%, and 96% on the training, validation, and test sets, respectively, with weighted average F1-scores exceeding 0.96. The regression model achieved a coefficient of determination (R^2) of 0.9940 and a root mean square error (RMSE) of 0.4051 on the test set. Both models demonstrate excellent performance and generalization, achieving full coverage from “qualitative state identification” to “quantitative wear assessment,” thereby providing reliable decision support for cutter maintenance and replacement. © 2026 by the authors.

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Data assimilation with machine learning for dynamical systems: Modelling indoor ventilation

Item Type Journal Article

Author C.E. Heaney

Author J. Tang

Author J. Yan

Author D. Guo

Author J. Ipock

Author S. Kaluvakollu

Author Y. Lin

Author D. Shao

Author B. Chen

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Author P. Kumar

Author C.C. Pain

Abstract Data assimilation is a method of combining physical observations with prior knowledge (for instance, a computational simulation) in order to produce an improved estimate of the state of a system; that is, improved over what the physical observations or the computational simulation, alone, could offer. Recently, machine learning techniques have been deployed in order to address the significant computational burden that is associated with the procedures involved in data assimilation. In this paper we propose an approach that uses a non-intrusive reduced-order model (NIROM) as a surrogate for a high-resolution model thereby saving computational effort. The mismatch between observations and the surrogate model is propagated forwards and backwards in time in a manner similar to 4D-variational data assimilation methods. The observations and prior are reconciled in a new way which takes full advantage of the neural network used in the NIROM and also means that there is no need to form the sensitivities explicitly when propagating the mismatch. Instead, the observations are part of the input and output of the network. The key advantages of this data assimilation approach are its simplicity (as we can avoid differentiating the forward model and bypass the use of an optimisation method), and its ability to integrate with control and uncertainty quantification methodologies. Modelling the air quality in a school classroom is the test case for our demonstration. After comparing the proposed data assimilation approach with 4D Variational Data Assimilation, we investigate two scenarios. The first of these is a dual-twin type experiment, for which the proposed approach is shown to perform very well, and the second is a test case which assimilates predictions from the NIROM with observations collected from a classroom in Houndsfield Primary School. © 2024 The Author(s)

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Data Augmentation and Clustering for Vehicle Make/Model Classification

Item Type Conference Paper

Author M. Nafzi

Author M. Brauckmann

Author T. Glasmachers

Editor Arai K.

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Abstract Vehicle shape information is very important in Intelligent Traffic Systems (ITS). In this paper, we present a way to exploit a training data set of vehicles released in different years and captured under different perspectives. Also the efficacy of clustering to enhance the make/model classification is presented. Both steps led to improved classification results and a greater robustness. Deeper convolutional neural network based on ResNet architecture has been designed for the training of the vehicle make/model classification. The unequal class distribution of training data produces an a priori probability. Its elimination, obtained by removing of the bias and through hard normalization of the centroids in the classification layer, improves the classification results. A developed application has been used to test the vehicle re-identification on video data manually based on make/model and color classification. This work was partially funded under the grant. © 2020, Springer Nature Switzerland AG.

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Data driven and neural network based approaches for optimization of film cooling parameters in gas turbines—A Comprehensive Review, Challenges and Future Trends

Item Type Journal Article

Author V.G. Krishna Anand

Author V. Baiju

Abstract Artificial intelligence and machine learning approaches are widely adopted for the optimization of multiple input parameters of complex thermal systems. The film cooling (FC) performance of gas turbines depends on various flow, geometric, and evaluation parameters. The optimization of these parameters is vital for the efficient design of FC systems. In this study, various data-driven and neural network-based optimization approaches applied for the optimization of geometric and flow parameters in gas turbines were summarized in detail. This study comprehensively reviews 216 articles spanning from 2005 to early 2025. The optimization approaches

were grouped into six categories: surrogate-based, machine learning-based, deep learning-based, Taguchi-based, multi-criteria decision-making-based, topology-based optimization, and other novel methods. Each optimization approach was elucidated extensively for its application to different gas turbine components, the predictive model used, input parameters, response parameter, generation of training samples, search algorithm, and factor level. The various parameters of the gas turbine FC used in the optimization are explained in detail. The uncertainty quantification method, uncertainty parameters, and distribution type applied to the gas turbine wall surface were compared and discussed in this study. In addition, the challenges involved in the implementation of optimization approaches and future research trends on gas turbine FC optimization are highlighted. © 2026 Elsevier Ltd.

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Data on consistency among different methods to assess atherosclerotic plaque echogenicity on standard ultrasound and intraplaque neovascularization on contrast-enhanced ultrasound imaging in human carotid artery

Item Type Journal Article**Author** M. Cattaneo**Author** D. Staub**Author** A.P. Porretta**Author** J.M. Gallino**Author** P. Santini**Author** C. Limoni**Author** R. Wyttenbach**Author** A. Gallino

Abstract Here we provide the correlation among different carotid ultrasound (US) variables to assess echogenicity n standard carotid US and to assess intraplaque neovascularization on contrast enhanced US. We recruited 45 consecutive subjects with an asymptomatic $\geq 50\%$ carotid artery stenosis. Carotid plaque echogenicity at standard US was visually graded according to Gray–Weale classification (GW) and measured by the greyscale median (GSM), a semi-automated computerized measurement performed by Adobe Photoshop®. On CEUS imaging IPNV was graded according to the visual appearance of contrast within the plaque according to three different methods: CEUS_A (1=absent; 2=present); CEUS_B a three-point scale (increasing IPNV from 1 to 3); CEUS_C a four-point scale (increasing IPNV from 0 to 3). We have also implemented a new simple quantification method derived from region of interest (ROI) signal intensity ratio as assessed by QLAB software. Further information is available in “Contrast-enhanced ultrasound imaging of intraplaque neovascularization and its correlation to plaque echogenicity in human carotid arteries atherosclerosis (M. Cattaneo, D. Staub, A.P. Porretta, J.M. Gallino, P. Santini, C. Limoni et al., 2016) [1]. © 2016 The Authors

Date 2016**DOI** 10.1016/j.dib.2016.09.041**URL** <https://www.scopus.com/pages/publications/85030562181?origin=resultslist>**Archive** Scopus**Language** English**Volume** 9**Publisher** Elsevier Inc.**Pages** 563-567**Publication** Data in Brief**ISSN** 23523409 (ISSN)**Journal Abbr** Data Brief**Date Added** 20/09/2026, 9:52:53 pm**Modified** 20/09/2026, 9:52:53 pm**Tags:**

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Notes:

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DATA PRUNING BY INFORMATION MAXIMIZATION

Item Type Conference Paper

Author H. Tan

Author S. Wu

Author W. Huang

Author S. Zhao

Author X. Qi

Abstract In this paper, we present InfoMax, a novel data pruning method, also known as coreset selection, designed to maximize the information content of selected samples while minimizing redundancy. By doing so, InfoMax enhances the overall informativeness of the coreset. The information of individual samples is measured by importance scores, which capture their influence or difficulty in model learning. To quantify redundancy, we use pairwise sample similarities, based on the premise that similar samples contribute similarly to the learning process. We formalize the coreset selection problem as a discrete quadratic programming (DQP) task, with the objective of maximizing the total information content, represented as the sum of individual sample contributions minus the redundancies introduced by similar samples within the coreset. To ensure practical scalability, we introduce an efficient gradient-based solver, complemented by sparsification techniques applied to the similarity matrix and dataset partitioning strategies. This enables InfoMax to seamlessly scale to datasets with millions of samples. Extensive experiments demonstrate the superior performance of InfoMax in various data pruning tasks, including image classification, vision-language pre-training, and instruction tuning for large language models. © 2025 13th International Conference on Learning Representations, ICLR 2025. All rights reserved.

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Data-driven analysis of EV public charging station demand across heterogeneous built environment

Item Type Journal Article

Author J. Tian

Author S. Li

Author K. Wang

Author Z. Zheng

Author Q. Zhang

Abstract Public charging stations (PCS) have become critical infrastructure in urban transportation systems. PCS charging demand patterns are driven by complex mechanisms that conventional methods struggle to quantify, while the influence of the built environment (BE) remains underexplored. An integrated empirical framework is proposed that combines machine learning and causal inference models to analyze PCS demand patterns and generate evidence-based policy implications. Specifically, XGBoost with partial dependence plots is employed to predict demand, identify key determinants, and reveal nonlinear relationships, followed by double machine learning to quantify the causal effects of critical policy-relevant factors. We construct a real-world charging dataset incorporating BE, temporal, and price attributes for empirical analysis. Results show that charging demand is shaped by public transport accessibility and the spatial configuration of PCS. Electricity price has a stronger negative impact in areas farther away from subway stations. These findings support optimized charging infrastructure planning and targeted policy interventions. © 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Data-Driven Approach to Analyzing Factors Influencing Construction Accident Severity Using SHAP Analysis

Item Type Journal Article

Author J. Son

Author J. Jeong

Author J. Jeong

Author L. Kumi

Author H. Mun

Abstract Despite advances in construction safety research, existing studies face critical limitations, including severe data imbalance in accident severity classification and lack of interpretable machine-learning models for factor contribution analysis. This study addresses these gaps by combining extreme gradient boosting (XGBoost) with Shapley additive explanation (SHAP) analysis to quantitatively evaluate key factors influencing construction accident severity based on workday loss. Advanced oversampling techniques resolved class imbalance among severity levels (fatal, very serious, serious, and minor), while hyperparameter tuning optimized model performance. The analysis identified the top five factors influencing accident severity: original cause material (55.76), accident month (46.38), project scale

(40.17), PET range (29.04), and age (21.53). The XGBoost-SHAP framework successfully demonstrated superior performance in accident prediction while providing interpretable factor contributions, validating workday loss as an effective severity quantification method. The findings enable risk assessment in large-scale projects and extreme environmental conditions, offering a scientific basis for developing targeted accident prevention strategies and optimizing safety resource allocations. © 2025 American Society of Civil Engineers.

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Accident prevention, Risk assessment, Accidents, Safety factor, Learning systems, Machine-learning, Machine learning, Factor analysis, Hyper-parameter, Hyperparameter tuning, Optimization, Disasters, Shapley, Shapley additive explanation, Accident severity, Additives, Construction accident severity, Construction accidents, Construction safety, Factor contribution

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Data-driven machine-learning quantifies differences in the voiding initiation network in neurogenic voiding dysfunction in women with multiple sclerosis

Item Type Journal Article

Author C. Karmonik

Author T. Boone

Author R. Khavari

Abstract Purpose: To quantify the relative importance of brain regions responsible for reduced functional connectivity (FC) in their Voiding Initiation Network in female multiple sclerosis (MS) patients with neurogenic lower urinary tract dysfunction (NLUTD) and voiding dysfunction (VD). A data-driven machine-learning approach is utilized for quantification. Methods: Twenty-seven ambulatory female patients with MS and NLUTD (group 1: voiders, n = 15 and group 2: VD, n = 12) participated in a functional magnetic resonance imaging (fMRI) voiding study. Brain activity was recorded by fMRI with simultaneous urodynamic testing. The Voiding Initiation Network was identified from averaged fMRI activation maps. Four machine-learning algorithms were employed to optimize the area under curve (AUC) of the receiver-operating characteristic curve. The optimal model was used to identify the relative importance of relevant brain regions. Results: The Voiding Initiation Network exhibited stronger FC for voiders in frontal regions and stronger disassociation in cerebellar regions. Highest AUC values were obtained with 'random forests' (0.86) and 'partial least squares' algorithms (0.89). While brain regions with highest relative importance (> 75%) included superior, middle, inferior frontal and cingulate regions, relative importance was larger than 60% for 186 of the 227 brain regions of the Voiding Initiation Network, indicating a global effect. Conclusions: Voiders and VD patients showed distinctly different FC in their Voiding Initiation Network. Machine-learning is able to identify brain centers contributing to these observed differences. Knowledge of these centers and their connectivity may allow phenotyping patients to centrally focused treatments such as cortical modulation. © 2019 Korean Continence Society

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Article, female, human, machine learning, Machine learning, algorithm, quantitative analysis, controlled study, nerve cell network, clinical article, phenotype, Functional magnetic resonance imaging, brain region, functional magnetic resonance imaging, functional connectivity, hippocampus, multiple sclerosis, Multiple sclerosis, temporal lobe, thalamus, fusiform gyrus, urodynamics, frontal lobe, insula, lower urinary tract symptom, micturition disorder, Neurogenic lower urinary tract dysfunction, subcallosal gyrus, substantia nigra

Notes:

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Export Date: 20 September 2026; Cited By: 20; Correspondence Address: C. Karmonik; Department of Urology, Houston Methodist Hospital, Houston, United States; email: ckarmonik@houstonmethodist.org

Data-driven nucleus subclassification on colon hematoxylin and eosin using style-transferred digital pathology

Item Type Journal Article

Author L.W. Remedios

Author S. Bao

Author S.W. Remedios

Author H.H. Lee

Author L.Y. Cai

Author T. Li

Author R. Deng

Author N.R. Newlin

Author A.M. Saunders

Author C. Cui

Author J. Li

Author Q. Liu

Author K.S. Lau

Author J.T. Roland

Author M.K. Washington

Author L.A. Coburn

Author K.T. Wilson

Author Y. Huo

Author B.A. Landman

Abstract Purpose: Cells are building blocks for human physiology; consequently, understanding the way cells communicate, co-locate, and interrelate is essential to furthering our understanding of how the body functions in both health and disease. Hematoxylin and eosin (H&E) is the standard stain used in histological analysis of tissues in both clinical and research settings. Although H&E is ubiquitous and reveals tissue microanatomy, the classification and mapping of cell subtypes often require the use of specialized stains. The recent CoNIC Challenge focused on artificial intelligence classification of six types of cells on colon H&E but was unable to classify epithelial subtypes (progenitor, enteroendocrine, goblet), lymphocyte subtypes (B, helper T, cytotoxic T), and connective subtypes (fibroblasts). We propose to use inter-modality learning to label previously un-labelable cell types on H&E. Approach: We took advantage of the cell classification information inherent in multiplexed immunofluorescence (MxIF) histology to create cell-level annotations for 14 subclasses. Then, we performed style transfer on the MxIF to synthesize realistic virtual H&E. We assessed the efficacy of a supervised learning scheme using the virtual H&E and 14 subclass labels. We evaluated our model on virtual H&E and real H&E. Results: On virtual H&E, we were able to classify helper T cells and

epithelial progenitors with positive predictive values of 0.34 ± 0.15 (prevalence 0.03 ± 0.01) and 0.47 ± 0.1 (prevalence 0.07 ± 0.02), respectively, when using ground truth centroid information. On real H&E, we needed to compute bounded metrics instead of direct metrics because our fine-grained virtual H&E predicted classes had to be matched to the closest available parent classes in the coarser labels from the real H&E dataset. For the real H&E, we could classify bounded metrics for the helper T cells and epithelial progenitors with upper bound positive predictive values of 0.43 ± 0.03 (parent class prevalence 0.21) and 0.94 ± 0.02 (parent class prevalence 0.49) when using ground truth centroid information. Conclusions: This is the first work to provide cell type classification for helper T and epithelial progenitor nuclei on H&E. © 2024 SPIE. All rights reserved.

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Data-Driven Tooth Selection Enhances Partial-Mouth Periodontal Screening

Item Type Journal Article

Author N. Lin

Author C. Wang

Author Y. Shan

Author Z. Wu

Abstract The high prevalence of periodontitis has imposed a significant global disease burden. Epidemiologic surveys rely on full-mouth periodontal examination (FMPE) or partial-mouth periodontal examination (PMPE). While FMPE is resource-intensive, the efficiency of PMPE remains questionable. Moreover, the value of subpopulation-tailored PMPE protocols is still unassessed. To address these gaps, we developed an interpretable, data-driven framework that ranks the importance of teeth using SHapley Additive exPlanations (SHAP) values from machine learning models. Using data from the National Health and Nutrition Examination Survey 2009–2014, XGBoost (XGB) and LightGBM (LGB) were trained on maximum interproximal probing depth and clinical attachment loss across 28 teeth from adults aged 35 y and older. Absolute SHAP values were aggregated separately for each model to calculate global tooth importance. The top 10 teeth were then evaluated and benchmarked against the Community Periodontal Index (CPI) and modified Ramfjord protocol. Primary outcomes included quadratic weighted kappa (QWK) for diagnostic agreement with FMPE and the inflation factor (IF) for prevalence estimation bias. The XGB-derived protocol (FDI 47, 27, 17, 26, 37, 16, 42, 46, 36, 41) achieved QWK = 0.85 and IF = 127.29% in the external validation set, with significantly lower IF than CPI (P = 0.003), indicating its potential to replace CPI. Further optimization for age- and gender-specific protocols occasionally yielded localized improvements, such as males aged 55 to 64 y (teeth 17, 16, 27, 47, 26, 33, 45, 37, 43, 25), while the unified XGB-derived protocol remained robust across all strata. These findings support the general use of a unified, data-driven protocol for large-scale epidemiological surveys. SHAP-guided tooth selection offers an interpretable and efficient alternative to traditional PMPE, bridging the gap between accuracy and feasibility in large-scale periodontal surveys. © International Association for Dental, Oral, and Craniofacial Research and American Association for Dental, Oral, and Craniofacial Research 2026

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DOI 10.1177/00220345261432570**URL** <https://www.scopus.com/pages/publications/105035294890?origin=resultslist>**Archive** Scopus**Language** English**Publisher** SAGE Publications Inc.**Publication** Journal of Dental Research**ISSN** 00220345 (ISSN)**Journal Abbr** J. Dent. Res.**Date Added** 20/09/2026, 9:51:35 pm**Modified** 20/09/2026, 9:51:35 pm**Tags:**

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Data-driven versus conventional N₂O EF quantification methods in wastewater; how can we quantify reliable annual EFs?

Item Type Journal Article**Author** V. Vasilaki**Author** S. Danishvar**Author** A. Mousavi**Author** E. Katsou

Abstract A long-term N₂O dataset from a full-scale biological process was analysed for knowledge discovery. Non-parametric, multivariate timeseries changepoint detection techniques were applied to operational variables (i.e. NH₄-N loads) in the system. The majority of changepoints, could be linked with the observed changes of the N₂O emissions profile. The results showed that even three-day sampling campaigns between changepoints have a high probability (>80%) to result to an emission factor (EF) quantification with ~10% error. The analysis revealed that support vector machine (SVM) classification models can be trained to detect operational behaviour of the system and the expected range of N₂O emission loads. The proposed approach can be applied when long-term online sampling is not feasible (due to budget or

equipment limitations) to identify N₂O emissions “hotspot” periods and guide towards the identification of operational periods requiring extensive investigation of N₂O pathways generation. © 2020

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Support vector machines, Budget control, Classification models, Quantification methods, Biological process, Change point detection, Changepoint detection, Long-term monitoring campaign, N₂O emissions, On-line sampling, Operational periods, Operational variables, Sampling campaigns, Support vector Machine classification model

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Data-Efficient Computational Pathology Platform for Faster and Cheaper Breast Cancer Subtype Identifications: Development of a Deep Learning Model

Item Type Journal Article

Author K. Bae

Author Y.S. Jeon

Author Y. Hwangbo

Author C.W. Yoo

Author N. Han

Author M. Feng

Abstract Background: Breast cancer subtyping is a crucial step in determining therapeutic options, but the molecular examination based on immunohistochemical staining is expensive and time-consuming. Deep learning opens up the possibility to predict the subtypes based on the morphological information from hematoxylin and eosin staining, a much cheaper and faster alternative. However, training the predictive model conventionally requires a large number of histology images, which is challenging to collect by a single institute. Objective: We aimed to develop a data-efficient computational pathology platform, 3DHistoNet, which is capable of learning from z-stacked histology images to accurately predict breast cancer subtypes with a small sample size. Methods: We retrospectively examined 401 cases of patients with primary breast carcinoma diagnosed between 2018 and 2020 at the Department of Pathology, National Cancer Center, South Korea. Pathology slides of the patients with breast carcinoma were prepared according to the standard protocols. Age, gender, histologic grade, hormone receptor (estrogen receptor [ER], progesterone receptor [PR], and androgen receptor [AR]) status, erb-B2 receptor tyrosine kinase 2 (HER2) status, and Ki-67 index were evaluated by reviewing medical charts and pathological records. Results: The area under the receiver operating characteristic curve and decision curve were analyzed to evaluate the performance of our 3DHistoNet platform for predicting the ER, PR, AR, HER2, and Ki67 subtype biomarkers with 5-fold cross-validation. We demonstrated that 3DHistoNet can predict all clinically important biomarkers (ER, PR, AR, HER2, and Ki67) with performance exceeding the conventional multiple instance learning models by a considerable margin (area under the receiver operating characteristic curve: 0.75-0.91 vs 0.67-0.8). We further showed that our z-stack histology scanning method can make up for insufficient training data sets without any additional cost incurred. Finally, 3DHistoNet offered an additional capability to generate attention maps that reveal correlations between Ki67 and histomorphological features, which renders the hematoxylin and eosin image in higher fidelity to the pathologist. Conclusions: Our stand-alone, data-efficient pathology platform that can both generate z-stacked images and predict key biomarkers is an appealing tool for breast cancer diagnosis. Its development would encourage morphology-based diagnosis, which is faster, cheaper, and less error-prone compared to the protein quantification method based on immunohistochemical staining. © Kideog Bae, Young Seok Jeon, Yul Hwangbo, Chong Woo Yoo, Nayoung Han, Mengling Feng.

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Data-Free Network Debiasing for Long-Tailed Visual Recognition

Item Type Conference Paper

Author J. Cai

Author Z. Wang

Author H. Fu

Author J. Chen

Author Y.-G. Jiang

Abstract Real-world data is often unbalanced and exhibits long-tailed distribution over classes. Vanilla classification models trained on imbalanced datasets inherently exhibit bias towards dominant classes. Existing debiasing methods mostly balance the data or the loss during training. Nevertheless, these data-acquiring methods are not suitable for situations where training data are unavailable. In this paper, we appeal to solutions without access to training data and propose a datafree debiasing (Free-D) method that serves as a plug-and-play module for any standard classification model. Specifically, our method adjusts both the feature representation via feature representation shifting and the classifier weight via class prior compensation in a data-free manner. We evaluate and compare our methods on four long-tailed visual recognition datasets, i.e., long-tailed CIFAR-10/-100, ImageNet-LT, and Places-LT. Extensive experiments demonstrate that the proposed data-free method achieves comparable results of other data-acquired methods. © 2022 IEEE.

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Dating skeletal remains with luminol-chemiluminescence. Validity, intra- and interobserver error

Item Type Journal Article

Author F. Ramsthaler

Author K. Kreutz

Author K. Zipp

Author M.A. Verhoff

Abstract Amongst numerous other morphological, chemical, physical, and histological dating methods, the assessment of the chemiluminescence of weathered organic traces on bone finds by means of the luminol reaction is common practice. Opinion differs on the diagnostic value of this procedure for differentiating between historical and recent skeletonized human remains. Design: Chemiluminescence (CL) was tested in randomized, blind experiments on pulverized bone samples with known postmortem intervals from four different periods (1st-3rd century, 1878-1902, 1965-1972,

1995-2003). In addition to determining the sensitivity and specificity of the CL-analysis, likelihood ratios and predictive values were calculated. The degree of interobserver and intraobserver agreement was determined using Kappa statistics. Results: Based on our sample, we found false positive results in approximately 7.5% of all examined historical cases and false negative results in approximately 15% of the recent samples. Altogether, a correct classification of 88.75% (overall classification) was found (sensitivity = 0.85, specificity = 0.925; likelihood ratio = 11.3; odds ratio = 69.8). Taking a-priori probabilities into consideration (prevalence for recent = 26%), a positive predictive value (PPV) of only 79.9% could be calculated, while the negative predictive value (NPV) lay at 94.6%. The results from subsequent Kappa statistics further illustrated that the quality of the results was, on the whole, observer independent ($\kappa = 0.93$; 95% CI = 0.84-1.0). Intraobserver bias calculation yielded a value of $\kappa = 0.975$ (95% CI = 0.92-1.0). Conclusion: The analysis of CL on dry bones in determining the postmortem interval is not valid as an only method. In other respects, the luminol test is reproducible, repeatable and cost-effective and could help in assessing the postmortem interval (PMI). The results of this study document the potential danger of forensically relevant cases being erroneously sorted out if further examinations do not take place. We would like to emphasize that still only a combination of different methods including the analysis of environmental conditions and the interpretation of the anthropological profile can allow an acceptable estimate of the PMI from skeletonized remains, even if the methodological spectrum is being expanded by different absolute dating methods using radionuclide techniques. © 2009 Elsevier Ireland Ltd. All rights reserved.

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Export Date: 20 September 2026; Cited By: 55; Correspondence Address: F. Ramsthaler; Center of Legal Medicine, Institute of Forensic Medicine, Johann Wolfgang Goethe University Frankfurt, 60596 Frankfurt Main, Kennedy Allee 104, Germany; email: ramsthaler@em.uni-frankfurt.de; CODEN: FSIND

De-migration-based supervised learning for interpolation and regularization of 3D offset classes

Item Type Journal Article

Author V. Hlebnikov

Author T.A.L. Greiner

Author V. Vinje

Author J.E. Lie

Author G. Poole

Abstract Regularization and interpolation of 3D offset classes prior to imaging are an important and challenging step in the marine seismic data processing flow. Here we describe how to perform this task using a deep neural network, and we explain how to overcome the challenge of creating a suitable training data set. The training data set is generated by de-migrating stacked pre-stack depth migration images. For each offset class volume, we de-migrate the pre-stack depth migrated stacked image into two configurations: (i) the original survey configuration consisting of the recorded source/receiver positions and (ii) an ‘Ideal’ survey configuration with constant offset and azimuth for each 3D offset class. The training creates a 3D convolutional encoder–decoder model that will regularize and interpolate seismic data. The convolutional encoder–decoder is trained on 3D sliding windows in each 3D offset cube to map from (i) to (ii), i.e. to map the original survey configuration with irregular and sparse sampling into the fully sampled regular offset cubes suitable for offset-based migration, such as Kirchhoff migration. Such migration algorithms rely on regular and sufficiently dense sampling to achieve constructive interference to image the structures and destructive interference to suppress migration noise. We test the new method on one synthetic and one field data example and show that it performs better than a standard regularization/interpolation method based on anti-leakage Fourier transform, especially for the smallest offset classes. On the synthetic data, we also demonstrate that the convolutional encoder–decoder method preserves the amplitude versus offset as well as the standard method. © 2022 CGG Services Norway AS. Geophysical Prospecting published by John Wiley & Sons Ltd on behalf of European Association of Geoscientists & Engineers.

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Decision aggregation in distributed classification by a transductive extension of maximum entropy/improved iterative scaling

Item Type Journal Article

Author D.J. Miller

Author Y. Zhang

Author G. Kesidis

Abstract In many ensemble classification paradigms, the function which combines local/base classifier decisions is learned in a supervised fashion. Such methods require common labeled training examples across the classifier ensemble. However, in some scenarios, where an ensemble solution is necessitated, common labeled data may not exist: (i) legacy/proprietary classifiers, and (ii) spatially distributed and/or multiple modality sensors. In such cases, it is standard to apply fixed (untrained) decision aggregation such as voting, averaging, or naive Bayes rules. In recent work, an alternative transductive learning strategy was proposed. There, decisions on test samples were chosen aiming to satisfy constraints measured by each local classifier. This approach was shown to reliably correct for class prior mismatch and to robustly account for classifier dependencies. Significant gains in accuracy over fixed aggregation rules were demonstrated. There are two main limitations of that work. First, feasibility of the constraints was not guaranteed. Second, heuristic learning was applied. Here, we overcome these problems via a transductive extension of maximum entropy/improved

iterative scaling for aggregation in distributed classification. This method is shown to achieve improved decision accuracy over the earlier transductive approach and fixed rules on a number of UC Irvine datasets.

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Notes:

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: D. J. Miller; Department of Electrical Engineering, Pennsylvania State University, University Park, PA 16802, United States; email: djmiller@engr.psu.edu

Decision support system for the diagnosis of Parkinson's disease

Item Type Conference Paper

Author A. Ericsson

Author M.N. Lonsdale

Author K. Astrom

Author L. Edenbrandt

Author L. Friberg

Abstract Recently new nuclear medical diagnostic imaging methods have been introduced by which it is possible to diagnose Parkinson's disease, PD, at an early stage. The binding of a specific receptor ligand [¹²³I]-FP-CIT in specific structures of the human brain is visualized by way of single photon emission computerized tomography, SPECT. The interpretation of the SPECT data can be accessed by visual analysis and manual quantification methods. We have developed a computer aided automatic decision support system in attempt to facilitate the diagnosis of Parkinson's disease from the acquired SPECT images. A rigid model

was used for the segmentation of the basal ganglia of the human brain. The aim of the study was to develop an automated method for quantification and classification of the images. The study comprises SPECT scans 89 patients, who underwent a [123I]-FP-CIT examination because of suspected Parkinson's disease. An experienced nuclear medicine physician interpreted the images and diagnosed 65 of the patients as most likely suffering from Parkinson's disease. The method included the following steps; (i) segmentation of basal ganglia by fitting a rigid 3D model to the image, (ii) extraction of 17 features based on image intensity distribution in the basal ganglia and a reference based on image intensity distribution outside the basal ganglia, (iii) classification using Support Vector Machine (SVM). The classification based on the automated method showed a true acceptance of 96.9% and a true rejection of 91.6%. The classification based on a manual quantification method gave a true acceptance of 98.5% and a true rejection of 100%. The method proposed here is fully automatic and it makes use of the full 3D data set in contrast to a method that is widely used at hospitals today which only uses a few 2D image slices. © Springer-Verlag Berlin Heidelberg 2005.

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Decision tree enhancer (DTE): Improving decision trees with optimization

Item Type Journal Article**Author** F.A. Lim-Apo**Author** F. Oliveira**Author** S. Hamacher

Abstract Decision trees are off-the-shelf machine learning models widely used for classification and regression tasks in medical, logistics, financial, and other critical areas where interpretability is a key factor. They can efficiently handle numerical and categorical variables, making them a versatile choice for various applications. However, traditional decision-tree training methods are based on greedy heuristics, which cannot provide guarantees regarding whether further improvements could be achieved. We propose Decision Tree Enhancer (DTE), which employs optimization as a post-training step to improve previously trained decision trees. Moreover, the proposed method precludes the need for a pre-processing step for continuous features such as discretization or bucketization, and can be applied regardless of the model used to first train the decision tree. Lastly, DTE's mathematical programming formulation enables, for example, the consideration of recall thresholds and class prioritization. Tested on 63 classification datasets from the UCI Machine Learning Repository, using tree depths from 1 to 5, four time limits (1, 5, 10, and 30 s), and 5 randomized train-test splits for cross-validation, the proposed post-training step demonstrated superior performance over CART (Classification And Regression Tree), for both in- and out-of-sample data. With a 30 s time limit, DTE was able to improve the weighted recall in 83.2% of the datasets with an average improvement of 9.0% in training and 5.0% in testing. © 2026 The Author(s)

Date 2026**DOI** 10.1016/j.knosys.2026.115597**URL** <https://www.scopus.com/pages/publications/105031303673?origin=resultslist>**Archive** Scopus**Language** English**Volume** 339**Publisher** Elsevier B.V.**Publication** Knowledge-Based Systems**ISSN** 09507051 (ISSN)**Journal Abbr** Knowl Based Syst**Date Added** 20/09/2026, 9:51:51 pm**Modified** 20/09/2026, 9:51:51 pm**Tags:**

Optimisations, Classification (of information), Learning systems, Machine learning, Key factors, Machine learning models, Statistical tests, Forestry, Decision trees, Interpretability, Sampling, Training methods, Decision tree, Heuristic methods, Categorical variables, Mathematical programming, Numerical variables, Off-the-shelf machine, Post-training optimization, Training optimizations

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Decoding the nature and complexity of extracellular mtDNA: Types and implications for health and disease

Item Type Journal Article

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Author A. Benavides-Almeida

Author A. Haro-Vinueza

Author J. Peña-Cisneros

Author Á.A. Pérez-Meza

Author J. Michelson

Author S. Peñaherrera

Author M. Picard

Abstract The mitochondrial DNA (mtDNA) is replicated and canonically functions within intracellular mitochondria, but recent discoveries reveal that the mtDNA has another exciting extracellular life. mtDNA fragments and mitochondria-containing vesicular structures are detected at high concentrations in cell-free forms, in different biofluids. Commonly referred to as cell-free mtDNA (cf-mtDNA), the field is currently without a comprehensive classification system that acknowledges the various biological forms of mtDNA and whole mitochondria existing outside the cell. This absence of classification hampers the creation of precise and consistent quantification methods across different laboratories, which is crucial for unraveling the molecular and biological characteristics of mtDNA. In this article, we integrate recent findings to propose a classification for different types of Extracellular mtDNA [ex-mtDNA]. The major biologically distinct types include: Naked mtDNA [N-mtDNA], mtDNA within non-mitochondrial Membranes [M-mtDNA], Extracellular mitochondria [exM-mtDNA], and mtDNA within Mitochondria enclosed in a Membrane [MM-mtDNA]. We outline the challenges associated with accurately quantifying these ex-mtDNA types, suggest potential physiological roles for each ex-mtDNA type, and explore how this classification could establish a foundation for future research endeavors and further analysis and definitions for ex-mtDNA. By proposing this classification of circulating mtDNA forms, we draw a parallel with the clinically recognized forms of cholesterol, such as HDL and LDL, to illustrate potential future significance in a similar manner. While not directly analogous, these mtDNA forms may one day be as biologically relevant in clinical interpretation as cholesterol fractions are currently. We also discuss how advancing methodologies to reliably quantify distinct ex-mtDNA forms could significantly enhance their utility as health or disease biomarkers, and how their application may offer innovative therapeutic approaches. © 2024 Elsevier B.V. and Mitochondria Research Society

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Decoupling convolution network for characterizing the metastatic lymph nodes of breast cancer patients

Item Type Journal Article
Author R. Zeng
Author X. Zhang
Author C. Zheng
Author J.-H. Du
Author Z. Gao
Author W. Jun

Author J. Shen

Author Y. Lu

Abstract Purpose: The dual-energy computed tomography (DECT) technique is an emerging imaging tool that can better characterize material features and has the potential to be a noninvasive means of predicting lymph node metastasis. The purpose of this study was to establish a DECT-specified quantitative approach based on a neural network to characterize the sentinel lymph node (SLN). Methods: With IRB approval, we retrospectively collected a total of 229 patients (100/229 metastasis) with biopsy proven breast cancer in this study. The chest and axillary spectral CT examinations were performed prior to the axillary lymph node (ALN) surgery. A decoupling convolution network with 11 ROIs from sequential keV (40 to 140 keV with 10 keV increment) was proposed to explicitly extract the spectral and spatial features in a DECT to predict the lymph node status. Focal loss was introduced as the loss function. The metric of the slope of the spectral Hounsfield unit curve measured at the venous phase was used as the baseline approach in comparison to our approach. In additional, a logistic model with radiomic features was also compared to our approach. The area under ROC curve (AUC) was used as the figure of merit to evaluate the classification performance. Results: By introducing spectral convolution and focal loss, AUC on test set could be improved by 0.15 and 0.01 separately. Compared to the slope of the spectral curve with the average AUC of 0.611 and radiomic model with AUC of 0.825, the proposed approach demonstrates a considerably better performance, with test set AUC value of 0.837, by using decoupling spectral and spatial convolution together with focal loss function. Conclusions: We presented a new decoupling neural network based quantification method for DECT analysis, which might have potential as a noninvasive tool to predict metastasis lymph node status for breast cancer in clinical practice. © 2021 American Association of Physicists in Medicine

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Deep artifact suppression for spiral real-time phase contrast cardiac magnetic resonance imaging in congenital heart disease

Item Type Journal Article

Author O. Jaubert

Author J. Steeden

Author J. Montalt-Tordera

Author S. Arridge

Author G.T. Kowalik

Author V. Muthurangu

Abstract Purpose: Real-time spiral phase contrast MR (PCMR) enables rapid free-breathing assessment of flow. Target spatial and temporal resolutions require high acceleration rates often leading to long reconstruction times. Here we propose a deep artifact suppression framework for fast and accurate flow quantification. Methods: U-Nets were trained for deep artifact suppression using 520 breath-hold gated spiral PCMR aortic datasets collected in congenital heart disease patients. Two spiral trajectories (uniform and perturbed) and two losses (Mean Absolute Error -MAE- and average

structural similarity index measurement -SSIM-) were compared in synthetic data in terms of MAE, peak SNR (PSNR) and SSIM. Perturbed spiral PCMR was prospectively acquired in 20 patients. Stroke Volume (SV), peak mean velocity and edge sharpness measurements were compared to Compressed Sensing (CS) and Cartesian reference. Results: In synthetic data, perturbed spiral consistently outperformed uniform spiral for the different image metrics. U-Net MAE showed better MAE and PSNR while U-Net SSIM showed higher SSIM based metrics. In-vivo, there were no significant differences in SV between any of the real-time reconstructions and the reference standard Cartesian data. However, U-Net SSIM had better image sharpness and lower biases for peak velocity when compared to U-Net MAE. Reconstruction of 96 frames took ~59 s for CS and 3.9 s for U-Nets. Conclusion: Deep artifact suppression of complex valued images using an SSIM based loss was successfully demonstrated in a cohort of congenital heart disease patients for fast and accurate flow quantification. © 2021

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Deep CNN with graph Laplacian regularization for multi-label image annotation

Item Type Conference Paper

Author J. Mojoo

Author K. Kurosawa

Author T. Kurita

Editor Cheriet F.

Editor Karray F.

Editor Campilho A.

Abstract To compensate for incomplete or imprecise tags in training samples, this paper proposes a learning algorithm for the convolutional neural network (CNN) for multi-label image annotation by introducing co-occurrence dependency between tags as a graph Laplacian regularization term. To exploit the co-occurrence dependency, we apply Hayashi's quantification method-type III to the tags in the training samples and use the distances between the acquired representative vectors to define the weights for graph Laplacian regularization. By introducing this regularization term, the possibility of co-occurrence between tags with high co-occurrence frequency can be increased. To confirm the effectiveness of the proposed algorithm, we have done experiments using Corel5k's dataset for multi-label image annotation. © Springer International Publishing AG 2017.

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Deep continual learning for medical call incidents text classification under the presence of dataset shifts

Item Type Journal Article

Author P. Ferri

Author V. Lomonaco

Author L.C. Passaro

Author A. Félix-De Castro

Author P. Sánchez-Cuesta

Author C. Sáez

Author J.M. García-Gómez

Abstract The aim of this work is to develop and evaluate a deep classifier that can effectively prioritize Emergency Medical Call Incidents (EMCI) according to their life-threatening level under the presence of dataset shifts. We utilized a dataset consisting of 1 982 746 independent EMCI instances obtained from the Health Services Department of the Region of Valencia (Spain), with a time span from 2009 to 2019 (excluding 2013). The dataset includes free text dispatcher observations recorded during the call, as well as a binary variable indicating whether the event was life-threatening. To evaluate the presence of dataset shifts, we examined prior probability shifts, covariate shifts, and concept shifts. Subsequently, we designed and implemented four deep Continual Learning (CL) strategies—cumulative learning, continual fine-tuning, experience replay, and synaptic intelligence—alongside three deep CL baselines—joint training, static approach, and single fine-tuning—based on DistilBERT models. Our results demonstrated evidence of prior probability shifts, covariate shifts, and concept shifts in the data. Applying CL techniques had a statistically significant ($\alpha=0.05$) positive impact on both backward and forward knowledge transfer, as measured by the F1-score, compared to non-continual approaches. We can argue that the utilization of CL techniques in the context of EMCI is effective in adapting deep learning classifiers to changes in data distributions, thereby maintaining the stability of model performance over time. To our knowledge, this study represents the first exploration of a CL approach using real

EMCI data. © 2024 The Author(s)

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Deep Cross-Branch Multi-Modal Fusion Network for early Alzheimer's diagnosis

Item Type Journal Article**Author** J. Li**Author** Y. Gao**Author** Z. Guan**Author** T. Cheng

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Author H. Yu

Author A. Yang

Author M. Xu

Author Y. Wang

Author P. Yang

Author T. Wang

Author G. Ma

Author B. Lei

Abstract Alzheimer's disease (AD) is a prevalent neurodegenerative disorder where early diagnosis is pivotal for effective intervention, yet it is hindered by subtle pathological feature differences among AD subtypes and severe class imbalance in medical imaging datasets. Existing structural magnetic resonance imaging (sMRI) and resting-state functional MRI (rs-fMRI) multimodal fusion methods for AD diagnosis mostly adopt simple concatenation or summation without fine-grained cross-modal alignment and interaction. To address these issues, we propose a Deep Cross-Branch Multi-Modal Fusion Network (DCMFNet) for early AD diagnosis. We first preprocess sMRI and rs-fMRI to extract ROI-based features, then perform dimension unification and normalization to realize cross-modal feature alignment. A novel Deep Cross-branch Multi-modal Feature Fusion (DCMF) module with three parallel branches and a dual-pathway cross-modal branch is designed to fully mine complementary and correlated cross-modal information, and the fused features are input into a Transformer encoder for classification. Moreover, we introduce the Logit Adjustment Cross-Entropy (LACE) loss to mitigate class imbalance by correcting decision boundaries based on class prior probabilities, enhancing the recognition of minor classes. The model is evaluated on a private clinical dataset. Experimental results show that DCMFNet outperforms traditional machine learning methods and state-of-the-art deep learning models in six binary AD subtype classification tasks, with the LACE loss and DCMF module effectively alleviating class imbalance and improving cross-modal feature representation. This work provides a reliable multimodal fusion framework for early AD diagnosis and reduces the diagnostic burden on healthcare professionals. © 2026 Elsevier B.V.

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Deep Domain Isolation and Sample Clustered Federated Learning for Semantic Segmentation

Item Type Conference Paper

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Author C. Lartizien

Author S. Duffner

Editor Bifet A.

Editor Davis J.

Editor Krilavicius T.

Editor Kull M.

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Editor Žliobaite I.

Abstract Empirical studies show that federated learning exhibits convergence issues in Non Independent and Identically Distributed (IID) setups. However, these studies only focus on label distribution shifts, or concept shifts (e.g. ambiguous tasks). In this paper, we explore for the first time the effect of covariate shifts between participants' data in 2D segmentation tasks, showing an impact way less serious than label shifts but still present on convergence. Moreover, current Personalized (PFL) and Clustered (CFL) Federated Learning methods intrinsically assume the homogeneity of the dataset of each participant and its consistency with future test samples by operating at the client level. We introduce a more general and realistic framework where each participant owns a mixture of multiple underlying feature domain distributions. To diagnose such pathological feature distributions affecting a model being trained in a federated fashion, we develop Deep Domain Isolation (DDI) to isolate image domains directly in the gradient space of the model. A federated Gaussian Mixture Model is fit to the sample gradients of each class, while the results are combined with spectral clustering on the server side to isolate decentralized sample-level domains. We leverage this clustering algorithm through a Sample Clustered Federated Learning (SCFL) framework, performing standard federated learning of several independent models, one for each decentralized image domain. Finally, we train a classifier enabling to associate a test sample to its corresponding domain cluster at inference time, offering a final set of models that are agnostic to any assumptions on the test distribution of each participant. We validate our approach on a toy segmentation dataset as well as different partitionings of a combination of Cityscapes and GTA5 datasets using an EfficientViT-B0 model, showing a significant performance gain compared to other approaches. Our code is available at https://github.com/MatthisManthe/DDI_SCFL. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Deep imbalanced domain adaptation for transfer learning fault diagnosis of bearings under multiple working conditions

Item Type Journal Article

Author Y. Ding

Author M. Jia

Author J. Zhuang

Author Y. Cao

Author X. Zhao

Author C.-G. Lee

Abstract The tremendous success of deep learning and transfer learning broadened the scope of fault diagnosis, especially the latter further improved the diagnosis accuracy under multiple working conditions. However, most existing attempts assume that label distribution is domain-invariant despite taking into account the different feature distributions. This does not accommodate the diversity of fault mode distributions under different operating conditions and weakens the generalization to imbalanced domain adaptation (IDA) scenarios. Therefore, this work proposed a novel deep imbalanced domain adaptation (DIDA) framework for fault diagnosis of bearings, aiming at the challenging scenario where feature shift and label shift exist simultaneously under different working conditions. Specifically, DIDA overcomes the class-imbalanced label shift and achieves a fine-grained latent space matching by cost-sensitive learning and categorical alignment. Besides, margin loss regularization is introduced to further optimize classification boundaries and improve cross-domain generalization for IDA fault diagnosis tasks. Finally, we simulated the IDA protocols on experimental datasets and conducted case studies under multiple working conditions, thus validating the effectiveness and superiority of the proposed framework. © 2022 Elsevier Ltd

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Deep Imbalanced Separation Network: A Holistic Fault Detection Framework Considering Class-Imbalance and Partial Label-Unknown

Item Type Journal Article

Author M. Qian

Author Y.-F. Li

Author H. Wu

Abstract The challenges of class-imbalance and partially unknown training labels often arise in fault detection tasks. When these two problems occur simultaneously, existing imbalanced classification methods cannot be directly used due to the absence of the label, and the class-imbalance would lead to severe bias prediction. In this study, we proposed a novel deep imbalance separation network (deepImSN) framework that is capable of dealing with fault detection problems with both class imbalance and partially unknown labels. This framework integrates the one-class learning concept into the positive-unlabeled (PU) learning theory for the first time. It alleviates the bias of the class-imbalance while making full use of the limited label information in the PU set to optimize the feature space and guide model training. The proposed deepImSN is designed to be used in different scenarios. It can accurately complete the fault detection task whether only part of fault samples or normal samples are

labeled, and the class-prior is known or unknown. Experimental results on real-world problems, such as high-speed rail wheels fault inspection and wafer map fault detection, demonstrate that deepImSN outperforms existing methods in various experimental conditions. © 2005-2012 IEEE.

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Deep learning based ensemble approach to predict student academic performance: Case study

Item Type Conference Paper

Author Y.K. Salal

Author S.M. Abdullaev

Abstract Traditional practices of education, medicine, and economics often use collective expert methods to find acceptable solutions to complex issues. The computer model of such methods is an ensemble classification which combines prediction of a number base classifier trained by various machine learning techniques. The

predictive sample in the educational process is always not balanced-it leads to a shift in estimates. The basic idea of this work is that the binary forecasting on unbalanced datasets is carried out by a Meta-classifier-an ensemble of different classification algorithm (Decision tree J48, Naive base NB, Multilayer perception MLP, k-Neighbour Nearest k-NN and Support Vector Machine SVM) which trained by stacking, bagging and boosting procedures on SMOTE, RUS, ROS balanced samples. The experiments conducted on moderately unbalanced dataset of 350 tuples of student data with using simple voting Meta-ensemble models, show that the best overall accuracy A and F1 measure of minority class were achieved by the Meta-ensembles containing 5-15 members classifiers trained on samples prepared by SMOTE. It is also proposed that these of Meta-ensembles can be used to resolve quantification learning problem-novel task to predict collective performance. As in our recent works the deterministic outputs of Meta-ensemble members can be transformed to probabilistic outputs and then to prediction of class frequency. After analyzing the results of quantification, it becomes obvious that the most optimal design of the meta-ensembles is to collect pre-boosted classifiers trained on the SMOTE sample. © 2020 IEEE.

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Deep Learning for Automated Extraction of Primary Sites from Cancer Pathology Reports

Item Type Journal Article

Author J.X. Qiu

Author H.-J. Yoon

Author P.A. Fearn

Author G.D. Tourassi

Abstract Pathology reports are a primary source of information for cancer registries which process high volumes of free-text reports annually. Information extraction and coding is a manual, labor-intensive process. In this study, we investigated deep learning and a convolutional neural network (CNN), for extracting ICD-O-3 topographic codes from a corpus of breast and lung cancer pathology reports. We performed two experiments, using a CNN and a more conventional term frequency vector approach, to assess the effects of class prevalence and inter-class transfer learning. The experiments were based on a set of 942 pathology reports with human expert annotations as the gold standard. CNN performance was compared against a more conventional term frequency vector space approach. We observed that the deep learning models consistently outperformed the conventional approaches in the class prevalence experiment, resulting in micro- and macro-F score increases of up to 0.132 and 0.226, respectively, when class labels were well populated. Specifically, the best performing CNN achieved a micro-F score of 0.722 over 12 ICD-O-3 topography codes. Transfer learning provided a consistent but modest performance boost for the deep learning methods but trends were contingent on the CNN method and cancer site. These encouraging results demonstrate the potential of deep learning for automated abstraction of pathology reports. © 2013 IEEE.

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Notes:

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Export Date: 20 September 2026; Cited By: 120; Correspondence Address: G.D. Tourassi; Biomedical Sciences Engineering, and Computing Group, Computational Sciences and Engineering Division, Health Data Sciences Institute, Oak Ridge National Laboratory, Oak Ridge, 37831, United States; email: tourassig@ornl.govs; CODEN: ITIBF

Deep learning framework for quantifying self-organization in *Myxococcus xanthus*

Item Type Journal Article

Author J. Zhang

Author E.A. Caro

Author P. Chen

Author T.T. Khan

Author P.A. Murphy

Author L.J. Shimkets

Author A.B. Patel

Author R.D. Welch

Author O.A. Igoshin

Abstract Under starvation, *Myxococcus xanthus* bacteria initiate a multicellular developmental program in which cells move to form fruiting bodies and differentiate into distinct cell types. Many genes affecting this process have been identified, and it is assumed that perturbing genes within the same pathway induces similar changes in the phenotype, although these changes may be subtle or obscured by pleiotropy. However, these pathways cannot be systematically mapped because there are no reliable methods for quantifying phenotype similarity. Here, we applied deep learning to quantify phenotype patterns and self-organization dynamics of 292 genetically

distinct strains. We integrated ResNet and StyleGAN2 into a Variational Autoencoder and trained it together with a Siamese network that learns phenotypic similarity. This end-to-end system encoded high-resolution microscopy images into 13-dimensional feature vectors, effectively capturing variation in aggregation patterns across time and strains. Human evaluation confirmed that our model's reconstructions were visually indistinguishable from real images and closely aligned with input phenotypes. Importantly, the feature space is interpretable: Individual dimensions correlate with biological features such as aggregate number and size, and extrapolation along these dimensions produces predictable morphological changes. Remarkably, our model revealed that developmental phenotypes and ultimate aggregation fate are predictable from the earliest images before visible aggregation begins. This predictability held across both genetic and environmental sources of variation, suggesting that subtle, early-stage phenotypic signatures carry critical information about developmental trajectories. These results demonstrate how machine learning can reveal hidden aspects of complex multicellular dynamics and provide methods for phenotypic analysis without manual annotation. Copyright © 2026 the Author(s).

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Deep learning to quantify pulmonary edema in chest radiographs

Item Type Journal Article

Author S. Horng

Author R. Liao

Author X. Wang

Author S. Dalal

Author P. Golland

Author S.J. Berkowitz

Abstract Purpose: To develop a machine learning model to classify the severity grades of pulmonary edema on chest radiographs. Materials and Methods: In this retrospective study, 369 071 chest radiographs and associated radiology reports from 64 581 patients (mean age, 51.71 years; 54.51% women) from the MIMIC-CXR chest radiograph dataset were included. This dataset was split into patients with and without congestive heart failure (CHF). Pulmonary edema severity labels from the associated radiology reports were extracted from patients with CHF as four different ordinal levels: 0, no edema; 1, vascular congestion; 2, interstitial edema; and 3, alveolar edema. Deep learning models were developed using two approaches: a semisupervised model using a variational autoencoder and a pretrained supervised learning model using a dense neural network. Receiver operating characteristic curve analysis was performed on both models. Results: The area under the receiver operating characteristic curve (AUC) for differentiating alveolar edema from no edema was 0.99 for the semisupervised model and 0.87 for the pretrained models. Performance of the algorithm was inversely related to the difficulty in categorizing milder states of pulmonary edema (shown as AUCs for semisupervised model and pretrained model, respectively): 2 versus 0, 0.88 and 0.81; 1 versus 0, 0.79 and 0.66; 3 versus 1, 0.93 and 0.82; 2 versus 1, 0.69 and 0.73; and 3 versus 2, 0.88 and 0.63. Conclusion: Deep learning models were trained on a large chest radiograph dataset and could grade the severity of pulmonary edema on chest radiographs with high performance. © RSNA, 2021.

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Deep learning-based analysis of deep ultraviolet (UV) microscopy images for label-free, point-of-care hematological analysis

Item Type Conference Paper

Author N. Kaza

Author L.R. Chiou

Author M.H. Lenk

Author J.J.M. Law

Author K. Mabry

Author F.E. Robles

Editor Cote G.L.

Editor Baba J.S.

Abstract Hematological analysis is based on assessing changes in the numbers of different blood cells and their morphological, molecular, and cytogenetic properties via a complete blood count. It is integral to diagnose and monitor a range of blood conditions and diseases, ranging from allergies and infections to different types of cancers. The conventional approach to hematology analysis requires time-consuming protocols, multiple expensive chemical reagents, complex equipment, and highly trained personnel for operation, and presents a significant burden to patients and healthcare systems. There is a need for simple, fast, low-cost alternatives such as label-free techniques that eliminate the need for staining or exogenous labels. We recently demonstrated label-free hematology analysis using deep-ultraviolet (UV) microscopy, a high-resolution imaging technique that yields quantitative molecular and structural information from biological samples. In this work, we present a fast, automated analysis pipeline to classify and count the different blood cell types in single-channel UV microscopy images using a low-cost, compact deep-UV microscope. Our previous work focused primarily on white blood cells; here, we further add platelets and red blood cells. We train a YOLOv7-style network to identify and count different blood cells in smear images acquired from a deep-UV microscopy system. Our deep-UV microscope in an LED-based, compact, and portable configuration and single-step analysis

pipeline could be further combined with UV-transparent PDMS-based microfluidic devices to develop a fully automated, low-cost, label-free hematology analyzer. © COPYRIGHT SPIE. Downloading of the abstract is permitted for personal use only.

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Deep Learning-Based Assessment of Lip Symmetry for Patients With Repaired Cleft Lip

Item Type Journal Article
Author K. Rosero
Author A.N. Salman
Author L.M. Harrison
Author A.A. Kane
Author C. Busso
Author R.R. Hallac

Abstract Objective: Post-surgical lip symmetry assessment is a key indicator of cleft repair success. Traditional methods rely on distances between anatomical landmarks, which are impractical for video analysis and overlook texture and appearance. We propose an artificial intelligence (AI) approach to automate this process, analyzing lateral lip morphology for a quantitative symmetry evaluation. Design: We utilize contrastive learning to quantify lip symmetry by measuring the similarity between the representations of the sides, which is subsequently used to classify the severity of asymmetry. Our model does not require patient images for training. Instead, we introduce dissimilarities in face images from open datasets using two methods: temporal misalignment for video frames and face transformations to simulate lip asymmetry observed in the target population. The model differentiates the left and right image representations to assess asymmetry. We evaluated our model on 146 images of patients with repaired cleft lip. Results: The deep learning model trained with face transformations categorized patient images into five asymmetry levels, achieving a weighted accuracy of 75% and a Pearson correlation of 0.31 with medical expert human evaluations. The model utilizing temporal misalignment achieved a weighted accuracy of 69% and a Pearson correlation of 0.27 for the same classification task. Conclusions: We propose an automated approach for assessing lip asymmetry in patients with repaired cleft lip by transforming facial images of control subjects to train a deep learning model, eliminating manual anatomical landmarks. Our promising results provide a more efficient and objective tool for evaluating surgical outcomes. © 2025, American Cleft Palate Craniofacial Association.

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Deep learning-based frameworks for the detection and classification of soniferous fish

Item Type Journal Article

Author Z. Huang

Author D. Ochs

Author M.C.P. Amorim

Author P.J. Fonseca

Author M. Goel

Author N.J. Nunes

Author M. Vieira

Author M. Lopes

Abstract Passive acoustic monitoring (PAM) is emerging as a valuable tool for assessing fish populations in natural habitats. This study compares two deep learning-based frameworks: (1) a multi-label segmentation-based classification system (SegClas) combining convolutional neural networks and long short term memory networks and, (2) an object detection approach (ObjDet) using a You Only Look Once based model to detect, classify, and count sounds produced by soniferous fish in the Tagus Estuary, Portugal. The target species—Lusitanian toadfish (*Halobatrachus didactylus*), meagre (*Argyrosomus regius*), and weakfish (*Cynoscion regalis*)—exhibit overlapping vocalization patterns, posing classification challenges. Results show both methods achieve high accuracy (over 96%) and F1 scores above 87% for species-level sound identification, demonstrating their effectiveness under varied noise conditions. ObjDet generally offers slightly higher classification performance (F1 up to 92%) and can annotate each vocalization for more precise counting. However, it requires bounding-box annotations and higher computational costs (inference time of ca. 1.95 s/h of recording). In contrast, SegClas relies on segment-level labels and provides faster inference (ca. 1.46 s/h). This study also compares both counting strategies, each offering distinct advantages for different ecological and operational needs. Our results highlight the potential of deep learning-based PAM for fish population assessment. © 2025 Acoustical Society of America.

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ISSN 00014966 (ISSN)**Issue** 2**Journal Abbr** J. Acoust. Soc. Am.**Date Added** 20/09/2026, 9:51:45 pm**Modified** 20/09/2026, 9:51:45 pm**Tags:**

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Deep Metric Learning-Based Classification for Pavement Distress Images

Item Type Journal Article**Author** Y. Li**Author** J. Wang**Author** B. Lü**Author** H. Yang**Author** X. Wu**Abstract** This study proposes a deep metric learning-based pavement distress classification method to address critical limitations in conventional approaches, including their dependency on large training datasets and inability to incrementally learn new

categories. To resolve high intra-class variance and low inter-class distinction in distress images, we design a CNN head with multi-cluster centroids trained via SoftTriple loss, simultaneously maximizing inter-class separation while establishing multiple intra-class centers. An adaptive weighting strategy combining sample similarity and class priors mitigates data imbalance, while soft-label techniques reduce labeling noise by evaluating similarity against support-set exemplars. Evaluations on the UAV-PDD2023 dataset demonstrate superior performance—3.2% higher macro-recall than supervised learning, and 6.7%/8.5% improvements in macro-F1/weighted-F1 over iCaRL incremental learning—validating the method’s effectiveness for real-world road inspection scenarios with evolving distress types and limited annotation. © 2025 by the authors.

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Deep neural network uncertainty quantification for LArTPC reconstruction

Item Type Journal Article

Author D. Koh

Author A. Mishra

Author K. Terao

Abstract We evaluate uncertainty quantification (UQ) methods for deep learning applied to liquid argon time projection chamber (LArTPC) physics analysis tasks. As deep learning applications enter widespread usage among physics data analysis, neural networks with reliable estimates of prediction uncertainty and robust performance against overconfidence and out-of-distribution (OOD) samples are critical for their full deployment in analyzing experimental data. While numerous UQ methods have been tested on simple datasets, performance evaluations for more complex tasks and datasets are scarce. We assess the application of selected deep learning UQ methods on the task of particle classification using the PiLArNet [1] monte carlo 3D LArTPC point cloud dataset. We observe that UQ methods not only allow for better rejection of prediction mistakes and OOD detection, but also generally achieve higher overall accuracy across different task settings. We assess the precision of uncertainty quantification using different evaluation metrics, such as distributional separation of prediction entropy across correctly and incorrectly identified samples, receiver operating characteristic curves (ROCs), and expected calibration error from observed empirical accuracy. We conclude that ensembling methods can obtain well calibrated classification probabilities and generally perform better than other existing methods in deep learning UQ literature. © 2023 IOP Publishing Ltd and Sissa Medialab.

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Tags:

Performance, Deep neural networks, Classification (of information), Learning systems, Forecasting, Uncertainty analysis, Pattern recognition, Data handling, Monte Carlo methods, Quantification methods, Uncertainty quantifications, Calibration, Calibration method, Analyse and statistical method, Analysis and statistical methods, Data processing methods, Fitting method, High energy physics detector, Liquefied gases, Pattern recognition, cluster finding, calibration and fitting method, Pattern recognition, cluster finding, calibration and fitting methods, Performance of high energy physic detector, Performance of High Energy Physics Detectors

Notes:

EID: 2-s2.0-85180556613

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: D. Koh; Stanford University, Stanford, Serra Mall, 450, United States; email: koh0207@stanford.edu

Deep Sense: Deep Learning for Early Staging the Onset of Diabetes Emanating Recognition of Activity Patterns

Item Type Journal Article

Author M.S. Alam

Author E.H. Bahadur

Author M.F.I. Khan

Author F.U. Mahmud

Author M.I.H. Siddiqui

Author A.K.M. Masum

Abstract This study presents a comprehensive framework for Human Activity Recognition (HAR) using smartphone sensor data, with a specific focus on identifying activities associated with diabetic risk factors. We investigate the efficacy of two deep learning architectures (Long Short-Term Memory (LSTM) networks and Graph Convolutional Networks (GCNs)) for activity recognition under conditions of limited training data. To address data scarcity, we employ Generative Adversarial Networks (GANs) to augment the training dataset with synthetically generated sensor data, enhancing model robustness beyond what is achievable with real sensor data alone. Continuous accelerometer and gyroscope data spanning daily activities were collected from experimental subjects over a 60-day period. This dataset was used to train and evaluate both LSTM and GCN models, with results demonstrating that the GCN architecture achieves superior performance in recognizing diabetes-related activities such as sedentary behavior, physical inactivity, and irregular meal patterns. Furthermore, we propose a novel risk quantification method that estimates diabetes risk by analyzing the duration and frequency of engagement in diabetes-related activities. We employ cosine similarity to measure the correspondence between activity patterns of diagnosed diabetic patients and experimental subjects, yielding a quantitative risk score. To validate the proposed framework, we conducted clinical HbA1c (A1C) assays on experimental subjects. One subject exhibited an A1C level of 6.1%, corresponding to a prediabetes diagnosis, which corroborated the high-risk classification predicted by our framework. These results demonstrate that the proposed HAR-based approach can accurately assess diabetes risk and classify individuals according to clinically validated diagnostic criteria, offering potential applications in continuous health monitoring and early intervention strategies. © 2025 Mohammed Shamsul Alam, Erfanul Hoque Bahadur, Md Fokrul Islam Khan, Farhad Uddin Mahmud, Md Ismail Hossain Siddiqui, Abdul Kadar Muhammad Masum. This open access article is distributed under a Creative Commons Attribution (CC-BY) 4.0 license.

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Deep Learning, Long Short-Term Memory Networks, Generative Adversarial Networks, Diabetes Risk Assessment, Graph Convolutional Networks, Health Monitoring, Human Activity Recognition, Smartphone Sensors

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Export Date: 20 September 2026; Cited By: 1; Funding details: Department of Computer Science and Engineering; International Islamic University Chittagong; IIUC; Correspondence Address: A.K.M. Masum; Department of Computer Science and Engineering, Southeast University, Dhaka, Bangladesh; email: akmmasum@yahoo.com

Deep transfer learning strategy based on TimesBlock-CDAN for predicting thermal environment and air conditioner energy consumption in residential buildings

Item Type Journal Article

Author L. Sun

Author Z. Hu

Author M. Mac

Author T. Imaizumi

Abstract The deployment of data-driven deep learning black-box models for thermal environment and air conditioner energy consumption modeling has gained popularity due to their high accuracy in the residential building sector. However, extensive data collection in real-world residential settings is both time-consuming and costly, complicating the development of these models. Although traditional domain adaptation methods enable transfer learning with minimal data from target buildings, significant discrepancies in insulation performance, thermal capacity, and air conditioner types between different buildings result in substantial label shifts between source and target domains. In this study, a domain adaptation strategy based on the TimesBlock-CDAN approach is proposed. This method integrates conditional features, including air conditioner heat input and the resulting variations in temperature, humidity, and energy consumption, into the adversarial domain

adaptation process, effectively minimizing the domain shift between the source and target domains and thereby enhancing the overall efficiency and accuracy of transfer learning. Specifically, two real-world tasks were designed: transfer learning between different setpoint temperatures within the same building, and transfer learning across different buildings. A comparative analysis of performance in single-step and iterative multi-step predictions within these two tasks demonstrates that the proposed approach improves predictive accuracy by 20% compared to models trained solely on target building data and shows superior performance compared to other transfer learning approaches. © 2024 The Author(s)

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Performance, machine learning, Transfer learning, Energy-consumption, energy use, Energy consumption, Domain adaptation, Learning strategy, Target domain, Air conditioner, air conditioning, Air conditioning, building, insulation, Residential building, Residential buildings, Thermal environment, Timesblock-CDAN, TimesBlock-CDAN

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Deep transfer learning to quantify pleural effusion severity in chest X-rays

Item Type Journal Article

Author T. Huang

Author R. Yang

Author L. Shen

Author A. Feng

Author L. Li

Author N. He
Author S. Li
Author L. Huang
Author J. Lyu

Abstract Purpose: The detection of pleural effusion in chest radiography is crucial for doctors to make timely treatment decisions for patients with chronic obstructive pulmonary disease. We used the MIMIC-CXR database to develop a deep learning model to quantify pleural effusion severity in chest radiographs. Methods: The Medical Information Mart for Intensive Care Chest X-ray (MIMIC-CXR) dataset was divided into patients 'with' or 'without' chronic obstructive pulmonary disease (COPD). The label of pleural effusion severity was obtained from the extracted COPD radiology reports and classified into four categories: no effusion, small effusion, moderate effusion, and large effusion. A total of 200 datasets were randomly sampled to manually check each item and determine whether the tags are correct. A professional doctor re-tagged these items as a verification cohort without knowing their previous tags. The learning models include eight common network structures including Resnet, DenseNet, and GoogleNET. Three data processing methods (no sampling, downsampling, and upsampling) and two loss algorithms (focal loss and cross-entropy loss) were used for unbalanced data. The Neural Network Intelligence tool was applied to train the model. Receiver operating characteristic curves, Area under the curve, and confusion matrix were employed to evaluate the model results. Grad-CAM was used for model interpretation. Results: Among the 8533 patients, 15,620 chest X-rays with clearly marked pleural effusion severity were obtained (no effusion, 5685; small effusion, 4877; moderate effusion, 3657; and large effusion, 1401). The error rate of the manual check label was 6.5%, and the error rate of the doctor's relabeling was 11.0%. The highest accuracy rate of the optimized model was 73.07. The micro-average AUCs of the testing and validation cohorts was 0.89 and 0.90, respectively, and their macro-average AUCs were 0.86 and 0.89, respectively. The AUC of the distinguishing results of each class and the other three classes were 0.95 and 0.94, 0.76 and 0.83, 0.85 and 0.83, and 0.87 and 0.93. Conclusion: The deep transfer learning model can grade the severity of pleural effusion. © 2022, The Author(s).

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Deep-learning-based probabilistic estimation of solar PV soiling loss

Item Type Journal Article

Author W. Zhang

Author S. Liu

Author O. Gandhi

Author C.D. Rodriguez-Gallegos

Author H. Quan

Author D. Srinivasan

Abstract Although the integration of solar photovoltaic (PV) systems is gaining widespread acceptance, the intermittency and instability of PV power generation lead to several operational challenges. PV power generation can be impacted by multiple environmental factors, such as the soiling of solar PV panels. There are some conventional methods proposed to deterministically estimate the solar power loss caused by soiling. However, the error of deterministic estimation cannot be eliminated due to the inherent volatility of solar power. Therefore, this paper proposes a probabilistic quantification method, namely SolarQRNN, to estimate the power loss by leveraging images captured by surveillance cameras. Specifically, the proposed model employs a novel quantile loss function and deep learning structures (backbone networks based on residual convolution units), which combines quantile regression and computer vision models for the first time. The proposed method has been extensively tested on a solar panel soiling image dataset. Test results indicate that SolarQRNN outperforms benchmark classification models with at least 51% improvements in evaluating metrics. © 2010-2012 IEEE.

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DeepWILD: Wildlife Identification, Localisation and estimation on camera trap videos using Deep learning

Item Type Journal Article

Author F. Simões

Author C. Bouveyron

Author F. Precioso

Abstract Videos and images from camera traps are more and more used by ecologists to estimate the population of species on a territory. It is a laborious work since experts have to analyse massive data sets manually. This takes also a lot of time to filter these videos when many of them do not contain animals or are with human presence. Fortunately, deep learning algorithms for object detection can help ecologists to identify multiple relevant species on their data and to estimate their population. In this study, we propose to go even further by using object detection model to detect, classify and count species on camera traps videos. To this end, we developed a 3-step process: (i) At the first stage, after splitting videos into images, we annotate images by associating bounding boxes to each label thanks to MegaDetector algorithm; (ii) then, we extend MegaDetector based on Faster R-CNN architecture with backbone Inception-ResNet-v2 in order to not only detect the 13 relevant classes but also to classify them; (iii) finally, we design a method to count individuals based on the

maximum number of bounding boxes detected. This final stage of counting is evaluated in two different contexts: first including only detection results (i.e. comparing our predictions against the right number of individuals, no matter their true class), then an evolved version including both detection and classification results (i.e. comparing our predictions against the right number in the right class). The results obtained during the evaluation of our model on the test data set are: (i) 73,92% mAP for classification, (ii) 96,88% mAP for detection with a ratio Intersection-Over-Union (IoU) of 0.5 (overlapping ratio between groundtruth bounding box and the detected one), and (iii) 89,24% mAP for detection at IoU = 0.75. Highly represented classes, like humans, have highest values of mAP around 81% whereas less represented classes in the train data set, such as dogs, have lowest values of mAP around 66%. Regarding the proposed counting method, we predicted a count either exact or ± 1 unit for 87% with detection results and for 48% with detection and classification results of our test data set. Our model is also able to detect empty videos. To the best of our knowledge, this is the first study in France about the use of object detection model on a French national park to locate, identify and estimate the population of species from camera trap videos. © 2023 Elsevier B.V.

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Defect classification and quantification method based on AC magnetic flux leakage time domain signal characteristics

Item Type Journal Article**Author** R. Wang**Author** Y. Chen**Author** H. Yu**Author** Z. Xu**Author** J. Tang**Author** B. Feng**Author** Y. Kang**Author** K. Song

Abstract Pipelines play a crucial role in industries such as petroleum and nuclear power, where non-destructive testing is essential. Magnetic flux leakage testing methods are widely used for pipeline inspection due to their ability to detect both internal and external defects. However, accurately classifying and evaluating the size of such defects poses challenges due to the complex coupling relationship between ferromagnetic materials and defect magnetic fields. This paper proposes a method for defect classification and quantification based on the time domain characteristics of AC magnetic flux leakage signals. Firstly, the paper explores the shielding effects caused by transient high magnetic permeability within the material on signals from internal defects. It analyzes the differences in signals between internal and external defects. Then, based on the nonlinear attributes of defect signals, the paper proposes a defect classification method based on derivatives analysis of the windowed time-domain signal. Moreover, the study finds that rising time and zero-crossing time can be used to assess the depth of internal and external defects separately, which can decouple the width and depth. Finally, experimental validation confirms the effectiveness of defects classification and quantification. This paper provides a feasible method for evaluating the defect of ferromagnetic materials. © 2024 Elsevier Ltd

Date 2025**DOI** 10.1016/j.ndteint.2024.103250**URL** <https://www.scopus.com/pages/publications/85207082679?origin=resultslist>**Archive** Scopus**Language** English**Volume** 149**Publisher** Elsevier Ltd**Publication** NDT and E International**ISSN** 09638695 (ISSN)**Journal Abbr** NDT E Int**Date Added** 20/09/2026, 9:51:56 pm**Modified** 20/09/2026, 9:51:56 pm**Tags:**

Pipelines, Classification, Nuclear fuels, Classification methods, Magnetic flux leakage, Magnetic leakage, Ferromagnetics, Ability testing, Pipeline corrosion, AC magnetic flux leakage testing, Defect classification, Leakage testing, Magnetic shielding, Magnetic shielding effect, Magnetic shielding effects, Nuclear industry, Rising time, Rising-time, Time-domain signal, Zero-crossing time

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Defining the dance: Quantification and classification of endoplasmic reticulum dynamics

Item Type Journal Article

Author C. Pain

Author V. Kriechbaumer

Abstract The availability of quantification methods for subcellular organelle dynamic analysis has increased rapidly over the last 20 years. The application of these techniques to contiguous subcellular structures that exhibit dynamic remodelling over a range of scales and orientations is challenging, as quantification of ‘movement’ rarely corresponds to traditional, qualitative classifications of types of organelle movement. The plant endoplasmic reticulum represents a particular challenge for dynamic quantification as it itself is an entirely contiguous organelle that is in a constant state of flux and gross remodelling, controlled by the actinomyosin cytoskeleton. © The Author(s) 2019. Published by Oxford University Press on behalf of the Society for Experimental Biology.

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Dynamics, Microscopy, Quantitative analysis, Movement, Endoplasmic reticulum (ER), ER remodeling

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Defining varus thrust assessment and measurement: A scoping review

Item Type Journal Article**Author** H. Shen**Author** N. D'Souza**Author** C.E. Hiller**Author** M.J. McKay**Author** K. Hu**Author** Z. Wang**Author** M. Simic

Abstract Objective: The primary aim of this scoping review was to identify and evaluate the methodologies used for classifying and quantifying varus thrust of the knee during walking. The secondary aim was to report outcomes of reliability, validity, prevalence, and quantitative cutoffs of varus thrust in existing studies. Methods: A systematic literature search was conducted across five electronic databases from inception to April 2024. Studies that reported varus thrust during gait in adults were included. Two independent reviewers extracted data, and data were synthesized based on thrust classification methods, prevalence, reliability, and validity results. Results: A total of 59 studies were included. Varus thrust was classified by observational methods ($k = 22$) with three different definitions used, three-dimensional motion capture systems ($k = 20$), and wearable three-dimensional motion sensors ($k = 17$). Reliability of both classification and quantification of varus thrust were reported by 27% of studies, with kappa values ranging from moderate to high (0.46–0.92). Only three studies reported diagnostic validity of their quantification methodology and their corresponding prevalence. 30 studies reported prevalence of varus thrust, however five different definitions were used across these studies, yielding prevalence values that ranged from 16% to 81.8%, with higher rates reported by three-dimensional motion sensor-based studies. Conclusion: In conclusion, this scoping review identified various methodologies for classifying and quantifying varus thrust, including observational methods and quantitative techniques. Reliability and validity assessments were reported inconsistently, as most studies did not report reliability and validity results. Prevalence of varus thrust ranged widely depending on the methodology used. © 2026 The Authors

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adult, human, Humans, range of motion, walking, Walking, reliability, reproducibility, Reproducibility of Results, validity, Review, prevalence, classification, information processing, methodology, data extraction, information retrieval, Medline, Scopus, physiology, knee joint, Knee Joint, knee osteoarthritis, Embase, Web of Science, pathophysiology, Biomechanical Phenomena, biomechanics, gait, Gait, gait disorder, knee instability, measurement, observational method, Range of Motion, Articular, scoping review, screening, search engine, varus knee, varus thrust, walking speed

Notes:

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Demographic trends of school-based musculoskeletal injuries between 2019 and 2023: An epidemiological and risk analysis

Item Type Journal Article**Author** E. Thompson**Author** S. Sofo**Author** D. Tomchuk**Author** K.R. Schneider**Author** A. Ramos**Author** S. Chung

Abstract Introduction Background Musculoskeletal injuries occur in various school-based settings. However, no information describes musculoskeletal injury risk in school-aged populations relative to age, race, season, body part, and injury classification. We sought to perform an epidemiological analysis and create estimated multinomial logistic regression (MLR) models to profile school-based musculoskeletal injuries seen and treated at United States emergency departments from 2019 through 2023. Methods We retrospectively searched the National Electronic Injury Surveillance System (NEISS) database from the United States Consumer Product Safety Commission (CPSC) between 2019 and 2023 for all school-based injuries in patients between 5 and 18 years of age. Through relative risk ratio calculations and MLR modeling, we created a comprehensive epidemiological analysis and predictive risk assessment profile of school-based musculoskeletal injuries between 2019 and 2023.

Results School-based musculoskeletal injuries declined during the 3-year COVID-19 timeframe and returned to above pre-COVID levels during 2023. Middle school-aged students experienced the most injuries, with high school students experiencing the least. A total of 54.6 % of injuries were to the upper extremity, with 38.5 % of injuries occurring in the lower extremity and 6.9 % in the trunk. Our estimated MLR models indicated that the most at-risk males for school-based musculoskeletal injuries were elementary students for upper and lower arm, wrist, and finger fractures during the summer, while middle and high school females demonstrated the highest risk of sustaining strains/sprains to the wrist and ankle during the summer and winter months. Conclusions From the NEISS database, we described the prevalence estimations of school-based musculoskeletal injuries reported to and treated by the United States emergency departments from 2019 through 2023. With our estimated MLR models, teachers, staff, school districts, administrators, healthcare providers, and other school leaders could create education and supervision initiatives to help prevent musculoskeletal injuries sustained in school settings. © 2025 Elsevier Ltd.

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Notes:

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Denoised generative fusion networks for noise-robust few-shot image classification

Item Type Journal Article

Author J. Wu

Author J. Chen

Author J. Luo

Author W. Yu

Author J. Hu

Author H. Li

Abstract Few-shot image classification (FSC) must often contend with inaccurate annotations and biased representations that arise when only a handful of training samples are available. These imperfections severely erode the reliability of existing FSC models. We propose Denoised Generative Fusion Networks (DGFN), a unified framework that tackles the twin challenges of label noise and representation bias in few-shot image classification. DGFN begins by pruning the support set with a principled uniqueness-aware composite distance that jointly considers intra-class coherence and inter-class separability, thus eliminating mislabeled or outlying samples without external supervision. A conditional variational auto-encoder is then trained on the cleaned support instances to hallucinate high-fidelity features, enriching class priors and effectively expanding the sample budget. Finally, an episodic fusion head adaptively weights real and synthetic features so that augmentation gains translate into reliable generalization. Extensive experiments on miniImageNet, tieredImageNet, CIFAR-FS, and CUB-FS show that DGFN boosts 5-way 1-shot accuracy by up to 4.6% over the state of the art and maintains its advantage under 20%~40% synthetic noise and cross-domain shifts, demonstrating its promise for safety-critical applications where clean, abundant annotations are unattainable. © The Author(s) 2025.

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Density-Fixing: Simple yet Effective Regularization Method based on the Class Priors

Item Type Conference Paper**Author** M. Kimura**Author** R. Izawa

Abstract Machine learning models suffer from overfitting, which is caused by a lack of labeled data. We proposed a framework of regularization methods, called density-fixing, that can be used commonly for supervised and semi-supervised learning to tackle this problem. Our proposed regularization method improves the generalization performance by forcing the model to approximate the class prior distribution or occurrence frequency. This regularization term is naturally derived from the formula of maximum likelihood estimation and is theoretically justified. We further investigated the asymptotic behavior of the proposed method and how the regularization term behaves when assuming a prior distribution of several classes in practice. We provide several theoretical analyses of the proposed method including asymptotic behavior. Our experimental results on multiple benchmark datasets are sufficient to support our argument, and we suggest that this simple and effective regularization method is useful in real-world machine learning problems. © 2021 IEEE.

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DePerio: Innovative Deep Learning-Based Framework for Periodontal Disease Diagnosis and Severity Evaluation Using Saliva Samples

Item Type Journal Article

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Abstract Recent research has identified salivary oral polymorphonuclear neutrophils (oPMNs) as reliable cellular biomarkers for monitoring the progression of periodontal disease (PD). While conventional diagnostic tools such as periodontal probing and radiographic imaging are effective at detecting advanced disease stages, they lack the sensitivity required for early diagnosis. oPMNs, derived from circulating white blood cells (WBCs), transmigrate through the oral epithelium and appear in saliva with diverse morphological characteristics. Although deep learning-based WBC quantification has been extensively explored using peripheral blood datasets, the unique morphology of oPMNs necessitates the development of new image datasets and retraining of models tailored to their detection. To address these challenges, we introduce DePerio—an AI-powered diagnostic pipeline that integrates a novel oPMN

isolation protocol with deep neural network (DNN) architectures. We evaluate the performance of multiple DNN models to achieve accurate detection and quantification of oPMNs. Validated against standard quantification methods for oPMNs and oral inflammatory load (OIL), DePerio achieved a detection error rate of less than 5%. In a clinical study involving 111 human saliva samples, spanning individuals from healthy to severe periodontitis cases, and successfully classified five distinct OIL levels. This robust and low-complexity platform provides a scalable and practical solution for early PD detection and longitudinal monitoring in clinical dental practice. © 2013 IEEE.

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Depression prevalence based on the Edinburgh Postnatal Depression Scale compared to Structured Clinical Interview for DSM Disorders classification: Systematic review and individual participant data meta-analysis

Item Type Journal Article

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Author L. Giardinelli
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Author J. Kohlhoff
Author L. Kusminskas
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Author I. Tendais
Author A. Töreki
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Abstract Objectives: Estimates of depression prevalence in pregnancy and postpartum are based on the Edinburgh Postnatal Depression Scale (EPDS) more than on any other method. We aimed to determine if any EPDS cutoff can accurately and consistently estimate depression prevalence in individual studies. Methods: We analyzed datasets that compared EPDS scores to Structured Clinical Interview for DSM (SCID) major depression status. Random-effects meta-analysis was used to compare prevalence with EPDS cutoffs versus the SCID. Results: Seven thousand three hundred and fifteen participants (1017 SCID major depression) from 29 primary studies were

included. For EPDS cutoffs used to estimate prevalence in recent studies (≥ 9 to ≥ 14), pooled prevalence estimates ranged from 27.8% (95% CI: 22.0%–34.5%) for EPDS ≥ 9 to 9.0% (95% CI: 6.8%–11.9%) for EPDS ≥ 14 ; pooled SCID major depression prevalence was 9.0% (95% CI: 6.5%–12.3%). EPDS ≥ 14 provided pooled prevalence closest to SCID-based prevalence but differed from SCID prevalence in individual studies by a mean absolute difference of 5.1% (95% prediction interval: –13.7%, 12.3%). Conclusion: EPDS ≥ 14 approximated SCID-based prevalence overall, but considerable heterogeneity in individual studies is a barrier to using it for prevalence estimation. © 2020 The Authors. International Journal of Methods in Psychiatric Research published by John Wiley & Sons Ltd.

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Depressive subtypes in an elderly cohort identified using latent class analysis

Item Type Journal Article

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Author M.L. Stek

Author R.C. van der Mast

Author D. Rhebergen

Abstract Background Clinical findings indicate heterogeneity of depressive disorders, stressing the importance of subtyping depression for research and clinical care. Subtypes of the common late life depression are however seldom studied. Data-driven methods may help provide a more empirically-based classification of late-life depression. Methods Data were used from the Netherlands Study of Depression in Older People (NESDO)

derived from 359 persons, aged 60 years or older, with a current diagnosis of major depressive disorder. Latent class analysis (LCA) was used to identify subtypes of depression, using ten CIDI-based depression items. Classes were then characterized using various sociodemographic and clinical characteristics. Results The most prevalent class, as identified by LCA, was a moderate-severe class (prevalence 46.5%), followed by a severe melancholic class (prevalence 38.4%), and a severe atypical class (prevalence 15.0%). The strongest distinguishing features between the three classes were appetite and weight and, to a lesser extent, psychomotor symptoms and loss of interest. Compared with the melancholic class, the severe atypical class had the highest prevalence of females, the lowest mean age, the highest BMI, and highest prevalence of both cardiovascular disease, and metabolic syndrome. Limitations The strongest distinguishing symptoms, appetite and weight, could be correlated. Further, only longitudinal studies could demonstrate whether the identified classes are stable on the long term. Discussion In older persons with depressive disorders, three distinct subtypes were identified, similar to subtypes found in younger adults. The strongest distinguishing features were appetite and weight; moreover, classes differed strongly on prevalence of metabolic syndrome and cardiovascular disease. These findings suggest differences in the involvement of metabolic pathways across classes, which should be considered when investigating the pathogenesis and (eventually) treatment of depression in older persons. © 2017

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Dermoscopic diagnosis of melanoma in a 4D space constructed by active contour extracted features

Item Type Journal Article

Author M. Mete

Author N.M. Sirakov

Abstract Dermoscopy, also known as epiluminescence microscopy, is a major imaging technique used in the assessment of melanoma and other diseases of skin. In this study we propose a computer aided method and tools for fast and automated diagnosis of malignant skin lesions using non-linear classifiers. The method consists of three main stages: (1) skin lesion features extraction from images; (2) features measurement and digitization; and (3) skin lesion binary diagnosis (classification), using the extracted features. A shrinking active contour (S-ACES) extracts color regions boundaries, the number of colors, and lesion's boundary, which is used to calculate the abrupt boundary. Quantification methods for measurements of asymmetry and abrupt endings in skin lesions are elaborated to approach the second stage of the method. The total dermoscopy score (TDS) formula of the ABCD rule is modeled as linear support vector machines (SVM). Further a polynomial SVM classifier is developed. To validate the proposed framework a dataset of 64 lesion images were selected from a collection with a ground truth. The lesions were classified as benign or malignant by the TDS based model and the SVM polynomial classifier. Comparing the results, we showed that the latter model has a better f-measure then the TDS-based model (linear classifier) in the classification of skin lesions into two groups, malignant and benign. © 2012 Elsevier Ltd.

Date 2012

DOI 10.1016/j.compmedimag.2012.06.002

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Notes:

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Export Date: 20 September 2026; Cited By: 29; Funding details: TAMU-C; Correspondence Address: M. Mete; Department of Computer Science, Texas A and M University-Commerce, Commerce, TX 75429, P.O. Box 3011, United States; email: mutlu.mete@tamuc.edu; CODEN: CMIGE

Design a Prototype of the Application System of Classification and Calculating Motor Vehicles on Highway

Item Type Conference Paper

Author M.T. Tamam

Author W. Dwiono

Author R.J.P. Safian

Editor Haryanto null

Abstract The growth rate of the number of motor vehicles has a tendency that is always increasing. The increase in the number of motorised vehicles which is not followed by the addition of roads is a cause of traffic jams on the highway. This study aims to design an application system that is used to determine the level of traffic density on certain road sections by using the background subtraction method to classify and count the number of vehicles passing on that road section. From the results of software testing, three types of vehicles can be classified, namely motorcycles, cars, and trucks/buses with an accuracy of 96%, 83%, and 87.15%, respectively. © 2020 IOP Publishing Ltd.

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Design and implementation of a knowledge-based system to improve maximum likelihood classification accuracy

Item Type Journal Article

Author A.A. Alesheikh

Author F. Sadeghi Naeeni Fard

Abstract Numerous classification algorithms have been developed, many of which are highly specific and only solve a reduced class of problems. Maximum likelihood classification (MLC) is the most widely used classification method. The underlying assumption in performing MLC is that the prior probability of land cover is equal. However, a priori occurrence probability has a crucial effect on classification results. The objective of this paper is to improve the accuracy of MLC using a priori information in a knowledge-based system. The mathematical formulations and the strategy are presented. Estimates of a prior probability through crop areas, crop calendar, and some a priori probability about agricultural practices have been used in augmenting the probability of pixels. An industrial agricultural field, Moghan Plain in northwestern Iran, has been selected for testing. A total of 176 ground truth points were identified and measured in the field, and 323 fields were used for accuracy evaluation. Prior probabilities were estimated based on transition matrices of five successive years. Overall classification accuracy based on conditional prior probabilities increased from 53.2% to 66.7% relative to MLC. Knowledge about harvesting time is formalized using a normalized difference vegetation index (NDVI) map from an advanced spaceborne thermal emission and reflection radiometer (ASTER) image. The overall accuracy was then increased to 72.3%. Object-based classification is used to determine the crop type of agricultural fields for which the geometry is constrained. Overall accuracy is then raised to 88.7%. © 2007, Taylor & Francis Group, LLC. All rights reserved.

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Notes:

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Export Date: 20 September 2026; Cited By: 11; Correspondence Address: A.A. Alesheikh; Faculty of Geodesy and Geomatics Engineering, K.N. Toosi University of Technology, Tehran, 19967-15433, ValyAsr Street, Mirdamad Cross, Iran; email: alesheikh@kntu.ac.ir

Design for a Sustainability Approach to Organic Solar Cell Design: The Use of Machine Learning to Quantify the Trade-off between Performance, Stability, and Environmental Impact

Item Type Journal Article

Author T.W. David

Author J. Kettle

Abstract Organic photovoltaics (OPVs) are considered one of the best-performing photovoltaic (PV) technologies from an environmental point of view. Many of the constituent materials possess low embodied energies, which can generally be processed and disposed of in a less energy-intensive manner than other PV technologies. There has been an enormous range of materials used in OPVs; however, identification of the optimal materials and device architectures that provide the best environmental profile within this large search space has yet to have been conducted. This is a non-trivial task because the selection of these materials not only impacts the environmental profile but also on the solar cell efficiency and its operational stability. Here, we have developed a methodology that enables rapid assessment of the trade-off between efficiency, stability, and embodied energy of an OPV using machine learning. To achieve this, a database of OPV data was used, which has been acquired from the literature between 2011 and 2020 and consists of 1580 device data points. Our results highlight the importance of focusing activity on particular transport layers, substrates, and active layer materials, which are discussed further in the manuscript. We demonstrate that the trained and validated models can predict, with a high degree of confidence, the efficiency, stability, and embodied energy of an OPV. The

methodology set out in this work provides a means of identifying optimum device configurations in a rapid manner such that the net energy production is maximized, whereas the environmental impact of OPVs is minimized. Materials which show promise toward delivering a positive net energy are PET + barrier layers, PET (substrates), NiOx, ZrOx, CrO2, ZnO, LiF, and MoO3 (transport layers). Active layer materials which show promise for delivering a positive net energy are DRCN7T, DR3TSBDT, ZnPc, PDPP4T-2F, PCDTBT (donors), and IT-4F, C61 (acceptors). © 2022 American Chemical Society. All rights reserved.

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Design of a hybrid classifier for natural textures in images from the Bayesian and fuzzy paradigms

Item Type Conference Paper

Author M. Guijarro

Author G. Pajares

Author R. Abreu**Author** L. Garmendia**Author** M. Santos

Abstract One objective for classifying textures in natural images is to achieve the best performance possible. The combination of multiple classifiers has been tested as a suitable technique. This is because the individual behaviors of the different classifiers are exploited by joining performances. Two main problems are to be addressed for combining different classifiers. First, which set of classifiers are to be selected? Second, which is the best combination method? Different strategies have been proposed focusing on both problems. We propose a new automatic approach based on two well tested classifiers, the non-parametric Parzen's windows and the Fuzzy clustering. A probability density function (PDF) is estimated through the Parzen's windows estimator to be embedded in the Bayesian framework. The decision is made by combining a likelihood value provided by the PDF and the a priori probability supplied by the Fuzzy clustering strategy through the membership degree. Experimental results for aerial images demonstrate the performance of the proposed design. ©2007 IEEE.

Date 2007**DOI** 10.1109/WISP.2007.4447562**URL** <https://www.scopus.com/pages/publications/51149118626?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IEEE Int. Symp. Intelligent Signal Process., WISP**Publisher** IEEE Computer Society**ISBN** 142440830X (ISBN); 978-142440830-6 (ISBN)**Proceedings Title** IEEE Int. Symp. Intelligent Signal Process., WISP**Conference Name** 2007 IEEE International Symposium on Intelligent Signal Processing, WISP**Date Added** 20/09/2026, 9:53:02 pm**Modified** 20/09/2026, 9:53:02 pm**Tags:**

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Design of personalised recommendation algorithm for e-commerce platform based on Naive Bayes model and Spark

Item Type Conference Paper

Author L. Bi

Abstract With the rapid growth of user scale and product categories in e-commerce platforms, traditional recommendation algorithms face challenges such as low efficiency of high-dimensional sparse data processing and insufficient real-time performance. With the goal of improving the computational efficiency of the recommendation system, this study proposes a personalised recommendation algorithm based on a Naive Bayes model and Spark. By integrating user behaviour data (including browsing, purchasing, collecting, etc.) and combining the parallelized data processing capability of Spark MLlib machine learning library, an efficient feature extraction and analysis process is constructed. Aiming at the high-dimensional characteristics of user behavioural features, the dynamic update strategy of a priori probability and conditional probability is designed to adapt to the dynamic changes of user interests. The experimental results show that the parallelisation improvement based on Spark architecture effectively solves the scalability bottleneck of the traditional Naive Bayes model in big data scenarios, and provides the e-commerce platform with a recommendation solution with both real-time and accuracy, which has certain application value. © 2025 Copyright held by the owner/author(s).

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Detailed monitoring of agricultural production based on the MODIS radiometer data

Item Type Conference Paper

Author V. Povkh

Author E. Vorobeychik

Author L. Shljakhova

Author G. Garbuzov

Author I. Beljaeva

Author I. Sytnyck

Abstract The opportunity of regular receiving of the MODIS/TERRA radiometer data in the direct broadcast mode makes it possible to use these data as perspective information source for the detailed analysis of agricultural production on a regional scale. The most suitable MODIS/TERRA data subset for this purpose is a pair of red and NIR spectral bands with 250 m spatial resolution. The specified spectral ranges are in accordance with the well known remote sensing algorithms applied to estimate vegetation properties. In the SRIA-Centre, Rostov-on-Don supported by the Rostov region administration, techniques have been tested for several years aimed at prompt checking of winter wheat crops by the MODIS/TERRA radiometer data. The type of crops selected for the related studies is due to the large size of individual fields of winter wheat in the region. Thus, any losses of accuracy resulted from the low spatial resolution of MODIS/TERRA radiometer data are diminished. To achieve the needed accuracy, the quality assessment of land vector layers is of great importance. These vector layers are used both for registration of space images and for field masks creation during image interpretation. The vector layers have been created on the basis of the Russian high resolution space photo of the KFA-1000 starting from 1999. Thematic analysis of MODIS/TERRA radiometer data is made with use of supervised classification techniques which require to define training classes prior to execution. Training classes are determined using the selected test sites (crop fields). The numerous regular inspected test sites are sorted in various agricultural zones of the region. GPS receivers serve to make analyses of the inspected fields and probe points location. The scaling test of winter wheat crops monitoring technology was carried out in 2002. Comparison of its results and farm reports has shown a sufficient reliability of remote sensing estimations on both sizes of the areas of agricultural production and crop condition assessment. All remote sensing processing procedures are conducted by ENVI 3.5 software.

Date 2004

DOI 10.1117/12.510638

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Extra Journal Abbreviation: Proc SPIE Int Soc Opt Eng
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Agriculture, Remote sensing, Vegetation, Data reduction, Estimation, Classification, Image analysis, Crops, MODIS, Radiometers, Optical resolving power, Crop monitoring

Notes:

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Detecting and Correcting for Label Shift with Black Box Predictors

Item Type Conference Paper

Author Z.C. Lipton

Author Y.-X. Wang

Author A.J. Smola

Editor Dy J.

Editor Krause A.

Abstract Faced with distribution shift between training and test set, we wish to detect and quantify the shift, and to correct our classifiers without test set labels. Motivated by medical diagnosis, where diseases (targets), cause symptoms (observations), we focus on label shift, where the label marginal $p(y)$ changes but the conditional $p(x|y)$ does not. We propose Black Box Shift Estimation (BBSE) to estimate the test distribution $p(y)$. BBSE exploits arbitrary black box predictors to reduce dimensionality prior to shift correction. While better predictors give tighter estimates, BBSE works even when predictors are biased, inaccurate, or uncalibrated, so long as their confusion matrices are invertible. We prove BBSE's consistency, bound its error, and introduce a statistical test that uses BBSE to detect shift. We also leverage BBSE to correct classifiers. Experiments demonstrate accurate estimates and improved prediction, even on high-dimensional datasets of natural images. © 2018 by the author(s).

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Proceedings Title Proc. Mach. Learn. Res.
Conference Name Proceedings of Machine Learning Research
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Classification (of information), Image enhancement, Statistical tests, Black boxes, Labeling, Diagnosis, Test sets, Training sets, Confusion matrix, Shift-and, Shift estimations, Marginals, Natural images, Black-box testing, High-dimensional dataset

Notes:

EID: 2-s2.0-105020572039

Export Date: 20 September 2026; Cited By: 277; Correspondence Address: Z.C. Lipton; Carnegie Mellon University, Pittsburgh, United States; email: zlipton@cmu.edu; Y.-X. Wang; Amazon AI, Palo Alto, United States; email: yuxiangw@amazon.com; A.J. Smola; Amazon AI, Palo Alto, United States; email: smola@amazon.com; Conference name: 35th International Conference on Machine Learning, ICML 2018; Conference date: 10 July 2018 through 15 July 2018; Conference code: 340699

Detecting and correcting for label shift with black box predictors

Item Type Conference Paper

Author Z.C. Lipton

Author Y.-X. Wang

Author A.J. Smola

Editor Dy J.

Editor Krause A.

Abstract Faced with distribution shift between training and test set, we wish to detect and quantify the shift, and to correct our classifiers without test set labels. Motivated by medical diagnosis, where diseases (targets), cause symptoms (observations), we focus on label shift, where the label marginal $p(y)$ changes but the conditional $p(x|y)$ does not. We propose Black Box Shift Estimation (BBSE) to estimate the test distribution $p(y)$. BBSE exploits arbitrary black box predictors to reduce dimensionality prior to shift correction. While better predictors give tighter estimates, BBSE works even when predictors are biased, inaccurate, or uncalibrated, so long as their confusion matrices are invertible. We prove BBSE's consistency, bound its error, and introduce a statistical test that uses BBSE to detect shift. We also leverage BBSE to correct classifiers. Experiments

demonstrate accurate estimates and improved prediction, even on high-dimensional datasets of natural images. © Copyright 2018 by the author(s).

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Language English

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Proceedings Title Int. Conf. Mach. Learn., ICML

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Artificial intelligence, Classification (of information), Learning systems, Image enhancement, Black boxes, Diagnosis, Test sets, Shift estimations, Natural images, Confusion matrices, High dimensional datasets

Notes:

EID: 2-s2.0-85057277412

Export Date: 20 September 2026; Cited By: 141; Correspondence Address: Z.C. Lipton; Carnegie Mellon University, Pittsburgh, United States; email: zlipton@crau.edu; Conference name: 35th International Conference on Machine Learning, ICML 2018; Conference date: 10 July 2018 through 15 July 2018; Conference code: 141700

Detecting Cochlear Synaptopathy Through Curvature Quantification of the Auditory Brainstem Response

Item Type Journal Article

Author J. Bao

Author S.L. Jegede

Author J.W. Hawks

Author B. Dade

Author Q. Guan

Author S. Middaugh

Author Z. Qiu

Author A. Levina

Author T.-H. Tsai

Abstract The sound-evoked electrical compound potential known as auditory brainstem response (ABR) represents the firing of a heterogenous population of auditory neurons in response to sound stimuli, and is often used for clinical diagnosis based on wave amplitude and latency. However, recent ABR applications to detect human cochlear synaptopathy have led to inconsistent results, mainly due to the high variability of ABR wave-1 amplitude. Here, rather than focusing on the amplitude of ABR wave 1, we evaluated the use of ABR wave curvature to detect cochlear synaptic loss. We first compared four curvature quantification methods using simulated ABR waves, and identified that the cubic spline method using five data points produced the most accurate quantification. We next evaluated this quantification method with ABR data from an established mouse model with cochlear synaptopathy. The data clearly demonstrated that curvature measurement is more sensitive and consistent in identifying cochlear synaptic loss in mice compared to the amplitude and latency measurements. We further tested this curvature method in a different mouse model presenting with otitis media. The change in curvature profile due to middle ear infection in otitis media is different from the profile of mice with cochlear synaptopathy. Thus, our study suggests that curvature quantification can be used to address the current ABR variability issue, and may lead to additional applications in the clinic diagnosis of hearing disorders. Copyright © 2022 Bao, Jegede, Hawks, Dade, Guan, Middaugh, Qiu, Levina and Tsai.

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Article, male, simulation, machine learning, nonhuman, controlled study, noise, quantitative study, animal experiment, mouse, ketamine, xylazine, CBA mouse, cochlear synaptopathy, cornea curvature, cubic spline method, deafferentation, distortion product otoacoustic emission, Down syndrome, evoked brain stem auditory response, experimental mouse, hearing impairment, hearing test, hidden hearing loss, lagrange polynomial method, mathematical phenomena, mouse model, noise-induced hearing loss, nonlinear least square method, numerical differentiation method, otitis media, Ts65Dn mouse, tympanometry

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Detecting Distributional Shift Responsible for Predictive Model's Failure

Item Type Conference Paper

Author D.S. Mukherjee

Author D. Bhandari

Author N.G. Yeri

Abstract Any predictive software deployed with this hypothesis that test data distribution will not differ from training data distribution. Real time scenario does not follow this rule, which results inconsistent and non-transferable observation in various cases. This makes the dataset shift, a growing concern. In this paper, we've explored the recent concept of Label shift detection and classifier correction with the help of Black Box shift detection(BBSD), Black Box shift estimation(BBSE) and Black Box shift correction(BBSC). Digits dataset from 'sklearn' and 'LogisticRegression' classifier have been used for this investigation. Knock out shift was clearly detected by applying Kolmogorov-Smirnov test for BBSD. Performance of the classifier got improved after applying BBSE and BBSC from 91% to 97% in terms of overall accuracy. © 2022 IEEE.

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Language English

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Export Date: 20 September 2026; Cited By: 0; Conference name: 2022 International Conference for Advancement in Technology, ICONAT 2022; Conference date: 21 January 2022 through 22 January 2022; Conference code: 177711

Detecting Key Indicators of Student Concentration in Mathematics Using TIMSS 2023 Data: An Explainable Machine Learning Approach

Item Type Conference Paper

Author A. Elouafi

Author I. Tammouch

Author S. Nouna

Author I. Laassar

Author A. Nouna

Editor El Bouanani F.

Editor Ayoub F.

Abstract In Morocco there is an enormous issue to keep students engaged and focused on a math class because this directly affects the way they learn and perform at schools overall. A 2022 to 26 June Strategic Roadmap for Education Reform clarified that the loss of concentration is a scourge that affects performance and participation. With the help of data from the TIMSS 2023 assessment, which focuses on students in the eighth grade, this study seeks to identify the major factors that affect the ability of students to stay focused in mathematics classes. Prior to analysis, we cleansed and processed the data by eliminating the absence of entries and by aggregating related categories. After that, we used an autoencoder to reduce the size of the data and help to reveal the hidden patterns in behavior and contexts in the classroom. Based on these capabilities, several models, such as random forests, decision trees, XGBoost, and an unstructured neural network, were developed to categorize students based on their attention level as either low or high. SHAP values revealed which variables were the most influential on the students' attention, making it easier to understand the findings. The results highlighted some persistent factors: the environment of the classroom and instructional delivery, technology accessibility, and the quality of interactions between teachers and students are all essential for maintaining focus. The research was able to show that explicable AI can be an extremely asset in research on education, providing explicit information and strategies that can be implemented by teachers and policymakers in the position to facilitate involvement in mathematics education. © 2025 IEEE.

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Conference Name 8th International Conference on Advanced Communication Technologies and Networking, CommNet 2025 - Proceedings

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Deep learning, Auto encoders, autoencoder, Learning systems, Forestry, Engineering education, deep learning, Explainable AI, Students, Decision trees, explainable AI, Random forests, random forest, Xgboost, XGBoost, Teaching, Education computing, Teachers', Machine learning approaches, Key indicator, SHAP value, SHAP values, Student concentration

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Detecting Opioid Use Disorder in Health Claims Data With Positive Unlabeled Learning

Item Type Journal Article

Author P. Kumar

Author F. Moomtaheen

Author S.A. Malec

Author J.J. Yang

Author C.G. Bologa

Author K.A. Schneider

Author Y. Zhu

Author M. Tohen

Author G. Villarreal

Author D.J. Perkins

Author E.M. Fielstein

Author S.E. Davis

Author M.E. Matheny

Author C.G. Lambert

Abstract Accurate detection and prevalence estimation of behavioral health conditions, such as opioid use disorder (OUD), are crucial for identifying at-risk individuals, determining treatment needs, monitoring prevention and intervention efforts, and recruiting

treatment-naïve participants for clinical trials. The availability of extensive health data, combined with advancements in machine learning (ML) frameworks, has enabled researchers to employ various ML techniques to predict or identify OUD within patient health data. Ideally, we could directly estimate the prevalence, or the proportion of a population with a condition over time. However, underdiagnosis and undercoding of conditions in patient health records make it challenging to determine the true prevalence of these conditions and to identify at-risk patients with less severe conditions who are more likely to be missed. Consequently, patients without diagnoses may comprise positive and negative examples for a given condition. Treating all undiagnosed (uncoded) patients as negative when applying ML methods can introduce bias into models, affecting their predictive power. To address this issue, we employed Positive Unlabeled Learning Selected Not At Random (PULSNAR), a Positive and Unlabeled (PU) learning technique, to estimate the probability of a given patient having OUD during a time window and the overall population prevalence of OUD. In a sample of 3,342,044 commercially insured US patients with at least one opioid prescription filled, PULSNAR estimated that 5.08% of patients have a cumulative prevalence of OUD over a 2-5 a observation period, compared to the 1.35% with a recorded OUD diagnosis, with 73.5% of cases not diagnosed/coded. The prevalence estimates provided by PULSNAR are consistent with those reported in other studies. © 2013 IEEE.

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Detecting Out-of-distribution Data through In-distribution Class Prior

Item Type Conference Paper

Author X. Jiang

Author F. Liu

Author Z. Fang

Author H. Chen

Author T. Liu

Author F. Zheng

Author B. Han

Editor Krause A.

Editor Brunskill E.

Editor Cho K.

Editor Engelhardt B.

Editor Sabato S.

Editor Scarlett J.

Abstract Given a pre-trained in-distribution (ID) model, the inference-time out-of-distribution (OOD) detection aims to recognize OOD data during the inference stage. However, some representative methods share an unproven assumption that the probability that OOD data belong to every ID class should be the same, i.e., these OOD-to-ID probabilities actually form a uniform distribution. In this paper, we show that this assumption makes the above methods incapable when the ID model is trained with class-imbalanced data. Fortunately, by analyzing the causal relations between ID/OOD classes and features, we identify several common scenarios where the OOD-to-ID probabilities should be the ID-class-prior distribution and propose two strategies to modify existing inference-time detection methods: 1) replace the uniform distribution with the ID-class-prior distribution if they explicitly use the uniform distribution; 2) otherwise, reweight their scores according to the similarity between the ID-class-prior distribution and the softmax outputs of the pre-trained model. Extensive experiments show that both strategies can improve the OOD detection performance when the ID model is pre-trained with imbalanced data, reflecting the importance of ID-class prior in OOD detection. The codes are available at https://github.com/tmlr-group/class_prior. © 2023 Proceedings of Machine Learning Research. All rights reserved.

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Machine learning, Detection methods, Probability distributions, Imbalanced data, Prior distribution, Uniform distribution, Inference stages, Causal relations, Distribution models, Distribution probability, Time detection, Time-outs

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Detecting Signs of Depression Using Social Media Texts Through an Ensemble of Ensemble Classifiers

Item Type Journal Article

Author R. Chiong

Author G.S. Budhi

Author E. Cambria

Abstract Artificial intelligence-based machine learning models have been widely used to explore and address various mental health-related problems in recent years, including depression. In this study, we present an ensemble approach to complement the 90 unique input features that we proposed in a previous study on depression detection using social media texts. Our proposed Ensemble of Ensemble Classifiers (EECs) combines many ensemble models, including Bagging Predictors, Random Forests, Adaptive Boosting and Gradient Boosting, as inner ensembles. These inner ensembles are arranged in a parallel fashion, where each of them is trained using

different subsets of data sampled from the training data via bootstrap sampling. After the models are trained, during the testing phase, the results of all inner ensembles are processed using two methods—majority vote or class priority threshold—to get the final result as an output. From the experiments, we find that EECs are accurate in detecting signs of depression in social media users by analysing their posts in social media platforms such as Twitter. Our approach outperforms other ensemble methods on the public datasets we used. Moreover, if set correctly, the parameters of EECs can further improve the performance of the proposed ensemble in detecting signs of depression. © 2010-2012 IEEE.

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Notes:

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Detecting Uncoded Self-Harm in Veterans' Electronic Health Records Using Positive and Unlabeled Learning: Retrospective Cohort Study

Item Type Journal Article

Author P. Kumar

Author A.D. Vízolay

Author R. Upadhayaya

Author F. Moomtaheen

Author D.R. Greer

Author C.G. Bologna

Author K.A. Schneider

Author S.E. Davis

Author M.E. Matheny

Author D. van der Goes

Author G. Villarreal

Author Y. Zhu

Author M. Tohen

Author S.A. Malec

Author J.J. Yang

Author E.M. Fielstein

Author C.G. Lambert

Abstract Background: Underdiagnosis and undercoding are common across mental health conditions, particularly suicide and self-harm. This leaves health care datasets lacking reliable negative examples needed for predictive modeling, phenotype prevalence estimation, and identification of individuals at elevated risk. We use positive and unlabeled (PU) learning to address this challenge. Objective: This study aims to identify US Veterans whose self-harm events were not explicitly captured through diagnostic codes in electronic health records (EHRs) and estimate the underlying prevalence using a novel PU learning algorithm. Methods: We performed a retrospective cohort study using Veterans Health Administration EHRs (from October 1, 1999, to August 31, 2019), selecting a random 25% sample of 1,329,120 Veterans out of 5,316,480 (1,193,563 males and 135,557 females) with at least 2 years of observation. The study cohort comprised 24,625 Veterans with coded self-harm and 1,304,495 uncoded, with the mean ages of 38.39 (SD 12.17) and 48.76 (SD 15.04) years, respectively. We applied our PULSNAR (positive unlabeled learning selected not at random) algorithm to estimate the proportion of individuals with uncoded self-harm. Covariates included age, medical conditions, procedures, and clinical observations. Four experts (raters) independently reviewed charts of 97 uncoded Veterans, each selected from 1% intervals of calibrated PULSNAR probabilities from 0.01 to 0.97. Agreement was assessed among raters, PULSNAR classifications, and consensus review decisions. Post hoc calibration was used to refine prevalence estimates. Results: Of the 159,049 covariates in the dataset, PULSNAR's Extreme Gradient Boosting (XGBoost) model identified 1302 (0.82%) as informative for classification. Only 1.85% (24,625/1,329,120) of Veterans had diagnostic codes indicating self-harm events, while PULSNAR estimated an overall prevalence of 10.46% (139,026/1,329,120) by identifying an additional $\alpha=8.77\%$ (114,404/1,304,495) of self-harm cases among the uncoded population. Of the 97 chart-reviewed patients, 39 had documented but uncoded self-harm. PULSNAR probabilities were post hoc calibrated such that their sum over the 97 cases equaled 39, which adjusted the combined coded and imputed prevalence downward from 10.46% to 7.91% (105,133/1,329,120). By applying this calibration to shift the probabilities of all uncoded Veterans, with bootstrapping for confidence intervals, PULSNAR estimates that coded self-harm represents only 23.4% (95% CI 17.76% to 31.51%) of all documented (coded+notes) self-harm. Conclusions: Under the "selected not at random" assumption, PULSNAR provides an innovative and scalable framework for estimating the clinically documented prevalence of mental health conditions and identifying the uncoded individuals with calibrated prediction, without requiring confirmed negative labels. This method offers an alternative to time-consuming chart reviews for detecting likely cases missing structured coding capture. By addressing diagnostic undercoding of mental health conditions in EHRs, this

approach has the potential to enhance the estimation of mental health prevalence and support screening, activation of automated clinical decision support, targeted intervention, better resource allocation, and research to improve outcomes in real-world settings. © Praveen Kumar, Alexandria D Vízolay, Rajesh Upadhyaya, Fariha Moomtaheen, Donald R Greer, Cristian G Bologa, Kristan A Schneider, Sharon E Davis, Michael E Matheny, David van der Goes, Gerardo Villarreal, Yiliang Zhu, Mauricio Tohen, Scott A Malec, Jeremy J Yang, Elliot M Fielstein, Christophe Gerard Lambert.

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Detecting, classifying, and counting blue whale calls with Siamese neural networks

Item Type Journal Article

Author M. Zhong

Author M. Torterotot

Author T.A. Branch

Author K.M. Stafford

Author J.-Y. Royer

Author R. Dodhia

Author J. Lavista Ferres

Abstract The goal of this project is to use acoustic signatures to detect, classify, and count the calls of four acoustic populations of blue whales so that, ultimately, the conservation status of each population can be better assessed. We used manual annotations from 350 h of audio recordings from the underwater hydrophones in the Indian Ocean to build a deep learning model to detect, classify, and count the calls from four acoustic song types. The method we used was Siamese neural networks (SNN), a class of neural network architectures that are used to find the similarity of the inputs by comparing their feature vectors, finding that they outperformed the more widely used convolutional neural networks (CNN). Specifically, the SNN outperform a CNN with 2% accuracy improvement in population classification and 1.7%-6.4% accuracy improvement in call count estimation for each blue whale population. In addition, even though we treat the call count estimation problem as a classification task and encode the number of calls in each spectrogram as a categorical variable, SNN surprisingly learned the ordinal relationship among them. SNN are robust and are shown here to be an effective way to automatically mine large acoustic datasets for blue whale calls. © 2021 Acoustical Society of America.

Date 2021

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Pages 3086-3094**Publication** Journal of the Acoustical Society of America**ISSN** 00014966 (ISSN)**Issue** 5**Journal Abbr** J. Acoust. Soc. Am.**Date Added** 20/09/2026, 9:52:16 pm**Modified** 20/09/2026, 9:52:16 pm**Tags:**

Deep learning, Network architecture, Convolutional neural networks, article, Large dataset, nonhuman, deep learning, Acoustics, convolutional neural network, Neural Networks, Computer, Manual annotation, Classification tasks, audio recording, Audio acoustics, animal, Animals, acoustics, vocalization, Vocalization, Animal, Conservation, Categorical variables, Accuracy Improvement, Acoustic signature, Balaenoptera, blue whale, Conservation status, Count estimation, Indian Ocean, Underwater acoustic communication, Underwater hydrophones

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Detection and classification of neurotoxins using a novel short-term plasticity quantification method

Item Type Journal Article**Author** G. Gholmieh**Author** S. Courellis**Author** S. Fakheri**Author** E. Cheung**Author** V. Marmarelis**Author** M. Baudry**Author** T. Berger

Abstract A tissue-based biosensor is described for screening chemical compounds that rapidly affect the nervous system. The proposed sensor is an extension of a previous work on cultured hippocampal slices [Biosens. Bioelectron. 16 (2001) 491]. The detection of the chemical compounds is based on a novel quantification method of short-term plasticity (STP) of the CA1 system in acute hippocampal slices, using random electrical impulse sequences as inputs and population spike (PS) amplitudes as

outputs. STP is quantified by the first and the second order kernels using a variant of the Volterra modeling approach. This approach is more specific and time-efficient than the conventional paired pulse and fixed frequency train methods [J. Neurosci. Methods 2 (2002) 111]. Describing the functional state of the biosensor, the kernels changed accordingly as chemical compounds were added. The second order kernel was decomposed into nine Laguerre functions. The corresponding Laguerre coefficients along with the first order kernel were used as features for classification purposes. The biosensor was tested using picrotoxin (100 μ M), trimethylpropane phosphate (10 μ M), tetraethylammonium (4 mM), valproate (5 mM), carbachol (5 mM), DAP5 (25 μ M), CNQX (3 μ M), and DNQX (0.15, 1.5, 3, 5 and 10 μ M). Each chemical compound gave a different feature profile corresponding to its pharmacological class. The first order kernel and the Laguerre coefficients formed the input to an artificial neural network (ANN) comprised of a single layer of perceptrons. The ANN was able to classify each tested compound into its respective class. © 2003 Elsevier Science B.V. All rights reserved.

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perception, article, Neural networks, quantitative analysis, nonhuman, controlled study, artificial neural network, Nonlinear analysis, animal tissue, Screening, Neurology, nerve cell plasticity, Biosensors, Plasticity, nonlinear system, Tissue, screening test, 6 cyano 7 nitro 2,3 quinoxalinedione, 6,7 dinitro 2,3 quinoxalinedione, biosensor, brain slice, CA1, carbachol, Kernels, neurotoxin, Neurotoxins, Paired pulse, picrotoxin, propane, Random train, Short-term plasticity, tetrammonium, valproic acid

Notes:

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Export Date: 20 September 2026; Cited By: 24; Funding details: Biomedical Simulation Resource; Office of Naval Research, ONR; Defense Advanced Research Projects Agency, DARPA; Correspondence Address: G. Gholmie; Department of Biomedical Engineering, University of Southern California, Hedco Neuroscience Bldg., Los Angeles, CA 90089-1451, 3614 Watt Way, United States; email: gholmie@usc.edu; CODEN: BBIOE

Detection and quantification of breast arterial calcifications on mammograms: a deep learning approach

Item Type Journal Article

Author N. Mobini

Author M. Codari

Author F. Riva

Author M.G. Ienco

Author D. Capra

Author A. Cozzi

Author S. Carriero

Author D. Spinelli

Author R.M. Trimboli

Author G. Baselli

Author F. Sardanelli

Abstract Objective: Breast arterial calcifications (BAC) are a sex-specific cardiovascular disease biomarker that might improve cardiovascular risk stratification in women. We implemented a deep convolutional neural network for automatic BAC detection and quantification. Methods: In this retrospective study, four readers labelled four-view mammograms as BAC positive (BAC+) or BAC negative (BAC-) at image level. Starting from a pretrained VGG16 model, we trained a convolutional neural network to discriminate BAC+ and BAC- mammograms. Accuracy, F1 score, and area under the receiver operating characteristic curve (AUC-ROC) were used to assess the diagnostic performance. Predictions of calcified areas were generated using the generalized gradient-weighted class activation mapping (Grad-CAM++) method, and their correlation with manual measurement of BAC length in a subset of cases was assessed using Spearman ρ . Results: A total 1493 women (198 BAC+) with a median age of 59 years (interquartile range 52–68) were included and partitioned in a training set of 410 cases (1640 views, 398 BAC+), validation set of 222 cases (888 views, 89 BAC+), and test set of 229 cases (916 views, 94 BAC+). The accuracy, F1 score, and AUC-ROC were 0.94, 0.86, and 0.98 in the training set; 0.96, 0.74, and 0.96 in the validation set; and 0.97, 0.80, and 0.95 in the test set, respectively. In 112 analyzed views, the Grad-CAM++ predictions displayed a strong correlation with BAC measured length ($\rho = 0.88$, $p < 0.001$). Conclusion: Our model showed promising performances in BAC detection and in quantification of BAC burden, showing a strong correlation with manual measurements. Clinical relevance statement: Integrating our model to clinical practice could improve BAC reporting without increasing clinical workload, facilitating large-scale studies on the impact of BAC as a biomarker of cardiovascular risk, raising awareness on women's cardiovascular health, and leveraging mammographic screening. Key Points: • We implemented a deep convolutional neural network (CNN) for BAC detection and quantification. • Our CNN had an area under the receiving operator curve of 0.95 for BAC detection in the test set composed of 916 views, 94 of which were BAC+. • Furthermore, our CNN showed a strong correlation with manual BAC measurements ($\rho = 0.88$) in a set of 112 views. © 2023, The Author(s).

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Artificial intelligence, adult, Article, female, Female, human, Humans, middle aged, Middle Aged, aged, Deep Learning, procedures, deep learning, major clinical study, convolutional neural network, prevalence, cardiovascular disease, area under the curve, receiver operating characteristic, transfer of learning, Retrospective Studies, retrospective study, diagnostic imaging, binary classification, mammography, artery calcification, breast disease, Breast Diseases, Cardiovascular Diseases, Mammography, multiclass classification, breast arterial calcification, Cardiovascular diseases, detection algorithm, Risk factors, Vascular calcification

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Detection and quantification of temperature sensor drift using probabilistic neural networks

Item Type Journal Article

Author M. Pereira

Author B. Glisic

Abstract Temperature effects are a major driver of strain and deformations in weather-exposed civil infrastructure, such as bridges and buildings. For such structures, long-term temperature data holds the potential for data-driven prediction of expected structural behavior, which in turn enables the detection of anomalous structural behavior. For this reason, structural health monitoring (SHM) strategies typically employ temperature sensors. However, the success of SHM is contingent on the quantity and quality of the available temperature data. Hence, accurate automatic methods for the detection and quantification of anomalies in temperature data are needed. In particular, gradual temperature sensor drifts are difficult to detect and can introduce

errors into the thermal compensation of strain sensors, which can be erroneously confounded with time-dependent structural behavior. Current data-driven methods use air temperature as predictor because it exhibits good correlation with temperatures in the structure. Sensor drift is typically quantified by analysis of the prediction residuals; however, these methods are not robust to outliers and can be affected by seasonal biases. In this work, a probabilistic neural network is used as the nonlinear data-driven temperature prediction model which enables the introduction of a sensible threshold to mitigate seasonal model bias. Furthermore, a novel drift detection method based on the evolution of parameters of a trinomial probability distribution is introduced, together with a robust drift quantification method. The performance of this method is assessed using real temperature data from a pedestrian bridge, spanning over seven years of the structure's life. © 2022 The Author(s)

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Tags:

Machine-learning, Machine learning, Forecasting, Neural-networks, Neural networks, Anomaly detection, Temperature sensors, Data validation, Probability distributions, Data driven, Probabilistics, Structural health monitoring, Fiber optics, Data-driven prediction, Fiber-optics, Long-term structural health monitoring, Probabilistic neural network, Probabilistic neural networks, Sensor drift, Temperature prediction, Temperature sensor drift

Notes:

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Detection of Blood Cancer Cells using Microscopic Image

Item Type Journal Article

Author J. Kirubakaran

Author P. Vedhanath

Author P. Lakshman

Author Y. Surya Ashok

Abstract For the automatic diagnosis and classification of leukemia and leukemoid reactions, the IDB2 (acute lymphoblastic leukemia-image database) dataset has been utilized. This paper focuses on an automated method to differentiate between leukemoid and leukemia reactions using images of blood smear. MobileNetV3 is employed to classify and count WBC types from segmented images. The BCCD (Blood Cell Count Detection) dataset, which contains 364 images of blood smear and 349 single WBC type images, has been used in this work. The image segmentation algorithm incorporates Fuzzy C-means clustering, the snake algorithm, and fusion rules. For classification, the VGG19 CNN (Convolutional Neural Network) architecture-based deep learning technique is implemented. Python and Google colab is used for designing and executing the algorithm respectively. © 2024 Inventive Research Organization.

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Python, MobileNetV3, Fuzzy C-Means, Google Colab, VGG 19

Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: J. Kirubakaran; Department of Electronics and Communication Engineering, Muthayammal Engineering College, India; email: kirubakaranj.rnd@gmail.com

Detection of Phenotypic Alterations Using High-Content Analysis of Whole-Slide Images

Item Type Journal Article

Author A. Shirinifard

Author S. Thiagarajan

Author P. Vogel

Author A. Sablauer

Abstract Tumors exhibit spatial heterogeneity, as manifested in immunohistochemistry (IHC) staining patterns. Current IHC quantification methods lose information by reducing this heterogeneity in each whole-slide image (WSI) or in selective fields of view to a single staining index. The aim of this study was to investigate the sensitivity of an IHC quantification method that uses this heterogeneity to reliably compare IHC staining patterns. We virtually partitioned WSIs by a grid of square tiles, and computed the staining index distributions to quantify heterogeneities. We used samples from these distributions as inputs to non-parametric statistical comparisons. We applied our grid method to fixed tumor samples from 26 tumors obtained from a double-blind preclinical study of a patient-derived orthotopic xenograft model of pediatric neuroblastoma in CD1 nude mice. We compared the results of our grid method to the results based on whole-slide indices, the current practice. We show that our grid method reliably detects phenotypic alterations that other tests based on whole-slide indices fail to detect. Based on robustness and increased sensitivity of statistical inference, we conclude that our method of whole-slide grid quantification is superior to existing whole-slide quantification techniques. © 2016, © The Author(s) 2016.

Date 2016

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Article, human, Humans, nonhuman, procedures, controlled study, content analysis, child, Child, statistical analysis, image processing, human cell, image analysis, Image Processing, Computer-Assisted, pathology, animal, Animals, error, metabolism, Biomarkers, Tumor, immunohistochemistry, Immunohistochemistry, tumor marker, heterogeneity, antineoplastic agent, animal experiment, animal model, animal tissue, mouse, staining, Data Interpretation, Statistical, vascularization, confidence interval, double blind procedure, cancer classification, Brain Neoplasms, rank sum test, Ki 67 antigen, statistical significance, priority journal, cisplatin, tissue section, sampling bias, sample size, molecular imaging, nonparametric test, randomized controlled trial (topic), neuroblastoma, Neuroblastoma, Antineoplastic Combined Chemotherapy Protocols, apoptosis, Apoptosis, cancer combination chemotherapy, cancer transplantation, CD34 antigen, cyclophosphamide, Cyclophosphamide, Double-Blind Method, doxorubicin, Doxorubicin, drug cytotoxicity, etoposide, Etoposide, health care quality, Heterografts, histone H3, Mice, Nude, multiple cycle treatment, Neoplasm Transplantation, neuroblastoma cell line, nude mouse, spatial patterns, vascular tumor, whole slide imaging, whole-slide quantification, xenograft

Notes:

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Detection, Classification and Counting of Moving Vehicles from Videos

Item Type Conference Paper

Author A. Sunny

Author N. Manohar

Editor Guru D.S.

Editor Kumar N.V.

Editor Javed M.

Abstract Detecting and Classifying vehicles into several categories demonstrates a considerable possibility in the field of intelligent transportation systems. In this paper, we aim to propose a Convolutional Neural network integrated with YOLO framework to detect, classify and count vehicles from videos considering different climatic conditions like rainy, foggy, night and sunny weathers. Experimental results show that on applying this proposed network on MIO vision Traffic Camera Dataset for training and testing the model with different traffic videos, the network has a good testing and classification ability taking into consideration of the complex features of the image dataset. Our algorithm achieves an average accuracy of 94.4% in the detection, classification and counting of vehicles depending on the visibility of the image in the frame. © The

Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2024.

Date 2024

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Intelligent vehicle highway systems, Vehicles, Deep learning, Intelligent systems, Classification (of information), Intelligent transportation systems, Convolution, Convolutional neural network, Convolutional neural networks, Statistical tests, YOLO, Detection, Classification, Ability testing, Climatic conditions, Counting, Moving vehicles, Traffic camera, Training and testing

Notes:

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Detection, classification, and counting blood cells using YOLOv8

Item Type Journal Article

Author D.T. Luong

Author D.T. Tuan

Author D.D. Anh

Author T.T. Hanh

Author H.T.L. Huong

Author T.X. n Thang

Abstract Distinguishing normal white blood cells from leukemia cells plays a role in assisting in the diagnosis of blood diseases. Up to now, many automated methods are applied to provide more time efficiency, timing, and accuracy such as YOLO, SVM, CNN,

and Faster CNN. In this research, we propose to use YOLOv8 to detect, classify, and count normal white blood cells and leukemia cells. The result of the experiment method with accuracy is 95.1% using the first number of blood images is 1500 images from AML public source and Hanoi Medical University collected by ourselves, and after augmentation is 3629 images that are used for training, validation, and testing. At the same time, we compare the accuracy of this method with studies using different versions of YOLO. © MIR Labs

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machine learning, classification, detection, Leukemia cell, Normal White Blood Cells, WBC counting

Notes:

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Export Date: 20 September 2026; Cited By: 8; Correspondence Address: D.T. Luong; School of Electrical and Electronics, Hanoi University of Science and Technology, Viet Nam; email: luong.duongtrong@hust.edu.vn

Determination of glycated hemoglobin using near-infrared spectroscopy combined with equidistant combination partial least squares

Item Type Journal Article

Author Y. Han

Author J. Chen

Author T. Pan

Author G. Liu

Abstract A novel near-infrared-spectroscopy-based quantification method for glycated hemoglobin (HbA1c), a major clinical diagnosis indicator of diabetes, was developed on the basis of simultaneous determination of hemoglobin (Hb) and absolute HbA1c content (Hb•HbA1c) in human hemolysate samples. Equidistant combination partial least squares (EC-PLS) method was proposed to perform wavelengths selection.

Competitive adaptive reweighted sampling PLS (CARS-PLS) and Monte Carlo uninformative variable elimination PLS (MC-UVE-PLS) methods were also conducted for comparison. A randomness and stability dependent rigorous process of calibration, prediction, and validation was performed to produce objective and stable models. The search range covered the unsaturated region (780-1880. nm, 2090-2330. nm). For Hb and Hb•HbA1c, only 6 and 14 wavelengths were selected with EC-PLS, 23 and 30 wavelengths were selected with CARS-PLS, and 100 and 120 wavelengths were selected with MC-UVE-PLS, respectively. The predicted values of relative percentage HbA1c were calculated from the predicted Hb and Hb•HbA1c values. The sensitivity and specificity for diabetes were 93.5% and 97.1% with EC-PLS, 91.3% and 94.1% with CARS-PLS, and 89.1% and 76.5% with MC-UVE-PLS, respectively. In three methods, EC-PLS not only employed the least wavelengths but also produced the best quantification accuracy for HbA1c. EC-PLS also achieved the highest classification accuracy for negative and positive samples for diabetes. The results confirm the feasibility of HbA1c quantification based on the simultaneous analysis of Hb and Hb•HbA1c with NIR spectroscopy. This technique is rapid and simple when compared with conventional methods, and is a promising tool for screening diabetes in large populations. © 2015 Elsevier B.V.

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Determining a mutual fund's equity class

Item Type Journal Article

Author F.L. Detzel

Abstract Mutual fund investors who use an asset allocation model must be able to readily identify each fund's equity class. Prior studies examine two approaches to classifying equity funds, factor loadings and portfolio characteristics. Their implementation is not feasible for many investors, however. I investigate a practicable alternative, Morningstar's characteristics-based style box. I find that it does not predict mutual fund returns as well as a discrete factor-loadings-based alternative during 1994 to 2004 overall, but predicts better in 2003 to 2004 after Morningstar changed its methodology. I also find that fund classifications drift considerably over the years. Actively-managed-fund investors need to monitor their portfolios regularly. © 2006 Academy of Financial Services. All rights reserved.

Date 2006

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Asset allocation, Equity class, Investment style, Mutual funds

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Export Date: 20 September 2026; Cited By: 8; Correspondence Address: F.L. Detzel; College of Business Administration, California State University San Marcos, San Marcos, 92096-0001, United States; email: ldetzel@csusm.edu

Developing a 3D mineral texture quantification method of drill core for geometallurgy

Item Type Journal Article

Author M.J. Voigt

Author J. Miller

Author L. Bbosa

Author R.A. Govender

Author D. Bradshaw

Author A. Mainza

Author M. Becker

Abstract Mineral texture is a critical factor which controls ore variability and is an important attribute in geometallurgy. In relation to downstream processes, it affects the fracture pattern during breakage, where rock strength is inherently a function of mineral texture. Because of the subjective nature of mineral texture, it has not been easy to quantify, especially in the context of a measurement suitable for use in geometallurgical programmes. The aim of this paper is to present the first steps in developing a 3D mineral texture quantification method for drill core and to assess its sensitivity to differences in rock strength using a case study. The methodology includes classifying the textural information using the 3D grey level co-occurrence matrices (GLCM) and X-ray computed tomography (XCT) coupled method. Rock strength tests were performed using the split Hopkinson pressure bar (SHPB). The case study investigates a heterogeneous polymetallic sulphide deposit and a homogeneous shale subdivided into three 'mineral textural types'. The variability is largely captured by the GLCM matrices, and preliminary trends can be observed where the shale is finer grained and has a higher yield strength in comparison with the coarser grained polymetallic sulphide ore. © 2019 South African Institute of Mining and Metallurgy. All rights reserved.

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Notes:

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Development and Evaluation of a Clinical Prediction Model for Acute Kidney Injury Risk in Elderly Patients with Heart Failure with Reduced Ejection Fraction

Item Type Journal Article

- Author** R. Yang
- Author** Q. Song
- Author** H. Ma
- Author** Y. Fan
- Author** S. Lin
- Author** T. Liang
- Author** J. Chen
- Author** N. Tan
- Author** L. Jiang

Abstract Background: Acute kidney injury (AKI) is a significant complication among patients with heart failure with reduced ejection fraction (HFrEF). Once AKI occurs, therapeutic options are limited to supportive care, underscoring the clinical importance of early identification of patients at risk. This study sought to identify independent predictors of AKI development in geriatric HFrEF patients and subsequently establish a clinically applicable bedside risk quantification. Method: A total of 1,496 elderly patients (≥60 years) diagnosed with HFrEF at Guangdong Provincial People's Hospital from January 2010 to December 2024 were enrolled according to predefined inclusion/exclusion criteria and stratified into an AKI group (n = 300) and a non-AKI group (n = 1196). Relevant parameters were screened using LASSO (Least Absolute Shrinkage and Selection Operator) regression, and risk factors for AKI in HFrEF patients were identified through univariate and multivariate

logistic regression analyses. A nomogram was developed based on multivariate logistic regression results, accompanied by a corresponding heatmap. The predictive accuracy of the nomogram was evaluated using receiver operating characteristic (ROC) curves and calibration plots, while its clinical utility was demonstrated through decision curve analysis (DCA). Result: This study identified four independent predictors of AKI in patients with HFrEF, including N-terminal pro-B-type natriuretic peptide (NT-proBNP), uric acid (UA), CRP (C-reaction protein) and urea. A nomogram was developed based on these factors. The evaluation results demonstrated that the model exhibited good diagnostic accuracy, with an area under the receiver operating characteristic curve (AUC) of 0.721 (95% CI: 0.658-0.785). The calibration curve and DCA further indicated that the model possesses favorable clinical applicability. Conclusion: This study developed and validated a nomogram model for predicting AKI in patients with HFrEF. Clinicians can utilize this model to optimize clinical decision-making based on individualized patient characteristics, enabling personalized risk stratification for therapeutic interventions and prognostic evaluations. © 2026, Ubiquity Press. All rights reserved.© 2026 The Author(s). This is an open-access article distributed under the terms of the Creative Commons Attribution 4.0 International License (CC-BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See <https://creativecommons.org/licenses/by/4.0/>.

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Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, risk assessment, Risk Assessment, machine learning, China, aged, Aged, Aged, 80 and over, very elderly, demographics, risk factor, Risk Factors, independent variable, predictive model, procedures, controlled study, major clinical study, epidemiology, complication, etiology, logistic regression analysis, area under the curve, receiver operating characteristic, evaluation study, cross validation, diagnostic accuracy, clinical decision making, cohort analysis, scoring system, Retrospective Studies, retrospective study, ROC Curve, physiology, pathophysiology, prognosis, Prognosis, nomogram, Nomograms, blood, multivariate logistic regression analysis, acute kidney failure, Acute Kidney Injury, amino terminal pro brain natriuretic peptide, brain natriuretic peptide, C reactive protein, Clinical prediction model, creatine kinase, development, heart failure, Heart Failure, heart failure with reduced ejection fraction, Heart Failure with Reduced Ejection Fraction, heart stroke volume, least absolute shrinkage and selection operator, Natriuretic Peptide, Brain, serum albumin, Stroke Volume, synergistic effect, troponin, univariate analysis, urea, uric acid

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Development and evaluation of a culture-independent method for source determination of fecal wastes in surface and storm waters using reverse transcriptase-PCR detection of FRNA coliphage genogroup gene sequences

Item Type	Journal Article
Author	J. Paar
Author	M.M. Doolittle
Author	M. Varma
Author	S. Sieftring
Author	K. Oshima
Author	R.A. Haugland
Abstract	A method, incorporating recently improved reverse transcriptase-PCR primer/probe assays and including controls for detecting interferences in RNA recovery and analysis, was developed for the direct, culture-independent detection of genetic

markers from FRNA coliphage genogroups I, II & IV in water samples. Results were obtained from an initial evaluation of the performance of this method in analyses of waste water, ambient surface water and stormwater drain and outfall samples from predominantly urban locations. The evaluation also included a comparison of the occurrence of the FRNA genetic markers with genetic markers from general and human-related bacterial fecal indicators determined by current or pending EPA-validated qPCR methods. Strong associations were observed between the occurrence of the putatively human related FRNA genogroup II marker and the densities of the bacterial markers in the stormwater drain and outfall samples. However fewer samples were positive for FRNA coliphage compared to either the general bacterial fecal indicator or the human-related bacterial fecal indicator markers particularly for ambient water samples. Together, these methods show promise as complementary tools for the identification of contaminated storm water drainage systems as well as the determination of human and non-human sources of contamination. © 2015.

Date 2015

DOI 10.1016/j.mimet.2015.02.015

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Article, human, Humans, comparative study, nonhuman, procedures, controlled study, classification, water pollution, genetics, isolation and purification, genotype, animal, Animals, gene sequence, virology, reverse transcription polymerase chain reaction, feces, Feces, surface water, Genotype, water sampling, virus detection, process development, priority journal, virus RNA, Bacteria (microorganisms), water analysis, Reverse Transcriptase Polymerase Chain Reaction, Bacterial fecal indicators, coliphage, Coliphages, Detection and quantification methods, FRNA coliphage, genetic marker, Genetic markers, storm water, Urban stormwater, viral contamination, virus culture, waste water, Water Pollution

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Export Date: 20 September 2026; Cited By: 4; Funding details: U.S. Environmental Protection Agency's New England Regional Applied Research Effort; CODEN: JMIMD

Development and Evaluation of a Volumetric Quantification Method for Fecal Particle Size Classification in Rainbow Trout Fed Different Diets

Item Type Journal Article

Author T.L. Welker

Author K. Overturf

Author F. Barrows

Abstract Alternative plant protein sources, such as soybean meal, can interfere with digestion due to the presence of anti-nutritional factors that cause a diarrhea-like condition in Rainbow Trout *Oncorhynchus mykiss*, resulting in very fine fecal particles that can adversely affect aquaculture systems and the environment. More research is needed to examine the impacts of feed formulation changes on fecal quality. The prevailing method (laser diffraction) used to measure fecal particle size (FPS) requires expensive equipment and is labor intensive. Methods for FPS measurement that are affordable and easy to implement are required. We developed a visual, volumetric method that can separate feces into three particle size-classes. Feces were collected from tanks fitted with radial flow separators, transferred to Imhoff cones, and allowed to settle. Three distinct layers were visually identified, and microscopic analysis was used to categorize them into three particle size-classes: larger than 1.2 mm (bottom layer or large particles), 0.6–1.2 mm (middle layer or mid-sized particles), and 0.5 mm or smaller (top layer or fines). Independent observations confirmed that the method is reproducible and without observer bias. In a second study, the method was validated and further tested during a 4-week trial evaluating the effects of three diets (fish meal, plant based, or plant based + guar gum) and two feeding methods (hand feeding to satiation or automated belt feeding) on FPS production. Diets were isonitrogenous and isoenergetic, containing protein at 40 g/kg and lipid at 20 g/kg. Addition of guar gum significantly reduced the production of fines and increased the production of large fecal particles in Rainbow Trout (150 g/fish) that were fed the plant-based diet, but feeding method had no effect on FPS. Published 2020. This article is a U.S. Government work and is in the public domain in the USA.

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Export Date: 20 September 2026; Cited By: 10; Funding details: Jason Frost; Michael Becerra; U.S. Department of Agriculture, USDA; Soy Aquaculture Alliance, SAA, (1365-512-5697); Soy Aquaculture Alliance, SAA; Correspondence Address: T.L. Welker; U.S. Department of Agriculture, Agricultural Research Service, Hagerman Fish Culture Experiment Station, Hagerman, 3059F National Fish Hatchery Road, 83332, United States; email: thomas.welker@usda.gov; CODEN: NAJAF

Development and use of a quantum dot probe to track multiple yeast strains in mixed culture

Item Type Journal Article**Author** F.S. Gustafsson**Author** M.D. Whiteside**Author** V. Jiranek**Author** D.M. Durall

Abstract *Saccharomyces cerevisiae* strains vary in their ability to develop and enhance sensory attributes of alcoholic beverages and are often found growing in mixed strain fermentations; however, quantifying individual strains is challenging due to quantification inaccuracies, low marker longevity, and compromised kinetics. We developed a fluorescent probe, consisting of glutathione molecules conjugated to a quantum dot (QD). Two *S. cerevisiae* strains were incubated with different coloured probes (QD attached to glutathione molecules, QD-GSH), fermented at multiple ratios, and quantified using confocal microscopy. The QD method was compared with a culture method using microsatellite DNA analysis (MS method). Probes were taken up by an ADP1 encoded transporter, transferred from mother cell to daughter cell, detectable in strains throughout fermentation, and were non-toxic. This resulted in a new quantification method that was more accurate and efficient than the MS method.

Date 2014**DOI** 10.1038/srep06971**URL** <https://www.scopus.com/pages/publications/84923344334?origin=resultslist>**Archive** Scopus**Language** English**Volume** 4**Publisher** Nature Publishing Group**Publication** Scientific Reports**ISSN** 20452322 (ISSN)**Journal Abbr** Sci. Rep.**Date Added** 20/09/2026, 9:52:54 pm**Modified** 20/09/2026, 9:52:54 pm

Tags:

classification, genetics, quantum dot, metabolism, fermentation, Fermentation, Quantum Dots, *Saccharomyces cerevisiae*, yeast, Yeasts

Notes:

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Development and validation of a UV spectrophotometric method for biowaiver studies in pyrazinamide tablets

Item Type	Journal Article
Author	V. Saldaña-Bobadilla
Author	C.P. Barreto-Villanueva
Author	F.J. Ganoza-Gasco
Author	E. Perez-Chauca
Author	O.E. Caballero-Aquiño
Author	A.E. Mantilla-Rodriguez
Author	P.M. Alva-Plasencia
Author	M.S. Aurora-Prado
Abstract	Context: For regulatory reasons, a biowaiver is characterized as the substitution or exemption of bioavailability and/or bioequivalence research with in vitro dissolution tests, according to the Biopharmaceutical Classification System (BCS). These studies need validated quantification methods to give reliable results. Aims: To develop and validate the UV spectrophotometric method for biowaiver studies in pyrazinamide tablets. Methods: The parameters for validation, such as specificity, linearity, range, accuracy, precision, stability of the analytical solution, filter test, limits of detection, and quantification, were assessed according to the standards set by the guidelines of the International Conference on Harmonization and the United States Pharmacopeia. The dissolution operating conditions were 900 mL of dissolution media at pH 1.2, 4.5, and 6.8, with the paddle method at 75 rpm and sampling times of 5, 10, 15, 20, and 30 minutes. Results: The wavelength of maximum absorbance was 268 nm for the three dissolution media. The method was linear in the range of 1.1112–13.3334 µg/mL (r2>0.998). The limits of detection ranged from 0.0157 to 0.0222 µg/mL, and the limits of quantification were between 0.0415 and 0.0546 µg/mL. The recovered values were between 99.4 and 103.0%. No interference from the placebo was observed. The relative standard deviation values were less than 2% for precision. For robustness, different conditions were evaluated as wavelengths, changes in concentration, and Whatman No. 40, 41, and 42 filters; they did not modify the method response. The samples were kept unchanged for a period of 24 hours at 25°C. Conclusions: The UV spectrophotometric method was linear, accurate, specific, and precise in the concentration range studied and in the dissolution media at pH levels of 1.2, 4.5, and 6.8, being suitable for biowaiver studies in pyrazinamide tablets. © 2024 Academic Association of Pharmaceutical Sciences from Antofagasta (ASOCIFA). All rights reserved.
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Short Title Desarrollo y validación de un método espectrofotométrico UV para estudios de bioexención en tabletas de pirazinamida
Language English
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Article, human, controlled study, practice guideline, validation study, measurement accuracy, limit of quantitation, pH, spectrophotometry, limit of detection, tablet, in vitro study, drug synthesis, ultraviolet spectrophotometry, placebo, antitubercular drugs, bioequivalence, drug bioavailability, drug dissolution, drug solubility, pyrazinamide, solubility

Notes:

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Export Date: 20 September 2026; Cited By: 3

Development of a global 30m impervious surface map using multisource and multitemporal remote sensing datasets with the Google Earth Engine platform

Item Type Journal Article

Author X. Zhang

Author L. Liu

Author C. Wu

Author X. Chen

Author Y. Gao

Author S. Xie

Author B. Zhang

Abstract The amount of impervious surface is an important indicator in the monitoring of the intensity of human activity and environmental change. The use of remote sensing techniques is the only means of accurately carrying out global mapping of impervious surfaces covering large areas. Optical imagery can capture surface reflectance characteristics, while synthetic-aperture radar (SAR) images can be used to provide information on the structure and dielectric properties of surface materials. In addition,

nighttime light (NTL) imagery can detect the intensity of human activity and thus provide important a priori probabilities of the occurrence of impervious surfaces. In this study, we aimed to generate an accurate global impervious surface map at a resolution of 30m for 2015 by combining Landsat 8 Operational Land Image (OLI) optical images, Sentinel-1 SAR images and Visible Infrared Imaging Radiometer Suite (VIIRS) NTL images based on the Google Earth Engine (GEE) platform. First, the global impervious and nonimpervious training samples were automatically derived by combining the GlobeLand30 land-cover product with VIIRS NTL and MODIS enhanced vegetation index (EVI) imagery. Then, the local adaptive random forest classifiers, allowing for a regional adjustment of the classification parameters to take into account the regional characteristics, were trained and used to generate regional impervious surface maps for each $5^{\circ} \times 5^{\circ}$ geographical grid using local training samples and multisource and multitemporal imagery. Finally, a global impervious surface map, produced by mosaicking numerous $5^{\circ} \times 5^{\circ}$ regional maps, was validated by interpretation samples and then compared with five existing impervious products (GlobeLand30, FROM-GLC, NUACI, HBASE and GHSL). The results indicated that the global impervious surface map produced using the proposed multisource, multitemporal random forest classification (MSMT_RF) method was the most accurate of the maps, having an overall accuracy of 95.1% and kappa coefficient (one of the most commonly used statistics to test interrater reliability; Olofsson et al., 2014) of 0.898 as against 85.6% and 0.695 for NUACI, 89.6% and 0.780 for FROM-GLC, 90.3% and 0.794 for GHSL, 88.4% and 0.753 for GlobeLand30, and 88.0% and 0.745 for HBASE using all 15 regional validation data. Therefore, it is concluded that a global 30m impervious surface map can accurately and efficiently be generated by the proposed MSMT_RF method based on the GEE platform. The global impervious surface map generated in this paper is available at <https://doi.org/10.5281/zenodo.3505079> (Zhang and Liu, 2019). © Author(s) 2020.

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Export Date: 20 September 2026; Cited By: 302; Correspondence Address: L. Liu; State Key Laboratory of Remote Sensing Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, 100094, China; email: liuly@radi.ac.cn; L. Liu; University of the Chinese Academy of Sciences, Beijing, 100049, China; email: liuly@radi.ac.cn

Development of a simultaneous quantification method for ten trichothecenes including deoxynivalenol-3-glucoside in feed

Item Type Journal Article

Author M. Nomura

Author K. Shidara

Author I. Yasuda

Author K. Aoyama

Author A. Takahashi

Author T. Ishibashi

Abstract An analytical method for the simultaneous quantitation of ten trichothecenes of type A (HT-2 toxin, T-2 toxin, diacetoxyscirpenol, and neosolaniol) and type B (3-acetyldeoxynivalenol, 15-acetyldeoxynivalenol, deoxynivalenol, deoxynivalenol-3-glucoside, nivalenol, and fusarenon-X) in feed has been developed using liquid chromatography with tandem mass spectrometry. Mycotoxins extracted twice from samples using aqueous acetonitrile were purified using a multifunctional clean-up column, followed by a phospholipid removal column. Trichothecenes were analysed using liquid chromatography atmospheric pressure chemical ionization tandem mass spectrometry. The extraction efficiency of the mycotoxins and the repeatability of some were improved by repeated extractions. Ionization enhancement (signal enhancement) of some mycotoxins was improved by using the phospholipid removal column at the clean-up step. Spike and recovery tests of trichothecenes were conducted on maize, barley, soybean meal, rapeseed meal, and formula feeds (for starting broiler chicks, suckling pigs, and beef cattle). The mean recovery values were 70.6–119% with relative standard deviations < 17%. The limit of quantification and the limit of detection of our method were 20 and 6 µg/kg, respectively, for 3-acetyldeoxynivalenol and 15-acetyldeoxynivalenol; 10 and 3 µg/kg, respectively, for T-2 toxin, deoxynivalenol, and fusarenon-X; and 5 and 2 µg/kg, respectively, for nivalenol and the remaining mycotoxins. © 2020, Society for Mycotoxin (Research Gesellschaft für Mykotoxinforschung e.V.) and Springer-Verlag GmbH Germany, part of Springer Nature.

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Notes:

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: M. Nomura; Food and Agricultural Materials Inspection Center, Saitama, Saitama Shintoshin National Government Building, Kensato Building, Shintoshin 2-1, Chuo-Ku, Saitama-Shi, 330-9731, Japan; email: masayo_nomura336@famic.go.jp

Development of an Environment-Aware Locomotion Mode Recognition System for Powered Lower Limb Prostheses

Item Type Journal Article

Author M. Liu

Author D. Wang

Author H. Helen Huang

Abstract This paper aimed to develop and evaluate an environment-aware locomotion mode recognition system for volitional control of powered artificial legs. A portable terrain recognition (TR) module, consisting of an inertia measurement unit and a laser distance meter, was built to identify the type of terrain in front of the wearer while walking. A decision tree was used to classify the terrain types and provide either coarse or refined information about the walking environment. Then, the obtained environmental information was modeled as a priori probability and was integrated with a neuromuscular-mechanical-fusion-based locomotion mode (LM) recognition system. The designed TR module and environment-aware LM recognition system was evaluated separately on able-bodied subjects and a transfemoral amputee online. The results showed that the TR module provided high quality environmental information: TR accuracy is above 98% and terrain transitions are detected over 500 ms before the time required to switch the prosthesis control mode. This enabled smooth locomotion mode transitions for the wearers. The obtained environmental information further improved the performance of LM recognition system, regardless of whether coarse or refined information was used. In addition, the environment-aware LM recognition system produced reliable online performance when the TR

output was relatively noisy, which indicated the potential of this system to operate in unconstructed environment. This paper demonstrated that environmental information should be considered for operating wearable lower limb robotic devices, such as prosthetics and orthotics. © 2001-2011 IEEE.

Date 2016

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Development of Tier 2 LC-MRM-MS protein quantification methods for liquid biopsies

Item Type Journal Article

Author N. Diederiks

Author C.J. Ravensbergen

Author M. Treep

Author M. van Wezel

Author M. Kuruc

Author L. Renee Ruhaak

Author R.A.E.M. Tollenaar

Author C.M. Cobbaert

Author Y.E.M. van der Burgt

Author W.E. Mesker

Abstract In the pursuit of personalized diagnostics and tailored treatments, quantitative protein tests contribute to a more precise definition of health and disease. The development of new quantitative protein tests should be driven by an unmet clinical need and performed in a collaborative effort that involves all stakeholders. With regard to the analytical part, mass spectrometry (MS)-based platforms are an excellent tool for quantification of specific proteins in body fluids, for example focused on cancer. The obtained readouts have great potential in determining tumor aggressiveness to facilitate treatment decisions, and can furthermore be used to monitor patient response. Internationally standardized TNM classifications of malignant tumors are beneficial for diagnosis, however treatment outcome and survival of cancer patients is poorly predicted. To this end, the importance of the tumor microenvironment has endorsed the introduction of the tumor-stroma ratio as a prognostic parameter in solid primary tumor types. Currently, the stromal content of tumor tissues is determined via routine diagnostic pathology slides. With the development of liquid chromatography (LC)-MS methods we aim at quantification of tumor-stroma specific proteins in body fluids. In this mini-review the analytical aspect of this developmental trajectory is further detailed. © 2022 THE AUTHORS

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Journal Abbr J. Mass. Spectrom. Adv. Clin. Lab

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Article, human, clinical decision making, cancer prognosis, measurement precision, liquid chromatography-mass spectrometry, protein analysis, tumor microenvironment, liquid biopsy, multiple reaction monitoring, cancer diagnosis, Liquid biopsy, Cancer proteomics, Clinical chemistry, glycosylation, phosphorylation, protein degradation, protein expression level, Protein quantification, triple quadrupole mass spectrometry, Tumor-stroma ratio, vein puncture

Notes:

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: Y.E.M. van der Burgt; Department of Clinical Chemistry and Laboratory Medicine, Leiden University Medical Center, ZA Leiden, Albinusdreef 2, 2333, Netherlands; email: y.e.m.van_der_burgt@lumc.nl

DFSA: A classification capability quantification method for human action recognition

Item Type Conference Paper

Author C. Xu

Author J. He

Author X. Zhang

Abstract Motion related human action recognition using wearable sensors can potentially enable various useful daily applications. So far, most studies view it as a stand-alone mathematical classification problem without considering the physical nature of human motions. Consequently, they suffer from data dependencies and encounter the disaster of dimension problem and the over-fitting issue, and their models are never human-readable. In this study, we start from a deep analysis on natural physical properties of human motions, and then propose a useful feature selection method to quantify each feature's classification contribution capability. On one hand, the 'disaster of dimension' problem can be avoided to some extent, due to the constrained dimension of key features; On the other hand, over-fitting issue can be eased since the knowledge implied in human motions are nearly invariant, which compensates the possible data inadequacy. The experiment results indicate that the proposed method performs superior to those adopted in related works, such as decision tree, k-NN, SVM, neural networks. © 2017 IEEE.

Date 2018

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Extra Journal Abbreviation: IEEE SmartWorld Ubiquitous Intell. Comput., Adv. Trust. Comput., Scalable Comput. Commun., Cloud Big Data Comput., Internet People Smart City Innov., SmartWorld/SCALCOM/UIC/ATC/CBDCOM/IOP/SCI - Conf. Proc.

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Pages 1-6**Proceedings Title** IEEE SmartWorld Ubiquitous Intell. Comput., Adv. Trust. Comput., Scalable Comput. Commun., Cloud Big Data Comput., Internet People Smart City Innov., SmartWorld/SCALCOM/UIC/ATC/CBDCOM/IOP/SCI - Conf. Proc.**Conference Name** 2017 IEEE SmartWorld Ubiquitous Intelligence and Computing, Advanced and Trusted Computed, Scalable Computing and Communications, Cloud and Big Data Computing, Internet of People and Smart City Innovation, SmartWorld/SCALCOM/UIC/ATC/CBDCOM/IOP/SCI 2017 - Conference Proceedings**Date Added** 20/09/2026, 9:53:00 pm**Modified** 20/09/2026, 9:53:00 pm**Tags:**

Big data, Nearest neighbor search, Smart city, Ubiquitous computing, Decision trees, Disasters, Disaster prevention, Quantification methods, Action recognition, Feature selection methods, Human-action recognition, action recognition, classification contribution capability, Data dependencies, Human-readable, physical attributes, Physical nature

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DGCNet: a clinical knowledge-driven model for the segmentation of reflux esophagitis in endoscopic images

Item Type Journal Article**Author** Z. Ge**Author** Y. Fang**Author** J. Chang**Author** Z. Yu**Author** Y. Qiao**Author** Y. Hu**Author** J. Zhang**Author** Z. Duan**Author** X. Yang

Abstract Background: Digestive endoscopy is vital for diagnosing reflux esophagitis (RE) and is crucial for assessing disease severity. Although some existing pixel-level precision segmentation models can achieve fine-grained lesion region recognition, artificial intelligence-assisted diagnosis of RE is still in its infancy. The mutable shape, uneven distribution, and indistinct inflammatory texture boundaries inherent to RE endoscopic features pose formidable challenges for accurate lesion recognition and segmentation. This study proposes a classification and segmentation model to improve the Los Angeles grade (LA-grade) accuracy and enable precise lesion identification and quantification. Methods: In this study, we compiled the largest RE endoscopic image dataset, conducted a detailed classification study on it, and tested current mainstream and classic models. Based on RE endoscopic features, we introduced a clinical knowledge-driven RE segmentation model: the Deep Graph Convolution Network (DGCNet). We devised two novel modules: the Point Preservation Module to enhance model adaptability to lesion edge features and the Edge Graph Extraction Module for multi-scale fusion of edge and internal lesion features. Results: For classification results, we tested four classic models. The Swin Transformer model achieved the best performance for various evaluation indicators, with a classification accuracy of $86.2\% \pm 2.1\%$ and an F1 Score of $82.9\% \pm 2.3\%$. This study proposed a DGCNet framework and compared it with six classic segmentation networks. The results showed that our proposed model is significantly better than the other models, with mean Dice reaching $87.8\% \pm 1.4\%$ and mean IoU of $79.2\% \pm 1.6\%$. Conclusions: DGCNet demonstrates exceptional segmentation performance, showing promise for enhancing the accuracy of endoscopic LA-grade classification and precise recognition and segmentation of lesions. © The Author(s) 2026. Published by Oxford University Press and Sixth Affiliated Hospital of Sun Yat-sen University. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (<https://creativecommons.org/licenses/by-nc/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

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Diagnostic accuracy of texture analysis and machine learning for quantification of liver fibrosis in MRI: correlation with MR elastography and histopathology

Item Type Journal Article

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Abstract Objectives: To compare the diagnostic accuracy of texture analysis (TA)–derived parameters combined with machine learning (ML) of non-contrast-enhanced T1w and T2w fat-saturated (fs) images with MR elastography (MRE) for liver fibrosis quantification. Methods: In this IRB-approved prospective study, liver MRIs of participants with suspected chronic liver disease who underwent liver biopsy between August 2015 and May 2018 were analyzed. Two readers blinded to clinical and histopathological findings performed TA. The participants were categorized into no or low-stage (0–2) and high-stage (3–4) fibrosis groups. Confusion matrices were calculated using a support vector machine combined with principal component analysis. The diagnostic accuracy of ML-based TA of liver fibrosis and MRE was assessed by area under the receiver operating characteristic curves (AUC). Histopathology served as reference standard. Results: A total of 62 consecutive participants (40 men; mean age $\pm$ standard deviation, 48 ± 13 years) were included. The accuracy of TA and ML on T1w was 85.7% (95% confidence interval [CI] 63.7–97.0) and 61.9% (95% CI 38.4–81.9) on T2w fs for classification of liver fibrosis into low-stage and high-stage fibrosis. The AUC for TA on T1w was similar to MRE (0.82 [95% CI 0.59–0.95] vs. 0.92 [95% CI 0.71–0.99], $p = 0.41$), while the AUC for T2w fs was significantly lower compared to MRE (0.57 [95% CI 0.34–0.78] vs. 0.92

[95% CI 0.71–0.99], $p = 0.008$). Conclusion: Our results suggest that liver fibrosis can be quantified with TA-derived parameters of T1w when combined with a ML algorithm with similar accuracy compared to MRE. Key Points: • Liver fibrosis can be categorized into low-stage fibrosis (0–2) and high-stage fibrosis (3–4) using texture analysis–derived parameters of T1-weighted images with a machine learning approach. • For the differentiation of low-stage fibrosis and high-stage fibrosis, the diagnostic accuracy of texture analysis on T1-weighted images combined with a machine learning algorithm is similar compared to MR elastography. © 2020, European Society of Radiology.

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Diagnostic Gain from Surgical Biopsy for Interstitial Lung Disease - When is it Worth the Risk?

Item Type Journal Article

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Abstract Background: History, clinical presentation, lung function testing, radiographs including HRCT and nonsurgical biopsy techniques in most cases provide sufficient information for classification of interstitial lung disease (ILD). However, in a small percentage it is not possible to establish the diagnosis so that lung biopsy may be required. We analyzed under which circumstances a reduction of invasive procedures is reasonable. Methods: Between January 1997 and December 2009 we examined 3399 specimens from 1299 patients with benign inflammatory and granulomatous diseases in whom ILD was clinically hypothesized. We compared the probability of disease according to Bayes before and after surgery which corresponds to the clinical diagnosis (a priori probability) and the final diagnosis (a posteriori probability). Additionally, procedures, operation related complications and the patients smoking habits were documented. Results: In 111 patients (8.5%) surgical evaluation was performed (14 mediastinoscopies, 97 thoracotomies/VATS biopsies). All mediastinoscopies substantiated a epitheloid cell granulomatosis. In 30% of all VATS procedures a prolonged air leak of more than 4 days was observed. One patient died and one had to get a new chest tube after removal. Changes of a priori/a posteriori probabilities was shown for non-smokers in Wegners granulomatosis (0.6 vs. 2.2%) and IPF (16.7 vs. 34.8%), for smokers in Langerhans cell histiocytosis (1.4 vs. 7.8%) and IPF (16.7 vs. 33.3%). In the majority of cases even a reduction of probability was seen. Conclusion: Considering complications and limited diagnostic gain, lung biopsies for diagnosis of ILD should be recommended only in selected patients. © 2016 Georg Thieme Verlag KG Stuttgart New York.

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Notes:

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Diagnostic performance of AI-based EEG interpretation versus human clinical experts for epilepsy detection: systematic review and meta-analysis

Item Type Journal Article
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Abstract Background : Electroencephalography (EEG) interpretation for epilepsy diagnosis faces persistent challenges including specialist shortages, variable interpretation accuracy, and limited accessibility. Artificial intelligence (AI)-based automated interpretation systems promise to address these limitations, yet their diagnostic performance compared to clinical experts remains incompletely characterized. Objective: To systematically evaluate and meta -analyze the diagnostic accuracy of AI-enabled EEG interpretation compared with human clinical experts for epilepsy detection. Methods : We conducted a systematic review following PRISMA-DTA and Cochrane MECIR standards, searching PubMed, Scopus, IEEE Xplore, and Google Scholar through June 2025. Studies directly comparing AI algorithms with human expert interpretation using identical EEG datasets were included. Quality assessment employed QUADAS-AI criteria. Bivariate meta -analysis was performed to estimate pooled sensitivity, specificity, diagnostic odds ratios, and likelihood ratios. It is important to note that AI algorithms evaluated in these studies were trained using expert-labeled EEG data through supervised learning approaches; our comparison focuses on diagnostic accuracy outcomes rather than algorithmic independence from expert knowledge. Results : Three studies encompassing 9,775 EEG examinations met inclusion criteria. AI demonstrated superior pooled sensitivity (0.83–0.85 vs. 0.77–0.80) and specificity (0.83–0.85 vs. 0.72–0.75) compared to human experts. The diagnostic odds ratio for AI was approximately double that of humans (12–13 vs. 6–7). AI exhibited consistently narrower confidence intervals, indicating greater interpretive reliability. For normal versus abnormal EEG classification, AI achieved 86–90% sensitivity with enhanced consistency compared to human evaluators. However, substantial heterogeneity (I² ' 75%) and methodological limitations were identified across studies. Conclusions: AI-based EEG interpretation demonstrates diagnostic performance equal or superior to human experts with enhanced consistency, supporting potential implementation as clinical triage tools. However, limited transparency, patient selection bias, and deployment feasibility constraints warrant further investigation before widespread clinical adoption. Multiple sources of uncertainty affect AI-based diagnostic systems in clinical applications. Internal uncertainties include model parameter uncertainty, threshold selection variability, and training data limitations. External uncertainties encompass population heterogeneity, EEG acquisition variability, and clinical context differences. Parametric uncertainties arise from model architecture choices, while non-parametric uncertainties reflect distribution-free variations in real-world data. The amounts and structures of these uncertainties are often unknown in operational settings, necessitating robust uncertainty quantification methods for safe clinical deployment. © 2026 The Author(s).

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Diameter-dependent classification of dermal vasculature using optical coherence tomography angiography

Item Type Journal Article

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Author M. Katsuyama

Author T. Yamashita

Author C. Katagiri

Author R.K. Wang

Author Y. Hara

Abstract Background: Dermal blood vessels beneath the epidermis play critical roles in epidermal homeostasis and are functionally divided into several types, such as capillaries. Optical coherence tomography angiography (OCTA) is a powerful tool for the non-invasive assessment of dermal vasculature. However, the classification of vessel types has been limited. To address this issue, we proposed an algorithm for diameter-dependent classification that preserves three-dimensional (3D) information using OCTA. Methods: OCTA data were acquired by a prototype swept-source-type optical coherence tomography (OCT) system, which was processed through several imaging filters: an optical microangiography (OMAG) imaging filter, a vesselness imaging filter, and a diameter map filter. All vessels were visually classified into

three types based on their diameters, as micro-vessels, intermediate vessels, and thick vessels. Aging-related alterations and their association with the epidermis were investigated for each vessel type. The measurements were conducted on the cheeks of 124 female subjects aged 20–79 years. Results: The 3D vascular structure was visualized by applying our proposed post-processing filters. Based on visual assessment, the thresholds for the diameters of the micro, intermediate and thick vessels were set at 80 and 160 μm . It was found that micro-vessels were predominantly located in the upper layer of the dermis and thick vessels in the deeper layer. Analysis of vessel metrics revealed that the volume density of the micro-vessels decreased significantly with age ($r=-0.36$, $P<0.001$) and was positively correlated with epidermal thickness ($r=0.50$, $P<0.001$). In contrast, the volume density of thick vessels significantly increased with age ($r=0.2$, $P<0.05$) and was not significantly correlated with epidermal thickness ($r=0.13$, $P\geq 0.05$). Conclusions: In this study, we proposed a 3D quantification method using OCTA for dermal blood vessels and various vessel metrics, such as vessel volume density. This proposed classification will be beneficial for determining the function of the dermal vasculature and its diagnostic applications. © Quantitative Imaging in Medicine and Surgery. All rights reserved.

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“Did you read about Berlin?” Terrorist attacks, online media reporting and support for refugees in Germany

Item Type Journal Article

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Abstract We analyze whether the Islamist terror attack on the Christmas market in Berlin in 2016 had an impact on public opinion toward immigration in general and, since the attacker has entered Germany to apply for asylum, toward refugees in particular. An analysis of this natural experiment reveals a negative shift regarding the latter, while no differences are observed for the former. To shed more light on the driver of attitude change, we combine these findings with a quantitative content analysis of online media reporting about refugees before and after the attack. Mass media have long been considered to have an impact on exclusionary attitudes toward ethnic minorities. However, empirical evidence on this relationship remains largely anecdotal. We draw upon unsupervised machine learning to quantify the developments in reporting in three popular German online news websites. Results reveal that the attack had significant impact on media reporting on these websites. However, the strong focus on the attack was only short lived, quickly decreasing already in the second week after the attack. Linking media data to the public opinion data reveals no clear connection between reporting and attitudes. In contrast to theoretical expectations, descriptive evidence even shows that both follow almost opposite trends, since people changed their attitudes only weeks after the attack. We discuss potential explanations of these, at first sight, counterintuitive findings. © 2020 Nomos Verlagsgesellschaft mbH und Co. All rights reserved.

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Differentiable sampling of molecular geometries with uncertainty-based adversarial attacks

Item Type Journal Article

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Abstract Neural network (NN) interatomic potentials provide fast prediction of potential energy surfaces, closely matching the accuracy of the electronic structure methods used to produce the training data. However, NN predictions are only reliable within well-learned training domains, and show volatile behavior when extrapolating. Uncertainty quantification methods can flag atomic configurations for which prediction confidence is low, but arriving at such uncertain regions requires expensive sampling of the NN phase space, often using atomistic simulations. Here, we exploit automatic differentiation to drive atomistic systems towards high-likelihood, high-uncertainty configurations without the need for molecular dynamics simulations. By performing adversarial attacks on an uncertainty metric, informative geometries that expand the training domain of NNs are sampled. When combined with an active learning loop, this approach bootstraps and improves NN potentials while decreasing the number of calls to the ground truth method. This efficiency is demonstrated on sampling of kinetic barriers, collective variables in molecules, and supramolecular chemistry in zeolite-molecule interactions, and can be extended to any NN potential architecture and materials system. © 2021, The Author(s).

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Differential and longitudinal immune gene patterns associated with reprogrammed microenvironment and viral mimicry in response to neoadjuvant radiotherapy in rectal cancer

- | | |
|-----------|---------------------|
| Item Type | Journal Article |
| Author | A. Wilkins |
| Author | E. Fontana |
| Author | G. Nyamundanda |
| Author | C. Ragulan |
| Author | Y. Patil |
| Author | D. Mansfield |
| Author | J. Kingston |
| Author | F. Errington-Mais |
| Author | D. Bottomley |
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| Author | M. Dillon |
| Author | S. Demaria |
| Author | D. Sebag-Montefiore |
| Author | K. Harrington |
| Author | N. West |
| Author | A. Melcher |

Author A. Sadanandam

Abstract Background Rectal cancers show a highly varied response to neoadjuvant radiotherapy/chemoradiation (RT/CRT) and the impact of the tumor immune microenvironment on this response is poorly understood. Current clinical tumor regression grading systems attempt to measure radiotherapy response but are subject to interobserver variation. An unbiased and unique histopathological quantification method (change in tumor cell density (δ TCD)) may improve classification of RT/CRT response. Furthermore, immune gene expression profiling (GEP) may identify differences in expression levels of genes relevant to different radiotherapy responses: (1) at baseline between poor and good responders, and (2) longitudinally from preradiotherapy to postradiotherapy samples. Overall, this may inform novel therapeutic RT/CRT combination strategies in rectal cancer. Methods We generated GEPs for 53 patients from biopsies taken prior to preoperative radiotherapy. TCD was used to assess rectal tumor response to neoadjuvant RT/CRT and δ TCD was subjected to k-means clustering to classify patients into different response categories. Differential gene expression analysis was performed using statistical analysis of microarrays, pathway enrichment analysis and immune cell type analysis using single sample gene set enrichment analysis. Immunohistochemistry was performed to validate specific results. The results were validated using 220 pretreatment samples from publicly available datasets at metalevel of pathway and survival analyses. Results δ TCD scores ranged from 12.4% to -47.7% and stratified patients into three response categories. At baseline, 40 genes were significantly upregulated in poor (n=12) versus good responders (n=21), including myeloid and stromal cell genes. Of several pathways showing significant enrichment at baseline in poor responders, epithelial to mesenchymal transition, coagulation, complement activation and apical junction pathways were validated in external cohorts. Unlike poor responders, good responders showed longitudinal (preradiotherapy vs postradiotherapy samples) upregulation of 198 immune genes, reflecting an increased T-cell-inflamed GEP, type-I interferon and macrophage populations. Longitudinal pathway analysis suggested viral-like pathogen responses occurred in post-treatment resected samples compared with pretreatment biopsies in good responders. Conclusion This study suggests potentially druggable immune targets in poor responders at baseline and indicates that tumors with a good RT/CRT response reprogrammed from immune "cold" towards an immunologically "hot" phenotype on treatment with radiotherapy. ©

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Differential diagnosis of neurodegenerative diseases using structural MRI data

Item Type Journal Article

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Abstract Different neurodegenerative diseases can cause memory disorders and other cognitive impairments. The early detection and the stratification of patients according to the underlying disease are essential for an efficient approach to this healthcare challenge. This emphasizes the importance of differential diagnostics. Most studies compare patients and controls, or Alzheimer's disease with one other type of dementia. Such a bilateral comparison does not resemble clinical practice, where a clinician is faced with a number of different possible types of dementia. Here we studied which features in structural magnetic resonance imaging (MRI) scans could best distinguish four types of dementia, Alzheimer's disease, frontotemporal dementia, vascular dementia, and dementia with Lewy bodies, and control subjects. We extracted an extensive set of features quantifying volumetric and morphometric characteristics from T1 images, and vascular characteristics from FLAIR images. Classification was performed using a multi-class classifier based on Disease State Index methodology. The classifier provided continuous probability indices for each disease to support clinical decision making. A dataset of 504 individuals was used for evaluation. The cross-validated classification accuracy was 70.6% and balanced accuracy was 69.1% for the five disease groups using only automatically determined MRI features. Vascular dementia patients could be detected with high sensitivity (96%) using features from FLAIR images. Controls (sensitivity 82%) and Alzheimer's disease patients (sensitivity 74%) could be accurately classified using T1-based features, whereas the most difficult group was the dementia with Lewy bodies (sensitivity 32%). These results were notable better than the classification accuracies obtained with visual MRI ratings (accuracy 44.6%, balanced accuracy 51.6%). Different quantification methods provided complementary information, and consequently, the best results were obtained by utilizing several quantification methods. The results prove that automatic quantification methods and computerized decision support methods are feasible for clinical practice and provide comprehensive information that may help clinicians in the diagnosis making. © 2016 The Authors.

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Differential Privacy for Positive and Unlabeled Learning with Known Class Priors

Item Type Conference Paper

Author A.T. Pham

Author R. Raich

Abstract Despite the increasing attention to big data, there are several domains where labeled data is scarce or too costly to obtain. For example, for data from information retrieval, gene analysis, and social network analysis, only training samples from the positive class are annotated while the remaining unlabeled training samples consist of both unlabeled positive and unlabeled negative samples. The specific positive and unlabeled (PU) data from those domains necessitates a mechanism to learn a two-class classifier from only one-class labeled data. Moreover, because data from those domains is highly sensitive and private, preserving training samples privacy is essential. This paper addresses the challenge of private PU learning by designing a differentially private algorithm for positive and unlabeled data. We first propose a learning framework for the PU setting when the class prior probability is known, with a theoretical guarantee of convergence to the optimal classifier. We then propose a privacy-preserving

mechanism for the designed framework where the privacy and utility are both theoretically and empirically proved. © 2018 IEEE.

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Language English

Extra Journal Abbreviation: IEEE Stat. Signal Process. Workshop, SSP

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Differentially Private Semi-Supervised Learning with Known Class Priors

Item Type Conference Paper

Author A.T. Pham

Author J. Xi

Editor Abe N.

Editor Liu H.

Editor Pu C.

Editor Hu X.

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Editor Qiao M.

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Editor Kossmann D.

Editor Liu B.

Editor Lee K.

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Editor He J.

Editor Saltz J.

Abstract Machine learning algorithms benefit greatly from a large amount of data in the big data regime, however, the labeling cost for all training samples is prohibitively high. Semi-supervised learning is a designed framework to overcome the labeling challenge. In particular, in semi-supervised learning, there are only a small number of labeled samples and unlabeled samples are used as side information to leverage the overall performance. Additionally, preserving the privacy of the data is another challenge in big data analysis. This paper addresses challenges by designing a privacy-preserving method for semi-supervised learning. Assuming that class priors are available, the proposed method can achieve the optimal classifier. Moreover, the privacy and utility of the proposed method are theoretically guaranteed. Experiments on real UCI datasets illustrate the effectiveness of our proposed method. © 2018 IEEE.

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Diffusion magnetic resonance imaging for kidney cyst volume quantification and non-cystic tissue characterisation in ADPKD

Item Type Journal Article

Author A. Caroli

Author G. Villa

Author P. Brambilla

Author M. Trillini

Author K. Sharma

Author S. Sironi

Author G. Remuzzi

Author N. Perico

Author A. Remuzzi

Abstract Objectives: Beyond total kidney and cyst volume (TCV), non-cystic tissue plays an important role in autosomal dominant polycystic kidney disease (ADPKD) progression. This study aims at presenting and preliminarily validating a diffusion MRI (DWI)-based TCV quantification method and providing evidence of DWI potential in characterising non-cystic tissue microstructure. Methods: T2-weighted MRI and DWI scans ($b = 0, 15, 50, 100, 200, 350, 500, 700, 1000$; 3 directions) were acquired from 35 ADPKD patients with CKD stage 1 to 3a and 15 healthy volunteers on a 1.5 T scanner. ADPKD classification was performed using the Mayo model. DWI scans were processed by mono- and segmented bi-exponential models. TCV was quantified on T2-weighted MRI by the reference semi-automatic method and automatically computed by thresholding the pure diffusivity (D) histogram. The agreement between reference and DWI-based TCV values and the differences in DWI-based parameters between healthy and ADPKD tissue components were assessed. Results: There was strong correlation between DWI-based and reference TCV ($\rho = 0.994, p < 0.001$). Non-cystic ADPKD tissue had significantly higher D, and lower pseudo-diffusion and flowing fraction than healthy tissue ($p < 0.001$). Moreover, apparent diffusion coefficient and D values significantly differed by Mayo imaging class, both in the whole kidney (Wilcoxon $p = 0.007$ and $p = 0.004$) and non-cystic tissue ($p = 0.024$ and $p = 0.007$). Conclusions: DWI shows potential in ADPKD to quantify TCV and characterise non-cystic kidney tissue microstructure, indicating the presence of microcysts and peritubular interstitial fibrosis. DWI could complement existing biomarkers for non-invasively staging, monitoring, and predicting ADPKD progression and evaluating the impact of novel therapies, possibly targeting damaged non-cystic tissue besides cyst expansion. Clinical relevance statement: This study shows diffusion-weighted MRI (DWI) potential to quantify total cyst volume and characterise non-cystic kidney tissue microstructure in ADPKD. DWI could complement existing biomarkers for non-invasively staging, monitoring, and predicting ADPKD progression and evaluating the impact of novel therapies, possibly targeting damaged non-cystic tissue besides cyst expansion. Key Points: • Diffusion magnetic resonance imaging shows potential to quantify total cyst volume in ADPKD. • Diffusion magnetic resonance imaging might allow to non-invasively characterise non-cystic kidney tissue microstructure. • Diffusion magnetic resonance imaging-based biomarkers significantly differ by Mayo imaging class, suggesting their possible prognostic value. © 2023, The Author(s), under exclusive licence to European Society of Radiology.

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Diffusion tensor imaging in primary brain tumors: Reproducible quantitative analysis of corpus callosum infiltration and contralateral involvement using a probabilistic mixture model

Item Type Journal Article
Author B. Stieltjes
Author M. Schlüter
Author B. Didingr
Author M.-A. Weber
Author H.K. Hahn

Author P. Parzer
Author J. Rexilius
Author O. Konrad-Verse
Author H.-O. Peitgen
Author M. Essig

Abstract Diffusion tensor imaging (DTI) has been advocated as a promising tool for delineation of the extent of tumor infiltration by primary brain tumors. First reports show conflicting results mainly due to difficulties in reproducible determination of DTI-derived parameters. A novel method based on probabilistic voxel classification for a user-independent analysis of DTI-derived parameters is presented and tested in healthy controls and patients with primary brain tumors. The proposed quantification method proved to be highly reproducible both in healthy controls and patients. Fiber integrity in the corpus callosum (CC) was measured using this quantification method, and the profiles of fractional anisotropy (FA) provided additional information of the possible extent of infiltration of primary brain tumors when compared to conventional imaging. This yielded additional information on the nature of ambiguous contralateral lesions in patients with primary brain tumors. The results show that DTI-derived parameters can be determined reproducibly and may have a strong impact on evaluation of contralateral extent of primary brain tumors. © 2006 Elsevier Inc. All rights reserved.

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Export Date: 20 September 2026; Cited By: 64; Correspondence Address: B. Stieltjes; German Cancer Research Center, Department of Diagnostic Radiology, 69120 Heidelberg, Im Neuenheimer Feld 280, Room N 155, Germany; email: bstieltjes@dkfz.de; CODEN: NEIME

Digital image analysis of picrosirius red staining: A robust method for multi-organ fibrosis quantification and characterization

Item Type Journal Article

Author G.E. Courtoy

Author I. Leclercq

Author A. Froidure

Author G. Schiano

Author J. Morelle

Author O. Devuyst

Author F. Huaux

Author C. Bouzin

Abstract Current understanding of fibrosis remains incomplete despite the increasing burden of related diseases. Preclinical models are used to dissect the pathogenesis and dynamics of fibrosis, and to evaluate anti-fibrotic therapies. These studies require objective and accurate measurements of fibrosis. Existing histological quantification methods are operator-dependent, organ-specific, and/or need advanced equipment. Therefore, we developed a robust, minimally operator-dependent, and tissue-transposable digital method for fibrosis quantification. The proposed method involves a novel algorithm for more specific and more sensitive detection of collagen fibers stained by picrosirius red (PSR), a computer-assisted segmentation of histological structures, and a new automated morphological classification of fibers according to their compactness. The new algorithm proved more accurate than classical filtering using principal color component (red-green-blue; RGB) for PSR detection. We applied this new method on established mouse models of liver, lung, and kidney fibrosis and demonstrated its validity by evidencing topological collagen accumulation in relevant histological compartments. Our data also showed an overall accumulation of compact fibers concomitant with worsening fibrosis and evidenced topological changes in fiber compactness proper to each model. In conclusion, we describe here a robust digital method for fibrosis analysis allowing accurate quantification, pattern recognition, and multi-organ comparisons useful to understand fibrosis dynamics. © 2020 by the authors. Licensee MDPI, Basel, Switzerland.

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Digital imaging applications and informatics in dermatology

Item Type Journal Article**Author** F. Giuste**Author** J.C. Vizcarra**Author** D.A. Gutman

Abstract In this chapter, we present the use of whole slide imaging (WSI) and dermoscopy in the field of dermatology. Image digitization has allowed for increasing computer-assisted clinical decision-making. An introduction to common digital imaging data sources such as WSI and dermoscopy is provided. We also review some commonly used image quantification methods and their potential applications in dermatology. Finally, we review how machine learning approaches utilize novel large dermatology image datasets. © 2019 Frontline Medical Communications

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Digital multiplexed analysis of circular RNAs in FFPE and fresh non-small cell lung cancer specimens

Item Type Journal Article

Author C. Pedraz-Valdunciel

Author S. Giannoukakos

Author N. Potie

Author A. Giménez-Capitán

Author C.-Y. Huang

Author M. Hackenberg

Author A. Fernandez-Hilario

Author J. Bracht

Author M. Filipiska

Author E. Aldeguer

Author S. Rodríguez

Author T.G. Bivona

Author S. Warren

Author C. Aguado

Author M. Ito

Author A. Aguilar-Hernández

Author M.A. Molina-Vila

Author R. Rosell

Abstract Although many studies highlight the implication of circular RNAs (circRNAs) in carcinogenesis and tumor progression, their potential as cancer biomarkers has not yet been fully explored in the clinic due to the limitations of current quantification methods. Here, we report the use of the nCounter platform as a valid technology for the analysis of circRNA expression patterns in non-small cell lung cancer (NSCLC) specimens. Under this context, our custom-made circRNA panel was able to detect circRNA expression both in NSCLC cells and formalin-fixed paraffin-embedded (FFPE) tissues. CircFUT8 was overexpressed in NSCLC, contrasting with circEPB41L2, circBNC2, and circSOX13 downregulation even at the early stages of the disease. Machine learning (ML) approaches from different paradigms allowed discrimination of NSCLC from nontumor controls (NTCs) with an 8-circRNA signature. An additional 4-circRNA signature was able to classify early-stage NSCLC samples from NTC, reaching a maximum area under the ROC curve (AUC) of 0.981. Our results not only present two circRNA signatures with diagnosis potential but also introduce nCounter processing following ML as a feasible protocol for the study and development of circRNA signatures for NSCLC. © 2022 The Authors. Molecular Oncology published by John Wiley & Sons Ltd on behalf of Federation of European Biochemical Societies.

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DIMER: EEG-based DNA-inspired multi-dimension feature emotion quantization recognition method

Item Type Journal Article

Author L. Hou

Author X. Li

Author D. Liu

Author Y. Zhou

Abstract A novel DNA-Inspired Multi-dimensional Emotion Recognition Network (DIMER) is proposed to address critical challenges in affective computing. Firstly, addressing the issue of incomplete emotional state quantification in the field of emotion recognition, a 16-level emotion quantification methodology based on valence, arousal, dominance, and liking is proposed, enabling fine-grained classification of emotional states. Secondly, to overcome the complexity of EEG feature extraction and poor multi-dimensional feature fusion performance, a dynamic complementary fusion approach for multi-dimensional features based on DNA interaction mechanisms is introduced. This approach conceptualizes EEG differential entropy features and their visualized representations as DNA double-strand carriers, quantifying the complementary degree between features by simulating hydrogen bond strength through feature affinity calculations. Finally, main chain-complementary chain feature associations are established based on DNA base-pairing principles, thereby achieving adaptive feature fusion. Experimental results on the DEAP emotion dataset demonstrate that subject-dependent experiments achieve 95.07% accuracy, outperforming MLP and LSTM by 1.82% and 18.16%, respectively, while subject-independent experiments attain 94.82% accuracy, surpassing MLP and LSTM by 15.66% and 33.33%, respectively. The proposed method effectively advances the development of high-precision multi-dimensional emotion recognition. © 2026

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Direct Quantification of Mixed Organic Acids Based on Spectral Image with Deep Learning

Item Type Journal Article

Author W. Wang

Author R. Luo

Author Q. Duan

Author Y. Feng

Author J. Chen

Author Y. Huang

Author S. Bi

Author F. Liu

Author J. Lee

Abstract Identification of organic acid is crucial indicator for field of environment, healthcare and industry, which involves the time-consuming and labor-intensive progress. The simultaneous detection for complex mixture becomes a new target when large-scale instruments is the mainstream measurement approach. With the burst development of machine learning in solving complex problem, the progress is being made for quick detection of mixtures. The next challenge is to provide sufficient data to satisfy the need of algorithm. In this work, we proposed the modified spectrometer technology combined with the deep learning to quantify the mixed organic acids. Organic acids interact with light in the various background to map the information in the form of images. Deep learning could establish the relationship between images and concentrations. According to the result, trained convolutional neural network could achieve the goal of simultaneous measurement of organic acids, which speeds up the implementation in the parallel detection and suggests new twists in the field of chemistry. © 2021 Wiley-VCH GmbH

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Discrete Box-Constrained Minimax Classifier for Uncertain and Imbalanced Class Proportions

Item Type Journal Article

Author C. Gilet

Author S. Barbosa

Author L. Fillatre

Abstract This paper aims to build a supervised classifier for dealing with imbalanced datasets, uncertain class proportions, dependencies between features, the presence of both numeric and categorical features, and arbitrary loss functions. The Bayes classifier suffers when prior probability shifts occur between the training and testing sets. A solution is to look for an equalizer decision rule whose class-conditional risks are equal. Such a classifier corresponds to a minimax classifier when it maximizes the Bayes risk. We develop a novel box-constrained minimax classifier which takes into account some constraints on the priors to control the risk maximization. We analyze the empirical Bayes risk with respect to the box-constrained priors for discrete inputs. We show that this risk is a concave non-differentiable multivariate piecewise affine function. A projected subgradient algorithm is derived to maximize this empirical Bayes risk over the box-constrained simplex. Its convergence is established and its speed is bounded. The optimization algorithm is scalable when the number of classes is large. The robustness of our classifier is studied on diverse databases. Our classifier, jointly applied with a clustering algorithm to process mixed attributes, tends to equalize the class-conditional risks while being not too pessimistic. © 1979-2012 IEEE.

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Classification (of information), Cluster Analysis, cluster analysis, algorithm, Algorithms, Clustering algorithms, Statistical tests, Optimization algorithms, Databases, Factual, factual database, Bayes theorem, Bayes Theorem, prior probability shift, Bayesian classifier, Minimax classifier, imbalanced datasets, Bayesian robustness, Categorical features, discrete bayes classifier, histogram rule, Imbalanced Data-sets, Non-differentiable, Piecewise affine functions, Projected subgradient algorithms, Risks, Supervised classifiers, uncertain class proportions, Γ I -minimax classifier

Notes:

EID: 2-s2.0-85098766728

Export Date: 20 September 2026; Cited By: 9; Correspondence Address: L. Fillatre; University Côte d'Azur, Route des Lucioles, 06900, France; email: lionel.fillatre@i3s.unice.fr; CODEN: ITPID

Discrete Minimax Regret Classifier for Sustainability Development**Item Type** Conference Paper**Author** C. Gilet**Author** L. Fillatre

Abstract While supervised classification is becoming increasingly used in sustainability development, the issue of prior probability shifts often occurs in these contexts and remains important to be addressed. Prior probability shift characterizes an evolution of the prior probabilities between test samples and the class proportions observed in training dataset. A robust approach to address this issue is the minimax regret strategy. The regret corresponds to the loss of performance associated with a classifier compared to the optimal performance reached by the Bayes classifier with respect to given prior probabilities. The minimax-regret strategy minimizes the maximum regret of a classifier that can be reached when prior probabilities shifts occur. This paper is devoted to learn a minimax regret classifier in a context of mixed attributes, that can be coupled with any pre-trained neural networks for sustainable classification tasks, and that can take into account given costs of classification errors provided by the experts of the application domain. We especially show that we can learn a minimax regret classifier by learning a usual minimax classifier when considering a specific loss function. Experiments on real databases for sustainability development illustrate the benefits of our proposed approach. © 2026 IEEE.

Date 2026**DOI** 10.1109/3SCEA68071.2026.11602990**URL** <https://www.scopus.com/pages/publications/105046078476?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IEEE Int. Conf. Smart Sustain. Syst. Comput. Eng. Appl., 3SCEA**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-833155668-6 (ISBN)**Pages** 99-108**Proceedings Title** IEEE Int. Conf. Smart Sustain. Syst. Comput. Eng. Appl., 3SCEA**Conference Name** 2026 IEEE International Conference on Smart Sustainable Systems for Computer and Engineering Applications, 3SCEA 2026**Date Added** 20/09/2026, 9:51:37 pm**Modified** 20/09/2026, 9:51:37 pm**Tags:**

Classification (of information), Sustainable development, Supervised learning, Neural networks, Statistical tests, Optimization, Classifiers, Learn+, Probability shifts, Prior probability, Prior probability shift, Discrete minimax classifier, Discrete Minimax Classifier, Minimax, Minimax regret, Minimax regret classifier, Minimax Regret Classifier, prior probability shift, Supervised classification, Sustainability development

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Agence Nationale de la Recherche, ANR, (projet DMC); Conference name: 2026 IEEE International Conference on Smart Sustainable Systems for Computer and Engineering Applications, 3SCEA 2026; Conference date: 19 April 2026 through 21 April 2026; Conference code: 226063

Discrete Perineural Invasion Focus Number in Quantification for T1-T2 Oral Squamous Cell Carcinoma

Item Type Journal Article

Author P.-Y. Wei

Author W.-Y. Li

Author S.-K. Tai

Abstract Objective: Perineural invasion (PNI) has been an established poor prognostic feature for T1-T2 oral squamous cell carcinoma (OSCC). Different presentations and amounts of PNI are commonly observed, but PNI is currently recorded as being present or absent. This study asked whether the quantification of PNI provides additional information for early OSCC. Study Design: Retrospective cohort study. Setting: Tertiary referral medical center. Subjects and Methods: Pathologic reevaluations were performed for 314 patients with T1-T2 OSCC who underwent curative surgery from June 2001 to August 2009. A novel parameter, PNI focus number, was defined for PNI quantification. With 5 PNI foci as the cutoff, patients were categorized into 3 groups: no PNI (0 PNI foci), low PNI (PNI foci, 1-5), and high PNI (PNI foci >5). Rate of cervical lymph node metastasis (LN+), 5-year disease-specific survival (DSS), and 5-year overall survival (OS) were analyzed among these groups. Results: PNI focus number independently predicted for LN+, poor DSS, and poor OS in multivariate analysis after controlling for T classification, lymphovascular invasion, differentiation, margin, and tumor thickness. The 5-year DSS demonstrated a dose-dependent decrease among the 3 groups (no PNI, 88.6%; low PNI, 75.2%; high PNI, 33.8%; $P < .001$). Moreover, the 5-year DSS of the high PNI group was significantly worse than that of the low PNI group. Conclusion: PNI focus number can be a novel parameter for PNI quantification in early OSCC. Although optimal quantification methods still require further investigation, this study offers clear clinical support for the nerve-tumor interaction hypothesis and advocates further mechanistic research for the exploration of PNI-related treatment concepts for OSCC. © American Academy of Otolaryngology–Head and Neck Surgery Foundation 2018.

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Tags:

adult, Adult, Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, young adult, Young Adult, aged, Aged, Aged, 80 and over, very elderly, major clinical study, prevalence, follow up, cohort analysis, Retrospective Studies, retrospective study, human tissue, pathology, cancer prognosis, immunohistochemistry, mortality, tumor recurrence, cancer grading, quantification, disease free survival, overall survival, cancer staging, lymph node metastasis, Lymphatic Metastasis, cancer surgery, Neoplasm Invasiveness, tumor invasion, disease specific survival, Carcinoma, Squamous Cell, chemoradiotherapy, Disease-Free Survival, lymph node dissection, Mouth Neoplasms, mouth squamous cell carcinoma, mouth tumor, neck dissection, Neoplasm Staging, oral squamous cell carcinoma, perineural invasion, perineural invasion focus number, squamous cell carcinoma, survival rate, Survival Rate

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Export Date: 20 September 2026; Cited By: 25; Funding details: Department of Oncology; Department of Otolaryngology; Division of Medical Oncology; Taipei Veterans General Hospital; Correspondence Address: S.-K. Tai; Department of Otolaryngology, Taipei Veterans General Hospital, Taipei, Taiwan; email: sktai@vghtpe.gov.tw; CODEN: OTOLD

Discriminant Insights: Applying Linear Discriminant Analysis to Tamil Vowel Usage for Social Awareness and Emotional Lyrics in Education

Item Type Conference Paper
Author B. Sharmila
Author S.P. Gnana Prakasam Louis Raja
Author R. Ganesan

Author N. Kalaiyarasi

Author N. Vijayan

Abstract Tamil lyrics occupy a central role in language and literature classrooms, where they are used to explore phonology, stylistic variation, and the interaction between linguistic form and social meaning. This paper presents an exploratory baseline tool rather than a production classifier: it investigates whether a compact and statistically transparent model can support pedagogical analysis by summarizing vowel usage patterns across two lyric categories - Social Awareness and Emotional lyrics - frequently discussed in teaching and cultural discourse. Each of 54 authors is represented by raw counts of the twelve Tamil vowels, Uyir Ezhuthukal. Linear Discriminant Analysis (LDA) is applied with equal class priors to derive a single canonical discriminant function. The model yields an eigenvalue of 0.411 and a canonical correlation of 0.540, indicating moderate separation. Wilks' lambda is 0.709 ($p=0.198$), confirming non-significant class separation and limiting strong inferential claims. Resubstitution accuracy is 74.1% and leave-one-out cross-validation accuracy is 55.6%, corroborating the baseline nature of the approach. Despite modest predictive performance, interpretable LDA coefficients, centroids, and diagnostic statistics constitute structured decision-support artifacts that augment - rather than replace - instructor judgment in classroom discussion, example selection, and hypothesis formation. Raw counts are used in this baseline to preserve stylistic intensity and enable classroom reproducibility; the pedagogical implications of this choice and directions toward normalized features are explicitly discussed. A teacher-oriented analytical workflow, a concrete classroom deployment example, and governance considerations for responsible educational deployment are outlined, drawing on principles from human-centered design and trustworthy AI. © 2026 IEEE.

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Conference Name 2026 IEEE International Conference on Human-Machine Systems, ICHMS 2026

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Tags:

Behavioral research, Artificial intelligence, Decision support systems, Engineering education, Linguistics, Teaching, Decision supports, Tamil, Discriminant analysis, AI in education, educational technology, Educational technology, human-AI decision support, Human-AI decision support, Interpretable model, interpretable models, Linear Discriminant Analysis, Linear discriminant analyze, Social awareness, stylometry, Stylometry, Uyir ezhuthukal, Uyir Ezhuthukal, vowel frequency, Vowel frequency

Notes:

EID: 2-s2.0-105045590853

Export Date: 20 September 2026; Cited By: 0; Conference name: 2026 IEEE International Conference on Human-Machine Systems, ICHMS 2026; Conference date: 1 July 2026 through 3 July 2026; Conference code: 226012

Discriminatively trained language models using support vector machines for language identification

Item Type Conference Paper**Author** L.-F. Zhai**Author** M.-H. Siu**Author** X. Yang**Author** H. Gish

Abstract In this paper, we explore the use of the Support Vector Machines (SVMs) to learn a discriminatively trained n-gram model for automatic language identification. Our focus is on practical considerations that make SVM technology more effective. We address the performance related issues of class priors, data imbalance, feature weighting, score normalization and combining multiple knowledge sources with SVMs. Using modified n-gram counts as features, we show that the SVM-trained n-grams are effective classifiers but they are sensitive to changes in prior class distributions. Using balanced prior distributions or score normalization procedures, the SVM-trained n-gram out-performed the traditional n-gram in parallel phoneme recognition with language model and GMM-UBM-based language identification systems by more than 30% relative error reduction on the OGI-TS corpus. © 2006 IEEE.

Date 2006**DOI** 10.1109/ODYSSEY.2006.248098**URL** <https://www.scopus.com/pages/publications/42749108714?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IEEE Odyssey Workshop Speaker Lang. Recognition**ISBN** 142440472X (ISBN); 978-142440472-8 (ISBN)**Proceedings Title** IEEE Odyssey Workshop Speaker Lang. Recognition**Conference Name** IEEE Odyssey 2006: Workshop on Speaker and Language Recognition**Date Added** 20/09/2026, 9:53:08 pm**Modified** 20/09/2026, 9:53:08 pm**Tags:**

Automation, Feature extraction, Support vector machines, Speech recognition, Mathematical models, Identification (control systems), Language models, Automatic language identification, n-gram models, Score normalization

Notes:

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Export Date: 20 September 2026; Cited By: 37; Correspondence Address: L.-F. Zhai; Hong Kong University of Science and Technology, Hong Kong, Clearwater Bay, Hong Kong; email: lfzhai@ust.hk; Conference name: IEEE Odyssey 2006: Workshop on Speaker and Language Recognition; Conference date: 28 June 2006 through 30 June 2006; Conference code: 69791

Discussion on “INTACT: a method for integration of longitudinal physical activity data from multiple sources” by Jingru Zhang, Erjia Cui, Hongzhe Li, and Haochang Shou

Item Type Journal Article

Author D. Fuller

Author K. Stanley

Abstract The INTACT (INtegration of Time varying data from weArable sensors for physiCal acTivity) method proposed by Zhang et al. represents a timely and important contribution to the harmonization of accelerometer data across studies. The authors provide a mathematically rigorous presentation and an open source R package. INTACT has clear utility for large scale data integration initiatives. INTACT draws inspiration from techniques used in domains where latent biological mechanisms are poorly defined. We argue that physical activity research is distinct in that the primary latent constructs, time and intensity, are well understood. Accelerometer summary metrics represent proxies for these constructs, raising questions about whether abstraction through an eigenmodel improves interpretation of the physiological meaning. The post harmonization “activity intensity” metric produced by INTACT lacks clear units, scale consistency, and it is unclear how the harmonized metric would be applied to activity classification methods. The method’s reliance on eigenfunctions is both a strength and a limitation. There are challenges for comparability across studies, and it will be important for applied users to conduct sensitivity analyses to evaluate eigenfunction selection. We also highlight the critical assumption of a shared eigenspace across datasets and populations, which may be violated in contexts with different physical activity patterns. Additional methodological development and clear interpretability will play an important role in the adoption of this method by the research community. © The Author(s) 2026. Published by Oxford University Press on behalf of The International Biometric Society. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site-for further information please contact journals.permissions@oup.com. This article is published and distributed under the terms of the Oxford University Press, Standard Journals Publication Model (<https://academic.oup.com/pages/standard-publication-reuse-rights>)

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Tags:

human, Humans, prevalence estimation, exercise, accelerometry, Accelerometry, activity intensity classification, Exercise, Longitudinal Studies, longitudinal study, physical activity, wearable electronic device, Wearable Electronic Devices

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EID: 2-s2.0-105043614304

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: D. Fuller; Department of Community Health and Epidemiology, College of Medicine, University of Saskatchewan, Saskatoon, Box 7, Health Science Building, 107 Wiggins Road, S7N 5E5, Canada; email: daniel.fuller@usask.ca; CODEN: BIOMA

Disentangling conditional redshift effects in Quasar–Galaxy photometric classification

Item Type Journal Article

Author R. Mattera

Abstract Redshift is widely used to improve photometric classification of astronomical sources, yet the origin of its contribution is rarely examined explicitly. In this work, we propose an experimental framework designed to disentangle whether redshift provides genuine conditional information or primarily reflects distance-driven class priors. We introduce a controlled comparison between original and redshift-matched samples, evaluated via a paired, repeated cross-validation scheme to ensure statistical robustness. Using spectroscopically confirmed sources from the Sloan Digital Sky Survey within an intermediate redshift range, we isolate the marginal redshift effects from non-linear feature interactions. Our analysis reveals that, after marginal balancing, redshift does not improve the performance of linear models, acting as a control validation of the design. In contrast, non-linear classifiers retain a residual, magnitude-dependent gain, which increases toward fainter sources. This demonstrates that our experimental design successfully isolates the conditional role of redshift, showing it contributes through non-linear interactions with photometric features rather than serving solely as a marginal proxy for distance. © 2026 The Author

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Tags:

Artificial intelligence, Classification (of information), Learning systems, Machine-learning, Machine learning, Balancing, Sampling, Cross validation, Astronomical sources, Feature relevance, Marginals, Photometric classification, Photometrics, Photometry, Red shift, Red Shift, Redshift, Redshift effect, Selection effect, Selection effects

Notes:

EID: 2-s2.0-105031911141

Export Date: 20 September 2026; Cited By: 0

Disentangling Label Distribution for Long-tailed Visual Recognition

Item Type Conference Paper

Author Y. Hong

Author S. Han

Author K. Choi

Author S. Seo

Author B. Kim

Author B. Chang

Abstract The current evaluation protocol of long-tailed visual recognition trains the classification model on the long-tailed source label distribution and evaluates its performance on the uniform target label distribution. Such protocol has questionable practicality since the target may also be long-tailed. Therefore, we formulate long-tailed visual recognition as a label shift problem where the target and source label distributions are different. One of the significant hurdles in dealing with the label shift problem is the entanglement between the source label distribution and the model prediction. In this paper, we focus on disentangling the source label distribution from the model prediction. We first introduce a simple but overlooked baseline method that matches the target label distribution by post-processing the model prediction trained by the cross-entropy loss and the Softmax function. Although this method surpasses state-of-the-art methods on benchmark datasets, it can be further improved by directly disentangling the source label distribution from the model prediction in the training phase. Thus, we propose a

novel method, Label distribution DisEntangling (LADE) loss based on the optimal bound of Donsker-Varadhan representation. LADE achieves state-of-the-art performance on benchmark datasets such as CIFAR-100-LT, Places-LT, ImageNet-LT, and iNaturalist 2018. Moreover, LADE outperforms existing methods on various shifted target label distributions, showing the general adaptability of our proposed method. © 2021 IEEE

Date 2021

DOI 10.1109/CVPR46437.2021.00656

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Archive Scopus

Language English

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Conference Name Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition

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'current, Performance, Forecasting, Computer vision, Benchmarking, Benchmark datasets, Classification models, Target labels, Label distribution, Simple++, Model prediction, Visual recognition, Evaluation protocol

Notes:

EID: 2-s2.0-85123196809

Export Date: 20 September 2026; Cited By: 316; Correspondence Address: B. Chang; Hyperconnect; email: buru.chang@hpcnt.com; Conference name: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2021; Conference date: 19 June 2021 through 25 June 2021; Conference code: 174911; CODEN: PIVRE

Dist-PU: Positive-Unlabeled Learning from a Label Distribution Perspective

Item Type Conference Paper

Author Y. Zhao

Author Q. Xu

Author Y. Jiang

Author P. Wen

Author Q. Huang

Abstract Positive-Unlabeled (PU) learning tries to learn binary classifiers from a few labeled positive examples with many unlabeled ones. Compared with ordinary semi-supervised learning, this task is much more challenging due to the absence of any known negative labels. While existing cost-sensitive-based methods have achieved state-of-the-art performances, they explicitly minimize the risk of classifying unlabeled data as negative samples, which might result in a negative-prediction preference of the classifier. To alleviate this issue, we resort to a label distribution perspective for PU learning in this paper. Noticing that the label distribution of unlabeled data is fixed when the class prior is known, it can be naturally used as learning supervision for the model. Motivated by this, we propose to pursue the label distribution consistency between predicted and ground-truth label distributions, which is formulated by aligning their expectations. Moreover, we further adopt the entropy minimization and Mixup regularization to avoid the trivial solution of the label distribution consistency on unlabeled data and mitigate the consequent confirmation bias. Experiments on three benchmark datasets validate the effectiveness of the proposed method. © 2022 IEEE.

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Classification (of information), Computer vision, Entropy, Semi-supervised learning, State of the art, Unlabeled data, Learn+, Label distribution, Negative samples, Cost-sensitive, Binary classifiers, Positive examples, Self- & semi- & meta- & unsupervised learning

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Distance-based image classification: Generalizing to new classes at near-zero cost

Item Type Journal Article

Author T. Mensink

Author J. Verbeek

Author F. Perronnin

Author G. Csurka

Abstract We study large-scale image classification methods that can incorporate new classes and training images continuously over time at negligible cost. To this end, we consider two distance-based classifiers, the k-nearest neighbor (k-NN) and nearest class mean (NCM) classifiers, and introduce a new metric learning approach for the latter. We also introduce an extension of the NCM classifier to allow for richer class representations. Experiments on the ImageNet 2010 challenge dataset, which contains over \$(106)\$ training images of 1,000 classes, show that, surprisingly, the NCM classifier compares favorably to the more flexible k-NN classifier. Moreover, the NCM performance is comparable to that of linear SVMs which obtain current state-of-the-art performance. Experimentally, we study the generalization performance to classes that were not used to learn the metrics. Using a metric learned on 1,000 classes, we show results for the ImageNet-10K dataset which contains 10,000 classes, and obtain performance that is competitive with the current state-of-the-art while being orders of magnitude faster. Furthermore, we show how a zero-shot class prior based on the ImageNet hierarchy can improve performance when few training images are available. © 1979-2012 IEEE.

Date 2013

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Issue 11**Journal Abbr** IEEE Trans Pattern Anal Mach Intell**Date Added** 20/09/2026, 9:52:50 pm**Modified** 20/09/2026, 9:52:50 pm**Tags:**

artificial intelligence, Artificial Intelligence, article, Transfer learning, algorithm, Algorithms, Models, Theoretical, procedures, theoretical model, Image classification, computer simulation, Computer Simulation, Text processing, zero-shot learning, transfer learning, methodology, computer assisted diagnosis, Image Interpretation, Computer-Assisted, Metric learning, image retrieval, Image retrieval, image enhancement, automated pattern recognition, Pattern Recognition, Automated, Membership functions, Image Enhancement, Class mean, K-nearest neighbors, k-nearest neighbors classification, large scale image classification, nearest class mean classification

Notes:

EID: 2-s2.0-84884582871

Export Date: 20 September 2026; Cited By: 420; CODEN: ITPID

Distinguish normal white blood cells from leukemia cells by detection, classification, and counting blood cells using YOLOv5

Item Type Conference Paper**Author** D.T. Luong**Author** D.D. Anh**Author** T.X. Thang**Author** H.T.L. Huong**Author** T.T. Hanh**Author** D.M. Khanh

Abstract Leukemia is a group of blood cancers that usually begin in the bone marrow and result in high numbers of abnormal blood cells. Distinguishing normal white blood cells from leukemia cells plays a role in the diagnosis of blood diseases. Over the years, manual methods can detect, classify, and count blood cells that are labor-intensive slow, time-consuming, subject to error, and costly. An automatic system is applied in the processing detection through microscopic images that will be fast, cheap, and high accuracy without special needs of equipment for lab testing. In this paper, we propose a method of detection, classification, and counting to distinguish five types of normal white blood cells (WBCs) from abnormal white blood cells (leukemia) using the You Only Look Once version 5 (YOLOv5) to aid the diagnosis of blood diseases. After using the YOLOv5 algorithm, we manually labeled the blood cells, and the leukemia cells labeled are the largest with 619 cells, while the number of normal WBCs labeled is a basophil, 115 neutrophils, 23 eosinophils, 80 lymphocytes, and 73

monocytes, respectively. The tested result with the average accuracy of detection and classification of the blood cells is 93.0%. © 2022 IEEE.

Date 2022

DOI 10.1109/ATiGB56486.2022.9984098

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Language English

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Tags:

Blood, Detection, Diagnosis, classification, Diseases, Cells, White blood cells, Bone, Bone marrow, Counting, Cytology, Blood cells, Blood cancer, counting, detection, distinguish, Distinguish, Leukaemia cells, Leukemia cell, Manual methods, Normal white blood cell, Normal White Blood Cells

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EID: 2-s2.0-85146249630

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Distributed heterogeneous compute infrastructure for the study of additive manufacturing systems

Item Type Journal Article

Author M. Thomas

Author M. Schram

Author K. Fox

Author J. Strube

Author N.S. Oblath

Author R. Rallo

Author Z.C. Kennedy

Author T. Varga

Author A.K. Battu

Author C.A. Barrett

Abstract We present the current status of a scalable computing framework to address the need of the multidisciplinary effort to study chemical dynamics. Specifically, we are enabling scientists to process and store experimental data, run large-scale computationally expensive high-fidelity physical simulations, and analyze these results using state-of-the-art data analytics, machine learning, and uncertainty quantification methods using heterogeneous computing resources. We present the results of this framework on a single metadata-driven workflow to accelerate an additive manufacturing use-case. Copyright © Materials Research Society 2020.

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Notes:

EID: 2-s2.0-85088303938

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Distributed optimization of classifier committee hyperparameters

Item Type Conference Paper

Author S. Aubakirov

Author P. Trigo

Author D. Ahmed-Zaki

Editor Bernardino J.

Editor Quix C.

Abstract In this paper, we propose an optimization workflow to predict classifiers accuracy based on the exploration of the space composed of different data features and the configurations of the classification algorithms. The overall process is described

considering the text classification problem. We take three main features that affect text classification and therefore the accuracy of classifiers. The first feature considers the words that comprise the inputtext; here we use the N-gram concept with different N values. The second feature considers the adoption of textual pre-processing steps such as the stop-word filtering and stemming techniques. The third feature considers the classification algorithms hyperparameters. In this paper, we take the well-known classifiers K-Nearest Neighbors (KNN) and Naive Bayes (NB) where K (from KNN) and a-priori probabilities (from NB) are hyperparameters that influence accuracy. As a result, we explore the feature space (correlation among textual and classifier aspects) and we present an approximation model that is able to predict classifiers accuracy. Copyright © 2018 by SCITEPRESS-Science and Technology Publications, Lda. All rights reserved.

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Tags:

Distributed computer systems, Nearest neighbor search, Classification (of information), Learning systems, Machine learning, Text classification, Text processing, Barium compounds, Classification algorithm, A-priori probabilities, K nearest neighbor (KNN), Pre-processing step, Distributed computing, Approximation model, Distributed optimization, Hyperparameters, Hyperparameters optimization

Notes:

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Distribution Alignment: A Unified Framework for Long-tail Visual Recognition

Item Type Conference Paper

Author S. Zhang

Author Z. Li

Author S. Yan

Author X. He

Author J. Sun

Abstract Despite the recent success of deep neural networks, it remains challenging to effectively model the long-tail class distribution in visual recognition tasks. To address this problem, we first investigate the performance bottleneck of the two-stage learning framework via ablative study. Motivated by our discovery, we propose a unified distribution alignment strategy for long-tail visual recognition. Specifically, we develop an adaptive calibration function that enables us to adjust the classification scores for each data point. We then introduce a generalized re-weight method in the two-stage learning to balance the class prior, which provides a flexible and unified solution to diverse scenarios in visual recognition tasks. We validate our method by extensive experiments on four tasks, including image classification, semantic segmentation, object detection, and instance segmentation. Our approach achieves the state-of-the-art results across all four recognition tasks with a simple and unified framework. © 2021 IEEE

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Archive Scopus

Language English

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Pages 2361-2370

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Tags:

Deep neural networks, Computer vision, Object detection, Semantics, Semantic Segmentation, Learning frameworks, Unified framework, Datapoints, Class distributions, Visual recognition, Long tail, Adaptive calibration, Calibration functions, Performance bottlenecks, Weight methods

Notes:

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Export Date: 20 September 2026; Cited By: 368; Funding details: BAAI; Beijing Academy of Artificial Intelligence; Shanghai NSF, (18ZR1425100); National Key Research and Development Program of China, NKRDPC, (2017YFA0700800); National Key Research and Development Program of China, NKRDPC; Conference name: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2021; Conference date: 19 June 2021 through 25 June 2021; Conference code: 174911; CODEN: PIVRE

Distribution Matching for Graph Quantification Under Structural Covariate Shift

Item Type Conference Paper

Author C. Damke

Author E. Hüllermeier

Editor Ribeiro R.P.

Editor Jorge A.M.

Editor Soares C.

Editor Gama J.

Editor Pfahringer B.

Editor Japkowicz N.

Editor Larrañaga P.

Editor Abreu P.H.

Abstract Graphs are commonly used in machine learning to model relationships between instances. Consider the task of predicting the political preferences of users in a social network; to solve this task one should consider, both, the features of each individual user and the relationships between them. However, oftentimes one is not interested in the label of a single instance but rather in the distribution of labels over a set of instances; e.g., when predicting the political preferences of users, the overall prevalence of a given opinion might be of higher interest than the opinion of a specific person. This label prevalence estimation task is commonly referred to as quantification learning (QL). Current QL methods for tabular data are typically based on the so-called prior probability shift (PPS) assumption which states that the label-conditional instance distributions should remain equal across the training and test data. In the graph setting, PPS generally does not hold if the shift between training and test data is structural, i.e., if the training data comes from a different region of the graph than the test data. To address such structural shifts, an importance sampling variant of the popular adjusted count quantification approach has previously been proposed. In this work, we extend the idea of structural importance sampling to the state-of-the-art KDEy quantification approach. We show that our proposed method adapts to structural shifts and outperforms standard quantification approaches. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

Date 2025

DOI 10.1007/978-3-032-05981-9_24

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Conference Name Lecture Notes in Computer Science
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Tags:

Artificial intelligence, Learning systems, Machine-learning, Training data, Probability distributions, Test data, Covariate shifts, Probability shifts, Prior probability, Importance sampling, Distribution matching, Quantification learning, Covariate Shift, Graph quantification, Graph Quantification, Quantification Learning, Structural shifts

Notes:

EID: 2-s2.0-105019311190

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: C. Damke; Institute of Informatics, LMU Munich, Munich, Germany; email: clemens.damke@ifi.lmu.de; Conference name: European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, ECML PKDD 2025; Conference date: 15 September 2025 through 19 September 2025; Conference code: 339289

Distribution-Free Uncertainty Quantification for Classification under Label Shift

Item Type Conference Paper
Author A. Podkopaev
Author A. Ramdas
Editor de Campos C.
Editor Maathuis M.H.
Abstract Trustworthy deployment of ML models requires a proper measure of uncertainty, especially in safety-critical applications. We focus on uncertainty quantification (UQ) for classification problems via two avenues - prediction sets using conformal prediction and calibration of probabilistic predictors by post-hoc binning - since these possess distribution-free guarantees for i.i.d. data. Two common ways of generalizing beyond the i.i.d. setting include handling covariate and label shift. Within the context of distribution-free UQ, the former has already received attention, but not the latter. It is known that label shift hurts prediction, and we first argue that it also hurts UQ, by showing degradation in coverage and calibration. Piggybacking on recent progress in addressing label shift (for better

prediction), we examine the right way to achieve UQ by reweighting the aforementioned conformal and calibration procedures whenever some unlabeled data from the target distribution is available. We examine these techniques theoretically in a distribution-free framework and demonstrate their excellent practical performance. © 2021 37th Conference on Uncertainty in Artificial Intelligence, UAI 2021. All Rights Reserved.

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Language English

Extra Journal Abbreviation: Conf. Uncertain. Artif. Intell., UAI

Publisher Association For Uncertainty in Artificial Intelligence (AUAI)

Pages 844-853

Proceedings Title Conf. Uncertain. Artif. Intell., UAI

Conference Name 37th Conference on Uncertainty in Artificial Intelligence, UAI 2021

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Tags:

Safety engineering, Artificial intelligence, Forecasting, Uncertainty analysis, Recent progress, Safety critical applications, Probability distributions, Uncertainty quantifications, Calibration, Probabilistics, Conformal predictions, Measure of uncertainty, Distribution-free, Re-weighting, IID data, Covariates

Notes:

EID: 2-s2.0-85111179588

Export Date: 20 September 2026; Cited By: 74; Funding details: Chirag Gupta; Conference name: 37th Conference on Uncertainty in Artificial Intelligence, UAI 2021; Conference date: 27 July 2021 through 30 July 2021; Conference code: 176565

Distribution-Free Uncertainty Quantification for Classification Under Label Shift

Item Type Conference Paper

Author A. Podkopaev

Author A. Ramdas

Editor de Campos C.

Editor Maathuis M.H.

Abstract Trustworthy deployment of ML models requires a proper measure of uncertainty, especially in safety-critical applications. We focus on uncertainty quantification (UQ) for classification problems via two avenues — prediction sets using conformal prediction and calibration of probabilistic predictors by post-hoc binning — since these possess distribution-free guarantees for i.i.d. data. Two common ways of generalizing beyond the i.i.d. setting include handling covariate

and label shift. Within the context of distribution-free UQ, the former has already received attention, but not the latter. It is known that label shift hurts prediction, and we first argue that it also hurts UQ, by showing degradation in coverage and calibration. Piggybacking on recent progress in addressing label shift (for better prediction), we examine the right way to achieve UQ by reweighting the aforementioned conformal and calibration procedures whenever some unlabeled data from the target distribution is available. We examine these techniques theoretically in a distribution-free framework and demonstrate their excellent practical performance. © 2021 Proceedings of Machine Learning Research. All Rights Reserved.

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Language English

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Safety engineering, Machine learning, Forecasting, Uncertainty analysis, Recent progress, Safety critical applications, Probability distributions, Uncertainty quantifications, Calibration, Probabilistics, Conformal predictions, Measure of uncertainty, Distribution-free, Re-weighting, IID data, Covariates

Notes:

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DISTRIBUTIONALLY ROBUST POST-HOC CLASSIFIERS UNDER PRIOR SHIFTS

Item Type Conference Paper

Author J. Wei

Author H. Narasimhan

Author E. Amid

Author W.-S. Chu

Author Y. Liu

Author A. Kumar

Abstract The generalization ability of machine learning models degrades significantly when the test distribution shifts away from the training distribution. We investigate the problem of training models that are robust to shifts caused by changes in the distribution of class-priors or group-priors. The presence of skewed training priors can often lead to the models overfitting to spurious features. Unlike existing methods, which optimize for either the worst or the average performance over classes or groups, our work is motivated by the need for finer control over the robustness properties of the model. We present an extremely lightweight post-hoc approach that performs scaling adjustments to predictions from a pre-trained model, with the goal of minimizing a distributionally robust loss around a chosen target distribution. These adjustments are computed by solving a constrained optimization problem on a validation set and applied to the model during test time. Our constrained optimization objective is inspired from a natural notion of robustness to controlled distribution shifts. Our method comes with provable guarantees and empirically makes a strong case for distributional robust post-hoc classifiers. An empirical implementation is available at <https://github.com/weijiaheng/Drops>. © 2023 11th International Conference on Learning Representations, ICLR 2023. All rights reserved.

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Language English

Extra Journal Abbreviation: Int. Conf. Learn. Represent., ICLR

Publisher International Conference on Learning Representations, ICLR

Proceedings Title Int. Conf. Learn. Represent., ICLR

Conference Name 11th International Conference on Learning Representations, ICLR 2023

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Tags:

Performance, Machine learning models, Overfitting, Constrained optimization, Scalings, Training model, Robustness (control systems), Generalization ability, Constrained optimization problems, Robustness properties, Spurious features, Validation sets

Notes:

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DLQMA: A Deep Learning Framework for Qualitative and Quantitative NMR Analysis of Complex Hydrocarbon Mixtures

Item Type Journal Article

Author W. Dong

Author X. Liu

Author D. Xun

Author Y. Wang

Author Y. Qiao

Abstract Nuclear magnetic resonance (NMR) is a key technique for structural and quantitative analysis of organic compounds, yet its application to complex mixtures is hindered by severe spectral overlap and the absence of efficient, standard-free quantification methods. DLQMA (Deep Learning for Qualitative and Quantitative Mixture Analysis) is introduced as a deep learning framework that simultaneously performs compound identification and concentration estimation directly from ¹H NMR spectra. Built on a pseudo-Siamese architecture with an added regression head, DLQMA enables end-to-end analysis without the need for manual spectral reconstruction or external standards. Performance was assessed using C8 hydrocarbon mixtures, a system characterized by severe spectral overlap and composed of multiple industrially relevant isomers with overlapping NMR signals. On this data set, DLQMA achieved 99.39% classification accuracy and a Pearson correlation of 0.98 for concentration predictions across 5,000 pairs of augmented validation spectra. The framework is further compatible with advanced NMR techniques such as 1D CSSF TOCSY, allowing automated interpretation of virtually separated subspectra. A standalone software implementation with an intuitive interface enables automated, high-throughput analysis of complex chemical mixtures. These advances have the potential to transform NMR spectroscopy into a fully automated tool for high-throughput analysis in chemical, petrochemical, and environmental applications. © 2025 American Chemical Society

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DOI 10.1021/acs.analchem.5c04983

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Notes:

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Export Date: 20 September 2026; Cited By: 2; Funding details: Chinese Academy of Sciences, CAS, (XDB1190100); Correspondence Address: Y. Wang; College of Chemistry and Chemical Engineering, Taiyuan University of Technology, Taiyuan, 79 Yingze West Street, Wanbailin District, 030024, China; email: wangyingxiong@tyut.edu.cn; Y. Qiao; State Key Laboratory of Coal Conversion, Institute of Coal Chemistry, Chinese Academy of Sciences, Taiyuan, 27 South Taoyuan Road, 030001, China; email: qiaoy@sxicc.ac.cn; CODEN: ANCHA

DNA barcoded competitive clone-initiating cell analysis reveals novel features of metastatic growth in a cancer xenograft model

Item Type Journal Article

Author S.M.M. Aalam

Author X. Tang

Author J. Song

Author U. Ray

Author S.J. Russell

Author S.J. Weroha

Author J. Bakkum-Gamez

Author V. Shridhar

Author M.E. Sherman

Author C.J. Eaves

Author D.J.H.F. Knapp

Author K.R. Kalari

Author N. Kannan

Abstract A problematic feature of many human cancers is a lack of understanding of mechanisms controlling organ-specific patterns of metastasis, despite recent progress in identifying many mutations and transcriptional programs shown to confer this potential. To address this gap, we developed a methodology that enables different aspects of the metastatic process to be comprehensively characterized at a clonal resolution. Our approach exploits the application of a computational pipeline to analyze and visualize clonal data obtained from transplant experiments in which a cellular DNA barcoding strategy is used to distinguish the separate clonal contributions of two or more competing cell populations. To illustrate the power of this methodology, we demonstrate its ability to discriminate the metastatic behavior in immunodeficient mice of a well-established human metastatic cancer cell line and

its co-transplanted LRRC15 knockdown derivative. We also show how the use of machine learning to quantify clone-initiating cell (CIC) numbers and their subsequent metastatic progeny generated in different sites can reveal previously unknown relationships between different cellular genotypes and their initial sites of implantation with their subsequent respective dissemination patterns. These findings underscore the potential of such combined genomic and computational methodologies to identify new clonally-relevant drivers of site-specific patterns of metastasis. © 2022 The Author(s).

Date 2022

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Journal Abbr NAR Cancer

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Article, female, human, machine learning, nonhuman, controlled study, dimensionality reduction, methodology, genotype, human cell, animal experiment, animal model, mouse, gene, fluorescence activated cell sorting, bioluminescence, cell DNA, clone initiating cell, DNA barcoding, ibuprofen, injection site, ketamine, lrcc15 gene, multidimensional scaling, progeny, tumor growth, tumor xenograft, xylazine

Notes:

EID: 2-s2.0-85141104253

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: N. Kannan; Division of Experimental Pathology and Laboratory Medicine, Department of Laboratory Medicine and Pathology, Mayo Clinic, Rochester, United States; email: Kannan.Nagarajan@mayo.edu

DNA variant classification—reconsidering “allele rarity” and “phenotype” criteria in ACMG/AMP guidelines

Item Type Journal Article

Author C.D. Davieson

Author K.E. Joyce

Author L. Sharma

Author C.L. Shovlin

Abstract Recent guidance suggested modified DNA variant pathogenicity assignments based on genome-wide allele rarity. Different a priori probabilities of pathogenicity operate where patients already have clinical diagnoses, and are found to have a very rare variant in a gene known to cause their disease, compared to predictive testing of a clinically unaffected individual. We tested new recommendations from the ClinGen Sequence Variant Interpretation Working Group for ClinVar-listed, loss-of-function variants meeting the very strong evidence of pathogenicity criterion [PVS1] in genes for 3 specific diseases where causal gene identification can modify clinical care of an individual- Von Willebrand disease, cystic fibrosis and hereditary haemorrhagic telangiectasia. Across these diseases, current rules leave 20/1,278 (1.6%) of loss-of-function variants as variants of uncertain significance (VUS that may not be reported to clinicians), and 207/1,278 (17.2%) as likely pathogenic. Applying the new ClinGen rule enabling PVS1 and the allele rarity criterion PM2 to delineate likely pathogenicity still left 8/1,278 (0.9%) as VUS (reflecting non-PVS1 calls by the submitters), and the majority of null alleles meeting PVS1 as merely likely pathogenic. We favour an approach whereby, for PVS1 variants in patients who personally meet the phenotypic PP4 criterion for a disease where casual variants are commonly family-specific, that PM2 is upgraded to permit a pathogenic call. Of 1,278 ClinVar-listed frameshift, nonsense and canonical splice site variants that met PVS1 in the 3 conditions, 16.0% (204/1,278) would be newly designated as pathogenic, avoiding misinterpretation outside of clinical genetics communities. We suggest further discussion around variant assessment across different clinical applications, potentially guided by PP4 alerts to distinguish personal versus family phenotypic history. © 2021 The Authors

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Journal Abbr Eur. J. Med. Genet.

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adult, Article, female, human, Humans, male, procedures, controlled study, major clinical study, practice guideline, population, DNA sequence, genetics, consensus development, phenotype, gene frequency, Gene Frequency, mutation, Mutation, DNA, Practice Guidelines as Topic, Phenotype, genetic variation, loss of function mutation, pathogenicity, allele, Consensus Development Conferences as Topic, cystic fibrosis, Cystic Fibrosis, gene identification, genetic screening, Genetic Testing, Likely pathogenic, Loss-of-function, monogenic disorder, null allele, Pathogenic, PM2, Population frequency, Rendu Osler Weber disease, Telangiectasia, Hereditary Hemorrhagic, Variant of uncertain significance, von Willebrand disease, von Willebrand Diseases

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Do NOT Classify and Count: Hybrid Attribute Control Success Evaluation

Item Type Conference Paper

Author F.M. Saaro

Author P. von Däniken

Author M. Cieliebak

Author J.M. Deriu

Editor Demberg V.

Editor Inui K.

Editor Marquez Villodre L.

Abstract Evaluating attribute control success in controllable text generation and related generation tasks typically relies on pretrained classifiers. We show that this widely used classify-and-count approach yields biased and inconsistent results, with estimates varying significantly across classifiers. We frame control success estimation as a quantification task and apply a hybrid Bayesian method that combines classifier predictions with a small number of human labels for calibration. To test our approach, we collected a two-modality test dataset consisting of 600 human-rated samples and 60,000 automatically rated samples. Our experiments show that our approach produces robust estimates of control success across both text and text-to-image generation tasks, offering a principled alternative to current evaluation practices. © 2026 Association for Computational Linguistics.

Date 2026

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Conference Name EACL 2026 - 19th Conference of the European Chapter of the Association for Computational Linguistics, Proceedings of the Conference, Vol. 1 - (Long Papers)
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Tags:

Computational linguistics, 'current, Classification (of information), Statistical tests, Bayesian networks, Text processing, Text generations, Bayesian methods, Hybrid attributes, Image generations, Robust estimate, Success evaluations

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Domain Adaptation for Time Series Under Feature and Label Shifts

Item Type Conference Paper
Author H. He
Author O. Queen
Author T. Koker
Author C. Cuevas
Author T. Tsiligkaridis
Author M. Zitnik
Editor Krause A.
Editor Brunskill E.
Editor Cho K.
Editor Engelhardt B.
Editor Sabato S.
Editor Scarlett J.

Abstract Unsupervised domain adaptation (UDA) enables the transfer of models trained on source domains to unlabeled target domains. However, transferring complex time series models presents challenges due to the dynamic temporal structure variations across domains. This leads to feature shifts in the time and frequency representations. Additionally, the label distributions of tasks in the source and target domains can differ significantly, posing difficulties in addressing label shifts and recognizing labels unique to the target domain. Effectively transferring complex time series models remains a formidable problem. We present RAINCOAT, the first model for both closed-set and universal domain adaptation on complex time series. RAINCOAT addresses feature and label shifts by considering both temporal and frequency features, aligning them across domains, and correcting for misalignments to facilitate the detection of private labels. Additionally, RAINCOAT improves transferability by identifying label shifts in target domains. Our experiments with 5 datasets and 13 state-of-the-art UDA methods demonstrate that RAINCOAT can improve transfer learning performance by up to 16.33% and can handle both closed-set and universal domain adaptation. © 2023 Proceedings of Machine Learning Research. All rights reserved.

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Domain adaptation methods for improving lab-to-field generalization of cocaine detection using wearable ECG

Item Type Conference Paper

Author A. Natarajan

Author G. Angarita

Author E. Gaiser

Author R. Malison

Author D. Ganesan

Author B.M. Marlin

Abstract Mobile health research on illicit drug use detection typically involves a two-stage study design where data to learn detectors is first collected in lab-based trials, followed by a deployment to subjects in a free-living environment to assess detector performance. While recent work has demonstrated the feasibility of wearable sensors for illicit drug use detection in the lab setting, several key problems can limit labtofield generalization performance. For example, lab-based data collection often has low ecological validity, the groundtruth event labels collected in the lab may not be available at the same level of temporal granularity in the field, and there can be significant variability between subjects. In this paper, we present domain adaptation methods for assessing and mitigating potential sources of performance loss in lab-to-field generalization and apply them to the problem of cocaine use detection from wearable electrocardiogram sensor data. © 2016 ACM.

Date 2016

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Conference Name UbiComp 2016 - Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing

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Classification (of information), Ubiquitous computing, Electrocardiography, Wearable technology, Classification, Domain adaptation, Laboratories, Covariate shifts, Generalization performance, Prior probability, Prior probability shift, Covariate shift, Wearable sensors, Cocaine detection, Detector performance, Ecological validity, Temporal granularity

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Domain Adaptation under Open Set Label Shift

Item Type Conference Paper

Author S. Garg

Author S. Balakrishnan

Author Z.C. Lipton

Editor Koyejo S.

Editor Mohamed S.

Editor Agarwal A.

Editor Belgrave D.

Editor Cho K.

Editor Oh A.

Abstract We introduce the problem of domain adaptation under Open Set Label Shift (OSLS) where the label distribution can change arbitrarily and a new class may arrive during deployment, but the class-conditional distributions $p(x|y)$ are domain-invariant. OSLS subsumes domain adaptation under label shift and Positive-Unlabeled (PU) learning. The learner's goals here are two-fold: (a) estimate the target label distribution, including the novel class; and (b) learn a target classifier. First, we establish necessary and sufficient conditions for identifying these quantities. Second, motivated by advances in label shift and PU learning, we propose practical methods for both tasks that leverage black-box predictors. Unlike typical open set domain adaptation problems, which tend to be

ill-posed and amenable only to heuristics, OSLs offers a well-posed problem amenable to more principled machinery. Experiments across numerous semi-synthetic benchmarks on vision, language, and medical datasets demonstrate that our methods consistently outperform open set domain adaptation baselines, achieving 10-25% improvements in target domain accuracy. Finally, we analyze the proposed methods, establishing finite-sample convergence to the true label marginal and convergence to optimal classifier for linear models in a Gaussian setup. © 2022 Neural information processing systems foundation. All rights reserved.

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Condition, Black boxes, Sampling, Domain adaptation, Learn+, Target labels, Label distribution, Shift-and, Conditional distribution, Ill posed, Practical method

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Domain adaptation under structural causal models

Item Type Journal Article

Author Y. Chen

Author P. Bühlmann

Abstract Domain adaptation (DA) arises as an important problem in statistical machine learning when the source data used to train a model is different from the target data used to test the model. Recent advances in DA have mainly been application-driven

and have largely relied on the idea of a common subspace for source and target data. To understand the empirical successes and failures of DA methods, we propose a theoretical framework via structural causal models that enables analysis and comparison of the prediction performance of DA methods. This framework also allows us to itemize the assumptions needed for the DA methods to have a low target error. Additionally, with insights from our theory, we propose a new DA method called CIRM that outperforms existing DA methods when both the covariates and label distributions are perturbed in the target data. We complement the theoretical analysis with extensive simulations to show the necessity of the devised assumptions. Reproducible synthetic and real data experiments are also provided to illustrate the strengths and weaknesses of DA methods when parts of the assumptions in our theory are violated. ©2021 Yuansi Chen and Peter Bühlmann.

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Domain adaptation with asymmetrically-relaxed distribution alignment

Item Type Conference Paper

Author Y. Wu

Author E. Winston

Author D. Kaushik

Author Z.C. Lipton

Abstract Domain adaptation addresses the common situation in which the target distribution generating our test data differs from the source distribution generating our training data. While absent assumptions, domain adaptation is impossible, strict conditions, e.g. covariate or label shift, enable principled algorithms. Recently-proposed domain-adversarial approaches consist of aligning source and target encodings, an approach often motivated as minimizing two (of three) terms in a theoretical bound on target error. Unfortunately, this minimization can cause arbitrary increases in the third term, a problem guaranteed to arise under shifting label distributions. We propose asymmetrically-relaxed distribution alignment, a new approach that overcomes some limitations of standard domain-adversarial algorithms. Moreover, we characterize precise assumptions under which our algorithm is theoretically principled and demonstrate empirical benefits on both synthetic and real datasets. Copyright © 2019 ASME

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Artificial intelligence, Machine learning, Software engineering, Training data, Domain adaptation, Label distribution, Source distribution, New approaches, Real data sets, Standard domains, Theoretical bounds

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Domain adaptation with conditional distribution matching and generalized label shift

Item Type Conference Paper

Author R.T. Des Combes

Author H. Zhao

Author Y.-X. Wang

Author G. Gordon

Abstract Adversarial learning has demonstrated good performance in the unsupervised domain adaptation setting, by learning domain-invariant representations. However, recent work has shown limitations of this approach when label distributions differ between the source and target domains. In this paper, we propose a new assumption, generalized label shift (GLS), to improve robustness against mismatched label distributions. GLS states that, conditioned on the label, there exists a representation of the input that is invariant between the source and target domains. Under GLS, we provide theoretical guarantees on the transfer performance of any classifier. We also devise necessary and sufficient conditions for GLS to hold, by using an estimation of the relative class weights between domains and an appropriate reweighting of samples. Our weight estimation method could be straightforwardly and generically applied in existing domain adaptation (DA) algorithms that learn domain-invariant representations, with small computational overhead. In particular, we modify three DA algorithms, JAN, DANN and CDAN, and evaluate their performance on standard and artificial DA tasks. Our algorithms outperform the base versions, with vast improvements for large label distribution mismatches. Our code is available at <https://tinyurl.com/y585xt6j>. © 2020 Neural information processing systems foundation. All rights reserved.

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Domain compensatory adversarial networks for partial domain adaptation

Item Type Journal Article

Author J. Huang

Author P. Zhang

Author Z. Zhou

Author K. Fan

Abstract Recently, domain adaptation has stimulated interest in the community of machine learning since it can improve the performance of the model in a new domain (target domain) by borrowing knowledge from a labeled source domain. At the same time, the presence of large-scale labeled datasets also raised significant attention in this scenario: the class labels in the new domain are only a subset of those in the source domain. We propose an adversarial-net-based method in this paper, called domain compensatory adversarial network (DCAN). The critical difficulty of this problem is to reduce the negative impact of source instances with weak discriminability and filter out outlier source classes by exploiting the prior probability of classes. DCAN can identify source instances with weak discriminability and reverse its domain label to compensate for the target label space, which alleviates the negative effect of mismatching label space. Besides, DCAN reweights outlier source classes with the class prior distributions of source data for both category classifier and domain classifier to promote positive transfer. Experiments have revealed the superiority of DCAN compared to the existing methods. © 2021, Springer Science+Business Media, LLC, part of Springer Nature.

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Domain Expansion and Boundary Growth for Open-Set Single-Source Domain Generalization

Item Type Journal Article

Author P. Jiao

Author N. Zhao

Author J. Chen

Author Y.-G. Jiang

Abstract Open-set single-source domain generalization aims to use a single-source domain to learn a robust model that can be generalized to unknown target domains with both domain shifts and label shifts. The scarcity of the source domain and the unknown data distribution of the target domain pose a great challenge for domain-invariant feature learning and unknown class recognition. In this article, we propose a novel learning approach based on domain expansion and boundary growth to expand the scarce source samples and enlarge the boundaries across the known classes that indirectly broaden the boundary between the known and unknown classes. Specifically, we achieve domain expansion by employing both background suppression and style augmentation on the source data to synthesize new samples. Then we force the model to distill consistent knowledge from the synthesized samples so that the model can learn domain-invariant information. Furthermore, we realize boundary growth across classes by using edge maps as an additional modality of samples when training multi-binary classifiers. In this way, it enlarges the boundary between the inliers and outliers, and consequently improves the unknown class recognition during open-set generalization. Extensive experiments show that our approach can achieve significant improvements and reach state-of-the-art performance on several cross-domain image classification datasets. © 1999-2012 IEEE.

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DOI 10.1109/TMM.2025.3535334**URL** <https://www.scopus.com/pages/publications/85216890575?origin=resultslist>**Archive** Scopus**Language** English**Volume** 27**Publisher** Institute of Electrical and Electronics Engineers Inc.**Pages** 3925-3938**Publication** IEEE Transactions on Multimedia**ISSN** 15209210 (ISSN)**Journal Abbr** IEEE Trans Multimedia**Date Added** 20/09/2026, 9:51:54 pm**Modified** 20/09/2026, 9:51:54 pm**Tags:**

Adversarial machine learning, Generalisation, Image classification, Domain Knowledge, Single source, Class boundary, Binary classifiers, Unknown class, Binary classifier, class boundary, domain expansion, Domain expansion, edge map, Edge map, open-set, Open-set, out of distribution, Out of distribution, single-source domain generalization, Single-source domain generalization

Notes:

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Domain-invariant adversarial learning with conditional distribution alignment for unsupervised domain adaptation

Item Type Journal Article**Author** X. Wang**Author** B. Sun**Author** H. Dong

Abstract Unsupervised domain adaption aims to reduce the divergence between the source domain and the target domain. The final objective is to learn domain-invariant features from both domains that get the minimised expected error on the target domain. The divergence between domains which is also called domain shift is mainly between the distributions of domains' samples. Additionally, the label shift is also a

tricky challenge in domain adaptation. In this study, domain-invariant adversarial learning with conditional distribution alignment is proposed to alleviate the effect of domain shift with label shift. To obtain the domain-invariant features, the proposed method modifies adversarial auto-encoder architecture and performs semi-supervised learning to enlarge the inter-class discrepancy. The marginal distribution is aligned in the adversarial learning process of extracting domain-invariant features. Meanwhile, the label information is incorporated in this way to align the conditional distribution. The proposed work also theoretically analyses the generalisation bound of the proposed model. Finally, the proposed method is evaluated based on several domain adaptation tasks, including digit classification and object recognition, and achieves state-of-the-art performance. © 2020 Institution of Engineering and Technology. All rights reserved.

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DortmundAI at LeQua 2022: Regularized SLD

Item Type Conference Paper

Author M. Senz

Author M. Bunse
Editor Faggioli G.
Editor Department of Information Engineering, Via Gradenigo 6/b, Padova University of Padova
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Editor Department of Information Engineering, Via Gradenigo 6/b, Padova University of Padova
Editor Hanbury A.
Editor Favoritenstrasse 9, Vienna Vienna University of Technology
Editor Potthast M.
Editor Augustusplatz 10, Leipzig University of Leipzig

Abstract The LeQua 2022 competition was conducted with the purpose of evaluating different quantification methods on text data. In the following, we present the solution of our team “DortmundAI”, which ranked first in the multi-class quantification task T1B. This solution is based on a modification of the well-known Saerens-Latinne-Decaestecker (SLD) method. Here, the SLD method, which is based on expectation maximization, is extended by a regularization technique. Additional experiments with the test data, which we took out after the competition closed, reveal that our excellent ranking stems primarily from an extensive hyperparameter tuning of the classifier. © 2022 Copyright for this paper by its authors.

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Downscaling multilayer soil moisture using parameterized depth profiles associated with environmental factors

Item Type Journal Article

Author M. Zhang

Author Y. Ge

Author Y. Ma

Author Y. Jin

Author Y. Chen

Author S. Liu

Abstract High-resolution multilayer soil moisture is valuable for water resources management, drought monitoring, and crop optimization. However, integrating process-based models into statistical downscaling methods to estimate multilayer soil moisture can increase both prediction uncertainty and computational complexity of parameterization. Here, three parameterized equations (Peak, Trough, and Balance) were proposed to quantify soil moisture profiles associated with environmental factors, and downscale multilayer soil moisture. Global (0–40 cm) and local (0–10, 10–20, and 20–40 cm) fitting strategies represented the vertical extent of environmental influences. Three machine learning models were employed to downscale satellite soil moisture and predict depth parameters, which were then used to generate sub-surface soil moisture. The results indicated that the local fitting strategy consistently outperformed the global strategy across all depths, reducing the mean absolute error (MAE) and root mean square error (RMSE) by 81.8 % and 80.4 %, and increasing the coefficient of determination (R^2) by 22.5 %. Furthermore, local fitting revealed distinct nonlinear and linear patterns within the 0–20 cm and 20–40 cm intervals, respectively. The prediction accuracy of sub-surface soil moisture was improved compared to surface results, with MAE and RMSE reduced by 30.2 %–60.7 % and 34.9 %–56.7 %, respectively. The modified local fitting strategy further enhanced prediction performance, with the Peak–Trough–Balance combination recommended as the optimal configuration. Future studies should account for variations in soil and vegetation cover, refine interval division by incorporating soil-forming processes, and adopt advanced uncertainty quantification methods to enhance its adaptability and robustness. This study provides a reference for multilayer soil moisture downscaling in large-scale regions with depth variability and environmental heterogeneity. © 2025 Elsevier B.V.

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DP-EVA: data-efficient fine-tuning framework via maximizing pretrained knowledge of large atomistic models for developing domain-specific machine-learning interatomic potentials

Item Type Journal Article

Author Z. Liu

Author Z. Deng

Author H. Zhao

Author H. Wang

Author M. Chen

Author H. Jiang

Abstract Machine-learning interatomic potentials (MLIPs) have greatly extended the temporal and spatial scales of atomistic simulations, enabling the theoretical study of complex processes at affordable computational cost compared with conventional density functional theory. Recently, large atomistic models (LAMs) have drawn intense interest, as their unified encoders embed extensive chemical knowledge and support fine-tuning methodologies for efficiently adapting models to domain-specific downstream tasks. While many active learning frameworks exist for building MLIP datasets from scratch, dedicated data-generation pipelines for fine-tuning pretrained

LAMs remain scarce. Here, we introduce the Deep Potential EVolution Accelerator (DP-EVA)—a data-efficient fine-tuning framework that maximizes the utilization of the pretrained knowledge of LAMs during data generation, accelerating the evolution of domain-specific fine-tuned models with minimal datasets. DP-EVA collects highly representative data through a dual-dimensional shallow-ensemble-based uncertainty quantification method based on parallel fine-tuning on the LAM decoder and a Dimensionality-Reduced Encoded Clusters with sTratified (DIRECT) sampling strategy based on the LAM encoder. Tests show that DP-EVA delivers optimal chemical space coverage in the task of drastically reducing the size of an existing MLIP dataset for iron-based Fischer–Tropsch synthesis. DP-EVA fills the gap of active learning frameworks suitable for fine-tuning LAMs toward domain-specific MLIPs and it is also open-source, Slurm-native, and agent-ready for the coming era of agentic scientific research. © The Author(s) 2026. Published by Oxford University Press on behalf of National Institute of Clean-and-Low-Carbon Energy. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (<https://creativecommons.org/licenses/by-nc/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

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Drift mining in data: A framework for addressing drift in classification

Item Type Journal Article

Author V. Hofer

Author G. Krempel

Abstract A novel statistical methodology for analysing population drift in classification is introduced. Drift denotes changes in the joint distribution of explanatory variables and class labels over time. It entails the deterioration of a classifier's performance and requires the optimal decision boundary to be adapted after some time. However, in the presence of verification latency a re-estimation of the classification model is impossible, since in such a situation only recent unlabelled data are available, and the true corresponding labels only become known after some lapse in time. For this reason a novel drift mining methodology is presented which aims at detecting changes over time. It allows us either to understand evolution in the data from an ex-post perspective or, ex-ante, to anticipate changes in the joint distribution. The proposed drift mining technique assumes that the class priors change by a certain factor from one time point to the next, and that the conditional distributions do not change within this time period. Thus, the conditional distributions can be estimated at a time where recent labelled data are available. In subsequent periods the unconditional distribution can be expressed as a mixture of the conditional distributions, where the mixing proportions are equal to the class priors. However, as the unconditional distributions can also be estimated from new unlabelled data, they can then be compared to the mixture representation by means of least-squares criteria. This allows for easy and fast estimation of the changes in class prior values in the presence of verification latency. The usefulness of this drift mining approach is demonstrated using a real-world dataset from the area of credit scoring. © 2012 Elsevier B.V. All rights reserved.

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Notes:

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Export Date: 20 September 2026; Cited By: 31; Correspondence Address: V. Hofer; Department of Statistics and Operations Research, University of Graz, A-8010 Graz, Universitätsstraße 15/E3, Austria; email: vera.hofer@uni-graz.at; CODEN: CSDAD

Driving mechanisms of water yield in the Yellow River Basin: Insights from an explainable machine learning approach

Item Type Journal Article

Author J. Hu

Author L. Wang

Author Y. Wang

Author Y. Liu

Author X. Su

Author L. Li

Author J. Li

Abstract Study region: Yellow River Basin, a semi-arid and semi-humid basin in China. Study focus: Water yield (WY) is a critical indicator for water resources assessment. However, existing studies face limitations in WY validation and the exploration of nonlinear driving mechanisms. This study innovatively combines InVEST-modeled WY (validated using surface water and GLEAM_AET data) with explainable machine learning to quantify the nonlinear effects of climatic, anthropogenic, topographic, and soil factors on WY. New hydrological insights for the region: This work obtained more accurate WY and analyzed the nonlinear action mode of driving factors: (1) the new method obtained WY with equal total volume and similar spatial distribution; (2) precipitation (PRE) is the main factor affecting the spatial distribution of WY, with a contribution rate of 37.11 %, and its effect shows obvious geographical differentiation: the effect size gradually increases from north to south. In humid areas (PRE > 400 mm), WY and PRE increase linearly, while in arid areas, the response is not significant; the synergistic effect of altitude and root-restricting layer depth dominates the terrain-soil interaction process; (3) In terms of temporal evolution, the contribution rate of PRE change reached 74.22 %, and its increment was linearly correlated with WY gain. Land use change affects WY by changing evapotranspiration and root depth. © 2025 The Authors

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DOI 10.1016/j.ejrh.2025.102610**URL** <https://www.scopus.com/pages/publications/105011079296?origin=resultslist>**Archive** Scopus**Language** English**Volume** 61**Publisher** Elsevier B.V.**Publication** Journal of Hydrology: Regional Studies**ISSN** 22145818 (ISSN)**Journal Abbr** J. Hydrol. Reg. Stud.**Date Added** 20/09/2026, 9:51:43 pm**Modified** 20/09/2026, 9:51:43 pm**Tags:**

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Drone-Based Sheep Recognition and Counting: A Comparative Study of Color-Shape, and YOLOv5 Algorithms

Item Type Conference Paper
Author J.R.C. Ticona
Author G.J.F. Chipana
Author G.H.C. Cabrera
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Author G.R.V. Ponce
Editor Iano Y.
Editor Saotome O.
Editor Arthur R.
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Editor Gomes de Oliveira G.

Abstract This study presents sheep detection and quantification methods for video frames captured by UAVs and examine their advantages over traditional UAV video recording. The implementation of two digital image processing approaches for pattern recognition is based on color-shape analysis and the YOLOv5 classifier to detect sheep in a field from a video source. Both methods are implemented in Python 3 using OpenCV tools, where morphological operations and feature classification algorithms are applied. These methods perform multiple operations on the video source and have proven to be effective for detecting sheep images under different lighting conditions, as well as at varying distances between the UAV and the sheep. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Dual-center RAPID-LSSVM: Radius-adaptive, probability and imbalance driven weighting for Alzheimer's diagnosis

Item Type Journal Article

Author M. Akhtar
Author A. Quadir
Author M. Tanveer
Author M. Arshad

Abstract Alzheimer's disease (AD) is a leading neurodegenerative disorder and the primary cause of dementia, where early, reliable diagnosis remains challenging. Although many machine learning methods have been developed for early AD detection, their performance often degrades under label noise, outliers, and class imbalance. To counter these issues, we propose RAPID, a Radius-Adaptive, Probability and Imbalance Driven flexible weighting mechanism, and integrate it into least-squares SVM to obtain two models: RAPID-LSSVM-I (mean-center) and RAPID-LSSVM-II (median-center). RAPID combines three complementary components. First, a radius-adaptive proximity weight that plateaus for samples near the class center and decays smoothly beyond a threshold, preserving the influence of boundary samples while improving robustness to central noise. Second, a local class-probability term that down-weights potentially mislabeled or ambiguous instances, and third, an imbalance-ratio term that compensates for class prior skew. The dual-center design enables either conventional mean centering or a median-based center that is resilient to outliers and asymmetric distributions. To validate the effectiveness of the proposed RAPID-LSSVM models, experiments are conducted on benchmark KEEL and UCI datasets under both clean and label-noise settings. Additionally, we tested the models on the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset for AD diagnosis. Empirical findings demonstrate the superiority of the RAPID-LSSVM models over baseline models, highlighting their potential in improving AD diagnosis and handling noisy data. © 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Dual-Contrastive Multiview Graph Attention Network for Industrial Fault Diagnosis under Domain and Label Shift

Item Type Journal Article

Author J. Zhu

Author S. Wu

Author Y. Xiao

Author B. Wang

Author R. Cai

Abstract Recently, domain generalization (DG) methods have been actively researched for the complex industrial fault diagnosis, which aims to learn generalized representations from historical working conditions to build a diagnosis model that can perform well on unseen working conditions. However, these methods ignore the interactions between monitoring variables, which may fail to learn the feature representation with topological structure in non-Euclidean space. In addition, these methods assume the same label distribution across historical and unseen working conditions, which is generally challenging in practice, as the probability of faults varies across different working conditions. This label shift problem can negatively impact the generalization performance. To address these issues, a novel dual-contrastive multiview graph attention network (DMGAT) is proposed in this article. Specifically, a multiview graph attention network (GAT) is designed to explore the intrinsic topological structure of the data, which learns an optimal graph structure that best serves DG by integrating both graph learning and graph convolution in a unified network architecture. In addition, a novel dual-weighted contrastive learning strategy is developed. The intradomain contrastive learning facilitates the extraction of expressive node features, while interdomain contrastive learning simultaneously considers the alignment and separation of semantic probability distributions to extract

shared feature representations for multiple source domains under both domain and label shifts. Furthermore, a label sampling probability is used to weight the interdomain contrastive loss and the source domain classification loss, to encourage the model to learn from minor classes in fault diagnosis. Experiments on two cases demonstrate the superiority of the proposed method. © 1963-2012 IEEE.

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Dual-stream algorithms for dementia detection: Harnessing structured and unstructured electronic health record data, a novel approach to prevalence estimation

Item Type Journal Article

Author T.A. Collyer

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Author V.K. Srikanth

Abstract INTRODUCTION: Identifying individuals with dementia is crucial for prevalence estimation and service planning, but reliable, scalable methods are lacking. We developed novel set algorithms using both structured and unstructured electronic health record (EHR) data, applying Diagnostic and Statistical Manual of Mental Disorders criteria for dementia case identification. METHODS: Our cohort (n = 1082) included individuals aged ≥ 60 with dementia identified through specialist clinics and a comparison group without dementia. Clinicians from Australia and the United States informed predictor selection. We developed algorithms through a biostatistics stream for structured data and a natural language processing (NLP) stream for text, synthesizing results via logistic regression. RESULTS: The final structured model retained 16 variables (area under the receiver operating characteristic curve [AUC] 0.853, specificity 72.2%, sensitivity 80.6%). NLP classifiers (logistic regression, support vector machine, and random forest models) performed comparably. The final, combined model outperformed all others (AUC = 0.951, $P < 0.001$ for comparison to structured model). DISCUSSION: Embedding text-derived insights within algorithms trained on structured medical data significantly enhances dementia identification capacity. Highlights: Algorithmic tools for detection of individuals with dementia are available; however, previous work has used heterogeneous case definitions which are not clinically meaningful, and has relied on proxies such as diagnostic codes or medications for case ascertainment. We used a novel, dual-stream algorithmic development approach, simultaneously and separately modeling a clinically meaningful outcome (diagnosis of dementia according to specialized clinical impression) using structured and unstructured electronic health record datasets. Our clinically grounded case definition supported the inclusion of key structured variables (such as dementia International Classification of Disease codes and medications) as modeling predictors rather than outcomes. Our algorithms, published in detail to support validation and replication, represent a major step forward in the use of routinely collected data for detection of diagnosed dementia. © 2025 The Author(s). Alzheimer's & Dementia published by Wiley Periodicals LLC on behalf of Alzheimer's Association.

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Dual-stream collaborative network (DSCN): Multimodal sentiment analysis via modality-invariant and modality-specific representation learning

Item Type Journal Article
Author S. Chen
Author D. Zhou
Author C. Wei
Author X. Li
Author H. Yao

Author H. Luo

Abstract Multimodal Sentiment Analysis (MSA) is a critical technology for human-machine interaction. It performs sentiment analysis by integrating complementary information from multiple modalities such as text, audio, and vision. However, most existing multimodal fusion methods regard the text as the primary modality and treat speech and visual modalities as auxiliary. This approach limits the potential for synergistic interaction among the three modalities. To address this limitation, this paper proposes a Dual-Channel Collaborative Transformer (DCCT), which enables collaborative interaction among the three modalities. DCCT incorporates a Self-modal Binding Channel (SBC) to enhance intra-modal consistency, and a Cross-modal Binding Channel (CBC) to complement the idiosyncratic features of other modalities. Then, a Feed-Forward Network (FFN) integrates the outputs of both channels for effective trimodal feature fusion. Furthermore, to mitigate the label shift problem in multi-task joint learning, we propose a framework that distinguishes between intensity and polarity tasks based on modality decomposition. By separating common and idiosyncratic features, we design dedicated tasks for intensity and polarity, and use a shared-private linear layer in the Trimodal Label Generator (TMLG) to capture the interrelation between the two aspects and facilitate task adaptation. Experiments conducted on the CMU-MOSI and CMU-MOSEI datasets show that the proposed Dual-Stream Collaborative Network (DSCN), which integrates DCCT and TMLG, achieves ACC-2 accuracy of 86.75 % and 86.69 %, respectively. These results outperform the best existing baseline by 0.65 % and 0.83 %. © 2025 Elsevier B.V.

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DWLR: Domain Adaptation under Label Shift for Wearable Sensor

Item Type Conference Paper

Author J. Li

Author Y. Yang

Author Y. Chen

Author J. Zhang

Author Z. Lai

Author L. Pan

Editor Larson K.

Abstract Wearable sensors play a crucial role in real-world scenarios, such as human activity recognition, sleep monitoring and electrocardiogram monitoring. However, deploying classifiers on them is challenged by distribution shifts across users and devices. Unsupervised domain adaptation (UDA) is proposed to address this, yet existing methods mostly focus on feature distribution shift, neglecting the potential misclassification due to label shift. In this paper, we propose Domain adaptation under label shift for Wearable sensor with Learnable Reweighting (DWLR) to handle both feature and label shifts. Specifically, DWLR employs learnable reweighting to align label distributions between source and target domains. It incorporates elements of information gain during the reweighting process to counter potential distribution shift that could emerge from over-reliance on data with high-confidence pseudo labels. Importantly, since wearable sensor data is time-series data, and can be subjected to distribution shifts originating from either the time domain, the frequency domain, or both, DWLR performs reweighting and alignment separately in these two domains to more robustly handle potential feature distribution shifts. Extensive experiments on three distinct wearable sensor datasets demonstrate the effectiveness of DWLR, yielding a remarkable average performance improvement of 5.85%. © 2024 International Joint Conferences on Artificial Intelligence. All rights reserved.

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Dynamic Evolution of Financial Technology and Artificial Intelligence for Economy Based on Bayesian Algorithm in Cyber-Physical System

Item Type Journal Article

Author X. Wang

Abstract With the rapid development of information technology and the popularity of big data, the demand for data analysis has become increasingly large. In recent years, data mining technology has received extensive attention. It is a technology that converts a large amount of data into useful information and finds the relationship between data and information. The classification algorithm is one of the important research contents in data mining. It mainly divides the data items in the dataset into a fixed category according to various classifiers. From the perspective of the development trend of the world economy, technological progress, consumption upgrading and institutional change are mutually reinforcing and integrating. In financial technology, the evolution of technology is reshaping the financial industry. Data intelligence technology based on machine learning and other algorithms is changing rapidly. Data is the largest means of production, the most important factor of production of artificial intelligence (AI), and the core factor of production. Big data, cloud computing, AI and other emerging technologies build different data structures for processing through deep learning and other learning algorithms to achieve accurate analysis, rapid decision-making and other characteristics. At the same time, it also provides financial institutions with more accurate and comprehensive data services. It has established long-term cooperation with financial institutions, promoting the accelerated integration of finance and technology and jointly creating a more intelligent financial system. Bayesian algorithm is a classical algorithm in classification algorithms. It is a possibility reasoning method based on probability statistics. Bayesian algorithm calculates a posteriori probability through a priori probability. Because of its small error rate, it has been widely used in various fields. This paper will analyze banking financial technology and AI economy based on cyber-physical systems and Bayesian algorithms. The research results showed that

Bank A added 1,275 small customers in the first quarter of 2020, accounting for 69.52% in 2019; in the first quarter of 2020, inclusive microfinance increased by 4.7 billion yuan, with a year-on-year growth of 35.34%, indicating that Bank A played an irreplaceable role in using financial technology in times of crisis. It points out a new development direction for the application of cyber-physical systems. © This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<http://creativecommons.org/licenses/by-nc/3.0/>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

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Dynamic Quantification With Constrained Error Under Unknown General Dataset Shift

Item Type Journal Article

Author B. Denham

Author E.M.-K. Lai

Author R. Sinha

Author M.A. Naeem

Abstract Quantification research has sought to accurately estimate class distributions under dataset shift. While existing methods perform well under assumed conditions of shift, it is not always clear whether such assumptions will hold in a given application. This work extends the analysis and experimental evaluation of our Gain-Some-Lose-Some (GSLs) model for quantification under general dataset shift and incorporates it into a method for dynamically selecting the most appropriate quantification method. Selection by a Kolmogorov-Smirnov test for any shift followed by a newly proposed 'Adjusted Kolmogorov-Smirnov' test for non-prior shift is found to best balance quantification and runtime performance. We also present a framework for constraining quantification prediction intervals to user-specified limits by requesting a smaller set of instance class labels from the user than required with confidence-based rejection. © 1989-2012 IEEE.

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DyS: A framework for mixture models in quantification

Item Type Conference Paper

Author A. Maletzke

Author D. dos Reis

Author E. Cherman

Author G. Batista

Abstract Quantification is an expanding research topic in Machine Learning literature. While in classification we are interested in obtaining the class of individual observations, in quantification we want to estimate the total number of instances that belong to each class. This subtle difference allows the development of several algorithms that incur smaller and more consistent errors than counting the classes issued by a classifier. Among such new quantification methods, one particular family stands out due to its accuracy, simplicity, and ability to operate with imbalanced training samples: Mixture Models (MM). Despite these desirable traits, MM, as a class of algorithms, lacks a more in-depth understanding concerning the influence of internal parameters on its performance. In this paper, we generalize MM with a base framework called DyS: Distribution y-Similarity. With this framework, we perform a thorough evaluation of the most critical design decisions of MM models. For instance, we assess 15 dissimilarity functions to compare histograms with varying numbers of bins from 2 to 110 and, for the first time, make a connection between quantification accuracy and test sample size, with experiments covering 24 public benchmark datasets. We conclude that, when tuned, Topsøe is the histogram distance function that consistently leads to smaller quantification errors and, therefore, is recommended to general use, notwithstanding Hellinger Distance's popularity. To rid MM models of the dependency on a choice for the number of histogram bins, we introduce two dissimilarity functions that can operate directly on observations. We show that SORD, one of such measures, presents performance that is slightly inferior to the tuned Topsøe, while not requiring the sensible parameterization of the number of bins. © 2019, Association for the Advancement of Artificial Intelligence (www.aaai.org).

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Early Detection of Crack Vulnerability in Foldable Displays Through Critical-Angle Curvature Analysis

Item Type Conference Paper

Author K. Park

Author J. Kim

Author D. Kim

Abstract Foldable displays have become increasingly popular due to their portability and the provision of larger screens when unfolded. However, the use of flexible materials makes them more susceptible to mechanical failures, particularly crack formation in the hinge area where stress concentration is highest during folding operations. This study presents a novel approach to predict crack formation in foldable displays by measuring the curvature of the hinge area at a specific critical angle—where the mechanical stress peaks—and analyzing this data using Principal Component Analysis (PCA). The curvature measurements were initially reduced using PCA, retaining two principal components that captured around 60% of the dataset's variance. While this facilitated visualization, it was insufficient for clear classification between defective and non-defective samples due to overlapping data points and information loss. To enhance predictive accuracy, we introduced an uncertainty quantification method based on a -nearest neighbors approach with distance-weighted contributions. This method calculates an uncertainty score for each sample, reflecting the confidence in classifying it as susceptible or resistant to cracking. By integrating the uncertainty analysis into an artificial intelligence (AI) model, we assigned a numerical score ranging from +1 (clearly non-defective) to -1 (high risk of cracking) to each sample. Applying this model to a larger dataset of 200 samples, we successfully quantified the risk of crack occurrence—a task previously unfeasible with conventional methods. The model demonstrated strong discriminative power, effectively distinguishing between at-risk and safe samples. This integrated

methodology combining precise curvature measurement, uncertainty analysis, and AI modelling provides a robust framework for early detection and prediction of crack formation in foldable displays. The approach enhances product reliability and contributes valuable insights into material behavior under stress, supporting improved quality control measures in manufacturing. Future work will focus on expanding the dataset and incorporating additional features to further refine the model's predictive capabilities. © 2025, John Wiley and Sons Inc. All rights reserved.

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Early radiomic experiences in classifying prostate cancer aggressiveness using 3D local binary patterns

Item Type Conference Paper

Author R. Sicilia

Author E. Cordelli

Author M. Merone
Author E. Luperto
Author R. Papalia
Author G. Iannello
Author P. Soda

Abstract Prostate cancer is the most common form of cancer in Western countries and there is the need to develop clinical decision support systems able to support physicians in the diagnosis of clinical relevant prostate cancer and avoid useless invasive prostate biopsies. In this respect, this paper introduces a radiomic approach that classifies the prostate cancer aggressiveness by combining Three Orthogonal Planes-Local Binary Pattern (TOP - LBP) with other texture measures. Furthermore, to combat the skewed nature of class priors, our proposal employs a data augmentation technique. The results achieved on 99 samples are up-and-coming, they favorably compare against conventional PI-RADS-based approach, and they show also the benefit given by the introduction of TOP-LBP in the radiomic signature. © 2019 IEEE.

Date 2019

DOI 10.1109/CBMS.2019.00078

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Language English

Extra Journal Abbreviation: Proc. IEEE Symp. Comput.-Based Med. Syst.

Volume 2019-June

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Artificial intelligence, Decision support systems, Learning systems, Machine-learning, Textures, Diagnosis, Data augmentation, Diseases, Clinical decision support systems, Urology, Precision medicine, Prostate cancers, Local binary patterns, Orthogonal plane, Prostate cancer, Radiomics, TOP-LBP, Western countries

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Easy Interpretation of Image Classification Results with Feature-Level Visualization

Item Type Conference Paper

Author D. Bhati

Author Y. Zhao

Author T.-H. Wu

Author M. Amiruzzaman

Author J. Yang

Abstract Given a picture classified as a Persian cat by an AI model, users may ask questions such as, “What are the contributions of the eyes and ears to the classification result?” or “Which features contribute the most?” While existing post-hoc XAI methods effectively explain model predictions at the pixel or patch level, they are limited in directly quantifying the contributions of human-interpretable semantic features. In this paper, we propose a visual analytics approach for feature-level interpretation of image classification results. Our contributions are twofold. First, we introduce a semantic contribution quantification method that builds upon existing pixel-level attribution techniques (e.g., Layer-wise Relevance Propagation, Grad-CAM). Specifically, we aggregate and normalize pixel-level relevance scores over predefined semantic regions (such as eyes, ears, and body) to compute comparable contribution scores for each semantic feature within an image. Second, we present an interactive visual interface that leverages these quantified semantic feature contributions to support exploration, comparison, and analysis of AI outputs across image collections. Through illustrative scenarios and expert feedback, we demonstrate that our approach provides an intuitive, scalable, and semantically meaningful means to interpret image classification explanations. ©2026 Society for Imaging Science and Technology.

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DOI 10.2352/EI.2026.38.1.VDA-349

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ECG based human identification using Second Order Difference Plots

Item Type Journal Article

Author G. Altan

Author Y. Kutlu

Author M. Yeniad

Abstract Background and objective: ECG is one of the biometric signals that has been studied in peer-reviewed over past years. The developments on the signal analysis methods show that the studies on the ECG would continue unabatedly. It has a common use on cardiac diseases with high rates of classification performances by integrating it with signal analysis methods. The aim of the study is to utilize the ECG for human identification. Methods: Second Order Difference Plot (SODP) is a non-linear time-series analysis method that allows determining the features using the statistical analysis of the wave distributions. The SODP features were extracted using different quantification methods for ECG-based human identification. A new quantification approach has been proposed on the SODP for ECG-based human identification. The proposed method, Logarithmic Grid Analysis, was compared with the existing quantification methods on the SODP. The region of the SODP was divided into sub-regions with logarithmically increasing distances and the numbers of data points in each logarithmic sub regions were calculated in the proposed method. Three different databases were used to test the validity of the method. These records have been tested with the conventional feature extraction methods on the SODP. The long-term ECG signals were divided into 5-s short-term ECG signals. Results: The Logarithmic Grid Analysis features that were counted from short-time ECG signals were classified with k-Nearest Neighbor algorithm using 10-fold cross validation, and the identification performance of the proposed model was evaluated. Consequently, high accuracy rates of 91.96%, 99.86% and 95.12% were achieved on ECG-based human identification using the Logarithmic Grid Analysis method on the SODP. Conclusions: The density score of data points at the center of the SODP is too high. This case increases the importance of the regions close the center in order to find the detailed and significant features from the SODP. The number of data points at the center has been extracted in more detail and the vertex areas of the major axes of the SODP can be interpreted in the aggregate sub-regions by using logarithmically increasing distances with a small number of feature size. © 2019 Elsevier B.V.

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Publication Computer Methods and Programs in Biomedicine
ISSN 01692607 (ISSN)
Journal Abbr Comput. Methods Programs Biomed.
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adult, Adult, Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, young adult, Young Adult, Nearest neighbor search, adolescent, Adolescent, aged, Aged, algorithm, Algorithms, Electrocardiography, procedures, feature extraction, Biometrics, Databases, Factual, factual database, Time series analysis, k nearest neighbor, Identification (control systems), Information dissemination, classifier, validation study, Classification performance, electrocardiogram, electrocardiography, Biomedical signal processing, biometry, Biometry, nonlinear system, Feature extraction methods, 10-fold cross-validation, Biometric, ECG, forensic anthropology, Forensic Anthropology, Identification, K nearest neighbor algorithm, Nonlinear time-series analysis, PQRST complex, second order difference plot, Second Order Difference Plot, Second orders, Signal analysis, SODP

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EID: 2-s2.0-85060255434

Export Date: 20 September 2026; Cited By: 32; Correspondence Address: G. Altan; Iskenderun Technical University, Turkey; email: gokhan.altan@iste.edu.tr; CODEN: CMPBE

ECGNet: edge and class guided semantic segmentation network for remote sensing urban scene images

Item Type Journal Article
Author H. Liu
Author M. Liu
Author S. Song
Author G. Guo
Author Z. Yuan
Author K. Chen

Author S. Yang

Author J. Yu

Author H. Zhang

Abstract Semantic segmentation of remote sensing images in urban scenes suffers from blurred multi-scale target boundaries, insufficient use of global context, and classification errors caused by high inter-class variance and low intra-class variance. Therefore, we propose a semantic segmentation network with edge and class guidance (ECGNet). First, ECGNet introduces multi-scale edge prior knowledge to address the problem of blurred target boundaries. Second, ECGNet applies synergistic class augmented attention to introduce class prior knowledge while retaining rich spatial dimensional localization information to alleviate the problem of classification errors caused by low intra-class variance and high inter-class variance. Finally, the multi-scale large receptive field attention in ECGNet simulates a large convolutional kernel to capture multi-scale global context information. Experiments conducted on the ISPRS Vaihingen and ISPRS Potsdam datasets show that the proposed method is competitive. © 2024 Society of Photo-Optical Instrumentation Engineers (SPIE).

Date 2024

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Journal Abbr J. Appl. Remote Sens.

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Edge-Prioritize IDS: Zero-Retraining Class Prioritization for Real-Time Edge Intrusion Detection

Item Type Journal Article

Author P. Pawar

Author G. Epiphaniou

Abstract Deploying deep neural networks-based intrusion detection systems on resource-constrained edge devices demands inference strategies that balance latency, energy, and accuracy under shifting threat landscapes. This paper presents Edge-Prioritize IDS, a class-prioritized early-exit framework that accelerates inference for high-risk attack classes without post-deployment retraining. A lightweight K-dimensional control vector encodes per-class runtime priorities and steers samples toward earlier exits via adaptive normalization and cost-sensitive training. Evaluation across five benchmarks NSL-KDD, CIC-IDS2017, UNSW-NB15, WISDM, and CIFAR-10 on an NVIDIA Jetson TX2 shows that Edge-Prioritize IDS preserves baseline accuracy (up to 99.6%) while reducing latency by up to 55% and energy by up to 50% for prioritized classes. Ablation studies isolate each component's contribution, and a controlled distribution-shift experiment demonstrates the sliding-window heuristic's ability to recover near-baseline latency within 500 samples under synthetic class-frequency drift. Once trained under the proposed framework, the model requires no additional retraining, firmware updates, or additional memory beyond the priority vector itself when runtime priorities change. © 2026 by the authors.

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Journal Abbr Information

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Network security, Classification (of information), Intrusion detection, Energy, Edge computing, Computer crime, Intrusion Detection Systems, IoT security, Runtimes, Network intrusion, intrusion detection system, edge computing, Firmware, Fighter aircraft, Edge detection, Cost-sensitive learning, class-aware classification, Class-aware classification, cost-sensitive learning, dynamic inference, Dynamic inference, early-exit network, Early-exit network, Prioritization

Notes:

EID: 2-s2.0-105040214149

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: P. Pawar; Warwick Manufacturing Group (WMG), University of Warwick, Coventry, CV4 7AL, United Kingdom; email: pruthviraj-rajendra.pawar@warwick.ac.uk

EEG feature extraction and machine learning for consumer preference analysis of whisky

Item Type Journal Article

Author Z. Zeng

Author X. Liu

Author Y. Xu

Author K. Tang

Abstract Understanding consumer preference for food and beverages is central to product development, yet subjective self-reports alone often lack objectivity. Electroencephalography (EEG) can reveal implicit neural signals of preference. However, existing flavor studies utilizing EEG are mostly confined to group difference analysis, cannot identify stable preference-related EEG markers or achieve objective preference prediction. To address this gap, this study integrated sensory evaluation, systematic EEG feature extraction and machine learning to quantify and predict consumer whisky preference. Twenty-four panelists were tested with 12 whisky samples via 9-point hedonic and 5-point JAR scales. Time-domain and frequency-domain features were extracted from EEG recordings, and four classification models were trained to predict preference. Compared with the accuracy of models built on raw EEG data ($\approx 51\%$), feature extraction significantly boosted performance, with the Random Forest model achieving the highest test accuracy of 82.6%. Feature weight analysis further identified potentially predictive EEG indicators, including frontal α asymmetry and α band power. This work verifies the predictive capacity of EEG features for machine learning modeling, thereby establishing a neurophysiologically based method for the objective evaluation. This scheme complements traditional subjective sensory assessment and delivers actionable technical guidance for industrial flavor tuning and product development. However, the high homogeneity of participants in age and background limits the generalizability of our experimental results. © 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license. <http://creativecommons.org/licenses/by-nc-nd/4.0/>

Date 2026**DOI** 10.1016/j.crfs.2026.101540**URL** <https://www.scopus.com/pages/publications/105047851907?origin=resultslist>**Archive** Scopus**Language** English**Volume** 13**Publisher** Elsevier B.V.**Publication** Current Research in Food Science**ISSN** 26659271 (ISSN)**Journal Abbr** Curr. Res.Food Sci.**Date Added** 20/09/2026, 9:51:44 pm**Modified** 20/09/2026, 9:51:44 pm**Tags:**

adult, female, human, human experiment, male, young adult, Feature extraction, machine learning, Machine learning, article, consumer, prediction, feature extraction, controlled study, random forest, self report, flavor, sensory evaluation, EEG, electroencephalogram, electroencephalography, Consumer preference, product development, Sensory evaluation, whiskey, Whisky

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K. Tang; Lab of Brewing Microbiology and Applied Enzymology, School of Biotechnology and Key Laboratory of Industrial Biotechnology of Ministry of Education, Jiangnan University, Wuxi, 214122, China; email: tandy81@jiangnan.edu.cn

EEG may serve as a biomarker in Huntington's disease using machine learning automatic classification

Item Type Journal Article**Author** O.F.F. Odish**Author** K. Johnsen**Author** P. van Someren**Author** R.A.C. Roos**Author** J.G. van Dijk

Abstract Reliable markers measuring disease progression in Huntington's disease (HD), before and after disease manifestation, may guide a therapy aimed at slowing or halting disease progression. Quantitative electroencephalography (qEEG) may provide a quantification method for possible (sub)cortical dysfunction occurring prior to or concomitant with motor or cognitive disturbances observed in HD. In this pilot study we construct an automatic classifier distinguishing healthy controls from HD gene carriers using qEEG and derive qEEG features that correlate with clinical markers

known to change with disease progression in HD, with the aim of exploring biomarker potential. We included twenty-six HD gene carriers (49.7 ± 8.5 years) and 25 healthy controls (52.7 ± 8.7 years). EEG was recorded for three minutes with subjects at rest. An EEG index was created by applying statistical pattern recognition to a large set of EEG features, which was subsequently tested using 10-fold cross-validation. The index resulted in a continuous variable ranging from 0 to 1: a low value indicating a state close to normal and a high value pointing to HD. qEEG features that correlate specifically with commonly used clinical markers in HD research were derived. The classification index had a specificity of 83%, a sensitivity of 83% and an accuracy of 83%. The area under the curve of the receiver operator characteristic curve was 0.9. qEEG analysis on subsets of electrophysiological features resulted in two highly significant correlations with clinical scores. The results of this pilot study suggest that qEEG may serve as a biomarker in HD. The indices correlating with modalities changing with the progression of the disease may lead to tools based on qEEG that help monitor efficacy in intervention studies. © 2018, The Author(s).

Date 2018

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Issue 1

Journal Abbr Sci. Rep.

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female, Female, human, Humans, male, Male, middle aged, Middle Aged, machine learning, Machine Learning, receiver operating characteristic, diagnostic imaging, ROC Curve, biological marker, Biomarkers, metabolism, Electroencephalography, electroencephalography, case control study, Case-Control Studies, Huntington chorea, Huntington Disease

Notes:

EID: 2-s2.0-85055853613

Export Date: 20 September 2026; Cited By: 51; Correspondence Address: O.F.F. Odish; Department of Neurology, University Medical Center Groningen, Groningen, Netherlands; email: o.f.f.odish@umcg.nl

EEG Signal Classification for BCI based on Neural Network

Item Type Conference Paper

Author K. Chenane

Author Y. Touati

Editor Schmidt H.

Editor Griol D.

Editor Wang H.

Editor Baumbach J.

Editor Zheng H.

Editor Callejas Z.

Editor Hu X.

Editor Dickerson J.

Editor Zhang L.

Abstract This paper proposes a method for classifying ElectroEncephaloGraphic (EEG) signals for Brain-Computer Interface (BCI). BCI is a system that allows people with disabilities to interact directly or indirectly with their environment only by using brain activity. In this study, we present a model for classifying EEG signals from five different mental activities (converse, read, watch video, rest and work). The method of signal acquisition is non-invasive, resulting in significant data loss and artifacts that must be reduced. In order to do this, a quantification method using Fast Fourier Transform (FFT) and Root Mean Square (RMS) for feature extraction are applied to improve the quality of the signal reception. We propose an approach to the neural network Self-Organizing Map (SOM) to classify these filtered signals. © 2018 IEEE.

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Proceedings Title Proc. - IEEE Int. Conf. Bioinform. Biomed., BIBM

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Tags:

Classification (of information), Brain, Neural networks, People with disabilities, neural networks, classification, Bioinformatics, Quantification methods, Biomedical signal processing, EEG, Electroencephalography, Brain computer interface, Neurophysiology, Fast Fourier transforms, Conformal mapping, Self organizing maps, BCI, BMI, brain activity, Brain activity, EEG signal classification, Electroencephalographic signals, FFT, Filtered signals, Root Mean Square, Signal acquisitions

Notes:

EID: 2-s2.0-85062508187

Export Date: 20 September 2026; Cited By: 7; Conference name: 2018 IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2018; Conference date: 3 December 2018 through 6 December 2018; Conference code: 144524

EEG-based personalized digital experience**Item Type** Conference Paper**Author** O. Sourina**Author** Y. Liu**Author** Q. Wang**Author** M.K. Nguyen

Abstract To make human computer interfaces more immersive and intuitive, a new dimension could be added. Real-time brain state recognition from EEG including emotion recognition and level of concentration recognition would make an access to information more adaptive and personalized. Modern EEG techniques give us an easy and portable way to monitor brain activities by using suitable signal processing and classification methods and algorithms. We proposed a new subject-dependent fractal-based approach to brain state recognition and innovative applications based on EEG-enable user's interaction. The algorithms of the "inner" brain state quantification including emotion recognition would advance research on human computer interaction bringing the proposed novel objective quantification methods and algorithms as new tools in medical, entertainment, and even digital art methodology applications, and allowing us an integration of the brain state quantification algorithms in the human computer interfaces. In this paper, we describe our fractal-based approach to the brain state recognition and its EEG-enable applications such as serious games, emotional avatar, music therapy, music player, and storytelling. © 2011 Springer-Verlag.

Date 2011**DOI** 10.1007/978-3-642-21663-3_64**URL** <https://www.scopus.com/pages/publications/79960291418?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Lect. Notes Comput. Sci.**Volume** 6766 LNCS**ISBN** 16113349 (ISSN); 978-364221662-6 (ISBN)**Pages** 591-599**Proceedings Title** Lect. Notes Comput. Sci.**Conference Name** Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)**Issue** PART 2**Date Added** 20/09/2026, 9:52:55 pm**Modified** 20/09/2026, 9:52:55 pm

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Brain, Human computer interaction, State estimation, Algorithms, Signal processing, HCI, Knowledge management, emotion recognition, Brain computer interface, Interfaces (computer), Fractal dimension, BCI, fractal dimension, music therapy, Music therapy, Musical instruments, Occupational therapy, serious game, Serious games, storytelling

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EEG-based seizure prediction via hybrid vision transformer and data uncertainty learning

Item Type Journal Article

Author Z. Deng

Author C. Li

Author R. Song

Author X. Liu

Author R. Qian

Author X. Chen

Abstract Feature embeddings derived from continuous mapping using the deep neural network are critical for accurate classification in seizure prediction tasks. However, the embeddings of individual electroencephalogram (EEG) samples learned through modern decoding algorithms may contain ambiguous and noisy representations, owing to the susceptibility of weak EEG signals to interference from other signals unrelated to EEG activity. To address this issue, we consider data uncertainty learning (DUL), which models each representation of the EEG sample as a Gaussian or Laplacian distribution to mitigate potential noise interference and enhance model robustness. Moreover, data uncertainty learning for transformer architectures is seldom explored due to the limitation of multi-headed self-attention mechanisms in processing local features and the vanishing of potential top-level gradients. In this study, we introduce a novel hybrid visual transformer (HViT) architecture, which enhances the processing capability of localized features in the transformer through the convolutional neural network (CNN). Concretely, we learn the mean of the distribution using the HViT, and an additional branch is designed to capture the variance of the Gaussian distribution or the scale of the Laplacian distribution during training. We also propose a learnable manner to learn constraint coefficients in the loss functions for different patients, resulting in better optimization across patients. In addition, we introduce a simple uncertainty quantification method for each alarm of the k-of-n continuous prediction strategy by utilizing the continuity of the EEG signals. Empirical evaluations on two publicly available epilepsy datasets

demonstrate the superiority of our DUL method and the effectiveness of the proposed HViT architecture. © 2023 Elsevier Ltd

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Effect of feature extraction for brain tumor classification based on short echo time 1H MR spectra

Item Type Journal Article

Author J. Luis

Author J.-B. Pouillet

Author J.M. Garcia-Gomez

Author A. Heerschap

Author M. Robles

Author J.A.K. Suykens

Author S. Van Huffel

Abstract This study examines the effect of feature extraction methods prior to automated pattern recognition based on magnetic resonance spectroscopy (MRS) for brain tumor diagnosis. Since individual inspection of spectra is time-consuming and requires specific spectroscopic expertise, the introduction of clinical decision support systems (DSSs) is expected to strongly promote the clinical use of MRS. This study focuses on the feature extraction step in the preprocessing protocol of MRS when using a DSS. On two independent data sets, encompassing single-voxel and multi-voxel data, it is observed that the use of the full spectra together with a kernel-based technique, handling high dimensional data, or using an automated pattern recognition method based on independent component analysis or Relief-F achieves accurate performances. In addition, these approaches have low cost and are easy to automate. When sophisticated quantification methods are used in a DSS, user interaction should be minimized. The computationally intensive quantification techniques do not tend to increase the performance in these circumstances. The results suggest to simplify the feature reduction step in the pre-processing protocol when using a DSS purely for classification purposes. This can greatly speed up the execution of classifiers and DSSs and may accelerate their introduction into clinical practice. © 2008 Wiley-Liss, Inc.

Date 2008

DOI 10.1002/mrm.21626

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Decision support systems, automation, Automation, human, Feature extraction, Classification (of information), Features extraction, Brain, article, Clustering algorithms, Extraction, Classification, decision support system, Magnetic resonance, Magnetic resonance imaging, Spectra's, proton nuclear magnetic resonance, brain tumor, Brain tumors, Tumors, tumor diagnosis, Independent component analysis, Magnetic resonance spectroscopy, Magnetism, tumor classification, Magnetic resonance spectroscopic imaging, Automated pattern recognition, Brain tumor classifications, Decision support system, Decision support system (DSS), Echo time, Magnetic resonance spectroscopic imaging (MRSI), Magnetic resonance spectroscopy (MRS)

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Effect of separate sampling on classification accuracy

Item Type Journal Article

Author M. Shahrokh Esfahani

Author E.R. Dougherty

Abstract Motivation: Measurements are commonly taken from two phenotypes to build a classifier, where the number of data points from each class is predetermined, not random. In this 'separate sampling' scenario, the data cannot be used to estimate the class prior probabilities. Moreover, predetermined class sizes can severely degrade classifier performance, even for large samples. Results: We employ simulations using both synthetic and real data to show the detrimental effect of separate sampling on a variety of classification rules. We establish propositions related to the effect on the expected classifier error owing to a sampling ratio different from the population class ratio. From these we derive a sample-based minimax sampling ratio and provide an algorithm for approximating it from the data. We also extend to arbitrary distributions the classical population-based Anderson linear discriminant analysis minimax sampling ratio derived from the discriminant form of the Bayes classifier. © 2013 The Author.

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Effect of separate sampling on classification and the minimax criterion

Item Type Conference Paper
Author M. Shahrokh Esfahani
Author E.R. Dougherty
Abstract It is commonplace in bioinformatics (and elsewhere) to build a classifier from sample data in which the sample sizes of the classes are not random; that is, they are selected prior to sampling. The result is that there is no estimate of the prior class probabilities available from the data. In this paper, we find an analytic result for the minimax solution for the class prior probabilities for a general Neyman-Pearson induced classifier. From that we derive Anderson's classical minimax prior probability 'estimate.' Using synthetic and real data, we demonstrate the degradation in classifier performance from using inaccurate values for the prior probabilities. © 2013 IEEE.
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DOI 10.1109/GENSIPS.2013.6735935
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Archive Scopus
Language English
Extra Journal Abbreviation: IEEE Int. Workshop on Genomic Signal Proces. Statistics, GENSIPS
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Signal processing, Probability, classification accuracy, Bioinformatics, Classification accuracy, Prior probability, Minimax, Class probabilities, Genes, Classifier performance, minimax, Minimax criterion, Neyman-pearson, Separate sampling, Synthetic and real data

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Effective Anomaly Detection Model Training with only Unlabeled Data by Weakly Supervised Learning Techniques

Item Type Conference Paper

Author W. Yang

Author K.-Y. Lam

Editor Gao D.

Editor Li Q.

Editor Guan X.

Editor Liao X.

Abstract Intrusion detection systems (IDS) play an important role in security monitoring to identify anomalous or suspicious activities. Traditional IDS could be signature-based (or rule-based) or anomaly-based (or analytics-based). With the objectives of detecting zero-day attacks, analytics-based IDS have attracted great interest of the cybersecurity community. Furthermore, machine learning (ML) techniques have been extensively explored for advancing analytics-based IDS. Many ML techniques have been studied to improve the efficiency of intrusion detection and some have shown good performance. However, traditional supervised learning algorithms need strong supervision information, fully correctly labeled (FCL) data, to train an accurate model. Whereas, with the rapid development of network and communication technologies, the volume of network traffic and system logs has increased drastically in recent years, especially with the introduction of Next Generation Broadband Network (NGBN) and 5G networks. This caused huge pressure on analytics-based IDS because, for ML to train predictive models, security-relevant data need to be labeled manually, hence leading to practical barriers to achieving effective IDS. In order to avoid being overly dependent on strong supervision information, weakly supervised learning techniques, which utilize incomplete, inexact, or possibly inaccurate labels, have been studied by cybersecurity researchers in that such weak supervision information are easier and cheaper to obtain than FCL data. This research aims to explore the feasibility of weakly supervised learning techniques in IDS tasks so as to reduce the reliance on a massive amount of strong supervision information, which will only continue to grow tremendously in the big data society. We also investigated the detection stability of the proposed scheme when inaccurate weak supervision information is provided. In this article, we propose an IDS model training scheme that is based on a weakly supervised learning algorithm, which requires only unlabeled data. Experiments have been performed on three publicly available IDS evaluation datasets. The results showed that the proposed scheme performs well and is even better than some supervised learning-based IDS (SL-IDS) models. Experimental results also indicated that the weakly supervised learning based IDS model is robust and can be applied in real world situations. Besides, we examined detection performance of the proposed method when it faces class-imbalanced training data and the experiment results show that it performs better than the

compared methods. © 2021, Springer Nature Switzerland AG.

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Cyber security, Network security, Intrusion detection, Learning algorithms, Supervised learning, Machine learning techniques, 5G mobile communication systems, Intrusion Detection Systems, Model training, Anomaly detection, Intrusion-Detection, Unlabeled data, System models, Network traffic, Weakly supervised learning, Class prior, Broadband networks

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Effective classification of the prevalence of *Schistosoma mansoni*

Item Type Journal Article

Author S.A. Mitchell

Author M. Pagano

Abstract Objective To present an effective classification method based on the prevalence of *Schistosoma mansoni* in the community. Methods We created decision rules (defined by cut-offs for number of positive slides), which account for imperfect sensitivity, both with a simple adjustment of fixed sensitivity and with a more complex adjustment of changing sensitivity with prevalence. To reduce screening costs while maintaining accuracy, we propose a pooled classification method. To estimate sensitivity, we use the De Vlas model for worm and egg distributions. We compare

the proposed method with the standard method to investigate differences in efficiency, measured by number of slides read, and accuracy, measured by probability of correct classification. Results Modelling varying sensitivity lowers the lower cut-off more significantly than the upper cut-off, correctly classifying regions as moderate rather than lower, thus receiving life-saving treatment. The classification method goes directly to classification on the basis of positive pools, avoiding having to know sensitivity to estimate prevalence. For model parameter values describing worm and egg distributions among children, the pooled method with 25 slides achieves an expected 89.9% probability of correct classification, whereas the standard method with 50 slides achieves 88.7%. Conclusions Among children, it is more efficient and more accurate to use the pooled method for classification of *S. mansoni* prevalence than the current standard method. © 2012 Blackwell Publishing Ltd.

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Humans, article, nonhuman, Estimation, accuracy, classification, sensitivity analysis, Classification, numerical model, classification algorithm, Child, accuracy assessment, Disease control, estimation method, Animals, Prevalence, Mass Screening, medical decision making, Sensitivity and Specificity, pooled testing, disease prevalence, cost benefit analysis, parasite prevalence, Chemoprevention, Schistosoma mansoni, disease control, Decision Support Techniques, annelid, Anthelmintics, cálculo prevalencia S. mansoni, Cálculo prevalencia S. mansoni, clasificación, Clasificación, control enfermedades, Control enfermedades, Cost Control, egg parasitoid, estimation, host-parasite interaction, Katz-Kato, lutte contre les maladies, Lutte contre les maladies, Parasite Egg Count, Pooled testing, prévalence, Prévalence, pruebas de grupo, Pruebas de grupo, S. mansoni, S. mansoni, S. mansoni prevalence estimation, S. mansoni prevalence estimation, Schistosomiasis mansoni, test poolé, Test poolé, worm

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Effectiveness of E-Learning with fifth-year medical students studying metabolic bone diseases

Item Type Journal Article

Author P. Woratanarat

Author T. Woratanarat

Author T. Woratanarat

Author W. Wajanavisit

Author S. Laohajaroensombat

Author A. Narthanarung

Author P. Aubcherye

Author N. Aubcherye

Author B. Noysang

Abstract Background: E-learning is a tool for enhancing knowledge which can potentially increase medical students' ability to learn and understand complex subjects. Objective: To assess the effectiveness of e-learning in increasing understanding of metabolic bone diseases by medical students during their orthopaedic rotation. Materials and Methods: A retrospective cohort study was conducted at the Department of Orthopaedics, Faculty of Medicine, Ramathibodi Hospital. Fifth-year medical students doing their rotation in the Orthopedics Department between August 2012 and July 2013 were enrolled in the study. All participants were instructed to study metabolic bone diseases using a lesson plan, handouts, and a textbook. They were also given the opportunity to volunteer to participate in an e-learning class prior to attending a one-hour case-based lecture. Students who missed the class were excluded from the present study. Baseline characteristics, in-class scores, multiple choice question [MCQ] grades, and orthopaedic grades of the participants were collected. Associations between study factors and outcomes were analyzed using multiple regression analysis. Results: A total of 154 medical students were included in the study. Their average age was 23 years, and 57% were female. Only 12 (8%) participated in the voluntary e-learning class. The e-learning group had higher in-class scores (9.00 ± 0.95) compared to the non e-learning group (7.70 ± 1.30) with p -value = 0.001. Cumulative GPA and participation in the voluntary e-learning class were factors included in the multiple regression analysis. After adjusting for cumulative GPA, attendance at the e-learning class was significantly associated with in-class scores (coefficient 1.20, 95% confidence interval 0.43, 1.96, p -value = 0.002). Participation in the e-learning class was not significantly correlated with either MCQ grades or orthopaedic grades. Conclusion: E-learning as an adjunct to traditional teaching methods can increase knowledge about metabolic bone disorders among medical students, but it does not affect either MCQ grades or orthopaedic grades. © 2018, Medical Association of Thailand. All rights reserved.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: P. Woratanarat; Department of Orthopaedics, Faculty of Medicine, Ramathibodi Hospital, Mahidol University, Bangkok, 10400, Thailand; email: pataraw@yahoo.com; CODEN: JMTHB

Efficiency of two-phase designs for prevalence estimation

Item Type Journal Article

Author R. McNamee

Abstract Background. Unbiased estimation of the prevalence of diseases and other conditions is important but can be expensive, especially for conditions which do not necessarily lead to contact with health services. A two-phase population survey may seem an attractive option when there is a relatively cheap, although fallible, test for disease status available: the test is used in the first phase of the survey but in the second, only a subsample are classified by the relatively expensive, gold standard. Previously the cost efficiency of such studies compared with simple, one-phase random sample designs was investigated empirically and some questions remain unclear. Methods. A simple formula for the maximum reduction in cost or standard error that can be achieved by two-phase sampling compared with simple random sampling is derived mathematically. A formula for the minimum reduction is also given and the influence of prevalence on efficiency explained. Results. The main result shows that the sensitivity and specificity of the first stage test set an absolute limit on the efficiency of two-phase designs; in particular, two-phase sampling can never be justified on efficiency grounds alone if the test is not accurate enough.

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Efficient data integration under prior probability shift

Item Type Journal Article

Author M.-Y. Huang

Author J. Qin

Author C.-Y. Huang

Abstract Conventional supervised learning usually operates under the premise that data are collected from the same underlying population. However, challenges may arise when integrating new data from different populations, resulting in a phenomenon known as dataset shift. This paper focuses on prior probability shift, where the distribution of the outcome varies across datasets but the conditional distribution of features given

the outcome remains the same. To tackle the challenges posed by such shift, we propose an estimation algorithm that can efficiently combine information from multiple sources. Unlike existing methods that are restricted to discrete outcomes, the proposed approach accommodates both discrete and continuous outcomes. It also handles high-dimensional covariate vectors through variable selection using an adaptive least absolute shrinkage and selection operator penalty, producing efficient estimates that possess the oracle property. Moreover, a novel semiparametric likelihood ratio test is proposed to check the validity of prior probability shift assumptions by embedding the null conditional density function into Neyman's smooth alternatives (Neyman, 1937) and testing study-specific parameters. We demonstrate the effectiveness of our proposed method through extensive simulations and a real data example. The proposed methods serve as a useful addition to the repertoire of tools for dealing dataset shifts. © The Author(s) 2024. Published by Oxford University Press on behalf of The International Biometric Society. All rights reserved.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M.-Y. Huang; Institute of Statistical Science, Academia Sinica, Taipei, 11529, Taiwan; email: myh0728@uw.edu; CODEN: BIOMA

Efficient integration of generative topic models into discriminative classifiers using robust probabilistic kernels

Item Type Journal Article

Author K.E. Ihou

Author N. Bouguila

Author W. Bouachir

Abstract We propose an alternative to the generative classifier that usually models both the class conditionals and class priors separately, and then uses the Bayes theorem to compute the posterior distribution of classes given the training set as a decision boundary. Because SVM (support vector machine) is not a probabilistic framework, it is really difficult to implement a direct posterior distribution-based discriminative classifier. As SVM lacks in full Bayesian analysis, we propose a hybrid (generative–discriminative) technique where the generative topic features from a Bayesian learning are fed to the SVM. The standard latent Dirichlet allocation topic model with its Dirichlet (Dir) prior could be defined as Dir–Dir topic model to characterize the Dirichlet placed on the document and corpus parameters. With very flexible conjugate priors to the multinomials such as generalized Dirichlet (GD) and Beta-Liouville (BL) in our proposed approach, we define two new topic models: the BL–GD and GD–BL. We take advantage of the geometric interpretation of our generative topic (latent) models that associate a K-dimensional manifold (K is the size of the topics) embedded into a V-dimensional feature space (word simplex) where V is the vocabulary size. Under this structure, the low-dimensional topic simplex (the subspace) defines a document as a single point on its manifold and associates each document with a single probability. The SVM, with its kernel trick, performs on these document probabilities in classification where it utilizes the maximum margin learning approach as a decision boundary. The key note is that points or documents that are close to each other on the manifold must belong to the same class. Experimental results with text documents and images show the merits of the proposed framework. © 2020, Springer-Verlag London Ltd., part of Springer Nature.

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Efficient Multimedia Frame-Skipping Architecture Using Deep Learning in Vehicular Networks

Item Type Journal Article

Author U. Ahmed

Author J.C.-W. Lin

Author G. Srivastava

Abstract With the development of 5G networks, vehicle-to-vehicle communication is helping to make travel safe. However, vehicle type detection in the multimedia feed remains a problem. This helps reduce processing time and enable more dynamic connectivity between different vehicles. The development of object classes requires more robust computer vision models and algorithms. However, the main difficulty still lies in image quality, which depends on the lighting conditions, viewing angle, and physical structure of the vehicles. This research mainly focuses on the development and deployment of a deep learning-based system for traffic congestion analysis. The model uses multiple video feeds and vehicle information to detect, classify, and count vehicles in the live traffic feed. The model is trained with a deep learning approach to align the video image and detect the object in top down multimedia. The dynamic skipping method helps to process a long video feed and accurately compares the video image with the viewer. The standard query for the vehicle can help in recognizing and creating the models in real-time traffic situations. The proposed model is suitable for many applications that require a specific area for monitoring real-time data analysis and multimedia routine tasks. © 2022 IEEE.

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Efficient Non-stationary Online Learning by Wavelets with Applications to Online Distribution Shift Adaptation

Item Type Conference Paper**Author** Y.-Y. Qian**Author** P. Zhao**Author** Y.-J. Zhang**Author** M. Sugiyama**Author** Z.-H. Zhou**Editor** Salakhutdinov R.**Editor** Kolter Z.**Editor** Heller K.**Editor** Weller A.**Editor** Oliver N.**Editor** Scarlett J.**Editor** Berkenkamp F.

Abstract Dynamic regret minimization offers a principled way for non-stationary online learning, where the algorithm's performance is evaluated against changing comparators. Prevailing methods often employ a two-layer online ensemble, consisting of a group of base learners with different configurations and a meta learner that combines their outputs. Given the evident computational overhead associated with two-layer algorithms, this paper investigates how to attain optimal dynamic regret without deploying a model ensemble. To this end, we introduce the notion of underlying dynamic regret, a specific form of the general dynamic regret that can encompass many applications of interest. We show that almost optimal dynamic regret can be obtained using a single-layer model alone. This is achieved by an adaptive restart equipped with wavelet detection, wherein a novel streaming wavelet operator is introduced to online update the wavelet coefficients via a carefully designed binary indexed tree. We apply our method to

the online label shift adaptation problem, leading to new algorithms with optimal dynamic regret and significantly improved computation/storage efficiency compared to prior arts. Extensive experiments validate our proposal. Copyright 2024 by the author(s)

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Efficient Prototype Consistency Learning in Semi-Supervised Medical Image Segmentation via Joint Uncertainty and Data Augmentation

Item Type Conference Paper

Author L. Li

Author Y. He

Author C.-M. Pun

Editor Cannataro M.

Editor Zheng H.

Editor Gao L.

Editor Cheng J.
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Editor Zumpano E.
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Editor Park T.

Abstract Recently, prototype learning has emerged in semi-supervised medical image segmentation and achieved remarkable performance. However, the scarcity of labeled data limits the expressiveness of prototypes in previous methods, potentially hindering the complete representation of prototypes for class embedding. To overcome this issue, we propose an efficient prototype consistency learning via joint uncertainty quantification and data augmentation (EPCL-JUDA) to enhance the semantic expression of prototypes based on the framework of Mean-Teacher. The concatenation of original and augmented labeled data is fed into student network to generate expressive prototypes. Then, a joint uncertainty quantification method is devised to optimize pseudo-labels and generate reliable prototypes for original and augmented unlabeled data separately. High-quality global prototypes for each class are formed by fusing labeled and unlabeled prototypes, which are utilized to generate prototype-to-features to conduct consistency learning. Notably, a prototype network is proposed to reduce high memory requirements brought by the introduction of augmented data. Extensive experiments on Left Atrium, Pancreas-NIH, Type B Aortic Dissection datasets demonstrate EPCL-JUDA's superiority over previous state-of-the-art approaches, confirming the effectiveness of our framework. The code will be released soon. © 2024 IEEE.

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Efficient quantification on large-scale networks

Item Type Journal Article

Author A. Micheli

Author A. Moreo

Author M. Podda

Author F. Sebastiani

Author W. Simoni

Author D. Tortorella

Abstract Network quantification (NQ) is the problem of estimating the proportions of nodes belonging to each class in subsets of unlabelled graph nodes. When prior probability shift is at play, this task cannot be effectively addressed by first classifying the nodes and then counting the class predictions. In addition, unlike non-relational quantification, NQ demands enhanced flexibility in order to capture a broad range of connectivity patterns, resilience to the challenge of heterophily, and scalability to large networks. In order to meet these stringent requirements, we introduce XNQ, a novel method that synergizes the flexibility and efficiency of the unsupervised node embeddings computed by randomized recursive Graph Neural Networks, with an Expectation-Maximization algorithm that provides a robust quantification-aware adjustment to the output probabilities of a calibrated node classifier. In an extensive evaluation, in which we also validate the design choices underpinning XNQ through comprehensive ablation experiments, we find that XNQ consistently and significantly improves on the best network quantification methods to date, thereby setting the new state of the art for this challenging task. XNQ also provides a training speed-up of up to 10x–100x over other methods based on graph learning. © The Author(s) 2025.

Date 2025

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Efficient quantifying track structure cracks using deep learning

Item Type Journal Article

Author H. Sun

Author L. Song

Author Z. Yu

Abstract High-speed railway ballastless track structure crack detection usually has a high demand for the efficiency of crack detection technology. To overcome the limitation that current crack quantification methods usually require multiple steps, this paper proposes an efficient quantification method for track structure cracks using deep learning. This method applies the deep neural network (DNN) to the direct prediction of crack severity index values by modifying DNNs used for image classification. This method adopts a deep learning-based multi-step crack quantification method to calculate crack severity index values, establishes a dataset for predicting track structure interlayer crack severity index values using crack width mean values as labels, establishes a dataset for predicting track structure complex crack severity

index values using crack width mean values and crack area values as labels, and utilizes the established datasets to train the modified DNNs. This method crops the track structure panorama in spatial order to obtain images, which not only facilitates DNN prediction but also enables the acquisition of more information such as crack distribution. Under the condition of using the data enhancement method, the mean absolute errors (MAEs) of the prediction results of the trained DNNs under the corresponding testing sets are 0.0191 and 0.0183, and the prediction results are in good agreement with the reference values. The image processing rates of the trained DNNs under the corresponding testing sets are all close to 75 images per second (resolution 512×512), which are 8.57 and 13.93 times as computationally efficient as the adopted deep learning-based multi-step crack quantification method. © 2025 The Author(s). Computer-Aided Civil and Infrastructure Engineering published by Wiley Periodicals LLC on behalf of Editor.

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Deep neural networks, machine learning, Image enhancement, quantitative analysis, structural analysis, prediction, Railroad transportation, artificial neural network, Quantification methods, Image acquisition, Multisteps, crack, Crack severity, Crack-width, High-speed railways, Index values, Mean values, Railroad tracks, Railroads, railway, Severity index, Testing sets, Track structure

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Efficient Uncertainty Quantification for Multilabel Text Classification

Item Type Conference Paper

Author J. Yu

Author A.I. Cristea

Author A. Harit

Author Z. Sun

Author O.T. Aduragba

Author L. Shi

Author N.A. Moubayed

Abstract Despite rapid advances of modern artificial intelligence (AI), there is a growing concern regarding its capacity to be explainable, transparent, and accountable. One crucial step towards such AI systems involves reliable and efficient uncertainty quantification methods. Existing approaches to uncertainty quantification in natural language processing (NLP) take a Bayesian Deep Learning approach. However, the latter is known to not be computationally efficient in testing time, thus hindering its applicability in real-life scenarios. This paper proposes a new focus on the efficiency of uncertainty quantification methods, evaluating them on four multi-label text classification tasks. Our novel methods of representing epistemic and aleatoric uncertainties enable efficient uncertainty quantification (around 13 to 45 times faster than existing approaches, depending on architecture) with posterior analysis in the (approximated) latent- and data space. We conduct extensive experiments and studies on diverse neural network architectures (LSTM, CNN and Transformer) to analyse their power. Our results prove the benefits of explicitly modelling uncertainty in neural networks. © 2022 IEEE.

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Deep learning, Classification (of information), Network architecture, Uncertainty analysis, Language processing, Natural language processing, Natural language processing systems, Natural languages, Artificial intelligence systems, Neural-networks, deep learning, Long short-term memory, Bayesian networks, natural language processing, neural network, Text processing, Multi-label text classification, Quantification methods, Uncertainty quantifications, Bayesian methods, uncertainty quantification, Bayesian method

Notes:

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Export Date: 20 September 2026; Cited By: 6; Correspondence Address: J. Yu; Durham University, Department of Computer Science, Durham, United Kingdom; email: jialin.yu@durham.ac.uk; Conference name: 2022 International Joint Conference on Neural Networks, IJCNN 2022 at the IEEE World Congress on Computational Intelligence, WCCI 2022; Conference date: 18 July 2022 through 23 July 2022; Conference code: 183333; CODEN: 85OFA

Efficient Video Object Classification Using Class Prior Adaptation Under Continuous Class-Balance Changes

Item Type Conference Paper

Author Y. Ouyang

Author S. Mizutani

Author Y. Kishino

Editor Yasumoto K.

Editor Keidar I.

Editor Yamaguchi H.

Editor Silvestri S.

Editor Gramoli V.

Abstract Video object classification is a fundamental task with broad applications such as anomaly detection in video streams and wildlife monitoring via camera traps. In practice, video classification is often decomposed into per-frame image classification. However, a major challenge arises from non-stationarity: class priors often vary over time, and a mismatch between training and test priors leads to performance degradation. Previous works have addressed this issue through runtime adaptation, where they exploit the temporal locality of the class occurrence and detect change points. One applies Bayesian filtering for adaptation against class prior change, while another pre-trains lightweight models under different priors and switches among them dynamically. Inspired by these approaches, we focus on class-balance changes over time. At a short timescale, class priors exhibit non-stationarity (e.g., different animals appear at different times of the day), while at a longer timescale, priors can show strong periodicity (e.g., seasonal animal activities). Unlike previous works based on change point detection, our approach leverages training data to predict priors in advance and dynamically adjusts the output of the original classifier at test time through a lightweight prior adaptation framework. Experiments on the wildlife ecological

dataset show that our prior adaptation effectively improves classification accuracy under time-dependent class-balance change. This lightweight and effective adaptation scheme is particularly suited to edge computing environments with constrained resources. © 2026 Copyright held by the owner/author(s).

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Egocentric Online Action Segmentation via Evidential Temporal Contextualization

Item Type Journal Article

Author X. Jiang

Author X. Xu

Author Z. Han

Author Z. Wang

Author J. Song

Author H.T. Shen

Abstract The Egocentric Online Action Segmentation (EOAS) task aims to temporally segment a streaming egocentric video into a set of actions. Due to the first-person perspective views and online constraints, the EOAS task raised two intrinsic difficulties: (1) The shot movements could impact the egocentric vision and make it difficult to observe overall activities. (2) The visual signals are processed in real-time, requiring models to understand the observed visual frame without any post-contextual cues after the observed frame. To address these two problems, we proposed a novel EOAS approach named Evidential Temporal Contextualization (ETC), which consists of two key modules: (1) Context-Aggregated TRansformer that predicts and aggregates potential post context to enhance representation of frame-level features, effectively addressing the insufficient temporal contextualization in streaming videos. (2) Evidential Action Contextualization that leverages evidential deep learning to quantify the uncertainty in frame-level classification and dynamically adjust the model training and inference process accordingly. Extensive experiments on three public EOAS benchmark datasets, i.e., GTEA, EgoProceL, and EgoPER, demonstrate our proposed ETC approach outperforms recent state-of-the-art methods. Additionally, we also conduct more ablation studies and generalize our proposed ETC method to exocentric videos, furtherly prove its superiority. © 1999-2012 IEEE.

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Elastic net-based cost-sensitive broad learning system with F-measure maximization and density harmonization

Item Type Journal Article

Author Y. Li

Author Y. Zhang

Author Y. Gao

Author Y. Wang

Author J. Jin

Author X. Sun

Author C.L.P. Chen

Abstract The broad learning system (BLS) is a shallow network architecture commonly adopted for classification and regression tasks. However, BLS and its variants struggle with imbalanced classification due to architectural redundancy from random feature projection and the lack of mechanisms for differentiating sample importance across classes, typically resulting in predictions biased toward majority classes. To overcome these challenges, we propose an elastic net-based cost-sensitive BLS with F-measure maximization and density harmonization (FDECBLs). This framework comprises two core innovations: (1) an elastic net regularization applied to singular values to mitigate redundancy and improve discriminative ability; (2) a dual-cost learning mechanism that combines a density-based cost scheme, which dynamically adjusts sample importance based on local neighborhood structures and global class imbalance information, with a penalty mechanism derived from F-measure maximization to impose class-specific costs for minority-class prioritization. The associated optimization problem is solved efficiently via the alternating direction method of multipliers, with theoretical convergence guarantees and empirically verified computational efficiency. Extensive experiments on 35 binary-class and 9 multi-class imbalanced datasets demonstrate the effectiveness and efficiency of FDECBLs. It attains leading average scores of 90.69% in F-measure, 92.77% in G-mean, and 93.24% in AUC on binary tasks, while consistently surpassing state-of-the-art baselines in multi-class settings. The code is available at <https://github.com/AdamMing/FDECBLs>. © 2026 Elsevier Ltd

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Electrostatic and Capillary Force-Enhanced Dynamic SERS Combined with Machine Learning for Rapid Quantification of Structurally Similar Triazole Pesticides in Tobacco

Item Type Journal Article**Author** X. He**Author** S. Chao**Author** S. Zhou**Author** R. Zhao**Author** C. Valsecchi**Author** Y. Lin**Author** H. Yang**Author** M. Fan

Abstract Rapid and accurate detection of pesticide residues is crucial for safeguarding both environmental and human health. Nonetheless, the quantitative detection of structurally similar pesticides, which are frequently employed concurrently in agricultural practices to manage a range of pests and diseases, presents significant challenges. In this study, we introduced a methodology that integrates dynamic surface-enhanced Raman spectroscopy (D-SERS) with machine learning to quantify the binary mixtures of triadimenol and triadimefon. Initially, halogen-modified nanoparticles (Ag-Cl NPs) were synthesized to serve as SERS substrates, and the D-SERS method identified an optimal hot spot duration of approximately 60 s through the combined effects of electrostatic and capillary forces. The SERS spectra acquired

at this hot spot exhibited high stability (relative deviation < 9.67%), enabling sensitive detection, with limits of detection (LODs) of 74 ppb for triadimenol and 68 ppb for triadimefon. Using these spectra, a PCA-SVM model was developed and trained to accurately quantify both pesticides, individually and in mixtures, in tobacco samples. This integrated D-SERS and machine learning platform provides a rapid, sensitive, and reliable analytical method for the simultaneous quantification of structurally similar multicomponent pesticides, offering a promising solution for addressing pesticide residue issues in agricultural products. © 2025 American Chemical Society.

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ELSA: Efficient Label Shift Adaptation through the Lens of Semiparametric Models

Item Type Conference Paper

Author Q. Tian

Author X. Zhang

Author J. Zhao
Editor Krause A.
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Editor Scarlett J.

Abstract We study the domain adaptation problem with label shift in this work. Under the label shift context, the marginal distribution of the label varies across the training and testing datasets, while the conditional distribution of features given the label is the same. Traditional label shift adaptation methods either suffer from large estimation errors or require cumbersome post-prediction calibrations. To address these issues, we first propose a moment-matching framework for adapting the label shift based on the geometry of the influence function. Under such a framework, we propose a novel method named Efficient Label Shift Adaptation (ELSA), in which the adaptation weights can be estimated by solving linear systems. Theoretically, the ELSA estimator is $\sqrt{n}$ -consistent (n is the sample size of the source data) and asymptotically normal. Empirically, we show that ELSA can achieve state-of-the-art estimation performances without post-prediction calibrations, thus, gaining computational efficiency. © 2023 Proceedings of Machine Learning Research. All rights reserved.

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Emissions Optimization Platform to Quantify and Reduce Rig-Based Emissions

Item Type Conference Paper

Author N. Minhas

Author A. Abdelmoaty

Author S. Al Dakhil

Author S. Alsemail

Author J. Razzak

Author M. Jenkins

Abstract Reducing emissions at each step of the Oil & Gas Value chain is needed to help meet the global goal of keeping the temperature increase below 1.5C° of preindustrial level. This paper presents a method to calculate emissions and optimize operations to reduce emissions at the well planning stage. Three specific rigs were chosen and the data from their operations and the diesel consumption over the last 18 months (about 1 and a half years) was used to build a model that links specific operation on the rig with its associated emissions. Part of the data was used for model building while the rest was used for model verification. The energy consumption of each piece of equipment on the rig was also used in the calculations to document their wattage. An optimization tool is developed to help optimize rig operations for emissions reduction by choosing the proper rig and the rest of the equipment. The diesel usage data was lower resolution (one data points per day) compared to the operations (hourly). So, the model development was particularly challenging. It was found that there is considerable room for energy optimization especially with regards to the use of diesel generators. A considerable amount of excess energy is produced which gets wasted. Also, the cementing and casing has the greatest impact on the emissions as both materials emit a huge amount of carbon. Optimization of the emissions on the well planning stage is much more effective than emission reduction strategies post planning as the room for optimization is significantly less. Our approach is based on state-of-the-art deep reinforcement learning (DRL) and data science techniques that have been proven to work on complex systems that have different inputs. DRL trains an algorithm with a reward system that supplies positive and negative feedback (degree of emissions) to an Artificial Intelligence (AI) agent for performing actions. These actions are the various strategies in rig operations the agent takes to reduce emissions, the lower the emissions the higher the rewards for the agent. In iterative process the agent narrows down the best policy (plan) for each rig that lowers the emission. The use of data science and machine learning in well planning for emissions reduction is a novel approach with expert feedback being constant in our methods. Using DRL (deep reinforcement learning) to quantify emissions in well planning is a revolutionary technique. It is the first time in the industry that DRL is being used to assist in emissions

reductions from upstream activities. We bring technology that has redefined the capabilities of AI in a variety of applications, including reservoir optimization and complex decision-making systems. © 2023, Society of Petroleum Engineers.

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Employing machine learning to quantify long-term climatological and regulatory impacts on groundwater availability in intensively irrigated regions

Item Type Journal Article

Author S. Nozari

Author R.T. Bailey

Author E.M.K. Haacker

Author Z.T. Zambreski

Author Z. Xiang

Author X. Lin

Abstract The steady overexploitation of the Ogallala Aquifer underlying the U.S. High Plains Region has put irrigated crop production at risk, particularly in the Southern and Central High Plains. To manage this issue properly, a data-driven modeling framework is developed and tested that is fast to employ and yet provides reliable

long-term groundwater level (GWL) forecasts as a function of climatological and anthropogenic factors. The modeling framework uses the random forests (RF) technique in combination with ordinary kriging, and is tested in Finney County in southwest Kansas. The introduction of groundwater withdrawal potential as a new surrogate for pumping intensity enables the RF model to capture decline in groundwater depletion rate as the system progresses towards aquifer depletion and/or as a result of well retirement policies. The RF model is executed from 2017 to 2099 for 20 different downscaled global climate models (GCMs) for the two representative concentration pathways (RCP) scenarios of 4.5 and 8.5. The results show the aquifer will cease to support irrigated agriculture in most of the county by 2060 under status quo management and average climate conditions. Moreover, climate will likely shift the aquifer's depletion time frame by 15 years or less in most of the study area. The long-term combined impact of well retirement plans and climate conditions on groundwater depletion trends imply well retirement policies do not lead to sustained groundwater savings. This study demonstrates the capacity of machine learning models to serve as a rapid assessment tool, informing policymakers about future groundwater availability in intensively irrigated regions and under different climate and management conditions. © 2022 Elsevier B.V.

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End-To-End Uncertainty Quantification with Analytical Derivatives for Design Under Uncertainty

Item Type Conference Paper

Author B.D. Phillips

Author J. Schmidt

Author R.D. Falck

Author D. Aretskin-Hariton

Abstract Uncertainty quantification (UQ) is a rapidly growing and evolving discipline, especially within the aerospace community. Performing analysis with UQ can provide decision makers with a wealth of information about a candidate design. However, the value of UQ is fully realized when the information gained during UQ analysis is leveraged in a feedback loop of a design optimization process, often referred to as design under uncertainty. Although design under uncertainty can be a powerful risk mitigation technique, there are a number of roadblocks that prevent its implementation. Two primary factors are computational costs and added complexity of the analysis. High fidelity simulations having on the order of tens of uncertain variables quickly become computationally infeasible. Also, implementing UQ into an existing multidisciplinary design and optimization (MDO) process often requires extensive knowledge of the UQ methods and careful treatment of the problem formulation. The objective of this work is to address these two primary roadblocks and enable practitioners to efficiently perform design under uncertainty with limited knowledge of the UQ discipline. Methods outlined in this paper demonstrate MDO incorporating UQ into the design process, leveraging an analytical derivative tool chain through the entire optimization. The proposed approach leverages machine learning techniques to generate a differentiable confidence interval output from polynomial chaos models. This technique, coupled with the incorporation of analytical derivatives through the Polynomial Chaos Expansion (PCE) process, eliminates the need to estimate derivatives, which are usually obtained from finite difference, complex step, or similar methods. Developing a differentiable confidence interval allows mixed uncertainty problems (both epistemic and aleatory) to be modeled. Without such modeling, these problems cannot accurately predict objective functions containing statistical quantities such as mean and variance. The addition of analytical derivatives to a UQ method based on polynomial chaos can decrease the computational costs of performing design under uncertainty by orders of magnitude in comparison with methods such as complex step. The method and codes developed are modular in nature and are a drop-in solution for design under uncertainty within existing MDO problems. Low-fidelity analytical multidisciplinary optimizations under uncertainty for a wing design in OpenMDAO are detailed in this paper. These demonstration cases includes both objective functions and constraints which are influenced by uncertain parameters. Results show an 87.7% reduction in computational costs compared to current methods. Comparison of deterministic and uncertain designs shows significant improvement for the quantity of interest when the optimizer is given information about the uncertain space. © 2024, American Institute of Aeronautics and Astronautics Inc, AIAA. All rights reserved.

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Enhanced classification based on probabilistic extreme learning machine in wastewater treatment process

Item Type Conference Paper

Author L.J. Zhao

Author X.K. Diao

Author D.C. Yuan

Author W. Tang

Abstract A binary classification method, Probabilistic Extreme Learning Machine (called P-ELM), is proposed to enhance the reliability of the classification of an unknown object. P-ELM method integrates ELM, density methods and Bayesian decision theory in order to take into account a priori probability of the process and the uncertainty of the ELM predictions. The P-ELM algorithm may inhibit uncertainty of the extreme learning machine prediction in the different trials of simulation due to the initialization of input weights and bias, which would damage the reliability of the classification for the new objects. Simulations results from a municipal wastewater treatment plant show that the proposed P-ELM binary classification method can provide the reliability of the classification for those samples near the boundaries of the classes and the reliability and accuracy outperform the ELM model. © 2011 Published by Elsevier Ltd.

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Reliability, Learning systems, Decision theory, Information science, Binary classification, Extreme learning machine, Wastewater, Wastewater treatment, A-priori probabilities, Bayesian decision, Bayesian decision theory, Binary classification methods, Computer simulation, Control theory, Density methods, ELM, Input weights, Municipal wastewater treatment plants, Probability ELM, Unknown objects, Wastewater treatment process

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Enhanced fusion prediction of composite material performance degradation via interval processes and machine learning

Item Type Journal Article

Author Z. Ding
Author X. Liang
Author X. Wu
Author H. Ning
Author D. Sun
Author A. Lee
Author F. Liu
Author S. Zhou
Author L. Zhao

Author N. Hu

Abstract Predicting the performance degradation of composite materials is of paramount importance for ensuring the safe operation of composite engineering structures. Traditional experimental methods involve protracted, costly accelerated ageing tests, while theoretical methods also subject to their own inherent limitations. Physics-based methods face parameter-tuning difficulties and incomplete degradation descriptions, while data-driven approaches are data-dependent and may have low accuracy. To overcome these challenges, this paper presents an innovative methodology that integrates interval degradation progress models with Bi-LSTM (Bidirectional Long Short-Term Memory) neural networks to enhance prediction performance. Specifically, the Grey System Theory (GST) is employed to quantify the interval process. By combining GST with physical information, a novel quantification method for the interval degradation process model is proposed. This model serves as a reference framework, enabling the mapping of experimental degradation data. The Bi-LSTM network then harnesses this nonlinear mapping to forecast the degradation trajectory of composite materials, leveraging its powerful sequential learning capabilities. The accuracy of the proposed method is verified using two different experimental datasets: stiffness degradation under hygrothermal conditions and fatigue damage under cyclic stress. The results clearly demonstrate that this fusion approach outperforms existing methods in terms of predictive accuracy, indicating its great potential for practical applications in the field of composite material performance degradation prediction. © 2026 Elsevier Ltd.

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Enhanced fuzzy Gaussian networks for sputum image based Mycobacterium detection

Item Type Journal Article

Author K.S. Mithra

Abstract The primary step involved in the diagnosis of Mycobacterium tuberculosis is the Sputum Smear Microscopy. The limitations of manual detection can be avoided with an automated technique which is carried out in this study. This paper uses the Enhanced Fuzzy Gaussian Networks (EFGN) for the detection of TB, by integrating the Gaussian model with the fuzzy and the neural network. In EFGN, the classified output obtained from the enhanced fuzzy and neural network is combined together depending on the Gaussian mixture model. The sputum smear microscopic image acts as the input, to which the process of thresholding is imposed to get the segmented result. Local Gradient Pattern (LGP), length, density, area, and histogram features are utilized to classify and count the bacilli objects using EFGN classifier. The performance is estimated based on the factors, such as Segmentation Accuracy (SA), and Mean Squared Error (MSE). © 2023 The Royal Photographic Society.

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: K.S. Mithra; Department of Computer Science and Applications, S. T. Hindu College, Nagercoil, Tamil Nadu, India; email: ksmithra1@gmail.com

Enhanced probabilistic classify and count methods for multi-label text quantification

Item Type Conference Paper

Author R. Levin

Author H. Roitman

Abstract In this work we address the problem of Multi-Label Text Quantification. To this end, for a given collection of documents, each was pre-classified with one or more labels by some multi-label classifier, our goal is to find an estimate of the cardinality of each actual label set, as accurate as possible. We present two enhanced Probabilistic Classify and Count (PCC) methods that focus on improving the quantification accuracy by employing another supervised learning phase. Using a real-world multi-label documents dataset, we report on an experimental evaluation that compares the estimated label counts produced by our solution (and several alternatives) to the actual label counts derived from labels assigned by human experts. Our results confirm that, using our solution, the quantification accuracy can be significantly improved. © 2017 Copyright held by the owner/author(s).

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Enhancement Technique Based on the Breast Density Level for Mammogram for Computer-Aided Diagnosis

Item Type Journal Article**Author** N.F. Razali**Author** I.S. Isa**Author** S.N. Sulaiman**Author** N.K. Abdul Karim**Author** M.K. Osman**Author** Z.H. Che Soh

Abstract Mass detection in mammograms has a limited approach to the presence of a mass in overlapping denser fibroglandular breast regions. In addition, various breast density levels could decrease the learning system's ability to extract sufficient feature descriptors and may result in lower accuracy performance. Therefore, this study is proposing a textural-based image enhancement technique named Spatial-based Breast Density Enhancement for Mass Detection (SbBDEM) to boost textural features of the overlapped mass region based on the breast density level. This approach determines the optimal exposure threshold of the images' lower contrast limit and optimizes the parameters by selecting the best intensity factor guided by the best Blind/Reference-less Image Spatial Quality Evaluator (BRISQUE) scores separately for both dense and non-dense breast classes prior to training. Meanwhile, a modified You Only Look Once v3 (YOLOv3) architecture is employed for mass detection by specifically assigning an extra number of higher-valued anchor boxes to the shallower detection head using the enhanced image. The experimental results show that the use of SbBDEM prior to training mass detection promotes superior performance with an increase in mean Average Precision (mAP) of 17.24% improvement over the non-enhanced trained image for mass detection, mass segmentation of 94.41% accuracy, and 96% accuracy for benign and malignant mass classification. Enhancing the mammogram images based on breast density is proven to increase the overall system's performance and can aid in an improved clinical diagnosis process. © 2023 by the authors.

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deep learning, breast cancer, breast density, CAD, image enhancement, textural

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Enhancements of Pass Play Quantification Method with Geometric Features of Formations

Item Type Conference Paper**Author** T. Mimura**Author** Y. Nakada

Abstract In the last two decades, the practical use of player position data has attracted a great deal of attention for analyzing and visualizing plays and tactics in many team sports including soccer, American football, basketball, and rugby. With this background, we proposed a machine learning-based method to quantify pass plays via multiple geometric feature values of formations computed from player position data. This method was successfully applied against a real soccer dataset, but there has still been room for enhancement. In this study, we propose two enhancements of the pass play quantification method to overcome its limitations. The first enhancement is an improvement in modeling by adding pass play feature values as inputs. The improved models are also validated by using a larger real soccer dataset than in our previous work. The second enhancement is related to the visualization for quantification results of the effectiveness and risk for potential pass plays from the detected ball possessor. This visualization is also applied to a real soccer dataset using our visualization tool. © 2022 IEEE.

Date 2022**DOI** 10.1109/ICEET56468.2022.10007279**URL** <https://www.scopus.com/pages/publications/85146492295?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Int. Conf. Eng. Emerg. Technol., ICEET**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 978-166549106-8 (ISBN)**Proceedings Title** Int. Conf. Eng. Emerg. Technol., ICEET**Conference Name** 8th International Conference on Engineering and Emerging Technologies, ICEET 2022**Date Added** 20/09/2026, 9:52:34 pm**Modified** 20/09/2026, 9:52:34 pm**Tags:**

Geometry, Visualization, Data visualization, Sports, Feature values, Formation analyse, formation analysis, Gaussian process classification, Gaussian process classifications, Geometric feature, Geometric feature value, geometric feature values, player position data, Player position data, Position data, Sport data analyse, Sport data visualization, Sports data, sports data analysis, sports data visualization

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Enhancing amide proton transfer imaging in ischemic stroke using a machine learning approach with partially synthetic data

Item Type Journal Article**Author** M. Viswanathan**Author** L. Yin**Author** Y. Kurmi**Author** A. Afzal**Author** Z. Zu

Abstract Amide proton transfer (APT) imaging, a technique sensitive to tissue pH, holds promise in the diagnosis of ischemic stroke. Achieving accurate and rapid APT imaging is crucial for this application. However, conventional APT quantification methods either lack accuracy or are time-consuming. Machine learning (ML) has recently been recognized as a potential solution to improve APT quantification. In this paper, we applied an ML model trained on a new type of partially synthetic data,

along with an optimization approach utilizing recursive feature elimination, to predict APT imaging in an animal stroke model. This partially synthetic datum is not a simple blend of measured and simulated chemical exchange saturation transfer (CEST) signals. Rather, it integrates the underlying components including all CEST, direct water saturation, and magnetization transfer effects partly derived from measurements and simulations to reconstruct the CEST signals using an inverse summation relationship. Training with partially synthetic data requires less in vivo data compared to training entirely with fully synthetic or in vivo data, making it a more practical approach. Since this type of data closely resembles real tissue, it leads to more accurate predictions than ML models trained on fully synthetic data. Results indicate that an ML model trained on this partially synthetic data can successfully predict the APT effect with enhanced accuracy, providing significant contrast between stroke lesions and normal tissues, thus clearly delineating lesions. In contrast, conventional quantification methods such as the asymmetric analysis method, three-point method, and multiple-pool model Lorentzian fit showed inadequate accuracy in quantifying the APT effect. Moreover, ML methods trained using in vivo data and fully synthetic data exhibited poor predictive performance due to insufficient training data and inaccurate simulation pool settings or parameter ranges, respectively. Following optimization, only 13 frequency offsets were selected from the initial 69, resulting in significantly reduced scan time. © 2024 The Author(s). NMR in Biomedicine published by John Wiley & Sons Ltd.

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Enhancing Anomaly Alert Prioritization Through Calibrated Standard Deviation Uncertainty Estimation With an Ensemble of Auto-Encoders

Item Type Journal Article

Author J.F. Masakuna

Author D.K. Nkashama

Author A. Soltani

Author M. Frappier

Author P.-M. Tardif

Author F. Kabanza

Abstract Deep auto-encoders (AEs) are widely employed deep learning methods in the field of anomaly detection across diverse domains (e.g., cybersecurity analysts managing large volumes of alerts, or medical practitioners monitoring irregular patient signals). In such contexts, practitioners often face challenges of scale and limited processing resources. To cope, strategies such as false positive reduction, human-in-the-loop review, and alert prioritization are commonly adopted. This paper explores the integration of uncertainty quantification (UQ) methods into alert prioritization for anomaly detection using ensembles of AEs. UQ models highlight doubtful classification decisions, enabling analysts to address the most certain alerts first, since higher certainty typically correlates with greater accuracy. Our study reveals a nuanced issue where applying UQ to ensembles of AEs can produce skewed distributions of large reconstruction errors (errors exceeding a pre-defined threshold), which may falsely suggest high uncertainty when standard deviation is used as the

metric. Conventionally, a high standard deviation indicates high uncertainty. However, contrary to intuition, large reconstruction errors often reflect AE is strongly confident that an input is anomalous—not uncertainty about it. Moreover, ensembles of AEs generate reconstruction errors with varying ranges, complicating interpretation. To address this, we propose an extension that calibrates the standard deviation distribution of uncertainties, mitigating erroneous prioritization. Evaluation on 10 benchmark datasets demonstrates that our calibration approach improves the effectiveness of UQ methods in prioritizing alerts, while maintaining favorable trade-offs across other key performance metrics. © 2004-2012 IEEE.

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Enhancing decision support in crop production: Analyzing conformal prediction for uncertainty quantification

Item Type Journal Article

Author M. Farag

Author A. Emam

Author J. Leonhardt

Author R. Roscher

Abstract Assessing the confidence of machine learning models is increasingly important for enhancing decision support in digital agriculture. Addressing this challenge involves identifying and classifying sources of uncertainty and applying a suitable quantification method. To fully realize the potential of uncertainty quantification in digital agriculture, it is necessary to explore and evaluate different techniques, demonstrating their effectiveness in diverse scenarios. In this paper, we focus on the conformal prediction (CP) framework, specifically inductive conformal prediction (ICP), as a non-parametric tool for uncertainty quantification. Conformal prediction has gained a foothold for its multiple advantages, including being model-agnostic – requiring no retraining or changes to model architecture – and computationally efficient. Moreover, CP operates under a distribution-free framework, making no assumptions about the data, and provides calibrated uncertainty estimates where empirical errors match pre-defined theoretical levels. These properties make it particularly well-suited for agricultural tasks, where datasets often exhibit variability across regions and lack reliable ground truth labels. Through ablation studies, we comprehensively analyze ICP's performance as a post-hoc approach for ResNet-18 and ViT-B/16, demonstrating that both achieve the required pre-defined error levels. We compare ICP performance against other methods in diverse agricultural tasks, including out-of-distribution detection and covariate shift. Our experiments highlight CP's unique benefits, such as validity, efficiency, and low computational overhead, making it a promising approach for developing more reliable agricultural machine learning systems. © 2025 The Authors

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Enhancing Email Security Through Accurate Phishing Detection Using Deep Transformer Models

Item Type Conference Paper

Author S. Irfan

Editor Kumar R.

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Editor Gupta M.

Abstract Phishing is also a significant issue in cybersecurity, and it needs innovative solutions to make it detected and prevented instantly. The development of email protection infrastructure with the help of ML and DL algorithms has taken the lead in combating phishing. This paper proposes an effective phishing email detecting system based on the DistilBERT-base-uncased model, which is a more computationally efficient version of BERT, yet highly accurate. The Kaggle phishing email data set has undergone several operations including cleaning, one-hot encoding and TF-IDF vectorization and feature scaling. The data was balanced to address the imbalance of classes prior to the model training. The proposed model has already shown important results with a testing accuracy of 98.45, precision of 97.67, recall of 100 and F1score of 98.61, which are superior to already available traditional ML (DT, NB, LR, RF, AdaBoost, NB+KNN) and DL (CNN, LSTM, NN, MLP, RoBERTA) models. These findings indicate that the model is highly contextual, generalizable, and resistant to phishing detection errors. The research reveals that a lightweight transformer is able to offer optimum performance in regards to accuracy and also is power efficient thereby making possibility of scalable and real-time e-mail security solutions. © 2026 IEEE.

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Enhancing evolutionary fuzzy systems for multi-class problems: Distance-based relative competence weighting with truncated confidences (DRCW-TC)

Item Type Journal Article

Author A. Fernández

Author M. Elkano

Author M. Galar

Author J.A. Sanz

Author S. Alshomrani

Author H. Bustince

Author F. Herrera

Abstract Classification problems with multiple classes suppose a challenge in Data Mining tasks. There is a difficulty inherent to the learning process when trying to find the most adequate discrimination functions among the different concepts within the dataset. Using Fuzzy Rule Based Classification Systems in general, and Evolutionary Fuzzy Systems in particular, provide the advantage of describing smoother borderline areas, thanks to the linguistic label-based representation. In multi-classification, the pairwise learning approach (One-vs-One) has gained a notorious attention. However, there is certain dependence between the goodness of the confidence degrees or scores of binary classifiers, and the final performance shown by the global model. Regarding this fact, the problem of non-competent classifiers is of special relevance. It occurs when a binary classifier outputs a positive score for a couple of classes unrelated with the input example, which may degrade the final accuracy. Precisely, the previously exposed properties of fuzzy classifiers make them more prone to the former condition. In this paper, we propose an extension of the distance-based combination strategy to overcome this non-competence problem. It is based on the truncation of

the confidence degrees of the classes prior to the distance-based tuning. This allows taking advantage of the good classification abilities of Evolutionary Fuzzy Systems, while diminishing the adverse effect of the aforementioned non-competence. Experimental results, using FARC-HD with overlap functions as the fuzzy learning algorithm, show that this new adaptation of the Distance-based Relative Competence Weighting model outperforms both the OVO and standard distance-based approaches, and it is competitive with robust classifiers such as Support Vector Machines. © 2016 Elsevier Inc. All rights reserved.

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Enhancing information standards for automated construction waste quantification and classification

Item Type Journal Article

Author S. Sivashanmugam

Author S. Rodriguez

Author F. Pour Rahimian

Author F. Elghaish

Author N. Dawood

Abstract Accurate quantification and detailed classification of construction waste are paramount to improving their management. Over the last decades, various quantification models have been developed to measure, manage, and report construction waste generation. A detailed understanding of those models is essential to explore their applications across the life-cycle stages of a built asset. Existing reviews primarily focused on analysing the functions of quantification methodologies, but the digital and information standards to automate the quantification process are under-explored in the existing literature. A review is adopted to analyse papers published from 2012 to 2022. Out of 279 articles retrieved, 71 papers meeting the eligibility criteria were included. A critical analysis of the models indicates that unified data structure, standard information, life-cycle approach and interoperability between the BIM and waste knowledge bases are vital to automate and reinforce the quantification efficiency. Based on the review findings, a conceptual framework is developed to demonstrate the quantification workflow for building projects. The outcomes from the review will facilitate researchers to identify the prevailing gaps and enhance the waste quantification system to meet digital construction demands. © 2023

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Enhancing NLU in Large Language Models Using Adversarial Noisy Instruction Tuning

Item Type Conference Paper

Author S. Bai

Author Q. Li

Author Z. Wang

Author N. Zhou

Author N. Yao

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Editor Shah J.

Editor Kolter Z.

Abstract Instruction tuning has emerged as an effective approach that notably improves large language models (LLMs) performance, showing particular promise in natural language generation tasks by producing more diverse, coherent, and task-relevant outputs. However, extending instruction tuning to natural language understanding (NLU) tasks presents significant challenges, primarily due to the difficulty in achieving high-precision responses and the scarcity of large-scale, high-quality instruction data necessary for effective tuning. In this work, we introduce Adversarial Noisy Instruction Tuning (ANIT) to improve NLU performance on LLMs. First, we leverage low-resource techniques to construct noisy instruction datasets. Second, we employ semantic distortion-aware techniques to quantify the intensity of noise within these instructions. Last, we devise an adversarial training method that incorporates a noise response strategy to achieve noisy instruction tuning. ANIT enhances LLMs capability to detect and accommodate semantic distortions in noisy instructions, thereby augmenting their comprehension of task objectives and ability to generate more accurate responses. We evaluate our approach across diverse noisy instructions and semantic distortion quantification methods on multiple NLU tasks. Comprehensive empirical results demonstrate that our method consistently outperforms existing approaches across various experimental settings. Copyright © 2025, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

Date 2025

DOI 10.1609/aaai.v39i22.34513

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Language English

Extra Journal Abbreviation: Proc. AAAI Conf. Artif. Intell.

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Enhancing Reliability in Heartbeat Sound Classification: A Monte Carlo Dropout Approach to Uncertainty Estimation in Deep Learning Models

Item Type Conference Paper

Author M.S.M. Rafi

Author M.M. Hossen

Author Md.S. Ali

Author S.R. Etu

Author M.S. Akter

Author T. Azhar

Abstract Reliable classification of heartbeat sounds is crucial for computer-aided diagnosis in cardiology. This paper presents a deep learning-based framework enhanced with Monte Carlo Dropout (MCD) to estimate predictive uncertainty - an oftenoverlooked aspect in medical AI systems. We evaluate three Convolutional Neural Network (CNN) architectures: a custom Swish-activated CNN, a Two-Stream CNN combining MFCC and Mel-spectrogram features, and a Two-Stream FractalNet. While the Swish CNN attained the highest classification accuracy 85.00%, the Two-Stream CNN demonstrated the most favorable uncertainty profile, with lower predictive entropy and better calibration. These results highlight the importance of not only achieving high accuracy but also understanding model confidence, especially in clinical decision-making scenarios. The use of Monte Carlo Dropout provides deeper insight into model behavior, making it a valuable tool for enhancing the safety and reliability of diagnostic systems. Future work will focus on expanding the dataset, integrating advanced uncertainty quantification methods, and enabling real-time deployment in medical settings. © 2025 IEEE.

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Language English

Extra Journal Abbreviation: Int. Conf. Comput. Inf. Technol., ICCIT
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Tags:

Deep learning, Reliability, Decision making, Learning systems, Convolutional neural network, Convolutional neural networks, Learning models, Computer aided diagnosis, Uncertainty estimation, Monte Carlo methods, Uncertainty quantifications, Model reliability, Uncertainty Quantification, Predictive uncertainty, Monte carlo dropout, Cardiology, Heartbeat sound classification, Heartbeat Sound Classification, Medical Diagnosis, Model Reliability, Monte Carlo Dropout, Predictive Uncertainty, Sound classification, Two-stream

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Export Date: 20 September 2026; Cited By: 0; Conference name: 28th International Conference on Computer and Information Technology, ICCIT 2025; Conference date: 19 December 2025 through 21 December 2025; Conference code: 223902

Enhancing the Discriminative Ability for Multi-Label Classification by Handling Data Imbalance

Item Type Conference Paper

Author J.-H. Lim

Author M.-S. Oh

Author S.-W. Lee

Abstract In computer vision, long-tailed multi-label visual recognition is a challenging problem due to the imbalance between classes and the recognition of rare classes. Previous methods for resolving data imbalance mostly originate from single-label classification, which can be obstructed by the label co-occurrence issue, and they attempt to minimize or compensate for it. In this paper, we propose a novel tail class priority sampling method for long-tailed multi-label classification that untangles both issues. Our method samples tail class more often and earlier in order to make the model learn tail classes before it has bias toward common classes. Due to label co-occurrence, other classes will be spontaneously learned in same iteration, ensuring a balanced representation of the head and medium classes. To further enhance the recognition performance, we modify to a bilateral structure that samples both original and proposed sampling distribution to better

represent the tail classes. We evaluate our proposed method on two widely used datasets in long-tailed version, COCO-LT and VOC-LT, and compare it with previous methods. The experimental results show that our method achieves a new state-of-the-art performance for tail classes on both datasets. Our method is applicable in various real-world scenarios, making rare class recognition achievable, and can be easily incorporated into conventional recognition frameworks. © 2023 IEEE.

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Tags:

Classification (of information), Computer vision, Data handling, Iterative methods, Multi-label classifications, Multi-labels, Learn+, Co-occurrence, Rare class, Visual recognition, Data imbalance, Discriminative ability, Priority samplings, Sampling method

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Enhancing Vessel Segment Extraction in Retinal Fundus Images Using Retinal Image Analysis and Six Sigma Process Capability Index

Item Type Journal Article

Author S.A. Badawi

Author M. Takruri

Author I. ElBadawi

Author I.A. Chaudhry

Author N.U. Mahar

Author A.K. Nileshwar

Author E. Mosalam

Abstract Retinal vessel segmentation, skeletonization, and the generation of vessel segments are considered significant steps in any automated system for measuring the vessel biomarkers of several disease diagnoses. Most of the current tortuosity quantification methods rely on precise vascular segmentation and skeletonization of the retinal vessels. Additionally, the existence of a reference dataset for accurate vessel segment images is an essential need for implementing deep learning solutions and an automated system for measuring the vessel biomarkers of several disease diagnoses, especially for optimized quantification of vessel tortuosity or accurate measurement of AV-nicking. This study aimed to present an improved method for skeletonizing and extracting the retinal vessel segments from the 504 images in the AV classification dataset. The study utilized the Six Sigma process capability index, sigma level, and yield to measure the vessels' tortuosity calculation improvement before and after optimizing the extracted vessels. As a result, the study showed that the sigma level for the vessel segment optimization improved from 2.7 to 4.39, the confirming yield improved from 88 percent to 99.77 percent, and the optimized vessel segments of the AV classification dataset retinal images are available in monochrome and colored formats. © 2023 by the authors.

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: S.A. Badawi; Center of Information, Communication and Networking Education and Innovation (ICONET), American University of Ras Al Khaimah, Ras Al Khaimah, 72603, United Arab Emirates; email: Sufian.badawi@seecs.edu.pk

Ensemble learning head-neck motion predictor with validation and application in cybersickness evaluation

Item Type Journal Article

Author K. Wu

Author F. Sun

Author C. Chen

Author H. Yang

Author W. Li

Author F. Mo

Abstract Despite growing concerns over cybersickness (CS) in virtual reality (VR) systems, effective methods for quantifying CS and evaluating VR scene comfort remain lacking. This study provided and validated a bio-inspired ensemble learning head-neck motion predictor (HMP) that simulated the human neurosensory-motor control loop through coordinating machine learning models. Building upon the HMP outputs, we developed and evaluated a cybersickness index (CSI) quantification method for VR scenes comfort assessment based on head-neck motion. Using motion data collected from thirty participants exposed to five representative VR scenes, the HMP was trained and evaluated with a subject-independent validation strategy to prevent participant data leakage, achieving a mean absolute error of 1.43 ± 0.59 (dimensionless average of angular and linear errors). CSI quantification effectively stratified scene discomfort levels, with VR scene 1 ($CSI = 3.47 \pm 0.82$) inducing severe symptoms, in stark contrast to the optimal comfort offered by VR scene 5 ($CSI = 0.36 \pm 0.15$). Crucially, spectral analysis identified 0.35 Hz as a biomechanical threshold, where sustained body sway frequencies above this value exponentially increased CS severity. Furthermore, the derived CS risk curve established quantitative design guidelines, proposing $CSI < 2.49$ as a safety benchmark for nausea prevention. While these thresholds were empirically obtained from the tested participant group and VR content set, they serve as foundational references that can inform future validation and generalization efforts. Overall, the proposed framework provides a practical and scalable pathway for mitigating CS and enhancing human-machine interaction performance in VR systems. © 2026 Elsevier Ltd.

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Ensemble Multi-Quantiles: Adaptively Flexible Distribution Prediction for Uncertainty Quantification

Item Type Journal Article

Author X. Yan

Author Y. Su

Author W. Ma

Abstract We propose a novel, succinct, and effective approach for distribution prediction to quantify uncertainty in machine learning. It incorporates adaptively flexible distribution prediction of $P(y|X=x)P(y|X=x)$ in regression tasks. This conditional distribution's quantiles of probability levels spreading the interval $(0,1)$ are boosted by additive models which are designed by us with intuitions and interpretability. We seek an adaptive balance between the structural integrity and the flexibility for $P(y|X=x)P(y|X=x)$, while Gaussian assumption results in a lack of flexibility for real data and highly flexible approaches (e.g., estimating the quantiles separately without a distribution structure) inevitably have drawbacks and may not lead to good generalization. This ensemble multi-quantiles approach called EMQ proposed by us is totally data-driven, and can gradually depart from Gaussian and discover the optimal conditional distribution in the boosting. On extensive regression tasks from UCI datasets, we show that EMQ achieves state-of-the-art performance comparing to many recent uncertainty quantification methods. Visualization results further illustrate the necessity and the merits of such an ensemble model. © 1979-2012 IEEE.

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Notes:

EID: 2-s2.0-85162859859

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: W. Ma; Renmin University of China, Institute of Statistics and Big Data, Beijing, 100872, China; email: mawenxuan@ruc.edu.cn; CODEN: ITPID

Ensembles of Classifiers and Quantifiers with Data Fusion for Quantification Learning

Item Type Conference Paper

Author A.B.S. Serapião

Author Z. Donyavi

Author G. Batista

Editor Bifet A.

Editor Lorena A.C.

Editor Ribeiro R.P.

Editor Gama J.

Editor Abreu P.H.

Abstract Quantification is a supervised Machine Learning task that estimates the class distribution in an unlabeled test set. Quantification has practical applications in various fields, including medical research, environmental monitoring, and quality control. For instance, medical research often estimates the prevalence of a particular disease in a population. Despite being a thriving research area, most existing quantification methods are limited to binary-class problems. Moreover, recent experimental evidence suggests that modern state-of-the-art quantifiers do not perform well for multi-class problems, which are prevalent in quantification. This paper proposes two novel multi-class ensemble quantifiers, FMC-SQ and FMC-MQ, that use data fusion methods at the classifier and quantifier levels. We conducted experiments with 12 state-of-the-art (single and ensemble) quantifiers

to evaluate our models on 31 multi-class datasets. Our experimental results indicate that FMC-MQ is the best-performing quantifier outperforming other single and ensemble methods. Also, aggregating quantifier outputs seem to be a more promising research direction than aggregating classification scores for quantification. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Notes:

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Ensembles of α -trees for imbalanced classification problems

Item Type Journal Article

Author Y. Park

Author J. Ghosh

Abstract This paper introduces two kinds of decision tree ensembles for imbalanced classification problems, extensively utilizing properties of α -divergence. First, a novel splitting criterion based on α -divergence is shown to generalize several well-known splitting criteria such as those used in C4.5 and CART. When the α -divergence splitting criterion is applied to imbalanced data, one can obtain decision trees that tend to be less correlated (α -diversification) by varying the value of α . This increased diversity in an ensemble of such trees improves AUROC values across a range of minority class priors. The second ensemble uses the same alpha trees as base classifiers, but uses a lift-aware stopping criterion during tree growth. The resultant ensemble produces a set of interpretable rules that provide higher lift values for a given coverage, a property that is much desirable in applications such as direct marketing. Experimental results across many class-imbalanced data sets, including BRFS, and MIMIC data sets from the medical community and several sets from UCI and KEEL are provided to highlight the effectiveness of the proposed ensembles over a wide range of data distributions and of class imbalance. © 1989-2012 IEEE.

Date 2014

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Notes:

EID: 2-s2.0-84890363076

Export Date: 20 September 2026; Cited By: 35; CODEN: ITKEE

Ensuring Fairness under Prior Probability Shifts

Item Type Conference Paper

Author A. Biswas

Author S. Mukherjee

Abstract Prior probability shift is a phenomenon where the training and test datasets differ structurally within population subgroups. This phenomenon can be observed in the yearly records of several real-world datasets, for example, recidivism records and medical expenditure surveys. If unaccounted for, such shifts can cause the predictions of a classifier to become unfair towards specific population subgroups. While the fairness notion called Proportional Equality (PE) accounts for such shifts, a procedure to ensure PE-fairness was unknown. In this work, we design an algorithm, called CAPE, that ensures fair classification under such shifts. We introduce a metric, called prevalence difference, which CAPE attempts to minimize in order to achieve fairness under prior probability shifts. We theoretically establish that this metric exhibits several properties that are desirable for a fair classifier. We evaluate the efficacy of CAPE via a thorough empirical evaluation on synthetic datasets. We also compare the performance of CAPE with several state-of-the-art fair classifiers on real-world datasets like COMPAS (criminal risk assessment) and MEPS (medical expenditure panel survey). The results indicate that CAPE ensures a high degree of PE-fairness in its predictions, while performing well on other important metrics. © 2021 ACM.

Date 2021

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Conference Name AIES 2021 - Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society

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Epidemiology of multiple childhood traumatic events: Child abuse, parental psychopathology, and other family-level stressors

Item Type Journal Article

Author C. Menard

Author K.J. Bandeen-Roche

Author H.D. Chilcoat

Abstract Background: Multiple family-level childhood stressors are common and are correlated. It is unknown if clusters of commonly co-occurring stressors are identifiable. The study was designed to explore family-level stressor clustering in the general population, to estimate the prevalence of exposure classes, and to examine the correlation of sociodemographic characteristics with class prevalence. Method: Data were collected from an epidemiological sample and analyzed using latent class regression. Results: A six-class solution was identified. Classes were characterized by low risk (prevalence = 23%), universal high risk (7 %), family conflict (11 %), household substance problems (22 %), non-nuclear family structure (24 %), parent's mental illness (13 %). Conclusions: Class prevalence varied with race and welfare status, not gender. Interventions for childhood stressors are person-focused; the analytic approach may uniquely inform resource allocation. © Steinkopff Verlag 2004.

Date 2004

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Epidemiology of urticaria including physical urticaria and angioedema in Korea

Item Type Journal Article

Author J.-H. Seo

Author J.-W. Kwon

Abstract Background/Aims: Despite the high burden and frequency of urticaria, its epidemiology is not well known. We investigated the epidemiology of various type of urticaria in Korea and changes in its annual prevalence over 5 years. Methods: We used data from the 2010 to 2014 Health Insurance Review and Assessment database, which covers the claims of 97.0% of the South Korean population. Patients aged > 10 years old were included in this study. The presence of urticaria was identified on physician-certified diagnoses using the International Classification of Diseases 10th Revision (ICD-10) codes that include various type of urticaria (L500–L509) and angioedema (T783). Epidemiology of all type of urticaria, physical urticaria and angioedema were investigated. Results: The prevalence of all-type urticaria over the 5 years was 4.5% with a peak in individuals, especially females, aged 30 to 59 years. The age-specific prevalence of all-type urticaria increased with age from the 10-to 19-year age group to the 70-to 79-year age group. The prevalence of dermatographism, cholinergic urticaria, and angioedema were 0.12%, 0.025%, and 0.027%, respectively. Cholinergic urticaria was most prevalent in those aged 10 to 29 years with male predominance. The annual prevalence of all-type urticaria, dermatographism, and angioedema increased over the 5 years. Conclusions: The prevalence of urticaria has increased annually in Korea. Cholinergic urticaria showed unique distribution in its age and gender, and angioedema showed remarkable increases in annual prevalence, although the prevalence estimation is still exploratory and diagnosis of urticaria based on ICD-10 codes need to be validated. © 2019 The Korean Association of Internal Medicine.

Date 2019

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Notes:

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Export Date: 20 September 2026; Cited By: 33; Correspondence Address: J.-W. Kwon; Department of Internal Medicine, Kangwon National University Hospital, Chuncheon, 156 Baengnyeo-ng-ro, 24289, South Korea; email: legend@hanmail.net; CODEN: KJIME

Erroneous and excessive treatment anomaly detection based on unsupervised learning

Item Type Journal Article

Author S. Zhu

Author C. Li

Author M. Liu

Author W. Liu

Abstract To provide an effective way to optimize the diagnosis and treatment process, in this research, our model of erroneous medical diagnosis is developed from the association rule learning by Apriori algorithm in order to correlate diseases and symptoms, and further expanded with an innovative numerical quantification method of prescription therapeutic function (called “Treatment Weight Effect Approach”). This is the first time an optimal expert prescription can be discovered from a large scale set of similar prescriptions without requiring explicit labels of training data. In this study, the idea of Hamming distance in K-Modes clustering algorithm is used to calculate the similarity of drugs, and then new prescriptions are compared with the best expert prescription to detect excess treatment. Additional experiments are conducted to verify the feasibility of the program with diagnosis and prescription treatment data. The testing results have shown that the model is successful in providing detection of erroneous and excessive treatments from the existing prescriptions and treatments. We address the critical issue of Treatment Anomalies due to Erroneous and Excessive Medications. We have overcome the traditional medical anomaly detection methods using data mining, Bayesian classification, decision trees and other supervised learning detection methods that were limited in scale and lacked data labels. © 2020, Politecnica University of Bucharest. All rights reserved.

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Estimating and Explaining Model Performance When Both Covariates and Labels Shift

Item Type Conference Paper

Author L. Chen

Author M. Zaharia

Author J. Zou

Editor Koyejo S.

Editor Mohamed S.

Editor Agarwal A.

Editor Belgrave D.

Editor Cho K.

Editor Oh A.

Abstract Deployed machine learning (ML) models often encounter new user data that differs from their training data. Therefore, estimating how well a given model might perform on the new data is an important step toward reliable ML applications. This is very challenging, however, as the data distribution can change in flexible ways, and we may not have any labels on the new data, which is often the case in monitoring settings. In this paper, we propose a new distribution shift model, Sparse Joint Shift (SJS), which considers the joint shift of both labels and a few features. This unifies and generalizes several existing shift models including label shift and sparse covariate shift, where only marginal feature or label distribution shifts are considered. We describe mathematical conditions under which SJS is identifiable. We further propose SEES, an algorithmic framework to characterize the distribution shift under SJS and to estimate a model's performance on new data without any labels. We conduct extensive experiments on several real-world datasets with various ML models. Across different datasets and distribution shifts, SEES achieves significant (up to an order of magnitude) shift estimation error improvements over existing approaches. © 2022 Neural information processing systems foundation. All rights reserved.

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Estimating belowground plant abundance with DNA metabarcoding

Item Type Journal Article**Author** S. Matesanz**Author** D.S. Pescador**Author** B. Pías**Author** A.M. Sánchez**Author** J. Chacón-Labela**Author** A. Illuminati**Author** M. de la Cruz**Author** J. López-Angulo**Author** N. Marí-Mena**Author** A. Vizcaíno**Author** A. Escudero

Abstract Most work on plant community ecology has been performed above ground, neglecting the processes that occur in the soil. DNA metabarcoding, in which multiple species are computationally identified in bulk samples, can help to overcome the logistical limitations involved in sampling plant communities belowground. However, a major limitation of this methodology is the quantification of species' abundances based on the percentage of sequences assigned to each taxon. Using root tissues of five dominant species in a semi-arid Mediterranean shrubland (*Bupleurum frutescens*, *Helianthemum cinereum*, *Linum suffruticosum*, *Stipa pennata* and *Thymus vulgaris*), we built pairwise mixtures of relative abundance (20%, 50% and 80% biomass), and implemented two methods (linear model fits and correction indices) to improve estimates of root biomass. We validated both methods with multispecies mixtures that simulate field-collected samples. For all species, we found a positive and highly significant relationship between the percentage of sequences and biomass in the mixtures ($R^2 = .44-.66$), but the equations for each species (slope and intercept) differed among them, and two species were consistently over- and under-estimated. The correction indices greatly improved the estimates of biomass percentage for all five species in the multispecies mixtures, and reduced the overall error from 17% to 6%. Our results show that, through the use of post-sequencing quantification methods on mock communities, DNA metabarcoding can be effectively used to determine not only species' presence but also their relative abundance in field samples of root mixtures. Importantly, knowledge of these aspects will allow us to study key, yet poorly understood, belowground processes. © 2019 John Wiley & Sons Ltd

Date 2019**DOI** 10.1111/1755-0998.13049**URL** <https://www.scopus.com/pages/publications/85069877905?origin=resultslist>**Archive** Scopus**Language** English**Volume** 19**Publisher** Blackwell Publishing Ltd**Pages** 1265-1277**Publication** Molecular Ecology Resources**ISSN** 1755098X (ISSN)**Issue** 5**Journal Abbr** Mol. Ecol. Resour.**Date Added** 20/09/2026, 9:52:57 pm**Modified** 20/09/2026, 9:52:57 pm

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procedures, classification, genetics, metagenomics, Metagenomics, plant root, Plant Roots, Plants, DNA, Plant, plant DNA, DNA barcoding, plant, DNA Barcoding, Taxonomic, biota, Biota, coexistence, DNA metabarcoding, Mediterranean shrubland, mock communities, plant abundance, rbcL region, root biomass, sequence

Notes:

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Export Date: 20 September 2026; Cited By: 35; Funding details: AllGenetics; Joaquín Vierna; Ministerio de Economía y Competitividad, MINECO, (CGL2015-66809-P); Correspondence Address: S. Matesanz; Área de Biodiversidad y Conservación, Universidad Rey Juan Carlos, Móstoles, Spain; email: silvia.matesanzgarcia@gmail.com

Estimating classification accuracy in positive-unlabeled learning: Characterization and correction strategies

Item Type Conference Paper

Author R. Ramola

Author S. Jain

Author P. Radivojac

Editor Altman R.B.

Editor Keith Dunker A.

Editor Hunter L.

Editor Ritchie M.D.

Editor Klein T.E.

Abstract Accurately estimating performance accuracy of machine learning classifiers is of fundamental importance in biomedical research with potentially societal consequences upon the deployment of bestperforming tools in everyday life. Although classification has been extensively studied over the past decades, there remain understudied problems when the training data violate the main statistical assumptions relied upon for accurate learning and model characterization. This particularly holds true in the open world setting where observations of a phenomenon generally guarantee its presence but the absence of such evidence cannot be interpreted as the evidence of its absence. Learning from such data is often referred to as positive-unlabeled learning, a form of semi-supervised learning where all labeled data belong to one (say, positive) class. To improve the best practices in the field, we here study the quality of estimated performance in positive-unlabeled learning in the biomedical domain. We provide evidence that such estimates can be wildly inaccurate, depending on the fraction of positive examples in the unlabeled data and the fraction of negative examples mislabeled as positives in the labeled data. We then present correction methods for four such measures and demonstrate that the knowledge or accurate estimates of class priors in the unlabeled data and noise in the labeled data are sufficient for the recovery of true classification performance. We provide theoretical support as well

as empirical evidence for the efficacy of the new performance estimation methods. © 2018 The Authors.

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Language English

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Notes:

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Estimating distributed autumn irrigation water use in a large irrigation district by combining machine learning with water balance models

Item Type Journal Article

Author X. Qian

Author H. Qi

Author S. Shang

Author H. Wan

Author R. Wang

Abstract Autumn irrigation (AI) after crop harvesting is vital for agricultural production in cold arid regions, typically requiring more water than regular irrigation during crop growth phases. Understanding distributed AI water use (AIWU) is key to optimizing irrigation strategies and refining agro-hydrological simulations. However, field-scale monitoring of irrigation water use is challenging, particularly in large districts, and quantification methods for distributed AIWU have seldom been investigated. Given this, we proposed a remote sensing-based model to estimate AIWU from water balance analysis, with key components derived through machine learning algorithms. Specifically, to address the unique challenge of short water ponding duration during AI, we selected MODIS as the primary data source, supplemented with Sentinel-2 data to enrich field-level details. A Sentinel-2 empirical bathymetry model, informed by in-situ measurements, served as a foundation for constructing a MODIS bathymetry model using random forest. This MODIS bathymetry model was essential for determining water ponding depths, which allowed for accurate estimation of infiltration rates when applied to the MODIS-based water index time series. By integrating these estimates into the water balance framework, we achieved a more precise estimate of AIWU. The proposed model was applied to Hetao Irrigation District (HID), the largest irrigation district in the arid region of China. The estimated steady infiltration rate and canal efficiency align with similar studies. The results indicate that the average AIWU per unit AI area in the HID remains stable during 2010–2020 with substantial spatial variability at the pixel level, providing new insights into the local hydrological cycle. The AIWU of HID has been decreasing by 19 million m³ per year during this period mainly due to the reduction in AI area, with a more prominent trend from 2010 to 2017. Our study deepens insights into the spatial–temporal dynamics of AIWU in arid regions, contributing to better irrigation management. © 2024 Elsevier B.V.

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Estimating group properties in online social networks with a classifier

Item Type Conference Paper

Author G. Berry

Author A. Sirianni

Author N. High

Author A. Kellum

Author I. Weber

Author M. Macy

Editor Koltsova O.

Editor Ignatov D.I.

Editor Staab S.

Abstract We consider the problem of obtaining unbiased estimates of group properties in social networks when using a classifier for node labels. Inference for this problem is complicated by two factors: The network is not known and must be crawled, and even high-performance classifiers provide biased estimates of group proportions. We propose and evaluate AdjustedWalk for addressing this problem. This is a three step procedure which entails: (1) walking the graph starting from an arbitrary node; (2) learning a classifier on the nodes in the walk; and (3) applying a post-hoc adjustment to classification labels. The walk step provides the information necessary to make inferences over the nodes and edges, while the adjustment step corrects for classifier bias in estimating group proportions. This process provides de-biased estimates at the cost of additional variance. We evaluate AdjustedWalk on four tasks: The proportion of nodes belonging to a minority group, the proportion of the minority group among high degree nodes, the proportion of within-group edges, and Coleman's homophily index. Simulated and empirical graphs show that this procedure performs well compared to optimal baselines in a variety of circumstances, while indicating that variance increases can be large for low-recall classifiers. © Springer Nature Switzerland AG 2018.

Date 2018**DOI** 10.1007/978-3-030-01129-1_5**URL** <https://www.scopus.com/pages/publications/85057321263?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Lect. Notes Comput. Sci.**Volume** 11185 LNCS**Publisher** Springer Verlag**ISBN** 03029743 (ISSN); 978-303001128-4 (ISBN)**Pages** 67-85**Proceedings Title** Lect. Notes Comput. Sci.**Conference Name** Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)**Date Added** 20/09/2026, 9:52:40 pm**Modified** 20/09/2026, 9:52:40 pm**Tags:**

Classification (of information), Social networking (online), Classifiers, Quantification learning, Classification errors, Classification error, Classification labels, Digital demography, High-degree nodes, Network sampling, Network samplings, On-line social networks, Unbiased estimates

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Estimating prevalence for limb-girdle muscular dystrophy based on public sequencing databases

Item Type Journal Article**Author** W. Liu**Author** S. Pajusalu**Author** N.J. Lake**Author** G. Zhou**Author** N. Ioannidis**Author** P. Mittal**Author** N.E. Johnson**Author** C.C. Wehl

Author B.A. Williams

Author D.E. Albrecht

Author L.E. Rufibach

Author M. Lek

Abstract Purpose: Limb-girdle muscular dystrophies (LGMD) are a genetically heterogeneous category of autosomal inherited muscle diseases. Many genes causing LGMD have been identified, and clinical trials are beginning for treatment of some genetic subtypes. However, even with the gene-level mechanisms known, it is still difficult to get a robust and generalizable prevalence estimation for each subtype due to the limited amount of epidemiology data and the low incidence of LGMDs. Methods: Taking advantage of recently published exome and genome sequencing data from the general population, we used a Bayesian method to develop a robust disease prevalence estimator. Results: This method was applied to nine recessive LGMD subtypes. The estimated disease prevalence calculated by this method was largely comparable with published estimates from epidemiological studies; however, it highlighted instances of possible underdiagnosis for LGMD2B and 2L. Conclusion: The increasing size of aggregated population variant databases will allow for robust and reproducible prevalence estimates of recessive disease, which is critical for the strategic design and prioritization of clinical trials. © 2019, American College of Medical Genetics and Genomics.

Date 2019

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Article, female, Female, human, Humans, male, Male, conceptual framework, prevalence, disease classification, genetics, whole genome sequencing, Bayes theorem, Bayes Theorem, Prevalence, gene frequency, genetic variability, rare disease, mutation, Mutation, Databases, Genetic, genetic database, disease prevalence, whole exome sequencing, chromosomal mapping, Chromosome Mapping, exome, Exome, limb girdle muscular dystrophy, limb-girdle muscular dystrophy, Muscular Dystrophies, Limb-Girdle

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Estimating probabilities of peptide database identifications to LC-FTICR-MS observations

Item Type Journal Article

Author K.K. Anderson

Author M.E. Monroe

Author D.S. Daly

Abstract Background: The field of proteomics involves the characterization of the peptides and proteins expressed in a cell under specific conditions. Proteomics has made rapid advances in recent years following the sequencing of the genomes of an increasing number of organisms. A prominent technology for high throughput proteomics analysis is the use of liquid chromatography coupled to Fourier transform ion cyclotron resonance mass spectrometry (LC-FTICR-MS). Meaningful biological conclusions can best be made when the peptide identities returned by this technique are accompanied by measures of accuracy and confidence. Methods: After a tryptically digested protein mixture is analyzed by LC-FTICR-MS, the observed masses and normalized elution times of the detected features are statistically matched to the theoretical masses and elution times of known peptides listed in a large database. The probability of matching is estimated for each peptide in the reference database using statistical classification methods assuming bivariate Gaussian probability distributions on the uncertainties in the masses and the normalized elution times. Results: A database of 69,220 features from 32 LC-FTICR-MS analyses of a tryptically digested bovine serum albumin (BSA) sample was matched to a database populated with 97% false positive peptides. The percentage of high confidence identifications was found to be consistent with other database search procedures. BSA database peptides were identified with high confidence on average in 14.1 of the 32 analyses. False positives were identified on average in just 2.7 analyses. Conclusion: Using a priori probabilities that contrast peptides from expected and unexpected proteins was shown to perform better in identifying target peptides than using equally likely a priori probabilities. This is because a large percentage of the target peptides were similar to unexpected peptides which were included to be false positives. The use of triplicate analyses with a "2 out of 3" reporting rule was shown to have excellent rejection of false positives. © 2006 Anderson et al; licensee BioMed Central Ltd.

Date 2006

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ISSN 14775956 (ISSN)**Journal Abbr** Proteome Sci.**Date Added** 20/09/2026, 9:53:03 pm**Modified** 20/09/2026, 9:53:03 pm**Tags:**

human, article, nonhuman, controlled study, false positive result, multivariate analysis, classification, probability, statistical analysis, uncertainty, protein analysis, liquid chromatography, normal distribution, protein database, bovine serum albumin, Bovinae, elution, ion cyclotron resonance mass spectrometry, peptide derivative, sample

Notes:

EID: 2-s2.0-33646177915

Export Date: 20 September 2026; Cited By: 12; Correspondence Address: D.S. Daly; Computer Science and Mathematics Division, Pacific Northwest National Laboratory, Richland, WA, Battelle Boulevard, United States; email: ds.daly@pnl.gov; CODEN: PSRCC

Estimating the class prior and posterior from noisy positives and unlabeled data

Item Type Conference Paper**Author** S. Jain**Author** M. White**Author** P. Radivojac**Editor** Garnett R.**Editor** Lee D.D.**Editor** von Luxburg U.**Editor** Guyon I.**Editor** Sugiyama M.

Abstract We develop a classification algorithm for estimating posterior distributions from positive-unlabeled data, that is robust to noise in the positive labels and effective for high-dimensional data. In recent years, several algorithms have been proposed to learn from positive-unlabeled data; however, many of these contributions remain theoretical, performing poorly on real high-dimensional data that is typically contaminated with noise. We build on this previous work to develop two practical classification algorithms that explicitly model the noise in the positive labels and utilize univariate transforms built on discriminative classifiers. We prove that these univariate transforms preserve the class prior, enabling estimation in the univariate space and avoiding kernel density estimation for high-dimensional data. The theoretical development and parametric and nonparametric algorithms proposed here constitute an important step towards wide-spread use of robust classification algorithms for positive-unlabeled data. © 2016 NIPS Foundation - All Rights Reserved.

Date 2016

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Tags:

Data mining, Clustering algorithms, Classification algorithm, Robust classification, Kernel Density Estimation, High dimensional data, Non-parametric algorithm, Posterior distributions, Discriminative classifiers, Theoretical development

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Estimating the class prior for positive and unlabelled data via logistic regression

Item Type Journal Article

Author M. Łazęcka

Author J. Mielniczuk

Author P. Teisseyre

Abstract In the paper, we revisit the problem of class prior probability estimation with positive and unlabelled data gathered in a single-sample scenario. The task is important as it is known that in positive unlabelled setting, a classifier can be successfully learned if the class prior is available. We show that without additional assumptions, class prior probability is not identifiable and thus the existing non-parametric estimators are necessarily biased in general if extra assumptions are not imposed. The magnitude of their bias is also investigated. The problem becomes identifiable when the probabilistic structure satisfies mild semi-parametric assumptions. Consequently, we propose a method based on a logistic fit and a concave minorization of its (non-concave) log-likelihood. The experiments conducted on artificial and benchmark datasets as well as on a large clinical database MIMIC indicate that the estimation errors for the proposed method are usually lower than for its competitors and that it is

robust against departures from logistic settings. © 2021, The Author(s).

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Notes:

EID: 2-s2.0-85107422414

Export Date: 20 September 2026; Cited By: 14; Correspondence Address: P. Teisseyre; Faculty of Mathematics and Information Sciences, Warsaw University of Technology, Warsaw, Poland; email: Pawel.Teisseyre@ipipan.waw.pl

Estimating the class prior from positive and unlabeled data

Item Type Journal Article

Author F. Shang

Author Q. Zhang

Author R. Sun

Abstract Positive and unlabeled (PU) learning is an important branch of semi-supervised learning. In the setting of positive and unlabeled learning, if the class prior is known, the classifier would be successfully learned. However, in practical applications, the class prior is unknown, and it can only be obtained from the PU data through the estimation method. In this paper, we proposed a novel method HDBEL (Hierarchical Division Based on Ensemble Learning) for estimating the label frequency from PU data. Label frequency serves as a proxy for the class prior. First, HDBEL algorithm looks for the attribute subdomains with large label probability through the hierarchical partition method based on ensemble learning and then estimates the

lower bound of label frequency by combining with the approximation formula of Hoeffding inequality. Finally, experiments on ten benchmark datasets showed that the HDBEL algorithm had higher accuracy than the existing technical methods, and the speed is seven times higher than the TICe algorithm. © 2020, Asian Association for Agricultural Engineering. All rights reserved.

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Learning systems, Ensemble learning, Approximation algorithms, Benchmark datasets, Semi-supervised learning, Unlabeled data, Positive and unlabeled learning, Estimation methods, Class prior, Hierarchical division, Hoeffding inequality, Lower bounds, Partition methods

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: R. Sun; College of Information and Electrical Engineering, China Agricultural University, Beijing, 100083, China; email: sunruizhi@cau.edu.cn; R. Sun; Scientific Research Base for Integrated Technologies of Precision Agriculture (Animal Husbandry), the Ministry of Agriculture, Beijing, 100083, China; email: sunruizhi@cau.edu.cn; CODEN: IAEJE

Estimating the class prior in positive and unlabeled data through decision tree induction

Item Type Conference Paper

Author J. Bekker

Author J. Davis

Abstract For tasks such as medical diagnosis and knowledge base completion, a classifier may only have access to positive and unlabeled examples, where the unlabeled data consists of both positive and negative examples. One way that enables learning from this type of data is knowing the true class prior. In this paper, we propose a simple yet effective method for estimating the class prior, by estimating

the probability that a positive example is selected to be labeled. Our key insight is that subdomains of the data give a lower bound on this probability. This lower bound gets closer to the real probability as the ratio of labeled examples increases. Finding such subsets can naturally be done via top-down decision tree induction. Experiments show that our method makes estimates which are equivalently accurate as those of the state of the art methods, and is an order of magnitude faster. Copyright © 2018, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Conference Name 32nd AAAI Conference on Artificial Intelligence, AAAI 2018

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Artificial intelligence, Data mining, Knowledge based systems, Decision trees, State-of-the-art methods, Probability, Diagnosis, Trees (mathematics), Unlabeled data, Lower bounds, Positive examples, Negative examples, Decision tree induction, Knowledge base, Sub-domains

Notes:

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Estimating the clinical prevalence of Wilson's disease in the UK

Item Type Journal Article

Author P. Wijayasiri

Author J. Hayre

Author E.S. Nicholson

Author P. Kaye

Author E.A. Wilkes

Author J. Evans
Author G.P. Aithal
Author G. Jones
Author F. Pearce
Author A.D. Aravinthan

Abstract Background & Aims: The clinical prevalence of Wilson's disease (WD) in the UK remains unknown. The estimated genetic prevalence in the UK, 142/million, is higher than the clinical prevalence (15/million) reported in other European studies. The aim of this study was to estimate the clinical prevalence of WD utilising readily available laboratory and clinical data. Method: Patients with WD who attended Nottingham University Hospital NHS Trust (NUH) between 2011 and 2018 were identified using multiple sources of case ascertainment: serum ceruloplasmin, 24-hour urinary copper, 'Wilson' in liver biopsy report, hospital prescription for penicillamine/trientine/zinc and admission coded with ICD-10 Code E83.0 (disorder of copper metabolism). Potential cases were identified using the Leipzig score, diagnosis was confirmed in hospital records and the point prevalence was calculated using the Office for National Statistics mid-2017 population estimates. Results: A total of 1,794 patients were identified from ≥ 1 source; 19 patients had WD, of whom 11 were from within the study catchment area and alive at the time of point prevalence estimation. Twenty-nine patients had a Leipzig score ≥ 2 without a diagnosis of WD, but none had WD on screening ($n = 16$). The overall prevalence of WD was 15.5/million; males 16.9/million and females 14.1/million. Conclusion: This is the first UK population-based study to assess the clinical prevalence of WD. The reported clinical prevalence is lower than the UK genetic prevalence, but comparable to the clinical prevalence reported in Europe. The case ascertainment approach used in this study may be cost-effective, and similar practises could be adopted nationally. Lay summary: Our study estimates the clinical prevalence of Wilson's disease, a rare genetic disorder of copper metabolism, in the UK. The estimated clinical prevalence in this study is markedly lower than the estimated UK genetic prevalence. © 2021

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adult, Article, female, human, male, middle aged, young adult, adolescent, United Kingdom, major clinical study, prevalence, cost effectiveness analysis, child, cohort analysis, population, scoring system, retrospective study, International Classification of Diseases, diagnostic procedure, sex ratio, clinical outcome, screening, school child, preschool child, protein blood level, Wilson disease, laboratory test, ceruloplasmin, Clinical prevalence, clinical study, copper, Copper metabolism disorder, Leipzig score, liver biopsy, Multiple sources of case ascertainment, penicillamine, point prevalence, prescription, trientine, urine level, Wilson's disease, zinc

Notes:

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Export Date: 20 September 2026; Cited By: 11; Funding details: Orphan UK; Correspondence Address: A.D. Aravinthan; Nottingham Digestive Diseases Centre, Translational Medical Sciences, School of Medicine, University of Nottingham, QMC Campus, Nottingham, Derby Road, NG7 2UH, United Kingdom; email: aloysious.aravinthan@nottingham.ac.uk

Estimating the Density Ratio between Distributions with High Discrepancy using Multinomial Logistic Regression

Item Type Journal Article

Author A. Srivastava

Author S. Han

Author K. Xu

Author B. Rhodes

Author M.U. Gutmann

Abstract Functions of the ratio of the densities p/q are widely used in machine learning to quantify the discrepancy between the two distributions p and q . For high-dimensional distributions, binary classification-based density ratio estimators have shown great promise. However, when densities are well separated, estimating the density ratio with a binary classifier is challenging. In this work, we show that the state-of-the-art density ratio estimators perform poorly on well separated cases and demonstrate that this is due to distribution shifts between training and evaluation time. We present an alternative method that leverages multi-class classification for density ratio estimation and does not suffer from distribution shift issues. The method uses a set of auxiliary densities $\{m_k\}_{K=1}$ and trains a multi-class logistic regression to classify the samples from p , q and $\{m_k\}_{K=1}$ into $K + 2$ classes. We show that if these auxiliary densities are constructed such that they overlap with p and q , then a multi-class logistic regression allows for estimating $\log p/q$ on the domain of any of the $K + 2$ distributions and resolves the distribution shift problems of the current state-of-the-art methods. We compare our method to state-of-the-art density ratio estimators on both synthetic and real datasets and demonstrate its superior performance on the tasks of density ratio estimation, mutual information estimation, and representation learning. Code: <https://www.blackswan.com/mdre/>. © 2023, Transactions on Machine Learning Research. All rights reserved.

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Estimating the minority class proportion with the ROC curve using Military Personality Inventory data of the ROK Armed Forces

Item Type Journal Article

Author M. Sun

Author K. Choi

Author S. Cho

Abstract The Republic of Korea Armed Forces includes maladjusted conscripts such as the mentally ill, the suicidal, the imprisoned, and those determined by the military commander to be maladjusted. To counteract these problems, it is necessary to identify the maladjusted conscripts to determine who among them would qualify for exemption from active military service or need special attention. We use the Military Personality Inventory (MPI) to make this prediction. Such a prediction presents a kind of class imbalance and class overlap problem, where the majority fulfil active service and the minority are maladjusted, the latter being discharged early from active service. Therefore, most classification algorithms are likely to show low classification performance. As an alternative, this study demonstrates the effective utilization of the receiver operating characteristics curve using MPI data to estimate the maladjusted proportion of persons sharing similar MPI test results. We confirm that the suggested method performs well using the real-world MPI data set. The suggested method is very useful to estimate the proportion of conscripts maladjusted to military life and can help in the management of such persons subject to conscription. © 2015 Taylor & Francis.

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class imbalance, prevalence estimation, class overlap, Military Personality Inventory, minority proportion, ROC curve

Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M. Sun; Center for Defense Management, Korea Institute for Defense Analyses, Seoul, South Korea; email: misunny@kida.re.kr

Estimation of a priori probabilities of sleep stages: A cycle-based approach

Item Type Conference Paper

Author A. Tataraidze

Author L. Anishchenko

Author L. Korostovtseva

Author M. Bochkarev

Author Y. Sviryayev

Author S. Ivashov

Abstract This paper presents a model for the estimation of a priori probabilities of sleep epoch classes based on the epoch location in a sleep cycle. These probabilities are used as additional features for sleep stage classification based on the analysis of respiratory effort. The model was validated with data of 685 subjects selected from the Sleep Heart Health Study dataset. The model improves a base algorithm by 8 percent points and demonstrates Cohen's kappa of 0.56 ± 0.12 . Our results will contribute to the development of screening tools for unobtrusive sleep structure estimation. © 2017 IEEE.

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Language English

Extra Journal Abbreviation: Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. EMBS

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Pages 3745-3748**Proceedings Title** Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. EMBS**Conference Name** Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS**Date Added** 20/09/2026, 9:52:41 pm**Modified** 20/09/2026, 9:52:41 pm**Tags:**

algorithm, Algorithms, Probability, polysomnography, probability, signal processing, Signal Processing, Computer-Assisted, sleep stage, Sleep Stages, Polysomnography

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Estimation of brain amyloid accumulation using deep learning in clinical [11C]PiB PET imaging

Item Type Journal Article**Author** C.N. Ladefoged**Author** L. Anderberg**Author** K. Madsen**Author** O.M. Henriksen**Author** S.G. Hasselbalch**Author** F.L. Andersen**Author** L. Højgaard**Author** I. Law

Abstract Introduction: Estimation of brain amyloid accumulation is valuable for evaluation of patients with cognitive impairment in both research and clinical routine. The development of high throughput and accurate strategies for the determination of amyloid status could be an important tool in patient selection for clinical trials and amyloid directed treatment. Here, we propose the use of deep learning to quantify amyloid accumulation using standardized uptake value ratio (SUVR) and classify amyloid status based on their PET images. Methods: A total of 1309 patients with cognitive impairment scanned with [11C]PiB PET/CT or PET/MRI were included. Two convolutional neural networks (CNNs) for reading-based amyloid status and SUVR prediction were trained using 75% of the PET/CT data. The remaining PET/CT (n = 300) and all PET/MRI (n = 100) data was used for evaluation. Results: The prevalence of amyloid positive patients was 61%. The amyloid status classification model reproduced the expert reader's classification with 99% accuracy. There was a high correlation between reference and predicted SUVR ($R^2 = 0.96$). Both reference

and predicted SUVR had an accuracy of 97% compared to expert classification when applying a predetermined SUVR threshold of 1.35 for binary classification of amyloid status. Conclusion: The proposed CNN models reproduced both the expert classification and quantitative measure of amyloid accumulation in a large local dataset. This method has the potential to replace or simplify existing clinical routines and can facilitate fast and accurate classification well-suited for a high throughput pipeline. © 2023, The Author(s).

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AI, adult, Article, human, Deep learning, aged, Convolutional neural network, reproducibility, very elderly, quantitative analysis, correlation analysis, prediction, deep learning, controlled study, major clinical study, Decision support, convolutional neural network, prevalence, Dementia, decision support system, diagnostic test accuracy study, diagnostic accuracy, cohort analysis, cognitive defect, disease classification, Alzheimer disease, Alzheimer's disease, computer assisted diagnosis, nuclear magnetic resonance imaging, medical expert, positron emission tomography-computed tomography, biological model, PET, positron emission tomography, dementia, high throughput analysis, stratification, Amyloid, amyloid protein, Automatic diagnosis, bioaccumulation, Pittsburgh compound B, standardized uptake value ratio, Stratification

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Export Date: 20 September 2026; Cited By: 8; Funding details: Alzheimer's Disease Neuroimaging Initiative, ADNI; Correspondence Address: C.N. Ladefoged; Department of Clinical Physiology and Nuclear Medicine, Rigshospitalet, University of Copenhagen, Copenhagen, Blegdamsvej 9, 2100, Denmark; email: claes.noehr.ladefoged@regionh.dk

Estimation of class priors for improving classification accuracy during deployment

Item Type Journal Article**Author** B. McIntosh**Author** N. Daba**Author** A. Mahalanobis

Abstract Conventional classifiers are trained and evaluated using “balanced” data sets, in which all classes are equally present. However, it is unlikely that the prior probability of occurrence will be the same for the classes that are encountered in each deployment scenario. Unfortunately, the exact class priors are generally unknown and can vary spatially and over time. We explore different methods for estimating the class priors based on the response of the classifier itself. We observe that the distribution of classifier's decisions made in response to a collection test image can be expressed as a linear combination of the distribution of its decisions made in response to each individual class. The weights of the linear combination are the prior probabilities with which the different classes occur in the test set. This can be used along with the classifier's precision and recall for each class to estimate the class priors. We then show that incorporating the estimated class priors in the overall decision scheme improves the classifier's accuracy in the context of the deployment. We first illustrate the potential improvements in classification accuracy using the CIFAR-100 and Tiny ImageNet data sets and then demonstrate the application of this technique for improving the classification and recognition of objects in satellite imagery using the well-known Xview dataset. © 2024 Elsevier Inc. All rights reserved.

Date 2024**DOI** 10.1016/B978-0-44-319077-3.00010-9**URL** <https://www.scopus.com/pages/publications/85199112930?origin=resultslist>**Archive** Scopus**Language** English**Publisher** Elsevier**Pages** 71-91**Publication** Advances in Machine Learning and Image Analysis for GeoAI**ISSN** 978-044319077-3 (ISBN); 978-044319078-0 (ISBN)**Journal Abbr** Advances in Machine Learning and Image Analysis for Geoai**Date Added** 20/09/2026, 9:51:52 pm**Modified** 20/09/2026, 9:51:52 pm**Tags:**

Classifier, Context, Performance accuracy, Unknown priors

Notes:

EID: 2-s2.0-85199112930

Export Date: 20 September 2026; Cited By: 0

Estimation of the number of Anisakis larvae in commercial fish using a descriptive model based on real-time PCR

Item Type Journal Article**Author** C. Godínez-González**Author** X. Roca-Geronès**Author** I. Montoliu**Author** R. Fisa

Abstract BACKGROUND: Seafood parasitization by Anisakis (Anisakidae) larvae has been reported in most of the oceans and seas worldwide. The presence of these nematodes in commonly consumed fish represents a potential hazard for consumers as they can provoke gastrointestinal symptoms and allergic reactions. In the present work, the capacity of a SYBR Green qPCR protocol to quantify Anisakis larvae in commercial fish was evaluated using experimentally spiked samples with different numbers (0–50) of *A. simplex* third-stage larvae (L3). To verify the agreement of the obtained results, 25 naturally infected fish specimens of Atlantic blue whiting underwent a parallel visual inspection. RESULTS: The logarithmic behavior of the Cq data obtained from the experimentally spiked samples allowed the development of a descriptive mathematical model that correlates the Cq value with the number of Anisakis larvae ($R^2 = 0.9908$, $CV = 2.37\%$). In the commercial blue whiting specimens there was a high correlation between the results of the molecular technique and the visual inspection ($R^2 = 0.9912$); the Bland–Altman analysis showed that 94% of the differences were within the limits of agreement (−4.98 and 6.68), indicating the reliability of the descriptive mathematical model based on the SYBR Green qPCR technique. CONCLUSION: The descriptive function presented based on the SYBR Green qPCR assay is promising as a sensitive and accurate tool for measuring the Anisakis larval load in commercial fish, with a potential application not only in the food industry but also in prevention programs for public health. © 2020 Society of Chemical Industry. © 2020 Society of Chemical Industry

Date 2021**DOI** 10.1002/jsfa.10718**URL** <https://www.scopus.com/pages/publications/85090789386?origin=resultslist>**Archive** Scopus**Language** English**Volume** 101**Publisher** John Wiley and Sons Ltd**Pages** 1085-1090**Publication** Journal of the Science of Food and Agriculture**ISSN** 00225142 (ISSN)**Issue** 3**Journal Abbr** J. Sci. Food Agric.**Date Added** 20/09/2026, 9:52:27 pm**Modified** 20/09/2026, 9:52:27 pm**Tags:**

procedures, classification, veterinary medicine, genetics, isolation and purification, food safety, animal, Animals, larva, Larva, parasitology, fish, Fishes, real time polymerase chain reaction, Real-Time Polymerase Chain Reaction, anisakiasis, Anisakiasis, Anisakis, fish disease, Fish Diseases, foodborne parasites, larval quantification method, molecular technique

Notes:

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Export Date: 20 September 2026; Cited By: 6; Funding details: CONACYT-Becas M?xico; CONACYT-Becas México, (383879); Generalitat de Catalunya; Universitat Oberta de Catalunya, UOC; Correspondence Address: R. Fisa; Laboratory of Parasitology, Department of Biology, Health and Environment, Faculty of Pharmacy and Food Sciences, University of Barcelona, Barcelona, Spain; email: rfisa@ub.edu; CODEN: JSFAA

Evaluating a Targeted Minimum Loss-Based Estimator for Capture-Recapture Analysis: An Application to HIV Surveillance in San Francisco, California

Item Type Journal Article**Author** P. Wesson**Author** M. Das**Author** M. Chen**Author** L. Hsu**Author** W. McFarland**Author** E. Kennedy**Author** N.P. Jewell

Abstract The capture-recapture method is a common tool used in epidemiology to estimate the size of "hidden" populations and correct the underascertainment of cases, based on incomplete and overlapping lists of the target population. Log-linear models are often used to estimate the population size yet may produce implausible and unreliable estimates due to model misspecification and small cell sizes. A novel targeted minimum loss-based estimation (TMLE) model developed for capture-recapture makes several notable improvements to conventional modeling: "targeting" the parameter of interest, flexibly fitting the data to alternative functional forms, and limiting bias from small cell sizes. Using simulations and empirical data from the San Francisco, California, Department of Public Health's human immunodeficiency virus (HIV) surveillance registry, we evaluated the performance of the TMLE model and compared results with those of other common models. Based on 2,584 people observed on 3 lists reportable to the surveillance registry, the TMLE model estimated the number of San Francisco residents living with HIV as of December 31, 2019, to be 13,523 (95% confidence interval: 12,222, 14,824). This estimate, compared with a "ground truth" of 12,507, was the most accurate and precise of all models examined. The TMLE model is a significant advancement in capture-recapture studies, leveraging modern statistical methods to improve estimation of the sizes of hidden populations. © 2024 The Author(s) 2023. Published by Oxford University Press on behalf of the Johns Hopkins Bloomberg School of Public Health.

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adult, Article, California, female, human, Humans, male, United States, machine learning, adolescent, spatiotemporal analysis, Bias, epidemiology, random forest, bias, child, cross validation, data extraction, statistical analysis, Linear Models, statistical model, prevalence estimation, confidence interval, Human immunodeficiency virus, biosurveillance, capture recapture analysis, capture-recapture method, geostatistical analysis, hidden populations, HIV, HIV Infections, human immunodeficiency virus, Human immunodeficiency virus infection, instrumentation, loglinear model, multinomial logistic regression, San Francisco, San Francisco [California], statistical bias, SuperLearner, targeted minimum loss based estimation model, targeted minimum loss-based estimation

Notes:

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Evaluating child-friendliness in the 15-minute community living circle: An example from Shanghai’s Dongyuan I Village

Item Type Journal Article
Author M. Liu
Author Y. Xu
Author Y. Chen
Author W. Yan

Abstract Under the pressing challenges posed by urbanization, such as road traffic, limited activity spaces, and environmental inequalities, ensuring children’s well-being in urban environments has become a critical issue. Children’s physical, cognitive, and social development is significantly influenced by their living environments, yet traditional urban planning often overlooks their unique needs. To address this gap, this study focuses on integrating child-friendliness into urban community planning, particularly through the lens of the “15-minute community living circle” model. This model emphasizes the creation of urban spaces where essential facilities are within a

15-minute walk, aligning with children's physical capabilities and daily needs. The study develops a comprehensive and quantitative evaluation system for assessing child-friendliness living circles, focusing on safety, comfort, and fun in travel and stay spaces. Using Shanghai's Dongyuan I Village as a case study, the research uses online questionnaires, field surveys, and machine learning to quantify child-friendliness. Results demonstrate the model's ability to provide micro-level evaluations and actionable insights for optimizing urban spaces. This research highlights the importance of prioritizing child-friendly principles in urban planning, offering valuable guidance for fostering safe, inclusive, and vibrant environments that support children's growth and contribute to sustainable urban development. © 2025 Urban Affairs Association.

Date 2026

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: Y. Xu; China Institute of FTZ Supply Chain, Shanghai Maritime University, Shanghai, 201306, China; email: 202030510221@stu.shmtu.edu.cn

Evaluating classifiers: Relation between area under the receiver operator characteristic curve and overall accuracy

Item Type Conference Paper

Author M.A. Mazurowski

Author G.D. Tourassi

Abstract In this study, we investigated the relation between two popular classifier performance measures: area under the receiver operator characteristic curve and overall accuracy. We also evaluated the impact of class imbalance and number of examples in test set on this relation. We perform a set of experiments in which we train multiple neural networks and test them in various, well controlled conditions. The experimental results show that given a large and balanced test set, increase in one performance measure is a very good indicator of increase in the other measure. Furthermore increasing the total number of examples, while keeping the positive class prevalence constant generally increases the correlation between the two measures. Our results also indicate that increasing the extent of class imbalance in the test set has a detrimental effect on this correlation. ©2009 IEEE.

Date 2009

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Learning systems, Neural networks, Class imbalance, Classifiers, Test sets, Performance measure, Classifier performance, Controlled conditions, Detrimental effects, Multiple neural networks, Receiver operator characteristic curves

Notes:

EID: 2-s2.0-70449421914

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: M. A. Mazurowski; Department of Radiology, Duke University, Medical Center, Durham, NC 27705, United States; email: maciej.mazurowski@duke.edu; Conference name: 2009 International Joint Conference on Neural Networks, IJCNN 2009; Conference date: 14 June 2009 through 19 June 2009; Conference code: 78384; CODEN: 85OFA

Evaluating methods for distinguishing between human-readable text and garbled text

Item Type Conference Paper

Author J. Henderson

Author D. Frazee

Author N. Siegel
Author C. Martin
Author A. Liu
Editor Markov Z.
Editor Russell I.

Abstract In some cybersecurity applications, it is useful to differentiate between human-readable text and garbled text (e.g., encoded or encrypted text). Automated methods are necessary for performing this task on large volumes of data. Which method is best is an open question that depends on the specific problem context. In this paper, we explore this open question via empirical tests of many automated categorization methods for differentiating human-readable versus garbled text under a variety of conditions (e.g., different class priors, different problem contexts, concept drift, etc.). The results indicate that the best approaches tend to be either variants of naïve Bayes or classifiers that use low-dimensional, structural features. The results also indicate that concept drift is one of the most problematic issues when classifying garbled text. © 2016, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

Date 2016

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Extra Journal Abbreviation: Proc. Int. Florida Artif. Intell. Res. Soc. Conf., FLAIRS

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Proceedings Title Proc. Int. Florida Artif. Intell. Res. Soc. Conf., FLAIRS

Conference Name Proceedings of the 29th International Florida Artificial Intelligence Research Society Conference, FLAIRS 2016

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Artificial intelligence, Automated categorizations, Structural feature, Automated methods, Evaluating method, Human-readable text, Low dimensional, nocv1, Problematic issues, Specific problems

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Evaluating Sentiment Quantification Methods in Brazilian Portuguese Corpora

Item Type Conference Paper

Author L.N. dos Santos Silva
Author D.F. Silva
Author H. de Medeiros Caseli
Editor Paes A.
Editor Verri F.A.N.

Abstract This paper evaluates sentiment quantification methods applied to Brazilian Portuguese corpora. Sentiment quantification, distinct from sentiment classification, estimates the distribution of sentiment classes (positive and negative) within a dataset. We investigate several quantification techniques, including the family Classify and Count (CC) and more sophisticated methods, such as Kernel Density Estimation (KDE) and Distribution y-Similarity (DyS). Our analysis uses five datasets, each containing different distributions of sentiment classes. Our experimental results indicate that KDE and DyS methods consistently outperform others, achieving the best average ranks in terms of quantification accuracy. Statistical tests, including the Friedman and Nemenyi tests, confirm significant performance differences among the methods, with KDE and DyS showing statistically significant improvements over the baseline CC method. These findings highlight the importance of choosing robust quantification techniques for accurate sentiment quantification in corpora across different domains. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Evaluating the Effects of C3 Inhibition on Geographic Atrophy Progression from Deep-Learning OCT Quantification: A Split-Person Study

Item Type Journal Article

Author D.J. Fu

Author V. Lipkova

Author B. Liefers

Author S. Ginton

Author L. Faes

Author A. McKeown

Author L. Scheibler

Author N. Pontikos

Author P.J. Patel

Author G. Zhang

Author P.A. Keane

Author K. Balaskas

Abstract Introduction: To evaluate the effect pegcetacoplan, a C3 and C3b inhibitor, on the rate of progression of geographic atrophy (GA) as assessed by spectral domain optical coherence tomography (SD-OCT) using a split-person study design and deep-learning quantification. Methods: A post hoc analysis of phase 2 FILLY trial data comparing study (treated monthly, treated every other month and sham-treated) and fellow (untreated) eyes in a split-person study design was performed. This analysis included 288 eyes from 144 patients with bilateral GA from the FILLY phase 2 trial (Clinical Trials identifier: NCT02503332). Only patients with bilateral GA and without evidence of choroidal neovascularisation in either eye were included. Patient study eyes were treated with sham injections or with pegcetacoplan monthly (PM) or every other month (PEOM) for 12 months. SD-OCT scans of study and fellow eyes taken at baseline and 12 months were used for the analysis. The main outcomes were the annual change in the area of retinal pigment epithelial and outer retinal atrophy (RORA), its constituent features (photoreceptor degeneration [PRD], retinal pigment epithelium [RPE] loss, hypertransmission) and intact macula as compared to the untreated fellow eye. Results: Annual GA growth was reduced in eyes treated with PM versus untreated fellow eyes for OCT features, including RORA (study eye 0.792 vs. fellow eye 1.13 mm²; P = 0.003), PRD (0.739 vs. 1.23 mm²; P = 0.015), RPE-loss (0.789 vs. 1.17 mm²; P = 0.007) and intact macula (− 0.735 vs. − 1.29 mm²; P = 0.011). Similar (but not statistically significant) trends were observed with the PEOM treatment or when GA was quantified with fundus autofluorescence (FAF). The sham treatment demonstrated no effect. Pearson correlation coefficients showed concordance in the enlargement rate of GA between the study and fellow eyes in the

sham ($R = 0.64$) and PEOM ($R = 0.68$) groups, but not in the PM group ($R = 0.21$).
Conclusions: Pegcetacoplan-treated eyes demonstrated a reduction in spatial GA progression compared to their untreated counterparts. This effect was more evident on OCT than with FAF. Trial Registration: Clinical Trials identifier: NCT02503332.
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Article, female, human, male, machine learning, Machine learning, aged, deep learning, major clinical study, post hoc analysis, correlation coefficient, pathophysiology, optical coherence tomography, phase 2 clinical trial, B scan, autofluorescence, AMD, complement component C3, complement component C3b, geographic atrophy, Geographic atrophy, growth rate, NCT02503332, OCT, pegcetacoplan, Pegcetacoplan, photoreceptor, retinal pigment epithelium, sham procedure, spectral domain optical coherence tomography, split person study, subretinal neovascularization

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Export Date: 20 September 2026; Cited By: 4; Funding details: NIHR Biomedical Research Center at Moorfields Eye Hospital NHS Foundation Trust; Wellcome Trust, WT, (206619); Wellcome Trust, WT; Correspondence Address: D.J. Fu; NIHR Biomedical Research Centre at Moorfields Eye Hospital NHS Foundation Trust–UCL Institute of Ophthalmology, London, 162 City Rd, EC1V 2PD, United Kingdom; email: d.fu@nhs.net

Evaluation and Optimization of Sample Handling Methods for Quantification of Short-Chain Fatty Acids in Human Fecal Samples by GC-MS

Item Type Journal Article

Author Y.-L. Hsu

Author C.-C. Chen

Author Y.-T. Lin
Author W.-K. Wu
Author L.-C. Chang
Author C.-H. Lai
Author M.-S. Wu
Author C.-H. Kuo

Abstract The gut microbiota has attracted a great deal of interest in recent years due to its association with many diseases. Short-chain fatty acids (SCFAs), the end products of dietary fiber fermentation by the intestinal microbiota, are among the most frequently discussed gut metabolites. As the sample handling method greatly affects the integrity of data, this study investigated the most important parameters that affect the bias of SCFA comparisons in human fecal studies. An accurate gas chromatography-mass spectrometry (GC-MS) method was first established and validated for quantifying six SCFAs, including acetic, propionic, butyric, isobutyric, isovaleric, and valeric acids. To remove interfering species, we used butanol to extract SCFAs from acidified fecal suspensions. The validated quantification method was then applied to evaluate fecal sample handling protocols. We found that lyophilization of fecal samples can not only minimize bias due to the water content but also provide better stability of SCFAs. Six SCFAs were stable and that their recoveries were higher than 90% after lyophilization. Lyophilization of a large fecal sample is extremely time-consuming, and 1 g of fecal sample is suggested for lyophilization to minimize sampling bias. The interindividual difference was significantly higher than the intra-individual difference when using 1 g of fecal sample to study SCFAs. Finally, an effective protocol from sample collection to GC-MS analysis was proposed. As SCFAs have been shown to play an important role in health maintenance and disease development, the proposed protocol is anticipated to be applicable to clinical studies to delineate the biological functions of each SCFA. © Copyright 2019 American Chemical Society.

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Evaluation measures for quantification: an axiomatic approach

Item Type Journal Article

Author F. Sebastiani

Abstract Quantification is the task of estimating, given a set σ of unlabelled items and a set of classes $C = \{c_1, \dots, c_l \mid C\}$, the prevalence (or “relative frequency”) in σ of each class $c_i \in C$. While quantification may in principle be solved by classifying each item in σ and counting how many such items have been labelled with c_i , it has long been shown that this “classify and count” method yields suboptimal quantification accuracy. As a result, quantification is no longer considered a mere byproduct of classification, and has evolved as a task of its own. While the scientific community has devoted a lot of attention to devising more accurate quantification methods, it has not devoted much to discussing what properties an evaluation measure for quantification (EMQ) should enjoy, and which EMQs should be adopted as a result. This paper lays down a number of interesting properties that an EMQ may or may not enjoy, discusses if (and when) each of these properties is desirable, surveys the EMQs that have been used so far, and discusses whether they enjoy or not the above properties. As a result of this investigation, some of the EMQs that have been used in the literature turn out to be severely unfit, while others emerge as closer to what the quantification community actually needs. However, a significant result is that no existing EMQ satisfies all the properties identified as desirable, thus indicating that more research is needed in order to identify (or synthesize) a truly adequate EMQ. © 2019, Springer Nature B.V.

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Supervised learning, Quantification, Supervised prevalence estimation, Evaluation measures

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Evaluation of a computerized method for denture biofilm quantification: Inter-examiner reproducibility

Item Type Journal Article

Author C.H. Silva-Lovato

Author A.M.G. Totti

Author H.F.O. Paranhos

Author V.G. Totti

Abstract Purpose: The objective of this work was to evaluate the accuracy and inter-examiner reproducibility of the computerized method Image Tool 2.02 (Windows, version 2.02, University of Texas Health Science Center, San Antonio, TX) in the quantification of complete denture biofilm. Materials and Methods: Two hundred digital photographs of the internal surface of complete upper dentures with biofilm dyed in 1% neutral red were used. The photographs were burned to a CD and sent to two researchers who received practical training and written instructions on how to use the method; then, in a 20-day period, the stained biofilm on the internal surface was measured, and ten photographs were analyzed daily. Each examiner independently quantified the percentage of stained area on each photograph. The inter-examiner reproducibility (i.e., agreement) was estimated by calculating an intra-class correlation coefficient (ICC) based on an ANOVA mixed model (SPSS software, Chicago, IL). Results: The ICC showed excellent agreement (ICC = 0.993, $p < 0.001$). The difference in the percentage of the stained area recorded by the two examiners was ≤ 1 point for 187 (93.5%) of the 200 photographs. Conclusions: The obtained results suggest that the computerized method aided by the Image Tool software (2.0) can be employed to quantify complete denture biofilm, seeing that it demonstrated inter-examiner reproducibility of the results. © 2009 by The American College of Prosthodontists.

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Evaluation of a Concept for Classification of Micromobility Vehicles Based on Thermal-Infrared Imaging and Neuromorphic Processing

Item Type Conference Paper

Author B. Stahl

Author R. Lange

Author J. Apfelbeck

Editor Kimata M.

Editor Shaw J.A.

Editor Valenta C.R.

Abstract Current trends of environmental awareness combined with a focus on personal fitness and health, motivates many people to switch from cars and public transport to micromobility solutions, namely bicycles, electric bicycles, cargo bikes, or scooters. To accommodate urban planning for these changes, cities, and communities need to know how many (micromobility) vehicles are on the road.

In this work, we design a concept for a compact, mobile, and energy-efficient system to classify and count micromobility vehicles. We aim for a battery-powered system, which can be installed in a short time by a single person and is non-invasive for the ground or traffic. For this purpose, we investigate a system architecture consisting of uncooled long-wave infrared (LWIR) imaging sensors and a neuromorphic co-processor. This processor can perform classifications in a fixed time of 16,2 μ s without requiring additional hardware for learning. It can adapt its knowledge base to new environments without external updates. Our system is aimed to achieve adequate accuracy only using a low-resolution sensor and moderate processing power. We will show that an object representation of only 256 Bytes is sufficient to achieve a four-class classification accuracy higher than 80%, with an overall power consumption of less than 7mW. We estimate a total operating time of at least two weeks using a common 12V 17Ah lead battery. © 2021 SPIE. All rights reserved.

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Evaluation of a Nondepleted Plasma Multiprotein-Based Model for Discriminating Psychiatric Disorders Using Multiple Reaction Monitoring-Mass Spectrometry: Proof-of-Concept Study

Item Type Journal Article

Author D. Shin

Author J. Lee

Author Y. Kim

Author J. Park

Author D. Shin

Author Y. Song

Author E.-J. Joo

Author S. Roh

Author K.Y. Lee

Author S. Oh

Author Y.M. Ahn

Author S.J. Rhee

Author Y. Kim

Abstract Psychiatric evaluation relies on subjective symptoms and behavioral observation, which sometimes leads to misdiagnosis. Despite previous efforts to utilize plasma proteins as objective markers, the depletion method is time-consuming. Therefore, this study aimed to enhance previous quantification methods and construct objective discriminative models for major psychiatric disorders using nondepleted plasma. Multiple reaction monitoring-mass spectrometry (MRM-MS) assays for quantifying 453 peptides in nondepleted plasma from 132 individuals [35 major depressive disorder (MDD), 47 bipolar disorder (BD), 23 schizophrenia (SCZ) patients, and 27 healthy controls (HC)] were developed. Pairwise discriminative models for MDD, BD, and SCZ, and a discriminative model between patients and HC were constructed by machine learning approaches. In addition, the proteins from nondepleted plasma-based discriminative models were compared with previously developed depleted plasma-based discriminative models. Discriminative models for MDD versus BD, BD versus SCZ, MDD versus SCZ, and patients versus HC were constructed with 11 to 13 proteins and showed reasonable performances (AUROC = 0.890-0.955). Most of the shared proteins between nondepleted and depleted plasma models had consistent directions of expression levels and were associated with neural signaling, inflammatory, and lipid metabolism pathways. These results suggest that multiprotein markers from nondepleted plasma have a potential role in psychiatric evaluation. © 2023 American Chemical Society

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adult, Article, female, human, Humans, male, machine learning, comparative study, demographics, quantitative analysis, controlled study, major clinical study, sensitivity analysis, receiver operating characteristic, network analysis, diagnostic accuracy, mental disease, bipolar disorder, clinical feature, DSM-5, schizophrenia, limit of quantitation, sensitivity and specificity, body mass, major depression, metabolism, nervous system inflammation, disease model, antidepressant agent, liquid chromatography-mass spectrometry, mass spectrometry, proteomics, sex, lipid, signal transduction, multiple reaction monitoring, Mass Spectrometry, protein expression, mini international neuropsychiatric interview, anticonvulsive agent, benzodiazepine derivative, Bipolar Disorder, Brief Psychiatric Rating Scale, cancer patient, Depressive Disorder, Major, discriminative model, fasting, Hamilton Anxiety Scale, hypnotic agent, lipid metabolism, lithium, major depressive disorder, Montgomery Asberg Depression Rating Scale, MRM-MS, neuroleptic agent, nondepleted plasma, plasma protein, protein S 100, retention time, retinoid X receptor, Schizophrenia, synaptogenesis, T lymphocyte receptor, treatment duration, Young Mania Rating Scale

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Evaluation of a Novel Automated Method for the Quantification of Inflammation in Renal Biopsies

Item Type Conference Paper**Author** V. Mattioli**Author** A. Bocchi**Author** E. Martinelli**Author** C. Pala

Author M. Delsante

Author U. Maggiore

Author R. Raheli

Abstract The development of robust software-based technologies for objective image interpretation has facilitated the integration of Artificial Intelligence (AI) tools into digital pathology workflows for diagnostic applications. In this work, a novel automated method for quantifying inflammation in renal tissue biopsies is presented. To this purpose, the open-source QuPath software is exploited to design a custom classifier specifically trained to handle images stained with a novel combination of Periodic Acid-Schiff (PAS) and Leukocyte Common Antigen (LCA) techniques. This dual-staining approach enhances the visualization of both extracellular structures and inflammatory cells, enabling a more accurate assessment of the inflammation status. The performance of the presented quantification method is compared against measurements performed by two expert professionals, also referred to as observers, employing statistical metrics including the correlation coefficient and the normalized Mean Squared Error (MSE). For a thorough inspection, a Passing-Bablok regression analysis is also performed, showing a good agreement between outcomes obtained from the proposed method and one of the two observers. Discrepancies between the measurements performed by the two observers are also highlighted. © 2025 IEEE.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: V. Mattioli; University of Parma, Department of Engineering and Architecture, Parma, Italy; email: veronica.mattioli@unipr.it; A. Bocchi; University of Parma, Department of Engineering and Architecture, Parma, Italy; email: andrea.bocchi@unipr.it; R. Raheli; University of Parma, Department of Engineering and Architecture, Parma, Italy; email: riccardo.raheli@unipr.it; Conference name: 2025 International Workshop on Biomedical Applications, Technologies and Sensors, BATS 2025; Conference date: 27 October 2025 through 28 October 2025; Conference code: 219702

Evaluation of a Urine Pooling Strategy for the Rapid and Cost-Efficient Prevalence Classification of Schistosomiasis

Item Type Journal Article

Author N.C. Lo

Author J.T. Coulibaly

Author E. Bendavid

Author E.K. N’Goran

Author J. Utzinger

Author J. Keiser

Author I.I. Bogoch

Author J.R. Andrews

Abstract Background: A key epidemiologic feature of schistosomiasis is its focal distribution, which has important implications for the spatial targeting of preventive chemotherapy programs. We evaluated the diagnostic accuracy of a urine pooling strategy using a point-of-care circulating cathodic antigen (POC-CCA) cassette test for detection of *Schistosoma mansoni*, and employed simulation modeling to test the classification accuracy and efficiency of this strategy in determining where preventive chemotherapy is needed in low-endemicity settings. Methodology: We performed a cross-sectional study involving 114 children aged 6–15 years in six neighborhoods in Azaguié Ahoua, south Côte d’Ivoire to characterize the sensitivity and specificity of the POC-CCA cassette test with urine samples that were tested individually and in pools of 4, 8, and 12. We used a Bayesian latent class model to estimate test characteristics for individual POC-CCA and quadruplicate Kato-Katz thick smears on stool samples. We then developed a microsimulation model and used lot quality assurance sampling to test the performance, number of tests, and total cost per school for each pooled testing strategy to predict the binary need for school-based preventive chemotherapy using a 10% prevalence threshold for treatment. Principal Findings: The sensitivity of the urine pooling strategy for *S. mansoni* diagnosis using pool sizes of 4, 8, and 12 was 85.9%, 79.5%, and 65.4%, respectively, when POC-CCA trace results were considered positive, and 61.5%, 47.4%, and 30.8% when POC-CCA trace results were considered negative. The modeled specificity ranged from 94.0–97.7% for the urine pooling strategies (when POC-CCA trace results were considered negative). The urine pooling strategy, regardless of the pool size, gave comparable and often superior classification performance to stool microscopy for the same number of tests. The urine pooling strategy with a pool size of 4 reduced the number of tests and total cost compared to classical stool microscopy. Conclusions/Significance: This study introduces a method for rapid and efficient *S. mansoni* prevalence estimation through examining pooled urine samples with POC-CCA as an

alternative to widely used stool microscopy. © 2016 Lo et al.

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Export Date: 20 September 2026; Cited By: 13; Correspondence Address: N.C. Lo; Division of Infectious Diseases and Geographic Medicine, Stanford University School of Medicine, Stanford, United States; email: nathan.lo@stanford.edu

Evaluation of an automatic lean meat percentage quantification method based on a partial volume model from computed tomography scans

Item Type Journal Article

Author P. Xiberta

Author A. Bardera

Author I. Boada

Author M. Gispert

Author A. Brun

Author M. Font-i-Furnols

Abstract The quality of a pig carcass is mainly measured by the lean meat percentage (LMP), which can be virtually estimated from computed tomography (CT) scans. Different strategies exist to classify the CT voxels into tissues such as fat, lean and bone, being the thresholding-based methods the most commonly used. However, these methods are usually affected by the partial volume effect, and also by data variability, which is implicit from different CT scanners and protocols, since no standard behaviour has been defined. The aim of this paper is to extend an LMP quantification method which uses a partial volume model by adding a new step to detect the animal skin, and thoroughly evaluate the new approach by analysing each of its steps. The evaluation is performed by comparing the whole pipeline of the proposed approach with a simple thresholding method and a thresholding method with bone filling and skin detection, which is an intermediate step of the new pipeline. Five experiments have been designed to test how accurate are the results of the method regarding the LMP values computed from the manual dissection, as well as the robustness to data variability. Two different manual dissection methodologies have been tested: the partial dissection, which estimates the LMP using the lean of the four main cuts of the carcass plus the tenderloin, and the total dissection, which uses the lean of the twelve main cuts. A total of 146 half carcasses have been used for this study (105 using the partial dissection methodology, and 41 using the total dissection one). To evaluate the experiments, the LMP values virtually obtained from the three methods have been compared mostly with the LMP values from the manual dissection, computing the coefficient of determination R^2 from the correlations, as well as the root mean square error of prediction by means of leave-one-out cross-validation. A statistical analysis is performed to resolve if two correlations are significantly different. The experiments' results confirm the high accuracy of the proposed approach for the LMP estimation, and mainly its high robustness to data variability. The experiments also disclose that the detection of the animal skin and its classification as a new tissue, instead of classifying it as lean, improve the results. The evaluated method has demonstrated to be as effective as the thresholding method with bone filling and skin detection, and more robust to data variability than the other evaluated methods. © 2018 Elsevier B.V.

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Evaluation of an average numerator relationship matrix model and a Bayesian hierarchical model for growth traits in Nellore cattle with uncertain paternity

Item Type Journal Article

Author L. Shiotsuki

Author F.F. Cardoso

Author J.A.I.I.V. Silva

Author G.J.M. Rosa

Author L.G. Albuquerque

Abstract The objective of this work was to compare a model based on the use of an average numerator relationship matrix (ANRM) and a hierarchical animal model (HIER) to indicate the most appropriate statistical procedure to better estimate the genetic value of Nellore animals that have unknown paternity. The data set contained records of 62,212 Nellore animals. The pedigree file contained a total of 75,088 animals. Two approaches were adopted for the treatment of uncertain paternity. In the model based on the use of the ANRM probabilities were attributed to each of the possible parents of the animals with uncertain paternity. The other method adopted in the present study, i.e., the HIER, considers uncertainty in the assignment of paternity of animals participating in the multiple-sire (MS) system. Within this context, a priori probabilities are assigned to each possible sire of animals with uncertain paternity, which are altered according to information present in the data for the generation of posterior probabilities. Univariate analyses were carried out under Bayesian approach via Markov Chain Monte Carlo (MCMC) methods, implementing a chain of 400,000 rounds where the first 10,000 rounds were discarded (burn-in period). Models were compared by deviance information criteria (DIC) and pseudo Bayes factors (PBF). The model that best fits the data for estimating genetic parameter of animals with uncertain paternity is the Bayesian hierarchical model. Nevertheless, for genetic evaluation, the choice between these models would have no impact on genetic value classification of animals for selection. © 2011 Elsevier B.V.

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Evaluation of broken steel wire rope using magnetic flux leakage and optimised convolutional neural network

Item Type Journal Article

Author G. Li

Author B. Cao

Author Y. Zhou

Author M. Fan

Author L. Yang

Abstract Wire rope has become a common component in mines because of its advantages such as high strength and flexibility. Nevertheless, prolonged utilisation in severe environments can result in damage due to breakage. To address this problem, this paper proposes a damage quantification method for multi-objective optimised convolutional neural networks incorporating tiny damage signal enhancement. Firstly, the signal characteristics of wire rope breakage damage are analysed by constructing a magnetic dipole model and exploring the applicability of axial and radial signals for quantitative identification. Secondly, an algorithm to enhance the radial damage signal of a small broken wire is proposed, which can enhance the peak-to-peak value of 1 broken wire by about 2.4 times, and improve the detection ability of small broken wire damage. Finally, a convolutional neural network (CNN) was constructed, and the structural parameters of the network were optimised using the crown

porcupine optimisation (CPO) algorithm for quantifying wire rope breakage damage. The experimental results show that the model proposed in this study exhibits excellent performance in terms of accuracy, precision, recall, and F1 score compared to multiple models. The accuracy is augmented to 99.17%, thereby enhancing the quantisation precision of broken wire defects. © 2025 Informa UK Limited, trading as Taylor & Francis Group.

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Evaluation of method bias for determining bacterial populations in bacterial community analyses

Item Type Journal Article

Author M. Takahashi

Author Y. Kita

Author A. Mizuno

Author N. Goto-Yamamoto

Abstract Various methods are used for analyzing a bacterial community. We recently developed a method for quantifying each bacterium constituting the microbiota by combining a next-generation sequencing (NGS) analysis with a quantitative polymerase chain reaction (NGS-qPCR) assay. Our NGS-qPCR method is useful for analyzing a comprehensive bacterial community because it enables the easy calculation of the amounts of each bacterium constituting the microbiota. However, it has not been confirmed whether the estimated bacterial community obtained using this NGS-qPCR method corresponds to the results obtained using conventional methods. Accordingly, we prepared model bacterial community samples and analyzed them by several methods (NGS-qPCR, species-specific qPCR, flow cytometry, total direct counting by epifluorescent microscopy [TDC], and plate count). The total bacterial cell densities determined by the PCR-based methods were largely consistent with those determined by the TDC method. There was a difference between the amounts of each bacterium analyzed by NGS-qPCR and species-specific qPCR, although the same trend was shown by both species-specific qPCR and NGS-qPCR. Our findings also demonstrated that there is a strong positive correlation between the cell densities of a specific bacterial group in craft beer samples determined by group-specific qPCR and NGS-qPCR, and there were no significant differences among quantification methods (we tested two bacterial groups: lactic acid bacteria and acetic acid bacteria). Thus, the NGS-qPCR method is a practical method for analyzing a comprehensive bacterial community based on a bacterial cell density. © 2017 The Society for Biotechnology, Japan

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Article, Bias, quantitative analysis, nonhuman, procedures, controlled study, Chains, classification, Beer, intermethod comparison, evaluation study, Bacteria, bacterium, flow cytometry, genetics, high throughput sequencing, High-Throughput Nucleotide Sequencing, isolation and purification, microbial community, polymerase chain reaction, Quantification methods, microbiology, cytology, metabolism, next generation sequencing, Quantification method, real time polymerase chain reaction, Real-Time Polymerase Chain Reaction, statistical bias, Polymerase chain reaction, cell density, mathematical computing, bacterial strain, lactic acid, bacterial count, acetic acid, Acetic Acid, acetic acid bacterium, bacterial cell, Bacterial community analysis, Bacterial population, beer, Colony Count, Microbial, epifluorescence microscopy, Lactic acid, Lactic Acid, lactic acid bacterium, Next-generation sequencing-quantitative polymerase chain reaction, plate count, Quantitative polymerase chain reaction, Relative abundance

Notes:

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Export Date: 20 September 2026; Cited By: 11; Correspondence Address: M. Takahashi; National Research Institute of Brewing (NRIB), Higashi-Hiroshima, 3-7-1 Kagamiyama, 739-0046, Japan; email: m.takahashi@nrrib.go.jp; CODEN: JBBIF

Evaluation of novel data-driven metrics of amyloid β deposition for longitudinal PET studies

Item Type Journal Article

Author A. Bollack

Author P.J. Markiewicz

Author A.M. Wink

Author L. Prosser

Author J. Lilja

Author P. Bourgeat

Author J.M. Schott

Author W. Coath

Author L.E. Collij

Author H.G. Pemberton

Author G. Farrar

Author F. Barkhof

Author D.M. Cash

Abstract Purpose: Positron emission tomography (PET) provides in vivo quantification of amyloid- β (A β) pathology. Established methods for assessing A β burden can be affected by physiological and technical factors. Novel, data-driven metrics have been developed to account for these sources of variability. We aimed to evaluate the performance of four of these amyloid PET metrics against conventional techniques,

using a common set of criteria. Methods: Three cohorts were used for evaluation: Insight 46 (N=464, [18F]florbetapir), AIBL (N=277, [18F]flutemetamol), and an independent test-retest data (N=10, [18F]flutemetamol). Established metrics of amyloid tracer uptake included the Centiloid (CL) and where dynamic data was available, the non-displaceable binding potential (BPND). The four data-driven metrics computed were the amyloid load (A β load), the A β -PET pathology accumulation index (A β index), the Centiloid derived from non-negative matrix factorisation (CLNMF), and the amyloid pattern similarity score (AMPSS). These metrics were evaluated using reliability and repeatability in test-retest data, associations with BPND and CL, variability of the rate of change and sample size estimates to detect a 25% slowing in A β accumulation. Results: All metrics showed good reliability. A β load, A β index and CLNMF were strong associated with the BPND. The associations with CL suggest that cross-sectional measures of CLNMF, A β index and A β load are robust across studies. Sample size estimates for secondary prevention trial scenarios were the lowest for CLNMF and A β load compared to the CL. Conclusion: Among the novel data-driven metrics evaluated, the A β load, the A β index and the CLNMF can provide comparable performance to more established quantification methods of A β PET tracer uptake. The CLNMF and A β load could offer a more precise alternative to CL, although further studies in larger cohorts should be conducted. © 2023

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Evaluation of the Effects of Climate Change on Surface Water Resources at the Ain Kwachia Dam Using Machine Learning Between 2008 and 2021

Item Type Conference Paper

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Author M. Batchi

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Author J. El Bouziani

Editor Rebai N.

Editor Moumen A.

Editor El Bouhaddioui M.

Abstract Surface water resources are subject to climate change, particularly in arid zones such as Morocco's central plateau, and to the massive demand placed on water by human activities, which calls for equitable management of water potential, In this sense, hydraulic engineering is a solution for better storage and management of surface water, as well as for mitigating and adapting to climate variability. This is the aim of the Ain Kwaachia dam on the Ykem river in the northwestern part of the central plateau of Morocco. While the use of remote sensing technology, based on the acquisition of satellite images, is a kind of machine learning to quantify spatio-temporal variations in water potential, so the processing and the analysis of these image data, based on a GIS geographic information system, and this, from the Land Use index for land cover, with an unsupervised classification from 2008, the year of commissioning of the dam subject of this research, and a supervised classification around the year 2021. The results obtained show that the variations affecting the study area have been noticeable in recent years, from 9.78 km² for 2016, to 4.23 km² for 2021, as a result of the repercussions of climate change, with falling rainfall on the one hand, and in the other hand rising temperatures. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Evaluation of uncertainty quantification methods in multi-label classification: A case study with automatic diagnosis of electrocardiogram

Item Type Journal Article

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Author L. Famiglini

Author A. Campagner

Author D. Folgado

Author R. Simão

Author F. Cabitza

Author H. Gamboa

Abstract Artificial Intelligence (AI) use in automated Electrocardiogram (ECG) classification has continuously attracted the research community's interest, motivated by their promising results. Despite their great promise, limited attention has been paid to the robustness of their results, which is a key element for their implementation in clinical

practice. Uncertainty Quantification (UQ) is a critical for trustworthy and reliable AI, particularly in safety-critical domains such as medicine. Estimating uncertainty in Machine Learning (ML) model predictions has been extensively used for Out-of-Distribution (OOD) detection under single-label tasks. However, the use of UQ methods in multi-label classification remains underexplored. This study goes beyond developing highly accurate models comparing five uncertainty quantification methods using the same Deep Neural Network (DNN) architecture across various validation scenarios, including internal and external validation as well as OOD detection, taking multi-label ECG classification as the example domain. We show the importance of external validation and its impact on classification performance, uncertainty estimates quality, and calibration. Ensemble-based methods yield more robust uncertainty estimations than single network or stochastic methods. Although current methods still have limitations in accurately quantifying uncertainty, particularly in the case of dataset shift, incorporating uncertainty estimates with a classification with a rejection option improves the ability to detect such changes. Moreover, we show that using uncertainty estimates as a criterion for sample selection in active learning setting results in greater improvements in classification performance compared to random sampling. © 2023 The Author(s)

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Evidence Mapping of ctDNA Reporting in Pancreatic Ductal Adenocarcinoma: Toward a Shared Quantitative Language for ctDNA

Item Type Journal Article

Author D. Croagh

Author S. Aslani

Abstract Background/Objectives: Circulating tumour DNA (ctDNA) assays enable non-invasive assessment of tumour burden and treatment response in oncology. However, quantitative ctDNA outputs (such as variant allele frequency, tumour fraction, and aggregate burden scores) remain difficult to interpret and compare across platforms. This evidence-mapping review evaluates current quantitative reporting approaches in pancreatic ductal adenocarcinoma (PDAC) and examines the potential role of KRAS mutant ctDNA as a biologically grounded reference metric. Methods: A systematic literature search was conducted across PubMed/MEDLINE and Scopus to identify studies reporting quantitative ctDNA metrics in PDAC. Eligible studies included those measuring plasma KRAS mutations and/or reporting variant allele frequency, tumour fraction, or multi-locus aggregate metrics. Additional relevant primary studies identified through broader manual searching of PubMed were assessed against the same prespecified eligibility and classification criteria before inclusion. Data were synthesised narratively, focusing on reporting frameworks, units of measurement, assay characteristics, and the interpretability of quantitative outputs across platforms. Results: Substantial heterogeneity was observed in ctDNA quantification methods and reporting standards. Ratio-based metrics such as variant allele frequency and tumour fraction were commonly used but varied according to assay design, plasma input volume, and background cell-free DNA levels. Few studies reported absolute mutant molecule counts per unit volume. Given that approximately 90–95% of PDACs harbour truncal activating KRAS mutations, plasma KRAS was consistently represented across platforms and demonstrated potential as a shared quantitative anchor. Limited standardisation was noted in distinguishing detectability from quantifiability based on sampling depth and counting statistics. Conclusions: Current ctDNA reporting in PDAC lacks a shared quantitative reference, limiting cross-study comparability. Reporting KRAS mutant molecules per millilitre and adopting an assay-agnostic framework distinguishing detection from quantification may improve interpretability, support harmonisation across platforms, and facilitate cumulative learning in pancreatic cancer ctDNA research. © 2026 by the authors.

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Evidence-based medicine and electrophysiology in schizophrenia

Item Type Journal Article

Author S. Galderisi

Author A. Mucci

Author U. Volpe

Author N. Boutros

Abstract In research on schizophrenia electrophysiological measures have been investigated to identify biomarkers of the disorder, indices enabling differential diagnosis among psychotic disorders, prognostic indicators or endophenotypes. The present systematic review will focus on the most largely studied electrophysiological indices, i.e., qualitative or quantitative (limited to spectral analysis) EEG and the P300 event-related potential. The PubMed clinical query was used with research methodology filters for each of the following categories: diagnosis/prognosis/aetiology and a broad sensitive search strategy. The key-words: SCHIZOPHRENIA AND EEG/P3/P300 were used. The search results were then narrowed by including the terms "human" and "English language", and cross-referenced. Systematic reviews and meta-analyses, when available, were also used for cross-referencing. Case reports and studies irrelevant to the topics and methodologies under examination were excluded. The remaining papers were screened to verify the eligibility for this systematic review. Inclusion criteria were: a) a diagnosis of schizophrenia confirmed by DSM-III/ICD-9 criteria (or later editions of the same classification systems); b) the inclusion of both a schizophrenia study group and an healthy control group (when appropriate, i.e., for P300 and quantitative EEG); c) qualitative or spectral EEG findings and amplitude measures for P300. The included studies were then reviewed to verify homogeneity of the results, as well as the presence of the information needed for the present systematic review and meta-analysis. Previous reviews and studies meeting the above requirements (n=22 for qualitative EEG; n=45 for spectral EEG and n=132 for P300) were classified according to the Oxford Centre for Evidence-based Medicine (EBM) levels of evidence criteria. For qualitative EEG as a diagnostic test, the majority of studies predated the introduction of DSM-III and were excluded from the review. Few post DSM-III studies investigated the usefulness of qualitative EEG in the differential

diagnosis between schizophrenia and psychosis due to general medical condition. None of them was Oxford CEBM level 3b (non-consecutive-study or cohort-study without consistently-applied reference standard) or better (exploratory or validating cohort-study). No meta-analysis could be conducted due to the lack of reliable quantification methods in the reviewed studies. For spectral EEG as a diagnostic test, most studies qualified as level 4 (case-control study with poor reference standard), and only 24% as level 3b or better. An increase of slow activity in patients is reported by most of these studies. As to meta-analyses examining 29 studies, with 32 independent samples for the delta band and 35 for the theta band, a moderate effect size was found and only 1 study yielded findings in the opposite direction for both measures. There was no identified source for the discrepancy. The analysis of moderator factors included medication, band frequency limits, spectral parameters and disease stage. The medication status was significant for the theta band but the effect was unclear as findings for drug-naïve and drug-free patients were in a different direction. Chronicity had a significant effect on both delta and theta bands, with slow activity increase larger in chronic than in first episode patients. For P3 amplitude reduction as a diagnostic index, 63% of the studies qualified as level 3b or better. Meta-analysis (52 studies, 60 independent samples) results demonstrated a large effect size. None of the studies reported opposite findings. The analysis of moderator factors, including medication status and disease stage, revealed no significant effect on data heterogeneity. In conclusion, the examined indices are good candidates but are not ready yet for clinical applications aimed to improve present diagnostic standards for schizophrenia. Further research carried out according to adequate methodological standards and based on large scale multi-center studies is mandatory.

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Evidential deep learning for interatomic potentials

Item Type Journal Article

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Author J. Ma

Author D. Zhou

Author Y. Li

Author X. Gao

Author X. Gong

Author W. Ouyang

Author S. Zhang

Author M. Su

Abstract Machine learning interatomic potentials have been widely used to facilitate large-scale molecular simulations with accuracy comparable to ab initio methods. To ensure the reliability of the simulation, the training dataset is iteratively expanded through active learning, where uncertainty serves as a critical indicator for identifying and collecting out-of-distribution data. However, existing uncertainty quantification methods tend to involve either expensive computations or compromise prediction accuracy. Here we show an evidential deep learning framework for interatomic potentials with a physics-inspired design. Our method provides uncertainty quantification without significant computational overhead or decreased prediction accuracy, consistently outperforming other methods across a variety of datasets. Furthermore, we demonstrate applications in exploring diverse atomic configurations, using examples including water and universal potentials. These results highlight the potential of our method as a robust and efficient alternative for uncertainty quantification in molecular simulations. © The Author(s) 2025.

Date 2026

DOI 10.1038/s41467-025-67663-y**URL** <https://www.scopus.com/pages/publications/105028491388?origin=resultslist>**Archive** Scopus**Language** English**Volume** 17**Publisher** Nature Research**Publication** Nature Communications**ISSN** 20411723 (ISSN)**Issue** 1**Journal Abbr** Nat. Commun.**Date Added** 20/09/2026, 9:51:35 pm**Modified** 20/09/2026, 9:51:35 pm**Tags:**

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Evidential deep learning to quantify classification uncertainty

Item Type Conference Paper**Author** M. Sensoy**Author** L. Kaplan**Author** M. Kandemir**Editor** Larochelle H.**Editor** Grauman K.**Editor** Cesa-Bianchi N.**Editor** Garnett R.**Editor** Grauman K.**Editor** Wallach H.

Editor Bengio S.

Abstract Deterministic neural nets have been shown to learn effective predictors on a wide range of machine learning problems. However, as the standard approach is to train the network to minimize a prediction loss, the resultant model remains ignorant to its prediction confidence. Orthogonally to Bayesian neural nets that indirectly infer prediction uncertainty through weight uncertainties, we propose explicit modeling of the same using the theory of subjective logic. By placing a Dirichlet distribution on the class probabilities, we treat predictions of a neural net as subjective opinions and learn the function that collects the evidence leading to these opinions by a deterministic neural net from data. The resultant predictor for a multi-class classification problem is another Dirichlet distribution whose parameters are set by the continuous output of a neural net. We provide a preliminary analysis on how the peculiarities of our new loss function drive improved uncertainty estimation. We observe that our method achieves unprecedented success on detection of out-of-distribution queries and endurance against adversarial perturbations. © 2018 Curran Associates Inc. All rights reserved.

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Evidential Fuzzy Rule-Based Machine Learning to Quantify Classification Uncertainty

Item Type Conference Paper

Author H. Shiraishi

Author H. Ishibuchi

Author M. Nakata

Editor Ochoa G.

Abstract Learning Fuzzy-Classifier Systems (LFCSSs), also known as evolutionary fuzzy rule-based machine learning, combine evolutionary algorithms with fuzzy rules to create interpretable models. However, traditional inference schemes lack mechanisms to quantify uncertainty in predictions. This paper introduces a novel class inference scheme based on the Dempster-Shafer Theory of Evidence that explicitly models epistemic uncertainty through an “I don’t know” state. By calculating belief masses for each class hypothesis and uncertainty, our approach enhances robustness in real-world applications. Experiments on real-world datasets demonstrate statistically significant improvements in classification performance compared to conventional approaches. Our method forms smoother decision boundaries and provides quantitative measures of prediction confidence, addressing a critical gap in reliable decision-making for fuzzy rule-based machine learning. © 2025 Copyright held by the owner/author(s).

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Evolutionary optimization of weights of a neuro-fuzzy classifier and the effects on benchmark data and complex chemical data

Item Type Conference Paper**Author** J. Paetz

Abstract The integral part of adaptive systems are learning algorithms that are often based on heuristics. Such algorithms are used for neural network, fuzzy system and neuro-fuzzy system training. The performance can be measured on benchmark data. In this contribution we evaluate the performance of a neuro-fuzzy system with respect to the adapted weights. A comparison is given between the performance using the trained set of weights and two sets of optimized weights. Evolutionary algorithms can be used for optimizing the trained weights and for optimizing the weights directly without using the adapted weights as a basis. Additionally, all weights are set to one to measure the influence of the weight values. The results show that the weights have only a small impact on performance using benchmark data, but a high impact when using more complex, higher dimensional chemical data with skewed a priori probabilities. © 2005 IEEE.

Date 2005**DOI** 10.1109/NAFIPS.2005.1548608**URL** <https://www.scopus.com/pages/publications/33744966602?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Annu Conf North Am Fuzzy Inf Process Soc NAFIPS**Volume** 2005**Pages** 615-620**Proceedings Title** Annu Conf North Am Fuzzy Inf Process Soc NAFIPS**Conference Name** Annual Conference of the North American Fuzzy Information Processing Society - NAFIPS**Date Added** 20/09/2026, 9:53:05 pm**Modified** 20/09/2026, 9:53:05 pm**Tags:**

Classification (of information), Learning algorithms, Benchmarking, Neural networks, Optimization, Adaptive systems, Evolutionary algorithms, Heuristic methods, Fuzzy sets, Benchmark data, Neuro-fuzzy system, Evolutionary optimization, Neuro-fuzzy classifier

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: J. Paetz; Department of Chemical and Pharmaceutical Sciences, J. W. Goethe-University Frankfurt am Main, 60439 Frankfurt am Main, Germany; email: paetz@bioinfonnatik.uni-fraikfurt.de; Conference name: NAFIPS 2005 - 2005 Annual Meeting of the North American Fuzzy Information Processing Society; Conference date: 26 June 2005 through 28 June 2005; Conference code: 67328

Evolving Intelligent Network Attack Classifier Under Label Distribution Shift

Item Type Journal Article**Author** M. Kim**Author** M. Joe**Author** M. Kwon

Abstract The next-generation internet is being reshaped by the growing intelligence and connectivity of artificial intelligence of things (AIoT) devices. This evolution emphasizes the importance of AI-enabled network intrusion detection systems (NIDS) in intelligent network environments. Two major technical challenges in developing such systems are addressing class imbalance and adapting to label distribution shifts after deployment. Class imbalance, caused by the dominance of normal traffic and the scarcity and uneven distribution of attack types, often results in low performance for minority attack type classes. In addition, evolving attack patterns necessitate efficient and continuous model post-training, despite the limited and unlabeled data available in the post-deployment phase. This study proposes a unified training framework consisting of a two-step pre-training scheme and an adaptive post-training scheme. In Step I of pre-training, balanced contrastive pair selection is performed to account for class imbalance, while Step II refines the representations of hard-to-distinguish samples located near decision boundaries using latent features. In the post-training phase, the framework leverages both model outputs and latent representations to generate reliable pseudo-labels for learning. A key insight of this work is that a well-structured latent representation space enables reliable pseudo-labeling during post-training. Extensive simulations across eight online label shift scenarios with three datasets demonstrate that the proposed method achieves up to an 8.6% improvement in accuracy and F1-score over eight state-of-the-art approaches, along with a significant reduction in post-training complexity. © 2026 IEEE.

Date 2026**DOI** 10.1109/TNSE.2026.3669948**URL** <https://www.scopus.com/pages/publications/105031846234?origin=resultslist>**Archive** Scopus**Language** English**Volume** 13**Publisher** IEEE Computer Society**Pages** 7448-7464**Publication** IEEE Transactions on Network Science and Engineering**ISSN** 23274697 (ISSN)

Journal Abbr IEEE Trans. Netw. Sci. Eng.

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Expand and Shrink: Federated Learning with Unlabeled Data Using Clustering

Item Type Journal Article

Author A. Kumar

Author A.K. Singh

Author S.S. Ali

Author B.J. Choi

Abstract The amalgamation of the Internet of Things (IoT) and federated learning (FL) is leading the next generation of data usage due to the possibility of deep learning with data privacy preservation. The FL architecture currently assumes labeled data samples from a client for supervised classification, which is unrealistic. Most research works in the literature focus on local training, update receiving, and global model updates. However, by principle, the labeling must be performed on the client side because the data samples cannot leave the source under the FL principle. In the literature, a few works have proposed methods for unlabeled data for FL using “class-prior probabilities” or “pseudo-labeling”. However, these methods make either unrealistic or uncommon assumptions, such as knowing class-prior probabilities are impractical or unavailable for each classification task and even more challenging in the IoT ecosystem. Considering these limitations, we explored the possibility of performing federated learning with unlabeled data by providing a clustering-based method of labeling the sample before training or federation. The proposed work will be suitable for every type of classification task. We performed different experiments on the client by varying the labeled data ratio, the number of clusters, and the client participation ratio. We achieved accuracy rates of (Formula presented.) and (Formula presented.) by using (Formula presented.) and (Formula presented.) of the truth

labels, respectively. © 2023 by the authors.

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Experiential Equity: Reconstructing Cities Through Intersecting Theories

Item Type Journal Article

Author R. Mazumder

Abstract Considering the unprecedented rate of global urbanisation, attention must be paid to the implications of city environments on human health. Urban infrastructures, like parks, greenspace, active transportation, and other forms of public space, are seen as health amenities. The concept of distributional equity acknowledges the disparities in

access to these health-supporting spaces; however, more granularity is needed in understanding how these spaces themselves are experienced, and the impact of marginalisation and other forms of oppression on the health benefits must be examined. Experiential equity is a framework that can be used to conceptualise and quantify the energetic cost of experiencing unpredictable urban environments, using the free energy principle to unite the energetic costs of the emotional and cognitive burden of marginalisation, and views these costs through an Intersectional Theoretical lens to establish the energetic disparities that exist depending on aspects of identity. The cognitive and emotional costs of uncertainty in urban environments can be understood through the Theory of Constructed Emotion, the model is also informed by environmental prediction and provides a link to the psychophysiology of Racism-Related Vigilance and Stereotype Threat. Applying the Person-Environment-Occupation Model and Affordance Theory, in conjunction, we can conceptualise the environment and examine the ways in which it can constrain or enable engagement with it. Leveraging machine learning to quantify aspects of the built environment and using neuroscientific methodologies like electroencephalography, electrodermal activity, and heart rate variability, we can attempt to metricise the energetic costs of societal and systemic oppression; the consideration of the environment allows potential insights into how spatial variables mediate these experiences and can have implications on road, park, and general urban policy and design. © The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2024, Corrected Publication 2024, 2025.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: R. Mazumder; Future Cities Institute, School of Planning, University of Waterloo, Waterloo, Canada; email: robin.mazumder@uwaterloo.ca

Experimental analysis of feature selection stability for high-dimension and low-sample size gene expression classification task

Item Type Conference Paper

Author D. Dernoncourt

Author B. Hanczar

Author J.-D. Zucker

Abstract Gene selection is a crucial step when building a classifier from microarray or metagenomic data. As the number of observations is small, the gene selection tends to be unstable. It is common that two gene subsets, obtained from different datasets but dealing with the same classification problem, do not overlap significantly. Although it is a crucial problem, few works have been done on the selection stability. In this paper, we first present some stability quantification methods, then we study the variations of those measures with various parameters (dimensionality, sample size, feature distribution, selection threshold) on both artificial and real data, as well as the resulting classification performance. Feature selection was performed with t-test and classification with linear discriminant analysis. We point out a strong empiric correlation between the dimensionality/sample size ratio and selection instability. © 2012 IEEE.

Date 2012

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Feature extraction, Classification (of information), Classification performance, Bioinformatics, Feature selection, Convergence of numerical methods, Quantification methods, Feature distribution, Sample sizes, Stability, Gene expression, Linear discriminant analysis, Small samples, Data sets, dimensionality/sample size ratio, Experimental analysis, Gene expression classification, Gene selection, Size ratio, small sample, stability

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Experimental Demonstration for Learning Analytics and Data Mining on e-assessment

Item Type Conference Paper

Author S. Bhurre

Author S. Prajapat

Author S. Raikwar

Author P. Goswami

Author S. Kothari

Author G. Khanuja

Author K. Yadav

Author P. Choure

Editor Naik N.

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Editor Prajapat S.

Abstract Every subject and course has a learning goal that we work towards during the teaching-learning process, whether it be online or offline. Under the current system, assignments, project analysis, semester exams, and internal exams are used to evaluate students' learning. Several factors, including participation in class, prior knowledge, engagement time, activity logs, forum posts, understanding level, etc., influence how well students learn throughout a course or session. All these factors will be difficult to analyze in offline mode. All of these activities determine whether or not the learning objective is met. This study examines an experiment that provided blended learning data on students' academic details, learning behaviors, and evaluation modules to determine the success of the two subjects. It discussed the details of each phase involved in the experiment. Students enrolled in integrated courses, MTech(PG-1) and MCA(PG-2) were analyzed in this study procedure through 4 weeks of offline classes. Moodle Gnomio was used for the online processes related to the Registration process, Quizzes, and discussion forums. A total of 6 quizzes for PG-1(M-Tech) and PG-2(MCA) were organized, and learning resources were provided before the exam. This study discusses detailed experiment setup, and Statistical analysis involving difficulty level of quizzes. This study proposed a framework for the dashboard that utilize this dataset and involves analysis related to student performance, grade distribution, inferring learning patterns, and course success. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Explainable Deep Learning Framework for Classification of Total Hip Arthroplasty Gaits

Item Type Conference Paper
Author R. Pantonial
Author M. Simic
Editor Chen Y.-W.
Editor Howlett R.J.
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Abstract Application of clinical gait analysis (CGA) in the evaluation of Hip Osteoarthritis (HOA) is a crucial step in the identification of functional abnormalities and surgical planning through total hip arthroplasty (THA). Measured kinematic

parameters provide understanding of the gait patterns that can be used for accurate diagnostics and personalized rehabilitation. Streamlined approach is essential for efficient assessment due to the substantial amount of data generated. This study aims to explore the use of Deep Learning (DL) methodology in the classification of gait patterns before and after THA. All kinematic parameters of both lower limbs are aggregated, and a 2-dimensional data representation is acquired as input to the DL model. The best classification accuracy is 98.2% from the test subset. Consequently, model prediction is explained through Local Interpretable Model-Agnostic Explanation (LIME) to determine input features that best distinguish between gait classes. Features are ranked in terms of their relevancy, and only the most common discriminating features are considered for further analysis. Generally, the sagittal hip angle is the most significant kinematic parameter across classes, albeit from different gait cycle events. All healthy and 75% of the HOA gaits in the test subset exhibit discrimination on the single limb support towards the swing phase. The sagittal angles of the knee and ankle during the same gait events are also discriminatory for HOA with 72% of the gaits exhibiting such feature. Additionally, the same kinematic angles are important for THA class prior to heel rise of the stance phase with 85% of the gaits examined. With features acquired directly from the input data, explanation is more transparent promoting trust and understanding of the DL model. High accuracy scores coupled with explainability showcases clinical relevance and a potential for aiding medical experts in crucial decision making for HOA patients before and after THA. © 2025 Elsevier B.V.. All rights reserved.

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Explainable long-term building energy consumption prediction using QLattice

Item Type Journal Article

Author S. Wenninger

Author C. Kaymakci

Author C. Wiethe

Abstract The global building sector is responsible for nearly 40% of total carbon emissions, offering great potential to move closer to set climate goals. Energy performance certificates designed to increase the energy efficiency of buildings require accurate predictions of building energy performance. With significant advances in information and communication technology, data-driven methods have been introduced into building energy performance research demonstrating high computational efficiency and prediction performance. However, most studies focus on prediction performance without considering the potential of explainable artificial intelligence. To bridge this gap, the novel QLattice algorithm, designed to satisfy both aspects, is applied to a dataset of over 25,000 German residential buildings for predicting annual building energy performance. The prediction performance, computation time, and explainability of the QLattice is compared to the established machine learning algorithms artificial neural network, support vector regression, extreme gradient boosting, and multiple-linear regression in a case study, variable importance analyzed, and appropriate applications proposed. The results show quite strongly that the QLattice should be further considered in the research of energy performance certificates and may be a potential alternative to established machine learning algorithms for other prediction tasks in energy research. © 2021 Elsevier Ltd

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Explainable machine learning to quantify the value of proximal remote sensing in latent energy flux estimation

Item Type Journal Article

Author S. Gaur

Author G. Aslan-Sungur (Rojda)

Author A. VanLoocke

Author D.T. Drewry

Abstract Proximal remote sensing has the potential to provide critical information on vegetation biophysical factors that can predict land-atmosphere exchange of water and energy. Latent energy (LE) flux is traditionally estimated using process-based models which rely on vegetation parameters that change during the growing season. Data-driven models have the potential to address these issues by offering flexible predictor selection and more efficient utilization of the information in predictor sets. These models require careful choice of predictors to avoid redundancy and allow robust cross-validation. In this study we present a systematic and comprehensive evaluation of machine learning (ML) models to assess the capability of meteorological and proximal sensing data for predicting LE at a half-hourly temporal resolution across multiple growing seasons for an agricultural system. The results presented here demonstrate that a model using four environmental predictors in combination with two proximal sensing variables can capture 88 % of the variability in LE. ML models using only three predictors (one meteorological and two proximal remote sensing) captured 81 % of LE variability, offering the best trade-off between performance and complexity. An ML model utilizing only two predictors, one proximal remote sensing variable and downwelling radiation, captured 77 % of LE variability. These results demonstrate the power of proximal remote sensing and meteorological observations to estimate land-atmosphere water vapor exchange, providing a solution where more direct methods such as eddy covariance are not available and for evaluations of agronomic management and genotypic variations. © 2025 The Authors

Date 2025**DOI** 10.1016/j.agwat.2025.109643**URL** <https://www.scopus.com/pages/publications/105010685183?origin=resultslist>**Archive** Scopus**Language** English**Volume** 317**Publisher** Elsevier B.V.**Publication** Agricultural Water Management**ISSN** 03783774 (ISSN)**Journal Abbr** Agric. Water Manage.**Date Added** 20/09/2026, 9:51:51 pm**Modified** 20/09/2026, 9:51:51 pm**Tags:**

Economic and social effects, Learning systems, machine learning, Machine-learning, Machine learning, Remote sensing, Remote-sensing, Machine learning models, evapotranspiration, remote sensing, Vegetation, trade-off, Agricultural machinery, Explainable machine learning, Cultivation, biophysics, downwelling, eddy covariance, energy balance, energy flux, Energy fluxes, Evapotranspiration, Growing season, Interfacial energy, Latent energy flux, Proximal remote sensing, Surface energy balance, Vegetation biophysic, Vegetation biophysics, water vapor

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Explainable multi-sensor geospatial modeling of forest biomass dynamics under compound climate extremes in subtropical monsoon regions

Item Type Journal Article**Author** B. Xu**Author** T. Dang**Author** X. Wu

Abstract Climate extremes are intensifying worldwide, yet the mechanisms by which forest biomass responds to compound climatic and anthropogenic pressures remain poorly resolved. Here, we integrate multi-sensor remote sensing with explainable machine learning to quantify the interactions between multiple drivers and changes in aboveground biomass density (Δ AGBD) across a subtropical monsoon region of China during 2000–2019. Annual Δ AGBD maps were derived from Landsat and GEDI, and hotspots of climatic extremes were delineated using ETCCDI (Expert Team on Climate Change Detection and Indices). Across 92 predictors, attribution indicates that—even within areas exposed to extremes—AGBD is not driven by extreme events alone. Bioclimatic and extreme-climate variables dominate overall variability, whereas topography and human disturbance strongly modulate their effects. Gradient analyses further show that climatic influences intensify with elevation and heat load, whereas anthropogenic impacts remain pronounced at low population densities and in areas distant from impervious surfaces. Collectively, these findings demonstrate that biomass dynamics under climatic extremes do not arise from a single climatic forcing but from the mutual regulation of climate, topography, and human pressure. This nonlinear, compound-mechanism framework provides a transferable basis for assessing ecosystem vulnerability and designing adaptive, resilience-oriented management strategies under intensifying climate extremes. © 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

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Explainable uncertainty quantifications for deep learning-based molecular property prediction

Item Type Journal Article

Author C.-I. Yang

Author Y.-P. Li

Abstract Quantifying uncertainty in machine learning is important in new research areas with scarce high-quality data. In this work, we develop an explainable uncertainty quantification method for deep learning-based molecular property prediction. This method can capture aleatoric and epistemic uncertainties separately and attribute the uncertainties to atoms present in the molecule. The atom-based uncertainty method provides an extra layer of chemical insight to the estimated uncertainties, i.e., one can analyze individual atomic uncertainty values to diagnose the chemical component that introduces uncertainty to the prediction. Our experiments suggest that atomic uncertainty can detect unseen chemical structures and identify chemical species whose data are potentially associated with significant noise. Furthermore, we propose a post-hoc calibration method to refine the uncertainty quantified by ensemble models for better confidence interval estimates. This work improves uncertainty calibration and provides a framework for assessing whether and why a prediction should be considered unreliable. Graphical Abstract: [Figure not available: see fulltext.]. © 2023, The Author(s).

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Explanations and a new objective function of the minimum error thresholding based on the Rayleigh distribution

Item Type Journal Article

Author B. Lei

Author Y. Ma

Author L. He

Author Z. Yang

Author Y. Tang

Abstract The minimum error thresholding (MET) method is based on the Bayes minimum classification error principle. Under different assumptions of the initial grayscale distribution of an image, MET criterion functions based on the mixed Gaussian distribution, mixed Poisson distribution, and shifted Rayleigh distribution have been proposed. Mathematical explanations have been presented for the MET criterion functions of mixed Gaussian and Poisson distributions from relative entropy and the joint maximum likelihood principle. However, the explanations for the MET criterion function of the shifted Rayleigh distribution (RMET) have not been presented. In this paper, the detailed joint maximum likelihood principle and relative entropy interpretations of the RMET method were presented, and the principles for calculating the class parameters in the RMET method were also explained. Moreover, since the class prior probability is unknown, it is more reasonable to use conditional likelihood estimation. Therefore, a new MET method of the shifted Rayleigh distribution is proposed based on the conditional maximum likelihood function. Performance tests on synthetic images, synthetic aperture radar (SAR) images, and natural images in the experiments confirm the effectiveness of the proposed new MET algorithm. © 2025 Elsevier Ltd

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Distribution functions, Gaussian distribution, Barium compounds, Errors, Maximum principle, Criterion functions, Image thresholding, Joint maximum likelihood, Likelihood function, Likelihood functions, Maximum entropy methods, Maximum likelihood estimation, Minimum error thresholding, Mixed Gaussian distributions, Mixed poisson distributions, Poisson distribution, Rayleigh distribution, Rayleigh distributions, Rayleigh number, Rayleigh scattering, Relative entropy, Thresholding methods

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Exploiting Feature Interactions for Malicious Website Detection with Overhead-accuracy Tradeoff

Item Type Conference Paper

Author S. Shen

Author C. Yu

Author K. Zhang

Author S. Ci

Abstract Malicious websites attempt to install malware on user's devices without permission, which can disrupt device operation, steal personal information, and even acquire access to the device for future attacks. Accurate detection of malicious website behaviors is crucial for network security but still faces challenges. Firstly, various types and semantics of website features are required to identity the wide range of malicious characteristics, leading to massive training data and computational overhead. Secondly, to reduce model dimensionality, a proper selection of website features is essential but difficult due to the complex relations among features that can affect each other's contribution to detection outcomes. In this paper, we propose a lightweight feature-based detection scheme against malicious websites considering the interaction measures among features and the overhead-accuracy tradeoff. Specifically, we systematically characterize the interactions among website features in a non-additive manner to indicate the aggregated impacts of feature subsets. Then we propose a quantification method to measure the feature interactions based on multivariate regression. With this method, important features are selected to substantially reduce the model dimension and computational complexity while maintaining desirable accuracy. Meanwhile, the proposed scheme provides an interpretable model that preserves the physical meanings of original features. It allows users to balance the overhead-accuracy tradeoff for detection model training through feature subset selection to fit the requirements and constraints of real applications. © 2021

IEEE.

Date 2021**DOI** 10.1109/ICC42927.2021.9500731**URL** <https://www.scopus.com/pages/publications/85115726392?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IEEE Int Conf Commun**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 15503607 (ISSN); 978-172817122-7 (ISBN)**Proceedings Title** IEEE Int Conf Commun**Conference Name** IEEE International Conference on Communications**Date Added** 20/09/2026, 9:52:23 pm**Modified** 20/09/2026, 9:52:23 pm**Tags:**

Network security, Feature extraction, Classification (of information), network security, Networks security, Malware, Complex networks, Semantics, Regression analysis, classification, Websites, Personal information, Features selection, feature selection, Additives, Feature interactions, Device operations, Measure theory, Non-additive measure, non-additive measure theory, Non-additive measure theory, User devices, Website features

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Exploiting physics-based machine learning to quantify geodynamic effects – insights from the Alpine region

Item Type Journal Article**Author** D. Degen**Author** A. Kumar**Author** M. Scheck-Wenderoth**Author** M. Cacace

Abstract Geodynamical processes are important to understand and assess the evolution of the Earth system as well as its natural resources. Given the wide range of characteristic spatial and temporal scales of geodynamic processes, their analysis routinely relies on computer-assisted numerical simulations. To provide reliable predictions such simulations need to consider a wide range of potential input parameters, material properties as they vary in space and time, in order to address associated uncertainties. To obtain any quantifiable measure of these uncertainties is challenging both because

of the high computational cost of the forward simulation and because data is typically limited to direct observations at the near surface and for the present-day state. To account for both of these challenges, we present how to construct efficient and reliable surrogate models that are applicable to a wide range of geodynamic problems using a physics-based machine learning method. In this study, we apply our approach to the case study of the Alpine region, as a natural example for a complex geodynamic setting where several subduction slabs as imaged by tomographic methods interact below a heterogeneous lithosphere. We specifically develop surrogates for two sets of observables, topography and surface velocity, to provide models that can be used in probabilistic frameworks to validate the underlying model structure and parametrization. We additionally construct models for the deeper crustal and mantle domains of the model, to improve the system understanding. For this last family of models, we highlight different construction methods to develop models to either allow evaluations in the entirety of the 3D model or only at specific depth intervals. © 2025 Denise Degen et al.

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Exploration of genomic, proteomic, and histopathological image data integration methods for clinical prediction

Item Type Conference Paper

Author A. Poruthoor

Author J.H. Phan

Author S. Kothari

Author M.D. Wang

Abstract The emergence of large multi-platform and multi-scale data repositories in biomedicine has enabled the exploration of data integration for holistic decision making. In this research, we investigate multi-modal genomic, proteomic, and histopathological image data integration for prediction of ovarian cancer clinical endpoints in The Cancer Genome Atlas (TCGA). Specifically, we study two data integration techniques, simple data concatenation and ensemble classification, to determine whether they can improve prediction of ovarian cancer grade or patient survival. Results indicate that integration via ensemble classification is more effective than simple data concatenation. We also highlight several key factors impacting data integration outcome such as predictability of endpoint, class prevalence, and unbalanced representation of features from different data modalities. © 2013 IEEE.

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Forecasting, Data integration, Image processing, Data processing, Multi-platform, Diseases, gene expression, bioinformatics, image processing, Bioinformatics, Gene expression, Multi-scale datum, Ensemble classification, Integration method, data integration, Histopathological images, Integration techniques, Ovarian cancers, Patient survivals

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Explore the RNA-sequencing and the next-generation sequencing in crops responding to abiotic stress

Item Type Journal Article**Author** A. Kido**Author** J.R.C. Ferreira-Neto**Author** E. Binneck**Author** M. da Silva**Author** W. da Silva**Author** A.M. Benko-Iseppon

Abstract Most of the environmental stresses damage crops provoking economic losses. Molecular characterization of genotypes identifying suitable genes helps breeders to develop abiotic stress-tolerant crops. Among the omics technologies, transcriptomics represents the most informative techniques for global gene expression profiling of an organism, organ/tissue, or even cell at a given snapshot in time. In this context the next-generation sequencing technologies revolutionized plant transcriptomics, particularly nonmodel plants of economic interest, enhancing the high-throughput data generated from biological samples. This chapter shows an updated RNA-Seq (sequencing of RNA) overview covering scientific reports published in the last 5 years. The assortment of the data-mined RNA-Seq studies concerning relevant issues identified the most studied crops, abiotic stresses, plant organs/tissues, stress-treatment exposition time, global or specific gene expression profiling, noncoding RNA approach, extra omics data, supplementary physiological data, studied accession(s), transcriptome assembly approach, gene expression validation process, relative quantification method, and reference gene(s) in the qPCR assays. Besides, an overall RNA-Seq analysis workflow, covering data generation, raw data processing, and data analysis, was provided, together with some information involving transcriptome assembly, quality of the transcriptome assembly, transcript quantification, differential expression analysis, and annotation/functional analysis. Finally, in the functional genomics topic, the integration of multiomic data stands out, improving our understanding of genetic networks, which helps us to envision machine learning strategies and genome editing applications involving agronomic characteristics. These approaches may help plant breeders identify new genes and related metabolic subpathways associated with relevant agronomic traits. © 2022 Elsevier Inc. All rights reserved.

Date 2022**DOI** 10.1016/B978-0-323-89778-5.00005-2**URL** <https://www.scopus.com/pages/publications/85143682187?origin=resultslist>**Archive** Scopus**Language** English**Publisher** Elsevier**Pages** 161-175

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Journal Abbr Bioinformatics in Agriculture: Next Generation Sequencing Era
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Tags:

drought, gene expression, crop, RNA-Seq, abiotic stress, bioinformatic, breeding, environmental stress, plant biotechnology, RT-qPCR, salinity, SuperSAGE, Transcriptomic

Notes:

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Exploring federated learning in optical music recognition

Item Type Journal Article

Author E. Ayllon

Author B. Serrano Sánchez

Author J. Calvo-Zaragoza

Abstract Optical Music Recognition (OMR) technology plays a crucial role in the preservation of cultural heritage by automating the digitization of music documents, enabling their storage in symbolic formats and their subsequent analysis through digital tools. Progress in this field is, however, constrained by limited data availability: access to or distribution of music collections is frequently restricted due to legal or ownership barriers. This work investigates a potential mitigation of this issue through Federated Learning (FL) strategies, which enable decentralized training and eliminate the need to release restricted collections. Our methodology assumes a setting in which client nodes operate on small, heterogeneous corpora, while evaluation is conducted on established OMR benchmark collections. We examine widely used FL aggregation techniques, such as FedAvg, FedProx, and SCAFFOLD; as well as modern methodologies, such as FedKT. In addition, we introduce two modules specifically designed for OMR: FedClassPrior, which integrates class-prior information to improve symbol balance, and FedNGram, which supports decentralized language modelling to exploit notational regularities during decoding. Experimental results show consistent accuracy improvements when using FL compared to purely local training. Furthermore, the proposed modules substantially narrow the performance gap between standard FL and the ideal scenario in which centralized training is feasible. © 2026 The Authors

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Correspondence Address: J. Calvo-Zaragoza; University of Alicante, Alicante, Carretera San Vicente del Raspeig S/N, 03690, Spain; email: jorge.calvo@ua.es; CODEN: NRCGE

Exploring Label Correlations for Quantification of ICD Codes

Item Type Conference Paper

Author I. Coutinho

Author B. Martins

Editor Bifet A.

Editor Lorena A.C.

Editor Ribeiro R.P.

Editor Gama J.

Editor Abreu P.H.

Abstract The International Classification of Diseases (ICD) has been adopted worldwide in the healthcare domain, e.g. to summarize the key information in clinical documents. Since manual ICD coding is very expensive, time-consuming, and error-prone, deep learning algorithms have been proposed to automate this task. However, the final goal of ICD coding often lays not in determining the codes associated with individual documents, but instead in quantifying the prevalence of each code within sets of documents. In this work, we experimentally assess different quantification methods in connection to ICD coding, including a simple learning-based approach that leverages associations between the codes, in order to predict their relative frequencies more accurately. Experiments show that the proposed approach can effectively explore existing associations between ICD codes, improving the quantification performance over baseline methods that deal

with each code independently. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Exploring Pelvic Symptom Dynamics in Relation to the Menstrual Cycle: Implications for Clinical Assessment and Management

Item Type Journal Article

Author M. Blanco-Diaz

Author A. Vielva-Gomez

Author M. Legasa-Susperregui

Author B. Perez-Dominguez

Author E.M. Medrano-Sánchez

Author E. Diaz-Mohedo

Abstract Background: Pelvic floor dysfunctions (PFDs) encompass an array of conditions with discrepant classification systems, hampering accurate prevalence estimation. Despite potentially affecting up to 25% of women during their lifetime, many remain undiagnosed, underestimating the true extent. Objectives: This cross-sectional study aimed to examine the impacts of the menstrual cycle on PFDs and dysfunctions. Secondary objectives included investigating differences between athletic and nonathletic women. Methods: An online questionnaire examined the effects of the menstrual cycle (MC) on 477 women's pelvic symptoms (aged 16–63 years), stratified by athletic status. This ad hoc instrument built upon a validated screening tool for female athletes. Results: Most participants reported symptom fluctuations across menstrual phases, with many modifying or reducing exercise participation. A concerning number experienced daily undiagnosed pelvic floor symptoms, emphasizing needs for comprehensive medical evaluation. Conclusions: Exacerbated pelvic symptoms showed complex relationships with menstruation, highlighting the importance of considering the MC in customized clinical management approaches. Symptoms demonstrated differential links to menstruation, indicating needs for individualized evaluation and tailored treatment plans based on symptom profiles and hormonal interactions. Educating professionals and patients remains essential to enhancing awareness, detection, and therapeutic outcomes. Further controlled longitudinal research should elucidate intricate relationships between menstrual cycles and pelvic symptom variability. © 2024 by the authors.

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Exploring the Impact of Alternatives of Object-Oriented Cohesion Measures on Machine Learning-Based Predictions of Inheritance Reusability

Item Type Journal Article

Author J. Al Dallal

Author B. Alkhazi

Abstract The cohesion of an object-oriented class refers to the relatedness of its methods and attributes. Constructors, destructors, and access methods are special types of methods featuring unique characteristics that can artificially affect class cohesion quantification. Methods within a class can also directly or transitively invoke each other, representing another cohesion aspect not considered by most existing cohesion measures. The impact of considering special methods (SPs) and transitive relations (TRs) in cohesion measurement on the abilities of the measures to predict inheritance reusability has yet to be investigated. In this paper, we empirically explored this effect. We applied a statistical technique to test the significance of the cohesion value changes across seven scenarios of ignoring or considering SPs and TRs. In addition, we applied a machine learning-based technique to build inheritance reusability prediction models using each of the considered measures and scenarios, evaluated the classification performance of the prediction models, and statistically compared the inheritance reusability prediction results. The results show that for most of the considered measures, the ignorance/consideration of SPs and TRs changed the cohesion values and the corresponding prediction significantly. Based on the study findings, when building inheritance reusability prediction models, software engineers are advised to 1) combine cohesion with other quality factors; 2) exclude the TRs from cohesion quantification; and 3) decide whether to consider or ignore SPs in cohesion quantification based on the selected measure(s) to be used in the prediction model, as this decision differs from one measure to another. © 2024 The Authors.

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Exploring the Trade-Off between generalist and specialized Models: A center-based comparative analysis for glioblastoma segmentation

Item Type Journal Article

Author F. Javier Gil-Terrón

Author P. Ferri

Author V. Montosa-i-Micó

Author M. Gómez Mahiques

Author C. Lopez-Mateu

Author P. Martí

Author J.M. García-Gómez

Author E. Fuster-Garcia

Abstract Introduction: Inherent variations between inter-center data can undermine the robustness of segmentation models when applied at a specific center (dataset shift). We investigated whether specialized center-specific models are more effective compared to generalist models based on multi-center data, and how center-specific data could enhance the performance of generalist models within a particular center using a fine-tuning transfer learning approach. For this purpose, we studied the dataset shift at center level and conducted a comparative analysis to assess the impact of data source on glioblastoma segmentation models. Methods & Materials: The three key components of dataset shift were studied: prior probability shift—variations in tumor size or tissue distribution among centers; covariate shift—inter-center MRI alterations; and concept shift—different criteria for tumor segmentation. BraTS 2021 dataset was used, which includes 1251 cases from 23 centers. Thereafter, 155 deep-learning models were developed and compared, including 1) generalist models trained with multi-center data, 2) specialized models using only center-specific data, and 3) fine-tuned generalist models using center-specific data. Results: The three key components of dataset shift were characterized. The amount of covariate shift was substantial, indicating large variations in MR imaging between different centers. Glioblastoma segmentation models tend to perform best when using data from the application center. Generalist models, trained with over 700 samples, achieved a

median Dice score of 88.98%. Specialized models surpassed this with 200 cases, while fine-tuned models outperformed with 50 cases. Conclusions: The influence of dataset shift on model performance is evident. Fine-tuned and specialized models, utilizing data from the evaluated center, outperform generalist models, which rely on data from other centers. These approaches could encourage medical centers to develop customized models for their local use, enhancing the accuracy and reliability of glioblastoma segmentation in a context where dataset shift is inevitable. © 2024 The Authors

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Exploring Turbulence and micro-scale mixing mechanisms for enhancing jet impingement heat transfer using micro-roughness elements: A data-driven and numerical analysis

Item Type Journal Article

Author K. Nagesha

Abstract This study uses machine learning to quantify micro-scale mixing mechanisms that enhance jet impingement heat transfer. Experiments and computational fluid dynamics (CFD) simulations tested flat, discrete protrusion, and continuous V-groove surfaces under Reynolds numbers(ReN) from 10,000 to 27,500 and nozzle-to-plate distances(NPD) between 2 and 5 times the diameter of the nozzle. An integrated approach combining experimental data, CFD, and four neural network models was used for comprehensive turbulence analysis. The neural networks, trained on the combined sixty datasets, showed high predictive accuracy with R-squared over 0.999 and Mean Absolute Error over 1e-7. The results highlight that micro-scale turbulence, characterized by Reynolds stress, dominates over operating parameters such as ReN and NPD in enhancing heat transfer. Discrete protrusions actively disrupt the thermal boundary layer, promoting vigorous mixing, while weaker turbulence and insulation effects in V-grooves contribute less. Percentage change analysis shows protrusions are more effective at extracting energy from jet and generating turbulence at smaller NPDs, but V-groove performance increases more strongly with rising distance. This data-driven analysis provides insight into surface roughness-induced mixing mechanisms and compares key turbulence parameters to assess thermal performance. The advanced understanding will aid in developing optimized designs for improved heat transfer in practical applications. © 2025 Elsevier Ltd

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Export Date: 20 September 2026; Cited By: 6; Correspondence Address: K. Nagesha; Indian Institute of Technology Madras, Chennai, Tamilunadu, India; email: knagesha33@gmail.com; CODEN: IHMTD

Extended a Priori Probability (EAPP): A Data-Driven Approach for Machine Learning Binary Classification Tasks

Item Type Journal Article

Author V.O. Castello

Author F.J. Perez-Benito

Author O.D.T. Catala

Author I.S. Igual

Author R. Llobet

Author J.-C. Perez-Cortes

Abstract The a priori probability of a dataset is usually used as a baseline for comparing a particular algorithm's accuracy in a given binary classification task. ZeroR is the simplest algorithm for this, predicting the majority class for all examples. However, this is an extremely simple approach that has no predictive power and does not describe other dataset features that could lead to a more demanding baseline. In this paper, we present the Extended A Priori Probability (EAPP), a novel semi-supervised baseline metric for binary classification tasks that considers not only the a priori probability but also some possible bias present in the dataset as well as other features that could provide a relatively trivial separability of the target classes. The approach is based on the area under the ROC curve (AUC ROC), known to be quite insensitive to class imbalance. The procedure involves multiobjective feature extraction and a clustering stage in the input space with autoencoders and a subsequent combinatory weighted assignation from clusters to classes depending on the distance to nearest clusters for each class. Class labels are then assigned to establish the combination that maximizes AUC ROC for each number of clusters considered. To avoid overfit in the combined feature extraction and clustering method, a cross-validation scheme is performed in each case. EAPP is defined for different numbers of clusters, starting from the inverse of the minority class proportion, which is useful for a fair comparison among diversely imbalanced datasets. A high EAPP usually relates to an easy binary classification task, but it also may be due to a significant coarse-grained bias in the dataset, when the task is previously known to be difficult. This metric represents a baseline beyond the a priori probability to assess the actual capabilities of binary classification models. © 2013 IEEE.

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Extending Adversarial Attacks to Produce Adversarial Class Probability Distributions

Item Type Journal Article**Author** J. Vadillo**Author** R. Santana**Author** J.A. Lozano

Abstract Despite the remarkable performance and generalization levels of deep learning models in a wide range of artificial intelligence tasks, it has been demonstrated that these models can be easily fooled by the addition of imperceptible yet malicious perturbations to natural inputs. These altered inputs are known in the literature as adversarial examples. In this paper, we propose a novel probabilistic framework to generalize and extend adversarial attacks in order to produce a desired probability distribution for the classes when we apply the attack method to a large number of inputs. This novel attack paradigm provides the adversary with greater control over the target model, thereby exposing, in a wide range of scenarios, threats against deep learning models that cannot be conducted by the conventional paradigms. We introduce four different strategies to efficiently generate such attacks, and illustrate our approach by extending multiple adversarial attack algorithms. We also experimentally validate our approach for the spoken command classification task and the Tweet emotion classification task, two exemplary machine learning problems in the audio and text domain, respectively. Our results demonstrate that we can closely approximate any probability distribution for the classes while maintaining a high fooling rate and even prevent the attacks from being detected by label-shift detection methods. ©2023 Jon Vadillo, Roberto Santana and Jose A. Lozano.

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Performance, Neural-networks, Learning models, Adversarial machine learning, Generalisation, Generative adversarial networks, Classification tasks, Deep Neural Networks, Robust classification, Class probabilities, Probability: distributions, Adversarial Examples, Class probability distribution, Class Probability Distributions, Linear Programming, Linear-programming, Robust Classification

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Extensions of transductive learning for distributed ensemble classification and application to biometric authentication

Item Type Journal Article

Author D.J. Miller

Author S. Pal

Author Y. Wang

Abstract We consider ensemble classification when there is no common labeled data for designing the function which aggregates classifier decisions. In recent work, we dubbed this problem distributed ensemble classification, addressing when local classifiers are trained on different (e.g., proprietary, legacy) databases or operate on different sensing modalities. Typically, fixed (untrained) rules of classifier combination such as voting methods are used in this case. However, these may perform poorly, especially when (i) the local class priors, used in training, differ from the true (test batch) priors and (ii) classifier decisions are statistically dependent. Alternatively, we proposed several transductive methods, optimizing the combining rule for objective functions measured on the test batch. We proposed both maximum likelihood (ML) and constraint-based (CB) objectives and found that CB achieved superior performance. Here, we develop CB extensions (i) for sequential

decisionmaking and (ii) exploiting additional class information contained in the local classifier feature vectors. The new sequential method is applied to biometric authentication. We demonstrate these new CB methods achieve better ensemble decision accuracy than methods which apply fixed rules in combining classifier decisions. © 2008 Elsevier B.V. All rights reserved.

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External Cervical Resorption: A Volumetric Analysis on Evolution of Defects over Time

Item Type Journal Article

Author A. Nosrat

Author O. Dianat

Author P. Verma

Author M.D. Levin

Author J.B. Price

Author A. Aminoshariae

Author F.A.P. Rizzante

Abstract Introduction: The aim of this study was to assess the temporal evolution of external cervical resorption (ECR) defects using a volumetric quantification method. Methods: Cone-beam computed tomographic (CBCT) images of patients diagnosed with ECR who chose not to receive treatment and attended recalls were collected. ECR defects were segmented in CBCT images at baseline and recall, and their volumes were quantified. The volumetric ratio of resorption defects/teeth was calculated. Three-dimensional classification of defects at baseline and recall and the prevalence of root surface perforations ≥ 1 mm were determined. The Wilcoxon matched pairs signed rank test, chi-square test, and linear regression models were used to analyze the data. Results: Fifteen patients with 20 teeth diagnosed with ECR and an average recall time of 21 months were included. Nine (45%) teeth showed a change in 3-dimensional classification at recall. The volume of resorption defects ($P = .0001$) and the volumetric ratio of resorption defects/teeth ($P = .0001$) increased over time. The prevalence of root surface perforations ≥ 1 mm was higher at recall ($n = 17$, 85%) compared with baseline ($n = 9$, 45%) ($P = .008$). Linear regression models showed significant associations between the resorption defect volume at recall compared with baseline ($P < .0001$; 95% confidence interval, 0.053–0.081) and the volumetric ratio of resorption defects/teeth at recall compared with baseline ($P < .0001$; 95% confidence interval, 0.205–0.356). There was no association between the volume of resorption defects at recall with the length of the recall period, sex, or age ($P > .05$). Conclusions: When left untreated, ECR defects can increase in size and develop more root surface perforations. ECR has a dynamic nature, and its volumetric increase over time does not result from uniform/linear expansion of the defects. © 2022 American Association of Endodontists

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External validation of a machine learning prediction model for massive blood loss during surgery for spinal metastases: a multi-institutional study using 880 patients

Item Type Journal Article

Author D.C. de Reus

Author R.H. Kuijten

Author P. Saha

Author D.A.A. Lastoria

Author A. Warr-Esser

Author C.F.C. Taylor

Author O.Q. Groot

Author D. Lui

Author J.-J. Verlaan

Author D.G. Tobert

Abstract Background context: A machine learning (ML) model was recently developed to predict massive intraoperative blood loss ($>2,500$ mL) during posterior decompressive surgery for spinal metastasis that performed well on external validation within the same region in China. Purpose: We sought to externally validate this model across new geographic regions (North America and Europe) and patient cohorts. Study design: Multiinstitutional retrospective cohort study. Patient sample: We retrospectively included patients 18 years or older who underwent decompressive surgery for spinal metastasis across three institutions in the United States, the United Kingdom and the Netherlands between 2016 and 2022. Inclusion and exclusion criteria were consistent with the development study with additional inclusion of (1) patients undergoing palliative decompression without stabilization, (2) patients with multiple myeloma and lymphoma, and (3) patients who continued anticoagulants perioperatively. Outcome measures: Model performance was assessed by comparing the incidence of massive intraoperative blood loss ($>2,500$ mL) in our cohort to the predicted risk generated by the ML model. Blood loss was quantified in 7 ways (including the formula from the development study) as no gold standard exists, and the method in the development paper was not clearly defined. We estimated blood loss using the anesthesia report, and calculated it using transfusion data, and preoperative and postoperative hematocrit levels. Methods: The following five input variables necessary for risk calculation by the ML model were manually collected: tumor type, smoking status, ECOG score, surgical process, and preoperative platelet count. Model performance was assessed on overall fit (Brier score), discriminatory

ability (area under the curve (AUC)), calibration (intercept & slope), and clinical utility (decision curve analysis)) for the total validation cohort, and for the North American and European cohorts separately. A subanalysis, excluding the additional included patient groups, assessed the predictive model's performance with the same inclusion and exclusion criteria as the development cohort. Results: A total of 880 patients were included with a massive blood loss incidence ranging from 5.3% to 18% depending on the quantification method used. Using the most favorable quantification method, the predictive model overestimated risk in our total validation cohort and scored poorly on overall fit (Brier score: 0.278), discrimination (AUC: 0.631 [95%CI: 0.583, 0.680]), calibration, (intercept: -2.082, [95%CI: -2.285, -1.879]), slope: 0.283 [95%CI: 0.173, 0.393]), and clinical utility, with net harm observed in decision curve analysis from 20%. Similar poor performance results were observed in the subanalysis excluding the additional included patients (n=676) and when analyzing the North American (n=539) and European (n=341) cohorts separately. Conclusions: To our knowledge, this is the first published external validation of a predictive ML model within orthopedic surgery to demonstrate poor performance. This poor performance might be attributed to overfitting and sampling bias as the development cohort had an insufficient sample size, and distributional shift as our cohort had key differences in predictive variables used by the model. These findings emphasize the importance of extensive validation in different geographical areas and addressing biases and pitfalls of ML model development before clinical implementation, as untested models may do more harm than good. © 2025 The Author(s)

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Extracellular Vesicles in Serum Carry Trop2 Protein as a Potential Molecular Indicator in Prostate Cancer

Item Type Journal Article

Author P.D. Sinawang

Author M.O. Ozen

Author S. Liu

Author E.-C. Hsu

Author D. Akin

Author E. Ding

Author R. Nolley

Author J.D. Brooks

Author T. Stoyanova

Author U. Demirci

Abstract Extracellular vesicles (EVs) are lipid nano-to-micro-sized vesicles increasingly studied for their role in intercellular communication and their potential as minimally invasive molecular indicators in various diseases. However, challenges remain in characterizing specific surface molecules on EVs due to cargo heterogeneity and the lack of convenient quantification methods. In this study, we show the isolation, characterization, detection, and quantification of Trop2-carrying EVs (EV-Trop2) in serum of prostate cancer patients. This work combines the unique advantages of our

EV isolation method with ELISA to enable surface-protein-specific EV analysis directly from serum. This is, to our knowledge, the first demonstration to isolate and quantify EV-Trop2 from prostate cancer patient serum to study its expression patterns in relation to prostate cancer status. Analysis of serum samples from three patient groups: high-risk prostate cancer (n = 22), low-risk prostate cancer (n = 23), and cancer-free groups (n = 21), revealed significantly different levels of EV-Trop2 expression between the high-risk and low-risk patient groups ($p = 0.0015$) and between high-risk patient and cancer-free groups ($p < 0.0001$). Multivariate modeling further showed that EV-Trop2 contributed to improved classifier metrics across the three sample groups. These findings highlight a strategy for probing EV-associated surface targets and suggest broader applicability of this approach across multiple cancers. © 2025 The Author(s). Journal of Extracellular Biology published by Wiley Periodicals LLC on behalf of International Society for Extracellular Vesicles.

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Extraction of landcover themes out of aerial orthoimages in mountainous areas using external information

Item Type Conference Paper

Author A. Le Bris

Author D. Boldo

Abstract In mountainous areas, the landcover extraction out of orthoimages through semi-automatic classification is limited by several factors (such as large shadow areas, radiometric similarities between different themes, inhomogeneous radiometry among regions of the same class...). Image information is not sufficient to separate the different classes. Nevertheless, good results can be obtained by dividing each landcover class "c" into two subclasses "c in shadow" / "c not in shadow" and introducing external information in the classification process. This information can be an older or more generalized database, geographic knowledge concerning the links between relief and landcover, or prior information concerning shadows. This external knowledge is then interpreted in terms of a priori probabilities and merged with radiometric information from the image in a MAP per region classification process. Besides, the results can also be improved by the use of combinations of channels calculated from the initial image.

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Classification (of information), Data fusion, Extraction, Image classification, Image analysis, Land cover, Landcover extraction, Landcoverknowledge based interpretation, MAP classification, Map classification method, Mountainous landcover extraction, National base map, Photogrammetry, Radiometry, Uncertainty management

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Extreme learning machine based sub-surface crack detection and quantification method for ACFM

Item Type Journal Article

Author S. Zhao

Author Y. Shen

Author J. Wang

Author R. Zhu

Author W. Zhai

Author H. Dong

Author Z. Mao

Author J. Gao

Abstract In this paper, a quantification method for sub-surface crack evaluation based on extreme learning machine (ELM) neural network is proposed to the application of alternating current field measurement (ACFM). According to simulation results of finite element model, the dimensions of cracks can be described by four features of the Bz and Bx curves. An optimized ELM feasible for ACFM detection is trained and validated based on the selected features. Taking the advantage of our proposed sub-surface crack sizing method, the experimental results indicate that the detection and quantification for sub-surface crack is conducted readily. The maximum errors of length and depth predicted by ELM can be limited to 5.65 % and 6.27 %, respectively. © 2021 Elsevier B.V.

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EyeLineNet: An Ensemble Deep Learning for Eye Health Monitoring with Attention Mechanisms and Grad-CAM Visualization

Item Type Conference Paper

Author A.M. Towhid

Author F. Yasmin

Author S. Hasan

Editor Jakhar A.K.

Editor Kumar A.

Editor Sehgal V.K.

Abstract A game changing innovation that delivers swift, precise eye care - protecting vision beforehand. The conventional approaches to diagnosing eye diseases are often inefficient and require expert intervention as well as financially burdensome, thus not optimal for widespread implementation. This study aims to discover a logical solution through the application of advanced intelligence approaches on a publicly available dataset containing 21,577 images categorized into ten classes. Prior to analysis, the dataset was processed and partitioned into training, testing and validation sets. However, the previous outcomes reached an accuracy of 89.64%, employing MobileNet model with Adam optimizer. On the contrary, our applied EfficientNetV2-M (92.50%) and ConvNeXt-Small (92.27%) with an attention mechanism to boost the accuracy through the ensemble of the models, obtained an accuracy of 93.5%, surpassing previous findings in effectiveness. Our method has the potential to be refined for clinical applications, assisting in prompt detection and intervention to prevent vision impairment. © 2025 IEEE.

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: A.M. Towhid; East West University, Dept. of Cse, Dhaka, Bangladesh; email: arafatmahmudtowhid@gmail.com; Conference name: 8th International Conference on Image Information Processing, ICIIP 2025; Conference date: 27 November 2025 through 29 November 2025; Conference code: 219631

Facial image analysis for fully automatic prediction of difficult endotracheal intubation

Item Type Journal Article**Author** G.L. Cuendet**Author** P. Schoettker**Author** A. Yüce**Author** M. Sorci**Author** H. Gao**Author** C. Perruchoud**Author** J.-P. Thiran

Abstract Goal: Difficult tracheal intubation is a major cause of anesthesia-related injuries with potential life threatening complications. Detection and anticipation of difficult airway in the preoperative period is, thus, crucial for the patients' safety. We propose an automatic face-analysis approach to detect morphological traits related to difficult intubation and improve its prediction. Methods: For this purpose, we have collected a database of 970 patients including photos, videos, and ground truth data. Specific statistical face models have been learned using the faces in our database providing an automated parametrization of the facial morphology. The most discriminative morphological features are selected through the importance ranking provided by the random forest algorithm. The random forest approach has also been used to train a classifier on these selected features. We compare a threshold tuning method based on class prior with two methods, which learn an optimal threshold on a training set for tackling the inherent imbalanced nature of the database. Results: Our fully automated method achieves an AUC of 81.0% in a simplified experimental setup, where only easy and difficult patients are considered. A further validation on the entire database has proven that our method is applicable for real-world difficult intubation prediction, with AUC = 77.9%. Conclusion: The system performance is in line with the state-of-

the-art medical diagnosis, based on ratings provided by trained anesthesiologists, whose assessment is guided by an extensive set of criteria. Significance: We present the first completely automatic and noninvasive difficult intubation detection system that is suitable for use in clinical settings. © 2015 IEEE.

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Facial palsy estimation algorithm

Item Type Conference Paper

Author W.L. Martínez-Angeles

Author O. González-Reynoso

Author M.A. García-Ramírez

Editor Rosado-Mendez I.M.

Editor Hernandez-Bojorquez M.

Editor Morales-Barcenas J.H.

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Editor Porras-Chaverri M.A.

Abstract The facial nerve is an array that encompasses motor and sensory fibers. It is one of the most common cranial nerves involved in disorders that considers congenital, traumatic, inflammatory and neoplastic disorders. Facial palsy is a common disease that results from the abnormal state of the facial nerve. In order to understand the full behavior, a quantification method for facial palsy is developed. The method that we propose uses the muscle trajectory information in order to perform a comparison between the right and left side of the face starting from the sagittal plane. To obtain such information, we developed an algorithm that will read a sensor array distributed across the face in order to read the state of the muscle in real time. Moreover, a statistical analysis is performed allowing us to know the state of the facial nerve. Accurate facial palsy measurement are key in the disorder treatment. According to the results found by our system, healthy faces must show a high correlation as well as a large standard deviation, correlation coefficient and a statistical mean. © 2021 American Institute of Physics Inc.. All rights reserved.

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Facilitating objective forest land use decisions by site classification and tree growth modeling: A case study from Vietnam

Item Type Journal Article

Author N.D. Cuong

Author M. Volker

Author M. Köhl

Abstract Onsite information pertaining to forest growth potential is a significant prerequisite for selecting suitable forest plantation locations and safeguards sustainable timber production and income generation. In the scope of forest landscape restoration, the selection of the most suitable sites for reforestation remains a major issue. The current study introduces an operational, objective, and straightforward methodology for the identification and prioritization of sites that are suitable for forest plantations. The methodology is based on an approach that combines land use suitability assessment with site-specific growth and yield predictions. The land use suitability assessment is based on a methodological framework presented by the Food and Agriculture Organization (FAO). In a study area in Vietnam four different suitability classes are defined for the species *Acacia mangium* (*A. mangium*). Field assessments in existing *A. mangium* plantations were utilized to develop yield models. Among the Korf, Gompertz, and Chapman-Richards growth equations Korf performed the best for all suitability classes. Prioritization of sites is realized by the analytical hi-erarchy process (AHP). Our study offers a pragmatic approach for selecting the most suitable sites for large-scale forest restoration activities. Integrating growth and yield predictions supports reforestation practices and promotes sustainable timber production. © SISEF.

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Factorizable Joint Shift in Multinomial Classification

Item Type Journal Article

Author D. Tasche

Abstract Factorizable joint shift (FJS) was recently proposed as a type of dataset shift for which the complete characteristics can be estimated from feature data observations on the test dataset by a method called Joint Importance Aligning. For the multinomial (multiclass) classification setting, we derive a representation of factorizable joint shift in terms of the source (training) distribution, the target (test) prior class probabilities and the target marginal distribution of the features. On the basis of this result, we propose alternatives to joint importance aligning and, at the same time, point out that factorizable joint shift is not fully identifiable if no class label information on the test dataset is available and no additional assumptions are made. Other results of the paper include correction formulae for the posterior class probabilities both under general dataset shift and factorizable joint shift. In addition, we investigate the consequences of assuming factorizable joint shift for the bias caused by sample selection. © 2022 by the author.

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Fairness and AI generalizability in medical image analysis

Item Type Conference Paper

Author N.D. Forkert

Editor Yang X.

Editor Hsu W.

Abstract The rapid evolution of artificial intelligence (AI) and in particular machine learning (ML) for healthcare applications has opened many new exciting opportunities to automatically analyze increasingly complex clinical data, especially related to medical imaging, one of the biggest data contributors in health care. However, while machine learning models have demonstrated considerable potential in the research setting to improve and accelerate diagnostic accuracy, reduce clinician workload, and enable integration of multi-modal data, their real-world deployment in clinical settings remains limited in many cases. A major obstacle is that ML models trained on medical images consistently fail to generalize well and perform poorly when applied to data from institutions, scanners, and patient populations that were not or poorly represented in the training set. These performance gaps often stem from biological and non-biological variations and biases that are only spuriously but not causally correlated with the medical task of interest. However, it still remains an open question how such biases propagate through deep learning model architectures and shape learned representations. This invited paper summarizes our recent efforts to address these challenges through controlled bias experiments and distributed learning methods. More precisely, the Simulated Bias in Artificial Medical Images (SimBA) framework is introduced, which enables the generation of realistic brain MRI datasets with known and fully controllable morphological and intensity-based biases, thereby facilitating counterfactual analyses of how biases are encoded and used by deep learning models. Using SimBA, we demonstrate that standard convolutional neural networks encode biases across all layers and that shortcut learning depends on various factors, including spatial proximity, bias salience, and class prevalence. We further discuss distributed training strategies as a scalable solution to overcome structural barriers to data sharing and enhance deep learning model generalizability. Together, this work provides foundational insights toward developing robust, interpretable, and equitable ML solutions for the analysis of medical imaging data and downstream computer-aided diagnosis tasks. © 2026 SPIE. All rights reserved.

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Modal analysis, Deep neural networks, Learning systems, Machine-learning, Artificial Intelligence, Population statistics, Learning models, Contrastive Learning, Bias, Machine Learning, Machine learning models, Health care, Health care application, Medical image processing, Medical computing, Fairness, Clinical research, Computer aided diagnosis, Clinical data, Data contributors, Diagnostic accuracy, Medical image analysis, Medical Imaging

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Fairness in Recommender Systems: Evaluation Approaches and Assurance Strategies

Item Type Journal Article

Author Y. Wu

Author J. Cao

Author G. Xu

Abstract With the wide application of recommender systems, the potential impacts of recommender systems on customers, item providers and other parties have attracted increasing attention. Fairness, which is the quality of treating people equally, is also becoming important in recommender system evaluation and algorithm design. Therefore, in the past years, there has been a growing interest in fairness measurement and assurance in recommender systems. Although there are several reviews on related topics, such as fairness in machine learning and debias in recommender systems, they do not present a systematic view on fairness in recommender systems, which is context aware and has a multi-sided meaning. Therefore, in this review, the concept of fairness is discussed in detail in the various

contexts of recommender systems. Specifically, a comprehensive framework to classify fairness metrics is proposed from four dimensions, i.e., Fairness for Whom, Demographic Unit, Time Frame, and Quantification Method. Then the strategies for eliminating unfairness in recommendations, fairness in different recommendation tasks and datasets are reviewed and summarized. Finally, the challenges and future work are discussed. © 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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Fairness through the lens of proportional equality

Item Type Conference Paper

Author A. Biswas

Author S. Mukherjee

Abstract Today, automated algorithms, such as machine learning classifiers, are playing an increasingly pivotal role in important societal decisions such as hiring, loan allocation, and criminal risk assessment. This motivates the need to probe the outcomes of a prediction model for discriminatory traits towards specific groups of individuals. In this context, one of the crucial challenges is to formally define a satisfactory notion of fairness. Our contribution in this paper is to formalize

Proportional Equality (PE) as a fairness notion. We additionally show that it is a more appropriate criterion than the existing popular notion called Disparate Impact (DI), which is used for evaluating the fairness of a classifier's outcomes.
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Risk assessment, Autonomous agents, Classification (of information), Machine learning, Multi agent systems, Classification, Prior probability, Discrimination, Fairness concepts, Gender bias, Prior probability shifts, Racial bias

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Fairness Transferability Subject to Bounded Distribution Shift

Item Type Conference Paper

Author Y. Chen

Author R. Raab

Author J. Wang

Author Y. Liu

Editor Koyejo S.

Editor Mohamed S.

Editor Agarwal A.

Editor Belgrave D.**Editor** Cho K.**Editor** Oh A.

Abstract Given an algorithmic predictor that is “fair” on some source distribution, will it still be fair on an unknown target distribution that differs from the source within some bound? In this paper, we study the transferability of statistical group fairness for machine learning predictors (i.e., classifiers or regressors) subject to bounded distribution shift. Such shifts may be introduced by initial training data uncertainties, user adaptation to a deployed predictor, dynamic environments, or the use of pre-trained models in new settings. Herein, we develop a bound that characterizes such transferability, flagging potentially inappropriate deployments of machine learning for socially consequential tasks. We first develop a framework for bounding violations of statistical fairness subject to distribution shift, formulating a generic upper bound for transferred fairness violations as our primary result. We then develop bounds for specific worked examples, focusing on two commonly used fairness definitions (i.e., demographic parity and equalized odds) and two classes of distribution shift (i.e., covariate shift and label shift). Finally, we compare our theoretical bounds to deterministic models of distribution shift and against real-world data, finding that we are able to estimate fairness violation bounds in practice, even when simplifying assumptions are only approximately satisfied. © 2022 Neural information processing systems foundation. All rights reserved.

Date 2022**URL** <https://www.scopus.com/pages/publications/85148318510?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Adv. neural inf. proces. syst.**Volume** 35**Publisher** Neural information processing systems foundation**ISBN** 10495258 (ISSN); 978-171387108-8 (ISBN)**Proceedings Title** Adv. neural inf. proces. syst.**Conference Name** Advances in Neural Information Processing Systems**Date Added** 20/09/2026, 9:52:29 pm**Modified** 20/09/2026, 9:52:29 pm**Tags:**

Algorithmics, Machine-learning, Dynamic environments, Training data, Shift-and, Data uncertainty, Source distribution, Upper Bound, User adaptation, Worked examples

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Fast Inner-Product Algorithms and Architectures for Deep Neural Network Accelerators

Item Type Journal Article

Author T.E. Pogue

Author N. Nicolici

Abstract We introduce a new algorithm called the Free-pipeline Fast Inner Product (FFIP) and its hardware architecture that improve an under-explored fast inner-product algorithm (FIP) proposed by Winograd in 1968. Unlike the unrelated Winograd minimal filtering algorithms for convolutional layers, FIP is applicable to all machine learning (ML) model layers that can mainly decompose to matrix multiplication, including fully-connected, convolutional, recurrent, and attention/transformer layers. We implement FIP for the first time in an ML accelerator then present our FFIP algorithm and generalized architecture which inherently improve FIP's clock frequency and, as a consequence, throughput for a similar hardware cost. Finally, we contribute ML-specific optimizations for the FIP and FFIP algorithms and architectures. We show that FFIP can be seamlessly incorporated into traditional fixed-point systolic array ML accelerators to achieve the same throughput with half the number of multiply-accumulate (MAC) units, or it can double the maximum systolic array size that can fit onto devices with a fixed hardware budget. Our FFIP implementation for non-sparse ML models with 8 to 16-bit fixed-point inputs achieves higher throughput and compute efficiency than the best-in-class prior solutions on the same type of compute platform. © 1968-2012 IEEE.

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: T.E. Pogue; McMaster University, Department of Electrical and Computer Engineering, Hamilton, L8S 4L8, Canada; email: pogue@mcmaster.ca; CODEN: ITCOB

Fast LCNN ica for unsupervised hyperspectral image classifier

Item Type Journal Article

Author I. Kopriva

Author H. Szu

Abstract Since in remote sensing each pixel could have its own unique radiation source s including man-made objects associated with different spectral reflectance matrix A , we could not average over neighborhood pixels. Instead, we solve pixel-by-pixel independent classes analysis (ica) without pixel average by Lagrange Constraint of the data measurement model and Gibbs' equal a priori probability assumption based on Shannon's Entropy $-\sum_{j=1}^M s_j \ln(s_j)$ with probability normalization condition $\sum_{j=1}^M s_j = 1$ for an arbitrary number of M classes that is bounded by the spectral data components N . We formulate the Fast Lagrangian method to maximize the Shannon entropy with the equality constraints in order to achieve $O(N)$ numerical complexity contrary to the $O(N^2)$ numerical complexity associated with the solution of the inverse problem required in the classical Lagrangian formulation. Trivial equal probability solution with uniformly distributed class vector s is avoided by introducing additional set of the inequality constraints. The unknown spectral reflectance matrix A is estimated blindly in non-parameterized form minimizing an LMS energy function $E(A)$. We apply the Riemannian metric to the gradient learning for $(\partial E / \partial A) A^T A$ reproducing the biological Hebbian rule in terms of a full rank vector outer product formula and demonstrate faster convergence than standard Euclidean gradient. Since the proposed Fast Lagrangian method has $O(N)$ numerical complexity we have achieved a real time hyperspectral remote sensing capability as platform moves, samples and processes. A FPGA firmware implementation for massive pixel parallel algorithm has been filed for patent. © Society of Photo-Optical Instrumentation Engineers.

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Notes:

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Fast nonparametric estimation of class proportions in the positive-unlabeled classification setting

Item Type Conference Paper

Author D. Zeiberg

Author S. Jain

Author P. Radivojac

Abstract Estimating class proportions has emerged as an important direction in positive-unlabeled learning. Well-estimated class priors are key to accurate approximation of posterior distributions and are necessary for the recovery of true classification performance. While significant progress has been made in the past decade, there remains a need for accurate strategies that scale to big data. Motivated by this need, we propose an intuitive and fast nonparametric algorithm to estimate class proportions. Unlike any of the previous methods, our algorithm uses a sampling strategy to repeatedly (1) draw an example from the set of positives, (2) record the minimum distance to any of the unlabeled examples, and (3) remove the nearest unlabeled example. We show that the point of sharp increase in the recorded distances corresponds to the desired proportion of positives in the unlabeled set and train a deep neural network to identify that point. Our distance-based algorithm is evaluated on forty datasets and compared to all currently available methods. We provide evidence that this new approach results in the most accurate performance and can be readily used on large datasets. Copyright © 2020, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Fast Recognition and Counting Method of Dragon Fruit Flowers and Fruits Based on Video Stream

Item Type Journal Article

Author X. Li

Author X. Wang

Author P. Ong

Author Z. Yi

Author L. Ding

Author C. Han

Abstract Dragon fruit (*Hylocereus undatus*) is a tropical and subtropical fruit that undergoes multiple ripening cycles throughout the year. Accurate monitoring of the flower and fruit quantities at various stages is crucial for growers to estimate yields, plan orders, and implement effective management strategies. However, traditional manual counting methods are labor-intensive and inefficient. Deep learning techniques have proven effective for object recognition tasks but limited research has been conducted on dragon fruit due to its unique stem morphology and the coexistence of flowers and fruits. Additionally, the challenge lies in developing a lightweight recognition and tracking model that can be seamlessly integrated into mobile platforms, enabling on-site quantity counting. In this study, a video stream inspection method was proposed to classify and count dragon fruit flowers, immature fruits (green fruits), and mature fruits (red fruits) in a dragon fruit plantation. The approach involves three key steps: (1) utilizing the YOLOv5 network for the identification of different dragon fruit categories, (2) employing the improved ByteTrack object tracking algorithm to assign unique IDs to each target and track their movement, and (3) defining a region of interest area for precise classification and counting of dragon fruit across categories. Experimental results demonstrate recognition accuracies of 94.1%, 94.8%, and 96.1% for dragon fruit flowers, green fruits, and red fruits, respectively, with an overall

average recognition accuracy of 95.0%. Furthermore, the counting accuracy for each category is measured at 97.68%, 93.97%, and 91.89%, respectively. The proposed method achieves a counting speed of 56 frames per second on a 1080ti GPU. The findings establish the efficacy and practicality of this method for accurate counting of dragon fruit or other fruit varieties.

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Fast TILs—A pipeline for efficient TILs estimation in non-small cell Lung cancer

Item Type Journal Article

Author N. Shvetsov

Author A. Sildnes

Author M. Tafavvoghi

Author L.-T.R. Busund

Author S.M. Dalen

Author K. Møllersen

Author L.A. Bongo

Author T.K. Kilvær

Abstract The prognostic relevance of tumor-infiltrating lymphocytes (TILs) in non-small cell Lung cancer (NSCLC) is well-established. However, manual TIL quantification in hematoxylin and eosin (H&E) whole slide images (WSIs) is laborious and prone to variability. To address this, we aim to develop and validate an automated computational pipeline for the quantification of TILs in WSIs of NSCLC. Such a

solution in computational pathology can accelerate TIL evaluation, thereby standardizing the prognostication process and facilitating personalized treatment strategies. We develop an end-to-end automated pipeline for TIL estimation in Lung cancer WSIs by integrating a patch extraction approach based on hematoxylin component filtering with a machine learning-based patch classification and cell quantification method using the HoVer-Net model architecture. Additionally, we employ randomized patch sampling to further reduce the processed patch amount. We evaluate the effectiveness of the patch sampling procedure, the pipeline's ability to identify informative patches and computational efficiency, and the clinical value of produced scores using patient survival data. Our pipeline demonstrates the ability to selectively process informative patches, achieving a balance between computational efficiency and prognostic integrity. The pipeline filtering excludes approximately 70% of all patch candidates. Further, only 5% of eligible patches are necessary to retain the pipeline's prognostic accuracy (c-index = 0.65), resulting in a linear reduction of the total computational time compared to the filtered patch subset analysis. The pipeline's TILs score has a strong association with patient survival and outperforms traditional CD8 immunohistochemical scoring (c-index = 0.59). Kaplan–Meier analysis further substantiates the TILs score's prognostic value. This study introduces an automated pipeline for TIL evaluation in Lung cancer WSIs, providing a prognostic tool with potential to improve personalized treatment in NSCLC. The pipeline's computational advances, particularly in reducing processing time, and clinical relevance demonstrate a step forward in computational pathology. © 2025 The Authors

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Fault detection in batch processes through variable selection integrated to multiway principal component analysis

Item Type Journal Article

Author F.A.P. Peres

Author T.N. Peres

Author F.S. Fogliatto

Author M.J. Anzanello

Abstract The main purpose of fault detection in batch process monitoring is to identify batches displaying atypical behavior in comparison to normal operating data. The current growth in the number of measurable variables due to process automation yields datasets in which the number of variables is much larger than the number of batches. That may compromise the performance of Multiway Principal Component Analysis (MPCA), which is the most popular quality control approach used in batch processes. To overcome that, new strategies to handle high-dimensional datasets become necessary. In this paper we propose the Pareto Variable Selection (PVS) – MPCA method to monitor batch processes described by high-dimensional datasets. The main idea of PVS-MPCA is to select process variables that promote the best classification of production batches in conforming or non-conforming classes, prior to the construction of T2 and Q control charts used to monitor batch performance. Our proposition was applied to a real dataset from a chocolate conching batch operation and compared to classical MPCA-based monitoring. PVS-MPCA promoted a reduction of 85.18% in false alarm rate retaining only 5 unfolded variables, in opposition to 2,864 unfolded variables used in classical MPCA. The missed detection rate was null, ensuring that only conforming batches were released to the production line. © 2019 Elsevier Ltd

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FCP-Pro: Federated conformal prediction algorithm based on prototype similarity

Item Type Journal Article

Author G. Li

Author Y. Zhang

Author Y. Wang

Author C. Wang

Abstract Research on trustworthy federated learning (FL) is still in its nascent stages, which severely limits the potential for FL to be applied in high-stakes scenarios, such as medical diagnosis, financial computing, and autonomous driving. To quantify uncertainty for the federated classification task, we extend conformal prediction to the federated learning system. First, we design the prototype-based adaptive prediction set (PAPS) score function for FL to compute the integrated prototype-based non-conformity score. It utilizes prototypes as proxies to protect privacy and reduce the size of the prediction set. Subsequently, we propose the federated conformal prediction algorithm based on prototype similarity (FCP-Pro) to overcome the data non-exchangeability and improve the prediction efficiency. It simultaneously enforces label shift correction and similarity calibration adjustment to the empirical distribution of federated calibration samples. The coverage guarantee of the proposed FCP-Pro algorithm is proved theoretically. To validate the effectiveness of the FCP-Pro algorithm, extensive experiments have been conducted on the CIFAR-10, CIFAR-100, and ImageNet datasets. The experimental results demonstrate that FCP-Pro significantly outperforms other state-of-the-art federated conformal prediction algorithms in terms of both empirical coverage and efficiency. © 2025 Elsevier Ltd

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Feasibility of single-beat detection of ventricular late potentials on standard ECG leads via interpretable gradient boosting

Item Type Journal Article

Author A. Novak

Author D. Barić

Author I. Zeljković

Author B. Penayo

Author N. Pavlović

Author Š. Manola

Abstract Background: Ventricular late potentials (VLPs) are markers of arrhythmogenic substrate, but conventional assessment using signal-averaged ECG (SAECG) requires prolonged acquisition and operator-dependent artifact handling, limiting scalability and ambulatory use. Single-beat detection of VLP-like activity from standard surface ECG remains insufficiently validated. Objective: To evaluate the technical feasibility of interpretable single-beat detection of VLP-like perturbations from standard ECG leads without signal averaging. Methods: Using the MIMIC-IV-ECG database, we analyzed 120,000 beats from leads II, V2, and V6. Because large public datasets with beat-level clinically adjudicated VLP labels are not currently available, physiologically constrained synthetic VLP-like signals were injected into a subset of

beats to create a controlled feasibility benchmark. For each beat, more than 200 features were extracted, including time-domain statistics, frequency-domain measures, wavelet coefficients, autocorrelation features, and localized windowed summaries. Ten classifiers were optimized using nested patient-wise cross-validation and evaluated in five settings: single-lead detection, cross-lead generalization, mixed-lead training, reduced training size, and class-imbalance robustness. Results: Gradient-boosted ensembles, particularly XGBoost and CatBoost, achieved strong discrimination on held-out single-beat data ($AUC > 0.99$; $F1 > 0.93$), while remaining stable with 10% of the training data and 5% positive-class prevalence. Performance was also robust in lead-transfer experiments. SHAP analysis identified localized entropy, dispersion, and related high-frequency descriptors in late post-R windows as the dominant predictors. Conclusion: These findings support the methodological feasibility of interpretable single-beat detection of VLP-like signatures from routine surface ECG under controlled synthetic conditions. Validation on clinically adjudicated cohorts and external datasets is required before clinical translation. © 2026 Elsevier B.V.

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Feature selection for high-dimensional imbalanced data

Item Type Journal Article**Author** L. Yin**Author** Y. Ge**Author** K. Xiao**Author** X. Wang**Author** X. Quan

Abstract Given its importance, the problem of classification in imbalanced data has attracted great attention in recent years. However, few efforts have been made to develop feature selection techniques for the classification of imbalanced data. This paper thus fills this critical void by introducing two approaches for the feature selection of high-dimensional imbalanced data. To this end, after introducing three traditional methods, we study and illustrate the challenges of feature selection in imbalanced data with Bayesian learning. Indeed, we reveal that the samples in the larger classes have a dominant influence on these feature selection methods. However, the samples in rare classes are essential for the learning performances of rare classes. Based on these observations, we provide a new feature selection approach based on class decomposition. Specifically, we partition the large classes into relatively smaller pseudo-subclasses and generate the pseudo-class labels accordingly. Feature selection is then performed on the new decomposed data for computing the goodness measurement of features. In addition, we also introduce a Hellinger distance-based method for feature selection. Hellinger distance is a measure of distribution divergence, which is strongly skew insensitive as the class prior information is not involved for computing the distance. Finally, we theoretically show the effectiveness of these two approaches with Bayesian learning on synthetic data. We also test and compare the performances of the proposed feature-selection methods on some real-world data sets. The experimental results show that both decomposition-based and Hellinger distance-based methods can outperform existing feature-selection methods with a clear margin on imbalanced data. © 2012 Elsevier B.V.

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Feature-based classification and segmentation of mitral regurgitation echocardiography images quantification using PISA method

Item Type Journal Article

Author P.A. Khayum

Author R.S. Babu

Abstract Echocardiography is the enormously admired scientific specification for the evaluation of valvular regurgitation and gives significant knowledge on the bareness of mitral regurgitation MR). MR is a general heart disease which does not cause indications till its final phase. A technique is advanced for jet area separation and quantification in MR assessment in arithmetical expressions. Previous to this separation method count preprocessing and some attributes are mined from the record to arrangement method. From the cataloguing method, support vector machine SVM) classifier developed to confidential echocardiogram images. Entire abnormal images to the modified region growing MRG) separation method to segment jet area of MR. This segmented jet area in MR quantification process passed out with the support of proximal isovelocity surface area PISA). This procedure is based on mass diverse limitations like blood flow rate, regurgitant fraction, EROA, etc. From the outcomes, this projected effort associated with the current method fuzzy with PISA quantification process, the projected work attained accuracy rate 99.05% in the study of jet area segmenting and quantification method. © 2019 Inderscience Enterprises Ltd..

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Feature-center-based quantification of second-coordination-shell effects in carbene-coordinated single-atom catalysts for NRR

Item Type Journal Article
Author D. Ye
Author S. Liu
Author D. Zheng
Author W. Lu
Author W. Liu
Author T. Hu
Author J. Xu

Abstract Second-coordination-shell effects have emerged as a crucial factor in regulating the performance of single-atom catalysts for the nitrogen reduction reaction (NRR), but quantifying their impact remains challenging due to their spatial distribution and lack of a unique descriptor. Here, we propose a feature-center-based spatial–physicochemical quantification method to describe second-coordination-shell effects in terms of composition and effective distribution, using a carbene-coordinated single-atom catalyst platform with varying ratios of N-heterocyclic carbene (NHC) and cyclic (alkyl)(amino)carbene (CAAC) ligands. High-throughput screening of 203 designed systems identified 32 computationally screened candidate systems, four of which exhibited low calculated limiting potentials of -0.15 to -0.17 V. Machine-learning analysis identified the first-ionization-energy feature-center-based descriptor (dIE) as an important catalyst-intrinsic descriptor for quantifying second-coordination-shell effects, highlighting the relevance of spatially distributed electronic characteristics to catalytic performance. The proposed representation further shows transferability to chemically related coordination environments. These results demonstrate the potential of the proposed framework for quantifying second-coordination-shell effects in carbene-coordinated single-atom catalysts and provide insights into the design of coordination-sensitive electrocatalysts. © 2026 Elsevier B.V.

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FEDERATED LEARNING FROM ONLY UNLABELED DATA WITH CLASS-CONDITIONAL-SHARING CLIENTS

Item Type Conference Paper

Author N. Lu

Author Z. Wang

Author X. Li

Author G. Niu

Author Q. Dou

Author M. Sugiyama

Abstract Supervised federated learning (FL) enables multiple clients to share the trained model without sharing their labeled data. However, potential clients might even be reluctant to label their own data, which could limit the applicability of FL in practice. In this paper, we show the possibility of unsupervised FL whose model is still a classifier for predicting class labels, if the class-prior probabilities are shifted while the class-conditional distributions are shared among the unlabeled data owned by the clients. We propose federation of unsupervised learning (FedUL), where the unlabeled data are transformed into surrogate labeled data for each of the clients, a modified model is trained by supervised FL, and the wanted model is recovered from the modified model. FedUL is a very general solution to unsupervised FL: it is compatible with many supervised FL methods, and the recovery of the wanted model can be theoretically guaranteed as if the data have been labeled. Experiments on benchmark and real-world datasets demonstrate the effectiveness of FedUL. Code is available at <https://github.com/lunanbit/FedUL>. © 2022 ICLR 2022 - 10th International Conference on Learning Representations. All rights reserved.

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FedERFT: Improving federated learning through feature-enriched regularization and post-aggregation fine-tuning

Item Type Journal Article

Author S. Zhu

Author C. Qiu

Author G. Sun

Abstract Federated learning (FL) is a privacy-preserving distributed machine learning paradigm that enables collaborative model training through local updates and model sharing. However, in realistic deployments the data are typically non-independent and identically distributed (non-IID), and in particular label shift severely limits the generalization ability and convergence efficiency of the global model. Existing model-averaging-based algorithms struggle in such heterogeneous settings, making them insufficient for many practical applications. To address these issues, we propose FedERFT, a federated learning framework with server-side staged optimization, which systematically improves model stability and generalization by combining feature generation with progressive fine-tuning. In the first stage, the server aggregates client models to obtain an initial global model. In the second stage, the global encoder is frozen and only the classification head is fine-tuned, allowing rapid adaptation to the feature distributions of different clients. In the final stage, both the global encoder and the classification head are jointly fine-tuned to further enhance overall performance. In addition, we design a feature generation strategy based on bidirectional Kullback-Leibler (KL) divergence minimization. This strategy applies random dropout and rescaling to client parameter increments to generate diversified feature representations, effectively aligning client feature distributions and substantially mitigating the training instability caused by data heterogeneity. Experiments on CIFAR-10, CIFAR-100, and Tiny-ImageNet show that FedERFT achieves a 1.5%-4.5% accuracy improvement across various non-IID scenarios, and exhibits particularly strong generalization and cross-client robustness under extreme heterogeneity. These results demonstrate that FedERFT provides an efficient, robust, and widely applicable solution for federated learning in non-IID environments. © 2026 Elsevier B.V.

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FedIIC: Towards Robust Federated Learning for Class-Imbalanced Medical Image Classification

Item Type Conference Paper**Author** N. Wu**Author** L. Yu**Author** X. Yang**Author** K.-T. Cheng**Author** Z. Yan**Editor** Greenspan H.**Editor** Greenspan H.**Editor** Madabhushi A.**Editor** Mousavi P.**Editor** Salcudean S.**Editor** Duncan J.**Editor** Syeda-Mahmood T.**Editor** Taylor R.

Abstract Federated learning (FL), training deep models from decentralized data without privacy leakage, has shown great potential in medical image computing recently. However, considering the ubiquitous class imbalance in medical data, FL can exhibit performance degradation, especially for minority classes (e.g. rare diseases). Existing methods towards this problem mainly focus on training a

balanced classifier to eliminate class prior bias among classes, but neglect to explore better representation to facilitate classification performance. In this paper, we present a privacy-preserving FL method named FedIIC to combat class imbalance from two perspectives: feature learning and classifier learning. In feature learning, two levels of contrastive learning are designed to extract better class-specific features with imbalanced data in FL. In classifier learning, per-class margins are dynamically set according to real-time difficulty and class priors, which helps the model learn classes equally. Experimental results on publicly-available datasets demonstrate the superior performance of FedIIC in dealing with both real-world and simulated multi-source medical imaging data under class imbalance. Code is available at <https://github.com/wnn2000/FedIIC>. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2023.

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FedMCLA: Multi-Cluster Label-Shift-Aware Backdoor Defense for Heterogeneous IoT Federations

Item Type Journal Article

Author X. Zhang

Author J. Kang

Abstract Federated intelligence enables heterogeneous Internet of Things (IoT) gateways to train without sharing raw data, but non-IID label skew lets malicious updates hide among overlapping benign modes. FedMCLA uses a client-specific label-shift signature to jointly drive fuzzy multi-reference consensus and client-conditioned benign-update reconstruction. The defense runs at a compute-capable aggregation coordinator with visible client updates. Vision models use their native classifier bias; the decoder-only federated large language model (FLLM) path requires a persistent vocabulary-sized output-bias augmentation, so it is not purely coordinator-side. FedMCLA is also incompatible with standard secure aggregation that reveals only an aggregate. Experiments cover vision and FLLM attacks, matched method-specific white-box attacks, contamination stressors, and deployment envelopes. Under matched defense-specific attacks on CIFAR-10, FedMCLA obtains 23.14% ASR, 63.36% benign accuracy (BA), and 7.53% benign error rate (BER), versus the next-lowest ASR of 42.68% for FedDLAD; with 50% same-round update exposure, Attack-MCLA+-E reaches 34.26% ASR. Across seven standard CIFAR-10 attacks, FedMCLA reduces mean ASR from 18.65% with DoBlock to 7.33%. On three Alpaca attacks, it achieves 9.12%–12.45% ASR while keeping MMLU within 1.2 percentage points of the undefended model. © 2014 IEEE.

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FedR2G: Federated long-tailed learning via geometric structure correction and generative rebalancing

Item Type Journal Article

Author J. Ren

Author G. Ye

Author Y. Zhang

Author Y. Xin

Author A. Wu

Author Y. Yuan

Author S. Zhao

Author Y. Li

Abstract Federated long-tailed learning is difficult because the model must learn from clients that are both heterogeneous and class-imbalanced. Tail classes may appear on only a few clients, and the prototypes used by generative augmentation can therefore be shifted by sample scarcity, label skew, and classifier bias. FedR2G addresses this problem by making the generation condition reliable before generating more tail samples. It first trains clients with a class-prior-scaled ETF directional classifier, which maintains a stable inter-class angular reference while strengthening weak tail-class responses. Clients then upload normalized class prototypes and local class counts. On the server, FedR2G uses these counts to calibrate latent class centers and performs tail-aware generation from the calibrated anchors. A latent consistency loss returns the public generative prior to client training. Experiments on CIFAR-10-LT, CIFAR-100-LT, and TinyImageNet-LT show consistent gains, especially in severe long-tailed settings. On CIFAR-100-LT with IF = 100, FedR2G reaches 37.64% Top-1 accuracy, improves Few-class accuracy by 3.28 percentage points over RUCR, and raises Macro-F1 by 2.69 percentage points. Ablations under a fixed StyleGAN-XL generator further show that the gains come from corrected semantic anchors and count-aware calibration, not only from the generator prior. FedR2G thus provides a practical way to connect geometric correction with generative rebalancing under federated long-tailed constraints. © 2026 Elsevier B.V.

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FedSea: Federated Learning via Selective Feature Alignment for Non-IID Multimodal Data

Item Type Journal Article

Author M. Tan

Author Y. Feng

Author L. Chu

Author J. Shi

Author R. Xiao

Author H. Tang

Author J. Yu

Abstract The growing demands for privacy protection challenge the joint training of one model by leveraging multiple datasets. Federated learning (FL) provides a new way to

overcome this challenge and has attracted many research interests, which enables multiple parties to collaboratively train a machine learning model without exchanging their local data. Despite some success, the non-independent and identically distributed (non-IID) data distributions in different parties remain challenging and easily damage the performance of FL methods, specifically for the heterogeneous multimodal data. Existing FL studies on non-IID data settings are often dedicated to the label space, neglecting the non-IID issues in feature space, thus limiting their performance when the parties with non-IID multimodal data. This paper proposes a new Federated learning method via Selective feature Alignment (FedSea) to align representations across multiple parties in the feature space. FedSea uses a domain adversarial learning framework consisting of an affine-transform-based generator and a gradient-reversal-based client discriminator to perform IID transformation and reduce data source distinguishability, respectively. An attention-based mask module and a feature IID confidence quantification method are introduced to effectively address the diverse feature non-IID levels across multimodal data. Comprehensive experiments are conducted on three widely-used public datasets and one large-scale industrial dataset, showing FedSea has: 1) better performance than state-of-the-art FL methods on both multimodal and single-modal datasets; 2) superior feature alignment ability on non-IID datasets, and 3) good model interpretability. © 1999-2012 IEEE.

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FedShift: Federated Learning with Classifier Shift for Class Imbalance

Item Type Conference Paper

Author Y. Shen

Author H. Wang

Author H. Lv

Abstract Federated learning enables collaborative model training without sharing raw data, but its performance degrades significantly under non-IID data. A particularly challenging case is label distribution skew, where clients have different class proportions and local updates drift away from the global objective. Existing methods such as FedProx, SCAFFOLD, FedDC, and FedDyn mitigate this issue only indirectly through regularization or variance reduction. We propose FedShift, a simple method that adds a client-specific shift to the classifier logits during local training to correct the mismatch between local and global label distributions. We show that, under label shift, this modification aligns the local optimum with the global optimum, and we further establish convergence guarantees under the strongly convex assumption. Experiments on CIFAR-10, CINIC-10, and Tiny-ImageNet show that FedShift consistently outperforms strong baselines in accuracy and improves communication efficiency by up to 4.34×. These results demonstrate that FedShift is an effective and practical solution for federated learning under label distribution skew. © 2026 IEEE.

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FedUA: An Uncertainty-Aware Distillation-Based Federated Learning Scheme for Image Classification

Item Type Journal Article

Author S.-M. Lee

Author J.-L. Wu

Abstract Recently, federated learning (FL) has gradually become an important research topic in machine learning and information theory. FL emphasizes that clients jointly engage in solving learning tasks. In addition to data security issues, fundamental challenges in this type of learning include the imbalance and non-IID among clients' data and the unreliable connections between devices due to limited communication bandwidths. The above issues are intractable to FL. This study starts from the uncertainty analysis of deep neural networks (DNNs) to evaluate the effectiveness of FL, and proposes a new architecture for model aggregation. Our scheme improves FL's performance by applying knowledge distillation and the DNN's uncertainty quantification methods. A series of experiments on the image classification task confirms that our proposed model aggregation scheme can effectively solve the problem of non-IID data, especially when affordable transmission costs are limited. © 2023 by the authors.

Date 2023

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FedUV: Uniformity and Variance for Heterogeneous Federated Learning

Item Type Conference Paper

Author H.M. Son

Author M.-H. Kim

Author T.-M. Chung

Author C. Huang

Author X. Liu

Abstract Federated learning is a promising framework to train neural networks with widely distributed data. However, performance degrades heavily with heterogeneously distributed data. Recent work has shown this is due to the final layer of the network being most prone to local bias, some finding success freezing the final layer as an orthogonal classifier. We investigate the training dynamics of the classifier by applying SVD to the weights motivated by the observation that freezing weights results in constant singular values. We find that there are differences when training in IID and non-IID settings. Based on this finding, we introduce two regularization terms for local training to continuously emulate IID settings: (1) variance in the dimension-wise probability distribution of the classifier and (2) hyperspherical uniformity of representations of the encoder. These regularizations promote local models to act as if it were in an IID setting regardless of the local data distribution, thus off-setting proneness to bias while being flexible to the data. On extensive experiments in both label-shift and feature-shift settings, we verify that our method achieves highest performance by a large margin especially in highly non-IID cases in addition to being scalable to larger models and datasets. © 2024 IEEE.

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Few-Shot Object Detection for SAR Images via Context-Aware and Robust Gaussian Flow Representation

- Item Type** Journal Article
Author P. Zhao
Author J. Chen
Author H. Wan
Author Y. Cao
Author S. Wang
Author Y. Zhang
Author Y. Li
Author Z. Huang
Author B. Wu

Abstract In recent decades, few-shot object detection in SAR imagery has gained prominence as a major research focus. The unique imaging mechanism of SAR causes the model to suffer from foreground–background imbalance and inaccurate extraction of class prototypes for novel class instances. Therefore, we propose an innovative few-shot object detection algorithm for SAR images via context-aware and robust Gaussian flow representation. First, we design the Context-Aware Enhancement module to address the foreground–context imbalance problem by refining representative support features into fine-grained prototypes, which are deeply fused with query features based on the prototype matching paradigm. Second, we devise the Manifold Class Distribution Estimation module to address the difficulty of class distribution estimation and the fluctuation of class centers of the sparse novel class. Furthermore, we formulate the Category-Balanced Difference Aggregation module to model the relationship between the base class and the novel class, addressing the sensitivity of the model to the variance of the novel class instances. Finally, we design the Cosine Decoupling Module so that the aggregated features are executed only on the classification branch without affecting the precise localization of the target. Experiments based on SAR-AIRcraft-1.0 and the self-constructed MSAR-AIR dataset indicate that the fine-grained detection and identification performance of the novel class of airplanes can reach 32.90% and 55.26%, respectively, in the 10-shot and 50-shot cases. In addition, our method enables cross-domain detection for different scenarios and sample types and exhibits excellent generalization performance in data-sparse scenarios. © 2025 by the authors.

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Few-shot Open-set Recognition by Transformation Consistency

Item Type Conference Paper

Author M. Jeong

Author S. Choi

Author C. Kim

Abstract In this paper, we attack a few-shot open-set recognition (FSOSR) problem, which is a combination of few-shot learning (FSL) and open-set recognition (OSR). It aims to quickly adapt a model to a given small set of labeled samples while rejecting unseen class samples. Since OSR requires rich data and FSL considers closed-set classification, existing OSR and FSL methods show poor performances in solving FSOSR problems. The previous FSOSR method follows the pseudo-unseen class sample-based methods, which collect pseudo-unseen samples from the other dataset or synthesize samples to model unseen class representations. However, this approach is heavily dependent on the composition of the pseudo samples. In this paper, we propose a novel unknown class sample detector, named SnaTCHer, that does not require pseudo-unseen samples. Based on the transformation consistency, our method measures the difference between the transformed prototypes and a modified prototype set. The modified set is composed by replacing a query feature and its predicted class prototype. SnaTCHer rejects samples with large differences to the transformed prototypes. Our method alters the unseen class distribution estimation problem to a relative feature transformation problem, independent of pseudo-unseen class samples. We investigate our SnaTCHer with various prototype transformation methods and observe that our method consistently improves unseen class sample detection performance without closed-set classification reduction. © 2021 IEEE

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Computer vision, Learning methods, Recognition methods, Class distribution estimation, Poor performance, Unknown class, Closed-set classifications, Estimation problem, Feature transformations, Synthesize Samples, Transformation methods

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Finding rare objects and building pure samples: Probabilistic quasar classification from low-resolution Gaia spectra

Item Type Journal Article

Author C.A.L. Bailer-Jones

Author K.W. Smith

Author C. Tiede

Author R. Sordo

Author A. Vallenari

Abstract We develop and demonstrate a probabilistic method for classifying rare objects in surveys with the particular goal of building very pure samples. It works by modifying the output probabilities from a classifier so as to accommodate our expectation (priors) concerning the relative frequencies of different classes of objects. We demonstrate our method using the Discrete Source Classifier (DSC), a supervised classifier currently based on support vector machines, which we are developing in preparation for the Gaia data analysis. DSC classifies objects using their very low resolution optical spectra. We look in detail at the problem of quasar classification, because identification of a pure quasar sample is necessary to define the Gaia astrometric reference frame. By varying a posterior probability threshold in DSC, we can trade off sample completeness and contamination. We show, using our simulated data, that it is possible to achieve a pure sample of quasars (upper limit on contamination of 1 in 40 000) with a completeness of 65 per cent at magnitudes of $G = 18.5$, and 50 per cent at $G = 20.0$, even when quasars have a frequency of only 1 in every 2000 objects. The star sample completeness is simultaneously 99 per cent with a contamination of 0.7 per cent. Including parallax and proper motion in the classifier barely changes the results. We further show that not accounting for class priors in the target population leads to serious misclassifications and poor predictions for sample completeness and contamination. We discuss how a classification model prior may, or may not, be influenced by the class distribution in the training data. Our method controls this prior and so allows a single model to be applied to any target population without having to tune the training data and retrain the model. © 2008 RAS.

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Notes:

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Finding rare objects and building pure samples: Probabilistic quasar classification with Gaia

Item Type Conference Paper
Author C.A.L. Bailer-Jones
Author K.W. Smith

Abstract We have developed a method for classifying rare objects in surveys with the particular goal of building very pure samples. It works by modifying the output probabilities from a classifier so as to accommodate our expectation (priors) concerning the relative frequencies of different classes of objects. We demonstrate our method using the Discrete Source Classifier, a supervised classifier currently based on Support Vector Machines, which we are developing in preparation for the Gaia data analysis. DSC classifies objects using their very low resolution optical spectra. We look in detail at the problem of quasar classification (partly because a pure quasar sample is necessary to define the Gaia astrometric reference frame). By varying a posterior probability threshold in DSC (Pt) we can trade off sample completeness and contamination. We show, using

our simulated data, that it is possible to achieve a pure sample of quasars (upper limit on contamination of 1 in 40 000) with a completeness of 65% at magnitudes of $G=18.5$, and 50% at $G=20.0$, even when quasars have a frequency of only 1 in every 2000 objects. The star sample completeness is simultaneously more than 99% with a contamination of 0.7%. Including parallax and proper motion in the classifier barely changes the results. Not accounting for class priors leads to serious misclassifications and poor predictions for sample completeness and contamination. Our method controls the prior and so allows a single model to be applied to any target population without having to tune the training data and retrain the model. © 2008 .American Institute of Physics.

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Fine-Grained Uncertainty Quantification via Collisions

Item Type Journal Article

Author J. Friedbaum

Author S. Adiga

Author R. Tandon

Abstract We propose a new and intuitive metric for aleatoric uncertainty quantification (UQ), the prevalence of class collisions defined as the same input being observed in different classes. We use the rate of class collisions to define the collision matrix, a novel and uniquely fine-grained measure of uncertainty. For a classification problem involving K classes, the $K \times K$ collision matrix S measures the inherent difficulty in distinguishing between each pair of classes. We discuss several applications of the

collision matrix, establish its fundamental mathematical properties, and show its relationship with existing UQ methods, including the Bayes error rate (BER). We also address the new problem of estimating the collision matrix using one-hot labeled data by proposing a series of innovative techniques to estimate S . First, we learn a pair-wise contrastive model which accepts two inputs and determines if they belong to the same class. We then show that this contrastive model (which is PAC learnable) can be used to estimate the row Gramian matrix of S , defined as $G = SST$. Finally, we show that under reasonable assumptions, G can be used to uniquely recover S , a new result on non-negative matrices which could be of independent interest. With a method to estimate S established, we demonstrate how this estimate of S , in conjunction with the contrastive model, can be used to estimate the posterior class probability distribution of any point. Experimental results are also presented to validate our methods of estimating the collision matrix and class posterior distributions on several datasets. © 2026 IEEE. All rights reserved.

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Fine-to-coarse ranking in ordinal and imbalanced domains: An application to liver transplantation

Item Type Conference Paper

Author M. Pérez-Ortiz

Author K. Fernandes

Author R. Cruz

Author J.S. Cardoso

Author J. Briceño

Author C. Hervás-Martínez

Editor Rojas I.

Editor Catala A.

Editor Joya G.

Abstract Nowadays imbalanced learning represents one of the most vividly discussed challenges in machine learning. In these scenarios, one or some of the classes in the problem have a significantly lower a priori probability, usually leading to trivial or non-desirable classifiers. Because of this, imbalanced learning has been researched to a great extent by means of different approaches. Recently, the focus has switched from binary classification to other paradigms where imbalanced data also arise, such as ordinal classification. This paper tests the application of learning pairwise ranking with multiple granularity levels in an ordinal and imbalanced classification problem where the aim is to construct an accurate model for donor-recipient allocation in liver transplantation. Our experiments show that approaching the problem as ranking solves the imbalance issue and leads to a competitive performance. © Springer International Publishing AG 2017.

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Finer-CAM: Spotting the Difference Reveals Finer Details for Visual Explanation

Item Type Conference Paper

Author Z. Zhang

Author J. Gu

Author A. Chowdhury

Author Z. Mai

Author D. Carlyn

Author T. Berger-Wolf

Author Y. Su

Author W.-L. Chao

Abstract Class activation map (CAM) has been widely used to highlight image regions that contribute to class predictions. Despite its simplicity and computational efficiency, CAM often struggles to identify discriminative regions that distinguish visually similar fine-grained classes. Prior efforts address this limitation by introducing more sophisticated explanation processes, but at the cost of extra complexity. In this paper, we propose Finer-CAM, a method that retains CAM's efficiency while achieving precise localization of discriminative regions. Our key insight is that the deficiency of CAM lies not in "how" it explains, but in "what" it explains. Specifically, previous methods attempt to identify all cues contributing to the target class's logit value, which inadvertently also activates regions predictive of visually similar classes. By explicitly comparing the target class with similar classes and spotting their differences, Finer-CAM suppresses features shared with other classes and emphasizes the unique, discriminative details of the target class. Finer-CAM is easy to implement, compatible with various CAM methods, and can be extended to multi-modal models for accurate localization of specific concepts. Additionally, Finer-CAM allows adjustable comparison strength, enabling users to selectively highlight coarse object contours or fine discriminative details. Quantitatively, we show that masking out the top 5% of activated pixels by Finer-CAM results in a larger relative confidence drop compared to baselines. The source code and demo are available at <https://github.com/Imageomics/Finer-CAM>. © 2025 IEEE.

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Fish feeding intensity assessment method using deep learning-based analysis of feeding splashes

Item Type Journal Article

Author Y. Wu

Author X. Wang

Author Y. Shi

Author Y. Wang

Author D. Qian

Author Y. Jiang

Abstract Assessment of fish feeding intensity provides effective feedback on fish starvation, which is important for improving feed utilisation and reducing water pollution. The vigorous gathering, jumping, or chasing of fish during feeding produces splashes of different sizes. The size of the splash reflects the intensity of fish feeding. However, the diverse splashes, reflected light spots on the surface of the water, and ripples created by fish movement pose inevitable problems for direct quantification methods. These methods require extensive data labelling and complex pre-processing to eliminate disturbances. Hence, this study proposes a method for assessing fish feeding intensity using a workflow involving several deep learning-based techniques such as an adaptive receptive field image feature extraction network based on a multi-kernel selection structure that becomes the teacher module for a semi-supervised splash detection module that uses a class activation map to create pseudo-labels for splashes. Feature extraction is done on the detected feeding splashes using a grey-level co-occurrence matrix and a grey-gradient co-occurrence matrix, which are then used to assess fish feeding intensity using a regression multi-classification model. The experimental results show that the feeding intensity assessment method proposed in this study achieves a mean accuracy of 98.22%, a false detection rate of 1.27%, and a missed detection rate of 2.75%; these results provide practical implications for the design of effective feeding systems that can improve feed utilisation, reduce water pollution, and enhance the welfare of farmed fish. © 2024 Elsevier B.V.

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Fisher consistency for prior probability shift

Item Type Journal Article

Author D. Tasche

Abstract We introduce Fisher consistency in the sense of unbiasedness as a desirable property for estimators of class prior probabilities. Lack of Fisher consistency could be used as a criterion to dismiss estimators that are unlikely to deliver precise estimates in test data sets under prior probability and more general data set shift. The usefulness of this unbiasedness concept is demonstrated with three examples of classifiers used for quantification: Adjusted Count, EM-algorithm and CDE-Iterate. We find that Adjusted Count and EM-algorithm are Fisher consistent. A counter-example shows that CDE-Iterate is not Fisher consistent and, therefore, cannot be trusted to deliver reliable estimates of class probabilities. © 2017 Dirk Tasche.

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Fitness Action Counting Based on MediaPipe

Item Type Conference Paper

Author X. Li

Author M. Zhang

Author J. Gu

Author Z. Zhang

Editor Chen X.

Editor Cao L.

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Editor Wang Y.

Editor Wang L.

Abstract AI fitness has become a new and practical way of fitness, but most of the mainstream fitness apps focus on guiding and planning fitness activities, ignoring the detection and evaluation of users' fitness movements. Aiming at this phenomenon, this paper proposed a method to classify and count basic fitness movements based on Google Mediapipe framework. The method consists of three steps: First, a single fitness action is divided into two detection states: up and down, and the corresponding picture samples are collected and trained. Secondly, based on the generated training set (csv file), KNN algorithm was used to identify and classify different fitness actions. Finally, the classification results are processed and the fitness actions are counted. The best recognition angle and threshold are obtained through the test accuracy. Compared with the mainstream human pose recognition frameworks such as Openpose and Alphapose, Mediapipe's BlazePose algorithm has lower performance requirements, faster recognition speed and a high level of accuracy, which is more suitable for personalized needs for smart fitness on mobile devices today. © 2022 IEEE.

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Tags:

Google+, Classification (of information), KNN, Training sets, Classification results, Test accuracy, Blazepose, BlazePose, Evaluation of users, Gesture recognition, Human pose recognition, Mediapipe, MediaPipe, Movement-based

Notes:

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Flashover Detection and Anomaly Prediction in Aerial Images of Insulator Strings in Complex Environments

Item Type Journal Article

Author L. Zhang

Author X. Li

Author J. Zhao

Author Y. Zhang

Author Q. Zhang

Abstract The normal operation of insulator strings affects the safety and stability of the power system, and insulator string flashover is one of the important faults. In this paper, considering the flashover characteristics of insulator strings in complex environments, noise is added to the collected insulator strings to simulate the actual complex environment, and then the collected data are rotationally transformed and filtered using an improved non-local mean filtering algorithm. To accurately locate insulator strings in complex environments, the geodesic active contour correction algorithm is employed to locate and segment the image. This algorithm is developed based on the level set model, replaces the image target boundary finding problem with the problem of solving a weighted arc length minimum, and introduces a penalty term into the level set energy generalization that constrains the distance function of the level set, which effectively solves the edge leakage problem. For the segmented image of insulator strings, grayscale features and wavelet features are extracted, and then the optimized random forest algorithm is utilized for prediction analysis, and the

diversity of weak learners in the random forest algorithm is increased not only through sample perturbation but also through attribute perturbation, leading to excellent generalization performance. Additionally, by combining the misclassification matrix and the a priori probability optimization algorithm, the random forest algorithm is optimized to achieve a better classification effect. Finally, the simulation of flashover discharge phenomenon characteristics and simulation experiments on a large number of insulator string images demonstrate that the proposed method has high practicality. © 2013 IEEE.

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Antennas, Image segmentation, Images segmentations, Random forests, Image edge detection, random forest, Edge detection, Problem solving, Active contours, Flashover, Geodesy, geodetic active contour, Geodetic active contour, Insulator, Insulator flashover, Noise, non-local mean filtering, Non-local means filtering, Partial discharges

Notes:

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: L. Zhang; North University of China, School of Electrical and Control Engineering, Taiyuan, 030051, China; email: zhangliping@nuc.edu.cn

Flexibility Provision From Urban Buildings to Low-Carbon Power Systems: Quantification, Aggregation and System Integration

Item Type Journal Article

Author H. Yuan

Author W. Tang

Abstract Achieving net-zero emissions requires a comprehensive transformation of energy infrastructure. The expansion of intermittent renewable generation together with increased demand, for example, from electric vehicles and heat pumps, exerts

significant stress on grid stability and reliability, making power system flexibility essential for maintaining power and energy balance and enhancing system resilience. Urban buildings, with integrated heating, cooling, on-site generation, storage and flexible demand, constitute a substantial, cost-effective flexibility resource. In this paper, building flexibility is first defined and its physical drivers are identified. Quantification methods at the individual building scale, including physics-based simulations, machine-learning models and hybrid approaches supported by recent experiments, are then surveyed. Aggregation frameworks that include Virtual Power Plants and Virtual Energy Storage Systems are examined, comparing centralised, decentralised and hierarchical control strategies, ICT requirements and market-integration pathways. The participation of aggregated building flexibility in wholesale markets and its provision of distribution-level services are analysed. Enabling measures such as dynamic tariffs, demand-response programmes, smart-readiness indicators and relevant standards are critically evaluated. Finally, future research needs in integration, standardisation and policy development are outlined. That leveraging urban building flexibility may be crucial for developing a reliable, cost-effective, low-carbon power system is suggested by our analysis. © 2025 The Author(s). IET Energy Systems Integration published by John Wiley & Sons Ltd on behalf of The Institution of Engineering and Technology and Tianjin University.

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DOI 10.1049/esi2.70017

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Energy, Smart grid, Integration, Cost effectiveness, Hierarchical systems, Power, Cost benefit analysis, Demand response, Smart power grids, Zero-carbon, Electric power transmission networks, building energy, Building energy, Demand response, demand responses, Distributed energy, distributed energy resources, Distributed Energy Resources, Electric power system stability, flexibility, Flexibility, Low carbon, smart grids, Urban buildings, virtual power plant, Virtual power plants, Virtual Power Plants

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Export Date: 20 September 2026; Cited By: 6; Funding details: Special Project for Research and Development in Key areas of Guangdong Province, (2023B0909040003); Special Project for Research and Development in Key areas of Guangdong Province; Correspondence Address: W. Tang; School of Electric Power Engineering, South China University of Technology, Guangzhou, China; email: wenhutang@scut.edu.cn

Flexible Distribution Alignment: Towards Long-Tailed Semi-supervised Learning with Proper Calibration

Item Type Conference Paper

Author E. Sanchez Aimar

Author N. Helgesen

Author Y. Xu

Author M. Kuhlmann

Author M. Felsberg

Editor Leonardis A.

Editor Ricci E.

Editor Roth S.

Editor Russakovsky O.

Editor Sattler T.

Editor Varol G.

Abstract Long-tailed semi-supervised learning (LTSSL) represents a practical scenario for semi-supervised applications, challenged by skewed labeled distributions that bias classifiers. This problem is often aggravated by discrepancies between labeled and unlabeled class distributions, leading to biased pseudo-labels, neglect of rare classes, and poorly calibrated probabilities. To address these issues, we introduce Flexible Distribution Alignment (FlexDA), a novel adaptive logit-adjusted loss framework designed to dynamically estimate and align predictions with the actual distribution of unlabeled data and achieve a balanced classifier by the end of training. FlexDA is further enhanced by a distillation-based consistency loss, promoting fair data usage across classes and effectively leveraging underconfident samples. This method, encapsulated in ADELLO (Align and Distill Everything All at Once), proves robust against label shift, significantly improves model calibration in LTSSL contexts, and surpasses previous state-of-the-art approaches across multiple benchmarks, including CIFAR100-LT, STL10-LT, and ImageNet127, addressing class imbalance challenges in semi-supervised learning. Our code is available at <https://github.com/emasa/ADELLO-LTSSL>. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

Volume 15112 LNCS**Publisher** Springer Science and Business Media Deutschland GmbH**ISBN** 03029743 (ISSN); 978-303172948-5 (ISBN)**Pages** 307-327**Proceedings Title** Lect. Notes Comput. Sci.**Conference Name** Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)**Date Added** 20/09/2026, 9:51:57 pm**Modified** 20/09/2026, 9:51:57 pm**Tags:**

Federated learning, Adversarial machine learning, Contrastive Learning, Self-supervised learning, Labeled data, Semi-supervised, Semi-supervised learning, Unlabeled data, Calibration, Class distributions, Distribution alignment, Balanced classifiers, Confidence calibration, Confidence Calibration, Data encapsulation, Data usage, Distribution Alignment, Long-tailed, Rare class, Semi-supervised Learning

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EID: 2-s2.0-85208545165

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Flow cytometric S-phase fraction measurement in breast carcinoma: Influence of software and histogram resolution

Item Type Journal Article**Author** M.-L. Jourdan**Author** M. Ferrero-Poüs**Author** F. Spyrtos**Author** S. Romain**Author** P.-M. Martin**Author** A. Chassevent

Abstract Background: S-phase fraction (SPF) measurement by flow cytometry is a clinically useful prognostic factor in patients with breast carcinoma. Standardized SPF determination is essential. As part of a multicenter study, we evaluated the influence of the choice of software and histogram resolution (256, 512, or 1,024 channels) on SPF quantification. Methods: One hundred thirty-three DNA histograms were analyzed in three laboratories with Modfit 5.2, Modfit LT, and Multicycle AV software. Strict rules for histogram interpretation and software management were

applied. The following five options were compared: MF 5.2 1024, MF 5.2 256, MF LT 256, MCAV 256, and MC AV 512. Results: In the DNA diploid and aneuploid groups, SPF distributions were not statistically different among the five options, Excellent quantitative correlations were obtained between pairs of options. When using tertiles as cutpoints for SPF classification, concordance rates ranged from 79.7% to 93.2% for DNA diploid samples and from 87.8% to 95.9% for DNA aneuploid samples, the best results being obtained with software working with a similar histogram resolution. Conclusions: Standardized use of commercially available software, including the choice of histogram resolution, provides comparable SPF results. © 2002 Wiley-Liss, Inc.

Date 2002

DOI 10.1002/cyto.10116

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female, Female, human, Humans, article, reliability, reproducibility, Reproducibility of Results, comparative study, quantitative analysis, controlled study, methodology, Comparative Study, evaluation, flow cytometry, human tissue, pathology, clinical trial, Breast Neoplasms, breast tumor, prognosis, Prognosis, DNA, histogram, multicenter study, standard, DNA determination, priority journal, breast carcinoma, carcinoma, Flow Cytometry, DNA, Neoplasm, diploidy, Software Design, computer program, cell cycle S phase, Breast carcinoma, Flow cytometry, Support, Non-U.S. Gov't, Human, aneuploidy, Carcinoma, cell cycle G0 phase, cell cycle G1 phase, DNA content, Histogram resolution, Ploidies, ploidy, S Phase, S-phase fraction, Software program

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Export Date: 20 September 2026; Cited By: 16; Correspondence Address: M.-L. Jourdan; Laboratoire de Cancérologie, INSERM EMI 0211, CHU Bretonneau, 37044 Tours Cedex, 2 Bd Tonnellé, France; email: m.jourdan@chu-tours.fr; CODEN: CYTOD

Flowcam: Quantification and classification of phytoplankton by imaging flow cytometry

Item Type Journal Article

Author N.J. Poulton

Abstract The ability to enumerate, classify, and determine biomass of phytoplankton from environmental samples is essential for determining ecosystem function and their role in the aquatic community and microbial food web. Traditional micro-phytoplankton quantification methods using microscopic techniques require preservation and are slow, tedious and very laborious. The availability of more automated imaging microscopy platforms has revolutionized the way particles and cells are detected within their natural environment. The ability to examine cells unaltered and without preservation is key to providing more accurate cell concentration estimates and overall phytoplankton biomass. The Flow Cam ® is an imaging cytometry tool that was originally developed for use in aquatic sciences and provides a more rapid and unbiased method for enumerating and classifying phytoplankton within diverse aquatic environments. © Springer Science+Business Media New York 2016.

Date 2016

DOI 10.1007/978-1-4939-3302-0_17

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Journal Abbr Methods Mol. Biol.

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Article, nonhuman, procedures, classification, aquatic environment, Image analysis, flow cytometry, chemistry, image analysis, cytology, biomass, imaging, devices, priority journal, Flow Cytometry, image cytometry, chlorophyll, Chlorophyll, cell size, Chlorophyll fluorescence, flow cytometer, Flow cytometry, FlowCam, Image Cytometry, microscope, phycoerythrin, Phycoerythrin, phytoplankton, Phytoplankton

Notes:

EID: 2-s2.0-84990040283

Export Date: 20 September 2026; Cited By: 54

FMAPLS: Bayesian Label Shift Estimation Based on Dynamic Dirichlet Parameter Adaptation

Item Type Journal Article

Author J. Hu

Author J.A. Barria

Abstract Label shift, a critical challenge in supervised learning, occurs when the class prior distribution of test data deviates from that of training data, leading to significant degradation in classifier performance. This issue is particularly impactful in real-world applications such as medical diagnosis and satellite imaging, where a reduction in classification accuracy can cause severe consequences. Existing methods, such as Maximum A Posterior Label Shift (MAPLS), rely on strict Dirichlet hyperparameter constraints, thereby imposing a limitation on their adaptability to dynamic learning environments and class-imbalanced scenarios. To address this problem, we propose Full Maximum A Posterior Label Shift (FMAPLS), a Bayesian label shift estimation framework that dynamically co-optimizes Dirichlet hyperparameters and class priors through Expectation-Maximization (EM) iterations. By eliminating rigid hyperparameter constraints and introducing a linear surrogate function, FMAPLS enhances both expressivity and robustness in prior estimation while reducing computational complexity. Extensive experiments on CIFAR100 under shuffled long-tail and Dirichlet imbalanced test priors demonstrate FMAPLS's superiority over state-of-the-art baselines, achieving up to 50% lower KL divergence and maintaining accuracy improvements by up to 0.5% in extreme imbalance settings. The results validate the efficacy of dynamic hyperparameter adaptation in estimating label shift, particularly under high class imbalance and distributional uncertainty. © 1994-2012 IEEE.

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Journal Abbr IEEE Signal Process Lett

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Tags:

ON dynamics, Classification (of information), Uncertainty analysis, Statistics, Medical imaging, Critical challenges, Hyper-parameter, Optimization, Diagnosis, Expectation maximization algorithm, Label shift, Maximum principle, Bayesian, Shift estimations, expectation-maximization algorithm, Dirichlet, Expectations maximization algorithms, imbalance data, Imbalance data, Parameter adaptation

Notes:

EID: 2-s2.0-105019545719

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: J.A. Barria; Imperial College London, Department of Electrical and Electronic Engineering, London, SW7 2AZ, United Kingdom; email: j.barria@imperial.ac.uk; CODEN: ISPLE

Food waste quantification; Methods and Benefits

Item Type Conference Paper

Author S. Anwar

Author M. Shahbaz

Author A. Saeed

Author R. Parsad

Abstract Food waste is a significant socioeconomic and environmental problem. Recent research has provided more evidence, particularly considering the UNO's sustainable development goals. Many factors accelerate food wastage, and most studies consider food waste quantification a technique to reduce food wastage, where multiple classical and manual methods have been applied. However, in most studies, food waste quantification is deployed to measure single or specific waste, not considering the additional contributing factors. Also, multiple studies highlighted the importance of different factors that helps in food waste reduction. The paper provides a comprehensive literature review linking many food waste factors for food waste quantification. In addition, we provide two diagrams that show methods and critical areas where food waste quantification helps in the policymaking process to reduce wasting food. © 2022 IEEE.

Date 2022

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Publisher IEEE Computer Society

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Tags:

Sustainable development, Computer vision, Internet of things, IoT, sustainable development, computer vision, Quantification methods, Waste classification, Waste quantifications, food loss, Food loss, Food wastages, Food waste, food waste quantification, Food waste quantification, quantification methods, Socio-economic problems, waste classification

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Forecasting CPU and Memory Usage in Cloud Workloads Using Task Metadata: A Machine Learning Approach on Google Cluster Data

Item Type Conference Paper

Author N. Kumar

Author S. Kukreja

Author Y. Kumar

Author Y. Choudhary

Editor Vijayan K V.K.

Editor Singh V.

Abstract Precise estimation of CPU and memory usage in cloud settings is crucial for improving resource allocation efficiency and reducing operational expenditures. The proposed machine learning framework in this work aims at predicting average CPU usage and average memory usage from task-level metadata of the Google 2019 Cluster Dataset with features of scheduling class, priority, requested resources, task duration, and failure status. Hybrid models that integrate tree-based techniques (LightGBM, Random Forest) and deep learning models (Long Short-Term Memory (LSTM), Multilayer Perceptron (MLP)) are compared to isolated models such as XGBoost and Convolutional Neural Network (CNN). Experiments prove that the hybrid LightGBM + MLP + LSTM model performs better with an RMSE of 0.08 for CPU prediction and 0.12 for memory consumption prediction and R2 scores of 0.94 and 0.91, respectively. Results indicate the importance of incorporating task metadata, especially resource requests and time, for enhancing scheduling efficiency. This eliminates provisioning errors by 25%, supports dynamic scaling of resources, and reduces infrastructure expenses in cloud data centers. © 2026 The Authors.

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Extra Journal Abbreviation: Procedia Comput. Sci.

Volume 283

Publisher Elsevier B.V.**ISBN** 18770509 (ISSN)**Pages** 1907-1916**Proceedings Title** Procedia Comput. Sci.**Conference Name** Procedia Computer Science**Date Added** 20/09/2026, 9:51:46 pm**Modified** 20/09/2026, 9:51:46 pm**Tags:**

Google+, Learning systems, Machine-learning, Resource allocation, Forecasting, Convolutional neural networks, Resources allocation, Cluster analysis, Metadata, Cloud Computing, Machine Learning, Hybrid model, Cloud-computing, Long short-term memory, Cloud computing, Decision trees, Random forests, Resource Allocation, Memory management, Short term memory, Multilayers perceptrons, Machine learning approaches, Hybrid Models, Cluster computing, Memory usage, Task metadata, Task Metadata

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Kukreja; SCSET, Bennett University, Greater Noida, U.P., 201310, India; email: kukreja.sonal@gmail.com; Conference name: 4th international conference on Machine Learning and Data Engineering, ICMLDE 2025; Conference date: 6 November 2025 through 8 November 2025; Conference code: 226014

Forecasting the Future Development in Quality and Value of Professional Football Players

Item Type Journal Article**Author** K. van Arem**Author** F. Goes-Smit**Author** J. Söhl

Abstract Featured Application: This paper studies what models are most suitable for forecasting future values of player performance metrics in association football (soccer). The resulting forecast statistics find applications in team management and player scouting at football clubs. As transfer decisions concern whether a player should play for a club in the future, the predictions of future performance metrics offer a forward-looking improvement over the traditional backward-looking assessments. Transfers in professional football (soccer) are risky investments because of the large transfer fees and high risks involved. Although data-driven models can be used to improve transfer decisions, existing models focus on describing players' historical progress, leaving their future performance unknown. Moreover, recent developments have called for the use of explainable models combined with methods for uncertainty quantification of predictions to improve applicability for practitioners. This paper assesses explainable machine learning models in a practitioner-oriented

way for the prediction of the future development in quality and transfer value of professional football players. To this end, the methods for uncertainty quantification are studied through the literature. The predictive accuracy is studied by training the models to predict the quality and value of players one year ahead, equivalent to one season. This is carried out by training them on two data sets containing data-driven indicators describing the player quality and player value in historical settings. In this paper, the random forest model is found to be the most suitable model because it provides accurate predictions as well as an uncertainty quantification method that naturally arises from the bagging procedure of the random forest model. Additionally, this research shows that the development of player performance contains nonlinear patterns and interactions between variables, and that time series information can provide useful information for the modeling of player performance metrics. The resulting models can help football clubs make more informed, data-driven transfer decisions by forecasting player quality and transfer value. © 2025 by the authors.

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Notes:

EID: 2-s2.0-105014462108

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: K. van Arem; Delft Institute of Applied Mathematics, Delft University of Technology, Delft, 2628 CD, Netherlands; email: k.w.vanarem@tudelft.nl

Fostering knowledge transfer and learning in shift work environments

Item Type Journal Article

Author R. Bosua

Author K. Venkitachalam

Abstract Shift work is a continuous 'round-the-clock' work practice that involves rotating work schedules with a vital process of 'handover' denoting a change of teams between shifts. Handover as an activity requires that outgoing shift teams pass on insights and responsibility to incoming shift teams. Knowledge transfer in shift work environments is therefore crucial to allow for a seamless continuation of work practices between shifts. Studies in shift work indicate that knowledge transfer between shifts often fails—that is, incoming workers tend to solve problems with inadequate information, have an incomplete understanding of significant events that occurred in prior shifts, while workers often attempt to solve the same problems across different shifts. This study investigates the challenges associated with shift handover and proposes knowledge transfer enablers that can make a difference to handover. In addition, these enablers can foster learning, a process often overlooked in shift environments. A qualitative research methodology was used to study three distinctive case organisations in the manufacturing and educational sectors, where the nature of day-to-day work is shift-bound and transfer issues were present during handover processes. Our findings suggest that three enablers facilitate knowledge transfer problems associated with shift work: (1) a purposeful knowledge codification and classification culture, (2) open access to established boundary objects and boundary spanners and (3) a unified information infrastructure to facilitate knowledge transfer during shifts and handover. © 2015 John Wiley & Sons, Ltd.

Date 2015

DOI 10.1002/kpm.1456

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Foundation models for clean energy forecasting: A comprehensive review

Item Type Journal Article

Author M.M. Ferdous

Author T. Dam

Author M.R. Sarkar

Author M. Uddin

Author S.G. Anavatti

Abstract As global energy systems transition to clean energy, accurate renewable generation forecasting is imperative for effective grid management. Foundation Models (FMs), which are large-scale, pre-trained neural networks capable of learning adaptable representations from large datasets, possess transformative potential for renewable energy forecasting by effectively analyzing intricate, high-dimensional time-series data and adapting to a variety of downstream tasks. This review examines FMs in renewable energy forecasting, with primary emphasis on wind and solar power generation. We systematically analyze how FMs address diverse forecasting scenarios spanning multiple temporal scales (ultra-short to long-term), spatial resolutions (individual installations to system-wide), data modalities (univariate to multi-modal), and operational contexts (zero-shot to supervised learning). The focus on generation forecasting rather than demand forecasting is motivated by the unique challenges of renewable generation, including weather dependency, spatial-temporal variability, and intermittency, which necessitate specialized multi-modal data integration (satellite imagery, numerical weather prediction, atmospheric modeling) distinct from the socioeconomic and behavioral patterns characterizing load prediction. An overview of FM architectures, pre-training strategies, fine-tuning methods, and data modalities used in renewable energy forecasting is presented. Particular emphasis is placed on large-scale, domain-specific Transformer architectures that capture spatial-temporal correlations, embed domain knowledge, and address the intermittent nature of renewable generation. Recent FM-based advancements in forecast accuracy are assessed, including multi-scale temporal prediction reconciliation and uncertainty quantification methods. Existing challenges and areas for improvement in long-term and multi-variate time series forecasting are reviewed. A distinction between theoretical developments and practical deployment considerations is established throughout the survey. The strengths and weaknesses of FMs are critically assessed, and future research directions in this rapidly emerging area are advanced. © 2025 Elsevier Ltd

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Export Date: 20 September 2026; Cited By: 19; Correspondence Address: M.M. Ferdaus; Department of Computer Science, University of New Orleans, New Orleans, 70148, United States; email: mferdaus@uno.edu; CODEN: RSERF

Foundations of statistical mechanics for unstable interactions

Item Type Journal Article

Author R. Hilfer

Abstract Traditional Boltzmann-Gibbs statistical mechanics does not apply to systems with unstable interactions, because for such systems the conventional thermodynamic limit does not exist. In unstable systems the ground state energy does not have an additive lower bound, i.e., no lower bound linearly proportional to the number N of particles or degrees of freedom. In this article unstable systems are studied whose ground state energy is bounded below by a regularly varying function of N with index $\sigma \geq 1$. The index $\sigma \geq 1$ of regular variation introduces a classification with respect to stability. Stable interactions correspond to $\sigma = 1$. A simple example for an unstable system with $\sigma = 2$ is an ideal gas with a nonvanishing constant two-body potential. The foundations of statistical physics are revisited, and generalized ensembles are introduced for unstable interactions in such a way that the thermodynamic limit exists. The extended ensembles are derived by identifying and postulating three basic properties as extended foundations for statistical mechanics: first, extensivity of thermodynamic systems, second, divisibility of equilibrium states, and third, statistical independence of isolated systems. The traditional Boltzmann-Gibbs postulate, resp. the hypothesis of equal a priori probabilities, is identified as a special case of the extended ensembles. Systems with unstable interactions are found to be thermodynamically normal and extensive. The formalism is applied to ideal gases with constant many-body potentials. The results show that, contrary to claims in the literature, stability of the interaction is not a necessary condition for the existence of a thermodynamic limit. As a second example the formalism is applied to the Curie-Weiss-Ising model with strong coupling. This model has index of stability $\sigma = 2$. Its thermodynamic potentials [originally obtained in R. Hilfer, *Physica A* 320, 429 (2003)10.1016/S0378-4371(02)01585-6] are confirmed up to a trivial energy shift. The strong coupling model shows a thermodynamic phase transition of order 1 representing a novel mean-field universality class. The disordered high temperature phase collapses into the ground state of the system. The metastable extension of the high temperature free energy to low temperatures ends at absolute zero in a phase transition of order $1/2$. Between absolute zero and the critical temperature of the first order transition all fluctuations are absent. © 2022 American Physical Society.

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Tags:

article, Statistical mechanics, probability, Temperature, Low bound, physics, high temperature, phase transition, gas, Absolute zero, Boltzmann-Gibbs statistical mechanics, critical temperature, degree of freedom, Degrees of freedom (mechanics), Foundations, Free energy, Ground state, Ground-state energies, Ideal gas, Ising model, Linearly proportional, low temperature, Phase transitions, Regular variations, Regularly varying functions, Thermodynamic limits, Unstable system

Notes:

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Export Date: 20 September 2026; Cited By: 3

Four decades of full-depth profiles reveal layer-resolved drivers of reservoir thermal regimes and episodic hypolimnetic warming

Item Type Journal Article

Author C. Mi

Author B. Gai

Author X. Kong

Author Y. Jiang

Author C.N. Chan

Author K. Rinke

Abstract Thermal structure shapes ecological dynamics in lakes and reservoirs. Yet full-profile temperature records over multi-decades remain scarce, constraining mechanistic understanding of depth-resolved thermal changes and subseasonal extremes (e.g., surface heat waves and late-season hypolimnetic warming). In this study, we focused on Rappbode Reservoir, Germany's largest drinking-water reservoir, and compiled four decades of high-resolution, full-depth temperature profiles with concurrent hydro-meteorological records that are rarely available for stratified systems. Building

on these data, we developed a novel two-step analytical framework that integrates long-term monitoring and process-based modelling to yield a high-resolution, internally consistent dataset of spatiotemporal temperature dynamics. We then applied interpretable machine learning to quantify depth-specific statistical attributions of hydro-meteorological and operational drivers, using process-based interpretation to assess mechanisms associated with late-stratification hypolimnetic warming. Our results suggested that driver contributions varied markedly with depth and stratification phase: stratification-strength metrics tied to atmospheric heat exchange (i.e., surface temperature, vertical temperature difference, Schmidt stability) showed the largest SHAP contributions from the 30 d antecedent moving average of shortwave radiation and air temperature. For hypolimnetic temperature, outflow discharge accounted for the largest contribution during late stratification. Additional process-based analyses supported a link between episodic hypolimnetic warming by up to 10 °C in four specific years and intensified deep withdrawals, which weakened the density gradient and promoted downward transport of warm upper-layer water into the hypolimnion. The dual-perspective framework developed here, which integrates process-based and machine-learning approaches, is broadly transferable for analyzing ecological processes and supporting evidence-based management in stratified waters. © 2026 Chenxi Mi et al.

Date 2026

DOI 10.5194/hess-30-4611-2026

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Archive Scopus

Language English

Volume 30

Publisher Copernicus Publications

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Publication Hydrology and Earth System Sciences

ISSN 10275606 (ISSN)

Issue 14

Journal Abbr Hydrol. Earth Syst. Sci.

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Learning systems, machine learning, Machine learning, data set, Germany, Ecology, Atmospheric temperature, Reservoirs (water), High resolution, analytical framework, Depth-profile, Depth-resolved, Ecological dynamics, Heat radiation, Lake sand reservoirs, Mechanistics, outflow, Process-based, shortwave radiation, stratification, Thermal changes, Thermal regimes, Thermal structure, warming

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Helmholtz-Zentrum für Umweltforschung, UFZ; National Key Research and Development Program of China, NKPs, (2023YFF0807203); Self-directed Research Project of the State Key Laboratory of Lake and Watershed Science for Water Security, (SKL2026-QN03); Chinese Academy of Sciences, CAS, (2024PG0017, 2024DC0005, 045GJHZ2023055FN); Science and Technology Projects of Xizang Autonomous Region, China, (XZ202501ZY0091); National Natural Science Foundation of China, NSFC, (42177062, 42577078); Correspondence Address: X. Kong; State Key Laboratory of Lake and Watershed Science for Water Security, Nanjing Institute of Geography and Limnology, Chinese Academy of Sciences, Nanjing, 211135, China; email: xzkong@niglas.ac.cn

Fourth International Workshop on Reproducible Research in Pattern Recognition, RRPR 2022

Item Type Journal Article

Editor Kerautret B.

Editor Colom M.

Editor Krähenbühl A.

Editor Lopresti D.

Editor Monasse P.

Editor Perret B.

Abstract The proceedings contain 9 papers. The special focus in this conference is on Reproducible Research in Pattern Recognition. The topics include: A Novel Pattern-Based Edit Distance for Automatic Log Parsing: Implementation and Reproducibility Notes; companion Paper: Deep Saliency Map Generators for Multispectral Video Classification; on Challenging Aspects of Reproducibility in Deep Anomaly Detection; On the Implementation of Baselines and Lightweight Conditional Model Extrapolation (LIMES) Under Class-Prior Shift; Combining Max-Tree and CNN for Segmentation of Cellular FIB-SEM Images; Automatic Forest Road Extraction from LiDAR Data Using Convolutional Neural Networks; Promoting Reproducibility of Research Results in International Events (Report from the 4th RRPR).

Date 2023

URL <https://www.scopus.com/pages/publications/85172249258?origin=resultslist>

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Language English

Volume 14068 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

ISSN 03029743 (ISSN); 978-303140772-7 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

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Export Date: 20 September 2026; Cited By: 0; Conference name: Fourth International Workshop on Reproducible Research in Pattern Recognition, RRPR 2022; Conference date: 21 August 2022 through 21 August 2022; Conference code: 299649

Framework for incorporating class priors into discriminative classification

Item Type Conference Paper

Author R. Jin

Author Y. Liu

Abstract Discriminative and generative methods provide two distinct approaches to machine learning classification. One advantage of generative approaches is that they naturally model the prior class distributions. In contrast, discriminative approaches directly model the conditional distribution of class given inputs, so the class priors are only implicitly obtained if the input density is known. In this paper, we propose a framework for incorporating class prior proportions into discriminative methods in order to improve their classification accuracy. The basic idea is to enforce that the distribution of class labels predicted on the test data by the discriminative model is consistent with the class priors. Therefore, the discriminative model has to not only fit the training data well but also predict class labels for the test data that are consistent with the class priors. Experiments on five different UCI datasets and one image database show that this framework is effective in improving the classification accuracy when the training data and the test data come from the same class proportions, even if the test data does not have exactly the same feature distribution as the training data. © Springer-Verlag Berlin Heidelberg 2005.

Date 2005

DOI 10.1007/11430919_66

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Archive Scopus

Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

Volume 3518 LNAI

Publisher Springer Verlag

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Pages 568-577

Proceedings Title Lect. Notes Comput. Sci.

Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Tags:

Classification (of information), Learning systems, Data mining, Framework, Image processing, Management information systems, Database systems, Discriminative classification, Input density

Notes:

EID: 2-s2.0-26944477755

Export Date: 20 September 2026; Cited By: 3; Correspondence Address: R. Jin; Department of Computer Science and Engineering, Michigan State University, East Lansing, MI 48824, United States; email: rongjin@cse.msu.edu; Conference name: 9th Pacific-Asia Conference on Advances in Knowledge Discovery and Data Mining, PAKDD 2005; Conference date: 18 May 2005 through 20 May 2005; Conference code: 67063

From manual scoring to machine learning: recent developments in image-based arbuscular mycorrhizal fungi root quantification

Item Type Journal Article**Author** A. Bouvette**Author** S. Basiru**Author** M. Hijri

Abstract Selecting an appropriate method to quantify arbuscular mycorrhizal fungi (AMF) colonization in plants can be challenging, particularly for non-experts, due to the wide diversity of available techniques, some of which date back to the 1980s. Despite rapid methodological advances, comprehensive reviews focusing specifically on recent image analysis approaches for AMF root quantification are still scarce in the scientific literature. Given the growing potential of image-based methods to improve the accuracy, reproducibility, and high throughput of AMF root colonization assessment, this review synthesizes conceptual structure and thematic map of studies focusing on AMF colonization from 2001 to 2026. In addition, we examine widely used traditional approaches, such as the gridline intersect method, and emerging tools, including Visual Basic for Application–Arbuscular Mycorrhizal Fungi (VBA-AMF), MycoPatt, WinRHIZO, ImageJ-based workflows, Zeiss Intellesis, and deep learning approaches such as Automatic Mycorrhizal Finder (AMFinder), “Tool for Analysing root images to calculate the Infection rate of arbuscular Mycorrhizal fungi” (TAIM) and Mask Region-based Convolutional Neural Network (Mask R-CNN). By critically comparing their principles, capabilities, and limitations, this review aims to guide researchers in selecting the most appropriate image analysis methods for AMF quantification according to their experimental objectives and technical constraints. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026.

Date 2026**DOI** 10.1007/s00572-026-01283-2**URL** <https://www.scopus.com/pages/publications/105043785292?origin=resultslist>**Archive** Scopus**Language** English**Volume** 36**Publisher** Springer Science and Business Media Deutschland GmbH**Publication** Mycorrhiza**ISSN** 09406360 (ISSN)**Issue** 4**Journal Abbr** Mycorrhiza

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Tags:

machine learning, Machine learning, thematic mapping, Machine Learning, procedures, Image analysis, accuracy assessment, image processing, Quantification methods, arbuscular mycorrhiza, Arbuscular mycorrhizal fungi, Digital microscopy, fungus, image analysis, Image Processing, Computer-Assisted, microbiology, microscopy, mycorrhiza, Mycorrhizae, physiology, plant root, Plant Roots, root colonization, Root colonization

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Desjardins Foundation; Natural Sciences and Engineering Research Council of Canada, CRSNG, (ALLRP-577245-2022, RGPIN-2024-04954); Fonds de recherche du Québec, FRQ, (369572, 2007662); Correspondence Address: M. Hijri; Institut de Recherche en Biologie Végétale (IRBV), Département de Sciences Biologiques, Université de Montréal, Montréal, H1X 2B2, Canada; email: Mohamed.Hijri@umontreal.ca; CODEN: MSCORE

From outcrop observations to dynamic simulations: an efficient workflow for generating ensembles of geologically plausible fracture networks and assessing their impact on flow and transport

Item Type Journal Article

Author E. Kamel Targhi

Author P.-O. Bruna

Author A. Daniilidis

Author G. Rongier

Author S. Geiger

Abstract Fractures are ubiquitous in geological formations and can often have an impact on subsurface applications such as geothermal energy, groundwater management or CO₂ storage. Quantifying the relationship between the uncertainties inherent to fracture networks and the corresponding flow behaviour for these applications remains an open challenge. Simulation studies that are based on outcrop analogues of fracture networks have yielded many new insights about heat and mass transfer in fractured geological formations but are restricted to a limited number of fracture network realizations, simplified assumptions about fracture network properties or deterministic models, making it difficult to analyse a wide range of uncertainties. This study introduces a flexible workflow that generates ensembles of geologically plausible fracture networks that can be based on statistical data from outcrop analogues. The fracture networks are generated using a computationally efficient approach that combines mechanical and statistical methods. The ensembles are then seamlessly linked to multi-purpose flow and transport simulations where the fractures are represented explicitly in a porous and permeable rock matrix. This approach can

enable new uncertainty quantification methods, supported by machine-learning-based emulators, to analyse how fracture network properties, such as fracture intensity, fracture aperture or fracture orientation, influence heat and mass transfer in fractured geological formations. The workflow is illustrated using two classic example applications pertinent to fracture network modelling – one based on outcrop data to assess thermal behaviour in geothermal systems, and one synthetic study to analyse the transition from matrix-dominated to fracture-dominated flow – and released as open-source code. © 2025 The Author(s). Published by The Geological Society of London for GSL and EAGE.

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From Perception to Regulation: Frontier Advances in Structural Mechanisms, Objective Quantification, and Debitting Strategies of Bitter Peptides

Item Type Journal Article

Author J. Gao

Author Y. Yuan

Author J. Li

Abstract The bitterness of protein hydrolysates is a critical sensory defect that severely compromises consumer acceptance and limits their application in functional foods. Bitter peptides, the primary source of this undesirable taste, often possess significant bioactivity, creating a challenging “flavor-function” duality. This conflict arises fundamentally because the structural mechanisms governing peptide-bitter taste receptors (TAS2R) interactions and the correlation between bitterness and bioactivity remain incompletely understood. Consequently, objective quantification methods lack standardization, and debittering strategies often lack precision, hindering the development of palatable functional products. This review aims to systematically

analyze the structural basis of bitterness, its relationship with bioactivity, objective quantification technologies, and advanced strategies for bitter peptide removal, masking, and modification. Key findings reveal that bitterness perception is intricately linked to specific hydrophobic motifs and spatial conformation, often overlapping with bioactive features. Emerging computational resources and instruments, such as comprehensive databases, quantitative structure–activity relationship (QSAR) and machine learning models, and high-sensitivity bionic electronic tongue, offer superior high-throughput capabilities and accelerate the elucidation of peptide-TAS2R binding mechanisms compared to traditional sensory evaluation. Furthermore, distinct debittering strategies are identified: Encapsulation and masking technologies are preferred when retaining the bioactivity of functional bitter peptides is essential. Ultimately, resolving the “flavor-function” duality requires a deeper understanding of the structural mechanisms governing peptide-TAS2R binding. This fundamental insight will provide the necessary basis for standardized prediction models and the design of precision debittering technologies, thereby enabling the optimized application of bitter peptides in the development of palatable and functional protein hydrolysates. © 2026 Institute of Food Technologists.

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DOI 10.1111/1750-3841.71062

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Journal Abbr J. Food. Sci.

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human, Humans, machine learning, review, predictive model, chemistry, biological activity, bitter peptides, bitter taste, data base, debittering strategies, encapsulation, flavor, functional food, Functional Food, functional properties, G protein coupled receptor, metabolism, peptide, Peptides, protein hydrolysate, Protein Hydrolysates, quantifying bitterness, quantitative structure activity relation, Quantitative Structure-Activity Relationship, Receptors, G-Protein-Coupled, regulatory mechanism, sensory dysfunction, sensory evaluation, standardization, structural characteristics, taste, Taste, taste bud, Taste Perception, taste receptor type 2, Taste Receptors, Type 2

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Export Date: 20 September 2026; Cited By: 1; Funding details: Key Research and Development Program of Ningxia, (2024BEE02022); Correspondence Address: J. Li; Key Laboratory of Geriatric Nutrition and Health (Beijing Technology and Business University), Ministry of Education, Beijing, China; email: lijinlong@btbu.edu.cn; CODEN: JFDSA

From Pixels to Probabilities: A Streamlined Interpretable Model for Digit Classification Using PCA Denoising and KDE

Item Type Conference Paper

Author Y. Wu

Author E.Y. Zhu

Author R. Ding

Author H. Li

Author P. Shan

Author J. Ding

Author T. Kang

Author Y. Qiao

Author C. Jiang

Abstract This paper presents a streamlined and interpretable framework for handwritten digit classification on the MNIST dataset. While deep learning models offer superior accuracy, they often function as 'black boxes' with high computational demands. To address this, we propose a modular pipeline that integrates Principal Component Analysis (PCA) as a denoising filter with mutual information-based pixel selection and nonparametric kernel density estimation (KDE). Crucially, we find ICA to be redundant within reconstruction-based pipelines, as PCA denoising provides sufficient feature preservation for KDE modeling. The framework further incorporates an optimized Jensen-Shannon divergence-based weighting scheme to resolve confusing digit pairs and a Bayesian MAP decision rule using empirical class priors. The resulting method achieves an accuracy of (83.36%) while maintaining full pixel-level transparency and high computational efficiency. This work demonstrates that efficient generative models can provide a rigorous and explainable alternative to discriminative models in scenarios where interpretability and uncertainty quantification are paramount. © 2026 IEEE.

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DOI 10.1109/SoutheastCon63549.2026.11475991

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Tags:

Deep learning, Classification (of information), Learning systems, Machine-learning, Uncertainty analysis, Computational efficiency, Statistics, Bayesian networks, Principal component analysis, Generative model, Character recognition, De-noising, Bayesian classification, Generative modeling, Handwritten digit recognition, Interpretable machine learning, Jensen-Shannon divergence, Kernel density estimation, Kernel Density Estimation, MNIST, Mutual information, Mutual informations, PCA denoising, Pixels, Principal component analyze denoising, Principal-component analysis

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Y. Wu; College of Computing, Georgia Institute of Technology, Atlanta, United States; email: ywu3134@gatech.edu; Conference name: 2026 IEEE SoutheastCon, SoutheastCon 2026; Conference date: 20 February 2026 through 15 March 2026; Conference code: 222644; CODEN: CPISD

From quantification to classification: comparative analysis of two software applications for machine learning–based prediction of early Parkinson’s disease using ¹²³I-Ioflupane metrics

Item Type Journal Article

Author L. Filippi

Author F. Bianconi

Author A. Marongiu

Author M.L. Fravolini

Author M. Minestrini

Author S. Nuvoli

Author A. Spanu

Author O. Schillaci

Author B. Palumbo

Abstract Purpose: To compare semiquantitative striatal indices produced by two widely used quantification platforms — BasGanV2™ and NeuroTrans3D (Oasis©) — for ¹²³I-Ioflupane DAT SPECT, to evaluate their diagnostic performance for Parkinson’s disease (PD) versus non-degenerative presentations, and to determine whether supervised machine-learning (ML) classifiers can clarify relative utility of the two methods. Methods: Retrospective analysis of 90 consecutive subjects (48 PD, 42 not-PD) undergoing routine ¹²³I-Ioflupane SPECT. Striatal binding ratios (caudate and putamen, left and right) were computed with both tools. Distributional differences were tested (Mann–Whitney U and Brown–Forsythe/Levene tests). Three supervised classifiers (Gaussian naïve Bayes, k-nearest neighbours, and SVM-RBF) were trained on four regional metrics from each tool using stratified shuffle-split (60:40 train: test)

repeated 50 times; performance was summarised as mean (95% percentile CI) accuracy, sensitivity, specificity and AUC. Results: Both quantification methods discriminated PD from not-PD (all regions $p < 0.0001$). NeuroTrans3D/Oasis© produced consistently higher mean binding values and significantly greater variance than BasGanV2™. ML models trained on either tool achieved excellent AUCs (0.917–0.960) and similar accuracy (≈ 86 –91%), with overlapping confidence intervals and no single tool/classifier combination clearly outperforming the others. Conclusion: BasGanV2™ and NeuroTrans3D (Oasis©) are comparably effective for distinguishing PD from non-degenerative cases when combined with straightforward ML classifiers, despite systematic differences in absolute values and dispersion between tools. ML aids comparison by quantifying discriminative performance but, in this dataset, does not indicate a clear superiority of one quantification pipeline over the other. © The Author(s) 2026.

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Rehabilitation, Quantitative analysis, Neurology, ^{123}I -Ioflupane, BasGanV2, Dopamine transporters, Movement disorders, Oasis©, Parkinson's disease, Personalized medicine

Notes:

EID: 2-s2.0-105037853710

Export Date: 20 September 2026; Cited By: 0; Funding details: Università degli Studi di Roma Tor Vergata; Ministero della Salute; Correspondence Address: L. Filippi; Department of Biomedicine and Prevention, University of Rome Tor Vergata, Rome, Via Montpellier, 1, 00133, Italy; email: luca.filippi@uniroma2.it

Full classification of permutation rational functions and complete rational functions of degree three over finite fields

Item Type Journal Article

Author A. Ferraguti

Author G. Micheli

Abstract Let q be a prime power, F_q be the finite field of order q and $F_q(x)$ be the field of rational functions over F_q . In this paper we classify and count all rational functions $\varphi \in F_q(x)$ of degree 3 that induce a permutation of $P_1(F_q)$. As a consequence of our classification, we can show that there is no complete permutation rational function of degree 3 unless $3 \mid q$ and φ is a polynomial. © 2020, Springer Science+Business Media, LLC, part of Springer Nature.

Date 2020

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Journal Abbr Des Codes Cryptography

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Notes:

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Fully automatic segmentation of multiple sclerosis lesions in brain MR FLAIR images using adaptive mixtures method and markov random field model

Item Type Journal Article

Author R. Khayati

Author M. Vafadust

Author F. Towhidkhah

Author S. Nabavi

Abstract In this paper, an approach is proposed for fully automatic segmentation of MS lesions in fluid attenuated inversion recovery (FLAIR) Magnetic Resonance (MR) images. The proposed approach, based on a Bayesian classifier, utilizes the adaptive mixtures method (AMM) and Markov random field (MRF) model to obtain and upgrade the class conditional probability density function (CCPDF) and the a priori probability of each class. To compare the performance of the proposed approach with those of previous approaches including manual segmentation, the similarity criteria of different slices related to 20 MS patients were calculated. Also, volumetric comparison of lesions volume between the fully automated segmentation and the gold standard was performed using correlation coefficient (CC). The results showed a better performance for the proposed approach, compared to those of previous works. © 2007 Elsevier Ltd. All rights reserved.

Date 2008

DOI 10.1016/j.combiomed.2007.12.005

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Language English

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Notes:

EID: 2-s2.0-39549119121

Export Date: 20 September 2026; Cited By: 145; Correspondence Address: F. Towhidkhah; Biomedical Engineering Faculty, Amirkabir University of Technology, Tehran, Iran; email: Towhidkhah@aut.ac.ir; CODEN: CBMDA

Functional surfaces in mechanical systems: Classification, fabrication, and characterisation

Item Type Journal Article

Author A. Godi

Author L. De Chiffre

Abstract This chapter deals with functional surfaces in mechanical systems. These surfaces are named in a multitude of ways; therefore, a classification is provided based on how the texture is designed. For a better clarification, a number of practical examples for each category are given. An overview of fabrication methods employed to produce such surfaces is presented. The numerous fabrication methods are classified based on the mechanism they employ for providing a texture. Three main categories (removing, moving, and adding material techniques) are identified, and for each of them, a number of processes are described with examples of the textures they can create. The last section of the chapter deals with surface metrology, a topic of central importance in the design and generation of surfaces for functional purposes. The process of surface characterisation is outlined, reviewing measuring instruments, classical and advanced filtering procedures, quantification methods, and eventually, providing traceability considerations. © 2014 by IGI Global. All rights reserved.

Date 2014

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Publication Surface Engineering Techniques and Applications: Research Advancements

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Journal Abbr Surf. Eng. Techniques and Appl.: Res. Adv.

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Quantification methods, Mechanical systems, Fabrication, Fabrication method, Filtering procedures, Functional surfaces, Measuring instruments, Mechanics, Surface characterisation, Surface metrology

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EID: 2-s2.0-84945987678

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Fundamentals of Uncertainty Quantification for Engineers

Item Type Book

Author Y. Wang

Author A. Tran

Author D.L. McDowell

Abstract Fundamentals of Uncertainty Quantification for Engineers: Methods and Models provides a comprehensive introduction to uncertainty quantification (UQ) accompanied by a wide variety of applied examples and implementation details to reinforce the concepts outlined in the book. Sections start with an introduction to the history of probability theory and an overview of recent developments of UQ methods in the domains of applied mathematics and data science. Major concepts of copula, Monte Carlo sampling, Markov chain Monte Carlo, polynomial regression, Gaussian process regression, polynomial chaos expansion, stochastic collocation, Bayesian inference, modelform uncertainty, multi-fidelity modeling, model validation, local and global sensitivity analyses, linear and nonlinear dimensionality reduction are included. Advanced UQ methods are also introduced, including stochastic processes, stochastic differential equations, random fields, fractional stochastic differential equations, hidden Markov model, linear Gaussian state space model, as well as non-probabilistic methods such as robust Bayesian analysis, Dempster-Shafer theory, imprecise probability, and interval probability. The book also includes example applications in multiscale modeling, reliability, fatigue, materials design, machine learning, and decision making. © 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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of Pages 1

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Tags:

Decision making, Dimensionality reduction, Learning systems, Uncertainty analysis, Gaussian distribution, Gaussian noise (electronic), Data mining, Stochastic systems, Bayesian networks, Inference engines, Sensitivity analysis, Markov chain Monte Carlo, Markov Chain Monte-Carlo, Markov chains, Quantification methods, Uncertainty quantifications, Stochastic models, Polynomials, Differential equations, Probabilistic logics, Applied mathematics, Gaussian process regression, Monte Carlo sampling, Polynomial regression, Probability theory, Regression polynomials

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Fuzzy decision tree learning for preoperative classification of adnexal masses

Item Type Conference Paper**Author** E. Ahmadi**Author** H. Javadi**Author** A. Khansefid**Author** A. Asadi**Author** M.M. Ebadzadeh**Author** D. Timmerman

Abstract The study problem was learning a fuzzy decision tree to classify patients with adnexal mass into either of benign or malignant class prior to surgery using patients' medical history, physical exam, laboratory tests, and ultrasonography. A learning algorithm was developed to learn a fuzzy decision tree in three steps. In the growing step, a binary decision tree was learned from a dataset of patients while fuzzy discretization was used in decision nodes testing continuous attributes. The best degree of fuzziness was automatically found by an algorithm based on optimization procedures. In the pruning step, the overfitted nodes were removed by an algorithm based on critical value post-pruning method. In the refitting step, the labels of the leaf nodes were optimized. The final resulted tree had 10 decision nodes and 11 leaf nodes. Performance testing of the tree gave AUC of ROC of 0.91 and mean squared error of 0.1. The tree was translated into a set of 11 fuzzy if-then rules and the clinical plausibility of the rules was assessed by domain experts. All rules were verified to be in agreement with medical knowledge in the domain. Despite the small learning set and the lack of some important input variables, this method gave accurate and, more importantly, clinically interpretable results.

Date 2011**URL** <https://www.scopus.com/pages/publications/79960169213?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: HEALTHINF - Proc. Int. Conf. Health Informatics**ISBN** 978-989842534-8 (ISBN)**Pages** 364-375**Proceedings Title** HEALTHINF - Proc. Int. Conf. Health Informatics**Conference Name** HEALTHINF 2011 - Proceedings of the International Conference on Health Informatics**Date Added** 20/09/2026, 9:52:47 pm**Modified** 20/09/2026, 9:52:47 pm**Tags:**

Learning algorithms, Statistical tests, Fuzzy systems, Decision trees, Fuzzy logic, Optimization, Information science, Trees (mathematics), Domain experts, Medical knowledge, Ultrasonography, Mean squared error, Data sets, Adnexal mass, Adnexal diseases/diagnosis, Binary decision trees, Continuous attribute, Critical value, Decision tree learning, Discretizations, Fuzzy decision trees, Fuzzy if-then rules, Input variables, Laboratory test, Medical history, Optimization procedures, Performance testing, Plant extracts, Pruning steps

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: E. Ahmadi; Digestive Diseases Research Center, Shariati Hospital, Tehran University of Medical Sciences, 14117 Tehran, Iran; email: emad.ahmadi@gmail.com; Conference name: International Conference on Health Informatics, HEALTHINF 2011; Conference date: 26 January 2011 through 29 January 2011; Conference code: 85509

G4 & the balanced metric family – a novel approach to solving binary classification problems in medical device validation & verification studies

Item Type Journal Article

Author A. Marra

Abstract Background: In medical device validation and verification studies, the area under the receiver operating characteristic curve (AUROC) is often used as a primary endpoint despite multiple reports showing its limitations. Hence, researchers are encouraged to consider alternative metrics as primary endpoints. A new metric called G4 is presented, which is the geometric mean of sensitivity, specificity, the positive predictive value, and the negative predictive value. G4 is part of a balanced metric family which includes the Unified Performance Measure (also known as P4) and the Matthews' Correlation Coefficient (MCC). The purpose of this manuscript is to unveil the benefits of using G4 together with the balanced metric family when analyzing the overall performance of binary classifiers. Results: Simulated datasets encompassing different prevalence rates of the minority class were analyzed under a multi-reader-multi-case study design. In addition, data from an independently published study that tested the performance of a unique ultrasound artificial intelligence algorithm in the context of breast cancer detection was also considered. Within each dataset, AUROC was reported alongside the balanced metric family for comparison. When the dataset prevalence and bias of the minority class approached 50%, all three balanced metrics provided equivalent interpretations of an AI's performance. As the prevalence rate increased / decreased and the data became more imbalanced, AUROC tended to overvalue / undervalue the true classifier performance, while the balanced metric family was resistant to such imbalance. Under certain circumstances where data imbalance was strong (minority-class prevalence < 10%), MCC was preferred for standalone assessments while P4 provided a stronger effect size when evaluating between-groups analyses. G4 acted as a middle ground for maximizing both standalone assessments and between-groups analyses. Conclusions: Use of AUROC as the primary endpoint in binary classification problems provides misleading results as the dataset becomes more

imbalanced. This is explicitly noticed when incorporating AUROC in medical device validation and verification studies. G4, P4, and MCC do not share this limitation and paint a more complete picture of a medical device’s performance in a clinical setting. Therefore, researchers are encouraged to explore the balanced metric family when evaluating binary classification problems. © The Author(s) 2024.

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Article, artificial intelligence, human, algorithm, case study, prevalence, accuracy, validation study, Binary classification, predictive value, sensitivity and specificity, binary classification, geometry, linear system, breast cancer, AUROC, G4, Matthew, MCC, Medical device validation, MRMC, P4, study design

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: A. Marra; Clinical Biostatistician at GE Healthcare, Chicago, United States; email: Andrew.Marra@gehealthcare.com

Gain-Some-Lose-Some: Reliable Quantification under General Dataset Shift

Item Type Conference Paper

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Author R. Sinha

Author M.A. Naeem

Editor Bailey J.

Editor Miettinen P.

Editor Koh Y.S.

Editor Tao D.

Editor Wu X.

Abstract When applying supervised learning to estimate class distributions of unlabelled samples (so-called quantification), dataset shift is an expected yet challenging problem. Existing quantification methods make strong assumptions on the nature of dataset shift that often will not hold in practice. We propose a novel Gain-Some-Lose-Some (GSLS) model that accounts for more general conditions of dataset shift. We present a method for fitting the GSLS model without any labelled instances from the target sample, and experimentally demonstrate that GSLS can produce reliable quantification prediction intervals under broader conditions of shift than existing quantification methods. © 2021 IEEE.

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Conference Name Proceedings - IEEE International Conference on Data Mining, ICDM

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GC-Flow: A Graph-Based Flow Network for Effective Clustering

Item Type Conference Paper

Author T. Wang

Author F. Mirzazadeh

Author X. Zhang

Author J. Chen

Editor Krause A.

Editor Brunskill E.
Editor Cho K.
Editor Engelhardt B.
Editor Sabato S.
Editor Scarlett J.

Abstract Graph convolutional networks (GCNs) are discriminative models that directly model the class posterior $p(y|x)$ for semi-supervised classification of graph data. While being effective, as a representation learning approach, the node representations extracted from a GCN often miss useful information for effective clustering, because the objectives are different. In this work, we design normalizing flows that replace GCN layers, leading to a generative model that models both the class conditional likelihood $p(x|y)$ and the class prior $p(y)$. The resulting neural network, GCFlow, retains the graph convolution operations while being equipped with a Gaussian mixture representation space. It enjoys two benefits: it not only maintains the predictive power of GCN, but also produces well-separated clusters, due to the structuring of the representation space. We demonstrate these benefits on a variety of benchmark data sets. Moreover, we show that additional parameterization, such as that on the adjacency matrix used for graph convolutions, yields additional improvement in clustering. © 2023 Proceedings of Machine Learning Research. All rights reserved.

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: T. Wang; Pennsylvania State University, United States; email: tkw5356@psu.edu; Conference name: 40th International Conference on Machine Learning, ICML 2023; Conference date: 23 July 2023 through 29 July 2023; Conference code: 191855

Gene expression of somatostatin receptor subtypes SSTR2a, SSTR3 and SSTR5 in peripheral blood of neuroendocrine lung cancer affected patients

Item Type Journal Article

Author L.A. Muscarella

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Author V.M. Fazio

Author G. Vendemiale

Abstract Background Somatostatin (SS) acts as a universal endocrine off-switch, and also inhibits the growth of neuroendocrine tumours through its specific receptors (SSTRs). Somatostatin receptors are G-protein-coupled receptors, which are encoded by five separate genes (SSTR1-5). Short peptide analogues demonstrate specific binding only for the subgroup consisting of SSTR2a, SSTR3 and SSTR5. Moreover, previous studies reported that expression of mRNA for SSTR2a correlated with therapeutic outcome in patients with carcinoid tumours treated with somatostatin analogs. Purpose To develop and apply a Real Time Quantitative PCR technique (RT-qPCR) to compare and contrast the mRNA levels of SSTR2a, SSTR3 and SSTR5 in Neuroendocrine Lung Cancer affected patients. Methods Peripheral blood samples from 21 neuroendocrine lung cancer affected patients (14 SCLC, 6 LC and 1LCNEC) subjected to scintigraphy with ¹¹¹In-DTPA-DPhe1- octreotide (OctreoScan) and 24 healthy blood donors were investigated by RT-qPCR. mRNA levels for SSTR2a, SSTR3 and SSTR5 were measured in peripheral blood samples with a relative quantification method using plasmid dilutions as calibration curves and GAPDH as reference gene. Results A statistically significant increase in target genes/ GAPDH copy number ratio was found for SSTR2a (median 38; IQR 22-141) and SSTR5 (median 51; IQR 19-499) in neuroendocrine lung cancer affected patients as compared with samples from healthy blood donors ($P \leq 0.0003$ and $P \leq 0.0005$). Since low levels of expression were detected in the control group for all three genes, optimal cut-off values were assessed using ROC curve analyses and were equal to 9.05 for SSTR2a and 16.97 for SSTR5. These cut off values resulted in a sensitivity of 86% (95%IC 65-95) for both markers and a specificity of 83% (95%IC 64-93%) and 79% (95%IC 60-91%) for SSTR2a and SSTR5 respectively. Comparison between OctreoScan results and RT-qPCR analysis demonstrated agreement in 76% of the cases. Conclusions Our results suggest that SSTR2a and SSTR5 mRNAs are detectable in peripheral blood of neuroendocrine lung cancer affected patients using real-time quantitative PCR, with a good agreement with OctreoScan. The high sensitivity of this non-invasive molecular technique suggests that this method could represent a useful tool in the clinical management of neuroendocrine lung cancers. © The Author(s) 2011.

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DOI 10.1007/s13402-011-0025-9**URL** <https://www.scopus.com/pages/publications/84860570657?origin=resultslist>**Archive** Scopus**Language** English**Volume** 34**Pages** 435-441**Publication** Cellular Oncology**ISSN** 22113436 (ISSN)**Issue** 5**Journal Abbr** Cell. Oncol.**Date Added** 20/09/2026, 9:53:01 pm**Modified** 20/09/2026, 9:53:01 pm**Tags:**

adult, female, Female, human, Humans, male, Male, middle aged, Middle Aged, article, quantitative analysis, controlled study, classification, gene expression, receiver operating characteristic, clinical article, gene dosage, genetics, blood sampling, ROC Curve, sensitivity and specificity, human tissue, pathology, cancer chemotherapy, messenger RNA, tumor marker, reverse transcription polymerase chain reaction, gene, case control study, Case-Control Studies, blood, real time polymerase chain reaction, blood analysis, Lung Neoplasms, lung tumor, cancer staging, yttrium 90, statistical significance, gene expression regulation, Gene Expression Regulation, Neoplastic, cancer patient, priority journal, Real-time quantitative PCR, Neoplasm Staging, Reverse Transcriptase Polymerase Chain Reaction, blood donor, cancer radiotherapy, carcinoid, glyceraldehyde 3 phosphate dehydrogenase, large cell neuroendocrine carcinoma, lung carcinoid, lung small cell cancer, Neuroendocrine lung cancer, neuroendocrine tumor, Neuroendocrine Tumors, pentetreotide in 111, receptor subtype, Receptors, Somatostatin, scintigraphy, somatostatin derivative, somatostatin receptor, somatostatin receptor 2, somatostatin receptor 2a gene, somatostatin receptor 3, somatostatin receptor 3 gene, somatostatin receptor 5, somatostatin receptor 5 gene, SSTRs, Tumor Markers, Biological

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Generalized Bayes Quantification Learning under Dataset Shift

Item Type Journal Article

Author J. Fiksel

Author A. Datta

Author A. Amouzou

Author S. Zeger

Abstract Quantification learning is the task of prevalence estimation for a test population using predictions from a classifier trained on a different population. Quantification methods assume that the sensitivities and specificities of the classifier are either perfect or transportable from the training to the test population. These assumptions are inappropriate in the presence of dataset shift, when the misclassification rates in the training population are not representative of those for the test population. Quantification under dataset shift has been addressed only for single-class (categorical) predictions and assuming perfect knowledge of the true labels on a small subset of the test population. We propose generalized Bayes quantification learning (GBQL) that uses the entire compositional predictions from probabilistic classifiers and allows for uncertainty in true class labels for the limited labeled test data. Instead of positing a full model, we use a model-free Bayesian estimating equation approach to compositional data using Kullback–Leibler loss-functions based only on a first-moment assumption. The idea will be useful in Bayesian compositional data analysis in general as it is robust to different generating mechanisms for compositional data and allows 0's and 1's in the compositional outputs thereby including categorical outputs as a special case. We show how our method yields existing quantification approaches as special cases. Extension to an ensemble GBQL that uses predictions from multiple classifiers yielding inference robust to inclusion of a poor classifier is discussed. We outline a fast and efficient Gibbs sampler using a rounding and coarsening approximation to the loss functions. We establish posterior consistency, asymptotic normality and valid coverage of interval estimates from GBQL, which to our knowledge are the first theoretical results for a quantification approach in the presence of local labeled data. We also establish finite sample posterior concentration rate. Empirical performance of GBQL is demonstrated through simulations and analysis of real data with evident dataset shift. Supplementary materials for this article are available online. © 2021 The Author(s). Published with license by Taylor & Francis Group, LLC.

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Generalized Universal Domain Adaptation

Item Type Journal Article**Author** W. Su**Author** Z. Han**Author** X. Liu**Author** Y. Yin

Abstract Universal Domain Adaptation (UniDA) stands as a pivotal framework for transferring knowledge between disparate domains under the challenge of domain shifts. It uniquely caters to mismatches in class space without necessitating prior knowledge. Despite its utility, UniDA's effectiveness is often compromised in practical scenarios by the label shift, which poses significant hurdles to its generalization capabilities. Moreover, the existing UniDA approaches lack true universality, as they tend to erroneously detect “unknown” class samples even in situations where no “unknown” class are present, leading to unnecessary errors. In this paper, we propose a new framework known as Generalized Universal Domain Adaptation (GUDA) to handle the new challenges posed by both label shifts and the lack of true universality. At the core of our GUDA lies the Generalized Universal Adaptation Network (GUAN), comprising three innovative modules: the dual centroid learning module, which explores the internal structure of “unknown” classes and allows the model to determine if the “unknown” classes exist; the dual centroid employment module, which promotes cohesive clustering and emphasizes learning on minority classes to recognize “unknown” classes without prior knowledge; and the weighted multi-class adversarial alignment module, which aligns source and target samples with class-specific adjustments, preserving class boundaries amidst label shifts. These modules enable GUAN to form adaptable clusters tailored to varied target settings, enhancing classification precision and showcasing the true universality of our GUDA approach. Through extensive experimentation on benchmark datasets, we have demonstrated that our method substantially surpasses existing leading approaches in performance. © 2024 Elsevier B.V.

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Generalized universal inference on risk minimizers

Item Type Journal Article

Author N. Dey

Author R. Martin

Author J.P. Williams

Abstract A common goal in statistics and machine learning is estimation of unknowns. Point estimates alone are of little value without an accompanying measure of uncertainty, but traditional uncertainty quantification methods, such as confidence sets and p-values, often require distributional or structural assumptions that may not be justified in modern applications. The present paper considers a very common case in machine learning, where the quantity of interest is the minimizer of a given risk (expected loss) function. We propose a generalization of universal inference specifically designed for inference on risk minimizers. Notably, our generalized universal inference attains finite-sample frequentist validity guarantees under a condition common in the statistical learning literature. One version of our procedure is also anytime-valid, i.e. it maintains the finite-sample validity properties regardless of the stopping rule used for the data collection process. Practical use of our proposal requires tuning, and we offer a data-driven procedure with strong empirical performance across a broad range of challenging statistical and machine learning examples. © The Royal Statistical Society 2025. This is an Open Access article distributed under the terms of the Creative Commons Attribution License (<https://creativecommons.org/licenses/by/4.0/>), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.

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Generated Preserved Adversarial Federated Learning for Enhanced Image Analysis (GPAF)

Item Type Journal Article

Author S. Lakrouni

Author S. Bah

Author M. Sebgui

Abstract Federated Learning (FL) has recently emerged as a promising paradigm that enables medical institutions to collaboratively train robust models without centralizing sensitive patient information. Data collected from different institutions represent distinct source domains. Consequently, discrepancies in feature distributions can significantly hinder a model's generalization to unseen domains. While domain generalization (DG) methods have been proposed to address this challenge, many may compromise data privacy in FL by requiring clients to transmit their local feature representations to the server. Furthermore, existing adversarial training methods, commonly used to align marginal feature distributions, fail to ensure the consistency of conditional distributions. This consistency is often critical for accurate predictions in unseen domains. To address these limitations, we propose GPAF, a privacy-preserving federated learning (FL) framework that mitigates both domain and label shifts in healthcare applications. GPAF aligns conditional distributions across clients in the latent space and restricts communication to model parameters. This design preserves class semantics, enhances privacy, and improves communication efficiency. At the server, a global generator learns a conditional feature distribution from clients' feedback. During local training, each client minimizes an adversarial loss to align its local conditional distribution with the global distribution, enabling the FL model to

learn robust, domain-invariant representations across all source domains. To evaluate the effectiveness of our approach, experiments on a medical imaging benchmark demonstrate that GPAF outperforms four FL baselines, achieving up to 17% higher classification accuracy and 25% faster convergence in non-IID scenarios. These results highlight GPAF's capability to generalize across domains while maintaining strict privacy, offering a robust solution for decentralized healthcare challenges.

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Generating prior probabilities for classifiers of brain tumours using belief networks

Item Type Journal Article

Author G.M. Reynolds

Author A.C. Peet

Author T.N. Arvanitis

Abstract Background. Numerous methods for classifying brain tumours based on magnetic resonance spectra and imaging have been presented in the last 15 years. Generally, these methods use supervised machine learning to develop a classifier from a database of cases for which the diagnosis is already known. However, little has been published on developing classifiers based on mixed modalities, e.g. combining imaging information with spectroscopy. In this work a method of generating probabilities of tumour class from anatomical location is presented. Methods. The method of "belief networks" is introduced as a means of generating probabilities that a tumour is any given type. The belief networks are constructed using a database of paediatric tumour cases consisting of data collected over five decades; the problems associated with using this data are discussed. To verify the usefulness of the networks, an application of the method is presented in which prior probabilities were generated and combined with a classification of tumours based solely on MRS data. Results. Belief networks were constructed from a database of over 1300 cases. These can be used to generate a probability that a tumour is any given type. Networks are presented for astrocytoma grades I and II, astrocytoma grades III and IV, ependymoma, pineoblastoma, primitive neuroectodermal tumour (PNET), germinoma, medulloblastoma, craniopharyngioma and a group representing rare tumours, "other". Using the network to generate prior probabilities for classification improves the accuracy when compared with generating prior probabilities based on class prevalence. Conclusion. Bayesian belief networks are a simple way of using discrete clinical information to generate probabilities usable in classification. The belief network method can be robust to incomplete datasets. Inclusion of a priori knowledge is an effective way of improving classification of brain tumours by non-invasive methods. © 2007 Reynolds et al; licensee BioMed Central Ltd.

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Generation of monoclonal pan-hemagglutinin antibodies for the quantification of multiple strains of influenza

Item Type Journal Article

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- Author C. Gadoury
- Author B. Jaentschke
- Author X. Li
- Author E. Petiot
- Author Y. Durocher
- Author J. Baardsnes
- Author M. Rosa-Calatrava
- Author S. Ansoerge
- Author A.A. Kamen

Abstract Vaccination is the most effective course of action to prevent influenza. About 150 million doses of influenza vaccines were distributed for the 2015-2016 season in the USA alone according to the Centers for Disease Control and Prevention. Vaccine dosage is calculated based on the concentration of hemagglutinin (HA), the main surface glycoprotein expressed by influenza which varies from strain to strain. Therefore yearly-updated strain-specific antibodies and calibrating antigens are required. Preparing these quantification reagents can take up to three months and significantly slows down the release of new vaccine lots. Therefore, to circumvent the need for strain-specific sera, two anti-HA monoclonal antibodies (mAbs) against a

highly conserved sequence have been produced by immunizing mice with a novel peptide-conjugate. Immunoblots demonstrate that 40 strains of influenza encompassing HA subtypes H1 to H13, as well as B strains from the Yamagata and Victoria lineage were detected when the two mAbs are combined to form a pan-HA mAb cocktail. Quantification using this pan-HA mAbs cocktail was achieved in a dot blot assay and results correlated with concentrations measured in a hemagglutination assay with a coefficient of correlation of 0.80. A competitive ELISA was also optimised with purified viral-like particles. Regardless of the quantification method used, pan-HA antibodies can be employed to accelerate process development when strain-specific antibodies are not available, and represent a valuable tool in case of pandemics. These antibodies were also expressed in CHO cells to facilitate large-scale production using bioreactor technologies which might be required to meet industrial needs for quantification reagents. Finally, a simulation model was created to predict the binding affinity of the two anti-HA antibodies to the amino acids composing the highly conserved epitope; different probabilities of interaction between a given amino acid and the antibodies might explain the affinity of each antibody against different influenza strains. © 2017 Manceur et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Article, female, human, Humans, simulation, nonhuman, prediction, controlled study, classification, immune response, animal, Animals, HEK293 cell line, HEK293 Cells, animal experiment, animal model, animal tissue, mouse, immunology, virus antibody, animal cell, enzyme linked immunosorbent assay, Enzyme-Linked Immunosorbent Assay, antibody specificity, surface plasmon resonance, Antibodies, Monoclonal, binding affinity, monoclonal antibody, Bioreactors, Influenza virus, epitope, virus like agent, virus strain, amino acid derivative, antibody affinity, antibody production, antigen antibody reaction, bioreactor, CHO cell line, CHO Cells, Cricetinae, Cricetulus, hamster, Hemagglutinin Glycoproteins, Influenza Virus, influenza, Influenza A virus, Influenza B virus, influenza vaccination, influenza vaccine, Influenza virus hemagglutinin, pandemic influenza, peptide vaccine, protein antibody, Surface Plasmon Resonance, virus hemagglutination

Notes:

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Export Date: 20 September 2026; Cited By: 8; CODEN: POLNC

Generative Calibration for In-context Learning**Item Type** Conference Paper**Author** Z. Jiang**Author** Y. Zhang**Author** C. Liu**Author** J. Zhao**Author** K. Liu

Abstract As one of the most exciting features of large language models (LLMs), in-context learning is a mixed blessing. While it allows users to fast-prototype a task solver with only a few training examples, the performance is generally sensitive to various configurations of the prompt such as the choice or order of the training examples. In this paper, we for the first time theoretically and empirically identify that such a paradox is mainly due to the label shift of the in-context model to the data distribution, in which LLMs shift the label marginal $p(y)$ while having a good label conditional $p(x|y)$. With this understanding, we can simply calibrate the in-context predictive distribution by adjusting the label marginal, which is estimated via Monte-Carlo sampling over the in-context model, i.e., generation of LLMs. We call our approach as generative calibration. We conduct exhaustive experiments with 12 text classification tasks and 12 LLMs scaling from 774M to 33B, generally find that the proposed method greatly and consistently outperforms the ICL as well as state-of-the-art calibration methods, by up to 27% absolute in macro-F1. Meanwhile, the proposed method is also stable under different prompt configurations. © 2023 Association for Computational Linguistics.

Date 2023**DOI** 10.18653/v1/2023.findings-emnlp.152**URL** <https://www.scopus.com/pages/publications/85183293339?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Find. Assoc. Comput. Linguist.: EMNLP**Publisher** Association for Computational Linguistics (ACL)**ISBN** 979-889176061-5 (ISBN)**Pages** 2312-2333**Proceedings Title** Find. Assoc. Comput. Linguist.: EMNLP**Conference Name** Findings of the Association for Computational Linguistics: EMNLP 2023**Date Added** 20/09/2026, 9:52:26 pm**Modified** 20/09/2026, 9:52:26 pm**Tags:**

Computational linguistics, Language model, Performance, Classification (of information), Text classification, Text processing, In contexts, Context learning, Context models, Monte Carlo methods, Data distribution, Training example, Calibration, Monte Carlo sampling, Predictive distributions

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Generative diffusion model enables quantification of calibration-free arterial spin labeling perfusion magnetic resonance imaging data in an Alzheimer's disease cohort

Item Type Journal Article

Author Q. Shou

Author S. Cen

Author N.-K. Chen

Author J.M. Ringman

Author H. Kim

Author C.R. Jack

Author B.J. Borowski

Author M.L. Senjem

Author A. Arani

Author D.J.J. Wang

Abstract INTRODUCTION: M0 images were missing in Siemens ASL data in Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, prohibiting cerebral blood flow (CBF) quantification. METHODS: A conditional latent diffusion model was trained and evaluated on in-house datasets, then applied to the Siemens data in ADNI-3. Regional CBF differences by Alzheimer's disease (AD) stages, their accuracy for AD classification, and CBF trajectory slopes were compared between generated data (Siemens) and acquired data (General Electric). RESULTS: The diffusion model generated M0 images with high fidelity (SSIM = 0.918 ± 0.023 , PSNR = 31.361 ± 2.537) and minimal CBF bias (mean difference is 0.21 ± 1.58 mL/100 g/min). Both generated and acquired CBF showed similar spatial patterns and decreasing trends with AD progression in specific AD-related regions. Generated CBF also improved accuracy in classifying AD stages compared to qualitative perfusion images. CONCLUSION: This study shows the potential of diffusion models for imputing missing modalities in large-scale studies exploring the use of ASL as a biomarker of AD. Highlights: Using latent diffusion model, we can generate M0 image from control image in arterial spin labeling (ASL) with high fidelity. The generated M0 can be used for cerebral blood flow (CBF) quantification

in Alzheimer's Disease Neuroimaging Initiative dataset. The performance of classification between Alzheimer's disease (AD) patients and cognitive normal people is better when using generated CBF maps than using non-quantitative perfusion images. ASL CBF decreases with AD progression in key AD-related brain regions. © 2025 The Author(s). Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring published by Wiley Periodicals, LLC on behalf of Alzheimer's Association.

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Notes:

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Genetic architecture of 11 organ traits derived from abdominal MRI using deep learning

Item Type Journal Article

Author Y. Liu

Author N. Basty

Author B. Whitcher

Author J.D. Bell

Author E.P. Sorokin

Author N. van Bruggen

Author E.L. Thomas

Author M. Cule

Abstract Cardiometabolic diseases are an increasing global health burden. While socioeconomic, environmental, behavioural, and genetic risk factors have been identified, a better understanding of the underlying mechanisms is required to develop more effective interventions. Magnetic resonance imaging (MRI) has been used to assess organ health, but biobank-scale studies are still in their infancy. Using over 38,000 abdominal MRI scans in the UK Biobank, we used deep learning to quantify volume, fat, and iron in seven organs and tissues, and demonstrate that imaging-derived phenotypes reflect health status. We show that these traits have a substantial heritable component (8–44%) and identify 93 independent genome-wide significant associations, including four associations with liver traits that have not previously been reported. Our work demonstrates the tractability of deep learning to systematically quantify health parameters from high-throughput MRI across a range of organs and tissues, and use the largest-ever study of its kind to generate new insights into the genetic architecture of these traits. © Liu et al.

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GeneWalk identifies relevant gene functions for a biological context using network representation learning

Item Type Journal Article

Author R. Ietswaart

Author B.M. Gyori

Author J.A. Bachman

Author P.K. Sorger

Author L.S. Churchman

Abstract A bottleneck in high-throughput functional genomics experiments is identifying the most important genes and their relevant functions from a list of gene hits. Gene Ontology (GO) enrichment methods provide insight at the gene set level. Here, we introduce GeneWalk (github.com/churchmanlab/genewalk) that identifies individual genes and their relevant functions critical for the experimental setting under examination. After the automatic assembly of an experiment-specific gene regulatory network, GeneWalk uses representation learning to quantify the similarity between vector representations of each gene and its GO annotations, yielding annotation significance scores that reflect the experimental context. By performing gene- and condition-specific functional analysis, GeneWalk converts a list of genes into data-driven hypotheses. © 2021, The Author(s).

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Notes:

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Geographic discrimination of Sauce-flavor rounded-base Baijiu using GC–MS/MS combined with multivariate analysis and machine learning

Item Type Journal Article

Author S. Luo

Author Y. Shen

Author W. Cheng

Author N. Wang

Author Y. Zhang

Author J. Wang

Author D. Chang

Author J. Li

Author X. Guo

Author H. Gao

Abstract A high-throughput targeted quantification method based on GC–MS/MS was established to determine 156 trace volatile aroma compounds (TVACs) in Sauce-flavor Baijiu. The method showed good linearity over 2–10,000 µg/L ($R^2 > 0.99$), low LOQs (< 25.76 µg/L), satisfactory precision ($RSD < 3.96\%$), acceptable recoveries (82.11–118.04%), and low matrix effects ($\leq \pm 15\%$). Compared with HS-SPME-GC–MS, it allowed high-throughput, more accurate quantification with simple pretreatment. Based on orthogonal partial least squares discriminant analysis (OPLS-DA) combined with $VIP > 1$ and $p < 0.05$, 40 potential differential compounds were screened from 46 SRBB samples originating from six regions. Random forest model was subsequently established and evaluated using stratified 5-fold cross-validation to preserve class distribution, followed by validation with an independent test set, achieving an accuracy of 90.00%. SHAP (Shapley additive explanations) analysis identified compounds contributing to differences among regions, providing a useful approach for regional discrimination and flavor profiling of Baijiu. Copyright © 2026. Published by Elsevier Ltd.

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Geographical aspects of enterobiasis in Estonia

Item Type Journal Article

Author K. Remm

Author M. Remm

Abstract This paper examines the geographical pattern of the prevalence of enterobiasis among children in Estonian counties, and methods for risk modelling. The methodological questions were as follows:(1)Is the prevalence of enterobiasis related to geographical region and settlement size?(2)Is the reliability of risk estimations related to the modelling method and sample size? Case-based predictions were more liable to result in over fitting in the case of small training samples than the classification tree models. The spatial autocorrelation of the prevalence was significant up to a distance of 20 km. The main reason for the differences in the predictions might simply be the regional differences in the prevalence. © 2009 Elsevier Ltd. All rights reserved.

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Export Date: 20 September 2026; Cited By: 3; Funding details: Ministry of Education, Ethiopia, MOE, (SF0180052s07); Correspondence Address: K. Remm; Institute of Ecology and Earth Sciences, University of Tartu, 51014 Tartu, Vanemuise 46, Estonia; email: kalle.remm@ut.ee; CODEN: HEPLF

Geometric Continuity and Consistency Learning for Self-Supervised Point Cloud Completion

Item Type Journal Article

Author J. Ma

Author S. Wang

Author Q. Zheng

Author X. Mai

Abstract Point cloud completion aims to infer the complete point clouds from incomplete ones. In real-world scenarios, where the paired data is absent, self-supervised methods have emerged as a promising solution. Although existing self-supervised methods perform well at relatively low resolutions, they suffer significant performance degradation at

higher resolution primarily because they focus on point cloud reconstruction at patch-level or point-level. In this paper, we propose a self-supervised method based on Geometric Continuity and Consistency Learning (GCCL) at multi-scale level to improve the accuracy of predicting local details and global shapes of point clouds. Specifically, to capture local details, we employ a patch-to-point strategy and a coarse-fine manner for geometric continuity learning. To constrain the global shapes, we construct multiple branches for mutual supervision and utilize class priors to build a memory queue for contrasting current features, enhancing the network focus on geometric consistency learning. We evaluate GCCL on multiple datasets, and the results show that our method outperforms existing self-supervised methods by a 4.4 improvement in CD- ℓ_2 on the synthetic PCN dataset and can generate more uniformly distributed completion results on real-world datasets. © 1999-2012 IEEE.

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Notes:

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GeoNet: Benchmarking Unsupervised Adaptation across Geographies

Item Type Conference Paper

Author T. Kalluri

Author W. Xu

Author M. Chandraker

Abstract In recent years, several efforts have been aimed at improving the robustness of vision models to domains and environments unseen during training. An important practical problem pertains to models deployed in a new geography that is under-represented in the training dataset, posing a direct challenge to fair and inclusive computer vision. In this paper, we study the problem of geographic robustness and make three main contributions. First, we introduce a large-scale dataset GeoNet for geographic adaptation containing benchmarks across diverse tasks like scene recognition (GeoPlaces), image classification (GeoImNet) and universal adaptation (GeoUniDA). Second, we investigate the nature of distribution shifts typical to the problem of geographic adaptation and hypothesize that the major source of domain shifts arise from significant variations in scene context (context shift), object design (design shift) and label distribution (prior shift) across geographies. Third, we conduct an extensive evaluation of several state-of-the-art unsupervised domain adaptation algorithms and architectures on GeoNet, showing that they do not suffice for geographical adaptation, and that large-scale pre-training using large vision models also does not lead to geographic robustness. Our dataset is publicly available at <https://tarun005.github.io/GeoNet>. © 2023 IEEE.

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Image classification, Training dataset, Under-represented, Large-scale datasets, Unsupervised adaptation, Geographics, Vision model, Dataset and evaluation, Datasets and evaluation, Geonets, Practical problems, Scene recognition

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Geospatial Analysis of Land Use Changes and Its Implications on Agricultural Production

Item Type Conference Paper

Author M.D. Sree

Author V. Sarada

Author S.K. Singh

Author V. Priyadarshini

Author S.S. Vasanthi

Author V. Balasubramanian

Abstract The conversion of land use patterns generates substantial effects on food cultivation which affects the stability of both food supply and economy. Specifically, this methodology utilizes advanced satellite imagery processing and machine learning to quantify land use transformations and their impacts on agriculture, and geospatial analysis to carry out this process. The pre-analysis processes including cloud mask technique combined with georeferencing and feature extraction yield quality satellite dataset information. ReliefF algorithm serves for feature selection by finding key spectral and temporal indicators that affect agricultural productivity. The research employs CNN+LSTM deep learning techniques which use both spatial feature extraction from CNN and temporal ability of LSTM for land use change classification. The research results show accurate land classification outcomes which enable exact agricultural transition detection. Post-classification validation methods through change detection analysis and statistical assessments help show how much and what patterns in agricultural land transformations appear. The applied research offers vital data-based information for sustainable land management that helps officials and agricultural organizations reduce detrimental land use change effects. © 2025 IEEE.

Date 2025

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GeT: Generative Target Structure Debiasing for Domain Adaptation

Item Type Conference Paper

Author C. Zhang

Author G.H. Lee

Abstract Domain adaptation (DA) aims to transfer knowledge from a fully labeled source to a scarcely labeled or totally unlabeled target under domain shift. Recently, semi-supervised learning-based (SSL) techniques that leverage pseudo labeling have been increasingly used in DA. Despite the competitive performance, these pseudo labeling methods rely heavily on the source domain to generate pseudo labels for the target domain and therefore still suffer considerably from source data bias. Moreover, class distribution bias in the target domain is also often ignored in the pseudo label generation and thus leading to further deterioration of performance. In this paper, we propose GeT that learns a non-bias target embedding distribution with high quality pseudo labels. Specifically, we formulate an online target generative classifier to induce the target distribution into distinctive Gaussian components weighted by their class priors to mitigate source data bias and enhance target class discriminability. We further propose a structure similarity regularization framework to alleviate target class distribution bias and further improve target class discriminability. Experimental results show that our proposed GeT is effective and achieves consistent improvements under various DA settings with and without class distribution bias. Our code is available at: <https://lulusindazc.github.io/getproject/>. © 2023 IEEE.

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Gi-MAPS: a quantitative engineering framework for AI-guided pediatric gut microbiome ecological interpretation and digital-twin simulation

Item Type Journal Article

Author X. Wang

Author W. Hu

Author R. Li

Author R. Sun

Author M.-T. Liong

Author Q. Yu

Author D. Chen

Abstract Background – Quantitative and reproducible microbiome analysis is limited by fragmented workflows lacking standardized anaerobic sampling, absolute quantification methods, and transparent AI inference. Patent-documented engineering integration is required for reliable microbiome analytics at population scale. Methods – Gi-MAPS was designed as an end-to-end analytical system integrating several core patented innovations, including (i) a press-activated anaerobic sample-preservation device that maintains ultra-low residual oxygen to protect obligate anaerobes during transport, (ii) a multiplex qPCR assay enabling simultaneous absolute quantification

of key HMO-utilizing *Bifidobacterium* species in a single reaction, and (iii) a CIT-Net-based digital-twin engine that supports forward simulation of gut microbiota ecological trajectories. These modules are coupled with explainable ensemble artificial intelligence models to form a fully quantitative and simulation-enabled microbiome analysis framework. Each subsystem was validated under granted patents to define engineering performance boundaries and reproducibility specifications. Results – System validation demonstrated <0.1% residual oxygen stability for anaerobic preservation, detection sensitivity down to five genomic copies per microliter, AUC > 0.97 for ecological maturity estimation, 89% accuracy for disease-risk classification, and 95% concordance for digital-twin forecasting. Execution-layer software copyright modules and filed patents extend automation, visualization, and future application domains. Conclusion – Gi-MAPS provides a patent-anchored, standardized engineering framework whose key novelties lie in oxygen-controlled anaerobic sampling, absolute microbial quantification via multiplex qPCR, and digital-twin ecological simulation, enabling quantitative, function-aware, and prospective microbiome analysis. It establishes a reproducible foundation enabling large-scale cohort deployment, longitudinal ecological monitoring, digital-twin simulation, and future multi-omics integration. Copyright © 2026 Wang, Hu, Li, Sun, Liong, Yu and Chen.

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Global Balanced Experts for Federated Long-Tailed Learning

Item Type Conference Paper

Author Y. Zeng

Author L. Liu

Author L. Liu

Author L. Shen

Author S. Liu

Author B. Wu

Abstract Federated learning (FL) is a prevalent distributed machine learning approach that enables collaborative training of a global model across multiple devices without sharing local data. However, the presence of long-tailed data can negatively deteriorate the model's performance in real-world FL applications. Moreover, existing re-balance strategies are less effective for the federated long-tailed issue when directly utilizing local label distribution as the class prior at the clients' side. To this end, we propose a novel Global Balanced Multi-Expert (GBME) framework to optimize a balanced global objective, which does not require additional information beyond the standard FL pipeline. In particular, a proxy is derived from the accumulated gradients uploaded by the clients after local training, and is shared by all clients as the class prior for re-balance training. Such a proxy can also guide the client grouping to train a multi-expert model, where the knowledge from different clients can be aggregated via the ensemble of different experts corresponding to different client groups. To further strengthen the privacy-preserving ability, we present a GBME-p algorithm with a theoretical guarantee to prevent privacy leakage from the proxy. Extensive experiments on long-tailed decentralized datasets demonstrate the effectiveness of GBME and GBME-p, both of which show superior performance to state-of-the-art methods. The code is available at here. © 2023 IEEE.

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Global Meta-Analysis Integrated with Machine Learning Assesses Context-Dependent Microplastic Effects on Soil Microbial Biomass Carbon and Nitrogen

Item Type Journal Article

Author Y. Xiang

Author M.C. Rillig

Author J. Peñuelas

Author L. Nizzetto

Author J. Sardans

Author J. Long

Author J. Zhang

Author R. Li

Author Y. Liu

Author Y. Luo

Author B. Yao

Author Y. Li

Abstract Microplastics (MPs) in soil can paradoxically stimulate microbial biomass in a highly context-dependent manner, potentially inducing decomposition and affecting carbon and nitrogen cycles. We conducted a global meta-analysis with 90 studies (710 observations of microbial biomass carbon (MBC), 354 of microbial biomass nitrogen (MBN)) integrated with machine learning to quantify MPs effects on soil microbial biomass. Field studies showed no significant effects, contrasting with controlled experiments where MPs increased MBC by 9.6% (95% CI: 7.2–11.9%) and MBN by

10.4% (6.8–14.0%). Biodegradable plastics (PBAT, PLA) induced stronger effects (36.1–67.6%) than conventional polymers (PE, PP, PS, PVC). Temperature emerged as the dominant factor, with a contrasting MPs effect on MBC (positive) and MBN (negative) at higher temperatures, suggesting potential decoupling of carbon and nitrogen cycles under warming conditions. Machine learning models (XGBoost, $R^2 = 0.62$) significantly outperformed linear regressions ($R^2 = 0.02–0.05$), revealing nonlinear responses and threshold effects. Stimulatory effects were most significant for medium-sized MPs (30–90 μm), at high concentrations ($>10 \text{ g kg}^{-1}$), and in soils with intermediate fertility, highlighting context-dependent risks to soil carbon and nitrogen cycling. © 2025 The Authors. Published by American Chemical Society

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Google Earth Engine-enabled Mangrove Monitoring: A Systematic Review

Item Type Journal Article

Author K. Patil

Author R. Pawar

Author N. Gangan

Author A. Kurmi

Abstract Mangrove ecosystems are central to global climate mitigation and coastal resilience, yet their monitoring remains inherently complex due to tidal inundation, spectral ambiguity and persistent cloud cover across tropical regions. The emergence of cloud-based geospatial platforms, particularly the Google Earth Engine (GEE), has transformed mangrove remote sensing from scene-based analysis to scalable, multi-sensor, time-series frameworks operating at planetary scale. This study presents a systematic review and critical synthesis of 162 peer-reviewed studies published between 2017 and 2025, examining the structural evolution, methodological foundations and ecological implications of GEE-enabled mangrove research. Bibliometric assessment indicates rapid expansion of the field, reflecting a transition from early methodological experimentation to widespread application. Thematic evolution reveals a clear shift from two-dimensional extent mapping to increasingly dynamic and process-oriented analyses, including disturbance monitoring, hydroperiod assessment, biomass estimation and blue carbon quantification. Methodologically, the literature is strongly anchored in ensemble machine learning approaches, particularly Random Forest, supported by multi-sensor integration of optical time-series (Landsat, Sentinel-2) and Synthetic Aperture Radar (SAR) to mitigate atmospheric and tidal constraints. Despite these advances, critical structural limitations persist. Research output and study-area focus remain geographically concentrated in a limited number of countries, whereas several ecologically significant mangrove regions remain underrepresented, indicating a mismatch between research capacity and ecosystem distribution. Furthermore, although classification accuracy has improved substantially, persistent uncertainties arising from tidal variability, spectral similarity, training data bias and limited model transferability constrain ecological interpretation. These findings indicate that GEE has successfully enabled large-scale mangrove monitoring but has not yet fully translated computational advances into ecological understanding or operational decision-making. Future research should prioritize cross-regional model validation, the integration of structural datasets such as spaceborne light detection and ranging (LiDAR), and the development of interpretable, process-based frameworks to better connect remote sensing outputs with ecosystem functioning. Strengthening these directions is essential to ensure that GEE-based monitoring systems effectively

contribute to global mangrove conservation, blue carbon accounting and climate-resilient coastal management. © The Author(s), under exclusive licence to Springer Nature Switzerland AG 2026.

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Gossen's first law in the modeling for demand side management: a thorough heat pump case study with deep learning based partial time series data generation

Item Type Journal Article

Author C. Li

Author G. Brecher

Author J. Kovačević

Author H.K. Çakmak

Author K. Förderer

Author J. Matthes

Author V. Hagenmeyer

Abstract Gossen's First Law describes the law of diminishing marginal utility. This paper aims to further verify the proposed hypothesis that Gossen's First Law also holds in the modeling for Demand Side Management (DSM) with a thorough heat pump case study. The proposed hypothesis states that in general the complexity-utility relationship in the field of DSM modeling could be represented by a diminishing marginal utility curve. On the other hand, in data based modeling, when utilizing a large dataset for validation, the data integrity is critical to the reliability of the results. However, the absence of partial time series data may occur during the measurement due to missing sensors or IT related issues. In this work, an extensive real-world open dataset of a ground source heat pump is utilized for the case study. In the raw data, one key variable namely the flow rate is missing. Thus, three different algorithms based on machine learning and deep learning architectures namely Random Forest (RF), Long Short-Term Memory (LSTM) and Transformer are applied to predict the flow rate by utilizing an open loop forecasting. The raw data are first pre-processed with a time interval of one hour and then used for training, validation and forecast. Furthermore, a modified persistence model as the baseline is also defined. The predicted flow rate using LSTM yields the lowest error of 7.47% nMAE and 10.56% nRMSE respectively. The forecast results are then utilized in the following step of modeling of a heat pump use case. With the introduced quantification method for complexity and a modified version for utility, we further verify the proposed hypothesis with a longer time horizon of 7 days. © The Author(s) 2024.

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Gradient-Based Quantification of Epistemic Uncertainty for Deep Object Detectors

Item Type Conference Paper

Author T. Riedlinger

Author M. Rottmann

Author M. Schubert

Author H. Gottschalk

Abstract The majority of uncertainty quantification methods for deep object detectors are based on the network output, such as sampling strategies like Monte-Carlo dropout or deep ensembles with straight-forward transfers to object detection. Here, we study gradient-based uncertainty features for object detection. We show that they contain information orthogonal to that of common, output-based uncertainty approximation methods. Meta classification and meta regression are used to produce confidence estimates using gradient features and other methods which are applicable to numerous object detection architectures. Our results show that gradient uncertainty itself performs on par with state-of-the-art methods across different detectors and datasets. We find that combined meta classifiers outperform standalone models. This suggests that sampling strategies may be supplemented by gradient-based uncertainty to obtain improved confidences, contributing to the probabilistic reliability of object detectors in down-stream applications. © 2023 IEEE.

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Deep learning, Feature extraction, Object recognition, Machine-learning, Network architecture, Uncertainty analysis, Object detection, Objects detection, Image segmentation, Uncertainty, Image recognition, Learning architectures, Monte Carlo methods, Formulation, formulations, Machine learning architecture, Machine learning architectures, Algorithm: image recognition and understanding (object detection, categorization, segmentation), Algorithms: Image recognition and understanding (object detection, categorization, segmentation), And algorithm (including transfer, low-shot, semi-, self-, and un-supervised learning), and algorithms (including transfer, low-shot, semi-, self-, and un-supervised learning), Gradient based, Un-supervised learning

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Graph positive-unlabeled learning via Bootstrapping Label Disambiguation

Item Type Journal Article

Author C. Liang

Author L. Wang

Author X. Feng

Author Y. Cheng

Author M. Li

Author S. Pan

Author H. Zhang

Abstract Graph positive-unlabeled learning is an important task that tries to learn binary classification models from only positive and unlabeled (PU) nodes. While state-of-the-art methods focus on training graph neural networks, they frequently rely on weak objective functions derived solely from a given class prior probability or inferred exclusively from the graph structure, leading their performance significantly lags behind that of fully labeled counterparts. In this paper, we fill this gap by treating unlabeled nodes as samples ambiguously labeled as both positive and negative, and by introducing a learning method called Bootstrap Label Disambiguation (BLD), which progressively resolves label ambiguities during the training of binary classifiers. BLD comprises a node representation learning module via bootstrapping and a novel central region-based label disambiguation strategy. The learning module leverages both previous representations and the derived positive centroid as targets to train positive-aligned representations, eliminating the need for a prior. Consequently,

the disambiguation strategy constructs a central-region to identify ambiguous nodes and steadily transforms them into effective supervision. Extensive experiments on a range of real-world datasets show that our BLD method significantly outperforms existing approaches and in many cases even surpasses fully labeled classification models. The source code is available at <https://github.com/yunyun85/BLD>. © 2025

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Graph-based label propagation with dissimilarity regularization

Item Type Conference Paper

Author H. Zheng

Author H.H.S. Ip

Abstract Recent studies have shown promising performance of graphbased semi-supervised learning. But one of major limitations of most graph-based semi-supervised learning approaches is that they did not explore the label dissimilarity knowledge. In this paper, we presented a novel graph-based label propagation framework that effectively incorporates similarity and dissimilarity information into semi-supervised classification. The class mass normalization is utilized to make the label decision rule match class priors. The function induction algorithm is also proposed to predict the labels of test data. In particular, by solving quadratic optimization, our approach can give rise to closed-form solution for classification functions of unlabeled data and out-of-sample data. The proposed framework has been extensively evaluated over three widely used datasets for image classification task. The experimental results in terms of classification accuracy demonstrate that the proposed method can achieve significant performance improvements with respect to the state-of-the-arts. © Springer International Publishing Switzerland 2013.

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GRAPH-BASED LIGHT-CURVE FEATURES FOR ROBUST TRANSIENT CLASSIFICATION

Item Type Journal Article

Author J.D. Petro-Ramos

Author D.J. Ruiz-Morales

Author D. Sierra-Porta

Abstract We investigate graph-based representations of astronomical light curves for transient classification on a quality-controlled, class-balanced subset of the MANTRA benchmark (minimum coverage $N_{\min} = 100$ epochs; $N = 1,705$ objects after filtering and Non-Tr. subsampling). Each series is mapped to three visibility-graph views—horizontal (HVG), directed (DHVG), and weighted (W-HVG)—from which we extract compact, length-aware network descriptors (degree/strength moments, clustering and motifs, assortativity, path/efficiency, and spectral summaries). Using object-level stratified five-fold validation and tree-based learners, the best configuration (LightGBM with HVG+DHVG+W-HVG features) attains a macro-F1 of 0.622 ± 0.010 and accuracy of 0.661 ± 0.010 on this subset. For context, the published MANTRA baseline reports $F1_{\text{macro}} = 0.528$ on the full dataset; because class priors differ after quality control, this reference is not a like-for-like comparison. Ablations show that weighted contrasts and directed asymmetry contribute complementary gains to undirected topology. Per-class analysis highlights strong performance for CV, HPM, and Non-Tr., with residual confusions concentrated in the AGN–Blazar–SN block. These results indicate that visibility graphs offer a simple, survey-agnostic bridge between irregular photometric time series and standard classifiers, yielding competitive multiclass performance without bespoke deep architectures. We release code and feature definitions together with the list of object IDs used in the evaluation subset to facilitate reproducibility and future extensions. © 2026, National University of Ireland Maynooth. All rights reserved.

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Graph-based PU learning for binary and multiclass classification without class prior

Item Type Journal Article

Author J. Yoo

Author J. Kim

Author H. Yoon

Author G. Kim

Author C. Jang

Author U. Kang

Abstract How can we classify graph-structured data only with positive labels? Graph-based positive-unlabeled (PU) learning is to train a binary classifier given only the positive labels when the relationship between examples is given as a graph. The problem is of great importance for various tasks such as detecting malicious accounts in a social network, which are difficult to be modeled by supervised learning when the true negative labels are absent. Previous works for graph-based PU learning assume that the prior distribution of positive nodes is known in advance, which is not true in many real-world cases. In this work, we propose GRAB (Graph-based Risk minimization with iterative Belief propagation), a novel end-to-end approach for graph-based PU learning that requires no class prior. GRAB runs marginalization and update steps iteratively. The marginalization step models the given graph as a Markov network and estimates the marginals of latent variables. The update step trains the binary classifier by utilizing the computed marginals in the objective function. We then generalize GRAB to multi-positive unlabeled (MPU) learning, where multiple positive classes exist in a dataset. Extensive experiments on five real-world datasets show that GRAB achieves the state-of-the-art performance, even when the true prior is given only to the competitors. © 2022, The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature.

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Graph-based semi-supervised learning as a generative model

Item Type Conference Paper

Author J. He

Author J. Carbonell

Author Y. Liu

Abstract This paper proposes and develops a new graph-based semi-supervised learning method. Different from previous graph-based methods that are based on discriminative models, our method is essentially a generative model in that the class conditional probabilities are estimated by graph propagation and the class priors are estimated by linear regression. Experimental results on various datasets show that the proposed method is superior to existing graph-based semi-supervised learning methods, especially when the labeled subset alone proves insufficient to estimate meaningful class priors.

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Graph-in-graph discriminative feature enhancement network for fine-grained visual classification

Item Type Journal Article

Author Y. Wang

Author C. Xu

Author Y. Wang

Author X. Wang

Author W. Ding

Abstract Fine-grained visual classification (FGVC) seeks to identify sub-classes within the same meta-class. Prior efforts mainly mine the features of discriminative parts to enhance classification performance. However, we argue that most of these works ignore the spatial details inside each part and the spatial correlations between parts when extracting local features and fusing global features, inhibiting the further improvement of feature quality, especially for the irregular discriminative parts. To alleviate this issue, we rethink the feature generation route from pixels to parts and to objects, and propose a novel graph-in-graph discriminative feature enhancement network (G2DFE-Net). Specifically, the G2DFE-Net consists of two nested graph convolutional networks, where an internal graph is first developed based on the spatial attention strategy to highlight details of the irregular discriminative regions. Then, a KNN-based external graph is introduced to capture the spatial context correlation among independent discriminative parts. With the collaboration of internal and external graph, G2DFE-Net boosts the class separability and compactness of global feature representation, thereby benefiting the accurate FGVC. We conduct thorough experiments on five benchmark datasets, and both quantitative and qualitative results confirm the superior accuracy of our G2DFE-Net compared to previous state-of-the-art algorithms. The code is available at <https://github.com/WangYuPeng1/G2DFE-Net>. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024.

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Graph-Smoothed Bayesian Black-Box Shift Estimator and Its Information Geometry

Item Type Conference Paper

Author M. Kimura

Abstract Label shift adaptation aims to recover target class priors when the labelled source distribution P and the unlabelled target distribution Q share $P(X | Y) = Q(X | Y)$ but $P(Y) \neq Q(Y)$. Classical black-box shift estimators invert an empirical confusion matrix of a frozen classifier, producing a brittle point estimate that ignores sampling noise and similarity among classes. We present Graph-Smoothed Bayesian BBSE (GS-B3SE), a fully probabilistic alternative that places Laplacian-Gaussian priors on both target log-priors and confusion-matrix columns, tying them together on a label-similarity graph. The resulting posterior is tractable with HMC or a fast block Newton-CG scheme. We prove identifiability, $N^{-1/2}$ contraction, variance bounds that shrink with the graph's algebraic connectivity, and robustness to Laplacian misspecification. We also reinterpret GS-B3SE through information geometry, showing that it generalizes existing shift estimators. © 2025 Neural information processing systems

foundation. All rights reserved.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Kimura; School of Mathematics and Statistics, The University of Melbourne, Australia; email: m.kimura@unimelb.edu.au; Conference name: 39th Conference on Neural Information Processing Systems, NeurIPS 2025; Conference date: 2 December 2025 through 7 December 2025; Conference code: 226431

Graphboot: Quantifying uncertainty in node feature learning on large networks

Item Type Journal Article

Author C.G. Akcora

Author Y.R. Gel

Author M. Kantarcioglu

Author V. Lyubchich

Author B. Thuraisingham

Abstract In recent years, as online social networks continue to grow in size, estimating node features, such as sociodemographics, preferences and health status, in a scalable and reliable way has become a primary research direction in social network mining. Although many techniques have been developed for estimating various node features, quantifying uncertainty in such estimations has received little attention. Furthermore, most existing methods study networks parametrically, which limits insights about necessary quantity of queried data, reliable feature estimation, and estimator uncertainty. Uncertainty quantification is critical for answering key questions, such as, given a limited availability of social network data, how much data should be queried from the network?, and which node features can be learned reliably? More importantly, how can we evaluate uncertainty of our estimators? Uncertainty quantification is not equivalent to network sampling but constitutes a key complementary concept to sampling and the associated reliability analysis. To our knowledge, this paper is the first work that sheds light on uncertainty quantification

and uncertainty propagation in social network feature mining. We propose a novel non-parametric bootstrap method for uncertainty analysis of node features in social network mining, derive its asymptotic properties, and demonstrate its effectiveness with extensive experiments. Furthermore, we develop a new metric based on dispersion of estimations, enabling analysts to assess how much more information is needed for increasing prediction reliability based on the estimated uncertainty. We demonstrate the effectiveness of our new uncertainty quantification methodology with extensive experiments on real life social networks, and a case study of mental health on Twitter. © 1989-2012 IEEE.

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GraphCast-Qtf: An improved weather prediction model based on uncertainty quantification methods

Item Type Conference Paper

Author L. Zheng

Author R. Xing

Author Y. Wang

Abstract In the field of climate prediction, although the traditional numerical weather forecasting (NWP) method has made significant progress in the processing of historical data and improving the prediction accuracy, it still has significant limitations in using a large amount of historical climate data and improving the credibility of the prediction results. With the rapid development of deep learning technologies, deep learning-based climate prediction models, such as GraphCast, have shown their potential in global mid-term (10-day) climate variable prediction. However, existing methods often ignore the quantification of prediction uncertainty, which is crucial to improving the prediction's reliability and practical application value. This study proposes an improved GraphCast model called GraphCast-Qtf, mainly by introducing uncertainty quantification mechanisms designed to improve the accuracy and credibility of climate predictions. This research is divided into data set construction, model construction, and system development. First, this study constructed a global climate dataset based on the ERA5 reanalysis data set of ECMWF, including historical weather states and associated climate variables from 1979 to the present. This dataset covers the global scope and provides highresolution climate variable data, providing rich input features for the training of deep learning models. Secondly, based on the GraphCast model, this study introduces the Bayesian neural network and Monte Carlo dropout technology to enable the model to output the confidence interval of the prediction value to quantify the uncertainty of the prediction results. These improvements enhance the model's ability to learn from the intrinsic changing laws of climate systems and enable the model to handle better and reflect the uncertainty of prediction outcomes. Finally, we developed a simple climate prediction system based on the GraphCast-Qtf model. The system provides mediumterm climate forecasts and assessments of their uncertainty. The system's front end is developed using JavaScript and the Vue framework. It is user-friendly and easy to operate, enabling users to provide scientific and reasonable decision support according to the prediction results and confidence intervals provided by the model. Through extensive experimental validation, the GraphCast-Qtf model performed better than traditional methods in terms of prediction accuracy and uncertainty quantification of multiple key climate variables. The application examples in the system development section further demonstrate the model's effectiveness and potential in practical climate prediction scenarios. © 2024 IEEE.

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Notes:

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GraphGS: Mitigating Pseudo-Label Shift in Imbalanced Node Classification via Balanced Feature Propagation and Minority Node Selection

Item Type Journal Article

Author Z. He

Author C. Zhu

Author C. Xie

Author H. Cai

Author Y. Yang

Abstract Graph-structured data in real-world applications often grapple with class imbalance, where underrepresented minority nodes result in biased predictions and diminished learning performance. Although pseudo-labeling methods offer a promising solution for addressing class imbalance, they remain susceptible to pseudo-label shift. This phenomenon occurs when class biases in pseudo-labels propagate during data augmentation, further compromising model effectiveness. To overcome these challenges, we present GraphGS, an innovative framework designed to alleviate pseudo-label shift in class-imbalanced graph datasets. GraphGS integrates two pivotal components: a feature balancer that curtails excessive information transfer from majority to minority nodes, safeguarding the unique characteristics of minority classes, and a node selector that identifies high-confidence minority nodes through similarity metrics. Together, these components enhance minority class representations without introducing noise, enabling more equitable and reliable training of Graph Neural Networks (GNNs). Comprehensive experiments on diverse real-world datasets underscore the robustness and adaptability of GraphGS. Notably, on the CiteSeer dataset, GraphGS achieves an outstanding 2.60% improvement in accuracy over state-of-the-art methods, marking a significant advance in minority class classification. © 2017 IEEE.

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GraphX NET- Chest X-Ray Classification Under Extreme Minimal Supervision

Item Type Conference Paper
Author A.I. Aviles-Rivero
Author N. Papadakis
Author R. Li
Author P. Sellars
Author Q. Fan
Author R.T. Tan
Author C.-B. Schönlieb
Editor Shen D.
Editor Yap P.-T.
Editor Liu T.
Editor Peters T.M.
Editor Khan A.
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Editor Essert C.

Editor Zhou S.

Abstract The task of classifying X-ray data is a problem of both theoretical and clinical interest. Whilst supervised deep learning methods rely upon huge amounts of labelled data, the critical problem of achieving a good classification accuracy when an extremely small amount of labelled data is available has yet to be tackled. In this work, we introduce a novel semi-supervised framework for X-ray classification which is based on a graph-based optimisation model. To the best of our knowledge, this is the first method that exploits graph-based semi-supervised learning for X-ray data classification. Furthermore, we introduce a new multi-class classification functional with carefully selected class priors which allows for a smooth solution that strengthens the synergy between the limited number of labels and the huge amount of unlabelled data. We demonstrate, through a set of numerical and visual experiments, that our method produces highly competitive results on the ChestX-ray14 data set whilst drastically reducing the need for annotated data. © 2019, Springer Nature Switzerland AG.

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Grounding Visual Explanations

Item Type Conference Paper

Author L.A. Hendricks

Author R. Hu

Author T. Darrell

Author Z. Akata

Editor Hebert M.

Editor Weiss Y.

Editor Ferrari V.

Editor Sminchisescu C.

Abstract Existing visual explanation generating agents learn to fluently justify a class prediction. However, they may mention visual attributes which reflect a strong class prior, although the evidence may not actually be in the image. This is particularly concerning as ultimately such agents fail in building trust with human users. To overcome this limitation, we propose a phrase-critic model to refine generated candidate explanations augmented with flipped phrases which we use as negative examples while training. At inference time, our phrase-critic model takes an image and a candidate explanation as input and outputs a score indicating how well the candidate explanation is grounded in the image. Our explainable AI agent is capable of providing counter arguments for an alternative prediction, i.e. counterfactuals, along with explanations that justify the correct classification decisions. Our model improves the textual explanation quality of fine-grained classification decisions on the CUB dataset by mentioning phrases that are grounded in the image. Moreover, on the FOIL tasks, our agent detects when there is a mistake in the sentence, grounds the incorrect phrase and corrects it significantly better than other models. © 2018, Springer Nature Switzerland AG.

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Notes:

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Export Date: 20 September 2026; Cited By: 30; Correspondence Address: L.A. Hendricks; UC Berkeley, Berkeley, United States; email: lisa_anne@eecs.berkeley.edu; Conference name: 15th European Conference on Computer Vision, ECCV 2018; Conference date: 8 September 2018 through 14 September 2018; Conference code: 219419

Guided Waves Benchmark Dataset and Classifier Comparison

Item Type Conference Paper
Author Z. Dworakowski
Author M. Heesch
Author J. Gorski
Author M. Dziendzikowski
Editor Rizzo P.
Editor Milazzo A.
Abstract The guided-wave-based damage detection methodology is a well-established branch of Structural Health Monitoring. There are, however, many different approaches to processing and interpretation of signals acquired using guided waves. This paper presents the results of the SYPIN project in which a large database acquired in a diverse set of experiments was assembled. The experiments included static and variable environmental conditions, composite and metallic specimens of simple and complex geometries. The set includes, in particular, results of experiments on real-life aircraft specimens. This dataset was then used to benchmark several damage detection and quantification methodologies, including various classifiers and feature extraction methods. The performance of the methods is evaluated in a series of experiments, including knowledge transfer between different specimens. The paper presents the results of this research and provides a brief description of the dataset, which is made available for the scientific community's use. © 2023, The Author(s), under

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H-MEAN: Hierarchical Multi-view Evidence Aggregation Network for Trustworthy Medical Image Classification

Item Type Conference Paper

Author Y. Li

Author Y. Chen

Author W. Liu

Author J. Qu

Author C. Ma

Author X. Yue

Editor Liu J.
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Editor Zou X.
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Editor Hu B.
Editor Xiong Y.

Abstract Clinical disease classification naturally follows a hierarchical reasoning process, where clinicians first distinguish coarse diagnostic categories (e.g., benign vs. malignant) before identifying specific subtypes to enhance diagnostic accuracy. While multi-view imaging offers complementary perspectives, most existing machine learning approaches perform flat decision fusion, ignoring this intrinsic hierarchy and handling inter-view conflicts and uncertainty primarily at the decision-fusion level. To address this gap, we propose the Hierarchical Multi-view Evidence Aggregation Network (H-MEAN), an end-to-end architecture that explicitly mimics clinical decision-making process. H-MEAN consists of three key modules: (i) Multi-View Fusion, which jointly encodes heterogeneous imaging views to capture complementary features; (ii) Hierarchical Tree Alignment, which structurally propagates diagnostic cues across granularity levels to enforce clinical consistency; and (iii) Hierarchical Evidential Aggregation, which leverages evidential deep learning to quantify uncertainty throughout the diagnostic hierarchy. Extensive experiments on two multi-view medical datasets demonstrate that H-MEAN consistently outperforms competitive methods, achieving higher classification accuracy, particularly in challenging settings with missing or noisy views. © 2025 IEEE.

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Habitat associations of two entomopathogenic nematodes: A quantitative study using real-time quantitative polymerase chain reactions

Item Type Journal Article

Author P. Torr

Author S.E. Spiridonov

Author S. Heritage

Author M.J. Wilson

Abstract Despite nematodes being the most abundant animals on earth, very few animal ecologists study them, probably because of the difficulties of identifying them to species by morphological methods. A group of nematodes that are important both ecologically and economically is the entomopathogenic nematodes, which play a key role in regulating soil food webs and are sold throughout the world as biological insecticides, yet for which very little is known of their population ecology. A novel detection and quantification method was developed for soil nematodes using real-time polymerase chain reaction (PCR), and the technique was used to estimate numbers of two closely related species of entomopathogenic nematodes, *Steinernema kraussei* and *S. affine* in 50 soil samples from 10 sites in Scotland representing two distinct habitats (woodland and grassland). There was a high degree of correlation between our molecular and traditional morphological estimates of population size and our data clearly showed that *Steinernema affine* occurred only in grassland areas, whereas *S. kraussei* was found in grassland and woodland samples to a similar degree. Real-time PCR offers a rapid and accurate method of detecting individual nematode species from soil samples without the need for a specialist taxonomist, and has much potential for use in studies of nematode population ecology. © 2007 The Authors.

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HairNet2: deep learning to quantify cotton leaf hairiness, a complex genetic and environmental trait

Item Type Journal Article
Author M. Farazi
Author W.C. Conaty
Author L. Egan
Author S.P.J. Thompson
Author I.W. Wilson
Author S. Liu
Author W.N. Stiller
Author L. Petersson

Author V. Rolland

Abstract Background: Cotton accounts for 80% of the global natural fibre production. Its leaf hairiness affects insect resistance, fibre yield, and economic value. However, this phenotype is still qualitatively assessed by visually attributing a Genotype Hairiness Score (GHS) to a leaf/plant, or by using the HairNet deep-learning model which also outputs a GHS. Here, we introduce HairNet2, a quantitative deep-learning model which detects leaf hairs (trichomes) from images and outputs a segmentation mask and a Leaf Trichome Score (LTS). Results: Trichomes of 1250 images were annotated (AnnCoT) and a combination of six Feature Extractor modules and five Segmentation modules were tested alongside a range of loss functions and data augmentation techniques. HairNet2 was further validated on the dataset used to build HairNet (CotLeaf-1), a similar dataset collected in two subsequent seasons (CotLeaf-2), and a dataset collected on two genetically diverse populations (CotLeaf-X). The main findings of this study are that (1) leaf number, environment and image position did not significantly affect results, (2) although GHS and LTS mostly correlated for individual GHS classes, results at the genotype level revealed a strong LTS heterogeneity within a given GHS class, (3) LTS correlated strongly with expert scoring of individual images. Conclusions: HairNet2 is the first quantitative and scalable deep-learning model able to measure leaf hairiness. Results obtained with HairNet2 concur with the qualitative values used by breeders at both extremes of the scale (GHS 1-2, and 5-5+), but interestingly suggest a reordering of genotypes with intermediate values (GHS 3-4+). Finely ranking mild phenotypes is a difficult task for humans. In addition to providing assistance with this task, HairNet2 opens the door to selecting plants with specific leaf hairiness characteristics which may be associated with other beneficial traits to deliver better varieties. © Crown 2024.

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Publisher BioMed Central Ltd

Publication Plant Methods

ISSN 17464811 (ISSN)

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Tags:

Deep learning, Machine learning, Neural network, Phenotyping, Cotton, HairNet, Leaf, Trichome

Notes:

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Export Date: 20 September 2026; Cited By: 4; Funding details: CSIRO Machine Learning; Commonwealth Scientific and Industrial Research Organisation, CSIRO; Artificial Intelligence Future Science Platform; Correspondence Address: V. Rolland; Agriculture and Food, Commonwealth Scientific and Industrial Research Organisation, Canberra, Clunnies Ross St, 2601, Australia; email: vivien.rolland@csiro.au

Handling concept drift via ensemble and class distribution estimation technique

Item Type Conference Paper

Author N. Limsetto

Author K. Waiyamai

Abstract In real world settings there is situation where class distribution of data may change after classifier is built resulting in performance degradation of classifier. Attempts to solve this problem from previous Class Distribution Estimation method (CDE method) yield quite interesting performance however we notice there is some flaw since CDE method still have some bias toward train data thus we decide to improve them with ensemble method. Our Class Distribution Estimation-Ensemble (CDE-EM) methods estimate class distribution from many models instead of one resulting in less bias than previous method. All methods are evaluated using accuracy on set of benchmark UCI data sets. Experimental results demonstrate that our methods yield better performance if class distribution of test data is different from train data. © 2011 Springer-Verlag.

Date 2011

DOI 10.1007/978-3-642-25856-5_2

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Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

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Pages 13-26

Proceedings Title Lect. Notes Comput. Sci.

Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Classification (of information), Data mining, Estimation, Classification, Ensemble methods, Real world setting, Performance degradation, Test data, Concept drifts, Quantification, Class distribution estimation, Class distributions, Concept drift, Data sets, Class distribution Ensemble method, Cost sensitive learning

Notes:

EID: 2-s2.0-84255186218

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: N. Limsetto; Department of Computer Engineering, Kasetsart University, BangKhen Campus, Bangkok 10900, Thailand; email: khim1204@gmail.com; Conference name: 7th International Conference on Advanced Data Mining and Applications, ADMA 2011; Conference date: 17 December 2011 through 19 December 2011; Conference code: 87846

Handling New Class in Online Label Shift

Item Type Journal Article**Author** Y.-Y. Qian**Author** Y. Bai**Author** Z.-Y. Zhang**Author** P. Zhao**Author** Z.-H. Zhou

Abstract In many real-world applications, data are continuously accumulated in open environments, and new classes may emerge over time. For instance, in disease diagnosis, the prevalence of a certain disease may vary seasonally, and new diseases can also emerge. This paper investigates the problem of learning from unlabeled data stream where the label distribution evolves over time, and meanwhile, previously unseen new classes may appear. To handle the emerging new classes in online label shift, we first design a novel risk estimator by unbiased risk rewriting and mixture proportion estimation, which enables the identification of new class data. Subsequently, we employ the online ensemble paradigm for model updating to handle unknown distribution shifts. Moreover, we introduce the sketching and ensemble pruning mechanisms to improve the efficiency of the algorithm, making it more lightweight and practical. The proposed approach enjoys a theoretical guarantee of dynamic regret, ensuring its effectiveness in adapting to the unknown distribution shifts and the emergence of new classes in streaming data. Experiments on diverse benchmark datasets and two real-world applications demonstrate the effectiveness of the algorithm. © 1989-2012 IEEE.

Date 2025**DOI** 10.1109/TKDE.2025.3583138**URL** <https://www.scopus.com/pages/publications/105009472194?origin=resultslist>**Archive** Scopus**Language** English**Volume** 37**Publisher** IEEE Computer Society**Pages** 5257-5270**Publication** IEEE Transactions on Knowledge and Data Engineering**ISSN** 10414347 (ISSN)**Issue** 9**Journal Abbr** IEEE Trans Knowl Data Eng**Date Added** 20/09/2026, 9:52:10 pm

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Tags:

Risk perception, Benchmarking, Real-world, Diagnosis, Diseases, Disease diagnosis, Data stream, Weakly supervised learning, Distribution shift, online label shift, Online label shift, distribution shift, Application data, Open environment, Learning from unlabeled data, new class, New class, weakly supervised learning

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Handling New Class in Online Label Shift

Item Type Conference Paper

Author Y.-Y. Qian

Author Y. Bai

Author Z.-Y. Zhang

Author P. Zhao

Author Z.-H. Zhou

Editor Chen G.

Editor Khan L.

Editor Gao X.

Editor Qiu M.

Editor Pedrycz W.

Editor Wu X.

Abstract In many real-world applications, data are continuously accumulated within open environments. For instance, in disease diagnosis, the prevalence of diseases can vary across seasons, and new types of diseases can emerge. This paper investigates the problem of learning from unlabeled data where the label distribution evolves over time, and meanwhile, previously unseen new class appears in the data stream. To handle the new class in online label shift, we first design a novel risk estimator by unbiased risk rewriting and mixture proportion estimation. Subsequently, we employ the online ensemble paradigm for model updating to handle unknown distribution shifts. The proposed approach enjoys a theoretical guarantee of dynamic regret, ensuring its effectiveness in adapting to the changing label distribution and the presence of the new class in streams.

Experiments conducted on diverse benchmark datasets and two real-world applications demonstrate the effectiveness of the proposed algorithm. © 2023 IEEE.

Date 2023

DOI 10.1109/ICDM58522.2023.00162

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. IEEE Int. Conf. Data Min. ICDM

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ISBN 23748486 (ISSN); 979-835030788-7 (ISBN)

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Proceedings Title Proc. IEEE Int. Conf. Data Min. ICDM

Conference Name Proceedings - IEEE International Conference on Data Mining, ICDM

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Tags:

Risk perception, Benchmarking, Real-world, E-learning, Diagnosis, Label distribution, Disease diagnosis, data stream, Data stream, Weakly supervised learning, Distribution shift, online label shift, Online label shift, distribution shift, Application data, Open environment, new class, New class, weakly supervised learning

Notes:

EID: 2-s2.0-85185407902

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Handling uncertainty in optimization enabled the deep Q network for traffic sign and criticality detection

Item Type Journal Article

Author S. Salim

Author J. J. S

Author S. B

Abstract Objective: This research aims to develop an efficient model for Traffic Sign Recognition (TSR) with uncertainty quantification. Methods: The process involves

pre-processing, sign localization, TSR, and criticality detection. After pre-processing with a Gaussian filter, sign localization is carried out using Segmentation U-Net (SegU-Net). Following TSR, a Deep Q Network (DQN) is trained using Gradient Descent-Teamwork Optimization Algorithm-based Deep Q Network (GD-TOA based DQN), which incorporates Gradient Descent model into Teamwork Optimization Algorithm (TOA), is used to classify traffic signs into three categories: mandatory, cautionary, and informatory. After classification, the system revalidates critical signs using Mahalanobis distance to compute confidence levels. A significant drop triggers a flag for human review if it has a high uncertainty. Results: Empirical findings demonstrate that presented methodology achieved an accuracy rate of 97.9%, a precision rate of 98.2%, a recall rate of 98.3%, F-measure of 98.2%, specificity of 97.5%, and balanced accuracy of 97.9%. Conclusions: The results have proven that proposed model is a very promising technique for criticality detection. © 2026 Taylor & Francis Group, LLC.

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Optimisations, optimization, Deep learning, Uncertainty analysis, Criticality (nuclear fission), Pattern recognition, Traffic signs, Uncertainty, deep learning, Optimization algorithms, Traffic sign recognition, Optimization, Localisation, Pre-processing, Gradient methods, criticality detection, Criticality detection, Gradient-descent, handling uncertainty, Handling uncertainty, Traffic sign

Notes:

EID: 2-s2.0-105046844807

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Salim; Sree Chitra Thirunal College Of Engineering, Pappanamcode, Thiruvananthapuram, India; email: soja@sctce.ac.in; CODEN: TIPRC

Harnessing Machine learning to quantify ecosystem services in coastal Wetlands: A case study of the Bohai economic Rim

Item Type Journal Article

Author T. Lu

Author Y.P. Wang

Author J. Jin

Author S. Zhu

Author J. Gao

Author H. Ding

Author Q. Wu

Author F. Wang

Author C. Gao

Abstract Coastal wetland ecosystems are pivotal for the sustainable economic development of coastal regions. While prior studies have explored interactions between wetlands and human activities across multiple spatiotemporal scales, current ecosystem service (ES) assessments predominantly rely on existing land cover data products with only ~ 80 % spatial accuracy, insufficient for capturing fine-scale ES spatial variations. This limitation hinders the accurate characterization of complex relationships in wetland environmental science and constrains the reliability of subsequent conservation and management decisions. To address this critical gap, this study aims to improve the accuracy of coastal wetland delineation and ES quantification to ≥ 90 % via an integrated technical method. In this work, the Bohai Economic Rim (BER), a region with intensive interactions between climate change and human activities, was selected as the study case for high-precision coastal wetland mapping and quantification of four key ES types (carbon storage, habitat quality, soil conservation, and water yield). By integrating knowledge-based rules implemented on the Google Earth Engine (GEE) platform with a machine learning algorithm, we identified ES spatial hotspots and cold spots, analyzed trade-off and synergy relationships among the four ESs, and further employed a random forest model to evaluate the relative importance of driving factors (including climate variables and human activity indicators). Results show the total ES index ranges from 0.02 to 0.82 (normalized to 0–1), with all six pairs of ESs exhibiting significant positive correlations, indicating strong synergistic effects. The random forest model reveals that forest proportion and precipitation are the top two influential factors shaping ES spatial variation. This study provides a scalable high-precision ES assessment framework for complex coastal wetland systems with accuracy improved from ~ 80 % to 93.76 %, offering valuable scientific insights and targeted policy recommendations for the conservation and sustainable management of coastal wetland ecosystems in the BER. © 2026 Elsevier B.V.

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Notes:

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Harnessing the power of PCR molecular fingerprinting methods and next generation sequencing for understanding structure and function in microbial communities

Item Type Journal Article

Author S. Phadke

Author A.F. Salvador

Author J.I. Alves

Author O. Bretschger

Author M.M. Alves

Author M.A. Pereira

Abstract Polymerase chain reaction (PCR) is central to methods in molecular ecology. Here, we describe PCR-dependent approaches useful for investigating microbial diversity and its function in various natural, human-associated, and built environment ecosystems. Protocols routinely used for DNA extraction, purification, cloning, and sequencing are included along with various resources for the statistical analysis following gel electrophoresis-based methods (DGGE) and sequencing. We also provide insights into eukaryotic microbiome analysis, sample preservation techniques, PCR troubleshooting, DNA quantification methods, and commonly used ordination techniques. © 2017, Springer Science+Business Media LLC.

Date 2017

DOI 10.1007/978-1-4939-7060-5_16

URL <https://www.scopus.com/pages/publications/85019662290?origin=resultslist>

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Language English

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ISSN 10643745 (ISSN)

Journal Abbr Methods Mol. Biol.

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human, Humans, nonhuman, procedures, classification, statistical analysis, Bacteria, bacterium, DNA extraction, DNA sequence, genetics, high throughput sequencing, High-Throughput Nucleotide Sequencing, isolation and purification, metagenomics, microbial community, polymerase chain reaction, microbial diversity, RNA 16S, RNA, Ribosomal, 16S, Polymerase Chain Reaction, screening, DNA, Escherichia coli, next generation sequencing, genetic variation, Genetic Variation, Canonical correspondence analysis (CCA), Denaturing gradient gel electrophoresis (DGGE), DNA fingerprinting, DNA Fingerprinting, DNA purification, electrophoresis, gel electrophoresis, microbial identification, Microbiome, molecular cloning, Next generation sequencing (NGS), Polymerase chain reaction (PCR), restriction endonuclease, restriction fragment length polymorphism, Restriction fragment length polymorphism (T-RFLP)

Notes:

EID: 2-s2.0-85019662290

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: M.A. Pereira; CEB–Centre of Biological Engineering, University of Minho, Braga, Campus de Gualtar, 4710-057, Portugal; email: alcina@deb.uminho.pt

Heavy Vehicle Classification Through Deep Learning

Item Type Conference Paper

Author P.-Y. Sun

Author W.-Y. Sun

Author Y. Jin

Author R.O. Sinnott

Editor Nepal S.

Editor Cao W.

Editor Nasridinov A.

Editor Bhuiyan M.Z.A.

Editor Guo X.

Editor Zhang L.-J.

Abstract Understanding the flow of traffic on road networks is increasingly important especially with the continued urbanization of the global population. Numerous hardware and software technologies have been applied to measure traffic volumes by Government agencies and/or organizations such as Google, however they are either expensive to deploy; limited in their ability to disambiguate the kinds of vehicles on the road network, or of increasing importance, they infringe on the privacy of individuals, e.g. tracking phones. In this paper we describe work applying deep learning technologies to identify and classify different vehicles on

the road network of Victoria with specific focus on heavy goods vehicles (trucks and trailers). Specifically, we present an approach to automatically detect, classify and count the unique classes of trucks and trailers that are found on the road network and the direction of travel. We apply and compare leading deep learning approaches including You Only Look Once version 3 (YOLOv3) and Single Shot Multi-Box Detector (SSD). This paper builds upon earlier work [1] which focused on data (video) from a single traffic junction in Melbourne. This work is based on a wider range of data (videos) from locations reflecting the diversity of road use including multi-lane motorways, rural roads and city roads. © 2020, Springer Nature Switzerland AG.

Date 2020

DOI 10.1007/978-3-030-59612-5_16

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Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

Volume 12402 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

ISBN 03029743 (ISSN); 978-303059611-8 (ISBN)

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Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Tags:

Motor transportation, Road vehicles, Automobiles, Big data, Deep learning, Roads and streets, Hardware and software, Learning technology, Learning approach, Global population, Trucks, SSD, Trailers, Truck trailers, YOLOv3, Deep SORT, Government agencies, Heavy goods vehicles, Traffic junctions, Traffic volumes

Notes:

EID: 2-s2.0-85092099869

Export Date: 20 September 2026; Cited By: 3; Funding details: Australian Research Council, ARC; Correspondence Address: P.-Y. Sun; School of Computing and Information Systems, The University of Melbourne, Melbourne, Australia; email: pssun@unimelb.edu.au; Conference name: 9th International Conference on Big Data, BigData 2020, held as part of the Services Conference Federation, SCF 2020; Conference date: 18 September 2020 through 20 September 2020; Conference code: 249009

HebbR2-Taffic: A novel application of neuro-fuzzy network for visual based traffic monitoring system

Item Type Journal Article**Author** S.-Y. Cho**Author** C. Quek**Author** S.-X. Seah**Author** C.-H. Chong

Abstract This paper presents a robust methodology that automatically counts moving vehicles along an expressway. The domain of interest for this paper is using both neuro-fuzzy network and simple image processing techniques to implement traffic flow monitoring and analysis. As this system is dedicated for outdoor applications, efficient and robust processing methods are introduced to handle both day and night analysis. In our study, a neuro-fuzzy network based on the Hebbian-Mamdani rule reduction architecture is used to classify and count the number of vehicles that passed through a three- or four-lanes expressway. As the quality of the video captured is corrupted under noisy outdoor environment, a series of preprocessing is required before the features are fed into the network. A vector of nine feature values is extracted to represent whether a vehicle is passing through a lane and this vector serves as input patterns would be used to train the neuro-fuzzy network. The vehicle counting and classification would then be performed by the well-trained network. The novel approach is benchmarked against the MLP and RBF networks. The results of using our proposed neuro-fuzzy network are very encouraging with a high degree of accuracy. © 2008 Elsevier Ltd. All rights reserved.

Date 2009**DOI** 10.1016/j.eswa.2008.08.043**URL** <https://www.scopus.com/pages/publications/58849091317?origin=resultslist>**Archive** Scopus**Language** English**Volume** 36**Publisher** Elsevier Ltd**Pages** 6343-6356**Publication** Expert Systems with Applications**ISSN** 09574174 (ISSN)**Issue** 3 PART 2**Journal Abbr** Expert Sys Appl**Date Added** 20/09/2026, 9:52:47 pm**Modified** 20/09/2026, 9:52:47 pm**Tags:**

Vehicles, Traffic control, Outdoor environment, Image processing, Fuzzy logic, Monitoring, Fuzzy inference, Fuzzy neural networks, Vehicle counting and classifications, Number of vehicles, Novel applications, High degree of accuracy, Neuro-fuzzy network, Processing, Simple image processing techniques, Traffic monitoring system, Traffic monitoring systems

Notes:

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Export Date: 20 September 2026; Cited By: 25; Correspondence Address: S.-Y. Cho; Centre for Computational Intelligence, School of Computer Engineering, Nanyang Technological University, Singapore 639798, Nanyang Avenue, Singapore; email: assycho@ntu.edu.sg; CODEN: ESAPE

Hellinger distance decision trees for PU learning in imbalanced data sets

Item Type Journal Article

Author C. Ortega Vázquez

Author S. vanden Broucke

Author J. De Weerd

Abstract Learning from positive and unlabeled data, or PU learning, is the setting in which a binary classifier can only train from positive and unlabeled instances, the latter containing both positive as well as negative instances. Many PU applications, e.g., fraud detection, are also characterized by class imbalance, which creates a challenging setting. Not only are fewer minority class examples compared to the case where all labels are known, there is also only a small fraction of unlabeled observations that would actually be positive. Despite the relevance of the topic, only a few studies have considered a class imbalance setting in PU learning. In this paper, we propose a novel technique that can directly handle imbalanced PU data, named the PU Hellinger Decision Tree (PU-HDT). Our technique exploits the class prior to estimate the counts of positives and negatives in every node in the tree. Moreover, the Hellinger distance is used instead of more conventional splitting criteria because it has been shown to be class-imbalance insensitive. This simple yet effective adaptation allows PU-HDT to perform well in highly imbalanced PU data sets. We also introduce PU Stratified Hellinger Random Forest (PU-SHRF), which uses PU-HDT as its base learner and integrates a stratified bootstrap sampling. Our empirical analysis shows that PU-SHRF substantially outperforms state-of-the-art PU learning methods for imbalanced data sets in most experimental settings. © The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature 2023.

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Journal Abbr Mach Learn

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Notes:

EID: 2-s2.0-85151236775

Export Date: 20 September 2026; Cited By: 12; Funding details: KBC Groep NV; Correspondence Address: C. Ortega Vázquez; Research Centre for Information Systems Engineering, Faculty of Economics and Business, KU Leuven, Leuven, Belgium; email: carloseduardo.ortegavazquez@kuleuven.be; CODEN: MALEE

Hepatitis E virus seroprevalence rate among Eastern Mediterranean and middle eastern countries; A systematic review and pooled analysis

Item Type Journal Article

Author M.H. Karbalaie Niya

Author M.S. Rezaee-Zavareh

Author A. Ranaei

Author S.M. Alavian

Abstract Context Hepatitis E virus (HEV) as a hepatotropic virus is one of the major global health concerns. Autochthonous HEV transmitted by oral fecal-route in poor sanitation conditions as well as vertical and rarely blood transfusion. HEV occurrence is more common in developing countries and recently increased in developed countries too. Middle East (ME) and Eastern Mediterranean region (EMR) of WHO have been an endemic region for HEV infection. In this regard, we aimed to design a systematic review and pooled analysis to determine seroprevalence of anti-HEV antibody in ME and EMR countries. Evidence acquisition By using PRISMA guideline, data were collected from papers identified through PubMed, Web of Science, Science Direct, Scopus and also from some national and regional databases from January 1990 to June 2016. Serum anti-HEV antibody (IgG) used for HEV prevalence estimation. HEV prevalence in the ME, WHO EMR countries, and in total, calculated by each country population size based on 2015 UN report. Results overall, 62 papers with a total sample size of 31,673 were fulfilled our eligibility criteria and included in our project. Considering anti-HEV antibody (IgG), prevalence of HEV infection in the countries of ME, WHO EMR and in total were 12.17% (95% CI: 11.79–12.57), 11.81% (95% CI: 11.43–12.21), and 11.87% (95% CI: 11.52–12.23) respectively. Conclusions HEV seroprevalence in WHO EMR and ME countries has high rate and more considerations are needed for the prevention and control of this infection especially in high-risk groups such as pregnant women. © 2017

Date 2017

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Tags:

female, Female, human, Humans, Jordan, systematic review, Israel, Saudi Arabia, risk factor, Risk Factors, Review, United Arab Emirates, Oman, prevalence, classification, information processing, pregnancy, Palestine, Pakistan, Epidemiology, Pregnancy, Iran, Prevalence, Systematic review, blood, immunology, Seroepidemiologic Studies, seroepidemiology, seroprevalence, population research, priority journal, Iraq, immunoglobulin G, Immunoglobulin G, pathogenicity, Developing Countries, developing country, sample size, Afghanistan, Bahrain, Cyprus, Djibouti, Eastern Mediterranean region, Egypt, Hepatitis Antibodies, hepatitis antibody, hepatitis E, Hepatitis E, Hepatitis E virus, Kuwait, Lebanon, Libyan Arab Jamahiriya, Mediterranean Region, Middle East, Morocco, Qatar, Somalia, Southern Europe, Sudan, Syrian Arab Republic, Tunisia, Yemen

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Export Date: 20 September 2026; Cited By: 16; Correspondence Address: S.M. Alavian; Baqiyatallah Research Center for Gastroenterology and Liver Diseases (BRCGL), Baqiyatallah University of Medical Sciences, Middle East Liver Disease (MELD) Center, Tehran, Iran; email: Alavian@thc.ir; CODEN: MIPAE

Heterogeneous Label Shift: Theory and Algorithm

Item Type Conference Paper
Author C. Xu
Author X. Tang
Author C. Hou
Editor Singh A.
Editor Fazel M.
Editor Hsu D.
Editor Lacoste-Julien S.
Editor Berkenkamp F.
Editor Maharaj T.

Editor Wagstaff K.

Editor Zhu J.

Abstract In open-environment applications, data are often collected from heterogeneous modalities with distinct encodings, resulting in feature space heterogeneity. This heterogeneity inherently induces label shift, making cross-modal knowledge transfer particularly challenging when the source and target data exhibit simultaneous heterogeneous feature spaces and shifted label distributions. Existing studies address only partial aspects of this issue, leaving the broader problem unresolved. To bridge this gap, we introduce a new concept of Heterogeneous Label Shift (HLS), targeting this critical but underexplored challenge. We first analyze the impact of heterogeneous feature spaces and label distribution shifts on model generalization and introduce a novel error decomposition theorem. Based on these insights, we propose a bound minimization HLS framework that decouples and tackles feature heterogeneity and label shift accordingly. Extensive experiments on various benchmarks for cross-modal classification validate the effectiveness and practical relevance of the proposed approach. © 2025 by the author(s).

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Language English

Extra Journal Abbreviation: Proc. Mach. Learn. Res.

Volume 267

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Conference Name Proceedings of Machine Learning Research

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Modal analysis, Cross-modal, Knowledge management, Knowledge transfer, Encodings, Label distribution, Feature space, Application data, Domain decomposition methods, Heterogeneous features, Open environment, Partial aspects, Space distribution

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EID: 2-s2.0-105023837901

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Hidden markov model classification based on empirical frequencies of observed symbols

Item Type Conference Paper

Author C. Keroglou

Author C.N. Hadjicostis

Editor Lesage J.-J.

Editor Faure J.-M.

Editor 61 av. du president Wilson, Cachan, 94235 Ecole Normale Superieure de Cachan

Editor Ribeiro Cury J.E.

Editor Lennartson B.

Abstract Given a sequence of observations, classification among two known hidden Markov models (HMMs) can be accomplished with a classifier that minimizes the probability of error (i.e., the probability of misclassification) by enforcing the maximum a posteriori probability (MAP) rule. For this MAP classifier, the a priori probability of error (before any observations are made) can be obtained, as a function of the length of the sequence of observations, by summing up the probability of error over all possible observation sequences of the given length, which is a computationally expensive task. In this paper, we obtain an upper bound on the probability of error of the MAP classifier. Our results are based on a suboptimal decision rule that ignores the order with which observations occur and relies solely on the empirical frequencies with which different symbols appear. We describe necessary and sufficient conditions under which this bound on the probability of error decreases exponentially with the length of the observation sequence. Apart from the usefulness of the suboptimal rule in bounding the probability of misclassification, its numerous advantages (such as low computational complexity, reduced storage requirements, and potential applicability to distributed or decentralized decision schemes) could prove a useful alternative to the MAP rule for HMM classification in many applications. Copyright © 2014 IFAC.

Date 2014

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Hidden markov model classification based on empirical frequencies of observed symbols

Item Type Conference Paper

Author C. Keroglou

Author C.N. Hadjicostis

Editor Lesage J.-J.

Editor Faure J.-M.

Editor Cury J.E.R.

Editor Lennartson B.

Abstract Given a sequence of observations, classification among two known hidden Markov models (HMMs) can be accomplished with a classifier that minimizes the probability of error (i.e., the probability of misclassification) by enforcing the maximum a posteriori probability (MAP) rule. For this MAP classifier, the a priori probability of error (before any observations are made) can be obtained, as a function of the length of the sequence of observations, by summing up the probability of error over all possible observation sequences of the given length, which is a computationally expensive task. In this paper, we obtain an upper bound on the probability of error of the MAP classifier. Our results are based on a suboptimal decision rule that ignores the order with which observations occur and relies solely on the empirical frequencies with which different symbols appear. We describe necessary and sufficient conditions under which this bound on the probability of error decreases exponentially with the length of the observation sequence. Apart from the usefulness of the suboptimal rule in bounding the probability of misclassification, its numerous advantages (such as low computational complexity, reduced storage requirements, and potential applicability to distributed or decentralized decision schemes) could prove a useful alternative to the MAP rule for HMM classification in many applications. © IFAC.

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Hierarchical Identification to Quantification Method to Detect Adulterants in Honey by Time Domain Nuclear Magnetic Resonance (TD-NMR)

Item Type Journal Article

Author L.F.R. Lemes

Author V.S. da Paz

Author J.T. de Lima

Author L.M.R. de Novais

Author C. Da Ros Montes D'Oca

Author F.L.F. Soares

Abstract This study introduces a hierarchical strategy designed to differentiate, classify, and quantify adulterants—namely, sugarcane molasses, inverted sugar, and corn syrup—in commercial honey samples analyzed through TD-NMR. The sample preparation process was simplified to involve only heating, ensuring crystallization was avoided. The DD-SIMCA, a one-class classifier, successfully distinguished between pure and adulterated samples, achieving a specificity of 100.00%. The next step involved

predicting the type of adulterant using PLS-DA analysis, which led to classification accuracy exceeding 97.0% across all three adulterants. Furthermore, the PLS model quantified the adulterant levels in the samples, showing RMSEV values below 0.5 (% w w-1) and an RPD greater than 6.38 for each adulterant. The elliptic joint confidence region (EJCR) test was conducted to assess the method's feasibility and reliability. Consequently, the proposed approach highlights the effectiveness of TD-NMR combined with chemometric methods as a valuable tool for verifying the authenticity of commercial honey and preventing fraudulent adulterations. © The Author(s) 2026.

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Hierarchical Integral Probability Metrics: A distance on random probability measures with low sample complexity

Item Type Conference Paper

Author M. Catalano

Author H. Lavenant

Editor Salakhutdinov R.

Editor Kolter Z.

Editor Heller K.

Editor Weller A.

Editor Oliver N.

Editor Scarlett J.

Editor Berkenkamp F.

Abstract Random probabilities are a key component to many nonparametric methods in Statistics and Machine Learning. To quantify comparisons between different laws of random probabilities several works are starting to use the elegant Wasserstein over Wasserstein distance. In this paper we prove that the infinite dimensionality of the space of probabilities drastically deteriorates its sample complexity, which is slower than any polynomial rate in the sample size. We propose a new distance that preserves many desirable properties of the former while achieving a parametric rate of convergence. In particular, our distance 1) metrizes weak convergence; 2) can be estimated numerically through samples with low complexity; 3) can be bounded analytically from above and below. The main ingredient are integral probability metrics, which lead to the name hierarchical IPM. Copyright 2024 by the author(s)

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HIFN-Transformer: Learnable Information-Theoretic Parameters for Interpretable Deep Classification

Item Type Journal Article

Author M. Tawfik

Abstract Deep learning models lack intrinsic interpretability and require manual tuning of information-theoretic hyperparameters. This paper presents HIFN-Transformer (HIFN-T), a framework extending the Variational Information Bottleneck through four jointly learnable per-layer parameters: information retention, entropy budget, magnitude scaling, and global information gates. The retention and budget parameters are co-adapted with the network weights via a stop-gradient decoupling that stabilizes training and substantially reduces the manual hyperparameter burden by eliminating per-layer β grid search; the global gates learn per-neuron importance weights, and a Transformer encoder captures complex feature interactions via multi-head self-attention. We prove HIFN generalizes standard VIB as a special case and characterize the role of the entropy budget as an early-training stabilizer that prevents variance explosion before convergence. We evaluate HIFN-T on five benchmarks spanning network intrusion detection (CICIDS-2017, NSL-KDD, GothamDataset2025), image classification (MNIST), and speech emotion recognition (EMO-DB), comparing against XGBoost, TabNet, FT-Transformer, and four additional baselines. HIFN-T delivers consistent improvements across all five. On the official NSL-KDD KDDTrain+ $\rightarrow$ KDDTest+ benchmark, HIFN-T with a 4-method consensus feature-selection front-end achieves 80.22% accuracy and 0.6898 macro F1, lying within the established 78–87% band of published deep models on this cross-distribution split. On the large-scale GothamDataset2025 IoT benchmark, we evaluate HIFN-T on the full 35,126,611-row release under natural class prevalence as a scalability demonstration on this recently published dataset. MNIST is included as a domain-generalization sanity check rather than a competitive benchmark. Gate-derived feature importance correlates strongly with SHAP analysis across all five datasets, validating intrinsic interpretability. We report all results as mean $\pm$ standard deviation over ten independent runs with Friedman and Wilcoxon signed-rank statistical significance tests. © 2013 IEEE.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Tawfik; Faculty of Computer and Information Technology, Sana'a University, Sanaa, Yemen; email: m.tawfik@su.edu.ye

High dimensional binary classification under label shift: phase transition and regularization

Item Type Journal Article

Author J. Cheng

Author M. Chen

Author H. Liu

Author T. Zhao

Author W. Liao

Abstract Label Shift has been widely believed to be harmful to the generalization performance of machine learning models. Researchers have proposed many approaches to mitigate the impact of the label shift, e.g., balancing the training data. However, these methods often consider the underparametrized regime, where the sample size is much larger than the data dimension. The research under the overparametrized regime is very limited. To bridge this gap, we propose a new asymptotic analysis of the Fisher Linear Discriminant classifier for binary classification with label shift. Specifically, we prove that there exists a phase transition phenomenon: Under certain overparametrized regime, the classifier trained using imbalanced data outperforms the counterpart with reduced balanced data. Moreover, we investigate the impact of regularization to the label shift: The aforementioned phase transition vanishes as the regularization becomes strong. © 2023, The Author(s), under exclusive licence to Springer Nature Switzerland AG.

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High Expression of THY1 in Intestinal Gastric Cancer as a Key Factor in Tumor Biology: A Poor Prognosis-Independent Marker Related to the Epithelial–Mesenchymal Transition Profile

Item Type Journal Article

Author P. Rohan

Author E.C. dos Santos

Author E. Abdelhay

Author R. Binato

Abstract Gastric cancer (GC) is an important cancer-related death worldwide. Among its histological subtypes, intestinal gastric cancer (IGC) is the most common. A previous work showed that increased expression of the THY1 gene was associated with poor overall survival in IGC. Furthermore, it was shown that IGC tumor cells with high expression of THY1 have a greater capacity for tumorigenesis and metastasis in vitro. This study aimed to identify molecular differences between IGC with high and low expression of THY1. Using a feature selection method, a group of 35 genes were found to be the most informative gene set for THY1high IGC tumors. Through a classification model, these genes differentiate THY1high from THY1low tumors with 100% of accuracy both in the test subset and the independent test set. Additionally, this group of 35 genes correctly clustered 100% of the samples. An extensive validation of this potential molecular signature in multiple cohorts successfully segregated between THY1high and THY1low IGC tumors (>95%), proving to be independent of the gene expression quantification methodology. These genes are involved in central processes to tumor biology, such as the epithelial–mesenchymal transition (EMT) and remodeling of the tumor tissue composition. Moreover, patients with THY1high IGC demonstrated poor survival and a more advanced

clinicopathological staging. Our findings revealed a molecular signature for IGC with high THY1 expression. This signature showed EMT and remodeling of the tumor tissue composition potentially related to the biology of IGC. Altogether, our results indicate that THY1high IGC tumors are a particular subset of tumors with a specific molecular and prognosis profile. © 2023 by the authors.

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High-Accuracy Mangrove Extraction and Degradation Diagnosis Using Time-Series Remote Sensing and Deep Learning: A Case Study of the Largest Delta in the Northern Beibu Gulf, China

Item Type Journal Article

Author X. Xie

Author R. Wang

Author Z. Dai

Author X. Liu

Abstract Mangrove extent has increased in many regions under strengthened conservation policies and large-scale restoration programs. Nevertheless, mangrove ecosystems continue to face multiple pressures, including limited total area, habitat degradation, biodiversity decline, and biological invasion, and localized deterioration in ecosystem structure and function has been increasingly reported. Despite extensive mapping efforts, the spatiotemporal dynamics of mangrove degradation—particularly in tidally influenced environments—remain insufficiently understood. Focusing on the Nanliu River Delta, the largest deltaic mangrove system in the Northern Beibu Gulf of China, this study integrates long-term Landsat time-series imagery (1990–2025) with deep learning to quantify both mangrove extent change and canopy degradation. To mitigate tidal inundation effects, a NDVI Pseudo-P75 compositing strategy was applied using Google Earth Engine (GEE), enabling consistent observation of mangrove canopies across tidal stages. Global Mangrove Watch v4 (GMW-V4) and HGMP2020 mangrove dataset for China were used as reference labels to train a ResNet34–UNet segmentation framework incorporating Digital Elevation Model (DEM) constraints. The model achieved high classification performance, with an IoU of 0.822 for mangroves and 0.981 for background, yielding a mean IoU of 0.902. The resulting maps, following manual verification, provided a robust basis for spatiotemporal and degradation analyses. Canopy condition was further assessed using the Enhanced Vegetation Index (EVI), which is less prone to saturation in high-biomass mangrove stands. Results show that mangrove area in the Nanliu River Delta expanded from 266 ha in 1990 to 1414 ha in 2025, with the annual expansion rate after 2005 being nearly seven times higher than that before 2005. Despite this net gain, a cumulative loss of 347.45 ha was recorded, primarily during 1990–2000, with approximately 70% converted to aquaculture and coastal engineering. Spatial analysis revealed that mangrove expansion occurred predominantly seaward, whereas both mangrove loss and canopy degradation exhibited an inverse J-shaped relationship with seawall proximity. More than 80% of mangrove loss occurred within 200 m of seawalls, indicating concentrated anthropogenic encroachment, while 75.6% of canopy degradation was observed within 350 m, potentially reflecting landward forest senescence. These results indicate a transition in dominant threats from permanent land conversion in the late 20th century to more subtle, internal functional

degradation in recent decades, underscoring the need to complement extent-based assessments with canopy condition monitoring in mangrove conservation and management. © 2026 by the authors.

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High-resolution and accelerated multi-parametric mapping with automated characterization of vessel disease using intravascular MRI

Item Type Journal Article

Author G. Wang

Author Y. Zhang

Author S.S. Hegde

Author P.A. Bottomley

Abstract Background: Atherosclerosis is prevalent in cardiovascular disease, but present imaging modalities have limited capabilities for characterizing lesion stage, progression and response to intervention. This study tests whether intravascular magnetic resonance imaging (IVMRI) measures of relaxation times (T1, T2) and proton density (PD) in a clinical 3 Tesla scanner could characterize vessel disease, and evaluates a practical strategy for accelerated quantification. Methods: IVMRI was performed in fresh human artery segments and swine vessels in vivo, using fast multi-parametric sequences, 1-2 mm diameter loopless antennae and 200-300 μm resolution. T1, T2 and PD data were used to train a machine learning classifier (support vector machine, SVM) to automatically classify normal vessel, and early or advanced disease, using histology for validation. Disease identification using the SVM was tested with receiver operating characteristic curves. To expedite acquisition of T1, T2 and PD data for vessel characterization, the linear algebraic method ('SLAM') was modified to accommodate the antenna's highly-nonuniform sensitivity, and used to provide average T1, T2 and PD measurements from compartments of normal and pathological tissue segmented from high-resolution images at acceleration factors of $R \leq 18$ -fold. The results were validated using compartment-average measures derived from the high-resolution scans. Results: The SVM accurately classified $\sim 80\%$ of samples into the three disease classes. The 'area-under-the-curve' was 0.96 for detecting disease in 248 samples, with T1 providing the best discrimination. SLAM T1, T2 and PD measures for $R \leq 10$ were indistinguishable from the true means of segmented tissue compartments. Conclusion: High-resolution IVMRI measures of T1, T2 and PD with a trained SVM can automatically classify normal, early and advanced atherosclerosis with high sensitivity and specificity. Replacing relaxometric MRI with SLAM yields good estimates of T1, T2 and PD an order-of-magnitude faster to facilitate IVMRI-based characterization of vessel disease. © 2017 The Author(s).

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High-Resolution Burned-Area Mapping and Vegetation Resilience in Heterogeneous Landscapes Using Sentinel-2 and Explainable Machine Learning

Item Type Journal Article

Author S. Lu

Author J. Shang

Author Z. Ouyang

Author C. Wei

Author F. Liu

Abstract Accurate wildfire impact assessment and understanding post-disturbance recovery are essential for land management in fire-prone regions. This study develops a Sentinel-2–based burned-area extraction framework and integrates NDVI time-series analysis with explainable machine learning to quantify vegetation resilience across

five fire-affected regions in China. The burned-area map achieves an overall accuracy of 99.8%, substantially outperforming MODIS products (77.9% and 92.7%) by better detecting fragmented patches in complex terrain. NDVI trajectories reveal three resilience pathways: compensatory recovery, stable recovery without compensation, and persistent degradation. Recovery times ranged from approximately 2 months to over 6 months, with some high-altitude areas showing no effective recovery. An XGBoost–SHAP model explains spatial recovery variability ($R^2 = 0.50\text{--}0.88$) and identifies a consistent shift from early climate control to later topographic regulation. Landscape heterogeneity promotes recolonization only within intermediate thresholds, temperature exhibits optimal windows, and precipitation shows diminishing returns. Topography acts primarily by redistributing hydrothermal conditions rather than as an independent driver. The results demonstrate strong spatial variability in ecosystem stability and highlight nonlinear interactions among climate, terrain, and landscape structure as key determinants of resilience. The proposed framework improves burned-area monitoring and supports targeted ecological restoration and adaptive land-use planning in heterogeneous landscapes. © 2026 by the authors.

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High-THC Cannabis Smoke Impairs Incidental Memory Capacity in Spontaneous Tests of Novelty Preference for Objects and Odors in Male Rats

Item Type Journal Article**Author** I.L. Barnard**Author** T.J. Onofrychuk**Author** A.D. Toderash**Author** V.N. Patel**Author** A.E. Glass**Author** J.C. Adrian**Author** R.B. Laprairie**Author** J.G. Howland

Abstract Working memory is an executive function that orchestrates the use of limited amounts of information, referred to as working memory capacity, in cognitive functions. Cannabis exposure impairs working memory in humans; however, it is unclear whether Cannabis facilitates or impairs rodent working memory and working memory capacity. The conflicting literature in rodent models may be at least partly because of the use of drug exposure paradigms that do not closely mirror patterns of human Cannabis use. Here, we used an incidental memory capacity paradigm where a novelty preference is assessed after a short delay in spontaneous recognition-based tests. Either object or odorbased stimuli were used in test variations with sets of identical [identical stimuli test (IST)] and different [different stimuli test (DST)] stimuli (three or six) for low-memory and high-memory loads, respectively. Additionally, we developed a human-machine hybrid behavioral quantification approach which supplements stopwatch-based scoring with supervised machine learning-based classification. After validating the spontaneous IST and DST in male rats, 6-item test versions with the hybrid quantification method were used to evaluate the impact of acute exposure to high- Δ^9 -tetrahydrocannabinol (THC) or high-CBD Cannabis smoke on novelty preference. Under control conditions, male rats showed novelty preference in all test variations. We found that high-THC, but not high-CBD, Cannabis smoke exposure impaired novelty preference for objects under a high-memory load. Odor-based recognition deficits were seen under both low-memory and high-memory loads only following high-THC smoke exposure. Ultimately, these data show that Cannabis smoke exposure impacts incidental memory capacity of male rats in a memory load-dependent, and stimuli-specific manner. © 2023 Barnard et al.

Date 2023**DOI** 10.1523/ENEURO.0115-23.2023**URL** <https://www.scopus.com/pages/publications/85178208087?origin=resultslist>**Archive** Scopus**Language** English**Volume** 10**Publisher** Society for Neuroscience**Publication** eNeuro**ISSN** 23732822 (ISSN)**Issue** 12**Journal Abbr** eNeuro**Date Added** 20/09/2026, 9:51:53 pm**Modified** 20/09/2026, 9:51:53 pm**Tags:**

adult, Article, human, Humans, male, Male, machine learning, nonhuman, controlled study, cognition, facial recognition, convolutional neural network, tetrahydrocannabinol, working memory, memory, cannabis, dronabinol, Dronabinol, animal, Animals, animal experiment, fragrance, Odorants, rat, cannabidiol, Cannabidiol, cannabinoid, cannabinoid receptor agonist, Cannabinoid Receptor Agonists, Cannabis, cannabis use, dorsal striatum, drug exposure, executive function, mental function assessment, novel object recognition test, odor, open field test, Rats, recognition memory, short term exposure, smoke, Smoke, spontaneous incidental memory test, task performance

Notes:

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Export Date: 20 September 2026; Cited By: 9; Funding details: College of Education, University of Saskatchewan; Natural Sciences and Engineering Research Council of Canada, NSERC; Correspondence Address: J.G. Howland; Department of Anatomy, Physiology, and Pharmacology, University of Saskatchewan, Saskatoon, S7N5E5, Canada; email: john.howland@usask.ca

High-throughput Glomeruli analysis of μ CT kidney images using tree priors and scalable sparse computation

Item Type Conference Paper

Author C.C. Shokiche

Author P. Baumann

Author R. Hlushchuk

Author V. Djonov

Author M. Reyes

Editor Ourselin S.

Editor Joskowicz L.

Editor Sabuncu M.R.

Editor Wells W.

Editor Unal G.

Abstract Kidney-related diseases have incrementally become one major cause of death. Glomeruli are the physiological units in the kidney responsible for the blood filtration. Therefore, their statistics including number and volume, directly describe the efficiency and health state of the kidney. Stereology is the current quantification method relying on histological sectioning, sampling and further 2D analysis, being laborious and sample destructive. New micro-Computed Tomography (μ CT) imaging protocols resolute structures down to capillary level. However large-scale glomeruli analysis remains challenging due to object identifiability, allotted memory resources and computational time. We present a methodology for high-throughput glomeruli analysis that incorporates physiological apriori information relating the kidney vasculature with estimates of glomeruli counts. We propose an effective sampling strategy that exploits scalable sparse segmentation of kidney regions for refined estimates of both glomeruli count and volume. We evaluated the proposed approach on a database

of μ CT datasets yielding a comparable segmentation accuracy as an exhaustive supervised learning method. Furthermore we show the ability of the proposed sampling strategy to result in improved estimates of glomeruli counts and volume without requiring a exhaustive segmentation of the μ CT image. This approach can potentially be applied to analogous organizations, such as for example the quantification of alveoli in lungs. © Springer International Publishing AG 2016.

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Export Date: 20 September 2026; Cited By: 1; Funding details: Kommission für Technologie und Innovation, CTI, (14055.1 PFIW-IW); Kommission für Technologie und Innovation, CTI; Correspondence Address: C.C. Shokiche; Institute for Surgical Technology and Biomechanics, University of Bern, Bern, Switzerland; email: carlos.correa@istb.unibe.ch

Higher depression risks in medium- than in highdensity urban form across Denmark

Item Type Journal Article

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Author K. Samuelsson

Author A.M. Closter

Author M. Davies

Author S. Barthel

Author C.B. Pedersen

Author A.V. Prishchepov

Author C.E. Sabel

Abstract Urban areas are associated with higher depression risks than rural areas. However, less is known about how different types of urban environments relate to depression risk. Here, we use satellite imagery and machine learning to quantify three-dimensional (3D) urban form (i.e., building density and height) over time. Combining satellite-derived urban form data and individual-level residential addresses, health, and socioeconomic registers, we conduct a case-control study (n = 75,650 cases and 756,500 controls) to examine the association between 3D urban form and depression in the Danish population. We find that living in dense inner-city areas did not carry the highest depression risks. Rather, after adjusting for socioeconomic factors, the highest risk was among sprawling suburbs, and the lowest was among multistory buildings with open space in the vicinity. The finding suggests that spatial land-use planning should prioritize securing access to open space in densely built areas to mitigate depression risks. © 2023 The Authors.

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Export Date: 20 September 2026; Cited By: 29; Correspondence Address: T.-H.K. Chen; School of the Environment, Yale University, New Haven, United States; email: karen.t.chen@yale.edu

Highly Reproducible, Vendor-Agnostic, Motion-Insensitive Liver PDFF Mapping at 0.55T, 1.5T, and 3T

Item Type Journal Article

Author J. Tang

Author D. Tamada

Author J.-F. Nielsen

Author J. Starekova

Author J.F. Heidenreich

Author F. Schön

Author A.A. Anagnostopoulos

Author A. Roshanshad

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Author S. Yee

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Author D. Rutkowski

Author J. Kammerman

Author J.H. Brittain

Author X. Zhong

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Author M. Zaitsev

Author A.L. Carrel

Author Y. Rathi

Author Y. Jiang

Author B. Bilgic

Author S.B. Reeder

Author D. Hernando

Abstract Purpose: To develop and validate a vendor-agnostic, motion-insensitive proton-density fat-fraction (PDFF) quantification method. Methods: Flip-angle-modulated (FAM) 2D chemical-shift-encoded (CSE) MRI for PDFF quantification was implemented in both the vendor-agnostic platform Pulseseq (“Pulseseq-FAM”) and one vendor-specific platform (“GE-specific FAM”). These implementations were distributed to four sites with twelve MR systems of three vendors (Siemens/GE/Philips) and field strengths (0.55T/1.5T/3T). A sequentially-shipped 16-vial phantom (PDFF = 0%–30%/T1water = 200–1400 ms) underwent confounder-corrected PDFF mapping with commercial 3D-CSE methods and GE-specific FAM as available on each system, and Pulseseq-FAM on every system. To assess bias, phantom PDFF measurements were compared to reference. Between-system variance was evaluated with linear mixed-effects modeling. Different volunteers were also imaged at each site to assess free-breathing PDFF mapping feasibility. A prospective single-site volunteer study was also conducted. Adult patients and children were imaged with breath-held 3D-CSE and free-breathing GE-specific and Pulseseq-FAM. Radiologists evaluated images for overall quality and motion artifacts. To assess bias, Pulseseq-FAM

PDF measurements were compared to 3D-CSE and GE-specific FAM. Test–retest repeatability was assessed by re-imaging after repositioning. Between-field-strength reproducibility was assessed at 1.5T and 3.0T. Results: In the multi-center study, Pulseq-FAM showed reduced T1-bias and between-system variability versus 3D-CSE in phantom PDF measurements, and free-breathing feasibility in volunteers. In the single-site volunteer study (N = 57), Pulseq-FAM improved image quality and motion artifacts versus 3D-CSE ($p < 0.01$). Pulseq-FAM showed excellent agreement with 3D-CSE (95% limits-of-agreement (LoA) = 3.4% PDF) and GE-specific FAM (LoA = 2.0%). Pulseq-FAM showed excellent repeatability (repeatability coefficient (RC) = 1.6% PDF) and between-field-strength reproducibility (reproducibility coefficient (RDC) = 2.4%) versus 3D-CSE (RC = 2.7%/RDC = 3.4%; differences $p < 0.05$). Conclusion: Pulseq-FAM enables accurate, reproducible, vendor-agnostic, and motion-insensitive PDF quantification in adults and children. © 2025 The Author(s). Magnetic Resonance in Medicine published by Wiley Periodicals LLC on behalf of International Society for Magnetic Resonance in Medicine.

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Highly sensitive automated method for DNA damage assessment: Gamma-H2AX foci counting and cell cycle sorting

Item Type Journal Article

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Author M. Terradas

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Author L. Tusell

Author A. Genescà

Abstract Phosphorylation of the H2AX protein is an early step in the double strand break (DSB) repair pathway; therefore, phosphorylated histone (γ H2AX) foci scoring is widely used as a measure for DSBs. Foci scoring is performed either manually or semi-automatically using hand-operated capturing and image analysis software. In general, both techniques are laborious and prone to artifacts associated with manual scoring. While a few fully automated methods have been described in the literature, none of them have been used to quantify γ H2AX foci in combination with a cell cycle phase analysis. Adding this feature to a rapid automated γ H2AX foci quantification method would reduce the scoring uncertainty that arises from the variations in the background level of the γ H2AX signal throughout the cell cycle. The method was set up to measure DNA damage induced in human mammary epithelial cells by irradiation under a mammogram device. We adapted a FISH (fluorescent in situ hybridization) Spot-counting system, which has a slide loader with automatic scanning and cell capture system throughout the thickness of each cell (z-stack), to meet our assay requirements. While scanning the sample, the system classifies the selected nuclei according to the signal patterns previously described by the user. For our purposes, a double staining immunofluorescence was carried out with antibodies to detect γ H2AX and pericentrin, an integral component of the centrosome. We could thus distinguish both the number of γ H2AX foci per cell and the cell cycle phase. Furthermore, restrictive settings of the program classifier reduced the "touching nuclei" problem described in other image analysis software. The automated scoring was faster than and as sensitive as its manually performed counterpart. This system is a reliable tool for γ H2AX radio-induced foci counting and provides essential information about the cell cycle stage. It thus offers a more complete and rapid assessment of DNA damage. © 2013 by the authors; licensee MDPI, Basel, Switzerland.

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Histopathology based AI model predicts anti-angiogenic therapy response in renal cancer clinical trial

Item Type Journal Article
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Author H. Zhong
Author V. Panwar
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Author D. Carrillo

Author A. Christie
Author D. Rakheja
Author Z. Modrusan
Author E.E. Kadel
Author N. Beig
Author M. Huseni
Author J. Brugarolas
Author P. Kapur
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Abstract Anti-angiogenic (AA) therapy is a cornerstone of metastatic clear cell renal cell carcinoma (ccRCC) treatment, but not everyone responds, and predictive biomarkers are lacking. CD31, a marker of vasculature, is insufficient, and the Angioscore, an RNA-based angiogenesis quantification method, is costly, associated with delays, difficult to standardize, and does not account for tumor heterogeneity. Here, we developed an interpretable deep learning (DL) model that predicts the Angioscore directly from ubiquitous histopathology slides yielding a visual vascular network (H&E DL Angio). H&E DL Angio achieves a strong correlation with the Angioscore across multiple cohorts (spearman correlations of 0.77 and 0.73). Using this approach, we found that angiogenesis inversely correlates with grade and stage and is associated with driver mutation status. Importantly, DL Angio expediently predicts AA response in both a real-world and IMmotion150 trial cohorts, out-performing CD31, and closely approximating the Angioscore (c-index 0.66 vs 0.67) at a fraction of the cost. © The Author(s) 2025.

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Holistic Transfer: Towards Non-Disruptive Fine-Tuning with Partial Target Data

Item Type Conference Paper

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Author H.-Y. Chen

Author Z. Mai

Author J. Zhong

Author V. Pahuja

Author T. Berger-Wolf

Author S. Gao

Author C. Stewart

Author Y. Su

Author W.-L. Chao

Editor Oh A.

Editor Neumann T.

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Editor Saenko K.

Editor Hardt M.

Editor Levine S.

Abstract We propose a learning problem involving adapting a pre-trained source model to the target domain for classifying all classes that appeared in the source data, using target data that covers only a partial label space. This problem is practical, as it is unrealistic for the target end-users to collect data for all classes prior to adaptation. However, it has received limited attention in the literature. To shed light on this issue, we construct benchmark datasets and conduct extensive experiments to uncover the inherent challenges. We found a dilemma - on the one hand, adapting to the new target domain is important to claim better performance; on the other hand, we observe that preserving the classification accuracy of classes missing in the target adaptation data is highly challenging, let alone improving them. To tackle this, we identify two key directions: 1) disentangling domain gradients from classification gradients, and 2) preserving class relationships. We present several effective solutions that maintain the accuracy of the missing classes and enhance the overall performance, establishing solid baselines for holistic transfer of pre-trained models with partial target data. © 2023 Neural information processing systems foundation. All rights reserved.

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Homeolog expression quantification methods for allopolyploids

Item Type Journal Article

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Author M. Hatakeyama

Author T. Tameshige

Author K.K. Shimizu

Author J. Sese

Abstract Genome duplication with hybridization, or allopolyploidization, occurs in animals, fungi and plants, and is especially common in crop plants. There is an increasing interest in the study of allopolyploids because of advances in polyploid genome assembly; however, the high level of sequence similarity in duplicated gene copies (homeologs) poses many challenges. Here we compared standard RNA-seq expression quantification approaches used currently for diploid species against subgenome-classification approaches which maps reads to each subgenome separately. We examined mapping error using our previous and new RNA-seq data in which a subgenome is experimentally added (synthetic allotetraploid *Arabidopsis kamchatica*) or reduced (allohexaploid wheat *Triticum aestivum* versus extracted allotetraploid) as ground truth. The error rates in the two species were very similar. The standard approaches showed higher error rates (>10% using pseudo-alignment with Kallisto) while subgenome-classification approaches showed much lower error rates (<1% using EAGLE-RC, <2% using HomeoRoq). Although downstream analysis may partly mitigate mapping errors, the difference in methods was substantial in hexaploid wheat, where Kallisto appeared to have systematic differences relative to other methods. Only approximately half of the differentially expressed homeologs detected using Kallisto overlapped with those by any other method in wheat. In general, disagreement in low-expression genes was responsible for most of the discordance between methods, which is consistent with known biases in Kallisto. We also observed that there exist uncertainties in genome sequences and annotation which can affect each method differently. Overall, subgenome-classification approaches tend to perform better than standard approaches with EAGLE-RC having the highest precision. © 2018 The Author(s) 2018. Published by Oxford University Press.

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How large a training set is needed to develop a classifier for microarray data?

Item Type Journal Article

Author K.K. Dobbin

Author Y. Zhao

Author R.M. Simon

Abstract Purpose: A common goal of gene expression microarray studies is the development of a classifier that can be used to divide patients into groups with different prognoses, or with different expected responses to a therapy. These types of classifiers are developed on a training set, which is the set of samples used to train a classifier. The question of how many samples are needed in the training set to produce a good classifier from high-dimensional microarray data is challenging. Experimental Design: We present a model-based approach to determining the sample size required to adequately train a classifier. Results: It is shown that sample size can be determined from three quantities: standardized fold change, class prevalence, and number of genes or features on the arrays. Numerous examples and important experimental design issues are discussed. The method is adapted to address ex post facto determination of whether the size of a training set used to develop a classifier was adequate. An interactive web site for performing the sample size calculations is provided. Conclusion: We showed that sample size calculations for classifier development from high dimensional microarray data are feasible, discussed numerous important considerations, and presented examples. © 2008 American Association for Cancer Research.

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Export Date: 20 September 2026; Cited By: 115; Correspondence Address: K. K. Dobbin; National Cancer Institute, EPN, Rockville, MD 20852, 6130 Executive Boulevard, United States; email: dobbinke@mail.nih.gov; CODEN: CCREF

How Painful are Lumbar Hernias? A Comprehensive Review of Intervention Strategies

Item Type Journal Article

Author R. Moreno-Gómez-Toledano

Author I. Méndez-Mesón

Author S. Aguado-Henche

Author A. Sebastián-Martín

Author M. Grande-Alonso

Abstract Purpose of Review: Low back pain (LBP) is considered an important issue of public health, with annual prevalence estimations almost achieving 60% of the worldwide population. Available treatments have a limited impact on this condition, although they allow to alleviate pain and recover the patient's quality of life. This review aims to go deeper on the understanding of this condition, providing an updated, brief, and concise whole picture of this common musculoskeletal problem. Recent Findings: Scientific literature, current clinical practice and clinical guidelines are summarized, focusing on three key aspects: classification of LBP, diagnosis of symptomatic lumbar hernia, and intervention strategies (conservative, surgical, and pharmacological). Benefits and drawbacks of each approach are tackled. Summary: The most appropriate intervention for LBP suffers is hitherto a conservative treatment based on therapeutic exercise, manual therapy and therapeutic education on the neurophysiological mechanisms of pain. Whether patient's condition is severe, does not improve with conservative treatment, or presents neurological symptoms, then surgical intervention is recommended. The efficiency of pharmaceutical approaches

for LBP lacks high-quality evidence-based studies, and still needs to be in-depth explored. Current treatments help to improve symptoms and patient's perspectives. However, further research in the field of herniated discs is essential in order to seek a therapy that could definitely cure or eliminate this condition. © The Author(s) 2025.

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Article, human, Humans, diagnosis, practice guideline, therapy, complication, nonsteroid antiinflammatory agent, disease severity, patient attitude, symptom, patient education, clinical practice, disease classification, physical examination, nuclear magnetic resonance imaging, medical research, ozone, computer assisted tomography, low back pain, agmatine, amitriptyline, anamnesis, antidepressant agent, antiinflammatory agent, conservative treatment, corticosteroid, endoscopic surgery, evidence based medicine, gabapentin, glucocorticoid, Intervertebral Disc Displacement, intervertebral disk hernia, kinesiotherapy, Low Back Pain, lumbar discectomy, lumbar disk hernia, Lumbar hernia, lumbar microdiscectomy, lumbar vertebra, Lumbar Vertebrae, manipulative medicine, Manual therapy, medical decision making, muscle relaxant agent, myelography, neurophysiology, opiate, oxygen, pain assessment, paracetamol, percutaneous discectomy, Pharmacology, Physiotherapy, risk benefit analysis, scientific literature, serotonin antagonist, spine radiography, spine scintiscanning, spine surgery, Surgery, Therapeutic exercise, thrombocyte rich plasma

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A. Sebastián-Martín; Universidad de Alcalá, School of Medicine and Health Sciences, Department of Surgery, Medical and Social Sciences, Area of Human Anatomy and Embryology, Universidad de Alcalá, Alcalá de Henares, University Campus - C/ 19 Av de Madrid Km 33 600, Madrid, 28871, Spain; email: alba.sebastian@uah.es; CODEN: CPHRG

How to evaluate uncertainty estimates in machine learning for regression?

Item Type Journal Article

Author L. Sluijterman

Author E. Cator

Author T. Heskes

Abstract As neural networks become more popular, the need for accompanying uncertainty estimates increases. There are currently two main approaches to test the quality of these estimates. Most methods output a density. They can be compared by evaluating their loglikelihood on a test set. Other methods output a prediction interval directly. These methods are often tested by examining the fraction of test points that fall inside the corresponding prediction intervals. Intuitively, both approaches seem logical. However, we demonstrate through both theoretical arguments and simulations that both ways of evaluating the quality of uncertainty estimates have serious flaws. Firstly, both approaches cannot disentangle the separate components that jointly create the predictive uncertainty, making it difficult to evaluate the quality of the estimates of these components. Specifically, the quality of a confidence interval cannot reliably be tested by estimating the performance of a prediction interval. Secondly, the loglikelihood does not allow a comparison between methods that output a prediction interval directly and methods that output a density. A better loglikelihood also does not necessarily guarantee better prediction intervals, which is what the methods are often used for in practice. Moreover, the current approach to test prediction intervals directly has additional flaws. We show why testing a prediction or confidence interval on a single test set is fundamentally flawed. At best, marginal coverage is measured, implicitly averaging out overconfident and underconfident predictions. A much more desirable property is pointwise coverage, requiring the correct coverage for each prediction. We demonstrate through practical examples that these effects can result in favouring a method, based on the predictive uncertainty, that has undesirable behaviour of the confidence or prediction intervals. Finally, we propose a simulation-based testing approach that addresses these problems while still allowing easy comparison between different methods. This approach can be used for the development of new uncertainty quantification methods. © 2024 The Authors

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human, simulation, machine learning, Machine learning, Forecasting, Uncertainty analysis, article, Neural-networks, Neural networks, Uncertainty, Machine Learning, prediction, artificial neural network, computer simulation, Computer Simulation, nerve cell network, Neural Networks, Computer, Quality control, Test sets, uncertainty, bootstrapping, Confidence interval, Prediction interval, Uncertainty estimates, confidence interval, Bootstrap, Dropout, Log likelihood, Regression

Notes:

EID: 2-s2.0-85186697287

Export Date: 20 September 2026; Cited By: 43; Correspondence Address: L. Sluijterman; Department of Mathematics, Radboud University, Nijmegen, P.O. Box 9010-59, 6500 GL, Netherlands; email: L.Sluijterman@math.ru.nl; CODEN: NNETE

How to optimize blue-green space to enhance urban diurnal cooling effects?

Item Type Journal Article

Author X. Li

Author W. Qian

Author Y. Chen

Abstract Rapid urbanization has intensified the urban heat island effect, exacerbating heatwaves and threatening public health and energy security. Most previous studies have focused on blue or green spaces separately, whereas the combined cooling effects of Blue-Green Spaces (BGSs) have not yet been fully researched. The role of their morphological structure and landscape configuration in regulating diurnal land surface temperature variations remains largely unexplored. Moreover, traditional studies have generally relied on linear models, whereas the relationship between BGS and land surface temperature is inherently nonlinear. Machine learning methods can efficiently capture these complex, nonlinear mechanisms, enabling more precise quantification. Therefore, this study employs machine learning to quantify the cooling effects of BGS in high-density urban areas of Nanjing. Results reveal significant differences between daytime and nighttime cooling. Large Core areas provide strong daytime cooling but may release heat at night, while high landscape connectivity enhances cooling potential across both periods. Interaction analyses show that, for a fixed patch size, increasing the Core area strengthens daytime cooling, and for a fixed core area, highly connected patches further amplify cooling effects. This study highlights the complex role of BGS in urban thermal regulation and its diurnal variations, revealing the synergistic cooling effects of BGS morphology and landscape configuration, and providing scientific guidance for urban planning and climate-adaptive city development. © 2025 Elsevier B.V.

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Learning systems, Machine-learning, Machine learning, Public health, Urban growth, Morphology, Land surface temperature, Atmospheric temperature, Cooling, Cooling effects, Surface properties, Blue-green space, Core area, Green spaces, Landscape configuration, Morphological structure, Morphological structures, Rapid urbanizations, Surface measurement, Urban Heat Island Effects

Notes:

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Export Date: 20 September 2026; Cited By: 16; Correspondence Address: Y. Chen; School of Architecture, Southeast University, Nanjing, 2 Sipailou, 210096, China; email: chenye@seu.edu.cn; CODEN: ENEBD

Human action recognition based on Adaptive Distance Generalization of Isometric Mapping

Item Type Conference Paper

Author H. Qu

Author J. Cheng

Abstract Manifold learning could effectively represent human action which has Non-linear characteristics. Isometric Mapping (ISOMAP) is a classic unsupervised algorithm of manifold learning. However, ISOMAP couldn't work well for the data with the class priori information. Moreover, the computational complexity of dimension reduction to the new data point is too high to be used in real time. Considering two shortages of ISOMAP, the Adaptive Distance Generalization of Isometric Mapping (ADGI) is proposed, using human action silhouette sequences as the features, in which the adaptive distance factor is introduced to combine with generalization of ISOMAP. Finally the nearest neighbor classifier is used for recognition. For the dimension reduction of human action features, ADGI is effective. Experiments in Weizmann database show the presented algorithm is better both in recognition ratio and in real time for human action recognition. © 2012 IEEE.

Date 2012

DOI 10.1109/CISP.2012.6469785

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Language English

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Algorithms, Signal processing, Mapping, Dimension reduction, Gesture recognition, adaptive distance, Adaptive distance, human action recognition, Human-action recognition, ISOMAP, manifold learning, Manifold learning, Nearest Neighbor classifier, Nonlinear characteristics, Unsupervised algorithms

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EID: 2-s2.0-84875025671

Export Date: 20 September 2026; Cited By: 3; Conference name: 2012 5th International Congress on Image and Signal Processing, CISP 2012; Conference date: 16 October 2012 through 18 October 2012; Conference code: 96020

HVAC Equipment Monitoring

Item Type Journal Article

Author Z. Zhao

Author X. Han

Author S. Yang

Author B. Yang

Abstract This chapter comprehensively examines non-contact monitoring technologies for heating, ventilation, and air conditioning (HVAC) systems, focusing on infrared thermography and computer vision (CV) applications. Infrared thermography (IRT) enables precise fault detection in mechanical, electrical, and HVAC equipment by capturing temperature anomalies, with advancements in image enhancement algorithms improving visualization and diagnostic accuracy. CV techniques automate frost/dew observation and enhance defrost control, utilizing machine learning to quantify frost thickness, coverage, and fractal dimension for optimized defrost timing. In condensation heat transfer measurement, CV technology integrates droplet dynamics with deep learning to predict heat transfer performance, reducing experimental errors via multi-angle imaging and extended video analysis. Applications span energy infrastructure monitoring and equipment operation performance assessment. Future directions emphasize multi-technology integration (IRT and CV and artificial intelligence), dynamic droplet feature extraction, and universal condensation models to minimize energy consumption and enhance system reliability. These innovations collectively advance intelligent HVAC management, supporting energy efficiency and sustainability across built environmental domains. © 2026 selection and editorial matter, Bin Yang, Zhihan Lyu, Xiaogang Cheng, and Yihang Liu; individual chapters, the contributors.

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Deep learning, Learning systems, Energy efficiency, Video analysis, Fault detection, Precise faults, Drops, Faults detection, Computer vision applications, Condensation, Conditioning equipment, Conditioning systems, Equipment monitoring, Fractal dimension, Heat transfer performance, Heating equipment, Heating ventilation and air conditioning, HVAC, Mechanical heating, Monitoring technologies, Non-contact monitoring, Thermography (imaging)

Notes:

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Hybrid Bayesian classifier for improved classification accuracy

Item Type Journal Article

Author U. Kumar

Author S.K. Raja

Author C. Mukhopadhyay

Author T.V. Ramachandra

Abstract The widely used Bayesian classifier is based on the assumption of equal prior probabilities for all the classes. However, inclusion of equal prior probabilities may not guarantee high classification accuracy for the individual classes. Here, we propose a novel technique Hybrid Bayesian Classifier (HBC) where the class prior probabilities are determined by unmixing a supplemental low spatial high spectral resolution multispectral (MS) data that are assigned to every pixel in a high spatial low spectral resolution MS data in Bayesian classification. This is demonstrated with two separate experiments first, class abundances are estimated per pixel by unmixing Moderate Resolution Imaging Spectroradiometer data to be used as prior probabilities, while posterior probabilities are determined from the training

data obtained from ground. These have been used for classifying the Indian Remote Sensing Satellite LISS-III MS data through Bayesian classifier. In the second experiment, abundances obtained by unmixing Landsat Enhanced Thematic Mapper Plus are used as priors, and posterior probabilities are determined from the ground data to classify IKONOS MS images through Bayesian classifier. The results indicated that HBC systematically exploited the information from two image sources, improving the overall accuracy of LISS-III MS classification by 6% and IKONOS MS classification by 9%. Inclusion of prior probabilities increased the average producer's and user's accuracies by 5.5% and 6.5% in case of LISS-III MS with six classes and 12.5% and 5.4% in IKONOS MS for five classes considered. © 2010 IEEE.

Date 2011

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Hybrid Deep Learning with Uncertainty Quantification for Robust Detection of Acute Lymphoblastic Leukemia in Blood Smear Images

Item Type Conference Paper

Author S.J. Mitu
Author M.F. Jaman
Author S.S. Shouma
Author M. Parvin

Abstract The diagnosis of acute lymphoblastic leukemia (ALL) is extremely difficult because of the very aggressive course wreaking severe havoc on the economy. A very rapid recognition is important in order to lower the mortality. In this work we present a complete deep learning pipeline with the aim of detecting leukemia from the blood smear microscopy as images. Different CNN architectures using pretrained convolutional networks are provided in this work, MobileNetV2, InceptionV3 and EfficientNetB3 are provided. Gated recurrent units (GRU) are also incorporated to extract not only spatial aspects, but temporal characteristics as well. A very important difficulty in this area is the great imbalance of the classes in open public data sets. We employ strategic data augmentation and leverage deep ensemble methods with uncertainty quantification to increase model confidence and improve interpretability. The preprocessing pipeline incorporates standard techniques such as resizing and normalization and different geometric transformations to improve robustness. We evaluate our methodology on the ALL-IDB1 and ALL-IDB2 datasets, each separately and in combination, to show consistent improvement over baseline applications. The EfficientNetB3-GRU variant having ensemble uncertainty estimation achieves 99.67% accuracy and 99.45% F1 quality measures on the combined dataset, outperforming competitive hybrid methods and published results. Our framework offers both high diagnostic quality and reliable confidence scores as a potential clinical aid. Our main contributions are the hybrid CNN-GRU design, methods to overcome data imbalance, and the introduction of uncertainty quantification methods to enhance automation of ALL diagnosis. © 2025 IEEE.

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Language English

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Tags:

Deep learning, Classification (of information), Data mining, Robust detection, Deep Learning, Image classification, Medical image processing, Blood, Class imbalance, Data augmentation, Diseases, Convolutional networks, Mathematical transformations, Data Augmentation, Class Imbalance, Computer aided diagnosis, Uncertainty quantifications, Uncertainty Quantification, Medical image classification, Acute lymphoblastic leukaemias, Acute Lymphoblastic Leukemia, Blood smears, Hybrid CNN-gated recurrent unit, Hybrid CNN-GRU, Medical Image Classification, Open Data

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S.J. Mitu; Daffodil International University, DIU NLP & ML Research Lab, Department of CSE, Dhaka, Bangladesh; email: jannatmitu.cse@diu.edu.bd; Conference name: 2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health, BECITHCON 2025; Conference date: 29 November 2025 through 30 November 2025; Conference code: 223832

Hybrid Modelling of Austempering in the Automotive Industry

Item Type Conference Paper

Author J. Wörner

Author T. Waldenmaier

Author L. Hagymási

Author V. Schulze

Abstract Austempering is a heat treatment process used in the automotive industry. In contrast to case hardening, austempering results in a bainitic microstructure with only slightly lower hardness and significantly higher fatigue strength. For economic reasons, the heat treatment is carried out in batch processes, usually with several components being quality tested. Extensive sensor data is available for each batch, but the variance is low due to the stability of the series process. At the same time, quality control using Vickers hardness testing is subject to measurement uncertainty, especially when only a few indentations are measured, as is common in production. Deviations within a batch further limit the prediction accuracy for an individual part. The prediction quality of a machine learning (ML) model is limited by the quality of the data. One way to better model the process, is to use simulation data in a hybrid model. Another approach is Uncertainty Quantification, which provides an indication of how confident the model is that the prediction is accurate. Many of these models are also more robust and have higher prediction accuracy. The aim of this work is the implementation of several Uncertainty Quantification methods for a hybrid model as well as the comparison of these models and a recommended action for hybrid models in production. The ML models with Uncertainty Quantification are better suited for data with a higher uncertainty (e.g. measurement uncertainty). These models learn next to the labels how reliable the predictions are. This is particularly important in production and can be used as an additional criterion in quality control. The developed model will be evaluated with production data to assess its performance. © 2023 28th IFHTSE 2023 Congress. All rights reserved.

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Publisher International Federation for Heat Treatment and Surface Engineering, IFHTSE
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Hybrid statistical-machine learning approach for analyzing legacy and new phosphorus losses from subsurface drainage systems

Item Type Journal Article

Author E. Aniekwensi

Author E. Ghane

Abstract Phosphorus (P) is essential for crop growth but leaches through subsurface drainage discharge, impacting water quality. This study's objectives are to (1) apply hybrid statistical-machine learning to quantify the contributions of incidental (new) and legacy (old) P in drainage discharge from organic site and inorganic site and (2) evaluate the effect of manure application timing on P loss. We collected data from two on-farm sites in southeast Michigan, USA. A linear regression equation was used to analyze P load based on drainage discharge and fertilizer application timing. The data were split into calibration and validation sets, and machine learning was used for training. The results showed strong model prediction performance. Organic fertilizers contributed approximately twice the observed total phosphorus (TP) loss (7.54 kg ha⁻¹ vs. 3.73 kg ha⁻¹) and nearly four times the dissolved reactive phosphorus (DRP) loss (4.90 kg ha⁻¹ vs. 1.05 kg ha⁻¹) compared to inorganic P loss, mainly due

to the greater P application rate and higher soil test P. When applied during winter months (December–January), organic fertilizer contributed to greater new P loss, whereas early fall applications (October–November) resulted in lower new P loss, showing the importance of application timing. At the organic site, legacy P was the dominant contributor to TP and DRP losses, accounting for 84% and 79% of losses, respectively. At the inorganic site, legacy P was responsible for 97% of TP loss and the entirety (100%) of DRP loss. In conclusion, legacy P was the dominant source of P loss through drainage discharge, and winter organic fertilizer application significantly increased new P loss. © 2026 The Author(s). Journal of Environmental Quality published by Wiley Periodicals LLC on behalf of American Society of Agronomy, Crop Science Society of America, and Soil Science Society of America.

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HYPER EVIDENTIAL DEEP LEARNING TO QUANTIFY COMPOSITE CLASSIFICATION UNCERTAINTY

Item Type Conference Paper

Author C. Li

Author K. Li

Author Y. Ou

Author L.M. Kaplan

Author A. Jøsang

Author J.-H. Cho

Author D.H. Jeong

Author F. Chen

Abstract Deep neural networks (DNNs) have been shown to perform well on exclusive, multi-class classification tasks. However, when different classes have similar visual features, it becomes challenging for human annotators to differentiate them. This scenario necessitates the use of composite class labels. In this paper, we propose a novel framework called Hyper-Evidential Neural Network (HENN) that explicitly models predictive uncertainty due to composite class labels in training data in the context of the belief theory called Subjective Logic (SL). By placing a grouped Dirichlet distribution on the class probabilities, we treat predictions of a neural network as parameters of hyper-subjective opinions and learn the network that collects both single and composite evidence leading to these hyper-opinions by a deterministic DNN from data. We introduce a new uncertainty type called vagueness originally designed for hyper-opinions in SL to quantify composite classification uncertainty for DNNs. Our results demonstrate that HENN outperforms its state-of-the-art counterparts based on four image datasets. The code and datasets are available at: <https://github.com/Hugo101/HyperEvidentialNN>. © 2024 12th International Conference on Learning Representations, ICLR 2024. All rights reserved.

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Tags:

Deep neural networks, Neural-networks, Uncertainty, Visual feature, Class labels, Probability distributions, Classification tasks, Different class, Multi-class classification, Predictive uncertainty, Subjective Logic, Model predictive

Notes:

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Hyperdimensional Uncertainty Quantification for Multimodal Uncertainty Fusion in Autonomous Vehicles Perception

Item Type Conference Paper

Author L. Chen

Author J. Wang

Author T. Mortlock

Author P. Khargonekar

Author M.A. Al Faruque

Abstract Uncertainty Quantification (UQ) is crucial for ensuring the reliability of machine learning models deployed in real-world autonomous systems. However, existing approaches typically quantify task-level output prediction uncertainty without considering epistemic uncertainty at the multimodal feature fusion level, leading to sub-optimal outcomes. Additionally, popular uncertainty quantification methods, e.g., Bayesian approximations, remain challenging to deploy in practice due to high computational costs in training and inference. In this paper, we propose HyperDUM, a novel deterministic uncertainty method (DUM) that efficiently quantifies feature-level epistemic uncertainty by leveraging hyper-dimensional computing. Our method captures the channel and spatial uncertainties through channel and patch -wise projection and bundling techniques respectively. Multimodal sensor features are then adaptively weighted to mitigate uncertainty propagation and improve feature fusion. Our evaluations show that HyperDUM on average outperforms the state-of-the-art (SOTA) algorithms by up to 2.01%/1.27% in 3D Object Detection and up to 1.29% improvement over baselines in semantic segmentation tasks under various types of uncertainties. Notably, HyperDUM requires $2.36\times$ less Floating Point Operations and up to $38.30\times$ less parameters than SOTA methods, providing an efficient solution for real-world autonomous systems. © 2025 IEEE.

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autonomous vehicles, Autonomous vehicles, Autonomous Vehicles, Object recognition, Learning systems, Computer vision, Object detection, Objects detection, Human computer interaction, Real-world, Semantics, Uncertainty, Semantic Segmentation, Multi-modal, Bayesian networks, Inference engines, Semantic segmentation, object detection, semantic segmentation, Uncertainty quantifications, Epistemic uncertainties, Three dimensional computer graphics, epistemic uncertainty quantification, Epistemic uncertainty quantification, multimodal sensor fusion, Multimodal sensor fusion

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Hyperspectral image classification and development of fluorescence index for single corn kernels infected with *Aspergillus flavus*

Item Type Journal Article

Author H. Yao

Author Z. Hruska

Author R. Kincaid

Author R.L. Brown

Author D. Bhatnagar

Author T.E. Cleveland

Abstract Aflatoxins are toxic secondary metabolites predominantly produced by the fungi *Aspergillus flavus* and *Aspergillus parasiticus*. Aflatoxin-contaminated corn is toxic to domestic animals when ingested in feed and is a known carcinogen associated with liver and lung cancer in humans. Consequently, aflatoxin levels in food and feed are regulated by the U.S. Food and Drug Administration (FDA), which allows 20 ppb (parts per billion) limits in food and 100 ppb in feed for interstate commerce. Currently, aflatoxin detection and quantification methods are based on analytical tests, including thin-layer chromatography (TCL) and high-performance liquid chromatography (HPLC). These analytical tests require the destruction of samples and are costly and time consuming. Thus, the ability to detect aflatoxin in a rapid, nondestructive way is crucial to the grain industry, particularly the corn industry. Hyperspectral imaging technology offers a non-invasive approach to screening for

food safety inspection and quality control based on spectral signatures. The focus of this study was to classify aflatoxin-contaminated single corn kernels using fluorescence hyperspectral imagery. Field-inoculated corn kernels were used in the study. Contaminated and control kernels under long-wavelength ultraviolet excitation were imaged using a visible near-infrared (VNIR) hyperspectral camera. The imaged kernels were chemically analyzed to provide reference information for image analysis. This article describes a procedure for processing corn kernels located in different images for statistical training and classification. Two classification algorithms (maximum likelihood and binary encoding) were used to classify each corn kernel as "control" or "contaminated" through pixel classification. The binary encoding approach had a slightly better performance, with accuracy equal to 87% or 88% when 20 ppb or 100 ppb, respectively, was used as the classification threshold. In addition, three narrow-band fluorescence indices were developed and tested in this study. It was found that the highest correlation was -0.81 with the normalized difference fluorescence index (NDFI). The two bands used for the NDFI were 437 and 537 nm. The use of key wavelengths for contamination detection would be helpful for developing rapid and non-invasive inspection systems. This study demonstrated the potential of using fluorescence hyperspectral imagery for aflatoxin contamination detection in corn kernels infected with *A. flavus*.

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Tags:

United States, Remote sensing, Image classification, analytical method, Encoding (symbols), Classification algorithm, fungus, image analysis, carcinogen, High performance liquid chromatography, food safety, fungal disease, Detection and quantifications, image classification, maize, toxin, pixel, cancer, Food and Drug Administration, Hyperspectral image classification, Hyperspectral image, fluorescence, Fluorescence, Spectroscopy, Contamination, Aflatoxin, Aflatoxins, agroindustry, Aspergillus, Chromatography, Corn, food industry, Hyper-spectral images, Hyperspectral imaging technologies, Index, secondary metabolite

Notes:

EID: 2-s2.0-84890839863

Export Date: 20 September 2026; Cited By: 39; Correspondence Address: H. Yao; Stennis Space Center, MS 39529, 1021 Balch Blvd., United States; email: haibo@gri.msstate.edu

Hyperspectral imagery classification based on sparse feature and markov random field

Item Type Journal Article

Author J. Yang

Author L. Wang

Author J. Qian

Abstract Aiming at the deficiencies that some implicit information is not well reflected in the traditional hyperspectral image classification methods, this paper presents a new hyperspectral image classification method based on Sparse Feature and Markov Random Field (SFMRF). The core idea of the SFMRF is firstly to express the hyperspectral image with the sparse feature, and then to apply the Support Vector Machine (SVM) to estimate the class conditional probability density functions, also, the MRF method is used to estimate the context-based class priors. Finally, the final class labels are determined by the Maximum a Posteriori (MAP) method. Theory analysis and simulation results on the Indian Pines hyperspectral data show that the proposed SFMRF method can greatly improve the classification accuracy. Copyright © 2015 Binary Information Press.

Date 2015

DOI 10.12733/jics20105904

URL <https://www.scopus.com/pages/publications/84930532758?origin=resultslist>

Archive Scopus

Language English

Volume 12

Publisher Binary Information Press

Pages 3073-3082

Publication Journal of Information and Computational Science

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Issue 8

Journal Abbr J. Inf. Comput. Sci.

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Tags:

Probability density function, Markov processes, Support vector machines, Classification (of information), Image classification, Classification, SVM, Classification accuracy, HyperSpectral, Hyperspectral, Spectroscopy, Conditional probability density, Markov Random Fields, Analysis and simulation, Hyperspectral imagery classifications, Implicit informations, Magnetorheological fluids, MRF, Sparse feature, Sparse features

Notes:

EID: 2-s2.0-84930532758

Export Date: 20 September 2026; Cited By: 1; Funding details: National Natural Science Foundation of China, (61275010); Correspondence Address: J. Yang; College of Information and Communications Engineering, Harbin Engineering University, Harbin, 150001, China; email: yang06081102@163.com

Hypophosphatemic Rickets in Colombia: A Prevalence-Estimation Model in Rare Diseases. 2018

Item Type Journal Article

Author J.A. Huertas-Quintero

Author N. Losada-Trujillo

Author D.A. Cuellar-Ortiz

Author H.M. Velasco-Parra

Abstract Background: Hypophosphatemic rickets is a rare, genetic syndrome with multisystem involvement. It causes skeletal abnormalities, painful enthesopathies, increased risk of fracture, and short stature; leading to a substantial burden of disease, disability, and worsening of quality of life. To improve health conditions of people living with this disease, it is essential to know its prevalence which is currently unknown in Colombia. This study aimed to estimate the prevalence of hypophosphatemic rickets in Colombia by using a mathematical model and national statistic records. Methods: We executed a model to estimate probabilities of transitions between health, disease, and death states (Markov chains). The model was fed with international prevalences taken from original studies (systematic review) and administrative records' data from SISPRO (a national health information system) using the International Classification of Diseases (ICD-10) E833 code, vital statistics, and census data. World Health Organization's (WHO) DISMOD II software was used to develop the model. Findings: The estimated overall prevalence of hypophosphatemic rickets in Colombia in 2018 was 2.03 cases per 100 000 people (981 affected people), with a sensitive range of 1.97 to 2.09. The estimated prevalence by sex was 2.61 (645 people) and 1.43 (336 people) cases per 100 000 women and men, respectively. Interpretation: Our overall estimated prevalence shows consistency with original international data. This is the first prevalence estimation of hypophosphatemic rickets in Colombia and will be relevant to support public health decisions for rare diseases and to provide a pre-test probability framework in clinical practice. DISMOD II and the model are useful tools to estimate the prevalence of rare and orphan diseases, when probabilistic studies cannot be carried out. There are limited bibliographic resources worldwide reporting prevalence values supported by original studies. Our study can be used as a cost-effective methodology reference in this regard, especially for Latin America.

Funding: Ultragenyx Pharmaceutical, as a donation. © 2021 The Authors

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Archive Scopus

Language English

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Publisher Elsevier Ltd

Publication The Lancet Regional Health - Americas

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Journal Abbr Lancet Reg. Health - Am.

Date Added 20/09/2026, 9:52:17 pm

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Tags:

prevalence, Rare diseases, Models, Statistical, Colombia, Osteomalacia, Rickets, Hypophosphatemic

Notes:

EID: 2-s2.0-85126961373

Export Date: 20 September 2026; Cited By: 3; Correspondence Address: J.A. Huertas-Quintero; Calle 44 # 59-75, 111321, Departamento de Medicina Interna, Hospital Universitario Nacional de Colombia, Bogotá, Colombia; email: jahuertasq@unal.edu.co

Hypoxia within tumor microenvironment characterizes distinct genomic patterns and aids molecular subtyping for guiding individualized immunotherapy

Item Type Journal Article

Author R. Shi

Author J. Sun

Author H. Zhou

Author T. Hu

Author Z. Gao

Author X. Wang

Author M. Li

Author Z. Zhou

Author Y. Shu

Abstract Assessing the hypoxic status within the tumor microenvironment (TME) is crucial for its significant clinical relevance in evaluating drug resistance and tailoring individualized strategies. In this study, we proposed a robust pan-cancer hypoxic quantification method utilizing multiple public databases, diverse bioinformatics, and statistical methods. All tumor samples were classified into four subtypes: non-

hypoxic/TMEhigh (C1), hypoxic/TMEhigh (C2), non-hypoxic/TMElow (C3), and hypoxic/TMElow (C4). We systematically analyzed multi-omics data and single-cell RNA-sequencing (scRNA-seq) data to reveal distinct immune landscape patterns and genomic characteristics among the four subtypes across pan-cancer. Furthermore, we employed multiple machine learning approaches to construct a hypoxic-TME model to enhance the predictive accuracy of immunotherapy response. Additionally, drug repositioning was implemented for cancer patients predicted as non-responders to immunotherapy. A pan-cancer analysis identified PDK1 as a hub gene linking tumor hypoxia, glycolysis, and immunotherapy resistance. In vivo experimental validation further confirmed that targeting PDK1 could improve the response to immunotherapy. Overall, our study may offer valuable insights for integrating hypoxic-TME classification into tumor staging and providing personalized strategies for cancer patients. © The Author(s) 2024.

Date 2024

DOI 10.1186/s40537-024-00945-2

URL <https://www.scopus.com/pages/publications/85195588589?origin=resultslist>

Archive Scopus

Language English

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Publication Journal of Big Data

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Issue 1

Journal Abbr J. Big Data

Date Added 20/09/2026, 9:52:04 pm

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Tags:

Diseases, Tumors, 'omics', Cancer patients, Drug-resistance, Formal languages, Genes, Genome, Genomic patterns, Hypoxia, Immunotherapy, Molecular subtyping, Multi-omic, Multi-omics, Precision medicine, Subtypings, Tumor microenvironment, Tumor microenvironments

Notes:

EID: 2-s2.0-85195588589

Export Date: 20 September 2026; Cited By: 5; Funding details: Jiangsu Provincial Medical Innovation Center of Jiangsu Province; Science, Technology and Education, (CXZX202204); Natural Science Foundation of Jiangsu Province, (BK20220732); Jiangsu Funding Program for Excellent Postdoctoral Talent, (2022ZB427); Postdoctoral International Exchange Program, (YJ20220433); Correspondence Address: Y. Shu; Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China; email: shuyongqian@csc.org.cn

Identification and classification of trucks and trailers on the road network through deep learning

Item Type Conference Paper

Author L. Chen

Author P.-Y. Sun

Author Y. Jia

Author R.O. Sinnott

Abstract Understanding the flow of traffic on road networks is a primary focus of road transport authorities. A range of technologies has been applied to measure throughput and potential congestion on the roads. However, such technologies are currently limited in disambiguating the kinds of vehicles on the road network. Certain vehicles types are especially important to distinguish from other traffic, e.g. trucks and trailers. These are larger vehicles that can have a major impact on the roads and surrounding areas, e.g. due to pollution or the excess noise that they can give rise to. In this paper, we present an approach to automatically detect, classify and count the unique classes of trucks and trailers that are found on the road network of Victoria using approaches based on deep learning. In particular, we consider two leading object detection systems: You Only Look Once version 3 (YOLOv3) and Single Shot Multi-Box Detector (SSD). We compare the models and the performance achieved for truck and trailer classification, including dealing with more common and less common truck and trailer types. We measure the efficiency and accuracy of the approaches based on the mean average precision achieved. © 2019 Association for Computing Machinery.

Date 2019

DOI 10.1145/3365109.3368781

URL <https://www.scopus.com/pages/publications/85077340948?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: BDCAT - Proc. IEEE/ACM Int. Conf. Big Data Comput., Appl. Technol.

Publisher Association for Computing Machinery, Inc

ISBN 978-145037016-5 (ISBN)

Pages 117-126

Proceedings Title BDCAT - Proc. IEEE/ACM Int. Conf. Big Data Comput., Appl. Technol.

Conference Name BDCAT 2019 - Proceedings of the 6th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies

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Tags:

Automobiles, Big data, Deep learning, Roads and streets, Object recognition, Object detection, Object Detection, Deep Learning, Road transports, Telecommunication traffic, Traffic, Trucks, Excess noise, Image Augmentation, Light trailers, Object detection systems, Road network, Single shots, SSD, Trailers, Truck trailers, Truck transportation, YOLOv3

Notes:

EID: 2-s2.0-85077340948

Export Date: 20 September 2026; Cited By: 7; Funding details: LIEF, (LE170100200); Roads Cooperation of Victoria; University of Melbourne; Conference name: 6th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies, BDCAT 2019; Conference date: 2 December 2019 through 5 December 2019; Conference code: 155676

Identification and counting of mature apple fruit based on BP feed forward neural network

Item Type Conference Paper

Author S. Lal

Author S.K. Behera

Author P.K. Sethy

Author A.K. Rath

Abstract Classification of fruits is an onerous and tedious task because of countless number of fruits. The traditional approach for detection and classification of fruit and its maturity level is based on the naked eye observation by the experts, which is both time consuming and causes eye fatigue. Advance techniques in image processing and machine learning helps to automatic classify and count the fruits, accurately, quickly and non-destructively. A method to automatic detect and classify apple fruit maturity level, whether it is mature or immature based on its color features has been proposed. Images of the apple are resized and Image Processing Techniques are applied for the extraction of apple color components (R, G, B). Artificial Neural Network is used as a classifier to identify and count the mature and immature apples using color components. The proposed model has an accuracy of 98.1%, when all the three attributes are used as an input. © 2017 IEEE.

Date 2017

DOI 10.1109/SSPS.2017.8071621

URL <https://www.scopus.com/pages/publications/85039906777?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. IEEE Int. Conf. Sens., Signal Process. Secur., ICSSS

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-150904929-5 (ISBN)

Pages 361-368

Proceedings Title Proc. IEEE Int. Conf. Sens., Signal Process. Secur., ICSSS

Conference Name Proceedings of 2017 3rd IEEE International Conference on Sensing, Signal Processing and Security, ICSSS 2017

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Modified 20/09/2026, 9:52:55 pm

Tags:

Artificial intelligence, Classification (of information), Features extraction, Learning systems, Neural networks, Machine Learning, Extraction, Image processing, Signal processing, classification, Traditional approaches, Backpropagation, Fruits, RGB, Image processing technique, Apple fruits, Back Propagation Feed Forward Neural Network, Color component, Color features, Eye fatigue, features extraction, Maturity levels

Notes:

EID: 2-s2.0-85039906777

Export Date: 20 September 2026; Cited By: 31; Conference name: 3rd IEEE International Conference on Sensing, Signal Processing and Security, ICSSS 2017; Conference date: 4 May 2017 through 5 May 2017; Conference code: 131366

Identification and Estimation of Causal Effects Defined by Shift Interventions

Item Type Conference Paper

Author N. Sani

Author J.J.R. Lee

Author I. Shpitser

Editor Peters J.

Editor Sontag D.

Abstract Causal inference quantifies cause effect relationships by means of counterfactual responses had some variable been artificially set to a constant. A more refined notion of manipulation, where a variable is artificially set to a fixed function of its natural value is also of interest in particular domains. Examples include increases in financial aid, changes in drug dosing, and modifying length of stay in a hospital. We define counterfactual responses to manipulations of this type, which we call shift interventions. We show that in the presence of multiple variables being manipulated, two types of shift interventions are possible. Shift interventions on the treated (SITs) are defined with respect to natural values, and are connected to effects of treatment on the treated. Shift interventions as policies (SIPs) are defined recursively with respect to values of responses to prior shift interventions, and are connected to dynamic treatment regimes. We give sound and complete identification algorithms for both types of shift interventions, and derive efficient semi-parametric estimators for the mean response to a shift intervention in a special case motivated by a healthcare problem. Finally, we demonstrate the utility of our method by using an electronic health record dataset to estimate the effect of extending the length of stay in the intensive care unit (ICU) in a hospital by an extra day on patient ICU readmission probability. © 2020 Proceedings of Machine Learning Research. All rights reserved.

Date 2020

URL <https://www.scopus.com/pages/publications/85126843441?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Mach. Learn. Res.

Volume 124**Publisher** ML Research Press**ISBN** 26403498 (ISSN)**Pages** 949-958**Proceedings Title** Proc. Mach. Learn. Res.**Conference Name** Proceedings of Machine Learning Research**Date Added** 20/09/2026, 9:52:47 pm**Modified** 20/09/2026, 9:52:47 pm**Tags:**

Machine learning, Causal inferences, Counterfactuals, Dynamic treatments, Semiparametric, Cause-effect relationships, Financial aids, Identification algorithms, Intensive care units, Length of stay, Sound and complete, TO effect

Notes:

EID: 2-s2.0-85126843441

Export Date: 20 September 2026; Cited By: 7; Conference name: 36th Conference on Uncertainty in Artificial Intelligence, UAI 2020; Conference date: 3 August 2020 through 6 August 2020; Conference code: 189147

Identification and quantification of feathers, down, and hair of avian and mammalian origin using matrix-assisted laser desorption/ionization time-of-flight mass spectrometry

Item Type Journal Article**Author** K. Hollemeyer**Author** W. Altmeyer**Author** E. Heinzle

Abstract We describe a fast method for the identification of the origin of native and chemically processed feathers, down, and hair and for distinguishing closely related species using enzyme digestion and MALDI-TOF mass spectrometry. Additionally we present two methods for the quantification of different identified bird and mammalian samples, respectively, in binary species mixtures. Without any prior cleaning or isolation of single proteins, enzymatical digests of feathers, down, and hair are performed. Fragments generated are analyzed by MALDI-TOF mass spectrometry, and peak groups of different selectivity are established for every animal species. For the identification of individual animal species, only unique species-specific peaks are accepted while for species classification of tinted or bleached hair, the group of semispecific peaks (SEMPs) is used. Samples from native animal species show a higher coincidence of digest peak masses of SEMPs than far-related species, indicating a phylogenetical relationship of the investigated structure proteins. Quantification of mixed binary avian samples is performed by identification of preweighed single feathers and down, followed by calculation of the gravimetric mass fractions. The composition of mixed binary mammalian samples is calculated

from correlation of the quotient of the relative peak intensities or areas of these species and the quotient of corresponding gravimetric mass fractions. High accuracy is achieved by both quantification methods. The fast methods are well suited for industrial quality control for example.

Date 2002

DOI 10.1021/ac020347f

URL <https://www.scopus.com/pages/publications/1842865461?origin=resultslist>

Archive Scopus

Language English

Volume 74

Pages 5960-5968

Publication Analytical Chemistry

ISSN 00032700 (ISSN)

Issue 23

Journal Abbr Anal. Chem.

Date Added 20/09/2026, 9:53:04 pm

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Tags:

human, Humans, article, quantitative analysis, nonhuman, calculation, accuracy, methodology, chemistry, protein, Proteins, animal, Animals, Biodiversity, phylogeny, Phylogeny, mass spectrometry, Mass spectrometry, protein structure, Biological organs, Mammals, Species classification, matrix assisted laser desorption ionization time of flight mass spectrometry, Spectrometry, Mass, Matrix-Assisted Laser Desorption-Ionization, species difference, Species Specificity, Birds, Enzymes, bird, Desorption, enzyme degradation, feather, Feathers, Gravimetric analysis, gravimetry, hair, Hair, hair analysis, Industrial applications, mammal, Mammalian origin, molecular phylogeny, protein isolation

Notes:

EID: 2-s2.0-1842865461

Export Date: 20 September 2026; Cited By: 43; Correspondence Address: K. Hollemeyer; Biochemical Engineering Institute, Saarland University, D-66123 Saarbrücken, Im Stadtwald, Germany; email: k.Hollemeyer@mx.uni-saarland.de; CODEN: ANCHA

Identification and quantification of individual volatile organic compounds in a binary mixture by SAW multisensor array and pattern recognition analysis

Item Type Journal Article

Author M. Penza

Author G. Cassano

Author F. Tortorella

Abstract We have developed a surface acoustic wave (SAW) multisensor array with five acoustic sensing elements configured as two-port resonator 433.92 MHz oscillators and a reference SAW element to recognize different individual components and determine their concentrations in a binary mixture of volatile organic compounds (VOCs) such as methanol and acetone, in the ranges 15-130 and 50-250 ppm, respectively. The SAW sensors have been specifically coated by various sensing thin films such as arachidic acid, carbowax, behenic acid, triethanolamine or acrylated polysiloxane, operating at room temperature. By using the relative frequency change as the output signal of the SAW multisensor array with an artificial neural network (ANN), a recognition system has been realized for the identification and quantification of tested VOCs. The features of the SAW multisensor array exposed to a binary component organic mixture of methanol and acetone have been extracted from the output signals of five SAW sensors by pattern recognition (PARC) techniques, such as principal component analysis (PCA). An organic vapour pattern classifier has been implemented by using a multilayer neural network with a backpropagation learning algorithm. The normalized responses of a reduced set of SAW sensors or selected principal components scores have been used as inputs for a feed-forward multilayer perceptron (MLP), resulting in a 70% correct recognition rate with the normalized responses of the four SAW sensors and in an enhanced 80% correct recognition rate with the first two principal components of the original data consisting of the normalized responses of the four SAW sensors. The prediction of the individual vapour concentrations has been tackled with PCA for features extraction and by using the first two principal components scores as inputs to a feed-forward MLP consisting of a gating network, which decides which of three specific subnets should be used to determine the output concentration: the first subnet for methanol only, the second subnet for acetone only and the third subnet for methanol and acetone in the binary mixture. Good 0.941 and 0.932 correlation coefficients for the predicted versus real concentrations of methanol and acetone, respectively, as individual components in a binary mixture have been obtained. The experimental results demonstrated that the proposed binary organic vapour mixture classifier is effective in the identification of the tested VOCs of methanol and acetone. Also, the combination of PCA and ANN techniques provides a rapid and accurate quantification method for the individual components' concentration in the tested binary mixture of methanol and acetone.

Date 2002

DOI 10.1088/0957-0233/13/6/304

URL <https://www.scopus.com/pages/publications/0036294043?origin=resultslist>

Archive Scopus

Language English

Volume 13

Publisher Institute of Physics Publishing

Pages 846-858

Publication Measurement Science and Technology

ISSN 09570233 (ISSN)

Issue 6

Journal Abbr Meas. Sci. Technol.

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Tags:

Feature extraction, Sensor data fusion, Learning algorithms, Pattern recognition, sensor, Artificial neural networks, Principal component analysis, Multilayer neural networks, Acoustic sensing, volatile organic compound, Backpropagation, Volatile organic compounds, Acoustic arrays, Acoustic surface wave devices, binary mixture, Binary mixture, Binary mixtures, measurement method, Multisensor array, Resonators, SAW chemical sensors, Thin films

Notes:

EID: 2-s2.0-0036294043

Export Date: 20 September 2026; Cited By: 32; Correspondence Address: M. Penza; ENEA, CR Brindisi, Materials and New Technologies Unit, Brindisi 72100, Italy; email: michele.penza@brindisi.enea.it; CODEN: MSTCE

Identification and quantification of the traditional chinese medicinal plant *Gentiana macrophylla* using taqman real-time PCR

Item Type Journal Article

Author C.-Y. Xue

Author D.-Z. Li

Author Q.-Z. Wang

Abstract *Gentiana macrophylla* Pall. is a commonly used antirheumatic herb. There are four species of *Gentiana* recorded as herbal drugs in the Chinese Pharmacopoeia. The other species are often marketed as *G. macrophylla*, and thus the therapeutic effects of *G. macrophylla* are not achieved. A novel one-step methodology based on real-time polymerase chain reaction (PCR) technology has been developed for the identification of *G. macrophylla*. This relative quantification methodology does not require a known amount of standard, allowing the analysis of many more samples together. The utilization of real-time PCR does not require sample handling, preventing contamination and resulting in much faster and higher throughput results. © Georg Thieme Verlag KG Stuttgart.

Date 2008

DOI 10.1055/s-0028-1088329

URL <https://www.scopus.com/pages/publications/58049214846?origin=resultslist>

Archive Scopus

Language English

Volume 74

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Journal Abbr *Planta Med.*

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Tags:

article, reliability, quantitative analysis, nonhuman, accuracy, Classification, DNA extraction, DNA, medicinal plant, nucleotide sequence, Plants, Medicinal, real time polymerase chain reaction, plant DNA, high throughput screening, species difference, quantitative assay, Species identification, Drugs, Chinese Herbal, Sequence Alignment, Chinese herb, DNA, Ribosomal, gentian, Gentiana, gentiana macrophylla, Gentiana macrophylla, Gentianaceae, plant identification, Real-time PCR, Reverse Transcriptase Polymerase Chain Reaction, technique

Notes:

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Export Date: 20 September 2026; Cited By: 11; Correspondence Address: C.-Y. Xue; Kunming Institute of Botany, Chinese Academy of Sciences, Kunming 650204 Yunnan, China; email: chyxue@mail.kib.ac.cn; CODEN: PLMEA

Identification of attacks against wireless sensor networks based on behaviour analysis

Item Type Journal Article

Author V. Korzhuk

Author A. Groznykh

Author A. Menshikov

Author M. Strecker

Abstract The paper describes the problem of identification of network layer attacks on wireless sensor networks for subsequent development of intrusion detection systems. 15 types of behaviour (including 14 states under attack and the normal state) and identification features are briefly specified. A technique for attacks identifying based on a combination of Random Forest algorithm and probability classifier is proposed. A study of the dependence of the accuracy of the probability classifier and the average number of used features on a confidence level and a priori probability of the normal state, which allows reducing the feature space to one feature, is described. For most attacks, there is a relative constancy of the dependence values in the range from 0 to 0.7 ± 0.15 , after which a decrease occurs, corresponding to the quality of identification. Recommendations to the algorithm parameters are formed and its evaluation is carried out. © 2019, Innovative Information Science and Technology Research Group. All rights reserved.

Date 2019

DOI 10.22667/JOWUA.2019.06.30.001

URL <https://www.scopus.com/pages/publications/85074815835?origin=resultslist>

Archive Scopus

Language English

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Journal Abbr "Journal of Wireless Mobile Networks, Ubiquitous Computing, and Dependable Applications"
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Tags:

Information security, Behavior analysis, Attack identification, Wireless sensor network

Notes:

EID: 2-s2.0-85074815835

Export Date: 20 September 2026; Cited By: 26; Correspondence Address: V. Korzhuk; Saint-Petersburg, Esenina str. 6-1, 194354, Russian Federation; email: vika@cit.ifmo.ru

Identification of diatom taxonomy by a combination of region-based full convolutional network, online hard example mining, and shape priors of diatoms

Item Type Journal Article

Author J. Deng

Author W. Guo

Author Y. Zhao

Author J. Liu

Author R. Lai

Author G. Gu

Author Y. Zhang

Author Q. Li

Author C. Liu

Author J. Zhao

Abstract Diatom test is one of the commonly used diagnostic methods for drowning in forensic pathology, which provides supportive evidence for drowning. However, in forensic practice, it is time-consuming and laborious for forensic experts to classify and count diatoms, whereas artificial intelligence (AI) is superior to human experts in processing data and carrying out classification tasks. Some AI techniques have focused on searching diatoms and classifying diatoms. But, they either could not classify diatoms correctly or were time-consuming. Conventional detection deep network has been used to overcome these problems but failed to detect the occluded diatoms and the diatoms similar to the background heavily, which could lead to false positives or false negatives. In order to figure out the problems above, an improved

region-based full convolutional network (R-FCN) with online hard example mining and the shape prior of diatoms was proposed. The online hard example mining (OHEM) was coupled with the R-FCN to boost the capacity of detecting the occluded diatoms and the diatoms similar to the background heavily and the priors of the shape of the common diatoms were explored and introduced to the anchor generation strategy of the region proposal network in the R-FCN to locate the diatoms precisely. The results showed that the proposed approach significantly outperforms several state-of-the-art methods and could detect the diatom precisely without missing the occluded diatoms and the diatoms similar to the background heavily. From the study, we could conclude that (1) the proposed model can locate the position and identify the genera of common diatoms more accurately; (2) this method can reduce the false positives or false negatives in forensic practice; and (3) it is a time-saving method and can be introduced. © 2021, The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature.

Date 2021

DOI 10.1007/s00414-021-02664-2

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Archive Scopus

Language English

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Pages 2519-2530

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procedures, Neural Networks, Computer, classification, Classification, diatom, Diatom test, Diatoms, Drowning, Forensic science, Taxonomy identification

Notes:

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Identification of physical properties and volatile and non-volatile compounds for discrimination between different Emmental-type cheeses: a preliminary study

Item Type Journal Article

Author K.I. Hartmann

Author A. Dunkel

Author H. Hillmann

Author D. Hansen

Author P. Schieberle

Author T. Hofmann

Author J. Hinrichs

Abstract Differentiation between Emmental cheese produced in diverse regions is of major interest to verify product authenticity and avoid product imitation. The objective of this study was to identify parameters that allow a differentiation between Southern German Emmental cheese and Emmental cheese produced in other regions based on physical properties and volatile and non-volatile compounds. Emmental cheese samples were collected from seven German dairies located in Bavaria and Baden-Wuerttemberg (regional Emmental cheese). Additionally, generic Emmental cheeses produced in Northern Germany, the Netherlands and Poland (European Emmental cheese) were added to the data set as well as Swiss Emmental and Allgäu Emmental cheeses. A large variety of characteristics was previously analysed by application of thermo-rheological methods and targeted quantification methods (e.g. GC-MS and dynamic oscillatory shear measurement). Stepwise discriminant analysis was used to identify parameters with high discriminant power. These were brightness L*, loss modulus G'' at 20 °C, loss factor tan δmax, 3-methylbutyric acid, acetic acid, propionic acid, γ-Glu-Leu, γ-Glu-Val and the peptide Ile-Val-Pro-Asn. The data set resulting from the nine parameters was subjected to linear discriminant analysis followed by a cross validation step. Regional Emmental cheese was classified correctly at 95%, whereas 75% of European Emmental cheese and 100% of Swiss and Allgäu Emmental cheeses were correctly classified. The results are preliminary but may provide, upon further verification, a useful tool for cheese makers to identify

Emmental cheese produced in different regions based on a small number of characteristic parameters and prevent product imitation. © 2015, INRA and Springer-Verlag France.

Date 2015

DOI 10.1007/s13594-015-0255-0

URL <https://www.scopus.com/pages/publications/84941961570?origin=resultslist>

Archive Scopus

Language English

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ISSN 19585586 (ISSN)

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Journal Abbr Dairy Sci. Technol.

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Classification (of information), Multivariant analysis, Classification, Discriminant analysis, Quantification methods, Volatile compounds, Volatile organic compounds, Linear discriminant analysis, Emmental cheese, Multivariate statistical analysis, Non-volatile compounds, Physical properties, Propionic acid, Rheological method, Stepwise discriminant analysis

Notes:

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Export Date: 20 September 2026; Cited By: 9

Identifying depressive subtypes in a large cohort study: Results from the Netherlands Study of Depression and Anxiety (NESDA)

Item Type Journal Article

Author F. Lamers

Author P. De Jonge

Author W.A. Nolen

Author J.H. Smit

Author F.G. Zitman

Author A.T.F. Beekman

Author B.W.J.H. Penninx

Abstract Objective: The heterogeneity of depression in the current classification system remains a point of discussion in the psychiatric field, despite previous efforts to

subclassify depressive disorders. Data-driven techniques may help to come to a more empirically based classification. This study aimed to identify depressive subtypes within a large cohort of subjects with depression. Method: Baseline data from 818 persons with a DSM-IV diagnosis of current major depressive disorder or minor depression who participated in the Netherlands Study of Depression and Anxiety were used. Respondents were recruited in the community, in primary care, and in specialized mental health care from September 2004 through February 2007. Latent classes were derived from latent class analysis using 16 depressive symptoms from the Composite International Diagnostic Interview and the Inventory of Depressive Symptomatology. Classes were characterized using demographic, clinical psychiatric, psychosocial, and physical health descriptors. Results: Three classes were identified: a severe melancholic class (prevalence, 46.3%), a severe atypical class (prevalence, 24.6%), and a class of moderate severity (prevalence, 29.1%). Both severe classes were characterized by more neuroticism (melancholic OR = 1.05 [95% CI, 1.01-1.10]; atypical OR = 1.07 [95% CI, 1.03-1.12]), more disability (melancholic OR = 1.07 [95% CI, 1.05-1.09]; atypical OR = 1.06 [95% CI, 1.04-1.07]), and less extraversion (melancholic OR = 0.95 [95% CI, 0.92-0.99]; atypical OR = 0.95 [95% CI, 0.92-0.99]) than the moderate class. Comparing the melancholic class with the atypical class revealed that the melancholic class had more smokers (atypical OR = 0.57 [95% CI, 0.39-0.84]) and more childhood trauma (atypical OR = 0.86 [95% CI, 0.74-1.00]), whereas the atypical class had more women (atypical OR = 1.52 [95% CI, 0.99-2.32]), a higher body mass index (atypical OR = 1.13 [95% CI, 1.09-1.17]), and more metabolic syndrome (atypical OR = 2.17 [95% CI, 1.38-3.42]). Conclusions: Both depression severity (moderate vs severe) and the nature of depressive symptoms (melancholic vs atypical) were found to be important differentiators between subtypes. Higher endorsement rates of somatic symptoms and more metabolic syndrome in the atypical class suggest the involvement of a metabolic component. © Copyright 2010 Physicians Postgraduate Press, Inc.

Date 2010

DOI 10.4088/JCP.09m05398blu

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Language English

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adult, Adult, female, Female, human, Humans, male, Male, Middle Aged, Adolescent, aged, Aged, article, Risk Factors, anxiety, Stress, Psychological, major clinical study, Anxiety, Depression, disease severity, disease classification, body mass, major depression, Severity of Illness Index, smoking, Diagnostic and Statistical Manual of Mental Disorders, Life Change Events, Prevalence, Cohort Studies, Comorbidity, Depressive Disorder, Major, priority journal, Body Mass Index, diagnostic and statistical manual of mental disorders, Questionnaires, Netherlands, Adult Survivors of Child Abuse, mental health care, Metabolic Syndrome X, primary medical care, Sex Factors

Notes:

EID: 2-s2.0-78650507190

Export Date: 20 September 2026; Cited By: 250; Correspondence Address: F. Lamers; Department of Psychiatry, VU University Medical Center Amsterdam, 1081 HL, Amsterdam, AJ Ernststraat 887, Netherlands; email: f.lamers@vumc.nl; CODEN: JCLPD

Identifying spatiotemporal patterns and driving factors of nitrogen leaching in Henan Province by integrating a process-based model with machine learning

Item Type Journal Article

Author S. Chen

Author P. Feng

Author F. Chen

Author J. Ren

Author H. Liang

Author Y. Gao

Author K. Hu

Abstract Understanding the spatiotemporal patterns and driving factors of nitrogen (N) leaching is crucial to mitigating groundwater nitrate contamination. This study integrated a process-based model with machine learning to quantify the spatiotemporal distribution of N leaching and identify its dominant drivers in Henan Province, a major wheat-maize rotation region in China. Thirty-year (1992–2022) simulations revealed a south-to-north decreasing gradient in annual N leaching, from 215 kg N ha⁻¹ in southern areas to 85 kg N ha⁻¹ in the north. Mean N leaching was substantially higher during the maize season (36–123 kg N ha⁻¹) than during the wheat season (42–100 kg N ha⁻¹), with the maize season contributing about 70% of annual N leaching in parts of Henan. Interannual variability was mainly driven by precipitation, with significantly greater maize-season leaching occurring in wet years. Water and N management were the dominant drivers across all temporal scales. Soil factors exerted stronger influence than meteorological conditions during the wheat season, but weaker influences during the maize season. Nitrogen application, precipitation, irrigation, relative humidity and field capacity (FC) collectively explained over 70% of leaching variability. Nitrogen application exhibited a softplus-type relationship with N leaching, whereas water-related variables displayed sigmoidal response patterns. Field capacity regulated the seasonal redistribution of N leaching. Relatively high FC promoted N accumulation during dry years, resulting in

higher N leaching during wet years. These findings provide a scientific basis for developing targeted water and N management and soil amendment to mitigate N leaching risks. © 2026 The Authors.

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environmental factor, simulation, Learning systems, machine learning, Machine learning, China, spatiotemporal analysis, Grain (agricultural product), wheat, nitrogen, Nitrogen, Soils, hydrological modeling, maize, Groundwater, agricultural modeling, crop rotation, field capacity, Field capacity, Groundwater pollution, Henan, leaching, Leaching, N-leaching, Nitrogen leaching, North China Plain, Partial dependence plot, Partial dependence plots, Pattern factors, Regional simulation, soil amendment, Spatiotemporal patterns, WHCNS model, Wheat-maize rotation

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: National Key Research and Development Program of China, NKRDP, (2021YFD1700904); China Agricultural University, CAU, (1191-00109011); Correspondence Address: K. Hu; College of Land Science and Technology China Agricultural University, Beijing, 100193, China; email: hukel@cau.edu.cn; CODEN: AWMAD

Identifying Undiagnosed High-Risk Suicidality Cases by Matching Patients With a Similar Comorbidity Burden: Retrospective Observational Study

Item Type Journal Article

Author L. Bode

Author R. Xu

Author M. Garber

Author K.D. Mandl

Author A.J. McMurry

Abstract Background: Suicide is the second leading cause of death for children and adolescents aged 6 to 18 years. Pediatric suicidality is underreported, which poses significant challenges for effective intervention and prevention strategies. Identifying populations at risk of suicidality can provide critical benefits in terms of study cohort selection, prevalence estimation, and clinical resource allocation. Objective: This study sought to (1) measure the prevalence of mental health comorbidities in pediatric suicidality and (2) identify undiagnosed high-risk suicidality cases by matching them with patients with a similar mental health comorbidity burden. Methods: Electronic health record data from a large academic pediatric hospital in Boston, Massachusetts, were analyzed for patients aged 6 to 18 years presenting to the emergency department between June 1, 2016, and June 1, 2022. Suicidality cases were defined using International Classification of Diseases, 10th Revision (ICD-10) codes for 3 suicidality subtypes: suicidal ideation, self-harm, and suicide attempt. Comorbidities of suicidality were calculated as the conditional probability of ICD-10 code pairs. After multiple hypothesis corrections, statistically significant comorbidities and patient encounter demographics were input as covariates into a propensity score matching (PSM) model. The accuracy of the PSM model was validated against chart review by 2 independent subject matter experts. Results: In total, 2.9% (2638/90,980) of emergency department encounters met an ICD-10–based case definition of suicidality during the study period. The prevalence of suicidality by subtype was 2.5% (2275/90,980) for ideation, 1.1% (1030/90,980) for self-harm, and 0.2% (177/90,980) for suicide attempt. Suicidality prevalence was more common for the female sex (1825/43,929, 4.2%) than for the male sex (813/47,045, 1.7%). Comorbidities of suicidality were statistically significant for 55 frequently co-occurring ICD-10 codes. Nearly half of these comorbidities (26/55, 47.3%) were not present in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, and nearly a quarter (12/55, 21.8%) consisted of ICD-10 codes for accidental rather than intentional self-harm. Increased probability of suicidality was observed for patients with personality disorder (84/190, 44.2%), gender dysphoria (143/333, 42.9%), bipolar disorder (162/448, 36.2%), depression (1791/5426, 33%), and schizophrenia spectrum disorders (133/411, 32.4%). On the basis of gold-standard chart review, 53.4% of propensity-matched noncases were unrecognized cases of suicidality. Conclusions: PSM using comorbidity profiles is an effective approach for identifying suicidality cases that lack ICD-10 codes for suicidality. © Louisa Bode, Rena Xu, Matthew Garber, Kenneth D Mandl, Andrew J Mcmurry.

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Tags:

medical informatics, suicidal ideation, comorbidity, self-injurious behavior, Diagnostic and Statistical Manual of Mental Disorders, propensity score, pediatrics, DSM, electronic health records, mental disorders

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Centers for Disease Control and Prevention, USCDC; U.S. Department of Health and Human Services, HEW; National Center for Advancing Translational Sciences, NCATS; US government; National Institutes of Health, NIH, (U01TR002623); Correspondence Address: A.J. McMurry; Computational Health Informatics Program, Boston Children's Hospital, Boston, 300 Longwood Ave, 02115, United States; email: Andrew.McMurry@childrens.harvard.edu

Image analysis of coffee seeds submitted to the LERCAFE test

Item Type Journal Article

Author B.G. Ribeiro

Author C.P. Abraão

Author M.C. Nery

Author R.M. Nascimento

Author J.C. Rezende

Author C.M.T. Fialho

Abstract The aim of this experiment was to quantify the stained areas of coffee seeds submitted to the LERCAFE test using image analysis. The seeds used were of the Catuaí Vermelho IAC 99 and Paradise cultivars. The physiological quality of the lots was assessed using germination tests, moisture content and a germination speed index. The LERCAFE test was conducted using seeds without parchment immersed in a solution of 3% sodium hypochlorite for 3 hours. Color photographs (RGB) with a resolution of 5 MPx were taken of each seed. The seeds were visually evaluated, and the functions generated from the analyses of the stained and non-stained regions were quantified by the Matlab R2009b program. Classification models were developed based on the Fisher Linear Discriminant Function and the evaluation of the adequacy of the models confusion matrix between the visual references and the classification generated by the linear functions. The image analysis for the creation of Fisher's linear discriminant function in the development of classifiers for the coffee seeds submitted to the LERCAFE test is potentially efficient; however, it remains necessary to test other discriminant functions and quantification methodologies. © 2016, Eduem - Editora da Universidade Estadual de Maringá. All rights reserved.

Date 2016

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Short Title Análise de imagens de sementes de café submetidas ao Teste LERCAFÉ

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Pixels, Determinants functions, Sodium hypochlorite

Notes:

EID: 2-s2.0-84976345194

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: C.M.T. Fialho; Universidade Federal dos Vales do Jequitinhonha e Mucuri, Jardim São Paulo, Teófilo Otoni, Rua do Cruzeiro, 1, 39803-371, Brazil; email: cintiamtfialho@yahoo.com.br

Image based inspection of concrete cracks using UAV photography

Item Type Conference Paper

Author J.C. Avendaño

Author J. Leander

Author R. Karoumi

Editor Casas J.-R.

Editor Frangopol D.M.

Editor Turmo J.

Abstract Traditional inspections are typically performed manually. They require specialized equipment, which usually is expensive, time-consuming and can often be a source of risk. Unmanned Aerial Vehicles (UAV) provide an alternative to overcome the challenges of traditional manual inspections. This has been studied as well as the use of machine learning and image analysis algorithms for detection and quantification of damages, in particular cracks. This paper presents a framework for inspections that combines data acquisition, crack detection, and the quantification of essential parameters of concrete cracks. From this method, the width and length of cracks are determined and compared with control measurements to estimate the accuracy of the method. The results show that using the pre-trained network to detect cracks, only a few to no false negatives are obtained. From the images classified as containing cracks, the quantification methodology is performed obtaining measurements of crack width down to 0.13 mm. © 2022 the Author(s).

Date 2023

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Archive	Scopus
Language	English
Extra	Journal Abbreviation: Bridge Saf., Maint., Manag., Life-Cycle, Resil. Sustain. - Proc. Int. Conf. Br. Maint., Saf. Manag.
Publisher	CRC Press/Balkema
ISBN	978-103234531-4 (ISBN)
Pages	1973-1978
Proceedings Title	Bridge Saf., Maint., Manag., Life-Cycle, Resil. Sustain. - Proc. Int. Conf. Br. Maint., Saf. Manag.
Conference Name	Bridge Safety, Maintenance, Management, Life-Cycle, Resilience and Sustainability - Proceedings of the 11th International Conference on Bridge Maintenance, Safety and Management, IABMAS 2022
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Aerial vehicle, Antennas, Unmanned aerial vehicles (UAV), Machine-learning, Machine learning, Data acquisition, Damage detection, Concretes, Detection and quantifications, Crack detection, Image-based, Concrete cracks, Control measurements, Image analysis algorithms, Length of crack, Manual inspection, Specialized equipment

Notes:

EID: 2-s2.0-85170399875

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Image Processing Techniques for Diagnosing Rice Plant Disease: A Survey

Item Type	Conference Paper
Author	P.K. Sethy
Author	N.K. Barpanda
Author	A.K. Rath
Author	S.K. Behera
Editor	Singh V.
Editor	Asari V.K.
Editor	Li K.-C.
Abstract	Over the past decades, rice crops are crucially admitted as one of the powerful energy streams for the production of resources. Rice plant diseases are considered as a raising factor behind the agricultural, economic and communal loss in the upcoming development of the agricultural field. Since last 10 years diagnosis of plant disease in approach to image processing techniques have remained keen are

of interest among the researcher. A number of disease detection, identification and quantification methods have been developed and applied in a wide variety of crops. This paper reviews related research papers from the period between 2007 and 2018 with a focus on the development of state of the art. The related studies are compared based image segmentation, feature extraction, feature selection and classification. This paper also outlines the current achievements, limitations, and suggestions for future research associated with the diagnosis of rice plant diseases. © 2020 The Authors. Published by Elsevier B.V.

Date 2020

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Artificial intelligence, Feature extraction, Classification (of information), Agricultural robots, Image segmentation, feature extraction, Research papers, Diagnosis, classification, State of the art, Crops, Quantification methods, feature selection, Agricultural fields, Disease detection, Energy streams, Feature selection and classification, Image processing technique, image segmentation, rice diseases

Notes:

EID: 2-s2.0-85084434968

Export Date: 20 September 2026; Cited By: 155; Correspondence Address: P.K. Sethy; Department of Electronics, Sambalpur University, Burla, Jyoti Vihar, 768019, India; email: prabirsethy.05@gmail.com; Conference name: 2019 International Conference on Computational Intelligence and Data Science, ICCIDS 2019; Conference date: 6 September 2019 through 7 September 2019; Conference code: 159537

Image-Based Quantitation of Host Cell–Toxoplasma gondii Interplay Using HRMAN: A Host Response to Microbe Analysis Pipeline

Item Type Journal Article

Author D. Fisch

Author A. Yakimovich

Author B. Clough**Author** J. Mercer**Author** E.-M. Frickel

Abstract Research on *Toxoplasma gondii* and its interplay with the host is often performed using fluorescence microscopy-based imaging experiments combined with manual quantification of acquired images. We present here an accurate and unbiased quantification method for host–pathogen interactions. We describe how to plan experiments and prepare, stain and image infected specimens and analyze them with the program HRMAN (Host Response to Microbe Analysis). HRMAN is a high-content image analysis method based on KNIME Analytics Platform. Users of this guide will be able to perform infection studies in high-throughput volume and to a greater level of detail. Relying on cutting edge machine learning algorithms, HRMAN can be trained and tailored to many experimental settings and questions. © Springer Science+Business Media, LLC, part of Springer Nature 2020.

Date 2020**DOI** 10.1007/978-1-4939-9857-9_21**URL** <https://www.scopus.com/pages/publications/85075418242?origin=resultslist>**Archive** Scopus**Language** English**Volume** 2071**Publisher** Humana Press Inc.**Pages** 411-433**Publication** Methods in Molecular Biology**ISSN** 10643745 (ISSN)**Journal Abbr** Methods Mol. Biol.**Date Added** 20/09/2026, 9:52:31 pm**Modified** 20/09/2026, 9:52:31 pm**Tags:**

Artificial intelligence, artificial intelligence, machine learning, Machine learning, Artificial Intelligence, algorithm, Algorithms, Machine Learning, nonhuman, procedures, controlled study, immune response, image analysis, human tissue, fluorescence microscopy, pipeline, High-content image analysis, host cell, host pathogen interaction, Host-Pathogen Interactions, Host–pathogen interaction, HRMAN, KNIME Analytics platform, microorganism, Microscopy, Fluorescence, pathogenicity, stain, *Toxoplasma*, *Toxoplasma gondii*

Notes:

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Imaging Amyloid and Tau in the Retina: Current Research and Future Directions

Item Type Journal Article

Author M.Y. Tang

Author M.S. Blazes

Author C.S. Lee

Author F. Costello

Author S. Prasad

Abstract Background:The retina is a key focus in the search for biomarkers of Alzheimer's disease (AD) because of its accessibility and shared development with the brain. The pathological hallmarks of AD, amyloid beta (A β), and hyperphosphorylated tau (pTau) have been identified in the retina, although histopathologic findings have been mixed. Several imaging-based approaches have been developed to detect retinal AD pathology in vivo. Here, we review the research related to imaging AD-related pathology in the retina and implications for future biomarker research.Evidence Acquisition:Electronic searches of published literature were conducted using PubMed and Google Scholar.Results:Curcumin fluorescence and hyperspectral imaging are both promising methods for detecting retinal A β , although both require validation in larger cohorts. Challenges remain in distinguishing curcumin-labeled A β from background fluorescence and standardization of dosing and quantification methods. Hyperspectral imaging is limited by confounding signals from other retinal features and variability in reflectance spectra between individuals. To date, evidence of tau aggregation in the retina is limited to histopathologic studies. New avenues of research are on the horizon, including near-infrared fluorescence imaging, novel A β labeling techniques, and small molecule retinal tau tracers. Artificial intelligence (AI) approaches, including machine learning models and deep learning-based image analysis, are active areas of investigation.Conclusions:Although the histopathological evidence seems promising, methods for imaging retinal A β require further validation, and in vivo imaging of retinal tau remains elusive. AI approaches may hold the greatest promise for the discovery of a characteristic retinal imaging profile of AD. Elucidating the role of A β and pTau in the retina will provide key insights into the complex processes involved in aging and in neurodegenerative disease. © 2023 Lippincott Williams and Wilkins. All rights reserved.

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Article, artificial intelligence, human, Humans, machine learning, Artificial Intelligence, aging, quantitative analysis, nonhuman, deep learning, histopathology, clinical research, Alzheimer disease, Alzheimer Disease, diagnostic imaging, hyperspectral imaging, image analysis, nuclear magnetic resonance imaging, pathology, biological marker, Biomarkers, data base, standardization, mild cognitive impairment, degenerative disease, positron emission tomography, curcumin, Montreal cognitive assessment, in vivo study, amyloid beta protein, Amyloid beta-Peptides, autofluorescence, Curcumin, fluorescence imaging, near infrared fluorescence imaging, Neurodegenerative Diseases, protein aggregation, retina, Retina, scanning laser ophthalmoscopy, tau protein

Notes:

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Export Date: 20 September 2026; Cited By: 14; Funding details: National Institutes of Health, NIH, (R01AG060942, U19AG066567); Research to Prevent Blindness, RPB; Correspondence Address: C.S. Lee; Department of Ophthalmology, University of Washington, Seattle, Box 359608, 325 Ninth Avenue, 98104, United States; email: leecs2@uw.edu; CODEN: JNEOE

Imaging and 3D morphological analysis of collagen fibrils

Item Type Journal Article

Author H. Altendorf

Author E. Decenci re

Author D. Jeulin

Author P.D.S. Peixoto

Author A. Deniset-Besseau

Author E. Angelini

Author G. Mosser

Author M.-C. Schanne-Klein

Abstract The recent booming of multiphoton imaging of collagen fibrils by means of second harmonic generation microscopy generates the need for the development and automation of quantitative methods for image analysis. Standard approaches sequentially analyse two-dimensional (2D) slices to gain knowledge on the spatial arrangement and dimension of the fibrils, whereas the reconstructed three-dimensional (3D) image yields better information about these characteristics. In this work, a 3D analysis method is proposed for second harmonic generation images of collagen fibrils, based on a recently developed 3D fibre quantification method. This analysis uses operators from mathematical morphology. The fibril structure is scanned with a directional distance transform. Inertia moments of the directional distances yield the main fibre orientation, corresponding to the main inertia axis. The collaboration of directional distances and fibre orientation delivers a geometrical estimate of the fibre radius. The results include local maps as well as global distribution of orientation and radius of the fibrils over the 3D image. They also bring a segmentation of the image into foreground and background, as well as a

classification of the foreground pixels into the preferred orientations. This accurate determination of the spatial arrangement of the fibrils within a 3D data set will be most relevant in biomedical applications. It brings the possibility to monitor remodelling of collagen tissues upon a variety of injuries and to guide tissues engineering because biomimetic 3D organizations and density are requested for better integration of implants. © 2012 Royal Microscopical Society.

Date 2012

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Tags:

Nonlinear optics, Image segmentation, article, nonhuman, Medical applications, calculation, controlled study, Morphology, Image analysis, mathematical analysis, image processing, image analysis, Biomimetics, density, protein structure, structure analysis, Mathematical morphology, Fibers, protein localization, priority journal, Collagen, Histology, Tissue, 3D morphological analysis, Biopolymers, collagen fibril, Collagen fibrils, Directional distances, excitation, Fiber segmentation, Fiber systems, Fibre orientation, Fibre segmentation, Fibre System, Harmonic analysis, Harmonic generation, Mathematical operators, Multiphoton imaging, Second harmonic microscopy, Second-harmonic microscopy, Spatial arrangements, three dimensional imaging, Three dimensional imaging, Three-dimensional imaging, Tissue engineering

Notes:

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Export Date: 20 September 2026; Cited By: 42; Correspondence Address: H. Altendorf; Department of Image Processing, Fraunhofer Institute of Industrial Mathematics, D-67663 Kaiserslautern, Fraunhofer-Platz 1, Germany; email: Hellen.Altendorf@mines-paristech.fr; CODEN: JMICA

Imbalanced domain adaptation Net for multi-class Electrocardiography signals classification

Item Type Journal Article

Author R. An
Author Z. Shang
Author Y. Liu
Author X. Wu
Author J. Ren
Author Z. Zhao

Abstract Arrhythmia detection using Electrocardiogram (ECG) signals plays a crucial role in cardiovascular disease diagnosis. Deep learning has shown great promise in developing diagnostic models for ECG classification. However, maintaining the performance of these models remains challenging in cross-domain scenarios, where ECG data might be collected from different devices, patient populations, or environments. Although existing unsupervised domain adaptation (UDA) methods have been utilized to improve the generalization of the model, many consider the discrepancies in feature distributions across domains but overlook variations in label distributions, also known as label shift. To tackle this problem, an imbalanced domain adaptation network (IDANet) is proposed for multi-class ECG diagnosis, accounting for both changes in feature distributions and variations in label distributions. Specifically, to address label shift, a class-sensitive re-weighting method with label-distribution-aware re-margining regularization is utilized which effectively enables the extraction of class-invariant features and facilitates the establishment of differentiated classification boundaries. Meanwhile, semantic alignment is applied to reduce the category-wise feature distribution discrepancies. Experiments on four ECG datasets show a considerable improvement (up to ~ 16 % in accuracy and ~ 17 % in F1 score) over other state-of-the-art UDA methods. These results highlight the promising potential of the proposed approach for future medical diagnostic applications. © 2025 Elsevier Ltd

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Publication Biomedical Signal Processing and Control

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Article, human, Electrocardiography, feature extraction, controlled study, adaptation, Domain adaptation, Label distribution, Label shift, Feature distribution, Adaptation methods, Arrhythmia detection, ECG diagnosis, Electrocardiogram diagnose, Electrocardiogram signal, electrocardiography, Imbalanced domain adaptation, Signal classification

Notes:

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Export Date: 20 September 2026; Cited By: 5; Funding details: Kunshan Municipal Government;
Correspondence Address: Z. Zhao; Global Health Research Center, Duke Kunshan University, Jiangsu,
China; email: zhaozhibin@xjtu.edu.cn

ImbSAM: A Closer Look at Sharpness-Aware Minimization in Class-Imbalanced Recognition

Item Type Conference Paper

Author Y. Zhou

Author Y. Qu

Author X. Xu

Author H. Shen

Abstract Class imbalance is a common challenge in real-world recognition tasks, where the majority of classes have few samples, also known as tail classes. We address this challenge with the perspective of generalization and empirically find that the promising Sharpness-Aware Minimization (SAM) fails to address generalization issues under the class-imbalanced setting. Through investigating this specific type of task, we identify that its generalization bottleneck primarily lies in the severe overfitting for tail classes with limited training data. To overcome this bottleneck, we leverage class priors to restrict the generalization scope of the class-agnostic SAM and propose a class-aware smoothness optimization algorithm named Imbalanced-SAM (ImbSAM). With the guidance of class priors, our ImbSAM specifically improves generalization targeting tail classes. We also verify the efficacy of ImbSAM on two prototypical applications of class-imbalanced recognition: long-tailed classification and semi-supervised anomaly detection, where our ImbSAM demonstrates remarkable performance improvements for tail classes and anomaly. Our code implementation is available at <https://github.com/cool-xuan/Imbalanced-SAM>. © 2023 IEEE.

Date 2023

DOI 10.1109/ICCV51070.2023.01042

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Language English

Extra Journal Abbreviation: Proc IEEE Int Conf Comput Vision

Publisher Institute of Electrical and Electronics Engineers Inc.

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Pages 11311-11321

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Conference Name Proceedings of the IEEE International Conference on Computer Vision

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Tags:

Performance, Computer vision, Real-world, Generalisation, Anomaly detection, Optimization algorithms, Class imbalance, Semi-supervised, Overfitting, Limited training data, Minimisation

Notes:

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Export Date: 20 September 2026; Cited By: 31; Correspondence Address: X. Xu; University of Electronic Science and Technology of China, Center for Future Media, School of Computer Science and Engineering, China; email: xing.xu@uestc.edu.cn; Conference name: 2023 IEEE/CVF International Conference on Computer Vision, ICCV 2023; Conference date: 2 October 2023 through 6 October 2023; Conference code: 196783; CODEN: PICVE

Impact of image analysis methodology on diagnostic and surgical classification of patients with thoracic aortic aneurysms

Item Type Journal Article

Author D.D. Mendoza

Author M. Kochar

Author R.B. Devereux

Author C.T. Basson

Author J.K. Min

Author K. Holmes

Author H.C. Dietz

Author D.M. Milewicz

Author S.A. Lemaire

Author R.E. Pyeritz

Author J.E. Bavaria

Author C.L. Maslen

Author H. Song

Author B.L. Kroner

Author K.A. Eagle

Author J.W. Weinsaft

Abstract Background: For patients with thoracic aortic aneurysms (TAA), aortic size on imaging is widely used to guide clinical decision making. This study examined the impact of methodological variance on aortic quantification. Methods: We studied enrollees in the National Registry of Genetically Triggered Thoracic Aortic Aneurysms and Cardiovascular Conditions. Aortic size on computed tomography was quantified by 2 linear methods; cross-sectional dimensions in axial (AX) and double oblique (DO) plane. Calculated area was compared to planimetry. Established cutoffs (area/height >10 cm²/m, diameter < 5cm) for prophylactic TAA repair were used to compare surgical eligibility by each method. Results: Fifty subjects were studied. Aortic size differed between AX and DO at all locations ($p \leq 0.001$), with magnitude greatest at the sinotubular junction (4.8 ± 1.1 vs 4.0 ± 1.0 cm, $p < 0.001$). The difference between AX and DO correlated with aortic angular displacement ($r = 0.37$, $p < 0.01$), which was threefold larger at the sinotubular junction (37 ± 12 degrees)

than the ascending aorta (12 ± 5 degrees; $p < 0.001$). At all locations, aortic area calculated using DO yielded smaller differences with planimetry than AX ($p < 0.05$). DO and planimetry yielded equal prevalence (24%) of subjects eligible for prophylactic TAA repair based on area-height cutoff, whereas AX prevalence was higher (44%; $p = 0.006$). Using a linear cutoff, AX yielded over a twofold greater prevalence of surgically eligible subjects (56%) than did DO (24%; $p < 0.001$). Conclusions: Established linear methods for aortic measurement yield different results that impact surgical eligibility. DO yielded improved agreement with planimetry and differed with AX in proportion to aortic geometric obliquity. Findings support DO measurements for imaging evaluation of subjects with TAA. © 2011 The Society of Thoracic Surgeons.

Date 2011

DOI 10.1016/j.athoracsur.2011.03.130

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Archive Scopus

Language English

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Journal Abbr Ann. Thorac. Surg.

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adult, female, human, male, article, quantitative analysis, cross-sectional study, major clinical study, clinical decision making, image analysis, patient coding, computer assisted tomography, geometry, anatomical variation, priority journal, artery diameter, disease registry, thoracic aorta aneurysm

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Export Date: 20 September 2026; Cited By: 73; Correspondence Address: J.W. Weinsaft; Division of Cardiology, Department of Medicine, Weill Cornell Medical College, New York, NY 10021, 525 E 68th St, United States; email: jww2001@med.cornell.edu; CODEN: ATHSA

Impact of low class prevalence on the performance evaluation of neural network based classifiers: Experimental study in the context of computer-assisted medical diagnosis

Item Type Conference Paper

Author M.A. Mazurowski

Author P.A. Habas

Author J.M. Zurada

Author G.D. Tourassi

Abstract This paper presents an experimental study on the impact of low class prevalence on the neural network based classifier performance as measured using Receiver Operator Characteristic (ROC) analysis. Two methods of dealing with the problem are investigated: oversampling and undersampling in the context of varying the class prevalence and the size of training datasets with uncorrelated and correlated features. The results show that the class imbalance can significantly decrease the classifier performance especially in the case of small training datasets. Furthermore, the oversampling method is shown to be more effective than the undersampling method in compensating the class imbalance. Statistically significant differences, however, are observed only in the cases with large total number of samples and very low prevalence. ©2007 IEEE.

Date 2007

DOI 10.1109/IJCNN.2007.4371266

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Language English

Extra Journal Abbreviation: IEEE Int. Conf. Neural. Netw. Conf. Proc.

ISBN 10987576 (ISSN); 142441380X (ISBN); 978-142441380-5 (ISBN)

Pages 2005-2009

Proceedings Title IEEE Int. Conf. Neural. Netw. Conf. Proc.

Conference Name IEEE International Conference on Neural Networks - Conference Proceedings

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Tags:

Artificial intelligence, Classification (of information), Neural networks, Vegetation, Image classification, Computer networks, Diagnosis, Classifiers, Joint conference

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: M. A. Mazurowski; Department of Electrical and Computer Engineering, University of Louisville, Louisville, KY 40292, United States; email: m.mazurowski@ieee.org; Conference name: 2007 International Joint Conference on Neural Networks, IJCNN 2007; Conference date: 12 August 2007 through 17 August 2007; Conference code: 72674; CODEN: ICNNF

Impact of quantitatively determined native thoracic aortic tortuosity on endoleak development after thoracic endovascular aortic repair

Item Type Journal Article

Author T. Ueda

Author H. Takaoka

Author B. Raman

Author J. Rosenberg

Author G.D. Rubin

Abstract **OBJECTIVE.** The objective of our study was to assess whether there is an association between native thoracic aortic curvature and the development of endoleaks after thoracic endovascular aortic repair. **MATERIALS AND METHODS.** Quantitative analysis of the native aortic lumen was performed on preprocedural CT angiograms of 40 patients with thoracic aortic aneurysm treated by thoracic endovascular aortic repair. The curvature of the median centerline was measured. Tortuosity indexes were calculated on the basis of the sum of the curvature values within the diseased segment and in the proximal and distal fixation zones. The association between the tortuosity index and endoleak was analyzed. **RESULTS.** Compared with patients without endoleaks, the tortuosity index of the proximal fixation zone was higher in patients with type Ia endoleak (9.5 vs 1.5 cm⁻¹, $p < 0.01$); the tortuosity index of the distal fixation zone was higher in type Ib endoleak patients (6.6 vs 0.5 cm⁻¹, $p < 0.05$); and the tortuosity indexes of the proximal fixation zone and of the diseased segment were higher in type III endoleak patients (11.0 vs 1.5 cm⁻¹, $p < 0.01$; and 15.8 vs 7.2 cm⁻¹, $p < 0.01$, respectively). Patients with a type III endoleak had longer diseased segments and larger mean diameters of the aneurysm than patients without endoleaks (148.6 vs 87.1 mm, $p < 0.01$; and 75.4 vs 63.2 mm, $p < 0.05$, respectively). Logistic regression analysis revealed that the risk of a type I or type III endoleak increased as the tortuosity index increased, with a 90% risk of endoleak at a tortuosity index of 10 cm⁻¹ in the proximal fixation zone. **CONCLUSION.** Quantification of aortic tortuosity using CT angiograms may help to predict whether an endoleak will develop after thoracic endovascular aortic repair, and this quantification method may become an important tool for risk stratification before thoracic endovascular aortic repair. © American Roentgen Ray Society.

Date 2011

DOI 10.2214/AJR.11.6819

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Language English

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Pages W1140-W1146

Publication American Journal of Roentgenology

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Journal Abbr Am. J. Roentgenol.

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adult, Adult, female, Female, human, Humans, male, Male, Middle Aged, risk assessment, aged, Aged, Aged, 80 and over, article, quantitative analysis, controlled study, clinical article, cohort analysis, disease classification, Retrospective Studies, retrospective study, image analysis, Contrast Media, disease association, computed tomographic angiography, Tomography, X-Ray Computed, Angiography, Postoperative Complications, priority journal, three dimensional imaging, preoperative evaluation, artery diameter, thoracic aorta aneurysm, Statistics, Nonparametric, computer analysis, Logistic Models, aorta reconstruction, Aorta, Thoracic, Aortic Aneurysm, Thoracic, blood vessel parameters, Blood Vessel Prosthesis, Blood Vessel Prosthesis Implantation, computed tomography scanner, endoleak, Endoleak, endovascular surgery, iodinated contrast medium, nonionic contrast medium, Quantitative assessment, thoracic aorta tortuosity, Thoracic endovascular aortic repair

Notes:

EID: 2-s2.0-82455194907

Export Date: 20 September 2026; Cited By: 40; Correspondence Address: T. Ueda; Department of Radiology, Saint Luke's International Hospital, Chuo-ku, Tokyo 104-8560, 9-1 Akashi-cho, Japan; email: takueda@luke.or.jp; CODEN: AJROA

Impact of the preprocessing stage on the performance of offline automatic vehicle counting using YOLO

Item Type Journal Article

Author D. Valencia

Author E. Munoz Espana

Author M. Munoz Anasco

Abstract Vehicle counting systems detect, classify, and count vehicles with sensors or image processing, providing valuable information for road management. Image processing systems provide detailed information on vehicle flow with adequate lighting conditions and a higher computational cost compared to sensor systems. The image processing systems with higher accuracy require higher computational cost. This feature limits the number of application cases in cities with low technology level. This research analyzes urban vehicle counting using an automatic image processing system using YOLOv5 in the vehicle detection-classification stage and the SORT algorithm in the tracking stage. The study used videos recorded from a pedestrian bridge in Popayan, Colombia, for an exploratory study of the influence of preprocessing operations on the performance of a low-tech vehicle counting system. The study performed a comparative statistical analysis to determine the impact of different settings on system performance. An ANOVA analysis evaluates the incidence of frame cut and reshape on YOLO processing. The results indicate that a 30% cut of the image area prior to YOLO processing produces the lowest weighted average error. In addition, the frame reshape only increases the processing time. The study proposes improvements in the performance of an offline automatic vehicle counting system from the video preprocessing stage. © 2003-2012 IEEE.

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Publication IEEE Latin America Transactions
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Offline, Highway administration, Highway traffic control, Street traffic control, Performance, Traffic control, Footbridges, Vehicles detection, Vehicle detection, Image enhancement, Object tracking, Image processing, Images processing, Object Tracking, Vehicle counting, Off road vehicles, Automatic vehicles, Computational costs, Counting system, Image processing system

Notes:

EID: 2-s2.0-85203655511

Export Date: 20 September 2026; Cited By: 8

Impact of updated pediatric hypertension criteria on prevalence estimates of hypertension among Chinese children

Item Type Journal Article

Author B. Luo

Author Y. Lin

Author S. Gao

Author Y. Lu

Author Y. Zhao

Author J. Xie

Author X. Zhang

Author H. Zhu

Abstract This study aimed to evaluate the impact of updated pediatric hypertension (HTN) criteria (the 2017 AAP Guidelines) on prevalence estimates of HTN and prehypertension among Chinese children compared to the 2004 Fourth Report. A total of 2093 children aged 7–15 years from five schools in Tianjin, China were selected using a multistage random cluster sampling method. The prevalence of HTN per the 2017 AAP Guidelines (10.1%) was significantly higher than that per the

Fourth Report (6.6%), whereas the prevalence of prehypertension per the 2017 AAP Guidelines (6.3%) was significantly lower than that per the Fourth Report (8.8%). From the 2004 Fourth Report to the 2017 AAP Guidelines, a total of 117 (5.6%) children were reclassified to have higher blood pressure. The two criteria had better consistency in the diagnosis of systolic abnormalities than in the diagnosis of diastolic abnormalities. The updated definitions for pediatric HTN have a substantive impact on the prevalence estimation among Chinese children, especially among boys, overweight children, and older children. © 2020, The Author(s), under exclusive licence to Springer Nature Limited.

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Notes:

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Impact of Urban Expansion and in Situ Greenery on Community-Wide Carbon Emissions: Method Development and Insights from 11 US Cities

Item Type Journal Article

Author M. Milnar

Author A. Ramaswami

Abstract Biogenic CO₂ emissions in cities are shaped by urban land cover change which can release carbon stocks, and, carbon sequestration by in situ vegetation. To date, these two processes have not been studied together and compared with transboundary fossil fuel-based CO₂ emissions of urban energy use. We leverage remote sensing and machine learning to quantify biogenic CO₂ emissions between 2006 and 2012, across 11 U.S cities, including central and suburban cities, in different climate zones. Results indicate that in situ carbon sequestration by greenery varied moderately across cities (-2.1 to -0.87 Mg CO₂ ha⁻¹ yr⁻¹), while emissions from the carbon stock change due to land conversion varied much more (-3.4 to 9.8 Mg CO₂ ha⁻¹ yr⁻¹), indicating that the latter dominates biogenic CO₂ emissions. Net biogenic CO₂ emissions were negative (carbon sink) in four cities, while large net positive emissions were present in rapidly growing suburbs. As a ratio of community-wide energy use for travel and buildings, biogenic CO₂ emissions were a small proportion in the core cities Denver (0.17%) and Minneapolis (0.33%) and as high as 38.2% in growing exurban communities. These results show that land cover change and greenery will be important policy levers in zero-carbon city planning. © 2020 American Chemical Society.

Date 2020

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land use, Article, human, United States, carbon emission, Remote sensing, Cities, city, controlled study, urban area, Carbon emissions, urban development, greenspace, city planning, carbon, Carbon, climate, Climate, carbon dioxide, Carbon dioxide, energy, carbon footprint, community care, carbon sequestration, Carbon sequestration, fossil fuel, Fossil fuels, Carbon Dioxide, Fossil Fuels, Land conversion, Land-cover change, Magnesium, Matthiola, Method development, Positive emissions, travel, Urban expansion, Urban land cover

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Export Date: 20 September 2026; Cited By: 35; Funding details: National Science Foundation, NSF, (SRN-1444745); National Science Foundation, NSF; Correspondence Address: A. Ramaswami; Department of Civil and Environmental Engineering, Princeton University, Princeton, 08544, United States; email: anu.ramaswami@princeton.edu; CODEN: ESTHA

Impacts of climate, cryosphere and vegetation changes on blue and green water in the Qinghai-Tibetan Plateau

Item Type Journal Article**Author** W. Kang**Author** G. Wang**Author** K. Li**Author** X. Sun**Author** B. Yong**Author** J. Sun**Author** L. Guo**Author** P. Huang**Author** J. Ma**Author** Z. Deng**Author** H. Wang**Author** S. Sun

Abstract The Qinghai-Tibetan Plateau is experiencing pronounced warming and wetting, which are reshaping regional hydrological regimes. However, the systemic impacts on blue and green water resources remain insufficiently understood. This study integrates hydrological modeling and machine learning to quantify how warming and wetting, vegetation greening, and cryosphere degradation drive changes in blue and green water in the source region of the Yangtze River. Results show that blue and green water resources increased by 37% and 10%, respectively, in the basin from 1982 to 1998 to 1999-2022. Precipitation increase dominates the blue water increase, contributing 85.2%; the green water increase is jointly driven by precipitation increase and temperature rise, with contributions of 57.2% and 30.8%, respectively. Vegetation greening decreases blue water and increases green water by enhancing evapotranspiration. Cryosphere degradation, including snowmelt increase and thickening of the active layer, contributes positively to both blue and green water. Notably, despite upward trends in blue and green water, the green water coefficient declines, indicating a shift in the regional water resource structure. This transition may enhance basin water conservation capacity and temporarily increase downstream water supply, yet it could raise downstream flood risk and promote swamping of alpine meadows over longer horizons. The findings clarify mechanisms governing water cycle evolution on the Tibetan Plateau under concurrent environmental changes and provide a scientific basis for regional water resource management. © 2026 Elsevier Ltd

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Tags:

land use, Article, machine learning, China, climate change, Climate change, evapotranspiration, precipitation, Vegetation, water management, Water management, water supply, controlled study, Floods, random forest, Climate Change, water conservation, Water Supply, vegetation, Rivers, water, climate, Climate, climate warming, topography, vegetation dynamics, hydrological modeling, Qinghai-Xizang Plateau, Tibet, Water conservation, altitude, Blue and green water, Blue water, cold, cryosphere, Cryosphere, Green water, hydrological cycle, Hydrological models, Hydrological regime, permafrost, Qinghai-Tibetan plateau, Qinghai-Tibetan Plateau, Regional water resources, river, snowmelt, snowmelt runoff, Soil and Water Assessment Tool model, Vegetation change, Vegetation greening, Water, water resource, Waters resources

Notes:

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Implementation of automatic crack evaluation using Crack Fundamental Element

Item Type Conference Paper
Author Y.J. Tsai
Author C. Jiang
Author Z. Wang

Abstract Crack evaluation is a vital component in the pavement condition survey. However, researchers have had difficulty automatically mimicking the in-field evaluation practice to achieve a fully automated pavement crack evaluation due to the diversity of real-world crack patterns and the complexity of crack definitions in an agency's survey protocol. The multi-scale crack analysis model based on a Crack Fundamental Element is developed to topologically represent complex crack patterns and provide systematic crack properties to support crack classification. Utilizing this model, this study implements an automatic crack classification and quantification method for use on the state highways in Georgia, United States. Both image-based validation and field validation are conducted with the pavement experts from the Georgia Department of Transportation (GDOT) on State Route 236 in Atlanta, and the proposed method achieves a classification accuracy of 92.2% for load cracking and 98.1% for B/T cracking. © 2014 IEEE.

Date 2014

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Language English

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Transportation routes, Surveys, Image processing, Topology, Classification accuracy, Quantification methods, Cracks, Pavements, Crack evaluation, Crack classification, Crack Classification, Crack Fundamental Element, Field validation, Georgia department of transportations, Pavement condition surveys, Survey protocols

Notes:

EID: 2-s2.0-84983109518

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Implementing Real-Time Retail Shelf Monitoring via YOLOv8 on a ROS-Based Mobile Robot

Item Type Conference Paper

Author D.J. Hoong

Author Z.H. Azizul

Author J.H. Chia

Abstract Maintaining optimal shelf inventory and product availability is critical in the contemporary retail landscape to maximise sales and customer satisfaction. The demand for service robots is increasing, but few existing works have explored retail product detection, classification, and counting on a robotic platform. This paper presents an automated, real-time system for monitoring retail products. The primary goal is to detect, classify, and count retail products on shelves to streamline inventory management and minimise stockouts. This work utilised the YOLOv8 object detection model, selected for its superior balance of speed and accuracy, and trained it on the Freiburg Groceries Dataset. The system was implemented on a robot based on ROS. The model achieved a mean Average Precision (mAP50) of 88.3%, and completed detection within 3 seconds, showcasing a robust and scalable solution for automated retail product management. © 2025 IEEE.

Date 2025

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Conference Name 2025 5th International Conference on Robotics, Automation, and Artificial Intelligence, RAAI 2025

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Real- time, Deep learning, Real time systems, Sales, Object recognition, Object detection, Objects detection, Object Detection, Intelligent robots, Customer satisfaction, Deep Learning, Information management, Robot Operating System, Inventory control, Interactive computer systems, Mobile robots, Product monitoring, YOLOv8, Service robots, Customers' satisfaction, Product availability, Retail product monitoring, Retail Product Monitoring, Robot operating system, Robot Operating System (ROS)

Notes:

EID: 2-s2.0-105035993428

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Importance Weight Estimation and Generalization in Domain Adaptation under Label Shift

Item Type Journal Article

Author K. Azizzadenesheli

Abstract We study generalization under labeled shift for categorical and general normed label spaces. We propose a series of methods to estimate the importance weights from labeled source to unlabeled target domain and provide confidence bounds for these estimators. We deploy these estimators and provide generalization bounds in the unlabeled target domain. © 1979-2012 IEEE.

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DOI 10.1109/TPAMI.2021.3086060

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Tags:

Artificial intelligence, Task analysis, Learning systems, machine learning, Machine-learning, Statistics, algorithm, Algorithms, Generalisation, Medical imaging, Predictive models, Diagnosis, Diseases, Domain adaptation, Target domain, Label space, Importance weights, Weights estimation, Medical diagnostic imaging, Hilbert spaces

Notes:

EID: 2-s2.0-85107387618

Export Date: 20 September 2026; Cited By: 14; Correspondence Address: K. Azizzadenesheli; Purdue University, The Computer Science Department, West Lafayette, 47907, United States; email: kaazizzad@gmail.com; CODEN: ITPID

Improved approach to robust speech recognition using minimum error classification

Item Type Journal Article

Author M.-T. Lin

Author A. Spanias

Author P. Loizou

Abstract An effective way of applying minimum error classification (MEC) to improve robustness in speech recognition is presented in this paper. In contrast to the traditional maximum likelihood (ML) training procedure that attempts to maximize the a priori probability of generating the training data set, MEC training attempts to minimize a function of the recognition error on the given training data set. In the MEC training procedure, the N-best algorithm is used to maximize the separation between the correct and competing models over confusable training tokens. The main focus of this paper is to investigate the effectiveness of MEC training when combined with four existing speech recognition algorithms under noisy and telephone mismatched environments. These algorithms are the weighted projection measure (WPM), the minimax approach (MA), the cepstral mean subtraction (CMS) method and the stochastic matching algorithms (SMAs). Experiments were performed using the Texas Instruments isolated digits database and the E-set words from the OGI Spelled and Spoken Telephone Corpus. The average word error rate reduction due to MEC training was 22.5% for isolated digit recognition and 8% for E-set word recognition.

Date 2000

DOI 10.1016/S0167-6393(99)00027-8

URL <https://www.scopus.com/pages/publications/0033889906?origin=resultslist>

Archive Scopus

Language English

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Pages 27-36

Publication Speech Communication

ISSN 01676393 (ISSN)

Issue 1

Journal Abbr Speech Commun

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Tags:

Markov processes, Speech recognition, Algorithms, Mathematical models, Error analysis, Speech analysis, Robustness (control systems), Maximum likelihood estimation, Cepstral mean substraction (CMS) method, Minimax approach (MA), Minimum error classification (MEC), Weighted projection measure (WPM)

Notes:

EID: 2-s2.0-0033889906

Export Date: 20 September 2026; Cited By: 5; CODEN: SCOMD

Improved Balanced Classification with Theoretically Grounded Loss Functions

Item Type	Conference Paper
Author	C. Cortes
Author	M. Mohri
Author	Y. Zhong
Abstract	The balanced loss is a widely adopted objective for multi-class classification under class imbalance. By assigning equal importance to all classes, regardless of their frequency, it promotes fairness and ensures that minority classes are not overlooked. However, directly minimizing the balanced classification loss is typically intractable, which makes the design of effective surrogate losses a central question. This paper introduces and studies two advanced surrogate loss families: Generalized Logit-Adjusted (GLA) loss functions and Generalized Class-Aware weighted (GCA) losses. GLA losses generalize Logit-Adjusted losses, which shift logits based on class priors, to the broader general cross-entropy loss family. GCA loss functions extend the standard class-weighted losses, which scale losses inversely by class frequency, by incorporating class-dependent confidence margins and extending them to the general cross-entropy family. We present a comprehensive theoretical analysis of consistency for both loss families. We show that GLA losses are Bayes-consistent, but only H-consistent for complete (i.e., unbounded) hypothesis sets. Moreover, their H-consistency bounds depend inversely on the minimum class probability, scaling at least as $1/p_{\min}$. In contrast, GCA losses are H-consistent for any hypothesis set that is bounded or complete, with H-consistency bounds that scale more favorably as $1/\sqrt{p_{\min}}$, offering significantly stronger theoretical guarantees in imbalanced settings. We report the results of experiments demonstrating that, empirically, both the GCA losses with calibrated class-dependent confidence margins and GLA losses can greatly outperform straightforward class-weighted losses as well as the LA losses. GLA generally performs slightly better in common benchmarks, whereas GCA exhibits a slight edge in highly imbalanced settings. Thus, we advocate for both GLA and GCA losses as principled, theoretically sound, and state-of-the-art surrogates for balanced classification under class imbalance. © 2025 Neural information processing systems foundation. All rights reserved.
Date	2025
URL	https://www.scopus.com/pages/publications/105048589029?origin=resultslist
Archive	Scopus
Language	English

Extra Journal Abbreviation: Adv. neural inf. proces. syst.
Volume 38
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ISBN 10495258 (ISSN); 979-833133827-5 (ISBN)
Pages 86424-86465
Proceedings Title Adv. neural inf. proces. syst.
Conference Name Advances in Neural Information Processing Systems
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EID: 2-s2.0-105048589029

Export Date: 20 September 2026; Cited By: 0; Conference name: 39th Conference on Neural Information Processing Systems, NeurIPS 2025; Conference date: 2 December 2025 through 7 December 2025; Conference code: 226431

Improved classification of Landsat Thematic Mapper data using modified prior probabilities in large and complex landscapes

Item Type Journal Article

Author L. Pedroni

Abstract The use of modified prior probabilities to exploit ancillary data and increase classification accuracy has been proposed before. However, this method has not been widely applied because it has heavy computing requirements and because obtaining prior probability estimates has presented practical problems. This article presents a procedure that generates large sets of prior probability estimates from class frequencies modelled with ancillary data and a Mahalanobis Distance selection of previously classified pixels. The method produces a pixel sample size that is large enough to estimate class frequencies in numerous strata, which is particularly desirable for the study of large and complex landscapes. A case study is presented in which the procedure made it possible to estimate 537 sets of prior probabilities for an entire Landsat Thematic Mapper (TM) scene of central Costa Rica. After modifying the class prior probabilities, the overall classification consistency of the training sites improved from 74.6% to 91.9%, while the overall classification accuracy of sites controlled in the field by independent studies improved from 68.7% to 89.0%. The classification accuracy was most improved for spectrally similar classes. The method improves classification accuracy in large and complex landscapes with spectrally mixed land-cover categories.

Date 2003

DOI 10.1080/01431160304998

URL <https://www.scopus.com/pages/publications/0037428335?origin=resultslist>

Archive Scopus

Language English

Volume 24

Publisher Taylor and Francis Ltd.

Pages 91-113**Publication** International Journal of Remote Sensing**ISSN** 01431161 (ISSN)**Issue** 1**Journal Abbr** Int. J. Remote Sens.**Date Added** 20/09/2026, 9:53:04 pm**Modified** 20/09/2026, 9:53:04 pm**Tags:**

satellite data, Remote sensing, data processing, Data processing, Probability, classification, Computational methods, Landsat thematic mapper, Thematic mapper (TM)

Notes:

EID: 2-s2.0-0037428335

Export Date: 20 September 2026; Cited By: 41; Funding details: INTERCOOPERATION; Swiss Organization for Development and Cooperation; Correspondence Address: L. Pedroni; Tropical Agr. Res./Higher Educ. Ctr., Turrialba 7170, Costa Rica; email: lpedroni@catie.ac.cr; CODEN: IJSED

Improved modulation classification using a factor-graph-based iterative receiver

Item Type Conference Paper**Author** D. Jakubisin**Author** R.M. Buehrer

Abstract We bring together two research topics which have been the focus of significant research individually: modulation classification and iterative receiver design. In this work, these topics are joined within the framework of factor graphs which provide a unified approach to representing a variety of algorithms, especially iterative algorithms. Specifically, in this paper we present a factor graph which incorporates modulation classification into the iterative receiver structure. The proposed iterative receiver applies message passing on the factor graph to approximate the optimal solution to joint modulation classification, demodulation, and decoding. This results in a classifier which treats feedback from the decoder as a priori probabilities for the coded bits. We show that the proposed receiver is able to achieve significant performance gains over a receiver which performs maximum likelihood classification separately from demodulation and decoding. © 2012 IEEE.

Date 2012**DOI** 10.1109/MILCOM.2012.6415729**URL** <https://www.scopus.com/pages/publications/84874314622?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc IEEE Mil Commun Conf MILCOM**Publisher** Institute of Electrical and Electronics Engineers Inc.

ISBN 21557578 (ISSN); 978-146731729-0 (ISBN)

Proceedings Title Proc IEEE Mil Commun Conf MILCOM

Conference Name Proceedings - IEEE Military Communications Conference MILCOM

Date Added 20/09/2026, 9:52:40 pm

Modified 20/09/2026, 9:52:40 pm

Tags:

Graph algorithms, Graph-based, Research topics, Maximum likelihood, Signal modulation, Demodulation, Factor graphs, Iterative algorithm, Iterative decoding, Iterative receiver designs, Iterative receivers, Message-passing, Modulation classification, Receiver structure, Unified approach

Notes:

EID: 2-s2.0-84874314622

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: D. Jakubisin; Mobile and Portable Radio Research Group (MPRG), Wireless VT, Virginia Tech, Blacksburg, VA, United States; email: jakubisin@vt.edu; Conference name: 2012 IEEE Military Communications Conference, MILCOM 2012; Conference date: 1 November 2012 through 1 November 2012; Conference code: 95770; CODEN: PMICE

Improved nearest centroid classifier with shrunken distance measure for null LDA method on cancer classification problem

Item Type Journal Article

Author A. Sharma

Author K.K. Paliwal

Abstract Null linear discriminant analysis (LDA) is a well-known dimensionality reduction technique for the small sample size problem. When the null LDA technique projects the samples to a lower dimensional space, the covariance matrices of individual classes become zero, i.e. all the projected vectors of a given class merge into a single vector. In this case, only the nearest centroid classifier (NCC) can be applied for classification. To improve the classification performance of NCC in the reduced-dimensional space, a shrunken distance based NCC technique is proposed that uses class-conditional a priori probabilities for distance computation. Experiments on several DNA microarray gene expression datasets using the proposed technique show very encouraging results for cancer classification. © 2010 The Institution of Engineering and Technology.

Date 2010

DOI 10.1049/el.2010.1927

URL <https://www.scopus.com/pages/publications/77956330302?origin=resultslist>

Archive Scopus

Language English

Volume 46

Pages 1251-1252

Publication Electronics Letters
ISSN 00135194 (ISSN)
Issue 18
Journal Abbr Electron. Lett.
Date Added 20/09/2026, 9:52:43 pm
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Tags:

Face recognition, Probability, Covariance matrices, Classifiers, Dimensionality reduction techniques, Classification performance, Vector spaces, Discriminant analysis, Covariance matrix, A-priori probabilities, Gene expression, Linear discriminant analysis, Cancer classification, Dimensional spaces, Distance computation, Distance measure, Distance-based, DNA micro-array, LDA technique, Nearest centroid classifiers, Reduced-dimensional, Single vectors, Small sample size problems

Notes:

EID: 2-s2.0-77956330302

Export Date: 20 September 2026; Cited By: 8; Correspondence Address: A. Sharma; Griffith University, Signal Processing Lab, Brisbane, QLD-4111, Australia; email: sharma_al@usp.ac.fj; CODEN: ELLEA

Improved quantification of amyloid burden and associated biomarker cut-off points: results from the first amyloid Singaporean cohort with overlapping cerebrovascular disease

Item Type Journal Article
Author T. Tanaka
Author M.C. Stephenson
Author Y.-H. Nai
Author D. Khor
Author F.N. Saridin
Author S. Hilal
Author S. Villaraza
Author B. Gyanwali
Author M. Ihara
Author H. Vrooman
Author A.A. Weekes
Author J.J. Totman
Author E.G. Robins
Author C.P. Chen
Author A. Reilhac

Abstract Purpose: The analysis of the [11C]PiB-PET amyloid images of a unique Asian cohort of 186 participants featuring overlapping vascular diseases raised the question about the validity of current standards for amyloid quantification under abnormal conditions. In this work, we implemented a novel pipeline for improved amyloid PET quantification of this atypical cohort. Methods: The investigated data correction and amyloid quantification methods included motion correction, standardized uptake value ratio (SUVR) quantification using the parcellated MRI (standard method) and SUVR quantification without MRI. We introduced a novel amyloid analysis method yielding 2 biomarkers: A β L which quantifies the global A β burden and ns that characterizes the non-specific uptake. Cut-off points were first determined using visual assessment as ground truth and then using unsupervised classification techniques. Results: Subject's motion impacts the accuracy of the measurement outcome but has however a limited effect on the visual rating and cut-off point determination. SUVR computation can be reliably performed for all the subjects without MRI parcellation while, when required, the parcellation failed or was of mediocre quality in 10% of the cases. The novel biomarker A β L showed an association increase of 29.5% with the cognitive tests and increased effect size between positive and negative scans compared with SUVR. ns was found sensitive to cerebral microbleeds, white matter hyperintensity, volume, and age. The cut-off points for SUVR using parcellated MRI, SUVR without parcellation, and A β L were 1.56, 1.39, and 25.5. Finally, k-means produced valid cut-off points without the requirement of visual assessment. Conclusion: The optimal processing for the amyloid quantification of this atypical cohort allows the quantification of all the subjects, producing SUVR values and two novel biomarkers: A β L, showing important increased in their association with various cognitive tests, and ns, a parameter sensitive to non-specific retention variations caused by age and cerebrovascular diseases. © 2019, Springer-Verlag GmbH Germany, part of Springer Nature.

Date 2020

DOI 10.1007/s00259-019-04642-8

URL <https://www.scopus.com/pages/publications/85076787099?origin=resultslist>

Archive Scopus

Language English

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Publication European Journal of Nuclear Medicine and Molecular Imaging

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Issue 2

Journal Abbr Eur. J. Nucl. Med. Mol. Imaging

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Tags:

Article, female, human, Humans, male, aged, reliability, quantitative analysis, age, cognition, major clinical study, Singapore, k means clustering, unsupervised machine learning, classification algorithm, diagnostic test accuracy study, outcome assessment, diagnostic accuracy, prospective study, cohort analysis, measurement accuracy, Alzheimer disease, Alzheimer Disease, Alzheimer’s disease, diagnostic imaging, sensitivity and specificity, nuclear magnetic resonance imaging, disease burden, neuroimaging, white matter lesion, disease association, biological marker, Biomarkers, positron emission tomography, Positron-Emission Tomography, Quantification, gaussian mixture model, Montreal cognitive assessment, Amyloid, Pittsburgh compound B, standardized uptake value ratio, cerebrovascular disease, Cerebrovascular disease, Mini Mental State Examination, amyloid, amyloidosis, amyloid beta protein, Amyloid beta-Peptides, hierarchical clustering, brain hemorrhage, Amyloid PET, Aniline Compounds, aniline derivative, brain size, Cerebrovascular Disorders, Clinical Dementia Rating, constants and coefficients, Cut-off point, effect size, head movement, non specific coefficient, reference value, Singaporean, verbal memory, visual memory

Notes:

EID: 2-s2.0-85076787099

Export Date: 20 September 2026; Cited By: 21; Funding details: National Medical Research Council, NMRC, (NMRC/CG/013/2013, NMRC/CG/NUHS/2010 - R-184-005-184-511, NMRC/ CIRG/1446/2016); National Medical Research Council, NMRC; Correspondence Address: T. Tanaka; Clinical Imaging Research Centre, National University of Singapore, Singapore, 14 Medical Drive, #B1-01, 117599, Singapore; email: tanakat@ncvc.go.jp; CODEN: EJNMA

Improved white blood cells classification based on pre-trained deep learning models

Item Type Journal Article

Author E.H. Mohamed

Author W.H. El-Behaidy

Author G. Khoriba

Author J. Li

Abstract Leukocytes, or white blood cells (WBCs), are microscopic organisms that fight against infectious disease, bacteria, viruses and others. The manual method to classify and count WBCs is tedious, time-consuming and may have inaccurate results, whereas the automated methods are costly. This research aims to automatically identify and classify WBCs in a microscopic image into four types with higher accuracy. BCCD is the used dataset in this study, which is a scaled-down blood cell detection dataset. BCCD is firstly preprocessed by passing through various processes such as segmentation and augmentation; then, it is passed to the proposed model. Our model combines the advantage of deep models in automatically extracting features with the higher classification accuracy of traditional machine learning classifiers. The proposed model consists of two main stages: a shallow tuning pre-trained model and

a traditional machine learning classifier on top of it. In this study, ten different pre-trained models with six types of machine learning are used. Moreover, the fully connected network (FCN) of pre-trained models is used as a baseline classifier for comparison. The evaluation process shows that the hybrid of MobileNet-224 as a feature extractor and logistic regression as classifier has a higher rank-1 accuracy of 97.03%. Besides, the proposed hybrid model outperformed the baseline FCN by 25.78% on average. © 2020 CCIS

Date 2020

DOI 10.24138/jcomss.v16i1.818

URL <https://www.scopus.com/pages/publications/85084797561?origin=resultslist>

Archive Scopus

Language English

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Pages 37-45

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ISSN 18456421 (ISSN)

Issue 1

Journal Abbr J.Comm. Softw. Sys.

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Tags:

Deep learning, Feature extraction, Classification, White blood cells (WBCs)

Notes:

EID: 2-s2.0-85084797561

Export Date: 20 September 2026; Cited By: 60; Correspondence Address: W.H. El-Behaidy; Computer Science Department, Faculty of Computers and Information, Helwan University, Cairo, Egypt; email: w_behaidy@fci.helwan.edu.eg

Improvement of Defects in Soil Moisture Monitoring of Wireless Sensor Network by Mobile Sensor Platform

Item Type Conference Paper

Author Z. Cheng

Author S. Yu

Author Z. Zhu

Author L. Zhang

Author Z. Zhang

Author Z. Leng

Author Y. Su

Abstract With the explosive development of the Internet of things technology, IOT technology is used in agricultural production. Wireless sensor network(WSN) is a universal form of agricultural Internet of Things. It realizes fixed-point acquisition of data and storage of data. However, when it is applied to soil moisture monitoring, its sensor is easily damaged, resulting in incomplete data collection. In the presence of these problems, this paper put forward a kind solution of mobile sensing platform combined with cloud platform. The sensor, smart car, GPRS module, and cloud platform constitute a vegetable greenhouse environmental information cloud platform system. The system can realize environmental information monitoring in vegetable greenhouses. The sensor and smart car are used to build up mobile sensing platform. The GPRS module is responsible for transmitting the soil data to the cloud service platform through network transmission. The cloud service platform can receive data and store, classify, and count data. © 2018 IEEE.

Date 2018

DOI 10.1109/ICICTA.2018.00043

URL <https://www.scopus.com/pages/publications/85057141559?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - Int. Conf. Intell. Comput. Technol. Autom., ICICTA

Publisher Institute of Electrical and Electronics Engineers Inc.

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Proceedings Title Proc. - Int. Conf. Intell. Comput. Technol. Autom., ICICTA

Conference Name Proceedings - 11th International Conference on Intelligent Computation Technology and Automation, ICICTA 2018

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Tags:

Digital storage, Classification (of information), Data acquisition, Internet of things, Data collection, Wireless sensor networks, Internet of things technologies, Soil moisture, Soil surveys, Data Collection, Agricultural productions, Vegetables, Environmental information, Distributed database systems, Greenhouses, Microwave measurement, Mobile Sensing Platform, Mobile sensing platforms, Mobile sensor platform, Moisture control, Network transmission, Soil moisture monitoring, Soil Moisture Monitoring

Notes:

EID: 2-s2.0-85057141559

Export Date: 20 September 2026; Cited By: 3; Correspondence Address: Y. Su; Anhui University, Hefei, 230601, China; email: ustcsyh@ahu.edu.cn; Conference name: 11th International Conference on Intelligent Computation Technology and Automation, ICICTA 2018; Conference date: 22 September 2018 through 23 September 2018; Conference code: 141725

Improving 18F-FDG PET Quantification Through a Spatial Normalization Method

Item Type Journal Article

Author D. Kim

Author S.K. Kang

Author S.A. Shin

Author H. Choi

Author J.S. Lee

Abstract Quantification of 18F-FDG PET images is useful for accurate diagnosis and evaluation of various brain diseases, including brain tumors, epilepsy, dementia, and Parkinson disease. However, accurate quantification of 18F-FDG PET images requires matched 3-dimensional T1 MRI scans of the same individuals to provide detailed information on brain anatomy. In this paper, we propose a transfer learning approach to adapt a pretrained deep neural network model from amyloid PET to spatially normalize 18F-FDG PET images without the need for 3-dimensional MRI. **Methods:** The proposed method is based on a deep learning model for automatic spatial normalization of 18F-FDG brain PET images, which was developed by fine-tuning a pretrained model for amyloid PET using only 103 18F-FDG PET and MR images. After training, the algorithm was tested on 65 internal and 78 external test sets. All T1 MR images with a 1-mm isotropic voxel size were processed with FreeSurfer software to provide cortical segmentation maps used to extract a ground-truth regional SUV ratio using cerebellar gray matter as a reference region. These values were compared with those from spatial normalization-based quantification methods using the proposed method and statistical parametric mapping software. **Results:** The proposed method showed superior spatial normalization compared with statistical parametric mapping, as evidenced by increased normalized mutual information and better size and shape matching in PET images. Quantitative evaluation revealed a consistently higher SUV ratio correlation and intraclass correlation coefficients for the proposed method across various brain regions in both internal and external datasets. The remarkably good correlation and intraclass correlation coefficient values of the proposed method for the external dataset are noteworthy, considering the dataset's different ethnic distribution and the use of different PET scanners and image reconstruction algorithms. **Conclusion:** This study successfully applied transfer learning to a deep neural network for 18F-FDG PET spatial normalization, demonstrating its resource efficiency and improved performance. This highlights the efficacy of transfer learning, which requires a smaller number of datasets than does the original network training, thus increasing the potential for broader use of deep learning-based brain PET spatial normalization techniques for various clinical and research radiotracers. COPYRIGHT © 2024 by the Society of Nuclear Medicine and Molecular Imaging.

Date 2024

DOI 10.2967/jnumed.123.267360

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Archive Scopus

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Volume 65

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Issue 10

Journal Abbr J. Nucl. Med.

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Tags:

Article, artificial intelligence, female, Female, human, Humans, male, Male, middle aged, Middle Aged, machine learning, aged, Aged, brain, Brain, Deep Learning, algorithm, image segmentation, procedures, training, deep learning, cognition, artificial neural network, methodology, cost effectiveness analysis, image processing, diagnostic imaging, image analysis, Image Processing, Computer-Assisted, Magnetic Resonance Imaging, nuclear magnetic resonance imaging, signal noise ratio, neuroimaging, functional magnetic resonance imaging, fluorodeoxyglucose f 18, positron emission tomography, Positron-Emission Tomography, brain cortex, dementia, quantification, epilepsy, Parkinson disease, gray matter, brain PET, brain tumor, Fluorodeoxyglucose F18, glucose metabolism, Mini Mental State Examination, single photon emission computed tomography, spatial normalization, spatial normalization method

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Improving classification under changes in class and within-class distributions

Item Type Conference Paper

Author R. Alaiz-Rodríguez

Author A. Guerrero-Curieses

Author J. Cid-Sueiro

Abstract The fundamental assumption that training and operational data come from the same probability distribution, which is the basis of most learning algorithms, is often not satisfied in practice. Several algorithms have been proposed to cope with classification problems where the class priors may change after training, but they can show a poor performance when the class conditional data densities also change. In this paper, we propose a re-estimation algorithm that makes use of unlabeled operational data to adapt the classifier behavior to changing scenarios.

We assume that (a) the classes may be decomposed in several (unknown) subclasses, and (b) the prior subclass probabilities may change after training. Experimental results with practical applications show an improvement over an adaptive method based on class priors, while preserving a similar performance when there are no subclass changes. © 2009 Springer Berlin Heidelberg.

Date 2009

DOI 10.1007/978-3-642-02478-8_16

URL <https://www.scopus.com/pages/publications/68749099982?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

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Proceedings Title Lect. Notes Comput. Sci.

Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

Issue PART 1

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Tags:

Learning systems, Learning algorithms, Neural networks, Probability distributions, Class distributions, Backpropagation, Poor performance, Estimation algorithm, Adaptive methods, Data density, Operational data

Notes:

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Export Date: 20 September 2026; Cited By: 9; Funding details: Ministerio de Economía y Competitividad, MINECO, (DPI2006-02550, TEC2008-01348/TEC); Correspondence Address: R. Alaiz-Rodríguez; Dpto. de Ingeniería Eléctrica y de Sistemas, Universidad de León, Campus de Vegazana, León 24071, Spain; email: rocio.alaiz@unileon.es; Conference name: 10th International Work-Conference on Artificial Neural Networks, IWANN 2009; Conference date: 10 June 2009 through 12 June 2009; Conference code: 77015

Improving CNN classifiers by estimating test-time priors

Item Type Conference Paper

Author M. Sulc

Author J. Matas

Abstract The problem of different training and test set class priors is addressed in the context of CNN classifiers. We compare two approaches to the estimation of the unknown test priors: an existing Maximum Likelihood Estimation (MLE) method

and a proposed Maximum a Posteriori (MAP) approach introducing a Dirichlet hyper-prior on the class prior probabilities. Experimental results show a significant improvement in the fine-grained classification tasks using known evaluation-time priors, increasing top-1 accuracy by 4.0% on the FGVC iNaturalist 2018 validation set and by 3.9% on the FGVCx Fungi 2018 validation set. Estimation of the unknown test set priors noticeably increases the accuracy on the PlantCLEF dataset, allowing a single CNN model to achieve state-of-the-art results and to outperform the competition-winning ensemble of 12 CNNs. The proposed MAP estimation increases the prediction accuracy by 2.8% on PlantCLEF 2017 and by 1.8% on FGVCx Fungi, where the MLE method decreases accuracy. © 2019 IEEE.

Date 2019

DOI 10.1109/ICCVW.2019.00402

URL <https://www.scopus.com/pages/publications/85082505931?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - Int. Conf. Comput. Vis. Workshop, ICCVW

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-172815023-9 (ISBN)

Pages 3220-3226

Proceedings Title Proc. - Int. Conf. Comput. Vis. Workshop, ICCVW

Conference Name Proceedings - 2019 International Conference on Computer Vision Workshop, ICCVW 2019

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Tags:

Classification (of information), Computer vision, Convolutional neural networks, Statistical tests, Classification tasks, State of the art, Fungi, Prior probability, Maximum likelihood estimation, Prediction accuracy, Probabilistic classifiers, Priors, Maximum a posteriori, Prior

Notes:

EID: 2-s2.0-85082505931

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Improving imbalance classification via ensemble learning based on two-stage learning

Item Type Journal Article

Author N. Liu

Author J. Wang

Author Y. Zhu

Author L. Wan

Author Q. Li

Abstract The excellent performance of deep neural networks on image classification tasks depends on a large-scale high-quality dataset. However, the datasets collected from the real world are typically biased in their distribution, which will lead to a sharp decline in model performance, mainly because an imbalanced distribution results in the prior shift and covariate shift. Recent studies have typically used a two-stage learning method consisting of two rebalancing strategies to solve these problems, but the combination of partial rebalancing strategies will damage the representational ability of the networks. In addition, the two-stage learning method is of little help in addressing the problem of covariate shift. To solve the above two issues, we first propose a sample logit-aware reweighting method called (SLA), which can not only repair the weights of majority class hard samples and minority class samples but will also integrate with logit adjustment to form a stable two-stage learning strategy. Second, to solve the covariate shift problem, inspired by ensemble learning, we propose a multi-domain expert specialization model, which can achieve a more comprehensive decision by averaging expert classification results from multiple different domains. Finally, we combine SLA and logit adjustment into a two-stage learning method and apply our model to the CIFAR-LT and ImageNet-LT datasets. Compared with the most advanced methods, our experimental results show excellent performance. Copyright © 2024 Liu, Wang, Zhu, Wan and Li.

Date 2023

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URL <https://www.scopus.com/pages/publications/85182456223?origin=resultslist>

Archive Scopus

Language English

Volume 17

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Improving Reliability of Seismic Stratigraphy Prediction: Integration of Uncertainty Quantification in Attention Mechanism Neural Network

Item Type Conference Paper

Author C.T. Ang

Author A.H. Elsheikh

Abstract Despite technological advancement, subsurface studies continue to encounter uncertainties caused by structural complexities and data noise, which can result in inaccurate seismic interpretation and drilling locations. Although machine learning holds great potential by enabling the simultaneous analysis of large datasets, its effectiveness is often compromised by data noise and ambiguity, which can degrade the accuracy of the algorithms. Hence, this research incorporates uncertainty quantification into attention mechanism neural network to produce more reliable outcomes in seismic interpretation and stratigraphic mapping. The methodology is also benchmarked against other uncertainty quantification methods such as dropout and randomized ensemble techniques, followed by an evaluation using the Brier score. Copyright 2024, Society of Petroleum Engineers.

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Language English

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Volume 2024-September

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Proceedings Title Proc SPE Annu Tech Conf Exhib

Conference Name Proceedings - SPE Annual Technical Conference and Exhibition

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Benchmarking, Neural-networks, Uncertainty, Technological advancement, Attention mechanisms, Uncertainty quantifications, Data noise, Gasoline, Seismic response, Seismic interpretation, Seismic stratigraphy, Stratigraphy, Structural data, Subsurface study

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Export Date: 20 September 2026; Cited By: 0; Funding details: PETRONAS; Heriot-Watt University, HWU; Conference name: 2024 SPE Annual Technical Conference and Exhibition, ATCE 2024; Conference date: 23 September 2024 through 25 September 2024; Conference code: 203353; CODEN: PSAEE

Improving Reliability of Seismic Stratigraphy Prediction: Integration of Uncertainty Quantification in Attention Mechanism Neural Network

Item Type Conference Paper

Author C.T. Ang

Author A.H. Elsheikh

Abstract Despite technological advancement, subsurface studies continue to encounter uncertainties caused by structural complexities and data noise, which can result in inaccurate seismic interpretation and drilling locations. Although machine learning holds great potential by enabling the simultaneous analysis of large datasets, its effectiveness is often compromised by data noise and ambiguity, which can degrade the accuracy of the algorithms. Hence, this research incorporates uncertainty quantification into attention mechanism neural network to produce more reliable outcomes in seismic interpretation and stratigraphic mapping. The methodology is also benchmarked against other uncertainty quantification methods such as dropout and randomized ensemble techniques, followed by an evaluation using the Brier score. Copyright 2024, Society of Petroleum Engineers.

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Language English

Extra Journal Abbreviation: SPE Annu. Tech. Conf. Proc.

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Proceedings Title SPE Annu. Tech. Conf. Proc.

Conference Name SPE Annual Technical Conference Proceedings

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Learning systems, Large datasets, Neural-networks, Neural networks, Uncertainty, Technological advancement, Attention mechanisms, Uncertainty quantifications, Data noise, Gasoline, Seismic interpretation, Seismic stratigraphy, Stratigraphy, Structural data, Subsurface study, Seismology, Petroleum engineering, Petroleum geology

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Improving System Utilization on Wireless HPC Systems with Torus Interconnects

Item Type Conference Paper**Author** K. Li**Author** Y. Qiu**Author** C. Jiang**Author** M. Malawski**Author** J. Nabrzyski

Abstract Recent advances in wireless technologies promote the concept of wireless HPC systems, in which the wired cable interconnect in a conventional HPC system is augmented with wireless links to configure the system for desired interconnection topology. In this paper, we study the problem of improving system utilization and job scheduling efficiency on HPC systems with torus interconnects. First, we design and analyze a novel wireless HPC system to improve system utilization, avoid inter-job communication interference, and achieve efficient job scheduling. Second, we present a topology-Aware job scheduling algorithm, which is applicable to both of our proposed wireless HPC system and a conventional 3D torus HPC system. Third, the simulation results validate the efficiency of our proposed wireless HPC system in improving system utilization. At last, we apply machine learning to quantify the impact of job arrival rate and the number of backfilling jobs on the system utilization of HPC systems with torus interconnects. © 2020 IEEE.

Date 2020**DOI** 10.1109/HPCC-SmartCity-DSS50907.2020.00009**URL** <https://www.scopus.com/pages/publications/85105255504?origin=resultslist>**Archive** Scopus**Language** English

Extra Journal Abbreviation: Proc. - IEEE Int. Conf. High Perform. Comput. Commun., IEEE Int. Conf. Smart City IEEE Int. Conf. Data Sci. Syst., HPCC-SmartCity-DSS

Publisher Institute of Electrical and Electronics Engineers Inc.**ISBN** 978-172817649-9 (ISBN)**Pages** 60-69

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Conference Name Proceedings - 2020 IEEE 22nd International Conference on High Performance Computing and Communications, IEEE 18th International Conference on Smart City and IEEE 6th International Conference on Data Science and Systems, HPCC-SmartCity-DSS 2020

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Tags:

Efficiency, Smart city, Data communication systems, Data Science, Scheduling, Topology, Communication interferences, Improving systems, Interconnection topologies, Job arrival rate, job scheduling, Job scheduling algorithms, system fragmentation, system utilization, System utilization, Topology aware, torus interconnect, Wireless HPC systems, Wireless technologies

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Improving the Output Quality of Official Statistics Based on Machine Learning Algorithms

Item Type Journal Article

Author Q.A. Meertens

Author C.G.H. Diks

Author H.J. Van Den Herik

Author F.W. Takes

Abstract National statistical institutes currently investigate how to improve the output quality of official statistics based on machine learning algorithms. A key issue is concept drift, that is, when the joint distribution of independent variables and a dependent (categorical) variable changes over time. Under concept drift, a statistical model requires regular updating to prevent it from becoming biased. However, updating a model asks for additional data, which are not always available. An alternative is to reduce the bias by means of bias correction methods. In the article, we focus on estimating the proportion (base rate) of a category of interest and we compare two popular bias correction methods: the misclassification estimator and the calibration estimator. For prior probability shift (a specific type of concept drift), we investigate the two methods analytically as well as numerically. Our analytical results are expressions for the bias and variance of both methods. As numerical result, we present a decision boundary for the relative performance of the two methods. Our results provide a better understanding of the effect of prior probability shift on output quality. Consequently, we may recommend a novel approach on how to use machine

learning algorithms in the context of official statistics. © 2022 Q.A. Meertens et al., published by Sciendo.

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prior probability shift, concept drift, misclassification bias, Output quality

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Export Date: 20 September 2026; Cited By: 5

Improving the performance of association classifiers by class prioritization

Item Type Journal Article

Author C. Chen

Author D. Chiang

Author C. Chen

Abstract Although different associative classification algorithms have been proposed, as pointed out by [8], none of the available associative algorithm considers the rule dependence problem that will directly influence the classification accuracy of the associative classifier. Since finding the optimal execution order of class association rules (CARs) is a combinational problem, in this paper, instead of finding the optimal execution order of CARs, we propose a polynomial time algorithm to re-rank the execution order of CARs by class priority to reduce the influence of the rule dependency problem. Consequently, the performance (the classification accuracy and recall rate) of the associative classification algorithm can be improved. The experimental results show that the association classifier with our method can get better classification result than that of the association classifier without considering the rule dependence problem. © 2011 by Binary Information Press.

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Algorithms, Optimization, Classification accuracy, Prioritization, Classification results, Polynomial approximation, Association rule, Association rules, Associative classification, Associative classification algorithms, Associative classifiers, Combinational problems, Polynomial-time algorithms, Ranking, Recall rate, Rule dependency

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: C. Chen; Department of Computer Science and Information Engineering, Tamkang University, Taipei, Taiwan; email: chchen@mail.tku.edu.tw

Improving the Performance of the Feature Drift Detector by Lasso Observation of Sample Feature Fluctuations

Item Type Journal Article

Author P. Porwik

Author T. Orczyk

Author N. Japkowicz

Abstract Feature drift is a subtype of data distribution drift that occurs when the statistical significance of input features changes over time, despite the overall decision boundary remaining stable. This phenomenon can cause a subtle degradation in model accuracy in streaming environments. In this paper, we propose a new model-aware method called feature importance-driven drift detection (FIDD). Rather than relying on classification error signals, FIDD tracks changes in feature importance rankings obtained from LASSO regression across neighbouring data fragments. As it observes the dynamics of feature importance instead of global label shifts, this method is particularly suited to detecting subtle shifts in data distribution. Experimental evaluation on both synthetic and real-world data (including different types of drift, such as abrupt, gradual, incremental and recurrent) shows that FIDD achieves higher accuracy consistently and produces significantly fewer false alarms than standard drift detectors (e.g., DDM, EDDM and ADWIN). Furthermore, FIDD

is robust to labelling noise and computationally efficient, which makes it a practical and interpretable solution for adaptive learning in real-time applications. © 2025 Piotr Porwik et al., published by University of Zielona Góra.

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Performance, Feature extraction, Classification (of information), Regression analysis, classification, Data accuracy, Data distribution, data stream, Data stream, drift detection, Drift detection, Drift detectors, feature drift, Feature drift, feature ranking, Feature ranking, Input features, Sample features, Statistical significance

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: P. Porwik; Institute of Computer Science, University of Silesia, Sosnowiec, ul. Będzińska 60, 41-00, Poland; email: piotr.porwik@us.edu.pl

Improving unsupervised domain adaptation with representative selection techniques

Item Type Conference Paper

Author I.-T. Chen

Author H.-T. Lin

Editor Kottke D.

Editor Kreml G.

Editor Lemaire V.

Editor Holzinger A.

Editor Calma A.

Abstract Domain adaptation is a technique that tackles the dataset shift scenario, where the training (source) data and the test (target) data can come from different distributions. Current research works mainly focus on either the covariate shift or the label shift settings, each making a different assumption on how the source and target data are related. Nevertheless, we observe that neither of the settings can perfectly match the needs of a real-world bio-chemistry application. We carefully study the difficulties encountered by those settings on the application and propose a novel method that takes both settings into account to improve the performance on the application. The key idea of our proposed method is to select examples from the source data that are similar to the target distribution of interest. We further explore two selection schemes, the hard-selection scheme that plugs similarity into a nearest-neighbor style approach, and the soft-selection scheme that enforces similarity by soft constraints. Experiments demonstrate that our proposed method not only achieves better accuracy for the bio-chemistry application but also shows promising performance on other domain adaptation tasks when the similarity can be concretely defined. © 2020 for this paper by its authors. Use permitted under CC BY 4.0.

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In Situ Sequencing: A High-Throughput, Multi-Targeted Gene Expression Profiling Technique for Cell Typing in Tissue Sections

Item Type Journal Article

Author M.M. Hilscher

Author D. Gyllborg

Author C. Yokota

Author M. Nilsson

Abstract Recent advances of image-based in situ mRNA quantification methods allow to visualize where in a tissue section a set of genes is expressed. It enables to map large numbers of genes in parallel and by capturing cellular boundaries allows to assign genes to cells. Here, we present a high-throughput, multi-targeted gene expression profiling technique called in situ sequencing that is capable of localizing hundreds of genes simultaneously and supports cell type classifications that follow transcriptome-based taxonomy. In situ sequencing is a targeted, amplified, and barcoded approach using padlock probes (PLPs) and rolling circle amplification (RCA). The current protocol relies on mRNA fixation, mRNA reverse transcription, residual mRNA degradation, and PLP hybridization. PLPs are amplified by RCA and labeled with fluorophore-conjugated probes, allowing their detection under conventional fluorescence microscopes. © 2020, Springer Science+Business Media, LLC, part of Springer Nature.

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DOI 10.1007/978-1-0716-0623-0_20

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human, Humans, procedures, taxonomy, gene expression, gene amplification, genetics, high throughput sequencing, isolation and purification, messenger RNA, RNA, Messenger, gene expression profiling, transcriptome, Transcriptome, tissue section, Single-cell, fluorescence in situ hybridization, Gene Expression, In Situ Hybridization, Fluorescence, In situ sequencing, microarray analysis, Microarray Analysis, molecular probe, nucleic acid amplification, Nucleic Acid Amplification Techniques, nucleic acid hybridization, Nucleic Acid Hybridization, Padlock probe, reverse transcription, RNA degradation, rolling circle amplification, Rolling circle amplification, Single-molecule, Spatial transcriptomics

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EID: 2-s2.0-85084604874

Export Date: 20 September 2026; Cited By: 13; Correspondence Address: M.M. Hilscher; Molecular Diagnostics, Science for Life Laboratory, Department of Biochemistry and Biophysics, Stockholm University, Solna, Sweden; email: markus.hilscher@scilifelab.se

In vitro cell migration quantification method for scratch assays

Item Type Journal Article**Author** A.V.P. Bobadilla**Author** J. Arévalo**Author** E. Sarró**Author** H.M. Byrne**Author** P.K. Maini**Author** T. Carraro**Author** S. Balocco**Author** A. Meseguer**Author** T. Alarcón

Abstract The scratch assay is an in vitro technique used to assess the contribution of molecular and cellular mechanisms to cell migration. The assay can also be used to evaluate therapeutic compounds before clinical use. Current quantification methods of scratch assays deal poorly with irregular cell-free areas and crooked leading edges which are features typically present in the experimental data. We introduce a new migration quantification method, called 'monolayer edge velocimetry', that permits analysis of low-quality experimental data and better statistical classification of migration rates than standard quantification methods. The new method relies on quantifying the horizontal component of the cell monolayer velocity across the leading edge. By performing a classification test on in silico data, we show that the method exhibits significantly lower statistical errors than standard methods. When applied to in vitro data, our method outperforms standard methods by detecting differences in the migration rates between different cell groups that the other methods could not detect. Application of this new method will enable quantification of migration rates from in vitro scratch assay data that cannot be analysed using existing methods. © 2019 The Author(s) Published by the Royal Society. All rights reserved.

Date 2019**DOI** 10.1098/rsif.2018.0709**URL** <https://www.scopus.com/pages/publications/85062771324?origin=resultslist>**Archive** Scopus**Language** English**Volume** 16**Publisher** Royal Society Publishing**Publication** Journal of the Royal Society Interface**ISSN** 17425689 (ISSN)**Issue** 151**Journal Abbr** J. R. Soc. Interface

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In-Depth Analysis of the Effective Factors in Green Supply Chain Management in the Offshore Industry

Item Type Journal Article

Author B.M. Aljahdali

Author Y. Alsubhi

Author H.T. Sulaimani

Author A.F. Alghanmi

Abstract The proposed study is based on the hybrid framework, which is a convergence between machine learning algorithms and fuzzy decision-making techniques to determine and rank the most important factors regarding the Green Supply Chain Management (GSCM) within the offshore industry under the influence of climate change. The research follows a three-phase methodology: (1) systematic literature review and expert consultation to establish the dimensions of relevancy in GSCM (2) hybrid fuzzy Delphi machine learning to quantify uncertainty and elicit expert opinion (3) an Analytic Network Process to establish interdependency and global priorities. The framework was applied to Saudi Arabia's Arabian Gulf offshore sector, where four primary GSCM dimensions and twelve operational indicators were validated with expert consensus levels between 0.82 and 0.93. Results show that climate change adaptation mechanisms represent the most influential dimension (global weight = 0.334), while Climate Risk Assessment Protocols rank as the top indicator (0.127). The hybrid model got an accuracy of 0.863 which was 34.4 percent higher than the traditional methods in predicting disruption and 44.3 percent in risk assessment accuracy. Three offshore validation indicated performance improvement

of 11 to 25. These results have shown that the process of combining machine learning and fuzzy logic makes GSCM decision-making much more effective, providing a pragmatic and climate-adaptive architecture to make offshore operations more sustainable and resilient. © 2026 University of Novi Sad. All rights reserved.

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Incomplete Multi-View Learning Under Label Shift

Item Type Journal Article

Author R. Fan

Author X. Ouyang

Author T. Luo

Author D. Hu

Author C. Hou

Abstract In image processing, images are usually composed of partial views due to the uncertainty of collection and how to efficiently process these images, which is called incomplete multi-view learning, has attracted widespread attention. The incompleteness and diversity of multi-view data enlarges the difficulty of annotation, resulting in the divergence of label distribution between the training and testing data, named as label shift. However, existing incomplete multi-view methods generally

assume that the label distribution is consistent and rarely consider the label shift scenario. To address this new but important challenge, we propose a novel framework termed as Incomplete Multi-view Learning under Label Shift (IMLLS). In this framework, we first give the formal definitions of IMLLS and the bidirectional complete representation which describes the intrinsic and common structure. Then, a multilayer perceptron which combines the reconstruction and classification loss is employed to learn the latent representation, whose existence, consistency and universality are proved with the theoretical satisfaction of label shift assumption. After that, to align the label distribution, the learned representation and trained source classifier are used to estimate the importance weight by designing a new estimation scheme which balances the error generated by finite samples in theory. Finally, the trained classifier reweighted by the estimated weight is fine-tuned to reduce the gap between the source and target representations. Extensive experimental results validate the effectiveness of our algorithm over existing state-of-the-arts methods in various aspects, together with its effectiveness in discriminating schizophrenic patients from healthy controls. © 1992-2012 IEEE.

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Increased chloroplast occupancy in bundle sheath cells of rice hap3H mutants revealed by Chloro-Count: a new deep learning-based tool

Item Type Journal Article

Author J. Lambret Frotte

Author P.P. Buarque de Gusmão

Author G. Smith

Author S.-F. Lo

Author S.-M. Yu

Author R.W. Hendron

Author S. Kelly

Author J.A. Langdale

Abstract There is an increasing demand to boost photosynthesis in rice to increase yield potential. Chloroplasts are the site of photosynthesis, and increasing their number and size is a potential route to elevate photosynthetic activity. Notably, bundle sheath cells do not make a significant contribution to overall carbon fixation in rice, and thus, various attempts are being made to increase chloroplast content specifically in this cell type. In this study, we developed and applied a deep learning tool, Chloro-Count, and used it to quantify chloroplast dimensions in bundle sheath cells of OsHAP3H gain- and loss-of-function mutants in rice. Loss of OsHAP3H increased chloroplast occupancy in bundle sheath cells by 50%. When grown in the field, mutants exhibited increased numbers of tillers and panicles. The implementation of Chloro-Count enabled precise quantification of chloroplasts in loss- and gain-of-function OsHAP3H mutants and facilitated a comparison between 2D and 3D quantification methods. Collectively, our observations revealed that a mechanism operates in bundle sheath cells to restrict chloroplast occupancy as cell dimensions increase. That mechanism is unperturbed in Oshap3H mutants but loss of OsHAP3H function leads to an increase in chloroplast numbers. The use of Chloro-Count also revealed that 2D quantification is compromised by the positioning of chloroplasts within the cell. © 2024 The Author(s). New Phytologist © 2024 New Phytologist Foundation.

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Inductive Conformal Prediction for Harvest-Readiness Classification of Cauliflower Plants: A Comparative Study of Uncertainty Quantification Methods

Item Type Conference Paper

Author M. Farag

Author J. Kierdorf

Author R. Roscher

Abstract Quantifying the uncertainty of machine learning models is a promising way to make better-informed decisions in digital agriculture. Efforts have been made to address this, ranging from understanding and segregating sources of uncertainty to utilizing diverse approaches for quantifying the cumulative amount. However, in order to fully realize the potential of uncertainty quantification in digital agriculture, more research is needed to compare and contrast different methods and determine which are most effective in different contexts. In this paper, we investigate inductive conformal prediction as another family of machine learning methods besides the commonly used softmax outputs and Monte Carlo dropout. Inductive conformal prediction constructs valid prediction sets by selecting a pre-defined level of predictive confidence in the system. In our experiments, we

analyze this method for an image-based harvest-readiness classification task of cauliflower plants, and compare the results to softmax outputs and uncertainties derived from Monte Carlo dropout. Inductive conformal prediction turns out as a complementary tool offering distinct advantages and providing another level of information for decision support. © 2023 IEEE.

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Decision support systems, Deep learning, Learning systems, Forecasting, Uncertainty analysis, Agriculture, Computer vision, Computer Vision, Uncertainty, Deep Learning, Machine learning models, E-learning, Comparatives studies, Informed decision, Monte Carlo methods, Quantification methods, Uncertainty quantifications, Uncertainty Quantification, Conformal predictions, Digital agriculture, Sources of uncertainty, Digital Agriculture

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Export Date: 20 September 2026; Cited By: 7; Conference name: 19th IEEE/CVF International Conference on Computer Vision Workshops, ICCVW 2023; Conference date: 2 October 2023 through 6 October 2023; Conference code: 195833

Inductive transfer learning for unlabeled target-domain via hybrid regularization

Item Type Journal Article

Author F. Zhuang

Author P. Luo

Author Q. He

Author Z. Shi

Abstract Recent years have witnessed an increasing interest in transfer learning. This paper deals with the classification problem that the target-domain with a different distribution from the source-domain is totally unlabeled, and aims to build an inductive model for unseen data. Firstly, we analyze the problem of class ratio drift in the previous work of transductive transfer learning, and propose to use a normalization method to move towards the desired class ratio. Furthermore, we develop a hybrid regularization framework for inductive transfer learning. It considers three factors, including the distribution geometry of the target-domain by manifold regularization, the entropy value of prediction probability by entropy regularization, and the class prior by expectation regularization. This framework is used to adapt the inductive model learnt from the source-domain to the target-domain. Finally, the experiments on the real-world text data show the effectiveness of our inductive method of transfer learning. Meanwhile, it can handle unseen test points. © 2009 Science in China Press and Springer-Verlag GmbH.

Date 2009

DOI 10.1007/s11434-009-0171-x

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Journal Abbr Chin. Sci. Bull.

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Transfer Learning, Hybrid Regularization, Inductive Learning, Transductive Learning

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Inference-enabled tracking of acute mental stress via multi-modal wearable physiological sensing: A proof-of-concept study

Item Type Journal Article

Author Y. Zhou
Author A.S. Mousavi
Author Y.R. Chalumuri
Author J.D. Parreira
Author M. Modak
Author J.A. Sanchez-Perez
Author A.H. Gazi
Author O.T. Inan
Author J.-O. Hahn

Abstract Objective: To develop a novel algorithm for tracking acute mental stress which can infer acute mental stress state from multi-modal digital signatures of physiological parameters compatible with wearable-enabled sensing. Methods: We derived prominent digital signatures of physiological responses to mental stress using cross-integration of multi-modal physiological signals including the electrocardiogram (ECG), photoplethysmogram (PPG), seismocardiogram (SCG), ballistocardiogram (BCG), electrodermal activity (EDA), and respiratory effort. Then, we developed an algorithm for tracking acute mental stress that can continuously classify stress vs no stress states by computing an aggregated likelihood computed with respect to a priori probability density distributions associated with the digital signatures of mental stress under stress and no stress states. Results: Our algorithm could adequately infer mental stress state (average classification accuracy: 0.85, sensitivity: 0.85, specificity: 0.86) using a small number of prominent digital signatures derived from cross-integration of multi-modal physiological signals. The digital signatures in our work significantly outperformed the digital signatures employed in the state-of-the-art in tracking acute mental stress. Its exploitation of collective inference allowed for improved inference of mental stress state relative to naïve data mining techniques. Conclusion: Our algorithm for tracking acute mental stress has the potential to make a leap in continuous, high-accuracy, and high-confidence inference of mental stress via convenient wearable-enabled physiological sensing. Significance: The ability to continuously monitor and track mental stress can collectively improve human wellbeing. © 2024 Nalecz Institute of Biocybernetics and Biomedical Engineering of the Polish Academy of Sciences

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adult, Article, female, human, human experiment, male, young adult, digital technology, algorithm, normal human, mental stress, classification, Mental stress, probability, cohort analysis, measurement accuracy, sensitivity and specificity, electrocardiogram, photoelectric plethysmography, ballistocardiography, breathing, Collective inference, Digital signatures, electrodermal response, Multi-modal physiological sensing, physiological process, proof of concept, Wearable sensing

Notes:

EID: 2-s2.0-85204348697

Export Date: 20 September 2026; Cited By: 10; Correspondence Address: J.-O. Hahn; Department of Mechanical Engineering, University of Maryland, College Park, United States; email: jhahn12@umd.edu

Inference-Time Selective Debiasing to Enhance Fairness in Text Classification Models

Item Type Conference Paper

Author G. Kuzmin

Author N. Yadav

Author I. Smirnov

Author T. Baldwin

Author A. Shelmanov

Editor Chiruzzo L.

Editor Ritter A.

Editor Wang L.

Abstract We propose selective debiasing - an inference-time safety mechanism designed to enhance the overall model quality in terms of prediction performance and fairness, especially in scenarios where retraining the model is impractical. The method draws inspiration from selective classification, where at inference time, predictions with low quality, as indicated by their uncertainty scores, are discarded. In our approach, we identify the potentially biased model predictions and, instead of discarding them, we remove bias from these predictions using LEACE - a post-processing debiasing method. To select problematic predictions, we propose a bias quantification approach based on KL divergence, which achieves better results than standard uncertainty quantification methods. Experiments on text classification datasets with encoder-based classification models demonstrate that selective debiasing helps to reduce the performance gap between post-processing methods and debiasing techniques from the at-training and pre-processing categories.¹ © 2025 Association for Computational Linguistics

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DOI 10.18653/v1/2025.naacl-short.9

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Extra Journal Abbreviation: Proc. Conf. Nations Americas Chapter Assoc. Comput. Linguist.: Hum. Lang. Technol., NAACL-HLT
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Conference Name Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies: Long Papers, NAACL-HLT 2025
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Notes:

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Export Date: 20 September 2026; Cited By: 1; Conference name: 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2025; Conference date: 29 April 2025 through 4 May 2025; Conference code: 215900

Inferring insect feeding patterns from sugar profiles: a comparison of statistical methods

Item Type Journal Article
Author M. Luquet
Author N. Parisey
Author M. Hervé
Author E. Desouhant
Author A.-M. Cortesero
Author A. Peñalver-Cruz
Author B. Lavandero
Author S. Anton
Author B. Jaloux

Abstract Investigations in nutritional ecology often require the identification of animal feeding patterns in natural conditions (what, where, and when do animals eat). Thus, methods are needed to trace not only individual resource uptake but also the relative use of different resources in a population or community. Recent biochemical developments allow predicting the use of sugar-rich resources from insects in the field. Individual feeding status (feeding history, food sources) is inferred by comparing insect sugar profiles with those of individuals fed on controlled diets. Individual assignments are then used to predict the relative consumption of different resources at the population or community level. As both steps may generate error, accurate prediction rules are needed. However, research from other domains (e.g., protein-marking studies) suggests that classical decision rules used for such tasks may sometimes induce bias. This study evaluated the performance of these rules and compared them to alternative methods on simulated, realistic datasets. It tested different methods for individual classification but also introduced methods for prevalence estimation, whose specific purpose is to estimate the relative frequency of different classes. Alternative methods substantially outperformed the traditional algorithms to predict insect individual feeding status and population class distribution (relative frequency of insects with different feeding status). This study provided a simple decision tool to choose a method according to dataset size, variance, and biochemical method used. Alternative methods should increase prediction confidence in future studies. Such approaches should easily be generalized to a wider range of systems. © 2020 The Royal Entomological Society

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Influence of Classification Task and Distribution Shift Type on OOD Detection in Fetal Ultrasound

Item Type Conference Paper

Author C.K. Wong

Author A.N. Christensen

Author C.I. Bercea

Author J.A. Schnabel

Author M.G. Tolsgaard

Author A. Feragen

Editor Gee J.C.

Editor Hong J.

Editor Sudre C.H.

Editor Golland P.

Editor Alexander D.C.

Editor Iglesias J.E.

Editor Venkataraman A.

Editor Kim J.H.

Abstract Reliable out-of-distribution (OOD) detection is important for safe deployment of deep learning models in fetal ultrasound amidst heterogeneous image characteristics and clinical settings. OOD detection relies on estimating a classification model's uncertainty, which should increase for OOD samples. While existing research has largely focused on uncertainty quantification methods, this work investigates the impact of the classification task itself. Through experiments with eight uncertainty quantification methods across four classification tasks on the same image dataset, we demonstrate that OOD detection performance significantly varies with the task, and that the best task depends on the defined ID-OOD criteria; specifically, whether the OOD sample is due to: i) an image characteristic shift or ii) an anatomical feature shift. Furthermore, we reveal that superior OOD detection does not guarantee optimal abstained prediction, underscoring the necessity to align task selection and uncertainty strategies with the specific downstream application in medical image analysis. Code: <https://github.com/wong-ck/ood-fetal-us>. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Deep learning, Feature extraction, Classification (of information), Learning models, Uncertainty, Medical image processing, Medical computing, Classification models, Classification tasks, Quantification methods, Uncertainty quantifications, uncertainty quantification, Clinical settings, fetal ultrasound, Fetal ultrasound, Image characteristics, OOD, Out-of-distribution

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Influence of different protein sources in the broiler diet on the presence of *Campylobacter* spp. in excreta and caecal content

Item Type Journal Article

Author C.F. Visscher

Author A. Abd El-Wahab

Author M.F.E. Ahmed

Author J. Hankel

Author V. Taube

Author J. Kamphues

Abstract The aim of the present study was to examine possible effects of a different protein supply on the presence of *Campylobacter* in excreta and caecal content of broiler chickens. Therefore, 220 one-day-old chickens were fed five different diets in the experimental period (day 8 onwards). In the control group (CON; n = 3 boxes), a soya bean-based industrially produced complete diet containing 5–10% whole wheat

and coccidiostats was fed, whereas in the remaining groups, four own-made complete diets with four protein sources and combinations (4×2 boxes; soya bean meal-based diet = SBM; rapeseed meal-containing diet = RSM; haemoglobin meal-containing diet = HGP; algae-containing diet = ALG) were offered. During the 6-week trial period, data concerning performance parameters, excreta and litter characteristics and microbiological analysis of excreta and caecal content at dissection were collected. The qualitative bacteriological investigation was based on the EN ISO 10272-1:2006, taken from the official collection of analysis methods in accordance with § 64 LFBG. For quantitative bacteriological tests, a 10-fold dilution series was made followed by culture-based quantification methods on approved selective media in duplicate. Comparing the results of the final dissection (day 44/45), nearly all samples of animals receiving the own-made complete diet were *Campylobacter* positive in the caecal content, whereas only seven of fifteen samples were positive in the CON group. With regard to the CFU in the caecal content, the counts of *Campylobacter* were significantly lower for animals fed the commercial diet (2.47 ± 3.06 vs. $7.36 \pm 0.66 \log_{10}$ CFU/g). When only the positive animals were evaluated, significant differences were also found between the CON group and the total of the other groups (5.30 ± 2.08 vs. $7.54 \pm 1.70 \log_{10}$ CFU/g). Although significant differences were seen in the results, no final recommendation can be given how to create a diet to reduce an infection with *Campylobacter*. Journal of Animal Physiology and Animal Nutrition © 2017 Blackwell Verlag GmbH

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Notes:

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Export Date: 20 September 2026; Cited By: 8; Correspondence Address: C.F. Visscher; Institute for Animal Nutrition, University of Veterinary Medicine Hannover, Foundation, Hannover, Germany; email: christian.visscher@tiho-hannover.de

Influence of iris color on corneal densitometry

Item Type Journal Article

Author A.R. Arizcuren

Author A. Consejo

Abstract Purpose: To determine whether iris pigmentation introduces a measurable bias in corneal densitometry (CD) values obtained via Scheimpflug imaging, and to develop an objective metric for iris color quantification. Methods: This observational study included 91 eyes from 47 healthy adults. CD was assessed as mean pixel intensity (MPI) from 25 Scheimpflug images per eye. Brightness artefacts from the iris were quantified using automated image processing. Iris color was objectively characterized from slit-lamp photographs using an objective single-value metric (IrisColor) derived from normalized CIELAB components. Associations among CD, iris brightness, and iris pigmentation were evaluated using Pearson correlation and linear mixed-effects models (LMMs). Results: CD correlated positively with iris brightness artefacts ($r = 0.47$, $\beta = 1.49$, $p < 0.001$), which in turn showed a strong negative correlation with IrisColor ($r = -0.83$, $\beta = -1.11$, $p < 0.001$). Lighter-colored irises (lower IrisColor values) exhibited statistically significantly higher CD values equivalent to a 6.6% relative overestimation. Groupwise comparisons confirmed that iris pigmentation significantly influences both CD and overall image brightness. Conclusion: Iris pigmentation induces a measurable bias in Scheimpflug-based CD estimates, primarily through increased brightness artefacts in light-colored eyes. The proposed IrisColor metric enables objective, continuous classification of iris color and could support future corrections for pigmentation-induced bias in CD-based diagnostics. © 2026 The Authors

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Influence of missing values on artificial neural network performance

Item Type Conference Paper

Author C.M. Ennett

Author M. Frize

Author C.R. Walker

Abstract The problem of databases containing missing values is a common one in the medical environment. Researchers must find a way to incorporate the incomplete data into the data set to use those cases in their experiments. Artificial neural networks (ANNs) cannot interpret missing values, and when a database is highly skewed, ANNs have difficulty identifying the factors leading to a rare outcome. This study investigates the impact on ANN performance when predicting neonatal mortality of increasing the number of cases with missing values in the data sets. Although previous work using the Canadian Neonatal Intensive Care Unit (NICU) Network's database showed that the ANN could not correctly classify any patients who died when the missing values were replaced with normal or mean values, this problem did not arise as expected in this study. Instead, the ANN consistently performed better than the constant predictor (which classifies all cases as belonging to the outcome with the highest training set a priori probability) with a 0.6-1.3% improvement over the constant predictor. The sensitivity of the models ranged from 14.5-20.3% and the specificity ranged from 99.2-99.7%. These results indicate that nearly 1 in 5 babies who will eventually die are correctly classified by the ANN, and very few babies were incorrectly identified as patients who will die. These findings are important for patient care, counselling of parents and resource allocation. © 2001 IMIA. All right reserved.

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INFLUENCE OF PRE-PROCESSING METHODS ON THE AUTOMATIC PRIORITY PREDICTION OF NATIVE-LANGUAGE END-USERS' MAINTENANCE REQUESTS THROUGH MACHINE LEARNING METHODS

Item Type Journal Article

Author M. D'Orazio

Author G. Bernardini

Author E. Di Giuseppe

Abstract Feedback and requests by occupants are relevant sources of data to improve building management, and building maintenance. Indeed, most predictable faults can be directly identified by occupants and communicated to facility managers through communications written in the end-users' native language. In this sense, natural language processing methods can support the request identification and attribution process if they are robust enough to extract useful information from these unstructured textual sources. Machine learning (ML) can support assessing and managing these data, especially in the case of many simultaneous communications. In this field, the application of pre-processing and ML methods to English-written databases has been widely provided, while efforts in other native languages are still

limited, impacting the real applicability. Moreover, the performance of combinations of methods for pre-processing, ML and classification classes attribution, has been limitedly investigated while comparing different languages. To fill this gap, this work hence explores the performance of automatic priority assignment of maintenance end-users' requests depending on the combined influence of: (a) different natural language pre-processing methods, (b) several supervised ML algorithms, (c) two priority classification rules (2-class versus 4-class), (d) the database language (i.e. the original database written in Italian, the native end-users' language; a translated database version in English, as standard reference). Analyses are performed on a database of about 12000 maintenance requests written in Italian concerning a stock of 23 buildings open to the public. A random sample of the sentences is supervised and labelled by 20 expert annotators following the best-worst method to attribute a priority score. Labelled sentences are then pre-processed using four different approaches to progressively reduce the number of unique words (potential predictors). Five different consolidated ML methods are applied, and comparisons involve accuracy, precision, recall and F1-score for each combination of pre-processing action, ML method and the number of priority classes. Results show that, within each ML algorithm, different pre-processing methods limitedly impact the final accuracy and average F1-score. In both Italian and English conditions, the best performance is obtained by NN, LR, SVM methods, while NB generally fails, and by considering the 2-class priority classification scale. In this sense, results confirm that facility managers can be effectively supported by ML methods for preliminary priority assessments in building maintenance processes, even when the requests database is written in end-users' native language. © 2024 The author(s).

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Influence of visual landscape on dwell and movement behaviours in Du Fu Thatched Cottage

Item Type Journal Article

Author X. Gong

Author L. Guo

Author Z. Zhu

Author J. Li

Author D. Zhang

Author J. Xu

Author W. Yong

Author Y. Zhou

Author M. Li

Abstract Within contemporary tourism contexts, classical-garden spatial narratives have become decoupled from visitor behaviours, underscoring the need to elucidate mechanisms for engagement. This study investigates the Du Fu Thatched Cottage in Xishu Gardens as a case study, integrating panoramic imaging, semantic segmentation and machine learning to quantify five visual-spatial metrics: naturalness, commemorativeness, complexity, scale and colour. Visitor dwell time and visit frequency were recorded via a panoramic roaming platform. The results reveal that the spatiotemporal distributions of visual attributes and visitor dwell and movement behaviours synergistically align along the central axis. These distributions exhibit positive correlations, with commemorative element visibility and water-body coverage showing the strongest associations and manifesting multidimensional threshold effects within the commemorative dimension. By examining both dwell and movement behaviours, this study uncovers synergistic mechanisms underlying heritage-space interactions and offers actionable recommendations for the sustainable conservation and management of classical gardens. © The Author(s) 2025.

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Information fusion strategy for aircraft engine health management

Item Type	Conference Paper
Author	L. Yu
Author	D. Cleary
Author	M. Osborn
Author	V. Rajiv
Abstract	Modern aircraft engines are equipped with sophisticated sensing instruments to enable proactive condition monitoring and effective health management capability. Development of intelligent systems that efficiently process sensor and operational data both onboard and off-board, to provide maintenance personnel with timely decision support, is the key to minimize flight service disruption and reduce engine ownership cost. The goal of this research is to develop a practical approach and strategy to leverage various available information sources and modeling techniques to streamline the engine health management process and maximize system accuracy and efficiency. This paper demonstrates a flexible fusion architecture that encapsulates the key elements of the engine monitoring and diagnostic process, i.e., sensor trend analysis module for anomaly detection, feature selection and fault isolation module for root cause identification, a decision module for diagnostic model fusion and action determination, and finally, a feedback module for knowledge validation and continuous learning. At the core of this engine health management system is a diagnostic fusion model designed around a common fault hierarchy which captures both a priori probabilities and interactions among various engine faults isolated by different classification models. The fusion model will resolve conflicting assessments from individual diagnostic models and provide a more accurate and comprehensive engine state estimate. Copyright © 2007 by ASME.
Date	2007
DOI	10.1115/GT2007-27174
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Extra	Journal Abbreviation: Proc. ASME Turbo Expo
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Conference Name	Proceedings of the ASME Turbo Expo
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Decision support systems, Feature extraction, Cost reduction, Sensors, Optimization, Information fusion, Computer aided design, Aircraft engines, Diagnostic model fusion, Health management, Timely decision support

Notes:

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Export Date: 20 September 2026; Cited By: 13; Correspondence Address: L. Yu; General Electric Global Research, Niskayuna, NY 12309, One Research Circle, United States; email: yuli@research.ge.com; Conference name: 2007 ASME Turbo Expo; Conference date: 14 May 2007 through 17 May 2007; Conference code: 70222

Information value of property description: A Machine learning approach

Item Type Journal Article

Author L. Shen

Author S. Ross

Abstract This paper employs machine learning to quantify the value of “soft” information contained in real estate property descriptions. Textual descriptions contain information that traditional hedonic attributes cannot capture. A one standard deviation increase in the uniqueness of a property based on this “soft” information leads to a 15% increase in property sale price in a hedonic price model and a 10% increase in a repeat sales price model. The effects in the hedonic model appear to arise through two channels: the unobserved quality of the housing unit, and the market power of the housing unit relative to competing properties. The effects in the repeat sales model appear to be driven entirely by the market power of the unit. Further, an annual hedonic price index ignoring our measure of unobserved quality overstates real estate prices by between 10% to 23% and mistimes the stabilization of housing prices following the Great Recession. Similar, but smaller effects, are observed for the repeat sales price index. © 2020 Elsevier Inc.

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Language English

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Journal Abbr J. Urban Econ.

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Natural language processing, Housing prices, Unsupervised machine learning, Price indexes, Property descriptions, Soft information

Notes:

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Export Date: 20 September 2026; Cited By: 35; Correspondence Address: L. Shen; Clemson University, United States; email: lilyshencal@gmail.com; CODEN: JUECD

Information-theoretic representation learning for positive-unlabeled classification

Item Type Journal Article

Author T. Sakai

Author G. Niu

Author M. Sugiyama

Abstract Recent advances in weakly supervised classification allow us to train a classifier from only positive and unlabeled (PU) data. However, existing PU classification methods typically require an accurate estimate of the class-prior probability, a critical bottleneck particularly for high-dimensional data. This problem has been commonly addressed by applying principal component analysis in advance, but such unsupervised dimension reduction can collapse the underlying class structure. In this letter, we propose a novel representation learning method from PU data based on the information-maximization principle. Our method does not require class-prior estimation and thus can be used as a preprocessing method for PU classification. Through experiments, we demonstrate that our method, combined with deep neural networks, highly improves the accuracy of PU class-prior estimation, leading to state-of-the-art PU classification performance. © 2020 Massachusetts Institute of Technology.

Date 2021

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Issue 1

Journal Abbr Neural Comp.

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Deep learning, Deep neural networks, Classification (of information), Learning systems, machine learning, algorithm, Algorithms, Machine Learning, Clustering algorithms, Neural Networks, Computer, Classification performance, Classification methods, Prior probability, Information theory, Supervised classification, High dimensional data, Information maximization, Dimension reduction, Pre-processing method

Notes:

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Initial interpretation and evaluation of a profile-based classification system for the anxiety and mood disorders: Incremental validity compared to DSM-IV categories

Item Type Journal Article

Author A.J. Rosellini

Author T.A. Brown

Abstract Limitations in anxiety and mood disorder diagnostic reliability and validity due to the categorical approach to classification used by the Diagnostic and Statistical Manual of Mental Disorders (DSM) have been long recognized. Although these limitations have led researchers to forward alternative classification schemes, few have been empirically evaluated. In a sample of 1,218 outpatients with anxiety and mood disorders, the present study examined the validity of Brown and Barlow's (2009) proposal to classify the anxiety and mood disorders using an integrated dimensional-categorical approach based on transdiagnostic emotional disorder vulnerabilities and phenotypes. Latent class analyses of 7 transdiagnostic dimensional indicators suggested that a 6-class (i.e., profile) solution provided the best model fit and was the most conceptually interpretable. Interpretation of the classes was further supported when compared with DSM diagnoses (i.e., within-class prevalence of diagnoses, using diagnoses to predict class membership). In addition, hierarchical multiple regression models were used to demonstrate the incremental validity of the profiles; class probabilities consistently accounted for unique variance in anxiety and mood disorder outcomes above and beyond DSM diagnoses. These results provide support for the potential development and utility of a hybrid dimensional-categorical profile approach to anxiety and mood disorder classification. In particular, the availability of dimensional indicators and corresponding profiles may serve as a useful complement to DSM diagnoses for both researchers and clinicians. © 2014 American Psychological Association.

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Journal Abbr Psychol. Assess.
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adult, Adult, Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, questionnaire, young adult, Young Adult, adolescent, Adolescent, aged, Aged, Surveys and Questionnaires, Aged, 80 and over, reliability, reproducibility, Reproducibility of Results, validity, very elderly, comparative study, controlled study, major clinical study, Latent class analysis, classification, Hybrid classification, self report, diagnostic test accuracy study, probability, disease classification, sex difference, anxiety disorder, Diagnostic and Statistical Manual of Mental Disorders, ethnicity, mood disorder, DSM-IV, Psychiatric Status Rating Scales, psychological rating scale, African American, Anxiety and mood disorders, Anxiety Disorders, Anxiety Disorders Interview Schedule Lifetime Version, Asian, Beck Depression Inventory, Caucasian, Dimensional-categorical classification, emotionality, latent structure analysis, Mood Disorders, semi structured interview

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Insomnia-LCA classifier: an open web application for insomnia subtype classification using latent class analysis

Item Type Journal Article
Author M. Carpi
Author D.R. Marques
Abstract The heterogeneity of insomnia presentations has long challenged research and clinical practice, motivating efforts to identify reliable disorder phenotypes. Person-centered, data-driven approaches such as latent class analysis (LCA) have provided new insights, suggesting that insomnia subtypes may differ not only in nocturnal symptoms but also in perceived impact and daytime distress. Despite this progress, LCA solutions often remain confined to the original datasets, limiting replication and applied use. To address this gap, we developed the insomnia-LCA classifier , an

open-source web application that assigns new Insomnia Severity Index (ISI) response profiles to one of four subtypes identified in a previously published LCA of Italian university students: no insomnia (NI), subthreshold insomnia (SI), high insomnia risk (HI), and predominant daytime symptoms (DS). Using the original model's class priors and item-level conditional response probabilities, the app computes posterior class probabilities from user-entered ISI responses, individually or in batch mode. Outputs include class probabilities and modal assignment, ISI total and subscale scores, and a visual comparison between the individual profile and subtype mean patterns. Reclassification of the original dataset showed near-perfect agreement with the latent class model (accuracy = 0.999; Cohen's kappa = 0.999), and synthetic profiles behaved as expected. The insomnia-LCA classifier provides a practical, reproducible tool for deploying and testing LCA-derived phenotypes in clinical research. © 2026 The Authors.

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DOI 10.1016/j.sleep.2026.108813

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Carpi; Department of Human Neuroscience, Sapienza University of Rome, Rome, Piazzale Aldo Moro 5, 00185, Italy; email: matteo.carpi@gmail.com; CODEN: SMLEA

InteBOMB: Integrating generic object tracking and segmentation with pose estimation for animal behavior analysis

Item Type Journal Article**Author** H. Zhai**Author** H.-Y. Yan**Author** J.-Y. Zhou**Author** J. Liu**Author** Q.-W. Xie**Author** L.-J. Shen**Author** X. Chen**Author** H. Han

Abstract Advancements in animal behavior quantification methods have driven the development of computational ethology, enabling fully automated behavior analysis. Existing multi-animal pose estimation workflows rely on tracking-by-detection frameworks for either bottom-up or top-down approaches, requiring retraining to accommodate diverse animal appearances. This study introduces InteBOMB, an integrated workflow that enhances top-down approaches by incorporating generic object tracking, eliminating the need for prior knowledge of target animals while maintaining broad generalizability. InteBOMB includes two key strategies for tracking and segmentation in laboratory environments and two techniques for pose estimation in natural settings. The “background enhancement” strategy optimizes foreground-background contrastive loss, generating more discriminative correlation maps. The “online proofreading” strategy stores human-in-the-loop long-term memory and dynamic short-term memory, enabling adaptive updates to object visual features. The “automated labeling suggestion” technique reuses the visual features saved during tracking to identify representative frames for training set labeling. Additionally, the “joint behavior analysis” technique integrates these features with multimodal data, expanding the latent space for behavior classification and clustering. To evaluate the framework, six datasets of mice and six datasets of nonhuman primates were compiled, covering laboratory and natural scenes. Benchmarking results demonstrated a 24% improvement in zero-shot generic tracking and a 21% enhancement in joint latent space performance across datasets, highlighting the effectiveness of this approach in robust, generalizable behavior analysis. Copyright ©2025 Editorial Office of Zoological Research, Kunming Institute of Zoology, Chinese Academy of Sciences.

Date 2025**DOI** 10.24272/j.issn.2095-8137.2024.268**URL** <https://www.scopus.com/pages/publications/86000674527?origin=resultslist>**Archive** Scopus**Short Title** InteBOMB: 面向动物行为分析的通用目标跟踪、分割和姿态预测集成化工具**Language** English**Volume** 46**Publisher** Science Press**Pages** 355-369**Publication** Zoological Research**ISSN** 20958137 (ISSN)**Issue** 2**Journal Abbr** Zoolog. Res.**Date Added** 20/09/2026, 9:51:48 pm**Modified** 20/09/2026, 9:51:48 pm

Tags:

procedures, body position, Background subtraction, image processing, Image Processing, Computer-Assisted, physiology, animal, Animals, Online learning, Mice, mouse, animal behavior, Behavior analysis, Behavior, Animal, Generic object tracking, Joint latent space, Pose estimation, Posture, primate, Primates, Selective labeling

Notes:

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Integrated machine learning approaches for flow cytometric quantification of myeloid-derived suppressor cells in acute sepsis

Item Type Journal Article

Author A.S. Bonavia

Author A. Samuelsen

Author J. Luthy

Author E.S. Halstead

Abstract Highly heterogeneous cell populations require multiple flow cytometric markers for appropriate phenotypic characterization. This exponentially increases the complexity of 2D scatter plot analyses and exacerbates human errors due to variations in manual gating of flow data. We describe a semi-automated workflow, based entirely on the Flowjo Graphical User Interface (GUI), that involves the stepwise integration of several, newly available machine learning tools for the analysis of myeloid-derived suppressor cells (MDSCs) in septic and non-septic critical illness. Supervised clustering of flow cytometric data showed correlation with, but significantly different numbers of, MDSCs as compared with the cell numbers obtained by manual gating. Neither quantification method predicted 30-day clinical outcomes in a cohort of 16 critically ill and septic patients and 5 critically ill and non-septic patients. Machine learning identified a significant decrease in the proportion of PMN-MDSC in critically ill and septic patients as compared with healthy controls. There was no difference between the proportion of these MDSCs in septic and non-septic critical illness. Copyright © 2022 Bonavia, Samuelsen, Luthy and Halstead.

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adult, Article, female, human, Humans, male, machine learning, aged, Machine Learning, workflow, clinical article, prospective study, flow cytometry, Sepsis, critical illness, Critical Illness, clinical immunology, clustering analysis, Flow Cytometry, high parameter flow cytometry, myeloid-derived suppressor cell, Myeloid-Derived Suppressor Cells, myeloid-derived suppressor cells (MDSC), renal replacement therapy, sepsis, sepsis - diagnostics

Notes:

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Export Date: 20 September 2026; Cited By: 4; Funding details: National Institute of General Medical Sciences, NIGMS, (K08GM138825); National Institute of General Medical Sciences, NIGMS; Correspondence Address: A.S. Bonavia; Division of Critical Care Medicine, Department of Anesthesiology and Perioperative Medicine, Penn State Milton S. Hershey Medical Center, Hershey, United States; email: abonavia@pennstatehealth.psu.edu

Integrated Networks for Viscoelastic FWI: Mapping from Q to Relaxation Variables and Quantifying Modeling Error

Item Type Journal Article

Author T. Zhang

Author D. Trad

Author K. Innanen

Abstract In the conventional approach to viscoelastic full waveform inversion (FWI) utilizing the generalized standard linear solid (GSLs) model, the quality factor (Q) is usually not directly inverted. Instead, this method converts Q into a suite of relaxation variables before inversion. Consequently, viscoelastic FWI entails the inversion of both elastic and relaxation models, with a subsequent conversion of relaxation variables back to the Q model to obtain final results. This method is partially due to the accurate intricate relationship between Q values and relaxation variables, where the mapping functions between these two sets are not simple inverses of each other. We introduce an approach that incorporates a multilayer perceptron (MLP) that is pretrained to learn complex mapping from Q values to relaxation variables. This MLP is then seamlessly integrated into a recurrent neural network (RNN)-based

GSLS viscoelastic FWI framework, establishing a computational graph, linking Q models and elastic models to data discrepancies, facilitating the direct inversion of both Q and elastic models. To address the inherent uncertainty in the MLP's mapping process, we apply Monte Carlo (MC) dropout within the neural network, quantifying the uncertainty of converting Q values into relaxation variables. Assuming a constant Q model, this uncertainty quantification method highlights the limited ability of relaxation variables to represent a constant Q model precisely. The feasibility of adapting our method for frequency-variant Q values appears straightforward. We also explore how this limitation, essentially a modeling error, influences the accuracy of forward modeling data. Our numerical results reveal that this modeling error has apparent impacts on the inversion outcomes for both elastic and attenuation models, underscoring the critical nature of this error in the viscoelastic FWI processes. © 1980-2012 IEEE.

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machine learning, Machine-learning, Machine learning, Neural-networks, Uncertainty, quality control, Inverse problems, Recurrent neural networks, artificial neural network, Mapping, numerical model, Multilayer neural networks, Multilayers perceptrons, error analysis, Monte Carlo analysis, model validation, Model errors, Elastic models, Full-waveform inversion, mapping method, modeling error, Q factor measurement, Q-values, Viscoelastic full waveform inversion, viscoelastic full waveform inversion (FWI), viscoelasticity, Viscoelastics, waveform analysis

Notes:

EID: 2-s2.0-85209371576

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Integrating artificial intelligence into radiological practice for automated pneumonia assessment

Item Type Journal Article

Author S. Benlakhdar

Author M. Rziza

Author R. Oulad Haj Thami

Abstract Background: Accurate assessment of lung involvement in coronavirus disease 2019 (COVID-19) patients is crucial for clinical decision-making. Traditional semi-quantitative and manual methods, while commonly used, may lack objectivity and reproducibility. This study aims to evaluate the clinical utility of artificial intelligence (AI)-based automated quantification of lung lesions and compare its performance with conventional semi-quantitative and manual approaches. Methods: This retrospective cross-sectional study was conducted at CHU Ibn Sina in Rabat from March 17 to May 11, 2020, and included 106 patients diagnosed with COVID-19 pneumonia. All patients underwent chest computed tomography (CT) using Siemens Somatom Definition AS+ equipment. Lung involvement was assessed using three methods: semi-quantitative scoring (Q20 and Q24), manual visual estimation by radiologists, and AI-based automated quantification. Concordance between the methods was evaluated using Lin's concordance correlation coefficient. Analysis time for each method was compared using the Kruskal-Wallis test. Results: AI-based quantification showed strong agreement with the semi-quantitative method, with a Lin's concordance correlation coefficient of 0.9271 [95% confidence interval (CI): 0.8975-0.9484]. The average analysis time was 65±8 s for AI and 28±5 s for the semi-quantitative method (P<0.001). The agreement in lesion classification between radiologists and AI reached 88%, indicating the potential reliability of AI in supporting clinical interpretation. Conclusions: AI-based automated quantification methods demonstrate high concordance with conventional techniques and provide consistent, objective, and reproducible assessments of COVID-19-related lung damage. Although AI requires more time than semi-quantitative approaches, its accuracy and reliability suggest it could complement or even replace traditional methods in certain clinical settings. Further large-scale studies are recommended to confirm these findings and assess the long-term clinical utility of AI in radiological evaluation. © AME Publishing Company, Taiwan Society for Therapeutic Radiology and Oncology.

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Integrating causal inference and machine learning to quantify climate-malaria relationships: Evidence of temperature and rainfall thresholds from Colombian municipalities

Item Type Journal Article

Author J.D. Gutiérrez

Abstract Rainfall and temperature are key climate determinants of malaria incidence; yet their causal exposure-response curves on malaria incidence across the entire Colombian territory remain unquantified. We estimated the effects of rainfall and temperature on malaria incidence at the municipal scale from 2007 to 2023. We conducted an ecological observational study in 969 Colombian municipalities located below 1,600 meters. The monthly Standardized Incidence Ratio (SIR) of malaria was calculated for each municipality. Directed acyclic graphs guided the identification of the appropriate adjustments needed to emulate the corresponding experimental design and avoid inducing bias, and the effect was estimated using a modified approach of the Targeted Maximum Likelihood Estimation (TMLE). Exposure-response curves were estimated for two outcomes: the current month and the moving average for the current and previous month. A total of 1,075,112 cases of malaria were reported. The results suggest a non-linear relationship between rainfall and temperature concerning the SIR of malaria, indicating an optimal temperature of 25 °C and approximately 37 mm of rainfall for the highest incidence. The negative control test revealed the presence of residual confounding bias ($p < 0.05$) in all estimates. Meanwhile, the estimations of the E-value indicated low to moderate tolerance (E-value = 1.14 - 1.48) to an unmeasured confounder. These findings support the integration of rainfall and temperature thresholds into early-warning systems for targeted malaria control. © 2026 Juan David Gutiérrez. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use,

distribution, and reproduction in any medium, provided the original author and source are credited.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: J.D. Gutiérrez; Facultad de Ciencias Médicas y de la Salud, Universidad de Santander, Instituto Masira, Bucaramanga, Santander, Colombia; email: jdgutierrez@udes.edu.co

Integrating InSAR and Channel Steepness for AI-Based Coseismic Landslide Modeling in the Nepal Himalaya

Item Type Journal Article

Author R. Silwal

Author G. Wang

Author S. KC

Author R. Rimal

Author S. Rawal

Abstract Earthquake-induced landslides in active orogens such as the Nepal Himalaya pose severe threats to lives, infrastructure, and post-disaster recovery. While machine learning (ML) and deep learning (DL) approaches to coseismic landslide susceptibility mapping have advanced considerably, spaceborne interferometric synthetic aperture radar (InSAR) products, particularly line-of-sight (LOS) displacement and coherence-based damage proxy maps (DPMs), remain underutilized in event-based frameworks. This study develops and evaluates a multi-factor coseismic landslide probability model that integrates InSAR-derived deformation metrics with geomorphic and hydrologic predictors to support rapid post-earthquake hazard assessment. Using the 25 April 2015 Mw 7.8 Gorkha earthquake as a case study, LOS displacement was derived from ALOS-2 PALSAR-2 ScanSAR interferometry, and the normalized channel steepness index (K_{sn}) was computed from a digital elevation model. Fourteen conditioning factors were used to train five architectures: Random Forest (RF), XGBoost, CNN, U-Net, and

DeepLabV3. Spatial autocorrelation was mitigated using a leave-one-basin-out three-fold spatial cross-validation strategy, with models evaluated on a patch-based domain comprising 655,360 pixels at a positive-class prevalence of 6.35%, establishing a no-skill AUC-PR baseline of 0.0635. InSAR integration consistently improved model performance under high class imbalance, increasing AUC-PR across all models by 7.8% to 17.3%. Random Forest achieved the highest AUC-PR (0.7940, nearly 12.5 times the baseline) and CSI (0.3027), providing the best balance between landslide recall (88.09%) and non-landslide specificity (88.68%) with the lowest false alarm rate (11.32%). XGBoost attained the highest AUC-ROC (0.9501) but exhibited lower recall (83.73%) and poorer calibration (Brier = 0.1397). Among DL models, DeepLabV3 produced the best-calibrated probabilities (Brier = 0.0693) and the highest CSI (0.2307), while U-Net offered the most balanced DL performance and CNN achieved the highest recall (92.40%) at the expense of elevated false alarms. Permutation feature importance identified Ksn as the dominant predictor, highlighting the strong tectono-geomorphic control on coseismic landslide occurrence. These results demonstrate that integrating InSAR-derived products substantially enhances landslide hazard assessment and supports more reliable rapid response in the Nepal Himalaya. © 2026 by the authors.

Date 2026

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Notes:

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Integrating meta-analysis and machine learning to decipher the impact of low-light stress on yield and grain components in staple crops

Item Type Journal Article

Author Y. Zhang

Author X. Zhang

Author X. Dong

Author M. Zhao

Author M. Wang

Author F. Zhang

Author Q. Yang

Author L. Tian

Author B. Feng

Abstract Purpose: Low-light stress poses a significant threat to global food security by impairing crop yield and grain quality. Therefore, investigating the impact of low-light stress on staple crops is imperative. Methods: In this study, we integrated meta-analysis (239 studies, 8019 observations) and machine learning to quantify the effects of low-light stress on crop yield and grain components across diverse environmental contexts. Results and conclusions: Our results demonstrated that low-light stress significantly reduced crop yield (CY: 54.13 %), thousand-grain weight (TGW: 11.38 %), grain number (GN: 29.70 %), and harvest index (HI: 26.70 %), while altering grain composition, including reduced starch content (SC: 7.40 %), starch yield (SY: 67.10 %), amylose content (AC: 6.53 %), amylopectin content (APC: 4.81 %), and protein yield (PY: 32.25 %), despite increasing protein content (PC: 11.72 %). Machine learning models identified crop type, low-light stage, degree, duration, climatic and soil conditions as key factors driving these responses. Under low-light stress, maize (cereal crops) exhibited the most severe reductions in CY, GN, and HI, while legume crops showed the greatest declines in PC and PY, and root and tuber crops experienced the largest decreases in SC and SY. The stage of low-light stress was the most critical factor influencing CY, TGW, GN, and HI, with the reproductive growth stage being the period when crops are most sensitive to low-light stress. Furthermore, the detrimental effects of low-light stress on crops aggravated with either increased degree or prolonged duration. With mean annual temperature (MAT) of 10–15°C or mean annual precipitation (MAP) of 400–800 mm, low-light stress caused the most pronounced reductions in CY, GN, and HI. When MAT > 15°C, the greatest reductions in SC, SY, AC, and PY were observed under low-light stress, with the highest increase in PC. Additionally, in sandy or neutral soils (pH 6.5–7.5), low-light stress exerted the most significant impacts on CY, GN, SC, and AC. Conversely, soils with high organic matter content (> 20 kg–1) help mitigate the negative effects of low-light stress on crops. Significance: These findings provide a scientific basis for designing targeted strategies to alleviate low-light stress and promoting sustainable agricultural production under changing climatic conditions. © 2025 Elsevier B.V.

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Publication Field Crops Research**ISSN** 03784290 (ISSN)**Journal Abbr** Field Crops Res.**Date Added** 20/09/2026, 9:51:46 pm**Modified** 20/09/2026, 9:51:46 pm**Tags:**

machine learning, climate change, Meta-analysis, protein, crop yield, meta-analysis, maize, agricultural production, alternative agriculture, cereal, climate conditions, food security, Grain composition, Low-light stress, Staple crops, starch, Yield, yield response

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EID: 2-s2.0-105023135598

Export Date: 20 September 2026; Cited By: 2; Funding details: Shaanxi Province Minor Grain Crops Industry Technology System, (NYKJ-2021-YL, 2021-LLRH-07); Correspondence Address: L. Tian; School of Agronomy, Anhui Agricultural University, Hefei, Anhui, 230036, China; email: tianlixin@ahau.edu.cn; B. Feng; College of Agronomy, Northwest A & F University, Yangling, Shaanxi, 712100, China; email: fengbaili@nwafu.edu.cn; CODEN: FCRED

Integrating mineral elements and metabolite features to distinguish Lotus seeds from different geographic origins

Item Type Journal Article**Author** H. Su**Author** Z. Li**Author** W. Yu**Author** T. Liu**Author** L. Luo

Abstract The characteristics of lotus seeds (LS) are influenced by variety and environment. However, it remains unknown the difference of metabolites and elements of LS from different origins. In this study, an accurate quantification method (97–107 %) for 20 mineral elements in LS was developed, and a metabolomic method was established to identify a total of 323 metabolites in LS. Mineral composition analysis revealed significant variations in the mineral element contents among LS samples from seven geographical regions. LS were rich in potassium (14,710 mg/kg), manganese (67.19 mg/kg), with a low level of sodium (210 mg/kg). A total of 10 mineral elements and 117 metabolites ($p < 0.05$ and $VIP > 1$) were identified as the potential geographical markers of LS by integration analysis. The linear discriminant analysis model showed high prediction accuracy. This study provides strong experimental evidence to maintain the authenticity and quality of LS in the food industry. © 2024 Elsevier Ltd

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Article, machine learning, China, nonhuman, controlled study, multivariate analysis, learning algorithm, confusion matrix, selenium, cross validation, chemistry, chemometrics, metabolism, Metabolomics, discriminant analysis, Discriminant Analysis, geography, metabolomics, chromium, metabolic fingerprinting, vanadium, Accurate quantifications, analytic method, arsenic, boron, cadmium, calcium ion, chemical fingerprinting, cobalt, cuprous ion, Different origins, geographic origin, Geographic origins, Geographical indication agricultural product, Geographical indication agricultural products, Geography, inductively coupled plasma mass spectrometry, iron, lead, Lotus, Lotus (genus), Lotus seeds, magnesium ion, manganese, metabolite, Metabolomic, microwave radiation, mineral, Mineral element, Minerals, Multi-element analysis, Multielement analysis, nickel, Origin tracing and identification, plant seed, potassium, Quality evaluation, Seeds, sodium, strontium, ultra performance liquid chromatography, zinc ion

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Integrating MPC with Learning-Based and Adaptive Methods to Enhance Safety, Performance and Reliability in Automated Insulin Delivery

Item Type Conference Paper
Author I. Hajizadeh
Author M.R. Askari

Author R. Kumar
Author V.M. Zavala
Author A. Cinar
Editor Findeisen R.
Editor Hirche S.
Editor Janschek K.
Editor Martin M.

Abstract In this work, an adaptive-learning model predictive control (AL-MPC) framework that integrates disturbance forecasting, uncertainty quantification, learning, and recursive subspace identification is developed. The proposed technique can be used for continuous systems affected by repetitive disturbances with unknown periods. The AL-MPC integrates online learning from historical data to anticipate impending disturbances and proactively counteract their effects to an adaptive MPC. This is done by using machine learning to quantify the significant disturbances from historical data and forecast their future evolution time series. Behavior patterns of the system can be identified from historical data, and the set-point, objective function weights, and constraints of the controller can be modified in advance for the anticipated time periods of the disturbance effects. AL-MPC is used to regulate glucose concentration (GC) in people with diabetes by automated insulin delivery. Simulation results indicate that the AL-MPC can regulate the BGC 75.4% of the simulation time in the target range (70-180) mg/dL without causing any hypoglycemia and hyperglycemia events. © 2020 Elsevier B.V.. All rights reserved.

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: A. Cinar; Department of Chemical and Biological Engineering, Illinois Institute of Technology, Chicago, United States; email: cinar@iit.edu; Conference name: 21st IFAC World Congress 2020; Conference date: 12 July 2020 through 17 July 2020; Conference code: 145388

Integrating multi-source remote sensing and ensemble machine learning to quantify the surface urban heat island in Oradea municipality, Romania

Item Type Journal Article

Author M. El Kihel

Author T. Caciora

Author C. Grava

Author R.-C. Țarcă

Author L. Saafadi

Author A. Ouallali

Abstract Land surface temperature (LST) is a key parameter for understanding thermal conditions in urban environments, where surface characteristics and anthropogenic activities strongly influence the intensity of the Urban Heat Island (UHI) phenomenon. Analysing the spatial and temporal variability of LST enables the assessment of how land-use and land-cover changes affect the local temperature distribution. This study proposes a machine-learning (ML) based framework for assessing the Surface Urban Heat Island (SUHI) effect in a medium-to-large city in Romania. Spectral indices derived from Landsat 8 imagery, including the Normalised Difference Vegetation Index (NDVI), Normalised Difference Built-up Index (NDBI), Normalised Difference Water Index (NDWI), Urban Index (UI), and Albedo, were integrated as explanatory variables. Non-parametric correlation analyses were applied to evaluate the relationships between these indices and LST. ML-derived LST estimates were validated against independent LST data from the Moderate Resolution Imaging Spectroradiometer (MODIS) instrument on the Terra satellite to assess model accuracy and robustness. The results indicate a significant increase in LST between 2013 and 2024, with an average warming of approximately 1.6 °C across the study area, largely driven by urban expansion and structural land-use changes. Among the evaluated models, the Random Forest algorithm achieved the highest predictive accuracy (RMSE $\approx$ 2.70 °C, MAPE $\approx$ 6.47%, $R^2 \approx$ 0.49), followed by Extreme Gradient Boosting (XGBoost) (RMSE $\approx$ 2.73 °C, MAPE $\approx$ 6.55%, $R^2 \approx$ 0.48) and the Support Vector Machine (SVM) (RMSE $\approx$ 2.79 °C, MAPE $\approx$ 6.78%, $R^2 \approx$ 0.45). Although all models successfully delineated the spatial patterns of SUHI, their ability to reproduce absolute LST values was limited, and some biases were observed in the estimation of thermal extremes. Overall, the findings demonstrate that integrating multi-source remote sensing data with ML techniques provides a reliable and effective approach for monitoring SUHI dynamics. © 2026 The Author(s)

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Machine learning, Land surface temperature, Spectral indices, Surface urban Heat Island, Urbanisation

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Integrating uncertainty quantification and ML for network robustness study: From metrics to surfaces

Item Type Journal Article

Author F.A. Lucay

Author W. Palma

Abstract Network robustness, defined as a system's capacity to retain functionality under failures or attacks, is conventionally studied using synthetic models such as Erdős-Rényi (ER), Barabási-Albert (BA), and Holme-Kim (HK). However, model parameters are often chosen arbitrarily, neglecting their influence on robustness metrics. This study integrates forward uncertainty quantification and machine learning to quantify parameter impact on measures —natural connectivity (NC), global efficiency (GE), algebraic connectivity (AC), spectral radius (SR), average path length (APL), and effective graph conductance (C). We analyse how these metrics vary across model configurations by constructing response surfaces based on influential parameters. Some key findings reveal distinct parameter dominance: in ER models, the number of nodes (n) and edge probability (p) govern NC, AC, and SR, while p predominantly shapes GE, APL, and C. For BA models, n and attachment edges variably influence all metrics. In KH models, n and random edge additions drive all robustness metrics with minimal impact from triangle-formation probability. The derived robustness surfaces offer practical tools for network design: identifying critical vulnerability thresholds, optimising topologies and benchmarking architectures for resilience. Copyright © 2025 EPLA. All rights, including for text and data mining, AI training, and similar technologies, are reserved.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: F.A. Lucay; Escuela de Ingeniería Química, Pontificia Universidad Católica de Valparaíso, Valparaíso, Chile; email: freddy.lucay@pucv.cl

Integrating weight with ensemble to handle changes in class distribution

Item Type Conference Paper

Author N. Limsetto

Author K. Waiyamai

Abstract Concept drift can be considered as a distribution mismatch problem where class distribution changes as a time passes. This problem is commonly found in classification task of data mining. Among the proposed solutions, the cost-based Class Distribution Estimation (CDE) shows the best performance in coping with difference in class distribution between train and test datasets. However there is still some problem, as CDE lost its performance when there is too much change in class distribution. In this paper, CDE-weight is proposed to reduce the impact of high change in class distribution. The idea is to use many models suitable with many class distributions along with dynamic weighting method that adjusts weight of each model according to its class distribution. Experimented results indicate that CDE-Weight methods are able to reduce the impact of misestimating and improve the classifier performance when train and test data are different. © 2014 Springer International Publishing Switzerland.

Date 2014

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Language English

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Pages 91-106**Proceedings Title** Lect. Notes Comput. Sci.**Conference Name** Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)**Date Added** 20/09/2026, 9:52:54 pm**Modified** 20/09/2026, 9:52:54 pm**Tags:**

Artificial intelligence, Classification (of information), Learning systems, Vehicle performance, Pattern recognition, Data mining, Classification, Ensemble methods, Cost estimating, Concept drifts, Quantification, Class distribution estimation, Cost-sensitive learning, Concept drift, Ensemble method

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: N. Limsetto; Department of Computer Engineering, Kasetsart University, BangKhen Campus, Bangkok 10900, Thailand; email: khim1204@gmail.com; Conference name: 10th International Conference on Machine Learning and Data Mining in Pattern Recognition, MLDM 2014; Conference date: 21 July 2014 through 24 July 2014; Conference code: 106686

Intelligent Assistant Diagnosis System of Osteosarcoma MRI Image Based on Transformer and Convolution in Developing Countries

Item Type Journal Article**Author** Z. Ling**Author** S. Yang**Author** F. Gou**Author** Z. Dai**Author** J. Wu

Abstract Osteosarcoma is a malignant bone tumor commonly found in adolescents or children, with high incidence and poor prognosis. Magnetic resonance imaging (MRI), which is the more common diagnostic method for osteosarcoma, has a very large number of output images with sparse valid data and may not be easily observed due to brightness and contrast problems, which in turn makes manual diagnosis of osteosarcoma MRI images difficult and increases the rate of misdiagnosis. Current image segmentation models for osteosarcoma mostly focus on convolution, whose segmentation performance is limited due to the neglect of global features. In this paper, we propose an intelligent assisted diagnosis system for osteosarcoma, which can reduce the burden of doctors in diagnosing osteosarcoma from three aspects. First, we construct a classification-image enhancement module consisting of resnet18 and DeepUPE to remove redundant images and improve image clarity, which can facilitate doctors' observation. Then, we experimentally compare the performance of serial, parallel, and hybrid fusion transformer and convolution, and propose a Double U-shaped visual transformer with convolution (DUconViT) for automatic

segmentation of osteosarcoma to assist doctors' diagnosis. This experiment utilizes more than 80,000 osteosarcoma MRI images from three hospitals in China. The results show that DUconViT can better segment osteosarcoma with DSC 2.6% and 1.8% higher than Unet and Unet++, respectively. Finally, we propose the pixel point quantification method to calculate the area of osteosarcoma, which provides more reference basis for doctors' diagnosis. © 2013 IEEE.

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Article, artificial intelligence, human, Humans, Features extraction, machine learning, adolescent, Adolescent, Convolution, Image segmentation, algorithm, Image enhancement, image segmentation, Images segmentations, Medical imaging, procedures, feature extraction, Transformer, artificial neural network, convolutional neural network, random forest, learning algorithm, support vector machine, transformer, child, Child, Computer aided diagnosis, image processing, diagnostic imaging, image analysis, Image Processing, Computer-Assisted, Magnetic resonance imaging, Magnetic Resonance Imaging, nuclear magnetic resonance imaging, signal noise ratio, human tissue, tumor volume, computer assisted tomography, histogram, scientific literature, residual neural network, Developing countries, Tumors, image enhancement, Diagnose system, Osteosarcoma, Bone Neoplasms, bone tumor, Developing Countries, developing country, Duconvit, DUconViT, Intelligent assisted diagnose, intelligent assisted diagnosis, Medical diagnostic imaging, osteosarcoma, Osteosarcomas

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Export Date: 20 September 2026; Cited By: 39; Correspondence Address: Z. Dai; Central South University, Department of Spine Surgery, The Second Xiangya Hospital, Changsha, 410011, China; email: f2daizhehao@csu.edu.cn; J. Wu; Monash University Melbourne, Research Center for Artificial Intelligence, Clayton, 3800, Australia; email: jiawu5110@163.com; CODEN: ITIBF

Intelligent data analysis techniques—machine learning and data mining

Item Type Journal Article

Author D. Simovici

Abstract This introductory chapter presents some of the main paradigms of intelligent data analysis provided by machine learning and data mining. After discussing several types of learning (supervised, unsupervised, semi-supervised, active and reinforcement learning) we examine several classes of learning algorithms (naive Bayes classifiers, decision trees, support vector machines, and neural networks) and the modalities to evaluate their performance. Examples of specific applications of algorithms are given using System R. © Springer International Publishing Switzerland 2015.

Date 2015

DOI 10.1007/978-3-319-16531-8_1

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Artificial intelligence, Learning systems, Reinforcement learning, Education, Learning algorithms, Supervised learning, Data mining, Neural networks, Entropy, Information analysis, Data handling, Decision trees, Active learning, Active Learning, Classifiers, Overfitting, Unsupervised learning, A-priori probabilities, Clustering, A-posteriori probabilities, Naive Bayes classifiers, A posteriori probabilities, A priori probabilities, Generalization, Impurities, Impurity, Naive bayes classifiers, Perceptrons

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Intelligent Quantification Method of Pipeline Flow Based on 'YOLO v5-FCN-AdaBoost'

Item Type Conference Paper

Author H. Yu

Author D. Di

Author J. Shi

Abstract The quantification of underground drainage pipeline flow is an important component of urban flood warning. In response to the high error rate and instability of current flow measurement devices, poor measurement accuracy, and the inability to accurately measure in complex drainage pipeline environments with confined spaces, strong corrosion, and low visibility, we have proposed, for the first time, an improved YOLO v5 algorithm, an Fully Convolutional Networks(FCN) algorithm based on dynamic pixel weighting strategy, and an AdaBoost regression algorithm combined with neural networks and machine learning to develop an intelligent quantification method of drainage pipeline flow. Based on this, the 'object recognition + image segmentation + flow regression analysis' steps are taken. Generative Adversarial Networks (GAN) are used to provide more samples for target detection. Through data collection, image enhancement, iterative expansion of the dataset, and continuous updating of the parameters of the neural network and machine learning models, intelligent quantification of drainage pipeline flow is ultimately achieved, ensuring real-time, high-precision, and intelligentization. © 2024 IEEE.

Date 2024

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Extra Journal Abbreviation: Int. Semin. Artif. Intell., Netw. Inf. Technol., AINIT

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Interpret the Predictions of Deep Networks Via Re-Label Distillation

Item Type Conference Paper

Author Y. Hua

Author S. Ge

Author D. Zhang

Abstract Interpreting the predictions of a black-box deep network can facilitate the reliability of its deployment. In this work, we propose a re-label distillation approach to learn a direct map from the input to the prediction in a self-supervision manner. The image is projected into a VAE subspace to generate some synthetic images by randomly perturbing its latent vector. Then, these synthetic images can be annotated into one of two classes by identifying whether their labels shift. After that, using the labels annotated by the deep network as teacher, a linear student model is trained to approximate the annotations by mapping these synthetic images to the classes. In this manner, these re-labeled synthetic images can well describe the local classification mechanism of the deep network, and the learned student can provide a more intuitive explanation towards the predictions. Extensive experiments verify the effectiveness of our approach qualitatively and quantitatively. © 2021 IEEE.

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Interpretable analysis of hybrid sturgeon behavioral response to Ammonia nitrogen stress based on single-frame spatial distribution quantification

Item Type Journal Article

Author Q. He

Author Z. Zhang

Author J. Chen

Author J. Mei

Author H. Qin

Author L. Song

Author D. Li

Author H. Yu

Author Y. Chen

Abstract Ammonia nitrogen stress causes massive mortality in industrial aquaculture, making real-time monitoring critical. However, existing methods for recognizing fish stress behaviors often focus on state classification while neglecting the interpretability of behavioral response patterns. To address the above issue, an interpretable behavioral response analysis method based on the spatial distribution quantification of single-frame images was proposed. Cultured sturgeon under ammonia nitrogen stress were used as the experimental subject. The proposed method first constructed a lightweight object detection model, MD-YOLO, to precisely extract fish spatial positions and provide high-quality coordinate data for spatial distribution quantification. Subsequently, seven biologically meaningful quantitative indicators were constructed to characterize the spatial distribution of fish schools from three perspectives: population aggregation, spatial coverage, and distribution uniformity. Unlike end-to-end deep learning methods, these indicators quantitatively characterize the spatial behavioral patterns of stressed fish, thereby significantly improving the interpretability of behavioral analysis. By combining these indicators into a feature

vector and utilizing a Random Forest (RF) classifier, accurate analysis of stress behavior was achieved. The experimental results showed that the MD-YOLO model outperformed the baseline model YOLOv10n by 1.1% and 4.4% in mAP@0.5 and mAP@0.75, respectively. Furthermore, the proposed method successfully characterized fish behavioral responses to ammonia nitrogen stress and achieved a recognition accuracy of 97.2%. The single-frame spatial distribution quantification method proposed in this study does not require cross-frame trajectory modeling, providing a high-precision, low-complexity, and interpretable solution for fish behavior monitoring in industrial aquaculture. © 2026 Elsevier B.V.

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Interpretable Educational AI Practice: Dynamic Optimization of Personalized Teaching Based on Neural Collaborative Filtering and Causal Reasoning

Item Type Conference Paper

Author M. Dong

Abstract This paper proposes an educational artificial intelligence framework that integrates neural collaborative filtering with causal reasoning, aiming to achieve explainable dynamic personalized teaching optimization. The system employs

deep neural networks to construct a nonlinear interaction model between students and resources for accurately predicting individual learning needs, while establishing a causal reasoning engine to dissect the effect mechanisms of teaching interventions, achieving counterfactual decision optimization. Key technologies include a hybrid architecture of generalized matrix factorization and multi-layer perceptrons to handle the complexity of educational scenarios, double machine learning to quantify individual treatment effects, and modular explanation templates to generate educationally logical decision attribution chains. Experimental validation demonstrates that this framework significantly enhances the explainability and long-term learning efficiency of teaching decisions while maintaining large-scale recommendation efficiency, providing a solution robust in both technology and ethical compliance. © 2025 Copyright held by the owner/author(s).

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Extra Journal Abbreviation: Proc. Int. Conf. Educ. Technol. Artif. Intell., ETAIC

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Optimisations, Deep neural networks, Efficiency, Learning systems, Artificial Intelligence, Neural-networks, Collaborative learning, Causal reasoning, Teaching, Causal reasoning engines, Dynamic optimization, Effect mechanism, Factorization, Individual learning, Integrated Framework, Integrated frameworks, Nonlinear interaction models, Personalized learning, Personalized Learning

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Dong; Officers College of PAP, Chengdu, Sichuan, China; email: dmm281551466@163.com; Conference name: 2025 International Conference on Educational Technology and Artificial Intelligence, ETAIC 2025; Conference date: 25 July 2025 through 27 July 2025; Conference code: 215024

Interpretable Ensemble Prediction of Thermally Induced Long-Term Pile Settlement from Limited Monitoring Data

Item Type Journal Article**Author** B. Sun**Author** C. Shi

Abstract Energy piles are a promising green building technology that combines traditional pile foundations with ground heat exchangers, providing both structural support and thermal regulation for buildings. Predicting long-term pile settlement induced by seasonal heating and cooling cycles is crucial for the safety and serviceability of energy piles. However, traditional numerical methods are limited by the inadequacies of soil constitutive models, and machine learning approaches typically require large datasets and lack interpretability. To address these challenges, this study proposes an interpretable sparse ensemble learning (ISEL) method that integrates finite element model outputs with limited monitoring data to dynamically predict thermally induced long-term pile displacement. The method employs orthogonal matching pursuit for adaptive model selection and adopts sparse Bayesian learning to quantify prediction uncertainty. The performance of the proposed framework is illustrated using centrifuge model tests, and sensitivity analyses are conducted to investigate the robustness of the model under operational uncertainty and varying thermal conditions. Results indicate that the ISEL method can accurately predict pile displacement, even with limited monitoring data. More importantly, the incorporation of physical laws significantly improves prediction reliability and efficiency, making it a handy tool for the design and analysis of energy piles subjected to cyclic thermal loading. © 2025 American Society of Civil Engineers.

Date 2025**DOI** 10.1061/JGGEFK.GTENG-13605**URL** <https://www.scopus.com/pages/publications/105011357935?origin=resultslist>**Archive** Scopus**Language** English**Volume** 151**Publisher** American Society of Civil Engineers (ASCE)**Publication** Journal of Geotechnical and Geoenvironmental Engineering**ISSN** 10900241 (ISSN)**Issue** 9**Journal Abbr** J. Geotech. Geoenviron. Eng.**Date Added** 20/09/2026, 9:51:50 pm**Modified** 20/09/2026, 9:51:50 pm**Tags:**

Learning systems, Forecasting, Uncertainty analysis, Large datasets, prediction, Ensemble learning, Learning methods, Sensitivity analysis, numerical model, Thermal modeling, uncertainty analysis, Soil testing, constitutive equation, Constitutive models, Energy piles, Model uncertainty, Modeling uncertainties, monitoring, Pile displacement, Pile foundations, pile response, Pile-settlement, Piles, Settlement of structures, Sparse learning, Thermal, Thermal cycle, Thermal cycles, Thermal cycling, Thermally induced

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Interpretable Minority Synthesis for Imbalanced Classification

Item Type Conference Paper

Author Y. He

Author F. Lin

Author X. Yuan

Author N.-F. Tzeng

Editor Zhou Z.-H.

Abstract This paper proposes a novel oversampling approach that strives to balance the class priors with a considerably imbalanced data distribution of high dimensionality. The crux of our approach lies in learning interpretable latent representations that can model the synthetic mechanism of the minority samples by using a generative adversarial network (GAN). A Bayesian regularizer is imposed to guide the GAN to extract a set of salient features that are either disentangled or intensionally entangled, with their interplay controlled by a prescribed structure, defined with human-in-the-loop. As such, our GAN enjoys an improved sample complexity, being able to synthesize high-quality minority samples even if the sizes of minority classes are extremely small during training. Empirical studies substantiate that our approach can empower simple classifiers to achieve superior imbalanced classification performance over the state-of-the-art competitors and is robust across various imbalance settings. Code is released in github.com/fudonglin/IMSIC. © 2021 International Joint Conferences on Artificial Intelligence. All rights reserved.

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Generative adversarial networks, Human-in-the-loop, Over sampling, Imbalanced data, Data distribution, Bayesian, Imbalanced classification, High dimensionality, Regularizer, Salient features, Synthetic mechanism

Notes:

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Interpretable Quantification of Scene-Induced Driver Visual Load: Linking Eye-Tracking Behavior to Road Scene Features via SHAP Analysis

Item Type Journal Article

Author J. Ni

Author Y. Shao

Author Y. Guo

Author Y. Gu

Abstract Road traffic accidents remain a major global public health concern, where complex urban driving environments significantly elevate drivers’ visual load and accident risks. Unlike existing research that adopts a macro perspective by considering multiple factors such as the driver, vehicle, and road, this study focuses on the driver’s visual load, a key safety factor, and its direct source—the driver’s visual environment. We have developed an interpretable framework combining computer vision and machine learning to quantify how road scene features influence oculomotor behavior and scene-induced visual load, establishing a complete and interpretable link between scene features, eye movement behavior, and visual load. Using the DR(eye)VE dataset, visual attention demand is established through occlusion experiments and confirmed to correlate with eye-tracking metrics. K-means clustering is applied to classify visual load levels based on discriminative oculomotor features, while semantic segmentation extracts quantifiable road scene features such as the Green Visibility Index, Sky Visibility Index and Street Canyon Enclosure. Among multiple machine learning models (Random Forest, Ada-Boost, XGBoost, and SVM), XGBoost demonstrates optimal performance in visual load detection. SHAP analysis reveals critical thresholds: the probability of high visual load increases when pole density exceeds 0.08%, signage surpasses 0.55%, or buildings account for more than 14%; while blink duration/rate decrease when street enclosure exceeds 38% or road congestion goes beyond 25%, indicating elevated visual load. The proposed framework provides actionable insights for urban design and driver assistance systems, advancing traffic safety through data-driven optimization of road environments. © 2025 by the authors.

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human factors, explainable AI, driving visual load, road traffic safety, SHAP analysis, urban design, urban scenario, visual attention demand

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Interpretable, Adaptive, and Robust Online Machine Learning for Data Streams - An Overview and Update

Item Type Conference Paper

Author Z. Chen

Editor Stahlbock R.

Editor Weiss G.M.

Editor Arabnia H.R.

Editor Deligiannidis L.

Abstract In this survey, we will summarize our recent representative research studies toward interpretable, adaptive, and robust online machine learning for data streams with multiple streaming natures, such as skewed distributions, concept drifts, and streaming or mix-typed features, which are four vital components towards intelligent online machine learning. I will also discuss some potential streaming applications for online anomaly detection, and future work on online semi-supervised learning, online label shift learning, online continuous learning, and large language model enhanced online machine learning for data streams. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2027.

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Learning systems, Learning algorithms, Anomaly detection, Data streams, Online systems, E-learning, Computer hardware description languages, Semi-supervised learning, Data stream, Concept drifts, Cost-sensitive, Cost-sensitive online learning, Density estimation, Local adaptive online density estimation, Local-adaptive, Online learning, Robust online learning, Sparse online learning

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Interpretable, calibrated neural networks for analysis and understanding of inelastic neutron scattering data

Item Type Journal Article

Author K.T. Butler

Author M.D. Le

Author J. Thiyaalingam

Author T.G. Perring

Abstract Deep neural networks (NNs) provide flexible frameworks for learning data representations and functions relating data to other properties and are often claimed to achieve 'super-human' performance in inferring relationships between input data and desired property. In the context of inelastic neutron scattering experiments, however, as in many other scientific scenarios, a number of issues arise: (i) scarcity of labelled experimental data, (ii) lack of uncertainty quantification on results, and (iii) lack of interpretability of the deep NNs. In this work we examine approaches to all three

issues. We use simulated data to train a deep NN to distinguish between two possible magnetic exchange models of a half-doped manganite. We apply the recently developed deterministic uncertainty quantification method to provide error estimates for the classification, demonstrating in the process how important realistic representations of instrument resolution in the training data are for reliable estimates on experimental data. Finally we use class activation maps to determine which regions of the spectra are most important for the final classification result reached by the network. © 2021 The Author(s). Published by IOP Publishing Ltd.

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Interpreting deep sports analytics: Valuing actions and players in the NHL

Item Type Conference Paper

Author G. Liu

Author W. Zhu

Author O. Schulte

Editor Zimmermann A.

Editor Davis J.

Editor Brefeld U.

Editor Van Haaren J.

Abstract Deep learning has started to have an impact on sports analytics. Several papers have applied action-value Q learning to quantify a team's chance of success, given the current match state. However, the black-box opacity of neural networks prohibits understanding why and when some actions are more valuable than others. This paper applies interpretable Mimic Learning to distill knowledge from the opaque neural net model to a transparent regression tree model. We apply Deep Reinforcement Learning to compute the Q function, and action impact under different game contexts, from 3M play-by-play events in the National Hockey League (NHL). The impact of an action is the change in Q-value due to the action. The play data along with the associated Q functions and impact are fitted by a mimic regression tree. We learn a general mimic regression tree for all players, and player-specific trees. The transparent tree structure facilitates understanding the general action values by feature influence and partial dependence plots, and player's exceptional characteristics by identifying player-specific relevant state regions. © Springer Nature Switzerland AG 2019.

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Notes:

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: G. Liu; Simon Fraser University, Burnaby, Canada; email: gla68@sfu.ca; Conference name: 5th International Workshop on Machine Learning and Data Mining for Sports Analytics, MLSA 2018, colocated with the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, ECML PKDD 2018; Conference date: 10 September 2018 through 10 September 2018; Conference code: 225199

Interpreting Deep Text Quantification Models

Item Type Conference Paper

Author Y.Q. Bang

Author M. Khaleel

Author W. Tavanapong

Editor Strauss C.

Editor Amagasa T.

Editor Kotsis G.

Editor Khalil I.

Editor Tjoa A.M.

Abstract Quantification learning is a relatively new deep learning task. Differing from a classic classification problem where the class of a single instance is predicted, a quantification model predicts the distribution of classes within a given set of instances. Quantification learning has applications in various domains. For example, in designing political campaign ads, it is important to know the proportion of different aspects voters care about. QuaNet is a recent deep learning quantification model that was shown to achieve good quantification performance. Like many deep learning models, there is no explanation about the contributions of different inputs QuaNet uses to predict a class distribution. In this study, we propose a method to provide such an explanation, which is important to increase users' trust in the model. Our method is the first work on interpreting deep learning quantification models. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Intra and inter-observer variability of transformation zone assessment in colposcopy: A qualitative and quantitative study

Item Type Journal Article

Author E. Vallikad

Author P.T. Siddartha

Author K.A. Kulkarni

Author C. Firtion

Author P. Keswarpu

Author P. Vajinepalli

Author S. Naik

Author L. Gupta

Abstract Introduction: Colposcopy is an important tool in the diagnosis of cervical precancer and early cancer. The assessment of women with abnormal cytology and selection of those who require further therapy or follow up depends on the colposcopic assessment of the Transformation Zone (TZ). Identification of the TZ is thus an important part of this examination. Intra and inter-observer variability is known to be relatively high in the colposcopic interpretation of abnormal features. However, there are hardly any studies on the observer variability in the assessment of the type of TZ. Aim: The present study was conducted with the aim to compare the intra and inter-observer variability of the TZ type classification and the Squamo-Columnar Junction (SCJ) visibility and to quantitatively measure the intra and inter-observer correlations of drawings of the TZ contours. Materials and Methods: Colposcopy images were obtained for a total of 170 cases. They were reviewed by three colposcopists independently. The colposcopists classified the TZ type and also marked the SCJ contours on the images. Each observer independently reviewed the cases at two different instances (few weeks apart) and the result was compiled for intra-observer variation. The intra and inter observer variability on the TZ type was compared using Cohen's Kappa. This was followed by a quantitative measurement of TZ observation variability using Hausdorff distance. Results: Inter-observer agreement for the TZ

type classification was moderate (Kappa= 0.53 to 0.66). Intra-observer agreement was moderate to strong (0.60 to 0.86). Conclusion: Colposcopic in vivo examination increases variability of identification of the TZ compared to analysis of recorded images. The disagreement in the TZ type was mostly for categories Type 2 vs. Type 3 followed by Type 1 vs. Type 2. A computerized quantification method can be used for quality control and training purpose in colposcopy. © 2017 Journal of Clinical and Diagnostic Research. All rights reserved.

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Intraobserver variability in grading severity of repeated identical cases of mitral regurgitation

Item Type Journal Article

Author N. Thomas

Author B. Unsworth

Author E.A. Ferenczi

Author J.E. Davies

Author J. Mayet

Author D.P. Francis

Abstract Background: In clinical practice, mitral regurgitation (MR) is often assessed from the visual impression of the color Doppler image. It is recognized that repeated scanning can give different images and that different observers may grade the same images differently. In this study, we focus on a more intrinsic source of variability-intraobserver variability in grading of identical images, presented more than once at the same sitting. Furthermore, we look for evidence that observer grading is influenced by the severity of the immediately preceding case viewed. Methods: Anonymous, 4-chamber color Doppler 2-dimensional video clips of MR were obtained from 60 unselected patients with MR. Six clips were identified by 2 observers as of contentious severity (either between mild and moderate or between moderate and severe). A 72-clip sequence was constructed from the 54 "uncontentious" selected cases of MR intermingled with 3 presentations of each of the 6 "contentious" images. Each contentious image was shown once without a designed order, once preceded by 3 clips of less severe MR, and once preceded by 3 clips of more severe MR. Results: Only 1 (8%) of 12 observers gave consistent gradings for the triply presented images. More than 90% (11/12) of the observers reported a different grading of the same clip of MR in at least 1 of the 6 cases. The MR severity was changed in 51.4% of the triplets of identical images shown. Of 12 reporters, 5 (42%) showed classification variability between severe and nonsevere grades in at least 1 of the 6 cases. The direction of change showed no sign of consistency between observers ($P = .375$). Conclusion: Even a skilled observer cannot be relied upon to give an identical grading to an identical simple video clip of MR, when re-presented surreptitiously within a few minutes. Interobserver variability cannot therefore simply be blamed on differential levels of skill. Because, even under these ideal and dramatically simplified conditions, visual assessment is so variable, the future emphasis of echocardiographic MR grading may lie in integrating qualitative analysis with simple quantification methods. © 2008 Mosby, Inc. All rights reserved.

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Intravenous Delayed Gadolinium-Enhanced MR Imaging of the Endolymphatic Space: A Methodological Comparative Study

Item Type Journal Article

Author R. Boegle

Author J. Gerb

Author E. Kierig

Author S. Becker-Bense

Author B. Ertl-Wagner

Author M. Dieterich

Author V. Kirsch

Abstract In-vivo non-invasive verification of endolymphatic hydrops (ELH) by means of intravenous delayed gadolinium (Gd) enhanced magnetic resonance imaging of the inner ear (iMRI) is rapidly developing into a standard clinical tool to investigate peripheral vestibulo-cochlear syndromes. In this context, methodological comparative studies providing standardization and comparability between labs seem even more important, but so far very few are available. One hundred eight participants [75 patients with Meniere's disease (MD; 55.2 ± 14.9 years) and 33 vestibular healthy controls (HC; 46.4 ± 15.6 years)] were examined. The aim was to understand (i) how variations in acquisition protocols influence endolymphatic space (ELS) MR-signals; (ii) how ELS quantification methods correlate to each other or clinical data; and finally, (iii) how ELS extent influences MR-signals. Diagnostics included neuro-otological assessment, video-oculography during caloric stimulation, head-impulse test, audiometry, and iMRI. Data analysis provided semi-quantitative (SQ) visual grading and automatic algorithmic quantitative segmentation of ELS area [2D, mm²] and volume [3D, mm³] using deep learning-based segmentation and volumetric local thresholding. Within the range of 0.1–0.2 mmol/kg Gd dosage and a 4 h $\pm$ 30 min time delay, SQ grading and 2D- or 3D-quantifications were independent of signal intensity (SI) and signal-to-noise ratio (SNR; FWE corrected, $p < 0.05$). The ELS quantification methods used were highly reproducible across raters or thresholds and correlated strongly (0.3–0.8). However, 3D-quantifications showed the least variability. Asymmetry indices and normalized ELH proved the most useful for

predicting quantitative clinical data. ELH size influenced SI (cochlear basal turn $p < 0.001$), but not SNR. SI could not predict the presence of ELH. In conclusion, (1) Gd dosage of 0.1–0.2 mmol/kg after 4 h $\pm$ 30 min time delay suffices for ELS quantification. (2) A consensus is needed on a clinical SQ grading classification including a standardized level of evaluation reconstructed to anatomical fixpoints. (3) 3D-quantification methods of the ELS are best suited for correlations with clinical variables and should include both ears and ELS values reported relative or normalized to size. (4) The presence of ELH increases signal intensity in the basal cochlear turn weakly, but cannot predict the presence of ELH. © Copyright © 2021 Boegle, Gerb, Kierig, Becker-Bense, Ertl-Wagner, Dieterich and Kirsch.

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Intrinsic Consistency Preservation with Adaptively Reliable Samples for Source-Free Domain Adaptation

Item Type Journal Article

Author J. Tian

Author A.E. Saddik

Author X. Xu

Author D. Li

Author Z. Cao

Author H.T. Shen

Abstract Unsupervised domain adaptation (UDA) aims to alleviate the domain shift by transferring knowledge learned from a labeled source dataset to an unlabeled target domain. Although UDA has seen promising progress recently, it requires access to data from both domains, making it problematic in source data-absent scenarios. In this article, we investigate a practical task source-free domain adaptation (SFDA) that alleviates the limitations of the widely studied UDA in simultaneously acquiring source and target data. In addition, we further study the imbalanced SFDA (ISFDA) problem, which addresses the intra-domain class imbalance and inter-domain label shift in SFDA. We observe two key issues in SFDA that: 1) target data form clusters in the representation space regardless of whether the target data points are aligned with the source classifier and 2) target samples with higher classification confidence are more reliable and have less variation in their classification confidence during adaptation. Motivated by these observations, we propose a unified method, named intrinsic consistency preservation with adaptively reliable samples (ICPR), to jointly cope with SFDA and ISFDA. Specifically, ICPR first encourages the intrinsic consistency in the predictions of neighbors for unlabeled samples with weak augmentation (standard flip-and-shift), regardless of their reliability. ICPR then generates strongly augmented views specifically for adaptively selected reliable samples and is trained to fix the intrinsic consistency between weakly and strongly augmented views of the same image concerning predictions of neighbors and their own. Additionally, we propose to use a prototype-like classifier to avoid the classification confusion caused by severe intra-domain class imbalance and inter-domain label shift. We demonstrate the effectiveness and general applicability of ICPR on six benchmarks of both SFDA and ISFDA tasks. The reproducible code of our proposed ICPR method is available at https://github.com/CFM-MSG/Code_ICPR. © 2024 IEEE.

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Data privacy, human, Job analysis, Task analysis, article, reliability, Reliability analysis, prediction, controlled study, diagnosis, Adaptation models, benchmarking, Class imbalance, classifier, adaptation, Domain adaptation, Consistency preservation, domain and label shift, Domain and label shift, imbalanced distribution, Imbalanced distribution, Inter-domain, Intra-domain, source-free data, Source-free data

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Intrinsic logit-based debiasing for class-imbalanced semi-supervised learning

Item Type Journal Article

Author Q. Tang

Author Y. Cheng

Author X. Guo

Author X. Wei

Author S. Zhang

Author W. Pedrycz

Abstract In Class-Imbalanced Semi-Supervised Learning (CISSL), classifiers suffer from severe confirmation bias, particularly when the class distribution of unlabeled set are unknown or mismatched with that of the labeled set. Existing debiasing methods often rely on impractical assumptions for classifier bias estimation, such as known class priors or risky external proxies. To overcome these limitations, we propose Intrinsic Logit-Based Debiasing (ILBD), a robust post-hoc framework that estimates bias directly from task-relevant data without external dependencies and class priors. ILBD constructs a comprehensive bias estimator by decomposing output logits into two distinct components: (1) Label-Free Intrinsic Bias, derived from non-target (Formula presented) logits by masking target-class maximum logits to capture the model's background distributional skew; and (2) Label-Guided Learning Bias, derived from target-class maximum logits to quantify confidence disparities between head and tail classes arising from varying learning difficulties. To ensure estimation reliability, we further employ a class-adaptive threshold to filter noisy pseudo-labels. By subtracting this estimated bias at inference, ILBD effectively rectifies the classifier's bias directly from the inherent statistical patterns of the training data, without the need for target-prior assumptions or external data. Extensive experiments on CIFAR-10/100-LT, STL-10-LT, and Small-ImageNet-127 demonstrate the effectiveness of ILBD. Our code is available at <https://github.com/aroid721/ILBD>. © 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining,

AI training, and similar technologies.

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Article, machine learning, Learning algorithms, reliability, Data mining, entropy, algorithm, quantitative analysis, correlation analysis, nonhuman, prediction, training, Self-supervised learning, Labeled data, sensitivity analysis, classifier, Semi-supervised learning, mathematical model, post hoc analysis, Learning classifiers, semi supervised machine learning, statistical analysis, calibration, Class distributions, Imbalanced Learning, class imbalanced semi supervised learning, Class-imbalanced learning, confirmation bias, Confirmation bias, De-biasing, intrinsic logit based debiasing, Logit adjustment, Long-tailed recognition, Target class

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Introduction of an information-theoretic method to predict recovery rates of active compounds for Bayesian in silico screening: Theory and screening trials

Item Type Journal Article

Author M. Vogt

Author J. Bajorath

Abstract We present the first method to predict compound recovery rates from descriptor statistics. A log-odds function is designed that models probability distributions of descriptor values of active and inactive molecules in chemical space and used to

determine the likelihood of database compounds to exhibit a specific activity. The divergence of probability models for active and inactive compounds is applied to evaluate the ability of the log-odds likelihood function to recover active compounds from a background database. The divergence measure, which is closely related to the Kullback - Leibler distance, is strongly correlated with recovery rates of Bayesian virtual screening calculations. It has thus been possible to predict compound recovery rates for different activity classes. Prior to practical virtual screening trials, one can also estimate how likely it would be to recover active compounds from a given screening database. © 2007 American Chemical Society.

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article, Statistics, Information retrieval, Bayesian networks, classification, Probability distributions, Computational methods, methodology, Drug products, Computational Biology, drug, Bayes theorem, Bayes Theorem, chemistry, statistics, Database systems, biology, Pharmaceutical Preparations, chemical model, Drug Evaluation, Preclinical, drug screening, Kullback Leibler distance, Models, Chemical, Probability models, Silico screening, Virtual screening calculations

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Intrusion Detection based on Non-negative Positive-unlabeled Learning

Item Type Conference Paper

Author S. Lv

Author Y. Liu

Author Z. Liu
Author W. Chao
Author C. Wu
Author B. Wang
Editor Sun M.
Editor Zhang H.

Abstract Due to the diversity of network traffic flow, intrusion detection is usually studied as an anomaly detection problem. In this paper, Positive-unlabeled with Non-negative Risk Estimator(nnPU) learning is introduced for intrusion detection. The cyber attacks is treated as positive samples in PU learning. A risk estimator is raised to estimates the binary classification loss. For data imbalance in intrusion detection, we improve the risk estimator of nnPU through focal loss(FL-nnPU). The dynamic weights in focal loss is used to balance the small class prior. The experiments result show that FL-nnPU have a close performance to binary classification, and it performs better than nnPU under data imbalance problems.
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Network security, Intrusion detection, Learning systems, Cyber-attacks, intrusion detection, Anomaly detection, Binary classification, Non negatives, positive-unlabeled learning, risk estimator, Risk estimators, Data imbalance, data imbalance, Dynamic weight, Network traffic flow, Pu learning

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Invasion in the Niger Delta: remote sensing of mangrove conversion to invasive *Nypa fruticans* from 2015 to 2020

Item Type Journal Article

Author A. Barenblitt

Author L. Fatoyinbo

Author N. Thomas

Author A. Stovall

Author C. de Sousa

Author C. Nwobi

Author L. Duncanson

Abstract Invasive species are a leading threat to biodiversity worldwide. *Nypa* palm (*Nypa fruticans*) has emerged as the predominant invasive species in the Niger Delta region of Nigeria. While endemic mangroves have high rates of carbon sequestration, stabilize coastlines and protect biodiversity, *Nypa* does not provide these services outside its native region of Southeast Asia. Oil exploration and urbanization in this region also exacerbate mangrove loss and *Nypa* spread. As *Nypa* is difficult to distinguish from endemic mangrove species in remotely sensed data, estimates of mangrove and ecosystem services losses in Nigeria are highly uncertain. Here, we analyse multisensor satellite data with machine learning to quantify the rapid expansion of *Nypa* from 2015 to 2020 in Nigeria. Using Landsat imagery and random forest classification, we quantify total potential *Nypa* extent in Nigeria in 2019. We then produced a *Nypa* extent map using iterative combinations of Sentinel-1 SAR, Sentinel-2 MSI and ALOS PALSAR. Random forest classifications using SAR data from ALOS and Sentinel-1 were best suited for mapping *Nypa* extent with similar accuracies (78% and 75%, respectively). Based on data availability and accuracy, we focussed our change analysis on Sentinel-1 SAR. Our results show ~28 000 ha of mangroves were converted to *Nypa* in Nigeria by 2020 and covered a larger extent than endemic mangroves, compounding the effect of the existing degradation and deforestation in the region. We also compared forest height and complexity estimates from Global Ecosystem Dynamics Investigation LiDAR to further distinguish between endemic mangroves and *Nypa* in three dimensions. *Nypa* structural variability, measured by top-of-canopy height, vegetation cover, plant area index, and foliage height diversity, was lower than that of mangroves. At current rates of *Nypa* expansion, the entire area of study would be invaded by *Nypa* by 2028, with potentially detrimental consequences to the ecosystem services provided by mangroves. © 2023 The Authors. Remote Sensing in Ecology and Conservation published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

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Investigating and suggesting the evaluation dataset for image classification model

Item Type Journal Article

Author S. Sivamani

Author S.I. Chon

Author D.Y. Choi

Author J.H. Park

Abstract Image processing systems are widespread with the digital transformation of artificial intelligence. Many researchers developed and tested several image classification models using machine learning and statistical techniques. Nevertheless, the current research seldom focuses on the quality assurance of these models. The existing methods lack to verify the quality assurance, with the lack of test cases to prepare the evaluation dataset to test the model, which can cause critical drawbacks in the nuclear field and defense system. In this article, we discuss and suggest the preparation of the evaluation dataset using improved test cases through Cause-Effect Graphing. The proposed method can generate the evaluation dataset with automated test cases through the quantification method, which consists of 1) image characteristic selection 2) creating the Cause-Effect graphing approach of the image with the feature, and 3) generate all possible test coverage. The testing is performed with the COCO dataset, which shows the declining prediction accuracy with the adjusted brightness and sharpness ranging between -75 to 75%, which indicates the negligence of the important characteristics in the existing test dataset. The experiment shows the prediction fails while sharpness is less than the 0%, and the brightness fails at -75% with less number of detection object between -50% and 75%. This indicates that characteristic changes affects the prediction accuracy and the number of detected objects in an image. Our approach proves the importance of the characteristic selection process for the overall image to generate a more efficient model and increase the accuracy of object detection. © 2020 Institute of Electrical and Electronics Engineers Inc.. All rights reserved.

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Investigating the antimicrobial and anticancer potential of culturable fungal endophytes isolated from the stems of *Kirkia acuminata* Oliv

Item Type Journal Article

Author M. Magagula

Author T.E. Motaung

Author Z. Mbita

Author K. Dithebe

Abstract Background: Fungal endophytes produce various structurally and chemically diverse bioactive secondary metabolites including those that are similar to their host plants. However, fungal endophytes from South African medicinal plants are relatively under-explored. The medicinal plant, *Kirkia acuminata* Oliv., is on the decline in the natural environment due to overharvesting. This necessitates the search for novel alternatives to sustainably obtain the plant's bioactive metabolites. Thus, fungal endophytes may serve as suitable candidates as they can produce host-similar bioactive compounds. Results: Eighteen morphologically distinct fungal endophytes were isolated from the surface-sterilised stems of *K. acuminata* Oliv. Sequencing of the internal transcribed spacer (ITS) region revealed that the isolates were distributed among three genera, namely *Diaporthe*, *Neofusicoccum* and *Pseudofusicoccum*. The

broth micro-dilution assay showed that 17 of the 18 ethyl acetate crude extracts exhibited inhibitory activity with minimum inhibitory concentration (MIC) values ranging from 0.31 to 2.5 mg/mL and 1.25 to 2.5 mg/mL against bacterial pathogens and *Candida albicans*, respectively. The 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) assay revealed that most of the crude extracts had dose-dependent cytotoxicity against non-cancerous human embryonic kidney (HEK-293) cells, with the crude extracts of the *N. parvum* KaS-3, *D. macadamiae* KaS-4, *P. olivaceum* KaS-5 and *D. neotheicola* KaS-6 isolates demonstrating safety against the non-cancerous cells. The alamarBlue assay revealed that the four non-cytotoxic crude extracts had moderate anticancer activity against cervical cancer ME-180 and melanoma A375 cancerous cell lines. Moreover, mycochemical analysis of the non-cytotoxic crude extracts using colourimetric quantification methods revealed that the observed cytotoxic effect could be attributed to the high total phenolic content in the crude extracts. Conclusion: The study highlights that the fungal endophytes inhabiting the stems of *K. acuminata* Oliv. produce secondary metabolites that may serve as leads for novel antimicrobial and non-toxic anticancer agents. © The Author(s) 2025.

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Investigating the Behavior and Reliability of Eccentrically Loaded Steel Connections Made with Bolts and Welds in Combination

Item Type Journal Article

Author M.F. Tamimi

Author M. Soliman

Author E. Stringer

Author B.W. Russell

Abstract This paper investigates the behavior of eccentrically loaded connections made by combining fillet welds and high-strength pretensioned bolts. New models are introduced for characterizing the load-deformation behavior of slip-resistant bolted connections with different faying surface conditions. These models are utilized to improve the ability of the instantaneous center of rotation (ICR) method to predict the capacity of eccentrically loaded connections utilizing friction as a load-transfer mechanism including bolted-only and/or bolted-welded combination connections.

Experimental results are used to quantify the accuracy of the ICR method in predicting the capacity of these combination connections considering different connecting elements to be participating in a single load resisting system. Finite-element (FE) analysis is used to investigate the load-transfer mechanisms within the combination connections. Reliability analysis is conducted using FE analysis assisted by machine learning to quantify the reliability level of the investigated connections and establish the resistance factors necessary to maintain the reliability level above prescribed thresholds. The reliability analysis shows that using the ICR method for designing these combination connections leads to a highly reliable connection for different types of faying surfaces. © 2023 American Society of Civil Engineers.

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Investigation of Different Free Image Analysis Software for High-Throughput Droplet Detection

Item Type Journal Article

Author I. Sanka

Author S. Bartkova

Author P. Pata

Author O.-P. Smolander

Author O. Scheler

Abstract Droplet microfluidics has revealed innovative strategies in biology and chemistry. This advancement has delivered novel quantification methods, such as droplet digital polymerase chain reaction (ddPCR) and an antibiotic heteroresistance analysis tool. For droplet analysis, researchers often use image-based detection techniques. Unfortunately, the analysis of images may require specific tools or programming skills to produce the expected results. In order to address the issue, we explore the potential use of standalone freely available software to perform image-based droplet detection. We select the four most popular software and classify them into rule-based and machine learning-based types after assessing the software's modules. We test and evaluate the software's (i) ability to detect droplets, (ii) accuracy and precision, and (iii) overall components and supporting material. In our experimental setting, we find that the rule-based type of software is better suited for image-based droplet detection. The rule-based type of software also has a simpler workflow or pipeline, especially aimed for non-experienced users. In our case, CellProfiler (CP) offers the most user-friendly experience for both single image and batch processing analyses. © 2021 The Authors. Published by American Chemical Society

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Investigation of Human Factors Using HFACS Framework—A Case Study for Unintended Reactor Trip Events in NPP

Item Type Conference Paper

Author M. Karthick

Author C. Senthil Kumar

Author T. Paul Robert

Editor Varde P.V.

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Editor Vinod G.

Abstract Safety remains to be the principal concern in nuclear industry, and layered protective measures are implemented to ensure safety to public and environment. Nevertheless, accidents in a nuclear plant do occur, and human factors are confirmed to be the prime cause of these accidents. Several studies have been carried out for human factor analysis, and still, there is a need for additional methods to model human errors. In this paper, human factor analysis and classification system (HFACS) is used to analyze the causal factors for an accident/incident. HFACS is a validated and reliable model for human factor analysis and provides a simple way to evaluate the causal factors in an accident. Although HFACS framework can be applied to analyze any accident events, in this paper, it is applied to review eighteen unintended reactor trip events due to human error from 2000 to 2006. In this study, experts analyze and rate the events against each hierarchal level in HFACS. Expert judgment is sought as the uncertainty and complexity involved in modeling human and organizational factors are very high. Homogeneity of experts is assessed using inter-rated reliability and Cohen's kappa value. Expert evaluation on human factors is quantified and based on the correlation in rating, most significant factor is identified. Further, based on the information on the reported events and the analysis, statistical tests are employed to confirm the associations between the hierarchal levels in the HFACS framework for reactor trip events. The results indicate four pairs of factors, viz. resource management, planned inappropriate actions, adverse mental states, and skill-based errors have strong associations between adjacent categories. Based on these associations, adverse mental states are deemed the most potent for accident occurrences. Further analysis indicated that slips of attention, memory failures for events such a failure to open or close the valves, and perception failures for visual illumination (i.e., 'skilled based errors') are the most common 'unsafe acts' committed and appear more frequently along with nuclear control room actions. The results reveal that human errors are caused mainly due to potential human factors associated with cognition. Few active failures are also identified. Thus, the analysis highlights the importance of attention toward possible human factors among others. Overall, HFACS tool proves useful in categorizing operator errors from reported events. HFACS framework and the quantification method help to determine which factors are more dominant and influential in the accident sequence and provide additional insight for the analysts into the significance and relative importance of each of the human factors. © 2020, Springer Nature Singapore Pte Ltd.

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INVESTIGATION OF MACHINE LEARNING METHOD WITH GENERALIZATION PERFORMANCE FOR AUTOMATIC HAMMERING TEST

Item Type Conference Paper

Author H. Shimbo

Author N. Hasegawa

Author T. Mizobuchi

Author T. Ozeki

Author J.-I. Nojima

Author H. Nakamura

Author S. Tomoto

Abstract The authors investigated the applicability of machine learning-based hammering test quantification methods to laser hammering systems. This method quantitatively evaluates the degree of defects using an anomaly detection technique with CNN (Convolutional Neural Network) features extraction from the imaged hammering sound. The quantitative evaluation and generalization performance for the conventional human hammering test sound were examined. The laser hammering system applies a high-power laser beam to the concrete surface for an extremely short period of time creating a physical impact. This technology enables hammering tests remotely and automatically. As a result of the study, it was shown that capability of quantitative evaluation of the degree of

defects up to around 15mm depth using laser hammering sounds. The capacity may be modifiable with the system parameters. © 2024 International Committee of the SCMT conferences. All rights reserved.

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Tags:

Network security, Performance, Feature extraction, Learning systems, Machine-learning, Convolutional neural networks, Machine Learning, Anomaly detection, Anomaly Detection, Quantitative evaluation, Machine learning methods, Quantification methods, Defects, Mahalanobis distances, High power lasers, Laser beams, Laser hammering, Laser hammering test, Laser Hammering Test, Mahalanobis' Distance, Scalogram

Notes:

EID: 2-s2.0-105019644459

Export Date: 20 September 2026; Cited By: 0; Conference name: 6th International Conference on Sustainable Construction Materials and Technologies, SCMT 2024; Conference date: 9 June 2024 through 14 June 2024; Conference code: 213643

Investigations on classifying pulsed eddy current signals with a neural network

Item Type Journal Article

Author Z. Liu

Author D.S. Forsyth

Author B.A. Lepine

Author I. Hammad

Author B. Farahbakhsh

Abstract This paper investigates the classification of pulsed eddy current signals with neural networks. In the experiments, a pulsed eddy current inspection was carried out to assess the condition of a multi-layered structure, which contains metal loss at various locations. The test specimen is an engineered specimen designed to assist in the development of inspection techniques for corrosion damage detection in aircraft

fuselage splices. Separate neural network classifiers were developed using features extracted in the time domain, the coefficients of a transfer function based on a system identification approach, and the coefficients from a wavelet transform. The experimental results show the potential of applying neural networks to classify pulsed eddy current signals and to quantify the material loss evaluation. The robustness of this approach is also discussed.

Date 2003

DOI 10.1784/insi.45.9.608.52940

URL <https://www.scopus.com/pages/publications/0141857448?origin=resultslist>

Archive Scopus

Language English

Volume 45

Pages 608-614

Publication Insight: Non-Destructive Testing and Condition Monitoring

ISSN 13542575 (ISSN)

Issue 9

Journal Abbr Insight Non Destr Test Cond Monit

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Tags:

Feature extraction, Multilayer neural networks, Quantification methods, Time domain analysis, Calibration, Wavelet transforms, Eddy current testing, Corrosion, Transfer functions, Fuselages

Notes:

EID: 2-s2.0-0141857448

Export Date: 20 September 2026; Cited By: 8; Correspondence Address: Z. Liu; Institute for Aerospace Research, National Research Council Canada, Building M-14, Ottawa, Ont., K1A 0R6, Montreal Road 1191, Canada; email: zheng.liu@nrcnrc.gc.ca; CODEN: ITMOE

Is Efficient PAC Learning Possible with an Oracle That Responds “Yes” or “No”?

Item Type Conference Paper

Author C. Daskalakis

Author N. Golowich

Editor Agrawan S.

Editor Roth A.

Abstract The empirical risk minimization (ERM) principle has been highly impactful in machine learning, leading both to near-optimal theoretical guarantees for ERM-based learning algorithms as well as driving many of the recent empirical successes in deep learning. In this paper, we investigate the question of whether the ability to perform ERM, which computes a hypothesis minimizing empirical

risk on a given dataset, is necessary for efficient learning: in particular, is there a weaker oracle than ERM which can nevertheless enable learnability? We answer this question affirmatively, showing that in the realizable setting of PAC learning for binary classification, a concept class can be learned using an oracle which only returns a single bit indicating whether a given dataset is realizable by some concept in the class. The sample complexity and oracle complexity of our algorithm depend polynomially on the VC dimension of the hypothesis class, thus showing that there is only a polynomial price to pay for use of our weaker oracle. Our results extend to the agnostic learning setting with a slight strengthening of the oracle, as well as to the partial concept, multiclass and real-valued learning settings. In the setting of partial concept classes, prior to our work no oracle-efficient algorithms were known, even with a standard ERM oracle. Thus, our results address a question of (Alon et al., 2021) who asked whether there are algorithmic principles which enable efficient learnability in this setting. © 2024 C. Daskalakis & N. Golowich.

Date 2024

URL <https://www.scopus.com/pages/publications/85203668457?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Mach. Learn. Res.

Volume 247

Publisher ML Research Press

ISBN 26403498 (ISSN)

Pages 1263-1307

Proceedings Title Proc. Mach. Learn. Res.

Conference Name Proceedings of Machine Learning Research

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Tags:

Machine-learning, Federated learning, Adversarial machine learning, Contrastive Learning, Learning settings, Concept class, Empirical risk minimization, Empirical risk minimization oracle, ERM oracle, Learnability, Near-optimal, One-inclusion graph, PAC learning, Partial concept class

Notes:

EID: 2-s2.0-85203668457

Export Date: 20 September 2026; Cited By: 2; Funding details: Hertz Foundation, HF; National Science Foundation, NSF, (CCF-1901292, DMS-2022448, DMS2134108); National Science Foundation, NSF; Conference name: 37th Annual Conference on Learning Theory, COLT 2024; Conference date: 30 June 2024 through 3 July 2024; Conference code: 202164

IS IMPORTANCE WEIGHTING INCOMPATIBLE WITH INTERPOLATING CLASSIFIERS?

Item Type Conference Paper

Author K.A. Wang

Author N.S. Chatterji

Author S. Haque

Author T. Hashimoto

Abstract Importance weighting is a classic technique to handle distribution shifts. However, prior work has presented strong empirical and theoretical evidence demonstrating that importance weights can have little to no effect on overparameterized neural networks. Is importance weighting truly incompatible with the training of overparameterized neural networks? Our paper answers this in the negative. We show that importance weighting fails not because of the overparameterization, but instead, as a result of using exponentially-tailed losses like the logistic or cross-entropy loss. As a remedy, we show that polynomially-tailed losses restore the effects of importance reweighting in correcting distribution shift in overparameterized models. We characterize the behavior of gradient descent on importance weighted polynomially-tailed losses with overparameterized linear models, and theoretically demonstrate the advantage of using polynomially-tailed losses in a label shift setting. Surprisingly, our theory shows that using weights that are obtained by exponentiating the classical unbiased importance weights can improve performance. Finally, we demonstrate the practical value of our analysis with neural network experiments on a subpopulation shift and a label shift dataset. When reweighted, our loss function can outperform reweighted cross-entropy by as much as 9% in test accuracy. Our loss function also gives test accuracies comparable to, or even exceeding, well-tuned state-of-the-art methods for correcting distribution shifts. © 2022 ICLR 2022 - 10th International Conference on Learning Representations. All rights reserved.

Date 2022

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Archive Scopus

Language English

Extra Journal Abbreviation: ICLR - Int. Conf. Learn. Represent.

Publisher International Conference on Learning Representations, ICLR

Proceedings Title ICLR - Int. Conf. Learn. Represent.

Conference Name ICLR 2022 - 10th International Conference on Learning Representations

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Tags:

Neural-networks, Entropy, Loss functions, Gradient methods, Re-weighting, Cross entropy, Importance weighting, Entropy loss, Importance weights, Test accuracy, Classic techniques, Overparameterization

Notes:

EID: 2-s2.0-85150364283

Export Date: 20 September 2026; Cited By: 7; Conference name: 10th International Conference on Learning Representations, ICLR 2022; Conference date: 25 April 2022 through 29 April 2022; Conference code: 186704

Is the Investment Worth it? Using Machine Learning to Quantify Advertising Impact

Item Type	Conference Paper
Author	M. Yin
Author	X. Pan
Author	Z. Yu
Abstract	The scientific quantification of advertising effectiveness on e-commerce platforms is a challenge for many advertisers. In this study, we employed machine learning and data mining techniques, specifically using CatBoost for calculating the original feature weights and OpenFE for calculating composite feature weights. Our findings suggest that, overall, the impact of advertising promotion on e-commerce platforms is minimal compared to other indicators such as store's total sales, ratings, discount intensity, and poster types. However, it is encouraging to note that as the popularity of a store increases, there is a greater need to increase advertising investment in order to gain a competitive advantage over rival stores. From this perspective, advertisers' expenditure is not entirely wasted. (Abstract) © 2024 IEEE.
Date	2024
DOI	10.1109/IMCEC59810.2024.10575552
URL	https://www.scopus.com/pages/publications/85198621959?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: IMCEC - IEEE Adv. Inf. Manag., Commun., Electron. Autom. Control Conf.
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	979-835031652-0 (ISBN)
Pages	105-109
Proceedings Title	IMCEC - IEEE Adv. Inf. Manag., Commun., Electron. Autom. Control Conf.
Conference Name	IMCEC 2024 - IEEE 6th Advanced Information Management, Communicates, Electronic and Automation Control Conference
Date Added	20/09/2026, 9:52:02 pm
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Tags:

Sales, Machine-learning, Machine learning, Data mining, Electronic commerce, Machine Learning, E-commerces, Competition, Data-mining techniques, Key words, Commerce platforms, advertising impact (key words), Advertizing, Advertizing effectiveness, Advertizing impact (key word), data-mining, Feature weight, feature weights, Machine data

Notes:

EID: 2-s2.0-85198621959

Export Date: 20 September 2026; Cited By: 0; Funding details: University Philosophy and Social Science Research General Project of Jiangsu Province, (2023SJYB0948); 2023 Science Association Soft Science Project of Wuxi City, (KX- 23-C006); Correspondence Address: M. Yin; Wuxi Taihu University, College of Business, Jiangsu, China; email: yinmj@wxu.edu.cn; Conference name: 6th IEEE Advanced Information Management, Communicates, Electronic and Automation Control Conference, IMCEC 2024; Conference date: 24 May 2024 through 26 May 2024; Conference code: 200771

Is uncertainty quantification in deep learning sufficient for out-of-distribution detection?

Item Type Conference Paper

Author A. Schwaiger

Author P. Sinhamahapatra

Author J. Gansloser

Author K. Roscher

Editor Espinoza H.

Editor McDermid J.

Editor Huang X.

Editor Castillo-Effen M.

Editor Chen X.C.

Editor Hernandez-Orallo J.

Editor hEigartaigh S.O.

Editor Mallah R.

Abstract Reliable information about the uncertainty of predictions from deep neural networks could greatly facilitate their utilization in safety-critical applications. Current approaches for uncertainty quantification usually focus on in-distribution data, where a high uncertainty should be assigned to incorrect predictions. In contrast, we focus on out-of-distribution data where a network cannot make correct predictions and therefore should always report high uncertainty. In this paper, we compare several state-of-the-art uncertainty quantification methods for deep neural networks regarding their ability to detect novel inputs. We evaluate them on image classification tasks with regard to metrics reflecting requirements important for safety-critical applications. Our results show that a portion of out-of-distribution inputs can be detected with reasonable loss in overall accuracy. However, current uncertainty quantification approaches alone are not sufficient for an overall reliable out-of-distribution detection. © 2020 for this paper by its

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Date 2020

URL <https://www.scopus.com/pages/publications/85089629284?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: CEUR Workshop Proc.

Volume 2640

Publisher CEUR-WS

ISBN 16130073 (ISSN)

Proceedings Title CEUR Workshop Proc.

Conference Name CEUR Workshop Proceedings

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Tags:

Safety engineering, Deep learning, Deep neural networks, Forecasting, Uncertainty analysis, Neural networks, Safety critical applications, State of the art, Uncertainty quantifications, Overall accuracies

Notes:

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Export Date: 20 September 2026; Cited By: 6; Funding details: ADA-Center; Bavarian Ministry of Economic Affairs, Regional Development and Energy; Center for Analytics; Center for Analytics-Data-Applications; Intel Collaborative Research Institute on Mobile Networking and Computing, ICRI-MNC; Correspondence Address: A. Schwaiger; Fraunhofer IKS, Fraunhofer Institute for Cognitive Systems, Germany; email: Adrian.Schwaiger@iks.fraunhofer.de; Conference name: 2020 Workshop on Artificial Intelligence Safety, AISafety 2020; Conference date: 5 January 2021 through 10 January 2021; Conference code: 161835

iTRAQ-based analysis of developmental dynamics in the soybean leaf proteome reveals pathways associated with leaf photosynthetic rate

Item Type Journal Article

Author J. Qin

Author J. Zhang

Author D. Liu

Author C. Yin

Author F. Wang

Author P. Chen

Author H. Chen

Author J. Ma

Author B. Zhang

Author J. Xu**Author** M. Zhang

Abstract Photosynthetic rate which acts as a vital limiting factor largely affects the potential of soybean production, especially during the senescence phase. However, the physiological and molecular mechanisms that underlying the change of photosynthetic rate during the developmental process of soybean leaves remain unclear. In this study, we compared the protein dynamics during the developmental process of leaves between the soybean cultivar Hobbit and the high-photosynthetic rate cultivar JD 17 using the iTRAQ (isobaric tags for relative and absolute quantification) method. A total number of 1269 proteins were detected in the leaves of these two cultivars at three different developmental stages. These proteins were classified into nine expression patterns depending on the expression levels at different developmental stages, and the proteins in each pattern were also further classified into three large groups and 20 small groups depending on the protein functions. Only 3.05–6.53 % of the detected proteins presented a differential expression pattern between these two cultivars. Enrichment factor analysis indicated that proteins involved in photosynthesis composed an important category. The expressions of photosynthesis-related proteins were also further confirmed by western blotting. Together, our results suggested that the reduction in photosynthetic rate as well as chloroplast activity and composition during the developmental process was a highly regulated and complex process which involved a serial of proteins that function as potential candidates to be targeted by biotechnological approaches for the improvement of photosynthetic rate and production. © 2016, Springer-Verlag Berlin Heidelberg.

Date 2016**DOI** 10.1007/s00438-016-1202-3**URL** <https://www.scopus.com/pages/publications/84962673505?origin=resultslist>**Archive** Scopus**Language** English**Volume** 291**Publisher** Springer Verlag**Pages** 1595-1605**Publication** Molecular Genetics and Genomics**ISSN** 16174615 (ISSN)**Issue** 4**Journal Abbr** Mol. Genet. Genomics**Date Added** 20/09/2026, 9:52:53 pm**Modified** 20/09/2026, 9:52:53 pm**Tags:**

Article, comparative study, nonhuman, procedures, controlled study, plant leaf, classification, protein, Photosynthesis, metabolism, plant protein, Plant Proteins, protein analysis, proteomics, Proteomics, chloroplast, Chloroplasts, photosynthesis, gene expression regulation, priority journal, soybean, Western blotting, gene regulatory network, Gene Regulatory Networks, Gene Expression Regulation, Plant, protein protein interaction, Plant Leaves, developmental stage, Gene Expression Regulation, Developmental, isobaric tags for relative and absolute quantification, iTRAQ, photosynthetic rate, Protein Interaction Maps, Proteome, Soybean, Soybeans

Notes:

EID: 2-s2.0-84962673505

Export Date: 20 September 2026; Cited By: 21; Funding details: China Scholarship Council Program, (201208130236); Key Research Foundation for Excellent Returned Overseas Chinese, (C2011006001); National Natural Science Foundation of China, NSFC, (31100880); National Natural Science Foundation of China, NSFC; Natural Science Foundation of Hebei Province, (C2012301020); Natural Science Foundation of Hebei Province; Natural Science Foundation for Distinguished Young Scholars of Hunan Province, (C2014301035); Natural Science Foundation for Distinguished Young Scholars of Hunan Province; Correspondence Address: M. Zhang; National Soybean Improvement Center Shijiazhuang Sub-Center, North China Key Laboratory of Biology and Genetic Improvement of Soybean Ministry of Agriculture, Cereal and Oil Crop Institute, Hebei Academy of Agricultural and Forestry Sciences, Shijiazhuang, 050031, China; email: mengchenzhang@hotmail.com; CODEN: MGGOA

JaxSGMC: Modular stochastic gradient MCMC in JAX

Item Type Journal Article

Author S. Thaler

Author P. Fuchs

Author A. Cukarska

Author J. Zavadlav

Abstract We present JaxSGMC, an application-agnostic library for stochastic gradient Markov chain Monte Carlo (SG-MCMC) in JAX. SG-MCMC schemes are uncertainty quantification (UQ) methods that scale to large datasets and high-dimensional models, enabling trustworthy neural network predictions via Bayesian deep learning. JaxSGMC implements several state-of-the-art SG-MCMC samplers to promote UQ in deep learning by reducing the barriers of entry for switching from stochastic optimization to SG-MCMC sampling. Additionally, JaxSGMC allows users to build custom samplers from standard SG-MCMC building blocks. Due to this modular structure, we anticipate that JaxSGMC will accelerate research into novel SG-MCMC schemes and facilitate their application across a broad range of domains. © 2024 The Author(s)

Date 2024

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Journal Abbr SoftwareX

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Markov processes, Deep learning, Learning systems, Machine-learning, Machine learning, Modulators, Large datasets, Stochastic systems, Bayesian inference, Bayesian networks, Inference engines, Optimization, Markov chain Monte Carlo, Markov Chain Monte-Carlo, Quantification methods, Uncertainty quantifications, Monte carlo schemes, SGMCMC, Stochastic gradient

Notes:

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Export Date: 20 September 2026; Cited By: 3; Funding details: Munich Data Science Institute Seed Fund; European Research Council, ERC, (101077842); European Research Council, ERC; Correspondence Address: J. Zavadlav; Multiscale Modeling of Fluid Materials, Department of Engineering Physics and Computation, TUM School of Engineering and Design, Technical University of Munich, Germany; email: julija.zavadlav@tum.de

Joint space narrowing progression quantification with joint angle correction in rheumatoid arthritis

Item Type Conference Paper

Author Y. Ou

Author P. Ambalathankandy

Author R. Furuya

Author S. Kawada

Author T. Kamishima

Author M. Ikebe

Abstract Rheumatoid arthritis is a form of autoimmune disease characterized by synovitis that can ultimately cause joint deformities and impaired functioning. The cartilage destruction is one of the most important indicators for diagnosis and treatment of Rheumatoid arthritis, and it is radiographically manifested as joint space narrowing. In this study, we propose a joint location detection method and a sub-pixel accurate method for quantifying joint space narrowing progression with a joint angle correction. The proposed joint location detection method can detect the location of 14 joints from a given hand radiographic image, the error of 89.13% joints is less than 3 pixels (spatial resolution: 0.175 mm/pixel). In our previous works, we measured joint space narrowing progression between a baseline and its follow-up finger joint images by using partial image phase only correlation. We found that the inconsistency of joint angles may lead to characteristic mismatch and thus affect the accuracy of joint space narrowing quantification. In this work, we introduce rotation invariant phase only correlation in joint space narrowing quantification for joint angle correction. In our experiment, the improved quantification method can effectively manage the mismatch due to the inconsistency of joint angles. © 2022 IEEE.

Date 2022

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Language English
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Publisher Institute of Electrical and Electronics Engineers Inc.
ISBN 978-166548903-4 (ISBN)

Proceedings Title BMEiCON - Biomed. Eng. Int. Conf.
Conference Name BMEiCON 2022 - 14th Biomedical Engineering International Conference
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Tags:

machine learning, Machine-learning, Machine learning, Location, Diseases, Computer aided diagnosis, Computer aided instruction, Frequency domain analysis, Pixels, Computer aided analysis, rheumatoid arthritis, Rheumatoid arthritis, Angle correction, computer-aided diagnosis, frequency domain analysis, Frequency-domain analysis, Joint angle, Joint locations, joint space narrowing, Joint space narrowing, Joints (anatomy), key point detection, Key point detection, Keypoints, Point detection, radiology

Notes:

EID: 2-s2.0-85147250252

Export Date: 20 September 2026; Cited By: 2; Funding details: Japan Society for the Promotion of Science, KAKEN, (18H05288, 21K07611); Japan Science and Technology Agency, JST, (JPMJSP2119); Conference name: 14th Biomedical Engineering International Conference, BMEiCON 2022; Conference date: 10 November 2022 through 13 November 2022; Conference code: 186085

Kernel density estimation for multiclass quantification

Item Type Journal Article

Author A. Moreo

Author P. González

Author J.J. del Coz

Abstract Several disciplines, like the social sciences, epidemiology, sentiment analysis, or market research, are interested in knowing the distribution of the classes in a population rather than the individual labels of the members thereof. Quantification is the supervised machine learning task concerned with obtaining accurate predictors of class prevalence, and to do so particularly in the presence of label shift. The distribution-matching (DM) approaches represent one of the most important families among the quantification methods that have been proposed in the literature so far. Current DM approaches model the involved populations using histograms of posterior probabilities. In this paper, we argue that their application to the multiclass setting is suboptimal since the histograms become class-specific, thus missing the opportunity to model inter-class information that may exist in the data. We propose a new representation mechanism based on multivariate densities that we model via kernel density estimation (KDE). The experiments we have carried out show our

method, dubbed KDEy, yields superior quantification performance compared to previous DM approaches and other state-of-the-art quantification systems. © The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature 2025.

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Journal Abbr Mach Learn

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Sentiment analysis, Self-supervised learning, Supervised machine learning, Market researches, Kernel density estimation, Kernel Density Estimation, Prevalence estimation, Learning tasks, Class prevalence estimation, Distribution matching, Mixture model, Mixture modeling, Multiclass quantification

Notes:

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Key Features and Considerations for Using Automated Long-Term Monitoring of Heat-Seeking Behavior of Mosquitoes

Item Type Journal Article

Author C. Sakuma

Author E. Maekawa

Author H. Kanuka

Abstract Mosquitoes are vectors of various deadly infectious diseases, including malaria and dengue fever. Because these pathogens are transmitted by mosquito blood-feeding behavior, it is important to understand how mosquitoes are attracted to their hosts and how they feed on blood. The simplest method is to observe their behavior with the naked eye or video recording. Furthermore, various devices have been invented to

assay mosquito behavior, such as olfactometers. Although each method has distinctive advantages, all have disadvantages, such as limitations in the number of individuals that can be assayed at one time and in observable durations, objective quantification methods, and others. To solve these problems, we have developed an automated device to quantify the carbon dioxide- activated heat-seeking behavior of *Anopheles stephensi* and *Aedes aegypti* with continuous monitoring for up to 1 week. This device, which is detailed in an accompanying protocol, can be applied to search for substances and molecules that affect heat-seeking behavior. It may also be applicable to other hematophagous insects. © 2023 Cold Spring Harbor Laboratory Press. All rights reserved.

Date 2023

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Archive Scopus

Language English

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Journal Abbr Cold Spring Harbor Protoc.

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Tags:

Article, human, Humans, machine learning, behavior, nonhuman, locomotion, videorecording, bioinformatics, animal, Animals, dengue, odor, Aedes, Aedes aegypti, Anopheles, Anopheles stephensi, feeding behavior, Feeding Behavior, heat seeking behavior, high temperature, Hot Temperature, malaria, mosquito, mosquito vector, Mosquito Vectors, sanguivore, static electricity

Notes:

EID: 2-s2.0-85176496100

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Kanuka; Department of Tropical Medicine, The Jikei University School of Medicine, Tokyo, 105-8461, Japan; email: kanuka@jikei.ac.jp

Key Findings on the Role of Corner Frequency in Site-Based Stochastic Ground Motion Models

Item Type Journal Article

Author M. Su

Author M. Dabaghi

Author M. Broccardo

Abstract Site-based stochastic ground motion models (GMMs) are essential for representing seismic input in uncertainty quantification methods for earthquake engineering. A key element controlling the long-period spectral behavior of these models is the high-pass filter's corner frequency (f_c), which must thus be included as a free parameter. In fact, our previous study demonstrated the need to optimize f_c to effectively control the long-period response spectra. This paper builds on and complements our previous study by presenting three key findings. First, for site-based stochastic GMMs, not only f_c 's values but also f_c 's correlation with other GM parameters (e.g., Arias intensity, predominant frequency, and duration) significantly impact the correlation of linear-response spectra $S_a(T)$ at different spectral periods. Second, using a modulated filtered-white-noise GMM with 11 parameters, including the optimized f_c , we effectively fit a large data set comprising 3,243 strong motion records. Specifically, a visual classification indicates that circa 93% of the recorded spectra $S_a(T)$ closely fall within the 2-sigma confidence bounds of the simulated $S_a(T)$. Third, the fitted f_c values contain unrealistic information when the records downloaded from strong-motion databases are overfiltered to remove signal noises. © 2025 American Society of Civil Engineers.

Date 2025

DOI 10.1061/AJRUA6.RUENG-1473

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Issue 3

Journal Abbr ASCE-ASME J. Risk Uncertain. Eng. Syst. Part A. Civ. Eng.

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Tags:

Uncertainty analysis, Quantification methods, Uncertainty quantifications, Stochastic models, Corner frequency, Free parameters, High-pass filter, Key elements, Long period response spectrum, Seismic input, Spectral behaviors, Stochastic ground motion model

Notes:

EID: 2-s2.0-105005186681

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: M. Su; Dept. of Civil Environmental and Mechanical Engineering, Univ. of Trento, Trento, 38123, Italy; email: maijia.su@unitn.it

Knowledge Graph-K-modes Clustering Algorithm for Archival Text Categorization

Item Type Journal Article

Author X.-J. Wu

Author H.-B. Zhang

Abstract Automated classification of electronic archive text's can improve document management efficiency, support rapid information retrieval, and enhance data analysis capabilities, thus helping to achieve efficient use of information resources and knowledge management. However, medium-scale archival text classification needs to deal with complex natural languages as well as deal with noise, ambiguity and sparsity in unstructured text data. In order to solve the above problems, this work proposes a knowledge graph-K-modes clustering algorithm-CTransR-Full-K-Modes for archival text classification. Firstly, existing knowledge representation learning modes and dissimilarity measures in clustering algorithms are analysed. Next, we provide a representation learning model called CTransR. This model is based on hyperparameter tuning and relation matrix projection. Its major purpose is to address the issue of poor recognition capacity in the current TransR model while dealing with distinct entities that have great resemblance. Allocate items and relations into distinct vector spaces. The hyperparameter tuning principle and a priori probability based on the use of relations are employed to distinguish relations that encode similarity, thus facilitating the process of knowledge fusion in archival knowledge mapping. Finally, a full attribute-valued K-Modes algorithm-Full-K-Modes for archival knowledge graphs is proposed to achieve clustering processing of archival texts using preclustering-based multi-attribute-valued clustering centres and their corresponding dissimilarity measures to mitigate the effects of local optimal cases. The experimental results show that the proposed CTransR-Full-K-Modes algorithm can effectively improve the accuracy of the clustering results of ordered-type categorical data, and achieve the purpose of automatic categorisation mining for medium-sized digital archives. © 2025 J. Network Intell. All rights reserved.

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Issue 2

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Resource allocation, Clusterings, Hyper-parameter, K-means clustering, Text classification, Labeled data, Knowledge graph, Knowledge graphs, Knowledge transfer, Image retrieval, Archive classification, Clustering algorithm, Dissimilarity measures, K-mode, K-Modes, K-modes algorithm, Knowledge organization, Online searching, Ordered categorical data, Transr, TransR

Notes:

EID: 2-s2.0-105007468691

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: X.-J. Wu; Archives China Jiliang University, Hangzhou, 310018, China; email: xuejw02@163.com

KULeuven at LeQua 2022: Model Calibration in Quantification Learning

Item Type	Conference Paper
Author	T. Popordanoska
Author	M.B. Blaschko
Editor	Faggioli G.
Editor	Department of Information Engineering, Via Gradenigo 6/b, Padova University of Padova
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Abstract	In many decision-making applications it is of particular interest to obtain aggregated properties about the data sample. Quantification learning (or prevalence estimation) is the task of training predictors via supervised learning to estimate the class distribution of an unlabelled test set. In this paper, we describe our perspective on the "LeQua 2022: Learning to Quantify" challenge, and our entry in "the vector task - T1A," which achieved 1st place on the leaderboard in terms of relative absolute error, and 2nd place in terms of absolute error. We explored the relationship between probabilistic calibration and quantification, concluding that calibration can help with quantification, but this typically requires an i.i.d. assumption or known distribution shift. © 2022 Copyright for this paper by its authors.
Date	2022
URL	https://www.scopus.com/pages/publications/85136979330?origin=resultslist
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Language	English
Extra	Journal Abbreviation: CEUR Workshop Proc.
Volume	3180
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ISBN 16130073 (ISSN)**Pages** 1905-1910**Proceedings Title** CEUR Workshop Proc.**Conference Name** CEUR Workshop Proceedings**Date Added** 20/09/2026, 9:52:21 pm**Modified** 20/09/2026, 9:52:21 pm**Tags:**

Decisions makings, Decision making, Supervised learning, Uncertainty, Bias, Property, bias, Model calibration, Probability distributions, prevalence estimation, Prevalence estimation, Quantification learning, quantification learning, Absolute error, Data sample, uncertainty calibration, Uncertainty calibration

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Export Date: 20 September 2026; Cited By: 2; Funding details: Onderzoeksprogramma Artificiële Intelligentie; Vlaamse regering; Correspondence Address: T. Popordanoska; Processing Speech and Images, Dept. of Electrical Engineering (ESAT), KU Leuven, Belgium; email: teodora.popordanoska@kuleuven.be; Conference name: 23rd Working Notes of Conference and Labs of the Evaluation Forum, CLEF 2022; Conference date: 5 September 2022 through 8 September 2022; Conference code: 181762

Label Shift Correction via Bidirectional Marginal Distribution Matching

Item Type Conference Paper**Author** R. Fan**Author** X. Ouyang**Author** H. Tao**Author** C. Hou

Abstract Due to the timeliness and uncertainty of data acquisition, label shift, which assumes that the source (training) and target (test) label distributions differ, occurs with the changing environment and reduces the generalization ability of traditional models. To correct the label shift, existing methods estimate the true label distribution by prediction of target data from a source classifier, which results in high variance, especially with large label shift. In this paper, we tackle this problem by proposing a novel approach termed as Label Shift Correction via Bidirectional Marginal Distribution Matching (BMDM). Our approach matches the label and feature marginal distributions simultaneously to ensure the stability of estimated class proportions. We prove theoretically that there is a unique optimal solution, i.e., true target label distribution, for our approach under mild conditions, and an efficient optimization strategy is also proposed. On this basis, in multi-shot scenario where label distribution changes continuously, we extend BMDM by designing a new distribution matching mechanism and constructing a regularization term that constrains the direction of label distribution change.

Extensive experimental results validate the effectiveness of our approach over existing state-of-the-arts methods. © 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Date	2024
DOI	10.1145/3637528.3671867
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Language	English
Extra	Journal Abbreviation: Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.
Publisher	Association for Computing Machinery
ISBN	2154817X (ISSN); 979-840070490-1 (ISBN)
Pages	735-746
Proceedings Title	Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.
Conference Name	Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining
Date Added	20/09/2026, 9:52:08 pm
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Tags:

Uncertainty, Traditional models, Label distribution, Label shift, Marginal distribution, label shift, Distribution matching, bidirectional, Bidirectional, Changing environment, distribution matching, Generalization ability, Optimal solutions

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Label Shift Estimation for Class-Imbalance Problem: A Bayesian Approach

Item Type	Conference Paper
Author	C. Ye
Author	R. Tsuchida
Author	L. Petersson
Author	N. Barnes
Abstract	As a type of distribution shift, label shift occurs when the source and target domains have different label distributions $P(Y)$ but identical conditional distributions of data given labels $P(X Y)$. Under a Bayesian framework, we

propose a novel Maximum A Posteriori (MAP) model and a novel posterior sampling model for the label shift problem. We prove the MAP objective admits a unique optimum and derive an EM algorithm that converges to the global optimum. We propose a novel Adaptive Prior Learning (APL) model to adaptively select prior parameters given data. We use the Markov Chain Monte Carlo (MCMC) method in our posterior sampling model to estimate and correct for label shift. Our methods can effectively resolve class imbalance problems on large-scale datasets without fine-tuning the classifier. Experiments show that our model outperforms existing methods on a variety of label shift settings. Our code is available at <https://github.com/ChangkunYe/MAPLS/>. © 2024 IEEE.

Date 2024

DOI 10.1109/WACV57701.2024.00111

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Language English

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Pages 1062-1071

Proceedings Title Proc. - IEEE Winter Conf. Appl. Comput. Vis., WACV

Conference Name Proceedings - 2024 IEEE Winter Conference on Applications of Computer Vision, WACV 2024

Date Added 20/09/2026, 9:52:12 pm

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Tags:

Classification (of information), Large datasets, Algorithms, Bayesian networks, Image recognition, Target domain, Label distribution, Monte Carlo methods, EM algorithms, Conditional distribution, Shift estimations, Image recognition and understanding, Class imbalance problems, Bayesian approaches, Bayesian frameworks, Posteriori, Sampling model

Notes:

EID: 2-s2.0-85191962468

Export Date: 20 September 2026; Cited By: 11; Conference name: 2024 IEEE Winter Conference on Applications of Computer Vision, WACV 2024; Conference date: 4 January 2024 through 8 January 2024; Conference code: 198735

Label Shift Estimation With Incremental Prior Update

Item Type Conference Paper

Author Y. Zhang

Author G. Batista

Author S.S. Kanhere

Abstract An assumption often made in supervised learning is that the training and testing sets have the same label distribution. However, in real-life scenarios, this assumption rarely holds. For example, medical diagnosis result distributions change over time and across locations; fraud detection models must adapt as patterns of fraudulent activity shift; the category distribution of social media posts changes based on trending topics and user demographics. In the task of label shift estimation, the goal is to estimate the changing label distribution $p(y)$ in the testing set, assuming the likelihood $p(x|y)$ does not change, implying no concept drift. In this paper, we propose a new approach for post-hoc label shift estimation, unlike previous methods that perform moment matching with confusion matrix estimated from a validation set or maximize the likelihood of the new data with an expectation-maximization algorithm. We aim to incrementally update the prior on each sample, adjusting each posterior for more accurate label shift estimation. The proposed method is based on intuitive assumptions on classifiers that are generally true for modern probabilistic classifiers. The proposed method relies on a weaker notion of calibration compared to other methods. As a post-hoc approach for label shift estimation, the proposed method is versatile and can be applied to any black-box probabilistic classifier. Experiments on CIFAR-10 and MNIST show that the proposed method consistently outperforms the current state-of-the-art maximum likelihood-based methods under different calibrations and varying intensities of label shift. Copyright © 2025 by SIAM.

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Language English

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Publisher Society for Industrial and Applied Mathematics Publications

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Proceedings Title SIAM Int. Conf. Data Mining, SDM

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Tags:

Classification (of information), Detection models, Pattern recognition, Social media, Fraud detection, Training sets, Label distribution, Expectation maximization algorithm, Maximum principle, Shift estimations, Testing sets, Maximum likelihood estimation, Change-over time, Training and testing, Probabilistic classifiers

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: Y. Zhang; University of New South Wales, School of Computer Science and Engineering, Australia; email: yunrui.zhang@unsw.edu.au; Conference name: 2025 SIAM International Conference on Data Mining, SDM 2025; Conference date: 1 May 2025 through 3 May 2025; Conference code: 209320

Label Shift Meets Online Learning: Ensuring Consistent Adaptation with Universal Dynamic Regret

Item Type Conference Paper

Author Y. Dai

Author S. Gu

Author R. Fan

Author C. Xu

Author C. Hou

Abstract Label shift, which investigates the adaptation of label distributions between the fixed source and target domains, has attracted significant research interests and broad applications in offline settings. In real-world scenarios, however, data often arrives as a continuous stream. Addressing label shift in online learning settings is paramount. Existing strategies, which tailor traditional offline label shift techniques to online settings, have degraded performance due to the inconsistent estimation of label distributions and violation of convex assumption for theoretical guarantee. In this paper, we propose a novel method to ensure consistent adaptation to online label shift. We construct a new convex risk estimator that is pivotal for both online optimization and theoretical analysis. Furthermore, we enhance an optimistic online algorithm as the base learner and refine the classifier using an ensemble method. Theoretically, we derive a universal dynamic regret which achieves minimax optimal. Extensive experiments on both real-world datasets and human motion task demonstrate the superiority of our method comparing existing methods. © 2025 IEEE.

Date 2025

DOI 10.1109/CVPR52734.2025.01434

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Language English

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Label-free DNA quantification by multi-frequency impedance cytometry and machine learning analysis

Item Type Conference Paper

Author J. Sui

Author N. Gandotra

Author P. Xie

Author Z. Lin

Author C. Scharfe

Author M. Javanmard

Abstract DNA quantification plays an important role in clinical diagnosis and research. A number of DNA quantification methods have been established, such as fluorescence imaging. In this work, we demonstrate a new label-free approach for DNA recognition by using multi-frequency impedance cytometry with machine learning. © 17CBMS-0001.

Date 2020

URL <https://www.scopus.com/pages/publications/85079606036?origin=resultslist>

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Language English

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Tags:

Learning systems, Machine learning, Diagnosis, Clinical research, DNA quantification, DNA, Label free, Clinical diagnosis, Cytometry, DNA quantification method, DNA recognition, Fluorescence imaging, Impedance cytometry, Label-free, Label-free approach, Multi frequency

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Export Date: 20 September 2026; Cited By: 2; Funding details: Robert Lorber from Microelectronics Research Laboratory; National Science Foundation, NSF, (IDBR 1556253); Mitsubishi Electric Research Laboratories, MERL; Correspondence Address: J. Sui; Department of Electrical and Computer Engineering, Rutgers University, United States; email: jianye.sui@rutgers.edu; Conference name: 21st International Conference on Miniaturized Systems for Chemistry and Life Sciences, MicroTAS 2017; Conference date: 22 October 2017 through 26 October 2017; Conference code: 157376

Label-free hyperspectral imaging and quantification methods for surgical margin assessment of tissue specimens of cancer patients

Item Type Conference Paper

Author B. Fei

Author G. Lu

Author M.T. Halicek

Author X. Wang

Author H. Zhang

Author J.V. Little

Author K.R. Magliocca

Author M. Patel

Author C.C. Griffith

Author M.W. El-Deiry

Author A.Y. Chen

Abstract Hyperspectral imaging (HSI) is a relatively new modality in medicine and can have many potential applications. In this study, we developed label-free hyperspectral imaging for tumor margin assessment. HSI data, hypercube (x, y, λ) , consists of a series of images of the same field of view that are acquired at different wavelengths. Every pixel in the hypercube has an optical spectrum. We collected surgical tissue specimens from 16 human subjects who underwent head and neck (H&N) cancer surgery. We acquired both HSI, autofluorescence images, and fluorescence images with 2-NBDG and proflavine from the specimens. Digitized histologic slides were examined by an H&N pathologist. We developed image preprocessing and classification methods for HSI data and differentiate cancer from benign tissue. The hyperspectral imaging and classification method was able to distinguish between cancer and normal tissue from oral cavity with an average accuracy of $90 \pm 8\%$, sensitivity of $89 \pm 9\%$, and specificity of $91 \pm 6\%$. This study suggests that label-free hyperspectral imaging has great potential for surgical margin assessment in tissue specimens of H&N cancer patients. Further development of the imaging technology and quantification methods is warranted

for its application in image-guided surgery. © 2017 IEEE.

Date 2017

DOI 10.1109/EMBC.2017.8037743

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Language English

Extra Journal Abbreviation: Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. EMBS

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Proceedings Title Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. EMBS

Conference Name Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS

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Label-free imaging of M1 and M2 macrophage phenotypes in the human dermis in vivo using two-photon excited FLIM

Item Type Journal Article

Author M. Kröger

Author J. Scheffel

Author E.A. Shirshin

Author J. Schleusener

Author M.C. Meinke

Author J. Lademann

Author M. Maurer

Author M.E. Darwin

Abstract Macrophages (MΦs) are important immune effector cells that promote (M1 MΦs) or inhibit (M2 MΦs) inflammation and are involved in numerous physiological and pathogenic immunoresponses. Their precise role and relevance, however, are not fully understood for lack of noninvasive quantification methods. Here, we show that two-photon excited fluorescence lifetime imaging (TPE-FLIM), a label-free noninvasive method, can visualize MΦs in the human dermis *in vivo*. We demonstrate *in vitro* that human dermal MΦs exhibit specific TPE-FLIM properties that distinguish them from the main components of the extracellular matrix and other dermal cells. We visualized MΦs, their phenotypes and phagocytosis in the skin of healthy individuals *in vivo* using TPE-FLIM. Additionally, machine learning identified M1 and M2 MΦs with a sensitivity of 0.88 ± 0.04 and 0.82 ± 0.03 and a specificity of 0.89 ± 0.03 and 0.90 ± 0.03 , respectively. In clinical research, TPE-FLIM can advance the understanding of the role of MΦs in health and disease. © Kröger et al.

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EID: 2-s2.0-85140415321

Export Date: 20 September 2026; Cited By: 30; Funding details: Donor Association for German Science and Humanities; Evelin Hagen; Loris Busch; Niklas Mahnke; Russian Science Foundation, RSF, (19-75-10077); Russian Science Foundation, RSF; Correspondence Address: M.E. Darwin; Charité – Universitätsmedizin Berlin, corporate member of Freie Universität Berlin, Humboldt-Universität zu Berlin, Berlin Institute of Health, Department of Dermatology, Venerology and Allergology, Berlin, Germany; email: maxim.darwin@protonmail.com

Label-free reflectance hyperspectral imaging for tumor margin assessment: A pilot study on surgical specimens of cancer patients

Item Type Journal Article

Author B. Fei

Author G. Lu

Author X. Wang

Author H. Zhang

Author J.V. Little

Author M.R. Patel

Author C.C. Griffith

Author M.W. El-Diery

Author A.Y. Chen

Abstract A label-free, hyperspectral imaging (HSI) approach has been proposed for tumor margin assessment. HSI data, i.e., hypercube (x, y, λ) , consist of a series of high-resolution images of the same field of view that are acquired at different wavelengths. Every pixel on an HSI image has an optical spectrum. In this pilot clinical study, a pipeline of a machine-learning-based quantification method for HSI data was implemented and evaluated in patient specimens. Spectral features from HSI data were used for the classification of cancer and normal tissue. Surgical tissue specimens were collected from 16 human patients who underwent head and neck (H&N) cancer surgery. HSI, autofluorescence images, and fluorescence images with 2-deoxy-2-[(7-nitro-2,1,3-benzoxadiazol-4-yl)amino]-D-glucose (2-NBDG) and proflavine were acquired from each specimen. Digitized histologic slides were examined by an H&N pathologist. The HSI and classification method were able to distinguish between cancer and normal tissue from the oral cavity with an average accuracy of $90\% \pm 8\%$, sensitivity of $89\% \pm 9\%$, and specificity of $91\% \pm 6\%$. For tissue specimens from the thyroid, the method achieved an average accuracy of $94\% \pm 6\%$, sensitivity of $94\% \pm 6\%$, and specificity of $95\% \pm 6\%$. HSI outperformed autofluorescence imaging or fluorescence imaging with vital dye (2-NBDG or proflavine). This study demonstrated the feasibility of label-free, HSI for tumor margin assessment in surgical tissue specimens of H&N cancer patients. Further development of the HSI technology is warranted for its application in image-guided surgery. © 2021 The Authors.

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Notes:

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LAMDA: Label Matching Deep Domain Adaptation

Item Type Conference Paper

Author T. Le

Author T. Nguyen

Author N. Ho

Author H. Bui

Author D. Phung

Abstract Deep domain adaptation (DDA) approaches have recently been shown to perform better than their shallow rivals with better modeling capacity on complex domains (e.g., image, structural data, and sequential data). The underlying idea is to learn domain invariant representations on a latent space that can bridge the gap between source and target domains. Several theoretical studies have established insightful understanding and the benefit of learning domain invariant features; however, they are usually limited to the case where there is no label shift, hence hindering its applicability. In this paper, we propose and study a new challenging setting that allows us to use a Wasserstein distance (WS) to not only quantify the data shift but also to define the label shift directly. We further develop a theory to demonstrate that minimizing the WS of the data shift leads to closing the gap between the source and target data distributions on the latent space (e.g., an intermediate layer of a deep net), while still being able to quantify the label shift with respect to this latent space. Interestingly, our theory can consequently

explain certain drawbacks of learning domain invariant features on the latent space. Finally, grounded on the results and guidance of our developed theory, we propose the Label Matching Deep Domain Adaptation (LAMDA) approach that outperforms baselines on real-world datasets for DA problems. Copyright © 2021 by the author(s)

Date 2021

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Language English

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Volume 139

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Proceedings Title Proc. Mach. Learn. Res.

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Air navigation, Machine learning, Domain adaptation, Learn+, Complex domains, Invariant representation, Invariant features, Wasserstein distance, Structural data, Datum shifts, Label Matching, Sequential data

Notes:

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Land Resources for Wind Energy Development Requires Regionalized Characterizations

Item Type Journal Article

Author T. Dai

Author J.M. Jose Valanarasu

Author Y. Zhao

Author S. Zheng

Author Y. Sun

Author V.M. Patel

Author S.M. Jordaan

Abstract Estimates of the land area occupied by wind energy differ by orders of magnitude due to data scarcity and inconsistent methodology. We developed a method that combines machine learning-based imagery analysis and geographic information systems and examined the land area of 318 wind farms (15,871 turbines) in the U.S. portion of the Western Interconnection. We found that prior land use and human modification in the project area are critical for land-use efficiency and land transformation of wind projects. Projects developed in areas with little human modification have a land-use efficiency of 63.8 ± 8.9 W/m² (mean $\pm$ 95% confidence interval) and a land transformation of 0.24 ± 0.07 m²/MWh, while values for projects in areas with high human modification are 447 ± 49.4 W/m² and 0.05 ± 0.01 m²/MWh, respectively. We show that land resources for wind can be quantified consistently with our replicable method, a method that obviates >99% of the workload using machine learning. To quantify the peripheral impact of a turbine, buffered geometry can be used as a proxy for measuring land resources and metrics when a large enough impact radius is assumed (e.g., >4 times the rotor diameter). Our analysis provides a necessary first step toward regionalized impact assessment and improved comparisons of energy alternatives. © 2024 American Chemical Society.

Date 2024

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LaSCal: Label-Shift Calibration without target labels**Item Type** Conference Paper**Author** T. Popordanoska**Author** G. Radevski**Author** T.T.M.B. Blaschko**Editor** Globerson A.**Editor** Mackey L.**Editor** Belgrave D.**Editor** Fan A.**Editor** Paquet U.**Editor** Tomczak J.**Editor** Zhang C.

Abstract When machine learning systems face dataset shift, model calibration plays a pivotal role in ensuring their reliability. Calibration error (CE) provides insights into the alignment between the predicted confidence scores and the classifier accuracy. While prior works have delved into the implications of dataset shift on calibration, existing CE estimators either (i) assume access to labeled data from the target domain, often unavailable in practice, or (ii) are derived under a covariate shift assumption. In this work we propose a novel, label-free, consistent CE estimator under label shift. Label shift is characterized by changes in the marginal label distribution $p(Y)$, with a constant conditional $p(X|Y)$ distribution between the source and target. We introduce a novel calibration method, called LaSCal, which uses the estimator in conjunction with a post-hoc calibration strategy, to perform unsupervised calibration on the target distribution. Our thorough empirical analysis demonstrates the effectiveness and reliability of the proposed approach across different modalities, model architectures and label shift intensities. © 2024 Neural information processing systems foundation. All rights reserved.

Date 2024**URL** <https://www.scopus.com/pages/publications/105000512086?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Adv. neural inf. proces. syst.**Volume** 37**Publisher** Neural information processing systems foundation

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Laser-Induced Breakdown Spectroscopy Quantitative Analysis Using a Bayesian Optimization-Based Tunable Softplus Backpropagation Neural Network

- Item Type** Journal Article
- Author** X. Xu
 - Author** S. Luo
 - Author** X. Zhang
 - Author** W. Xu
 - Author** R. Shu
 - Author** J. Wang
 - Author** X. Liu
 - Author** P. Li
 - Author** C. Li
 - Author** L. Li

Abstract Laser-induced breakdown spectroscopy (LIBS) has played a critical role in Mars exploration missions, substantially contributing to the geochemical analysis of Martian surface substances. However, the complex nonlinearity of LIBS processes can considerably limit the quantification accuracy of conventional LIBS chemometric methods. Hence chemometrics based on artificial neural network (ANN) algorithms have become increasingly popular in LIBS analysis due to their extraordinary ability in nonlinear feature modeling. The hidden layer activation functions are key to ANN model performance, yet common activation functions usually suffer from problems such as gradient vanishing (e.g., Sigmoid and Tanh) and dying neurons (e.g., ReLU).

In this study, we propose a novel LIBS quantification method, named the Bayesian optimization-based tunable Softplus backpropagation neural network (BOTS-BPNN). Based on a dataset comprising 1800 LIBS spectra collected by a laboratory duplicate of the MarSCoDe instrument onboard the Zhurong Mars rover, we have revealed that a BPNN model adopting a tunable Softplus activation function can achieve higher prediction accuracy than BPNN models adopting other common activation functions if the tunable Softplus parameter β is properly selected. Moreover, the way to find the proper β value has also been investigated. We demonstrate that the Bayesian optimization method surpasses the traditional grid search method regarding both performance and efficiency. The BOTS-BPNN model also shows superior performance over other common machine learning models like random forest (RF). This work indicates the potential of BOTS-BPNN as an effective chemometric method for analyzing Mars in situ LIBS data and sheds light on the use of chemometrics for data analysis in future planetary explorations. © 2025 by the authors.

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LEAP: Linear equations for classifier accuracy prediction under prior probability shift

Item Type Journal Article

Author L. Volpi

Author A. Moreo

Author F. Sebastiani

Abstract The standard technique for predicting the accuracy that a classifier will have on unseen data (classifier accuracy prediction—CAP) is cross-validation (CV). However, CV relies on the assumption that the training data and the test data are sampled from the same distribution, an assumption that is often violated in many real-world scenarios. When such violations occur (i.e., in the presence of dataset shift), the estimates returned by CV are unreliable. The contribution of this paper is three-fold. First, we propose a CAP method specifically designed to work under prior probability shift (PPS), an instance of dataset shift in which the training and test distributions are characterized by different class priors. This method estimates the entries of the contingency table of the test data (thus allowing to estimate the value of any specific evaluation measure) by solving a system of independent linear equations, with n the number of classes. Second, we show that the equations that the cells of the contingency table must satisfy are actually more than, which gives rise to an overconstrained problem, and present a family of methods each based on a different selection of such equations. Third, we observe that, since a key step of the above methods involves predicting the class priors of the test data, one can exploit intuitions from the field of class prior estimation (a.k.a. “quantification”). Our experiments show that, when combined with state-of-the-art quantification techniques, under PPS our methods tend to outperform existing CAP methods. © The Author(s) 2025.

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Issue 12**Journal Abbr** Mach Learn**Date Added** 20/09/2026, 9:51:36 pm**Modified** 20/09/2026, 9:51:36 pm**Tags:**

Classification (of information), Forecasting, Statistical tests, Data accuracy, Probability distributions, Test data, Label shift, Probability shifts, Accuracy prediction, Classifier accuracy prediction, Cross validation, Dataset shifts, Linear equations, Prior probability, Prior probability shift, Quantification, Test facilities

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Learn to classify and count: A unified framework for object classification and counting

Item Type Conference Paper**Author** Z. Song**Author** Q. Qiu

Abstract In this paper, we focus on the problem of object classification and counting in natural scenes. The existing methods of instance detection can be utilized to classify and count. These works depend on the bounding box annotations, which requires time-consuming human labeling effort. Meanwhile, traditional methods of object counting are mostly based on the class-insensitive density map, therefore they are only able to count one specific category. We propose a unified framework for object classification and counting to overcome these drawbacks of existing works. We employ a two-stream convolutional neural network (CNN), which consists of classification and counting branches. Considering the correlation between classification and counting tasks, we propose a novel correlation loss function to coordinate representations learned by both two branches of network. Without hand-labeled bounding box annotations as supervision information, our end-to-end model can recognize the category of objects in the image and count object numbers per category. The proposed method is evaluated on PASCAL VOC 2007 dataset in multi-label classification and object counting tasks. We have achieved improved performance on both two tasks, which shows the effectiveness of our proposed method. © 2018 Association for Computing Machinery.

Date 2018

DOI 10.1145/3191442.3191459**URL** <https://www.scopus.com/pages/publications/85047333117?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: ACM Int. Conf. Proc. Ser.**Publisher** Association for Computing Machinery**ISBN** 978-145036367-9 (ISBN)**Pages** 110-114**Proceedings Title** ACM Int. Conf. Proc. Ser.**Conference Name** ACM International Conference Proceeding Series**Date Added** 20/09/2026, 9:52:48 pm**Modified** 20/09/2026, 9:52:48 pm**Tags:**

Classification (of information), Convolution, Convolutional neural network, Image segmentation, Neural networks, Object classification, Multi label classification, Object counting, Multi-label Classification, Unified framework, Convolutional Neural Networks (CNN), Coordinate representations, Correlation loss, Correlation Loss, Object Counting, Two-stream Convolutional Neural Network

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Export Date: 20 September 2026; Cited By: 1; Conference name: 2018 International Conference on Image and Graphics Processing, ICIGP 2018; Conference date: 24 February 2018 through 26 February 2018; Conference code: 136347

Learning automata based classifier

Item Type Journal Article**Author** S.-H. Zahiri

Abstract In this paper a new classifier has been designed based on the learning automata. This classifier can efficiently approximate the decision hyperplanes in the feature space without need to know the class distributions and the a priori probabilities. The performance of the proposed classifier has been tested on different kinds of benchmarks with nonlinear, overlapping class boundaries and different feature space dimensions. Extensive experimental results on these data sets are provided to show that the performance of the proposed classifier is comparable to, sometimes better than multi-layer perceptron, k-nearest neighbor classifier, genetic classifier, and particle swarm classifier. Also the comparative results are provided to show the effectiveness of the proposed method in comparison to similar researches. Furthermore the effect of the number of training points on the performance of the designed classifier is investigated. It is found that as the number of training data increases, the performance of the classifier tends to the performance of Bayes classifier which is an optimal one. © 2007 Elsevier B.V. All rights reserved.

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Export Date: 20 September 2026; Cited By: 31; Correspondence Address: S.-H. Zahiri; Department of Electrical Engineering, Faculty of Engineering, Birjand University, Birjand, P.O. Box 97175-376, Iran; email: hzahiri@birjand.ac.ir

Learning Bayesian classifiers from positive and unlabeled examples

Item Type Journal Article

Author B. Calvo

Author P. Larrañaga

Author J.A. Lozano

Abstract The positive unlabeled learning term refers to the binary classification problem in the absence of negative examples. When only positive and unlabeled instances are available, semi-supervised classification algorithms cannot be directly applied, and thus new algorithms are required. One of these positive unlabeled learning algorithms is the positive naive Bayes (PNB), which is an adaptation of the naive Bayes induction algorithm that does not require negative instances. In this work we propose two ways of enhancing this algorithm. On one hand, we have taken the concept behind PNB one step further, proposing a procedure to build more complex Bayesian classifiers in the absence of negative instances. We present a new algorithm (named positive tree augmented naive Bayes, PTAN) to obtain tree augmented naive Bayes models in the positive unlabeled domain. On the other hand, we propose a new Bayesian approach to deal with the a priori probability of the positive class that models the uncertainty over this parameter by means of a Beta distribution. This

approach is applied to both PNB and PTAN, resulting in two new algorithms. The four algorithms are empirically compared in positive unlabeled learning problems based on real and synthetic databases. The results obtained in these comparisons suggest that, when the predicting variables are not conditionally independent given the class, the extension of PNB to more complex networks increases the classification performance. They also show that our Bayesian approach to the a priori probability of the positive class can improve the results obtained by PNB and PTAN. © 2007 Elsevier B.V. All rights reserved.

Date 2007

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Learning classifiers on positive and unlabeled data with policy gradient

Item Type Conference Paper

Author T. Li

Author C.-C. Wang

Author Y. Ma

Author P. Ortal

Author Q. Zhao
Author B. Stenger
Author Y. Hirate
Editor Wang J.
Editor Shim K.
Editor Wu X.

Abstract Existing algorithms aiming to learn a binary classifier from positive (P) and unlabeled (U) data generally require estimating the class prior or label noises ahead of building a classification model. However, the estimation and classifier learning are normally conducted in a pipeline instead of being jointly optimized. In this paper, we propose to alternatively train the two steps using reinforcement learning. Our proposal adopts a policy network to adaptively make assumptions on the labels of unlabeled data, while a classifier is built upon the output of the policy network and provides rewards to learn a better strategy. The dynamic and interactive training between the policy maker and the classifier can exploit the unlabeled data in a more effective manner and yield a significant improvement on the classification performance. Furthermore, we present two different approaches to represent the actions sampled from the policy. The first approach considers continuous actions as soft labels, while the other uses discrete actions as hard assignment of labels for unlabeled examples. We validate the effectiveness of the proposed method on two benchmark datasets as well as one e-commerce dataset. The result shows the proposed method is able to consistently outperform state-of-the-art methods in various settings. © 2019 IEEE.

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: T. Li; Rakuten Institute of Technology, Tokyo, Japan; email: tianyu.li@rakuten.com; Conference name: 19th IEEE International Conference on Data Mining, ICDM 2019; Conference date: 8 November 2019 through 11 November 2019; Conference code: 157223

Learning From Incomplete and Inaccurate Supervision

Item Type Journal Article

Author Z.-Y. Zhang

Author P. Zhao

Author Y. Jiang

Author Z.-H. Zhou

Abstract In plenty of real-life tasks, strongly supervised information is hard to obtain, and thus weakly supervised learning has drawn considerable attention recently. This paper investigates the problem of learning from incomplete and inaccurate supervision, where only a limited subset of training data is labeled but potentially with noise. This setting is challenging and of great importance but rarely studied in the literature. We notice that in many applications, the limited labeled data are with certain structures, which paves us a way to design effective methods. Specifically, we observe that labeled data are usually with one-sided noise such as the bug detection task, where the identified buggy codes are indeed with defects, while codes checked many times or newly fixed may still have other flaws. Furthermore, when there occurs two-sided noise in the labeled data, we exploit the class-prior information of unlabeled data, which is typically available in practical tasks. We propose novel approaches for the incomplete and inaccurate supervision learning tasks and effectively alleviate the negative influence of label noise with the help of a vast number of unlabeled data. Both theoretical analysis and extensive experiments justify and validate the effectiveness of the proposed approaches. © 1989-2012 IEEE.

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Notes:

EID: 2-s2.0-85101732712

Export Date: 20 September 2026; Cited By: 22; Correspondence Address: Y. Jiang; National Key Laboratory for Novel Software Technology, Nanjing University, Nanjing, 210023, China; email: jiangy@lamda.nju.edu.cn; CODEN: ITKEE

Learning from positive and unlabeled data without explicit estimation of class prior

Item Type Conference Paper

Author C. Zhang

Author Y. Hou

Author Y. Zhang

Abstract Learning a classifier from positive and unlabeled data may occur in various applications. It differs from the standard classification problems by the absence of labeled negative examples in the training set. So far, two main strategies have typically been used for this issue: the likely negative examples-based strategy and the class prior-based strategy, in which the likely negative examples or the class prior is required to be obtained in a preprocessing step. In this paper, a new strategy based on the Bhattacharyya coefficient is put forward, which formalizes this learning problem as an optimization problem and does not need a preprocessing step. We first show that with the given positive class conditional probability density function (PDF) and the mixture PDF of both the positive class and the negative class, the class prior can be estimated by minimizing the Bhattacharyya coefficient of the positive class with respect to the negative class. We then show how to use this result in an implicit mixture model of restricted Boltzmann machines to estimate the positive class conditional PDF and the negative class conditional PDF directly to obtain a classifier without the explicit estimation of the class prior. Many experiments on real and synthetic datasets illustrated the superiority of the proposed approach. Copyright © 2020, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Learning from positive and unlabeled examples -Finite size sample bounds

Item Type Conference Paper
Author F. Mansouri
Author S. Ben-David
Abstract PU (Positive Unlabeled) learning is a variant of supervised classification learning in which the only labels revealed to the learner are of positively labeled instances. PU learning arises in many real-world applications. Most existing work relies on the simplifying assumptions that the positively labeled training data is drawn from the restriction of the data generating distribution to positively labeled instances and/or that the proportion of positively labeled points (a.k.a. the class prior) is known apriori to the learner. This paper provides a theoretical analysis of the statistical complexity of PU learning under a wider range of setups. Unlike most prior work, our study does not assume that the class prior is known to the learner. We prove upper and lower bounds on the required sample sizes (of both the positively labeled and the unlabeled samples). © 2025 Neural information processing systems foundation. All rights reserved.
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Learning Groundwater Contaminant Diffusion-Sorption Processes With a Finite Volume Neural Network

Item Type Journal Article

Author T. Praditia

Author M. Karlbauer

Author S. Otte

Author S. Oladyshkin

Author M.V. Butz

Author W. Nowak

Abstract Improved understanding of complex hydrosystem processes is key to advance water resources research. Nevertheless, the conventional way of modeling these processes suffers from a high conceptual uncertainty, due to almost ubiquitous simplifying assumptions used in model parameterizations/closures. Machine learning (ML) models are considered as a potential alternative, but their generalization abilities remain limited. For example, they normally fail to predict accurately across different boundary conditions. Moreover, as a black box, they do not add to our process understanding or to discover improved parameterizations/closures. To tackle this issue, we propose the hybrid modeling framework FINN (finite volume neural network). It merges existing numerical methods for partial differential equations (PDEs) with the learning abilities of artificial neural networks (ANNs). FINN is applied on discrete control volumes and learns components of the investigated system equations, such as numerical stencils, model parameters, and arbitrary closure/constitutive relations. Consequently, FINN yields highly interpretable results. We demonstrate FINN's potential on a diffusion-sorption problem in clay. Results on numerically generated data show that FINN outperforms other ML models when tested under modified boundary conditions, and that it can successfully differentiate between the usual, known sorption isotherms. Moreover, we also equip FINN with uncertainty quantification methods to lay open the total uncertainty of scientific learning, and then apply it to a laboratory experiment. The results show that FINN performs better than calibrated PDE-based models as it is able to flexibly learn and model sorption isotherms without being restricted to choose among available

parametric models. © 2022. The Authors.

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Learning individual roles from video in a smart home

Item Type Conference Paper

Author O. Brdiczka

Author J. Maisonnasse

Author P. Reignier

Author J.L. Crowley

Abstract This paper addresses learning and recognition of individual roles from video data in a smart home environment. The proposed approach is part of a framework for acquiring a high-level contextual model for human behaviour in an intelligent environment. The proposed methods for role learning and recognition are based on Bayesian models. The input is the targets and their properties generated and tracked by a robust video tracking system in the environment. The output is the roles "walking", "standing", "sitting", "interacting with table", "sleeping" for each target. A Bayesian classifier produced good results for a framewise classification of these roles, while a hidden Markov model had even better performance taking into account a priori probabilities of roles and role transitions. A support vector machine produced best classification results. The classifiers had, however, problems to distinguish ambiguous roles like "walking" and "standing" in the environment. The obtained results permit to pass to the next step in future work: learning and recognizing relations and situation.

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Notes:

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Learning Label Shift Correction for Test-Agnostic Long-Tailed Recognition

Item Type Conference Paper

Author T. Wei

Author Z. Mao

Author Z.-H. Zhou

Author Y. Wan

Author M.-L. Zhang

Editor Salakhutdinov R.

Editor Kolter Z.

Editor Heller K.

Editor Weller A.

Editor Oliver N.

Editor Scarlett J.

Editor Berkenkamp F.

Abstract Long-tail learning primarily focuses on mitigating the label distribution shift between long-tailed training data and uniformly distributed test data. However, in real-world applications, we often encounter a more intricate challenge where the test label distribution is agnostic. To address this problem, we first theoretically establish the substantial potential for reducing the generalization error if we can precisely estimate the test label distribution. Motivated by the theoretical insight, we introduce a simple yet effective solution called label shift correction (LSC). LSC estimates the test label distribution within the proposed framework of generalized black box shift estimation, and adjusts the predictions from a pre-trained model to align with the test distribution. Theoretical analyses confirm that accurate estimation of test label distribution can effectively reduce the generalization error. Extensive experimental results demonstrate that our method significantly outperforms previous state-of-the-art approaches, especially when confronted with non-uniform test label distribution. Notably, the proposed method is general and complements existing long-tail learning approaches, consistently improving their performance. The source code is available at <https://github.com/Stomach-ache/label-shift-correction>. Copyright 2024 by the author(s)

Date 2024

URL <https://www.scopus.com/pages/publications/85203831494?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Mach. Learn. Res.

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Proceedings Title Proc. Mach. Learn. Res.

Conference Name Proceedings of Machine Learning Research

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Tags:

Real-world, Adversarial machine learning, Contrastive Learning, Black boxes, Training data, Label distribution, Simple++, Test data, Shift estimations, Generalization Error, Black-box testing, Effective solution, Long tail

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Export Date: 20 September 2026; Cited By: 6; Funding details: Big Data Computing Center of Southeast University; National Natural Science Foundation of China, NSFC, (62206049, 62225602); National Natural Science Foundation of China, NSFC; Correspondence Address: T. Wei; School of Computer Science and Engineering, Southeast University, Nanjing, China; email: weit@seu.edu.cn; Conference name: 41st International Conference on Machine Learning, ICML 2024; Conference date: 21 July 2024 through 27 July 2024; Conference code: 201670

Learning Minority Class prior to Minority Oversampling

Item Type Conference Paper

Author P. Sadhukhan

Abstract The success of minority oversampling in dealing with class-imbalanced dataset is well manifested by existing approaches. But that do not guarantee the true class of the synthetic minority points. We address the given context in this paper, Learning Minority Class prior to Minority Oversampling (LMCMO). To guarantee the class information of synthetic minority points, we estimate the minority spaces before generating the synthetic minority points. The performance efficiency of the LMCMO oversampled dataset is tested on C4.5 decision tree and Linear Support Vector Machine (SVM) classifier. Empirical evaluations on 21 datasets using four diversified metrics indicate substantial improvement in Linear SVM outcomes of the proposed method over four competing methods. A modest but still significant gain is achieved by our method over other methods on classification using C4.5 decision tree. © 2019 IEEE.

Date 2019

DOI 10.1109/IJCNN.2019.8852188

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc Int Jt Conf Neural Networks

Volume 2019-July

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-172811985-4 (ISBN)

Proceedings Title Proc Int Jt Conf Neural Networks

Conference Name Proceedings of the International Joint Conference on Neural Networks

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Tags:

Support vector machines, Classification (of information), Decision trees, Over sampling, oversampling, Empirical evaluations, Class imbalance learning, Linear Support Vector Machines, C4.5 decision trees, learning minority class, Performance efficiency, synthetic minority points

Notes:

EID: 2-s2.0-85073249601

Export Date: 20 September 2026; Cited By: 8; Correspondence Address: P. Sadhukhan; Machine Intelligence Unit, Indian Statistical Institute, Kolkata, India; email: payel0410@gmail.com; Conference name: 2019 International Joint Conference on Neural Networks, IJCNN 2019; Conference date: 14 July 2019 through 19 July 2019; Conference code: 152291; CODEN: 85OFA

Learning self-driven collective dynamics with graph networks

Item Type Journal Article

Author R. Wang

Author F. Fang

Author J. Cui

Author W. Zheng

Abstract Despite decades of theoretical research, the nature of the self-driven collective motion remains indigestible and controversial, while the phase transition process of its dynamic is a major research issue. Recent methods propose to infer the phase transition process from various artificially extracted features using machine learning. In this thesis, we propose a new order parameter by using machine learning to quantify the synchronization degree of the self-driven collective system from the perspective of the number of clusters. Furthermore, we construct a powerful model based on the graph network to determine the long-term evolution of the self-driven collective system from the initial position of the particles, without any manual features. Results show that this method has strong predictive power, and is suitable for various noises. Our method can provide reference for the research of other physical systems with local interactions. © 2022, The Author(s).

Date 2022

DOI 10.1038/s41598-021-04456-5

URL <https://www.scopus.com/pages/publications/85122822882?origin=resultslist>

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Volume 12

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Publication Scientific Reports

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Issue 1

Journal Abbr Sci. Rep.

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Tags:

machine learning, article, learning, noise

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EID: 2-s2.0-85122822882

Export Date: 20 September 2026; Cited By: 11; Funding details: National Natural Science Foundation of China, NSFC, (11702289, 2020XXX013); National Key Research and Development Program of China, NKRDPC; Correspondence Address: W. Zheng; Institute of Public-Safety and Big Data, College of Data Science, Taiyuan University of Technology, Taiyuan, 030060, China; email: zhengwen@tyut.edu.cn

Learning the Regulatory Code of Gene Expression

Item Type Journal Article

Author J. Zrimec

Author F. Buric

Author M. Kokina

Author V. Garcia

Author A. Zelezniak

Abstract Data-driven machine learning is the method of choice for predicting molecular phenotypes from nucleotide sequence, modeling gene expression events including protein-DNA binding, chromatin states as well as mRNA and protein levels. Deep neural networks automatically learn informative sequence representations and interpreting them enables us to improve our understanding of the regulatory code governing gene expression. Here, we review the latest developments that apply shallow or deep learning to quantify molecular phenotypes and decode the cis-regulatory grammar from prokaryotic and eukaryotic sequencing data. Our approach is to build from the ground up, first focusing on the initiating protein-DNA interactions, then specific coding and non-coding regions, and finally on advances that combine multiple parts of the gene and mRNA regulatory structures, achieving unprecedented performance. We thus provide a quantitative view of gene expression regulation from nucleotide sequence, concluding with an information-centric overview of the central dogma of molecular biology. © Copyright © 2021 Zrimec, Buric, Kokina, Garcia and Zelezniak.

Date 2021

DOI 10.3389/fmolb.2021.673363

URL <https://www.scopus.com/pages/publications/85109049972?origin=resultslist>

Archive Scopus

Language English

Volume 8

Publisher Frontiers Media S.A.

Publication Frontiers in Molecular Biosciences

ISSN 2296889X (ISSN)

Journal Abbr Front. Mol. Biosci.

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Tags:

machine learning, deep neural networks, chromatin accessibility, cis-regulatory grammar, gene expression prediction, gene regulatory structure, mRNA & protein abundance, regulatory genomics

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Learning to identify container contents through tactile vibration signatures

Item Type Conference Paper

Author C.L. Chen

Author J.O. Snyder

Author P.J. Ramadge

Abstract We examine using a simple contact sensor coupled with standard machine learning algorithms to classify and count objects shaken in a container. The contact sensor measures the resulting vibrations, and these signatures are used to learn a classifier that maps vibration signatures to known object categories. A linear support vector machine trained on labeled vibration signatures achieves a mean binary classification accuracy of 99% over 66 pairs of objects and a mean multi-class classification accuracy of 94% over 12 classes. It is also shown that useful tasks such as approximate counting of objects over the range 1 to 10 is possible. We see potential applications of these ideas in service robots engaged in cleanup and inventory control in labs, workshops, stores, warehouses and homes. © 2016 IEEE.

Date 2017

DOI 10.1109/SIMPAR.2016.7862373

URL <https://www.scopus.com/pages/publications/85015817584?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: IEEE Int. Conf. Simul., Model., Program. Auton. Robot., SIMPAR

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-150904616-4 (ISBN)

Pages 43-48

Proceedings Title IEEE Int. Conf. Simul., Model., Program. Auton. Robot., SIMPAR

Conference Name 2016 IEEE International Conference on Simulation, Modeling, and Programming for Autonomous Robots, SIMPAR 2016

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Tags:

Robot programming, Learning systems, Learning algorithms, Robots, Inventory control, Service robots, Containers, Binary classification, Multi-class classification, Object categories, Linear Support Vector Machines, Approximate counting, Contact sensors, Standard machines, Vibration signature

Notes:

EID: 2-s2.0-85015817584

Export Date: 20 September 2026; Cited By: 19; Conference name: 2016 IEEE International Conference on Simulation, Modeling, and Programming for Autonomous Robots, SIMPAR 2016; Conference date: 13 December 2016 through 16 December 2016; Conference code: 126624

Learning to Quantify

Item Type Journal Article
Author A. Esuli
Author A. Fabris
Author A. Moreo
Author F. Sebastiani
Date 2023
DOI 10.1007/978-3-031-20467-8
URL <https://www.scopus.com/pages/publications/85195529714?origin=resultslist>
Archive Scopus
Language English
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Publication Information Retrieval Series
ISSN 18717500 (ISSN)
Journal Abbr Inf. Retr. Ser.
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Tags:

Machine Learning, Data Science, Data Mining, Information Retrieval, Supervised Learning, Class Prior Estimation, Open Access, Prevalence Estimation

Notes:

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Export Date: 20 September 2026; Cited By: 14; Funding details: Istituto di Scienza e Tecnologie dell'Informazione, ISTI; SoBigData.it; FAIR; Ministero dell'Istruzione, dell'Università e della Ricerca, MIUR; Italian Ministry for University and Research; Horizon 2020 Framework Programme, H2020, (871042); Horizon 2020 Framework Programme, H2020; European Commission, EC, (951911, ICT-48-2020); European Commission, EC

Learning to Quantify Emphysema Extent: What Labels Do We Need?

Item Type Journal Article

Author S.N. Orting

Author J. Petersen

Author L.H. Thomsen

Author M.M.W. Wille

Author M. De Bruijne

Abstract Accurate assessment of pulmonary emphysema is crucial to assess disease severity and subtype, to monitor disease progression, and to predict lung cancer risk. However, visual assessment is time-consuming and subject to substantial inter-rater variability while standard densitometry approaches to quantify emphysema remain inferior to visual scoring. We explore if machine learning methods that learn from a large dataset of visually assessed CT scans can provide accurate estimates of emphysema extent and if methods that learn from emphysema extent scoring can outperform algorithms that learn only from emphysema presence scoring. Four Multiple Instance Learning classifiers, trained on emphysema presence labels, and five Learning with Label Proportions classifiers, trained on emphysema extent labels, are compared. Performance is evaluated on 600 low-dose CT scans from the Danish Lung Cancer Screening Trial and we find that learning from emphysema presence labels, which are much easier to obtain, gives equally good performance to learning from emphysema extent labels. The best performing Multiple Instance Learning and Learning with Label Proportions classifiers, achieve intra-class correlation coefficients around 0.90 and average overall agreement with raters of 78% and 79% compared to an inter-rater agreement of 83%. © 2013 IEEE.

Date 2020

DOI 10.1109/JBHI.2019.2932145

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Language English

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Risk assessment, adult, Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, machine learning, Machine learning, Large dataset, algorithm, Algorithms, Machine Learning, procedures, controlled study, major clinical study, dimensionality reduction, learning, Diagnosis, Diseases, support vector machine, classifier, interrater reliability, correlation coefficient, disease severity, Inter-rater agreements, disease exacerbation, Machine learning methods, diagnostic imaging, image analysis, computer assisted diagnosis, Image Interpretation, Computer-Assisted, pathology, Computerized tomography, computer assisted tomography, principal component analysis, chronic obstructive lung disease, Tomography, X-Ray Computed, x-ray computed tomography, Disease Progression, Biological organs, maximum likelihood method, Lung, chest CT, computer prediction, Correlation coefficient, Disease progression, Emphysema, label proportions learning, learning with label proportions, lung, Lung cancer screening, lung emphysema, multiple instance learning, Multiple instance learning, Pulmonary emphysema, Pulmonary Emphysema, Visual assessments

Notes:

EID: 2-s2.0-85083024312

Export Date: 20 September 2026; Cited By: 3; Funding details: Netherlands Organization for Scientific Research; Natur og Univers, Det Frie Forskningsråd, FNU, DFF; Nederlandse Organisatie voor Wetenschappelijk Onderzoek, NWO; Danmarks Frie Forskningsfond, DFF; Correspondence Address: S.N. Orting; Department of Computer Science, University of Copenhagen, Copenhagen, 2100, Denmark; email: silas@di.ku.dk; CODEN: ITIBF

Learning to quantify: Estimating class prevalence via supervised learning

Item Type Conference Paper

Author A. Moreo

Author F. Sebastiani

Date 2019

DOI 10.1145/3331184.3331389

URL <https://www.scopus.com/pages/publications/85073788895?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: SIGIR - Proc. Int. ACM SIGIR Conf. Res. Dev. Inf. Retr.

Publisher Association for Computing Machinery, Inc

ISBN 978-145036172-9 (ISBN)

Pages 1415-1416

Proceedings Title SIGIR - Proc. Int. ACM SIGIR Conf. Res. Dev. Inf. Retr.

Conference Name SIGIR 2019 - Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval

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Learning to Quantify: Methods and Applications (LQ 2021)

Item Type Conference Paper

Author J.J. Del Coz

Author P. González

Author A. Moreo

Author F. Sebastiani

Abstract Learning to Quantify (LQ) is the task of training class prevalence estimators via supervised learning. The task of these estimators is to estimate, given an unlabelled set of data items D and a set of classes $C = \{c_1, \dots, c_{|C|}\}$, the prevalence (i.e., relative frequency) of each class c_i in D . LQ is interesting in all applications of classification in which the final goal is not determining which class (or classes) individual unlabelled data items belong to, but estimating the distribution of the unlabelled data items across the classes of interest. Example disciplines whose interest in labelling data items is at the aggregate level (rather than at the individual level) are the social sciences, political science, market research, ecological modelling, and epidemiology. While LQ may in principle be solved by classifying each data item in D and counting how many such items have been labelled with c_i , it has been shown that this "classify and count" (CC) method yields suboptimal quantification accuracy. As a result, quantification is now no longer considered a mere byproduct of classification and has evolved as a task of its own. The goal of this workshop is bringing together all researchers interested in methods, algorithms, and evaluation measures and methodologies for LQ, as well as practitioners interested in their practical application to managing large quantities of data. © 2021 Owner/Author.

Date 2021

DOI 10.1145/3459637.3482040

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Archive Scopus

Language English

Extra Journal Abbreviation: Int Conf Inf Knowledge Manage

Publisher Association for Computing Machinery

ISBN 21550751 (ISSN); 978-145038446-9 (ISBN)

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Proceedings Title Int Conf Inf Knowledge Manage

Conference Name International Conference on Information and Knowledge Management, Proceedings

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Classification (of information), Frequency estimation, Labelings, Market Research, C (programming language), Unlabeled data, Market researches, Dataset shifts, Quantification, quantification, dataset shift, Relative frequencies, Individual levels, Data items, Political science, Training class

Notes:

EID: 2-s2.0-85119211989

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Learning to sample: Counting with complex queries

Item Type Conference Paper**Author** B. Walenz**Author** S. Sintos**Author** S. Roy**Author** J. Yang

Abstract We study the problem of efficiently estimating counts for queries involving complex filters, such as user-defined functions, or predicates involving self-joins and correlated subqueries. For such queries, traditional sampling techniques may not be applicable due to the complexity of the filter preventing sampling over joins, and sampling after the join may not be feasible due to the cost of computing the full join. The other natural approach of training and using an inexpensive classifier to estimate the count instead of the expensive predicate suffers from the difficulties in training a good classifier and giving meaningful confidence intervals. In this paper we propose a new method of learning to sample where we combine the best of both worlds by using sampling in two phases. First, we use samples to learn a probabilistic classifier, and then use the classifier to design a stratified sampling method to obtain the final estimates. We theoretically analyze algorithms for obtaining an optimal stratification, and compare our approach with a suite of natural alternatives like quantification learning, weighted and stratified sampling, and other techniques from the literature. We also provide extensive experiments in diverse use cases using multiple real and synthetic datasets to evaluate the quality, efficiency, and robustness of our approach.

Date 2020**DOI** 10.14778/3368289.3368302**URL** <https://www.scopus.com/pages/publications/85092086539?origin=resultslist>**Archive** Scopus

Language English

Extra Journal Abbreviation: Proc. VLDB Endow.

Volume 13

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Conference Name Proceedings of the VLDB Endowment

Issue 3

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Tags:

Synthetic datasets, Confidence interval, Probabilistic classifiers, Method of learning, Natural approaches, Sampling technique, Stratified sampling, User Defined Functions

Notes:

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Learning under nonstationarity: Covariate shift and class-balance change

Item Type Journal Article

Author M. Sugiyama

Author M. Yamada

Author M.C. du Plessis

Abstract One of the fundamental assumptions behind many supervised machine-learning algorithms is that training and test data follow the same probability distribution. However, this important assumption is often violated in practice, for example, because of an unavoidable sample selection bias or nonstationarity of the environment. Owing to violation of the assumption, standard machine-learning methods suffer a significant estimation bias. In this article, we consider two scenarios of such distribution change-the covariate shift where input distributions differ and class-balance change where class-prior probabilities vary in classification-and review semi-supervised adaptation techniques based on importance weighting. © 2013 Wiley Periodicals, Inc.

Date 2013

DOI 10.1002/wics.1275

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Archive Scopus

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Learning systems, Probability distributions, Covariate shifts, Density ratio estimation, Density-ratio estimations, Covariate shift, Importance weighting, Class-balance change, Divergence approximation

Notes:

EID: 2-s2.0-84886749447

Export Date: 20 September 2026; Cited By: 23; Correspondence Address: M. Sugiyama; Department of Computer Science, Tokyo Institute of Technology, Tokyo, Japan; email: sugi@cs.titech.ac.jp

Learning-based epsilon most stringent test for Gaussian samples classification

Item Type Conference Paper
Author L. Fillatre
Author I. Nikiforov
Abstract This paper studies the problem of classifying some Gaussian samples into one of two parametric probabilistic models, also called sources, when the parameter and the a priori probability of each source are unknown. Each source is governed by an univariate normal distribution whose mean is unknown. A training sequence is available for each source in order to compensate the lack of prior information. An almost optimal most stringent test is proposed to solve this classification problem subject to a constrained false alarm probability. This learning-based test minimizes its maximum shortcoming with respect to the most powerful test which knows exactly the parameters of the sources. It also guarantees a prescribed false alarm probability whatever the size of the training sequences. The threshold, the probability of false alarm and the probability of correct detection are calculated analytically. © 2017 IEEE.
Date 2017
DOI 10.1109/ISIT.2017.8006752
URL <https://www.scopus.com/pages/publications/85034017387?origin=resultslist>
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Language English
Extra Journal Abbreviation: IEEE Int Symp Inf Theor Proc
Volume 0

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Conference Name IEEE International Symposium on Information Theory - Proceedings
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Tags:

Errors, Normal distribution, A-priori probabilities, Prior information, Alarm systems, False alarm probability, Gaussian sample, Most stringent tests, Parametric probabilistic model, Probability of false alarm, Sample classification, Training sequences, Univariate

Notes:

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Export Date: 20 September 2026; Cited By: 0; Conference name: 2017 IEEE International Symposium on Information Theory, ISIT 2017; Conference date: 25 June 2017 through 30 June 2017; Conference code: 129916; CODEN: PISTF

Left Ventricle Quantification Challenge: A Comprehensive Comparison and Evaluation of Segmentation and Regression for Mid-Ventricular Short-Axis Cardiac MR Data

- Item Type** Journal Article
- Author** W. Xue
 - Author** J. Li
 - Author** Z. Hu
 - Author** E. Kerfoot
 - Author** J. Clough
 - Author** I. Oksuz
 - Author** H. Xu
 - Author** V. Grau
 - Author** F. Guo
 - Author** M. Ng
 - Author** X. Li
 - Author** Q. Li
 - Author** L. Liu
 - Author** J. Ma
 - Author** E. Grinias
 - Author** G. Tziritas
 - Author** W. Yan

Author A. Atehortua

Author M. Garreau

Author Y. Jang

Author A. Debus

Author E. Ferrante

Author G. Yang

Author T. Hua

Author S. Li

Abstract Automatic quantification of the left ventricle (LV) from cardiac magnetic resonance (CMR) images plays an important role in making the diagnosis procedure efficient, reliable, and alleviating the laborious reading work for physicians. Considerable efforts have been devoted to LV quantification using different strategies that include segmentation-based (SG) methods and the recent direct regression (DR) methods. Although both SG and DR methods have obtained great success for the task, a systematic platform to benchmark them remains absent because of differences in label information during model learning. In this paper, we conducted an unbiased evaluation and comparison of cardiac LV quantification methods that were submitted to the Left Ventricle Quantification (LVQuan) challenge, which was held in conjunction with the Statistical Atlases and Computational Modeling of the Heart (STACOM) workshop at the MICCAI 2018. The challenge was targeted at the quantification of 1) areas of LV cavity and myocardium, 2) dimensions of the LV cavity, 3) regional wall thicknesses (RWT), and 4) the cardiac phase, from mid-ventricle short-axis CMR images. First, we constructed a public quantification dataset Cardiac-DIG with ground truth labels for both the myocardium mask and these quantification targets across the entire cardiac cycle. Then, the key techniques employed by each submission were described. Next, quantitative validation of these submissions were conducted with the constructed dataset. The evaluation results revealed that both SG and DR methods can offer good LV quantification performance, even though DR methods do not require densely labeled masks for supervision. Among the 12 submissions, the DR method LDAMT offered the best performance, with a mean estimation error of 301 mm² for the two areas, 2.15 mm for the cavity dimensions, 2.03 mm for RWTs, and a 9.5% error rate for the cardiac phase classification. Three of the SG methods also delivered comparable performances. Finally, we discussed the advantages and disadvantages of SG and DR methods, as well as the unsolved problems in automatic cardiac quantification for clinical practice applications. © 2013 IEEE.

Date 2021

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Archive Scopus

Language English

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Journal Abbr IEEE J. Biomedical Health Informat.

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Article, human, Humans, Deep neural network, machine learning, segmentation, Heart, reliability, algorithm, image segmentation, training, artificial neural network, deep neural network, computer model, learning algorithm, support vector machine, validation process, intermethod comparison, Quantification methods, diagnostic imaging, image analysis, Magnetic resonance, Magnetic Resonance Imaging, nuclear magnetic resonance imaging, Phase classification, heart arrhythmia, heart ventricle, Heart Ventricles, catheter ablation, quantification, heart left ventricle, regression, heart failure, heart, task performance, Automatic quantification, cardiac muscle, cardiovascular magnetic resonance, heart ventricle remodeling, heart cycle, cine magnetic resonance imaging, Magnetic Resonance Imaging, Cine, sudden cardiac death, arterial wall thickness, Cardiac magnetic resonance, Cardiac quantification, Comprehensive comparisons, congestive cardiomyopathy, constrictive pericarditis, Evaluation of segmentation, left ventricle, peak systolic velocity, pericarditis, Quantitative validation

Notes:

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Legitimate ground-truth-free metrics for deep uncertainty classification scoring

Item Type Conference Paper

Author A. Pignet

Author C. Regniez

Author J. Klein

Editor Li Y.

Editor Mandt S.

Editor Agrawal S.

Editor Khan E.

Abstract Despite the increasing demand for safer machine learning practices, the use of Uncertainty Quantification (UQ) methods in production remains limited. This limitation is exacerbated by the challenge of validating UQ methods in absence of UQ ground truth. In classification tasks, when only a usual set of test data is at

hand, several authors suggested different metrics that can be computed from such test points while assessing the quality of quantified uncertainties. This paper investigates such metrics and proves that they are theoretically well-behaved and actually tied to some uncertainty ground truth which is easily interpretable in terms of model prediction trustworthiness ranking. Equipped with those new results, and given the applicability of those metrics in the usual supervised paradigm, we argue that our contributions will help promoting a broader use of UQ in deep learning. Copyright 2025 by the author(s).

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LeQua@CLEF2022: Learning to Quantify

Item Type Conference Paper

Author A. Esuli

Author A. Moreo

Author F. Sebastiani

Editor Hagen M.

Editor Verberne S.

Editor Macdonald C.

Editor Seifert C.**Editor** Balog K.**Editor** Nørvåg K.**Editor** Setty V.

Abstract LeQua 2022 is a new lab for the evaluation of methods for “learning to quantify” in textual datasets, i.e., for training predictors of the relative frequencies of the classes of interest in sets of unlabelled textual documents. While these predictions could be easily achieved by first classifying all documents via a text classifier and then counting the numbers of documents assigned to the classes, a growing body of literature has shown this approach to be suboptimal, and has proposed better methods. The goal of this lab is to provide a setting for the comparative evaluation of methods for learning to quantify, both in the binary setting and in the single-label multiclass setting. For each such setting we provide data either in ready-made vector form or in raw document form. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.

Date 2022**DOI** 10.1007/978-3-030-99739-7_47**URL** <https://www.scopus.com/pages/publications/85128788707?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Lect. Notes Comput. Sci.**Volume** 13186 LNCS**Publisher** Springer Science and Business Media Deutschland GmbH**ISBN** 03029743 (ISSN); 978-303099738-0 (ISBN)**Pages** 374-381**Proceedings Title** Lect. Notes Comput. Sci.**Conference Name** Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)**Date Added** 20/09/2026, 9:52:31 pm**Modified** 20/09/2026, 9:52:31 pm**Tags:**

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Lesion Volume Quantification Using Two Convolutional Neural Networks in MRIs of Multiple Sclerosis Patients

Item Type Journal Article

Author M. de Oliveira

Author M. Piacenti-Silva

Author F.C.G. da Rocha

Author J.M. Santos

Author J.S. Cardoso

Author P.N. Lisboa-Filho

Abstract Background: Multiple sclerosis (MS) is a neurologic disease of the central nervous system which affects almost three million people worldwide. MS is characterized by a demyelination process that leads to brain lesions, allowing these affected areas to be visualized with magnetic resonance imaging (MRI). Deep learning techniques, especially computational algorithms based on convolutional neural networks (CNNs), have become a frequently used algorithm that performs feature self-learning and enables segmentation of structures in the image useful for quantitative analysis of MRIs, including quantitative analysis of MS. To obtain quantitative information about lesion volume, it is important to perform proper image preprocessing and accurate segmentation. Therefore, we propose a method for volumetric quantification of lesions on MRIs of MS patients using automatic segmentation of the brain and lesions by two CNNs. Methods: We used CNNs at two different moments: the first to perform brain extraction, and the second for lesion segmentation. This study includes four independent MRI datasets: one for training the brain segmentation models, two for training the lesion segmentation model, and one for testing. Results: The proposed brain detection architecture using binary cross-entropy as the loss function achieved a 0.9786 Dice coefficient, 0.9969 accuracy, 0.9851 precision, 0.9851 sensitivity, and 0.9985 specificity. In the second proposed framework for brain lesion segmentation, we obtained a 0.8893 Dice coefficient, 0.9996 accuracy, 0.9376 precision, 0.8609 sensitivity, and 0.9999 specificity. After quantifying the lesion volume of all patients from the test group using our proposed method, we obtained a mean value of 17,582 mm³. Conclusions: We concluded that the proposed algorithm achieved accurate lesion detection and segmentation with reproducibility corresponding to state-of-the-art software tools and manual segmentation. We believe that this quantification method can add value to treatment monitoring and routine clinical evaluation of MS patients. © 2022 by the authors. Licensee MDPI, Basel, Switzerland.

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Leveraging advanced ensemble models to increase building energy performance prediction accuracy in the residential building sector

Item Type Journal Article

Author K. Konhäuser

Author S. Wenninger

Author T. Werner

Author C. Wiethe

Abstract Accurate predictions for buildings' energy performance (BEP) are crucial for retrofitting investment decisions and building benchmarking. With the increasing data availability and popularity of machine learning across disciplines, research started investigating machine learning for BEP predictions. While stand-alone machine learning models showed first promising results, a comprehensive analysis of advanced ensemble models to increase prediction accuracy is missing for annual BEP predictions. We implement and thoroughly tune twelve machine learning models to bridge this research gap, ranging from stand-alone to homogeneous and heterogeneous ensemble learning models. We benchmark their prediction accuracy based on an extensive real-world dataset of over 25,000 German residential buildings. The results provide strong evidence that ensemble models substantially outperform stand-alone machine learning models both on average and in the case of the best-performing model. All models are tested for robustness and systematic bias by evaluating their prediction performance along different building age classes, living

space bins, and several error measures. Extreme gradient boosting as an ensemble model exhibits the highest prediction accuracy, followed by a multilayer perceptron ahead of further ensemble models. We conclude that ensemble models for annual BEP prediction are advantageous compared to stand-alone models and outperform their results in most cases. © 2022 Elsevier B.V.

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Leveraging explainable AI for informed building retrofit decisions: Insights from a survey

Item Type Journal Article

Author D. Leuthe

Author J. Mirlach

Author S. Wenninger

Author C. Wiethe

Abstract Accurate predictions of building energy consumption are essential for reducing the energy performance gap. While data-driven energy quantification methods based on machine learning deliver promising results, the lack of Explainability prevents their

widespread application. To overcome this, Explainable Artificial Intelligence (XAI) was introduced. However, to this point, no research has examined how effective these explanations are concerning decision-makers, i.e., property owners. To address this, we implement three transparent models (Linear Regression, Decision Tree, QLattice) and apply four XAI methods (Partial Dependency Plots, Accumulated Local Effects, Local Interpretable Model-Agnostic Explanations, Shapley Additive Explanations) to an Artificial Neural Network using a real-world dataset of 25,000 residential buildings. We evaluate their Prediction Accuracy and Explainability through a survey with 137 participants considering the human-centered dimensions of explanation satisfaction and perceived fidelity. The results quantify the Explainability-Accuracy trade-off in building energy consumption forecasting and how it can be counteracted by choosing the right XAI method to foster informed retrofit decisions. For research, we set the foundation for further increasing the Explainability of data-driven energy quantification methods and their human-centered evaluation. For practice, we encourage using XAI to reduce the acceptance gap of data-driven methods, whereby the XAI method should be selected carefully, as the Explainability within the methods varies by up to 10 %. © 2024 The Author(s)

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Leveraging Google Earth Engine and Machine Learning to Estimate Evapotranspiration in a Commercial Forest Plantation

Item Type Journal Article

Author S. Gokool

Author R. Kunz

Author A. Clulow

Author M. Toucher

Abstract Estimation of actual evapotranspiration (ETa) based on reference evapotranspiration (ETo) and the crop coefficient (Kc) remains one of the most widely used ETa estimation approaches. However, its application in non-agricultural and natural environments has been limited, largely due to the lack of well-established Kc coefficients in these environments. Alternate Kc estimation approaches have thus been proposed in such instances, with techniques based on the use of leaf area index (LAI) estimates being quite popular. In this study, we utilised satellite-derived estimates of LAI acquired through the Google Earth Engine geospatial cloud computing platform and machine learning to quantify the water use of a commercial forest plantation situated within the eastern region of South Africa. Various machine learning-based models were trained and evaluated to predict Kc as a function of LAI, with the Kc estimates derived from the best-performing model then being used in conjunction with in situ measurements of ETo to estimate ETa. The ET estimates were then evaluated through comparisons against in situ measurements. An ensemble machine learning model showed the best performance, yielding RMSE and R2 values of 0.05 and 0.68, respectively, when compared against measured Kc. Comparisons between estimated and measured ETa yielded RMSE and R2 values of 0.51 mm d⁻¹ and 0.90, respectively. These results were quite promising and further demonstrate the potential of geospatial cloud computing and machine learning-based approaches to provide a robust and efficient means of handling large volumes of data so that they can be optimally utilised to assist planning and management decisions. © 2024 by the authors.

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Leveraging Industry 4.0: Deep Learning, Surrogate Model, and Transfer Learning with Uncertainty Quantification Incorporated into Digital Twin for Nuclear System

Item Type Journal Article

Author M. Rahman

Author A.H. Khan

Author S. Anowar

Author M. Al-Imran

Author R. Verma

Author D. Kumar

Author K. Kobayashi

Author S. Alam

Abstract Industry 4.0 targets the conversion of the traditional industries into intelligent ones through technological revolution. This revolution is only possible through innovation, optimization, interconnection, and rapid decision-making capability. Numerical models are believed to be the key components of Industry 4.0, facilitating quick decision-making through simulations instead of costly experiments. However, numerical investigation of precise, high-fidelity models for optimization or decision-making is usually time-consuming and computationally expensive. In such instances, data-driven surrogate models are excellent substitutes for fast computational analysis and the probabilistic prediction of the output parameter for new input parameters. The emergence of Internet of Things (IoT) and machine learning (ML) has made the concept of surrogate modeling even more viable. However, these surrogate models contain intrinsic uncertainties, originate from modeling defects, or both. These uncertainties, if not quantified and minimized, can produce a skewed result. Therefore, proper implementation of uncertainty quantification techniques is crucial during optimization, cost reduction, or safety enhancement processes analysis. This chapter begins with a brief overview of the concept of surrogate modeling, transfer learning, IoT, and digital twins. After that, a detailed overview of uncertainties, uncertainty quantification frameworks, and specifics of uncertainty quantification methodologies for a surrogate model linked to a digital twin is presented. Finally, the use of uncertainty quantification approaches in the nuclear industry has been

addressed. © Springer Nature Switzerland AG 2023.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Alam; Department of Nuclear Engineering and Radiation Science, Missouri University of Science and Technology, Rolla, United States; email: alams@mst.edu

Leveraging Internet of Things for Accurate Load Forecasting in Power Systems: Opportunities, Challenges, and Applications

Item Type Conference Paper

Author N. Pradeep

Author S.K. Girish

Author G.S. Sheshadri

Author H.M. Manjunatha

Editor Mandal D.

Editor Eshwarappa M.N.

Editor Frnda J.

Abstract Maintaining the efficient and dependable operation of power systems requires load forecasting. Integrating Internet of Things (IoT) technologies provides a great opportunity to improve load forecasting accuracy by utilizing detailed data from smart meters, environmental sensors, and grid monitoring devices. This study examines the utilization of IoT-enabled load forecasting models, which encompass machine learning and deep learning techniques, as well as ensemble and hybrid models, to capture intricate patterns and nonlinear associations in

energy usage. This paper focuses on the attributes of IoT-enabled load forecasting, including real-time data processing, scalability, and adaptability. These features allow for a wide range of applications in short-term, medium-term, and long-term forecasting, demand response, renewable energy integration, and energy trading strategies. Although there are notable advantages, there are also problems that need to be acknowledged and resolved, including those related to the accuracy of data, privacy concerns, the ability of different systems to work together, and the complexity of computing processes. Case studies and applications exemplify the effective utilization of IoT-enabled load forecasting in many areas such as short-term grid operations, demand response programs, renewable energy integration, and energy trading. The study explores future research directions, such as advanced data analytics methodologies, edge computing architectures, blockchain integration, and uncertainty quantification methods. Suggestions are given for promoting cooperation among individuals or groups involved in addressing difficulties and promoting the widespread acceptance of smart grids and sustainable energy practices. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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Leveraging machine learning predicted confidence for boosting assay submission and decision-making efficiencies

Item Type Journal Article

Author D. Bassani

Author M. Reutlinger

Author H. Fischer

Abstract Machine learning (ML) has become very popular, and its benefits are widely recognized within the scientific community. The ability of ML approaches to leverage large datasets to find patterns among composite single data points has made these approaches widespread across different fields. Small molecule pharmaceutical research has experienced the advantages of these methods for tasks such as molecular property prediction, secondary pharmacology analysis, de novo generation, and compound clustering. Coupling efficient ML models with robust uncertainty quantification methods gives the additional advantage of discriminating among predictions to identify the ones that can be truly reliable. Herein, the Roche experience with ML uncertainty quantification for influencing decision making in pharmacokinetic assay submission is described. After setting up an initial threshold for error acceptance via nonadditivity analysis, the combined efforts of ML and experimental scientists developed an optimal uncertainty threshold in ML models. By excluding compounds with predicted properties that were within a confidence level equal to or greater than the agreed one, the analysis highlights how significant numbers of molecules could potentially be excluded from assay submission (up to 25 % of the normal submission rate), allowing significant time and cost savings for the organization. © 2025 The Authors

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Leveraging machine learning to quantify the effect of porosity on mode I delamination in ceramic matrix composites

Item Type Journal Article

Author J. Gunst

Author E. Vaghefi

Author E. Mirkoochi

Author G. Deshpande

Author B. Prorok

Author S. Gururaja

Abstract This work examines the effect of pores on mode I delamination in plain woven ceramic matrix composite (CMC) in the presence of process-induced porosity under a double cantilever beam (DCB) test. A cohesive zone modeling (CZM) approach within a finite element (FE) numerical framework has been used to simulate mode I crack growth. Pore distributions along the crack path have been explicitly modeled at varying porosities between 10%–30%, varying pore geometry, orientation, and location. A Monte Carlo simulation approach has been used to generate reaction load-displacement plots and resistance curves (R-curves) for the stochastic porous distribution within the FE-CZM numerical approach. A subset of the FE-CZM results is used to train a long-short-term memory (LSTM) machine learning (ML) model that considers the pore parameters as variables. The trained ML model is then used to predict reaction load-displacement plots for the remaining porous microstructures, yielding a fairly good match with a considerable speed up in run time. In addition to rapidly predicting material behavior when given a specific pore distribution, the ML model quantifies the impact of different pore parameters, such as size and location, on mode I delamination behavior in woven CMCs. © 2025 Elsevier Ltd

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Correspondence Address: S. Gururaja; Aerospace Engineering, Auburn University, Auburn, United States; email: suhasini.gururaja@auburn.edu; CODEN: COMSE

LIBS basics, part III: Deriving the analytical answer - Calibrated solutions with LIBS

Item Type Journal Article
Author S. Buckley
Abstract There are obviously many methods to come to both quantitative calibration and qualitative classification with LIBS. I am often asked what to expect in terms of overall performance. With real-world samples, typical relative standard deviation (RSD) numbers are in the range of 2-3% for many types of analysis. For the most homogenous samples, RSDs less than 1% can be achieved, for example, on obsidian glass samples or gemstones. With both classification and quantification methods, one must be sure to avoid over-fitting and rigorously test any model before deployment. Following the general guidelines here, with a bit of experimentation, should get you a good way toward successful implementation of LIBS.
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Life-Cycle Cost Analysis of Exclusive Lanes for Autonomous Trucks Considering the Risk of Intrusion into Adjacent Lanes under Crosswinds: Joint Decision of Lane Width and Lateral Control Modes

Item Type Journal Article

Author J. Chen

Author F. Chen

Abstract To quantify the impact of lane width and truck lateral control modes on the life-cycle cost (LCC) and operational safety of exclusive lanes for autonomous trucks (ATs), this study proposes a LCC framework that considers the risk of ATs intruding into adjacent lanes under crosswind. Based on the maximum lateral deviation data obtained from XFlow-TruckSim, the gradient boosting decision trees (GBDT) model, which performed best among five machine learning methods, was utilized as the risk quantification method. The study considered three lateral control modes for ATs: central distribution, uniform distribution, and normal distribution, evaluating both their safety and the damage they cause to the exclusive lanes. Furthermore, with lane width and truck lateral control modes as decision variables, a multiobjective optimization model was established to incorporate vehicle risk into the life-cycle costs analysis of the exclusive lanes. The trend of risk probability was generally similar to the variation in total life-cycle costs, primarily due to the overwhelming importance of user safety cost. Adopting a exclusive lane width of 3.54 m for AT operation under crosswind is a favorable decision. For already constructed exclusive lanes with a width less than 3.05 m, it is preferable for ATs to adopt a uniform distribution lateral control mode. When the lane width exceeds 3.21 m, centrally traveling ATs become a better choice. If the lane width is between 3.05 and 3.21 m, central travel is generally superior, and it is least recommended for ATs to follow a

uniform distribution lateral control mode. The findings provide crucial guidance for the design of exclusive lanes for ATs and the lateral control modes adopted in their operation postconstruction. The study addresses the conflicting challenge of increasing lane width, which can raise institutional costs while reducing the risk of lateral vehicle accidents. © 2025 American Society of Civil Engineers.

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Autonomous vehicles, Risk perception, Traffic safety, Risk assessment, Magnetic levitation vehicles, Automobiles, risk assessment, travel behavior, unmanned vehicle, Cost reduction, Risk analysis, transportation safety, Multiobjective optimization, traffic management, Adaptive boosting, life cycle analysis, Cost benefit analysis, Linear programming, Maximum likelihood estimation, Autonomous truck, Autonomous truck (AT), Control modes, cost analysis, Exclusive lane, Lane width, Lateral control, Lateral control mode, Life cycle costs analysis, Life-cycle cost, Life-cycle cost (LCC), Normal distribution, trucking, Trucks, Uniform distribution

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Lightning UQ Box: Uncertainty Quantification for Neural Networks

Item Type Journal Article

Author N. Lehmann

Author N.M. Gottschling

Author J. Gawlikowski

Author A.J. Stewart

Author S. Depeweg

Author E. Nalisnick

Abstract Although neural networks have shown impressive results in a multitude of application domains, the “black box” nature of deep learning and lack of confidence estimates have led to scepticism, especially in domains like medicine and physics where such estimates are critical. Research on uncertainty quantification (UQ) has helped elucidate the reliability of these models, but existing implementations of these UQ methods are sparse and difficult to reuse. To this end, we introduce Lightning UQ Box, a PyTorch-based Python library for deep learning-based UQ methods powered by PyTorch Lightning. Lightning UQ Box supports classification, regression, semantic segmentation, and pixelwise regression applications, and UQ methods from a variety of theoretical motivations. With this library, we provide an entry point for practitioners new to UQ, as well as easy-to-use components and tools for scalable deep learning applications. ©2025 Nils Lehmann, Nina Maria Gottschling, Jakob Gawlikowski, Adam J. Stewart, Stefan Depeweg, and Eric Nalisnick.

Date 2025

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Archive Scopus

Language English

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Publication Journal of Machine Learning Research

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Journal Abbr J. Mach. Learn. Res.

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Deep learning, Deep neural networks, Uncertainty analysis, Neural-networks, Semantics, Deep Learning, Black boxes, Bayesian networks, Quantification methods, Uncertainty quantifications, Bayesian, Bayesian deep learning, Uncertainty Quantification, Conformal predictions, Conformal Prediction, Applications domains, Bayesian Deep Learning, Lightning protection, Pytorch, PyTorch

Notes:

EID: 2-s2.0-105018916957

Export Date: 20 September 2026; Cited By: 3

Lightweight Conditional Model Extrapolation for Streaming Data under Class-Prior Shift

Item Type Conference Paper

Author P. Tomaszewska

Author C.H. Lampert

Abstract We introduce LIMES, a new method for learning with non-stationary streaming data, inspired by the recent success of meta-learning. The main idea is not to attempt to learn a single classifier that would have to work well across all occurring data distributions, nor many separate classifiers, but to exploit a hybrid strategy: we learn a single set of model parameters from which a specific classifier for any specific data distribution is derived via classifier adaptation. Assuming a multiclass classification setting with class-prior shift, the adaptation step can be performed analytically with only the classifier's bias terms being affected. Another contribution of our work is an extrapolation step that predicts suitable adaptation parameters for future time steps based on the previous data. In combination, we obtain a lightweight procedure for learning from streaming data with varying class distribution that adds no trainable parameters and almost no memory or computational overhead compared to training a single model. Experiments on a set of exemplary tasks using Twitter data show that LIMES achieves higher accuracy than alternative approaches, especially with respect to the relevant real-world metric of lowest within-day accuracy. © 2022 IEEE.

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DOI 10.1109/ICPR56361.2022.9956195

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Int. Conf. Pattern Recognit.

Volume 2022-August

Publisher Institute of Electrical and Electronics Engineers Inc.

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Conference Name Proceedings - International Conference on Pattern Recognition

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Tags:

Classification (of information), Learning systems, Metalearning, Learn+, Lime, Data distribution, Continual learning, meta-learning, Class-prior shift, class-prior shift, Conditional models, continual learning, Extrapolation, incremental training, Incremental training, Model extrapolation, non-stationary data, Nonstationary data, Streaming data

Notes:

EID: 2-s2.0-85143624053

Export Date: 20 September 2026; Cited By: 2; Conference name: 26th International Conference on Pattern Recognition, ICPR 2022; Conference date: 21 August 2022 through 25 August 2022; Conference code: 184755; CODEN: PICRE

Lipid droplet distribution quantification method based on lipid droplet detection by constrained reinforcement learning

Item Type Journal Article

Author Y. Harada

Author H. Nishida

Author K. Matsuura

Abstract We previously proposed the lipid droplet detection by reinforcement learning (LiDRL) method using a limited dataset of pathological images. The method automatically detects lipid droplets using reinforcement learning to optimize filter combinations based on their size and grayscale contrast. In this study, we aimed to detect lipid droplets reliably and analyze their distribution patterns across pathological tissue images. For this purpose, we improved the environmental and agent-side functions in LiDRL to obtain a revised method. These improvements increased the stability and robustness of the system, enabling consistent extraction of lipid droplets of similar sizes across all rank levels in the pathological tissue images. We quantified the lipid droplet distribution using average probability density and entropy and visualized it as a heat map. This analysis facilitates the extraction of lipid droplet characteristics that could serve as indicators of liver disease. © 2025 Harada et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Language English

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Journal Abbr Plos One.

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Article, human, human experiment, Humans, entropy, algorithm, Algorithms, procedures, training, deep learning, controlled study, reinforcement learning (machine learning), accuracy, learning, diagnostic test accuracy study, probability, histopathology, image processing, sensitivity and specificity, image analysis, Image Processing, Computer-Assisted, human tissue, pathology, metabolism, liver, fat droplet, Lipid Droplets, Liver, liver disease

Notes:

EID: 2-s2.0-105017185081

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Nishida; Diagnostic Pathology, Faculty of Medicine, Oita University, Oita, Japan; email: nharuto@oita-u.ac.jp; CODEN: POLNC

“Lipidomics”: Mass spectrometric and chemometric analyses of lipids

Item Type Journal Article

Author Z. Wu

Author G.I. Bagarolo

Author S. Thoröe-Boveleth

Author J. Jankowski

Abstract Lipids are ubiquitous in the human organism and play essential roles as components of cell membranes and hormones, for energy storage or as mediators of cell signaling pathways. As crucial mediators of the human metabolism, lipids are also involved in metabolic diseases, cardiovascular and renal diseases, cancer and/or hepatological and neurological disorders. With rapidly growing evidence supporting the impact of lipids on both the genesis and progression of these diseases as well as patient wellbeing, the characterization of the human lipidome has gained high interest and importance in life sciences and clinical diagnostics within the last 15 years. This is mostly due to technically advanced molecular identification and quantification methods, mainly based on mass spectrometry. Mass spectrometry has become one of the most powerful tools for the identification of lipids. New lipidic mediators or biomarkers of diseases can be analysed by state-of-the art mass spectrometry techniques supported by sophisticated bioinformatics and biostatistics. The lipidomic approach has developed dramatically in the realm of life sciences and clinical diagnostics due to the available mass spectrometric methods and in particular due to the adaptation of biostatistical methods in recent years. Therefore, the current knowledge of lipid extraction methods, mass-spectrometric approaches, biostatistical data analysis, including workflows for the interpretation of lipidomic high-throughput data, are reviewed in this manuscript. © 2020 The Authors

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DOI 10.1016/j.addr.2020.06.009

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Living apart-together: Microhabitat differentiation of cryptic nematode species in a saltmarsh habitat

Item Type Journal Article

Author R.M. Guden

Author A.-M. Vafeiadou

Author N. De Meester

Author S. Derycke

Author T. Moens

Abstract Coexistence of highly similar species is at odds with ecological theory of competition; coexistence, then, requires stabilizing mechanisms such as differences in ecological niche. In the bacterivore nematode *Litoditis marina* species complex, which occurs associated with macro-algae, four cryptic lineages (Pm I-IV) co-occur in the field along the south-western coast and estuaries of The Netherlands. Here we investigate the temporal and/or spatial niche differentiation in their natural environment using a qPCR-based detection and relative quantification method. We collected different algal species (i.e. two *Fucus* species and *Ulva* sp.) and separated algal structures (i.e. receptacula, thalli, non-fertile tips and bladders) at different

sampling months and times (i.e. twice per sampling month), to examine differences in microhabitat use between coexisting *L. marina* species. Results demonstrate that the cryptic species composition varied among different algal species and algal structures, which was also subject to temporal shifts. Pm I dominated on *Fucus* spp., Pm II showed dominance on *Ulva* sp., while Pm III overall had the lowest frequencies. Microhabitat partitioning was most pronounced between the two cryptic species which had similar microbiomes (Pm I and Pm II), and less so between the two species which had significantly different microbiomes (Pm I and Pm III), suggesting that species which share the same microhabitats may avoid competition through resource partitioning. The interplay of microhabitat differentiation and temporal dynamics among the cryptic species of *L. marina* implies that there is a complex interaction between biotic components and abiotic factors which contributes to their coexistence in the field. © 2018 Guden et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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human, article, nonhuman, classification, biodiversity, physiology, animal, Animals, Biodiversity, nematode, habitat, Wetlands, real time polymerase chain reaction, sampling, species composition, microbiome, competition, differentiation, *Fucus*, microhabitat, Netherlands, resource partitioning, Rhabditoidea, *Ulva*, wetland

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LiWXAI, Interpretable and Efficient LiDAR-based Weather Condition Detection and Identification using GLCM and AI-based Methodology

Item Type Conference Paper

Author A. Jacquemart

Author S.-S. Ieng

Author M. Hadj-Bachir

Author D. Gruyer

Editor Nedevschi S.

Editor Potolea R.

Editor Slavescu R.R.

Abstract This paper presents an innovative and interpretable LiDAR-based architecture for weather condition detection and identification, leveraging the Gray-Level Co-occurrence Matrix (GLCM) and a dedicated AI model. By exploiting the unique spatial characteristics of LiDAR point cloud data, the proposed approach enhances the accuracy and robustness of detecting three distinct weather conditions: clear, rainy, and foggy. The LiDAR point cloud is preprocessed and projected into a discretized map, from which GLCMs are computed and used as input for model training. In addition, interpretability is a core feature of the method, addressed through two complementary techniques. First, an ante-hoc analysis estimates the 2D conditional probability distribution of GLCM features given the weather class prior to training. Second, a post-hoc visualization is enabled via a reshaped weight map that reveals the information learned by the model. The model is evaluated on the RADIATE dataset and benchmarked against three state-of-the-art weather classifiers. Results demonstrate the effectiveness of the LiDAR-GLCM approach and its strong potential for real-time, embedded weather recognition systems. © 2025 IEEE.

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - IEEE Int. Conf. Intell. Comput. Commun. Process. Conf., ICCP

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 979-833155896-3 (ISBN)

Proceedings Title Proc. - IEEE Int. Conf. Intell. Comput. Commun. Process. Conf., ICCP

Conference Name Proceedings - 2025 IEEE 21st International Conference on Intelligent Computer Communication and Processing Conference, ICCP 2025

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LLM Performance Predictors: Learning When to Escalate in Hybrid Human-AI Moderation Systems

Item Type Conference Paper

Author O. Bachar

Author O. Levi

Author S. Mishra

Author A. Levi

Author M.S. Minhas

Author J. Miller

Author O. Ben-Porat

Author E. Sheetrit

Author J. Morra

Abstract As LLMs are increasingly integrated into human-in-the-loop content moderation systems, a central challenge is deciding when their outputs can be trusted versus when escalation for human review is preferable. We propose a novel framework for supervised LLM uncertainty quantification, learning a dedicated meta-model based on LLM Performance Predictors (LPPs) derived from LLM outputs: log-probabilities, entropy, and novel uncertainty attribution indicators. We demonstrate that our method enables cost-aware selective classification in real-world human-AI workflows: escalating high-risk cases while automating the rest. Experiments across state-of-the-art LLMs, including both off-the-shelf (Gemini, GPT) and open-source (Llama, Qwen), on multimodal and multilingual moderation tasks, show significant improvements over existing uncertainty estimators in accuracy-cost trade-offs. Beyond uncertainty estimation, the LPPs enhance explainability by providing new insights into failure conditions (e.g., ambiguous content vs. under-specified policy). This work establishes a principled framework for uncertainty-aware, scalable, and responsible human-AI moderation workflows. © 2026 International Foundation for Autonomous Agents and Multiagent Systems.

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Language English

Extra Journal Abbreviation: AAMAS - Proc. Int. Conf. Auton. Agents Multiagent Syst.

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Language model, Large language model, Autonomous agents, Economic and social effects, Uncertainty, Human-in-the-loop, Large Language Models, Multi agent systems, Human-AI collaboration, Human-AI Collaboration, Content moderation, Content Moderation, Work-flows, Uncertainty quantifications, Cost estimating, Meta model, Performance predictor, Uncertainty Quantification

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: O. Bachar; Zefr, Los Angeles, United States; email: or.bachar@post.runi.ac.il; Conference name: 25th International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2026; Conference date: 25 May 2026 through 29 May 2026; Conference code: 224081

LLM-Based Multi-Agent System and Simplicial Self-Supervised Learning Model for Regional Cancer Prevalence Estimation Using Satellite Imagery

Item Type Conference Paper

Author J.-A. Yang

Author Y. Chen

Author C.P. Tribby

Author H. Lee

Author L. Erhunmwunsee

Author T. Benmarhnia

Author C.A. Thompson

Author Y.R. Gel

Author M.M. Jankowska

Editor Mokbel M.

Editor Shekar S.

Editor Zufle A.

Editor Chiang Y.-Y.

Editor Damiani M.L.

Editor Youssef M.

Abstract Traditional cancer rate estimations are often limited in spatial resolutions and lack considerations of environmental factors. Satellite imagery has become a vital data source for monitoring diverse urban environments, supporting applications across environmental, socio-demographic, and public health domains. However, while deep learning (DL) tools, particularly convolutional neural networks, have demonstrated strong performance in extracting features from high-resolution imagery, their reliance on local spatial cues often limits their ability to capture complex, non-local, and higher-order structural information. To overcome this limitation, we propose a novel LLM-based multi-agent coordination system for satellite image analysis, which integrates visual and contextual reasoning through a simplicial contrastive learning framework (Agent-SNN). Our Agent-SNN

contains two augmented superpixel-based graphs and maximizes mutual information between their latent simplicial complex representations, thereby enabling the system to learn both local and global topological features. The LLM-based agents generate structured prompts that guide the alignment of these representations across modalities. Experiments with satellite imagery of Los Angeles and San Diego demonstrate that Agent-SNN achieves significant improvements over state-of-the-art baselines in regional cancer prevalence estimation tasks. © 2025 Copyright held by the owner/author(s).

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Language English

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Loading classification proposal for fatigue design of automotive chassis-parts: A relevant process for variable amplitude and multi-input load cases

Item Type Journal Article

Author E. Bellec

Author C. Doudard

Author M.L. Facchinetti

Author S. Calloch

Author S. Moyne

Abstract It is essential for car manufacturers to ensure their vehicle robustness under a daily usage. As no detailed method nor standard prevails, each manufacturer is expected to setup its own methodology for safety parts life-assessment. The early stages of a renewed methodology are proposed in this study. This is based on proving-grounds measurements analysis to quantify the impact of various loads. The method, developed from the load measured at the vehicle wheels, considers the multi-input load case and its variable amplitude content. Measured time-series are analyzed and partitioned into two main categories depending on the time-correlation between the load axes. The driver-induced loads, Driven-Road, (DR) loads, are separated from the effects of random road conditions, Random-Road, (RR) loads. Then, for partition validation, a suitable life quantification method is applied in parallel to the two types of loading. On one hand, an assumption based on strong physical meaning eases the Rainflow Counting method application on the DR load. On the other hand, the use of spectral methods is explored to handle the random loads. This paper details the partition process of the time-series and each of the subsequent validation steps of the method to apply such a partition. Such a validation is performed via a comparison of the pseudo-damage between a referenced signal and the one issued from the partition. © 2022 Elsevier Ltd

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Local similarity discriminant analysis

Item Type Conference Paper

Author L. Cazzanti

Author M.R. Gupta

Abstract We propose a local, generative model for similarity-based classification. The method is applicable to the case that only pairwise similarities between samples are available. The classifier models the local class-conditional distribution using a maximum entropy estimate and empirical moment constraints. The resulting exponential class conditional-distributions are combined with class prior probabilities and misclassification costs to form the local similarity discriminant analysis (local SDA) classifier. We compare the performance of local SDA to a non-local version, to the local nearest centroid classifier, the nearest centroid classifier, k-NN, and to the recently-developed potential support vector machine (PSVM). Results show that local SDA is competitive with k-NN and the computationally-demanding PSVM while offering the advantages of a generative classifier.

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Conference Name ACM International Conference Proceeding Series

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Long term analysis of alpine swamp meadow: area and water balance

Item Type Journal Article

Author J. Yang

Author X. Jin

Author Y. Jin

Author X. Mao

Abstract Background and aims: Alpine swamp meadows play a vital role in water conservation and maintaining ecological balance. However, the response mechanisms of its area and hydrological functions under global climate change remain unclear, particularly the impact of permafrost degradation on water storage capacity, which urgently requires quantification. Methods: We integrated multi—temporal Landsat data (2000—2023) and phenological features to construct a classification framework for alpine swamp meadows. A multi—source remote sensing—based water balance assessment method was developed. Random forest importance evaluation and piecewiseSEM were employed to quantify the impacts and pathways of multidimensional driving factors on changes in alpine swamp meadow area and water storage. Results: The phenology—based classification method effectively extracted alpine swamp meadows with a mean producer's accuracy of 92.84%, user's accuracy of 92.14%, and a Kappa coefficient of 0.95. The study found that the spatial expansion of alpine swamp meadows in the watershed showed an "initial decrease followed by an increase" trend, while the water storage capacity continued to decline, indicating a significant decoupling between the two. Conclusion: Under climate change, increased precipitation and reduced snow cover albedo have led to the expansion of alpine swamp meadows, while enhanced evapotranspiration and the degradation of permafrost aquicludes have caused a systematic decline in their water storage capacity. These findings provide a scientific basis for assessing the health of alpine ecosystems and managing water resources under climate change. © The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025.

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Qinghai Lake Basin, Alpine swamp meadow, Area change, Driving factor analysis, Remote sensing techniques, Water balance

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Long-term vehicle motion prediction

Item Type Conference Paper

Author C. Hermes

Author C. Wohler

Author K. Schenk

Author F. Kummert

Abstract Future driver assistance systems will have to cope with complex traffic situations, especially in the road crossing scenario. To detect potentially hazardous situations as early as possible, it is therefore desirable to know the position and motion of the ego-vehicle and vehicles around it for several seconds in advance. For this purpose, we propose in this study a long-term prediction approach based on a combined trajectory classification and particle filter framework. As a measure for the similarity between trajectories, we introduce the quaternion-based rotationally invariant longest common subsequence (QRLCS) metric. The trajectories are classified by a radial basis function (RBF) classifier with an architecture that is able to process trajectories of arbitrary non-uniform length. The particle filter framework simultaneously tracks and assesses a large number of motion hypotheses (~ 102), where the class-specific probabilities estimated by the RBF classifier are used as a-priori probabilities for the hypotheses of the particle filter. The hypotheses are clustered with a mean-shift technique and are assigned a likelihood value. Motion prediction is performed based on the cluster centre with the highest likelihood. While traditional motion prediction based on curve radius and acceleration is inaccurate especially during turning manoeuvres, we show that our approach achieves a reasonable motion prediction even for long prediction intervals of 3 s for these complex motion patterns. © 2009 IEEE.

Date 2009

DOI 10.1109/IVS.2009.5164354

URL <https://www.scopus.com/pages/publications/70449602543?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: IEEE Intell Veh Symp Proc

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 19310587 (ISSN); 978-142443504-3 (ISBN)

Pages 652-657

Proceedings Title IEEE Intell Veh Symp Proc

Conference Name IEEE Intelligent Vehicles Symposium, Proceedings

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Tags:

Risk perception, Risk assessment, Magnetic levitation vehicles, Motor transportation, Road crossing, Trajectories, Advanced driver assistance systems, Traffic situations, Particle filter, Radial basis function networks, Classification filters, Future driver assistance systems, Long-term prediction, Motion prediction, Radial basis function classifiers, Trajectory classification, Vehicle motion

Notes:

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LooM: An anonymity quantification method in pervasive computing environments

Item Type Journal Article

Author M. Imada

Author M. Ohta

Author M. Yamaguchi

Author S.Y. Kim

Abstract Purpose: The paper's aim is to present a novel anonymity quantification method, LooM (loosely managed privacy protection method) for achieving privacy protection in pervasive computing environments. Design/methodology/approach: The main feature is that the method quantitatively controls anonymity by a single value (disclosure threshold value) using a classification algorithm of the decision tree. The value is not affected by the set size of users or the distribution of users' private information. The effectiveness of this method is confirmed by simulation using sample databases of attribute-value pairs. Proposes a model of privacy information disclosure that achieves a balance between users' privacy protection requirements and service providers' disclosure requirements and applies web questionnaire survey data to this model. Findings: LooM can be applicable to a variety of pervasive computing and communication services handling a huge amount of data containing privacy information. Originality/value: The paper proposes a model of privacy information disclosure that achieves a balance between users' privacy protection requirements and service providers' disclosure requirements. © Emerald Group Publishing Limited.

Date 2008

DOI 10.1108/17427370810873147

URL <https://www.scopus.com/pages/publications/84893389645?origin=resultslist>

Archive Scopus

Language English

Volume 4

Pages 110-123

Publication International Journal of Pervasive Computing and Communications

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Issue 1

Journal Abbr Int. J. Pervasive Comput. Commun.

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Tags:

Privacy, Computer networks

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EID: 2-s2.0-84893389645

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Imada; NTT Network Innovation Labs, Tokyo, Japan; email: imada.miyuki@lab.ntt.co.jp

LooM: An anonymity quantification method in pervasive computing environments

Item Type Conference Paper

Author M. Imada

Author M. Ohta

Author M. Yamaguchi

Abstract We propose a novel anonymity quantification method for privacy protection in pervasive computing environments. Its main feature is that it can quantitatively control anonymity by a single value (disclosure threshold value) using a classification algorithm of the decision tree. The value is not affected by the user set size or the contents of private information. This property was confirmed by evaluation using sample databases. In order to decide the disclosure threshold value for controlling anonymity, we established a model of privacy information disclosure that achieves a balance (equilibrium point) between users' privacy protection requirements and service providers' disclosure requirements. Applying web questionnaire data to this model, we found the equilibrium point for each service. © 2006 IEEE.

Date 2006

DOI 10.1109/AINA.2006.210

URL <https://www.scopus.com/pages/publications/33751102679?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Int. Conf. Adv. Inf. Netw. Appl. AINA

Volume 2

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 1550445X (ISSN); 0769524664 (ISBN); 978-076952466-5 (ISBN)

Pages 92-96

Proceedings Title Proc. Int. Conf. Adv. Inf. Netw. Appl. AINA

Conference Name Proceedings - International Conference on Advanced Information Networking and Applications, AINA

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Tags:

Feature extraction, Privacy protection, Algorithms, Decision theory, Computation theory, Security of data, Data structures, Information theory, World Wide Web, Anonymity quantification, Control anonymity, Pervasive computing

Notes:

EID: 2-s2.0-33751102679

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: M. Imada; NTT Network Innovation Laboratories, Musashino-shi, Tokyo, 180-8585, Midori-cho 3-9-11, Japan; email: imada@ma.onlab.ntt.co.jp; Conference name: 20th International Conference on Advanced Information Networking and Applications, AINA 2006; Conference date: 18 April 2006 through 20 April 2006; Conference code: 68528

LotBoNC: Novel Botnet Traffic Classification under Long-tailed Distributions

Item Type Conference Paper

Author H. Hu

Author Z. Li

Author C. Doerr

Abstract Botnets are a persistent cyber threat, leveraging global infrastructures to launch large-scale attacks. Yet, most existing classification methods are evaluated under balanced and closed-set assumptions, which fail to capture real-world conditions. In practice, botnet traffic is both long-tailed and open-world: unknown variants continually emerge, and rare threats are buried under dominant traffic, often evading detection. To reflect real-world conditions, we define a deployment-oriented setting where unlabeled traffic follows a long-tailed distribution, with dominant known classes in the head and rare novel botnet variants in the tail. We propose LotBoNC, a unified framework for encrypted traffic classification under long-tailed open-world conditions. LotBoNC first performs self-supervised pre-training to learn transferable representations, then applies entropy-regularized optimal transport to assign pseudo-labels aligned with estimated class priors. An EM-style loop iteratively refines prototypes and priors, improving class

separation between frequent and rare categories. We evaluate LotBoNC on three public encrypted traffic datasets with diverse long-tailed scenarios. LotBoNC consistently outperforms prior state-of-the-art methods and accurately classifies known botnets and discovers unseen botnet variants in diverse, unbalanced open-world scenarios. © 2026 Copyright held by the owner/author(s).

Date 2026

DOI 10.1145/3779208.3785276

URL <https://www.scopus.com/pages/publications/105042465490?origin=resultslist>

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Language English

Extra Journal Abbreviation: ASIA CCS - Proc. ACM ASIA Conf. Comput. Commun. Secur.

Publisher Association for Computing Machinery, Inc

ISBN 979-840072356-8 (ISBN)

Pages 1013-1027

Proceedings Title ASIA CCS - Proc. ACM ASIA Conf. Comput. Commun. Secur.

Conference Name ASIA CCS 2026 - Proceedings of the 21st ACM ASIA Conference on Computer and Communications Security

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Tags:

Artificial intelligence, Network security, Condition, Classification (of information), Intrusion detection, Learning systems, Machine-learning, Networks security, Human computer interaction, Real-world, Computer crime, Network Security, Machine Learning, Intrusion-Detection, Telecommunication traffic, Traffic classification, Open world, Out-of-distribution, Bot (Internet), Botnet, Botnets, Intrusion Detection, Long-tailed distributions, Out-of-Distribution

Notes:

EID: 2-s2.0-105042465490

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: C. Doerr; Hasso Plattner Institute, Potsdam, Germany; email: christian.doerr@hpi.de; Conference name: 21st ACM Asia Conference on Computer and Communications Security, AsiaCCS 2026; Conference date: 1 June 2026 through 5 June 2026; Conference code: 224474

Low Complexity, Low Probability Patterns and Consequences for Algorithmic Probability Applications

Item Type Journal Article

Author M. Alaskandarani

Author K. Dingle

Abstract Developing new ways to estimate probabilities can be valuable for science, statistics, engineering, and other fields. By considering the information content of different output patterns, recent work invoking algorithmic information theory inspired arguments has shown that a priori probability predictions based on pattern complexities can be made in a broad class of input-output maps. These algorithmic probability predictions do not depend on a detailed knowledge of how output patterns were produced, or historical statistical data. Although quantitatively fairly accurate, a main weakness of these predictions is that they are given as an upper bound on the probability of a pattern, but many low complexity, low probability patterns occur, for which the upper bound has little predictive value. Here, we study this low complexity, low probability phenomenon by looking at example maps, namely a finite state transducer, natural time series data, RNA molecule structures, and polynomial curves. Some mechanisms causing low complexity, low probability behaviour are identified, and we argue this behaviour should be assumed as a default in the real-world algorithmic probability studies. Additionally, we examine some applications of algorithmic probability and discuss some implications of low complexity, low probability patterns for several research areas including simplicity in physics and biology, a priori probability predictions, Solomonoff induction and Occam's razor, machine learning, and password guessing. © 2023 Mohammad Alaskandarani and Kamaludin Dingle.

Date 2023

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Language English

Volume 2023

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Journal Abbr Complexity

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Forecasting, Information theory, Prediction-based, Information contents, Upper Bound, A-priori probabilities, Algorithmic information theory, Algorithmic probability, Input output map, Lower complexity, Lower probabilities, Parallel processing systems, Polynomial approximation, Statistical datas

Notes:

EID: 2-s2.0-85160210358

Export Date: 20 September 2026; Cited By: 7; Correspondence Address: K. Dingle; Centre for Applied Mathematics and Bioinformatics (CAMB), Department of Mathematics and Natural Sciences, Gulf University for Science and Technology, Mubarak Al-Abdullah, Kuwait; email: dinglek@caltech.edu

Low-Resource Text Processing with Lightweight Multi-modal Agents: Enhanced Entity Recognition based on Few-Shot Learning

Item Type Conference Paper

Author Y. Zhang

Abstract In the context of low-resource text processing, the scarcity of labeled samples, insufficient semantic expression capabilities, and limited model knowledge calibration abilities are common practical challenges. The named entity recognition task generally faces issues such as decreased generalization performance and insufficient prediction stability. To address this core problem, this paper proposes a lightweight multimodal intelligent entity recognition framework for low-resource text processing. This method first constructs a multi-scale semantic plane representation mechanism, extending the traditional one-dimensional sequence encoding to a two-dimensional span-level semantic modeling structure. This paradigm can enhance the model's ability to express entity boundaries and context-dependent relationships. On this basis, this paper designs the Adaptive Biaffine Span Synthesis (ABSS) algorithm. This algorithm achieves context information fusion through multi-scale expansion convolution and dynamically estimates the importance of different semantic scales using the Scale-Gating Calibration Network (SGCN). Subsequently, this paper proposes a few-shot knowledge injection mechanism based on the Label Semantic Field (LSF). This mechanism realizes dynamic semantic alignment between text span representation and label knowledge by continuously modeling the label description information. At the same time, it enhances the classification ability through the Contrast-Calibrated Interaction (CCI) function. To solve the conflict between internal parameter knowledge and external context knowledge in low-resource environments, this paper constructs a lightweight knowledge reasoning intelligent agent. This paradigm introduces a knowledge conflict quantification method based on Jensen-Shannon divergence and an adaptive uncertainty decision-making mechanism based on information gain. This paper conducted systematic experiments on datasets such as Few-NERD INTRA, Few-NERD INTER, MIT-Restaurant Low-Resource, and a self-built dataset. The experimental results showed that in the 1-shot task of Few-NERD INTER, the method proposed in this paper achieved an F1 value of 42.05%. This result was 6.43 percentage points higher than the optimal baseline AugGPT-NER. Under the conditions of 5-shot and 10-shot, the F1 performance reached 53.96% and 62.38% respectively. On the noisy SME dataset, the method proposed in this paper outperformed StructShot by 8.09 percentage points in the 1-shot scenario. The ablation experiment results indicated that the model performance decreased by 7.75 percentage points after removing the label semantic field module. Meanwhile, the knowledge conflict analysis results showed that the uncertainty-aware reasoning mechanism proposed in this paper could significantly improve the recognition stability of high-conflict samples. © 2026 IEEE.

Date 2026

DOI 10.1109/ICOSAAS68663.2026.11649291

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Archive Scopus

Language English

Extra Journal Abbreviation: Int. Conf. Intell. Sustain. AI Syst., ICOSAAS - Conf. Proc.

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 979-833157974-6 (ISBN)

Proceedings Title Int. Conf. Intell. Sustain. AI Syst., ICOSAAS - Conf. Proc.
Conference Name International Conference on Intelligent and Sustainable AI Systems, ICOSAAS 2026 - Conference Proceedings
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Tags:

Classification (of information), Decision making, Uncertainty analysis, Convolution, Semantics, Text processing, Few-shot learning, Multi agent systems, Information fusion, Intelligent agents, Few-Shot Learning, Percentage points, Label semantics, Text-processing, Low-resource, Low-Resource, Text Processing, Entity recognition, Calibration, Entity Recognition, Lightweight multimodal agent, Lightweight Multimodal Agents, Mechanism-based, Multi-modal agent, Multi-scales

Notes:

EID: 2-s2.0-105048464393

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Y. Zhang; Taren Times Technology Group Co., Ltd., Beijing, 100098, China; email: zhangyanzhong5@163.com; Conference name: 2026 International Conference on Intelligent and Sustainable AI Systems, ICOSAAS 2026; Conference date: 20 July 2026 through 22 July 2026; Conference code: 227004

LUMA: A Benchmark Dataset for Learning from Uncertain and Multimodal Data

Item Type Conference Paper

Author G. Bezirganyan

Author S. Sellami

Author L. Berti-Équille

Author S. Fournier

Abstract Multimodal Deep Learning enhances decision-making by integrating diverse information sources, such as texts, images, audio, and videos. To develop trustworthy multimodal approaches, it is essential to understand how uncertainty impacts these models. We propose LUMA, a unique multimodal dataset, featuring audio, image, and textual data from 50 classes, specifically designed for learning from uncertain data. It extends the well-known CIFAR 10/100 dataset with audio samples extracted from three audio corpora, and text data generated using the Gemma-7B Large Language Model (LLM). The LUMA dataset enables the controlled injection of varying types and degrees of uncertainty to achieve and tailor specific experiments and benchmarking initiatives. LUMA is also available as a Python package including the functions for generating multiple variants of the dataset with controlling the diversity of the data, the amount of noise for each modality, and adding out-of-distribution samples. A baseline pre-trained model is also provided alongside three uncertainty quantification methods: Monte-Carlo Dropout, Deep Ensemble, and Reliable Conflicting Multi-View Learning. This comprehensive dataset and its tools are intended to promote and support the

development, evaluation, and benchmarking of trustworthy and robust multimodal deep learning approaches. We anticipate that the LUMA dataset will help the research community to design more trustworthy and robust machine learning approaches for safety critical applications. The code and instructions for downloading and processing the dataset can be found at: <https://github.com/bezorganyan/LUMA>. © 2025 Copyright held by the owner/author(s).

Date 2025

DOI 10.1145/3726302.3730302

URL <https://www.scopus.com/pages/publications/105011819704?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: SIGIR - Proc. Int. ACM SIGIR Conf. Res. Dev. Inf. Retr.

Publisher Association for Computing Machinery, Inc

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Pages 3782-3791

Proceedings Title SIGIR - Proc. Int. ACM SIGIR Conf. Res. Dev. Inf. Retr.

Conference Name SIGIR 2025 - Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval

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Tags:

Computational linguistics, Safety engineering, Deep learning, Decisions makings, Decision making, Learning systems, Uncertainty analysis, Benchmarking, Large datasets, Data mining, Multi-modal, Information sources, dataset, Benchmark datasets, Text images, Multi-modal approach, Dataset, Audio acoustics, Uncertainty quantifications, uncertainty quantification, Audio and video, multimodal deep learning, Multimodal deep learning

Notes:

EID: 2-s2.0-105011819704

Export Date: 20 September 2026; Cited By: 3; Conference name: 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2025; Conference date: 13 July 2025 through 18 July 2025; Conference code: 210356

Lumen segmentation and stenosis quantification of atherosclerotic carotid arteries in CTA utilizing a centerline intensity prior

Item Type Journal Article

Author H. Tang

Author T. Van Walsum

Author R. Hameeteman

Author R. Shahzad

Author L.J. Van Vliet

Author W.J. Niessen

Abstract Purpose: The degree of stenosis is an important biomarker in assessing the severity of cardiovascular disease. The purpose of our work is to develop and evaluate a semiautomatic method for carotid lumen segmentation and subsequent carotid artery stenosis quantification in CTA images. Methods: The authors present a semiautomatic stenosis detection and quantification method following lumen segmentation. The lumen of the carotid arteries is segmented in three steps. First, centerlines of the internal and external carotid arteries are extracted with an iterative minimum cost path approach in which the costs are based on a measure of medialness and intensity similarity to lumen. Second, the lumen boundary is delineated using a level set procedure which is steered by gradient information, regional intensity information, and spatial information. Special effort is made in adding terms based on local centerline intensity prior so as to exclude all possible plaque tissues from the segmentation. Third, side branches in the segmented lumen are removed by applying a shape constraint to the envelope of the maximum inscribed spheres of the segmentation. From the segmented lumen, the authors detect and quantify the cross-sectional area-based and cross-sectional diameter-based stenosis degrees according to the North American Symptomatic Carotid Endarterectomy Trial criterion. Results: The method is trained and tested on a publicly available database from the cls2009 challenge. For the segmentation, the authors obtain a Dice similarity coefficient of 90.2% and a mean absolute surface distance of 0.34 mm. For the stenosis quantification, the authors obtain an average error of 15.7% for cross-sectional diameter-based stenosis and 19.2% for cross-sectional area-based stenosis quantification. Conclusions: With these results, the method ranks second in terms of carotid lumen segmentation accuracy, and first in terms of carotid artery stenosis quantification. © 2013 American Association of Physicists in Medicine.

Date 2013

DOI 10.1118/1.4802751

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Language English

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automation, human, Image segmentation, article, procedures, controlled study, Iterative methods, methodology, process optimization, image processing, disease classification, measurement accuracy, Image Processing, Computer-Assisted, computed tomographic angiography, Angiography, calcium, internal carotid artery, Atherosclerosis, analytic method, kernel method, carotid artery, angiography, atherosclerosis, Cardio-vascular disease, radiography, priority journal, three dimensional imaging, artery diameter, blood vessel wall, Calcium, cardiovascular procedure, Carotid Arteries, carotid artery bifurcation, carotid artery disease, Carotid Artery Diseases, carotid artery obstruction, Carotid artery stenosis, carotid artery stenosis quantification, carotid atherosclerosis, carotid lumen segmentation, Constriction, Pathologic, Cross sectional diameters, envelope of the maximum inscribed spheres, External carotid arteries, external carotid artery, geodesic active contour, Geodesic Active Contour, gradient magnitude, Gradient magnitude, Maximum inscribed spheres, regional intensity, stenosis, occlusion and obstruction

Notes:

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Export Date: 20 September 2026; Cited By: 20; Correspondence Address: H. Tang; Biomedical Imaging Group Rotterdam, Departments of Radiology and Medical Informatics, Erasmus MC, 3000 CA Rotterdam, P.O. Box 2040, Netherlands; email: tanghui.seu@gmail.com; CODEN: MPHYA

M-Count: an application that uses machine learning object detection and color thresholding to count settled mussel larvae

- Item Type Journal Article
- Author L.W. Miller
- Author N. Nune
- Author T.B. LeFevre
- Author C.N. Hermanson
- Author J.D. Daddona
- Author A. Valdez
- Author A. Kannan
- Author W. Chouyyok
- Author G.T. Bonheyo
- Author R.S. Addleman
- Author C.J. Larimer

Abstract Quantification of biofouling is a complex task that often involves counting both isolated and tightly packed groups of organisms on a test surface. Mussel larvae, which settle as both individuals and as clumps, are an important fouling organism because adult mussel colonies form via the settlement of larvae, making the settlement or repellence of mussel larvae a good indicator of a surface's antifouling performance. Manual quantification methods are time-consuming, and existing

automatic machine learning-based methods are poorly suited for use by coding non-experts and often lack the ability to detect both isolated and grouped organisms in one workflow. The objective of this work was to develop a machine learning-based application that is user-friendly and well-suited to the quantification of biofouling. We developed M-Count, an application that combines a neural network object detection model for individual organism detection and a color thresholding algorithm for grouped organism detection. M-Count was demonstrated on the quantification of mussel larvae on sample surface images obtained from a previously developed mussel larvae fouling assay. This study revealed three important characteristics of M-Count: speed, consistency, and accuracy. The primary benefit of M-Count is its speed, being 60× faster than manual counting. The secondary benefit of M-Count is consistency, as it performs the same task repeatedly without bias. Finally, these benefits are obtained while maintaining good accuracy, with the normalized average maximum residual being 0.220 for M-Count and 0.209 for manual counting. This journal is © The Royal Society of Chemistry, 2026

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Machine Learning Aided Holistic Handover Optimization for Emerging Networks

Item Type Conference Paper

Author M.U.B. Farooq

Author M. Manalastas

Author S.M.A. Zaidi

Author A. Abu-Dayya

Author A. Imran

Abstract In the wake of network densification and multi-band operation in emerging cellular networks, mobility and handover management is becoming a major bottleneck. The problem is further aggravated by the fact that holistic mobility management solutions for different types of handovers, namely inter-frequency and intra-frequency handovers, remain scarce. This paper presents a first mobility management solution that concurrently optimizes inter-frequency related A5 parameters and intra-frequency related A3 parameters. We analyze and optimize five parameters namely A5-time to trigger (TTT), A5-threshold1, A5-threshold2, A3-TTT, and A3-offset to jointly maximize three critical key performance indicators (KPIs): edge user reference signal received power (RSRP), handover success rate (HOSR) and load between frequency bands. In the absence of tractable analytical models due to system level complexity, we leverage machine learning to quantify the KPIs as a function of the mobility parameters. An XGBoost based model has the best performance for edge RSRP and HOSR while random forest outperforms others for load prediction. An analysis of the mobility parameters provides several insights: 1) there exists a strong coupling between A3 and A5 parameters; 2) an optimal set of parameters exists for each KPI; and 3) the optimal parameters vary for different KPIs. We also perform a SHAP based sensitivity to help resolve the parametric conflict between the KPIs. Finally, we formulate a maximization problem, show it is non-convex, and solve it utilizing simulated annealing (SA). Results indicate that ML-based SA-aided solution is more than 14x faster than the brute force approach with a slight loss in optimality. © 2022 IEEE.

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Archive Scopus

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Volume 2022-May

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Conference Name IEEE International Conference on Communications

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Tags:

Optimisations, Machine-learning, Machine learning, Benchmarking, 5G mobile communication systems, 5G, Machine Learning, Decision trees, 6G, Simulated annealing, 5g, 6g, Hand over, Handover optimization, Key performance indicators, Mobility management, Mobility Management, Received power, Reference signals

Notes:

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Machine learning algorithm based battery modeling and management method: A Cyber-Physical System perspective

Item Type Conference Paper

Author S. Li

Author H. He

Author J. Li

Author P. Yin

Author H. Wang

Abstract In recent years, in order to realize the accurate state monitoring and management of battery, the development of a flexible, self-reconfigurable and reliable model has become one of the most crucial technologies for electric vehicles. This paper mainly focuses on the battery management issues in new energy vehicles, in which the concept of artificial intelligence and grid-connected vehicle is introduced. Firstly, the concept of Cyber-Physical system (CPS) is applied in battery management issues in our work for a better use of battery data. To establish a precise battery model in cloud, the Support vector regression (SVR) algorithm, a classical artificial intelligence algorithm, is used in our work to model the battery. Finally, a rain-flow cycle counting algorithm-based battery degradation quantification method is proposed to deal with the influence of battery aging phenomenon during modeling the battery. © 2019 IEEE.

Date 2019

DOI 10.1109/CVCI47823.2019.8951635

URL <https://www.scopus.com/pages/publications/85078756186?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Conf. Veh. Control Intell., CVCI

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Proceedings Title Conf. Veh. Control Intell., CVCI

Conference Name 3rd Conference on Vehicle Control and Intelligence, CVCI 2019

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Tags:

Big data, electric vehicle, Electric vehicles, Learning systems, machine learning, Machine learning, Cyber Physical System, Learning algorithms, Embedded systems, Artificial intelligence algorithms, Battery degradation, Battery management systems, Information management, cyber-physical system, New energy vehicles, Control system synthesis, big data, Quantification methods, Cyber-physical systems (CPS), grid-connected vehicle, Grid-connected vehicles, Lithium-ion batteries, lithium-ion battery, Self-reconfigurable, Support vector regression (SVR)

Notes:

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Machine Learning Algorithms Predict Achievement of Clinically Significant Outcomes After Orthopaedic Surgery: A Systematic Review

Item Type Journal Article

Author K.N. Kunze

Author L.M. Krivich

Author I.M. Clapp

Author B.M. Bodendorfer

Author B.U. Nwachukwu

Author J. Chahla

Author S.J. Nho

Abstract Purpose: To determine what subspecialties have applied machine learning (ML) to predict clinically significant outcomes (CSOs) within orthopaedic surgery and to determine whether the performance of these models was acceptable through assessing discrimination and other ML metrics where reported. Methods: The PubMed, EMBASE, and Cochrane Central Register of Controlled Trials databases were queried for articles that used ML to predict achievement of the minimal clinically important difference (MCID), patient acceptable symptomatic state (PASS), or substantial clinical benefit (SCB) after orthopaedic surgical procedures. Data pertaining to demographic characteristics, subspecialty, specific ML algorithms, and algorithm performance were analyzed. Results: Eighteen articles met the inclusion

criteria. Seventeen studies developed novel algorithms, whereas one study externally validated an established algorithm. All studies used ML to predict MCID achievement, whereas 3 (16.7%) predicted SCB achievement and none predicted PASS achievement. Of the studies, 7 (38.9%) concerned outcomes after spine surgery; 6 (33.3%), after sports medicine surgery; 3 (16.7%), after total joint arthroplasty (TJA); and 2 (11.1%), after shoulder arthroplasty. No studies were found regarding trauma, hand, elbow, pediatric, or foot and ankle surgery. In spine surgery, concordance statistics (C-statistics) ranged from 0.65 to 0.92; in hip arthroscopy, 0.51 to 0.94; in TJA, 0.63 to 0.89; and in shoulder arthroplasty, 0.70 to 0.95. Most studies reported C-statistics at the upper end of these ranges, although populations were heterogeneous. Conclusions: Currently available ML algorithms can discriminate the propensity to achieve CSOs using the MCID after spine, TJA, sports medicine, and shoulder surgery with a fair to good performance as evidenced by C-statistics ranging from 0.6 to 0.95 in most analyses. Less evidence is available on the ability of ML to predict achievement of SCB, and no evidence is available for achievement of the PASS. Such algorithms may augment shared decision-making practices and allow clinicians to provide more appropriate patient expectations using individualized risk assessments. However, these studies remain limited by variable reporting of performance metrics, CSO quantification methods, and adherence to predictive modeling guidelines, as well as limited external validation. Level of Evidence: Level III, systematic review of Level III studies. © 2021 Arthroscopy Association of North America

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Machine Learning and Intelligent Diagnostics in Dental and Orofacial Pain Management: A Systematic Review

Item Type Journal Article

Author T.H. Farook

Author N.B. Jamayet

Author J.Y. Abdullah

Author M.K. Alam

Abstract Purpose. The study explored the clinical influence, effectiveness, limitations, and human comparison outcomes of machine learning in diagnosing (1) dental diseases, (2) periodontal diseases, (3) trauma and neuralgias, (4) cysts and tumors, (5) glandular disorders, and (6) bone and temporomandibular joint as possible causes of dental and orofacial pain. Method. Scopus, PubMed, and Web of Science (all databases) were searched by 2 reviewers until 29th October 2020. Articles were screened and narratively synthesized according to PRISMA-DTA guidelines based on predefined eligibility criteria. Articles that made direct reference test comparisons to human clinicians were evaluated using the MI-CLAIM checklist. The risk of bias was assessed by JBI-DTA critical appraisal, and certainty of the evidence was evaluated using the GRADE approach. Information regarding the quantification method of dental pain and disease, the conditional characteristics of both training and test data cohort in the machine learning, diagnostic outcomes, and diagnostic test comparisons with clinicians, where applicable, were extracted. Results. 34 eligible articles were found for data synthesis, of which 8 articles made direct reference comparisons to human clinicians. 7 papers scored over 13 (out of the evaluated 15 points) in the MI-CLAIM approach with all papers scoring 5+ (out of 7) in JBI-DTA appraisals. GRADE approach revealed serious risks of bias and inconsistencies with most studies containing more positive cases than their true prevalence in order to facilitate machine learning. Patient-perceived symptoms and clinical history were generally found to be less reliable than radiographs or histology for training accurate machine learning models. A low agreement level between clinicians training the models was suggested to have a negative impact on the prediction accuracy. Reference comparisons found nonspecialized clinicians with less than 3 years of experience to be disadvantaged against trained models. Conclusion. Machine learning in dental and orofacial healthcare has shown respectable results in diagnosing diseases with symptomatic pain and with improved future iterations and can be used as a diagnostic aid in the clinics. The current review did not internally analyze the machine learning models and their respective algorithms, nor consider the confounding variables and factors responsible for shaping the orofacial disorders responsible for eliciting pain. © 2021 Taseef Hasan Farook et al.

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Export Date: 20 September 2026; Cited By: 46; Correspondence Address: J.Y. Abdullah; Craniofacial Imaging and Additive Manufacturing Laboratory, School of Dental Sciences, Universiti Sains Malaysia, Health Campus, Kelantan, 16150, Malaysia; email: johariyap@usm.my; M.K. Alam; College of Dentistry, Jouf University, Sakaka, Saudi Arabia; email: dralam@gmail.com; CODEN: PRMAF

Machine learning approaches for brain morphology analysis in MR images

Item Type Conference Paper

Author S. Kobashi

Author S.B. Alam

Author M. Nii

Abstract Brain shape deforms by aging and/or brain disorders. In order to reduce the radiologists' subjective evaluation, automated quantification method of brain magnetic resonance (MR) images is required. This paper introduces a machine learning algorithm into automated quantification of the brain morphology condition. The first application is to estimate the brain age, and the second application is to estimate Mini-Mental State Examination (MMSE). The method is applied to OASIS database. © 2016, Fuji Technology Press. All rights reserved.

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Machine Learning Assisted Image Analysis for Microalgae Prediction

Item Type Journal Article

Author K. Meenatchi Sundaram

Author S. Sravan Kumar

Author A. Deshpande

Author S. Chinnadurai

Author K. Rajendran

Abstract Microalgae-based wastewater treatment has resulted in a paradigm shift toward nutrient removal and simultaneous resource recovery. However, traditionally used microalgal biomass quantification methods are time-consuming and costly, limiting their large-scale use. The aim of this study is to develop a simple and cost-effective image-based method for microalgae quantification, replacing cumbersome traditional techniques. In this study, preprocessed microalgae images and associated optical density data were utilized as inputs. Three feature extraction methods were compared alongside eight machine learning (ML) models, including linear regression (LR), random forest (RF), AdaBoost, gradient boosting (GB), and various neural networks. Among these algorithms, LR with principal component analysis achieved an R^2 value of 0.97 with the lowest error of 0.039. Combining image analysis and ML removes the need for expensive equipment in microalgae quantification. Sensitivity analysis was performed by varying the train-test splitting ratio. Training time was included in the evaluation, and accounting for energy consumption in the study leads to the achievement of high model performance and energy-efficient ML model utilization. © 2024 American Chemical Society.

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ISSN 26900645 (ISSN)**Issue** 2**Journal Abbr** ACS. ES. T. Eng.**Date Added** 20/09/2026, 9:51:38 pm**Modified** 20/09/2026, 9:51:38 pm**Tags:**

Machine learning, Prediction, Image analysis, Wastewater, Microalgae

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Machine learning enhanced predictions of ICRF heating: Overcoming numerical limitations via data curation

Item Type Journal Article**Author** A. Sanchez-Villar**Author** Z. Bai**Author** N. Bertelli**Author** E.W. Bethel**Author** J. Hillairet**Author** T. Perciano**Author** S. Shiraiwa**Author** G.M. Wallace**Author** J.C. Wright

Abstract In this work, we present the development of robust surrogate models for Ion Cyclotron Range of Frequencies (ICRF) and High-Harmonic Fast Wave (HHFW) heating predictions in fusion plasmas. Building upon our previous efforts to achieve real-time capable models, we identify the cause of the outliers found using TORIC in certain HHFW heating scenarios. The outliers are observed to be spurious ion Bernstein wave (IBW)-like modes caused by a wavelength control algorithm designed to address challenging scenarios with high perpendicular wavenumbers. The effect arises from the modulation in the perpendicular susceptibility, which can induce sign reversal and IBW-like propagation for scenarios featuring normalized ion Larmor radius $\lambda_i \gg 1$. We use TORIC with this algorithm disabled to generate a novel HHFW-NSTX database that is free of outliers. Surrogate models trained on this database, including Random Forest Regressor (RFR), Multi-Layer Perceptrons, and Gaussian Process Regressors (GPR), demonstrate the ability to accurately predict HHFW heating profiles, with regression scores of $R^2 \in [0.93 - 0.99]$. Additionally we demonstrate that it is possible to generalize predictions beyond training data by the use of both RFR and GPR models, enabling the prediction of scenarios previously

limited to the original model. GPR models also provide uncertainty quantification, offering insights into model confidence. This work introduces a comprehensive Verification, Validation, and Uncertainty Quantification methodology for surrogate modeling, applicable not only to ICRF heating but also to other RF heating challenges and fusion physics problems. Beyond accelerated inference, these models show effective extrapolation capabilities, providing an alternative for addressing numerical challenges. © 2025 Author(s).

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Machine learning framework for revealing the intelligent transportation combined strategies for energy-saving potential

Item Type Journal Article

Author Z. Jia

Author H. Mao

Author D. Lu

Author J. Zhang

Author L. Zhang

Author S. Meng

Author Y. Zhou

Author Y. Guo

Author W. Fu

Abstract The combined strategy of vehicle electrification and traffic intelligence is a viable option for decarbonizing road transport and improving energy efficiency. However, the difficulty in quantifying the energy saving potential hinders the sustainability of this strategy. This study combines Markov chains and machine learning to quantify the energy saving potential of new energy vehicles (NEVs) under all-mode Intelligent Transportation Speed Guidance System (ITSGS). The combined strategy of Multi-Intersection ITSGS (MITSGS) and NEVs (plug-in hybrid electric vehicle, PHEV and battery electric vehicle, BEV) has significant energy saving potentials of up to 21% and 79%, respectively, when compared to Internal Combustion Engine Vehicle (ICEV) without ITSGS. For the PHEV, the energy consumption model and reported results correspond to charge-sustaining operation. The energy consumption of other combination strategies is as follows. When without ITSGS, the average energy consumption of ICEV, PHEV and BEV is 0.62 ± 0.07 kWh/km, 0.56 ± 0.05 kWh/km and 0.15 ± 0.01 kWh/km, respectively. When with single-intersection ITSGS (SITSGS), these values are 0.56 ± 0.04 kWh/km, 0.52 ± 0.04 kWh/km, 0.14 ± 0.01 kWh/km. When with MITSGS, these values are 0.54 ± 0.04 kWh/km, 0.49 ± 0.04 kWh/km, 0.13 ± 0.01 kWh/km. ITSGS reduces energy consumption by improving vehicle driving behavior at intersections. The framework harmonizes the energy consumption assessment scales of full-mode ITSGS and NEVs combination strategies, which is helpful for the promotion and popularization of combination strategies. This framework uniquely harmonizes scenario-based driving cycles with machine learning to enable consistent, quantifiable evaluation of energy saving potentials across different ITSGS modes and vehicle types. © 2026 Taylor & Francis Group, LLC.

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Machine learning in the prediction of cardiac epicardial and mediastinal fat volumes

Item Type Journal Article

Author É.O. Rodrigues

Author V.H.A. Pinheiro

Author P. Liatsis

Author A. Conci

Abstract We propose a methodology to predict the cardiac epicardial and mediastinal fat volumes in computed tomography images using regression algorithms. The obtained results indicate that it is feasible to predict these fats with a high degree of correlation, thus alleviating the requirement for manual or automatic segmentation of both fat volumes. Instead, segmenting just one of them suffices, while the volume of the other may be predicted fairly precisely. The correlation coefficient obtained by the Rotation Forest algorithm using MLP Regressor for predicting the mediastinal fat based on the epicardial fat was 0.9876, with a relative absolute error of 14.4% and a root relative squared error of 15.7%. The best correlation coefficient obtained in the prediction of the epicardial fat based on the mediastinal was 0.9683 with a relative absolute error of 19.6% and a relative squared error of 24.9%. Moreover, we analysed the feasibility of using linear regressors, which provide an intuitive interpretation of the underlying approximations. In this case, the obtained correlation coefficient was 0.9534 for predicting the mediastinal fat based on the epicardial, with a relative absolute error of 31.6% and a root relative squared error of 30.1%. On the prediction of the epicardial fat based on the mediastinal fat, the correlation coefficient was 0.8531, with a relative absolute error of 50.43% and a root relative squared error of

52.06%. In summary, it is possible to speed up general medical analyses and some segmentation and quantification methods that are currently employed in the state-of-the-art by using this prediction approach, which consequently reduces costs and therefore enables preventive treatments that may lead to a reduction of health problems. © 2017 Elsevier Ltd

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Machine learning of three-dimensional subsurface geological model for a reclamation site in Hong Kong

Item Type Journal Article

Author C. Shi

Author Y. Wang

Abstract Land reclamation from ocean is a major solution to deal with land shortage in coastal megacities such as Hong Kong. The primary geotechnical risk associated with land reclamation is consolidation of fine-grained materials, e.g., soft marine deposit, and a sound understanding of spatial distribution of three-dimensional (3D) subsurface soil layer boundaries, or interfaces, and their stratigraphic connectivity to surrounding drainage boundaries is a prerequisite for an effective reclamation design. In practice, accurate delineation of 3D subsurface stratigraphic boundaries is challenging due to a lack of effective tools for building 3D subsurface geological domains from limited site-specific data while taking full account of geological uncertainty. In this study, a novel stratigraphic modelling and uncertainty quantification method, called 3D iterative convolution eXtreme Gradient Boosting (IC-XGBoost3D), is adopted for automatically developing 3D subsurface geological domains from limited measurements. IC-XGBoost3D roots in deep learning and learns typical stratigraphic features from a pair of perpendicular training images reflecting local prior geological knowledge. The method is physics-informed and data-driven and can efficiently build and update subsurface geological models from limited site-specific data with quantified uncertainty. The method is applied to develop the 3D subsurface geological domain of a reclamation site in Hong Kong. The model performance is evaluated statistically using leave-one-out cross-validation. Results indicate that complex depositional stratigraphic patterns of fine-grained materials at the reclamation site can reasonably be replicated. Effects of measurement data number on the model performance are investigated, and useful insights are gained for developing subsurface geological domains of reclamation sites. © 2022, Springer-Verlag GmbH Germany, part of Springer Nature.

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Machine Learning Reveals Adaptive COVID-19 Narratives in Online Anti-Vaccination Network

Item Type Conference Paper

Author R. Sear

Author R. Leahy

Author N.J. Restrepo

Author Y. Lupu

Author N. Johnson

Editor Yang Z.

Editor von Briesen E.

Abstract The COVID-19 pandemic sparked an online “infodemic” of potentially dangerous misinformation. We use machine learning to quantify COVID-19 content from opponents of establishment health guidance, in particular vaccination. We quantify this content in two different ways: number of topics and evolution of keywords. We find that, even in the early stages of the pandemic, the anti-vaccination community had the infrastructure to more effectively garner support than their pro-vaccination counterparts by exhibiting a broader array of discussion topics. This provided an advantage in terms of attracting new users seeking COVID-19 guidance online. We also find that our machine learning framework can pick up on the adaptive nature of discussions within the anti-vaccination community, tracking distrust of authorities, opposition to lockdown orders, and an interest in early vaccine trials. Our approach is scalable and hence tackles the urgent problem facing social media platforms of having to analyze huge volumes of online health misinformation. With vaccine booster shots being approved and vaccination rates stagnating, such an automated approach is key in understanding how to combat the misinformation that slows the eradication of the pandemic. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.

Date 2022**DOI** 10.1007/978-3-030-96188-6_12**URL** <https://www.scopus.com/pages/publications/85128726347?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Springer Proc. Complex.**Publisher** Springer Science and Business Media B.V.**ISBN** 22138684 (ISSN); 978-303096187-9 (ISBN)**Pages** 164-175**Proceedings Title** Springer Proc. Complex.**Conference Name** Springer Proceedings in Complexity**Date Added** 20/09/2026, 9:52:23 pm**Modified** 20/09/2026, 9:52:23 pm**Tags:**

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Machine learning techniques to increase the performance of indirect methane quantification from a single, stationary sensor

Item Type Journal Article**Author** R.S. Heltzel**Author** D.R. Johnson**Author** M.T. Zaki**Author** A.K. Gebreslase**Author** O.I. Abdul-Aziz

Abstract Researchers are searching for ways to better quantify methane emissions from natural gas infrastructure. Current indirect quantification techniques (IQTs) allow for more frequent or continuous measurements with fewer personnel resources than direct methods but lack accuracy and repeatability. Two IQTs are Other Test Method (OTM) 33A and Eddy Covariance (EC). We examined a novel approach to improve the accuracy of single sensor IQT whereby the results from both OTM and EC were combined with two machine learning (ML) models, a random forest (RF) and a neural network (NN). Then, models were enhanced with feature reduction and hyperparameter tuning and compared to traditional quantification methods. The NN and RF

improved upon the default OTM by an average of 44% and 78%, respectively. When compared to traditional OTM estimates with low Data Quality Indicators (DQIs), RF and NN models reduced 1σ errors from $\pm 66\%$ to $\pm 13\%$ and $\pm 34\%$, respectively. Models also reduced the standard deviation of estimates with 93% and 85% of estimates falling within $\pm 50\%$ of the known release rate. This approach can be deployed with single sensor systems at well sites to improve confidence in reported emissions, reducing the number of anomalous overestimates that would trigger unnecessary site evaluations. Additional improvements could be realized by expanding training datasets with more methane release rates. Further, deployment of such models in a variety of situations could enhance their ability help close the gap between bottom-up inventory and top-down studies by enabling continuous monitoring of temporal emissions that could identify with improved confidence, atypically higher emissions. Accurate remote single sensor systems are key in developing an improved understanding of methane emissions to enable industry to identify and reduce methane emissions. © 2022

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Machine learning to detect, classify, and count blackbirds damaging agriculture using drone-based imagery: Supporting AI-driven automation for deployment of damage management tools

Item Type Journal Article**Author** J.L. Duttonhefner**Author** A.A. ElSaid**Author** P.E. Klug

Abstract As human populations expand and land use changes, human-wildlife conflicts are increasing, requiring cost-effective management tools that balance human well-being and wildlife conservation. Hazing devices are often used to mitigate conflicts between wildlife and agriculture with drones serving as both frightening devices and monitoring systems. Artificial intelligence (AI) enables automated detection, identification, and counting of wildlife, allowing for field-based Internet of Things (IoT) systems to selectively deploy tools when target species reach a critical threshold. We acquired drone imagery of mixed-species blackbird flocks – dominated by red-winged blackbirds (RWBL) *Agelaius phoeniceus* – damaging sunflowers (*Helianthus annuus*) in North Dakota (September–October 2021–2022). We trained a ResNet-18 Convolutional Neural Network (CNN) model to A) detect flocks (accuracy = 95.0 %); and Faster Region-based Convolutional Neural Network (Faster-RCNN) models to B) detect individual blackbirds (accuracy = 65.7 %, precision = 97.6 %), C) classify individual blackbirds by species and for RWBL sex and age class, and D) count blackbirds (% difference: birds = 37.5 %; male RWBL = 39.6 %; female RWBL = 43.1 %). The model correctly classified RWBL to species (87.6 %), sex, and age class (adult males = 89.8 %; hatch-year males = 27.6 %; females = 80.0 %). The RWBL were misclassified as other blackbirds that inflict crop damage, not non-target species. Variability in background (e.g., sky, green vegetation, tan vegetation) and complexity (e.g., contrast, texture), along with bird camouflage, required background removal to enhance performance when moving drones captured still images of moving targets. Camera orientation (e.g., depth perception, target overlap) and image quality (e.g., blurred, shadowed objects) affected detection, classification, and counts. Automated deployment reduces labor and wildlife habituation, increasing longevity and efficacy of management tools. Copyright © 2025. Published by Elsevier B.V.

Date 2025**DOI** 10.1016/j.ecoinf.2025.103495**URL** <https://www.scopus.com/pages/publications/105021128334?origin=resultslist>**Archive** Scopus**Language** English**Volume** 92**Publisher** Elsevier B.V.**Publication** Ecological Informatics**ISSN** 15749541 (ISSN)**Journal Abbr** Ecol. Informatics**Date Added** 20/09/2026, 9:51:51 pm**Modified** 20/09/2026, 9:51:51 pm**Tags:**

Drones, artificial intelligence, automation, Automation, machine learning, Monitoring, UAS, image analysis, management practice, agricultural application, Background removal, Blackbird, crop damage, Faster-RCNN, songbird

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Machine learning to quantify habitual physical activity in children with cerebral palsy

Item Type Journal Article**Author** B.I. Goodlich**Author** E.L. Armstrong**Author** S.A. Horan**Author** E. Baque**Author** C.P. Carty**Author** M.N. Ahmadi**Author** S.G. Trost

Abstract Aim: To investigate whether activity-monitors and machine learning models could provide accurate information about physical activity performed by children and adolescents with cerebral palsy (CP) who use mobility aids for ambulation. Method: Eleven participants (mean age 11y [SD 3y]; six females, five males) classified in Gross Motor Function Classification System (GMFCS) levels III and IV, completed six physical activity trials wearing a tri-axial accelerometer on the wrist, hip, and thigh. Trials included supine rest, upper-limb task, walking, wheelchair propulsion, and cycling. Three supervised learning algorithms (decision tree, support vector machine [SVM], random forest) were trained on features in the raw-acceleration signal. Model-performance was evaluated using leave-one-subject-out cross-validation accuracy. Results: Cross-validation accuracy for the single-placement models ranged from 59% to 79%, with the best performance achieved by the random forest wrist model (79%). Combining features from two or more accelerometer placements significantly improved classification accuracy. The random forest wrist and hip model achieved an overall accuracy of 92%, while the SVM wrist, hip, and thigh model achieved an overall accuracy of 90%. Interpretation: Models trained on features in the raw-acceleration signal may provide accurate recognition of clinically relevant physical activity behaviours in children and adolescents with CP who use mobility aids for ambulation in a controlled setting. What this paper adds: Machine learning may assist clinicians in evaluating the efficacy of surgical and therapy-based interventions. Machine learning may help researchers better understand the short- and long-term benefits of physical activity for children with more severe motor impairments. © 2020 Mac Keith Press

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Export Date: 20 September 2026; Cited By: 34; Correspondence Address: B.I. Goodlich; School of Allied Health Sciences, Griffith University, Gold Coast, Australia; email: benjamin.goodlich@griffithuni.edu.au; CODEN: DMCNA

Machine Learning to Quantify In Situ Humoral Selection in Human Lupus Tubulointerstitial Inflammation

Item Type Journal Article
Author A.J. Kinloch
Author Y. Asano
Author A. Mohsin
Author C. Henry
Author R. Abraham
Author A. Chang
Author C. Labno
Author P.C. Wilson
Author M.R. Clark

Abstract In human lupus nephritis, tubulointerstitial inflammation (TII) is associated with in situ expansion of B cells expressing anti-vimentin antibodies (AVAs). The mechanism by which AVAs are selected is unclear. Herein, we demonstrate that AVA somatic hypermutation (SHM) and selection increase affinity for vimentin. Indeed, germline reversion of several antibodies demonstrated that higher affinity AVAs can be selected from both low affinity B cell germline clones and even those that are strongly reactive with other autoantigens. While we demonstrated affinity maturation, enzyme-linked immunosorbent assays (ELISAs) suggested that affinity maturation might be a consequence of increasing polyreactivity or even non-specific binding. Therefore, it was unclear if there was also selection for increased specificity. Subsequent multi-color confocal microscopy studies indicated that while TII AVAs often appeared polyreactive by ELISA, they bound selectively to vimentin fibrils in whole cells or inflamed renal tissue. Using a novel machine learning pipeline (CytoSkaler) to quantify the cellular distribution of antibody staining, we demonstrated that TII AVAs were selected for both enhanced binding and specificity in situ. Furthermore, reversion of single predicted amino acids in antibody variable regions indicated that we could use CytoSkaler to capture both negative and positive selection events. More broadly, our data suggest a new approach to assess and define antibody polyreactivity based on quantifying the distribution of binding to native and contextually relevant antigens. © Copyright © 2020 Kinloch, Asano, Mohsin, Henry, Abraham, Chang, Labno, Wilson and Clark.

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Machine learning to quantify physical activity in children with cerebral palsy: Comparison of group, group-personalized, and fully-personalized activity classification models

Item Type Journal Article

Author M.N. Ahmadi

Author M.E. O'neil

Author E. Baque

Author R.N. Boyd

Author S.G. Trost

Abstract Pattern recognition methodologies, such as those utilizing machine learning (ML) approaches, have the potential to improve the accuracy and versatility of accelerometer-based assessments of physical activity (PA). Children with cerebral palsy (CP) exhibit significant heterogeneity in relation to impairment and activity limitations; however, studies conducted to date have implemented “one-size fits all” group (G) models. Group-personalized (GP) models specific to the Gross Motor Function Classification (GMFCS) level and fully-personalized (FP) models trained on individual data may provide more accurate assessments of PA; however, these

approaches have not been investigated in children with CP. In this study, 38 children classified at GMFCS I to III completed laboratory trials and a simulated free-living protocol while wearing an ActiGraph GT3X+ on the wrist, hip, and ankle. Activities were classified as sedentary, standing utilitarian movements, or walking. In the cross-validation, FP random forest classifiers (99.0–99.3%) exhibited a significantly higher accuracy than G (80.9–94.7%) and GP classifiers (78.7–94.1%), with the largest differential observed in children at GMFCS III. When evaluated under free-living conditions, all model types exhibited significant declines in accuracy, with FP models outperforming G and GP models in GMFCS levels I and II, but not III. Future studies should evaluate the comparative accuracy of personalized models trained on free-living accelerometer data. © 2020 by the authors. Licensee MDPI, Basel, Switzerland.

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Machine learning to quantify techno-functional properties - A case study for gel stiffness with pea ingredients

Item Type Journal Article

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Author A. Garre

Author T. Nissink

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Author A. van der Padt

Author R. Boom

Abstract Mildly refined ingredients are included more easily in food products when selected based on techno-functional properties instead of composition. We assess different machine learning methods that quantitatively link relevant techno-functional properties to the composition and processing history of the ingredient in a case study using the gel stiffness (Young's modulus) by conventionally and mildly refined ingredients of yellow pea. Linear (multiple, log transformed and polynomial) and non-linear models (spline regression, decision trees, and neural networks) were explored. The final model selection was based on 1) the statistical model metrics (RMSE, R2, and MAE) of the training and independent test set and 2) expert knowledge to evaluate the plausibility of the model predictions. In this case, neural networks can describe the gel stiffness of yellow pea ingredients most accurately. The approach that we follow can be applied to other techno-functional properties to improve the chain sustainability while ensuring the full functionality of the products. © 2022

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Machine learning-assisted analysis of ignition and combustion properties in as-milled and annealed Al-Zr composite powders

Item Type Journal Article

Author M.R. Flickinger

Author S. Raguraman

Author A.L. Polk

Author C. Goodman

Author M. Bokhoor

Author R. Knio

Author M. Kruppa

Author M.A. Foster

Author T.P. Weihs

Abstract Micron-scale metal-based composite powders are promising for energetic applications due to their tailored ignition and combustion properties. In particular, ball-milled Al/Zr composites exhibit lower ignition thresholds than pure aluminum, driven by exothermic intermetallic formation reactions and have demonstrated enhanced combustion properties. However, the extent to which this heat release governs ignition and combustion remains unclear, especially when progressively removed through annealing. To systematically investigate this effect, we synthesized Al/Zr powders (3Al:Zr, Al:Zr, and Al:3Zr) via ball milling, annealed them in argon up to 1000 °C to partially complete the formation reactions, and characterized their ignition and combustion behavior. Ignition thresholds were measured using a hot wire method across different environments, while high-speed hyperspectral imaging tracked single-particle burn durations and temperatures. A machine-learning approach utilizing convolutional neural networks (CNNs) was developed to quantify the frequency of microexplosions. Results show that annealing – and thus reducing available reaction heat – increases ignition thresholds, most significantly for Al-rich compositions. In contrast, Zr-rich powders exhibit little change in ignition thresholds due to oxidation aiding ignition. Despite removing the available heat that drives ignition, average combustion temperatures range from 2400 K to 3000 K and increased with annealing for Al- and Zr-rich powders. Average maximum temperatures are 100 to 400 K higher. The frequency of microexplosions remains high (>46%) and increases with annealing for all but the Al-rich powders. These findings suggest that while homogeneous Al/Zr powders (e.g., atomized) may exhibit higher ignition thresholds, they can achieve comparable combustion performance once ignited. **Novelty and Significance Statement** While previous studies have explored how powder chemistry and milling conditions affect the ignition behavior of ball-milled composites, none have directly examined how removing exothermic heat from intermixing and formation reactions influences ignition thresholds and combustion behavior. This is the first study to investigate multiple Al/Zr

stoichiometries in ball-milled composite powders that were annealed to form pre-existing intermetallics, and to evaluate how this controlled intermetallic formation affects combustion performance by comparing as-milled and annealed powders at each stoichiometry. Microexplosion frequency of single particles was quantified using a novel convolutional neural network pipeline; to our knowledge, this is the first application of deep learning to quantify microexplosions in single-particle or plume experiments. © 2025

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Machine learning-assisted intelligent interpretation of distributed fiber optic sensor data for automated monitoring of pipeline corrosion

Item Type Journal Article

Author Y. Liu

Author X. Tan

Author Y. Bao

Abstract Distributed fiber optic sensor (DFOS) offers unique capabilities of monitoring corrosion for long pipelines. However, manually interpreting DFOS data is labor-intensive and time-consuming. To address this challenge, this paper presents a machine learning approach for real-time automatic interpretation of DFOS data used to monitor both uniform and non-uniform corrosion in pipeline. A machine learning model is developed to automatically detect corrosion based on DFOS data, and a corrosion quantification method is developed based on the output of the machine learning model. The proposed approaches are evaluated using laboratory experiments in terms of accuracy and robustness to pipeline diameter, spatial resolution of DFOS, type of fiber optic cable, and sensor installation methods. The results show that the F1 score for corrosion detection and the R2 value for corrosion quantification are 0.986 and 0.953, respectively. This research will facilitate pipeline corrosion monitoring by enabling automatic distributed sensor data interpretation. © 2024 Elsevier Ltd

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Machine learning-based acoustic emission technique for corrosion-induced damage monitoring in reinforced concrete structures

Item Type Journal Article

Author A. Thirumalaiselvi

Author S. Sasmal

Abstract Early-stage detection of corrosion in reinforcement embedded inside concrete, generally considered as the major problem in structures under extreme marine environments, can help to schedule timely maintenance/repair and to avoid any catastrophic failure. However, quantitative evaluation of the degree of deterioration, especially at a very early stage, i.e., before its appearance on the surface, is extremely difficult due to the complex process in heterogeneous material. This study aims to characterize the critical damage stages in reinforced concrete specimens under accelerated conditions using the Acoustic emission (AE) damage quantification methods supported by machine learning (ML) techniques. The stages of corrosion-induced damage are predicted by a two-step predictive model that involve both regression and classification tasks. Three supervised regression models — support vector regression (SVR), relevance vector machine (RVM) and kernel ridge regression (KRR) — are first adopted to evaluate the amount of current. Then the output from regression model is given as an input to the classification problem to predict the discrete class label output. Characterized damage stages are validated based on mass loss and crack width measurements. The results show that the corrosion initiation (at day 3) and crack formation (at day 7) can be determined from the sudden change in acoustic parameters. Further, a double layer (supervised-unsupervised) model is proposed which is found to be more effective in correctly detecting the progression of damage. The findings indicate that the proposed approach that combines corrosion detection by analysing AE signals with ML technique can accurately and robustly predict corrosion severity. © 2024 Elsevier Ltd

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Machine learning-based approach reveals essential features for simplified TSPO PET quantification in ischemic stroke patients

Item Type Journal Article

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Author S. Hein

Author A. Schindler

Author M. Duering

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Author S. Lindner

Author M. Schidlowski

Author P. Bartenstein

Author N. Albert

Author M. Brendel

Author S.I. Ziegler

Abstract Introduction: Neuroinflammation evaluation after acute ischemic stroke is a promising option for selecting an appropriate post-stroke treatment strategy. To assess neuroinflammation in vivo, translocator protein PET (TSPO PET) can be used. However, the gold standard TSPO PET quantification method includes a 90 min scan and continuous arterial blood sampling, which is challenging to perform on a routine basis. In this work, we determine what information is required for a simplified

quantification approach using a machine learning algorithm. **Materials and Methods:** We analyzed data from 18 patients with ischemic stroke who received 0–90 min [18F]GE-180 PET as well as T1-weighted (T1w), FLAIR, and arterial spin labeling (ASL) MRI scans. During PET scans, five manual venous blood samples at 5, 15, 30, 60, and 85 min post injection (p.i.) were drawn, and plasma activity concentration was measured. Total distribution volume (VT) was calculated using Logan plot with the full dynamic PET and an image-derived input function (IDIF) from the carotid arteries. IDIF was scaled by a calibration factor derived from all the measured plasma activity concentrations. The calculated VT values were used for training a random forest regressor. As input features for the model, we used three late PET frames (60–70, 70–80, and 80–90 min p.i.), the ASL image reflecting perfusion, the voxel coordinates, the lesion mask, and the five plasma activity concentrations. The algorithm was validated with the leave-one-out approach. To estimate the impact of the individual features on the algorithm's performance, we used Shapley Additive Explanations (SHAP). Having determined that the three late PET frames and the plasma activity concentrations were the most important features, we tested a simplified quantification approach consisting of dividing a late PET frame by a plasma activity concentration. All the combinations of frames/samples were compared by means of concordance correlation coefficient and Bland-Altman plots. **Results:** When using all the input features, the algorithm predicted VT values with high accuracy ($87.8 \pm 8.3\%$) for both lesion and non-lesion voxels. The SHAP values demonstrated high impact of the late PET frames (60–70, 70–80, and 80–90 min p.i.) and plasma activity concentrations on the VT prediction, while the influence of the ASL-derived perfusion, voxel coordinates, and the lesion mask was low. Among all the combinations of the late PET frames and plasma activity concentrations, the 70–80 min p.i. frame divided by the 30 min p.i. plasma sample produced the closest VT estimate in the ischemic lesion. **Conclusion:** Reliable TSPO PET quantification is achievable by using a single late PET frame divided by a late blood sample activity concentration. © 2022

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Machine Learning-based Modeling and Uncertainty Quantification for Radar Cross Section of a Cone-like Target

Item Type Conference Paper

Author D. Guo

Author J. Zhai

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Author H. Yin

Author Y. Zhu

Abstract Radar cross section (RCS) plays an important role in the recognition of targets. RCS varies dramatically with the incident angle and the size of targets, and it is difficult to accurately predict the RCS values. In this paper, an efficient modeling and uncertainty quantification method based on the support vector machine and the k-nearest neighbor is proposed for the RCS prediction of cone-like targets. The proposed method is compared with two uncertainty quantification methods, an ensemble based on lower upper bound estimation and a neural network with dropout. The root mean square error, the prediction interval coverage probability, the mean prediction interval width and the computation time are used as the

performance metrics, and the experimental results demonstrate that the proposed method is superior to the compared methods in accuracy and efficiency. © 2022 IEEE.

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Machine Learning-Based Monitoring of Chemical Contamination from Drilling Leaks

Item Type Journal Article

Author J. Sun

Author Y. Ai

Author K. Tang

Author Z. Zhang

Abstract With the increasing requirements for environmental protection in oil and gas field development, the leakage during drilling and the chemical pollution caused by it have become the key technical difficulties restricting sustainable development. In this study, a set of intelligent monitoring and risk assessment system integrating multi-

source data is constructed, based on three uncertainty quantification methods, namely, Dropout Bayesian Approximation, Density Network Integration, and Mixed Density Network (MDN), to deeply mine and distribute the drilling parameter and chemical detection data of 17 wells. To address the limitations of the traditional methods, which are difficult to take into account the nonlinearity, multiple peaks and uncertainty, SHAP value and BPCA are introduced for feature selection and dimensionality reduction, and the MDN model is optimised based on maximum likelihood and entropy regularisation strategies. The experimental results show that the MDN outperforms other methods in terms of negative log-likelihood (NLL), mean square error (MSE) and confidence interval coverage (PICP), and achieves accurate fitting of the pollution concentration distribution and risk warning. Finally, combining the pollution probability distribution and well control scheduling rules output from the model, the parameter optimisation and operation recommendation scheme is proposed, which achieves an early warning accuracy of more than 93% in the field deployment, and provides strong support for green drilling and environmental protection. © Springer Science+Business Media, LLC, part of Springer Nature 2025.

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Machine learning-based prediction models unleash the enhanced production of fucoxanthin in *Isochrysis galbana*

Item Type Journal Article

Author J. Manochkumar

Author A. Jonnalagadda

Author A.K. Cherukuri

Author B. Vannier

Author D. Janjaroen

Author R. Chandrasekaran

Author S. Ramamoorthy

Abstract Introduction: The marine microalga *Isochrysis galbana* is prolific producer of fucoxanthin, which is a xanthophyll carotenoid with substantial global market value boasting extensive applications in the food, nutraceutical, pharmaceutical, and cosmetic industries. This study presented a novel integrated experimental approach coupled with machine learning (ML) models to predict the fucoxanthin content in *I. galbana* by altering the type and concentration of phytohormone supplementation, thus overcoming the multiple methodological limitations of conventional fucoxanthin quantification. Methods: A novel integrated experimental approach was developed, analyzing the effect of varying phytohormone types and concentrations on fucoxanthin production in *I. galbana*. Morphological analysis was conducted to assess changes in microalgal structure, while growth rate and fucoxanthin yield correlations were explored using statistical analysis and machine learning models. Several ML models were employed to predict fucoxanthin content, with and without hormone descriptors as variables. Results: The findings revealed that the Random Forest (RF) model was highly significant with a high (Formula presented.) of 0.809 and (Formula presented.) of 0.776 when hormone descriptors were excluded, and the inclusion of hormone descriptors further improved prediction accuracy to (Formula presented.) of 0.839, making it a useful tool for predicting the fucoxanthin yield. The model that fitted the experimental data indicated methyl jasmonate (0.2 mg/L) as an effective phytohormone. The combined experimental and ML approach demonstrated rapid, reliable, and cost-efficient prediction of fucoxanthin yield. Discussion: This study highlights the potential of machine learning models, particularly Random Forest, to optimize parameters influencing microalgal growth and fucoxanthin production. This approach offers a more efficient alternative to conventional methods, providing valuable insights into improving fucoxanthin production in microalgal cultivation. The findings suggest that leveraging diverse ML models can enhance the predictability and efficiency of fucoxanthin production, making it a promising tool for industrial applications. Copyright © 2024 Manochkumar, Jonnalagadda, Cherukuri, Vannier, Janjaroen, Chandrasekaran and Ramamoorthy.

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Machine learning-driven design of high-entropy alloys: Phase prediction, performance optimization, and challenges

Item Type Journal Article

Author F. Yu

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Author A. Liu

Author E. Zhu

Author W. Zhang

Abstract High-entropy alloys (HEAs) have become a revolutionary breakthrough in the field of materials design due to their excellent mechanical properties, thermal stability, and corrosion resistance. However, their large compositional space poses a significant challenge to traditional design paradigms. This paper elucidates the fundamentals of machine learning (ML) -assisted HEAs design from a synergistic perspective integrating intrinsic HEAs mechanisms and ML methodology, systematically reviewing progress in ML application for HEAs research with a focus on three pivotal aspects: phase structure prediction, performance optimization, and critical bottleneck breakthrough. This review demonstrates that the deep integration of ML and HEAs research enables efficient construction of quantitative “component-structure-property” correlations while exhibiting significant advantages in mitigating data scarcity and enhancing model interpretability. Nevertheless, key challenges persist—model opacity, experimental validation delays, and multi-objective

optimization complexity. Future research should prioritize developing inverse material design frameworks, establishing uncertainty quantification methods, and constructing digital twin validation platforms to bridge computational prediction and experimental verification. The deep fusion of data-driven techniques with thermodynamic parameters, material intrinsic features, and engineering process parameters will propel HEAs into a performance-driven intelligent customization era, delivering breakthrough solutions for biomedical, energy, and aerospace applications.
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Machine Learning-Driven Quantification of CO2 Plume Dynamics at Illinois Basin Decatur Project Sites Using Microseismic Data

Item Type Journal Article

Author I. Iyegbekedo

Author E. Fathi

Author T.R. Carr

Author F. Belyadi

Abstract This study utilizes machine learning to quantify CO₂ plume extents by analyzing microseismic data from the Illinois Basin Decatur Project (IBDP). Leveraging a unique dataset of well logs, microseismic records, and CO₂ injection metrics, this work aims to predict the temporal evolution of subsurface CO₂ saturation plumes. The findings illustrate that machine learning can predict plume dynamics, revealing vertical clustering of microseismic events over distinct time periods within certain proximities to the injection well, consistent with an invasion percolation model. The buoyant CO₂ plume partially trapped within sandstone intervals periodically breaches localized barriers or baffles, which act as leaky seals and impede vertical migration until buoyancy overcomes gravity and capillary forces, leading to breakthroughs along vertical zones of weakness. Between different unsupervised clustering techniques, K-Means and DBSCAN were applied and analyzed in detail, where K-means outperformed DBSCAN in this specific study by indicating the combination of the highest Silhouette Score and the lowest Davies–Bouldin Index. The predictive capability of machine learning models in quantifying CO₂ saturation plume extension is significant for real-time monitoring and management of CO₂ sequestration sites. The models exhibit high accuracy, validated against physical models and injection data from the IBDP, reinforcing the viability of CO₂ geological sequestration as a climate change mitigation strategy and enhancing advanced tools for safe management of these operations. © 2024 by the authors.

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Machine Learning-Enabled Quantification of Fucosylated Human Milk Oligosaccharides in Human Breast Milk by Benchtop 1H NMR Spectroscopy

Item Type Journal Article

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Author T. Aizawa

Abstract Human milk oligosaccharides (HMOs) are bioactive components of human breast milk (HBM), but their concentrations vary with maternal secretor phenotype and lactation stage, making individual HMO quantification analytically demanding. Here, we present a reference-guided benchtop 60 MHz NMR workflow integrating chemometrics and machine learning to quantify major fucosylated HMOs in HBM. A total of 111 HBM samples from 37 donors across lactation stages were analyzed. Using 800 MHz NMR-derived concentrations as references, predictive models were developed from benchtop NMR spectra for 2'-fucosyllactose (2'-FL), 3-fucosyllactose (3-FL), and lacto-N-fucopentaose I (LNFP-I). Unlike conventional peak-fitting-based NMR quantification, this workflow recovered HMO-specific quantitative information from highly overlapped 60 MHz carbohydrate signals that were not directly resolvable in authentic HBM. Elastic Net yielded practical models, particularly for lower-abundance 3-FL and LNFP-I. These results support benchtop NMR combined with machine learning as an accessible platform for HMO screening. © 2026 American Chemical Society

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Machine Learning–Driven prediction of wettability in brine–co₂–mineral systems: A step toward reliable geological carbon storage

Item Type Journal Article
Author M. Ahmadi

Abstract Understanding wettability behavior in brine–CO₂–mineral systems is fundamental to predicting the efficiency and long-term security of geological carbon storage. In this study, we combine statistical characterization, mechanistic interpretation, and advanced machine learning to quantify the thermodynamic, geochemical, and mineralogical controls on contact angle variability. Exploratory analyses reveal strong mineral-specific heterogeneity, with quartz and feldspar exhibiting predominantly hydrophilic behavior, mica displaying intermediate tendencies, and calcite showing the largest variability, consistent with its sensitivity to geochemical reactions and surface chemistry. Pressure emerges as a dominant driver of wettability alteration, but its effect is nonlinear and mineral-dependent, with increasing pressure promoting CO₂-wet tendencies in some regimes while showing weaker or even opposite trends in others due to coupled temperature, salinity, and mineralogical influences. Machine learning models, including ensemble learners and neural approaches, demonstrate strong predictive capacity for contact angle estimation (R² up to 0.91), with stacking ensembles achieving the highest accuracy and most balanced residual patterns. Feature attribution using Shapley Additive Explanations (SHAP) consistently highlights contact angle at ambient condition (θ₀) and pressure as the most influential predictors, while mineral type, temperature, and salinity exert complementary roles in shaping wettability outcomes. Importantly, SHAP interaction analyses reveal synergistic effects, whereby elevated temperature and salinity amplify the predictive importance of pressure, underscoring the nonlinear, coupled nature of wettability controls. These findings establish a robust, interpretable framework that integrates physicochemical mechanisms with interpretable machine learning, providing robust predictive insights into wettability evolution under reservoir-relevant conditions. The outcomes advance fundamental understanding of multiphase interfacial processes and offer practical tools for risk assessment, capacity estimation, and secure implementation of carbon storage strategies. © 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Machine-learning and scRNA-Seq-based diagnostic and prognostic models illustrating survival and therapy response of lung adenocarcinoma

Item Type Journal Article

Author Q. Cheng

Author W. Zhao

Author X. Song

Author T. Jin

Abstract Lung cancer is a major cause accounting for cancer-related mortalities, with lung adenocarcinoma (LUAD) being the most prevalent subtype. Given the high clinical and cellular heterogeneities of LUAD, accurate diagnosis and prognosis are crucial to avoid overdiagnosis and overtreatment. Taking full advantage of scRNA-Seq data to resolve the tumor heterogeneities, we explored the overall landscape of LUAD microenvironment. Utilizing the stage-specific tumor cell markers, we have developed highly accurate diagnostic and prognostic models with elevated sensitivity and specificity. The diagnostic model, developed through random forest algorithms with a thirteen-gene signature, achieved an accuracy of 96.4% and an AUC of 0.993. These metrics were further demonstrated by benchmarking with available models and scoring systems in independent cohorts. Concurrently, the prognostic model, formulated via Cox regression with a six-gene signature, effectively predicted overall survival, with elevated risk scores associated with increased fractions of cancer-associated fibroblasts, and higher likelihood of immune escape and T-cell exclusion. Subsequently, two nomograms were developed to predict survival and drug responses, facilitating their integration into clinical practice. Overall, this study underscores the potential of our models for efficient, rapid, and cost-effective diagnosis and prognosis of LUAD, adaptable to multiple expression profiling platforms and quantification methods. (Figure presented.) © The Author(s), under exclusive licence to Springer Nature Limited 2024.

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Machine-learning based study on the on-site renewable electrical performance of an optimal hybrid PCMs integrated renewable system with high-level parameters' uncertainties

Item Type Journal Article

Author Y. Zhou

Author S. Zheng

Author G. Zhang

Abstract The uncertainty and sensitivity analyses for multivariables of the optimal system based on deterministic parameters are necessary, as multivariables are full of uncertainties in the real operation. However, the generic methodology for multi-dimensional uncertainties quantification is rare, and the energy performance simulation is normally at high computational cost, especially considering a huge amount of parameters' uncertainties. In this study, the on-site renewable electricity generation of an optimal hybrid renewable system based on deterministic parameters, was investigated, under high-level parameters' uncertainties. A generic uncertainty quantification methodology was proposed using the two-dimensional Markov Chain Monte Carlo to quantify multi-dimensional uncertainties. A machine-learning based data-driven model, using the supervised machine learning with high computational efficiency, was developed to predict the on-site renewable electricity generation, and thereafter used for the uncertainty and sensitivity analyses. Compared with the deterministic scenario parameters, the cases with the scenario uncertainties can increase the peak power and the total amount of the on-site renewable electricity generation. This study proposes a novel generic uncertainty quantification methodology, together with a machine-learning based data-driven model for conducting the uncertainty analysis of an optimal renewable system based on deterministic parameters, which are important for the promotion of renewable and sustainable buildings. © 2019 Elsevier Ltd

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MAESTRO: A Multi-Scale Ensemble Framework with GAN-Based Data Refinement for Robust Malicious Tor Traffic Detection

Item Type Journal Article

Author J. Geng

Author Y. Xie

Author J. Li

Author X. Yu

Author L. He

Abstract Malicious Tor traffic data contains deep domain-specific knowledge, which makes labeling challenging, and the lack of labeled data degrades the accuracy of learning-based detectors. Real-world deployments also exhibit severe class imbalance, where malicious traffic constitutes a small minority of network flows, which further reduces detection performance. In addition, Tor’s fixed 512-byte cell architecture removes packet-size diversity that many encrypted-traffic methods rely on, making feature extraction difficult. This paper proposes an efficient three-stage framework, MAESTRO v1.0, for malicious Tor traffic detection. In Stage 1, MAESTRO extracts multi-scale behavioral signatures by fusing temporal, positional, and directional embeddings at cell, direction, and flow granularities to mitigate feature homogeneity; it then compresses these representations with an autoencoder into compact latent features. In Stage 2, MAESTRO introduces an ensemble-based quality quantification method that combines five complementary anomaly detection models to produce robust discriminability scores for adaptive sample weighting, helping the classifier to emphasize high-quality samples. MAESTRO also trains three specialized GANs per minority class and applies strict five-model ensemble validation to synthesize diverse high-fidelity samples, addressing extreme class imbalance. We evaluate MAESTRO under systematic imbalance settings, ranging from the natural distribution to an extreme 1% malicious ratio. On the CCS’22 Tor malware dataset, MAESTRO

achieves 92.38% accuracy, 64.79% recall, and 73.70% F1-score under the natural distribution, improving F1-score by up to 15.53% compared with state-of-the-art baselines. Under the 1% malicious setting, MAESTRO maintains 21.1% recall, which is 14.1 percentage points higher than the best baseline, while conventional methods drop below 10%. © 2026 by the authors.

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Magnetic Resonance Spectroscopy Deep Learning with Magnetic Resonance Background Generator Enables In Vivo Metabolite Quantification of Hepatic Encephalopathy

Item Type Journal Article

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Author Z. Huang

Author J. Zhang

Author C. Qian

Author X. Jiang

Author Q. Xu

Author H. Chen

Author D. Guo

Author X. Qu

Abstract Objective: Short-TE Proton Magnetic Resonance Spectroscopy (SPMRS) allows non-invasive, radiation-free detection of biomolecules including key brain metabolic biomarkers for neurological diseases. However, SPMRS also contains signals from macromolecules, lipids and baseline. Existing Deep Learning (DL) quantification methods for SPMRS overlook baseline interference. Methods: To address this, we proposed a novel DL quantification incorporating macromolecules, lipids and baseline (called background signal). First, background signal generators were built by capturing feature distributions of 200 in vivo background signals. Then, the synthetic SPMRS spectra were generated, composed of synthetic metabolite components based on the quantum mechanical and exponential models, background signal generated by the background generator and noise. Finally, the synthetic SPMRS were used to train the advanced DL quantification model QNet (our earlier study). Results: The proposed DL method was validated on a large-scale healthy human brain dataset (BigGABA) and our collected data from different manufacturers (totaling 216 spectra). Furthermore, it was applied to classification of minimal hepatic encephalopathy (totaling 176 spectra), highlighting its clinical potential. Conclusion: We proposed and validated a novel deep learning-based quantification method for SPMRS which is explicitly taken into account the background effect. Significance: The proposed method is incorporated the background effect into the DL framework of SPMRS quantification, and DL quantification of SPMRS is utilized for the clinical diagnosis of minimal hepatic encephalopathy. © 1964-2012 IEEE.

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Journal Abbr IEEE Trans. Biomed. Eng.

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Deep learning, Learning systems, Brain, Diagnosis, Neurodegenerative diseases, Quantification methods, Magnetic resonance, Biomedical signal processing, Biomolecules, Metabolites, Background signal interference, Background signals, Hepatic encephalopathies, Hepatic encephalopathy, In-vivo, Metabolic quantification, Metabolism, Proton magnetic resonance spectroscopies, Short-TE proton magnetic resonance spectroscopy, Spectra's

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Magnetic Resonance Spectroscopy Signal Analysis Based on Fingerprinting Dictionary Approaches

Item Type Conference Paper

Author I. Venglovskyi

Editor Benedicenti L.

Editor Liu Z.

Abstract Magnetic resonance spectroscopy (MRS) is a technique applicable in medical diagnosis or research, which has the unique capability to give non-invasive access to the biochemical content (metabolites) of scanned organs. Up to recent times, all the proposed methods solved metabolite quantification as an optimization problem attempting to minimize the difference between the data and a given parameterized model function. This paper proposes quantification of metabolites in MR spectroscopic imaging using a fingerprinting method, whose function is based on the creation of a dictionary of linear combinations of metabolite signals. Experimental results demonstrate the accuracy of the proposed method, compared to data obtained by a standard quantification method (QUEST), on concentration estimates of 8 metabolites from signals with macromolecule background and noise. The prototype results indicate that the concept of MR fingerprinting dictionary, useful also for preparing data for machine learning, can serve as an alternative method for metabolite quantification by NMR signal analysis. © 2021 Avestia Publishing. All rights reserved.

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DOI 10.11159/icbes21.114

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artificial intelligence, magnetic resonance spectroscopy, fingerprinting dictionary, spectroscopic imaging

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Manifold Embedding Induced by Multidimensional Scaling and Its Application to Alzheimer's Disease and Mild Cognitive Impairment

Item Type Journal Article

Author H. Park

Abstract Quantifying shape information related to a disease such as Alzheimer's disease (AD) is an important area of brain research. Neuroimaging data are highly dimensional and are thus cumbersome to analyze. Manifold learning techniques, which find a low-dimensional representation for highdimensional data, have been applied to brain magnetic resonance imaging (MRI). A shape quantification method based on multidimensional scaling (MDS), a well-known manifold learning technique, was proposed. The method successfully distinguished between AD and normal patients. We extended the MDS-based quantification method by 1) applying it to distinguish patients with mild cognitive impairment (MCI) from normal patients, and 2) showing the effectiveness of the induced low-dimensional embedding space in predicting key clinical variables such as mini-mental-state exam scores and clinical diagnosis by using a standard multiple linear regression. Distinguishing patients with MCI from normal patients is also important as it is related to early detection of AD. We were able to 1) classify not only AD/normal patients but also MCI/normal patients better than we could with the traditional classification based on the hippocampus volume, and to 2) show good statistical power for predicting key clinical variables. Our test group consisted of 25 normal, 25 AD, and 25 MCI patients.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Park; Department of Biomedical Engineering, Gachon University of Medicine and Science, Incheon 406-799, South Korea; email: hyunjinp@gachon.ac.kr

Manual and semi-automatic quantification of in vivo ^1H -MRS data for the classification of human primary brain tumors

Item Type Journal Article

Author S. Cuellar-Baena

Author L.M.T.S. Morais

Author F. Cendes

Author A.V. Faria

Author G. Castellano

Abstract In vivo proton magnetic resonance spectroscopy (^1H -MRS) is a technique capable of assessing biochemical content and pathways in normal and pathological tissue. In the brain, ^1H -MRS complements the information given by magnetic resonance images. The main goal of the present study was to assess the accuracy of ^1H -MRS for the classification of brain tumors in a pilot study comparing results obtained by manual and semi-automatic quantification of metabolites. In vivo single-voxel ^1H -MRS was performed in 24 control subjects and 26 patients with brain neoplasms that included meningiomas, high-grade neuroglial tumors and pilocytic astrocytomas. Seven metabolite groups (lactate, lipids, N-acetyl-aspartate, glutamate and glutamine group, total creatine, total choline, myo-inositol) were evaluated in all spectra by two methods: a manual one consisting of integration of manually defined peak areas, and the advanced method for accurate, robust and efficient spectral fitting (AMARES), a semi-automatic quantification method implemented in the jMRUI software. Statistical methods included discriminant analysis and the leave-one-out cross-validation method. Both manual and semi-automatic analyses detected differences in metabolite content between tumor groups and controls ($P < 0.005$). The classification accuracy obtained with the manual method was 75% for high-grade neuroglial tumors, 55% for meningiomas and 56% for pilocytic astrocytomas, while for the semi-automatic method it was 78, 70, and 98%, respectively. Both methods classified all control subjects correctly. The study demonstrated that ^1H -MRS accurately differentiated normal from tumoral brain tissue and confirmed the superiority of the semi-automatic quantification method.

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female, human, male, article, diagnostic test accuracy study, clinical article, limit of quantitation, proton nuclear magnetic resonance, lipid, metabolite, in vivo study, brain tumor, Brain tumors, AMARES, diagnostic value, creatine, glutamic acid, inositol, n acetylaspartic acid, clinical assessment tool, tumor classification, accurate robust and efficient spectral fitting, astrocytoma, choline, glutamine, lactic acid, meningioma, Proton magnetic resonance spectroscopy, Tumor classification

Notes:

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: G. Castellano; Departamento de Raios Cósmicos e Cronologia, Instituto de Física Gleb Wataghin, UNICAMP, 13083-970 Campinas, SP, Brazil; email: gabriela@ifi.unicamp.br; CODEN: RBPMB

Many-to-Many Prediction for Effective Modeling of Frequent Label Transitions in Time Series

Item Type Conference Paper

Author A. Katrompas

Author V. Metsis

Editor Karim E.

Editor Nikanfar S.

Editor Pavel H.R.

Abstract Time-series classification is vital in health monitoring and human activity recognition, as well as in areas such as financial forecasting, process control, and a wide array of forecasting tasks. Traditional time-series models segment data into windows and assign one label per window, often missing label transitions within those windows. This paper presents a novel many-to-many time-series

model and post-processing using hybrid recurrent neural networks with attention mechanisms, which more effectively captures label transitions over traditional many-to-one models. Further, unlike typical other many-to-many models, our approach doesn't require a decoder. Instead, it employs an RNN, generating a label for every input time step. During inference, a weighted voting scheme consolidates overlapping predictions into one label per time step. Experiments show our model remains effective on time-series with sparse label shifts, but particularly excels in detecting frequent transitions. This model is ideal for tasks demanding accurate pinpointing of rapid label changes in time-series data, such as gesture recognition, making it ideal for fast-paced human activity recognition.

1 © 2024 ACM.

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Conference Name ACM International Conference Proceeding Series

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Deep learning, Learning systems, Machine-learning, Forecasting, Pattern recognition, Neural-networks, Deep Learning, Neural Networks, Recurrent Neural Networks, Machine Learning, Recurrent neural networks, Human activity recognition, Attention mechanisms, Attention Mechanisms, Time series, Times series, Human Activity Recognition, Times series models, Many to many, Self attention, Self Attention, Time step, Time-Series

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Mapping the Brazilian savanna's natural vegetation: A SAR-optical uncertainty-aware deep learning approach

Item Type Journal Article

Author P. Silva Filho

Author C. Persello

Author R.V. Mareto

Author R. Machado

Abstract The Brazilian savanna (Cerrado) is considered a hotspot for conservation. Despite its environmental and social importance, the biome has suffered a rapid transformation process due to human activities. Mapping and monitoring the remaining vegetation is essential to guide public policies for biodiversity conservation. However, accurately mapping the Cerrado's vegetation is still an open challenge. Its diverse but spectrally similar physiognomies are a source of confusion for state-of-the-art (SOTA) methods. This study proposes a deep learning model to map the natural vegetation of the Cerrado at the regional to biome level, fusing Synthetic Aperture Radar (SAR) and optical data. The proposed model is designed to deal with uncertainties caused by the different resolutions of the input Sentinel-1/2 images (10 m) and the reference data, derived from Landsat images (30 m). We designed a multi-resolution label-propagation (MRLP) module that infers maps at both resolutions and uses the class scores from the 30 m output as features for the 10 m classification layer. We train the model with the proposed calibrated dual focal loss function in a 2-stage hierarchical manner. Our results reached an overall accuracy of 70.37%, representing an increase of 15.64% compared to a SOTA random forest (RF) model. Moreover, we propose an uncertainty quantification method, which has shown to be useful not only in validating the model, but also in highlighting areas of label noise in the reference. The developed codes and dataset are available on Github. © 2024 The Authors

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Mapping the spatial heterogeneity of nitrogen species in road dust through integrated multi-source remote sensing and interpretable machine learning

Item Type Journal Article

Author M.N. Ashraf

Author S. Huang

Author A. Basheer

Abstract Nitrogen accumulation in road dust constitutes a significant yet underexplored threat to environmental quality, particularly in regions where rapid urbanization intersects with intensive agriculture. The spatial distribution patterns and governing factors of individual nitrogen species within this environmental matrix remain poorly characterized. This study presents an integrated framework coupling field-based chemical speciation analysis, multi-source remote sensing, and interpretable machine learning to quantify and map the spatial distribution of total nitrogen (TN), nitrate-nitrogen (NO_3^- -N), ammonium-nitrogen (NH_4^+ -N), and organic nitrogen (Org-N) in road dust across South Tianjin, China. Laboratory analysis of 41 road dust samples collected along urban–rural transects revealed pronounced spatial heterogeneity, with TN concentrations spanning 289–3782 mg/kg; the highest values were consistently associated with agricultural land use. Org-N constituted the dominant nitrogen fraction, accounting for more than 88% of TN across all land-use categories. Predictive models constructed with the Extra Trees algorithm demonstrated strong generalization capability, yielding cross-validated coefficients of determination (R^2) of 0.891 for TN, 0.887 for Org-N, 0.846 for NH_4^+ -N, and 0.806 for NO_3^- -N. Among all predictors, spatial distance from the urban core emerged as the most influential variable, explaining 32–46% of model variance across nitrogen species. Model interpretation via SHapley Additive exPlanations (SHAP) and permutation importance further identified surface runoff, precipitation, and population density as potential drivers of nitrogen variability, while Monte Carlo uncertainty analysis indicated greater predictive uncertainty for Org-N relative to inorganic nitrogen forms. Spatially explicit predictions delineated agricultural districts in eastern and southern South Tianjin as critical nitrogen hotspots. The proposed remote sensing–machine learning framework effectively discriminates between diffuse agricultural nitrogen inputs and localized urban emission sources, offering actionable guidance for source-specific pollution management and sustainable land-use governance in rapidly urbanizing regions. © 2026 Elsevier Inc.

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land use, Article, environmental factor, Roads and streets, sustainability, Learning systems, machine learning, Machine-learning, Environmental Monitoring, Machine learning, China, spatiotemporal analysis, Population statistics, Remote sensing, Remote-sensing, Machine Learning, Forestry, precipitation, remote sensing, predictive model, procedures, governance approach, rural area, urban area, dimensionality reduction, population density, machine learning algorithm, Multi-Sources, spatial distribution, environmental monitoring, Monte Carlo method, Monte Carlo methods, explainable machine learning, Shapley additive explanation, spatial analysis, nitrogen, Nitrogen, Land use, Interpretable machine learning, Agricultural machinery, agricultural land, Agricultural pollution, Agricultural runoff, ammonia, biogeochemical cycle, chemical analysis, Chemical speciation, dust, Dust, Multi-source remote sensing, nitrate, nitrogen derivative, Nitrogen speciation, organic nitrogen, pollutant source, Population distribution, Remote Sensing Technology, Road dust, Road dusts, Spatial prediction, Spatial variables measurement, speciation (chemistry), surface runoff, Tianjin, urban region, Urban–agricultural gradient

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Markov random field based background subtraction method for foreground detection under moving background scene

Item Type Conference Paper
Author T.-H. Tsai
Author C.-Y. Lin

Abstract Detection of moving objects in video streams is the first relevant step of information extraction in many computer vision applications. Aside from the intrinsic usefulness of being able to segment video streams into moving and background components, detecting moving objects provides a focus of attention for recognition, classification, and activity analysis, making these later steps more efficient. In this paper, a background subtraction based on Bayesian estimation is proposed. The basic of our solution is a Bayesian likelihood test which can distinguish between foreground variation and dynamic background variation. The prior knowledge about the likelihood test is brought to bear by appropriately specified a priori probability as Markov random field. Based on this approach, decision thresholds vary depending on context, thus improving detection performance substantially. We compare our method with other modeling techniques and report experimental results, both in term of detection accuracy, for color video sequences that represent typical situations critical for video surveillance systems. Quantitative evaluation and comparison with the existing methods show that the proposed algorithm provides much improved results. © 2010 IEEE.

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Security systems, Computer vision, Algorithms, Statistical tests, Bayesian networks, Monitoring, Probability, Background subtraction, Computer hardware description languages, Detection accuracy, Quantitative evaluation, Bayesian, Detection performance, Video streaming, Computer vision applications, Bayesian estimations, A-priori probabilities, Color video, Markov Random Fields, Decision threshold, Computer applications, Activity analysis, Background scenes, Background subtraction method, Detecting moving objects, Detection of moving object, Detector circuits, Dynamic background, Existing method, Focus of Attention, Foreground detection, Information Extraction, Markov random field, Modeling technique, Prior knowledge, Video streams, Video surveillance systems, Visual surveillance

Notes:

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Conference name: 4th International Conference on Genetic and Evolutionary Computing, ICGEC 2010; Conference date: 13 December 2010 through 15 December 2010; Conference code: 84151

Master Your Metrics with Calibration

Item Type Conference Paper

Author W. Siblini

Author J. Fréry

Author L. He-Guelton

Author F. Oblé

Author Y.-Q. Wang

Editor Berthold M.R.

Editor Feelders A.

Editor Krempl G.

Abstract Machine learning models deployed in real-world applications are often evaluated with precision-based metrics such as F1-score or AUC-PR (Area Under the Curve of Precision Recall). Heavily dependent on the class prior, such metrics make it difficult to interpret the variation of a model's performance over different subpopulations/subperiods in a dataset. In this paper, we propose a way to calibrate the metrics so that they can be made invariant to the prior. We conduct a large number of experiments on balanced and imbalanced data to assess the behavior of calibrated metrics and show that they improve interpretability and provide a better control over what is really measured. We describe specific real-world use-cases where calibration is beneficial such as, for instance, model monitoring in production, reporting, or fairness evaluation. © 2020, The Author(s).

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Masticatory performance in postmaxillectomy patients with edentulous maxillae fitted with obturator prostheses

Item Type Journal Article

Author T. Ono

Author H. Kohda

Author K. Hori

Author T. Nokubi

Abstract Purpose: Oral cancer develops frequently in older populations. In Japan, the incidence of oral and pharyngeal cancer is 9,201 per year, with such tumors accounting for 1.74% of all cancers. Although obturator prostheses play an important role in the rehabilitation of postmaxillectomy patients, clinicians sometimes experience difficulty in placing obturator prostheses in edentulous maxillae, and criteria for the objective evaluation of treatment outcome have yet to be established. The present study aimed to investigate postsurgical factors influencing the masticatory performance of postmaxillectomy patients with edentulous maxillae fitted with obturator prostheses. Materials and Methods: The extent of the hard palate defect, status of the mandibular dentition, mouth-opening distance, and maximum occlusal force were investigated, and masticatory performance was measured using a testing gummy jelly in 27 postmaxillectomy patients with edentulous maxillae fitted with obturator prostheses. The influence of these items was evaluated quantitatively, and theoretical masticatory performance for each subject was calculated using multiple-regression analysis (quantification method type 1). Results: Although average masticatory performance was almost equivalent to that of healthy, independent older patients with occlusal support classified as Eichner C, considerable individual variation was noted among subjects. The order of strength of influence on masticatory performance was identified by category weight: extent of hard palate defect > status of posterior mandibular teeth > maximum occlusal force > mouth-opening distance. A relatively high correlation ($R^2 = 0.78$, $P < .01$, Pearson correlation coefficient) was evident between theoretical and actual values. Conclusion: Masticatory performance could be predicted by evaluating postsurgical factors in patients with edentulous maxillae fitted with obturator prostheses.

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female, Female, human, Humans, male, Male, middle aged, Middle Aged, aged, Aged, Aged, 80 and over, article, classification, physiology, pathology, pathophysiology, Range of Motion, Articular, edentulousness, mandible, maxilla, joint characteristics and functions, Bite Force, Dental Prosthesis Design, hard palate, Jaw, Edentulous, Jaw, Edentulous, Partially, Mandible, mastication, Mastication, Maxilla, Palatal Obturators, Palate, Hard, temporomandibular joint, Temporomandibular Joint, tooth prosthesis

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Export Date: 20 September 2026; Cited By: 30; Correspondence Address: T. Ono; Osaka University, Graduate School of Dentistry, Suita City, Osaka 565-0871, 1-8 Yamadaoka, Japan; email: ono@dent.osaka-u.ac.jp

Match: A Maximum-Likelihood Approach for Classification under Label Shift

Item Type Conference Paper

Author Z. Donyavi

Author F. Li

Author Y. Zhang

Author D.F. Silva

Author G.E. Batista

Abstract Machine learning models often suffer from performance degradation when dealing with class distributions that differ from the training distribution, a scenario commonly referred to as label shift. Addressing this challenge, this paper introduces Match, a novel adjustment approach that maximizes the likelihood of predicted probabilities under class prevalence constraints. Unlike existing methods such as retraining with instance re-weighting and the Bayes update rule, Match ensures that the adjusted class distribution aligns precisely with the prevalence estimates from quantifiers. By formulating the adjustment process as a

binary integer linear optimization problem, Match benefits from efficient mixed-integer solvers. Extensive experiments demonstrate that Match outperforms the state-of-the-art in classifier adjustment with statistical significance, particularly in handling scenarios with imbalanced distributions. © 2025 ACM.

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Conference Name Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining

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Artificial intelligence, Classification (of information), Learning systems, Machine learning models, linear optimization, Domain adaptation, domain adaptation, Performance degradation, Label shift, label shift, Linear programming, Quantification, quantification, Binary integers, Class distributions, Integer linear programming, Integer programming, Linear optimization, Maximum likelihood, Maximum-likelihood approach, Mixed-integer linear programming, Re-weighting

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Matching Distributions Algorithms Based on the Earth Mover's Distance for Ordinal Quantification

Item Type Journal Article

Author A. Castaño

Author P. González

Author J.A. González

Author J.J. Del Coz

Abstract The goal of quantification learning is to induce models capable of accurately predicting the class distribution for new bags of unseen examples. These models only return the prevalence of each class in the bag because prediction of individual examples is irrelevant in these tasks. A prototypical application of ordinal quantification is to predict the proportion of opinions that fall into each category from one to five stars. Ordinal quantification has hardly been studied in the literature, and in fact, only one approach has been proposed so far. This article presents a comprehensive study of ordinal quantification, analyzing the applicability of the most important algorithms devised for multiclass quantification and proposing three new methods that are based on matching distributions using Earth mover's distance (EMD). Empirical experiments compare 14 algorithms on synthetic and benchmark data. To statistically analyze the obtained results, we further introduce an EMD-based scoring function. The main conclusion is that methods using a criterion somehow related to EMD, including two of our proposals, obtain significantly better results. © 2012 IEEE.

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Job analysis, Task analysis, machine learning, Proposal, Machine learning, Forecasting, Learning algorithms, Prediction algorithms, Predictive models, Probability distributions, Matchings, prevalence estimation, Prevalence estimation, Quantification learning, Probability: distributions, Ordinal quantification, Earth mover's distance, Earth mover's distance (EMD), ordinal quantification, quantification learning

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Export Date: 20 September 2026; Cited By: 8; Funding details: Ministerio de Economía y Competitividad, MINECO; European Regional Development Fund, ERDF, (PID2019-110742RB-I00); European Regional Development Fund, ERDF; Correspondence Address: J.J. Del Coz; University of Oviedo, Artificial Intelligence Center, Gijón, 33204, Spain; email: juanjo@uniovi.es

MATES: a deep learning-based model for locus-specific quantification of transposable elements in single cell

Item Type Journal Article

Author R. Wang

Author Y. Zheng

Author Z. Zhang

Author K. Song

Author E. Wu

Author X. Zhu

Author T.P. Wu

Author J. Ding

Abstract Transposable elements (TEs) are crucial for genetic diversity and gene regulation. Current single-cell quantification methods often align multi-mapping reads to either ‘best-mapped’ or ‘random-mapped’ locations and categorize them at the subfamily levels, overlooking the biological necessity for accurate, locus-specific TE quantification. Moreover, these existing methods are primarily designed for and focused on transcriptomics data, which restricts their adaptability to single-cell data of other modalities. To address these challenges, here we introduce MATES, a deep-learning approach that accurately allocates multi-mapping reads to specific loci of TEs, utilizing context from adjacent read alignments flanking the TE locus. When applied to diverse single-cell omics datasets, MATES shows improved performance over existing methods, enhancing the accuracy of TE quantification and aiding in the identification of marker TEs for identified cell populations. This development facilitates the exploration of single-cell heterogeneity and gene regulation through the lens of TEs, offering an effective transposon quantification tool for the single-cell genomics community. © The Author(s) 2024.

Date 2024

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human, Humans, machine learning, article, Deep Learning, data set, nonhuman, procedures, deep learning, controlled study, numerical model, gene expression, genetics, genomics, animal, Animals, genetic variability, animal experiment, cell, genetic variation, single cell analysis, Single-Cell Analysis, gene locus, transcriptomics, cell heterogeneity, cell population, DNA Transposable Elements, DNA transposon, gene control, Genetic Loci, Genomics, transposon

Notes:

EID: 2-s2.0-85206122947

Export Date: 20 September 2026; Cited By: 14; Correspondence Address: J. Ding; School of Computer Science, McGill University, Montreal, Canada; email: jun.ding@mcgill.ca; T.P. Wu; Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, United States; email: tao.wu@bcm.edu

Matt Finish- Emerging Trend in Automotive Sector for Visual Application

Item Type	Conference Paper
Author	V. Kumar
Author	R. Lalwani
Author	B. Jayanthan
Abstract	Exterior paint look is one the important tie braker on vehicle aesthetic appeal when compared with equal contenders. Time to time, globally new trend emerges to make the vehicle look attractive, one of the technologies is matt finish. To adapt with changing trends, current Indian market trend is moving from glossy finish paint to Matte finish paint by low gloss and attractive texture characteristics, which can provide soft light and pleasant feel. Matt finish helps to enhance the shape and features of the painted surface, feature lines clarity and enhanced visual appeal with absorption of light rather than reflection by glossy finish. Any new technology comes with challenges and limitations compared to traditional approach of vehicle painting and handling in field. In this paper, we have discussed on development of matt finish paint for automotive sector as per OEM manufacturing setup, definition of gloss for matt finish, quantification method, classification of gloss, mechanism, method, as well as the factors that affect the gloss of films. Finally, acceptance criteria for all the tests are defined as per Automotive standards. Also, explicitly defines the technical requirements and Robust technical evaluation methodologies of various automotive short- and long-term tests. All the performance results were compared for screening of product and to meet the first-time right implementation target. © 2025 Mahindra & Mahindra Ltd.
Date	2025
DOI	10.4271/2025-28-0038
URL	https://www.scopus.com/pages/publications/86000007445?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: SAE Techni. Paper.

Publisher SAE International**ISBN** 01487191 (ISSN)**Proceedings Title** SAE Techni. Paper.**Conference Name** SAE Technical Papers**Date Added** 20/09/2026, 9:51:36 pm**Modified** 20/09/2026, 9:51:36 pm**Tags:**

'current, Automobiles, Automotive industry, Acceptance tests, Automotive sector, Emerging trends, Aesthetic appeals, Changing trends, Exterior paints, Indian market, Market trends, Texture characteristics, Visual applications

Notes:

EID: 2-s2.0-86000007445

Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 International Conference on Advances in Design, Materials, Manufacturing and Surface Engineering for Mobility, ADMMS 2025; Conference date: 7 February 2025 through 8 February 2025; Conference code: 207112

Maximizing model generalization under feature and label shifts for structural damage detection via Bayesian theory

Item Type Journal Article**Author** X. Wang**Author** J. Jiao**Author** X. Zhou**Author** Y. Xia

Abstract Machine learning models face performance degradation when migrating to datasets with different distributions from training data. This challenge limits their applications because recollecting labeled data to retrain models is expensive and time-consuming. Domain generalization (DG) aims to learn a predictive model that can extract invariant features across source domains and then generalize to related but unseen target domains. However, most existing feature-invariant DG methods rely on unrealistic assumptions (e.g., stable feature distribution or no label shift) and simplify the problem to learn either invariant feature marginal or conditional distributions. In practice, both feature and label shifts may exist, rendering these assumptions invalid. This study develops a novel DG method to simultaneously consider conditional, marginal, and label shifts. With Bayes' theorem, DG is interpreted as a posterior distribution alignment problem, which is derived by the likelihood function, evidence, and prior. The likelihood function and evidence correspond to feature conditional and marginal distributions, respectively, which are estimated by exploiting variational Bayesian inference. The conditional distributions across domains are aligned by minimizing the Kullback–Leibler divergence, and the marginal distributions across domains are aligned through moment minimization. The label prior shift is estimated by the label smoothing mechanism and class-wise

prototype learning. Consequently, DG is achieved by aligning the label posterior distribution according to the Bayesian equation. Numerical and experimental examples demonstrate that the developed method outperforms state-of-the-art DG methods in damage detection of both mechanical and civil structures. The method can be extended to DG tasks in other fields. © 2024

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Journal Abbr Mech Syst Signal Process

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Markov processes, Machine-learning, Machine learning, Generalisation, Inference engines, Learn+, Domain generalization, Marginal distribution, Conditional distribution, Invariant features, Feature and label shift, Feature and label shifts, Structural damage detection, Variational Bayesian inference, Variational Bayesian inferences

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Export Date: 20 September 2026; Cited By: 8; Funding details: University Grants Committee of Hong Kong, (15201920, 15217522); Correspondence Address: Y. Xia; Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, Kowloon, China; email: ceyxia@polyu.edu.hk; CODEN: MSSPE

Maximum likelihood in cost-sensitive learning: Model specification, approximations, and upper bounds

Item Type Journal Article

Author J.P. Dmochowski

Author P. Sajda

Author L.C. Parra

Abstract The presence of asymmetry in the misclassification costs or class prevalences is a common occurrence in the pattern classification domain. While much interest has been devoted to the study of cost-sensitive learning techniques, the relationship between cost-sensitive learning and the specification of the model set in a parametric estimation framework remains somewhat unclear. To that end, we differentiate

between the case of the model including the true posterior, and that in which the model is misspecified. In the former case, it is shown that thresholding the maximum likelihood (ML) estimate is an asymptotically optimal solution to the risk minimization problem. On the other hand, under model misspecification, it is demonstrated that thresholded ML is suboptimal and that the risk-minimizing solution varies with the misclassification cost ratio. Moreover, we analytically show that the negative weighted log likelihood (Elkan, 2001) is a tight, convex upper bound of the empirical loss. Coupled with empirical results on several real-world data sets, we argue that weighted ML is the preferred cost-sensitive technique. © 2010 Shyam Visweswaran and Gregory F. Cooper.

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Notes:

EID: 2-s2.0-79551493545

Export Date: 20 September 2026; Cited By: 58; Correspondence Address: J. P. Dmochowski; Department of Biomedical Engineering, City College of New York - City University of New York, New York, NY 10031, United States; email: JDMOCHOWSKI@CCNY.CUNY.EDU

Maximum likelihood with bias-corrected calibration is hard-To-beat at label shift adaptation

Item Type Conference Paper

Author A.M. Alexandari

Author A. Kundaje

Author A. Shrikumar

Editor Daume H.

Editor Singh A.

Abstract Label shift refers to the phenomenon where the prior class probability $p(y)$ changes between the training and test distributions, while the conditional probability $p(x|y)$ stays fixed. Label shift arises in settings like medical diagnosis, where a classifier trained to predict disease given symptoms must be adapted to scenarios where the baseline prevalence of the disease is different. Given estimates of $p(y|x)$ from a predictive model, Saerens et al. proposed an efficient maximum likelihood algorithm to correct for label shift that does not require model retraining, but a limiting assumption of this algorithm is that $p(y|x)$ is calibrated, which is not true of modern neural networks. Recently, Black Box Shift Learning (BBSL) and Regularized Learning under Label Shifts (RLLS) have emerged as state-of-the-art techniques to cope with label shift when a classifier does not output calibrated probabilities, but both methods require model retraining with importance weights and neither has been benchmarked against maximum likelihood. Here we (1) show that combining maximum likelihood with a type of calibration we call bias-corrected calibration outperforms both BBSL and RLLS across diverse datasets and distribution shifts, (2) prove that the maximum likelihood objective is concave, and (3) introduce a principled strategy for estimating source-domain priors that improves robustness to poor calibration. This work demonstrates that the maximum likelihood with appropriate calibration is a formidable and efficient baseline for label shift adaptation. See sec. 6 for links to code, video, and blogpost. © ICML 2020. All rights reserved.

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Conference Name 37th International Conference on Machine Learning, ICML 2020

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Machine learning, Predictive analytics, Black boxes, Predictive modeling, State-of-the-art techniques, Diagnosis, Conditional probabilities, Calibration, Maximum likelihood, Class probabilities, Importance weights, Maximum likelihood algorithm

Notes:

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Maximum Likelihood with Bias-Corrected Calibration is Hard-To-Beat at Label Shift Adaptation

Item Type Conference Paper

Author A.M. Alexandari

Author A. Kundaje

Author A. Shrikumar

Editor Daumé III H.

Editor Singh A.

Abstract Label shift refers to the phenomenon where the prior class probability $p(y)$ changes between the training and test distributions, while the conditional probability $p(x|y)$ stays fixed. Label shift arises in settings like medical diagnosis, where a classifier trained to predict disease given symptoms must be adapted to scenarios where the baseline prevalence of the disease is different. Given estimates of $p(y|x)$ from a predictive model, Saerens et al. proposed an efficient maximum likelihood algorithm to correct for label shift that does not require model retraining, but a limiting assumption of this algorithm is that $p(y|x)$ is calibrated, which is not true of modern neural networks. Recently, Black Box Shift Learning (BBSL) and Regularized Learning under Label Shifts (RLLS) have emerged as state-of-the-art techniques to cope with label shift when a classifier does not output calibrated probabilities, but both methods require model retraining with importance weights and neither has been benchmarked against maximum likelihood. Here we (1) show that combining maximum likelihood with a type of calibration we call bias-corrected calibration outperforms both BBSL and RLLS across diverse datasets and distribution shifts, (2) prove that the maximum likelihood objective is concave, and (3) introduce a principled strategy for estimating source-domain priors that improves robustness to poor calibration. This work demonstrates that the maximum likelihood with appropriate calibration is a formidable and efficient baseline for label shift adaptation. See sec. 6 for links to code, video, and blogpost. © 2020 by the author(s).

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Notes:

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MC-SQ and MC-MQ: Ensembles for Multi-Class Quantification

Item Type Journal Article

Author Z. Donyavi

Author A.B.S. Serapiao

Author G. Batista

Abstract Quantification research proposes methods to estimate the class distribution in an independent sample. Quantification methods find applications in areas that rely on estimated aggregated quantities, such as epidemiology, sentiment analysis, political research, and ecological surveillance. For instance, epidemiologists are often concerned with the dynamics of the number of disease cases across space and time. Thus, while classification predicts individual subjects, quantifiers are the methods that directly estimate the number of cases. Although quantification is a thriving area of research, with numerous approaches proposed in the last decade, most focus has been on binary-class quantifiers. One common approach for multi-class quantification is the one-versus-all (OVA) approach, but empirical evidence suggests its performance is suboptimal. This paper's first contribution is to elucidate why OVA quantifiers struggle to perform well in multi-class settings due to a distribution shift. To circumvent this problem, our second proposal is two new multi-class quantifiers based on ensemble learning that significantly improve performance for binary and multi-class settings. Our comprehensive experimental setup with 37 state-of-the-art (single and ensemble) quantifiers shows that our ensembles are the best-performing quantifiers and rank first in a recent quantification competition. © 1989-2012 IEEE.

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MC-SQ: A Highly Accurate Ensemble for Multi-class Quantification

Item Type Conference Paper

Author Z. Donyavi

Author A. Serapio

Author G. Batista

Abstract Quantification research proposes methods to estimate the class distribution in an independent sample. Many areas, such as epidemiology, sentiment analysis, political research and ecological surveillance, rely on quantification methods to estimate aggregated quantities. For instance, epidemiologists are often concerned with the dynamics of the number of disease cases across space and time. Thus, while classification predicts individual subjects, quantification is the class of methods that directly estimate the number of cases. Quantification is a thriving research area, and the community has proposed several approaches in the last decade. Nevertheless, most quantification research has focused on binary-class quantifiers, expecting these approaches to extend to multi-class using the one-

versus-all (OVA) approach. However, there is enough empirical evidence indicating the performance of OVA multi-class quantifiers is subpar. This paper has two main contributions. First, we demonstrate why OVA quantifiers are doomed to underperform in multi-class settings due to a distribution shift they cannot handle. Second, we propose a new class of quantifiers based on ensemble learning that boosts the performance of the base quantifiers in the binary and, more importantly, multi-class settings. In one of the most comprehensive experimental setups ever attempted in quantification research, we show that our ensembles are the best-performing quantifiers compared with 33 state-of-the-art (single and ensemble) quantifiers and rank first in a recent quantification competition. Copyright © 2023 by SIAM.

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McGET: A rapid image-based method to determine the morphological characteristics of gravels on the Gobi desert surface

Item Type Journal Article

Author Y. Mu

Author F. Wang

Author B. Zheng

Author W. Guo

Author Y. Feng

Abstract The relationship between morphological characteristics (e.g. gravel size, coverage, angularity and orientation) and local geomorphic features (e.g. slope gradient and aspect) of desert has been used to explore the evolution process of Gobi desert. Conventional quantification methods are time-consuming, inefficient and even prove impossible to determine the characteristics of large numbers of gravels. We propose a rapid image-based method to obtain the morphological characteristics of gravels on the Gobi desert surface, which is called the “morphological characteristics gained effectively technique” (McGET). The image of the Gobi desert surface was classified into gravel clusters and background by a machine-learning “classification and regression tree” (CART) algorithm. Then gravel clusters were segmented into individual gravel clasts by separating objects in images using a “watershed segmentation” algorithm. Thirdly, gravel coverage, diameter, aspect ratio and orientation were calculated based on the basic principles of 2D computer graphics. We validated this method with two independent datasets in which the gravel morphological characteristics were obtained from 2728 gravels measured in the field and 7422 gravels measured by manual digitization. Finally, we applied McGET to derive the spatial variation of gravel morphology on the Gobi desert along an alluvial-proluvial fan located in Hami, Xinjiang, China. The validated results show that the mean gravel diameter measured in the field agreed well with that calculated by McGET for large gravels ($R^2 = 0.89$, $P < 0.001$). Compared to manual digitization, the McGET accuracies for gravel coverage, gravel diameter and aspect ratio were 97%, 83% and 96%, respectively. The orientation distributions calculated were consistent across two different methods. More importantly, McGET significantly shortens the time cost in obtaining gravel morphological characteristics in the field and laboratory. The spatial variation results show that the gravel coverage ranged from 88% to 65%, the gravel diameter was unimodally distributed and ranged from 19 mm to 13 mm. Most gravels were bladed or rod-like, with a mean aspect ratio of 1.57, and had no preferred orientation on the surveyed Gobi desert. From the center to the edge of the fan, gravel coverage decreased 2.2% per 100 m elevation decrease ($R^2 = 0.69$, $P < 0.001$), mean gravel diameter decreased 0.5 mm per 100 m elevation decrease ($R^2 = 0.52$, $P < 0.001$), and mean aspect ratio slightly increased 0.004 per 100 m elevation decrease ($R^2 = 0.26$, $P < 0.05$). These results imply that surface washing was the main process on the investigated Gobi desert. This study demonstrates that the new method can quickly and accurately calculate the gravel coverage, diameter, aspect ratio and orientation from the images of Gobi desert. © 2017 Elsevier B.V.

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Measurement of Adolescent Drug Use

Item Type Journal Article

Author A.R. Morral

Author D.F. McCaffrey

Author S. Chien

Abstract There is widespread agreement that estimates of adolescent drug use prevalence from the National Household Survey of Drug Abuse (NHSDA) and Monitoring the Future (MTF) are subject to considerable measurement error. Nevertheless, some have suggested that trends over time in these prevalence estimates probably reflect true trends in drug use, since underreporting may be assumed to be constant over time. A recent National Research Council report criticizes this assumption on logical grounds. The present study examines adolescent drug use responses on the NHSDA and MTF for evidence of “drug omission,” “jargon confusion” and “conceptual confusion,” three types of misreporting expected to vary in magnitude with changes in drug use practices and changes in survey items. Results demonstrate that adolescent drug users are significantly more likely than adults to report use of drugs not listed in the NHSDA. Among adolescents who wrote in the “other” drugs they used, 66% and 86% of hallucinogen and inhalant responses showed confusion over the meaning of the pharmacological terms used in the NHSDA. Almost 20% of MTF respondents who report lifetime use of Rohypnol or ecstasy, when specifically queried about these drugs, deny lifetime use of any substances in the drug classes intended to assess use of Rohypnol and ecstasy. MTF respondents reporting lifetime use of PCP underreport use of hallucinogens at rates that vary substantially over time, from a high of 45% (in 1986), to a low of just 8% (in 1998). The implications of these findings for adolescent drug use prevalence estimation and survey design are discussed. © 2003 Taylor and Francis Group, LLC.

Date 2003

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human, adolescent, prevalence, drug use, health survey, measurement, psychedelic agent, conference paper, drug classification, 3,4 methylenedioxymethamphetamine, amphetamine, analgesic agent, barbituric acid derivative, central stimulant agent, flunitrazepam, narcotic agent, sedative agent, tranquilizer

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Measurement of Drivers' Confidence Levels in Left-Turn Decisions Using Machine Learning

Item Type Conference Paper

Author Y. Liu

Author J. Liu

Author H. Li

Author L.-Z. Yang

Abstract Decision confidence in driving, especially in complex operations such as left turns, is crucial for understanding human cognitive behavior and improving driving safety. At present, the lack of confidence quantification methods for dynamic real-time driving tasks limits the development of intelligent driving systems. This study proposes a method based on machine learning, which for the first time utilizes driving performance data (such as speed, acceleration, steering wheel angle) to measure driver confidence levels in left turn decisions. In a simulated driving environment, data from 33 participants were used to train seven machine learning models. Among them, the stacked classifier performed the best within the 0-5 second time window after stepping on the accelerator, with an accuracy of 90.97% and an F1 score of 90.91%, better than the 5 - 10 second

time window. The study identifies the standard deviation of forward direction position, maximum speed and mean speed as key features for confidence measurement. These findings provide the possibility for real-time assessment of driver confidence and have significant potential for improving human-machine interaction in intelligent driving systems and enhancing road safety. © 2025 IEEE.

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Behavioral research, Accident prevention, Intelligent vehicle highway systems, Motor transportation, Intelligent systems, Real time systems, Learning systems, Machine-learning, Machine learning, Traffic control, Automobile drivers, Driver training, Machine Learning, Cognitive behavior, Complex operations, Confidence levels, Decision confidence, Decision Confidence, Driving performance, Driving Performance, Intelligent driving systems, Left turns, Measurements of, Time windows

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Measuring Cross-National Variations in Religiosity and Attitudes Toward Science and Technology Using Machine Learning

Item Type Journal Article

Author J.J. Lee

Abstract This study uses resampling methods and machine learning to measure how religio-scientific groups are distributed across regions, countries/territories, and religious groups. Across 76 societies (N = 143,092), the distribution of class membership is as follows: traditional (31.9 percent), modern (23.7 percent), post-secular (30.3 percent), and postmodern (14.1 percent). Although most societies are dominated by a single class, there is evidence of significant heterogeneity within societies in class prevalence. Those with post-secular views are both religious and feel favorably toward science; however, when faced with a conflict between religion and science they tend to support religion. Ultimately, societies with large traditional and post-secular classes are significantly more likely to support religion given a conflict with science; in contrast, the reverse is true for societies with large modern and postmodern classes. © 2023 MSS.

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Tags:

methodology, Global and transnational sociology, religion, sociology of science, knowledge, and technology

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: J.J. Lee; Department of Sociology, Northwestern University, Evanston, 1810 Chicago Avenue, 60208, United States; email: johnlee2021@u.northwestern.edu

Measuring daytime and nighttime variations in leisure service efficiency of community green spaces for older adults

Item Type Journal Article

Author C. Chen

Author H. Li

Author X. Wen

Author W. Zhang

Author J. Zhang

Author E.H.K. Yung

Author L. Meng

Abstract Community green spaces (CGSs) offer a range of leisure services (LSs) for older adults. However, the relationship between LS resources and their effectiveness, particularly regarding efficiency, across day and night scenarios and different geographical contexts remains underexplored. Using Guangzhou, China, as a case study, this research integrates human-perspective CGS imagery, crowdsourced data, and deep learning to quantify LS resources, effectiveness, and efficiency in both daytime and nighttime contexts. The results show that areas with abundant resources or high effectiveness do not necessarily exhibit high efficiency. High-efficiency areas during the day are primarily community parks and high-quality residential green spaces, but shift at night to roadside green spaces and residential green spaces in older urban areas. Conversely, roadside green spaces and residential green spaces within urban villages show low efficiency during the day, whereas at night, community parks and high-quality residential areas exhibit lower efficiency. The study recommends increasing daytime LS resources in urban villages and older neighborhoods and enhancing nighttime resources in community parks and high-quality residential areas. In addition, the findings highlight the nonlinear impact of LS resources on efficiency, with brightness identified as the most influential factor. This study proposes a theoretical model for assessing LS efficiency for older adults and provides practical recommendations for optimizing CGS resource allocation to support creation of age-friendly environments. © 2025 Elsevier Inc.

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adult, Deep learning, Efficiency, machine learning, China, Rural areas, urban area, High quality, Older adults, greenspace, Leisure, resource allocation, Guangdong, Housing, Green spaces, Age-friendly, Age-friendly environment, Age-friendly environments, Community park, Guangzhou, Health benefits, leisure industry, Natural environment, Service resources, Street view image, Street view images

Notes:

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Measuring predictive reliability under temporal class-prior drift: A forward cohort-based evaluation framework with application to student retention prediction

Item Type Journal Article

Author A.P. Utomo

Author P. Purwanto

Author B. Surarso

Abstract Evaluating predictive reliability under temporally evolving data distributions remains a critical measurement challenge in real-world machine learning systems. In educational contexts, student populations change over time, introducing temporal class-prior drift and generalization instability that are often overlooked by conventional evaluation protocols based on random data splitting or cross-validation. This study proposes a measurement-oriented evaluation framework for assessing predictive reliability under forward cohort-based temporal conditions. The framework integrates discrimination performance, validation–test generalization gaps, and probability calibration to provide a deployment-aligned assessment of model robustness. The approach is demonstrated through sequential student retention prediction across thirteen semester-level stages using longitudinal academic data, comparing a sequential deep learning model (Long Short-Term Memory, LSTM) with non-sequential ensemble baselines (Random Forest and XGBoost). Empirical results reveal substantial temporal variability in predictive reliability. Intermediate prediction stages exhibit the largest generalization gaps, indicating reduced temporal generalization stability under forward cohort evaluation, while temporal class-prior evolution contributes to instability across cohorts. Calibration analysis further shows that probability reliability follows a stage-dependent pattern, deteriorating in intermediate stages and partially recovering in later stages, even when discrimination performance remains competitive. These findings demonstrate that predictive reliability in non-stationary environments cannot be adequately assessed using discrimination metrics alone. Instead, a multi-dimensional measurement framework is required to capture temporal generalization stability and calibration behavior, providing a more realistic evaluation of predictive models under deployment conditions. © 2026 Elsevier Ltd

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MEDIC - a method for predicting residual service life and refurbishment investment budgets

Item Type Journal Article

Author F. Flourentzou

Author E. Brandt

Author C. Wetzel

Abstract In the European project, EPIQR deterioration of building materials and components is described by the use of a classification system with four classes for the qualitative condition of, e.g., a facade or window. The prediction of the probable period of passing into the next (worse) deterioration state is of high interest as this is directly connected to higher refurbishment costs. MEDIC, which stands for 'Methode d'Evaluation de scenarios de Degradation probables d'Investissements Correspondants' is a proposal for a new method to predict the future degradation state of the building. It calculates for each element regarded in EPIQR the probability to pass from one class to another during time. The prediction is based on the combination of the a priori probability based on experience from a large number of previous investigations/refurbishments and the current state of the object under study.

Date 2000

DOI 10.1016/S0378-7788(99)00031-6

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Costs, Buildings, Probability, Retrofitting, Building degradation, Building diagnosis, Energy performance indoor environmental quality retrofit (IPIQR), Residual service life, Service life

Notes:

EID: 2-s2.0-0034135302

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Meta-analytic connectivity modeling revisited: Controlling for activation base rates

Item Type Journal Article

Author R. Langner

Author C. Rottschy

Author A.R. Laird

Author P.T. Fox

Author S.B. Eickhoff

Abstract Co-activation of distinct brain regions is a measure of functional interaction, or connectivity, between those regions. The co-activation pattern of a given region can be investigated using seed-based activation likelihood estimation meta-analysis of functional neuroimaging data stored in databases such as BrainMap. This method reveals inter-regional functional connectivity by determining brain regions that are consistently co-activated with a given region of interest (the "seed") across a broad range of experiments. In current implementations of this meta-analytic connectivity modeling (MACM), significant spatial convergence (i.e. consistent co-activation) is distinguished from noise by comparing it against an unbiased null-distribution of random spatial associations between experiments according to which all gray-matter voxels have the same chance of convergence. As the a priori probability of finding activation in different voxels markedly differs across the brain, computing such a quasi-rectangular null-distribution renders the detection of significant convergence more likely in those voxels that are frequently activated. Here, we propose and test a modified MACM approach that takes this activation frequency bias into account. In this new specific co-activation likelihood estimation (SCALE) algorithm, a null-

distribution is generated that reflects the base rate of reporting activation in any given voxel and thus equalizes the a priori chance of finding across-study convergence in each voxel of the brain. Using four exemplary seed regions (right visual area V4, left anterior insula, right intraparietal sulcus, and subgenual cingulum), our tests corroborated the enhanced specificity of the modified algorithm, indicating that SCALE may be especially useful for delineating distinct core networks of co-activation. © 2014 Elsevier Inc.

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Meta-DermDiagnosis: Few-shot skin disease identification using meta-learning

Item Type Conference Paper

Author K. Mahajan

Author M. Sharma

Author L. Vig

Abstract Annotated images for diagnosis of rare or novel diseases are likely to remain scarce due to small affected patient population and limited clinical expertise to annotate images. Deep networks employed for image based diagnosis need to be robust enough to quickly adapt to novel diseases with few annotated images. Further, in case of the frequently occurring long-tailed class distributions in skin lesion and other disease classification datasets, conventional training approaches lead to poor generalization on classes at the tail end of the distribution due to biased class priors. This paper focuses on the problems of disease identification and quick model adaptation in such data-scarce and long-tailed class distribution scenarios by exploiting recent advances in meta-learning. This involves training a neural network on few-shot image classification tasks based on an initial set of class labels / head classes of the distribution, prior to adapting the model for classification on a set of unseen / tail classes. We named the proposed method Meta-DermDiagnosis because it utilizes meta-learning based few-shot learning techniques such as the gradient based Reptile and distance metric based Prototypical networks for identification of diseases in skin lesion datasets. We evaluate the effectiveness of our approach on publicly available skin lesion datasets, namely the ISIC 2018, Derm7pt and SD-198 datasets and obtain significant performance improvement over pretrained models with just a few annotated examples. Further, we incorporate Group Equivariant convolutions (G-convolutions) for the Meta-DermDiagnosis network to improve disease identification performance as these images generally do not have any prevailing global orientation / canonical structure and G-convolutions make the network equivariant to any discrete transformations like rotation, reflection and translation. © 2020 IEEE.

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Notes:

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Meta-learning for few-shot land cover classification

Item Type Conference Paper

Author M. Ruswurm

Author S. Wang

Author M. Korner

Author D. Lobell

Abstract The representations of the Earth's surface vary from one geographic region to another. For instance, the appearance of urban areas differs between continents, and seasonality influences the appearance of vegetation. To capture the diversity within a single category, such as urban or vegetation, requires a large model capacity and, consequently, large datasets. In this work, we propose a different perspective and view this diversity as an inductive transfer learning problem where few data samples from one region allow a model to adapt to an unseen region. We evaluate the modelagnostic meta-learning (MAML) algorithm on classification and segmentation tasks using globally and regionally distributed datasets. We find that few-shot model adaptation outperforms pre-training with regular gradient descent and fine-tuning on the (1) Sen12MS dataset and (2) DeepGlobe dataset when the source domain and target domain differ. This indicates that model optimization with meta-learning may benefit tasks in the Earth sciences whose data show a high degree of diversity from region to region, while traditional gradient-based supervised learning remains suitable in the absence of a feature or label shift. © 2020 IEEE.

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Pages 788-796**Proceedings Title** IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recogn. Workshops**Conference Name** IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops**Date Added** 20/09/2026, 9:52:36 pm**Modified** 20/09/2026, 9:52:36 pm**Tags:**

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Meta-Learning Without Data via Unconditional Diffusion Models

Item Type Journal Article**Author** Y. Wei**Author** Z. Hu**Author** L. Shen**Author** Z. Wang**Author** L. Li**Author** Y. Li**Author** C. Yuan

Abstract Although few-shot learning aims to address data scarcity, it still requires large, annotated datasets for training, which are often unavailable due to cost and privacy concerns. Previous studies have utilized pre-trained diffusion models, either to synthesize auxiliary data besides limited labeled samples, or to employ diffusion models as zero-shot classifiers. However, they are limited to conditional diffusion models needing class prior information (e.g., carefully crafted text prompts) about unseen tasks. To overcome this, we leverage unconditional diffusion models without needs for class information to train a meta-model capable of generalizing to unseen tasks. The framework contains (1) a meta-learning without data approach that uses synthetic data during training; and (2) a diffusion model-based data augmentation to calibrate the distribution shift during testing. During meta-training, we implement a self-taught class-learner to gradually capture class concepts, guiding unconditional diffusion models to generate a labeled pseudo dataset. This pseudo dataset is then used to jointly train the class-learner and the meta-model, allowing for iterative refinement and clear differentiation between classes. During meta-testing, we

introduce a data augmentation that employs the diffusion models used in meta-training, to narrow the gap between meta-training and meta-testing task distribution. This enables the meta-model trained on synthetic images to effectively classify real images in unseen tasks. Comprehensive experiments showcase the superiority and adaptability of our approach in four real-world scenarios. Code available at <https://github.com/WalkerWorldPeace/MLWDUDM>. © 1991-2012 IEEE.

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Pages 11874-11885

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Metabolic detection of malignant brain gliomas through plasma lipidomic analysis and support vector machine-based machine learning

Item Type Journal Article

Author J. Zhou
Author N. Ji
Author G. Wang
Author Y. Zhang
Author H. Song
Author Y. Yuan
Author C. Yang
Author Y. Jin
Author Z. Zhang
Author L. Zhang
Author Y. Yin

Abstract Background: Most malignant brain gliomas (MBGs) are associated with dismal outcomes, mainly due to their late diagnosis. Current diagnostic methods for MBGs are based on imaging and histological examination, which limits their early detection. Here, we aimed to identify reliable plasma lipid biomarkers for non-invasive diagnosis for MBGs. Methods: Untargeted lipidomic analysis was firstly performed using a discovery cohort (n=107). The data were processed by a support vector machine (SVM)-based discriminating model to retrieve a panel of candidate biomarkers. Then, a targeted quantification method was developed, and the SVM-based diagnostic model was constructed using a training cohort (n=750) and tested using a test cohort (n=225). Finally, the performance of the diagnostic model was further evaluated in an independent validation cohort (n=920) enrolled from multiple medical centers. Findings: A panel of 11 plasma lipids was identified as candidate biomarkers with an accuracy of 0.999. The diagnostic model developed achieved a high performance in distinguishing MBGs patients from normal controls with an area under the receiver-operating characteristic curve (AUC) of 0.9877 and 0.9869 in the training and test cohorts, respectively. In the validation cohort, the 11 lipid panel still achieved an accuracy of 0.9641 and an AUC of 0.9866. Interpretation: The present study demonstrates the applicability and robustness of utilizing a machine learning algorithm to analyze lipidomic data for efficient and reliable biomarker screening. The 11 lipid biomarkers show great potential for the non-invasive diagnosis of MBGs with high throughput. Funding: A full list of funding bodies that contributed to this study can be found in the Acknowledgments section. © 2022 The Authors

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Notes:

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Metabolic spatial connectivity in amyotrophic lateral sclerosis as revealed by independent component analysis

Item Type

Journal Article

Author

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F. Nobili

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S. Morbelli

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P. Fania

Author

A. Cistaro

Author

A. Chiò

Abstract

Objectives: Positron emission tomography (PET) and volume of interest (VOI) analysis have recently shown in amyotrophic lateral sclerosis (ALS) an accuracy of 93% in differentiating patients from controls. The aim of this study was to disclose by spatial independent component analysis (ICA) the brain networks involved in ALS pathological processes and evaluate their discriminative value in separating patients from controls. Experimental design: Two hundred fifty-nine ALS patients and 40 age-

and sex-matched control subjects underwent brain 18F-2-fluoro-2-deoxy-D-glucose PET (FDG-PET). Spatial ICA of the preprocessed FDG-PET images was performed. Intensity values were converted to z-scores and binary masks were used as data-driven VOIs. The accuracy of this classifier was tested versus a validated system processing intensity signals in 27 brain meta-VOIs. A support vector machine was independently applied to both datasets and the 'leave-one-out' technique verified the general validity of results. Principal observations: The 8 components selected as pathophysiologically meaningful discriminated patients from controls with 99.0% accuracy, the discriminating value of bilateral cerebellum/midbrain alone representing 96.3%. Among the meta-VOIs, right temporal lobe alone reached an accuracy of 93.7%. Conclusions: Spatial ICA identified in a very large cohort of ALS patients distinct spatial networks showing a high discriminatory value, improving substantially on the previously obtained accuracy. The cerebellar/midbrain component accounted for the highest accuracy in separating ALS patients from controls. Spatial ICA and multivariate analysis perform better than univariate semi-quantification methods in identifying the neurodegenerative features of ALS and pave the way for inclusion of PET in clinical trials and early diagnosis. *Hum Brain Mapp* 37:942-953, 2016. © 2015 Wiley Periodicals, Inc.

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Notes:

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Export Date: 20 September 2026; Cited By: 43; Correspondence Address: M. Pagani; Institute of Cognitive Sciences and Technologies, CNR, Rome, Via Palestro 32, 00185, Italy; email: marco.pagani@istc.cnr.it; CODEN: HBMAE

MetaQuant: A framework for metaheuristic based quantification

Item Type Journal Article

Author A.B.S. Serapião

Author Z. Donyavi

Author G. Batista

Abstract Metaheuristics are widely employed in various domains; however, their application in quantification learning remains largely unexplored. This study introduces a pioneering approach where the quantification task is treated as a global optimization problem and formulated as a linear regression model solvable by any metaheuristic algorithm. This novel methodology addresses the limitations of traditional quantifiers solved using deterministic methods, such as being trapped in local optima, encountering the curse of dimensionality, and necessitating non-convex cost functions. The paper presents MetaQuant, a novel framework for metaheuristic-based quantification algorithms, and conducts an extensive evaluation comparing 48 nature-inspired metaheuristic algorithms implementing quantifiers with 20 traditional quantifiers. The evaluation encompasses 31 multi-class datasets and reveals compelling results. The findings demonstrate the robustness and exceptional performance of metaheuristic quantifiers, with all 48 MetaQuantifiers outperforming 19 state-of-the-art quantifiers and 18 MetaQuantifiers surpassing the best traditional quantifier. This study contributes towards a promising new line of research in quantification learning. © 2025 Elsevier B.V.

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Methane Retrieval Algorithms Based on Satellite: A Review

Item Type Journal Article

Author Y. Jiang

Author L. Zhang

Author X. Zhang

Author X. Cao

Abstract As the second most predominant greenhouse gas, methane-targeted emission mitigation holds the potential to decelerate the pace of global warming. Satellite remote sensing is an important monitoring tool, and we review developments in the satellite detection of methane. This paper provides an overview of the various types of satellites, including the various instrument parameters, and describes the different types of satellite retrieval algorithms. In addition, the currently popular methane point source quantification method is presented. Based on existing research, we delineate the classification of methane remote sensing satellites into two overarching categories: area flux mappers and point source imagers. Area flux mappers primarily concentrate on the assessment of global or large-scale methane concentrations, with a

further subclassification into active remote sensing satellites (e.g., MERLIN) and passive remote sensing satellites (e.g., TROPOMI, GOSAT), contingent upon the remote sensing methodology employed. Such satellites are mainly based on physical models and the carbon dioxide proxy method for the retrieval of methane. Point source imagers, in contrast, can detect methane point source plumes using their ultra-high spatial resolution. Subcategories within this classification include multispectral imagers (e.g., Sentinel-2, Landsat-8) and hyperspectral imagers (e.g., PRISMA, GF-5), contingent upon their spectral resolution disparities. Area flux mappers are mostly distinguished by their use of physical algorithms, while point source imagers are dominated by data-driven methods. Furthermore, methane plume emissions can be accurately quantified through the utilization of an integrated mass enhancement model. Finally, a prediction of the future trajectory of methane remote sensing satellites is presented, in consideration of the current landscape. This paper aims to provide basic theoretical support for subsequent scientific research. © 2024 by the authors.

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Methodological issues in international comparison of interval breast cancers

Item Type Journal Article

Author J.-L. Bulliard

Author P. Sasieni

Author C. Klabunde

Author J.-P. De Landtsheer

Author B.C. Yankaskas

Author J. Fracheboud

Abstract International comparisons of interval cancers (IC) are important to better understand the relationship between programmes' performance and screening practices. In this respect, differences in (i) definition, (ii) identification and (iii) quantification of IC have received little attention. To examine these 3 comparability issues and activities involving IC, an assessment was conducted among member countries of the International Breast Cancer Screening Network, and the impact of accuracy of identification and quantification practices was estimated using 1996-98 data from the Dutch breast cancer screening programme. Information was obtained from 19 screening programmes in 18 countries, 16 of which acknowledged the coexistence of opportunistic screening. IC data were collected to evaluate performance of the screening programme (100% of programmes) and the radiologists (89%); 53% of programmes had a designated review process for IC. Most programmes (84%) agreed with the European Guidelines definition of IC, but a case situation exercise evidenced substantial discrepancy in classification of cancers that occurred after a positive screen. Completeness of identification of IC appears to contribute most to international variation, and cannot be easily controlled for in methodologically rigorous comparisons. Statistically significant differences of about 4% were measured between quantification methods for IC. An operational definition of IC is proposed to enhance international comparability. Valid comparisons of IC are possible with careful attention to the definition but true differences in IC frequency across screening programmes should exceed 10% to be possibly indicative of real differences between programmes. © 2006 Wiley-Liss, Inc.

Date 2006

DOI 10.1002/ijc.21941

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Export Date: 20 September 2026; Cited By: 46; Correspondence Address: J.-L. Bulliard; Unité d'Épidémiologie du Cancer, Institut Universitaire de Médecine Sociale et Préventive, 1005 Lausanne, rue du Bugnon 17, Switzerland; email: Jean-Luc.Bulliard@chuv.ch; CODEN: IJCNA

Methodology for quantifying groundwater abstractions for agriculture via remote sensing and GIS

Item Type Journal Article

Author S. Castaño

Author D. Sanz

Author J.J. Gómez-Alday

Abstract Among the difficulties and uncertainties that arise when determining water balance is the calculation of groundwater abstraction. This factor is particularly important in aquifers whose extension and heavy agricultural use make direct quantification methods unfeasible (i.e. flow meters and power consumption data). This study presents a method of quantifying groundwater abstractions for irrigation based on the analysis of multitemporal and multispectral satellite images. The process begins with a highly detailed classification of irrigated crops; these data are entered in a Geographic Information System, overlain with a correct estimate of the irrigation requirements of the crop, and corrected in accordance with the agricultural practices of the area. The results reveal the spatial and temporal distribution of the groundwater volume abstracted and used for agriculture. This methodology has been applied in the Mancha Oriental Hydrogeological System (Spain, 7,260 km²), where abstractions for agriculture comprise more than 90% of the hydrological resources consumed. In this context, accuracies of over 95% have been obtained with a cost sixty times lower than that of traditional methods. © Springer Science+Business Media B.V. 2009.

Date 2010

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Methodology to Estimate Low Grade Enhanced Geothermal System (EGS) Reserves Using Heat Extracted from Depleted Wells in High Temperature Shale Gas Formations

Item Type Conference Paper**Author** C. Ehlig-Economides**Author** M. Nikolaou**Author** M.A. Khan

Abstract Eventually there will be low-production gas wells in the Haynesville and Eagle Ford shale formations that are no longer monetized. The objective of this study is to assess the potential for repurposing multiple transverse fracture horizontal wells (MTFHWs) in these formations as low-grade enhanced/engineered geothermal systems. We use DrillingInfo and other available data, including formation temperature, to categorize existing gas wells as active, depleted, and still on production or abandoned, and to estimate an abandonment time for those still on production. Once the categorization is done, further study applies machine learning to quantify the current and future EGS potential of these wells. Finally, the results from the preceding studies are used to evaluate how best to distribute power plants for effective use of these resources, and to estimate how much power generation is possible. The result of this research is a methodology for

forecast of EGS power generation potential of low-grade EGS reserves in the Eagle Ford and Haynesville shale gas formations. This forecast addresses both technical and economic aspects. In effect, this work also provides a methodology for assessing the economic potential of EGS from existing MTFHWs in other formations having sufficiently high temperature for low-grade geothermal heat extraction. This research provides input to support potential EGS development in Texas that could provide clean and affordable continuous (unfluctuating) power for small municipalities or industrial applications and/or as a contribution to the regional grid power supply. Copyright 2022, Society of Petroleum Engineers.

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Extra Journal Abbreviation: SPE Annu. Tech. Conf. Proc.

Publisher Society of Petroleum Engineers (SPE)

ISBN 978-161399859-5 (ISBN)

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Conference Name SPE Annual Technical Conference Proceedings

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Export Date: 20 September 2026; Cited By: 0; Conference name: 2022 SPE Annual Technical Conference and Exhibition, ATCE 2022; Conference date: 3 October 2022 through 5 October 2022; Conference code: 213273

Methodology to Estimate Low Grade Enhanced Geothermal System (EGS) Reserves Using Heat Extracted from Depleted Wells in High Temperature Shale Gas Formations

Item Type Conference Paper

Author C. Ehlig-Economides

Author M. Nikolaou

Author M.A. Khan

Abstract Eventually there will be low-production gas wells in the Haynesville and Eagle Ford shale formations that are no longer monetized. The objective of this study is to assess the potential for repurposing multiple transverse fracture horizontal wells (MTFHWs) in these formations as low-grade enhanced/engineered geothermal systems. We use DrillingInfo and other available data, including formation temperature, to categorize existing gas wells as active, depleted, and still on production or abandoned, and to estimate an abandonment time for those still on production. Once the categorization is done, further study applies machine learning to quantify the current and future EGS potential of these wells. Finally, the results from the preceding studies are used to evaluate how best to distribute power plants for effective use of these resources, and to estimate how much power generation is possible. The result of this research is a methodology for forecast of EGS power generation potential of low-grade EGS reserves in the Eagle Ford and Haynesville shale gas formations. This forecast addresses both technical and economic aspects. In effect, this work also provides a methodology for assessing the economic potential of EGS from existing MTFHWs in other formations having sufficiently high temperature for low-grade geothermal heat extraction. This research provides input to support potential EGS development in Texas that could provide clean and affordable continuous (unfluctuating) power for small municipalities or industrial applications and/or as a contribution to the regional grid power supply. Copyright © 2022, Society of Petroleum Engineers.

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Methods for Predicting Multichannel Images Classification Efficiency

Item Type Conference Paper

Author I. Vasilyeva

Author V. Lukin

Editor Ageyev D.

Abstract A method for predicting probability characteristics for algorithms of supervised classification of multichannel images is proposed. Implementation of quasi-Bayesian image recognition strategy in conditions of incomplete a priori information about a satellite image is considered. Total weighted probability of correct class recognition is taken as a criterion of effectiveness of decision rule. Weights are interpreted as estimates of a priori probability for occurrence of precedents of the corresponding classes. The procedure for estimating a priori probabilities of classes by Monte Carlo method based on the results of fuzzy classification of samples with fixed size is described; the empirical dependence of the relative error of estimates on the sample size is given. © 2020 IEEE.

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Notes:

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Methods of basic research applied to urinary and genital systems during the human fetal period

Item Type Journal Article

Author L.A. Favorito

Author F.J.B. Sampaio

Abstract In this chapter, we review the most important techniques of fetal development and morphometry applied to translational research in pediatric urology. Fetal classification and determination of fetal age, the histologic techniques, the quantification methods, and the macroscopic techniques of fetal organ study such as injection/corrosion are demonstrated in details. © The Author(s) 2021. All rights reserved.

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Methods of measurement and evaluation of natural antioxidant capacity/activity (IUPAC technical report)

Item Type Journal Article

Author R. Apak

Author S. Gorinstein

Author V. Böhm

Author K.M. Schaich

Author M. Özyürek

Author K. Güçlü

Abstract The chemical diversity of natural antioxidants (AOXs) makes it difficult to separate, detect, and quantify individual antioxidants from a complex food/biological matrix. Moreover, the total antioxidant power is often more meaningful to evaluate health beneficial effects because of the cooperative action of individual antioxidant species. Currently, there is no single antioxidant assay for food labeling because of the lack of standard quantification methods. Antioxidant assays may be broadly classified as the electron transfer (ET)- and hydrogen atom transfer (HAT)-based assays. The results obtained are hardly comparable because of the different mechanisms, redox potentials, pH and solvent dependencies, etc. of various assays. This project will aid the identification and quantification of properties and mutual effects of antioxidants, bring a more rational basis to the classification of antioxidant assays with their constraints and challenges, and make the results more comparable and understandable. In this regard, the task group members convey their own experiences in various methods of antioxidants measurement. © 2013 IUPAC.

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EID: 2-s2.0-84877251040

Export Date: 20 September 2026; Cited By: 521; Correspondence Address: R. Apak; Department of Chemistry, Istanbul University, Avcilar, 34320 Istanbul, Turkey; email: rapak@istanbul.edu.tr; CODEN: PACHA

MHFNet: Multimodal hybrid fusion framework for misaligned SAR-Optical ship detection

Item Type Journal Article

Author W. Zhang

Author X. Zhang

Author X. Xu

Author S. Wei

Author J. Shi

Author Y. Wang

Author T. Zeng

Abstract Multimodal ship detection using optical and synthetic aperture radar (SAR) imagery is highly desirable for robust maritime monitoring. However, most existing fusion-based methods assume spatially aligned optical–SAR pairs, an assumption that rarely holds in practice due to differences in sensor geometry, acquisition timing, and registration errors. Such misalignment severely undermines the effectiveness of conventional fusion strategies. To address this challenge, we propose MHFNet, an alignment-aware hybrid fusion framework that systematically integrates improvements at the feature, loss, and decision levels. First, we introduce a Feature Alignment and Fusion Module (FAFM) that leverages deformable convolution and content-guided attention to achieve robust cross-modal feature alignment. Second, we design an Uncertainty-Aware Shape IoU Loss (US-Loss) that downweights unreliable training samples caused by cross-modal label shift while enhancing bounding box regression with shape-aware constraints. Third, we develop a Distance Matching Probabilistic Fusion (DMP-Fusion) algorithm that resolves inconsistencies in multimodal detection outputs through distance-based matching and Bayesian inference. Extensive experiments conducted on two benchmark datasets (QXS and Suez) show that MHFNet consistently outperforms state-of-the-art methods, improving average precision by up to + 4.77 % and F1-score by + 4.93 %. These results demonstrate that MHFNet offers a robust and generalizable solution to multimodal ship detection under realistic cross-modal misalignment conditions. The code and datasets are released on <https://github.com/wsdmw1/MHFNet>. © 2025 International Society for Photogrammetry and Remote Sensing, Inc. (ISPRS)

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Micro-flow imaging for in-situ and real-time enumeration and identification of microplastics in water

Item Type Journal Article

Author N. Chorán

Author B. Örmeci

Abstract Microplastics (MPs) are emerging contaminants that have recently gained global attention. Current identification and quantification methods are known to be time-consuming, labor-intensive, and lack consensus on protocol standardization. This study explored the potential of micro-flow imaging (MFI) technology for rapid and in-situ identification and enumeration of MPs in water using two (2) MFI-based particle counters. Advantages, limitations, and recommendations for using MFI for MPs analysis were discussed. MPs with diverse physical (i.e., microbeads, fragments, fibers, and films) and surface (i.e., reflectivity, microporosity, color) characteristics were analyzed to understand the detection capabilities and limitations of MFI technology. Results demonstrated that MFI effectively automates most manually obtained particle features, such as size, color, object intensity and shape descriptors. It imparts consistency and reduces the subjective nature of results, thus enabling reliable comparison of the generated data. The particles can be further categorized

based on their circularity and aspect ratio providing further insight into the shape and potential erosion of MPs in the environment. Transparent particles, often missed with other techniques such as microscopy, were detected by the MFI technology. The ability to assign particle IDs to MPs was an important advantage of the MFI technology that enabled the further investigation of selected MPs of interest. The limitations of the MFI technology were apparent in samples with high particle concentrations, with reflective MPs, and in the presence of bubbles. The color of the background against which the image was captured also influenced the detection accuracy. Procedural modifications during sample analysis and improvements in image analysis can assist in overcoming these challenges. MFI requires minimal sample preparation and gives real-time imaging data, making it a prime candidate for field monitoring in surface water systems in addition to laboratory analysis. With the potential application of machine learning and similar developments in the future, MFI-based particle counters are well-positioned to meet an important need in in-flow and real-time identification and enumeration of MPs. Copyright © 2023 Choran and Örmeci.

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Microplastic incorporation into soil aggregates of arable land

Item Type Journal Article

Author M. Groß

Author M. Braun

Author J. Leonhardt
Author C. Bogner
Author R. Debastiani
Author L. Hennig
Author R. Hurley
Author M. Mail
Author V. Martínez-Hernández
Author L. Nizzetto
Author R. Roscher
Author P.E. Redondo-Hasselerharm
Author T. Scherer
Author V. Schlierenkamp
Author S. Selonon
Author H. Soinne
Author W. Amelung

Abstract Agricultural mulch films are widely applied to increase crop productivity but are a significant source of microplastics (MPs) in the environment. We investigated the incorporation of conventional and biodegradable MPs into soil aggregates from agricultural fields in Finland, Germany and Spain, hypothesising that (i) MPs become increasingly occluded within soil aggregates over time, (ii) the extent of occlusion differs between conventional and biodegradable MPs, and (iii) occluded MPs differ in morphology and surface characteristics from free MPs. Arable fields were spiked with MPs derived from either a biodegradable polybutylene adipate terephthalate-starch blend (PBAT/starch) or conventional linear low-density polyethylene (LLDPE) mulch films. Soil samples were collected after two barley-growing seasons, separated into free and occluded fractions, and analysed using digital microscopy and deep learning to quantify MP abundance, size and shape. Surface properties were assessed using scanning electron microscopy. Total MP abundance declined between cropping seasons for both polymer types, while the proportion of aggregate-occluded MPs increased, reaching 76% for PBAT/starch and 65% for LLDPE. Soil exposure altered the size, shape and surface properties of MPs in a polymer-specific manner. Occluded PBAT/starch MPs were larger and more solid than their free counterparts, consistent with protection against weathering and fragmentation within aggregates. Occluded LLDPE MPs were smaller than free particles but also exhibited higher solidity, likely reflecting preferential incorporation of particles with smoother, more uniform edges. We conclude that soil aggregate occlusion is a key mechanism governing MP fate in agricultural soils, promoting particle stability while simultaneously reducing mobility and bioavailability. © 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license. <http://creativecommons.org/licenses/by/4.0/>

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Microplastics in Different Environmental Matrices: Co-Contaminants and its Monitoring Techniques

Item Type Journal Article

Author A. Kumar

Author K. Ram

Abstract Microplastics (MPs) are tiny plastic particles ranging from 1 to 5000 µm in size and are of particular concern nowadays because of their ecological risk & health concerns. MPs are prevalent environmental contaminants and have been reported in various ecosystems such as soil, air, water, and road dust. However, a thorough understanding of their distribution across various environmental matrices remains elusive. Therefore, this review aims to fill this knowledge gap by examining the abundance of MPs and different techniques utilized for their identification and characterization including sampling, extraction, identification, characterization, and quantification methods. MPs possess a large surface area and exhibit hydrophobic properties, providing a substrate for environmental pollutants such as Polyaromatic hydrocarbons (PAHs) and organochlorine pesticides (OCPs). The primary mechanisms through which pollutants attached to MPs are hydrophobic interactions,

electrostatic attraction, hydrogen bonding, halogen bonding, and π - π interactions. Therefore, a better understanding of these interactions is crucial for determining the fate and transport of MPs and co-contaminants in the environment. Researchers commonly deploy techniques such as stereomicroscopy and spectroscopy for the characterization and quantification of MPs. However, size and chemical properties of MPs pose a challenge in their quantification, thereby necessitating the development of standardized analytical methods. Use of bio-indicators for MPs biomonitoring has gained popularity in recent years as an affordable option over conventional techniques however research in this area is limited. With this intention, the present review is designed to provide information about the status of the field and advance our knowledge of MPs in various environmental matrices. © The Author(s), under exclusive licence to Springer Nature Switzerland AG 2024.

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Microplastics in terrestrial ecosystems: Soil contamination and its implications for pedology

Item Type Journal Article**Author** C.B. Andrade**Author** M.R. Francelino**Author** A. Ten Caten

Abstract The environmental persistence of plastics, their slow degradation and constant fragmentation pose risks to soil health. Microplastic (MP) contamination is already a real global environmental issue, with increasing evidence of its widespread presence in all terrestrial ecosystems, affecting physical and chemical properties, microbial communities, and pedogenetic processes. Soil serves as a significant sink for MPs, primarily through agricultural practices, landfills, waste mismanagement, wastewater treatment and atmospheric deposition. Furthermore, plastics can contribute to the formation of "rocks" (e.g., plastiglomerates and antropoquinas) that could redefine soil classification systems. Existing classification frameworks, including the World Reference Base for Soil Resources, Australian Soil Classification, U.S. Soil Taxonomy, and Brazilian Soil Classification System (SiBCS), differ in how they address the presence of plastics, highlighting a lack of consensus in recognizing anthropogenic artifacts within soils. Despite its increasing prevalence, plastics are often overlooked in soil classification systems, prompting the need for new qualifiers. Furthermore, advancements in detection and quantification methods, such as Soil Spectroscopy and Digital Soil Mapping, are essential to improve our understanding of MP behavior in soil matrices. Pedologists play a crucial role in quantifying MP impacts, developing soil management strategies, and contributing to global environmental policy. Addressing soil MP pollution requires international cooperation, transdisciplinary research, and robust methodological approaches to safeguard soil ecosystems and human health. Research on soil MP pollution has the potential to alter our understanding of human influence on soil properties and functions, with implications for the genesis and classification of Anthrosols. This review highlights important aspects of this phenomenon, with a greater focus on the Brazilian context. Soil scientists are encouraged to pay attention to this matter and contribute to filling current knowledge gaps. © 2026, Revista Brasileira de Ciencia do Solo. All rights reserved.

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MicroRNA expression signature in gastrointestinal stromal tumour & their molecular & histological features

Item Type Journal Article

Author A. Ghani

Author H. Singh

Author H. Kumar

Author K. Vaiphei

Abstract Background & objectives: In gastrointestinal stromal tumour (GIST), not only genetic abnormalities are responsible for adverse clinical events, but epigenetic modifications also play a crucial role. MicroRNA (miRNA) dysregulation plays a significant role in carcinogenesis as miRNAs serve as natural silencer for their targets. Our study aimed to explore the miRNAs expression and its association with molecular and histopathological characteristics of GIST. Methods: Fifty GIST samples, including 45 formalin fixed paraffin embedded (FFPE) and fresh tissues were included. Peripheral non-tumour tissues were used as controls. All the cases were confirmed using immunohistochemistry. RNA was extracted using miRNA-specific kit, and the expression was performed using RT-qPCR. The data were evaluated using AriaMx software version 1.5 (Agilent, US). MiRNAs expression was analyzed by using the relative quantification method ($\Delta\Delta CT$). Results: miR-221, miR-222, miR-494 and miR-34a showed significant down-regulation in tumours relative to non-tumour tissues. The expression levels of these miRNAs were significantly down-regulated in c-KIT (proto-oncogene encoding the tyrosine kinase transmembrane receptor)-positive tumours compared to c-KIT-negative. Further analysis revealed that reduced expression was associated with spindle subtypes and gastric localization. However,

there was no significant correlation with other histological features. Additionally, miR-221/222, and miR-494 were down-regulated in most of the KIT exon 11 mutant subtypes, while miRNA-34a was associated with platelet derived growth factor receptor alpha (PDGFRA) mutations. Interpretation & conclusions: The present study showed that the down-regulation of these miRNAs may help better molecular classification and characterization of GISTs. Our results offer new insight into the association between miRNAs and histological features, enabling a more thorough understanding of GISTs at the molecular level. © 2024, Indian Journal of Medical Research.

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Microvascular Metrics on Diabetic Retinopathy Severity: Analysis of Diabetic Eye Images from Real-World Data

Item Type Journal Article

Author C. Cuscó

Author P. Esteve-Bricullé

Author A. Almazán-Moga

Author J. Fernández-Carneado

Author B. Ponsati

Abstract Objective: To quantify microvascular lesions in a large real-world data (RWD) set, based on single central retinal fundus images of diabetic eyes from different origins, with the aim of validating its use as a precision tool for classifying diabetic retinopathy (DR) severity. Design: Retrospective meta-analysis across multiple fundus image datasets. Sample size: The study analyzed 2445 retinal fundus images from diabetic patients across four diverse RWD international datasets, including populations from Spain, India, China and the US. Intervention: The quantification of specific microvascular lesions: microaneurysms (MAs), hemorrhages (Hmas) and hard exudates (HEs) using advanced automated image analysis techniques on central retinal images to validate reliable metrics for DR severity assessment. The images were pre-classified in the DR severity levels as defined by the International Clinical Diabetic Retinopathy (ICDR) scale. Main Outcome Measures: The primary variables measured were the number of MAs, Hmas, red lesions (RLs) and HEs. These counts were related with DR severity levels using statistical methods to validate the relationship between lesion counts and disease severity. Results: The analysis revealed a robust and statistically significant increase ($p < 0.001$) in the number of microvascular lesions and the DR severity across all datasets. Tight data distributions were reported for MAs, Hmas and RLs, supporting the reliability of lesion quantification for accurately assessing DR severity. HEs also followed a similar pattern, but with a broader dispersion of data. Data used in this study are consistent with the definition of the DR severity levels established by the ICDR guidelines. Conclusions: The statistically significant increase in the number of microvascular lesions across DR severity validate the use of lesion quantification in a single central retinal field as a key biomarker for disease classification and assessment. This quantification method demonstrates an improvement over traditional assessment scales, providing a quantitative microvascular metric that enhances the precision of disease classification and patient monitoring. The inclusion of a numerical component allows for the detection of subtle variations within the same severity level, offering a deeper understanding of disease progression. The consistency of results across diverse datasets not only confirms the method's reliability but also its applicability in a global healthcare setting. © 2024 by the authors.

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Mid-infrared spectra of dried and roasted cocoa (*Theobroma cacao* L.): A dataset for machine learning-based classification of cocoa varieties and prediction of theobromine and caffeine content

Item Type Journal Article**Author** G.A. Collazos-Escobar**Author** A.F. Bahamón-Monje**Author** N. Gutiérrez-Guzmán

Abstract This paper presents a comprehensive dataset of mid-infrared spectra for dried and roasted cocoa beans (*Theobroma cacao* L.), along with their corresponding theobromine and caffeine content. Infrared data were acquired using Attenuated Total Reflectance-Fourier Transform Infrared (ATR-FTIR) spectroscopy, while High-Performance Liquid Chromatography (HPLC) was employed to accurately quantify theobromine and caffeine in the dried cocoa beans. The theobromine/caffeine relationship served as a robust chemical marker for distinguishing between different cocoa varieties. This dataset provides a basis for further research, enabling the integration of mid-infrared spectral data with HPLC (as a standard) to fine-tune machine learning and deep learning models that could be used to simultaneously predict the theobromine and caffeine content, as well as cocoa variety in both dried and roasted cocoa samples using a non-destructive approach based on spectral data. The tools developed from this dataset could significantly advance automated processes in the cocoa industry and support decision-making on an industrial scale, facilitating real-time quality control of cocoa-based products, improving cocoa variety classification, and optimizing bean selection, blending strategies, and product formulation, while reducing the need for labor-intensive and costly quantification methods. The dataset is organized into Excel sheets and structured according to experimental conditions and replicates, providing a valuable framework for further analysis, model development, and calibration of multivariate statistical models. © 2024 The Author(s)

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Middleware Systems for Efficient and Robust Distributed Machine Learning

Item Type Conference Paper

Author R.A. Bhope

Abstract Distributed machine learning faces fundamental challenges arising from heterogeneous data, shifting client populations, and volatile inference workloads. This dissertation advances the thesis that middleware - not the models themselves - must serve as an active, adaptive decision layer that unifies training-time and inference-time robustness. On the training side, FLIPS performs privacy-preserving participant selection via label-distribution clustering within TEEs, improving fairness and convergence under non-IID data, while ShiftEx detects covariate and label shifts in streaming environments and dynamically creates or reuses experts to accelerate adaptation with minimal retraining. On the inference side, OptiSeq improves the reliability of LLM-based API sequencing by selecting example orderings using inference-time log-probability reasoning, and VOILA formulates multimodal fidelity selection as a value-of-information optimization,

achieving large cost reductions with calibrated accuracy guarantees. Together, these systems establish adaptive middleware as a unifying approach for improving efficiency, robustness, and accuracy across modern distributed ML deployments. © 2025 Copyright held by the owner/author(s).

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Migraine and white matter hyperintensities: Protocol for a systematic review and meta-analysis

Item Type Journal Article

Author C. Gollion

Author C.-C. Chiang

Author M.J. Lee

Author R. Dominguez-Moreno

Author C.J. Schankin

Author A. Harriott

Author L.M. Sanders

Author M. Romoli

Author C.D. Chong

Author E.S. Kristoffersen

Author A. Hougaard

Abstract White matter hyperintensities (WMHs) are common MRI findings and established markers of cerebral small-vessel disease associated with stroke, cognitive decline, and vascular risk factors. Several studies have suggested that individuals with migraine, particularly migraine with aura (MA), may have an increased burden of WMHs compared with individuals without migraine. Early population-based studies and meta-analyses reported a higher prevalence of WMHs in migraine, whereas more recent large neuroimaging cohorts have reported conflicting findings or no association. Differences in study design, migraine ascertainment, MRI acquisition parameters, WMH quantification methods, and adjustment for vascular risk factors complicate interpretation of the existing literature. An updated systematic review focusing on methodologically comparable population-based cohorts is therefore warranted. The primary objective of this systematic review and meta-analysis is to determine whether adults with migraine have a higher prevalence or burden of WMHs on MRI than non-migraine controls. Secondary objectives include evaluating associations according to migraine subtype (MA vs. migraine without aura and episodic vs. chronic migraine), sex, WMH location, age, and MRI acquisition or WMH quantification methods, as well as examining longitudinal WMH progression where available. We will include population-based cross-sectional or longitudinal cohort studies of adults undergoing brain MRI with sequences suitable for WMH detection. Migraine exposure may be defined using International Classification of Headache Disorders criteria, structured interviews, validated questionnaires, physician diagnosis, or ICD codes. Controls must originate from the same source population. Searches will be conducted in PubMed/MEDLINE, EMBASE, Web of Science, and Scopus. Study selection, data extraction, and risk-of-bias assessment will be performed independently by two reviewers using Covidence, and risk of bias will be evaluated using the ROBINS-E tool. Because WMH burden is reported using heterogeneous metrics across studies, the primary synthesis will use a precision-weighted direction-of-effect approach, in which each study contributes the direction of the association, weighted by statistical precision. Where sufficiently comparable outcomes are available, random-effects meta-analysis will be performed. Sensitivity analyses will explore the influence of diagnostic criteria, MRI field strength, and confounder adjustment. This review will provide an updated and methodologically rigorous synthesis of the evidence on the association between migraine and WMHs. © The Author(s) 2026. This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access page (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A. Hougaard; Department of Neurology, Copenhagen University Hospital – Herlev and Gentofte, Herlev, Borgmester Ib Juuls Vej 1, DK-2730, Denmark; email: ahougaard@dadlnet.dk

Mind the class weight bias: Weighted maximum mean discrepancy for unsupervised domain adaptation

Item Type Conference Paper

Author H. Yan

Author Y. Ding

Author P. Li

Author Q. Wang

Author Y. Xu

Author W. Zuo

Abstract In domain adaptation, maximum mean discrepancy (MMD) has been widely adopted as a discrepancy metric between the distributions of source and target domains. However, existing MMD-based domain adaptation methods generally ignore the changes of class prior distributions, i.e., class weight bias across domains. This remains an open problem but ubiquitous for domain adaptation, which can be caused by changes in sample selection criteria and application scenarios. We show that MMD cannot account for class weight bias and results in

degraded domain adaptation performance. To address this issue, a weighted MMD model is proposed in this paper. Specifically, we introduce class-specific auxiliary weights into the original MMD for exploiting the class prior probability on source and target domains, whose challenge lies in the fact that the class label in target domain is unavailable. To account for it, our proposed weighted MMD model is defined by introducing an auxiliary weight for each class in the source domain, and a classification EM algorithm is suggested by alternating between assigning the pseudo-labels, estimating auxiliary weights and updating model parameters. Extensive experiments demonstrate the superiority of our weighted MMD over conventional MMD for domain adaptation. © 2017 IEEE.

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Mind the Gap: Bridging Prior Shift in Realistic Few-shot Crop-type Classification

Item Type Conference Paper

Author J. Reuss
Author E. Gikalo
Author M. Körner
Editor Polidori L.
Editor Gomes A.
Editor Faure J.
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Editor Lichti D.
Editor Jabari S.

Abstract Real-world agricultural distributions often suffer from severe class imbalance, typically following a long-tailed distribution. Labeled datasets for crop-type classification are inherently scarce and remain costly to obtain. When working with such limited data, training sets are frequently constructed to be artificially balanced-in particular in the case of few-shot learning-failing to reflect real-world conditions. This mismatch induces a shift between training and test label distributions, degrading real-world generalization. To address this, we propose Dirichlet Prior Augmentation (DirPA), a novel method that simulates an unknown label distribution skew of the target domain proactively during model training. Specifically, we model the real-world distribution as Dirichlet-distributed random variables, effectively performing a prior augmentation during few-shot learning. Our experiments show that DirPA successfully shifts the decision boundary and stabilizes the training process by acting as a dynamic feature regularizer. © Copyright: 2026 Joana Reuss et al.

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Minimax classifiers based on neural networks

Item Type Journal Article**Author** R. Alaiz-Rodríguez**Author** A. Guerrero-Curieses**Author** J. Cid-Sueiro

Abstract The problem of designing a classifier when prior probabilities are not known or are not representative of the underlying data distribution is discussed in this paper. Traditional learning approaches based on the assumption that class priors are stationary lead to sub-optimal solutions if there is a mismatch between training and future (real) priors. To protect against this uncertainty, a minimax approach may be desirable. We address the problem of designing a neural-based minimax classifier and propose two different algorithms: a learning rate scaling algorithm and a gradient-based algorithm. Experimental results show that both succeed in finding the minimax solution and it is also pointed out the differences between common approaches to cope with this uncertainty in priors and the minimax classifier. © 2004 Pattern Recognition Society. Published by Elsevier Ltd. All rights reserved.

Date 2005**DOI** 10.1016/j.patcog.2004.05.007**URL** <https://www.scopus.com/pages/publications/4644273126?origin=resultslist>**Archive** Scopus**Language** English**Volume** 38**Pages** 29-39**Publication** Pattern Recognition**ISSN** 00313203 (ISSN)**Issue** 1**Journal Abbr** Pattern Recogn.**Date Added** 20/09/2026, 9:53:05 pm**Modified** 20/09/2026, 9:53:05 pm**Tags:**

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Export Date: 20 September 2026; Cited By: 11; Correspondence Address: R. Alaiz-Rodríguez; Depto. de Ing. Eléctrica, Esc. de Ingenierías Indust./Info., Universidad de León, 24071 León, Spain; email: dierar@unileon.es; CODEN: PTNRA

Minimax confidence intervals for the Sliced Wasserstein distance***Item Type** Journal Article**Author** T. Manole**Author** S. Balakrishnan**Author** L. Wasserman

Abstract Motivated by the growing popularity of variants of the Wasserstein distance in statistics and machine learning, we study statistical inference for the Sliced Wasserstein distance—an easily computable variant of the Wasserstein distance. Specifically, we construct confidence intervals for the Sliced Wasserstein distance which have finite-sample validity under no assumptions or under mild moment assumptions. These intervals are adaptive in length to the regularity of the underlying distributions. We also bound the minimax risk of estimating the Sliced Wasserstein distance, and as a consequence establish that the lengths of our proposed confidence intervals are minimax optimal over appropriate distribution classes. To motivate the choice of these classes, we also study minimax rates of estimating a distribution under the Sliced Wasserstein distance. These theoretical findings are complemented with a simulation study demonstrating the deficiencies of the classical bootstrap, and the advantages of our proposed methods. We also show strong correspondences between our theoretical predictions and the adaptivity of our confidence interval lengths in simulations. We conclude by demonstrating the use of our confidence intervals in the setting of simulator-based likelihood-free inference. In this setting, contrasting popular approximate Bayesian computation methods, we develop uncertainty quantification methods with rigorous frequentist coverage guarantees. © 2022, Institute of Mathematical Statistics. All rights reserved.

Date 2022**DOI** 10.1214/22-EJS2001**URL** <https://www.scopus.com/pages/publications/85128580385?origin=resultslist>**Archive** Scopus**Language** English**Volume** 16**Publisher** Institute of Mathematical Statistics**Pages** 2252-2345**Publication** Electronic Journal of Statistics**ISSN** 19357524 (ISSN)**Issue** 1**Journal Abbr** Electron. J. Stat.**Date Added** 20/09/2026, 9:52:16 pm**Modified** 20/09/2026, 9:52:16 pm

Tags:

Optimal transport, likelihood-free inference, minimax lower bound, nonparametric inference, Sliced Wasserstein distance

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Minimax optimal approaches to the label shift problem in non-parametric settings

Item Type Journal Article

Author S. Maity

Author Y. Sun

Author M. Banerjee

Abstract We study the minimax rates of the label shift problem in non-parametric classification. In addition to the unsupervised setting in which the learner only has access to unlabeled examples from the target domain, we also consider the setting in which a small number of labeled examples from the target domain is available to the learner. Our study reveals a difference in the difficulty of the label shift problem in the two settings, and we attribute this difference to the availability of data from the target domain to estimate the class conditional distributions in the latter setting. We also show that a class proportion estimation approach is minimax rate-optimal in the unsupervised setting. ©2022 Subha Maity, Yuekai Sun, and Moulinath Banerjee.

Date 2022

URL <https://www.scopus.com/pages/publications/85148050977?origin=resultslist>

Archive Scopus

Language English

Volume 23

Publisher Microtome Publishing

Publication Journal of Machine Learning Research

ISSN 15324435 (ISSN)

Journal Abbr J. Mach. Learn. Res.

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Tags:

Supervised learning, Transfer learning, Optimization, transfer learning, Binary classification, Target domain, Conditional distribution, Minimax, Nonparametrics, Semisupervised classification (SSC), Minimax rates, non-parametric classification, Nonparametric classification, Optimal approaches, semi-supervised classification

Notes:

EID: 2-s2.0-85148050977

Export Date: 20 September 2026; Cited By: 14

Minimax strategies for training classifiers under unknown priors

- Item Type** Conference Paper
- Author** R. Alaiz-Rodríguez
- Author** J. Cid-Sueiro
- Editor** Bourlard H.
- Editor** Adali T.
- Editor** Bengio S.
- Editor** Larsen J.
- Editor** Douglas S.
- Abstract** Most supervised learning algorithms are based on the assumption that the training data set reflects the underlying statistical model of the real data. However, this stationarity assumption is not always satisfied in practice: quite frequently, class prior probabilities are not in accordance with the class proportions in the training data set. The minimax approach is based on selecting the classifier that minimize the error probability under the worst case conditions. We propose a two-step learning algorithm to train a neural network in order to estimate the minimax classifier that is robust to changes in the class priors. During the first step, posterior probabilities based on training data priors are estimated. During the second step, class priors are modified in order to minimize a cost function that is asymptotically equivalent to the worst-case error rate. This procedure is illustrated on a softmax-based neural network. Several experimental results show the advantages of the proposed method with respect to other approaches. © 2002 IEEE.
- Date** 2002
- DOI** 10.1109/NNSP.2002.1030036
- URL** <https://www.scopus.com/pages/publications/84953710219?origin=resultslist>
- Archive** Scopus
- Language** English
- Extra** Journal Abbreviation: Neural Networks Signal Process Proc IEEE
- Volume** 2002-January
- Publisher** Institute of Electrical and Electronics Engineers Inc.
- ISBN** 0780376161 (ISBN)
- Pages** 249-258

Proceedings Title Neural Networks Signal Process Proc IEEE
Conference Name Neural Networks for Signal Processing - Proceedings of the IEEE Workshop
Date Added 20/09/2026, 9:53:04 pm
Modified 20/09/2026, 9:53:04 pm

Tags:

Condition, Neural-networks, Adversarial machine learning, Contrastive Learning, Self-supervised learning, Training dataset, Prior probability, Minimax, Statistic modeling, Cost functions, Error probabilities, Minimax approach, Minimax strategy, Stationarity

Notes:

EID: 2-s2.0-84953710219

Export Date: 20 September 2026; Cited By: 0; Conference name: 12th IEEE Workshop on Neural Networks for Signal Processing, NNSP 2002; Conference date: 6 September 2002 through 6 September 2002; Conference code: 116302; CODEN: 85QHA

Minimising quantifier variance under prior probability shift

Item Type Conference Paper

Author D. Tasche

Editor Cong G.

Editor Ramanath M.

Abstract For the binary prevalence quantification problem under prior probability shift, we determine the asymptotic variance of the maximum likelihood estimator. We find that it is a function of the Brier score for the regression of the class label on the features under the test data set distribution. This observation suggests that optimising the accuracy of a base classifier, as measured by the Brier score, on the training data set helps to reduce the variance of the related quantifier on the test data set. Therefore, we also point out training criteria for the base classifier that imply optimisation of both of the Brier scores on the training and the test data sets. © 2021 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0). CEUR Workshop Proceedings (CEUR-WS.org)

Date 2021

URL <https://www.scopus.com/pages/publications/85122885299?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: CEUR Workshop Proc.

Volume 3052

Publisher CEUR-WS

ISBN 16130073 (ISSN)

Proceedings Title CEUR Workshop Proc.

Conference Name CEUR Workshop Proceedings

Date Added 20/09/2026, 9:52:23 pm**Modified** 20/09/2026, 9:52:23 pm**Tags:**

Classification (of information), Statistical tests, Data set, Test data, Probability shifts, Prior probability, Prior probability shift, Class distribution estimation, Maximum likelihood estimation, Brier score, Crame Rao bounds, Cramér-Rao bound, Cramer-Rao bounds, Maximum likelihood estimator, Quantifier

Notes:

EID: 2-s2.0-85122885299

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: D. Tasche; Independent Researcher, Zurich, Switzerland; email: dirk.tasche@gmx.net; Conference name: 2021 International Conference on Information and Knowledge Management Workshops, CIKMW 2021; Conference date: 1 November 2021 through 5 November 2021; Conference code: 175585

Minimizing error in measurement of error: A proposed method for calculation of error in a two-dimensional motor task

Item Type Journal Article**Author** J. Kim**Author** S. Chung**Author** L.K. Tennant**Author** R.N. Singer**Author** C.M. Janelle

Abstract Traditional one-dimensional error scores are still consistently used in research on motor learning to quantify two-dimensional error; however, the inherent differences in two-dimensional tasks rejider that application inappropriate and often misleading. Consequently, the purpose of this paper was to propose a novel method of presenting errors, which more precisely represents the accuracy, direction, and variability of error in two-dimensional settings. Although closely related to several alternatives for representing errors, the methodology used and the results obtained provide a more accurate procedure for pinpointing critical trends in what have been commonly referred to as AE (absolute error), VE (variable error), CE (constant error), and E (total variability). The proposed measurements of AVTZ (adjusted variable error), DE (directional error), TSE (total spread of error), and RE (radial error) provide composite error scores carrying a variety of information about performance on two-dimensional tasks. Formulas and examples are provided to facilitate computation and enhance understanding of the proposed scores.

Date 2000**DOI** 10.2466/pms.2000.90.1.253**URL** <https://www.scopus.com/pages/publications/0034136322?origin=resultslist>**Archive** Scopus

Language English
Volume 90
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Publication Perceptual and Motor Skills
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Journal Abbr Percept. Mot. Skills
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human, Humans, article, Statistics, classification, physiology, statistics, task performance, motor performance, Task Performance and Analysis, Motor Skills

Notes:

EID: 2-s2.0-0034136322

Export Date: 20 September 2026; Cited By: 8; Correspondence Address: R.N. Singer; University of Florida, Dept. of Exercise and Sport Sciences, 100 Florida Gym, Gainesville, FL 32611, United States; email: rsinger@hhp.ufl.edu; CODEN: PMOSA

Minimizing Estimated Risks on Unlabeled Data: A New Formulation for Semi-Supervised Medical Image Segmentation

Item Type Journal Article

Author F. Wu

Author X. Zhuang

Abstract Supervised segmentation can be costly, particularly in applications of biomedical image analysis where large scale manual annotations from experts are generally too expensive to be available. Semi-supervised segmentation, able to learn from both the labeled and unlabeled images, could be an efficient and effective alternative for such scenarios. In this work, we propose a new formulation based on risk minimization, which makes full use of the unlabeled images. Different from most of the existing approaches which solely explicitly guarantee the minimization of prediction risks from the labeled training images, the new formulation also considers the risks on unlabeled images. Particularly, this is achieved via an unbiased estimator, based on which we develop a general framework for semi-supervised image segmentation. We validate this framework on three medical image segmentation tasks, namely cardiac segmentation on ACDC2017, optic cup and disc segmentation on REFUGE dataset and 3D whole heart segmentation on MM-WHS dataset. Results show that the proposed estimator is effective, and the segmentation method achieves superior performance and demonstrates great potential compared to the other state-of-the-art approaches. Our code and data will be released via <https://zmiclalab.github.io/projects.html>, once the manuscript is accepted for publication. © 1979-2012 IEEE.

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Publication IEEE Transactions on Pattern Analysis and Machine Intelligence
ISSN 01628828 (ISSN)
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Journal Abbr IEEE Trans Pattern Anal Mach Intell
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Tags:

Risk perception, Risk assessment, Risk management, Deep learning, Task analysis, Risks management, Heart, Image segmentation, Supervised learning, Risk analysis, algorithm, Algorithms, Images segmentations, Medical imaging, deep learning, Biomedical imaging, Semi-supervised learning, Image analysis, image processing, diagnostic imaging, Image Processing, Computer-Assisted, Medical image segmentation, Prior probability, Risk minimization, class-prior probability, Class-prior probability, heart, medical image segmentation, Minimisation, risk minimization

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Minimum-Risk Recalibration of Classifiers

Item Type Conference Paper
Author Z. Sun
Author D. Song
Author A. Hero
Editor Oh A.
Editor Neumann T.
Editor Globerson A.
Editor Saenko K.
Editor Hardt M.

Editor Levine S.

Abstract Recalibrating probabilistic classifiers is vital for enhancing the reliability and accuracy of predictive models. Despite the development of numerous recalibration algorithms, there is still a lack of a comprehensive theory that integrates calibration and sharpness (which is essential for maintaining predictive power). In this paper, we introduce the concept of minimum-risk recalibration within the framework of mean-squared-error (MSE) decomposition, offering a principled approach for evaluating and recalibrating probabilistic classifiers. Using this framework, we analyze the uniform-mass binning (UMB) recalibration method and establish a finite-sample risk upper bound of order $\tilde{O}(B/n + 1/B^2)$ where B is the number of bins and n is the sample size. By balancing calibration and sharpness, we further determine that the optimal number of bins for UMB scales with $n^{1/3}$, resulting in a risk bound of approximately $O(n^{-2/3})$. Additionally, we tackle the challenge of label shift by proposing a two-stage approach that adjusts the recalibration function using limited labeled data from the target domain. Our results show that transferring a calibrated classifier requires significantly fewer target samples compared to recalibrating from scratch. We validate our theoretical findings through numerical simulations, which confirm the tightness of the proposed bounds, the optimal number of bins, and the effectiveness of label shift adaptation. © 2023 Neural information processing systems foundation. All rights reserved.

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Extra Journal Abbreviation: Adv. neural inf. proces. syst.

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Predictive models, Sampling, Mean square error, Calibration, Recalibrations, Upper Bound, Probabilistic classifiers, Error decomposition, Finite samples, Mean squared error, Minimum risks, Optimal number, Predictive power

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Minor mining, major influence: Economic implications and policy challenges of artisanal gold mining

Item Type Journal Article

Author M. Yu

Author Y. Wang

Author M. Umair

Abstract This study presents an in-depth analysis of the economic policies and implications surrounding artisanal gold mining, reframing the discussion from geographical specifics to a broader policy and economic context. Utilizing advanced geospatial data from the Landsat image collection through “Google Earth Engine,” the research evaluates the economic impact and policy challenges posed by artisanal gold mining over a period from 2011 to 2021. The approach combines change detection methodologies with machine learning to quantify the economic repercussions of land cover changes due to mining. The analysis reveals significant economic transformations, with an average of 2700 ha of land affected annually, culminating in around 48,000 ha over the decade, highlighting a critical period from 2013 to 2020. Importantly, the study identifies policy gaps, especially in the management of protected areas where approximately 800 ha of mining occurred. The findings underscore the urgent need for policy interventions to mitigate the adverse economic and environmental impacts of unregulated mining. The study contributes to a broader understanding of the economic ramifications of artisanal mining and serves as a call to action for policymakers to address the challenges of sustainable mining practices and economic development. © 2024

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Language English

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Publication Resources Policy

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Journal Abbr Resour. Policy

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Environmental protection, Economic and social effects, Environmental impact, Decision making, environmental impact, Economic analysis, In-depth analysis, land cover, Economic geology, spatial data, Conservation, Artisanal gold minings, artisanal mining, Artisanal mining, economic development, economic impact, Economic implications, Economic policies, economic policy, Extent mapping, Extent mappings, Geo-spatial data, Geospatial data, gold mine, Gold mines, Gold mining, Mercury, mercury (element), Mercury (metal), Mining, Reframing

Notes:

EID: 2-s2.0-85189107991

Export Date: 20 September 2026; Cited By: 117; Funding details: National Office for Philosophy and Social Sciences, NPOPSS; Correspondence Address: Y. Wang; Institute for Industrial System Modernization, Zhejiang University of Technology, Hangzhou, 310023, China; email: zjut8747@163.com

Mitigating Diagnostic Errors in Lung Cancer Classification: A Multi-Eyes Principle to Uncertainty Quantification

Item Type Journal Article

Author R. Zahari

Author J. Cox

Author B. Obara

Abstract In radiology, particularly in lung cancer diagnosis, diagnostic errors and cognitive biases pose substantial challenges. These issues, including perceptual errors, interpretive mistakes, and cognitive biases such as anchoring and premature closure, are often unnoticed by experienced radiologists. To address these challenges, we propose the Multi-Eyes principle approach, which utilises multiple deep learning models to reduce bias and potentially improve diagnostic accuracy. Inspired by the Four-Eyes principle in business and cybersecurity, this methodology employs various 3D and 2D (for validation) deep learning architectures and three uncertainty quantification techniques: Monte Carlo Dropout, Deep Ensemble, and Ensemble Monte Carlo Dropout. Each model functions as an independent reviewer, similar to blind reviews. With entropy selected as the uncertainty measurement, it is averaged, followed by ensemble averaging of predictions. The effectiveness of this approach was demonstrated using the LIDC-IDRI dataset for lung cancer classification. Statistical analysis of the uncertainty's distribution reveals that with more models, uncertainty in incorrect predictions becomes more peaked and left skewed, indicating consensus on uncertainty levels. This results in accuracy and F1 score improvements, even with the best performing model, addressing overconfidence in single-model systems. These findings highlight the potential of the Multi-Eyes principle to significantly improve diagnostic performance in computer-aided diagnostic systems. Future research may explore different uncertainty quantification methods and feedback mechanisms for further advancement. © 2013 IEEE.

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DOI 10.1109/JBHI.2024.3446040

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Issue 11

Journal Abbr IEEE J. Biomedical Health Informat.

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Article, computer security, human, Humans, prevention and control, Intelligent systems, Uncertainty, Deep Learning, algorithm, prediction, predictive model, training, deep learning, artificial neural network, Predictive models, Accuracy, nerve cell network, accuracy, Diagnosis, classification, Diseases, learning algorithm, memory, area under the curve, diagnostic test accuracy study, predictive value, diagnostic accuracy, F1 score, statistical analysis, transfer of learning, scoring system, Monte Carlo method, Monte Carlo Method, uncertainty, Uncertainty quantifications, diagnostic imaging, Monte Carlo dropout, uncertainty quantification, computer assisted tomography, discriminant analysis, lung cancer, overconfidence, Biological organs, Lung cancer, Monte carlo dropout, Radiology, Lung Neoplasms, lung tumor, diagnostic error, Three-dimensional display, Deep ensemble, Diagnostic Errors, Lung, Lung Cancer, Multi-eye, multi-eyes

Notes:

EID: 2-s2.0-85201744817

Export Date: 20 September 2026; Cited By: 2; Funding details: Government of Brunei Darussalam; Correspondence Address: B. Obara; Newcastle University, School of Computing and Biosciences Institute, Newcastle Upon Tyne, NE1 7RU, United Kingdom; email: boguslaw.obara@newcastle.ac.uk; CODEN: ITIBF

Mitigating Overfitting in Supervised Classification from Two Unlabeled Datasets: A Consistent Risk Correction Approach

Item Type Conference Paper

Author N. Lu

Author T. Zhang

Author G. Niu

Author M. Sugiyama

Editor Chiappa S.

Editor Calandra R.

Abstract The recently proposed unlabeled-unlabeled (UU) classification method allows us to train a binary classifier only from two unlabeled datasets with different class priors. Since this method is based on the empirical risk minimization, it works as if it is a supervised classification method, compatible with any model and optimizer. However, this method sometimes suffers from severe overfitting, which we would like to prevent in this paper. Our empirical finding in applying the original UU method is that overfitting often co-occurs with the empirical risk going negative, which is not legitimate. Therefore, we propose to wrap the terms

that cause a negative empirical risk by certain correction functions. Then, we prove the consistency of the corrected risk estimator and derive an estimation error bound for the corrected risk minimizer. Experiments show that our proposal can successfully mitigate overfitting of the UU method and significantly improve the classification accuracy. Copyright © 2020 by the author(s)

Date 2020

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Archive Scopus

Language English

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Volume 108

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Conference Name Proceedings of Machine Learning Research

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Tags:

Risk perception, Classification (of information), Supervised learning, Overfitting, Different class, Classification methods, Supervised classification, Binary classifiers, Empirical risk minimization, Correction approaches, Empirical findings, Empirical risks, Optimizers

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Mitigating Source Label Dependency in Time-Series Domain Adaptation under Label Shifts

Item Type Conference Paper

Author J. Na

Author Y. Nam

Author J. Kang

Author J.-G. Lee

Abstract Time-series unsupervised domain adaptation (TS-UDA) is essential in fields such as healthcare and manufacturing, where data often consists of distinct entities, such as individual patients or machinery. This heterogeneity leads to

discrepancies not only in feature distributions but also in label distributions, posing a significant challenge for domain adaptation. However, prior studies have mostly focused on alleviating covariate shifts, resulting in predicted target labels that are often biased toward the source domain's label distribution. To address this issue, we propose Time-series domain Adaptation for mitigating Label Shifts (TA4LS), a novel label refinement approach. TA4LS leverages the consistency between predicted labels and clustering information obtained from the unique characteristics that differentiate each label in the target domain. Furthermore, our approach as a plug-in module achieves performance improvements across diverse existing unsupervised domain adaptation methods, particularly in scenarios with significant discrepancies between source and target label distributions. In experiments on four benchmark datasets with label shifts, TA4LS demonstrates superior performance across six unsupervised domain adaptation methods and six label shift handling modules. © 2025 Association for Computing Machinery. All rights reserved.

Date 2025

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Volume 2

Publisher Association for Computing Machinery

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Pages 2150-2161

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Conference Name Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining

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Information systems, Performance, Learning systems, Machine learning, Benchmarking, Transfer learning, Information use, Knowledge management, Medical computing, transfer learning, Domain adaptation, Target labels, Label distribution, Time series, Times series, time series, Adaptation methods, In-field, Label dependencies, unsupervised domain adaptation, Unsupervised domain adaptation

Notes:

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Mitigating Word Bias in Zero-shot Prompt-based Classifiers

Item Type Conference Paper

Author A. Liusie

Author P. Manakul

Author M.J.F. Gales

Editor Park J.C.

Editor Arase Y.

Editor Hu B.

Editor Lu W.

Editor Wijaya D.

Editor Purwarianti A.

Editor Krisnadhi A.A.

Abstract Prompt-based classifiers are an attractive approach for zero-shot classification. However, the precise choice of the prompt template and label words can largely influence performance, with semantically equivalent settings often showing notable performance difference. This discrepancy can be partly attributed to word biases, where the classifier may be biased towards classes. To address this problem, it is possible to optimise classification thresholds on a labelled data set, however, this mitigates some of the advantages of prompt-based classifiers. This paper instead approaches this problem by examining the expected marginal probabilities of the classes. Here, probabilities are reweighted to have a uniform prior over classes, in an unsupervised fashion. Further, we draw a theoretical connection between the class priors and the language models' word prior, and offer the ability to set a threshold in a zero-resource fashion. We show that matching class priors correlates strongly with the oracle upper bound performance and demonstrate large consistent performance gains for prompt settings over a range of NLP tasks. © 2023 Asian Federation of Natural Language Processing.

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Conference Name IJCNLP-AAACL 2023 - 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics, Findings of the Association for Computational Linguistics: IJCNLP-AAACL 2023
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Tags:

Language model, Performance, Classification (of information), Natural language processing systems, Labeled data, Shot classification, Zero-shot learning, Data set, Matchings, Upper Bound, Consistent performance, Marginal probability, Performance Gain

Notes:

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Mitotic HEp-2 cells recognition under class skew

Item Type Conference Paper

Author G. Percannella

Author P. Soda

Author M. Vento

Abstract Indirect immunofluorescence (IIF) is the recommended method to diagnose the presence of antinuclear autoantibodies in patient serum. A main step of the diagnostic procedure requires to detect mitotic cells in the well under examination. However, such cells rarely occur in comparison to other cells and, hence, traditional recognition algorithms fail in this task since they cannot cope with large differences between the number of samples in each class, resulting in a low predictive accuracy over the minority class. In this paper we present a system for mitotic cells recognition based on multiobjective optimisation, which is able to handle their low a priori probability. It chooses between the output of a classifier trained on the original skewed distribution and the output of a classifier trained according to a learning method addressing the course of imbalanced data. This choice is driven by a parameter whose value maximises, on a validation set, two objective functions, i.e. the global accuracy and the accuracies for each class. The approach has been evaluated on an annotated dataset of mitotic cells and successfully compared to five learning methods applying four different

classification paradigms. © 2011 Springer-Verlag.

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DOI 10.1007/978-3-642-24088-1_37

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Publisher Springer Verlag

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Issue PART 2

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Tags:

Contrastive Learning, Multiobjective optimization, Learning methods, Predictive accuracy, Recognition algorithm, Antinuclear auto-antibodies, Cell recognition, Class skew, Diagnostic procedure, Indirect immunofluorescence, Mitotic cells, Number of samples

Notes:

EID: 2-s2.0-80052973463

Export Date: 20 September 2026; Cited By: 22; Correspondence Address: G. Percannella; Dipartimento di Ingegneria dell'Informazione Ed Ingegneria Elettrica, Università di Salerno, Italy; email: pergen@unisa.it; Conference name: 16th International Conference on Image Analysis and Processing, ICIAP 2011; Conference date: 14 September 2011 through 16 September 2011; Conference code: 86563

Mitral regurgitation severity analysis based on features and optimal HE (OHE) with quantification using PISA method

Item Type Journal Article

Author P.A. Khayum

Author R.P.S. Babu

Abstract Heart disease is the foremost reason for death and also the main source of incapability in the developed nations. Mitral regurgitation (MR) is a typical heart disease that does not bring about manifestations until its end position. In view of the hidden etiologies of heart distress, functional MR can be partitioned into two subgroups, ischemic and no ischemic MR. A procedure is progressed for jet area separation and quantification in MR evaluation in arithmetical expressions. Thus, a strategy that depends on

echocardiography recordings, image processing methods, and artificial intelligence could be useful for clinicians, particularly in marginal cases. In this research paper, MR segmentation is analyzed by the optimal histogram equalization (OHE) system used to segment the jet area. For a better execution of the work, threshold in HE was improved with the help of the krill herd optimization (KHO) strategy. With the MR quantification procedure, this segmented jet area was supported by the proximal isovelocity surface area (PISA); in this procedure, a few parameters in the segmentation were evaluated. From the results, this proposed methodology accomplishes better accuracy in the segmented and quantification method in contrast with the existing examination. © 2019 De Gruyter. All rights reserved.

Date 2021

DOI 10.1515/jisys-2017-0116

URL <https://www.scopus.com/pages/publications/85037710731?origin=resultslist>

Archive Scopus

Language English

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Publication Journal of Intelligent Systems

ISSN 03341860 (ISSN)

Issue 5

Journal Abbr J Intell Syst

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Tags:

Artificial intelligence, Classification (of information), Image processing, Diseases, Classification, Graphic methods, Echocardiography, Doppler echocardiography, Quantification, histogram, quantification, Cardiology, Histogram, Dimensional stability, features, Features, Mitral regurgitation

Notes:

EID: 2-s2.0-85037710731

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: P.A. Khayum; Department of ECE, G. Pulla Reddy Engineering College, Kurnool, AP, India; email: pinjariabdulkhayum2001@gmail.com; CODEN: JISYE

Mixed Mode Latent Class Analysis: An Examination of Fit Index Performance for Classification

Item Type Journal Article

Author G.B. Morgan

Abstract This Monte Carlo study examines the performance of fit indices commonly used by applied researchers interested in finite mixture modeling for the purposes of classification. Conditions for the simulation study were selected to reflect conditions found in applied educational and psychological research. The factors included in the investigation were metric level of indicators, sample size, and class prevalence. All models contained a combination of categorical and continuous indicators. All categorical indicators were dichotomous, and continuous indicators were normally distributed. The fit indices examined were Akaike's information criterion, Bayesian information criterion (BIC), sample size-adjusted Bayesian information criterion (SSBIC), integrated classification likelihood criterion with Bayesian-type approximation, and Lo–Mendell–Rubin likelihood ratio test. Overall, SSBIC tended to identify the correct solution with higher frequency than other indices. BIC tended to identify the correct solution with higher frequency than the other indices in models with more continuous than categorical indicators, or when rare classes were absent. © Taylor & Francis Group, LLC.

Date 2015

DOI 10.1080/10705511.2014.935751

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Archive Scopus

Language English

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Publication Structural Equation Modeling

ISSN 10705511 (ISSN)

Issue 1

Journal Abbr Struct. Equ. Model.

Date Added 20/09/2026, 9:52:40 pm

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Tags:

Classification (of information), Statistical tests, Latent class analysis, latent class analysis, Normal distribution, Model Selection, model selection, Bayesian information criterion, Finite mixture modeling, fit indices, Integrated classification, mixed mode data, Mixed-mode data, Psychological research

Notes:

EID: 2-s2.0-84918538954

Export Date: 20 September 2026; Cited By: 195

Mixed-dual-head Meets Box Priors: A Robust Framework for Semi-supervised Segmentation

Item Type Conference Paper

Author C. Chen

Author T. Liu

Author W. Tan

Author S. Pu

Abstract As it is costly to densely annotate large scale datasets for supervised semantic segmentation, extensive semi-supervised methods have been proposed. However, the accuracy, stability and flexibility of existing methods are still far from satisfactory. In this paper, we propose an effective and flexible framework for semi-supervised semantic segmentation using a small set of fully labeled images and a set of weakly labeled images with bounding box labels. In our framework, position and class priors are designed to guide the annotation network to predict accurate pseudo masks for weakly labeled images, which are used to train the segmentation network. We also propose a mixed-dual-head training method to reduce the interference of label noise while enabling the training process more stable. Experiments on PASCAL VOC 2012 show that our method achieves state-of-the-art performance and can achieve competitive results even with very few fully labeled images. Furthermore, the performance can be further boosted with extra weakly labeled images from COCO dataset. © 2022 IEEE.

Date 2022

DOI 10.1109/WACV51458.2022.00265

URL <https://www.scopus.com/pages/publications/85126138573?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - IEEE/CVF Winter Conf. Appl. Comput. Vis., WACV

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-166540915-5 (ISBN)

Pages 2592-2602

Proceedings Title Proc. - IEEE/CVF Winter Conf. Appl. Comput. Vis., WACV

Conference Name Proceedings - 2022 IEEE/CVF Winter Conference on Applications of Computer Vision, WACV 2022

Date Added 20/09/2026, 9:52:36 pm

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Tags:

Deep learning, Computer vision, Supervised learning, Semantics, Semantic Segmentation, Large dataset, Segmentation, Semantic segmentation, Semi-supervised learning, Few-shot, Semi-supervised segmentations, Un-supervised learning, Transfer, Labeled images, Dual-head, Grouping and shape deep learning, Grouping and Shape Deep Learning, Semi- and Un- supervised Learning

Notes:

EID: 2-s2.0-85126138573

Export Date: 20 September 2026; Cited By: 1; Conference name: 22nd IEEE/CVF Winter Conference on Applications of Computer Vision, WACV 2022; Conference date: 4 January 2022 through 8 January 2022; Conference code: 177326

Mixture-of-Experts for Open Set Domain Adaptation: A Dual-Space Detection Approach

Item Type Journal Article

Author Z. Du

Author J. An

Author Y. Tu

Author J. Hong

Author D. Wu

Abstract Open set domain adaptation (OSDA) copes with the distribution and label shifts between the source and target domains simultaneously, performing accurate classification for known classes while identifying unknown class samples in the target domain. Most existing OSDA approaches, depending on the final image feature space of deep models, require manually-tuned thresholds, and may easily misclassify unknown samples as known classes. Mixture-of-experts (MoE) could be a remedy. Within an MoE, different experts handle distinct input features, producing unique expert routing patterns for various classes in a routing feature space. As a result, unknown class samples may display different expert routing patterns to known classes. This article proposes dual-space detection, which exploits the inconsistencies between the image feature space and the routing feature space to detect unknown class samples without any threshold. A graph router is further introduced to better make use of the spatial information among the image patches. Experiments on three datasets validated the effectiveness and superiority of our approach. © 2020 IEEE.

Date 2025

DOI 10.1109/TAI.2025.3560590

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Archive Scopus

Language English

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Publisher Institute of Electrical and Electronics Engineers Inc.

Pages 3207-3216

Publication IEEE Transactions on Artificial Intelligence

ISSN 26914581 (ISSN)

Issue 12

Journal Abbr IEEE. Trans. Artif. Intell.

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Tags:

Routings, Domain adaptation, Target domain, Mixture of experts, Feature space, Unknown class, Dual spaces, Mixture-of-experts (MoEs), Open set domain adaptation, open set domain adaptation (OSDA), Routing feature, routing feature., Space detection

Notes:

EID: 2-s2.0-105002752837

Export Date: 20 September 2026; Cited By: 3; Funding details: STI, (2021ZD0201300); Correspondence Address: D. Wu; Huazhong University of Science and Technology, Key Laboratory of the Ministry of Education for Image Processing and Intelligent Control, School of Artificial Intelligence and Automation, Wuhan, 430074, China; email: drwu@hust.edu.cn

MobileNetV3-DETR with locality-biased decoding for chip-leg defects on embedded AOI

Item Type Journal Article

Author Z. Zhang

Author H. Liu

Author B. Huang

Abstract We present a scene-specific MobileNetV3-DETR for chip-leg defect inspection, where targets are tiny, dense, and arranged on a regular lattice. Instead of stacking generic tricks, we formalize three complementary mechanisms: (A1) a task-aligned FPN pruning criterion that selects the minimal pyramid levels {P3–P5} to match the empirical defect-size distribution; (A2) a lattice-aware relative positional encoding that biases attention toward physically plausible row/column offsets; and (A3) a defect-prior adaptive Top-K sparse attention that allocates the decoder’s attention budget by local response quantiles with a device-aware cap $K_{\max}$. In training, a defect-aware assignment (D-AA) re-weights the Hungarian classification term by smoothed class priors, improving recall of rare, safety-critical defects without changing the inference graph. Under a unified embedded protocol on Jetson Nano (TensorRT FP16, 512×512 , batch = 1), the model uses ~ 7.2 M parameters and ~ 29.9 GFLOPs, reaches 89.5% mAP@50 and an average of 55.25% mAP@[0.5:0.95], and delivers ≈ 60 ms per image. Ablations demonstrate independent gains from A1–A3, while confusion-matrix analysis confirms reduced errors on the hardest pairs (e.g., bent_leg vs. damaged_leg; dirty vs. scratches), indicating an improved accuracy–latency balance for inline AOI deployment. © The Author(s) 2026.

Date 2026

DOI 10.1007/s44163-026-01179-2

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Archive Scopus

Language English

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Publisher Springer Nature

Publication Discover Artificial Intelligence

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Issue 1

Journal Abbr Discov. Artif. Intell.

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Tags:

Budget control, Image coding, Jetson nano, Jetson Nano, Decoding, Signal encoding, Encoding (symbols), Encodings, Defects, Chip-leg defect detection, Defect detection, DETR, Embedded AOI, Fighter aircraft, Locality-biased decoding, Mobilenetv3, MobileNetV3, Relative positional encoding, Sparse attention

Notes:

EID: 2-s2.0-105041149224

Export Date: 20 September 2026; Cited By: 0; Funding details: University Natural Science Research Project of Anhui Province, (2024AH051383); Correspondence Address: Z. Zhang; Department of Electronic Information and Media, Chizhou Vocational and Technical College, Chizhou, 247000, China; email: zhiminzhang8975@163.com

Model and score adaptation for biometric systems: Coping with device interoperability and changing acquisition conditions

Item Type Conference Paper

Author N. Poh

Author J. Kittler

Author S. Marcel

Author D. Matrouf

Author J.-F. Bonastre

Abstract The performance of biometric systems can be significantly affected by changes in signal quality. In this paper, two types of changes are considered: change in acquisition environment and in sensing devices. We investigated three solutions: (i) model-level adaptation, (ii) score-level adaptation (normalisation), and (iii) the combination of the two, called "compound" adaptation. In order to cope with the above changing conditions, the model-level adaptation attempts to update the parameters of the expert systems (classifiers). This approach requires the authenticity of the candidate samples used for adaptation be known (corresponding to supervised adaptation), or can be estimated (unsupervised adaptation). In comparison, the score-level adaptation merely involves postprocessing the expert output, with the objective of rendering the associated decision threshold to be dependent only on the class priors despite the changing acquisition conditions. Since the above adaptation strategies treat the underlying biometric experts/classifiers as a black-box, they can be applied to any unimodal or multimodal biometric system, thus facilitating system-level integration and performance optimisation. Our contributions are: (i) proposal of compound adaptation; (ii) investigation and comparison of two different quality-dependent

score normalisation strategies; and, (iii) empirical comparison of the merit of the above three solutions on the BANCA face (video) and speech database. © 2010 IEEE.

Date 2010

DOI 10.1109/ICPR.2010.306

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Int. Conf. Pattern Recognit.

Publisher Institute of Electrical and Electronics Engineers Inc.

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Proceedings Title Proc. Int. Conf. Pattern Recognit.

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Tags:

Condition, Performance, Biometrics, Expert systems, Normalisation, Unsupervised adaptation, Decision threshold, Biometric systems, Device interoperability, Sensing devices, Signal quality, Three solutions

Notes:

EID: 2-s2.0-78149481153

Export Date: 20 September 2026; Cited By: 26; Funding details: European Commission, EC; Seventh Framework Programme, FP7, (214324); Seventh Framework Programme, FP7; Correspondence Address: N. Poh; CVSSP, FEPS, University of Surrey, GU2 7XH, United Kingdom; email: n.poh@surrey.ac.uk; CODEN: PICRE

Model Uncertainty and Correctability for Directed Graphical Models*

Item Type Journal Article

Author P. Birmpa

Author J. Feng

Author M.A. Katsoulakis

Author L. Rey-Bellet

Abstract Probabilistic graphical models are a fundamental tool in probabilistic modeling, machine learning, and artificial intelligence. They allow us to integrate in a natural way expert knowledge, physical modeling, heterogeneous and correlated data, and quantities of interest. For exactly this reason, multiple sources of model uncertainty are inherent within the modular structure of the graphical model. In this paper we develop information-theoretic, robust uncertainty quantification methods and nonparametric stress tests for directed graphical models to assess the effect and the

propagation through the graph of multisourced model uncertainties to quantities of interest. These methods allow us to rank the different sources of uncertainty and correct the graphical model by targeting its most impactful components with respect to the quantities of interest. Thus, from a machine learning perspective, we provide a mathematically rigorous approach to correctability that guarantees a systematic selection for improvement of components of a graphical model while controlling potential new errors created in the process in other parts of the model. We demonstrate our methods in two physicochemical examples, namely, quantum scale-informed chemical kinetics and materials screening to improve the efficiency of fuel cells. © 2022 Society for Industrial and Applied Mathematics and American Statistical Association.

Date 2022

DOI 10.1137/21M1434453

URL <https://www.scopus.com/pages/publications/85145594200?origin=resultslist>

Archive Scopus

Language English

Volume 10

Publisher Society for Industrial and Applied Mathematics Publications

Pages 1461-2512

Publication SIAM-ASA Journal on Uncertainty Quantification

ISSN 21662525 (ISSN)

Issue 4

Journal Abbr SIAM/ASA J. Uncertain. Quantif.

Date Added 20/09/2026, 9:52:21 pm

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Tags:

Machine-learning, Machine learning, Uncertainty analysis, Statistical tests, Bayesian networks, Bayesian networks, Directed graphs, Sensitivity analysis, Systematic errors, sensitivity analysis, Graphic methods, Stress test, Uncertainty quantifications, uncertainty quantification, Modeling uncertainties, correctability, Correctability, Fuel cells, Graphical model, information inequalities, Information inequality, Probabilistic graphical models, Quantity of interest, stress tests

Notes:

EID: 2-s2.0-85145594200

Export Date: 20 September 2026; Cited By: 0; Funding details: Air Force Office of Scientific Research, AFOSR, (FA-9550-18-1-0214, FA9550-21-1-0354); National Science Foundation, NSF, (CISE-1934846, DMS-2008970); Defense Advanced Research Projects Agency, DARPA, (W911NF1520122)

Modeling and characterizing software vulnerabilities

Item Type Journal Article

Author N. Bhatt**Author** A. Anand**Author** V.S.S. Yadavalli**Author** V. Kumar

Abstract With the association of software security assurance in the development of code based systems; software developers are relying on the Vulnerability discovery models to mitigate the breaches by estimating the total number of vulnerabilities, before they're exploited by the intruders. Vulnerability Discovery Models (VDMs) provide the quantitative classification of the flaws that exists in a software that will be discovered after a software is released. In this paper, we develop a vulnerability discovery model that accumulate the vulnerabilities due to the influence of previously discovered vulnerabilities. We further evaluate the proportion of previously discovered vulnerabilities along with the fraction additional vulnerabilities detected. The quantification methodology presented in this article has been accompanied with an empirical illustration on popular operating systems' vulnerability data. © 2017, International Journal of Mathematical, Engineering and Management Sciences.

Date 2017**DOI** 10.33889/ijmems.2017.2.4-022**URL** <https://www.scopus.com/pages/publications/85052436169?origin=resultslist>**Archive** Scopus**Language** English**Volume** 2**Publisher** International Journal of Mathematical, Engineering and Management Sciences**Pages** 288-299**Publication** International Journal of Mathematical, Engineering and Management Sciences**ISSN** 24557749 (ISSN)**Issue** 4**Journal Abbr** Int. J. Math. Eng. Manag. Sci.**Date Added** 20/09/2026, 9:52:40 pm**Modified** 20/09/2026, 9:52:40 pm**Tags:**

Software security, Vulnerability categorization, Vulnerability discovery modeling

Notes:

EID: 2-s2.0-85052436169

Export Date: 20 September 2026; Cited By: 41; Correspondence Address: A. Anand; Department of Operational Research, University of Delhi, Delhi, 110007, India; email: adarsh.anand86@gmail.com

Modeling Groundwater Resources in Data-Scarce Regions for Sustainable Management: Methodologies and Limits

Item Type Journal Article

Author I. Borzi

Abstract Groundwater modeling in data-scarce regions faces significant challenges due to the lack of comprehensive, high-quality data, impacting model accuracy. This systematic review of Scopus-indexed papers identifies various approaches to address these challenges, including coupled hydrological-groundwater models, machine learning techniques, distributed hydrological models, water balance models, 3D groundwater flow modeling, geostatistical techniques, remote sensing-based approaches, isotope-based methods, global model downscaling, and integrated modeling approaches. Each methodology offers unique advantages for groundwater assessment and management in data-poor environments, often combining multiple data sources and modeling techniques to overcome limitations. However, these approaches face common challenges related to data quality, scale transferability, and the representation of complex hydrogeological processes. This review emphasizes the importance of adapting methodologies to specific regional contexts and data availability. It underscores the value of combining multiple data sources and modeling techniques to provide robust estimates for sustainable groundwater management. The choice of method ultimately depends on the specific objectives, scale of the study, and available data in the region of interest. Future research should focus on improving the integration of diverse data sources, enhancing the representation of complex hydrogeological processes in simplified models, and developing robust uncertainty quantification methods tailored for data-scarce conditions. © 2025 by the author.

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DOI 10.3390/hydrology12010011

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Publication Hydrology

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Issue 1

Journal Abbr Hydrology

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Tags:

uncertainty analysis, data-scarce aquifer, groundwater data lack, groundwater modeling, sustainable groundwater resource management, ungauged aquifer

Notes:

EID: 2-s2.0-85216034315

Export Date: 20 September 2026; Cited By: 28; Correspondence Address: I. Borzi; Department of Mathematics, Computer Sciences, Physics and Earth Sciences, The University of Messina, Messina, 98166, Italy; email: iolanda.borzi@unime.it

Modeling Uncertainty With Engression: A Deep Generative Time-Series Approach

Item Type Journal Article

Author B. Kraft

Author S. Stalder

Author W.H. Aeberhard

Author N.H. Ruiz

Author N. Meinshausen

Author X. Shen

Author L. Gudmundsson

Abstract Deep learning enables precise environmental predictions across spatial and temporal scales. However, reliable uncertainty estimation with generative capabilities remains crucial for actionable forecasting and simulation, yet robust quantification methods remain challenging. Recently, engression, a generative approach for model-agnostic uncertainty quantification, has been proposed. We evaluate its feasibility for environmental time-series modeling, specifically applying it to rainfall-runoff prediction using long short-term memory (LSTM) networks. As a benchmark, we use quantile regression, a generalization of the mean absolute error. In our analysis, we focus on the quality of the uncertainty estimates and realism of generated sequences, while including a comparison to point-prediction performance as a sanity check. Our results show that LSTM-engression is an effective and easy-to-use method for generative modeling, outperforming quantile regression in uncertainty quantification. Qualitative analysis of the generated time series indicates high dynamic fidelity. These findings highlight LSTM-engression as a promising approach for incorporating uncertainty into environmental modeling. © 2026. The Author(s).

Date 2026

DOI 10.1029/2025GL120122

URL <https://www.scopus.com/pages/publications/105028375188?origin=resultslist>

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Publication Geophysical Research Letters

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Issue 2

Journal Abbr Geophys. Res. Lett.

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Tags:

machine learning, Prediction models, Uncertainty analysis, spatiotemporal analysis, Benchmarking, Weather forecasting, qualitative analysis, Long short-term memory, Time series analysis, long short-term memory (LSTM), Short term memory, Generative model, regression analysis, generative modeling, Time series, Uncertainty quantifications, Times series, time series, uncertainty quantification, uncertainty analysis, Times series models, engression, Engression, environmental modeling, model validation, Rainfall runoff, rainfall-runoff modeling, rainfall-runoff prediction, Rainfall-runoff prediction, Runoff prediction, time-series modeling

Notes:

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Modeling with Confidence: Leveraging Conformal Prediction for Calibrated Machine Learning Based Mechanical and Petrophysical Models

Item Type	Conference Paper
Author	W.B. Saleh
Author	B. Zhang
Abstract	Machine learning (ML) is revolutionizing reservoir characterization practices by directly using well log and drilling data. However, deterministic predictions of ML models can be misleading and result in expensive mistakes. Hence, uncertainties in ML-predicted mechanical and/or petrophysical properties need to be well quantified. By leveraging a distribution-free and computationally efficient uncertainty quantification method called Conformal Prediction (CP), we can derive calibrated ML models and quantify the uncertainties in their predictions. Several ML models are developed for permeability and elastic modulus prediction in a geothermal site in south Saskatchewan. The Catboost model outperforms other models achieving an R2 of 0.91 and 0.92 for permeability and elastic modulus, respectively. A Conformal Prediction is then built on the selected ML models to complement the predictions with valid measures of prediction intervals with 95% coverage. For a test well, where two different lab-measured permeabilities exist, more than 90% of measured permeabilities fall within the 95% prediction interval. Triaxial geomechanical test results are also comfortably within the bounds of the 95% interval. This suggests that these models provide reliable predictions with limited uncertainties. This paper underscores the crucial role of uncertainty quantification of ML-based prediction models. The study demonstrates how quantifying uncertainty can enhance our confidence in ML-predicted reservoir properties for rigorous subsurface reservoir characterization. Copyright © 2025 ARMA, American Rock Mechanics Association.
Date	2025

DOI 10.56952/ARMA-2025-0602**URL** <https://www.scopus.com/pages/publications/105019492824?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: US Rock Mech./Geomech. Symp.**Publisher** American Rock Mechanics Association (ARMA)**Proceedings Title** US Rock Mech./Geomech. Symp.**Conference Name** 59th US Rock Mechanics/Geomechanics Symposium**Date Added** 20/09/2026, 9:52:06 pm**Modified** 20/09/2026, 9:52:06 pm**Tags:**

Learning systems, Machine-learning, Machine learning, Prediction models, Forecasting, Uncertainty analysis, Rock mechanics, Uncertainty, Machine learning models, Petroleum reservoir evaluation, Uncertainty quantifications, Conformal predictions, Prediction interval, Well logging, Elastic moduli, Mechanical modeling, Mechanical permeability, Petrophysical models, Petrophysics, Reservoir characterization, Well log data

Notes:

EID: 2-s2.0-105019492824

Export Date: 20 September 2026; Cited By: 3; Funding details: University of Alberta; Natural Resources Canada, RNCAN; Canada First Research Excellence Fund, CFREF, (T02-A03); Canada First Research Excellence Fund, CFREF; Conference name: 59th US Rock Mechanics/Geomechanics Symposium; Conference date: 8 June 2025 through 11 June 2025; Conference code: 213441

Molecular monitoring of Dinophysis species assemblage in mussel farms in the Northwestern Adriatic Sea

Item Type Journal Article**Author** G. Ravera**Author** M. Cangini**Author** S. Capellacci**Author** S. Dall'Ara**Author** G. Prioli**Author** M. Marini**Author** E. Manini**Author** A. Penna**Author** S. Casabianca

Abstract Several Dinophysis species can produce potent lipophilic toxins that pose a risk to human health when contaminated seafood is consumed, especially filter-feeding bivalve mussels. In the mussel farms of the Northwestern Adriatic Sea, seawater and seafood are regularly monitored for the presence of Dinophysis species and their

associated toxins, but the current methodological approaches, such as light microscopy determinations, require a long time to make results available to local authorities. A molecular qPCR-based assay can be used to quantify various toxic Dinophysis species in a shorter timeframe. However, this approach is not currently employed in official testing activities. In this study, field samples were collected monthly or bi-weekly over one year from various mussel farms along the Northwestern Adriatic coast. The abundance of Dinophysis species in the seawater was determined using both traditional microscopy and qPCR assays. In addition, the concentration of lipophilic toxins for DSP in mussel flesh was quantified using LC-MS/MS focusing on the okadaic acid group. Dinophysis spp. site-specific single cells were isolated and analysed by qPCR yielding a mean rDNA copy number per cell of $1.21 \times 10^4 \pm 1.81 \times 10^3$. The qPCR assay gave an efficiency of 98 % and detected up to 10 copies of the rDNA target gene. The qPCR and light microscopy determinations in environmental samples showed a significant positive correlation (Spearman $r_s = 0.57$, $p\text{-value} < 0.001$) with a ratio of 2.24 between the two quantification methods, indicating that light microscopy estimates were generally 44.6 % lower than those obtained by the qPCR assay. The qPCR approach showed several advantages such as rapidity, sensitivity and efficiency over conventional microscopy analysis, showing its potential future role in phytoplankton monitoring under the Official Controls Regulations for shellfish. © 2024

Date 2024

DOI 10.1016/j.hal.2024.102686

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Language English

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Journal Abbr Harmful Algae

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Tags:

Environmental Monitoring, procedures, Monitoring, classification, environmental monitoring, genetics, chemistry, animal, Animals, bivalve, Bivalvia, aquaculture, Aquaculture, Adriatic sea, dinoflagellate, Dinoflagellida, Dinophysis spp., DSP (diarrhetic shellfish poisoning), LM (light microscopy), marine toxin, Marine Toxins, qPCR, sea water, Seawater

Notes:

EID: 2-s2.0-85198513720

Export Date: 20 September 2026; Cited By: 3; Funding details: School of University of Urbino; Correspondence Address: S. Casabianca; CoNISMa, National Inter-University Consortium for Marine Sciences, Roma, Piazzale Flaminio, 6, 00184, Italy; email: silvia.casabianca@uniurb.it; CODEN: HAALD

Monitoring respiratory muscles effort during mechanical ventilation

Item Type Journal Article

Author J.P. Van Oosten

Author E. Akoumianaki

Author A.H. Jonkman

Abstract Purpose of review To summarize basic physiological concepts of breathing effort and outline various methods for monitoring effort of inspiratory and expiratory muscles. Recent findings Esophageal pressure (Pes) measurement is the reference standard for respiratory muscle effort quantification, but various noninvasive screening tools have been proposed. Expiratory occlusion pressures (P0.1 and Pocc) could inform about low and high effort and the resulting lung stress, with Pocc outperforming P0.1 in identifying high effort. The pressure muscle index during an inspiratory hold could unveil inspiratory muscle effort, however obtaining a reliable inspiratory plateau can be difficult. Surface electromyography has the potential for inspiratory effort estimation, yet this is technically challenging for real-time assessment. Expiratory muscle activation is common in the critically ill warranting their assessment, that is, via gastric pressure monitoring. Expiratory muscle activation also impacts inspiratory effort interpretation which could result in both under- and overestimation of the resulting lung stress. There is likely a future role for machine learning applications to automate breathing effort monitoring at the bedside. Summary Different tools are available for monitoring the respiratory muscles' effort during mechanical ventilation - from noninvasive screening tools to more invasive quantification methods. This could facilitate a lung and respiratory muscle-protective ventilation approach. © 2024 The Author(s). Published by Wolters Kluwer Health, Inc.

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Morphological characterization and process optimization of gas-atomized powders based on synchrotron X-ray computed tomography and machine learning

Item Type Journal Article

Author S. Luo

Author Y. Ouyang

Author Z. Tang

Author M. Wang

Author Y. Wu

Author H. Wang

Author H. Wang

Abstract Metal powder is a key raw material for additive manufacturing (AM), with its quality directly determining the performance of AM components. Gas atomization (GA) is a dominant powder production method due to its high efficiency and cost-effectiveness, yet it frequently generates defective powders (e.g., hollow, satellite, and irregular particles) that significantly degrade the mechanical and fatigue properties of AM parts. Existing powder quality quantification methods primarily focus on particulate and bulk properties, but overlook the correlation between powder morphology and the GA process—failing to provide effective feedback for process optimization. This study aims to comprehensively evaluate GAed powders via morphological characteristic analysis. Synchrotron X-ray computed tomography (SXCT) and 3D reconstruction are used to characterize powders, which are classified into five types (irregular, satellite, normal, hollow, multi-defect) based on the formation mechanisms of defective powders in the GA process. By using 2598 powder samples and 9 morphological parameters, the machine learning models are trained. Through principal component and correlation analyses, 4 key defect-distinguishing parameters are identified, and all models achieved high accuracy. To assess GAed powders, the

particle defect distribution (PDD) is built for quality evaluation. Guided by these findings, an improved nozzle is designed, manufactured, and used for powder preparation. PDD results indicated about 300% increase in the mass ratio of normal powders with high sphericity. © 2026 Elsevier B.V.

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MOST: Multi-Source Domain Adaptation via Optimal Transport for Student-Teacher Learning

Item Type Conference Paper

Author T. Nguyen

Author T. Le

Author H. Zhao

Author Q.H. Tran**Author** T. Nguyen**Author** D. Phung**Editor** de Campos C.**Editor** Maathuis M.H.

Abstract Multi-source domain adaptation (DA) is more challenging than conventional DA because the knowledge is transferred from several source domains to a target domain. To this end, we propose in this paper a novel model for multi-source DA using the theory of optimal transport and imitation learning. More specifically, our approach consists of two cooperative agents: a teacher classifier and a student classifier. The teacher classifier is a combined expert that leverages knowledge of domain experts that can be theoretically guaranteed to handle perfectly source examples, while the student classifier acting on the target domain tries to imitate the teacher classifier acting on the source domains. Our rigorous theory developed based on optimal transport makes this cross-domain imitation possible and also helps to mitigate not only the data shift but also the label shift, which are inherently thorny issues in DA research. We conduct comprehensive experiments on real-world datasets to demonstrate the merit of our approach and its optimal transport based imitation learning viewpoint. Experimental results show that our proposed method achieves state-of-the-art performance on benchmark datasets for multi-source domain adaptation including Digits-five, Office-Caltech10, and Office-31 to the best of our knowledge. © 2021 37th Conference on Uncertainty in Artificial Intelligence, UAI 2021. All Rights Reserved.

Date 2021**URL** <https://www.scopus.com/pages/publications/85124325554?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Conf. Uncertain. Artif. Intell., UAI**Publisher** Association For Uncertainty in Artificial Intelligence (AUAI)**Pages** 225-235**Proceedings Title** Conf. Uncertain. Artif. Intell., UAI**Conference Name** 37th Conference on Uncertainty in Artificial Intelligence, UAI 2021**Date Added** 20/09/2026, 9:52:33 pm**Modified** 20/09/2026, 9:52:33 pm**Tags:**

Artificial intelligence, Benchmarking, Students, Domain adaptation, Target domain, Student teachers, Teachers', Domain experts, Multi-Sources, Optimal transport, Cooperative agents, Imitation learning, Teacher learning

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MOST: Multi-Source Domain Adaptation via Optimal Transport for Student-Teacher Learning

Item Type Conference Paper

Author T. Nguyen

Author T. Le

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Author Q.H. Tran

Author T. Nguyen

Author D. Phung

Editor de Campos C.

Editor Maathuis M.H.

Abstract Multi-source domain adaptation (DA) is more challenging than conventional DA because the knowledge is transferred from several source domains to a target domain. To this end, we propose in this paper a novel model for multi-source DA using the theory of optimal transport and imitation learning. More specifically, our approach consists of two cooperative agents: a teacher classifier and a student classifier. The teacher classifier is a combined expert that leverages knowledge of domain experts that can be theoretically guaranteed to handle perfectly source examples, while the student classifier acting on the target domain tries to imitate the teacher classifier acting on the source domains. Our rigorous theory developed based on optimal transport makes this cross-domain imitation possible and also helps to mitigate not only the data shift but also the label shift, which are inherently thorny issues in DA research. We conduct comprehensive experiments on real-world datasets to demonstrate the merit of our approach and its optimal transport based imitation learning viewpoint. Experimental results show that our proposed method achieves state-of-the-art performance on benchmark datasets for multi-source domain adaptation including Digits-five, Office-Caltech10, and Office-31 to the best of our knowledge. © 2021 Proceedings of Machine Learning Research. All Rights Reserved.

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MOT: Masked Optimal Transport for Partial Domain Adaptation

Item Type	Conference Paper
Author	Y.-W. Luo
Author	C.-X. Ren
Abstract	As an important methodology to measure distribution discrepancy, optimal transport (OT) has been successfully applied to learn generalizable visual models under changing environments. However, there are still limitations, including strict prior assumption and implicit alignment, for current OT modeling in challenging real-world scenarios like partial domain adaptation, where the learned transport plan may be biased and negative transfer is inevitable. Thus, it is necessary to explore a more feasible OT methodology for real-world applications. In this work, we focus on the rigorous OT modeling for conditional distribution matching and label shift correction. A novel masked OT (MOT) methodology on conditional distributions is proposed by defining a mask operation with label information. Further, a relaxed and reweighting formulation is proposed to improve the robustness of OT in extreme scenarios. We prove the theoretical equivalence between conditional OT and MOT, which implies the well-defined MOT serves as a computation-friendly proxy. Extensive experiments validate the effectiveness of theoretical results and proposed model. © 2023 IEEE.
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Volume	2023-June
Publisher	IEEE Computer Society
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'current, Deep learning, Machine-learning, Federated learning, Adversarial machine learning, Contrastive Learning, Domain adaptation, Learn+, Conditional distribution, Optimal transport, Changing environment, Machine learning (other than deep learning), Transport modelling, Visual model

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Movement imagery classification on the basis of single-trial EEGs

Item Type Conference Paper**Author** H. Maki**Author** H. Takayanagi**Author** T. Yamazaki**Author** K.-I. Kamijo**Author** T. Yamanoi

Abstract In order to develop a new brain-computer interface (BCI) using EEGs, the independent component analysis (ICA), multiple equivalent current dipole source localization (ECDL) method and then Hayashi's quantification method II were applied to single-trial EEGs during movement imagery. Up to the 20th independent component, each of them contained one or two positive (or negative) peaks in terms of 32-channel EEGs reconstructed on the electrode surface by the deflation procedure, and these peaks were analyzed by 1- or 2-dipole source localization method. Moreover, the quantification method based on these ECDL results yielded a good discrimination between right- and left-hand movement imageries, even if regardless of the time course of the localization results, instead of only for one male subject. ©2007 IEEE.

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Motion estimation, Medical imaging, Image classification, Electroencephalography, Independent component analysis, Equivalent current dipole source localization (ECDL) method, Hayashi's quantification method, Movement imageries

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Moving shadow and object detection in traffic scenes

Item Type Journal Article

Author I. Mikić

Author P.C. Cosman

Author G.T. Kogut

Author M.M. Trivedi

Abstract We present an algorithm for segmentation of traffic scenes that distinguishes moving objects from their moving cast shadows. A fading memory estimator calculates mean and variance of all three color components for each background pixel. Given the statistics for a background pixel, simple rules for calculating its statistics when covered by a shadow are used. Then, MAP classification decisions are made for each pixel. In addition to the color features, we examine the use of neighborhood information to produce smoother classification. We also propose the use of temporal information by modifying class a priori probabilities based on predictions from the previous frame. © 2000 IEEE.

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MPED: Quantifying Point Cloud Distortion Based on Multiscale Potential Energy Discrepancy

Item Type Journal Article

Author Q. Yang

Author Y. Zhang

Author S. Chen

Author Y. Xu

Author J. Sun

Author Z. Ma

Abstract In this article, we propose a new distortion quantification method for point clouds, the multiscale potential energy discrepancy (MPED). Currently, there is a lack of effective distortion quantification for a variety of point cloud perception tasks. Specifically, in human vision tasks, a distortion quantification method is used to predict human subjective scores and optimize the selection of human perception task parameters, such as dense point cloud compression and enhancement. In machine vision tasks, a distortion quantification method usually serves as loss function to guide the training of deep neural networks for unsupervised learning tasks (e.g., sparse point cloud reconstruction, completion, and upsampling). Therefore, an effective distortion quantification should be differentiable, distortion discriminable, and have low computational complexity. However, current distortion quantification cannot satisfy all three conditions. To fill this gap, we propose a new point cloud

feature description method, the point potential energy (PPE), inspired by classical physics. We regard the point clouds are systems that have potential energy and the distortion can change the total potential energy. By evaluating various neighborhood sizes, the proposed MPED achieves global-local tradeoffs, capturing distortion in a multiscale fashion. We further theoretically show that classical Chamfer distance is a special case of our MPED. Extensive experiments show that the proposed MPED is superior to current methods on both human and machine perception tasks. Our code is available at <https://github.com/Qi-Yangsjtu/MPED>. © 1979-2012 IEEE.

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MRI Voxel-Based Morphometry to Classify Brain Structural Alterations in Parkinson's Disease

Item Type Conference Paper

Author S. Datta

Author M.K. Acharaya

Author B. Chakrabarty

Author N. Ghosh

Abstract The introduction of voxel-based morphometry (VBM) has profoundly impacted neuroimaging, offering a robust and objective approach to analyze brain structure using MRI. This study presents a customized pipeline for VBM methodology and its applications in both healthy elderly individuals and patients with Parkinson's disease (PD). The VBM process mainly involves spatial normalization to a standard template, segmentation into gray matter (GM) and white matter (WM), smoothing, and rigorous statistical analysis. VBM has proven invaluable in detecting subtle brain structural changes in various neurological and psychiatric conditions, as well as identifying gross structural abnormalities. The study highlights VBM's crucial role in advancing neuroimaging research, making it an indispensable tool in the field. While VBM has significantly advanced our understanding of brain morphology, careful attention to the algorithmic implementation is essential to ensure the accuracy of conclusions about anatomical differences related to disease states. This study underscores that combining VBM with complementary neuroimaging techniques can greatly enhance our understanding of normal brain processes and improve comparisons between patient and control groups. This study deals with 12 healthy controls (HCs) and 30 PD patients. It is concluded that a more effective biomarker could be prepared with the help of suggested pipeline of automated VBM without custom templates and custom tissue class priors. © 2024 IEEE.

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MrIML: Multi-response interpretable machine learning to model genomic landscapes

Item Type Conference Paper

Author N.M. Fountain-Jones

Author C.P. Kozakiewicz

Author B.R. Forester

Author E.L. Landguth

Author S. Carver

Author M. Charleston

Author R.B. Gagne

Author B. Greenwell

Author S. Kraberger

Author D.R. Trumbo

Author M. Mayer

Author N.J. Clark

Author G. Machado

Abstract We introduce a new R package “MrIML” (“Mister iml”; Multi-response Interpretable Machine Learning). MrIML provides a powerful and interpretable framework that enables users to harness recent advances in machine learning to quantify multilocus genomic relationships, to identify loci of interest for future landscape genetics studies, and to gain new insights into adaptation across environmental gradients. Relationships between genetic variation and environment are often nonlinear and interactive; these characteristics have been challenging to address using traditional landscape genetic approaches. Our package helps capture this complexity and offers functions that fit and interpret a wide range of highly flexible models that are routinely used for single-locus landscape genetics studies but are rarely extended to estimate response functions for multiple loci. To demonstrate the package's broad functionality, we test its ability to recover landscape relationships from simulated genomic data. We also apply the package to two empirical case studies. In the first, we model genetic variation of North American balsam poplar (*Populus balsamifera*, Salicaceae) populations across environmental gradients. In the second case study, we recover the landscape and host drivers of feline immunodeficiency virus genetic variation in bobcats (*Lynx rufus*). The ability to model thousands of loci collectively and compare models from linear regression to extreme gradient boosting, within the same analytical framework, has the potential to be transformative. The MrIML framework is also extendable and not limited to modelling genetic variation; for example, it can quantify the environmental drivers of microbiomes and coinfection dynamics. © 2021 John Wiley & Sons Ltd

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Multi-atlas segmentation and quantification of muscle, bone and subcutaneous adipose tissue in the lower leg using peripheral quantitative computed tomography

Item Type Journal Article

Author S. Makrogiannis

Author A. Okorie

Author A. Di Iorio

Author S. Bandinelli

Author L. Ferrucci

Abstract Accurate and reproducible tissue identification is essential for understanding structural and functional changes that may occur naturally with aging, or because of a chronic disease, or in response to intervention therapies. Peripheral quantitative computed tomography (pQCT) is regularly employed for body composition studies, especially for the structural and material properties of the bone. Furthermore, pQCT

acquisition requires low radiation dose and the scanner is compact and portable. However, pQCT scans have limited spatial resolution and moderate SNR. pQCT image quality is frequently degraded by involuntary subject movement during image acquisition. These limitations may often compromise the accuracy of tissue quantification, and emphasize the need for automated and robust quantification methods. We propose a tissue identification and quantification methodology that addresses image quality limitations and artifacts, with increased interest in subject movement. We introduce a multi-atlas image segmentation (MAIS) framework for semantic segmentation of hard and soft tissues in pQCT scans at multiple levels of the lower leg. We describe the stages of statistical atlas generation, deformable registration and multi-tissue classifier fusion. We evaluated the performance of our methodology using multiple deformable registration approaches against reference tissue masks. We also evaluated the performance of conventional model-based segmentation against the same reference data to facilitate comparisons. We studied the effect of subject movement on tissue segmentation quality. We also applied the top performing method to a larger out-of-sample dataset and report the quantification results. The results show that multi-atlas image segmentation with diffeomorphic deformation and probabilistic label fusion produces very good quality over all tissues, even for scans with significant quality degradation. The application of our technique to the larger dataset reveals trends of age-related body composition changes that are consistent with the literature. Because of its robustness to subject motion artifacts, our MAIS methodology enables analysis of larger number of scans than conventional state-of-the-art methods. Automated analysis of both soft and hard tissues in pQCT is another contribution of this work. Copyright © 2022 Makrogiannis, Okorie, Di Iorio, Bandinelli and Ferrucci.

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Multi-Attribute Decision Making (MADM) Model Based on Bayesian Neural Network Classification with Comparative Scoring Quantification Method Can Solve Intangible Value Valuation Issues in Benefit-Cost Analysis (BCA)

Item Type Conference Paper

Author Z. Wang

Author H.-H. Wang

Editor Huang K.-W.

Editor Cao Q.

Editor Su R.

Abstract This paper is Methodology research of Benefit-Cost Analysis (BCA) under machine learning-driven Classification Bayesian Neural Network. BCA is currently the predominant decision-making tool for corporate capital investment. Due to its inherent single-metric nature, BCA lacks complete quantifiability and responsiveness in terms of intangible benefits and ethics. In this paper the author presents how advanced AI scoring algorithm enables Multi-Attribute Decision Making (MADM) approach to dissolve monetization failure issue stemming from ethical concerns in Benefit-Cost Analysis (BCA). In the current context of increasing emphasis on Corporate Social Responsibility (CSR), ethical considerations have become vital for businesses. Therefore, it is advisable to utilize machine learning leveraging the business big data to train more sophisticated and stable multi-dimensional decision-making models, then integrate them into the framework of BCA, thereby evolving the BCA methodology to overcome its drawbacks. The three main problems associated with BCA discussed are: (1) intangible value is difficult to monetize, and (2) human life-related benefits should not be monetized due to the ethical principle. And (3) the normative and distributional problem of BCA that cannot take into account both the efficiency criterion and the equity criterion in one decision-making process. For the three problems of BCA, author introduces a Bayesian Neural Network Scoring algorithm based Analytic Hierarchy Process (AHP) to achieve Multi-Attribute Decision Making (MADM) and explains why it can solve the above three problems that are difficult for BCA to handle solely. Furthermore, arguing the limitations of such AHP-MADM methodology and puts forward the improvement methods including the Expected Utility Theory [1], and VIKOR method based on “Prospect Theory” [2, 3] which allow each organization to have its own unique decision preferences. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Multi-class Classification from Multiple Unlabeled Datasets with Partial Risk Regularization

Item Type Conference Paper

Author Y. Tang

Author N. Lu

Author T. Zhang

Author M. Sugiyama

Editor Khan E.

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Abstract Recent years have witnessed a great success of supervised deep learning, where predictive models were trained from a large amount of fully labeled data. However, in practice, labeling such big data can be very costly and may not even be possible for privacy reasons. Therefore, in this paper, we aim to learn an

accurate classifier without any class labels. More specifically, we consider the case where multiple sets of unlabeled data and only their class priors, i.e., the proportions of each class, are available. Under this problem setup, we first derive an unbiased estimator of the classification risk that can be estimated from the given unlabeled sets and theoretically analyze the generalization error of the learned classifier. We then find that the classifier obtained as such tends to cause overfitting as its empirical risks go negative during training. To prevent overfitting, we further propose a partial risk regularization that maintains the partial risks with respect to unlabeled datasets and classes to certain levels. Experiments demonstrate that our method effectively mitigates overfitting and outperforms state-of-the-art methods for learning from multiple unlabeled sets. © 2022 Y. Tang, N. Lu, T. Zhang & M. Sugiyama.

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Multi-Class Machine Learning to Quantify the Impact of Nitrogen Management Practices on Grassland Biomass

Item Type Journal Article

Author S. Raubitzek

Author M. Hartlieb

Author P. König

Author J. Hinderling

Author K. Mallinger

Abstract Grassland biomass yield reflects a complex interaction of management intensity and environmental factors, yet quantifying the relative role of practices such as mowing and fertilization remains challenging. In this study, we introduce a multi-class machine learning framework to predict above-ground biomass on 150 permanent grassland plots across eight years (2009–2016) in Germany’s Biodiversity Exploratories and to evaluate the influence of key management variables. Following rigorous data cleaning, imputation of missing nitrogen values, feature standardization, and encoding of categorical practices, we trained CatBoost classifiers optimized via Bayesian hyperparameter search and mitigated class imbalance with ADASYN oversampling. We assessed model performance under binary, three-class, four-class, and five-class quantile-based categorizations, achieving test accuracies of 0.76, 0.57, 0.42, and 0.38, respectively. Across all schemes, mowing frequency and mineral nitrogen input emerged as the dominant predictors, while secondary variables such as drainage and conditioner use contributed as well. These results demonstrate that broad biomass categories can be forecast reliably from standardized management records, whereas finer distinctions necessitate additional environmental information or automated sensing to capture nonlinear effects and reduce reporting bias. This work shows both the potential and the limits of machine learning for informing sustainable grassland management and explainability thereof. Frequent mowing and higher mineral nitrogen inputs explained most of the predictable variation, enabling a 76% accurate separation of low and high biomass categories. Predictive accuracy fell below 60% for finer class resolutions, indicating that management records alone are insufficient for detailed yield forecasts without complementary environmental data. © 2025 by the authors.

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MULTI-CLASS UNCERTAINTY CALIBRATION VIA MUTUAL INFORMATION MAXIMIZATION-BASED BINNING

Item Type Conference Paper

Author K. Patel

Author W. Beluch

Author B. Yang

Author M. Pfeiffer

Author D. Zhang

Abstract Post-hoc multi-class calibration is a common approach for providing high-quality confidence estimates of deep neural network predictions. Recent work has shown that widely used scaling methods underestimate their calibration error, while alternative Histogram Binning (HB) methods often fail to preserve classification accuracy. When classes have small prior probabilities, HB also faces the issue of severe sample-inefficiency after the conversion into K one-vs-rest class-wise calibration problems. The goal of this paper is to resolve the identified issues of HB in order to provide calibrated confidence estimates using only a small holdout calibration dataset for bin optimization while preserving multi-class ranking accuracy. From an information-theoretic perspective, we derive the I-Max concept for binning, which maximizes the mutual information between labels and quantized logits. This concept mitigates potential loss in ranking performance due to lossy quantization, and by disentangling the optimization of bin edges and representatives allows simultaneous improvement of ranking and calibration performance. To improve the sample efficiency and estimates from a small calibration set, we propose a shared class-wise (sCW) calibration strategy, sharing one calibrator among similar classes (e.g., with similar class priors) so that the training sets of their class-wise calibration problems can be merged to train the single calibrator. The combination of sCW and I-Max binning outperforms the state of the art calibration methods on various evaluation metrics across different benchmark datasets and models, using a small calibration set (e.g., 1k samples for ImageNet). © 2021 ICLR 2021 - 9th International Conference on Learning Representations. All rights reserved.

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Multi-classification recognition and quantitative characterization of surface defects in belt grinding based on YOLOv7

Item Type Journal Article

Author B. Zhu

Author G. Xiao

Author Y. Zhang

Author H. Gao

Abstract Due to the complex and varied surface defects of the belt grinding, the automatic recognition and quantitative characterization of the surface defects is still an issue that needs to be resolved. To characterize surface defects in belt grinding, this study proposed a multi-classification recognition and quantification method. The YOLOv7 algorithm was utilized to detect defects, which would obtain classification and location information of defects. Aiming at the defects with continuous geometric boundaries, an object detection strategy based on Hough straight-line image correction was designed to quantitatively describe the characteristics of the defects in the grinding direction. In order to capture the quantitative information of burn defects without continuous geometric boundaries, an adaptive threshold segmentation approach with correction factors was developed. The results demonstrated that the YOLOv7 performance on our dataset reaches the mAP of 0.907. The suggested quantitative characterization method can effectively quantitatively characterize and analyze the defects. © 2023 Elsevier Ltd

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Multi-cohort Sleep EEG-based Depression Detection Using Multi-modal Data from the National Sleep Research Resource

Item Type Conference Paper
Author M. Mohtashamian
Author X. Li
Author L. Rasmy
Author L. Cui
Abstract Depression is a highly prevalent mental health disorder that significantly impairs quality of life and functional capacity. Accurate and timely identification is essential to support effective clinical intervention and long-term management. This study examines the potential of integrating multiple cohorts from the National Sleep Research Resource (NSRR) for depression detection based on multi-modal sleep electroencephalogram (EEG), demographic, and clinical data. We integrated three NSRR cohorts comprising 1,288 individuals (209 depressed and 1,079 non-depressed). Overnight sleep EEG recordings from C3/C4 electrode channels were used to characterize sleep-related neurophysiological patterns, together with demographic and clinical variables. To optimize predictive performance, feature selection was performed using Least Absolute Shrinkage

and Selection Operator (LASSO), followed by evaluation using multiple classical machine learning and deep learning models. Predictive models were developed and evaluated using a rigorous validation framework to assess generalizability across heterogeneous datasets. Our best-performing models achieved an area under the receiver operating characteristic curve (AUROC) of up to 0.688 with area under the precision-recall curve (AUPRC) up to 0.327 on the imbalanced cohort (positive-class prevalence ≈ 0.162), exceeding the random baseline expected under class imbalance. Although conservative relative to small-sample studies, these results reflect improved generalizability enabled by the large cohort size, multi-cohort design, and a strictly leakage-free evaluation in which feature selection is performed within each training fold. Overall, the findings demonstrate the feasibility of using multi-modal data from NSRR to support scalable and generalizable approaches for sleep EEG-based depression detection. © 2026 IEEE.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: X. Li; The University of Texas Health Science Center at Houston, McGovern Medical School, Houston, United States; email: xiaojin.li@uth.tmc.edu; L. Cui; The University of Texas Health Science Center at Houston, McWilliams School of Biomedical Informatics, Houston, United States; email: licong.cui@uth.tmc.edu; Conference name: 14th IEEE International Conference on Healthcare Informatics, ICHI 2026; Conference date: 1 June 2026 through 3 June 2026; Conference code: 226950

Multi-instance positive and unlabeled learning with bi-level embedding

Item Type Journal Article

Author X. Tang

Author C. Xu

Author T. Luo

Author C. Hou

Abstract Multiple Instance Learning (MIL) is a widely studied learning paradigm which arises from real applications. Existing MIL methods have achieved prominent performances under the premise of plenty annotation data. Nevertheless, sufficient labeled data is often unattainable due to the high labeling cost. For example, the task in web image identification is to find similar samples among a large size of unlabeled dataset through a small number of provided target pictures. This leads to a particular scenario of Multiple Instance Learning with insufficient Positive and superabundant Unlabeled data (PU-MIL), which is a hot research topic in MIL recently. In this paper, we propose a novel method called Multiple Instance Learning with Bi-level Embedding (MILBLE) to tackle PU-MIL problem. Unlike other PU-MIL method using only simple single-level mapping, the bi-level embedding strategy are designed to customize specific mapping for positive and unlabeled data. It ensures the characteristics of key instance are not erased. Moreover, the weighting measure adopted in positive data can extracts the uncontaminated information of true positive instances without interference from negative ones. Finally, we minimize the classification error loss of mapped examples based on class-prior probability to train the optimal classifier. Experimental results show that our method has better performance than other state-of-the-art methods. © 2022 - IOS Press. All rights reserved.

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Multi-Label Quantification

Item Type Journal Article

Author A. Moreo

Author M. Francisco

Author F. Sebastiani

Abstract Quantification, variously called supervised prevalence estimation or learning to quantify, is the supervised learning task of generating predictors of the relative frequencies (a.k.a. prevalence values) of the classes of interest in unlabelled data samples. While many quantification methods have been proposed in the past for binary problems and, to a lesser extent, single-label multiclass problems, the multi-label setting (i.e., the scenario in which the classes of interest are not mutually exclusive) remains by and large unexplored. A straightforward solution to the multi-label quantification problem could simply consist of recasting the problem as a set of independent binary quantification problems. Such a solution is simple but naïve, since the independence assumption upon which it rests is, in most cases, not satisfied. In these cases, knowing the relative frequency of one class could be of help in determining the prevalence of other related classes. We propose the first truly multi-label quantification methods, i.e., methods for inferring estimators of class prevalence values that strive to leverage the stochastic dependencies among the classes of interest in order to predict their relative frequencies more accurately. We show empirical evidence that natively multi-label solutions outperform the naïve approaches by a large margin. The code to reproduce all our experiments is available online. © 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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MULTI-LEVEL AUGMENTATION CONSISTENCY LEARNING AND SAMPLE SELECTION FOR SEMI-SUPERVISED DOMAIN GENERALIZATION

Item Type Conference Paper

Author M. Yu

Author Y. Zhang

Author X. Li

Author R. Zhang

Author H. Jiang

Author J. Gao

Author Z. Liu

Abstract Semi-supervised domain generalization (SSDG) aims to build a domain-generalized model using partially labeled data from source domains. Mainstream SSDG methods follow the augmentation consistency in FixMatch. However, the extraction of domain-invariant features may be challenging due to the absence of feature-based operations, further leading to overfitting of the classifier. To this end, we propose Multi-level Augmentation Consistency Learning (MACMatch), which improves the generalization of feature extractor and classifier through feature-based augmentation consistency. On the other hand, existing methods assume labeled data are class-balanced and domain-balanced, which is easily violated in practice. Based on this, we introduce Representativity and Diversity-based Sample Selection (RDSS), which models data as graphs to evaluate reasonable samples for labeling, relaxing the assumption for labeled data. Experiments on PACS and OfficeHome demonstrate that MACMatch outperforms state-of-the-art SSDG methods. Furthermore, MACMatch with RDSS achieves competitive results without domain and class priori assumptions. Code is available at <https://github.com/Y-J-Zhang/MACMatch-RDSS>. ©2024 IEEE.

Date 2024**DOI** 10.1109/ICASSP48485.2024.10446462**URL** <https://www.scopus.com/pages/publications/105001479229?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: ICASSP IEEE Int Conf Acoust Speech Signal Process Proc**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 15206149 (ISSN)**Pages** 6550-6554**Proceedings Title** ICASSP IEEE Int Conf Acoust Speech Signal Process Proc**Conference Name** ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings**Date Added** 20/09/2026, 9:51:54 pm**Modified** 20/09/2026, 9:51:54 pm**Tags:**

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Multi-objective Learning to Overcome Catastrophic Forgetting in Time-series Applications

Item Type Journal Article**Author** R.A. Mahmoud**Author** H. Hajj

Abstract One key objective of artificial intelligence involves the continuous adaptation of machine learning models to new tasks. This branch of continual learning is also referred to as lifelong learning (LL), where a major challenge is to minimize catastrophic forgetting, or forgetting previously learned tasks. While previous work on catastrophic forgetting has been focused on vision problems; this work targets time-series data. In addition to choosing an architecture appropriate for time-series sequences, our work addresses limitations in previous work, including the handling of distribution shifts in class labels. We present multi-objective learning with three loss functions to minimize catastrophic forgetting, prediction error, and errors in generalizing across label shifts, simultaneously. We build a multi-task autoencoder

network with a hierarchical convolutional recurrent architecture. The proposed method is capable of learning multiple time-series tasks simultaneously. For cases where the model needs to learn multiple new tasks, we propose sequential learning, starting with tasks that have the best individual performances. This solution was evaluated on four benchmark human activity recognition datasets collected from mobile sensing devices. A wide set of baseline comparisons is performed, and an ablation analysis is run to evaluate the impact of the different losses in the proposed multi-objective method. The results demonstrate an up to 4% performance improvement in catastrophic forgetting compared to the use of loss functions in state-of-the-art solutions while demonstrating minimal losses compared to upper bound methods of traditional fine-tuning (FT) and multi-task learning (MTL). © 2022 Association for Computing Machinery.

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Multi-Source Remote Sensing Data Fusion for Aboveground Biomass Estimation in Tropical Forests: Recent Advances, Challenges, and Future Trends

Item Type Journal Article

Author Q. Meng

Author Y.-B. Leau

Author J. Shi

Author J. Zhou

Abstract Accurate estimation of aboveground biomass (AGB) in tropical forests is critical for global carbon cycle research and climate policies such as REDD+, but single remote sensing sources face limitations: optical sensors suffer from saturation and cloud interference, SAR has saturation and terrain-induced distortions, and lidar (e.g., GEDI) provides high-precision vertical structures but sparse coverage. Multi-source data fusion - including data-level, feature-level, decision-level fusion, and spatio-temporal scaling - leverages the complementary strengths of optical, SAR, and LiDAR data to mitigate these issues, machine learning (e.g., RF, XGBoost) and deep learning (e.g., CNN, Transformer) to improve performance, and hybrid models to balance learning ability and robustness. Key challenges including data heterogeneity, geometric/radiometric inconsistencies, saturation in dense forests, cloud-induced optical data loss, and sparse GEDI footprints are addressed through validation frameworks (involving forest maps, NFI, airborne LiDAR) and uncertainty quantification methods (e.g., spatial cross-validation, Monte Carlo simulations). Regional case studies (Amazon, Congo Basin, Southeast Asia) show optimal strategies for specific environments, while future directions focus on collaborative GEDI with P-band SAR, dynamic monitoring through temporal fusion, physical based deep learning, federated/transfer learning, interpretable AI, and standardized REDD+ protocols. Based on a systematic review of AGB estimation studies in tropical forests in the past five years, this paper summarizes the development progress of multi-source remote sensing data fusion and advanced algorithms, identifies the existing technical bottlenecks, and proposes a future-oriented comprehensive optimization framework to improve the accuracy and robustness of AGB estimation. © 2013 IEEE.

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Multi-Target Detection in Underwater Sensor Networks Based on Bayesian Deep Learning

Item Type Journal Article**Author** X. Du**Author** Y. Wen**Author** J. Yan**Author** Y. Zhang**Author** X. Luo**Author** X. Guan

Abstract Underwater target detection and its development have an important role in advancing marine science and technology. However, the complex and dynamic underwater environment poses challenges for detecting non-cooperative targets. This paper focuses on the problem of detecting and recognizing multiple non-cooperative targets in USNs. Specifically, the generative model is firstly utilized to learn the probability distribution of underwater signals, and then Bayesian fusion of active and passive measurements is utilized to achieve target detection. Along with this, a Bayesian deep learning classification framework is employed to categorize multiple targets. Compared to the traditional statistical detection methods, our method excels in handling underwater complexity and dynamics. In addition, unlike traditional deep learning, our classification framework combines Bayesian inference with deep learning to quantify environmental uncertainty. This approach helps the model perform more robust detection and improves the management of noise and uncertainty. Experimental and simulation analysis demonstrate the effectiveness of Bayesian deep learning methods in solving the challenges of underwater target detection. These findings highlight the potential of our approach in enhancing sensing and surveillance capabilities in complex underwater environments. © 2013 IEEE.

Date 2025**DOI** 10.1109/TNSE.2025.3535572**URL** <https://www.scopus.com/pages/publications/105003803223?origin=resultslist>**Archive** Scopus**Language** English**Volume** 12**Publisher** IEEE Computer Society**Pages** 1581-1596**Publication** IEEE Transactions on Network Science and Engineering**ISSN** 23274697 (ISSN)

Issue 3**Journal Abbr** IEEE Trans. Netw. Sci. Eng.**Date Added** 20/09/2026, 9:52:01 pm**Modified** 20/09/2026, 9:52:01 pm**Tags:**

Markov processes, Deep learning, Contrastive Learning, Decisions fusion, Active learning, Bayesian, Bayesian deep learning, Active measurement, Classification framework, decision fusion, multi-target detection, Multi-target detection, Non-cooperative target, passive measurement, Passive measurements, Underwater environments, Underwater target detection

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Multi-Task Attention-Guided Deep Learning for Simultaneous Breast Cancer Detection and Density Estimation in Mammography

Item Type Journal Article**Author** G. Esen**Author** M. Nurtas**Author** L. la Paglia**Author** J. Amankulov**Author** B. Matkerim**Author** A. Altaibek

Abstract Early detection of breast cancer is critical for improving patient outcomes, and deep learning (DL) has shown promise in mammographic analysis. However, many existing systems tackle a single task (e.g., malignancy classification) and provide limited interpretability, which hinders clinical adoption. We present a multi-task, attention-guided framework that jointly predicts malignancy (benign vs. malignant) and BI-RADS breast density (A–D) from a single mammogram while producing interpretable attention maps. The architecture couples a pretrained convolutional backbone with lightweight channel–spatial attention and two task-specific heads. We evaluate six backbones (ResNet-50, DenseNet-121, EfficientNet-B0/B3/B7, MobileNetV3-Large) under a rigorous evaluation protocol with stratified hold-out split (64%/16%/20%) and 5-fold cross-validation. To support reliable deployment, we report comprehensive metrics including discrimination measures (ROC-AUC, PR-AUC, F1-score, sensitivity, specificity), calibration metrics (Brier score, Expected

Calibration Error), and ablation studies contrasting single-task vs. multi-task learning and models with vs. without attention mechanisms. Across all evaluation protocols, multi-task learning consistently outperforms single-task baselines by 2.8% in pathology accuracy and 3.5% in density estimation, indicating that density cues provide complementary information for malignancy assessment. Among backbones, EfficientNet-B3 attains the strongest discrimination (AUC= 0.962, accuracy= 93.6%) on hold-out test with excellent calibration (Brier= 0.010), while MobileNetV3-Large achieves competitive performance (AUC= 0.940, accuracy= 91.2%) with only 4.26M parameters—an optimal accuracy–efficiency trade-off for resource-constrained deployment. Cross-validation results demonstrate robust generalization (AUC = 0.954 ± 0.003 , accuracy = $93.0 \pm 0.3\%$). Attention maps highlight diagnostically relevant regions including architectural distortions and suspicious calcifications, offering interpretable visual rationale aligned with radiological assessment. We discuss limitations including dataset size (410 images), class imbalance in rare density categories, and calibration challenges (ECE= 0.365–0.419), and outline directions for external validation on multi-institutional cohorts and integration of uncertainty quantification methods. © 2013 IEEE.

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Notes:

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Multi-Time Scale Reserve Requirement Quantification Method Based on Empirical Mode Decomposition

Item Type Conference Paper

Author D. Wang

Author L. Zhang

Author Q. Lin

Author K. Wang

Author S. Qiu

Author X. Yu

Author Y. Wang

Author Z. Lin

Abstract With the increasing penetration of renewable energy there is an urgent need to consider the influence of renewable generation forecast errors on the reserve requirement. In the paper, referring to the multi-time scale classification system for reserve resources based on their response times, a multi-time scale reserve requirement quantification method is proposed based on based on Empirical Mode Decomposition (EMD). In this method, the net load forecast error samples are decomposed into multiple fluctuation components. Based on the dominant period, the fluctuation components are subjected to envelope reconstruction, thereby quantifying the multi-time scale reserve requirement. Finally, case studies on net load forecast error curve samples are presented, and the result shows that the proposed model can effectively optimize the allocation of multitime scale reserve requirement, reduces reserve overestimation. Through the calculation of reserve requirements at various renewable penetration rate, proactive adjustments to reserve requirement can be made, guiding grid planning efforts. © 2026 IEEE.

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Multiclass microarray data classification based on confidence evaluation

Item Type Journal Article

Author H.L. Yu

Author S. Gao

Author B. Qin

Author J. Zhao

Abstract Microarray technology is becoming a powerful tool for clinical diagnosis, as it has potential to discover gene expression patterns that are characteristic for a particular disease. To date, this possibility has received much attention in the context of cancer research, especially in tumor classification. However, most published articles have concentrated on the development of binary classification methods while neglected ubiquitous multiclass problems. Unfortunately, only a few multiclass classification approaches have had poor predictive accuracy. In an effort to improve classification accuracy, we developed a novel multiclass microarray data classification method. First, we applied a "one versus rest-support vector machine" to classify the samples. Then the classification confidence of each testing sample was evaluated according to its distribution in feature space and some with poor confidence were extracted. Next, a novel strategy, which we named as "class priority estimation method based on centroid distance", was used to make decisions about categories for those poor confidence samples. This approach was tested on seven benchmark multiclass microarray datasets, with encouraging results, demonstrating effectiveness and feasibility. ©FUNPEC-RP.

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Export Date: 20 September 2026; Cited By: 9; Correspondence Address: H. L. Yu; School of Computer Science and Engineering, Jiangsu University of Science and Technology, Zhenjiang, China; email: yuhualong@just.edu.cn

Multidimensional analysis and predictive modeling of cognitive decline risk in the United States using propensity score matching and machine learning

Item Type Journal Article

Author J. Li

Author G. Chen

Author Y. Liu

Abstract Cognitive decline, an early indicator of neurodegenerative disorders, presents a growing public health challenge. This study aimed to integrate causal inference and machine learning to quantify the causal impact of high-risk status on cognitive function, explore geographic and temporal heterogeneity, and develop predictive models for early identification of at-risk individuals in the United States. We analyzed 22,182 records from the behavioral risk factor surveillance system. Causal effects were estimated using propensity score matching and doubly robust estimation to mitigate confounding. Geographic and temporal heterogeneity were assessed through stratified analyses. Predictive models were developed using random forest and XGBoost, trained on 80% of the dataset and evaluated on 20%, with performance assessed by accuracy, precision, recall, and F1 score. Propensity score matching estimated an average treatment effect (ATE) of 19.92 points (95% confidence interval: 19.70–20.15), and doubly robust estimation yielded an ATE of 16.95 points (95% confidence interval: 16.77–17.14), confirming a significant causal link between high-risk status and cognitive decline. Regional heterogeneity was pronounced, with

US territories showing the highest ATE (59.40). Temporal analysis from 2015 to 2022 revealed no significant overall trend ($P = .54$), although annual fluctuations were observed. The random forest model achieved the best predictive performance (accuracy = 0.77, $F1 = 0.67$), outperforming XGBoost in balancing precision and recall. The findings demonstrate a significant causal relationship between high-risk status and cognitive decline, with marked geographic disparities. The integration of causal inference with machine learning provides a robust framework for both understanding risk factors and building practical screening tools. These results offer actionable insights for targeting public health interventions and underscore the potential of data-driven approaches in cognitive health management. Copyright © 2026 the Author(s). Published by Wolters Kluwer Health, Inc.

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Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, risk assessment, Risk Assessment, United States, machine learning, aged, Aged, Machine Learning, risk factor, Risk Factors, geographic distribution, predictive model, procedures, diagnosis, cognition, risk prediction, epidemiology, health outcome, prevalence, random forest, etiology, data accuracy, health care access, temporal analysis, machine learning algorithm, cognitive defect, behavioral risk factor surveillance system, Behavioral Risk Factor Surveillance System, causal attribution, causal inference, circannual rhythm, cognitive decline, Cognitive Dysfunction, Guam, high risk population, low risk population, propensity score, Propensity Score, propensity score matching, Puerto Rico, treatment effect heterogeneity, Virgin Islands (U.S.)

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Multilevel Feature Score Learning for Few-Shot Open-Set Recognition of Hyperspectral Images

Item Type Journal Article

Author Z. Zhang

Author Z. Li

Author Y. Wang

Author X. Gong

Author J. Guo

Author J. Tian

Abstract Few-shot open-set recognition (FSOSR) for hyperspectral images (HSIs) is a challenging task of great practical value. It aims to classify a sample to one of the known classes by few examples while being able to reject the sample from unknown classes. Prior research mainly focuses on high-level semantic features and ignores mid-level and low-level detailed features. Furthermore, recent endeavors in generative few-shot open-set recognition (FSOSR) have the limitation that the generated samples do not accurately reflect the original data distribution and introduce additional noise. To solve these problems, a novel FSOSR method for HSI based on multilevel feature score learning (MFSL) is proposed, which considers multilevel features and combines semantic information with detailed information. By designing the multilayer score fusion (MSF) module, the similarity of multilevel features and prototypes is converted into scores, so that the score of known samples is lower and the score of unknown samples is higher. The intra–inter similarity contrast enhancement (ISCE) module is also introduced to enhance the discrimination of features, increasing the intraclass similarity while decreasing the interclass similarity by adjusting the similarity of known samples and prototypes. Experiments on two HSI datasets verify the effectiveness of MFSL. © 2004-2012 IEEE.

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Multimodal algorithm-optimized chirped ultrasound thermography for enhanced quantitative detection of metal fatigue cracks

Item Type Journal Article

Author R. Li

Author F. Wang

Author Y. Liu

Author F. Yang

Author S. Sfarra

Author L. Liu

Author Y. Yang

Author V.S. Ghali

Author G.T. Vesala

Author R. Mulaveesala

Author H. Yue

Author J. Liu

Abstract Metal fatigue cracks under cyclic loading conditions cannot be ignored. Ultrasound-induced thermography has been emerged as an effective technique for crack detection, however, single-frequency ultrasound excitation often produces weak thermal signals, especially for small or shallow cracks, limiting identification accuracy. To overcome these challenges, Multimodal Algorithm-enhanced Chirped

Ultrasound-Induced Thermography (MA-CUIT) was proposed for quantitative crack characterization. Firstly, metal cracks three-dimensional thermal-wave model under chirp excitation was established to analyze temperature distribution and thermal diffusion. Subsequently, a comprehensive feature extraction framework integrating frequency-domain, time-domain, and machine learning methods was developed under chirp ultrasound excitation, with systematic optimization of frequency modulation parameters to determine optimal excitation ranges. MA-CUIT experimental platform was established for quantification of real crack defects. The results indicate that MA-CUIT, achieved exceptional crack sizing accuracy with relative errors of 5.9% (FFT phase optimized), 6.7% (PLSR 1st coefficient optimized), and 6.3% (DOD phase optimized), establishing MA-CUIT optimized by FFT phase analysis as the most precise quantification method. © 2025

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Multimodal out-of-distribution individual uncertainty quantification enhances binding affinity prediction for polypharmacology

Item Type Journal Article

Author A. Badkul

Author L. Xie

Author S. Zhang

Author L. Xie

Abstract Polypharmacology, a single drug that targets multiple proteins, holds promise for addressing unmet medical needs. Achieving accurate, reliable and scalable predictions of protein–ligand binding affinity across multiple proteins is crucial to realizing the potential of polypharmacology. Machine learning offers a powerful tool for multitarget binding affinity prediction. However, three major challenges remain: generalizing predictions to out-of-distribution compounds that are structurally different from those in the training data; quantifying the uncertainty of predictions in out-of-distribution scenarios where the assumption underlying existing methods does not hold; and scaling to billions of compounds, which remains unattainable for current structure-based methods. Here, to overcome these challenges, we propose a model-agnostic anomaly detection-based individual uncertainty quantification method: embedding Mahalanobis Outlier Scoring and Anomaly Identification via Clustering (eMOSAIC). eMOSAIC features the divergence between the multimodal representations of known cases and unseen instances and quantifies individual prediction uncertainty on a compound-by-compound basis. We integrate eMOSAIC with a multimodal deep neural network for multitarget ligand binding affinity predictions, leveraging a structure-informed large protein language model. Comprehensive validation in out-of-distribution settings demonstrates that eMOSAIC significantly outperforms state-of-the-art sequence-based and structure-based methods as well as existing uncertainty quantification approaches. These findings underscore eMOSAIC’s potential to advance real-world polypharmacology and other applications that require robust predictions and scalable solutions. © The Author(s), under exclusive licence to Springer Nature Limited 2025.

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Multioriented video scene text detection through bayesian classification and boundary growing

Item Type Journal Article

Author P. Shivakumara

Author R.P. Sreedhar

Author T.Q. Phan

Author S. Lu

Author C.L. Tan

Abstract Multioriented text detection in video frames is not as easy as detection of captions or graphics or overlaid texts, which usually appears in the horizontal direction and has high contrast compared to its background. Multioriented text generally refers to scene text that makes text detection more challenging and interesting due to unfavorable characteristics of scene text. Therefore, conventional text detection methods may not give good results for multioriented scene text detection. Hence, in this paper, we present a new enhancement method that includes the product of Laplacian and Sobel operations to enhance text pixels in videos. To classify true text pixels, we propose a Bayesian classifier without assuming a priori probability about the input frame but estimating it based on three probable matrices. Three different ways of clustering are performed on the output of the enhancement method to obtain the three probable matrices. Text candidates are obtained by intersecting the output of the Bayesian classifier with the Canny edge map of the input frame. A boundary growing method is introduced to traverse the multioriented scene text lines using text candidates. The boundary growing method works based on the concept of nearest neighbors. The robustness of the method has been tested on a variety of datasets that include our own created data (nonhorizontal and horizontal text data) and two publicly available data, namely, video frames of Hua and complex scene text data of ICDAR 2003 competition (camera images). Experimental results show that the performance of the proposed method is encouraging compared with results of existing methods in terms of recall, precision, F-measures, and computational times. © 2012 IEEE.

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Multiple imputation of missing data using efficient machine learning approach

Item Type Journal Article

Author S. Kanchana

Author A.S. Thanamani

Abstract Missing values are an unavoidable problem in dealing with most of the real world data sources, and the missing values in a dataset may affect the quality of the supervised learning process. This paper focuses machine learning techniques Naïve Bayesian Classifier method for the imputation of missing values in large datasets. Single value imputation produce biased results, whereas multiple imputation generates right value to replace. The study was conducted to implement several algorithms in unsupervised techniques like Mean, Median, Standard Deviation, and Regression and supervised machine learning techniques like Naïve Bayesian classifier provides different methods to calculate Class prior probability, Posterior probability, Predictor probability and Maximum Likelihood. Both methods are trying to find hidden structure in unlabeled data. The performance of this method has been compared by using Correlation statistics analysis which produces the imputed values are positively related or negatively related or not related with each other. To evaluate the performance, the standard machine learning repository dataset has been used. We

experimentally shows that our approach significantly outperforms some standard machine learning methods for handling missing values in classification tasks. © Research India Publications.

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Multiple Summary Knowledge-Based Disease Prevalence Estimation for Group Testing Data

Item Type Journal Article

Author W. Yang

Author J. Ding

Author W. Xiong

Author S. Wang

Author Q. Li

Abstract Group testing is widely recognized for its ability to reduce time and cost in screening for rare diseases. However, with limited samples, the parameter estimation from group-testing data might be inefficient. In some scenarios, the external summary statistics are available. To make full use of them, we propose a multiple-source knowledge transfer method. The resulting prevalence estimator is consistent and more efficient than the conventional estimator based on the target group-testing data alone. We establish the asymptotic normality of the proposed estimator. In addition, we develop a data-driven screening procedure to identify the transferable sources. We extend this framework to Dorfman's two-stage group testing design and show that integrating the retesting data yields efficiency gains. The asymptotic normality of the corresponding estimator is also established. Extensive simulations and a real-data

application support the theoretical results and demonstrate the favorable finite-sample performance of the proposed method. © 2026 John Wiley & Sons Ltd.

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Multiscale 1-DCNN for Damage Localization and Quantification Using Guided Waves With Novel Data Fusion Technique and New Self-Attention Module

Item Type Journal Article

Author Y. Liao

Author Y. Wang

Author X. Zeng

Author M. Wu

Author X. Qing

Abstract This article proposes an innovative damage localization and size quantification method named as MSCNNSAM based on a new multiscale convolutional neural network (MSCNN) and novel small-weighted zero-setting self-attentive module (SAM) in carbon fiber reinforced plastic structures. First, an improved piecewise aggregate approximation algorithm (IPAA) is developed to compress the guided wave signal and extract a series of damage indexes (DI). Considering the different effects of the damage location on the different sensing paths, a new method of damage information targeting enhancement and multipath data fusion is proposed. Then, a novel MSCNN architecture is also proposed for the inherent multiscale characteristics of the guided wave signal, which takes the multipath fused data as input and uses regression and classification methods to directly predict the location and size of the damage. Finally, to further improve the performance of the MSCNN, a SAM is proposed to effectively avoid the influence of low-information channel features and improve the damage feature extraction capability of the network. The proposed method is evaluated through experiments on a guided wave testing platform. Experimental results and a comprehensive comparison analysis with respect to the state-of-the-art damage localization and quantification. methods have demonstrated the superiority of the proposed MSCNNSAM. © 2005-2012 IEEE.

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Multiscale characterization of heart failure

Item Type Journal Article

Author F. Sahli Costabal

Author J.S. Choy

Author K.L. Sack

Author J.M. Guccione

Author G.S. Kassab

Author E. Kuhl

Abstract Dilated cardiomyopathy is a progressive irreversible disease associated with contractile dysfunction and heart failure. During dilated cardiomyopathy, elevated diastolic wall strains trigger mechanotransduction pathways that initiate the addition of sarcomeres in series and an overall increase in myocyte length. At the whole organ level, this results in a chronic dilation of the ventricles, an increase in end diastolic and end systolic volumes, and a decrease in ejection fraction. However, how exactly changes in sarcomere number translate into changes in myocyte morphology, and how these cellular changes translate into ventricular dilation remains incompletely understood. Here we combined a chronic animal study, continuum growth modeling, and machine learning to quantify correlations between sarcomere dynamics, myocyte morphology, and ventricular dilation. In an eight-week long volume overload study of six pigs, we found that the average sarcomere number increased by +3.8%/week, from 47 to 62, resulting in a myocyte lengthening of +3.3%/week, from 85 to 108 μm , while the sarcomere length and myocyte width remained unchanged. At the same time, the average end diastolic volume increased by +6.0%/week. Using continuum growth modeling and Bayesian inference, we correlated alterations on the subcellular, cellular, and organ scales and found that the serial sarcomere number explained 88% of myocyte lengthening, which, in turn, explained 54% of cardiac dilation. Our results demonstrate that sarcomere number and myocyte length are closely correlated and constitute the major determinants of dilated heart failure. We anticipate our study to be a starting point for more sophisticated multiscale models of heart failure. Our study suggests that altering sarcomere turnover—and with it myocyte morphology and ventricular dimensions—could be a potential therapeutic target to attenuate or reverse the progression of heart failure. **Statement of Significance:** Heart failure is a significant global health problem that affects more than 25 million people worldwide and increases in prevalence as the population ages. Heart failure has been studied excessively at various scales; yet, there is no compelling concept to connect knowledge from the subcellular, cellular, and organ level across the scales. Here we combined a chronic animal study, continuum growth modeling, and machine learning to quantify correlations between sarcomere dynamics, myocyte morphology, and ventricular dilation. We found that the serial sarcomere number explained 88% of myocyte lengthening, which, in turn, explained 54% of cardiac dilation. Our results show that sarcomere number and myocyte length are closely correlated and constitute the major determinants of dilated heart failure. This suggests that altering the sarcomere turnover—and with it myocyte morphology and ventricular dimensions—

could be a potential therapeutic target to attenuate or reverse heart failure. © 2019 Acta Materialia Inc.

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N-BEATS-GAN: A risk-aware financial time series forecasting with generative adversarial networks

Item Type Journal Article

Author M. Dai

Author M.Y. Seow

Author R.S. Filho

Author A. Sclip

Author K. Chawla

Author R. Siow Mong Goh

Author J. Chatteraj

Abstract Time series forecasting is essential for informed decision-making in financial markets, but it poses challenges due to the stochastic nature of financial data. Existing forecasting methods, including classical statistical models and machine learning approaches, often struggle with the complexities of multi-series, multi-output predictions. This paper introduces a deep learning framework, N-BEATS-GAN, which combines the strengths of Generative Adversarial Networks (GANs) for risk modelling with the efficient trend predictive capabilities of the deep learning model N-BEATS. The method integrates GAN loss with the Transformation Invariant Loss function with Distance Equilibrium (TILDE-Q) loss to address amplitude and phase distortions in time series forecasting and accelerate model convergence. A dataset derived from MSCI (Morgan Stanley Capital International) Country Index data across ten countries is used for validation. Predictive accuracy is assessed using Mean Absolute Error, Mean Squared Error, Continuous Ranked Probability Score, and a risk quantification method is proposed to evaluate the capacity to capture financial risk. Experimental results demonstrate that N-BEATS-GAN significantly outperforms StockMixer, N-BEATS, DLinear, and PAGAN in both prediction accuracy and risk quantification. N-BEATS-GAN overcomes the limitations of existing models, enabling effective multi-series, multi-output forecasting while accounting for inherent risk in financial time series, highlighting its potential to enhance forecasting and risk assessment in financial markets. © 2025 Elsevier B.V.

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Nanoparticles: A global vision. Characterization, separation, and quantification methods. Potential environmental and health impact

Item Type Journal Article

Author A. López-Serrano

Author R.M. Olivas

Author J.S. Landaluze

Author C. Cámara

Abstract Nanotechnology is a multidisciplinary science that includes scientific and technological activities at molecular and atomic scales (1-100 nm). Scientific principles and new properties can be understood and controlled when working at this scale. Nanotechnology is advancing rapidly due to the great progress achieved in various fields including electronics, mechanics, medicine, cosmetics, food, etc. This increased use of nano-sized materials leads to the release of a substantial amount of nanoparticles into the environment. To date there is a lack of standardized procedures to assess their safety and impact on the environment. Specific toxicological studies as well as characterization and quantification of nanoparticles are required to establish regulations to control field application of nanoscale materials. A classification of nanoparticles and the techniques employed in their characterization, separation, and quantification are summarized and described in this review. A global perspective on nanoparticle exposure and environmental effect is also discussed. © 2014 The Royal Society of Chemistry.

Date 2014

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Export Date: 20 September 2026; Cited By: 261; Correspondence Address: R.M. Olivas; Department of Analytical Chemistry, Faculty of Chemistry, University Complutense de Madrid, 28040 Madrid, Spain; email: rimunoz@ucm.es

Nanoplastics and co-pollutants negatively affect the photosynthesis of photosynthetic organisms: insights from meta-analysis and machine learning

Item Type Journal Article

Author C. Wu

Author Y. Wang

Author J. Wu

Author D. Zhou

Author J. Ren

Author J. Yang

Author X. Li

Author G. Liu

Author Y. Wu

Author X. Yan

Abstract Nanoplastics are increasingly recognized as emerging stressors that impair photosynthesis in photosynthetic organisms, and their co-occurrence with environmental pollutants can further intensify phytotoxic effects. To address the fragmented understanding of these combined impacts, we integrated a meta-analysis of 1067 experimental observations with explainable machine learning to quantify how nanoplastics and co-pollutants jointly influence photosynthetic processes. The results showed predominantly negative effects across light reactions and rubisco enzyme activity, with organism species, nanoplastics dose, size, and exposure duration identified as major determinants of photosynthetic inhibition. High-sensitivity indicators of photosynthetic stress were identified, and corn emerged as a particularly sensitive organism. By linking physicochemical properties and exposure conditions to quantitative photosynthetic outcomes, this study provides a clearer and more accessible framework for predicting phytotoxicity and supports improved ecological risk assessment and management strategies. © 2026 Elsevier B.V.

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NC 2 E: boosting few-shot learning with novel class center estimation

Item Type Journal Article

Author Z. Wu

Author C. Shen

Author K. Guo

Author E. Luo

Author L. Wang

Abstract Accurate class distribution estimation is expected to solve the problem of the poor generalization ability that exists in few-shot learning models due to data shortages. However, the reliability of class distributions estimates based on limited samples and knowledge is questionable, especially for similar classes. We find that the distribution calibration method is inaccurate in estimating similar classes due to limited knowledge being reused through double-validation experiments. To address this issue, we propose a novel class center estimation (NC2E) method, which consists of a two-stage center estimation (TCE) algorithm and a class centroid estimation (CCE) algorithm. The class centers estimated by TCE in two stages are closer to the truth,

and its superiority is demonstrated by error theory. CCE searches for the centroid of the base class iteratively and is used as the basis for the novel class calibration. Sufficient simulation samples are generated based on the estimated class distribution to augment the training data. The experimental results show that, compared with the distribution calibration method, the proposed method achieves an approximately 1% performance improvement on the miniImageNet and CUB datasets; an approximately 1.45% performance improvement for similar class classification; and an approximately 6.06% performance improvement for non-similar class classification. © 2022, The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature.

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Network intrusion detection with cascading classification

Item Type Conference Paper

Author S. Marcek

Author M. Drozda

Editor Al-Dabass D.

Editor Sauli Z.

Editor Zakaria Z.

Abstract The KDD99 network intrusion contest and the related intrusion data sets attracted increased attention of the research community. The success rate of contest participants was evaluated in terms of the obtained classification cost. The classification cost of the contest winner was 0.2331, the best approach prior to our work carries the classification cost of 0.2224. We show that a simple approach based on cascading classification leads to the classification cost of 0.2079. Cascading classification is in our case done by applying 2-nearest-neighbor classification. The samples which could not be predicted with 2-nearest-neighbor classification (4-6%) are further classified with a clustering approach with class priority. This clustering approach when applied in isolation under performs other approaches. However, when applied in cascading classification, it can take advantage of the reduced number of samples. We argue that cascading classification is a viable alternative in scenarios where less complex machine learning approaches are favorable, for example due to possible performance degradation in resource constrained devices such as mobile phones, embedded systems or sensors. © 2014 IEEE.

Date 2015

DOI 10.1109/ISMS.2014.17

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Intelligent systems, Nearest neighbor search, Intrusion detection, Network intrusion detection, Embedded systems, Research communities, Performance degradation, Classifier learning, Cascading classification, Clustering approach, Exact Euclidean Locality Sensitive Hashing (E2LSH), K-nearest neighbor classification, K-nearest-neighbor classification, KDD99 classifier learning contest, Locality sensitive hashing, Resourceconstrained devices

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Network quantification despite biased labels

Item Type Conference Paper

Author L. Tang

Author H. Gao

Author H. Liu

Abstract The increasing availability of participatory web and social media presents enormous opportunities to study human relations and collective behaviors. Many applications involving decision making want to obtain certain generalized properties about the population in a network, such as the proportion of actors given a category, instead of the category of individuals. While data mining and machine learning researchers have developed many methods for link-based classification or relational learning, most are optimized to classify individual nodes in a network. In order to accurately estimate the prevalence of one class in a network, some quantification method has to be used. In this work, two kinds of approaches are presented: quantification based on classification or quantification based on link analysis. Extensive experiments are conducted on several representative network data, with interesting findings reported concerning efficacy and robustness of different quantification methods, providing insights to further quantify the ebb and flow of online collective behaviors at macro-level. © 2010 ACM.

Date 2010

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Network Security Evaluation and Optimal Active Defense based on Attack and Defense Game Model

Item Type Conference Paper**Author** F. Li

Abstract With the rapid development of network technology, information spreads rapidly through the network. High-speed wide area network and broadband metropolitan area network have been widely used in enterprise communications, promoting the improvement of enterprise information level in modern society, and enhancing the market competitiveness of enterprises. However, it is worth noting that while Internet technology and the Internet are widely used, the activities of viruses and hackers are increasingly rampant, and various new network attacks and viruses emerge in an endless stream, bringing unprecedented security risks to network security. The target of attackers is to penetrate the network and steal valuable data. The goal of the defender is to prevent attackers from intruding into the network. Attacks can be divided into two types: reconnaissance and violent attack. Reconnaissance is when an attacker tries to find information about the target, while a violent attack attempts all possible passwords or key combinations until a valid password or password is found. Violent attacks are more likely to succeed against poorly protected systems because they do not require time to test each password or key combination. In order to evaluate the network information system security and active defense, the network defense graph model, the classification of attack and defense strategies and the cost quantification method, the network attack and defense game model and the optimal active defense selection algorithm based on the above models are proposed. Finally, a typical network example is used to analyze the application of the above models and algorithms in network security evaluation and optimal active defense. The analysis results show that the proposed models and methods are feasible and effective. © 2023 IEEE.

Date 2023**DOI** 10.1109/ICDCECE57866.2023.10151226**URL** <https://www.scopus.com/pages/publications/85164251275?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IEEE Int. Conf. Distrib. Comput. Electr. Circuits Electron., ICDCECE**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-835034745-6 (ISBN)**Proceedings Title** IEEE Int. Conf. Distrib. Comput. Electr. Circuits Electron., ICDCECE

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Network security, Authentication, Classification (of information), Networks security, Computer viruses, Personal computing, Competition, Wide area networks, Viruses, Network attack, Active defense, attack and defense game model, Attack and defense game model, Game models, Network defense, network security evaluation, Network security evaluation, Network technologies, optimal active defense, Optimal active defense, Security evaluation

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Network-based disease fingerprinting with neuroinflammation PET imaging

Item Type Journal Article

Author L. Barzon

Author L. Maccioni

Author M. Carranza Mellana

Author J.J. Schubert

Author L. Brusafferri

Author O. Cousins

Author I. Rosenzweig

Author Y. Mizuno

Author T.R. Marques

Author N.A. Harrison

Author T. Fryer

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Author C. Pariante

Author D. Sharp

Author G. Scott

Author J.B. Pereira

Author O. Howes

Author V. Sossi

Author B. Bodini
Author B. Stankoff
Author M.L. Loggia
Author F.E. Turkheimer
Author M. Veronese

Abstract Neuroinflammation is a hallmark of numerous neurodegenerative, psychiatric, and chronic pain disorders and can be assessed in vivo with 18 kDa translocator protein (TSPO) positron emission tomography (PET). However, conventional quantification methods of TSPO PET are limited and often overlook the spatial relationships between regional signals. The application of network-based approaches to TSPO PET imaging may provide a novel framework to capture disease-specific neuroinflammatory patterns. To address this question, here we developed a data-driven, network-based approach to generate individual brain-wide TSPO PET matrices, employing Euclidean distance to quantify inter-regional pharmacokinetics similarity. We applied this approach to a large multicenter dataset of 528 PET scans utilizing three different TSPO tracers ([11C]-PBR28, [18F]-DPA714, [11C]-PK11195), including healthy controls and patients with different diseases such as multiple sclerosis, traumatic brain injury, schizophrenia, depression, and chronic low back pain. Statistical modelling and machine learning classifiers were applied to evaluate the impact of experimental and biological factors on TSPO similarity patterns and to investigate their potential for capturing disease-specific signatures. TSPO similarity patterns demonstrated high biological specificity and reproducibility, with strong test–retest correlations (mean Spearman’s $\rho = 0.84$). Average precision of disease classification exceeded chance performance by 23–89% across conditions and was driven by condition-specific regional hubs whose topological distributions closely mirrored disease pathophysiology. This specificity was further supported by minimal overlap in feature importance values across conditions. Altogether, our findings show that network-based analysis of human TSPO PET data can detect disease-specific neuroinflammatory signatures. Such methodologies underscore the biological significance of TSPO PET and enhance its translational value, supporting precision medicine strategies for neuroinflammatory disorders. © The Author(s) 2026.

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Neural minimax classifiers

Item Type Conference Paper
Author R. Alaiz-Rodríguez
Author J. Cid-Sueiro
Editor Dorronsoro J.R.
Editor Dorronsoro J.R.

Abstract Many supervised learning algorithms are based on the assumption that the training data set reflects the underlying statistical model of the real data. However, this stationarity assumption may be partially violated in practice: for instance, if the cost of collecting data is class dependent, the class priors of the training data set may be different from that of the test set. A robust solution to this problem is selecting the classifier that minimize the error probability under the worst case conditions. This is known as the minimax strategy. In this paper we propose a mechanism to train a neural network in order to estimate the minimax classifier that is robust to changes in the class priors. This procedure is illustrated on a softmax-based neural network, although it can be applied to other structures. Several experimental results show the advantages of the proposed methods with respect to other approaches. © Springer-Verlag Berlin Heidelberg 2002.

Date 2002

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Condition, Data collection, Neural-networks, Neural networks, Adversarial machine learning, Contrastive Learning, Algorithms, Statistical tests, Self-supervised learning, Test sets, Training dataset, Minimax, Statistic modeling, Training data sets, Statistical modeling, Error probabilities, Minimax strategy, Stationarity, Heidelberg, Robust solutions

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Neural networks involving the medial temporal structures in temporal lobe epilepsy

Item Type Journal Article

Author F. Bartolomei

Author F. Wendling

Author J.-J. Bellanger

Author J. Régis

Author P. Chauvel

Abstract Objectives: In a previous study using the averaged coherence technique to study interactions between medial/limbic and lateral/neocortical regions, we observed that epileptogenic networks in temporal lobe epilepsy seizures (TLES) could be divided into 4 subtypes, i.e. medial (M), medial-lateral (ML), lateral-medial (LM), and lateral (L). In the ML and LM subtypes, medial structures and the anterior temporal neocortex are co-activated at the onset of seizures. However, using this approach, we were unable to determine the direction of coupling and may have overlooked non-linear variations in interdependency. The purpose of the present study using non-linear regression for analysis of stereoelectroencephalographic (SEEG) signal pairs

was to measure the degree and direction of coupling between medial and neocortical areas during TLES in patients with the M, ML, and LM subtypes. Methods: Eighteen patients with drug-resistant TLEs who underwent SEEG recording were studied. We used a non-linear correlation method as a measure of the degree and the direction of coupling on SEEG signal pairs. Patients with pure lateral TLEs were not studied. We analyzed the functional coupling between 3 regions of the temporal lobe: the anterior temporal neocortex, the amygdala, and the anterior hippocampus. A physiological model of EEG generation was used to validate the non-linear quantification method and assess its applicability to real SEEG signals. Results: Results are first based on a physiological model of EEG data in which both degree and direction of coupling are explicitly represented, thus allowing construction of the neural systems inside which causality relationships are controlled and generation of multichannel EEG signals from these systems. These signals provide an objective way of studying the performance of non-linear regression analysis on real signals. In medial networks (10 patients), the ictal discharge is limited to the medial limbic structures and may propagate secondarily to the cortex. Quantified results demonstrated no significant coupling between medial and lateral structures at the beginning of the seizures. Conversely, almost constant unidirectional or bidirectional coupling was observed between hippocampus and amygdala. In medial-lateral (5 patients) and lateral-medial (3 patients) networks, the initial ictal discharge includes both limbic and neocortical regions. A rapid 'tonic' discharge is observed over the temporal neocortex at the onset of seizure. Quantitative analysis showed an initial increase in the non-linear correlation coefficient between neocortex and medial structures. Quantification of the coupling direction demonstrated influence of medial over lateral structures (medial-lateral) or of the lateral neocortex over medial structures (lateral-medial). Conclusions: These results confirm the existence of several generic and organized networks involving the medial structures during TLE seizures. © 2001 Elsevier Science Ireland Ltd. All rights reserved.

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Notes:

EID: 2-s2.0-0034886379

Export Date: 20 September 2026; Cited By: 240; Correspondence Address: F. Bartolomei; Service de Neurophysiologie Clinique, Unité d'Epileptologie, 13005 Marseille, CHU Timone-264 Rue St Pierre, France; email: fbartol@aol.com; CODEN: CNEUF

Neural stochastic differential equations network as uncertainty quantification method for EEG source localization

Item Type Journal Article

Author R.S. Wabina

Author C. Silpasuwanchai

Abstract EEG source localization remains a challenging problem given the uncertain conductivity values of the volume conductor models (VCMs). As uncertain conductivities vary across people, they may considerably impact the forward and inverse solutions of the EEG, leading to an increase in localization mistakes and misdiagnoses of brain disorders. Calibration of conductivity values using uncertainty quantification (UQ) techniques is a promising approach to reduce localization errors. The widely-known UQ methods involve Bayesian approaches, which utilize prior conductivity values to derive their posterior inference and estimate their optimal calibration. However, these approaches have two significant drawbacks: solving for posterior inference is intractable, and choosing inappropriate priors may lead to increased localization mistakes. This study used the Neural Stochastic Differential equations Network (SDE-Net), a combination of dynamical systems and deep learning techniques that utilizes the Wiener process to minimize conductivity uncertainties in the VCM and improve the inverse problem. Results revealed that SDE-Net generated a lower localization error rate in the inverse problem compared to Bayesian techniques. Future studies may employ new stochastic dynamical systems-based techniques as a UQ technique to address further uncertainties in the EEG Source Localization problem. Our code can be found here: <https://github.com/rrwabina/SDENet-UQ-ESL>. © 2023 IOP Publishing Ltd.

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adult, Article, human, human experiment, Humans, Deep learning, Learning systems, machine learning, Uncertainty analysis, Uncertainty, normal human, Stochastic systems, procedures, Inverse problems, deep learning, controlled study, Bayesian networks, Dynamical systems, Neural Networks, Computer, electric potential, Localisation, Bayes theorem, Bayes Theorem, Quantification methods, uncertainty, Uncertainty quantifications, Bayesian methods, calibration, image quality, nuclear magnetic resonance imaging, signal noise ratio, T1 weighted imaging, neuroimaging, uncertainty quantification, Calibration, electroencephalogram, Electroencephalography, electroencephalography, epilepsy, electric conductivity, water content, Deep bayesian method, deep bayesian methods, Differential equations, discretization, EEG source localization, Electric Conductivity, impedance measurement, Localization errors, magnetic resonance electrical impedance tomography, neural stochastic differential equations network, SDE-Net, Stochastic differential equation network, Stochastic differential equations, Volume conductor model

Notes:

EID: 2-s2.0-85148678600

Export Date: 20 September 2026; Cited By: 3; Correspondence Address: R.S. Wabina; Center for Health and Wellness Technology, Asian Institute of Technology (AIT), Pathum Thani, Khlong Luang, Thailand; email: romen.wabina@ait.asia

NeuralUQ: A Comprehensive Library for Uncertainty Quantification in Neural Differential Equations and Operators

Item Type Journal Article

Author Z. Zou

Author X. Meng

Author A.F. Psaros

Author G.E. Karniadakis

Abstract Uncertainty quantification (UQ) in machine learning is currently drawing increasing research interest, driven by the rapid deployment of deep neural networks across different fields, such as computer vision and natural language processing, and by the

need for reliable tools in risk-sensitive applications. Recently, various machine learning models have also been developed to tackle problems in the field of scientific computing with applications to computational science and engineering (CSE). Physics-informed neural networks and deep operator networks are two such models for solving partial differential equations (PDEs) and learning operator mappings, respectively. In this regard, a comprehensive study of UQ methods tailored specifically for scientific machine learning (SciML) models has been provided in [A. F. Psaros et al., J. Comput. Phys., 477 (2023), art. 111902]. Nevertheless, and despite their theoretical merit, implementations of these methods are not straightforward, especially in large-scale CSE applications, hindering their broad adoption in both research and industry settings. In this paper, we present an open-source Python library (<https://github.com/Crunch-UQ4MI>), termed NeuralUQ and accompanied by an educational tutorial, for employing UQ methods for SciML in a convenient and structured manner. The library, designed for both educational and research purposes, supports multiple modern UQ methods and SciML models. It is based on a succinct workflow and facilitates flexible employment and easy extensions by the users. We first present a tutorial of NeuralUQ and subsequently demonstrate its applicability and efficiency in four diverse examples, involving dynamical systems and high-dimensional parametric and time-dependent PDEs. Copyright © by SIAM.

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New development for detecting partial discharge using an UHF method and its application to power apparatus in Japan

Item Type Conference Paper

Author M. Hanai

Author F. Endo

Author S. Okabe

Author T. Kato

Author H. Hama

Author M. Nagao

Abstract Insulation diagnosis has been widely applied worldwide in factories and on-site tests to assure the quality of power equipment. Detection of electromagnetic signals in the ultra-high-frequency (UHF) band from a partial discharge (PD) has been widely studied as a highly sensitive insulation diagnostic method for gas-insulated switchgear (GIS). Many of these studies are based on measurements with actual devices and a few systematic theoretical examinations of the basic characteristics of individual sensors and electromagnetic wave propagation [1]. The most frequently authorized method for detecting PD applies the conventional test standard defined by IEC 60270. The conventional method focuses on the kHz band, although the UHF method, which has been recommended by "CIGRE TF15 / 33.03.05" [2], uses wide-frequency components from the MHz to GHz ranges. Besides factory and on-site tests for quality assurance, PD monitoring systems have been developed and the UHF method is also applied by manufacturers. The UHF method, however, raises technical difficulties for quality assurance and product development, i.e., the measurement principle is totally different from IEC 60270; calibration is complicated due to attenuation and reflection; and, TE/TM modes propagate inside GIS. Many research studies using the UHF method are cited in this paper to solve the above difficulties with comparative calibration using the UHF method and IEC 60270 [3-5], sensitivity check of PD-detecting coupler in UHF range [6-9], risk assessment based on estimation of defect size [10-13], and advanced classification and quantification method for PD [14, 15].

Date 2006

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Proceedings Title Int. Conf. Large High Voltage Electr. Syst., CIGRE

Conference Name 41st International Conference on Large High Voltage Electric Systems 2006, CIGRE 2006

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Monitoring, Diagnosis, Quality assurance, Sensitivity, Calibration, Discharge (fluid mechanics), Partial discharges, Method, Discharge, Electric switchgear, Electromagnetic wave propagation, IEC 60270, IEC60270, Partial, Pattern, Phase, Product development, Resolved, UHF, UHF devices

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EID: 2-s2.0-84876782069

Export Date: 20 September 2026; Cited By: 10; Correspondence Address: M. Hanai; Toshiba Corporation, Japan; email: masahiro.hanai@toshiba.co.jp; Conference name: 41st International Conference on Large High Voltage Electric Systems 2006, CIGRE 2006; Conference date: 27 August 2006 through 1 September 2006; Conference code: 96752

New methodology for facilitating food wastage quantification. Identifying gaps and data inconsistencies

Item Type Journal Article

Author H. Barco

Author I. Oribe-Garcia

Author M.V. Vargas-Viedma

Author C.E. Borges

Author C. Martín

Author A. Alonso-Vicario

Abstract The Food and Agriculture Organization of the United Nations estimated that about 1.3 billion tons of food produced for human consumption was lost or wasted globally. Thus, the reduction of the current food loss and waste along the agrifood chain is becoming a priority, both for optimization of resources and reduction waste generation costs. For this purpose, the first step is to quantify the food wastage generation to be able to identify corrective measures. However, in spite of the considerable efforts already undertaken to establish common methodologies to measure the food wastage at different geographical scales, there are still some gaps and inconsistencies. In this regard, the information gathering is labour-intensive because of the different actors involved. The creation of new methodologies and tools capable of automatically identifying these agents would be of great value so as to subsequently apply the more appropriate quantification methodologies. This work aims at providing a new methodology to facilitate this process thanks to the previous identification and classification of the potential food wastage generators. As a result, it provides baseline information for one of the earliest steps of the food wastage quantification process, which is the establishment of the scope of the food wastage inventory. The baseline data needed is taken from the Statistical classification of economic activities in the European Community (NACE), particularly from the most disaggregated level called “classes”. This information has been combined with data from the trading income tax at municipal scale thanks to the use of Geographic

Information Systems (GIS) and the common codes for NACE classes, generating a visual tool for the localization of points with potential of food-wastage generation and their weight of each economic activity over the agrifood chain. The proposed methodology has been implemented for the real case of the municipality of Zamudio (Spain) and it has allowed the identification of the different entities linked with economic activities that are potential generators of food wastage, the weight of each activity over the entire agrifood chain, and the geographical location of these entities in the territory. Furthermore, this methodology was used to compare the nature and number of these activities in another municipality (Karrantza, Spain) and it has also been applied to the entire region of the Basque Country (Spain). © 2018 Elsevier Ltd

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Export Date: 20 September 2026; Cited By: 13; Funding details: Horizon 2020 Framework Programme, H2020, (688995); Correspondence Address: H. Barco; DeustoTech - Fundación Deusto, Bilbao, Avda. Universidades, 24, 48007, Spain; email: hector.barco@deusto.es; CODEN: JEVMA

New results on diagnosis by fuzzy pattern recognition

Item Type Conference Paper

Author M.S. Bouguelid

Author M.S. Mouchaweh

Author P. Billaudel

Abstract We use the classification method Fuzzy Pattern Matching (FPM) to realize the industrial and medical diagnosis. FPM is marginal, i.e., its global decision is based on the selection of one of the intermediate decisions. Each intermediate decision is based on one attribute. Thus, FPM does not take into account the correlation between attributes. Additionally, FPM considers the shape of classes as convex one. Finally the classes are considered as equi-important by FPM. These drawbacks make FPM unusable for many real world applications. In this paper, we propose improving FPM to solve these drawbacks. Several synthetic and real data sets are used to show the performances of the Improved FPM (IFPM) with respect to classical one as well as to the well known classification method K Nearest Neighbours (KNN). KNN is known to be preferment in the case of data represented by correlated attributes or by classes with different a priori probabilities and non convex shape.

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Language English

Volume ICSO

Pages 167-172

Conference Name ICINCO 2007 - 4th International Conference on Informatics in Control, Automation and Robotics, Proceedings

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Pattern recognition, Robotics, Diagnosis, Classification methods, Pattern matching, A-priori probabilities, Synthetic and real data, K nearest neighbours (k-NN), Real-world application, Fuzzy pattern matching, Fuzzy pattern recognition, Medical diagnosis, Multi-criteria decision, Nearest neighbours techniques, New results, Non-convex shapes

Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M. S. Bouguelid; Centre de Recherche en Science et Technologie de L 'Information (CReSTIC), Université de Reims-Champagne-Ardenne Moulin de la Housse, 51687 Reims, BP 1039, France; email: ms.bouguelid@univ-reims.fr; Conference name: 4th International Conference on Informatics in Control, Automation and Robotics, ICINCO 2007; Conference date: 9 May 2007 through 12 May 2007; Conference code: 76374

New Trends in Quantitative Nuclear Cardiology Methods

Item Type Journal Article

Author J. Gomez
Author R. Doukky
Author G. Germano
Author P. Slomka

Abstract Purpose of Review: The use of quantitative analysis in single photon emission computed tomography (SPECT) and positron emission tomography (PET) has become an integral part of current clinical practice and plays a crucial role in the detection and risk stratification of coronary artery disease. Emerging technologies, new protocols, and new quantification methods have had a significant impact on the diagnostic performance and prognostic value of nuclear cardiology imaging while reducing the need for clinician oversight. In this review, we aim to describe recent advances in automation and quantitative analysis in nuclear cardiology. Recent Findings: Recent publications have shown that fully automatic processing is feasible, limiting human input to specific cases where aberrancies are detected by the quality control software. Furthermore, there is evidence indicating that fully quantitative analysis of myocardial perfusion imaging is feasible and can achieve at least similar diagnostic accuracy as visual interpretation by an expert clinician. In addition, the use of fully automated quantification in combination with machine learning algorithms can provide incremental diagnostic and prognostic value over the traditional method of expert visual interpretation. Summary: Emerging technologies in nuclear cardiology focus on automation and the use of artificial intelligence as part of the interpretation process. This review highlights the benefits and limitations of these applications and outlines future directions in the field. © 2018, Springer Science+Business Media, LLC, part of Springer Nature.

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Language English

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Tags:

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NLSC: A noise-robust label shift correction framework via three-head training and class-adaptive cleaning

Item Type Journal Article

Author X. Wu

Author R. Fan

Author T. Luo

Author C. Hou

Abstract Label shift occurs when the conditional distributions remain consistent between source and target domains, but the marginal label distributions differ. For instance, during the early stage of the COVID-19 outbreak, the proportion of pneumonia cases compared to common cold cases in hospitals may have been relatively low. This ratio could shift dramatically in later stages of the pandemic, with pneumonia cases becoming predominant, even though the symptomatic presentation of each disease remained consistent. Existing label shift methods typically aim to adapt a classifier's output to match the target domain's label distribution, assuming the source domain has clean labels. However, real-world scenarios often involve label noise in the source domain. For example, during COVID-19's early phase, mild and confusable symptoms frequently led to misdiagnoses of COVID-19 as the common cold, introducing label noise. Such noise compromises the effectiveness of traditional methods, necessitating novel approaches. To address this, we analyze classifier error bounds under label shift correction using noisy source data. Based on this analysis, we propose a Noise-robust Label Shift Correction (NLSC) framework. NLSC employs a Three-Head Architecture Training (THAT) strategy for robust feature learning and a Class-Adaptive Threshold Cleaning (CATC) strategy for source data purification. Extensive experiments confirm that our method outperforms existing state-of-the-art techniques, particularly in real-world scenarios with high source domain noise rates. © 2025 Elsevier Inc.

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Notes:

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No-reference image quality assessment of T2-weighted magnetic resonance images in prostate cancer patients

Item Type Conference Paper**Author** S. Masoudi**Author** S. Harmon**Author** S. Mehralivand**Author** N. Lay**Author** U. Bagci**Author** B.J. Wood**Author** P.A. Pinto**Author** P. Choyke**Author** B. Turkbey

Abstract No-reference image quality assessment in magnetic resonance (MR) imaging is a challenging task due to the variable nature of these images and lack of standard quantification methods, which makes the interpretation to be almost always subjective. In this study, we propose an architecture where we: (i) extended the no-reference image quality assessment problem of MRI into a full-reference image quality assessment using unpaired generative adversarial network (GAN) and (ii) employed a weakly supervised trained deep classifier to determine the quality of MR images by comparing each image with its synthetic higher quality reference image. Using this approach, we achieved 11.28% improvement in the accuracy of our MR image quality assessment algorithm on an independent data test with FPR in detecting low quality images, reduced from 13% to 9.6%. © 2021 IEEE.

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Tags:

Computer vision, Image enhancement, Image quality, Medical imaging, Generative adversarial networks, High quality, Diseases, Quantification methods, Magnetic resonance imaging, Generative adversarial network, Cancer patients, Image quality assessment, Full references, Magnetism, No-reference image quality assessment, No-reference images, Prostate cancers, Reference image, Resonance, T2 weighted

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Node trust prediction framework in mobile Ad Hoc networks

Item Type Conference Paper
Author H. Xia
Author G.-D. Wang
Author Z.-K. Pan
Abstract Trust quantification methods depends on the deployment region, network applications, level of security required. Considering the environment of mobile ad hoc networks, in this study we propose a new trust model. This new model is divided into two parts: trust assessment and trust prediction. The process of node trust assessment is based on node's historical behaviors, in which the trust decision factors include the subjective reputation and indirect reputation. Then utilizing the obtained trust data sequence, we propose an improved dynamic grey-Markov chain prediction measure to predict node's trust for future decision making. In which we eliminates grey bias via using unbiased GM(1,1) model. Moreover, in order to improve anti-jamming performance, we introduce the fuzzy classification method to classify the fuzzy states into some reasonable groups. Experimental results have been conducted to evaluate the effectiveness of the

proposed model. It is proved in theory and practice that the prediction results not only are more reliable but also can help the decision maker with grasping the change tendency of trust and making proper decisions. © 2016 IEEE.

Date 2016

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Language English

Extra Journal Abbreviation: Proc. - IEEE Int. Conf. Trust, Secur. Priv. Comput. Commun., IEEE Int. Conf. Big Data Sci. Eng. IEEE Int. Symp. Parallel Distrib. Process. Appl., IEEE TrustCom/BigDataSE/ISPA

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Conference Name Proceedings - 15th IEEE International Conference on Trust, Security and Privacy in Computing and Communications, 10th IEEE International Conference on Big Data Science and Engineering and 14th IEEE International Symposium on Parallel and Distributed Processing with Applications, IEEE TrustCom/BigDataSE/ISPA 2016

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Notes:

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Non-Destructive Detection of Mycotoxin Contamination in Maize Silage Based on Machine Vision

Item Type Journal Article

Author X. Zheng

Author H. Tian

Author K. Zhao

Author L. Guo

Author D. Wan

Author Y. Yu

Author C. Zhuo

Author S. Wang

Abstract Mycotoxin contamination by aflatoxin B1 (AFB1) and deoxynivalenol (DON) in maize silage threatens feed safety, requiring rapid, non-destructive monitoring tools. This study developed a visible-light machine vision approach combined with machine learning to quantify AFB1 and DON and classify contamination levels. A total of 210 silage samples were imaged, and 111 RGB-based color and texture features were extracted, followed by correlation analysis and model-based feature selection. Using the 10 selected features, support vector regression (SVR) achieved the best quantitative performance for AFB1 ($R^2 = 0.9945$, $RMSE = 3.18 \mu\text{g}\cdot\text{kg}^{-1}$), while XGBoost performed best for DON ($R^2 = 0.9816$, $RMSE = 45.50 \mu\text{g}\cdot\text{kg}^{-1}$). For classification, random forest and XGBoost correctly identified AFB1 contamination levels with an Accuracy of 90.48%, whereas SVM achieved 97.62% Accuracy for DON. Comparison of correlation-based and model-based feature importance confirmed the complementary value of statistically significant and nonlinearly predictive features. Biological interpretation suggested that color and texture responses may reflect fungal pigmentation/browning and multiscale surface-structure alterations, respectively, with selected texture descriptors changing earlier than color descriptors during aerobic exposure. The proposed low-cost RGB imaging strategy, coupled with multi-feature fusion, offers a promising approach for high-throughput preliminary mycotoxin screening in feed production. © 2026 by the authors.

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Non-Destructive Quantification of Mycelial Biocomposite Growth Over Time

Item Type Journal Article

Author L.E. Pierce

Author A. Folley

Author L.R. White

Author B. Craig

Author D. Johnstone

Author C.E. Shelmerdine

Author S. Zier

Author M. El Hajam

Author A. Kordijazi

Author M. Tajvidi

Author C. Howell

Abstract Mycelial biocomposites are sustainable alternatives to nonbiodegradable materials in building and packaging. Efficient manufacturing requires accurate, non-destructive quantification of growth over time, yet existing methods are often destructive or imprecise. This study develops and evaluates several non-destructive quantification methods for wood-flour biocomposites by using images to define mycelial density levels, low (no visible growth, removable surface hyphal coverage $< 7.8\%/cm^2$), medium (light growth, $7.8\%–26.7\%/cm^2$), and high (dense coverage, $> 26.7\%/cm^2$), and tracking changes in each level over time. The first method, the manual creation of masks for each growth level, provided rapid but coarse classification, estimating 64.1% high growth after 16 days with 3.5% user variability. An algorithmic masking approach improved detail detection, increasing estimated high-growth coverage to 81.7% but also variability to 9.3%. A fully automated deep-learning model proved fastest and most consistent, yielding 77.8% high-growth coverage and eliminating intra-user variability (0%). The deep-learning method was then applied to assess the effects of substrate supplements, revealing distinct growth patterns for each—differences not captured by traditional methods. These results demonstrate the effectiveness of automated quantification for mycelial biocomposites, enabling reproducible, high-resolution, non-destructive monitoring of growth and providing a foundation for more precise engineering and wider adoption of these materials. © 2025 Wiley Periodicals LLC.

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Article, automation, Deep learning, Learning systems, Computer vision, algorithm, nonhuman, procedures, deep learning, controlled study, computer vision, Image analysis, image processing, image analysis, Image Processing, Computer-Assisted, microbiology, flour, Non destructive, Nondestructive examination, In-buildings, density, Biocomposite, Biological materials, fungus hyphae, growth, development and aging, High growth, Image analyze, Image-analysis, mycelial density, Mycelial density, mycelial growth, Mycelial material, mycelial materials, mycelium, Mycelium, Packaging materials, temporal growth, Temporal growth, Trametes versicolor, wood, Wood

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Non-exchangeable Conformal Prediction with Optimal Transport: Tackling Distribution Shifts with Unlabeled Data

Item Type Conference Paper

Author A.H.C. Correia

Author C. Louizos

Abstract Conformal prediction is a distribution-free uncertainty quantification method that has gained popularity in the machine learning community due to its finite-sample guarantees and ease of use. Its most common variant, dubbed split conformal prediction, is also computationally efficient as it boils down to collecting statistics

of the model predictions on some calibration data not yet seen by the model. Nonetheless, these guarantees only hold if the calibration and test data are exchangeable, a condition that is difficult to verify and often violated in practice due to so-called distribution shifts. The literature is rife with methods to mitigate the loss in coverage in this non-exchangeable setting, but these methods require some prior information on the type of distribution shift to be expected at test time. In this work, we study this problem via a new perspective, through the lens of optimal transport, and show that it is possible to estimate the loss in coverage and mitigate arbitrary distribution shifts, offering a principled and broadly applicable solution. © 2025 Neural information processing systems foundation. All rights reserved.

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Non-intrusive soil carbon content quantification methods using machine learning algorithms: A comparison of microwave and millimeter wave radar sensors

Item Type Journal Article

Author D. An

Author Y. Chen

Abstract Agricultural and forestry biomass can be converted to biochar through pyrolysis gasification, making it a significant carbon source for soil. Applying biochar to soil is a carbon-negative process that helps combat climate change, sustain soil biodiversity, and regulate water cycling. However, quantifying soil carbon content conventionally is time-consuming, labor-intensive, imprecise, and expensive, making it difficult to accurately measure in-field soil carbon's effect on storage water and nutrients. To address this challenge, this paper for the first time, reports on extensive lab tests

demonstrating non-intrusive methods for sensing soil carbon and related smart biochar applications, such as differentiating between biochar types from various biomass feedstock species, monitoring soil moisture, and biochar water retention capacity using portable microwave and millimeter wave sensors, and machine learning. These methods can be scaled up by deploying the sensor in-field on a mobility platform, either ground or aerial. The paper provides details on the materials, methods, machine learning workflow, and results of our investigations. The significance of this work lays the foundation for assessing carbon-negative technology applications, such as soil carbon content accounting. We validated our quantification method using supervised machine learning algorithms by collecting real soil mixed with known biochar contents in the field. The results show that the millimeter wave sensor achieves high sensing accuracy (up to 100%) with proper classifiers selected and outperforms the microwave sensor by approximately 10%–15% accuracy in sensing soil carbon content. © 2023 The Authors

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Non-stationary data sequence classification using online class priors estimation

Item Type Journal Article

Author C. Yang

Author J. Zhou

Abstract Online classification is important for real time data sequence classification. Its most challenging problem is that the class priors may vary for non-stationary data sequences. Most of the current online-data-sequence-classification algorithms assume that the class labels of some new-arrived data samples are known and retrain the classifier accordingly. Unfortunately, such assumption is often violated in real applications. But if we were able to estimate the class priors on the test data sequence accurately, we could adjust the classifier without retraining it while preserving a reasonable accuracy. There has been some work on the class priors estimation to classify static data sets using the offline iterative EM algorithm, which has been proved to be quite effective to adjust the classifier. Inspired by the offline iterative EM algorithm for static data sets, in this paper, we propose an online incremental EM algorithm to estimate the class priors along the data sequence. The classifier is adjusted accordingly to keep pace with the varying distribution. The proposed online algorithm is more computationally efficient because it scans the sequence only once. Experimental results show that the proposed algorithm indeed performs better than the conventional offline iterative EM algorithm when the class priors are non-stationary. © 2008 Elsevier Ltd. All rights reserved.

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Nondestructive detection of carotenoids in chrysanthemum and identification of drought using hyperspectral and machine learning

Item Type Journal Article

Author J. Xia

Author F. Chen

Author G. Bai

Author Y. Zhou

Author C. Bi

Author H. Liang

Author H. Huang

Author W. Liu

Abstract Carotenoids, a class of naturally occurring pigments ubiquitously distributed across plant and animal kingdoms, serve vital biological functions in photoprotection and antioxidant defense mechanisms. Within plant systems, these compounds critically modulate photosynthetic efficiency under drought stress conditions. While carotenoid content fluctuations offer a reliable biomarker for evaluating plant drought responses, conventional chemical quantification methods remain labor-intensive and unsuitable for large-scale agricultural monitoring. This investigation employs hyperspectral imaging technology (operating in the 859–1705 nm spectral range) to acquire spectral signatures from *Chrysanthemum × morifolium* (Ramat.) Hemsl. samples, subsequently applying five spectral preprocessing techniques to optimize data quality. Three feature selection algorithms (Successive Projections Algorithm, Max-Relevance Min-Redundancy, and Competitive Adaptive Reweighted Sampling) were employed to identify diagnostically significant wavelengths, while three regression models (Random Forest Regression, Partial Least Squares Regression, and Gradient Boosting Machine) were constructed for carotenoid prediction. SHapley Additive exPlanations analysis was performed to assess the interpretability and relative contributions of different feature selection approaches. Experimental results demonstrate that the SPA-Random Forest Regression hybrid model achieved optimal prediction performance (RC = 0.9545, RMSEC = 0.0096, RP = 0.9228, RSMEP = 0.0134). Notably, the Random Forest Regression algorithm attained 99.03% classification accuracy in drought stress discrimination. This methodology establishes the viability of integrating hyperspectral imaging with machine learning as a non-destructive analytical framework for simultaneous carotenoid quantification and plant drought stress assessment. © 2026 Elsevier B.V.

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Noninvasive assessment of organ-specific and shared pathways in multi-organ fibrosis using T1 mapping

Item Type Journal Article

- Author V. Nauffal
- Author M.D.R. Klarqvist
- Author M.C. Hill
- Author D.F. Pace
- Author P. Di Achille
- Author S.H. Choi
- Author J.T. Rämö
- Author J.P. Pirruccello
- Author P. Singh
- Author S. Kany
- Author C. Hou
- Author K. Ng
- Author A.A. Philippakis
- Author P. Batra
- Author S.A. Lubitz
- Author P.T. Ellinor

Abstract Fibrotic diseases affect multiple organs and are associated with morbidity and mortality. To examine organ-specific and shared biologic mechanisms that underlie fibrosis in different organs, we developed machine learning models to quantify T1 time, a marker of interstitial fibrosis, in the liver, pancreas, heart and kidney among 43,881 UK Biobank participants who underwent magnetic resonance imaging. In phenome-wide association analyses, we demonstrate the association of increased organ-specific T1 time, reflecting increased interstitial fibrosis, with prevalent diseases across multiple organ systems. In genome-wide association analyses, we identified 27, 18, 11 and 10 independent genetic loci associated with liver, pancreas, myocardial and renal cortex T1 time, respectively. There was a modest genetic

correlation between the examined organs. Several loci overlapped across the examined organs implicating genes involved in a myriad of biologic pathways including metal ion transport (SLC39A8, HFE and TMPRSS6), glucose metabolism (PCK2), blood group antigens (ABO and FUT2), immune function (BANK1 and PPP3CA), inflammation (NFKB1) and mitosis (CENPE). Finally, we found that an increasing number of organs with T1 time falling in the top quintile was associated with increased mortality in the population. Individuals with a high burden of fibrosis in ≥ 3 organs had a 3-fold increase in mortality compared to those with a low burden of fibrosis across all examined organs in multivariable-adjusted analysis (hazard ratio = 3.31, 95% confidence interval 1.77–6.19; $P = 1.78 \times 10^{-4}$). By leveraging machine learning to quantify T1 time across multiple organs at scale, we uncovered new organ-specific and shared biologic pathways underlying fibrosis that may provide therapeutic targets. © The Author(s), under exclusive licence to Springer Nature America, Inc. 2024.

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Normative Alignment of Recommender Systems via Internal Label Shift

Item Type Conference Paper

Author J. Kruse

Author K. Lindschow

Author M.R. Andersen

Author R. Shimizu

Author J. McAuley

Author P.-A. Mattei

Author J. Frellsen

Abstract Recommender systems optimized solely for user engagement often fail to meet broader normative objectives such as fairness, diversity, or editorial values. We introduce NAILS (Normative Alignment of recommender systems via Internal Label Shift), a simple and scalable method for aligning recommendation outputs with target distributions over item-level attributes (e.g., categories). NAILS modifies the user-conditional item distribution to induce a specified marginal distribution over attributes, leveraging existing user-item preferences without retraining the model. To achieve this, we recast the problem as a form of label shift applied internally within a hierarchical classification framework. Adopting a stakeholder-centric perspective, NAILS enables alignment with global normative goals. Empirically, we show that NAILS consistently improves attribute-level alignment with minimal impact on user engagement, providing a practical mechanism for value-driven recommendation. Our code is available at <https://github.com/johanneskruse/nails>. © 2025 Copyright held by the owner/author(s).

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Novel algorithm for classification of medical images

Item Type Conference Paper

Author B. Bhushan

Author M. Juneja

Abstract Content-based image retrieval (CBIR) methods in medical image databases have been designed to support specific tasks, such as retrieval of medical images. These methods cannot be transferred to other medical applications since different imaging modalities require different types of processing. To enable content-based queries in diverse collections of medical images, the retrieval system must be familiar with the current Image class prior to the query processing. Further, almost all of them deal with the DICOM imaging format. In this paper a novel algorithm based on energy information obtained from wavelet transform for the classification of medical images according to their modalities is described. For this two types of wavelets have been used and have been shown that energy obtained in either case is quite distinct for each of the body part. This technique can be successfully applied to different image formats. The results are shown for JPEG imaging format. © 2010 American Institute of Physics.

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Novel Algorithm to Detect, Classify, and Count Mussel Larvae in Seawater Samples Using Computer Vision

Item Type Journal Article

Author P. Orgeira-Crespo

Author C. Gabín-Sánchez

Author F. Aguado-Agelet

Author G. Rey-González

Abstract Featured Application: Automatic detection, classification, and accounting of mussel larvae in their different stages of growth using computer vision. The European Union's mussel production industry is dependent on obtaining mussel larvae as seed for cultivation, a process traditionally monitored through labor-intensive manual sampling and microscopic counting prone to human error and time-consuming procedures. To address these challenges, our research presents a computer vision-based methodology for accurately identifying, classifying, and quantifying mussel larvae individuals across various developmental stages from microscopic images of water samples. Utilizing a neural network architecture derived from the YOLO method, our approach integrates convolutional, pooling, and fully connected layers to automate detection, classification, and accounting tasks. Through training with manually labeled samples and employing data augmentation techniques, we established a robust framework capable of processing diverse larval specimens effectively. Our research not only streamlines mussel larvae monitoring processes but also underscores the potential of computer vision techniques to enhance efficiency and accuracy in aquaculture industries. © 2024 by the authors.

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Novel approach for quantification for severity estimation of blight diseases on leaves of tomato plant

Item Type Journal Article

Author A.S. Charak

Author A. Sinha

Author T. Jain

Abstract This study uses digital image processing and machine learning to quantify the infection patterns on tomato leaves due to blight diseases. Quantification, also known as severity measurement, is a technique to determine how much a leaf is diseased by calculating a numeric value. This value could be a fraction representing how much the diseased region is present on the leaf compared to the entire leaf region, or it can be a percentage value too. There are two main approaches to measuring disease severity; the first technique involves visual estimation using references like standard area diagrams. The second approach involves taking a digital image of the leaf, separating the diseased regions from the healthy regions, and then calculating the area of those two regions. The approach we took is similar. We first took the digital image and segmented the diseased and healthy regions. For quantification, we calculated the ratio of total pixels representing the diseased region to the total number of pixels representing the leaf. While finding ways to improve the accuracy of the segmentation algorithm, we also discovered our segmentation technique which automatically segments the diseased regions of the leaves from the healthy areas using k-means clustering. The clustering-segmentation algorithm did give good results for the sample images to which it was applied. The main thing about the clustering-segmentation algorithm is that it tends to be automatic compared to some

of the semi-automatic segmentation approaches that have been discovered till now. We could reproduce the validated quantification results as other authors achieved in the recent work, which also validated our methodology. © 2022 John Wiley & Sons Ltd.

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Correspondence Address: A. Sinha; Department of Computer Science and Engineering, Manipal
University Jaipur, Jaipur, Rajasthan, India; email: adityasinha.1988@gmail.com; CODEN: EXSYE

Novel fluorescence spectroscopy method coupled with N-PLS-R and PLS-DA models for the quantification of cannabinoids and the classification of cannabis cultivars

Item Type Journal Article

Author M. Birenboim

Author D. Kenigsbuch

Author J.A. Shimshoni

Abstract Introduction: Cannabis sativa L. inflorescences are rich in secondary metabolites, particularly cannabinoids. The most common techniques for elucidating cannabinoid composition are expensive technologies, such as high-pressure liquid chromatography (HPLC). Objectives: We aimed to develop and evaluate the

performance of a novel fluorescence spectroscopy-based method coupled with N-way partial least squares regression (N-PLS-R) and partial least squares discriminant analysis (PLS-DA) models to replace the expensive chromatographic methods for preharvest cannabinoid quantification. Methodology: Fresh medicinal cannabis inflorescences were collected and ethanol extracts were prepared. Their excitation–emission spectra were measured using fluorescence spectroscopy and their cannabinoid contents were determined by HPLC-PDA. Subsequently, N-PLS-R and PLS-DA models were applied to the excitation–emission matrices (EEMs) for cannabinoid concentration prediction and cultivar classification, respectively. Results: The N-PLS-R model was based on a set of EEMs ($n = 82$) and provided good to excellent quantification of (–)- Δ^9 -trans-tetrahydrocannabinolic acid, cannabidiolic acid, cannabigerolic acid, cannabichromenic acid, and (–)- Δ^9 -trans-tetrahydrocannabinol (R^2_{CV} and $R^2_{pred} > 0.75$; $RPD > 2.3$ and $RPIQ > 3.5$; $RMSECV/RMSEC$ ratio < 1.4). The PLS-DA model enabled a clear distinction between the four major classes studied (sensitivity, specificity, and accuracy of the prediction sets were all ≥ 0.9). Conclusions: The fluorescence spectral region (excitation 220–400 nm, emission 280–550 nm) harbors sufficient information for accurate prediction of cannabinoid contents and accurate classification using a relatively small data set. © 2023 John Wiley & Sons Ltd.

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Article, Forecasting, Spectrum analysis, nonhuman, Least squares approximations, controlled study, cross validation, Discriminant analysis, cannabis, dronabinol, measurement accuracy, limit of quantitation, chemistry, least square analysis, Least-Squares Analysis, sensitivity and specificity, Partial least square regression, partial least squares regression, high performance liquid chromatography, limit of detection, Metabolites, unclassified drug, Secondary metabolites, measurement repeatability, discriminant analysis, Partial least squares discriminant analyses (PLSDA), Mass spectrometry, Chromatographic analysis, cannabinoid, Cannabis, cannabichromenic acid, cannabidiolic acid, cannabigerolic acid, cannabinoids, Cannabinoids, Cannabis sativa, Cannabis sativa L, chemotype, Discriminant analysis model, Excitation emission matrices, Excitation–emission matrix, excitation–emission matrix (EEM), fluorescence, Fluorescence, Fluorescence spectroscopy, Hallucinogens, High pressure liquid chromatography, inflorescence, medical cannabis, N-way partial least square regression, N-way partial least squares regression (N-PLS-R), Partial least square discriminant analyse, partial least squares discriminant analysis (PLS-DA), psychedelic agent, spectrofluorometry, Spectrometry, Fluorescence, Spectroscopic analysis, tetrahydrocannabinolic acid

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NRU-HSE at SemEval-2016 task 4: Comparative analysis of two iterative methods using quantification library

Item Type Conference Paper

Author N. Karpov

Author A. Porshnev

Author K. Rudakov

Abstract In many areas, such as social science, politics or market research, people need to track sentiment and their changes over time. For sentiment analysis in this field it is more important to correctly estimate proportions of each sentiment expressed in the set of documents (quantification task) than to accurately estimate sentiment of a particular document (classification). Basically, our study was aimed to analyze the effectiveness of two iterative quantification techniques and to compare their effectiveness with baseline methods. All the techniques are evaluated using a set of synthesized data and the SemEval-2016 Task4 dataset. We made the quantification methods from this paper available as a Python open source library. The results of comparison and possible limitations of the quantification techniques are discussed. © 2016 Association for Computational

Linguistics.

Date 2016

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Language English

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Computational linguistics, Sentiment analysis, Synthesised, Comparative analyzes, Market Research, Market researches, Document Classification, Quantification methods, Change-over time, Baseline methods, Open-source libraries, Social science research

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Nuclear magnetic resonance analysis of extra virgin olive oil: classification through secoiridoids

Item Type Journal Article

Author C.G. Tsiafoulis

Author C. Liaggou

Author A. Garoufis

Author P. Magiatis

Author I.G. Roussis

Abstract BACKGROUND: Extra virgin olive oil (EVOO), a natural product with a multidisciplinary role, has been and is continuing to be studied from several points of view. Among them, its chemical analysis is of major importance and several methods have been used. Nuclear magnetic resonance (NMR) spectroscopy has inherent

advantages, among them monitoring the chemical constituents without the need for a separation technique and without, for instance, possible carry-over effects. Additionally, several magnetic resonance spectroscopic techniques can provide a novel powered insight into the nature and properties of a sample under study. Moreover, -omics procedure can reveal new information and can lead to the classification of populations under study. The main objective of the present work was the possible classification of the EVOO samples based on their aldehyde content using a proposed unreferenced ¹H-NMR spectroscopic quantification method combined with a metabolomic approach. Moreover, the study of the impact of such elevated aldehyde content on several spectra regions of importance in the proton NMR spectra led to the proposal of a possible new isomer indicator. RESULTS: Univariate analysis of 12 EVOO samples showed that oleacein, oleocanthal, elenolic acid, hydroxytyrosol/hydroxytyrosol derivatives and tyrosol/tyrosol derivatives strongly differentiated two classes of EVOO: OEH (for high aldehyde EVOO content) and OE (for non-high aldehyde content). Moreover, we pointed out the 'impact' of such elevated secoiridoid and derivatives content, through their moieties' units, on a range of several resonances of the ¹H-NMR spectrum. The metabolomic approach demonstrated the classification of EVOO samples based on their secoiridoid and derivatives content. Multivariate analysis showed a strong influence on the discrimination of the EVOO classes based on the protons resonating at the aldehyde region of the ¹H-NMR spectrum; the aldehyde protons corresponding to 5S,4R-ligstrodiol and 5S,4R-oleuropeindiol, oleacein, oleocanthal, elenolic acid, p-HPEA-EA, 3,4-DHPEA-EA, 5S,4R- and 5S,4S-ligstrodiol and the proton corresponding to a new compound were reported for the first time. This isomer compound, reported for the first time, could serve as a possible indicator for EVOO classification. CONCLUSIONS: An unreferenced quantification method was proposed and EVOO samples were classified into two classes: OEH and OE, according to their aldehyde content, gaining thus probably higher nutrient and possible pharmacological value. Moreover, we point out the 'impact' of such elevated aldehyde content on several spectral regions of the ¹H spectrum. Finally, a new compound was detected in the OEH samples and is reported for the first time. This compound could possibly be an indicator. © 2023 The Authors. Journal of The Science of Food and Agriculture published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry. © 2023 The Authors. Journal of The Science of Food and Agriculture published by John Wiley & Sons Ltd on behalf of Society of Chemical Industry.

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article, nonhuman, controlled study, multivariate analysis, classification, proton, aldehyde, chemical analysis, nuclear magnetic resonance spectroscopy, proton nuclear magnetic resonance, univariate analysis, aldehyde content, extra virgin olive oil, extra virgin olive oil (EVOO), hydroxytyrosol, indicator, isomer, metabolomic analysis, natural product, nuclear magnetic resonance, pharmacology, secoiridoid, secoiridoids, separation technique, tyrosol, unreferenced quantification method

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Nutrient fluxes are underestimated in the Qinghai Lake Basin, the largest endorheic basin of the Asian water tower

Item Type Journal Article

Author C. Niu

Author Y. Zheng

Author Y. Chen

Author C. Wu

Author G. Li

Author X. Zhou

Author H. Li

Author Q. Wang

Author Z. Shen

Author L. Chen

Abstract Earth's fragile high-altitude ecosystems are facing challenges from nutrient pollution, where data scarcity and complex topography create significant knowledge gaps. This study conducted the first large-scale survey and established a comprehensive inventory of human activities in the Qinghai Lake basin, the largest lake on the Qinghai-Tibet Plateau. We developed a hybrid framework that integrated detailed surveys, process-based modeling, and interpretable machine learning to quantify nitrogen and phosphorus fluxes despite data gaps. Notably, ungauged basins around the lake contribute substantially (34% of phosphorus and 21% of nitrogen) to the lake load, highlighting a serious underestimation of a blind spot in global monitoring networks. Contemporary anthropogenic emissions contributed only 41.12% of TN flux and 45.22% of TP flux. Further machine learning analysis indicates that non-direct human factors (runoff, climate, and geographical conditions) have a dominant influence on nutrient movement. These findings provide a replicable framework for data-poor regions and call for shifting management paradigms from anthropocentric to ecosystem-based approaches across high-altitude areas. © 2026 Elsevier Ltd

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NYUS.2: an automated machine learning prediction model for the large-scale real-time simulation of grapevine freezing tolerance in North America

Item Type Journal Article
Author H. Wang
Author G.D. Moghe
Author A.P. Kovaleski
Author M. Keller
Author T.E. Martinson

Author A.H. Wright
Author J.L. Franklin
Author A. Hébert-Haché
Author C. Provost
Author M. Reinke
Author A. Atucha
Author M.G. North
Author J.P. Russo
Author P. Helwi
Author M. Centinari
Author J.P. Londo

Abstract Accurate and real-time monitoring of grapevine freezing tolerance is crucial for the sustainability of the grape industry in cool climate viticultural regions. However, on-site data are limited due to the complexity of measurement. Current prediction models underperform under diverse climate conditions, which limits the large-scale deployment of these methods. We combined grapevine freezing tolerance data from multiple regions in North America and generated a predictive model based on hourly temperature-derived features and cultivar features using AutoGluon, an automated machine learning engine. Feature importance was quantified by AutoGluon and SHAP (SHapley Additive exPlanations) value. The final model was evaluated and compared with previous models for its performance under different climate conditions. The final model achieved an overall 1.36°C root-mean-square error during model testing and outperformed two previous models using three test cultivars at all testing regions. Two feature importance quantification methods identified five shared essential features. Detailed analysis of the features indicates that the model has adequately extracted some biological mechanisms during training. The final model, named NYUS.2, was deployed along with two previous models as an R shiny-based application in the 2022–23 dormancy season, enabling large-scale and real-time simulation of grapevine freezing tolerance in North America for the first time. © The Author(s) 2024. Published by Oxford University Press on behalf of Nanjing Agricultural University.

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OASIS: Open-world Adaptive Self-supervised and Imbalanced-aware System

Item Type Conference Paper

Author M. Kim

Author M. Joe

Author M. Kwon

Abstract The expansion of machine learning into dynamic environments presents challenges in handling open-world problems where label shift, covariate shift, and unknown classes emerge concurrently. Post-training methods have been explored to address these challenges, adapting models to newly emerging data. However, these methods struggle when the initial pre-training is performed on class-imbalanced datasets, limiting generalization to minority classes. To address this, we propose OASIS, an Open-world Adaptive Self-supervised and Imbalanced-aware System. OASIS consists of two learning phases: pre-training and post-training. The pre-training phase aims to improve the classification performance of samples near class boundaries via a novel borderline sample refinement step. Notably, the borderline sample refinement step critically improves the robustness of the decision boundary in the representation space. Through this robustness of the pre-trained model, OASIS generates reliable pseudo-labels, adapting the model against open-world problems in the post-training phase. Extensive experiments demonstrate that our method significantly outperforms state-of-the-art post-training techniques in both accuracy and efficiency across diverse open-world scenarios. © 2025 Copyright held by the owner/author(s).

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Information systems, Human engineering, Learning systems, Machine-learning, Dynamic environments, Self-supervised learning, Pre-training, Sampling, Semi-supervised learning, semi-supervised learning, borderline sample refinement, Borderline sample refinement, online post-training, Online post-training, Open world, open-world problem, Open-world problem, Refinement step, Training phasis

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Object recognition using proportion-based prior information: Application to fisheries acoustics

Item Type Journal Article

Author R. Lefort

Author R. Fablet

Author J.-M. Boucher

Abstract This paper addresses the inference of probabilistic classification models using weakly supervised learning. The main contribution of this work is the development of learning methods for training datasets consisting of groups of objects with known relative class priors. This can be regarded as a generalization of the situation addressed by Bishop and Ulusoy (2005), where training information is given as the presence or absence of object classes in each set. Generative and discriminative classification methods are conceived and compared for weakly supervised learning, as well as a non-linear version of the probabilistic discriminative models. The considered models are evaluated on standard datasets and an application to fisheries acoustics is reported. The proposed proportion-based training is demonstrated to outperform model learning based on presence/absence information and the potential of the non-linear discriminative model is shown. © 2010 Elsevier B.V. All rights reserved.

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Object recognition, Supervised learning, Learning methods, Classification models, Model learning, Classification methods, Weakly supervised learning, Fisheries, Training data sets, Discriminative models, Data sets, Discriminative classification model, Fisheries acoustics, Generative classification model, Non-linear, Nonlinear versions, Object class, Presence/absence, Prior information, Probabilistic classification models

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Export Date: 20 September 2026; Cited By: 6; Correspondence Address: R. Lefort; Ifremer/STH (French Research Institute for Exploitation of the Sea), Technople Brest Iroise, 29280 Plouzane, France; email: riwal.lefort@telecom-bretagne.eu

Objective class definitions using correlation of similarities between remotely sensed and environmental data

Item Type Journal Article

Author S. Andrefouet

Author M. Claereboudt

Abstract This letter presents a natural approach for selecting a subset of environmental variables to define classes prior to mapping them efficiently using remotely sensed data. The procedure is based on correlation calculations between a set of 'environmental' similarity matrices and an 'image' similarity matrix. The correlation is a modified Spearman rank correlation. Using a SPOT-HRV image and seven environmental variables, an application in a coral reef environment showed that only three environmental variables are relevant to defining classes prior to mapping. © 2000 Taylor & Francis Group, LLC.

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Objective evaluation of deep uncertainty predictions for COVID-19 detection

Item Type Journal Article

Author H. Asgharnezhad

Author A. Shamsi

Author R. Alizadehsani

Author A. Khosravi

Author S. Nahavandi

Author Z.A. Sani

Author D. Srinivasan

Author S.M.S. Islam

Abstract Deep neural networks (DNNs) have been widely applied for detecting COVID-19 in medical images. Existing studies mainly apply transfer learning and other data representation strategies to generate accurate point estimates. The generalization power of these networks is always questionable due to being developed using small datasets and failing to report their predictive confidence. Quantifying uncertainties associated with DNN predictions is a prerequisite for their trusted deployment in medical settings. Here we apply and evaluate three uncertainty quantification techniques for COVID-19 detection using chest X-Ray (CXR) images. The novel concept of uncertainty confusion matrix is proposed and new performance metrics for the objective evaluation of uncertainty estimates are introduced. Through comprehensive experiments, it is shown that networks pertained on CXR images outperform networks pretrained on natural image datasets such as ImageNet. Qualitatively and quantitatively evaluations also reveal that the predictive uncertainty estimates are statistically higher for erroneous predictions than correct predictions. Accordingly, uncertainty quantification methods are capable of flagging risky

predictions with high uncertainty estimates. We also observe that ensemble methods more reliably capture uncertainties during the inference. DNN-based solutions for COVID-19 detection have been mainly proposed without any principled mechanism for risk mitigation. Previous studies have mainly focused on generating single-valued predictions using pretrained DNNs. In this paper, we comprehensively apply and comparatively evaluate three uncertainty quantification techniques for COVID-19 detection using chest X-Ray images. The novel concept of uncertainty confusion matrix is proposed and new performance metrics for the objective evaluation of uncertainty estimates are introduced for the first time. Using these new uncertainty performance metrics, we quantitatively demonstrate when we could trust DNN predictions for COVID-19 detection from chest X-rays. It is important to note the proposed novel uncertainty evaluation metrics are generic and could be applied for evaluation of probabilistic forecasts in all classification problems. © 2022, The Author(s).

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Obukhov Length Estimation From Spaceborne Radars

Item Type Journal Article

Author O. O'Driscoll

Author A. Mouche

Author B. Chapron

Author M. Kleinherenbrink

Author P. López-Dekker

Abstract Two air-sea interaction quantification methods are employed on synthetic aperture radar (SAR) scenes containing atmospheric-turbulence signatures. Quantification performance is assessed on Obukhov length L , an atmospheric surface-layer stability metric. The first method correlates spectral energy at specific turbulence-spectrum wavelengths directly to L . Improved results are obtained from the second method, which relies on a machine-learning algorithm trained on a wider array of SAR-derived parameters. When applied on scenes containing convective signatures, the second method is able to predict approximately 80% of observed variance with respect to validation. Estimated wind speed provides the bulk of predictive power while parameters related to the kilometer-scale distribution of spectral energy contribute to a significant reduction in prediction errors, enabling the methodology to be applied on a scene-by-scene basis. Differences between these physically based estimates and parameterized numerical models may guide the latter's improvement. © 2023. The Authors.

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Old Rules in a New Game: Mapping Uncertainty Quantification to Quantum Machine Learning

Item Type Conference Paper

Author M. Wendlinger

Author K. Tschärke

Author P. Debus

Editor Culhane C.

Editor Byrd G.

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Abstract One of the key obstacles in traditional deep learning is the reduction in model transparency caused by increasingly intricate model functions, which can lead to problems such as overfitting and excessive confidence in predictions. With the advent of quantum machine learning offering possible advances in computational power and latent space complexity, we notice the same opaque behavior. Despite significant research in classical contexts, there has been little advancement in addressing the black-box nature of quantum machine learning. Consequently, we approach this gap by building upon existing work in classical uncertainty quantification and initial explorations in quantum Bayesian modeling to theoretically develop and empirically evaluate techniques to map classical uncertainty quantification methods to the quantum machine learning domain. Our findings emphasize the necessity of leveraging classical insights into uncertainty quantification to include uncertainty awareness in the process of designing new quantum machine learning models. © 2025 IEEE.

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ON INFORMATION GAIN, KULLBACK-LEIBLER DIVERGENCE, ENTROPY PRODUCTION AND THE INVOLUTION KERNEL

Item Type Journal Article

Author A.O. Lopes

Author J.K. Mengue

Abstract It is well known that in Information Theory and Machine Learning the Kullback-Leibler divergence, which extends the concept of Shannon entropy, plays a fundamental role. Given an a priori probability kernel $\hat{v}$ and a probability π on the measurable space $X \times Y$ we consider an appropriate definition of entropy of π relative to $\hat{v}$, which is based on previous works. Using this concept of entropy we obtain a natural definition of information gain for general measurable spaces which coincides with the mutual information given from the K-L divergence in the case $\hat{v}$ is identified with a probability v on X . This will be used to extend the meaning of specific information gain and dynamical entropy production to the model of thermodynamic formalism for symbolic dynamics over a compact alphabet (TFCA model). Via the concepts of involution kernel and dual potential, one can ask if a given potential is symmetric - the relevant information is available in the potential. In the affirmative case, its corresponding equilibrium state has zero entropy production. © 2022 American Institute of Mathematical Sciences. All rights reserved.

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Export Date: 20 September 2026; Cited By: 10; Funding details: Conselho Nacional de Desenvolvimento Científico e Tecnológico, CNPq; Correspondence Address: A.O. Lopes; Instituto de Matemática e Estatística - UFRGS, Porto Alegre, 91509-900, Brazil; email: arturoscar.lopes@gmail.com

On Intra-page and Inter-page Semantic Analysis of Web Pages

Item Type Conference Paper

Author J. Wang

Author J. Wang

Author G. Wu

Author H. Tsuda

Editor Ji D.H.

Editor Lua K.T.

Abstract To make real Web information more machine processable, this paper presents a new approach to intra-page and inter-page semantic analysis of Web pages. Our approach consists of Web pages structure analysis and semantic clustering for intra-page semantic analysis, and machine learning based link semantic analysis for inter-page analysis. Based on the automatic repetitive patterns discovery in structure level and clustering in semantic level, we explore the intra-page semantic structure of Web pages and extend the processing unit from the whole page to a finer granularity, i.e., semantic information blocks within pages. After observing the various hyperlinks, we synthesize the Web inter-page semantic and

define an information organizing oriented hyperlink semantic category. Considering the presentation of the hyperlink carrier and intra-page semantic structure, we propose corresponding feature selection and quantification methods, and then exploit the C4.5 decision-tree method to classify hyperlink semantic type and analyze the inter-page semantic structure. In our experiments, the results suggest that our approach is feasible for machine processing. © 2003 Chinese and Oriental Languages Information Processing Society. All right reserved.

Date 2003

URL <https://www.scopus.com/pages/publications/105044126281?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Pac. Asia Conf. Lang., Inf. Comput., PACLIC

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Pages 408-415

Proceedings Title Proc. Pac. Asia Conf. Lang., Inf. Comput., PACLIC

Conference Name Proceedings of the 17th Pacific Asia Conference on Language, Information and Computation, PACLIC 2003

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Tags:

Semantics, Semantic Web, Websites, Hyperlinks, Hypertext systems, Semantic analysis, Latent semantic analysis, New approaches, Processable, Semantic structures, Structure analysis, Structure semantic, Web information, Web page structure, Web-page

Notes:

EID: 2-s2.0-105044126281

Export Date: 20 September 2026; Cited By: 1; Conference name: 17th Pacific Asia Conference on Language, Information and Computation, PACLIC 2003; Conference date: 1 October 2003 through 3 October 2003; Conference code: 224392

On learning multcategory classification with sample queries

Item Type Journal Article

Author J. Ratsaby

Abstract Consider the pattern recognition problem of learning multcategory classification from a labeled sample, for instance, the problem of learning character recognition where a category corresponds to an alphanumeric letter. The classical theory of pattern recognition assumes labeled examples appear according to the unknown underlying pattern-class conditional probability distributions where the pattern classes are picked randomly according to their a priori probabilities. In this paper we pose the following question: Can the learning accuracy be improved if labeled examples are independently randomly drawn according to the underlying class conditional probability distributions but the pattern classes are chosen not necessarily

according to their a priori probabilities? We answer this in the affirmative by showing that there exists a tuning of the sub-sample proportions which minimizes a loss criterion. The tuning is relative to the intrinsic complexity of the Bayes-classifier. As this complexity depends on the underlying probability distributions which are assumed to be unknown, we provide an algorithm which learns the proportions in an on-line manner utilizing sample querying which asymptotically minimizes the criterion. In practice, this algorithm may be used to boost the performance of existing learning classification algorithms by apportioning better sub-sample proportions.

Date 2003

DOI 10.1016/S0890-5401(03)00083-X

URL <https://www.scopus.com/pages/publications/0042695679?origin=resultslist>

Archive Scopus

Language English

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Publisher Academic Press

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Tags:

Risk assessment, Learning systems, Learning algorithms, Pattern recognition, Computational complexity, Vectors, Probability distributions, Random processes, Character recognition, Convergence of numerical methods, Boolean functions, Multicategory classification, Asymptotic stability, On-line learning algorithm, Online learning algorithm, Sample queries, Stochastic gradient descent learning, Structural risk minimization

Notes:

EID: 2-s2.0-0042695679

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: J. Ratsaby; Department of Computer Science, University College London, London WC1E 6BT, Gower Street, United Kingdom; email: j.ratsaby@cs.ucl.ac.uk; CODEN: INFCE

On Machine Learning-Based Techniques for Future Sustainable and Resilient Energy Systems

Item Type Journal Article

Author J. Wang

Author P. Pinson

Author S. Chatzivasileiadis

Author M. Panteli

Author G. Strbac

Author V. Terzija

Abstract Permanently increasing penetration of converter-interfaced generation and renewable energy sources (RESs) makes modern electrical power systems more vulnerable to low probability and high impact events, such as extreme weather, which could lead to severe contingencies, even blackouts. These contingencies can be further propagated to neighboring energy systems over coupling components/technologies and consequently negatively influence the entire multi-energy system (MES) (such as gas, heating and electricity) operation and its resilience. In recent years, machine learning-based techniques (MLBTs) have been intensively applied to solve various power system problems, including system planning, or security and reliability assessment. This paper aims to review MES resilience quantification methods and the application of MLBTs to assess the resilience level of future sustainable energy systems. The open research questions are identified and discussed, whereas the future research directions are identified. © 2010-2012 IEEE.

Date 2023

DOI 10.1109/TSTE.2022.3194728

URL <https://www.scopus.com/pages/publications/85135747137?origin=resultslist>

Archive Scopus

Language English

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Publisher Institute of Electrical and Electronics Engineers Inc.

Pages 1230-1243

Publication IEEE Transactions on Sustainable Energy

ISSN 19493029 (ISSN)

Issue 2

Journal Abbr IEEE Trans. Sustainable Energy

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Tags:

Security, Learning systems, machine learning, Machine-learning, Energy conservation, Machine learning, Extreme events, resilience, Reliability analysis, Resilience, Power, Power system, Couplings, Electric power system planning, Electric power transmission networks, multi-energy systems, Multi-energy systems, Power system dynamics, Power systems reliability, Power systems stability, Renewable energy resources, Stability analyze, sustainable energy systems, Sustainable energy systems

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On naive Bayesian fusion of dependent classifiers

Item Type Journal Article

Author H. Altınçay

Abstract In classifier combination, the relative values of a posteriori probabilities assigned to different hypotheses are more important than the accuracy of their estimates. Because of this, the independence requirement in naive Bayesian fusion should be examined from combined accuracy point of view. In this study, it is investigated whether there is a set of dependent classifiers which provides a better combined accuracy than independent classifiers when naive Bayesian fusion is used. For this purpose, two classes and three classifiers case is initially considered where the pattern classes are not equally probable. Taking into account the increased complexity in formulations, equal a priori probabilities are considered in the general case where N classes and K classifiers are used. The analysis carried out has shown that the combination of dependent classifiers using naive Bayesian fusion may provide much better combined accuracies compared to independent classifiers. © 2005 Elsevier B.V. All rights reserved.

Date 2005

DOI 10.1016/j.patrec.2005.05.003

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Archive Scopus

Language English

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Publication Pattern Recognition Letters

ISSN 01678655 (ISSN)

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Journal Abbr Pattern Recogn. Lett.

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Computational complexity, Probability, Classifiers, Best dependent classifiers, Classifier fusion, Independent classifiers, Naive Bayesian fusion

Notes:

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Export Date: 20 September 2026; Cited By: 23; Correspondence Address: H. Altınçay; Department of Computer Engineering, Eastern Mediterranean University, Gazi Mağusa, Northern Cyprus via Mersin 10, Turkey; email: hakan.altincay@emu.edu.tr

On Target Shift in Adversarial Domain Adaptation

Item Type Conference Paper

Author Y. Li

Author M. Murias

Author S. Major

Author G. Dawson

Author D.E. Carlson

Editor Chaudhuri K.

Editor Sugiyama M.

Abstract Discrepancy between training and testing do-mains is a fundamental problem in the gener-alization of machine learning techniques. Re-cently, several approaches have been proposed to learn domain invariant feature representa-tions through adversarial deep learning. How-ever, label shift, where the percentage of data in each class is different between domains, has received less attention. Label shift naturally arises in many contexts, especially in behav-ioral studies where the behaviors are freely chosen. In this work, we propose a method called Domain Adversarial nets for Target Shift (DATS) to address label shift while learn-ing a domain invariant representation. This is accomplished by using distribution match-ing to estimate label proportions in a blind test set. We extend this framework to handle multiple domains by developing a scheme to upweight source domains most similar to the target domain. Empirical results show that this framework performs well under large la-bel shift in synthetic and real experiments, demonstrating the practical importance. © 2019 by the author(s).

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Mach. Learn. Res.

Volume 89

Publisher ML Research Press

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Proceedings Title Proc. Mach. Learn. Res.

Conference Name Proceedings of Machine Learning Research

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Tags:

Deep learning, Learning systems, Learning algorithms, Machine learning techniques, Adversarial machine learning, Generative adversarial networks, Labeling, Personnel training, Domain adaptation, Learn+, Test sets, Blind test, Invariant representation, Invariant features, Training and testing, Address labels, Do-mains

Notes:

EID: 2-s2.0-105020690102

Export Date: 20 September 2026; Cited By: 6; Funding details: Marcus Foundation; National Institute of Child Health and Human Development, NICHD, (P50-HD093074); National Institute of Child Health and Human Development, NICHD; Conference name: 22nd International Conference on Artificial Intelligence and Statistics, AISTATS 2019; Conference date: 16 April 2019 through 18 April 2019; Conference code: 340719

On target shift in adversarial domain adaptation

Item Type Conference Paper

Author Y. Li

Author M. Murias

Author S. Major

Author G. Dawson

Author D.E. Carlson

Abstract Discrepancy between training and testing domains is a fundamental problem in the generalization of machine learning techniques. Recently, several approaches have been proposed to learn domain invariant feature representations through adversarial deep learning. However, label shift, where the percentage of data in each class is different between domains, has received less attention. Label shift naturally arises in many contexts, especially in behavioral studies where the behaviors are freely chosen. In this work, we propose a method called Domain Adversarial nets for Target Shift (DATS) to address label shift while learning a domain invariant representation. This is accomplished by using distribution matching to estimate label proportions in a blind test set. We extend this framework to handle multiple domains by developing a scheme to upweight source domains most similar to the target domain. Empirical results show that this framework performs well under large label shift in synthetic and real experiments, demonstrating the practical importance. © 2019 by the author(s).

Date 2019

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Language English

Extra Journal Abbreviation: AISTATS - Int. Conf. Artif. Intell. Stat.

Volume 89

Publisher PLMR

Proceedings Title AISTATS - Int. Conf. Artif. Intell. Stat.

Conference Name AISTATS 2019 - 22nd International Conference on Artificial Intelligence and Statistics

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Tags:

Deep learning, Machine learning techniques, Generalisation, Domain adaptation, Learn+, Feature representation, Invariant representation, Invariant features, Training and testing, Address labels, Behavioural studies

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On the characterization of continuous fibres fracture by quantifying acoustic emission and acousto-ultrasonics waveforms

Item Type Journal Article

Author Y.Z. Pappas

Author A. Kontsos

Author T.H. Loutas

Author V. Kostopoulos

Abstract The aim of the present work is the classification of the characteristics of elastic waves, which are generated by fibre failures during quasi-static tensile fibre bundle tests and captured by the use of acoustic emission (AE) method. In addition, elastic waves generated out of the fibre bundle gauge length and propagated through the bundle at different stages of the loading process using an acousto-ultrasonic (AU) technique are also classified. To this target, a large number of tests were conducted on organic, ceramic and carbon fibre bundles according to DIN 53942. An in-house developed analysis and quantification methodology of the captured AE and AU waveforms is proposed in order to identify the frequency content of the fibre failure event, to characterize the medium of propagation and to investigate the effect of the acquisition system on the monitored signals. In fact, the application of the proposed analysis on the results of the conducted mechanical tests leads to the development of a useful database, concerning the ranges of AE features and the representative sets of frequency values that correspond to fibre fracture. Furthermore, the proposed database offers valuable knowledge for the role of material parameters, such as fibre structure and properties, on the characteristics of the recorded waveforms, constituting in this way a valuable tool that enables a better understanding of elastic waves initiation and propagation through continuous fibres. © 2003 Published by Elsevier Ltd.

Date 2004
DOI 10.1016/j.ndteint.2003.11.004
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Language English
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Tags:

Sensitivity analysis, Frequency domain analysis, Time domain analysis, Nondestructive examination, Database systems, Acoustic emission, Fracture, Transducers, Acoustic emissions, Acoustic wave propagation, Acousto-ultrasonics, Acousto-ultrasonics technique, Carbon fibers, Database, Elastic waves, Fiber failures, Fibre failure, Parametric sensitivity analysis, Wave propagation, Waveform analysis

Notes:

EID: 2-s2.0-2442626717

Export Date: 20 September 2026; Cited By: 36; Correspondence Address: V. Kostopoulos; Applied Mechanics Laboratory, Department of Mechanical Engineering, University of Patras, 265 00 Rio Patras, Greece; email: kostopoulos@mech.upatras.gr; CODEN: NDTIE

On the damage behaviour in dual-phase DP800 steel deformed in single and combined strain paths

Item Type Journal Article
Author M.A. Wollenweber
Author S. Medghalchi
Author L.R. Guimarães
Author N. Lohrey
Author C.F. Kusche
Author U. Kerzel
Author T. Al-Samman
Author S. Korte-Kerzel

Abstract Ductile damage in dual-phase steels deteriorates the mechanical properties, resulting in a lowered crashworthiness and a shorter life-time under cyclic loading. However, most works focus on the observation of damage under simple strain paths, even though work pieces are often subjected to combined strain paths during forming. In this work, we set out to characterise the damage behaviour for a DP800 steel subjected to combinations of tensile and bending loads. To this end, we use an automated approach based on machine learning to quantify damage in high-resolution SEM panoramic images, coupled with finite element modelling to determine the stress state, and high-throughput nanoindentation to determine the strain hardening behaviour. We find a strong connection between loading direction and damage formation in both steps of the forming process. By focussing on different possible combinations of positive and negative triaxiality, we show that the combination of two deformation modes with positive triaxiality leads to a promotion of damage formation. A negative triaxiality applied after a positive triaxiality leads to void closure, reducing the number of damage sites and total void area. Importantly, deformation with negative triaxiality carried out before deformation with positive triaxiality inhibits damage nucleation and growth due to strain hardening. © 2023 The Authors

Date 2023

DOI 10.1016/j.matdes.2023.112016

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Publication Materials and Design

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Journal Abbr Mater. Des.

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Tags:

Machine-learning, Machine learning, Machine Learning, Dual-phases steels, Crashworthiness, Damage, Damage behaviour, Damage formation, Dual phase, Dual phasis, Dual-phase steel, Nano indentation, Nanoindentation, Plasticity, Strain hardening, Strain paths, Triaxiality

Notes:

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On the Impact of Perceptual Compression on Deep Learning

Item Type Conference Paper

Author G. Friedland

Author R. Jia

Author J. Wang

Author B. Li

Author N. Mundhenk

Abstract This paper proposes a fundamental answer to a frequently asked question in multimedia evaluation and data set creation: Do artifacts from perceptual compression contribute to error in the machine learning process and if so, how much? Our approach to the problem is an information reinterpretation of the Helmholtz free energy formula to explain the relationship between content and noise when using sensors (such as cameras or microphones) to capture multimedia data. The reinterpretation guides a bit-measurement of the noise contained in images, audio, and video by combining a classifier with perceptual compression, such as JPEG or MP3. Our experiments on CIFAR-10, ImageNet, and CSAIL Places as well as Fraunhofer's IDMT-SMT-Audio-Effects dataset indicate that, at the right quality level, perceptual compression is actually not harmful but contributes to a significant reduction of complexity of the machine learning process. That is, our noise quantification method can be used to speed up the training of deep learning classifiers significantly while maintaining, or sometimes even improving, overall classification accuracy. © 2020 IEEE.

Date 2020

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - Int. Conf. Multimed. Inf. Process. Retr., MIPR

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-172814272-2 (ISBN)

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Conference Name Proceedings - 3rd International Conference on Multimedia Information Processing and Retrieval, MIPR 2020

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Tags:

Deep learning, Learning systems, Neural Networks, Audio, Learning classifiers, Classification accuracy, datasets, Audio acoustics, Acoustic noise, Free energy, Audio effects, Compression, Frequently asked questions, Images, Information, Multimedia data, Noise quantification, Perceptual compression, Quality levels

Notes:

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On the minimal supervision for training any binary classifier from only unlabeled data

Item Type Conference Paper

Author N. Lu

Author G. Niu

Author A.K. Menon

Author M. Sugiyama

Abstract Empirical risk minimization (ERM), with proper loss function and regularization, is the common practice of supervised classification. In this paper, we study training arbitrary (from linear to deep) binary classifier from only unlabeled (U) data by ERM. We prove that it is impossible to estimate the risk of an arbitrary binary classifier in an unbiased manner given a single set of U data, but it becomes possible given two sets of U data with different class priors. These two facts answer a fundamental question-what the minimal supervision is for training any binary classifier from only U data. Following these findings, we propose an ERM-based learning method from two sets of U data, and then prove it is consistent. Experiments demonstrate the proposed method could train deep models and outperform state-of-the-art methods for learning from two sets of U data. © 7th International Conference on Learning Representations, ICLR 2019. All Rights Reserved.

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Extra Journal Abbreviation: Int. Conf. Learn. Represent., ICLR

Publisher International Conference on Learning Representations, ICLR

Proceedings Title Int. Conf. Learn. Represent., ICLR

Conference Name 7th International Conference on Learning Representations, ICLR 2019

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Tags:

Risk perception, Classification (of information), Learning systems, State-of-the-art methods, Learning methods, Unlabeled data, Different class, Loss functions, Supervised classification, Binary classifiers, Empirical risk minimization

Notes:

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On the surplus accuracy of data-driven energy quantification methods in the residential sector

Item Type Journal Article

Author L. Wederhake

Author S. Wenninger

Author C. Wiethe

Author G. Fridgen

Abstract Increasing trust in energy performance certificates (EPCs) and drawing meaningful conclusions requires a robust and accurate determination of building energy performance (BEP). However, existing and by law prescribed engineering methods, relying on physical principles, are under debate for being error-prone in practice and ultimately inaccurate. Research has heralded data-driven methods, mostly machine learning algorithms, to be promising alternatives: various studies compare engineering and data-driven methods with a clear advantage for data-driven methods in terms of prediction accuracy for BEP. While previous studies only investigated the prediction accuracy for BEP, it yet remains unclear which reasons and cause–effect relationships lead to the surplus prediction accuracy of data-driven methods. In this study, we develop and discuss a theory on how data collection, the type of auditor, the energy quantification method, and its accuracy relate to one another. First, we introduce cause–effect relationships for quantifying BEP method-agnostically and investigate the influence of several design parameters, such as the expertise of the auditor issuing the EPC, to develop our theory. Second, we evaluate and discuss our theory with literature. We find that data-driven methods positively influence cause–effect relationships, compensating for deficits due to auditors’ lack of expertise, leading to high prediction accuracy. We provide recommendations for future research and practice to enable the informed use of data-driven methods. © 2022, The Author(s).

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Energy, Energy efficiency, Machine learning, Forecasting, Learning algorithms, Data quality, Quantification methods, Building energy performance, Energy quantification methods, Prediction accuracy, Building energy, Energy quantification method, Building energy data, Data-driven methods, Energy data, Prediction accuracy theory

Notes:

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On the Use of m-Probability-Estimation and Imprecise Probabilities in the Naïve Bayes Classifier

Item Type Journal Article
Author J.G. Castellano
Author S. Moral-García
Author C.J. Mantas
Author J. Abellán
Abstract Within the field of supervised classification, the naïve Bayes (NB) classifier is a very simple and fast classification method that obtains good results, being even comparable with much more complex models. It has been proved that the NB model is strongly dependent on the estimation of conditional probabilities. In the literature, it had been shown that the classical and Laplace estimations of probabilities have some drawbacks and it was proposed a NB model that takes into account the a priori probabilities in order to estimate the conditional probabilities, which was called m-probability-estimation. With a very scarce experimentation, this approximation based on m-probability-estimation demonstrated to provide better results than NB with classical and Laplace estimations of probabilities. In this research, a new naïve Bayes variation is proposed, which is based on the m-probability-estimation version and takes into account imprecise probabilities in order to calculate the a priori probabilities. An exhaustive experimental research is carried out, with a large number of data sets and different levels of class noise. From this experimentation, we can conclude that the proposed NB model and the m-probability-estimation approach

provide better results than NB with classical and Laplace estimation of probabilities. It will be also shown that the proposed NB implies an improvement over the m-probability-estimation model, especially when there is some class noise. © 2020 World Scientific Publishing Company.

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Issue 4

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Supervised learning, Probability, Barium compounds, Classifiers, Naive bayes, Conditional probabilities, Naive Bayes models, Probability estimation, Laplace transforms, A-priori probabilities, Noisy data, imprecise probabilities, Imprecise probabilities, m-estimate, M-estimates, m-probability-estimation, M-probability-estimation, naïve Bayes, Naive Bayes classifiers, noisy data

Notes:

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On the value of uncertainty quantification in deep learning based breast cancer molecular subtype classification

Item Type Journal Article

Author G.A.D. Tchokponhoue

Author A. Idri

Abstract Cancer is still a major global health issue and one of the leading causes of death globally. Despite significant progress in the field of machine learning (ML) and deep learning (DL) in the medical field and especially in cancer research, uncertainty quantification has been largely under-investigated, resulting in the development of decision systems with questionable reliability and confidence, which are essential for

effective integration and adoption in a clinical context. In this study, we evaluate and discuss five uncertainty quantification (UQ) methods: Single Deterministic, Monte Carlo Dropout (MCD), Bayesian Neural Network based on Variational Inference (BNN), Deep Ensemble (DE), and Bootstrap Ensemble (BG) for the identification of breast cancer molecular subtype using a deep neural network (DNN). These empirical evaluations were based, in addition to the classical classification performance criteria such as accuracy, recall, precision and f1 score, on other advanced criteria related to sensibility, interpretability, robustness, and out-of-distribution (OOD) detection. Moreover, we introduce and evaluate new algorithms for synthetic OOD data generation, crucial for assessing the reliability of quantified uncertainties. Results indicated that Ensemble-based UQ methods generally provided more robust uncertainty estimates and performed best in terms of sensitivity. Stochastic methods, on the other hand, achieved superior calibration, with Bayesian Neural Networks in particular demonstrating strong robustness to adversarial attacks, an attribute that may enhance the generalization capacity of classifiers. Overall, we found that uncertainty estimates not only provide reliable confidence measures but also enhance the transparency of model predictions, supporting safer clinical deployment. © 2025 Elsevier B.V.

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Notes:

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On Truthing Issues in Supervised Classification

Item Type Journal Article

Author J.K. Su

Abstract Ideal supervised classification assumes known correct labels, but various truthing issues can arise in practice: noisy labels; multiple, conflicting labels for a sample; missing labels; and different labeler combinations for different samples. Previous work introduced a noisy-label model, which views the observed noisy labels as random variables conditioned on the unobserved correct labels. It has mainly focused on estimating the conditional distribution of the noisy labels and the class prior, as well as estimating the correct labels or training with noisy labels. In a complementary manner, given the conditional distribution and class prior, we apply estimation theory to classifier testing, training, and comparison of different combinations of labelers. First, for binary classification, we construct a testing model and derive approximate marginal posteriors for accuracy, precision, recall, probability of false alarm, and F-score, and joint posteriors for ROC and precision-recall analysis. We propose minimum mean-square error (MMSE) testing, which employs empirical Bayes algorithms to estimate the testing-model parameters and then computes optimal point estimates and credible regions for the metrics. We extend the approach to multi-class classification to obtain optimal estimates of accuracy and individual confusion-matrix elements. Second, we present a unified view of training that covers probabilistic (i.e., discriminative or generative) and non-probabilistic models. For the former, we adjust maximum-likelihood or maximum a posteriori training for truthing issues; for the latter, we propose MMSE training, which minimizes the MMSE estimate of the empirical risk. We also describe suboptimal training that is compatible with existing infrastructure. Third, we observe that mutual information lets one express any labeler combination as an equivalent single labeler, implying that multiple mediocre labelers can be as informative as, or more informative than, a single expert labeler. Experiments demonstrate the effectiveness of the methods and confirm the implication. © 2023 Massachusetts Institute of Technology.

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On using the modularity of recurrence network communities to detect change-point behaviour

Item Type Journal Article

Author D.M. Walker

Author A. Zaitouny

Author D.C. Corrêa

Abstract The behaviour of a dynamical system can be examined from time series using the technique of recurrence plots through visualization and quantification methods. One such method is the quadrant scan in which a recurrence plot of a time series is converted into a second time series whose local maxima can be identified and interpreted as possible transitions in dynamic behaviour. A recurrence plot can also be represented as a complex network. A quadrant scan can similarly be thought of as a partition of the network vertices into two communities. Here, we argue that different dynamic behavioural regimes can be realized as network communities and the quality of such partitions can be assessed using modularity. Thereby, community modularity can be used as an alternative to the quadrant scan. We investigate this correspondence with respect to two promising arenas for quadrant scan uptake, namely, concept drift detection from machine learning and tipping point or failure and damage monitoring in system maintenance. We also examine two additional data sets to highlight the potential of the methods. These are a geophysical time series of earth tremor data and a physiological time series of an electrocardiogram. © 2021 Elsevier Ltd

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Export Date: 20 September 2026; Cited By: 12; Funding details: Australian Research Council, ARC, (DP180100718, IC180100030); Correspondence Address: D.C. Corrêa; Department of Mathematics and Statistics, The University of Western Australia, Crawley, 6009, Australia; email: debora.correa@uwa.edu.au; CODEN: ESAPE

One of the major challenges of masking the bitter taste in medications: an overview of quantitative methods for bitterness

Item Type Journal Article

Author P. Wang

Author H. Li

Author Y. Wang

Author F. Dong

Author H. Li

Author X. Gui

Author Y. Ren

Author X. Gao

Author X. Li

Author R. Liu

Abstract Objective: The aim of the present study was to carry out a systematic research on bitterness quantification to provide a reference for scholars and pharmaceutical developers to carry out drug taste masking research. Significance: The bitterness of medications poses a significant concern for clinicians and patients. Scientifically measuring the intensity of drug bitterness is pivotal for enhancing drug palatability and broadening their clinical utility. Methods: The current study was carried out by conducting a systematic literature review that identified relevant papers from indexed databases. Numerous studies and research are cited and quoted in this article to summarize the features, strengths, and applicability of quantitative bitterness assessment methods. Results: In our research, we systematically outlined the classification and key advancements in quantitative research methods for assessing

drug bitterness, including in vivo quantification techniques such as traditional human taste panel methods, as well as in vitro quantification methods such as electronic tongue analysis. It focused on the quantitative methods and difficulties of bitterness of natural drugs with complex system characteristics and their difficulties in quantification, and proposes possible future research directions. Conclusion: The quantitative methods of bitterness were summarized, which laid an important foundation for the construction of a comprehensive bitterness quantification standard system and the formulation of accurate, efficient and rich taste masking strategies. Copyright © 2024 Wang, Li, Wang, Dong, Li, Gui, Ren, Gao, Li and Liu.

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active pharmaceuticals ingredients, bitterness, quantitative method, traditional Chinese medicine, traditional human taste panel method

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Export Date: 20 September 2026; Cited By: 12; Correspondence Address: X. Li; Department of Pharmacy, The First Affiliated Hospital of Henan University of Chinese Medicine, Zhengzhou, China; email: xuelinli450000@163.com; R. Liu; Department of Pharmacy, The First Affiliated Hospital of Henan University of Chinese Medicine, Zhengzhou, China; email: liuruixin7@163.com

One-class quantification

Item Type Conference Paper

Author D. dos Reis

Author A. Maletzke

Author E. Cherman

Author G. Batista

Editor Bonchi F.

Editor Gärtner T.

Editor Hurley N.

Editor Ifrim G.

Editor Berlingerio M.

Abstract This paper proposes one-class quantification, a new Machine Learning task. Quantification estimates the class distribution of an unlabeled sample of instances. Similarly to one-class classification, we assume that only a sample of examples of a single class is available for learning, and we are interested in counting the cases of such class in a test set. We formulate, for the first time, one-class quantification methods and assess them in a comprehensible open-set evaluation. In an open-set problem, several “subclasses” represent the negative class, and we cannot assume to have enough observations for all of them at training time. Therefore, new classes may appear after deployment, making this a challenging setup for existing quantification methods. We show that our proposals are simple and more accurate than the state-of-the-art in quantification. Finally, the approaches are very efficient, fitting batch and stream applications. Code related to this paper is available at: <https://github.com/denismr/One-class-Quantification>. © 2019, Springer Nature Switzerland AG.

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One-step Label Shift Adaptation via Robust Weight Estimation

Item Type Conference Paper

Author R. Fan

Author X. Ouyang

Author T. Luo

Author L. Zhang

Author C. Hou

Editor Kwok J.

Abstract Label shift is a prevalent phenomenon encountered in open environments, characterized by a notable discrepancy in the label distributions between the source (training) and target (test) domains, whereas the conditional distributions given the labels remain invariant. Existing label shift methods adopt a two-step strategy: initially computing the importance weight and subsequently utilizing it to calibrate the target outputs. However, this conventional strategy overlooks the intricate interplay between output adjustment and weight estimation. In this paper, we introduce a novel approach termed as One-step Label Shift Adaptation (OLSA). Our methodology jointly learns the predictive model and the corresponding weights through a bi-level optimization framework, with the objective of minimizing an upper bound on the target risk. To enhance the robustness of our proposed model, we incorporate a debiasing term into the upper-level classifier training and devise a regularization term for the lower-level weight estimation. Furthermore, we present theoretical analyses about the generalization bounds, offering guarantees for the model's performance. Extensive experimental results substantiate the efficacy of our proposal. © 2025 International Joint Conferences on Artificial Intelligence. All rights reserved.

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One-stop automated diagnostic system for active sacroiliitis in three-dimensional magnetic resonance images using artificial intelligent models: a retrospective study

Item Type Journal Article

- Author** X. Ji
- Author** Z. Li
- Author** L. Zeng
- Author** A. Wang
- Author** J. Dong
- Author** L. Sun
- Author** S. Yang
- Author** J. Zhu
- Author** F. Huang
- Author** T. Li
- Author** J. Zhang
- Author** K. Li

Abstract Background: Axial spondyloarthritis (axSpA) can potentially progress to ankylosing spondylitis, and diagnostic delay may lead to irreversible structural damage. Although sacroiliac joint magnetic resonance imaging (MRI) can detect bone marrow edema (BME) for early diagnosis, the complex etiologies and reliance on expert interpretation impede efficient identification in chronic low back pain populations.

Therefore, we developed an artificial intelligence (AI) system that integrates MRI analysis and automates report generation to facilitate the detection of active sacroiliitis and BME quantification. **Methods:** This retrospective study analyzed 691 patients (540 with axSpA and 151 with non-spondyloarthritis) from the Chinese People's Liberation Army General Hospital (2011–2023). Data were split into training and testing cohorts (4:1 ratio), with five-fold cross-validation applied to the training set. The system comprises four modules: (1) image preprocessing (intensity normalization), (2) coarse-to-fine 3D U-Net-based quadrant segmentation, (3) ResNet18-driven edema recognition (depth/intensity classification), and (4) diagnostic report generation using the QWEN large language model. **Results:** Among 691 patients (335 active sacroiliitis-positive and 356 negative), the AI system achieved Spondyloarthritis Research Consortium of Canada (SPARCC) scores of (15.12 ± 10.73) vs. (0.88 ± 1.13) in positive versus negative groups, respectively. The quadrant segmentation module achieved DICE similarity coefficients above 0.7 on the training, validation, and testing datasets. The edema inflammation, depth, and intensity classifiers exhibited good performance on the testing dataset, with balanced accuracies of 77.54%, 76.26%, and 80.91%, and area under the curve values of 0.85, 0.87, and 0.92, respectively. The intraclass correlation coefficient between SPARCC scores by our system and those by rheumatologists was 0.84 on the testing dataset. At the patient-level, our system achieved 90.81% sensitivity for diagnosis of active sacroiliitis on the testing dataset. **Conclusion:** The developed automated system enhances axSpA diagnostic efficiency by automating the identification of active sacroiliitis and enabling quantitative assessment of BME. © 2025

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Online Adaptation to Label Distribution Shift

Item Type Conference Paper

Author R. Wu

Author C. Guo

Author Y. Su

Author K.Q. Weinberger

Editor Ranzato M.

Editor Beygelzimer A.

Editor Dauphin Y.

Editor Liang P.S.

Editor Wortman Vaughan J.

Abstract Machine learning models often encounter distribution shifts when deployed in the real world. In this paper, we focus on adaptation to label distribution shift in the online setting, where the test-time label distribution is continually changing and the model must dynamically adapt to it without observing the true label. Leveraging a novel analysis, we show that the lack of true label does not hinder estimation of the expected test loss, which enables the reduction of online label shift adaptation to conventional online learning. Informed by this observation, we propose adaptation algorithms inspired by classical online learning techniques such as Follow The Leader (FTL) and Online Gradient Descent (OGD) and derive their regret bounds. We empirically verify our findings under both simulated and real world label distribution shifts and show that OGD is

particularly effective and robust to a variety of challenging label shift scenarios.
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Online Continual Learning via Logit Adjusted Softmax

Item Type Journal Article

Author Z. Huang

Author T. Li

Author C. Yuan

Author Y. Wu

Author X. Huang

Abstract Online continual learning is a challenging problem where models must learn from a non-stationary data stream while avoiding catastrophic forgetting. Inter-class imbalance during training has been identified as a major cause of forgetting, leading to model prediction bias towards recently learned classes. In this paper, we

theoretically analyze that inter-class imbalance is entirely attributed to imbalanced class-priors, and the function learned from intra-class intrinsic distributions is the optimal classifier that minimizes the class-balanced error. To that end, we present that a simple adjustment of model logits during training can effectively resist prior class bias and pursue the corresponding optimum. Our proposed method, Logit Adjusted Softmax, can mitigate the impact of inter-class imbalance not only in class-incremental but also in realistic scenarios that sum up class and domain incremental learning, with little additional computational cost. We evaluate our approach on various benchmarks and demonstrate significant performance improvements compared to prior arts. For example, our approach improves the best baseline by 4.6% on CIFAR10. Codes are available at. © 2024, Transactions on Machine Learning Research. All rights reserved.

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Online Feature Updates Improve Online (Generalized) Label Shift Adaptation

Item Type Conference Paper

Author R. Wu

Author S. Datta

Author Y. Su

Author D. Baby

Author Y.-X. Wang

Author K.Q. Weinberger

Editor Globerson A.

Editor Mackey L.

Editor Belgrave D.

Editor Fan A.

Editor Paquet U.

Editor Tomczak J.

Editor Zhang C.

Abstract This paper addresses the prevalent issue of label shift in an online setting with missing labels, where data distributions change over time and obtaining timely labels is challenging. While existing methods primarily focus on adjusting or updating the final layer of a pre-trained classifier, we explore the untapped potential of enhancing feature representations using unlabeled data at test-time. Our novel method, Online Label Shift adaptation with Online Feature Updates (OLS-OFU), leverages self-supervised learning to refine the feature extraction process, thereby improving the prediction model. By carefully designing the algorithm, theoretically OLS-OFU maintains the similar online regret convergence to the results in the literature while taking the improved features into account. Empirically, it achieves substantial improvements over existing methods, which is as significant as the gains existing methods have over the baseline (i.e., without distribution shift adaptations). © 2024 Neural information processing systems foundation. All rights reserved.

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Online Label Shift: Optimal Dynamic Regret meets Practical Algorithms

Item Type Conference Paper

Author D. Baby

Author S. Garg

Author T.-C. Yen

Author S. Balakrishnan

Author Z.C. Lipton

Author Y.-X. Wang

Editor Oh A.

Editor Neumann T.

Editor Globerson A.

Editor Saenko K.

Editor Hardt M.

Editor Levine S.

Abstract This paper focuses on supervised and unsupervised online label shift, where the class marginals $Q(y)$ varies but the class-conditionals $Q(x|y)$ remain invariant. In the unsupervised setting, our goal is to adapt a learner, trained on some offline labeled data, to changing label distributions given unlabeled online data. In the supervised setting, we must both learn a classifier and adapt to the dynamically evolving class marginals given only labeled online data. We develop novel algorithms that reduce the adaptation problem to online regression and guarantee optimal dynamic regret without any prior knowledge of the extent of drift in the label distribution. Our solution is based on bootstrapping the estimates of online regression oracles that track the drifting proportions. Experiments across numerous simulated and real-world online label shift scenarios demonstrate the superior performance of our proposed approaches, often achieving 1-3% improvement in accuracy while being sample and computationally efficient. Code is publicly available at this url. © 2023 Neural information processing systems foundation. All rights reserved.

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Online optimization methods for the quantification problem

Item Type Conference Paper**Author** P. Kar**Author** S. Li**Author** H. Narasimhan**Author** S. Chawla**Author** F. Sebastiani

Abstract The estimation of class prevalence, i.e., of the fraction of a population that belongs to a certain class, is an important task in data analytics, and finds applications in many domains such as the social sciences, market research, epidemiology, and others. For example, in sentiment analysis the goal is often not to estimate whether a specific text conveys a positive or a negative sentiment, but rather to estimate the overall distribution of positive and negative sentiments, e.g., in a certain time frame. A popular way of performing the above task, often dubbed quantification, is to use supervised learning in order to train a prevalence estimator from labeled data. In the literature there are several performance metrics for measuring the success of such prevalence estimators. In this paper we propose the first online stochastic algorithms for directly optimizing these quantification-specific performance measures. We also provide algorithms that optimize hybrid performance measures that seek to balance quantification and classification performance. Our algorithms present a significant advancement in the theory of multivariate optimization; we show, via a rigorous theoretical analysis, that they exhibit optimal convergence. We also report extensive experiments on benchmark and real data sets which demonstrate that our methods significantly outperform existing optimization techniques used for these performance measures. © 2016 ACM.

Date 2016**DOI** 10.1145/2939672.2939832**URL** <https://www.scopus.com/pages/publications/84984992237?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.**Volume** 13-17-August-2016**Publisher** Association for Computing Machinery

ISBN 978-145034232-2 (ISBN)**Pages** 1625-1634**Proceedings Title** Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.**Conference Name** Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining**Date Added** 20/09/2026, 9:52:57 pm**Modified** 20/09/2026, 9:52:57 pm**Tags:**

Sentiment analysis, Benchmarking, Population statistics, Data mining, Social sciences computing, Stochastic systems, Negative sentiments, Classification performance, Performance measure, Performance metrics, Multivariate optimization, Online optimization, Optimization techniques, Stochastic algorithms

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Open Set Label Shift with Test Time Out-of-Distribution Reference

Item Type Conference Paper**Author** C. Ye**Author** R. Tsuchida**Author** L. Petersson**Author** N. Barnes

Abstract Open set label shift (OSLS) occurs when label distributions change from a source to a target distribution, and the target distribution has an additional out-of-distribution (OOD) class. In this work, we build estimators for both source and target open set label distributions using a source domain in-distribution (ID) classifier and an ID/OOD classifier. With reasonable assumptions on the ID/OOD classifier, the estimators are assembled into a sequence of three stages: 1) an estimate of the source label distribution of the OOD class, 2) an EM algorithm for Maximum Likelihood estimates (MLE) of the target label distribution, and 3) an estimate of the target label distribution of OOD class under relaxed assumptions on the OOD classifier. The sampling errors of estimates in 1) and 3) are quantified with a concentration inequality. The estimation result allows us to correct the ID classifier trained on the source distribution to the target distribution without retraining. Experiments on a variety of open set label shift settings demonstrate the effectiveness of our model. Our code is available at <https://github.com/ChangkunYe/OpenSetLabelShift>. © 2025 IEEE.

Date 2025**DOI** 10.1109/CVPR52734.2025.02851

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Language English
Extra Journal Abbreviation: Proc IEEE Comput Soc Conf Comput Vision Pattern Recognit
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Tags:

label shift, em algorithm, open set classification

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: C. Ye; Australian National University, Australia; email: changkun.ye@anu.edu.au; Conference name: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025; Conference date: 11 June 2025 through 15 June 2025; Conference code: 211514; CODEN: PIVRE

Open-Set Hypothesis Transfer with Semantic Consistency

Item Type Journal Article

Author Z. Feng

Author C. Xu

Author D. Tao

Abstract Unsupervised open-set domain adaptation (UODA) is a realistic problem where unlabeled target data contain unknown classes. Prior methods rely on the coexistence of both source and target domain data to perform domain alignment, which greatly limits their applications when source domain data are restricted due to privacy concerns. In this paper we address the challenging hypothesis transfer setting for UODA, where data from source domain are no longer available during adaptation on target domain. Specifically, we propose to use pseudo-labels and a novel consistency regularization on target data, where using conventional formulations fails in this open-set setting. Firstly, our method discovers confident predictions on target domain and performs classification with pseudo-labels. Then we enforce the model to output consistent and definite predictions on semantically similar transformed inputs, discovering all latent class semantics. As a result, unlabeled data can be classified into discriminative classes coincided with either source classes or unknown classes. We theoretically prove that under perfect semantic transformation, the proposed objective that enforces consistency can recover the information of true labels in

prediction. Experimental results show that our model outperforms state-of-the-art methods on UODA benchmarks. © 1992-2012 IEEE.

Date 2021

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URL <https://www.scopus.com/pages/publications/85110566346?origin=resultslist>

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Forecasting, article, Semantics, theoretical study, Privacy concerns, prediction, State-of-the-art methods, semantics, Domain adaptation, Target domain, Unlabeled data, domain adaptation, Unknown class, consistency regularization, Open-set, pseudo-label, Semantic consistency, Semantic transformation

Notes:

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Export Date: 20 September 2026; Cited By: 17; Funding details: Australian Research Council, ARC, (DE180101438, DP210101859, FL-170100117, DP-180103424); Correspondence Address: C. Xu; School of Computer Science, Faculty of Engineering, The University of Sydney, Sydney, 2008, Australia; email: c.xu@sydney.edu.au; CODEN: IIPRE

OpenSlot: Mixed Open-Set Recognition With Object-Centric Learning

Item Type Journal Article

Author X. Yin

Author F. Pan

Author G. An

Author Y. Huo

Author Z. Xie

Author S.-E. Yoon

Abstract Existing open-set recognition (OSR) studies typically assume that each image contains only one class label, with the unknown test set (negative) having a disjoint label space from the known test set (positive), a scenario referred to as full-label shift. This paper introduces the mixed OSR problem, where test images contain multiple

class semantics, with both known and unknown classes co-occurring in the negatives, leading to a more complex super-label shift that better reflects real-world scenarios. To tackle this challenge, we propose the OpenSlot framework, based on object-centric learning, which uses slot features to represent diverse class semantics and generate class predictions. The proposed anti-noise slot (ANS) technique helps mitigate the impact of noise (invalid or background) slots during classification training, addressing the semantic misalignment between class predictions and ground truth. We evaluate OpenSlot on both mixed and conventional OSR benchmarks. Without elaborate designs, our method not only excels existing approaches in detecting super-label shifts across OSR tasks, but also achieves state-of-the-art performance on conventional benchmarks. Meanwhile, OpenSlot can localize class objects without using bounding boxes during training, demonstrating competitive performance in open-set object detection and potential for generalization. © 1999-2012 IEEE.

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Object recognition, Object detection, Objects detection, Class labels, Test sets, Label space, Class prediction, Disjoint labels, Mixed open-set recognition, object-centric learning, Object-centric learning, open-set object detection, Open-set object detection, Test images

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Opportunistic Osteoporosis Screening from Routine Knee Radiographs Using a Multi-Stage CNN Framework with External Validation

Item Type Journal Article

Author N. Sinnathakorn

Author C. Fahpinyo

Author W. Chalamjiak

Author S. Suantai

Abstract Background/Objectives: Osteoporosis is a major public health concern associated with increased fracture risk and reduced quality of life if not detected at an early stage. Automated analysis of knee X-ray images using artificial intelligence has shown promising potential for opportunistic osteoporosis screening. This study aims to develop and evaluate a multi-stage deep learning and machine learning framework for osteoporosis classification, with particular emphasis on external validation, calibration drift, and cross-domain generalization performance. Methods: Knee X-ray images were categorized into three classes: Normal, Osteopenia, and Osteoporosis. Deep features were extracted using pretrained convolutional neural networks, including ResNet18, EfficientNetB0, and DenseNet121. The extracted features were subsequently classified using multiple machine learning models, including Neural Network, Efficient Linear, Support Vector Machine, and Naive Bayes classifiers. Two data augmentation strategies were investigated: targeted minority-class augmentation and full $3\times$ dataset expansion with class balancing. Model performance was evaluated using accuracy, precision, recall, F1-score, and AUC on internal validation, independent test sets, and external validation datasets. Additional analyses included reliability calibration assessment, isotonic recalibration, and class-prior boosting with cross-validated threshold optimization to address external domain shift. Results: EfficientNetB0 and DenseNet121 consistently outperformed ResNet18 across most evaluation metrics. Under the balanced augmentation strategy, EfficientNetB0 combined with Efficient Linear demonstrated strong and stable performance, while DenseNet121 paired with a Neural Network achieved the highest overall classification performance. External validation revealed a substantial discrepancy between AUC and threshold-based metrics, indicating the presence of calibration drift and class-prior mismatch across imaging domains. Reliability analysis showed severe probability collapse in the Osteopenia class during external testing. Post-hoc recalibration improved probability reliability, while class-prior boosting substantially increased Osteopenia sensitivity and improved balanced accuracy and macro F1-score under external validation conditions. Conclusions: The proposed framework demonstrates the feasibility of combining pretrained CNN-based deep feature extraction with machine learning classifiers for osteoporosis classification from knee X-ray images. The findings further highlight that maintaining model performance under external testing conditions may require not only strong feature extraction capability but also adaptive recalibration and deployment-aware threshold optimization to address calibration drift and cross-domain variability. While the results are encouraging, the present study should be considered a proof-of-concept investigation. Although the framework was evaluated using an independent public external dataset, further validation using larger and more diverse multi-center clinical cohorts is necessary to establish generalizability and clinical utility before routine clinical implementation can be considered. © 2026 by the authors.

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Article, artificial intelligence, human, machine learning, deep learning, feature extraction, controlled study, major clinical study, neural network, convolutional neural network, support vector machine, confusion matrix, diagnostic test accuracy study, post hoc analysis, Bayesian learning, predictive value, World Health Organization, sensitivity and specificity, image analysis, image quality, screening, anatomical concepts, bone density, CNN framework, dual energy X ray absorptiometry, efficient linear, external validation, Gaussian noise, knee radiography, knee X-ray, osteopenia, osteoporosis, radiodiagnosis, residual neural network

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Optimal Bayesian classifier for land cover classification using Landsat TM data

Item Type Conference Paper
Author Yuanxin Zhu
Author Yunxin Zhao
Author Kannappan Palaniappan
Author Xiaobo Zhou
Author Xinhua Zhuang
Abstract An optimal Bayesian classifier using mixture distribution class models with joint learning of loss and prior probability functions is proposed for automatic land cover classification. The probability distribution for each land cover class is more realistically modeled as a population of Gaussian mixture densities. A novel two-stage learning algorithm is proposed to learn the Gaussian mixture model parameters for each land cover class and the optimal Bayesian classifier that minimizes the loss due to misclassification. In the first stage, the Gaussian

mixture model parameters for a given land cover class is learned using the Expectation-Maximization algorithm. The Minimum Description Length principle is used to automatically determine the number of Gaussian components required in the mixture model without overfitting. In the second stage, the loss functions and the a priori probabilities are jointly learned using a multiclass perceptron algorithm. Preliminary results indicate that modeling the multispectral, multitemporal remotely sensed radiance data for land cover using a Gaussian mixture model is superior to using unimodal Gaussian distributions. Higher classification accuracies for eight typical land cover categories over one full Landsat scene in central Missouri are demonstrated.

Date 2000

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Archive Scopus

Language English

Extra Journal Abbreviation: Dig Int Geosci Remote Sens Symp (IGARSS)

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Learning algorithms, Satellites, Remote sensing, Vegetation, Mathematical models, Optimization, Land cover, Expectation maximization algorithm, Minimum description length, Optimal Bayesian classifier

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Optimal classification and generalized prevalence estimates for diagnostic settings with more than two classes

Item Type Journal Article

Author R.A. Luke

Author A.J. Kearsley

Author P.N. Patrone

Abstract An accurate multiclass classification strategy is crucial to interpreting antibody tests. However, traditional methods based on confidence intervals or receiver operating characteristics lack clear extensions to settings with more than two classes. We address this problem by developing a multiclass classification based on probabilistic

modeling and optimal decision theory that minimizes the convex combination of false classification rates. The classification process is challenging when the relative fraction of the population in each class, or generalized prevalence, is unknown. Thus, we also develop a method for estimating the generalized prevalence of test data that is independent of classification of the test data. We validate our approach on serological data with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) naïve, previously infected, and vaccinated classes. Synthetic data are used to demonstrate that (i) prevalence estimates are unbiased and converge to true values and (ii) our procedure applies to arbitrary measurement dimensions. In contrast to the binary problem, the multiclass setting offers wide-reaching utility as the most general framework and provides new insight into prevalence estimation best practices. © 2023

Date 2023

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Journal Abbr Math. Biosci.

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human, Humans, Classification (of information), article, nonhuman, Decision theory, diagnosis, COVID-19, epidemiology, prevalence, SARS-CoV-2, Severe acute respiratory syndrome coronavirus 2, classification, Diseases, Multi-class classification, Coronavirus, Test data, Confidence interval, Prevalence, Prevalence estimation, Probabilistic models, antibody, Antibody testing, COVID-19 Testing, Receiver operating characteristics, Antibodies, Diagnostic, Diagnostics, disease prevalence, Modeling decisions, multiclass classification, Multiclass classification, Optimal classification

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OPTIMAL TRANSPORT MEETS NOISY LABEL ROBUST LOSS AND
MIXUP REGULARIZATION FOR DOMAIN ADAPTATION

Item Type Conference Paper

Author K. Fatras

Author H. Naganuma

Author I. Mitliagkas

Editor Chandar S.

Editor Pascanu R.

Editor Precup D.

Abstract It is common in computer vision to be confronted with domain shift: images which have the same class but different acquisition conditions. In domain adaptation (DA), one wants to classify unlabeled target images using source labeled images. Unfortunately, deep neural networks trained on a source training set perform poorly on target images which do not belong to the training domain. One strategy to improve these performances is to align the source and target image distributions in an embedded space using optimal transport (OT). However OT can cause negative transfer, i.e. aligning samples with different labels, which leads to overfitting especially in the presence of label shift between domains. In this work, we mitigate negative alignment by explaining it as a noisy label assignment to target images. We then mitigate its effect by appropriate regularization. We propose to couple the MixUp regularization (Zhang et al., 2018a) with a loss that is robust to noisy labels in order to improve domain adaptation performance. We show in an extensive ablation study that a combination of the two techniques is critical to achieve improved performance. Finally, we evaluate our method, called MIXUNBOT, on several benchmarks and real-world DA problems. © Proceedings of Machine Learning Research 2022.

Date 2022

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Extra Journal Abbreviation: Proc. Mach. Learn. Res.

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Conference Name Proceedings of Machine Learning Research

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Condition, Performance, Deep neural networks, Image enhancement, Image classification, Domain adaptation, Training sets, Regularisation, Noisy labels, Optimal transport, Source images, Labeled images, Target images

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Optimal Trunk-Reservation by Policy Learning

Item Type Conference Paper

Author A. Massaro

Author F. De Pellegrini

Author L. Maggi

Abstract In the framework of queuing theory with multiclass jobs, trunk-reservation is an admission control technique to handle job class priority in an online fashion, and serve as many high priority jobs as possible. This is achieved by rejecting jobs with sufficiently low priority when the buffer space becomes a scarce resource, i.e., when the queue may soon overflow. Mathematically, the objective is to maximize the long-term reward of the admitted jobs, where each job is assigned a reward which is monotonic with respect to the priority class it belongs to. In this paper we study online learning of optimal trunk-reservation policies when the system parameters, i.e., class arrival rates and service time, are unknown. Starting from a Markov Decision Process (MDP) formulation, we leverage the stairway structure of the optimal policy to define Integer Gradient Ascent (IGA), a reinforcement learning (RL) algorithm based on policy-gradient methods, specifically tailored to the mathematical properties of the problem at hand. We provide theoretical results on the convergence properties of IGA. Extensive numerical experiments characterize its behavior and confirm that it outperforms standard RL techniques in terms of convergence rate. © 2019 IEEE.

Date 2019

DOI 10.1109/INFOCOM.2019.8737552

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Language English

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Conference Name Proceedings - IEEE INFOCOM

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Markov processes, Machine learning, reinforcement learning, Reinforcement learning, Learning algorithms, Queueing theory, Gradient methods, Structural optimization, Mathematical properties, Policy gradient, Convergence properties, Markov Decision Processes, Numerical experiments, policy gradient, Policy gradient methods, reinforce, Trunk reservations, trunk-reservation

Notes:

EID: 2-s2.0-85068232461

Export Date: 20 September 2026; Cited By: 7; Conference name: 2019 IEEE Conference on Computer Communications, INFOCOM 2019; Conference date: 29 April 2019 through 2 May 2019; Conference code: 148846; CODEN: PINFE

Optimization of Spare Parts Varieties Based on Stochastic DEA Model

Item Type Journal Article

Author M. Wen

Author T. Zu

Author M. Guo

Author R. Kang

Author Y. Yang

Abstract Accurate inventory management starts with the scientific and rational classification of numerous varieties of spare parts. This paper presents a stochastic data envelopment analysis (DEA) model to address the problem of optimization of spares varieties under uncertainty. An index system is proposed in terms of product life-cycle process, which contains five design indexes, four operation indexes and five support indexes. Then, the quantification method of the index system is briefly discussed in preparation for mathematical calculation. A stochastic spares optimization model (SSOM) is proposed based on stochastic DEA with the constraints of 14 factors of the index system. The SSOM could be converted into equivalent deterministic models by probability theory, which overcomes the difficulty in solving non-linear programming. A numerical example is given to illustrate the proposed method in terms of ability to provide reasonable inventory management policies. © 2013 IEEE.

Date 2018

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optimization, Product design, Uncertainty analysis, Life cycle, Stochastic systems, Inventory control, Nonlinear programming, Optimization, Quantification methods, Stochastic models, Numerical methods, Optimization modeling, data envelopment analysis, Data envelopment analysis, Deterministic models, Mathematical calculations, spare parts, Spare parts, stochastic, Stochastic data envelopment analysis, Uncertain environment, Uncertain environments

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Optimization of the area under the ROC curve

Item Type Conference Paper

Author C.L. Castro

Author A.P. Braga

Abstract In this paper, we propose a new binary classification algorithm (AUCtron), based on gradient descent learning, that directly optimizes AUC (Area Under the ROC Curve). We compare it with a linear classifier and with AUCsplit proposed by [5]. The AUCtron algorithm implicitly considers class prior probabilities in the decision criteria. Our results demonstrated that AUC is a sensitive enough metric that when used in small and imbalanced data sets may lead to a better separation. © 2008 IEEE.

Date 2008

DOI 10.1109/SBRN.2008.25

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - Brazilian Symp. Neural Netw., SBRN

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Conference Name Proceedings - 10th Brazilian Symposium on Neural Networks, SBRN 2008

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Learning algorithms, Neural networks, Curve fitting, ROC curves, Linear classifiers, Imbalanced data sets, Prior probabilities, Binary classifications, Gradient descents

Notes:

EID: 2-s2.0-58049211597

Export Date: 20 September 2026; Cited By: 18; Correspondence Address: C. L. Castro; Universidade Federal de Minas Gerais, Depto. Engenharia Eletrônica, Campus UFMG, Pampulha 30.161-970, Belo Horizonte, MG, Av. Antônio Carlos, 6.627, Brazil; email: crislcastro@gmail.com; Conference name: 10th Brazilian Symposium on Neural Networks, SBRN 2008; Conference date: 26 October 2008 through 30 October 2008; Conference code: 74640

Optimized Distributed Automatic Modulation Classification in Wireless Sensor Networks Using Information Theoretic Measures

Item Type Journal Article

Author S. Hakimi

Author G. Abed Hodtani

Abstract Automatic modulation classification of digital signals is essential for intelligent communication systems. This paper addresses the distributed classification of digital amplitude-phase modulated signals in a system of multiple sensors, which observe the unknown signal corrupted with the additive white Gaussian noise. The sensors are connected to a fusion center through block-fading orthogonal multiple access channels. We introduce a new method, where: 1) for classification, an information-theoretic similarity measure known as correntropy is exploited by each local sensor; 2) for transmitting local decisions, an estimation of a priori probabilities is used by local sensors; and 3) for optimizing power allocation to each sensor, the Bhattacharyya distance is employed as the objective function by the fusion center. The proposed scheme improves the classification accuracy through exploiting channel diversity, which, in turn, enhances the overall performance. Simulation results validate the theoretically claimed improvement in the performance. © 2001-2012 IEEE.

Date 2017

DOI 10.1109/JSEN.2017.2688240

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Tags:

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Notes:

EID: 2-s2.0-85018943601

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Optimizing community-level surveillance data for pediatric asthma management

Item Type Journal Article

Author W.O. Benka-Coker

Author S.L. Gale

Author S.J. Brandt

Author J.R. Balmes

Author S. Magzamen

Abstract Community-level approaches for pediatric asthma management rely on locally collected information derived primarily from two sources: claims records and school-based surveys. We combined claims and school-based surveillance data, and examined the asthma-related risk patterns among adolescent students. Symptom data collected from school-based asthma surveys conducted in Oakland, CA were used for case identification and determination of severity levels for students (high and low). Survey data were matched to Medicaid claims data for all asthma-related health care encounters for the year prior to the survey. We then employed recursive partitioning to develop classification trees that identified patterns of demographics and healthcare utilization associated with severity. A total of 561 students had complete matched data; 86.1% were classified as high-severity, and 13.9% as low-severity asthma. The classification tree consisted of eight subsets: three indicating high severity and five indicating low severity. The risk subsets highlighted varying combinations of non-specific demographic and socioeconomic predictors of asthma prevalence, morbidity and severity. For example, the subset with the highest class-prior probability (92.1%) predicted high-severity asthma and consisted of students without prescribed rescue medication, but with at least one in-clinic nebulizer treatment. The predictive accuracy of the tree-based model was approximately 66.7%, with an estimated 91.1% of high-severity cases and 42.3% of low-severity cases correctly predicted. Our analysis draws on the strengths of two complementary datasets to provide community-level information on children with asthma, and demonstrates the utility of recursive partitioning methods to explore a combination of features that convey asthma severity. © 2018 The Authors

Date 2018

DOI 10.1016/j.pmedr.2018.02.004**URL** <https://www.scopus.com/pages/publications/85042213810?origin=resultslist>**Archive** Scopus**Language** English**Volume** 10**Publisher** Elsevier Inc.**Pages** 55-61**Publication** Preventive Medicine Reports**ISSN** 22113355 (ISSN)**Journal Abbr** Preventive Med. Reports**Date Added** 20/09/2026, 9:52:45 pm**Modified** 20/09/2026, 9:52:45 pm**Tags:**

Article, female, human, male, risk assessment, risk factor, controlled study, major clinical study, prevalence, Classification, health care, patient care, disease severity, child, health survey, disease association, asthma, pediatrics, community care, Disease management, priority journal, health care utilization, Asthma, disease surveillance, medicaid, morbidity, Risk stratification, Statistical data analysis

Notes:

EID: 2-s2.0-85042213810

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Optimizing text quantifiers for multivariate loss functions

Item Type Journal Article**Author** A. Esuli**Author** F. Sebastiani

Abstract We address the problem of quantification, a supervised learning task whose goal is, given a class, to estimate the relative frequency (or prevalence) of the class in a dataset of unlabeled items. Quantification has several applications in data and text mining, such as estimating the prevalence of positive reviews in a set of reviews of a given product or estimating the prevalence of a given support issue in a dataset of transcripts of phone calls to tech support. So far, quantification has been addressed by learning a general-purpose classifier, counting the unlabeled items that have been assigned the class, and tuning the obtained counts according to some heuristics. In this article, we depart from the tradition of using general-purpose classifiers and use instead a supervised learning model for structured prediction, capable of generating

classifiers directly optimized for the (multivariate and nonlinear) function used for evaluating quantification accuracy. The experiments that we have run on 5,500 binary high-dimensional datasets (averaging more than 14,000 documents each) show that this method is more accurate, more stable, and more efficient than existing state-of-the-art quantification methods. © 2015 ACM.

Date 2015

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Language English

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Classification (of information), Supervised learning, Optimization, Text classification, Text mining, Loss functions, Quantification, Prevalence estimation, Prior estimation, Kullback-Leibler divergence, Kullback Leibler divergence

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Ordinal Quantification Through Regularization

Item Type Conference Paper

Author M. Bunse

Author A. Moreo

Author F. Sebastiani

Author M. Senz

Editor Amini M.-R.

Editor Canu S.

Editor Fischer A.

Editor Guns T.

Editor Kralj Novak P.

Editor Tsoumakas G.

Abstract Quantification, i.e., the task of training predictors of the class prevalence values in sets of unlabelled data items, has received increased attention in recent years. However, most quantification research has concentrated on developing algorithms for binary and multiclass problems in which the classes are not ordered. We here study the ordinal case, i.e., the case in which a total order is defined on the set of $n > 2$ classes. We give three main contributions to this field. First, we create and make available two datasets for ordinal quantification (OQ) research that overcome the inadequacies of the previously available ones. Second, we experimentally compare the most important OQ algorithms proposed in the literature so far. To this end, we bring together algorithms that are proposed by authors from very different research fields, who were unaware of each other's developments. Third, we propose three OQ algorithms, based on the idea of preventing ordinally implausible estimates through regularization. Our experiments show that these algorithms outperform the existing ones if the ordinal plausibility assumption holds. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

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Proceedings Title Lect. Notes Comput. Sci.

Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Unlabeled data, Ordinal classification, Regularisation, Quantification, Prevalence estimation, Quantification research, Supervised prevalence estimation, Data items, Multiclass problem, Binary problems

Notes:

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Outcome of HLA-oriented therapy for gastric cancer in retrospective and prospective study

Item Type Journal Article

Author K. Ogoshi

Author Y. Koyanagi

Author K. Tsuji

Author K. Isono

Abstract There are no strategies to predict a patient's response to therapy. In a previous report, we classified Japanese patients into four groups according to the incidence of HLA antigens, by using the quantification method III. We examined the HLA antigens of patients before surgery and evaluated their outcome according to the HLA classification by this quantification method. We also reported that HLA-oriented therapy was better than old fashioned therapy. The aim of the present study was to evaluate whether HLA classification is useful in the treatment of individual cancer patients, by using a consecutive retrospective study series of 1932 patients from Jan 1977 to May 1996 and a consecutive prospective study series of 582 patients from Jun 1996 to Aug 2005, all of whom received gastrectomies. There were significant differences between male patients who agreed to get the recommended therapy (the patients in agreement) and those who did not (the patients in disagreement), those with lymph node metastasis, those with undifferentiated adenocarcinoma, and those in stage 3A. Moreover, stage 2 and 3A patients benefitted from HLA-oriented therapy in a prospective study. In conclusion, our results showed that it is beneficial to examine HLA antigens preoperatively, that HLA-oriented therapy is promising, and that we must determine predictive factors for response to therapy in the future. We concluded that HLA antigens are promising predictive markers for cancer treatment.

Date 2008

DOI 10.4993/acrt.16.36

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female, human, male, article, controlled study, major clinical study, fluorouracil, histopathology, sex difference, treatment outcome, cancer staging, cancer survival, lymph node metastasis, fluoropyrimidine derivative, gastrectomy, Gastric cancer, HLA, HLA antigen, HLA system, HLA-oriented therapy, krestin, mitomycin C, stomach adenocarcinoma, tegafur, UFT, undifferentiated carcinoma

Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: K. Ogoshi; Department of Gastroenterological Surgery, School of Medicine, Tokai University, Bohseidai Isehara, Kanagawa, 259-1193, Japan; email: ogoshi@is.icc.u-tokai.ac.jp; CODEN: ACRTC

Pain, sleep disturbances, and functional limitations in people living with hiv/ aids-associated distal sensory peripheral neuropathy

Item Type Journal Article

Author R. Sandoval

Author T. Roddey

Author T.P. Giordano

Author K. Mitchell

Author C. Kelley

Abstract Background: Pain, sleep, and functional disturbances are a common occurrence in people living with HIV/AIDS-related distal sensory peripheral neuropathy (PLWHA-DSPN) yet lack group classification and quantification. Methods: A total of 46 PLWHA-DSPN were recruited, as part of a 2-group intervention study, to complete the Neuropathic Pain Scale and the Pittsburgh Sleep Quality Index (PSQI) questionnaires. The participant's performance during a forward reach task and walking distance in 6 minutes was recorded as a measure of function. Results: The pain (60.77 ± 17.85) and sleep (14.62 ± 4.28) scores denote marked pain and sleep disturbances, compared to seronegative, age-matched individuals. The ambulation distance was limited (243.99 ± 141.04 m) and inversely associated with the PSQI-sleep efficiency subscale ($r_s = .35$, $P > .05$). The average reaching distances measured (36.07 ± 7.37 cm) were similar to seronegative, age-matched individuals. Pain, sleep, and functional measures exhibited significant associations. Conclusions: The data collected suggest that PLWHA-DSPN report moderate-to-severe pain and significant sleep disturbances and exhibit limited ambulation distances. © The Author(s) 2014.

Date 2014

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Notes:

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Pairwise Alignment Improves Graph Domain Adaptation

Item Type Conference Paper
Author S. Liu
Author D. Zou
Author H. Zhao
Author P. Li
Editor Salakhutdinov R.
Editor Kolter Z.

Editor Heller K.
Editor Weller A.
Editor Oliver N.
Editor Scarlett J.
Editor Berkenkamp F.

Abstract Graph-based methods, pivotal for label inference over interconnected objects in many real-world applications, often encounter generalization challenges, if the graph used for model training differs significantly from the graph used for testing. This work delves into Graph Domain Adaptation (GDA) to address the unique complexities of distribution shifts over graph data, where interconnected data points experience shifts in features, labels, and in particular, connecting patterns. We propose a novel, theoretically principled method, Pairwise Alignment (Pair-Align) to counter graph structure shift by mitigating conditional structure shift (CSS) and label shift (LS). Pair-Align uses edge weights to recalibrate the influence among neighboring nodes to handle CSS and adjusts the classification loss with label weights to handle LS. Our method demonstrates superior performance in real-world applications, including node classification with region shift in social networks, and the pileup mitigation task in particle colliding experiments. For the first application, we also curate the largest dataset by far for GDA studies. Our method shows strong performance in synthetic and other existing benchmark datasets. Copyright 2024 by the author(s)

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Conference Name Proceedings of Machine Learning Research

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Performance, Real-world, Generalisation, Model training, Graph theory, Domain adaptation, Shift-and, Datapoints, Graph data, Graph-based methods, Pairwise alignment

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Pan-cancer computational histopathology reveals mutations, tumor composition and prognosis

Item Type Journal Article

Author Y. Fu

Author A.W. Jung

Author R.V. Torne

Author S. Gonzalez

Author H. Vöhringer

Author A. Shmatko

Author L.R. Yates

Author M. Jimenez-Linan

Author L. Moore

Author M. Gerstung

Abstract We use deep transfer learning to quantify histopathological patterns across 17,355 hematoxylin and eosin-stained histopathology slide images from 28 cancer types and correlate these with matched genomic, transcriptomic and survival data. This approach accurately classifies cancer types and provides spatially resolved tumor and normal tissue distinction. Automatically learned computational histopathological features correlate with a large range of recurrent genetic aberrations across cancer types. This includes whole-genome duplications, which display universal features across cancer types, individual chromosomal aneuploidies, focal amplifications and deletions, as well as driver gene mutations. There are widespread associations between bulk gene expression levels and histopathology, which reflect tumor composition and enable the localization of transcriptomically defined tumor-infiltrating lymphocytes. Computational histopathology augments prognosis based on histopathological subtyping and grading, and highlights prognostically relevant areas such as necrosis or lymphocytic aggregates. These findings show the remarkable potential of computer vision in characterizing the molecular basis of tumor histopathology. © 2020, The Author(s), under exclusive licence to Springer Nature America, Inc.

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Article, female, human, Humans, male, machine learning, aged, Deep Learning, prevalence, learning algorithm, validation process, outcome assessment, gene expression, diagnostic accuracy, histopathology, human tissue, clinical outcome, gene mutation, cancer prognosis, immunohistochemistry, prognosis, Prognosis, cancer grading, mutation, Mutation, melanoma, uterine cervix cancer, liver cell carcinoma, DNA methylation, overall survival, neoplasm, cancer staging, lymphocytic infiltration, glioblastoma, RNA sequence, transcriptomics, Neoplasms, hematoxylin, chromatin immunoprecipitation, colon adenocarcinoma, copy number variation, esophagus carcinoma, Hematoxylin, squamous cell lung carcinoma, tumor associated leukocyte, tumor suppressor gene, uvea melanoma, whole exome sequencing

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Parathyroid gland identification and angiography classification using simple machine learning methods

Item Type Journal Article
Author P.D. Mcentee
Author J.E. Greevy
Author F. Triponez
Author M.S. Demarchi
Author R.A. Cahill

Abstract Background: Near-infrared indocyanine green angiography allows experienced surgeons to reliably evaluate parathyroid gland vitality during thyroid and parathyroid operations in order to predict postoperative function. To facilitate equal performance between surgeons, we developed an automatic computational quantification method using computer vision that portrays expert interpretation of visualized parathyroid gland near-infrared indocyanine green angiographic fluorescence signals. Methods: Near-infrared indocyanine green-parathyroid gland angiography video recordings (Fluobeam® LX, Fluoptics, Grenoble - part of Getinge - Göteborg) from patients undergoing endocrine cervical surgery in a high-volume unit were used for model development. Computation (MATLAB, Mathworks, Ireland) included segmentation-identification of the parathyroid gland (by autofluorescence), image stabilization (by linear translation) and adjusted time-fluorescence intensity profile generation. Relative upslope and maximum intensity ratios then trained a simple logistic regression model based on expert interpretation and outcome (including hypoparathyroidism), with subsequent unseen testing for validation. Results: The model was trained on 37 patient videos (45 glands, 29 judged well perfused by parathyroid gland angiography experts), achieving feature data separation with 100% accuracy, and tested on 22 unseen videos (27 glands, 15 judged well perfused), including four in real time. Segmentation-guided parathyroid gland detection correctly identified all parathyroid glands during unseen testing along with three additional non-parathyroid gland regions (90% positive predictive value). Subsequent time-fluorescence intensity profile extraction with vitality prediction was shown feasible in all cases within 5 min, with a 96.3% model accuracy (sensitivity and specificity were 93.3 and 100% respectively) when compared with expert judgement. Conclusion: Automatic parathyroid gland perfusion quantification using simple machine learning computational methods discriminates parathyroid gland perfusion in concordance with expert surgeon interpretation, providing a means for near-infrared indocyanine green-parathyroid gland signal evaluation. © 2024 The Author(s).

Date 2024

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Notes:

EID: 2-s2.0-85208057770

Export Date: 20 September 2026; Cited By: 14; Correspondence Address: R.A. Cahill; UCD Centre for Precision Surgery, UCD, Dublin, 47 Eccles Street, Ireland; email: ronan.cahill@ucd.ie

Partial AUC maximization in a linear combination of dichotomizers

Item Type Journal Article

Author M.T. Ricamato

Author F. Tortorella

Abstract Classifier combination is a useful and common methodology to design an effective classification system. A large number of combination rules has been proposed hitherto, mostly aimed at minimizing the error rate. Recently, some methods have been presented that are devoted to maximize the area under the ROC curve (AUC), a more suitable performance measure when dealing with two-class problems with imprecise environment and/or imbalanced class priors. However, there are several applications that do not operate in the complete range of the ROC curve, but only in particular regions of it. In these cases, it is better to analyze the performance only in a part of the curve and to use the partial AUC (pAUC). This paper presents a new method that aims at maximizing the pAUC by means of linear combination of classifiers. The effectiveness of the proposed method has been proved on two biometric databases. © 2011 Elsevier Ltd. All rights reserved.

Date 2011

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Notes:

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Export Date: 20 September 2026; Cited By: 24; Correspondence Address: F. Tortorella; DAEIMI, Università Degli Studi di Cassino, 03043 Cassino, via G. Di Biasio 43, Italy; email: tortorella@unicas.it; CODEN: PTNRA

Partial Domain Adaptation via Importance Sampling-Based Shift Correction

Item Type Journal Article**Author** C.-J. Guo**Author** C.-X. Ren**Author** Y.-W. Luo**Author** X.-L. Xu**Author** H. Yan

Abstract Partial domain adaptation (PDA) is a challenging task in real-world machine learning scenarios. It aims to transfer knowledge from a labeled source domain to a related unlabeled target domain, where the support set of the source label distribution subsumes the target one. Previous PDA works managed to correct the label distribution shift by weighting samples in the source domain. However, the simple reweighing technique cannot explore the latent structure and sufficiently use the labeled data, and then models are prone to over-fitting on the source domain. In this work, we propose a novel importance sampling-based shift correction (IS2C) method, where new labeled data are sampled from a built sampling domain, whose label distribution is supposed to be the same as the target domain, to characterize the latent structure and enhance the generalization ability of the model. We provide theoretical guarantees for IS2C by proving that the generalization error can be sufficiently dominated by IS2C. In particular, by implementing sampling with the mixture distribution, the extent of shift between source and sampling domains can be connected to generalization error, which provides an interpretable way to build IS2C. To improve knowledge transfer, an optimal transport-based independence criterion is proposed for conditional distribution alignment, where the computation of the criterion can be adjusted to reduce the complexity from $O(n^3)$ to $O(n^2)$ in realistic PDA scenarios. Extensive experiments on PDA benchmarks validate the theoretical results and demonstrate the effectiveness of our IS2C over existing methods. © 1992-2012 IEEE.

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Tags:

Learning systems, machine learning, article, Error analysis, benchmarking, Labeled data, Knowledge management, Domain Knowledge, Knowledge transfer, adaptation, Domain adaptation, Target domain, Label distribution, Label shift, label shift, importance sampling, Importance sampling, sampling, conditional shift, Conditional shift, Generalization Error, generalization error analysis, Generalization error analyze, Latent structures, Partial domain adaptation, Sampling-based, Signal sampling

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Passenger Comfort Quantification for Automated Vehicle Based on Stacking of Psychophysics Mechanism and Encoder-Transformer Model

Item Type Journal Article

Author W. Zhu

Author Z. Xi

Author C. Hu

Author B. Zhao

Author Y. Niu

Abstract Passenger comfort is a crucial aspect that influences humans' acceptance of automated vehicles. The passenger comfort score (PCS) is closely related to the passengers' psychological states, however, comfort quantification methods based on the passengers' psychophysics mechanism are rare. This research pioneers a

passenger comfort quantification model (PCQM) specifically designed for automated vehicles, demonstrating the model's ability to accurately quantify subjective PCS under urban LCS. Three significant contributions form the basis of this study: 1) A dataset dedicated to comfort quantification is collected. A novel PCQM based on ensemble learning of psychophysics mechanism based sub-model and encoder-transformer based sub-model is proposed. The psychophysics mechanism model is derived from Stevens' power law. 2) As a subjective indicator, the self-reported score (SRS), which is the indicator of PCS contains considerable noise. The PCQM addresses the issue of substantial noise prevalent in the subjective SRS by incorporating a semi-supervised learning strategy, which enhances data consistency and suppresses noise. 3) The efficacy of the proposed PCQM is corroborated via deployment on an automated vehicle, where the model's real-time predictions strongly align with SRS from onboard passengers. © 2000-2011 IEEE.

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Patterns of Trauma Exposure in Childhood and Adolescence and Their Associations With Behavioral Well-Being

Item Type Journal Article

Author C.M. Connell

Author S.L. Pittenger

Author J.M. Lang

Abstract The majority of youth living in the United States experience a potentially traumatic event (PTE) by 18 years of age, with many experiencing multiple PTEs. Variation in the nature and range of PTE exposure differentially impacts youth functioning, although this association is poorly understood. We used latent class analysis (LCA) to identify patterns of PTE exposure from caregiver and youth report in a treatment-seeking sample of children and adolescents (N = 701) and examined how these patterns predict youths' behavioral health outcomes. We identified four classes based on both caregiver and youth reports of PTE exposure, with the best-fitting model representing a constrained measurement model across reporters; these included high polyvictimization, moderate polyvictimization (general), moderate polyvictimization (interpersonal), and low polyvictimization classes. Prevalence of classes varied across reporters, and agreement in classification based on caregiver and youth report was mixed. Despite these differences, we observed similar patterns of association between caregiver- and youth-reported classes and their respective ratings of posttraumatic stress disorder and depressive symptoms, as well as both caregiver and therapist ratings of problem behavior, with Cohen's d effect size estimates of significant differences ranging from d = 0.25 to d = 0.51. The PTE exposure classes did not differ with respect to ratings of child functioning. Findings highlight the importance of gathering information from multiple informants. © 2018 International Society for Traumatic Stress Studies

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Paying attention to cardiac surgical risk: An interpretable machine learning approach using an uncertainty-aware attentive neural network

Item Type Journal Article

Author J.C. Penny-Dimri

Author C. Bergmeir

Author C.M. Reid

Author J. Williams-Spence

Author A.D. Cochrane

Author J.A. Smith

Abstract Machine learning (ML) is increasingly applied to predict adverse postoperative outcomes in cardiac surgery. Commonly used ML models fail to translate to clinical practice due to absent model explainability, limited uncertainty quantification, and no flexibility to missing data. We aimed to develop and benchmark a novel ML approach, the uncertainty-aware attention network (UAN), to overcome these common limitations. Two Bayesian uncertainty quantification methods were tested, generalized variational inference (GVI) or a posterior network (PN). The UAN models were compared with an ensemble of XGBoost models and a Bayesian logistic regression model (LR) with imputation. The derivation datasets consisted of 153,932 surgery events from the Australian and New Zealand Society of Cardiac and Thoracic Surgeons (ANZSCTS) Cardiac Surgery Database. An external validation consisted of 7343 surgery events which were extracted from the Medical Information Mart for Intensive Care (MIMIC) III critical care dataset. The highest performing model on the external validation dataset was a UAN-GVI with an area under the receiver operating characteristic curve (AUC) of 0.78 (0.01). Model performance improved on high confidence samples with an AUC of 0.81 (0.01). Confidence calibration for aleatoric uncertainty was excellent for all models. Calibration for epistemic uncertainty was more variable, with an ensemble of XGBoost models performing the best with an AUC of 0.84 (0.08). Epistemic uncertainty was improved using the PN

approach, compared to GVI. UAN is able to use an interpretable and flexible deep learning approach to provide estimates of model uncertainty alongside state-of-the-art predictions. The model has been made freely available as an easy-to-use web application demonstrating that by designing uncertainty-aware models with innately explainable predictions deep learning may become more suitable for routine clinical use. © 2023 Penny-Dimri et al.

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human, machine learning, article, Australia, Uncertainty, New Zealand, Machine Learning, prediction, deep learning, controlled study, artificial neural network, attention, medical information, Neural Networks, Computer, attention network, receiver operating characteristic, Bayes theorem, Bayes Theorem, uncertainty, calibration, animal, Animals, Cardiac Surgical Procedures, heart surgery, intensive care, Lepidoptera, surgical risk, thoracic surgeon

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Export Date: 20 September 2026; Cited By: 5; Funding details: National Health and Medical Research Council, NHMRC, (1092642, 1136372); National Health and Medical Research Council, NHMRC; Correspondence Address: J.C. Penny-Dimri; Department of Surgery, School of Clinical Sciences at Monash Health, Monash University, Melbourne, Australia; email: jahan.penny-dimri@monash.edu; CODEN: POLNC

Pear flower cluster quantification using RGB drone imagery

Item Type Journal Article

Author Y. Vanbrabant

Author S. Delalieux

Author L. Tits

Author K. Pauly

Author J. Vandermaesen

Author B. Somers

Abstract High quality fruit production requires the regulation of the crop load on fruit trees by reducing the number of flowers and fruitlets early in the growing season, if the bearing is too high. Several automated flower cluster quantification methods based on proximal and remote imagery methods have been proposed to estimate flower cluster numbers, but their overall performance is still far from satisfactory. For other methods, the performance of the method to estimate flower clusters within a tree is unknown since they were only tested on images from one perspective. One of the main reported bottlenecks is the presence of occluded flowers due to limitations of the top-view perspective of the platform-sensor combinations. In order to tackle this problem, the multi-view perspective from the Red–Green–Blue (RGB) colored dense point clouds retrieved from drone imagery are compared and evaluated against the field-based flower cluster number per tree. Experimental results obtained on a dataset of two pear tree orchards (N = 144) demonstrate that our 3D object-based method, a combination of pixel-based classification with the stochastic gradient boosting algorithm and density-based clustering (DBSCAN), significantly outperforms the state-of-the-art in flower cluster estimations from the 2D top-view ($R^2 = 0.53$), with $R^2 > 0.7$ and $RRMSE < 15\%$. © 2020 by the authors.

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Drones, Remote sensing, Classification, Clustering, Colored dense point clouds, Flower clusters, Pear orchards, RGB, RPAS

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Penalized logistic regression using functional connectivity as covariates with an application to mild cognitive impairment

Item Type Journal Article

Author J.-H. Jung

Author S.-J. Ji

Author E. Lee

Author H. Zhu

Author J.G. Ibrahim

Author Y. Fan

Abstract There is an emerging interest in brain functional connectivity (FC) based on functional Magnetic Resonance Imaging in Alzheimer's disease (AD) studies. The complex and high-dimensional structure of FC makes it challenging to explore the association between altered connectivity and AD susceptibility. We develop a pipeline to refine FC as proper covariates in a penalized logistic regression model and classify normal and AD susceptible groups. Three different quantification methods are proposed for FC refinement. One of the methods is dimension reduction based on common component analysis (CCA), which is employed to address the limitations of the other methods. We applied the proposed pipeline to the Alzheimer's Disease Neuroimaging Initiative (ADNI) data and deduced pathogenic FC biomarkers associated with AD susceptibility. The refined FC biomarkers were related to brain regions for cognition, stimuli processing, and sensorimotor skills. We also demonstrated that a model using CCA performed better than others in terms of classification performance and goodness-of-fit. © 2020 The Korean Statistical Society, and Korean International Statistical Society. All rights reserved.

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DOI 10.29220/CSAM.2020.27.6.603

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Journal Abbr Commun. Stat. Appl. Methods

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mild cognitive impairment, Alzheimers disease, common component analysis, penalized logistic regression, resting-state functional magnetic resonance imaging

Notes:

EID: 2-s2.0-85098544987

Export Date: 20 September 2026; Cited By: 1; Funding details: Chungnam National University, CNU; Correspondence Address: E. Lee; Department of Information and Statistics, Chungnam National University, Yuseong-gu, 99 Daehak-ro, 34134, South Korea; email: eunjee.cnu@gmail.com

Performance Analysis of Vehicle Counting and Classification Using YOLO at a Four-Way Intersection Based on Indonesian Road Capacity Manual

Item Type	Conference Paper
Author	M.A. Murtaqib
Author	F.A. Alunjati
Author	U. Elviani
Author	F. Hidayat
Author	I.G.B.B. Nugraha
Author	S.H. Supangkat
Abstract	This study introduces a vehicle volume analysis system utilizing video analytics technology to improve traffic counting accuracy at intersections using CRISP-DM methodology. By leveraging the YOLO (You Only Look Once) algorithm, it can detect, classify, and count vehicles using footage from public CCTV cameras based on Indonesian Road Capacity Manual. This research focuses on a four-way intersection in Sedayu, Yogyakarta, Indonesia, and involves preparing a dataset by extracting and labeling video frames. This research train and evaluate several YOLO models-YOLOv5s, YOLOv7, YOLOv8s, and Gelan-c from YOLOv9-to assess their effectiveness in real-world traffic scenarios. The results demonstrate that YOLOv8s delivers superior performance compared to the other versions, achieving a precision of 0.971 and a recall of 0.988 across all vehicle classes. This high accuracy highlights the potential of YOLOv8s for improving traffic volume counting at busy intersections. However, challenges remain, including false positives, double detections, and misclassifications, which tend to occur under certain traffic conditions, such as low visibility or dense vehicle clustering. Despite these issues, YOLOv8s shows promise for enhancing the efficiency and precision of traffic analysis systems. The contribution of this paper is to support the evaluation of intersection performance based on turn movement count in accordance with the Indonesian Road Capacity Manual, where currently the evaluation is still carried out manually. Future research will focus on addressing these challenges by refining the detection process and incorporating additional techniques, such as advanced post-processing methods and model enhancements, to increase robustness in various traffic environments. © 2024 IEEE.
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DOI	10.1109/ICICYTA64807.2024.10913190
URL	https://www.scopus.com/pages/publications/105001666130?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: Int. Conf. Intell. Cybern. Technol. Appl., ICICyTA
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	979-833150649-0 (ISBN)

Pages 495-500
Proceedings Title Int. Conf. Intell. Cybern. Technol. Appl., ICICyTA
Conference Name 2024 International Conference on Intelligent Cybernetics Technology and Applications, ICICyTA 2024
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Tags:

Performances analysis, Advanced traffic management systems, Highway traffic control, Magnetic levitation vehicles, Motor transportation, Road vehicles, Street traffic control, Vehicles detection, Vehicle detection, YOLO, Analysis system, video analytics, Video analytics, CRISP-DM, Road capacity, road intersection, Road intersections, Vehicle counting and classifications, vehicle detection, Volume analysis, You only look once

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Export Date: 20 September 2026; Cited By: 2; Funding details: Lembaga Pengelola Dana Pendidikan, LPDP; Indonesian Endowment Fund for Education; Ministry of Education, Culture, Research, and Technology of the Republic of Indonesia; Institut Teknologi Bandung, ITB, (2963/E3/AL.04/2024, B07/KS.00/2024); Institut Teknologi Bandung, ITB; Conference name: 4th International Conference on Intelligent Cybernetics Technology and Applications, ICICyTA 2024; Conference date: 17 December 2024 through 19 December 2024; Conference code: 207547

Performance evaluation of fuzzy rule-based systems with class priority for medical diagnosis problems

Item Type Conference Paper
Author T. Nakashima
Author Y. Yokota
Author G. Schaefer
Author H. Ishibuchi
Abstract In this paper we examine the performance of fuzzy rule-based systems with classification priority for medical diagnosis problems. The assumption in this paper is that a classification priority is given a priori for each class in a pattern classification problem. Our fuzzy rule-based system consists of a set of fuzzy if-then rules that are automatically generated from a set of given training patterns. The consequent class of fuzzy if-then rules are decided based on the number of covered training patterns for each class. We apply the fuzzy classifier with class priority to two medical diagnosis problems: appendix diagnosis and breast cancer diagnosis, and compare its performance with that of a conventional fuzzy rule-based systems. © 2007 ECMS.
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Language English
Extra Journal Abbreviation: Eur. Conf. Model. Simul.: Simul. United Europe, ECMS
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Automatically generated, Computer aided diagnosis, Fuzzy inference, Fuzzy rules, Fuzzy if-then rules, Medical problems, Breast cancer diagnosis, Diagnosis problem, Fuzzy classifiers, Pattern classification problems, Training patterns

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: T. Nakashima; Osaka Prefecture University, Naka-ku, Sakai Osaka 599-8531, Gakuen-cho 1-1, Japan; email: nakashi@cs.osakafu-u.ac.jp; Conference name: 21st European Conference on Modelling and Simulation, ECMS 2007; Conference date: 4 June 2007 through 6 June 2007; Conference code: 88762

Performance metrics for marine mammal signal detection and classification

Item Type Journal Article

Author J.A. Hildebrand

Author K.E. Frasier

Author T.A. Helble

Author M.A. Roch

Abstract Automatic algorithms for the detection and classification of sound are essential to the analysis of acoustic datasets with long duration. Metrics are needed to assess the performance characteristics of these algorithms. Four metrics for performance evaluation are discussed here: receiver-operating-characteristic (ROC) curves, detection-error-trade-off (DET) curves, precision-recall (PR) curves, and cost curves. These metrics were applied to the generalized power law detector for blue whale D calls [Helble, Ierley, D'Spain, Roch, and Hildebrand (2012). J. Acoust. Soc. Am. 131(4), 2682-2699] and the click-clustering neural-net algorithm for Cuvier's beaked whale echolocation click detection [Frasier, Roch, Soldevilla, Wiggins, Garrison, and Hildebrand (2017). PLoS Comp. Biol. 13(12), e1005823] using data prepared for the 2015 Detection, Classification, Localization and Density Estimation Workshop. Detection class imbalance, particularly the situation of rare occurrence, is common

for long-term passive acoustic monitoring datasets and is a factor in the performance of ROC and DET curves with regard to the impact of false positive detections. PR curves overcome this shortcoming when calculated for individual detections and do not rely on the reporting of true negatives. Cost curves provide additional insight on the effective operating range for the detector based on the a priori probability of occurrence. Use of more than a single metric is helpful in understanding the performance of a detection algorithm. © 2022 Author(s).

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Perfusion Quantification from Endoscopic Videos: Learning to Read Tumor Signatures

Item Type Conference Paper

Author S. Zhuk
Author J.P. Epperlein
Author R. Nair
Author S. Tirupathi
Author P. Mac Aonghusa
Author D.F. O'Shea
Author R. Cahill
Editor Martel A.L.
Editor Abolmaesumi P.
Editor Stoyanov D.
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Abstract Intra-operative (this work was partially supported by Disruptive Technologies Innovation Fund, Ireland, project code DTIF2018 240 CA) identification of malignant versus benign or healthy tissue is a major challenge in fluorescence guided cancer surgery. We propose a perfusion quantification method for computer-aided interpretation of subtle differences in dynamic perfusion patterns which can be used to distinguish between normal tissue and benign or malignant tumors intra-operatively by using multispectral endoscopic videos. The method exploits the fact that vasculature arising from cancer angiogenesis gives tumors differing perfusion patterns from the surrounding normal tissues. Experimental evaluation of our method on a cohort of colorectal cancer surgery endoscopic videos suggests that it discriminates between healthy, cancerous and benign tissues with 95% accuracy. © 2020, Springer Nature Switzerland AG.

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Persistent Homology for the Quantitative Evaluation of Architectural Features in Prostate Cancer Histology

Item Type Journal Article

Author P. Lawson

Author A.B. Sholl

Author J.Q. Brown

Author B.T. Fasy

Author C. Wenk

Abstract The current system for evaluating prostate cancer architecture is the Gleason grading system which divides the morphology of cancer into five distinct architectural patterns, labeled 1 to 5 in increasing levels of cancer aggressiveness, and generates a score by summing the labels of the two most dominant patterns. The Gleason score is currently the most powerful prognostic predictor of patient outcomes; however, it suffers from problems in reproducibility and consistency due to the high intra-observer and inter-observer variability amongst pathologists. In addition, the Gleason system lacks the granularity to address potentially prognostic architectural features beyond Gleason patterns. We evaluate prostate cancer for architectural subtypes using techniques from topological data analysis applied to prostate cancer glandular architecture. In this work we demonstrate the use of persistent homology to capture architectural features independently of Gleason patterns. Specifically, using persistent homology, we compute topological representations of purely graded prostate cancer histopathology images of Gleason patterns 3,4 and 5, and show that persistent homology is capable of clustering prostate cancer histology into architectural groups through a ranked persistence vector. Our results indicate the ability of persistent homology to cluster prostate cancer histopathology images into unique groups with dominant architectural patterns consistent with the continuum of Gleason patterns. In addition, of particular interest, is the sensitivity of persistent homology to identify specific sub-architectural groups within single Gleason patterns, suggesting that persistent homology could represent a robust quantification method for prostate cancer architecture with higher granularity than the existing semi-quantitative measures. The capability of these topological representations to segregate prostate cancer by architecture makes them an ideal

candidate for use as inputs to future machine learning approaches with the intent of augmenting traditional approaches with topological features for improved diagnosis and prognosis. © 2019, The Author(s).

Date 2019

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PET/CT-derived whole-body composition and survival in resectable NSCLC: double machine learning-based adjusted association analysis of intermuscular adiposity burden and sex-specific metabolic phenotypes

Item Type Journal Article

Author W. Zhai

Author R. Wang

Author M. Ye

Author X. Li

Author Y. Wang

Author T. Zhou

Author X. Zhou

Author Q. Jin

Author Z. Zhang

Author L. Fan

Abstract Purpose: To evaluate whether whole-body PET/CT-derived body composition features are associated with survival in patients with resectable non-small cell lung cancer (NSCLC), using double machine learning to quantify adjusted associations with restricted mean survival time. Methods: This retrospective multicenter study included 769 patients with stage $\leq$ IIIA NSCLC who underwent preoperative 18F-fluorodeoxyglucose positron emission tomography/computed tomography (18F-FDG PET/CT) and curative resection. Center 1 was used for model development (n = 555) and Center 2 for external validation (n = 214). Automated whole-body segmentation was used to extract volumetric, attenuation-based and metabolic features of skeletal muscle and adipose tissue. Restricted mean survival time-based double machine learning with generalized propensity score weighting estimated DML-adjusted differences in restricted mean survival time for overall survival (OS) and progression-free survival (PFS). Ridge-penalized Cox models were further used to evaluate prognostic performance and incremental value. Results: Higher intermuscular adipose tissue (IMAT) volume index was associated with shorter survival (DML-adjusted RMST difference, -4.29 months for OS and - 2.74 months for PFS per 1-SD higher IMAT volume index). Higher TAT SUR (Mean) showed an exploratory favorable adjusted association with longer OS (+ 4.42 months). Sex-stratified analyses suggested stronger adverse adipose-volume associations in male patients and stronger favorable adipose-metabolic associations in female patients. Model analyses showed moderate external discrimination, whereas incremental-value metrics were modest and endpoint-dependent. Conclusion: Whole-body PET/CT body-composition phenotyping may provide prognostic information in resectable NSCLC. Higher IMAT burden was associated with shorter survival, whereas TAT SUR (Mean) showed an exploratory favorable adjusted association with OS. A local software framework may support reproducible feature extraction and research-oriented risk stratification. © The Author(s) 2026.

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Photocatalytic nitrogen reduction to ammonia: Insights into the role of defect engineering in photocatalysts

Item Type Journal Article**Author** H. Shen**Author** M. Yang**Author** L. Hao**Author** J. Wang**Author** J. Strunk**Author** Z. Sun

Abstract Engineering of defects in semiconductors provides an effective protocol for improving photocatalytic N₂ conversion efficiency. This review focuses on the state-of-the-art progress in defect engineering of photocatalysts for the N₂ reduction toward ammonia. The basic principles and mechanisms of thermal catalyzed and photon-induced N₂ reduction are first concisely recapped, including relevant properties of the N₂ molecule, reaction pathways, and NH₃ quantification methods. Subsequently, defect classification, synthesis strategies, and identification techniques are compendiously summarized. Advances of in situ characterization techniques for monitoring defect state during the N₂ reduction process are also described. Especially, various surface defect strategies and their critical roles in improving the N₂ photoreduction performance are highlighted, including surface vacancies (i.e., anionic vacancies and cationic vacancies), heteroatom doping (i.e., metal element doping and nonmetal element doping), and atomically defined surface sites. Finally, future opportunities and challenges as well as perspectives on further development of defect-engineered photocatalysts for the nitrogen reduction to ammonia are presented. It is expected that this review can provide a profound guidance for more specialized design of defect-engineered catalysts with high activity and stability for nitrogen photochemical fixation. [Figure not available: see fulltext.] © 2021, The Author(s).

Date 2022**DOI** 10.1007/s12274-021-3725-0**URL** <https://www.scopus.com/pages/publications/85114172950?origin=resultslist>**Archive** Scopus**Language** English**Volume** 15**Publisher** Tsinghua University**Pages** 2773-2809

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Physical and Physiological Responses during Competition

Item Type Journal Article

Author J. Yanci

Author A. Iturricastillo

Author M. Henríquez

Author D. Castillo

Abstract The aim of this chapter is to analyse the physical and physiological responses of football players with CP in competitive matches. To achieve this, a narrative review was conducted, including the principal studies that explore the physical and physiological responses of this specific group of para-footballers. In the introduction section, the importance of analysing the match physical and physiological responses of players with CP and their implications for training, classification and competition is described. In the physical and physiological responses sections, the most relevant material and quantification methods are presented, considering the available data compiled and examined from recent literature. It is known that contextual variables can influence the match physical responses of players; therefore, these different

contextual variables are taken into account. To conclude the chapter, the practical applications section is presented, along with future research needs in the field. © 2025 selection and editorial matter, Iván Peña-González, Raúl Reina and Manuel Moya-Ramón; individual chapters, the contributors.

Date 2024

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Physics-informed neural networks for capturing the true relationships between parameters to predict the dynamic triaxial strength of rocks in cold environments

Item Type Journal Article

Author T. Ma

Author C. Chen

Author L. Shen

Author K. Luo

Author Z. Jiang

Author S. Xie

Author Z. Yang

Author H. Chen

Author K. Peng

Abstract Due to freeze–thaw cycles, rocks in cold regions are prone to fatigue damage, making the accurate prediction of their dynamic triaxial strength critical for mining engineering in cold regions. A physics-informed neural network (PINN) approach was proposed, integrating rock mechanics theory with neural networks to improve prediction accuracy, interpretability, and generalization. To enhance reliability, an uncertainty-aware PINN (UAPINN) model was developed by incorporating uncertainty quantification. To assess UAPINN model performance, six machine learning models were developed, and their performance was evaluated using metrics such as R2, MAE, RMSE and VAF. A feature importance quantification method

based on Bayesian neural network uncertainty analysis was introduced. The results show that the UAPINN model outperforms traditional models, with the performance ranking: UAPINN > PINN > LightGBM > CatBoost > GRU > ELM > RF > SVR. Feature importance analysis reveals that the most influential parameters for dynamic triaxial strength are confining pressure > P-wave velocity > strain rate > freeze–thaw cycles > water absorption > dry density. This study provides an efficient method for predicting rock strength in cold-region engineering applications. © 2025 Elsevier Ltd

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PIETRA: Physics-Informed Evidential Learning for Traversing Out-of-Distribution Terrain

Item Type Journal Article

Author X. Cai

Author J. Queeney

Author T. Xu

Author A. Datar

Author C. Pan
Author M. Miller
Author A. Flather
Author P.R. Osteen
Author N. Roy
Author X. Xiao
Author J.P. How

Abstract Self-supervised learning is a powerful approach for developing traversability models for off-road navigation, but these models often struggle with inputs unseen during training. Existing methods utilize techniques like evidential deep learning to quantify model uncertainty, helping to identify and avoid out-of-distribution terrain. However, always avoiding out-of-distribution terrain can be overly conservative, e.g., when novel terrain can be effectively analyzed using a physics-based model. To overcome this challenge, we introduce Physics-Informed Evidential Traversability (PIETRA), a self-supervised learning framework that integrates physics priors directly into the mathematical formulation of evidential neural networks and introduces physics knowledge implicitly through an uncertainty-aware, physics-informed training loss. Our evidential network seamlessly transitions between learned and physics-based predictions for out-of-distribution inputs. Additionally, the physics-informed loss regularizes the learned model, ensuring better alignment with the physics model. Extensive simulations and hardware experiments demonstrate that PIETRA improves both learning accuracy and navigation performance in environments with significant distribution shifts. © 2016 IEEE.

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Pipeline defect size quantification via multi-branch online distillation with magnetic flux leakage priors and adaptive fusion

Item Type Journal Article

Author J. Li

Author H. Geng

Author W. Shao

Author B. Liu

Author L. Yang

Abstract Due to the complex industrial inspection environment, the limited number of labeled defect samples, and the insufficient exploitation of inherent defect characteristics in existing quantification methods, achieving accurate and reliable high-precision three-dimensional size quantification of defects directly from magnetic flux leakage signals remains challenging. To address the above issues, this paper proposes a pipeline magnetic flux leakage (MFL) defect three-dimensional sizing method based on multi-branch online knowledge distillation. This method takes the collected three-axis MFL data as input, extracts information in a targeted manner via multiple branches, and leverages the Magnetic-prior Network (Net-M) branch to effectively fuse prior knowledge and features. The adaptive fusion subnet (AFS) is introduced to adaptively fuse extracted data features, thereby enhancing the model's robustness and interpretability in few-shot scenarios and under practical signal disturbances. In addition, a dual supervised learning mechanism is incorporated to ensure stable model training, and finally, collaborative regression outputs of defect length, width, and depth are obtained. Extensive experiments conducted on the MFL defect data from the pull-through tests of the $\Phi 1016$ pipeline show that the average quantification errors of the proposed method are 0.67 mm in length, 1.51 mm in width, and 0.60 mm in depth. The proposed method improves the defect sizing accuracy by >10%, and its detection precision meets the requirements of ultra-high-definition pipeline inline inspection (ILI) equipment. It enables intelligent batch-level quantification of pipeline defect dimensions and maintains high performance with a total dataset size of 50 and across different pipe diameters. © 2026 The Authors.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Geng; School of Information Science and Engineering, Shenyang University of Technology, Shenyang, 110870, China; email: 1025284323@qq.com

Pitfalls in Quantification Assessment

Item Type Conference Paper**Author** W. Hassan**Author** A. Maletzke**Author** G. Batista**Editor** Cong G.**Editor** Ramanath M.

Abstract Quantification is a research area that develops methods that estimate the class attribute prevalence in an independent sample. Like the other fields in Machine Learning, quantification researchers often use experimental assessment to evaluate and compare the performance of their proposals. Therefore, the design of the experimental protocols is critical to the research in quantification. Currently, two protocols have dominated the assessment of quantifiers: the artificial-prevalence protocol (APP) and the natural-prevalence protocol (NPP). APP is the most employed since it allows the use of classification datasets, plenty available. However, APP has a shortcoming: the synthetic class prevalence of the test sets. This paper discusses the practical consequences of this shortcoming and shows simple examples of quantifiers that exploit these limitations to improve their performance artificially. We propose a baseline quantifier, lazy, and radar charts as tools to identify situations where the proposed quantifiers are performing poorly. © 2021 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0). CEUR Workshop Proceedings (CEUR-WS.org)

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Plant recognition by AI: Deep neural nets, transformers, and kNN in deep embeddings

Item Type Journal Article

Author L. Picek

Author M. Šulc

Author Y. Patel

Author J. Matas

Abstract The article reviews and benchmarks machine learning methods for automatic image-based plant species recognition and proposes a novel retrieval-based method for recognition by nearest neighbor classification in a deep embedding space. The image retrieval method relies on a model trained via the Recall@k surrogate loss. State-of-the-art approaches to image classification, based on Convolutional Neural Networks (CNN) and Vision Transformers (ViT), are benchmarked and compared with the proposed image retrieval-based method. The impact of performance-enhancing techniques, e.g., class prior adaptation, image augmentations, learning rate scheduling, and loss functions, is studied. The evaluation is carried out on the PlantCLEF 2017, the ExpertLifeCLEF 2018, and the iNaturalist 2018 Datasets—the largest publicly available datasets for plant recognition. The evaluation of CNN and ViT classifiers shows a gradual improvement in classification accuracy. The current state-of-the-art Vision Transformer model, ViT-Large/16, achieves 91.15% and 83.54% accuracy on the PlantCLEF 2017 and ExpertLifeCLEF 2018 test sets,

respectively; the best CNN model (ResNeSt-269e) error rate dropped by 22.91% and 28.34%. Apart from that, additional tricks increased the performance for the ViT-Base/32 by 3.72% on ExpertLifeCLEF 2018 and by 4.67% on PlantCLEF 2017. The retrieval approach achieved superior performance in all measured scenarios with accuracy margins of 0.28%, 4.13%, and 10.25% on ExpertLifeCLEF 2018, PlantCLEF 2017, and iNat2018–Plantae, respectively. Copyright © 2022 Pícek, Šulc, Patel and Matas.

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Plant recognition by inception networks with test-time class prior estimation

Item Type Conference Paper

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Author L. Pícek

Author J. Matas

Editor Ferro N.

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Editor Soulier L.

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Editor Department of Information Engineering (DEI), Via Gradenigo 6/B, Padova
University of Padua

Abstract The paper describes an automatic system for recognition of 10, 000 plant species from one or more images. The system finished 1st in the ExpertLifeCLEF 2018 plant identification challenge with 88.4% accuracy and performed better than 5 of the 9 participating plant identification experts. The system is based on the Inception-ResNet-v2 and Inception-v4 Convolutional Neural Network (CNN) architectures. Performance improvements were achieved by: adjusting the CNN predictions according to the estimated change of the class prior probabilities, replacing network parameters with their running averages, and test-time data augmentation.

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PlasmaCell CAD: A computer-aided diagnosis software tool for plasma cell recognition and characterization in microscopic images

Item Type Journal Article

Author R. Kasbgar

Author A. Vard

Abstract Background and objective: In the traditional diagnostic process for multiple myeloma cancer, a pathologist screens prepared blood samples using a microscope to detect, classify, and count plasma cells. This manual approach is time-consuming, exhausting, and prone to human errors. Consequently, medical experts and researchers are highly interested in any tool that partially or entirely automates this process. To achieve this goal, we developed a software tool called PlasmaCell CAD to analyze effective cells for diagnosing multiple myeloma cancers through microscopic images. Methods: In the proposed software, to detect and segment cells, we exploit the Mask-RCNN model that has been enhanced by leveraging the circlet transform for the anchor generation. Also, we use the SVM classifier to identify normal and abnormal plasma cells in this software. Moreover, we designed and developed a graphical user interface (GUI) for the PlasmaCell CAD so that users would be able to work with it more easily. Results: we considered the performance of the proposed software on both a publicly available dataset and a locally collected dataset. The experimental results demonstrated the capability and efficiency of PlasmaCell CAD software in segmenting and classifying plasma cells as well as its ease of use. Conclusions: PlasmaCell CAD is a free software tool that can easily be downloaded and installed on any computers running Windows. PlasmaCell CAD provides a user-friendly GUI with several image processing and visualization facilities for the user that can accelerate the diagnosis process. In light of promising results, PlasmaCell CAD software can be useful to pathologists in helping to diagnose multiple myeloma cancer. © 2025 Elsevier B.V.

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POETIC-NET: an augmented contextual feature representation via graph centrality relation for emotion recognition of English poetry

Item Type Journal Article

Author P.K. Kazipeta

Author V.P.K. Sistla

Author V.K.K. Kolli

Abstract The task of extracting emotion from poetry data is one that has received little attention in the literature. Poetry has a complex structure and emotional richness, in contrast to prose data such as tweets, product reviews, emails, and chats. This disparity makes the prediction of emotions from the poetry a challenging endeavor. Many prior studies focus on capturing emotions derived from local relations while overlooking a poem's global relations. This study introduces Poetic-Net, a multi-channel neural architecture for effective emotion recognition in poetry that leverages both structural and contextual insights. The model integrates Global Centrality Relation (GCR) features extracted via graph-based learning to quantify each word's contribution to a poem's central theme, and contextual features derived by passing BERT-pretrained embeddings through a sequence of Bi-LSTM, CNN, and Attention layers. By fusing GCR and deep contextual representations, Poetic-Net becomes a robust system that captures subtle and complex emotional patterns inherent in poetic text. Poetic-Net classifies poetry into five emotion classes viz. "Happy," "Fear," "Sad," "Angry," and "Surprise". Due to the unavailability of public datasets on emotion-annotated English poetry, an English poetry dataset ABIEMO (American,

British, and Indian Poetry datasets for Emotion detection) is developed to evaluate the performance of the proposed model. The proposed Poetic-Net is evaluated with five-word representations (GloVe, FastText, Word2vec-CBOW, Word2vec-SkipGram, and BERT) against baseline classification models. The proposed Poetic-Net model outperformed all the baseline models with an accuracy of 92.89% as both global and contextual features contributed significantly in feature extraction for improved performance. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: V.P.K. Sistla; Department of Computer Science and Engineering, Vignan's Foundation for Science, Technology and Research, Guntur (Dt), Vadlamudi, Andhra Pradesh, India; email: svrphanikumar@yahoo.com

Poisoning-Assisted Membership Inference in Federated Learning

Item Type Journal Article

Author X. Luan

Author Y. Bi

Author K. Zhang

Author Z. Huang

Author Z. Su

Author T.H. Luan

Author B. Hu

Abstract Federated learning (FL) is a distributed machine learning approach that allows multiple clients to collaboratively train a global model without sharing their local raw data. Despite this advantage, FL remains vulnerable to membership inference attacks (MIAs), which can compromise the membership privacy of federated clients. Nevertheless, existing MIAs exhibit significant performance degradation when the malicious clients lack knowledge of the target model and the training data of benign clients. In this paper, we propose a novel scheme, Temporal Evolution of the Poisoning Effects (TEPE), to conduct MIAs against black-box FL models without any prior knowledge of the target models or benign clients' training data. Specifically, we design a two-stage inference method: the poisoning process and the scoring process. For the poisoning process, we propose three poisoning strategies for three different auditing requirements. For the scoring process, we design a novel quantification method and an unsupervised inference model to extract membership features within the effects of poisoning attacks. The proposed attack leverages the intuition that if a sample is used by FL benign clients, its prediction may not be readily altered by poisoning attacks. Finally, we propose a modified random response algorithm (RR-Group) to detect our attacks, which can guarantee the performance of FL models and effectively reduce the attack performance of TEPE. Extensive experiments demonstrate that TEPE achieves competitive inference accuracy and F1-score compared to the most MIAs, while evading two representative defenses, except for our proposed detection method RR-Group. © 2004 IEEE.

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Polytect: an automatic clustering and labeling method for multicolor digital PCR data

Item Type Journal Article

Author Y. Chen

Author W. De Spiegelaere

Author W. Trypsteen

Author J. Vandesompele

Author G. Wils

Author D. Gleerup

Author A. Lievens

Author O. Thas

Author M. Vynck

Abstract Digital polymerase chain reaction (dPCR) is a state-of-the-art targeted quantification method of nucleic acids. The technology is based on massive partitioning of a reaction mixture into individual PCR reactions. The resulting partition-level end-point fluorescence intensities are used to classify partitions as positive or negative, i.e. containing or not containing the target nucleic acid(s). Many automatic dPCR partition classification methods have been proposed, but they are limited to the analysis of single- or dual-color dPCR data. While general-purpose or flow cytometry clustering methods can be directly applied to multicolor dPCR data, these methods do not exploit the approximate prior knowledge on cluster center locations available in dPCR data. We present Polytect, a method that relies on crude cluster results from flowPeaks, previously shown to offer good partition classification performance, and subsequently refines flowPeaks' results by automatic cluster merging and cluster labeling, exploiting the prior knowledge on cluster center locations. Comparative analyses with established methods such as flowPeaks, dpcp, and ddPCRclust reveal that Polytect often surpasses established methods, both on empirical and simulated data. Polytect manages to merge excess clusters, while also successfully identifying empty clusters when fewer than the maximally observable number of clusters are present. On par with recent developments in instruments, Polytect extends beyond two-color data. The method is available as an R package and R Shiny app (<https://digpcr.shinyapps.io/Polytect/>). © 2025 The Author(s).

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Pornographic image classification based on top down color-saliency based BoW representation

Item Type Conference Paper

Author C. Tian

Author X. Zhang

Author X. Gao

Author W. Wei

Abstract Since color is an important visual clue of the pornographic image, this study presents a new framework for pornographic image classification based on the fusion of color and shape information for the bag of words representation. This framework contains three fusion patterns: The early fusion, late fusion and top down color-saliency based fusion, which are compared intensively. Based on the comparison, the top down color-saliency fusion based pornographic image classification method is proposed by using the statistical class prior of each color word to weight the shape word. In the late fusion and color-saliency based fusion, color name is adopt to represent the color information. To verify the effectiveness of spatial constrain on the words, we also compared the shape features quantized by vector quantization and locality-constrained linear coding. The experimental results show that our model combines the shape and color information properly and it is superior over the popular methods to distinguish the normal and pornographic-like images from the pornographic ones. © 2014 IEEE.

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POSFed: Tackling Non-IID Challenges in One-Shot Federated Learning via Personalization

Item Type Conference Paper

Author Y. Zhang

Author X. Xiao

Author J. Guo

Author Z. Tang

Author Q. Ni

Author W. Wu

Editor Ding W.

Editor Vreeken J.

Editor Lu C.-T.

Editor Gunopulos D.

Editor Wu X.

Abstract Federated Learning (FL) enables collaborative model training across distributed clients without requiring the exchange of raw data. However, existing One-Shot FL (OSFL) methods, designed for communication efficiency by reducing federated rounds to one, suffer substantial performance degradation when faced with highly non-IID data across clients, primarily due to critical distribution shifts: label shift, feature shift, and concept shift. In this paper, we introduce POSFed, a new personalized three-stage approach, to systematically address these fundamental limitations: (1) Each client locally generates robust and label-agnostic synthetic datasets via self-supervising learning, ensuring essential knowledge is captured despite local distribution shifts; (2) The server aggregates all synthetic datasets to train a global feature extractor, capturing generalizable and transferable representations across heterogeneous client data; and (3) Each client efficiently adapts the feature extractor by learning a personalized classification head on its own data, enabling effective local customization and mitigating both feature and concept shifts. Extensive experiments across multiple benchmarks demonstrate that POSFed significantly outperforms state-of-the-art methods, achieving performance comparable to multi-round personalized approaches while using only one communication round. By ensuring both superior personalization and practical communication efficiency, POSFed establishes a feasible paradigm for FL under more realistic and heterogeneous conditions. Code is available at <https://github.com/I643204431IPOSFed>. © 2025 IEEE.

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Posistive-Unlabeled Learning via Optimal Transport and Margin Distribution

Item Type Conference Paper

Author N. Cao

Author T. Zhang

Author X. Shi

Author H. Jin

Editor De Raedt L.

Editor De Raedt L.

Abstract Positive-unlabeled (PU) learning deals with the circumstances where only a portion of positive instances are labeled, while the rest and all negative instances are unlabeled, and due to this confusion, the class prior can not be directly available. Existing PU learning methods usually estimate the class prior by training a nontraditional probabilistic classifier, which is prone to give an overestimation. Moreover, these methods learn the decision boundary by optimizing the minimum margin, which is not suitable in PU learning due to its sensitivity to label noise. In this paper, we enhance PU learning methods from the above two aspects. More specifically, we first explicitly learn a transformation from unlabeled data to positive data by entropy regularized optimal transport to achieve a much more precise estimation for class prior. Then we switch to optimizing the margin distribution, rather than the minimum margin, to obtain a label noise insensitive classifier. Extensive empirical studies on both synthetic and real-world data sets demonstrate the superiority of our proposed method. © 2022 International Joint Conferences on Artificial Intelligence. All rights reserved.

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Conference Name IJCAI International Joint Conference on Artificial Intelligence

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Tags:

Artificial intelligence, Learning systems, Metadata, Learning methods, Unlabeled data, Learn+, Negative instances, Positive instances, Optimal transport, Positive data, Probabilistic classifiers, Decision boundary, Margin distributions

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Positive and unlabeled learning for anomaly detection with multi-features

Item Type Conference Paper

Author J. Zhang

Author Z. Wang

Author J. Yuan

Author Y.-P. Tan

Abstract Anomaly detection is of great interest to big data applications, and both supervised and unsupervised learning have been applied for anomaly detection. However, it still remains a challenging problem because: (1) for supervised learning, it is difficult to acquire training data for anomaly samples; while (2) for unsupervised learning, the performance may not be satisfactory due to the lack of training data. To address the limitations, we propose a hybrid solution by using both normal (positive) data and unlabeled data (could be positive or negative) for semi-supervised anomaly detection. Particularly, we introduce a new framework based on Positive and Unlabeled (PU) Learning using multi-features to detect anomalies. We extend previous PU learning methods to (1) better address unbalanced class problem which is typical for anomaly detection, and (2) handle multiple features for anomaly detection. An iterative algorithm is proposed to learn the anomaly classifier incrementally from the labeled normal data and also unlabeled data. Our proposed method is verified on three benchmark datasets and one synthetic dataset. Experimental results show that our method outperforms existing methods under different class priors and different proportions of given positive classes. © 2017 ACM.

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Proceedings Title MM - Proc. ACM Multimed. Conf.
Conference Name MM 2017 - Proceedings of the 2017 ACM Multimedia Conference
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Tags:

Big data, Feature extraction, Intrusion detection, Supervised learning, Anomaly detection, Iterative methods, Semi-supervised learning, Unsupervised learning, Positive and unlabeled learning, PU learning, Different proportions, Semi- supervised learning, Pu learning, Iterative algorithm, Big data applications, Supervised and unsupervised learning

Notes:

EID: 2-s2.0-85035226669

Export Date: 20 September 2026; Cited By: 19; Conference name: 25th ACM International Conference on Multimedia, MM 2017; Conference date: 23 October 2017 through 27 October 2017; Conference code: 131373

Positive Unlabeled Learning Selected Not At Random (PULSNAR): class proportion estimation without the selected completely at random assumption

Item Type Journal Article

Author P. Kumar

Author C.G. Lambert

Abstract Positive and unlabeled (PU) learning is a type of semi-supervised binary classification where the machine learning algorithm differentiates between a set of positive instances (labeled) and a set of both positive and negative instances (unlabeled). PU learning has broad applications in settings where confirmed negatives are unavailable or difficult to obtain, and there is value in discovering positives among the unlabeled (e.g., viable drugs among untested compounds). Most PU learning algorithms make the selected completely at random (SCAR) assumption, namely that positives are selected independently of their features. However, in many real-world applications, such as healthcare, positives are not SCAR (e.g., severe cases are more likely to be diagnosed), leading to a poor estimate of the proportion, α , of positives among unlabeled examples and poor model calibration, resulting in an uncertain decision threshold for selecting positives. PU learning algorithms vary; some estimate only the proportion, α , of positives in the unlabeled set, while others calculate the probability that each specific unlabeled instance is positive, and some can do both. We propose two PU learning algorithms to estimate α , calculate

calibrated probabilities for PU instances, and improve classification metrics: i) PULSCAR (positive unlabeled learning selected completely at random), and ii) PULSNAR (positive unlabeled learning selected not at random). PULSNAR employs a divide-and-conquer approach to cluster SNAR positives into subtypes and estimates α for each subtype by applying PULSCAR to positives from each cluster and all unlabeled. In our experiments, PULSNAR outperformed state-of-the-art approaches on both synthetic and real-world benchmark datasets. © 2024 Kumar and Lambert

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Journal Abbr PeerJ Comput. Sci.

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Machine-learning, Machine learning, Federated learning, Adversarial machine learning, Contrastive Learning, Self-supervised learning, Class imbalance, Semi-supervised learning, Unsupervised learning, PULSNAR, Positive-unlabeled learning, Probability calibration, Mixture proportions, Noisy labels, Class prior, Mixture proportion estimation, Noisy label learning, Positive unlabeled learning selected completely at random, Positive unlabeled learning selected not at random, Proportion estimations, PULSCAR, SCAR, Selected completely at random, SNAR

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Positive unlabeled learning with class-prior approximation

Item Type Conference Paper

Author S. Chang

Author B. Du

Author L. Zhang

Editor Bessiere C.

Abstract The positive unlabeled (PU) learning aims to train a binary classifier from a set of positive labeled samples and other unlabeled samples. Much research has been done on this special branch of weakly supervised classification problems. Since only part of the positive class is labeled, the classical PU model trains the classifier assuming the class-prior is known. However, the true class prior is usually difficult to obtain and must be learned from the given data, and the traditional methods may not work. In this paper, we formulate a convex formulation to jointly solve the class-prior unknown problem and train an accurate classifier with no need of any class-prior assumptions or additional negative samples. The class prior is estimated by pursuing the optimal solution of gradient thresholding and the classifier is simultaneously trained by performing empirical unbiased risk. The detailed derivation and theoretical analysis of the proposed model are outlined, and a comparison of our experiments with other representative methods prove the superiority of our method. © 2020 Inst. Sci. inf., Univ. Defence in Belgrade. All rights reserved.

Date 2020

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Language English

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Risk perception, Artificial intelligence, Negative samples, Unlabeled samples, Supervised classification, Binary classifiers, Thresholding, Optimal solutions, Model trains

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Positive-unlabeled classification under class prior shift and asymmetric error

Item Type Conference Paper

Author N. Charoenphakdee

Author M. Sugiyama

Abstract Bottlenecks of binary classification from positive and unlabeled data (PU classification) are the requirements that given unlabeled patterns are drawn from the same distribution as the test distribution, and the penalty of the false positive error is identical to the false negative error. However, such requirements are often not fulfilled in practice. In this paper, we generalize PU classification to the class prior shift and asymmetric error scenarios. Based on the analysis of the Bayes optimal classifier, we show that given a test class prior, PU classification under class prior shift is equivalent to PU classification with asymmetric error. Then, we propose two different frameworks to handle these problems, namely, a risk minimization framework and density ratio estimation framework. Finally, we demonstrate the effectiveness of the proposed frameworks through experiments. Copyright © 2019 by SIAM.

Date 2019

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Tags:

Risk perception, Classification (of information), Data mining, Binary classification, Unlabeled data, Errors, False positive, Density-ratio estimations, Risk minimization, Asymmetric errors, Bayes-optimal, Equivalence classes, False negative errors

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Positive-unlabeled classification under class-prior shift: a prior-invariant approach based on density ratio estimation

Item Type Journal Article

Author S. Nakajima

Author M. Sugiyama

Abstract Learning from positive and unlabeled (PU) data is an important problem in various applications. Most of the recent approaches for PU classification assume that the class-prior (the ratio of positive samples) in the training unlabeled dataset is identical to that of the test data, which does not hold in many practical cases. In addition, we usually do not know the class-priors of the training and test data, thus we have no clue on how to train a classifier without them. To address these problems, we propose a novel PU classification method based on density ratio estimation. A notable advantage of our proposed method is that it does not require the class-priors in the training phase; class-prior shift is incorporated only in the test phase. We theoretically justify our proposed method and experimentally demonstrate its effectiveness. © 2022, The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature.

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Export Date: 20 September 2026; Cited By: 8; Funding details: , (20H04206); Correspondence Address: S. Nakajima; The University of Tokyo, Tokyo, Japan; email: nakajima@alumni.u-tokyo.ac.jp; CODEN: MALEE

Positive-Unlabeled Hyperspectral Image Classification using nnPU Loss and Masked Autoencoder

Item Type Conference Paper

Author O.S. Melashchenko

Abstract Hyperspectral image classification has seen significant progress with deep learning, especially CNN- and Transformer-based architectures that capture spectral-spatial relationships. However, two fundamental challenges remain: (i) the scarcity of labeled data due to the cost of pixel-level annotation, and (ii) the difficulty of fully utilizing abundant unlabeled data without overfitting. Positive-unlabeled learning addresses these issues by exploiting unlabeled data alongside a small set of positive labels, but its accuracy depends heavily on the quality of the unlabeled samples and the precision of class prior estimation. In this work, we investigate the case where the prior is known and focus on improving representation learning within positive-unlabeled frameworks. We propose a joint masked autoencoder and positive-unlabeled learning method for hyperspectral image classification. Experimental results show that our approach achieves consistent gains over standard positive-unlabeled baselines under scenarios with limited labeled data. These results highlight the effectiveness of combining masked autoencoder-based representations with positive-unlabeled classification. Code is at: <https://github.com/melalex/hsi-classification-playground> © 2025 IEEE.

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Deep learning, Auto encoders, Learning systems, deep learning, Image classification, Self-supervised learning, Labeled data, Semi-supervised learning, Unlabeled data, representation learning, Representation learning, semi-supervised learning, Hyperspectral imaging, positive-unlabeled learning, Positive-unlabeled learning, Hyperspectral image classification, hyperspectral image classification, masked autoencoder, Masked autoencoder, self-learning, Self-learning

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: O.S. Melashchenko; Institute for Applied System Analysis, Kyiv, Ukraine; email: alexander.melashchenko@gmail.com; Conference name: 8th IEEE International Conference on Methods and Systems of Navigation and Motion Control, MSNMC 2025; Conference date: 21 October 2025 through 24 October 2025; Conference code: 221101

Positive-unlabeled learning for anomaly detection based on dual-branch generative adversarial networks

Item Type Journal Article

Author Q. Wei

Author A. Li

Author J. Hu

Author C. Han

Author Y. Chi

Author Y. Shi

Abstract Anomaly detection under positive-unlabeled (PU) learning remains challenging as only a few anomalies are labeled, whereas the majority of data remain unlabeled and may be contaminated by unknown anomalies. Existing PU learning approaches often rely on imprecise class-prior estimation or biased synthetic sample generation, potentially leading to unstable optimization and degraded detection performance. To mitigate these limitations, we propose PUGANomaly, a novel PU learning method for anomaly detection based on dual-branch generative adversarial networks (GANs). PUGANomaly integrates anomaly detection and PU learning within a unified architecture. By incorporating a self-training mechanism and a joint anomaly scoring strategy, PUGANomaly iteratively refines pseudo-negative samples and model parameters from anomaly similarity and normal dissimilarity perspectives. Furthermore, we design a set of PU-specific loss functions to explicitly enforce increased separation between normal and anomalous representations. To further strengthen discriminative capability, an auxiliary classifier is incorporated to learn the decision boundary between labeled anomalies and pseudo-normal samples, and the outputs are subsequently fused with reconstruction-based anomaly scores through an ensemble learning module. Extensive experiments conducted on the MNIST, FashionMNIST, and Alzheimer datasets demonstrate that PUGANomaly consistently and statistically outperforms state-of-the-art GAN-based PU learning models on anomaly detection tasks, achieving superior detection accuracy and robustness under limited supervision. © 2026 Elsevier B.V.

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Positive-Unlabeled Learning With Label Distribution Alignment

Item Type Journal Article

Author Y. Jiang

Author Q. Xu

Author Y. Zhao

Author Z. Yang

Author P. Wen

Author X. Cao

Author Q. Huang

Abstract Positive-Unlabeled (PU) data arise frequently in a wide range of fields such as medical diagnosis, anomaly analysis and personalized advertising. The absence of any known negative labels makes it very challenging to learn binary classifiers from such data. Many state-of-the-art methods reformulate the original classification risk with individual risks over positive and unlabeled data, and explicitly minimize the risk of classifying unlabeled data as negative. This, however, usually leads to classifiers with a bias toward negative predictions, i.e., they tend to recognize most unlabeled data as negative. In this paper, we propose a label distribution alignment formulation for PU learning to alleviate this issue. Specifically, we align the distribution of predicted labels with the ground-truth, which is constant for a given class prior. In this way, the proportion of samples predicted as negative is explicitly controlled from a global perspective, and thus the bias toward negative predictions could be intrinsically eliminated. On top of this, we further introduce the idea of functional margins to enhance the model's discriminability, and derive a margin-based learning framework named Positive-Unlabeled learning with Label Distribution Alignment (PULDA). This framework is also combined with the class prior

estimation process for practical scenarios, and theoretically supported by a generalization analysis. Moreover, a stochastic mini-batch optimization algorithm based on the exponential moving average strategy is tailored for this problem with a convergence guarantee. Finally, comprehensive empirical results demonstrate the effectiveness of the proposed method. © 1979-2012 IEEE.

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Notes:

EID: 2-s2.0-85173062942

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Positive-Unlabeled Learning-Based Hybrid Deep Network for Intelligent Fault Detection

Item Type Journal Article

Author M. Qian

Author Y.-F. Li

Author T. Han

Abstract Intelligent fault detection methods based on deep learning have been developed rapidly in recent years. However, most of these methods are based on supervised learning which requires a fully labeled training set. It is difficult to obtain massive labeled samples in real applications incredibly accurately labeled fault samples from an operating system. The lack of labels and label noise becomes a great challenge for fault detection. To tackle this problem, in this article, we propose a positive-unlabeled learning based hybrid network (PUHN). It only needs part of the normal operating samples to be labeled. All other samples (including the rest of the normal samples and all fault samples) are unlabeled, which greatly reduces the labeling cost. PUHN consists of three modules: a nonnegative risk positive-unlabeled (PU) network for training the classifier, a feature extraction module, and a clustering layer for improving data separability and estimating the class priors of PU learning. The three are optimized as a whole and the corresponding optimization strategy is designed. The monitoring data of 24 wind turbines are used to verify the effectiveness and robustness of the proposed method. The experimental results indicate that the proposed method is superior to the benchmark methods, and the performance is significantly better than the supervised learning method when there exists label noise. © 2005-2012 IEEE.

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Risk perception, Risk assessment, Deep learning, Features extraction, Supervised learning, Benchmarking, Labelings, Detection methods, Non negatives, Training sets, Fault detection, Faults detection, Positive-unlabeled learning, positive-unlabeled (PU) learning, Real applications, Fault sample, Hybrid network, wind turbine, Wind turbines

Notes:

EID: 2-s2.0-85129301665

Export Date: 20 September 2026; Cited By: 29; Correspondence Address: Y.-F. Li; Department of Industrial Engineering, Tsinghua University, Beijing, 100084, China; email: liyanfu@tsinghua.edu.cn

PostgreSQL anomalous query detector

Item Type Conference Paper

Author B. Shebaro

Author A. Sallam

Author A. Kamra

Author E. Bertino

Abstract We propose to demonstrate the design, implementation, and the capabilities of an anomaly detection (AD) system integrated with a relational database management system (DBMS). Our AD system is trained by extracting relevant features from the parse-tree representation of the SQL commands, and then uses the DBMS roles as the classes for the bayesian classifier. In the detection phase, the maximum apriori probability role is chosen by the classifier which, if not matching the role associated with the SQL command, raises an alarm. We have implemented such system in the PostgreSQL DBMS, integrated with the statistics collection and the query processing mechanism of the DBMS. During the demonstration, our audience will be given the choice of training our system using either synthetic role-based SQL query traces based on probability sampling, or by entering their own set of training queries. In the subsequent detection mode, the audience can test the detection capabilities of the system by submitting arbitrary SQL commands. We will also allow the audience to generate arbitrary work loads to measure the overhead of the training phase and the detection phase of our AD mechanism on the performance of the DBMS. © 2013 Authors.

Date 2013

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Language English

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Conference Name ACM International Conference Proceeding Series

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Intrusion detection, Detection capability, Query processing, intrusion detection, Anomaly detection, anomaly detection, Detectors, Bayesian classifier, A-priori probabilities, Database systems, DBMS, Detection phase, Probability sampling, Relational database management systems, Relevant features

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Power control strategy for distributed multiple-hypothesis detection

Item Type Journal Article

Author H.-S. Kim

Author N.A. Goodman

Abstract We introduce a local-channel power control strategy applicable to multiple-hypothesis distributed detection systems communicating over slow-fading orthogonal multiple-access channels. In earlier work, it was demonstrated that performance could be improved by adjusting transmit power to maximize the J-divergence measure of a binary detection system. The local power control strategy introduced here further improves performance by exploiting a priori probabilities and local sensor statistics. Moreover, the local power optimization can be combined with additional power control based on the state of the propagation channel. We extend the optimization to systems performing multiple-hypothesis detection, and evaluate outage probability for these systems. Various numerical results are shown. © 2010 IEEE.

Date 2010

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Export Date: 20 September 2026; Cited By: 13; Correspondence Address: H.-S. Kim; Department of Electrical and Computer Engineering, University of Arizona, Tucson, AZ 85721, United States; email: hskim@ece.arizona.edu; CODEN: ITPRE

Practical Uncertainty Quantification in Orbital Mechanics

Item Type Journal Article

Author M. Vasile

Abstract The chapter provides an overview of methods to quantify uncertainty in orbital mechanics. It also provides an initial classification of these methods with particular attention to whether the quantification method requires a knowledge of the system model or not. For some methods the chapter provides applications examples and numerical comparisons on selected test cases. © 2019, Springer Nature Switzerland AG.

Date 2019

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Uncertainty quantification, Uncertainty propagation, Orbital mechanics

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Pre- and post-harvest spectral estimation of carnosic acid and rosmarinic acid in rosemary

Item Type Journal Article**Author** A. Mishra**Author** A. Krief**Author** M.M. Sahoo**Author** A. Schachter**Author** I. Gonda**Author** N. Dudai**Author** T. Trigano**Author** I. Herrmann

Abstract Rosemary extracts, including carnosic and rosmarinic acids (CA and RA, respectively), are known for their antimicrobial and antioxidant capabilities. Traditional quantification methods of CA and RA (later on called selected phytochemicals) are often destructive and time-consuming. This study presents a spectral, non-destructive, and time-efficient approach for estimating selected phytochemicals in pre- and post-harvest stages. We acquired spectral data from field-grown rosemary plants, dry leaves, and powder as well as UAV-borne hyperspectral imagery. The analysis included a transformation sequence (second derivative, Yeo–Johnson, and standardization), followed by partial least squares regression (PLSR). To mimic real-life scenarios, we investigated a training–testing strategy denoted by “leave-one-day-out”, systematically excluding each day's data from training. For CA estimation, the PLSR model achieved a coefficient of determination (R^2) of 0.75 with a relative root mean square error (RRMSE) of 10.42% at the canopy level, 0.80 (RRMSE: 8.91%) for dry leaves, and 0.76 (RRMSE: 9.09%) for powder. RA estimation was challenging at the canopy level with an R^2 of 0.52 (RRMSE: 13.42%), but improved in post-harvest samples, reaching R^2 of 0.79 (RRMSE: 10.0%) for dry leaves and 0.75 (RRMSE: 9.78%) for powder. These results demonstrated the efficiency of the proposed approach. It offers a reliable alternative to traditional methods, with potential applications in agriculture and post-harvest industry. © 2026 The Authors

Date 2026**DOI** 10.1016/j.compag.2026.111501**URL** <https://www.scopus.com/pages/publications/105029078406?origin=resultslist>**Archive** Scopus**Language** English**Volume** 244**Publisher** Elsevier B.V.**Publication** Computers and Electronics in Agriculture**ISSN** 01681699 (ISSN)**Journal Abbr** Comput. Electron. Agric.**Date Added** 20/09/2026, 9:51:30 pm**Modified** 20/09/2026, 9:51:30 pm**Tags:**

Learning systems, machine learning, Machine-learning, Machine learning, Metadata, Remote sensing, Spectrum analysis, Least squares approximations, Mean square error, regression analysis, HyperSpectral, antioxidant, Carnosic acid, Dry leaves, estimation method, Harvesting, HPLC, Hyperspectral, multispectral image, Partial least square regression, Phytochemical, Phytochemicals, phytochemistry, Plants (botany), PLSR, Postharvest, Root mean square errors, Yeo–johnson transformation, Yeo–Johnson transformation

Notes:

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Export Date: 20 September 2026; Cited By: 1; Funding details: Shamoon College of Engineering research program; Robert H. Smith Faculty of Agriculture, Food and Environment; Hebrew University of Jerusalem, HUJI; Ministry of Agriculture and Rural Development, MOARD, (20-01-0026); Ministry of Agriculture and Rural Development, MOARD; Correspondence Address: T. Trigano; Department of Electrical Engineering, Sami Shamoon College of Engineering, Ashdod, Jabotinski St, 84, 77245, Israel; email: thomast@ac.sce.ac.il; CODEN: CEAGE

Precision phenotyping for curating research cohorts of patients with unexplained post-acute sequelae of COVID-19

Item Type Journal Article

- Author A. Azhir
- Author J. Hügel
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- Author Z.H. Strasser
- Author J.G. Klann
- Author S.N. Murphy
- Author H. Estiri

Abstract Background: Scalable identification of patients with post-acute sequelae of COVID-19 (PASC) is challenging due to a lack of reproducible precision phenotyping algorithms, which has led to suboptimal accuracy, demographic biases, and underestimation of the PASC. Methods: In a retrospective case-control study, we

developed a precision phenotyping algorithm for identifying cohorts of patients with PASC. We used longitudinal electronic health records data from over 295,000 patients from 14 hospitals and 20 community health centers in Massachusetts. The algorithm employs an attention mechanism to simultaneously exclude sequelae that prior conditions can explain and include infection-associated chronic conditions. We performed independent chart reviews to tune and validate the algorithm. Findings: The PASC phenotyping algorithm improves precision and prevalence estimation and reduces bias in identifying PASC cohorts compared to the ICD-10-CM code U09.9. The algorithm identified a cohort of over 24,000 patients with 79.9% precision. Our estimated prevalence of PASC was 22.8%, which is close to the national estimates for the region. We also provide in-depth analyses, encompassing identified lingering effects by organ, comorbidity profiles, and temporal differences in the risk of PASC. Conclusions: PASC precision phenotyping boasts superior precision and prevalence estimation while exhibiting less bias in identifying patients with PASC. The cohort derived from this algorithm will serve as a springboard for delving into the genetic, metabolomic, and clinical intricacies of PASC, surmounting the constraints of prior PASC cohort studies. Funding: This research was funded by the US National Institute of Allergy and Infectious Diseases (NIAID). © 2024 The Author(s)

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Precision-Recall Operating Characteristic (P-ROC) curves in imprecise environments

Item Type	Conference Paper
Author	T.C.W. Landgrebe
Author	P. Paclik
Author	R.P.W. Duin
Author	A.P. Bradley
Abstract	Traditionally, machine learning algorithms have been evaluated in applications where assumptions can be reliably made about class priors and/or misclassification costs. In this paper, we consider the case of imprecise environments, where little may be known about these factors and they may well vary significantly when the system is applied. Specifically, the use of precision-recall analysis is investigated and compared to the more well known performance measures such as error-rate and the receiver operating characteristic (ROC). We argue that while ROC analysis is invariant to variations in class priors, this invariance in fact hides an important factor of the evaluation in imprecise environments. Therefore, we develop a generalised precision-recall analysis methodology in which variation due to prior class probabilities is incorporated into a multi-way analysis of variance (ANOVA). The increased sensitivity and reliability of this approach is demonstrated in a remote sensing application. © 2006 IEEE.
Date	2006
DOI	10.1109/ICPR.2006.941
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Archive	Scopus
Language	English
Extra	Journal Abbreviation: Proc. Int. Conf. Pattern Recognit.
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Analysis of variance (ANOVA), Learning systems, Learning algorithms, Remote sensing, Probability, Sensitivity analysis, Error analysis, Misclassification costs, Imprecise environments, Receiver operating characteristics (ROC)

Notes:

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Export Date: 20 September 2026; Cited By: 88; Correspondence Address: T.C.W. Landgrebe; Elect. Eng., Maths and Comp. Sc., Delft University of Technology, Netherlands; email: t.c.w.landgrebe@ewi.tudelft.nl; Conference name: 18th International Conference on Pattern Recognition, ICPR 2006; Conference date: 20 August 2006 through 24 August 2006; Conference code: 69443; CODEN: PICRE

Predicting and classifying deoxynivalenol in wheat flour using ATR-FTIR spectroscopy and explainable machine learning

Item Type Journal Article

- Author** J. Liu
- Author** K. Yao
- Author** C. Chen
- Author** D. Li
- Author** D. Xu
- Author** R. Guo
- Author** W. Yao
- Author** Y. Deng
- Author** W. Xue
- Author** Q. Wu
- Author** T. Li
- Author** X. He
- Author** P.K. Singh

Abstract Deoxynivalenol (DON) is a major hazardous component in wheat kernels infected by Fusarium head blight. Given the lack of both reliable and cost-effective quantification methods, this study aims to use Fourier-transform infrared spectroscopy with attenuated total reflectance (ATR-FTIR) for predicting DON concentrations. Four machine learning models, including eXtreme Gradient Boosting (XGBoost), Random Forest (RF), Tabular Prior-data Fitted Network (TabPFN), and Categorical Boosting (CatBoost), were firstly developed using key spectral features selected by recursive feature elimination (RFE). TabPFN model outperformed others in quantitative

prediction, achieving a coefficient of determination (R^2) of 0.86 on hold-out test set. SHapley Additive exPlanations (SHAP) were firstly applied to enhance model interpretability and transparency, identifying wavenumbers at 996 cm^{-1} , 1083 cm^{-1} , 1135 cm^{-1} , and 1574 cm^{-1} as having higher influence on model performance. For binary classification of DON levels (<1 or ≥ 1 mg/kg), Catboost model achieved superior results, yielding a recall of 0.91, accuracy of 0.88, and an F2 score of 0.91. This study highlights the effectiveness of machine learning based analysis of FTIR spectra. Furthermore, the proposed approach herein could be a feasible way for monitoring DON levels in wheat flour, with potential applications at both laboratory and industrial scales to ensure food safety. © 2026 The Authors

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Accident prevention, Article, Learning systems, machine learning, Machine learning, Forecasting, Machine Learning, quantitative analysis, Cost effectiveness, prediction, controlled study, Adaptive boosting, Random forests, SHAP, random forest, Xgboost, XGBoost, intermethod comparison, Shapley, chemistry, Fourier transform infrared spectroscopy, ATR FTIR, attenuated total reflectance Fourier transform infrared spectroscopy, binary classification, biomonitoring, Categorical Boosting, chemical compound, Deoxynivalenol, DON, explainable machine learning, eXtreme Gradient Boosting, feature selection, flour, Flour, food contamination, Food Contamination, food safety, Food safety, FTIR, FTIR spectroscopy, fungal disease, Fusarium head blight, Fusarium head blights, Grain (agricultural product), machine prediction model, recursive feature elimination, Shapley additive explanation, spectral reflectance, Spectroscopy, Fourier Transform Infrared, Supersaturation, TabPFN, Tabular prior-data fitted network, Tabular Priordata Fitted Network, trichothecene, Trichothecenes, Triticum, vomitoxin, wheat, Wheat, wheat flour, Wheat flours

Notes:

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Predicting cognitive decline with deep learning of brain metabolism and amyloid imaging

Item Type Journal Article

Author H. Choi

Author K.H. Jin

Abstract For effective treatment of Alzheimer's disease (AD), it is important to identify subjects who are most likely to exhibit rapid cognitive decline. We aimed to develop an automatic image interpretation system based on a deep convolutional neural network (CNN) which can accurately predict future cognitive decline in mild cognitive impairment (MCI) patients using flurodeoxyglucose and florbetapir positron emission tomography (PET). PET images of 139 patients with AD, 171 patients with MCI and 182 normal subjects obtained from Alzheimer's Disease Neuroimaging Initiative database were used. Deep CNN was trained using 3-dimensional PET volumes of AD and normal controls as inputs. Manually defined image feature extraction such as quantification using predefined region-of-interests was unnecessary for our approach. Furthermore, it used minimally processed images without spatial normalization which has been commonly used in conventional quantitative analyses. Cognitive outcome of MCI subjects was predicted using this network. The prediction accuracy of the conversion of mild cognitive impairment to AD was compared with the conventional feature-based quantification approach. Accuracy of prediction (84.2%) for conversion to AD in MCI patients outperformed conventional feature-based quantification approaches. ROC analyses revealed that performance of CNN-based approach was significantly higher than that of the conventional quantification methods ($p < 0.05$). Output scores of the network were strongly correlated with the longitudinal change in cognitive measurements ($p < 0.05$). These results show the feasibility of deep learning as a practical tool for developing predictive neuroimaging biomarker. © 2018 Elsevier B.V.

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Publication Behavioural Brain Research

ISSN 01664328 (ISSN)**Journal Abbr** Behav. Brain Res.**Date Added** 20/09/2026, 9:52:55 pm**Modified** 20/09/2026, 9:52:55 pm**Tags:**

Article, female, Female, human, Humans, male, Male, Deep learning, aged, Aged, brain, Brain, Convolutional neural network, procedures, controlled study, artificial neural network, major clinical study, convolutional neural network, classification, support vector machine, Support Vector Machine, image processing, cognitive defect, Alzheimer disease, Alzheimer Disease, diagnostic imaging, sensitivity and specificity, image analysis, computer assisted diagnosis, Image Interpretation, Computer-Assisted, Imaging, Three-Dimensional, neuroimaging, metabolism, prognosis, Prognosis, mild cognitive impairment, Cognitive Dysfunction, fluorodeoxyglucose f 18, positron emission tomography, Positron-Emission Tomography, radiopharmaceutical agent, Radiopharmaceuticals, imaging, principal component analysis, Sensitivity and Specificity, Alzheimer's disease, Amyloid, amyloid protein, Fluorodeoxyglucose F18, Mini Mental State Examination, amyloid, priority journal, florbetapir, Aniline Compounds, aniline derivative, Clinical Dementia Rating, three dimensional imaging, brain metabolism, Brain PET, ethylene glycol derivative, Ethylene Glycols, functional assessment, institutional review, neuropsychology

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Predicting floor heave risk in road tunnels with machine learning

Item Type Journal Article**Author** X. Ou

Author Y. Zhou
Author Y. Kong
Author T. Qu
Author S. Xu
Author W. Liao
Author C. Tang
Author X. Zhang

Abstract Floor heave is a common defect in mountainous tunnels. It is critical but challenging to predict the risk of floor heave, as traditional methods often fail to characterize this phenomenon effectively. This study proposes a data-driven approach utilizing a support vector machine (SVM) optimized by the sparrow search algorithm (SSA) to address the issue. The model was developed and validated using a dataset collected from 100 tunnels. Shapley value analysis was conducted to identify the key features influencing floor heave defects. Moreover, a committee-based uncertainty quantification method is presented to evaluate the reliability of each prediction. The results show that: (1) Data feature engineering and SSA play pivotal roles in expediting the convergence of the SVM model. (2) Groundwater and high in situ stress are key factors contributing to tunnel floor heave. (3) In comparison to backpropagation (BP) neural networks, the SSA-SVM demonstrates superior robustness in handling imperfect and limited data. (4) The committee-based uncertainty quantification method is proven effective to evaluate the trustworthiness of each prediction. This data-driven surrogate model offers an effective strategy for understanding the factors that impact tunnel floor defects and accurately predicting tunnel floor heave deformation. © 2025 Institute of Rock and Soil Mechanics, Chinese Academy of Sciences

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Predicting incomplete gene microarray data with the use of supervised learning algorithms

Item Type Journal Article

Author B. Twala

Author M. Phorah

Abstract Motivation: With the wealth of sequence data and the huge amount of data generated from molecular technologies, the issue of gene classification/ prediction has become a central challenge in the field of microarray data analysis. This has led to the application of many well-established supervised learning (SL) algorithms in an attempt to provide more accurate and automatic diagnosis class (cancer/non cancer) prediction. Virtually all research on SL addresses the task of learning to classify complete domain instances. However, in some research situations we often have to classify instances given incomplete vectors, which can affect the predictive accuracy of learned classifiers. The task of learning an accurate incomplete data classifier from instances raises a number of new issues some of which have not been properly addressed by bioinformatics research. Thus, an effective missing value estimation method is required for improving predictive accuracy. Results: The essence of the approach is the proposal that prediction using supervised learning can be improved in probabilistic terms given incomplete microarray data. This imputation approach is based on the a priori probability of each value determined from the instances at that node of a decision tree (PDT) that have specified values. The proposed approach exploits the total probability and Bayes' theorems and it has three versions. We evaluate our approach with other supervised learning techniques including C5.0, classification and regression trees (CART), k-nearest neighbour (k-NN), linear discrimination (LD) naïve Bayes classifier (NBC), Repeated Incremental Pruning to Produce Error Reduction (RIPPER) and support vector machines (SVMs), from the point of view of their effect or tolerance of incomplete test data. Eight cancer related gene expression datasets are utilized for this task. Experimental results are provided to illustrate the efficiency and the robustness of the proposed algorithm. © 2010 Elsevier B.V. All rights reserved.

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Notes:

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Export Date: 20 September 2026; Cited By: 15; Funding details: Bangladesh Council of Scientific and Industrial Research, (MDSARR1); Correspondence Address: B. Twala; Department of Electrical and Electronic Engineering Science, Faculty of Engineering and Built Environment, University of Johannesburg, Auckland Park 2006, Johannesburg, P.O. Box 524, South Africa; email: btwala1@csir.co.za

Predicting mechanical properties of low-alloy steels using features extracted from Electron Backscatter Diffraction characterization

Item Type Journal Article**Author** Y. Li**Author** J. Zhao**Author** X. Li**Author** Z. Xing**Author** Q. Duan**Author** X. Liang**Author** X. Wang

Abstract Machine learning (ML) approaches have recently been increasingly employed to establish quantitative relationships between material composition, processing, microstructure, and properties. However, the complexities of microstructure pose challenges for straightforward modeling, thereby complicating research efforts. This study introduces a series of multi-parametric quantification methods based on Electron Backscatter Diffraction (EBSD) data tailored to the microstructural characteristics of low-alloy steels. These methods include quantification of boundary densities across various misorientation angles, distinct types of boundaries, and geometrically necessary dislocation densities. Through thermomechanical simulation and micro-tensile testing of low-alloy steels, data on yield and ultimate tensile

strengths were obtained, alongside EBSD-based extraction of microstructure characteristics. Several ML methods, including Random Forest (RF), Gradient Boosting Decision Trees (GBDT), and Extreme Gradient Boosting (XGBoost), were utilized to predict yield strength (YS) and ultimate tensile strength (UTS) using the aforementioned microstructural features. The GBDT model outperformed other algorithms, demonstrating high accuracy in predicting YS, UTS, and elongation. The model achieved an Mean Squared Error (MSE) of 972.18, an Mean Absolute Error (MAE) of 24.75 and an Coefficient of Determination (R2) of 0.864 for YS, and an MSE of 812.28, an MAE of 22.87 and an R2 of 0.823 for UTS. These results confirm GBDT's effectiveness in predicting mechanical properties from microstructural data. This successful integration of ML with multi-parametric description microstructural features underscores its potential in facilitating material design and development processes. © 2024 The Authors

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Export Date: 20 September 2026; Cited By: 6; Funding details: University of Science and Technology Beijing, USTB; Baosteel Industrial Brain Project, (B0602875); Correspondence Address: X. Li; Collaborative Innovation Center of Steel Technology, University of Science and Technology Beijing, Beijing, 30 Xueyuan Road, 100083, China; email: xiuchengli@ustb.edu.cn; Z. Xing; Central Research Institute of Baosteel, Shanghai, 889 Fujin Road, 201999, China; email: xingzhao@baosteel.com; X. Wang; Collaborative Innovation Center of Steel Technology, University of Science and Technology Beijing, Beijing, 30 Xueyuan Road, 100083, China; email: wxm@mater.ustb.edu.cn

Predicting protein functions using positive-unlabeled ranking with ontology-based priors

Item Type Journal Article

Author F. Zhapa-Camacho

Author Z. Tang

Author M. Kulmanov

Author R. Hoehndorf

Abstract Automated protein function prediction is a crucial and widely studied problem in bioinformatics. Computationally, protein function is a multilabel classification problem where only positive samples are defined and there is a large number of unlabeled annotations. Most existing methods rely on the assumption that the unlabeled set of protein function annotations are negatives, inducing the false negative issue, where potential positive samples are trained as negatives. We introduce a novel approach named PU-GO, wherein we address function prediction as a positive-unlabeled ranking problem. We apply empirical risk minimization, i.e. we minimize the classification risk of a classifier where class priors are obtained from the Gene Ontology hierarchical structure. We show that our approach is more robust than other state-of-the-art methods on similarity-based and time-based benchmark datasets. © The Author(s) 2024. Published by Oxford University Press.

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Predicting short-term interruptions of antiretroviral therapy from summary adherence data: Development and test of a probability model

Item Type Journal Article

Author R.A. Harris

Author J.E. Haberer

Author N. Musinguzi

Author K.-M. Chang

Author C.B. Schechter

Author C.A. Doubeni

Author R. Gross

Abstract Antiretroviral therapy (ART) for HIV is vulnerable to unplanned treatment interruptions—consecutively missed doses over a series of days—which can result in virologic rebound. Yet clinicians lack a simple, valid method for estimating the risk of interruptions. If the likelihood of ART interruption could be derived from a convenient-to-gather summary measure of medication adherence, it might be a valuable tool for both clinical decision-making and research. We constructed an a priori probability model of ART interruption based on average adherence and tested its predictions using data collected on 185 HIV-infected, treatment-naïve individuals over the first 90 days of ART in a prospective cohort study in Mbarara, Uganda. The outcome of interest was the presence or absence of a treatment gap, defined as >72 hours without a dose. Using the pre-determined value of 0.50 probability as the cut point for predicting an interruption, the classification accuracy of the model was 73% (95% CI = 66%– 79%), the specificity was 87% (95% CI = 79%– 93%), and the sensitivity was 59% (95% CI = 48%– 69%). Overall model performance was satisfactory, with an area under the receiver operator characteristic curve (AUROC) of 0.85 (95% CI = 0.80–0.91) and Brier score of 0.20. The study serves as proof-of-concept that the probability model can accurately differentiate patients on the continuum of risk for short-term ART interruptions using a summary measure of adherence. The model may also aid in the design of targeted interventions. © This is an open access article, free of all copyright, and may be freely reproduced, distributed, transmitted, modified, built upon, or otherwise used by anyone for any lawful purpose. The work is made available under the Creative Commons CC0 public domain dedication.

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Predicting the cytotoxicity of nanomaterials through explainable, extreme gradient boosting

Item Type Journal Article

Author A. Conti

Author L. Campagnolo

Author S. Diciotti

Author A. Pietroiusti

Author N. Toschi

Abstract Nanoparticles (NPs) are a wide class of materials currently used in several industrial and biomedical applications. Due to their small size (1-100 nm), NPs can easily enter the human body, inducing tissue damage. NP toxicity depends on physical and

chemical NP properties (e.g., size, charge and surface area) in ways and magnitudes that are still unknown. We assess the average as well as the individual importance of NP atomic descriptors, along with chemical properties and experimental conditions, in determining cytotoxicity endpoints for several nanomaterials. We employ a multicenter cytotoxicity nanomaterial database (12 different materials with first and second dimensions ranging between 2.70 and 81.2 nm and between 4.10 and 4048 nm, respectively). We develop a regressor model based on extreme gradient boosting with hyperparameter optimization. We employ Shapley additive explanations to obtain good cytotoxicity prediction performance. Model performances are quantified as statistically significant Spearman correlations between the true and predicted values, ranging from 0.5 to 0.7. Our results show that i) size in situ and surface areas larger than 200 nm and 50 m²/g, respectively, ii) primary particles smaller than 20 nm; iii) irregular (i.e., not spherical) shapes and iv) positive Z-potentials contribute the most to the prediction of NP cytotoxicity, especially if lactate dehydrogenase (LDH) assays are employed for short experimental times. These results were moderately stable across toxicity endpoints, although some degree of variability emerged across dose quantification methods, confirming the complexity of nano-bio interactions and the need for large, systematic experimental characterization to reach a safer-by-design approach. © 2022 Informa UK Limited, trading as Taylor & Francis Group.

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Prediction Accuracy & Reliability: Classification and Object Localization Under Distribution Shift

Item Type Journal Article

Author F. Diet

Author M. Kassem Sbeyti

Author M. Karg

Abstract Natural distribution shiftDistribution shift causes a performance deterioration for the visual perception computed by convolutional neural networks (CNNs). This can lead to safety-critical situations, when deploying CNNs to real-world traffic data. To facilitate a quantitative analysis of this deterioration, this work addresses: (1) investigating the effect of natural distribution shiftDistribution shift and weather augmentations on the performance of detection and confidence estimation, (2) evaluating model accuracy and confidence prediction for both object classificationClassification and its localization, and (3) benchmarking two common uncertaintyUncertainty quantification methods - EnsemblesEnsemble and compute-efficient variants of Monte-Carlo (MC) DropoutDropout. For this purpose, a novel dataset has been curated from publicly available autonomous driving datasets. The in-distribution (ID) data is based on cutouts of individual objects, for which both class and bounding box annotations are available. The six distribution-shift datasets cover adverse weather scenarios, simulated rain and fog, corner cases, and out-of-distribution data. A granular analysis of CNNs under distribution shiftDistribution shift allows to quantize the impact of different types of shifts on both, task performance and confidence estimation. Our novel key findings are: ConvNeXt-Tiny is more robust to distribution shiftDistribution shift than EfficientNet-B0; heavy rain degrades classificationClassification accuracy stronger than localization, whereas heavy fog has the opposite effect; integrating MC-Dropout into selected layers only has the potential to enhance task performance and confidence estimation, with the identification of these layers depending on the type of distribution shiftDistribution shift and the considered task. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Intelligent systems, Rain, Convolutional neural network, Convolutional neural networks, Uncertainty, Weather forecasting, Classification, Ensemble, Multilayer neural networks, Localisation, Shift-and, Distribution shift, Confidence estimation, Dropout, Regression, Natural distribution

Notes:

EID: 2-s2.0-85205021459

Export Date: 20 September 2026; Cited By: 3; Correspondence Address: F. Diet; Continental AG, Hannover, Germany; email: fabi.nk97@yahoo.de

Prediction and outlier detection in classification problems

Item Type Journal Article

Author L. Guan

Author R. Tibshirani

Abstract We consider the multi-class classification problem when the training data and the out-of-sample test data may have different distributions and propose a method called BCOPS (balanced and conformal optimized prediction sets). BCOPS constructs a prediction set $C(x)$ as a subset of class labels, possibly empty. It tries to optimize the out-of-sample performance, aiming to include the correct class and to detect outliers x as often as possible. BCOPS returns no prediction (corresponding to $C(x)$ equal to the empty set) if it infers x to be an outlier. The proposed method combines supervised learning algorithms with conformal prediction to minimize a misclassification loss averaged over the out-of-sample distribution. The constructed prediction sets have a finite sample coverage guarantee without distributional assumptions. We also propose a method to estimate the outlier detection rate of a given procedure. We prove asymptotic consistency and optimality of our proposals under suitable assumptions and illustrate our methods on real data examples. © 2022 The Authors. Journal of the Royal Statistical Society: Series B (Statistical Methodology) published by John Wiley & Sons Ltd on behalf of Royal Statistical Society.

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ISSN 13697412 (ISSN)**Issue** 2**Journal Abbr** J. R. Stat. Soc. Ser. B Stat. Methodol.**Date Added** 20/09/2026, 9:52:20 pm**Modified** 20/09/2026, 9:52:20 pm**Tags:**

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Prediction for Tunnelling-Induced Ground Settlement in Multilayered Soils: An Improved Gradient Boosting Approach

Item Type Journal Article**Author** H. An**Author** Y. Chen**Author** J. Lei**Author** H. Luo**Author** E.J. Chen

Abstract During the construction of a shield tunnel, it will disturb the surrounding ground and affect the use and structural safety of buildings around the tunnel. The geometric parameters of the tunnel, the operating parameters of the shield machine, and the geological parameters will affect the degree of disturbance. However, the existing theories and models are difficult to comprehensively consider the interaction of these factors, and it is difficult to accurately predict the response of the formation to solve the above problems. The research is based on the machine learning algorithm to establish a prediction model of stratum settlement caused by shield tunneling, which provides a new idea for real time prediction of the ground response caused by shield tunneling and risk reduction. The main results of this research are as follows: (1) propose a novel quantification method for geological parameters that can comprehensively consider the physical and mechanical properties of the rock, soil layers, and the geometric characteristics of depth and thickness and (2) establish a more robust proxy model and use the k-fold cross-validation method to enhance its performance. Copyright © 2025 Hongbin An et al. Advances in Civil Engineering published by John Wiley & Sons Ltd.

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Prediction Interval in Renewable Energy Forecasting: A Comprehensive Review of Uncertainty Quantification Methods

Item Type Journal Article

Author N. Sakib

Author M.A. Hosen

Author B. Khan

Author B. Gunn

Author M. Johnstone

Abstract Robust and reliable forecasting of renewable energy generation is essential for managing grid stability, optimising energy resources, and meeting global climate targets, such as the European Union's goal of achieving Net-Zero greenhouse gas emissions by 2050. Renewable energy sources, including solar, wind, and tidal, are inherently variable and difficult to predict, while hydro is comparatively more stable but still subject to seasonal and climatic fluctuations. This variability poses significant challenges to grid operators and policymakers striving to integrate more renewable energy into the power system. Prediction Interval (PI), which quantifies uncertainty by providing a probabilistic range of potential outcomes, offer a solution to this problem by enhancing the reliability and accuracy of renewable energy forecasts. This paper reviews the methodologies for estimating prediction interval, including statistical, machine learning, and deep learning approaches, and demonstrates how these techniques can address the key challenges in renewable energy forecasting. By offering a structured comparison of prediction interval estimation methods across different renewable energy sectors, the paper highlights their potential to reduce uncertainty, improve decision-making for grid integration,

and assist policymakers in developing strategies to mitigate risks from energy variability. Moreover, the review identifies critical gaps in current research and proposes future directions that could further refine prediction interval estimation methods for greater accuracy and applicability. The insights provided in this review can aid scientists, engineers, and decision-makers in advancing renewable energy forecasting techniques, thereby accelerating the transition toward a more sustainable and resilient energy system. © 2013 IEEE.

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Export Date: 20 September 2026; Cited By: 15; Correspondence Address: N. Sakib; Deakin University, Institute for Intelligent Systems, Waurn Ponds, 3216, Australia; email: s.sakib@deakin.edu.au

Prediction of driver alertness levels on mountain roads using machine learning models: A naturalistic driving study in China

Item Type Journal Article

Author T. Liu

Author D. Xie

Author T. Liu

Author J. Shan

Author C. Tang

Abstract Objective: A method for evaluating driver alertness on mountain roads was developed to enhance dynamic safety monitoring in high-risk sections. An indicator system integrating human and environmental factors was established, with 13 variables used for alertness classification and 17 initial variables applied for quantification. Methods: Field tests were conducted in Guizhou, China, where data on drivers' heart rates, eye movements, and demographics were collected. Kernel principal component analysis (KPCA) was employed to extract four representative factors from 13 driving-related indicators. K-means clustering was employed to categorize drivers into high- and low-alertness groups. Logistic regression scoring, XGBoost, and Tabular Prior-Data Fitted Network (TabPFN) models were developed to assess driver alertness probabilities. Results: Superior performance was demonstrated by the XGBoost model, achieving an area under the receiver operating characteristic curve (AUC) value of 0.97 compared with 0.80 for logistic regression scoring model, indicating improved classification accuracy and robustness. TabPFN further enhanced performance, yielding the highest results (AUC: 0.98; KS: 0.85; F1: 0.90), confirming its robustness and adaptability in small-sample, high-dimensional data settings. Conclusion: Driver alertness levels were effectively quantified by integrating human and environmental factors. TabPFN model was found to be more effective than logistic regression scoring and XGBoost, making it the preferred approach for monitoring driver alertness on mountain roads. © 2025 Taylor & Francis Group, LLC.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: T. Liu; College of Traffic & Transportation, Chongqing Jiaotong University, Chongqing, China; email: liutangzhi@cqjtu.edu.cn; CODEN: TIPRC

Prediction of Tail Water Level under the Influence of Backwater Effect Based on Deep Learning Models: A Case Study in the Xiangjiaba Hydropower Station

Item Type Journal Article

Author S. Zhang

Author S. Xie

Author Y. Wang

Author Y. Xu

Author Z. Zhang

Author B. Jia

Abstract Accurate forecasting of the tail water level (TWL) is of great importance for the safe and economic operation of hydropower stations. The prediction accuracy is significantly influenced by the backwater effect of downstream tributaries and the operation of adjacent hydropower stations, but the explicit quantification method of the backwater effect is lacking. In this study, a deep-learning-model-based forecasting method for TWL predictions under the backwater effect is developed and applied in the Xiangjiaba (XJB) hydropower station, which is influenced by the backwater effect of downstream tributaries, including the Hengjiang River (HJR) and the Minjiang River (MJR). Firstly, the random forest algorithm was used to analyze the influence of HJR and MJR flows with different lag times on the TWL prediction error of the XJB hydropower station. The results show that the time lags of the backwater effect of HJR and MJR run offs on the TWL of the XJB are 5~7 h and 1~2 h, respectively. Then, the run off thresholds of the HJR and MJR for impacting the TWL of the XJB station are obtained through scenario comparison, and the results show that the run off thresholds of the HJR and the MJR are 700 m³/s and 7000 m³/s, respectively. Finally, based on the analysis of the time lag and the threshold of the backwater effect, a deep learning model (LSTM)-based TWL forecasting method is established and applied to predict the TWL of the XJB station. The results show that the forecasting model has a good predictive performance, with 98.22% of absolute errors less than 20 cm. The mean absolute error over the validation dataset is 5.27 cm, and the maximum absolute error is 63.35 cm. Compared with the LSTM-based prediction model without considering the backwater effect, the mean absolute error decreased by 31%, and the maximum absolute error decreased by 71%. © 2023 by the authors.

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Predictive factors of success following radio-frequency stylet (RFS) ablation of incompetent perforating veins (IPV)

Item Type Journal Article
Author A.P. Hingorani
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Author A. Shiferson
Author N. Patel

Author K. Gopal

Author T. Jacob

Abstract Objective: To evaluate potential predictive factors associated with success or failure of incompetent perforating veins (IPVs) treated with radio-frequency stylet (RFS). Methods: Over the last 12 months in this observational study, 38 consecutive patients with various degrees of venous insufficiency and IPVs underwent 48 office-based radio-frequency ablation procedures (1 - C 3; 7 - C 4; 10 - C 5; 30 - C 6) in 44 limbs. There were 21 females and 17 males with a mean age of 67 ± 17 years (38-93 years) who had a total of 93 IPVs (40 calf; 53 ankle). Eighteen patients (47%) had ipsilateral great saphenous vein (GSV) radio-frequency closures performed prior to current procedure. The venous flow pattern was classified by spectral waveform analysis as "normal" (spontaneous with respiratory phasicity) in 33 patients and "pulsatile" (with bidirectional cardiac phasicity) in five patients. Follow-up duplex scans were performed from 3 to 7 days postprocedure. Statistical analyses were performed for determining correlation between the various factors such as, age, pulsatile flow, CEAP class, prior GSV ablation, vein diameter, reflux, and patency. Results: The mean number of ablated IPVs was 1.94 ± 0.38 ranging from 1-3. Immediate success rate was 88% (82 cases, 32 patients). IPVs had a duplex measured mean diameter of 3.8 ± 1.1 mm (2-6.6 mm). Eleven IPVs remained patent in six patients. There was no significant difference between the patent and the obliterated IPV groups concerning age ($P = 0.75$), prior GSV ablation ($P = .19$), IPV diameter ($P = .08$) and CEAP classification. Conversely, four of the five procedures (80%) performed in patients with "pulsatile" venous flow failed, while only two of the remaining 43 procedures (4.7%) in patients with "normal" venous flow failed ($P < .001$). Conclusion: These data show that a pulsatile venous flow pattern is a significant predictor of failure following RFS for IPVs. © 2009 Society for Vascular Surgery.

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Export Date: 20 September 2026; Cited By: 41; Correspondence Address: A.P. Hingorani; Division of Vascular Surgery, Maimonides Medical Center, Brooklyn, NY, United States; email: ahingorani@maimonidesmed.org; CODEN: JVSUE

Predictive value of late gadolinium enhancement cardiovascular magnetic resonance in patients with persistent atrial fibrillation: Dual-centre validation of a standardized method

- Item Type Journal Article
- Author T.F. Althoff
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- Author D. Padilla-Cueto
- Author H. Lehrmann
- Author P. Garre
- Author S. Schoechlin
- Author E. Ferro
- Author E. Invers
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- Author A. Porta-Sanchez
- Author J.M. Tolosana
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- Author I. Roca-Luque
- Author E. Arbelo

Author F.-J. Neumann
Author D. Westermann
Author M. Sitges
Author J. Brugada
Author T. Arentz
Author L. Mont
Author A. Jadidi

Abstract Aims: With recurrence rates up to 50% after pulmonary vein isolation (PVI) in persistent atrial fibrillation (AF), predictive tools to improve patient selection are needed. Patient selection based on left atrial late gadolinium enhancement (LGE) cardiovascular magnetic resonance (CMR) has been proposed previously (UTAH-classification). However, this approach has not been widely established, in part owed to the lack of standardization of the LGE quantification method. We have recently established a standardized LGE-CMR method enabling reproducible LGE-quantification. Here, the ability of this method to predict outcome after PVI was evaluated. Methods and results: This dual-centre study (n = 219) consists of a prospective derivation cohort (n = 37, all persistent AF) and an external validation cohort (n = 182; 66 persistent, 116 paroxysmal AF). All patients received an LGE-CMR prior to first-Time PVI-only ablation. LGE was quantified based on the signal-intensity-ratio relative to the blood pool, applying a uniform LGE-defining threshold of >1.2. In patients with persistent AF in the derivation cohort, left atrial LGE-extent above a cut-off value of 12% was found to best predict relevant low-voltage substrate (≥ 2 cm two with <0.5 mV during sinus rhythm) and arrhythmia-free survival 12 months post-PVI. When applied to the external validation cohort, this cut-off value was also predictive of arrhythmia-free survival for both, the total cohort and the subgroup with persistent AF (LGE < 12%: 80% and 76%; LGE > 12%: 55% and 44%; P = 0.007 and P = 0.029, respectively). Conclusion: This dual-centre study established and validated a standardized, reproducible LGE-CMR method discriminating PVI responders from non-responders, which may improve choice of therapeutic approach or ablation strategy for patients with persistent AF. © 2022 The Author(s). Published by Oxford University Press on behalf of the European Society of Cardiology.

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Pregnancy outcome in women with congenital heart disease: A single-center experience

Item Type Journal Article

Author D. Ntiloudi

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Author M.A. Bazmpani

Author D. Parcharidou

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Author S. Hadjimiltiades

Author H. Karvounis

Author G. Giannakoulas

Abstract Objective: Pregnancies in patients with adult congenital heart disease (ACHD) are often complicated. We aimed to highlight the nature and the rate of these complications in a single-center patient population. Methods: We retrospectively studied all the pregnancies of women who presented on an outpatient basis, and all pregnancies were reviewed for maternal and fetal outcomes. Results: Of 117 pregnancies from 52 ACHD patients (age at pregnancy 28.3 ± 6.6 years), 10 were therapeutically aborted. A proportion of 41.1% of the remaining 107 pregnancies were complicated either with cardiac (3.7%), obstetric (15.0%), or fetal/neonatal (11.2%) adverse events or with spontaneous abortion (17.8%). Hospitalization during pregnancy was required in 10 patients. Elective cesarean sections were initially planned for 31% of the 87 pregnancies, which were finally completed, while 8% required an emergency cesarean section, mostly for obstetric reasons. NYHA class deterioration after pregnancy was detected in 9.3% of our cohort. Modified WHO class III/IV, prior medication use, and higher body mass index (BMI) were related to cardiac complications during pregnancy or NYHA deterioration. Conclusion: Pregnancies in ACHD patients are at high risk for complications. Advanced modified WHO class, prior medication use, and high BMI were related to adverse cardiac events. © 2017 Hellenic Society of Cardiology

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adult, Adult, Article, female, Female, human, Humans, risk assessment, Risk Assessment, procedures, major clinical study, classification, complication, pregnancy, Abortion, Induced, gestational age, induced abortion, Pregnancy, newborn, Retrospective Studies, retrospective study, body mass, heart arrhythmia, thromboembolism, dyspnea, Infant, Newborn, high risk pregnancy, heart failure, length of stay, congenital heart disease, congenital heart malformation, Heart Defects, Congenital, Body Mass Index, statistics and numerical data, Abortion, Spontaneous, Adult congenital heart disease, bleeding, Cardiac complications, cesarean section, Cesarean Section, elective surgery, Elective Surgical Procedures, Greece, heart atrium septum defect, heart death, Infant, Newborn, Diseases, Neonatal complications, newborn disease, Obstetric complications, placenta previa, preeclampsia, pregnancy complication, Pregnancy Complications, Cardiovascular, pregnancy diabetes mellitus, pregnancy outcome, Pregnancy, High-Risk, pulmonary hypertension, solutio placentae, spontaneous abortion

Notes:

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Export Date: 20 September 2026; Cited By: 26; Correspondence Address: G. Giannakoulas; Department of Cardiology, AHEPA University Hospital, Thessaloniki, St. Kiriakidi 1, 54636, Greece; email: giannak@med.auth.gr; CODEN: HLKEA

Premonitory symptoms in migraine: a systematic review and meta-analysis of observational studies reporting prevalence or relative frequency

Item Type Journal Article
Author A.K. Eigenbrodt
Author R.H. Christensen
Author H. Ashina
Author A. Iljazi
Author C.E. Christensen
Author T.J. Steiner
Author R.B. Lipton

Author M. Ashina

Abstract Background: Observational studies on the prevalence of premonitory symptoms in people with migraine, preceding the headache pain (or aura) phase, have shown conflicting results. We conducted a systematic review and meta-analysis to estimate the prevalence, and relative frequency among clinic populations, of premonitory symptoms in people with migraine, overall and of the multifarious individual symptoms, and to review the methodologies used to assess them. Methods: We searched PubMed and Embase for studies published from database inception until 31st of May 2022. Two investigators independently screened titles, abstracts, and full texts. We retrieved observational studies that reported the prevalence/relative frequency of one or more premonitory symptoms in people with migraine. Two investigators independently extracted data and assessed risk of bias. Results were pooled using random-effects meta-analysis. Our main outcomes were the percentage of people with migraine who experienced at least one premonitory symptom and the percentages who experienced different individual premonitory symptoms. To describe our outcomes, we used the terms prevalence for data from population-based samples and relative frequency for data from clinic-based samples. We also descriptively and critically assessed the methodologies used to assess these symptoms. Results: The pooled estimated prevalence in population-based studies of at least one premonitory symptom was 29% (95% CI: 8–63; I² 99%) and the corresponding pooled estimated relative frequency in clinic-based studies was 66% (95% CI: 45–82; I² 99%). The data from clinic-based studies only supported meta-analysis of 11 of 96 individual symptoms, with relative frequency estimates ranging from 11 to 49%. Risk of bias was determined as high in 20 studies, moderate in seven, and low in two. Conclusions: The substantial between-study heterogeneity demands cautious interpretation of our estimates. Studies showed wide methodological variations, and many lacked rigor. Overall, the evidence was insufficient to support reliable prevalence estimation or characterization of premonitory symptoms. More data are needed, of better quality, to confirm the existence of a distinctive premonitory phase of migraine, and its features. Methodological guidelines based on expert consensus are a prerequisite. © 2022, The Author(s).

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Article, human, Humans, systematic review, prevalence, depression, headache, nausea, pain, Pain, symptom, fatigue, Epidemiology, international classification of headache disorder, migraine, Migraine Disorders, Prevalence, meta analysis, Clinical characteristics, food craving, Headache, International classification of headache disorders, irritability, mental concentration, mood change, phonophobia, photophobia, Prodrome, stiff neck, yawning

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Prevalence Estimation and Genetic Characterization of Porcine Parainfluenza Virus 1 (PPIV-1) in Hungary and the First Report of the Virus in Slovakia

Item Type Journal Article

Author B. Igriczi

Author L. Dénes

Author A. Czétényi

Author T. Révész

Author Z. Somogyi

Author G. Balka

Abstract In the last few decades, many new paramyxoviruses have been discovered, causing diverse, mostly respiratory diseases in animals and humans. The porcine parainfluenza virus 1 (PPIV-1, species Porcine respirovirus 1), which has been reported in many countries worldwide, was found in both healthy and clinically ill pigs showing respiratory signs. Here, we report the expected prevalence and genetic diversity of PPIV-1 in Hungarian pig herds and the detection in one Slovakian pig farm, which is the first report of evidence for the presence of the virus in the country. To estimate the prevalence in Hungary 211 oral fluid samples were collected from 23 large-scale swine herds in a systematic way and tested by real-time quantitative RT-PCR. The presence of the virus was detected in 10 of the 23 Hungarian farms (43%) included in our study. One hundred eighty-one nasal swab samples were collected cross-sectionally from three Hungarian and one Slovakian PPIV-1-positive herd and PPIV-1 was most prevalent in 6-week-old pigs on farms located in Hungary and in the 2-week-old pigs on the Slovakian farm. Phylogenetic analysis of three Hungarian and two Slovakian PPIV-1 F-gene sequences showed high-nucleotide identity (>93%) and all belonged to Clade I, together with the other European strains. © 2024 Barbara Igriczi et al.

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Article, nonhuman, cross-sectional study, epidemiology, prevalence, classification, veterinary medicine, genetics, isolation and purification, animal, Animals, Prevalence, gene sequence, genetic variability, virology, phylogeny, Phylogeny, sequence analysis, animal experiment, animal tissue, Hungary, genetic variation, pig, Slovakia, Swine, swine disease, Swine Diseases, capillary electrophoresis, Genetic Variation, Human parainfluenza virus 1, nose smear, Paramyxoviridae, phylogenetic tree, real time reverse transcription polymerase chain reaction, Respirovirus, Respirovirus infection, Respirovirus Infections, RNA extraction, Sanger sequencing, swine influenza virus, viral genetics, virus detection, weaning

Notes:

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Export Date: 20 September 2026; Cited By: 6; Funding details: Recovery and Resilience Facility; Correspondence Address: G. Balka; Department of Pathology, University of Veterinary Medicine, Budapest, István Str 2., 1078, Hungary; email: balka.gyula@univet.hu

Prevalence estimation and optimal classification methods to account for time dependence in antibody levels

Item Type Journal Article

Author P. Bedekar

Author A.J. Kearsley

Author P.N. Patrone

Abstract Serology testing can identify past infection by quantifying the immune response of an infected individual providing important public health guidance. Individual immune responses are time-dependent, which is reflected in antibody measurements. Moreover, the probability of obtaining a particular measurement from a random sample changes due to changing prevalence (i.e., seroprevalence, or fraction of individuals exhibiting an immune response) of the disease in the population. Taking into account these personal and population-level effects, we develop a mathematical

model that suggests a natural adaptive scheme for estimating prevalence as a function of time. We then combine the estimated prevalence with optimal decision theory to develop a time-dependent probabilistic classification scheme that minimizes the error associated with classifying a value as positive (history of infection) or negative (no such history) on a given day since the start of the pandemic. We validate this analysis by using a combination of real-world and synthetic SARS-CoV-2 data and discuss the type of longitudinal studies needed to execute this scheme in real-world settings. © 2022

Date 2023

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optimization, Article, human, Humans, validity, calculation, experimental study, Optimization, COVID-19, coronavirus disease 2019, epidemiology, prevalence, SARS-CoV-2, classification, mathematical model, probability, mathematical analysis, estimation method, Prevalence, measurement, Prevalence estimation, infection, Antibodies, Viral, antibody, antibody detection, Antibody testing, COVID-19 Testing, decision theory, medical history, probabilistic neural network, Seroepidemiologic Studies, seroepidemiology, seroprevalence, Time-dependent classification, virus antibody

Notes:

EID: 2-s2.0-85144352589

Export Date: 20 September 2026; Cited By: 9; Funding details: National Institute of Standards and Technology, NIST, (70NANB18H162); Correspondence Address: P. Bedekar; Department of Applied Mathematics and Statistics, Johns Hopkins University, Baltimore, 3400 N. Charles Street, 21218, United States; email: pbedekal@jh.edu; CODEN: JTBIA

Prevalence Estimation and Validation of New Instruments in Psychiatric Research: An Application of Latent Class Analysis and Sensitivity Analysis

Item Type Journal Article

Author B.W. Pence

Author W.C. Miller

Author B.N. Gaynes

Abstract Prevalence and validation studies rely on imperfect reference standard (RS) diagnostic instruments that can bias prevalence and test characteristic estimates. The authors illustrate 2 methods to account for RS misclassification. Latent class analysis (LCA) combines information from multiple imperfect measures of an unmeasurable latent condition to estimate sensitivity (Se) and specificity (Sp) of each measure. With simple algebraic sensitivity analysis (SA), one uses researcher-specified RS misclassification rates to correct prevalence and test characteristic estimates and can succinctly summarize a range of scenarios with Monte Carlo simulation. The authors applied LCA to a validation study of a new substance use disorder (SUD) screener and a larger prevalence study. With a traditional validation study analysis in which an error-free RS (Structured Clinical Interview for DSM-IV Axis I Disorders [SCID]; M. H. First, R. L. Spitzer, M. Gibbon, & J. Williams, 1990) is assumed, the authors estimated the screener had 86% Se and 75% Sp. Validation study estimates from LCA were 91% Se, 81% Sp (screener), 73% Se, and 98% Sp (SCID). SA in the prevalence study suggested the prevalence of SUD was underestimated by 22% because SCID was assumed to be error-free. LCA and SA can assist investigators in relaxing the unrealistic assumption of perfect RSs in reporting prevalence and validation study results. © 2009 American Psychological Association.

Date 2009

DOI 10.1037/a0015686

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Notes:

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Export Date: 20 September 2026; Cited By: 7; Funding details: National Institute of Mental Health, NIMH, (K23MH001951); National Institute of Mental Health, NIMH; Correspondence Address: B.W. Pence; Department of Community and Family Medicine, Duke Global Health Institute, Health Inequalities Program, United States; email: bpence@aya.yale.edu; CODEN: PYASE

Prevalence estimation of anorectal malformations using German diagnosis related groups system

Item Type Journal Article

Author E. Jenetzky

Abstract Without major registries, true prevalence of rare congenital diseases is not known. Estimations for occurrence of anorectal malformation (ARM) are based on monitoring centers and epidemiological studies. The new German economical system for payment of inpatient care (G-DRG) obligates the report of each hospitalization, including diagnoses and procedures. These codes and classifications originally developed for morbidity statistics are now misused for economical purposes. Is there an epidemiological use? We present a new method to estimate national wide prevalence rate of congenital malformations exemplarily for imperforated anus. Due to the new German DRG-system, treatment data collections of the years 2002-2005 are freely accessible. This method is applicable if a life saving surgery is mandatory for newborns and has to be ciphered by specific codes. Overall, in German hospitals 1,012 children below 1 year of age are surgically treated with a reconstructive anorectal surgery during the period of 4 years. In the same time 2,817,388 babies are born in Germany. Hence the national wide prevalence rate is about 3.6 (95% CI: 3.4-3.8) per 10,000 or 1:2,784 for all different types of anal atresia requiring surgery. Main ICD-10 diagnosis Q42 was given twice this rate, probably due to at least two hospitalizations in the newborn period. The economic data of the G-DRG system can be used to estimate yearly prevalence of some rare congenital diseases in Germany, in case of specific surgical procedures. It may be a useful supplement to smaller regional registries because of larger size. Further calculations for other epidemiological questions have to be faced. © 2007 Springer-Verlag.

Date 2007

DOI 10.1007/s00383-007-2023-6

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human, Humans, article, prevalence, Germany, Retrospective Studies, hospital, Prevalence, Infant, Newborn, Rectum, priority journal, surgical technique, Registries, Anal Canal, anorectal malformation, Anorectal malformations, anus atresia, Anus Diseases, Birth defects, Congenital malformations, Diagnosis Related Groups, Diagnosis-Related Groups, Incidence, plastic surgery, Rectal Diseases, Self-help

Notes:

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Export Date: 20 September 2026; Cited By: 45; Correspondence Address: E. Jenetzky; Institute for Medical Biometry and Informatics, University of Heidelberg, Heidelberg 69120, Im Neuenheimer Feld 305, Germany; email: jenetzky@imbi.uni-heidelberg.de; CODEN: PSUIE

Prevalence Estimation of Dementia/Alzheimer’s Disease Using Health and Retirement Study Database in the United States

Item Type Journal Article

Author A.A.T. Monfared

Author N. Hummel

Author A. Chandak

Author A. Khachatryan

Author R. Zhang

Author Q. Zhang

Abstract Background: Updated prevalence estimates along the continuum of Alzheimer’s disease (AD) can foster a more nuanced and effective approach to managing AD within the current healthcare landscape. Objectives: This study aims to estimate the prevalence and severity distribution of dementia/AD (including mild, moderate, and severe stages) and all-cause mild cognitive impairment (MCI) in the United States using data from the Health and Retirement Study (HRS). Design: Retrospective study. Setting: Data from the bi-annual HRS surveys involving in-depth interviews of a representative sample of Americans aged >50 years. Participants: Dementia/AD and all-cause MCI patients from the 4 most recent HRS surveys (2014, 2016, 2018 and 2020). Measurements: AD was identified based on diagnosis (self-report). Cognitive performance (modified Telephone Interview of Cognitive Status [TICS-m]) scores in the dementia/AD range were also captured; all-cause MCI was similarly identified using the TICS-m. Dementia/AD and MCI prevalence, as well as the distribution by dementia/AD stage (mild, moderate, or severe), were estimated. Sampling weights developed by HRS were applied to ensure the sample’s representativeness of the target population and unbiased estimates for population parameters. Results: Across the four HRS surveys, the total number of HRS respondents ranged from 15,000 to 21,000 (unweighted); 7,000 to 14,000 had TICS-m scores. The estimated prevalence of AD (all severity categories combined) in the 2014, 2016, 2018, and 2020 HRS surveys was 1.2%, 1.2%, 1.3% and 1.0%, respectively using the diagnosis-based approach; using the cognitive performance-based approach, 23–27% patients had

scores in the dementia/AD ranges across the 4 surveys. The estimated prevalence of all-cause MCI was consistently 23% in each survey. In the 2020 survey, the distribution of mild, moderate, and severe disease stages was 34%, 45%, and 21%, respectively, in patients self-reporting an AD diagnosis, and 55%, 40%, and 5%, respectively in all patients meeting TICS-m threshold for dementia/AD. Conclusion: The prevalence of AD diagnosis based on self-report was approximately 1% across the 4 most recent HRS surveys and may reflect the proportion of patients who have actively sought healthcare for AD. Among HRS survey respondents with cognitive scores available, over 20% were in the dementia/AD range. The distribution of disease by stage differed for self-reported AD diagnosis vs dementia/AD based on cognitive scores. Discordance in estimates of dementia/AD and stage distributions underscores a need for better understanding of clinical practice patterns in AD diagnosis, use of clinical assessment tools, and severity classification in the United States. Accurate patient identification is needed, especially early in the AD disease continuum, to allow for timely and appropriate initiation of new anti-amyloid treatments. © The Authors 2024.

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Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, United States, aged, Aged, Aged, 80 and over, very elderly, diagnosis, major clinical study, Databases, Factual, factual database, epidemiology, prevalence, Dementia, self report, health care, disease severity, health survey, cognitive defect, Retrospective Studies, retrospective study, Alzheimer disease, Alzheimer Disease, Alzheimer's disease, severity of illness index, Severity of Illness Index, Prevalence, mild cognitive impairment, Cognitive Dysfunction, dementia, all cause mortality, American, mental performance, severity distribution

Notes:

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Prevalence estimation of essential tremor in Hungary between 2010 and 2020 based on the National Health Insurance Fund Database

Item Type Journal Article

Author A.K. Papp

Author Á.J. Berki

Author P. Vinnai

Author A. Ajtay

Author D. Bereczki

Author L. Erőss

Author G. Tamás

Abstract The published prevalence of essential tremor is variable worldwide and lacking in Central Europe. We aimed to estimate its prevalence in Hungary and to explore the locally applied therapeutic approaches. We collected data from the National Health Insurance Fund database and pharmacy database registered between 2010 and 2020. Matching the specified codes of the International Classification of Diseases and the individually tailored combination of interventions, we attempted to exclude Parkinson's disease and other tremor evoking conditions. We estimated the period prevalence of essential tremor age standardized to the European Standard Population at 378–388/100,000. After excluding patients with possible Parkinsonian syndromes, we found that 36.4% of patients with tremor did not take any medication during the study period. Most of the rest used alprazolam followed by propranolol for the longest period of time; the alprazolam and propranolol combination was the most preferred. Deep brain stimulation and ablative surgery were chosen for less than 0.5% of the patients. Our strict methods probably underestimate the essential tremor prevalence in Hungary, which, however, does not differ considerably from international results. Given the limitations of medication therapy, expanding and improving neurosurgical interventions may help improve the quality of life of patients with essential tremor. © The Author(s) 2025.

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Prevalence Estimation of Protected Health Information in Swedish Clinical Text

Item Type Journal Article**Author** A. Henriksson**Author** M. Kvist**Author** H. Dalianis**Editor** Randell R.**Editor** Cornet R.**Editor** Scott P.J.**Editor** Cornet R.**Editor** Peek N.**Editor** McCowan C.

Abstract Obscuring protected health information (PHI) in the clinical text of health records facilitates the secondary use of healthcare data in a privacy-preserving manner. Although automatic de-identification of clinical text using machine learning holds much promise, little is known about the relative prevalence of PHI in different types of clinical text and whether there is a need for domain adaptation when learning predictive models from one particular domain and applying it to another. In this study, we address these questions by training a predictive model and using it to estimate the prevalence of PHI in clinical text written (1) in different clinical specialties, (2) in different types of notes (i.e., under different headings), and (3) by persons in different professional roles. It is demonstrated that the overall PHI density is 1.57%; however, substantial differences exist across domains. © 2017 European Federation for Medical Informatics (EFMI) and IOS Press.

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: A. Henriksson; Department of Computer and Systems Sciences, Stockholm University, Sweden; email: aron.henriksson@dsv.su.se

Prevalence of attention-deficit hyperactivity disorder (ADHD): Systematic review and meta-analysis

Item Type Journal Article

Author S. Popit

Author K. Serod

Author I. Locatelli

Author M. Stuhec

Abstract Background The estimates of attention-deficit hyperactivity disorder (ADHD) prevalence across various studies are significantly variable, contributing to uncertainty in ADHD prevalence estimation. Previous systematic reviews and meta-analyses have attributed this variability primarily to the methodological characteristics of the studies, including the diagnostic criteria, source of information, and impairment requirement for the diagnosis. Methods Review identified studies reporting ADHD prevalence in representative samples of children and adults in Europe and worldwide. Studies that were conducted in the general population were included. We focused on studies that report ADHD prevalence based on clinical diagnosis (clinical diagnostic criteria based on the Diagnostic and Statistical Manual

of Mental Disorders and International Classification of Diseases criteria, other diagnostic tools, such as various scales or interviews based on clinical diagnostic criteria). PubMed/Medline was searched to identify relevant articles published until 2024/2/01. The study was registered in PROSPERO (CRD42020200220) and followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 guidelines for systematic review and meta-analysis. Results In total, 117 studies were subjected to full evaluation. In the meta-analysis, 103 studies representing 159 independent datapoints were included. The overall prevalence of ADHD in register studies was 1.6%, 95% CI [0.9; 3.0], in survey studies 5.0%, 95% CI [2.9; 8.6], in one-stage clinical studies 4.2%, 95% CI [2.9; 6.0], and in two-stage clinical studies 4.8%, 95% CI [4.0; 5.8]. Conclusions Exact comparisons among studies with different diagnostic criteria and types of sampling can impact prevalence estimates. When comparing data from methodologically different studies, these factors need to be considered. © The Author(s), 2024. Published by Cambridge University Press on behalf of European Psychiatric Association.

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adult, female, human, male, adolescent, systematic review, Europe, Review, epidemiology, prevalence, Africa, child, DSM-5, North America, attention deficit hyperactivity disorder, underdiagnosis, meta-analysis, meta analysis, ICD-10, Asia, attention-deficit hyperactivity disorder (ADHD), Child Behavior Checklist, DSM-IV, ICD-9, mini international neuropsychiatric interview, South America, undertreatment

Notes:

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Export Date: 20 September 2026; Cited By: 74; Correspondence Address: M. Stuhec; Department of Pharmacology, Faculty of Medicine Maribor, University of Maribor, Maribor, Slovenia; email: matejstuhec@gmail.com; CODEN: EUPSE

Prevalence of cerebral palsy in Quebec: Alternative approaches

Item Type Journal Article**Author** M. Oskoui**Author** L. Joseph**Author** L. Dagenais**Author** M. Shevell

Abstract Aim: To provide an estimate of the period prevalence of cerebral palsy (CP) in the province of Quebec. Methods: Children with CP were identified from three consecutive birth cohorts (1999-2001) from the Quebec CP Registry, covering 6 of the 17 administrative health regions of the province. Two inferential approaches were applied for period prevalence estimation, frequentist and bayesian. Results: 228 children were identified with CP. Using a frequentist approach, the overall prevalence of CP was 1.84 per 1,000 children aged 9-11 years living in those areas in 2010 (95% CI 1.60-2.08). Using a bayesian approach taking into account the uncertainty about the registry's sensitivity in capturing all cases, the overall prevalence is higher at 2.30 per 1,000 children with a 95% CI (1.99-2.65). Conclusion: Using a bayesian approach to adjust for the registry's known high specificity and lower sensitivity, the prevalence estimate is in concordance with worldwide estimates and estimates using administrative databases in western Canadian provinces. Future studies are needed to validate the diagnosis of CP within administrative databases and to evaluate possible regional trends across Canada in both prevalence and health service utilization, which may highlight disparities in healthcare delivery. Copyright © 2013 S. Karger AG, Basel.

Date 2013**DOI** 10.1159/000345120**URL** <https://www.scopus.com/pages/publications/84873052116?origin=resultslist>**Archive** Scopus**Language** English**Volume** 40**Pages** 264-268**Publication** Neuroepidemiology**ISSN** 14230208 (ISSN)**Issue** 4**Journal Abbr** Neuroepidemiology**Date Added** 20/09/2026, 9:52:57 pm**Modified** 20/09/2026, 9:52:57 pm

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Article, female, Female, human, Humans, male, Male, article, major clinical study, prevalence, follow up, child, Child, health care delivery, cohort analysis, Canada, cognitive defect, health service, Bayes theorem, Bayes Theorem, sensitivity and specificity, Prevalence, sensory dysfunction, Infant, cerebral palsy, Cerebral Palsy, Child, Preschool, preschool child, epilepsy, Sensitivity and Specificity, Gross Motor Function Classification System, health care utilization, disease registry, Cerebral palsy, childhood disease, health care disparity, language disability, population size, Quebec, register, Registries, Registry

Notes:

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Export Date: 20 September 2026; Cited By: 57; Correspondence Address: M. Oskoui; Department of Pediatrics, Neurology and Neurosurgery, McGill University, FRCPC Montreal Children's Hospital, Montreal, QC H3H 1P3, 2300 Tupper Street, Canada; email: Maryam.oskoui@mcgill.ca; CODEN: NEEPD

Prevalence of financial fraud and scams among older adults in the United States: A systematic review and meta-analysis

Item Type Journal Article

Author D. Burnes

Author C.R. Henderson

Author C. Sheppard

Author R. Zhao

Author K. Pillemer

Author M.S. Lachs

Abstract Background. The financial exploitation of older adults was recently recognized by the Centers for Disease Control and Prevention as a serious public health problem. Knowledge of the prevalence of elder financial exploitation is mostly limited to the category of financial abuse, which occurs in relationships involving an expectation of trust. Little is known about the other major category of elder financial exploitation-elder financial fraud and scams, which is perpetrated by strangers. A valid estimate of elder financial fraud-scam prevalence is necessary as a foundation for research and prevention efforts. Objectives. To estimate the prevalence of elder financial fraud-scam victimization in the United States based on a systematic review and meta-analysis. Search Methods. Multiple investigators independently screened titles and abstracts and reviewed relevant full-text records from PubMed, Medline, PsycINFO, Criminal Justice Abstracts, Social Work Abstracts, and AgeLine databases. Selection Criteria. To maximize the validity and generalizability of prevalence estimation, we restricted eligibility to general populationbased studies (English speaking, 1990 onward) using state- or national-level probability sampling and collecting data directly from older adults. Data Collection and Analysis. Information on elder financial fraud-scam prevalence and study-level characteristics was extracted independently by 2 investigators. Meta-analysis of elder financial fraud-scam prevalence used generalized mixed models with individual studies as levels of a random classification factor. Main Results. We included 12 studies involving a total of 41 711 individuals in the meta-analysis. Overall pooled elder financial fraud-scam prevalence (up to 5-year period) across studies was 5.6% (95% confidence interval [CI] = 4.0%, 7.8%), with a 1-year period prevalence of 5.4% (95% CI = 3.2%, 7.6%). Studies using a series of questions describing specific fraud-scam events to measure victimization found a significantly higher prevalence (7.1%; 95% CI = 4.8%, 9.4%) than studies using a single, generalquestion self-report assessment approach (3.6%; 95% CI = 1.8%, 5.4%). Author's Conclusions. Elder financial fraud and scams is a common problem, affecting approximately 1 of every 18 cognitively intact, community-dwelling older adults each year; it requires further attention from researchers, clinicians, and policymakers. Elder financial fraud-scam prevalence findings in this study likely underestimate the true population prevalence. We provide methodological recommendations to limit older adult participation and reporting bias

in future population-based research. Public Health Implications. Elder financial exploitation victimization is associated with mortality, hospitalization, and poor physical and mental health. Health care professionals working with older adults likely routinely encounter patients who are fraud-scam victims. Validation of instruments to screen for elder financial fraud and scams in clinical settings is an important area of future research. Without effective primary prevention strategies, the absolute scope of this problem will escalate with the growing population of older adults.

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Prevalence of mixed mania using 3 definitions

Item Type Journal Article

Author E. Vieta

Author C. Morralla

Abstract Objectives: Mixed episodes are a combination of depressive and manic symptoms in bipolar disorder (BD). We want to identify the proportion of patients who have depressive symptoms during an acute episode and also the validity of current methods for its diagnosis. Material and method: Cross-sectional multicentre study of patients with type I BD who are admitted to specialized units. 368 patients in 76 centres were included. The patients should have a well established diagnosis of BD and need

hospitalisation. The severity of the disorder and clinical status were evaluated upon admission and discharge using CGI-BP-M clinical impression scales, the Hamilton depression scale (HAMD-17) and the Young mania rating scale (YMRS). Upon admission, the necessary criteria for diagnosing a mixed type episode were recorded according to DSM-IV-TR, ICD-10 and McElroy criteria. Clinical judgment of the current type of episode was also recorded. Results: Prevalence estimations for mixed episodes were: 12.9% according to DSM-IV-TR (n = 45), 9% according to ICD-10 (n = 31), 16.7% according to McElroy criteria (n = 58), and 23.2% according to clinical judgment (n = 81). Statistically significant differences were found between the estimated prevalence rates (Cochrane's Q-test, $p < 0.0001$), with the maximum concordance level found between the McElroy and ICD-10 (Kappa = 0.66, 95% CI, 0.54-0.77). The DSM-IV-TR criteria only present moderate concordance with ICD-10 (Kappa = 0.65, 95% CI, 0.52 to 0.78) and McElroy criteria (Kappa = 0.62, 95% CI, 0.50 to 0.74). Conclusions: The definition of mixed episodes for BD must be revised to improve consensus and, consequently, therapeutic management. Current diagnostic systems, based on DSM-IV and IDC-10, only capture a limited proportion of patients suffering from mixed episodes, giving rise to important limitations concerning the therapeutic management of BP patients. © 2010 Elsevier B.V. All rights reserved.

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Prevalence-Insensitive Evaluation of Diagnostic Systems Under Class Imbalance: The Harmonic Mean of Per-Class Sensitivity

Item Type Journal Article

Author J.S. Aguilar-Ruiz

Abstract In safety-critical diagnosis under class imbalance, widely used global metrics—e.g., accuracy, F-score, and MCC—conflate class-conditional behavior with class priors, often yielding overly optimistic assessments of overall performance, thereby hindering stable comparisons across datasets. We formalize the prevalence sensitivity of these indices and advance a prevalence-insensitive evaluation built solely from class-conditional rates (binary TPR/TNR; multiclass per-class sensitivities). We analyze the arithmetic, geometric, and harmonic means as class-symmetric aggregators and provide a geometric characterization of their isocurves. Empirically, on 18 UCI datasets, accuracy exceeded the harmonic mean by $\sim 3\%$ on average, but the gap widened on skewed sets (e.g., cardiocography: (Formula presented.) vs. (Formula presented.), (Formula presented.)). On a 6756-sample, 35-class tumor dataset, high overall agreement ((Formula presented.), Cohen's (Formula presented.)) coexisted with dispersed per-class sensitivities ((Formula presented.), (Formula presented.), (Formula presented.)) and moderate MCP ((Formula presented.)), evidencing weak minority-class performance despite strong accuracy. These results show that reliance on accuracy or (Formula presented.) can overstate performance precisely where it matters most, inflating perceived safety and obscuring rare-class failure modes. Prevalence-insensitive, class-symmetric means—especially the harmonic mean of per-class sensitivity—yield conservative, comparable summaries better aligned with risk-aware evaluation and deployment in critical systems. © 2025 by the author.

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Prevalence, morphologic features and proliferation indices of breast carcinoma molecular classes using immunohistochemical surrogate markers

Item Type Journal Article

Author R. Bhargava

Author J. Striebel

Author S. Beriwal

Author J.C. Flickinger

Author A. Onisko

Author G. Ahrendt

Author D.J. Dabbs

Abstract There is dearth of studies that provide a practical working formulation of breast cancer gene expression analysis for the surgical pathologist. ER, PR, HER2 were used, as surrogate markers to classify 205 breast carcinomas into molecular classes. Ki-67 labeling index was calculated using an image analysis system. The data was analyzed for molecular class prevalence, and inter-relationships amongst morphologic parameters, Ki-67 index, and molecular classes. Of the 205 tumors, 113 (55%) were classified as luminal A (strong ER+, HER2 negative), 34 (17%) as luminal B (weak to moderate ER+, HER2 negative), 32 (15%) as triple negative (negative for ER/PR and HER2), 8 (4%) as ERBB2 (negative for ER/PR but HER2+), 10 (5%) as luminal A-HER2 hybrid (strong ER+ and HER2+), and 8 (4%) as luminal B-HER2 hybrid (weak to moderate ER+ and HER2+). The average Ki-67 index was lowest in luminal A (15.8%), intermediate for ERBB2 (27.8%) and highest for triple negative tumors (>50%). Multivariate logistic regression analyses found the following associations: ERBB2 tumors with apocrine differentiation ($p=0.0031$); Triple negative tumors with high Ki-67 index ($p<0.0001$) and CK5 positivity ($p<0.0001$); HER2 negative-low receptor positive tumors (luminal B) with increased lymph node involvement ($p=0.0141$). The immunohistologic criteria were validated on a different set of 359 cases treated with neoadjuvant chemotherapy, which showed a pathologic complete response predominantly in ERBB2 and triple negative tumors. Immunohistochemistry is a reliable surrogate tool to classify breast carcinoma according to the gene expression profile classification.

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PrimSeq: A deep learning-based pipeline to quantitate rehabilitation training

Item Type Journal Article

Author A. Parnandi

Author A. Kaku

Author A. Venkatesan

Author N. Pandit

Author A. Wirtanen

Author H. Rajamohan

Author K. Venkataramanan

Author D. Nilsen

Author C. Fernandez-Granda

Author H. Schambra

Abstract Stroke rehabilitation seeks to accelerate motor recovery by training functional activities, but may have minimal impact because of insufficient training doses. In animals, training hundreds of functional motions in the first weeks after stroke can substantially boost upper extremity recovery. The optimal quantity of functional motions to boost recovery in humans is currently unknown, however, because no practical tools exist to measure them during rehabilitation training. Here, we present PrimSeq, a pipeline to classify and count functional motions trained in stroke rehabilitation. Our approach integrates wearable sensors to capture upper-body motion, a deep learning model to predict motion sequences, and an algorithm to tally motions. The trained model accurately decomposes rehabilitation activities into elemental functional motions, outperforming competitive machine learning methods. PrimSeq furthermore quantifies these motions at a fraction of the time and labor costs

of human experts. We demonstrate the capabilities of PrimSeq in previously unseen stroke patients with a range of upper extremity motor impairment. We expect that our methodological advances will support the rigorous measurement required for quantitative dosing trials in stroke rehabilitation. © 2022 Parnandi et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Principled efficient triple-weighting for AUC-oriented imbalanced covariate shift

Item Type Journal Article

Author Y. Zhang

Author X. Zheng

Author T. Guo

Author D. Zhang

Author G. Wu

Author Y. Yin

Abstract Unlike traditional machine learning tasks under the independent and identically distributed (i.i.d.) assumption, Covariate Shift (CS) tackles the learning problem where training and test (instance) marginal distributions differ while conditional distributions are the same. Further, datasets (or distributions) are often class-imbalanced in practice, where AUC is a widely adopted imbalance-aware evaluation metric. However, few works have investigated the problem of AUC-targeted

imbalanced CS learning. In this work, we first provide a theoretical analysis of the ultimate learning goal, revealing key influencing factors. Inspired by the theoretical insights, we propose a new efficient triple-weighting method with univariate surrogate losses. Specifically, the proposed objective jointly incorporates three complementary weighting factors: the instance-level density ratio for correcting covariate shift, the class-wise average density ratios for aligning positive and negative sample contributions under the test distribution, and the imbalance-aware degree for alleviating class-prior bias. This design enables the model to optimize an AUC-oriented objective under the target distribution while avoiding the high computational cost of pairwise AUC losses. The proposed framework can be naturally instantiated with density-ratio-based covariate shift methods, such as static importance weighting and dynamic importance weighting. Theoretically, we provide favorable consistency and generalization guarantees. Finally, experimental results illustrate its effectiveness and robustness compared with related baselines. © 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Prior Shift Using the Ratio Estimator

Item Type Conference Paper

Author A. Vaz

Author R. Izbicki

Author R.B. Stern
Editor Louzada F.
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Editor Izbicki R.

Abstract Several machine learning applications use classifiers as a way of quantifying the prevalence of positive class labels in a target dataset, a task named quantification. For instance, a naive way of determining what proportion of people like a given product with no labeled reviews is to (i) train a classifier based on the Google Shopping reviews to predict whether a user likes a product given its review, and then (ii) apply this classifier to Facebook/Google+ posts about that product. It is well known that such a two-step approach, named Classify and Count, fails because of dataset shift, and thus, several improvements have been recently proposed under an assumption named prior shift. Unfortunately, these methods only explore the relationship between the covariates and the response via classifiers. Moreover, the literature lacks in the theoretical foundation to improve these techniques. We propose a new family of estimators named Ratio Estimator which is able to explore the relationship between the covariates and the response using any function $g: X \rightarrow \mathbb{R}$ and not only classifiers. We show that for some choices of g , our estimator matches standard estimators used in the literature. We also explore alternative ways of constructing functions g that lead to estimators with good performance, and compare them using real datasets. Finally, we provide a theoretical analysis of the method. © 2018, Springer International Publishing AG, part of Springer Nature.

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Prioritizing conservation areas based on contributions to rarity and beta diversity in Mediterranean forest ecosystems in Türkiye

Item Type Journal Article

Author K. Özkan

Author Ö. Şentürk

Author M. Arslan

Author M.G. Negiz

Abstract The depletion of forest biodiversity arising from insufficient conservation of resources due to wood extraction, overgrazing, fire, and land use presents a significant challenge in the Mediterranean region. It becomes crucial to identify priority conservation areas for safeguarding biodiversity. We used plant species data taken from 800 plots in the Kuyucak mountain district, a typical Mediterranean forest ecosystem in Türkiye and maps of seven potential environmental predictors: elevation, aspect, slope, head index, topographical position index, landform characteristics, and bedrock geology. To assign priority areas for the conservation, local contribution to beta diversity (LCBD) and relative contribution to total rarity (RIRR) were examined with community metrics and environmental predictors, respectively. Correlation results showed that LCBD was positively related to rare species richness but negatively related to common species richness. RIRR was significantly associated with all the community metrics. According to the results of Random Forest regression models, elevation was the most important variable of LCBD, followed by slope, heat index, and pebblestone. For RIRR, the most contributed variables were elevation, heat index, limestone, and slope, respectively. The LCBD and RIRR Random Forest regression models were extrapolated along the Kuyucak mountain district, resulting in the creation of potential distribution maps for both LCBD and RIRR. Subsequently, from these maps, two distinct conservation value maps have been developed for the Kuyucak mountain district based on four conservation priority classes (Priority Class 1, Priority Class 2, Priority Class 3, and Priority Class 4): one employing a conservation cost approach and the other applying an ecology-centered approach. The results suggest that areas with elevated values of both LCBD and RIRR values are primarily situated below 1100 m on steep valley slopes, characterized by the presence of karstic limestone and intricate terrain. Compared to other areas, prioritizing the protection of such areas can provide greater benefits in contributing to ecological uniqueness. © 2024, Editura Silvica. All rights reserved.

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Prioritizing unit testing effort using software metrics and machine learning classifiers

Item Type Conference Paper

Author F. Toure

Author M. Badri

Abstract Unit testing plays a crucial role in object-oriented software quality assurance. Unfortunately, software testing is often conducted under severe pressure due to limited resources and tight time constraints. Therefore, testing efforts have to be focused, particularly on critical classes. As a consequence, testers do not usually cover all software classes. Prioritizing unit testing effort is a crucial task. We previously investigated a unit testing prioritization approach based on software information histories. We analyzed different attributes of ten open-source Java software systems tested using the JUnit framework. We used machine learning classifiers (Multivariate Logistic Regression and Naïve Bayes) to obtain, for each system, a set of classes to be tested. The obtained sets of candidate classes have been compared to the sets of classes for which JUnit test cases have been actually developed by testers. The cross system validation (CSV) technique results showed, among others, that the sets of candidate classes suggested by machine learning classifiers properly reflect the testers' selection. In this paper, we extend

our previous work by investigating more classifiers and using leave one system out validation (LOSOV) technique. This LOSOV technique uses a combination of training datasets from different systems. The obtained results indicate that: (1) the new classifiers correctly suggest classes to be tested, and (2) tested classes are particularly well predicted in the case of large-size systems. © 2018 Universitat zu Koln. All rights reserved.

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PRIS kidult anti-SPAM solution at the TREC 2005 spam track: Improving the performance of naive bayes for spam detection

Item Type Conference Paper

Author Y. Zhen

Author X. Weiran

Author C. Bo

Author H. Jiani

Author G. Jun

Abstract Recently, the spam already constituted a serious problem for both e-mail users and Internet Service Providers (ISP). Solutions to the abuse of spam would be both technical and legal regulatory. This paper reports our solution for the TREC 2005 spam track, in which we consider the use of Naive Bayes spam filter for its desirable properties (simplicity, low time and memory requirements, etc.). Then the approaches to modify the Naive Bayes by simply introducing weight and classifier assemble based on dynamic threshold are proposed, which can help to improve the accuracy of a Naive Bayes spam classifier dramatically. Additionally, we discuss some steps that must be adopted naturally thought before, such as stop list, word stemming, feature selection, class prior probabilities. The theory analysis implies these steps are not necessarily the best way to extend the Bayesian classifier, and these were also verified empirically. Many of these techniques appear to be counterintuitive but can be explained by the statistical properties of e-mail itself. Experiment results of TREC 2005 spam track demonstrate the effectiveness of the proposed method.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Y. Zhen; PRIS Lab, School of Information Engineering, Beijing University of Posts and Telecommunications, Beijing, 100876, China; email: yangzhen@pris.edu.cn; Conference name: 14th Text REtrieval Conference, TREC 2005; Conference date: 15 November 2005 through 18 November 2005; Conference code: 95379; CODEN: NSPUE

Privacy aware crowd-counting using thermal cameras

Item Type Conference Paper

Author R. Tse

Author T. Wang

Author M. Im

Author G. Pau

Editor Jiang X.

Editor Fujita H.

Abstract Visual analytics has been in the limelight as a powerful tool to support large scale management of places, people, and activities. Harnessing the power of machine learning is possible to quickly identify critical issues across thousands of cameras. Several stakeholders have voiced concerns about privacy. Visual analytics techniques can be used in facial recognition thus enabling fine-grain user tracking. This paper addresses such privacy concerns for some specific scenarios. It explores the feasibility of visual analytics in using low-cost/low-resolution thermal cameras thus delivering context-awareness information yet protecting user's privacy. This paper proposes a model able to classify and count humans, in indoor hallway settings, using low-resolution thermal pictures. The model is designed to work with YOLOv3 and leverages the power of deep-learning. Results show that it is possible to classify and count humans with over 90% accuracy based on the images from a low-cost 80x60 pixel thermal camera. The results were evaluated against the ground truth checked by a human agent and recorded through a regular camera. The study exposed possibilities and limits offered by low-cost thermal cameras and identifies the potential application scenarios. The dataset including both real and thermal images used for the training and the testing will be made available to the scientific community. © 2020 SPIE.

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Extra Journal Abbreviation: Proc SPIE Int Soc Opt Eng

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Publisher SPIE

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Proceedings Title Proc SPIE Int Soc Opt Eng

Conference Name Proceedings of SPIE - The International Society for Optical Engineering

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Deep learning, Learning systems, Visualization, Costs, Application scenario, Face recognition, Privacy concerns, Statistical tests, Cameras, Facial recognition, Visual analytics, Critical issues, Lime, Infrared devices, Cnn, Context- awareness, Human detection, Privacy-friendly, Scale management, Scientific community, Thermal camera, Yolov3

Notes:

EID: 2-s2.0-85087935356

Export Date: 20 September 2026; Cited By: 7; Funding details: Macao Polytechnic Institute, (RP/ESAP-01/2019); Conference name: 12th International Conference on Digital Image Processing, ICDIP 2020; Conference date: 19 May 2020 through 22 May 2020; Conference code: 161141; CODEN: PSISD

Privacy risk assessment method incorporating sensitivity and correlation with empirical study

Item Type Journal Article

Author R. Geng

Author T. Zhang

Author S. Li

Author Y. Xu

Abstract Introduction. User-generated content (UGC) has emerged as a prominent vector for privacy breaches, especially due to the context-dependence of data sensitivity and vulnerabilities introduced by data correlations. These challenges highlight the growing limitations of traditional assessment method s. Method. This study proposes a privacy risk quantification method integrating both attribute sensitivity and inter-attribute association, with an experimental validation conducted on the ‘Friend Identification ’ section of the <https://muchong.com> . A BERT-BiLSTM-CRF deep learning model is utilized for the automatic identification of attributes from unstructured text. Using a predefined privacy data lexicon, attribute sensitivity is quantified, and pointwise mutual information (PMI) is introduced to measure attribute associations. Combined with a privacy subject identification factor, these elements collectively quantify privacy ris k values, followed by risk level classification. Results. Ablation experiments and manual validation have confirmed the feasibility of the proposed scheme, demonstrating its capability to identify, assess, and classify privacy risks in unstructured textual data with broad applicability. Conclusion. The study validates the proposed solution theoretically, technically, and empirically, overcoming the limitations of traditional isolated-field evaluation paradigms. The method can be extended to high-sensitivity domains such as healthcare and finance, providing a basis for dynamic, risk-informed classification policies. © 2026, University of Boras. All rights reserved.

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Export Date: 20 September 2026; Cited By: 0; Funding details: National Social Science Fund of China, NSSFC, (22BTQ074, 2025 - CXRC - 24); National Social Science Fund of China, NSSFC; Correspondence Address: S. Li; School of Information Management, ZhengZhou University, ZhengZhou, China; email: lisentao99@gs.zzu.edu.cn

Probabilistic Contrastive Test-Time Adaptation

Item Type	Conference Paper
Author	L. You
Author	J. Lu
Author	X. Huang
Abstract	Test-time adaptation (TTA) enhances generalization against out-of-distribution data during inference. Recent advances in TTA leverage some techniques such as contrastive learning and entropy minimization to improve the discriminability and robustness of models in target domains. However, existing methods often overlook simultaneous distribution shifts of sample and label, such as long-tail distributions, and contrastive learning approaches may require substantial storage for sample pairs. In this paper, we propose a novel Probabilistic Contrastive Test-time Adaptation (PCTA) method based on Expectation Maximization (EM), which is used to estimate the von Mises Fisher (vMF) distribution of test samples to capture both sample distribution and class proportions. The estimated distributions are used for probabilistic contrastive learning to adapt feature representations and optimize classification through class-weighted entropy minimization. Experimental results show that PCTA significantly enhances the performance across various distribution shifts and outperforms state-of-the-art methods in different scenarios involving both sample and label shifts. Code is available at https://github.com/youlj109/PCTA . ©2025 IEEE.
Date	2025
DOI	10.1109/ICASSP49660.2025.10890260
URL	https://www.scopus.com/pages/publications/105009587616?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: ICASSP IEEE Int Conf Acoust Speech Signal Process Proc
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	15206149 (ISSN)
Proceedings Title	ICASSP IEEE Int Conf Acoust Speech Signal Process Proc
Conference Name	ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings
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Tags:

Artificial intelligence, Digital storage, Learning systems, Contrastive Learning, Generalisation, Signal processing, Probability distributions, Expectation maximization algorithm, Label shift, Maximum principle, Test time, Probabilistics, Distribution shift, Maximum entropy methods, Test-time adaptation, Discriminability, Distribution Shifts, Entropy minimization, Label Shifts, Probabilistic contrastive learning, Probabilistic Contrastive Learning, Robustness of model, Test-Time Adaptation

Notes:

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Probabilistic deep learning regression framework for post-hazard hurricane wind damage assessment of buildings

Item Type Journal Article

Author A.A. Ewis

Author O. Nofal

Abstract Hurricanes significantly affect coastal communities, causing substantial disruption to their socio-economic systems. Therefore, post-hurricane rapid assessment is essential for effective disaster response and recovery. Traditional human-based assessments through field reconnaissance are time-consuming, leading to delays in critical decision-making. Therefore, machine learning and deep learning have emerged as powerful alternatives enabling automated, near-real-time damage evaluation. By leveraging data from past expert-classified and systematically organized events, new models can predict damage either immediately after or even during a hurricane event. This study introduces a probabilistic deep learning regression framework that predicts component-level building damage percentages using multi-view geo-coded post-disaster imagery. Unlike conventional classification models, the proposed approach estimates continuous damage ratios for eight key envelope components, enabling more detailed and loss-relevant characterization of structural damage. A pretrained convolutional backbone is integrated with a regression head to fuse features extracted from multiple building views, while a systematic hyperparameter sensitivity analysis is performed to optimize model performance. To enhance the reliability of the developed model, uncertainty quantification was incorporated using Monte Carlo dropout and bootstrapping techniques, producing confidence intervals for each prediction. Post-disaster images from Hurricane Harvey (2017), covering 436 buildings, were used as a case study to build this model. Results demonstrate that the distribution of training data strongly influences prediction accuracy across components. Both uncertainty quantification methods performed well in capturing

mean values, though they diverged in their confidence intervals. Bootstrapping offered superior uncertainty characterization, albeit with a higher computational cost compared to the Monte Carlo approach. © 2026 Elsevier Ltd.

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Notes:

EID: 2-s2.0-105039822097

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: A.A. Ewis; Florida International University, Miami, United States; email: amoha209@fiu.edu; CODEN: RESSE

Probabilistic facies discrimination from simultaneous seismic-inversion results in clastic reservoirs in southwestern Venezuela

Item Type Journal Article

Author J. Reverón

Author J. Roomer

Abstract Facies volumes were obtained in southwestern Venezuela reservoirs integrating rock-physics analysis and seismic-inversion results. Facies volumes showed a sandstone-shale spatial distribution that matches log data and the conceptual sedimentologic model of the reservoir. After seismic-gather conditioning, applying a simple but robust processing sequence, the signal-to-noise ratio was increased. Two angle stacks were generated to perform a simultaneous seismic inversion to obtain P-and S-wave impedance volumes. Parallel to the inversion process, rock-physics analysis was done

using rock-physics templates (RPT) for constant-cement models (CCM) to discriminate between sandstone and shale. P-and S-wave impedance bivariate probabilistic density functions (PDF) were computed from well-log data as an a priori probability. Finally, these PDFs were used to classify facies from P-and S-wave impedance volumes obtained from simultaneous inversion, using a Bayesian probabilistic approach. This classification produced two facies volumes, sandstone and shale, which honored the well information and rock-physics models. © 2014 © 2014 by The Society of Exploration Geophysicists.

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DOI 10.1190/tle33070784.1

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Journal Abbr Leading Edge

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Tags:

Probability density function, Classification (of information), Signal to noise ratio, geophysics, Well logging, A-priori probabilities, data inversion, seismic data, Seismic inversion, Seismology, Shale, Clastic reservoirs, facies, Inversion process, P-wave, Probabilistic approaches, Probabilistic density function, Rock physics analysis, Rock physics model, S-wave, sandstone, Sandstone, sedimentology, Shear waves, Venezuela

Notes:

EID: 2-s2.0-84904540721

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: J. Reverón; Repsol Exploration, United States; email: jorgeluis.reveron@repsol.com; CODEN: LEEDF

Probabilistic inversion of Rayleigh-wave dispersion data: An application to Mt. Etna, Italy

Item Type Journal Article

Author L. Cauchie

Author G. Saccorotti

Abstract We present a methodology for determining the elastic properties of the shallow crust from inversion of surface wave dispersion characteristics through a fully nonlinear procedure. Using volcanic tremor data recorded by a small-aperture seismic array on Mount Etna, we measured the surface waves dispersion curves with the multiple signal classification technique. The large number of measurements allows the determination of an a priori probability density function without the need of making any assumption about the uncertainties on the observations. Using this information, we successively conducted the inversion of phase velocities using a probabilistic approach. Using a wave-number integration method, we calculated the predicted dispersion function for thousands of 1-D models through a systematic grid search investigation of shear-wave velocities in individual layers. We joined this set of theoretical dispersion curves to the experimental probability density function (PDF), thus obtaining the desired structural model in terms of an a posteriori PDF of model parameters. This process allowed the representation of the objective function, showing the non-uniqueness of the solutions and providing a quantitative view of the uncertainties associated with the estimation of each parameter. We then compared the solution with the surface wave group velocities derived from diffuse noise Green's functions calculated at pairs of widely spaced (~5-10 km) stations. In their gross features, results from the two different approaches are comparable, and are in turn consistent with the models presented in several earlier studies. © 2012 Springer Science+Business Media B.V.

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Italy, probability, data inversion, seismic data, Catania [Sicily], crustal structure, Dispersion curves, elastic property, Etna, Etna volcano, Green function, Nonlinear inversion, probability density function, Rayleigh wave, Sicily, Surface waves, Volcanic tremor, volcano

Notes:

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Export Date: 20 September 2026; Cited By: 12; Funding details: Irish Research Council for Science, Engineering and Technology, IRCSET; Correspondence Address: L. Cauchie; School of Geological Sciences, University College of Dublin, Dublin 4, Belfield, Ireland; email: lena.cauchie@ucdconnect.ie; CODEN: DXUEF

Probing Ink Evaporation-Deposition Behavior for Scalable Catalyst-Layer Fabrication in PEM Electrolysis: Systematic Droplet Metrology Coupled with Explainable Machine Learning

Item Type Journal Article

Author Z. Huang

Author X. Fu

Author R. Qi

Abstract Nonuniform catalyst-ink deposition during droplet drying, such as coffee-ring formation, remains a key bottleneck for scalable fabrication of proton exchange membrane (PEM) membrane-electrode assemblies (MEAs), as it compromises catalyst-layer uniformity, pore structure, and mass transport in PEM electrolysis. This work integrates systematic single-droplet metrology with explainable machine learning to quantify the evaporation-deposition behavior of Ir-based catalyst inks, including IrO₂ and ATO-supported formulations with PFSA ionomers. Quantitative morphology descriptors were established by combining optical microscopy with 3d-profilometry-assisted interpretation to characterize thickness nonuniformity and radial deposition distribution, including normalized variance, thickness-partitioned area fractions, and an edge-to-center grayscale ratio (ECCR). A data set spanning ionomer equivalent weight (EW = 720–1100), solid content (SC = 1–4 wt %), ionomer-to-catalyst mass ratio (I/Ir = 0.1–0.4), isopropanol volume fraction (ϕ_{IPA}), and substrate temperature (T_{sub} = 50–90 °C) was used to train a multioutput gradient-boosting decision tree model. Together with SHAP analysis, the model captures trend-level relationships between formulation/process parameters and morphology metrics (training R^2 = 0.74–0.88) and identifies $\phi_{\text{IPA}} \times \text{EW}$ and $\phi_{\text{IPA}} \times T_{\text{sub}}$ as dominant interaction terms governing ECCR and deposition-pattern transitions, whereas coverage and thick-region fractions are more sensitive to I/Ir and $\phi_{\text{IPA}} \times \text{SC}$. Based on these descriptors, an empirical deposition uniformity index (DUI) was further defined to rank relative deposition uniformity and guide process-window prediction. At EW = 1100, the model predicts a robust core evaporation window of $\phi_{\text{IPA}} = 0.568 \pm 0.013$ and $T_{\text{sub}} = 60 \pm 4.4$ °C. Under $T_{\text{sub}} = 60$ °C, a favorable formulation region is further identified in the SC–I/Ir plane, corresponding to SC = 2.56–3.34 wt % and I/Ir = 0.26–0.34. Independent validation experiments confirmed that samples within the predicted window generally exhibited more uniform deposition and more stable electrochemical responses than out-of-window controls. In contrast, varying relative humidity from 30% to 80% RH and droplet volume from 0.04 to 0.4 μL caused only limited changes in the final deposition pattern. Overall, this work presents an integrated framework that combines droplet-scale morphology quantification, explainable learning, and process-window prediction, providing practical guidance for catalyst-ink formulation and scalable PEM catalyst-layer fabrication. © 2026 American Chemical Society

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Learning systems, Machine learning, Forecasting, article, Data mining, Forestry, temperature, prediction, controlled study, Decision trees, benchmarking, learning, Gradient boosting, proton, explainable machine learning, decision tree, J+ catalyst, 2 propanol, catalyst, Catalyst activity, catalyst layer fabrication, Catalyst layer fabrication, Catalysts layers, coffee, Coffee rings, coffee-ring formation, Coffee-ring formation, Deposition, Drop breakup, Drop formation, droplet drying, Droplet drying, Drying, electrode, Electrodes, electrolysis, evaporation, Gradient boosting decision tree, gradient boosting decision trees (GBDT), ink, Ink, Ionomers, Ir-based catalyst ink, light microscopy, membrane, Membrane electrode assemblies, Membrane–electrode assembly, membrane–electrode assembly (MEA), Membranes, Pore structure, process window, Process window, Profilometry, Proton exchange membrane fuel cells (PEMFC), Ring formation, spray coating, thickness, ultrasonic spray deposition, Ultrasonic spray deposition, ultrasound

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Procedure for combining field measurements and machine learning to quantify impact of different track parameters on ride quality of railway tracks

Item Type Journal Article

Author A. Roghani

Author R. Pall

Author E. Toma

Abstract Ride quality in terms of vibration is a fundamental factor affecting passengers' satisfaction. Every year, passenger carriers invest significantly in various aspects of their system, including track and infrastructure, to improve ride quality. The assessment of ride quality and understanding the extent of the impact of different parameters on its magnitude is essential for railway operators to make informed decisions regarding capital expenditures. This paper presents a methodology for using machine learning techniques to find the correlation between various parameters (including train speed, weather conditions, presence of track features, and composition of the track substructure) and ride quality (quantified using measurements from accelerometers mounted on a rail car). The statistical model was developed using field measurements collected over a 50 km section of VIA Rail's track in Canada. This paper describes the collected field data, the development of the statistical model, and discusses the importance of each parameter on the accuracy of the model. © IMechE 2021.

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Notes:

EID: 2-s2.0-85102685619

Export Date: 20 September 2026; Cited By: 1; Funding details: Infrastructure Canada; National Research Council Canada, NRC; Correspondence Address: A. Roghani; National Research Council Canada Automotive and Surface Transportation Research Centre, Ottawa, Canada; email: Alireza.roghani@nrc-cnrc.gc.ca; CODEN: PMFTE

Proceedings - 2014 International Conference on Virtual Reality and Visualization, ICVRV 2014

Item Type Journal Article

Editor Zhang G.

Editor Shen X.

Editor Luo X.

Editor Zhou Z.

Editor Zhang X.

Abstract The proceedings contain 78 papers. The topics discussed include: color calibration with isolated distortion pixels removal based on human visual characteristics; real-time light field rendering for multi-projector 3D Display; animation of refitted 3D garment models for reshaped bodies; nonlinear revision control for web-based 3D scene editor; multiplayer collaborative training system based on mobile ar innovative interaction technology; spiculation quantification method based on edge gradient orientation histogram; crowd simulation for evacuation behaviors based on multi-agent system and cellular automaton; figure-aware tracking under occlusion from monocular videos; single-view dressed human modeling via morphable template; an improved algorithm for QR code image binarization; structure-based object classification and recognition for 3d scenes in point clouds; and effects on performance of analytical tools for visually demanding tasks through direct and indirect touch interaction in an immersive visualization.

Date 2015

URL <https://www.scopus.com/pages/publications/84962135789?origin=resultslist>

Archive Scopus

Language English

Publisher Institute of Electrical and Electronics Engineers Inc.

Publication Proceedings - 2014 International Conference on Virtual Reality and Visualization, ICVRV 2014

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Process security assessment: Operational space classification and process security index

Item Type Journal Article

Author H.H. Lou

Author R. Muthusamy

Author Y. Huang

Abstract Many chemical processes can be operationally risky, environmentally harmful and potentially dangerous when abnormal or destructive situations occur. This is due to the fact that chemical processes are often operated under high pressures, at high temperatures, and with fast material flows and complex manufacturing mechanisms. In the extreme, catastrophes such as explosions, toxic release and loss of life will occur unexpectedly and rapidly, particularly when a terrorist who has a sufficient technical background in chemical operations attacks a plant. To assure process security, systematic and effective process security-bearing operational strategies must be developed. This paper introduces a process operational space classification method and a process operational security index. The method and the index can be an effective quantitative tool for characterizing and analysing process security, especially when the process experiences various disturbances set by saboteurs who may be technically very knowledgeable. The efficacy of the quantification method is demonstrated by its application to an exothermic batch reactor.

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Volume 81

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Issue 6

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Notes:

EID: 2-s2.0-0346149867

Export Date: 20 September 2026; Cited By: 27; Correspondence Address: H.H. Lou; Department of Chemical Engineering, Lamar University, Beaumont, TX 77710, United States; email: louhh@hal.lamar.edu; CODEN: PSEPE

ProFiMed: Prompt-Based Federated Medical Image Classification via Pruned Optimal Transport and Residual Cross-Correlation Self-Attention

Item Type Conference Paper

Author S. Dai

Author H. Yang

Author S. Yu

Editor Liu J.

Editor Huang J.

Editor Wang X.

Editor Zhang F.

Editor Zou X.

Editor Tian T.

Editor Hu X.

Editor Hu B.

Editor Xiong Y.

Abstract Federated learning (FL) enables collaborative modelling of medical images without sharing raw data, yet the severe feature- and label-shift across hospitals compromises the performance of a single global model. Recent promptbased FL frameworks such as FedOTP mitigate this issue by splitting learnable prompts into a shared and a client-specific part, but still (i) rely on entropy-regularised unbalanced optimal transport (UOT) that oversmooths tiny lesions, and (ii) inherit CLIP's location-invariant self-attention, which blurs pathological details. To overcome these limitations we introduce ProFiMed, a communication-efficient framework that combines pruned UOT with a Residual Correlative Self-Attention (RCSA) block. Pruned UOT clips the OT cost matrix and reformulates the transport as a KL projection, yielding noise-resilient vision-text alignment at $O(V)$ complexity. RCSA fuses cross-layer cross-correlation attention with correlative self-attention, re-injecting position sensitivity into the final Transformer block without updating any backbone weights. Under a 16-shot protocol on five public medical benchmarks (CT, MRI, histology and dermoscopy) our method surpasses the state-of-the-art FedOTP backbone by 5.67 pp on average and outperforms six strong plug-in variants, while keeping the same privacy and communication budget. Extensive ablations reveal that pUOT and RCSA are complementary, robust to threshold and hyper-parameter choices, and scalable to heterogeneous hospital distributions. © 2025 IEEE.

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Tags:

Language model, Performance, Learning systems, Computer vision, Federated learning, Collaborative modeling, Budget control, Medical image processing, Hospitals, Learning frameworks, vision-language models, Global models, Vision-language model, Computerized tomography, Cross-correlations, Medical image classification, medical imaging, optimal transport, Optimal transport, self-attention, Self-attention

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Yu; School of Artificial Intelligence, South China Normal University, Guangzhou, China; email: yusongsen@m.scnu.edu.cn; Conference name: 2025 IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2025; Conference date: 15 December 2025 through 18 December 2025; Conference code: 219682

Prognostic aspects of 18F-FDG PET kinetics in patients with metastatic colorectal carcinoma receiving FOLFOX chemotherapy

Item Type Journal Article

Author A. Dimitrakopoulou-Strauss

Author L.G. Strauss

Author C. Burger

Author A. Rühl

Author G. Irngartinger

Author W. Stremmel

Author J. Rudi

Abstract We evaluated quantitative measurement series (MS) with 18F-FDG and PET and compared different quantification methods for prediction of individual survival in patients with metastatic colorectal cancer receiving chemotherapy with 5-fluorouracil, folinic acid, and oxaliplatin (FOLFOX). Methods: The study comprised 25 patients. All patients were examined before the onset of FOLFOX therapy and after completion of the first and fourth cycles. SUV, fractal dimension (FD), a 2-compartment model with computation of k_1 , k_2 , k_3 , and k_4 , and vascular fraction

(VB) were used for data evaluation. Survival data served as a reference for the PET data. Discriminant analysis (DA), regression, and best-subset analysis were applied to the data. Results: Twenty of 25 patients died up to 801 d after the first PET study. A cutoff of 1 y (364 d) was used to classify the patients into 2 a priori groups, namely the short- and long-term survival groups. DA was used to predict the 2 categories using SUV and kinetic parameters of 18F-FDG metabolism as predictor variables. SUV provided a correct classification rate (CCR) ranging from 62% to 69%. SUV of the third MS resulted in a CCR of 69% as a single parameter. The best results were yielded by the use of kinetic parameters (k_1 , k_3 , VB, and FD) as predictor variables. CCR was 78% using kinetic 18F-FDG parameters of the first and third MS, in comparison with 69% for the corresponding SUVs. A multiple linear regression model was applied to the data to assess the relationship between individual survival and the PET data. The best-subset method revealed a correlation coefficient of 0.850 for the kinetic parameters of the first (k_3 , k_4 , VB, and FD) and third (k_1 , k_2 , k_4 , and VB) MS. Conclusion: The combination of kinetic parameters of the first and the third MS is acceptable for classification into a short or long survival class. Furthermore, even an individual prognosis of survival can be achieved using kinetic 18F-FDG parameters of the first and third MS.

Date 2004

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Notes:

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Export Date: 20 September 2026; Cited By: 89; Correspondence Address: A. Dimitrakopoulou-Strauss; Medical PET Group - Biological Imaging (E0601), German Cancer Research Center, D-69120 Heidelberg, Im Neuenheimer Feld 280, Germany; email: ads@ads-lgs.com; CODEN: JNMEA

Prognostic Power of Quantitative Assessment of Functional Mitral Regurgitation and Myocardial Scar Quantification by Cardiac Magnetic Resonance

Item Type Journal Article**Author** T.K.M. Wang**Author** D. Kocyigit**Author** H. Choi**Author** C.M. Anthony**Author** N. Chan**Author** J. Bullen**Author** Z.B. Popović**Author** S.R. Kapadia**Author** A. Krishnaswamy**Author** B.P. Griffin**Author** S.D. Flamm**Author** W.H.W. Tang**Author** D.H. Kwon

Abstract **BACKGROUND:** The severity classification of functional mitral regurgitation (FMR) remains controversial despite adverse prognosis and rapidly evolving interventions. Furthermore, it is unclear if quantitative assessment with cardiac magnetic resonance can provide incremental risk stratification for patients with ischemic cardiomyopathy (ICM) or non-ICM (NICM) in terms of FMR and late gadolinium enhancement (LGE). We evaluated the impact of quantitative cardiac magnetic resonance parameters on event-free survival separately for ICM and NICM, to assess prognostic FMR thresholds and interactions with LGE quantification. **METHODS:** Patients (n=1414) undergoing cardiac magnetic resonance for cardiomyopathy (ejection fraction<50%) assessment from April 1, 2001 to December 31, 2017 were evaluated. The primary end point was all-cause death, heart transplant, or left ventricular assist device implantation during follow-up. Multivariable Cox analyses were conducted to determine the impact of FMR, LGE, and their interactions with event-free survival. **RESULTS:** There were 510 primary end points, 395/782 (50.5%) in ICM and 114/632 (18.0%) in NICM. Mitral regurgitation-fraction per 5% increase was independently associated with the primary end point, hazards ratios (95% CIs) of 1.04 (1.01-1.07; P=0.034) in ICM and 1.09 (1.02-1.16; P=0.011) in NICM. Optimal mitral regurgitation-fraction threshold for moderate and severe FMR were $\geq 20\%$ and $\geq 35\%$, respectively, in both ICM and NICM, based on the prediction of the primary outcome. Similarly, optimal LGE thresholds were $\geq 5\%$ in ICM and $\geq 2\%$ in NICM. Mitral regurgitation-fraction $\times$ LGE emerged as a significant

interaction for the primary end point in ICM ($P=0.006$), but not in NICM ($P=0.971$).
CONCLUSIONS: Mitral regurgitation-fraction and LGE are key quantitative cardiac magnetic resonance biomarkers with differential associations with adverse outcomes in ICM and NICM. Optimal prognostic thresholds may provide important clinical risk prognostication and may further facilitate the ability to derive selection criteria to guide therapeutic decision-making. © 2023 Lippincott Williams and Wilkins. All rights reserved.

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Proposal and evaluation of methods using the quantification theory and machine learning for detecting C&C server used in a botnet

Item Type Conference Paper**Author** S. Okayasu**Author** R. Sasaki**Editor** Ahamed S.I.**Editor** Chang C.K.**Editor** Chu W.**Editor** Crnkovic I.**Editor** Hsiung P.-A.**Editor** Huang G.**Editor** Yang J.

Abstract In recent years, the damage caused by botnets has increased and become a big problem. To solve this problem, we proposed a method to detect unjust C&C servers by using Hayashi's quantification theory class II. This method is able to detect unjust C&C servers, even if they are not included in a blacklist. However, it was predicted that the detection rate for this method decreases with passing time. Therefore, we have been continuing the investigation of the detection rate and adjusting the optimal detection method in different time periods. This paper deals with the results of an investigation for 2014. In addition, we newly introduce a method using a support vector machine (SVM) for comparison with quantification theory class II. We found that the detection rates by using quantification theory class II and those by the SVM are both very good, with very little difference in accuracy between them. © 2015 IEEE.

Date 2015**DOI** 10.1109/COMPSAC.2015.165**URL** <https://www.scopus.com/pages/publications/84962128028?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc Int Comput Software Appl Conf**Volume** 3**Publisher** IEEE Computer Society**ISBN** 07303157 (ISSN); 978-146736563-5 (ISBN)**Pages** 24-29**Proceedings Title** Proc Int Comput Software Appl Conf**Conference Name** Proceedings - International Computer Software and Applications Conference**Date Added** 20/09/2026, 9:53:00 pm

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Support vectors machine, Machine-learning, SVM, Quantification methods, Bot (Internet), Botnet, Botnets, Detection rates, Evaluation of methods, C&C server, C&C Server, Class II, DNS, Hayashi quantification method, Hayashi's quantification methods

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Protein spot detection and quantification in 2-DE gel images using machine-learning methods

Item Type Journal Article

Author P. Tsakanikas

Author E.S. Manolakos

Abstract Two-dimensional gel electrophoresis (2-DE) is the most established protein separation method used in expression proteomics. Despite the existence of sophisticated software tools, 2-DE gel image analysis still remains a serious bottleneck. The low accuracies of commercial software packages and the extensive manual calibration that they often require for acceptable results show that we are far from achieving the goal of a fully automated and reliable, high-throughput gel processing system. We present a novel spot detection and quantification methodology which draws heavily from unsupervised machine-learning methods. Using the proposed hierarchical machine learning-based segmentation methodology reduces both the number of faint spots missed (improves sensitivity) and the number of extraneous spots introduced (improves precision). The detection and quantification performance has been thoroughly evaluated and is shown to compare favorably (higher F-measure) to a commercially available software package (PDQuest). The whole image analysis pipeline that we have developed is fully automated and can be used for high-throughput proteomics analysis since it does not require any manual intervention for recalibration every time a new 2-DE gel image is to be analyzed. Furthermore, it can be easily parallelized for high performance and also applied without any modification to prealigned group average gels. © 2011 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

Date 2011

DOI 10.1002/pmic.201000601

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Protein-protein interactions classification from text via local learning with class priors

Item Type Conference Paper

Author Y. He

Author C. Lin

Abstract Text classification is essential for narrowing down the number of documents relevant to a particular topic for further pursuit, especially when searching through large biomedical databases. Protein-protein interactions are an example of such a topic with databases being devoted specifically to them. This paper proposed a semi-supervised learning algorithm via local learning with class priors (LL-CP) for biomedical text classification where unlabeled data points are classified in a vector space based on their proximity to labeled nodes. The algorithm has been evaluated on a corpus of biomedical documents to identify abstracts containing information about protein-protein interactions with promising results. Experimental results show that LL-CP outperforms the traditional semisupervised learning algorithms such as SVM and it also performs better than local learning without incorporating class priors. © 2010 Springer-Verlag Berlin Heidelberg.

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: Y. He; School of Engineering, Computing and Mathematics, University of Exeter, Exeter EX4 4QF, North Park Road, United Kingdom; email: y.he@exeter.ac.uk; Conference name: 14th International conference on Applications of Natural Language to Information Systems, NLDB 2009; Conference date: 24 June 2009 through 26 June 2009; Conference code: 83268

Proteomic analysis of eutopic and ectopic endometriotic tissues based on isobaric peptide tags for relative and absolute quantification (iTRAQ) method

Item Type Journal Article

Author P. Marianowski

Author I. Szymusik

Author J. Malejczyk

Author M. Hibner

Author M. Wielgoś

Abstract **OBJECTIVE:** The present study aimed at performing proteomic analysis of matched eutopic endometrium and ovarian endometrioid cysts from women with endometriosis in order to discover any abnormal protein expression related to the disease. **DESIGN AND SETTING:** The study included 8 women with stage III/IV endometriosis according to revised American Fertility Society (rAFS) classification

and one woman with no signs of the disease as a reference. Proteomic analysis was performed using a novel isobaric tag-based methodology for relative and absolute peptide quantification (iTRAQ) coupled with multidimensional liquid chromatography and tandem mass spectrometry. RESULTS: The selection of 419 proteins was found in all endometriosis specimens. Using normal eutopic endometrium from woman without endometriosis as a reference, some proteins expressions were significantly increased in all endometriosis samples. They included collagen $\alpha 1$ (XIV), calmodulin, collagen α (VI), plexin, integrin $\alpha V\beta 3$, transgelin, desmin, and vimentin. The comparison of these proteins' expression in paired eutopic and ovarian endometriosis samples has revealed that only vimentin was significantly increased in ovarian endometrioma. CONCLUSIONS: It was confirmed that endometriosis is associated with different expression of proteins in endometriotic samples. Nevertheless, further studies seem to be necessary as they may reveal possible markers that would be useful in clinical diagnosis of the disease. © 2013 Neuroendocrinology Letters.

Date 2013

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Notes:

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Proteomics-driven innovations in plant-based foods: Current advances, emerging technologies, and future perspectives

Item Type Journal Article

Author F. Boukid

Abstract Despite rapid market growth, plant-based foods such as meat analogs, plant-based milk, yogurt alternatives, and fermented products still fall short of matching the sensory, structural, and nutritional qualities of animal-based counterparts, primarily due to simple ingredient substitution that fails to reproduce the molecular structure, interactions, and functional properties required for optimal texture, flavor, and nutritional performance. Proteomics, using advanced mass spectrometry (MS) and label-free quantification methods, provides an approach to analyze plant protein composition, structure, interactions, and modifications, enabling targeted functional improvements. This review describes how proteomic workflows inform formulation across three areas. First, protein compositional and structural characterization employs techniques such as liquid chromatography-tandem mass spectrometry (LC-MS/MS) and differential scanning calorimetry (DSC) coupled with MS to map protein composition and structural behavior, supporting decisions on protein sources, fractionation, and purification. Second, indirect proteomic methods coupled with other non-proteomic complementary tools are used to determine structure–function relationships induced by processing to examine processing-induced crosslinking, enzymatic modifications, and lipid–protein interactions that influence texture. Third, targeted MS methods, including selected reaction monitoring (SRM) and parallel reaction monitoring (PRM), are applied to profile off-flavor compounds and identify protein modification sites relevant to sensory and nutritional properties. By integrating proteomic data with processing strategies, this review outlines how proteomics can be used to examine key functional attributes related to texture, flavor, and nutritional quality in plant-based foods. Significance: This review highlights the pivotal role of proteomics in advancing next-generation plant-based foods. Proteomic analysis enables an in-depth understanding of plant protein structure, composition, interactions, and bioactivity, providing critical insights for the development of functionally enhanced and consumer-acceptable alternatives. By integrating proteomics with AI, machine learning, multi-omics approaches, and cutting-edge analytical tools such as spatial proteomics and mass spectrometry imaging, the review demonstrates how protein functionality, flavor, texture, nutrition, and allergenicity can be optimized. Furthermore, it emphasizes the potential of proteomics to accelerate innovation in personalized nutrition, support sustainable and circular food systems, improve food safety, and reduce waste by valorizing plant-based by-products. This work serves as a roadmap for researchers and industry stakeholders seeking to leverage proteomics to design novel, high-quality, and sustainable plant-based protein products. © 2026 Elsevier B.V.

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Prototype-guided cyclic federated learning with knowledge distillation

Item Type Journal Article

Author Q. Meng

Author Y. Zhao

Author X. Li

Author S. Chen

Abstract In this paper, we propose a novel federated learning (FL) framework, termed Prototype-Guided Cyclic Federated Learning with Knowledge Distillation (PCFedKD), to address the challenges of data heterogeneity in distributed machine learning. Unlike traditional FL models, PCFedKD incorporates global and local prototypes to enable personalized model training while ensuring efficient global knowledge sharing. The framework leverages a cyclic training approach, where clients sequentially engage in knowledge distillation (KD) to preserve and refine knowledge across training rounds. KD enables the efficient transfer of knowledge among clients, improves learning accuracy, and reduces communication overhead.

Additionally, the use of prototypes provides class-specific representations of data, which help align global knowledge with local client-specific data distributions. Global prototypes capture shared knowledge among clients, while local prototypes guide client-specific personalization, enabling improved model adaptation and performance. To summarize, PCFedKD establishes a lightweight and privacy-preserving FL paradigm that jointly enhances global consensus and local personalization through prototype-guided cyclic distillation. Extensive experiments demonstrate that our method significantly outperforms benchmark models, achieves higher accuracy, faster convergence, and improved scalability across diverse datasets, including those with feature and label shifts. The results highlight the practical applicability and robustness of the proposed framework in real-world heterogeneous FL scenarios. © 2026 Elsevier B.V.

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Provable Discriminative Hyperspherical Embedding for Out-of-Distribution Detection

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Proximal detecting invertebrate pests on crops using a deep residual convolutional neural network trained by virtual images

Item Type Journal Article

Author H. Liu

Author J.S. Chahl

Abstract Detecting invertebrate pests on crops at early stages is essential for pest management. Traditionally, traps were used to sample pests and then human experts undertook classification and counting to estimate the levels of infestation, which is subjective,

error-prone and labour intensive. Recently, semi-automatic pest detection is possible by using computer vision technologies to classify and count pest samples in laboratories or insect traps, however, the decision made by the laboratory-based or trap-based approaches are still too late for more optimised pest management decisions. Today, precision agriculture needs detection of pests on crops so that real-time actions can be taken or optimised decision can be made based on accurate information of time and location pest occurs. In this study, we used computer vision and machine learning technologies to detect invertebrates on crops in the field. We first evaluated the performances of the state-of-art convolutional neural networks (CNNs) and proposed a standard training pipeline. Facing the challenge of rapidly developing comprehensive training data, we used a novel method to generate a virtual database which was successfully used to train a deep residual CNN with an accuracy of 97.8% in detecting four species of pests in farming environments. The proposed method can be applied to a robotic system for proximal detection of invertebrate pests on crops in real-time. © 2021 Elsevier B.V.

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PSDC: A Prototype-Based Shared-Dummy Classifier Model for Open-Set Domain Adaptation

Item Type Journal Article

Author Z. Liu

Author G. Chen

Author Z. Li

Author Y. Kang

Author S. Qu

Author C. Jiang

Abstract Open-set domain adaptation (OSDA) aims to achieve knowledge transfer in the presence of both domain shift and label shift, which assumes that there exist additional unknown target classes not presented in the source domain. To solve the OSDA problem, most existing methods introduce an additional unknown class to the source classifier and represent the unknown target instances as a whole. However, it is unreasonable to treat all unknown target instances as a group since these unknown instances typically consist of distinct categories and distributions. It is challenging to identify all unknown instances with only one additional class. In addition, most existing methods directly introduce marginal distribution alignment to alleviate distribution shift between the source and target domains, failing to learn discriminative class boundaries in the target domain since they ignore categorical discriminative information in the adaptation. To address these problems, in this article, we propose a novel prototype-based shared-dummy classifier (PSDC) model for the OSDA. Specifically, our PSDC introduces an auxiliary dummy classifier to calibrate the source classifier and simultaneously develops a weighted adaptation procedure to align class-wise prototypes for adaptation. We further design a pseudo-unknown learning algorithm to reduce the open-set risk. Extensive experiments on Office-31, Office-Home, and VisDA datasets show that the proposed PSDC can outperform existing methods and achieve the new state-of-the-art performance. The code will be made public. © 2013 IEEE.

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Pseudo-label enhancement for weakly supervised object detection using self-supervised vision transformer

Item Type

Journal Article

Author

K. Yang

Author

Y. Wu

Author

J. Li

Author

C. Yin

Author

X. Li

Abstract

Weakly supervised object detection (WSOD) using image-level labels has gained attention in the computer vision community. Most advanced WSOD approaches generate instance-level labels based on class activation maps (CAMs), which are limited by incomplete box labels and sparse information regarding class labels. These inaccurate pseudo-labels mislead the training of the subsequent detection module, thus hampering detection performance. To address these issues, we propose a novel pseudo-label enhancement (PLE) framework for a one-stage WSOD method that consists of a CAM reassembly (CAMR) and a composite scoring module (CSM). The CAMR explores the collaboration between class priors from CAMs and the semantic-aware grouping ability of self-supervised vision transformers (ViTs). Specifically, CAMR initially propagates object localization cues from CAMs to feature maps obtained using a self-supervised ViT. Subsequently, CAMR utilizes the patch affinity of these feature maps to extract more integral object information and improve the instance localization accuracy and completeness. Subsequently, the reassembled CAMs are refined using a multi-threshold strategy to generate bounding-box pseudo-labels. The CSM replaces one-hot labels with soft-class labels that leverage localization precision and classification confidence to counter information sparseness limitations. The proposed PLE is evaluated using the PASCAL VOC and MS-COCO datasets. The experimental results demonstrate that the performance of PLE surpasses that of the advanced methods. On average, PLE improves by 7.1% mAP on the VOC 2007 test set and 6.0% mAP on the MS-COCO 2017 validation set compared with the one-stage WSOD method. © 2025

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Psychiatric Treatments with Short-Duration Psychedelics and AI-Driven Behavioral Monitoring

Item Type Journal Article

Author R.B. Kargbo

Abstract Integrating advanced pharmaceutical innovations and artificial intelligence (AI) offers transformative potential for psychiatric care. This Patent Highlight reviews novel therapeutic strategies, including the synergistic use of monoamine antidepressants and short-duration psychedelics, alongside AI-driven behavioral efficacy tracking. The combination of selective serotonin reuptake inhibitors (SSRIs) with short-acting psychedelics, such as N,N-dimethyltryptamine (DMT) and psilocybin, provides rapid and sustained improvements in treatment-resistant depression and anxiety. Meanwhile, AI-enhanced behavioral monitoring leverages motion tracking and machine learning to quantify treatment outcomes in animal models, accelerating drug development. Together, these approaches redefine therapeutic paradigms, offering personalized and effective treatments for psychiatric disorders. Published 2025 by American Chemical Society

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PU-KBS: A Robust Positive and Unlabeled Learning Framework with Key Band Selection for One-Class Hyperspectral Image Classification

Item Type Journal Article

Author Z. Liu

Author H. Zhao

Author X. Wang

Author S. Wang

Author J. Li

Author Y. Zhong

Abstract Positive and unlabeled (PU) learning is aimed at building a binary classifier to distinguish the target from the background using only the known positive samples, which is an advanced solution for the hyperspectral target detection (HTD) task. However, when PU learning (PUL) meets complex hyperspectral scenarios, there are two main challenges: 1) How to estimate the class prior accurately? The class prior, i.e., the target proportion, is an important prior for PUL to learn the discriminant boundary, but it is difficult to estimate in hyperspectral imagery, due to the interclass spectral similarity and 2) How to remove redundancy and improve the discriminative features of the target? The diagnostic spectral feature extraction is important for the weakly supervised PUL models as it can help with separating the target from the background. In this article, to tackle these challenges, a robust PUL framework with key band selection (PU-KBS) is proposed, which is modeled as an end-to-end and class prior free PUL framework, where the accurate class prior and the most discriminative key band subset are jointly initialized and iteratively updated until

reaching the optimal result by evolutionary search. Meanwhile, a deep PUL detector is introduced for guiding the subsequent search direction and discriminative deep feature extraction. The proposed PU-KBS framework was verified using different hyperspectral datasets, where accurate class prior estimation, diagnostic spectral characteristics, and robust detection results could be obtained simultaneously by the PU-KBS framework. Furthermore, the improvement in band selection interpretability and detection performance was proven experimentally. © 1980-2012 IEEE.

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PULNS: Positive-Unlabeled Learning with Effective Negative Sample Selector

Item Type Conference Paper

Author C. Luo

Author P. Zhao

Author C. Chen

Author B. Qiao

Author C. Du

Author H. Zhang

Author W. Wu

Author S. Cai

Author B. He

Author S. Rajmohan

Author Q. Lin

Abstract Positive-unlabeled learning (PU learning) is an important case of binary classification where the training data only contains positive and unlabeled samples. The current state-of-the-art approach for PU learning is the cost-sensitive approach, which casts PU learning as a cost-sensitive classification problem and relies on unbiased risk estimator for correcting the bias introduced by the unlabeled samples. However, this approach requires the knowledge of class prior and is subject to the potential label noise. In this paper, we propose a novel PU learning approach dubbed PULNS, equipped with an effective negative sample selector, which is optimized by reinforcement learning. Our PULNS approach employs an effective negative sample selector as the agent responsible for selecting negative samples from the unlabeled data. While the selected, likely negative samples can be used to improve the classifier, the performance of classifier is also used as the reward to improve the selector through the REINFORCE algorithm. By alternating the updates of the selector and the classifier, the performance of both is improved. Extensive experimental studies on 7 real-world application benchmarks demonstrate that PULNS consistently outperforms the current state-of-the-art methods in PU learning, and our experimental results also confirm the effectiveness of the negative sample selector underlying PULNS. Copyright © 2021, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Pure stress urinary incontinence: Analysis of prevalence, estimation of costs, and financial impact

Item Type Journal Article

Author E. Rubilotta

Author M. Balzarro

Author A. D'Amico

Author M.A. Cerruto

Author S. Bassi

Author C. Bovo

Author V. Iacovelli

Author D. Bianchi

Author W. Artibani

Author E. Finazzi Agrò

Abstract Background: The prevalence of pure stress urinary incontinence (P-SUI) and the role of urodynamic investigation (UDI) prior to surgery for stress urinary incontinence (SUI) is debated. Since the exact prevalence of P-SUI is not clear, its clinical and economic impact is not well defined. The aims of this study were to evaluate the prevalence of P-SUI in a population of women who underwent UDI for urinary incontinence (UI), also assessing: 1) the correspondence between clinical diagnosis of P-SUI and urodynamic findings; 2) the analysis of costs in terms of UDI and eventually post-UDI avoided surgical procedures. Methods: A single cohort of women who underwent UDI for UI between January 2012 and July 2016 was prospectively collected and retrospectively analyzed. Clinical P-SUI was defined by the strict criteria of the International Continence Society. For each patient, history, physical examination and UDI were collected. The correspondence between clinical and urodynamic findings of P-SUI was analyzed. The rate of clinical P-SUI changed after performing UDI and the number of unnecessary intervention after UDI were reported. A wide cost analysis of UDIs, and the amount of surgical procedures that were believed unnecessary after UDI was reported. Results: Stress urinary incontinence was present in 323/544 (59.4%) patients. The prevalence of clinical P-SUI was 20.7% (67/323), while the prevalence of complicated SUI (C-SUI) was 79.3% (256/323). After UDI, diagnosis of P-SUI decreased to 18.3% (59/232). In 10.2% of cases (6/59) the scheduled middle urethral sling (MUS) was suppressed after the UDI results because 3/6 cases had detrusor overactivity and urge incontinence, in 2/6 cases SUI was treated with a conservative management, in 1/6 case an important voiding dysfunction was detected. Considering the national reimbursement in our country, the cost of each UDI was 296.5 euros and the total amount was 17,493.5 euros. So far the surgery-related savings covered 61.7-105.0% of the costs of total number of UDIs performed in the uncomplicated patients.

Conclusions: The prevalence of clinical P-SUI is relevant, involving about 20% of women with clinical SUI. Although the correspondence between clinical and urodynamic diagnosis was high, we demonstrated that UDI may help in some cases to avoid an inappropriate surgical treatment. Therefore, UDI prior to SUI surgery should be considered to achieve a correct diagnosis and a proper therapeutic strategy. © 2019 The Author(s).

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PyAMARES, an Open-Source Python Library for Fitting Magnetic Resonance Spectroscopy Data

Item Type Journal Article

Author J. Xu

Author M. Vaeggemose

Author R.F. Schulte

Author B. Yang

Author C.-Y. Lee

Author C. Laustsen

Author V.A. Magnotta

Abstract Background/Objectives: Magnetic resonance spectroscopy (MRS) is a valuable tool for studying metabolic processes in vivo. While numerous quantification methods exist, the advanced method for accurate, robust, and efficient spectral fitting (AMARES) is among the most used. This study introduces pyAMARES, an open-source Python implementation of AMARES, addressing the need for a flexible, user-friendly, and versatile MRS quantification tool within the Python ecosystem. Methods: PyAMARES was developed as a Python library, implementing the AMARES algorithm with additional features such as multiprocessing capabilities and customizable objective functions. The software was validated against established AMARES implementations (OXSA and jMRUI) using both simulated and in vivo MRS data. Monte Carlo simulations were conducted to assess robustness and accuracy across various signal-to-noise ratios and parameter perturbations. Results: PyAMARES utilizes spreadsheet-based prior knowledge and fitting parameter settings, enhancing flexibility and ease of use. It demonstrated comparable performance to existing software in terms of accuracy, precision, and computational efficiency. In addition to conventional AMARES fitting, pyAMARES supports fitting without prior knowledge, frequency-selective AMARES, and metabolite residual removal from mobile macromolecule (MM) spectra. Utilizing multiple CPU cores significantly enhances the performance of pyAMARES. Conclusions: PyAMARES offers a robust, flexible, and user-friendly solution for MRS quantification within the Python ecosystem. Its open-source nature, comprehensive documentation, and integration with popular data science tools enhance reproducibility and collaboration in MRS research. PyAMARES bridges the gap between traditional MRS fitting methods and modern machine learning frameworks, potentially accelerating advancements in metabolic studies and clinical applications. © 2024 by the authors.

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QuAcc: Using Quantification to Predict Classifier Accuracy Under Prior Probability Shift

Item Type Journal Article

Author L. Volpi

Author A. Moreo

Author F. Sebastiani

Abstract Using cross-validation to predict the accuracy of a classifier on unseen data can be done reliably only in the absence of dataset shift, i.e., when the training data and the unseen data are IID. In this work we deal instead with the problem of predicting classifier accuracy on unseen data affected by prior probability shift (PPS), an important type of dataset shift. We propose QuAcc, a method built on top of “quantification” algorithms robust to PPS, i.e., algorithms devised for estimating the prevalence values of the classes in unseen data affected by PPS. QuAcc is based on the idea of viewing the cells of the contingency table (on which classifier accuracy is computed) as classes, and of estimating, via a quantification algorithm, their prevalence values on the unseen data labelled by the classifier. We perform systematic experiments in which we compare the prediction error incurred by QuAcc with that of state-of-the-art classifier accuracy prediction (CAP) methods. © The Author(s) 2025

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Quadruply stochastic gradients for large scale nonlinear semi-supervised AUC optimization

Item Type Conference Paper

Author W. Shi

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Author X. Li

Author X. Geng

Author H. Huang

Editor Kraus S.

Abstract Semi-supervised learning is pervasive in real-world applications, where only a few labeled data are available and large amounts of instances remain unlabeled. Since AUC is an important model evaluation metric in classification, directly optimizing AUC in semi-supervised learning scenario has drawn much attention in the machine learning community. Recently, it has been shown that one could find an unbiased solution for the semi-supervised AUC maximization problem without knowing the class prior distribution. However, this method is hardly scalable for nonlinear classification problems with kernels. To address this problem, in this paper, we propose a novel scalable quadruply stochastic gradient algorithm (QSG-S2AUC) for nonlinear semi-supervised AUC optimization. In each iteration of the stochastic optimization process, our method randomly samples a positive instance, a negative instance, an unlabeled instance and their random features to compute the gradient and then update the model by using this quadruply stochastic gradient to approach the optimal solution. More importantly, we prove that QSGS2AUC can converge to the optimal solution in $O(1/t)$, where t is the iteration number. Extensive experimental results on a variety of benchmark datasets show that QSG-S2AUC is far more efficient than the existing state-of-the-art algorithms for semi-supervised AUC maximization, while retaining the similar generalization performance. © 2019 International Joint Conferences on Artificial Intelligence. All rights reserved.

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Qualitative analysis of the hodgkin-huxley model of neuron excitability based on classification rules

Item Type Conference Paper**Author** V. Martsenyuk**Author** Y. Rudyak**Author** A. Sverstiuk**Author** Z. Mayhruk**Author** A. Horkunenko**Author** M. Kasianchuk**Editor** Fisun M.**Editor** Krainyk Y.

Abstract The proposed multivariate method of qualitative analysis of models of electrophysiological processes is an approach that allows to solve problems of classification of excitability of cells which cannot be solved by other traditional methods, for example stability theory, or boundary cycles. As a whole, the method combines the Monte Carlo approach for the formation of training kits and the data mining classification algorithms: the sequential coverage method with the generation of classification rules and the induction method of the decision tree. The advantages of the classification rules, which can be built on the 5th step of the algorithm, are that they correspond to the natural reflection of knowledge in the thinking of people and are more expressive. In addition, the sequential coverage algorithm is easier to implement and debug than the recursive algorithms of decision trees, and its computational complexity is simpler than that of finite state machines. The decision tree induction method, being more complex to implement, allows visualization and a priori probability values for the type of cell excitability based on the relationship between the initial values and the velocity parameters in the Hodgkin-Huxley model. The developed method consists of 5 stages. The approach is proven in the form of software in the Java-

class package. Copyright © 2019 for this paper by its authors.

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Qualitative and quantitative analysis of chlorinated solvents using Raman spectroscopy and machine learning

Item Type Conference Paper

Author J. Conroy

Author A.G. Ryder

Author M.N. Leger

Author K. Hennessey

Author M.G. Madden

Abstract The unambiguous identification and quantification of hazardous materials is of increasing importance in many sectors such as waste disposal, pharmaceutical manufacturing, and environmental protection. One particular problem in waste disposal and chemical manufacturing is the identification of solvents into chlorinated or non-chlorinated. In this work we have used Raman spectroscopy as

the basis for a discrimination and quantification method for chlorinated solvents. Raman spectra of an extensive collection of solvent mixtures (200+) were collected using a JY-Horiba LabRam, infinity with a 488 nm excitation source. The solvent mixtures comprised of several chlorinated solvents: dichloromethane, chloroform, and 1,1,1-trichloroethane, mixed with solvents such as toluene, cyclohexane and/or acetone. The spectra were then analysed using a variety of chemometric techniques (Principal Component Analysis and Principal Component Regression) and machine learning (Neural Networks and Genetic Programming). In each case models were developed to identify the presence of chlorinated solvents in mixtures at levels of ~5%, to identify the type of chlorinated solvent and then to accurately quantify the amount of chlorinated solvent.

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Quality evaluation of Qiongyu Runfei Paste based on HPLC fingerprinting combined with chemometrics

Item Type Journal Article

Author X. PAN

Author Q. LIANG

Author R. SA

Author M. MENG

Author X. XU

Author L. ZHAO

Abstract OBJECTIVE To establish a high-performance liquid chromatography (HPLC) fingerprint and a multi-component quantification method for Qiongyu Runfei Paste, and to construct a multidimensional quality evaluation approach by integrating chemical pattern recognition techniques, so as to provide a reference for quality control. METHODS An Agilent Eclipse Plus C18 column (4.6 mm × 250 mm, 5 μm) was used at 30 °C. The mobile phase consisted of acetonitrile (A) and 0.1% phosphoric acid aqueous solution (B) with gradient elution. The flow rate was 1.0 mL·min⁻¹, the detection wavelength was 290 nm, and the injection volume was 10 μL. The contents of six marker compounds, namely gallic acid, neochlorogenic acid, chlorogenic acid, cryptochlorogenic acid B, cryptochlorogenic acid A, and cryptochlorogenic acid C, were determined. Using the Similarity Evaluation System for Chromatographic Fingerprint of Traditional Chinese Medicine (2012), HPLC fingerprints of 10 batches of samples were established, common peaks were identified, and similarity analysis was performed. Based on the peak areas and relative peak areas of the common fingerprint peaks, hierarchical cluster analysis (HCA), principal component analysis (PCA), and partial least squares-discriminant analysis (PLS-DA) were applied to evaluate the quality of the 10 batches of Qiongyu Runfei Paste. RESULTS A total of 15 common peaks were identified in the HPLC fingerprint of Qiongyu Runfei Paste, and six components were identified as gallic acid, neochlorogenic acid, chlorogenic acid, cryptochlorogenic acid B, cryptochlorogenic acid A, and cryptochlorogenic acid C. The similarity values of the fingerprints of all 10 batches were greater than 0.97. HCA showed that the 10 batches could be clustered into two groups according to component content and proportion. PCA and PLS-DA further confirmed this classification and identified seven components, including chlorogenic acid, cryptochlorogenic acid A, and cryptochlorogenic acid C, as the main marker compounds contributing to inter-batch quality differences. CONCLUSION The established fingerprint and content determination methods showed high accuracy and good reproducibility. Combined with chemometric techniques, they can be used for holistic quality evaluation and control of Qiongyu Runfei Paste. © (2026), (Editorial Office of Chinese Journal of Hospital Pharmacy). All rights reserved.

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Quantification in data streams: Initial results

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Author A.G. Maletzke

Author D.M. Dos Reis

Author G.E.A.P.A. Batista

Abstract In the last decades, learning from data streams has attracted the attention of researchers and practitioners due to its large number of applications. These applications have motivated the research community to propose a significant number of methods that can be used to solve problems in diverse tasks, more prominently in classification, prediction, and clustering. However, a relevant task known as quantification has remained largely unexplored. The quantification goal is to provide an estimate of the class prevalence in an unlabeled set. In this paper, we discuss the relevance and challenges of quantification for data streams and compare how it differs from the batch setting, in which quantification has attracted more attention from the research community. We propose an algorithm to estimate the class distribution in a data stream and frame our algorithm in the active learning framework. In addition, we define two other approaches as baseline and topline strategies for this problem. The experimental results demonstrate that our algorithm has significantly higher quantification accuracy than the baseline and almost as large as the topline while requiring a fraction of the true labels requested by the latter approach. © 2017 IEEE.

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Conference Name Proceedings - 2017 Brazilian Conference on Intelligent Systems, BRACIS 2017

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Intelligent systems, Active Learning, Research communities, data stream, Data stream, Concept drifts, quantification, Class distributions, concept drift, verification latency, Number of methods, Verification latencies

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EID: 2-s2.0-85049514236

Export Date: 20 September 2026; Cited By: 10; Correspondence Address: A.G. Maletzke; Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos, SP, Brazil; email: andregustavo@usp.br; Conference name: 6th Brazilian Conference on Intelligent Systems, BRACIS 2017; Conference date: 2 October 2017 through 5 October 2017; Conference code: 134134

Quantification in social networks

Item Type Conference Paper**Author** L. Milli**Author** A. Monreale**Author** G. Rossetti**Author** D. Pedreschi**Author** F. Giannotti**Author** F. Sebastiani**Editor** Gaussier E.**Editor** Cao L.**Editor** Gallinari P.**Editor** Kwok J.**Editor** Pasi G.**Editor** Zaiane O.

Abstract In many real-world applications there is a need to monitor the distribution of a population across different classes, and to track changes in this distribution over time. As an example, an important task is to monitor the percentage of unemployed adults in a given region. When the membership of an individual in a class cannot be established deterministically, a typical solution is the classification task. However, in the above applications the final goal is not determining which class the individuals belong to, but estimating the prevalence of each class in the unlabeled data. This task is called quantification. Most of the work in the literature addressed the quantification problem considering data presented in conventional attribute format. Since the ever-growing availability of web and social media we have a flourish of network data representing a new important source of information and by using quantification network techniques we could quantify collective behavior, i.e., the number of users that are involved in certain type of activities, preferences, or behaviors. In this paper we exploit the

homophily effect observed in many social networks in order to construct a quantifier for networked data. Our experiments show the effectiveness of the proposed approaches and the comparison with the existing state-of-the-art quantification methods shows that they are more accurate. © 2015 IEEE.

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Quantification Method for Assessment of Asteroid Resource Accessibility

Item Type Conference Paper

Author K.I. Alvarado

Author M.C.F. Bazzocchi

Abstract Asteroid mining and retrieval have been explored as viable alternatives to supplying resources from Earth for space missions. Various methods for determining the accessibility of near-Earth asteroids have been presented; however, this method focuses on the quantity and types of resources that are available within a certain energy constraint given a redirection time frame. Lambert's problem is solved to determine the optimal transfer orbit between the Earth and asteroid for rendezvous. A genetic algorithm is used to determine the minimum energy for transfer with varying flight durations and launch dates. The physical properties and spectral classifications of known asteroids are used to estimate the quantity and chemical composition of accessible resources. This

method is applied to asteroids that have assigned taxonomic classifications. An overall distribution of asteroid types and energy requirements is presented for Earth and investigated for a sample asteroid scenario. © 2022 IEEE.

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DOI 10.1109/AERO53065.2022.9843629

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Genetic algorithms, Quantification methods, Asteroids, Earth (planet), Energy constraint, Flight duration, Lambert's problem, Minimum energy, Near-earth asteroids, Optimal transfers, Orbits, Property classification, Space missions, Time frame

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Quantification Method for Rivet Hole Cracks in an Aircraft Fuselage Using Guided Waves: A BTBD-Theory-Hybrid Space-Time Cross Fusion SpatNet

Item Type Journal Article

Author H. Sun

Author Q. Feng

Author S. Huang

Author J. Li

Author L. Peng

Author S. Li

Author J. Tu

Abstract In harsh and complex aerospace environments, stress concentration causes fatigue cracks to easily form at the edges of rivet holes in an aircraft fuselage, threatening overall aircraft safety. Although testing methods based on high-order focusing ultrasonic guided waves are sensitive to cracks, there remains a lack of high-precision

quantification methods. Existing signal processing and machine learning methods have shortcomings. For example, the former cannot achieve intelligent detection and the latter has black box problems with inexplicability. To solve the above-mentioned problems, we propose a space-time cross-fusion bin-to-bin distance-theory-hybrid deep neural network (DNN) called SpatNet that combines the physical knowledge of defect quantification with a DNN to effectively detect and quantify rivet hole cracks in an aluminum alloy fuselage. In addition, we analyze the attenuation coefficients k_R, G, B and window length w_l , which can describe the degree of physical knowledge introduction, and compare the proposed SpatNet with other classical methods. The results show that the reasonable introduction of physical knowledge can effectively improve the prediction accuracy of a DNN and reduce network dependence on large-scale datasets. Related research on SpatNet based on the fusion of physical knowledge and neural network space-time cross-fusion has the potential to promote the further development of the intelligent ultrasonic testing field. © 2025 IEEE.

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Aircraft accidents, Neural-networks, Bin-to-bin distance -theory-hybrid, Bin-to-bin distance (BTBD)-theory-hybrid, Bins, Fatigue crack propagation, Fatigue testing, High-order, High-order guided wave, high-order guided waves, Higher-order, Rivet hole, Rivet hole crack, rivet hole cracks, Riveting, space-time cross fusion, Space-time cross fusion, Spacetime, Stress analysis, Stress concentration, ultrasonic focusing, Ultrasonic focusing, Ultrasonic testing, Ultrasonic waves

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Quantification method of ctDNA using cell-free DNA methylation profile for noninvasive screening and monitoring of colon cancer

Item Type Journal Article

Author H. Ryu

Author J.-H. Kim

Author Y.J. Kim

Author H. Jeon

Author B.-C. Kim

Author Y. Jeon

Author Y. Kim

Author H. Bak

Author Y. Kang

Author C. Kim

Author H. Um

Author J.-H. Ahn

Author H. Hyun

Author B.C. Kim

Author I. Song

Author S. Jeon

Author J. Bhak

Author E.C. Han

Abstract Background: Colon cancer ranks as the second most lethal form of cancer globally. In recent years, there has been active investigation into using the methylation profile of circulating tumor DNA (ctDNA), derived from blood, as a promising indicator for diagnosing and monitoring colon cancer. Results: We propose a liquid biopsy-based epigenetic method developed by utilizing 49 patients and 260 healthy controls methylation profile data to screen and monitor colon cancer. Our method initially identified 901 colon cancer-specific hypermethylated (CaSH) regions in the tissues of the 49 cancer patients. We then used these CaSH regions to accurately quantify the amount of circulating tumor DNA (ctDNA) in the blood samples of these same patients, utilizing cell-free DNA methylation profiles. Notably, the methylation profiles of ctDNA in the blood exhibited high sensitivity (82%) and specificity (93%) in distinguishing patients with colon cancer from the control group, with an area under the curve of 0.903. Furthermore, we confirm that our method for ctDNA quantification is effective for monitoring cancer patients and can serve as a valuable tool for postoperative prognosis. Conclusions: This study demonstrated a successful application of the quantification of ctDNA among cfDNA using the original cancer tissue-derived CaSH region for screening and monitoring colon cancer. © The Author(s) 2024.

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Publication Clinical Epigenetics**ISSN** 18687075 (ISSN)**Issue** 1**Journal Abbr** Clin. Epigenetics**Date Added** 20/09/2026, 9:51:54 pm**Modified** 20/09/2026, 9:51:54 pm**Tags:**

adult, Adult, Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, machine learning, aged, Aged, procedures, controlled study, diagnosis, follow up, clinical article, genetics, blood sampling, cell free nucleic acid, Cell-Free Nucleic Acids, sensitivity and specificity, human tissue, Biomarkers, Tumor, cancer prognosis, prognosis, Prognosis, tumor marker, DNA, unclassified drug, postoperative period, DNA methylation, DNA Methylation, case control study, Case-Control Studies, Sensitivity and Specificity, blood, circulating tumor DNA, liquid biopsy, blood analysis, cancer diagnosis, cancer epidemiology, cell free DNA, Circulating Tumor DNA, colon cancer, Colon cancer, colon tumor, Colonic Neoplasms, ctDNA, DNA determination, early cancer diagnosis, Early Detection of Cancer, Epigenesis, Genetic, Epigenetic diagnosis, epigenetics, genetic epigenesis, Liquid biopsy, Liquid Biopsy, non invasive procedure, Postoperative monitoring, process development, tumor burden

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Export Date: 20 September 2026; Cited By: 14; Funding details: ministry of Science and Technology of Korea and Ulsan Metropolitan city; Dongnam Institute of Radiological & Medical Sciences; Ministry of Commerce, Industry, and Energy, (20023198); MSIT, (50605–2023); Correspondence Address: S. Jeon; Clinomics, Inc., Ulsan, 44919, South Korea; email: jsw0061@gmail.com; J. Bhak; Clinomics, Inc., Ulsan, 44919, South Korea; email: jongbhak@genomics.org; E.C. Han; Division of Colorectal Surgery, Department of Surgery, Dongnam Institute of Radiological and Medical Sciences, Busan, 46033, South Korea; email: eonchulhan@dirams.re.kr

Quantification method of tubing defects based on machine learning algorithm and magnetic flux leakage signals

Item Type Journal Article**Author** M. Shi**Author** M. Ni**Author** L. Qin**Author** Y. Liang**Author** Z. Huang

Abstract Tubing is the pipeline that transports crude oil and natural gas from the oil and gas layer to the surface of the earth. Due to the harsh operating environment, the tubing will suffer from etch pits, scratches, cracks, perforations, and even direct fractures of different degrees of defective conditions. If tubing defects are not detected and quantified in a timely manner, the continued use of tubing will result in tubing leakage and failure. Magnetic flux leakage (MFL) testing as a nondestructive testing method enables the identification and quantitative analysis of defects in metal tubing. To improve the quantification accuracy of defects in the wellhead MFL testing of tubing defects during workover operations, this paper proposes a multi-output least-squares support vector regression machine (MLSSVR) model optimized based on the simulated annealing algorithm. The size of tubing defects can be quantified by establishing the mapping between the characteristic quantity of MFL signals and the defect size. The experimental results of MFL testing of tubing defects show that the root mean square error (RMSE) of the diameter of tubing defects of the simulated annealing algorithm optimized multi-output least-squares support vector regression (SA-MLSSVR) machine model proposed in this paper is 0.4562 mm, and the RMSE of the depth of tubing defects is 0.1504 mm. Compared with the non-optimized MLSSVR model, the overall RMSE of tubing defects is reduced by 36.48%. The SA-MLSSVR model only needs one-ninth of the time to achieve the same quantification accuracy as the particle swarm optimized multi-output least-squares support vector regression machine model. © 2023 Author(s).

Date 2023

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Quantification methods for human and large animal leukocytes using DNA dyes by flow cytometry

Item Type Journal Article

Author I.L. Pieper

Author G. Radley

Author C.H.H. Chan

Author Y. Friedmann

Author G. Foster

Author C.A. Thornton

Abstract Ovine and bovine blood is used heavily within the development of blood-handling medical devices, such as heart pumps (left ventricular assist devices, LVADs), for which blood cell damage needs to be monitored during in vitro testing. Hematology analyzers provide cell counts but no information about cell viability. The anthraquinone DNA dyes CyTRAK Orange™ and DRAQ7™ have practical and spectral properties rendering them suitable for multicolor assays. Compared to other DNA dyes such as Vybrant Dycycle, CyTRAK Orange enables a faster staining protocol and does not require incubation at +37°C. Compared to traditional viability dyes such as propidium iodide and 7AAD, DRAQ7's unique spectral profile of excitation in both blue and red lasers and far-red emission enables identification of dual positive dead cell events and frees up detectors for use with other reagents. CyTRAK Orange and DRAQ7 could be used in combination with absolute counting bead standards to provide cell counts and viability but the combination of these dyes has previously only been used for microscopy on rodent cells. The purpose of this study was to evaluate the use of these dyes in combination in large animal blood samples for flow cytometry. A viability and cell counting protocol for bovine, ovine, and human leukocytes using CyTRAK Orange and DRAQ7 was prepared. Four different counting bead standards were evaluated using the Navios and FACSria cytometers and compared to counts obtained from hematology analyzers. CyTRAK Orange successfully detected CD45+ leukocytes in all species. The DRAQ7 single-stained dead cell counts correlated well with the CyTRAK Orange/DRAQ7 double-stained dead cell counts in human and bovine blood, but not in ovine blood, which could be related to the blood source. In conclusion, for human and bovine blood, this method works well for viability counts using different flow cytometers and bead standards. © 2016 International Society for Advancement of Cytometry. © 2016 International Society for Advancement of Cytometry

Date 2016

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Export Date: 20 September 2026; Cited By: 11; Funding details: Medical Research Council, MRC, (MC_PC_13039); Correspondence Address: I.L. Pieper; Institute of Life Science, Swansea University Medical School, Swansea, SA2 8PP, United Kingdom; email: inalaura@caloncardio.com; CODEN: CPAYA

Quantification of *Actaea racemosa* L. (black cohosh) from some of its potential adulterants using qPCR and dPCR methods

Item Type Journal Article

Author J. Shanmughanandhan

Author D. Shanmughanandhan

Author S. Ragupathy

Author T.A. Henry

Author S.G. Newmaster

Abstract The demand for popular natural health products (NHPs) such as Black Cohosh is increasing considerably, which in turn challenges quality assurance (QA) throughout the supply chain. To detect and quantify the target species present in a given NHP, DNA-based molecular techniques such as Real-time quantitative PCR (qPCR) and digital PCR (dPCR) are standard tools in the food and pathogen testing industries.

There is a gap in the literature concerning validated quantitative PCR methods for botanicals that can be utilized for QA and good manufacturing practices. The objective of this study is to develop an efficient quantification method using qPCR and dPCR techniques for the detection and quantification of *Actaea racemosa* (Black cohosh) NHPs from its potential adulterants. These developed methods are validated for applicability on commercial NHPs. Species-specific hydrolysis probe assays were designed to analyze the black cohosh NHPs using qPCR and dPCR techniques. The results confirmed that the developed qPCR and dPCR methods are highly precise for identifying and quantifying black cohosh NHPs, indicating their potential applicability in future routine industrial and laboratory testing. This enables a single qPCR test to determine not only the presence of a specific botanical, but also the amount when mixed with an adulterant. © 2021, The Author(s).

Date 2021

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procedures, classification, genetics, polymerase chain reaction, Polymerase Chain Reaction, medicinal plant, Plants, Medicinal, *Actaea*, *Cimicifuga*, DNA contamination, DNA Contamination, DNA, Plant, ethnobotany, Ethnobotany, plant DNA

Notes:

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: D. Shanmughanandhan; NHP Research Alliance, College of Biological Sciences, University of Guelph, Guelph, 50 Stone Road East, N1G 2W1, Canada; email: shanmugd@uoguelph.ca

Quantification of anomalies in rats' spinal cords using autoencoders

Item Type Journal Article

Author M.E. Tschuchnig

Author D. Zillner

Author P. Romanelli

Author D. Hercher**Author** P. Heimel**Author** G.J. Oostingh**Author** S. Couillard-Després**Author** M. Gadermayr

Abstract Computed tomography (CT) scans and magnetic resonance imaging (MRI) of spines are state-of-the-art for the evaluation of spinal cord lesions. This paper analyses micro-CT scans of rat spinal cords with the aim of generating lesion progression through the aggregation of anomaly-based scores. Since reliable labelling in spinal cords is only reasonable for the healthy class in the form of untreated spines, semi-supervised deviation-based anomaly detection algorithms are identified as powerful approaches. The main contribution of this paper is a large evaluation of different autoencoders and variational autoencoders for aggregated lesion quantification and a resulting spinal cord lesion quantification method that generates highly correlating quantifications. The conducted experiments showed that several models were able to generate 3D lesion quantifications of the data. These quantifications correlated with the weakly labelled true data with one model, reaching an average correlation of 0.83. We also introduced an area-based model, which correlated with a mean of 0.84. The possibility of the complementary use of the autoencoder-based method and the area feature were also discussed. Additionally to improving medical diagnostics, we anticipate features built on these quantifications to be useful for further applications like clustering into different lesions. © 2021

Date 2021**DOI** 10.1016/j.compbimed.2021.104939**URL** <https://www.scopus.com/pages/publications/85116963495?origin=resultslist>**Archive** Scopus**Language** English**Volume** 138**Publisher** Elsevier Ltd**Publication** Computers in Biology and Medicine**ISSN** 00104825 (ISSN)**Journal Abbr** Comput. Biol. Med.**Date Added** 20/09/2026, 9:52:15 pm**Modified** 20/09/2026, 9:52:15 pm**Tags:**

Article, human, male, Auto encoders, autoencoder, Learning systems, Cluster Analysis, cluster analysis, comparative study, Autoencoder, algorithm, Algorithms, Anomaly detection, correlation analysis, nonhuman, qualitative analysis, controlled study, Labelings, E-learning, Diagnosis, Semi-supervised learning, State of the art, cross validation, disease exacerbation, Digital pathologies, Digital pathology, limit of quantitation, diagnostic imaging, Magnetic resonance imaging, Magnetic Resonance Imaging, nuclear magnetic resonance imaging, animal, Animals, pathophysiology, Computerized tomography, outlier detection, animal experiment, animal model, rat, Computed tomography scan, Rats, Paper analysis, Lesion quantification, micro-computed tomography, Micro-CT, Rat spinal cord, spinal cord, Spinal Cord, spinal cord malformation, Spinal-cord

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Quantification of Breast Arterial Calcification in Mammograms Using a UNet-Based Deep Learning for Detecting Cardiovascular Disease

Item Type Journal Article**Author** W. Li**Author** Q. Zhang**Author** D. Black**Author** H. Ding**Author** C. Iribarren**Author** A. Shojazadeh**Author** S. Molloy

Abstract Background: Breast arterial calcification (BAC) is increasingly recognized as a significant indicator of cardiovascular risk, necessitating improvements in detection and quantification methods through mammographic screening. Purpose: To develop and validate a deep-learning model capable of detecting, segmenting, and quantifying BAC in mammograms, improving mammographic screening for cardiovascular risk assessment. Materials and Methods: We conducted a retrospective study using mammograms from 369 patients. The study utilized a modified U-Net architecture that incorporates Hausdorff loss, Dice loss, and Binary Cross-Entropy (BCE) loss for segmentation and subsequent quantification. The model's performance was assessed using the Dice score, BCE loss for segmentation accuracy, linear fit, and Bland–Altman analysis for quantification accuracy. Results: Our model achieved high segmentation accuracy with Dice scores of 0.90 for the training set and 0.89 for the validation set. Quantification reliability was validated through Bland–Altman analysis, showing a mean difference of −0.98 mg of calcium in the training set. The model also demonstrated high classification accuracy with F1 scores of 0.97 and 0.93 for validation and training sets, respectively, in BAC detection. Conclusion: The deep-learning framework substantially improves BAC detection, segmentation, and quantification in mammograms, advancing the accuracy and efficiency of cardiovascular risk screening. This study supports the potential for integrated dual-purpose screening in women's healthcare. © 2025 The Association of University Radiologists

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Notes:

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: S. Molloi; Department of Radiological Sciences, School of Medicine, University of California, Irvine, 92697, United States; email: symolloi@uci.edu; CODEN: ARADF

Quantification of eosinophilic area and its potential molecular feature in clear cell renal cell carcinoma

Item Type Journal Article
Author N. Wen
Author X. Li
Author J. Lu
Author L. Pan
Author P. Yang
Author Y. Zhang
Author K. Chen
Author Y. Cao

Abstract Objective: Previous studies have acknowledged the presence of eosinophilic cytoplasm in clear cell renal cell carcinoma, yet the precise quantification method and potential molecular attributes in clear cell renal cell carcinoma remain elusive. This study endeavours to precisely quantify the eosinophilic attribute and probe into the molecular mechanisms governing its presence in clear cell renal cell carcinoma. Methods: Data from cohorts of clear cell renal cell carcinoma patients who underwent nephrectomy, comprising The Cancer Genome Atlas cohort (n = 475) and Sun Yat-sen University Cancer Center cohort (n = 480), were aggregated to assess the eosinophilic attribute. Additionally, Omics data from Clinical Proteomic Tumor Analysis Consortium (CPTAC) (n = 58) were leveraged to explore the potential molecular features associated with eosinophilic clear cell renal cell carcinoma. Employing receiver operating characteristic curve analysis, the proportion of tumour cells with eosinophilic cytoplasm was determined, leading to the classification of each cohort into distinct groups: a clear group (<5%) and an eosinophilic group ($\geq 5\%$). Results: In both cohorts, the eosinophilic feature consistently correlated with higher International Society of Urological Pathology (ISUP) grade, elevated tumor stage, and the presence of necrosis. Furthermore, the Kaplan-Meier method demonstrated that patients in the eosinophilic group exhibited shorter overall survival or disease-free survival compared with those in the clear group, a pattern reaffirmed in various stratified survival analyses. Intriguingly, within The Cancer Genome Atlas cohort, the pathological characterization of cell cytoplasm (eosinophilic vs. clear) emerged as an independent risk factor for overall survival (hazard ratio = 2.507 [95% confidence interval: 1.328-4.733], $P = 0.005$) or disease-free survival (hazard ratio = 1.730 [95% confidence interval: 1.062-2.818], $P = 0.028$) via Cox regression analysis. Moreover, multi-Omics data unveiled frequent BAP1 mutations and down-regulation of Erythroblast Transformation-Specific-Related Gene associated with the eosinophilic feature in clear cell renal cell carcinoma. Additionally, patients with low expression of Erythroblast Transformation-Specific-Related Gene showed worse overall survival ($P < 0.001$). Conclusions: The quantification of the eosinophilic feature serves as a robust predictor of clinical prognosis in clear cell renal cell carcinoma. Furthermore, the manifestation of this feature may be linked to BAP1 mutations and the down-regulation of Erythroblast Transformation-Specific-Related Gene in clear cell renal cell carcinoma. Significantly, the expression levels of Erythroblast Transformation-Specific-Related Gene manifest as an exemplary prognostic marker, providing exceptional predictive accuracy for the clinical prognosis in clear cell renal cell carcinoma. © 2024 The Author(s). Published by Oxford University Press. All rights reserved.

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Export Date: 20 September 2026; Cited By: 1; Funding details: National Natural Science Foundation of China, NSFC, (81872085); National Natural Science Foundation of China, NSFC; Correspondence Address: K. Chen; Department of Pathology, Sun Yat-sen University Cancer Center, Guangzhou, China; email: chenkm@sysucc.org.cn; Y. Cao; Department of Pathology, Sun Yat-sen University Cancer Center, Guangzhou, China; email: caoyun@sysucc.org.cn; CODEN: JJCOA

Quantification of fibrosis and osteosclerosis in myeloproliferative neoplasms: A computer-assisted image study

Item Type Journal Article

Author C.J. Teman

Author A.R. Wilson

Author S.L. Perkins

Author K. Hickman

Author J.T. Prchal

Author M.E. Salama

Abstract Evaluation of bone marrow fibrosis and osteosclerosis in myeloproliferative neoplasms (MPN) is subject to interobserver inconsistency. Performance data for currently utilized fibrosis grading systems are lacking, and classification scales for osteosclerosis do not exist. Digital imaging can serve as a quantification method for fibrosis and osteosclerosis. We used digital imaging techniques for trabecular area assessment and reticulin fiber quantification. Patients with all Philadelphia negative MPN subtypes had higher trabecular volume than controls ($p \leq 0.0015$). Results suggest that the degree of osteosclerosis helps differentiate primary myelofibrosis from other MPN. Numerical quantification of fibrosis highly correlated with subjective scores, and interobserver correlation was satisfactory. Digital imaging provides accurate quantification for osteosclerosis and fibrosis. © 2010 Elsevier Ltd.

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Quantification of Flavonoid Contents in Holy Basil Using Hyperspectral Imaging and Deep Learning Approaches

Item Type Journal Article
Author A. Suratane
Author P. Chutimanukul
Author K. Plaimas

Abstract Holy basil (*Ocimum tenuiflorum* L.) is a medicinal herb rich in bioactive flavonoids with therapeutic properties. Traditional quantification methods rely on time-consuming and destructive extraction processes, whereas hyperspectral imaging provides a rapid, non-destructive alternative by analysing spectral signatures. However, effectively linking hyperspectral data to flavonoid levels remains a challenge for developing early detection tools before harvest. This study integrates deep learning with hyperspectral imaging to quantify flavonoid contents in 113 samples from 26 Thai holy basil cultivars collected across diverse regions of Thailand. Two deep learning architectures, ResNet1D and CNN1D, were evaluated in combination with feature extraction techniques, including wavelet transformation and Gabor-like filtering. ResNet1D with wavelet transformation achieved optimal performance, yielding an area under the receiver operating characteristic curve (AUC) of 0.8246 and an accuracy of 0.7702 for flavonoid content classification. Cross-validation demonstrated the model's robust predictive capability in identifying antioxidant-rich samples. Samples with the highest predicted flavonoid content were identified, and cultivars exhibiting elevated levels of both flavonoids and phenolics were highlighted across various regions of Thailand. These findings demonstrate the predictive capability of hyperspectral data combined with deep learning for phytochemical assessment. This approach offers a valuable tool for non-destructive quality evaluation and supports cultivar selection for higher phytochemical content in breeding programs and agricultural applications. © 2025 by the authors.

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Quantification of hematoma and perihematoma edema volumes in intracerebral hemorrhage study: Design considerations in an artificial intelligence validation (QUANTUM) study

Item Type Journal Article

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Author J. Patrie

Author S. Ng

Author D. Ding

Author T. Rizvi

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Author M.Z. Hussein

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Author L.M. Shabo

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Author D. Roh

Author A. Lignelli-Dipple

Author J. Claassen

Author B.B. Worrall

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Author M.S. Park

Author E. Sander Connolly

Author S. Mukherjee

Author A.M. Southerland

Author C.-J. Chen

Abstract Background: Hematoma and perihematoma edema volumes are important radiographic markers in spontaneous intracerebral hemorrhage. Accurate, reliable, and efficient quantification of these volumes will be paramount to their utility as

measures of treatment effect in future clinical studies. Both manual and semi-automated quantification methods of hematoma and perihematoma edema volumetry are time-consuming and susceptible to inter-rater variability. Efforts are now underway to develop a fully automated algorithm that can replace them. A (QUANTUM) study to establish inter-quantification method measurement equivalency, which deviates from the traditional use of measures of agreement and a comparison hypothesis testing paradigm to indirectly infer quantification method measurement equivalence, is described in this article. The Quantification of Hematoma and Perihematoma Edema Volumes in Intracerebral Hemorrhage study aims to determine whether a fully automated quantification method and a semi-automated quantification method for quantification of hematoma and perihematoma edema volumes are equivalent to the hematoma and perihematoma edema volumes of the manual quantification method. Methods/Design: Hematoma and perihematoma edema volumes of supratentorial intracerebral hemorrhage on 252 computed tomography scans will be prospectively quantified in random order by six raters using the fully automated, semi-automated, and manual quantification methods. Primary outcome measures for hematoma and perihematoma edema volumes will be quantified via computed tomography scan on admission (<24 h from symptom onset) and on day 3 (72 ± 12 h from symptom onset), respectively. Equivalence hypothesis testing will be conducted to determine if the hematoma and perihematoma edema volume measurements of the fully automated and semi-automated quantification methods are within 7.5% of the hematoma and perihematoma edema volume measurements of the manual quantification reference method. Discussion: By allowing direct equivalence hypothesis testing, the Quantification of Hematoma and Perihematoma Edema Volumes in Intracerebral Hemorrhage study offers advantages over radiology validation studies which utilize measures of agreement to indirectly infer measurement equivalence and studies which mistakenly try to infer measurement equivalence based on the failure of a comparison two-sided null hypothesis test to reach the significance level for rejection. The equivalence hypothesis testing paradigm applied to artificial intelligence application validation is relatively uncharted and warrants further investigation. The challenges encountered in the design of this study may influence future studies seeking to translate artificial intelligence medical technology into clinical practice. © The Author(s) 2022.

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Quantification of heterogeneity as a biomarker in tumor imaging: A systematic review

Item Type Journal Article

Author L. Alic

Author W.J. Niessen

Author J.F. Veenland

Abstract Background: Many techniques are proposed for the quantification of tumor heterogeneity as an imaging biomarker for differentiation between tumor types, tumor grading, response monitoring and outcome prediction. However, in clinical practice these methods are barely used. This study evaluates the reported performance of the described methods and identifies barriers to their implementation in clinical practice. Copyright:; Methodology: The Ovid, Embase, and Cochrane Central databases were searched up to 20 September 2013. Heterogeneity analysis methods were classified into four categories, i.e., non-spatial methods (NSM), spatial grey level methods (SGLM), fractal analysis (FA) methods, and filters and transforms (F&T). The performance of the different methods was compared.; Principal Findings: Of the 7351 potentially relevant publications, 209 were included. Of these studies, 58% reported the use of NSM, 49% SGLM, 10% FA, and 28% F&T. Differentiation between tumor types, tumor grading and/or outcome prediction was the goal in 87% of the studies. Overall, the reported area under the curve (AUC) ranged from 0.5 to 1 (median 0.87). No relation was found between the performance and the quantification methods used, or between the performance and the imaging modality. A negative correlation was found between the tumor-feature ratio and the AUC, which is presumably caused by overfitting in small datasets. Cross-validation was reported in 63% of the classification studies. Retrospective analyses were conducted in 57% of the studies without a clear description.; Conclusions: In a research setting, heterogeneity quantification methods can differentiate between tumor types, grade tumors, and predict outcome and monitor treatment effects. To translate these methods to clinical practice, more prospective studies are required that use external datasets for

validation: these datasets should be made available to the community to facilitate the development of new and improved methods. © 2014 Alic et al.

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Quantification of Ineffective Water Circulation in Thick Reservoir Based on Machine Learning

Item Type Conference Paper

Author X.-Z. Yin

Author J.-H. Guo

Author L. Sun

Author L.-Y. Wei
Author Y.-Q. Wang
Author Z.-G. Wang
Author H.-T. Wang
Editor Lin J.

Abstract Being water flooded for more than 6 decades, the Lasaxing Oilfield in Daqing is now developing in the extra-high water cut period and the ineffective water circulation (IWC) worsens year by year. Due to the heterogeneity and the gravity effect, there always exists high permeability channels at the bottom of the thick reservoir, which affects the water flooding efficiency greatly. As to the quantification of the IWC in thick reservoir, a novel method based on machine learning is promoted in this paper. Firstly, the reservoir classification model based on machine learning is given. 7 parameters of 60000 well-layer samples of the typical block are analyzed by means of PCA, and the clustering model is then constructed based on K-Means and manual intervention. The reservoir is then classified into 9 categories according to the interlayer position, the reservoir thickness, the reservoir permeability and etc. Secondly, the quantification method of the IWC in thick reservoir based on inspecting wells and machine learning is promoted. The reservoirs with highly water flooded section, which implies the IWC thickness, are taken as learning samples. The method of SVM and KNN is selected in the model training, which lead to a conformity rate of 75%. Thirdly, the method promoted is implemented in typical well groups, which shows that the percentage of ineffective thickness is 20–60%. The low-price and high-efficient profile controlling agent is then used in the stimulation pilot, which lead to an oil increasing of 3890t. The methodology promoted in this paper has the advantage of practical, short time of evaluating and easy-to-use, and presents an effective in the identification of the IWC. The method could also be used in other similar oilfields, which offers valuable reference for the effective development of a mature field at the extra-high water cut period. © 2022, The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd.

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Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: X.-Z. Yin; School of Computer and Information Technology, Northeast Petroleum Univer-Sity, Daqing, China; email: yinxiaoze81@163.com; Conference name: 11th International Field Exploration and Development Conference, IFEDC 2021; Conference date: 20 October 2021 through 22 October 2021; Conference code: 283119

Quantification of Large Transmural Biopsies Reveals Heterogeneity in Innervation Patterns in Chronic Myocardial Infarction

Item Type Journal Article

Author H.S. Chen

Author L.M. Voortman

Author J.C. van Munsteren

Author L.J. Wisse

Author B.J. Tofig

Author S.B. Kristiansen

Author C.A. Glashan

Author M.C. DeRuiter

Author K. Zeppenfeld

Author M.R.M. Jongbloed

Abstract Background: Abnormal cardiac innervation plays an important role in arrhythmogenicity after myocardial infarction (MI). Data regarding reperfusion models and innervation abnormalities in the medium to long term after MI are sparse. Histologic quantification of the small-sized cardiac nerves is challenging, and transmural analysis has not been performed. Objectives: This study sought to assess cardiac innervation patterns in transmural biopsy sections in a porcine reperfusion model of MI (MI-R) using a novel method for nerve quantification. Methods: Transmural biopsy sections from 4 swine (n = 83) at 3 months after MI-R and 3 controls (n = 38) were stained with picrosirius red (fibrosis) and beta-III-tubulin (autonomic nerves). Biopsy sections were classified as infarct core, border zone, or remote zone. Each biopsy section was analyzed with a custom software pipeline, allowing calculation of nerve density and classification into innervation types at the 1 × 1-mm resolution level. Relocation of the classified squares to the original biopsy

position enabled transmural quantification and innervation heterogeneity assessment. Results: Coexisting hyperinnervation, hypoinnervation, and denervation were present in all transmural MI-R biopsy sections. The innervation heterogeneity was greatest in the infarct core (median: 0.14; IQR: 0.12-0.15), followed by the border zone (median: 0.05; IQR: 0.04-0.07; $P = 0.02$) and remote zone (median: 0.02; IQR: 0.02-0.03; $P < 0.0001$). Only in the border zone was a positive linear relation between fibrosis and innervation heterogeneity observed ($R = 0.79$; $P < 0.0001$). Conclusions: This novel method allows quantification of nerve density and heterogeneity in large transmural biopsy sections. In the chronic phase after MI-R, alternating innervation patterns were identified within the same biopsy section. Persistent innervation heterogeneity, in particular in the border zone biopsy sections, may contribute to late arrhythmogenicity. © 2023 American College of Cardiology Foundation

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: M.R.M. Jongbloed; Center for Congenital Heart Disease Amsterdam-Leiden (CAHAL), Department of Cardiology and Anatomy and Embryology, Leiden University Medical Center, Leiden, Postal zone: S-1-P, PO Box 9600, 2300, Netherlands; email: m.r.m.jongbloed@lumc.nl

Quantification of Left Atrium Fibrosis from LGE MRI in Atrial Fibrillation

Item Type Conference Paper

Author A. Andalo

Author C. Fabbri

Author M. Valinoti

Author A. Masci

Author C. Corsi

Abstract Late gadolinium enhanced magnetic resonance imaging (LGE-MRI) is a recent technique used for fibrosis distribution assessment in the left atrium (LA) wall. Unfortunately, LGE-MRI analysis does not rely on a standardized image processing protocol. The aim of the study was to compare different methods to quantify fibrosis in the LA in the 3D domain. LGE-MRI from 60 AF patients were analyzed applying five different approaches for fibrosis segmentation: an histogram-based (H-6SD), the image intensity ratio (IIR), the blood pool normalization (BP), the Chan-Vese (CV) and the graph-cut (GC) method. For each method we quantified the percentage of fibrosis with respect to the entire LA wall mass. Results showed these five approaches can be divided in two groups; the group composed by the H-6SD, CV and GC is characterized similar results (mean coefficient of variation =0.3), while the results obtained applying BP and IIR strongly depend on the quality of the acquisition (mean coefficient of variation=0.62). Utah stage classification resulted in a wide disagreement (22/60 patients, 37%) among BP and IIR. The two approaches which best matched the Utah classification were the CV and GC (49/60 patients, 82%). This study confirmed the evaluation of the quantification method appears critical and further research is needed to define a satisfactory standard for atrial fibrosis segmentation. © 2018 Creative Commons Attribution.

Date 2018

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Quantification of metabolites in magnetic resonance spectroscopic imaging using machine learning

Item Type Conference Paper

Author D. Das

Author E. Coello

Author R.F. Schulte

Author B.H. Menze

Editor Maier-Hein L.

Editor Franz A.

Editor Jannin P.

Editor Duchesne S.

Editor Descoteaux M.

Editor Collins D.L.

Abstract Magnetic Resonance Spectroscopic Imaging (MRSI) is a clinical imaging modality for measuring tissue metabolite levels in-vivo. An accurate estimation of spectral parameters allows for better assessment of spectral quality and metabolite concentration levels. The current gold standard quantification method is the LCModel - a commercial fitting tool. However, this fails for spectra having poor signal-to-noise ratio (SNR) or a large number of artifacts. This paper introduces a framework based on random forest regression for accurate estimation of the output parameters of a model based analysis of MR spectroscopy data. The goal of our proposed framework is to learn the spectral features from a training set comprising of different variations of both simulated and in-vivo brain spectra and then use this learning for the subsequent metabolite quantification. Experiments involve training and testing on simulated and in-vivo human brain spectra. We estimate parameters such as concentration of metabolites and compare our results with that from the LCModel. © Springer International Publishing AG 2017.

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Quantification of pancreatic exocrine function with secretin-enhanced magnetic resonance cholangiopancreatography: Normal values and short-term effects of pancreatic duct drainage procedures in chronic pancreatitis. Initial results

Item Type Journal Article

Author M.A. Bali

Author A. Sztantics

Author T. Metens

Author M. Arvanitakis

Author M. Delhayé

Author J. Devière

Author C. Matos

Abstract The aim of this study was to quantify pancreatic exocrine function in normal subjects and in patients with chronic pancreatitis (CP) before and after pancreatic duct drainage procedures (PDDP) with dynamic secretin-enhanced magnetic resonance (MR) cholangiopancreatography (S-MRCP). Pancreatic exocrine secretions [quantified by pancreatic flow output (PFO) and total excreted volume (TEV)] were quantified twice in ten healthy volunteers and before and after treatment in 20 CP patients (18 classified as severe, one as moderate, and one as mild according to the Cambridge classification). PFO and TEV were derived from a linear regression between MR-calculated volumes and time. In all subjects, pancreatic exocrine fluid

volume initially increased linearly with time during secretin stimulation. In controls, the mean PFO and TEV were 6.8 ml/min and 97 ml; intra-individual deviations were 0.8 ml/min and 16 ml. In 10/20 patients with impaired exocrine secretions before treatment, a significant increase of PFO and TEV was observed after treatment ($P<0.05$); 3/20 patients presented post-procedural acute pancreatitis and a reduced PFO. The S-MRCP quantification method used in the present study is reproducible and provides normal values for PFO and TEV in the range of those obtained from previous published intubation studies. The initial results in CP patients have demonstrated non-invasively a significant short-term improvement of PFO and TEV after PDDP. © Springer-Verlag 2005.

Date 2005

DOI 10.1007/s00330-005-2819-5

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Archive Scopus

Language English

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Journal Abbr Eur. Radiol.

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adult, Adult, female, Female, human, Humans, male, Male, Middle Aged, Aged, article, Reproducibility of Results, quantitative analysis, controlled study, disease severity, clinical article, Prospective Studies, disease classification, Image Processing, Computer-Assisted, Magnetic Resonance Imaging, Time Factors, clinical trial, Drainage, linear regression analysis, chronic pancreatitis, Chronic pancreatitis, Pancreatitis, Chronic, postoperative period, Radiographic Image Interpretation, Computer-Assisted, Observer Variation, controlled clinical trial, diagnostic value, priority journal, pancreas duct, Pancreatic Juice, Reference Values, acute pancreatitis, Cholangiopancreatography, Magnetic Resonance, Gastrointestinal Agents, intubation, magnetic resonance cholangiopancreatography, MR cholangiopancreatography, MR quantification, non invasive measurement, pancreas function, pancreas secretion, Pancreas, Exocrine, Pancreatic Ducts, Pancreatic Function Tests, secretin, Secretin, surgical drainage, volunteer

Notes:

EID: 2-s2.0-26044462395

Export Date: 20 September 2026; Cited By: 79; Correspondence Address: M.A. Bali; Department of Radiology, Hôpital Erasme, Université Libre de Bruxelles, 1070 Brussels, Route de Lennik 808, Belgium; email: mbali@ulb.ac.be; CODEN: EURAE

Quantification of pass plays based on geometric features of formations in team sports

Item Type Conference Paper

Author T. Mimura

Author Y. Nakada

Abstract In recent years, geometric features of formations such as players' dominant regions and their adjacency have been frequently computed and utilized for team sports' analysis. Such geometric features of formations have also been successfully applied to real game data in many team sports. In this paper, we propose a novel quantification method of pass plays by combining multiple geometric features of formations from two viewpoints: (i) effectiveness of leading to a shoot, and (ii) risk of being robbed a ball. The proposed method extracts many feature values including passers' and receivers' geometric feature values and then constructs quantification models from the two viewpoints based on statistical classification methods using the extracted feature values. Thus, the proposed method enables users to effectively understand pass plays and tactics used through visualization tools of players and ball position data. For validation, this method is applied to real players' position data during a soccer game. © 2019 Copyright is held by the owner/author(s).

Date 2019

DOI 10.1145/3368926.3369673

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Archive Scopus

Language English

Extra Journal Abbreviation: ACM Int. Conf. Proc. Ser.

Publisher Association for Computing Machinery

ISBN 978-145037245-9 (ISBN)

Pages 306-313

Proceedings Title ACM Int. Conf. Proc. Ser.

Conference Name ACM International Conference Proceeding Series

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Tags:

Classification (of information), Geometry, Data visualization, Classification models, Sports, Quantification methods, Quantification model, Geometric feature, Classification model, Geometric features of formations, Pass play evaluation, Statistical classification methods, Team sports, Visualization tools

Notes:

EID: 2-s2.0-85077818473

Export Date: 20 September 2026; Cited By: 13; Conference name: 10th International Symposium on Information and Communication Technology, SoICT 2019; Conference date: 4 December 2019 through 6 December 2019; Conference code: 156141

Quantification of patient-derived 3D cancer spheroids in high content screening images

Item Type Conference Paper

Author M.-S. Kang

Author S.-M. Rhee

Author J.-H. Seo

Author M.-H. Kim

Editor Farkas D.L.

Editor Nicolau D.V.

Editor Leif R.C.

Abstract We present a cell image quantification method for image-based drug response prediction from patient-derived glioblastoma cells. Drug response of each person differs at the cellular level. Therefore, quantification of a patient-derived cell phenotype is important in drug response prediction. We performed fluorescence microscopy to understand the features of patient-derived 3D cancer spheroids. A 3D cell culture simulates the in-vivo environment more closely than 2D adherence culture, and thus, allows more accurate cell analysis. Furthermore, it allows assessment of cellular aggregates. Cohesion is an important feature of cancer cells. In this paper, we demonstrate image-based quantification of cellular area, fluorescence intensity, and cohesion. To this end, we first performed image stitching to create an image of each well of the plate with the same environment. This image shows colonies of various sizes and shapes. To automatically detect the colonies, we used an intensity based classification algorithm. The morphological features of each cancer cell colony were measured. Next, we calculated the location correlation of each colony that is appeal of the cell density in the same well environment. Finally, we compared the features for drug-treated and untreated cells. This technique could potentially be applied for drug screening and quantification of the effects of the drugs. © 2017 SPIE.

Date 2017

DOI 10.1117/12.2250594

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Language English

Extra Journal Abbreviation: Progr. Biomed. Opt. Imaging Proc. SPIE

Volume 10068

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Proceedings Title Progr. Biomed. Opt. Imaging Proc. SPIE

Conference Name Progress in Biomedical Optics and Imaging - Proceedings of SPIE

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Diseases, Cells, Biomolecules, Cytology, Cancer cells, Fluorescence, Cell culture, Histology, Tissue, Image quantification, Molecular biology, 3D spheroids, Adhesion, Cancer cell, Fluorescence microscopy, High content screening, High-content screening, Patient-derived

Notes:

EID: 2-s2.0-85020399725

Export Date: 20 September 2026; Cited By: 0; Conference name: Imaging, Manipulation, and Analysis of Biomolecules, Cells, and Tissues XV; Conference date: 30 January 2017 through 1 February 2017; Conference code: 127881

Quantification of play behaviour in calves using automated ultra-wideband location data and its association with age, weaning and health status

Item Type Journal Article**Author** J.A. Vázquez-Diosdado**Author** C. Doidge**Author** E.V. Bushby**Author** F. Occhiuto**Author** J. Kaler

Abstract Play behaviour can act as an indicator of positive animal welfare. Previous attempts to predict play behaviour in farmed calves are limited because of the classification methods used, which lead to overestimation, and the short time periods that calves are observed. The study aimed to automatically classify and quantify play behaviour in farmed calves using location data from ultra-wide band sensors and to investigate factors associated with play behaviour. Location data were collected from 46 calves in three cohorts for a period of 18 weeks. Behavioural observations from video footage were merged with location data to obtain a total of 101.36 h of labelled data. An AdaBoost ensemble learning algorithm was implemented to classify play behaviour. To account for overestimation, generally seen in low-prevalence behaviours, an adjusted count technique was applied to the outputs of the classifier. Two generalized linear mixed models were fitted to investigate factors (e.g. age, health) associated with duration of play and number of play instances per day. Our algorithm identified play behaviour with > 94% accuracy when evaluated on the test set with no animals used for training, and 16% overestimation, which was computed based on the predicted number of samples of play versus the number of samples labelled as play on the test set. The instances and duration of play behaviour per day significantly decreased with age and sickness, whilst play behaviour significantly increased during and after weaning. The instances of play also significantly decreased as mean temperature increased. We suggest that the quantification method that we used could be used to detect and monitor other low prevalence behaviours (e.g. social grooming) from location data, including indicators of positive welfare. © The

Author(s) 2024.

Date 2024

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Language English

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Tags:

Machine learning, algorithm, Algorithms, health status, animal, Animals, Quantification, animal behavior, Behavior, Animal, Farms, weaning, agricultural worker, animal welfare, Animal Welfare, bovine, Calf, Cattle, Health Status, Play, Positive welfare, Precision livestock technology, Weaning

Notes:

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Export Date: 20 September 2026; Cited By: 11; Funding details: Biotechnology and Biological Sciences Research Council, BBSRC, (BB/W020459/1); Correspondence Address: J. Kaler; School of Veterinary Medicine and Science, Sutton Bonington Campus, University of Nottingham, Leicestershire, LE12 5RD, United Kingdom; email: jasmeet.kaler@nottingham.ac.uk

Quantification of prostate cancer Gleason pattern 4 to predict oncological outcome

Item Type Journal Article

Author J. Olivier

Author E. Vertosick

Author L. Delobel

Author T. Bommelaere

Author P. Puech

Author X. Leroy

Author A. Vickers

Author A. Villers

Abstract Objectives: To determine if quantifying Gleason pattern 4 (GP4) is superior to Grade Group (GG) in predicting any and distant metastatic recurrence after radical prostatectomy (RP) for prostate cancer, and compare various GP4 quantification

methods for predicting metastatic recurrence using preoperative targeted biopsy and magnetic resonance imaging (MRI) data. Patients and Methods: We conducted a retrospective study of patients who underwent MRI-guided biopsies and RP from 2009 to 2018. Patients with GG 2–4 without GP5 disease on biopsy and/or RP specimen were included. The predictors compared were biopsy GG, percentage of GP4 in biopsy cores, millimetres of GP4 in biopsy cores, and volume of GP4 based on MRI lesion volume. These methods were also compared to the Cancer of the Prostate Risk Assessment (CAPRA), International Staging Collaboration for Cancer of the Prostate (STAR-CAP), European Association of Urology (EAU), and National Comprehensive Cancer Network (NCCN) risk classifications. The C-index for each model was calculated to evaluate discrimination performance. Results: A total of 446 patients were analysed, with a median follow-up of 6.9 years for patients without an event; 46 patients developed any metastasis. For any metastatic recurrence based on biopsy findings, the CAPRA score (C-Index = 0.72) showed the highest discrimination among risk scores, surpassing biopsy GG (C-Index = 0.70), but was outperformed by percentage GP4 (C-Index = 0.74), millimetres GP4 (C-Index = 0.77), and volume of GP4 (C-Index = 0.80). Conclusion: For patients with GG 2–4 prostate cancer containing GP4, preoperative GP4 volume estimation using MRI and targeted biopsy outperforms Gleason scoring classification and standard risk scores in predicting any and distant metastatic recurrence. Further research is warranted on the best methods to quantify GP4 before incorporation in treatment decision-making. © 2025 The Author(s). BJU International published by John Wiley & Sons Ltd on behalf of BJU International.

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Language English

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Issue 6

Journal Abbr BJU Int.

Date Added 20/09/2026, 9:51:34 pm

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adult, Article, human, Humans, male, Male, middle aged, Middle Aged, risk assessment, Risk Assessment, aged, Aged, prediction, controlled study, surgery, major clinical study, follow up, scoring system, Retrospective Studies, retrospective study, diagnostic imaging, Magnetic Resonance Imaging, nuclear magnetic resonance imaging, human tissue, pathology, clinical outcome, Neoplasm Recurrence, Local, tumor recurrence, biopsy, cancer grading, cancer of the prostate risk assessment, exploratory factor analysis, Gleason score, image guided biopsy, Image-Guided Biopsy, lesion volume, magnetic resonance imaging, MRI guided biopsy, Neoplasm Grading, neoplasm metastasis, oncological outcome, positron emission tomography-computed tomography, precision medicine, prostate, Prostate, prostate cancer, prostate tumor, prostatectomy, Prostatectomy, prostatic neoplasms, Prostatic Neoplasms, radical prostatectomy

Notes:

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Export Date: 20 September 2026; Cited By: 6; Funding details: National Institutes of Health, NIH; National Cancer Institute, NCI, (P30 CA008748); National Cancer Institute, NCI; Correspondence Address: J. Olivier; Service d'urologie, Hôpital Claude Huriez, CHU de Lille, Lille, 59037, France; email: jonathan.olivier@chu-lille.fr; CODEN: BJINF

Quantification of sulfate-reducing bacteria in industrial wastewater, by real-time polymerase chain reaction (PCR) using *dsrA* and *apsA* genes

Item Type Journal Article

Author E. Ben-Dov

Author A. Brenner

Author A. Kushmaro

Abstract Real-time polymerase chain reaction (PCR) is considered a highly sensitive method for the quantification of microbial organisms in environmental samples. This study was conducted to evaluate real-time PCR with SybrGreen detection as a quantification method for sulfate-reducing bacteria (SRB) in industrial wastewater produced by several chemical industries. We designed four sets of primers and developed standard curves based on genomic DNA of *Desulfovibrio vulgaris* from pure culture and on plasmids containing dissimilatory sulfate reductase (*dsrA*) or adenosine-5'-phosphosulfate reductase (*apsA*) genes of SRB. All the standard curves, two for *dsrA* and two for *apsA* genes, had a linear range between 0.95×10^2 and 9.5×10^6 copies/ μL and between 1.2×10^3 and 1.2×10^7 copies/ μL , respectively. The theoretical copy numbers of the tenfold dilutions of *D. vulgaris* genomic DNA were best estimated (between 2.7 to 10.5 times higher than theoretical numbers) by the standard curve with DSR1F and RH3-dsr-R primers. To mimic the effect of foreign DNA in environmental samples, serial dilutions of *D. vulgaris* genomic DNA were mixed with *Escherichia coli* chromosomal DNA (40 ng per assay). This influenced neither PCR amplification nor the quantification of target DNA. Industrial wastewater was sampled during a 15-month period and analyzed for the presence of SRB, based on *dsrA* gene amplification. SRB displayed a higher abundance during the summer

(about 107-108 targets mL⁻¹) and lower during the winter (about 104-105 targets mL⁻¹). The results indicate that our real-time PCR approach can be used for detection of uncultured SRB and will provide valuable information related to the abundance of SRB in durable environmental samples, such as complex and saline industrial wastewaters. © 2007 Springer Science+Business Media, LLC.

Date 2007

DOI 10.1007/s00248-007-9233-2

URL <https://www.scopus.com/pages/publications/35148888601?origin=resultslist>

Archive Scopus

Language English

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Journal Abbr Microb. Ecol.

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Tags:

article, classification, methodology, DNA sequence, genetics, Sequence Analysis, DNA, polymerase chain reaction, microbiology, sewage, metabolism, Polymerase Chain Reaction, unclassified drug, Escherichia coli, nucleotide sequence, phylogeny, Phylogeny, Sewage, Bacteria (microorganisms), Base Sequence, molecular genetics, Molecular Sequence Data, sequence homology, adenylylsulfate reductase, bacterial protein, Bacterial Proteins, Desulfovibrio vulgaris, enzymology, Gram negative anaerobic bacteria, hydrogen sulfite reductase, Hydrogensulfite Reductase, oxidation reduction reaction, Oxidation-Reduction, oxidoreductase, Oxidoreductases Acting on Sulfur Group Donors, Sequence Homology, Nucleic Acid, Sulfur-Reducing Bacteria

Notes:

EID: 2-s2.0-35148888601

Export Date: 20 September 2026; Cited By: 120; Correspondence Address: A. Kushmaro; Department of Biotechnology Engineering, Faculty of Engineering Sciences, Ben-Gurion University of the Negev, Be'er-Sheva 84105, P.O. Box 653, Israel; email: arielkus@bgu.ac.il; CODEN: MCBEB

Quantification of tongue colour using machine learning in Kampo medicine

Item Type Journal Article

Author T. Kawanabe

Author N.D. Kamarudin

Author C.Y. Ooi

Author F. Kobayashi

Author X. Mi

Author M. Sekine

Author A. Wakasugi

Author H. Odaguchi

Author T. Hanawa

Abstract Introduction The evaluation of tongue colour has been an important approach to examine human health in Kampo medicine (traditional Japanese medicine) because the change in tongue colour may suggest physical or mental disorders. Several tongue colour quantification methods have been published to objectify clinical information among East Asian countries. However, reliable tongue colour analysis results among Japanese test persons are limited because of a lack of quantitative evaluation of tongue colour. We aimed to use advances in digital imaging processing to quantify and verify clinical data tongue colour diagnosis by characterising differences in tongue features. Methods The DS01-B tongue colour information acquisition system was used to extract tongue images of 1080 Japanese test subjects. Evaluation of tongue colour, body and coating was performed by 10 experienced Kampo medicine physicians. The acquired images were classified into five tongue body colour categories and six tongue coating colour categories based on evaluations from 10 physicians with extensive Kampo medicine experience. K-means clustering algorithm was applied as a machine learning (the study of pattern recognition by computational learning) method to the acquired images to quantify tongue body and coating colour information. Results Tongue body (n = 550) and tongue coating (n = 516) colour samples were classified and analysed. Clusters consisting of five tongue body colour categories and six tongue coating colour categories were experimentally described in the CIELAB colour space. Statistical differences were evident among the clinically primary tongue colours. Conclusions Clinically important tongue colour differences in Kampo medicine can be visualised by applying machine learning to tongue images taken under stable conditions. This has implications for developing globally unified, reliable tongue colour diagnostic criteria which could be used to explore the relevance between clinical status and tongue colour. © 2016 The Author(s)

Date 2016

DOI 10.1016/j.eujim.2016.04.002

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Journal Abbr Eur. J. Integr. Med.

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Tags:

Article, human, machine learning, Machine learning, cluster analysis, algorithm, major clinical study, K-means clustering, Chinese medicine, pattern recognition, image processing, color, image analysis, image quality, physician, priority journal, digital imaging, Kampo medicine, tongue, Tongue colour diagnosis, Traditional Japanese medicine

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Quantification of *Trichoderma afroharzianum*, *Trichoderma harzianum* and *Trichoderma gamsii* inoculants in soil, the wheat rhizosphere and in planta suppression of the crown rot pathogen *Fusarium pseudograminearum*

Item Type Journal Article

Author B.E. Stummer

Author Q. Zhang

Author X. Zhang

Author R.A. Warren

Author P.R. Harvey

Abstract Aims: Develop quantitative assays (qPCR) to determine the detection threshold limits, colonization and persistence of *Trichoderma gamsii*, *Trichoderma afroharzianum* and *T. harzianum* inoculants in cropping soils, the wheat rhizosphere and their in planta suppressive efficacy against the crown rot pathogen *Fusarium pseudograminearum*. Methods and Results: *Trichoderma* qPCR primers were designed from the internal transcribed spacer region of 5.8S rDNA and from sequences of DNA fragments diagnostic for each inoculant genotype. The minimum detection thresholds of qPCR assays varied between 1×10^3 (log 3) and 8×10^4 (log 4.9) conidia (genome) equivalents per gram of soil for multi- and single-copy target sequences, respectively and were independent of soil type. There was a strong correlation ($r^2 = 0.974$) between culture-dependent and culture-independent (qPCR) quantification methods. In wheat bioassays, *Trichoderma* inoculants colonized rhizosphere soils and wheat roots at 56–112 days postemergence to a depth of 20 cm but were more abundant ($P < 0.001$) at 0–10 cm root depth, means ranging from 2×10^2 (log 2.3) to 4×10^5 (log 5.6) conidia equivalents per gram of rhizosphere soil or root tissue. Inoculants reduced ($P < 0.001$) *F. pseudograminearum* biomass in wheat crown and root tissue by up to 5754-fold and increased ($P = 0.008$) plant biomass, relative to the pathogen control. Conclusions: The qPCR assays provided sensitive and accurate assessment of wheat root and rhizosphere soil colonization of *Trichoderma* inoculants. Strains persisted through to grain maturity at levels shown to significantly suppress *F. pseudograminearum* in planta. Significance and Impact of the Study: The qPCR assays developed here were used to determine the wheat rhizosphere dynamics of *T. harzianum*, *T. afroharzianum* and *T. gamsii* inoculants and their suppressive efficacy against *F. pseudograminearum* in planta. These assays can

be applied to monitor inoculant dynamics in suppressing crown rot and other wheat root diseases in the field. © 2020 The Society for Applied Microbiology

Date 2020

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Article, prevention and control, quantitative analysis, nonhuman, Diagnosis, classification, detection method, DNA sequence, genetics, isolation and purification, genotype, fungus, microbiology, physiology, plant root, Plant Roots, fungal disease, Triticum, wheat, soil, Soil Microbiology, Soils, Fungi, growth, development and aging, real time polymerase chain reaction, agricultural inoculation, bioassay, biocontrol, biological control agent, Biological Control Agents, colonization, conidium, Crown rot, Detection threshold, DNA sequences, DNA, Fungal, Edible Grain, food grain, fungal DNA, fungi, Fusarium, Fusarium pseudograminearum, genome, Hypocrea lixii, Marker, markers, plant disease, Plant disease, plant diseases, Plant Diseases, Polymerase chain reaction, quantitative assay, rhizosphere, Rhizosphere, Rhizospheres soil, ribosome DNA, Root tissues, Tissue, Trichoderma, Trichoderma gamsii, Trichoderma harzianum, Triticum aestivum, Ttrichoderma afroharzianum, Wheat roots

Notes:

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Quantification of workload and characterisation of key performance factors in elite adolescent female volleyball players using machine learning

Item Type Journal Article

Author T. Bouzigues

Author R. Candau

Author S. Äyrämö

Author O. Maurelli

Author J. Prioux

Abstract This study investigates key performance indicators (KPIs) influencing physical jumping performance in elite-level female volleyball players. This study aims to investigate three hypotheses: (1) the quantification of training load with “Stress Training Response” score is better in explaining and predicting jump performance than classic quantification methods such as mean or sum, (2) high intense exercises stand as primary variables explaining jump performance, and (3) non-linear models are better than linear models to explain and predict jump performance. Nineteen elite-level female volleyball players were monitored over a 190-day season. Various training-related parameters, including workload (external and internal) quantification, player positions, and menstrual cycle data, were collected. Machine learning techniques were employed to analyse and predict jump performance based on these variables. The Random Forest model outperformed other models in describing jump performance ($R^2 = 0.64$). Key performance indicators identified included workload dynamics, age, and the percentage of intense jumps made during the season. Prediction on a new dataset demonstrated promising results (Mean Absolute Error (MAE) = 4.95 cm (Confidence Interval (CI) 4.56 cm and 5.42 cm), $R^2 = 0.55$ (CI 0.45 and 0.62)). Findings suggest that intense training prior to performance enhances jump performance, with older players exhibiting superior jumping performance. © 2024 Cardiff Metropolitan University.

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Journal Abbr Int. J. Perform. Anal. Sport

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age, high intensity, Random forest, training load

Notes:

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Quantification Over Time

Item Type Conference Paper

Author F. Li

Author H.H. Gharakheili

Author G. Batista

Editor Bifet A.

Editor Davis J.

Editor Krilavicius T.

Editor Kull M.

Editor Ntoutsis E.

Editor Žliobaite I.

Abstract Quantification is the supervised machine learning task that estimates the class distribution in a sample. Therefore, quantification applications typically involve predicting aggregated quantities, such as the prevalence of positive comments about a product, personality or company on a set of social media posts. However, quantification analysis is more informative when performed over time, such as when we are interested in tracking public opinion on social media and relating changes in opinion with relevant events. The vast majority of the literature considers quantification as a standalone task, assuming the output of quantifiers to be independent even when applied to temporal data. This paper proposes a new quantification task, Quantification over Time (QoT), that allies quantification with time series forecasting methods. We propose an approach based on the Kalman filter, which can help improve the performance of standalone quantifications and a general framework that includes both ours and SOTA methods. In an experimental comparison with several textual datasets and numeral datasets, we show that our method outperforms existing methods for QoT in the literature, such as a simple composition of the classify and count method with moving averages and ReadMe2 as a standalone quantifier. We also show that our proposal can outperform several baselines, including recently proposed quantifiers used as standalone approaches. Codes are available at <https://github.com/frieli11/quantification-over-time>. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Intelligent systems, Public opinions, Social aspects, Learning systems, Machine-learning, Machine learning, Social networking (online), Supervised learning, Social media, Kalman filters, Kalman filter, Supervised machine learning, Time series, Times series, time series, Quantification, class distribution estimation, Class distribution estimation, quantification, Learning tasks, Class distributions, Quantification analysis

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EID: 2-s2.0-105023435203

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Quantification trees

Item Type Conference Paper

Author L. Milli

Author A. Monreale

Author G. Rossetti

Author F. Giannotti

Author D. Pedreschi

Author F. Sebastiani

Abstract In many applications there is a need to monitor how a population is distributed across different classes, and to track the changes in this distribution that derive from varying circumstances, an example such application is monitoring the percentage (or "prevalence") of unemployed people in a given region, or in a given age range, or at different time periods. When the membership of an individual in a class cannot be established deterministically, this monitoring activity requires classification. However, in the above applications the final goal is not determining which class each individual belongs to, but simply estimating the prevalence of each class in the unlabeled data. This task is called quantification. In a supervised learning framework we may estimate the distribution across the classes in a test set from a training set of labeled

individuals. However, this may be sub optimal, since the distribution in the test set may be substantially different from that in the training set (a phenomenon called distribution drift). So far, quantification has mostly been addressed by learning a classifier optimized for individual classification and later adjusting the distribution it computes to compensate for its tendency to either under-or over-estimate the prevalence of the class. In this paper we propose instead to use a type of decision trees (quantification trees) optimized not for individual classification, but directly for quantification. Our experiments show that quantification trees are more accurate than existing state-of-the-art quantification methods, while retaining at the same time the simplicity and understandability of the decision tree framework. © 2013 IEEE.

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DOI 10.1109/ICDM.2013.122

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Learning frameworks, Self-supervised learning, State of the art, Unlabeled data, Different class, Test sets, Training sets, Quantification methods, Monitoring activities, Time-periods, Understandability

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Export Date: 20 September 2026; Cited By: 47; Conference name: 13th IEEE International Conference on Data Mining, ICDM 2013; Conference date: 7 December 2013 through 10 December 2013; Conference code: 102826

Quantification under prior probability shift: The ratio estimator and its extensions

Item Type Journal Article

Author A.F. Vaz

Author R. Izbicki

Author R.B. Stern

Abstract The quantification problem consists of determining the prevalence of a given label in a target population. However, one often has access to the labels in a sample from the training population but not in the target population. A common assumption in this situation is that of prior probability shift, that is, once the labels are known, the distribution of the features is the same in the training and target populations. In this paper, we derive a new lower bound for the risk of the quantification problem under the prior shift assumption. Complementing this lower bound, we present a new approximately minimax class of estimators, ratio estimators, which generalize several previous proposals in the literature. Using a weaker version of the prior shift assumption, which can be tested, we show that ratio estimators can be used to build confidence intervals for the quantification problem. We also extend the ratio estimator so that it can: (i) incorporate labels from the target population, when they are available and (ii) estimate how the prevalence of positive labels varies according to a function of certain covariates. © 2019 Afonso Fernandes Vaz, Rafael Izbicki and Rafael Bassi Stern.

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Language English

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Risk perception, Probability distributions, Data set, Prior probability, Prior probability shift, Quantification, Domain shift, Semi- supervised learning, Semisupervised learning, Data set shift

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Quantification using permutation-invariant networks based on histograms

Item Type Journal Article

Author O. Pérez-Mon

Author A. Moreo

Author J.J.D. Coz

Author P. González

Abstract Quantification, also known as class prevalence estimation, is the supervised learning task in which a model is trained to predict the prevalence of each class in a given bag of examples. This paper investigates the application of deep neural networks for tasks of quantification in scenarios where it is possible to apply a symmetric supervised approach that eliminates the need for classification as an intermediate step, thus directly addressing the quantification problem. Additionally, it discusses existing permutation-invariant layers designed for set processing and assesses their suitability for quantification. Based on our analysis, we propose HistNetQ, a novel neural architecture that relies on a permutation-invariant representation based on histograms that is especially suited for quantification problems. Our experiments carried out in two standard competitions, which have become a reference in the quantification field, show that HistNetQ outperforms other deep neural network architectures designed for set processing, as well as the current state-of-the-art quantification methods. Furthermore, HistNetQ offers two significant advantages over traditional quantification methods: i) it does not require the labels of the training examples but only the prevalence values of a collection of training bags, making it applicable to new scenarios; and ii) it is able to optimize any custom quantification-oriented loss function. © The Author(s) 2024.

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DOI 10.1007/s00521-024-10721-1

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EID: 2-s2.0-85212125119

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Quantification via gaussian latent space representations

Item Type Journal Article

Author O. Pérez-Mon

Author J. José del Coz

Author P. González

Abstract Quantification, or prevalence estimation, is the task of predicting the prevalence of each class within an unknown bag of examples. Most existing quantification methods in the literature rely on prior probability shift assumptions to create a quantification model that uses the predictions of an underlying classifier to make optimal prevalence estimates. In this work, we present an end-to-end neural network that uses Gaussian distributions in latent spaces to obtain invariant representations of bags of examples. This approach addresses the quantification problem using deep learning, enabling the optimization of specific loss functions relevant to the problem and avoiding the need for an intermediate classifier, tackling the quantification problem as a direct optimization problem. Our method achieves state-of-the-art results, both against traditional quantification methods and other deep learning approaches for quantification. The code needed to reproduce all our experiments is publicly available at <https://github.com/AICGijon/gmnet>. © 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license. <http://creativecommons.org/licenses/by/4.0/>

Date 2026

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Article, human, Humans, Deep learning, Classification (of information), Learning systems, machine learning, Machine-learning, Machine learning, Gaussian distribution, Gaussians, Neural-networks, Neural networks, Deep Learning, prediction, deep learning, feature extraction, artificial neural network, Optimization, convolutional neural network, prevalence, nerve cell network, Neural Networks, Computer, classifier, classification algorithm, Classification Algorithms, Computational methods, probability, recurrent neural network, Optimal systems, Quantification methods, Probability shifts, Prior probability, Quantification, Prevalence estimation, gaussian latent space, normal distribution, Normal Distribution, Quantification model

Notes:

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Quantification-oriented learning based on reliable classifiers

Item Type Journal Article

Author J. Barranquero

Author J. Díez

Author J. José Del Coz

Abstract Real-world applications demand effective methods to estimate the class distribution of a sample. In many domains, this is more productive than seeking individual predictions. At a first glance, the straightforward conclusion could be that this task, recently identified as quantification, is as simple as counting the predictions of a classifier. However, due to natural distribution changes occurring in real-world problems, this solution is unsatisfactory. Moreover, current quantification models based on classifiers present the drawback of being trained with loss functions aimed at classification rather than quantification. Other recent attempts to address this issue suffer certain limitations regarding reliability, measured in terms of classification abilities. This paper presents a learning method that optimizes an alternative metric that combines simultaneously quantification and classification performance. Our proposal offers a new framework that allows the construction of binary quantifiers that are able to accurately estimate the proportion of positives, based on models with reliable classification abilities. © 2014 Elsevier Ltd. All rights reserved.

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Reliability, Forecasting, Pattern recognition, Classification performance, Quantification, Class distribution estimation, Quantification model, Natural distribution, Performance metrics, Individual prediction, Classification ability, Multivariate predictions

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QuantificationLib: A Python library for quantification and prevalence estimation

Item Type Journal Article

Author A. Castaño

Author J. Alonso

Author P. González

Author P. Pérez

Author J.J. del Coz

Abstract QuantificationLib is an open-source Python library that provides a comprehensive set of algorithms for quantification learning. Quantification, also known as prevalence estimation, is a supervised machine-learning task where the objective is to train a model that is able to predict the distribution of classes in a set of unseen examples or bags. This library offers a wide variety of quantification methods suited for easy prototyping and experimentation, applicable to a wide range of quantification applications. © 2024 The Authors

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DOI 10.1016/j.softx.2024.101728

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Supervised learning, Python, Supervised machine learning, High level languages, Open-source, Quantification methods, Prevalence estimation, Learning tasks, Quantification learning, Ordinal quantification

Notes:

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Quantified groundwater–surface water dynamics across Tigris–Euphrates basin through integrated modelling

Item Type Journal Article**Author** Q. Alkhawlan**Author** S. Algburi**Author** S. Sabeeh**Author** M. Alsubih**Author** S. Islam**Author** R. Abad Khan

Abstract Water scarcity in Iraq's Tigris–Euphrates basin is intensifying under climate change, upstream regulation, and increasing demand, creating pressure on groundwater and surface water sustainability. This study develops an integrated framework that merges Gravity Recovery and Climate Experiment groundwater anomalies, Global Land Data Assimilation System soil moisture fields, Moderate Resolution Imaging Spectroradiometer evapotranspiration, hydrometric observations, and machine learning to quantify groundwater–surfacewater exchanges. An extreme gradient boosting model predicts groundwater levels with R^2 of 0.92 and root mean square error of 0.15 m. Results indicate groundwater storage declines ranging from -1.4 cm year $^{-1}$ in the north to -0.9 cm year $^{-1}$ in the south during 2002–2023, with seasonal deficits amplifying depletion. Exchange flux analysis reveals groundwater discharge sustaining rivers during droughts and flood-induced recharge during high flows. Scenario projections to 2050 show moderated depletion under RCP4.5 but accelerated decline under RCP8.5, emphasizing the need for adaptive abstraction

control and managed recharge strategies. © 2026 IAHS.

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Quantifying consciousness through intrinsic probability density function

Item Type Journal Article

Author N.E. Huang

Author W.-S. Yuan

Author A.C. Yang

Author T.B.J. Kuo

Author W.-X. Tang

Author H. Kang

Author M. Wagner

Author W.-K. Liang

Abstract Consciousness remains a multifaceted phenomenon that is difficult to be measured by traditional quantification methods. Here we propose the intrinsic probability density function (iPDF) as a quantitative method to evaluate the dynamic inter-cortical interactions that underlie conscious states. First, the method utilizes empirical mode decomposition to derive intrinsic mode functions (IMFs) from EEG signals. Then, the method generates scale-dependent probability density functions for successive partial sums of IMFs that can capture subtle variations in neural modulation patterns. We tested the iPDF analysis across various consciousness states such as general anesthesia, distinct sleep stages (wakefulness, REM, and deep sleep), sensory conditions (eyes open versus eyes closed), and between dementia patients and healthy subjects. Our findings reveal that active neural interactions or modulations during wakefulness and REM sleep are characterized by super-Gaussian iPDF patterns. By contrast, the reduced interactions observed in anesthesia and deep sleep yield near-Gaussian iPDF profiles. We also present a classification model built on iPDF features that achieved an accuracy of approximately 87 % in distinguishing dementia patients from health controls, demonstrating the iPDF as a potential biomarker in clinical screening. This study supports the idea that consciousness emerges from complex, scale-dependent neural processes and presents a robust, quantitative framework that may enhance both our theoretical understanding and practical assessment of various states of consciousness. © 2025 The Authors

Date 2025

DOI 10.1016/j.biopsycho.2025.109101

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adult, Adult, Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, aged, Aged, brain, Brain, theoretical study, quantitative analysis, controlled study, major clinical study, probabilistic reasoning, Probability, Dementia, probability, measurement accuracy, physiology, pathophysiology, screening, electroencephalogram, Electroencephalography, brain cortex, cognition assessment, Cognitive states, Conscious states, consciousness, Consciousness, consciousness level, dementia, electroencephalography, EMD, Emperical Mode Decomposition, empirical research, general anesthesia, Intercortical communications, Intrinsic Probability Density Function, IPDF, molecular interaction, nerve cell membrane potential, Neural interactions, Neural modulations, REM sleep, sensory processing, Sleep, sleep stage, Sleep Stages, slow wave sleep, wakefulness, Wakefulness

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Quantifying construction and demolition waste: An analytical review

Item Type Journal Article**Author** Z. Wu**Author** A.T.W. Yu**Author** L. Shen**Author** G. Liu

Abstract Quantifying construction and demolition (C&D) waste generation is regarded as a prerequisite for the implementation of successful waste management. In literature, various methods have been employed to quantify the C&D waste generation at both regional and project levels. However, an integrated review that systemically describes and analyses all the existing methods has yet to be conducted. To bridge this research gap, an analytical review is conducted. Fifty-seven papers are retrieved based on a set of rigorous procedures. The characteristics of the selected papers are classified according to the following criteria - waste generation activity, estimation level and quantification methodology. Six categories of existing C&D waste quantification methodologies are identified, including site visit method, waste generation rate method, lifetime analysis method, classification system accumulation method, variables modelling method and other particular methods. A critical comparison of the identified methods is given according to their characteristics and implementation constraints. Moreover, a decision tree is proposed for aiding the selection of the most appropriate quantification method in different scenarios. Based on the analytical review, limitations of previous studies and recommendations of potential future research directions are further suggested. © 2014 Elsevier Ltd.

Date 2014**DOI** 10.1016/j.wasman.2014.05.010**URL** <https://www.scopus.com/pages/publications/84905591201?origin=resultslist>**Archive** Scopus**Language** English**Volume** 34**Publisher** Elsevier Ltd**Pages** 1683-1692**Publication** Waste Management**ISSN** 0956053X (ISSN)**Issue** 9

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Notes:

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Export Date: 20 September 2026; Cited By: 367; Correspondence Address: A.T.W. Yu; Department of Building and Real Estate, Hong Kong Polytechnic University, Kowloon, HKG, Hong Kong; email: bsannyu@polyu.edu.hk; CODEN: WAMAE

Quantifying counts and costs via classification

Item Type Journal Article

Author G. Forman

Abstract Many business applications track changes over time, for example, measuring the monthly prevalence of influenza incidents. In situations where a classifier is needed to identify the relevant incidents, imperfect classification accuracy can cause substantial bias in estimating class prevalence. The paper defines two research challenges for machine learning. The 'quantification' task is to accurately estimate the number of positive cases (or class distribution) in a test set, using a training set that may have a substantially different distribution. The 'cost quantification' variant estimates the total cost associated with the positive class, where each case is tagged with a cost attribute, such as the expense to resolve the case. Quantification has a very different utility model from traditional classification research. For both forms of quantification, the paper describes a variety of methods and evaluates them with a suitable methodology, revealing which methods give reliable estimates when training data is scarce, the testing class distribution differs widely from training, and the positive class is rare, e.g., 1% positives. These strengths can make quantification practical for business use, even where classification accuracy is poor. © 2008 Springer Science+Business Media, LLC.

Date 2008

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Quantifying COVID-19 Content in the Online Health Opinion War Using Machine Learning

Item Type Journal Article

Author R.F. Sear

Author N. Velasquez

Author R. Leahy

Author N.J. Restrepo

Author S.E. Oud

Author N. Gabriel

Author Y. Lupu

Author N.F. Johnson

Abstract A huge amount of potentially dangerous COVID-19 misinformation is appearing online. Here we use machine learning to quantify COVID-19 content among online opponents of establishment health guidance, in particular vaccinations ('anti-vax'). We find that the anti-vax community is developing a less focused debate around

COVID-19 than its counterpart, the pro-vaccination ('pro-vax') community. However, the anti-vax community exhibits a broader range of 'flavors' of COVID-19 topics, and hence can appeal to a broader cross-section of individuals seeking COVID-19 guidance online, e.g. individuals wary of a mandatory fast-tracked COVID-19 vaccine or those seeking alternative remedies. Hence the anti-vax community looks better positioned to attract fresh support going forward than the pro-vax community. This is concerning since a widespread lack of adoption of a COVID-19 vaccine will mean the world falls short of providing herd immunity, leaving countries open to future COVID-19 resurgences. We provide a mechanistic model that interprets these results and could help in assessing the likely efficacy of intervention strategies. Our approach is scalable and hence tackles the urgent problem facing social media platforms of having to analyze huge volumes of online health misinformation and disinformation. © 2013 IEEE.

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machine learning, Machine learning, Health, E-learning, Social media platforms, topic modeling, COVID-19, social computing, Vaccines, Community IS, Herd immunities, Intervention strategy, mechanistic model, Mechanistic modeling, Urgent problems

Notes:

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Export Date: 20 September 2026; Cited By: 101; Correspondence Address: N.F. Johnson; Institute for Data Democracy and Politics, George Washington University, Washington, 20052, United States; email: neiljohnson@gwu.edu

Quantifying Deep Learning Model Uncertainty in Conformal Prediction

Item Type Conference Paper

Author H. Karimi

Author R. Samavi

Editor Soh H.

Editor Geib C.

Editor Petrick R.

Abstract Precise estimation of predictive uncertainty in deep neural networks is a critical requirement for reliable decisionmaking in machine learning and statistical modeling, particularly in the context of medical AI. Conformal Prediction (CP) has emerged as a promising framework for representing the model uncertainty by providing well-calibrated confidence levels for individual predictions. However, the quantification of model uncertainty in conformal prediction remains an active research area, yet to be fully addressed. In this paper, we explore state-of-the-art CP methodologies and their theoretical foundations. We propose a probabilistic approach in quantifying the model uncertainty derived from the produced prediction sets in conformal prediction and provide certified boundaries for the computed uncertainty. By doing so, we allow model uncertainty measured by CP to be compared by other uncertainty quantification methods such as Bayesian (e.g., MC-Dropout and DeepEnsemble) and Evidential approaches. © 2023, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Quantifying Elevation Changes Under Engineering Measures Using
Multisource Remote Sensing and Interpretable Machine Learning: A Case
Study of the Chinese Loess Plateau

Item Type Journal Article**Author** S. Zhou**Author** Q. Zhu**Author** S. Li

Abstract Highlights: What are the main findings? Interpretable machine learning models identify the relative importance and spatial patterns of environmental and anthropogenic factors affecting elevation change. What are the implications of the main findings? The transparent analytical framework supports land management and targeted implementation of erosion control measures on the Loess Plateau. This approach bridges remote sensing analytics and geomorphological processes, facilitating large-scale environmental monitoring and sustainable erosion management. Understanding the effectiveness of engineering measures in mitigating surface erosion is crucial for sustainable land management. However, studies explicitly quantifying the combined effects of large-scale engineering measures and environmental factors remain limited. In this study, multisource remote sensing data were integrated with interpretable machine learning to quantify and analyze the regional influence of erosion control measures. We constructed a comprehensive indicator system encompassing spectral, textural, and topographic variables derived from high-resolution satellite imagery and DEM data. To address model transparency and enhance the interpretability of the results, we employed an interpretable machine learning framework capable of both accurate prediction and explicit attribution of feature importance. The results indicate that the implementation of engineering measures substantially reduces erosion intensity across the study area. Spatial heterogeneity in erosion mitigation effectiveness was closely associated with the distribution patterns of engineering measures and site-specific environmental conditions. Basins with a high proportion of check dams showed average elevation gains of up to 2.5 m compared with those without check dams, and terraces contributed to elevation increases of ~1.9 m in typical loess hilly regions. The interpretable machine learning model achieved $R^2 = 0.62$ at Basin 1 (average area ~100 km²) and $R^2 = 0.73$ at Basin 2 (~600 km²), demonstrating reliable predictive capability. The findings not only validate the role of engineering interventions in erosion mitigation but also provide a transparent analytical framework that connects remote sensing analytics with process-based geomorphological understanding. © 2025 by the authors.

Date 2025**DOI** 10.3390/rs17203451**URL** <https://www.scopus.com/pages/publications/105020260900?origin=resultslist>**Archive** Scopus**Language** English**Volume** 17**Publisher** Multidisciplinary Digital Publishing Institute (MDPI)**Publication** Remote Sensing**ISSN** 20724292 (ISSN)**Issue** 20**Journal Abbr** Remote Sens.**Date Added** 20/09/2026, 9:51:48 pm**Modified** 20/09/2026, 9:51:48 pm**Tags:**

Learning systems, machine learning, Machine-learning, Machine learning, Remote sensing, Remote-sensing, Machine learning models, Multisource data, Land use, Landforms, Erosion, Satellite imagery, Control measures, elevation change, Elevation change, Engineering measures, Environmental factors, Environmental management, Erosion control, Geomorphology, Loess Plateau, multisource data, Sediments

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Export Date: 20 September 2026; Cited By: 3; Funding details: National Natural Science Foundation of China, NNSFC, (42401507); Natural Science Foundation of Jiangsu Province, (BK20240600); Correspondence Address: S. Li; State Key Laboratory of Climate System Prediction and Risk Management, Nanjing Normal University, Nanjing, 210023, China; email: lisijin@njnu.edu.cn

Quantifying fire severity: a brief review and recommendations for improvement

Item Type Journal Article

Author G. Yang

Author D. Han

Author X. Di

Author L. Sun

Author Y. Weng

Abstract Introduction: Quantifying fire severity is an important aspect of studying the response mechanism of terrestrial ecosystems to wildfire, and it is of great significance to fire ecology. In this paper we comprehensively introduce and compare the classification and quantification methods for fire severity; we discuss the development and application status of various methods, and we elucidate their existing problems. Results: 1) According to features of the burned area, fire severity can be classified as light, moderate, and heavy. 2) Using composite burn index (CBI) to quantify and record the fire severity. 3) In quantifying fire severity with vegetation change, there are certain limitations and theoretical problems to be solved. 4) Remote sensing could very well be an important means of measuring fire severity in the future, but there are still many problems that need to be solved before the remote sensing index can become a global fire severity indicator. Discussion and Conclusion: Only by clarifying the relationship between fire behavior, fire severity, time related variables and the pre- and post-fire ecosystem can the existing models be perfected or new, better fire severity measurement models be proposed for broad applications. © 2021 The Author(s). Published by Taylor & Francis Group and Science Press on behalf of the Ecological Society of China.

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fire behavior, composite burn index, ecosystem response, fire intensity, Fire severity, remote sensing index

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Quantifying homeliness: Bridging the semantic gap in China's elderly care facility design

Item Type Journal Article

Author J. He

Author J. Jiang

Author Y. Hou

Author Y. Zhao

Author Y. Qi

Author D. Zhou

Author J. Dai

Abstract In response to global aging, creating “homeliness”-oriented elderly care facilities has become a core goal to improve the elderly’s quality of life and emotional attachment to the environment. However, “homeliness” remains a subjective concept, lacking objective metrics to guide spatial design. This study introduces a novel methodology combining OneFormer-based image semantic segmentation and machine learning, to quantify the perceptual quality of spatial environments, and translate the findings into actionable design strategies for elderly care facilities. Through field surveys and interviews across 24 home care and 20 institutional care facilities in Xi'an, combined with a systematic literature review, we first established a quantifiable spatial assessment system based on elderly living patterns and emotional needs. Using OneFormer for spatial image semantic segmentation, we extracted and analyzed

environmental indicators from spatial imagery, identifying key qualitative and quantitative differences between the home care and institutional care settings. Multiple machine learning models, interpreted via SHAP values, were then applied to elucidate the relationships between spatial elements and perceived “homeliness.” Based on “homeliness” characteristics, the study proposes spatial reconstruction strategies (unit types, layout) and targeted design advice for residential spaces, communal activity spaces and corridors, to build elderly care environments that enhance the well-being of the elderly. © 2026 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY license. <http://creativecommons.org/licenses/by/4.0/>

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Learning systems, Machine learning, Semantics, Semantic Segmentation, Design, Semantic segmentation, SHAP, Image analysis, Elderly care, Facilities, Facility designs, Home care, Homeliness, Image semantic segmentation, Image semantics, Institutional care, Semantic gap

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Quantifying Inflammatory Response and Drug-Aided Resolution in an Atopic Dermatitis Model with Deep Learning

Item Type Journal Article

Author D.A. Greenfield

Author A. Feizpour

Author C.L. Evans

Abstract Noninvasive quantification of dermal diseases aids efficacy studies and paves the way for broader enrollment in clinical studies across varied demographics. Related to atopic dermatitis, accurate quantification of the onset and resolution of inflammatory flare ups in the skin remains challenging because the commonly used macroscale cues do not necessarily represent the underlying inflammation at the cellular level. Although atopic dermatitis affects over 10% of Americans, the genetic underpinnings and cellular-level phenomena causing the physical manifestation of the disease require more clarity. Current gold standards of quantification are often invasive, requiring biopsies followed by laboratory analysis. This represents a gap in our ability to diagnose and study skin inflammatory disease as well as develop improved topical therapeutic treatments. This need can be addressed through noninvasive imaging methods and the use of modern quantitative approaches to streamline the generation of relevant insights. This work reports the noninvasive image-based quantification of inflammation in an atopic dermatitis mouse model on the basis of cellular-level deep learning analysis of coherent anti-Stokes Raman scattering and stimulated Raman scattering imaging. This quantification method allows for timepoint-specific disease scores using morphological and physiological measurements. The outcomes we show set the stage for applying this workflow to future clinical studies. © 2023 The Authors

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Article, female, comparative study, Deep Learning, quantitative analysis, nonhuman, deep learning, controlled study, inflammation, morphology, disease severity, clinical assessment, pilot study, transfer of learning, gold standard, diagnostic imaging, image analysis, binary classification, pathology, animal, Animals, animal experiment, animal model, Mice, mouse, treatment outcome, skin, isoflurane, Administration, Topical, atopic dermatitis, coherent anti Stokes Raman spectroscopy, Dermatitis, Atopic, ear thickness, hairless mouse, Inflammation, Janus kinase inhibitor, leave one out cross validation, Skin, statistical significance, topical drug administration, water content

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Export Date: 20 September 2026; Cited By: 8; Correspondence Address: C.L. Evans; Wellman Center for Photomedicine, Harvard Medical School/Massachusetts General Hospital, Charlestown, CNY149-3210, 13th Street, 02129, United States; email: evans.conor@mgh.harvard.edu; CODEN: JIDEA

Quantifying Interdisciplinarity in Scientific Articles Using Deep Learning Toward a TRIZ-Based Framework for Cross-Disciplinary Innovation

Item Type Journal Article

Author N. Douard

Author A. Samet

Author G. Giakos

Author D. Cavallucci

Abstract Interdisciplinary research (IDR) is essential for addressing complex global challenges that surpass the capabilities of any single discipline. However, measuring interdisciplinarity remains challenging due to conceptual ambiguities and inconsistent methodologies. To overcome these challenges, we propose a deep learning approach that quantifies interdisciplinarity in scientific articles through semantic analysis of titles and abstracts. Utilizing the Semantic Scholar Open Research Corpus (S2ORC), we leveraged metadata field tags to categorize papers as either interdisciplinary or monodisciplinary, establishing the foundation for supervised learning in our model. Specifically, we preprocessed the textual data and employed a Text Convolutional Neural Network (Text CNN) architecture to identify semantic patterns indicative of interdisciplinarity. Our model achieved an F1 score of 0.82, surpassing baseline machine learning models. By directly analyzing semantic content and incorporating metadata for training, our method addresses the limitations of previous approaches that rely solely on bibliometric features such as citations and co-authorship. Furthermore, our large-scale analysis of 136 million abstracts revealed that approximately 25% of the literature within the specified disciplines is interdisciplinary. Additionally, we outline how our quantification method can be integrated into a TRIZ-based (Theory of Inventive Problem Solving) methodological framework for cross-disciplinary innovation, providing a foundation for systematic knowledge transfer and inventive problem solving across domains. Overall, this approach not only offers a scalable measurement of interdisciplinarity but also contributes to a framework for facilitating innovation through structured cross-domain knowledge integration. © 2025 by the authors.

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Quantifying irregular pulsation of intracranial aneurysms using 4D-CTA

Item Type Journal Article

Author H. Xie

Author H. Yu

Author H. Wu

Author J. Wang

Author S. Wu

Author J. Zhang

Author H. Zhao

Author M. Yuan

Author J. Benitez Mendieta

Author H. Anbananthan

Author C. Winter

Author C. Zhu

Author Z. Li

Abstract Recent studies have suggested that irregular pulsation of intracranial aneurysm during the cardiac cycle may be potentially associated with aneurysm rupture risk. However, there is a lack of quantification method for irregular pulsations. This study aims to quantify irregular pulsations by the displacement and strain distribution of the intracranial aneurysm surface during the cardiac cycle using four-dimensional CT angiographic image data. Four-dimensional CT angiography was performed in 8 patients. The image data of a cardiac cycle was divided into approximately 20 phases, and irregular pulsations were detected in four intracranial aneurysms by visual observation, and then the displacement and strain of the intracranial aneurysm was quantified using coherent point drift and finite element method. The displacement and strain were compared between aneurysms with irregular and normal pulsations in two different ways (total and stepwise). The stepwise first principal strain was

significantly higher in aneurysms with irregular than normal pulsations (0.20 ± 0.01 vs 0.16 ± 0.02 , $p=0.033$). It was found that the irregular pulsations in intracranial aneurysms usually occur during the consecutive ascending or descending phase of volume changes during the cardiac cycle. In addition, no statistically significant difference was found in the aneurysm volume changes over the cardiac cycle between the two groups. Our method can successfully quantify the displacement and strain changes in the intracranial aneurysm during the cardiac cycle, which may be proven to be a useful tool to quantify intracranial aneurysm deformability and aid in aneurysm rupture risk assessment. © 2024 The Authors

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Risk assessment, adult, Adult, Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, risk assessment, machine learning, aged, Aged, Heart, algorithm, image segmentation, time series analysis, procedures, Image data, retrospective study, Quantification methods, diagnostic imaging, image analysis, contrast medium, image quality, three-dimensional imaging, pathophysiology, Computerized tomography, radiation dose, computed tomographic angiography, Intracranial aneurysms, 4d-CTA, 4D-CTA, Aneurysm rupture, aneurysm volume, Cardiac cycles, Computed Tomography Angiography, Displacement distribution, finite element analysis, four dimensional computed tomography, Four-Dimensional Computed Tomography, heart cycle, intracranial aneurysm, Intracranial aneurysm, Intracranial Aneurysm, Irregular pulsation, pulsatile flow, Pulsatile Flow, pulse wave, RR interval, Rupture risk, Strain, viscosity, Volume change

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Quantifying microstickies via a new agglomeration technique

Item Type Conference Paper

Author Y. Ben

Author M. Ricard

Author G. Dorris

Abstract Despite the recognized importance of microstickies to the various deposition problems experienced in pulping, papermaking and printing operations, there is still no consensus on a proper method to quantify them. One proposed method is to agglomerate them in a pulp sample by a long cold storage or by exposure to reduced pressure conditions followed by a classification step to isolate them from the pulp, as macrostickies. In this paper, we tested this approach but using a more effective agglomeration method to convert microstickies into macrostickies. Using model systems to validate the microstickies quantification method, we describe how a relatively short boiling treatment along with controlled mixing promotes and speeds up the agglomeration process.

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Testing, Printing, Image analysis, Quantification methods, Quantification, Deposition, Agglomeration, Boiling, Boiling liquids, Cold storage, Macro stickies, Method, Micro-stickies, Microstickies, Paper Making, Printing operations, Problem Solving, Pulp, Pulping, Reduced pressure, Stickies, Test.

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Quantifying multi-dimensional services of water ecosystems and breakpoint-based spatial radiation of typical regulating services considering the hierarchical clustering-based classification

Item Type Journal Article

Author X. Guan

Author Y. Xu

Author Y. Meng

Author W. Xu

Author D. Yan

Abstract This study proposes a set of water ecosystem services (WES) research system, including classification, benefit quantification and spatial radiation effect, with the goal of promoting harmonious coexistence between humans and nature, as well as providing a theoretical foundation for optimizing water resources management. Hierarchical cluster analysis was applied to categorize WES taking in to account the four nature constraints of product nature, energy flow relationships, circularity, and human social utility. A multi-dimensional benefit quantification methodology system for WES was constructed by combining the emergy theory with multidisciplinary methods of ecology, economics, and sociology. Based on the theories of spatial autocorrelation and breaking point, we investigated the spatial radiation effects of typical services in the cyclic regulation category. The proposed methodology has been applied to Luoyang, China. The results show that the Resource Provisioning (RP) and Cultural Addition (CA) services change greatly over time, and drive the overall WES to increase and then decrease. The spatial and temporal distribution of water resources is uneven, with WES being slightly better in the southern region than the northern region. Additionally, spatial radiation effects of typical regulating services are most prominent in S County. This finding suggests the establishment of scientific and rational intra-basin or inter-basin water management systems to expand the beneficial impacts of water-rich areas on neighboring regions. © 2023 Elsevier Ltd

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Quantifying rainfall teleconnections with climate modes in agro climatic zones of India

Item Type Journal Article

Author S. Sah

Author R.N. Singh

Author B. Das

Author G. Chaturvedi

Author A.K. Singh

Author K.S. Reddy

Abstract Understanding the spatiotemporal teleconnections of rainfall in India with large-scale modes of climate variability is crucial for agricultural planning, water resource management, and climate adaptation. This study investigates the long-term teleconnections of monsoon rainfall with the Indian Ocean Dipole (IOD) and El Niño-Southern Oscillation (ENSO) across different Agro Climatic Zones (ACZs) of India from 1901 to 2022 using correlation, partial correlation and interpretable machine learning to quantify their impacts in different ACZs. Results revealed significant correlations and partial correlations between monsoon rainfall with the June-July-August-September (JJAS), October-November-December (OND), and annual (AN) mean values of ENSO indices, while the correlation with IOD was mostly non-significant. The January-February (JF) and March-April-May (MAM) mean values of both ENSO and IOD indices were poorly correlated with rainfall. The JJAS mean values of indices consistently showed the strongest correlations and partial correlations with monsoon rainfall across all ACZs. Comparing the periods 1903–1962 and 1963–2022 revealed evolving teleconnections, suggesting a weakening influence in the ACZs of western and central parts of India and a strengthening influence in the ACZs of southern, eastern, and northern parts of the country. We also identified key climate indices using the Boruta algorithm to train a random forest model and quantified their impacts on rainfall variability using accumulated local effect plots for each ACZ, which revealed that ENSO has a more dominant impact on rainfall than the IOD in most ACZs. This analysis provides insights into regional climate dynamics and will aid in improved planning and management of water resources. © The Author(s), under exclusive licence to Springer-Verlag GmbH Austria, part of Springer Nature 2025.

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Quantifying riverbank failure risk in tidal river deltas: A hybrid physics-based and data-driven framework

Item Type Journal Article

Author L. Cong Chinh

Author N.T. Pham

Author T. Khac Thac

Abstract The complex, nonlinear synergy between three-dimensional hydrodynamics and transient geotechnical processes governs riverbank erosion in tidal river deltas. Traditional assessments often decouple these mechanisms, limiting predictive accuracy under combined extreme forcings. This study proposes a novel hybrid deterministic-stochastic framework that integrates in-situ observations, physics-based modelling, and machine learning to quantify riverbank instability. First, a 3D RANS hydrodynamic model (Delft3D) was one-way coupled with transient seepage and limit-equilibrium models (GeoStudio) to generate model-derived reference stability states. Using the Buckingham Pi theorem, the physical variables were reduced to two critical dimensionless groups representing hydrodynamic toe scour (driven by secondary flows) and geotechnical susceptibility (driven by rapid drawdown pore-pressure lag). To formulate a generalised predictive metric, a Monte Carlo simulation was executed to generate a diverse synthetic dataset, upon which Logistic Regression was applied to calibrate a novel Composite Erosion Index (IE). The index was calibrated on a synthetic dataset comprising 500 Monte Carlo scenarios, each labelled by the coupled deterministic models. The data-driven calibration achieved a classification accuracy of 96.7% with respect to these model-derived reference labels, quantifying the fidelity of the closed-form index to the physics-based simulations rather than field-predictive skill, and isolated a decision boundary ($IE \geq 1.0$) that normalises the disparate magnitudes of fluid shear and soil resistance. Application to the highly dynamic Vam Nao River in the Vietnamese Mekong Delta indicated that catastrophic mass failures are most likely to occur when severe secondary currents and rapid tidal drawdown coincide in space and time. The proposed index provides a scalable screening tool for assessing geomorphic hazards and supporting the design of Nature-based Solutions (NbS) for riverbank stabilisation in complex deltaic environments. The calibrated coefficients are site-specific, and independent validation against dated field-failure inventories remains a priority for future work. © 2026 The Authors.

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QUANTIFYING ROBUSTNESS IN TALL (TIMBER) BUILDINGS: A CASE STUDY

Item Type Conference Paper

Author K. Voulpiotis

Author S. Schär

Author A. Frangi

Editor Nyrud A.Q.

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Abstract Robustness research has become popular, however very little is known on its explicit quantification. This paper summarises a quantification method previously published by the main author and proceeds in demonstrating its step-by-step application with a case study tall timber building. A hypothetical 15-storey timber building is designed for normal loads and four improved options are designed to account for abnormal loads in order to increase the building's robustness. A detailed, nonlinear dynamic Finite Element model is set up in Abaqus® to model three ground floor column removal scenarios, and a Random Forest classifier is set up to propagate uncertainties and efficiently calculate the probability of certain collapse classes occurring and the importance of each input parameter. The results show how design improvements at the whole building scale (e.g., strong floors) have a higher impact on robustness performance than just improving the strength and ductility of some selected connections, although these results are exclusive to the building studied. The whole procedure is put in context of the practicing engineer, with a suggestion for a calculation-free, purely

qualitative robustness framework. The case study reinforces the importance of a sound conceptual design for achieving robustness in tall timber buildings. © 13th World Conference on Timber Engineering, WCTE 2023. All rights reserved.

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machine learning, Machine-learning, Machine learning, Case-studies, Conceptual design, Quantification methods, Structural design, Floors, Finite element analyse, Architectural design, ABAQUS, Abnormal loads, Disproportionate collapse, Dynamic finite element model, Finite Element Analysis, Finite element method, Normal loads, Random forest classification, Random Forest classification, Timber, Timber buildings, Wooden buildings

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Quantifying robustness in tall timber buildings: A case study

Item Type Journal Article

Author K. Voulpiotis

Author S. Schär

Author A. Frangi

Abstract Robustness research has become popular, however very little is known on its explicit quantification. This paper summarises a quantification method previously published by the main author and proceeds in demonstrating its step-by-step application with a case study tall timber building. A hypothetical 15-storey post-and-beam timber

building with a central core is designed for normal loads, and four improved options are designed to account for abnormal loads in order to increase the building's robustness. A detailed, nonlinear, dynamic Finite Element model is set up in Abaqus® to model three ground floor column removal scenarios, and a Random Forest classifier is set up to propagate uncertainties, to efficiently estimate the probability of certain collapse classes occurring, and to calculate the importance of each input parameter. The results show how design improvements at the whole building scale (e.g., strong floors) have a higher impact on robustness performance than just improving the strength and ductility of some selected connections, although these results are exclusive to the building studied. The case study reinforces the importance of a sound conceptual design for achieving robustness in tall timber buildings. © 2022 The Authors

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Quantifying Soil Microbiome Abundance by Metatranscriptomics and Complementary Molecular Techniques—Cross-Validation and Perspectives

Item Type Journal Article**Author** M.B. Dahl**Author** S. Brachmann**Author** A. Söllinger**Author** M. Schnell**Author** L. Ahlers**Author** M. Wutkowska**Author** K.J. Hoff**Author** N. Nath**Author** V. Groß**Author** H. Wang**Author** M. Weil**Author** M. Piecha**Author** M. Schaffer**Author** C. Jensen**Author** A.W. Kuss**Author** C. Gall**Author** E. Wimmer**Author** T. Pribasnig**Author** A.T. Tveit**Author** B.D. Sigurdsson**Author** C. Schleper**Author** A. Richter**Author** T. Urich

Abstract Linking meta-omics and biogeochemistry approaches in soils has remained challenging. This study evaluates the use of an internal RNA extraction standard and its potential for making quantitative estimates of a given microbial community size (biomass) in soil metatranscriptomics. We evaluate commonly used laboratory protocols for RNA processing, metatranscriptomic sequencing and quantitative reverse transcription polymerase chain reaction (qRT-PCR). Metatranscriptomic profiles from soil samples were generated using two library preparation protocols and prepared in triplicates. RNA extracted from pure cultures of *Saccharolobus solfataricus* was added to the samples as an internal nucleic acid extraction standard (NAEstd). RNA reads originating from NAEstd were identified with a 99.9% accuracy. A remarkable replication consistency between triplicates was seen (average Bray–Curtis dissimilarity 0.03 ± 0.02), in addition to a clear library preparation bias. Nevertheless, the between-sample pattern was not affected by library type. Estimates of 16S rRNA transcript abundance derived from qRT-PCR experiments, NAEstd and a previously published quantification method of metatranscriptomics (hereafter qMeTra) were compared with microbial biomass carbon (MBC) and nitrogen (MBN) extracts. The derived biomass estimates differed by orders of magnitude. While most estimates were significantly correlated with each other, no correlation was observed between NAEstd and MBC extracts. We discuss how simultaneous changes in community size and the soils nucleic acid retention strength might hamper accurate biomass estimation. Adding NAEstd has the potential to shed important light on nucleic acid retention in the substance matrix (e.g., soil) during extraction. © 2025 The Author(s). Molecular Ecology Resources published by John Wiley & Sons Ltd.

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Quantifying technology-industry spillover effects based on patent citation network analysis of unmanned aerial vehicle (UAV)

Item Type Journal Article

Author D.H. Kim

Author B.K. Lee

Author S.Y. Sohn

Abstract Unmanned aerial vehicle (UAV) technologies have been fast developing over the past 20 years and are expected to generate extensive spillovers into other industry sectors. However, no previous studies have investigated such spillover effects. In this study, we propose the framework of two-mode network analysis to quantify the spillover effects of UAV technology into various industries using patent citation data of the United States Patent and Trademark Office. A two-mode matrix consists of rows corresponding to UAV technologies and columns corresponding to beneficiary industries, and the value depicts the spillover probability obtained using International

Patent Classification codes and the technology/industry concordance table. The out- and in-degree centralities of the spillover network are used to identify strong spillover-generating UAV technologies and strong spillover-receiving industries, respectively. We observed that the weapon industry received extensive spillover effects during the period 2005-2009. Based on Mann-Kendall tests, the spillover effects of UAV-related software technologies exhibited a consistently upward trend during both the last 10 and 20 years. The past significant trend of spillovers can help us to forecast future trends. The proposed quantification method can be readily applied to investigate other specific technology-industry spillover patterns. © 2016 Elsevier Inc.

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Unmanned aerial vehicle, Unmanned aerial vehicles (UAV), Vehicles, United States, unmanned vehicle, technological development, Patents and inventions, Software testing, network analysis, Quantification methods, spillover effect, intellectual property rights, International patents, MERIT concordance, Patent citation analysis, Patent citation network analysis, Software technology, Technology concordance, Technology industry, Technology industry spillovers, United States Patent and Trademark Office

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Quantifying terrestrial carbon in the context of climate change: a review of common and novel technologies and methods

Item Type Journal Article**Author** S. Gameiro**Author** M.E. Ferreira**Author** L.F.C. Ruiz**Author** G.L. Galford**Author** M. Zeraatpisheh**Author** V.F. Nascimento**Author** R.G. Collevatti

Abstract Background: Understanding carbon dynamics in Earth's ecosystem is necessary for mitigating climate change. With recent advancements in technologies, it is important to understand both how carbon quantification in soil and vegetation is measured and how it can be improved. Therefore, this study conducted a bibliometric and bibliographic review of the most common carbon quantification methodologies. Results: Among the most widely used techniques, the Walkley-Black method and Elemental Analysis stand out for measuring below-ground carbon, while forest inventories are prominent for assessing above-ground carbon. Additionally, we found that the United States and China have the largest number of publications on this topic, with forest and agricultural areas being the most studied, followed by grasslands and mangroves. However, it should be noted that despite being indirect techniques, remote sensing, regression analysis, and machine learning have increasingly been used to generate geo-environmental carbon models for various areas. Landsat satellite images are the most widely used in remote sensing, followed by LiDAR digital models. Conclusions: These results demonstrate that while new technologies do yet not replace analytical techniques, they are valuable allies working in conjunction with the current carbon quantification process. © The Author(s) 2025.

Date 2025**DOI** 10.1186/s13021-025-00316-1**URL** <https://www.scopus.com/pages/publications/105012885156?origin=resultslist>**Archive** Scopus**Language** English**Volume** 20**Publisher** BioMed Central Ltd**Publication** Carbon Balance and Management**ISSN** 17500680 (ISSN)**Issue** 1**Journal Abbr** Carbon Balance Manage.**Date Added** 20/09/2026, 9:51:50 pm**Modified** 20/09/2026, 9:51:50 pm**Tags:**

United States, China, Remote sensing, remote sensing, Landsat, PRISMA, Analytical techniques, carbon sequestration, Carbon stock, satellite altimetry, soil-vegetation interaction

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Quantifying the effect of vortex generator installation on wind power production: An academia-industry case study

Item Type Journal Article

Author H. Hwangbo

Author Y. Ding

Author O. Eisele

Author G. Weinzierl

Author U. Lang

Author G. Pechlivanoglou

Abstract This paper presents an academia-industry joint study concerning effective methods to estimate and quantify the effect of vortex generator installation on wind power production. This problem has presented a challenge to the wind industry, because (a) vortex generator installation may lead to a moderate 1–5% extra power production, but this level of improvement is difficult to be accurately detected; and (b) it is equally difficult to validate the estimated effect of vortex generator installation because a controlled experiment is practically impossible to conduct to provide a credible baseline. An academic institute and a wind technology company team up to tackle this challenge. The two teams develop their own version of quantification methods, which are profoundly different. The academic method uses 10-min data and makes use of both power and environmental data, whereas the company method uses high-frequency data via primarily a direct power comparison approach that relies less on the environmental data. When applying the respective methods to two inland wind farms, each of which presents four pairs of turbines, the quantification results from the two methods are surprisingly consistent. We believe the consistent outcome presents a strong case of cross validation, testifying to the respective method's capability and credibility. © 2017 Elsevier Ltd

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Learning systems, machine learning, Machine learning, quantitative analysis, Statistical methods, Statistical analysis, statistical analysis, Wind power, Quantification methods, model validation, Vortex flow, wind farm, wind power, instrumentation, wind turbine, Academic institutes, Controlled experiment, frequency analysis, Industry case studies, installation, Installation, Power curve, Power curves, Power versus power, Vortex generator, Vortex generators, Vorticity, Wind power production

Notes:

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Quantifying the effects of urban fabric and vegetation combination pattern to mitigate particle pollution in near-road areas using machine learning

Item Type Journal Article**Author** X. Liu**Author** X.-Q. Shi**Author** Z.-R. Peng**Author** H.-D. He

Abstract There are many intertwined factors such as urban fabric, vegetation combination, and meteorological conditions that influence the spatial distribution of fine particles (PM_{2.5}, aerodynamic diameter less than 2.5 μm) in near-road urban areas, and it is difficult yet important to disentangle them. Identifying the relative importance of those factors could facilitate designing urban fabric and roadside vegetation combination patterns to mitigate the negative impacts of particles. This study first proposes an effective and reliable integrated method of the ENVI-met microclimate model and interpretable machine learning to quantify the multifactorial effects. Data from ENVI-met models with different built environments and meteorological conditions are used to train and test the machine learning (ML) models. All ML models show high accuracy in predicting PM_{2.5} concentrations with R² high than 0.87. Results indicate that height level and study area are the two most important features, explaining 22.8% and 20.4% of the variation in PM_{2.5} concentration, respectively. The feature pairwise interactions account for 45.8% of ambient particle concentrations. The combination of trees and hedges in urban fabrics with longer building length than width could mitigate PM_{2.5} concentrations in residential areas. The integrated methods in this study could be extrapolated to future research to perform source identification analysis. © 2023 Elsevier Ltd

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Roads and streets, machine learning, Machine-learning, Machine learning, Machine learning models, Vegetation, urban area, spatial distribution, Spatial distribution, ENVI-met, Integrated control, Integrated method, Meteorological condition, microclimate, Particles (particulate matter), particulate matter, Particulate matter, Particulate Matter, PM 2.5, roadside environment, Source contribution, Source contributions, tree, Urban fabric, Urban fabrics, Vegetation combination

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Quantifying the Impact of Urban Air Quality on Individual Well-being: A Multimodal Approach for Stress Classification

Item Type Conference Paper

Author K.I. Tusó

Author E.N. Capuchino

Author R.L. Nierva

Author N. Orillaza

Author M. Rosales

Author J.R. Hizon

Abstract Urban air pollution contributes to respiratory diseases, cardiovascular conditions, and mental stress. This study explores the relationship between individual exposure to air pollution and well-being, by integrating quantitative and qualitative methods, focusing on psychological stress levels through multimodal

data collection. It integrates real-time physiological measurements (e.g., heart rate), environmental sensors (e.g., PM2.5 sensor), and self-reported psychological stress to develop a classification framework for stress episodes based on exposure patterns. The study employs Geographic Information Systems (GIS), Pearson's R correlation, and machine learning techniques such as Random Forest and k-Nearest Neighbors (kNN) to analyze spatial-temporal exposure data and assess the correlations between urban air quality and individual well-being. Stationary air quality monitoring nodes provide confidence in the measured data through the wearable air quality monitor. Findings show that environmental and physiological data can classify momentary psychological stress, showing 98.56% accuracy with Random Forest model, which can be built on top of emotion analysis for expansion of well-being classification. This study fills research gaps by combining real-time data collection and machine learning to quantify the complex relationship between urban air quality and individual well-being. © 2025 IEEE.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K.I. Tuso; University of the Philippines, UP Manila Philippine General Hospital, Electrical and Electronics Engineering Institute, Quezon City, 1101, Philippines; email: kathleen.issandra.tuso@eee.upd.edu.ph; Conference name: 2025 International Conference on Advanced Machine Learning and Data Science, AMLDS 2025; Conference date: 19 July 2025 through 21 July 2025; Conference code: 212911

Quantifying the partial volume effect in PET using Benford's Law

Item Type Journal Article

Author K. Wells

Author J. Chiverton

Author M. Partridge

Author M. Barry

Author H. Kadhem

Author B. Ott

Abstract Partial Volume (PV) correction techniques in PET or SPECT represents a key step in image quantification methods. The PV effect arises because of the blurring induced by the imaging system's Point Spread Function (PSF), producing intra-voxel mixing of the signals arising from different functional tissue classes. Quantification of this effect is often required to recover the mixing components within a group of voxels, from whence the true tissue concentration in a given volume or region can be estimated. In this work we consider a probabilistic methodology that uses a phenomenological distribution known as Benford's Law to quantify the partial volume effect. We establish for the first time, that the probability distribution of voxels subjected to the PV effect in discrete volumetric data can be well described by Benford's Law. The probabilistic framework devised here can be applied generically across different imaging modalities including PET and SPECT. Results from simulated data are presented, along with a PET phantom study utilizing registered processed CT data as ground truth, to determine the quality of the resulting probabilistic voxel classification scheme. For a water filled hot insert using a 5:1 insert:background activity concentration, we find an overall voxel RMS error of 3% (compared to ground truth) in the estimated voxel mixing vectors. This error rises to 8% for a cold air-filled insert in a warm background. © 2007 IEEE.

Date 2007

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Export Date: 20 September 2026; Cited By: 7; Funding details: Engineering and Physical Sciences Research Council, EPSRC; Correspondence Address: K. Wells; School of Electronics and Physical Sciences, University of Surrey, Guildford, Surrey GU2 5XH, United Kingdom; email: k.wells@surrey.ac.uk; CODEN: IETNA

Quantifying the spatial quality of urban streets with open street view images: A case study of the main urban area of Fuzhou

Item Type Journal Article

Author Q. Rui

Author H. Cheng

Abstract With the advancement of global urbanization, urban streetscapes have become a critical part of urban public spaces. As one of the important ways to enhance the living environment, relevant government departments have accelerated the construction of urban streetscapes; however, certain issues, such as emphasis on surface and light protection, still persist. The visual effects of street landscapes hardly meet people's daily and spiritual needs. Therefore, assessing the quality of urban streetscapes has scientific and practical significance for improving urban planning and construction and creating a new intelligent and green city. In this study, the spatial distribution of vehicle interference index, spatial feasibility index, road area index, green visual index, sky visibility index, spatial enclosure index, color richness index, and visual entropy evaluation indices were drawn from the framework of combining deep learning and machine learning to quantify the spatial quality of urban streets by integrating a large number of streetscape images. The spatial quality of street landscape was evaluated according to the ELO scoring mechanism and random forest method, taking the main urban area of Fuzhou City as an example. The results showed that the overall street landscape quality of Fuzhou City was high, and the spatial distribution showed a pattern of concentrated high values in the center and dispersed high values in the periphery. Therefore, based on quantifying the spatial quality of urban streets, strategies, such as the pavement layout of pedestrian and vehicular paths, control of the continuity and permeability of visual landscape elements, and plant color matching, should be regulated. The method proposed in this study can provide novel insights for urban builders to improve the construction of cities. © 2023

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Quantifying trends accurately despite classifier error and class imbalance

Item Type Conference Paper

Author G. Forman

Abstract This paper promotes a new task for supervised machine learning research: quantification - the pursuit of learning methods for accurately estimating the class distribution of a test set, with no concern for predictions on individual cases. A variant for cost quantification addresses the need to total up costs according to categories predicted by imperfect classifiers. These tasks cover a large and important family of applications that measure trends over time. The paper establishes a research methodology, and uses it to evaluate several proposed methods that involve selecting the classification threshold in a way that would spoil the accuracy of individual classifications. In empirical tests, Median Sweep methods show outstanding ability to estimate the class distribution, despite wide disparity in testing and training conditions. The paper addresses shifting class priors and costs, but not concept drift in general. Copyright 2006 ACM.

Date 2006

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Quantifying uncertainty in foraminifera classification: How deep learning methods compare to human experts

Item Type Journal Article

Author I. Martinsen

Author S.A. Sørensen

Author S. Ortega

Author F. Godtliebsen

Author M. Tejedor

Author E. Myrvoll-Nilsen

Abstract Foraminifera are shell-bearing microorganisms that are commonly found in marine deposits on the seabed. They are important indicators in many analyses, are used in climate change research, monitoring marine environments, evolutionary studies, and are also frequently used in the oil and gas industry. Although some research has focused on automating the classification of foraminifera images, few have addressed the uncertainty in these classifications. Although foraminifera classification is not a safety-critical task, estimating uncertainty is crucial to avoid misclassifications that could overlook rare and ecologically significant species that are informative indicators of the environment in which they lived. Uncertainty estimation in deep learning has gained significant attention and many methods have been developed. However, evaluating the performance of these methods in practical settings remains a challenge. To create a benchmark for uncertainty estimation in the classification of foraminifera, we administered a multiple choice questionnaire containing classification tasks to four senior geologists. By analyzing their responses, we generated human-derived uncertainty estimates for a test set of 260 images of foraminifera and sediment grains. These uncertainty estimates served as a baseline for comparison when training neural networks in classification. We then trained multiple deep neural networks using a range of uncertainty quantification methods to classify and state the uncertainty about the classifications. The results of the deep learning

uncertainty quantification methods were then analyzed and compared with the human benchmark, to see how the methods performed individually and how the methods aligned with humans. Our results show that human-level performance can be achieved with deep learning and that test-time data augmentation and ensembling can help improve both uncertainty estimation and classification performance. Our results also show that human uncertainty estimates are helpful indicators for detecting classification errors and that deep learning-based uncertainty estimates can improve calibration and classification accuracy. © 2025 The Authors

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Quantifying Uncertainty in Land-Use/Land-Cover Classification Accuracy: A Stochastic Simulation Approach

Item Type Journal Article

Author K.-S. Cheng

Author J.-Y. Ling

Author T.-W. Lin

Author Y.-T. Liu

Author Y.-C. Shen

Author Y. Kono

Abstract In numerous applications of land-use/land-cover (LULC) classification, the classification rules are determined using a set of training data; thus, the results are inherently affected by uncertainty in the selection of those data. Few studies have assessed the accuracy of LULC classification with this consideration. In this article, we provide a general expression of various measures of classification accuracy with regard to the sample data set for classifier training and the sample data set for the evaluation of the classification results. We conducted stochastic simulations for LULC classification of a two-feature two-class case and a three-feature four-class case to show the uncertainties in the training sample and reference sample confusion matrices. A bootstrap simulation approach for establishing the 95% confidence interval of the classifier global accuracy was proposed and validated through rigorous stochastic simulation. Moreover, theoretical relationships among the producer accuracy, user accuracy, and overall accuracy were derived. The results demonstrate that the sample size of class-specific training data and the a priori probabilities of individual LULC classes must be jointly considered to ensure the correct determination of LULC classification accuracy. © Copyright © 2021 Cheng, Ling, Lin, Liu, Shen and Kono.

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Quantifying uncertainty in machine learning for nuclear binding energy

Item Type Journal Article

Author M. Huang

Author K.A. Wendt

Author N.F. Schunck

Author E.M. Holmbeck

Abstract Techniques from artificial intelligence and machine learning are increasingly employed in nuclear theory; however, the uncertainties that arise from the complex parameter manifold encoded by the neural networks are often overlooked. Epistemic uncertainties arising from training the same network multiple times for an ensemble of initial weight sets offer a first insight into the confidence of machine learning predictions, but they often come with a high computational cost. Instead, we apply a single-model uncertainty quantification method called Δ -UQ that gives epistemic uncertainties with one-time training. We demonstrate our approach on a two-feature model of nuclear binding energies per nucleon with proton and neutron number pairs as inputs. We show that Δ -UQ can produce reliable and self-consistent epistemic uncertainty estimates and can be used to assess the degree of confidence in predictions made with deep neural networks. © 2025 American Physical Society

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Quantitative ^{99m}Tc -DPD-SPECT/CT assessment of cardiac amyloidosis

Item Type Journal Article

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Author P. Fragoso Costa
Author D. Kersting
Author W. Jentzen
Author M. Weber
Author P. Lüdike
Author A. Carpinteiro
Author S. Oubari
Author T. Hagenacker
Author A. Thimm
Author T. Rassaf
Author K. Herrmann
Author M. Papathanasiou
Author C. Rischpler

Abstract Introduction: Transthyretin (ATTR) amyloidosis is responsible for the majority of cardiac amyloidosis (CA) cases and can be reliably diagnosed with bone scintigraphy and the visual Perugini score. We aimed to implement a quantification method of cardiac amyloid deposits in patients with suspected cardiac amyloidosis and to compare performance to visual scoring. Methods and materials: 136 patients received ^{99m}Tc-DPD-bone scintigraphy including SPECT/CT of the thorax in case of suspicion of cardiac amyloidosis. Imaging phantom studies were performed to determine the scaling factor for standardized uptake value (SUV) quantification from SPECT/CT. Myocardial tracer uptake was quantified in a whole heart volume of interest. Results: Forty-five patients were diagnosed with CA. A strong relationship between cardiac SUVmax and Perugini score was found (Spearman r 0.75, $p < 0.0001$). Additionally, tracer uptake in bone decreased with increasing cardiac SUVmax and Perugini score ($p < 0.0001$). ROC analysis revealed good performance of the SUVmax for the detection of ATTR-CA with AUC of 0.96 ± 0.02 ($p < 0.0001$) with sensitivity 98.7% and specificity 87.2%. Conclusion: We demonstrate an accessible and accurate quantitative SPECT approach in CA. Quantitative assessment of the cardiac tracer uptake may improve diagnostic accuracy and risk classification. This method may enable monitoring and assessment of therapy response in patients with ATTR amyloidosis. © 2022, The Author(s).

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Article, female, human, Humans, male, aged, Heart, quantitative analysis, procedures, clinical article, cohort analysis, retrospective study, blood sampling, sensitivity and specificity, urine sampling, echocardiography, unclassified drug, radiopharmaceutical agent, imaging, Tomography, X-Ray Computed, x-ray computed tomography, quantification, heart, single photon emission computed tomography, 3,3 diphosphono 1,2 propanodicarboxylic acid tc 99m, AL, AL amyloidosis, amyloid, amyloidosis, Amyloidosis, ATTR, bone scintiscanning, Cardiac amyloidosis, Cardiomyopathies, cardiomyopathy, familial amyloidosis, heart amyloidosis, heart ejection fraction, heart failure with preserved ejection fraction, heart volume, maximum standardized uptake value, mean standardized uptake value, Perugini, Prealbumin, Single Photon Emission Computed Tomography Computed Tomography, single photon emission computed tomography-computed tomography, SPECT, standardized uptake value, Tomography, Emission-Computed, Single-Photon, transthyretin

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Quantitative assessment of retinopathy using multi-parameter image analysis

Item Type Journal Article

Author Z. Ghanian

Author K. Staniszewski

Author N. Jamali

Author R. Sepehr

Author S. Wang

Author C.M. Sorenson

Author N. Sheibani

Author M. Ranji

Abstract A multi-parameter quantification method was implemented to quantify retinal vascular injuries in microscopic images of clinically relevant eye diseases. This method was applied to wholemount retinal trypsin digest images of diabetic Akita/+, and bcl-2 knocked out mice models. Five unique features of retinal vasculature were extracted to monitor early structural changes and retinopathy, as well as quantifying the disease progression. Our approach was validated through simulations of retinal images. Results showed fewer number of cells ($P = 5.1205e-05$), greater population ratios of endothelial cells to pericytes (PCs) ($P = 5.1772e-04$; an indicator of PC loss), higher fractal dimension ($P = 8.2202e-05$), smaller vessel coverage ($P = 1.4214e-05$), and greater number of acellular capillaries ($P = 7.0414e-04$) for diabetic retina as

compared to normal retina. Quantification using the present method would be helpful in evaluating physiological and pathological retinopathy in a high-throughput and reproducible manner. © 2016 Journal of Medical Signals & Sensors.

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Article, male, segmentation, quantitative analysis, image segmentation, nonhuman, controlled study, support vector machine, validation process, Classification, diagnostic accuracy, image processing, sensitivity and specificity, image analysis, microscopy, image quality, fluorescence microscopy, retinopathy, animal experiment, animal model, mouse, endothelium cell, diabetic retinopathy, contrast enhancement, fluorescence imaging, priority journal, disease course, cell count, brightfield imaging, cell loss, fractals, hyperglycemia, image cytometry, imaging and display, pericyte, systematic error

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Quantitative autistic traits ascertained in a national survey of 22 529 Japanese schoolchildren

Item Type Journal Article

Author Y. Kamio

Author N. Inada

Author A. Moriwaki

Author M. Kuroda

Author T. Koyama

Author H. Tsujii

Author Y. Kawakubo

Author H. Kuwabara

Author K.J. Tsuchiya

Author Y. Uno

Author J.N. Constantino

Abstract Objective: Recent epidemiologic studies worldwide have documented a rise in prevalence rates for autism spectrum disorders (ASD). Broadening of diagnostic criteria for ASD may be a major contributor to the rise in prevalence, particularly if superimposed on an underlying continuous distribution of autistic traits. This study sought to determine the nature of the population distribution of autistic traits using a quantitative trait measure in a large national population sample of children. Method: The Japanese version of the Social Responsiveness Scale (SRS) was completed by parents on a nationally representative sample of 22 529 children, age 6-15. Results: Social Responsiveness Scale scores exhibited a skewed normal distribution in the Japanese population with a single-factor structure and no significant relation to IQ within the normal intellectual range. There was no evidence of a natural 'cutoff' that would differentiate populations of categorically affected children from unaffected children. Conclusion: This study provides evidence of the continuous nature of autistic symptoms measured by the SRS, a validated quantitative trait measure. The findings reveal how paradigms for diagnosis that rest on arbitrarily imposed categorical cutoffs can result in substantial variation in prevalence estimation, especially when measurements used for case assignment are not standardized for a given population. © 2012 John Wiley & Sons A/S. Published by John Wiley & Sons Ltd.

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female, Female, human, Humans, male, Male, adolescent, Adolescent, article, Personality, Japan, Risk Factors, major clinical study, Questionnaire, Diagnosis, Classification, Adolescent Behavior, child, Child, autism, Autistic Disorder, Japanese, Prevalence, Mass Screening, school child, preschool child, social interaction, Age Distribution, mental patient, Sex Distribution, priority journal, quantitative trait, Psychiatric Status Rating Scales, Autism, Asperger syndrome, Autism Diagnostic Interview Revised, Child Behavior, Child Development, child psychiatry, genetic susceptibility, high school, intelligence quotient, Interpersonal Relations, mental deficiency, Peer Group, pervasive developmental disorder not otherwise specified, population distribution, primary school, Psychometrics, social adaptation, Social Behavior, Wechsler Intelligence Scale

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Quantitative classification and threshold determination of rock blasting damage based on blasting crater tests and numerical simulation

Item Type Journal Article

Author M. Zhao

Author C. Zhou

Author Á. Prado

Author N. Jiang

Author X. Meng

Author Y. Yang

Abstract The RHT constitutive model has been widely applied in numerical simulations of rock blasting. However, a clear quantitative relationship between its damage variable D and macroscopic fragmentation degree is still lacking. In this study, single-hole blasting experiments were combined with LS-DYNA simulations to investigate crater morphology, fragment size distribution, and damage evolution. A damage quantification method based on damaged volume ratio was proposed, establishing a direct quantitative correlation between the RHT damage variable D and fragmentation degree. Based on this relationship, blast-induced damage is classified into four levels: undamaged/slight damage ($0 \leq D \leq 0.3$), moderate damage ($0.3 \leq D \leq 0.8$), severe damage ($0.8 \leq D \leq 0.9$), and crushed damage ($0.9 \leq D \leq 1$). The proposed damage–fragmentation correlation provides a quantitative reference for interpreting numerical damage fields and offers practical guidance for improving blasting damage prediction and fragmentation control in numerical simulations. Copyright © 2026. Published by Elsevier Ltd.

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Damage detection, Rock mechanics, Copyrights, Blast damage classification, Blast Damage Classification, Blasting, Damage classification, Damage threshold, Damage Threshold, Damage variables, Fragmentation analysis, Fragmentation Analysis, Fragmentation degree, Numerical models, Numerical Simulation, Quantitative classifications, RHT constitutive model, RHT Constitutive Model, Rock blasting, Rock bursts, Threshold determination

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Quantitative color profiling of digital images with earth mover's distance using the R package colordistance

Item Type Journal Article

Author H.I. Weller

Author M.W. Westneat

Abstract Biological color may be adaptive or incidental, seasonal or permanent, species- or population-specific, or modified for breeding, defense or camouflage. Although color is a hugely informative aspect of biology, quantitative color comparisons are notoriously difficult. Color comparison is limited by categorization methods, with available tools requiring either subjective classifications, or expensive equipment, software, and expertise. We present an R package for processing images of organisms (or other objects) in order to quantify color profiles, gather color trait data, and compare color palettes on the basis of color similarity and amount. The package treats image pixels as 3D coordinates in a “color space,” producing a multidimensional

color histogram for each image. Pairwise distances between histograms are computed using earth mover's distance, a technique borrowed from computer vision, that compares histograms using transportation costs. Users choose a color space, parameters for generating color histograms, and a pairwise comparison method to produce a color distance matrix for a set of images. The package is intended as a more rigorous alternative to subjective, manual digital image analyses, not as a replacement for more advanced techniques that rely on detailed spectrophotometry methods unavailable to many users. Here, we outline the basic functions of colordistance, provide guidelines for the available color spaces and quantification methods, and compare this toolkit with other available methods. The tools presented for quantitative color analysis may be applied to a broad range of questions in biology and other disciplines. © 2019 Weller and Westneat.

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Notes:

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Quantitative method for intestinal eosinophils in inflammatory bowel disease using artificial intelligence-based software

Item Type Journal Article

Author Y. Hamamoto

Author Y. Nakazawa

Author N. Kinoshita

Author K. Kakimoto

Author T. Miyazaki

Author H. Nishikawa

Author Y. Hirose

Abstract Inflammatory bowel disease (IBD) is a condition of unknown etiology. Eosinophilia is sometimes prominent in pathological specimens of the small and large intestines of patients with IBD, but no consensus exists on the clinical significance of eosinophils in IBD. Eosinophil quantification methods include manual counting by pathologists and software analysis. In this study, we aimed to establish a quantitative technique for tissue eosinophils in small and large intestinal pathological specimens of IBD using artificial intelligence-based software and confirm its pathological validity. Small and large intestine specimens were collected from 40 patients. Hematoxylin and eosin-stained slides were prepared and scanned using WSI. In the eosinophil-rich field of view for the 39 cases, the eosinophil counts of the two pathologists and the software were performed. The software was trained in advance using the remnant case. Significant positive correlations were found between the eosinophil counts of the two pathologists' average and the software (Spearman's rank correlation coefficient = 0.919, $P = 1.62 \times 10^{-16}$) and of the two pathologists (Spearman's rank correlation coefficient = 0.767, $P = 0.0000000122$). Moreover, the average eosinophil counts of the two pathologists were predicted using the eosinophil count software. These results will enable future investigation of the clinicopathological significance (differential diagnosis, treatment response, and prognosis) of the total tissue eosinophil count in IBD, as well as reproducible assessment of disease activity in terms of eosinophils. © 2026 The Authors.

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Quantitative performance evaluation algorithms for pavement distress segmentation

Item Type Journal Article**Author** V. Kaul**Author** Y. Tsai**Author** R.M. Mersereau

Abstract Algorithms for pavement distress image segmentation are crucial to developing an automatic pavement distress detection and classification system. Many algorithms for pavement distress segmentation have been developed in the past decade; however, the lack of good methods to evaluate their performance quantitatively hinders the focused development of better segmentation algorithms. In this paper, a novel method is developed to quantitatively evaluate the performance of different pavement distress segmentation algorithms. This method uses the buffered Hausdorff distance to estimate the deviation of the cracks in the automatically segmented image from the ground truth cracks. The proposed method captures the local effectiveness of segmentation methods around the crack region without compromising its robustness to isolated pixel deviations caused by noise. Besides real pavement images, synthetic images simulating extreme pavement distress conditions are used to evaluate the capability of the proposed method and show its merits. The proposed method outperforms four other possible quantification methods and demonstrates its superior capability in providing a better score separation to distinguish the performance of different segmentation algorithms.

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Performance, Image segmentation, Images segmentations, Detection system, Classification system, Cracks, Performances evaluation, Pavement distress, Pavement distress detection, Pavement distresses classifications, Pavements, Segmentation algorithms, Evaluation algorithm

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Export Date: 20 September 2026; Cited By: 19; Correspondence Address: Y. Tsai; School of Civil and Environmental Engineering, Georgia Institute of Technology, Savannah, GA 31407, 210 Technology Circle, United States; email: james.tsai@ce.gatech.edu; CODEN: TRRED

Quantitative relationships between soil landscape index and classification grain

Item Type Journal Article

Author Y. Pan

Author X. Lu

Author D. Yu

Author J. Huang

Author X. Sun

Author Z. Xu

Abstract Purpose: Understanding space grain impact on soil landscape patterns is helpful to discover specific spatial heterogeneity of soil landscape. The objectives of this study were to reveal the effect of soil classification grain on soil landscape indices and establish a quantitative characterization function between soil landscape index and soil classification grain then we can reveal the sensitivity of soil landscape indices to soil classification grain and the significance of the function with respect to practical applications. Materials and methods: A soil and land use vector map of Dongxiang County was made at a scale of 1:50,000. The vector map units were classified into five hierarchies of soil classification grain, such as soil great group (G1), subgroup (G2), family (G3), species (G4), and species + land use (G5). Using Fragstats 4.2 software, 40 soil landscape indices of the five soil classification grains were calculated from the vector map and its derived raster maps, at different grain sizes ranging from 10 m × 10 m to 6 km × 6 km. A numerically investigated method was developed to quantify and analyze values of the different soil classification grains, quantitative relationships, and sensitivities of soil landscape indices to soil classification grains. Results and discussion: The effects of soil classification grain varied according to different soil landscape indices. The quantitative values of G1-G5 soil classification grains were quantified to be 0.7 km × 0.7 km, 0.5 km × 0.5 km, 0.2 km × 0.2 km, 0.15 km × 0.15 km, and 0.06 km × 0.06 km, respectively. The quantitative relationships of the abovementioned 28 landscape indices to soil classification grain were described by linear, logarithmic, reciprocal, quadratic, and power function with different sensitivities. The 13 landscape indices and their functions of the quantitative relationships to soil classification grain, which exhibited high sensitivity coefficients (PEAR > 0.90), should be adopted as a new approach in the future to quantitatively assess and forecast the spatial heterogeneity of soil landscape. Conclusions: The soil classification grain strongly affects the spatial heterogeneity of the soil landscape. The quantification method of soil classification

grains and the quantitative expression functions between soil landscape indices and the classification grain developed in this study could potentially provide a basic tool to quantitatively assess and forecast the spatial heterogeneity of soil landscapes. Among them, the landscape indices with high sensitivity to soil classification grains should be preferentially adopted within the new approach. © 2018, Springer-Verlag GmbH Germany, part of Springer Nature.

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Quantitative study on the discriminative value of fingerprint minutiae

Item Type Journal Article

Author S. Wu

Author J. Li

Author J. Huang

Author Y. Tang

Author Y. Luo

Author X. Li

Author Y. Qu

Author J. Xue

Author B. Yu

Abstract Traditional fingerprint identification primarily relies on the number of matching minutiae between the questioned and reference prints, where identity is determined by whether the match count exceeds a fixed threshold. However, this “minimum matching pair threshold method” lacks statistical validation. This study establishes a large-scale fingerprint data analysis framework based on artificial intelligence and machine learning to quantify the discriminative value of fingerprint minutiae. A YOLOv12 hybrid model is designed for high-precision minutiae detection. Using 619,338 fingerprints from four pattern classes, the occurrence frequencies of six types of minutiae are statistically analyzed, providing the foundation for quantitative modeling. A minutiae matching method is further proposed to perform large-scale matching according to specified minutia types and quantities. The approach integrates three modules: positional matching, local ridge-flow similarity comparison, and image similarity evaluation. 772 million pairs of non-mated fingerprints are analyzed to empirically refine traditional identification methods and quantify feature-level discriminative value. Results indicate that fingerprint matching stability depends not only on minutiae count but also on fingerprint pattern and minutia types. Building upon Shannon's information theory, a quantitative model for evaluating minutiae discriminative value is established based on feature frequency and matching performance. After residual correction, the model achieves a coefficient of determination (R^2) of 0.958, demonstrating high explanatory power and robustness. © 2026 American Academy of Forensic Sciences.

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artificial intelligence, human, Humans, machine learning, Artificial Intelligence, Machine Learning, image processing, Image Processing, Computer-Assisted, dermatoglyphics, Dermatoglyphics, discriminative value, fingerprint minutiae, fingerprint pattern class, forensic fingerprint identification, image similarity, matching minutiae, quantitative model, residual correction, Shannon information theory

Notes:

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Quantitative wavelength analysis and image classification for intraoperative cancer diagnosis with hyperspectral imaging

Item Type Conference Paper**Author** G. Lu**Author** X. Qin**Author** D. Wang**Author** Z.G. Chen**Author** B. Fei**Editor** Webster R.J.**Editor** Yaniv Z.R.

Abstract Complete surgical removal of tumor tissue is essential for postoperative prognosis after surgery. Intraoperative tumor imaging and visualization are an important step in aiding surgeons to evaluate and resect tumor tissue in real time, thus enabling more complete resection of diseased tissue and better conservation of healthy tissue. As an emerging modality, hyperspectral imaging (HSI) holds great potential for comprehensive and objective intraoperative cancer assessment. In this paper, we explored the possibility of intraoperative tumor detection and visualization during surgery using HSI in the wavelength range of 450 nm - 900 nm in an animal experiment. We proposed a new algorithm for glare removal and cancer detection on surgical hyperspectral images, and detected the tumor margins in five mice with an average sensitivity and specificity of 94.4% and 98.3%, respectively. The hyperspectral imaging and quantification method have the potential to provide an innovative tool for image-guided surgery. © 2015 SPIE.

Date 2015**DOI** 10.1117/12.2082284**URL** <https://www.scopus.com/pages/publications/84943520718?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Progr. Biomed. Opt. Imaging Proc. SPIE**Volume** 9415**Publisher** SPIE**ISBN** 16057422 (ISSN); 978-162841505-6 (ISBN)**Proceedings Title** Progr. Biomed. Opt. Imaging Proc. SPIE**Conference Name** Progress in Biomedical Optics and Imaging - Proceedings of SPIE

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Quantitatively understanding the ozone formation characteristics in areas with heavy precursor emission through machine learning simulations

Item Type Journal Article**Author** Q. Wang**Author** S. Yao**Author** C. Zhu**Author** S. Huang**Author** X. Ou**Author** J. Zhao**Author** S. Zhang**Author** W. Li**Author** J. Chen

Abstract The understanding of ozone (O₃) formation processes in polluted industrial areas remains limited. A machine learning (ML) approach combining random forest (RF) and artificial neural network (ANN) algorithms was proposed, showing a strong predictive performance of 0.940 in R². Analysis of variable importance suggests that NO₂ concentration, temperature (T), and relative humidity (RH) are the most important variables. Focusing on prevalent conditions, we established a quantified function for predicting O₃ concentration based on these important variables, which demonstrates a robust predictive capability. The results suggesting that the influence of NO₂ concentration on O₃ formation is primarily controlled near linearly by T and quadratically by RH. These results improve the understanding of O₃ formation in those NO_x-sensitive and VOC-oversaturated areas, thereby providing further support for identifying high O₃ concentration periods and important influence variables, which help to mitigate the increase in O₃ concentration in similar areas. Additionally,

the RF-ANN model provided a deep understanding of O₃ formation characteristics, while the quantification method with controlled variables demonstrated their effectiveness and timeliness in supporting local O₃ concentration control strategies.
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Quantum Computing and Quality Quantifiers Shaping the Future of Intelligent Systems

Item Type Conference Paper

Author D. Bhanu

Author M. Grover

Author M. Preetha

Author A. Mishra

Author S. Varalakshmi

Author V. Marathe

Abstract Quantum computing and quality quantifiers promote advanced computing. Their utilization will develop future smart systems. Quantum-Enhanced Quality Quantification (Q2) is introduced in this paper. Q2 combines quantum computing's built-in capabilities and quality quantification methodologies to

assess and enhance smart systems' performance, reliability, and morality. The proposed Q2 technique uses the quantum-enhanced quality integration (QEIQ), quality quantification (QQA), and quantum error correction (QEC) algorithms. Quantum error correction (QEC) uses stabilizer codes and syndromes to rectify quantum computing errors without affecting the quantum state. The Quality Quantification Algorithm rates system performance based on dependability, moral conformity, correctness, speed, and error rate. The QEIQ method uses new discoveries in quality measurement and error-corrected quantum data to make sure that smart systems offer solutions that are both technically correct and morally right. Six quantum computing methods were compared against Q2 to determine its performance: Quantum quantum, supreme accorrectiondes for quantum computing, quantum machine learning algorithms, quantum annealing, quality quantifiers for ethical AI, and quantum simulations in material science. Speed, accuracy, error rate, development, reliability, and morality determined success. The data reveal that Q2 outperforms traditional techniques in every manner. Precision, working speed, mistake rates, scalability, reliability, and morality improve. These findings demonstrate that Q2 can transform intelligent systems by improving technology and ensuring ethical behavior. © 2024 IEEE.

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Quantum-Inspired Neural Stem Cell Optimization for histopathological breast cancer detection: A clinically-informed approach

Item Type Journal Article

Author A. Kumar

Author F. Ahmad

Author B. Alam

Abstract Breast cancer remains a leading cause of cancer-related morbidity among women worldwide, where early and reliable diagnosis is critical for improving clinical outcomes. This study proposes a hybrid classification framework integrating DenseNet201 for hierarchical feature extraction, a Quantum-Inspired Neural Stem Cell Optimization (QNSCO) algorithm for feature selection, and a Support Vector Machine (SVM) for final classification. The QNSCO algorithm reformulates classical stem cell optimization within a probabilistic amplitude-based representation to enhance exploration–exploitation balance in high-dimensional feature spaces. The framework was evaluated on the BreakHis histopathological dataset across four magnification levels (40×, 100×, 200×, and 400×) under binary and multi-class settings using patient-wise stratified validation and repeated cross-validation protocols. The proposed DenseNet201–QNSCO–SVM model achieved classification accuracies of 99.88% (binary) and 99.01% (multi-class), while maintaining strict fold-level isolation to prevent information leakage. External validation under zero-shot and few-shot transfer settings further demonstrated stable generalization under magnification shifts and class prevalence variations. These results indicate that the proposed framework provides a reproducible and computationally structured optimization approach for histopathological image analysis, with potential applicability in computer-aided diagnostic research. Copyright © 2026. Published by Elsevier Ltd.

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Notes:

EID: 2-s2.0-105043714340

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: F. Ahmad; Department of Computer Engineering, Faculty of Engineering, Jamia Millia Islamia, Delhi, India; email: fahmad1@jmi.ac.in; CODEN: CPEEB

QuaPy: A Python-Based Framework for Quantification

Item Type Conference Paper

Author A. Moreo

Author A. Esuli

Author F. Sebastiani

Abstract QuaPy is an open-source framework for performing quantification (a.k.a. supervised prevalence estimation), written in Python. Quantification is the task of training quantifiers via supervised learning, where a quantifier is a predictor that estimates the relative frequencies (a.k.a. prevalence values) of the classes of interest in a sample of unlabelled data. While quantification can be trivially performed by applying a standard classifier to each unlabelled data item and counting how many data items have been assigned to each class, it has been shown that this "classify and count" method is outperformed by methods specifically designed for quantification. QuaPy provides implementations of a number of baseline methods and advanced quantification methods, of routines for quantification-oriented model selection, of several broadly accepted evaluation measures, and of robust evaluation protocols routinely used in the field. QuaPy also makes available datasets commonly used for testing quantifiers, and offers visualization tools for facilitating the analysis and interpretation of the results. The software is open-source and publicly available under a BSD-3 licence via <https://github.com/HLT-ISTI/QuaPy>, and can be installed via pip (<https://pypi.org/project/QuaPy/>). © 2021 ACM.

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Supervised learning, Open source frameworks, Open source software, Open systems, Python, HTTP, High level languages, Unlabeled data, Open-source, Quantification, Prevalence estimation, quantification, supervised learning, Learning to quantify, Supervised prevalence estimation, learning to quantify, Relative frequencies, supervised prevalence estimation, Data items, Baseline methods, open source, python

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Quasar and galaxy classification in Gaia data release 2

Item Type Journal Article

Author C.A.L. Bailer-Jones

Author M. Fouesneau

Author R. Andrae

Abstract We construct a supervised classifier based on Gaussian Mixture Models to probabilistically classify objects in Gaia data release 2 (GDR2) using only photometric and astrometric data in that release. The model is trained empirically to classify objects into three classes - star, quasar, galaxy - for $G \geq 14.5$ mag down to the Gaia magnitude limit of $G = 21.0$ mag. Galaxies and quasars are identified for the training set by a cross-match to objects with spectroscopic classifications from the Sloan Digital Sky Survey. Stars are defined directly from GDR2. When allowing for the expectation that quasars are 500 times rarer than stars, and galaxies 7500 times rarer than stars (the class imbalance problem), samples classified with a threshold probability of 0.5 are predicted to have purities of 0.43 for quasars and 0.28 for galaxies, and completenesses of 0.58 and 0.72, respectively. The purities can be increased up to 0.60 by adopting a higher threshold. Not accounting for this expected low frequency of extragalactic objects (the class prior) would give both erroneously optimistic performance predictions and severely impure samples. Applying our model

to all 1.20 billion objects in GDR2 with the required features, we classify 2.3 million objects as quasars and 0.37 million objects as galaxies (with individual probabilities above 0.5). The small number of galaxies is due to the strong bias of the satellite detection algorithm and on-ground data selection against extended objects. We infer the true number of quasars and galaxies - as these classes are defined by our training set - to be 690 000 and 110 000, respectively (± 50 per cent). The aim of this work is to see how well extragalactic objects can be classified using only GDR2 data. Better classifications should be possible with the low resolution spectroscopy (BP/RP) planned for GDR3. © 2019 The Author(s)

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RACENet: Real-time Adaptive Class-aware Early-exit Networks for Edge Devices

Item Type Conference Paper

Author M. Ayyat

Author M. Osman

Author T. Nadeem

Abstract As the integration of edge and IoT devices continues to surge, the need for streamlined machine learning solutions, notably Deep Neural Networks (DNNs), becomes paramount. However, the inherent computational demands and significant memory requirements of DNNs present hurdles for seamless edge integration. Early-Exit Networks (EENs) have emerged as a solution by adding early exits to DNNs, allowing for early inference and facilitating dynamic deployment on devices with varying capabilities. Despite their advantages, current EEN models uniformly handle samples from all classes, which is suboptimal in many edge scenarios where different classes require varying processing speeds and response times. Our research introduces RACENet, a novel architecture with adaptive early-exit capabilities that adjusts class prioritization in real-time. This flexibility enables RACENet to respond to evolving class priorities during runtime. Extensive evaluations on vision tasks, and network intrusion detection demonstrate that RACENet maintains accuracy while dynamically managing class priorities and accelerating inference for high-priority classes, all without adding additional computational burdens on constrained edge devices. © 2025 IEEE.

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Radio Galaxy Zoo: using semi-supervised learning to leverage large unlabelled data sets for radio galaxy classification under data set shift

Item Type Journal Article

Author I.V. Slijepcevic

Author A.M.M. Scaife

Author M. Walmsley

Author M. Bowles

Author O. Ivy Wong

Author S.S. Shabala

Author H. Tang

Abstract In this work, we examine the classification accuracy and robustness of a state-of-the-art semi-supervised learning (SSL) algorithm applied to the morphological classification of radio galaxies. We test if SSL with fewer labels can achieve test accuracies comparable to the supervised state of the art and whether this holds when incorporating previously unseen data. We find that for the radio galaxy classification problem considered, SSL provides additional regularization and outperforms the baseline test accuracy. However, in contrast to model performance metrics reported on computer science benchmarking data sets, we find that improvement is limited to a narrow range of label volumes, with performance falling off rapidly at low label volumes. Additionally, we show that SSL does not improve model calibration, regardless of whether classification is improved. Moreover, we find that when different underlying catalogues drawn from the same radio survey are used to provide the labelled and unlabelled data sets required for SSL, a significant drop in classification performance is observed, highlighting the difficulty of applying SSL techniques under data set shift. We show that a class-imbalanced unlabelled data pool negatively affects performance through prior probability shift, which we suggest may explain this performance drop, and that using the FrCrossed D sign $\odot$ chet distance between labelled and unlabelled data sets as a measure of data set shift can provide a prediction of model performance, but that for typical radio galaxy data sets with labelled sample volumes of $O(10^3)$, the sample variance associated with this technique is high and the technique is in general not sufficiently robust to replace a train-test cycle.

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Notes:

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Random Balance ensembles for multiclass imbalance learning

Item Type Journal Article

Author J.J. Rodríguez

Author J.-F. Díez-Pastor

Author Á. Arnaiz-González

Author L.I. Kuncheva

Abstract Random Balance strategy (RandBal) has been recently proposed for constructing classifier ensembles for imbalanced, two-class data sets. In RandBal, each base classifier is trained with a sample of the data with a random class prevalence, independent of the a priori distribution. Hence, for each sample, one of the classes will be undersampled while the other will be oversampled. RandBal can be applied on its own or can be combined with any other ensemble method. One particularly successful variant is RandBalBoost which integrates Random Balance and boosting. Encouraged by the success of RandBal, this work proposes two approaches which extend RandBal to multiclass imbalance problems. Multiclass imbalance implies that at least two classes have substantially different proportion of instances. In the first approach proposed here, termed Multiple Random Balance (MultiRandBal), we deal with all classes simultaneously. The training data for each base classifier are sampled with random class proportions. The second approach we propose decomposes the multiclass problem into two-class problems using one-vs-one or one-vs-all, and builds an ensemble of RandBal ensembles. We call the two versions of the second

approach OVO-RandBal and OVA-RandBal, respectively. These two approaches were chosen because they are the most straightforward extensions of RandBal for multiple classes. Our main objective is to evaluate both approaches for multiclass imbalanced problems. To this end, an experiment was carried out with 52 multiclass data sets. The results suggest that both MultiRandBal, and OVO/OVA-RandBal are viable extensions of the original two-class RandBal. Collectively, they consistently outperform acclaimed state-of-the art methods for multiclass imbalanced problems. © 2019 Elsevier B.V.

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Random forest model for feature-based Alzheimer's disease conversion prediction from early mild cognitive impairment subjects

Item Type Journal Article

Author M. Velazquez

Author Y. Lee

Abstract Alzheimer's Disease (AD) conversion prediction from the mild cognitive impairment (MCI) stage has been a difficult challenge. This study focuses on providing an individualized MCI to AD conversion prediction using a balanced random forest model that leverages clinical data. In order to do this, 383 Early Mild Cognitive Impairment (EMCI) patients were gathered from the Alzheimer's Disease Neuroimaging Initiative (ADNI). Of these patients, 49 would eventually convert to AD (EMCI_C), whereas the remaining 334 did not convert (EMCI_NC). All of these patients were split randomly into training and testing data sets with 95 patients reserved for testing. Nine clinical features were selected, comprised of a mix of demographic, brain volume, and cognitive testing variables. Oversampling was then performed in order to balance the initially imbalanced classes prior to training the model with 1000 estimators. Our results showed that a random forest model was effective (93.6% accuracy) at predicting the conversion of EMCI patients to AD based on these clinical features. Additionally, we focus on explainability by assessing the importance of each clinical feature. Our model could impact the clinical environment as a tool to predict the conversion to AD from a prodromal stage or to identify ideal candidates for clinical trials. © 2021 Velazquez et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Notes:

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Export Date: 20 September 2026; Cited By: 57; Correspondence Address: M. Velazquez; Department of Computer Science, University of Missouri - Kansas City, Kansas City, United States; email: mv3md@mail.umkc.edu; CODEN: POLNC

Rank classification method for cascade reservoirs considering scale, benefits, and risk consequences

Item Type Journal Article

Author T. Wang

Author Z. Li

Author W. Ge

Author Y. Zhang

Author Y. Jiao

Author L. Jing

Author P. van Gelder

Abstract Cascade reservoirs are essential in water resource utilization, flood control, and disaster mitigation. They have been built in several rivers worldwide. The rank classification for cascade reservoirs has significant practical implications for resource distribution decisions and risk management strategies and is also essential for ensuring the security of the basin. Internationally, reservoirs are typically classified based on project scale and dam failure consequences, whereas the risk transmission and superposition effect renders the classification method of reservoirs not directly applicable to cascade reservoirs. Therefore, to address this issue, this study proposed a rank classification method for cascade reservoirs. First, based on the scale, benefits, and risk consequences of dam failure, an index system for the rank classification of cascade reservoirs was established. Second, by constructing social risk criteria for cascade reservoirs and establishing a link among “project rank-reliability index-annual failure probability,” the allowable dam failure losses for different project ranks were determined. Thereafter, considering risk transmission and superposition, the rank classification standard and index quantification method for cascade reservoirs were proposed. Finally, five cascade reservoirs were selected for a feasibility study. The proposed method provides a supplement to the current rank classification standard of water conservancy projects in China and can also serve as a reference for risk assessment and classification management of cascade reservoirs in other countries. © 2023 Elsevier B.V.

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Rapid classification and quantification of *Euryales semen* (*Euryale ferox* Salisb.) from different origins and varieties using multispectral fingerprints combined with machine learning methods

Item Type Journal Article

Author D.-X. Yu

Author C.-Y. Dai

Author C. Qu

Author Y. Wu

Author Z. Liu

Author Y.-Q. Zhao

Author H. Yan

Author Q.-N. Wu

Abstract *Euryales semen* (ES, *Euryale ferox* Salisb.) is a highly nutritious food, which is rich in starch. The quality of ES is usually affected by differences in geographic origin and variety. This study proposed a rapid classification and quantification method based on near-infrared, mid-infrared, and Raman spectroscopy combined with intelligent algorithms to achieve origin traceability and quality evaluation of ES. Traditional chemometrics initially classified ES from different regions and varieties using multispectral data. Machine learning algorithms combined with single and

fused datasets were successfully improved the performance of classification models. Random forest (RF) classifier constructed using the feature-level data fusion strategy exhibited optimal performance, with an accuracy of 94.74 % in origin traceability and 100 % in varietal traceability. Moreover, multispectral fingerprints were applied to rapidly determine the contents of total starch in ES. RF regression combined with fused dataset (near-infrared and mid-infrared fingerprints) produced optimal results, with the highest values for the ratio of prediction to deviation and coefficient of determination for prediction of 2.60 and 0.82, respectively. Overall, the proposed combination of multispectral fingerprints with intelligent algorithms shows great potential for origin determination and quality assessment of ES as well as of other foods and herbs. © 2025 Elsevier Inc.

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Rapid detection of defect structures in grapheme by the machine learning

Item Type Journal Article

Author Y. Wu

Author L. Li

Author B. Wang

Author Z. Zhu

Author T. Gao

Author Q. Xie

Author Q. Chen

Author Q. Xiao

Abstract Graphene has long been considered to have a wide application perspective in many fields, such as electronic information, chemical industry and aerospace. However, the actual graphene materials contain a variety of defects that affect their properties. Accurate and rapid identification of defect structure types is of great significance for the application of graphene. First, the image preprocessing technology is used to improve the classification precision. Next, use the convolutional neural network to classify the type of vacancy defect and perform rapid analytical tasks to accurately identify graphene defect structures. This study aims to contribute to this growing area of research in graphene to classify and count the vacancy defects automatically and quickly in monitoring its surface structures. © 2022 World Scientific Publishing Co. Pte Ltd. All rights reserved.

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Deep learning, Convolutional networks, Graphene, Image processing.

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: Q. Xiao; Institute of Advanced Type Optoelectronic Materials and Technology, College of Big Data and Information Engineering, Guizhou University, Guiyang, 550025, China; email: gaotinghong@sina.com; CODEN: MPLBE

Rapid quantification of active pharmaceutical ingredient for sugar-free Yangwei granules in commercial production using FT-NIR spectroscopy based on machine learning techniques

Item Type Journal Article

Author J. Zhao

Author G. Tian

Author Y. Qiu

Author H. Qu

Abstract Rapid quantification methods for sugar-free Yangwei granules were developed based on near-infrared (NIR) spectroscopy combined with machine learning approaches as a quality control strategy for Chinese medicine granules (CMGs). Different machine learning approaches—i.e., interval partial least squares optimized by the genetic algorithm (GA-iPLS), the backpropagation artificial neural network (BP-ANN), and the particle swarm optimization-support vector machine (PSO-SVM)—were used to develop prediction models for three active pharmaceutical ingredients (APIs), namely, albiflorin, paeoniflorin, and benzoylpaeoniflorin. The partial least squares (PLS) algorithm was used for linear model calibration and comparison of the prediction performance of these developed models. The performance of the final models was assessed by the correlation coefficient (R), root mean square error of calibration set (RMSEC), and root mean square error of prediction set (RMSEP). All models performed well in model fitting and provided satisfactory prediction accuracy. The results indicate that the machine learning approaches are more stable, predictable, and suitable for CMGs when a high-accuracy analysis is required. In summary, NIR spectroscopy coupled with machine learning techniques is a suitable tool for the straightforward quantification of CMGs. © 2020 Elsevier B.V.

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Rationalization of thermal injury quantification methods: Application to skin burns

Item Type Journal Article

Author B.L. Viglianti

Author M.W. Dewhirst

Author J.P. Abraham

Author J.M. Gorman

Author E.M. Sparrow

Abstract Classification of thermal injury is typically accomplished either through the use of an equivalent dosimetry method (equivalent minutes at 43 °C, CEM43 °C) or through a thermal-injury-damage metric (the Arrhenius method). For lower-temperature levels, the equivalent dosimetry approach is typically employed while higher-temperature applications are most often categorized by injury-damage calculations. The two methods derive from common thermodynamic/physical chemistry origins. To facilitate the development of the interrelationships between the two metrics, application is made to the case of skin burns. This thermal insult has been quantified by numerical simulation, and the extracted time-temperature results served for the evaluation of the respective characterizations. The simulations were performed for skin-surface exposure temperatures ranging from 60 to 90 °C, where each surface temperature was held constant for durations extending from 10 to 110 s. It was demonstrated that values of CEM43 at the basal layer of the skin were highly correlated with the depth of injury calculated from a thermal injury integral. Local

values of CEM43 were connected to the local cell survival rate, and a correlating equation was developed relating CEM43 with the decrease in cell survival from 90% to 10%. Finally, it was shown that the cell survival/CEM43 relationship for the cases investigated here most closely aligns with isothermal exposure of tissue to temperatures of ~50 °C. © 2013 Elsevier Ltd and ISBI.

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human, Humans, simulation, article, controlled study, evaluation study, human cell, human tissue, skin, Skin, high temperature, Hot Temperature, heat transfer, low temperature, burn, Burns, cell damage, cell death, cell survival, Cell Survival, CEM43°C, exposure, heat, Hyperthermia, physical chemistry, protein denaturation, skin injury, skin temperature, Skin Temperature, Thermal ablation, thermal conductivity, Thermal dosimetry, thermal injury, Thermal injury, Thermal injury index, thermodynamics

Notes:

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Export Date: 20 September 2026; Cited By: 60; Correspondence Address: J.P. Abraham; School of Engineering, University of St. Thomas, St. Paul, MN 55105, United States; email: jpabraham@stthomas.edu; CODEN: BURND

RB-CCR: Radial-Based Combined Cleaning and Resampling algorithm for imbalanced data classification

Item Type Journal Article

Author M. Koziarski

Author C. Bellinger

Author M. Woźniak

Abstract Real-world classification domains, such as medicine, health and safety, and finance, often exhibit imbalanced class priors and have asynchronous misclassification costs. In such cases, the classification model must achieve a high recall without significantly impacting precision. Resampling the training data is the standard approach to improving classification performance on imbalanced binary data. However, the state-of-the-art methods ignore the local joint distribution of the data or correct it as a post-processing step. This can causes sub-optimal shifts in the training distribution, particularly when the target data distribution is complex. In this paper, we propose Radial-Based Combined Cleaning and Resampling (RB-CCR). RB-CCR utilizes the concept of class potential to refine the energy-based resampling approach of CCR. In particular, RB-CCR exploits the class potential to accurately locate sub-regions of the data-space for synthetic oversampling. The category sub-region for oversampling can be specified as an input parameter to meet domain-specific needs or be automatically selected via cross-validation. Our 5×2 cross-validated results on 57 benchmark binary datasets with 9 classifiers show that RB-CCR achieves a better precision-recall trade-off than CCR and generally out-performs the state-of-the-art resampling methods in terms of AUC and G-mean. © Crown 2021.

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Re-assessing the “Classify and Count” Quantification Method

Item Type Conference Paper

Author A. Moreo

Author F. Sebastiani

Editor Hiemstra D.

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Abstract Learning to quantify (a.k.a. quantification) is a task concerned with training unbiased estimators of class prevalence via supervised learning. This task originated with the observation that “Classify and Count” (CC), the trivial method of obtaining class prevalence estimates, is often a biased estimator, and thus delivers suboptimal quantification accuracy. Following this observation, several methods for learning to quantify have been proposed and have been shown to outperform CC. In this work we contend that previous works have failed to use properly optimised versions of CC. We thus reassess the real merits of CC and its variants, and argue that, while still inferior to some cutting-edge methods, they deliver near-state-of-the-art accuracy once (a) hyperparameter optimisation is performed, and (b) this optimisation is performed by using a truly quantification-oriented evaluation protocol. Experiments on three publicly available binary sentiment classification datasets support these conclusions. © 2021, Springer Nature Switzerland AG.

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Re-Thinking Classification Confidence with Model Quality Quantification

Item Type Conference Paper

Author Y. Pan

Author H. Zhao

Abstract Deep neural networks using for real-world classification task require high reliability and robustness. However, the Softmax output by the last layer of network is often over-confident. We propose a novel confidence estimation method by considering model quality for deep classification models. Two metrics, MQ-Repres and MQ-Discri are developed accordingly to evaluate the model quality, and also provide a new confidence estimation called MQ-Conf for online inference. We demonstrate the capability of the proposed method by the 3D semantic segmentation tasks using three different deep networks. Through confusion analysis and feature visualization we show the rationality and reliability of the model quality quantification method.¹¹This work is supported by the National Natural Science Foundation of China under Grant U22A2061 and High-performance Computing Platform of Peking University. © 2023 IEEE.

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Real-time EEG-based emotion recognition using Riemannian quantification learning

Item Type Journal Article

Author B. Lv

Author Y. Yu

Author F. Zhuo

Author F. Tang

Abstract Emotion recognition via electroencephalogram (EEG) has obtained significant attention in many fields, including brain-computer interfaces. It is crucial to accurately identify emotional states in real-time scenarios with few-channel EEG signals and minimal user training data. This study tries to use symmetric positive definite (SPD) matrices to capture the characteristics of emotional EEG signals. We propose a classifier, termed as Riemannian robust soft learning vector quantization with the log-Euclidean metric learning (RRSLVQ-LEML), to distinguish the SPD matrices. To provide thorough analyses, we first evaluated the offline classification performance with channel selection and training data reduction across three public datasets, i.e., SEED, SEED-IV, and SEED-V. The results demonstrated that the proposed model outperformed the SPD network (SPDnet) significantly ($p < 0.05$). The classification accuracies on the three datasets using selected universal channels (22 channels) and a limited number of training samples (150 samples for each category) were $88.33\% \pm 5.77\%$, $76.05\% \pm 7.85\%$, and $70.62\% \pm 11.66\%$, respectively. We further validated the proposed model under the selected channel setup and a small amount of training in the online scenario by recruiting seven participants to conduct experiments for four-category emotion recognition. The average online classification accuracy was $77.28\% \pm 6.68\%$, which exhibited a strong linear correlation with the offline results ($r = 0.82$). These findings indicate that the RRSLVQ-LEML model is suitable for real-time emotion recognition with few EEG channels and minimal training samples. Furthermore, the offline performance evaluated by the RRSLVQ-LEML model is transferable to the online scenario. © 2025 The Authors

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Real-Time PCR Quantification of Chloroplast DNA Supports DNA Barcoding of Plant Species

Item Type Journal Article
Author H.S. Kikkawa
Author K. Tsuge

Author R. Sugita

Abstract Species identification from extracted DNA is sometimes needed for botanical samples. DNA quantification is required for an accurate and effective examination. If a quantitative assay provides unreliable estimates, a higher quantity of DNA than the estimated amount may be used in additional analyses to avoid failure to analyze samples from which extracting DNA is difficult. Compared with conventional methods, real-time quantitative PCR (qPCR) requires a low amount of DNA and enables quantification of dilute DNA solutions accurately. The aim of this study was to develop a qPCR assay for quantification of chloroplast DNA from taxonomically diverse plant species. An absolute quantification method was developed using primers targeting the ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit (rbcL) gene using SYBR Green I-based qPCR. The calibration curve was generated using the PCR amplicon as the template. DNA extracts from representatives of 13 plant families common in Japan. This demonstrates that qPCR analysis is an effective method for quantification of DNA from plant samples. The results of qPCR assist in the decision-making will determine the success or failure of DNA analysis, indicating the possibility of optimization of the procedure for downstream reactions. © 2016, Springer Science+Business Media New York.

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Real-world adverse event analysis of Adagrasib: Insights from the FAERS database

Item Type Journal Article

Author S. Jian

Author S. Liang

Author X. Ma

Author S. Chen

Author X. Zhang

Author X. Lan

Author P. Zuo

Author E. Hou

Author S. Tu

Abstract Adagrasib is a potent and orally available small-molecule inhibitor that irreversibly targets the KRAS G12C mutant isoform. This innovative therapeutic agent represents a significant advancement in the field of oncology, offering a targeted approach to treating cancers driven by this specific mutation. The precision and potency of Adagrasib make it a promising candidate for improving patient outcomes in KRAS G12C-mutated cancers. However, alongside its therapeutic potential, it is crucial to

thoroughly understand and evaluate the safety profile of Adagrasib. The safety assessment includes a detailed examination of the adverse events (AEs) associated with its use. Comprehensive evaluation of these AEs is essential to ensure that the benefits of the drug outweigh any potential risks to patients. This retrospective pharmacovigilance study evaluated AEs associated with Adagrasib using data from the FDA adverse event reporting system database. We collected adverse drug reaction data for Adagrasib from the fourth quarter of 2022 to the first quarter of 2025. After standardizing the data, we applied several signal quantification methods, including the reporting odds ratio, proportional reporting ratio, Bayesian confidence propensity for neural networks, and Multi-item Gamma Poisson Shrinker (MGPS), for analysis. In this analysis of 556 adverse drug event reports where Adagrasib was identified as the primary suspect, we discovered 38 preferred terms across 20 system organ classifications. The 3 most prevalent system organ classifications were general disorders and administration site conditions, gastrointestinal disorders, and nervous system disorders. Notably, this study uncovered several new, previously unlabeled adverse reactions, including skin hyperpigmentation and seizures. These findings highlight the importance of ongoing pharmacovigilance to ensure comprehensive understanding of a drug's safety profile. These findings highlight the need for continued monitoring and understanding of Adagrasib potential risks. Further research is necessary to confirm these associations and address any previously unrecognized safety concerns. Copyright © 2025 the Author(s). Published by Wolters Kluwer Health, Inc.

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Recent advances in laser-induced breakdown spectroscopy quantification: From fundamental understanding to data processing

Item Type Journal Article

Author Z. Wang

Author M.S. Afgan

Author W. Gu

Author Y. Song

Author Y. Wang

Author Z. Hou

Author W. Song

Author Z. Li

Abstract Laser-induced breakdown spectroscopy (LIBS) is regarded as the future superstar for chemical analysis, but the relatively high measurement uncertainty and error remain the persistent challenges for its technological development as well as wide applications. In the present work, mechanisms of measurement uncertainty generation and basic principle of signal uncertainty and matrix effects impacting quantification performance were explained. Furthermore, methods for raw signal improvement

including sample preparation, system optimization, and especially plasma modulation, which modulates the laser-induced plasma evolution process for higher signal repeatability and signal-to-noise ratio, were reviewed and discussed. Different LIBS mathematical quantification methods including calibration-free methods and calibration methods, which were classified into physical-principle based calibration model, data-driven based calibration model, and hybrid model, were discussed and compared. Overall, a framework of quantification improvement strategy including key steps and main way-out was summarized and recommended for LIBS future development. © 2021 Elsevier B.V.

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Export Date: 20 September 2026; Cited By: 281; Funding details: National Natural Science Foundation of China, NSFC, (51906124, 61675110); National Natural Science Foundation of China, NSFC; National Key Research and Development Program of China, NKRDPC, (2016YFC0302102); National Key Research and Development Program of China, NKRDPC; Correspondence Address: Z. Wang; State Key Lab of Power Systems, International Joint Laboratory on Low Carbon Clean Energy, Innovation, Department of Energy and Power Engineering, Tsinghua University, Beijing, 100084, China; email: zhewang@tsinghua.edu.cn; CODEN: TTAED

Recent developments in capabilities for analysing chlorinated paraffins in environmental matrices: A review

Item Type Journal Article

Author L.M. Van Mourik

Author P.E.G. Leonards

Author C. Gaus

Author J. De Boer

Abstract Concerns about the high production volumes, persistency, bioaccumulation potential and toxicity of chlorinated paraffin (CP) mixtures, especially short-chain CPs (SCCPs), are rising. However, information on their levels and fate in the environment is still insufficient, impeding international classifications and regulations. This knowledge gap is mainly due to the difficulties that arise with CP analysis, in particular the chromatographic separation within CPs and between CPs and other compounds. No fully validated routine analytical method is available yet and only semi-quantitative analysis is possible, although the number of studies reporting new and improved methods have rapidly increased since 2010. Better cleanup procedures that remove interfering compounds, and new instrumental techniques, which distinguish between medium-chain CPs (MCCPs) and SCCPs, have been developed. While gas chromatography coupled to an electron capture negative ionisation mass spectrometry (GC/ECNI-MS) remains the most commonly applied technique, novel and promising use of high resolution time of flight MS (TOF-MS) has also been reported. We expect that recent developments in high resolution TOF-MS and Orbitrap technologies will further improve the detection of CPs, including long-chain CPs (LCCPs), and the group separation and quantification of CP homologues. Also, new CP quantification methods have emerged, including the use of mathematical algorithms, multiple linear regression and principal component analysis. These quantification advancements are also reflected in considerably improved interlaboratory agreements since 2010. Analysis of lower chlorinated paraffins (<C15) remains, however, challenging and better approaches to analysing these homologues are needed. Furthermore, suitable quantification standards would facilitate improving the quality of CP analysis. © 2015 Elsevier Ltd.

Date 2015

DOI 10.1016/j.chemosphere.2015.05.045

URL <https://www.scopus.com/pages/publications/84983146538?origin=resultslist>

Archive Scopus

Language English

Volume 136

Publisher Elsevier Ltd**Pages** 259-272**Publication** Chemosphere**ISSN** 00456535 (ISSN)**Journal Abbr** Chemosphere**Date Added** 20/09/2026, 9:52:58 pm**Modified** 20/09/2026, 9:52:58 pm**Tags:**

Classification (of information), Environmental Monitoring, quantitative analysis, quality control, procedures, Extraction, Review, environment, Principal component analysis, Quality control, Linear regression, environmental monitoring, calibration, gas chromatography, Gas Chromatography-Mass Spectrometry, mass fragmentography, water, unclassified drug, principal component analysis, mass spectrometry, Mass spectrometry, Liquid chromatography, Environmental Pollutants, Mass Spectrometry, pollutant, toxicity, analytic method, bioaccumulation, Environment, Negative ions, paraffin, Principal Component Analysis, analysis, Gas chromatography, air, Analysis, Biochemistry, chlorinated hydrocarbon, Chlorinated paraffin (CP), Chlorinated paraffins, chromatography, Chromatography, Gas, ecotoxicology, environmental hazard, Hydrocarbons, Chlorinated, Inductively coupled plasma, long chain chlorinated paraffin, medium chain chlorinated paraffin, Medium-chain chlorinated paraffin (MCCP), Medium-chain chlorinated paraffins, numerical method, organochlorine derivative, Paraffin, Paraffins, pollution monitoring, Polychlorinated n-alkanes, Polychlorinated-n-alkane (PCA), short chain chlorinated paraffin, Short-chain chlorinated paraffin (SCCP), Short-chain chlorinated paraffins

Notes:

EID: 2-s2.0-84983146538

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Recent Rapid Increase of Cover Crop Adoption Across the U.S. Midwest Detected by Fusing Multi-Source Satellite Data

Item Type Journal Article**Author** Q. Zhou**Author** K. Guan**Author** S. Wang**Author** C. Jiang**Author** Y. Huang**Author** B. Peng

Author Z. Chen
Author S. Wang
Author J. Hipple
Author D. Schaefer
Author Z. Qin
Author S. Stroebel
Author J. Coppess
Author M. Khanna
Author Y. Cai

Abstract Cover crops have critical significance for agroecosystem sustainability and have long been promoted in the U.S. Midwest. Knowledge of cover cropping variations and impacts of government policies remains very limited. We developed an accurate and cost-effective approach utilizing satellite fusion data, environmental variables, and machine learning to quantify cover cropping in corn and soybean fields from 2000 to 2021 in the U.S. Midwest. We found that cover crop adoption in most counties was stagnant from 2000 to 2011, but has significantly increased from 2011 to 2021. The adoption of 2021 is four times that of 2011, which was highly correlated to the funding for conservation programs. However, the percentage of cover crop adoption in the U.S. Midwest is still low (7.2%). Our work fills a critical gap in quantifying long-term field-level cover crop adoption at large regions and highlights the potential importance of incentive programs to promote sustainable agricultural practices. © 2022. The Authors.

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DOI 10.1029/2022GL100249

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Issue 22

Journal Abbr Geophys. Res. Lett.

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Tags:

environmental factor, sustainability, United States, machine learning, Machine-learning, Sustainable development, satellite data, Cost effectiveness, detection method, Crops, Multi-Sources, agricultural land, alternative agriculture, agricultural ecosystem, agricultural policy, Agroecosystem sustainability, Corn fields, Cost-effective approach, cover crop, Cover crops, Environmental variables, Highly-correlated, incentive, Midwest, Satellite data, soybean, Soybean fields

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Recognition and Counting of an Object Using Yolo and CNN

Item Type Conference Paper

Author R. Vallikannu

Author V. Kiruthika

Author M. Kaviya

Editor Sharma D.K.

Editor Peng S.

Editor Sharma R.

Editor Zaitsev D.A.

Abstract Object identification and classification are vital in automation domains. It has gained conscience in the recent days owing to the growing domain of image processing and automation based on the growth of multiple sectors implementing automations. Object detection, identification and counting mechanisms are gaining momentum. Random urgencies and the trending algorithms of machine learning and deep learning strategies have made the object detection domain even more vulnerable. This paper implements a strategy to detect, classify and count objects based on image classification machine learning algorithms using Yolo. The proposed work considered training data collection using the app dataset, pre-processing using the DWT and GLCM, feature extraction using supervised ML algorithm, prune the available data and classification based on the proposed prediction models using R-CNN algorithms. The proposed methodology provided better accuracy with less computational overheads than the existing methods. © 2022, The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd.

Date 2022

DOI 10.1007/978-981-16-8721-1_28

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Archive Scopus

Language English

Extra Journal Abbreviation: Lect. Notes Networks Syst.

Volume 373

Publisher Springer Science and Business Media Deutschland GmbH

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Pages 289-296

Proceedings Title Lect. Notes Networks Syst.

Conference Name Lecture Notes in Networks and Systems

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Object detection, Python, Object counting, Computer vision (CV), Convolution neural network (CNN), You look only once (YOLOv3)

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: R. Vallikannu; Department of ECE, Hindustan Institute of Technology and Science, Chennai, 603103, India; email: vallikannu@hindustanuniv.ac.in; Conference name: 5th International Conference on Microelectronics and Telecommunication Engineering, ICMETE 2021; Conference code: 274209

Recognition of hand gestures tracked by a dataglove: Exploiting Hidden Markov Models discriminative training and environment description to improve recognition performance

Item Type Conference Paper

Author V. Lippi

Author E. Ruffaldi

Author C.A. Avizzano

Author M. Bergamasco

Abstract Sequence classification based on Hidden Markov Models (HMMs) is widely employed in gesture recognition. Usually, HMMs are trained to recognize sequences by adapting their parameters through the Baum-Welch algorithm, based on Maximum Likelihood (ML). Several articles have pointed out that ML can lead to poor discriminative performances among gestures. This happens because ML is not optimal for this purpose until the modellized process is actually an HMM. In this paper we present a gesture recognition system featuring a discriminative training algorithm based on Maximal Mutual Information (MMI) and the integration of environment information. The environment is described through a set of fuzzy clauses, on the basis of which a priori probabilities are computed. Adaptive systems such as unsupervised neural networks are used to build a codebook of symbols representing the hand's states. An experiment on a set of meaningful gestures performed during the interaction with a virtual environment is then used to evaluate the performance of this solution.

Date 2009

URL <https://www.scopus.com/pages/publications/74549214350?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Artif. Neural Netw. Intelligent Inf. Process. - Proc. Int. Workshop Artif. Neural Netw. Intelligent Inf. Process. - ANNIIP Co

ISBN 978-989674002-3 (ISBN)

Pages 33-42

Proceedings Title Artif. Neural Netw. Intelligent Inf. Process. - Proc. Int. Workshop Artif. Neural Netw. Intelligent Inf. Process. - ANNIIP Co

Conference Name Artificial Neural Networks and Intelligent Information Processing - Proc. 5th Int. Workshop on Artificial Neural Networks and Intelligent Information Processing - ANNIIP 2009, held with ICINCO 2009

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Tags:

Virtual reality, Neural networks, Data processing, Adaptive systems, Codebooks, Mutual informations, Maximum likelihood, A-priori probabilities, Gesture recognition, Hidden Markov models, Baum-Welch algorithms, Dataglove, Discriminative training, Environment information, Gesture recognition system, Hand gesture, Recognition performance, Sequence classification, Unsupervised neural networks, Virtual environments

Notes:

EID: 2-s2.0-74549214350

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: V. Lippi; PERCRO Laboratory, Scuola Superiore Sant'anna, Pisa, Italy; email: v.lippi@sssup.it; Conference name: 5th International Workshop on Artificial Neural Networks and Intelligent Information Processing - ANNIIP 2009 In Conjunction with ICINCO 2009; Conference date: 1 July 2009 through 1 July 2009; Conference code: 79038

Recognition of the Amazonian flora by Inception Networks with Test-time Class Prior Estimation CMP submission to PlantCLEF 2019

Item Type Conference Paper

Author L. Picek

Author M. Šulc

Author J. Matas

Editor Cappellato L.

Editor Department of Information Engineering (DEI), Via Gradenigo 6/B, Padova University of Padua

Editor Ferro N.

Editor Department of Information Engineering (DEI), Via Gradenigo 6/B, Padova University of Padua

Editor Losada D.E.

Editor Muller H.

Abstract The paper describes an automatic system for recognition of 10,000 plant species, with focus on species from the Guiana shield and the Amazon rain forest. The proposed system achieves the best results on the PlantCLEF 2019 test set with 31.9% accuracy. Compared against human experts in plant recognition, the system performed better than 3 of the 5 participating human experts and achieved 41.0% accuracy on the subset for expert evaluation. The proposed system is based on the Inception-v4 and Inception-ResNet-v2 Convolutional Neural Network (CNN) architectures. Performance improvements were achieved by: adjusting the CNN predictions according to the estimated change of the class prior probabilities, replacing network parameters with their running averages, test-time data augmentation, filtering the provided training set and adding additional training images from GBIF. © 2019 CEUR-WS. All rights reserved.

Date 2019

URL <https://www.scopus.com/pages/publications/85070527849?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: CEUR Workshop Proc.

Volume 2380

Publisher CEUR-WS

ISBN 16130073 (ISSN)

Proceedings Title CEUR Workshop Proc.

Conference Name CEUR Workshop Proceedings

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Classification (of information), Learning systems, Computer vision, Computer Vision, Convolution, Convolutional neural network, Neural networks, Convolutional Neural Networks, Image enhancement, Machine Learning, Fine grained, Classification, Automatic systems, Fine-grained, Prior probability, Class Prior Estimation, Network parameters, Plant recognition, Plant Recognition, Amazon rain forest, Expert evaluation

Notes:

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Reconciling Binary Replicates: Beyond the Average

Item Type Journal Article

Author H. Lorenzo

Author P. Pudlo

Author M. Royer-Carenzi

Abstract Binary observations are often repeated to improve data quality, creating technical replicates. Several scoring methods are commonly used to infer the actual individual state and obtain a probability for each state. The common practice of averaging replicates has limitations, and alternative methods for scoring and classifying individuals are proposed. Additionally, an indecisive response might be wiser than classifying all individuals based on their replicates in the medical context, where 1 indicates a particular health condition. Building on the inherent limitations of the averaging approach, three alternative methods are examined: the median, maximum penalized likelihood estimation, and a Bayesian algorithm. The theoretical analysis suggests that the proposed alternatives outperform the averaging approach, especially the Bayesian method, which incorporates uncertainty and provides credible intervals. Simulations and real-world medical datasets are used to demonstrate the practical implications of these methods for improving diagnostic accuracy and disease prevalence estimation. © 2026 The Author(s). Statistics in Medicine published by John Wiley & Sons Ltd.

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Article, human, Humans, simulation, machine learning, algorithm, Algorithms, risk factor, prediction, controlled study, false positive result, computer simulation, Computer Simulation, prevalence, data quality, mathematical model, diagnostic test accuracy study, follow up, probability, diagnostic accuracy, statistical analysis, scoring system, false negative result, Bayes theorem, Bayes Theorem, Monte Carlo method, sensitivity and specificity, statistical model, prevalence estimation, Bayesian algorithm, Data Interpretation, Statistical, expectation-maximization algorithm, expectation-maximization algorithm, Likelihood Functions, mammography, median, medical diagnosis, Models, Statistical, reconciling binary replicate, technical replicates

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: P. Pudlo; UMR 7373, CNRS, Centrale Marseille, I2M, Aix Marseille Univ, Marseille, France; email: pierre.pudlo@univ-amu.fr; CODEN: SMEDD

Recovering true classifier performance in positive-unlabeled learning

Item Type Conference Paper

Author S. Jain

Author M. White

Author P. Radivojac

Abstract A common approach in positive-unlabeled learning is to train a classification model between labeled and unlabeled data. This strategy is in fact known to give an optimal classifier under mild conditions; however, it results in biased empirical estimates of the classifier performance. In this work, we show that the typically used performance measures such as the receiver operating characteristic curve, or the precisionrecall curve obtained on such data can be corrected with the knowledge of class priors; i.e., the proportions of the positive and negative examples in the unlabeled data. We extend the results to a noisy setting where some of the examples labeled positive are in fact negative and show that the correction also requires the knowledge of the proportion of noisy examples in the labeled positives. Using state-of-the-art algorithms to estimate the positive class prior and the proportion of noise, we experimentally evaluate two correction approaches and demonstrate their efficacy on real-life data. © Copyright 2017, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

Date 2017

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Archive Scopus

Language English

Extra Journal Abbreviation: AAAI Conf. Artif. Intell., AAAI

Publisher AAAI press

Pages 2066-2072

Proceedings Title AAAI Conf. Artif. Intell., AAAI

Conference Name 31st AAAI Conference on Artificial Intelligence, AAAI 2017

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Artificial intelligence, Classification models, Labeled and unlabeled data, Performance measure, Correction approaches, Receiver operating characteristic curves, State-of-the-art algorithms, Classifier performance, Optimal classifiers

Notes:

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Rectifying Bias in Ordinal Observational Data Using Unimodal Label Smoothing

Item Type Conference Paper

Author S. Haas

Author E. Hüllermeier

Editor De Francisci Morales G.

Editor Bonchi F.

Editor Perlich C.

Editor Ruchansky N.

Editor Kourtellis N.

Editor Baralis E.

Abstract This paper proposes a novel approach for modeling observational data in the form of expert ratings, which are commonly given on an ordered (numerical or ordinal) scale. In practice, such ratings are often biased, due to the expert's preferences, psychological effects, etc. Our approach aims to rectify these biases, thereby preventing machine learning methods from transferring them to models trained on the data. To this end, we make use of so-called label smoothing, which allows for redistributing probability mass from the originally observed rating to other ratings, which are considered as possible corrections. This enables the incorporation of domain knowledge into the standard cross-entropy loss and leads to flexibly configurable models. Concretely, our method is realized for ordinal ratings and allows for arbitrary unimodal smoothings using a binary smoothing relation. Additionally, the paper suggests two practically motivated smoothing heuristics to address common biases in observational data, a time-based smoothing to handle concept drift and a class-wise smoothing based on class priors to mitigate data imbalance. The effectiveness of the proposed methods is demonstrated on four real-world goodwill assessment data sets of a car manufacturer with the aim of automating goodwill decisions. Overall, this paper presents a promising approach for modeling ordinal observational data that can improve decision-making processes and reduce reliance on human expertise. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

Date 2023

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Language English

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Conference Name Lecture Notes in Computer Science

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Decision making, Machine-learning, Machine learning, Domain Knowledge, Ordinal classification, Unimodal, Label smoothing, Expert ratings, Observational data, Ordinal regression, Ordinal scale, Prescriptive machine learning, Unimodal distribution

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EID: 2-s2.0-85174443319

Export Date: 20 September 2026; Cited By: 6; Correspondence Address: S. Haas; BMW Group, Munich, Germany; email: stefan.sh.haas@bmwgroup.com; Conference name: 23rd Joint European Conference on Machine Learning and Knowledge Discovery in Databases, ECML PKDD 2023; Conference date: 18 September 2023 through 22 September 2023; Conference code: 301649

Reduce false positives for human detection by a priori probability in videos

Item Type Conference Paper

Author L. Wang

Author X. Zhao

Author Y. Liu

Abstract In this work, we address the problem of reducing the false positives for human detection in videos. We employ the motion cue to build a foreground probability model. Then the mean expectation of the pixel-level foreground probability is computed to assign a priori probability to the sliding window in detection. We combine the response of Deformable Part Models and the mean probability expectation to form the features and train a linear classifier. The proposed approach is threshold-free, and reduces the false positives in human detection by the foreground cues. As well, we describe an integral probability image for fast computation of the mean probability expectation. Experimental results show that the proposed method achieve superior performance over the baseline of Deformable Part Models. © 2015 IEEE.

Date 2016

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - IAPR Asian Conf. Pattern Recognit., ACPR

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-147996100-9 (ISBN)**Pages** 584-588**Proceedings Title** Proc. - IAPR Asian Conf. Pattern Recognit., ACPR**Conference Name** Proceedings - 3rd IAPR Asian Conference on Pattern Recognition, ACPR 2015**Date Added** 20/09/2026, 9:53:01 pm**Modified** 20/09/2026, 9:53:01 pm**Tags:**

Pattern recognition, Probability, False positive, A-priori probabilities, Human detection, Linear classifiers, Deformable part models, Deformation, Probability image, Probability modeling, Fast computation

Notes:

EID: 2-s2.0-84978832160

Export Date: 20 September 2026; Cited By: 2; Conference name: 3rd IAPR Asian Conference on Pattern Recognition, ACPR 2015; Conference date: 3 November 2016 through 6 November 2016; Conference code: 122205

Reduce false positives for object detection by a priori probability in videos

Item Type Journal Article**Author** L. Wang**Author** X. Zhao**Author** Y. Liu

Abstract In this work, we address the problem of reducing the false positives for object detection in videos. We employ the motion cue to build a foreground probability model. Then the mean expectation of the pixel-level foreground probability is computed to assign a priori probability to the sliding window in detection. The proposed foreground model is evaluated with the detection framework of Deformable Part Models (DPM). We combine the response of DPM detector and the mean probability expectation to form the features and train a linear classifier. The proposed approach is threshold-free, and reduces the false positives in object detection by the foreground cues. Besides, we describe an integral probability image for fast computation of the mean probability expectation. Experimental results show that the proposed method achieve superior performance over the baseline of Deformable Part Models. © 2016 Elsevier B.V.

Date 2016**DOI** 10.1016/j.neucom.2016.03.082**URL** <https://www.scopus.com/pages/publications/84973455069?origin=resultslist>**Archive** Scopus**Language** English**Volume** 208**Publisher** Elsevier B.V.

Pages 325-332**Publication** Neurocomputing**ISSN** 09252312 (ISSN)**Journal Abbr** Neurocomputing**Date Added** 20/09/2026, 9:52:46 pm**Modified** 20/09/2026, 9:52:46 pm**Tags:**

Article, artificial intelligence, Object recognition, Object detection, conceptual framework, Pedestrian detection, Probability, videorecording, Detection framework, classifier, intermethod comparison, probability, analytical error, image analysis, image quality, statistical model, Foreground modeling, motion, linear system, A-priori probabilities, priority journal, mathematical computing, Linear classifiers, association, computer analysis, Deformable part models, Deformation, Probability image, Probability modeling

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Export Date: 20 September 2026; Cited By: 4; Funding details: National Natural Science Foundation of China, NSFC, (61273285, 61375019); Correspondence Address: X. Zhao; Key Laboratory of System Control and Information Processing Department of Automation, Shanghai Jiao Tong University, Shanghai, 800 Dongchuan RD, 200240, China; email: zhaoxu@sjtu.edu.cn; CODEN: NRCGE

REDUCR: Robust Data Downsampling using Class Priority Reweighting

Item Type Conference Paper**Author** W. Bankes**Author** G. Hughes**Author** I. Bogunovic**Author** Z. Wang**Editor** Globerson A.**Editor** Mackey L.**Editor** Belgrave D.**Editor** Fan A.**Editor** Paquet U.**Editor** Tomczak J.**Editor** Zhang C.

Abstract Modern machine learning models are becoming increasingly expensive to train for real-world image and text classification tasks, where massive web-scale data is collected in a streaming fashion. To reduce the training cost, online batch selection techniques have been developed to choose the most informative datapoints. However, many existing techniques are not robust to class imbalance and distributional shifts, and can suffer from poor worst-class generalization

performance. This work introduces REDUCR, a robust and efficient data downsampling method that uses class priority reweighting. REDUCR reduces the training data while preserving worst-class generalization performance. REDUCR assigns priority weights to datapoints in a class-aware manner using an online learning algorithm. We demonstrate the data efficiency and robust performance of REDUCR on vision and text classification tasks. On web-scraped datasets with imbalanced class distributions, REDUCR significantly improves worst-class test accuracy (and average accuracy), surpassing state-of-the-art methods by around 15%. © 2024 Neural information processing systems foundation. All rights reserved.

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Extra Journal Abbreviation: Adv. neural inf. proces. syst.

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Conference Name Advances in Neural Information Processing Systems

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Region-based evidential deep learning to quantify uncertainty and improve robustness of brain tumor segmentation

Item Type Journal Article

Author H. Li

Author Y. Nan

Author J. Del Ser

Author G. Yang

Abstract Despite recent advances in the accuracy of brain tumor segmentation, the results still suffer from low reliability and robustness. Uncertainty estimation is an efficient solution to this problem, as it provides a measure of confidence in the segmentation results. The current uncertainty estimation methods based on quantile regression, Bayesian neural network, ensemble, and Monte Carlo dropout are limited by their high computational cost and inconsistency. In order to overcome these challenges,

Evidential Deep Learning (EDL) was developed in recent work but primarily for natural image classification and showed inferior segmentation results. In this paper, we proposed a region-based EDL segmentation framework that can generate reliable uncertainty maps and accurate segmentation results, which is robust to noise and image corruption. We used the Theory of Evidence to interpret the output of a neural network as evidence values gathered from input features. Following Subjective Logic, evidence was parameterized as a Dirichlet distribution, and predicted probabilities were treated as subjective opinions. To evaluate the performance of our model on segmentation and uncertainty estimation, we conducted quantitative and qualitative experiments on the BraTS 2020 dataset. The results demonstrated the top performance of the proposed method in quantifying segmentation uncertainty and robustly segmenting tumors. Furthermore, our proposed new framework maintained the advantages of low computational cost and easy implementation and showed the potential for clinical application. © 2022, The Author(s).

Date 2023

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Notes:

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Regional patterns and drivers of 15N fertilizer fate across China's wheat systems: Integrating meta-analysis and machine learning

Item Type Journal Article

Author Y. Yan

Author S. Zhang

Author Y. Zhang

Author M. Xu

Author J. Xu

Author Y. Li

Author B. Zhao

Author L. Yuan

Abstract Context or problem: Balancing wheat production with improved nitrogen (N) use efficiency and reduced N loss depends on a clear understanding of fertilizer-N fate. However, integrative syntheses of 15N evidence across China's diverse wheat-growing regions remain scarce. Objective and methods: Using a compiled database of 421 observations from 86 published 15N field tracer studies in China's wheat systems, we integrated meta-analysis with machine learning to quantify the effects of climate, soil, and N management factors on wheat yield and 15N fate, and to predict region-specific attainable 15N uptake potential. Results and conclusions: Wheat yield under N fertilization (52.5–489.0 kg N ha⁻¹) averaged 6874.02 kg ha⁻¹ nationwide, with fertilizer-15N exhibiting uptake, residual, and loss percentages of 37.29%, 31.53%, and 31.35%, respectively. Regionally, wheat yield ranked as Central China (CC) > Southern China (SC) > Northern China (NC), whereas fertilizer-15N uptake% ranked as CC > NC > SC. Random forest (RF) analysis showed different dominant drivers for yield (SOM, MAT, and BD) and fertilizer-15N uptake% (TN, MAP, and soil pH), suggesting that the inconsistent regional patterns between yield and fertilizer-15N uptake% may be partly attributable to differences in soil fertility and hydroclimatic conditions. Spatial prediction was used to estimate attainable fertilizer-15N uptake, defined as RF-predicted 15N uptake% based on gridded predictors representing broad spatial gradients. Regional attainable uptake was 46.76%, 51.67%, and 49.86% in NC, CC, and SC, respectively, with attainable–observed fertilizer-15N uptake differences of 10.24, 12.36, and 17.45 %age points. Implications: These findings provide a scientific basis for region-specific optimization of N management to enhance 15N uptake under local climatic and soil conditions, supporting sustainable precision agriculture. © 2026 Elsevier B.V.

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Regularization-based methods for ordinal quantification

Item Type Journal Article

Author M. Bunse

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Author F. Sebastiani

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Abstract Quantification, i.e., the task of predicting the class prevalence values in bags of unlabeled data items, has received increased attention in recent years. However, most quantification research has concentrated on developing algorithms for binary and multi-class problems in which the classes are not ordered. Here, we study the ordinal case, i.e., the case in which a total order is defined on the set of $n > 2$ classes. We give three main contributions to this field. First, we create and make available two datasets for ordinal quantification (OQ) research that overcome the inadequacies of the previously available ones. Second, we experimentally compare the most important OQ algorithms proposed in the literature so far. To this end, we bring together algorithms proposed by authors from very different research fields, such as data mining and astrophysics, who were unaware of each others' developments. Third, we propose a novel class of regularized OQ algorithms, which outperforms existing algorithms in our experiments. The key to this gain in performance is that our regularization prevents ordinally implausible estimates, assuming that ordinal distributions tend to be smooth in practice. We informally verify this assumption for several real-world applications. © The Author(s) 2024.

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DOI 10.1007/s10618-024-01067-2

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Notes:

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Regularized learning for domain adaptation under label shifts

Item Type Conference Paper

Author K. Azizzadenesheli

Author F. Yang

Author A. Liu

Author A. Anandkumar

Abstract We propose Regularized Learning under Label shifts (RLLS), a principled and a practical domain-adaptation algorithm to correct for shifts in the label distribution between a source and a target domain. We first estimate importance weights using labeled source data and unlabeled target data, and then train a classifier on the weighted source samples. We derive a generalization bound for the classifier on the target domain which is independent of the (ambient) data dimensions, and instead only depends on the complexity of the function class. To the best of our knowledge, this is the first generalization bound for the label-shift problem where the labels in the target domain are not available. Based on this bound, we propose a regularized estimator for the small-sample regime which accounts for the uncertainty in the estimated weights. Experiments on the CIFAR-10 and MNIST datasets show that RLLS improves classification accuracy, especially in the low sample and large-shift regimes, compared to previous methods. © 7th International Conference on Learning Representations, ICLR 2019. All Rights Reserved.

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Language English
Extra Journal Abbreviation: Int. Conf. Learn. Represent., ICLR
Publisher International Conference on Learning Representations, ICLR
Proceedings Title Int. Conf. Learn. Represent., ICLR
Conference Name 7th International Conference on Learning Representations, ICLR 2019
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Rehabilitation exergames: Use of motion sensing and machine learning to quantify exercise performance in healthy volunteers

Item Type Journal Article

Author R.H. Osgouei

Author D. Soulsby

Author F. Bello

Abstract Background: Performing physiotherapy exercises in front of a physiotherapist yields qualitative assessment notes and immediate feedback. However, practicing the exercises at home lacks feedback on how well patients are performing the prescribed tasks. The absence of proper feedback might result in patients performing the exercises incorrectly, which could worsen their condition. We present an approach to generate performance scores to enable tracking the progress by both the patient at home and the physiotherapist in the clinic. Objective: This study aims to propose the use of 2 machine learning algorithms, dynamic time warping (DTW) and hidden Markov model (HMM), to quantitatively assess the patient's performance with respect to a reference. Methods: Movement data were recorded using a motion sensor (Kinect V2), capable of detecting 25 joints in the human skeleton model, and were compared with those of a reference. A total of 16 participants were recruited to

perform 4 different exercises: shoulder abduction, hip abduction, lunge, and sit-to-stand exercises. Their performance was compared with that of a physiotherapist as a reference. Results: Both algorithms showed a similar trend in assessing participant performance. However, their sensitivity levels were different. Although DTW was more sensitive to small changes, HMM captured a general view of the performance, being less sensitive to the details. Conclusions: The chosen algorithms demonstrated their capacity to objectively assess the performance of physical therapy. HMM may be more suitable in the early stages of a physiotherapy program to capture and report general performance, whereas DTW could be used later to focus on the details. The scores enable the patient to monitor their daily performance. They can also be reported back to the physiotherapist to track and assess patient progress, provide feedback, and adjust the exercise program if needed. ©Rachel K Nelligan, Rana S Hinman, Pek Ling Teo, Kim L Bennell.

Date 2020

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Machine learning, Performance assessment, Dynamic time warping, Hidden Markov model, Motion sensing, Rehabilitation exergames, Similarity score

Notes:

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Export Date: 20 September 2026; Cited By: 30; Funding details: Higher Education Funding Council for England, HEFCE; European Regional Development Fund, FEDER; Correspondence Address: R.H. Osgouei; Imperial College Centre for Engagement and Simulation Science (ICCESS), Faculty of Medicine, Department of Surgery and Cancer Imperial College London, Room H 3 8, ICCESS, London, Chelsea and Westminster 369 Fulham Road, SW10 9NH, United Kingdom; email: r.haghighi-osgouei@imperial.ac.uk

Relative aortic valve stiffness: a new parameter quantifying aortic stenosis

Item Type Journal Article

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Author A. Papadis

Author I. Blaser

Author N. Nozica

Author M. Kassar

Author R. Ludwig

Author D. Berger

Author M. Hänggi

Author M. Mauri

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Abstract Aortic stenosis (AS) quantification relies on transvalvular gradients or aortic valve opening area (AVA), two measures that inherently depend on the instantaneous transvalvular flow rate (Q). We present a new quantification method for AS based on new modeling where AVA saturates with increasing Q, contradicting the assumption of linear Q-AVA relation currently predominant in the literature. The objective of this study is the development of a new method for the quantification of AS. In 141 patient undergoing transesophageal echocardiography, AVA was obtained frame-by-frame over an entire systole from 3-D echocardiograms using multiplanar reconstruction. Q was retrieved by Doppler- and 3-D echocardiography of the left ventricular outflow tract. A sigmoid Q-AVA relation was fitted with a machine learning algorithm. The measured and the predicted AVA were compared between a parametrized linear and sigmoid Q-AVA relation model. A relative aortic valve stiffness (rAVS) was calculated, and an isostiffness nomogram was constructed. Cox proportional hazard modeling was used to look for predictors of rehospitalization-free survival. Compared with the linear model, the sigmoid model consistently better predicted measured AVA in all AS severity groups. Within severity groups, rAVS remained robust and constant, a property not shared by the linear model. A receiver operating characteristic (ROC)-analysis revealed a rAVS value of 1.51 as the best cut-off for distinguishing severe from nonsevere AS. Cox proportional hazard modeling showed its value as an independent predictor of rehospitalization-free survival. We present a new method for the quantification of AS that could help simplify the diagnosis of patients with low-flow, low-gradient AS. Copyright © 2026 The Authors.

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Article, female, Female, human, Humans, male, Male, middle aged, Middle Aged, machine learning, aged, Aged, Aged, 80 and over, very elderly, Machine Learning, risk factor, predictive model, controlled study, major clinical study, diagnostic test accuracy study, disease severity, receiver operating characteristic, predictive value, machine learning algorithm, diagnostic imaging, sensitivity and specificity, image reconstruction, severity of illness index, Severity of Illness Index, pathophysiology, echocardiography, Predictive Value of Tests, aortic stenosis, aortic valve, Aortic Valve, aortic valve stenosis, Aortic Valve Stenosis, biological model, cardiovascular parameters, Doppler echocardiography, Echocardiography, Doppler, Echocardiography, Three-Dimensional, Echocardiography, Transesophageal, heart left ventricle outflow tract, hemodynamics, Models, Cardiovascular, nomogram, Nomograms, proportional hazards model, rehospitalization free survival, relative aortic valve stiffness, stenosis quantification, survival, systole, three dimensional echocardiography, transesophageal echocardiography, valvular heart disease

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Relative quantification of plasma N-glycans in type II congenital disorder of glycosylation patients by mass spectrometry

Item Type Journal Article

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Author N.D.C. Fontes

Author S.C.L. Santos

Author D.J. Lefeber

Author C. Bloch

Author J.M. Brum

Author G.D. Brand

Abstract Background: Type II Congenital Disorders of Glycosylation (CDG-II) are a group of diseases with challenging diagnostics characterized by defects in the processing of glycans in the Golgi apparatus. Mass Spectrometry (MS) has been a valuable tool in the definition of CDG-II subtypes. While some CDG-II subtypes are associated with specific N-glycan structures, others only produce changes in relative levels, reinforcing the demand for quantification methods. Methods: Plasma samples from control individuals were pooled, derivatized with deuterated iodomethane (I-CD 3),

and used as internal standards for controls and patients whose glycans were derivatized with iodomethane (I-CH₃), followed by MALDI MS, LC-MS and -MS/MS analyses. Results: Total N-glycans from fifteen CDG-II patients were evaluated, and 4 cases with molecular diagnosis were considered in detail: 2ATP6V0A2-CDG siblings, and 2 MAN1B1-CDG patients, one of them carrying a previously undescribed p.Gly536Val mutation. Conclusions: Our methodology offers a feasible alternative to the current methods for CDG-II diagnosis by MS, which quantify glycan structures as fractions of the total summed signal across a mass spectrum, a strategy that lowers the variability of minor components. Moreover, given its sensitivity for less concentrated yet biologically relevant structures, it might assist the uncovering of novel diagnostic glycans in other CDG-II subtypes. © 2019 Elsevier B.V.

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Reliability assessment framework based on feature separability for pathological cell image classification under prior bias

Item Type Journal Article

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Author M. Fujita

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Author Y. Kakeji

Abstract Background and objective: Prior probability shift between training and deployment datasets can compromise deep learning-based medical image classification, and standard posterior reweighting is not uniformly beneficial. We developed a feature-separability-based reliability assessment framework to identify when prior correction is likely to improve pathological cell classification. Methods: We analysed 303 colorectal cancer specimens with CD103/CD8 dual immunostaining, generating 171,638 annotated cell images across 19 cell types. After class-capped, imbalance-reduced sampling, 16 cell types were evaluated using ResNet models trained under controlled prior-bias conditions. Feature separability was quantified using cosine similarity-based likelihood quality scores derived from intra-class and inter-class relationships in learned embedding spaces. Associations with F1 performance were examined together with prior-bias magnitude and sample adequacy. Results: Feature separability was the dominant predictor of F1 performance in the multivariable model ($\beta=1.650$, $p<0.001$), whereas prior-bias magnitude showed a much smaller effect ($\beta=0.004$, $p=0.018$). A likelihood-quality threshold of 0.294 provided an empirical operating point for identifying cases likely to benefit from correction. Overall, 78.0% of cases showed no improvement after correction, indicating that correction should not be applied uniformly. Supplementary ViT and UNI analyses showed the same broad separability-reliability relationship. Conclusion: Feature separability, rather than prior-bias magnitude alone, determined when prior correction improved classification performance. The proposed framework supports selective correction decisions during labelled validation, with task-specific thresholds requiring recalibration for other datasets. Copyright © 2026. Published by Elsevier B.V.

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Reliability assessment of guided wave damage localization with deep learning uncertainty quantification methods

Item Type Journal Article

Author I.D. Khurjekar

Author J.B. Harley

Abstract Guided wave-based structural health monitoring is an attractive option for detecting structural defects in an automated manner. In this work, we focus on the task of damage localization. Deep learning methods have been shown to have superior performance for damage localization. Yet, environmental variations introduce uncertainty in the system and reduce its reliability. For this reason, it is crucial to assess the reliability of estimates taken from structural health monitoring systems. In this work, we estimate the localization reliability from a single snapshot of sparse array guided wave measurements instead of reporting values averaged over an entire set of test measurements. The assessment strategy can be added to any deep learning localization model and produces both a localization and uncertainty estimate. The

deep learning model is trained using only guided wave simulations. We assess the uncertainty using both simulated and experimental data with temperature variations. Multiple deep learning-based uncertainty quantification methods are analyzed. Results demonstrate correlations between uncertainty, temperature variations, and the presence of synthetic damage. We also compare with reliability derived from delay-and-sum localization. We find that a deep ensemble learning strategy provides the most reliable damage localization and uncertainty quantification. © 2024 Elsevier Ltd

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Notes:

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Reliable predictions in the wild: A confidence-aware prior injection for mitigating distributional risks

Item Type Journal Article

Author J. Zhang

Author P. Li

Author G. Zhang

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Author P. Gong

Abstract Few-shot classifiers deployed in safety-critical environments are highly vulnerable to distributional shifts, particularly when facing simultaneous label shift and open-set contamination. These conditions compromise decision integrity, causing models to misinterpret potential outliers or open-set samples as high-confidence known classes. A conventional defense is to estimate the test-time prior and inject it globally into classifier logits. However, we identify that this uniform correction introduces a brittleness in the inference pipeline: it creates a trade-off where strengthening the prior to correct in-distribution shifts inadvertently over-corrects ambiguous or adversarial-like queries, increasing the risk of harmful prediction flips. To harden few-shot systems against these distributional risks, we introduce Piecewise- π , a lightweight inference-time defense module. Unlike rigid global injection, Piecewise- π employs a confidence-aware mechanism that partitions queries based on their reliability. It applies a defensive, conservative correction to low-confidence queries (which are statistically likely to be outliers) while enforcing stronger calibration only on high-confidence, verified predictions. Experiments on CIFAR and STL benchmarks under severe imbalance and contamination demonstrate that Piecewise- π significantly improves the risk-coverage trade-off and system reliability without retraining the backbone. Behavioral analysis confirms that Piecewise- π effectively minimizes harmful decision reversals, offering a computationally efficient strategy to secure classifier performance in dynamic, open environments. © 2026 Elsevier Ltd.

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Remote sensing and image analysis of macro-plastic litter: A review

Item Type Journal Article

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Author K. Topouzelis

Author A. Isobe

Abstract Effective reduction of oceanic plastic pollution requires scalable and objective monitoring methods that go beyond traditional human-based surveys. This review synthesizes recent advances in remote sensing and AI-driven image analysis for detecting macro-plastic litter. Peer-reviewed studies published up to 2024 were systematically selected from the Scopus database, focusing on applications of remote sensing platforms including webcams, drones, balloons, aircraft, and satellites for monitoring plastic litter in coastal, riverine, and other aquatic environments. Quantification methods ranged from manual annotation to deep learning-based models. Although machine learning has been increasingly adopted since around 2020, manual screening and rule-based approaches remain prevalent, reflecting the complexity of litter types and shapes. The review revealed considerable variability in quantification metrics—such as litter-covered area, volume, weight, and item count per unit area—which complicates cross-study comparisons and data harmonization. While remote sensing enhances spatial coverage, consistency, and repeatability, it faces persistent challenges, including environmental interference, limited resolution, and inconsistent protocols. Our findings highlight the urgent need for methodological standardization and harmonization of quantification units across platforms and geographic regions. Among available metrics, litter-covered area and item count per unit area are most suitable for cross-platform comparison. Continued development and integration of such technologies hold strong potential to facilitate science-based policymaking and long-term monitoring of plastic transport from land to ocean. © 2025 The Authors

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Remote Sensing Monitoring for Carbon Neutrality: High-Precision Tracking of Greenhouse Gas Emissions and Carbon Sink Dynamics

Item Type Conference Paper

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Author H. Qi

Author H. Zhang

Editor He B.-J.

Editor Prasad D.

Editor Cheshmehzangi A.

Editor Ma M.

Abstract In order to meet the 1.5 °C temperature control target of the Paris Agreement and the implementation requirements of carbon tariff policy, this paper builds a multi-modal remote sensing monitoring framework for carbon neutrality, breaking through the technical bottlenecks of traditional methods in emission traceability accuracy (error > 5%) and dynamic timeliness of carbon sinks (quarterly update). The framework design includes: a multi-modal spatio-temporal fusion model integrating Landsat8, Sentinel-5P TROPOMI and ERA5 data, and using the radiative transfer equation to correct the scale difference, the spatial positioning accuracy of point source emissions is less than 1 km; A dynamic carbon sink assessment algorithm based on the coupling model of NDVI and soil organic carbon was proposed. The parameter sensitivity (NDVI contribution 62.3%) was optimized by machine learning to quantify the interannual change rate of carbon sink. Edge computing monitoring system based on federated learning architecture, realizing hourly data update and response delay < 2 h. The multi-dimensional test verification of the Beijing-Tianjin-Hebei Demonstration zone showed that the F1-score of the industrial point source reached 0.89 (23.6% higher than that of the traditional method), the RMSE of the carbon sink decreased to 0.34 MtC/km² (65% less error), and the system covered 12 provinces and administrative regions and identified 1235 hidden emission sources. Significant contribution to research innovation: pioneered the three-level fusion paradigm of “radiative transfer, deep learning and federal computing” to support accurate carbon tariff accounting; Develop a dynamic monitoring system for subkilometer carbon sinks to meet the needs of global carbon budget verification under the Paris Agreement; Prototype a scalable global monitoring network to provide the technical tools to achieve UNFCCC goals. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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DOI 10.1007/978-3-032-11910-0_3

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K. Fan; The First Institute of Surveying and Mapping of Hebei Province, Shijiazhuang, Hebei, China; email: kenan-fan@163.com; Conference name: International Conference on Climate-resilient and Low-carbon Cities, iCCRLCC 2025; Conference date: 22 August 2025 through 24 August 2025; Conference code: 346789

Resampling ensemble algorithm for class imbalance problem using optimization algorithm

Item Type Journal Article**Author** S. Lavanya**Author** S. Palaniswami**Author** M. Divyabharathi

Abstract Learning classifiers from imbalanced data has attracted a significant amount of interest in recent years. The imbalanced class in the dataset problem is a special type of classification problem where the class priors are highly unequal and imbalanced. Resampling ensemble algorithm is developed focused on the classification problems for imbalanced datasets. The small classes are over sampled by the use of SMOTE (Synthetic Minority Oversampling Technique) algorithm and large classes are under-sampled where the samples are to be selected randomly from the majority class data. The optimization technique has been introduced on the oversampling and undersampling process in imbalanced classification. The aim is to study the use of Optimization Technique to improve the performance of imbalanced classification. A novel approach called as BAT Algorithm for over and under sampling (BAOUS) technique where the accuracy rate of all the classes is optimized at the same time. From the experimental result, the system is used to reduce the time complexity as well as it will enhance the accuracy rate of the imbalance classification result. © Research India Publications.

Date 2015**URL** <https://www.scopus.com/pages/publications/84942304130?origin=resultslist>**Archive** Scopus**Language** English**Volume** 10**Publisher** Research India Publications**Pages** 11520-11526**Publication** International Journal of Applied Engineering Research**ISSN** 09734562 (ISSN)**Issue** 13**Journal Abbr** Int. J. Appl. Eng. Res.**Date Added** 20/09/2026, 9:52:43 pm**Modified** 20/09/2026, 9:52:43 pm**Tags:**

Class imbalance, Classification, Classifiers, Features selection, Bagging, BAT algorithm, Imbalanced data-sets, Skewed distribution

Notes:

EID: 2-s2.0-84942304130

Export Date: 20 September 2026; Cited By: 1

Research and Application of Multilevel Wellbore Quality Quantification Method Based on Random Forest Algorithm

Item Type Conference Paper

Author Y. Xiao

Author T. Zhou

Author H. Li

Author X. Kong

Author L. Song

Author S. Diao

Author J. Liu

Author Z. Zhong

Author Y. Jin

Author B. Li

Editor Lin J.

Abstract The quality of drilling wellbore is crucial for oilfield companies to evaluate the construction acceptance of drilling operations, evaluate the wellbore quality before cementing operations, and smoothly implement subsequent casing and electrical logging tools. At present, the evaluation of wellbore quality only requires dispersed trajectory parameters of different well sections (well inclination angle, dog leg angle, target range), and only considers whether the trajectory parameters are within the qualified range during completion acceptance. A quantitative scoring method for wellbore quality has not yet been developed, which is not conducive to the evaluation of well quality by oilfield enterprises during drilling completion acceptance. In this paper, the key parameters such as deviation measurement interval, dogleg degree, maximum deviation and target hit are comprehensively considered. Based on the comprehensive method of mathematical statistics and Matlab programming calculation, a multi-level quantitative scoring model of well bore quality is established. The key parameters in the multi-level quantitative model of well bore quality are determined by using the main control factor analysis method of random forest algorithm. Taking the A oil field as an example, a quantitative scoring was conducted on the quality of 74 drilled wells in 2021 to evaluate the effectiveness of the operation team in controlling well quality. The analysis results showed that for the A oil field, the importance of factors affecting well quality was as follows: maximum inclination > Target radius > dog leg = survey interval. Crew 1 had better control over well quality, and the proportion of high-quality wells in 2021 decreased compared to 2020, The analysis results of

the multi-level well bore quality quantification method based on random forest algorithm proposed in this paper are in line with the field work practice, which provides a theoretical basis for well bore quality quantification, and is of great significance for the process management and efficiency improvement in oil fields. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2024.

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machine learning, modeling, Factor analysis, Statistics, algorithm, Quality control, Multilevels, Quantification methods, Parameter estimation, Oil well logging, Infill drilling, Bore quality, borehole, Boreholes, Evaluation of main control factor, Evaluation of main control factors, Main control factor, MATLAB, oil field, Oil field equipment, oil well, Quantitative model, Quantitative models, Quantitative scoring, Random forest algorithm, Wellbore, Wellbore quality

Notes:

EID: 2-s2.0-85188465973

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: T. Zhou; CNPC Engineering Technology R&D Company Limited, Beijing, 102206, China; email: zhoutuodr@cnpc.com.cn; Conference name: 13th International Field Exploration and Development Conference, IFEDC 2023; Conference date: 19 September 2023 through 21 September 2023; Conference code: 308259

Research into the Influence Mechanisms of Visual-Comfort and Landscape Indicators of Urban Green Spaces

Item Type Journal Article

Author Y. Meng

Author J. Shi

Author M. Lyu

Author D. Sun

Author H. Fukuda

Abstract Urban green spaces play a crucial role in providing social services and enhancing residents' mental health. It is essential for sustainable urban planning to explore the relationship between urban green spaces and human perceptions, particularly their visual comfort. However, most current research has analyzed green spaces using two-dimensional indicators (remote sensing), which often overlook human visual perceptions. This study combined two-dimensional and three-dimensional methods to evaluate urban green spaces. Additionally, the study employed machine learning to quantify residents' visual comfort in green-space environments and explored the relationship between green spaces and human visual perceptions. The results indicated that Kitakyushu exhibited a moderate FCV and an extremely low Green View Index (GVI). Yahatanishi-ku was characterized as having the highest visual comfort. Tobata-ku demonstrated the lowest visual comfort. Natural, GVI, openness, enclosure, vegetation diversity, landscape diversity, and NDBI were positively correlated with visual comfort. FCV and ENVI were negatively correlated with visual comfort. Vegetation diversity had the most impact on improving visual comfort. By integrating remote sensing and street-view data, this study introduces a methodology to ensure a more holistic assessment of green spaces. Urban planners could use it to better identify areas with insufficient green space or areas that require improvement in terms of green-space quality. Meanwhile, it could be helpful in providing valuable input for formulating more effective green-space policies and improving overall urban environmental quality. The study provides a scientific foundation for urban planners to improve the planning and construction of healthy and sustainable cities. © 2024 by the authors.

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machine learning, remote sensing, urban development, greenspace, research, landscape change, image analysis, image classification, green spaces, street-view image, visual analysis, visual comfort

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Export Date: 20 September 2026; Cited By: 12; Funding details: Project of Liaoning Provincial Department of Education, (JYTMS20231583); Correspondence Address: H. Fukuda; Department of Architecture, The University of Kitakyushu, Kitakyushu, 808-0135, Japan; email: fukuda@kitakyu-u.ac.jp

Research on an Intelligent Prediction Method for the Carbon Emissions of Prefabricated Buildings During the Construction Stage, Based on Modular Quantification

Item Type Journal Article

Author Y. Yang

Author X. Cai

Author X. Ma

Author G. Yao

Author T. Lei

Author H. Tan

Author Y. Wang

Abstract Prefabricated buildings are widely utilized due to their effectiveness in reducing carbon emissions. The construction stage has a significantly higher carbon emission rate than the other stages of their life cycle, but this is difficult to accurately quantify and predict due to the high variability. This study clarifies the system boundary of carbon emissions and the parameters of influence in carbon emissions predictions. The carbon emission quantification model was improved by using the process analysis method and the carbon emission factor method, and a modular calculation formula was proposed. Based on the machine learning algorithm, a carbon emissions prediction model for prefabricated buildings' construction stage was established and hyperparameter optimization was conducted. A sample database for predicting prefabricated buildings' carbon emissions during the construction stage was established using a modular quantification method, and the thin plate spline interpolation algorithm was introduced to expand this. The prediction results of carbon emission prediction models using four algorithms, SVR, BPNN, ELM, and RF, were compared and analyzed by RMSE and R2. The results show that the model based on BPNN has the highest prediction accuracy when determining the carbon emissions of prefabricated building during the construction stage, and this method can provide a more accurate reference for subsequent quantitative research on carbon emissions from prefabricated buildings. © 2025 by the authors.

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ISSN 20755309 (ISSN)**Issue** 12**Journal Abbr** Buildings**Date Added** 20/09/2026, 9:51:37 pm**Modified** 20/09/2026, 9:51:37 pm**Tags:**

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Research on Data Security Analysis and Label Recognition Technology Based on Big Data Business Scenario

Item Type Conference Paper**Author** D. Liu**Author** D. Kong**Author** X. Liu**Author** L. Ma**Author** Y. Chang**Author** H. Zhang**Author** R. Wang**Author** H. Yu**Editor** Li W.**Editor** Zhang X.

Abstract Data is the inevitable product of social high informatization, and its security risk is an indispensable part of information security. In the case of accelerating the construction of ubiquitous power Internet of things, this paper deeply studies the

data security analysis and label recognition technology based on big data business scenarios. Based on the process of active and passive discovery, a complete catalog of data assets is established by means of database scanning, server scanning, terminal scanning, terminal monitoring, network monitoring and database access monitoring. The threat tree is used to model various threats to data security. On the basis of data asset cataloging and threat modeling, this paper studies the data security risk analysis and quantification method by comprehensively judging the threats faced by the business system, as well as the vulnerability and asset value of the system itself. Based on the heuristic integrated learning strategy of automatic semantic tagging for massive network resources, metadata is extracted through active and passive collection, and the semantic relationship between tags is extracted and processed by natural language processing and machine learning technology. Finally, automatic labels are generated through analysis capabilities such as business correlation analysis, organizational correlation analysis, data distribution analysis, data classification and grade. © 2020 IEEE.

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Notes:

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Research on Generative Language Models Aiding Project Quantification Methods

Item Type Conference Paper

Author F. Gao

Abstract With the construction of digital projects, the Work Breakdown Structure (WBS)-based review system plays an important role in the feasibility study review of projects. Therefore, the main objective is to design a set of methods and a review system for quantifying the influence factors of feasibility study review projects based on digital technology. This is intended to address the issue in the existing project review system where the verification of influence factors mainly relies on experts' experience, making it difficult to achieve objectivity and accuracy. This technology will use Natural Language Understanding (NLU) technology to analyze the textual descriptions of project functions, identify key entities and actions, and then match them with pre-defined rule abstraction models to determine the corresponding fixed influence factors. In addition, multiple methods will be employed to quantify floating influence factors, including traditional machine learning prediction models, large language model-based semantic similarity-weighted influence factors, and generative large model-based automatic code generation evaluation. © 2025 Author(s).

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Computational linguistics, Language model, Deep learning, Model-based OPC, Learning systems, Machine-learning, Natural language processing systems, Semantics, Deep Learning, Machine Learning, Quantification methods, Quantification Methods, Structure-based, Feasibility studies, Generative language model, Generative Language Models, Review systems, Work breakdown structure

Notes:

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Research on Interaction Potential of Electric Vehicles in Power Grid Dispatching and Operation Scenarios

Item Type Conference Paper

Author P. Liu

Author X. Yang

Author B. Cao

Author Z. Wang

Author E. Zhou

Editor S. Shmaliy Y.

Editor Nayyar A.

Abstract As the number of electric vehicles increases year by year, electric vehicles can be used as energy storage equipment to optimize the grid load curve. Studying the interactive potential evaluation and quantification methods of electric vehicles as adjustable resources to participate in grid mobilization and operation has far-reaching significance for the optimal operation of electric vehicles and grid load. This paper is based on the historical operation data of all electric vehicles in Shanghai, collected by the Internet of Vehicles detection platform. First, classify electric vehicles as adjustable resources in interaction, and build corresponding indicator system based on classification. Secondly, the evaluation method and quantitative method are given according to the classification. Finally, an example is simulated based on the data. The simulation results show that the electric vehicles in Shanghai, as an interactive and adjustable resource for grid operation, have the potential to optimize the grid operation curve in peak shaving and valley filling. © 2023, The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd.

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Digital storage, Electric vehicles, Electric vehicle, Curve fitting, Evaluation methods, Power grids, Electric load dispatching, Grid operation, Interaction potentials, Load dispatching, Network interaction, Peak cutting and valley filling, Power grid dispatching, Power grid load dispatching operation, Train network interaction

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Export Date: 20 September 2026; Cited By: 0; Funding details: State Grid Co., LTD, (520940210005); Correspondence Address: P. Liu; National Engineering Research Center of Electric Vehicles, Beijing Institute of Technology, Beijing, 10081, China; email: lp@bitnei.cn; Conference name: 7th International Conference on Computing, Control and Industrial Engineering, CCIE 2023; Conference date: 25 February 2023 through 26 February 2023; Conference code: 298689

Research on Intrusion Detection Based on Interpretable Machine Learning

Item Type Conference Paper

Author M. Chen

Author B. Ma

Author H. Jiang

Abstract Intrusion Detection Systems (IDS) grapple with challenges such as delayed updates and a high false positive rate in traditional rule-based approaches under high-dimensional network traffic data. Although deep learning has significantly improved detection accuracy, the conflict between its "black box" nature and traceability requirements still persists. This paper focuses on port scanning attack detection and provides a "dual-wheel-driven" solution of "architecture optimization and interpretability enhancement" to build a high-precision interpretable learning framework. To tackle the vanishing gradient and overfitting problems of traditional Deep Neural Networks (DNN), an enhanced DNN model architecture is designed: through dynamic residual structure, "expansion-compression" design, and adaptive regularization strategy, the learning ability of high-dimensional network traffic features is improved. Moreover, to address the common "black-box" problem in current models, this paper introduces a feature importance quantification method based on gradient-weighting, which integrates

dynamic prediction confidence weights to enhance the interpretability analysis of model decisions. In this study, the CICIDS-2017 dataset is chosen for training and evaluation. The results indicated that the enhanced DNN model had significantly better accuracy on the test set than traditional DNN and traditional machine learning models. More importantly, the enhanced DNN model can also identify key attack features, providing high-precision and transparent methodological support for cybersecurity defense. © 2025 Copyright held by the owner/author(s).

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Notes:

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Research on Landslide Risk Assessment of Mountainous Highways

Item Type Journal Article
Author H.-J. Wen
Author F.-Y. Yan
Author J.-Y. Zhao
Author Y. Liu

Abstract Aiming at the problems that there is a lack of a comprehensive assessment framework for landslide risks of mountainous expressways and existing studies only focus on landslide susceptibility, which makes it difficult to fully support risk management and control, this study intends to construct a scientific landslide risk assessment system for mountainous expressways. Based on Geographic Information System (GIS) technology, this study proposes a three-dimensional assessment framework integrating “landslide susceptibility of slope units-vulnerability of expressways-exposure”. Specifically, firstly, combined with the landslide triggering mechanism and engineering geological survey data, 15 landslide influencing factors including elevation, slope gradient and slope position were selected, and a LightGBM landslide susceptibility evaluation model was constructed using the semi-supervised learning method. Secondly, by quantifying the traffic volume after road completion and the economic losses that may be caused by the damage of different structures, 5 vulnerability indicators were determined, including life loss, vehicle loss, direct loss of expressway structures, repair costs and indirect loss caused by traffic interruption. The entropy weight method was used to calculate the weight of each indicator to complete the vulnerability evaluation. Finally, the exposure of the expressway was characterized by the area of the slope unit where the road affected by landslides is located and the average distance from the slope unit to the road. By multiplying susceptibility, vulnerability and exposure, this study realized the quantitative comprehensive assessment of landslide risks of mountainous expressways. This study breaks through the limitation of traditional studies that only focus on landslide susceptibility, and constructs a “susceptibility-vulnerability-exposure” trinity comprehensive assessment framework for landslide risks of mountainous expressways. Moreover, the selection and quantification methods of indicators in each dimension are more in line with practical engineering scenarios. The research results can provide methodological support for the comprehensive assessment of landslide risks, and at the same time offer a scientific basis for the mitigation, control and risk management decision-making of landslide risks of mountainous expressways. © (2024), (Chang'an University). All Rights Reserved.

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Research on Network Security Evaluation and Optimal Active Defense based on Attack and Defense Game Model in Big Data Era

Item Type Conference Paper

Author X. Ge

Abstract Due to the increasing progress of socialist modernization and the increasing progress of social productivity, the national computer industry has developed rapidly in recent years, and has become the main driving force to promote China's economic development. Facing the increasingly complex network environment, the traditional network security technology can not effectively resist the existing network attacks. Low defense and high memory consumption are two disadvantages of traditional network information security assessment. In essence, network security is the process of mutual game between network attackers and network defenders. In the process of mutual game, if one party benefits, it is the winner. In order to carry out the security evaluation and active defense of network information system, the network defense graph model, the classification of attack and defense strategies and their cost quantification method, the network attack and defense game model and the optimal active defense selection algorithm based on the above model are proposed. Based on the attack and defense game theory, this paper analyzes the network security evaluation and optimal active defense methods by establishing a model © 2022 ACM.

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Notes:

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Research on Offline Classification and Counting Algorithm of Long Fitness Video

Item Type Conference Paper

Author Z. Zheng

Author Z. Ma

Abstract With the holding of sports events such as the Winter Olympics, "fitness" has once again become a key topic of widespread concern. Scientific fitness has become a new requirement when people engage in fitness activities today. Scientific and efficient fitness guidance is often inseparable from the support of personal exercise data. At present, the motion data quantification method for motion recognition and counting through wearable devices such as sensors has problems such as poor convenience and poor visualization. This paper uses deep learning methods to study motion classification and counting, which makes up for the shortcomings of traditional quantification methods. The purpose of this paper is to identify and count five common types of exercises in long videos: squats, sit-ups, push-ups, pull-ups, and jumping jacks. The main work and innovations are as follows: (1) This paper proposes an algorithm for motion classification and counting based on long videos. The algorithm first uses the temporal action detection model MPGCN proposed in this paper to locate and classify valid segments of long videos, and then uses the ATRepNet motion counting model

proposed in this paper to count the cropped valid segments. (2) The large difference between the action duration and the action amplitude in the long video leads to the deterioration of the effect of the time-series action detection model. In view of this, this paper designs a MS multi-scale graph convolution layer which is integrated into the PGCN model for graph convolution multi-scale information fusion, and the MPGCN model is proposed. Finally, the mean Average Precision (mAP) of the localization and classification of dual-stream features in the public dataset is improved by 1.2%. (3) The complex video background in this scene leads to poor counting accuracy. In response to this, this paper improves the hybrid attention mechanism and integrates it into the counting model RepNet, and proposes the ATRepNet model. Finally, on the self-made test dataset, the counting accuracy is improved by 1.5%. © 2022 IEEE.

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Conference Name 2022 7th International Conference on Image, Vision and Computing, ICIVC 2022

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Deep learning, Classification (of information), Learning systems, Detection models, Convolution, Health, Statistical tests, Data visualization, Motion estimation, Deterioration, Quantification methods, Action counting, Action Counting, Action recognition, Action Recognition, Counting models, Motion classification, Offline classification, Olympics, Sports events, Temporal action detection, Temporal Action Detection

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Export Date: 20 September 2026; Cited By: 1; Conference name: 7th International Conference on Image, Vision and Computing, ICIVC 2022; Conference date: 26 July 2022 through 28 July 2022; Conference code: 182902

RESEARCH ON PULSE EDDY CURRENT DETECTION TECHNIQUE OF THINNING DEFECTS IN THE ALUMINUM-STEEL STRUCTURE OF WING WALL PLATE

Item Type Journal Article**Author** T. Chen**Author** Y. Yin**Author** C. Lv**Author** X. Song**Author** Z. Deng**Author** Q. Wu

Abstract Owing to the variation in metal activity, the aluminum-steel structure of wing wall panels is particularly susceptible to galvanic corrosion. Because of the concealed nature of the defects and the significant disparity in the physical properties of the materials, detecting thinning defects in aluminum-steel structures proves highly challenging. To tackle these issues, this study introduces a pulsed eddy current detection technique capable of accurately identifying various types of thinning defects within aluminum-steel structures. First, finite element simulation software is employed to establish the detection model, constructing the mapping relationship between the characteristic signal and thinning through simulation signal analysis. Second, the defect identification classification space and the inversion curve for thinning are developed. Finally, a pulsed eddy current detection system is developed to validate the proposed defect classification and quantification method. Experimental results demonstrate that the method effectively identifies and quantifies various thinning defects in the aluminum-steel structures of wing wall panels. ©2025 American Society for Nondestructive Testing.

Date 2025**DOI** 10.32548/2025.me-04482**URL** <https://www.scopus.com/pages/publications/85217945526?origin=resultslist>**Archive** Scopus**Language** English**Volume** 83**Publisher** American Society for Nondestructive Testing**Pages** 44-54**Publication** Materials Evaluation**ISSN** 00255327 (ISSN)**Issue** 2**Journal Abbr** Mater. Eval.**Date Added** 20/09/2026, 9:51:45 pm**Modified** 20/09/2026, 9:51:45 pm**Tags:**

Quantitative assessments, Aluminum alloys, quantitative assessment, Aluminum coated steel, Aluminum corrosion, Aluminum steel, Aluminum-steel structure, aluminum-steel structures, Eddy current detection, Eddy current testing, Galvanic corrosion, Metal activity, pulsed eddy current, Pulsed eddy current, Steel corrosion, Steel testing, Thinning defect, thinning defects, Thinnings, Wall panels, Wing wall panel, wing wall panels

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: State Administration for Market Regulation Science and Technology Plan Project, (2024MK084); National Natural Science Foundation of China, NNSFC, (52205564); Natural Science Foundation of Hubei Province, (2022CFB898); Correspondence Address: T. Chen; The Key Laboratory of Modern Manufacturing Quality in Hubei Province, School of Mechanical Engineering, Hubei University of Technology, Wuhan, 430068, China; email: chentao@hnu.edu.cn; CODEN: MAEVA

Research on Risk Quantification Methods for Connected Autonomous Vehicles Based on CNN-LSTM

Item Type Journal Article

Author K. Wang

Author D. Qu

Author D. Shao

Author L. Wei

Author Z. Zhang

Abstract Quantifying and predicting driving risks for connected autonomous vehicles (CAVs) is critical to ensuring the safe operation of traffic in complex environments. This study first establishes a car-following model for CAVs based on molecular force fields. Subsequently, using a convolutional neural network and long short-term Memory (CNN-LSTM) deep-learning model, the future trajectory of the target vehicle is predicted. Risk is quantified by employing models that assess both the collision probability and collision severity, with deep-learning techniques applied for risk classification. Finally, the High-D dataset is used to predict the vehicle trajectory, from which the speed and acceleration of a target vehicle are derived to forecast driving risks. The results indicate that the CNN-LSTM model, when compared with standalone CNN and LSTM models, demonstrates a superior generalization performance, a higher sensitivity to risk changes, and an accuracy rate exceeding 86% for medium- and high-risk predictions. This improved accuracy and efficacy contribute to enhancing the overall safety of connected vehicle platoons. © 2024 by the authors.

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Research on safety risk assessment index system for low-altitude flight in urban dense areas

Item Type Conference Paper

Author W. Xiaoning

Author J. Zhe

Author F. Tiegang

Abstract With the rapid proliferation of low-altitude aircraft in urban dense areas, their safety risks have become a critical constraint on industry development. Based on a four-dimensional framework encompassing technology, environment, operations, and management, this paper proposes a set of safety risk identification and classification indicator systems for low-altitude flight in urban dense areas. Through literature analysis, historical data mining, and expert assessment, we systematically identified multiple typical risk sources and revealed the chain propagation mechanisms of multi-dimensional risks. Furthermore, we constructed a three-level indicator system comprising 'Target Layer-Criterion Layer-Indicator Layer,' clarifying quantification methods and data sources. The study demonstrates that this system can provide a structured analytical tool for airspace management, route planning, and emergency response, facilitating the shift in low-altitude flight safety management from passive handling to proactive prevention and control. © 2025 IEEE.

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DOI 10.1109/SRSE67406.2025.11357312

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Language English

Extra Journal Abbreviation: Int. Conf. Syst Reliab. Saf. Eng., SRSE

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 979-833155470-5 (ISBN)

Pages 618-622**Proceedings Title** Int. Conf. Syst Reliab. Saf. Eng., SRSE**Conference Name** 2025 7th International Conference on System Reliability and Safety Engineering, SRSE 2025**Date Added** 20/09/2026, 9:52:09 pm**Modified** 20/09/2026, 9:52:09 pm**Tags:**

Accident prevention, Risk assessment, Safety risks, Air traffic control, Urban planning, Flight control systems, Fighter aircraft, Environmental management, Aviation, Altitude control, Assessment index system, Critical constraints, Digital avionics, Grading, Indicator System, Indicators systems, Low altitudes, Low-altitude flight, Low-Altitude Flight, Multidimensional coupling, Multidimensional Coupling, Risk grading, Risk Grading, Safety risk assessments, Urban dense area, Urban Dense Areas

Notes:

EID: 2-s2.0-105032847650

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: W. Xiaoning; Aerospace Times Low Altitude Technology Co., Ltd., Chongqing, China; email: 1402967052@qq.com; Conference name: 7th International Conference on System Reliability and Safety Engineering, SRSE 2025; Conference date: 20 November 2025 through 23 November 2025; Conference code: 219251

Research on the Auxiliary Diagnosis Method of Parkinson's Disease Based on FMRI

Item Type Conference Paper**Author** M. Zuo**Author** M. Nie**Author** Q. Zhang

Abstract With the accelerating global aging population, the incidence of Parkinson's disease (PD) has been increasing annually, verwhelming families and society. However, traditional clinical diagnostic methods suffer from high subjectivity and elevated misdiagnosis rates. To address these challenges in PD diagnosis, this study proposes an enhanced Spatio-Temporal Linear Attention Transformer (STLAT) deep learning model based on the classical Transformer encoder. By extracting temporal and spatial features from resting-state functional magnetic resonance imaging (rs-fMRI), we introduce a multi-scale attention mechanism and an explainability quantification method to achieve accurate classification of PD patients and visualized diagnostic decision-making. Our approach constructs a standardized preprocessing pipeline to extract spatiotemporal features of brain regions and innovatively designs a spatiotemporal cross-attention module to capture dynamic functional connectivity patterns. Furthermore, we incorporate the integrated gradients saliency quantification (IG-SQ) method to interpret the decision rationale of the neural network. Experimentally, the method is proved to be correct up to 98.60% in PD binary classification. Through explainability

analysis, we successfully localize key pathological regions, including the basal ganglia and the default mode network (DMN). This study provides an explainable neuroimaging biomarker analysis framework for objective PD diagnosis and pathological mechanism research. © 2025 IEEE.

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Language English

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Export Date: 20 September 2026; Cited By: 0; Funding details: Beijing Municipal Institutions; National Natural Science Foundation of China, NNSFC, (62433002); National Natural Science Foundation of China, NNSFC; Beijing Scholars Program, (099); Correspondence Address: Q. Zhang; National Engineering Research Center for Agri-product Quality Traceability, BeiJing Technology and Business University, Beijing, China; email: zqc1982@126.com; Conference name: 5th International Conference on Artificial Intelligence, Big Data and Algorithms, CAIBDA 2025; Conference date: 20 June 2025 through 22 June 2025; Conference code: 213671

Research on the freezing phenomenon of quantum correlation by machine learning

Item Type Journal Article

Author X. Li

Author Q. Zhu

Author Y. Huang

Author Y. Hu

Author Q. Meng

Author C. Su

Author Q. Yang

Author S. Wu

Author X. Liu

Abstract Quantum correlation shows a fascinating nature of quantum mechanics and plays an important role in some physics topics, especially in the field of quantum information. Quantum correlations of the composite system can be quantified by resorting to geometric or entropy methods, and all these quantification methods exhibit the peculiar freezing phenomenon. The challenge is to find the characteristics of the quantum states that generate the freezing phenomenon, rather than only study the conditions which generate this phenomenon under a certain quantum system. In essence, this is a classification problem. Machine learning has become an effective method for researchers to study classification and feature generation. In this work, we prove that the machine learning can solve the problem of X form quantum states, which is a problem of physical significance. Subsequently, we apply the density-based spatial clustering of applications with noise (DBSCAN) algorithm and the decision tree to divide quantum states into two different groups. Our goal is to classify the quantum correlations of quantum states into two classes: one is the quantum correlation with freezing phenomenon for both Rènyi discord ($\alpha = 2$) and the geometric discord (Bures distance), the other is the quantum correlation of non-freezing phenomenon. The results demonstrate that the machine learning method has reasonable performance in quantum correlation research. © 2020 Tech Science Press. All rights reserved.

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Research on the Spatiotemporal-Coupled High-Resolution Remote Sensing Land Use Classification Method

Item Type Journal Article

Author J. Yang

Author X. Hu

Author W. Ma

Author J. Luo

Author T. Wu

Author Z. Shi

Author H. Yu

Author P. Jin

Author Q. Tan

Author Y. Xu

Abstract Highlights: What are the main findings? A “spatiotemporal coupling” classification paradigm is proposed, which effectively mitigates the performance degradation in high-resolution remote sensing image classification caused by temporal differences through temporal segmentation and dedicated feature extractor strategies. A parcel-oriented deep texture feature extraction and quantification method is designed and implemented, transforming pixel-level features into statistically descriptive vectors with clear geographical significance, significantly enhancing feature discriminability and interpretability. What are the implications of the main findings? This framework provides a scalable solution for fine-grained land cover recognition using multi-temporal, high-resolution remote sensing imagery, particularly suitable for agricultural areas with significant seasonal surface cover variations. The proposed technical workflow of “feature extraction–entity constraint–statistical quantification” promotes the effective integration of deep learning models with Geographic Information System (GIS) management needs, offering practical reference value for real-world applications. High-spatial-resolution remote sensing imagery provides a data foundation for fine-grained land use classification. However, due to long revisit cycles and susceptibility to cloud cover, large-area imagery often suffers from temporal inconsistency, which severely limits the classification accuracy of traditional unified models. To address this issue, this study proposes a geographic entity-oriented, spatiotemporally coupled land use classification method for high-resolution remote sensing imagery, with agricultural land (including paddy fields, dry farmland and gardens) as an example for validation. In this method, the study area is first divided into multiple sub-regions based on image acquisition time, ensuring temporal consistency within each sub-region. A dedicated deep texture feature

extraction model is then constructed for each sub-region. This model is adapted from the advanced CAPTN texture recognition network: its classification head is removed, and a multi-scale feature fusion module is introduced, transforming it into an encoder focused on extracting spatial texture feature maps. Additionally, a self-supervised loss function combining masked feature reconstruction and cross-view consistency is designed to improve the quality of the learned texture features. During the prediction stage, the corresponding feature extractor is invoked based on the temporal phase of the imagery to generate a full-region texture feature map. This feature map is then cropped using land parcel vectors, and statistical feature vectors describing the texture attributes of each parcel are formed by calculating the mean and standard deviation of the features within each parcel. Finally, a Random Forest classifier is employed to determine the land parcel categories. This study uses the Jiangjin District of Chongqing City as the experimental area. The results show that, compared to training a unified deep learning model directly on full-region multi-temporal imagery or using traditional texture features, the proposed spatiotemporally coupled classification framework achieves significant improvements in overall accuracy and Kappa coefficient, reaching 92.3% and 0.89, respectively. © 2026 by the authors.

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Deep learning, Feature extraction, Classification (of information), Learning systems, Feature map, Agriculture, Image segmentation, Textures, Image enhancement, Remote sensing, Remote-sensing, Extraction, Image classification, Image resolution, Classification methods, Image acquisition, Geographic information systems, Feature extractor, High spatial resolution, high spatial resolution remote sensing, High spatial resolution remote sensing, Image texture, Land use, land use classification, Landuse classifications, parcel, Parcel, spatiotemporal coupling, Spatiotemporal coupling, texture features, Texture features

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Research progress on accounting methods for domestic and international sulfur hexafluoride greenhouse gas emissions reduction

Item Type	Conference Paper
Author	Y. Zhao
Author	W. Dong
Author	F. Ma
Author	F. Zhu
Author	W. Liu
Author	Y. Song
Author	S. Zhu
Author	H. Chen
Abstract	Sulfur hexafluoride (SF6) is a high-quality ultra-high voltage insulating dielectric material with unique insulation and arc-extinguishing properties, and is widely used in power companies. However, sulfur hexafluoride is the strongest known greenhouse gas and is extremely difficult to degrade, so its emissions into the atmosphere need to be strictly limited. In the process of carbon emission reduction, the first step is to conduct carbon accounting, and accounting methods for SF6 emissions at home and abroad are gradually being supplemented and improved. Although major breakthroughs have been made in various aspects of SF6 carbon accounting at home and abroad, there is a lack of comprehensive discussion of the current main progress. This article first summarizes the current process of SF6 carbon accounting under two major categories and four methods at home and abroad. After that, the boundaries of each category, quantification methods, and accounting methods under the four major methods were further compared in detail. Finally, the positive impact that these methods can bring to the company's operations is analysed from the perspective of inventory reporting, certification application and domestic carbon market transaction profits. This article systematically explains the current research progress from the aspects of method classification, practical operation and final practical application, in order to provide theoretical support for further promoting new accounting methods in the future. © Published under licence by IOP Publishing Ltd.
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DOI	10.1088/1742-6596/2831/1/012032
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'current, Risk management, Kyoto Protocol, High quality, Property, Greenhouse gas emissions, Greenhouses gas, Zero-carbon, Carbon sequestration, Emission control, Accounting methods, Bioremediation, Carbon accounting, Carbon capture and utilization, Emissions into the atmosphere, Greenhouse gas emission reduction, Low emission, Power company, SF6 insulation, Ultra high voltage

Notes:

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Residential Energy Consumption Forecasting in Electric Utilities: An Approach Based on Random Forests and Time Series

Item Type Journal Article
Author J.L. Hernández
Author D. De Yong
Author F. Magnago
Author S. Bragnolo
Author J. Amaya

Abstract Forecasting the monthly electricity consumption of residential users is a critical task for improving energy planning, demand management, and the efficient integration of renewable energy sources into the electrical system. This study predicts the consumption of a single residential user based on historical data, including monthly consumption and average temperature records from November 2018 to December 2024. Five forecasting approaches are compared: moving averages, ARIMA, standard Random Forest, Random Forest with lag variables, and Random Forest with hyperparameter optimization using RandomizedSearchCV. The models performance is evaluated using MAE, MSE, and RMSE metrics over the last 12 months of the analyzed period, with 95% confidence intervals calculated via bootstrapping for both the validation phase and the estimation for January 2025. The results show that Random Forest models with lag variables and hyperparameter optimization outperform traditional methods such as moving averages and ARIMA in terms of accuracy. Additionally, the use of confidence intervals provides a more robust

assessment of prediction reliability. It is concluded that the combined use of machine learning techniques, selection of relevant historical variables, and uncertainty quantification methods offers an effective tool for anticipating residential electricity consumption behavior. This approach can be valuable for electric utilities and policymakers seeking datadriven, reliable, and reproducible decisions. © 2026 IEEE.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Bragagnolo; Cidtee, Facultad Regional Cordoba, Universidad Tecnologica Nacional, Cordoba, Argentina; email: sbragagnolo@frc.utn.edu.ar

Resistant starch-enriched bread: Fibre content, analytical method comparison, and in-vitro digestibility effects

Item Type Journal Article

Author R. Sempio

Author L. Nyhan

Author E. Zannini

Author J. Walter

Author E.K. Arendt

Abstract Modern diets are often deficient in dietary fibre, increasing chronic disease risk. Reformulating staple foods like bread with resistant starch (RS) offers a practical solution. This study evaluated the partial replacement of flour with high concentration of RS-II, RS-III, RS-IV, and RS2-IV in bread, analysing their structural, nutritional, and digestibility impacts. Generally, breads enriched with RS showed a decrease in specific volume ranged between 32% and 42%, and an increase in hardness of up to 475%. Additionally, two fibre quantification methods were compared: AOAC 991.43 and AOAC 2017.16, as established by the AOAC (Association of Official Analytical Collaboration). All RS-enriched formulations met the “high in fibre” claim under AOAC 991.43, providing 17–31.5% of the daily fibre intake per 50 g portion versus 4.6% in the white control formulation. However, AOAC 2017.16 classified only RS-II and RS-III as “high in fibre,” while the two types of RS IV qualified as a “source of fibre”. Despite this, RS type IV showed the greatest reduction in reducing sugar release during in-vitro digestion (32% lower maltose release compared to the control), highlighting its potential to modulate postprandial glycaemic response. The study clarifies method-dependent fibre classification discrepancies and demonstrates RS-IV unique functional benefits compared to the other RS ingredients. These findings align with global reformulation policies, offering actionable strategies to enhance fibre intake through staple foods. Future research should focus on in-vivo studies and prioritise consumer acceptance and long-term health outcomes of RS-fortified products. © 2026 The Author(s).

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Dietary fibre, Fibre methods, Gluten network, High in fibre product, Starch digestion

Notes:

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Export Date: 20 September 2026; Cited By: 1; Funding details: Science Foundation Ireland, SFI, (19/RP/6853); Correspondence Address: E.K. Arendt; School of Food and Nutritional Sciences, University College Cork, Cork, Ireland; email: e.arendt@ucc.ie

Results of Approbation of the Method for Predicting the Classification Accuracy of Multichannel Images

Item Type Conference Paper

Author V. Lukin

Author I. Vasilyeva

Author B. Vozel

Abstract The results of verification of the method for predicting the classification accuracy for three-channel remote sensing images are presented. This study is carried out based on pixel-by-pixel classification according to the maximum likelihood criterion. As a criterion for classification accuracy, the weighted total probability of correct recognition of classes is taken, where the weight coefficients are a priori probabilities of the occurrence of objects from the corresponding classes. The results of comparison of statistical estimates of a priori probabilities of classes and their predictions by the Monte Carlo method using 'hard' and fuzzy classification are presented. Empirical dependences of the relative errors of prediction of the probabilities of correct recognition on the sample size are obtained. It is shown that predictions of the total probability of correct recognition, provided that the random sample is representative and of sufficient volume, produce the specified reliability. At the same time, the volume of processed data is significantly (at least by 1-2 orders of magnitude) reduced. This, in turn, leads to the corresponding reduction in the complexity of calculations and the time spent on obtaining predictions for the desired criteria. © 2022 IEEE.

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RETHINKING CLASS-PRIOR ESTIMATION FOR POSITIVE-UNLABELED LEARNING

Item Type Conference Paper

Author Y. Yao

Author T. Liu

Author B. Han

Author M. Gong

Author G. Niu

Author M. Sugiyama

Author D. Tao

Abstract Given only positive (P) and unlabeled (U) data, PU learning can train a binary classifier without any negative data. It has two building blocks: PU class-prior estimation (CPE) and PU classification; the latter has been well studied while the former has received less attention. Hitherto, the distributional-assumption-free CPE methods rely on a critical assumption that the support of the positive data distribution cannot be contained in the support of the negative data distribution. If this is violated, those CPE methods will systematically overestimate the class prior; it is even worse that we cannot verify the assumption based on the data. In this paper, we rethink CPE for PU learning-can we remove the assumption to make CPE always valid? We show an affirmative answer by proposing Regrouping CPE (ReCPE) that builds an auxiliary probability distribution such that the support of the positive data distribution is never contained in the support of the negative data distribution. ReCPE can work with any CPE method by treating it as the base method. Theoretically, ReCPE does not affect its base if the assumption already holds for the original probability distribution; otherwise, it reduces the positive bias of its base. Empirically, ReCPE improves all state-of-the-art CPE methods on various datasets, implying that the assumption has indeed been violated here. © 2022 ICLR 2022 - 10th International Conference on Learning Representations. All rights reserved.

Date 2022

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Archive Scopus

Language English

Extra Journal Abbreviation: ICLR - Int. Conf. Learn. Represent.

Publisher International Conference on Learning Representations, ICLR

Proceedings Title ICLR - Int. Conf. Learn. Represent.

Conference Name ICLR 2022 - 10th International Conference on Learning Representations

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Tags:

State of the art, Data distribution, Binary classifiers, Estimation methods, Probability: distributions, Positive data, Building blockes, Positive bias

Notes:

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Rethinking the Invariant Feature Learning: Variational Bayesian Inference for Domain Generalization

Item Type Conference Paper

Author X. Liu

Author B. Hu

Author L. Jin

Author X. Han

Author F. Xing

Author J. Ouyang

Author J. Lu

Author G. El Fakhri

Author J. Woo

Editor Zhou Z.-H.

Abstract In this work, we propose a domain generalization (DG) approach to learn on several labeled source domains and transfer knowledge to a target domain that is inaccessible in training. Considering the inherent conditional and label shifts, we would expect the alignment of $p(x|y)$ and $p(y)$. However, the widely used domain invariant feature learning (IFL) methods relies on aligning the marginal concept shift w.r.t. $p(x)$, which rests on an unrealistic assumption that $p(y)$ is invariant across domains. We thereby propose a novel variational Bayesian inference framework to enforce the conditional distribution alignment w.r.t. $p(x|y)$ via the prior distribution matching in a latent space, which also takes the marginal label shift w.r.t. $p(y)$ into consideration with the posterior alignment. Extensive experiments on various benchmarks demonstrate that our framework is robust to the label shift and the cross-domain accuracy is significantly improved, thereby achieving superior performance over the conventional IFL counterparts. © 2021 International Joint Conferences on Artificial Intelligence. All rights reserved.

Date 2021

DOI 10.24963/ijcai.2021/122**URL** <https://www.scopus.com/pages/publications/105001497243?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IJCAI Int. Joint Conf. Artif. Intell.**Publisher** International Joint Conferences on Artificial Intelligence**ISBN** 10450823 (ISSN); 978-099924119-6 (ISBN)**Pages** 881-887**Proceedings Title** IJCAI Int. Joint Conf. Artif. Intell.**Conference Name** IJCAI International Joint Conference on Artificial Intelligence**Date Added** 20/09/2026, 9:52:26 pm**Modified** 20/09/2026, 9:52:26 pm**Tags:**

Machine learning, Benchmarking, Generalisation, Bayesian networks, Inference engines, Learning methods, Alignment, Target domain, Learn+, Conditional distribution, Prior distribution, Invariant features, Variational Bayesian inferences, Concept shifts, Feature learning

Notes:

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Export Date: 20 September 2026; Cited By: 11; Funding details: Hong Kong Polytechnic University, PolyU; Natural Science Foundation of Jiangsu Province, (BK20200238); Natural Science Foundation of Jiangsu Province; Conference name: 30th International Joint Conference on Artificial Intelligence, IJCAI 2021; Conference date: 19 August 2021 through 27 August 2021; Conference code: 177242

Revealing the built environment impacts of idling emissions from electric ride-hailing vehicles through explainable machine learning

Item Type Journal Article**Author** F. Lu**Author** S. Wang**Author** H. Bi

Abstract Understanding how built environment characteristics influence electric vehicles (EV) idling emissions is crucial for designing low-carbon transportation systems, yet this stationary emission source remains largely overlooked. This study fills this gap by integrating high-resolution GPS trajectory data, grid carbon intensity, and explainable machine learning to quantify the nonlinear impacts of the built environment on EV idling emissions. We developed a LightGBM-SHAP framework that achieves robust predictive performance ($R^2 = 0.889$) and systematically identified nonlinear thresholds. Three key findings emerged. First, across the seven observation days, lower daily mean temperatures were generally associated with higher estimated daily CO₂ emissions under the adopted temperature-dependent emission-accounting framework. Second, road network length, residential land, transportation facilities,

and population density collectively accounted for nearly 80% of total SHAP importance, with multiple built environment features exhibiting nonlinear threshold effects. Third, feature significance exhibited spatiotemporal heterogeneity, with identical built environment configurations exerting markedly different impacts across varying time periods and spatial scales. These findings should be interpreted as predictive associations rather than causal planning standards, offering diagnostic evidence for future policy evaluation. © 2026

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Archive Scopus

Language English

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Publisher Elsevier Ltd

Publication Transportation Research Part A: Policy and Practice

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Motor transportation, electric vehicle, Electric vehicles, Learning systems, machine learning, Machine-learning, Machine learning, carbon emission, transportation system, Population statistics, prediction, Carbon, Explainable machine learning, Built environment, Electric ride-hailing vehicle, Electric ride-hailing vehicles, Environment impact, GPS, Idling emission, Idling emissions, Non-linear thresholds, Nonlinear threshold effect, nonlinearity, Threshold effect, Vehicle idling

Notes:

EID: 2-s2.0-105047320163

Export Date: 20 September 2026; Cited By: 0; Funding details: National Key Research and Development Program of China, NKRDP, (2020YFB1712802); National Key Research and Development Program of China, NKRDP; Correspondence Address: S. Wang; Yanshan University, Qinhuangdao City, Hebei Province, China; email: 1972579652@qq.com

Revealing the Proximate Long-Tail Distribution in Compositional Zero-Shot Learning

Item Type Conference Paper

Author C. Jiang

Author H. Zhang

Editor Wooldridge M.

Editor Dy J.

20/09/2026, 9:54 pm

Visual feature, Zero-shot learning, Modeling performance, State of the art, Long-tailed distributions, Class imbalance problems, Classifier training, Long-tail distribution

Notes:

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Export Date: 20 September 2026; Cited By: 21; Correspondence Address: H. Zhang; School of Computer Science and Engineering, Nanjing University of Science and Technology, China; email: zhanghf@njjust.edu.cn; Conference name: 38th AAAI Conference on Artificial Intelligence, AAAI 2024; Conference date: 20 February 2024 through 27 February 2024; Conference code: 198370

Review and Performance Evaluation of Uncertainty Quantification in Data-Driven AI-Assisted Measurements

Item Type Journal Article

Author S. Shirmohammadi

Author F. Wang

Author C.-H. Hsu

Abstract As artificial intelligence (AI) becomes more prevalent in measurement systems and synthetic instruments, quantifying the uncertainty of AI-assisted measurements becomes a crucial and necessary part of the measurement process. In this article, we take a holistic approach toward both measurement science and AI's formulation and implementation of uncertainty, and we review and categorize data-driven AI-assisted uncertainty quantification methods with a novel taxonomy. We also provide a one-stop shop identifying AI literature practices that are noncompliant with measurement standards, allowing readers to spot such noncompliances and understand their practical impact. Furthermore, specifically for classification-assisted measurements, we test the most common epistemic Type A uncertainty quantification methods with 12 diverse datasets, and we evaluate their indirect measurement accuracy, one of the most important metrics for engineering applications, as well as precision, recall, and F1 score, each in both macro and weighted modes. Finally, we study the multiobservation variation of misclassification probability (MP) and experimentally show that in some cases it can be an indication of uncertainty: an interesting fact considering MP itself is not uncertainty. © 2022 IEEE.

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DOI 10.1109/OJIM.2025.3621742

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Language English

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Tags:

Artificial intelligence, Classification (of information), Uncertainty analysis, Uncertainty, Data accuracy, Data driven, Quantification methods, Uncertainty quantifications, uncertainty quantification, AI-assisted measurements, artificial intelligence (AI) uncertainty, Artificial intelligence uncertainty, Artificial intelligence-assisted measurement, Measurement system, measurement uncertainty, Measurement uncertainty, Misclassification probability, Performances evaluation, Precision engineering

Notes:

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Export Date: 20 September 2026; Cited By: 10; Funding details: National Science and Technology Council, NSTC, (113-2811-E-007-012); Correspondence Address: S. Shirmohammadi; University of Ottawa, School of Eecs, Ottawa, K1N 6N5, Canada; email: shervin.s@uottawa.ca

Review on macropore flow in soil

Item Type Journal Article

Author Z.X. Gao

Author X.X. Xu

Author J.N. Zhao

Author C.P. Zhao

Author S.N. Zhang

Abstract Macropore flow is a natural and ubiquitous phenomenon in soil environment. Research on macropore flow involves studying mechanisms of soil water movement from homogeneous to heterogeneous fields, so it has become research focus in soil physical science. Understanding of mechanisms on macropore flow would provide useful information for irrigation, fertilization, crops planting and pollution tendency simulating. This paper introduces present research situation of macropore flow from classification standards, the influencing factors and quantitative methods, the deficiency of present research and research directions are discussed. The purpose of this paper is to provide some new ideas for deeply studying macropore flow.

Date 2014

DOI 10.5846/stxb201301220126

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Publisher Science Press

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ISSN 10000933 (ISSN)

Issue 11**Journal Abbr** Shengtai Xuebao Acta Ecol. Sin.**Date Added** 20/09/2026, 9:52:55 pm**Modified** 20/09/2026, 9:52:55 pm**Tags:**

environmental factor, quantitative analysis, Quantification methods, Prognosis, soil water, water flow, Influencing factors, macropore, Macropore flow

Notes:

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Export Date: 20 September 2026; Cited By: 22; Correspondence Address: X. X. Xu; Institute of Soil and Water Conservation, Chinese Academy of Science and Ministry of Water Resources, Yangling 712100, China; email: xuxuexuan@nwsuaf.edu.cn

Risk factor identification and nomogram-based predictive model development for recurrence of pituitary adenomas following transsphenoidal surgery: a dual-center retrospective study

Item Type Journal Article**Author** X. Yang**Author** W. Yang**Author** P. Hou**Author** Y. Chen**Author** J. Li**Author** J. Zhong**Author** S. Wang**Author** H. Chen

Abstract Objective – To identify independent risk factors for pituitary adenoma recurrence after transsphenoidal surgery and construct and validate a nomogram for recurrence risk quantification. Methods – We retrospectively collected clinical data of 469 patients with pituitary adenomas from Fuzhou 900 Hospital and Huashan Hospital of Fudan University. LASSO regression was applied to screen potential predictive factors, followed by multivariate Logistic regression to identify independent prognostic factors. Nomograms were subsequently constructed to evaluate the model's discriminative ability, calibration, and clinical utility. Results – Recurrence occurred in 64 cases (13.65%). Multivariate analysis confirmed modified Knosp classification (grades 3B–4), clival invasion, and subtotal resection as independent risk factors ($P < 0.05$). The model had AUC 0.905 [95%CI (0.8254, 0.9847)], C-index 0.905, sensitivity 85%, specificity 86.4%, with good calibration and clinical net benefits (DCA: 0.05–0.90 threshold). Conclusion – The nomogram, which is the first to integrate clival invasion with modified Knosp classification and extent of resection, effectively predicts pituitary adenoma recurrence after transsphenoidal surgery and shows promising clinical utility. © 2026 Yang, Yang, Hou, Chen, Li,

Zhong, Wang and Chen.

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Archive Scopus

Language English

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tumor recurrence, nomograms, pituitary adenomas, prognostic models, transsphenoidal surgery

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RISK QUANTIFICATION IN MANUFACTURING INDUSTRY INVESTMENTS: A STOCHASTIC APPROACH WITH ARTIFICIAL INTELLIGENCE

Item Type Journal Article

Author R.V. da Silva

Author E.M. Frazzon

Abstract Manufacturing investment decisions are often hindered by significant uncertainty. This paper introduces a conceptual model that integrates stochastic simulation with machine learning to quantify investment risk. Our hybrid approach employs a Monte Carlo simulation using time series data of fixed, variable, and investment costs as inputs. To enhance the simulation's realism, Long Short-Term Memory (LSTM) recurrent neural networks forecast the trend components of these series, while a Vector Autoregression (VAR) model captures their inter-correlations. This framework generates a multitude of potential scenarios, each evaluated through a mathematical model of the supply chain to produce a distinct cash flow. The subsequent application of investment metrics, such as Net Present Value (NPV) and Discounted Payback, to the distribution of these cash flows enables a comprehensive statistical analysis of the investment's risk profile, thereby providing robust support for strategic decision-

making. © 2025 by authors and Galileo Institute of Technology and Education of the Amazon (ITEGAM).

Date 2025

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Investments, Supply Chain, Decision-Making Methods, Risk Evaluation, Techno-Economic Feasibility

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RiskTree: Decision trees for asset and process risk assessment quantification in big data platforms

Item Type Journal Article

Author H. Zhan

Author J. Yang

Author Z. Guo

Author J. Cao

Author D. Zhang

Author X. Zhao

Author W. You

Author H. Li

Abstract Currently, big data platforms are widely applied across various industries. These platforms are characterized by large scale, diverse forms, high update frequency, and rapid data flow, making it challenging to directly apply existing risk quantification methods to them. Additionally, the composition of big data platforms varies among enterprises due to factors such as industry, economic capability, and technical proficiency. To address this, we first developed a risk quantification assessment

process tailored to different types of big data platforms, taking into account relevant laws, regulations, and standards. Subsequently, we developed RiskTree, a risk quantification system for big data platforms, which supports automated detection of configuration files, traffic, and vulnerabilities. For situations where automated detection is not feasible or permitted, we provide a customized questionnaire system to collect assets and data processing procedures. We utilize a knowledge graph (KG) to integrate and analyze the collected data. Finally, we apply a random forest algorithm to compute risk index weights, risk values, and risk levels, enabling the quantification of risks on big data platforms. To validate the proposed process, we conducted experiments on an educational big data platform. The results demonstrate that the risk index system presented in this paper objectively and comprehensively reflects the risks faced by big data platforms. Furthermore, the proposed risk assessment process not only effectively identifies and quantifies risks but also provides highly interpretable evaluation results. © 2024 The Author(s).

Date 2024

DOI 10.1051/sands/2024009

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Language English

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Machine learning, Quantitative risk assessment, Big data platform

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RKHS bayes discriminant: A subspace constrained nonlinear feature projection for signal detection

Item Type Journal Article

Author U. Ozertem

Author D. Erdogmus

Abstract Given the knowledge of class probability densities, a priori probabilities, and relative risk levels, Bayes classifier provides the optimal minimum-risk decision rule. Specifically, focusing on the two-class (detection) scenario, under certain symmetry assumptions, matched filters provide optimal results for the detection problem. Noticing that the Bayes classifier is in fact a nonlinear projection of the feature vector to a single-dimensional statistic, in this paper, we develop a smooth nonlinear projection filter constrained to the estimated span of class conditional distributions as does the Bayes classifier. The nonlinear projection filter is designed in a reproducing kernel Hilbert space leading to an analytical solution both for the filter and the optimal threshold. The proposed approach is tested on typical detection problems, such as neural spike detection or automatic target detection in synthetic aperture radar (SAR) imagery. Results are compared with linear and kernel discriminant analysis, as well as classification algorithms such as support vector machine, AdaBoost and LogitBoost. © 2009 IEEE.

Date 2009

DOI 10.1109/TNN.2009.2021473

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Language English

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Notes:

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RLSbench: Domain Adaptation Under Relaxed Label Shift

Item Type Conference Paper

Author S. Garg

Author N. Erickson

Author J. Sharpnack

Author A.J. Smola

Author S. Balakrishnan

Author Z.C. Lipton

Editor Krause A.

Editor Brunskill E.

Editor Cho K.

Editor Engelhardt B.

Editor Sabato S.

Editor Scarlett J.

Abstract Despite the emergence of principled methods for domain adaptation under label shift, their sensitivity to shifts in class conditional distributions is precariously under explored. Meanwhile, popular deep domain adaptation heuristics tend to falter when faced with label proportions shifts. While several papers modify these heuristics in attempts to handle label proportions shifts, inconsistencies in evaluation standards, datasets, and baselines make it difficult to gauge the current best practices. In this paper, we introduce RLSBENCH, a large-scale benchmark for relaxed label shift, consisting of 500 distribution shift pairs spanning vision, tabular, and language modalities, with varying label proportions. Unlike existing benchmarks, which primarily focus on shifts in class-conditional $p(x|y)$, our benchmark also focuses on label marginal shifts. First, we assess 13 popular domain adaptation methods, demonstrating more widespread failures under label proportion shifts than were previously known. Next, we develop an effective two-step meta-algorithm that is compatible with most domain adaptation heuristics: (i) pseudo-balance the data at each epoch; and (ii) adjust the final classifier with target label distribution estimate. The meta-algorithm improves existing domain adaptation heuristics under large label proportion shifts, often by 2-10% accuracy points, while conferring minimal effect ($\sim 0.5\%$) when label proportions do not shift. We hope that these findings and the availability of RLSBENCH will encourage researchers to rigorously evaluate proposed methods in relaxed label shift settings. Code is publicly available at <https://github.com/acmi-lab/RLSbench>. © 2023 Proceedings of Machine Learning Research. All rights reserved.

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Conference Name Proceedings of Machine Learning Research

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RNA-Seq Can Be Used to Quantify Gene Expression Levels for Use in the GARDskin Assay

Item Type Journal Article

Author R. Gradin

Author J. Andersson

Author A. Forreryd

Author H. Johansson

Abstract Non-animal methods for identification and characterization of skin sensitizers are continuously evolving, advancing towards more effective, accurate, and informational assays. The GARDskin assay is a scientifically and regulatory recognized assay for the assessment of skin sensitizers. It currently relies on targeted gene expression measurement to derive hazard classifications. With the progression of next generation sequencing technologies, whole transcriptome analysis provides an interesting alternative to the currently implemented targeted gene expression approach. RNA-seq was evaluated for its use in the GARDskin assay as a gene expression quantification method. Based on 24 paired samples acquired on both RNA-seq and the NanoString nCounter platform (the currently standard GARDskin acquisition method), gene expression profiles were found to be highly similar. Comparisons of treatment effects yielded a Spearman's correlation coefficient of 0.95 and a Lin's concordance correlation coefficient of 0.87. RNA-seq data was also used to classify the sensitizing hazard of 24 treatments using the standard GARDskin analysis pipeline. The classifications corresponded completely with references, rendering correct classifications for all treatments. In conclusion, it was found that the RNA-seq data strongly resembled NanoString nCounter data, and that it could be used to derive reliable hazard classifications in the GARDskin assay. © 2025 by the authors.

Date 2026

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Article, correlation coefficient, high throughput sequencing, assay, gene expression level, gene expression profiling, NanoString, next generation sequencing, non-animal methods, RNA sequencing, RNA-seq, skin sensitization

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: R. Gradin; Senzagen AB, Lund, 22381, Sweden; email: robin.gradin@senzagen.com

Robust and accurate quantification of biomarkers of immune cells in lung cancer micro-environment using deep convolutional neural networks

Item Type Journal Article

Author L. Aprupe

Author G. Litjens

Author T.J. Brinker

Author J. Van Der Laak

Author N. Grabe

Abstract Recent years have seen a growing awareness of the role the immune system plays in successful cancer treatment, especially in novel therapies like immunotherapy. The characterization of the immunological composition of tumors and their microenvironment is thus becoming a necessity. In this paper we introduce a deep learning-based immune cell detection and quantification method, which is based on supervised learning, i.e., the input data for training comprises labeled images. Our approach objectively deals with staining variation and staining artifacts in immunohistochemically stained lung cancer tissue and is as precise as humans. This is evidenced by the low cell count difference to humans of 0.033 cells on average. This method, which is based on convolutional neural networks, has the potential to provide a new quantitative basis for research on immunotherapy. © Copyright 2019 Aprupe et al.

Date 2019

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Tags:

Article, human, Deep learning, deep learning, artificial neural network, convolutional neural network, nerve cell network, human cell, human tissue, biological marker, immunohistochemistry, tumor marker, artifact, lung cancer, CD8 antigen, tumor microenvironment, Lung cancer, T lymphocyte, lung adenocarcinoma, lung parenchyma, cancer tissue, CD20 antigen, cell count, Biomarker quantification, cancer immunology, Cancer micro-environment, CD3 antigen, feature detection, Immune cells, immunocompetent cell, lung resection

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Robust and scalable uncertainty estimation with conformal prediction for machine-learned interatomic potentials

Item Type Journal Article

Author Y. Hu

Author J. Musielewicz

Author Z.W. Ulissi

Author A.J. Medford

Abstract Uncertainty quantification (UQ) is important to machine learning (ML) force fields to assess the level of confidence during prediction, as ML models are not inherently physical and can therefore yield catastrophically incorrect predictions. Established a-posteriori UQ methods, including ensemble methods, the dropout method, the delta method, and various heuristic distance metrics, have limitations such as being computationally challenging for large models due to model re-training. In addition, the uncertainty estimates are often not rigorously calibrated. In this work, we propose combining the distribution-free UQ method, known as conformal prediction (CP), with the distances in the neural network's latent space to estimate the uncertainty of energies predicted by neural network force fields. We evaluate this method (CP+latent) along with other UQ methods on two essential aspects, calibration, and sharpness, and find this method to be both calibrated and sharp under the assumption of independent and identically-distributed (i.i.d.) data. We show that the method is relatively insensitive to hyperparameters selected, and test the limitations of the method when the i.i.d. assumption is violated. Finally, we demonstrate that this method can be readily applied to trained neural network force fields with traditional and graph neural network architectures to obtain estimates of uncertainty with low computational costs on a training dataset of 1 million images to showcase its scalability and portability. Incorporating the CP method with latent distances offers a calibrated, sharp and efficient strategy to estimate the uncertainty of neural network

force fields. In addition, the CP approach can also function as a promising strategy for calibrating uncertainty estimated by other approaches. © 2022 The Author(s). Published by IOP Publishing Ltd.

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Robust Blood Cell Detection and Counting Using Efficient Channel Attention-Integrated YOLOv10

Item Type Conference Paper

Author R.R. Sohana

Author S. Ahmed

Author S.B. Shahid

Author R.H. Hridoy

Abstract Automated blood cell analysis plays a crucial role in early disease diagnosis. Traditional microscopic examination remains labor-intensive and prone to subjective variability. In this study, we have proposed an enhanced YOLOv10-

based detection framework integrated with Efficient Channel Attention (ECA) to classify and count red blood cells (RBCs), white blood cells (WBCs), and platelets from peripheral blood smear images. The ECA mechanism improves channel-wise feature selection without introducing computational overhead, enabling more accurate discrimination of morphologically similar cell types. Additionally, we have introduced an improved preprocessing pipeline, Contrast Limited Adaptive Histogram Equalization (CLAHE) that enhances local contrast in an image along with other techniques. The final model achieves an accuracy of approximately 97.00% across all classes, demonstrating superior robustness in both detection and categorization. These results highlight the potential of attention-enhanced lightweight architectures for clinical hematological analysis and pave the way for reliable and rapid diagnostic support. © 2025 IEEE.

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DOI 10.1109/WIECON-ECE69386.2025.11525973

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Extra Journal Abbreviation: IEEE Int. WIE Conf. Electr. Comput. Eng./WIECON-ECE

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Proceedings Title IEEE Int. WIE Conf. Electr. Comput. Eng./WIECON-ECE

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Robust fine-grained echocardiographic view classification with supervised contrastive learning

Item Type Journal Article**Author** P. Naidoo**Author** P. Fernandes**Author** N. Dadashi Serej**Author** C.C. Stowell**Author** C.H. Manisty**Author** D.P. Francis**Author** M. Zolgharni

Abstract Accurate classification of echocardiographic views is fundamental for automated cardiac analysis. However, clinical practice relies on a large, heterogeneous set of fine-grained acquisitions that introduce substantial inter-observer variability. Existing studies have primarily focused on limited view sets, often collapsing specialised views into broad categories, which limits their clinical relevance. We address this limitation by introducing TTE47, the first publicly available benchmark comprising 47 clinically meaningful views annotated independently by three experts. This dataset enables the rigorous quantification of inter-observer agreement and establishes a foundation for reproducible, clinically relevant evaluation. To tackle the dual challenges of subtle inter-class distinctions and structured label variability, we propose a novel supervised contrastive learning framework incorporating a tailored loss function. Our method outperforms cross-entropy and standard supervised contrastive baselines, achieving leading performance among evaluated methods on TTE47 and surpassing prior work on TMED-2 without dataset-specific pretraining, using a model pretrained on TTE47. Beyond accuracy, we introduce clustering-based metrics, Detection Rate and Label Recovery Precision, that measure semantic coherence and the model's ability to resist annotation variability. Results show that the learned feature space aligns more strongly with underlying anatomical structure than with any single annotator's style, enabling resilience to label shifts and maintaining robustness comparable to human-level disagreement. By integrating multi-expert evaluation, robust representation learning, and interpretable feature-space analysis, this work establishes a scalable and clinically relevant framework for fine-grained echo view classification. Our findings highlight the potential of contrastive pretraining to standardise interpretation, mitigate subjectivity, and enhance the reliability of AI-assisted echocardiography in diverse clinical settings. © 2026 The Authors.

Date 2026**DOI** 10.1016/j.media.2026.104006**URL** <https://www.scopus.com/pages/publications/105031441424?origin=resultslist>**Archive** Scopus**Language** English**Volume** 110**Publisher** Elsevier B.V.**Publication** Medical Image Analysis**ISSN** 13618415 (ISSN)**Journal Abbr** Med. Image Anal.**Date Added** 20/09/2026, 9:51:48 pm**Modified** 20/09/2026, 9:51:48 pm**Tags:**

adult, human, Humans, Classification (of information), Learning systems, article, reliability, reproducibility, Reproducibility of Results, Supervised learning, Semantics, entropy, Contrastive Learning, algorithm, Algorithms, procedures, Learning frameworks, benchmarking, Pre-training, Fine grained, Contrastive learning, learning, classification, supervised machine learning, interrater reliability, feature learning (machine learning), clinical practice, Loss functions, computer assisted diagnosis, Image Interpretation, Computer-Assisted, clinical significance, echocardiography, Echocardiography, Inter class, anatomical concepts, Clinical practices, Function evaluation, Supervised Machine Learning, Cross entropy, Interobserver variability, observer bias, Observer variability, observer variation, Observer Variation, View classification

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Robust Label Shift Correction via Denoising Expectation-Maximization

Item Type Conference Paper

Author X. Wu

Author R. Fan

Author C. Hou

Editor Kittler J.

Editor Xiong H.

Editor Lin W.

Editor Yang J.

Editor Chen X.

Editor Lu J.

Editor Yu J.

Editor Zheng W.

Abstract The label shift hypothesis posits that the source and target domains exhibit differing label priors while maintaining identical conditional probabilities. This core assumption enables models to adapt to domain distribution discrepancies through loss re-weighting or classifier output calibration, making it a prominent focus in open-environment machine learning. However, the prevalent existence of label noise severely degrades the classification efficacy of source classifiers, fundamentally undermining traditional label shift adaptation approaches. To address the dual challenges of label shift and label noise, we propose the Denoising Expectation-Maximization (DNEM) framework, which enhances model robustness against distribution shifts and label noise via iterative estimation of noise transition matrices and optimization of importance weights.

Leveraging the memorization effect in deep learning, DNEM dynamically adjusts the learning rate for noise matrix estimation, maintaining conservative updates during initial training phases to preserve knowledge from relatively clean labels. Extensive experiments on multiple benchmark datasets demonstrate that DNEM consistently outperforms state-of-the-art methods under high noise rates and diverse shift conditions. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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Robust model selection for positive and unlabeled learning with constraints

Item Type Journal Article

Author T. Wei

Author H. Wang

Author W. Tu

Author Y. Li

Abstract Positive and unlabeled (PU) learning is the problem in which training data contains only PU samples. Although PU learning is widely used in real-world applications, its model selection remains challenging. Specifically, traditional model selection methods are often highly sensitive to the class prior as well as the data size, resulting in human overhead in hyperparameter optimization. In this paper, we present a method called ODE (robust model selection) for robust model selection in PU learning. Two novel model evaluators based on the integral probability metric and area under the curve, which are free of the class prior, are introduced, and a variance reduction method is further employed to improve the quality of model selection. In addition, we perform model selection under user-defined constraints and propose a fast halving-style searching algorithm to efficiently identify the most promising model configuration. Extensive empirical studies demonstrate that our proposed method performs more robustly and is more computationally efficient than many state-of-the-art methods. © 2022, Science China Press and Springer-Verlag GmbH Germany, part of Springer Nature.

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Robust multi-class transductive learning with graphs

Item Type Conference Paper

Author W. Liu

Author S.-F. Chang

Abstract Graph-based methods form a main category of semisupervised learning, offering flexibility and easy implementation in many applications. However, the performance of these methods is often sensitive to the construction of a neighborhood graph, which is non-trivial for many realworld problems. In this paper, we propose a novel framework that builds on learning the graph given labeled and unlabeled data. The paper has two major contributions. Firstly, we use a nonparametric algorithm to learn the entire adjacency matrix of a symmetry-favored k-NN graph, assuming that the matrix is doubly stochastic. The nonparametric algorithm makes the constructed graph highly robust to noisy samples and capable of approximating underlying submanifolds or clusters. Secondly, to address multi-class semi-supervised classification, we formulate a constrained label propagation problem on the learned graph by incorporating class priors, leading to a simple closed-form solution. Experimental results on both synthetic and real-world datasets show that our approach is significantly better than the state-of-the-art graph-based semi-supervised learning algorithms in terms of accuracy and robustness. ©2009 IEEE.

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Robust Segmentation-Free Algorithm for Homogeneity Quantification in Images

Item Type Journal Article

Author F. Milano

Author A. Chevrier

Author G. De Crescenzo

Author M. Lavertu

Abstract Objective: Homogeneity is a notion used to describe images in various fields and is often linked to critical aspects of those fields. However, this term is rarely defined in the literature and no gold standard exists for its quantification. A few quantification algorithms have been proposed, but they lack both simplicity and robustness. As a result, the scientific community uses the notion of homogeneity in subjective analysis, preventing objective comparison of a large number of data or of different studies. The main objectives of this manuscript are to propose a definition of homogeneity and an algorithm for its quantification. Method: This algorithm, called MASQH, rely on a multi-scale, statistical and segmentation-free approach and outputs a single homogeneity index, which makes it robust and easy to use. Results: The performance and reliability of the method are demonstrated through three case studies: Firstly, on synthetic images to study the behavior and assess the relevance of the algorithm in diverse situations and hence, in various potential fields. Secondly, on histological images derived from experimental chitosan-platelet-rich-plasma hybrid biomaterial, where the quantitative results are compared to a qualitative classification provided by an expert in the field. Thirdly, on experimental nanocomposites images for which results are compared to two other homogeneity quantification algorithms from the field of nanocomposites. Conclusion and significance: By quantifying homogeneity, the MASQH method may help to compare disparate studies in the literature and quantitatively demonstrate the impact of homogeneity in various fields. The MASQH method is freely available online. © 1992-2012 IEEE.

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Robust Thresholding Strategies for Highly Imbalanced and Noisy Data

Item Type Conference Paper

Author J.M. Johnson

Author T.M. Khoshgoftaar

Editor Wani M.A.

Editor Sethi I.K.

Editor Shi W.

Editor Qu G.

Editor Raicu D.S.

Editor Jin R.

Abstract Many studies have shown that non-default decision thresholds are required to maximize classification performance on highly imbalanced data sets. Thresholding strategies include using a threshold equal to the prior probability of the positive class or identifying an optimal threshold on training data. It is not clear, however, how these thresholding strategies will generalize to imbalanced data sets that contain class label noise. When class noise is present, the positive class prior is influenced by the class label noise, and a threshold that is optimized on noisy training data may not generalize to test data. We employ four thresholding strategies: two thresholds that are optimized on training data and two thresholds that depend on the positive class prior. Threshold strategies are evaluated on a range of noise levels and noise distributions using the Random Forest, Multilayer Perceptron, and XGBoost learners. While all four thresholding strategies significantly outperform the default threshold with respect to the Geometric Mean (G-Mean), three of the four thresholds yield unstable true positive rates (TPR) and true negative rates (TNR) in the presence of class noise. Results show that setting the threshold equal to the prior probability of the noisy positive class consistently performs best according to G-Mean, TPR, and TNR.

This is the first evaluation of thresholding strategies for imbalanced and noisy data, to the best of our knowledge, and our results contradict related works that have suggested optimizing thresholds on training data as the best approach. © 2021 IEEE.

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Robust Uncertainty Quantification Using Conformalised Monte Carlo Prediction

Item Type Conference Paper

Author D. Bethell

Author S. Gerasimou

Author R. Calinescu

Editor Wooldridge M.

Editor Dy J.

Editor Natarajan S.

Abstract Deploying deep learning models in safety-critical applications remains a very challenging task, mandating the provision of assurances for the dependable operation of these models. Uncertainty quantification (UQ) methods estimate the model's confidence per prediction, informing decision-making by considering the effect of randomness and model misspecification. Despite the advances of state-of-the-art UQ methods, they are computationally expensive or produce conservative prediction sets/intervals. We introduce MC-CP, a novel hybrid UQ method that combines a new adaptive Monte Carlo (MC) dropout method with conformal prediction (CP). MC-CP adaptively modulates the traditional MC dropout at runtime to save memory and computation resources, enabling predictions to be consumed by CP, yielding robust prediction sets/intervals. Throughout comprehensive experiments, we show that MC-CP delivers significant improvements over comparable UQ methods, like MC dropout, RAPS and CQR, both in classification and regression benchmarks. MC-CP can be easily added to existing models, making its deployment simple. The MC-CP code and replication package is available at <https://github.com/team-daniel/MC-CP>. Copyright © 2024, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Robustness evaluation of deep learning models based on local prediction consistency

Item Type Conference Paper

Author Z. Shi

Author C. Zhong

Author Y. Yokota

Author W. Xia

Author J. Sun

Editor Wani M.A.

Editor Khoshgoftaar T.M.

Editor Wang D.

Editor Wang H.

Editor Seliya N.

Abstract It is important to estimate the performance gap of a given deep learning model on the target data set, since discrepancy or bias between source and target domains is a common and fundamental problem in the practice of machine learning techniques. Without any assumptions on data bias, such as label shift or covariate shift and without target data labels, we propose a robustness estimation method based on prediction consistency evaluation between source and target data in the neighborhood of the source samples. Considering outliers and whether the user provided model is fully trained, a variety of variant methods are also tried, including setting neighborhood threshold to average intra-class distance for each category and relative robustness. Furthermore, the time complexity of this method is $O(n \log n)$, which is applicable for large datasets. Experiments on the handwritten digit recognition and Japanese handwriting recognition show that the proposed methods are effective. © 2019 IEEE.

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ROC analysis with multiple classes and multiple tests: Methodology and its application in microarray studies

Item Type Journal Article

Author J. Li

Author J.P. Fine

Abstract The accuracy of a single diagnostic test for binary outcome can be summarized by the area under the receiver operating characteristic (ROC) curve. Volume under the surface and hypervolume under the manifold have been proposed as extensions for multiple class diagnosis (Scurfield, 1996, 1998). However, the lack of simple inferential procedures for such measures has limited their practical utility. Part of the difficulty is that calculating such quantities may not be straightforward, even with a single test. The decision rule used to generate the ROC surface requires class probability assessments, which are not provided by the tests. We develop a method based on estimating the probabilities via some procedure, for example, multinomial logistic regression. Bootstrap inferences are proposed to account for variability in estimating the probabilities and perform well in simulations. The ROC measures are compared to the correct classification rate, which depends heavily on class prevalences. An example of tumor classification with microarray data demonstrates that this property may lead to substantially different analyses. The ROC-based analysis yields notable decreases in model complexity over previous analyses. © The Author 2008. Published by Oxford University Press. All rights reserved.

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Notes:

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Role of garbage classification in air pollution improvement of a municipal solid waste disposal base

Item Type Journal Article
Author S. Gao
Author L. Meng
Author X. Ge
Author Y. Li
Author Y. Yang
Author Y. Duan
Author Q. Fu
Author S. Zhang
Author X. Yang
Author W. Fei

Author Z. Jiao

Abstract Assessing the impact of garbage classification on air pollution in rapidly developing economic regions is particularly necessary in light of the current policy trends. We evaluated the resulting changes in air pollution levels before and after garbage classification based on a three-year field observation using high-resolution online instruments in a municipal solid waste comprehensive disposal base. A targeted scheme is proposed to analyze the variation characteristics of major sources in a complex multi-source environment (Environments with multiple sources of pollution). We found that the concentrations of hydrogen sulfide (H₂S) and volatile organic compounds (VOCs) in ambient air decreased by 48% and 43% respectively after garbage classification, and their spike frequency was controlled at approximately 0.01. The average H₂S and VOC concentrations were generally low during the daytime, but relatively high during the nighttime because of meteorology and photochemistry. The ozone formation potential of the base decreased by 56%. Alternatively, emissions from transportation and combustion sources increased, while landfill pollution decreased owing to changes in the method of waste disposal. This finding suggests that garbage classification is an effective means of improving the air quality. This new scheme has a good potential in actual research scenarios. © 2023

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Root Cause Analysis of Degradation in Protonic Ceramic Electrochemical Cell with Interfacial Electrical Sensors Using Data-Driven Machine Learning

Item Type Journal Article

Author W. Wu

Author C. Wang

Author W. Bian

Author B. Hua

Author J.Y. Gomez

Author C.J. Orme

Author W. Tang

Author F.F. Stewart

Author D. Ding

Abstract Protonic ceramic electrochemical cells (PCECs) offer promising paths for energy storage and conversion. Despite considerable achievements made, PCECs still face challenges such as physiochemical compatibility between components and suboptimal solid–solid contact at the interfaces between the electrolytes and electrodes. In this study, a novel approach is proposed that combines in situ electrochemical characterization of interfacial electrical sensor embedded PCECs and machine learning to quantify the contributions of different cell components to total degradation, as well as to predict the remaining useful life. The experimental results suggest that the overpotential induced by the oxygen electrode is 48% less than that of oxygen electrode/electrolyte interfacial contact for up to 1171 h. The data-driven machine learning simulation predicts the RUL of up to 2132 h. The root cause of degradation is overpotential increase induced by oxygen electrode, which accounts for 82.9% of total cell degradation. The success of the failure diagnostic model is demonstrated by its consistency with degradation modes that do not manifest in electrolysis fade during early real operations. This synergistic approach provides valuable insights into practical failure diagnosis of PCECs and has the potential to revolutionize their development by enabling improved performance prediction and material selection for enhanced durability and efficiency. © 2023 The Authors. Advanced Science published by Wiley-VCH GmbH.

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Digital storage, machine learning, Machine-learning, Machine learning, Forecasting, Root cause analysis, Data driven, Electrodes, Machine components, degradation prediction, Degradation predictions, Electrical sensors, Electrolysis, Electrolytic cells, Overpotential, Oxygen, Oxygen electrode, Protonic, protonic ceramic electrochemical cell, Protonic ceramic electrochemical cell, Regenerative fuel cells, root cause analysis, steam electrolysis, Steam electrolysis

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Rotation-Invariant Self-Supervised Representation Learning for Object Detection in Autonomous Driving

Item Type Conference Paper

Author A. Marathe

Author S. Roy

Author D. Singh

Author R. Walambe

Author K. Kotecha

Abstract This paper describes a self supervised representation learning approach that can perform robust object detection in out-of-distribution rotated images for autonomous driving task. Keeping in mind the limitations of vehicle sensors we incorporate self-supervised rotation invariance for correcting rotational corruptions. We present a comprehensive analysis of observed label shifts, validate the results using holdout sets and provide explainability for the

performance using GradCAM approach. We demonstrate results on real-world adversarial examples: (Autonomous Scenes - DAWN dataset) that significantly outperform common methods. The proposed framework outperforms the rotation-affected benchmark by 14.73 points of mAP for the OD task in autonomous driving applications. ©2024 IEEE.

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S2MetNet: A novel dataset and deep learning benchmark for methane point source quantification using Sentinel-2 satellite imagery

Item Type Journal Article

Author A. Radman

Author M. Mahdianpari

Author D.J. Varon

Author F. Mohammadimanesh

Abstract Methane, as a crucial greenhouse gas, plays a significant role in global warming, contributing to approximately one-quarter of the observed climate change since pre-industrial times. Consequently, the detection and quantification of major methane emitters are vital in addressing this issue effectively. Satellite sensors with shortwave infrared (SWIR) spectral bands provide valuable information for monitoring methane emissions. For example, Sentinel-2 multispectral data have the capability to detect methane plumes of large point sources. As such, a wide range of quantification approaches have been developed to quantify methane source rates based on this dataset. Most of the existing methods, however, require auxiliary data, such as wind speed, and have large uncertainties. In this study, we introduce a novel approach based on deep learning models to enhance the precision of methane quantification using Sentinel-2 data without the reliance on external data sources. To train the proposed deep learning model, a comprehensive benchmark dataset has been generated, using Sentinel-2 data. This dataset is created by integrating simulated plumes and background noise extracted from real Sentinel-2 images. This approach ensures the integration of realistic environmental conditions within the simulated data, enhancing the robustness and reliability of our proposed model. The generated benchmark dataset is utilized in different deep learning architectures, namely VGG-19, ResNet-50, Inception-v3, DenseNet-121, Swin-T, and EfficientNet-V2L, to estimate methane source rate. The performance of deep models has been evaluated in three learning strategies, namely from scratch, transfer-learning, and fine-tuning. The fine-tuned EfficientNet-V2L achieves the highest accuracy with root-mean-square error (RMSE), mean absolute percentage error (MAPE), and Pearson R of 2101 kg h⁻¹, 10.05%, and 95.70%, respectively. More importantly, the proposed model demonstrates superior performance compared to conventional physical-based quantification methods (e.g., integrated mass enhancement) and recently developed deep learning model techniques (e.g., MethaNet). In particular, the proposed model exhibits an improvement of approximately 1287 kg h⁻¹ in terms of RMSE, a 3.92% reduction in MAPE, and a 5.01% enhancement in R compared to the IME method. These results highlight the advancements achieved by the proposed approach in accurately quantifying methane emissions using Sentinel-2 imagery. The generated benchmark dataset and the developed deep learning model presented in this study serve as a fundamental resource and constructive framework for future research, promoting extensive implementation across various methane monitoring scenarios on different satellites and in distinct geographic regions, which delivering greater effectiveness to global methane emission monitoring initiatives. © 2023 Elsevier Inc.

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Safety assessment of basiliximab using real-world adverse event data from the FDA Adverse Event Reporting System Database: A retrospective observational study

Item Type Journal Article

Author S. Chen

Author X. Ma

Author J. Zhang

Abstract This study analyzed adverse drug events (ADEs) associated with basiliximab, sourced from the Food and Drug Administration Adverse Event Reporting System (FAERS) database, spanning the first quarter of 2004 to the fourth quarter of 2023. We collected ADE data for basiliximab from 2004 Q1 to 2023 Q4. After standardization, we employed several signal quantification methods for analysis, such as the Reporting Odds Ratio (ROR), Proportional Reporting Ratio (PRR), Bayesian Confidence Propensity for Neural Networks (BCPNN), and empirical bayes geometric mean (EBGM). In this analysis of 1520 ADEs reports citing basiliximab as the primary suspect, we identified 295 preferred terms across 24 system organ classifications (SOCs). The 3 most prevalent SOCs were investigated (n = 1403, ROR 2.84, PRR 2.54, IC 1.34, EBGM 2.54), infections and infestations (n = 1198, ROR 2.85, PRR 2.59, IC 1.37, EBGM 2.59), and renal and urinary disorders (n = 903, ROR 6.01, PRR 5.48, IC 2.45, EBGM 5.47). Increased blood creatinine and pyrexia were the most frequently reported adverse events (AEs) associated with basiliximab, and cytomegalovirus infection also demonstrated significant signal intensity. Notably, this study revealed some adverse reactions beyond basiliximab drug instructions, such as mitral valve calcification, diastolic dysfunction, pelvic fluid collection, testicular swelling, soft tissue necrosis, and muscle necrosis. Although basiliximab offers therapeutic benefits, it carries the risk of several adverse reactions. Clinicians should monitor patients for signs of increased serum creatinine level, fever, cytomegalovirus infection, anaphylactic shock, mitral valve calcification, diastolic

dysfunction, pelvic fluid collection, testicular swelling, soft tissue necrosis, muscle necrosis, and other events during clinical use. © 2024 Lippincott Williams and Wilkins. All rights reserved.

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E1, protein blood level, proteinuria, prothrombin time, Pseudomonas infection, pyelonephritis, ranitidine, Recombinant Fusion Proteins, rejection, respiratory distress, respiratory failure, rituximab, scrotal swelling, septic shock, shock, soft tissue disease, splenomegaly, Staphylococcus infection, stenosis, sulfamethoxazole, systemic inflammatory response syndrome, tacrolimus, thrombocytopenia, thrombotic microangiopathy, tissue necrosis, torasemide, transplant complication, United States Food and Drug Administration, urea blood level, uremia, ureter disease, uric acid blood level, urine volume, urosepsis, vein thrombosis, viremia, virus infection, wound complication, wound dehiscence, wound infection

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Safety Certificate against Latent Variables with Partially Unidentifiable Dynamics

Item Type	Conference Paper
Author	H. Jing
Author	Y. Nakahira
Editor	Singh A.
Editor	Fazel M.
Editor	Hsu D.
Editor	Lacoste-Julien S.
Editor	Berkenkamp F.
Editor	Maharaj T.
Editor	Wagstaff K.
Editor	Zhu J.
Abstract	Existing control techniques often assume access to complete dynamics or perfect simulators with fully observable states, which are necessary to verify whether the system remains within a safe set (forward invariance) or safe actions are persistently feasible at all times. However, many systems contain latent variables that make their dynamics partially unidentifiable or cause distribution shifts in the observed statistics between offline and online data, even when the underlying mechanistic dynamics are unchanged. Such “spurious” distribution shifts can break many techniques that use data to learn system models or safety certificates. To address this limitation, we propose a technique for designing probabilistic safety certificates for systems with latent variables. A key technical enabler is the formulation of invariance conditions in probability space, which can be constructed using observed statistics in the presence of distribution shifts due to latent variables. We use this invariance condition to construct a safety certificate that can be implemented efficiently in real-time control. The proposed safety

certificate can persistently find feasible actions that control long-term risk to stay within tolerance. Stochastic safe control and (causal) reinforcement learning have been studied in isolation until now. To the best of our knowledge, the proposed work is the first to use causal reinforcement learning to quantify long-term risk for the design of safety certificates. This integration enables safety certificates to efficiently ensure longterm safety in the presence of latent variables. The effectiveness of the proposed safety certificate is demonstrated in numerical simulations. © 2025, ML Research Press. All rights reserved.

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Sag Measurement and Quantification in Transmission Lines: A Review

Item Type Conference Paper

Author S.P. Rao

Author F. Rahman

Author P. Ranganathan

Abstract Sags are a serious issue and play a critical role for its cause due to multiple environmental factors such as extreme weather events: wind speed, blizzard, and heat loss. This paper provides a comprehensive review of sag measurements and its quantification methods. Current sag measurement and monitoring approaches are quantified using optical sensors, phasor measurement units, image processing techniques, smart grid technologies, and machine learning algorithms. The paper also proposes a futuristic sag measuring system that is hierarchical and categorized into a sensing, communication, control, and storage levels irrespective of the deployed sensing technology. © 2024 IEEE.

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: S.P. Rao; University of North Dakota, College of Engineering and Mines (CEM), Grand Forks, United States; email: sriram.prabhakara@und.edu; Conference name: 2024 IEEE International Conference on Electro Information Technology, eIT 2024; Conference date: 30 May 2024 through 1 June 2024; Conference code: 201490

SAINTq: Scoring protein-protein interactions in affinity purification – mass spectrometry experiments with fragment or peptide intensity data

Item Type Journal Article

Author G. Teo

Author H. Koh
Author D. Fermin
Author J.-P. Lambert
Author J.D.R. Knight
Author A.-C. Gingras
Author H. Choi

Abstract SAINT (Significance Analysis of INteractome) is a probabilistic method for scoring bait-prey interactions against negative controls in affinity purification – mass spectrometry (AP-MS) experiments. Our published SAINT algorithms use spectral counts or protein intensities as the input for calculating the probability of true interaction, which enables objective selection of high-confidence interactions with false discovery control. With the advent of new protein quantification methods such as Data Independent Acquisition (DIA), we redeveloped the scoring method to utilize the reproducibility information embedded in the peptide or fragment intensity data as a key scoring criterion, bypassing protein intensity summarization required in the previous SAINT workflow. The new software package, SAINTq, addresses key issues in the interaction scoring based on intensity data, including treatment of missing values and selection of peptides and fragments for scoring each prey protein. We applied SAINTq to two independent DIA AP-MS data sets profiling the interactome of MEPCE and EIF4A2 and that of 14-3-3 β , and benchmarked the performance in terms of recovering previously reported literature interactions in the iRefIndex database. In both data sets, the SAINTq analysis using the fragment-level intensity data led to the most sensitive detection of literature interactions at the same level of specificity. This analysis outperforms the analysis using protein intensity data summed from fragment intensity data that is equivalent to the model in SAINTexpress. © 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

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Sample pooling approaches simulated under resource scarcity, lapses in testing capacity, and rapid processing demands for surveillance testing: a data-driven performance comparison

Item Type Journal Article

Author C. Burgess

Author A. Curran

Author A. Ceci

Author C.V. Finkielstein

Author K. Lahmers

Abstract Background: Sample pooling is a critical strategy to meet increased testing demand and conserve resources in surveillance testing. Much of its effectiveness depends on how well optimized the pool size is to the prevalence of infection in the sampled population, which can be difficult to anticipate in many circumstances. Multiple methods exist to better optimize pooling, with unique trade-offs. Methods: Pooling optimization methods were simulated to examine trade-offs between surveillance priorities and operational characteristics using SARS-CoV-2 surveillance data and workflows generated by the Virginia Tech Molecular Diagnostics Laboratory under varying capacity conditions. All in-house validation procedures were designed and established exclusively under CLIA to ensure full control of the analytical framework and to accurately reflect true capacity constraints. We used binary surveillance data to run Monte Carlo simulations (MCS) comparing conservative and large fixed pools, historical prevalence optimization (HPO), prevalence estimation testing (PET), truly optimized pooling, and individual testing. Median test counts from the MCS fed a discrete-event simulation (DES) that assessed processing time at different lab capacities under surveillance and outbreak conditions. We then used the combined performance results to build a classification tree to guide method selection under

different testing priorities and constraints. Results: MCS results indicated that small pools (4 samples), HPO, and PET resulted in test counts that were not statistically different from truly optimized pooling ($p > 0.05$). The DES showed that pooling methods generally performed comparably to individual testing in processing time at low laboratory capacity, but individual testing became faster as capacity increased. Across capacity conditions, individual testing processed fewer than 500 daily samples more quickly, yet it demanded more hands-on time than pooling. Large-scale surveillance favored pooled methods, which were quicker under most conditions, while outbreak scenarios often favored individual testing when capacity wasn't highly limited. Machine learning analysis highlighted surveillance priorities and sample intake as key determinants in selecting the best pooling optimization method for the given circumstance. Conclusion: This study demonstrates the importance of maintaining multiple pooling optimization approaches and adapting strategies to match evolving demands and potential constraints in surveillance testing. Clinical trial number: Not applicable. © The Author(s) 2025.

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Notes:

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Sample-based Uncertainty Quantification with a Single Deterministic Neural Network

Item Type Conference Paper

Author T. Kanazawa

Author C. Gupta

Editor Back T.

Editor van Stein B.

Editor Wagner C.

Editor Garibaldi J.

Editor Lam H.K.

Editor Cottrell M.

Editor Doctor F.

Editor Filipe J.

Editor Warwick K.

Editor Kacprzyk J.

Abstract Development of an accurate, flexible, and numerically efficient uncertainty quantification (UQ) method is one of fundamental challenges in machine learning. Previously, a UQ method called DISCO Nets has been proposed (Bouchacourt et al., 2016) that trains a neural network by minimizing the so-called energy score on training data. This method has shown superior performance on a hand pose estimation task in computer vision, but it remained unclear whether this method works as nicely for regression on tabular data, and how it competes with more recent advanced UQ methods such as NGBoost. In this paper, we propose an improved neural architecture of DISCO Nets that admits a more stable and smooth training. We benchmark this approach on miscellaneous real-world tabular datasets and confirm that it is competitive with or even superior to standard UQ baselines. We also provide a new elementary proof for the validity of using the energy score to learn predictive distributions. Further, we point out that DISCO Nets in its original form ignore epistemic uncertainty and only capture aleatoric uncertainty. We propose a simple fix to this problem. Copyright © 2022 by SCITEPRESS - Science and Technology Publications, Lda.

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Energy, Uncertainty analysis, Computer vision, Neural networks, Uncertainty, Quantification methods, Uncertainty quantifications, Uncertainty Quantification, Epistemic uncertainties, Aleatoric uncertainty, Aleatoric Uncertainty, CRPS, DISCO net, DISCO Nets, Energy score, Energy Score, Ensemble forecasting, Ensemble Forecasting, Epistemic Uncertainty

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Sample-based Uncertainty Quantification with a Single Deterministic Neural Network

Item Type Conference Paper

Author T. Kanazawa

Author C. Gupta

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Editor Filipe J.

Editor Warwick K.

Abstract Development of an accurate, flexible, and numerically efficient uncertainty quantification (UQ) method is one of fundamental challenges in machine learning. Previously, a UQ method called DISCO Nets has been proposed (Bouchacourt et al., 2016) that trains a neural network by minimizing the so-

called energy score on training data. This method has shown superior performance on a hand pose estimation task in computer vision, but it remained unclear whether this method works as nicely for regression on tabular data, and how it competes with more recent advanced UQ methods such as NGBoost. In this paper, we propose an improved neural architecture of DISCO Nets that admits a more stable and smooth training. We benchmark this approach on miscellaneous real-world tabular datasets and confirm that it is competitive with or even superior to standard UQ baselines. We also provide a new elementary proof for the validity of using the energy score to learn predictive distributions. Further, we point out that DISCO Nets in its original form ignore epistemic uncertainty and only capture aleatoric uncertainty. We propose a simple fix to this problem. © 2023 by SCITEPRESS – Science and Technology Publications, Lda.

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Satellite-Derived Approaches for Coal Mine Methane Estimation: A Review

Item Type Journal Article

Author A. Chauhan

Author S. Raval

Abstract Highlights: What are the main findings? Satellite observations have advanced from coarse (~10 km) to fine (25 m) resolution, greatly improving plume detection and narrowing gaps between bottom-up and top-down methane estimates. Significant uncertainties persist due to wind variability, terrain heterogeneity, retrieval-algorithm limitations, and limited temporal sampling, leading to inconsistent quantification, especially for diffuse surface coal-mine emissions. What are the implications of the main findings? Reliable site-level methane estimates require integrating satellites with ground and aerial validation, improved atmospheric modelling, and machine-learning methods that address diffuse and variable coal mine emissions. Reducing these uncertainties is essential for accurate reporting and effective mitigation in coal mining regions, thereby strengthening global climate action. Methane emissions from coal mines, especially surface operations, are spatially diffuse, presenting significant challenges for accurate quantification. Satellites such as TROPOMI, GHGSat, PRISMA, GaoFen-5, and GOSAT have been extensively used for detecting methane emissions at various scales, from individual point sources to regional and global assessments. Despite various advancements, methane quantification via satellite observations remains subject to several challenges. Various quantification methods for the same observation can produce variable results. Also, meteorological conditions, terrain complexity, and surface heterogeneity introduce uncertainties in emission estimates. The selection of wind speed and direction, along with retrieval-algorithm limitations, can lead to significant discrepancies in reported emissions. Additionally, satellite-based observations capture emissions only at specific overpass times, which may introduce temporal uncertainties compared to inventories derived from continuous emission estimations. This study provides a comprehensive review of satellite-based coal mine methane (CMM) monitoring, evaluating current methodologies, their limitations, and recent technological advancements. We discussed the potential of emerging machine-learning techniques, improved atmospheric modelling, and integrated observational approaches to enhance methane emission quantification. By refining satellite-based monitoring techniques and addressing existing challenges, this research will support the development of more accurate emission inventories and effective mitigation strategies for the coal mining sector. © 2025 by the authors.

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SC-JNMF: Single-cell clustering integrating multiple quantification methods based on joint non-negative matrix factorization

Item Type Journal Article

Author M. Shiga

Author S. Seno

Author M. Onizuka

Author H. Matsuda

Abstract Single-cell RNA-sequencing is a rapidly evolving technology that enables us to understand biological processes at unprecedented resolution. Single-cell expression analysis requires a complex data processing pipeline, and the pipeline is divided into two main parts: The quantification part, which converts the sequence information into gene-cell matrix data; the analysis part, which analyzes the matrix data using statistics and/or machine learning techniques. In the analysis part, unsupervised cell clustering plays an important role in identifying cell types and discovering cell diversity and subpopulations. Identified cell clusters are also used for subsequent analysis, such as finding differentially expressed genes and inferring cell trajectories. However, single-cell clustering using gene expression profiles shows different results depending on the quantification methods. Clustering results are greatly affected by the quantification method used in the upstream process. In other words, even if the original RNA-sequence data is the same, gene expression profiles processed by different quantification methods will produce different clusters. In this article, we propose a robust and highly accurate clustering method based on joint non-negative matrix factorization (joint-NMF) by utilizing the information from multiple gene expression profiles quantified using different methods from the same RNA-sequence data. Our joint-NMF can extract common factors among multiple gene expression profiles by applying each NMF under the constraint that one of the factorized matrices is shared among multiple NMFs. The joint-NMF determines more robust and accurate cell clustering results by leveraging multiple quantification methods compared to conventional clustering methods, which use only a single gene expression profile. Additionally, we showed the usefulness of discovering marker genes with the extracted features using our method. Copyright © 2021 Shiga et al.

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Scalability of continuous active learning for reliable high-recall text classification

Item Type Conference Paper**Author** G.V. Cormack**Author** M.R. Grossman

Abstract For finite document collections, continuous active learning ("CAL") has been observed to achieve high recall with high probability, at a labeling cost asymptotically proportional to the number of relevant documents. As the size of the collection increases, the number of relevant documents typically increases as well, thereby limiting the applicability of CAL to low-prevalence high-stakes classes, such as evidence in legal proceedings, or security threats, where human effort proportional to the number of relevant documents is justified. We present a scalable version of CAL ("S-CAL") that requires $O(\log N)$ labeling effort and $O(N \log N)$ computational effort - where N is the number of unlabeled training examples - to construct a classifier whose effectiveness for a given labeling cost compares favorably with previously reported methods. At the same time, S-CAL offers calibrated estimates of class prevalence, recall, and precision, facilitating both threshold setting and determination of the adequacy of the classifier. © 2016 Copyright held by the owner/author(s).

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Scalability vs. Utility: Do We Have to Sacrifice One for the Other in Data Importance Quantification?

Item Type Conference Paper

Author R. Jia
Author F. Wu
Author X. Sun
Author J. Xu
Author D. Dao
Author B. Kailkhura
Author C. Zhang
Author B. Li
Author D. Song

Abstract Quantifying the importance of each training point to a learning task is a fundamental problem in machine learning and the estimated importance scores have been leveraged to guide a range of data workflows such as data summarization and domain adaption. One simple idea is to use the leave-one-out error of each training point to indicate its importance. Recent work has also proposed to use the Shapley value, as it defines a unique value distribution scheme that satisfies a set of appealing properties. However, calculating Shapley values is often expensive, which limits its applicability in real-world applications at scale. Multiple heuristics to improve the scalability of calculating Shapley values have been proposed recently, with the potential risk of compromising their utility in real-world applications. How well do existing data quantification methods perform on existing workflows? How do these methods compare with each other, empirically and theoretically? Must we sacrifice scalability for the utility in these workflows when using these methods? In this paper, we conduct a novel theoretical analysis comparing the utility of different importance quantification methods, and report extensive experimental studies on existing and proposed workflows such as noisy label detection, watermark removal, data summarization, data acquisition, and domain adaptation. We show that Shapley value approximation based on a KNN surrogate over pretrained feature embeddings obtains comparable utility with existing algorithms while achieving significant scalability improvement, often by orders of magnitude. Our theoretical analysis also justifies its advantage over the leave-one-out error. The code is available at <https://github.com/AIsecure/Shapley-Study>. © 2021 IEEE

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Scalable Quasi-Bayesian Inference for Instrumental Variable Regression

Item Type Conference Paper

Author Z. Wang

Author Y. Zhou

Author T. Ren

Author J. Zhu

Editor Ranzato M.

Editor Beygelzimer A.

Editor Dauphin Y.

Editor Liang P.S.

Editor Wortman Vaughan J.

Abstract Recent years have witnessed an upsurge of interest in employing flexible machine learning models for instrumental variable (IV) regression, but the development of uncertainty quantification methodology is still lacking. In this work we present a scalable quasi-Bayesian procedure for IV regression, building upon the recently developed kernelized IV models. Contrary to Bayesian modeling for IV, our approach does not require additional assumptions on the data generating process, and leads to a scalable approximate inference algorithm with time cost comparable to the corresponding point estimation methods. Our algorithm can be further extended to work with neural network models. We analyze the theoretical properties of the proposed quasi-posterior, and demonstrate through empirical evaluation the competitive performance of our method. © 2021 Neural information processing systems foundation. All rights reserved.

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Scale-Dependent hydroxyl radical generation and energy efficiency in vortex diode hydrodynamic cavitation: machine learning insights toward industrial-scale applications in water treatment

Item Type Journal Article**Author** X. Pang**Author** V.P. Sarvothaman**Author** S.R. Kulkarni**Author** W.L. Roberts**Author** V.V. Ranade

Abstract Hydrodynamic cavitation (HC) reactors are increasingly applied in the remediation of organic pollutants in water, leveraging intense shear and hydroxyl radical ($\text{OH}\cdot$) generation to accelerate degradation processes. However, scaling up HC devices remains a challenge in environmental engineering due to poorly understood effects at different scales of operation. Vortex-based HC (VDs) provide superior cavitation efficiency compared to traditional orifice and venturi devices and therefore this study examines the scale-dependent generation of $\text{OH}\cdot$ in VDs. Coumarin dosimetry was employed as the quantification method for $\text{OH}\cdot$ generation. Available published experimental datasets were analysed: (i) varying inlet pressures from 100 to 400 kPa for throat diameters of 6 and 12 mm, and (ii) throat diameters ranging from 6 to 38 mm (nominal capacities of 5 to 200 L/min) at a fixed pressure drop of 280 kPa. After normalization of the available data, seven machine learning (ML) models were trained to establish relationships between operating conditions and $\text{OH}\cdot$ generation

performance. eXtreme Gradient Boosting (XGB) and artificial neural networks (ANN) outperformed the others, with higher R² and lower RMSE. After using SHAP interpretation, these two models were used to elucidate scale effects on both radical yield and energy efficiency, resulting in actionable design guidelines for VDs at different scales. By combining experimental dosimetry with predictive ML, this work advances the fundamental understanding and practical implementation of cavitation-based advanced oxidation technologies, particularly for efficient and energy-optimized treatment of organic pollutants in wastewater. © 2026 The Authors

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Scaling Smart Cities with Federated Learning: Balancing Accuracy and Privacy for Building Energy Performance Prediction

Item Type Journal Article**Author** J. Delgado Fernández**Author** L. Willburger**Author** C. Wiethe**Author** S. Wenninger**Author** G. Fridgen

Abstract The building energy sector is a significant contributor to carbon emissions, thereby playing a crucial role in driving global sustainability efforts to achieve the net-zero targets outlined in the Paris Climate Agreement. Precise predictions of building energy performance are imperative for effective planning and investment decisions aimed at enhancing energy efficiency. While data-driven methods, primarily leveraging machine learning techniques, offer promising predictive capabilities, they heavily rely on large datasets for accurate assessments. However, a prevalent challenge arises as energy consultants and agencies often lack expansive datasets, and if they do, they are reluctant to share their data. To overcome these hurdles, the study implements a decentralized, privacy-preserving machine learning approach known as federated learning. This approach was applied to a dataset encompassing over 25,000 residential buildings featuring diverse construction attributes and energy sources. The simulation involved mimicking different energy agencies by segmenting geographic regions. The study compared the prediction performance of federated learning with that of a model accessing the entire dataset and a fully isolated local model. The findings demonstrate that federated learning achieves a 12% improvement in prediction performance compared to the isolated model. This outcome underscores federated learning's capacity to leverage the full potential of scaling data-driven methodologies, providing a pathway to unlock new business models in both research and practice, while aligning with net-zero aspirations. © The Author(s) 2025.

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Screening mammography beyond breast cancer: breast arterial calcifications as a sex-specific biomarker of cardiovascular risk

Item Type Journal Article

Author R.M. Trimboli

Author M. Codari

Author M. Guazzi

Author F. Sardanelli

Abstract Purpose: To highlight the importance of quantitative breast arterial calcifications (BAC) assessment for an effective stratification of cardiovascular (CV) risk in women, for whom current preventive strategies are inadequate. BAC, easily detectable on mammograms, are associated with CV disease and represent a potential imaging biomarker for CV disease prevention in women. Method: We summarized the available evidence on this topic. Results: Age, parity, diabetes, and hyperlipidemia were found to positively correlate with BAC. Women with BAC have a higher CV risk than those without BAC: the relative risk was reported to be 1.4 for transient ischemic attack/stroke, 1.5 for thrombosis, 1.8 for myocardial infarction; the reported hazard ratio was 1.32 for coronary artery disease (CAD), 1.52 for heart failure, 1.29 for CV death, 1.44 for death from CAD. However, BAC do not alarm radiologists; when reported, they are commonly mentioned as “present”, not impacting on CV decision-making. Of 18 published studies, 9 reported only presence/absence of BAC, 4 used a semi-quantitative scale, and 5 a continuous scale (with manual, automatic or semiautomatic segmentation). Various appearance, topological complexity, and vessels overlap make BAC quantification difficult to standardize. Nevertheless, machine learning approaches showed promising results in BAC quantification on mammograms. Conclusions: There is a strong rationale for mammography to become a dual test for breast cancer screening and CV disease prevention. However, robust and automated quantification methods are needed for a deeper insight on the association between BAC and CV disease, to stratifying CV risk and define personalized preventive actions. © 2019 Elsevier B.V.

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Screening of specific quantitative peptides of beef by LC–MS/MS coupled with OPLS-DA

Item Type Journal Article**Author** C. Kang**Author** Y. Zhang**Author** M. Zhang**Author** J. Qi**Author** W. Zhao**Author** J. Gu**Author** W. Guo**Author** Y. Li

Abstract A rapid, simple, and efficient analysis methodology for screening specific quantitative peptides of beef was established based on liquid chromatography-tandem mass spectrometry (LC–MS/MS) coupled with orthogonal partial least squares-

discriminant analysis (OPLS-DA). The OPLS-DA model was built to select species-specific peptides that make a significant contribution to classification. Peptides with statistical significance were selected based on the variable importance in the projection (VIP) values and univariate P values. After the workflow of the statistical process, three specific quantitative peptides were identified by using homemade products with different beef contents. A quantification method for selected specific quantitative peptides was established by using LC–MS/MS. The quantitative results were applied to commercialized beef products. The developed method has high sensitivity, specificity, and repeatability. The results of this study proved that the integration of LC–MS/MS coupled with OPLS-DA is an efficient method for screening specific quantitative peptides and identification of the authenticity of meat products. © 2022

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Seeking regularity from irregularity: unveiling the synthesis-nanomorphology relationships of heterogeneous nanomaterials using unsupervised machine learning

Item Type Journal Article

Author L. Yao

Author H. An

Author S. Zhou

Author A. Kim

Author E. Luijten

Author Q. Chen

Abstract Nanoscale morphology of functional materials determines their chemical and physical properties. However, despite increasing use of transmission electron microscopy (TEM) to directly image nanomorphology, it remains challenging to quantify the information embedded in TEM data sets, and to use nanomorphology to link synthesis and processing conditions to properties. We develop an automated, descriptor-free analysis workflow for TEM data that utilizes convolutional neural networks and unsupervised learning to quantify and classify nanomorphology, and thereby reveal synthesis-nanomorphology relationships in three different systems. While TEM records nanomorphology readily in two-dimensional (2D) images or three-dimensional (3D) tomograms, we advance the analysis of these images by identifying and applying a universal shape fingerprint function to characterize nanomorphology. After dimensionality reduction through principal component analysis, this function then serves as the input for morphology grouping through unsupervised learning. We demonstrate the wide applicability of our workflow to both 2D and 3D TEM data sets, and to both inorganic and organic nanomaterials, including tetrahedral gold nanoparticles mixed with irregularly shaped impurities, hybrid polymer-patched gold nanoprisms, and polyamide membranes with irregular and heterogeneous 3D crumple structures. In each of these systems, unsupervised nanomorphology grouping identifies both the diversity and the similarity of the nanomaterial across different synthesis conditions, revealing how synthetic parameters guide nanomorphology development. Our work opens possibilities for enhancing synthesis of nanomaterials through artificial intelligence and for understanding and controlling complex nanomorphology, both for 2D systems and in the far less explored case of 3D structures, such as those with embedded voids or hidden interfaces. © 2022 The Royal Society of Chemistry.

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Embedded systems, Property, Principal component analysis, Morphology, Data set, Unsupervised learning, Gold nanoparticles, Descriptors, Chemical analysis, Chemical and physical properties, Functional materials, High resolution transmission electron microscopy, Nanomorphologies, Nanoscale morphology, Processing condition, Synthesis and processing, Synthesis conditions, Unsupervised machine learning

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Segmentation and quantification of atherosclerotic plaques in optical coherence tomography

Item Type Journal Article

Author V.V. Danilov

Author V.V. Laptev

Author K.Y. Klyshnikov

Author I.S. Bessonov

Author N.V. Litvinyuk

Author E.A. Ovcharenko

Author N.A. Kochergin

Abstract Introduction: Cardiovascular disease remains a leading cause of mortality worldwide, with atherosclerosis at its core. Accurate identification of vulnerable plaques is critical for preventing acute cardiovascular events. While optical coherence tomography (OCT) provides high-resolution imaging of plaque features, manual analysis is labor-intensive and operator-dependent. This study addresses the need for automated, accurate segmentation of atherosclerotic plaques in OCT pullbacks using advanced machine learning (ML) techniques. Methods: A comprehensive multi-center dataset of OCT pullbacks, encompassing 103 patients and annotated for key

plaque morphological features (lumen, fibrous cap, lipid core, and vasa vasorum), was used to tune, train and evaluate nine ML models, including U-Net, U-Net++, DeepLabV3, DeepLabV3+, FPN, LinkNet, PSPNet, PAN, and MA-Net. To address dataset imbalances and optimize performance for each plaque feature, we introduced a hybrid segmentation strategy: single-class models were deployed for highly prevalent features (e.g., lumen) and underrepresented classes (e.g., vasa vasorum), while a multi-class model targeted morphologically complex features with overlapping boundaries (e.g., fibrous cap and lipid core). Hyperparameter tuning was performed using Bayesian optimization, and segmentation accuracy was assessed with the Dice Similarity Coefficient (DSC) and other metrics. Results: The models achieved high segmentation accuracy for lumen (DSC: 0.987), fibrous cap (DSC: 0.736), and lipid core (DSC: 0.751), demonstrating the potential of leveraging ML techniques to enhance OCT's diagnostic capabilities. While lumen segmentation showed exceptional precision, the moderate accuracy for fibrous cap and lipid core highlights challenges with complex morphologies. Satisfactory results for vasa vasorum (DSC: 0.610) suggest areas for further refinement. By integrating these models into a weighted ensemble, taking into account class prevalence and model confidence, the combined system achieved a weighted DSC of 0.882 across all plaque features, a significant improvement over individual models. These findings confirm the hybrid strategy's ability to balance computational efficiency with accuracy through rigorous optimization, tailored model selection, and ensemble integration. Conclusion: This study presents a robust ML-driven framework for automated OCT segmentation that uses a hybrid approach and weighted ensemble learning to address class imbalance and feature complexity, significantly improving the accuracy and efficiency of atherosclerotic plaque analysis. The findings suggest potential clinical implications, including improved detection of high-risk plaques and enhanced decision-making in cardiovascular care. However, further prospective validation is required before clinical adoption. Future research should focus on expanding datasets, integrating multimodal imaging, and refining models for real-time clinical use, paving the way for transformative advancements in cardiovascular diagnostics. © 2025 The Authors

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Segmentation of MS lesions using active contour model, adaptive mixtures method and MRF model

Item Type Conference Paper

Author A. Bijar

Author R. Khayati

Abstract This paper presents an approach for fully automatic segmentation of MS lesions in fluid attenuated inversion recovery (FLAIR) Magnetic Resonance (MR) images. The proposed method estimates a gaussian mixture model with three components as cerebrospinal fluid (CSF), normal tissue and MS lesions. To estimate this model, a region based Active Contour Model (ACM) is used to find the best initial values of model parameters. Then, Adaptive Mixture Method and Markov Random Field (MRF) model are utilized to obtain and upgrade the class conditional probability density function and the apriori probability of each class. After estimation of Model parameters and apriori probabilities, brain tissues are classified using Bayesian Classification. To evaluate the result of proposed method, the similarity criteria of different slices related to 20 MS patients are calculated and compared with other methods which include manual segmentation. Also, volume of segmented lesions are computed and compared with gold standard using correlation coefficient. The proposed method has better performance in comparison with previous works which are reported here. © 2011 University of Zagreb.

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Extra Journal Abbreviation: ISPA - Int. Symp. Image Signal Process. Anal.
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Proceedings Title ISPA - Int. Symp. Image Signal Process. Anal.
Conference Name ISPA 2011 - 7th International Symposium on Image and Signal Processing and Analysis
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Tags:

Probability density function, Brain, Image segmentation, Estimation, Signal processing, Probability, Gold standards, Magnetic resonance, Bayesian classification, Gaussian Mixture Model, Automatic segmentations, A-priori probabilities, Active contours, Correlation coefficient, Multiple sclerosis, Mixture method, Conditional probability density, Mixtures, Tissue, Model parameters, Active contour model, Active Contours, Adaptive Mixture Method, Brain tissue, Cerebro spinal fluids, Fluid attenuated inversion recoveries, Initial values, Manual segmentation, Markov Random Field model, Markov Random Field Model, Markov random field models, Multiple Sclerosis, Normal tissue, Region based active contours, Similarity criteria, Three component

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Export Date: 20 September 2026; Cited By: 6; Correspondence Address: A. Bijar; Department of Biomedical Engineering, Shahed University, Tehran, Iran; email: ahmad_bijar@live.com; Conference name: 7th International Symposium on Image and Signal Processing and Analysis, ISPA 2011; Conference date: 4 September 2011 through 6 September 2011; Conference code: 87698

Segmentation of sperm's Acrosome, Nucleus and Mid-piece in microscopic images of stained human semen smear

Item Type Conference Paper
Author A. Bijar
Author M. Mikaeili
Author A.P. Benavent
Author R. Khayati
Abstract The measurement or evaluation and clinical significance of human sperm morphology has always been and still is a controversial aspect of the semen analysis for the determination of a male's fertility potential. The evaluation of sperm size, shape and morphological smear characteristics should be assessed by carefully observing a stained sperm sample under a microscope. In order to avoid

subjectivity, numerous studies that incorporate image analysis techniques in the assessment of sperm morphology have been proposed. The primary step of all these methods is segmentation of sperm's parts. In this paper, we have proposed a new method for segmentation of sperm's Acrosome, Nucleus and Mid-piece. Sperm's Acrosome, Nucleus and Midpiece are segmented through a method based on a Bayesian classifier which utilizes the adaptive mixtures method (AMM) and Markov random field (MRF) model to obtain and upgrade the class conditional probability density function (CCPDF) and the apriori probability of each class. To compare the performance of the proposed approach with those of previous approaches including manual segmentation, the Accuracy, Sensitivity and Specificity were calculated. © 2012 IEEE.

Date 2012

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Language English

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Segmentation priors from local image properties: Without using bias field correction, location-based templates, or registration

Item Type Journal Article

Author A. Vovk

Author R.W. Cox

Author J. Stare

Author D. Suput

Author Z.S. Saad

Abstract We present a novel approach for generating information about a voxel's tissue class membership based on its signature—a collection of local image textures estimated over a range of neighborhood sizes. The approach produces a form of tissue class priors that can be used to initialize and regularize image segmentation. The signature-based approach is a departure from current location-based methods, which derive tissue class likelihoods based on a voxel's location in standard template space. To use location-based priors, one needs to register the volume in question to the template space, and estimate the image intensity bias field. Two optimizations, over more than a thousand parameters, are needed when high order nonlinear registration is employed. In contrast, the signature-based approach is independent of volume orientation, voxel position, and largely insensitive to bias fields. For these reasons, the approach does not require the use of population derived templates. The prior information is generated from variations in image texture statistics as a function of spatial scale, and an SVM approach is used to associate signatures with tissue types. With the signature-based approach, optimization is needed only during the training phase for the parameter estimation stages of the SVM hyperplanes, and associated PDFs; a training process separate from the segmentation step. We found that signature-based priors were superior to location-based ones aligned under favorable conditions, and that signature-based priors result in improved segmentation when replacing location-based ones in FAST (Zhang et al., 2001), a widely used segmentation program. The software implementation of this work is freely available as part of AFNI <http://afni.nimh.nih.gov>. © 2010.

Date 2011

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SEISMIC RISK CLASSIFICATION OF NON-STRUCTURAL ELEMENTS

Item Type	Journal Article
Author	G.J. O'reilly
Author	G.M. Calvi
Abstract	Non-structural elements (NSEs) are those not forming part of a building's structural load-bearing system but are nevertheless subjected to dynamic forces and deformations during ground shaking. In terms of their role in building cost, NSEs make up approximately 82%, 87% and 92% of the total monetary investment for office, hotel and hospital buildings, respectively. Additionally, it has been shown that for a typical school building in Italy, NSEs comprise the majority (>60%) of the direct monetary losses induced at frequent levels of ground shaking. These two points alone highlight the critical nature of NSEs both from an initial investment and potential monetary loss perspective. The ability to quantify the seismic risk associated with structural and non-structural elements is a critical aspect of earthquake engineering. While methods to improve the understanding of structural response to earthquake shaking and how to quantify their risk have been studied, NSEs have recently emerged as a crucial aspect to address given their pertinence in overall building performance. This paper describes the formulation of a risk quantification methodology for NSEs whereby the mean annual frequency of exceeding an NSE's damage state is computed and rated as part of a risk classification scheme. The basis of the methodology is described followed by example implementations, where the details surrounding hazard, structural and non-structural response are quantified consistently, ensuring that uncertainties are also incorporated to be in line with modern performance-based earthquake engineering. This relates to both storey drift-sensitive and floor acceleration-sensitive non-structural elements that typically comprise most building fixtures and fittings in addition to the contents themselves. The result is a simple but effective methodology that may be used to directly quantify

and rank the risk associated with NSEs from a life safety, functionality and economic loss perspective. It may be implemented as part of a prioritisation scheme for building retrofit or also serve as a commercial instrument for NSE manufacturers, whereby their products can be marketed as having a certain rating class, to demonstrate superior seismic resilience with respect to others. © The 17th World Conference on Earthquake Engineering.

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Assessment, Structural dynamics, Earthquake effects, Risk classification, assessment, Architectural design, Building structural, College buildings, design, Dynamic deformation, Dynamic response, Emergency rooms, Ground-shaking, Load-bearing systems, non-structural elements, Non-structural elements, risk classification, Seismic design, seismic performance, Seismic Performance, Seismic risk, Structural response

Notes:

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Export Date: 20 September 2026; Cited By: 0

Selected machine learning of HOMO-LUMO gaps with improved data-efficiency

Item Type Journal Article

Author B. Mazouin

Author A.A. Schöpfer

Author O.A. von Lilienfeld

Abstract Despite their relevance for organic electronics, quantum machine learning (QML) models of molecular electronic properties, such as HOMO-LUMO-gaps, often struggle to achieve satisfying data-efficiency as measured by decreasing prediction errors for increasing training set sizes. We demonstrate that partitioning training sets into different chemical classes prior to training results in independently trained QML models with overall reduced training data needs. For organic molecules drawn from

previously published QM7 and QM9-data-sets we have identified and exploited three relevant classes corresponding to compounds containing either aromatic rings and carbonyl groups, or single unsaturated bonds, or saturated bonds. The selected QML models of band-gaps (considered at GW and hybrid DFT levels of theory) reach mean absolute prediction errors of ~ 0.1 eV for up to an order of magnitude fewer training molecules than for QML models trained on randomly selected molecules. Comparison to Δ -QML models of band-gaps indicates that selected QML exhibit superior data-efficiency. Our findings suggest that selected QML, e.g. based on simple classifications prior to training, could help to successfully tackle challenging quantum property screening tasks of large libraries with high fidelity and low computational burden. © 2022 RSC.

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Selecting and Interpreting Multiclass Loss and Accuracy Assessment Metrics for Classifications with Class Imbalance: Guidance and Best Practices

Item Type Journal Article

Author S. Farhadpour

Author T.A. Warner

Author A.E. Maxwell

Abstract Evaluating classification accuracy is a key component of the training and validation stages of thematic map production, and the choice of metric has profound implications for both the success of the training process and the reliability of the final accuracy assessment. We explore key considerations in selecting and interpreting loss and assessment metrics in the context of data imbalance, which arises when the classes have unequal proportions within the dataset or landscape being mapped. The challenges involved in calculating single, integrated measures that summarize classification success, especially for datasets with considerable data imbalance, have led to much confusion in the literature. This confusion arises from a range of issues, including a lack of clarity over the redundancy of some accuracy measures, the importance of calculating final accuracy from population-based statistics, the effects of class imbalance on accuracy statistics, and the differing roles of accuracy measures when used for training and final evaluation. In order to characterize classification success at the class level, users typically generate averages from the class-based measures. These averages are sometimes generated at the macro-level, by taking averages of the individual-class statistics, or at the micro-level, by aggregating values within a confusion matrix, and then, calculating the statistic. We show that the micro-averaged producer's accuracy (recall), user's accuracy (precision), and F1-score, as well as weighted macro-averaged statistics where the class prevalences are used as weights, are all equivalent to each other and to the overall accuracy, and thus, are redundant and should be avoided. Our experiment, using a variety of loss metrics for training, suggests that the choice of loss metric is not as complex as it might appear to be, despite the range of choices available, which include cross-entropy (CE), weighted CE, and micro- and macro-Dice. The highest, or close to highest, accuracies in our experiments were obtained by using CE loss for models trained with balanced data, and for models trained with imbalanced data, the highest accuracies were obtained by using weighted CE loss. We recommend that, since weighted CE loss used with balanced training is equivalent to CE, weighted CE loss is a good all-round choice. Although Dice loss is commonly suggested as an alternative to CE loss when classes are imbalanced, micro-averaged Dice is similar to overall accuracy, and thus, is particularly poor for training with imbalanced data. Furthermore, although macro-Dice resulted in models with high accuracy when the training used balanced data, when the training used imbalanced data, the accuracies were lower than for weighted CE. In summary, the significance of this paper lies in its provision of readers with an overview of accuracy and loss metric terminology, insight regarding the redundancy of some measures, and guidance regarding best practices. © 2024 by the authors.

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Selecting informative conformal prediction sets with false coverage rate control

Item Type Journal Article**Author** U. Gazin**Author** R. Heller**Author** A. Marandon**Author** E. Roquain

Abstract In supervised learning, including regression and classification, conformal methods provide prediction sets for the outcome/label with finite sample coverage for any machine learning predictor. We consider here the case where such prediction sets come after a selection process. The selection process requires that the selected prediction sets be ‘informative’ in a well-defined sense. We consider both the classification and regression settings where the analyst may consider as informative only the sample with prediction sets small enough, excluding null values, or obeying other appropriate ‘monotone’ constraints. We develop a unified framework for building such informative conformal prediction sets while controlling the false coverage rate (FCR) on the selected sample. While conformal prediction sets after selection have been the focus of much recent literature in the field, the new introduced procedures, called InfoSP and InfoSCOP, are to our knowledge the first ones providing FCR control for informative prediction sets. We show the usefulness of our resulting procedures on real and simulated data. © The Royal Statistical Society 2025. All rights reserved.

Date 2025**DOI** 10.1093/jrsssb/qkae120**URL** <https://www.scopus.com/pages/publications/105016737688?origin=resultslist>**Archive** Scopus**Language** English**Volume** 87

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Selective Mixup Helps with Distribution Shifts, But Not (Only) because of Mixup

Item Type Conference Paper

Author D. Teney

Author J. Wang

Author E. Abbasnejad

Editor Salakhutdinov R.

Editor Kolter Z.

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Editor Weller A.

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Editor Scarlett J.

Editor Berkenkamp F.

Abstract Context. Mixup is a highly successful technique to improve generalization by augmenting training data with combinations of random pairs. Selective mixup is a family of methods that apply mixup to specific pairs e.g. combining examples across classes or domains. Despite remarkable performance on benchmarks with distribution shifts, these methods are still poorly understood. Findings. We find that an overlooked aspect of selective mixup explains some of its success in a completely new light. The non-random selection of pairs affects the training distribution and improves generalization by means completely unrelated to the mixing. For example in binary classification, mixup across classes implicitly resamples the data to uniform class distribution — a classical solution to label

shift. We verify empirically that this resampling explains some of the improvements reported in prior work. Theoretically, the effect relies on a “regression toward the mean”, an accidental property we find in several datasets. Outcomes. We now better understand why selective mixup works. This lets us predict a yet-unknown failure mode and conditions where the method is detrimental. We also use the equivalence to resampling to design variants with better combinations of mixing and resampling. Copyright 2024 by the author(s)

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Self-Assessment and Robust Anomaly Detection with Bayesian Deep Learning

Item Type Conference Paper

Author G. Carannante

Author D. Dera

Author O. Aminul

Author N.C. Bouaynaya

Author G. Rasool

Abstract Deep Learning (DL) models have achieved or even surpassed human-level accuracy in several areas, including computer vision and pattern recognition. The state-of-art performance of DL models has raised the interest in using them in real-world applications, such as disease diagnosis and clinical decision support

systems. However, the challenge remains the lack of trustworthiness and reliability of these DL models. The detection of incorrect decisions or flagging suspicious input samples is essential for the reliability of machine learning models. Uncertainty estimation in the output decision is a key component in establishing the trustworthiness and reliability of these models. In this work, we use Bayesian techniques to estimate the uncertainty in the model's output and use this uncertainty to detect distributional shifts linked to both input perturbations and labels shifts. We use the learned uncertainty information (i.e., the variance of the predictive distribution) in two different ways to detect anomalous input samples: 1) a static threshold based on average uncertainty of a model evaluated on the clean test data, and 2) a statistical threshold based on the significant increase in the average uncertainty of the model evaluated on corrupted (anomalous) samples. Our extensive experiments demonstrate that both approaches can detect anomalous samples. We observe that the proposed thresholding techniques can distinguish misclassified examples in the presence of noise, adversarial attacks, anomalies or distributional shifts. For example, when considering corrupted versions of MNIST and CIFAR-10 datasets, the rate of detecting misclassified samples is almost twice as compared to Monte-Carlo-based approaches. © 2022 International Society of Information Fusion.

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Self-Organizing Map-Based Resilience Quantification and Resilient Control of Distribution Systems Under Extreme Events

Item Type Journal Article

Author K. Utkarsh

Author F. Ding

Abstract Due to climate change, extreme weather events are occurring more frequently and with increasing impact. This trend poses a significant challenge for distribution system operators (DSO) to ensure that there is uninterrupted power supply to critical loads in their networks. To embed resilience into DSO's decision-making, resilience needs to be first quantified and then integrated into the system-level optimization. Therefore, this paper first develops a novel self-organizing map (SOM) based method (called SomRes) to quantify the time-varying resilience index of a system that can leverage the powerful classification property of SOMs and removes some of the disadvantages of subjective weight assignment methods. Using SomRes, a resilient resource allocation and operational dispatch algorithm is further developed to enhance system resilience against extreme events by considering the SomRes resilience index directly as the feedback. The proposed resilience quantification approach is benchmarked with a state-of-the-art approach and the efficacy of the proposed resilient dispatch algorithm is demonstrated through several deterministic and statistical case studies on the IEEE 123-bus distribution system. Simulation studies show that the proposed SomRes quantification method is an appropriate indicator of system resilience, and the resilient resource allocation and dispatch strategy can significantly reduce critical load shedding under varying event propagation scenarios. © 2010-2012 IEEE.

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Notes:

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Export Date: 20 September 2026; Cited By: 35; Correspondence Address: F. Ding; Power Systems Engineering Center, National Renewable Energy Laboratory, Golden, 80401, United States; email: fei.ding@nrel.gov

Self-supervised Learning for Electroencephalogram: A Systematic Survey

Item Type Journal Article

Author W. Weng

Author Y. Gu

Author S. Guo

Author Y. Ma

Author Z. Yang

Author Y. Liu

Author Y. Chen

Abstract Electroencephalography (EEG) is a non-invasive technique to record bioelectrical signals. Integrating supervised deep learning techniques with EEG signals has recently facilitated automatic analysis across diverse EEG-based tasks. However, the label issues of EEG signals have constrained the development of EEG-based deep models. Obtaining EEG annotations is difficult and requires domain experts to guide collection and labeling, and the variability of EEG signals among different subjects causes significant label shifts. To solve the above challenges, self-supervised learning (SSL) has been proposed to extract representations from unlabeled samples through well-designed pretext tasks. This article concentrates on integrating SSL frameworks with temporal EEG signals to achieve efficient representations and proposes a systematic survey of the SSL for EEG signals. In this article, (1) We introduce the concept and theory of self-supervised learning and typical SSL frameworks. (2) We provide a comprehensive survey of SSL for EEG analysis, including taxonomy, methodology, and technical details of the existing EEG-based SSL frameworks, and discuss the differences between these methods. (3) We investigate the adaptation of the SSL approach to various downstream tasks, including the task description and related benchmark datasets, and further explore its application in large-scale pre-trained foundation models for EEG signals. (4) Finally, we discuss the potential directions for future SSL-EEG research. © 2025 Copyright held by the owner/

author(s).

Date 2025**DOI** 10.1145/3736574**URL** <https://www.scopus.com/pages/publications/105011392024?origin=resultslist>**Archive** Scopus**Language** English**Volume** 57**Publisher** Association for Computing Machinery**Publication** ACM Computing Surveys**ISSN** 03600300 (ISSN)**Issue** 12**Journal Abbr** ACM Comput Surv**Date Added** 20/09/2026, 9:51:47 pm**Modified** 20/09/2026, 9:51:47 pm**Tags:**

Deep learning, Learning systems, Learning algorithms, Supervised learning, Benchmarking, Labelings, Learning frameworks, Self-supervised learning, contrastive learning, representation learning, Representation learning, Domain experts, Unlabeled samples, Biomedical signal processing, Electrophysiology, electroencephalogram, Electroencephalography, Automatic analysis, Bioelectric phenomena, Bioelectrical signals, Learning techniques, Noninvasive technique, Technical details

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SEM Image Processing Assisted by Deep Learning to Quantify Mesoporous γ -Alumina Spatial Heterogeneity and Its Predicted Impact on Mass Transfer

Item Type Journal Article**Author** A. Głowska**Author** E. Jolimaitre**Author** A. Hammoumi**Author** M. Moreaud**Author** L. Sorbier

Author C. de Faria Barros

Author V. Lefebvre

Author M.-O. Coppens

Abstract The pore network architecture of porous heterogeneous catalyst supports has a significant effect on the kinetics of mass transfer occurring within them. Therefore, characterizing and understanding structure-transport relationships is essential to guide new designs of heterogeneous catalysts with higher activity and selectivity and superior resistance to deactivation. This study combines classical characterization via N₂ adsorption and desorption and mercury porosimetry with advanced scanning electron microscopy (SEM) imaging and processing approaches to quantify the spatial heterogeneity of γ -alumina (γ -Al₂O₃), a catalyst support of great industrial relevance. Based on this, a model is proposed for the spatial organization of γ -Al₂O₃, containing alumina inclusions of different porosities with respect to the alumina matrix. Using original, advanced SEM image analysis techniques, including deep learning semantic segmentation and porosity measurement under gray-level calibration, the inclusion volume fraction and interphase porosity difference were identified and quantified as the key parameters that served as input for effective tortuosity factor predictions using effective medium theory (EMT)-based models. For the studied aluminas, spatial porosity heterogeneity impact on the effective tortuosity factor was found to be negligible, yet it was proven to become significant for an inclusion content of at least 30% and an interphase porosity difference of over 20%. The proposed methodology based on machine-learning-supported image analysis, in conjunction with other analytical techniques, is a general platform that should have a broader impact on porous materials characterization. © 2024 The Authors. Published by American Chemical Society

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Semantic Density: Uncertainty Quantification for Large Language Models through Confidence Measurement in Semantic Space

Item Type Conference Paper**Author** X. Qiu**Author** R. Miikkulainen**Editor** Globerson A.**Editor** Mackey L.**Editor** Belgrave D.**Editor** Fan A.**Editor** Paquet U.**Editor** Tomczak J.**Editor** Zhang C.

Abstract With the widespread application of Large Language Models (LLMs) to various domains, concerns regarding the trustworthiness of LLMs in safety-critical scenarios have been raised, due to their unpredictable tendency to hallucinate and generate misinformation. Existing LLMs do not have an inherent functionality to provide the users with an uncertainty/confidence metric for each response it generates, making it difficult to evaluate trustworthiness. Although several studies aim to develop uncertainty quantification methods for LLMs, they have fundamental limitations, such as being restricted to classification tasks, requiring additional training and data, considering only lexical instead of semantic information, and being prompt-wise but not response-wise. A new framework is proposed in this paper to address these issues. Semantic density extracts uncertainty/confidence information for each response from a probability distribution perspective in semantic space. It has no restriction on task types and is "off-the-shelf" for new models and tasks. Experiments on seven state-of-the-art LLMs, including the latest Llama 3 and Mixtral-8x22B models, on four free-form question-answering benchmarks demonstrate the superior performance and robustness of semantic density compared to prior approaches. © 2024 Neural information processing systems foundation. All rights reserved.

Date 2024**URL** <https://www.scopus.com/pages/publications/105000504286?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Adv. neural inf. proces. syst.

Volume 37**Publisher** Neural information processing systems foundation**ISBN** 10495258 (ISSN)**Proceedings Title** Adv. neural inf. proces. syst.**Conference Name** Advances in Neural Information Processing Systems**Date Added** 20/09/2026, 9:52:10 pm**Modified** 20/09/2026, 9:52:10 pm**Notes:**

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Semantic-Constraint Matching for transformer-based weakly supervised object localization

Item Type Journal Article**Author** Y. Cao**Author** Y. Su**Author** W. Wang**Author** Y. Liu**Author** Q. Wu**Author** Q. Wu

Abstract Weakly supervised object localization (WSOL) strives to localize objects with only image-level supervision. WSOL often faces challenges such as incomplete localization due to classifier bias and over-localization in real scenes where objects and backgrounds are strongly associated or structurally similar. While the latest Transformer-based methods effectively enhance localization performance by leveraging long-range feature dependencies, they may inadvertently amplify divergent background activation and remain susceptible to classification bias. To this end, we proposed a novel Semantic-Constraint Matching (SeCM) plug-in module tailored for transformer-based approaches. In detail, a local patch shuffle strategy is first introduced to disentangle partial contextual linkages, thereby creating image pairs. Then a semantic matching module extracts co-object knowledge from the primal-shuffled image pairs, drives the network to identify the association of foreground with semantic label to suppress divergent activation. Moreover, to alleviate incomplete localization and prevent excessive suppression of activation, we propose leveraging multi-modal class-specific textual representations to guide object localization by complementing intra-class priori diverse knowledge. Extensive experimental results conducted on CUB-200-2011 and ILSVRC datasets show that our method can achieve the new state-of-the-art performance. © 2024 Elsevier Ltd

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Semantic-Guided Multi-Vector Feature Representation for Few-Shot Image Classification

Item Type Conference Paper

Author X. Feng

Author L. Dong

Abstract Deep learning has achieved remarkable success across various fields, but its performance relies heavily on large-scale annotated data, limiting its application in real-world scenarios. Few-Shot Learning (FSL) provides an effective approach to address this data scarcity problem. Focusing on the key issues of feature extraction and feature representation in few-shot image classification tasks, this paper proposes a semantic-guided multi-vector feature representation method based on metric learning. In the feature extraction stage, an efficient Shuffle Attention (SA) mechanism is embedded into the ResNet-12 network to enhance the extraction of spatial and channel features from images. In terms of feature representation, semantic word vectors of categories are utilized to generate visual activation maps, which capture both the global and local features of the images. Furthermore, a base class prior knowledge transfer module is incorporated to transfer prior knowledge from base classes to novel classes, thereby enhancing the discriminability of novel class features. Experimental results on the

miniImageNet, CUB, and CIFAR-FS datasets demonstrate that the proposed model improves classification accuracy in both 5-way 1-shot and 5-way 5-shot tasks, with average increases of 1.25% and 1.11%, respectively. © 2026 IEEE.

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Large-scales, Deep learning, Performance, Feature extraction, Classification (of information), Features extraction, Learning systems, Semantics, Multi-modal, Image enhancement, Extraction, Image classification, Prior-knowledge, Few-shot learning, Multimodal, Knowledge management, Few-Shot Learning, Knowledge transfer, Image representation, Feature representation, Base class, Image Classification, Images classification, Metric learning, Metric Learning

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Semi-supervised Sentiment Analysis of Portuguese Tweets with Random Walk in Feature Sample Networks

Item Type Conference Paper

Author P. Gengo

Author F.A.N. Verri

Editor Cerri R.

Editor Prati R.C.

Abstract Nowadays, a huge amount of data is generated daily around the world and many machine learning tasks require labeled data, which sometimes is not available. Manual labeling such amount of data may consume a lot of time and resources. One way to overcome this limitation is to learn from both labeled and unlabeled data, which is known as semi-supervised learning. In this paper, we use a positive-unlabeled (PU) learning technique called Random Walk in Feature-Sample Networks (RWFSN) to perform semi-supervised sentiment analysis, which is an important machine learning that can be achieved by classifying the polarity of texts, in Brazilian Portuguese tweets. Although RWFSN reaches excellent performance in many PU learning problems, it has two major limitations when applied in our problem: it assumes that samples are long texts (many features) and that the class prior probabilities are known. We leverage the technique by augmenting the data representation in the feature space and by adding a validation set to better estimate the class priors. As a result, we identified unlabeled samples of the positive class with precision around at 70% in higher labeled ratio, but with high standard deviation, showing the impact of data variance in results. Moreover, given the properties of the RWFSN method, we provide interpretability of the results by pointing out the most relevant features of the task. © 2020, Springer Nature Switzerland AG.

Date 2020

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Machine-learning, Sentiment analysis, Learning algorithms, Supervised learning, Labeled data, Random processes, Learn+, Positive-unlabeled learning, Learning tasks, Manual labeling, Feature samples, Random walk, Random Walk, Semi-supervised classification, Semisupervised classification (SSC)

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EID: 2-s2.0-85094139624

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: P. Gengo; Data Science Team, Itaú Unibanco, São Paulo, Brazil; email: pedro.gengo.lourenco@gmail.com; Conference name: 9th Brazilian Conference on Intelligent Systems, BRACIS 2020; Conference date: 20 October 2020 through 23 October 2020; Conference code: 250329

Semi-supervised Bayesian classification by vector features with continuous and discrete components

Item Type Conference Paper

Author V.O. Vasilyev

Author A.V. Dobrovidov

Abstract A solution for the traditional Bayesian classification problem in non-traditional conditions is proposed, when the distributions and a priori probabilities of classes are unknown, but a trained sample from the zero class (labeled positive) and mixed sample (unlabeled) are available. Mixed sample will be employed in the learning to restore mixed distribution and as a test sample for constructed classifier. The case with vector features containing continuous and discrete components is considered. To restore unknown distributions nonparametric kernel techniques with data-driven bandwidth are used. A new algorithm for estimating the prior probability of zero class is given using positive labeled and unlabeled samples. This allows to find a good approximation of optimal threshold for the modified Bayesian classification algorithm. Numerical verification confirms the effectiveness of the proposed classification technique even in cases of strong overlapping of class distributions. © 2017 IEEE.

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Language English

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Classification (of information), Restoration, Approximation algorithms, Probability distributions, Bayesian classification, Class distributions, A-priori probabilities, Optimal threshold, Classification technique, Discrete components, Mixed distribution, Numerical verification

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Semi-supervised clinical text classification with Laplacian SVMs: An application to cancer case management

Item Type Journal Article**Author** V. Garla**Author** C. Taylor**Author** C. Brandt

Abstract Objective: To compare linear and Laplacian SVMs on a clinical text classification task; to evaluate the effect of unlabeled training data on Laplacian SVM performance. Background: The development of machine-learning based clinical text classifiers requires the creation of labeled training data, obtained via manual review by clinicians. Due to the effort and expense involved in labeling data, training data sets in the clinical domain are of limited size. In contrast, electronic medical record (EMR) systems contain hundreds of thousands of unlabeled notes that are not used by supervised machine learning approaches. Semi-supervised learning algorithms use both labeled and unlabeled data to train classifiers, and can outperform their supervised counterparts. Methods: We trained support vector machines (SVMs) and Laplacian SVMs on a training reference standard of 820 abdominal CT, MRI, and ultrasound reports labeled for the presence of potentially malignant liver lesions that require follow up (positive class prevalence 77%). The Laplacian SVM used 19,845 randomly sampled unlabeled notes in addition to the training reference standard. We evaluated SVMs and Laplacian SVMs on a test set of 520 labeled reports. Results: The Laplacian SVM trained on labeled and unlabeled radiology reports significantly outperformed supervised SVMs (Macro-F1 0.773 vs. 0.741, Sensitivity 0.943 vs. 0.911, Positive Predictive value 0.877 vs. 0.883). Performance improved with the number of labeled and unlabeled notes used to train the Laplacian SVM (pearson's $\rho=0.529$ for correlation between number of unlabeled notes and macro-F1 score). These results suggest that practical semi-supervised methods such as the Laplacian SVM can leverage the large, unlabeled corpora that reside within EMRs to improve clinical text classification. © 2013 Elsevier Inc.

Date 2013**DOI** 10.1016/j.jbi.2013.06.014**URL** <https://www.scopus.com/pages/publications/84883822654?origin=resultslist>**Archive** Scopus**Language** English**Volume** 46**Publisher** Academic Press Inc.**Pages** 869-875**Publication** Journal of Biomedical Informatics**ISSN** 15320464 (ISSN)

Issue 5**Journal Abbr** J. Biomed. Informatics**Date Added** 20/09/2026, 9:52:39 pm**Modified** 20/09/2026, 9:52:39 pm**Tags:**

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Semi-supervised deep learning for high-dimensional uncertainty quantification

Item Type Conference Paper**Author** Z. Wang**Author** M. Li

Abstract Conventional uncertainty quantification methods usually lacks the capability of dealing with high-dimensional problems due to the curse of dimensionality. This paper presents a semisupervised learning framework for dimension reduction and reliability analysis. An autoencoder is first adopted for mapping the high-dimensional space into a low-dimensional latent space, which contains a distinguishable failure surface. Then a deep feedforward neural network (DFN) is utilized to learn the mapping relationship and reconstruct the latent space, while the Gaussian process (GP) modeling technique is used to build the surrogate model of the transformed limit state function. During the training process of the

DFN, the discrepancy between the actual and reconstructed latent space is minimized through semisupervised learning for ensuring the accuracy. Both labeled and unlabeled samples are utilized for defining the loss function of the DFN. Evolutionary algorithm is adopted to train the DFN, then the Monte Carlo simulation method is used for uncertainty quantification and reliability analysis based on the proposed framework. The effectiveness is demonstrated through a mathematical example. Copyright © 2020 ASME.

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Semi-supervised intracranial aneurysm segmentation via reliable weight selection

Item Type Journal Article

Author W. Cao

Author X. Chen

Author J. Lv

Author L. Shao

Author W. Si

Abstract Early detection of intracranial aneurysms (IA) is crucial for risk reduction and treatment planning. Computed tomography angiography (CTA) has been widely applied in IA screening due to its rapid, cost-effective, and high-resolution capabilities. However, manual segmentation of IA from CTA images is labor-intensive and time-consuming, highlighting the urgent need for automated segmentation methods to improve diagnostic efficiency. This paper presents a reliable weight selection mean teacher framework for IA segmentation, leveraging semi-supervised learning to enhance network performance with limited labeled data. The framework consists of a student network and two teacher networks, while the teacher networks generate adversarial noise to guide the student network in learning. A novel reliable sample weight selection module employs ensemble learning to quantify prediction weights adaptively to minimize errors from individual models. Additionally, we integrate xLSTM module into the segmentation network to capture the contextual features between frames in the input data, while adversarial noise is introduced to bolster model robustness. Experiments show that our method improves Dice scores by at least 4.8% compared to the fully-supervised segmentation model baseline and at least 1.1% compared to other state-of-the-art semi-supervised methods under 20% labeled data. The source code and the dataset are available from <https://github.com/jinxiaokuang/RWS-MT>. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2024.

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Semi-supervised learning of class balance under class-prior change by distribution matching

Item Type Conference Paper

Author M.C. Du Plessis

Author M. Sugiyama

Abstract In real-world classification problems, the class balance in the training dataset does not necessarily reflect that of the test dataset, which can cause significant estimation bias. If the class ratio of the test dataset is known, instance re-weighting or resampling allows systematical bias correction. However, learning the class ratio of the test dataset is challenging when no labeled data is available from the test domain. In this paper, we propose to estimate the class ratio in the test dataset by matching probability distributions of training and test input data. We demonstrate the utility of the proposed approach through experiments. Copyright 2012 by the author(s)/owner(s).

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Language English

Extra Journal Abbreviation: Proc. Int. Conf. Mach. Learn., ICML

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Conference Name Proceedings of the 29th International Conference on Machine Learning, ICML 2012

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Export Date: 20 September 2026; Cited By: 38; Correspondence Address: M.C. Du Plessis; Department of Computer Science, Tokyo Institute of Technology, Tokyo, Japan; email: CHRISTO@SG.CS.TITECH.AC.JP; Conference name: 29th International Conference on Machine Learning, ICML 2012; Conference date: 26 June 2012 through 1 July 2012; Conference code: 93029

Semi-supervised learning of class balance under class-prior change by distribution matching

Item Type Journal Article**Author** M.C. Du Plessis**Author** M. Sugiyama

Abstract In real-world classification problems, the class balance in the training dataset does not necessarily reflect that of the test dataset, which can cause significant estimation bias. If the class ratio of the test dataset is known, instance re-weighting or resampling allows systematical bias correction. However, learning the class ratio of the test dataset is challenging when no labeled data is available from the test domain. In this paper, we propose to estimate the class ratio in the test dataset by matching probability distributions of training and test input data. We demonstrate the utility of the proposed approach through experiments. © 2013 Elsevier Ltd.

Date 2014**DOI** 10.1016/j.neunet.2013.11.010**URL** <https://www.scopus.com/pages/publications/84888799714?origin=resultslist>**Archive** Scopus**Language** English**Volume** 50**Publisher** Elsevier Ltd**Pages** 110-119**Publication** Neural Networks**ISSN** 08936080 (ISSN)**Journal Abbr** Neural Netw.**Date Added** 20/09/2026, 9:52:56 pm**Modified** 20/09/2026, 9:52:56 pm**Tags:**

Humans, Classification (of information), article, Supervised learning, Algorithms, Statistical tests, data analysis, Data Mining, Computer Simulation, Probability, learning algorithm, Classification, Probability distributions, Semi-supervised learning, information processing, probability, statistical analysis, analytical error, Training dataset, Learning, Bias correction, Density ratio, Selection bias, Distribution matching, selection bias, distribution matching, priority journal, Pattern Recognition, Automated, -divergence, class balance, class prior change, Class-prior change, F-divergence

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Export Date: 20 September 2026; Cited By: 72; Correspondence Address: M.C. Du Plessis; Tokyo Institute of Technology, Meguro-ku, Tokyo, 152-8552, 2-12-1-W8-74, O-okayama, Japan; email: christo@sg.cs.titech.ac.jp; CODEN: NNETE

Semi-Supervised Learning Using Gaussian Fields and Harmonic Functions

Item Type Conference Paper**Author** X. Zhu**Author** Z. Ghahramani**Author** J. Lafferty

Abstract An approach to semi-supervised learning is proposed that is based on a Gaussian random field model. Labeled and unlabeled data are represented as vertices in a weighted graph, with edge weights encoding the similarity between instances. The learning problem is then formulated in terms of a Gaussian random field on this graph, where the mean of the field is characterized in terms of harmonic functions, and is efficiently obtained using matrix methods or belief propagation. The resulting learning algorithms have intimate connections with random walks, electric networks, and spectral graph theory. We discuss methods to incorporate class priors and the predictions of classifiers obtained by supervised learning. We also propose a method of parameter learning by entropy minimization, and show the algorithm's ability to perform feature selection. Promising experimental results are presented for synthetic data, digit classification, and text classification tasks.

Date 2003**URL** <https://www.scopus.com/pages/publications/1942484430?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc. 20th Internat. Conf. Mach. Learn.**Volume** 2**ISBN** 1577351894 (ISBN)**Pages** 912-919**Proceedings Title** Proc. 20th Internat. Conf. Mach. Learn.**Conference Name** Proceedings, Twentieth International Conference on Machine Learning**Date Added** 20/09/2026, 9:53:06 pm**Modified** 20/09/2026, 9:53:06 pm**Tags:**

Learning systems, Entropy, Algorithms, Graph theory, Semi-supervised learning, Random processes, Computational methods, Problem solving, Harmonic analysis, Functions, Circuit theory, Energy minimization, Gaussian fields, Harmonic functions

Notes:

EID: 2-s2.0-1942484430

Export Date: 20 September 2026; Cited By: 3717; Correspondence Address: X. Zhu; School of Computer Science, Carnegie Mellon University, Pittsburgh, PA 15213, United States; email: ZHUXJ@CS.CMU.EDU; Conference name: Proceedings, Twentieth International Conference on Machine Learning; Conference date: 21 August 2003 through 24 August 2003; Conference code: 62778

Semi-supervised learning with local and global consistency

Item Type Journal Article

Author J. Gui

Author R. Hu

Author Z. Zhao

Author W. Jia

Abstract In a lot of practical machine learning applications, such as web page classification, protein shape classification, unlabelled instances are easy to obtain, but labelled instances are rather too expensive to get. Thus, recently, semi-supervised learning (SSL) methods including graph-based algorithms have attracted many interests from researchers. However, most of these algorithms used the Gaussian function to calculate weights of the edge of the graph. In this paper, we proposed a novel weight for graph-based semi-supervised algorithms. In this new algorithm, the label information is added from problem into SSL algorithm, and the geodesic distance is utilized instead of Euclidean distance to calculate the distance between two instances. Furthermore, class prior knowledge is also added from problem into the target function. In this paper, we focus on learning with local and global consistency. We found that the effect of class prior knowledge maybe different between under low-label rate and high-label rate. Experiments on two University of California Irvine (UCI) data sets and United States Postal Service handwritten digit recognition show that our proposed algorithm is really effective. © 2013, Taylor & Francis.

Date 2014

DOI 10.1080/00207160.2013.831082

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Language English

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Publication International Journal of Computer Mathematics

ISSN 00207160 (ISSN)

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Journal Abbr Int J Comput Math

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Tags:

Learning systems, Graph algorithms, Websites, Semi-supervised learning, Graphic methods, Character recognition, semi-supervised learning, Handwritten digit recognition, Machine learning applications, Label information, Geodesy, Prior knowledge, Geodesic distances, class prior knowledge, geodesic distance, label information, Semi-supervised algorithm, Semi-supervised learning (SSL), United States Postal Service

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Semi-supervised learning with local and global consistency by geodesic distance and sparse representation

Item Type Conference Paper

Author J. Gui

Author Z. Zhao

Author R. Hu

Author W. Jia

Abstract In many practical data mining applications, such as web categorization, key gene selection, etc., generally, unlabeled training examples are readily available, but labeled ones are fairly expensive to obtain. Therefore, in recent years, semi-supervised learning algorithms such as graph-based methods have attracted much attention. However, most of these traditional methods adopted a Gaussian function to calculate the edge weights of the graph. In this paper, a novel weight for semi-supervised graph-based methods is proposed. The new method adds the label information from problem into the target function, and adopts the geodesic distance rather than Euclidean distance as the measure of the difference between two data points when conducting the calculation. In addition, we also add class prior knowledge from problem into semi-supervised learning algorithm. Here we address the problem of learning with local and global consistency (LGC). It was found that the effect of class prior knowledge was probably different between under high label rate and low label rate. Furthermore, we integrate sparse representation (SR) in LGC algorithm. Experiments on a UCI data set show that our proposed method outperforms the original algorithms. © Springer-Verlag 2013.

Date 2013

DOI 10.1007/978-3-642-36669-7_16

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Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)
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Tags:

Learning algorithms, Supervised learning, Semi-supervised learning, Classification errors, Label information, Geodesy, Prior knowledge, Classification error, Class prior knowledge, Geodesic distance, Geodesic distances, Sparse representation

Notes:

EID: 2-s2.0-84892940870

Export Date: 20 September 2026; Cited By: 2; Conference name: 3rd Sino-Foreign-Interchange Workshop on Intelligent Science and Intelligent Data Engineering, IScIDE 2012; Conference date: 15 October 2012 through 17 October 2012; Conference code: 102166

Semi-supervised sufficient dimension reduction under class-prior change

Item Type Conference Paper
Author H. Kawakubo
Author M. Sugiyama
Abstract Sufficient dimension reduction (SDR) is a popular framework for supervised dimension reduction, aiming at reducing the dimensionality of input data while information on output data is maximally maintained. On the other hand, in many recent supervised classification learning tasks, it is conceivable that the balance of samples in each class varies between the training and testing phases. Such a phenomenon, referred to as class-prior change, causes existing SDR methods to perform undesirably particularly when the training data is highly imbalanced. In this paper, we extend the state-of-the-art SDR method called leastsquares gradients for dimension reduction (LSGDR) to be able to cope with such class-prior change under the semi-supervised learning setup where unlabeled test data are available in addition to labeled training data. Through experiments, we demonstrate the usefulness of our proposed method. © 2016 IEEE.
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Language English
Extra Journal Abbreviation: TAAI - Conf. Technol. Appl. Artif. Intell., Proc.

Publisher Institute of Electrical and Electronics Engineers Inc.

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Conference Name TAAI 2016 - 2016 Conference on Technologies and Applications of Artificial Intelligence, Proceedings

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Tags:

Artificial intelligence, Deep neural networks, Learning algorithms, Supervised learning, Data reduction, State of the art, Supervised classification, Training and testing, Dimension reduction, Semi-supervised learning, Labeled training data, Sufficient dimension reduction, Supervised dimension reductions

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EID: 2-s2.0-85017626087

Export Date: 20 September 2026; Cited By: 0; Conference name: 2016 Conference on Technologies and Applications of Artificial Intelligence, TAAI 2016; Conference date: 25 November 2016 through 27 November 2016; Conference code: 126910

Semiquantitative optic nerve grading scheme for determining axonal loss in experimental optic neuropathy

Item Type Journal Article

Author B.C. Chauhan

Author T.L. LeVatte

Author K.L. Garnier

Author F. Tremblay

Author I.-H. Pang

Author A.F. Clark

Author M.L. Archibald

Abstract PURPOSE. To describe and evaluate a semiquantitative optic nerve grading scheme for assessing axonal loss in endothelin (ET)-1-induced chronic optic neuropathy. METHODS. Optic nerve cross-sections from both eyes of 39 Brown Norway rats unilaterally treated with various concentrations of ET-1 or physiological saline solution via a surgically implanted osmotic minipump were processed for light and transmission electron microscopy (TEM). The optic nerve damage grade, between 0 (no damage) and 10 (total damage), was based on the number of zones of approximately equal damage and the mean percentage of damage within each zone. Grading was performed under light microscopy by three observers and compared with axonal survival determined with TEM using two quantification methods: the sampling method, in which ~10% of the section was counted, and the full-count

method, in which the whole section was counted ($n = 12$). Axonal survival was expressed as a ratio of axon counts in the experimental to control eye. Before these comparisons, the interand intraobserver agreement rates were determined in another group of 85 and 12 ET-1-treated animals, respectively. RESULTS. The interobserver κ was 0.66 (95% confidence interval [CI]: 0.58-0.74) for all eyes and 0.55 (95% CI: 0.43- 0.67) for the experimental eyes only. The intraobserver κ was 0.80, 0.81, and 0.80 for all 24 eyes and 0.60, 0.64, and 0.71 for experimental eyes only. The correlation between damage grade in the experimental eye and axonal survival using the TEM sampling method (Spearman's $\rho = -0.677$ for all animals and -0.827 for the subset of animals with full counts only) was lower than that with the full-count method (Spearman's $\rho = -0.926$). When axonal survival was less than 0.7, the sampling method always underestimated the extent of damage. CONCLUSIONS. The grading scheme had good inter- and intraobserver agreement, and high correlation with the TEM methods. It is a practical and time-saving method, requiring less than 1 minute per nerve and is an alternative to sampling methods that can yield significant errors. Copyright © Association for Research in Vision and Ophthalmology.

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Export Date: 20 September 2026; Cited By: 49; Correspondence Address: B.C. Chauhan; Department of Ophthalmology and Visual Sciences, Dalhousie University, Queen Elizabeth II Health Sciences Centre, Halifax, NS B3H 2Y9, Canada; email: bal@dal.ca; CODEN: IOVSD

Semmeldetector: Application of Machine Learning in Commercial Bakeries

Item Type Conference Paper

Author T.H. Schmitt

Author M. Bundscherer

Author T. Bocklet

Editor Arif Wani M.

Editor Boicu M.

Editor Sayed-Mouchaweh M.

Editor Abreu P.H.

Editor Gama J.

Abstract The Semmeldetector, is a machine learning application that utilizes object detection models to detect, classify and count baked goods in images. Our application allows commercial bakers to track unsold baked goods, which allows them to op-timize production and increase resource efficiency. We compiled a dataset comprising 1151 images that distinguishes between 18 different types of baked goods to train our detection models. To facilitate model training, we used a Copy-Paste augmentation pipeline to expand our dataset. We trained the state-of-the-art object detection model YOLOv8 on our detection task. We tested the impact of different training data, model scale, and online image augmentation pipelines on model performance. Our overall best performing model, achieved an AP_{0.5} of 89.1 % on our test set. Based on our results, we conclude that machine learning can be a valuable tool even for unforeseen industries like bakeries, even with very limited datasets. © 2023 IEEE.

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Tags:

Object recognition, Learning systems, machine learning, Machine-learning, Machine learning, Object detection, Objects detection, Detection models, Pipelines, YOLOv8, Industrial automation, object detection, Machine learning applications, Baked good, baked goods, food inspection, Food inspection, image composition, Image composition, industrial automation, Resource efficiencies

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Sensor placement optimization for shm systems under uncertainty

Item Type Conference Paper

Author R.F. Guratzsch

Author S. Mahadevan

Author C.L. Pettit

Author M.M. Derriso

Abstract Structural Health Monitoring (SHM) systems that report in real-time a flight vehicle's condition are central to meeting the goals of increasing flight vehicle safety and reliability, while reducing operating and maintenance costs. The structural response of flight vehicles is inherently random, requiring deterministic finite element analyses to be augmented with uncertainty quantification methods to compute the response statistics and structural damage probability. To detect damage with maximum probability, sensors must be placed optimally. This requires combining a probabilistic finite element method (FEM) with damage detection algorithms and optimization techniques. This study develops a methodology to integrate the above disciplines into a sensor placement optimization (SPO) methodology for SHM systems under uncertainty. To achieve this, the structural component under consideration is analyzed via FEM and uncertainty of model input quantities is included in the analysis via random processes and fields. In the next two steps probabilistic FEM analyses are performed to determine the model output variability and using these results, damage detection procedures such as feature extraction and state classification are applied to assess the current structural state of the component. Repeating these two steps using both healthy and damaged structural models helps quantify the reliability of a given sensor layout. Finally, SPO is achieved to maximize the reliability of damage detection. The sensor layout design of a thermal protection system component is used as a numerical example. Copyright © 2005 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved.

Date 2005

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Language English
Extra Journal Abbreviation: Collect Tech Pap AIAA ASME ASCE AHS Struct Struct Dyn Mater
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Conference Name Collection of Technical Papers - AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference
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Accident prevention, Damage detection, Condition monitoring, Sensors, Optimization, Aircraft, Finite element method, Sensor Placement Optimization (SPO), Structural Health Monitoring (SHM), Thermal protection

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Export Date: 20 September 2026; Cited By: 6; Conference name: 46th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference; Conference date: 18 April 2005 through 21 April 2005; Conference code: 66173; CODEN: CPSCD

Sensory-Driven Design Method for Personalizing Ride Comfort Tailored to Individual Vibration Perception

Item Type Conference Paper
Author H. Kikuchi
Author K. Inaba
Abstract This study aims to develop a design method that tailors the ride comfort and design variables of vehicle components according to individual differences in vibration perception. In conventional development, variations in vibration perception have been recognized; however, quantification methods remain undeveloped, preventing designs from being adapted to individual driver perceptions. The two unresolved problems include the uniformization of vibration perception in sensory performance modeling, which predicts sensory scores from vehicle vibrations, and design approaches that focus on minimizing vehicle vibrations without considering vibration perception. First, the authors' previous study quantified the existence of individual differences in vibration perception

through sensory scores obtained from ride simulator experiments involving 24 non-expert drivers using vibrations derived from a uniform vibration perception. Hierarchical clustering identified four perception groups; however, the reduced sample sizes resulting from the classification necessitated an uncertainty evaluation in the sensory performance modeling. This study formulated personalized sensory performance models for each group using an ensemble learning method with bootstrap resampling data to enable an uncertainty assessment. These models demonstrated sufficient prediction accuracy for sensory scores through cross-validation. Second, these models were integrated into a sensory-driven design method previously proposed by the authors to maximize sensory scores. This integrated method was validated by designing a unique electronically controlled damper system that yielded personalized solutions for ride comfort and design variables tailored to the vibration perception of each group. The practicality of the method lies in the personalization of vehicle performances with the potential for enhanced tailoring in future Software Defined Vehicles. © 2025 SAE International. All Rights Reserved.

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Language English
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Proceedings Title SAE Techni. Paper.
Conference Name SAE Technical Papers
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Learning systems, Uncertainty analysis, Vehicle performance, Design, Sensory analysis, Hierarchical clustering, Design method, Software design, Quantification methods, Design variables, Individual Differences, Performance Modeling, Ride comforts, Sensory perception, Sensory scores, Uniformization, Vehicle components, Vehicle vibrations, Vibrations (mechanical)

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EID: 2-s2.0-105008227022

Export Date: 20 September 2026; Cited By: 1; Conference name: 2025 SAE World Congress Experience, WCX 2025; Conference date: 8 April 2025 through 10 April 2025; Conference code: 209275

Sentiment Quantification of User-Generated Content

Item Type Journal Article

Author F. Sebastiani

Abstract Prevalence of c in set S : Percentage of items in S that belong to class c and also known as the “relative frequency” of c or the “prior probability” (or simply “prior”) of c Quantification: Estimation of the prevalence of each class $c \in C$ in a set S of unlabeled items (or estimation of the distribution of S across the classes in C), synonym of “supervised prevalence estimation” and “class prior estimation,” and also previously referred to as “counting.” Sentiment classification: A classification task whereby items (e.g., tweets, product reviews, comments, answers to open-ended questions) are classified based on the sentiment they convey (or opinion they express) about a certain entity or topic. It may take the form of binary classification (when the available classes are $C = \{\text{POSITIVE}, \text{NEGATIVE}, \dots\}$). © Springer Science+Business Media New York 2014.

Date 2018

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Classification (of information), Sentiment analysis, Classifieds, Frequency estimation, User-generated, Sentiment classification, Probability distributions, Classification tasks, Open-ended questions, C (programming language), Product reviews, Prior probability, Prevalence estimation, Relative frequencies, Estimating prevalence of sentiment class in user-generated content, Estimating prevalence of sentiment classes in user-generated content

Notes:

EID: 2-s2.0-105009575162

Export Date: 20 September 2026; Cited By: 1

Sequential Harmful Shift Detection Without Labels

Item Type Conference Paper

Author S.I. Amoukou

Author T. Bewley

Author S. Mishra

Author F. Lecue
Author D. Magazzeni
Author M. Veloso
Editor Globerson A.
Editor Mackey L.
Editor Belgrave D.
Editor Fan A.
Editor Paquet U.
Editor Tomczak J.
Editor Zhang C.

Abstract We introduce a novel approach for detecting distribution shifts that negatively impact the performance of machine learning models in continuous production environments, which requires no access to ground truth data labels. It builds upon the work of Podkopaev and Ramdas [2022], who address scenarios where labels are available for tracking model errors over time. Our solution extends this framework to work in the absence of labels, by employing a proxy for the true error. This proxy is derived using the predictions of a trained error estimator. Experiments show that our method has high power and false alarm control under various distribution shifts, including covariate and label shifts and natural shifts over geography and time. © 2024 Neural information processing systems foundation. All rights reserved.

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Conference Name Advances in Neural Information Processing Systems

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: S.I. Amoukou; J.P. Morgan AI Research, United States; email: salim.ibrahimamoukou@jpmorgan.com; Conference name: 38th Conference on Neural Information Processing Systems, NeurIPS 2024; Conference date: 9 December 2024 through 15 December 2024; Conference code: 207061

Sequential label shift detection in classification data: An application to dengue fever

Item Type Journal Article

Author C. Evans

Author M.G. Sell

Abstract Classifiers have been developed to help diagnose dengue fever in patients presenting with febrile symptoms. However, classifier predictions often rely on the assumption that new observations come from the same distribution as training data. If the population prevalence of dengue changes, as would happen with a dengue outbreak, it is important to raise an alarm as soon as possible, so that appropriate public health measures can be taken and also so that the classifier can be re-calibrated. In this paper, we consider the problem of detecting such a change in distribution in sequentially-observed, unlabeled classification data. We focus on label shift changes to the distribution, where the class priors shift but the class conditional distributions remain unchanged. We reduce this problem to the problem of detecting a change in the one-dimensional classifier scores, leading to simple nonparametric sequential changepoint detection procedures. Our procedures leverage classifier training data to estimate the detection statistic, and converge to their parametric counterparts in the size of the training data. In simulated outbreaks with real dengue data, we show that our method outperforms other detection procedures in this label shift setting. © 2024 Evans, G'Sell. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Sequential surrogate modeling for inverse design with conformal inference-based uncertainty quantification

Item Type Journal Article

Author H. Wu

Author Y. Xu

Abstract Inverse design poses challenges due to its nonlinear, high-dimensional, and often ill-posed nature. Machine learning has emerged as a promising alternative, offering cost-effective alternatives to computationally expensive models but can yield inconsistent or non-physical results without proper uncertainty quantification. This research introduces a novel framework that integrates uncertainty quantification into inverse design using sequential machine learning-based surrogate modeling. The framework operates through two key processes: a trainee model that identifies candidate design inputs expected to produce the desired outputs, and an assessor that applies conformal inference, guided by a user-defined confidence level, to evaluate and filter these candidates. By eliminating unreliable predictions, the framework delivers robust and accurate design solutions. Notably, the proposed approach is model-agnostic and does not rely on predefined prior distributions or model-specific likelihoods, distinguishing it from Bayesian-based uncertainty quantification methods. We validate the approach through one mathematical example and three engineering case studies. Results demonstrate that our method provides more robust and accurate solutions compared to conventional single-model approaches for inverse design. © 2025 Elsevier Ltd.

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Learning systems, Machine-learning, Machine learning, Uncertainty analysis, Cost effectiveness, Inverse problems, Bayesian networks, High-dimensional, Higher-dimensional, Uncertainty quantifications, Uncertainty quantification, Computer aided design, Conformal inference, Design automation, Design automations, Ill-posed natures, Inverse designs, Sequential surrogate modeling, Surrogate modeling

Notes:

EID: 2-s2.0-105020914080

Export Date: 20 September 2026; Cited By: 5; Correspondence Address: Y. Xu; University of Texas at Dallas, Richardson, United States; email: yanwen.xu@utdallas.edu; CODEN: RESSE

Seroprevalence for norovirus genogroup II, IV and VI in dogs

Item Type Journal Article

Author B. Di Martino

Author F. Di Profio

Author I. Melegari

Author V. Sarchese

Author I. Massirio

Author G. Palermo

Author G. Romito

Author E. Lorusso

Author G. Lanave

Author L. Bodnar

Author C. Buonavoglia

Author F. Marsilio

Author K.Y. Green

Author V. Martella

Abstract Molecular and serological data suggest that noroviruses (NoVs) might be transmitted between humans and domestic carnivores. In this study we screened an age-stratified collection of canine sera (n = 516) by using an ELISA assay based on virus-like particles (VLPs) of human NoVs GII.4 and GIV.1 and carnivore NoVs GIV.2 and GVI.2. Antibodies against GII.4 and GIV.1 human NoVs and GIV.2 and GVI.2 NoVs from carnivores were identified in dog sera (13.0%, 67/516) suggesting their exposure to homologous and heterologous NoVs. Analysis of the trends of age-class prevalence showed a gradual increase in the positive rate from 9.0% and 7.0%, in young dogs <1 year of age to 15.0% in dogs older than 12 years, for GII.4 and GVI.2 NoVs, respectively. A significant difference in the IgG distribution by age classes was observed for GIV.1 NoVs, with the highest rate of antibodies (7.0%) in the age

group <1 year and the lowest (1.0%) in the age-classes 7–9 ($P = 0.049$). High correlation between the reactivity to GII.4 and GVI.2 NoVs was observed, likely due to conserved epitopes in the capsid structure. © 2017 Elsevier B.V.

Date 2017

DOI 10.1016/j.vetmic.2017.03.006

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Article, human, Humans, nonhuman, controlled study, classification, age distribution, trend study, isolation and purification, genotype, animal, Animals, virology, animal experiment, protein structure, blood, immunology, Antibodies, Viral, Seroepidemiologic Studies, seroepidemiology, seroprevalence, virus antibody, enzyme linked immunosorbent assay, Enzyme-Linked Immunosorbent Assay, Age Distribution, groups by age, Antibodies, Caliciviridae Infections, calicivirus infection, capsid protein, Capsid Proteins, carnivore, dog, dog disease, Dog Diseases, Dogs, epitope, gastroenteritis, Gastroenteritis, Genotypes GII.4, GIV.1, GIV.2 and GVI.2, Norovirus, Noroviruses (NoVs), screening test, serum, veterinary, virus capsid, virus like agent, virus particle, virus strain

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Export Date: 20 September 2026; Cited By: 9; Funding details: National Institute of Allergy and Infectious Diseases, NIAID, (ZIAAI000897); National Institute of Allergy and Infectious Diseases, NIAID; Correspondence Address: B. Di Martino; Faculty of Veterinary Medicine, Università degli Studi di Teramo, Italy; email: bdimartino@unite.it; CODEN: VMICD

Shaping the Anthropocene: Understanding and Evaluating Human Impact on the Global Ecosystem

Item Type Journal Article

Author Y. Ding

Abstract In the Anthropocene epoch, human activities have extensively altered the global ecological system, leading to substantial shifts in species distribution and survival patterns due to industrialization and global warming. Quantifying these environmental impacts is essential; for example, the 2023 Global Footprint Network estimates humanity's ecological footprint to be 1.71 Earths, underscoring the urgent need for sustainable solutions. This study aims to identify and summarize primary quantification methods for assessing human impacts on the environment. Using the Web of Science database, this paper analyzes 1000 relevant papers to identify four primary methodologies: the ecological footprint, Human Appropriation of Net Primary Production (HANPP), the Planetary Boundary framework, and the Living Planet Index (LPI). This paper provides a detailed evaluation of each approach, assessing their strengths and limitations in measuring the effects of human activities on the global ecosystem. Additionally, this study explores the potential of machine learning models to enhance future predictions. By analyzing and comparing these methods, this paper offers insights that are critical for informing policies and practices aimed at reducing human-induced environmental degradation. © 2025 Wiley Periodicals LLC.

Date 2025

DOI 10.1002/tqem.70104

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Language English

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machine learning, environmental impact, global warming, Anthropocene, anthropogenic effect, ecological challenges, ecological footprint, global ecological system, human impact, industrialization

Notes:

EID: 2-s2.0-105006681027

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Y. Ding; School of Geography and the Environment, University of Oxford, Oxford, United Kingdom; email: armstrongding@163.com

Shapley Value Approximation Based on k-Additive Games

Item Type Conference Paper

Author G.D. Pelegrina

Author P. Kolpaczki
Author E. Hüllermeier
Editor Koenig S.
Editor Jenkins C.
Editor Taylor M.E.

Abstract The Shapley value is the prevalent solution for fair division problems in which a payout is to be divided among multiple agents. By adopting a game-theoretic view, the idea of fair division and the Shapley value can also be used in machine learning to quantify the individual contribution of features or data points to the performance of a predictive model. Despite its popularity and axiomatic justification, the Shapley value suffers from a computational complexity that scales exponentially with the number of entities involved, and hence requires approximation methods for its reliable estimation. We propose SVakADD, a novel approximation method that fits a k-additive surrogate game. By taking advantage of k-additivity, we are able to elicit the exact Shapley values of the surrogate game and then use these values as estimates for the original fair division problem. The efficacy of our method is evaluated empirically and compared to competing methods. © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

Date 2026

DOI 10.1609/aaai.v40i20.38770

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. AAAI Conf. Artif. Intell.

Volume 40

Publisher Association for the Advancement of Artificial Intelligence

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Proceedings Title Proc. AAAI Conf. Artif. Intell.

Conference Name Proceedings of the AAAI Conference on Artificial Intelligence

Issue 20

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Performance, Learning systems, Machine-learning, Game theory, Multi agent systems, Shapley value, Approximation theory, Additives, Additive games, Agents, Approximation methods, Datapoints, Fair divisions, Game-theoretic, Multiple agents, Value approximation

Notes:

EID: 2-s2.0-105034254098

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Shapley-based fault variable identification in MSPC

Item Type Journal Article

Author J. Lim

Author J. Kim

Author S. Lee

Author S. Lee

Abstract In multivariate statistical process control, Hotelling's (Formula presented.) control chart (HT chart) is a powerful tool for detecting shifts in a process's overall mean. However, when a process falls out of control, the HT chart only signals an issue without identifying the specific variables responsible for the deviation. To address this limitation, traditional approaches decompose the overall (Formula presented.) statistic into (Formula presented.) statistics of individual variables or higher-order interaction terms. While informative, this approach becomes computationally challenging and less effective as the number of variables increases. In this study, we propose a new procedure based on the Shapley value, widely applied in machine learning to quantify variable importance in predictions. In this approach, we consider the (Formula presented.) statistic as a value function for each subset of variables and employ Shapley sampling and KernelSHAP to estimate Shapley values in high dimensional data. Most importantly, we develop a procedure to identify variable causing the out of control signal using Shapley values, demonstrating that Shapley values serve as an e-value. We numerically compare the performance of this new Shapley-based procedure with two existing procedures: the Mason-Tracy-Young and the adaptive step-down procedures. Finally, we apply these procedures to detect changes in the monthly temperature of Seoul after the year 2000. © 2026 Taylor & Francis Group, LLC.

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Journal Abbr Qual Eng

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Tags:

Shapley values, e-value, fault variable identification, Hotelling's chart

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EID: 2-s2.0-105033245705

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Lee; Department of Statistics and Data Science, Dankook University, Gyeonggi, South Korea; email: silee@dankook.ac.kr; CODEN: QUENE

Shift-Aware Stacking Ensemble with Conformal Prediction for Brain Stroke Detection

Item Type Conference Paper

Author S. Batchala

Author T. Kamble

Author V. Katkam

Author B. Mallikarjuna

Abstract Machine learning models for clinical risk prediction often lose accuracy when used in new environments because of domain shift, where the training (source) and testing (target) data differ. This paper introduces a shift-aware framework for brain stroke detection that adapts a model trained on a large cardiovascular disease dataset to a stroke-specific population. The proposed multistage approach handles both covariate and label shifts through careful data preprocessing, feature alignment, and domain shift measurement using a domain classifier. The study reduces covariate shift through importance weighting, and trains stacking ensemble of Gradient Boosting models such as LightGBM, XGBoost, CatBoost, and Random Forest, on the weighted data in order to increase predictive accuracy. The ensemble's results are then measured with Isotonic Regression and filtered using conformal prediction to provide reliable, distribution free uncertainty estimates. On the target stroke dataset, the method achieves a ROC AUC of 0.972 and a PR AUC of 0.976, introducing effective knowledge transfer over domains. By combining calibration and conformal prediction, this framework gives trustworthy confidence measures, making it suitable for clinical use where reliability is essential. © 2025 IEEE.

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Archive Scopus

Language English

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Conference Name IC-DECON 2025 - 2025 International Conference on Data, Energy and Communication Network, Proceedings

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Classification (of information), Learning systems, Forecasting, Uncertainty analysis, Brain, Large datasets, Population statistics, Diseases, Stacking ensemble, Stacking Ensemble, Stackings, Knowledge management, Domain Knowledge, Knowledge transfer, Domain adaptation, Domain Adaptation, Uncertainty quantifications, Covariate shifts, Uncertainty Quantification, Conformal predictions, Importance weighting, Covariate Shift, Conformal Prediction, Brain strokes, Importance Weighting, Stroke detection, Stroke prediction, Stroke Prediction

Notes:

EID: 2-s2.0-105037324120

Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 International Conference on Data, Energy and Communication Network, IC-DECON 2025; Conference date: 12 December 2025 through 13 December 2025; Conference code: 221930

Shifts: A Dataset of Real Distributional Shift Across Multiple Large-Scale Tasks

Item Type	Conference Paper
Author	A. Malinin
Author	N. Band
Author	Y. Gal
Author	M.J.F. Gales
Author	A. Ganshin
Author	G. Chesnokov
Author	A. Noskov
Author	A. Ploskonosov
Author	L. Prokhorenkova
Author	I. Provilkov
Author	V. Raina
Author	V. Raina
Author	D. Roginskiy
Author	M. Shmatova
Author	P. Tigas
Author	B. Yangel
Editor	Vanschoren J.
Editor	Yeung S.

Abstract There has been significant research done on developing methods for improving robustness to distributional shift and uncertainty estimation. In contrast, only limited work has examined developing standard datasets and benchmarks for assessing these approaches. Additionally, most work on uncertainty estimation and robustness has developed new techniques based on small-scale regression or image classification tasks. However, many tasks of practical interest have different modalities, such as tabular data, audio, text, or sensor data, which offer significant challenges involving regression and discrete or continuous structured prediction. Thus, given the current state of the field, a standardized large-scale dataset of tasks across a range of modalities affected by distributional shifts is necessary. This will enable researchers to meaningfully evaluate the plethora of recently developed uncertainty quantification methods, as well as assessment criteria and state-of-the-art baselines. In this work, we propose the Shifts Dataset for evaluation of uncertainty estimates and robustness to distributional shift. The dataset, which has been collected from industrial sources and services, is composed of three tasks, with each corresponding to a particular data modality: tabular weather prediction, machine translation, and self-driving car (SDC) vehicle motion prediction. All of these data modalities and tasks are affected by real, “in-the-wild” distributional shifts and pose interesting challenges with respect to uncertainty estimation. In this work we provide a description of the dataset and baseline results for all tasks. © 2021 Neural information processing systems foundation. All rights reserved.

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Archive Scopus

Language English

Extra Journal Abbreviation: Adv. neural inf. proces. syst.

Publisher Neural information processing systems foundation

ISBN 10495258 (ISSN)

Proceedings Title Adv. neural inf. proces. syst.

Conference Name Advances in Neural Information Processing Systems

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Tags:

Large-scales, Spatio-temporal data, Classification tasks, Text data, Uncertainty estimation, Small scale, Shift-and, Shift estimations, Images classification, Audio sensors, Tabular data

Notes:

EID: 2-s2.0-105000269818

Export Date: 20 September 2026; Cited By: 35; Funding details: Intel Corporation; Turing Institute; CUED; Conference name: 35th Conference on Neural Information Processing Systems - Track on Datasets and Benchmarks, NeurIPS Datasets and Benchmarks 2021; Conference date: 6 December 2021 through 14 December 2021; Conference code: 207502

Sicca syndrome in the same family – The importance of anti-Ro60, anti-Ro52 and anti-La antibody profiling in Sjögren's syndrome

Item Type Journal Article

Author A. Koszarny

Author M. Majdan

Abstract Sicca syndrome, which is typical for Sjögren's syndrome (SS), both primary (pSS) and secondary (sSS), is relatively often comorbid with other autoimmune diseases. The current classification criteria for SS published in 2016 include only anti-SSA (anti-Ro) autoantibody, while the latest literature proposes that anti-Ro60/anti-Ro52 autoantibody profiles should be used instead, as these two types of antibodies correlate with specific clinical symptoms and laboratory test findings. The paper presents the case of a 41-year-old woman suffering from pSS and her three daughters, who were under observation for rheumatic disorders due to sicca symptoms, especially pSS, as well as a discussion on separate determination of anti-Ro60 and anti-Ro52 autoantibodies based on current literature in the PubMed database. When testing with antinuclear antibodies, the Ro60+Ro52+La+ autoantibody profile most closely matches for pSS. Further research is needed to find marker antibodies for SS and quantification methods. © 2021 Termedia Publishing House Ltd.. All rights reserved.

Date 2021

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Tags:

adult, female, human, adolescent, Review, arthralgia, systemic lupus erythematosus, antinuclear antibody, follow up, clinical article, unclassified drug, connective tissue disease, immunofluorescence, Anti-Ro52 antibody, Anti-Ro60 antibody, Anti-SSA antibody, autoimmune thyroiditis, case report, colecalciferol, conjunctival hyperemia, dry eye, eye drops, hydroxychloroquine, hyperchloremia, La antibody, livedo reticularis, morning stiffness, Primary Sjögren's syndrome, Ro antibody, Ro52 antibody, Ro60 antibody, Schirmer test, Sjogren syndrome, undifferentiated connective tissue disease, vasculitis, vitamin D deficiency, vitamin supplementation, Wegener granulomatosis

Notes:

EID: 2-s2.0-85114399414

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: A. Koszarny; Department of Rheumatology and Connective Tissue Diseases, Medical University of Lublin, Lublin, 8 Jaczewskiego St., 20-954, Poland; email: arekkoszarny@wp.pl; CODEN: RMTOA

Side-Channel Analysis-Based Model Extraction on Intelligent CPS: An Information Theory Perspective

Item Type Conference Paper

Author Q. Pan

Author J. Wu

Author X. Lin

Author J. Li

Editor Zheng J.

Editor Liu X.

Editor Luan T.H.

Editor Jayaraman P.P.

Editor Dai H.

Editor Mitra K.

Editor Qin K.

Editor Ranjan R.

Editor Wen S.

Abstract The intelligent cyber-physical system (CPS) has been applied in various fields, covering multiple critical infras-tructures and human daily life support areas. CPS Security is a major concern and of critical importance, especially the security of the intelligent control component. Side-channel analysis (SCA) is the common threat exploiting the weaknesses in system operation to extract information of the intelligent CPS. However, existing literature lacks the systematic theo-retical analysis of the side-channel attacks on the intelligent CPS, without the ability to quantify and measure the leaked information. To address these issues, we propose the SCA-based model extraction attack on intelligent CPS. First, we design an efficient and novel SCA-based model extraction framework, including the threat model, hierarchical attack process, and the multiple micro-space parallel search enabled weight extraction algorithm. Secondly, an information theory-empowered analy-sis model for side-channel attacks on intelligent CPS is built. We propose a mutual information-based quantification method and derive the capacity of side-channel attacks on intelligent CPS, formulating the amount of information leakage through side channels. Thirdly, we develop the theoretical bounds of the leaked information over multiple attack queries based on the data processing inequality and properties of entropy. These convergence bounds provide theoretical means to estimate the amount of information leaked. Finally, experimental evaluation, including real-world experiments, demonstrates the effective-ness of the proposed SCA-based model extraction algorithm and the information theory-based analysis method in intelligent CPS. © 2021 IEEE.

Date 2021

DOI 10.1109/iThings-GreenCom-CPSCoM-SmartData-Cybermatics53846.2021.00050

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Conference Name Proceedings - IEEE Congress on Cybermatics: 2021 IEEE International Conferences on Internet of Things, iThings 2021, IEEE Green Computing and Communications, GreenCom 2021, IEEE Cyber, Physical and Social Computing, CPSCom 2021 and IEEE Smart Data, SmartData 2021

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Cybersecurity, Machine learning, Cyber Physical System, Embedded systems, Extraction, Extract informations, Information theory, information theory, Amount of information, Control components, Daily-life supports, intelligent CPS, Intelligent cybe-physical system, Model extraction, Side channel attack, Side-channel analyse-based model extraction, Side-channel analysis, side-channel analysis-based model extraction, Side-channel attacks, Systems operation

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Simplified and representative bacterial community of maize roots

Item Type Journal Article

Author B. Niu

Author J.N. Paulson

Author X. Zheng

Author R. Kolter

Abstract Plant-associated microbes are important for the growth and health of their hosts. As a result of numerous prior studies, we know that host genotypes and abiotic factors influence the composition of plant microbiomes. However, the high complexity of these communities challenges detailed studies to define experimentally the mechanisms underlying the dynamics of community assembly and the beneficial effects of such microbiomes on plant hosts. In this work, from the distinctive microbiota assembled by maize roots, through host-mediated selection, we obtained a greatly simplified synthetic bacterial community consisting of seven strains (*Enterobacter cloacae*, *Stenotrophomonas maltophilia*, *Ochrobactrum pituitosum*, *Herbaspirillum frisingense*, *Pseudomonas putida*, *Curtobacterium pusillum*, and *Chryseobacterium indologenes*) representing three of the four most dominant phyla found in maize roots. By using a selective culture-dependent method to track the abundance of each strain, we investigated the role that each plays in community assembly on roots of axenic maize seedlings. Only the removal of *E. cloacae* led to the complete loss of the community, and *C. pusillum* took over. This result suggests that *E. cloacae* plays the role of keystone species in this model ecosystem. In planta and in vitro, this model community inhibited the phytopathogenic fungus *Fusarium verticillioides*, indicating a clear benefit to the host. Thus, combined with the selective culture-dependent quantification method, our synthetic seven-species community representing the root microbiome has the potential to serve as a useful system to explore how bacterial interspecies interactions affect root microbiome assembly and to dissect the beneficial effects of the root microbiota on hosts under laboratory conditions in the future.

Date 2017

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Archive Scopus

Language English

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Article, nonhuman, controlled study, classification, Bacteria, bacterium, genetics, isolation and purification, microbial community, Microbiota, microflora, microbiology, plant root, Plant Roots, RNA 16S, Soil Microbiology, phylogeny, Phylogeny, maize, RNA sequence, species composition, priority journal, bacterial colonization, bacterial strain, Biological control, Chryseobacterium indologenes, Community assembly, community structure, Curtobacterium pusillum, Enterobacter cloacae, fungal colonization, Fusarium verticillioides, Herbaspirillum, Herbaspirillum frisingense, keystone species, Maize, Ochrobactrum, Ochrobactrum pituitosum, population abundance, Pseudomonas putida, seedling, Stenotrophomonas maltophilia, Synthetic community, Zea mays

Notes:

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Export Date: 20 September 2026; Cited By: 675; Funding details: National Cancer Institute, NCI, (U01CA190234); National Cancer Institute, NCI; Correspondence Address: R. Kolter; Department of Microbiology and Immunobiology, Harvard Medical School, Boston, 02115, United States; email: roberto_kolter@hms.harvard.edu; CODEN: PNASA

SimPro: A Simple Probabilistic Framework Towards Realistic Long-Tailed Semi-Supervised Learning

Item Type Conference Paper

Author C. Du

Author Y. Han

Author G. Huang

Editor Salakhutdinov R.

Editor Kolter Z.

Editor Heller K.

Editor Weller A.

Editor Oliver N.

Editor Scarlett J.

Editor Berkenkamp F.

Abstract Recent advancements in semi-supervised learning have focused on a more realistic yet challenging task: addressing imbalances in labeled data while the class distribution of unlabeled data remains both unknown and potentially mismatched. Current approaches in this sphere often presuppose rigid assumptions regarding the class distribution of unlabeled data, thereby limiting the adaptability of models to only certain distribution ranges. In this study, we propose a novel approach, introducing a highly adaptable framework, designated as SimPro, which does not rely on any predefined assumptions about the distribution of unlabeled data. Our framework, grounded in a probabilistic model, innovatively refines the expectation-maximization (EM) algorithm by explicitly decoupling the modeling of conditional and marginal class distributions. This separation facilitates a closed-form solution for class distribution estimation during the maximization phase, leading to the formulation of a Bayes classifier.

The Bayes classifier, in turn, enhances the quality of pseudo-labels in the expectation phase. Remarkably, the SimPro framework not only comes with theoretical guarantees but also is straightforward to implement. Moreover, we introduce two novel class distributions broadening the scope of the evaluation. Our method showcases consistent state-of-the-art performance across diverse benchmarks and data distribution scenarios. Our code is available at <https://github.com/LeapLabTHU/SimPro>. Copyright 2024 by the author(s)

Date 2024

URL <https://www.scopus.com/pages/publications/85203821284?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Mach. Learn. Res.

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Conference Name Proceedings of Machine Learning Research

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'current, Contrastive Learning, Self-supervised learning, Labeled data, Semi-supervised learning, Unlabeled data, Simple++, Expectation maximization algorithm, Class distributions, Probabilistic models, Probabilistic framework, Bayes Classifier, Expectations maximization algorithms

Notes:

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siMS score- method for quantification of metabolic syndrome, confirms co-founding factors of metabolic syndrome

Item Type Journal Article

Author V. Dimitrijevic-Sreckovic

Author H. Petrovic

Author D. Dobrosavljevic

Author E. Colak

Author N. Ivanovic

Author D. Gostiljac

Author S. Ilic

Author D. Nikolic

Author J. Gacic

Author I. Soldatovic

Abstract Background: Adipose tissue is a dynamic endocrine organ, a highly active metabolic tissue, and an important source of cytokines. Inflammatory factors play an important role in visceral obesity associated with insulin resistance (IR), metabolic syndrome (MS), hypertension, non-alcoholic fatty liver disease (NAFLD), diabetes mellitus type 2 (DM2), endothelial dysfunction (ED) and atherosclerosis. Objectives: To examine correlation of siMS score, as a quantification method for metabolic syndrome (MS), with insulin resistance, glucoregulation parameters, as with other co-founding factors of MS, inflammation and thrombosis factors, microalbuminuria, uric acid, fatty liver index (FLI) and homocysteine. Methods: The study included 451 obese individuals with pre-metabolic syndrome (pre-MS) and MS (age 16–75, body mass index (BMI) > 25kg/m²) classified into two groups: I-age 10–30 (167 patients); II-age 31–75 (284 patients). International Diabetes Federation (IDF) classification was applied for diagnosing metabolic syndrome. Patients with less than three criteria indicated below were considered pre-metabolic syndrome. siMS risk score was used. Results: siMS score increased with age: I- 3.03 ± 0.87 , II- 3.27 ± 0.90 . siMS score correlated with associated factors of MS: hyperinsulinemia and IR, ALT, gama-GT, FLI, uric acid in both groups and CRP ($p < 0.01$) in group I. Correlations in II group: siMS score with PAI-1 ($p = 0.01$), microalbuminuria ($p = 0.006$), homocysteine ($p = 0.076$). Conclusion: Correlation of siMS score with HOMA-IR confirmed that hyperinsulinism and insulin resistance are in the basis of MS. Correlation of siMS score with parameters of NAFLD, CRP, PAI-1, uric acid, microalbuminuria and homocysteine indicates that they are significant co-founding factors of MS. Correlation of siMS score with PAI-1, microalbuminuria, homocysteine, indicates higher risk for progression of endothelial dysfunction and atherosclerosis with age. Copyright © 2023 Dimitrijevic-Sreckovic, Petrovic, Dobrosavljevic, Colak, Ivanovic, Gostiljac, Ilic, Nikolic, Gacic and Soldatovic.

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adult, Article, female, human, male, middle aged, adolescent, aged, correlation analysis, controlled study, major clinical study, alanine aminotransferase, inflammation, non insulin dependent diabetes mellitus, obesity, body weight, scoring system, limit of quantitation, spectrophotometry, body mass, hypertension, high density lipoprotein cholesterol, low density lipoprotein cholesterol, triacylglycerol, adipose tissue, diastolic blood pressure, systolic blood pressure, tumor necrosis factor, vascularization, insulin, insulin resistance, waist circumference, oral glucose tolerance test, C reactive protein, urea, uric acid, aspartate aminotransferase, creatinine, abdominal obesity, anthropometry, atherosclerosis, blood glucose monitoring, body height, cofounding factor, creatinine clearance, cytokine, endothelial dysfunction, fatty liver index, gamma glutamyltransferase, glycoregulation, glucoregulation parameters, glucose blood level, homeostasis model assessment, homocysteine, hyperinsulinemia, iliac bone, immune nephelometric method, impaired fasting glucose, impaired glucose tolerance, insulinemia, interleukin 6, International Diabetes Federation classification, lipid peroxidation, liver function, metabolic syndrome, metabolic syndrome X, microalbuminuria, non-alcoholic fatty liver disease, nonalcoholic fatty liver, oxidative stress, plasminogen activator inhibitor 1, plasminogen substrate essay, quantification method, siMS score, thrombosis factor

Notes:

EID: 2-s2.0-85146903022

Export Date: 20 September 2026; Cited By: 9; Correspondence Address: V. Dimitrijevic-Sreckovic; University Clinical Center of Serbia, Clinic of Endocrinology, Diabetes and Metabolic Diseases, Belgrade, Serbia; email: vesnadsendo@gmail.com

Simultaneous determination of 19 flavonoids in commercial trollflowers by using high-performance liquid chromatography and classification of samples by hierarchical clustering analysis

Item Type Journal Article

Author Z. Song

Author Y. Hashi

Author H. Sun

Author Y. Liang

Author Y. Lan

Author H. Wang

Author S. Chen

Abstract The flowers of Trollius species, named Jin Lianhua in Chinese, are widely used traditional Chinese herbs with vital biological activity that has been used for several decades in China to treat upper respiratory infections, pharyngitis, tonsillitis, and bronchitis. We developed a rapid and reliable method for simultaneous quantitative analysis of 19 flavonoids in trollflowers by using high-performance liquid

chromatography (HPLC). Chromatography was performed on Inertsil ODS-3 C18 column, with gradient elution methanol-acetonitrile-water with 0.02% (v/v) formic acid. Content determination was used to evaluate the quality of commercial trollflowers from different regions in China, while three *Trollius* species (*Trollius chinensis* Bunge, *Trollius ledebouri* Reichb, *Trollius buddae* Schipcz) were explicitly distinguished by using hierarchical clustering analysis. The linearity, precision, accuracy, limit of detection, and limit of quantification were validated for the quantification method, which proved sensitive, accurate and reproducible indicating that the proposed approach was applicable for the routine analysis and quality control of trollflowers. © 2013 Published by Elsevier B.V. All rights reserved.

Date 2013

DOI 10.1016/j.fitote.2013.09.006

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Cluster Analysis, cluster analysis, article, quantitative analysis, Chinese medicine, limit of quantitation, HPLC, phytochemistry, high performance liquid chromatography, limit of detection, flavonoid, Flavonoids, unclassified drug, flower, Flowers, Chromatography, High Pressure Liquid, priority journal, Drugs, Chinese Herbal, Hierarchical clustering analysis, drug identification, 2" beta levo arabinopyranosylvitexin, 2" o (2''' methylbutyryl)orientin, 2" o (2"methylbutyryl)vitexin, 2" o (3''' methoxylcaffeoyl)vitexin, 2" o (3''',4''' dimethoxylbenzoyl)vitexin, 2" o (3''',4'''dimethoxylbenzoyl) orientin, 2" o malonylvitexin, 4' methoxyl 2" o (2''' methylbutyryl)orientin, 4' methoxyl 2" o (2''' methylbutyryl)vitexin, 6" o acetyl orientin, 6" o acetyl vitexin, 6" o beta dextro glucosyl orientin, acacetin 7 o rutinoside, drug determination, drug structure, formic acid, hyperin, orientin, quercetin 3 o arabinosylglucoside, quercetin3 o xylosylglucoside, Ranunculaceae, *Trollius buddae*, *Trollius chinensis*, *Trollius ledebouri*, *Trollius* species, vitexin, vitexin 2" o alpha dextro xylopyranoside

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Simultaneous multi-source and multi-temporal land cover classification using a compound maximum likelihood classifier

Item Type Conference Paper

Author M.S. Reis

Author L.V. Dutra

Author M.I.S. Escada

Abstract The most widely used change detection method is to classify remote sensing images independently for each date, and stack them to form a class sequence vector. However, impossible transitions within the sequences might occur and errors might be accumulated. To solve this, we propose a novel algorithm called Compound Maximum Likelihood (CML), based on the Maximum Likelihood classifier (ML). In CML information from all images is used jointly by considering the a priori probability of each class sequence. The algorithm was tested for Synthetic Aperture Radar and optical images classification in a study area in Pará state, within the Brazilian Amazon. CML presented either similar or very improved accuracy index values over ML land cover classifications. © 2017 National Institute for Space Research INPE. All Rights Reserved.

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Conference Name Proceedings of the Brazilian Symposium on GeoInformatics

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Remote sensing, Remote sensing images, Synthetic aperture radar, Image classification, Maximum likelihood, A-priori probabilities, Land cover classification, Change detection, Novel algorithm, Multi-temporal, Brazilian Amazon, Geometrical optics, Maximum likelihood classifiers

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Single aflatoxin contaminated corn kernel analysis with fluorescence hyperspectral image

Item Type Conference Paper**Author** H. Yao**Author** Z. Hruska**Author** R. Kincaid**Author** A. Ononye**Author** R.L. Brown**Author** T.E. Cleveland

Abstract Aflatoxins are toxic secondary metabolites of the fungi *Aspergillus flavus* and *Aspergillus parasiticus*, among others. Aflatoxin contaminated corn is toxic to domestic animals when ingested in feed and is a known carcinogen associated with liver and lung cancer in humans. Consequently, aflatoxin levels in food and feed are regulated by the Food and Drug Administration (FDA) in the US, allowing 20 ppb (parts per billion) limits in food and 100 ppb in feed for interstate commerce. Currently, aflatoxin detection and quantification methods are based on analytical tests including thin-layer chromatography (TCL) and high performance liquid chromatography (HPLC). These analytical tests require the destruction of samples, and are costly and time consuming. Thus, the ability to detect aflatoxin in a rapid, nondestructive way is crucial to the grain industry, particularly to corn industry. Hyperspectral imaging technology offers a non-invasive approach toward screening for food safety inspection and quality control based on its spectral signature. The focus of this paper is to classify aflatoxin contaminated single corn kernels using fluorescence hyperspectral imagery. Field inoculated corn kernels were used in the study. Contaminated and control kernels under long wavelength ultraviolet excitation were imaged using a visible near-infrared (VNIR) hyperspectral camera. The imaged kernels were chemically analyzed to provide reference information for image analysis. This paper describes a procedure to process corn kernels located in different images for statistical training and classification. Two classification algorithms, Maximum Likelihood and Binary Encoding, were used to classify each corn kernel into "control" or "contaminated" through pixel classification. The Binary Encoding approach had a slightly better performance with accuracy equals to 87% or 88% when 20 ppb or 100 ppb was used as classification threshold, respectively. © 2010 Copyright SPIE - The International Society for Optical Engineering.

Date 2010**DOI** 10.1117/12.850327

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Commerce, Agriculture, Health, Remote sensing, Inspection, Image analysis, Encoding (symbols), Quantification methods, High performance liquid chromatography, Food safety, Grain (agricultural product), Animals, Metabolites, Non destructive, Secondary metabolites, image classification, Maximum likelihood, Fluorescence, Lung Cancer, Non-invasive, Contamination, Aflatoxins, Aspergillus, Chromatography, Hyper-spectral images, aflatoxin, Analytical tests, Aspergillus flavus, Aspergillus parasiticus, Binary encodings, corn, Corn kernels, Domestic animals, fluorescence hyperspectral image, Food and drug administration, Food safety inspection, Grain industry, Hyper-spectral cameras, Hyperspectral imagery, Hyperspectral Imaging, Interstate commerce, Long wavelength, Parts per billion, Pixel classification, Spectral signature, Two classification, Ultra violet excitation, Visible near-infrared

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Single Deterministic Neural Network with Hierarchical Gaussian Mixture Model for Uncertainty Quantification

Item Type Conference Paper
Author C. Ji
Author D. Gong
Editor Yu S.
Editor Zhang J.
Editor Zhang Z.

Editor Tan T.
Editor Yuen P.C.
Editor Guo Y.
Editor Han J.
Editor Lai J.

Abstract We propose a method to train a deterministic deep network for uncertainty quantification (UQ) with a single forward pass. Traditional Monte Carlo or ensemble based UQ methods largely leverage the variation of neural network weights to introduce uncertainty. We propose a hierarchical Gaussian mixture model (GMM) based nonlinear classifier to shape the extracted feature more flexibly and express the uncertainty by the entropy of the predicted posterior distribution. We perform large-scale training with this hierarchical GMM based loss function and introduce a natural gradient descent algorithm to update the parameters of the hierarchical GMM. With a single deterministic neural network, our uncertainty quantification approach performs well when training and testing on large datasets. We show competitive performance scores on several benchmark datasets and the out-of-distribution detection task on notable challenging dataset pairs such as CIFAR-10 vs. STL10/SVHN, and CIFAR100 vs. STL10/SVHN. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Deep neural networks, Model-based OPC, Uncertainty analysis, Gaussian distribution, Benchmarking, Neural-networks, Uncertainty, Large dataset, Gradient methods, Monte Carlo methods, Quantification methods, Uncertainty quantifications, Uncertainty quantification, Gaussian Mixture Model, Deterministics, Well testing, Deterministic deep neural network, Hierarchical gaussian mixture model, Hierarchical Gaussian mixture model, Network weights

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: C. Ji; Kuang-Chi Institute of Advanced Technology, Shenzhen, China; email: chunlin.ji@kuang-chi.org; Conference name: 5th Chinese Conference on Pattern Recognition and Computer Vision, PRCV 2022; Conference date: 4 November 2022 through 7 November 2022; Conference code: 285979

Single-step extraction & plasmonic sensing of sulphur in fuels using machine learning analysis of RGB images

Item Type Journal Article

Author D. Gallina

Author M.C. Valeriano

Author V.H. Paschoal

Author M.B. Mamián-López

Abstract Background: Mercaptans are compounds of great concern because they are one of the main sources of sulphur in fuels and can reduce engine performance. Diesel-biodiesel blends have partially replaced diesel fuel as part of efforts to promote greener energy. In Brazil, the current legislation permits up to 10 ppm of sulphur in diesel. Considering the impact of sulphur compounds produced within combustion engines, careful monitoring of fuels is essential to ensure a greener future, and methodologies to easily detect sulphur are needed, especially if they can be adapted for in situ measurements and are suitable for upcoming regulatory changes. Results: In this study, gold nanoparticles embedded in glycerol were used to extract, preconcentrate, and detect sulphur in biodiesel samples, with a mercaptan as sulphur source. The color changes due to nanoparticle aggregation were measured and processed using a smartphone app and machine learning to quantify sulphur. We demonstrate that this approach is suitable for determining this species in biodiesel at levels as low as 1.39 ppm, with the AuNPs@GLY-to-fuel ratio being easily adjustable, achieving lower limits of detection. Also, through SERS spectroscopy, we have demonstrated that the working principle behind the detection is the strong interaction between the thiol moiety and gold. This methodology can be easily customized, offering an interesting advantage from a practical perspective, as sulphur limit values vary worldwide and are influenced by the energy transition, which is a priority for public policies. Significance: The primary feature of our system is that it combines extraction, preconcentration, and sensing in a single step, eliminating the need for sample preparation or conditioning, and bringing the analysis to the field. The herein-developed sensor is low-cost, user-friendly, and can be customized and adapted to probe sulphur in commercial diesel, as verified here by SERS spectroscopy. © 2025 Elsevier B.V.

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SketchML: Accelerating distributed machine learning with data sketches

Item Type Conference Paper

Author J. Jiang

Author F. Fu

Author T. Yang

Author B. Cui

Editor Das G.

Editor Jermaine C.

Editor Eldawy A.

Editor Bernstein P.

Abstract To address the challenge of explosive big data, distributed machine learning (ML) has drawn the interests of many researchers. Since many distributed ML algorithms trained by stochastic gradient descent (SGD) involve communicating gradients through the network, it is important to compress the transferred gradient. A category of low-precision algorithms can significantly reduce the size

of gradients, at the expense of some precision loss. However, existing low-precision methods are not suitable for many cases where the gradients are sparse and nonuniformly distributed. In this paper, we study is there a compression method that can efficiently handle a sparse and nonuniform gradient consisting of key-value pairs? Our first contribution is a sketch based method that compresses the gradient values. Sketch is a class of algorithms using a probabilistic data structure to approximate the distribution of input data. We design a quantile-bucket quantification method that uses a quantile sketch to sort gradient values into buckets and encodes them with the bucket indexes. To further compress the bucket indexes, our second contribution is a sketch algorithm, namely MinMaxSketch. MinMaxSketch builds a set of hash tables and solves hash collisions with a MinMax strategy. The third contribution of this paper is a delta-binary encoding method that calculates the increment of the gradient keys and stores them with fewer bytes. We also theoretically discuss the correctness and the error bound of three proposed methods. To the best of our knowledge, this is the first effort combining data sketch with ML. We implement a prototype system in a real cluster of our industrial partner Tencent Inc., and show that our method is up to $10\times$ faster than existing methods. © 2018 Association for Computing Machinery.

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Sleep staging using cardiorespiratory signals

Item Type Journal Article

Author S.J. Redmond

Author P. De Chazal

Author C. O'Brien

Author S. Ryan

Author W.T. McNicholas

Author C. Heneghan

Abstract Question of study: We recently investigated the possibility of obtaining simplified Sleep- Wake-REM sleep stage information from subjects being assessed for Obstructive Sleep Apnea Syndrome (OSAS), using only electrocardiogram and respiration signals. The utility of such a system may be limited somewhat by the presence of OSAS in the patient group (in various degrees of severity). This study examines the effectiveness of such a system when applied to a subject group in which Sleep Disordered Breathing (SDB) is absent. Patients and methods: The study examined a database of 31 male subjects (Age = 42.0 ± 7.4 years, BMI = 30.7 ± 3.0 kg/m²). There was no significant presence of SDB in any of the subjects (AHI = 1.4 ± 1.2 events/h). A full polysomnography recording was obtained for each subject, including EEG, submental EMG and EOG for sleep staging. An automated sleep-staging system based solely on a single electrocardiogram signal and an inductance plethysmogram estimate of respiratory effort was developed. Features providing useful discrimination of sleep states were derived and the performance of both linear and quadratic discriminant classifiers were compared in correctly labeling 30-second epochs. The use of a time-dependent a priori probability in the classifier models was also investigated. Results: The best performance obtained was achieved by a linear discriminant classifier model using a time-dependent a priori probability. For a 3-class (W, S, R) system an agreement of $\kappa = 0.45$ was seen, which increases to $\kappa = 0.57$ when a simplified 2-class (W, S/R) system is considered. This corresponds to an epoch sleep-wake classification accuracy of 89%. Conclusions: Cardiorespiratory signals can provide sleep-wake staging accuracy which is comparable to actigraphy. Classification accuracy is not significantly altered by the presence or absence of sleep disturbed breathing. Cardiorespiratory-based sleep staging may be a useful addition to home sleep apnea monitoring systems. © 2007 Steinkopff-Verlag.

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SMOTE: Synthetic minority over-sampling technique

Item Type Journal Article

Author N.V. Chawla

Author K.W. Bowyer

Author L.O. Hall

Author W.P. Kegelmeyer

Abstract An approach to the construction of classifiers from unbalanced datasets is described. A dataset is imbalanced if the classification categories are not approximately equally represented. Often real-world data sets are predominately composed of "normal" examples with only a small percentage of "abnormal" or "interesting" examples. It is also the case that the cost of misclassifying an abnormal (interesting) example as a normal example is often much higher than the cost of the reverse error. Under-sampling of the majority (normal) class has been proposed as a good means of increasing the sensitivity of a classifier to the minority class. This paper shows that a combination of our method of over-sampling the minority (abnormal) class and under-sampling the majority (normal) class can achieve better classifier performance (in ROC space) than only under-sampling the majority class. This paper also shows that a combination of our method of over-sampling the minority class and under-sampling the majority class can achieve better classifier performance (in ROC space) than varying the loss ratios in Ripper or class priors in Naive Bayes. Our method of over-sampling the minority class involves creating synthetic minority class examples. Experiments are performed using C4.5, Ripper and a Naive Bayes classifier. The method is evaluated using the area under the Receiver Operating Characteristic curve (AUC) and the ROC convex hull strategy. © 2002 AI Access Foundation and Morgan Kaufmann Publishers. All rights reserved.

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Export Date: 20 September 2026; Cited By: 30446; Correspondence Address: N.V. Chawla; Department of Computer Science and Engineering, ENB 118, University of South Florida, Tampa, FL 33620-5399, 4202 E. Fowler Ave, United States; email: CHAWLA@CSEE.USF.EDU; CODEN: JAIRF

Softmin discrete minimax classifier for imbalanced classes and prior probability shifts

Item Type Journal Article

Author C. Gilet

Author M. Guyomard

Author S. Destercke

Author L. Fillatre

Abstract This paper proposes a new approach for dealing with imbalanced classes and prior probability shifts in supervised classification tasks. Coupled with any feature space partitioning method, our criterion aims to compute an almost-Bayesian randomized equalizer classifier for which the maxima of the class-conditional risks are minimized. Our approach belongs to the historically well-studied field of randomized minimax criteria. Our new criterion can be considered as a self-sufficient classifier, or can be easily coupled with any pretrained Convolutional Neural Networks and Decision Trees to address the issues of imbalanced classes and prior probability shifts. Numerical experiments compare our criterion to several state-of-the-art algorithms and show the relevance of our approach when it is necessary to well classify the minority classes and to equalize the risks per class. Experiments on the CIFAR-100 database show that our criterion scales well when the number of classes is large. © 2023, The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature.

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: C. Gilet; Université de Technologie de Compiègne, CNRS, Heudiasyc Laboratory, Compiègne, 60200, France; email: cyprien.gilet@hds.utc.fr; L. Fillatre; Université Côte d'Azur, CNRS, I3S Laboratory, Sophia-Antipolis, 06900, France; email: lionel.fillatre@i3s.unice.fr; CODEN: MALEE

SoLar: Sinkhorn Label Refinery for Imbalanced Partial-Label Learning

Item Type Conference Paper

Author H. Wang

Author M. Xia

Author Y. Li

Author Y. Mao

Author L. Feng

Author G. Chen

Author J. Zhao

Editor Koyejo S.

Editor Mohamed S.

Editor Agarwal A.

Editor Belgrave D.

Editor Cho K.

Editor Oh A.

Abstract Partial-label learning (PLL) is a peculiar weakly-supervised learning task where the training samples are generally associated with a set of candidate labels instead of single ground truth. While a variety of label disambiguation methods have been proposed in this domain, they normally assume a class-balanced scenario that may not hold in many real-world applications. Empirically, we observe degenerated performance of the prior methods when facing the combinatorial challenge from the long-tailed distribution and partial-labeling. In this work, we first identify the major reasons that the prior work failed. We subsequently propose SoLar, a novel Optimal Transport-based framework that allows to refine the disambiguated labels towards matching the marginal class prior distribution. SoLar additionally incorporates a new and systematic mechanism for estimating the long-tailed class prior distribution under the PLL setup. Through extensive experiments, SoLar exhibits substantially superior results on standardized benchmarks compared to the previous state-of-the-art PLL methods. Code and data are available at: <https://github.com/hbzju/SoLar>. © 2022 Neural information processing systems foundation. All rights reserved.

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Performance, Real-world, Labelings, Ground truth, Training sample, Prior distribution, Learning tasks, Long-tailed distributions, Weakly supervised learning, Disambiguation method

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SPARSE JOINT SHIFT IN MULTINOMIAL CLASSIFICATION

Item Type Journal Article

Author D. Tasche

Abstract Sparse joint shift (SJS) was recently proposed as a tractable model for general dataset shift which may cause changes to the marginal distributions of features and labels as well as the posterior probabilities and the class-conditional feature distributions. Fitting SJS for a target dataset without label observations may produce valid predictions of labels and estimates of class prior probabilities. We present new results on the transmission of SJS from sets of features to larger sets of features, a conditional correction formula for the class posterior probabilities under the target distribution, identifiability of SJS, and the relationship between SJS and covariate shift. In addition, we point out inconsistencies in the algorithms which were proposed for estimating the characteristics of SJS, as they could hamper the search for optimal solutions, and suggest potential improvements. © 2024 American Institute of Mathematical Sciences. All rights reserved.

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label shift, prior probability shift, class prior estimation, covariate shift, dataset shift, distribution shift, posterior correction, Sparse joint shift

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Export Date: 20 September 2026; Cited By: 1

Spatial appreciation of the floor plan complexity in housing price appraisal

Item Type Journal Article

Author S. Bae

Author S. An
Author G. Choi
Author K. Ahn

Abstract This study explores the relationship between structural feature of apartment interiors and housing prices across four different regions. This study first constructs floor plan complexity indices based on Shannon entropy and Delentropy and compares both entropy measures in seizing structural complexity of images. Subsequently, we adopt sophisticated instruments to scrutinize the linear and nonlinear effects of floor plan complexity on housing prices. A series of hedonic price models and machine learning models with the SHapley Additive exPlanation are utilized in capturing the impacts and importance of floor plan complexity indices on housing prices. This study exhibits three notable findings: (1) Delentropy can serve as a useful tool for capturing the structural complexity of floor plan images due to its capability of considering the spatial distribution of pixels; (2) floor plan complexity is significantly and positively associated with housing prices across all the surveyed areas; and (3) the importance of floor plan complexity varies upon local contexts. Our study proposes a quantification method that fully represents the images into the numerical data to analyze the relationship between floor plans and housing prices. This methodological approach serves as a significant analytical tool for site selection, enabling more informed decision-making through comprehensive market analysis in the context of real estate development. From a long-term perspective, local governments can support efficient management for the sustainable development of the real estate market. © 2025 Elsevier Ltd.

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Notes:

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Spatial context learning approach to automatic segmentation of pleural effusion in chest computed tomography images

Item Type Conference Paper

Author A. Mansoor

Author R. Casas

Author M.G. Linguraru

Editor Tourassi G.D.

Editor Armato S.G.

Abstract Pleural effusion is an abnormal collection of fluid within the pleural cavity. Excessive accumulation of pleural fluid is an important bio-marker for various illnesses, including congestive heart failure, pneumonia, metastatic cancer, and pulmonary embolism. Quantification of pleural effusion can be indicative of the progression of disease as well as the effectiveness of any treatment being administered. Quantification, however, is challenging due to unpredictable amounts and density of fluid, complex topology of the pleural cavity, and the similarity in texture and intensity of pleural fluid to the surrounding tissues in computed tomography (CT) scans. Herein, we present an automated method for the segmentation of pleural effusion in CT scans based on spatial context information. The method consists of two stages: first, a probabilistic pleural effusion map is created using multi-atlas segmentation. The probabilistic map assigns a priori probabilities to the presence of pleural uid at every location in the CT scan. Second, a statistical pattern classification approach is designed to annotate pleural regions using local descriptors based on a priori probabilities, geometrical, and spatial features. Thirty seven CT scans from a diverse patient population containing confirmed cases of minimal to severe amounts of pleural effusion were used to validate the proposed segmentation method. An average Dice coefficient of 0.82685 and Hausdorff distance of 16.2155 mm was obtained. © 2016 SPIE.

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Image segmentation, Medical imaging, Segmentation, Diagnosis, Diseases, Computer aided diagnosis, Computerized tomography, Quantification, Computed tomography scan, Computed tomography images, Pleural effusion, Chest computed tomography (CT), Congestive heart failures, Pulmonary diseases, Spatial context, Spatial context learning, Statistical pattern classification, Tomography

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: A. Mansoor; Sheikh Zayed Institute for Pediatric Surgical Innovation, Childrens National Medical Center, Washington, United States; email: awais.mansoor@gmail.com; Conference name: Medical Imaging 2016: Computer-Aided Diagnosis; Conference date: 28 February 2016 through 2 March 2016; Conference code: 123696

Spatial delineation of the compound flood transition zone using deep learning

Item Type Journal Article

Author F. Yarveysi

Author F.G. Diaz

Author H. Moftakhari

Author H. Moradkhani

Abstract Coastal and hydrologic floods are distinct yet interconnected phenomena, driven by oceanic and terrestrial processes, respectively. Their interaction—known as compound flooding—occurs when storm surge, heavy precipitation, and river flow coincide, significantly amplifying flood impacts in coastal riverine regions. These interactions give rise to a transition zone, where coastal and hydrologic flood processes converge, resulting in complex, prolonged inundation that is challenging to predict using traditional hydrodynamic models. Accurately delineating this zone is essential for improving flood risk assessment and mitigation strategies. In this study, we employ deep learning to quantify the relative contributions of terrestrial hydrologic and coastal flood drivers, enabling spatial delineation of the transition zone within Galveston Bay in Texas. This data-driven approach addresses the limitations of conventional models and supports more effective flood-resilience planning for vulnerable coastal communities. Our results reveal spatial patterns of flood driver dominance, with storm tide influencing coastal zones and river flow playing a greater role inland. The use of SHapley Additive exPlanations (SHAP) enables the delineation of a transition zone where no single driver dominates, underscoring the importance of compound flood modeling in such areas. This framework offers a scalable and interpretable solution for identifying high-risk zones, enhancing the precision of flood risk assessments, and informing targeted mitigation efforts in coastal regions. © 2025

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Risk assessment, risk assessment, Deep learning, United States, Learning systems, machine learning, Machine-learning, Machine Learning, Stream flow, Floods, Flood control, Rivers, Coastal flood, Coastal floods, Coastal transition zone, coastal zone, Coastal zones, flood, Flood risk assessments, Floodings, Galveston Bay, Hydrologic modeling, Hydrologic models, hydrological modeling, River flow, spatial data, Spatial delineation, Storm surges, Storms, Texas, Transition zones

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Spatial Leakage in Classifying NASA FIRMS Thermal Anomalies as Wildfire Incidents: A Leakage-Controlled Evaluation of Radiometric, Temporal, and Spatiotemporal Features

Item Type Journal Article

Author A. Soltan

Author A. González-Martínez

Abstract NASA's Fire Information for Resource Management System (FIRMS) provides near-real-time thermal anomaly detections from VIIRS, but not all detections correspond to wildfire incidents: industrial heat, agricultural burning, and sensor artifacts produce false alarms that contribute to alert fatigue for emergency-management analysts. We study whether contextual machine learning (ML) features improve wildfire-incident classification from FIRMS detections, and—more importantly—whether reported gains survive leakage-controlled evaluation. We construct a labeled dataset by matching 521,395 VIIRS SNPP detections across CONUS in 2024 to 3766 NIFC

2024 wildfire perimeters, yielding 131,771 (25.3%) wildfire-matched and 389,624 candidate non-wildfire detections spanning 1067 distinct wildfire incidents. We benchmark five operational baselines and six classifiers under four validation regimes (random, event-aware, 5° spatial-block, and temporal holdout) with and without raw geographic coordinates. A naive random split inflates LightGBM to $F1 = 0.985$, but a leakage-controlled event-aware split reduces it to $F1 = 0.767$, and a spatial-block holdout to $F1 = 0.627$. Feature attribution shows geographic coordinates account for 88.9% of model gain—the summed share of LightGBM’s total split-gain attributed to the three coordinate features within the full-feature model; removing coordinates improves spatial-block generalization from $F1 = 0.627$ to 0.818, demonstrating that raw coordinates drive memorization of where 2024 fires occurred rather than transferable discrimination. We further show that spatiotemporal clustering must be causal: a model using full-partition clustering appears strong ($F1 = 0.908$) but leaks future detections, whereas a properly causal trailing-window version ties plain LightGBM in-distribution ($F1 = 0.762$). Combining causal clustering with no raw coordinates is the most robust configuration under spatial transfer (spatial-block $F1 = 0.868$ vs. 0.627 for the coordinate model). Bootstrap 95% confidence intervals show these gaps far exceed statistical uncertainty, and sensitivity analyses show the conclusions are robust to the spatial-block size and to the clustering-window choice. Under natural class prevalence (14%), precision falls to 0.69, and results are sensitive to the labeling buffer. All ML models nonetheless far exceed FIRMS high-confidence thresholding ($F1 = 0.128$). We argue that spatial leakage—not raw accuracy—is the central methodological issue for FIRMS wildfire-incident classification, and recommend coordinate-free, causal spatiotemporal-clustering features evaluated under spatial holdout. The system is intended as an analyst-prioritization decision-support layer, not autonomous incident confirmation. © 2026 by the authors.

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Spatial mapping of the DNA adducts in cancer

Item Type Journal Article

Author K.L. Krieger

Author E.K. Mann

Author K.J. Lee

Author E. Bolterstein

Author D. Jebakumar

Author M.M. Ittmann

Author V.L. Dal Zotto

Author M. Shaban

Author A. Sreekumar

Author N.R. Gassman

Abstract DNA adducts and strand breaks are induced by various exogenous and endogenous agents. Accumulation of DNA damage is implicated in many disease processes, including cancer, aging, and neurodegeneration. The continuous acquisition of DNA damage from exogenous and endogenous stressors coupled with defects in DNA repair pathways contribute to the accumulation of DNA damage within the genome and genomic instability. While mutational burden offers some insight into the level of DNA damage a cell may have experienced and subsequently repaired, it does not quantify DNA adducts and strand breaks. Mutational burden also infers the identity of the DNA damage. With advances in DNA adduct detection and quantification methods, there is an opportunity to identify DNA adducts driving mutagenesis and correlate with a known exposome. However, most DNA adduct detection methods require isolation or separation of the DNA and its adducts from the context of the nuclei. Mass spectrometry, comet assays, and other techniques precisely quantify lesion types but lose the nuclear context and even tissue context of the DNA damage. The growth in spatial analysis technologies offers a novel opportunity to leverage DNA damage detection with nuclear and tissue context. However, we lack a wealth of techniques capable of detecting DNA damage in situ. Here, we review the limited existing in situ DNA damage detection methods and examine their potential to offer spatial analysis of DNA adducts in tumors or other tissues. We also offer a perspective on the need for spatial analysis of DNA damage in situ and highlight Repair Assisted Damage Detection (RADD) as an in situ DNA adduct technique with the potential to integrate with spatial analysis and the challenges to be addressed. © 2023 The Authors

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Article, human, Humans, machine learning, Damage detection, nonhuman, deep learning, genetics, protein, spatial analysis, malignant neoplasm, DNA, assay, staining, neoplasm, antibody, fluorescence, protein expression, oxidative stress, adductomics, cancer tissue, Digital spatial profiling, DNA adduct, DNA adducts, DNA Adducts, DNA damage, DNA Damage, DNA ligase, DNA repair, DNA Repair, Drosophila, hydroxyurea, mutagenesis, Mutagenesis, mutant, Neoplasms, ovary tumor, programmed death 1 ligand 1, Spatial analysis, Spatial DNA adductomics, stroma, wild type

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Spatial structure characteristics detecting of landform based on improved 3D Lacunarity model

Item Type Journal Article

Author Y. Tao

Author G. Tang

Author J. Strobl

Abstract The spatial structure characteristics of landform are the foundation of geomorphologic classification and recognition. This paper proposed a new method on quantifying spatial structure characteristics of terrain surface based on improved 3D Lacunarity model. Lacunarity curve and its numerical integration are used in this model to improve traditional classification result that different morphological types may share the close value of indexes based on global statistical analysis. Experiments at four test areas with different landform types show that improved 3D Lacunarity model can effectively distinguish different morphological types per texture analysis. Higher sensitivity in distinguishing the tiny differences of texture characteristics of terrain surface shows that the quantification method by 3D Lacunarity model and its numerical integration presented in this paper could contribute to improving the accuracy of landform classifications and relative studies. © 2012 Science Press,

Northeast Institute of Geography and Agroecology, CAS and Springer-Verlag Berlin Heidelberg.

Date 2012

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remote sensing, spatial analysis, terrain, geomorphology, 3D Lacunarity model, digital elevation model, Digital elevation model (DEM), landform, Landform, Spatial pattern, Terrain texture

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Spatial variation of waterborne Escherichia coli - Implications for routine water quality monitoring

Item Type Journal Article

Author R.S. Quilliam

Author K. Clements

Author C. Duce

Author S.B. Cottrill

Author S.K. Malham

Author D.L. Jones

Abstract Escherichia coli are often used as faecal indicator bacteria (FIB) to provide a measure of microbial pollution in recreational and shellfish harvesting waters. However, although model forecasts for predicting the concentrations of FIB in surface waters are becoming more robust, they suffer from an inconsistency in quantification methods and an understanding of the spatial variation of FIB within a water course.

The aim of this study was to investigate the transverse spatial variation in *E. coli* numbers (as an indicator of faecal pollution) across the estuary of the River Conwy, UK. Water samples were collected from four transverse transects across the estuary. Spatial variation of *E. coli* was significantly different from one side of the river to the other, although was not correlated with depth or the physiochemical properties of the water. Subsequently, microbial water quality classifications on the two opposite banks suggested very different levels of pollution coming down the river. This work has shown that the side of the river that routine water monitoring samples are taken from can make a significant difference to the classification of microbial water quality. This has important implications for sampling strategies and the use of microbial source tracking (MST) techniques. © IWA Publishing 2011.

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Export Date: 20 September 2026; Cited By: 32; Correspondence Address: R.S. Quilliam; School of Environment, Natural Resources and Geography, Bangor University, Bangor, Gwynedd LL57 2UW, United Kingdom; email: r.s.quilliam@bangor.ac.uk

Spatio-temporal contextual approaches to mapping land cover change based on remote sensing imagery

Item Type Conference Paper

Author J. Zhang

Author J. Zhang

Author Y. Tang

Author J. You

Abstract This paper explores data-driven methods for quantifying and incorporating spatio-temporal contextual information in the mapping of land cover change. In remote sensing, area classes of land cover are typically mapped via statistical manipulation of feature-space measurement, e.g., reflectance data, and other ancillary data. Contextual information has been known to have the potential of increasing the accuracy of land cover classification and change detection, on the ground that land cover often exhibits spatial and temporal correlations and, as such, should be properly accommodated. In Bayesian methods, a priori probabilities of class occurrences can be considered as contextual information, which are combined with class-conditional probability densities to arrive at discriminant decisions with minimized misclassification. These prior probabilities may be made to vary locally to honor variability in the strengths of spatial dependence in class occurrences. For deriving local prior joint probabilities in land cover co-occurrences over time, a modified Expectation and Maximization (EM) algorithm was developed, in which a local window size can be adjusted in the light of spatial dependences inferred from class probability densities computed from spectral data. Empirical studies were performed using bi-temporal Landsat TM image subsets in Wuhan, which confirmed the comparative benefits of incorporating localized prior probabilities in land cover change detection. © 2009 SPIE.

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Probability density function, Classification (of information), Signal detection, Remote sensing, Bayesian networks, Probability, Mapping, Contextual information, Land cover, Bayesian classification, Landforms, Prior probability, Space optics, Prior probabilities, Geo-statistics, Geostatistics

Notes:

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Spatio-Temporal Multi-Task Learning for Cardiac MRI Left Ventricle Quantification

Item Type Journal Article

Author S. Vesal

Author M. Gu

Author A. Maier

Author N. Ravikumar

Abstract Quantitative assessment of cardiac left ventricle (LV) morphology is essential to assess cardiac function and improve the diagnosis of different cardiovascular diseases. In current clinical practice, LV quantification depends on the measurement of myocardial shape indices, which is usually achieved by manual contouring of the endo- and epicardial. However, this process subjected to inter and intra-observer variability, and it is a time-consuming and tedious task. In this article, we propose a spatio-temporal multi-task learning approach to obtain a complete set of measurements quantifying cardiac LV morphology, regional-wall thickness (RWT), and additionally detecting the cardiac phase cycle (systole and diastole) for a given 3D Cine-magnetic resonance (MR) image sequence. We first segment cardiac LVs using an encoder-decoder network and then introduce a multitask framework to regress 11 LV indices and classify the cardiac phase, as parallel tasks during model optimization. The proposed deep learning model is based on the 3D spatio-temporal convolutions, which extract spatial and temporal features from MR images. We demonstrate the efficacy of the proposed method using cine-MR sequences of 145 subjects and comparing the performance with other state-of-the-art quantification methods. The proposed method obtained high prediction accuracy, with an average mean absolute error (MAE) of 129 mm², 1.23 mm, 1.76 mm, Pearson correlation coefficient (PCC) of 96.4%, 87.2%, and 97.5% for LV and myocardium (Myo) cavity regions, 6 RWTs, 3 LV dimensions, and an error rate of 9.0% for phase classification. The experimental results highlight the robustness of the proposed method, despite varying degrees of cardiac morphology, image appearance, and low contrast in the cardiac MR sequences. © 2013 IEEE.

Date 2021

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ISSN 21682194 (ISSN)**Issue** 7**Journal Abbr** IEEE J. Biomedical Health Informat.**Date Added** 20/09/2026, 9:52:18 pm**Modified** 20/09/2026, 9:52:18 pm**Tags:**

Quantitative assessments, Article, human, Humans, Deep learning, Learning systems, machine learning, Heart, entropy, quantitative analysis, nonhuman, deep learning, artificial neural network, Diagnosis, cardiovascular disease, Morphology, Multi-task learning, Quantification methods, diagnostic imaging, image quality, Magnetic resonance, Magnetic resonance imaging, Magnetic Resonance Imaging, nuclear magnetic resonance imaging, Phase classification, heart infarction, Correlation methods, heart ventricle, Heart Ventricles, systole, hippocampus, heart left ventricle, thickness, Prediction accuracy, heart, heart rate, kidney cortex, heart cycle, Mean absolute error, Cardiac MRI, anticoagulation, cardiac segmentation, Cardio-vascular disease, cine magnetic resonance imaging, diastole, heart atrium appendage, heart function, Intra-observer variability, left ventricle quantification, Magnetic Resonance Imaging, Cine, myocardial infraction, Pearson correlation coefficients, radiography, Radiography, sudden cardiac death, warfarin

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Species-specific detection and quantification of common barnacle larvae from the Japanese coast using quantitative real-time PCR.

Item Type Journal Article**Author** N. Endo**Author** K. Sato**Author** K. Matsumura**Author** E. Yoshimura**Author** Y. Odaka**Author** Y. Nogata

Abstract Species-specific detection and quantification methods for barnacle larvae using quantitative real-time polymerase chain reaction (qPCR) were developed. Species-specific primers for qPCR were designed for 13 barnacle species in the mitochondrial 12S ribosomal RNA gene region. Primer specificity was examined by PCR using template DNA extracted from each of the 13 barnacle species, other unidentified

barnacle species, and field collected zooplankton samples. The resulting PCR products comprised single bands following agarose gel electrophoresis when the templates corresponded to primers. The amplifications were highly species-specific even for the field plankton samples. The field plankton samples were subjected to qPCR assay. The calculated DNA contents for each barnacle species were closely correlated with the number of larvae measured by microscopic examination. The method could be applied to quantify barnacle larvae in natural plankton samples.

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article, Japan, classification, Classification, methodology, genetics, polymerase chain reaction, sequence alignment, chemistry, animal, Animals, biofouling, Biofouling, larva, Larva, Polymerase Chain Reaction, nucleotide sequence, sequence analysis, growth, development and aging, species difference, Species Specificity, Sequence Analysis, RNA, DNA Primers, primer DNA, ribosome RNA, Base Sequence, molecular genetics, Molecular Sequence Data, Oceans and Seas, RNA, Ribosomal, RNA, ribosomal, 12S, sea, Sequence Alignment, Thoracica

Notes:

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Export Date: 20 September 2026; Cited By: 11; Correspondence Address: N. Endo; email: n-endo@criepi.denken.or.jp

Spectral-spatial classification of hyperspectral images via spatial translation-invariant wavelet-based sparse representation

Item Type Journal Article

Author L. He

Author Y. Li

Author X. Li

Author W. Wu

Abstract For hyperspectral image (HSI) classification, it is challenging to adopt the methodology of sparse-representation-based classification. In this paper, we first propose an ℓ_1 -minimization-based spectral-spatial classification method for HSIs via a spatial translation-invariant wavelet (STIW)-based sparse representation (STIW-SR), wherein both the spectrum dictionary and the analyzed signal are formed with STIW features. Due to the capability of a STIW to reduce both the observation noise and the spatial nonstationarity while maintaining the ideal spectra, which is proved with our signal-interference-noise spectrum model involved, it is expected that the pixels in the same class congregate in a lower dimensional subspace, and the separations among class-specific subspaces are enhanced, thus yielding a highly discriminative sparse representation. Then, we develop an approach to evaluate the sparsity recoverability of an ℓ_1 -minimization on HSIs in a probabilistic framework. This approach takes into account not only the recovery probability under the given support length of the ℓ_0 -norm solution but also the a priori probability of the support length; consequently, it overcomes the inability of traditional mutual/cumulative coherence conditions to address high-coherence HSIs. This paper reveals that the higher sparsity recoverability of a STIW-SR leads to its higher classification accuracy and that the increasing coherence does not necessarily lead to a reduced sparsity recovery probability, and this paper verifies the connection between ℓ_0 - and ℓ_1 -minimizations on HSIs. Experimental results from real-world HSIs suggest that our classification method significantly outperforms several representative spectral-spatial classifiers and support vector machines. © 1980-2012 IEEE.

Date 2015

DOI 10.1109/TGRS.2014.2363682

URL <https://www.scopus.com/pages/publications/84921061644?origin=resultslist>

Archive Scopus

Language English

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Publisher Institute of Electrical and Electronics Engineers Inc.

Pages 2696-2712

Publication IEEE Transactions on Geoscience and Remote Sensing

ISSN 01962892 (ISSN)

Issue 5

Journal Abbr IEEE Trans Geosci Remote Sens

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Tags:

Image classification, Vectors, Probability, data quality, spatial analysis, image classification, pixel, wavelet, Wavelet analysis, Independent component analysis, spectral analysis, Signal interference, Spectroscopy, Sparse representation, Hyperspectral image (HSI), Hyperspectral images, Interference suppression, Recoverability, signal, sparse representation, sparsity recoverability, Spatial translation, spatial translation-invariant wavelet (STIW), spectral-spatial classification, Spectral-spatial classification

Notes:

EID: 2-s2.0-84921061644

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Spiculation Quantification Method Based on Edge Gradient Orientation Histogram

Item Type Conference Paper

Author G. Zhang

Author N. Xiao

Author W. Guo

Editor Shen X.

Editor Zhang X.

Editor Zhou Z.

Editor Zhang G.

Editor Luo X.

Abstract A new approach to quantify the lung nodule speculation levels in CT (computed tomography) images was proposed. Firstly, two-dimensional image of nodule was generated by using the spiral scan technology. Secondly, the lung nodule was segmented in the two-dimensional image using the dynamic programming. Thirdly, based on the expanded regions from the segmented boundary, the speculation was segmented by use of threshold segmentation. Fourthly, the feature region where speculation may exist was extracted on the region of speculation boundary. Finally, the edge gradient orientation histogram as a new quantitative index was extracted to quantify the speculation levels of the lung nodule. The experimental results indicate that this index can quantify the speculation levels accurately with better classification results. © 2014 IEEE.

Date 2015

DOI 10.1109/ICVRV.2014.26

URL <https://www.scopus.com/pages/publications/84962178578?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - Int. Conf. Virtual Real. Vis., ICVRV

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-147996854-1 (ISBN)

Pages 86-91

Proceedings Title Proc. - Int. Conf. Virtual Real. Vis., ICVRV

Conference Name Proceedings - 2014 International Conference on Virtual Reality and Visualization, ICVRV 2014

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Tags:

Virtual reality, Visualization, Dynamic programming, Quantification methods, Computerized tomography, Biological organs, Classification results, CT (computed tomography), Edge gradient, Edge gradient orientation histogram, Lung nodule, Lung nodules, Spiculation features, Threshold segmentation, Two dimensional images

Notes:

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Spline based inhomogeneity correction for 11C-PIB PET segmentation using expectation maximization

Item Type Conference Paper

Author P. Raniga

Author P. Bourgeat

Author V. Villemagne

Author G. O'Keefe

Author C. Rowe

Author S. Ourselin

Abstract With the advent of biomarkers such as 11C-PIB and the increase in use of PET, automated methods are required for processing and analyzing datasets from research studies and in clinical settings. A common preprocessing step is the calculation of standardized uptake value ratio (SUVR) for inter-subject normalization. This requires segmented grey matter (GM) for VOI refinement. However 11C-PIB uptake is proportional to amyloid build up leading to inhomogeneities in intensities, especially within GM. Inhomogeneities present a challenge for clustering and pattern classification based approaches to PET segmentation as proposed in current literature. In this paper we modify a MR image segmentation technique based on expectation maximization for 11C-PIB PET segmentation. A priori probability maps of the tissue types are used to initialize and enforce anatomical constraints. We developed a Bézier spline based inhomogeneity correction techniques that is embedded in the segmentation algorithm and minimizes inhomogeneity resulting in better segmentations of 11C-PIB PET images. We compare our inhomogeneity with a global polynomial correction technique and validate our approach using co-registered MRI segmentations. © Springer-Verlag Berlin Heidelberg 2007.

Date 2007

DOI 10.1007/978-3-540-75757-3_28

URL <https://www.scopus.com/pages/publications/38149133733?origin=resultslist>

Archive Scopus

Language English**Extra** Journal Abbreviation: Lect. Notes Comput. Sci.**Volume** 4791 LNCS**Publisher** Springer Verlag**ISBN** 03029743 (ISSN); 978-354075756-6 (ISBN)**Pages** 228-235**Proceedings Title** Lect. Notes Comput. Sci.**Conference Name** Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)**Issue** PART 1**Date Added** 20/09/2026, 9:53:04 pm**Modified** 20/09/2026, 9:53:04 pm**Tags:**

artificial intelligence, female, Female, human, Humans, male, Male, Artificial Intelligence, aged, Aged, brain, Brain, Image segmentation, article, reproducibility, Reproducibility of Results, Pattern recognition, algorithm, Algorithms, validation study, methodology, government, Alzheimer disease, Alzheimer Disease, sensitivity and specificity, computer assisted diagnosis, Image Interpretation, Computer-Assisted, Magnetic resonance imaging, statistical model, Biomarkers, scintiscanning, unclassified drug, positron emission tomography, Positron-Emission Tomography, radiopharmaceutical agent, Radiopharmaceuticals, artifact, Artifacts, Sensitivity and Specificity, Likelihood Functions, image enhancement, automated pattern recognition, Pattern Recognition, Automated, Aniline Compounds, aniline derivative, Image Enhancement, diagnostic agent, 2 (4' (methylamino)phenyl) 6 hydroxybenzothiazole, 2-(4'-(methylamino)phenyl)-6-hydroxybenzothiazole, Expectation maximization, Polyethylene terephthalates, Polynomial correction, thiazole derivative, Thiazoles

Notes:

EID: 2-s2.0-38149133733

Export Date: 20 September 2026; Cited By: 8; Conference name: 10th International Conference on Medical Image Computing and Computer-Assisted Intervention, MICCAI 2007; Conference date: 29 October 2007 through 2 November 2007; Conference code: 71013

Spring Festival and COVID-19 Lockdown: Disentangling PM Sources in Major Chinese Cities

Item Type Journal Article**Author** Q. Dai**Author** L. Hou**Author** B. Liu**Author** Y. Zhang

Author C. Song
Author Z. Shi
Author P.K. Hopke
Author Y. Feng

Abstract Responding to the 2020 COVID-19 outbreak, China imposed an unprecedented lockdown producing reductions in air pollutant emissions. However, the lockdown driven air pollution changes have not been fully quantified. We applied machine learning to quantify the effects of meteorology on surface air quality data in 31 major Chinese cities. The meteorologically normalized NO₂, O₃, and PM_{2.5} concentrations changed by −29.5%, +31.2%, and −7.0%, respectively, after the lockdown began. However, part of this effect was also associated with emission changes due to the Chinese Spring Festival, which led to ~14.1% decrease in NO₂, ~6.6% increase in O₃ and a mixed effect on PM_{2.5} in the studied cities that largely resulted from festival associated fireworks. After decoupling the weather and Spring Festival effects, changes in air quality attributable to the lockdown were much smaller: −15.4%, +24.6%, and −9.7% for NO₂, O₃, and PM_{2.5}, respectively. © 2021. The Authors.

Date 2021

DOI 10.1029/2021GL093403

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Language English

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Issue 11

Journal Abbr Geophys. Res. Lett.

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Tags:

machine learning, China, atmospheric pollution, COVID-19, meteorology, Nitrogen oxides, pollutant source, Air quality, particulate matter, air quality, Air pollutant emission, Air quality data, Applied machine learning, Chinese cities, Chinese springs, Emission change, emission control, festival, meteorological normalization, Mixed effects, PM_{2.5} concentration, source, spring festival

Notes:

EID: 2-s2.0-85107735215

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Ssiqa: Multi-task learning for non-reference ct image quality assessment with self-supervised noise level prediction

Item Type Conference Paper

Author A.-A.-Z. Imran

Author D. Pal

Author B. Patel

Author A. Wang

Abstract Reduction of CT radiation dose is important due to the potential effects on patients. But lowering dose incurs degradation in the reconstructed image quality, furthering compromise in the diagnostic and image-based analyses performance. Considering the patient health risks, high quality reference images cannot be easily obtained, making the assessment challenging. Therefore, automatic no-reference image quality assessment is desirable. Leveraging an innovative self-supervised regularization in a convolutional neural network, we propose a novel, fully automated, no-reference CT image quantification method namely self-supervised image quality assessment (SSIQA). Extensive experimentation via in-domain (abdomen CT) and cross-domain (chest CT) evaluations demonstrates SSIQA is accurate in quantifying CT image quality, generalized across the scan types, and consistent with the established metrics and different relative dose levels. © 2021 IEEE.

Date 2021

DOI 10.1109/ISBI48211.2021.9434044

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Archive Scopus

Language English

Extra Journal Abbreviation: IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recogn.

Volume 2021-April

Publisher IEEE Computer Society

ISBN 19457928 (ISSN); 978-166541246-9 (ISBN)

Pages 1962-1965

Proceedings Title IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recogn.

Conference Name Proceedings - International Symposium on Biomedical Imaging

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Tags:

Risk assessment, % reductions, Learning systems, Machine learning, Health risks, Image quality, Medical imaging, Self-supervised learning, Multitask learning, Diagnosis, Computerized tomography, Radiation effects, Noise levels, Dosimetry, CT, CT Image, G-SSIM, Image quality assessment, Potential effects, Radiation dose, Reconstructed image

Notes:

EID: 2-s2.0-85107215423

Export Date: 20 September 2026; Cited By: 17; Funding details: GE Healthcare; Conference name: 18th IEEE International Symposium on Biomedical Imaging, ISBI 2021; Conference date: 13 April 2021 through 16 April 2021; Conference code: 169147

Stacked Ensemble Learning with SHAP-Prioritized Intervention Ranking for Pavement Maintenance Automation

Item Type Journal Article

Author A. Gupta

Author P. Kumar

Abstract This study presents an automated decision-support framework for pavement maintenance and rehabilitation (M&R) prioritization using machine learning and explainable artificial intelligence (AI). A stacked ensemble classifier—combining tuned Light gradient boosting machine (LightGBM), ExtraTrees, support vector machine (SVM), and multilayer perceptron (MLP) as base learners with XGBoost as the meta-learner—was developed using nine pavement condition indicators with a dataset of 1166 points. The model predicts discrete M&R classes with strong predictive performance on the studied dataset, achieving an accuracy of 98.29%, macro F1-score of 0.98, precision of 0.963, and recall of 0.964. Model transparency was achieved through SHAP (SHapley Additive exPlanations), which identified Current million standard axles (MSA), Roughness International Roughness Index (IRI), and Structural Number as dominant predictors. A SHAP-informed composite scoring system enabled intra-class prioritization of road segments into Urgent, Medium, and Low categories. Sensitivity analysis confirmed that rankings for urgent interventions remained stable under $\pm 20\%$ feature contributions variations. The framework offers a robust and interpretable framework for network-level pavement management, supporting data-driven intervention planning with improved transparency and objectivity. © Chinese Society of Pavement Engineering 2026.

Date 2026

DOI 10.1007/s42947-026-00866-8

URL <https://www.scopus.com/pages/publications/105046128431?origin=resultslist>

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Language English

Publisher Springer

Publication International Journal of Pavement Research and Technology

ISSN 19966814 (ISSN)

Journal Abbr Int. J. Pavement Res. Technol.

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Artificial intelligence, Decision support systems, Highway administration, Automation, Learning systems, Transparency, Maintenance, Information management, Ensemble learning, Sensitivity analysis, SHAP, Shapley, Shapley additive explanation, Pavement maintenance and rehabilitation, Pavement maintenance and rehabilitation (M&R), Pavement maintenance and rehabilitations, Pavement overlays, Priority ranking system, Priority rankings, Ranking system, Sensitivity analyzes, Stacked ensemble learning

Notes:

EID: 2-s2.0-105046128431

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A. Gupta; Scientists, CSIR-Central Road Research Institute, New Delhi, 110025, India; email: aakash.crri@csir.res.in

Staircase Network: structural language identification via hierarchical attentive units

Item Type Conference Paper

Author T.N. Trong

Author V. Hautamäki

Author K. Jokinen

Abstract Language recognition system is typically trained directly to optimize classification error on the target language labels, without using the external, or meta-information in the estimation of the model parameters. However labels are not independent of each other, there is a dependency enforced by, for example, the language family, which affects negatively on classification. The other external information sources (e.g. audio encoding, telephony or video speech) can also decrease classification accuracy. In this paper, we attempt to solve these issues by constructing a deep hierarchical neural network, where different levels of meta-information are encapsulated by attentive prediction units and also embedded into the training progress. The proposed method learns auxiliary tasks to obtain robust internal representation and to construct a variant of attentive units within the hierarchical model. The final result is the structural prediction of the target language and a closely related language family. The algorithm reflects a “staircase” way of learning in both its architecture and training, advancing from the fundamental audio encoding to the language family level and finally to the target language level. This process not only improves generalization but also tackles the issues of imbalanced class priors and channel variability in the deep neural network model. Our experimental findings show that the proposed architecture outperforms the state-of-the-art i-vector approaches on both small and big language corpora by a significant margin. © ODYSSEY 2018. All rights reserved.

Date 2018

DOI 10.21437/Odyssey.2018-9

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Archive Scopus

Language English

Extra Journal Abbreviation: Speak. Lang. Recognit. Workshop, ODYSSEY

Publisher The International Society for Computers and Their Applications (ISCA)

Pages 60-67

Proceedings Title Speak. Lang. Recognit. Workshop, ODYSSEY

Conference Name Speaker and Language Recognition Workshop, ODYSSEY 2018

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Tags:

Deep neural networks, Classification (of information), Network architecture, Modeling parameters, Hierarchical systems, Information sources, Signal encoding, Language identification, External informations, Encoding (symbols), Recognition systems, Target language, Neural network models, Classification errors, Audio encoding, Language recognition, Meta information, Stairs

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Export Date: 20 September 2026; Cited By: 1; Funding details: Nvidia; Academy of Finland, AKA, (313970); Conference name: 2018 Speaker and Language Recognition Workshop, ODYSSEY 2018; Conference date: 26 June 2018 through 29 June 2018; Conference code: 179973

Standardized and semiautomated workflow for 3D characterization of liberated particles

Item Type Journal Article

Author S. Gupta

Author J.R. da Assuncao Godinho

Author K. Gotkowski

Author F. Isensee

Abstract MSPaCMAN is the recently developed workflow that does the mineralogical quantification of individual particles using its histograms while considering the effects of partial volume artefacts in interphases at particle level detail. This paper demonstrates and validates the new developments in the MSPaCMAN workflow, aiming to minimize user bias and enhance the accuracy of MSPaCMAN. Here, in the new developments of MSPaCMAN workflow, firstly, the recently developed deep learning method, namely ParticleSeg3D, is employed to distinguish particles from the background. Secondly, the particle's size and shape information are considered along with its histogram to classify and quantify the mineral phases in liberated particles. After the new developments, the detection limit of MSPaCMAN to characterize small and thin liberated particles is enhanced. Experimental results demonstrate the effectiveness of the MSPaCMAN's updated workflow. It was found that the mineralogical composition calculated by the MSPaCMAN's updated workflow was precise, with the highest coefficient of variance of 6.56%. Moreover, the mineralogical composition determined by MSPaCMAN's updated workflow had low

variance across three different reconstruction parameters and two different voxel sizes (5.5 μm and 10 μm). Comparisons with other quantification methods highlight the accuracy of MSPaCMan's updated workflow to determine the mineralogical compositions. When analyzing a test sample consisting of quartz, calcite, fluorite, and lepidolite, MSPaCMan's updated workflow achieved the highest mineralogical deviation of only 7.73% from the reference mineralogy. In contrast, the random forest algorithm resulted in a deviation of 51.47%, while the manual thresholding method yielded a deviation of 41.46%. Overall, these findings emphasize the reliability and accuracy of MSPaCMan's updated workflow in quantifying mineralogical composition. © 2023

Date 2024

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Language English

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Article, automation, Deep learning, Classification (of information), Learning systems, reliability, algorithm, image segmentation, Forestry, deep learning, controlled study, workflow, accuracy, random forest, Graphic methods, correlation coefficient, Work-flows, particle size, three-dimensional imaging, limit of detection, Computerized tomography, X-ray computed tomography, unclassified drug, histogram, 3D characterization, Calcite, calcium carbonate, fluorite, Individual particles, lepidolite, liberated particle, Mica, Mineral classification, mineralization, Mineralogical compositions, MSPaCMan, nnU-net, Nnu-net, Partial volumes, Particle segmentation, physical phase, silicon dioxide

Notes:

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Export Date: 20 September 2026; Cited By: 2; Funding details: Helmholtz Imaging; Deutsche Forschungsgemeinschaft, DFG, (470202518); Deutsche Forschungsgemeinschaft, DFG; Correspondence Address: S. Gupta; Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Freiberg, Chemnitz Straße 40, 09599, Germany; email: s.gupta@hzdr.de; CODEN: POTEB

Statin therapy and mortality among patients hospitalized with heart failure and preserved left ventricular function - A preliminary report

Item Type Journal Article

Author M. Roik

Author M.H. Starczewska

Author Z. Huczek

Author J. Kochanowski

Author G. Opolski

Abstract Background - A number of medications such as beta-blockers, ACE-inhibitors, angiotensin receptor blockers and aldosterone antagonists, improve survival in patients with heart failure (HF) and reduced ejection fraction. No therapy has been proved to be beneficial for patients with heart failure and preserved ejection fraction (PLVEF). Objectives - The aim of the study was to assess the effect of statin therapy on all-cause mortality and cardiovascular rehospitalization rate in patients with HF and PLVEF during one-year follow-up. Methods - We evaluated 146 patients with HF and PLVEF (ejection fraction $\geq 45\%$). Patients were divided into the statin therapy group ($n = 103$, mean age 69 ± 11 y, 52 men) and the group without statins ($n = 43$, mean age 66 ± 16 y, 25 men). We analysed the effects of the statin treatment prescribed to patients at discharge. Patients were followed up for one year. Results - The age, gender, NYHA functional class, prevalence of co-morbidities (renal dysfunction, COPD, diabetes mellitus) did not differ between the groups ($P = \text{NS}$). Patients receiving statin therapy more frequently had an ischaemic aetiology of HF (79% vs. 39.5%; $P < 0.001$) and hypertension (76% vs. 58%; $P < 0.05$). No differences in ejection fraction were observed neither by echocardiography (58% vs. 55%; $P = \text{NS}$) nor by basic laboratory data. Patients who received statins were often additionally treated with beta-blockers (91% vs. 70%; $P < 0.005$), aspirin (77 vs. 44%; $P < 0.01$), thienopyridines (22% vs. 5%; $P < 0.01$) and less frequently with oral anticoagulants (5% vs. 23%; $P < 0.005$). In the group receiving statins a significantly lower mortality (4% vs. 21%; $P < 0.001$) and rehospitalization rate (43% vs. 69%; $P < 0.05$) was documented. After adjustment for all univariate predictors of the occurrence of study primary end-points, statin therapy was shown to be associated with significant and independent reduction in all-cause mortality ($\text{HR} = 0.24$ [95%CI: 0.07 - 0.90] $P < 0.05$) and cardiovascular rehospitalization rate ($\text{HR} = 0.55$ [95%CI: 0.33 - 0.92] $P < 0.05$). After propensity matching statin therapy remained an independent factor reducing one-year mortality rate ($\text{HR} = 0.11$ [95%CI: 0.01 - 0.99] $P < 0.05$). Conclusions - This study showed that statin therapy may have beneficial effects on mortality and rehospitalization rates among patients with HF and PLVEF. It also suggests a potential role for statins as a new therapeutic option in patients with HF and PLVEF, but these observations need to be confirmed in large randomized trials.

Date 2008

DOI 10.2143/AC.63.6.2033384

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Language English

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adult, female, Female, human, Humans, male, Male, Middle Aged, aged, Aged, article, controlled study, age, major clinical study, hydroxymethylglutaryl coenzyme A reductase inhibitor, follow up, hospital patient, Hospitalization, disease classification, comorbidity, diabetes mellitus, hypertension, mortality, echocardiography, chronic obstructive lung disease, acetylsalicylic acid, beta adrenergic receptor blocking agent, calcium channel blocking agent, heart failure, Heart Failure, Stroke Volume, dipeptidyl carboxypeptidase inhibitor, heart left ventricle ejection fraction, laboratory test, hospital readmission, Proportional Hazards Models, prescription, amiodarone, anticoagulant agent, digoxin, fibric acid derivative, Heart failure with preserved ejection fraction, heart left ventricle function, Kaplan-Meiers Estimate, kidney dysfunction, loop diuretic agent, low molecular weight heparin, Morbidity, Mortality, spironolactone, statin, Statin therapy, thienopyridine derivative, Ventricular Function, Left

Notes:

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Export Date: 20 September 2026; Cited By: 20; Correspondence Address: M. Roik; Department of Cardiology, Medical University, 02-097 Warszawa, Banacha 1a st, Poland; email: mfroik@amwaw.edu.pl; CODEN: ACCAA

Statistical 3D object classification and localization with context modeling

Item Type Conference Paper

Author M. Grzegorzek

Author E. Izquierdo

Abstract This contribution presents a probabilistic approach for automatic classification and localization of 3D objects in 2D multi-object images taken from a real world environment. In the training phase, statistical object models and statistical context models are learned separately. For the object modeling, the recognition system extracts local feature vectors from training images using the wavelet transformation and models them statistically by density functions. Since in contextual environments a-priori probabilities for occurrence of different objects cannot be assumed to be equal, statistical context modeling is introduced in this work. The a-priori occurrence probabilities are learned in the training phase and stored in so-called context models. In the recognition phase, the system determines the unknown number of objects in a multi-object scene first. Then, the object classification and localization are performed. Recognition results for experiments made on a real dataset with 3240 test images compare the performance of the system with and without consideration of the context modeling. © 2007 EURASIP.

Date 2007

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Language English
Extra Journal Abbreviation: European Signal Proces. Conf.
Publisher European Signal Processing Conference, EUSIPCO
ISBN 22195491 (ISSN); 978-161738876-7 (ISBN); 978-839213402-2 (ISBN)
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Conference Name European Signal Processing Conference
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Tags:

Object modeling, Multiobject, Context models, Object localization, Training phasis, Wavelet transforms, 3D modeling, Automatic classification, 3D object, 3d object classifications, Automatic localization, Probabilistics approach

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EID: 2-s2.0-58849158802

Export Date: 20 September 2026; Cited By: 5; Correspondence Address: M. Grzegorzek; Multimedia and Vision Research Group, Queen Mary, University of London, E1 4NS London, Mile End Road, United Kingdom; email: marcin.grzegorzek@elec.qmul.ac.uk; Conference name: 15th European Signal Processing Conference, EUSIPCO 2007; Conference date: 3 September 2007 through 7 September 2007; Conference code: 91100

Statistical analysis to assess automated level of suspicion scoring methods in breast ultrasound

Item Type Conference Paper

Author M. Galperin

Abstract A well-defined rule-based system has been developed for scoring 0-5 the Level of Suspicion (LOS) based on qualitative lexicon describing the ultrasound appearance of breast lesion. The purposes of the research are to asses and select one of the automated LOS scoring quantitative methods developed during preliminary studies in benign biopsies reduction. The study has used Computer Aided Imaging System (CAIS) to improve the uniformity and accuracy of applying the LOS scheme by automatically detecting, analyzing and comparing breast masses. The overall goal is to reduce biopsies on the masses with lower levels of suspicion, rather that increasing the accuracy of diagnosis of cancers (will require biopsy anyway). On complex cysts and fibroadenoma cases experienced radiologists were up to 50% less certain in true negatives than CAIS. Full correlation analysis was applied to determine which of the proposed LOS quantification methods serves CAIS accuracy the best. This paper presents current results of applying statistical analysis for automated LOS scoring

quantification for breast masses with known biopsy results. It was found that First Order Ranking method yielded most the accurate results. The CAIS system (Image Companion®, Data Companion® software) is developed by Almen Laboratories and was used to achieve the results.

Date 2003

DOI 10.1117/12.479890

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc SPIE Int Soc Opt Eng

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Tags:

Statistical methods, Segmentation, Classification, Image analysis, Computer aided diagnosis, Mammography, Computer-aided, Cancer, Tumors, Ultrasonic imaging, ROC, Benign masses, Breast ultrasound, Diagnostic imaging, Level of suspicion scoring, Statistical similarity

Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: M. Galperin; Almen Laboratories, Inc., Escondido, CA 92025, 2105 Miller Ave., United States; email: mgalperin@almenlabs.com; Conference name: Medical Imaging 2003: Image Perception, Observer Performance, and Technology Assessment; Conference date: 18 February 2003 through 20 February 2003; Conference code: 61436; CODEN: PSISD

Statistical analysis, machine learning modeling, and text analytics of aggregation attachment efficiency: Mono and binary particle systems

Item Type Journal Article

Author A. Gomez-Flores

Author S.A. Bradford

Author G. Hong

Author H. Kim

Abstract The aggregation attachment efficiency (α) is the fraction of particle–particle collisions resulting in aggregation. Despite significant research, α predictions have not accounted for the full complexity of systems due to constraints imposed by particle types, dispersed matter, water chemistry, quantification methods, and

modeling. Experimental α values are often case-specific, and simplified systems are used to rule out complexity. To address these challenges, statistical analysis was performed on α databases to identify gaps in current knowledge, and machine learning (ML) was used to predict α under various particle types and conditions. Moreover, text analytics was employed to support knowledge from statistics and ML, as well as gain insight into the ideas communicated by current literature. Most studies investigated α in mono-particle systems, but binary or higher systems require more investigation. Furthermore, our work highlights that numerous variables, interactions, and mechanisms influence α behavior, making its investigation complex and difficult for both experiments and modeling. Consequently, future research should incorporate more particle types, shapes, coatings, and surface heterogeneities, and aim to address overlooked variables and conditions. Therefore, building a comprehensive α database can enable the development of more accurate empirical models for prediction. © 2023 Elsevier B.V.

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'current, Condition, Efficiency, machine learning, Machine-learning, Machine learning, complexity, Forecasting, correlation, article, Topic Modeling, Machine learning models, prediction, Statistical methods, Topic modeling, Text analytics, statistical analysis, Agglomeration, Attachment efficiency, data acquisition, efficiency measurement, Missing data imputation, Missing data imputations, Particles system, Word correlation

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Statistical and wavelet based texture features for fish oocytes classification

Item Type Conference Paper

Author E. González-Rufino

Author P. Carrión

Author A. Formella

Author M. Fernández-Delgado

Author E. Cernadas

Abstract The study of biology and population dynamics of fish species requires the estimation of fecundity parameters in individual fish in many fisheries laboratories. The traditional procedure used in fisheries research is to classify and count the oocytes manually on a subsample of known weight of the ovary, and to measure few oocytes under a binocular microscope. With an adequate interactive tool, this process might be done on a computer. However, in both cases the task is very time consuming, with the obvious consequence that fecundity studies are not conducted routinely. In this work we develop a computer vision system for the classification of oocytes using texture features in histological images. The system is structured in three stages: 1) extraction of the oocyte from the original image; 2) calculation of a texture feature vector for each oocyte; and 3) classification of the oocytes using this feature vector. A statistical evaluation of the proposed system is presented and discussed. © 2011 Springer-Verlag.

Date 2011

DOI 10.1007/978-3-642-21257-4_50

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Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Classification (of information), Pattern recognition, Neural networks, Image analysis, Image texture, Fish, Fisheries, Wavelet analysis, Binocular microscopes, Classification trees, Computer vision system, Fish oocyte, Fish oocytes, Statistical classifier, Statistical classifiers, Statistical evaluation, Texture classification, Texture feature vector

Notes:

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Export Date: 20 September 2026; Cited By: 1; Funding details: Xunta de Galicia; Correspondence Address: E. González-Rufino; Computer Science Department, Univ. de Vigo, Campus As Lagoas, 32004 Ourense, Spain; email: nrufino@uvigo.es

Statistical classification of buried unexploded ordnance using nonparametric prior models

Item Type Journal Article

Author A. Aliamiri

Author J. Stalnaker

Author E.L. Miller

Abstract We used kernel density estimation (KDE) methods to build a priori probability density functions (pdfs) for the vector of features that are used to classify unexploded ordnance items given electromagnetic-induction sensor data. This a priori information is then used to develop a new suite of estimation and classification algorithms. As opposed to the commonly used maximumlikelihood parameter estimation methods, here we employ a maximum a posteriori (MAP) estimation algorithm that makes use of KDE-generated pdfs. Similarly, we use KDE priors to develop a suite of classification schemes operating in both "feature" space as well as "signal/data" space. In terms of feature-based methods, we construct a support vector machine classifier and its extension to support M-ary classification. The KDE pdfs are also used to synthesize a MAP feature-based classifier. To address the numerical challenges associated with the optimal data-space Bayesian classifier, we have used several approximation techniques, including Laplacian approximation and generalized likelihood ratio tests employing the priors. Using both simulations and real field data, we observe a significant improvement in classification performance due to the use of the KDE-based prior models. © 2007 IEEE.

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Probability density function, Classification (of information), Algorithms, Classification, Density estimation, Maximum likelihood estimation, Computer simulation, Electromagnetic induction, Electromagnetic induction (EMI), Laplacian approximation, Maximum a posteriori (MAP) estimation, Ordnance, Subsurface sensing, Unexploded ordnance (UXO)

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Stay Open: Calibrating Weights Continuously for Detecting Out-of-Distribution Objects on the Fly

Item Type Conference Paper

Author X. Xiang

Author Q. Zhou

Editor Lin Z.

Editor Zha H.

Editor Cheng M.-M.

Editor He R.

Editor Liu C.-L.

Editor Ubul K.

Editor Silamu W.

Editor Zhou J.

Abstract Open-world object detection (OWOD) has recently obtained significant interest due to its practical applications. The core issue lies in a model's ability to identify new object categories and incrementally acquire knowledge about them, all while retaining information about previously learned classes. Prior solutions have typically relied on either strong or weak supervision for the novel-class (i.e., out-of-distribution) data during detection, an approach that might not be applicable in real-world scenarios. This paper constructs a new setup where novel classes are only encountered on the fly at the inference stage. In response to this challenge, an innovative OWOD detector is tailored for the open-set incremental learning environment. The proposed methodology incorporates label smoothing as a mechanism to prevent the detector from prematurely and confidently assigning novel classes to existing categories, thereby facilitating the discovery of novel classes. Extensive experiments conducted on our realistic setup demonstrate the efficacy of our method for discovering novel classes in the new setup. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Object recognition, Object detection, Objects detection, Real-world scenario, Adversarial machine learning, Open world, Inference stages, Incremental learning, label smoothing, Label smoothing, novel class discovery, Novel class discovery, Object categories, On-the-fly, open-world object detection, Open-world object detection

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Stochastic Binary Network for Universal Domain Adaptation

Item Type Conference Paper
Author S.K. Jain
Author S. Das
Abstract Universal domain adaptation (UniDA) is the unsupervised domain adaptation with label shift. UniDA aims to classify unlabeled target samples into one of the "known" categories or into a single "unknown" category. Its main challenge lies in detecting private classes from both domains and performing alignment between the common classes. Current methods employ various techniques and loss functions to address these challenges. However, these methods commonly represent classifiers as point weight vectors, which are prone to overfitting by the source domain samples due to the lack of supervision from the target domain.

Consequently, these classifiers struggle to separate target samples into known and unknown categories effectively. To address this, we introduce a novel framework called Stochastic Binary Network for Universal Domain Adaptation (STUN). STUN uses a Stochastic binary classifier for each class, whose weight is modeled as Gaussian distribution, enabling to sample an arbitrary number of classifiers while keeping the model size same as of two classifiers. Consistency between these sampled classifiers is used to derive the confidence scores for both source and target samples, which facilitates the alignment of common classes using weighted adversarial learning. Finally, we use deep discriminative clustering to formulate a loss function for solving the problem of fragmented feature distributions in the target domain. Extensive ablation studies and state-of-the-art results across three standard benchmark datasets show the efficacy of our framework. © 2024 IEEE.

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EID: 2-s2.0-85191991841

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Stochastic modeling of group testing with random total sample size

Item Type Journal Article

Author B. Mohammed

Author A. Sghir

Author S. El Idrissi

Author A. Maleb

Abstract In the classical Dorfman group testing procedure Dorfman (1943), a fixed population of size N is divided into k pools of equal size n [jls-end-space/]. Each pool is tested: a negative result clears all n individuals, whereas a positive result triggers individual testing within the pool. Put (Formula presented) where p denotes the infection prevalence and q_n is the probability that a pool is positive under perfect classification. The expected number of tests, denoted by XN [jls-end-space/], is given by (Formula presented) However, most applied presentations focus mainly on the expected number of tests and do not quantify the stochastic risk of exceeding a finite laboratory capacity when the number of pools, the total sample size, or the diagnostic accuracy is uncertain. This gap is important for operational screening programs, where a design that is optimal in expectation may still overload the laboratory with non-negligible probability. First, this paper investigates the distribution of XN [jls-end-space/], allowing us to derive the threshold probability (Formula presented) for exceeding a given testing capacity x [jls-end-space/]. Second, we further extend the model by considering random population sizes N and incorporating test misclassification through sensitivity (Formula presented) and specificity (Formula presented). In this setting, (Formula presented) We also add variance formulae, a normal tail approximation, a pool-size-dependent sensitivity (Formula presented), and a Monte Carlo validation showing that the analytical formulae match simulated operating characteristics. © 2026 Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license. <http://creativecommons.org/licenses/by-nc-nd/4.0/>

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A. Sghir; Mohammed First University, Faculty of Sciences, Department of Mathematics, Stochastic and Deterministic Modeling Laboratory, Oujda, Morocco; email: a.sghir@ump.ac.ma

Stochastic seismic inversion and Bayesian facies classification applied to porosity modeling and igneous rock identification

Item Type Journal Article

Author F.J.D. Fernandes

Author L. Teixeira

Author A.F.M. Freire

Author W.M. Lupinacci

Abstract We apply stochastic seismic inversion and Bayesian facies classification for porosity modeling and igneous rock identification in the presalt interval of the Santos Basin. This integration of seismic and well-derived information enhances reservoir characterization. Stochastic inversion and Bayesian classification are powerful tools because they permit addressing the uncertainties in the model. We used the ES-MDA algorithm to achieve the realizations equivalent to the percentiles P10, P50, and P90 of acoustic impedance, a novel method for acoustic inversion in presalt. The facies were divided into five: reservoir 1, reservoir 2, tight carbonates, clayey rocks, and igneous rocks. To deal with the overlaps in acoustic impedance values of facies, we included geological information using a priori probability, indicating that structural highs are reservoir-dominated. To illustrate our approach, we conducted porosity modeling using facies-related rock-physics models for rock-physics inversion in an area with a well drilled in a coquina bank and evaluated the thickness and extension of an igneous intrusion near the carbonate-salt interface. The modeled porosity and the classified seismic facies are in good agreement with the ones observed in the wells. Notably, the coquinas bank presents an improvement in the porosity towards the top. The a priori probability model was crucial for limiting the clayey rocks to the structural lows. In Well B, the hit rate of the igneous rock in the three scenarios is higher than 60%, showing an excellent thickness-prediction capability. © 2023 The Authors

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modeling, Stochastic systems, Stochastics, Bayesian analysis, classification, Stochastic models, Bayesian, Bayesian classification, Porosity, porosity, A-priori probabilities, Acoustic impedance, Carbonate reservoir, Carbonate reservoirs, carbonate rock, Clayey rocks, data inversion, igneous rock, Igneous rocks, Porosity modeling, Porosity models, Pre salts, reservoir characterization, reservoir rock, seismic data, Seismic inversion, Seismology, Stochastic inversion, stochasticity

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Stochastic variational inference improves quantification of multiple timepoint arterial spin labelling perfusion MRI

Item Type Journal Article

Author T.F. Kirk

Author G.G. Kenyon

Author M.S. Craig

Author M.A. Chappell

Abstract Multiple-timepoint arterial spin labelling MRI is a non-invasive imaging technique that permits measurement of both cerebral blood flow and arterial transit time, the latter of which is an emerging biomarker of interest for cerebrovascular health. Quantification of arterial spin labelling data is challenging due to the low signal to noise ratio and non-linear tracer kinetics of this technique. In this work, we introduce a new quantification method called SSVB that addresses limitations in existing methods and demonstrate its performance using simulations and acquisition data. Simulations showed that the method is more accurate, particularly for estimating arterial transit time, and more robust to noise than existing techniques. On high spatial resolution data acquired at 3 T, the method produced less noisy parameter maps than the comparator method and captured greater variation in arterial transit time on a cross-sectional cohort. Copyright © 2025 Kirk, Kenyon, Craig and Chappell.

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Article, female, human, male, simulation, machine learning, correlation analysis, correlation coefficient, quantitative study, limit of quantitation, nuclear magnetic resonance imaging, signal noise ratio, white matter, spatial analysis, brain blood flow, linear regression analysis, quantification, arterial spin labeling, arterial spin label (ASL) MRI, arterial transit time, arterial transit time (ATT), cerebral blood flow (CBF), multiple timepoints, perfusion, sampling, spatial resolution, statistical concepts, stochastic variational inference

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Strain ratio vs strain histogram for the elastographic evaluation of non-calcific chronic pancreatitis (CP) — A prospective, single-center, comparative study

Item Type Journal Article

Author Y.D. Nova

Author J.E. Domínguez Muñoz

Author J. Lariño-Noia

Author P. Míguez-Sánchez

Author J. Iglesias-García

Abstract Introduction: diagnosis of chronic pancreatitis (CP) at early stages remains challenging. Endoscopic ultrasound (EUS) guided strain elastography (SE) has improved the diagnostic yield of CP. The aim of our study was to evaluate the diagnostic yield of SE quantification methods, strain ratio (SR) and strain histogram (SH) to detect non-calcific CP. Methods: a prospective, comparative study was performed with patients undergoing EUS-guided SE for non-calcific CP evaluation. Procedures were performed with linear echoendoscopes. The number of EUS criteria

according to the Rosemont classification were evaluated. SH analysis targeted the pancreatic body area, while SR compared this area (A) to a soft extrapancreatic reference area (B), using B/A as the final ratio. Data are shown as percentages and means (95 % CI) and analyzed using ANOVA, linear regression and Spearman tests. Diagnostic accuracy of SR and SH were evaluated using the Rosemont classification as a reference. STARD criteria were followed. Results: two hundred and eighty-four patients were included (mean age 49.8 years, range 17-85; 142 males). Among them, 43 (15.1 %) had a normal pancreas, 110 (38.7 %) had indeterminate findings, and 131 (46.1 %) had findings suggestive of CP. A SR > 2.42 had a 91.7 % sensitivity and 88.1 % specificity (ROC = 0.968). An SH < 116.1 had an 88.0 % sensitivity and 97.7 % specificity (ROC = 0.964). EUS criteria correlated with pancreatic fibrosis severity by SR ($r = 0.785$, $p < 0.0001$) and SH ($r = -0.742$, $p < 0.0001$). Age correlated with EUS criteria ($r = 0.127$, $p = 0.03$) but not with SE. Conclusions: the quantification of the degree of pancreatic fibrosis by SR and SH during pancreatic EUS-guided SE had a high diagnostic accuracy for CP and was not influenced by age. © Copyright 2026 y Creative Commons (CC-BY-NC-ND). Sociedad Española de Patología Digestiva (SEPD) y © Arán Ediciones S.L.

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Straw-Derived Biochar Alters Soil Phosphorus Availability and Inorganic Phosphorus Fractions: A Global Synthesis

Item Type Journal Article

Author J. Xu

Author S. Liu

Author C. Liang

Author H. Liu

Author Y. Liu

Abstract Straw-derived biochar is increasingly used for residue recycling and soil fertility improvement, but its effects on soil inorganic phosphorus (P) fractions and their relationships with available P (AP) remain unclear. Here, we synthesized 1170 paired observations from 51 studies using meta-analysis, partial least squares path modeling, and machine learning to quantify straw biochar effects on soil AP, inorganic P fractions, microbial biomass P (MBP), and phosphatase activities, and to identify important variables related to these responses. Straw biochar increased soil AP by 77.0% and altered the distribution of inorganic P fractions. The relationships between inorganic P fractions and AP differed: Fe-bound P and occluded P showed positive and negative associations with AP, respectively, whereas Ca-bound P and Al-bound P showed no significant associations with AP. Fe-bound P and Al-bound P increased more strongly in soils with low initial AP, while higher biochar total P corresponded to weaker Fe-bound P accumulation but greater Ca-bound P accumulation. Straw biochar showed a positive response in MBP, while acid and neutral phosphatase activities showed decreasing trends, suggesting distinct microbial P cycling responses involving changes in microbial biomass P and phosphatase activities. These findings provide a P-fraction perspective for evaluating straw biochar as a site-specific amendment for soil P management. © 2026 by the authors.

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Strawberry Fruit Deformity Detection and Symmetry Quantification Using Deep Learning and Geometric Feature Analysis

Item Type Journal Article**Author** L. Jiang**Author** Y. Wang**Author** H. Yan**Author** Y. Yin**Author** C. Wu

Abstract The external appearance of strawberry fruits serves as a critical criterion for their commercial value and grading standards. However, current research primarily emphasizes ripeness and surface defects, with limited attention given to the quantitative analysis of geometric characteristics such as deformity and symmetry. To address this gap, this study proposes a comprehensive evaluation framework that integrates deep learning-based segmentation with geometric analysis for strawberry appearance quality assessment. First, an enhanced YOLOv11 segmentation model incorporating a Squeeze-and-Excitation attention mechanism was developed to enable high-precision extraction of individual fruits, achieving Precision, Recall, AP50, and F1 scores of 91.11%, 87.46%, 92.90%, and 88.45%, respectively. Second, a deformity quantification method was designed based on the number of deformity points (Nd), deformity rate (Rd), and spatial distance metrics (Gmin and Gmax). Experimental results demonstrated significant differences in Rd and Gmax between deformed and normal strawberries, indicating strong classification capability. Finally, principal component analysis (PCA) was employed to extract the primary axis direction, and morphological symmetry was quantitatively evaluated using Intersection over Union (IoU) and Area Difference Ratio (AreaD_Ratio). The results revealed that most samples fell within an IoU range of 0.6–0.8 and AreaD_Ratio below 0.4, indicating noticeable inter-individual differences in fruit symmetry. This study aims to establish a three-stage analytical framework—segmentation, deformity quantification, and symmetry evaluation—for assessing strawberry appearance quality, with the goal of supporting key applications in automated grading and precision quality inspection. © 2025 by the authors.

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deformity detection, segmentation model, strawberry fruit, symmetry quantification analysis

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Strengthening Machine Learning Reproducibility for Image Classification

Item Type Journal Article

Author G. Shao

Author H. Zhang

Author J. Shao

Author K. Woeste

Author L. Tang

Abstract Machine learning (ML) reproducibility needs to be informed with reliable evaluation measures. However, routine image classification is evaluated using metrics that are highly sensitive to class prevalence. Consequently, the reproducibility of ML models remains unclear due to class imbalance-induced noise. We suggest regularly using class imbalance-resistant evaluation metrics, including balanced accuracy, area under precision-recall curve, and image classification efficacy, for the evaluation of the reproducibility of ML models. Each of these evaluation metrics is conceptually consistent with and logically complements the others, and their joint use can help explain different aspects of classification performance at the whole-class level and individual class level. These metrics can be used for the validation, testing, and/or transfer of ML classifiers. Comprehensive analysis using these metrics as a routine approach strengthens the reproducibility of ML models. © Guofan Shao, et al.

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Deep learning, Efficacy, Class imbalance, classification, Discriminative power

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: G. Shao; Department of Forestry and Natural Resources, Purdue University, West Lafayette, United States; email: shao@purdue.edu; L. Tang; Key Laboratory of Urban Environment and Health, Institute of Urban Environment, Chinese Academy of Sciences, Xiamen, China; email: lntang@iue.ac.cn

Structural Factor Analysis of Lexical Complexity Constructs and Measures: A Quantitative Measure-Testing Process on Specialised Academic Texts

Item Type Journal Article

Author M. Nasser

Author P. McCarthy

Abstract This study evaluates 22 lexical complexity measures that represent the three constructs of density, diversity and sophistication. The selection of these measures stems from an extensive review of the SLA linguistics literature. All measures were subjected to qualitative screening for indicators/predictors of lexical proficiency/development and criterion validity based on the body of scholarship. This study's measure-testing process begins by dividing the selected measures into two groups, similarly calculated and dissimilarly calculated, based on their quantification methods and the results of correlation tests. Using a specialized corpus of postgraduate academic texts, a Structural Factor Analysis (SFA) comprising a Confirmatory Factor Analysis (CFA) and Exploratory Factor Analysis (EFA) is then conducted. The purpose of SFA is to 1) verify and examine the lexical classifications proposed in the literature, 2) evaluate the relationship between various lexical constructs and their representative measures, 3) identify the indices that best represent each construct and 4) detect possible new structures/dimensions. Based on the analysis of the corpus, the study discusses the construct-distinctiveness of lexical complexity constructs, as well as strong indicators of each conceptual/mathematical group among the measures. Finally, a unique and smaller set of measures representative of each construct is suggested for future studies that require measure selection. © 2023 Informa UK Limited, trading as Taylor & Francis Group.

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Correspondence Address: M. Nasser; Statistical Modelling and Software development consultant at
the American University of Sharjah (AUS), Hampshire, United Kingdom; email:
mxn309@alumni.bham.ac.uk

Study of the effects of mineral salts on the biofilm formation on polypropylene fibers using three quantification methods

Item Type Journal Article

Author L. Bystrianský

Author M. Hujslová

Author M. Gryndler

Abstract The microbial biofilms are ubiquitous in nature and represent important biological entities that affect various aspects of human life. As such, they attracted considerable attention during last decades, with the factors affecting the biofilm development being among the frequently studied topics. In our work, the biofilm was cultivated on the surface of polypropylene fibers in a nutrient medium inoculated by the suspension of two unsterile soils. The effects of ionic strength and valence of salt on the amount of the produced biofilm and on composition of biofilm microbial communities were investigated. The effect of valence was significant in some OTUs: *Arthrobacter*/*Pseudarthrobacter*/*Paenarthrobacter* and *Bacillus* with positive response to monovalent salt (KCl) and *Streptomyces*, *Lysinibacillus*, *Pseudomonas*, and *Ensifer* with positive response to divalent salt (MgSO₄). The significant preference for a certain concentration of salts was observed in the case of OTUs *Agrobacterium*, *Bacillus* (both 100 mM), and *Brevundimonas* (30 mM). A new quantification method based on measuring of oxidizable organic carbon in biofilm biomass, based on dichromate oxidation, was used. We compared the results obtained using this method with results of crystal violet destaining and measuring of extracted DNA

concentration as proxies of the biofilm biomass. The dichromate oxidation is simple, inexpensive, and fast, and our results show that it may be more sensitive than crystal violet destaining. The highest biomass values tended to associate with high concentrations of the divalent salt. This trend was not observed in treatments where the monovalent salt was added. Our data confirm the importance of inorganic ions for biofilm composition and biomass accumulation. © 2020, Institute of Microbiology, Academy of Sciences of the Czech Republic, v.v.i.

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classification, Bacteria, bacterium, genetics, isolation and purification, Microbiota, microflora, chemistry, microbiology, drug effect, biomass, Biomass, Soil Microbiology, Quantification, growth, development and aging, mineral, Minerals, polypropylene, Bacteriological Techniques, biofilm, Biofilm, Biofilms, Culture Media, culture medium, Dichromate oxidation, inorganic salt, Ionic strength, microbiological examination, Polypropylenes, Salts, Valence

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Study on deep learning methods for coal burst risk prediction based on mining-induced seismicity quantification

Item Type Journal Article

Author X. Cheng

Author W. Qiao

Author H. He

Abstract The assessment of Coal burst risk (CBR) is the premise of bump disaster prevention and control. It is the implementation criterion to guide various rock burst prevention and control measures. The existing static prediction and evaluation methods for CBR cannot be effectively combined with the results of underground dynamic monitoring. This study proposed a mining-induced seismicity information quantification method based on the fractal theory. Deep learning methods were used to construct a deep learning framework of coal burst risk (DLFR) based on the fractal dimension of microseismic information. Gray correlation analysis (GRA), information gain ratio (IGR), and Pearson correlation coefficient are used to screen and compare factors. Statistical evaluation indicators such as macro-F1, accuracy rate, and fitness curve were used to evaluate model performance. Taking the Gaojiapu coal mine as a case study, the performance of deep learning models such as BP Neural Network (BP), Support Vector Machine (SVM) and its optimized model based on particle swarm optimization (PSO) algorithm under this framework is discussed. The research results' reliability and validity are verified by comparing the predicted results with the actual results. The research results show that the prediction results of CBR in DLFR are consistent with the actual results, and the model is reliable and effective. The mining-induced seismicity quantification can solve the problem of insufficient training samples for the CBR. With this, different pressure relief measures can be formulated based on the results of the CBR predictions to achieve "graded" precise prevention and control. © 2023, The Author(s).

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Subclassification, survival prediction and drug target analyses of chemotherapy-naïve muscle-invasive bladder cancer with a molecular screening

Item Type Journal Article

Author S. Rinaldetti

Author E. Rempel

Author T.S. Worst

Author M. Eckstein

Author A. Steidler

Author C.A. Weiss

Author C. Bolenz

Author A. Hartmann

Author P. Erben

Abstract Background: Transcriptome expression studies identified distinct muscle invasive bladder cancer (MIBC) subtypes closely related with breast cancer subclasses. Here we developed a sensitive quantification method for MIBC subclassification (luminal, basal, p53-like). In addition, the subtype specific expression of drug targets has been investigated. Methods: Absolute quantification (nCounter) of a 64-gene panel was performed on MIBC patients (n=47) treated exclusively with radical cystectomy (RC). In conjunction of 170 MIBCs from 3 independent cohorts, a minimal set of consensus genes has been established. Survival of the consensus subtypes has been assessed by multivariate analysis. Relevant drug targets were tested for their subtype specificity in a clustering independent assessment. Results: A reduced 36-gene panel stably clustered into 3 subtypes throughout the cohorts (luminal, basal, infiltrated). Patients treated by RC only, showed worst 8-year disease specific survival (DSS) for the luminal subtype in contrast to the infiltrated subtype (17% vs. 73%, p=0.011). In multivariate analyses, the risk stratification based on luminal versus not-luminal MIBC proved to be an independent predictor for DSS superior to the TNM system in patients with RC. Drug targets (e.g. ERBB2, FGFR, AR, PDGFRB) showed a distinct subtype attribution. The subtypes based on this nCounter screening could further be validated by the TCGA cohort. Conclusion: This MIBC subtype screening predicted survival and allowed an analysis of subtype specific drug targets, thus being a powerful tool for the translation of personalized MIBC treatment concepts. © Rinaldetti et al.

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Subspace learning via local probability distribution for hyperspectral image classification

Item Type Journal Article

Author H. Luo

Author Y.Y. Tang

Author L. Yang

Abstract The computational procedure of hyperspectral image (HSI) is extremely complex, not only due to the high dimensional information, but also due to the highly correlated data structure. The need of effective processing and analyzing of HSI has met many difficulties. It has been evidenced that dimensionality reduction has been found to be a powerful tool for high dimensional data analysis. Local Fisher's liner discriminant analysis (LFDA) is an effective method to treat HSI processing. In this paper, a novel approach, called PD-LFDA, is proposed to overcome the weakness of LFDA. PD-LFDA emphasizes the probability distribution (PD) in LFDA, where the maximum distance is replaced with local variance for the construction of weight matrix and the class prior probability is applied to compute the affinity matrix. The proposed approach increases the discriminant ability of the transformed features in low dimensional space. Experimental results on Indian Pines 1992 data indicate that the proposed approach significantly outperforms the traditional alternatives. © 2015 Huiwu Luo et al.

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Dimensionality reduction, Clustering algorithms, Image classification, Probability, Probability distributions, Discriminant analysis, Matrix algebra, Prior probability, Hyperspectral image classification, Spectroscopy, Hyperspectral images, Subspace learning, Computational procedures, High-dimensional data analysis, Low-dimensional spaces

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SUN: Stochastic UNSupervised Learning for Data Noise and Uncertainty Reduction

Item Type Journal Article

Author N. Christakis

Author D. Drikakis

Abstract Unsupervised learning methods significantly benefit various practical applications by effectively identifying intrinsic patterns within unlabelled data. However, inherent data noise and uncertainties often compromise model reliability, result interpretability, and the overall effectiveness of unsupervised learning strategies, particularly in complex fields such as biomedical, engineering, and physics research. To address these critical challenges, this study proposes SUN (Stochastic UNsupervised learning), a novel approach that integrates probabilistic unsupervised techniques—specifically Gaussian Mixture Models—into the RUN-ICON unsupervised learning algorithm to achieve optimal clustering, systematically reduce data noise, and quantify inherent uncertainties. The SUN methodology strategically leverages probabilistic modelling for robust classification and detection tasks, explicitly targeting particle dispersion scenarios related to environmental pollution and airborne viral transmission, with implications for minimising public health risks. By combining advanced uncertainty quantification methods and innovative unsupervised denoising techniques, the proposed study aims to provide more reliable and interpretable insights than conventional methods while alleviating issues such as computational complexity and reproducibility that limit traditional mathematical modelling. This research contributes to enhanced trustworthiness and interpretability of unsupervised learning systems, offering a robust methodological framework for handling significant uncertainty in complex real-world data environments. © 2025 by the authors.

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Supervised learning to quantify amyloidosis in whole brains of an alzheimer's disease mouse model acquired with optical projection tomography

Item Type Journal Article**Author** D. Nguyen**Author** V. Uhlmann**Author** A.L. Planchette**Author** P.J. Marchand**Author** D. Van De Ville**Author** T. Lasser**Author** A. Radenovic

Abstract Alzheimer's disease (AD) is characterized by amyloidosis of brain tissues. This phenomenon is studied with genetically-modified mouse models. We propose a method to quantify amyloidosis in whole 5x FAD mouse brains, a model of AD. We use optical projection tomography (OPT) and a random forest voxel classifier to segment and measure amyloid plaques. We validate our method in a preliminary cross-sectional study, where we measure 6136 ± 1637 , 8477 ± 3438 , and 17267 ± 4241 plaques (AVG $\pm$ SD) at 11, 17, and 31 weeks. Overall, this method can be used in the evaluation of new treatments against AD. © 2019 Optical Society of America under the terms of the OSA Open Access Publishing Agreement.

Date 2019**DOI** 10.1364/BOE.10.003041**URL** <https://www.scopus.com/pages/publications/85067984405?origin=resultslist>**Archive** Scopus**Language** English**Volume** 10**Publisher** OSA - The Optical Society**Pages** 3041-3060**Publication** Biomedical Optics Express**ISSN** 21567085 (ISSN)**Issue** 6**Journal Abbr** Biomed. Opt. Express**Date Added** 20/09/2026, 9:52:47 pm**Modified** 20/09/2026, 9:52:47 pm**Tags:**

Article, female, male, Machine learning, algorithm, image segmentation, nonhuman, controlled study, age, artificial neural network, Decision trees, Random forests, random forest, supervised machine learning, Diseases, support vector machine, classifier, diagnostic test accuracy study, receiver operating characteristic, disease exacerbation, diagnostic accuracy, histopathology, image processing, measurement accuracy, Alzheimer disease, sensitivity and specificity, image analysis, animal experiment, animal model, animal tissue, mouse, brain tissue, amyloidosis, Tomography, Brain tissue, amyloid plaque, Amyloid plaques, Cross-sectional study, Mouse brain, Mouse models, Optical projection tomography, Optical projectors, optical tomography, Whole brains

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Supervised machine learning models for accurate prediction of micro pump displacement

Item Type Journal Article

Author F. Abid

Author M. Haddar

Author A. Elhami

Author M. Haddar

Abstract This study presents a supervised learning framework for predicting the maximum membrane displacement of a shape memory alloy (SMA) antagonistic micro pump subjected to material and geometric uncertainties. SMA micro pumps are indeed promising candidates for precise fluid control in microfluidics, but their nonlinear and hysteretic thermomechanical behavior makes parametric analyses and uncertainty evaluation using finite element analysis (FEA) methods particularly computationally expensive. To overcome this difficulty, a high-fidelity dataset comprising 500 design points was generated by three-dimensional simulation using a uniform randomized experimental design. Eight input parameters were investigated: five properties of the NiTi material (Young's modulus E , Poisson's ratio, hardening parameter h , yield strength R , and martensite end temperature M_f) and three geometric dimensions (thickness, membrane radius and spacer length). After preprocessing, five regression algorithms XGBoost, Gradient Boosting, LightGBM, Random Forest and Support Vector Regression were trained and optimized using RandomizedSearchCV and 5-fold cross-validation. Among these, XGBoost offered the best predictive performance, with,,, and across the entire test set. SHAP analysis showed that membrane thickness and radius are the most influential factors corroborating the physical consistency of the learned correlations. Learning curves and residual diagnostics confirmed the robustness of the XGBoost model. The proposed surrogate

model significantly reduces computation time relative to direct finite element methods based uncertainty quantification methods or reliability based design optimization while maintaining high predictive accuracy. These results demonstrate that machine learning based surrogate modeling constitutes an efficient and reliable complement to FEA for the design of SMA micro pumps in advanced microsystems. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026.

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Notes:

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Supervised statistical learning prediction of soybean varieties and cultivation sites using rapid uplc-ms separation, method validation, and targeted metabolomic analysis of 31 phenolic compounds in the leaves

Item Type Journal Article

Author C.-S. Rha

Author E.K. Jang

Author Y.D. Hong

Author W.S. Park

Abstract Soybean (*Glycine max*; SB) leaf (SL) is an abundant non-conventional edible resource that possesses value-adding bioactive compounds. We predicted the attributes of SB based on the metabolomes of an SL using targeted metabolomics. The SB was planted in two cities, and SLs were regularly obtained from the SB plant. Nine flavonol glycosides were purified from SLs, and a validated simultaneous quantification method was used to establish rapid separation by ultrahigh-performance liquid chromatography-mass detection. Changes in 31 targeted compounds were monitored, and the compounds were discriminated by various supervised machine learning (ML) models. Isoflavones, quercetin derivatives, and flavonol derivatives were discriminators for cultivation days, varieties, and cultivation sites, respectively, using the combined criteria of supervised ML models. The neural model exhibited higher prediction power of the factors with high fitness and low misclassification rates while other models showed lower. We propose that a set of phytochemicals of SL is a useful predictor for discriminating characteristics of edible plants. © 2021 by the authors. Licensee MDPI, Basel, Switzerland.

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Notes:

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: C.-S. Rha; AMOREPACIFIC R&D Center, Yongin, 17074, South Korea; email: teaman@amorepacific.com

Support vector machine learning-based cerebral blood flow quantification for arterial spin labeling MRI

Item Type Journal Article

Author Z. Wang

Abstract Purpose: To develop a multivariate machine learning classification-based cerebral blood flow (CBF) quantification method for arterial spin labeling (ASL) perfusion MRI. Methods: The label and control images of ASL MRI were separated using a machine-learning algorithm, the support vector machine (SVM). The perfusion-weighted image was subsequently extracted from the multivariate (all voxels) SVM classifier. Using the same pre-processing steps, the proposed method was compared with standard ASL CBF quantification method using synthetic data and in-vivo ASL images. Results: As compared with the conventional univariate approach, the proposed ASL CBF quantification method significantly improved spatial signal-to-noise-ratio (SNR) and image appearance of ASL CBF images. Conclusion: the multivariate machine learning-based classification is useful for ASL CBF quantification. Hum Brain Mapp 35:2869-2875, 2014. © 2013 Wiley Periodicals, Inc. © 2014 Wiley Periodicals, Inc.

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DOI 10.1002/hbm.22445

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Language English

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adult, Adult, female, Female, human, Humans, male, Male, young adult, Young Adult, brain, Brain, article, algorithm, Algorithms, controlled study, data processing, Support vector machine, learning algorithm, support vector machine, clinical article, image processing, Alzheimer disease, Alzheimer Disease, Image Processing, Computer-Assisted, physiology, image quality, signal noise ratio, cerebrovascular accident, pathology, white matter, Signal-To-Noise Ratio, functional magnetic resonance imaging, brain blood flow, vascularization, arterial spin labeling, perfusion weighted imaging, thickness, cerebrospinal fluid, gray matter, priority journal, Arteries, artery, brain perfusion, Support Vector Machines, Arterial spin labeling, brain circulation, brain mapping, Brain Mapping, Cerebral blood flow, Cerebrovascular Circulation, magnetic resonance angiography, Magnetic Resonance Angiography, moyamoya disease, Moyamoya Disease, spin labeling, Spin Labels, stroke patient

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Export Date: 20 September 2026; Cited By: 26; Funding details: National Institutes of Health, (R03DA023496, RR02305); National Institute on Deafness and Other Communication Disorders, NIDCD, (R21DC011074); National Institute on Deafness and Other Communication Disorders, NIDCD; Correspondence Address: Z. Wang; Treatment Research Center, University of Pennsylvania, Philadelphia, PA 19104, 3900 Chestnut St., United States; email: zewang@mail.med.upenn.edu; CODEN: HBMAE

Supporting Delay-Sensitive IoT Applications: A Machine Learning Approach

Item Type Conference Paper

Author A. Alnoman

Abstract In this paper, a supervised machine learning approach, namely, the decision tree is used to classify IoT applications based on their delay requirements. The decision-tree is trained and tested using simulated datasets to classify tasks into delay-sensitive and delay-insensitive based on the application features such as type and location. Delay-sensitive tasks are generally related to applications such as medical, manufacturing, and connected vehicles that require high service quality and short response time. Once delay-sensitive tasks are recognized, a prioritized scheduling mechanism is implemented to reduce the queueing delay at edge devices. Here, a two-class priority queueing system is used to model the scheduling mechanism at the edge device. Results show the effectiveness of machine learning in identifying delay-sensitive tasks that will experience shorter queueing delay at the edge device to enable high quality edge computing services. © 2020 IEEE.

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Classification (of information), Learning systems, Machine learning, Internet of things, IoT, Supervised learning, Queueing theory, edge computing, Decision trees, Scheduling, Supervised machine learning, Machine learning approaches, decision tree, Simulated datasets, Computing services, delay-sensitive, IOT applications, Priority queueing, Queueing networks, Scheduling mechanism, Short response time, Turing machines

Notes:

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Export Date: 20 September 2026; Cited By: 6; Conference name: 2020 IEEE Canadian Conference on Electrical and Computer Engineering, CCECE 2020; Conference date: 30 August 2020 through 2 September 2020; Conference code: 165133; CODEN: CCCEF

Surface Electromyography Quantification Methods for Evaluating Muscle Activity in Dysphagia

Item Type Journal Article**Author** A.S. Noor**Author** H. Susanti**Author** M.I. Mandasari

Abstract Quantitative evaluation of stroke patients with the risk of swallowing disorder or dysphagia is required to support diagnosis and further rehabilitation planning. Fluoroscopy X-ray imaging usually is used for swallowing diagnosis, though it gives radiation exposure to patients. Therefore, quantification of muscle coordination patterns involved in swallowing based on surface electromyography (sEMG) was introduced. However, an adequate quantification of sEMG for dysphagia diagnosis still lacks standardization. In this work, potential sEMG signal features, namely the contraction duration (DUR), the time to peak of maximum contraction (TTP), and the total RMS power (TP), were further investigated to evaluate the swallowing processes in healthy subjects and post-stroke patients. The experimental scheme instructed the participant, i.e. 20 healthy subjects and 20 patients, to swallow 3 mL of water in normal swallowing mode and swallow saliva in dry swallowing mode. The proposed signal processing procedure helps to establish the feature extraction of the

three features mentioned earlier. For dysphagia assessment, with the support of our proposed signal processing procedure, DUR and TTP can be used together to improve diagnosis reliability. The characteristic of both features in healthy subjects was shorter than in post-stroke patients. Also, the TP feature is useful as additional information to evaluate the role of suprahyoid (SUP) and infrahyoid (INF) muscle groups which are very important in the swallowing process. These results are promising to provide a reliable set of features in the time domain for swallowing analysis. Notably, this can also be utilized as a feature for supporting the automatic classification of dysphagia diagnosis. © 2021 Penerbit Universiti Kebangsaan Malaysia. All rights reserved.

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automation, signal processing, disease treatment, stroke, Dysphagia, muscle, Signals processing, Surface electromyography (sEMG), Swallowing disorder

Notes:

EID: 2-s2.0-85122234311

Export Date: 20 September 2026; Cited By: 0; Funding details: Institut Teknologi Bandung, ITB; Correspondence Address: Suprijanto; Medical Instrumentation Laboratory, Instrumentation and Control Research Group, Faculty of Industrial Technology, Institut Teknologi, Bandung, Indonesia; email: supri@tf.itb.ac.id

Surface parameters and bedrock properties covary across a mountainous watershed: Insights from machine learning and geophysics

Item Type Journal Article

Author S. Uhlemann

Author B. Dafflon

Author H.M. Wainwright

Author K.H. Williams

Author B. Minsley

Author K. Zamudio

Author B. Carr

Author N. Falco

Author C. Ulrich

Author S. Hubbard

Abstract Bedrock property quantification is critical for predicting the hydrological response of watersheds to climate disturbances. Estimating bedrock hydraulic properties over watershed scales is inherently difficult, particularly in fracture-dominated regions. Our analysis tests the covariability of above- and belowground features on a watershed scale, by linking borehole geophysical data, near-surface geophysics, and remote sensing data. We use machine learning to quantify the relationships between bedrock geophysical/hydrological properties and geomorphological/vegetation indices and show that machine learning relationships can estimate most of their covariability. Although we can predict the electrical resistivity variation across the watershed, regions of lower variability in the input parameters are shown to provide better estimates, indicating a limitation of commonly applied geomorphological models. Our results emphasize that such an integrated approach can be used to derive detailed bedrock characteristics, allowing for identification of small-scale variations across an entire watershed that may be critical to assess the impact of disturbances on hydrological systems. Copyright © 2022 The Authors

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DOI 10.1126/sciadv.abj2479

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Notes:

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Export Date: 20 September 2026; Cited By: 29; Correspondence Address: S. Uhlemann; Earth and Environmental Sciences Area, Lawrence Berkeley National Laboratory, Berkeley, 94720, United States; email: suhlemann@lbl.gov

Surrogate model uncertainty quantification for active learning reliability analysis

Item Type Journal Article

Author Y. PANG

Author S. ZHANG

Author P. LIANG

Author M. WANG

Author Z. GONG

Author X. SONG

Author Z. KAN

Abstract Surrogate models offer an efficient approach to tackle the computationally intensive evaluation of performance functions in reliability analysis. Nevertheless, the approximations inherent in surrogate models necessitate the consideration of surrogate model uncertainty in estimating failure probabilities. This paper proposes a new reliability analysis method in which the uncertainty from the Kriging surrogate model is quantified simultaneously. This method treats surrogate model uncertainty as an independent entity, characterizing the estimation error of failure probabilities. Building upon the probabilistic classification function, a failure probability uncertainty is proposed by integrating the difference between the traditional indicator function and the probabilistic classification function to quantify the impact of surrogate model uncertainty on failure probability estimation. Furthermore, the proposed uncertainty quantification method is applied to a newly designed reliability analysis approach termed SUQ-MCS, incorporating a proposed median approximation function for active learning. The proposed failure probability uncertainty serves as the stopping criterion of this framework. Through benchmarking, the effectiveness of the proposed uncertainty quantification method is validated. The empirical results present the competitive performance of the SUQ-MCS method relative to alternative approaches. © 2024

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Sustainability Quantification and Valuation. I: Definitions, Metrics, and Valuations for Decision Making

Item Type Journal Article

Author D. Webb

Author B.M. Ayyub

Abstract The term sustainability has many definitions in varied contexts. The development of metrics requires an examination of these definitions and other related concepts in terms of their hierarchical and nesting relationships. An understanding of these relationships could offer a basis for introducing requirements and ultimately the adoption of working sustainability definitions for construction and manufacturing. A comparative examination of these definitions resulted in recommended general and construction-centric definitions. Sustainability definitions that meet a set of requirements offer appropriate bases for developing metrics and valuations methods. Classification of sustainability indicators follows naturally from its definition. A probabilistic quantification framework is proposed to produce a rational indicator set. Converting indicators to a probability of meeting or exceeding sustainability needs incorporates uncertainty explicitly and puts the quantification method into a well-established mathematical framework. Using simulation, full statistical and probabilistic analyses are available for use on the results in characterizing the uncertainty in the results. The concept of consequences resulting from sustainability impacts establishes economic valuation. Each indicator level has a corresponding loss or gain that can be monetized into an economic consequence. A simulation method is developed that allows for uncertainty characterization under a wide range of dependency assumptions. Indicator levels are simulated and the probability of

meeting or exceeding sustainability needs is calculated along with corresponding consequence values. Then, a final need probability and consequence value for the entire structure or manufacturing process is obtained. The complete result can be analyzed using any statistical or economic techniques desired including benefit-cost, trade-offs, directional cosines, or sensitivity analysis. © 2016 American Society of Civil Engineers.

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DOI 10.1061/AJRUA6.0000893

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Notes:

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Export Date: 20 September 2026; Cited By: 17; Funding details: National Institute of Standards and Technology, NIST; Correspondence Address: D. Webb; Dept. of Mechanical Engineering, Center for Technology and Systems Management, Univ. of Maryland College Park, College Park, 20742, United States; email: dwebb1@umd.edu

Synthesis using fuzzy set theory and a Dempster-Shafer-based approach to compromise decision-making with multiple-attributes applied to risk control options selection

Item Type Journal Article

Author H.S. Sii

Author T. Ruxton

Author J. Wang

Abstract In decision-making there are always some parameters that are inaccurately known, meaning that the decision-making process must be based on subjective opinions. One of the useful aspects of fuzzy set theory is its ability to represent mathematically a class of decision problems called multiple objective decisions, with involvement of many vague (and thus fuzzy) goals and constraints. The objective of the fuzzy set decision methodology is to obtain an optimal decision in the sense that some set of goals is attained while observing (i.e., not violating) a simultaneous set of constraints. This paper presents a decision-making model as applied to multiple-attribute decision-making problems, which may be useful to decision-makers in many real-world problems. The proposed decision-making framework is a synthesis using an approach developed by Zadeh and Bellman in 1970 [1] and later extended by Yager, in 1977 [2], together with the Dempster-Shafer theory of evidence [3]. While dealing with a multiple-attribute decision-making problem the decision-maker may have to compromise with available alternatives, none of which may satisfy exactly his ultimate ideal best. The decision mechanism is constrained by the uncertainty and imprecision inherent in the determination of the relative importance of each attribute element and the classification of existing alternatives. The classification of alternatives is addressed through expert evaluation of the degree to which each attribute element is contained in each available alternative. The implementation of pairwise comparison and a ratio-scale quantification method by the decision-maker is performed to determine the relative importance of each attribute element. Then the 'belief and plausibility' that an alternative will satisfy the decision-maker's ideal 'best' are derived and combined to rank the available alternatives. Two examples in maritime safety engineering are presented to demonstrate the proposed framework.

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Synthesizing Proton-Density Fat Fraction and R2* from 2-Point Dixon MRI with Generative Machine Learning

Item Type Conference Paper**Author** S. Anand**Author** K. Xu**Author** C. O'Dushlaine**Author** S. Mukherjee

Abstract Magnetic Resonance Imaging (MRI) is the gold standard for measuring fat and iron content non-invasively in the body via measures known as Proton Density Fat Fraction (PDFF) and R2*, respectively. However, conventional PDFF and R2* quantification methods operate on MR images voxel-wise and require at least three measurements to estimate three quantities: water, fat, and R2*. Alternatively, the two-point Dixon MRI protocol is widely used and fast because it acquires only two measurements; however, these cannot be used to estimate three quantities voxel-wise. Leveraging the fact that neighboring voxels have similar values, we propose using a generative machine learning approach to learn PDFF and R2* from Dixon MRI. We use paired Dixon-IDEAL data from UK Biobank in the liver and a Pix2Pix conditional GAN [1] to demonstrate the first large-scale R2* imputation from two-point Dixon MRIs. Using our proposed approach, we synthesize PDFF and R2* maps that show significantly greater correlation with ground-truth than conventional voxel-wise baselines. © 2025 IEEE.

Date 2025**DOI** 10.1109/ISBI60581.2025.10981042**URL** <https://www.scopus.com/pages/publications/105005826397?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recogn.**Publisher** IEEE Computer Society**ISBN** 19457928 (ISSN); 979-833152052-6 (ISBN)**Proceedings Title** IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recogn.**Conference Name** Proceedings - International Symposium on Biomedical Imaging**Date Added** 20/09/2026, 9:51:33 pm**Modified** 20/09/2026, 9:51:33 pm

Tags:

machine learning, Machine-learning, Gold standards, Generative model, Quantification methods, Magnetic resonance imaging, Proton density, Neuroimaging, water, fat, Fat contents, generative models, Iron content, liver, Magnetic Resonance Imaging (MRI), MR-images, Two-point

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Systematic Classification of Curvature and Feature Descriptor of 3D Shape and Its Application to “Complexity” Quantification Methods

Item Type Journal Article**Author** K. Matsuyama**Author** T. Shimizu**Author** T. Kato

Abstract Generative design is a system that automates part of the design process, but it cannot evaluate psychological issues related to shapes, such as “beauty” and “liking”. Designers therefore evaluate and choose the generated shapes based on their experience. Among the design features, “complexity” is considered to influence “aesthetic preference”. Although feature descriptors calculated from curvature can be used to quantify “complexity”, the selection guidelines for curvature and feature descriptors have not been adequately discussed. Therefore, this study aimed to conduct a systematic classification of curvature and a feature descriptor of 3D shapes and to apply the results to the “complexity” quantification. First, we surveyed the literature on curvature and feature descriptors and conducted a systematic classification. To quantify “complexity”, we used five curvatures (Gaussian curvature, mean curvature, Casorati curvature, shape index, and curvature index) and a feature descriptor (entropy of occurrence probability) obtained from the classification and compared them with the sensory evaluation values of “complexity”. The results showed that the determination coefficient between the quantified and sensory evaluation values of “complexity” was highest when the mean curvature was used. In addition, the Casorati curvature tended to show the highest signal-to-noise ratio (i.e., a high determination coefficient irrespective of the parameters set in the entropy calculation). These results will foster the development of generative design of 3D shapes using psychological evaluation. © 2023 by the authors.

Date 2023**DOI** 10.3390/e25040624**URL** <https://www.scopus.com/pages/publications/85153959702?origin=resultslist>**Archive** Scopus**Language** English**Volume** 25**Publisher** MDPI**Publication** Entropy**ISSN** 10994300 (ISSN)**Issue** 4**Journal Abbr** Entropy**Date Added** 20/09/2026, 9:52:04 pm

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complexity, entropy, curvature, feature descriptor, generative design

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Systematic review and meta-analysis of automated methods for quantifying enlarged perivascular spaces in the brain

Item Type Journal Article

Author J.M.J. Waymont

Author M.D.C. Valdés Hernández

Author J. Bernal

Author R. Duarte Coello

Author R. Brown

Author F.M. Chappell

Author L. Ballerini

Author J.M. Wardlaw

Abstract Research into magnetic resonance imaging (MRI)-visible perivascular spaces (PVS) has recently increased, as results from studies in different diseases and populations are cementing their association with sleep, disease phenotypes, and overall health indicators. With the establishment of worldwide consortia and the availability of large databases, computational methods that allow to automatically process all this wealth of information are becoming increasingly relevant. Several computational approaches have been proposed to assess PVS from MRI, and efforts have been made to summarise and appraise the most widely applied ones. We systematically reviewed and meta-analysed all publications available up to September 2023 describing the development, improvement, or application of computational PVS quantification methods from MRI. We analysed 67 approaches and 60 applications of their implementation, from 112 publications. The two most widely applied were the use of a morphological filter to enhance PVS-like structures, with Frangi being the choice preferred by most, and the use of a U-Net configuration with or without residual connections. Older adults or population studies comprising adults from 18 years old onwards were, overall, more frequent than studies using clinical samples. PVS were mainly assessed from T2-weighted MRI acquired in 1.5T and/or 3T scanners, although combinations using it with T1-weighted and FLAIR images were also abundant. Common associations researched included age, sex, hypertension, diabetes, white matter hyperintensities, sleep and cognition, with occupation-related, ethnicity, and genetic/hereditary traits being also explored. Despite promising improvements to overcome barriers such as noise and differentiation from other confounds, a need

for joined efforts for a wider testing and increasing availability of the most promising methods is now paramount. © 2024 The Author(s)

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human, Humans, Deep learning, machine learning, brain, Brain, systematic review, algorithm, image segmentation, procedures, deep learning, cognition, Review, Image processing, convolutional neural network, k nearest neighbor, random forest, computer model, support vector machine, sleep, mathematical model, image processing, diagnostic imaging, Image Processing, Computer-Assisted, Magnetic Resonance Imaging, nuclear magnetic resonance imaging, neuroimaging, hypertension, Neuroimaging, meta analysis, geometry, Systematic review, epilepsy, ischemic stroke, automatic image segmentation, Automatic image segmentation, Cerebral small vessel disease, cerebrovascular disease, Cerebrovascular disease, Frangi filter, frangi filtering, genetic disorder, glymphatic system, Glymphatic System, nerve degeneration, perivascular space, Perivascular spaces, Thresholding, Virchow-Robin spaces, visual threshold

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Systematic Review of Quantitative Risk Quantification Methods in Construction Accidents

Item Type Journal Article

Author L. Kumi

Author J. Jeong

Author J. Jeong

Abstract Construction accidents pose significant risks to workers and the public, affecting industry productivity and reputation. While several reviews have discussed risk assessment methods, recent advancements in artificial intelligence (AI), big data analytics, and real-time decision support systems have created a need for an updated synthesis of the quantitative methodologies applied in construction safety. This study systematically reviews the literature from the past decade, following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. A thorough search identified studies utilizing statistical analysis, mathematical modeling, simulation, and artificial intelligence (AI). These methods were categorized and analyzed based on their effectiveness and limitations. Statistical approaches, such as correlation analysis, examined relationships between variables, while mathematical models, like factor analysis, quantified risk factors. Simulation methods, such as Monte Carlo simulations, explored risk dynamics and AI techniques, including machine learning, enhanced predictive modeling, and decision making in construction safety. This review highlighted the strengths of handling large datasets and improving accuracy, but also noted challenges like data quality and methodological limitations. Future research directions are suggested to address these gaps. This study contributes to construction safety management by offering an overview of best practices and opportunities for advancing quantitative risk assessment methodologies. © 2024 by the authors.

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t-RAIN: Robust generalization under weather-aliasing label shift attacks

Item Type Conference Paper

Author A. Marathe

Author S. Prabhu

Abstract In the classical supervised learning settings, classifiers are fit with the assumption of balanced label distributions and produce remarkable results on the same. In the real world, however, these assumptions often bend and in turn adversely impact model performance. Identifying bad learners in skewed target distributions is even more challenging. Thus achieving model robustness under these "label shift" settings is an important task in autonomous perception. In this paper, we analyze the impact of label shift on the task of multi-weather classification for autonomous vehicles. We use this information as a prior to better assess pedestrian detection in adverse weather. We model the classification performance as an indicator of robustness under 4 label shift scenarios and study the behavior of multiple classes of models. We propose t-RAIN a similarity mapping technique for synthetic data augmentation using large scale generative models and evaluate the performance on DAWN dataset. This mapping boosts model test accuracy by 2.1, 4.4, 1.9, 2.7 % in no-shift, fog, snow, dust shifts respectively. We present state-of-the-art pedestrian detection results on real and synthetic weather domains with best performing 82.69 AP (snow) and 62.31 AP (fog) respectively. © 2023 IEEE.

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Tackling the cybersecurity workforce gap with tailored cybersecurity study programs in Central and Eastern Europe

Item Type Conference Paper

Author M. Zivanovic

Author I. Lendák

Author R. Popovic

Abstract Digitalization of society brought improvement in many aspects of life but it also brought new cybersecurity challenges. The number of sophisticated, targeted cyber attacks is increasing, which requires constant improvements in Cybersecurity education. Despite this pressing need, the cybersecurity workforce gap is getting bigger. This paper presents a new approach for dynamic cybersecurity curriculum development that utilizes keyword extraction from various sources such as job ads, courses, and curricula with machine learning to quantify curriculum alignment with cybersecurity industry demands and address the workforce gap. The analysis illustrates curricula in the Central East Europe (CEE) region, maps cyber security job ads to curricula, and quantifies coverage of courses, industry, and reference framework topics based on keyword matching. The case study conducted with curricula from CEE illustrates coverage according to the ENISA's European Cybersecurity Skills Framework (ECSF) roles and optimization progress after adjustment application. The results demonstrate the importance of dynamic curriculum updates for academic institutions including cybersecurity workforce gap reduction and lack of real progress towards alignment with ECSF. © 2024 ACM.

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Cyber security, Cybersecurity, Network security, Personnel, Machine-learning, Cyber-attacks, Curricula, Keywords extraction, Pressung, New approaches, Central and Eastern Europe, Constant improvement, Curriculum development, Cyber-security educations

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Tags and titles of videos you watched tell your gender

Item Type Conference Paper

Author T. Feng

Author Y. Guo

Author Y. Chen

Author X. Tan

Author T. Xu

Author B. Shen

Author W. Zhu

Abstract In online video systems, viewer demographic information (gender, age, etc.) is of huge commercial value for delivering targeted advertising and video recommendations, but generally not available directly. This paper targets inferring viewers' gender based on implicit watching history in the large-scale online video systems. To tackle the sparsity problem without filtering out any cold users or videos, we not only introduce video tags as features, but also use an efficient Chinese word segmentation method to extract hot key-words from video titles as features. Moreover, users' viewing behavior distribute lognormally, hence we apply a logarithmic transformation on the inference matrixes and further find key features via principal components analysis (PCA). We then solve the gender inference as a classification problem and define some modified evaluation metrics adapt to the imbalance classification problem. We compare a set of classifiers including Class prior, EM, SVM, Logistic regression, Partially supervised soft-label and belief-based mixture and find that Logistic regression is the best. The inference results show that our algorithms can obtain high FF1 values for all classes. The highest value of PPTV dataset can reach nearly 0.75. And inference based on key-words results in a 14.63% increase of F1 contrast to the ratings of MovieLens. © 2014 IEEE.

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Tags:

Computational linguistics, Regression analysis, Inference engines, Logistic regression, Online systems, Social sciences, Principal component analysis, Key words, Logistic regressions, PCA, key words, Logarithmic scale, tags

Notes:

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Target Distribution Agnostic Domain Adaptation for in-the-Wild Image Classification under Both Domain and Label Shifts

Item Type Conference Paper

Author A. Zheng

Author J.-N. Hwang

Author R. Hussein

Author F. Wallace

Author K. Magrane

Author L. Shiosaka

Abstract Although significant advancements have been made in Unsupervised Domain Adaptation (UDA), existing methods are typically validated on curated public datasets that do not adequately represent the complexities of real-world applications, such as imbalanced (long-tailed) label distributions and simultaneous domain and label shifts. To bridge this gap, we introduce the Target Distribution Agnostic Adaptation Network (TDAAN), a novel framework designed to facilitate robust adaptation from source to target domains under significant label shifts. Additionally, we present a new domain adaptation dataset,

MS-DA, which focuses on marine species and incorporates natural domain and label shifts, a feature sorely lacking in current domain adaptation research. Our experiments show that TDAAN not only significantly outperforms the baseline UDA method but also surpasses the performance of leading UDA methods on the MS-DA dataset. Remarkably, TDAAN maintains competitive performance on standard UDA benchmarks, proving its efficacy even in scenarios with minimal label shifts. These results position TDAAN as a superior method for UDA, particularly in real-world applications characterized by complex and diverse data distributions. Our code is available at <https://github.com/SEFSC/FATES-ATI-DomainAdaptationLabelShift>. © 2025 IEEE.

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Language English

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Conference Name Proceedings - IEEE International Conference on Multimedia and Expo

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Classification (of information), Benchmarking, Complex networks, Real-world, Signal processing, classification, Domain Knowledge, Domain adaptation, Target domain, Public dataset, domain adaptation, Label distribution, Label shift, Adaptation methods, label shift, Images classification, in-the-wild, In-the-wild, Marine species

Notes:

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Techniques for evaluating the performance of landslide susceptibility models

Item Type Journal Article

Author P. Frattini

Author G. Crosta

Author A. Carrara

Abstract Evaluating the performance of landslide susceptibility models is needed to ensure their reliable application to risk management and land-use planning. When results from multiple models are available, a comparison of their performance is necessary to select the model which performs better. In this paper, different techniques to evaluate model performance are discussed and tested using shallow landslide/debris-flow susceptibility models recently presented in the literature (Carrara, A., Crosta, G.B., Frattini, P., 2008. Comparing models of debris-flow susceptibility in the alpine environment. *Geomorphology* 94 (3-4), 353-378). Moreover, an evaluation technique based on the minimization of costs that may arise from the adoption of the model as a land management regulatory tool, is presented. The results of the application show that simple statistics such as Accuracy, Threat score, Gilbert's skill score, Pierce's skill score, Heidke skill score, and Yule's Q are problematic as they need to split the classified objects into two classes (e.g., stable/unstable) by defining an a-priori value of cutoff susceptibility, which is often not trivial. ROC curves and Success-Rate curves are cutoff-independent and can be used to efficiently visualize and compare the performance of models, but do not explicitly include classification costs. In addition, Success-Rate curves, under certain conditions, can be misleading when applied to grid-cell models. Cost curves include costs and a-priori probabilities, and are suitable for landslide susceptibility model performance evaluation from a practical point of view. © 2009 Elsevier B.V. All rights reserved.

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Risk management, modeling, Costs, Risk analysis, Planning, Accuracy, Model evaluation, statistical analysis, Landslides, Land use, performance assessment, landslide, Landslide susceptibility, ROC curves, Cost curves, Misclassification costs, ROC curve, Cost curve, Debris, debris flow

Notes:

EID: 2-s2.0-75849134535

Export Date: 20 September 2026; Cited By: 439; Correspondence Address: P. Frattini; Dipartimento di Scienze Geologiche e Geotecnologie, Università degli Studi di Milano Bicocca, Milano, Italy; email: paolo.frattini@unimib.it; CODEN: EGGOA

Temporal distribution shift in real-world pharmaceutical data: Implications for uncertainty quantification in QSAR models

Item Type Journal Article

Author H.R. Friesacher

Author E. Svensson

Author S. Winiwarter

Author L. Mervin

Author A. Arany

Author O. Engkvist

Abstract The estimation of uncertainties associated with predictions from quantitative structure–activity relationship (QSAR) models can accelerate the drug discovery process by identifying promising experiments and allowing an efficient allocation of resources. Several computational tools exist that estimate the predictive uncertainty in machine learning models. However, deviations from the i.i.d. setting have been shown to impair the performance of these uncertainty quantification methods. We use a real-world pharmaceutical dataset to address the pressing need for a comprehensive, large-scale evaluation of uncertainty quantification approaches in the context of realistic distribution shifts over time. We investigate the performance of several popular uncertainty estimation methods for classification models, including ensemble-based and Bayesian approaches. Furthermore, we use this real-world setting to systematically assess the distribution shifts in label and descriptor space and their impact on the capability of the uncertainty quantification methods. Our study reveals significant shifts over time in both label and descriptor space and a clear connection between the magnitude of the shift and the nature of the assay. Moreover, we show that pronounced distribution shifts impair the performance of popular uncertainty quantification methods used in QSAR models. This work highlights the challenges of identifying uncertainty quantification techniques that remain reliable under distribution shifts introduced by real-world data. © 2025 The Authors

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Publisher Elsevier B.V.

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machine learning, article, prediction, Drug discovery, probability, uncertainty, calibration, drug development, Uncertainty quantification, quantitative structure activity relation, Probability calibration, Temporal evaluation, Distribution shift

Notes:

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Temporal Test-Time Adaptation with State-Space Models

Item Type Journal Article

Author M. Schirmer

Author D. Zhang

Author E. Nalisnick

Abstract Distribution shifts between training and test data are inevitable over the lifecycle of a deployed model, leading to performance decay. Adapting a model on test samples can help mitigate this drop in performance. However, most test-time adaptation methods have focused on synthetic corruption shifts, leaving a variety of distribution shifts underexplored. In this paper, we focus on distribution shifts that evolve gradually over time, which are common in the wild but challenging for existing methods, as we show. To address this, we propose STAD, a Bayesian filtering method that adapts a deployed model to temporal distribution shifts by learning the time-varying dynamics in the last set of hidden features. Without requiring labels, our model infers time-evolving class prototypes that act as a dynamic classification head. Through experiments on real-world temporal distribution shifts, we show that our method excels in handling small batch sizes and label shift. © 2025, Transactions on Machine Learning Research. All rights reserved.

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Terminal Traffic Situation Prediction Model under the Influence of Weather Based on Deep Learning Approaches

Item Type Journal Article

Author L. Yuan

Author Y. Zeng

Author H. Chen

Author J. Jin

Abstract In order to quantify the degree of influence of weather on traffic situations in real time, this paper proposes a terminal traffic situation prediction model under the influence of weather (TSPM-W) based on deep learning approaches. First, a feature set for predicting traffic situations is constructed based on data such as weather, traffic demand, delay conditions, and flow control strategies. When constructing weather data, a terminal area weather quantification method (TAWQM) is proposed to quantify various weather feature values. When constructing the traffic situation label, fuzzy C-means clustering (FCM) is used to perform cluster analysis on the traffic situation, and the traffic situation is marked as bad, average, or good. Accordingly, the multi-source data is fused as the input vector, based on the combined prediction model of convolutional neural network (CNN) and gated recurrent unit (GRU), TSPM-W is constructed. Finally, based on the historical operation data of the Guangzhou Baiyun International Airport terminal area, the proposed data set is used to predict the traffic situation time series at intervals of 1 h, 3 h, and 6 h. The comparative experimental results show that the proposed time series prediction model has higher prediction accuracy than other existing prediction methods. The proposed dataset is able to more accurately predict the traffic situation in the terminal area. © 2022 by the authors.

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time series prediction of traffic situation, traffic situation classification, traffic situation in terminal area, weather impact

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Test Sample Selection for Handwriting Recognition Through Language Modeling

Item Type Conference Paper

Author A. Rosello

Author E. Ayllon

Author J.J. Valero-Mas

Author J. Calvo-Zaragoza

Editor Pinho A.J.

Editor Georgieva P.

Editor Teixeira L.F.

Editor Sánchez J.A.

Abstract When it comes the need to automatically recognize handwritten information, there may be a scenario in which the capture device (such as the camera of a mobile phone) retrieves several samples—a burst—of the same content. Unlike general classification scenarios, combining different inputs is not straightforward for state-of-the-art handwriting recognition approaches based on neural end-to-end formulations. Moreover, since not all images within the burst may depict enough quality for contributing in the overall recognition task, it is hence necessary to select the appropriate subset. In this work, we propose a pilot study which addresses this scenario in which there exists a burst of pictures of the same manuscript to transcribe. Within this context, we present a straightforward strategy to select a single candidate out of the elements of the burst which achieves the best possible transcription. Our hypothesis is that the best source image is that whose transcription most likely matches a Language Model (LM) estimated on the application domain—in this work implemented by means of n-grams. Our strategy, therefore, processes all the images of the burst and selects the transcription with the highest a priori probability according to the LM. Our experiments recreate this scenario in two typical benchmarks of Handwritten Text Recognition (HTR) and Handwritten Music Recognition (HMR), validating the goodness of our proposal and leaving room for promising future work. © 2022, Springer Nature Switzerland AG.

Date 2022

DOI 10.1007/978-3-031-04881-4_1

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Test-Time Adaptation with Calibration of Medical Image Classification Nets for Label Distribution Shift

Item Type Conference Paper

Author W. Ma

Author C. Chen

Author S. Zheng

Author J. Qin

Author H. Zhang

Author Q. Dou

Editor Wang L.

Editor Dou Q.

Editor Fletcher P.T.

Editor Speidel S.

Editor Li S.

Abstract Class distribution plays an important role in learning deep classifiers. When the proportion of each class in the test set differs from the training set, the performance of classification nets usually degrades. Such a label distribution shift

problem is common in medical diagnosis since the prevalence of disease vary over location and time. In this paper, we propose the first method to tackle label shift for medical image classification, which effectively adapt the model learned from a single training label distribution to arbitrary unknown test label distribution. Our approach innovates distribution calibration to learn multiple representative classifiers, which are capable of handling different one-dominating-class distributions. When given a test image, the diverse classifiers are dynamically aggregated via the consistency-driven test-time adaptation, to deal with the unknown test label distribution. We validate our method on two important medical image classification tasks including liver fibrosis staging and COVID-19 severity prediction. Our experiments clearly show the decreased model performance under label shift. With our method, model performance significantly improves on all the test datasets with different label shifts for both medical image diagnosis tasks. Code is available at <https://github.com/med-air/TTADC>. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Notes:

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Text mining analysis of teaching evaluation questionnaires for the selection of outstanding teaching faculty members

Item Type Journal Article

Author C.-W. Tseng

Author J.-J. Chou

Author Y.-C. Tsai

Abstract This paper was conducted in collaboration with the Office of Institutional Research at National Ilan University in Taiwan to analyze textual opinions found in teaching evaluation questionnaires and applied the analysis results to assisting the selection of outstanding teaching faculty members. The selection of outstanding teachers requires that selection committee members spend a large amount of time reviewing written data. Therefore, this paper develops a set of systems for the analysis of textual opinions in teaching evaluation questionnaires, providing reference materials for the selection committee. The teaching evaluation questionnaire is a form of educational data. This paper analyzes these data using educational data mining. In text mining, text sentiment analysis is a common textual data quantification method that can analyze the sentiment tendency of a text author. This paper uses a text sentiment analysis to quantify the students' textual opinions and to provide the selection committee with the sentiment tendency of students' comments on teaching faculty members. We analyze text sentiment separately for different classifiers by using the Chinese text sentiment analysis kit SnowNLP. We compare the efficacy of classifiers that do not take time series factors into consideration (naïve Bayes and fully connected neural network) to those that do [recurrent neural network (RNN), long short-term memory (LSTM) RNN, and attention RNN]. We found that classifiers that consider time series factors are more effective at analyzing text sentiment. Furthermore, adding LSTM cells and an attention mechanism to a traditional RNN classifier effectively improved its efficacy on long-sequence tasks. As a result, we chose the attention LSTM classifier-with a positive sentiment recognition rate of 97% and a negative sentiments recognition rate of 87%- A s our preferred text sentiment classifier. Finally, we set up an analytics server that will be modularized to facilitate its integration into the systems of different schools. © 2018 IEEE.

Date 2018

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Language English

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Notes:

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Text mining student reports**Item Type** Conference Paper**Author** M.B. Bakaric**Author** M. Matetic**Author** S. Sisovic**Editor** Rachev B.**Editor** 43 Mountain St., Woburn, MA QUERBIE Inc.**Editor** Smrikarov A.**Editor** Department of Computer Systems and Technologies, 8 Studentska St., Ruse University of Ruse

Abstract This paper is a preliminary attempt at gaining useful knowledge from student reports, but also at inspecting their predictive power on grades or course success rate. First a descriptive statistical analysis is done in order to get a basic understanding of patterns found in reports. Then a frequency word analysis and contrastive analysis are done in order to identify the most frequent words per class and those words that occur distinctively between different classes. Prior to the classification task, correlations between the length of the final report and the final grade, and between the scores obtained on each reported activity

individually and the final grade are inspected. Finally, four different classifiers are evaluated on the task of classifying texts based on the type. The same four classifiers are used to determine the predictive power that reports have on course final grades. Copyright © 2015 ACM.

Date 2015

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Language English

Extra Journal Abbreviation: ACM Int. Conf. Proc. Ser.

Volume 1008

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Conference Name ACM International Conference Proceeding Series

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Classification (of information), Data mining, Text processing, Text mining, Classification, Classification tasks, Different class, Predictive power, Contrastive analysis, Word frequencies, Word frequency

Notes:

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Text quantification

Item Type Conference Paper

Author F. Sebastiani

Editor Pardalos P.

Editor Ignatov D.I.

Editor Koltsov S.

Editor Koltsova O.

Editor Volkovich Y.

Editor Markov I.

Editor Braslavski P.

Abstract In a number of applications involving text classification in recent years it has been pointed out that the final goal is not determining which class (or classes) individual unlabeled documents belong to, but determining the prevalence (or “relative frequency”) of each class in the unlabeled data. The latter task is known as text quantification (or prevalence estimation, or class prior estimation). The goal of this course was to introduce the audience to the problem of quantification, techniques that have been proposed for solving it, metrics used to evaluate them, applications in fields such as information retrieval, machine learning, and data mining, and to the open problems in the area. © Springer International Publishing Switzerland 2016.

Date 2016

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Language English

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Notes:

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Textural classification of mammographic parenchymal patterns with the SONNET selforganizing neural network

Item Type Journal Article

Author D. Howard

Author S.C. Roberts

Author C. Ryan

Author A. Brezulianu

Abstract In nationwide mammography screening, thousands of mammography examinations must be processed. Each consists of two standard views of each breast, and each mammogram must be visually examined by an experienced radiologist to assess it for any anomalies. The ability to detect an anomaly in mammographic texture is important to successful outcomes in mammography screening and, in this study, a large number of mammograms were digitized with a highly accurate scanner; and textural features were derived from the mammograms as input data to a SONNET selforganizing neural network. The paper discusses how SONNET was used to produce a taxonomic organization of the mammography archive in an unsupervised manner. This process is subject to certain choices of SONNET parameters, in these numerical experiments using the craniocaudal view, and typically produced $O(10)$, for example, 39 mammogram classes, by analysis of features from $O(10^3)$ mammogram images. The mammogram taxonomy captured typical subtleties to discriminate mammograms, and it is submitted that this may be exploited to aid the detection of mammographic anomalies, for example, by acting as a preprocessing stage to simplify the task for a computational detection scheme, or by ordering mammography examinations by mammogram taxonomic class prior to screening in order to encourage more successful visual examination during screening. The resulting taxonomy may help train screening radiologists and conceivably help to settle legal cases concerning a mammography screening examination because the taxonomy can reveal the frequency of mammographic patterns in a population. Copyright © 2008 Daniel Howard et al.

Date 2008

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female, Female, human, Humans, article, reproducibility, Reproducibility of Results, algorithm, Algorithms, artificial neural network, methodology, sensitivity and specificity, image analysis, computer assisted diagnosis, Breast Neoplasms, breast tumor, breast cancer, Sensitivity and Specificity, mammography, Mammography, Radiographic Image Interpretation, Computer-Assisted, cancer screening, radiography, automated pattern recognition, Pattern Recognition, Automated, computer program, Neural Networks (Computer), computer assisted radiography, digital radiography

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EID: 2-s2.0-48749106814

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: D. Howard; QinetiQ, Malvern, Worcestershire WR14 3PS, St Andrews Road, United Kingdom; email: dr.daniel.howard@gmail.com; CODEN: JBBOA

Textural properties of microglial activation in Alzheimer's disease as measured by (R)-[11C] PK11195 PET

Item Type Journal Article

Author M.L. Pais

Author L. Jorge

Author R. Martins

Author N. Canário

Author A.C. Xavier

Author R. Bernardes

Author A. Abrunhosa

Author I. Santana

Author M. Castelo-Branco

Abstract Alzheimer's disease is the most common form of dementia worldwide, accounting for 60–70% of diagnosed cases. According to the current understanding of molecular pathogenesis, the main hallmarks of this disease are the abnormal accumulation of amyloid plaques and neurofibrillary tangles. Therefore, biomarkers reflecting these underlying biological mechanisms are recognized as valid tools for an early diagnosis of Alzheimer's disease. Inflammatory mechanisms, such as microglial activation, are known to be involved in Alzheimer's disease onset and progression. This activated state of the microglia is associated with increased expression of the translocator protein 18 kDa. On that account, PET tracers capable of measuring this signature, such as (R)-[11C]PK11195, might be instrumental in assessing the state and evolution of Alzheimer's disease. This study aims to investigate the potential of Gray Level Co-occurrence Matrix-based textural parameters as an alternative to conventional quantification using kinetic models in (R)-[11C] PK11195 PET images. To achieve this goal, kinetic and textural parameters were computed on (R)-[11C]PK11195 PET images of 19 patients with an early diagnosis of Alzheimer's disease and 21 healthy controls and submitted separately to classification using a linear support vector machine. The classifier built using the textural parameters showed no inferior performance compared to the classical kinetic approach, yielding a slightly larger classification accuracy (accuracy of 0.7000, sensitivity of 0.6957, specificity of 0.7059 and balanced accuracy of 0.6967). In conclusion, our results support the notion that textural parameters may be an alternative to conventional quantification using kinetic models in (R)-[11C]PK11195 PET images. The proposed quantification method makes it possible to use simpler scanning procedures, which increase patient comfort and convenience. We further speculate that textural parameters may also provide an alternative to kinetic analysis in (R)-[11C]PK11195 PET neuroimaging studies involving other neurodegenerative disorders. Finally, we recognize that the potential role of this tracer is not in diagnosis but rather in the assessment and progression of the diffuse and dynamic distribution of inflammatory cell density in this disorder as a promising therapeutic target. © The Author(s) 2023. Published by

Oxford University Press on behalf of the Guarantors of Brain.

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Export Date: 20 September 2026; Cited By: 1; Funding details: European Regional Development Fund, ERDF; Fundação para a Ciência e a Tecnologia, FCT, (DSAIPA/DS/0041/2020, UID/04950P/2020, CEECINST/00041/2018, UID/04950B/2020); Fundação para a Ciência e a Tecnologia, FCT; Santa Casa Neuroscience Mantero Belard Prize, (MB-1049-2015); BIGDATIMAGE, (CENTRO-01-0145-FEDER-000016); MEDPERSYST, (POCI-01-0145-FEDER-016428); Programa Operacional Temático Factores de Competitividade, POFC, (POCI-01-0145-FEDER-007440); Programa Operacional Temático Factores de Competitividade, POFC; Correspondence Address: M. Castelo-Branco; ICNAS/CIBIT, Pólo das Ciências da Saúde da Universidade de Coimbra Azinhaga de Santa Comba, Coimbra, 3000-548, Portugal; email: mcbranco@fmed.uc.pt

Texture analysis for automated classification of geologic structures

Item Type Conference Paper

Author V. Shankar

Author J.J. Rodriguez

Author M.E. Gettings

Abstract Texture present in aeromagnetic anomaly images offers an abundance of useful geological information for discriminating between rock types, but current analysis of such images still relies on tedious, human interpretation. This study is believed to be the first effort to quantitatively assess the performance of texture-based digital image analysis for this geophysical exploration application. We computed several texture measures and determined the best subset using automated feature selection techniques. Pattern classification experiments measured the ability of various texture measures to automatically predict rock types. The classification

accuracy was significantly better than a priori probability and prior weights-of-evidence results. The accuracy rates and choice of texture measures that minimize the error rate are reported. © 2006 IEEE.

Date 2006

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Language English

Extra Journal Abbreviation: Proc IEEE Southwest Symp Image Anal Interpret

Volume 2006

ISBN 1424400694 (ISBN); 978-142440069-0 (ISBN)

Pages 81-85

Proceedings Title Proc IEEE Southwest Symp Image Anal Interpret

Conference Name Proceedings of the IEEE Southwest Symposium on Image Analysis and Interpretation

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Tags:

Pattern recognition, Textures, Information analysis, Probability, Image analysis, Bit error rate, Rocks, Priori probability, Automated classification, Aeromagnetic anomaly images, Geophysical exploration, Structural geology

Notes:

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Export Date: 20 September 2026; Cited By: 8; Conference name: 7th IEEE Southwest Symposium on Image Analysis and Interpretation; Conference date: 26 March 2006 through 28 March 2006; Conference code: 68549

The analysis of discrete data by machine learning : - Taking momentum in tennis match as an example

Item Type Conference Paper

Author J. Feng

Author Z. Wang

Author J. Xu

Abstract In tennis matches, comebacks often occur, and the existence of rules such as breaking and holding serve increases the probability of such situations. When a comeback happens, there is often a shift or fluctuation in "momentum." However, tennis data is mostly discrete, making it difficult to quantify the fluctuations in "momentum." We will use the random forest algorithm model in machine learning to quantify these fluctuations and assess the main influencing factors of the fluctuations through the method of feature importance index. Subsequently, we will summarize the factors affecting the "momentum" of different athletes and

analyze the data through the comprehensive scoring index method, observing the correlation between "momentum" factors. After that, we will conduct a sensitivity analysis of the model and find that the model remains stable even when adjusting 0-5 factors. Finally, we will provide relevant suggestions based on the influencing factors of "momentum" fluctuations summarized for different athletes. © 2024 IEEE.

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DOI	10.1109/TCS64526.2024.11025325
URL	https://www.scopus.com/pages/publications/105009062282?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: Int. Conf. Inf. Technol. Contemp. Sports, TCS
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	979-833152986-4 (ISBN)
Pages	95-98
Proceedings Title	Int. Conf. Inf. Technol. Contemp. Sports, TCS
Conference Name	2024 4th International Conference on Information Technology and Contemporary Sports, TCS 2024
Date Added	20/09/2026, 9:52:02 pm
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Tags:

Distributed computer systems, Intelligent systems, Learning systems, Machine-learning, Machine learning, Machine Learning, Breakings, Random forests, Sensitivity analysis, System theory, Sensitivity analyzes, Random forest algorithm, Algorithm model, Discrete data, Importance index, Momentum, Momentum factor, Momentum fluctuations, Random Forest Algorithm, Tennis, Volatility

Notes:

EID: 2-s2.0-105009062282

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: J. Feng; Beijing University of Technology, BJUT, Beijing, China; email: Feng230503110@163.com; Conference name: 4th International Conference on Information Technology and Contemporary Sports, TCS 2024; Conference date: 13 December 2024 through 15 December 2024; Conference code: 209516

The Application of Deep Learning to Quantify SAT/VAT in Human Abdominal Area

Item Type	Conference Paper
Author	W. Zgallai
Author	T. Brown
Author	A. Murtada
Author	S. Ali

Author A. Haji
Author K. Khalil
Author M. Omran
Author E.Z. Dalah
Author F.M.E. Al-Islam
Author A.K. Obaideen

Abstract MRI imaging is less risky to humans than CT scans, but is more difficult to extract information from due to the ambiguous gray levels. The objective of the paper is to describe a novel technique that has been developed to automate the measurement of the abdominal subcutaneous adipose tissue (SAT) and visceral adipose tissue (VAT) from MRI images for obese patients before and after fasting. Ground truth has been established with clinicians' inspection. Three hundred thirty images have been utilized in this study where deep learning and Convolutional Neural Networks have been employed to quantify SAT and VAT. This work would minimize the time doctors spend analyzing MRI images. © 2019 IEEE.

Date 2019

DOI 10.1109/ICASET.2019.8714313

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Language English

Extra Journal Abbreviation: Adv. Sci. Eng. Technol. Int. Conf., ASET

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-153868271-5 (ISBN)

Proceedings Title Adv. Sci. Eng. Technol. Int. Conf., ASET

Conference Name 2019 Advances in Science and Engineering Technology International Conferences, ASET 2019

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Tags:

Artificial intelligence, Deep learning, Learning systems, Machine learning, Convolution, Convolutional neural network, Convolutional neural networks, Neural networks, Extract informations, Novel techniques, Ground truth, Magnetic resonance imaging, MRI, Computerized tomography, Adipose tissue, Tissue, MRI Image, Mri imaging, SAT, Subcutaneous adipose tissues, VAT

Notes:

EID: 2-s2.0-85067076525

Export Date: 20 September 2026; Cited By: 6; Conference name: 2019 Advances in Science and Engineering Technology International Conferences, ASET 2019; Conference date: 26 March 2019 through 10 April 2019; Conference code: 148116

The association between estimated glucose disposal rate and self-reported diabetic retinopathy: evidence from two independent cohorts and machine learning

Item Type Journal Article

Author C. Yuan

Author Y. Hao

Author J. Wang

Author C. Wang

Author Z. Jiang

Abstract Background – Insulin resistance plays a key role in the pathogenesis of diabetic retinopathy (DR). Although established insulin resistance markers have been shown to predict a variety of complications, the association between the estimated glucose disposal rate (eGDR) and prevalence of DR remains incompletely characterized. This study aims to examine the relationship between eGDR and DR prevalence. Methods – This cross-sectional study analyzed complete participant data (N = 1, 536) from the 2007–2018 National Health and Nutrition Examination Survey (NHANES) for all relevant information. The relationship between the insulin resistance index and self-reported DR prevalence was evaluated by using multivariate logistic regression and a restricted cubic spline (RCS) model. Subgroup analysis was conducted to assess heterogeneity across groups, and two sensitivity analyses were performed to assess the robustness of the results. In machine learning, the Boruta algorithm is applied for feature selection. The selected features are subsequently utilized by XGBoost and random forest models for DR prevalence estimation. Use the Shapley additive explanations (SHAP) value to explain the independent contribution of eGDR. In the clinical cohort, we recruited patients who visited the Second Affiliated Hospital of Anhui Medical University from September 1, 2025, to December 30, 2025. A total of 297 participants who met the inclusion criteria were finally enrolled. Multivariable logistic regression and RCS curves were used to validate the findings from the NHANES analysis. Results – In the fully adjusted model, eGDR and self-reported DR prevalence show a significant negative linear correlation (OR = 0.79, 95% CI: 0.67–0.93, P = 0.0049). Subgroup and sensitivity analyses confirm the stability of this negative association. The Boruta algorithm identifies eGDR as a robust and important feature. Both the XGBoost (AUC = 0.773) and random forest (AUC = 0.764) models show moderate predictive performance, and eGDR has high variable importance. SHAP analysis indicates that eGDR, together with body mass index and income poverty, is a key determinant of self-reported DR prevalence. The results of the clinical cohort are like NHANES. Conclusion – This cross-sectional study suggested that lower eGDR is associated with a higher prevalence of self-reported DR. Accordingly, eGDR may serve as a potential marker for risk stratification rather than a causal or preventive factor. Prospective longitudinal research is necessary to confirm these findings and to explore whether a causal relationship exists. Copyright © 2026 Yuan, Hao, Wang, Wang and Jiang.

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Language English

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Publication Frontiers in Endocrinology

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Journal Abbr Front. Endocrinol.

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Tags:

adult, Article, female, human, male, middle aged, questionnaire, machine learning, algorithm, correlation analysis, cross-sectional study, major clinical study, prevalence, random forest, sensitivity analysis, area under the curve, diagnostic test accuracy study, obesity, receiver operating characteristic, prospective study, statistical analysis, cohort analysis, body mass, feature selection, Shapley additive explanation, alcohol consumption, cerebrovascular accident, diabetes mellitus, hypertension, ischemic heart disease, smoking, high density lipoprotein cholesterol, triacylglycerol, fasting blood glucose level, angina pectoris, antidiabetic agent, congestive heart failure, diabetic retinopathy, eGDR, estimated glucose disposal rate, hemoglobin A1c, HOMA index, insulin, insulin resistance, insulin sensitivity, multivariate logistic regression analysis, pathogenesis, waist circumference

Notes:

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The changing patterns and correlates of population-level polysubstance use in Australian youth: a multi-group latent class analysis of nationally representative samples spanning 12 years

Item Type Journal Article

Author G. Chan

Author J. Connor

Author W. Hall

Author J. Leung

Abstract Aims: To (1) identify population-level classes of polysubstance use among young Australians between 2004 and 2016, (2) test if these classes changed over the same period, in terms of class prevalence and probabilities of substance use within each class, and (3) identify demographic and health-related correlates of polysubstance use. Design: Repeated cross-sectional nationally representative household surveys. Setting: All Australian states/territories. Participants: Young adult samples (aged 18–30 years; 58% females) from the National Drug Strategy Household Surveys (n = 20 350). Measurements: Outcomes were the extent of past-year use of 10 licit (e.g. alcohol), and illicit substances (e.g. cannabis) were used to derive polysubstance use classes. The correlates were gender, age, psychological distress, general health, language background, personal income, education level, remoteness of residence and socio-economic index for area of residence. Findings: Three polysubstance use classes were consistently identified between 2004 and 2016 (SSABIC: 188349): minimal users (MU, ~60%), mainly tobacco, alcohol and cannabis users (TAC, ~30%) and extended range polysubstance users (POLY, ~10%). There were substantial changes in use of different substances within each class over the study period. For example, smoking decreased in all classes ($P < 0.05$), while harmful alcohol use only decreased in the first two classes ($P < 0.05$). Factors associated with TAC and POLY were similar over the study period. These included: being male and having an English-speaking background, a high level of psychological distress, suboptimal health and high personal income. Living in an affluent area was associated with reduced likelihood of being TAC, but an increased likelihood of being POLY. Conclusion: At the population-level among young Australians between 2004 and 2016, six in 10 did not engage in polysubstance use; four in 10 used a limited range of substances (mainly alcohol, tobacco and cannabis) and one in 10 used an extended range of substances. Over time, the types of substance within the extended polysubstance use class changed substantially. © 2019 Society for the Study of Addiction

Date 2020

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Journal Abbr Addiction

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adult, Adult, female, Female, human, Humans, male, Male, questionnaire, young adult, Young Adult, adolescent, Adolescent, Surveys and Questionnaires, Australia, cross-sectional study, Latent class analysis, classification, Cross-Sectional Studies, drug dependence, Substance-Related Disorders, cannabis, latent class analysis, Latent Class Analysis, smoking, time factor, Time Factors, Cannabis, Alcohol Drinking, drinking behavior, Illicit Drugs, Polydrug, polysubstance, Smoking, socio-economic status, trend, young adults

Notes:

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Export Date: 20 September 2026; Cited By: 28; Funding details: Australian Institute of Health and Welfare; NDSHS; Correspondence Address: G. Chan; Centre for Youth Substance Abuse Research, The University of Queensland, Brisbane, Australia; email: c.chan4@uq.edu.au; CODEN: ADICE

The cost-effectiveness of reclassification sampling for prevalence estimation

Item Type Journal Article

Author A. Bekmetjev

Author D. VanBruggen

Author B. McLellan

Author B. DeWinkle

Author E. Lunderberg

Author N. Tintle

Abstract Background: Typically, a two-phase (double) sampling strategy is employed when classifications are subject to error and there is a gold standard (perfect) classifier available. Two-phase sampling involves classifying the entire sample with an imperfect classifier, and a subset of the sample with the gold-standard. Methodology/Principal Findings: In this paper we consider an alternative strategy termed reclassification sampling, which involves classifying individuals using the imperfect classifier more than one time. Estimates of sensitivity, specificity and prevalence are provided for reclassification sampling, when either one or two binary classifications of each individual using the imperfect classifier are available. Robustness of estimates and design decisions to model assumptions are considered. Software is provided to compute estimates and provide advice on the optimal sampling strategy. Conclusions/Significance: Reclassification sampling is shown to be cost-effective (lower standard error of estimates for the same cost) for estimating prevalence as compared to two-phase sampling in many practical situations. © 2012 Bekmetjev et al.

Date 2012

DOI 10.1371/journal.pone.0032058

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Language English

Volume 7

Publication PLoS ONE**ISSN** 19326203 (ISSN)**Issue** 2**Journal Abbr** PLoS ONE**Date Added** 20/09/2026, 9:52:45 pm**Modified** 20/09/2026, 9:52:45 pm**Tags:**

human, Humans, article, Software, prevalence, classification, support vector machine, Classification, classifier, methodology, cost effectiveness analysis, mathematical analysis, economics, sensitivity and specificity, Prevalence, Sensitivity and Specificity, sampling, cost benefit analysis, Cost-Benefit Analysis, computer program, Support Vector Machines, reclassification sampling, sampling error

Notes:

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Export Date: 20 September 2026; Cited By: 1; Funding details: National Human Genome Research Institute, NHGRI, (R15HG004543); National Human Genome Research Institute, NHGRI; Correspondence Address: N. Tintle; Department of Mathematics, Statistics and Computer Science, Sioux Center, Dordt College, Iowa, United States; email: ntintle@dordt.edu

The course of infection of *Encephalitozoon cuniculi* genotype I in mice possess combination of features reported in genotypes II and III

Item Type Journal Article**Author** B. Sak**Author** K. Brdíčková**Author** N. Holubová**Author** D. Květoňová**Author** L. Hlásková**Author** M. Kváč

Abstract Out of three genotypes of *Encephalitozoon cuniculi* (I–III) available for experimental studies, *E. cuniculi* genotype I remains the less characterized. This study describes for the first time individual phases of microsporidiosis caused by *E. cuniculi* genotype I and efficacy of albendazole treatment in immunocompetent BALB/c and C57Bl/6 mice and immunodeficient SCID, CD4^{−/−} and CD8^{−/−} mice using molecular detection and quantification methods. We demonstrate asymptomatic infection despite an intense dissemination of microsporidia into most organs within the first weeks post infection, followed by a chronic infection characterized by significant microsporidia persistence in immunocompetent, CD4^{−/−} and CD8^{−/−} mice and a lethal outcome for SCID mice. Albendazole application led to loss *E. cuniculi* genotype I infection in immunocompetent mouse strains, decreased spore burden by half in CD4^{−/−} and CD8^{−/−} mice, and prolongation of survival of SCID mice. These results showed *Encephalitozoon cuniculi* genotype I infection extend and albendazole sensitivity was comparable to *E. cuniculi* genotype II, but the infection onset speed

and mortality rate was similar to *E. cuniculi* genotype III. These imply that differences in the course of infection and the response to treatment depend not only on immunological status of the host, but also on the genotype causing the infection. © 2021 Elsevier Inc.

Date 2021

DOI 10.1016/j.exppara.2021.108101

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Article, nonhuman, controlled study, classification, genetics, genotype, polymerase chain reaction, treatment response, animal, Animals, antiinfective agent, Polymerase Chain Reaction, survival, Anti-Infective Agents, animal experiment, animal tissue, C57BL mouse, Mice, Mice, Inbred C57BL, mouse, parasitology, CD8 antigen, CD8+ T lymphocyte, immunology, infection, Genotype, real time polymerase chain reaction, CD4 antigen, CD4+ T lymphocyte, animal cell, mortality rate, Real-Time Polymerase Chain Reaction, priority journal, albendazole, Albendazole, Bagg albino mouse, C57BL 6 mouse, CD4 Antigens, CD8 Antigens, drug sensitivity, Encephalitozoon cuniculi, Encephalitozoon cuniculi genotype I, encephalitozoonosis, Encephalitozoonosis, Experimental infection, feces analysis, immunocompetence, Immunocompetence, knockout mouse, Laboratory mice, Mice, Inbred BALB C, Mice, Knockout, Mice, SCID, RT PCR quantification, SCID mouse, survival time

Notes:

EID: 2-s2.0-85103423580

Export Date: 20 September 2026; Cited By: 3; Funding details: Grantová Agentura České Republiky, GA ČR, (17-12871S, 20-10706S); Correspondence Address: B. Sak; Institute of Parasitology, České Budějovice, Branišovská 31, 37005, Czech Republic; email: casio@paru.cas.cz; CODEN: EXPAA

The double-edged effects of regional environmental governance: evidence from policy tools of belt and road initiative

Item Type Journal Article

Author H. Hu

Abstract Regional environmental governance plays a pivotal role in global climate cooperation; however, the effectiveness of specific policy tools, especially their interactions, remains underexplored. The Belt and Road Initiative (BRI), with over 140 participating countries, provides a unique empirical context to assess how regional environmental governance affects carbon dynamics. Here, construct a novel dataset of 282 BRI agreements (2014–2024) and apply a multi-period difference-in-differences model, complemented by machine learning to quantify causal effects and identify synergistic interactions. Results indicate a dual impact: while BRI environmental governance significantly reduces carbon intensity, it simultaneously increases overall carbon footprints due to infrastructure expansion and trade growth. Further, there is substantial heterogeneity in the effectiveness of individual instruments, which demonstrates that certain policy combinations amplify mitigation in developed regions but exacerbate emissions in developing economies. These findings highlight the urgent need to tailor regional governance frameworks to local contexts and long-term sustainability goals. Our research makes three contributions: (1) it introduces a text-based empirical strategy to identify and classify environmental policy tools embedded in international agreements; (2) it provides the first integrated causal and machine-learning-based assessment of policy heterogeneity and synergy under the BRI framework; and (3) it advances methodological innovation in evaluating transnational environmental governance using multi-period panel data with staggered treatment. © The Author(s) 2025.

Date 2025

DOI 10.1007/s10668-025-06906-w

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Tags:

Belt and road initiative, Policy combinations, Policy tools, Regional environmental governance

Notes:

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Correspondence Address: H. Hu; University of Lodz, Lodz, ul. Prez, prez. Gabriela Narutowicza 68,
90-136, Poland; email: haisheng.hu@edu.uni.lodz.pl; CODEN: EDSNB

The effect of class imbalance on case selection for case-based classifiers, with emphasis on computer-aided diagnosis systems

Item Type Conference Paper

Author J.M. Malof

Author M.A. Mazurowski

Author G.D. Tourassi

Abstract In this paper the effect of class imbalance in the case base of a case-based classifier is investigated as it pertains to case base reduction and the resulting classifier performance. A k-nearest neighbor algorithm is used as a classifier and the Random Mutation Hill Climbing (RMHC) algorithm is used for case base reduction. The effects at various levels of positive class prevalence are tested in a binary classification problem. The results indicate that class imbalance is detrimental to both case base reduction and classifier performance. Selection with RMHC generally improves the classification performance regardless of the case base prevalence. ©2009 IEEE.

Date 2009

DOI 10.1109/IJCNN.2009.5178759

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Language English

Extra Journal Abbreviation: Proc Int Jt Conf Neural Networks

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Learning systems, Neural networks, Class imbalance, Classifiers, Classification performance, Computer aided diagnosis, Mammography, Imbalance, Classifier performance, K nearest neighbor algorithm, Binary classification problems, Case base, Case base reduction, Cased-based learning, Computer-aided decision, Computer-aided diagnosis system, Hill climbing, Random mutation

Notes:

EID: 2-s2.0-70449451136

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: J. M. Malof; Department of Electrical and Computer Engineering, University of Louisville, Louisville, KY 40292, United States; email: jmmalo03@gmail.com; Conference name: 2009 International Joint Conference on Neural Networks, IJCNN 2009; Conference date: 14 June 2009 through 19 June 2009; Conference code: 78384; CODEN: 85OFA

The Effects of Artificial Diets on the Expression of Molecular Marker Genes Related to Honey Bee Health

Item Type Journal Article**Author** O. Frunze**Author** H. Kim**Author** J.-H. Lee**Author** H.-W. Kwon

Abstract Honey bees are commonly used to study metabolic processes, yet the molecular mechanisms underlying nutrient transformation, particularly proteins and their effects on development, health, and diseases, still evoke varying opinions among researchers. To address this gap, we investigated the digestibility and transformation of water-soluble proteins from four artificial diets in long-lived honey bee populations (*Apis mellifera ligustica*), alongside their impact on metabolism and DWV relative expression ratio, using transcriptomic and protein quantification methods. Diet 2, characterized by its high protein content and digestibility, was selected for further analysis from the other studied diets. Subsequently, machine learning was employed to identify six diet-related molecular markers: SOD1, Trxr1, defensin2, JHAMT, TOR1, and vg. The expression levels of these markers were found to resemble those of honey bees who were fed with Diet 2 and bee bread, renowned as the best natural food. Notably, honey bees exhibiting chalkbrood symptoms (Control-N) responded differently to the diet, underscoring the unique nutritional effects on health-deficient bees. Additionally, we proposed a molecular model to elucidate the transition of long-lived honey bees from diapause to development, induced by nutrition. These findings carry implications for nutritional research and beekeeping, underscoring the vital role of honey bees in agriculture. © 2024 by the authors.

Date 2024**DOI** 10.3390/ijms25084271**URL** <https://www.scopus.com/pages/publications/85191366408?origin=resultslist>**Archive** Scopus**Language** English**Volume** 25**Publisher** Multidisciplinary Digital Publishing Institute (MDPI)**Publication** International Journal of Molecular Sciences**ISSN** 16616596 (ISSN)**Issue** 8**Journal Abbr** Int. J. Mol. Sci.**Date Added** 20/09/2026, 9:52:06 pm**Modified** 20/09/2026, 9:52:06 pm**Tags:**

Article, machine learning, agriculture, nonhuman, controlled study, gene expression, nutrition, veterinary medicine, pollen, genetics, animal, Animals, biological marker, Biomarkers, drug effect, functional food, metabolism, animal experiment, principal component analysis, field experiment, transcriptome, Transcriptome, Animal Feed, animal food, apiculture, Apis mellifera, artificial diet, bee, bee bread, Bees, deformed wing virus, diapause, diet, Diet, dietary supplement, digestion, early spring, gene expression regulation, Gene Expression Regulation, gene expression system, honeybee, innate immunity, insect development, insect model, insect protein, Insect Proteins, marker gene, molecular marker, molecular model, nectar, nutritional science, protein content, protein intake, spring, superoxide dismutase, total antioxidant capacity, transcriptomics, tryptophan, water solubility, water-soluble proteins, winter honey bees

Notes:

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The epidemiology of chronic venous insufficiency and varicose veins

Item Type Journal Article

Author J.L. Beebe-Dimmer

Author J.R. Pfeifer

Author J.S. Engle

Author D. Schottenfeld

Abstract Chronic venous disease is a common condition presenting to physicians in Western Europe and the United States. This article provides a comprehensive review of the published literature in the English language, from 1942 to the present, and focuses on the prevalence of chronic venous insufficiency and varicose veins, as well as the involved risk factors. Prevalence estimates vary widely by geographic location, with the highest reported rates in Western countries. Reports of prevalence of chronic venous insufficiency vary from < 1% to 40% in females and from < 1% to 17% in males. Prevalence estimates for varicose veins are higher, <1% to 73% in females and 2% to 56% in males. The reported ranges in prevalence estimations presumably reflect differences in the population distribution of risk factors, accuracy in application of diagnostic criteria, and the quality and availability of medical diagnostic and treatment resources. Established risk factors include older age, female gender, pregnancy, family history of venous disease, obesity, and occupations associated with orthostasis. Yet, there are several factors that are not well documented, such as diet, physical activity and exogenous hormone use, which may be important in the development of chronic venous disease and its clinical manifestations. © 2004 Elsevier Inc. All rights reserved.

Date 2005
DOI 10.1016/j.annepidem.2004.05.015
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Language English
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human, review, risk factor, geographic distribution, prevalence, standing, pregnancy, obesity, Epidemiology, age distribution, disease classification, sex difference, Prevalence, pathogenesis, physical activity, occupation, diet, priority journal, body composition, vein disease, family history, chronic vein insufficiency, Chronic Venous Insufficiency, heredity, Varicose Veins, varicosis

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Export Date: 20 September 2026; Cited By: 888; Correspondence Address: J.L. Beebe-Dimmer; School of Public Health I, Ann Arbor, MI 48109, 109 South Observatory Street, United States; email: jbeebe@med.umich.edu; CODEN: ANNPE

The Gene Taxonomic Prevalence (GeTPrev) Pipeline for Scalable Gene Prevalence Estimation Across Bacterial Taxa

Item Type Journal Article

Author W. Wu

Author A.M. Dickey

Author J.L. Vipham

Author T.M. Arthur

Author J.W. Schmidt

Abstract The National Center for Biotechnology Information (NCBI) stores over 1 million bacterial genome sequences, with no tools capable of estimating the prevalence of specific nucleotide sequences within or across taxa. To address this gap, we developed the Gene Taxonomic Prevalence (GeTPrev) pipeline. GeTPrev estimates the presence of user-specified genes in bacterial genome collections across taxa. Implemented in Bash, GeTPrev integrates BLAST-based sequence alignment with

two operational modes tailored to different analytical needs. The default mode performs a one-pass search against curated complete genome databases formatted for BLAST. A “Heavy” mode expands the search to include both complete and draft genomes for a broader representation of genomic diversity. GeTPrev is managed through a Conda environment and designed for compatibility with high-performance computing (HPC) systems, enabling efficient batch analysis of large genome datasets. GeTPrev supports the construction of user-defined gene taxonomic targets. However, prebuilt complete genome databases of seven Enterobacteriaceae genera are included with the pipeline to support rapid analysis. GeTPrev functionality and flexibility were demonstrated by nine example applications. GeTPrev offers a practical solution for gene-centric analysis in microbial genomics, molecular epidemiology, and food safety surveillance. © 2026

Date 2026

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Journal Abbr J. Food Protection

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Tags:

Article, nonhuman, taxonomy, prevalence, classification, information processing, bioinformatics, Bacteria, bacterium, genetics, genomics, sequence alignment, High-throughput analysis, genetic variability, Escherichia coli, Staphylococcus aureus, bacterial genome, Bacterial genomes, bacterial taxonomy, basic local alignment search tool, Bioinformatics pipeline, Databases, Genetic, default mode network, Enterobacter, Enterobacteriaceae, food safety monitoring, gene, gene centric analysis, Gene prevalence, gene taxonomic prevalence pipeline, genetic analysis, genetic database, Genome, Bacterial, genomic diversity, Klebsiella pneumoniae, large genome dataset, microbial genomics, molecular epidemiology, pipeline, Salmonella enterica, Salmonella enterica serovar Typhimurium, scalable gene prevalence estimation, Staphylococcus epidermidis, tetracycline resistance, ubiquitin protein ligase E3

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EID: 2-s2.0-105040960511

Export Date: 20 September 2026; Cited By: 0; Funding details: U.S. Department of Agriculture, USDA; Correspondence Address: W. Wu; United States Meat Animal Research Center, United States Department of Agriculture Agricultural Research Service, Clay Center, 844 Road 313, 68933, United States; email: weifan.wu@ufl.edu; CODEN: JFPRD

The Groups and Nilpotent Lie Rings of Order p^8 with Maximal Class

Item Type Journal Article

Author S. Lee

Author M. Vaughan-Lee

Abstract We classify and count the nilpotent Lie rings of order p^8 with maximal class for (Formula presented.). This also provides a classification of the groups of order p^8 with maximal class for (Formula presented.) via the Lazard correspondence. We also record the number of nilpotent Lie rings/groups of order p^n with maximal class for (Formula presented.) from currently known data and discuss its asymptotic behavior as n grows and its potential connection to Higman's PORC conjecture. © 2022 Taylor & Francis Group, LLC.

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DOI 10.1080/10586458.2022.2062074

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Publication Experimental Mathematics

ISSN 10586458 (ISSN)

Issue 2

Journal Abbr Exp. Math.

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Tags:

Higman's PORC conjecture, Lie p -ring, maximal class, p -group

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EID: 2-s2.0-85132658386

Export Date: 20 September 2026; Cited By: 1; Funding details: National Research Foundation of Korea, NRF; MEST, (2019R1A6A1A10073437); Correspondence Address: S. Lee; The Research Institute of Basic Sciences, Seoul National University, Seoul, 08826, South Korea; email: seungjai.lee@snu.ac.kr

The Hitchhiker's Guide to Prior-Shift Adaptation

Item Type Conference Paper

Author T. Sipka

Author M. Sulc

Author J. Matas

Abstract In many computer vision classification tasks, class priors at test time often differ from priors on the training set. In the case of such prior shift, classifiers must be adapted correspondingly to maintain close to optimal performance. This paper analyzes methods for adaptation of probabilistic classifiers to new priors and for estimating new priors on an unlabeled test set. We propose a novel method to address a known issue of prior estimation methods based on confusion matrices, where inconsistent estimates of decision probabilities and confusion matrices lead to negative values in the estimated priors. Experiments on fine-grained image classification datasets provide insight into the best practice of prior shift estimation and classifier adaptation, and show that the proposed method achieves state-of-the-art results in prior adaptation. Applying the best practice to two tasks with naturally imbalanced priors, learning from web-crawled images and plant species classification, increased the recognition accuracy by 1.1% and 3.4% respectively. © 2022 IEEE.

Date 2022

DOI 10.1109/WACV51458.2022.00209

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Extra Journal Abbreviation: Proc. - IEEE/CVF Winter Conf. Appl. Comput. Vis., WACV

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Proceedings Title Proc. - IEEE/CVF Winter Conf. Appl. Comput. Vis., WACV

Conference Name Proceedings - 2022 IEEE/CVF Winter Conference on Applications of Computer Vision, WACV 2022

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Tags:

Deep learning, Classification (of information), Computer vision, Object detection, Objects detection, Deep Learning, Image resolution, Semi-supervised learning, Few-shot, Un-supervised learning, Evaluation and comparison of vision algorithm, Evaluation and Comparison of Vision Algorithms, Learning dataset, Object detection/recognition/categorization, Object Detection/Recognition/Categorization, Semi- and un- supervised learning dataset, Semi- and Un- supervised Learning Datasets, Transfer, Vision algorithms, Web crawler

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The impact of the #MeToo movement on language at court A text-based causal inference approach

Item Type Journal Article

Author H. Langen

Abstract This study assesses the effect of the #MeToo movement on the language used in judicial opinions on sexual violence related cases from 51 U.S. state and federal appellate courts. The study introduces various indicators to quantify the extent to which actors in courtrooms employ language that implicitly shifts responsibility away from the perpetrator and onto the victim. One indicator measures how frequently the victim is mentioned as the grammatical subject, as research in the field of psychology suggests that victims are assigned more blame the more often they are referred to as the grammatical subject. The other two indices designed to gauge the level of victim-blaming capture the sentiment of and the context in sentences referencing the perpetrator. Additionally, judicial opinions are transformed into bag-of-words and tf-idf vectors to facilitate the examination of the evolution of language over time. The causal effect of the #MeToo movement is estimated by means of a Difference-in-Differences approach comparing the development of the language in opinions on sexual offenses and other crimes against persons as well as a Panel Event Study approach. The results do not clearly identify a #MeToo-movement-induced change in the language in court but suggest that the movement may have accelerated the evolution of court language slightly, causing the effect to materialize with a significant time lag. Additionally, the study considers potential effect heterogeneity with respect to the judge's gender and political affiliation. The study combines causal inference with text quantification methods that are commonly used for classification as well as with indicators that rely on sentiment analysis, word embedding models and grammatical tagging. © 2024 Henrika Langen. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Article, female, Female, human, Humans, male, Male, psychology, United States, social media, language, society, controlled study, natural language processing, sentiment analysis, public opinion, Language, embedding, word processing, politics, word embedding, victim, Sex Offenses, sexual assault, sexual crime, sexual violence, incidence, court, crime victim, Crime Victims, criminal justice, difference in differences approach, me too movement, social support, text vectorization, US justice system, victim blaming

Notes:

EID: 2-s2.0-85193274737

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Langen; Department of Economics, University of Fribourg, Fribourg, Switzerland; email: rikalangen@t-online.de; CODEN: POLNC

The implementation of intelligent systems in automating vehicle detection on the road

Item Type Journal Article

Author D.D. Budiarjo

Author A. Hendrawan

Author P.T. Pungkasanti

Abstract Indonesia is a country with a high population, especially in big cities. The road always crowded with various types of vehicles. Sometimes the growth of vehicles is not matched by road construction. During peak hours, too many vehicles can cause traffic jams on the road. The road is needed to be widened to accommodate the number of vehicles that pass each day. In order for road widening to be precise at locations that frequently occur in traffic jams, data on the number and classification of vehicles passing is required. Therefore, a system that can calculate and recognize the type of vehicle that passes is needed. The development of various studies on artificial intelligence especially about object detection can classify and calculate the type of vehicle. In this study, the authors used the YOLO object detection system using a convolution neural network (CNN) method to classify and count vehicles that pass automatically. The author uses a dataset of 600 images with 4 classes which are car, truck, bus, and motorbikes that pass through the road. The results showed that the YOLO object detection system can recognize objects consistently with accuracy more than 80% on CCTV video that installed on the road. © 2021, Institute of Advanced Engineering and Science. All rights reserved.

Date 2021

DOI 10.11591/ijai.v10.i3.pp571-575

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Publisher Institute of Advanced Engineering and Science

Pages 571-575**Publication** IAES International Journal of Artificial Intelligence**ISSN** 20894872 (ISSN)**Issue** 3**Journal Abbr** IAES Int. J. Artif. Intell.**Date Added** 20/09/2026, 9:52:37 pm**Modified** 20/09/2026, 9:52:37 pm**Tags:**

Artificial intelligence, Object detection, Convolutional neural network

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: A. Hendrawan; Department of Information Technology and Communication, Universitas Semarang, Indonesia; email: ariahendrawan@usm.ac.id

The importance of the test set size in quantification assessment

Item Type Conference Paper**Author** A. Maletzke**Author** W. Hassan**Author** D. dos Reis**Author** G. Batista**Editor** Bessiere C.

Abstract Quantification is a task similar to classification in the sense that it learns from a labeled training set. However, quantification is not interested in predicting the class of each observation, but rather measure the class distribution in the test set. The community has developed performance measures and experimental setups tailored to quantification tasks. Nonetheless, we argue that a critical variable, the size of the test sets, remains ignored. Such disregard has three main detrimental effects. First, it implicitly assumes that quantifiers will perform equally well for different test set sizes. Second, it increases the risk of cherry-picking by selecting a test set size for which a particular proposal performs best. Finally, it disregards the importance of designing methods that are suitable for different test set sizes. We discuss these issues with the support of one of the broadest experimental evaluations ever performed, with three main outcomes. (i) We empirically demonstrate the importance of the test set size to assess quantifiers. (ii) We show that current quantifiers generally have a mediocre performance on the smallest test sets. (iii) We propose a meta-learning scheme to select the best quantifier based on the test size that can outperform the best single quantification method. © 2020 Inst. Sci. inf., Univ. Defence in Belgrade. All rights reserved.

Date 2020**URL** <https://www.scopus.com/pages/publications/85097344225?origin=resultslist>**Archive** Scopus

Language English

Extra Journal Abbreviation: IJCAI Int. Joint Conf. Artif. Intell.

Volume 2021-January

Publisher International Joint Conferences on Artificial Intelligence

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Proceedings Title IJCAI Int. Joint Conf. Artif. Intell.

Conference Name IJCAI International Joint Conference on Artificial Intelligence

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Artificial intelligence, Classification (of information), Metalearning, Training sets, Quantification methods, Class distributions, Performance measure, Critical variables, Designing methods, Experimental evaluation

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The incidence of different removable partial denture cases related to age, gender and arch in a distinct population in Iraq

Item Type Journal Article

Author M.M. Refaat

Author N.A. Al-Shareeda

Author A.M. Hussain

Abstract Background: The classification of partially dentate cases allow identification the relation of teeth to edentulous ridges combinations, it investigate the partially edentulism pattern. Kennedy's classification system permit immediate visualization and recognition of different edentulous spaces in the dental arch. Aims of the study: Assessment the prevalence of the partially edentulous patients in Basrah city/south of Iraq and determines the correlation with age, arches and gender. Materials and Methods: A cross-sectional study is applied for 1000 partially edentulous male and female patients above 20 years age. Data collection based on face to face interview and clinical examination. Results: According to the collected data, Kennedy's cases of Class III were the most popular for both genders. The least was Class IV being less common pattern, its prevalence for males more than females. Maxilla is more than mandible in partially edentulism cases. Conclusion: Kennedy most common

classification for both maxilla and mandible dentalarches was Class III. The various classes prevalence not affected by gender, while patient age had a significant effect.
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Tags:

adult, Article, female, human, male, aged, city, correlation analysis, cross-sectional study, major clinical study, prevalence, interview, population, sex difference, incidence, groups by age, clinical examination, edentulousness, Iraq, Kennedy classification, mandible, maxilla, Missing teeth, Tooth-tissue supported

Notes:

EID: 2-s2.0-85092634957

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A.M. Hussain; College of Dentistry, University of Basrah, Basrah, Iraq; email: abdullah.hussain@uobasrah.edu.iq

The influence of building energy performance prediction accuracy on retrofit rates

Item Type Journal Article

Author C. Wiethe

Author S. Wenninger

Abstract The residential building sector bears enormous CO2 emission reduction potential. However, uncertainty connected to financial savings of retrofits, stemming largely from inaccurate building energy performance predictions, forms a significant investment barrier. So far, literature does not sufficiently cover the relationship between prediction accuracy and retrofit rates. Thus, this paper aims to provide guidance for policymakers by setting up an agent-based building stock model to derive this relationship for the German residential building sector assuming rational decision-making. Results indicate that higher prediction accuracies positively affect the retrofit rate. Using data-driven prediction methods established in research

significantly increases the retrofit rate by over 70%, from around 0.98% – 1.68% compared to the legally prescribed engineering method. This equals CO2 emission reductions of almost 45 Mt by 2050 for Germany, leading to a surplus in consumer rent of 310.19 mn €, while investments in retrofits increase by about 1.195 bn €. The government benefits from tax payments and saves opportunity costs. The findings allow to estimate the effect of revising current legislation on building energy performance predictions and thereby clearly guide policymakers. © 2023 Elsevier Ltd

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Journal Abbr Energy Policy

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Notes:

EID: 2-s2.0-85151256557

Export Date: 20 September 2026; Cited By: 10; Correspondence Address: C. Wiethe; Branch Business & Information Systems Engineering of the Fraunhofer FIT, Augsburg, 86159, Germany; email: christian.wiethe@fim-rc.de; CODEN: ENPYA

The influence of SAR image quantization method on detection precision

Item Type Conference Paper

Author B. Sun

Author Z. Zuo

Author P. Wang

Abstract In this paper, we combine deep learning with radar image processing to explore the influence of different quantization methods on the final detection performance of the radar image subjected to strong points after different quantification methods. Considering problems caused by the characteristics of SAR image data, the LeNet network model in deep learning was used to train and verify the quantified radar images respectively. The impact of different quantization methods on SAR image classification and detection was analyzed. The most friendly way to quantify the actual radar images was explored. Radar image target detection based on depth learning provides the basis for exploration. © 2018 IEEE.

Date 2018

DOI 10.1109/IGARSS.2018.8518783

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Language English

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Conference Name International Geoscience and Remote Sensing Symposium (IGARSS)

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Tags:

Deep learning, Synthetic aperture radar, Detection, Quantisation, Quantification methods, Detection performance, Radar imaging, Tracking radar, Quantitative method, Quantitative methods, Detection precision, Image compression, Image quantization, Radar image processing, SAR Image, SAR Images

Notes:

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The land use-climate change-biodiversity nexus in the perceptions of European islands stakeholders

Item Type Journal Article

Author A. Moustakas

Author I. Christoforidi

Author G. Zittis

Author N. Demirel
Author M. Fois
Author S. Zotos
Author E. Gallou
Author V. Stamatiadou
Author E. Tzirkalli
Author C. Zoumides
Author K. Košić
Author A. Christopoulou
Author A. Dragin
Author D. Łowicki
Author A. Gil
Author B. Almeida
Author P. Chrysos
Author M.V. Balzan
Author M.D.C. Mansoldo
Author R. Ólafsdóttir
Author C.K. Ayhan
Author L. Atay
Author M. Tase
Author V. Stojanović
Author M.M. Ladičorbić
Author J.P. Díaz
Author F.J. Expósito
Author S. Quiroga
Author M.Á.C. Cano
Author H. Wang
Author C. Suárez
Author P. Manolaki
Author I.N. Vogiatzakis

Abstract To promote climate adaptation and mitigation strategies, it is crucial to understand the perspectives and knowledge gaps of stakeholders involved in functions affected by or addressing land use and climate changes. A large number of stakeholders across 21 European islands were consulted regarding their views on climate change and land use change issues affecting ecosystem services on their island. Climate change characteristics perceptions included variables such as temperature, precipitation, humidity, extremes, and wind. Land use change characteristics perceptions included deforestation, coastal degradation, habitat protection, renewable energy facilities, wetlands and other variables. Other environmental and societal problem perceptions such as invasive species, water or energy scarcity, problems in infrastructures or austerity were also included. Climate and land use change impact perceptions were analysed with machine learning to quantify their importance on the perception outcome. For example if a stakeholder perceives that pollution, coastal degradation, deforestation, precipitation decrease, and increase of humidity are occurring on the island, and austerity is the biggest problem how likely is that the impact of climate change or land use change will be quantified by the stakeholder as negative, unclear, neutral, or positive? The predominant climatic change characteristic is related with temperature, and the predominant land use change characteristic with deforestation.

Water-related problems are top priorities for stakeholders. Energy-related problems, such as energy deficiency but also wind and solar energy facilities problems, rank high as combined climate change and land use change risks. Stakeholders generally perceive climate change impacts on ecosystem services as negative, with natural habitat destruction and biodiversity loss identified as the top variables. Land use change impacts are also negative but also more complex to explain, with a higher number of explanatory variables associated with the impact outcome. Stakeholders have common perceptions regarding climate change and land use change impacts on the benefits of biodiversity despite the geographic disparity. Stakeholders differentiate between factors related to climate change impacts and land use change impacts. Water, energy, and renewable energy related issues pose serious concerns to island stakeholders and management measures are needed to address them. © 2025 The Authors

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The long road to calibrated prediction uncertainty in computational chemistry

Item Type Journal Article**Author** P. Pernot

Abstract Uncertainty quantification (UQ) in computational chemistry (CC) is still in its infancy. Very few CC methods are designed to provide a confidence level on their predictions, and most users still rely improperly on the mean absolute error as an accuracy metric. The development of reliable UQ methods is essential, notably for CC to be used confidently in industrial processes. A review of the CC-UQ literature shows that there is no common standard procedure to report or validate prediction uncertainty. I consider here analysis tools using concepts (calibration and sharpness) developed in meteorology and machine learning for the validation of probabilistic forecasters. These tools are adapted to CC-UQ and applied to datasets of prediction uncertainties provided by composite methods, Bayesian ensembles methods, and machine learning and a posteriori statistical methods. © 2022 Author(s).

Date 2022**DOI** 10.1063/5.0084302**URL** <https://www.scopus.com/pages/publications/85126908485?origin=resultslist>**Archive** Scopus**Language** English**Volume** 156**Publisher** American Institute of Physics Inc.**Publication** Journal of Chemical Physics**ISSN** 00219606 (ISSN)**Issue** 11**Journal Abbr** J Chem Phys**Date Added** 20/09/2026, 9:52:29 pm**Modified** 20/09/2026, 9:52:29 pm**Tags:**

machine learning, Machine learning, Forecasting, Uncertainty analysis, article, prediction, validation process, Analysis tools, Quantification methods, uncertainty, Uncertainty quantifications, calibration, mean absolute error, meteorology, Computational chemistry, Prediction uncertainty, Confidence levels, Mean absolute error, Calibrated prediction, cheminformatics, Common standards, Industrial processes, infancy, Standard procedures

Notes:

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Export Date: 20 September 2026; Cited By: 29; Correspondence Address: P. Pernot; Institut de Chimie Physique, UMR8000 CNRS, Université Paris-Saclay, Orsay, 91405, France; email: pascal.pernot@universite-paris-saclay.fr; CODEN: JCPSA

The Magnitude and Impact of Food Allergens and the Potential of AI-Based Non-Destructive Testing Methods in Their Detection and Quantification

Item Type Journal Article

Author A.A. Adedeji

Author P.V. Priyesh

Author A.A. Odugbemi

Abstract Reaction to food allergens is on the increase and so is the attending cost on consumers, the food industry, and society at large. According to FDA, the “big-eight” allergens found in foods include wheat (gluten), peanuts, egg, shellfish, milk, tree nuts, fish, and soybeans. Sesame was added to the list in 2023, making the target allergen list nine instead of eight. These allergenic foods are major ingredients in many food products that can cause severe reactions in those allergic to them if found at a dose that can elicit a reaction. Defining the level of contamination that can elicit sensitivity is a work in progress. The first step in preventing an allergic reaction is reliable detection, then an effective quantification method. These are critical steps in keeping contaminated foods out of the supply chain of foods with allergen-free labels. The conventional methods of chemical assay, DNA-PCR, and enzyme protocols like enzyme-linked immunosorbent assay are effective in allergen detection but slow in providing a response. Most of these methods are incapable of quantifying the level of allergen contamination. There are emerging non-destructive methods that combine the power of sensors and machine learning to provide reliable detection and quantification. This review paper highlights some of the critical information on the types of prevalent food allergens, the mechanism of an allergic reaction in humans, the measure of allergenic sensitivity and eliciting doses, and the conventional and emerging AI-based methods of detection and quantification—the merits and downsides of each type. © 2024 by the authors.

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The Mechanism and Empirical Study on the Empowerment of Chinese Cultural Product Exports through the Digital Economy under the RCEP Institutional Framework

Item Type Conference Paper

Author H. Zhou

Author Q. Wu

Abstract Based on the panel data of China and RCEP partner countries from 2007 to 2022, this paper constructs a digital economy measurement system by employing principal component analysis, a core dimensionality reduction algorithm in machine learning, to quantify the level of digital economy development. The extended trade gravity model is used to analyse the impact of digital economy development on China's cultural product exports. The study leverages core computer technologies, including big data analytics for processing trade and consumption data, and incorporates discussions on the roles of AI-driven content adaptation and blockchain-based transaction systems. The study finds that: the development of the digital economy in RCEP partner countries has a significant positive effect on China's cultural product exports. This underscores the critical role of underlying digital infrastructure and data-driven decision-making. The heterogeneity test shows that the improvement of digital economy level in developing countries has a more prominent role in promoting China's cultural product exports, suggesting that algorithm and platform design should consider regional disparities. Accordingly, it puts forward policy suggestions such as strengthening intelligent digital infrastructure and AI-powered content adaptation tools, establishing blockchain-based copyright and transaction mechanisms, focusing on digital cultural exports to RCEP developing country markets, participating in formulating international technical standards for digital cultural assets, and building a national-level cloud-native digital cultural export platform, so as to promote the high-quality development of China's cultural product export trade and provide insights for the development of computational economic models. © 2025 Copyright held by the owner/author(s).

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Tags:

Economic and social effects, Big data, Block-chain, Blockchain, Product design, Dimensionality reduction, Learning systems, Artificial Intelligence, Learning algorithms, International trade, Digital economy, Data analytics, Big data analytics, Principal component analysis, Advanced Analytics, Big data analytic, Content adaptation, Cultural product export, Cultural product exports, Developing countries, Digital infrastructures, Partner country, Product exports, RCEP

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Zhou; Guangzhou Xinhua University, Guangzhou, Guangdong, China; email: zhouhuimin@xhsysu.edu.cn; Conference name: 2025 2nd International Conference on Digital Economy and Computer Science, DECS 2025; Conference date: 17 October 2025 through 19 October 2025; Conference code: 216832

The Music of Rivers: The Mathematics of Waves Reveals Global Structure and Drivers of Streamflow Regime

- Item Type** Journal Article
- Author** B.C. Brown
 - Author** A.H. Fulerton
 - Author** D. Kopp
 - Author** F. Tromboni
 - Author** A.J. Shogren
 - Author** J.A. Webb
 - Author** C. Ruffing
 - Author** M. Heaton
 - Author** L. Kuglerová
 - Author** D.C. Allen
 - Author** L. McGill
 - Author** J.P. Zarnetske
 - Author** M.R. Whiles
 - Author** J.B. Jones
 - Author** B.W. Abbott

Abstract River flows change on timescales ranging from minutes to millennia. These vibrations in flow are tuned by diverse factors globally, for example, by dams suppressing multi-day variability or vegetation attenuating flood peaks in some

ecosystems. The relative importance of the physical, biological, and human factors influencing flow is an active area of research, as is the related question of finding a common language for describing overall flow regime. Here, we addressed both topics using a daily river discharge data set for over 3,000 stations across the globe from 1988 to 2016. We first studied similarities between common flow regime quantification methods, including traditional flow metrics, wavelets, and Fourier analysis. Across all these methods, the flow data showed low-dimensional structure (i.e., simple and consistent patterns), suggesting that fundamental mechanisms are constraining flow regime. One such pattern was that day-to-day variability was negatively correlated with year-to-year variability. Additionally, the low-dimensional structure in river flow data correlated closely with only a small number of catchment characteristics, including catchment area, precipitation, and temperature—but notably not biome, dam surface area, or number of dams. We discuss these findings in a framework intended to be accessible to the many communities engaged in river research and management, while stressing the importance of letting structure in data guide both mechanistic inference and interdisciplinary discussion. © 2023. American Geophysical Union. All Rights Reserved.

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machine learning, Machine-learning, Machine learning, Information management, Stream flow, Flood control, Rivers, flood, River flow, streamflow, Eco-hydrology, ecohydrology, catchment, Catchments, Flow data, Flow regimes, Fourier analysis, Global structure, Low dimensional structure, river discharge, river flow, Runoff, streamflow regime, Streamflow regimes, Timeserie, timeseries, wavelet, Wavelet, wavelet analysis

Notes:

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The optimal classifier for disease diagnosis by roc analysis

Item Type Journal Article

Author T. Zhang

Author W. Guo

Author G. Wang

Author X. Wang

Author Y. Zhang

Author W. Hong

Abstract Accuracy is commonly used as an index to evaluate the performance of classifiers. However, accuracy has several deficiencies, such as the sensitivity to class prior distribution and misclassification costs. Especially in the disease diagnosis, the classifier selected according to accuracy is not necessarily the optimal classifier. While the Receiver Operating Characteristic Curve (ROC) and the Area Under ROC (AUC) analysis technology are insensitive to the class distribution and different misclassification cost. Besides, they have advantages of direct-viewing, and easy to understand. This experiment is testing and analyzing the UCI datasets (Parkinson, diabetes, liver-disorder, breast-cancer) using different classifiers (LDC, KNNC, SVC), and comparing with the general indicator, to find out the optimal classifier to each group of data sets. At the same time, experiment shows the advantages of ROC and AUC in the evaluation of medical decision-making. © 2013 ISSN 2185-2766.

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Accuracy, Optimization, Diagnosis, Classifiers, Cost benefit analysis, Prior distribution, Class distributions, Classifier, Medical decision making, Receiver operating characteristic curves, Misclassification costs, Performance of classifier, Experiments, Optimal classifiers, Receiver operating characteristic curve (roc)

Notes:

EID: 2-s2.0-84883236608

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: T. Zhang; College of Information Engineering, Yanshan University, Qinhuangdao 066004, No. 438, Hebei Ave, China; email: zhtao@ysu.edu.cn

The potential for fuel reduction to reduce wildfire intensity in a warming California

Item Type Journal Article

Author P.T. Brown

Author S.J. Strenfel

Author R.B. Bagley

Author C.B. Clements

Abstract Increasing fuel aridity due to climate warming has and will continue to increase wildfire danger in California. In addition to reducing global greenhouse gas emissions, one of the primary proposals for counteracting this increase in wildfire danger is a widespread expansion of hazardous fuel reductions. Here, we quantify the potential for fuel reduction to reduce wildfire intensity using empirical relationships derived from historical observations with a novel combination of spatiotemporal resolution (0.375 km, instantaneous) and extent (48 million acres, 9 years). We use machine learning to quantify relationships between sixteen environmental conditions (including ten fuel characteristics and four temperature-affected aridity characteristics) and satellite-observed fire radiative power. We use the derived relationships to create fire intensity potential (FIP) maps for sixty historical weather snapshots at a 2 km and hourly resolution. We then place these weather snapshots in differing background climatological temperature and fuel characteristic conditions to quantify their independent and combined influence on FIP. We find that in order to offset the effect of climate warming under the SSP2-4.5 emissions scenario, fuel reduction would need to be maintained perpetually on ~3 million acres (or 600 000 acres per year, 1% of our domain, at a 5 year return frequency) by 2050 and ~8 million acres (or 1.6 million acres per year, 3% of our domain, at a 5 year return frequency) by 2090. Overall, we find substantial potential for fuel reduction to negate the effects of climate warming on FIP. © 2025 The Author(s). Published by IOP Publishing Ltd.

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California, United States, machine learning, climate change, wildfire, aridity, climate modeling, Climate modeling, Climate warming, environmental conditions, Fire intensity, fuel aridity, Fuel aridity, fuel consumption, Fuel reduction, global warming, greenhouse gas, Greenhouse gas emissions, hazard assessment, Hazardous fuel reduction, hazardous fuels reductions, mechanical thinning, Mechanical thinning, Premixed flames, prescribed burning, Prescribed burning, Residual fuels, wildfire intensity, Wildfire intensity

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The potential of wastewater monitoring as a novel surveillance tool for early warning of Bordetella pertussis outbreaks

Item Type Journal Article

- Author S. Fu
- Author X. Du
- Author Z. Xu
- Author H. Li
- Author H. Yao
- Author S. Tang
- Author W. Zhao
- Author M. Hao
- Author Z. Qiu
- Author Z. Wang

Abstract The prolonged latency period and delayed reporting of clinical surveillance data have led to a sluggish response to Bordetella pertussis outbreaks in China. Wastewater-based surveillance (WBS) provides real-time, cost-effective monitoring of various viruses. To confirm whether this approach can be applied to B. pertussis, 271 patients infected with B. pertussis were enrolled to evaluate the fecal shedding patterns of the bacterium. Concurrently, we collected wastewater samples in a typical northern (Xi'an) and southern city (Nanchang) in China, from September 2023 to July 2024. The concentrations of B. pertussis in the sewage were detected using quantitative polymerase chain reaction (qPCR). Additionally, metagenomic sequencing with pre-enrichment was employed to trace genotypes of B. pertussis. We report that 40.2% of cases tested positive for B. pertussis in fecal samples. Sewage surveillance identified two waves of B. pertussis infection in Xi'an and one wave of B. pertussis outbreak in

Nanchang. The concentration of *B. pertussis* in sewage showed a strong correlation with the dynamics of diagnosed cases. Notably, each peak of sewage *B. pertussis* concentration occurred 10 days prior to clinical surveillance, underscoring the value of WBS in early warning. Additionally, both clinical and wastewater surveillance demonstrated consistency in the prevalence estimation of *B. pertussis* infection cases. Finally, relative to clinical surveillance, wastewater sequencing also revealed the same genotypes of *B. pertussis*. This study presents the first comprehensive analysis of the fecal shedding of *B. pertussis*, thereby establishing the potential for early detection of *B. pertussis* outbreaks through WBS. © 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group, on behalf of Shanghai Shangyixun Cultural Communication Co., Ltd.

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The prevalence of traumatic experiences and PTSD according to DSM-5 and ICD-11 in the German general population

Item Type Journal Article

Author A. Pettrich

Author Y. Nesterko

Author H. Glaesmer

Abstract Aims The Diagnostic and Statistical Manual of Mental Disorders - 5th Edition (DSM-5) and International Classification of Diseases - 11th Revision (ICD-11) employ different post-traumatic stress disorder (PTSD) criteria, necessitating updated prevalence estimates. Most of the existing evidence is still based on ICD-Tenth Revision and DSM-Fourth Edition criteria, leading to varied estimates across populations. This study provides current PTSD prevalence rates in the German general population, comparing DSM-5 and ICD-11 criteria and examines variations by age and gender. Methods In a 2016 cross-sectional survey of 2404 adults (18-94years) representative of the German general population, participants completed the Life-Events-Checklist for DSM-5 (LEC-5) for trauma exposure and the PTSD Checklist for DSM-5 (PCL-5) for PTSD symptoms. Probable PTSD diagnoses were based on DSM-5-, ICD-11-algorithms and suggested cut-off scores. Chi-square and McNemar's tests were used to test differences in prevalence rates by diagnostic framework, age and gender. Results Of the total sample, 47.2% (n=1135) reported experiencing at least one lifetime traumatic event (TE), with transportation accidents (7.3%) and life-threatening injuries (4.9%) being most common. Probable PTSD prevalence was 4.7% under both DSM-5 and ICD-11 criteria, and 2.6% based on a conservative cut-off normed for prevalence estimation. Gender and age were not significantly associated with TE exposure or PTSD prevalence, though trauma types varied: female participants more often reported sexual violence and severe suffering, while more male participants reported physical assaults and various types of accidents. DSM-5 and ICD-11 diagnostic algorithms had substantial yet not perfect agreement ($\kappa=0.62$). Particularly within the re-experiencing symptoms, cluster agreement was only moderate ($\kappa=0.57$). The cut-off method aligned more closely with DSM-5 ($\kappa=0.60$) than ICD-11 algorithm ($\kappa=0.42$). Conclusions This study provides updated PTSD prevalence estimates for the German general population and underscores differences between DSM-5 and ICD-11 in identifying cases, particularly with respect to re-experiencing symptoms. These findings emphasize that while overall PTSD prevalence rates under DSM-5 and ICD-11 criteria are similar, the diagnostic frameworks identify partially distinct cases, reflecting differences in

symptom definitions. This highlights the need to carefully consider the impact of evolving diagnostic criteria when interpreting prevalence estimates and comparing results across studies. © The Author(s), 2025. Published by Cambridge University Press.

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Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: A. Pettrich; Department of Medical Psychology and Medical Sociology, University of Leipzig, Leipzig, Germany; email: Amelie.Pettrich@medizin.uni-leipzig.de

The pulse waveform quantification method basing on contour and derivative

Item Type Journal Article

Author J. Jin

Author H. Zhang

Author X. Geng

Author Y. Zhang

Author T. Ye

Abstract Objective: Pulse waveform contains abundant physiological and pathological information. The condition of surrounding arteries can be reflected sensitively by the contour and derivative changes of pulse waves. In order to express these changes objectively, the pulse wave needs to be quantified. Methods: This study provides a novel quantification method for pulse waveform in the entire cardiac cycle. It involves two new quantification parameters k_1 and k_2 to display the waveform change caused by the superimposition of wave reflection in the systolic reflex period, which is the most significant changes period. In this method, multi parameters were fused by Kalman filter to obtain an optimal estimation, involving the new parameters and other parameters: k_0 for the early systolic period, C_1 and C_2 for diastole period, and K for pulse pressure. Results: Use correlation analysis to verify the effectiveness of new parameters that the coefficient is 0.7 between them and the typical augmentation index (AIx). The quantification results of 462 single-cycle pulse waves have consistent change trends with aging in 25–75 different age groups. For respiration analysis, the correlation coefficients are all greater than 0.6, even achieved 0.8 in six multi-cycle data between Kalman optimal estimation and breath wave. Conclusion: This method has quantified the waveform change with physiological status, and these quantification parameters can display the detail of each period. Significance: It will be used to verify waveform recognition accuracy and has a vast potential to detect diseases. © 2022 The Author(s)

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Article, human, Condition, machine learning, Physiological models, Kalman filters, procedures, controlled study, Kalman filter, Quantification methods, physiology, breathing rate, systole, diastolic blood pressure, Quantification method, Heart Rate, heart rate, Cardiac cycles, heart cycle, pulse wave, diastole, collagen, aortic root, arterial pressure, Arteries, artery, artery blood flow, artery compliance, artery resistance, augmentation index, blood pressure, Blood Pressure, curve fitting, Derivative, Diastole, elastin, heart ventricle contraction, New parameters, Optimal estimations, pulse pressure, Pulse wave, Pulse Wave Analysis, pulse wave velocity, Pulse waveforms, Pulse-wave, Quantification parameter, Quantification parameters, radial artery, systemic vascular resistance, Systole, vascular resistance, vascular smooth muscle, Wave reflections, waveform, Waveform changes

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The research on improved iterative control algorithm for maximum entropy model in electronic technology

Item Type Conference Paper

Author W. Yongqin

Author G. Yinjing

Author W. Na

Author Z. Rui

Abstract In the distributed processing, where common labeled data may be not available for designing classifier ensemble, however, an ensemble solution is necessary, traditional fixed decision aggregation could not account for class prior mismatch or classifier dependencies in electronic technology. Previous transductive learning strategies have several drawbacks, e.g., feasibility of the constraints was not guaranteed and heuristic learning was applied. We overcome these problems by developing improved iterative scaling (IIS) algorithm for optimal solution. This method is shown to achieve improved decision accuracy over the earlier approaches in electronic technology. © 2012 Springer-Verlag Berlin Heidelberg.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: W. Yongqin; College of Information and Electrical Engineering, Shan Dong University Science and Technology, Qingdao Economic and Technological Development Zone, Jinan, Qianwan Port Road Number 579, China; email: ieee2010@foxmail.com; Conference name: 2012 International Conference on Mechanical and Electronic Engineering, ICMEE 2012; Conference date: 23 June 2012 through 24 June 2012; Conference code: 93050

The Robust Semantic Segmentation UNCV2023 Challenge Results

Item Type	Conference Paper
Author	X. Yu
Author	Y. Zuo
Author	Z. Wang
Author	X. Zhang
Author	J. Zhao
Author	Y. Yang
Author	L. Jiao
Author	R. Peng
Author	X. Wang
Author	J. Zhang
Author	K. Zhang
Author	F. Liu
Author	R. Alcover-Couso
Author	J.C. Sanmiguel
Author	M. Escudero-Viñolo
Author	H. Tian
Author	K. Matsui
Author	T. Wang
Author	F. Adan
Author	Z. Gao
Author	X. He
Author	Q. Bouniot
Author	H. Moghaddam
Author	S.N. Rai
Author	F. Cermelli
Author	C. Masone
Author	A. Pilzer
Author	E. Ricci
Author	A. Bursuc
Author	A. Solin
Author	M. Trapp

Author R. Li
Author A. Yao
Author W. Chen
Author I. Simpson
Author N.D.F. Campbell
Author G. Franchi

Abstract This paper outlines the winning solutions employed in addressing the MUAD uncertainty quantification challenge held at ICCV 2023. The challenge was centered around semantic segmentation in urban environments, with a particular focus on natural adversarial scenarios. The report presents the results of 19 submitted entries, with numerous techniques drawing inspiration from cutting-edge uncertainty quantification methodologies presented at prominent conferences in the fields of computer vision and machine learning and journals over the past few years. Within this document, the challenge is introduced, shedding light on its purpose and objectives, which primarily revolved around enhancing the robustness of semantic segmentation in urban scenes under varying natural adversarial conditions. The report then delves into the top-performing solutions. Moreover, the document aims to provide a comprehensive overview of the diverse solutions deployed by all participants. By doing so, it seeks to offer readers a deeper insight into the array of strategies that can be leveraged to effectively handle the inherent uncertainties associated with autonomous driving and semantic segmentation, especially within urban environments. © 2023 IEEE.

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Extra Journal Abbreviation: Proc. - IEEE/CVF Int. Conf. Comput. Vis. Workshops, ICCVW

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The Robustness of the gray level co-occurrence matrices and x-ray computed tomography method for the quantification of 3d mineral texture

Item Type Journal Article

Author M. Voigt

Author J.A. Miller

Author A.N. Mainza

Author L.C. Bam

Author M. Becker

Abstract Mineral textural quantification methods have become critical in both geosciences and mineral processing as mineral texture is a critical factor contributing to ore variability. However, the lack of objective mineral texture classification has made quantification difficult. The aim of this study is therefore to investigate the robustness of applying the gray level co-occurrence matrices (GLCM) to 3-dimensional (3D) gray scale images measured by X-ray computed tomography (XCT) for the quantification of mineral texture in 3D. The data quality of the GLCM outputs like statistics, heat maps and histograms in response to changes in XCT conditions such as artefacts, resolution, and calibration was tested. The response of the GLCM outputs with respect to different mineral texture types with anisotropic features and inter-sample variability was also explored. The methodology included testing core sizes of 26, 19, 14, and 6 mm diameter. Calibration was tested using copper and tungsten wires. The study demonstrated the versatility of the method for different sample types. Inter-sample calibration and optimal scanning conditions (quality and integrity) were also demonstrated, and a basic link between the 3D GLCM statistical descriptors with the mineral texture features of rocks was established. The 3D mineral texture method can potentially bypass the XCT segmentation process for direct automation of 3D mineral texture information. © 2020 by the authors. Licensee MDPI, Basel, Switzerland.

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The role of prior and optimal team decision in binary classification

Item Type Conference Paper

Author K. Krishnamoorthy

Author M. Pachter

Abstract Optimal team decision making subject to error-prone team members with different capabilities has been studied extensively - particularly in the context of binary classification. The over-arching goal is to correctly classify an object as either being a True Target or a False Target. Each team member comes with pre-specified Type I and II error rates and is asked whether or not he determines the object to be a True Target. Based on the member responses, a collective decision that minimizes the probability of misclassification (or some other suitable metric) is made. This is a widely researched topic, having applications in pattern recognition, organizational decision making, social (dichotomous) choice situations, reliability studies etc.; however, in past work, the obvious connection to information theory is missing. In this work, we establish the optimal team decision rules that minimizes the probability of misclassification, by direct application of Bayesian decision theory. In doing so, we bring out the key role played by the a priori probability (prior) that the object is a True Target. In particular, for a homogeneous team composition, we establish the criteria whence the majority voting scheme is indeed optimal. Whereupon, it immediately follows that the higher the prior, the fewer the number of affirmative votes needed to classify the object as a True Target. © 2016 American Automatic Control Council (AACC).

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Decision making, Pattern recognition, Decision theory, Probability, Binary classification, Information theory, Reliability theory, A-priori probabilities, Bayesian decision theory, Probability of misclassification, Collective decision, Organizational decision making, Team composition, Type i and ii errors

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K. Krishnamoorthy; InfoSciTex Corporation, AFRL Contractor, Dayton, 45431, United States; email: krishnak@ucla.edu; Conference name: 2016 American Control Conference, ACC 2016; Conference date: 6 July 2016 through 8 July 2016; Conference code: 123142; CODEN: PRACE

The Role of Uncertainty Quantification for Trustworthy AI

Item Type Journal Article

Author J. Deuschel

Author A. Foltyn

Author K. Roscher

Author S. Scheele

Abstract The development of AI systems involves a series of steps, including data acquisition and preprocessing, model selection, training, evaluation, and deployment. However, each of these steps involves certain assumptions that introduce inherent uncertainty, which can result in inaccurate outcomes and reduced confidence in the system. To enhance confidence and comply with the EU AI Act, we recommend using Uncertainty Quantification methods to estimate the belief in the correctness of a model's output. To make these methods more accessible, we provide insights into the possible sources of uncertainty and offer an overview of the different available methods. We categorize these methods based on when they are used in the process, accounting for various application requirements. We distinguish between three types: data-based, architecture-modifying and post-hoc methods, and share our personal experiences with each.

Date 2024

DOI 10.1007/978-3-031-64832-8_5

URL <https://www.scopus.com/pages/publications/85211889843?origin=resultslist>

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Language English
Publisher Springer Nature
Pages 95-116
Publication Unlocking Artificial Intelligence: From Theory to Applications
ISSN 978-303164832-8 (ISBN); 978-303164831-1 (ISBN)
Journal Abbr Unlocking Artificial Intelligence: From Theory to Applications
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Tags:

Machine Learning, Trustworthy AI, Uncertainty Quantification, Safety Critical Systems

Notes:

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The solid component within part-solid nodules: 3-dimensional quantification, correlation with the malignant grade of nonmucinous pulmonary adenocarcinomas, and comparisons with 2-dimentional measures and semantic features in low-dose computed tomography

Item Type Journal Article

Author J. Liu

Author C. Xie

Author Y. Li

Author H. Xu

Author C. He

Author H. Qing

Author P. Zhou

Abstract Background: There is no consensus on 3-dimensional (3D) quantification method for solid component within part-solid nodules (PSNs). This study aimed to find the optimal attenuation threshold for the 3D solid component proportion in low-dose computed tomography (LDCT), namely the consolidation/tumor ratio of volume (CTRV), basing on its correlation with the malignant grade of nonmucinous pulmonary adenocarcinomas (PAs) according to the 5th edition of World Health Organization classification. Then we tested the ability of CTRV to predict high-risk nonmucinous PAs in PSNs, and compare its performance with 2-dimensional (2D) measures and semantic features. Methods: A total of 313 consecutive patients with 326 PSNs, who underwent LDCT within one month before surgery and were pathologically diagnosed with nonmucinous PAs, were retrospectively enrolled and were divided into training and testing cohorts according to scanners. The CTRV were automatically generated by setting a series of attenuation thresholds from – 400 to 50

HU with an interval of 50 HU. The Spearman's correlation was used to evaluate the correlation between the malignant grade of nonmucinous PAs and semantic, 2D, and 3D features in the training cohort. The semantic, 2D, and 3D models to predict high-risk nonmucinous PAs were constructed using multivariable logistic regression and validated in the testing cohort. The diagnostic performance of these models was evaluated by the area under curve (AUC) of receiver operating characteristic curve. Results: The CTRV at attenuation threshold of -250 HU (CTRV- 250HU) showed the highest correlation coefficient among all attenuation thresholds ($r = 0.655$, $P < 0.001$), which was significantly higher than semantic, 2D, and other 3D features (all $P < 0.001$). The AUCs of CTRV- 250HU to predict high-risk nonmucinous PAs were 0.890 (0.843–0.927) in the training cohort and 0.832 (0.737–0.904) in the testing cohort, which outperformed 2D and semantic models (all $P < 0.05$). Conclusions: The optimal attenuation threshold was - 250 HU for solid component volumetry in LDCT, and the derived CTRV- 250HU might be valuable for the risk stratification and management of PSNs in lung cancer screening. © 2023, The Author(s).

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adult, Article, female, human, Humans, male, Semantics, quantitative analysis, correlation analysis, prediction, procedures, controlled study, major clinical study, semantics, logistic regression analysis, intermethod comparison, diagnostic test accuracy study, correlation coefficient, receiver operating characteristic, single blind procedure, World Health Organization, cohort analysis, Retrospective Studies, retrospective study, diagnostic imaging, sensitivity and specificity, three-dimensional imaging, human tissue, volumetry, pathology, cancer grading, computer assisted tomography, radiation dose, Tomography, X-Ray Computed, x-ray computed tomography, cancer classification, cancer risk, early cancer diagnosis, Early Detection of Cancer, Adenocarcinoma of Lung, bronchography, cancer screening, consolidation tumor ratio of volume, lobulation, low dose computed tomography, Low-dose computed tomography, lung adenocarcinoma, lung consolidation, Lung Neoplasms, lung nodule, lung tumor, Malignant grade, margin, nonmucinous pulmonary adenocarcinoma, oncological parameters, part solid nodule, Part-solid nodule, pleural indentation, preoperative period, Pulmonary adenocarcinoma, respiratory tract parameters, semantic feature, shape, solid component, Solid component, spiculation, two dimensional measure, vacuole sign, vascular convergence sign

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The Story behind the Mask: A Narrative Review on Hypomimia in Parkinson’s Disease

Item Type Journal Article

Author E. Bianchini

Author D. Rinaldi

Author M. Alborghetti

Author M. Simonelli

Author F. D’Audino

Author C. Onelli

Author E. Pegolo

Author F.E. Pontieri

Abstract Facial movements are crucial for social and emotional interaction and well-being. Reduced facial expressions (i.e., hypomimia) is a common feature in patients with Parkinson’s disease (PD) and previous studies linked this manifestation to both motor symptoms of the disease and altered emotion recognition and processing. Nevertheless, research on facial motor impairment in PD has been rather scarce and only a limited number of clinical evaluation tools are available, often suffering from poor validation processes and high inter- and intra-rater variability. In recent years, the availability of technology-enhanced quantification methods of facial movements, such as automated video analysis and machine learning application, led to increasing interest in studying hypomimia in PD. In this narrative review, we summarize the current knowledge on pathophysiological hypotheses at the basis of hypomimia in PD, with particular focus on the association between reduced facial expressions and emotional processing and analyze the current evaluation tools and management strategies for this symptom, as well as future research perspectives. © 2024 by the authors.

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human, machine learning, review, anxiety, emotion, facial recognition, Review, videorecording, depression, knowledge, outcome assessment, visual information, clinical feature, Positive and Negative Syndrome Scale, psychotherapy, emotion recognition, clinical evaluation, neuroimaging, pathophysiology, facial expression, recognition, degenerative disease, REM sleep, electromyography, motor dysfunction, Parkinson's disease, amimia, amygdala, apathy, aphasia, blinking, bradykinesia, donepezil, emotional network, face, facial bradykinesia, facial expressions, fluorescence activated cell sorting, hypomimia, levodopa, Montreal cognitive assessment, mood disorder, motor control, neuromuscular function, neuropsychological assessment, Parkinson disease, physiotherapy, social interaction, speech therapy, Unified Parkinson Disease Rating Scale, wellbeing

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Export Date: 20 September 2026; Cited By: 27; Correspondence Address: F.E. Pontieri; Department of Neuroscience, Mental Health and Sensory Organs (NESMOS), Sapienza University of Rome, Rome, 00189, Italy; email: francesco.pontieri@uniroma1.it

The structure of dual Grassmann codes

Item Type Journal Article

Author P. Beelen

Author F. Piñero

Abstract In this article we study the duals of Grassmann codes, certain codes coming from the Grassmannian variety. Exploiting their structure, we are able to count and classify all their minimum weight codewords. In this classification the lines lying on the Grassmannian variety play a central role. Related codes, namely the affine Grassmann codes, were introduced more recently in Beelen et al. (IEEE Trans Inf Theory 56(7):3166–3176, 2010), while their duals were introduced and studied in Beelen et al. (IEEE Trans Inf Theory 58(6):3843–3855, 2010). In this paper we also classify and count the minimum weight codewords of the dual affine Grassmann codes. Combining the above classification results, we are able to show that the dual of a Grassmann code is generated by its minimum weight codewords. We use these properties to establish that the increase of value of successive generalized Hamming weights of a dual Grassmann code is 1 or 2. © 2015, Springer Science+Business

Media New York.

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DOI 10.1007/s10623-015-0085-3

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The use of detailed biotope data for linking biodiversity with ecosystem services in Finland

Item Type Journal Article

Author P. Vihervaara

Author T. Kumpula

Author A. Ruokolainen

Author A. Tanskanen

Author B. Burkhard

Abstract It has been widely accepted that ecosystem services (ESs) should be taken into account in natural resource management decisions. Hence, there is an increasing need for innovative quantification methods and tools to evaluate ESs on different landscape scales, and under varying land-use forms. Integrating biodiversity

protection with the provision of ESs is a key element for sustainable land-use planning. Geographic Information Systems (GIS) and spatial analysis, together with various environmental data, provide a suitable foundation for ESs evaluations. Recent advances in earth observation technologies have supported land-cover-based ESs mapping on global, regional and local scales. Global and regional land-cover maps can help in coarse assessments of some biophysical characteristics of the environment, but they cannot provide exact information about local biodiversity and biotope types that form the base of ESs supply. Therefore, more detailed tools such as aerial photographs and field surveys are needed. High-quality biotope data are usually fragmentary or absent for private land in Finland, but are available for most state-owned commercial forests and protection areas. We tested the use of biotope data derived from aerial photographs and an extensive field inventory to map ESs in natural protection areas in northern Finland. We argue that protection areas, where large and long-term databases have been collected, offer excellent study sites to develop further the methodology for integrating coarse-scale remote-sensing data, such as CORINE, with more detailed ecological and structural data collected from aerial photographs and ground surveys. In addition, the use of detailed biotope data supports the linkage of biodiversity information with landscapes' capacities to provide ESs. Different data sources will improve the management of protection areas, thereby optimizing multiple land-use objectives. © 2012 Taylor & Francis.

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satellite data, innovation, ecosystem service, biodiversity, Finland, GIS, land use change, habitat, resource management, nature reserve, biotope classification, biotype, conservation, ecosystem service mapping, environmental management, habitat conservation, natural resource

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The use of machine learning for inverse uncertainty quantification in TRACE code based on Marviken experiment

Item Type Journal Article

Author P. Domitr

Author M. Włostowski

Abstract Inverse Uncertainty Quantification (IUQ) methods are gaining recognition in the application of Best Estimate Plus Uncertainty methodologies. Typically, the code input parameters constituted the sources of uncertainties that cannot be estimated other than by user expertise. The IUQ methods aim at providing a more robust and scientific estimation of the distributions of those parameters. In this paper, an approach has been proposed to use machine learning to quantify the uncertainty of critical flow parameters available in the TRACE code. Serving as the means to the ultimate goal of IUQ, the machine learning algorithm has also been utilized in the qualitative assessment of discrepancies between the experiment and calculations. First, a set of Monte Carlo calculations were performed for selected Marviken Critical Flow Tests with permutated values of input parameters of TRACE code. Then, utilizing a supervised machine learning model and a set of features describing the mass flow progression, a Random Forest classifier was used to assign the results to one of the defined classes of prediction accuracy. This analysis has been done using two different training datasets to present the influence of the training data selection on the classification. The machine learning classification resulted in creating a class for which the results with the best fit to the experiment were collected. Derivation of this class created an opportunity to propose ranges and probability distribution functions for CHM12 and CHM22 TRACE internal variables, which is a goal for IUQ methods. Lognormal distribution has been proposed for CHM12 with a mode equal to 0.87, while normal distribution was proposed for CHM22 with a mean equal to 0.9. Variables C1RC2 and C2RC2 proved to be uninfluential on the results. The PDFs for CHM12 and CHM22 have been verified on all 26 Marviken tests. The article proposes a method that can be used in best-estimate analyses to obtain reliable distributions of input variables. © 2021 Elsevier B.V.

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Journal Abbr Nucl Eng Des

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Distribution functions, Classification (of information), Machine learning, Uncertainty analysis, Learning algorithms, Supervised learning, Uncertainty, Inverse problems, Decision trees, Quantification methods, Uncertainty quantifications, Input parameter, Parameter estimation, Normal distribution, Sources of uncertainty, Critical flow, Inverse uncertainty quantification, TRACE, Best estimates, Marviken

Notes:

EID: 2-s2.0-85117084759

Export Date: 20 September 2026; Cited By: 6; Correspondence Address: P. Domitr; Warsaw University of Technology, Institute of Heat Engineering, Warsaw, Nowowiejska 21/25, 00-665, Poland; email: pawel.domitr.dokt@pw.edu.pl; CODEN: NEDEA

The washing system based on RFID research

Item Type Conference Paper

Author Z. Yajuan

Author W. Jun

Author R. Hongwei

Author H. Yan

Author Z. Yong

Abstract Hospital has huge amount of bedding and clothing usages and complicated processes in cleaning and disinfection, some even have special requirements in the process of cleaning and transportation such as a separate isolation to prevent pollution. Nowadays, hospital bedding and clothing inventory and classification are mainly done by manual operation, it is hard to avoid cross-contamination in the unpack process, at the same time the manual counting cost large amount of labor but has low accuracy and efficiency. To avoid secondary pollution as well as cross-contamination and improve the counting efficiency and accuracy, we can use RFID (Radio Frequency Identification) chips to identify the bedding and clothing to realize the automatic counting function during transportation. This paper describes an automatic clothing inventory and classification system dependent on a ultra-high frequency (UHF) RFID chip, which can automatically classify and count the clothes, record washing, disinfection and transportation processes and usage condition of clothes through cloud server. Besides, it can also support the remote tracking, fee settlement of clothing cleaning, and disinfection and transportation. These processes can be performed by handheld devices and special devices. It makes hospital clothing cleaning, disinfection and transportation become traceable. © 2015 IEEE.

Date 2015

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The Youth Online Exploitation–Extremism Interface: A Conceptual Framework for Anti-Trafficking Identification and Digital Child Protection

Item Type Journal Article

Author H.N. Landress

Abstract Technology-facilitated child sexual exploitation increasingly intersects with coercive digital abuse and violent online subcultures that reward humiliation, self-harm, and abuse-material production as status markers. Conventional anti-trafficking identification is poorly suited to these cases, where exploitation operates through image-based leverage, reputational control, and platform migration rather than physical transportation or confinement. This Brief Report introduces the Youth Online Exploitation–Extremism Interface Framework, a conceptual-practice model with seven interacting domains: youth vulnerability, digital access points, coercive mechanisms, abuse-material production, group and status reinforcement, violence-aesthetic overlay, and response pathways. The domains are analytically separable but empirically entangled and recursive, so information in one domain can revise interpretation of another. The framework draws on 174 publicly documented incidents (2014 to 2026) in the PERIL Nihilistic Violent Extremism Tracker, serving as illustrative anchors rather than a representative sample for prevalence estimation. It supports identification, documentation, and survivor-centered referral across anti-trafficking, child protection, school safety, and survivor-serving settings. It is not a legal trafficking test, extremism classification, or risk-prediction instrument. Contribution to UN Sustainable Development Goals. This report supports UN

Sustainable Development Goal 16, Target 16.2 (ending abuse, exploitation, trafficking, and violence against children). By giving anti-trafficking, child-protection, school-safety, and platform-safety practitioners a shared vocabulary for trafficking-adjacent online exploitation in violent online subcultures, the framework can improve early identification, documentation, and survivor-centered referrals. It also contributes to SDG 5 (Target 5.2, eliminating violence and exploitation against women and girls), foregrounding youth-facing sexual coercion and image-based sexual abuse as distinct yet co-occurring harms requiring a multi-system response. © 2026 Taylor & Francis Group, LLC.

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Theoretical comparisons of positive-unlabeled learning against positive-negative learning

Item Type Conference Paper

Author G. Niu

Author M.C. Du Plessis

Author T. Sakai

Author Y. Ma

Author M. Sugiyama

Editor Garnett R.

Editor Lee D.D.

Editor von Luxburg U.

Editor Guyon I.

Editor Sugiyama M.

Abstract In PU learning, a binary classifier is trained from positive (P) and unlabeled (U) data without negative (N) data. Although N data is missing, it sometimes outperforms PN learning (i.e., ordinary supervised learning). Hitherto, neither theoretical nor experimental analysis has been given to explain this phenomenon. In this paper, we theoretically compare PU (and NU) learning against PN learning based on the upper bounds on estimation errors. We find simple conditions when PU and NU learning are likely to outperform PN learning, and we prove that, in terms of the upper bounds, either PU or NU learning (depending on the class-prior probability and the sizes of P and N data) given infinite U data will improve on PN learning. Our theoretical findings well agree with the experimental results on artificial and benchmark data even when the experimental setup does not match the theoretical assumptions exactly. © 2016 NIPS Foundation - All Rights Reserved.

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Prior probability, Binary classifiers, Upper Bound, Estimation errors, Pu learning, Benchmark data, Experimental analysis

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Theoretical insights into multiclass classification: A high-dimensional asymptotic view

Item Type Conference Paper

Author C. Thrampoulidis

Author S. Oymak

Author M. Soltanolkotabi

Abstract Contemporary machine learning applications often involve classification tasks with many classes. Despite their extensive use, a precise understanding of the statistical properties and behavior of classification algorithms is still missing, especially in modern regimes where the number of classes is rather large. In this paper, we take a step in this direction by providing the first asymptotically precise analysis of linear multiclass classification. Our theoretical analysis allows us to precisely characterize how the test error varies over different training algorithms, data distributions, problem dimensions as well as number of classes, inter/intra class correlations and class priors. Specifically, our analysis reveals that the classification accuracy is highly distribution-dependent with different algorithms achieving optimal performance for different data distributions and/or training/features sizes. Unlike linear regression/binary classification, the test error in multiclass classification relies on intricate functions of the trained model (e.g., correlation between some of the trained weights) whose asymptotic behavior is difficult to characterize. This challenge is already present in simple classifiers, such as those minimizing a square loss. Our novel theoretical techniques allow us to overcome some of these challenges. The insights gained may pave the way for a precise understanding of other classification algorithms beyond those studied in this paper. © 2020 Neural information processing systems foundation. All rights reserved.

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Classification (of information), Classification tasks, Multi-class classification, Classification algorithm, Classification accuracy, Machine learning applications, Asymptotic behaviors, Statistical properties, Theoretical technique

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Thermal bridging in buildings: A critical review of mechanisms, mitigation strategies, energy impacts, and research needs

Item Type Journal Article

Author M. Ahsan

Author M.A. Youssef

Author C. Van Dongen

Author D. Thompson

Author G. Stanier

Abstract The increasingly stringent energy and emission targets have made thermal bridging a critical consideration for building construction. Thermal bridging is excessive, localized heat flow through discontinuities in the insulation layer, such as at window and door assemblies, building envelope junctions, or balconies, that significantly increases space-heating and cooling demand, depresses interior surface temperatures, and promotes moisture-related durability problems. This paper presents a state-of-the-art critical review of thermal bridging in buildings, with emphasis on its mechanisms, quantification, and mitigation. The classifications, causes, and locations of thermal bridges within the building envelope are synthesized, and methods for evaluating them, including the Equivalent Wall Method (EWM), the Equivalent U-Value Method (EUVM), and 2D/3D numerical modelling, are compared. Element-specific mitigation strategies for openings, exposed structural components, walls, junctions, and balconies are reviewed, highlighting their influence on effective envelope R-values, building energy use, and condensation risk. The literature shows that thermal bridges account for a substantial fraction of heat loss in highly insulated buildings and become proportionally more important as envelope requirements tighten. Key knowledge gaps are identified, including the scarcity of long-term experimental validation and the need for holistic, constructible solutions that integrate structural, architectural, and thermal performance. Practical Application: Thermal bridging can materially affect heating and cooling loads, interior surface temperatures, and moisture risk in high-performance envelopes. This review provides building professionals with a comprehensive foundation to identify high-impact details (openings, slab edges, balconies, and junctions), select appropriate quantification methods (Ψ/χ values, EUVM, and 2D/3D modelling) for design and compliance, and prioritize constructible mitigation strategies, such as, but not limited to, continuous insulation, thermally broken connections, and improved detailing and quality control. The review supports a more holistic, durability-focused decision-making process through efficient HVAC system sizing and smart envelope design. © The Author(s) 2026. This article is distributed under the terms of the Creative Commons Attribution 4.0 License (<https://creativecommons.org/licenses/by/4.0/>) which permits any use, reproduction and distribution of the work without further permission provided the

original work is attributed as specified on the SAGE and Open Access page (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

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THREATENING DEGREE ASSESSMENT METHOD OF SONAR TARGETS BASED ON MULTI-SOURCE INFORMATION FUSION

Item Type Conference Paper

Author B. Zhou

Author X. Song

Author Q. Wang

Author Y. Chen

Editor Papadakis J.S.

Abstract Multi-source information fusion can expand the detection range of a sonar system and improve the reliability and robustness of target recognition. To solve the problems of large transmission bandwidth, uncertain priori probability of target threatening degree discrimination, and inability to perform fusion processing within the framework of probability theory in the sonar target association using full spectrum feature information, a threatening degree assessment method for sonar targets based on multi-source information fusion is proposed in this paper. The method combines the target information output from low-frequency array, high-frequency array and non-acoustic sensor, to perform threatening degree assessment of low-frequency sonar target through multi-source information preprocessing, target track association, association information fusion and target classification. For target track association, an improved gray association algorithm is proposed, which can distinguish two different targets with the same track change trend but far away from each other. In the absence of a priori probability of threatening target discrimination, a multi-source information fusion method based on Dempster-Shafer evidence theory is proposed to solve the problem that probability theory cannot be effectively applied to information fusion processing, so as to eliminate non-threatening targets. Finally the Template matching technique is used to further improve the accuracy of target threatening degree assessment. The proposed method is verified by sea trial data of multi-source information, including low-frequency array data, high-frequency array data and automatic identification system(AIS) data. This work can meet the needs of low-frequency sonar target threatening degree assessment. © 2019, I.A.C.M, Foundation for Research and Technology - Hellas. All rights reserved.

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Dempster-Shafer(D-S) evidence theory, multi-source information fusion, sonar target detection, template matching, threatening degree assessment method

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Three-Dimensional Echocardiographic Assessment of Left Heart Chamber Size and Function with Fully Automated Quantification Software in Patients with Atrial Fibrillation

Item Type Journal Article

Author K. Otani

Author A. Nakazono

Author I.S. Salgo

Author R.M. Lang

Author M. Takeuchi

Abstract Background Echocardiographic determination of left heart chamber volumetric parameters by using manual tracings during multiple beats is tedious in atrial fibrillation (AF). The aim of this study was to determine the usefulness of fully automated left chamber quantification software with single-beat three-dimensional transthoracic echocardiographic data sets in patients with AF. Methods Single-beat full-volume three-dimensional transthoracic echocardiographic data sets were prospectively acquired during consecutive multiple cardiac beats (≥ 10 beats) in 88 patients with AF. In protocol 1, left ventricular volumes, left ventricular ejection fraction, and maximal left atrial volume were validated using automated quantification against the manual tracing method in identical beats in 10 patients. In protocol 2, automated quantification-derived averaged values from multiple beats were compared with the corresponding values obtained from the indexed beat in all patients. Results Excellent correlations of left chamber parameters between automated quantification and the manual method were observed ($r = 0.88\text{--}0.98$) in protocol 1. The time required for the analysis with the automated quantification method (5 min) was significantly less compared with the manual method (27 min) ($P < .0001$). In protocol 2, there were excellent linear correlations between the averaged left chamber parameters and the corresponding values obtained from the indexed beat ($r = 0.94\text{--}0.99$), and test-retest variability of left chamber parameters was low (3.5%–4.8%). Conclusions Three-dimensional transthoracic echocardiography with fully automated quantification software is a rapid and reliable way to measure averaged values of left heart chamber parameters during multiple consecutive beats. Thus, it is a potential new approach for left chamber quantification in patients with AF in daily routine practice. © 2016 American Society of Echocardiography

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Three-dimensional microvascular network reconstruction from in vivo images with adaptation of the regional inhomogeneity in the signal-to-noise ratio

Item Type Journal Article
Author T. Sugashi
Author H. Yuki
Author T. Niizawa
Author H. Takuwa
Author I. Kanno

Author K. Masamoto

Abstract Objective: Quantification of angiographic images with two-photon laser scanning fluorescence microscopy (2PLSM) relies on proper segmentation of the vascular images. However, the images contain inhomogeneities in the signal-to-noise ratio (SNR) arising from regional effects of light scattering and absorption. The present study developed a semiautomated quantification method for volume images of 2PLSM angiography by adjusting the binarization threshold according to local SNR along the vessel centerlines. Methods: A phantom model made with fluorescent microbeads was used to incorporate a region-dependent binarization threshold. Results: The recommended SNR for imaging was found to be 4.2–10.6 that provide the true size of imaged objects if the binarization threshold was fixed at 50% of SNR. However, angiographic images in the mouse cortex showed variable SNR up to 45 over the depths. To minimize the errors caused by variable SNR and a spatial extent of the imaged objects in an axial direction, the microvascular networks were three-dimensionally reconstructed based on the cross-sectional diameters measured along the vessel centerline from the XY-plane images with adapted binarization threshold. The arterial volume was relatively constant over depths of 0–500 μm , and the capillary volume (1.7% relative to the scanned volume) showed the larger volumes than the artery (0.8%) and vein (0.6%). Conclusions: The present methods allow consistent segmentation of microvasculature by adapting the local inhomogeneity in the SNR, which will be useful for quantitative comparison of the microvascular networks, such as under disease conditions where SNR in the 2PLSM images varies over space and time. © 2021 John Wiley & Sons Ltd

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Three-year monitoring of paratuberculosis in dairy cattle by pooled faecal culture and individual prevalence estimation

Item Type Journal Article

Author A. Prieto

Author J.M. Díaz-Cao

Author P. Díaz

Author A. Pérez-Creo

Author G. López-Lorenzo

Author L.Á. Quintela

Author C. López

Author P. Morondo

Author P. Díez-Baños

Author G. Fernández

Abstract The objective of this study was to assess the implementation of a three-year *Mycobacterium avium* subsp. paratuberculosis monitoring programme using pooled faecal culture in small and medium-sized dairy herds to classify them as infected or non-infected and apply proper hygiene and biosecurity measures. Over a three-year period, 35 dairy herds were analysed annually by faecal culture of ten pooled samples. In addition, proper hygiene and biosecurity protocols were implemented in the farms after the first testing round. Considering a herd as infected with at least one culture positive in any of the three years, the accumulated percentage of infected herds was 25.7%, 40% and 45.7%, for each year respectively. Assuming that all infected herds had been detected at the end of the study, the percentage of infected

herds detected each year was 56.25% and 87.5% for the first and second year, respectively. Using frequentist and Bayesian approaches, the estimated individual prevalence revealed a downward trend from 3.30-3.65% in the first year to 1.66-1.86% in the third year. The results of this study indicate that pooled faecal culture allowed for proper classification of the herds and can be a useful tool for monitoring dairy herds against paratuberculosis. In addition, statistical analysis of pooled faecal culture results can be used to evaluate the evolution of individual prevalence in the population and therefore the function of the implemented control programmes. © Walter de Gruyter GmbH 2018. All rights reserved.

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Threshold optimization and random undersampling for imbalanced credit card data

Item Type Journal Article

Author J.L. Leevy

Author J.M. Johnson

Author J. Hancock

Author T.M. Khoshgoftaar

Abstract Output thresholding is well-suited for addressing class imbalance, since the technique does not increase dataset size, run the risk of discarding important instances, or modify an existing learner. Through the use of the Credit Card Fraud Detection

Dataset, this study proposes a threshold optimization approach that factors in the constraint True Positive Rate (TPR) $\geq$ True Negative Rate (TNR). Our findings indicate that an increase of the Area Under the Precision–Recall Curve (AUPRC) score is associated with an improvement in threshold-based classification scores, while an increase of positive class prior probability causes optimal thresholds to increase. In addition, we discovered that best overall results for the selection of an optimal threshold are obtained without the use of Random Undersampling (RUS). Furthermore, with the exception of AUPRC, we established that the default threshold yields good performance scores at a balanced class ratio. Our evaluation of four threshold optimization techniques, eight threshold-dependent metrics, and two threshold-agnostic metrics defines the uniqueness of this research. © 2023, The Author(s).

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Threshold-based quantification of fatty degeneration in the supraspinatus muscle on MRI as an alternative method to Goutallier classification and single-voxel MR spectroscopy

Item Type Journal Article

Author D. Lee
Author K.-T. Hong
Author W. Lee
Author E.K. Khil
Author G.Y. Lee
Author J.-A. Choi
Author Y. Song

Abstract Background: Conventional fat quantification methods for rotator cuff muscles have various limitations, such as inconsistent reliabilities of the Goutallier grades and need for advanced techniques in quantitative MRI sequences. We aimed to examine a threshold-based fat quantification method in the supraspinatus muscle on standard T1-weighted MR images and compare the threshold-based method with Goutallier grades and MR spectroscopy. Methods: We retrospectively examined 38 symptomatic patients, who underwent T1 and T2-weighted fast spin-echo MR imaging and a single voxel spin-echo MR spectroscopy. The supraspinatus muscle and fossa were manually segmented in T1-weighted sagittal images and clustering-based thresholding was applied to quantify the fat fractions in the segmented areas using custom MATLAB software. Threshold-based fat fractions were compared with the Goutallier grades and MR spectroscopy fat/water ratios. A one-way analysis of variance and Pearson correlation were tested in the MATLAB software. Results: Inter-observer reliability of threshold-based fat fractions for the supraspinatus muscle and fossa were 0.977 and 0.990 respectively, whereas the reliability of the Goutallier grading was 0.798. Threshold-based fat fractions in the supraspinatus fossa were significantly different between various Goutallier grades (one-way ANOVA, $p < 0.001$). Threshold-based fat fractions in the supraspinatus muscle strongly correlated with the MR spectroscopy fat/water ratio (Pearson correlation R-square = 0.83). Conclusions: Threshold-based fat quantification on standard T1-weighted MR images was highly reliable and produced comparable results to conventional Goutallier grades and MR spectroscopy fat/water ratios and could serve as an alternative method for accurate fat quantification in rotator cuff muscles. © 2020 The Author(s).

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Notes:

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Thresholding strategies for deep learning with highly imbalanced big data

Item Type Journal Article

Author J.M. Johnson

Author T.M. Khoshgoftaar

Abstract A variety of data-level, algorithm-level, and hybrid methods have been used to address the challenges associated with training predictive models with class-imbalanced data. While many of these techniques have been extended to deep neural network (DNN) models, there are relatively fewer studies that emphasize the significance of output thresholding. In this chapter, we relate DNN outputs to Bayesian a posteriori probabilities and suggest that the Default threshold of 0.5 is almost never optimal when training data is imbalanced. We simulate a wide range of class imbalance levels using three real-world data sets, i.e. positive class sizes of 0.03–90%, and we compare Default threshold results to two alternative thresholding strategies. The Optimal threshold strategy uses validation data or training data to search for the classification threshold that maximizes the geometric mean. The Prior threshold strategy requires no optimization, and instead sets the classification threshold to be the prior probability of the positive class. Multiple deep architectures are explored and all experiments are repeated 30 times to account for random error. Linear models and visualizations show that the Optimal threshold is strongly correlated with the positive class prior. Confidence intervals show that the Default threshold only performs well when training data is balanced and Optimal thresholds perform significantly better when training data is skewed. Surprisingly, statistical

results show that the Prior threshold performs consistently as well as the Optimal threshold across all distributions. The contributions of this chapter are twofold: (1) illustrating the side effects of training deep models with highly imbalanced big data and (2) comparing multiple thresholding strategies for maximizing class-wise performance with imbalanced training data. © 2021, The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd.

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Tied Prototype Model for Few-Shot Medical Image Segmentation

Item Type Conference Paper

Author H. Kim

Author S. Hansen

Author M. Kampffmeyer

Editor Gee J.C.

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Editor Alexander D.C.

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Editor Kim J.H.

Abstract Common prototype-based medical image few-shot segmentation (FSS) methods model foreground and background classes using class-specific prototypes. However, given the high variability of the background, a more promising direction is to focus solely on foreground modeling, treating the background as an anomaly—an approach introduced by ADNet. Yet, ADNet faces three key limitations: dependence on a single prototype per class, a focus on binary classification, and fixed thresholds that fail to adapt to patient and organ variability. To address these shortcomings, we propose the Tied Prototype Model (TPM), a principled reformulation of ADNet with tied prototype locations for foreground and background distributions. Building on its probabilistic foundation, TPM naturally extends to multiple prototypes and multi-class segmentation while effectively separating non-typical background features. Notably, both extensions lead to improved segmentation accuracy. Finally, we leverage naturally occurring class priors to define an ideal target for adaptive thresholds, boosting segmentation performance. Taken together, TPM provides a fresh perspective on prototype-based FSS for medical image segmentation. The code can be found at <https://github.com/hjk92g/TPM-FSS>. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Time series of Landsat-based bimonthly and annual spectral indices for continental Europe for 2000–2022

Item Type Journal Article

Author X. Tian

Author D. Consoli

Author M. Witjes

Author F. Schneider

Author L. Parente

Author M. Şahin

Author Y.-F. Ho

Author R. Minařík

Author T. Hengl

Abstract The production and evaluation of the analysis-ready and cloud-optimized (ARCO) data cube for continental Europe (including Ukraine, the UK, and Türkiye), derived from the Landsat analysis-ready dataset version 2 (ARD V2) produced by Global Land Analysis and Discovery (GLAD) team and covering the period from 2000 to 2022, is described. The data cube consists of 17 TB of data at a 30 m resolution and includes bimonthly, annual, and long-term spectral indices on various thematic topics, including surface reflectance bands, normalized difference vegetation index (NDVI), soil adjusted vegetation index (SAVI), fraction of absorbed photosynthetically active radiation (FAPAR), normalized difference snow index (NDSI), normalized difference water index (NDWI), normalized difference tillage index (NDTI), minimum normalized difference tillage index (minNDTI), bare soil fraction (BSF), number of seasons (NOS), and crop duration ratio (CDR). The data cube was developed with the intention to provide a comprehensive feature space for environmental modeling and mapping. The quality of the produced time series was assessed by (1) assessing the accuracy of gap-filled bimonthly Landsat data with artificially created gaps; (2) visual examination for artifacts and inconsistencies; (3) plausibility checks with ground survey data; and (4) predictive modeling tests, examples with soil organic carbon (SOC) and land cover (LC) classification. The time series reconstruction demonstrates high accuracy, with a root mean squared error (RMSE) smaller than 0.05, and R2 higher than 0.6, across all bands. The visual examination indicates that the product is complete and consistent, except for winter periods in northern latitudes and high-altitude areas, where high cloud and snow density introduce significant gaps and hence many artifacts remain. The plausibility check further shows that the indices logically and statistically capture the processes. The BSF index showed a strong negative correlation (−0.73) with crop coverage data, while the minNDTI index had a moderate positive correlation (0.57) with the Eurostat tillage practice survey data.

The detailed temporal resolution and long-term characteristics provided by different tiers of predictors in this data cube proved to be important for both soil organic carbon regression and LC classification experiments based on 60 723 LUCAS observations: long-term characteristics (tier 4) were particularly valuable for predictive mapping of SOC and LC, coming out on top of variable importance assessment. Crop-specific indices (NOS and CDR) provided limited value for the tested applications, possibly due to noise or insufficient quantification methods. The data cube is made available at <https://doi.org/10.5281/zenodo.10776891> (Tian et al., 2024) under a CC-BY license and will be continuously updated. © 2025 Xuemeng Tian et al.

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Time-Aware Crime Prediction: A Multi-Output AI Framework for Operational Risk Prioritization

Item Type Journal Article

Author C. Eroglu

Author H. Cakir

Abstract This study proposes an artificial intelligence (AI)-supported operational crime prediction framework that integrates temporal, spatial, and harm-aware risk modeling for short-term operational prioritization. Unlike conventional binary crime prediction

approaches, the proposed multi-output framework simultaneously estimates crime occurrence probability, expected crime volume, crime harm, and crime category. Experimental results demonstrate that the joint use of temporal and spatial features substantially improves predictive performance, increasing Precision-Recall Area Under the Curve (PR-AUC) from 0.1849 to 0.2346 despite severe class imbalance (positive-class prevalence = 0.0715), while Receiver Operating Characteristic Area Under the Curve (ROC-AUC) improved from 0.7236 to 0.7496. The stacking approach produced more balanced and generalizable performance compared to individual models. The framework also exhibited strong probabilistic calibration, achieving an Expected Calibration Error (ECE) of 0.0012 and a Brier score of approximately 0.06. The multi-output modules further achieved $RMSE \approx 0.445$ for expected crime count estimation and $RMSE \approx 4.02$ for expected harm prediction. Under realistic operational evaluation settings, the model achieved precision@500 of 0.7460, precision@5000 of 0.5418, and lift@5000 of 7.57, while maintaining stable performance under strict out-of-fold and rolling as-of backtesting conditions (PR-AUC ≈ 0.260 and PR-AUC ≈ 0.251 , respectively). The findings indicate that the proposed framework provides both improved predictive capability and practical utility for operational risk prioritization and decision-support processes in short-term crime forecasting. © 2013 IEEE.

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Artificial intelligence, Decision support systems, Risk assessment, Crime, Learning systems, machine learning, Machine-learning, Prediction models, Forecasting, Risk analysis, ensemble learning, Ensemble learning, Stackings, Multi-output, stacking, Areas under the curves, Calibration, Crime prediction, harm-based analysis, Harm-based analyze, risk ranking, Risk ranking, spatio-temporal modeling, Spatio-temporal models, top-k evaluation, Top-k evaluation

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: C. Eroglu; Gazi University, Graduate School of Informatics, Department of Digital Forensics, Ankara, Çankaya, 06680, Turkey; email: cem.eroglul@gazi.edu.tr

Token-Level Density-Based Uncertainty Quantification Methods for Eliciting Truthfulness of Large Language Models

Item Type Conference Paper

Author A. Vazhentsev

Author L. Rvanova

Author I. Lazichny

Author A. Panchenko

Author M. Panov

Author T. Baldwin

Author A. Shelmanov

Editor Chiruzzo L.

Editor Ritter A.

Editor Wang L.

Abstract Uncertainty quantification (UQ) is a prominent approach for eliciting truthful answers from large language models (LLMs). To date, information-based and consistency-based UQ have been the dominant UQ methods for text generation via LLMs. Density-based methods, despite being very effective for UQ in text classification with encoder-based models, have not been very successful with generative LLMs. In this work, we adapt Mahalanobis Distance (MD) - a well-established UQ technique in classification tasks - for text generation and introduce a new supervised UQ method. Our method extracts token embeddings from multiple layers of LLMs, computes MD scores for each token, and uses linear regression trained on these features to provide robust uncertainty scores. Through extensive experiments on eleven datasets, we demonstrate that our approach substantially improves over existing UQ methods, providing accurate and computationally efficient uncertainty scores for both sequence-level selective generation and claim-level fact-checking tasks. Our method also exhibits strong generalization to out-of-domain data, making it suitable for a wide range of LLM-based applications. © 2025 Association for Computational Linguistics.

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Tomato Health Monitoring System: Tomato Classification, Detection, and Counting System Based on YOLOv8 Model With Explainable MobileNet Models Using Grad-CAM++

Item Type Journal Article

Author L.-D. Quach

Author K.N. Quoc

Author A.N. Quynh

Author H.T. Ngoc

Author N. Thai-Nghe

Abstract Fruits and vegetables (especially, tomatoes) healthy detection are important tasks for smart agriculture. Several works have been published in tomato detection, however, there is little research on using explainable AI to detect, classify and count tomato fruit status. In this work, we propose a Tomatoes Health Check System by evaluating MobileNet models based on the physiological tomato dataset. Our research conducts experiments to evaluate the accuracy of the MobileNets, MobileNetV2 and MobileNetV3 models based on the evaluation metrics; the highest accuracy of 96.69% belongs to the MobileNetV3 model. The proposed method we suggest is to utilize Grad-CAM++ for a visual explanation of predictions made by models belonging to the MobileNets family. Subsequently, we calculate Intersection over Union metrics at various thresholds (0%, 25%, and 50%) based on each heatmap or region of importance. To assess model reliability, Grad-CAM++ is used to explain and evaluate reliability, with MobileNetV2 achieving the highest values at 100.00% (d=0), 100.00% (d=0.25), and 98.89% (d=0.5). An evaluation experiment combines the YOLOv8 and MobileNetV2 algorithms using the Simple Online and Real-time Tracking (SORT) algorithm to detect, classify, and count tomatoes based on physiological characteristics in videos. Finally, the research results are utilized to develop an application system. © 2013 IEEE.

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Notes:

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Export Date: 20 September 2026; Cited By: 74; Correspondence Address: N. Thai-Nghe; Can Tho University, Can Tho, 10000, Viet Nam; email: ntnghe@cit.ctu.edu.vn

Tools for Analyzing Signature-Based Hardware Solutions for Cyber Security Systems

Item Type Journal Article

Author S.Ya. Hilgurt

Author A.M. Davydenko

Author T.V. Matovka

Author M.P. Prygara

Abstract When creating signature-based cybersecurity systems for network intrusion detection (NIDS), spam filtering, protection against viruses, worms, etc., developers have to use hardware devices such as field programmable gate arrays (FPGA), since software solutions can no longer support the necessary speeds. There are many different approaches to build hardware circuits for pattern matching (where patterns are the parts of signatures). Choosing the optimal technical solution for certain conditions is not a trivial task. Developers of such hardware tend to act intuitively, heuristically. In this article, we provide tools to help them intelligently build cybersecurity systems using FPGAs. For the qualitative analysis of FPGA-based matching schemes, the classification of efficiency criteria and related indicators is considered. This classification was compiled by studying a large number of practical developments of FPGA-based cybersecurity systems, primarily NIDS. A method of rapid calculating numerical characteristics of the FPGA-based signature system components is

proposed as a quantitative assessment tool. This method based on the use of so-called estimation functions allows avoiding the time-consuming execution of the digital circuit synthesis procedure. A number of experiments were carried out with the most promising matching schemes, allowing evaluating the above-mentioned tools. The rapid quantification method allows developers of hardware-accelerated cybersecurity systems to even apply it at each iteration within the optimization procedure cycle. © 2023 River Publishers.

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Tooth segmentation in mixed dentition CBCT via a three-stage class-prior and morphology-aware framework

Item Type Journal Article

Author J. Zhang

Author X. Li

Author Y. Hu
Author W. Tang
Author X. Wang
Author J. Guo

Abstract Accurate three-dimensional (3D) tooth segmentation in pediatric cone-beam computed tomography (CBCT) scans remains challenging during the mixed dentition stage due to its complex anatomical characteristics. Existing methods designed for permanent dentition often fail to generalize effectively, primarily owing to the increased number of teeth, crowding, intricate spatial arrangements, and significant morphological variations inherent to mixed dentition. To overcome these limitations, we propose a novel three-stage segmentation framework that incorporates class-prior guidance and morphology-aware refinement. In the first stage, a 3D morphological constraint graph network (MCG-Net) is introduced, integrating a class prior constraint graph module (CPCGM) to dynamically model inter-class dependencies based on spatial and developmental priors of mixed dentition, and a morphological constraint module (MCM) to enhance structural precision through boundary-aware supervision. In the second stage, a 3D segmentation and classification refinement network (SCR-Net) is developed to jointly refine single-tooth segmentation and adjust the first-stage class predictions via multi-task learning, thereby mitigating the effects of tooth crowding and morphological variability. In the final stage, a class-weighted strategy based on voxel count is designed to reduce class conflicts in overlapping regions, improving segmentation accuracy for smaller teeth. The proposed framework is validated on a self-constructed mixed dentition dataset and a publicly available permanent dentition dataset. Quantitative and qualitative results demonstrate that our method achieves superior segmentation accuracy and robustness for mixed dentition, while maintaining competitive performance on permanent dentition. © 2026 Elsevier B.V.

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Toward a Rigorous Classification of Loss of Productivity Quantification Methods

Item Type Journal Article

Author T. Zhao

Abstract Lost productivity is a common cause of cost overrun on construction projects and often results in construction claims. There are several different loss of productivity quantification methods and variants of those methods. Due to lack of adequate classification, the use of these methods is not well disciplined. For example, the implementation of the measured mile method can vary significantly between analysts or projects. This paper proposes a more rigorous classification system for loss of productivity quantification methods. This work first defines key dimensions of the hierarchy/matrix of the classification system, including productivity type, measurement perspective, data reliance, similarity requirement, analysis technique, work composition and divisions, causation, and others, and then places various loss of productivity quantification method implementations to different positions in the classification system. With this rigorous classification of loss of productivity methods, the variations between different method implementations can be clearly identified and the exact methodology can be applied to a construction claim in a well-defined and unambiguous manner. © 2025 American Society of Civil Engineers.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: T. Zhao; Delta Consulting Group, Inc., Woodbridge, 4330 Prince William Pkwy, Ste 301, 22192, United States; email: tzhao@delta-cgi.com

Toward Robust Semi-Supervised Distribution Alignment Against Label Distribution Shift With Noisy Annotations

Item Type Journal Article

Author B. Chen

Author Z. Ye

Author Y. Liu

Author X. Fang

Author G. Lu

Author S. Xie

Author X. Li

Abstract Deep learning-based AI models typically require a large amount of high-quality annotated data to achieve optimal performance. However, the label distribution shift caused by noisy annotations can lead to perturbations in the classification boundary, reducing the robustness and generalization capabilities of deep learning models. To mitigate this issue, we transform the problem of learning from noisy labels into a semi-supervised learning problem, and propose a novel Semi-Supervised Distribution Alignment (SSDA) framework that strategically integrates noise-robust distribution alignment within a unified semi-supervised learning paradigm for combating noisy labels. By leveraging the similarity distribution between historical predictions, the proposed SSDA approach benefits from a flexible multi-historical regression modeling strategy, which aims to identify high-confidence samples/pairs and recalibrate the label shift through pseudo-labels. Furthermore, our approach employs

a comprehensive multi-granularity distribution adaptation strategy, incorporating both instance-wise and class-aware distribution alignment to quantitatively minimize semantic discrepancies across different mixed feature domains. In this way, our SSDA approach ultimately achieves more resilient and generalizable performance against label noise, even in the presence of substantial noise. Extensive experiments conducted on multiple simulated and real-world noisy benchmark datasets consistently demonstrate the superiority and effectiveness of our SSDA method compared to existing state-of-the-art baselines. © 1999-2012 IEEE.

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Toward Robust Uncertainty Estimation with Random Activation Functions

Item Type Conference Paper

Author Y. Stoyanova

Author S. Ghandi
Author M. Tavakol
Editor Williams B.
Editor Chen Y.
Editor Neville J.

Abstract Deep neural networks are in the limelight of machine learning with their excellent performance in many data-driven applications. However, they can lead to inaccurate predictions when queried in out-of-distribution data points, which can have detrimental effects especially in sensitive domains, such as healthcare and transportation, where erroneous predictions can be very costly and/or dangerous. Subsequently, quantifying the uncertainty of the output of a neural network is often leveraged to evaluate the confidence of its predictions, and ensemble models have proved to be effective in measuring the uncertainty by utilizing the variance of predictions over a pool of models. In this paper, we propose a novel approach for uncertainty quantification via ensembles, called Random Activation Functions (RAFTs) Ensemble, that aims at improving the ensemble diversity toward a more robust estimation, by accommodating each neural network with a different (random) activation function. Extensive empirical study demonstrates that RAFTs Ensemble outperforms state-of-the-art ensemble uncertainty quantification methods on both synthetic and real-world datasets in a series of regression tasks. Copyright © 2023, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Toward the classification of social media use disorder: Clinical characterization and proposed diagnostic criteria

Item Type Journal Article

Author T. Moretta

Author E. Wegmann

Abstract Empirical studies and theoretical models highlight that problematic use of social media can lead to significant functional impairments in several domains, such as social, relational, occupational, and psychological functioning, as well as physical health. However, social media use disorder is not currently recognized as an official disorder in major diagnostic systems, limiting comparability among studies, precise prevalence estimation, and ad-hoc preventive and treatment program development. The present work aims to classify social media use disorder as a pathological condition sharing main mechanisms and maladaptive patterns with addictive behaviors. We discuss diagnostic criteria for social media use disorder by integrating peculiar features of this maladaptive behavior with criteria for substance use disorders and behavioral addictions from major diagnostic systems (i.e., DSM-5 and ICD-11). Moreover, we address some controversies related to the classification of social media use disorder as a behavioral addiction and highlight literature findings indicating social media use disorder shares main alterations in mechanisms and processes characterizing addiction (i.e., maladaptive activation of reward systems and impairment of inhibitory control mechanisms). Despite the lack of studies including clinical populations and the need for future research to validate the proposed criteria and refine knowledge of the mechanisms underlying this condition, our work provides a structured framework for classifying and identifying social media use disorder. © 2025 The Author(s)

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Towards a classification of holographic multi-partite entanglement measures

Item Type Journal Article

Author A. Gadde

Author V. Krishna

Author T. Sharma

Abstract In this paper, we systematically study the measures of multi-partite entanglement with the aim of constructing those measures that can be computed in probe approximation in the holographic dual. We classify and count general measures as invariants of local unitary transformations. After formulating these measures in terms of permutation group elements, we derive conditions that a probe measure should satisfy and find a large class of solutions. These solutions are generalizations of the multi-entropy introduced in [1]. We derive their holographic dual with the assumption that the replica symmetry is unbroken in the bulk and check our prescription with explicit computations in 2d CFTs. Analogous to the multi-entropy, the holographic dual of these measures is given by the weighted area of the minimal brane-web but with branes having differing tensions. We discuss the replica symmetry assumption and also how the already known entanglement measures, such as entanglement negativity and reflected entropy fit in our framework. © 2023, The Author(s).

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Towards a fast and generalized microplastic quantification method in soil using terahertz spectroscopy

Item Type Journal Article

Author S. Zhao

Author Y. Zhang

Author Z. Qiu

Author Y. He

Author Y. Zhang

Abstract Extensive investigation of microplastic abundance in soil environment calls for rapid, accurate, efficient and harmonized quantification methods. Development of rapid quantification method requires made-to-measure soil samples with additions of standard polymers. Existing rapid quantification methods ignore the gap between standard polymers in laboratory and household microplastics in soil environment. Here, terahertz (0.6–1.67 thz) and NIR (950–1660 nm) spectroscopy were compared to explore a fast, accurate and potentially generalizable microplastic quantification method in soil. Soil sample was spiked with two standard polymers (polyvinyl chloride (PVC) and polystyrene (PS)) and their additive-containing household microplastics. Two standard sample sets and two household sample sets were prepared in concentrations ranging from 0.5 to 10%. Nine commonly used preprocessing methods and three machine learning algorithms were coupled to develop methods. Models were constructed by training sets from standard sample sets. When models transferred to household samples, prediction error (RMSE) of proposed terahertz method (Wdenosie_PLSR) only increased by 0.4% for PVC and 0.19% for PS, yet that of the NIR method increased by 1.49% and 1.16% respectively. The proposed terahertz method presented a detection limit around 1.12% and the NIR method showed a detection limit around 3.24%. Overall, our results suggest that compared with NIR method, the proposed terahertz method is not only more accurate but also demonstrate stronger generalizability to bridge the gaps between standard PVC/PS polymers and household PVC/PS microplastics. We also

propose MMD heatmap for diagnosing spectral preprocessing methods to further improve method efficiency. © 2022 Elsevier B.V.

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Towards a New Understanding of the Training of Neural Networks with Mislabeled Training Data

Item Type Conference Paper

Author H. Gish

Author J. Silovsky

Author M.-L. Sung

Author M.-H. Siu

Author W. Hartmann

Author Z. Jiang

Abstract We investigate the problem of machine learning with mislabeled training data. We try to make the effects of mislabeled training better understood through analysis of the basic model and equations that characterize the problem. This includes results about the ability of the noisy model to make the same decisions as the clean model and the effects of noise on model performance. In addition to providing better insights we also are able to show that the Maximum Likelihood (ML) estimate of the parameters of the noisy model determine those of the clean model. This property is obtained through the use of the ML invariance property and leads to an approach to developing a classifier when training has been mislabeled: namely train the classifier on noisy data and adjust the decision threshold based on the noise levels and/or class priors. We show how our approach to mislabeled training works with multi-layered perceptrons (MLPs). © 2020 IEEE.

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Towards adaptive unknown authentication for universal domain adaptation by classifier paradox

Item Type Journal Article

Author Y. Wang

Author Y. Liu

Author S. Chen

Abstract Universal domain adaptation (UniDA) is a general unsupervised domain adaptation setting, which addresses both domain and label shifts in adaptation. Its main challenge lies in how to identify target samples in unshared or unknown classes. Previous methods commonly strive to depict sample “confidence” along with a threshold for rejecting unknowns, and align feature distributions of shared classes across domains. However, it is still hard to pre-specify a “confidence” criterion and threshold which are adaptive to different tasks, and a mis-prediction of unknowns further incurs mis-alignment of features in shared classes. In this paper, we propose a new UniDA method with adaptive Unknown Authentication by Classifier Paradox (UACP), considering that samples with paradoxical predictions are probably unknowns belonging to none of the source classes. In UACP, a composite classifier is jointly designed with two types of predictors. That is, a multi-class (MC) predictor classifies samples to one of the multiple source classes, while a binary one-vs-all predictor further verifies the prediction by MC predictor. Samples with verification failure or paradox are identified as unknowns. Further, instead of feature alignment for shared classes, implicit domain alignment is conducted in output space such that samples across domains share the same decision boundary, though with feature discrepancy. Empirical results validate UACP under both open-set and universal UDA settings. © The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature 2022. Springer Nature or its licensor holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

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Towards explainable spoofed speech attribution and detection: A probabilistic approach for characterizing speech synthesizer components

Item Type Journal Article**Author** J. Mishra**Author** M. Chhibber**Author** H.-J. Shim**Author** T.H. Kinnunen

Abstract We propose an explainable probabilistic framework for characterizing spoofed speech by decomposing it into probabilistic attribute embeddings. Unlike raw high-dimensional countermeasure embeddings, which lack interpretability, the proposed probabilistic attribute embeddings aim to detect specific speech synthesizer components, represented through high-level attributes and their corresponding values. We use these probabilistic embeddings with four classifier back-ends to address two downstream tasks: spoofing detection and spoofing attack attribution. The former is the well-known bonafide-spoof detection task, whereas the latter seeks to identify the source method (generator) of a spoofed utterance. We additionally use Shapley values, a widely used technique in machine learning, to quantify the relative contribution of each attribute value to the decision-making process in each task. Results on the ASVspoof2019 dataset demonstrate the substantial role of waveform generator, conversion model outputs, and inputs in spoofing detection; and inputs, speaker, and duration modeling in spoofing attack attribution. In the detection task, the probabilistic attribute embeddings achieve 99.7% balanced accuracy and 0.22% equal error rate (EER), closely matching the performance of raw embeddings (99.9% balanced accuracy and 0.22% EER). Similarly, in the attribution task, our embeddings achieve 90.23% balanced accuracy and 2.07% EER, compared to 90.16% and 2.11% with raw embeddings. These results demonstrate that the proposed framework is both

inherently explainable by design and capable of achieving performance comparable to raw CM embeddings. © 2025 The Authors

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Publisher Academic Press

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Journal Abbr Comput Speech Lang

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Towards invariant and interpretable representations for domain generalization in time series classification

Item Type Journal Article

Author Y. Chen

Author Z. Weng

Author B.H. Gwee

Author Q. Cao

Author S.G. Razul

Author Z. Lin

Abstract Time series classification is crucial in many real-world applications, but models often suffer significant performance degradation under domain shifts between training and testing data. Existing Domain Generalization (DG) methods for time series mainly suppress domain-specific information via adversarial training, but they remain

insufficient to obtain category-consistent representations. To address these challenges, we propose Joint Inference for Time Series Representation (JITSR), a variationally motivated novel framework for domain-generalized time series classification. Through joint variational modeling of time series and labels, JITSR connects patch reconstruction, categorical prediction, and distributional alignment as complementary learning objectives. The alignment term encourages category-consistent representations across domains without requiring explicit domain annotations. Specifically, JITSR segments time series into patches and maps them to latent distributions, which are quantized by a learnable distributional codebook. A dual-path alignment mechanism then aligns sample-level semantic distributions with codebook-level class priors. Meanwhile, attention-based codeword weighting identifies the temporal patterns that contribute to classification, providing interpretable evidence for the learned representations. Extensive experiments on diverse time series datasets demonstrate that JITSR consistently outperforms state-of-the-art DG methods and learns more category-consistent representations across domains. © 2026 Elsevier Ltd

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Y. Chen; School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, Singapore; email: yongming001@e.ntu.edu.sg; CODEN: PTNRA

Towards Label Shift Adaptation for Robust IoT Device Identification

Item Type Conference Paper

Author S. Azizi

Author F.V. Calderan

Author N. Okui

Author M. Nakahara

Author A. Kubota

Author M.G. Quiles

Author G.E. Batista

Author H.H. Gharakheili

Editor Mashima D.

Editor Fawaz K.

Abstract Identifying IoT devices and managing their behaviors by analyzing network traffic is crucial to improving observability and security in modern network operations. Extensive prior research has focused on training machine learning models that identify IoT devices based on their network traffic collected at various granularities and vantage points. However, an under-explored challenge is the operational robustness of these models to shifts in class prevalence during inference, which is an inevitable consequence of the dynamic nature of IoT communications. By analyzing 640,000 IPFIX flow records collected from 12 real households, this paper makes three specific contributions. (1) We demonstrate that the distribution of IoT classes changes significantly over time in flow traces, matching the drop in classification performance. (2) By applying a Bayesian adjustment to the model's output scores using weights provided by an oracle, we recover many misclassified predictions, which also provides evidence for the hypothesis that the observed performance drop has been caused by the label shift. (3) As a practical alternative to the oracle, we explore the use of quantification methods to estimate class distributions in unlabeled data and show that they can successfully help adapt traffic classifiers to label shifts. © 2025 Copyright is held by the owner/author(s). Publication rights licensed to ACM.

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Conference Name CPSIoTSec 2025 - Proceedings of the 7th Joint Workshop on CPS and IoT Security and Privacy, Part of ACM CCS 2025

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Towards robust adversarial defense on perturbed graphs with noisy labels

Item Type Journal Article

Author D. Li

Author H. Xia

Author C. Hu

Author R. Zhang

Author Y. Du

Author X. Feng

Abstract Graph Neural Networks (GNNs) demonstrate powerful capabilities in graph representation learning tasks. However, real-world graphs are often perturbed and come with noisy labels, resulting in poor performance of GNNs in this scenario. To bridge this gap, we study a novel problem of developing robust graph adversarial defense on perturbed graphs with noisy labels, and propose a graph adversarial defense named Comprehensive Adversarial Defense (CARD). Initially, we utilize unsupervised singular value decomposition to separate the graph, preserving its main portion while isolating the mixed effects of perturbed graphs and noisy labels. Next, we employ an edge repairer to further restore the graph, thereby enhancing the robustness of GNNs against perturbed graphs. Subsequently, we perform fine-grained dynamic weighting of node-level losses during the training phase to prevent overfitting, and employ label alignment to address the significant label shift issue, thereby reducing the impact of noisy labels. Ultimately, we organize sufficient experiments under four benchmark datasets, four types of perturbed graphs, and two types of noisy labels. The empirical results validate the superior performance of CARD on the above adversarial scenarios. © 2025 Elsevier Ltd

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Tracing the evolution of household carbon emission research by machine learning

- Item Type** Journal Article
- Author** Y. Long
- Author** L. Huang
- Author** S. Montagna
- Author** Z. Chen
- Author** X. Ding
- Author** Y. Shigetomi
- Author** T. Zhao
- Author** K. Asatani
- Author** I. Sakata
- Author** Y. Yoshida

Abstract As a major driver of global production, household consumption warrants increased attention due to its potential for decarbonization. This study systematically reviews 4647 journal papers to uncover evolving research themes and regional disparities in household carbon emissions research. We identified a marked increase in publications after 2015 and, using natural language processing, defined 7 distinct research clusters. While earlier studies focused primarily on energy consumption, more recent research has broadened to encompass a range of topics, including food systems, supply-chain emissions, and renewable energy. Our regional analysis reveals varied levels of research engagement across countries. The Global South is increasingly involved,

focusing on cooking facilities and household energy use, whereas Europe emphasizes policy-oriented mitigation. North America, led by the United States, focuses on transportation emissions and residential energy-efficiency modeling, while Asian countries examine urban-rural disparities and household inequality. Drawing on these thematic and geographical insights, this study identifies current knowledge gaps, such as the need for cross-regional comparisons and supply chain integration. These insights offer actionable directions to promote sustainable consumption practices and support global climate goals. © 2025

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Traffic class prioritization-based slotted-CSMA/CA for IEEE 802.15.4 MAC in intra-WBANs

Item Type Journal Article

Author F. Masud

Author A.H. Abdullah

Author A. Altameem

Author G. Abdul-Salaam

Author F. Muchtar

Abstract This paper proposes an improved Traffic Class Prioritization based Carrier Sense Multiple Access/Collision Avoidance (TCP-CSMA/CA) scheme for prioritized channel access to heterogeneous-natured Bio-Medical Sensor Nodes (BMSNs) for IEEE 802.15.4 Medium Access Control (MAC) in intra-Wireless Body Area Networks (WBANs). The main advantage of the scheme is to provide prioritized channel access to heterogeneous-natured BMSNs of different traffic classes with reduced packet delivery delay, packet loss, and energy consumption, and improved throughput and packet delivery ratio (PDR). The prioritized channel access is achieved by assigning a distinct, minimized and prioritized backoff period range to each traffic class in every backoff during contention. In TCP-CSMA/CA, the BMSNs are distributed among four traffic classes based on the existing patient's data classification. The Backoff Exponent (BE) starts from 1 to remove the repetition of the backoff period range in the third, fourth, and fifth backoffs. Five moderately designed backoff period ranges are proposed to assign a distinct, minimized, and prioritized backoff period range to each traffic class in every backoff during contention. A comprehensive verification using NS-2 was carried out to determine the performance of the TCP-CSMA/CA in terms of packet delivery delay, throughput, PDR, packet loss ratio (PLR) and energy consumption. The results prove that the proposed TCP-CSMA/CA scheme performs better than the IEEE 802.15.4 based PLA-MAC, eMC-MAC, and PG-MAC as it achieves a 47% decrease in the packet delivery delay and a 63% increase in the PDR. © 2019 by the authors. Licensee MDPI, Basel, Switzerland.

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Training context-dependent DNN acoustic models using probabilistic sampling

Item Type Conference Paper

Author T. Grósz

Author G. Gosztolya

Author L. Tóth

Editor Lacerda F.

Editor Strombergsson S.

Editor Wlodarczak M.

Editor Heldner M.

Editor Gustafson J.

Editor House D.

Abstract In current HMM/DNN speech recognition systems, the purpose of the DNN component is to estimate the posterior probabilities of tied triphone states. In most cases the distribution of these states is uneven, meaning that we have a markedly different number of training samples for the various states. This imbalance of the training data is a source of suboptimality for most machine learning algorithms, and DNNs are no exception. A straightforward solution is to re-sample the data, either by upsampling the rarer classes or by downsampling the more common classes. Here, we experiment with the so-called probabilistic sampling method that applies downsampling and up-sampling at the same time. For this, it defines a new class distribution for the training data, which is a linear combination of the original and the uniform class distributions. As an extension to previous studies, we propose a new method to re-estimate the class priors, which is required to remedy the mismatch between the training and the test data distributions introduced by re-sampling. Using probabilistic sampling and the proposed modification we report 5% and 6% relative error rate reductions on the TED-LIUM and on the AMI corpora, respectively. Copyright © 2017 ISCA.

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Transductive Few-Shot Learning With High-Order Representation for Cross-Domain Hyperspectral Image Classification

Item Type Journal Article

Author H. Pan

Author R. Zhao

Author C. Shi

Author H. Ge

Author Y. Jin

Author M. Liu

Abstract Cross-domain hyperspectral image classification (HSIC) under few-shot conditions remains a formidable challenge due to severe data scarcity and pronounced cross-domain discrepancies. Recent advances in transductive few-shot learning (TFSL) offer a promising paradigm for tackling this problem. However, effective feature encoding, which is fundamental to TFSL, remains challenging for HSIs due to their complex high-order spectral-spatial correlations. Moreover, most existing TFSL methods assume a uniform class prior over query samples, which introduces severe bias under domain distribution mismatch and class imbalance, leading to significant performance degradation. To this end, we propose a novel method termed TFSL with high-order transformer-few shot learning (HOT-FSL), which mainly consists of three key components: high-order transformer encoder (HOTE), prototype-conditioned gated interaction module (PCGI), and conditional transport-based prototype-oriented unbiased transfer (CTPUT). The proposed HOTE employs factorized high-order attention with Kronecker-structured formulations to jointly model spatial-spectral dependencies and long-range interactions in HSI. PCGI models class prototypes from the support set and adaptively injects them into query features via contrastive routing,

thereby enhancing feature discriminability and class consistency. During inference, CTPUT employs conditional transport (CT) with forward-backward navigators and an expectation-maximization (EM)-based solver to iteratively learn unbiased class statistics and refine prototypes under imbalanced query distributions. Comprehensive experimental results on three widely used benchmarks consistently validate the superior performance of the proposed method. The code of this article is available at <https://github.com/1546342746/HOT-FSL> © 2026 IEEE.

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Transductive methods for distributed ensemble classification

Item Type Conference Paper

Author D.J. Miller

Author S. Pal

Abstract We consider ensemble classification for the case when there is no common labeled training data for designing the function which aggregates individual classifier decisions. We dub this problem distributed ensemble classification, addressing e.g. when individual classifiers are trained on different (e.g. proprietary, legacy) databases or operate (perhaps remotely) on different sensing modalities. Typically, fixed, principled (untrained) rules of classifier combination such as voting methods are used in this case for aggregating decisions. Alternatively, we take a transductive approach, optimizing the combining rule for an objective function measured on the unlabeled batch of test data. We propose specific maximum likelihood (ML) objectives that are shown to yield well-known forms of aggregation, albeit with iterative, EM-based adjustment to account for possible mismatch between the class priors used by individual classifiers and those reflected in the new data batch. We also propose an information-theoretic method which outperforms the ML methods and addresses some problem instances where the ML methods are not applicable. On benchmark data from the UC Irvine machine learning repository, all our methods give improvements in accuracy over the use of fixed rules when there is prior mismatch. © 2006 IEEE.

Date 2006

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Notes:

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Transductive methods for the distributed ensemble classification problem

Item Type Journal Article

Author D.J. Miller

Author S. Pal

Abstract We consider ensemble classification for the case where there is no common labeled training data for jointly designing the individual classifiers and the function that aggregates their decisions. This problem, which we call distributed ensemble classification, applies when individual classifiers operate (perhaps remotely) on different sensing modalities and when combining proprietary or legacy classifiers. The conventional wisdom in this case is to apply fixed rules of combination such as voting methods or rules for aggregating probabilities. Alternatively, we take a transductive approach, optimizing the combining rule for an objective function measured on the unlabeled batch of test data. We propose maximum likelihood (ML) objectives that are shown to yield well-known forms of probabilistic aggregation, albeit with iterative, expectation-maximization-based adjustment to account for mismatch between class priors used by individual classifiers and those reflected in the new data batch. These methods are extensions, for the ensemble case, of the work of Saerens, Latinne, and Decaestecker (2002). We also propose an information-theoretic method that generally outperforms the ML methods, better handles classifier redundancies, and addresses some scenarios where the ML methods are not applicable. This method also well handles the case of missing classes in the test batch. On UC Irvine benchmark data, all our methods give improvements in classification accuracy over the use of fixed rules when there is prior mismatch. © 2007 Massachusetts Institute of Technology.

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Notes:

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: D.J. Miller; Department of Electrical Engineering, Pennsylvania State University, University Park, PA 16802-2701, United States; email: millerdj@ee.psu.edu

Transductive Model Selection under Prior Probability Shift

Item Type Conference Paper

Author L. Volpi

Author A. Moreo

Author F. Sebastiani

Editor Folstad A.

Editor Apostolou D.

Editor Taylor S.

Editor Palumbo A.

Editor Tsalapati E.

Editor Stamatellos G.

Editor Catelli R.

Abstract Transductive learning is a supervised machine learning task in which, unlike in traditional inductive learning, the unlabelled data that require labelling are a finite set and are available at training time. Similarly to inductive learning contexts, transductive learning contexts may be affected by dataset shift, i.e., may be such that the assumption according to which the training data and the unlabelled data are independently and identically distributed (IID), does not hold. We here propose a method, tailored to transductive classification contexts, for performing model selection (i.e., hyperparameter optimisation) when the data exhibit prior probability shift, an important type of dataset shift typical of anti-causal learning problems. In our proposed method the hyperparameters can be optimised directly on the unlabelled data to which the trained classifier must be applied; this is unlike traditional model selection methods, that are based on performing cross-validation on the labelled training data. By tailoring model selection to the actual test distribution, our approach contributes to the trustworthiness of AI systems, as it enables more reliable and robust classifier deployment under changed conditions. We provide experimental results that show the benefits brought about by our method. © 2025 Copyright for this paper by its authors.

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Classification (of information), Learning systems, Supervised learning, Federated learning, Optimization, Unlabeled data, Hyper-parameter optimizations, Probability shifts, Accuracy prediction, Classifier accuracy prediction, Dataset shifts, Prior probability, Prior probability shift, Dataset shift, Hyperparameter optimisation, Model selection, Model Selection, Transductive learning

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: L. Volpi; Istituto di Scienza e Tecnologie dell'Informazione, Consiglio Nazionale delle Ricerche, Pisa, Via Giuseppe Moruzzi 1, 56124, Italy; email: lorenzo.volpi@isti.cnr.it; Conference name: 2025 The European Workshop on Trustworthy AI, TRUST-AI 2025; Conference date: 25 October 2025 through 26 October 2025; Conference code: 216525

Transfer estimation and the applications in data stream classification

Item Type Conference Paper

Author Z. Zhang

Author J. Zhou

Abstract Transfer estimation is a new parameter estimation method, which utilizes samples from not only the target distribution but also related ones. It is based on traditional estimators and can be used to solve the problem of small sample estimation. The key problem in transfer estimation is how to set the weighting coefficients when designing the transfer estimators. In this paper, we will propose a new algorithm to solve this problem. First, we will propose a common rule. Following the proposed rule, we can formulate the weighting coefficient setting problem as a constrained optimization problem. We will introduce an alternating optimized method and get a new algorithm of transfer estimation. The estimation of evolving class priors in data stream classification is a typical and important small sample estimation problem. In this paper, we will apply the proposed transfer estimation algorithm to the class prior estimation. Experiments on benchmark data sets will be performed, which show that the proposed algorithm can improve the performance on both class prior estimation and the final data stream classification. © 2011 SPIE.

Date 2011

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Language English

Extra Journal Abbreviation: Proc SPIE Int Soc Opt Eng

Volume 8004**ISBN** 0277786X (ISSN); 978-081948578-6 (ISBN)**Proceedings Title** Proc SPIE Int Soc Opt Eng**Conference Name** Proceedings of SPIE - The International Society for Optical Engineering**Date Added** 20/09/2026, 9:52:47 pm**Modified** 20/09/2026, 9:52:47 pm**Tags:**

Classification (of information), Computer vision, Benchmarking, Data mining, Algorithms, Data communication systems, Estimation, Computer hardware description languages, Constrained optimization, Data stream, Parameter estimation, Problem solving, Estimation algorithm, Small samples, Estimation problem, New parameters, Data stream classification, Benchmark data, Class priors estimation, Constrained optimization problems, Setting problems, Transfer estimation, Weighting coefficient

Notes:

EID: 2-s2.0-84055212323

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Z. Zhang; Department of Automation, Tsinghua National Laboratory for Information Science and Technology (TNlist), Tsinghua University, Beijing, 100084, China; email: zhangzh06@mails.tsinghua.edu.cn; Conference name: MIPPR 2011: Pattern Recognition and Computer Vision; Conference date: 4 November 2011 through 6 November 2011; Conference code: 87789; CODEN: PSISD

Transfer estimation of evolving class priors in data stream classification

Item Type Journal Article**Author** Z. Zhang**Author** J. Zhou

Abstract Data stream classification is a hot topic in data mining research. The great challenge is that the class priors may evolve along the data sequence. Algorithms have been proposed to estimate the dynamic class priors and adjust the classifier accordingly. However, the existing algorithms do not perform well on prior estimation due to the lack of samples from the target distribution. Sample size has great effects in parameter estimation and small-sample effects greatly contaminate the estimation performance. In this paper, we propose a novel parameter estimation method called transfer estimation. Transfer estimation makes use of samples not only from the target distribution but also from similar distributions. We apply this new estimation method to the existing algorithms and obtain an improved algorithm. Experiments on both synthetic and real data sets show that the improved algorithm outperforms the existing algorithms on both class prior estimation and classification. © 2010 Elsevier Ltd. All rights reserved.

Date 2010**DOI** 10.1016/j.patcog.2010.03.021

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Notes:

EID: 2-s2.0-79960376903

Export Date: 20 September 2026; Cited By: 25; Correspondence Address: J. Zhou; State Key Laboratory on Intelligent Technology and Systems, Department of Automation, Tsinghua University, Beijing 100084, China; email: jzhou@tsinghua.edu.cn; CODEN: PTNRA

Transfer Learning in Astronomy: A New Machine-Learning Paradigm

Item Type Conference Paper

Author R. Vilalta

Abstract The widespread dissemination of machine learning tools in science, particularly in astronomy, has revealed the limitation of working with simple single-task scenarios in which any task in need of a predictive model is looked in isolation, and ignores the existence of other similar tasks. In contrast, a new generation of techniques is emerging where predictive models can take advantage of previous experience to leverage information from similar tasks. The new emerging area is referred to as transfer learning. In this paper, I briefly describe the motivation behind the use of transfer learning techniques, and explain how such techniques can be used to solve popular problems in astronomy. As an example, a prevalent problem in astronomy is to estimate the class of an object (e.g., Supernova Ia) using a generation of photometric light-curve datasets where data abounds, but class labels are scarce; such analysis can benefit from spectroscopic data where class labels are known with high confidence, but the data sample is small. Transfer learning provides a robust and practical solution to leverage information from one domain to improve the accuracy of a model built on a different domain.

In the example above, transfer learning would look to overcome the difficulty in the compatibility of models between spectroscopic data and photometric data, since data properties such as size, class priors, and underlying distributions, are all expected to be significantly different. © Published under licence by IOP Publishing Ltd.

Date 2018

DOI 10.1088/1742-6596/1085/5/052014

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Learning systems, Transfer learning, Predictive analytics, Predictive modeling, Predictive models, Different domains, Photometry, Learning techniques, Spectroscopic analysis, Astronomy, Atmospheric optics, Photometric data, Practical solutions, Spectroscopic data, Underlying distribution

Notes:

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Export Date: 20 September 2026; Cited By: 9; Funding details: Center for Advanced Computing and Data Systems; Texas Institute for Measurement, Evaluation; University of Houston, UH; Correspondence Address: R. Vilalta; Department of Computer Science, University of Houston, Philip G. Hoffman Hall, Houston, 3551 Cullen Blvd., 77204-3010, United States; email: rvilalta@uh.edu; Conference name: 18th International Workshop on Advanced Computing and Analysis Techniques in Physics Research, ACAT 2017; Conference date: 21 August 2017 through 25 August 2017; Conference code: 141166

Transfer Learning-Based UAV Swarm Threat Perception and Evaluation Scheme in Air-Ground-Integrated Networks

Item Type Conference Paper

Author Y. Hou

Author S. Liu

Author L. Zhou

Author R. Zhang

Author Q. Shi

Abstract With the advancement of 5G and artificial intelligence technologies, unmanned aerial vehicle (UAV) has emerged as crucial elements in modern warfare. However, their small size and slow speed make them difficult to detect and easily mistaken for birds or other flying objects, posing challenges for defenders to respond promptly. Moreover, existing methods for evaluating UAV threats heavily rely on subjective expert scoring, leading to inefficiency, uncertainty, and ambiguity. In this paper, we propose a comprehensive approach for UAV swarm threat perception and evaluation in integrated air-ground networks. By integrating historical observation data, we establish an indicator quantification method to evaluate the perception of UAV swarm threats. Initially, we employ convolutional neural networks for intent recognition, allowing for coarse-grained classification of enemy UAVs. Subsequently, we utilize transfer learning to in-corporate the trained intent recognition model into the threat evaluation framework, significantly enhancing the efficiency and accuracy of UAV threat evaluation. Experimental results confirm the superior performance of our UAV swarm threat perception and evaluation model compared to existing approaches. © 2023 IEEE.

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Language English

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Tags:

Aerial vehicle, Antennas, Unmanned aerial vehicle, Unmanned aerial vehicles (UAV), unmanned aerial vehicle, Convolutional neural networks, Transfer learning, 5G mobile communication systems, transfer learning, Air grounds, Air-ground-integrated network, air-ground-integrated networks, Evaluation scheme, Integrated networks, Intent recognition, Threat evaluation, threat perception and evaluation, Threat perception and evaluation

Notes:

EID: 2-s2.0-85189342359

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: R. Zhang; School of Software Engineering, Tongji University, Shanghai, China; email: rongqingz@tongji.edu.cn; Conference name: 2023 China Automation Congress, CAC 2023; Conference date: 17 November 2023 through 19 November 2023; Conference code: 198194

Transmission biology of Raspberry latent virus, the first aphid-borne reovirus

Item Type Journal Article

Author D.F. Quito-Avila

Author D. Lightle

Author J. Lee

Author R.R. Martin

Abstract Raspberry latent virus (RpLV) is a newly characterized reovirus found in commercial raspberry fields in the Pacific Northwest (PNW). Thus far, all members of the plant reoviruses are transmitted in a replicative, persistent manner by several species of leafhoppers or planthoppers. After several failed attempts to transmit RpLV using leafhoppers, the large raspberry aphid, commonly found in the PNW, was tested as a vector of the virus. The virus was transmitted to new, healthy raspberry plants when inoculated with groups of at least 50 viruliferous aphids, suggesting that aphids are vectors of RpLV, albeit inefficient ones. Using absolute and relative quantification methods, it was shown that the virus titer in aphids continued to increase after the acquisition period even when aphids were serially transferred onto fresh, healthy plants on a daily basis. Transmission experiments determined that RpLV has a 6-day latent period in the aphid before it becomes transmissible; however, it was not transmitted transovarially to the next generation. To our knowledge, this is the first report of a plant reovirus transmitted by an aphid. Phylogenetic analyses showed that RpLV is related most closely to but distinct from Rice ragged stunt virus (RRSV), the type member of the genus Oryzavirus. Moreover, the conserved nucleotide termini of the genomic segments of RpLV did not match those of RRSV or other plant reoviruses, allowing us to suggest that RpLV is probably the type member of a new genus in the Reoviridae comprising aphid-transmitted reoviruses. © 2012 The American Phytopathological Society.

Date 2012

DOI 10.1094/PHYTO-12-11-0331

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Journal Abbr Phytopathology

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Notes:

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Export Date: 20 September 2026; Cited By: 27; Correspondence Address: D.F. Quito-Avila; Department of Botany and Plant Pathology, Oregon State University, Corvallis 97331, United States; email: quitod@onid.orst.edu; CODEN: PHYNB

TRAPL: Transformer-Based Patch Learning for Enhancing Semantic Representations Using Aggregated Features to Estimate Patch-Class Distribution

Item Type Conference Paper

Author S.R. Jyhne

Author P.-A. Andersen

Author I. Oveland

Author M. Goodwin

Editor Bramer M.

Editor Stahl F.

Abstract We introduce TRAPL, a Transformer-based Patch Learning technique that enhances semantic representations in segmentation models. TRAPL leverages aggregated features for precise patch-class distribution estimation, gathering features at key layers in the Transformer architecture. The method integrates an auxiliary objective with a convolution-based classifier, enabling robust semantic learning at the patch level. Our experiments demonstrate significant improvements in Intersection-over-Union (IoU) performance across models and datasets. TRAPL is compatible with both flat and hierarchical Transformers, ensuring minimal computational load during training and no extra overhead during inference. Our evaluations across state-of-the-art models and benchmarks demonstrate TRAPL's effectiveness for improving Transformer-based semantic segmentation. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

Date 2025

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Extra Journal Abbreviation: Lect. Notes Comput. Sci.

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Performance, Semantics, Semantic Segmentation, Adversarial machine learning, Contrastive Learning, Transformer, Semantic segmentation, Distribution transformers, Representation learning, Semantic representation, Representation Learning, Class distribution estimation, Class distributions, Learning techniques, Segmentation models, Semantic learning

Notes:

EID: 2-s2.0-85211435270

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S.R. Jyhne; Centre for Artificial Intelligence Research, Grimstad, Norway; email: sandsjyhne@gmail.com; Conference name: 44th SGAI International Conference on Innovative Techniques and Applications of Artificial Intelligence, SGAI 2024; Conference date: 17 December 2024 through 19 December 2024; Conference code: 323609

Tree-driven optimization inference strategy for highly resource-demanding data acquisition: Application to chemical sensors

Item Type Conference Paper

Author M. Pavone

Author R. Alesii

Author S. Tennina

Author G. Pola

Abstract Radio-frequency-based chemical sensors are a promising approach for noninvasive, low-cost identification of substances across a wide range of applications, including medical and biological contexts. However, their practical deployment is often limited by high costs or long acquisition times required to collect high-resolution spectral features, which adversely affect energy consumption, responsiveness, and the ability to capture fast-evolving chemical phenomena in healthcare applications. In this work, we address this limitation by proposing an acquisition-aware classification framework that exploits the intrinsic sequential structure of tree-based learning algorithms. Building on a previously introduced wireless chemical-sensing platform based on spectral analysis, we investigate the combined use of feature selection and decision-tree-based inference to reduce both data acquisition and inference costs. The proposed

approach models the sensing process as a query-driven system in which radio-frequency features are acquired conditionally as a classification tree is traversed. As a result, the number of acquired features required for inference is bounded by the depth of the tree rather than by the dimensionality of the full feature space. On a preliminary dataset spanning 3-6 GHz (1000 candidate features) and multiple substances, our results indicate that accurate classification can be achieved with a small subset of features, yielding a substantial reduction of expected acquisition/inference time in balanced operating conditions (>30%), with further gains when class priors are available. Furthermore, the proposed inference strategy allows additional savings when prior knowledge about class occurrence probabilities is available. These findings highlight the potential of acquisition-aware, tree-based models as an effective solution for real-world radio-frequency chemical sensing systems operating under strict time and energy constraints. © 2026 IEEE.

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Language English

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Optimisations, Classification (of information), Data acquisition, Energy utilization, Energy-consumption, Spectrum analysis, Forestry, Decision trees, Tree-based, Optimization, High costs, Cost benefit analysis, Acquisition time, Chemical detection, Chemical phenomenas, Chemical sensors, High resolution, Low-costs, Radio waves, Radiofrequencies, Spectral feature

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Item Type Journal Article

Author C.-Y. Chang

Author H.-J. Jen

Author W.-S. Su

Abstract Objective: To investigate the academic use of artificial intelligence (AI) in nursing. Background: A bibliometric analysis combined with the VOSviewer software quantification method has been utilized for a literature analysis. In recent years, this approach has attracted the interest of scholars in various research fields. Thus far, there is no publication using bibliometric analysis combined with the VOSviewer software to analyse the applications of AI in nursing. Method: A bibliometric analysis methodology was used to search for relevant articles published between 1984 and March 2022. Six databases, Embase, Scopus, PubMed, CINAHL, WoS and MEDLINE, were included to identify relevant studies, and data such as the year of publication, journals, country, institutional source, field and keywords were analysed. Results: Most relevant articles were published from institutions in the United States. The League of European Research Universities has published most research studies that use AI and nursing. Scholars have mainly focused on nursing, medical informatics, computer science AI, healthcare sciences services and physics particles fields. Commonly used keywords were machine learning, care, AI, natural language processing, prediction and nurse. Conclusion: Research articles were mainly published in Nurse Education Today. Research topics such as AI-assisted medical recording and medical decision making were also identified. According to this study, AI in nursing has the potential to attract more attention from researchers and nursing managers. Additional high-quality research beyond the scope of medical education, as well as on cross-domain collaboration, is warranted to explore the acceptability and effective implementation of AI technologies. Implications for Nursing Management: This study provides scholars and nursing managers with structured information regarding the use of AI in nursing based on scientific and technological developments across different fields and institutions. The application of AI can improve nursing management, nursing quality, safety management and team communication, as well as encourage future international collaboration. © 2022 John Wiley & Sons Ltd.

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Trends in concept and nosology of autism spectrum disorder: A review

Item Type Journal Article

Author S. Gyawali

Author B.N. Patra

Abstract Autism Spectrum Disorder is a neurodevelopmental disorder characterized by persistent deficits in social communication, social interaction and restricted, repetitive patterns of behavior, interests or activities. The concept of autism has changed since its inception, from childhood schizophrenia to neuro-variation. These changes in concept have been accompanied by changes in the diagnostic threshold through which the 'case' of autism is identified. The occurrence of multiple changes in its diagnostic criteria over last 80 odd years opens up the possibility of challenges being posed to the existence of the disorder as it is today, with a possibility of newer conceptualization of autism coming up in the future. The potential consequences of the changes in its nosology and concept, such as those on the management and on prevalence estimation are some of the essential issues which need attention. In the current paper, we evaluate the evolution of the concept and nosology of autism with an overview of the accompanying impact of these changes. © 2019 Elsevier B.V.

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Notes:

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Export Date: 20 September 2026; Cited By: 29; Correspondence Address: S. Gyawali; Department of Psychiatry, All India Institute of Medical Sciences, New Delhi, Room no. 4096, Ansari Nagar, 110029, India; email: shreeya.gyawali@gmail.com

TriDroid: a triage and classification framework for fast detection of mobile threats in android markets

Item Type Journal Article

Author A. Amira

Author A. Derhab

Author E.M.B. Karbab

Author O. Nouali

Author F.A. Khan

Abstract The Android platform is highly targeted by malware developers, which aim to infect the maximum number of mobile devices by uploading their malicious applications to different app markets. In order to keep a healthy Android ecosystem, app-markets check the maliciousness of newly submitted apps. These markets need to (a) correctly detect malicious app, and (b) speed up the detection process of the most likely dangerous applications among an overwhelming flow of submitted apps, to quickly mitigate their potential damages. To address these challenges, we propose TriDroid, a market-scale triage and classification system for Android apps. TriDroid prioritizes apps analysis according to their risk likelihood. To this end, we categorize the submitted apps as: botnet, general malware, and benign. TriDroid starts by performing a (1) Triage process, which applies a fast coarse-grained and less-accurate analysis on a continuous stream of the submitted apps to identify their corresponding queue in a three-class priority queuing system. Then, (2) the Classification process extracts fine-grained static features from the apps in the priority queue, and applies three-class machine learning classifiers to confirm with high accuracy the classification decisions of the triage process. In addition to the priority queuing model, we also propose a multi-server queuing model where the classification of each app category is run on a different server. Experiments on a dataset with more than 24K malicious and 3K benign applications show that the priority model offers a trade-off between waiting time and processing overhead, as it requires only one server compared to the multi-server model. Also it successfully prioritizes malicious apps analysis, which allows a short waiting time for dangerous applications compared to the FIFO policy. © 2020, Springer-Verlag GmbH Germany, part of Springer Nature.

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Risk assessment, Economic and social effects, Learning systems, Machine learning, Damage detection, Queueing theory, Malware, Data mining, Mobile security, Classification system, Classification framework, Accurate analysis, Android (operating system), Android platforms, Android security, App triage, Classification decision, Classification process, Detection process, Malware detection, Processing overhead

Notes:

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Trustworthy AI in Video Surveillance: The IMMAGINA Project

Item Type Conference Paper
Author E. Ledda
Author L. Putzu
Author R. Delussu
Author G. Fumera
Author F. Roli
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Editor Colantonio S.

Editor Via G. Moruzzi 1, Pisa Institute of Information Science and Technologies "A. Faedo" of the National Research Council of Italy

Abstract The increasing adoption of machine learning and deep learning models in critical applications raises the issue of ensuring their trustworthiness, which can be addressed by quantifying the uncertainty of their predictions. However, the black-box nature of many such models allows only to quantify uncertainty through ad hoc superstructures, which require to develop and train a model in an uncertainty-aware fashion. However, for applications where previously trained models are already in operation, it would be interesting to develop uncertainty quantification approaches acting as lightweight “plug-ins” that can be applied on top of such models without modifying and re-training them. In this contribution we present a research activity of the Pattern Recognition and Applications Lab of the University of Cagliari related to a recently proposed post hoc uncertainty quantification method, we named dropout injection, which is a variant of the well-known Monte Carlo dropout, and does not require any re-training nor any further gradient descent-based optimization; this makes it a promising, lightweight solution for integrating uncertainty quantification on any already-trained neural network. We are investigating a theoretically grounded solution to make dropout injection as effective as Monte Carlo dropout through a suitable rescaling of its uncertainty measure; we are also evaluating its effectiveness in the computer vision tasks of crowd counting and density estimation for intelligent video surveillance, thanks to our participation in a project funded by the European Space Agency. © 2022 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

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Trustworthy clinical AI solutions: A unified review of uncertainty quantification in Deep Learning models for medical image analysis

Item Type Journal Article

Author B. Lambert

Author F. Forbes

Author S. Doyle

Author H. Dehaene

Author M. Dojat

Abstract The full acceptance of Deep Learning (DL) models in the clinical field is rather low with respect to the quantity of high-performing solutions reported in the literature. End users are particularly reluctant to rely on the opaque predictions of DL models. Uncertainty quantification methods have been proposed in the literature as a potential solution, to reduce the black-box effect of DL models and increase the interpretability and the acceptability of the result by the final user. In this review, we propose an overview of the existing methods to quantify uncertainty associated with DL predictions. We focus on applications to medical image analysis, which present specific challenges due to the high dimensionality of images and their variable quality, as well as constraints associated with real-world clinical routine. Moreover, we discuss the concept of structural uncertainty, a corpus of methods to facilitate the alignment of segmentation uncertainty estimates with clinical attention. We then discuss the evaluation protocols to validate the relevance of uncertainty estimates. Finally, we highlight the open challenges for uncertainty quantification in the medical field. © 2024 The Author(s)

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Export Date: 20 September 2026; Cited By: 226; Correspondence Address: M. Dojat; Univ. Grenoble Alpes, Inserm, U1216, Grenoble Institut des Neurosciences, Grenoble, 38000, France; email: Michel.Dojat@inserm.fr; CODEN: AIMEE

Trustworthy Deep Neural Network for Inferring Anticancer Synergistic Combinations

Item Type Journal Article
Author M.A. Alsherbiny
Author I. Radwan
Author N. Moustafa
Author D.J. Bhuyan
Author M. El-Waisi
Author D. Chang
Author C.G. Li

Abstract The lack of a gold standard synergy quantification method for chemotherapeutic drug combinations warrants the consideration of different synergy metrics to develop efficient predictive models. Furthermore, neglecting combination sensitivity may lead to biased synergistic combinations, which are ineffective in cancer treatment. In this paper, we propose a deep learning-based model, SynPredict, which effectively predicts synergy in five synergy metrics together with the combination sensitivity score. SynPredict assesses the impact of multimodal fusion architectures of the input data, including the gene expression data of cancer cells, along with the representative chemical features of drugs in pairwise combinations. Both ONEIL and ALMANAC anticancer combination datasets are employed comparatively. The impact of the training datasets was more significant and consistent across most synergy models than input data fusion architectures. Synpredict outperforms the state-of-the-art predictive models, including DeepSynergy, AuDNN synergy, TranSynergy and DrugComb, with up to 74% decline in the mean square error. We highlight the pivotal need to consider a multiplex of synergy metrics and the combined sensitivity in the predictive models. © 2013 IEEE.

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Trustworthy Open Set Domain Generalization Network for Unknown Fault Diagnosis Under Unseen Conditions

Item Type Journal Article

Author Z. Fan

Author B. Yan

Author C. Xu

Author S. Lu

Author K. Zhong

Abstract Advanced intelligent fault diagnosis (IFD) methods based on domain generalization (DG) leverage multisource sensor data to overcome the limitations the limitations of domain adaptation (DA)-based models regarding target data demand in the training stage, which resolves domain shift problems under unseen conditions in industrial instrumentation. However, previous studies primarily focused on the closed-set diagnosis with same label space shared by the training and testing data, which struggle to address the problem of new fault identification under intricate industrial dynamics. To overcome this obstacle, a trustworthy evidential open-set DG network (EOSDGN) is proposed for open-set fault diagnosis under unseen conditions. In the EOSDGN method, an evidential deep classifier is constructed to quantify the uncertainty of predicted results, an evidential domain discriminator is employed to integrate the data from diverse source domains, and an evidential uncertainty calibration is established to reconcile the misleading evidence and class-based evidence assignments. The EOSDGN model effectively addresses both domain and label shift challenges by integrating domain-invariant features extraction and uncertainty quantification of the predicted probabilities, which enables efficient classification of known faults while simultaneously facilitating the identification of unknown faults. The effectiveness of the EOSDGN model has been substantiated using public and practical datasets. Experimental findings demonstrate that the EOSDGN model surpasses the performance of the state-of-the-art models. © 1963-2012 IEEE.

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Condition, Deep learning, Classification (of information), Uncertainty analysis, Uncertainty, Generalisation, Statistical tests, Failure analysis, Network models, Domain Knowledge, Domain generalization, domain generalization (DG), Electric fault currents, Fault detection, Fault diagnosis method, Faults diagnosis, Intelligent fault diagnosis, Open-set fault diagnose, open-set fault diagnosis

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Tunnel surrounding rock joint recognition and rock mass integrity evaluation based on an improved YOLOv8-seg model

Item Type Journal Article

Author W. Dong

Author W. Zou

Author L. Xu

Author Z. Ren

Abstract Rock fractures significantly influence rock classification and engineering parameter determination. Traditional joint extraction methods are time-consuming, subjective, and error-prone, hindering intelligent tunnel engineering and compromising construction safety. This study proposes an automated rock integrity quantification method using the GAM-BiFPN-YOLOv8-seg model. A standardized process for collecting tunnel face images was established, creating a diverse joint image dataset. Incorporating global attention mechanisms and weighted bidirectional feature pyramid networks into the YOLOv8-seg framework improved detection of subtle joint features and multi-scale fusion, enhancing joint extraction accuracy. After binarization and applying the Zhang-Suen thinning algorithm, joint traces were combined with surrounding joint counts to automate rock integrity quantification. Integrating the relationship between surrounding rock joint counts and rock mass integrity indices, the method achieved a 6.34% improvement in mAP50 and an 8.42% improvement in mAP50:95 compared to the original YOLOv8-seg network, with marginal gains in detection speed. During engineering validation at the NEOM New City Tunnel project, the proposed method demonstrated consistent performance with human decision-making in quantifying tunnel face joints. The method provides an effective solution for the accurate assessment of rock mass integrity and has broad

application prospects in the field of tunnel engineering and geological engineering. ©
The Author(s) 2026.

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EID: 2-s2.0-105034175619

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Z. Ren; School of Future Technology, Xi'an University of Architecture and Technology, Xi'an, Shaanxi, 710055, China; email: zhenzhongr@163.com

Tutorial: Supervised Learning for Prevalence Estimation

Item Type Conference Paper

Author A. Moreo

Author F. Sebastiani

Editor Cuzzocrea A.

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Editor Christiansen H.

Abstract Quantification is the task of estimating, given a set of unlabelled items and a set of classes, the relative frequency (or “prevalence”). Quantification is important in many disciplines (such as e.g., market research, political science, the social

sciences, and epidemiology) which usually deal with aggregate (as opposed to individual) data. In these contexts, classifying individual unlabelled instances is usually not a primary goal, while estimating the prevalence of the classes of interest in the data is. Quantification may in principle be solved via classification, i.e., by classifying each item in and counting, for all, how many such items have been labelled with. However, it has been shown in a multitude of works that this “classify and count” (CC) method yields suboptimal quantification accuracy, one of the reasons being that most classifiers are optimized for classification accuracy, and not for quantification accuracy. As a result, quantification has come to be no longer considered a mere byproduct of classification, and has evolved as a task of its own, devoted to designing methods and algorithms that deliver better prevalence estimates than CC. The goal of this tutorial is to introduce the main supervised learning techniques that have been proposed for solving quantification, the metrics used to evaluate them, and the most promising directions for further research. © 2019, Springer Nature Switzerland AG.

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Tweet sentiment quantification: An experimental re-evaluation

Item Type Journal Article**Author** A. Moreo**Author** F. Sebastiani

Abstract Sentiment quantification is the task of training, by means of supervised learning, estimators of the relative frequency (also called "prevalence") of sentiment-related classes (such as Positive, Neutral, Negative) in a sample of unlabelled texts. This task is especially important when these texts are tweets, since the final goal of most sentiment classification efforts carried out on Twitter data is actually quantification (and not the classification of individual tweets). It is well-known that solving quantification by means of "classify and count" (i.e., by classifying all unlabelled items by means of a standard classifier and counting the items that have been assigned to a given class) is less than optimal in terms of accuracy, and that more accurate quantification methods exist. Gao and Sebastiani 2016 carried out a systematic comparison of quantification methods on the task of tweet sentiment quantification. In hindsight, we observe that the experimentation carried out in that work was weak, and that the reliability of the conclusions that were drawn from the results is thus questionable. We here re-evaluate those quantification methods (plus a few more modern ones) on exactly the same datasets, this time following a now consolidated and robust experimental protocol (which also involves simulating the presence, in the test data, of class prevalence values very different from those of the training set). This experimental protocol (even without counting the newly added methods) involves a number of experiments 5,775 times larger than that of the original study. Due to the above-mentioned presence, in the test data, of samples characterised by class prevalence values very different from those of the training set, the results of our experiments are dramatically different from those obtained by Gao and Sebastiani, and provide a different, much more solid understanding of the relative strengths and weaknesses of different sentiment quantification methods. © 2022 Moreo, Sebastiani. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Date 2022**DOI** 10.1371/journal.pone.0263449**URL** <https://www.scopus.com/pages/publications/85136162729?origin=resultslist>**Archive** Scopus**Language** English**Volume** 17**Publisher** Public Library of Science**Publication** PLoS ONE**ISSN** 19326203 (ISSN)**Issue** 9 September**Journal Abbr** PLoS ONE**Date Added** 20/09/2026, 9:52:36 pm**Modified** 20/09/2026, 9:52:36 pm**Tags:**

attitude, human, Humans, article, Attitude, reliability, reproducibility, Reproducibility of Results, social media, Social Media, prevalence, intermethod comparison, information processing, Data Collection

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Twise at SemEval-2017 Task 4: Five-point Twitter Sentiment Classification and Quantification

- Item Type** Conference Paper
- Author** G. Balikas
- Abstract** The paper describes the participation of the team “Twise” in the SemEval-2017 challenge. Specifically, I participated at Task 4 entitled “Sentiment Analysis in Twitter” for which I implemented systems for five-point tweet classification (Subtask C) and five-point tweet quantification (Subtask E) for English tweets. In the feature extraction steps the systems rely on the vector space model, morpho-syntactic analysis of the tweets and several sentiment lexicons. The classification step of Subtask C uses a Logistic Regression trained with the one-versus-rest approach. Another instance of Logistic Regression combined with the classify-and-count approach is trained for the quantification task of Subtask E. In the official leaderboard the system is ranked 5/15 in Subtask C and 2/12 in Subtask E. © 2017 Association for Computational Linguistics
- Date** 2017
- DOI** 10.18653/v1/s17-2127
- URL** <https://www.scopus.com/pages/publications/85122581635?origin=resultslist>
- Archive** Scopus
- Language** English
- Extra** Journal Abbreviation: Proc. Annu. Meet. Assoc. Comput Linguist.
- Publisher** Association for Computational Linguistics (ACL)
- ISBN** 0736587X (ISSN); 978-194562655-5 (ISBN)
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Classification (of information), Features extraction, Sentiment analysis, Social networking (online), Semantics, Regression analysis, Sentiment classification, Logistics regressions, Syntactics, Vector spaces, Sentiment lexicons, Subtask, Syntactic analysis, Vector space models

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: G. Balikas; University of Grenoble Alps, CNRS, Grenoble INP, LIG, France; email: georgios.balikas@imag.fr; Conference name: 11th International Workshop on Semantic Evaluations, SemEval 2017, co-located with the 55th Annual Meeting of the Association for Computational Linguistics, ACL 2017; Conference date: 3 August 2017 through 4 August 2017; Conference code: 173837

Two-group classification via a biobjective margin maximization model

Item Type Journal Article**Author** E. Carrizosa**Author** B. Martin-Barragan

Abstract In this paper we propose a biobjective model for two-group classification via margin maximization, in which the margins in both classes are simultaneously maximized. The set of Pareto-optimal solutions is described, yielding a set of parallel hyperplanes, one of which is just the solution of the classical SVM approach. In order to take into account different misclassification costs or a priori probabilities, the ROC curve can be used to select one out of such hyperplanes by expressing the adequate tradeoff for sensitivity and specificity. Our result gives a theoretical motivation for using the ROC approach in case misclassification costs in the two groups are not necessarily equal. © 2005 Elsevier B.V. All rights reserved.

Date 2006**DOI** 10.1016/j.ejor.2005.06.059**URL** <https://www.scopus.com/pages/publications/33744968620?origin=resultslist>**Archive** Scopus**Language** English**Volume** 173**Pages** 746-761**Publication** European Journal of Operational Research**ISSN** 03772217 (ISSN)**Issue** 3**Journal Abbr** Eur J Oper Res**Date Added** 20/09/2026, 9:53:03 pm**Modified** 20/09/2026, 9:53:03 pm**Tags:**

Support vector machines, Classification (of information), Data mining, Mathematical models, Optimization, Classification, Computer programming, Parallel processing systems, ROC curve, Biobjective, Multiple objective programming

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Two-phase sampling for simultaneous prevalence estimation and case detection

Item Type Journal Article

Author R. McNamee

Abstract Two-phase designs for estimation of prevalence, where the first-phase classification is fallible and the second is accurate but relatively expensive, are not necessarily justified on efficiency grounds. However, they might be advantageous for dual-purpose studies, for example where prevalence estimation is followed by a clinical trial or case-control study, if they can identify cases of disease for the second study in a cost-effective way. Alternatively, they may be justified on ethical grounds if they can identify more, previously undetected but treatable cases of disease, than a simple random sample design. An approach to sampling is proposed, which formally combines the goals of efficient prevalence estimation and case detection by setting different notional study costs for investigating cases and noncases. Two variants of the method are compared with an "ethical" two-phase scheme proposed by Shrout and Newman (1989, *Biometrics* 45, 549-555), and with the most efficient scheme for prevalence estimation alone, in terms of the standard error of the prevalence estimate, the expected number of cases, and the fraction of cases among second-phase subjects, given a fixed budget. One variant yields the highest fraction and expected number of cases but also the largest standard errors. The other yields a higher fraction than Shrout and Newman's scheme and a similar number of cases but appears to do so more efficiently.

Date 2004

DOI 10.1111/j.0006-341X.2004.00229.x

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: R. McNamee; Biostatistics Group, Sch. of Epidemiol. and Hlth. Sci., University of Manchester, Manchester M13 9PT, Oxford Road, United Kingdom; email: rmcnamee@man.ac.uk; CODEN: BIOMA

U-TraCE: a conformal prediction approach to uncertainty quantification in black-box models

Item Type Journal Article

Author E. J Marchi

Author M. Liebl

Abstract Modern machine learning (ML) systems, particularly deep neural networks, have become increasingly complex and opaque, making reliable uncertainty quantification (UQ) essential. Conventional UQ methods often prove inadequate for black-box model evaluation scenarios, either because model internals are not accessible or because the resulting uncertainty is expressed in terms of a related predictor rather than the model under evaluation. We introduce U-TraCE (Uncertainty Tracking for Complex Estimators), a novel UQ method built on the Conformal Prediction (CP) framework that provides formal and traceable uncertainty bounds for opaque ML systems. Unlike existing CP approaches, which describe uncertainty through confidence intervals or prediction set sizes at a fixed uncertainty level, U-TraCE yields an explicit numerical bound on the error probability of the standalone black-box model. U-TraCE was evaluated on diverse tasks, including MNIST classification and cardiac magnetic resonance imaging (MRI) segmentation, representing both benchmark and real-world domains. Across these settings, it produced conservative yet well-calibrated uncertainty bounds that closely follow empirical uncertainty, even with limited calibration data. By bridging theoretical UQ and operational model validation, U-TraCE establishes a foundation for objective and automated conformity assessment of complex ML systems, supporting trustworthy deployment in safety-critical environments. © 2026 The Author(s). Published by IOP Publishing Ltd.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Liebl; Department of Metrology for Digital Transformation, Physikalisch-Technische Bundesanstalt (PTB), Berlin, Abbestr. 2-12, 10587, Germany; email: maik.liebl@ptb.de

UGPL: Uncertainty-Guided Progressive Learning for Evidence-Based Classification in Computed Tomography

Item Type Conference Paper

Author S. Venkatraman

Author S. Pavan Kumar

Author R.R. Madavan

Author S. Chandrakala

Abstract Accurate classification of computed tomography (CT) images is essential for diagnosis and treatment planning, but existing methods often struggle with the subtle and spatially diverse nature of pathological features. Current approaches typically process images uniformly, limiting their ability to detect localized abnormalities that require focused analysis. We introduce UGPL, an uncertainty-guided progressive learning framework that performs a global-to-local analysis by first identifying regions of diagnostic ambiguity and then conducting detailed examination of these critical areas. Our approach employs evidential deep learning to quantify predictive uncertainty, guiding the extraction of informative patches through a non-maximum suppression mechanism that maintains spatial diversity. This progressive refinement strategy, combined with an adaptive fusion mechanism, enables UGPL to integrate both contextual information and fine-grained details. Experiments across three CT datasets demonstrate that UGPL consistently outperforms state-of-the-art methods, achieving improvements of 3.29%, 2.46%, and 8.08% in accuracy for kidney abnormality, lung cancer, and

COVID-19 detection, respectively. Our analysis shows that the uncertainty-guided component provides substantial benefits, with performance dramatically increasing when the full progressive learning pipeline is implemented. Our code is available at <https://github.com/shravan-18/UGPL>. © 2025 IEEE.

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Language	English
Extra	Journal Abbreviation: Proc. - IEEE/CVF Int. Conf. Computer Vis. Workshops, ICCV-W
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	979-833158988-2 (ISBN)
Pages	969-979
Proceedings Title	Proc. - IEEE/CVF Int. Conf. Computer Vis. Workshops, ICCV-W
Conference Name	Proceedings - 2025 IEEE/CVF International Conference on Computer Vision Workshops, ICCV-W 2025
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Tags:

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Ultra-Sensitive and Validated method for Trace Quantification of Carcinogenic Benzene in Cetirizine Dihydrochloride

- Item Type** Journal Article
- Author** M.M. Gharat
- Author** P.T. Roy
- Author** A.N. Gosar
- Author** T.A. Shaikh
- Author** N.A. Mirgane

Abstract Benzene (BENZ) is classified as a Class 1 residual solvent according to ICH Q3C guidelines due to its established classification as a Group 1 human carcinogen. During the chemical production of Cetirizine Dihydrochloride (CTZ) API, Cetirizine Dihydrochloride tablet (CTZT), Levocetirizine Dihydrochloride (LCTZ) API, and Levocetirizine Dihydrochloride tablet (LCTZT), BENZ can appear as a residual process impurity with carcinogenic risk; development of a highly sensitive quantification method is a critical requirement for pharmaceutical quality control. This study focused on developing and validating a method for detecting trace-level BENZ in CTZ and LCTZ drug substance and drug product. The method demonstrated exceptional sensitivity, with a least detectable concentration of 0.04 ppm and a quantifiable concentration of 0.12 ppm, significantly lower than the ICH-mandated safety limit of 2 ppm. Linearity and accuracy studies yielded an outstanding percentage of samples spiked BEN in the drug substance and drug product of CTZ and LCTZ and found within acceptance limit, approving the method's validity in presence of the drug atmosphere. Conventional headspace GC-FID/GC-MS are established techniques for benzene analysis; the proposed HPLC method offers a simpler, cost-effective, and validated method, making it well suited for routine quality control laboratories where GC facilities may not be readily available. © 2026 John Wiley & Sons Ltd.

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Article, reproducibility, Reproducibility of Results, Cost effectiveness, procedures, controlled study, Quality control, Cost benefit analysis, Drug products, chemistry, benzene, Benzene, benzene (BENZ), carcinogen, carcinogens, Carcinogens, cetirizine, Cetirizine, Cetirizine dihydrochloride, cetirizine dihydrochloride (CTZ), concentration (parameter), Controlled drug delivery, drug contamination, Drug Contamination, Drug products plants, Drug substances, gas chromatography, Gas Chromatography-Mass Spectrometry, high performance liquid chromatography, High performance liquid chromatography, high performance liquid chromatography (HPLC), High-performance liquid chromatography, levocetirizine, Levocetirizine dihydrochloride, levocetirizine dihydrochloride (LCTZ), limit of detection, Limit of Detection, Linear Models, mass fragmentography, statistical model, tablet, Ultra sensitive method, Validated methods

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Ultrasound Operator Variance Classification for Agency in Artificial Intelligence Support of Cyber-Physical Systems

Item Type Conference Paper

Author C. Zhu

Author T.E. Doyle

Author M.D. Noseworthy

Abstract Ultrasound (US) is the most widely used medical imaging modality due to its low cost, portability, real time imaging ability and use of non-ionizing radiation. However, unlike other imaging modalities such as CT or MRI, it is a heavily operator dependent, requiring trained expertise to leverage these benefits. More broadly, the natural interaction between human and computer in general cyber-physical systems would benefit from the support of artificial intelligence (AI) that has the agency to adapt its response based on operator variance. The focus of this paper will be on US operator variance as a first step in demonstrating the concept of AI agency. Recently there has been an explosion of interest in AI across the medical community and many are turning to the growing trend of deep learning (DL) models to assist in diagnosis. However, deep learning models do not perform as well when training data is not fully representative of the problem. Due to this difference in training and deployment, model performance suffers which can lead to misdiagnosis. This issue is known as dataset shift. Two aims to address dataset shift were proposed. The first was to quantify how US operator skill and hardware affects acquired images. The second was to use this skill quantification method to screen and match data to deep learning models to improve performance. A CAE Healthcare BLUE phantom with various mock lesions was scanned by three operators using three different US systems (Siemens S3000, Clarius L15, and Ultrasonix SonixTouch) producing 39013 images. DL models were trained on a specific set to classify the presence of a simulated tumour and tested with data from differing sets. PCA for dimension reduction was applied then K-Means clustering was used to separate images into clusters. This clustering algorithm was then used to screen incoming images during deployment to best match input to an appropriate DL model which is trained specifically to classify that type of operator or hardware. Results showed a noticeable difference when models were given data from differing datasets with the largest accuracy drop being 81.26% to 31.26%. © 2022 IEEE.

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Volume 2022-September

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Deep learning, Learning systems, Cyber Physical System, Cybe-physical systems, Cyber-physical systems, Embedded systems, Learning models, cyber-physical system, Medical imaging, K-means clustering, Diagnosis, Magnetic resonance imaging, Dataset shifts, Low-costs, Ultrasonics, ultrasound, Imaging modality, Ionizing radiation, Operator variance, AI agency, Artificial intelligence agency, Intelligence agencies, Intelligence support, operator variance

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Unambiguous discrimination between linearly dependent equidistant states with multiple copies

Item Type Journal Article

Author W.-H. Zhang

Author G. Ren

Abstract Linearly independent quantum states can be unambiguously discriminated, but linearly dependent ones cannot. For linearly dependent quantum states, however, if C copies of the single states are available, then they may form linearly independent states, and can be unambiguously discriminated. We consider unambiguous discrimination among $N = D + 1$ linearly dependent states given that C copies are available and that the single copies span a D -dimensional space with equal inner products. The maximum unambiguous discrimination probability is derived for all C with equal a priori probabilities. For this classification of the linearly dependent equidistant states, our result shows that if C is even then adding a further copy fails to increase the maximum discrimination probability. © 2018, Springer Science+Business Media, LLC, part of Springer Nature.

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Unbiased Recommender Learning from Implicit Feedback via Weakly Supervised Learning

Item Type Conference Paper

Author H. Wang

Author Z. Chen

Author H. Wang

Author Y. Tan

Author L. Pan

Author T. Liu

Author X. Chen

Author H. Li

Author Z. Lin

Editor Singh A.

Editor Fazel M.

Editor Hsu D.

Editor Lacoste-Julien S.

Editor Berkenkamp F.

Editor Maharaj T.

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Editor Zhu J.

Abstract Implicit feedback recommendation is challenged by the missing negative feedback essential for effective model training. Existing methods of ten resort to negative sampling, a technique that assumes unlabeled interactions as negative samples. This assumption risks misclassifying potential positive samples within the unlabeled data, thereby undermining model performance. To address this issue, we introduce WeaklyRec, a model-agnostic framework that reframes implicit feedback recommendation as a weakly supervised learning task, eliminating the need for negative samples. However, its unbiasedness hinges on the accurate estimation of the class prior. To address this challenge, we propose Progressive Proximal Transport (PPT), which estimates the class prior by minimizing the proximal transport cost between positive and unlabeled samples. Experiments on three real-world datasets validate the efficacy of WeaklyRec in terms of improved recommendation quality. Code is available at <https://github.com/HowardZJU/weakrec>. © 2025 by the author(s).

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Uncertainty analysis on support vector machine for measuring organizational factors in probabilistic risk assessment of nuclear power plants

Item Type Journal Article

Author J. Yang

Author J. Kim

Author P. Farshadmanesh

Author T. Sakurahara

Author S. Reihani

Author C. Blake

Author Z. Mohaghegh

Abstract This paper is a product of a line of research by the authors to explicitly incorporate organizational factors into the probabilistic risk assessment (PRA) of complex socio-technical systems. This explicit incorporation helps (i) assess the system risk due to organizational and managerial weaknesses, (ii) identify the critical organizational root causes of failure scenarios, aiding in effective corrective action, and (iii) avoid the potential of underestimating the system risk involving human error and organizational factors. To facilitate the measurement of organizational factors contributing to the PRA scenarios, the previous studies by the authors developed the Data-Theoretic methodology, where “data analytics” are guided by a “theory” of underlying causation to prevent misleading results from solely data-oriented approaches. The Data-Theoretic methodology consists of two submodules: (a) DT-BASE that develops a detailed causal model based on a theory-building process and is equipped with a baseline quantification utilizing analyst interpretation of information extracted from relevant references; and (b) DT-SITE that conducts data analytics (text mining using machine learning) to quantify the organizational causal elements based on industry event databases. This article investigates uncertainty analysis for the uncertainties associated with machine learning-based data mining in DT-SITE using a Support Vector Machine (SVM) classifier for industry event databases. This article conducts a literature review and proposes a categorization scheme of the SVM uncertainties to establish a theoretical foundation for identifying uncertainty sources associated with the SVM classifier. The implementation of uncertainty analysis for DT-SITE is then illustrated using a nuclear power plant case study. The potential uncertainty sources related to the SVM classifier used in DT-SITE are identified using the categorization scheme developed based on the literature review. The uncertainty analysis procedure for the SVM classifier in DT-SITE is proposed, and the linkage to the current uncertainty analysis process in PRA is discussed. The proposed uncertainty analysis procedure for the SVM classifier in DT-SITE is applied to the illustrative case study to assess the impact of one of the potential uncertainty sources (i.e., the selection of document sections included in the dataset) on the risk-informed decision-making. © 2022 Elsevier Ltd

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Uncertainty assessment of proarrhythmia predictions derived from multi-level in silico models

Item Type Journal Article

Author K. Kopańska

Author P. Rodríguez-Belenguer

Author J. Llopis-Lorente

Author B. Trenor

Author J. Saiz

Author M. Pastor

Abstract In silico methods can be used for an early assessment of arrhythmogenic properties of drug candidates. However, their use for decision-making is conditioned by the possibility to estimate the predictions' uncertainty. This work describes our efforts to develop uncertainty quantification methods for the predictions produced by multi-level proarrhythmia models. In silico models used in this field usually start with experimental or predicted IC50 values that describe drug-induced ion channel blockade. Using such inputs, an electrophysiological model computes how the ion channel inhibition, exerted by a drug in a certain concentration, translates to an altered shape and duration of the action potential in cardiac cells, which can be represented as arrhythmogenic risk biomarkers such as the APD90. Using this framework, we identify the main sources of aleatory and epistemic uncertainties and propose a method based on probabilistic simulations that replaces single-point estimates predicted using multiple input values, including the IC50s and the electrophysiological parameters, by distributions of values. Two selected variability types associated with these inputs are then propagated through the multi-level model to estimate their impact on the uncertainty levels in the output, expressed by means of

intervals. The proposed approach yields single predictions of arrhythmogenic risk biomarkers together with value intervals, providing a more comprehensive and realistic description of drug effects on a human population. The methodology was tested by predicting arrhythmogenic biomarkers on a series of twelve well-characterised marketed drugs, belonging to different arrhythmogenic risk classes. © 2023, The Author(s).

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Uncertainty estimation of deep learning models for atrial fibrillation detection from Holter recordings: A benchmark study

Item Type Journal Article

Author M.M. Rahman

Author M.W. Rivolta

Author F. Badilini

Author R. Sassi

Abstract With the development of deep learning (DL)-based methods, automated atrial fibrillation (AF) detection from electrocardiograms (ECGs) has recently gained much attention. Although the performance of DL has been encouraging, the susceptibility of DL models to overfitting would benefit from the exploration of uncertainty quantification (UQ) to ensure safe integration into clinical practice. However, there has been limited exploration of UQ methods in the context of DL models for AF detection using Holter ECG recordings, and a comprehensive comparison of various UQ techniques remains absent. This study addressed this gap by introducing a benchmark study wherein 11 distinct UQ methods were rigorously evaluated and compared across three public Holter repositories: IRIDIA-AF, Long-Term AF, and MIT-BIH AF datasets. A residual DL model was used for the UQ methods, which is one of the most common architectures in this domain for its ability to capture complex patterns within ECG data. The findings revealed that batch-ensemble (BE) and packed-ensemble (PE) outperformed other UQ methods concerning both performance, as quantified by sensitivity, specificity and expected calibration error, and computational efficiency. In addition, when we implemented reject inference to discard ECG segments where the model confidence was not sufficiently high, BE and PE still showed to reject the least number of samples, while retaining the highest detection performance. © 2025

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Uncertainty Nonlinearly Guided Learning Framework for Full-Wave Inverse Scattering

Item Type Journal Article

Author S. He

Author L. Jin

Author L. Pan

Author X. Li

Author Z. Wei

Abstract Recently, deep learning schemes have been widely used in electromagnetic society. However, different from traditional methods constrained by physics models, the predicted results of these "black-box" neural networks are not reliable, and the prediction errors of networks can only be obtained when ground truths are known, which severely limits the applications of learning-based solvers. Consequently, uncertainty quantifications (UQs) are introduced in learning-based electromagnetic solvers to provide a confidence level. In this work, by incorporating the properties of physical setup in inverse scattering problems (ISPs), an equivariance-based uncertainty quantification (EUQ) method is first proposed, where the diversities caused by different antennas are utilized to quantify the model uncertainty with virtual experiments. More importantly, the quantified uncertainties further serve as nonlinear guidances in ISP learning solvers, which makes it flexible to adaptively choose targeted reconstruction regions. It is shown by intensive tests that the proposed EUQ has a better correlation with the true error compared with traditional UQ methods. The learning framework nonlinearly guided by EUQ also shows much better performance and achieves better predictions in target selected areas in both numerical and experimental tests. © 1963-2012 IEEE.

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Uncertainty prediction for prominence classification with chroma features

Item Type Conference Paper
Author J. Linke
Author S. Steger
Author P. Steinwender
Author G. Kubin
Author F. Pernkopf
Author B. Schuppler
Editor Rao B.D.

Editor Trancoso I.

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Editor Mehta N.B.

Abstract This paper presents methods for prominence classification in conversational speech. Most existing tools rely on prosodic features extracted at syllable- or phone-level, performing well on read speech. This is not the case for conversational speech, where the quality of automatic segmentation is significantly worse. We introduce entropy-based chroma features, requiring only word-level segmentations. They perform equally well as a random forest classifier with prosodic features (requiring phone-level segmentation), with accuracies in the range of the human inter-rater agreement. We further use Bayesian deep learning to quantify the epistemic and aleatoric uncertainty of the prediction for prosodic and chroma features. Whereas the aleatoric uncertainty is, as expected, consistent with inter-rater agreement and similarly high for both feature sets, the epistemic uncertainty is lower for the classifier based on chroma features, indicating higher classification consistency across the corpus. © 2025 IEEE.

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Uncertainty propagation from sensor data to deep learning models in autonomous driving

Item Type Journal Article

Author Y. Wang

Author T. Wang

Author T. Yue

Abstract Context: Deep learning has been widely used in Autonomous Driving Systems (ADS). Though significant progress has been made regarding their efficiency and accuracy, uncertainty remains a critical factor affecting ADS safety. Such uncertainties are often due to environmental noise and/or imperfect algorithm structures. Studies on uncertainty quantification mostly focus on single classification tasks and overlook how uncertainties propagate from the perception to downstream decision-making, studying of which is critical, as the interplay between perception and decision-making can significantly impact the overall safety of ADS. Objectives: We quantify and understand the uncertainty propagation from sensor data to deep learning models, as well as its impact on ADS safety. Methods: We present an empirical study that quantifies both aleatoric and epistemic uncertainties and assesses how such uncertainties propagate and impact ADS safety under various sensor noise conditions. We also investigate the suitability of two epistemic uncertainty quantification methods (i.e., MC Dropout and Deep Ensembles) to ADS tasks and their cost-effectiveness in selecting highly-uncertain samples. Results: Results show that increased noise can significantly increase uncertainty and degrade model performance, thereby compromising decision-making and potentially impacting ADS safety. Both MC Dropout and Deep Ensembles effectively measure the model's epistemic uncertainty, with MC Dropout showing higher correlation with ADS safety, and saving time and computational costs. Moreover, there are significant differences in the highly-uncertain samples they identified. Conclusion: Our results show the importance of considering uncertainty propagation to ensure the ADS safety. Compared to Deep Ensembles, MC Dropout's efficiency makes it a more suitable choice in the context of ADS. © 2025 Elsevier B.V.

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Uncertainty quantification and explainable AI in orthopaedic imaging: A timely call to action

Item Type Journal Article

Author A.P. Tafti

Author Q. Gu

Author J.F. Plate

Abstract Artificial intelligence (AI) has made a big leap in orthopaedic imaging, with deep learning models achieving remarkable accuracy in tasks such as knee osteoarthritis classification and grading, fracture detection, and implant assessment. Yet accuracy in AI models alone is insufficient for clinical trust, adoption, and uptake. Orthopaedic decision-making often carries high risk settings, where any misclassification or overconfidence can have significant consequences for treatment recommendations and patient outcomes. Despite this reality, most current AI models operate as “close boxes”, providing predictions without clarifying their reasoning or quantifying uncertainty. This forum article argues that the integration of uncertainty quantification and explainable AI is no longer optional, but a timely call to action for the orthopaedic community. Uncertainty quantification methods can highlight when predictions are unreliable, prompting confirmatory testing or human oversight, while

explainable AI techniques provide transparency into model reasoning, enabling surgeons and radiologists to better interpret AI outputs. Together, these advances are essential components of trustworthy AI, bridging the gap between technical innovation and real-world orthopaedic practice. By embracing uncertainty-aware and explainable AI models, orthopaedic imaging can move beyond accuracy toward accountability, responsibility, and safer integration into clinical workflows. The time to act is now. © 2025 Delhi Orthopedic Association. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Uncertainty Quantification and Quality Control for Heatmap-Based Landmark Detection Models

Item Type Journal Article

Author Y. Feng

Author J. Yang

Author L. Tang

Author S. Sun

Author Y. Wang

Abstract Uncertainty quantification is a vital aspect of explainable artificial intelligence that fosters clinician trust in medical applications and facilitates timely interventions, leading to safer and more reliable outcomes. Although deep learning models have reached clinically acceptable accuracy in anatomical landmark detection, their predictions remain susceptible to contextual noise due to the small size of the target structures, making uncertainty quantification more challenging than in classification and segmentation tasks. This paper presents an end-to-end uncertainty quantification method tailored for heatmap-based anatomical landmark detection models, designed to improve both interpretability and controllability in clinical applications. Leveraging Dempster-Shafer Theory and Subjective Logic Theory, we implement probability assignment and uncertainty quantification through a single forward pass to ensure computational efficiency. We introduce an evidence map that captures the strength of landmark evidence, alongside an uncertainty map that calibrates predicted probabilities within the Subjective Logic framework. The interaction between these two components, facilitated by a cross-attention mechanism, further improves landmark detection accuracy and enhances the effectiveness of uncertainty quantification. Experimental results demonstrate that the proposed method maintains detection accuracy, even in noisy environments, while outperforming state-of-the-art methods in terms of uncertainty quantification and quality control. Furthermore, the model effectively identifies out-of-distribution data solely through calibrated probabilities when encountering inconsistencies in multi-center data and novel data, underscoring its potential for clinical applications. © 1982-2012 IEEE.

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Distribution functions, human, Humans, male, intelligence, Detection models, article, Damage detection, Learning models, Uncertainty, Deep Learning, algorithm, Algorithms, quality control, prediction, procedures, Model predictive control, deep learning, diagnosis, explainable artificial intelligence, learning, electric potential, noise, Detection accuracy, probability, Heatmaps, image processing, uncertainty, Uncertainty quantifications, Image Processing, Computer-Assisted, uncertainty quantification, anatomical landmark, Normal distribution, Anatomical landmarks, Clinical application, Computer control, device comparison, environmental noise, Evidence learning, landmark detection, Landmark detection, Quality Control, Size distribution, Statistical process control, subjective logic, Subjective Logic

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Uncertainty Quantification for Agent Based Models: A Tutorial

Item Type Conference Paper

Author L. Kimpton

Author P. Challenor

Author J. Salter

Editor Czupryna M.

Editor Kaminski B.

Editor Verhagen H.

Abstract We explore the application of uncertainty quantification methods to agent-based models (ABMs) using a simple sheep and wolf predator-prey model. This work serves as a tutorial on how techniques like emulation can be powerful tools in this context. We also highlight the importance of advanced statistical methods in effectively utilising computationally expensive ABMs. Specifically, we implement stochastic Gaussian processes, Gaussian process classification, sequential design, and history matching to address uncertainties in model input parameters and outputs. Our results show that these methods significantly enhance the robustness, accuracy, and predictive power of ABMs. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Tags:

Design, Classification, Uncertainty quantification, Agent based models, Gaussian processes, History matching

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: L. Kimpton; University of Exeter, Exeter, United Kingdom; email: l.m.kimpton@exeter.ac.uk; Conference name: 19th Conference of the European Social Simulation Association, ESSA 2024; Conference date: 16 September 2024 through 20 September 2024; Conference code: 341799

Uncertainty Quantification for Conditional Treatment Effect Estimation under Dynamic Treatment Regimes

- Item Type** Conference Paper
- Author** L. Deng
- Author** H. Xiong
- Author** F. Wu
- Author** S. Kapoor
- Author** S. Ghosh
- Author** Z. Shahn
- Author** L.-W.H. Lehman
- Editor** Hegselmann S.
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Abstract In medical decision-making, clinicians must choose between different time-varying treatment strategies. Counterfactual prediction via g-computation enables comparison of alternative outcome distributions under such treatment strategies. While deep learning can better model high-dimensional data with complex temporal dependencies, incorporating model uncertainty into predicted conditional counterfactual distributions remains challenging. We propose a principled approach to model uncertainty in deep learning implementations of g-computations using approximate Bayesian posterior predictive distributions of counterfactual outcomes via variational dropout and deep ensembles. We evaluate these methods by comparing their counterfactual predictive calibration and performance in decision-making tasks, using two simulated datasets from mechanistic models and a real-world sepsis dataset. Our findings suggest that the proposed uncertainty quantification approach improves both calibration and decision-making performance, particularly in minimizing risks of worst-case adverse clinical outcomes under alternative dynamic treatment regimes. To our knowledge, this is the first work to propose and compare multiple uncertainty quantification methods in machine learning models of g-computation in estimating conditional treatment effects under dynamic treatment regimes. © 2024 L. Deng, H. Xiong, F. Wu, S. Kapoor, S. Ghosh, Z. Shahn, & L.H. Lehman.

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Conference Name Proceedings of Machine Learning Research

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Notes:

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Uncertainty quantification for joint demand prediction of multi-modal ride-sourcing services using spatiotemporal Mixture-of-Expert neural network

Item Type Journal Article

Author X. Liu

Author Y. Duan

Author Y.-A. Geng

Author Y. Wang

Author Q. Li

Author X. Yan

Author Z. Gao

Abstract Given the sparse uncertainty and highly imbalanced distribution of origin-destination demand in ride-sourcing, probabilistic prediction methods offer rich information to enhance operation decision-making reliability. However, existing uncertainty quantification methods face critical limitations: parametric approaches struggle with irregularities of real-world demand distributions. In contrast, nonparametric methods based on quantile regression are scarcely implemented and suffer from quantile crossing, compromising probabilistic consistency. Furthermore, current approaches seldom consider the crucial inter-service correlations among ride-sourcing services. This paper introduces the Spatiotemporal Mixture-of-Experts Residual-based Multi-quantile Regressive Network (ST-MoE-RMQRN). The implementation is available at <https://github.com/crimsonduanyu/MoE-UQ>. This novel hybrid framework integrates distribution-free quantile estimation with service correlation modeling. Our three main contributions are: 1) a Residual-based Multi-quantile Regressor that employs monotonicity-constrained residual fitting to ensure non-crossing quantiles while preserving distributional flexibility; 2) an environment-aware MoE architecture that captures service correlations through multiple shared-expert networks, addressing potential demand switches among service modals with sensitivity to temporal heterogeneity and exogenous contextual factors such as weather conditions; 3) comprehensive validation on two real-world ride-sourcing datasets demonstrating that our approach consistently outperforms recent probabilistic predictive models in both deterministic and probabilistic forecast metrics, including the Spatiotemporal Zero-Inflated Negative Binomial Graph Neural Network (STZINB-GNN). Additionally, data-wise and component-wise ablation studies confirm the effectiveness of the entire framework and each part of it, showing that explicitly modeling inter-service dependencies substantially improves both deterministic accuracy and probabilistic calibration. Our model is highly efficient in representing extreme demand fluctuations and data-sparse scenarios, which promise to address existing operational inefficiencies faced by transitioning Transportation Network Companies from reactive to anticipatory decision-making. © 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Uncertainty Quantification for Large Language Models

- Item Type** Conference Paper
- Author** M. Panov
- Author** A. Shelmanov
- Author** R. Vashurin
- Author** A. Vazhentsev
- Author** E. Fadeeva
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Abstract Large language models (LLMs) power many NLP applications; yet, they can produce fluent but incorrect content (hallucinations), which threatens reliability and user trust. This tutorial introduces uncertainty quantification (UQ) for text generation: methods that attach an explicit reliability signal to model outputs and enable practical safeguards such as hallucination detection and selective generation. We begin with core uncertainty concepts and explain why techniques that work well for classification do not directly transfer to autoregressive generation. We then survey representative white-box and black-box approaches, from entropy- and probability-based scores to learned probes that leverage internal representations. Retrieval-augmented generation (RAG) has become a core design pattern for LLM applications. Incorporating retrieved evidence introduces both new challenges and valuable structures for uncertainty estimation. In the ECIR edition of the tutorial, we focus on UQ techniques tailored to RAG pipelines and briefly discuss how uncertainty can guide agentic workflows. Practical demonstrations are done using LM-Polygraph (<https://github.com/IINemo/lm-polygraph>), an open-source toolkit that consolidates more than forty recent UQ and calibration methods and provides a large-scale benchmark, making it easy to reproduce results and integrate UQ into applications with minimal code. Overall, the tutorial is intended to lower the barrier to entry for researchers and developers who want to evaluate existing UQ methods, design improved ones, and deploy uncertainty-aware LLM systems. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Panov; MBZUAI, Abu Dhabi, United Arab Emirates; email: panov.maxim@gmail.com; Conference name: 48th European Conference on Information Retrieval, ECIR 2026; Conference date: 29 March 2026 through 2 April 2026; Conference code: 349479

Uncertainty Quantification for Machine Learning Output Assurance using Anomaly-based Dataset Dissimilarity Measures

Item Type Conference Paper

Author G. Incorvaia

Author D. Hond

Author H. Asgari

Abstract The use of Machine Learning (ML) models as predictive tools has increased dramatically in recent years. However, data-driven systems (such as ML models) exhibit a degree of uncertainty in their predictions. In other words, they could produce unexpectedly erroneous predictions if the uncertainty stemming from the data, choice of model and model parameters is not taken into account. In this paper, we introduce a novel method for quantifying the uncertainty of the performance levels attained by ML classifiers. In particular, we investigate and characterize the uncertainty of model accuracy when classifying out-of-distribution data that is statistically dissimilar from the data employed during training. A main element of this novel Uncertainty Quantification (UQ) method is a measure of the dissimilarity between two datasets. We introduce an innovative family of data dissimilarity measures based on anomaly detection algorithms, namely the Anomaly-based Dataset Dissimilarity (ADD) measures. These dissimilarity measures process feature representations that are derived from the activation values of neural networks when supplied with dataset items. The proposed UQ method for classification performance employs these dissimilarity measures to estimate the classifier accuracy for out-of-distribution datasets, and to give an uncertainty band for those estimates. A numerical analysis of the efficacy of the UQ method is conducted. The obtained results generally demonstrate that the amplitude of the uncertainty band tends to increase as the data dissimilarity measure increases. Overall, this research contributes to the verification and run-time performance prediction of systems composed of ML-based elements. © 2023 IEEE.

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Classification (of information), Machine-learning, Machine learning, Forecasting, Uncertainty analysis, Neural networks, Uncertainty, Machine learning models, Anomaly detection, Anomaly Detection, Classification, Quantification methods, Uncertainty quantifications, Uncertainty Quantification, Artificial Neural Networks, Dissimilarity measures, Numerical methods, Data dissimilarity measure, Data Dissimilarity Measures, Performance prediction, Performance Prediction, Predictive tools

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Uncertainty Quantification for Markov Random Fields

Item Type Journal Article

Author P. Birmpa

Author M.A. Katsoulakis

Abstract We present an information-based uncertainty quantification method for general Markov random fields (MRFs). MRFs are structured, probabilistic graphical models over undirected graphs and provide a fundamental unifying modeling tool for statistical mechanics, probabilistic machine learning, and artificial intelligence. Typically, MRFs are complex and high-dimensional with nodes and edges (connections) built in a modular fashion from simpler, low-dimensional probabilistic models and their local connections; in turn, this modularity allows one to incorporate available data to MRFs and efficiently simulate them by leveraging their graph-theoretic structure. Learning graphical models from data and/or constructing them from physical modeling and constraints necessarily involves uncertainties inherited from data, modeling choices, or numerical approximations. These uncertainties in the MRF can be manifested either in the graph structure or the probability distribution functions and necessarily will propagate in predictions for quantities of interest. Here we quantify such uncertainties using tight, information-based bounds on the predictions of quantities of interest; these bounds take advantage of the graphical structure of MRFs and are capable of handling the inherent high dimensionality of such graphical models. We demonstrate our methods in MRFs for medical diagnostics and statistical mechanics models. In the latter, we develop uncertainty quantification bounds for finite-size effects and phase diagrams, which constitute two of the typical prediction goals of statistical mechanics modeling. © 2021 Society for

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Uncertainty Quantification for Metamodels

Item Type Conference Paper

Author M. Okánik

Author A. Trantas

Author M. Pepijn de Bakker

Author E. Lazovik

Editor Vantini S.

Editor Fontana M.

Editor Solari A.

Editor Bostrom H.

Editor Carlsson L.

Editor Carlsson L.

Abstract In the realm of computational science, metamodels serve as indispensable tools for approximating complex systems, facilitating the exploration of scenarios where traditional modelling may prove computationally infeasible. However, the inherent uncertainties within these metamodels, particularly those driven by Machine Learning (ML), necessitate rigorous quantification to ensure reliability and robustness in decision-making processes. One alternative of obtaining uncertainty estimates is using ML models that have a native notion of uncertainty, such as the Bayesian Neural Networks (BNNs), however its repeated sampling necessary to approximate the output distribution is computationally demanding and might defeat the purpose of building metamodels in the first place. In datasets with multidimensional input space and a limited amount of training examples, error estimates provided by BNNs often have poor quality. This study explores alternative empirical approaches to uncertainty quantification, based on knowledge extraction from output space as opposed to input space. Leveraging patterns of magnitude of error committed by the metamodel in output space, we obtain significant improvement of adaptivity of prediction intervals, both over pure Conformal Prediction (CP) and BNNs. Our findings underscore the potential of integrating diverse uncertainty quantification methods to fortify reliability of metamodels, highlighting their robust and quantifiable confidence in model predictions. © 2024 M. Okánik, A. Trantas, M.P.d. Bakker & E. Lazovik.

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Uncertainty quantification for probabilistic machine learning in earth observation using conformal prediction

Item Type Journal Article

Author G. Singh

Author G. Moncrieff

Author Z. Venter

Author K. Cawse-Nicholson

Author J. Slingsby

Author T.B. Robinson

Abstract Machine learning is increasingly applied to Earth Observation (EO) data to obtain datasets that contribute towards international accords. However, these datasets contain inherent uncertainty that needs to be quantified reliably to avoid negative consequences. In response to the increased need to report uncertainty, we bring attention to the promise of conformal prediction within the domain of EO. Unlike previous uncertainty quantification methods, conformal prediction offers statistically valid prediction regions while concurrently supporting any machine learning model and data distribution. To support the need for conformal prediction, we reviewed EO datasets and found that only 22.5% of the datasets incorporated a degree of uncertainty information, with unreliable methods prevalent. Current open implementations require moving large amounts of EO data to the algorithms. We introduced Google Earth Engine native modules that bring conformal prediction to the data and compute, facilitating the integration of uncertainty quantification into existing traditional and deep learning modelling workflows. To demonstrate the versatility and scalability of these tools we apply them to valued EO applications spanning local to global extents, regression, and classification tasks. Subsequently, we discuss the opportunities arising from the use of conformal prediction in EO. We anticipate that accessible and easy-to-use tools, such as those provided here, will drive wider adoption of rigorous uncertainty quantification in EO, thereby enhancing the reliability of downstream uses such as operational monitoring and decision-making. © The Author(s) 2024.

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Uncertainty Quantification for Safe and Reliable Autonomous Vehicles: A Review of Methods and Applications

Item Type Journal Article**Author** K. Wang**Author** C. Shen**Author** X. Li**Author** J. Lu

Abstract In the past decade, deep learning has been widely applied across various fields. However, its applicability in open-world scenarios is often limited due to the lack of quantifying uncertainty in both data and models. In recent years, a multitude of uncertainty quantification (UQ) approaches for neural networks have emerged and found applications in safety-critical domains such as autonomous vehicles and medical analysis. This paper aims to review the latest advancements in UQ methods and investigate their application specifically in the field of computer vision and autonomous vehicles. Initially, we identify several key qualifications, namely practicability, robustness, accuracy, scalability, and efficiency (referred to as PRASE), and employ them as evaluation criteria throughout this study. By considering these criteria as uniform measurements, we meticulously evaluate and compare the performance of different types of UQ methods, including Bayesian methods, ensemble methods, and single deterministic methods. Furthermore, we delve into the discussion of their application in diverse tasks within the autonomous vehicle domain, such as semantic segmentation, object detection, depth estimation, and end-to-end control. Through comprehensive analysis and comparison, we identify a range of challenges and propose future research directions in this field. Our findings shed light on the importance of addressing uncertainty quantification in deep learning

models and provide insights into enhancing the reliability and performance of autonomous vehicles in real-world scenarios. © 2000-2011 IEEE.

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Uncertainty Quantification for Text Classification

Item Type Conference Paper

Author D. Zhang

Author M. Sensoy

Author M. Makrehchi

Author B. Taneva-Popova

Editor Kamps J.

Editor Goeuriot L.
Editor Crestani F.
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Abstract This half-day tutorial introduces modern techniques for practical uncertainty quantification specifically in the context of multi-class and multi-label text classification. First, we explain the usefulness of estimating aleatoric uncertainty and epistemic uncertainty for text classification models. Then, we describe several state-of-the-art approaches to uncertainty quantification and analyze their scalability to big text data: Virtual Ensemble in GBDT, Bayesian Deep Learning (including Deep Ensemble, Monte-Carlo Dropout, Bayes by Backprop, and their generalization Epistemic Neural Networks), as well as Evidential Deep Learning (including Prior Networks and Posterior Networks). Next, we discuss typical application scenarios of uncertainty quantification in text classification (including in-domain calibration, cross-domain robustness, and novel class detection). Finally, we list popular performance metrics for the evaluation of uncertainty quantification effectiveness in text classification. Practical hands-on examples/exercises are provided to the attendees for them to experiment with different uncertainty quantification methods on a few real-world text classification datasets such as CLINC150. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Deep learning, Classification (of information), Uncertainty analysis, Uncertainty, Bayesian networks, Text classification, Text processing, Class labels, Multi-label text classification, Text data, State-of-the-art approach, Uncertainty quantifications, Uncertainty quantification, Epistemic uncertainties, Modern techniques, Text classification models

Notes:

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: D. Zhang; Thomson Reuters Labs, London, United Kingdom; email: dell.z@ieee.org; Conference name: 45th European Conference on Information Retrieval, ECIR 2023; Conference date: 2 April 2023 through 6 April 2023; Conference code: 292029

Uncertainty Quantification for Text Classification

Item Type Conference Paper

Author D. Zhang

Author M. Sensoy

Author M. Makrehchi

Author B. Taneva-Popova

Author L. Gui

Author Y. He

Abstract This full-day tutorial introduces modern techniques for practical uncertainty quantification specifically in the context of multi-class and multi-label text classification. First, we explain the usefulness of estimating aleatoric uncertainty and epistemic uncertainty for text classification models. Then, we describe several state-of-the-art approaches to uncertainty quantification and analyze their scalability to big text data: Virtual Ensemble in GBDT, Bayesian Deep Learning (including Deep Ensemble, Monte-Carlo Dropout, Bayes by Backprop, and their generalization Epistemic Neural Networks), Evidential Deep Learning (including Prior Networks and Posterior Networks), as well as Distance Awareness (including Spectral-normalized Neural Gaussian Process and Deep Deterministic Uncertainty). Next, we talk about the latest advances in uncertainty quantification for pre-trained language models (including asking language models to express their uncertainty, interpreting uncertainties of text classifiers built on large-scale language models, uncertainty estimation in text generation, calibration of language models, and calibration for in-context learning). After that, we discuss typical application scenarios of uncertainty quantification in text classification (including in-domain calibration, cross-domain robustness, and novel class detection). Finally, we list popular performance metrics for the evaluation of uncertainty quantification effectiveness in text classification. Practical hands-on examples/exercises are provided to the attendees for them to experiment with different uncertainty quantification methods on a few real-world text classification datasets such as CLINC150. © 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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Language English

Extra Journal Abbreviation: SIGIR - Proc. Int. ACM SIGIR Conf. Res. Dev. Inf. Retr.

Publisher Association for Computing Machinery, Inc

ISBN 978-145039408-6 (ISBN)**Pages** 3426-3429**Proceedings Title** SIGIR - Proc. Int. ACM SIGIR Conf. Res. Dev. Inf. Retr.**Conference Name** SIGIR 2023 - Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval**Date Added** 20/09/2026, 9:51:57 pm**Modified** 20/09/2026, 9:51:57 pm**Tags:**

Computational linguistics, Language model, Deep learning, Classification (of information), Learning systems, Uncertainty analysis, Uncertainty, Bayesian networks, Text classification, Text processing, text classification, language models, Class labels, Multi-label text classification, State-of-the-art approach, Uncertainty quantifications, uncertainty quantification, Calibration, Epistemic uncertainties, Modern techniques, Text classification models

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Uncertainty Quantification in Deep Learning

Item Type Conference Paper**Author** L. Kong**Author** H. Kamarthi**Author** P. Chen**Author** B.A. Prakash**Author** C. Zhang

Abstract Deep neural networks (DNNs) have achieved enormous success in a wide range of domains, such as computer vision, natural language processing and scientific areas. However, one key bottleneck of DNNs is that they are ignorant about the uncertainties in their predictions. They can produce wildly wrong predictions without realizing, and can even be confident about their mistakes. Such mistakes can cause misguided decisions-sometimes catastrophic in critical applications, ranging from self-driving cars to cyber security to automatic medical diagnosis. In this tutorial, we present recent advancements in uncertainty quantification for DNNs and their applications across various domains. We first provide an overview of the motivation behind uncertainty quantification, different sources of uncertainty, and evaluation metrics. Then, we delve into several representative

uncertainty quantification methods for predictive models, including ensembles, Bayesian neural networks, conformal prediction, and others. We go on to discuss how uncertainty can be utilized for label-efficient learning, continual learning, robust decision-making, and experimental design. Furthermore, we showcase examples of uncertainty-aware DNNs in various domains, such as health, robotics, and scientific machine learning. Finally, we summarize open challenges and future directions in this area. © 2023 Owner/Author.

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Conference Name Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining

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Uncertainty quantification in hydrogen production systems design: A review

Item Type Journal Article

Author M.K. Marvin

Author V.I. Fagorite

Author E.O. Agbasi

Author E.N. Aminaho

Author M.D. Aminu

Abstract The increasing demand for hydrogen as an energy carrier highlights the need for efficient and scalable production systems. The performance and feasibility of these technologies are, however, strongly influenced by uncertainties in system parameters, operational conditions, and model inputs, which often limit large-scale deployment. This review provides a critical evaluation of uncertainty quantification techniques applied to hydrogen production systems. We examine key computational modelling approaches that integrate uncertainty quantification methods to improve predictive accuracy, guide process optimization, and support informed decision-making. Despite these advances, it is clearly indicative that many studies continue to employ simplified probability distributions due to the scarcity of high-quality empirical data, particularly for emerging technologies such as solid oxide electrolysis cells, photoelectrochemical systems, and advanced biomass gasifiers. Also, a lack of standardized uncertainty protocols and inconsistencies in hydrogen production modeling assumptions present challenges to reproducibility and cross-study comparison. Addressing this limitation requires greater emphasis on experimental validation and systematic data collection, which will strengthen the credibility of model-based insights and accelerate the scale-up of hydrogen production technologies. Harmonizing modeling frameworks and reporting practices across the hydrogen research community will also be critical to improving benchmarking and ensuring that findings translate effectively into investment and policy strategies. To this end, we aim to highlight pathways toward more reliable, cost-effective, and sustainable hydrogen production systems. © 2026 Elsevier Ltd

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M.K. Marvin; Department of Chemical Engineering, Faculty of Engineering, University of Maiduguri, Maiduguri, PMB 1069, Nigeria; email: marvin.mahlon@gmail.com; CODEN: RSERF

Uncertainty quantification in machine learning for engineering design and health prognostics: A tutorial

Item Type Journal Article

Author V. Nemani

Author L. Biggio

Author X. Huan

Author Z. Hu

Author O. Fink

Author A. Tran

Author Y. Wang

Author X. Zhang

Author C. Hu

Abstract On top of machine learning (ML) models, uncertainty quantification (UQ) functions as an essential layer of safety assurance that could lead to more principled decision making by enabling sound risk assessment and management. The safety and reliability improvement of ML models empowered by UQ has the potential to significantly facilitate the broad adoption of ML solutions in high-stakes decision settings, such as healthcare, manufacturing, and aviation, to name a few. In this tutorial, we aim to provide a holistic lens on emerging UQ methods for ML models with a particular focus on neural networks and the applications of these UQ methods in tackling engineering design as well as prognostics and health management problems. Towards this goal, we start with a comprehensive classification of uncertainty types, sources, and causes pertaining to UQ of ML models. Next, we provide a tutorial-style description of several state-of-the-art UQ methods: Gaussian process regression, Bayesian neural network, neural network ensemble, and deterministic UQ methods focusing on spectral-normalized neural Gaussian process. Established upon the mathematical formulations, we subsequently examine the soundness of these UQ methods quantitatively and qualitatively (by a toy regression example) to examine their strengths and shortcomings from different dimensions. Then, we review quantitative metrics commonly used to assess the quality of predictive uncertainty in classification and regression problems. Afterward, we discuss the increasingly important role of UQ of ML models in solving challenging problems in engineering design and health prognostics. Two case studies with source codes available on GitHub are used to demonstrate these UQ methods and compare their performance in the life prediction of lithium-ion batteries at the early stage (case study 1) and the remaining useful life prediction of turbofan engines (case study 2). © 2023 Elsevier Ltd

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Uncertainty Quantification in Machine Learning Solutions for Near-Field Microwave Imaging

Item Type Conference Paper
Author K. Narendra
Author B. Martin
Author J. Carneiro

Author C. Gilmore

Author I. Jeffrey

Abstract Three machine-learning-based uncertainty quantification methods for microwave imaging are reviewed and compared. Measurement data from a model of a 2-D near-field microwave imaging chamber is passed to a data-to-image neural network to produce a reconstruction of a target. First, we consider two methods that create an uncertainty map of the reconstructed target. We demonstrate that uncertainty maps correspond well with the true error when the test example is well represented by the targets in the training set, but the maps do not produce a meaningful representation of the true error when the test target lies out-of-range of the training data (e.g. the target is of drastically different shape or physical properties). In the third method, two autoencoder-based approaches (autoencoding the data versus autoencoding the image) are reviewed to demonstrate their ability to detect whether a given test example is in-range for a given model. The value of combining uncertainty maps with autoencoder-based approaches is discussed. © 2025 IEEE.

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Uncertainty quantification in medical image synthesis

Item Type Journal Article**Author** R. Barbano**Author** S. Arridge**Author** B. Jin**Author** R. Tanno

Abstract Machine learning approaches to medical image synthesis have shown outstanding performance, but often do not convey uncertainty information. In this chapter, we survey uncertainty quantification methods in medical image synthesis and advocate the use of uncertainty for improving clinicians' trust in machine learning solutions. First, we describe basic concepts in uncertainty quantification and discuss its potential benefits in downstream applications. We then review computational strategies that facilitate inference, and identify the main technical and clinical challenges. We provide a first comprehensive review to inform how to quantify, communicate and use uncertainty in medical synthesis applications. © 2022 Elsevier Inc. All rights reserved.

Date 2022**DOI** 10.1016/B978-0-12-824349-7.00033-5**URL** <https://www.scopus.com/pages/publications/85137549921?origin=resultslist>**Archive** Scopus**Language** English**Publisher** Elsevier**Pages** 601-641**Publication** Biomedical Image Synthesis and Simulation: Methods and Applications**ISSN** 978-012824349-7 (ISBN); 978-012824350-3 (ISBN)**Journal Abbr** Biomedical Image Synthesis and Simulation: Methods and Applications**Date Added** 20/09/2026, 9:52:28 pm**Modified** 20/09/2026, 9:52:28 pm**Tags:**

Deep learning, Uncertainty quantification, Bayesian neural networks, Approximate inference, Medical image synthesis

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Uncertainty Quantification in Molecular Machine Learning for Property Predictions under Data Shifts

Item Type Journal Article**Author** R. Parrondo-Pizarro**Author** J. Lanini**Author** R. Rodríguez-Pérez

Abstract Drug discovery and medicinal chemistry efforts are increasingly influenced by machine learning (ML), with compound property prediction as a central application. ML models have demonstrated strong performance in predicting various compound properties from chemical structure. However, these models can exhibit varying levels of prediction error, making uncertainty quantification (UQ) essential for informed decisions. Standard UQ metrics include the distance to the molecules in the training set and prediction variance, obtained through methods such as model ensembles or Bayesian modeling. Although several UQ methodologies have been developed in recent years, no single approach consistently outperformed others. Herein, we present a comprehensive benchmark of UQ strategies for ML-based prediction of absorption, distribution, metabolism, and excretion (ADME) properties, using both in-house and public data sets. We employed the recently introduced UNIQUE (UNCertaInty QUantification bEnchmarking) framework and evaluated UQ method performance under data shifts. Our findings indicate data-based UQ metrics (e.g., chemical distance), and model-based UQ metrics (e.g., predicted value and variance) may capture complementary aspects of uncertainty. Their combination through error models, designed to predict the original ML model's error, yielded higher-quality uncertainty estimates. These error models emerged as a promising strategy for enhancing UQ, showing robustness in under various degrees and types of data shift. Taken together, our work highlights the potential of combining diverse UQ metrics and error modeling to improve reliability in molecular property prediction. By establishing standardized evaluation setups and assessing UQ under data shifts, we provide a foundation for future UQ method development and benchmarking in the field. © 2026 The Authors. Published by American Chemical Society

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Uncertainty Quantification in Pairwise Difference Learning for Classification

Item Type Journal Article**Author** M.K. Belaid**Author** M. Rabus**Author** E. Hüllermeier

Abstract Instead of learning a mapping from instances to outcomes in the standard way, the key idea of pairwise difference learning (PDL) is to learn a function that takes two instances as input and predicts the difference between their respective outcomes. Given a function of this kind, predictions for a query instance are derived from every training example and then aggregated. We consider pairwise difference learning for classification (PDC) and propose two uncertainty quantification methods that faithfully represent and measure the classifier's predictive uncertainty. Building on recent work on decomposing total uncertainty into aleatoric and epistemic components, we propose different ways of extending corresponding measures to the setting of PDC, depending on how and to what extent the underlying base learner represents uncertainty. In large-scale empirical studies, we analyze both the predictive accuracy and the uncertainty awareness of the methods. © The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature 2025.

Date 2026**DOI** 10.1007/s10994-025-06949-0**URL** <https://www.scopus.com/pages/publications/105026248858?origin=resultslist>**Archive** Scopus**Language** English**Volume** 115**Publisher** Springer**Publication** Machine Learning**ISSN** 08856125 (ISSN)**Issue** 1**Journal Abbr** Mach Learn**Date Added** 20/09/2026, 9:51:34 pm**Modified** 20/09/2026, 9:51:34 pm**Tags:**

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Uncertainty quantification in skin cancer classification using three-way decision-based Bayesian deep learning

Item Type Journal Article

Author M. Abdar

Author M. Samami

Author S. Dehghani Mahmoodabad

Author T. Doan

Author B. Mazoure

Author R. Hashemifesharaki

Author L. Liu

Author A. Khosravi

Author U.R. Acharya

Author V. Makarenkov

Author S. Nahavandi

Abstract Accurate automated medical image recognition, including classification and segmentation, is one of the most challenging tasks in medical image analysis. Recently, deep learning methods have achieved remarkable success in medical image classification and segmentation, clearly becoming the state-of-the-art methods. However, most of these methods are unable to provide uncertainty quantification (UQ) for their output, often being overconfident, which can lead to disastrous consequences. Bayesian Deep Learning (BDL) methods can be used to quantify uncertainty of traditional deep learning methods, and thus address this issue. We apply three uncertainty quantification methods to deal with uncertainty during skin cancer image classification. They are as follows: Monte Carlo (MC) dropout, Ensemble MC (EMC) dropout and Deep Ensemble (DE). To further resolve the remaining uncertainty after applying the MC, EMC and DE methods, we describe a novel hybrid dynamic BDL model, taking into account uncertainty, based on the Three-Way Decision (TWD) theory. The proposed dynamic model enables us to use different UQ methods and different deep neural networks in distinct classification phases. So, the elements of each phase can be adjusted according to the dataset under consideration. In this study, two best UQ methods (i.e., DE and EMC) are applied in

two classification phases (the first and second phases) to analyze two well-known skin cancer datasets, preventing one from making overconfident decisions when it comes to diagnosing the disease. The accuracy and the F1-score of our final solution are, respectively, 88.95% and 89.00% for the first dataset, and 90.96% and 91.00% for the second dataset. Our results suggest that the proposed TWDBDL model can be used effectively at different stages of medical image analysis. © 2021 Elsevier Ltd

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Uncertainty Quantification in Table Structure Recognition

Item Type Conference Paper

Author K. Ajayi

Author L. Zhang

Author Y. He

Author J. Wu

Abstract Quantifying uncertainties for machine learning models is a critical step to reduce human verification effort by detecting predictions with low confidence. This paper proposes a method for uncertainty quantification (UQ) of table structure recognition (TSR). The proposed UQ method is built upon a mixture-of-expert approach termed Test-Time Augmentation (TTA). Our key idea is to enrich and diversify the table representations, to spotlight the cells with high recognition uncertainties. To evaluate the effectiveness, we proposed two heuristics to differentiate highly uncertain cells from normal cells, namely, masking and cell complexity quantification. Masking involves varying the pixel intensity to deem the detection uncertainty. Cell complexity quantification gauges the uncertainty of each cell by its topological relation with neighboring cells. The evaluation results based on standard benchmark datasets demonstrate that the proposed method is effective in quantifying uncertainty in TSR models. To our best knowledge, this study is the first of its kind to enable UQ in TSR tasks. Our code and data are available at: <https://github.com/lamps-lab/UQTTA.git>. © 2024 IEEE.

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Uncertainty quantification in the Permian Basin using conventional and modified bootstrap methodology

Item Type Journal Article

Author C.O. Okoli

Author S.D. Goddard

Author O.O. Awoleke

Abstract Various uncertainty quantification methodologies are presented using a combination of several deterministic decline curve analysis models and two bootstrapping algorithms. These probabilistic models are applied to 126 sample wells from the Permian basin. Results are presented for 12–72 months of production hindcast given an average well production history of 103 months. Based on the coverage rate and the forecast error (with the coverage rate being more significant in our choice of the best probabilistic models) and using up to one-half of the available production history for a group of sample wells from the Permian Basin, we find that the CBM-SEPD combination is the best probabilistic model for the Central Basin Platform, the MBM-Arps combination is the best probabilistic model for the Delaware Basin, the CBM-Arps is the best probabilistic model for the Midland Basin, and the best probabilistic model for the overall Permian Basin is the CBM-Arps when early time data is used as hindcast and CBM-SEPD for when one-quarter to one-half of the data is used as hindcast. When three-quarters or more of the available production history is used for analysis, the MBM-SEPD probabilistic model is the best combination in terms of both coverage rate and forecast error for all the sub-basins in the Permian. The novelty of this work lies in its extension of bootstrapping methods to other decline curve analysis models. This work also offers the engineer guidance on the best choice of probabilistic model whilst attempting to forecast production from the Permian Basin. © 2023 The Authors

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Uncertainty quantification method and application of landslide susceptibility zoning based on fuzzy mathematics

Item Type Journal Article

Author L. Huang

Author G. WEI

Author Z. Li

Author W. Li

Author Y. Zhang

Abstract Landslide susceptibility zoning boundary uncertainty is an important issue that affects the reliable interpretation of landslide susceptibility assessment (LSA). To address the downstream uncertainty introduced by the discretization of continuous susceptibility indices, this study proposes an uncertainty quantification framework integrating fuzzy logic and information entropy theory. A case study in Kang County, Gansu Province, China, the results indicate that: (1) the Random Forest (RF) model achieved satisfactory predictive performance ($AUC = 0.830$), providing a reliable basis for downstream uncertainty quantification; (2) areas with high uncertainty are mainly distributed in the transition zones between different susceptibility levels and in the northeastern part of the county, whereas low uncertainty areas are predominantly distributed in the southern region, with a pronounced clustering pattern in the southwestern; and (3) comparative validation using the Support Vector Machine (SVM) model demonstrates that the spatial patterns of uncertainty are highly consistent across different models, indicating the robustness and applicability of the proposed framework. The results suggest that susceptibility zonation based on classification methods such as Jenks inevitably introduces boundary fuzziness, and the proposed framework can serve as an effective complement to conventional LSA by identifying low reliability areas and providing support for the optimization of susceptibility zonation. © The Author(s) 2026.

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Uncertainty quantification of fluidized beds using a data-driven framework

Item Type Journal Article

Author V.M.K. Kottedda

Author J.A. Stephens

Author W. Spotz

Author V. Kumar

Author A. Kommu

Abstract We carried out a nondeterministic analysis of flow in a fluidized bed. The flow in the fluidized bed is simulated with National Energy Technology Laboratory's open source multiphase fluid dynamics suite MFiX. It does not possess tools for uncertainty quantification. Therefore, we developed a C++ wrapper to integrate an uncertainty

quantification toolkit developed at Sandia National Laboratory with MFiX. The wrapper exchanges uncertain input parameters and key output parameters among Dakota and MFiX. However, a data-driven framework is also developed to obtain reliable statistics as it is not feasible to get them with MFiX integrated into Dakota, Dakota-MFiX. The data generated from Dakota-MFiX simulations, with the Latin Hypercube method of sampling size 500, is used to train a machine learning algorithm. The trained and tested deep neural network algorithm is integrated with Dakota via the wrapper to obtain low order statistics of the bed height and pressure drop across the bed. In addition, it can be used to obtain statistics with various distributions of the uncertain input variables and various uncertainty quantification methods. In addition, sensitivity analysis is carried out for 9 input parameters and two output parameters. © 2019 Elsevier B.V.

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Uncertainty quantification of geochemical data imputation using Monte Carlo dropout

Item Type Journal Article

Author V. Puzyrev

Author P. Duuring

Abstract Machine learning models have shown their promise in geochemical data imputation tasks. However, being black-box solvers, these models require more confidence in their predictions. Using uncertainty quantification methods for deep neural networks can increase the reliability of their predictions. In this paper, we use Monte Carlo Dropout to estimate uncertainty in geochemical data imputation. Multiple forward passes with different dropout configurations yield a predictive distribution for the unknown analytes. The mean of this distribution is used as the prediction, while the standard deviation expresses the uncertainty of the neural networks. Two different scenarios, namely the WACHEM and WAMEX databases containing multi-element geochemical data for rock samples, illustrate the predictive accuracy of the method and its capability to measure the associated uncertainty. Dropout values of 0.1–0.2 were identified as a good balance in prediction accuracy and model uncertainty. © 2025

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Uncertainty quantification of graph convolution neural network models of evolving processes

Item Type Journal Article

Author J. Hauth

Author C. Safta

Author X. Huan

Author R.G. Patel

Author R.E. Jones

Abstract The application of neural network models to scientific machine learning tasks has proliferated in recent years. In particular, neural networks have proved to be adept at modeling processes with spatial–temporal complexity. Nevertheless, these highly parameterized models have garnered skepticism in their ability to produce outputs with quantified error bounds over the regimes of interest. Hence there is a need to find uncertainty quantification methods that are suitable for neural networks. In this work we present comparisons of the parametric uncertainty quantification of neural networks modeling complex spatial–temporal processes with Hamiltonian Monte Carlo and Stein variational gradient descent and its projected variant. Specifically we apply these methods to graph convolutional neural network models of evolving systems modeled with recurrent neural network and neural ordinary differential equations architectures. We show that Stein variational inference is a viable alternative to Monte Carlo methods with some clear advantages for complex neural network models. For our exemplars, Stein variational interference gave similar pushed forward uncertainty profiles through time compared to Hamiltonian Monte Carlo, albeit with generally more generous variance. Projected Stein variational gradient descent also produced similar uncertainty profiles to the non-projected counterpart, but large reductions in the active weight space were confounded by the stability of the neural network predictions and the convoluted likelihood landscape. © 2024 Elsevier B.V.

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Uncertainty Quantification of Hyperspectral Image Denoising Frameworks Based on Sliding-Window Low-Rank Matrix Approximation

Item Type Journal Article

Author J. Song

Author S. Xia

Author J. Wang

Author M. Patel

Author D. Chen

Abstract Sliding-window-based low-rank matrix approximation (LRMA) is a technique widely used in hyperspectral images (HSIs) denoising or completion. However, the uncertainty quantification of the restored HSI has not been addressed to date. Accurate uncertainty quantification of the denoised HSI facilitates applications such as multisource or multiscale data fusion, data assimilation, and product uncertainty quantification since these applications require an accurate approach to describe the statistical distributions of the input data. Therefore, we propose a prior-free closed-form element-wise uncertainty quantification method for LRMA-based HSI restoration. Our closed-form algorithm overcomes the difficulty of handling uncertainty in HSI patch mixing caused by the sliding-window strategy used in the conventional LRMA process. The proposed approach only requires the uncertainty of the observed HSI and provides the uncertainty result relatively rapidly and with similar computational complexity as the LRMA technique. We conduct extensive experiments to validate the estimation accuracy of the proposed closed-form uncertainty approach. The method is robust to at least 10% random impulse noise at the cost of 10%-20% of additional processing time compared to the LRMA. The experiments indicate that the proposed closed-form uncertainty quantification method is more applicable to real-world applications than the baseline Monte Carlo test, which is computationally expensive. © 1980-2012 IEEE.

Date 2022**DOI** 10.1109/TGRS.2021.3065570**URL** <https://www.scopus.com/pages/publications/85103238655?origin=resultslist>**Archive** Scopus**Language** English**Volume** 60**Publisher** Institute of Electrical and Electronics Engineers Inc.**Publication** IEEE Transactions on Geoscience and Remote Sensing**ISSN** 01962892 (ISSN)**Journal Abbr** IEEE Trans Geosci Remote Sens**Date Added** 20/09/2026, 9:52:27 pm**Modified** 20/09/2026, 9:52:27 pm**Tags:**

Uncertainty analysis, quantitative analysis, Data fusion, Restoration, Data assimilation, accuracy assessment, Monte Carlo methods, Uncertainty quantifications, Matrix algebra, multispectral image, Uncertainty quantification, uncertainty analysis, image classification, Hyperspectral images (HSIs), Image denoising, Impulse noise, Low-rank matrix approximation (LRMA), Low-rank matrix approximations, Multi-scale datum, Product uncertainties, Sliding window-based, Sliding-window strategies, Spectroscopy, Statistical distribution

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Uncertainty Quantification of Microblock-Based Resource Models and Sequencing of Sampling

Item Type Journal Article**Author** G.T. Nwaila**Author** E.J.M. Carranza**Abstract** Spatial models are fundamental across the mineral value chain, forming the basis for exploration and extraction. Geodata science and increasingly bigger data permit alternatives to traditional mineral resource estimation methods, particularly in spatial data interpolation. Interpolation has been formulated as a machine learning (ML)

task, providing new capabilities, such as automated deployment and remote real-time monitoring. However, a significant gap exists regarding how uncertainty propagates through ML workflows. This paper introduces an uncertainty propagation method to a ML-based interpolation method called microblocking that propagates epistemic uncertainty. Our method adheres to the data science framework and is fully ML-based. Epistemic uncertainty is the dominant uncertainty in geosciences, because data sparsity is created by both complex dynamics of physical systems and sampling limitations. Our uncertainty estimates are block-specific and can guide sampling and other activities. Biasing sampling toward blocks with high economic potential and high uncertainty enables the most cost-effective sequencing of sampling. A rapid, ML-based uncertainty quantification method provides a modern data-driven (feedback-based) framework to extraction guidance, built on big data, geodata science, and real-time mineral resource modeling. We compare our method with typical kriging uncertainty estimates and demonstrates that our results are more block-specific and broader in scope (more comprehensive). In an industry where financial stakes are significant, a thorough understanding of uncertainty can improve investor confidence. The method not only improves scientific rigor, but is also engineered to fit increasingly bigger data across the mineral value chain, and caters to the conservative nature of the mineral industry, where method validation occurs at a slower pace. © The Author(s) 2025.

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UNCERTAINTY QUANTIFICATION OF PHYSICS-BASED LABEL-FREE DEEP LEARNING AND PROBABILISTIC PREDICTION OF EXTREME EVENTS

Item Type	Conference Paper
Author	H. Li
Author	J. Yin
Author	X. Du
Abstract	Surrogate models from machine learning regression have been increasingly used in engineering analysis and design. Since surrogate models are usually built using data from solving expensive physical models, label-free machine learning methodologies have been developed to reduce the computational cost. Understanding and quantifying the model (epistemic) uncertainty of surrogate models is critical for their applications with quantified confidence. It is, however, much more computationally expensive, or even impossible to quantifying the model uncertainty for label-free machine learning. In this work we propose an uncertainty quantification method for the epistemic uncertainty of physics-based label-free regression. The method is used after a surrogate model has already been built by deep neural network based on the data of only input variables without labels (data of responses) and a system of physical equations. A surrogate model of the neural network regression model error is built with Gaussian Process regression using the existing training points and the derivatives of the system of physical equations at the training points. The error model is then used to compensate the error of the neural network surrogate model, therefore producing more accurate predictions. With higher accuracy, the proposed method is applied to probabilistic prediction of extreme events where both (aleatory) data uncertainty and model uncertainty coexist, and higher accuracy is required. Its application to time-dependent reliability prediction of a four-bar linkage mechanism demonstrates the high accuracy of the proposed method. Copyright © 2022 by ASME.
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Extra	Journal Abbreviation: Proc. ASME Des. Eng. Tech. Conf.
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Deep learning, Deep neural networks, Learning systems, Machine-learning, Forecasting, Uncertainty analysis, Gaussian distribution, Gaussian noise (electronic), reliability, Extreme events, Uncertainty, Regression analysis, deep learning, High-accuracy, Errors, Uncertainty quantifications, Surrogate modeling, Gaussian process, Gaussian Processes, Label free, extreme events, Physics-based

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Uncertainty Quantification of Soil Organic Carbon Estimation from Remote Sensing Data with Conformal Prediction

Item Type Journal Article

Author N. Kakhani

Author S. Alamdar

Author N.M. Kebonye

Author M. Amani

Author T. Scholten

Abstract Soil organic carbon (SOC) contents and stocks provide valuable insights into soil health, nutrient cycling, greenhouse gas emissions, and overall ecosystem productivity. Given this, remote sensing data coupled with advanced machine learning (ML) techniques have eased SOC level estimation while revealing its patterns across different ecosystems. However, despite these advances, the intricacies of training reliable and yet certain SOC models for specific end-users remain a great challenge. To address this, we need robust SOC uncertainty quantification techniques. Here, we introduce a methodology that leverages conformal prediction to address the uncertainty in estimating SOC contents while using remote sensing data. Conformal prediction generates statistically reliable uncertainty intervals for predictions made by ML models. Our analysis, performed on the LUCAS dataset in Europe and incorporating a suite of relevant environmental covariates, underscores the efficacy of integrating conformal prediction with another ML model, specifically random forest. In addition, we conducted a comparative assessment of our results against prevalent uncertainty quantification methods for SOC prediction, employing different evaluation metrics to assess both model uncertainty and accuracy. Our methodology showcases the utility of the generated prediction sets as informative indicators of uncertainty. These sets accurately identify samples that pose prediction challenges, providing valuable insights for end-users seeking reliable predictions in the complexities of SOC estimation. © 2024 by the authors.

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Uncertainty Quantification of Stochastic Defects in Materials

Item Type Book

Author L. Chu

Abstract Uncertainty Quantification of Stochastic Defects in Materials investigates the uncertainty quantification methods for stochastic defects in material microstructures. It provides effective supplementary approaches for conventional experimental observation with the consideration of stochastic factors and uncertainty propagation. Pursuing a comprehensive numerical analytical system, this book establishes a fundamental framework for this topic, while emphasizing the importance of stochastic and uncertainty quantification analysis and the significant influence of microstructure defects on the material macro properties. Key Features Consists of two parts: One exploring methods and theories and the other detailing related examples Defines stochastic defects in materials and presents the uncertainty quantification for defect location, size, geometrical configuration, and instability Introduces general Monte Carlo methods, polynomial chaos expansion, stochastic finite element

methods, and machine learning methods Provides a variety of examples to support the introduced methods and theories Applicable to MATLAB® and ANSYS software This book is intended for advanced students interested in material defect quantification methods and material reliability assessment, researchers investigating artificial material microstructure optimization, and engineers working on defect influence analysis and nondestructive defect testing. © 2022 Taylor & Francis Group, LLC.

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Uncertainty quantification using Bayesian neural networks in classification: Application to biomedical image segmentation

Item Type Journal Article

Author Y. Kwon

Author J.-H. Won

Author B.J. Kim

Author M.C. Paik

Abstract Most recent research of deep neural networks in the field of computer vision has focused on improving performances of point predictions by developing network architectures or learning algorithms. Reliable uncertainty quantification accompanied by point estimation can lead to a more informed decision, and the quality of prediction can be improved. In this paper, we invoke a Bayesian neural network and propose a natural way of quantifying uncertainties in classification problems by decomposing the moment-based predictive uncertainty into two parts: aleatoric and epistemic uncertainty. The proposed method takes into account the discrete nature of the outcome, yielding the correct interpretation of each uncertainty. We demonstrate that the proposed uncertainty quantification method provides additional insights into the point prediction using two Ischemic Stroke Lesion Segmentation Challenge datasets and the Digital Retinal Images for Vessel Extraction dataset. © 2019 Elsevier

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Date 2020**DOI** 10.1016/j.csda.2019.106816**URL** <https://www.scopus.com/pages/publications/85069976043?origin=resultslist>**Archive** Scopus**Language** English**Volume** 142**Publisher** Elsevier B.V.**Publication** Computational Statistics and Data Analysis**ISSN** 01679473 (ISSN)**Journal Abbr** Comput. Stat. Data Anal.**Date Added** 20/09/2026, 9:52:27 pm**Modified** 20/09/2026, 9:52:27 pm**Tags:**

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Uncertainty quantification-based robust deep learning for building energy systems considering distribution shift problem

Item Type Journal Article**Author** X. Liang**Author** Z. Liu**Author** J. Wang**Author** X. Jin**Author** Z. Du

Abstract Artificial intelligence becomes one of the key technologies in building energy conservation, where deep learning algorithms provide feasible solution for the modeling problem of complex building energy systems. However, existing deep learning models are lack of robustness under distribution shift scenarios, which greatly limits their real-world application. To tackle this problem, first, we enrich the

concept of distribution shift problem in building energy domain. And a cluster-based dataset splitting method is proposed to simulate the distribution shift scenarios. Second, we adopt uncertainty quantification methods to improve the overall robustness of deep learning model. Comprehensive data experiments are conducted based on the reference modeling problem of chiller, and five widely-used uncertainty quantification methods are compared under distribution shift scenarios. Both the model precision performance and robustness performance are investigated. The experimental result demonstrates that the deep ensemble (DE) model is a simple but efficient method to solve the distribution shift problem, where its precision performance and robustness performance outperformed the other methods, followed by Bayesian neural network (BNN). In addition, the benefit of robust deep learning model is also studied. It is shown that the deep learning model with higher robustness performance can better filter out the unreliable predictions, which is essential to the online active learning strategy. The proposed research framework might provide a solid foundation for the real-world application of deep learning model. © 2023 Elsevier Ltd

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Uncertainty study on thermal and energy performances of a deterministic parameters based optimal aerogel glazing system using machine-learning method

Item Type Journal Article

Author Y. Zhou

Author S. Zheng

Abstract Uncertainty and sensitivity analyses of deterministic parameters based optimal aerogel glazing system are necessary due to multi-dimensional uncertainties in the real working condition, whereas thermal and energy performances of aerogel glazing system, in the academia, are normally characterized by deterministic parameters. In this study, a generic uncertainty quantification methodology was proposed using the two-dimensional Markov Chain Monte Carlo to quantify both aleatory and epistemic uncertainties of scenario parameters in the aerogel glazing system. A surrogate model, trained by mathematical heat and optical models using the machine-learning based data-driven method, was developed to predict the thermal and energy performances under multi-level scenario uncertainties. Results showed that, the developed surrogate model is efficient to deal with computational complexity of sophisticated light and heat transfer processes. When considering scenario uncertainties, the annual value of heat flux is reduced from 237.2 to 185.3 kWh/(m².a) by 21.9%, and the annual value of total heat gain is reduced from 267.2 to 209.5 kWh/m².a by 21.6%. This study proposes a generic methodology for multi-dimensional uncertainties' quantification and a surrogate model for thousands of cases-based uncertainty analysis. Approaches for the stochastic uncertainty analysis on aerogel glazing system were presented, which can promote the optimal design in buildings. © 2019 Elsevier Ltd

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Uncertainty-Aware AI for Enhanced Chronic Kidney Disease Diagnosis: A Review of Explainable and Reliable Models

Item Type Conference Paper

Author B. Anoch

Author L. Parthiban

Abstract Chronic kidney disease is a progressive condition affecting millions globally, often progressing to end-stage renal disease without early detection and management. Traditional diagnostic methods, relying on serum biomarkers and manual image interpretation, can be time-consuming, subjective, and variable. Artificial intelligence, particularly deep learning, has emerged as a transformative tool, automating CKD detection, classification, and risk prediction using medical imaging modalities like ultrasound, CT, and MRI. This review comprehensively examines AI-driven CKD diagnosis, focusing on deep learning-based image analysis, segmentation techniques, classification models, and uncertainty quantification methods. We explore various architectures, including CNNs, U-Net, Vision Transformers, and hybrid AI models, highlighting their effectiveness in automating kidney segmentation and CKD severity classification. We also discuss uncertainty-aware AI models, emphasizing the importance of model reliability and interpretability for clinical deployment. Despite advancements, challenges remain, including limited annotated datasets, model generalization issues, interpretability concerns, domain shifts, and ethical considerations. Future research should prioritize developing large-scale multi-institutional datasets, integrating multi-modal AI approaches, enhancing explainability through interpretable AI models, and establishing standardized regulatory frameworks for AI-driven CKD diagnosis © 2025 IEEE.

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Uncertainty-Aware Deep Learning and Structural Feature Analysis for Reliable Nephrotoxicity Prediction

Item Type Journal Article

Author J.-W. Liu

Author K.-Y. Liu

Author Y.-C. Deng

Author S.-H. Shi

Author X.-Z. Fu

Author Y.-P. Jiang

Author J. Fang

Author Q. Zhang

Author D.-J. Jiang

Author S. Liu

Author D.-S. Cao

Abstract Nephrotoxicity remains a critical safety concern in drug development and clinical practice. Despite their significance, existing computational models for nephrotoxicity prediction face challenges related to limited precision and reliability. To address these challenges, this study constructed the largest publicly available database to date, comprising 1831 high-quality nephrotoxicity-related compounds. Using this dataset, we developed classification models employing both traditional machine learning algorithms and graph-based deep learning methods. Our results demonstrate that the Directed Message Passing Neural Network model, combined with molecular graphs and ChemoPy2D descriptors, outperformed other models, achieving a mean Kappa value of 70.3%. To improve the reliability of the model, we implemented an uncertainty quantification method to define the model's applicability domain and quantify the confidence of prediction. On the representative model, this approach significantly enhanced predictive performance within the applicability domain, yielding a Kappa metric of 90.4%. Notably, our model also achieved the highest performance on an external test set compared to existing models. Finally, we performed multiscale feature analysis to provide actionable insights for safer drug design. This analysis integrated dataset -centric methods to identify structural alerts and beneficial transformations, alongside model-centric techniques to reveal the key global and local features driving nephrotoxicity. The established prediction models, combined with uncertainty quantification and structural feature insights derived in this study, offer valuable tools for the development of safer and more effective drugs.
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Uncertainty-aware deep learning-based CAD system for breast cancer classification using ultrasound and mammography images

Item Type Journal Article

Author M. Chegini

Author A. Mahlooji Far

Abstract Breast cancer is one of the most common types of cancer in women. Early and accurate diagnosis of breast cancer can increase the treatment chances and decrease the mortality rate. Thus, the development of accurate and reliable Computer-Aided Diagnosis (CAD) systems using breast cancer images is an immediate priority for early diagnosis. Deep learning methods have had a widespread role in CAD systems. In spite of the advantages, most deep learning-based CADs cannot quantify the uncertainty of their predictions. Uncertainty in the predicted results from a CAD system might endanger people's life in these crucial steps for breast cancer classification. Therefore, in this study, the Monte Carlo Dropout method is utilised as an uncertainty quantification method. Furthermore, differently from most existing studies, we present an uncertainty-aware deep learning model which can classify accurately from different types of images, namely, mammograms and ultrasound. The results show the proposed model achieved the accuracy of 99.95% in mammogram dataset, namely, DDSM and 98.40% and 91.34% in ultrasound datasets, namely, BUSM and BUSI, respectively. Finally, since we found the proposed model shows high uncertainty in misclassifications, we suggested to use the uncertainty values to find appropriate subsets for further inspection to increase the performance. © 2023 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

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Uncertainty-aware Evaluation of Time-Series Classification for Online Handwriting Recognition with Domain Shift

Item Type Conference Paper

Author A. Klač

Author S.M. Lorenz

Author M.W. Lauer-Schmaltz

Author D. Rügamer

Author B. Bischl

Author C. Mutschler

Author F. Ott

Editor Sioutis M.

Editor Long Z.

Editor Stell J.

Editor Renz J.

Abstract For many applications, analyzing the uncertainty of a machine learning model is indispensable. While research of uncertainty quantification (UQ) techniques is very advanced for computer vision applications, UQ methods for spatio-temporal data are less studied. In this paper, we focus on models for online handwriting recognition, one particular type of spatio-temporal data. The data is observed from a sensor-enhanced pen with the goal to classify written characters. We conduct a broad evaluation of aleatoric (data) and epistemic (model) UQ based on two prominent techniques for Bayesian inference, Stochastic Weight Averaging-Gaussian (SWAG) and Deep Ensembles. Next to a better understanding of the model, UQ techniques can detect out-of-distribution data and domain shifts when combining right-handed and left-handed writers (an underrepresented group). © 2022 Copyright for this paper by its authors.

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Uncertainty-aware fragility modeling of urban building exteriors subjected to hurricane-induced windborne debris with conditional generative adversarial nets

Item Type Journal Article

Author Z. Xiong

Author G. Jia

Author Y. Dong

Author Y. Guo

Abstract Building exteriors are highly susceptible to windborne debris damage during hurricanes, a concern that is growing in urgency amid climate change. Fragility functions that estimate the probability of exterior failure as a function of hurricane intensity are commonly used to assess this risk. However, developing the fragility function often requires numerous calls of expensive physics-based models. To address this challenge, this study proposes a machine learning-based approach that leverages conditional generative adversarial networks (cGAN) and its generative capability to efficiently and accurately develop fragility functions. cGAN model is introduced to model the complex high-dimensional distribution of the maximum impact momentum of windborne debris on the building exterior conditioned on hurricane intensity measures. To ensure the accuracy and reliability of cGAN, a stochastic simulation-guided training strategy is proposed that treats data preparation as an augmented stochastic simulation problem. This approach enhances exploration of the input space and enables the formulation of a novel training performance metric tailored toward fragility modeling. Once trained, the cGAN model can rapidly generate extensive damage realizations under various hurricane scenarios, which facilitates the construction of fragility functions. Furthermore, Monte Carlo dropout is introduced to efficiently quantify the uncertainty of the fragility function established using cGAN. Such uncertainty provides end users with a measure of confidence in the derived fragility estimates. An illustrative example is presented to validate the efficacy of the proposed method. Results show that the proposed performance metric is effective in guiding the selection of a reliable cGAN model that provides satisfactory damage and fragility estimates, and the proposed uncertainty quantification method can help provide rational confidence in the estimates. © 2025 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license. <http://creativecommons.org/licenses/by-nc-nd/4.0/>

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Uncertainty-aware generative models for inferring document class prevalence

Item Type Conference Paper

Author K.A. Keith

Author B. O'Connor

Editor Riloff E.

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Abstract Prevalence estimation is the task of inferring the relative frequency of classes of unlabeled examples in a group—for example, the proportion of a document collection with positive sentiment. Previous work has focused on aggregating and adjusting discriminative individual classifiers to obtain prevalence point estimates. But imperfect classifier accuracy ought to be reflected in uncertainty over the predicted prevalence for scientifically valid inference. In this work, we present (1) a generative probabilistic modeling approach to prevalence estimation, and (2) the construction and evaluation of prevalence confidence intervals; in particular, we demonstrate that an off-the-shelf discriminative classifier can be given a generative re-interpretation, by backing out an implicit individual-level likelihood function, which can be used to conduct fast and simple group-level Bayesian inference. Empirically, we demonstrate our approach provides better confidence interval coverage than an alternative, and is dramatically more robust to shifts in the class prior between training and testing.¹ © 2018 Association for Computational Linguistics

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Under pressure: Integrating machine learning to quantify clutch moments in American football

Item Type Journal Article

Author A. Aich

Author D. Bhattacharjee

Author H. Saikia

Abstract This study introduces a novel pressure index framework designed to quantify the intensity of crucial moments in an American football game. Our model integrates game-specific metrics such as time remaining, score differential, field position, and play context to establish quantitative measures of pressure on offensive and defensive units. The two proposed indices, viz, the composite offensive pressure index (PIO3) and defensive pressure index (PID) dynamically capture the evolving situational stress during a game. To validate and assess their predictive power, we employ machine learning techniques, including Logistic Regression, Random Forest, and Gradient Boosting models. Using play-by-play data from the 2023 NFL season, the results demonstrate that the ensemble learning methods outperform logistic regression, with Gradient Boosting achieving the highest predictive precision (AUC = 0.944) in forecasting drive success. This machine learning-driven framework provides new insights into situational dynamics and performance under pressure such as identifying pivotal clutch moments and quantifying critical drive transitions offering potential applications in advanced performance analysis, coaching strategies, and sports analytics research. © The Author(s) 2025

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Undersampling is a Minimax Optimal Robustness Intervention in Nonparametric Classification

Item Type Journal Article

Author N.S. Chatterji

Author S. Haque

Author T.B. Hashimoto

Abstract While a broad range of techniques have been proposed to tackle distribution shift, the simple baseline of training on an undersampled balanced dataset often achieves close to state-of-the-art-accuracy across several popular benchmarks. This is rather surprising, since under-sampling algorithms discard excess majority group data. To understand this phenomenon, we ask if learning is fundamentally constrained by a lack of minority group samples. We prove that this is indeed the case in the setting of nonparametric binary classification. Our results show that in the worst case, an algorithm cannot outperform undersampling unless there is a high degree of overlap between the train and test distributions (which is unlikely to be the case in real-world datasets), or if the algorithm leverages additional structure about the distribution shift. In particular, in the case of label shift we show that there is always an undersampling algorithm that is minimax optimal. In the case of group-covariate shift we show that there is an undersampling algorithm that is minimax optimal when the overlap between the group distributions is small. We also perform an experimental case study on a label shift dataset and find that in line with our theory, the test accuracy of robust neural network classifiers is constrained by the number of minority samples. © 2023, Transactions on Machine Learning Research. All rights reserved.

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Understanding Hessian Alignment for Domain Generalization

Item Type Conference Paper
Author S. Hemati
Author G. Zhang
Author A. Estiri
Author X. Chen
Abstract Out-of-distribution (OOD) generalization is a critical ability for deep learning models in many real-world scenarios including healthcare and autonomous vehicles. Recently, different techniques have been proposed to improve OOD generalization. Among these methods, gradient-based regularizers have shown promising performance compared with other competitors. Despite this success, our understanding of the role of Hessian and gradient alignment in domain generalization is still limited. To address this shortcoming, we analyze the role of the classifier's head Hessian matrix and gradient in domain generalization using recent OOD theory of transferability. Theoretically, we show that spectral norm between the classifier's head Hessian matrices across domains is an upper bound of the transfer measure, a notion of distance between target and source domains. Furthermore, we analyze all the attributes that get aligned when we encourage similarity between Hessians and gradients. Our analysis explains the success of many regularizers like CORAL, IRM, V-REx, Fish, IGA, and Fishr as they regularize part of the classifier's head Hessian and/or gradient. Finally, we propose two simple yet effective methods to match the classifier's head Hessians and gradients in an efficient way, based on the Hessian Gradient Product (HGP) and Hutchinson's method (Hutchinson), and without directly calculating Hessians. We validate the OOD generalization ability of proposed methods in different scenarios, including transferability, severe correlation shift, label shift and diversity shift. Our results show that Hessian alignment methods achieve promising performance on various OOD benchmarks. The code is available here.
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Understanding the complexity of near-infrared quantification of highly porous patient-tailored drug products by utilizing chemometrics and stimulated Raman imaging

Item Type Journal Article

Author I. Ahola

Author T. Tomberg

Author C. Cornett

Author C. Strachan

Author J. Rantanen

Author N. Genina

Abstract Additively manufactured drug products, typically produced using small-scale, on-demand batch mode, require rapid and non-destructive quantification methods. A tunable modular design (TMD) approach combining porous polymeric freeze-dried modules and an additive manufacturing method, inkjet printing, was proposed in an earlier study to fabricate accurate and patient-tailored doses of an antidepressant citalopram hydrobromide. This approach addresses the unmet medical needs associated with antidepressant tapering. Non-destructive quantification of printed porous structures is challenging due to the presence of residual solvents and frequent fluctuation of the material density. These shortcomings were mitigated by utilizing a

spinning near-infrared spectroscopy (NIRS) measurement setup and a post-print drying step. A machine learning algorithm (ML), specifically support vector regression, was implemented to lessen potential non-linearities caused by the complex structure of TMD drug products. The non-linear support vector regression models performed better than linear partial least squares (PLS) models when modeling the entire sample set (prediction error improved by 19 %). By dividing the TMD samples into subtypes and creating individual models for each subtype improved model performance: linear PLS models performed better or equally to non-linear models. It was hypothesized that this outcome was due to the structural differences between different TMD sample subtypes that was later confirmed by stimulated Raman scattering (SRS) microscopy. It was demonstrated that for complex porous drug products ML algorithms can improve NIRS model performance when a single universal robust model is preferred, and SRS is a powerful tool to explain the challenges that printing onto porous drug products can introduce to the NIRS quantification. © 2025 The Authors

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Understanding the Disparities of PM_{2.5} Air Pollution in Urban Areas via Deep Support Vector Regression

Item Type Journal Article

Author Y. Xia

Author T. McCracken

Author T. Liu

Author P. Chen

Author A. Metcalf

Author C. Fan

Abstract In densely populated urban areas, PM_{2.5} has a direct impact on the health and quality of residents' life. Thus, understanding the disparities of PM_{2.5} is crucial for ensuring urban sustainability and public health. Traditional prediction models often overlook the spillover effects within urban areas and the complexity of the data, leading to inaccurate spatial predictions of PM_{2.5}. We propose Deep Support Vector Regression (DSVR) that models the urban areas as a graph, with grid center points as the nodes and the connections between grids as the edges. Nature and human activity features of each grid are initialized as the representation of each node. Based on the graph, DSVR uses random diffusion-based deep learning to quantify the spillover effects of PM_{2.5}. It leverages random walk to uncover more extensive spillover relationships between nodes, thereby capturing both the local and nonlocal spillover effects of PM_{2.5}. And then it engages in predictive learning using the feature vectors that encapsulate spillover effects, enhancing the understanding of PM_{2.5} disparities and connections across different regions. By applying our proposed model in the northern region of New York for predictive performance analysis, we found that DSVR consistently outperforms other models. During periods of PM_{2.5} surges, the R-square of DSVR reaches as high as 0.729, outperforming non-spillover models by 2.5 to 5.7 times and traditional spatial metric models by 2.2 to 4.6 times. Therefore, our proposed model holds significant importance for understanding disparities of PM_{2.5} air pollution in urban areas, taking the first steps toward a new method that considers both the spillover effects and nonlinear feature of data for prediction. © 2024 American Chemical Society.

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Export Date: 20 September 2026; Cited By: 14; Funding details: Clemson University Division of Research; SERCEES; Correspondence Address: Y. Xia; School of Mathematics, Southwest Jiaotong University, Chengdu, Sichuan province, 611756, China; email: xyl2024001@my.swjtu.edu.cn; C. Fan; School of Civil and Environmental Engineering and Earth Sciences, Clemson University, Clemson, 29634, United States; email: cfan@clemson.edu; CODEN: ESTHA

Understanding the Role of Layer Normalization in Label-Skewed Federated Learning

Item Type Journal Article

Author G. Zhang

Author M. Beitollahi

Author A. Bie

Author X. Chen

Abstract Layer normalization (LN) is a widely adopted deep learning technique especially in the era of foundation models. Recently, LN has been shown to be surprisingly effective in federated learning (FL) with non-i.i.d. data. However, exactly why and how it works remains mysterious. In this work, we reveal the profound connection between layer normalization and the label shift problem in federated learning. To understand layer normalization better in FL, we identify the key contributing

mechanism of normalization methods in FL, called feature normalization (FN), which applies normalization to the latent feature representation before the classifier head. Although LN and FN do not improve expressive power, they control feature collapse and local overfitting to heavily skewed datasets, and thus accelerates global training. Empirically, we show that normalization leads to drastic improvements on standard benchmarks under extreme label shift. Moreover, we conduct extensive ablation studies to understand the critical factors of layer normalization in FL. Our results verify that FN is an essential ingredient inside LN to significantly improve the convergence of FL while remaining robust to learning rate choices, especially under extreme label shift where each client has access to few classes. Our code is available at https://github.com/huawei-noah/Federated-Learning/tree/main/Layer_Normalization. © 2024, Transactions on Machine Learning Research. All rights reserved.

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Undoing the Damage of Label Shift for Cross-domain Semantic Segmentation

Item Type Conference Paper

Author Y. Liu

Author J. Deng

Author J. Tao

Author T. Chu

Author L. Duan

Author W. Li

Abstract Existing works typically treat cross-domain semantic segmentation (CDSS) as a data distribution mismatch problem and focus on aligning the marginal distribution or conditional distribution. However, the label shift issue is unfortunately overlooked, which actually commonly exists in the CDSS task, and often causes a classifier bias in the learnt model. In this paper, we give an in-depth analysis and show that the damage of label shift can be overcome by aligning the data conditional distribution and correcting the posterior probability.

To this end, we propose a novel approach to undo the damage of the label shift problem in CDSS. In implementation, we adopt class-level feature alignment for conditional distribution alignment, as well as two simple yet effective methods to rectify the classifier bias from source to target by remolding the classifier predictions. We conduct extensive experiments on the benchmark datasets of urban scenes, including GTA5 to Cityscapes and SYNTHIA to Cityscapes, where our proposed approach outperforms previous methods by a large margin. For instance, our model equipped with a self-training strategy reaches 59.3% mIoU on GTA5 to Cityscapes, pushing to a new state-of-the-art. The code will be available at https://github.com/manmanjun/Undoing_UDA. © 2022 IEEE.

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Conference Name Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition

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Unifying domain gap in Federated Learning: A geometric approach

Item Type Journal Article

Author M. Lei

Author M. Ding

Author C. Ma

Author Z. Huang

Author M. Gao

Author J. Xu

Abstract Federated Learning (FL) is a framework designed to enable distributed and privacy-preserving learning. It typically involves a central server coordinating multiple clients to collaboratively train a global model without the need for sharing private data. Recent studies have revealed that this parameter-only communication strategy can be hindered by non-independent and identically distributed (non-iid) data across clients, leading to statistical heterogeneity. This issue necessitates the development of Personalized Federated Learning (PFL), which aims to enhance the performance of individual clients while still allowing for the learning of a global model. Most existing work on PFL has focused on label shift in supervised tasks, with relatively little attention given to feature shift, another common problem in real-world applications of federated learning. For instance, client data may vary in terms of illumination, angles, and quality for natural images, or in imaging protocols for medical images. To address this issue, we propose a general yet powerful framework that leverages the geometric interpretation of deep neural networks. Geometrically, a deep neural network partitions the feature space into a Power Diagram (PD) or Voronoi Diagram (VD). Regularized losses are introduced to mitigate feature shift from the perspective of PD and VD. Additionally, a personalized generator network is formulated for each client to rectify feature distribution and adapt the input data from its own domain to an "averaged" or unified domain that generalizes well across all clients. Our method demonstrates competitive performance compared to current state-of-the-art methods. © 2024 Elsevier B.V.

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: J. Xu; Department of Computer Science and Engineering, University at Buffalo, Buffalo, 338 Davis Hall, 14260, United States; email: jinhui@buffalo.edu; CODEN: NRCGE

UniLeiden at LeQua 2022: The first step in understanding the behaviour of the median sweep quantifier using continuous sweep

Item Type Conference Paper

Author K. Kloos

Author Q.A. Meertens

Author J.D. Karch

Editor Faggioli G.

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Editor Hanbury A.

Editor Favoritenstrasse 9, Vienna Vienna University of Technology

Editor Potthast M.

Editor Augustusplatz 10, Leipzig University of Leipzig

Abstract This paper presents the continuous sweep quantifier, a smoothed adaptation of the median sweep quantifier. Previous research has shown that the median sweep quantifier is a good quantifier. However, it is not well understood why it performs well because it is hard to derive its theoretical properties. The continuous sweep quantifier is a modification of the median sweep quantifier that enables computing theoretical results. The continuous sweep quantifier 1) uses kernel estimates instead of the empirical distribution, 2) constructs decision boundaries instead of applying discrete decision rules, and 3) uses the mean instead of the median. We show that a simplified adaptation of the continuous sweep quantifier performs similarly to the median sweep quantifier in terms of bias and variance on the LeQua 2022 dataset. The continuous sweep quantifier can therefore be used to provide insights into the median sweep quantifier by computing theoretical expressions for bias and variance. © 2022 Copyright for this paper by

its authors.

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Conference Name CEUR Workshop Proceedings

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K. Kloos; Leiden University, Faculty of Social Sciences, Institute of Psychology, Department Methodology and Statistics, Leiden, Wassenaarseweg 52, 2333 AK, Netherlands; email: k.kloos@fsw.leidenuniv.nl; Conference name: 23rd Working Notes of Conference and Labs of the Evaluation Forum, CLEF 2022; Conference date: 5 September 2022 through 8 September 2022; Conference code: 181762

Unintended readmissions and reoperations within 30 and 90 days following adult spinal deformity surgery

Item Type Journal Article

Author S.-H. Seo

Author S.-J. Hyun

Author J.-K. Lee

Author K.-J. Kim

Abstract **OBJECTIVE** This study investigated the rates and etiologies of unintended readmissions and reoperations within 30 and 90 days after adult spinal deformity (ASD) surgery. The authors aimed to identify the risk factors for readmission and reoperation by analyzing patient demographic and surgical characteristics. **METHODS** This retrospective cohort study included 307 consecutive patients who

underwent surgery for ASD from 2012 to 2022 at a single academic institution. Data were collected on patient demographic characteristics, comorbidities, operative details, and postoperative complications. Chi-square and multivariable logistic regression models were used to identify the risk factors associated with 30- and 90-day readmissions and reoperations. **RESULTS** The mean $\pm$ SD age at surgery was 66.6 ± 10.5 years, and the majority (80.8%) of patients were female. The 30-day and 90-day readmission rates were 11.7% and 15.3%, respectively. Multivariable regression for 30-day readmissions revealed that length of hospital stay (LOS) after index surgery of > 20 days (OR 2.48) and surgical factors such as vertebral column resection (VCR) (OR 4.26) and pelvic fixation (OR 4.38) were risk factors. Other factors such as the American Society of Anesthesiologists Physical Status Classification System (ASA) class, prior spine surgery, and age were not associated with 30-day readmissions. Ninety-day readmission was associated with high ASA class (OR 2.37) and LOS > 20 days (OR 2.82). The 30- and 90-day reoperation rates were 7.8% and 10.1%, respectively. The variables associated with 30-day reoperations were intraoperative VCR (OR 3.34) and LOS > 20 days (OR 9.38). Ninety-day reoperations were associated with dural tears (OR 3.33) and LOS > 20 days (OR 3.68). **CONCLUSIONS** This study provides valuable insights into the incidence of unintended readmission and reoperation within 30 and 90 days after ASD surgery in an Asian population. By identifying the associated risk factors, healthcare providers can customize surgical strategies and optimize perioperative care to effectively mitigate these events. ©AANS 2025, except where prohibited by US copyright law.

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: S.-J. Hyun; Seoul National University Bundang Hospital, Seoul National University College of Medicine, Seongnam, Gyeonggi, South Korea; email: hyunsj@snu.ac.kr

UniOOD: A Unified Framework for Domain Generalization and Out-of-Distribution Detection in Time Series

Item Type Conference Paper

Author Y. Chen

Author W. Zheng

Author B.H. Gwee

Author Q. Cao

Author S.G. Razul

Author Z. Lin

Abstract Time series modeling faces two challenges: out-of-distribution (OOD) generalization to correctly classify sequences from distribution shift domains, and OOD detection to identify anomalous or previously unseen sequences. Existing methods usually address these separately, which limits their applicability in real-world scenarios. In this work, we propose UniOOD, a unified framework that tackles both challenges simultaneously. UniOOD divides sequences into patches, encodes each as a latent Gaussian distribution, and aggregates them into a sequence-level posterior via learned weighting. By aligning sequence posteriors with learnable class priors while maintaining inter-class separation, UniOOD learns representations that are both generalizable and discriminative. The distance between posteriors and class priors provides an intrinsic OOD score, enabling detection of unseen sequences without prior exposure. Experiments show UniOOD achieves state-of-the-art classification, strong OOD generalization, and effective OOD detection within a single unified model. © 2026 IEEE.

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Universal Domain Adaptation for Simulation-Assisted SAR Target Recognition Based on Adversarial Uncertainty and Neighbor Relation

Item Type Journal Article

Author F. Gao

Author W. Ma

Author C. Yuan

Author J. Wang

Author J. Sun

Author A. Hussain

Abstract In the field of synthetic aperture radar (SAR) target recognition, leveraging computer-generated labeled simulated SAR data to aid in the recognition of unlabeled real SAR

data has garnered significant attention. However, the domain shift between the simulated and real domain contravenes the assumption of independent and identically distributed in deep learning (DL). Moreover, the categories of real SAR data may not align precisely with those of the simulated SAR data, and the prior knowledge of the label shift typically remains unavailable. To align the common category samples and identify target-private unknown category samples in the presence of both domain shift and potential label shift, we introduce a two-stage universal domain adaptation method for simulation-assisted SAR target recognition based on adversarial uncertainty and neighbor relation (AUNR). The proposed AUNR introduces Dirichlet distribution-based evidential DL, which not only performs classification tasks but also characterizes decision uncertainty, enabling the model to express the concept of 'Unknow.' In the first stage, alongside supervised learning, we leverage data augmentation-based contrastive learning to achieve robust feature representation. In the second stage, adversarial learning based on model and data uncertainty is employed to separate common and unknown class samples, and contrastive learning based on neighbor relation aligns samples in two domains according to their geometric relationships. To the best of our knowledge, this is the first study on UniDA for SAR target recognition. Experimental results demonstrate that our method outperforms previous State-of-the-Art methods in various label shift scenarios, which indicates the effectiveness of the proposed method. © 1965-2011 IEEE.

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Universal Test-time Adaptation through Weight Ensembling, Diversity Weighting, and Prior Correction

Item Type Conference Paper

Author R.A. Marsden

Author M. Döbler

Author B. Yang

Abstract Since distribution shifts are likely to occur during testtime and can drastically decrease the model's performance, online test-time adaptation (TTA) continues to update the model after deployment, leveraging the current test data. Clearly, a method proposed for online TTA has to perform well for all kinds of environmental conditions. By introducing the variable factors domain non-stationarity and temporal correlation, we first unfold all practically relevant settings and define the entity as universal TTA. We want to highlight that this is the first work that covers such a broad spectrum, which is indispensable for the use in practice. To tackle the problem of universal TTA, we identify and highlight several challenges a self-training based method has to deal with: 1) model bias and the occurrence of trivial solutions when performing entropy minimization on varying sequence lengths with and without multiple domain shifts, 2) loss of generalization which exacerbates the adaptation to multiple domain shifts and the occurrence of catastrophic forgetting, and 3) performance degradation due to shifts in class prior. To prevent the model from becoming biased, we leverage a dataset and model-agnostic certainty and diversity weighting. In order to maintain generalization and prevent catastrophic forgetting, we propose to continually weightaverage the source and adapted model. To compensate for disparities in the class prior during test-time, we propose an adaptive prior correction scheme that reweights the model's predictions. We evaluate our approach, named ROID, on a wide range of settings, datasets, and models, setting new standards in the field of universal TTA. Code is available at: <https://github.com/mariodoebler/testtime-adaptation> © 2024 IEEE.

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Notes:

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Unknown fault recognition and diagnosis under domain and class shift via neighborhood invariance learning and cross-domain mixup

Item Type Journal Article

Author Y. He

Author W. Li

Author K. Xu

Author W. Yao

Author Z. Zuo

Author G. Zhang

Author Q. Wang

Author X. Kong

Abstract The application of transfer learning methods has effectively mitigated the challenges posed by operational condition variations in the fault diagnosis field. However, the assumption of label space consistency between the source and target domain has hindered the broader applicability of those methods. In industrial scenarios, the fault types across source and target domains often exhibit partial overlap, with potential emergence of novel, unknown fault classes in the target domain. This complex transfer scenario, characterized by both domain shift and label shift, defines the challenge of universal domain adaptation. To address this, we propose a universal domain adaptation diagnosis framework specifically designed for cross-domain identification of unknown faults unknown. The proposed method first employs the neighborhood invariance learning to optimize the separability of samples in the target domain. Then, a category prototype enhanced inter-domain mixup strategy generates pseudo-unknown samples, while an open-set classifier assigns the probabilities to unknown fault samples. Finally, a category consistency constraint is introduced to

refine the decision boundary between known and unknown fault categories. Multiple comparison experiments were conducted on the SDUST dataset to evaluate the effectiveness of the proposed method in these extreme transfer scenarios. The experimental results demonstrate that the method proposed in this manuscript achieves superior performance compared to state-of-the-art universal domain adaptation methods. © 2025 Elsevier Ltd

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Unmanned Aerial Multi-Object Dynamic Frame Detection and Skipping Using Deep Learning on the Internet of Drones

Item Type Journal Article

Author U. Ahmed

Author J.C.-W. Lin

Author G. Srivastava

Abstract The Internet of Drones (IoD) has a revolutionary impact on monitoring and preserving the environment. Traffic regulations face enormous challenges due to rapid growth in the number of vehicles. In IoD, multiple-aerial-drone video sensing infrastructure can increase detected objects. However, the main difficulty lies in picture quality due to lighting conditions, the angle of view, and the physical structure of vehicles. This research mainly focuses on the development and deployment of a deep-learning-based system to analyze traffic congestion. The model uses multiple drone video feeds and vehicle information to detect, classify, and count a transport vehicle in a live traffic feed. The model is trained with a deep learning approach to first align the video frame and then detect the object in a top-down aerial drone video. The dynamic skipping method helps process a long video feed and accurately compares the video frame to the viewer, and then, using the standard vehicle query (i.e., make, model and year of manufacture), detect traffic scenarios in real time. The proposed model has many applications requiring a particular area to monitor real-time data analysis and drone routine tasks. © 2018 IEEE.

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Export Date: 20 September 2026; Cited By: 5

Unraveling the efficiency-equity trade-off in urban heat island mitigation by blue–green–grey space optimization in China

Item Type Journal Article**Author** Z. Liu

Abstract Rapid urbanization has shifted the urban climate management paradigm from passive expansion control to the active optimization of existing biophysical properties. However, conventional mitigation frameworks often overlook the physical interdependence of urban surfaces and the inequity inherent in cooling resource distribution. This study integrates interpretable machine learning to quantify seasonal thermal contributions and a novel 3D Convex Hull model to determine the physically constrained mitigation ceilings of Blue–Green–Grey (BGG) spaces across 199 major prefecture-level cities in China (2000–2024) at a 1-km spatial resolution. The proposed method achieved good random hold-out performance ($R^2 = 0.83$) and revealed a distinct spatiotemporal differentiation of cooling mechanisms: Green space is statistically associated with the largest share of model-attributed cooling and exhibits a clear phenological curve, especially in the ecologically restored Loess Plateau. Blue space provides a unique dual-seasonal thermal buffer, peaking in both summer and winter. Grey space contribution is largely radiation-dependent, becoming highly pronounced in summer. Overall, BGG optimization can achieve a local cooling capacity of 1.01–5.46 °C, with a critical efficiency-equity trade-off under various intervention scenarios. While maximizing cooling efficiency yields a superior mitigation capacity (2.23 °C), it exacerbates spatial inequality. Conversely, cooling footprint expansion (1.83 °C) enhances spatial equity by reducing the Gini index by 0.3. These findings provide practical support for climate-resilient urban development. © 2026 Elsevier B.V.

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Unraveling the Impact of Heterophilic Structures on Graph Positive-Unlabeled Learning

Item Type Conference Paper

Author Y. Wu

Author J. Yao

Author B. Han

Author L. Yao

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Editor Scarlett J.

Editor Berkenkamp F.

Abstract While Positive-Unlabeled (PU) learning is vital in many real-world scenarios, its application to graph data still remains under-explored. We unveil that a critical challenge for PU learning on graph lies on the edge heterophily, which directly violates the irreducibility assumption for Class-Prior Estimation (class prior is essential for building PU learning algorithms) and degenerates the latent label inference on unlabeled nodes during classifier training. In response to this challenge, we introduce a new method, named Graph PU Learning with Label Propagation Loss (GPL). Specifically, GPL considers learning from PU nodes along with an intermediate heterophily reduction, which helps mitigate the negative impact of the heterophilic structure. We formulate this procedure as a bilevel optimization that reduces heterophily in the inner loop and efficiently learns a classifier in the outer loop. Extensive experiments across a variety of datasets have shown that GPL significantly outperforms baseline methods, confirming its effectiveness and superiority. Copyright 2024 by the author(s)

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Language English

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% reductions, ITS applications, Real-world scenario, Federated learning, Adversarial machine learning, Contrastive Learning, Critical challenges, Unsupervised learning, Classifier training, Graph data, Bi-level optimization, Label propagation, Named graphs, Propagation loss

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Unraveling the relationship between white matter lesions in MRI and migraine: a systematic review

Item Type Journal Article

- Author V.V. Viuniski
- Author M. de Lima Ruffini
- Author A.M. de Souza
- Author R.P. dos Santos
- Author D.H.G.F. Ribeiro
- Author L.L. Faoro
- Author R.G. Londero
- Author J.A. Duarte
- Author F. Reis

Abstract Background White matter hyperintensities (WMH) are commonly detected on brain magnetic resonance imaging (MRI) scans of migraine patients, but their clinical relevance and underlying mechanisms remain uncertain. Objective To systematically review the relationship between WMH and migraine, focusing on prevalence, progression, and associations with clinical and demographic characteristics. Methods We conducted a systematic review of observational studies published between 1990 and May 2025, including adult patients with migraine (with or without aura) who underwent brain MRIs with at least 1.5T scanners. Data extraction was performed by two independent reviewers, with disagreements resolved by a third. Study quality was assessed using the Newcastle-Ottawa scale for observational studies. Results A total of 25 studies were included, comprising approximately 3,600 participants, of whom 1,725 had migraine. Most participants were women and reported age means or medians typically between 30 and 60 years. Frequently, WMHs were observed in

migraine patients, particularly in those with aura, longer disease duration, and higher headache frequency. No consistent association was found between WMH and comorbidities. Significant heterogeneity in imaging protocols, lesion quantification methods, and study design limited data comparability and precluded meta-analysis. Conclusion Migraine patients often present with WMHs, but their clinical significance remains unclear. Future studies should employ standardized MRI protocols, volumetric lesion quantification, and consistent migraine phenotyping to clarify its pathophysiological role in migraine and potential implications for diagnosis and management. © 2025 The Author(s).

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Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: J.A. Duarte; Universidade Federal do Rio Grande do Sul, Faculdade de Medicina, Departamento de Medicina Interna, Porto Alegre, RS, Brazil; email: juduarte@hcpa.edu.br; CODEN: ANPIA

Unsupervised Adaptation from FDG to PSMA PET/CT for 3D Lesion Detection under Label Shift

Item Type Conference Paper

Author X. Liu

Author M. Xia

Author Y. Chemli

Author G. El Fakhri

Author C. Liu

Author J. Ouyang

Abstract In this work, we propose an unsupervised domain adaptation (UDA) framework for 3D volumetric lesion detection that adapts a detector trained on labeled FDG PET/CT to unlabeled PSMA PET/CT. Beyond covariate shift, cross tracer adaptation also exhibits label shift in both lesion size composition and the number of lesions per subject. We introduce self-training with two mechanisms that explicitly model and compensate for this label shift. First, we adaptively adjust the detection anchor shapes by re-estimating target domain box scales from selected pseudo labels and updating anchors with an exponential moving average. This increases positive anchor coverage for small PSMA lesions and stabilizes box regression. Second, instead of a fixed confidence threshold for pseudo-label selection, we allocate size bin-wise quotas according to the estimated target domain histogram over lesion volumes. The self-training alternates between supervised learning with prior-guided pseudo labeling on PSMA and supervised learning on labeled FDG. On AutoPET 2024, adapting from 501 labeled FDG studies to 36918 F-PSMA studies, the proposed method improves both AP and FROC over the source-only baseline and conventional self-training without label-shift mitigation, indicating that modeling target lesion prevalence and size composition is an effective path to robust cross-tracer detection. © 2026 IEEE.

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Conference Name Proceedings - International Symposium on Biomedical Imaging

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Computer vision, Supervised learning, Medical imaging, Self-supervised learning, Domain adaptation, Target domain, Self-training, Label shift, Unsupervised domain adaptation, Adaptation framework, Biomedical engineering, Label Shift, Lesion detection, Lesion Detection, PET/CT, Trace analysis, Unsupervised adaptation, Unsupervised Domain Adaptation, Volumetrics

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Unsupervised and Interpretable Domain Adaptation to Rapidly Filter Tweets for Emergency Services

Item Type Conference Paper

Author J. Krishnan

Author H. Purohit

Author H. Rangwala

Editor Atzmuller M.

Editor Coscia M.

Editor Missaoui R.

Abstract During the onset of a natural or man-made crisis event, public often share relevant information for emergency services on social web platforms such as Twitter. However, filtering such relevant data in real-time at scale using social media mining is challenging due to the short noisy text, sparse availability of relevant data, and also, practical limitations in collecting large labeled data during an ongoing event. In this paper, we hypothesize that unsupervised domain adaptation through multi-task learning can be a useful framework to leverage data from past crisis events for training efficient information filtering models during the sudden onset of a new crisis. We present a novel method to classify relevant social posts during an ongoing crisis without seeing any new data from this event (fully unsupervised domain adaptation). Specifically, we construct a customized multi-task architecture with a multi-domain discriminator for crisis analytics: multi-task domain adversarial attention network (MT-DAAN). This model consists of dedicated attention layers for each task to provide model interpretability; critical for real-world applications. As deep networks struggle with sparse datasets, we show that this can be improved by sharing a base layer for multitask learning and domain adversarial training. The framework is validated with the public datasets of TREC incident streams that provide labeled Twitter posts (tweets) with relevant classes (Priority, Factoid, Sentiment) across 10 different crisis events such as floods and earthquakes. Evaluation of domain adaptation for crisis events is performed by choosing one target event as the test set and training on the rest. Our results show that the multi-task model outperformed its single-task counterpart. For the qualitative evaluation of

interpretability, we show that the attention layer can be used as a guide to explain the model predictions and empower emergency services for exploring accountability of the model, by showcasing the words in a tweet that are deemed important in the classification process. Finally, we show a practical implication of our work by providing a use-case for the COVID-19 pandemic. © 2020 IEEE.

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Tags:

Learning systems, Social networking (online), Emergency services, Social Media, Interpretability, Information filtering, Text Classification, Text mining, Social media minings, Domain adaptation, Filtration, Multi-task learning, Multi-task model, Unsupervised Domain Adaptation, Model prediction, Classification process, Crisis Analytics, Multi domains, Qualitative evaluations

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Unsupervised Characterization of Temporal Dataset Shifts as an Early Indicator of AI Performance Variations: Evaluation Study Using the Medical Information Mart for Intensive Care-IV Dataset

Item Type Journal Article

Author D. Fernández-Narro

Author P. Ferri

Author A. Gutiérrez-Sacristán

Author J.M. García-Gómez

Author C. Sáez

Abstract Background: Reusing long-term data from electronic health records is essential for training reliable and effective health artificial intelligence (AI). However, intrinsic changes in health data distributions over time—known as dataset shifts, which include concept, covariate, and prior shifts—can compromise model performance, leading to model obsolescence and inaccurate decisions. Objective: In this study, we investigate whether unsupervised, model-agnostic characterization of temporal dataset shifts using data distribution analyses through Information Geometric Temporal (IGT) projections is an early indicator of potential AI performance variations before model development. Methods: Using the real-world Medical Information Mart for Intensive Care-IV (MIMIC-IV) electronic health record database, encompassing data from over 40,000 patients from 2008 to 2019, we characterized its inherent dataset shift patterns through an unsupervised approach using IGT projections and data temporal heatmaps. We trained and evaluated annually a set of random forests and gradient boosting models to predict in-hospital mortality. To assess the impact of shifts on model performance, we checked the association between the temporal clusters found in both IGT projections and the intertime embedding of model performances using the Fisher exact test. Results: Our results demonstrate a significant relationship between the unsupervised temporal shift patterns, specifically covariate and concept shifts, identified using the IGT projection method and the performance of the random forest and gradient boosting models ($P<.05$). We identified 2 primary temporal clusters that correspond to the periods before and after ICD-10 (International Statistical Classification of Diseases, Tenth Revision) implementation. The transition from ICD-9 (International Classification of Diseases, Ninth Revision) to ICD-10 was a major source of dataset shift, associated with a performance degradation. Conclusions: Unsupervised, model-agnostic characterization of temporal shifts via IGT projections can serve as a proactive monitoring tool to anticipate performance shifts in clinical AI models. By incorporating early shift detection into the development pipeline, we can enhance decision-making during the training and maintenance of these models. This approach paves the way for more robust, trustworthy, and self-adapting AI systems in health care. ©David Fernández-Narro, Pablo Ferri, Alba Gutiérrez-Sacristán, Juan M García-Gómez, Carlos Sáez. Originally published in JMIR Medical Informatics (<https://medinform.jmir.org>), 03.Dec.2025.

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Unsupervised classification of biomedical abstracts using lexical association

Item Type Conference Paper**Author** J. Read**Author** J. Webster**Author** A.C. Fang

Abstract The task of text classification is the assignment of labels that describe texts' characteristics, such as topic, genre or sentiment. Supervised machine learning techniques such as Support Vector Machines or the simple but effective Naïve Bayes have been successfully applied to this task. However, it is not always practical to acquire a sufficient corpus of labelled examples to train these methods. For these cases we describe an unsupervised method for text classification based on two hypotheses. Firstly, we propose that the class of a document may be determined by calculating its constituent features' similarity with prototypical examples of each class. Secondly, we note the importance of class priors in Naïve Bayes classifiers, and hypothesize that class distributions might be estimated using the relative frequency of prototype words. Performing experiments on a corpus of biomedical abstracts with topic information derived from the Medical Subject Headings (MeSH), we investigate the characteristics of the method when used in conjunction with basic, linguistic and knowledge-based features, and find that the performance of the unsupervised method is approximately 80% that of Naïve Bayes. Our research is significant in that it highlights a candidate method with good potential for further improvement when training on unlabelled data. © 2010 by Jonathon Read, Jonathan Webster, and Alex Chengyu Fang.

Date 2010**URL** <https://www.scopus.com/pages/publications/84863867245?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: PACLIC 24 - Proc. 24th Pacific Asia Conf. Lang., Inf. Comput.**Publisher** Institute for Digital Enhancement of Cognitive Development, Waseda University.**ISBN** 978-490516600-9 (ISBN)**Pages** 261-270**Proceedings Title** PACLIC 24 - Proc. 24th Pacific Asia Conf. Lang., Inf. Comput.**Conference Name** PACLIC 24 - Proceedings of the 24th Pacific Asia Conference on Language, Information and Computation**Date Added** 20/09/2026, 9:52:54 pm**Modified** 20/09/2026, 9:52:54 pm

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Unsupervised Domain Adaptation for Well Log Lithology Classification with Incomplete Source Domain Labels

Item Type Conference Paper

Author L. Zheng

Author J. Chang

Author Z. Yu

Author S. Dong

Author H. Duan

Author G. Cai

Author Y. Kang

Author W. Lv

Abstract Multi source domain adaptation has recently emerged as an effective approach for well logging lithology identification under cross well distribution shifts. However, most existing approaches are built on the assumption that all source domains are fully labeled. In practice, source wells may be only partially annotated due to limited expert resources and operational constraints. Such partial annotation often leads to degraded performance, especially when the available labels are inconsistent or incomplete across different source wells. To tackle these challenges, we propose SR2-MSDA, a semi-supervised multi-source domain adaptation framework. SR2-MSDA employs a dynamic source weighting strategy that adaptively adjusts source domain weights based on their proximity to the target in feature space, and leverages unlabeled source samples through a teacher-student consistency scheme. In addition, to mitigate label heterogeneity across domains, we propose modulating the adversarial loss by integrating the target class prior with a class level gating mechanism for each source sample. Experiments on real well log data from an eastern oilfield show that SR2-MSDA consistently improves Accuracy and Macro-F1 by 4%-6% over multi source domain adaptation methods that utilize only labeled source samples. © 2026 IEEE.

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Unsupervised Domain Adaptive Object Detection with Class Label Shift Weighted Local Features

Item Type Conference Paper
Author A. Tan
Author N. Hanselmann
Author S. Ding
Author F. Tombari
Author M. Cordts
Editor Karlinsky L.
Editor Michaeli T.
Editor Nishino K.

Abstract Due to the high transferability of features extracted from early layers (called local features), aligning marginal distributions of local features has achieved compelling results in unsupervised domain adaptive object detection. However, such marginal feature alignment suffers from the class label shift between source and target domains. Existing class label shift correction methods focus on image classification, and cannot be directly applied to object detection due to objects' co-occurrence. Meanwhile, one property of local features is that they have small receptive fields and can be easily mapped back to specific areas of input images. Therefore, to handle object co-occurrence scenarios, we propose to leverage this property to decompose the source feature maps and compute the source domain class distribution at the pixel level. The decomposition is based on each feature pixel's receptive field overlap with ground-truth bounding boxes. In the target domain, where no labels are available, we estimate this distribution using predicted bounding boxes and thus get the estimated class label shift between domains. This estimated shift is further used to re-weight source local features during the feature alignment. To the best of our knowledge, this is the first work trying to explicitly correct class label shift in unsupervised domain adaptive object detection. Experimental results demonstrate that this approach can systematically improve several recent domain adaptive object detectors, such as SW and HTCNet on benchmark datasets with different degrees of class label shift. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Unsupervised domain adaptive tumor region recognition for Ki67 automated assisted quantification

Item Type Journal Article

Author Q. He

Author Y. Liu

Author F. Pan

Author H. Duan

Author J. Guan

Author Z. Liang

Author H. Zhong

Author X. Wang

Author Y. He

Author W. Huang

Author T. Guan

Abstract Purpose: Ki67 is a protein associated with tumor proliferation and metastasis in breast cancer and acts as an essential prognostic factor. Clinical work requires recognizing tumor regions on Ki67-stained whole-slide images (WSIs) before quantitation. Deep learning has the potential to provide assistance but largely relies on massive annotations and consumes a huge amount of time and energy. Hence, a novel tumor region recognition approach is proposed for more precise Ki67 quantification. Methods: An unsupervised domain adaptive method is proposed, which combines adversarial and self-training. The model trained on labeled hematoxylin and eosin (H&E) data and unlabeled Ki67 data can recognize tumor regions in Ki67 WSIs. Based on the UDA method, a Ki67 automated assisted quantification system is developed, which contains foreground segmentation, tumor region recognition, cell counting, and WSI-level score calculation. Results: The proposed UDA method achieves high performance in tumor region recognition and Ki67 quantification. The AUC reached 0.9915, 0.9352, and 0.9689 on the validation set and internal and external test sets, respectively, substantially exceeding baseline (0.9334, 0.9167, 0.9408) and rivaling the fully supervised method (0.9950, 0.9284, 0.9652). The evaluation of automated quantification on 148 WSIs illustrated statistical agreement with pathological reports. Conclusion: The model trained by the proposed method is capable of accurately recognizing Ki67 tumor regions. The proposed UDA method can be readily extended to other types of immunohistochemical staining images. The results of automated assisted quantification are accurate and interpretable to provide assistance to both junior and senior pathologists in their interpretation. © 2022, CARS.

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Article, automation, female, Female, human, Humans, Deep learning, image segmentation, procedures, deep learning, unsupervised machine learning, evaluation study, histopathology, image processing, Image Processing, Computer-Assisted, computer assisted diagnosis, human tissue, pathology, metabolism, Breast Neoplasms, breast tumor, staining, Breast cancer, cell counting, invasive breast cancer, Ki 67 antigen, Ki-67 Antigen, Staining and Labeling, tumor diagnosis, Tumor region recognition, Unsupervised domain adaption

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Unsupervised extraction of flood-induced backscatter changes in SAR data using markov image modeling on irregular graphs

Item Type Conference Paper
Author S. Martinis
Author A. Twele
Author S. Voigt

Abstract The near real-time provision of precise information about flood dynamics from synthetic aperture radar (SAR) data is an essential task in disaster management. A novel tile-based parametric thresholding approach under the generalized Gaussian assumption is applied on normalized change index data to automatically solve the three-class change detection problem in large-size images with small class a priori probabilities. The thresholding result is used for the initialization of a hybrid Markov model which integrates scale-dependent and spatiocontextual information into the labeling process by combining hierarchical with noncausal Markov image modeling. Hierarchical maximum a posteriori (HMAP) estimation using the Markov chains in scale, originally developed on quadrees, is adapted to hierarchical irregular graphs. To reduce the computational effort of the iterative optimization process that is related to noncausal Markov models, a Markov random field (MRF) approach is defined, which is applied on a restricted region of the lowest level of the graph, selected according to the HMAP labeling result. The experiments that were performed on a bitemporal TerraSAR-X StripMap data set from South West England during and after a large-scale flooding in 2007 confirm the effectiveness of the proposed change detection method and show an increased classification accuracy of the hybrid MRF model in comparison to the sole application of the HMAP estimation. Additionally, the impact of the graph structure and the chosen model parameters on the labeling result as well as on the performance is discussed. © 2006 IEEE.

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Language English

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Notes:

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Unsupervised Learning under Latent Label Shift

Item Type Conference Paper

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Editor Cho K.

Editor Oh A.

Abstract What sorts of structure might enable a learner to discover classes from unlabeled data? Traditional approaches rely on feature-space similarity and heroic assumptions on the data. In this paper, we introduce unsupervised learning under Latent Label Shift (LLS), where we have access to unlabeled data from multiple domains such that the label marginals $p_{\mathbf{y}}(\mathbf{y})$ can shift across domains but the class conditionals $p_{\mathbf{x}|\mathbf{y}}(\mathbf{x}|\mathbf{y})$ do not. This work instantiates a new principle for identifying classes: elements that shift together group together. For finite input spaces, we establish an isomorphism between LLS and topic modeling: inputs correspond to words, domains to documents, and labels to topics. Addressing continuous data, we prove that when each label's support contains a separable region, analogous to an anchor word, oracle access to $p_{\mathbf{d}}(\mathbf{d})$ suffices to identify $p_{\mathbf{y}}(\mathbf{y})$ and $p_{\mathbf{x}|\mathbf{y}}(\mathbf{x}|\mathbf{y})$ up to permutation. Thus motivated, we introduce a practical algorithm that leverages domain-discriminative models as follows: (i) push examples through domain discriminator $p_{\mathbf{d}}(\mathbf{d})$; (ii) discretize the data by clustering examples in $p_{\mathbf{d}}(\mathbf{d})$ space; (iii) perform non-negative matrix factorization on the discrete data; (iv) combine the recovered $p_{\mathbf{y}|\mathbf{d}}(\mathbf{y}|\mathbf{d})$ with the discriminator outputs $p_{\mathbf{d}}(\mathbf{d})$ to compute $p_{\mathbf{x}|\mathbf{y}}(\mathbf{x}|\mathbf{y})$ @ $\mathbf{d}$. With semi-synthetic experiments, we show that our algorithm can leverage domain information to improve upon competitive unsupervised classification methods. We reveal a failure mode of standard unsupervised classification methods when data-space similarity does not indicate true groupings, and show empirically that our method better handles this case. Our results establish a deep connection between distribution shift and topic modeling, opening promising lines for future work. © 2022 Neural information processing systems foundation. All rights reserved.

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Extra Journal Abbreviation: Adv. neural inf. proces. syst.

Volume 35**Publisher** Neural information processing systems foundation**ISBN** 10495258 (ISSN); 978-171387108-8 (ISBN)**Proceedings Title** Adv. neural inf. proces. syst.**Conference Name** Advances in Neural Information Processing Systems**Date Added** 20/09/2026, 9:52:19 pm**Modified** 20/09/2026, 9:52:19 pm**Tags:**

Classification (of information), Topic Modeling, Clustering algorithms, Unlabeled data, Unsupervised learning, Discriminators, Classification methods, Traditional approaches, Shift-and, Feature space, Marginals, Unsupervised classification, Input space, Multiple domains, Non-negative matrix factorization

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Unsupervised optimal deep transfer learning for classification under general conditional shift

Item Type Journal Article**Author** J. Lang**Author** Y. Liu

Abstract Classifiers trained on labeled source data may yield misleading results when applied to unlabeled target data drawn from a different distribution. Transfer learning can rectify this by transferring knowledge from source to target data, but its effectiveness frequently relies on stringent assumptions, such as label shift or strong separation conditions. We introduce a novel general conditional shift assumption, which encompasses label shift as a special case and facilitates the identifiability of both the target distribution and the shift function without requiring a separation condition. Our classifier is constructed by integrating deep neural networks (DNNs) and a pseudo-maximum likelihood approach. We establish asymptotic error bounds for our DNN-based classifier and estimators of the conditional probabilities (Formula presented) for source data and the target label distribution (Formula presented), in terms of the intrinsic dimension of (Formula presented). Notably, the excess risk of the proposed classifier achieves the optimal minimax rate, up to a logarithmic factor. Our method not only eliminates the need to estimate the shift function, but also alleviates the

curse of dimensionality when (Formula presented) exhibits a low-dimensional structure. Numerical simulations, along with an analysis of an Alzheimer's disease dataset, underscore its exceptional performance. © The Author(s) 2026. Published by Oxford University Press on behalf of The International Biometric Society. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site-for further information please contact journals.permissions@oup.com. This article is published and distributed under the terms of the Oxford University Press, Standard Journals Publication Model (<https://academic.oup.com/pages/standard-publication-reuse-rights>)

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Updates on the OECD NEA Working Party on Scientific Issues and Uncertainty Analysis of Reactor Systems (WPRS)

Item Type Conference Paper

Author O. Buss

Author H. Ferroukhi

Author K. Ivanov

Abstract Under the guidance of the Nuclear Science Committee of the OECD Nuclear Energy Agency (NEA), the Working Party on Scientific Issues and Uncertainty Analysis of Reactor Systems (WPRS) deals with reactor physics, radiation transport and dosimetry, core thermal-hydraulics (T/H), fuel performance, and associated multi-physics aspects for present and future nuclear power systems. It studies the modelling and simulation (M&S) of reactor systems, along with verification, validation and uncertainty quantification (VVUQ). It acts as a platform for international collaboration sustaining a vibrant community of practice, fostering periodic and synergistic interactions among its participants. Main foci of the WPRS activities are recommendations and policy advice in its domain of expertise, international benchmarking activities, data preservation, and education. The WPRS runs 13 active benchmarks covering a wide variety of systems ranging from conventional reactor systems to advanced reactor designs, including micro- and small modular reactors (SMRs). Since PHYSOR 2022, three notable new activities have been launched: a multi-stage T/H code validation benchmark based on data measured at the High Temperature Test Facility (HTTF) at Oregon State University, US, a comprehensive lead fast reactor benchmark based on the ALFRED design, and a Task Force on Artificial Intelligence and Machine Learning (AI/ML) for Scientific Computing in Nuclear Engineering, which will evaluate the performance of AI/ML in multi-physics M&S of reactor systems. In the domain of data preservation WPRS is overseeing four database projects including the well-known International Handbook of Evaluated Reactor Physics Benchmark Experiments (IRPhE) Project with 169 different experimental series that were performed at 57 different nuclear facilities. Another significant project is the Shielding Integral Benchmark Archive and Database (SINBAD), which is pioneering the implementation of open source software methodologies into the benchmark database development. On the education side, WPRS has successfully organized the first International School on Simulation of Nuclear Reactor Systems (SINUS) this year. It delivered a hybrid, hands-on-training encompassing multi-physics simulations and associated uncertainty quantification methodologies. The article provides an overview of the WPRS activities emphasizing new activities that have been launched since PHYSOR 2022 and the international co-operation opportunities. © 2024 AMERICAN NUCLEAR SOCIETY. All rights reserved.

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Benchmarking, Uncertainty, Open source software, Curricula, Personnel training, Nuclear energy, Benchmark, Benchmarks, Bosons, Charged particles, Educational nuclear reactors, Fast reactors, Hadrons, High temperature reactors, International Co-operation, International co., Multi-physics, Nuclear Energy Agency, Nuclear reactor simulators, Particle beams, Radioactive waste transportation, Reactor cores, Reactor physics, Reactor Physics, Reactor refueling, Reactor shielding, Reactor systems, Research laboratories, Scientific issues, Scientific uncertainty, Small nuclear reactors, Uncertainty Analysis, Wind tunnels, Working parties

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Upgrading Reliability in Molecular Property Prediction by Robust Quantification of Uncertainty from Machine Learning Models

Item Type Journal Article

Author A. Kötter

Author K. Singh

Author H. Matter

Author G. Hessler

Author C. Grebner

Abstract Reliable methods to quantify the predictive uncertainty of machine learning (ML) models can significantly increase the impact of molecular property prediction and are routinely used in applications like active learning and ML-guided property optimization. Poor predictive accuracy of ML models is often related to (i) regions of the chemical space, which are characterized by large property differences for structurally similar molecules, and (ii) a lack of representation of test molecules in the training data. Here, we analyze the relationship between these error sources and the predictive uncertainty of popular uncertainty quantification (UQ) methods on molecular activity data sets. We find that several UQ methods struggle to identify poorly predicted compounds in regions of steep structure–activity relationships (SAR). We also demonstrate that the evaluation scenario, as defined by data splitting into training and test sets, significantly impacts observed UQ performance. Based on

our findings we introduce a simple but strong and very robust method for UQ that offers significant improvements over previous approaches in several evaluation scenarios and demonstrate its usefulness in an exploratory active learning setting. © 2025 The Authors. Published by American Chemical Society

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: A. Kötter; Digital R&D Large Molecule Research, Sanofi-Aventis Deutschland GmbH, Frankfurt am Main, 65926, Germany; email: alexander.koetter@sanofi.com; CODEN: JCISD

Upper and Lower Tight Error Bounds for Feature Omission with an Extension to Context Reduction

Item Type Journal Article

Author R. Schluter

Author E. Beck

Author H. Ney

Abstract In this work, fundamental analytic results in the form of error bounds are presented that quantify the effect of feature omission and selection for pattern classification in general, as well as the effect of context reduction in string classification, like automatic speech recognition, printed/handwritten character recognition, or statistical machine translation. A general simulation framework is introduced that supports discovery and proof of error bounds, which lead to the error bounds presented here. Initially derived tight lower and upper bounds for feature omission are generalized to feature selection, followed by another extension to context reduction of string class priors (aka language models) in string classification. For string classification, the quantitative effect of string class prior context reduction on symbol-level Bayes error is presented. The tightness of the original feature omission bounds seems lost in this case, as further simulations indicate. However, combining both feature omission and context reduction, the tightness of the bounds is retained. A central result of this work is the proof of the existence, and the amount of a statistical threshold w.r.t. the introduction of additional features in general pattern classification, or the increase of context in string classification beyond which a decrease in Bayes error is guaranteed. © 1979-2012 IEEE.

Date 2019

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Urban landfill site selection using AHP and SA W in GIS environment (case study: Kohkiluye-O-Boyer Ahmad Province, Iran)

Item Type Journal Article

Author H.R. Jafari

Author Y. Rafii

Author M. Ramezani Mehrian

Author H. Nasiri

Abstract Introduction: Environmental conservation and optimal utilization of natural resources, is considered as the requirements of sustainable development at all levels, including regional levels. Sanitary urban landfill like any other engineering project needs basic information and accurate planning. Today, geographic information system (GIS) is widely utilized in environmental planning and engineering problems. Selecting a suitable location for a landfill requires considering several factors. Thus, the necessity of using spatial information technology and its integration with other planning and management issues will be inevitable. Multi-criteria decision-making systems include models and methods that have been provided by different people to make the process of planning easier for planners. These systems gradually improve their performance by others. Analytical Hierarchy Process (AHP) and simple Additive weighting (SAW) are systems of multi-criteria decision making that have been used in this paper. Materials and Methods: Criteria to this case were selected based on existing resources and environmental principles. These criteria include: distance from city, main roads, faults, main waterways and villages, rainfall, slope, elevation, land cover and Geology. Based on the weight that was assigned to layers in each of the two methods 10 weighted layers were created using GIS. In the next step, constraint map of the burial areas was prepared. Finally, the prepared maps were incorporated and multiplied in constraint layers in GIS, according to acquisitive weights in both AHP and SAW. Fig. 1 shows this process simply. Results and Discussion: According to the results of this process and classifying output maps, the priorities of areas for landfill was identified in Arc GIS. Given that in this study, two models were used to compare the quality of the output map, it is normal that the results obtained from these two models are different. After combining layers and their integration with the constraint map, output maps (Figures 2 and 3) were prepared for each of the two methods. Final map were classified in five categories based on the current classification as follows: 1. Restrained 2. Relatively poor 3. Relatively good 4. Appropriate and 5. Perfect (Figure Presented) Due to differences in methods of weighting of criteria and alternatives, the outcomes of AHP and SAW are to some extent different. In AHP method, perfect, Appropriate, relatively good and relatively poor Priorities respectively are 19%, 18%, 2% and 1%. While in SAW, perfect, Appropriate and relatively poor Priorities respectively are 22%, 18% and 1%. The restriction in both methods was 59%. (Figure Presented) Conclusion: Pair-wise comparison is more appropriate for determining the weight than their direct

determination (SAW), because in the pair-wise comparisons (AHP) classes' weight for each criterion is obtained based on a class priority rather than all classes of the criterion, but this process is not done in the direct determination of weight (SAW). After the production of output layers by the two methods, can be concluded with a superficial analysis that the AHP model presented more conservative results than the SAW. Finally, it can be claimed that the integrated system of GIS and multi-criteria decision-making can be a useful tool for environmental planning.

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Use of an improved high-throughput absolute abundance quantification method to characterize soil bacterial community and dynamics

Item Type Journal Article

Author L. Yang

Author J. Lou

Author H. Wang

Author L. Wu

Author J. Xu

Abstract High-throughput sequencing has dramatically expanded our understanding of bacterial communities based on the information of the species types and their relative abundances. Recently, researchers have also become aware of a deficiency in not considering the absolute abundance in this technique. Combining two or more different methods has typically been used to achieve absolute quantification of microbial communities. However, making a combination of different methods not

only is time-consuming but also involves potential uncertainty due to variations in the experimental conditions. To simplify the experimental procedure and improve the high-throughput absolute abundance quantification (HAAQ) of a soil bacterial community, we propose an HAAQ method that uses an internal standard strain (ISS) HAAQ-GFP to simultaneously obtain both the relative and absolute abundances in the soil bacterial community. The results showed that a soil bacterial community and its dynamics can be better characterized by the HAAQ method when the optimal concentrations of ISS HAAQ-GFP (105 to 107 cells g⁻¹) were used, and a 16S rRNA gene copy number adjustment was applied. Based on the HAAQ method, we first found that soil bacterial absolute abundances at the genus level fitted well to the partial log-normal distribution function, and most genera concentrations were in the range of 103.5 to 106.5 cells g⁻¹ in the test soils. Our case studies also indicated that more comprehensive descriptions of soil bacterial communities and their dynamics can be achieved by both the relative and absolute abundances than by the relative abundance alone. The improved HAAQ method can be potentially applied to other microbial ecological studies and to stimulating the development of quantitative bacterial ecology studies. © 2018 Elsevier B.V.

Date 2018

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Use of ensemble based on GA for imbalance problem

Item Type Conference Paper

Author L. Cleofas

Author R.M. Valdovinos

Author V. García

Author R. Alejo

Abstract In real-world applications, it has been observed that class imbalance (significant differences in class prior probabilities) may produce an important deterioration of the classifier performance, in particular with patterns belonging to the less represented classes. One method to tackle this problem consists to resample the original training set, either by over-sampling the minority class and/or under-sampling the majority class. In this paper, we propose two ensemble models (using a modular neural network and the nearest neighbor rule) trained on datasets under-sampled with genetic algorithms. Experiments with real datasets demonstrate the effectiveness of the methodology here proposed. © 2009 Springer Berlin Heidelberg.

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Use of salt-marsh site classification for mangrove forest development and reforestation in the coastal area of thai binh province in the context of climate change

Item Type Conference Paper

Author H.T.L. Giang

Author D.Q. Manh

Author N.Q. Huy

Editor Viet N.T.

Editor Tung T.T.

Editor Xiping D.

Abstract This Salt-marsh site classification is an important scientific basis for the effective and sustainable management of salt-marsh in the coastal area. The main aim of this research is to use of salt-marsh site classification to select suitable mangrove species for mangrove forest development and reforestation in the coastal area of Thai Binh province. Quantification method and six indicators of salt-marsh site classification such as type of salt-marsh, current status of mangrove forest and salt-marsh, exposed time of mudflat, soil maturity, sand percentage, altitude of mudflat and mapping of salt-marsh site classification of Thai Binh province (Dong Long commune, Tien Hai district, Thai Binh province, the scale of 1/5,000). Planting techniques for *Kandelia obovata* Sheue, H.Y. Liu & J. Yong, and *Sonneratia caseolaris* (L.) Engl such as cutting the top bud at the early stage of planting, using 3 bamboo stakes to stabilize trees were used. Survival rate increased 27% after 3 years of planting compared with control. The growth of mangrove forests in the experiment formula is very different, especially the site type group is very convenient, convenient and difficult, the survival rate of the experiment formula in the field type is favorable and very convenient from 60-70% after 3 years of planting mangrove forests, while the difficulty in survival is 10-20%. The probability of accretion in the experimental models is very clear, in the site type group is very favorable after 36 months of mangroves planting from 9.1 to 10.0 cm, clearly higher than the control (3,9cm). In the site-wise group, the increase of accretion from 6.8 to 7.1 cm was significantly higher than control (3.0 cm). In the difficult site group, the average accretion volume in the

following 36 months of planting was 4.3-4.5 cm, clearly higher than the control (2.1 cm). © Springer Nature Singapore Pte Ltd. 2020.

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Using a unified measure function for heuristics, discretization, and rule quality evaluation in Ant-Miner

Item Type Conference Paper

Author K.M. Salama

Author F.E.B. Otero

Abstract Ant-Miner is a classification rule discovery algorithm that is based on Ant Colony Optimization (ACO) metaheuristic. cAnt-Miner is the extended version of the algorithm that handles continuous attributes on-the-fly during the rule construction process, while μ Ant-Miner is an extension of the algorithm that selects the rule class prior to its construction, and utilizes multiple pheromone types, one for each permitted rule class. In this paper, we combine these two algorithms to derive a new approach for learning classification rules using ACO. The proposed approach is based on using the measure function for 1) computing

the heuristics for rule term selection, 2) a criteria for discretizing continuous attributes, and 3) evaluating the quality of the constructed rule for pheromone update as well. We explore the effect of using different measure functions for on the output model in terms of predictive accuracy and model size. Empirical evaluations found that hypothesis of different functions produce different results are acceptable according to Friedman's statistical test. © 2013 IEEE.

Date 2013

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Language English

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Artificial intelligence, Algorithms, Quality control, Empirical evaluations, Predictive accuracy, Function evaluation, Quality evaluation, Continuous attribute, Ant Colony Optimization (ACO), Classification Rule Discovery, Classification rules, Construction process, Miners

Notes:

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Using Adversarial Training to Improve Uncertainty Quantification

Item Type Journal Article

Author K. Huang

Author M. Xu

Author Y. Wang

Abstract The success of adversarial attack methods suggests a small input change may mislead a trained machine-learning model. For example, changing one pixel of an image may cause the trained model to misclassify this updated image. Uncertainty quantification is crucial for detecting misclassifications; hence, precise uncertainty quantification, meaning uncertainty estimates that closely align with prediction correctness, is essential. We assume that misclassified samples should exhibit high uncertainty while correctly classified samples should exhibit low uncertainty. To evaluate the performance of uncertainty quantification, we investigate the task of uncertainty-

based misclassification detection under adversarial attack conditions. Our findings suggest that existing uncertainty quantification methods are unable to accurately identify misclassified predictions resulting from adversarial attacks due to training issues. We propose a simple adversarial training strategy for improving uncertainty quantification. Our results show that adversarial training improves the reliability of uncertainty quantification by better aligning uncertainty with prediction correctness. Specifically, we observe consistent improvements in misclassification detection performance, measured by AUC-ROC and AUC-PR, across clean and adversarial samples. © 2020 IEEE.

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Performance, Learning systems, Uncertainty analysis, Computer vision, Uncertainty, Adversarial attack, Adversarial machine learning, Classifieds, Machine learning models, Generative adversarial networks, Attack methods, Misclassifications, Uncertainty quantifications, uncertainty quantification, Uncertainty estimates, misclassification detection, Misclassification detection, Praseodymium compounds, Small inputs

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: National Science Foundation, NSF, (2430746, 2430747); Correspondence Address: Y. Wang; University of Tennessee at Chattanooga, Department of Computer Science and Engineering, Chattanooga, 37403, United States; email: yingfeng-wang@utc.edu

Using Bayesian networks to clarify interpretation of exposure–response regression coefficients: blood lead–mortality association as an example

Item Type Journal Article

Author L.A. Cox

Abstract We examine how Bayesian network (BN) learning and analysis methods can help to meet several methodological challenges that arise in interpreting significant regression coefficients in exposure–response regression modeling. As a motivating example, we consider the challenge of interpreting positive regression coefficients for blood lead level (BLL) as a predictor of mortality risk for nonsmoking men. We first note that practices such as dichotomizing or categorizing continuous confounders (e.g. income), omitting potentially important socioeconomic confounders (e.g. education), and assuming specific parametric regression model forms leave unclear to what extent a positive regression coefficient reflects these modeling choices, rather than a direct dependence of mortality risk on exposure. Therefore, significant exposure–response coefficients in parametric regression models do not necessarily reveal the extent to which reducing exposure-related variables (e.g. BLL) alone, while leaving fixed other correlates of exposure and mortality risks (e.g. education, income, etc.) would reduce adverse outcome risks (e.g. mortality risks). We then consider how BN structure-learning and inference algorithms and nonparametric estimation methods (partial dependence plots) can be used to clarify dependencies between variables, variable selection, confounding, and quantification of joint effects of multiple factors on risk, including possible high-order interactions and nonlinearities. We conclude that these details must be carefully modeled to determine whether a data set provides evidence that exposure itself directly affects risks; and that BN and nonparametric effect estimation and uncertainty quantification methods can complement regression modeling and help to improve the scientific basis for risk management decisions and policy-making by addressing these issues. © 2020 Informa UK Limited, trading as Taylor & Francis Group.

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Journal Abbr Crit. Rev. Toxicol.

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adult, human, human experiment, Humans, male, risk assessment, machine learning, review, algorithm, education, risk management, Bayesian network, Lead, outcome assessment, Bayes theorem, Bayes Theorem, uncertainty, lead, pollution, environmental exposure, mortality risk, adverse outcome, blood lead level, Causality, Environmental Exposure, Environmental Pollution, lead blood level, nonlinear system, partial dependence plots

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Using deep learning to quantify neuronal activation from single-cell and spatial transcriptomic data

Item Type Journal Article

Author E. Bahl

Author S. Chatterjee

Author U. Mukherjee

Author M. Elsadany

Author Y. Vanrobaeys

Author L.-C. Lin

Author M. McDonough

Author J. Resch

Author K.P. Giese

Author T. Abel

Author J.J. Michaelson

Abstract Neuronal activity-dependent transcription directs molecular processes that regulate synaptic plasticity, brain circuit development, behavioral adaptation, and long-term memory. Single cell RNA-sequencing technologies (scRNAseq) are rapidly developing and allow for the interrogation of activity-dependent transcription at cellular resolution. Here, we present NEUROeSTIMator, a deep learning model that integrates transcriptomic signals to estimate neuronal activation in a way that we demonstrate is associated with Patch-seq electrophysiological features and that is robust against differences in species, cell type, and brain region. We demonstrate this method's ability to accurately detect neuronal activity in previously published studies of single cell activity-induced gene expression. Further, we applied our model in a spatial transcriptomic study to identify unique patterns of learning-induced activity across different brain regions in male mice. Altogether, our findings establish NEUROeSTIMator as a powerful and broadly applicable tool for measuring neuronal activation, whether as a critical covariate or a primary readout of interest. © 2024, This is a U.S. Government work and not under copyright protection in the US; foreign copyright protection may apply.

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Using machine learning to quantify drivers of aerosol pollution trend in China from 2015 to 2022

Item Type Journal Article

Author Y. Ji
Author Y. Zhang
Author D. Liu
Author K. Zhang
Author P. Cai
Author B. Zhu
Author B. Zhang
Author J. Xian
Author H. Wang
Author X. Ge

Abstract Atmospheric aerosol pollution, such as fine particulate matter (PM2.5) and inhalable particulate matter (PM10), is one of the most important environmental problems in China. To mitigate particulate air pollution, the Chinese government has been implementing a series of clean air actions. Here, we conducted a comprehensive analysis on surface air pollutants data obtained from the China air quality observation network using a random forest (RF) approach, to evaluate the impact of clean air actions on aerosol pollution from 2015 to 2022. Overall, the observed PM2.5 and PM10 showed evidently decreasing trends during 2015–2022 in each season, with the largest trends in winter. Based on the RF model analysis, we further quantified the

seasonal-dependent trends in PM_{2.5} and PM₁₀ concentrations driven by anthropogenic emissions over China, which were approximately -3.84 and -6.14 , -2.82 and -4.71 , -2.58 and -4.45 , and -2.77 and -4.06 $\mu\text{g m}^{-3} \text{ yr}^{-1}$ for winter, spring, summer, and autumn, respectively. Furthermore, anthropogenic emissions were estimated to contribute 34.44 and 54.98, 24.10 and 37.22, 23.48 and 36.20, and 21.04 and 30.69 $\mu\text{g m}^{-3}$ to declines in annual mean PM_{2.5} and PM₁₀ concentrations during the eight years in megacity clusters of eastern China for winter, spring, summer, and autumn, respectively. These seasonal-dependent trend analyses reveal the largest reduction in ambient aerosol pollution due to anthropogenic emission abatements during cold season and further demonstrate substantial effectiveness of the clean air actions for improvement in particulate air quality in China. Correlation analysis on PM_{2.5} with the major meteorological parameters and the RF model built-in feature importance suggested that a combination of high relative humidity, shallow boundary layer heights, and low wind speeds could promote wintertime aerosol pollution in short-time scales over northern China (e.g., the North China plain). © 2023 Elsevier Ltd

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Using machine learning to quantify sources of light-absorbing water-soluble humic-like substances (HULISws) in Northeast China

Item Type Journal Article

Author Y. Hong

Author F. Cao

Author M.-Y. Fan

Author Y.-C. Lin

Author M. Bao

Author Y. Xue

Author J. Wu

Author M. Yu

Author X. Wu

Author Y.-L. Zhang

Abstract Water-soluble humic-like substances (HULISws) are a group of heterogeneous organic compounds in the atmosphere, greatly impacting climate change and human health. As the most significant biomass burning (BB) and fossil fuel burning polluted area in China, Northeast China Plain could generate more light absorbing HULISws to the atmosphere. In this work, daily fine particle (PM_{2.5}) samples were collected in Changchun, located in Northeast China Plain, from October 17th to November 29th, 2016, to investigate the potential sources of optical properties of HULISws. Here, the whole sampling period was divided into three sub-periods using the density of fire spots and the timeline of the central heating: the non-heating (October 17th to October 24th), early heating (October 25th to November 13th), and normal heating (November 14th to November 29th) periods. The mean mass concentrations of HULISws were 4.2 ± 1.2 , 8.6 ± 3.6 , and $2.4 \pm 1.3 \mu\text{g m}^{-3}$ during the non-heating, early heating, and normal heating periods, respectively. The positive matrix factorization (PMF) model results suggested that the contribution of primary BB, secondary BB, and fossil fuel burning emissions were 46%, 24%, and 19% during the non-heating period, 22%, 29%, and 17% during the early heating period, and 13%, 16%, and 50% during the normal heating period, respectively. Combining the PMF results with the random forest (RF) algorithm, the contribution of each source to the optical properties was quantified. Here, BB still dominated the optical properties of HULISws in Northeast China. During the study period, the primary BB and secondary BB contributed 25% and 34% in the light absorption coefficient (Abs),

22% and 33% in the mass absorption exponent (MAE), and 17% and 33% in the absorption Ångström exponent (AAE), respectively. Other sources like cooking and fossil combustion contributed 25% and 35% in Abs, 35% and 26% in MAE, and 16% and 33% in AAE, respectively. © 2022 Elsevier Ltd

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Using machine learning to quantify structural MRI neurodegeneration patterns of Alzheimer's disease into dementia score: Independent validation on 8,834 images from ADNI, AIBL, OASIS, and MIRIAD databases

Item Type Journal Article**Author** K. Popuri**Author** D. Ma**Author** L. Wang**Author** M.F. Beg

Abstract Biomarkers for dementia of Alzheimer's type (DAT) are sought to facilitate accurate prediction of the disease onset, ideally predating the onset of cognitive deterioration. T1-weighted magnetic resonance imaging (MRI) is a commonly used neuroimaging modality for measuring brain structure in vivo, potentially providing information enabling the design of biomarkers for DAT. We propose a novel biomarker using structural MRI volume-based features to compute a similarity score for the individual's structural patterns relative to those observed in the DAT group. We employed ensemble-learning framework that combines structural features in most discriminative ROIs to create an aggregate measure of neurodegeneration in the brain. This classifier is trained on 423 stable normal control (NC) and 330 DAT subjects, where clinical diagnosis is likely to have the highest certainty. Independent validation on 8,834 unseen images from ADNI, AIBL, OASIS, and MIRIAD Alzheimer's disease (AD) databases showed promising potential to predict the development of DAT depending on the time-to-conversion (TTC). Classification performance on stable versus progressive mild cognitive impairment (MCI) groups achieved an AUC of 0.81 for TTC of 6 months and 0.73 for TTC of up to 7 years, achieving state-of-the-art results. The output score, indicating similarity to patterns seen in DAT, provides an intuitive measure of how closely the individual's brain features resemble the DAT group. This score can be used for assessing the presence of AD structural atrophy patterns in normal aging and MCI stages, as well as monitoring the progression of the individual's brain along with the disease course. © 2020 The Authors. Human Brain Mapping published by Wiley Periodicals LLC.

Date 2020**DOI** 10.1002/hbm.25115**URL** <https://www.scopus.com/pages/publications/85087288569?origin=resultslist>**Archive** Scopus**Language** English**Volume** 41**Publisher** John Wiley and Sons Inc.**Pages** 4127-4147**Publication** Human Brain Mapping**ISSN** 10659471 (ISSN)**Issue** 14**Journal Abbr** Hum. Brain Mapp.**Date Added** 20/09/2026, 9:52:28 pm**Modified** 20/09/2026, 9:52:28 pm**Tags:**

Article, female, human, male, machine learning, aged, quantitative analysis, ensemble learning, major clinical study, classifier, disease severity, follow up, disease classification, sex difference, Alzheimer disease, nuclear magnetic resonance imaging, neuroimaging, feature selection, white matter, biological marker, data base, magnetic resonance imaging, mild cognitive impairment, dementia, Alzheimer's disease, nerve degeneration, cerebrospinal fluid, gray matter, amyloid beta protein, priority journal, disease course, brain size, brain atrophy, cross-database independent validation, dementia of Alzheimer's type, dementia score, disease progression, longitudinal diagnostic stratification, probabilistic classifier, prognosis prediction

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Using machine learning to quantify the impacts of genetically modified crops on US midwest corn yields

Item Type Journal Article

Author P.M. Johnson

Author R. Bennartz

Author J.V. Camp

Abstract Global food security is becoming increasingly stressed by growing populations and climate change. To compensate for these stresses, crop yields must increase throughout the upcoming century. One of the more prominently featured solutions entails genetically modified crops, but their impacts on yields are contested. Here, we leverage machine learning techniques to examine the effects genetically modified crops have had on US corn yields. In particular, a principal components analysis conducted on US Midwest county yields reveals that the commercialization of genetically modified corn accentuated preexisting spatial disparities in production

and explains approximately 6–12% of the region's inter-county variation in yields from 1980 to 2015. Additionally, counterfactual yield trajectories predicted by Bayesian structural time series models using non-genetically modified crops as synthetic controls suggest that the adoption of this biotechnology amounted to an approximate 13% increase in overall US corn yields from 1996 to 2015. © 2019 Elsevier Ltd

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Using Machine Learning to Quantify the Multimedia Risk Due to Fuzzing

Item Type Journal Article

Author G.S. Kashyap

Author K. Malik

Author S. Wazir

Author R. Khan

Abstract There have been considerable advancements in multimedia technologies over the past 5 years. It has been observed that state-of-the-art multimedia systems face three broad categories of challenges: (1) Dependency on continuous network connections, (2) Data-sharing applications & collaboration, and (3) Security issues. Among these,

security vulnerability poses a major threat to modern multimedia systems. Therefore, it is imperative to carefully investigate the security issues that can endanger wireless and mobile communications. At present, multimedia security research mainly focuses on wireless traffic monitoring, wireless system attacks, and wireless and mobile security. In this paper, we have used the network attack-type, “Reconnaissance”, which contains two types of malicious activities: (1) OS scanning, and (2) Fuzzing. The goal of this paper is to quantify multimedia security risks due to Fuzzing by using various types of machine learning models. The highest accuracy i.e., 96.8%, is obtained using the XGBoost classifier, which is good compared to the existing models present in the literature. This is the first paper, to the best of our knowledge, that uses machine learning methods to differentiate between benign and malignant Fuzzing attacks. © 2021, The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature.

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Using Machine Learning to Quantify the Robustness of Network Controllability

Item Type Conference Paper

Author A. Dhiman

Author P. Sun

Author R. Kooij

Editor Renault E.

Editor Boumerdassi S.

Editor Mühlethaler P.

Abstract This paper presents machine learning based approximations for the minimum number of driver nodes needed for structural controllability of networks under link-based random and targeted attacks. We compare our approximations with existing analytical approximations and show that our machine learning based approximations significantly outperform the existing closed-form analytical approximations in case of both synthetic and real-world networks. Apart from targeted attacks based upon the removal of so-called critical links, we also propose analytical approximations for out-in degree-based attacks. © 2021, Springer Nature Switzerland AG.

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Export Date: 20 September 2026; Cited By: 19; Correspondence Address: P. Sun; Delft University of Technology, Delft, Netherlands; email: P.Sun-1@tudelft.nl; Conference name: 3rd International Conference on Machine Learning for Networking, MLN 2020; Conference date: 24 November 2020 through 26 November 2020; Conference code: 256129

Using photographs and deep neural networks to understand flowering phenology and diversity in mountain meadows

Item Type Journal Article

Author A. John

Author E.J. Theobald

Author N. Cristea

Author A. Tan

Author J. Hille Ris Lambers

Abstract Mountain meadows are an essential part of the alpine–subalpine ecosystem; they provide ecosystem services like pollination and are home to diverse plant communities. Changes in climate affect meadow ecology on multiple levels, for example, by altering growing season dynamics. Tracking the effects of climate change on meadow diversity through the impacts on individual species and overall growing season dynamics is critical to conservation efforts. Here, we explore how to combine crowd-sourced camera images with machine learning to quantify flowering species richness across a range of elevations in alpine meadows located in Mt. Rainier National Park, Washington, USA. We employed three machine-learning techniques (Mask R-CNN, RetinaNet and YOLOv5) to detect wildflower species in images taken during two flowering seasons. We demonstrate that deep learning techniques can detect multiple species, providing information on flowering richness in photographed meadows. The results indicate higher richness just above the tree line for most of the species, which is comparable with patterns found using field studies. We found that the two-stage detector Mask R-CNN was more accurate than single-stage detectors like RetinaNet and YOLO, with the Mask R-CNN network performing best overall with mean average precision (mAP) of 0.67 followed by RetinaNet (0.5) and YOLO (0.4). We found that across the methods using anchor box variations in multiples of 16 led to enhanced accuracy. We also show that detection is possible even when pictures are interspersed with complex backgrounds and are not in focus. We found differential detection rates depending on species abundance, with additional challenges related to similarity in flower characteristics, labeling errors and occlusion issues. Despite these potential biases and limitations in capturing flowering abundance and location-specific quantification, accuracy was notable considering the complexity of flower types and picture angles in this dataset. We, therefore, expect that this approach can be used to address many ecological questions that benefit from automated flower detection, including studies of flowering phenology and floral resources, and that this approach can, therefore, complement a wide range of ecological approaches (e.g., field observations, experiments, community science, etc.). In all, our study suggests that ecological metrics like floral richness can be efficiently monitored by combining machine learning with easily accessible publicly curated datasets (e.g., Flickr, iNaturalist). © 2024 The Authors. Remote Sensing in Ecology and Conservation published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

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Using procrustes distance and shape space for Automatic Target Recognition

Item Type Journal Article

Author G.J. Meyer

Author S.C. Gustafson

Author G.D. Arnold

Abstract Equating objects based on shape similarity (for example scaled Euclidean transformations) is often desirable to solve the Automatic Target Recognition (ATR) problem. The Procrustes distance is a metric that captures the shape of an object independent of the following transformations: translation, rotation, and scale. The Procrustes metric assumes that all objects can be represented by a set of landmarks (i.e. points), that they have the same number of points, and that the points are ordered (i.e., the exact correspondence between the points is known from one object to the next). Although this correspondence is not known for many ATP, problems, computationally feasible methods for examining all possible combinations are being explored. Additionally, most objects can be mapped to a shape space where translation, rotation, and scaling are removed, and distances between object points in this space can then form another useful metric. To establish a decision boundary in any classification problem, it is essential to know the a priori probabilities in the appropriate space. This paper analyzes basic objects (triangles) in two-dimensional space to assess how a known distribution in Euclidean space maps to the shape space. Any triangles whose three coordinate points are uniformly distributed within a two-dimensional box transforms to a bivariate independent normal distribution with mean (0, 0) and standard deviations of 2 in Kendall shape space (two points of the triangle are mapped to (-1/2, 0) and (1/2, 0)). The Central Limit Theorem proves that the limit of sums of finite variance distributions approaches the normal distribution. This is a reasonable model of the relationship between the three Euclidean coordinates relative to the single Kendall shape space coordinate. This paper establishes the relationship

between different objects in the shape space and the Procrustes distance, which is an established shape metric, between these objects. Ignoring reflections (because it is a special case), the Procrustes distance is isometric to the shape space coordinates. This result demonstrates that both Kendall coordinates and Procrustes distance are useful features for ATR. © 2002 SPIE · 0277-786X/02/\$15.00.

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Using samples with label noise for robust continual learning

Item Type Journal Article

Author H. Nie

Author S. Fan

Author Y. Liu

Author Q. Yao

Author Z. Wang

Abstract Recent studies have shown that effectively leveraging samples with label noise can enhance model robustness by uncovering more reliable feature patterns. While existing methods, such as label correction methods and loss correction techniques, have demonstrated success in utilizing noisy labels, they assume that noisy and clean samples (samples with correct annotations) share the same label space. However, this assumption does not hold in continual machine learning, where new categories and tasks emerge over time, leading to label shift problems that are specific to this setting. As a result, existing methods may struggle to accurately estimate the ground truth

labels for noisy samples in such dynamic environments, potentially exacerbating label noise and further degrading performance. To address this critical gap, we propose a Shift-Adaptive Noise Utilization (SANU) method, designed to transform samples with label noise into usable samples for continual learning. SANU introduces a novel source detection mechanism that identifies the appropriate label space for noisy samples, leveraging a meta-knowledge representation module to improve the generalization of the detection process. By re-annotating noisy samples through label guessing and label generation strategies, SANU adapts to label shifts, turning noisy data into useful inputs for training. Experimental results across three continual learning datasets demonstrate that SANU effectively mitigates the label shift problem, significantly enhancing model performance by utilizing re-annotated samples with label noise. © 2025

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Using smartphones and machine learning to quantify Parkinson disease severity the mobile Parkinson disease score

Item Type Journal Article

Author A. Zhan

Author S. Mohan

Author C. Tarolli

Author R.B. Schneider

Author J.L. Adams

Author S. Sharma

Author M.J. Elson

Author K.L. Spear

Author A.M. Glidden

Author M.A. Little

Author A. Terzis

Author E. Ray Dorsey

Author S. Saria

Abstract IMPORTANCE: Current Parkinson disease (PD) measures are subjective, rater-dependent, and assessed in clinic. Smartphones can measure PD features, yet no smartphone-derived rating score exists to assess motor symptom severity in real-world settings. OBJECTIVES: To develop an objective measure of PD severity and test construct validity by evaluating the ability of the measure to capture intraday symptom fluctuations, correlate with current standard PD outcome measures, and respond to dopaminergic therapy. DESIGN, SETTING, AND PARTICIPANTS: This observational study assessed individuals with PD who remotely completed 5 tasks (voice, finger tapping, gait, balance, and reaction time) on the smartphone application. We used a novel machine-learning-based approach to generate a mobile Parkinson disease score (mPDS) that objectively weighs features derived from each smartphone activity (eg, stride length from the gait activity) and is scaled from 0 to 100 (where higher scores indicate greater severity). Individuals with and without PD additionally completed standard in-person assessments of PD with smartphone assessments during a period of 6 months. MAIN OUTCOMES AND MEASURES: Ability of the mPDS to detect intraday symptom fluctuations, the correlation between the mPDS and standard measures, and the ability of the mPDS to respond to

dopaminergic medication. **RESULTS:** The mPDS was derived from 6148 smartphone activity assessments from 129 individuals (mean [SD] age, 58.7 [8.6] years; 56 [43.4%] women). Gait features contributed most to the total mPDS (33.4%). In addition, 23 individuals with PD (mean [SD] age, 64.6 [11.5] years; 11 [48%] women) and 17 without PD (mean [SD] age 54.2 [16.5] years; 12 [71%] women) completed in-clinic assessments. The mPDS detected symptom fluctuations with a mean (SD) intraday change of 13.9 (10.3) points on a scale of 0 to 100. The measure correlated well with the Movement Disorder Society Unified Parkinson Disease's Rating Scale total ($r = 0.81$; $P < .001$) and part III only ($r = 0.88$; $P < .001$), the Timed Up and Go assessment ($r = 0.72$; $P = .002$), and the Hoehn and Yahr stage ($r = 0.91$; $P < .001$). The mPDS improved by a mean (SD) of 16.3 (5.6) points in response to dopaminergic therapy. **CONCLUSIONS AND RELEVANCE:** Using a novel machine-learning approach, we created and demonstrated construct validity of an objective PD severity score derived from smartphone assessments. This score complements standard PD measures by providing frequent, objective, real-world assessments that could enhance clinical care and evaluation of novel therapeutics. © 2018 American Medical Association. All rights reserved.

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Using statistical measures for feature ranking

Item Type Journal Article

Author E.G. Mansoori

Abstract Feature ranking is a fundamental preprocess for feature selection, before performing any data mining task. Essentially, when there are too many features in the problem, dimensionality reduction through discarding weak features is highly desirable. In this paper, we have developed an efficient feature ranking algorithm for selecting the more relevant features prior to derivation of classification predictors. Regardless the ranking criteria which rely on the training error of a predictor based on a feature, our approach is distance-based, employing only the statistical distribution of classes in each feature. It uses a scoring function as ranking criterion to evaluate the correlation measure between each feature and the classes. This function comprises three measures for each class: the statistical between-class distance, the interclass overlapping measure, and an estimate of class impurity. In order to compute the statistical parameters, used in these measures, a normalized form of histogram, obtained for each class, is employed as its a priori probability density. Since the proposed algorithm examines each feature individually, it provides a fast and cost-effective method for feature ranking. We have tested the effectiveness of our approach on some benchmark data sets with high dimensions. For this purpose, some top-ranked features are selected and are used in some rule-based classifiers as the target data mining task. Comparing with some popular feature ranking methods, the experimental results show that our approach has better performance as it can identify the more relevant features eventuate to lower classification error. © 2013 World Scientific Publishing Company.

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Export Date: 20 September 2026; Cited By: 19; Correspondence Address: E.G. Mansoori; School of Electrical and Computer Engineering, Shiraz University, Shiraz, Iran; email: mansoori@shirazu.ac.ir; CODEN: IJPIE

Using topological constraints as context for the joint classification of image regions in a traffic environment

Item Type Conference Paper

Author G. Lorenz

Abstract This paper presents an approach using context knowledge in form of topological constraints to improve the classification result for an automated scene analysis. The approach consists of an initial segmentation, in which the image is divided into a set of disjoint regions based on their respective color values and a subsequent joint classification, where the generated image regions are assigned to object classes. The classification is performed according to extracted feature measurements and context knowledge about the spatial relationships between the different object classes. The classification task is formulated as an optimization problem using a maximum a posteriori estimation rule. The classification criteria are combined using the Bayesian theorem, where feature measurements and context are coded as conditional probability density and a priori probability, respectively. A Markov random field model is used to get an analytical expression for the symbolic context knowledge. Through its associated Gibbs distribution a systematic way for designing the appropriate functional form of context is found. The optimization task is solved by applying an evolutionary algorithm. Due to a problem-related formulation of the necessary operators this has been proven to be a very efficient search strategy. The performance of the presented approach is demonstrated on synthetic and real world images, showing typical scenarios of road traffic on motorways.

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Using Uncertainty Quantification to Characterize and Improve Out-of-Domain Learning for PDEs

Item Type Conference Paper

Author S.C. Mouli

Author D.C. Maddix

Author S. Alizadeh

Author G. Gupta

Author A. Stuart

Author M.W. Mahoney

Author Y. Wang

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Editor Kolter Z.

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Abstract Existing work in scientific machine learning (SciML) has shown that data-driven learning of solution operators can provide a fast approximate alternative to classical numerical partial differential equation (PDE) solvers. Of these, Neural Operators (NOs) have emerged as particularly promising. We observe that several uncertainty quantification (UQ) methods for NOs fail for test inputs that are even moderately out-of-domain (OOD), even when the model approximates the

solution well for in-domain tasks. To address this limitation, we show that ensembling several NOs can identify high-error regions and provide good uncertainty estimates that are well-correlated with prediction errors. Based on this, we propose a cost-effective alternative, DIVERSENO, that mimics the properties of the ensemble by encouraging diverse predictions from its multiple heads in the last feed-forward layer. We then introduce OPERATOR-PROBCONSERV, a method that uses these well-calibrated UQ estimates within the PROBCONSERV framework to update the model. Our empirical results show that OPERATOR-PROBCONSERV enhances OOD model performance for a variety of challenging PDE problems and satisfies physical constraints such as conservation laws. Copyright 2024 by the author(s)

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Using unsupervised machine learning to quantify physical activity from accelerometry in a diverse and rapidly changing population

Item Type Journal Article

Author C.B. Thornton

Author N. Kolehmainen

Author K. Nazarpour

Abstract Accelerometers are widely used to measure physical activity behaviour, including in children. The traditional method for processing acceleration data uses cut points to define physical activity intensity, relying on calibration studies that relate the magnitude of acceleration to energy expenditure. However, these relationships do not generalise across diverse populations and hence they must be parametrised for each subpopulation (e.g., age groups) which is costly and makes studies across diverse populations and over time difficult. A data-driven approach that allows physical activity intensity states to emerge from the data, without relying on parameters derived from external populations, offers a new perspective on this problem and potentially improved results. We applied an unsupervised machine learning approach, namely a hidden semi-Markov model, to segment and cluster the raw accelerometer data recorded (using a waist-worn ActiGraph GT3X+) from 279 children (9–38 months old) with a diverse range of developmental abilities (measured using the Paediatric Evaluation of Disability Inventory–Computer Adaptive Testing measure). We benchmarked this analysis with the cut points approach, calculated using thresholds from the literature which had been validated using the same device and for a population which most closely matched ours. Time spent active as measured by this unsupervised approach correlated more strongly with PEDI-CAT measures of the child's mobility (R^2 : 0.51 vs 0.39), social-cognitive capacity (R^2 : 0.32 vs 0.20), responsibility (R^2 : 0.21 vs 0.13), daily activity (R^2 : 0.35 vs 0.24), and age (R^2 : 0.15 vs 0.1) than that measured using the cut points approach. Unsupervised machine learning offers the potential to provide a more sensitive, appropriate, and cost-effective approach to quantifying physical activity behaviour in diverse populations, compared to the current cut points approach. This, in turn, supports research that is more inclusive of diverse or rapidly changing populations. Copyright: © 2023 Thornton et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Using Whole Knee Cartilage Damage Index to Predict Knee Osteoarthritis: A Two-year Longitudinal Study

Item Type Conference Paper

Author Y. Du

Author J. Shan

Author R. Almajalid

Author T. Alon

Author M. Zhang

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Editor Callejas Z.

Editor Hu X.

Editor Dickerson J.

Editor Zhang L.

Abstract Knee osteoarthritis (OA) affects 10% of the population over 55 years old, and is a major cause of work absence, early retirement, and joint replacement. The purpose of this study is to explore the possibility of using a recently proposed knee osteoarthritis biomarker and machine learning method to predict OA progression. The biomarker, named cartilage damage index (CDI), was extracted from 3D knee MR images by measuring representative locations. In total, the CDI measured 60 locations on the whole knee cartilage, including 18 from femur, 18 from tibia, and 24 from patella. The CDI values at these 60 locations in the baseline year were used as feature input to the artificial neural network (ANN). We trained two types of model to predict OA and OA severity change, respectively. The label of each sample was 'OA' or 'non-OA', based on the severity level in two years when training the model for OA prediction; and the label of each sample was 'change' or 'no-change' based on the severity level change in two years, when training the model for OA change prediction. Separate ANN models were trained for three OA severity measures, i.e., lateral Joint Space Narrow (JSN), medial JSN and Kellgren and Lawrence (KL) grade. Besides using all the 60 informative locations, we tested different combinations of CDI locations at sub-regions. The best prediction result achieved in this work was AUC (area under ROC curve) = 0.912, using the whole knee CDI (all 60 locations) to predict lateral JSN. Experiment results showed that CDI can be used

as a reliable cartilage quantification method and help to predict OA progression in two years. © 2018 IEEE.

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Utilization of deep learning to quantify fluid volume of neovascular age-related macular degeneration patients based on swept-source OCT imaging: The ONTARIO study

Item Type Journal Article

Author S.K. Sodhi

Author A. Pereira

Author J.D. Oakley

Author J. Golding

Author C. Trimboli

Author D.B. Russakoff

Author N. Choudhry

Abstract Purpose To evaluate the predictive ability of a deep learning-based algorithm to determine long-term best-corrected distance visual acuity (BCVA) outcomes in neovascular age-related macular degeneration (nARMD) patients using baseline swept-source optical coherence tomography (SS-OCT) and OCT-angiography (OCT-A) data. Methods In this phase IV, retrospective, proof of concept, single center study, SS-OCT data from 17 previously treated nARMD eyes was used to assess retinal layer thicknesses, as well as quantify intraretinal fluid (IRF), subretinal fluid (SRF), and serous pigment epithelium detachments (PEDs) using a novel deep learning-based, macular fluid segmentation algorithm. Baseline OCT and OCT-A morphological features and fluid measurements were correlated using the Pearson correlation coefficient (PCC) to changes in BCVA from baseline to week 52. Results Total retinal fluid (IRF, SRF and PED) volume at baseline had the strongest correlation to improvement in BCVA at month 12 (PCC = 0.652, $p = 0.005$). Fluid was subsequently sub-categorized into IRF, SRF and PED, with PED volume having the next highest correlation (PCC = 0.648, $p = 0.005$) to BCVA improvement. Average total retinal thickness in isolation demonstrated poor correlation (PCC = 0.334, $p = 0.189$). When two features, mean choroidal neovascular membranes (CNVM) size and total fluid volume, were combined and correlated with visual outcomes, the highest correlation increased to PCC = 0.695 ($p = 0.002$). Conclusions In isolation, total fluid volume most closely correlates with change in BCVA values between baseline and week 52. In combination with complimentary information from OCT-A, an improvement in the linear correlation score was observed. Average total retinal thickness provided a lower correlation, and thus provides a lower predictive outcome than alternative metrics assessed. Clinically, a machine-learning approach to analyzing fluid metrics in combination with lesion size may provide an advantage in personalizing therapy and predicting BCVA outcomes at week 52. Copyright: © 2022 Sodhi et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Utilizing 3D fast spin echo anatomical imaging to reduce the number of contrast preparations in T1ρ quantification of knee cartilage using learning-based methods

Item Type	Journal Article
Author	J. Zhong
Author	C. Huang
Author	Z. Yu
Author	F. Xiao
Author	T. Blu
Author	S. Li
Author	T.-Y.M. Ong
Author	K.-W.K. Ho
Author	Q. Chan
Author	J.F. Griffith
Author	W. Chen
Abstract	Purpose: To propose and evaluate an accelerated (Formula presented.) quantification method that combines (Formula presented.) -weighted fast spin echo (FSE) images and proton density (PD)-weighted anatomical FSE images, leveraging deep learning

models for (Formula presented.) mapping. The goal is to reduce scan time and facilitate integration into routine clinical workflows for osteoarthritis (OA) assessment. Methods: This retrospective study utilized MRI data from 40 participants (30 OA patients and 10 healthy volunteers). A volume of PD-weighted anatomical FSE images and a volume of (Formula presented.) -weighted images acquired at a non-zero spin-lock time were used as input to train deep learning models, including a 2D U-Net and a multi-layer perceptron (MLP). (Formula presented.) maps generated by these models were compared with ground truth maps derived from a traditional non-linear least squares (NLLS) fitting method using four (Formula presented.) -weighted images. Evaluation metrics included mean absolute error (MAE), mean absolute percentage error (MAPE), regional error (RE), and regional percentage error (RPE). Results: The best-performed deep learning models achieved RPEs below 5% across all evaluated scenarios. This performance was consistent even in reduced acquisition settings that included only one PD-weighted image and one (Formula presented.) -weighted image, where NLLS methods cannot be applied. Furthermore, the results were comparable to those obtained with NLLS when longer acquisitions with four (Formula presented.) -weighted images were used. Conclusion: The proposed approach enables efficient (Formula presented.) mapping using PD-weighted anatomical images, reducing scan time while maintaining clinical standards. This method has the potential to facilitate the integration of quantitative MRI techniques into routine clinical practice, benefiting OA diagnosis and monitoring. © 2025 The Author(s). Magnetic Resonance in Medicine published by Wiley Periodicals LLC on behalf of International Society for Magnetic Resonance in Medicine.

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Utilizing an Ultrasonic Inspection System Operating Inside an Autoclave and Machine Learning to Quantify Porosity within Composites During Cure

Item Type Journal Article

Author T.B. Hudson

Author G.R. Chung

Author J.J. Pinakidis

Author P.J. Follis

Author T. Sreekantamurthy

Author F.L. Palmieri

Abstract Composite materials are increasingly being utilized in aerospace applications for their high stiffness and strength to weight ratios and fatigue resistance. However, defects in the composite may arise during cure (e.g. porosity, delamination, fiber waviness), and current technology only allows for post-cure evaluation (e.g. microscopy, ultrasonic inspection). A high-temperature ultrasonic scanning system was developed for deployment in an autoclave, which can detect porosity in composites during the cure process. This study focused on the implementation of machine learning techniques to help generate a model that can quantify porosity, in addition to detection and

localization that has previously been demonstrated. Two, six-hour-long experiments were conducted on curing of 762 mm × 305 mm (30 in. × 12 in.) composite panels with a [0/45/90/-45]_{4s} layup and varying regions of high and low pressure due to its tapered geometry in contact with a flat caul plate. The first experiment utilized a thick (12.7 mm) caul plate and the second utilized a thin (3.2 mm) caul plate. During experimentation, within the scan area (406 mm × 13 mm), data was recorded and stored for ultrasonic amplitude. Additional variables were measured or predicted including temperature, autoclave pressure, number of plies, slope of the composite panel surface with respect to the transducer, viscosity, and glass transition temperature. The pre-processed data was entered into the Regression Learner Application in MATLAB®,¹ and a rational quadratic Gaussian process regression (GPR) was chosen for the machine learning algorithm. The model was then trained on a larger data set to make it more robust and capable of predictions using a function callout. The result was a machine learning algorithm that can reliably quantify porosity in a composite panel during cure based on measured amplitude response and generate images for intuitive visualization. This tool can be further trained with more experimentation and potentially employed for real-time porosity detection and quantification of composite components during cure in an autoclave. Practical use of this technology is the potential to dynamically control processing parameters (e.g. autoclave pressure) in real-time to reduce the level of porosity within the laminate to acceptable limits (e.g. 2% by volume). © 2023 American Society for Nondestructive Testing.

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Validated reverse transcription droplet digital PCR serves as a higher order method for absolute quantification of Potato virus Y strains

Item Type Journal Article

Author N. Mehle

Author D. Dobnik

Author M. Ravnikar

Author M. Pompe Novak

Abstract RNA viruses have a great potential for high genetic variability and rapid evolution that is generated by mutation and recombination under selection pressure. This is also the case of Potato virus Y (PVY), which comprises a high diversity of different recombinant and non-recombinant strains. Consequently, it is hard to develop reverse transcription real-time quantitative PCR (RT-qPCR) with the same amplification efficiencies for all PVY strains which would enable their equilibrate quantification; this is specially needed in mixed infections and other studies of pathogenesis. To achieve this, we initially transferred the PVY universal RT-qPCR assay to a reverse transcription droplet digital PCR (RT-ddPCR) format. RT-ddPCR is an absolute quantification method, where a calibration curve is not needed, and it is less prone to inhibitors. The RT-ddPCR developed and validated in this study achieved a dynamic range of quantification over five orders of magnitude, and in terms of its sensitivity, it was comparable to, or even better than, RT-qPCR. RT-ddPCR showed lower measurement variability. We have shown that RT-ddPCR can be used as a reference tool for the evaluation of different RT-qPCR assays. In addition, it can be used for quantification of RNA based on in-house reference materials that can then be used as calibrators in diagnostic laboratories. © 2018, Springer-Verlag GmbH Germany, part of Springer Nature.

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Validation of a deep learning-based image analysis system to diagnose subclinical endometritis in dairy cows

Item Type Journal Article

Author H. Sadeghi

Author H.-S. Braun

Author B. Pantì

Author G. Opsomer

Author O.B. Pascottini

Abstract The assessment of polymorphonuclear leukocyte (PMN) proportions (%) of endometrial samples is the hallmark for subclinical endometritis (SCE) diagnosis. Yet, a non-biased, automated diagnostic method for assessing PMN% in endometrial cytology slides has not been validated so far. We aimed to validate a computer vision software based on deep machine learning to quantify the PMN% in endometrial cytology slides. Uterine cytobrush samples were collected from 116 postpartum Holstein cows. After sampling, each cytobrush was rolled onto three different slides. One slide was stained using Diff-Quick, while a second was stained using Naphthol (golden standard to stain PMN). One single observer evaluated the slides twice at different days under light microscopy. The last slide was stained with a fluorescent dye, and the PMN% were assessed twice by using a fluorescence microscope connected to a smartphone. Fluorescent images were analyzed via the Oculyze Monitoring Uterine Health (MUH) system, which uses a deep learning-based

algorithm to identify PMN. Substantial intra-method repeatabilities (via Spearman correlation) were found for Diff-Quick, Naphthol, and Oculyze MUH ($r = 0.67$ to 0.76). The intra-method agreements (via Kappa value) at $\geq 1\%$ PMN ($\kappa = 0.44$ to 0.47) were lower than at >5 ($\kappa = 0.69$ to 0.78) or $>10\%$ ($\kappa = 0.67$ to 0.85) PMN cut-offs. The inter-method repeatabilities (via Lin's correlation) were also substantial, and values between Diff-Quick and Oculyze MUH, Naphthol and Diff-Quick, and Naphthol and Oculyze MUH were 0.68 , 0.69 , and 0.77 , respectively. The agreements among evaluation methods at $\geq 1\%$ PMN were weak ($\kappa = 0.06$ to 0.28), while it increased at >5 ($\kappa = 0.48$ to 0.81) or $>10\%$ ($\kappa = 0.50$ to 0.65) PMN cut-offs. To conclude, deep learning-based algorithms in endometrial cytology are reliable and useful for simplifying and reducing the diagnosis bias of SCE in dairy cows. © 2022 Sadeghi et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Validation of an automated morphological MRI-based 123I-FP-CIT SPECT evaluation method

Item Type Journal Article

Author G. Perlaki

Author S. Szekeres

Author G. Orsi

Author L. Papp

Author B. Suha

Author S.A. Nagy

Author T. Doczi

Author J. Janszky

Author K. Zambo

Author N. Kovacs

Abstract Introduction Dopamine transporter imaging with 123I-FP-CIT single photon emission computed tomography (SPECT) is helpful for the differential diagnosis between Parkinsonian syndrome (PS) and essential tremor (ET). Although visual assessment and time-consuming manual evaluation techniques are readily available, a fully objective and automated dopamine transporter quantification technique is always preferable, at least in research and follow-up investigations. Our aim was to develop a novel automated magnetic resonance imaging (MRI)-based evaluation technique of dopamine transporter SPECT images and to compare its diagnostic accuracy with those of the gold-standard visual grading and manual dopamine transporter binding quantification methods. Methods 123I-FP-CIT SPECT and MRI sessions were conducted in 33 patients with PS (15 men; mean age: 60.3 ± 9.7 years) and 15 patients with ET (8 men; mean age: 54.7 ± 16.3 years). Striatal dopamine transporter binding was visually classified by 2 independent experts as normal or abnormal grade I, II and III. Caudal and putaminal specific uptake ratios were calculated by both automated MRI-based and manual evaluation techniques. Results We found almost perfect agreement ($\kappa = 0.829$) between the visual scores by the 2 observers. The automated method showed strong correlation with the visual and manual evaluation techniques and its diagnostic accuracy (sensitivity = 97.0%; specificity = 93.3%) was also comparable to these methods. The automatically determined uptake parameters showed negative correlation with the clinical severity of parkinsonism. Based on ordinal regression modelling, the automated MRI-based method could reliably determine the visual grading scores. Conclusion The novel MRI-based evaluation of 123I-FP-CIT SPECT images is useful for the differentiation of PS from ET. © 2016 Elsevier Ltd

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Validation of inertial sensing-based wearable device for tremor and bradykinesia quantification

Item Type Journal Article
Author H. Dai

Author G. Cai

Author Z. Lin

Author Z. Wang

Author Q. Ye

Abstract Neurologists judge the severity of Parkinsonian motor symptoms according to clinical scales, and their judgments exist inconsistent because of differences in clinical experience. Correspondingly, inertial sensing-based wearable devices (ISWDs) produce objective and standardized quantifications. However, ISWDs indirectly quantify symptoms by parametric modeling of angular velocities and linear accelerations and trained by the judgments of several neurologists through supervised learning algorithms. Hence, the ISWD outputs are biased along with the scores provided by neurologists. To investigate the effectiveness ISWDs for Parkinsonian symptoms quantification, technical verification and clinical validation of both tremor and bradykinesia quantification methods were carried out. A total of 45 Parkinson's disease patients and 30 healthy controls performed the tremor and finger-tapping tasks, which were tracked simultaneously by an ISWD and a 6-axis high-precision electromagnetic tracking system (EMTS). The Unified Parkinson's Disease Rating Scale (UPDRS) prescribed parameters obtained from the EMTS, which directly provides linear and rotational displacements, were compared with the scores provided by both the ISWD and seven neurologists. EMTS-based parameters were regarded as the ground truth and were employed to train several common machine learning (ML) algorithms, i.e., support vector machine (SVM), k-nearest neighbors (KNN), and random forest (RF) algorithms. Inconsistency among the scores provided by the neurologists was proven. Besides, the quantification performance (sensitivity, specificity, and accuracy) of the ISWD employed with ML algorithms were better than that of the neurologists. Furthermore, EMTS can be utilized to both modify the quantification algorithms of ISWDs and improve the assessment skills of young neurologists. © 2013 IEEE.

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Validation of Information Products of Airborne Hyperspectral Imagery Processing

Item Type Journal Article

Author V.V. Kozoderov

Author T.V. Kondranin

Author E.V. Dmitriev

Author V.P. Kamentsev

Abstract Abstract: In regards to the problem of airborne hyperspectral imagery processing, most attention is paid to the construction of an automated classifier which implements computational procedures of recognizing forest vegetation of different species composition and age. The validation of the information products obtained as a result of processing the data of hyperspectral remote sensing is based on finding a maximum of a posteriori probability for selected classes of objects using the contemporary views of Markov random fields allowing a priori probabilities for each element of hyperspectral apparatus resolution to be calculated. The necessity of taking into account the noise component of the apparatus to enhance the accuracy of

solving the applied problems under study is shown. Examples of comparing the data of hyperspectral imagery processing with the data of ground-based forest inventory for a selected territory are demonstrated. © 2019, Pleiades Publishing, Ltd.

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Varaps: a python package for estimating SARS-CoV-2 lineages proportions from pooled sequencing data (ANRS0160)

Item Type Journal Article

Author E.H. Djaout

Author N. Cluzel

Author V. Marechal

Author G. Nuel

Author M. Courbariaux

Abstract Background : Wastewater-based epidemiology has been investigated as a very effective way of monitoring SARS-CoV-2 variants. This can be achieved through accurate lineage deconvolution of wastewater sequencing data. Variants Ratios from Pooled Sequencing (VaRaPS) is a Python package designed for this purpose, utilizing pooled sequencing data and lineage mutation profiles to estimate their proportions. Results : VaRaPS re-implements core algorithms from the literature, achieving significant improvements in computational speed and efficiency. Comparative analyzes with simulated and synthetic data sets demonstrate its superior performance in lineage prevalence estimation, underscored by its user-oriented design for broader accessibility. Conclusions : By improving speed and accuracy in SARS-CoV-2 variant analysis, VaRaPS offers valuable insights into viral evolution, supporting ongoing surveillance efforts in the post-pandemic landscape. © The Author(s) 2025.

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Variations in Vegetation Cooling Efficiency Across 20 Major Cities Worldwide: The Interplay of Built-Up Fraction and Surface Properties

Item Type Journal Article

Author L. Hu

Author H. Neidhardt

Author A. Braun

Author V. Hochschild

Abstract Urban vegetation is widely regarded as a key strategy for mitigating urban heat, but its cooling performance is not constant and varies across climatic zones and urban structures. Most existing evidence derives from single-city or single-climate studies, leaving the differences in vegetation cooling with respect to building density and background climate insufficiently quantified globally. This study examined vegetation cooling efficiency (CE), defined as the absolute slope of the relationship between NDVI and normalized land surface temperature (LST), across 20 major cities spanning tropical, arid, temperate, and continental climate zones during 2000–2020. We combined city-level regression, built-up fraction stratification, and interpretable machine learning to quantify vegetation CE and its variation across climate and urban density gradients. CE varied roughly eightfold across cities (≈ 0.07 – 0.60), with the strongest responses in arid and continental cities such as Dubai and Almaty. Increasing built-up fraction systematically weakened the NDVI–LST relationship, turning near-neutral or slightly positive in the most compact temperate cores (80%–100% built-up). The machine learning model reproduced these patterns (out-of-sample $R^2 = 0.757$), identifying NDVI and evapotranspiration as dominant drivers. These findings indicate that vegetation cooling is strongly context-dependent, underscoring the need for climate-specific and morphology-based perspectives on urban greening rather than generalized evaluations. © 2026 by the authors.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: L. Hu; Geoinformatics Group, Department of Geosciences, University of Tübingen, Tübingen, Rümelinstraße 19-23, 72072, Germany; email: ling.hu@student.uni-tuebingen.de

Variogram Roughness Method for Casting Surface Characterization

Item Type Journal Article

Author D.W. Schimpf

Author F.E. Peters

Abstract Casting surface specifications are set based on aesthetics, functionality, or a combination of both. To classify casting surfaces, visual inspections are performed by an operator who compares the casting surface to pictures or comparator plates (e.g., metal, plastic) that represent a certain roughness level. This inspection process is highly subjective, and disagreements arise on the acceptance of a casting between the casting producer and buyer. To minimize these disagreements and use developments in 3D scanning, this study aims to develop a digital surface characterization method. The method developed and implemented in this study utilizes underlying geometry estimation, abnormality detection, and a new roughness characterization formula based on a variogram to determine a surface roughness value. Tests were done to compare the new roughness characterization formula with existing quantification methods (i.e., Sa, Sq) and to compare the results of the method with human operators. The tests indicated that the variogram roughness was able to differentiate between the roughness levels of the current surface roughness standards GAR-C9 and SCRATA. In addition, the results are repeatable as well as reproducible and agree with operator judgment based on a ranking comparison between the operator and the digital method. Overall, the digital surface roughness method has the potential to improve the communication between casting suppliers and designers and make the surface roughness classification more reliable and repeatable. © 2020, American Foundry Society.

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Vehicle classification and lane categorization

Item Type Conference Paper

Author S. Ramesh

Author V. Upadhyaya

Abstract This research was carried out in order to use image processing techniques to detect, track, classify, and count vehicles in a real-time traffic video. Additionally this research also focuses on occlusion detection and handling, which involves detecting and handling the states when one vehicle superimposes the other or when two superimposed vehicles split, and lane categorization i.e. dividing the vehicles in the moving traffic scene into their corresponding lanes of the road in which they are travelling in. The challenge taken up was to achieve all this using basic image processing techniques and algorithms and not compromise on the desired output, most importantly keeping the single frame processing time less and the speed of processed video playback to be comparable to that real-time video playback speeds. It is to note that all of the above is achieved through the techniques demonstrated in this research. The process cycle followed from frame extraction to the final classification of vehicles involves algorithms that are all highly region based and do the necessary processing only on the region of interest, only as and when required. Thus saving processing power and time. The final results obtained could easily be molded to use in any application in surveillance, traffic monitoring etc. by just adding the required decision-making statements in it. © 2016 IEEE.

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Notes:

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Vehicle Type Classification and Counting Using YOLOv4 Algorithm

Item Type Conference Paper

Author S.I.I.C. Imperial

Author A.L.L. Lucas

Author M.V. Caya

Abstract The study implements a system that detects, classify and count vehicles based on their body type. Classifying and counting has proven to be beneficial when monitoring and managing traffics. However, there are few of studies that focuses on classifying and counting vehicles based on their car types. Implementing the YOLOv4 for classification and counting for the car types coupe, pickup, sedan, sports utility vehicle (SUV) and van, obtained an accuracy of 92.13% for

classification and 89.14% for counting. The system was able to successfully classify and count vehicles based on their car type under one system compared to other system that only counts vehicles without classifying the car types. © 2022 IEEE.

Date 2022

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Language English

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Vehicles, Vehicles detection, YOLOv4, vehicles, Vehicle counting, vehicle counting, vehicle detection, Counting Vehicles, Raspberry pi, Raspberry Pi, Sport utility vehicles, Type classifications, vehicle type, Vehicle types

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Verifiably Robust Conformal Prediction

Item Type Conference Paper

Author L. Jeary

Author T. Kuipers

Author M. Hosseini

Author N. Paoletti

Editor Globerson A.

Editor Mackey L.

Editor Belgrave D.

Editor Fan A.

Editor Paquet U.

Editor Tomczak J.

Editor Zhang C.

Abstract Conformal Prediction (CP) is a popular uncertainty quantification method that provides distribution-free, statistically valid prediction sets, assuming that training and test data are exchangeable. In such a case, CP's prediction sets are guaranteed to cover the (unknown) true test output with a user-specified probability. Nevertheless, this guarantee is violated when the data is subjected to adversarial attacks, which often result in a significant loss of coverage. Recently, several approaches have been put forward to recover CP guarantees in this setting. These approaches leverage variations of randomised smoothing to produce conservative sets which account for the effect of the adversarial perturbations. They are, however, limited in that they only support ℓ_2 -bounded perturbations and classification tasks. This paper introduces VRCP (Verifiably Robust Conformal Prediction), a new framework that leverages recent neural network verification methods to recover coverage guarantees under adversarial attacks. Our VRCP method is the first to support perturbations bounded by arbitrary norms including ℓ_1 , ℓ_2 , and ℓ_∞ , as well as regression tasks. We evaluate and compare our approach on image classification tasks (CIFAR10, CIFAR100, and TinyImageNet) and regression tasks for deep reinforcement learning environments. In every case, VRCP achieves above nominal coverage and yields significantly more efficient and informative prediction regions than the SotA. © 2024 Neural information processing systems foundation. All rights reserved.

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Language English

Extra Journal Abbreviation: Adv. neural inf. proces. syst.

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Verifiably robust conformal prediction for probabilistic guarantees under adversarial attacks

Item Type Journal Article

Author L. Jeary

Author T. Kuipers

Author M. Hosseini

Author N. Paoletti

Abstract Conformal Prediction (CP) is a popular uncertainty quantification method that provides distribution-free, statistically valid prediction sets, assuming that training and test data are exchangeable. In such a case, CP's prediction sets are guaranteed to cover the (unknown) true test output with a user-specified probability. Nevertheless, this guarantee is violated when the data is subjected to adversarial attacks, which often result in a significant loss of coverage. Recently, several approaches have been put forward to recover CP guarantees in this setting. These approaches leverage variations of randomised smoothing to produce conservative sets which account for the effect of the adversarial perturbations. They are, however, limited in that they only support ℓ_2 -bounded perturbations and classification tasks. This paper introduces VRCP (Verifiably Robust Conformal Prediction), a new framework that leverages recent neural network verification methods to recover coverage guarantees under adversarial attacks. We also demonstrate how VRCP can be used to mitigate poisoning attacks. Our VRCP method is the first to support perturbations bounded by arbitrary norms including ℓ_1 , ℓ_2 , and ℓ_∞ , as well as regression tasks. We evaluate and compare our approach on image classification tasks (CIFAR10, CIFAR100, and TinyImageNet) and regression tasks for deep reinforcement learning environments. In every case, VRCP achieves above nominal coverage and yields significantly more efficient and informative prediction regions than the SotA. © 2025 The Authors

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: M. Hosseini; Department of Informatics, King's College London, London, Strand, United Kingdom; email: mehran.hosseini@manchester.ac.uk; CODEN: PTNRA

Vertical and horizontal approaches to the making of racial statistics in Britain

Item Type Journal Article

Author L. Brown

Abstract ABSTRACT: The release of data from the 2011 census fuelled a media storm over Britain's ethnic and immigrant composition, while at the same time a much less visible series of public debates developed over the scope, organization and purpose of government information-gathering centred on how the existing census could be replaced with 'Big data'. It is therefore particularly timely to explore the political choices, ontological shifts and statistical challenges shaping the array of enumeration projects that have developed in contemporary Britain to identify, classify, and count immigrants and their descendants. This article analyses the relationships between holistic (horizontal) and single-purpose (vertical) approaches to racial statistics, and how these were affected by the standardizations of classification in 1962 and 1991. A range of state archives and contemporary accounts are used to examine the material practices and organizational tensions that fuelled the divergence, transfer and interaction of these attempts at racial legibility. © 2016 Informa UK Limited, trading as Taylor & Francis Group.

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Vertically transmitted bacterivorous nematodes are consistent nest inhabitants in the Azteca-Cecropia ant-plant mutualism

Item Type Journal Article

Author V. Barrajon-Santos

Author M. Nepel

Author W. Sudhaus

Author B. Hausmann

Author D. Woebken

Author V.E. Mayer

Abstract Nematodes play key roles in natural and agricultural ecosystems. They contribute to organic matter transformation and the stability of soil food webs. Beyond their free-living forms, many nematode lineages have evolved in close associations with insects, ranging from mutualistic and commensal to parasitic interactions. Recent studies have revealed that nematodes are common in tropical ant-plant mutualisms and are particularly relevant in ant-made organic matter piles, or “patches”, within ant nests. To investigate nematode community dynamics during ant colony growth and their consistency across closely related ant species, we analysed patches from 65 ant colonies of the Azteca-Cecropia ant-plant mutualism using 18S rRNA metabarcoding combined with morphology-based quantification methods. Bacterivorous nematodes from the order Rhabditida were present in all samples, regardless of the ant or plant species and the colony developmental stage. Members of Tylenchida and Dorylaimida were also detected, though sporadically. Our results support the previously proposed vertical transmission of bacterivorous nematodes from mother to daughter colonies as well as horizontal transmission among patches within the same ant colony. Moreover, nematode community composition remained stable throughout colony development but varied between ant species. These findings demonstrate that nematodes constitute a persistent and functionally important component of this ant-plant mutualism. © The Author(s) 2026.

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Video frames text detection through Bayesian classification and boundary growing method

Item Type Conference Paper

Author A. Nancy

Author D. Jayapriya

Abstract Text detection in natural scene images is important for content-based image analysis. It consists of different component analysis for text detection video frames. Multioriented text detection in video frames is one of the text detection in videos frames. Multi-oriented text detection is not easy as detection of captions or graphics which is usually appears in horizontal direction and has high contrast compared to its background. Multi-oriented text detection generally refers to scene text that makes text detection more challenging. Therefore conventional text detection may not give good results for multi-oriented video frames text detection. Hence, in this paper, we present a new enhancement method that includes the product of Laplacian and Sobel operations to enhance text pixels in videos. To classify true text pixels, we propose a Bayesian classifier without assuming a priori probability about the input frame but estimating it based on three probable matrices. Three different ways of clustering are performed on the output of the enhancement method to obtain the three probable matrices. Text candidates are obtained by intersecting the output of the Bayesian classifier with the Canny edge map of the input frame. The robustness of the methods are analyzed our own datasets under different video frames. Finally, it coarse-to-fine detection locates text regions efficiently. © 2014 IEEE.

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Tags:

Image segmentation, Embedded systems, Text processing, Video scene, Pixels, Bayesian classifier, Laplace transforms, Laplacians, boundary growing, Boundary growing, Laplacian-Sobel product (LSP), maximum gradient difference, Maximum gradient differences, multioriented video scene text detection, text candidate detection

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Visible and near-infrared spectroscopy coupled with machine learning to quantify oxalate-extractable Al and Fe and identify andic soil properties for Japanese forest soils

Item Type Journal Article

Author S. Ishizuka

Author A. Imaiya

Author Y. Sakai

Author N. Yamashita

Abstract The combined content of oxalate-extractable aluminum (Alo) and 1/2 oxalate-extractable iron (Feo), expressed as $Alo + 1/2Feo$, is a key factor for identifying andic soil properties. Specifically, a value of ≥ 20 g kg⁻¹ for this variable is needed for andic soil properties in soils with less glass content. In this study, we thus developed a model to estimate the Alo and Feo contents in soils using visible and near-infrared spectroscopy coupled with machine learning. We evaluated the performance of three algorithms, namely partial least square regression, random forest (RF), and convolutional neural network (CNN). The CNN model outperformed the other models in estimating the Alo ($R^2 = 0.921$) and Feo ($R^2 = 0.723$) contents, with mean absolute errors of 2.3 and 2.0 g kg⁻¹ for Alo and Feo, respectively. In evaluating

whether the $\text{Al}_0 + 1/2\text{Fe}_0$ content of 2245 soil samples was $\geq 20 \text{ g kg}^{-1}$, the RF and CNN models produced only 124 and 88 misclassifications, respectively. Our method can serve as a time-efficient and chemical-free approach for identifying andic soil properties. © 2026 Japanese Society of Soil Science and Plant Nutrition.

Date 2026

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machine learning, andic soil properties, oxalate-extractable Al, oxalate-extractable Fe, Visible and near-infrared reflectance spectroscopy

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Vision-Based Structural Health Monitoring: A Survey

Item Type Conference Paper

Author H. Dabaja

Author H. Noura

Author M. Ouladsine

Editor Aitouche A.

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Abstract The field of Vision-based Structural Health Monitoring (SHM) utilizes computer vision (CV), imaging techniques, and machine learning to quantify the integrity of structures. This survey sheds light on this field presenting an overview over

key techniques and advancements. In particular the survey presents classical image processing, digital image correlation (DIC), 3D modeling, and deep learning methods. Edge detection, thresholding, and segmentation can be considered as the classical damage identification tools, whereas DIC provides accurate deformation and strain measurements. Recent advancements in 3D modeling and deep learning have further boosted the efficiency and accuracy of crack detection, structural evaluation, and monitoring. Advanced deep learning methods such as convolutional neural networks (CNNs) and transfer learning (TL) have proven their effectiveness in automating crack detection and segmentation tasks under challenging and noisy conditions. This survey outlines strengths, weaknesses, and applicability to SHM with a view to transforming SHM into a non-contact, inexpensive, and fully automated monitoring system. © 2025 IEEE.

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Notes:

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Visualization and quantification of coral reef soundscapes using
CoralSoundExplorer software

Item Type Journal Article**Author** L. Minier**Author** J. Rouch**Author** B. Sabbagh**Author** F. Bertucci**Author** E. Parmentier**Author** D. Lecchini**Author** F. Sèbe**Author** N. Mathevon**Author** R. Emonet

Abstract Despite hosting some of the highest concentrations of biodiversity and providing invaluable goods and services in the oceans, coral reefs are under threat from global change and other local human impacts. Changes in living ecosystems often induce changes in their acoustic characteristics, but despite recent efforts in passive acoustic monitoring of coral reefs, rapid measurement and identification of changes in their soundscapes remains a challenge. Here we present the new open-source software CoralSoundExplorer, which is designed to study and monitor coral reef soundscapes. CoralSoundExplorer uses machine learning approaches and is designed to eliminate the need to extract conventional acoustic indices. To demonstrate CoralSoundExplorer's functionalities, we use and analyze a set of recordings from three coral reef sites, each with different purposes (undisturbed site, tourist site and boat site), located on the island of Bora-Bora in French Polynesia. We explain the CoralSoundExplorer analysis workflow, from raw sounds to ecological results, detailing and justifying each processing step. We detail the software settings, the graphical representations used for visual exploration of soundscapes and their temporal dynamics, along with the analysis methods and metrics proposed. We demonstrate that CoralSoundExplorer is a powerful tool for identifying disturbances affecting coral reef soundscapes, combining visualizations of the spatio-temporal distribution of sound recordings with new quantification methods to characterize soundscapes at different temporal scales. © 2025 Minier et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Date 2025**DOI** 10.1371/journal.pcbi.1012050**URL** <https://www.scopus.com/pages/publications/105002480795?origin=resultslist>**Archive** Scopus**Language** English**Volume** 21**Publisher** Public Library of Science**Publication** PLoS Computational Biology**ISSN** 1553734X (ISSN)**Issue** 4 April**Journal Abbr** PLoS Comput. Biol.**Date Added** 20/09/2026, 9:51:45 pm**Modified** 20/09/2026, 9:51:45 pm**Tags:**

Article, environmental factor, Ecosystems, tourism, machine learning, Environmental Monitoring, spatiotemporal analysis, Machine Learning, clustering algorithm, ecosystem, Ecosystem, human impact (environment), nonhuman, procedures, controlled study, workflow, Software, Acoustics, convolutional neural network, benchmarking, unsupervised machine learning, computer model, biodiversity, software, dynamics, Macroinvertebrates, bioinformatics, Computational Biology, audio recording, temporal analysis, Abiotic, environmental monitoring, physiology, animal, Animals, Fourier transform, pressure, global change, acoustics, Passive acoustic monitoring, acoustic analysis, Acoustic characteristic, acoustic noise, Anthozoa, Audio recordings, coral reef, Coral reef, Coral Reefs, ecology, French Polynesia, Global change, Global warming, Good and services, heterogeneous anthropic pressure, Human impact, Open-source softwares, Polynesia, Rapid identification, Rapid measurement, Reefs, silhouette indices, sound, Sound, Sound recording, soundscape, Soundscapes

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Wafer Pattern Counting, Detection and Classification Based on Encoder-Decoder CNN Structure

Item Type Conference Paper

Author Y. Lin

Abstract This paper designs an automatic wafer pattern counting pipeline based on a convolutional neural network (CNN) based structure, also called the WPCCNN network pipeline. The study will utilize deep learning algorithms to detect, binary classify and count wafer patterns. In the sample dataset, over two hundred wafers have been scanned by industrial computed tomography, containing 11 different patterns for the dataset images. Each image includes three processed steps. Moreover, it utilizes the lightweight CNN structure to demonstrate detection, classification, and estimated counting [1, 3]. Besides, the study also uses encoder and decoder structure on the CNN algorithm to obtain the closest expected output. Compared to traditional object counting methods, such as localization and density estimation, using this new method to count objects will be more accurate, faster, and more accessible [1-3]. The experiment results indicate that our algorithm is highly accurate with the paring between the original patterns and the labeled markers. The average counting accuracy is 99.6% in a single wafer. © 2022 IEEE.

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Notes:

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Conference name: 2022 IEEE International Conference on Electro Information Technology, eIT 2022;
Conference date: 19 May 2022 through 21 May 2022; Conference code: 180722

Wafer Pattern Counting, Detection and Classification Based on Encoder-Decoder CNN Structure

Item Type Conference Paper

Author Y. Lin

Abstract This paper designs an automatic wafer pattern counting pipeline based on a convolutional neural network (CNN) based structure, also called the WPCCNN network pipeline. The study will utilize deep learning algorithms to detect, binary classify and count wafer patterns. In the sample dataset, over two hundred wafers have been scanned by industrial computed tomography, containing 11 different patterns for the dataset images. Each image includes three processed steps. Moreover, it utilizes the lightweight CNN structure to demonstrate detection, classification, and estimated counting [1], [3]. Besides, the study also uses encoder and decoder structure on the CNN algorithm to obtain the closest expected output. Compared to traditional object counting methods, such as localization and density estimation, using this new method to count objects will be more accurate, faster, and more accessible [1] -[3]. The experiment results

indicate that our algorithm is highly accurate with the paring between the original patterns and the labeled markers. The average counting accuracy is 99.6% in a single wafer. © 2022 IEEE.

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Notes:

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Wasserstein Gradient Boosting: A Framework for Distribution-Valued Supervised Learning

Item Type Conference Paper

Author T. Matsubara

Editor Globerson A.

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Editor Paquet U.

Editor Tomczak J.

Editor Zhang C.

Abstract Gradient boosting is a sequential ensemble method that fits a new weaker learner to pseudo residuals at each iteration. We propose Wasserstein gradient boosting, a novel extension of gradient boosting, which fits a new weak learner to alternative pseudo residuals that are Wasserstein gradients of loss functionals of probability distributions assigned at each input. It solves distribution-valued supervised learning, where the output values of the training dataset are probability distributions. In classification and regression, a model typically returns, for each input, a point estimate of a parameter of a noise distribution specified for a response variable, such as the class probability parameter of a categorical distribution specified for a response label. A main application of Wasserstein gradient boosting in this paper is tree-based evidential learning, which returns a distributional estimate of the response parameter for each input. We empirically demonstrate the competitive performance of the probabilistic prediction by Wasserstein gradient boosting in comparison with existing uncertainty quantification methods. © 2024 Neural information processing systems foundation. All rights reserved.

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Extra Journal Abbreviation: Adv. neural inf. proces. syst.

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Watershed segmentation for vehicle classification and counting

Item Type Journal Article

Author P. Abinaya

Author K.S. Ravichandran

Author B. Santhi

Abstract A robust video based system for the traffic surveillance system on the highway for vehicle detection, vehicle classification and counting for effective traffic analysis using only a single standard camera. The key goal of the proposed work is to successfully detect, track, classify and count the vehicle in partial occlusion and

connected together by shadow on the highways. Marker-controlled watershed segmentation method is initially used for the extraction of the foreground regions from the highway scene. For tracking Gabor filter is applied which is used to measure vehicle path in video sequences. For effective vehicle classification support vector machine is utilized. An experiment result shows the considerable performance of watershed segmentation in vehicle detection in the occluded and connected together by shadow in the highways environment.

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Publication International Journal of Engineering and Technology

ISSN 09754024 (ISSN)

Issue 2

Journal Abbr Intern. J. Eng. Technol.

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Support vector machine, Vehicle classification, Vehicle tracking, Watershed segmentation

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: P. Abinaya; School of Computing, SASTRA University, Thanjavur-613402, India; email: abirpk@gmail.com

Wave condition prediction and uncertainty quantification based on SG-MCMC and deep learning model

Item Type Journal Article

Author M. Yu

Author Z. Wang

Author W. Lu

Author D. Song

Abstract Deep learning methods have increasingly gained popularity in wave prediction; however, previous works predominantly focused on point estimates without quantifying the uncertainty of predictions. In certain scenarios, generating probabilistic forecasts with credible intervals is crucial for vessel navigation planning and marine infrastructure construction, etc. Therefore, this paper attempts to conduct a benchmark study on wave prediction uncertainty from both frequentist and Bayesian perspectives. A wave prediction model based on the ConvLSTM combined with various uncertainty quantification methods has been established, and a unified

evaluation framework has been constructed using statistical decision theory. We utilized the fifth-generation atmospheric reanalysis dataset (ERA5) from the European Centre for Medium-Range Weather Forecasts as experimental data, employing key wave parameters (significant wave height, mean wave period, and mean wave direction) for model training and testing. Through extensive repeated experiments, we found that Bayesian methods generally provide more accurate average predictions, with the SG-MCMC method performing the best, achieving RMSE values of 0.37 m and 0.49 s for HS and TM at the 24-hour prediction, respectively, and CC values of 93.65 % and 90.65 %. On the other hand, confidence intervals obtained from frequentist methods offer broader coverage for data variations, with the MIS regression method performing the best, yielding the lowest MIS scores of only 2.92 m and 1.93 s for HS and TM at the 24-hour prediction, respectively. Indeed, this result underscores a limitation of current methods: the confidence intervals do not adequately reflect the accuracy of the predictions. In contrast, the SG-MCMC method demonstrates superior performance by excelling in both mean prediction and confidence limits. © 2025 Elsevier Ltd

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Publisher Elsevier Ltd

Publication Ocean Modelling

ISSN 14635003 (ISSN)

Journal Abbr Ocean Model.

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Deep learning, machine learning, Prediction models, Marine navigation, Learning models, Weather forecasting, prediction, Bayesian analysis, model, Uncertainty quantifications, Confidence interval, Bayesian, Uncertainty quantification, uncertainty analysis, Condition prediction, MCMC method, ocean wave, SG-MCMC, Wave conditions, Wave predictions

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Weakly supervised classification of objects in images using soft random forests

Item Type Conference Paper

Author R. Lefort

Author R. Fablet

Author J.-M. Boucher

Abstract The development of robust classification model is among the important issues in computer vision. This paper deals with weakly supervised learning that generalizes the supervised and semi-supervised learning. In weakly supervised learning training data are given as the priors of each class for each sample. We first propose a weakly supervised strategy for learning soft decision trees. Besides, the introduction of class priors for training samples instead of hard class labels makes natural the formulation of an iterative learning procedure. We report experiments for UCI object recognition datasets. These experiments show that recognition performance close to the supervised learning can be expected using the propose framework. Besides, an application to semi-supervised learning, which can be regarded as a particular case of weakly supervised learning, further demonstrates the pertinence of the contribution. We further discuss the relevance of weakly supervised learning for computer vision applications. © 2010 Springer-Verlag.

Date 2010

DOI 10.1007/978-3-642-15561-1_14

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Weakly supervised deep learning model with size constraint for prostate cancer detection in multiparametric MRI and generalization to unseen domains

Item Type Conference Paper

Author R. Trombetta

Author O. Rouvière

Author C. Lartizien

Editor Burgos N.

Editor Petitjean C.

Editor Vakalopoulou M.

Editor Christodoulidis S.

Editor Coupe P.

Editor Delingette H.

Editor Lartizien C.

Editor Mateus D.

Abstract Fully supervised deep models have shown promising performance for many medical segmentation tasks. Still, the deployment of these tools in clinics is limited by the very time-consuming collection of manually expert-annotated data. Moreover, most of the state-of-the-art models have been trained and validated on moderately homogeneous datasets. It is known that deep learning methods are often greatly degraded by domain or label shifts and are yet to be built in such a way as to be robust to unseen data or label distributions. In the clinical setting, this problematic is particularly relevant as the deployment institutions may have different scanners or acquisition protocols than those from which the data has been collected to train the model. In this work, we propose to address these two challenges on the detection of clinically significant prostate cancer (csPCa) from bi-parametric MRI. We evaluate the method proposed by (Kervadec et al., 2018), which introduces a size constraint loss to produce fine semantic cancer lesions segmentations from weak circle scribbles annotations. Performance of the model is based on two public (PI-CAI and Prostate158) and one private databases. First, we show that the model achieves on-par performance with strong fully supervised baseline models, both on in-distribution validation data and unseen test images. Second, we observe a performance decrease for both fully supervised and weakly supervised models when tested on unseen data domains. This confirms the crucial need for efficient domain adaptation methods if deep learning models are aimed to be deployed in a clinical environment. Finally, we show that ensemble predictions from multiple trainings increase generalization performance. © 2024 CC-BY 4.0, R. Trombetta, O. Rouvière & C. Lartizien.

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Deep learning, Performance, Deep reinforcement learning, Federated learning, Learning models, Semantic Segmentation, Adversarial machine learning, Contrastive Learning, Generalisation, Self-supervised learning, Semi-supervised learning, Domain generalization, Weakly supervised learning, Medical segmentations, Multiparametric MRI, Prostate cancer detection, Size constraint, Urology

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Weight Entropy-Maximised Evidential Metamodel for Post Hoc Uncertainty Quantification

Item Type Conference Paper
Author G. Bala
Author D. Ganatra
Author A. Sethi
Editor Wu J.
Editor Pan J.
Editor Zhu Z.
Editor Luo L.
Editor Li Y.
Editor Xu M.
Editor Jin Y.
Editor Rueckert D.

Abstract Reliable uncertainty quantification (UQ) is crucial for deploying deep learning models in safety-critical domains such as medical imaging. Existing post hoc UQ methods either rely on multi-pass inference or suffer from limited expressiveness due to their dependence on final-layer embeddings. In this work, we propose evidential meta model, a lightweight post-hoc framework that enhances Dirichlet evidential modeling by extracting features from multiple layers of a frozen classifier. This multilayer strategy enriches the metamodel input with both low-level textures and high-level semantics, enabling more accurate modeling of aleatoric and epistemic uncertainty. To further boost epistemic fidelity, we incorporate Max-WEnt regularization, which maximizes the entropy of learnable scaling weights applied within the meta-model. This promotes internal hypothesis diversity without modifying the base network or incurring test-time overhead. Across seven benchmarks including medical datasets (BACH, DIV2K, HAM10000, BreakHis) and natural image tasks (SVHN, Fashion-MNIST, ImageNet-C) our evidential metamodel consistently improves AUROC and calibration over both the base model and prior post-hoc UQ methods. Ablation studies confirm the complementary benefits of multilayer features and Max-WEnt. Our approach offers a robust and efficient solution for trustworthy AI in clinical and other high stakes settings. © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Weight-height relationships and central obesity in 7-year-old to 10-year-old Polish urban children: A comparison of different BMI and WHtR standards

Item Type Journal Article

Author P. Tomaszewski

Author P. Zmijewski

Author K. Milde

Author E. Sienkiewicz-Dianzenza

Abstract Background: An increase in overweight and obesity rates among children is a major social problem; however, interpretation and comparability of estimations may be affected by the reference values and cut-off points used. The aim of the study was to assess the prevalence of underweight, overweight, obesity and central obesity in 7-year-old to 10-year-old urban children and to compare the results obtained through various standards of BMI or waist-to-height ratio (WHtR) indicators. Methods: The research was conducted on a sample group of 367 girls and 424 boys aged 6.5-10.5 years, randomly chosen from a number of primary schools in Warsaw, Poland. In all participants, basic somatic features were assessed, and based on BMI and WHtR values, participants were then classified according to different standards. The prevalence of underweight, overweight, obesity and central obesity in boys and girls was compared using the chi-square test; fractions obtained from various BMI and WHtR standards were compared through a test for proportions; and the conformity of classification methods was assessed using Cohen's kappa coefficient. Results: Approximately 9 % of girls and 6 % of boys were underweight, 15 % of all participants was classified as overweight, and approximately 4 % of girls and 6 % of boys aged 7-10 were obese. Central obesity was diagnosed in 18.6-20.9 % of all participants, while another 7.1 % of girls and 7.5 % of boys displayed symptoms of excessive fat deposition characterized by elevated body fat percentages. Even though the prevalence estimations varied depending on the standards used, the overall classification compliance reached 86-94 % with a Cohen's kappa coefficient ranging from 0.676 to 0.841. Conclusion: The prevalence of underweight, overweight and obesity among urban children is comparable to estimates for the general population. Of particular concern, in terms of health and proper physical development, is the problem of central obesity that affects one out of five children. The use of reference values representing body fat percentage seems justified as it allows for a more precise diagnosis of weight-related disorders, including the particularly threatening abdominal obesity. © 2015 Tomaszewski et al.

Date 2015

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Weighted and Class-Specific Maximum Mean Discrepancy for Unsupervised Domain Adaptation

Item Type Journal Article

Author H. Yan

Author Z. Li

Author Q. Wang

Author P. Li

Author Y. Xu

Author W. Zuo

Abstract Although maximum mean discrepancy (MMD) has achieved great success in unsupervised domain adaptation (UDA), most of existing UDA methods ignore the issue of class weight bias across domains, which is ubiquitous and evidently gives rise to the degradation of UDA performance. In this work, we propose two improved MMD metrics, i.e., weighted MMD (WMMD) and class-specific MMD (CMMD), to alleviate the adverse effect caused by the changes of class prior distributions between source and target domains. In WMMD, class-specific auxiliary weights are deployed to reweigh the source samples. In CMMD, we calculate the MMD for each class of source and target samples. Since the class labels of target samples are unknown for UDA problem, we present a classification expectation-maximization algorithm to estimate the pseudo-labels of target samples on the fly and update the model parameters using estimated labels. The proposed methods can be flexibly incorporated into deep convolutional neural networks to form WMMD and CMMD based domain adaptation networks, which we called WDAN and CDAN, respectively. By combining WMMD with CMMD, we present a CWMMD based domain adaptation network (CWDAN) to further improve classification performance. Experiments show that, both WMMD and CMMD benefit the classification accuracy, and our CWDAN can achieve compelling UDA performance in comparison with MMD and the state-of-the-art UDA methods. © 1999-2012 IEEE.

Date 2020

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Deep neural networks, Convolutional neural networks, Image segmentation, Image recognition, convolutional neural network, Domain adaptation, State of the art, Classification performance, Classification accuracy, Adverse effect, Maximum principle, Prior distribution, expectation-maximization algorithms, Expectation-maximization algorithms, Model parameters, unsupervised domain adaption

Notes:

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Weighted fuzzy similarity relations in case-based reasoning: A case study in classification

Item Type Conference Paper

Author S. Sandri

Author J. Mendonça

Author F. Martins-Bedê

Author R. Guimarães

Author O. Carvalho

Abstract This paper describes a fuzzy similarity relation approach to Case-Based Reasoning. Residuated implication operators are used to create a fuzzy resemblance relation between cases, modeling the CBR basic principle the more

similar the problem descriptions are, "the more similar the solution descriptions are" as a fuzzy gradual rule. We take the classification of Schistosomiasis prevalence estimation in a region of Brazil as case study, in order to investigate the effects in such a framework of weighting cases individually in classification tasks, considering a set of training strategies. © 2012 IEEE.

Date 2012

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Case-studies, Classification tasks, Prevalence estimation, Basic principles, Case based reasoning, Casebased reasonings (CBR), Fuzzy similarity relation, Gradual rules, Implication operators, Problem description, Residuated implications

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Weighted support vector machine for extremely imbalanced data

Item Type Journal Article

Author J. Mun

Author S. Bang

Author J. Kim

Abstract Based on an asymptotically optimal weighted support vector machine (SVM) that introduces label shift, a systematic procedure is derived for applying oversampling and weighted SVM to extremely imbalanced datasets with a cluster-structured positive class. This method formalizes three intuitions: (i) oversampling should reflect the structure of the positive class; (ii) weights should account for both the imbalance and oversampling ratios; (iii) synthetic samples should carry less weight than the original samples. The proposed method generates synthetic samples from the estimated positive class distribution using a Gaussian mixture model. To prevent

overfitting to excessive synthetic samples, different misclassification penalties are assigned to the original positive class, synthetic positive class, and negative class. The proposed method is numerically validated through simulations and an analysis of Republic of Korea Army artillery training data. © 2024 Elsevier B.V.

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Adversarial machine learning, Over sampling, Imbalanced dataset, Imbalanced data, Label shift, Asymptotically optimal, Bayes rule, Bayes' rule, Cost-sensitive learning, Gaussian mixture, Gaussian-mixtures, Imbalanced classification, Oversampling, Weighted support vector machine

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What Causes the Failure of Explicit to Implicit Discourse Relation Recognition?

Item Type Conference Paper
Author W. Liu
Author S. Wan
Author M. Strube
Editor Duh K.
Editor Gomez H.
Editor Bethard S.

Abstract We consider an unanswered question in the discourse processing community: why do relation classifiers trained on explicit examples (with connectives removed) perform poorly in real implicit scenarios? Prior work claimed this is due to linguistic dissimilarity between explicit and implicit examples but provided no empirical evidence. In this study, we show that one cause for such failure is a label shift after connectives are eliminated. Specifically, we find that the discourse relations expressed by some explicit instances will change when connectives disappear. Unlike previous work manually analyzing a few examples, we present empirical evidence at the corpus level to prove the existence of such shift. Then, we analyze why label shift occurs by considering factors such as the syntactic role played by connectives, ambiguity of connectives, and more. Finally, we investigate two strategies to mitigate the label shift: filtering out noisy data and joint learning with connectives. Experiments on PDTB 2.0, PDTB 3.0, and the GUM dataset demonstrate that classifiers trained with our strategies outperform strong baselines. ©2024 Association for Computational Linguistics.

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Conference Name Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL 2024

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What is Flagged in Uncertainty Quantification? Latent Density Models for Uncertainty Categorization

Item Type Conference Paper

Author H. Sun

Author B. van Breugel

Author J. Crabbé

Author N. Seedat

Author M. van der Schaar

Editor Oh A.

Editor Neumann T.

Editor Globerson A.

Editor Saenko K.

Editor Hardt M.

Editor Levine S.

Abstract Uncertainty Quantification (UQ) is essential for creating trustworthy machine learning models. Recent years have seen a steep rise in UQ methods that can flag suspicious examples, however, it is often unclear what exactly these methods identify. In this work, we propose a framework for categorizing uncertain examples flagged by UQ methods in classification tasks. We introduce the confusion density matrix-a kernel-based approximation of the misclassification density-and use this to categorize suspicious examples identified by a given uncertainty method into three classes: out-of-distribution (OOD) examples, boundary (Bnd) examples, and examples in regions of high in-distribution misclassification (IDM). Through extensive experiments, we show that our framework provides a new and distinct perspective for assessing differences between uncertainty quantification methods, thereby forming a valuable assessment benchmark. © 2023 Neural information processing systems foundation. All rights reserved.

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When Invariant Representation Learning Meets Label Shift: Insufficiency and Theoretical Insights

Item Type Journal Article**Author** Y.-W. Luo**Author** C.-X. Ren

Abstract As a crucial step toward real-world learning scenarios with changing environments, dataset shift theory and invariant representation learning algorithm have been extensively studied to relax the identical distribution assumption in classical learning setting. Among the different assumptions on the essential of shifting distributions, generalized label shift (GLS) is the latest developed one which shows great potential to deal with the complex factors within the shift. In this paper, we aim to explore the limitations of current dataset shift theory and algorithm, and further provide new insights by presenting a comprehensive understanding of GLS. From theoretical aspect, two informative generalization bounds are derived, and the GLS learner are proved to be sufficiently close to optimal target model from the Bayesian perspective. The main results show the insufficiency of invariant representation learning, and prove the sufficiency and necessity of GLS correction for generalization, which provide theoretical supports and innovations for exploring generalizable model under dataset shift. From methodological aspect, we provide a unified view of existing shift correction frameworks, and propose a kernel embedding-based correction algorithm (KECA) to minimize the generalization error and achieve successful knowledge transfer. Both theoretical results and extensive experiment evaluations demonstrate the sufficiency and necessity of GLS correction for addressing dataset shift and the superiority of proposed algorithm. © 1979-2012 IEEE.

Date 2024**DOI** 10.1109/TPAMI.2024.3417214**URL** <https://www.scopus.com/pages/publications/85196757769?origin=resultslist>**Archive** Scopus**Language** English**Volume** 46**Publisher** IEEE Computer Society**Pages** 9407-9422**Publication** IEEE Transactions on Pattern Analysis and Machine Intelligence**ISSN** 01628828 (ISSN)**Issue** 12**Journal Abbr** IEEE Trans Pattern Anal Mach Intell

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Optimisations, female, human, Learning systems, article, algorithm, learning, learning algorithm, Knowledge management, Domain adaptation, feature learning (machine learning), Representation learning, domain adaptation, Invariant representation, Dataset shifts, Information theory, Dataset shift, generalization error bound, Generalization error bounds, generalized label shift, Generalized label shift, invariant representation learning, Invariant representation learning, Real-world learning, Upper Bound

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When the Personal Becomes Political: Unpacking the Dynamics of Sexual Violence and Gender Justice Discourses Across Four Social Media Platforms

Item Type Journal Article**Author** J. Suk**Author** Y. Zhang**Author** Z. Yue**Author** R. Wang**Author** X. Dong**Author** D. Yang**Author** R. Lian

Abstract We propose a three-pronged framework to study discourses surrounding social media activism initiated by networked counterpublics: personalized expressions that share stories and support, demands for changes that address systematic problems, and contentions between various actors and perspectives. Situating our analysis in discourses related to sexual violence and gender justice activism on Twitter, Facebook, Instagram, and Reddit, we use supervised machine learning to quantify three discourses—networked acknowledgment, calls to action, and feminism contention—and apply time series analysis to model their interrelations. Results show that networked acknowledgment stimulated both calls to action and feminism contention and that calls to action predicted feminism contention across all platforms.

These discourses were more sensitive to real-world events on Twitter and Facebook, but more ephemeral and cyclical on Instagram and more persistent and coupled on Reddit. Our findings speak to the opportunities and challenges in social media activism and underscore cross-platform similarities and differences. © The Author(s) 2023.

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Which Uncertainty Quantification Method in Deep Learning Models for Knee Osteoarthritis Diagnosis?

Item Type Conference Paper

Author M.D. Fall

Author E.C. Faye

Abstract This article focuses on uncertainty quantification (UQ) in deep learning methods for knee osteoarthritis (OA) diagnosis. Knee OA is one of the leading causes of physical disability worldwide, with a high personal and socio-economic burden. However, the detection of knee OA relies on visual assessment, which is highly

subjective. Recently, deep learning has led to remarkable improvements in the detection of knee OA. However, the deep learning methods proposed in the current literature seem to be more concerned with model performance than with UQ. In this paper, we explore several UQ methods that aim to complement existing deep learning approaches. In particular, we consider conformal prediction, Monte Carlo dropout and evidential deep learning. The advantages and disadvantages of each method are highlighted in the context of the particular application. The aim is to provide a simple means of expressing uncertainty while allowing greater model transparency for clinical users. © 2025 IEEE.

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Deep learning, Classification (of information), Learning systems, Learning models, Medical imaging, Learning methods, Classification, Computer aided diagnosis, Socio-economics, Monte Carlo methods, Quantification methods, Uncertainty quantifications, Osteoarthritis, Uncertainty quantification, Economic burden, Knee osteoarthritis, Physical disability, X-ray

Notes:

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White Blood Cells Classification Model using Automatic Soft Computing

Item Type Conference Paper

Author P.P. Mahale

Author R.K. Deshmukh

Author P. Pise

Author S.R. Rangari

Abstract Accurate classification of blood cells, especially WBCs, is crucial for disease diagnosis and management. Traditional optical microscopic imaging provides limited information, but hyperspectral imaging (combining spatial and spectral data) offers richer insights. There are challenges and opportunities in white blood cell (WBC) classification using both Convolutional Neural Networks (CNN) including the following: (1) Important data, such as representative characteristics at various frequencies and orientations, may be lost due to the CNN model's usage of local receptive fields, shared weights, and subsamples. (2) To avoid over fitting, deep neural networks need a large pool of labeled training models. and (3) the higher computational overhead is yet another challenge. Therefore, these challenges motivate us the need of the proposed study. The scope of the proposed study is limited to the classification of WBC cells. Framework to classify and count the leukocytes in microscopic images and developed framework supports the pathologist and boosts the CAD system accuracy. The process of analyzing leukocytes in microscopic blood images is the major concern of this research. In this thesis, first we work on WBCs segmentation from microscopic blood images. © 2024 IEEE.

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Deep learning, Deep neural networks, Convolutional neural network, Convolutional neural networks, Image segmentation, Neural-networks, deep learning, Classification models, Neural network models, White blood cells, Blood images, Convolutional neural network, Convolutional Neural network, Leucocytes, WBC, White blood cell classification

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: P.P. Mahale; Kalinga University, Raipur, Department of Information Technology, AISSMS's Institute of Information Technology, Pune, India; email: pragati.mahale@aissmsioit.org; Conference name: 2024 Asia Pacific Conference on Innovation in Technology, APCIT 2024; Conference date: 26 July 2024 through 27 July 2024; Conference code: 202724

Whole slide image representation in bone marrow cytology

Item Type Journal Article

Author Y. Mu

Author H.R. Tizhoosh

Author T. Dehkharghanian

Author C.J.V. Campbell

Abstract One of the goals of AI-based computational pathology is to generate compact representations of whole slide images (WSIs) that capture the essential information needed for diagnosis. While such approaches have been applied to histopathology, few applications have been reported in cytology. Bone marrow aspirate cytology is the basis for key clinical decisions in hematology. However, visual inspection of aspirate specimens is a tedious and complex process subject to variation in interpretation, and hematopathology expertise is scarce. The ability to generate a compact representation of an aspirate specimen may form the basis for clinical decision-support tools in hematology. In this study, we leverage our previously published end-to-end AI-based system for counting and classifying cells from bone marrow aspirate WSIs, which enables the direct use of individual cells as inputs rather than WSI patches. We then construct bags of individual cell features from each WSI, and apply multiple instance learning to extract their vector representations. To evaluate the quality of our representations, we conducted WSI retrieval and classification tasks. Our results show that we achieved a mAP@10 of 0.58 ± 0.02 in WSI-level image retrieval, surpassing the random-retrieval baseline of 0.39 ± 0.1 . Furthermore, we predicted five diagnostic labels for individual aspirate WSIs with a weighted-average F1 score of 0.57 ± 0.03 using a k-nearest-neighbors (k-NN) model, outperforming guessing using empirical class prior probabilities (0.26 ± 0.02). We present the first example of exploring trainable mechanisms to generate compact, slide-level representations in bone marrow cytology with deep learning. This method has the potential to summarize complex semantic information in WSIs toward improved diagnostics in hematology, and may eventually support AI-assisted computational pathology approaches. © 2023 The Author(s)

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Decision support systems, human, Humans, Deep learning, Nearest neighbor search, article, Semantics, procedures, deep learning, controlled study, k nearest neighbor, learning, decision support system, probability, Digital pathologies, digital pathology, Digital pathology, Pathology, image processing, Image representation, Image Processing, Computer-Assisted, computer assisted diagnosis, Image Interpretation, Computer-Assisted, cytology, Whole slide images, Bone, Bone marrow, Cytology, bone marrow cell, Haematology, bone marrow, Bone Marrow, bone marrow aspiration, Bone Marrow Cells, Clinical decision, Compact representation, cytodiagnosis, Cytodiagnosis, hematology, Image representations, image retrieval, Image retrieval, Individual cells, Slide-level representation, Slide-level representations

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Whole-body tumor burden in PET/CT expert review

Item Type Journal Article

Author D.F. Santos

Author M.E. Takahashi

Author M. Camacho

Author M.C.L. de Lima

Author B.J. Amorim

Author E.M. Rohren

Author E. Etchebehere

Abstract Introduction: PET/CT whole-body tumor burden (WBTB), as a measure for overall burden of cancer, has been shown bear a strong correlation with prognosis. In the last decade, there has been significant progress in WBTB determination because of software advances and the increasing availability of positron-emitting radiopharmaceuticals. However, the determination of tumor burden with PET/CT is still a challenge especially in widespread metastatic disease. Methods: In this non-systematic review, we will discuss the current role of determination of WBTB in cancer such as non-small cell lung cancer, lymphoma, breast cancer, among others and with a variety of radiotracers. Furthermore, we will address imaging techniques and quantification methods available and challenges. Results: Many types of segmentation methods and different thresholds according to tumor types and radiotracers can be applied. These variations may show different WBTB results, but in general, despite variations, WBTB determination for staging purposes in lung cancer, breast cancer, lymphoma, melanoma, prostate cancer and neuroendocrine tumors have shown to bear a strong correlation with patient prognosis. Conclusion: PET/CT whole-body tumor burden has an invaluable potential to assess prognosis. The accelerated radiopharmaceutical development will provide molecules and

mechanisms to determine WBTB with advanced imaging qualification tools to further adjust radiotherapeutic doses in oncology. WBTB will most likely only become routinely accessible in clinical practice when fully automated programs become available and standardized. © 2022, The Author(s), under exclusive licence to Italian Association of Nuclear Medicine and Molecular Imaging.

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Notes:

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Why and how post-mastectomy edema should be quantified in patients with breast cancer

Item Type Journal Article

Author C. Galland

Author J.F. Auvert

Author A. Flahault

Author M. Vayssairat

Abstract In the present study, we advocate quantifying post-mastectomy edema in patients with breast cancer in millilitres and not in centimetre, contrarily to the usual routine practice. The water displacement and perimetric quantification methods exhibited both excellent reproducibility with an intraclass correlation coefficient > 0.99 and an accuracy coefficient $< 0.3\%$.

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adult, female, Female, human, Humans, male, article, reproducibility, Reproducibility of Results, quantitative analysis, controlled study, accuracy, classification, correlation coefficient, clinical article, hospitalization, clinical practice, Severity of Illness Index, water, Breast Neoplasms, breast tumor, breast cancer, Water, edema, anthropometry, postoperative complication, Edema, priority journal, reference value, bioassay, Anthropometry, Breast neoplasms, Endpoint Determination, Human, Lymphedema, mastectomy, Mastectomy, perimetry, Reference Values

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Why do we think? The dynamics of spontaneous thought reveal its functions

Item Type Journal Article

Author J.N. Mildner

Author D.I. Tamir

Abstract Spontaneous thought—mind wandering, daydreaming, and creative ideation—makes up most of everyday cognition. Is this idle thought, or does it serve an adaptive function? We test two hypotheses about the functions of spontaneous thought: First, spontaneous thought improves memory efficiency. Under this hypothesis, spontaneous thought should prioritize detailed, vivid episodic simulations. Second, spontaneous thought helps us achieve our goals. Under this hypothesis, spontaneous thought should prioritize content relevant to ongoing goal pursuits, or current concerns. We use natural language processing and machine learning to quantify the dynamics of thought in a large sample ($N = 3,359$) of think aloud data. Results suggest that spontaneous thought both supports memory optimization and keeps us focused on current concerns. © The Author(s) 2024.

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cognition, natural language processing, memory, goal pursuit, spontaneous thought

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Why Engineers Should Not Attempt to Quantify GSI

Item Type Journal Article

Author B. Yang

Author D. Elmo

Abstract In the past decade, there has been an increasing trend of digitalizing rock engineering processes. However, this process has not been accompanied by a critical analysis of the very same empirical methods that many complex numerical and digital methods

are founded upon. As engineers, we are taught to use and trust numbers. Indeed, we would not be able to define the factor of the safety of a structure without numbers. However, what happens when those numbers are nothing but numerical descriptions of qualitative assessments? In this paper we present a critical review of the many attempts presented in the literature to quantify GSI (geological strength index). To the authors' knowledge, this paper represents the first time that all the different GSI tables and quantification methods that have been proposed over the past two decades are collated and compared critically. In our critique, we argue against the paradigm whereby the quantification process adds the experience factor for inexperienced engineers. Furthermore, we discuss the limitations of the notion that GSI quantification methods could transform subjectivity into objectivity since the parameters under considerations are not quantitative measurements. Relying on empirically defined quantitative equivalences raises important questions, particularly when these quantitative equivalences are being used to define so-called accurate rock mass classification input for design purposes. © 2022 by the authors.

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digitalisation, geological strength index (GSI), quantified GSI, rock engineering

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Why is quantification an interesting learning problem?

Item Type Journal Article

Author P. González

Author J. Díez

Author N. Chawla

Author J.J. del Coz

Abstract There are real applications that do not demand to classify or to make predictions about individual objects, but to estimate some magnitude about a group of them. For instance, one of these cases happens in sentiment analysis and opinion mining. Some applications require to classify opinions as positives or negatives, but there are also others, even more useful sometimes, that just need an estimation of which is the proportion of each class during a concrete period of time. “How many tweets about our new product were positive yesterday?” Practitioners should apply quantification algorithms to tackle this kind of problems, instead of just using off-the-shelf classification methods, because classifiers are suboptimal in the context of quantification tasks. Unfortunately, quantification learning is still relatively an under explored area in machine learning. The goal of this paper is to show that quantification learning is an interesting open problem. To support its benefits, we shall show an application to analyze Twitter comments in which even the most simple quantification methods outperform classification approaches. © 2016, Springer-Verlag Berlin Heidelberg.

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Wilson disease in the USA: epidemiology and real-world patient characteristics based on a retrospective observational health claims study

Item Type Journal Article

Author S. Fang

Author C. Strader

Author H. Costantino

Author K.H. Weiss

Author P. Hedera

Abstract Objectives To describe the epidemiology, patient characteristics and comorbidities in patients with Wilson disease (WD) in the USA. Design Retrospective, population-based study. Setting The study used the US Komodo claims database containing records regarding medical claims for over 120 million individuals. Participants Patients with WD were identified via ICD-10 (10th revision of the International Classification of Diseases) code during the study period 2016-2019 and no age restriction was applied. A further stratification by disease subtype ('hepatic', 'neurologic' and 'psychiatric') was performed. Main outcome measures WD prevalence was reported by age, sex and US census regions/divisions. Adjusted prevalence was calculated using age-specific prevalence standardised to the USA (2010 US census) and to the world (WHO 2000-2025) to enable comparisons across countries, using direct standardisation of prevalence estimates by age group. Results Overall, 2115 patients with WD were identified during the study period. Among them, 56.8% had hepatic symptoms, 57.0% neurologic symptoms and 47.4% psychiatric symptoms. The most frequent manifestations in hepatic patients were liver signs and symptoms (90.8%), in neurologic patients cognitive defects (50.7%) and in psychiatric patients mood disorders (86.4%). The mean age in the overall cohort was 39.9 years. Prevalence estimation was based on 1481 patients with WD between 2017 and 2019. The 2017-2019 crude period prevalence was 21.2 patients per million (95% CI: 20.1 to 22.3), with similar prevalence observed for both sexes. Conclusions This study provides important real-world data on the diagnosed prevalence of WD in the USA and revealed the comorbidities associated with various disease subtypes, thereby providing a comprehensive basis for guiding physicians and policy makers in the management of this chronic disease. © Author(s) (or their employer(s)) 2024. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

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Correspondence Address: S. Fang; Alexion, AstraZeneca Rare Disease, Boston, United States; email: shona.fang@alexion.com

WOT-Class: Weakly Supervised Open-world Text Classification

Item Type Conference Paper

Author T. Wang

Author Z. Wang

Author W. Liu

Author J. Shang

Abstract State-of-the-art weakly supervised text classification methods, while significantly reduced the required human supervision, still requires the supervision to cover all the classes of interest. This is never easy to meet in practice when humans explore new, large corpora without complete pictures. In this paper, we work on a novel yet important problem of weakly supervised open-world text classification, where supervision is only needed for a few examples from a few known classes and the machine should handle both known and unknown classes in test time. General open-world classification has been studied mostly using image classification; however, existing methods typically assume the availability of sufficient known-class supervision and strong unknown-class prior knowledge (e.g., the number and/or data distribution). We propose a novel framework WOT-Class that lifts those strong assumptions. Specifically, it follows an iterative process of (a) clustering text to new classes, (b) mining and ranking indicative words for each class, and (c) merging redundant classes by using the overlapped indicative words as a bridge. Extensive experiments on 7 popular text

classification datasets demonstrate that WOT-Class outperforms strong baselines consistently with a large margin, attaining 23.33% greater average absolute macro-F1 over existing approaches across all datasets. Such competent accuracy illuminates the practical potential of further reducing human effort for text classification. © 2023 Copyright held by the owner/author(s).

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Classification (of information), Large dataset, Human supervision, Iterative methods, Text classification, Text processing, text classification, State of the art, C (programming language), Weak supervision, weak supervision, Text classification methods, Test time, Open world, Unknown class, open-world learning, Open-world learning, Large corpora

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WOUND MANAGEMENT, HEALING, AND EARLY PROSTHETIC REHABILITATION: PART 3 - A SCOPING REVIEW OF CHEMICAL BIOMARKERS

Item Type Journal Article

Author H. Williams-Reid

Author A. Johannesson

Author A.A. Buis

Abstract **BACKGROUND:** Poor post-amputation healing delays prosthetic fitting, adversely affecting mortality, quality of life, and cardiovascular health. Current residual limb assessments are subjective and lack standardized guidelines, emphasizing the need for objective biomarkers to improve healing and prosthesis readiness assessments. **OBJECTIVE(S):** This review aimed to identify predictive, diagnostic, and indicative chemical biomarkers of healing of the tissues and structures found in the residual limbs of adults with amputation. **METHODOLOGY:** This scoping review followed Joanna Briggs Institute (JBI) and PRISMA-ScR guidelines. Searches using the terms “biomarkers,” “wound healing,” and “amputation” were performed across Web of Science, Ovid Medline, Ovid Embase, Scopus, Cochrane, PubMed, and CINAHL databases. Inclusion criteria were: 1) References to chemical biomarkers and healing; 2) Residuum tissue healing; 3) Repeatable methodology with ethical approval. Included articles were evaluated for quality of evidence (QualSyst tool) and level of evidence (JBI classification). Sources were categorized by study (e.g., randomized controlled trial or bench research), wound (diabetic, amputation, other), and model (human, murine, other) type. Chemical biomarkers repeated across study categories, and quantification methods were reported on. **FINDINGS:** From 3,306 titles and abstracts screened, 646 underwent full-text review, and 203 met the criteria for data extraction, with 76% classified as strong quality. 38 chemical biomarkers were identified across 4 to 50 sources, with interleukins (predictive, indicative, and diagnostic) and HbA1c (predictive) most prevalent, appearing in 50 and 48 sources, respectively. Other biomarkers included predictive blood markers (e.g., cholesterol, white blood cell counts), indicative growth factors, bacteria presence (predictive), proteins (predictive, indicative, and diagnostic, e.g., matrix metalloproteinases), and cellular markers (indicative and diagnostic, e.g., Ki-67, alpha-smooth muscle actin [α -SMA]). **CONCLUSION:** Predictive biomarkers identify comorbidities that may hinder healing, aiding in pre-amputation risk assessment for poor recovery. Indicative biomarkers monitor key biological healing processes, such as angiogenesis (the formation of new blood vessels), wound contraction, and inflammation. Diagnostic biomarkers provide direct insights into tissue composition and cellular-level healing. Integrating these biomarkers into post-amputation assessments enables continuous monitoring of the healing process while accounting for comorbidities, enhancing the objectivity of post-surgical healing management and ensuring more effective, personalized rehabilitation strategies. © 2025 Canadian Online Publication Group. All rights reserved.

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Wrapper positive Bayesian network classifiers

Item Type Journal Article

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Abstract In the information retrieval framework, there are problems where the goal is to recover objects of a particular class from big sets of unlabelled objects. In some of these problems, only examples from the class we want to recover are available. For such problems, the machine learning community has developed algorithms that are able to learn binary classifiers in the absence of negative examples. Among them, we can find the positive Bayesian network classifiers, algorithms that induce Bayesian network classifiers from positive and unlabelled examples. The main drawback of these algorithms is that they require some previous knowledge about the a priori probability distribution of the class. In this paper, we propose a wrapper approach to tackle the learning when no such information is available, setting this probability at the optimal value in terms of the recovery of positive examples. The evaluation of classifiers in positive unlabelled learning problems is a non-trivial question. We have also worked on this problem, and we have proposed a new guiding metric to be used in the search for the optimal a priori probability of the positive class that we have called the pseudo F. We have empirically tested the proposed metric and the wrapper classifiers on both synthetic and real-life datasets. The results obtained in this empirical comparison show that the wrapper Bayesian network classifiers provide competitive results, particularly when the actual a priori probability of the positive class is high. © 2012 Springer-Verlag London Limited.

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Yield gap and resource utilization efficiency of three major food crops in the world – A review

Item Type Journal Article

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Abstract Yield gap analysis could provide management suggestions to increase crop yields, while the understandings of resource utilization efficiency could help judge the rationality of the management. Based on more than 110 published papers and data from Food and Agriculture Organization (FAO, www.fao.org/faostat) and the Global Yield Gap and Water Productivity Atlas (www.yieldgap.org), this study summarized the concept, quantitative method of yield gap, yield-limiting factors, and resource utilization efficiency of the three major food crops (wheat, maize and rice). Currently, global potential yields of wheat, maize and rice were 7.7, 10.4 and 8.5 t ha⁻¹,

respectively. However, actual yields of wheat, maize and rice were just 4.1, 5.5 and 4.0 t ha⁻¹, respectively. Climate, nutrients, moisture, crop varieties, planting dates, and socioeconomic conditions are the most mentioned yield-limiting factors. In terms of resource utilization, nitrogen utilization, water utilization, and radiation utilization efficiencies are still not optimal, and this review has summarized the main improvement measures. The current research focuses on quantitative potential yield and yield gap, with a rough explanation of yield-limiting factors. Subsequent research should use remote sensing data to improve the accuracy of the regional scale and use machine learning to quantify the role of yield-limiting factors in yield gaps and the impact of change crop management on resource utilization efficiency, so as to propose reasonable and effective measures to close yield gaps. © 2021 CAAS. Publishing services by Elsevier B.V

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Zono-Conformal Prediction: Zonotope-Based Uncertainty Quantification for Regression and Classification Tasks

Item Type Journal Article

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Abstract Conformal prediction is a popular uncertainty quantification method that augments a base predictor to return sets of predictions with statistically valid coverage guarantees. However, current methods are often computationally expensive and data-intensive, as they require constructing an uncertainty model before calibration. Moreover, existing approaches typically represent the prediction sets with intervals, which limits their ability to capture dependencies in multi-dimensional outputs. We address these limitations by introducing zono-conformal prediction, a novel approach inspired by interval predictor models and reachset-conformant identification that constructs prediction zonotopes with assured coverage. By placing zonotopic uncertainty sets directly into the model of the base predictor, zono-conformal predictors can be identified via a single, data-efficient linear program. While we can apply zono-conformal prediction to arbitrary nonlinear base predictors, we focus on feed-forward neural networks in this work. Aside from regression tasks, we also construct optimal zono-conformal predictors in classification settings where the output of an uncertain predictor is a set of possible classes. We provide probabilistic coverage guarantees and present methods for detecting outliers in the identification data. In extensive numerical experiments, we show that zono-conformal predictors are less conservative than interval predictor models and standard conformal prediction methods, while achieving a similar coverage over the test data. © 2025 Laura Lützow, Michael Eichelbeck, Mykel J. Kochenderfer, Matthias Althoff.

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